NISTIR XXXX Draft

Face Recognition Technology Evaluation (FRTE)

Part 1: Verification

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This publication is available free of charge from:
https://www.nist.gov/programs-projects/face-recognition-vendor-test-frvt-ongoing

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Howard Lutnick, Secretary

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DISCLAIMER

Specific hardware and software products identified in this report were used in order to perform the evaluations described in this document. In no case does identification of any commercial product, trade name, or vendor, imply recommendation or endorsement by the National Institute of Standards and Technology, nor does it imply that the products and equipment identified are necessarily the best available for the purpose.

INSTITUTIONAL REVIEW BOARD

The National Institute of Standards and Technology's Research Protections Office reviewed the protocol for this project and determined it is not human subjects research as defined in Department of Commerce Regulations, 15 CFR 27, also known as the Common Rule for the Protection of Human Subjects (45 CFR 46, Subpart A).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

FRTE STATUS

This report is a draft NIST Interagency Report, and is open for comment. It is the thirty seventh edition of the report since the first was published in June 2017. Prior editions of this report are maintained on the FRTE website, and may contain useful information about older algorithms and datasets no longer used in FRTE.

FRTE remains open: All tracks of the FRTE are open to new algorithm submissions.

2026-10-06 changes since 2026-09-01:

- We have added results for first algorithms from eight developers: Akor IT Services LLC, IDBIO, Nmetics Private Ltd, Northern Point Technology Pvt Ltd, Toss, Universidade Federal de Goiás, Universal Ubiquitous AI Co Ltd, and Verismart
- We have added results for new algorithms from thirteen returning developers: Akiyama S.A, Cognitec Systems GmbH, Onfido, FacePhi, Innov8tif Solutions, Lishui Shuyan Huirong Technology Co, Metsakuur, Neurotechnology, Paravision, PNPSecure, STCON LLC, Toshiba, and Viettel Artificial Intelligence and Data Services Center

2026-09-01 changes since 2026-07-31:

- We have added results for first algorithms from eight developers: Intpixel Labs Private Ltd, Integrated Software and Technologies Ltd, Norwegian University of Science and Technology, OCR Studio L.L.C.-FZ, OORU Digital Private Ltd, Security Pitch Co Ltd, Softtech Yazilim Teknolojileri Arastirma Gelistirme ve, and UniverseAI.
- We have added results for new algorithms from thirteen returning developers: Advance.AI, InfoCert, Innovatrics, Iris ID Systems, Miaxis Biometrics, Regula Forensics, ROC, Sensetime Group, Sum and Substance Ltd, Verigram, Viante.AI, Vision-Box, and Vsoft.

2026-07-31 changes since 2026-06-15:

- We have added results for first algorithms from two developers: Kaizen Secure Voiz, and Veertec SAL (Offshore).
- We have added results for new algorithms from fourteen returning developers: CAKE by VPBank, Compal Electronics, Dermalog, Facehawk Ltd, FRP LLC, Intozi Tech Pvt Ltd, IQSEC, Nominder, RealSense, Sparktek International Pte Ltd, STCON LLC, TAHAKOM, Technology Control Company, Techainer, and Veridium.

2026-06-15 changes since 2026-05-08:

- We have added results for first algorithms from five developers: AIPRACTIX, Bynn Intelligence Inc, ScanalitiX, Stellar Technologies, and TrueSight Laboratories Tech LLC.
- We have added results for new algorithms from nine returning developers: Cyberlink Corp, Enqura Ltd, Hyundai Motor Company, ID3 Technology, IDENTY, Matrix Comsec Pvt Ltd, Sightropic, Viettel Artificial Intelligence and Data Services Center, and Visica AI Solutions.

2026-05-08 changes since 2026-03-30:

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ We have added results for first algorithms from four developers: Ethiopian Labor Market Information System, Face Int, Lishui Shuyan Huirong Technology Co, and YuanLi Technology.
- ▷ We have added results for new algorithms from twenty-one returning developers: Advance.AI, BioID GmbH, Tripleize, Dralmey Solutions, FaceLocate, HyperVerge Inc, InfoCert, ioNetworks Inc, Innov8tif Solutions, Kneron Inc, Mobai, Onfido, PAPIL11 S.R.O., PT Technology Biometric Indonesia, Shanghai Jiao Tong University, TNG Digital Sdn Bhd, TOPPAN Security Co, Unissey, UZINFOCOM, Viante.AI, and Vsoft.

2026-03-30 changes since 2026-03-05:

- ▷ We have added results for first algorithms from three developers: Ice House, Shufti, and Smarg Technologies Pvt Ltd
- ▷ We have added results for new algorithms from four returning developers: AI Implementation Group, Daon, PNPSecure, and Sensetime Group

2026-03-05 changes since 2026-03-02:

- ▷ We have updated the missing results for VisionBox.

2026-03-02 changes since 2026-01-22:

- ▷ We have added results for first algorithms from two developers: FPT Telecom JSC, and Iris ID Systems
- ▷ We have added results for new algorithms from six returning developers: Biometrik Systems Private Limited, GTEL OTS JSC, ID3 Technology, NextMeta Labs, Nominder, and ROC.

2025-01-22 changes since 2025-12-19:

- ▷ We have added results for first algorithms from five developers: CAKE by VPBank, RealSense, Sum and Substance Ltd, Transmit Security Ltd, and Vaidio.
- ▷ We have added results for new algorithms from eleven returning developers: Beijing DeepSense Technologies, Compal Electronics, Candour Biometrics, Clearview AI Inc, IE SARSENOV, Miaxis Biometrics, Onfido, Paravision, Shanghai Jiao Tong University, STCON LLC, and Toshiba.

2025-12-19 changes since 2025-09-30:

- ▷ We have added results for first algorithms from one developers: Sparktek International Pte Ltd.
- ▷ We have added results for new algorithms from 13 returning developers: BioID GmbH, Chunghwa Telecom, Dermalog, Hyundai Motor Company, IDENTY, Innovatrics, InterLink labs, Matrix Comsec Pvt Ltd, Sightropic, Trust Stamp, Viettel Artificial Intelligence and Data Services Center, Viettel High Technology, and Vsoft.

2025-09-30 changes since 2025-08-13:

- ▷ We have added results for first algorithms from three developers: Biometrik Systems Private Limited, OOTO AI, and Toss.
- ▷ We have added results for new algorithms from four returning developers: Jaak IT, Online Mobile Services JSC, Regula Forensics, and Unissey

2025-08-13 changes since 2025-07-15:

- ▷ We have added results for first algorithms from four developers: BTS Digital, IE SARSENOV, UZINFOCOM, and Vsoft.
- ▷ We have added results for new algorithms from seventeen returning developers: 2060.io, Beijing DeepSense Technologies, ID R&D Inc, Innovatrics, IntelliVIX, Metsakuur, Mobai, NextMeta Labs, Neurotechnology, Onfido, QazSmartVision.AI, ROC, STCON LLC, Toshiba, VinBigData, Vision-Box, and Visica AI Solutions.

2025-07-15 changes since 2025-05-28:

- ▷ We have added results for first algorithms from three developers: Matrix Comsec Pvt Ltd, PNPSecure, and Urielsoft Co. Ltd.
- ▷ We have added results for new algorithms from eleven returning developers: Daon, Dermalog, Hyundai Motor Company, Incode Technologies Inc, Intel Research Group, Kloudspot Inc, Omnigarde Ltd, Paravision, Universidade de Coimbra, Viante.AI, and Vietnam Posts and Telecommunications Group.

2025-05-28 changes since 2025-04-25:

- ▷ We have added results for new algorithms from seven returning developers: BioID GmbH, Cubox, DiDi ChuXing Technology, InterLink labs, IDENTY, PAPIL11 S.R.O., and Trust Stamp.

2025-04-25 changes since 2025-03-18:

- ▷ We have added results for first algorithms from seven developers: ArVİs Teknoloji San.Tic.A.Ş., Compal Electronics, FCI Vietnam, GTEL OTS JSC, Shenzhen Polytechnic University, Sightropic, and TNG Digital Sdn Bhd.
- ▷ We have added results for new algorithms from eleven returning developers: Awidit Systems, Cyberextruder, FRP LLC, Kasikorn Labs, NCS Pte Ltd, Nominder, Shanghai Jiao Tong University, STCON LLC, Tech5 SA, Veridas Digital Authentication Solutions S.L., and Viettel High Technology.

2025-03-18 changes since 2025-03-05:

- ▷ We have added results for first algorithms from one developers: FaiceTech Ltd.
- ▷ We have added results for new algorithms from ten returning developers: Beijing DeepSense Technologies, BioID Technologies SA, EI Networks Private Ltd, Hangzhuo AIlu Network Information Technology, Innovatrics, Private Identity LLC, Samsung S1 Corp, TigerIT Bangladesh Ltd, Via Technologies Inc, and Viettel Cyberspace Center.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

2025-03-05 changes since 2025-01-30:

- ▷ We have added results for first algorithms from three developers: 2060.io, MB Bank, and Visica AI Solutions
- ▷ We have added results for new algorithms from twelve returning developers: AFR Engine, AI Implementation Group, Authme, Hyundai Motor Company, FacePhi, Inspur (Beijing) Electronic Information Industry Co, LightVision Inc, QazSmartVision.AI, ROC, Toshiba, Techainer, and VinBigData.

2025-01-30 changes since 2024-12-18:

- ▷ We have added results for first algorithms from four developers: Innov8tif Solutions, Maverick Research, PT Technology Biometric Indonesia, and TTI Group.
- ▷ We have added results for new algorithms from ten returning developers: Coretech Knowledge Inc, Intel Research Group, IQSEC, i2v Systems, PAPIL11 S.R.O., Paravision, Taiwan Mobile, Touchless ID, useB, and Viante.AI.

2024-12-18 changes since 2024-11-08:

- ▷ We have added results for first algorithms from six developers: BioID GmbH, InterLink labs, KYB-AI Ltd, Keyless Technologies, Pyramid Cyber Security + Forensic (P),and Syscake S.r.l..
- ▷ We have added results for new algorithms from sixteen returning developers: Alchera Inc, Awidit Systems, Cubox, DeCloak Intellegences, Beijing DeepSense Technologies, Innovatrics, Kasikorn Labs, Mitek Systems, Mukh Technologies, NCS Pte Ltd, NextMeta Labs, Nominder, PT Autentika Digital Indonesia, Sensetime Group, Shanghai Jiao Tong University, and STCON LLC.

2024-11-08 changes since 2024-10-07:

- ▷ We have added results for first algorithms from two developers: Hyderabad-Centre for Development of Advanced Computing, and TAHAKOM.
- ▷ We have added results for new algorithms from ten returning developers: Candour Biometrics, Cyberextruder, DICIO, First Credit Bureau Kazakhstan, Metsakuur, Mobbeel Solutions, Samsung S1 Corp, Vietnam Payment Solution, VinBigData, and Yuan High-Tech Development.

2024-10-07 changes since 2024-09-16:

- ▷ We have added results for first algorithms from three developers: Enqura, Kloudspot, and Penacomp Systemy Informatyczne.
- ▷ We have added results for new algorithms from fourteen returning developers: Chunghwa Telecom, Cyberlink, Dactionable Technologies, Daon, FPT Information System, Hangzhuo AIlu Network Information Technology, Kakao Bank, Panasonic R+D Center Singapore, Paravision, Private Identity, QazSmartVision.AI, Toshiba, Vietnam Posts and Telecommunications Group, and UNICC-Solution Architecture Section.

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2024-09-16 changes since 2024-08-09:

- ▷ We have added results for first algorithms from one developer: Taiwan Mobile.
- ▷ We have added results for new algorithms from eleven returning developers: Cognitec Systems GmbH, Coretech Knowledge Inc, FaceOnLive Inc, Idemia, Intel Research Group, Luxand Inc, Ovision, PAPIL11 S.R.O., Unissey, useB, and Vivotek.

2024-08-09 changes since 2024-07-03:

- ▷ We have added results for first algorithms from seven developers: Affinsys AI Ltd, AI Implementation Group, Celloscope Limited, Hyundai Motor Company, ID R&D Inc, LightVision Inc and NextMeta Labs.
- ▷ We have added results for new algorithms from eleven returning developers: AFR Engine, Antheus Technologia, CTBC Bank, Innovatrics, Mitek Systems, Neurotechnology, Nominder, Seventh Sense Artificial Intelligence, Universidade de Coimbra, University of Surrey-CVSSP, Viettel Cyberspace Center, Vision-Box and Yoti.

2024-07-03 changes since 2024-05-30:

- ▷ We have include all algorithms evaluated since the ongoing FRTE evaluation started in 2017. Previously we listed only results for each developer's two most recent implementations. The main results page has been revised with a "by developer" tab which lists only the most accurate algorithm from each developer, where accuracy is defined by the lowest FNMR value for the visa-border dataset.
- ▷ We have added results for first algorithms from two developers: Taiwan Computer Vision and Telexper International.
- ▷ We have added results for new algorithms from fifteen returning developers: Alchera Inc, Beijing DeepSense Technologies, Cyberlink Corp, Dermalog, FacePhi, Facia.ai, FRP LLC, NCS Pte Ltd, Omnigarde Ltd, Online Mobile Services JSC, Qnap Security, ROC, STCON LLC, Videmo Intelligente Videoanalyse, and Viettel Group.

2024-05-30 changes since 2024-04-17:

- ▷ We have added results for first algorithms from three developers: ALTTEKGlobal, Fourthline, and IQSEC
- ▷ We have added results for new algorithms from thirteen returning developers: Candour Biometrics, Chunghwa Telecom, Clearview AI Inc, Cognitec Systems GmbH, Incode Technologies Inc, ioNetworks Inc, HyperVerge Inc, Mobbeel Solutions, Panasonic R+D Center Singapore, QazSmartVision.AI, Vietnam Posts and Telecommunications Group, VinBigData and Yuan High-Tech Development
- ▷ We have retired results for eleven algorithms per our policy to only list results for eleven algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2024-04-17 changes since 2024-03-26:

FNMR(T) "False non-match rate"
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- ▷ We have added results for first algorithms from four developers: Ovision, Vietnam National Cyber Security Technology Corporation, Vivotek, and Vocalize.
- ▷ We have added results for new algorithms from four returning developers: Intel Research Group, Kakao Bank, Luxand, and Turkcell Technology.
- ▷ We have retired results for nine algorithms per our policy to only list results for four algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2024-03-26 changes since 2024-02-21:

- ▷ We have added results for first algorithms from six developers: FaceLocate, FPT Education, FPT Information System, Kogniza Technology, i2v Systems, and Seamfix.
- ▷ We have added results for new algorithms from eleven returning developers: AFR Engine, Aware, Beijing Hisign Technology, Cu-Face, Dactionable Technologies, First Credit Bureau Kazakhstan, Innovatrics, Nominder, Securif AI, Veridium, and Viettel Cyberspace Center.
- ▷ We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2024-02-21 changes since 2024-01-22:

- ▷ We discontinued the visa-visa benchmark in FRTE since Feburary 14th 2024. For highly performing algorithms, the only remaining false negative error in that set were in babies and in a few greyscale poor quality half-tone scans of paper photos.
- ▷ We have added results for first algorithms from four developers: Online Mobile Services JSC, Openedge Technologies, TNI Technoology, and Yoti.
- ▷ We have added results for new algorithms from nine returning developers: CMC Institute of Science and Technology, City and County of Honolulu, Dermalog, Euronovate SA, Maxvision Technology, Megvii/Face++, Neurotechnology, Toshiba, useB, and Universidade de Coimbra.
- ▷ We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2024-01-22 changes since 2023-12-15:

- ▷ We have added results for first algorithms from six developer: GPS Vietnam Trading, iCOMM Media & Tech, PAPIL11 S.R.O., QazSmartVision.AI, Sparsh CCTV, and VNIS Joint Stock.
- ▷ We have added results for new algorithms from eight returning developers: Cyberlink Corp, Intellivision, Hangzhuo AIlu Network Information Technology, Lebentech Biometrics, Momentum Digital, Regular Biometrics Solutions, ROC, and Via Technologies Inc.
- ▷ We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-12-15 changes since 2023-11-21:

- ▷ We have added results for first algorithms from five developer: CMC University, Element System Solutions Company, Fraud.com, Techainer, and Vietnam Payment Solutions.

- ▷ We have added results for new algorithms from thirteen returning developers: Accurascan, Adera Global, Alchera, Aware, Clearview AI, Cognitec Systems, Incode Technologies Inc, Inspur (Beijing) Electronic Information, IntelliVIX, ioNetworks, NHN Corp, PT Autentika Digital Indonesia, and Qnap Security.
- ▷ We have retired results for twelve algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-11-21 changes since 2023-10-27:

- ▷ We have added results for new algorithms from ten returning developers: Coretech Knowledge Inc, Beijing DeepSense Technologies, General Interface Solutions Holding Ltd, Glory, Kakao Brain, Nominder, Smarvist Teknoloji, STCON LLC, Suprema AI Inc, and Yuan High-Tech Development
- ▷ We have retired results for eight algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-10-27 changes since 2023-09-29:

- ▷ We have added results for first algorithms from two developer: Authme and Dactionable Technologies.
- ▷ We have added results for new algorithms from twelve returning developers: Intel Research Group, Guangzhou Pixel Solutions, Kasikorn Labs, Kakao Bank, Luxand, Megvii/Face++, Panasonic R+D Center Singapore, Private Identity LLC, Recognito, Seventh Sense Artificial Intelligence, UNICC-Solution Architecture Section, and Universidade de Coimbra.
- ▷ We have retired results for eight algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-09-29 changes since 2023-09-09:

- ▷ We have added results for first algorithms from one developer: Intozi Tech.
- ▷ We have added results for new algorithms from seven returning developers: Euronovate, Maxvision Technology, Miaxis Biometrics, Mukh Technologies, Toshiba, University of Surrey-CVSSP, and useB.
- ▷ We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-09-09 changes since 2023-08-16:

- ▷ We have renamed and split FRVT into two sets of technology evaluations: The Face Recognition Technology Evaluation (FRTE) is the umbrella for the ongoing 1:1, 1:N and twins disambiguation tracks; the Face Analysis Technology Evaluation (FATE) holds the morph and attacked detection tracks, the quality assessment tracks, and the new age estimation tracks.
- ▷ We have added results for first algorithms from two developers: Facehawk and Regular Biometrics Solutions.
- ▷ We have added results for new algorithms from eight returning developers: CMC Institute of Science and Technology, Canon, Cu-Face, EI Networks, ICM Airport Technics, Securif AI, Securifai, Vietnam Posts and Telecommunications Group, and Vision Intelligence Center of Meituan

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- ▷ We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-08-16 changes since 2023-07-21:

- ▷ We have added results for first algorithms from one developers: Facia.ai.
- ▷ We have added results for new algorithms from fifteen returning developers: Accurascan, Alchera, FOO, Hangzhuo AIlu Network Information Technology, HyperVerge, ioNetworks, Innovatrics, Nominder, Rank One Computing, Neurotechnology, Tech5, Tripleize, Unissey, UX Labs, and Verigram.
- ▷ We have retired results for ten algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-07-21 changes since 2023-06-16:

- ▷ We have added results for first algorithms from five developers: DeCloak Intellegences, Innominds Software SEZ India, Trust Stamp, Vcortex Labs, and Viettel Cyberspace Center
- ▷ We have added results for new algorithms from fourteen returning developers: AFR Engine, Alice Biometrics, Euronovate SA, IMDS Software, Idemia, IntelliVIX, Intellivision, NHN, Pangiam, STCON, Samtech InfoNet, Veridium, VinBigData, and Vision-Box
- ▷ We have retired results for eleven algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- ▷ We have stopped using wild images in FRVT since May 1st 2023. All results for that set have been removed from the website, and will be removed from future PDF reports.

2023-06-16 changes since 2023-04-20:

- ▷ We have added results for first algorithms from seven developers: EI Networks Private Ltd, General Interface Solutions Holding Ltd, Recognito, Serendipity Ltd, ST Engineering, and useB.
- ▷ We have added results for new algorithms from eighteen returning developers: Advance.AI, AYF Technology, Cyberlink Corp, Incode Technologies Inc, Intel Research Group, Intellibrain Technological Projects, Maxvision Technology, Megvii/Face++, Omnigarde Ltd, Paravision, Qnap Security, Seventh Sense Artificial Intelligence, Shanghai Jiao Tong University, Veridas Digital Authentication Solutions S.L., Universidade de Coimbra, Yuan High-Tech Development, UNICC-Solution Architecture Section, and Verihubs,verihubs-inteligensia.
- ▷ We have retired results for 15 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-04-20 changes since 2023-04-04:

- ▷ We have added results for first algorithms from one developers: IDENTY.
- ▷ We have added results for new algorithms from three returning developers: Metsakuur, Autentika Digital Indonesia, and Verigram.
- ▷ FRVT will re-open 2023-05-01.
- ▷ We have retired results for 3 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-04-04 changes since 2023-03-09:

- ▷ We have added results for first algorithms from six developers: Aratek Biometrics City and County of Honolulu, FOO, Intelligent Control Technology NCS, and Swsam Solutions.
- ▷ We have added results for new algorithms from nine returning developers: Cubox, Glory, InsightFace AI, Miaxis Biometrics, Mukh Technologies, Turkcell Technology, Samsung S1, Via Technologies, and Vision Intelligence Center of Meituan
- ▷ We have retired results for 7 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-03-09 changes since 2023-02-01:

- ▷ We have added results for first algorithms from eight developers: Biometric LLC (biometric.vision) KZ, Candour Biometrics, Fast Enterprises, KakaoBank, Mitek Systems, Nominder, Private Identity, and UNICC-Solution Architecture Section.
- ▷ We have added results for new algorithms from 22 returning developers: AFR Engine, Biocube Matrics, CMC Institute of Science and Technology, Cloudwalk - Moontime Smart Technology, Cyberlink Corp, Beijing DeepSense Technologies, First Credit Bureau Kazakhstan, Enface, Hangzhuo AIlu Network Information Technology, Herta Security, IMDS Software, Inspur (Beijing) Electronic Information Industry Co, Intellivision, MicroFocus, Neurotechnology, Pangiam, NSENSE Corp, STCON LLC, Touchless ID, University of Surrey-CVSSP, Vision-Box, and YooniK.
- ▷ We have retired results for 14 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- ▷ We have introduced new set of non-frontal portrait to border comparisons. The new images are described in section 2.3 and their use in seciton 3.2.

2023-02-01 changes since 2022-12-15:

- ▷ We have added results for first algorithms from four developers: CU-Face, Korea ID, Onfido, and TrueID-VNG.
- ▷ We have added results for new algorithms from 21 returning developers: Alchera, Armatura, Cogent-Thales, Dermalog, Didi ChuXing Global Face, Gorilla, Hyperverge, Innovatrics, Intel Research, IntelliVIX, Intema-LGL, Kasikorn Labs, Paravision, Rank One Computing, Sensetime Group, Suprema AI, Tech5, Unissey, U. Coimbra Visteam, Vixvizion (Imagus), and Yuan High-Tech Development.
- ▷ We have retired results for 20 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- ▷ We have introduced new set of non-frontal portrait to border comparisons. The new images are described in section 2.3 and their use in seciton 3.2.

2022-12-15 changes since 2022-11-06:

- ▷ We have added results for first algorithms from four developers: Miaxis Biometrics, PT Autentika Digital Indonesia, PT Qlue Performa Indonesia, and STCON.

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FMR(T) "False match rate"

- ▷ We have added results for new algorithms from 14 returning developers: Adera Global, Aiseemu Technology, Chunghwa Telecom,chtface, FRP, Griaule, Line Corporation, Maxvision Technology, Mukh Technologies, Papilon Savunma, Qnap Security, Realnetworks, Securif AI, SQIsoft, and Veridium.
- ▷ We have retired results for 10 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-11-06 changes since 2022-09-26:

- ▷ We have added results for first algorithms from six developers: AFR Engine, CMC Institute of Science and Technology, Saga Densan Center, Turkcell Technology, UXLabs, and Wise AI SDN BHD.
- ▷ We have added results for new algorithms from 14 returning developers: Coretech Knowledge, Cloudwalk - Moontime, Cloudmatrix, Deepglint, Guangzhou Pixel Solutions, Hangzhuo AIlu Network Information Technology, NEO Systems, One More Security, Palit Microsystems, Panasonic R+D Center Singapore, Samsung S1, Seventh Sense Artificial Intelligence, Touchless ID, and Veridas Digital Authentication Solutions S.L.
- ▷ We have retired results for 10 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-09-26 changes since 2022-08-30:

- ▷ We have added results for first algorithms from three developers: Codeline, First Credit Bureau Kazakhstan, and InfoCert.
- ▷ We have added results for new algorithms from 14 returning developers: Advancegroup, Armatura LLC, Beijing Hisign Technology, Cybercore, Cyberlink Corp, Herta Security, ICM Airport Technics, InsightFace AI, Metsakuur, NSENSE Corp, Samsung-SDS, Videmo Intelligente Videoanalyse, Vietnam Posts and Telecommunications Group, and Vision Intelligence Center of Meituan.
- ▷ We have retired results for 11 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-08-30 changes since 2022-07-29:

- ▷ We have added results for first algorithms from two developers: Aximetria, Intellibrain Technological Projects
- ▷ We have added results for new algorithms from twelve returning developers: Alchera Inc, Dermalog, Idemia, Incode Technologies Inc, Intellivision, Kasikorn Labs, Megvii/Face++, Techsign, TuringTech.vip, Universidade de Coimbra, Verijelais, Vixvizon
- ▷ We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-07-29 changes since 2022-06-27:

- ▷ We have added results for first algorithms from seven developers: FRP LLC (Hawaii), IMDS Software, Inspur (Beijing) Electronic Information Industry, Intema - LGL Group, PAPAGO, Qaz Biometric Systems, and VIDA-Digital Identity

- We have added results for new algorithms from nine returning developers: Cyberextruder, Glory, Maxvision Technology, Rank One Computing, Securif AI, Suprema AI, Suprema ID, Toshiba, and Yuan High-Tech Development.
- We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-07-29 changes since 2022-06-27:

- We have added results for first algorithms from seven developers: FRP LLC (Hawaii), IMDS Software, Inspur (Beijing) Electronic Information Industry, Intema - LGL Group, PAPAGO, Qaz Biometric Systems, and VIDA-Digital Identity
- We have added results for new algorithms from nine returning developers: Cyberextruder, Glory, Maxvision Technology, Rank One Computing, Securif AI, Suprema AI, Suprema ID, Toshiba, and Yuan High-Tech Development.
- We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-06-27 changes since 2022-06-03:

- We have added results for first algorithms from two developers: Krungthai Bank, and Smartbiometrik.
- We have added results for new algorithms from thirteen returning developers: Aiseemu, Corsight, Digidata, Griaule, Guangzhou Pixel Solutions, Hangzhuo AI Network Information Technology, Neurotechnology, Real Networks, Samsung S1, Sensetime Group, Smart Engines, Verihubs Inteligensia, and VinBigData.
- We have retired results for eight algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-06-03 changes since 2022-05-05:

- We have added results for first algorithms from seven developers: Jaak IT, Metsakuur, Palit Microsystems, Smarvist Teknoloji, and Touchless ID.
- We have added results for new algorithms from sixteen returning developers: Cyberlink, FaceOnLive, Kakao Enterprise, Line Corporation (Line Clova), Multi-Modality Intelligence, NEO Systems, and Unissey
- We have retired results for four algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- We have moved the results for the twenty human-difficult pairs used in the May 2018 paper *Face recognition accuracy of forensic examiners, superrecognizers, and face recognition algorithms* by Phillips et al. [1]. to the algorithm-specific report cards (example: PDF).
- Likewise, we have added figures showing impostor distribution shifts across demographics to the report card.

2022-05-05 changes since 2022-03-18:

- ▷ We have added results for first algorithms from seven developers: Accurascan, DICIO, FacePhi, Pangiam, University of Surrey-CVSSP, and Veridium.
- ▷ We have added results for new algorithms from sixteen returning developers: ACI Software, Canon Inc, Cloudwalk - Moontime Smart Technology, Cybercore,

2022-05-05 changes since 2022-03-18:

- ▷ We have added results for first algorithms from seven developers: Accurascan, DICIO, FacePhi, Pangiam, University of Surrey-CVSSP, and Veridium.
- ▷ We have added results for new algorithms from sixteen returning developers: ACI Software, Canon Inc, Cloudwalk - Moontime Smart Technology, Cybercore, Cyberextruder, Gemalto Cogent, HyperVerge Inc, KuKe3D Technology, Megvii/Face++, Mobbeel Solutions, Panasonic R+D Center Singapore, Qnap Security, Samsung-SDS, Vietnam Posts and Telecommunications Group, Viettel Group, and Vision Intelligence Center of Meituan.
- ▷ We have retired results for 12 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-03-18 changes since 2022-02-23:

- ▷ We have added support for the detection of multiple people in a single image (see Section 1.2). Specifically the API allows an algorithm to extract features from one or more faces it detects in an image. NIST scores such cases as a correct match when any detected face matches the reference photo, and as a false positive when either face matches a non-mated reference photo. The expected effect of doing this will be to improve reported false non-match rates, and to minimally elevate false match rates. This technique was only applied to images of type "border" and "kiosk".
- ▷ We have added results for first algorithms from four developers: IntelliVIX, Kasikorn Labs, Lebentech Biometrics, and Wicket.
- ▷ We have added results for new algorithms from 10 returning developers: Chunghwa Telecom, Cloudmatrix, Beijing DeepSense Technologies, FarBar Inc, Imagus Technology Pty, Intellivision, Maxvision Technology, NHN Corp, Seventh Sense Artificial Intelligence, and Verigram.
- ▷ We have retired results for 4 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-02-23 changes since 2022-01-24:

- ▷ We have added results for first algorithms from four developers: AFIS and Biometrics Consulting, Digidata, Graymatics, Hangzhuo AIlu Network Information Technology, KnowUTech LLC, Sukshi Technology Innovation, T4iSB, and TuringTech.vip
- ▷ We have added results for new algorithms from 18 returning developers: Cognitec Systems GmbH, GeoVision Inc, Glory, Herta Security, Intel Research Group, InsightFace AI, Kakao Enterprise, N-Tech Lab, Omnigarde Ltd, Papilon Savunma, Paravision, Realnetworks Inc, Reveal Media Ltd, Shenzhen Inst Adv Integrated Tech CAS, Suprema AI Inc, Toshiba, Universidade de Coimbra, and Yuan High-Tech Development
- ▷ We have retired results for 14 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

2022-01-24 changes since 2022-01-20:

- ▷ We have added results for new algorithms from one returning developer: Vocord.

2022-01-20 changes since 2021-12-18:

- ▷ We have added results for first algorithms from four developers: Armatura, Beyne.AI, One More Security, and VinBigData
- ▷ We have added results for new algorithms from 19 returning developers: AuthenMetric, BOE Technology Group, Cybercore, Cyberlink, Dahua Technology, FaceTag Co, Innovatrics, Megvii, Mobbeel Solutions, Neurotechnology, Oz Forensics, Rank One Computing, Regula Forensics, Samsung S1, Securif AI, Sensetime Group, TigerIT Americas, Videmo Intelligente Videoanalyse, and YooniK.
- ▷ We have retired results for 14 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

: **2021-12-16** changes since 2021-11-22:

- ▷ We have added results for first algorithms from five developers: Alfabeta, Cloudmatrix, Euronovate SA, FaceOnLive Inc, and Mobipin Technology.
- ▷ We have added results for new algorithms from ten returning developers: ACI Software, ITMO University, NEO Systems, Guangzhou Pixel Solutions, Panasonic R+D Center Singapore, Qnap Security, Scanovate, Tevian, Unissey, and Vietnam Posts and Telecommunications Group.
- ▷ We have retired results for eight algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- ▷ We have revamped the figure showing performance on 20 pairs of open-source images. It now color-codes false negatives and postives against a default threshold value.

2021-11-22 changes since 2021-10-28:

- ▷ We have added results to the website for kiosk-collected images where the design and geometry configuration mean that many images have considerable downward pitch angle. In some images, the face is partially cropped. Some images have other background faces.
- ▷ We have stopped using child exploitation images in FRVT, as we lost access to the imagery. All results for that set have been removed from the website, and will be removed from future PDF reports.
- ▷ We have added results for first algorithms from seven new developers: CUDO Communication, Daon, KuKe3D Technology, Mantra Softech India, Maxvision Technology, Multi-Modality Intelligence, and Samsung-SDS.
- ▷ We have added results for new algorithms from seven returning developers: Acer Incorporated, Cloudwalk-Moontime Smart Technology, Gorilla Technology, ID3 Technology, Incode Technologies, NSENSE Corp., and SQIsoft.
- ▷ We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

2021-10-28 changes since 2021-09-08:

- ▷ We have substantially revised the algorithm-specific report cards that are linked from the FRVT results page. (Example: HTML).
- ▷ We have added results for first algorithms from eight new developers: Beijing Mendaxia Technology, Beijing Hisign Technology, Biocube Matrics, Clearview AI, Reveal Media, Toppan ID Gate, Verigram, and Viettel High Technology.
- ▷ We have added results for new algorithms from thirty returning developers: 20Face, 3divi, Canon Inc Chunghwa Telecom, Corsight, Decatur Industries, Deepglint, Dermalog, FaceTag, Fiberhome Telecommunication Technologies, GeoVision, ICM Airport Technics, Imagus Technology, InsightFace AI, Kakao Enterprise, Kookmin University, Line Corporation, N-Tech Lab, NotionTag Technologies, Realnetworks, Suprema ID, Taiwan-Certificate Authority, Toshiba, Tripleize, Trueface.ai, Veridas Digital Authentication, Visidon, VisionLabs, YooniK, and Yuan High-Tech Development.
- ▷ We have retired results for twenty algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-09-08 changes since 2021-08-02:

- ▷ We have added results for first algorithms from seven new developers: Griaule, SQIsoft, Qnap Security, Techsign, Smart Engines, Verihubs, and Wuhan Tianyu Information Industry.
- ▷ We have added results for new algorithms from sixteen returning developers: ADVANCE.AI, AuthenMetric, CloudSmart Consulting, Code Everest Pvt, Cognitec Systems, Thales Gemalto Cogent, Intel Research Group, Omnigarde, Oz Forensics, Rank One Computing, Samsung S1 Corp, Securif AI, Tevian, TigerIT Americas, Universidade de Coimbra, and Vigilant Solutions
- ▷ We have retired results for eleven algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-08-02 changes since 2021-06-25:

- ▷ We have added results for first algorithms from eight new developers: Bee the Data, Closeli Inc, Coretech Knowledge Inc, Deepsense (France), ioNetworks Inc, Kakao Pay Corp, Seventh Sense Artificial Intelligence, and SK Telecom.
- ▷ We have added results for new algorithms from fifteen returning developers: Alchera Inc, Adera Global PTE, Aware, Bresee Technology, Cyberlink Corp, Expasoft LLC, Fujitsu Research and Development Center, Gorilla Technology, Idemia, Neurotechnology, NEO Systems, NHN Corp, Paravision, Panasonic R+D Center Singapore, and Shenzhen University-Macau University of Science and Technology.
- ▷ We have retired results for twelve algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-06-25 changes since 2021-05-21:

- ▷ We have added results for first algorithms from six new developers: Alice Biometrics, BOE Technology Group, Fincore, Neosecu, Sodec App, and Yuntu Data and Technology.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- We have added results for new algorithms from seven returning developers: Incode Technologies, HyperVerge, Mobbeel Solutions, Guangzhou Pixel Solutions, Remark Holdings, Sensetime, and Vietnam Posts and Telecommunications Group.
- We have retired results for four algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-05-21 changes since 2021-04-26:

- We have added results for first algorithms from five new developers: Ekin Smart City Technologies, Suprema ID, Tripleize, Taiwan-Certificate Authority, and Vision Intelligence Center of Meituan.
- We have added results for new algorithms from eight returning developers: ID3 Technology, Imagus Technology, Momentum Digital, N-Tech Lab, NSENSE, Shanghai Jiao Tong University, Vision-Box, and Yuan High-Tech Development
- We have retired results for seven algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-04-26 changes since 2021-04-16:

- We have added results for first algorithms from three new developers: Quantasoft, Rendip, and NEO Systems.
- We have added results for new algorithms from four returning developers: 3Divi, Realnetworks, Veridas Digital Authentication Solutions, and Universidade de Coimbra.
- We have retired results for three algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-04-16 changes since 2021-03-19:

- We have added results for first algorithms from six new developers: 20Face, Beijing DeepSense Technologies, BitCenter UK, Enface, FaceTag, InsightFace AI, Line Corporation, Lema Labs, Nanjing Kiwi Network Technology, Omnigarde, Regula Forensics, and Suprema.
- We have added results for new algorithms from ten returning developers: CloudSmart Consulting, Dermalog, GeoVision, Neurotechnology, Panasonic R+D Center Singapore, Samsung S1, Securif AI, Trueface.ai, Vigilant Solutions, and Visidon.
- We have retired results for ten algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-03-19 changes since 2021-03-05:

- We have added results for first algorithms from six new developers: Ajou University, AuthenMetric, Code Everest, Corsight, Papilon Savunma, and NHN Corp
- We have added results for new algorithms from seven returning developers: Alchera, Deepglint, Fiberhome Telecommunication Technologies, Kakao Enterprise, Kookmin University, Megvii/Face++, and NotionTag Technologies.

- We have updated many of the hyperlinked HTML report-cards to include seven figures on demographic dependence. Figures of this kind first appeared, and are documented in, the December 2019 document, NIST Interagency Report 8280 on demographic differentials in face recognition. The figures quantify false negative dependence on demographics using "visa-border" comparisons, and false positive dependence using comparisons of "application" photos that uniformly of quality and similar to visa photos.

2021-03-05 changes since 2021-01-19:

- We have added results for first algorithms from three new developers: IVA Cognitive, Mobbeel, and MoreDian Technology.
- We have added results for new algorithms from returning developers: Ability Enterprise - Andro Video, ACI Software, Adera Global, AnyVision, BioID Technologies, China Electronics Import-Export, Cognitec Systems, Fujitsu Research and Development Center, Glory, Guangzhou Pixel Solutions, Hengrui AI Technology, Incode Technologies, Intel Research, iQIYI, Mobai, Oz Forensics, Paravision, VisionLabs, and Xforward AI Technology.
- We have added a new "resources" tab to the main webpage. It includes sortable columns for data related to speed, model size, storage, and memory consumption.
- We have retired results for 13 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-01-19 changes since 2020-12-18:

- This report adds results for first algorithms from four developers: Herta Security, Irex AI, Shenzhen University-Macau University of Science and Technology, and Vietnam Posts and Telecommunications Group. See Table 22 for more information.
- The report also includes results for thirteen developers who have previously submitted algorithms: Bresee Technology, Canon (previously Canon Information Technology (Beijing)), Cyberlink, CSA IntelliCloud Technology, Dahua Technology, ID3 Technology, Imagus Technology (Vixvizion), Moontime Smart Technology, N-Tech Lab, Thales Cogent, Veridas Digital Authentication Solutions, Vocord, and Yuan High-Tech Development.
- We have retired results for ten algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2020-12-18 changes since 2020-10-09:

- This report adds results for first algorithms from ten developers: BitCenter UK, CloudSmart Consulting, Cubox, Institute of Computing Technology, Naver Corp, Minivision, NSENSE Corp, Viettel Group, Visage Technologies, and Xiamen University. See Table 22 for more information.
- The report also includes results for eighteen developers who have previously submitted algorithms: ADVANCE.AI, Awidit Systems, Chosun University, Dermalog, GeoVision, ICM Airport Technics, Idemia, Institute of Information Technologies, Kakao Enterprise, Neurotechnology, Panasonic R+D Center Singapore, Rank One Computing, Sensetime Group, Shanghai Jiao Tong University, TigerIT Americas LLC, Vigilant Solutions, Winsense, and YooniK

- We have retired results for twelve algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

Changes since September 18, 2020:

- This report adds results for first algorithms from five developers: Aigen, Cortica, Kookmin University, Securif AI and Vinai.
- The report also includes results for three developers who have previously submitted algorithms: Fujitsu Laboratories, Hengrui AI, and X-Forward AI.
- In the per-algorithm report-cards linked from tables and the main webpage, we have added a chart to showing reduction in error rates over the course of FRVT i.e. from 2017 onwards for all algorithms supplied by that developer. Similarly we have added a chart showing error rate reductions for our test of protective face mask verification.
- We plan to continue evaluating algorithms on various mask datasets. We hold that algorithms should be capable of detecting masks and verifying identity of all combinations of masked and unmasked faces. We have accordingly increased the amount to time allowed to extract those features from 1.0 to 1.5 seconds.

Changes since August 25, 2020:

- This report adds results for first algorithms from eight new developers. Akurat Satu Indonesia, Cybercore, Decatur Industries, Innef Labs, Satellite Innovation/Eocortex, Expasoft, and Mobai.
- The report includes results for seven developers who have previously submitted algorithms: 3Divi, BioID Technologies, Incode Technologies, Innovatrics, iSAP Solution, Synology, and Tevian.
- We have retired results for five algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

Changes since July 27, 2020:

- We have introduced per-algorithm report sheets. These are HTML documents linked from the accuracy tables in this report (i.e. Table 90) and on the FRVT 1:1 homepage. The sheets contain interactive graphics allowing, for example, mouseover exploration of FNMR(T) and FMR(T). Some of their content had previously appeared in this document.
- This report adds results for algorithms from six new developers. ACI Software, Bresee Technology, Fiberhome Telecommunication Technologies, Imageware Systems, Oz Forensics, and Pensees.
- The report includes results for thirteen developers who have previously submitted algorithms: Canon Information Technology (Beijing), Cyberlink, Dahua Technology, Gorilla Technology, ID3 Technology, Intel Research Group, iQIYI Inc, Momentum Digital, Netbridge Technology, Tech5 SA, Shenzhen AiMall Tech, Vigilant Solutions, and VisionLabs.
- We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

Changes since May 18, 2020:

- The report is the first FRVT update since the pandemic closed it from March to June 2020.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- This report includes results for algorithms from nine new developers: GeoVision Inc, Su Zhou NaZhi-TianDi Intelligent Technology, YooniK, AYF Technology, PXL Vision AG, Yuan High-Tech Development, Beihang University-ERCACAT, ICM Airport Technics, and Staqu Technologies
- This report includes results for algorithms from 15 returning developers Acer Incorporated, Antheus Technologia, Chosun University, Chunghwa Telecom, Idemia, Moontime Smart Technology, Neurotechnology, Guangzhou Pixel Solutions, Panasonic R+D Center Singapore, Rank One Computing, Scanovate, Shanghai Universiy - Shanghai Film Academy, Synesis, Trueface.ai, and Veridas Digital Authentication Solutions
- We have retired results for ten algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- We separated timing and other resource consumption from the main participation table. The new Table 56 includes template generation durations for four kinds of images, not just mugshots.
- We have published a separate report, NIST Interagency Report 8311 on accuracy of pre-pandemic algorithms on subjects wearing face masks. We plan to track improvements in accuracy on masked images going forward. In particular, we invite submission of algorithms that can detect whether a person is wearing a mask, extract features from the full face or the exposed periocular region, and do appropriate comparison. We do not intend to evaluate algorithms that assume 100% of images will be of masked individuals.

Changes since March 25, 2020:

- The report is a maintenance release - it does not add any new algorithms, and FRVT has been closed to new algorithms since mid March 2020.
- We modified the primary accuracy summary, Table 90, as follows:
  - For visa images, the column for FNMR at FMR = 0.0001 has been removed. The visa images are so highly controlled that the error rates for the most accurate algorithms are dominated by false rejection of very young children and by the presence of a few noisy greyscale images. For now, two visa columns remain: FNMR at FMR= 10^{-6} and, for matched covariates, FNMR at FMR= 10^{-4}.
  - We have inserted a new column labelled "BORDER" giving accuracy for comparison of moderately poor webcam border-crossing photos that exhibit pose variations, poor compression, and low contrast due to strong background illumination. The accuracies are the worst from all cooperative image datasets used in FRVT.
- Accordingly, we updated the failure-to-template rates in Table 116.
- We withdrew a figure showing how false matches are concentrated in certain visa images used in cross-comparison, because it didn't attempt to include demographic information.

Changes since February 27, 2020:

- The report adds results algorithms from two new developers: Beijing Alleyes Technology, and the Chinese University of Hong Kong. Results for newly submitted algorithms from two other developers will appear in the next report.
- The report adds results for algorithms from thirteen returning developers: ASUSTek Computer, Aware, Cyberlink Corp, Gorilla Technology, Innovative Technology, Kakao Enterprise, Lomonosov Moscow State University, Panasonic R+D Center Singapore, Shenzhen AiMall Technology, Shenzhen Intellifusion Technologies, Synology, Tech5 SA, and Via Technologies.

- Per policy to only list results for two algorithms per developer, we have dropped results for algorithms from Aware, Cyberlink, Gorilla Technology, Kakao Enterprise, Lomonosov Moscow State University, Panasonic R+D Center Singapore, and Tech5 SA.

Changes since January 20, 2020:

- The report adds results for five new developers: Ability Enterprise (Andro Video), Chosun University, Fujitsu Research and Development Center, University of Coimbra, and Xforward AI Technology.
- The report adds results for algorithms from six returning developers: AlphaSSTG, Incode Technologies, Kneron. Shanghai Jiao Tong University, Vocord, and X-Laboratory.
- We have corrected template comparison timing numbers for algorithms submitted September 2019 to January 2020. The values reported previously were slower due to a software bug.
- We have dropped results for algorithms from Vocord and Incode per policy to only list results for two algorithms per developer.
- The FRVT 1:1 homepage has been updated with latest accuracy results.
- The FRVT 1:N homepage now includes an update to the September 2019 NIST Interagency Report 8271. The new report adds results for one-to-many search algorithms submitted to NIST from June 2019 to January 2020.

Changes since January 6, 2020:

- Section 2 has been updated to better describe the Visa and Border images. The caption for Table 90 has been updated to better relate the accuracy values to particular image comparisons.
- The report adds results for five new developers: Acer, Advance.AI, Expasoft, Netbridge Technology, and Videmo Intelligente Videoanalyse.
- The report adds results for algorithms from 7 returning developers: China Electronics Import-Export Corp, Intel Research Group, ITMO University, Neurotechnology, N-Tech Lab, Rokid, and VisionLabs.
- We have dropped results from this edition of the report per policy to only list results for two algorithms per developer: N-Tech Lab, Neurotechnology, ITMO, Visionlabs, and CEIEC.
- The FRVT homepage has been updated with latest accuracy results.

Changes since November 11, 2019:

- Table 56 has been updated to include runtime memory usage. This is the first time such a quantity has been reported. The value is the peak size of the resident set size logged during enrollment of single images.
- We have migrated summary results table to a new platform that supports sortable tables: https://pages.nist.gov/frvt/html/frvt11.html
- The report adds results for four new developers: Antheus Technologia, BioID Technologies SA, Canon Information Tech. (Beijing), Samsung S1 (listed in the tables as S1), and Taiwan AI Labs.
- The report adds results for algorithms from 13 returning developers: Anke Investments, Chunghwa Telecom, Deepglint, Institute of Information Technologies, iQIYI, Kneron, Ping An Technology, Paravision, KanKan Ai, Rokid Corporation, Shanghai Universiy - Shanghai Film Academy, Veridas Digital Authentication Solutions, and Videonetics Technology.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- We have dropped results from this edition of the report per policy to only list results for two algorithms per developer: remarkai-000, veridas-001, sensetime-001, iit-000, anke-003, and everai-002. Results for these are available in prior editions of this report linked from the FRVT page.
- We issued NIST Interagency Report 8280: FRVT Part 3: Demographics on 2019-12-19. It includes results for many of the algorithms covered by this report.

Changes since October 16, 2019:

- The report adds results for ten new developers: Ai-Union Technology, ASUSTek Computer, DiDi ChuXing Technology, Innovative Technology, Luxand, MVision, Pyramid Cyber Security + Forensic, Scanovate, Shenzhen AiMall Tech, and TUPU Technology.
- The report adds results for 12 returning developers: CTBC Bank Glory Gorilla Technology Guangzhou Pixel Solutions Imagus Technology Incode Technologies Lomonosov Moscow State University Rank One Computing Samtech InfoNet Shanghai Ulucu Electronics Technology Synesis, and Winsense.
- We have dropped results from this edition of the report per policy to only list results for two algorithms per developer: glory-000, gorilla-002, incode-003, rankone-006, and synesis-004.
- Results for five recently submitted algorithms will appear in the next report.

Changes since September 11, 2019:

- The report adds results for five new participants: Awidit Systems (Awiros), Momemtum Digital (Sertis), Trueface AI, Shanghai Jiao Tong University, and X-Laboratory.
- The reports adds results for five new algorithms from returning developers: Cyberlink, Hengrui AI Technology, Idemia, Panasonic R+D Singapore, and Tevian. This causes three algorithm, to be de-listed from the report per policy to list results for two algorithms per developer.

Changes since July 31 2019:

- The HTML table on the FRVT 1:1 homepage has been updated to include a column for cross-domain Visa-Border verification. Results for this new dataset appeared in the July 29 report under the name "CrossEV" - these are now renamed "Visa-Border".
- The FRVT 1:1 homepage lists algorithms according to lowest mean rank accuracy:
  Rank($\text{FNMR}_{\text{VISA}}$ at FMR = 0.000001) +
  Rank($\text{FNMR}_{\text{VISA-BORDER}}$ at FMR = 0.000001) +
  Rank($\text{FNMR}_{\text{MUGSHOT}}$ at FMR = 0.00001 after 14 years) +
  Rank($\text{FNMR}_{\text{WILD}}$ at FMR = 0.00001)
  This ordering rewards high accuracy across all datasets.
- The main results in Table 90 is now in landscape format to accomodate extra columns for the Visa-Border set, and mugshot comparisons after at least 12 years.
- The report adds results for nine new participants: Alpha SSTG, Intel Research, ULSee, Chungwa Telecon, iSAP Solution, Rokid, Shenzhen EI Networks, CSA Intellicloud, Shenzhen Intellifusion Technologies.
- The reports adds results for six new algorithms from returning developers: Innovatrics, Dahua Technology, Tech5 SA, Intellivision, Nodeflux and Imperial College, London. One algorithm, from Imperial has been retired, per policy to list results for two algorithms per developer.
- The cross-country false match rate heatmaps have been replotted to reveal more structure by listing countries by region instead of alphabetically.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- The next version of this report will be posted around October 18, 2019.

Changes since July 3 2019:

- The HTML table on the FRVT 1:1 homepage has been updated to list the 20 most accurate developers rather than algorithms, choosing the most accurate algorithm from each developer based on visa and mugshot results. Also, the algorithms are ordered in terms of lowest mean rank across mugshot, visa and wild datasets, rewarding broad accuracy over a good result on one particular dataset.
- This report includes results for a new dataset - see the column labelled "visa-border" in Table 5. It compares a new set of high quality visa-like portraits with a set webcam border-crossing photos that exhibit moderately poor pose variations and background illumination. The two new sets are described in sections 2.2 and 2.4. The comparisons are "cross-domain" in that the algorithm must compare "visa" and "wild" images. Results for other algorithms will be added in future reports as they become available.
- This report adds results for algorithms from 9 developers submitted in early July 2019. These are from 3DiVi, Camvi, EverAI-Paravision, Facesoft, Farbar (F8), Institute of Information Technologies, Shanghai U. Film Academy, Via Technologies, and Ulucu Electronics Tech. Six of these are new participants.
- Several other algorithms have been submitted and are being evaluated. Results will be released in the next report, scheduled for September 5. That report will include results for new datasets.
- Older algorithms from Everai, Camvi and 3DiVi, have been retired, per the policy to list only two algorithms per developer.

Changes since June 20 2019:

- This report adds results for algorithms from 18 developers submitted in early June 2019. These are from CTBC Bank, Deep Glint, Thales Cogent, Ever AI Paravision, Gorilla Technology, Imagus, Incode, Kneron, N-Tech Lab, Neurotechnology, Notiontag Technologies, Star Hybrid, Videonetics, Vigilant Solutions, Winsense, Anke Investments, CEIEC, and DSK. Nine of these are new participants.
- Several other algorithms have been submitted and are being evaluated. Results will be released in the next report, scheduled for August 1.
- Older algorithms from Everai, Thales Cogent, Gorilla Technology, Incode, Neurotechnology, N-Tech Lab and Vigilant Solutions have been retired, per the policy to list only two algorithms per developer.

Changes since April 2019:

- This report adds results for nine algorithms from nine developers submitted in early June 2019. These are from Tencent Deepsea, Hengrui, Kedacom, Moontime, Guangzhou Pixel, Rank One Computing, Synesis, Sensetime and Vocord.
- Another 23 algorithms have been submitted and are being evaluated. Results will be released in the next report, scheduled for July 3.
- Older algorithms for Rank One, Synesis, and Vocord have been retired, per the policy to list only two algorithms per developer.

Changes since February 2019:

- This report adds results for 49 algorithms from 42 developers submitted in early March 2019.
- This report omits results for algorithms that we retired. We retired for three reasons: 1. The developer submitted a new algorithm, and we only list two. 2. The algorithm needs a GPU, and we no longer allow GPU-based algorithms. 3. Inoperable algorithms.
- Previous results for retired algorithms are available in older editions of this report linked here.
- The mugshot database used from February 2017 to January 2019 has been replaced with an extract of the mugshot database documented in NIST Interagency Report 8238, November 2018. The new mugshot set is described in section 2.5 and is adopted because:

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷▷ It has much better identity label integrity, so that false non-match rates are substantially lower than those reported in FRVT 1:1 reports to date - see Figure 294.
- ▷▷ It includes images collected over a 17 year period such that ageing can be much better characterized - - see Figure 807.

- ▷ Using the new mugshot database, Figure 807 shows accuracy for four demographic groups identified in the biographic metadata that accompanies the data: black females, black males, white females and white males.
- ▷ The report added a figure (now moved to web) with results for the twenty human-difficult pairs used in the May 2018 paper *Face recognition accuracy of forensic examiners, superrecognizers, and face recognition algorithms* by Phillips et al. [1].
- ▷ The report uses an update to the wild image database that corrects some ground truth labels.
- ▷ Some results for the child exploitation database are not complete. They are typically updated less frequently than for other image sets.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Contents

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650 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 817
651 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 818
652 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 819
653 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 820
654 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 821
655 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 822
656 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 823
657 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 824
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659 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 826
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664 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 831
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669 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 836
670 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 837
671 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 838
672 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 839
673 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 840
674 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 841
675 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 842
676 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 843
677 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 844
678 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 845
679 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 846
680 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 847
681 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 848
682 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 849
683 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 850
684 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 851
685 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 852
686 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 853
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688 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 855
689 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 856

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

690 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 857
691 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 858
692 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 859
693 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 860
694 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 861
695 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 862
696 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 863
697 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 864
698 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 865
699 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 866
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702 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 869
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704 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 871
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707 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 874
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709 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 876
710 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 877
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716 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 883
717 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 884
718 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 885
719 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 886
720 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 887
721 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 889
722 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 890
723 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 891
724 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 892
725 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 893
726 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 894
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731 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 899
732 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 900
733 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 901
734 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 902
735 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 903
736 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 904
737 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 905
738 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 906
739 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 907
740 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 908
741 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 909
742 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 910
743 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 911
744 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 912
745 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 913
746 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 914

FNMR(T) "False non-match rate"
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747 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 915
748 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 916
749 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 917
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802 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 970
803 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 971

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805 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 973
806 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 974
807 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 975
808 EFFECT OF SUBJECT AGE ON FNMR . . . 977
809 EFFECT OF SUBJECT AGE ON FNMR . . . 978
810 EFFECT OF SUBJECT AGE ON FNMR . . . 979
811 EFFECT OF SUBJECT AGE ON FNMR . . . 980
812 EFFECT OF SUBJECT AGE ON FNMR . . . 981
813 EFFECT OF SUBJECT AGE ON FNMR . . . 982
814 EFFECT OF SUBJECT AGE ON FNMR . . . 983
815 EFFECT OF SUBJECT AGE ON FNMR . . . 984
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830 EFFECT OF SUBJECT AGE ON FNMR . . . 999
831 EFFECT OF SUBJECT AGE ON FNMR . . . 1000
832 EFFECT OF SUBJECT AGE ON FNMR . . . 1001
833 EFFECT OF SUBJECT AGE ON FNMR . . . 1002
834 EFFECT OF SUBJECT AGE ON FNMR . . . 1003
835 EFFECT OF SUBJECT AGE ON FNMR . . . 1004
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838 EFFECT OF SUBJECT AGE ON FNMR . . . 1007
839 EFFECT OF SUBJECT AGE ON FNMR . . . 1008
840 EFFECT OF SUBJECT AGE ON FNMR . . . 1009
841 EFFECT OF SUBJECT AGE ON FNMR . . . 1010
842 EFFECT OF SUBJECT AGE ON FNMR . . . 1011
843 EFFECT OF SUBJECT AGE ON FNMR . . . 1012
844 EFFECT OF SUBJECT AGE ON FNMR . . . 1013
845 EFFECT OF SUBJECT AGE ON FNMR . . . 1014
846 EFFECT OF SUBJECT AGE ON FNMR . . . 1015
847 EFFECT OF SUBJECT AGE ON FNMR . . . 1016
848 EFFECT OF SUBJECT AGE ON FNMR . . . 1017
849 EFFECT OF SUBJECT AGE ON FNMR . . . 1018
850 EFFECT OF SUBJECT AGE ON FNMR . . . 1019
851 EFFECT OF SUBJECT AGE ON FNMR . . . 1020
852 EFFECT OF SUBJECT AGE ON FNMR . . . 1021
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854 EFFECT OF SUBJECT AGE ON FNMR . . . 1023
855 EFFECT OF SUBJECT AGE ON FNMR . . . 1024

FNMR(T) "False non-match rate"
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856 EFFECT OF SUBJECT AGE ON FNMR . . . 1025
857 EFFECT OF SUBJECT AGE ON FNMR . . . 1026
858 EFFECT OF SUBJECT AGE ON FNMR . . . 1027
859 EFFECT OF SUBJECT AGE ON FNMR . . . 1028
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861 EFFECT OF SUBJECT AGE ON FNMR . . . 1030
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879 EFFECT OF SUBJECT AGE ON FNMR . . . 1048
880 EFFECT OF SUBJECT AGE ON FNMR . . . 1049
881 EFFECT OF SUBJECT AGE ON FNMR . . . 1050
882 EFFECT OF SUBJECT AGE ON FNMR . . . 1051
883 EFFECT OF SUBJECT AGE ON FNMR . . . 1052
884 EFFECT OF SUBJECT AGE ON FNMR . . . 1053
885 EFFECT OF SUBJECT AGE ON FNMR . . . 1054
886 EFFECT OF SUBJECT AGE ON FNMR . . . 1055
887 EFFECT OF SUBJECT AGE ON FNMR . . . 1056
888 EFFECT OF SUBJECT AGE ON FNMR . . . 1057
889 EFFECT OF SUBJECT AGE ON FNMR . . . 1058
890 EFFECT OF SUBJECT AGE ON FNMR . . . 1059
891 EFFECT OF SUBJECT AGE ON FNMR . . . 1060
892 EFFECT OF SUBJECT AGE ON FNMR . . . 1061
893 EFFECT OF SUBJECT AGE ON FNMR . . . 1062
894 EFFECT OF SUBJECT AGE ON FNMR . . . 1063
895 EFFECT OF SUBJECT AGE ON FNMR . . . 1064
896 EFFECT OF SUBJECT AGE ON FNMR . . . 1065
897 EFFECT OF SUBJECT AGE ON FNMR . . . 1066
898 EFFECT OF SUBJECT AGE ON FNMR . . . 1067
899 IMPOSTOR COUNTS FOR CROSS COUNTRY FMR CALCULATIONS . . . 1071

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1	KR	Urielsoft Co. Ltd	urielsoft-000	000	2025-06-30
2	EE	2060.io	2060io-001	001	2025-01-22
3	EE	2060.io	2060io-002	002	2025-07-23
4	NL	20Face	20face-000	000	2021-04-12
5	NL	20Face	20face-001	001	2021-09-29
6	US	3Divi	3divi-000	000	2017-03-16
7	US	3Divi	3divi-001	001	2017-06-22
8	US	3Divi	3divi-002	002	2017-10-20
9	US	3Divi	3divi-003	003	2018-10-09
10	US	3Divi	3divi-004	004	2019-07-22
11	US	3Divi	3divi-005	005	2020-08-28
12	US	3Divi	3divi-006	006	2021-04-14
13	US	3Divi	3divi-007	007	2021-09-27
14	TH	ACI Software	acisw-003	003	2020-08-03
15	TH	ACI Software	acisw-006	006	2021-02-25
16	TH	ACI Software	acisw-007	007	2021-11-15
17	TH	ACI Software	acisw-008	008	2022-03-22
18	US	AFIS and Biometrics Consulting	afisbiometrics-000	000	2022-01-27
19	US	AFR Engine	afrengine-000	000	2022-09-29
20	US	AFR Engine	afrengine-001	001	2023-01-26
21	US	AFR Engine	afrengine-002	002	2023-06-27
22	US	AFR Engine	afrengine-003	003	2024-03-15
23	US	AFR Engine	afrengine-004	004	2024-07-22
24	US	AFR Engine	afrengine-005	005	2025-02-18
25	UZ	AI Implementation Group	aigroup-000	000	2024-07-17
26	UZ	AI Implementation Group	aigroup-001	001	2025-02-18
27	UZ	AI Implementation Group	aigroup-002	002	2026-03-12
28	NV	AIPRACTIX	aipractix-000	000	2026-05-12
29	VN	ALTTEKGlobal	alttekglobal-000	000	2024-04-12
30	TW	ASUSTek Computer Inc	asusaics-000	000	2019-10-24
31	TW	ASUSTek Computer Inc	asusaics-001	001	2020-02-25
32	CN	AYF Technology	ayftech-001	001	2020-07-06
33	CN	AYF Technology	ayftech-003	003	2023-05-15
34	TW	Ability Enterprise - Andro Video	androvideo-000	000	2021-01-25
35	IN	Accurascan	accurascan-001	001	2023-0-15
36	IN	Accurascan	accurascan-002	002	2023-08-01
37	IN	Accurascan	accurascan-003	003	2023-12-06
38	TW	Acer Incorporated	acer-000	000	2020-01-08
39	TW	Acer Incorporated	acer-001	001	2020-06-30
40	TW	Acer Incorporated	acer-002	002	2021-11-10
41	SG	Adera Global PTE	adera-001	001	2019-06-17
42	SG	Adera Global PTE	adera-002	002	2021-02-16
43	SG	Adera Global PTE	adera-003	003	2021-07-12
44	SG	Adera Global PTE	adera-004	004	2022-11-14
45	SG	Adera Global PTE	adera-005	005	2023-11-17
46	SG	Advance.AI	advance-002	002	2019-12-19
47	SG	Advance.AI	advance-003	003	2021-08-05
48	SG	Advance.AI	advance-004	004	2022-09-06
49	SG	Advance.AI	advance-005	005	2023-05-25
50	SG	Advance.AI	advance-006	006	2026-04-03
51	SG	Advance.AI	advance-007	007	2026-08-07
52	AE	Affinsys AI Ltd	affinsys-ai-000	000	2024-07-11
53	TH	Ai First	aifirst-001	001	2019-11-21
54	TW	AiUnion Technology	aiunionface-000	000	2019-10-22
55	TH	Aigen	aigen-001	001	2020-10-06
56	TH	Aigen	aigen-002	002	2021-03-15
57	CN	Aiseemu Technology	aiseemu-001	001	2022-06-16
58	CN	Aiseemu Technology	aiseemu-002	002	2022-11-18
59	KR	Ajou University	ajou-001	001	2021-03-08
60	br	Akiyama S.A	akiyama-003	003	2026-08-24
61	AE	Akor IT Services LLC	akor-000	000	2026-09-16
62	ID	Akurat Satu Indonesia	ptakuratsatu-000	000	2020-09-11
63	KR	Alchera Inc	alchera-000	000	2019-03-01
64	KR	Alchera Inc	alchera-001	001	2019-03-01
65	KR	Alchera Inc	alchera-002	002	2021-03-05
66	KR	Alchera Inc	alchera-003	003	2021-07-13
67	KR	Alchera Inc	alchera-004	004	2022-08-12
68	KR	Alchera Inc	alchera-005	005	2023-01-04
69	KR	Alchera Inc	alchera-006	006	2023-07-18
70	KR	Alchera Inc	alchera-007	007	2023-11-22

Table 1: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
71	KR	Alchera Inc	alchera-008	008	2024-06-17
72	KR	Alchera Inc	alchera-009	009	2024-11-14
73	ID	Alfabeta	alfabeta-001	001	2021-12-02
74	ES	Alice Biometrics	alice-000	000	2021-06-15
75	ES	Alice Biometrics	alice-001	001	2023-07-05
76	RU	Alivia / Innovation Sys	isystems-000	000	2018-01-11
77	RU	Alivia / Innovation Sys	isystems-001	001	2018-06-12
78	RU	Alivia / Innovation Sys	isystems-002	002	2018-10-18
79	IN	AllGoVision	allgovision-000	000	2019-03-01
80	CN	AlphaSSTG	alphaface-001	001	2019-09-03
81	CN	AlphaSSTG	alphaface-002	002	2020-02-20
82	GB	Amplified Group	amplifiedgroup-001	001	2019-03-01
83	CN	Anke Investments	anke-000	000	2018-06-26
84	CN	Anke Investments	anke-001	001	2018-06-26
85	CN	Anke Investments	anke-002	002	2018-10-19
86	CN	Anke Investments	anke-003	003	2019-02-27
87	CN	Anke Investments	anke-004	004	2019-06-27
88	CN	Anke Investments	anke-005	005	2019-11-21
89	BR	Antheus Technologia	akiyama-000	000	2019-12-05
90	BR	Antheus Technologia	akiyama-001	001	2020-06-25
91	BR	Antheus Technologia	akiyama-002	002	2024-07-15
92	GB	AnyVision	anyvision-001	001	2018-01-31
93	GB	AnyVision	anyvision-002	002	2018-01-31
94	GB	AnyVision	anyvision-004	004	2018-06-15
95	GB	AnyVision	anyvision-005	005	2021-02-03
96	TR	ArVİs Teknoloji San.Tic.A.Ş.	arvis-000	000	2025-04-08
97	CN	Aratek Biometrics Co Ltd	aratek-001	001	2023-03-27
98	US	Armatura LLC	armatura-001	001	2022-01-04
99	US	Armatura LLC	armatura-002	002	2022-09-16
100	US	Armatura LLC	armatura-003	003	2023-01-13
101	CN	AuthenMetric	authenmetric-002	002	2021-03-10
102	CN	AuthenMetric	authenmetric-003	003	2021-08-09
103	CN	AuthenMetric	authenmetric-004	004	2022-01-03
104	TW	Authme	authme-001	001	2023-10-17
105	TW	Authme	authme-002	002	2025-02-10
106	US	Aware	aware-000	000	2017-10-27
107	US	Aware	aware-001	001	2017-10-27
108	US	Aware	aware-002	002	2018-06-15
109	US	Aware	aware-003	003	2018-10-19
110	US	Aware	aware-004	004	2019-03-01
111	US	Aware	aware-005	005	2020-02-27
112	US	Aware	aware-006	006	2021-07-03
113	US	Aware	aware-007	007	2023-11-15
114	US	Aware	aware-008	008	2024-03-15
115	IN	Awidit Systems	awiros-001	001	2019-09-23
116	IN	Awidit Systems	awiros-002	002	2020-10-28
117	IN	Awidit Systems	awiros-003	003	2024-11-01
118	IN	Awidit Systems	awiros-004	004	2025-03-31
119	CH	Aximetria	aximetria-001	001	2022-08-10
120	JP	Ayonix	ayonix-000	000	2017-06-22
121	CN	BOE Technology Group	boetech-001	001	2021-06-22
122	CN	BOE Technology Group	boetech-002	002	2021-12-21
123	KZ	BTS Digital	btsdigital-001	001	2025-07-31
124	ES	Bee the Data	beethedata-000	000	2021-07-26
125	CN	Beihang University-ERCACAT	ercacat-001	001	2020-07-06
126	CN	Beijing Alleyes Technology	alleyes-000	000	2020-03-09
127	CN	Beijing DeepSense Technologies	deepsense-000	000	2021-03-19
128	CN	Beijing DeepSense Technologies	deepsense-001	001	2022-03-11
129	CN	Beijing DeepSense Technologies	deepsense-002	002	2023-02-22
130	CN	Beijing DeepSense Technologies	deepsense-003	003	2023-10-23
131	CN	Beijing DeepSense Technologies	deepsense-004	004	2024-06-21
132	CN	Beijing DeepSense Technologies	deepsense-005	005	2024-10-28
133	CN	Beijing DeepSense Technologies	deepsense-006	006	2025-03-07
134	CN	Beijing DeepSense Technologies	deepsense-007	007	2025-07-08
135	CN	Beijing DeepSense Technologies	deepsense-008	008	2026-01-13
136	CN	Beijing Hisign Technology	hisign-001	001	2021-09-24
137	CN	Beijing Hisign Technology	hisign-002	002	2022-09-09
138	CN	Beijing Hisign Technology	hisign-003	003	2024-03-15
139	CN	Beijing Mendaxia Technology	mendaxiatech-000	000	2021-09-15
140	CN	Beijing Vion Technology Inc	vion-000	000	2018-10-19

Table 2: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
141	DE	Betaface	betaface-000	000	2018-05-29
142	DE	Betaface	betaface-002	002	2018-05-29
143	KZ	Beyne.AI	beyneai-000	000	2022-01-03
144	DE	BioID GmbH	bioid-001	001	2024-11-13
145	DE	BioID GmbH	bioid-002	002	2025-04-25
146	DE	BioID GmbH	bioid-003	003	2025-11-26
147	DE	BioID GmbH	bioid-004	004	2026-04-13
148	CH	BioID Technologies SA	bioidtechswiss-000	000	2019-11-15
149	CH	BioID Technologies SA	bioidtechswiss-001	001	2020-08-28
150	CH	BioID Technologies SA	bioidtechswiss-002	002	2021-02-17
151	CH	BioID Technologies SA	bioidtechswiss-003	003	2025-03-12
152	IN	Biocube Matrics	biocube-001	001	2021-09-08
153	IN	Biocube Matrics	biocube-002	002	2023-02-09
154	KZ	Biometric LLC	biometric-vision-000	000	2023-01-25
155	IN	Biometrik Systems Private Limited	biometrik-001	001	2025-09-18
156	IN	Biometrik Systems Private Limited	biometrik-002	002	2026-01-20
157	GB	BitCenter UK	farfaces-001	001	2021-04-09
158	CN	Bitmain	bm-001	001	2018-10-17
159	CN	Bresee Technology	bresee-000	000	2020-08-07
160	CN	Bresee Technology	bresee-001	001	2020-12-30
161	CN	Bresee Technology	bresee-002	002	2021-06-30
162	US	Bynn Intelligence Inc	bynn-000	000	2026-05-29
163	VN	CAKE by VPBank	cakevn-001	001	2026-01-12
164	VN	CAKE by VPBank	cakevn-002	002	2026-06-21
165	VN	CMC Institute of Science and Technology	cist-001	001	2022-10-20
166	VN	CMC Institute of Science and Technology	cist-002	002	2023-02-09
167	VN	CMC Institute of Science and Technology	cist-003	003	2023-08-14
168	VN	CMC Institute of Science and Technology	cist-004	004	2024-02-15
169	VN	CMC Univerisity	cmcuni-001	001	2023-12-05
170	CN	CSA IntelliCloud Technology	intellicloudai-001	001	2019-08-13
171	CN	CSA IntelliCloud Technology	intellicloudai-002	002	2020-12-17
172	TW	CTBC Bank	ctbcbank-000	000	2019-06-28
173	TW	CTBC Bank	ctbcbank-001	001	2019-10-28
174	TW	CTBC Bank	ctbcbank-002	002	2024-07-10
175	KR	CUDO Communication	cudocommunication-001	001	2021-10-20
176	US	Camvi Technologies	camvi-001	001	2017-09-13
177	US	Camvi Technologies	camvi-002	002	2018-10-19
178	US	Camvi Technologies	camvi-003	003	2019-03-01
179	US	Camvi Technologies	camvi-004	004	2019-07-12
180	FI	Candour Biometrics	candour-001	001	2023-02-10
181	FI	Candour Biometrics	candour-002	002	2024-03-07
182	FI	Candour Biometrics	candour-003	003	2024-10-23
183	FI	Candour Biometrics	candour-004	004	2026-01-08
184	CN	Canon Inc	canon-002	002	2020-12-29
185	JP	Canon Inc	canon-003	003	2021-09-15
186	JP	Canon Inc	canon-004	004	2022-04-25
187	JP	Canon Inc	canon-005	005	2023-09-05
188	CN	Canon Inc	cib-000	000	2019-12-11
189	CN	Canon Inc	cib-001	001	2020-08-05
190	BD	Celloscope Limited	celloscope-001	001	2024-07-15
191	CN	China Electronics Import-Export Corp	ceiec-001	001	2019-03-01
192	CN	China Electronics Import-Export Corp	ceiec-002	002	2019-06-12
193	CN	China Electronics Import-Export Corp	ceiec-003	003	2020-01-06
194	CN	China Electronics Import-Export Corp	ceiec-004	004	2021-01-18
195	CN	China University of Petroleum	upc-001	001	2019-06-05
196	CN	Chinese University of Hong Kong	cuhkee-001	001	2020-03-18
197	KR	Chosun University	chosun-000	000	2020-02-12
198	KR	Chosun University	chosun-001	001	2020-07-01
199	KR	Chosun University	chosun-002	002	2020-11-25
200	TW	Chunghwa Telecom	chtface-001	001	2019-08-06
201	TW	Chunghwa Telecom	chtface-002	002	2019-12-07
202	TW	Chunghwa Telecom	chtface-003	003	2020-06-24
203	TW	Chunghwa Telecom	chtface-004	004	2021-10-08
204	TW	Chunghwa Telecom	chtface-005	005	2022-03-09
205	TW	Chunghwa Telecom	chtface-006	006	2022-11-03
206	TW	Chunghwa Telecom	chtface-007	007	2024-04-23
207	TW	Chunghwa Telecom	chtface-008	008	2024-09-27
208	TW	Chunghwa Telecom	chtface-009	009	2025-10-21
209	US	City and County of Honolulu	cchonolulu-000	000	2023-03-27
210	US	City and County of Honolulu	cchonolulu-001	001	2024-01-18

Table 3: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
211	US	Clearview AI Inc	clearviewai-000	000	2021-09-22
212	US	Clearview AI Inc	clearviewai-001	001	2023-12-05
213	US	Clearview AI Inc	clearviewai-002	002	2024-05-22
214	US	Clearview AI Inc	clearviewai-003	003	2026-01-09
215	CN	Closeli Inc	closeli-001	001	2021-07-15
216	US	CloudSmart Consulting LLC	csc-001	001	2020-11-20
217	US	CloudSmart Consulting LLC	csc-002	002	2021-03-24
218	US	CloudSmart Consulting LLC	csc-003	003	2021-08-26
219	TW	Cloudmatrix	cloudmatrix-000	000	2021-10-22
220	TW	Cloudmatrix	cloudmatrix-001	001	2022-02-16
221	TW	Cloudmatrix	cloudmatrix-002	002	2022-10-17
222	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-000	000	2019-03-01
223	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-001	001	2019-06-04
224	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-002	002	2019-10-08
225	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-003	003	2020-09-25
226	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-004	004	2021-02-10
227	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-000	000	2019-06-03
228	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-002	002	2020-07-02
229	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-003	003	2020-12-22
230	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-004	004	2021-11-09
231	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-005	005	2022-03-29
232	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-006	006	2022-10-20
233	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-007	007	2023-02-21
234	IN	Code Everest Pvt	facex-001	001	2021-03-08
235	IN	Code Everest Pvt	facex-002	002	2021-08-24
236	KR	Codeline	codeline-000	000	2022-09-13
237	DE	Cognitec Systems GmbH	cognitec-000	000	2018-10-19
238	DE	Cognitec Systems GmbH	cognitec-001	001	2019-03-01
239	DE	Cognitec Systems GmbH	cognitec-002	002	2021-02-24
240	DE	Cognitec Systems GmbH	cognitec-003	003	2021-07-30
241	DE	Cognitec Systems GmbH	cognitec-004	004	2022-02-10
242	DE	Cognitec Systems GmbH	cognitec-005	005	2023-12-01
243	DE	Cognitec Systems GmbH	cognitec-006	006	2024-04-29
244	DE	Cognitec Systems GmbH	cognitec-007	007	2024-08-29
245	DE	Cognitec Systems GmbH	cognitec-008	008	2026-09-24
246	TW	Compal Electronics	compal-000	000	2025-03-20
247	TW	Compal Electronics	compal-001	001	2025-12-08
248	TW	Compal Electronics	compal-002	002	2026-07-09
249	TW	Coretech Knowledge Inc	coretech-000	000	2021-07-12
250	TW	Coretech Knowledge Inc	coretech-001	001	2022-09-29
251	TW	Coretech Knowledge Inc	coretech-002	002	2023-11-02
252	TW	Coretech Knowledge Inc	coretech-003	003	2024-08-07
253	TW	Coretech Knowledge Inc	coretech-004	004	2024-12-23
254	IL	Corsight	corsight-001	001	2021-03-11
255	IL	Corsight	corsight-002	002	2021-09-01
256	IL	Corsight	corsight-003	003	2022-06-09
257	IL	Cortica	cor-001	001	2020-09-24
258	TW	Cu-Face	cu-face-002	002	2023-01-05
259	TW	Cu-Face	cu-face-003	003	2023-08-28
260	TW	Cu-Face	cu-face-004	004	2024-03-13
261	KR	Cubox	cubox-001	001	2020-12-07
262	KR	Cubox	cubox-002	002	2021-08-24
263	KR	Cubox	cubox-003	003	2023-03-07
264	KR	Cubox	cubox-004	004	2024-12-05
265	KR	Cubox	cubox-005	005	2025-05-14
266	JP	Cybercore	cybercore-000	000	2020-08-26
267	JP	Cybercore	cybercore-001	001	2021-12-15
268	JP	Cybercore	cybercore-002	002	2022-04-25
269	JP	Cybercore	cybercore-003	003	2022-08-31
270	US	Cyberextruder	cyberextruder-001	001	2017-08-02
271	US	Cyberextruder	cyberextruder-002	002	2018-01-30
272	US	Cyberextruder	cyberextruder-003	003	2022-03-16
273	US	Cyberextruder	cyberextruder-004	004	2022-07-20
274	US	Cyberextruder	cyberextruder-005	005	2024-10-01
275	US	Cyberextruder	cyberextruder-006	006	2025-03-19
276	TW	Cyberlink Corp	cyberlink-000	000	2019-03-01
277	TW	Cyberlink Corp	cyberlink-001	001	2019-03-01
278	TW	Cyberlink Corp	cyberlink-002	002	2019-06-12
279	TW	Cyberlink Corp	cyberlink-003	003	2019-10-07
280	TW	Cyberlink Corp	cyberlink-004	004	2020-02-27

Table 4: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
281	TW	Cyberlink Corp	cyberlink-005	005	2020-07-31
282	TW	Cyberlink Corp	cyberlink-006	006	2021-01-08
283	TW	Cyberlink Corp	cyberlink-007	007	2021-07-16
284	TW	Cyberlink Corp	cyberlink-008	008	2022-01-07
285	TW	Cyberlink Corp	cyberlink-009	009	2022-05-12
286	TW	Cyberlink Corp	cyberlink-010	010	2022-09-16
287	TW	Cyberlink Corp	cyberlink-011	011	2023-01-30
288	TW	Cyberlink Corp	cyberlink-012	012	2023-06-02
289	TW	Cyberlink Corp	cyberlink-013	013	2023-12-15
290	TW	Cyberlink Corp	cyberlink-014	014	2024-05-23
291	TW	Cyberlink Corp	cyberlink-015	015	2024-09-26
292	TW	Cyberlink Corp	cyberlink-016	016	2026-06-02
293	MX	DICIO	dicio-001	001	2022-03-22
294	MX	DICIO	dicio-002	002	2024-10-03
295	CN	DSK	dsk-000	000	2019-06-28
296	VN	Dactionable Technologies	datech-000	000	2023-10-16
297	VN	Dactionable Technologies	datech-001	001	2024-02-21
298	VN	Dactionable Technologies	datech-002	002	2024-09-11
299	CN	Dahua Technology	dahua-000	000	2018-10-19
300	CN	Dahua Technology	dahua-001	001	2018-10-19
301	CN	Dahua Technology	dahua-002	002	2019-03-01
302	CN	Dahua Technology	dahua-003	003	2019-08-14
303	CN	Dahua Technology	dahua-004	004	2019-12-18
304	CN	Dahua Technology	dahua-005	005	2020-08-13
305	CN	Dahua Technology	dahua-006	006	2020-12-30
306	CN	Dahua Technology	dahua-007	007	2021-12-20
307	IE	Daon	daon-000	000	2021-11-03
308	IE	Daon	daon-001	001	2024-09-26
309	IE	Daon	daon-002	002	2025-07-01
310	IE	Daon	daon-003	003	2026-03-12
311	TW	DeCloak Intellegences	decloakface-001	001	2023-06-13
312	TW	DeCloak Intellegences	decloakface-002	002	2024-10-28
313	US	Decatur Industries Inc	decatur-000	000	2020-08-18
314	US	Decatur Industries Inc	decatur-001	001	2021-09-27
315	CN	Deepglint	deepglint-001	001	2019-06-21
316	CN	Deepglint	deepglint-002	002	2019-11-15
317	CN	Deepglint	deepglint-003	003	2021-03-03
318	CN	Deepglint	deepglint-004	004	2021-09-17
319	CN	Deepglint	deepglint-005	005	2022-10-17
320	FR	Deepsense	dps-000	000	2021-07-16
321	DE	Dermalog	dermalog-001	001	2017-02-22
322	DE	Dermalog	dermalog-002	002	2017-02-22
323	DE	Dermalog	dermalog-003	003	2017-07-10
324	DE	Dermalog	dermalog-004	004	2017-10-26
325	DE	Dermalog	dermalog-005	005	2018-02-02
326	DE	Dermalog	dermalog-006	006	2018-10-18
327	DE	Dermalog	dermalog-007	007	2020-11-25
328	DE	Dermalog	dermalog-008	008	2021-03-25
329	DE	Dermalog	dermalog-009	009	2021-10-06
330	DE	Dermalog	dermalog-010	010	2022-07-25
331	DE	Dermalog	dermalog-011	011	2022-12-12
332	DE	Dermalog	dermalog-012	012	2024-01-19
333	DE	Dermalog	dermalog-013	013	2024-06-18
334	DE	Dermalog	dermalog-014	014	2025-06-17
335	DE	Dermalog	dermalog-015	015	2025-11-26
336	DE	Dermalog	dermalog-016	016	2026-06-25
337	CN	DiDi ChuXing Technology	didiglobalface-001	001	2019-10-23
338	CN	DiDi ChuXing Technology	didiglobalface-002	002	2023-01-09
339	CN	DiDi ChuXing Technology	didiglobalface-003	003	2025-05-01
340	IN	Digidata	digidata-000	000	2022-01-27
341	IN	Digidata	digidata-001	001	2022-06-10
342	GB	Digital Barriers	digitalbarriers-000	000	2017-05-31
343	GB	Digital Barriers	digitalbarriers-001	001	2017-07-20
344	GB	Digital Barriers	digitalbarriers-002	002	2019-03-01
345	UY	Dralmey Solutions	dralmey-001	001	2026-03-27
346	IN	EI Networks Private Ltd	einetworksindia-000	000	2023-05-03
347	IN	EI Networks Private Ltd	einetworksindia-001	001	2023-09-05
348	IN	EI Networks Private Ltd	einetworksindia-002	002	2025-02-21
349	TR	Ekin Smart City Technologies	ekin-002	002	2021-05-04
350	TH	Element System Solutions Company Ltd	element-000	000	2023-11-14

Table 5: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
351	AE	Enface	enface-000	000	2021-04-09
352	AE	Enface	enface-001	001	2021-12-17
353	AE	Enface	enface-002	002	2023-02-27
354	GB	Enqura Ltd	enqura-001	001	2024-09-25
355	GB	Enqura Ltd	enqura-002	002	2026-05-19
356	ET	Ethiopian Labor Market Information System	ethiopianlmis-000	000	2026-03-18
357	CH	Euronovate SA	euronovate-001	001	2021-11-15
358	CH	Euronovate SA	euronovate-002	002	2023-06-28
359	CH	Euronovate SA	euronovate-003	003	2023-09-19
360	CH	Euronovate SA	euronovate-004	004	2024-01-19
361	RU	Expasoft LLC	expasoft-000	000	2020-01-06
362	RU	Expasoft LLC	expasoft-001	001	2020-09-03
363	RU	Expasoft LLC	expasoft-002	002	2021-07-26
364	VN	FCI Vietnam	fci-000	000	2025-04-01
365	LB	FOO	foomobi-001	001	2023-03-13
366	LB	FOO	foomobi-002	002	2023-08-07
367	VN	FPT Education	fedu-001	001	2024-03-15
368	VN	FPT Information System	fpt-000	000	2024-02-26
369	VN	FPT Information System	fpt-001	001	2024-09-16
370	VN	FPT Telecom JSC	fpttelecom-000	000	2026-02-03
371	US	FRP LLC	frpkauai-001	001	2022-07-18
372	US	FRP LLC	frpkauai-002	002	2022-11-21
373	US	FRP LLC	frpkauai-003	003	2024-05-30
374	US	FRP LLC	frpkauai-004	004	2025-03-25
375	US	FRP LLC	frpkauai-005	005	2026-06-22
376	IL	Face Int	faceint-001	001	2026-03-27
377	CA	FaceLocate	facelocate-001	001	2024-02-22
378	CA	FaceLocate	facelocate-002	002	2026-03-25
379	DE	FaceOnLive Inc	faceonlive-001	001	2021-11-23
380	DE	FaceOnLive Inc	faceonlive-002	002	2022-04-11
381	DE	FaceOnLive Inc	faceonlive-003	003	2024-08-12
382	ES	FacePhi	facephi-000	000	2022-04-06
383	ES	FacePhi	facephi-001	001	2024-05-24
384	ES	FacePhi	facephi-002	002	2025-02-11
385	GB	FaceSoft	facesoft-000	000	2019-07-10
386	KR	FaceTag Co	facetag-000	000	2021-03-22
387	KR	FaceTag Co	facetag-001	001	2021-08-17
388	KR	FaceTag Co	facetag-002	002	2022-01-06
389	GB	Facehawk Ltd	facehawk-000	000	2023-08-22
390	GB	Facehawk Ltd	facehawk-001	001	2026-06-17
391	GB	Facia.ai	facia-001	001	2023-07-21
392	GB	Facia.ai	facia-002	002	2024-06-21
393	GB	FaiceTech Ltd	faicetech-000	000	2025-03-11
394	TW	FarBar Inc	f8-001	001	2019-07-11
395	TW	FarBar Inc	f8-002	002	2022-03-02
396	US	Fast Enterprises	fastenterprises-000	000	2023-03-01
397	CN	Fiberhome Telecommunication Technologies	fiberhome-nanjing-002	002	2020-08-10
398	CN	Fiberhome Telecommunication Technologies	fiberhome-nanjing-003	003	2021-03-12
399	CN	Fiberhome Telecommunication Technologies	fiberhome-nanjing-004	004	2021-09-14
400	GB	Fincore Ltd	fincore-000	000	2021-06-07
401	KZ	First Credit Bureau Kazakhstan	firstcreditkz-001	001	2022-08-22
402	KZ	First Credit Bureau Kazakhstan	firstcreditkz-002	002	2023-02-21
403	KZ	First Credit Bureau Kazakhstan	firstcreditkz-003	003	2024-02-20
404	KZ	First Credit Bureau Kazakhstan	firstcreditkz-004	004	2024-10-02
405	NL	Fourthline	fourthlinecom-001	001	2024-04-15
406	GB	Fraud.com	fraudcom-000	000	2023-12-01
407	CN	Fudan University	fdu-000	000	2017-11-22
408	CN	Fudan University	fdu-001	001	2018-01-11
409	CN	Fujitsu Research and Development Center	fujitsulab-000	000	2020-02-04
410	CN	Fujitsu Research and Development Center	fujitsulab-001	001	2020-09-30
411	CN	Fujitsu Research and Development Center	fujitsulab-002	002	2021-02-24
412	CN	Fujitsu Research and Development Center	fujitsulab-003	003	2021-07-12
413	VN	GPS Vietnam Trading	gpstechvn-000	000	2024-01-02
414	VN	GTEL OTS JSC	gtelots-000	000	2025-04-14
415	VN	GTEL OTS JSC	gtelots-001	001	2026-02-02
416	US	Gemalto Cogent	cogent-000	000	2018-01-12
417	US	Gemalto Cogent	cogent-001	001	2018-05-29
418	US	Gemalto Cogent	cogent-002	002	2018-10-19
419	US	Gemalto Cogent	cogent-003	003	2019-03-01
420	US	Gemalto Cogent	cogent-004	004	2019-06-14

Table 6: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
421	US	Gemalto Cogent	cogent-005	005	2020-12-29
422	US	Gemalto Cogent	cogent-006	006	2021-07-28
423	US	Gemalto Cogent	cogent-007	007	2022-04-11
424	US	Gemalto Cogent	cogent-008	008	2023-01-03
425	TW	General Interface Solutions Holding Ltd	gistouch-000	000	2023-05-08
426	TW	General Interface Solutions Holding Ltd	gistouch-001	001	2023-11-09
427	TW	GeoVision Inc	geo-000	000	2020-06-29
428	TW	GeoVision Inc	geo-001	001	2020-10-30
429	TW	GeoVision Inc	geo-002	002	2021-04-01
430	TW	GeoVision Inc	geo-003	003	2021-09-15
431	TW	GeoVision Inc	geo-004	004	2022-02-10
432	JP	Glory	glory-000	000	2018-06-06
433	JP	Glory	glory-001	001	2018-06-08
434	JP	Glory	glory-002	002	2019-11-12
435	JP	Glory	glory-003	003	2021-01-15
436	JP	Glory	glory-004	004	2022-02-08
437	JP	Glory	glory-005	005	2022-07-08
438	JP	Glory	glory-006	006	2023-03-23
439	JP	Glory	glory-007	007	2023-11-10
440	TW	Gorilla Technology	gorilla-000	000	2018-02-13
441	TW	Gorilla Technology	gorilla-001	001	2018-05-25
442	TW	Gorilla Technology	gorilla-002	002	2018-10-17
443	TW	Gorilla Technology	gorilla-003	003	2019-06-19
444	TW	Gorilla Technology	gorilla-004	004	2019-11-04
445	TW	Gorilla Technology	gorilla-005	005	2020-03-11
446	TW	Gorilla Technology	gorilla-006	006	2020-07-31
447	TW	Gorilla Technology	gorilla-007	007	2021-06-28
448	TW	Gorilla Technology	gorilla-008	008	2021-11-08
449	TW	Gorilla Technology	gorilla-009	009	2022-12-14
450	US	Graymatics	graymatics-001	001	2022-01-13
451	US	Griaule	griaule-000	000	2021-08-20
452	US	Griaule	griaule-001	001	2022-05-31
453	US	Griaule	griaule-002	002	2022-12-02
454	CN	Guangzhou Pixel Solutions	pixelall-002	002	2019-06-06
455	CN	Guangzhou Pixel Solutions	pixelall-003	003	2019-10-15
456	CN	Guangzhou Pixel Solutions	pixelall-004	004	2020-07-02
457	CN	Guangzhou Pixel Solutions	pixelall-005	005	2021-02-05
458	CN	Guangzhou Pixel Solutions	pixelall-006	006	2021-06-17
459	CN	Guangzhou Pixel Solutions	pixelall-007	007	2021-12-01
460	CN	Guangzhou Pixel Solutions	pixelall-008	008	2022-06-16
461	CN	Guangzhou Pixel Solutions	pixelall-009	009	2022-10-26
462	CN	Guangzhou Pixel Solutions	pixelall-010	010	2023-10-18
463	CN	Hangzhuo AIlu Network Information Technology	hzailu-001	001	2022-01-27
464	CN	Hangzhuo AIlu Network Information Technology	hzailu-002	002	2022-06-02
465	CN	Hangzhuo AIlu Network Information Technology	hzailu-003	003	2022-10-11
466	CN	Hangzhuo AIlu Network Information Technology	hzailu-004	004	2023-02-27
467	CN	Hangzhuo AIlu Network Information Technology	hzailu-005	005	2023-08-10
468	CN	Hangzhuo AIlu Network Information Technology	hzailu-006	006	2023-12-28
469	CN	Hangzhuo AIlu Network Information Technology	hzailu-007	007	2024-09-06
470	CN	Hangzhuo AIlu Network Information Technology	hzailu-008	008	2025-02-27
471	ES	Herta Security	hertasecurity-000	000	2021-01-05
472	ES	Herta Security	hertasecurity-001	001	2022-01-18
473	ES	Herta Security	hertasecurity-002	002	2022-09-02
474	ES	Herta Security	hertasecurity-003	003	2023-01-27
475	CN	Hikvision Research Institute	hik-001	001	2019-03-01
476	IN	Hyderabad-Centre for Development of Advanced Computing	cdachyd-001	001	2024-10-01
477	IN	HyperVerge Inc	hyperverge-001	001	2020-12-13
478	IN	HyperVerge Inc	hyperverge-002	002	2021-05-27
479	IN	HyperVerge Inc	hyperverge-003	003	2022-04-11
480	IN	HyperVerge Inc	hyperverge-004	004	2022-12-14
481	IN	HyperVerge Inc	hyperverge-005	005	2023-08-08
482	IN	HyperVerge Inc	hyperverge-006	006	2024-05-02
483	IN	HyperVerge Inc	hyperverge-007	007	2026-04-10
484	KR	Hyundai Motor Company	hyundai-roboticslab-000	000	2024-07-12
485	KR	Hyundai Motor Company	hyundai-roboticslab-001	001	2025-02-03
486	KR	Hyundai Motor Company	hyundai-roboticslab-002	002	2025-06-03
487	KR	Hyundai Motor Company	hyundai-roboticslab-003	003	2025-11-14
488	KR	Hyundai Motor Company	hyundai-roboticslab-004	004	2026-06-04
489	AU	ICM Airport Technics	icm-002	002	2020-11-13
490	AU	ICM Airport Technics	icm-003	003	2021-09-06

Table 7: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
491	AU	ICM Airport Technics	icm-004	004	2022-09-07
492	AU	ICM Airport Technics	icm-005	005	2023-09-05
493	US	ID R&D Inc	idrnd-000	000	2024-07-26
494	US	ID R&D Inc	idrnd-001	001	2025-07-31
495	FR	ID3 Technology	id3-001	001	2017-08-04
496	FR	ID3 Technology	id3-002	002	2017-08-04
497	FR	ID3 Technology	id3-003	003	2018-10-05
498	FR	ID3 Technology	id3-004	004	2019-03-01
499	FR	ID3 Technology	id3-005	005	2020-08-04
500	FR	ID3 Technology	id3-006	006	2020-12-17
501	FR	ID3 Technology	id3-007	007	2021-05-17
502	FR	ID3 Technology	id3-008	008	2021-11-10
503	FR	ID3 Technology	id3-009	009	2026-02-05
504	FR	ID3 Technology	id3-010	010	2026-06-05
505	US	IDBIO	idbio-000	000	2026-09-08
506	GB	IDENTY	identy-000	000	2023-04-04
507	GB	IDENTY	identy-001	001	2025-05-21
508	GB	IDENTY	identy-002	002	2025-11-14
509	GB	IDENTY	identy-003	003	2026-05-15
510	KZ	IE SARSENOV	asr-000	000	2025-07-30
511	KZ	IE SARSENOV	asr-001	001	2025-12-01
512	CA	IMDS Software	imds-software-001	001	2022-07-06
513	CA	IMDS Software	imds-software-002	002	2023-02-10
514	CA	IMDS Software	imds-software-003	003	2023-06-20
515	MX	IQSEC	iqsec-001	001	2024-04-12
516	MX	IQSEC	iqsec-002	002	2025-01-08
517	MX	IQSEC	iqsec-003	003	2026-07-20
518	RU	ITMO University	itmo-001	001	2017-06-12
519	RU	ITMO University	itmo-002	002	2017-08-07
520	RU	ITMO University	itmo-003	003	2018-01-23
521	RU	ITMO University	itmo-004	004	2018-06-13
522	RU	ITMO University	itmo-005	005	2018-10-19
523	RU	ITMO University	itmo-006	006	2019-03-01
524	RU	ITMO University	itmo-007	007	2020-01-06
525	RU	ITMO University	itmo-008	008	2021-11-19
526	RU	IVA Cognitive	ivacognitive-001	001	2021-01-29
527	FR	Idemia	idemia-003	003	2018-10-19
528	FR	Idemia	idemia-004	004	2019-03-01
529	FR	Idemia	idemia-005	005	2019-10-11
530	FR	Idemia	idemia-006	006	2020-07-06
531	FR	Idemia	idemia-007	007	2020-12-04
532	FR	Idemia	idemia-008	008	2021-07-07
533	FR	Idemia	idemia-009	009	2022-07-27
534	FR	Idemia	idemia-010	010	2023-06-30
535	FR	Idemia	idemia-011	011	2024-08-06
536	FR	Idemia	morpho-000	000	2017-07-11
537	FR	Idemia	morpho-001	001	2017-07-10
538	FR	Idemia	morpho-002	002	2017-12-28
539	US	Imageware Systems	iws-000	000	2020-08-12
540	GB	Imperial College London	imperial-000	000	2019-03-01
541	GB	Imperial College London	imperial-001	001	2019-03-01
542	GB	Imperial College London	imperial-002	002	2019-08-28
543	US	Incode Technologies Inc	incode-000	000	2018-06-01
544	US	Incode Technologies Inc	incode-001	001	2018-06-13
545	US	Incode Technologies Inc	incode-002	002	2018-10-18
546	US	Incode Technologies Inc	incode-003	003	2019-03-01
547	US	Incode Technologies Inc	incode-004	004	2019-06-12
548	US	Incode Technologies Inc	incode-005	005	2019-10-17
549	US	Incode Technologies Inc	incode-006	006	2020-02-20
550	US	Incode Technologies Inc	incode-007	007	2020-08-25
551	US	Incode Technologies Inc	incode-008	008	2021-01-19
552	US	Incode Technologies Inc	incode-009	009	2021-06-22
553	US	Incode Technologies Inc	incode-010	010	2021-10-22
554	US	Incode Technologies Inc	incode-011	011	2022-08-10
555	US	Incode Technologies Inc	incode-012	012	2023-05-05
556	US	Incode Technologies Inc	incode-013	013	2023-12-08
557	US	Incode Technologies Inc	incode-014	014	2024-04-08
558	US	Incode Technologies Inc	incode-015	015	2025-06-12
559	IT	InfoCert	infocert-001	001	2022-09-08
560	IT	InfoCert	infocert-002	002	2026-03-18

Table 8: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
561	IT	InfoCert	infocert-003	003	2026-07-20
562	IN	Innef Labs	innefulabs-000	000	2020-09-04
563	IN	Innominds Software SEZ India Pvt	innominds-001	001	2023-06-23
564	MY	Innov8tif Solutions	innov8tif-000	000	2024-12-10
565	MY	Innov8tif Solutions	innov8tif-001	001	2026-04-22
566	MY	Innov8tif Solutions	innov8tif-002	002	2026-09-07
567	GB	Innovative Technology	innovativetechnologyltd-001	001	2019-10-22
568	GB	Innovative Technology	innovativetechnologyltd-002	002	2020-02-26
569	SK	Innovatrics	innovatrics-000	000	2017-07-25
570	SK	Innovatrics	innovatrics-001	001	2017-07-25
571	SK	Innovatrics	innovatrics-002	002	2018-02-05
572	SK	Innovatrics	innovatrics-003	003	2018-06-14
573	SK	Innovatrics	innovatrics-004	004	2018-10-19
574	SK	Innovatrics	innovatrics-005	005	2019-03-01
575	SK	Innovatrics	innovatrics-006	006	2019-08-13
576	SK	Innovatrics	innovatrics-007	007	2020-08-19
577	SK	Innovatrics	innovatrics-008	008	2021-12-15
578	SK	Innovatrics	innovatrics-009	009	2022-01-19
579	SK	Innovatrics	innovatrics-010	010	2023-08-03
580	SK	Innovatrics	innovatrics-011	011	2024-03-08
581	SK	Innovatrics	innovatrics-012	012	2024-07-10
582	SK	Innovatrics	innovatrics-013	013	2024-11-13
583	SK	Innovatrics	innovatrics-014	014	2025-03-13
584	SK	Innovatrics	innovatrics-015	015	2025-07-24
585	SK	Innovatrics	innovatrics-016	016	2025-11-28
586	SK	Innovatrics	innovatrics-017	017	2026-07-30
587	CN	InsightFace AI	insightface-000	000	2021-03-17
588	CN	InsightFace AI	insightface-001	001	2021-09-27
589	CN	InsightFace AI	insightface-002	002	2022-01-31
590	CN	InsightFace AI	insightface-003	003	2022-08-23
591	CN	InsightFace AI	insightface-004	004	2023-03-31
592	CN	Inspur (Beijing) Electronic Information Industry Co	inspur-000	000	2022-07-19
593	CN	Inspur (Beijing) Electronic Information Industry Co	inspur-001	001	2023-02-24
594	CN	Inspur (Beijing) Electronic Information Industry Co	inspur-002	002	2023-12-08
595	CN	Inspur (Beijing) Electronic Information Industry Co	inspur-003	003	2025-02-11
596	CN	Institute of Computing Technology	icthtc-000	000	2020-11-29
597	RU	Institute of Information Technologies	iit-000	000	2019-03-01
598	RU	Institute of Information Technologies	iit-001	001	2019-07-05
599	RU	Institute of Information Technologies	iit-002	002	2019-12-04
600	RU	Institute of Information Technologies	iit-003	003	2020-12-01
601	BD	Integrated Software and Technologies Ltd	istl-001	001	2026-08-07
602	IS	Intel Research Group	intelresearch-000	000	2019-07-08
603	IS	Intel Research Group	intelresearch-001	001	2020-01-14
604	IS	Intel Research Group	intelresearch-002	002	2020-07-24
605	IS	Intel Research Group	intelresearch-003	003	2021-01-18
606	IS	Intel Research Group	intelresearch-004	004	2021-08-24
607	IS	Intel Research Group	intelresearch-005	005	2022-02-13
608	IS	Intel Research Group	intelresearch-006	006	2022-12-19
609	IS	Intel Research Group	intelresearch-007	007	2023-05-30
610	IS	Intel Research Group	intelresearch-008	008	2023-10-03
611	IS	Intel Research Group	intelresearch-009	009	2024-03-26
612	IS	Intel Research Group	intelresearch-010	010	2024-08-20
613	IS	Intel Research Group	intelresearch-011	011	2024-12-24
614	IS	Intel Research Group	intelresearch-012	012	2025-07-03
615	KR	IntelliVIX	intellivix-001	001	2022-02-25
616	KR	IntelliVIX	intellivix-002	002	2022-07-14
617	KR	IntelliVIX	intellivix-003	003	2022-12-12
618	KR	IntelliVIX	intellivix-004	004	2023-06-21
619	KR	IntelliVIX	intellivix-005	005	2023-11-14
620	KR	IntelliVIX	intellivix-006	006	2025-07-31
621	AE	Intellibrain Technological Projects	g42-intellibrain-001	001	2022-07-27
622	AE	Intellibrain Technological Projects	g42-intellibrain-002	002	2023-05-10
623	CN	Intelligent Control Technology Co Ltd - IGearx	igearx-face-000	000	2023-03-28
624	US	Intellivision	intellivision-001	001	2017-10-10
625	US	Intellivision	intellivision-002	002	2019-08-23
626	US	Intellivision	intellivision-003	003	2022-03-07
627	US	Intellivision	intellivision-004	004	2022-07-28
628	US	Intellivision	intellivision-005	005	2023-02-06
629	US	Intellivision	intellivision-006	006	2023-07-14
630	US	Intellivision	intellivision-007	007	2023-12-19

Table 9: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
631	LU	Intema-LGL Group	intema-000	000	2022-07-15
632	LU	Intema-LGL Group	intema-001	001	2023-01-11
633	US	InterLink labs	interlinklabs-001	001	2024-11-08
634	US	InterLink labs	interlinklabs-002	002	2025-04-21
635	US	InterLink labs	interlinklabs-003	003	2025-11-18
636	IN	Intozi Tech Pvt Ltd	intozi-001	001	2023-09-25
637	IN	Intozi Tech Pvt Ltd	intozi-002	002	2026-06-21
638	IN	Intpixel Labs Private Ltd	vogic-000	000	2026-07-30
639	US	IrexAI	irex-000	000	2020-12-17
640	US	Iris ID Systems	irisid-001	001	2026-01-30
641	US	Iris ID Systems	irisid-002	002	2026-08-05
642	IL	Is It You	isityou-000	000	2017-06-26
643	MX	Jaak IT	jaakit-001	001	2022-05-20
644	MX	Jaak IT	jaakit-002	002	2025-09-11
645	GB	KYB-AI Ltd	kbyai-000	000	2024-11-29
646	IN	Kaizen Secure Voiz	kaizenvoiz-001	001	2026-06-08
647	KR	Kakao Bank	kakaobank-000	000	2023-02-27
648	KR	Kakao Bank	kakaobank-001	001	2023-10-10
649	KR	Kakao Bank	kakaobank-002	002	2024-04-01
650	KR	Kakao Bank	kakaobank-003	003	2024-09-30
651	KR	Kakao Brain	kakao-001	001	2019-03-01
652	KR	Kakao Brain	kakao-002	002	2019-06-19
653	KR	Kakao Brain	kakao-003	003	2020-02-26
654	KR	Kakao Brain	kakao-004	004	2020-10-28
655	KR	Kakao Brain	kakao-005	005	2021-03-09
656	KR	Kakao Brain	kakao-007	007	2022-01-12
657	KR	Kakao Brain	kakao-008	008	2022-05-12
658	KR	Kakao Brain	kakao-009	009	2023-11-07
659	KR	Kakao Pay Corp	kakaopay-001	001	2021-07-06
660	TH	Kasikorn Labs	kasikornlabs-000	000	2022-03-02
661	TH	Kasikorn Labs	kasikornlabs-001	001	2022-07-26
662	TH	Kasikorn Labs	kasikornlabs-002	002	2022-12-13
663	TH	Kasikorn Labs	kasikornlabs-003	003	2023-10-05
664	TH	Kasikorn Labs	kasikornlabs-004	004	2024-11-04
665	TH	Kasikorn Labs	kasikornlabs-005	005	2025-04-04
666	SG	Kedacom International Pte	kedacom-000	000	2019-06-03
667	GB	Keyless Technologies	keyless-000	000	2024-11-21
668	US	Kloudspot Inc	kloudspot-001	001	2024-09-03
669	US	Kloudspot Inc	kloudspot-002	002	2025-06-16
670	US	Kneron Inc	kneron-003	003	2019-07-01
671	US	Kneron Inc	kneron-005	005	2020-02-21
672	US	Kneron Inc	kneron-006	006	2026-04-10
673	US	KnowUTech LLC	knowutech-000	000	2022-02-13
674	US	Kogniza Technology	kogniza-001	001	2024-03-01
675	KR	Kookmin University	kookmin-001	001	2020-09-28
676	KR	Kookmin University	kookmin-002	002	2021-03-05
677	KR	Korea Identification Inc	koreaid-001	001	2022-12-12
678	TH	Krungthai	krungthai-002	002	2022-06-21
679	CN	KuKe3D Technology	kuke3d-001	001	2021-10-28
680	CN	KuKe3D Technology	kuke3d-002	002	2022-04-14
681	MX	Lebentech Biometrics	lebentech-000	000	2022-02-16
682	MX	Lebentech Biometrics	lebentech-001	001	2023-12-12
683	IN	Lema Labs	lemalabs-001	001	2021-04-13
684	KR	LightVision Inc	lightvision-000	000	2024-06-27
685	KR	LightVision Inc	lightvision-001	001	2025-02-18
686	JP	Line Corporation	lineclova-000	000	2021-03-31
687	JP	Line Corporation	lineclova-001	001	2021-09-26
688	JP	Line Corporation	lineclova-002	002	2022-05-18
689	JP	Line Corporation	lineclova-003	003	2022-11-28
690	CN	Lishui Shuyan Huirong Technology Co	lssytech-000	000	2026-03-18
691	CN	Lishui Shuyan Huirong Technology Co	lssytech-001	001	2026-09-16
692	RU	Lomonosov Moscow State University	intsysmsu-000	000	2019-06-18
693	RU	Lomonosov Moscow State University	intsysmsu-001	001	2019-10-22
694	RU	Lomonosov Moscow State University	intsysmsu-002	002	2020-03-12
695	IN	Lookman Electroplast Industries	lookman-002	002	2018-06-13
696	IN	Lookman Electroplast Industries	lookman-004	004	2019-06-03
697	US	Luxand Inc	luxand-000	000	2019-11-07
698	US	Luxand Inc	luxand-001	001	2023-10-17
699	US	Luxand Inc	luxand-003	003	2024-03-25
700	US	Luxand Inc	luxand-004	004	2024-08-01

Table 10: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
701	VN	MB Bank	mbbankvn-000	000	2025-02-12
702	RU	MVision	mvision-001	001	2019-11-12
703	IN	Mantra Softech India	mantra-000	000	2021-10-28
704	ID	Matrix Comsec Pvt Ltd	matrixcomsec-000	000	2025-06-11
705	ID	Matrix Comsec Pvt Ltd	matrixcomsec-001	001	2025-11-26
706	ID	Matrix Comsec Pvt Ltd	matrixcomsec-002	002	2026-05-21
707	AE	Maverick Research	maverickresearch-000	000	2024-12-26
708	CN	Maxvision Technology	maxvision-000	000	2021-10-27
709	CN	Maxvision Technology	maxvision-001	001	2022-03-03
710	CN	Maxvision Technology	maxvision-002	002	2022-07-12
711	CN	Maxvision Technology	maxvision-003	003	2022-11-14
712	CN	Maxvision Technology	maxvision-004	004	2023-05-08
713	CN	Maxvision Technology	maxvision-005	005	2023-09-21
714	CN	Maxvision Technology	maxvision-006	006	2024-01-19
715	CN	Megvii/Face++	megvii-000	000	2018-02-13
716	CN	Megvii/Face++	megvii-001	001	2018-06-15
717	CN	Megvii/Face++	megvii-002	002	2018-10-19
718	CN	Megvii/Face++	megvii-003	003	2021-03-08
719	CN	Megvii/Face++	megvii-004	004	2021-11-19
720	CN	Megvii/Face++	megvii-005	005	2022-03-28
721	CN	Megvii/Face++	megvii-006	006	2022-08-08
722	CN	Megvii/Face++	megvii-007	007	2023-05-17
723	CN	Megvii/Face++	megvii-008	008	2023-10-12
724	CN	Megvii/Face++	megvii-009	009	2024-02-12
725	KR	Metsakuur	metsakuurcompany-001	001	2022-05-12
726	KR	Metsakuur	metsakuurcompany-002	002	2022-09-14
727	KR	Metsakuur	metsakuurcompany-003	003	2023-04-04
728	KR	Metsakuur	metsakuurcompany-004	004	2024-10-24
729	KR	Metsakuur	metsakuurcompany-005	005	2025-07-31
730	KR	Metsakuur	metsakuurcompany-006	006	2026-09-17
731	CN	Miaxis Biometrics	miaxis-001	001	2022-11-15
732	CN	Miaxis Biometrics	miaxis-002	002	2023-03-22
733	CN	Miaxis Biometrics	miaxis-003	003	2023-09-19
734	CN	Miaxis Biometrics	miaxis-004	004	2025-12-09
735	CN	Miaxis Biometrics	miaxis-005	005	2026-07-24
736	GB	MicroFocus	microfocus-001	001	2018-06-13
737	GB	MicroFocus	microfocus-002	002	2018-10-17
738	GB	MicroFocus	microfocus-003	003	2023-02-23
739	GB	Microfocus	microfocus-000	000	2018-06-13
740	CN	Minivision	minivision-000	000	2020-10-28
741	GB	Mitek Systems	mitek-000	000	2023-01-27
742	GB	Mitek Systems	mitek-001	001	2024-07-02
743	GB	Mitek Systems	mitek-002	002	2024-11-19
744	NO	Mobai	mobai-000	000	2020-08-26
745	NO	Mobai	mobai-001	001	2021-02-17
746	NO	Mobai	mobai-002	002	2025-07-07
747	NO	Mobai	mobai-003	003	2026-04-13
748	ES	Mobbeel Solutions	mobbl-000	000	2021-01-28
749	ES	Mobbeel Solutions	mobbl-001	001	2021-06-16
750	ES	Mobbeel Solutions	mobbl-002	002	2021-12-16
751	ES	Mobbeel Solutions	mobbl-003	003	2022-04-19
752	ES	Mobbeel Solutions	mobbl-004	004	2024-05-20
753	ES	Mobbeel Solutions	mobbl-005	005	2024-10-15
754	KR	Mobipin Technology	mobipintech-000	000	2021-11-23
755	TH	Momentum Digital	sertis-000	000	2019-10-07
756	TH	Momentum Digital	sertis-001	001	2020-07-30
757	TH	Momentum Digital	sertis-002	002	2021-05-13
758	TH	Momentum Digital	sertis-003	003	2023-12-27
759	CN	MoreDian Technology	moredian-000	000	2021-02-24
760	US	Mukh Technologies	mukh-001	001	2022-03-22
761	US	Mukh Technologies	mukh-002	002	2022-11-01
762	US	Mukh Technologies	mukh-003	003	2023-03-15
763	US	Mukh Technologies	mukh-004	004	2023-09-13
764	US	Mukh Technologies	mukh-005	005	2024-12-05
765	CN	Multi-Modality Intelligence	multimodality-000	000	2021-10-19
766	CN	Multi-Modality Intelligence	multimodality-001	001	2022-05-16
767	RU	N-Tech Lab	ntechlab-000	000	2017-03-13
768	RU	N-Tech Lab	ntechlab-001	001	2017-05-10
769	RU	N-Tech Lab	ntechlab-002	002	2017-08-23
770	RU	N-Tech Lab	ntechlab-003	003	2017-12-29

Table 11: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
771	RU	N-Tech Lab	ntechlab-004	004	2018-06-14
772	RU	N-Tech Lab	ntechlab-005	005	2018-10-19
773	RU	N-Tech Lab	ntechlab-006	006	2019-03-01
774	RU	N-Tech Lab	ntechlab-007	007	2019-06-25
775	RU	N-Tech Lab	ntechlab-008	008	2020-01-06
776	RU	N-Tech Lab	ntechlab-009	009	2020-12-30
777	RU	N-Tech Lab	ntechlab-010	010	2021-04-30
778	RU	N-Tech Lab	ntechlab-011	011	2021-09-13
779	RU	N-Tech Lab	ntechlab-012	012	2022-01-20
780	SG	NCS Pte Ltd	ncssg-001	001	2023-03-24
781	SG	NCS Pte Ltd	ncssg-002	002	2024-06-25
782	SG	NCS Pte Ltd	ncssg-003	003	2024-10-30
783	SG	NCS Pte Ltd	ncssg-004	004	2025-04-03
784	CA	NEO Systems	neosystems-001	001	2021-03-02
785	CA	NEO Systems	neosystems-002	002	2021-07-03
786	CA	NEO Systems	neosystems-003	003	2021-11-11
787	CA	NEO Systems	neosystems-004	004	2022-05-02
788	KR	NHN Corp	nhn-001	001	2021-03-15
789	KR	NHN Corp	nhn-002	002	2021-07-15
790	KR	NHN Corp	nhn-003	003	2022-02-22
791	KR	NHN Corp	nhn-004	004	2023-06-26
792	KR	NHN Corp	nhn-005	005	2023-12-06
793	KR	NSENSE Corp	nsensecorp-001	001	2020-10-20
794	KR	NSENSE Corp	nsensecorp-002	002	2021-05-06
795	KR	NSENSE Corp	nsensecorp-003	003	2021-10-29
796	KR	NSENSE Corp	nsensecorp-004	004	2022-09-08
797	KR	NSENSE Corp	nsensecorp-005	005	2023-02-08
798	CN	Nanjing Kiwi Network Technology	kiwitech-000	000	2021-03-19
799	KR	Naver Corp	clova-000	000	2020-10-21
800	KR	Neosecu Co	openface-001	001	2021-06-15
801	TW	Netbridge Technology Incoporation	netbridgetech-001	001	2020-01-08
802	TW	Netbridge Technology Incoporation	netbridgetech-002	002	2020-08-11
803	LT	Neurotechnology	neurotechnology-000	000	2017-03-22
804	LT	Neurotechnology	neurotechnology-001	001	2017-08-07
805	LT	Neurotechnology	neurotechnology-002	002	2017-11-02
806	LT	Neurotechnology	neurotechnology-003	003	2018-02-02
807	LT	Neurotechnology	neurotechnology-004	004	2018-10-19
808	LT	Neurotechnology	neurotechnology-005	005	2019-03-01
809	LT	Neurotechnology	neurotechnology-006	006	2019-06-26
810	LT	Neurotechnology	neurotechnology-008	008	2020-01-08
811	LT	Neurotechnology	neurotechnology-009	009	2020-07-07
812	LT	Neurotechnology	neurotechnology-010	010	2020-11-26
813	LT	Neurotechnology	neurotechnology-011	011	2021-03-26
814	LT	Neurotechnology	neurotechnology-012	012	2021-07-26
815	LT	Neurotechnology	neurotechnology-013	013	2022-01-07
816	LT	Neurotechnology	neurotechnology-015	015	2022-06-07
817	LT	Neurotechnology	neurotechnology-016	016	2023-02-01
818	LT	Neurotechnology	neurotechnology-017	017	2023-07-28
819	LT	Neurotechnology	neurotechnology-018	018	2024-02-09
820	LT	Neurotechnology	neurotechnology-019	019	2024-07-31
821	LT	Neurotechnology	neurotechnology-020	020	2025-08-01
822	LT	Neurotechnology	neurotechnology-021	021	2026-09-17
823	IN	NextMeta Labs	nextmetalabs-001	001	2024-07-02
824	IN	NextMeta Labs	nextmetalabs-002	002	2024-12-04
825	IN	NextMeta Labs	nextmetalabs-003	003	2025-07-08
826	IN	NextMeta Labs	nextmetalabs-004	004	2026-01-27
827	IN	Nmetics Private Ltd	nmetics-001	001	2026-09-16
828	US	Noblis	noblis-000	000	2017-07-05
829	ID	Nodeflux	nodeflux-000	000	2019-03-01
830	ID	Nodeflux	nodeflux-001	001	2019-03-01
831	ID	Nodeflux	nodeflux-002	002	2019-08-13
832	LT	Nominder	nominder-000	000	2023-02-06
833	LT	Nominder	nominder-001	001	2023-07-19
834	LT	Nominder	nominder-002	002	2023-11-04
835	LT	Nominder	nominder-003	003	2024-03-06
836	LT	Nominder	nominder-004	004	2024-07-11
837	LT	Nominder	nominder-005	005	2024-11-12
838	LT	Nominder	nominder-006	006	2025-04-07
839	LT	Nominder	nominder-007	007	2026-02-10
840	LT	Nominder	nominder-008	008	2026-07-20

Table 12: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
841	SG	Northern Point Technology Pvt Ltd	npt-000	000	2026-09-10
842	NO	Norwegian University of Science and Technology	ntnu-000	000	2026-08-18
843	IN	NotionTag Technologies Private Limited	notiontag-000	000	2019-06-12
844	IN	NotionTag Technologies Private Limited	notiontag-001	001	2021-03-04
845	IN	NotionTag Technologies Private Limited	notiontag-002	002	2021-09-17
846	AE	OCR Studio L.L.C.-FZ	ocrstudio-001	001	2026-07-24
847	IN	OORU Digital Private Ltd	oorudigital-000	000	2026-07-22
848	US	OOTO AI	ooto-001	001	2025-09-11
849	US	Omnigarde Ltd	omnigarde-000	000	2021-04-05
850	US	Omnigarde Ltd	omnigarde-001	001	2021-08-23
851	US	Omnigarde Ltd	omnigarde-002	002	2022-01-19
852	US	Omnigarde Ltd	omnigarde-003	003	2023-05-10
853	US	Omnigarde Ltd	omnigarde-004	004	2024-06-24
854	US	Omnigarde Ltd	omnigarde-005	005	2025-06-02
855	KR	One More Security	omface-000	000	2021-12-15
856	KR	One More Security	omface-001	001	2022-10-21
857	GB	Onfido	onfido-000	000	2022-12-13
858	GB	Onfido	onfido-001	001	2025-07-17
859	GB	Onfido	onfido-002	002	2025-12-16
860	GB	Onfido	onfido-003	003	2026-04-22
861	GB	Onfido	onfido-004	004	2026-08-28
862	VN	Online Mobile Services JSC	momovn-001	001	2024-01-26
863	VN	Online Mobile Services JSC	momovn-002	002	2024-06-04
864	VN	Online Mobile Services JSC	momovn-003	003	2025-09-19
865	IN	Openedge Technologies Pvt	openedge-000	000	2024-01-29
866	AE	Ovision	ovisionllc-001	001	2024-04-03
867	AE	Ovision	ovisionllc-002	002	2024-08-05
868	RU	Oz Forensics LLC	oz-001	001	2020-07-29
869	RU	Oz Forensics LLC	oz-002	002	2021-01-18
870	RU	Oz Forensics LLC	oz-003	003	2021-08-09
871	RU	Oz Forensics LLC	oz-004	004	2021-12-13
872	TW	PAPAGO Inc	papago-001	001	2022-07-19
873	CZ	PAPIL11 S.R.O.	papil11-000	000	2024-01-04
874	CZ	PAPIL11 S.R.O.	papil11-001	001	2024-08-06
875	CZ	PAPIL11 S.R.O.	papil11-002	002	2025-01-02
876	CZ	PAPIL11 S.R.O.	papil11-003	003	2025-04-29
877	CZ	PAPIL11 S.R.O.	papil11-004	004	2026-03-31
878	KR	PNPSecure	pnpsecure-000	000	2025-06-05
879	KR	PNPSecure	pnpsecure-001	001	2026-03-03
880	KR	PNPSecure	pnpsecure-002	002	2026-09-10
881	ID	PT Autentika Digital Indonesia	autentika-000	000	2022-12-05
882	ID	PT Autentika Digital Indonesia	autentika-001	001	2023-04-05
883	ID	PT Autentika Digital Indonesia	autentika-002	002	2023-11-21
884	ID	PT Autentika Digital Indonesia	autentika-003	003	2024-11-12
885	ID	PT Ice House	icehouse-000	000	2026-02-27
886	ID	PT Qlue Performa Indonesia	qluevision-001	001	2022-11-15
887	ID	PT Technology Biometric Indonesia	techbiometricid-001	001	2025-01-02
888	ID	PT Technology Biometric Indonesia	techbiometricid-002	002	2026-04-09
889	CH	PXL Vision AG	pxl-001	001	2020-06-30
890	TW	Palit Microsystems	palit-000	000	2022-05-16
891	TW	Palit Microsystems	palit-001	001	2022-09-26
892	SG	Panasonic R+D Center Singapore	psl-000	000	2018-06-18
893	SG	Panasonic R+D Center Singapore	psl-001	001	2018-10-12
894	SG	Panasonic R+D Center Singapore	psl-002	002	2019-02-28
895	SG	Panasonic R+D Center Singapore	psl-003	003	2019-10-01
896	SG	Panasonic R+D Center Singapore	psl-004	004	2020-03-03
897	SG	Panasonic R+D Center Singapore	psl-005	005	2020-07-06
898	SG	Panasonic R+D Center Singapore	psl-006	006	2020-11-13
899	SG	Panasonic R+D Center Singapore	psl-007	007	2021-03-19
900	SG	Panasonic R+D Center Singapore	psl-008	008	2021-07-21
901	SG	Panasonic R+D Center Singapore	psl-009	009	2021-12-08
902	SG	Panasonic R+D Center Singapore	psl-010	010	2022-04-19
903	SG	Panasonic R+D Center Singapore	psl-011	011	2022-10-06
904	SG	Panasonic R+D Center Singapore	psl-012	012	2023-10-13
905	SG	Panasonic R+D Center Singapore	psl-013	013	2024-04-18
906	SG	Panasonic R+D Center Singapore	psl-014	014	2024-09-11
907	US	Pangiam	pangiam-000	000	2022-04-04
908	US	Pangiam	pangiam-001	001	2023-02-21
909	US	Pangiam	pangiam-002	002	2023-06-21
910	TR	Papilon Savunma	papsav1923-001	001	2021-03-10

Table 13: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
911	TR	Papilon Savunma	papsav1923-002	002	2022-01-20
912	TR	Papilon Savunma	papsav1923-003	003	2022-11-25
913	US	Paravision	everai-000	000	2018-10-19
914	US	Paravision	everai-001	001	2018-10-19
915	US	Paravision	everai-002	002	2019-03-01
916	US	Paravision	paravision	003	2019-07-01
917	US	Paravision	paravision-004	004	2019-12-11
918	US	Paravision	paravision-006	006	2021-02-01
919	US	Paravision	paravision-008	008	2021-06-30
920	US	Paravision	paravision-010	010	2022-02-02
921	US	Paravision	paravision-011	011	2022-12-12
922	US	Paravision	paravision-013	013	2023-05-08
923	US	Paravision	paravision-015	015	2024-09-09
924	US	Paravision	paravision-017	017	2025-01-13
925	US	Paravision	paravision-018	018	2025-06-12
926	US	Paravision	paravision-020	020	2025-12-11
927	US	Paravision	paravision-022	022	2026-09-24
928	SG	Pensees Pte	pensees-001	001	2020-08-17
929	PL	Pentacomp Systemy Informatyczne	pentacomp-000	000	2024-09-26
930	CN	Ping An Technology	pa-002	002	2017-09-25
931	CN	Ping An Technology	pa-003	003	2017-05-25
932	US	Private Identity LLC	privid-001	001	2023-02-06
933	US	Private Identity LLC	privid-002	002	2023-10-03
934	US	Private Identity LLC	privid-003	003	2024-09-24
935	US	Private Identity LLC	privid-004	004	2025-03-05
936	IN	Pyramid Cyber Security + Forensic (P)	pyramid-000	000	2019-11-04
937	IN	Pyramid Cyber Security + Forensic (P)	pyramid-001	001	2024-11-26
938	KZ	Qaz Biometric Systems	qazbs-000	000	2022-06-22
939	KZ	QazSmartVision.AI	qazsmartvisionai-000	000	2024-01-11
940	KZ	QazSmartVision.AI	qazsmartvisionai-001	001	2024-05-21
941	KZ	QazSmartVision.AI	qazsmartvisionai-002	002	2024-09-25
942	KZ	QazSmartVision.AI	qazsmartvisionai-003	003	2025-01-30
943	KZ	QazSmartVision.AI	qazsmartvisionai-004	004	2025-07-18
944	TW	Qnap Security	qnap-000	000	2021-08-09
945	TW	Qnap Security	qnap-001	001	2021-12-09
946	TW	Qnap Security	qnap-002	002	2022-04-15
947	TW	Qnap Security	qnap-003	003	2022-12-09
948	TW	Qnap Security	qnap-004	004	2023-05-05
949	TW	Qnap Security	qnap-005	005	2023-11-16
950	TW	Qnap Security	qnap-006	006	2024-06-14
951	CZ	Quantasoft	quantasoft-003	003	2021-04-19
952	US	ROC	rankone-000	000	2017-03-21
953	US	ROC	rankone-001	001	2017-04-12
954	US	ROC	rankone-002	002	2017-08-18
955	US	ROC	rankone-003	003	2017-11-28
956	US	ROC	rankone-004	004	2018-06-12
957	US	ROC	rankone-005	005	2018-10-09
958	US	ROC	rankone-006	006	2019-02-27
959	US	ROC	rankone-007	007	2019-06-03
960	US	ROC	rankone-008	008	2019-11-12
961	US	ROC	rankone-009	009	2020-06-26
962	US	ROC	rankone-010	010	2020-11-05
963	US	ROC	rankone-011	011	2021-08-27
964	US	ROC	rankone-012	012	2021-12-27
965	US	ROC	rankone-013	013	2022-07-09
966	US	ROC	rankone-014	014	2022-12-21
967	US	ROC	rankone-015	015	2023-07-20
968	US	ROC	roc-016	016	2023-12-19
969	US	ROC	roc-017	017	2024-06-24
970	US	ROC	roc-018	018	2025-01-30
971	US	ROC	roc-019	019	2025-07-21
972	US	ROC	roc-020	020	2026-02-06
973	US	ROC	roc-021	021	2026-08-05
974	US	RealSense	realsense-001	001	2025-12-01
975	US	RealSense	realsense-002	002	2026-06-09
976	US	Realnetworks Inc	realnetworks-000	000	2018-05-30
977	US	Realnetworks Inc	realnetworks-001	001	2018-10-19
978	US	Realnetworks Inc	realnetworks-002	002	2019-02-28
979	US	Realnetworks Inc	realnetworks-003	003	2019-06-12
980	US	Realnetworks Inc	realnetworks-004	004	2021-04-15

Table 14: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
981	US	Realnetworks Inc	realnetworks-005	005	2021-09-27
982	US	Realnetworks Inc	realnetworks-006	006	2022-02-09
983	US	Realnetworks Inc	realnetworks-007	007	2022-06-14
984	US	Realnetworks Inc	realnetworks-008	008	2022-11-10
985	AE	Recognito	recognito-000	000	2023-05-24
986	AE	Recognito	recognito-001	001	2023-09-27
987	US	Regula Forensics	regula-000	000	2021-04-13
988	US	Regula Forensics	regula-001	001	2021-12-14
989	US	Regula Forensics	regula-002	002	2025-09-15
990	US	Regula Forensics	regula-003	003	2026-08-20
991	AM	Regular Biometrics Solutions	rebs-000	000	2023-08-22
992	AM	Regular Biometrics Solutions	rebs-001	001	2023-12-22
993	CN	Remark Holdings	remarkai-000	000	2019-03-01
994	CN	Remark Holdings	remarkai-001	001	2019-03-01
995	CN	Remark Holdings	remarkai-002	002	2019-11-21
996	CN	Remark Holdings	remarkai-003	003	2021-06-22
997	SG	Rendip	rendip-000	000	2021-04-19
998	GB	Reveal Media Ltd	revealmedia-005	005	2021-09-24
999	GB	Reveal Media Ltd	revealmedia-006	006	2022-01-26
1000	CN	Rokid Corporation	rokid-000	000	2019-08-01
1001	CN	Rokid Corporation	rokid-001	001	2019-12-13
1002	KR	SK Telecom	sktelecom-000	000	2021-07-09
1003	KR	SQIsoft	sqisoft-001	001	2021-07-27
1004	KR	SQIsoft	sqisoft-002	002	2021-11-03
1005	KR	SQIsoft	sqisoft-003	003	2022-10-26
1006	SG	ST Engineering	stengg-000	000	2023-06-06
1007	SA	STCON LLC	stcon-000	000	2022-11-02
1008	SA	STCON LLC	stcon-001	001	2023-03-03
1009	SA	STCON LLC	stcon-002	002	2023-07-05
1010	SA	STCON LLC	stcon-003	003	2023-11-07
1011	SA	STCON LLC	stcon-004	004	2024-06-18
1012	SA	STCON LLC	stcon-005	005	2024-11-13
1013	SA	STCON LLC	stcon-006	006	2025-03-17
1014	SA	STCON LLC	stcon-007	007	2025-07-18
1015	SA	STCON LLC	stcon-008	008	2025-12-30
1016	SA	STCON LLC	stcon-009	009	2026-05-01
1017	SA	STCON LLC	stcon-010	010	2026-09-11
1018	DE	Saffe	saffe-001	001	2018-10-19
1019	DE	Saffe	saffe-002	002	2019-03-01
1020	JP	Saga Densan Center Co Ltd	sdc-000	000	2022-10-18
1021	KR	Samsung S1 Corp	s1-001	001	2019-12-06
1022	KR	Samsung S1 Corp	s1-002	002	2021-03-24
1023	KR	Samsung S1 Corp	s1-003	003	2021-08-24
1024	KR	Samsung S1 Corp	s1-004	004	2022-01-04
1025	KR	Samsung S1 Corp	s1-005	005	2022-06-17
1026	KR	Samsung S1 Corp	s1-006	006	2022-10-17
1027	KR	Samsung S1 Corp	s1-007	007	2023-03-20
1028	KR	Samsung S1 Corp	s1-008	008	2024-10-16
1029	KR	Samsung S1 Corp	s1-009	009	2025-02-25
1030	KR	Samsung-SDS	samsungsds-000	000	2021-10-28
1031	KR	Samsung-SDS	samsungsds-001	001	2022-04-18
1032	KR	Samsung-SDS	samsungsds-002	002	2022-09-16
1033	IN	Samtech InfoNet Limited	samtech-000	000	2017-05-02
1034	IN	Samtech InfoNet Limited	samtech-001	001	2019-10-15
1035	IN	Samtech InfoNet Limited	samtech-002	002	2023-07-06
1036	RU	Satellite Innovation/Eocortex	eocortex-000	000	2020-08-26
1037	IN	ScanalitiX	scanalitix-001	001	2026-05-20
1038	IL	Scanovate	scanovate-001	001	2019-11-12
1039	IL	Scanovate	scanovate-002	002	2020-06-26
1040	IL	Scanovate	scanovate-003	003	2021-11-15
1041	NG	Seamfix	seamfix-001	001	2024-03-08
1042	RO	Securif AI	securifai-001	001	2020-10-06
1043	RO	Securif AI	securifai-002	002	2021-03-19
1044	RO	Securif AI	securifai-003	003	2021-08-03
1045	RO	Securif AI	securifai-004	004	2021-12-21
1046	RO	Securif AI	securifai-005	005	2022-05-16
1047	RO	Securif AI	securifai-006	006	2022-11-14
1048	RO	Securif AI	securifai-007	007	2023-08-17
1049	RO	Securif AI	securifai-008	008	2024-02-26
1050	TH	Security Pitch Co Ltd	securitypitch-000	000	2026-08-19

Table 15: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1051	CN	Sensetime Group	sensetime-001	001	2018-10-19
1052	CN	Sensetime Group	sensetime-002	002	2018-10-19
1053	CN	Sensetime Group	sensetime-003	003	2019-06-04
1054	CN	Sensetime Group	sensetime-004	004	2020-11-20
1055	CN	Sensetime Group	sensetime-005	005	2021-05-24
1056	CN	Sensetime Group	sensetime-006	006	2021-12-28
1057	CN	Sensetime Group	sensetime-007	007	2022-06-17
1058	CN	Sensetime Group	sensetime-008	008	2023-01-04
1059	CN	Sensetime Group	sensetime-010	010	2024-12-03
1060	CN	Sensetime Group	sensetime-011	011	2026-03-17
1061	CN	Sensetime Group	sensetime-012	012	2026-08-10
1062	UZ	Serendipity Ltd	serendipity-000	000	2023-05-31
1063	SG	Seventh Sense Artificial Intelligence	seventhsense-000	000	2021-06-29
1064	SG	Seventh Sense Artificial Intelligence	seventhsense-001	001	2022-03-04
1065	SG	Seventh Sense Artificial Intelligence	seventhsense-002	002	2022-10-17
1066	SG	Seventh Sense Artificial Intelligence	seventhsense-003	003	2023-05-31
1067	SG	Seventh Sense Artificial Intelligence	seventhsense-005	005	2023-10-05
1068	SG	Seventh Sense Artificial Intelligence	seventhsense-006	006	2024-07-09
1069	US	Shaman Software	shaman-000	000	2017-12-05
1070	US	Shaman Software	shaman-001	001	2018-01-13
1071	CN	Shanghai Jiao Tong University	sjtu-001	001	2019-09-27
1072	CN	Shanghai Jiao Tong University	sjtu-002	002	2020-02-12
1073	CN	Shanghai Jiao Tong University	sjtu-003	003	2020-11-02
1074	CN	Shanghai Jiao Tong University	sjtu-004	004	2021-05-13
1075	CN	Shanghai Jiao Tong University	sjtu-005	005	2023-05-05
1076	CN	Shanghai Jiao Tong University	sjtu-006	006	2024-12-06
1077	CN	Shanghai Jiao Tong University	sjtu-007	007	2025-04-16
1078	CN	Shanghai Jiao Tong University	sjtu-008	008	2025-12-11
1079	CN	Shanghai Jiao Tong University	sjtu-009	009	2026-04-13
1080	CN	Shanghai Ulucu Electronics Technology	uluface-002	002	2019-07-10
1081	CN	Shanghai Ulucu Electronics Technology	uluface-003	003	2019-11-12
1082	CN	Shanghai University - Shanghai Film Academy	shu-001	001	2019-06-17
1083	CN	Shanghai University - Shanghai Film Academy	shu-002	002	2019-12-10
1084	CN	Shanghai University - Shanghai Film Academy	shu-003	003	2020-06-24
1085	CN	Shanghai Yitu Technology	yitu-000	000	2017-05-23
1086	CN	Shanghai Yitu Technology	yitu-001	001	2018-06-12
1087	CN	Shanghai Yitu Technology	yitu-002	002	2018-10-19
1088	CN	Shanghai Yitu Technology	yitu-003	003	2019-03-01
1089	CN	Shenzhen AiMall Tech	aimall-001	001	2019-11-12
1090	CN	Shenzhen AiMall Tech	aimall-002	002	2020-03-12
1091	CN	Shenzhen AiMall Tech	aimall-003	003	2020-08-12
1092	CN	Shenzhen EI Networks	einetworks-000	000	2019-08-13
1093	CN	Shenzhen Inst Adv Integrated Tech CAS	siat-002	002	2018-06-13
1094	CN	Shenzhen Inst Adv Integrated Tech CAS	siat-004	004	2019-03-01
1095	CN	Shenzhen Inst Adv Integrated Tech CAS	siat-005	005	2022-02-08
1096	CN	Shenzhen Intellifusion Technologies	intellifusion-001	001	2019-08-22
1097	CN	Shenzhen Intellifusion Technologies	intellifusion-002	002	2020-03-18
1098	CN	Shenzhen Polytechnic University	szpu-000	000	2025-03-28
1099	CN	Shenzhen University-Macau University of Science and Technology	sztu-000	000	2020-12-17
1100	CN	Shenzhen University-Macau University of Science and Technology	sztu-001	001	2021-07-13
1101	GB	Shufti	shufti-001	001	2026-03-17
1102	QA	Sightropic	sightropic-000	000	2025-03-26
1103	QA	Sightropic	sightropic-001	001	2025-11-25
1104	QA	Sightropic	sightropic-002	002	2026-06-04
1105	IN	Smarg Technologies Pvt Ltd	smargtech-001	001	2026-03-17
1106	RU	Smart Engines	smartengines-000	000	2021-08-25
1107	RU	Smart Engines	smartengines-001	001	2022-05-31
1108	ES	Smartbiometrik	smartbiometrik-001	001	2022-05-16
1109	TR	Smarvist Teknoloji	smartvist-000	000	2022-05-10
1110	TR	Smarvist Teknoloji	smartvist-001	001	2023-11-02
1111	DE	Smilart	smilart-001	001	2018-02-06
1112	DE	Smilart	smilart-002	002	2018-02-06
1113	DE	Smilart	smilart-003	003	2019-03-01
1114	TR	Sodec App Inc	sodec-000	000	2021-06-02
1115	TR	Softtech Yazilim Teknolojileri Arastirma Gelistirme ve	softtech-000	000	2026-07-30
1116	CN	Sparktek International Pte Ltd	sparktek-000	000	2025-11-24
1117	CN	Sparktek International Pte Ltd	sparktek-001	001	2026-06-23
1118	IN	Sparsh CCTV	sparsh-001	001	2024-01-02
1119	IN	Staqu Technologies	staqu-000	000	2020-07-15
1120	CN	Star Hybrid Limited	starhybrid-001	001	2019-06-19

Table 16: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1121	US	Stellar Technologies	stellartecs-001	001	2026-05-25
1122	CN	Su Zhou NaZhiTianDi intelligent technology	nazhiai-000	000	2020-06-25
1123	IN	Sukshi Technology Innovation	sukshi-000	000	2022-02-13
1124	GB	Sum and Substance Ltd	sumsub-000	000	2025-12-15
1125	GB	Sum and Substance Ltd	sumsub-001	001	2026-07-27
1126	KR	Suprema AI Inc	suprema-000	000	2021-03-31
1127	KR	Suprema AI Inc	suprema-001	001	2021-09-23
1128	KR	Suprema AI Inc	suprema-002	002	2022-02-11
1129	KR	Suprema AI Inc	suprema-003	003	2022-07-20
1130	KR	Suprema AI Inc	suprema-004	004	2023-01-09
1131	KR	Suprema AI Inc	suprema-005	005	2023-10-25
1132	KR	Suprema ID Inc	supremaid-001	001	2021-05-04
1133	KR	Suprema ID Inc	supremaid-002	002	2022-06-24
1134	GB	Swsam Solutions	swsam-001	001	2023-03-13
1135	RU	Synesis	synesis-000	000	2018-02-09
1136	RU	Synesis	synesis-001	001	2018-05-31
1137	RU	Synesis	synesis-003	003	2018-10-19
1138	RU	Synesis	synesis-004	004	2019-03-01
1139	RU	Synesis	synesis-005	005	2019-06-06
1140	RU	Synesis	synesis-006	006	2019-10-10
1141	RU	Synesis	synesis-007	007	2020-06-24
1142	TW	Synology Inc	synology-000	000	2019-10-23
1143	TW	Synology Inc	synology-001	001	2020-02-26
1144	TW	Synology Inc	synology-002	002	2020-08-20
1145	IT	Syscake S.r.l.	syscake-000	000	2024-11-26
1146	BR	T4iSB	t4isb-000	000	2022-01-28
1147	SA	TAHAKOM	tahakom-001	001	2024-10-17
1148	SA	TAHAKOM	tahakom-002	002	2026-06-15
1149	MY	TNG Digital Sdn Bhd	tngd-001	001	2025-03-17
1150	MY	TNG Digital Sdn Bhd	tngd-002	002	2026-03-18
1151	VN	TNI Tech Co	tnitech-000	000	2024-02-05
1152	AE	TOPPAN Security Co	toppansecurity-001	001	2026-03-30
1153	VN	TTI Group	ttigroup-000	000	2024-12-16
1154	CN	TUPU Technology	tuputech-000	000	2019-10-11
1155	TW	Taiwan AI Labs	ailabs-001	001	2019-12-18
1156	TW	Taiwan Computer Vision	tcv-001	001	2024-05-30
1157	TW	Taiwan Mobile	taiwanmobile-001	001	2024-08-06
1158	TW	Taiwan Mobile	taiwanmobile-002	002	2024-12-11
1159	TW	Taiwan-Certificate Authority Incorporation	twface-000	000	2021-05-14
1160	TW	Taiwan-Certificate Authority Incorporation	twface-001	001	2021-09-14
1161	CH	Tech5 SA	tech5-000	000	2018-10-09
1162	CH	Tech5 SA	tech5-001	001	2018-10-19
1163	CH	Tech5 SA	tech5-002	002	2019-03-01
1164	CH	Tech5 SA	tech5-003	003	2019-08-19
1165	CH	Tech5 SA	tech5-004	004	2020-03-09
1166	CH	Tech5 SA	tech5-005	005	2020-07-24
1167	CH	Tech5 SA	tech5-007	007	2022-12-30
1168	CH	Tech5 SA	tech5-008	008	2023-08-11
1169	CH	Tech5 SA	tech5-009	009	2025-04-17
1170	VN	Techainer	techainer-001	001	2023-11-30
1171	VN	Techainer	techainer-002	002	2025-02-14
1172	VN	Techainer	techainer-003	003	2026-06-08
1173	SA	Technology Control Company	tcc-000	000	2026-02-10
1174	SA	Technology Control Company	tcc-001	001	2026-07-13
1175	TR	Techsign	techsign-000	000	2021-08-25
1176	TR	Techsign	techsign-001	001	2022-07-01
1177	TW	Telexper International	telexper-000	000	2024-06-21
1178	CN	Tencent Deepsea Lab	deepsea-001	001	2019-06-03
1179	RU	Tevian	tevian-000	000	2017-12-10
1180	RU	Tevian	tevian-001	001	2018-01-19
1181	RU	Tevian	tevian-002	002	2018-06-14
1182	RU	Tevian	tevian-003	003	2018-10-19
1183	RU	Tevian	tevian-004	004	2019-03-01
1184	RU	Tevian	tevian-005	005	2019-09-21
1185	RU	Tevian	tevian-006	006	2020-09-11
1186	RU	Tevian	tevian-007	007	2021-08-06
1187	RU	Tevian	tevian-008	008	2021-12-06
1188	US	TigerIT Bangladesh Ltd	tiger-001	001	2018-01-23
1189	US	TigerIT Bangladesh Ltd	tiger-002	002	2018-06-13
1190	US	TigerIT Bangladesh Ltd	tiger-003	003	2018-10-16

Table 17: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1191	US	TigerIT Bangladesh Ltd	tiger-004	004	2020-12-01
1192	US	TigerIT Bangladesh Ltd	tiger-005	005	2021-07-29
1193	US	TigerIT Bangladesh Ltd	tiger-006	006	2021-12-13
1194	US	TigerIT Bangladesh Ltd	tiger-007	007	2025-03-05
1195	RU	Tinkoff Bank	tinkoff-001	001	2021-05-13
1196	CN	TongYi Transportation Technology	tongyi-004	004	2018-10-19
1197	CN	TongYi Transportation Technology	tongyi-005	005	2019-06-12
1198	CN	TongYi Transportation Technology	tongyitrans-001	001	2017-04-01
1199	CN	TongYi Transportation Technology	tongyitrans-002	002	2017-07-15
1200	CN	TongYi Transportation Technology	tongyitrans-003	003	2018-05-15
1201	TW	Toppan ID Gate	toppanidgate-000	000	2021-09-28
1202	JP	Toshiba	toshiba-000	000	2018-01-11
1203	JP	Toshiba	toshiba-001	001	2018-01-11
1204	JP	Toshiba	toshiba-002	002	2018-10-19
1205	JP	Toshiba	toshiba-003	003	2019-03-01
1206	JP	Toshiba	toshiba-004	004	2021-09-27
1207	JP	Toshiba	toshiba-005	005	2022-02-09
1208	JP	Toshiba	toshiba-006	006	2022-06-29
1209	JP	Toshiba	toshiba-007	007	2023-09-13
1210	JP	Toshiba	toshiba-008	008	2024-02-01
1211	JP	Toshiba	toshiba-009	009	2024-09-16
1212	JP	Toshiba	toshiba-010	010	2025-02-03
1213	JP	Toshiba	toshiba-011	011	2025-07-08
1214	JP	Toshiba	toshiba-012	012	2026-01-09
1215	JP	Toshiba	toshiba-013	013	2026-09-16
1216	KR	Toss	toss-000	000	2025-09-22
1217	KR	Toss	toss-001	001	2026-08-28
1218	ES	Touchless ID	touchlessid-000	000	2022-05-02
1219	ES	Touchless ID	touchlessid-001	001	2022-09-21
1220	ES	Touchless ID	touchlessid-002	002	2023-01-23
1221	ES	Touchless ID	touchlessid-003	003	2023-07-12
1222	ES	Touchless ID	touchlessid-004	004	2024-12-27
1223	Il	Transmit Security Ltd	transmitsecurity-001	001	2025-12-04
1224	JP	Tripleize	aize-001	001	2021-04-23
1225	JP	Tripleize	aize-002	002	2021-10-08
1226	JP	Tripleize	aize-003	003	2023-08-10
1227	JP	Tripleize	aize-006	006	2026-04-08
1228	VN	TrueID-VNG	trueidvng-001	001	2023-01-05
1229	AE	TrueSight Laboratories Tech LLC	tsight-000	000	2026-05-22
1230	US	Trueface.ai	trueface-000	000	2019-10-08
1231	US	Trueface.ai	trueface-001	001	2020-07-20
1232	US	Trueface.ai	trueface-002	002	2021-03-29
1233	US	Trueface.ai	trueface-003	003	2021-09-30
1234	GB	Trust Stamp	truststamp-001	001	2023-07-14
1235	GB	Trust Stamp	truststamp-002	002	2025-05-23
1236	GB	Trust Stamp	truststamp-003	003	2025-11-19
1237	CN	TuringTech.vip	turingtechvip-001	001	2022-02-03
1238	CN	TuringTech.vip	turingtechvip-002	002	2022-07-27
1239	TR	Turkcell Technology	turkcell-000	000	2022-10-11
1240	TR	Turkcell Technology	turkcell-001	001	2023-03-15
1241	TR	Turkcell Technology	turkcell-002	002	2024-03-18
1242	CN	ULSee Inc	ulsee-001	001	2019-07-31
1243	UN	UNICC-Solution Architecture Section	unicc-001	001	2023-02-17
1244	UN	UNICC-Solution Architecture Section	unicc-002	002	2023-06-07
1245	UN	UNICC-Solution Architecture Section	unicc-003	003	2023-10-16
1246	UN	UNICC-Solution Architecture Section	unicc-004	004	2024-09-23
1247	TW	UXLabs	uxlabs-001	001	2022-09-19
1248	TW	UXLabs	uxlabs-003	003	2023-07-28
1249	UZ	UZINFOCOM	uzinfocom-000	000	2025-07-28
1250	UZ	UZINFOCOM	uzinfocom-001	001	2026-04-08
1251	HU	Ultinous	ultinous-000	000	2017-12-18
1252	HU	Ultinous	ultinous-001	001	2018-02-02
1253	FR	Unissey	unissey-001	001	2021-11-29
1254	FR	Unissey	unissey-002	002	2022-04-29
1255	FR	Unissey	unissey-003	003	2022-12-19
1256	FR	Unissey	unissey-004	004	2023-07-19
1257	FR	Unissey	unissey-005	005	2024-08-06
1258	FR	Unissey	unissey-006	006	2025-09-11
1259	FR	Unissey	unissey-007	007	2026-03-25
1260	CN	Universal Ubiquitous AI Co Ltd	uniubi-000	000	2026-09-24

Table 18: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1261	SG	UniverseAI	universeai-000	000	2026-08-11
1262	BR	Universidade Federal de Goiás	ceiaufgbr-000	000	2026-08-06
1263	PT	Universidade de Coimbra	visteam-000	000	2020-01-14
1264	PT	Universidade de Coimbra	visteam-001	001	2021-03-16
1265	PT	Universidade de Coimbra	visteam-002	002	2021-08-20
1266	PT	Universidade de Coimbra	visteam-003	003	2022-01-31
1267	PT	Universidade de Coimbra	visteam-004	004	2022-08-01
1268	PT	Universidade de Coimbra	visteam-005	005	2023-01-04
1269	PT	Universidade de Coimbra	visteam-006	006	2023-05-25
1270	PT	Universidade de Coimbra	visteam-007	007	2023-10-18
1271	PT	Universidade de Coimbra	visteam-008	008	2024-02-06
1272	PT	Universidade de Coimbra	visteam-009	009	2024-07-26
1273	PT	Universidade de Coimbra	visteam-010	010	2025-05-28
1274	GB	University of Surrey-CVSSP	surrey-cvssp-000	000	2022-03-25
1275	GB	University of Surrey-CVSSP	surrey-cvssp-001	001	2022-09-22
1276	GB	University of Surrey-CVSSP	surrey-cvssp-002	002	2023-02-16
1277	GB	University of Surrey-CVSSP	surrey-cvssp-003	003	2023-09-13
1278	GB	University of Surrey-CVSSP	surrey-cvssp-004	004	2024-07-11
1279	US	VCognition	vcog-002	002	2017-06-12
1280	VN	VNIS Joint Stock	vnis-000	000	2024-01-08
1281	US	Vaidio	vaidio-001	001	2025-12-10
1282	US	Vcortex Labs	vcortex-001	001	2023-06-14
1283	LB	Veertec SAL (Offshore)	veertec-001	001	2026-07-07
1284	ES	Veridas Digital Authentication Solutions S.L.	veridas-001	001	2019-03-01
1285	ES	Veridas Digital Authentication Solutions S.L.	veridas-002	002	2019-03-01
1286	ES	Veridas Digital Authentication Solutions S.L.	veridas-003	003	2019-11-27
1287	ES	Veridas Digital Authentication Solutions S.L.	veridas-004	004	2020-07-21
1288	ES	Veridas Digital Authentication Solutions S.L.	veridas-005	005	2020-12-15
1289	ES	Veridas Digital Authentication Solutions S.L.	veridas-006	006	2021-04-15
1290	ES	Veridas Digital Authentication Solutions S.L.	veridas-007	007	2021-09-02
1291	ES	Veridas Digital Authentication Solutions S.L.	veridas-008	008	2022-10-17
1292	ES	Veridas Digital Authentication Solutions S.L.	veridas-009	009	2023-05-08
1293	ES	Veridas Digital Authentication Solutions S.L.	veridas-010	010	2025-03-18
1294	GB	Veridium	veridium-000	000	2022-03-28
1295	GB	Veridium	veridium-001	001	2022-11-03
1296	GB	Veridium	veridium-002	002	2023-06-30
1297	GB	Veridium	veridium-003	003	2024-02-21
1298	GB	Veridium	veridium-004	004	2026-07-07
1299	KZ	Verigram	verigram-000	000	2021-09-06
1300	KZ	Verigram	verigram-001	001	2022-03-09
1301	KZ	Verigram	verigram-002	002	2023-04-03
1302	KZ	Verigram	verigram-003	003	2023-08-09
1303	KZ	Verigram	verigram-004	004	2026-08-10
1304	ID	Verihubs	verihubs-inteligensia-000	000	2021-07-27
1305	ID	Verihubs	verihubs-inteligensia-001	001	2022-06-16
1306	ID	Verihubs	verihubs-inteligensia-002	002	2023-06-07
1307	ID	Verijelas	verijelas-000	000	2022-08-01
1308	IN	Verismart	verismart-001	001	2026-08-25
1309	TW	Via Technologies Inc	via-000	000	2019-07-08
1310	TW	Via Technologies Inc	via-001	001	2020-01-08
1311	TW	Via Technologies Inc	via-004	004	2023-02-23
1312	TW	Via Technologies Inc	via-005	005	2023-12-26
1313	TW	Via Technologies Inc	via-006	006	2025-02-24
1314	AE	Viante.AI	viante-000	000	2023-06-09
1315	AE	Viante.AI	viante-001	001	2024-12-30
1316	AE	Viante.AI	viante-002	002	2025-06-17
1317	AE	Viante.AI	viante-003	003	2026-03-27
1318	AE	Viante.AI	viante-004	004	2026-08-05
1319	DE	Videmo Intelligente Videoanalyse	videmo-000	000	2019-12-19
1320	DE	Videmo Intelligente Videoanalyse	videmo-001	001	2021-12-22
1321	DE	Videmo Intelligente Videoanalyse	videmo-002	002	2022-08-31
1322	DE	Videmo Intelligente Videoanalyse	videmo-003	003	2024-06-18
1323	IN	Videonetics Technology Pvt	videonetics-001	001	2019-06-19
1324	IN	Videonetics Technology Pvt	videonetics-002	002	2019-11-21
1325	VN	Vietnam National Cyber Security Technology Corporation	ncsgroup-000	000	2024-03-18
1326	VN	Vietnam Payment Solution	vnpay-000	000	2023-12-05
1327	VN	Vietnam Payment Solution	vnpay-001	001	2024-10-15
1328	VN	Vietnam Posts and Telecommunications Group	vnpt-001	001	2021-01-08
1329	VN	Vietnam Posts and Telecommunications Group	vnpt-002	002	2021-06-08
1330	VN	Vietnam Posts and Telecommunications Group	vnpt-003	003	2021-12-01

Table 19: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1331	VN	Vietnam Posts and Telecommunications Group	vnpt-004	004	2022-04-15
1332	VN	Vietnam Posts and Telecommunications Group	vnpt-005	005	2022-08-24
1333	VN	Vietnam Posts and Telecommunications Group	vnpt-006	006	2023-08-18
1334	VN	Vietnam Posts and Telecommunications Group	vnpt-007	007	2024-04-29
1335	VN	Vietnam Posts and Telecommunications Group	vnpt-008	008	2024-09-20
1336	VN	Vietnam Posts and Telecommunications Group	vnpt-009	009	2025-06-04
1337	VN	Viettel Artificial Intelligence	viettelai-000	000	2023-06-26
1338	VN	Viettel Artificial Intelligence	viettelai-001	001	2023-11-20
1339	VN	Viettel Artificial Intelligence	viettelai-002	002	2024-07-17
1340	VN	Viettel Artificial Intelligence	viettelai-003	003	2025-02-27
1341	VN	Viettel Artificial Intelligence and Data Services Center	viettelai-004	004	2025-11-26
1342	VN	Viettel Artificial Intelligence and Data Services Center	viettelai-005	005	2026-05-11
1343	VN	Viettel Artificial Intelligence and Data Services Center	viettelai-006	006	2026-09-11
1344	VN	Viettel Group	vts-000	000	2020-11-04
1345	VN	Viettel Group	vts-001	001	2022-04-20
1346	VN	Viettel Group	vts-002	002	2024-06-04
1347	VN	Viettel High Technology	viettelhightech-000	000	2021-08-04
1348	VN	Viettel High Technology	viettelhightech-001	001	2025-03-18
1349	VN	Viettel High Technology	viettelhightech-002	002	2025-10-15
1350	US	Vigilant Solutions	vigilantsolutions-000	000	2017-03-30
1351	US	Vigilant Solutions	vigilantsolutions-001	001	2017-06-13
1352	US	Vigilant Solutions	vigilantsolutions-002	002	2017-09-28
1353	US	Vigilant Solutions	vigilantsolutions-003	003	2018-01-23
1354	US	Vigilant Solutions	vigilantsolutions-004	004	2018-06-14
1355	US	Vigilant Solutions	vigilantsolutions-005	005	2018-10-11
1356	US	Vigilant Solutions	vigilantsolutions-006	006	2019-03-01
1357	US	Vigilant Solutions	vigilantsolutions-007	007	2019-06-27
1358	US	Vigilant Solutions	vigilantsolutions-008	008	2020-08-03
1359	US	Vigilant Solutions	vigilantsolutions-009	009	2020-12-07
1360	US	Vigilant Solutions	vigilantsolutions-010	010	2021-04-07
1361	US	Vigilant Solutions	vigilantsolutions-011	011	2021-08-07
1362	VN	VinAI Research VietNam	vinai-000	000	2020-09-24
1363	VN	VinBigData	vinbigdata-001	001	2022-01-06
1364	VN	VinBigData	vinbigdata-002	002	2022-06-07
1365	VN	VinBigData	vinbigdata-003	003	2023-06-23
1366	VN	VinBigData	vinbigdata-004	004	2024-05-08
1367	VN	VinBigData	vinbigdata-005	005	2024-10-16
1368	VN	VinBigData	vinbigdata-006	006	2025-02-19
1369	VN	VinBigData	vinbigdata-007	007	2025-07-24
1370	SE	Visage Technologies	visage-000	000	2020-12-09
1371	BR	Visica AI Solutions	visica-001	001	2025-02-14
1372	BR	Visica AI Solutions	visica-002	002	2025-07-15
1373	BR	Visica AI Solutions	visica-003	003	2026-05-06
1374	FI	Visidon	vd-000	000	2018-06-13
1375	FI	Visidon	vd-001	001	2019-02-26
1376	FI	Visidon	vd-002	002	2021-04-12
1377	FI	Visidon	vd-003	003	2021-10-12
1378	CN	Vision Intelligence Center of Meituan	meituan-000	000	2021-05-14
1379	CN	Vision Intelligence Center of Meituan	meituan-001	001	2022-03-25
1380	CN	Vision Intelligence Center of Meituan	meituan-002	002	2022-09-14
1381	CN	Vision Intelligence Center of Meituan	meituan-003	003	2023-03-06
1382	CN	Vision Intelligence Center of Meituan	meituan-004	004	2023-08-16
1383	PT	Vision-Box	visionbox-000	000	2019-02-26
1384	PT	Vision-Box	visionbox-001	001	2019-03-01
1385	PT	Vision-Box	visionbox-002	002	2021-04-29
1386	PT	Vision-Box	visionbox-003	003	2023-02-01
1387	PT	Vision-Box	visionbox-004	004	2023-06-30
1388	PT	Vision-Box	visionbox-005	005	2024-07-29
1389	PT	Vision-Box	visionbox-006	006	2025-07-29
1390	PT	Vision-Box	visionbox-007	007	2026-02-03
1391	PT	Vision-Box	visionbox-008	008	2026-07-27
1392	RU	VisionLabs	visionlabs-001	001	2017-06-12
1393	RU	VisionLabs	visionlabs-002	002	2017-09-08
1394	RU	VisionLabs	visionlabs-003	003	2018-02-02
1395	RU	VisionLabs	visionlabs-004	004	2018-06-15
1396	RU	VisionLabs	visionlabs-005	005	2018-10-19
1397	RU	VisionLabs	visionlabs-006	006	2019-03-01
1398	RU	VisionLabs	visionlabs-007	007	2019-06-12
1399	RU	VisionLabs	visionlabs-008	008	2020-01-06
1400	RU	VisionLabs	visionlabs-009	009	2020-07-27

Table 20: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1401	RU	VisionLabs	visionlabs-010	010	2021-01-25
1402	RU	VisionLabs	visionlabs-011	011	2021-10-13
1403	TW	Vivotek	vivotek-001	001	2024-04-03
1404	TW	Vivotek	vivotek-002	002	2024-08-15
1405	AU	Vixvizon	vixvizion-000	000	2019-06-19
1406	AU	Vixvizon	vixvizion-001	001	2019-10-22
1407	AU	Vixvizon	vixvizion-002	002	2020-12-31
1408	AU	Vixvizon	vixvizion-003	003	2021-05-18
1409	AU	Vixvizon	vixvizion-004	004	2021-09-20
1410	AU	Vixvizon	vixvizion-005	005	2022-03-03
1411	AU	Vixvizon	vixvizion-006	006	2022-08-11
1412	AU	Vixvizon	vixvizion-007	007	2023-01-17
1413	US	Vocalize	vocalize-001	001	2024-03-27
1414	RU	Vocord	vocord-001	001	2017-04-21
1415	RU	Vocord	vocord-002	002	2017-06-07
1416	RU	Vocord	vocord-003	003	2018-01-23
1417	RU	Vocord	vocord-004	004	2018-06-15
1418	RU	Vocord	vocord-005	005	2018-10-19
1419	RU	Vocord	vocord-006	006	2019-03-01
1420	RU	Vocord	vocord-007	007	2019-06-06
1421	RU	Vocord	vocord-008	008	2020-01-31
1422	RU	Vocord	vocord-009	009	2020-12-28
1423	RU	Vocord	vocord-010	010	2021-12-20
1424	BR	Vsoft	vsoft-001	001	2025-08-01
1425	BR	Vsoft	vsoft-002	002	2025-11-28
1426	BR	Vsoft	vsoft-003	003	2026-03-30
1427	BR	Vsoft	vsoft-004	004	2026-07-31
1428	US	Wicket	wicket-000	000	2022-02-14
1429	CN	Winsense	winsense-000	000	2019-06-17
1430	CN	Winsense	winsense-001	001	2019-10-16
1431	CN	Winsense	winsense-002	002	2020-11-20
1432	MY	Wise AI SDN BHD	wiseai-001	001	2022-10-25
1433	CN	Wuhan Tianyu Information Industry	wuhantianyu-001	001	2021-08-05
1434	CN	X-Laboratory	x-laboratory-000	000	2019-09-03
1435	CN	X-Laboratory	x-laboratory-001	001	2020-01-21
1436	CN	Xforward AI Technology	xforwardai-000	000	2020-02-06
1437	CN	Xforward AI Technology	xforwardai-001	001	2020-09-25
1438	CN	Xforward AI Technology	xforwardai-002	002	2021-02-10
1439	CN	Xiamen Meiya Pico Information	meiya-001	001	2019-03-01
1440	CN	Xiamen University	xm-000	000	2020-10-19
1441	PT	YooniK	yoonik-000	000	2020-06-24
1442	PT	YooniK	yoonik-001	001	2020-10-26
1443	PT	YooniK	yoonik-002	002	2021-09-06
1444	PT	YooniK	yoonik-003	003	2022-01-06
1445	PT	YooniK	yoonik-004	004	2023-02-10
1446	GB	Yoti	yoti-001	001	2024-02-13
1447	GB	Yoti	yoti-002	002	2024-07-03
1448	TW	Yuan High-Tech Development	yuan-000	000	2020-06-30
1449	TW	Yuan High-Tech Development	yuan-001	001	2021-01-08
1450	TW	Yuan High-Tech Development	yuan-002	002	2021-05-17
1451	TW	Yuan High-Tech Development	yuan-003	003	2021-09-17
1452	TW	Yuan High-Tech Development	yuan-004	004	2022-01-14
1453	TW	Yuan High-Tech Development	yuan-005	005	2022-06-22
1454	TW	Yuan High-Tech Development	yuan-006	006	2022-12-14
1455	TW	Yuan High-Tech Development	yuan-007	007	2023-05-11
1456	TW	Yuan High-Tech Development	yuan-008	008	2023-11-06
1457	TW	Yuan High-Tech Development	yuan-009	009	2024-04-24
1458	TW	Yuan High-Tech Development	yuan-010	010	2024-10-22
1459	CN	YuanLi Technology	yuanli-001	001	2026-03-23
1460	CN	Yuntu Data and Technology	ytu-000	000	2021-06-16
1461	CN	Zhuhai Yisheng Electronics Technology	yisheng-000	000	2017-08-17
1462	CN	Zhuhai Yisheng Electronics Technology	yisheng-001	001	2017-08-17
1463	CN	Zhuhai Yisheng Electronics Technology	yisheng-002	002	2018-02-01
1464	CN	Zhuhai Yisheng Electronics Technology	yisheng-004	004	2018-06-12
1465	IN	i2v Systems	i2v-001	001	2024-03-11
1466	IN	i2v Systems	i2v-002	002	2025-01-15
1467	VN	iCOMM Media and Tech	icomm-000	000	2023-12-26
1468	CN	iQIYI Inc	iqface-000	000	2019-06-04
1469	CN	iQIYI Inc	iqface-001	001	2019-12-11
1470	CN	iQIYI Inc	iqface-003	003	2021-02-23

Table 21: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1471	TW	iSAP Solution Corporation	isap-001	001	2019-08-07
1472	TW	iSAP Solution Corporation	isap-002	002	2020-09-01
1473	TW	ioNetworks Inc	ionetworks-000	000	2021-07-20
1474	TW	ioNetworks Inc	ionetworks-001	001	2023-08-01
1475	TW	ioNetworks Inc	ionetworks-002	002	2023-12-08
1476	TW	ioNetworks Inc	ionetworks-003	003	2024-05-09
1477	TW	ioNetworks Inc	ionetworks-004	004	2026-04-22
1478	KR	useB	useb-000	000	2023-05-08
1479	KR	useB	useb-001	001	2023-09-12
1480	KR	useB	useb-002	002	2024-01-19
1481	KR	useB	useb-003	003	2024-08-09
1482	KR	useB	useb-004	004	2024-12-23

Table 22: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1	2060io-001	567279	43224	247 629	1468 10488 ± 0	1043 958 ± 22	767 983 ± 38	774 1045 ± 205	1092 1724 ± 310	1118 3474 ± 1170	1143 7105 ± 42	1139 7071 ± 47
2	2060io-002	579607	42980	659 1559	1469 10488 ± 0	354 398 ± 0	242 418 ± 1	305 525 ± 1	502 786 ± 1	-	1139 7000 ± 91	1133 6965 ± 85
3	20face-000	117155	324083	610 1417	579 2048 ± 0	163 232 ± 1	96 223 ± 1	77 226 ± 4	62 222 ± 1	44 224 ± 1	1392 44880 ± 134	1391 44462 ± 163
4	20face-001	226824	324119	639 1512	1241 4096 ± 0	208 279 ± 2	117 266 ± 1	91 266 ± 1	80 267 ± 1	61 267 ± 0	1087 5553 ± 54	1085 5541 ± 65
5	3divi-000	169360	139460	-	110 512 ± 0	217 285 ± 52	-	-	-	-	111 378 ± 20	115 375 ± 19
6	3divi-001	190867	139473	-	1220 4096 ± 0	200 274 ± 47	-	-	-	-	241 636 ± 19	253 634 ± 16
7	3divi-002	190867	175117	-	1225 4096 ± 0	207 279 ± 48	-	-	-	-	276 692 ± 22	300 707 ± 26
8	3divi-003	191636	55903	-	1279 4096 ± 0	674 650 ± 90	-	-	-	-	231 627 ± 11	240 623 ± 32
9	3divi-004	263670	50794	-	705 2048 ± 0	1072 984 ± 131	-	-	-	-	333 794 ± 35	346 801 ± 40
10	3divi-005	264098	53870	-	624 2048 ± 0	1081 993 ± 83	885 1136 ± 88	991 1309 ± 137	941 1348 ± 131	980 1671 ± 166	329 790 ± 20	337 791 ± 23
11	3divi-006	273866	52656	550 1252	487 2048 ± 0	684 654 ± 1	464 651 ± 0	435 660 ± 1	401 678 ± 2	388 759 ± 13	325 775 ± 19	329 770 ± 22
12	3divi-007	483115	24723	563 1288	756 2048 ± 0	627 615 ± 1	420 616 ± 1	387 623 ± 1	364 644 ± 1	362 727 ± 5	288 707 ± 31	306 712 ± 25
13	accurascan-001	14057	24437	410 944	142 512 ± 0	140 204 ± 27	91 215 ± 17	74 220 ± 16	70 237 ± 18	84 339 ± 19	1266 15634 ± 54	1268 15647 ± 77
14	accurascan-002	14057	24437	412 945	108 512 ± 0	100 173 ± 23	69 188 ± 14	61 191 ± 14	55 208 ± 16	72 317 ± 14	1227 12255 ± 69	1229 12231 ± 72
15	accurascan-003	998397	27018	769 1996	854 2048 ± 0	716 676 ± 1	494 677 ± 7	478 700 ± 8	471 751 ± 16	529 950 ± 47	168 540 ± 10	171 536 ± 10
16	acer-000	109735	88323	-	566 2048 ± 0	157 222 ± 0	102 233 ± 4	82 238 ± 4	78 262 ± 18	95 356 ± 46	433 1065 ± 40	460 1109 ± 35
17	acer-001	36650	66086	137 417	136 512 ± 0	136 199 ± 0	103 237 ± 28	79 229 ± 26	72 242 ± 37	57 259 ± 21	808 2453 ± 44	813 2461 ± 62
18	acer-002	43922	624858	71 192	965 2048 ± 0	115 184 ± 0	67 184 ± 0	52 185 ± 0	43 185 ± 0	30 186 ± 0	938 3370 ± 47	943 3350 ± 54
19	acisw-003	282029	35664	-	1476 18467 ± 8	164 232 ± 1	119 267 ± 22	269 488 ± 28	650 990 ± 24	1106 2977 ± 129	1478 847908 ± 16757	1478 851850 ± 17018
20	acisw-006	282029	36107	-	1475 18465 ± 8	155 219 ± 0	100 227 ± 0	194 410 ± 1	538 838 ± 1	1083 2532 ± 10	1474 548137 ± 16513	1474 549586 ± 9238
21	acisw-007	267619	36111	89 287	471 2048 ± 0	215 283 ± 0	138 293 ± 3	197 414 ± 0	169 404 ± 0	174 484 ± 1	521 1316 ± 22	525 1297 ± 23
22	acisw-008	171703	39359	489 1102	904 2048 ± 0	358 400 ± 1	191 362 ± 28	161 369 ± 9	98 300 ± 2	81 336 ± 5	523 1327 ± 19	534 1337 ± 32
23	adera-001	0	79272	-	1178 2560 ± 0	42 97 ± 0	-	-	-	-	613 1604 ± 71	624 1649 ± 56
24	adera-002	0	749797	-	1415 5120 ± 0	1366 1394 ± 11	1056 1381 ± 1	1049 1393 ± 1	988 1403 ± 1	896 1464 ± 2	744 2163 ± 32	743 2158 ± 28
25	adera-003	0	749778	400 917	1417 5120 ± 0	1352 1381 ± 12	1058 1385 ± 1	1050 1394 ± 1	983 1401 ± 1	898 1469 ± 1	742 2148 ± 34	738 2130 ± 32
26	adera-004	0	959123	713 1748	1427 6144 ± 0	1275 1246 ± 1	943 1204 ± 1	924 1230 ± 2	831 1207 ± 2	749 1254 ± 1	674 1840 ± 34	675 1828 ± 31
27	adera-005	0	959113	707 1734	1423 6144 ± 0	994 914 ± 1	672 879 ± 3	625 880 ± 2	575 886 ± 2	502 910 ± 2	690 1896 ± 14	686 1874 ± 20
28	advance-002	257173	20434	-	956 2048 ± 0	879 811 ± 2	609 803 ± 2	472 696 ± 2	423 699 ± 4	352 718 ± 1	399 987 ± 10	405 988 ± 45
29	advance-003	258867	78699	190 518	697 2048 ± 0	576 586 ± 0	-	350 583 ± 0	312 588 ± 0	252 591 ± 1	673 1813 ± 17	666 1788 ± 26
30	advance-004	803133	954494	275 681	687 2048 ± 0	1156 1099 ± 20	859 1107 ± 15	814 1093 ± 21	737 1103 ± 21	670 1138 ± 21	700 1935 ± 35	704 1936 ± 32
31	advance-005	277607	126218	243 624	778 2048 ± 0	556 573 ± 1	380 574 ± 2	341 573 ± 2	297 574 ± 2	261 601 ± 2	596 1517 ± 16	597 1518 ± 26
32	advance-006	472567	136481	228 605	386 2048 ± 0	710 673 ± 0	480 670 ± 0	449 673 ± 0	405 684 ± 0	380 745 ± 0	344 808 ± 4	353 808 ± 1
33	advance-007	1059702	136627	717 1768	555 2048 ± 0	1257 1224 ± 0	962 1228 ± 0	929 1234 ± 0	861 1236 ± 0	763 1277 ± 0	335 796 ± 2	344 800 ± 1
34	affinsys-ai-000	38016	46159	43 142	140 512 ± 0	15 48 ± 0	11 52 ± 0	24 106 ± 0	21 110 ± 0	35 194 ± 0	713 2007 ± 25	715 2006 ± 28
35	afisbiometrics-000	545886	32882	478 1085	101 512 ± 0	1251 1219 ± 1	884 1135 ± 1	852 1137 ± 2	773 1137 ± 1	677 1147 ± 1	543 1400 ± 29	540 1357 ± 32
36	afrengine-000	151875	382842	61 178	1230 4096 ± 0	52 107 ± 0	33 112 ± 0	101 284 ± 2	422 697 ± 2	1116 3299 ± 17	1413 54329 ± 140	1414 56195 ± 256
37	afrengine-001	410712	391289	437 1015	1267 4096 ± 0	1058 970 ± 0	766 982 ± 1	709 979 ± 2	656 996 ± 0	554 990 ± 4	1412 54240 ± 169	1413 56125 ± 179
38	afrengine-002	579618	239125	440 1018	469 2048 ± 0	928 851 ± 3	827 1065 ± 16	725 993 ± 7	553 855 ± 5	468 857 ± 9	503 1259 ± 13	516 1270 ± 23
39	afrengine-003	579563	239117	438 1017	612 2048 ± 0	1150 1088 ± 1	696 911 ± 15	799 1067 ± 5	535 833 ± 3	553 989 ± 2	517 1294 ± 48	522 1284 ± 29
40	afrengine-004	579587	239117	439 1018	419 2048 ± 0	955 875 ± 2	733 951 ± 119	748 1018 ± 5	582 895 ± 1	592 1043 ± 6	501 1254 ± 17	512 1258 ± 23
41	afrengine-005	694908	239155	441 1018	587 2048 ± 0	906 831 ± 10	754 972 ± 52	588 834 ± 2	578 891 ± 9	618 1065 ± 1	510 1276 ± 19	518 1273 ± 20
42	aifirst-001	224157	808777	167 485	346 2048 ± 0	582 587 ± 2	375 568 ± 2	351 584 ± 3	321 601 ± 6	384 755 ± 5	450 1099 ± 14	454 1087 ± 45
43	aigen-001	256958	595227	509 1154	703 2048 ± 0	1400 1448 ± 9	1110 1451 ± 8	1145 1759 ± 6	1170 2594 ± 4	1147 5691 ± 44	988 3772 ± 57	986 3736 ± 56
44	aigen-002	205300	1316138	392 896	720 2048 ± 0	575 586 ± 24	389 582 ± 4	653 920 ± 4	1098 1758 ± 5	1144 5427 ± 17	974 3678 ± 44	976 3646 ± 48

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 23: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
45	aigroup-000	357973	390046	130 398	885 2048 ± 0	294 351 ± 0	168 337 ± 1	147 353 ± 1	129 356 ± 1	112 385 ± 1	1188 9603 ± 216	1190 9689 ± 373
46	aigroup-001	298974	390068	218 587	567 2048 ± 0	510 535 ± 0	339 531 ± 0	308 533 ± 0	275 544 ± 5	255 595 ± 0	1182 9526 ± 170	1185 9562 ± 213
47	aigroup-002	271124	396396	389 888	1147 2078 ± 0	201 275 ± 0	128 276 ± 0	100 283 ± 2	105 318 ± 5	136 419 ± 0	1432 66379 ± 111	1432 66387 ± 124
48	ailabs-001	1054663	338989	549 1252	757 2048 ± 0	696 664 ± 4	587 774 ± 50	861 1145 ± 12	1136 1972 ± 74	1141 5205 ± 272	1461 104034 ± 661	1461 103415 ± 7722
49	aimall-001	128820	57257	-	257 1024 ± 0	267 326 ± 1	-	-	-	-	1016 4127 ± 72	1000 3909 ± 166
50	aimall-002	370156	25210	665 1576	671 2048 ± 0	839 776 ± 4	715 927 ± 27	671 940 ± 21	623 955 ± 34	568 1003 ± 75	1444 72811 ± 7399	1440 71216 ± 6286
51	aimall-003	504324	171935	752 1913	255 1024 ± 0	692 662 ± 1	557 740 ± 51	525 752 ± 62	462 741 ± 46	430 807 ± 47	1371 34565 ± 93	1373 34598 ± 118
52	aipractix-000	419503	113764	637 1504	1367 4112 ± 0	533 554 ± 0	361 553 ± 0	329 556 ± 0	280 548 ± 0	245 581 ± 0	761 2243 ± 37	767 2257 ± 40
53	aiseemu-001	0	1005354	887 2701	1288 4096 ± 0	1086 1001 ± 1	795 1017 ± 0	745 1014 ± 5	683 1022 ± 2	611 1059 ± 4	1056 4864 ± 25	1059 4855 ± 32
54	aiseemu-002	0	1216980	956 3453	1221 4096 ± 0	1309 1298 ± 5	1016 1303 ± 4	995 1313 ± 2	932 1329 ± 0	818 1348 ± 2	1061 4917 ± 37	1061 4916 ± 37
55	aiunionface-000	241642	840295	132 402	758 2048 ± 0	663 637 ± 13	573 754 ± 41	754 1025 ± 28	809 1179 ± 29	972 1639 ± 47	437 1072 ± 19	449 1080 ± 47
56	aize-001	268456	168970	650 1546	594 2048 ± 0	406 437 ± 10	264 440 ± 8	319 542 ± 17	474 756 ± 27	954 1583 ± 53	702 1937 ± 22	695 1919 ± 23
57	aize-002	257106	182517	217 586	452 2048 ± 0	437 467 ± 1	296 479 ± 1	530 756 ± 1	1042 1477 ± 1	1134 4617 ± 41	194 597 ± 16	211 598 ± 14
58	aize-003	257130	33045	202 546	654 2048 ± 0	120 187 ± 0	70 188 ± 0	50 182 ± 0	42 183 ± 0	36 195 ± 0	225 621 ± 37	203 592 ± 1
59	aize-006	276515	27909	175 497	1131 2060 ± 0	475 501 ± 0	321 504 ± 0	287 505 ± 0	248 508 ± 0	268 612 ± 0	959 3515 ± 35	962 3496 ± 40
60	ajou-001	363257	31734	148 442	609 2048 ± 0	502 530 ± 0	345 536 ± 0	311 535 ± 0	283 549 ± 0	244 577 ± 0	193 597 ± 19	208 596 ± 13
61	akiyama-000	119453	41994	31 116	175 520 ± 0	53 109 ± 1	68 187 ± 1	57 189 ± 1	46 195 ± 1	48 236 ± 2	1134 6901 ± 268	1131 6936 ± 103
62	akiyama-001	119453	41962	32 118	172 520 ± 0	60 120 ± 1	115 265 ± 13	253 468 ± 22	848 1223 ± 27	1086 2660 ± 87	1117 6218 ± 47	1112 6216 ± 45
63	akiyama-002	271192	189635	378 875	811 2048 ± 0	1160 1104 ± 1	887 1138 ± 1	855 1140 ± 1	776 1142 ± 0	732 1240 ± 1	348 817 ± 5	357 824 ± 16
64	akiyama-003	0	784249	539 1211	1372 4124 ± 0	554 570 ± 0	376 568 ± 0	336 569 ± 0	298 574 ± 0	325 685 ± 0	1006 3981 ± 133	1005 3933 ± 120
65	akor-000	250711	95296	636 1497	296 1280 ± 0	801 738 ± 0	554 739 ± 0	514 739 ± 0	466 747 ± 0	422 795 ± 2	58 246 ± 5	79 283 ± 10
66	alchera-000	258450	18848	-	870 2048 ± 0	580 587 ± 13	525 706 ± 48	1042 1389 ± 58	1173 2722 ± 123	1164 9011 ± 455	919 3189 ± 32	907 3031 ± 142
67	alchera-001	174013	18848	-	583 2048 ± 0	647 627 ± 11	-	-	-	-	935 3342 ± 81	934 3243 ± 47
68	alchera-002	405409	22275	-	852 2048 ± 0	1054 968 ± 1	758 976 ± 2	708 979 ± 1	648 988 ± 1	582 1025 ± 2	954 3488 ± 63	953 3430 ± 63
69	alchera-003	487718	24613	600 1376	818 2048 ± 0	932 854 ± 3	653 862 ± 2	615 870 ± 1	571 882 ± 2	508 918 ± 1	945 3426 ± 57	947 3383 ± 53
70	alchera-004	1001019	388616	556 1271	808 2048 ± 0	1064 975 ± 0	738 955 ± 0	696 960 ± 0	649 989 ± 0	679 1152 ± 1	962 3529 ± 54	969 3530 ± 63
71	alchera-005	1001019	388616	557 1272	735 2048 ± 0	1056 969 ± 1	770 987 ± 3	715 985 ± 3	659 998 ± 0	688 1162 ± 2	951 3481 ± 59	952 3422 ± 57
72	alchera-006	1000967	493456	614 1426	425 2048 ± 0	496 522 ± 2	348 540 ± 3	321 544 ± 2	294 569 ± 3	375 739 ± 3	759 2233 ± 38	765 2253 ± 77
73	alchera-007	1000948	493456	620 1430	597 2048 ± 0	501 530 ± 2	364 556 ± 2	334 566 ± 2	301 581 ± 2	390 762 ± 4	749 2178 ± 35	746 2169 ± 44
74	alchera-008	1002024	493460	621 1430	356 2048 ± 0	585 589 ± 3	370 563 ± 4	380 614 ± 3	332 614 ± 1	432 810 ± 3	1069 5129 ± 35	1213 11226 ± 120
75	alchera-009	2086479	454616	901 2763	967 2048 ± 0	1422 1486 ± 4	1130 1507 ± 4	1079 1438 ± 9	1055 1508 ± 11	1028 2008 ± 4	771 2284 ± 44	771 2278 ± 69
76	alfabeta-001	128232	21780	18 74	95 512 ± 0	197 271 ± 0	127 276 ± 0	243 459 ± 2	574 886 ± 2	1084 2547 ± 9	138 470 ± 25	142 458 ± 20
77	alice-000	1741293	19355	946 3329	1299 4096 ± 0	1034 950 ± 2	720 933 ± 1	682 949 ± 1	679 1011 ± 3	754 1264 ± 8	1256 14975 ± 201	1258 14890 ± 229
78	alice-001	1428001	17074	281 690	709 2048 ± 0	1145 1080 ± 1	831 1075 ± 0	817 1095 ± 1	778 1143 ± 1	879 1435 ± 1	1308 21885 ± 278	1313 21921 ± 265
79	alleyes-000	507636	997090	372 857	797 2048 ± 0	847 784 ± 1	751 970 ± 61	704 974 ± 62	615 943 ± 69	606 1057 ± 23	518 1298 ± 34	527 1303 ± 51
80	allgovision-000	172509	155862	209 561	515 2048 ± 0	340 384 ± 8	222 395 ± 17	196 413 ± 14	219 471 ± 14	345 710 ± 21	1352 29903 ± 406	1353 29735 ± 194
81	alphaface-001	259849	81636	193 527	764 2048 ± 0	619 612 ± 1	414 613 ± 3	375 612 ± 1	339 619 ± 1	293 640 ± 2	408 1008 ± 10	414 1002 ± 19
82	alphaface-002	768995	70692	622 1434	912 2048 ± 0	648 628 ± 2	562 746 ± 19	523 751 ± 18	494 779 ± 22	442 828 ± 40	387 945 ± 25	392 935 ± 17
83	alttekglobal-000	649718	3060239	270 673	504 2048 ± 0	1137 1066 ± 2	904 1158 ± 2	891 1184 ± 1	798 1168 ± 1	711 1204 ± 2	532 1370 ± 23	544 1377 ± 26
84	amplifiedgroup-001	0	47053	20 81	211 866 ± 2	39 93 ± 0	-	-	-	-	1421 57803 ± 4210	1415 56365 ± 1196
85	androvideo-000	174847	585063	134 405	814 2048 ± 0	203 277 ± 0	134 285 ± 0	118 314 ± 0	140 372 ± 1	276 620 ± 0	876 2860 ± 28	874 2847 ± 22
86	anke-000	323402	39031	-	1144 2072 ± 0	546 566 ± 12	-	-	-	-	178 567 ± 20	174 545 ± 29
87	anke-001	642061	37364	-	1143 2072 ± 0	520 541 ± 4	-	-	-	-	167 540 ± 16	169 530 ± 20
88	anke-002	798686	28364	-	1142 2072 ± 0	383 419 ± 30	-	-	-	-	155 501 ± 17	155 492 ± 17

Notes	
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4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
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Table 24: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
89	anke-003	340160	198638	-	1063 2056 ± 0	880 811 ± 23	-	-	-	-	121 425 ± 28	132 437 ± 32
90	anke-004	349388	410776	284 706	1084 2056 ± 0	643 625 ± 1	439 627 ± 2	413 635 ± 3	374 653 ± 2	548 982 ± 8	237 633 ± 22	251 632 ± 34
91	anke-005	328553	429160	501 1134	1050 2056 ± 0	587 590 ± 2	399 594 ± 5	366 601 ± 3	360 638 ± 4	438 821 ± 24	271 685 ± 19	287 687 ± 26
92	anyvision-001	662659	518032	-	241 1024 ± 0	174 244 ± 0	-	-	-	-	1449 74285 ± 212	1448 74145 ± 200
93	anyvision-002	662659	520039	-	228 1024 ± 0	179 248 ± 0	-	-	-	-	1448 74069 ± 188	1447 74019 ± 198
94	anyvision-004	401001	630797	490 1102	234 1024 ± 0	299 355 ± 1	-	-	-	-	688 1891 ± 51	676 1829 ± 85
95	anyvision-005	190979	116595	455 1048	246 1024 ± 0	1073 985 ± 1	782 997 ± 1	739 1004 ± 1	655 995 ± 1	559 995 ± 1	306 733 ± 14	319 733 ± 16
96	aratek-001	254521	15993	793 2135	859 2048 ± 0	183 251 ± 1	126 275 ± 0	88 252 ± 0	85 278 ± 0	65 286 ± 0	656 1776 ± 20	655 1744 ± 38
97	armatura-001	0	374608	962 3518	830 2048 ± 0	739 688 ± 1	505 689 ± 1	469 693 ± 1	432 708 ± 3	385 756 ± 13	68 270 ± 17	74 268 ± 11
98	armatura-002	0	1258644	542 1222	1430 6144 ± 0	1415 1476 ± 3	1115 1458 ± 5	1120 1535 ± 5	1053 1505 ± 12	962 1605 ± 26	614 1605 ± 26	614 1589 ± 25
99	armatura-003	0	836082	664 1572	1422 6144 ± 0	1113 1028 ± 1	808 1032 ± 1	757 1027 ± 0	690 1036 ± 1	590 1041 ± 3	1410 51850 ± 56	1410 51835 ± 48
100	arvis-000	529004	705484	380 878	265 1024 ± 0	144 209 ± 5	92 215 ± 1	136 342 ± 4	376 655 ± 13	1052 2158 ± 55	952 3484 ± 41	956 3461 ± 48
101	asr-000	461347	236001	415 951	1111 2056 ± 0	414 450 ± 0	286 462 ± 1	237 454 ± 0	199 453 ± 1	177 486 ± 0	496 1226 ± 14	510 1251 ± 23
102	asr-001	461347	236016	414 949	1096 2056 ± 0	761 701 ± 0	526 706 ± 0	481 704 ± 0	434 709 ± 0	426 800 ± 0	706 1951 ± 28	712 1982 ± 23
103	asusaics-000	257418	245320	230 605	573 2048 ± 0	459 484 ± 13	325 506 ± 21	600 850 ± 26	1105 1789 ± 61	1150 6305 ± 188	1078 5455 ± 78	1079 5422 ± 112
104	asusaics-001	257418	245330	224 595	1246 4096 ± 0	920 842 ± 17	790 1008 ± 20	1038 1377 ± 28	1167 2423 ± 90	1158 7284 ± 277	1175 8618 ± 42	1175 8638 ± 136
105	autentika-000	266093	3200425	753 1922	731 2048 ± 0	532 553 ± 1	406 605 ± 1	372 609 ± 2	325 608 ± 1	275 618 ± 2	1445 72833 ± 577	1441 71829 ± 541
106	autentika-001	0	3198534	684 1650	838 2048 ± 0	731 683 ± 2	697 912 ± 4	644 908 ± 3	595 919 ± 3	493 893 ± 4	1441 71804 ± 600	1439 70747 ± 582
107	autentika-002	0	3198136	690 1676	684 2048 ± 0	325 374 ± 1	326 511 ± 1	306 528 ± 1	267 534 ± 1	226 546 ± 1	1400 46941 ± 1172	1396 46351 ± 822
108	autentika-003	0	3198136	683 1648	895 2048 ± 0	403 437 ± 7	422 617 ± 14	352 584 ± 3	329 611 ± 8	269 614 ± 2	1402 47827 ± 1020	1398 46728 ± 916
109	authenmetric-002	449944	91489	-	711 2048 ± 0	1019 942 ± 1	732 950 ± 1	694 960 ± 1	628 960 ± 1	557 991 ± 2	637 1712 ± 20	647 1719 ± 19
110	authenmetric-003	293599	39492	375 866	779 2048 ± 0	1080 992 ± 1	787 1006 ± 1	738 1003 ± 2	666 1002 ± 1	588 1036 ± 1	652 1757 ± 19	656 1755 ± 19
111	authenmetric-004	381165	39492	540 1214	957 2048 ± 0	987 910 ± 1	694 909 ± 1	648 915 ± 1	597 921 ± 2	528 950 ± 1	642 1724 ± 14	636 1691 ± 29
112	authme-001	1015124	37791	788 2097	1276 4096 ± 0	1032 949 ± 1	749 969 ± 7	695 960 ± 6	629 961 ± 1	537 966 ± 4	870 2836 ± 34	871 2827 ± 54
113	authme-002	1016682	37819	783 2079	1304 4096 ± 0	1045 959 ± 1	744 959 ± 7	731 995 ± 5	635 966 ± 5	551 985 ± 6	864 2796 ± 14	862 2775 ± 34
114	aware-000	240240	27639	-	327 1572 ± 0	685 655 ± 26	-	-	-	-	1012 4030 ± 83	900 2984 ± 152
115	aware-001	240240	27639	-	325 1572 ± 0	688 656 ± 26	-	-	-	-	886 2902 ± 51	870 2810 ± 111
116	aware-002	271312	36134	-	326 1572 ± 0	567 582 ± 14	-	-	-	-	345 810 ± 32	354 812 ± 28
117	aware-003	377729	27975	-	1204 3108 ± 0	845 783 ± 10	-	-	-	-	538 1392 ± 42	531 1334 ± 80
118	aware-004	427829	28219	-	1154 2084 ± 0	976 900 ± 10	857 1105 ± 10	923 1226 ± 25	1076 1585 ± 72	1104 2937 ± 166	512 1279 ± 50	523 1287 ± 100
119	aware-005	300017	26320	554 1265	324 1572 ± 0	970 886 ± 23	814 1038 ± 21	839 1121 ± 22	935 1337 ± 58	1053 2195 ± 144	574 1475 ± 63	560 1427 ± 115
120	aware-006	298543	14124	409 944	80 352 ± 0	1192 1148 ± 3	896 1146 ± 2	893 1190 ± 2	911 1306 ± 20	991 1754 ± 84	833 2598 ± 42	832 2559 ± 60
121	aware-007	135312	18711	127 388	91 480 ± 0	776 713 ± 1	536 716 ± 1	505 731 ± 3	501 785 ± 8	561 999 ± 30	426 1058 ± 23	455 1091 ± 37
122	aware-008	135312	18715	126 387	92 480 ± 0	812 751 ± 1	552 737 ± 0	515 739 ± 3	528 825 ± 9	570 1007 ± 27	460 1120 ± 25	481 1152 ± 35
123	awiros-001	15499	87480	23 88	100 512 ± 0	43 97 ± 6	25 98 ± 4	32 138 ± 6	64 225 ± 7	236 556 ± 8	440 1079 ± 44	437 1050 ± 45
124	awiros-002	289016	203723	205 556	590 2048 ± 0	451 479 ± 0	318 500 ± 0	310 534 ± 0	338 618 ± 0	525 946 ± 1	708 1966 ± 31	708 1957 ± 25
125	awiros-003	292302	598824	421 976	427 2048 ± 0	394 428 ± 2	266 442 ± 1	235 454 ± 2	192 442 ± 1	151 444 ± 1	828 2558 ± 20	833 2580 ± 42
126	awiros-004	489913	598874	568 1295	606 2048 ± 0	476 502 ± 0	331 517 ± 0	300 519 ± 3	256 518 ± 1	198 512 ± 0	822 2527 ± 17	828 2537 ± 20
127	aximetria-001	408902	487912	858 2539	350 2048 ± 0	1097 1013 ± 1	799 1023 ± 21	760 1029 ± 5	661 999 ± 2	635 1091 ± 5	1029 4401 ± 94	1033 4490 ± 80
128	ayftech-001	195423	43580	303 731	148 512 ± 0	370 408 ± 23	293 476 ± 52	563 814 ± 108	1109 1827 ± 384	1143 5412 ± 1029	219 615 ± 16	373 885 ± 44
129	ayftech-003	526164	127257	679 1640	481 2048 ± 0	1088 1003 ± 5	796 1017 ± 6	743 1006 ± 6	677 1009 ± 7	614 1063 ± 5	520 1301 ± 20	526 1300 ± 25
130	ayonix-000	58505	5252	15 69	279 1036 ± 0	2 18 ± 2	-	-	-	-	224 621 ± 23	236 620 ± 26
131	beethedata-000	227849	1087592	206 557	580 2048 ± 0	433 465 ± 0	289 467 ± 0	254 468 ± 0	217 467 ± 0	165 467 ± 0	736 2121 ± 34	736 2110 ± 38
132	betaface-000	0	1060599	-	1161 2116 ± 0	913 837 ± 9	-	-	-	-	594 1514 ± 28	599 1520 ± 66

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Table 25: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
133	betaface-002	0	1060599	-	1160 2116 ± 0	942 870 ± 13	-	-	-	-	573 1474 ± 50	588 1488 ± 41
134	beyneai-000	256958	591433	771 1998	679 2048 ± 0	416 451 ± 8	273 449 ± 1	537 767 ± 7	1080 1603 ± 25	1136 4669 ± 124	981 3730 ± 57	978 3668 ± 54
135	biocube-001	25030	6192987	154 457	1282 4096 ± 0	213 282 ± 22	137 292 ± 24	302 521 ± 57	406 684 ± 59	768 1282 ± 68	1307 21787 ± 96	1311 21812 ± 109
136	biocube-002	69898	10651580	160 468	1214 4096 ± 0	901 826 ± 4	637 838 ± 4	882 1175 ± 4	1176 2933 ± 1074	1165 9144 ± 1090	1319 23074 ± 94	1321 23036 ± 99
137	bioid-001	361549	144	246 629	1453 7680 ± 0	121 190 ± 1	78 192 ± 1	102 290 ± 2	286 556 ± 7	935 1528 ± 25	1065 5051 ± 35	1068 5042 ± 19
138	bioid-002	364015	149	608 1415	1452 7680 ± 0	119 187 ± 2	73 190 ± 3	106 295 ± 4	289 560 ± 13	970 1630 ± 43	1067 5052 ± 34	1067 5031 ± 23
139	bioid-003	530171	156	658 1558	1471 15368 ± 0	568 582 ± 2	395 587 ± 2	484 706 ± 3	654 994 ± 7	1046 2123 ± 33	1122 6396 ± 25	1118 6377 ± 29
140	bioid-004	530301	156	662 1568	1472 15368 ± 0	604 603 ± 1	412 611 ± 1	420 640 ± 1	447 726 ± 2	607 1058 ± 3	1136 6948 ± 25	1132 6948 ± 19
141	bioidtechswiss-000	758466	119955	-	33 256 ± 0	653 630 ± 2	-	-	-	-	1370 34416 ± 137	1372 34403 ± 126
142	bioidtechswiss-001	1178769	120811	626 1455	147 512 ± 0	1052 966 ± 4	993 1270 ± 270	982 1294 ± 96	993 1409 ± 157	999 1793 ± 79	834 2610 ± 25	836 2624 ± 32
143	bioidtechswiss-002	744786	114842	427 990	125 512 ± 0	998 917 ± 2	717 930 ± 2	683 952 ± 2	617 947 ± 3	608 1058 ± 11	748 2177 ± 29	747 2170 ± 31
144	bioidtechswiss-003	518317	10378	1020 4730	786 2048 ± 0	1143 1075 ± 1	889 1139 ± 7	1184 2207 ± 9	1124 1891 ± 11	1139 4787 ± 21	1405 50218 ± 204	1406 50184 ± 163
145	biometric-vision-000	445487	72132	398 915	800 2048 ± 0	912 836 ± 3	630 831 ± 3	613 869 ± 5	624 955 ± 8	829 1370 ± 19	1249 14437 ± 161	1249 14325 ± 167
146	biometrik-001	23995	68818	28 101	99 512 ± 0	1397 1446 ± 3	1106 1448 ± 3	1089 1449 ± 3	1026 1456 ± 3	912 1485 ± 3	409 1016 ± 5	409 999 ± 17
147	biometrik-002	171539	84430	114 353	564 2048 ± 0	1427 1499 ± 3	1127 1499 ± 3	1112 1501 ± 3	1051 1503 ± 3	921 1507 ± 2	524 1328 ± 19	530 1321 ± 19
148	bm-001	287734	38076	48 148	1 64 ± 0	411 444 ± 88	-	-	-	-	687 1887 ± 31	688 1877 ± 26
149	boetech-001	261376	88710	971 3658	774 2048 ± 0	196 271 ± 1	121 268 ± 1	92 273 ± 0	88 286 ± 1	74 318 ± 1	1435 68519 ± 1921	1434 67648 ± 822
150	boetech-002	294347	88710	778 2057	503 2048 ± 0	243 305 ± 4	142 296 ± 1	109 302 ± 1	103 313 ± 1	92 348 ± 2	1437 68921 ± 2137	1438 69473 ± 2104
151	bresee-000	287880	22872	-	342 2048 ± 0	1317 1309 ± 1	1193 2126 ± 1	1005 1330 ± 0	920 1312 ± 0	812 1333 ± 0	1393 45317 ± 228	1403 48256 ± 425
152	bresee-001	287880	23227	543 1223	623 2048 ± 0	1256 1223 ± 3	949 1216 ± 1	1008 1331 ± 1	854 1227 ± 1	825 1360 ± 1	1379 37240 ± 655	1382 37167 ± 584
153	bresee-002	313627	30902	760 1951	451 2048 ± 0	806 743 ± 4	892 1143 ± 2	862 1146 ± 2	782 1148 ± 2	698 1176 ± 2	657 1778 ± 22	661 1775 ± 23
154	btsdigital-001	362015	234409	416 955	526 2048 ± 0	791 728 ± 2	535 713 ± 1	500 726 ± 4	442 716 ± 2	371 738 ± 0	912 3123 ± 22	917 3134 ± 37
155	bynn-000	271083	37637	131 400	988 2049 ± 0	1460 1834 ± 0	1166 1835 ± 0	1157 1840 ± 0	1111 1838 ± 0	1009 1875 ± 0	372 899 ± 7	385 915 ± 1
156	cakevn-001	271412	587471	306 739	443 2048 ± 0	583 587 ± 0	391 584 ± 0	353 585 ± 0	314 589 ± 0	310 664 ± 0	1452 78258 ± 72	1450 78329 ± 90
157	cakevn-002	1015434	73945	544 1228	628 2048 ± 0	979 905 ± 0	691 905 ± 0	643 907 ± 0	589 910 ± 0	504 916 ± 0	910 3115 ± 28	913 3086 ± 11
158	camvi-001	118759	21678	-	267 1024 ± 0	111 181 ± 6	-	-	-	-	143 481 ± 16	153 487 ± 23
159	camvi-002	236278	225285	305 737	264 1024 ± 0	719 677 ± 7	544 726 ± 36	614 869 ± 28	764 1129 ± 43	1095 2785 ± 113	216 612 ± 26	218 603 ± 20
160	camvi-003	285657	225320	-	243 1024 ± 0	810 750 ± 3	-	-	-	-	180 571 ± 23	184 565 ± 26
161	camvi-004	280733	615819	401 919	640 2048 ± 0	824 759 ± 10	652 861 ± 17	717 986 ± 34	899 1279 ± 51	1102 2891 ± 158	388 948 ± 40	397 963 ± 31
162	candour-001	150086	97632	165 477	739 2048 ± 0	1370 1400 ± 1	1071 1396 ± 1	1062 1408 ± 1	992 1407 ± 1	897 1464 ± 1	1205 10725 ± 133	1205 10712 ± 127
163	candour-002	149560	65228	129 392	372 2048 ± 0	80 144 ± 0	42 143 ± 0	34 145 ± 0	33 152 ± 1	39 206 ± 0	1160 7798 ± 56	1160 7781 ± 51
164	candour-003	339247	65570	353 833	1195 3072 ± 0	333 380 ± 0	220 392 ± 0	169 378 ± 0	159 393 ± 1	152 445 ± 0	1369 34039 ± 45	1371 34020 ± 44
165	candour-004	1192219	65226	853 2490	1265 4096 ± 0	1482 2243 ± 0	1195 2247 ± 0	1185 2223 ± 0	1160 2270 ± 1	1066 2330 ± 0	850 2714 ± 20	859 2755 ± 23
166	canon-002	446491	130232	-	1224 4096 ± 0	1315 1308 ± 2	1026 1315 ± 1	1002 1326 ± 2	938 1345 ± 1	891 1452 ± 1	1116 6211 ± 25	1110 6194 ± 25
167	canon-003	2550850	101378	1038 5487	1433 6180 ± 0	1286 1263 ± 3	989 1263 ± 1	971 1283 ± 1	925 1320 ± 1	909 1482 ± 2	1050 4783 ± 17	1051 4780 ± 19
168	canon-004	2399160	114188	1061 7037	1435 6200 ± 0	1030 948 ± 4	736 955 ± 3	693 959 ± 3	639 977 ± 3	617 1064 ± 2	1147 7172 ± 63	1144 7169 ± 51
169	canon-005	2521818	80309	1056 6700	1384 4148 ± 0	1255 1223 ± 21	955 1221 ± 2	925 1230 ± 4	859 1234 ± 5	809 1329 ± 3	1112 6142 ± 99	1113 6226 ± 146
170	cchonolulu-000	727461	37284	136 416	114 512 ± 0	65 123 ± 13	39 126 ± 10	31 129 ± 10	28 134 ± 10	43 223 ± 10	643 1726 ± 35	637 1697 ± 35
171	cchonolulu-001	733744	37284	140 426	493 2048 ± 0	92 162 ± 21	79 197 ± 23	292 513 ± 96	750 1118 ± 267	1050 2142 ± 584	1030 4418 ± 37	1024 4361 ± 35
172	cdachyd-001	238593	1941345	38 135	134 512 ± 0	1441 1642 ± 2	1156 1749 ± 1	1197 6467 ± 3	1195 17252 ± 10	1186 58681 ± 45	1216 11594 ± 85	1221 11639 ± 130
173	ceiaufgbr-000	254585	44874	153 453	817 2048 ± 0	180 250 ± 0	110 251 ± 0	87 252 ± 0	75 251 ± 0	56 253 ± 0	342 803 ± 4	350 804 ± 4
174	ceiec-001	159618	46136	-	225 1024 ± 0	251 314 ± 3	-	-	-	-	1317 22831 ± 108	1320 22813 ± 120
175	ceiec-002	269063	90975	-	368 2048 ± 0	622 612 ± 17	-	-	-	-	750 2188 ± 57	777 2301 ± 56
176	ceiec-003	260371	88707	144 430	700 2048 ± 0	889 817 ± 4	676 883 ± 57	633 897 ± 60	584 899 ± 72	523 944 ± 72	765 2256 ± 38	760 2241 ± 54

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2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 26: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
177	ceiec-004	263476	67011	135 408	724 2048 ± 0	1103 1024 ± 1	801 1027 ± 1	756 1027 ± 1	688 1030 ± 1	602 1055 ± 1	676 1844 ± 26	678 1836 ± 20
178	celloscope-001	38016	59158	40 138	120 512 ± 0	16 49 ± 0	12 52 ± 0	56 188 ± 0	249 509 ± 1	-	856 2733 ± 35	855 2730 ± 35
179	chosun-000	167093	694	-	740 2048 ± 0	409 441 ± 1	-	-	-	-	398 983 ± 20	403 983 ± 29
180	chosun-001	765615	707	172 491	945 2048 ± 0	844 783 ± 2	624 826 ± 4	1138 1662 ± 13	1184 3679 ± 67	1179 11694 ± 243	404 998 ± 25	426 1035 ± 11
181	chosun-002	234001	31875	151 451	802 2048 ± 0	178 248 ± 3	123 273 ± 3	1110 1495 ± 14	1190 7920 ± 90	1187 80302 ± 1349	227 623 ± 17	254 634 ± 13
182	chtface-001	94088	369529	-	577 2048 ± 0	151 212 ± 10	-	-	-	-	764 2256 ± 26	762 2252 ± 91
183	chtface-002	363153	369529	-	857 2048 ± 0	572 584 ± 14	533 712 ± 41	769 1038 ± 42	1115 1861 ± 75	1135 4661 ± 232	768 2264 ± 26	759 2234 ± 103
184	chtface-003	363153	369529	-	825 2048 ± 0	593 594 ± 16	541 720 ± 33	779 1050 ± 41	1122 1884 ± 90	1145 5606 ± 334	735 2110 ± 37	755 2219 ± 65
185	chtface-004	409656	311027	616 1428	812 2048 ± 0	273 332 ± 0	157 323 ± 1	128 329 ± 1	118 335 ± 1	107 377 ± 1	644 1727 ± 17	648 1720 ± 16
186	chtface-005	408364	311100	607 1415	738 2048 ± 0	262 322 ± 0	151 316 ± 1	123 325 ± 2	112 324 ± 1	132 411 ± 2	693 1907 ± 19	692 1898 ± 23
187	chtface-006	733645	610439	836 2424	423 2048 ± 0	495 522 ± 1	328 514 ± 1	312 536 ± 2	290 561 ± 1	334 693 ± 2	718 2034 ± 41	725 2049 ± 29
188	chtface-007	733528	610443	830 2393	604 2048 ± 0	222 292 ± 0	136 290 ± 0	113 307 ± 1	115 328 ± 1	186 501 ± 1	606 1570 ± 53	610 1561 ± 34
189	chtface-008	810179	583222	1072 8766	371 2048 ± 0	1123 1043 ± 8	797 1019 ± 9	750 1022 ± 14	694 1039 ± 16	620 1066 ± 23	593 1513 ± 14	605 1535 ± 44
190	chtface-009	810179	342213	1076 9371	961 2048 ± 0	1404 1455 ± 0	1117 1468 ± 16	1113 1501 ± 26	1049 1501 ± 0	942 1544 ± 15	754 2213 ± 13	758 2226 ± 29
191	cib-000	340288	69515	-	396 2048 ± 0	1082 993 ± 40	-	-	-	-	1327 24340 ± 60	1338 25972 ± 97
192	cib-001	436253	133766	-	439 2048 ± 0	679 651 ± 2	528 707 ± 13	492 716 ± 15	450 728 ± 3	437 820 ± 5	989 3783 ± 38	989 3765 ± 37
193	cist-001	0	300551	215 582	923 2048 ± 0	1059 972 ± 0	761 977 ± 1	712 981 ± 0	644 983 ± 0	612 1061 ± 0	892 2947 ± 20	894 2940 ± 22
194	cist-002	0	1186073	704 1730	1210 3200 ± 0	1065 976 ± 0	776 992 ± 0	807 1085 ± 1	897 1279 ± 0	1041 2101 ± 1	1114 6165 ± 22	1111 6207 ± 38
195	cist-003	0	1267350	888 2706	1421 5248 ± 0	1304 1291 ± 1	1015 1300 ± 1	1022 1352 ± 3	904 1287 ± 2	794 1313 ± 1	1161 7838 ± 63	1161 7823 ± 40
196	cist-004	0	1603990	864 2563	1420 5248 ± 0	1273 1243 ± 1	994 1271 ± 1	955 1264 ± 3	872 1252 ± 1	770 1287 ± 1	924 3216 ± 25	931 3220 ± 32
197	clearviewai-000	342491	211852	895 2745	496 2048 ± 0	1373 1402 ± 1	1073 1403 ± 1	1069 1412 ± 1	1000 1420 ± 1	864 1418 ± 1	611 1592 ± 37	609 1561 ± 37
198	clearviewai-001	613532	511807	589 1353	648 2048 ± 0	662 635 ± 2	458 642 ± 1	433 659 ± 1	385 664 ± 1	318 674 ± 1	351 827 ± 10	360 829 ± 20
199	clearviewai-002	713008	509504	655 1556	365 2048 ± 0	819 755 ± 2	586 773 ± 2	534 765 ± 3	497 783 ± 2	417 790 ± 1	361 864 ± 34	362 842 ± 29
200	clearviewai-003	712800	540093	581 1344	614 2048 ± 0	1164 1109 ± 0	845 1091 ± 0	810 1088 ± 0	730 1090 ± 0	659 1126 ± 0	865 2812 ± 45	872 2831 ± 50
201	closeli-001	420342	9851	321 773	1307 4096 ± 0	917 839 ± 1	640 843 ± 1	593 841 ± 1	544 845 ± 1	475 865 ± 1	1077 5404 ± 17	1078 5400 ± 25
202	cloudmatrix-000	309939	542141	-	878 2048 ± 0	815 754 ± 10	567 750 ± 2	527 754 ± 4	482 764 ± 1	419 793 ± 2	1403 49192 ± 206	1404 49275 ± 176
203	cloudmatrix-001	10390	542121	12 65	734 2048 ± 0	57 114 ± 1	34 117 ± 0	28 118 ± 0	25 123 ± 1	25 169 ± 1	1406 50263 ± 212	1407 50243 ± 237
204	cloudmatrix-002	256635	693318	452 1036	907 2048 ± 0	352 395 ± 1	228 398 ± 1	184 399 ± 1	167 402 ± 1	144 437 ± 20	1404 49578 ± 120	1405 49602 ± 180
205	cloudwalk-hr-000	284600	105489	-	1114 2057 ± 0	599 600 ± 2	-	-	-	-	1275 16747 ± 238	1277 16627 ± 220
206	cloudwalk-hr-001	346156	104483	-	1116 2057 ± 0	701 665 ± 3	-	-	-	-	1282 17816 ± 260	1287 17878 ± 464
207	cloudwalk-hr-002	390059	327494	-	1120 2057 ± 0	983 908 ± 3	-	-	-	-	1315 22530 ± 416	1309 21651 ± 533
208	cloudwalk-hr-003	383739	144263	424 983	1119 2057 ± 0	610 606 ± 0	396 588 ± 0	360 594 ± 0	330 612 ± 1	227 547 ± 2	1138 6982 ± 80	1135 6972 ± 84
209	cloudwalk-hr-004	502916	520169	604 1399	980 2049 ± 0	951 873 ± 1	670 877 ± 1	621 876 ± 1	566 879 ± 1	500 902 ± 3	1219 11652 ± 127	1219 11608 ± 123
210	cloudwalk-mt-000	372169	282036	-	987 2049 ± 0	786 724 ± 12	-	-	-	-	629 1678 ± 47	617 1614 ± 85
211	cloudwalk-mt-002	290753	145340	-	974 2049 ± 0	557 573 ± 1	537 717 ± 78	477 700 ± 66	467 749 ± 96	405 770 ± 80	1148 7205 ± 204	1145 7211 ± 244
212	cloudwalk-mt-003	490365	494959	-	977 2049 ± 0	1007 923 ± 1	701 918 ± 1	664 926 ± 1	599 925 ± 1	520 936 ± 1	1218 11620 ± 179	1222 11661 ± 128
213	cloudwalk-mt-004	1384602	512628	-	607 2048 ± 0	1005 923 ± 2	704 919 ± 1	651 918 ± 0	596 919 ± 0	514 927 ± 1	1220 11744 ± 170	1220 11631 ± 126
214	cloudwalk-mt-005	846026	573253	967 3615	905 2048 ± 0	1217 1179 ± 3	939 1200 ± 3	911 1209 ± 3	852 1226 ± 5	725 1229 ± 3	1235 12525 ± 225	1234 12394 ± 152
215	cloudwalk-mt-006	563322	480071	960 3499	653 2048 ± 0	1359 1385 ± 0	1064 1392 ± 1	1052 1398 ± 1	981 1397 ± 4	884 1444 ± 2	937 3364 ± 96	941 3324 ± 83
216	cloudwalk-mt-007	563322	480071	961 3499	750 2048 ± 0	1350 1379 ± 2	1092 1425 ± 7	1075 1427 ± 6	996 1417 ± 2	883 1443 ± 7	897 3000 ± 68	897 2971 ± 74
217	clova-000	198420	6824	-	673 2048 ± 0	404 437 ± 0	251 431 ± 0	217 435 ± 0	197 452 ± 2	196 508 ± 7	662 1794 ± 16	669 1795 ± 19
218	cmcuni-001	0	604227	519 1177	915 2048 ± 0	1202 1161 ± 1	928 1187 ± 1	899 1194 ± 5	824 1200 ± 1	714 1207 ± 1	916 3153 ± 22	922 3158 ± 26
219	codeline-000	361659	138388	534 1194	867 2048 ± 0	1402 1453 ± 0	1114 1456 ± 2	1094 1456 ± 0	1028 1457 ± 0	910 1483 ± 1	745 2171 ± 69	751 2194 ± 84
220	cogent-000	633812	18848	-	85 439 ± 0	484 511 ± 7	-	-	-	-	1183 9541 ± 54	1182 9500 ± 51

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Table 27: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
221	cogent-001	530508	18869	-	[89] 455 ± 0	[536] 557 ± 7	-	-	-	-	[1201] 10477 ± 278	[1203] 10392 ± 75
222	cogent-002	696959	18881	-	[337] 1979 ± 0	[1017] 941 ± 0	-	-	-	-	[1247] 14394 ± 134	[1251] 14385 ± 176
223	cogent-003	698290	40430	-	[221] 973 ± 0	[1041] 952 ± 0	-	-	-	-	[1233] 12496 ± 75	[1224] 11822 ± 163
224	cogent-004	722919	389164	-	[338] 1983 ± 0	[1075] 987 ± 50	-	-	-	-	[1265] 15536 ± 75	[1270] 15964 ± 708
225	cogent-005	1876796	75276	-	[1176] 2523 ± 0	[1252] 1221 ± 2	[967] 1236 ± 1	[974] 1289 ± 2	[1001] 1420 ± 4	[961] 1602 ± 5	[1334] 24854 ± 69	[1333] 24858 ± 71
226	cogent-006	1078167	58108	[656] 1557	[287] 1062 ± 0	[833] 768 ± 0	[598] 789 ± 1	[585] 831 ± 2	[603] 930 ± 1	[541] 971 ± 1	[668] 1802 ± 17	[670] 1797 ± 23
227	cogent-007	621565	72316	[749] 1888	[196] 550 ± 0	[1327] 1329 ± 2	[1039] 1333 ± 5	[1012] 1337 ± 4	[944] 1353 ± 5	[848] 1390 ± 4	[101] 355 ± 8	[112] 367 ± 14
228	cogent-008	856817	73587	[802] 2176	[197] 550 ± 0	[1380] 1412 ± 1	[1090] 1419 ± 2	[1074] 1426 ± 3	[1013] 1437 ± 3	[906] 1476 ± 1	[124] 436 ± 14	[134] 441 ± 23
229	cognitec-000	474759	27371	-	[1027] 2052 ± 0	[158] 224 ± 1	-	-	-	-	[992] 3835 ± 108	[991] 3782 ± 83
230	cognitec-001	476809	27487	-	[1028] 2052 ± 0	[223] 293 ± 17	-	-	-	-	[1020] 4253 ± 59	[1018] 4102 ± 167
231	cognitec-002	394088	62354	-	[1044] 2052 ± 0	[126] 192 ± 6	[93] 219 ± 6	[81] 233 ± 8	[71] 241 ± 6	[71] 314 ± 10	[930] 3250 ± 41	[933] 3241 ± 48
232	cognitec-003	471458	62502	[341] 817	[1005] 2052 ± 0	[317] 366 ± 9	[231] 403 ± 9	[193] 408 ± 9	[180] 424 ± 9	[197] 509 ± 13	[944] 3417 ± 51	[954] 3433 ± 53
233	cognitec-004	705645	62678	[213] 580	[1021] 2052 ± 0	[430] 463 ± 9	[312] 497 ± 9	[286] 504 ± 10	[258] 521 ± 10	[281] 631 ± 12	[903] 3028 ± 197	[911] 3059 ± 238
234	cognitec-005	326804	198231	[344] 819	[1031] 2052 ± 0	[248] 311 ± 10	[194] 364 ± 12	[162] 370 ± 10	[152] 390 ± 10	[184] 498 ± 16	[782] 2332 ± 48	[785] 2330 ± 40
235	cognitec-006	368383	40008	[359] 844	[680] 2048 ± 0	[280] 335 ± 7	[207] 381 ± 8	[180] 390 ± 7	[161] 396 ± 8	[159] 455 ± 11	[780] 2316 ± 60	[764] 2252 ± 50
236	cognitec-007	368401	40132	[360] 844	[665] 2048 ± 0	[426] 458 ± 20	[403] 601 ± 17	[436] 660 ± 17	[520] 815 ± 23	[886] 1445 ± 45	[722] 2051 ± 26	[723] 2045 ± 34
237	cognitec-008	407825	39616	[396] 914	[767] 2048 ± 0	[515] 538 ± 20	[430] 622 ± 23	[402] 631 ± 23	[394] 672 ± 24	[366] 735 ± 26	[1055] 4854 ± 124	[1069] 5052 ± 287
238	compal-000	1003818	186200	[813] 2242	[266] 1024 ± 0	[1280] 1255 ± 1	[1008] 1293 ± 2	[984] 1294 ± 2	[908] 1300 ± 2	[799] 1322 ± 4	[1367] 33116 ± 349	[1369] 33182 ± 252
239	compal-001	619796	186054	[950] 3395	[983] 2049 ± 0	[936] 861 ± 0	[661] 871 ± 0	[601] 851 ± 0	[554] 857 ± 0	[489] 887 ± 0	[1466] 149573 ± 724	[1466] 149664 ± 582
240	compal-002	620531	185671	[935] 3134	[991] 2049 ± 0	[957] 877 ± 0	[666] 873 ± 1	[632] 892 ± 1	[568] 879 ± 0	[481] 874 ± 0	[506] 1264 ± 13	[517] 1272 ± 16
241	cor-001	1194948	11240	[548] 1249	[1128] 2060 ± 0	[756] 699 ± 3	[654] 863 ± 76	[610] 865 ± 80	[560] 872 ± 89	[531] 952 ± 39	[1471] 270145 ± 2259	[1471] 282686 ± 11788
242	coretech-000	186423	43964	[99] 318	[132] 512 ± 0	[603] 602 ± 15	[469] 659 ± 12	[854] 1139 ± 24	[784] 1149 ± 25	[691] 1165 ± 23	[90] 333 ± 14	[92] 321 ± 13
243	coretech-001	235361	305490	[640] 1523	[813] 2048 ± 0	[740] 688 ± 7	[512] 695 ± 7	[616] 870 ± 17	[567] 879 ± 15	[483] 877 ± 15	[228] 625 ± 25	[262] 641 ± 25
244	coretech-002	339881	296194	[944] 3300	[876] 2048 ± 0	[427] 459 ± 1	[275] 451 ± 1	[602] 851 ± 1	[562] 875 ± 2	[547] 980 ± 2	[806] 2443 ± 17	[805] 2436 ± 13
245	coretech-003	339885	296194	[743] 1861	[406] 2048 ± 0	[886] 815 ± 2	[636] 838 ± 3	[962] 1273 ± 6	[896] 1279 ± 8	[833] 1373 ± 6	[800] 2416 ± 13	[801] 2415 ± 19
246	coretech-004	339885	296194	[736] 1828	[519] 2048 ± 0	[392] 427 ± 1	[269] 443 ± 2	[597] 847 ± 4	[555] 858 ± 2	[524] 945 ± 4	[802] 2436 ± 35	[803] 2432 ± 20
247	corsight-001	1437763	31525	-	[1134] 2064 ± 0	[1305] 1291 ± 3	[1002] 1285 ± 1	[978] 1293 ± 1	[909] 1303 ± 2	[838] 1379 ± 3	[1470] 249340 ± 1713	[1470] 248929 ± 1909
248	corsight-002	1474921	32093	[750] 1892	[1150] 2080 ± 0	[1303] 1290 ± 1	[1004] 1287 ± 1	[975] 1290 ± 1	[913] 1307 ± 2	[847] 1388 ± 4	[1335] 24953 ± 637	[1329] 24263 ± 578
249	corsight-003	1413063	32198	[680] 1642	[1152] 2080 ± 0	[1238] 1202 ± 2	[930] 1190 ± 5	[905] 1199 ± 3	[862] 1236 ± 3	[819] 1349 ± 7	[1348] 28754 ± 434	[1350] 28279 ± 446
250	csc-001	0	240698	-	[188] 544 ± 0	[237] 302 ± 1	[147] 303 ± 0	[112] 304 ± 0	[99] 309 ± 1	[87] 341 ± 4	[99] 354 ± 8	[107] 344 ± 11
251	csc-002	0	519768	[4] 30	[191] 544 ± 0	[446] 473 ± 0	[311] 494 ± 0	[262] 481 ± 1	[234] 490 ± 1	[202] 514 ± 5	[105] 367 ± 11	[114] 371 ± 10
252	csc-003	0	400435	[5] 31	[193] 544 ± 0	[470] 499 ± 0	[316] 500 ± 1	[282] 502 ± 0	[247] 508 ± 1	[216] 535 ± 4	[114] 393 ± 8	[124] 397 ± 7
253	ctbcbank-000	257208	599238	[212] 570	[434] 2048 ± 0	[550] 568 ± 43	[408] 606 ± 38	[465] 690 ± 53	[435] 711 ± 50	[448] 831 ± 51	[964] 3551 ± 87	[1054] 4805 ± 209
254	ctbcbank-001	275511	599238	[226] 603	[486] 2048 ± 0	[681] 652 ± 35	[594] 781 ± 30	[620] 875 ± 43	[583] 898 ± 51	[585] 1030 ± 47	[998] 3926 ± 45	[1001] 3924 ± 56
255	ctbcbank-002	550523	253310	[477] 1084	[117] 512 ± 0	[753] 694 ± 0	[529] 707 ± 0	[501] 727 ± 0	[465] 744 ± 0	[386] 758 ± 0	[1045] 4656 ± 146	[1081] 5486 ± 179
256	cu-face-002	812008	38655	[292] 723	[1324] 4096 ± 0	[1128] 1054 ± 1	[825] 1059 ± 0	[791] 1060 ± 0	[707] 1063 ± 1	[622] 1070 ± 0	[1420] 57287 ± 1750	[1419] 57027 ± 945
257	cu-face-003	1015124	37791	[790] 2116	[1277] 4096 ± 0	[1026] 945 ± 1	[737] 955 ± 3	[688] 957 ± 2	[619] 949 ± 3	[535] 963 ± 2	[861] 2769 ± 19	[865] 2782 ± 41
258	cu-face-004	1016682	37791	[784] 2083	[1273] 4096 ± 0	[1101] 1022 ± 1	[806] 1031 ± 1	[734] 998 ± 3	[715] 1070 ± 1	[579] 1021 ± 2	[878] 2884 ± 38	[878] 2864 ± 48
259	cubox-001	369627	75427	[259] 652	[402] 2048 ± 0	[980] 907 ± 1	[689] 902 ± 1	[638] 903 ± 0	[594] 917 ± 0	[517] 931 ± 0	[535] 1379 ± 37	[554] 1417 ± 38
260	cubox-002	542254	90975	[764] 1966	[381] 2048 ± 0	[1003] 921 ± 1	[706] 921 ± 1	[657] 922 ± 1	[606] 933 ± 1	[567] 1003 ± 1	[714] 2008 ± 72	[710] 1969 ± 57
261	cubox-003	1694397	209684	[959] 3491	[710] 2048 ± 0	[1218] 1179 ± 0	[929] 1189 ± 0	[904] 1198 ± 0	[833] 1208 ± 0	[755] 1264 ± 1	[774] 2289 ± 112	[774] 2297 ± 68
262	cubox-004	1304512	209634	[897] 2747	[718] 2048 ± 0	[1048] 963 ± 1	[769] 986 ± 1	[721] 992 ± 0	[674] 1008 ± 0	[628] 1087 ± 0	[575] 1476 ± 62	[573] 1460 ± 42
263	cubox-005	225237	209600	[601] 1387	[632] 2048 ± 0	[311] 362 ± 0	[202] 371 ± 0	[165] 373 ± 0	[155] 390 ± 0	[157] 453 ± 0	[649] 1748 ± 25	[646] 1718 ± 40
264	cudocommunication-001	385258	341277	[475] 1080	[366] 2048 ± 0	[1008] 925 ± 1	[710] 923 ± 1	[666] 928 ± 1	[604] 932 ± 0	[536] 964 ± 1	[823] 2534 ± 20	[827] 2537 ± 20

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	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
265	cuhkee-001	787853	74917	856 2515	1000 2052 ± 0	1067 977 ± 31	-	-	-	-	852 2719 ± 60	866 2783 ± 56
266	cybercore-000	86008	55441	-	154 512 ± 0	686 655 ± 3	506 689 ± 71	428 649 ± 6	370 648 ± 8	321 680 ± 6	1254 14800 ± 75	1269 15757 ± 782
267	cybercore-001	166096	7791	-	772 2048 ± 0	461 487 ± 0	304 486 ± 0	270 488 ± 0	232 487 ± 0	189 502 ± 0	1411 52119 ± 111	1411 52127 ± 111
268	cybercore-002	166096	7374	865 2565	901 2048 ± 0	463 489 ± 1	317 500 ± 4	281 500 ± 1	240 499 ± 2	213 528 ± 1	1231 12389 ± 123	1233 12352 ± 112
269	cybercore-003	289176	7969	1003 4311	1298 4096 ± 0	921 844 ± 2	648 855 ± 4	609 864 ± 4	558 862 ± 4	485 878 ± 2	1094 5744 ± 41	1096 5737 ± 31
270	cyberextruder-001	121211	13629	-	31 256 ± 0	974 893 ± 25	-	-	-	-	442 1083 ± 16	448 1079 ± 19
271	cyberextruder-002	168909	13924	-	534 2048 ± 0	504 532 ± 6	-	-	-	-	669 1803 ± 14	663 1779 ± 22
272	cyberextruder-003	253300	12354	149 444	145 512 ± 0	348 390 ± 1	215 388 ± 1	181 393 ± 1	164 399 ± 1	142 435 ± 1	41 198 ± 4	39 189 ± 8
273	cyberextruder-004	169301	12354	96 302	7 128 ± 0	142 206 ± 0	87 208 ± 0	69 209 ± 0	68 229 ± 0	54 249 ± 1	24 145 ± 14	27 155 ± 14
274	cyberextruder-005	269610	45367	231 606	128 512 ± 0	96 168 ± 2	54 169 ± 1	66 205 ± 3	93 295 ± 5	457 835 ± 9	23 142 ± 8	24 144 ± 10
275	cyberextruder-006	184816	45435	125 382	32 256 ± 0	37 91 ± 1	22 89 ± 0	20 97 ± 0	17 95 ± 0	19 130 ± 1	9 98 ± 7	8 101 ± 11
276	cyberlink-000	221801	91440	-	1006 2052 ± 0	284 338 ± 6	-	-	-	-	733 2102 ± 40	734 2093 ± 32
277	cyberlink-001	222009	91440	-	1022 2052 ± 0	387 425 ± 29	-	-	-	-	721 2051 ± 32	728 2060 ± 31
278	cyberlink-002	222311	95086	-	1001 2052 ± 0	689 656 ± 22	-	-	-	-	767 2264 ± 71	840 2649 ± 195
279	cyberlink-003	470949	104369	-	1023 2052 ± 0	386 423 ± 1	-	-	-	-	790 2366 ± 38	791 2373 ± 45
280	cyberlink-004	340894	104662	-	1377 4140 ± 0	773 712 ± 1	-	-	-	-	979 3693 ± 51	998 3898 ± 71
281	cyberlink-005	341680	111358	-	1378 4140 ± 0	783 721 ± 1	572 752 ± 2	528 754 ± 2	472 755 ± 1	418 791 ± 3	976 3680 ± 37	1016 4021 ± 97
282	cyberlink-006	341666	102456	-	1441 6212 ± 0	745 690 ± 1	519 702 ± 0	480 703 ± 0	437 712 ± 0	376 741 ± 0	67 270 ± 13	76 271 ± 13
283	cyberlink-007	380046	102446	-	1437 6212 ± 0	789 725 ± 1	549 732 ± 1	509 734 ± 1	459 736 ± 1	400 767 ± 1	81 304 ± 19	85 304 ± 16
284	cyberlink-008	380047	102470	-	1440 6212 ± 0	792 729 ± 1	542 725 ± 0	502 727 ± 0	456 732 ± 0	389 760 ± 0	64 263 ± 17	68 255 ± 13
285	cyberlink-009	588443	102201	697 1707	1386 4164 ± 0	1355 1384 ± 2	1070 1395 ± 2	1051 1398 ± 2	984 1401 ± 2	892 1456 ± 2	79 299 ± 17	86 304 ± 16
286	cyberlink-010	1590818	102180	973 3678	1462 8260 ± 0	1288 1265 ± 2	1024 1314 ± 5	981 1294 ± 2	892 1273 ± 2	787 1305 ± 2	140 476 ± 23	148 472 ± 14
287	cyberlink-011	1397496	102171	925 3029	1439 6212 ± 0	1233 1196 ± 3	932 1193 ± 2	901 1195 ± 3	841 1215 ± 6	728 1231 ± 5	987 3768 ± 75	985 3735 ± 57
288	cyberlink-012	1200318	102281	904 2789	1436 6212 ± 0	601 601 ± 1	428 621 ± 5	374 610 ± 2	333 614 ± 2	300 649 ± 5	786 2343 ± 26	792 2377 ± 34
289	cyberlink-013	893519	102285	762 1964	1438 6212 ± 0	606 603 ± 1	404 602 ± 2	370 606 ± 2	331 613 ± 2	295 642 ± 1	84 307 ± 7	88 306 ± 5
290	cyberlink-014	1190911	102304	911 2876	1463 8260 ± 0	781 720 ± 1	551 736 ± 5	510 735 ± 10	453 731 ± 3	391 762 ± 3	110 378 ± 8	116 377 ± 13
291	cyberlink-015	1190911	102309	910 2875	1464 8260 ± 0	794 730 ± 1	559 741 ± 6	511 735 ± 12	455 732 ± 2	411 778 ± 1	818 2497 ± 26	822 2493 ± 51
292	cyberlink-016	1503336	105850	965 3591	1467 10308 ± 0	1461 1846 ± 1	1177 1908 ± 1	1161 1863 ± 1	1118 1864 ± 0	1044 2115 ± 3	443 1085 ± 4	442 1069 ± 19
293	dahua-000	283642	171917	-	412 2048 ± 0	312 363 ± 7	-	-	-	-	109 373 ± 16	110 364 ± 17
294	dahua-001	283642	171917	-	461 2048 ± 0	313 363 ± 6	-	-	-	-	106 367 ± 10	109 354 ± 16
295	dahua-002	526452	171518	-	478 2048 ± 0	649 628 ± 7	-	-	-	-	132 461 ± 23	137 454 ± 20
296	dahua-003	605337	141332	-	815 2048 ± 0	509 535 ± 2	-	-	-	-	250 648 ± 20	247 631 ± 34
297	dahua-004	832455	141070	-	780 2048 ± 0	797 735 ± 3	-	-	-	-	304 730 ± 25	301 707 ± 44
298	dahua-005	1586899	169478	-	1248 4096 ± 0	1382 1418 ± 34	-	-	-	-	390 957 ± 23	400 969 ± 19
299	dahua-006	831641	119261	1031 5068	414 2048 ± 0	1369 1398 ± 2	1072 1397 ± 1	1059 1404 ± 1	985 1402 ± 1	853 1402 ± 1	60 249 ± 13	64 250 ± 11
300	dahua-007	1578737	119418	1063 7237	1275 4096 ± 0	1365 1393 ± 2	1054 1373 ± 1	1039 1378 ± 1	965 1378 ± 1	839 1379 ± 2	107 367 ± 102	130 434 ± 108
301	daon-000	280726	2307	773 2010	1135 2065 ± 0	542 562 ± 3	388 581 ± 5	548 791 ± 9	539 838 ± 15	601 1055 ± 32	1268 16052 ± 88	1271 16041 ± 85
302	daon-001	296035	69405	220 587	773 2048 ± 0	728 682 ± 0	473 661 ± 0	464 688 ± 0	404 683 ± 0	303 657 ± 0	341 803 ± 23	355 815 ± 22
303	daon-002	295360	66546	299 731	932 2048 ± 0	828 765 ± 0	583 767 ± 1	539 768 ± 6	491 776 ± 1	404 770 ± 0	355 840 ± 17	366 858 ± 26
304	daon-003	931820	66767	720 1776	1300 4096 ± 0	1479 2048 ± 0	1188 2046 ± 0	1178 2050 ± 0	1145 2069 ± 0	1034 2047 ± 0	751 2194 ± 59	753 2203 ± 28
305	datech-000	617605	139575	1040 5599	1305 4096 ± 0	555 573 ± 32	390 583 ± 44	359 593 ± 37	1194 14123 ± 9564	1182 15514 ± 2343	1162 7857 ± 29	1162 7852 ± 25
306	datech-001	835404	139615	1062 7179	1291 4096 ± 0	1100 1017 ± 83	821 1057 ± 65	794 1062 ± 88	743 1109 ± 81	782 1299 ± 92	1165 7944 ± 31	1166 7981 ± 56
307	datech-002	854184	139615	1066 7446	1312 4096 ± 0	1148 1086 ± 115	881 1130 ± 119	832 1113 ± 123	830 1204 ± 107	861 1413 ± 145	1163 7860 ± 20	1163 7857 ± 26
308	decatur-000	350495	171271	394 907	1332 4100 ± 0	1105 1024 ± 2	-	-	-	-	1213 11439 ± 80	1215 11418 ± 112

Notes	
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4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
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Table 29: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
309	decatur-001	342866	253734	638 1508	1030 2052 ± 0	1159 1103 ± 2	826 1064 ± 2	795 1063 ± 2	710 1067 ± 2	626 1084 ± 2	212 610 ± 19	217 602 ± 8
310	decloakface-001	29693	170719	120 368	397 2048 ± 0	131 196 ± 58	82 201 ± 52	78 229 ± 53	96 298 ± 60	215 532 ± 61	1044 4638 ± 19	1044 4646 ± 22
311	decloakface-002	1692	143986	80 260	619 2048 ± 0	68 127 ± 0	41 135 ± 0	42 157 ± 0	61 218 ± 0	161 458 ± 2	1027 4380 ± 31	1028 4383 ± 45
312	deepglint-001	569802	206788	-	1264 4096 ± 0	784 721 ± 4	-	-	-	-	975 3680 ± 35	967 3517 ± 182
313	deepglint-002	459642	272878	-	1231 4096 ± 0	720 677 ± 2	623 826 ± 74	598 848 ± 42	545 849 ± 55	487 886 ± 27	1240 13633 ± 87	1239 12905 ± 440
314	deepglint-003	838065	262081	-	1424 6144 ± 0	1201 1159 ± 1	895 1145 ± 1	863 1148 ± 1	783 1148 ± 1	690 1163 ± 1	1278 17227 ± 41	1280 17210 ± 51
315	deepglint-004	1073382	261571	933 3116	488 2048 ± 0	1410 1470 ± 1	1121 1474 ± 1	1107 1485 ± 1	1039 1474 ± 1	916 1492 ± 2	1104 5961 ± 34	1102 5955 ± 29
316	deepglint-005	960326	213877	920 2991	651 2048 ± 0	1379 1408 ± 1	1098 1431 ± 2	1073 1424 ± 3	1004 1424 ± 3	887 1446 ± 2	1130 6765 ± 38	1127 6765 ± 40
317	deepsea-001	147497	336250	115 358	252 1024 ± 0	652 630 ± 7	570 752 ± 37	519 746 ± 30	448 727 ± 32	435 820 ± 32	545 1401 ± 37	576 1467 ± 50
318	deepsense-000	357113	936618	-	340 2048 ± 0	697 664 ± 3	460 645 ± 1	434 660 ± 2	410 687 ± 2	431 808 ± 3	141 480 ± 22	143 459 ± 34
319	deepsense-001	73173	1288355	1034 5314	116 512 ± 0	1187 1142 ± 2	909 1164 ± 3	889 1183 ± 3	827 1201 ± 3	801 1323 ± 2	788 2356 ± 35	787 2354 ± 42
320	deepsense-002	73173	1266131	1032 5280	212 896 ± 0	1335 1350 ± 1	1014 1299 ± 8	1000 1324 ± 0	940 1348 ± 0	905 1474 ± 4	955 3505 ± 32	966 3515 ± 40
321	deepsense-003	73173	1359497	1052 6555	215 901 ± 0	909 832 ± 2	725 940 ± 2	595 845 ± 4	643 983 ± 1	545 976 ± 7	663 1797 ± 25	668 1795 ± 19
322	deepsense-004	560774	1052093	1065 7310	216 916 ± 0	1284 1259 ± 2	1010 1294 ± 9	958 1268 ± 7	881 1261 ± 8	813 1334 ± 7	685 1884 ± 19	689 1884 ± 25
323	deepsense-005	560774	1052093	1064 7309	217 916 ± 0	1388 1430 ± 1	990 1264 ± 1	940 1246 ± 1	875 1254 ± 1	822 1353 ± 4	697 1920 ± 26	702 1932 ± 40
324	deepsense-006	643809	1052098	1067 7719	218 916 ± 0	1332 1348 ± 2	1043 1347 ± 2	892 1184 ± 4	960 1369 ± 3	902 1473 ± 4	680 1860 ± 13	682 1853 ± 22
325	deepsense-007	643809	1070446	1068 7769	220 920 ± 0	1205 1163 ± 1	915 1173 ± 3	894 1190 ± 2	836 1209 ± 3	790 1310 ± 4	677 1847 ± 11	684 1860 ± 28
326	deepsense-008	643809	1070340	1069 7770	219 920 ± 0	1436 1571 ± 1	1140 1572 ± 0	1135 1610 ± 1	1093 1729 ± 1	1056 2216 ± 1	1034 4469 ± 109	1034 4498 ± 68
327	dermalog-001	0	287736	-	282 1043 ± 0	51 106 ± 1	-	-	-	-	1312 22160 ± 166	1316 22131 ± 131
328	dermalog-002	0	287738	-	281 1043 ± 0	30 81 ± 0	-	-	-	-	1313 22169 ± 114	1315 22105 ± 146
329	dermalog-003	0	309430	-	283 1043 ± 0	61 121 ± 22	-	-	-	-	1318 22957 ± 93	1319 22808 ± 131
330	dermalog-004	0	245770	-	9 128 ± 0	84 149 ± 5	-	-	-	-	149 490 ± 26	147 471 ± 26
331	dermalog-005	0	317687	-	5 128 ± 0	70 130 ± 11	-	-	-	-	153 499 ± 22	157 500 ± 22
332	dermalog-006	0	452387	-	4 128 ± 0	505 532 ± 12	-	-	-	-	156 506 ± 23	144 459 ± 23
333	dermalog-007	0	503332	-	141 512 ± 0	167 239 ± 9	-	-	-	-	369 887 ± 23	290 690 ± 25
334	dermalog-008	0	937895	-	119 512 ± 0	363 404 ± 2	235 410 ± 3	211 424 ± 5	183 430 ± 5	171 477 ± 5	136 468 ± 31	98 328 ± 13
335	dermalog-009	0	319363	597 1369	121 512 ± 0	292 349 ± 0	181 351 ± 0	145 352 ± 0	130 357 ± 0	116 389 ± 0	146 487 ± 34	120 385 ± 29
336	dermalog-010	0	525908	978 3789	143 512 ± 0	661 635 ± 0	456 640 ± 1	418 639 ± 4	367 647 ± 3	331 691 ± 5	128 444 ± 13	105 341 ± 26
337	dermalog-011	0	278395	290 716	11 128 ± 0	287 343 ± 0	173 345 ± 0	139 347 ± 0	124 351 ± 0	101 363 ± 0	78 299 ± 19	67 253 ± 14
338	dermalog-012	0	293431	293 723	10 128 ± 0	112 181 ± 0	59 179 ± 0	49 182 ± 0	44 186 ± 0	53 242 ± 0	27 163 ± 10	43 196 ± 16
339	dermalog-013	0	293658	1001 4263	6 128 ± 0	114 183 ± 2	62 180 ± 2	48 181 ± 1	41 183 ± 1	37 205 ± 1	35 186 ± 8	52 221 ± 7
340	dermalog-014	0	756422	537 1201	227 1024 ± 0	141 204 ± 0	80 197 ± 0	63 198 ± 1	51 199 ± 0	51 241 ± 0	120 424 ± 17	117 377 ± 17
341	dermalog-015	0	909348	846 2452	261 1024 ± 0	1471 1929 ± 1	1175 1891 ± 0	1173 1955 ± 1	1127 1899 ± 1	1014 1904 ± 1	103 360 ± 25	73 267 ± 14
342	dermalog-016	0	910481	845 2452	258 1024 ± 0	1469 1913 ± 2	1176 1908 ± 1	1165 1918 ± 1	1128 1917 ± 1	1016 1927 ± 1	162 523 ± 23	141 457 ± 25
343	dicio-001	61751	119517	22 84	173 520 ± 0	516 538 ± 0	371 563 ± 10	647 915 ± 3	1107 1800 ± 7	1142 5286 ± 30	867 2818 ± 20	868 2807 ± 31
344	dicio-002	571566	85868	566 1290	1125 2060 ± 0	924 846 ± 1	577 761 ± 0	542 775 ± 1	499 784 ± 1	447 831 ± 0	1118 6242 ± 74	1116 6256 ± 80
345	didiglobalface-001	259849	70680	192 527	662 2048 ± 0	620 612 ± 1	453 633 ± 3	411 634 ± 3	371 650 ± 15	313 666 ± 4	395 973 ± 20	404 988 ± 20
346	didiglobalface-002	260054	161508	337 812	433 2048 ± 0	640 622 ± 1	452 633 ± 1	423 642 ± 2	379 659 ± 4	361 726 ± 15	174 560 ± 10	185 567 ± 13
347	didiglobalface-003	1001649	83986	558 1274	343 2048 ± 0	803 740 ± 2	555 740 ± 2	538 767 ± 5	492 779 ± 4	466 853 ± 17	602 1544 ± 13	606 1539 ± 17
348	digidata-000	133370	30249	79 259	790 2048 ± 0	309 361 ± 0	186 360 ± 0	154 361 ± 0	133 363 ± 0	109 380 ± 0	731 2084 ± 37	720 2039 ± 42
349	digidata-001	254564	33036	118 367	816 2048 ± 0	538 559 ± 1	368 561 ± 1	331 562 ± 1	291 564 ± 1	263 602 ± 1	1199 10308 ± 102	1200 10301 ± 121
350	digitalbarriers-000	157794	427472	-	1059 2056 ± 0	50 104 ± 0	-	-	-	-	1237 13232 ± 166	1241 13226 ± 146
351	digitalbarriers-001	236915	427537	-	1086 2056 ± 0	225 294 ± 1	-	-	-	-	1230 12311 ± 164	1232 12347 ± 197
352	digitalbarriers-002	83002	598577	756 1930	1058 2056 ± 0	145 209 ± 11	109 250 ± 19	195 411 ± 37	513 808 ± 72	1057 2236 ± 123	1238 13409 ± 228	1242 13267 ± 206

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	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
353	dps-000	0	2211812	606 1405	1272 4096 ± 0	939 868 ± 2	684 893 ± 6	1086 1445 ± 9	1174 2910 ± 38	1167 9345 ± 17	572 1473 ± 37	581 1479 ± 37
354	dralmey-001	571556	82266	565 1290	1130 2060 ± 0	1225 1186 ± 0	908 1162 ± 0	878 1170 ± 0	810 1180 ± 0	734 1243 ± 0	1244 13996 ± 56	1247 13971 ± 60
355	dsk-000	11967	782905	77 252	109 512 ± 0	241 304 ± 47	154 317 ± 33	736 1001 ± 96	1171 2660 ± 170	1175 10451 ± 832	1146 7152 ± 115	1141 7134 ± 111
356	einetworks-000	372608	219883	382 880	1055 2056 ± 0	671 645 ± 3	-	-	-	-	1058 4876 ± 66	1070 5156 ± 77
357	einetworksindia-000	372375	479472	578 1322	1108 2056 ± 0	328 378 ± 1	211 382 ± 1	176 383 ± 1	163 397 ± 0	160 456 ± 3	1261 15171 ± 41	1262 15138 ± 62
358	einetworksindia-001	562398	481792	729 1808	1051 2056 ± 0	452 479 ± 0	299 481 ± 1	265 485 ± 1	238 494 ± 1	223 541 ± 1	1259 15147 ± 63	1261 15134 ± 74
359	einetworksindia-002	1214466	143768	850 2462	942 2048 ± 0	1149 1087 ± 1	835 1082 ± 1	797 1065 ± 0	734 1098 ± 3	678 1150 ± 2	1241 13785 ± 34	1244 13767 ± 56
360	ekin-002	51434	278	41 139	1196 3072 ± 0	1226 1186 ± 13	921 1180 ± 12	887 1181 ± 11	822 1191 ± 11	715 1207 ± 8	1023 4294 ± 80	1089 5569 ± 112
361	element-000	721679	24895	561 1287	450 2048 ± 0	357 399 ± 1	239 417 ± 1	212 428 ± 9	227 478 ± 14	328 688 ± 45	183 583 ± 10	194 585 ± 13
362	enface-000	369598	153781	269 671	222 1024 ± 0	534 555 ± 4	366 558 ± 4	447 669 ± 6	647 987 ± 15	1068 2349 ± 54	1142 7059 ± 62	1137 6980 ± 65
363	enface-001	370710	173609	268 669	242 1024 ± 0	530 550 ± 4	363 555 ± 3	446 668 ± 7	641 981 ± 15	1078 2416 ± 59	1129 6734 ± 68	1128 6766 ± 69
364	enface-002	858356	98741	1074 9004	226 1024 ± 0	1108 1026 ± 1	813 1038 ± 3	787 1057 ± 7	754 1123 ± 4	814 1338 ± 8	599 1529 ± 23	603 1530 ± 35
365	enqura-001	0	6136892	551 1263	407 2048 ± 0	156 221 ± 5	94 220 ± 3	83 239 ± 3	77 262 ± 3	66 288 ± 3	1 0 ± 0	1 0 ± 0
366	enqura-002	724175	1441517	376 870	1115 2057 ± 0	423 456 ± 1	281 457 ± 1	266 486 ± 1	340 619 ± 1	936 1532 ± 3	1350 29132 ± 232	1352 28946 ± 171
367	eocortex-000	255937	59432	73 224	510 2048 ± 0	242 305 ± 22	170 341 ± 25	224 440 ± 47	215 464 ± 45	201 513 ± 44	382 923 ± 11	388 918 ± 11
368	ercacat-001	811623	58012	906 2816	1040 2052 ± 0	1125 1052 ± 3	-	-	-	-	826 2551 ± 62	823 2501 ± 81
369	ethiopianlmis-000	309182	549917	166 481	601 2048 ± 0	963 882 ± 2	680 887 ± 1	575 824 ± 2	540 838 ± 2	501 903 ± 1	798 2400 ± 34	798 2407 ± 28
370	euronovate-001	0	1774966	583 1346	295 1177 ± 0	1117 1034 ± 2	910 1165 ± 3	871 1160 ± 3	806 1177 ± 3	694 1172 ± 2	1454 81294 ± 591	1452 81631 ± 931
371	euronovate-002	0	5476674	343 818	1403 4249 ± 0	224 293 ± 0	143 296 ± 0	108 298 ± 0	100 309 ± 0	85 339 ± 0	1365 32673 ± 391	1366 32587 ± 391
372	euronovate-003	0	5637394	533 1193	1171 2201 ± 0	234 300 ± 0	146 300 ± 0	110 303 ± 0	101 310 ± 0	86 340 ± 0	1381 37908 ± 968	1385 38285 ± 938
373	euronovate-004	0	1335993	456 1049	1170 2201 ± 0	219 290 ± 0	139 294 ± 0	105 294 ± 0	94 295 ± 0	67 294 ± 0	1395 46290 ± 490	1395 46199 ± 484
374	everai-000	229700	144802	-	737 2048 ± 0	421 455 ± 1	-	-	-	-	113 385 ± 10	119 382 ± 13
375	everai-001	449149	144802	-	952 2048 ± 0	762 701 ± 1	-	-	-	-	115 395 ± 11	125 404 ± 23
376	everai-002	561727	281965	-	1297 4096 ± 0	821 758 ± 0	-	-	-	-	248 644 ± 14	241 624 ± 35
377	everai-paravision-003	539802	118876	-	1259 4096 ± 0	713 674 ± 4	-	-	-	-	281 699 ± 20	307 713 ± 47
378	expasoft-000	15341	240451	-	617 2048 ± 0	22 68 ± 0	15 71 ± 0	11 75 ± 0	10 71 ± 0	8 72 ± 1	659 1779 ± 26	657 1757 ± 97
379	expasoft-001	39057	983064	44 142	633 2048 ± 0	23 70 ± 0	16 74 ± 0	12 77 ± 0	11 73 ± 0	9 74 ± 0	627 1660 ± 35	631 1676 ± 48
380	expasoft-002	38760	59825	58 170	390 2048 ± 0	8 34 ± 0	5 34 ± 0	5 34 ± 0	4 34 ± 0	2 34 ± 0	1178 8870 ± 78	1179 8838 ± 77
381	f8-001	272977	19668	559 1276	950 2048 ± 0	897 822 ± 39	-	-	-	-	1264 15262 ± 139	1267 15277 ± 212
382	f8-002	28278	215616	21 84	727 2048 ± 0	11 39 ± 0	7 41 ± 0	10 75 ± 0	50 197 ± 1	340 702 ± 1	1253 14765 ± 131	1256 14790 ± 133
383	facehawk-000	120009	7737	37 129	208 776 ± 1	355 399 ± 0	230 402 ± 0	198 414 ± 1	157 391 ± 2	147 440 ± 3	1208 10931 ± 155	1208 10915 ± 149
384	facehawk-001	173511	35863	110 343	1141 2070 ± 0	310 362 ± 0	193 363 ± 0	163 371 ± 0	144 375 ± 0	145 437 ± 0	568 1465 ± 19	578 1475 ± 11
385	faceint-001	110744	649612	916 2903	824 2048 ± 0	130 195 ± 0	75 191 ± 0	62 195 ± 0	53 203 ± 3	46 232 ± 10	557 1434 ± 11	562 1431 ± 19
386	facelocate-001	4739	11106	842 2449	19 168 ± 0	7 34 ± 1	6 38 ± 0	14 84 ± 1	22 112 ± 4	38 206 ± 13	1185 9565 ± 91	1189 9629 ± 131
387	facelocate-002	15198	486568	84 271	113 512 ± 0	152 214 ± 1	104 239 ± 0	205 422 ± 2	213 463 ± 5	241 571 ± 15	213 611 ± 8	237 620 ± 17
388	faceonlive-001	0	71529	95 300	1089 2056 ± 0	109 179 ± 0	60 179 ± 0	59 190 ± 0	59 217 ± 0	88 343 ± 1	432 1064 ± 37	425 1033 ± 35
389	faceonlive-002	155220	141019	429 997	529 2048 ± 0	846 783 ± 1	604 797 ± 2	551 794 ± 2	514 809 ± 3	498 901 ± 2	1242 13798 ± 197	1243 13743 ± 127
390	faceonlive-003	518321	560984	1017 4691	254 1024 ± 0	1116 1032 ± 4	844 1091 ± 3	1171 1942 ± 6	1091 1723 ± 5	1130 4168 ± 3	1269 16149 ± 41	1272 16135 ± 54
391	facephi-000	148904	5219	1084 20248	473 2048 ± 0	944 871 ± 2	674 881 ± 3	624 880 ± 4	577 888 ± 4	527 949 ± 12	1014 4067 ± 53	1017 4047 ± 53
392	facephi-001	138133	282874	584 1347	138 512 ± 0	216 285 ± 0	122 272 ± 2	104 291 ± 5	95 296 ± 3	179 489 ± 5	920 3194 ± 41	925 3184 ± 54
393	facephi-002	1031362	112529	1049 6074	453 2048 ± 0	1271 1240 ± 5	979 1251 ± 5	947 1253 ± 21	907 1298 ± 5	986 1713 ± 19	1224 11961 ± 81	1227 11970 ± 83
394	facesoft-000	370120	10612	329 796	600 2048 ± 0	715 675 ± 18	479 669 ± 3	460 686 ± 3	397 675 ± 5	326 687 ± 2	760 2239 ± 28	770 2277 ± 96
395	facetag-000	1232331	4022	417 965	204 684 ± 0	298 355 ± 17	201 369 ± 8	720 989 ± 33	1166 2408 ± 91	1160 7930 ± 316	1442 72003 ± 625	1442 71912 ± 612
396	facetag-001	1258247	4022	-	1339 4100 ± 0	1078 991 ± 3	780 995 ± 3	749 1018 ± 3	714 1069 ± 5	769 1284 ± 8	933 3323 ± 28	939 3287 ± 34

Notes	
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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 31: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
397	facetag-002	819806	4021	285 706	911 2048 ± 0	524 544 ± 1	354 544 ± 0	318 542 ± 0	278 545 ± 0	232 554 ± 0	645 1730 ± 25	652 1733 ± 25
398	facex-001	305074	930372	947 3336	490 2048 ± 0	385 422 ± 4	256 434 ± 4	301 520 ± 7	460 737 ± 13	979 1670 ± 27	682 1871 ± 23	680 1846 ± 29
399	facex-002	305074	928334	924 3011	420 2048 ± 0	389 426 ± 5	250 429 ± 4	295 516 ± 8	452 730 ± 12	989 1738 ± 36	233 631 ± 25	233 614 ± 19
400	facia-001	296496	745403	390 889	910 2048 ± 0	1155 1099 ± 1	862 1109 ± 0	831 1113 ± 0	759 1125 ± 3	740 1250 ± 1	1428 62245 ± 397	1425 62249 ± 496
401	facia-002	585227	694738	751 1907	1045 2052 ± 0	1330 1347 ± 4	1042 1337 ± 3	1116 1509 ± 7	1129 1919 ± 11	1124 3982 ± 34	1309 21963 ± 400	1312 21869 ± 318
402	faicetech-000	247687	34022	586 1349	393 2048 ± 0	1221 1179 ± 1	936 1199 ± 1	896 1193 ± 1	828 1202 ± 0	733 1240 ± 4	190 594 ± 4	205 593 ± 1
403	farfaces-001	346494	44581	81 262	112 512 ± 0	1220 1179 ± 1	922 1180 ± 1	885 1180 ± 0	817 1185 ± 1	718 1209 ± 2	359 855 ± 25	367 860 ± 31
404	fastenterprises-000	273365	235899	445 1030	922 2048 ± 0	693 662 ± 1	483 671 ± 0	443 664 ± 0	396 673 ± 3	327 688 ± 5	990 3806 ± 96	1365 32064 ± 185
405	fci-000	1486985	454513	934 3123	702 2048 ± 0	1408 1464 ± 12	1126 1494 ± 9	1095 1457 ± 10	1034 1462 ± 11	933 1525 ± 5	779 2309 ± 8	781 2314 ± 14
406	fdu-000	202296	1307	-	183 534 ± 0	33 84 ± 0	-	-	-	-	639 1713 ± 28	644 1715 ± 20
407	fdu-001	345583	1329	-	330 1808 ± 0	71 131 ± 0	-	-	-	-	516 1288 ± 26	520 1281 ± 22
408	fedu-001	466605	496375	657 1557	782 2048 ± 0	589 592 ± 1	432 623 ± 1	389 624 ± 0	345 623 ± 0	298 647 ± 0	789 2362 ± 13	788 2361 ± 22
409	fiberhome-nanjing-002	207398	596827	-	248 1024 ± 0	1248 1217 ± 2	1009 1294 ± 30	993 1312 ± 12	1041 1477 ± 14	806 1326 ± 11	608 1582 ± 57	607 1560 ± 51
410	fiberhome-nanjing-003	352895	1482309	373 859	701 2048 ± 0	1182 1136 ± 7	882 1134 ± 4	850 1132 ± 3	775 1139 ± 3	681 1154 ± 5	449 1097 ± 38	452 1083 ± 42
411	fiberhome-nanjing-004	443779	1482313	473 1077	1284 4096 ± 0	1322 1321 ± 5	1017 1304 ± 3	990 1307 ± 2	915 1308 ± 3	805 1326 ± 5	511 1276 ± 40	514 1265 ± 38
412	fincore-000	256615	19409	197 536	704 2048 ± 0	482 508 ± 3	322 505 ± 0	288 508 ± 1	252 513 ± 2	217 535 ± 1	653 1765 ± 31	659 1763 ± 22
413	firstcreditkz-001	553811	24803	494 1113	409 2048 ± 0	873 808 ± 0	781 997 ± 0	792 1061 ± 1	804 1174 ± 1	995 1774 ± 54	374 904 ± 20	377 903 ± 23
414	firstcreditkz-002	977538	24879	759 1951	394 2048 ± 0	1287 1265 ± 1	996 1275 ± 4	988 1299 ± 7	937 1341 ± 17	927 1515 ± 40	318 761 ± 28	330 774 ± 32
415	firstcreditkz-003	1145884	24907	797 2144	749 2048 ± 0	1228 1188 ± 79	1125 1492 ± 77	1190 2566 ± 293	1187 4939 ± 888	1181 14753 ± 3518	221 616 ± 10	231 613 ± 16
416	firstcreditkz-004	1205746	38973	871 2587	424 2048 ± 0	1134 1060 ± 5	966 1236 ± 4	1015 1343 ± 6	1096 1752 ± 15	1123 3765 ± 42	164 525 ± 16	164 521 ± 14
417	foomobi-001	0	219961	66 189	664 2048 ± 0	302 357 ± 0	216 388 ± 2	853 1138 ± 1	1178 2974 ± 10	1172 10101 ± 10	1401 47409 ± 453	1401 47413 ± 421
418	foomobi-002	0	219961	65 189	918 2048 ± 0	233 299 ± 0	158 323 ± 0	690 957 ± 1	1168 2496 ± 2	1162 8549 ± 7	1355 30257 ± 724	1354 30010 ± 320
419	fourthlinecom-001	0	437655	831 2410	560 2048 ± 0	1345 1372 ± 5	984 1257 ± 4	972 1283 ± 2	868 1243 ± 3	735 1244 ± 4	797 2391 ± 17	797 2403 ± 26
420	fpt-000	19013	170633	72 219	897 2048 ± 0	59 118 ± 0	36 122 ± 1	29 122 ± 0	24 120 ± 0	28 178 ± 2	994 3871 ± 31	996 3883 ± 29
421	fpt-001	256529	142347	225 600	933 2048 ± 0	602 602 ± 0	409 606 ± 0	388 623 ± 1	344 621 ± 0	289 637 ± 0	1000 3934 ± 10	1003 3931 ± 14
422	fpttelecom-000	266955	273536	168 485	777 2048 ± 0	1438 1587 ± 0	1146 1589 ± 0	1134 1593 ± 0	1078 1591 ± 0	960 1599 ± 0	1164 7881 ± 40	1164 7901 ± 48
423	fraudcom-000	354020	561861	418 966	1004 2052 ± 0	540 561 ± 2	347 539 ± 2	315 541 ± 1	287 557 ± 5	254 594 ± 6	766 2258 ± 10	766 2256 ± 29
424	frpkauai-001	507771	24807	997 4143	803 2048 ± 0	741 689 ± 1	509 691 ± 0	473 697 ± 2	440 714 ± 6	409 775 ± 31	313 752 ± 29	327 764 ± 23
425	frpkauai-002	519141	24803	495 1113	865 2048 ± 0	868 799 ± 0	771 987 ± 0	777 1046 ± 1	792 1163 ± 2	994 1769 ± 4	376 907 ± 20	374 886 ± 28
426	frpkauai-003	1135621	24939	804 2184	415 2048 ± 0	1031 948 ± 3	863 1110 ± 2	921 1225 ± 5	959 1369 ± 16	1027 2007 ± 34	163 524 ± 14	168 529 ± 19
427	frpkauai-004	513748	3521	547 1248	72 296 ± 0	259 317 ± 1	153 316 ± 3	121 317 ± 1	110 322 ± 1	103 370 ± 9	1158 7700 ± 81	1157 7659 ± 78
428	frpkauai-005	1250936	39141	861 2544	1026 2052 ± 0	1439 1592 ± 2	1165 1814 ± 2	1164 1910 ± 6	1149 2094 ± 12	1103 2906 ± 39	484 1181 ± 71	491 1182 ± 53
429	fujitsulab-000	0	205894	-	139 512 ± 0	381 419 ± 1	-	-	-	-	970 3613 ± 37	974 3621 ± 29
430	fujitsulab-001	0	548917	-	1358 4104 ± 0	1310 1300 ± 5	1007 1291 ± 2	986 1295 ± 2	910 1304 ± 2	841 1382 ± 2	874 2853 ± 16	875 2847 ± 14
431	fujitsulab-002	0	1088887	692 1685	1353 4104 ± 0	1267 1237 ± 2	956 1222 ± 2	931 1236 ± 1	870 1251 ± 2	808 1327 ± 2	869 2836 ± 25	869 2809 ± 44
432	fujitsulab-003	662263	318209	1055 6691	1359 4104 ± 0	1037 951 ± 20	726 941 ± 19	685 952 ± 19	637 971 ± 20	593 1045 ± 21	875 2855 ± 16	876 2849 ± 19
433	g42-intellibrain-001	1031335	235521	1087 25154	66 269 ± 0	1066 976 ± 6	756 975 ± 1	732 997 ± 2	712 1068 ± 3	828 1362 ± 8	1100 5878 ± 96	1100 5865 ± 71
434	g42-intellibrain-002	1043920	132389	1083 19747	67 269 ± 0	995 915 ± 2	705 920 ± 2	679 946 ± 2	660 998 ± 4	724 1224 ± 8	1251 14703 ± 56	1255 14658 ± 97
435	geo-000	114483	304983	-	392 2048 ± 0	927 851 ± 1	-	-	-	-	1465 137544 ± 6225	1465 137925 ± 6548
436	geo-001	258516	70163	-	432 2048 ± 0	1144 1079 ± 0	832 1076 ± 0	802 1076 ± 0	723 1078 ± 0	639 1102 ± 0	918 3163 ± 47	921 3158 ± 38
437	geo-002	369903	98667	442 1021	666 2048 ± 0	855 791 ± 1	600 793 ± 0	552 794 ± 0	505 795 ± 1	427 803 ± 1	943 3407 ± 45	951 3422 ± 65
438	geo-003	371712	102175	-	826 2048 ± 0	1297 1283 ± 1	1006 1290 ± 1	973 1285 ± 1	906 1292 ± 1	785 1302 ± 1	403 997 ± 13	412 1001 ± 20
439	geo-004	168980	107714	562 1287	546 2048 ± 0	1289 1268 ± 1	998 1279 ± 1	965 1274 ± 0	879 1259 ± 1	778 1296 ± 1	415 1023 ± 20	423 1028 ± 22
440	gistouch-000	155338	444582	201 541	943 2048 ± 0	266 325 ± 1	162 326 ± 1	134 337 ± 1	127 354 ± 1	182 495 ± 1	1181 9505 ± 604	1180 9443 ± 388

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Table 32: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
441	gistouch-001	426635	444545	685 1651	917 2048 ± 0	1010 930 ± 3	703 919 ± 3	670 937 ± 4	636 966 ± 2	646 1113 ± 1	1171 8475 ± 133	1173 8460 ± 173
442	glory-000	0	98176	-	84 418 ± 0	94 165 ± 2	-	-	-	-	1140 7003 ± 84	1136 6978 ± 71
443	glory-001	0	144786	-	329 1726 ± 0	350 393 ± 2	-	-	-	-	1189 9607 ± 128	1184 9539 ± 182
444	glory-002	0	385177	-	1158 2106 ± 0	595 594 ± 3	556 740 ± 3	681 948 ± 3	1154 2168 ± 6	32 191 ± 15	1131 6787 ± 85	1121 6551 ± 249
445	glory-003	0	536910	-	1399 4234 ± 0	462 489 ± 0	372 565 ± 0	506 732 ± 0	1121 1876 ± 2	1163 8941 ± 20	1106 6020 ± 90	1106 6003 ± 72
446	glory-004	0	999639	809 2200	1393 4182 ± 0	738 688 ± 0	575 759 ± 1	674 941 ± 1	1151 2134 ± 4	1168 9360 ± 47	1063 4982 ± 66	1064 4990 ± 63
447	glory-005	0	999999	843 2450	1391 4182 ± 0	765 703 ± 1	597 789 ± 0	702 972 ± 1	1156 2200 ± 25	1170 9679 ± 22	1073 5224 ± 93	1072 5176 ± 81
448	glory-006	0	1004784	839 2438	1392 4182 ± 0	766 704 ± 1	589 774 ± 0	700 968 ± 2	1162 2286 ± 18	1174 10381 ± 151	1098 5853 ± 77	1099 5819 ± 74
449	glory-007	0	1004846	840 2442	1394 4182 ± 0	342 385 ± 0	252 431 ± 0	344 575 ± 14	1035 1464 ± 24	1154 6782 ± 62	957 3506 ± 51	960 3489 ± 66
450	gorilla-000	98290	1122916	-	1397 4204 ± 0	184 256 ± 14	-	-	-	-	996 3875 ± 29	995 3872 ± 51
451	gorilla-001	93768	1101838	-	1165 2156 ± 0	90 160 ± 14	-	-	-	-	946 3429 ± 145	940 3288 ± 51
452	gorilla-002	93869	1281839	-	294 1132 ± 0	261 322 ± 14	-	-	-	-	851 2715 ± 68	834 2585 ± 84
453	gorilla-003	94409	1281823	-	293 1132 ± 0	279 334 ± 25	-	-	-	-	871 2840 ± 42	879 2865 ± 87
454	gorilla-004	186952	1274131	-	1169 2192 ± 0	345 389 ± 3	-	-	-	-	860 2768 ± 44	858 2745 ± 44
455	gorilla-005	100684	1297614	-	1168 2192 ± 0	368 407 ± 3	262 437 ± 2	229 448 ± 4	223 473 ± 4	250 589 ± 15	843 2678 ± 42	861 2770 ± 112
456	gorilla-006	172743	1318812	-	1401 4240 ± 0	420 454 ± 3	303 484 ± 3	277 497 ± 5	274 543 ± 25	515 928 ± 60	986 3755 ± 38	987 3737 ± 44
457	gorilla-007	441058	708166	691 1683	1442 6288 ± 0	590 592 ± 1	398 592 ± 1	368 603 ± 1	348 625 ± 2	356 722 ± 9	977 3686 ± 37	983 3709 ± 36
458	gorilla-008	450175	707000	732 1812	1465 8338 ± 0	597 595 ± 1	397 590 ± 0	365 600 ± 1	343 621 ± 2	355 720 ± 9	1039 4530 ± 44	1037 4524 ± 38
459	gorilla-009	329584	297395	575 1304	1402 4242 ± 0	975 899 ± 2	709 922 ± 1	637 901 ± 3	598 924 ± 4	587 1032 ± 12	775 2294 ± 74	778 2301 ± 66
460	gpstechvn-000	108365	345975	876 2646	476 2048 ± 0	230 296 ± 1	144 297 ± 1	111 304 ± 0	104 316 ± 0	110 382 ± 0	805 2442 ± 11	807 2440 ± 16
461	graymatics-001	13095	70406	35 128	1228 4096 ± 0	125 191 ± 1	83 203 ± 1	357 592 ± 5	1087 1698 ± 9	1156 7150 ± 34	1386 39874 ± 309	1388 39762 ± 295
462	griaule-000	0	598214	460 1055	1039 2052 ± 0	377 416 ± 6	245 425 ± 7	541 770 ± 14	1095 1749 ± 43	1152 6406 ± 189	1007 3987 ± 42	1007 3938 ± 38
463	griaule-001	0	412061	555 1271	1014 2052 ± 0	1206 1164 ± 1	852 1096 ± 5	820 1099 ± 4	772 1136 ± 2	924 1509 ± 2	1003 3948 ± 23	1010 3957 ± 32
464	griaule-002	0	1320474	745 1870	1038 2052 ± 0	896 822 ± 1	711 924 ± 4	642 907 ± 1	693 1038 ± 21	874 1430 ± 9	1008 4005 ± 32	1014 4012 ± 31
465	gtelots-000	252631	59244	204 552	111 512 ± 0	848 785 ± 1	602 794 ± 0	560 810 ± 0	526 823 ± 2	482 875 ± 0	1310 22029 ± 130	1314 21988 ± 93
466	gtelots-001	442881	58709	419 966	1167 2176 ± 0	1109 1027 ± 0	804 1028 ± 0	768 1037 ± 0	713 1069 ± 0	692 1167 ± 1	151 493 ± 31	156 496 ± 38
467	hertasecurity-000	0	780014	-	40 256 ± 0	45 99 ± 0	26 98 ± 0	23 100 ± 0	20 107 ± 0	20 139 ± 0	293 710 ± 31	275 667 ± 28
468	hertasecurity-001	0	944427	528 1188	150 512 ± 0	290 346 ± 0	174 345 ± 0	141 349 ± 0	126 354 ± 0	115 388 ± 0	655 1770 ± 45	649 1726 ± 48
469	hertasecurity-002	0	944582	522 1182	153 512 ± 0	458 484 ± 7	295 478 ± 3	259 480 ± 3	239 495 ± 3	209 520 ± 3	773 2289 ± 40	769 2267 ± 48
470	hertasecurity-003	0	944583	521 1181	118 512 ± 0	467 495 ± 7	298 480 ± 3	267 486 ± 3	235 491 ± 3	210 520 ± 4	1156 7619 ± 65	1156 7597 ± 94
471	hik-001	667866	9290	1053 6597	306 1408 ± 0	677 651 ± 0	478 667 ± 8	454 677 ± 16	409 686 ± 13	369 737 ± 12	147 488 ± 19	150 477 ± 22
472	hisign-001	732412	167488	660 1560	1149 2080 ± 0	1313 1306 ± 1	1029 1320 ± 1	996 1315 ± 1	921 1312 ± 1	803 1325 ± 1	42 201 ± 10	36 185 ± 13
473	hisign-002	1014906	102652	792 2131	1151 2080 ± 0	864 797 ± 0	606 800 ± 5	555 800 ± 0	508 801 ± 0	428 803 ± 1	55 232 ± 11	46 207 ± 11
474	hisign-003	2232829	156367	672 1599	1148 2080 ± 0	1121 1040 ± 5	836 1083 ± 4	806 1085 ± 4	729 1088 ± 4	636 1093 ± 4	28 167 ± 8	25 150 ± 10
475	hyperverge-001	260819	88624	-	951 2048 ± 0	727 682 ± 20	513 695 ± 17	902 1196 ± 37	1165 2400 ± 68	1157 7178 ± 204	1108 6026 ± 40	1104 5984 ± 38
476	hyperverge-002	2951900	198832	766 1975	224 1024 ± 0	1014 938 ± 1	724 939 ± 1	672 941 ± 1	616 945 ± 1	544 975 ± 1	1107 6023 ± 37	1103 5966 ± 40
477	hyperverge-003	1167779	282156	882 2687	268 1024 ± 0	1416 1477 ± 2	1129 1503 ± 3	1118 1520 ± 3	1059 1525 ± 4	950 1565 ± 3	177 566 ± 11	181 561 ± 8
478	hyperverge-004	4924393	282156	984 3908	726 2048 ± 0	1411 1471 ± 2	1099 1434 ± 1	1087 1446 ± 6	1019 1445 ± 3	915 1491 ± 3	402 996 ± 10	411 1000 ± 19
479	hyperverge-005	3413303	284050	881 2686	223 1024 ± 0	744 689 ± 0	507 690 ± 0	468 693 ± 0	426 702 ± 6	382 752 ± 5	133 465 ± 2	146 464 ± 10
480	hyperverge-006	3901500	284050	781 2070	250 1024 ± 0	832 768 ± 1	585 770 ± 2	536 766 ± 1	489 773 ± 1	443 828 ± 6	134 466 ± 2	145 463 ± 4
481	hyperverge-007	2223041	169168	800 2159	256 1024 ± 0	1325 1327 ± 1	1012 1296 ± 0	970 1280 ± 0	903 1286 ± 0	817 1348 ± 0	258 659 ± 19	285 684 ± 26
482	hyundai-roboticslab-000	181134	191546	508 1149	692 2048 ± 0	327 376 ± 3	198 368 ± 1	160 367 ± 2	142 374 ± 0	128 408 ± 1	866 2815 ± 47	867 2790 ± 50
483	hyundai-roboticslab-001	361330	106869	346 820	960 2048 ± 0	304 358 ± 1	189 361 ± 1	155 362 ± 2	143 374 ± 3	121 394 ± 0	911 3115 ± 34	914 3117 ± 45
484	hyundai-roboticslab-002	181140	106752	342 817	516 2048 ± 0	291 347 ± 0	184 355 ± 0	150 357 ± 0	132 362 ± 3	114 387 ± 0	926 3243 ± 111	935 3245 ± 114

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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 33: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
485	hyundai-roboticslab-003	484277	106656	775 2027	645 2048 ± 0	1456 1800 ± 0	1163 1792 ± 0	1152 1802 ± 0	1123 1889 ± 0	1001 1810 ± 0	1177 8841 ± 42	1178 8819 ± 54
486	hyundai-roboticslab-004	484277	106656	774 2026	514 2048 ± 0	1464 1882 ± 1	1170 1842 ± 2	1156 1839 ± 0	1110 1837 ± 0	1007 1871 ± 0	1207 10743 ± 201	1207 10767 ± 219
487	hzailu-001	0	372018	-	1075 2056 ± 0	1177 1126 ± 1	879 1128 ± 1	848 1130 ± 1	768 1132 ± 1	684 1159 ± 1	371 894 ± 19	376 899 ± 22
488	hzailu-002	1515880	74047	1019 4722	1057 2056 ± 0	1195 1150 ± 5	877 1127 ± 6	846 1129 ± 7	774 1137 ± 7	695 1172 ± 3	441 1079 ± 53	443 1070 ± 31
489	hzailu-003	1923030	222185	1026 4858	1201 3080 ± 0	1361 1389 ± 5	1038 1331 ± 7	1010 1334 ± 2	942 1349 ± 6	870 1424 ± 8	578 1483 ± 35	575 1464 ± 31
490	hzailu-004	1938475	275639	990 4023	1200 3080 ± 0	1425 1495 ± 8	1124 1490 ± 6	1109 1493 ± 8	1050 1502 ± 8	955 1583 ± 17	530 1361 ± 32	555 1419 ± 44
491	hzailu-005	1573726	275643	943 3292	1418 5128 ± 0	734 684 ± 10	489 673 ± 7	385 622 ± 1	403 682 ± 6	349 715 ± 2	679 1853 ± 51	677 1829 ± 34
492	hzailu-006	2996729	275643	1048 6070	1449 7176 ± 0	1426 1498 ± 11	1123 1483 ± 15	1080 1439 ± 8	1016 1441 ± 10	949 1564 ± 40	801 2434 ± 19	810 2452 ± 32
493	hzailu-007	2906424	275643	1042 5897	1451 7176 ± 0	1387 1430 ± 19	1107 1448 ± 10	1004 1328 ± 3	1052 1504 ± 17	877 1434 ± 4	815 2481 ± 28	818 2488 ± 34
494	hzailu-008	2906977	275643	1043 5898	1450 7176 ± 0	1401 1452 ± 13	1133 1557 ± 10	1121 1545 ± 12	1065 1561 ± 20	938 1533 ± 16	817 2492 ± 62	820 2489 ± 66
495	i2v-001	276820	26600	340 815	793 2048 ± 0	782 721 ± 0	546 729 ± 0	504 731 ± 0	454 731 ± 0	395 764 ± 0	1215 11523 ± 176	1217 11533 ± 142
496	i2v-002	339732	122488	701 1721	941 2048 ± 0	1393 1438 ± 1	1122 1476 ± 1	1083 1442 ± 1	1044 1480 ± 9	937 1533 ± 1	257 656 ± 35	221 604 ± 28
497	icehouse-000	287028	178880	388 887	536 2048 ± 0	651 629 ± 1	426 620 ± 1	427 648 ± 1	413 690 ± 4	534 961 ± 11	379 910 ± 11	393 938 ± 16
498	icm-002	621586	903	-	551 2048 ± 0	1115 1031 ± 7	-	-	-	-	1326 24052 ± 118	1327 24049 ± 124
499	icm-003	1513988	940	188 516	715 2048 ± 0	726 681 ± 6	484 672 ± 4	490 714 ± 11	536 837 ± 41	840 1381 ± 131	1328 24351 ± 161	1328 24227 ± 146
500	icm-004	2012129	1089	459 1052	659 2048 ± 0	382 419 ± 6	233 407 ± 6	236 454 ± 15	324 603 ± 51	934 1527 ± 235	1252 14730 ± 154	1254 14521 ± 152
501	icm-005	877530	1097	496 1116	570 2048 ± 0	116 185 ± 3	64 181 ± 3	67 208 ± 9	109 321 ± 36	577 1014 ± 185	1197 10245 ± 65	1199 10245 ± 69
502	icomm-000	807818	139431	1057 6728	1321 4096 ± 0	608 605 ± 26	411 610 ± 30	369 605 ± 27	327 610 ± 23	294 641 ± 32	1172 8507 ± 25	1174 8516 ± 26
503	icthtc-000	172459	1471004	513 1165	691 2048 ± 0	283 338 ± 11	169 338 ± 9	219 437 ± 16	429 705 ± 24	987 1719 ± 44	1076 5284 ± 63	1077 5290 ± 54
504	id3-001	225574	34433	-	177 520 ± 0	166 238 ± 19	-	-	-	-	427 1058 ± 32	436 1049 ± 28
505	id3-002	225574	34433	-	176 520 ± 0	455 482 ± 34	-	-	-	-	452 1100 ± 59	432 1048 ± 32
506	id3-003	265951	49731	-	61 264 ± 0	255 316 ± 19	-	-	-	-	525 1330 ± 25	538 1354 ± 28
507	id3-004	171526	49725	-	63 264 ± 0	518 541 ± 11	-	-	-	-	463 1135 ± 23	485 1156 ± 32
508	id3-005	153439	22125	-	64 264 ± 0	450 479 ± 1	-	-	-	-	1110 6072 ± 35	1109 6077 ± 38
509	id3-006	210116	7706	425 984	174 520 ± 0	732 683 ± 0	841 1088 ± 1	895 1192 ± 1	834 1209 ± 1	758 1270 ± 1	1086 5547 ± 34	1088 5563 ± 34
510	id3-007	185030	7728	-	65 264 ± 0	1098 1016 ± 1	890 1139 ± 1	1194 3911 ± 2	1192 10752 ± 12	1188 135182 ± 1912	1082 5500 ± 29	1082 5486 ± 35
511	id3-008	242416	8151	463 1059	59 264 ± 0	891 819 ± 0	946 1209 ± 2	987 1297 ± 2	933 1329 ± 1	875 1433 ± 1	1092 5658 ± 44	1093 5624 ± 40
512	id3-009	128638	9510	458 1051	58 264 ± 0	491 520 ± 1	332 518 ± 1	294 515 ± 2	257 519 ± 1	204 514 ± 2	266 682 ± 34	277 677 ± 44
513	id3-010	128638	9510	462 1056	60 264 ± 0	512 535 ± 2	344 535 ± 2	455 677 ± 2	781 1147 ± 4	1100 2889 ± 5	253 651 ± 34	238 621 ± 16
514	idbio-000	2	325054	330 796	179 522 ± 0	923 845 ± 3	642 847 ± 2	596 847 ± 2	547 853 ± 2	509 918 ± 1	1041 4553 ± 22	1042 4623 ± 16
515	idemia-003	427244	106432	-	77 352 ± 0	321 368 ± 6	-	-	-	-	1126 6654 ± 75	1057 4835 ± 90
516	idemia-004	406924	106436	-	79 352 ± 0	244 306 ± 5	-	-	-	-	1089 5592 ± 518	1084 5533 ± 426
517	idemia-005	509824	116761	-	200 588 ± 0	486 514 ± 15	-	-	-	-	1127 6657 ± 54	1124 6616 ± 53
518	idemia-006	570566	115633	-	203 668 ± 0	724 679 ± 4	714 926 ± 54	503 728 ± 7	478 760 ± 29	446 830 ± 31	1072 5223 ± 80	1073 5193 ± 72
519	idemia-007	353242	67485	-	90 468 ± 0	337 384 ± 0	217 389 ± 0	183 393 ± 1	170 405 ± 2	148 441 ± 8	927 3243 ± 63	927 3202 ± 63
520	idemia-008	374017	69922	532 1192	76 348 ± 0	424 457 ± 1	284 461 ± 0	249 466 ± 1	226 476 ± 2	200 513 ± 10	906 3080 ± 41	908 3046 ± 56
521	idemia-009	1066728	70572	886 2700	202 636 ± 0	1241 1207 ± 1	952 1218 ± 1	918 1222 ± 2	847 1222 ± 3	765 1280 ± 10	1093 5664 ± 84	1091 5597 ± 90
522	idemia-010	996690	66244	824 2306	83 412 ± 0	598 596 ± 1	450 632 ± 22	390 624 ± 18	352 633 ± 7	353 719 ± 5	635 1699 ± 37	639 1703 ± 38
523	idemia-011	472694	65098	951 3401	201 612 ± 0	887 816 ± 25	646 853 ± 9	432 658 ± 2	500 785 ± 3	364 732 ± 8	831 2583 ± 29	837 2627 ± 48
524	identy-000	271397	14217	162 474	537 2048 ± 0	742 689 ± 0	520 702 ± 0	488 712 ± 3	476 757 ± 0	540 970 ± 1	256 655 ± 25	258 637 ± 23
525	identy-001	385977	17850	527 1187	431 2048 ± 0	838 772 ± 1	639 841 ± 1	785 1056 ± 1	1068 1567 ± 5	1128 4081 ± 9	579 1484 ± 16	582 1480 ± 17
526	identy-002	1136161	40480	832 2414	1194 3072 ± 0	1429 1548 ± 0	1144 1584 ± 0	1124 1549 ± 0	1097 1757 ± 1	1059 2246 ± 1	900 3020 ± 41	905 3007 ± 19
527	identy-003	1245276	74623	625 1444	1193 3072 ± 0	1448 1730 ± 1	1155 1739 ± 0	1151 1794 ± 1	1138 1996 ± 3	1093 2775 ± 11	487 1204 ± 11	498 1198 ± 5
528	idrnd-000	1069136	69522	949 3357	1354 4104 ± 0	878 810 ± 1	625 828 ± 11	582 830 ± 10	565 877 ± 5	469 857 ± 9	1214 11504 ± 47	1216 11494 ± 62

Notes	
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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 34: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
529	idrnd-001	1080396	107151	955 3437	1356 4104 ± 0	594 594 ± 1	417 615 ± 6	382 615 ± 6	336 616 ± 6	292 640 ± 8	651 1757 ± 8	658 1759 ± 16
530	igearx-face-000	665623	298245	387 885	805 2048 ± 0	934 860 ± 1	649 855 ± 4	594 844 ± 0	570 881 ± 0	513 924 ± 3	620 1628 ± 45	627 1657 ± 41
531	iit-000	237317	33898	-	233 1024 ± 0	134 197 ± 8	-	-	-	-	601 1537 ± 81	521 1282 ± 20
532	iit-001	269176	52055	-	367 2048 ± 0	758 699 ± 4	-	-	-	-	428 1060 ± 48	445 1074 ± 54
533	iit-002	259579	52070	301 731	559 2048 ± 0	487 514 ± 1	340 531 ± 2	323 547 ± 1	306 583 ± 1	365 733 ± 2	414 1023 ± 7	415 1011 ± 66
534	iit-003	261288	53791	347 821	719 2048 ± 0	454 482 ± 0	308 493 ± 0	290 509 ± 0	272 541 ± 0	308 661 ± 0	89 324 ± 17	95 326 ± 8
535	imds-software-001	373399	352623	324 785	444 2048 ± 0	434 465 ± 1	742 958 ± 6	849 1131 ± 5	769 1134 ± 2	654 1119 ± 10	1222 11885 ± 120	1223 11779 ± 174
536	imds-software-002	373632	352627	350 833	435 2048 ± 0	441 469 ± 1	735 954 ± 2	816 1094 ± 5	708 1064 ± 4	680 1153 ± 4	1210 11100 ± 155	1211 11090 ± 145
537	imds-software-003	373648	352628	318 769	792 2048 ± 0	258 317 ± 5	466 654 ± 4	549 793 ± 6	487 770 ± 4	463 848 ± 2	1125 6624 ± 66	1122 6603 ± 77
538	imperial-000	370120	10623	331 796	522 2048 ± 0	703 669 ± 1	492 675 ± 3	457 683 ± 17	399 676 ± 2	329 689 ± 2	738 2130 ± 32	726 2052 ± 100
539	imperial-001	370260	10623	-	581 2048 ± 0	705 671 ± 0	-	-	-	-	732 2090 ± 28	729 2062 ± 74
540	imperial-002	472327	16134	735 1826	499 2048 ± 0	552 569 ± 1	387 581 ± 15	343 575 ± 5	299 576 ± 2	248 588 ± 3	770 2278 ± 90	739 2131 ± 44
541	incode-000	23697	18014	-	235 1024 ± 0	132 196 ± 9	-	-	-	-	1070 5175 ± 81	1060 4907 ± 77
542	incode-001	155416	18020	-	544 2048 ± 0	471 499 ± 9	-	-	-	-	1113 6154 ± 96	1108 6045 ± 182
543	incode-002	73239	32435	-	395 2048 ± 0	211 281 ± 15	-	-	-	-	603 1544 ± 38	587 1488 ± 69
544	incode-003	170632	42857	-	1253 4096 ± 0	339 384 ± 11	-	-	-	-	699 1928 ± 44	687 1876 ± 81
545	incode-004	260224	51685	-	388 2048 ± 0	453 479 ± 23	-	-	-	-	694 1913 ± 60	808 2443 ± 114
546	incode-005	256242	51670	-	675 2048 ± 0	545 566 ± 1	-	-	-	-	634 1695 ± 38	628 1663 ± 85
547	incode-006	266095	63518	-	474 2048 ± 0	445 472 ± 1	-	-	-	-	660 1788 ± 41	671 1798 ± 59
548	incode-007	266103	63524	-	736 2048 ± 0	442 470 ± 1	329 516 ± 0	296 518 ± 1	264 530 ± 2	264 603 ± 4	666 1799 ± 35	667 1789 ± 59
549	incode-008	266103	21014	-	770 2048 ± 0	474 500 ± 0	309 493 ± 0	276 496 ± 0	246 506 ± 1	219 537 ± 0	436 1070 ± 28	450 1081 ± 34
550	incode-009	266103	21014	-	416 2048 ± 0	477 503 ± 0	307 490 ± 1	278 498 ± 0	245 505 ± 0	220 537 ± 0	453 1102 ± 28	463 1113 ± 29
551	incode-010	627808	21014	860 2543	370 2048 ± 0	1223 1180 ± 2	917 1178 ± 1	888 1182 ± 1	813 1184 ± 1	722 1221 ± 1	479 1164 ± 32	474 1144 ± 32
552	incode-011	477280	21781	632 1483	595 2048 ± 0	947 872 ± 0	669 875 ± 0	626 881 ± 1	581 892 ± 1	521 939 ± 0	458 1117 ± 31	461 1109 ± 37
553	incode-012	477371	21786	630 1479	426 2048 ± 0	492 521 ± 0	324 506 ± 0	283 503 ± 0	253 514 ± 0	242 571 ± 0	804 2441 ± 25	815 2466 ± 37
554	incode-013	483720	41434	682 1647	591 2048 ± 0	511 535 ± 1	359 548 ± 1	328 555 ± 4	277 545 ± 1	273 618 ± 1	474 1152 ± 37	482 1154 ± 41
555	incode-014	2365157	41516	1059 6927	1458 8196 ± 0	1136 1064 ± 4	843 1089 ± 2	822 1100 ± 5	740 1107 ± 7	702 1190 ± 3	619 1627 ± 42	621 1640 ± 59
556	incode-015	1808779	70449	1051 6430	1432 6148 ± 0	1129 1056 ± 2	924 1182 ± 12	805 1081 ± 2	727 1084 ± 1	701 1187 ± 2	534 1373 ± 22	536 1352 ± 35
557	infocert-001	1204340	38972	634 1490	385 2048 ± 0	952 874 ± 1	681 891 ± 1	778 1050 ± 5	1038 1473 ± 2	1112 3174 ± 8	1068 5055 ± 108	1066 5008 ± 100
558	infocert-002	186823	19501	252 641	636 2048 ± 0	805 742 ± 0	553 738 ± 0	513 738 ± 0	461 738 ± 0	372 739 ± 0	888 2917 ± 114	892 2930 ± 50
559	infocert-003	186823	17115	189 517	829 2048 ± 0	826 763 ± 0	578 764 ± 0	532 764 ± 0	481 763 ± 0	398 766 ± 0	413 1022 ± 7	420 1023 ± 19
560	innefulabs-000	370588	162172	146 439	928 2048 ± 0	1090 1006 ± 3	800 1025 ± 3	762 1030 ± 4	696 1041 ± 2	668 1135 ± 3	1096 5782 ± 41	1097 5741 ± 45
561	innominds-001	102207	777689	498 1124	880 2048 ± 0	275 333 ± 21	180 350 ± 19	131 335 ± 16	121 339 ± 17	90 346 ± 18	816 2482 ± 23	821 2491 ± 29
562	innov8tif-000	170383	2163199	635 1494	1268 4096 ± 0	281 335 ± 23	178 348 ± 21	645 912 ± 64	1172 2661 ± 142	1177 10665 ± 528	186 591 ± 11	193 582 ± 11
563	innov8tif-001	118502	1241515	182 508	1236 4096 ± 0	954 875 ± 83	698 915 ± 73	1189 2360 ± 185	1188 6410 ± 390	1183 24097 ± 1477	323 775 ± 31	331 778 ± 14
564	innov8tif-002	258255	1108662	257 648	1133 2064 ± 0	432 464 ± 17	291 471 ± 15	339 572 ± 23	302 581 ± 24	279 628 ± 23	997 3917 ± 72	1002 3924 ± 72
565	innovativetechnologyltd-001	177232	335757	108 341	541 2048 ± 0	397 433 ± 7	272 446 ± 8	223 439 ± 4	198 452 ± 4	176 485 ± 7	684 1877 ± 42	698 1924 ± 97
566	innovativetechnologyltd-002	173939	372324	395 912	670 2048 ± 0	690 661 ± 2	543 726 ± 4	710 981 ± 27	657 997 ± 40	399 766 ± 3	675 1841 ± 50	683 1857 ± 59
567	innovatrics-000	0	143500	-	15 146 ± 0	560 578 ± 5	-	-	-	-	1062 4964 ± 63	1045 4665 ± 262
568	innovatrics-001	0	143500	-	68 284 ± 0	670 645 ± 5	-	-	-	-	1083 5506 ± 131	1063 4975 ± 308
569	innovatrics-002	0	249195	-	182 530 ± 0	86 153 ± 0	-	-	-	-	1246 14384 ± 260	1253 14497 ± 207
570	innovatrics-003	0	538170	-	181 530 ± 0	181 250 ± 0	-	-	-	-	1267 15709 ± 134	1265 15193 ± 292
571	innovatrics-004	0	496988	-	290 1076 ± 0	349 391 ± 0	-	-	-	-	1173 8573 ± 274	1165 7929 ± 244
572	innovatrics-005	0	664175	-	291 1076 ± 0	559 577 ± 1	-	-	-	-	1276 16880 ± 194	1281 17232 ± 385

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	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
573	innovatrics-006	0	466269	-	185 538 ± 0	893 820 ± 5	605 799 ± 4	557 805 ± 3	506 796 ± 9	490 890 ± 15	1099 5855 ± 204	1075 5266 ± 118
574	innovatrics-007	0	493269	-	289 1064 ± 0	1421 1485 ± 7	1161 1785 ± 184	1181 2078 ± 24	1150 2123 ± 15	1054 2210 ± 42	1105 5978 ± 88	1095 5690 ± 102
575	innovatrics-008	307323	59842	618 1429	184 538 ± 0	840 778 ± 6	581 767 ± 3	540 770 ± 3	509 803 ± 3	465 853 ± 10	901 3021 ± 66	845 2673 ± 88
576	innovatrics-009	624485	105187	758 1947	1375 4136 ± 0	1172 1116 ± 1	860 1107 ± 5	828 1104 ± 5	744 1110 ± 5	676 1146 ± 6	1066 5051 ± 54	1048 4733 ± 102
577	innovatrics-010	604731	111234	699 1715	1376 4136 ± 0	614 611 ± 3	447 632 ± 2	496 723 ± 1	451 730 ± 1	402 768 ± 6	1015 4077 ± 42	994 3857 ± 65
578	innovatrics-011	643997	50017	711 1745	1395 4184 ± 0	816 754 ± 2	582 767 ± 3	535 765 ± 3	498 783 ± 4	476 867 ± 3	1017 4135 ± 57	1006 3937 ± 91
579	innovatrics-012	0	680831	723 1789	1411 4721 ± 0	863 797 ± 3	635 837 ± 3	553 795 ± 1	586 904 ± 4	565 1001 ± 3	1453 80879 ± 981	1451 80521 ± 655
580	innovatrics-013	0	1665828	988 3931	1398 4212 ± 0	1231 1195 ± 2	964 1230 ± 11	920 1223 ± 4	856 1233 ± 5	810 1329 ± 3	1457 85968 ± 990	1455 85796 ± 836
581	innovatrics-014	0	2870004	1050 6138	1444 6303 ± 0	1102 1022 ± 6	807 1031 ± 2	780 1051 ± 1	700 1052 ± 5	689 1163 ± 11	1424 60311 ± 644	1421 60186 ± 499
582	innovatrics-015	0	5116879	1081 15251	1445 6327 ± 0	1334 1350 ± 5	1077 1405 ± 4	1032 1371 ± 3	990 1405 ± 21	919 1502 ± 17	1431 65175 ± 1027	1427 64294 ± 1390
583	innovatrics-016	0	3130540	1060 6982	1443 6303 ± 0	1481 2084 ± 0	1192 2097 ± 0	1182 2103 ± 1	1153 2139 ± 0	1055 2215 ± 0	1366 32998 ± 309	1368 32967 ± 351
584	innovatrics-017	2456649	28156	1033 5307	1400 4237 ± 0	1434 1569 ± 0	1143 1579 ± 0	1129 1570 ± 0	1072 1574 ± 0	956 1584 ± 0	1368 33161 ± 158	1370 33197 ± 143
585	insightface-000	806953	16606	-	1320 4096 ± 0	1093 1009 ± 1	798 1019 ± 2	746 1017 ± 2	682 1020 ± 2	586 1032 ± 2	658 1778 ± 31	660 1773 ± 35
586	insightface-001	776777	16606	980 3858	832 2048 ± 0	1341 1366 ± 2	1051 1368 ± 3	1034 1372 ± 3	964 1375 ± 5	845 1386 ± 4	459 1119 ± 29	459 1108 ± 34
587	insightface-002	800572	16606	-	840 2048 ± 0	1367 1396 ± 2	1060 1389 ± 4	1058 1403 ± 3	986 1402 ± 2	862 1413 ± 3	480 1169 ± 40	466 1118 ± 38
588	insightface-003	1016917	26668	654 1550	763 2048 ± 0	1142 1073 ± 0	847 1092 ± 2	800 1070 ± 1	726 1082 ± 1	638 1101 ± 1	196 597 ± 16	207 595 ± 17
589	insightface-004	928816	26668	727 1800	408 2048 ± 0	1308 1298 ± 1	1047 1356 ± 2	994 1313 ± 1	918 1311 ± 0	820 1349 ± 1	176 565 ± 7	179 559 ± 5
590	inspur-000	364844	91926	336 808	1244 4096 ± 0	1342 1367 ± 1	1037 1331 ± 2	1030 1368 ± 2	1037 1465 ± 1	1006 1861 ± 3	1193 9831 ± 37	1193 9860 ± 40
591	inspur-001	364862	91934	352 833	553 2048 ± 0	1358 1385 ± 1	1049 1361 ± 4	1055 1401 ± 2	1048 1495 ± 1	1013 1891 ± 1	1091 5641 ± 42	1092 5622 ± 40
592	inspur-002	447948	91934	434 1009	389 2048 ± 0	1130 1056 ± 5	820 1056 ± 7	819 1096 ± 7	1102 1776 ± 23	885 1444 ± 12	1035 4504 ± 32	1035 4515 ± 35
593	inspur-003	444594	91934	448 1033	548 2048 ± 0	1259 1225 ± 2	937 1199 ± 3	935 1241 ± 1	922 1316 ± 1	974 1645 ± 5	1038 4525 ± 16	1036 4523 ± 26
594	intellicloudai-001	220831	868246	261 655	794 2048 ± 0	439 468 ± 2	279 456 ± 1	248 466 ± 3	236 492 ± 1	282 632 ± 2	425 1056 ± 4	438 1051 ± 72
595	intellicloudai-002	259047	58559	964 3585	1350 4100 ± 0	925 847 ± 1	643 847 ± 2	599 849 ± 1	549 853 ± 1	484 878 ± 4	350 822 ± 28	356 818 ± 23
596	intellifusion-001	271872	289387	315 762	892 2048 ± 0	827 764 ± 38	588 774 ± 39	554 797 ± 42	511 803 ± 34	429 805 ± 33	455 1112 ± 28	468 1128 ± 41
597	intellifusion-002	762731	385841	408 941	1308 4096 ± 0	1033 950 ± 2	853 1096 ± 42	809 1088 ± 33	799 1168 ± 31	693 1171 ± 10	638 1713 ± 57	629 1665 ± 87
598	intellivision-001	43692	11649	-	1065 2056 ± 0	21 62 ± 2	-	-	-	-	829 2573 ± 91	829 2544 ± 38
599	intellivision-002	43692	14505	-	1064 2056 ± 0	263 322 ± 1	183 355 ± 2	164 372 ± 1	178 422 ± 2	259 600 ± 1	1239 13525 ± 134	1238 12782 ± 278
600	intellivision-003	64023	133748	332 800	1100 2056 ± 0	365 407 ± 3	226 398 ± 2	203 418 ± 2	196 450 ± 1	251 591 ± 4	1209 11069 ± 56	1210 11066 ± 75
601	intellivision-004	117727	131310	187 515	1067 2056 ± 0	272 330 ± 0	164 330 ± 0	140 347 ± 0	148 382 ± 0	203 514 ± 0	1211 11197 ± 63	1212 11165 ± 72
602	intellivision-005	372608	257259	279 686	1092 2056 ± 0	634 618 ± 0	431 623 ± 0	410 634 ± 0	381 659 ± 0	406 771 ± 0	1296 19244 ± 68	1298 19270 ± 77
603	intellivision-006	287736	262742	291 723	1060 2056 ± 0	153 214 ± 0	90 214 ± 0	75 221 ± 0	83 272 ± 0	96 357 ± 1	1260 15153 ± 32	1263 15142 ± 38
604	intellivision-007	192454	262742	138 419	1105 2056 ± 0	82 148 ± 0	49 157 ± 0	44 166 ± 0	40 180 ± 0	60 265 ± 0	1257 15100 ± 50	1259 15095 ± 45
605	intellivix-001	256654	111858	363 846	949 2048 ± 0	329 378 ± 1	206 379 ± 1	173 381 ± 1	150 384 ± 1	137 421 ± 3	451 1100 ± 16	462 1109 ± 22
606	intellivix-002	361566	116162	517 1175	379 2048 ± 0	1042 956 ± 0	730 947 ± 6	705 976 ± 0	645 984 ± 4	632 1089 ± 1	1354 30096 ± 128	1359 31287 ± 140
607	intellivix-003	234409	116167	573 1301	970 2048 ± 0	982 908 ± 0	700 916 ± 1	669 930 ± 0	630 961 ± 1	665 1129 ± 3	1353 30025 ± 137	1358 31190 ± 131
608	intellivix-004	361668	128097	531 1189	729 2048 ± 0	390 427 ± 0	261 437 ± 0	226 444 ± 0	207 461 ± 0	239 565 ± 2	588 1498 ± 37	580 1478 ± 16
609	intellivix-005	774405	128093	974 3692	347 2048 ± 0	711 673 ± 1	459 643 ± 3	430 654 ± 3	389 666 ± 3	403 769 ± 5	576 1481 ± 65	567 1451 ± 26
610	intellivix-006	792822	128093	976 3753	762 2048 ± 0	700 665 ± 1	468 657 ± 2	440 662 ± 2	400 677 ± 2	416 784 ± 1	583 1486 ± 13	584 1482 ± 16
611	intelresearch-000	388229	307985	-	462 2048 ± 0	978 902 ± 6	-	-	-	-	1053 4800 ± 152	1040 4561 ± 97
612	intelresearch-001	353997	272602	-	399 2048 ± 0	729 682 ± 4	-	-	-	-	965 3553 ± 57	957 3462 ± 161
613	intelresearch-002	452850	86454	-	405 2048 ± 0	769 707 ± 2	599 790 ± 33	545 788 ± 26	533 831 ± 29	472 862 ± 22	1018 4204 ± 91	1019 4153 ± 93
614	intelresearch-003	401343	85085	518 1177	714 2048 ± 0	1263 1232 ± 3	968 1237 ± 2	936 1242 ± 2	884 1263 ± 2	802 1324 ± 3	1032 4443 ± 75	1026 4374 ± 77
615	intelresearch-004	646918	85290	744 1866	458 2048 ± 0	1321 1319 ± 2	1031 1322 ± 3	1006 1330 ± 3	939 1345 ± 3	859 1411 ± 5	1047 4696 ± 63	1047 4692 ± 66
616	intelresearch-005	398137	85290	512 1164	938 2048 ± 0	1326 1328 ± 1	1041 1334 ± 2	1017 1344 ± 2	945 1356 ± 2	868 1423 ± 4	1037 4524 ± 87	1030 4461 ± 74

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	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
617	intelresearch-006	445223	101126	470 1069	357 2048 ± 0	1000 918 ± 1	673 881 ± 0	631 892 ± 0	591 913 ± 1	573 1008 ± 3	1111 6137 ± 410	1107 6024 ± 109
618	intelresearch-007	523218	101150	536 1197	759 2048 ± 0	586 590 ± 1	392 585 ± 1	354 588 ± 1	322 603 ± 4	315 668 ± 6	983 3738 ± 42	988 3752 ± 53
619	intelresearch-008	523218	101214	535 1196	721 2048 ± 0	657 634 ± 1	442 630 ± 1	403 632 ± 1	353 635 ± 2	347 711 ± 4	973 3661 ± 34	975 3641 ± 44
620	intelresearch-009	290599	120716	646 1535	856 2048 ± 0	539 560 ± 1	476 664 ± 3	417 638 ± 3	402 678 ± 13	333 692 ± 4	991 3829 ± 207	979 3679 ± 71
621	intelresearch-010	307886	120767	686 1655	641 2048 ± 0	1197 1153 ± 1	905 1158 ± 2	872 1160 ± 2	795 1166 ± 3	730 1237 ± 5	993 3850 ± 26	993 3822 ± 75
622	intelresearch-011	317134	120757	715 1758	535 2048 ± 0	1265 1236 ± 4	953 1219 ± 3	927 1231 ± 1	863 1237 ± 3	793 1312 ± 6	995 3873 ± 47	997 3890 ± 57
623	intelresearch-012	317135	120715	714 1758	513 2048 ± 0	1295 1280 ± 2	983 1256 ± 4	960 1270 ± 3	902 1284 ± 3	811 1331 ± 5	985 3744 ± 32	984 3722 ± 44
624	intema-000	1532392	19488	481 1088	156 513 ± 0	1095 1010 ± 0	785 1001 ± 4	728 994 ± 0	653 993 ± 5	604 1056 ± 1	380 910 ± 29	382 906 ± 32
625	intema-001	1122562	19536	627 1458	158 513 ± 0	1338 1354 ± 1	1028 1318 ± 5	1011 1336 ± 4	931 1328 ± 2	834 1375 ± 0	396 977 ± 31	402 980 ± 31
626	interlinklabs-001	155897	1013105	308 746	428 2048 ± 0	268 328 ± 3	175 346 ± 8	137 343 ± 7	125 352 ± 3	135 418 ± 11	560 1441 ± 10	566 1450 ± 14
627	interlinklabs-002	286341	1013147	595 1364	1234 4096 ± 0	407 438 ± 1	258 435 ± 4	218 435 ± 5	191 439 ± 0	193 505 ± 2	846 2699 ± 35	847 2678 ± 25
628	interlinklabs-003	528962	951057	652 1549	1293 4096 ± 0	796 734 ± 0	566 750 ± 0	520 746 ± 0	486 769 ± 0	518 932 ± 0	958 3513 ± 50	963 3504 ± 26
629	intozi-001	21863	74210	42 140	722 2048 ± 0	34 85 ± 0	21 86 ± 0	16 86 ± 0	13 87 ± 0	11 87 ± 0	1186 9579 ± 252	1183 9530 ± 167
630	intozi-002	172046	95119	361 844	879 2048 ± 0	1435 1571 ± 0	1145 1585 ± 1	1133 1580 ± 0	1077 1585 ± 0	977 1665 ± 1	343 805 ± 14	348 803 ± 4
631	intsysmsu-000	650193	171481	-	959 2048 ± 0	507 535 ± 20	-	-	-	-	210 610 ± 22	232 613 ± 31
632	intsysmsu-001	384409	172480	327 789	791 2048 ± 0	626 614 ± 2	418 615 ± 2	422 642 ± 2	468 750 ± 3	685 1159 ± 4	226 621 ± 8	226 611 ± 31
633	intsysmsu-002	765921	172298	326 786	263 1024 ± 0	591 593 ± 1	601 793 ± 2	578 827 ± 1	563 875 ± 104	776 1293 ± 3	169 549 ± 25	176 548 ± 29
634	ionetworks-000	287609	51236	113 352	846 2048 ± 0	395 430 ± 0	259 435 ± 0	216 433 ± 0	184 432 ± 0	150 444 ± 0	1135 6913 ± 102	1142 7150 ± 160
635	ionetworks-001	39860	333205	93 298	821 2048 ± 0	408 439 ± 2	253 433 ± 2	245 461 ± 2	200 454 ± 2	178 488 ± 2	1120 6335 ± 103	1117 6358 ± 91
636	ionetworks-002	337801	73038	712 1746	509 2048 ± 0	743 689 ± 1	487 672 ± 5	445 665 ± 3	393 669 ± 3	309 664 ± 6	1133 6884 ± 154	1129 6808 ± 102
637	ionetworks-003	309918	68642	676 1631	992 2050 ± 0	775 713 ± 0	497 681 ± 1	444 665 ± 0	391 668 ± 0	316 671 ± 0	1202 10649 ± 225	1204 10658 ± 106
638	ionetworks-004	302266	135028	734 1813	993 2050 ± 0	1062 975 ± 0	695 910 ± 0	656 921 ± 0	592 916 ± 0	506 917 ± 1	1345 27437 ± 130	1347 27472 ± 114
639	iqface-000	268819	596337	283 704	1412 4750 ± 32	513 538 ± 26	310 494 ± 2	320 543 ± 3	457 734 ± 4	850 1393 ± 4	1477 636433 ± 38446	1477 632654 ± 85615
640	iqface-001	1145590	760941	-	1413 4752 ± 35	605 603 ± 1	-	-	-	-	1475 568761 ± 3659	1475 570335 ± 5122
641	iqface-003	370803	963398	348 821	1414 4763 ± 37	498 529 ± 1	342 532 ± 2	364 599 ± 8	546 850 ± 2	982 1694 ± 2	1476 575924 ± 2601	1476 576653 ± 2051
642	iqsec-001	172239	41022	368 850	507 2048 ± 0	186 258 ± 6	112 254 ± 4	258 474 ± 9	757 1124 ± 24	1129 4155 ± 119	1330 24586 ± 143	1336 25631 ± 157
643	iqsec-002	172239	41022	370 852	562 2048 ± 0	173 243 ± 6	114 259 ± 4	284 504 ± 10	814 1184 ± 25	1127 4081 ± 124	1331 24688 ± 149	1337 25737 ± 142
644	iqsec-003	293643	15245	242 622	1370 4116 ± 0	949 873 ± 0	664 872 ± 0	617 872 ± 0	569 880 ± 0	516 929 ± 0	1031 4441 ± 20	1031 4462 ± 41
645	irex-000	741899	47419	786 2090	1203 3080 ± 0	929 852 ± 2	645 850 ± 1	619 874 ± 2	613 939 ± 1	739 1249 ± 5	45 201 ± 11	48 208 ± 8
646	irisid-001	480885	100974	430 1000	975 2049 ± 0	1053 966 ± 0	753 971 ± 0	758 1028 ± 0	791 1163 ± 0	976 1662 ± 1	689 1892 ± 13	721 2040 ± 17
647	irisid-002	735508	101002	580 1334	979 2049 ± 0	1449 1732 ± 3	1154 1730 ± 2	1144 1752 ± 2	1130 1923 ± 2	1077 2410 ± 2	1150 7367 ± 16	1150 7410 ± 48
648	isap-001	99049	204201	1 18	1266 4096 ± 0	1 0 ± 0	-	-	-	-	131 459 ± 17	140 456 ± 11
649	isap-002	256765	49931	90 288	502 2048 ± 0	834 769 ± 3	802 1027 ± 2	622 877 ± 2	479 761 ± 1	503 912 ± 2	904 3045 ± 94	898 2973 ± 66
650	isityou-000	48010	36621	29 110	1477 19200 ± 0	55 113 ± 5	-	-	-	-	1469 237517 ± 1318	1469 237374 ± 1279
651	istl-001	271118	27325	210 561	1127 2060 ± 0	596 595 ± 0	402 598 ± 0	362 598 ± 0	323 603 ± 0	265 605 ± 0	728 2073 ± 19	731 2072 ± 34
652	isystems-000	209270	713870	-	244 1024 ± 0	87 154 ± 4	-	-	-	-	116 398 ± 14	123 396 ± 16
653	isystems-001	274621	639268	486 1091	630 2048 ± 0	220 291 ± 9	-	-	-	-	171 557 ± 16	182 564 ± 22
654	isystems-002	358984	803389	669 1595	374 2048 ± 0	898 822 ± 8	-	-	-	-	311 749 ± 31	248 632 ± 28
655	itmo-001	1923215	62191	-	1482 37997 ± 0	1044 959 ± 18	-	-	-	-	1349 28901 ± 1492	1348 27517 ± 186
656	itmo-002	1923215	62191	-	1385 4162 ± 0	615 611 ± 17	-	-	-	-	1153 7423 ± 96	1153 7451 ± 94
657	itmo-003	579961	82630	-	1159 2114 ± 0	28 79 ± 2	-	-	-	-	1195 10038 ± 180	1196 10030 ± 134
658	itmo-004	487023	79012	-	1157 2100 ± 0	401 436 ± 0	-	-	-	-	1166 8219 ± 100	1170 8214 ± 130
659	itmo-005	482155	81168	-	1389 4173 ± 0	823 759 ± 1	-	-	-	-	1236 13214 ± 164	1237 12576 ± 257
660	itmo-006	599187	96762	-	1164 2121 ± 0	884 814 ± 1	631 831 ± 26	583 830 ± 17	532 830 ± 3	530 952 ± 38	1338 26154 ± 148	1339 26217 ± 260

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	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480X720	960X1440	1600X2400	3000X4500	GENUINE	IMPOSTOR
661	itmo-007	415979	245376	807 2199	864 2048 ± 0	804 741 ± 2	-	-	-	-	825 2551 ± 50	826 2529 ± 80
662	itmo-008	726866	318238	599 1376	1323 4096 ± 0	1133 1060 ± 1	822 1058 ± 1	788 1059 ± 1	718 1072 ± 4	642 1104 ± 1	967 3578 ± 25	973 3580 ± 28
663	ivacognitive-001	256958	62791	420 975	470 2048 ± 0	1306 1292 ± 3	1005 1289 ± 4	976 1292 ± 4	905 1292 ± 3	798 1321 ± 4	1019 4228 ± 41	1020 4226 ± 41
664	iws-000	30875	3063	19 77	106 512 ± 0	204 277 ± 5	133 283 ± 1	273 494 ± 3	646 984 ± 3	1107 2987 ± 39	405 999 ± 40	407 992 ± 22
665	jaakit-001	99024	24754	76 251	123 512 ± 0	26 76 ± 0	17 77 ± 0	13 79 ± 0	12 81 ± 0	13 93 ± 0	812 2466 ± 57	814 2465 ± 66
666	jaakit-002	179256	75037	85 277	497 2048 ± 0	822 758 ± 0	576 760 ± 0	663 925 ± 0	923 1318 ± 1	1096 2799 ± 2	386 943 ± 7	396 963 ± 5
667	kaizenvoiz-001	254811	74894	128 388	618 2048 ± 0	297 354 ± 0	187 360 ± 0	289 508 ± 0	-	-	947 3439 ± 19	964 3506 ± 31
668	kakao-001	107616	285676	-	237 1024 ± 0	332 379 ± 1	-	-	-	-	383 930 ± 22	394 948 ± 38
669	kakao-002	479406	412822	-	531 2048 ± 0	807 747 ± 6	-	-	-	-	641 1720 ± 62	643 1715 ± 83
670	kakao-003	414379	113944	-	858 2048 ± 0	958 878 ± 3	-	-	-	-	737 2128 ± 34	741 2134 ± 60
671	kakao-004	414411	135270	-	877 2048 ± 0	933 856 ± 1	650 855 ± 0	606 857 ± 0	556 858 ± 0	471 860 ± 1	707 1957 ± 28	707 1953 ± 37
672	kakao-005	414316	152216	-	1003 2052 ± 0	1140 1068 ± 1	830 1073 ± 1	803 1079 ± 0	720 1077 ± 1	631 1089 ± 1	726 2067 ± 26	722 2043 ± 34
673	kakao-007	526993	129545	938 3150	964 2048 ± 0	1040 952 ± 1	745 961 ± 1	692 958 ± 1	632 962 ± 1	539 968 ± 1	424 1056 ± 16	431 1047 ± 28
674	kakao-008	734583	104820	983 3879	448 2048 ± 0	1181 1135 ± 3	898 1148 ± 3	864 1150 ± 3	786 1156 ± 1	697 1175 ± 1	308 736 ± 23	316 727 ± 22
675	kakao-009	766989	104820	992 4042	517 2048 ± 0	592 593 ± 1	405 603 ± 4	378 613 ± 2	337 617 ± 2	287 636 ± 4	130 457 ± 20	135 448 ± 17
676	kakaobank-000	570796	98479	883 2690	1294 4096 ± 0	1264 1234 ± 1	980 1254 ± 7	942 1248 ± 3	973 1389 ± 3	1003 1843 ± 11	1052 4790 ± 35	1053 4803 ± 35
677	kakaobank-001	780621	98829	806 2189	972 2048 ± 0	573 584 ± 0	410 608 ± 1	371 607 ± 5	408 684 ± 1	511 922 ± 2	1344 27092 ± 174	1346 27065 ± 158
678	kakaobank-002	780621	98833	849 2459	819 2048 ± 0	1089 1004 ± 1	792 1010 ± 0	753 1025 ± 5	725 1078 ± 1	795 1313 ± 1	1342 26972 ± 118	1344 27022 ± 139
679	kakaobank-003	1409862	98833	999 4186	900 2048 ± 0	1230 1194 ± 2	960 1224 ± 6	928 1231 ± 2	886 1269 ± 5	939 1535 ± 5	1343 26986 ± 183	1345 27026 ± 151
680	kakaopay-001	397864	179869	278 685	1255 4096 ± 0	413 448 ± 0	350 542 ± 0	316 542 ± 0	273 542 ± 0	229 553 ± 0	236 633 ± 22	244 630 ± 22
681	kasikornlabs-000	256471	61000	260 655	694 2048 ± 0	984 908 ± 36	671 878 ± 22	701 969 ± 39	812 1184 ± 54	1073 2382 ± 145	1359 31669 ± 188	1360 31714 ± 182
682	kasikornlabs-001	256471	61037	273 676	896 2048 ± 0	989 912 ± 38	658 868 ± 10	741 1005 ± 50	805 1176 ± 44	1074 2387 ± 147	1356 30759 ± 198	1356 30867 ± 174
683	kasikornlabs-002	256431	61063	236 613	563 2048 ± 0	999 917 ± 35	693 907 ± 13	697 963 ± 13	926 1320 ± 45	1085 2629 ± 178	1358 31025 ± 180	1357 31054 ± 186
684	kasikornlabs-003	256442	61063	311 751	667 2048 ± 0	680 652 ± 5	467 656 ± 5	529 755 ± 11	673 1007 ± 27	1058 2240 ± 168	1325 24022 ± 121	1326 23998 ± 97
685	kasikornlabs-004	466064	61063	454 1045	463 2048 ± 0	1036 950 ± 7	752 970 ± 4	763 1031 ± 11	962 1372 ± 30	1070 2368 ± 156	1322 23867 ± 56	1324 23919 ± 91
686	kasikornlabs-005	466202	61063	444 1028	787 2048 ± 0	1012 936 ± 8	740 956 ± 14	786 1057 ± 14	866 1243 ± 28	1062 2285 ± 145	1324 23900 ± 44	1325 23923 ± 56
687	kbyai-000	502373	34357	835 2417	73 296 ± 0	1239 1202 ± 3	1194 2167 ± 3	1188 2293 ± 9	1169 2586 ± 13	1122 3689 ± 20	1281 17785 ± 51	1285 17801 ± 125
688	kedacom-000	245292	37401	1086 23574	71 292 ± 0	480 506 ± 3	357 547 ± 10	381 614 ± 9	311 588 ± 10	311 665 ± 24	269 684 ± 14	282 682 ± 16
689	keyless-000	551302	27472	511 1157	354 2048 ± 0	630 615 ± 0	550 736 ± 5	1183 2122 ± 2	1148 2086 ± 1	1042 2110 ± 4	377 909 ± 8	379 905 ± 16
690	kiwitech-000	369711	21375	356 841	520 2048 ± 0	588 591 ± 0	400 594 ± 0	361 595 ± 1	319 596 ± 0	267 609 ± 0	650 1755 ± 20	654 1734 ± 16
691	kloudspot-001	261662	887420	384 882	568 2048 ± 0	269 330 ± 1	156 319 ± 6	125 325 ± 1	106 318 ± 0	73 318 ± 0	1085 5539 ± 48	1087 5555 ± 66
692	kloudspot-002	635738	460698	648 1540	1229 4096 ± 0	1178 1128 ± 1	878 1128 ± 2	841 1123 ± 0	788 1158 ± 2	666 1132 ± 0	1079 5456 ± 154	1080 5460 ± 108
693	kneron-003	58366	1747	64 188	968 2048 ± 0	209 281 ± 3	131 280 ± 1	119 315 ± 13	135 365 ± 7	723 1224 ± 30	1075 5237 ± 63	1076 5274 ± 99
694	kneron-005	375374	13633	155 457	441 2048 ± 0	490 518 ± 2	334 522 ± 4	330 556 ± 5	475 757 ± 19	992 1760 ± 25	698 1922 ± 11	699 1926 ± 20
695	kneron-006	590352	63741	605 1402	751 2048 ± 0	563 579 ± 0	386 580 ± 0	363 599 ± 0	446 726 ± 1	764 1278 ± 4	762 2247 ± 22	763 2252 ± 28
696	knowutech-000	808045	32886	576 1312	309 1536 ± 0	1385 1419 ± 2	1053 1372 ± 1	1037 1377 ± 1	966 1382 ± 2	844 1386 ± 2	984 3743 ± 31	981 3693 ± 38
697	kogniza-001	0	394421	564 1288	310 1536 ± 0	1300 1285 ± 2	1018 1304 ± 4	969 1280 ± 2	895 1278 ± 2	804 1326 ± 2	894 2968 ± 37	896 2971 ± 20
698	kookmin-001	234001	31875	-	853 2048 ± 0	56 114 ± 1	31 110 ± 1	27 116 ± 1	27 128 ± 1	27 172 ± 1	232 629 ± 35	235 616 ± 11
699	kookmin-002	371771	30734	349 827	585 2048 ± 0	1119 1038 ± 2	816 1047 ± 1	775 1045 ± 1	706 1061 ± 1	650 1116 ± 1	243 638 ± 19	256 636 ± 20
700	koreaid-001	256261	20152	731 1810	361 2048 ± 0	341 384 ± 2	218 390 ± 1	227 444 ± 2	285 556 ± 6	423 795 ± 5	242 636 ± 11	255 636 ± 10
701	krungthai-002	2360957	15033	526 1185	353 2048 ± 0	245 308 ± 0	150 314 ± 5	115 309 ± 0	107 319 ± 0	100 362 ± 0	898 3014 ± 20	899 2980 ± 22
702	kuke3d-001	403462	68786	195 530	1309 4096 ± 0	883 814 ± 2	612 811 ± 2	564 814 ± 2	516 814 ± 1	455 834 ± 1	1123 6412 ± 57	1119 6413 ± 51
703	kuke3d-002	270544	1227855	345 820	744 2048 ± 0	478 504 ± 3	320 504 ± 1	291 511 ± 1	260 523 ± 2	247 585 ± 1	891 2943 ± 22	895 2966 ± 38
704	lebentech-000	0	10360	2 26	94 512 ± 0	3 22 ± 0	1 22 ± 0	1 22 ± 0	1 23 ± 0	1 23 ± 0	339 801 ± 42	358 825 ± 51

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	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
705	lebentech-001	0	144456	661 1567	561 2048 ± 0	101 174 ± 0	58 175 ± 0	64 200 ± 1	66 227 ± 9	125 397 ± 20	145 485 ± 17	149 473 ± 28
706	lemalabs-001	748400	198794	896 2745	834 2048 ± 0	877 810 ± 0	613 812 ± 0	562 813 ± 0	522 819 ± 0	461 844 ± 1	1223 11930 ± 35	1226 11913 ± 37
707	lightvision-000	182569	62181	404 930	465 2048 ± 0	374 411 ± 1	208 381 ± 0	170 379 ± 1	151 386 ± 3	124 396 ± 2	580 1484 ± 13	579 1477 ± 13
708	lightvision-001	266840	62181	673 1604	661 2048 ± 0	500 530 ± 2	336 523 ± 2	317 542 ± 1	281 548 ± 10	218 537 ± 1	564 1456 ± 10	571 1457 ± 14
709	lineclova-000	264443	407003	-	483 2048 ± 0	579 586 ± 0	413 612 ± 0	373 609 ± 1	328 611 ± 0	274 618 ± 1	857 2753 ± 19	857 2745 ± 23
710	lineclova-001	944355	407058	812 2233	622 2048 ± 0	910 833 ± 10	629 830 ± 3	580 828 ± 4	541 838 ± 8	452 833 ± 4	845 2696 ± 23	846 2677 ± 35
711	lineclova-002	475779	406756	587 1352	449 2048 ± 0	1299 1284 ± 1	997 1275 ± 2	966 1275 ± 1	893 1273 ± 2	766 1281 ± 2	859 2765 ± 10	860 2767 ± 31
712	lineclova-003	585149	410482	703 1725	638 2048 ± 0	1396 1444 ± 1	1103 1438 ± 1	1082 1439 ± 2	1015 1440 ± 1	888 1446 ± 2	880 2890 ± 23	885 2899 ± 29
713	lookman-002	138200	25410	1082 16518	194 548 ± 0	99 173 ± 1	-	-	-	-	211 610 ± 19	228 612 ± 22
714	lookman-004	244775	37401	1085 23548	195 548 ± 0	481 507 ± 5	355 545 ± 12	379 613 ± 12	316 590 ± 11	302 656 ± 16	362 871 ± 29	370 878 ± 29
715	lssytech-000	1316125	197944	1015 4622	1199 3076 ± 0	1199 1154 ± 0	873 1124 ± 0	847 1129 ± 0	763 1128 ± 0	663 1128 ± 0	307 733 ± 17	313 722 ± 13
716	lssytech-001	3239801	232221	1079 10421	1396 4184 ± 0	1403 1454 ± 1	1109 1450 ± 0	1119 1521 ± 2	1023 1452 ± 0	895 1462 ± 0	498 1237 ± 19	492 1183 ± 13
717	luxand-000	0	57908	596 1366	280 1040 ± 0	367 407 ± 23	255 433 ± 11	228 444 ± 14	216 464 ± 14	237 562 ± 25	353 828 ± 28	359 828 ± 32
718	luxand-001	85194	72189	249 636	678 2048 ± 0	396 431 ± 0	267 443 ± 0	225 443 ± 2	194 448 ± 0	208 519 ± 0	556 1433 ± 16	563 1433 ± 16
719	luxand-003	127425	72189	366 847	686 2048 ± 0	767 706 ± 4	514 697 ± 1	475 698 ± 3	418 694 ± 0	415 782 ± 3	562 1443 ± 17	561 1431 ± 17
720	luxand-004	127425	72189	367 848	837 2048 ± 0	718 677 ± 0	498 682 ± 0	466 691 ± 0	430 705 ± 0	397 766 ± 0	561 1441 ± 10	564 1440 ± 11
721	mantra-000	471458	62566	309 749	1036 2052 ± 0	375 413 ± 18	306 487 ± 19	272 494 ± 18	250 511 ± 18	258 598 ± 19	915 3151 ± 51	916 3127 ± 63
722	matrixcomsec-000	114103	177320	83 264	745 2048 ± 0	40 94 ± 0	24 94 ± 0	19 95 ± 0	38 173 ± 1	21 140 ± 1	222 616 ± 22	245 630 ± 28
723	matrixcomsec-001	179657	168204	504 1141	851 2048 ± 0	95 166 ± 0	53 168 ± 0	45 169 ± 0	37 171 ± 0	33 191 ± 0	66 268 ± 53	72 261 ± 62
724	matrixcomsec-002	93740	168203	239 618	512 2048 ± 0	81 147 ± 0	45 146 ± 0	37 147 ± 0	30 148 ± 0	29 178 ± 0	331 791 ± 13	336 790 ± 16
725	maverickresearch-000	256262	20268	104 331	835 2048 ± 0	1083 995 ± 1	783 999 ± 1	735 999 ± 2	658 997 ± 0	581 1023 ± 2	1303 20268 ± 108	1305 20254 ± 121
726	maxvision-000	133114	56426	-	130 512 ± 0	306 359 ± 0	185 356 ± 0	152 359 ± 0	128 356 ± 0	104 370 ± 1	810 2461 ± 20	811 2452 ± 17
727	maxvision-001	256146	61793	926 3037	940 2048 ± 0	202 275 ± 3	125 274 ± 2	93 277 ± 4	86 280 ± 4	75 325 ± 3	294 714 ± 13	309 717 ± 13
728	maxvision-002	171894	60623	716 1765	807 2048 ± 0	98 172 ± 0	56 171 ± 0	47 172 ± 0	39 174 ± 0	42 221 ± 0	303 725 ± 5	314 725 ± 5
729	maxvision-003	234062	61252	844 2452	925 2048 ± 0	447 474 ± 0	290 468 ± 0	255 471 ± 0	224 475 ± 0	207 519 ± 0	814 2467 ± 28	819 2488 ± 23
730	maxvision-004	394121	61583	940 3217	766 2048 ± 0	668 644 ± 1	463 648 ± 0	425 647 ± 1	358 637 ± 1	336 695 ± 1	691 1897 ± 10	693 1903 ± 14
731	maxvision-005	309326	61739	795 2138	906 2048 ± 0	1092 1007 ± 1	974 1245 ± 5	747 1017 ± 1	695 1040 ± 4	781 1299 ± 3	705 1950 ± 38	706 1948 ± 34
732	maxvision-006	315987	61739	939 3172	542 2048 ± 0	1276 1247 ± 3	945 1208 ± 3	913 1211 ± 5	840 1215 ± 1	757 1268 ± 4	703 1938 ± 17	700 1929 ± 35
733	mbbankvn-000	472410	139431	1000 4227	676 2048 ± 0	435 466 ± 17	297 480 ± 26	260 481 ± 27	228 479 ± 21	187 501 ± 29	1028 4388 ± 14	1027 4380 ± 17
734	megvii-000	1358915	16014	-	376 2048 ± 0	850 785 ± 0	-	-	-	-	1299 19816 ± 158	1301 19805 ± 174
735	megvii-001	1361523	16486	-	589 2048 ± 0	522 543 ± 0	-	-	-	-	1074 5228 ± 32	1074 5252 ± 60
736	megvii-002	1809564	16491	-	1331 4100 ± 0	669 644 ± 0	-	-	-	-	1408 50630 ± 183	1402 47591 ± 716
737	megvii-003	4430290	42790	-	1260 4096 ± 0	1243 1210 ± 1	958 1223 ± 0	1026 1356 ± 4	1075 1582 ± 7	1089 2727 ± 23	1467 225342 ± 3574	1468 225413 ± 6344
738	megvii-004	3962505	44019	-	1325 4097 ± 0	1302 1287 ± 1	1052 1369 ± 2	992 1310 ± 2	968 1384 ± 3	880 1436 ± 5	1398 46801 ± 204	1399 46832 ± 207
739	megvii-005	1378009	44038	991 4025	985 2049 ± 0	1320 1319 ± 5	976 1247 ± 6	934 1240 ± 2	869 1245 ± 2	780 1298 ± 3	1363 32025 ± 121	1364 32008 ± 114
740	megvii-006	1554938	44038	1004 4346	973 2049 ± 0	1301 1287 ± 3	1003 1286 ± 0	1047 1393 ± 5	924 1319 ± 1	827 1360 ± 1	1361 31845 ± 100	1362 31872 ± 118
741	megvii-007	1366028	44035	702 1723	984 2049 ± 0	1194 1148 ± 1	899 1153 ± 0	879 1172 ± 4	803 1172 ± 2	773 1288 ± 2	1302 20191 ± 1753	1303 20191 ± 1620
742	megvii-008	1644071	44035	785 2086	981 2049 ± 0	1318 1313 ± 1	1074 1404 ± 4	1028 1363 ± 2	952 1362 ± 3	866 1420 ± 2	1300 20059 ± 1598	1307 20448 ± 1633
743	megvii-009	1644071	44039	819 2294	976 2049 ± 0	1323 1322 ± 1	1035 1328 ± 1	1014 1339 ± 3	1147 2075 ± 3	856 1404 ± 4	1301 20081 ± 1621	1306 20260 ± 1577
744	meituan-000	259514	333178	-	480 2048 ± 0	402 436 ± 4	265 441 ± 1	395 626 ± 5	733 1098 ± 15	1110 3126 ± 53	244 638 ± 17	252 633 ± 16
745	meituan-001	615387	333249	493 1113	820 2048 ± 0	1099 1017 ± 4	789 1008 ± 3	744 1010 ± 2	678 1010 ± 3	575 1011 ± 4	254 654 ± 10	272 658 ± 14
746	meituan-002	686111	244091	808 2200	1239 4096 ± 0	1126 1052 ± 0	840 1086 ± 1	796 1064 ± 2	705 1060 ± 5	615 1063 ± 1	431 1064 ± 10	444 1070 ± 16
747	meituan-003	860010	244091	903 2772	1258 4096 ± 0	1423 1487 ± 1	1132 1518 ± 0	1101 1469 ± 6	1043 1478 ± 0	900 1471 ± 7	454 1112 ± 16	456 1097 ± 20
748	meituan-004	850507	244091	899 2759	1249 4096 ± 0	861 796 ± 1	620 822 ± 0	572 820 ± 11	518 815 ± 11	439 824 ± 0	468 1149 ± 10	477 1148 ± 5

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5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 39: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
749	meiya-001	280055	264913	180 507	989 2049 ± 0	639 622 ± 12	-	-	-	-	1168 8356 ± 615	1168 8134 ± 97
750	mendaxiatech-000	1941475	45484	1008 4397	1326 4097 ± 0	1272 1243 ± 2	982 1255 ± 1	1035 1373 ± 2	1079 1598 ± 3	1087 2689 ± 8	1399 46906 ± 275	1400 46872 ± 217
751	metsakuurcompany-001	445177	1091558	666 1588	1053 2056 ± 0	561 578 ± 1	394 587 ± 3	355 590 ± 1	380 659 ± 1	467 854 ± 1	1174 8600 ± 192	1169 8155 ± 298
752	metsakuurcompany-002	0	957558	423 980	1098 2056 ± 0	1069 980 ± 1	763 978 ± 1	706 976 ± 2	671 1005 ± 1	640 1103 ± 2	1176 8766 ± 326	1177 8786 ± 324
753	metsakuurcompany-003	0	957562	433 1004	1081 2056 ± 0	1051 965 ± 2	750 970 ± 5	789 1060 ± 12	669 1005 ± 6	657 1121 ± 1	1167 8234 ± 145	1167 8111 ± 280
754	metsakuurcompany-004	0	1035754	461 1056	1095 2056 ± 0	1027 946 ± 10	651 861 ± 6	623 877 ± 3	601 926 ± 4	556 991 ± 5	1119 6281 ± 160	1115 6232 ± 191
755	metsakuurcompany-005	0	305952	267 661	576 2048 ± 0	444 472 ± 0	294 477 ± 0	274 494 ± 1	262 527 ± 1	412 778 ± 2	940 3376 ± 22	946 3383 ± 40
756	metsakuurcompany-006	0	532431	436 1010	554 2048 ± 0	673 648 ± 0	465 651 ± 0	426 648 ± 0	368 647 ± 0	299 649 ± 0	401 993 ± 2	435 1049 ± 8
757	miaxis-001	0	215019	102 325	124 512 ± 0	206 279 ± 0	130 278 ± 0	95 278 ± 1	87 285 ± 0	68 297 ± 0	363 872 ± 14	372 881 ± 22
758	miaxis-002	0	216484	103 328	144 512 ± 0	221 292 ± 0	135 290 ± 1	107 297 ± 0	97 298 ± 0	98 359 ± 1	434 1065 ± 48	428 1040 ± 54
759	miaxis-003	0	439726	227 604	682 2048 ± 0	658 634 ± 0	434 624 ± 2	377 613 ± 0	261 524 ± 0	312 665 ± 0	724 2053 ± 20	724 2048 ± 28
760	miaxis-004	729303	89655	524 1183	986 2049 ± 0	1474 1941 ± 0	1183 1947 ± 0	1170 1940 ± 0	1182 3153 ± 0	1024 1963 ± 0	1460 103338 ± 44	1460 103361 ± 65
761	miaxis-005	738575	172996	1009 4416	978 2049 ± 0	1451 1741 ± 1	1159 1776 ± 13	1148 1771 ± 4	1101 1774 ± 1	1015 1925 ± 5	982 3738 ± 17	990 3774 ± 45
762	microfocus-000	103483	27242	-	25 256 ± 0	191 263 ± 18	-	-	-	-	46 202 ± 5	44 203 ± 7
763	microfocus-001	104524	27242	-	34 256 ± 0	192 264 ± 18	-	-	-	-	49 215 ± 8	50 217 ± 10
764	microfocus-002	96288	27362	60 176	37 256 ± 0	187 259 ± 18	-	-	-	-	93 337 ± 34	56 230 ± 25
765	microfocus-003	169603	27689	181 507	35 256 ± 0	464 490 ± 19	330 517 ± 30	526 752 ± 55	761 1126 ± 121	1088 2713 ± 345	20 121 ± 5	20 117 ± 5
766	minivision-000	836697	16597	1002 4263	1316 4096 ± 0	1118 1035 ± 1	811 1033 ± 2	767 1035 ± 1	692 1037 ± 1	609 1059 ± 2	813 2466 ± 26	812 2460 ± 25
767	mitek-000	105584	44643	86 277	238 1024 ± 0	185 256 ± 21	116 265 ± 10	167 375 ± 24	276 545 ± 42	637 1098 ± 124	872 2846 ± 88	863 2777 ± 74
768	mitek-001	519928	69522	721 1777	1068 2056 ± 0	323 371 ± 0	225 397 ± 0	192 407 ± 4	160 394 ± 5	133 416 ± 2	1128 6682 ± 28	1126 6694 ± 48
769	mitek-002	534528	68069	719 1773	1074 2056 ± 0	314 365 ± 0	203 372 ± 0	158 366 ± 2	145 377 ± 2	120 394 ± 1	407 1005 ± 19	398 963 ± 23
770	mobai-000	365451	80573	325 786	1426 6144 ± 0	829 766 ± 8	660 869 ± 6	909 1205 ± 31	1119 1867 ± 45	1120 3549 ± 190	1273 16458 ± 333	1276 16423 ± 1473
771	mobai-001	265297	60164	196 534	947 2048 ± 0	618 612 ± 3	416 614 ± 3	461 687 ± 9	576 886 ± 31	984 1707 ± 103	537 1386 ± 25	546 1377 ± 26
772	mobai-002	254811	33242	631 1482	929 2048 ± 0	300 356 ± 0	197 366 ± 0	153 359 ± 4	137 367 ± 2	140 428 ± 0	1169 8387 ± 126	1171 8381 ± 105
773	mobai-003	1046772	427612	963 3555	1262 4096 ± 0	1444 1691 ± 7	1153 1698 ± 3	1140 1693 ± 1	1088 1705 ± 2	996 1786 ± 1	1462 105417 ± 231	1462 105475 ± 169
774	mobbl-000	182052	58727	-	748 2048 ± 0	188 261 ± 16	118 267 ± 22	166 375 ± 92	375 655 ± 273	1035 2059 ± 1129	1226 12061 ± 142	1228 12050 ± 133
775	mobbl-001	231160	58706	74 226	781 2048 ± 0	113 183 ± 32	66 184 ± 25	148 354 ± 76	525 823 ± 396	1094 2781 ± 1166	1221 11832 ± 109	1225 11851 ± 88
776	mobbl-002	242920	60119	-	799 2048 ± 0	694 663 ± 6	471 660 ± 5	439 662 ± 5	383 663 ± 5	320 676 ± 5	1217 11616 ± 78	1218 11588 ± 97
777	mobbl-003	172248	60960	87 282	466 2048 ± 0	698 664 ± 6	472 661 ± 5	441 663 ± 5	386 665 ± 6	332 691 ± 5	1234 12506 ± 111	1235 12509 ± 100
778	mobbl-004	432204	970955	377 872	460 2048 ± 0	218 288 ± 0	132 281 ± 0	97 281 ± 1	90 288 ± 2	69 312 ± 2	934 3327 ± 22	945 3369 ± 48
779	mobbl-005	352570	983522	422 978	862 2048 ± 0	1154 1098 ± 1	874 1125 ± 2	842 1123 ± 2	797 1167 ± 2	823 1354 ± 1	1081 5498 ± 62	1083 5492 ± 53
780	mobipintech-000	370514	303291	515 1173	558 2048 ± 0	1274 1245 ± 1	965 1234 ± 1	956 1264 ± 1	950 1360 ± 1	985 1707 ± 1	1250 14506 ± 214	1252 14433 ± 197
781	momovn-001	246938	1013076	319 769	753 2048 ± 0	165 237 ± 1	120 268 ± 7	114 308 ± 10	92 293 ± 5	118 390 ± 3	559 1438 ± 13	565 1443 ± 13
782	momovn-002	320036	1013104	310 750	888 2048 ± 0	171 243 ± 3	107 248 ± 0	85 245 ± 1	76 253 ± 0	78 328 ± 2	563 1454 ± 8	570 1456 ± 16
783	momovn-003	528962	951056	651 1548	1286 4096 ± 0	831 766 ± 1	591 777 ± 0	544 783 ± 0	517 815 ± 1	594 1045 ± 1	1005 3979 ± 23	1012 3984 ± 29
784	moredian-000	525259	21374	451 1035	785 2048 ± 0	752 694 ± 0	515 698 ± 0	476 699 ± 0	424 700 ± 0	348 713 ± 1	670 1803 ± 11	664 1779 ± 23
785	morpho-000	100806	107129	-	3 116 ± 0	54 109 ± 1	-	-	-	-	400 993 ± 31	410 1000 ± 34
786	morpho-001	100806	368573	-	2 116 ± 0	12 41 ± 1	-	-	-	-	416 1030 ± 38	429 1042 ± 37
787	morpho-002	383021	110995	-	78 352 ± 0	373 411 ± 7	-	-	-	-	971 3620 ± 94	817 2481 ± 65
788	mukh-001	866223	451194	681 1647	230 1024 ± 0	1348 1375 ± 17	1062 1390 ± 12	1060 1406 ± 8	977 1394 ± 10	826 1360 ± 11	123 433 ± 14	131 435 ± 14
789	mukh-002	693809	454936	476 1081	608 2048 ± 0	1055 968 ± 1	708 921 ± 12	691 957 ± 2	622 954 ± 6	532 953 ± 5	214 612 ± 13	225 611 ± 17
790	mukh-003	556405	180996	741 1853	477 2048 ± 0	1465 1896 ± 3	1184 1948 ± 1	1175 2003 ± 0	1137 1981 ± 0	1030 2016 ± 0	195 597 ± 13	219 604 ± 16
791	mukh-004	405118	181085	588 1353	847 2048 ± 0	537 558 ± 1	374 567 ± 1	337 569 ± 1	295 569 ± 3	253 593 ± 1	284 704 ± 1	298 703 ± 2
792	mukh-005	555239	264759	582 1346	637 2048 ± 0	400 436 ± 3	305 486 ± 6	275 495 ± 7	263 527 ± 16	257 597 ± 25	286 705 ± 1	299 706 ± 2

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	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
793	multimodality-000	0	503924	609 1417	492 2048 ± 0	380 416 ± 0	244 420 ± 0	206 423 ± 0	182 427 ± 0	164 463 ± 0	357 848 ± 25	343 800 ± 28
794	multimodality-001	185719	545045	602 1390	1217 4096 ± 0	1229 1190 ± 2	912 1169 ± 2	875 1165 ± 2	796 1167 ± 2	699 1177 ± 2	553 1424 ± 35	549 1384 ± 42
795	mvision-001	227502	149531	294 723	152 512 ± 0	747 691 ± 21	521 702 ± 19	474 697 ± 24	433 708 ± 29	344 710 ± 27	461 1123 ± 40	484 1154 ± 38
796	nazhiai-000	547484	16141	890 2716	467 2048 ± 0	733 683 ± 3	503 687 ± 2	589 835 ± 27	543 840 ± 31	456 834 ± 34	758 2230 ± 34	740 2133 ± 81
797	ncsgroup-000	95110	345861	453 1041	693 2048 ± 0	89 157 ± 0	47 153 ± 0	39 151 ± 0	34 157 ± 0	45 231 ± 0	799 2407 ± 16	800 2411 ± 22
798	ncssg-001	148743	490151	364 847	646 2048 ± 0	1013 936 ± 2	729 946 ± 3	677 944 ± 2	612 938 ± 5	542 972 ± 2	968 3579 ± 50	955 3459 ± 34
799	ncssg-002	148859	415290	413 945	629 2048 ± 0	346 389 ± 1	224 397 ± 0	186 401 ± 1	171 407 ± 0	166 469 ± 0	1433 66390 ± 246	1431 66363 ± 179
800	ncssg-003	148876	414848	314 761	931 2048 ± 0	378 416 ± 1	263 437 ± 2	220 437 ± 1	193 443 ± 2	183 497 ± 1	1332 24709 ± 68	1331 24768 ± 134
801	ncssg-004	521184	483011	818 2272	935 2048 ± 0	736 687 ± 1	504 687 ± 0	485 706 ± 5	417 693 ± 1	363 731 ± 21	1333 24815 ± 117	1332 24799 ± 117
802	neosystems-001	575295	349959	-	849 2048 ± 0	1183 1137 ± 4	855 1098 ± 1	1147 1767 ± 4	1099 1769 ± 3	993 1765 ± 4	1288 18557 ± 189	1293 18640 ± 192
803	neosystems-002	599441	349942	-	500 2048 ± 0	1180 1135 ± 2	1174 1855 ± 3	1186 2258 ± 5	1157 2238 ± 3	1060 2247 ± 3	1289 18752 ± 167	1292 18610 ± 213
804	neosystems-003	599442	349942	-	863 2048 ± 0	1188 1143 ± 2	1168 1836 ± 7	1187 2260 ± 3	1161 2273 ± 6	1061 2273 ± 3	1294 19130 ± 223	1297 19167 ± 186
805	neosystems-004	243546	352623	194 529	771 2048 ± 0	265 324 ± 0	532 711 ± 3	576 827 ± 7	551 854 ± 2	505 916 ± 2	1248 14437 ± 176	1250 14355 ± 173
806	netbridgetech-001	133108	205875	183 508	1251 4096 ± 0	35 85 ± 1	19 83 ± 0	15 84 ± 0	15 92 ± 0	15 113 ± 4	1179 9280 ± 74	1181 9446 ± 512
807	netbridgetech-002	257687	49931	94 299	572 2048 ± 0	916 838 ± 6	638 838 ± 2	591 839 ± 1	542 839 ± 3	470 859 ± 3	881 2893 ± 65	909 3050 ± 123
808	neurotechnology-000	62129	33905	-	1448 7148 ± 0	617 611 ± 48	-	-	-	-	1450 74288 ± 2194	1445 72879 ± 2640
809	neurotechnology-001	280771	35716	-	1190 2718 ± 0	962 881 ± 46	-	-	-	-	1440 69356 ± 684	1436 69140 ± 579
810	neurotechnology-002	280771	36036	-	1189 2718 ± 0	566 581 ± 6	-	-	-	-	1436 68738 ± 748	1435 68905 ± 993
811	neurotechnology-003	339962	36125	-	1419 5214 ± 0	565 580 ± 5	-	-	-	-	1446 73589 ± 2548	1444 72837 ± 1546
812	neurotechnology-004	293384	54399	-	41 256 ± 0	359 401 ± 0	-	-	-	-	51 219 ± 8	58 231 ± 13
813	neurotechnology-005	270450	58088	-	30 256 ± 0	356 399 ± 0	-	-	-	-	56 238 ± 10	62 237 ± 7
814	neurotechnology-006	525541	54522	-	155 512 ± 0	721 678 ± 56	-	-	-	-	160 513 ± 14	170 535 ± 26
815	neurotechnology-008	119718	53462	-	29 256 ± 0	285 339 ± 0	-	-	-	-	135 467 ± 19	152 486 ± 26
816	neurotechnology-009	137515	53213	-	38 256 ± 0	344 388 ± 1	-	-	-	-	161 513 ± 10	161 512 ± 16
817	neurotechnology-010	362144	50226	-	165 514 ± 0	860 796 ± 1	603 796 ± 1	565 815 ± 3	548 853 ± 1	569 1006 ± 11	19 120 ± 8	17 115 ± 7
818	neurotechnology-011	364137	51141	-	163 514 ± 0	867 798 ± 1	607 802 ± 1	577 827 ± 3	561 873 ± 2	610 1059 ± 15	16 114 ± 11	16 114 ± 8
819	neurotechnology-012	147830	51395	-	39 256 ± 0	338 384 ± 0	213 387 ± 0	191 404 ± 1	188 435 ± 1	246 583 ± 7	18 119 ± 7	18 116 ± 7
820	neurotechnology-013	474749	85552	1025 4825	164 514 ± 0	1084 1000 ± 1	788 1006 ± 2	751 1022 ± 2	702 1053 ± 2	703 1195 ± 8	14 109 ± 4	13 110 ± 4
821	neurotechnology-015	474782	86045	1022 4819	167 515 ± 0	1112 1028 ± 3	810 1033 ± 3	781 1055 ± 4	732 1097 ± 4	786 1304 ± 18	21 130 ± 2	21 130 ± 4
822	neurotechnology-016	661450	89972	994 4074	166 515 ± 0	1328 1339 ± 3	1045 1351 ± 2	1031 1369 ± 4	974 1390 ± 4	923 1508 ± 21	31 179 ± 20	30 171 ± 20
823	neurotechnology-017	371650	98242	829 2385	36 256 ± 0	483 508 ± 2	327 513 ± 0	299 519 ± 1	266 533 ± 1	277 622 ± 15	4 80 ± 1	4 79 ± 1
824	neurotechnology-018	698669	98271	1005 4367	168 515 ± 0	787 725 ± 1	563 747 ± 1	521 747 ± 4	485 769 ± 2	454 833 ± 22	6 85 ± 2	7 87 ± 7
825	neurotechnology-019	365649	103237	826 2321	26 256 ± 0	644 625 ± 1	437 625 ± 0	408 633 ± 3	407 684 ± 2	378 742 ± 17	2 71 ± 2	2 72 ± 4
826	neurotechnology-020	807529	103241	1030 5010	161 513 ± 0	1235 1201 ± 4	934 1196 ± 5	900 1195 ± 8	851 1224 ± 7	832 1373 ± 22	11 102 ± 1	9 101 ± 1
827	neurotechnology-021	372239	147289	730 1810	27 256 ± 0	1368 1396 ± 0	1067 1394 ± 0	1066 1410 ± 5	1014 1438 ± 0	943 1545 ± 41	25 147 ± 1	26 152 ± 4
828	nextmetalabs-001	361607	4729276	245 627	887 2048 ± 0	991 914 ± 0	716 928 ± 0	668 928 ± 0	618 948 ± 0	510 919 ± 0	1059 4876 ± 185	1050 4774 ± 72
829	nextmetalabs-002	571398	5046464	497 1123	578 2048 ± 0	1283 1259 ± 2	963 1229 ± 1	948 1253 ± 0	877 1257 ± 14	777 1296 ± 1	1051 4786 ± 103	1049 4753 ± 97
830	nextmetalabs-003	1105548	5046426	823 2301	658 2048 ± 0	1249 1218 ± 1	951 1217 ± 2	922 1225 ± 6	832 1207 ± 2	746 1253 ± 1	1057 4874 ± 57	1056 4833 ± 85
831	nextmetalabs-004	1105548	4174065	579 1332	447 2048 ± 0	1473 1940 ± 1	1185 1949 ± 0	1166 1919 ± 0	1131 1925 ± 1	1029 2011 ± 2	1232 12441 ± 88	1236 12539 ± 154
832	nhn-001	336391	817674	-	1257 4096 ± 0	1110 1027 ± 3	805 1029 ± 1	761 1029 ± 1	697 1044 ± 1	633 1090 ± 1	1417 56650 ± 260	1418 56639 ± 210
833	nhn-002	363471	817674	274 679	1216 4096 ± 0	1186 1141 ± 3	886 1138 ± 2	857 1141 ± 2	785 1151 ± 6	710 1203 ± 2	1416 56608 ± 579	1417 56549 ± 606
834	nhn-003	933665	432730	615 1427	1302 4096 ± 0	1260 1229 ± 2	988 1261 ± 1	954 1263 ± 3	898 1279 ± 2	836 1375 ± 3	1407 50560 ± 105	1408 50592 ± 142
835	nhn-004	931648	432753	909 2864	1233 4096 ± 0	623 612 ± 0	448 632 ± 0	415 636 ± 0	398 676 ± 6	436 820 ± 0	1279 17323 ± 99	1282 17334 ± 133
836	nhn-005	1278729	432776	912 2890	1223 4096 ± 0	881 811 ± 0	481 671 ± 13	429 652 ± 0	445 726 ± 12	491 890 ± 3	1283 18123 ± 400	1288 18214 ± 250

Notes	
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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 41: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
837	nmetics-001	1304721	145561	799 2157	882 2048 ± 0	1432 1559 ± 0	1137 1564 ± 0	1130 1570 ± 1	1066 1561 ± 1	971 1634 ± 1	550 1407 ± 115	541 1357 ± 174
838	noblis-000	713573	253204	-	286 1061 ± 0	102 174 ± 20	-	-	-	-	527 1343 ± 45	543 1373 ± 56
839	nodeflux-000	347079	797842	-	588 2048 ± 0	521 542 ± 1	-	-	-	-	932 3283 ± 51	938 3285 ± 56
840	nodeflux-001	262553	797842	-	656 2048 ± 0	177 247 ± 1	-	-	-	-	925 3242 ± 81	937 3255 ± 93
841	nodeflux-002	774668	690213	158 466	403 2048 ± 0	770 708 ± 4	531 709 ± 4	491 716 ± 5	441 716 ± 7	368 736 ± 3	950 3475 ± 62	950 3408 ± 143
842	nominder-000	547519	102693	709 1741	43 258 ± 0	764 701 ± 0	517 701 ± 2	495 721 ± 2	444 720 ± 3	424 796 ± 1	85 314 ± 5	54 226 ± 45
843	nominder-001	332018	102774	514 1172	42 258 ± 0	118 186 ± 0	71 189 ± 1	54 188 ± 0	52 202 ± 1	62 271 ± 1	54 226 ± 1	31 172 ± 41
844	nominder-002	249049	116931	740 1852	46 258 ± 0	364 405 ± 1	246 425 ± 2	210 424 ± 1	187 434 ± 0	191 505 ± 0	10 101 ± 11	10 102 ± 14
845	nominder-003	249964	116931	834 2417	47 258 ± 0	714 675 ± 2	500 683 ± 5	453 677 ± 2	425 701 ± 3	410 775 ± 3	7 87 ± 4	5 84 ± 4
846	nominder-004	249964	114221	833 2416	49 258 ± 0	818 755 ± 2	568 751 ± 4	517 740 ± 3	484 768 ± 3	462 846 ± 1	5 81 ± 1	6 86 ± 8
847	nominder-005	289014	114221	885 2699	45 258 ± 0	1211 1170 ± 4	933 1196 ± 5	873 1161 ± 6	846 1221 ± 8	815 1343 ± 7	8 94 ± 14	19 117 ± 7
848	nominder-006	304867	114225	879 2660	44 258 ± 0	800 736 ± 2	561 746 ± 3	531 759 ± 3	480 762 ± 2	477 869 ± 5	13 105 ± 5	12 108 ± 10
849	nominder-007	249970	101106	405 932	48 258 ± 0	1146 1085 ± 0	829 1071 ± 0	821 1099 ± 1	748 1117 ± 2	729 1235 ± 5	29 171 ± 5	41 190 ± 7
850	nominder-008	484231	102193	693 1688	162 514 ± 0	1476 1982 ± 1	1186 1953 ± 0	1168 1937 ± 1	1135 1954 ± 3	1037 2074 ± 6	30 172 ± 2	42 191 ± 2
851	notiontag-000	92753	406791	-	199 584 ± 0	527 548 ± 64	358 548 ± 35	1091 1450 ± 99	1185 3771 ± 251	1180 13146 ± 792	1391 44672 ± 269	1392 44593 ± 358
852	notiontag-001	92753	427967	255 642	198 584 ± 0	1009 929 ± 35	846 1092 ± 39	1193 3709 ± 81	1191 10233 ± 180	1185 31181 ± 521	1388 43636 ± 286	1389 43724 ± 330
853	notiontag-002	271987	967207	918 2965	1162 2120 ± 0	419 453 ± 2	277 453 ± 3	234 453 ± 3	203 458 ± 2	167 471 ± 3	1304 20278 ± 194	1304 20195 ± 186
854	npt-000	466008	22512	351 833	1132 2064 ± 0	1061 974 ± 0	779 994 ± 0	727 994 ± 0	651 990 ± 1	571 1007 ± 0	704 1947 ± 29	713 1983 ± 34
855	nsensecorp-001	187421	258593	-	344 2048 ± 0	282 336 ± 0	167 335 ± 0	132 336 ± 0	116 335 ± 0	82 337 ± 0	1396 46605 ± 93	1397 46613 ± 112
856	nsensecorp-002	187421	122407	-	958 2048 ± 0	276 333 ± 0	165 333 ± 0	133 337 ± 0	120 338 ± 0	94 351 ± 0	1394 45965 ± 213	1394 45988 ± 158
857	nsensecorp-003	199895	117041	286 710	364 2048 ± 0	691 661 ± 0	474 664 ± 0	438 662 ± 1	378 659 ± 1	306 659 ± 0	1390 44658 ± 51	1393 44654 ± 72
858	nsensecorp-004	513276	139178	689 1675	593 2048 ± 0	1391 1433 ± 0	1105 1445 ± 7	1092 1450 ± 3	1047 1487 ± 5	-	795 2388 ± 42	794 2385 ± 63
859	nsensecorp-005	411845	138437	545 1240	939 2048 ± 0	1405 1455 ± 0	1112 1453 ± 0	1090 1449 ± 0	1030 1457 ± 2	914 1490 ± 4	740 2131 ± 51	745 2165 ± 66
860	ntechlab-000	191530	22085	-	1191 2906 ± 1	205 278 ± 13	-	-	-	-	1357 30787 ± 142	1355 30846 ± 77
861	ntechlab-001	691296	22848	-	1446 6744 ± 1	581 587 ± 11	-	-	-	-	1434 67692 ± 833	1433 67486 ± 244
862	ntechlab-002	894169	25023	-	1447 6744 ± 1	271 330 ± 11	-	-	-	-	1455 82508 ± 213	1453 82524 ± 220
863	ntechlab-003	1422738	24649	-	1192 2906 ± 1	69 128 ± 3	-	-	-	-	1373 35327 ± 117	1376 35371 ± 123
864	ntechlab-004	3752378	115782	-	1212 3482 ± 1	890 818 ± 11	-	-	-	-	1384 38929 ± 81	1387 38916 ± 111
865	ntechlab-005	1726214	116068	-	336 1940 ± 0	774 712 ± 1	-	-	-	-	367 882 ± 42	371 880 ± 29
866	ntechlab-006	7901590	43009	-	1187 2600 ± 0	809 749 ± 1	-	-	-	-	423 1055 ± 93	363 844 ± 48
867	ntechlab-007	2509686	52474	-	1211 3348 ± 0	857 792 ± 3	-	-	-	-	489 1209 ± 59	515 1267 ± 65
868	ntechlab-008	1138002	52483	-	303 1300 ± 0	535 556 ± 1	-	-	-	-	448 1095 ± 45	434 1049 ± 51
869	ntechlab-009	1748043	43730	-	335 1940 ± 0	1171 1115 ± 2	866 1114 ± 1	837 1119 ± 1	765 1130 ± 2	706 1202 ± 4	218 614 ± 20	206 595 ± 28
870	ntechlab-010	698591	217167	-	297 1280 ± 0	1216 1177 ± 2	920 1180 ± 2	903 1197 ± 2	850 1224 ± 1	807 1326 ± 3	118 405 ± 13	126 416 ± 31
871	ntechlab-011	786933	209458	1058 6867	298 1280 ± 0	1193 1148 ± 2	891 1142 ± 1	870 1159 ± 1	816 1185 ± 1	775 1290 ± 3	32 179 ± 11	33 173 ± 11
872	ntechlab-012	570796	212350	1036 5450	1183 2560 ± 0	1316 1309 ± 1	1033 1323 ± 1	1007 1331 ± 1	951 1360 ± 1	894 1460 ± 3	48 211 ± 8	49 211 ± 7
873	ntnu-000	293085	433109	687 1665	1473 16896 ± 0	1395 1440 ± 0	1104 1441 ± 0	1077 1430 ± 0	1022 1452 ± 0	904 1473 ± 1	664 1799 ± 53	635 1690 ± 42
874	ocrstudio-001	9010	57372	13 66	24 240 ± 0	32 83 ± 0	20 83 ± 0	17 88 ± 0	19 104 ± 0	31 190 ± 2	1204 10725 ± 62	1209 11030 ± 130
875	omface-000	45945	844976	50 154	239 1024 ± 0	117 185 ± 1	85 206 ± 2	65 203 ± 1	47 195 ± 1	34 193 ± 1	142 481 ± 42	139 456 ± 20
876	omface-001	146370	1799745	47 145	231 1024 ± 0	129 194 ± 2	95 222 ± 2	68 209 ± 0	58 216 ± 1	47 233 ± 1	1287 18369 ± 19	1290 18366 ± 32
877	omnigarde-000	264057	32882	-	253 1024 ± 0	1023 944 ± 0	679 887 ± 0	630 888 ± 1	580 892 ± 0	499 902 ± 0	841 2671 ± 35	835 2620 ± 29
878	omnigarde-001	200523	32882	159 466	151 512 ± 0	1015 941 ± 0	677 883 ± 1	629 886 ± 1	579 891 ± 1	495 898 ± 0	547 1405 ± 31	547 1379 ± 26
879	omnigarde-002	368860	32882	313 760	251 1024 ± 0	1311 1303 ± 1	975 1246 ± 1	943 1249 ± 1	874 1253 ± 1	753 1261 ± 1	855 2727 ± 34	850 2686 ± 32
880	omnigarde-003	427751	35153	312 756	171 517 ± 0	708 673 ± 0	446 631 ± 1	409 633 ± 1	359 637 ± 1	296 647 ± 1	504 1259 ± 22	508 1244 ± 16

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Table 42: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
881	omnigarde-004	995019	35158	746 1873	170 517 ± 0	1236 1201 ± 1	987 1261 ± 1	910 1207 ± 1	839 1215 ± 0	726 1229 ± 0	509 1274 ± 22	506 1241 ± 26
882	omnigarde-005	1316067	35158	862 2545	180 525 ± 0	1210 1170 ± 1	906 1159 ± 0	876 1169 ± 7	818 1186 ± 1	704 1199 ± 1	585 1490 ± 40	583 1481 ± 38
883	onfido-000	273478	959781	397 914	860 2048 ± 0	428 459 ± 17	276 451 ± 15	233 451 ± 14	209 462 ± 15	192 505 ± 18	618 1617 ± 50	620 1637 ± 53
884	onfido-001	344394	1021960	898 2756	760 2048 ± 0	143 208 ± 4	86 208 ± 4	72 213 ± 4	63 223 ± 3	55 250 ± 4	366 874 ± 28	365 852 ± 20
885	onfido-002	852775	252949	432 1003	1313 4096 ± 0	1234 1199 ± 5	941 1202 ± 5	906 1203 ± 5	837 1213 ± 5	741 1250 ± 5	837 2625 ± 60	838 2628 ± 84
886	onfido-003	852775	252949	435 1010	1292 4096 ± 0	1240 1203 ± 5	940 1202 ± 5	908 1205 ± 5	838 1214 ± 5	744 1252 ± 5	811 2463 ± 14	843 2669 ± 37
887	onfido-004	855472	1085274	996 4116	1431 6144 ± 0	1440 1635 ± 5	1148 1639 ± 6	1136 1630 ± 6	1082 1650 ± 5	981 1681 ± 6	844 2686 ± 112	880 2869 ± 50
888	oorudigital-000	1027828	75456	233 609	1138 2066 ± 0	695 664 ± 0	475 664 ± 0	467 693 ± 0	473 756 ± 0	621 1066 ± 8	615 1606 ± 13	626 1652 ± 23
889	ooto-001	382081	95138	755 1923	1197 3072 ± 0	1022 944 ± 0	734 953 ± 1	733 997 ± 0	626 957 ± 0	563 999 ± 0	1064 4993 ± 17	1065 5004 ± 50
890	openedge-000	447048	272235	598 1376	1082 2056 ± 0	319 367 ± 0	232 404 ± 1	143 352 ± 1	123 350 ± 0	113 385 ± 0	1258 15122 ± 41	1260 15110 ± 48
891	openface-001	0	40111	27 100	699 2048 ± 0	83 148 ± 1	48 154 ± 0	157 365 ± 3	173 409 ± 9	271 616 ± 31	209 608 ± 14	220 604 ± 13
892	ovisionllc-001	192059	252531	468 1066	229 1024 ± 0	104 175 ± 0	57 175 ± 0	51 183 ± 0	48 196 ± 0	52 241 ± 0	879 2889 ± 38	887 2901 ± 47
893	ovisionllc-002	247778	96584	629 1478	259 1024 ± 0	194 266 ± 0	113 256 ± 0	86 250 ± 0	79 263 ± 1	70 313 ± 0	858 2763 ± 72	853 2727 ± 29
894	oz-001	303723	238311	-	1373 4125 ± 0	1191 1147 ± 3	923 1182 ± 3	963 1273 ± 4	1081 1617 ± 7	1101 2890 ± 19	1468 228011 ± 5455	1467 220746 ± 5572
895	oz-002	716022	170261	-	1136 2065 ± 0	1135 1064 ± 3	913 1171 ± 3	1192 2953 ± 6	1189 7352 ± 13	1184 26658 ± 29	1464 131108 ± 1408	1464 126758 ± 913
896	oz-003	484147	519652	1080 11575	1048 2053 ± 0	1346 1375 ± 12	1059 1388 ± 3	1149 1773 ± 16	1142 2039 ± 6	1113 3209 ± 5	1447 73905 ± 456	1446 73892 ± 444
897	oz-004	373982	1075452	1070 7845	1047 2053 ± 0	908 832 ± 7	662 871 ± 6	635 899 ± 10	722 1078 ± 12	965 1608 ± 10	1426 61654 ± 418	1424 61749 ± 450
898	pa-002	442564	308954	-	1474 18436 ± 0	72 132 ± 7	-	-	-	-	907 3088 ± 32	910 3051 ± 28
899	pa-003	714102	1464254	-	485 2048 ± 0	907 831 ± 31	-	-	-	-	392 962 ± 23	383 914 ± 38
900	palit-000	428754	144958	592 1360	1301 4096 ± 0	553 570 ± 1	384 578 ± 1	346 576 ± 3	304 583 ± 1	270 614 ± 1	756 2227 ± 16	757 2226 ± 16
901	palit-001	173886	145564	219 587	937 2048 ± 0	159 227 ± 0	97 224 ± 1	76 224 ± 1	67 229 ± 3	59 262 ± 2	469 1150 ± 16	470 1135 ± 23
902	pangiam-000	464252	24512	989 3963	861 2048 ± 0	646 627 ± 5	424 618 ± 4	383 615 ± 3	341 620 ± 3	291 639 ± 3	17 118 ± 7	15 113 ± 7
903	pangiam-001	1015455	37259	663 1569	506 2048 ± 0	1060 972 ± 1	765 982 ± 3	711 981 ± 2	652 990 ± 7	599 1052 ± 2	22 136 ± 11	22 139 ± 13
904	pangiam-002	486035	39834	538 1205	884 2048 ± 0	253 316 ± 0	152 316 ± 0	120 317 ± 0	108 319 ± 1	77 326 ± 2	15 111 ± 13	11 103 ± 2
905	papago-001	669274	52817	827 2341	845 2048 ± 0	1290 1272 ± 6	1013 1296 ± 7	985 1295 ± 6	900 1281 ± 3	816 1345 ± 3	1263 15236 ± 169	1264 15184 ± 142
906	papil11-000	1146378	24895	805 2184	362 2048 ± 0	1020 942 ± 3	861 1108 ± 2	907 1204 ± 6	958 1369 ± 10	1031 2033 ± 35	215 612 ± 7	229 613 ± 11
907	papil11-001	1221825	38957	873 2618	695 2048 ± 0	1038 951 ± 2	883 1135 ± 2	917 1220 ± 4	970 1386 ± 10	1063 2292 ± 69	157 506 ± 22	163 519 ± 20
908	papil11-002	1208976	38985	859 2543	383 2048 ± 0	997 916 ± 2	850 1094 ± 2	874 1163 ± 1	912 1307 ± 4	1049 2140 ± 10	173 560 ± 22	189 577 ± 22
909	papil11-003	1263123	39005	878 2657	586 2048 ± 0	1120 1038 ± 1	957 1222 ± 1	979 1293 ± 2	1010 1431 ± 3	1064 2298 ± 8	158 508 ± 10	159 501 ± 10
910	papil11-004	560851	18192	491 1104	74 296 ± 0	494 522 ± 0	335 522 ± 0	303 524 ± 0	265 531 ± 0	234 556 ± 2	1320 23204 ± 63	1322 23224 ± 68
911	papsav1923-001	279210	52652	163 474	891 2048 ± 0	645 626 ± 1	440 628 ± 1	401 630 ± 1	369 648 ± 2	379 744 ± 3	302 725 ± 25	318 731 ± 28
912	papsav1923-002	491185	24727	503 1139	998 2052 ± 0	858 792 ± 1	762 978 ± 1	771 1042 ± 1	789 1158 ± 1	973 1641 ± 19	488 1209 ± 29	500 1206 ± 38
913	papsav1923-003	515576	24803	492 1113	873 2048 ± 0	862 797 ± 0	772 987 ± 1	772 1043 ± 1	808 1178 ± 1	1000 1809 ± 7	373 903 ± 26	380 905 ± 34
914	paravision-004	556670	145440	-	1227 4096 ± 0	904 829 ± 2	634 834 ± 6	586 832 ± 2	534 833 ± 4	453 833 ± 2	309 737 ± 31	310 718 ± 38
915	paravision-006	542423	202234	-	1232 4096 ± 0	760 700 ± 1	523 704 ± 0	483 705 ± 0	428 704 ± 0	342 705 ± 0	87 321 ± 16	89 315 ± 14
916	paravision-008	542190	204400	603 1399	1235 4096 ± 0	759 699 ± 0	516 700 ± 0	479 701 ± 0	427 702 ± 1	339 702 ± 0	92 337 ± 17	100 330 ± 13
917	paravision-010	688291	205854	798 2150	1337 4100 ± 0	659 634 ± 0	455 635 ± 0	414 635 ± 0	354 635 ± 0	286 635 ± 1	607 1577 ± 35	612 1571 ± 32
918	paravision-011	781138	95589	837 2424	1333 4100 ± 0	930 852 ± 0	663 871 ± 1	607 858 ± 1	552 854 ± 0	480 873 ± 1	616 1608 ± 35	618 1625 ± 32
919	paravision-013	1537125	95602	1016 4648	1330 4100 ± 0	1079 991 ± 9	777 993 ± 5	723 992 ± 4	665 1002 ± 5	566 1002 ± 7	577 1481 ± 26	589 1490 ± 25
920	paravision-015	1584003	164658	1023 4820	1348 4100 ± 0	1344 1371 ± 12	1142 1577 ± 1	1111 1500 ± 4	1020 1449 ± 1	930 1521 ± 7	555 1432 ± 11	568 1455 ± 32
921	paravision-017	1584003	164639	1024 4820	1329 4100 ± 0	1307 1295 ± 11	897 1147 ± 4	867 1158 ± 8	794 1166 ± 7	673 1140 ± 12	566 1461 ± 10	572 1459 ± 16
922	paravision-018	1596372	164639	1027 4909	1341 4100 ± 0	1414 1474 ± 11	1119 1472 ± 12	1105 1475 ± 11	1031 1459 ± 11	931 1523 ± 16	600 1532 ± 68	598 1520 ± 40
923	paravision-020	1646513	155644	958 3487	1328 4100 ± 0	1478 2043 ± 0	1187 2043 ± 0	1176 2048 ± 0	1141 2038 ± 0	1040 2100 ± 0	734 2110 ± 10	735 2110 ± 17
924	paravision-022	1646513	155659	957 3460	1344 4100 ± 0	1480 2070 ± 0	1191 2057 ± 2	1180 2056 ± 0	1144 2068 ± 1	1039 2098 ± 0	725 2063 ± 72	749 2181 ± 38

Notes	
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4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
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Table 43: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
925	pensees-001	1619431	408932	754 1922	1459 8200 ± 0	1162 1108 ± 3	1108 1448 ± 17	1081 1439 ± 10	1036 1464 ± 5	944 1546 ± 9	914 3151 ± 34	919 3143 ± 25
926	pentacomp-000	261914	192048	264 658	844 2048 ± 0	418 452 ± 1	270 444 ± 0	241 458 ± 4	201 456 ± 5	163 461 ± 3	444 1085 ± 25	451 1081 ± 38
927	pixelall-002	0	169030	-	1182 2560 ± 0	123 191 ± 1	-	-	-	-	495 1223 ± 56	503 1230 ± 47
928	pixelall-003	0	707030	-	1177 2560 ± 0	757 699 ± 8	-	-	-	-	481 1174 ± 28	471 1139 ± 68
929	pixelall-004	0	550919	-	1179 2560 ± 0	417 451 ± 11	274 449 ± 3	232 451 ± 1	204 459 ± 2	188 502 ± 4	464 1139 ± 65	495 1186 ± 31
930	pixelall-005	0	1001355	-	1416 5120 ± 0	1170 1112 ± 3	867 1115 ± 1	838 1120 ± 1	755 1124 ± 2	674 1143 ± 2	505 1259 ± 29	507 1243 ± 23
931	pixelall-006	0	746305	-	1180 2560 ± 0	1104 1024 ± 3	803 1028 ± 2	764 1033 ± 1	689 1032 ± 1	600 1054 ± 2	314 754 ± 14	312 722 ± 10
932	pixelall-007	0	444912	-	552 2048 ± 0	1107 1026 ± 4	815 1038 ± 2	811 1089 ± 2	728 1087 ± 2	658 1124 ± 2	289 708 ± 14	296 701 ± 19
933	pixelall-008	0	992249	710 1742	1456 8192 ± 0	1412 1471 ± 3	1076 1405 ± 4	1064 1409 ± 4	995 1413 ± 3	872 1426 ± 4	665 1799 ± 50	674 1807 ± 48
934	pixelall-009	0	1009114	705 1732	1455 8192 ± 0	1420 1484 ± 3	1069 1395 ± 3	1053 1400 ± 4	975 1391 ± 3	876 1433 ± 3	678 1848 ± 13	679 1842 ± 19
935	pixelall-010	0	1469903	884 2693	1454 8192 ± 0	1198 1153 ± 3	838 1084 ± 4	825 1103 ± 4	731 1093 ± 3	653 1118 ± 4	692 1906 ± 13	691 1898 ± 29
936	pnpsecure-000	9726	146266	24 93	768 2048 ± 0	9 35 ± 0	8 42 ± 0	35 145 ± 0	174 409 ± 1	951 1565 ± 5	696 1918 ± 11	694 1916 ± 16
937	pnpsecure-001	1015928	59243	779 2060	944 2048 ± 0	712 674 ± 0	518 701 ± 0	628 885 ± 0	1024 1452 ± 1	1115 3286 ± 3	836 2621 ± 19	841 2649 ± 41
938	pnpsecure-002	1015393	146387	796 2139	809 2048 ± 0	964 883 ± 0	659 869 ± 0	611 867 ± 0	600 926 ± 0	574 1009 ± 0	839 2640 ± 11	852 2711 ± 11
939	privid-001	0	76008	46 143	1479 30720 ± 0	415 450 ± 0	278 454 ± 0	250 467 ± 1	222 472 ± 1	190 504 ± 1	1277 17041 ± 74	1279 17053 ± 100
940	privid-002	0	219733	447 1031	616 2048 ± 0	1285 1261 ± 3	901 1156 ± 4	866 1155 ± 2	790 1162 ± 3	784 1302 ± 4	272 686 ± 22	280 680 ± 20
941	privid-003	0	188341	464 1062	445 2048 ± 0	1389 1430 ± 2	1102 1437 ± 6	1098 1459 ± 7	1021 1451 ± 7	918 1499 ± 18	301 725 ± 7	311 720 ± 13
942	privid-004	0	992936	574 1304	626 2048 ± 0	1312 1303 ± 1	999 1282 ± 2	1001 1324 ± 3	953 1363 ± 0	846 1387 ± 0	296 717 ± 29	294 699 ± 19
943	psl-000	260619	1537815	-	532 2048 ± 0	650 629 ± 11	-	-	-	-	71 280 ± 17	78 280 ± 17
944	psl-001	382035	1461947	-	1076 2056 ± 0	849 785 ± 16	-	-	-	-	77 298 ± 13	83 292 ± 14
945	psl-002	804934	1462048	-	999 2052 ± 0	972 888 ± 9	-	-	-	-	610 1590 ± 48	469 1133 ± 78
946	psl-003	1159643	1434859	-	1163 2120 ± 0	937 865 ± 3	-	-	-	-	421 1052 ± 14	421 1025 ± 51
947	psl-004	771727	1560145	-	1205 3144 ± 0	918 839 ± 3	-	-	-	-	264 680 ± 34	276 670 ± 22
948	psl-005	774031	1560153	-	1206 3144 ± 0	1004 922 ± 37	-	-	-	-	283 703 ± 32	302 708 ± 26
949	psl-006	1211577	524525	-	1208 3144 ± 0	1394 1438 ± 6	1100 1434 ± 4	1078 1434 ± 4	1011 1434 ± 5	881 1437 ± 5	74 291 ± 34	75 268 ± 25
950	psl-007	954351	524521	-	1207 3144 ± 0	1378 1408 ± 5	1087 1417 ± 3	1071 1418 ± 3	998 1419 ± 2	867 1422 ± 3	65 265 ± 22	70 258 ± 17
951	psl-008	954351	524525	-	1209 3144 ± 0	1381 1412 ± 4	1086 1415 ± 3	1070 1416 ± 2	997 1418 ± 2	865 1418 ± 2	63 259 ± 22	66 252 ± 22
952	psl-009	411027	411504	-	1387 4168 ± 0	1353 1382 ± 2	1055 1381 ± 1	1041 1383 ± 1	967 1383 ± 2	843 1385 ± 1	86 316 ± 14	82 289 ± 14
953	psl-010	411027	591157	1035 5362	1388 4168 ± 0	1374 1403 ± 9	1066 1393 ± 3	1046 1392 ± 3	978 1395 ± 3	851 1396 ± 3	100 354 ± 53	99 329 ± 29
954	psl-011	814579	606050	1029 4959	1461 8248 ± 0	1324 1324 ± 2	1032 1323 ± 8	1003 1326 ± 8	930 1324 ± 8	800 1322 ± 4	630 1680 ± 37	634 1688 ± 40
955	psl-012	1318147	622004	1014 4535	1380 4144 ± 0	1168 1111 ± 2	871 1122 ± 2	826 1104 ± 2	751 1119 ± 1	651 1117 ± 1	184 583 ± 17	196 589 ± 17
956	psl-013	1318151	622036	1011 4525	1379 4144 ± 0	1176 1119 ± 1	856 1105 ± 1	830 1110 ± 1	739 1104 ± 1	660 1126 ± 1	429 1062 ± 62	457 1098 ± 51
957	psl-014	915965	622044	1013 4525	1381 4144 ± 0	1163 1109 ± 1	888 1139 ± 1	827 1104 ± 1	749 1118 ± 1	648 1114 ± 1	462 1132 ± 105	465 1114 ± 37
958	ptakuratsatu-000	0	585434	585 1347	186 538 ± 0	953 875 ± 3	655 863 ± 48	667 928 ± 9	627 958 ± 17	619 1066 ± 26	1102 5900 ± 103	1094 5687 ± 167
959	pxl-001	110116	78231	53 168	133 512 ± 0	47 101 ± 5	30 104 ± 5	58 189 ± 12	172 408 ± 27	899 1470 ± 144	1090 5598 ± 45	1090 5590 ± 68
960	pyramid-000	372608	219883	333 804	1103 2056 ± 0	569 583 ± 2	-	-	-	-	1145 7147 ± 59	1155 7586 ± 425
961	pyramid-001	1040862	143739	789 2098	605 2048 ± 0	876 809 ± 0	611 806 ± 0	566 815 ± 4	537 837 ± 0	486 881 ± 2	1228 12265 ± 48	1230 12276 ± 51
962	qazbs-000	362015	805258	449 1034	689 2048 ± 0	1314 1307 ± 1	973 1243 ± 0	941 1248 ± 9	873 1253 ± 1	759 1270 ± 0	1071 5181 ± 62	1071 5167 ± 93
963	qazsmartvisionai-000	560570	99515	828 2371	102 512 ± 0	754 694 ± 1	530 708 ± 7	493 717 ± 5	436 711 ± 5	360 723 ± 7	334 795 ± 29	332 781 ± 34
964	qazsmartvisionai-001	1417181	102251	919 2987	192 544 ± 0	1158 1103 ± 20	880 1129 ± 3	843 1124 ± 5	779 1145 ± 7	667 1134 ± 16	332 793 ± 29	349 804 ± 32
965	qazsmartvisionai-002	1365926	102252	914 2895	190 544 ± 0	1139 1068 ± 6	875 1125 ± 4	835 1114 ± 6	709 1066 ± 1	645 1113 ± 15	338 800 ± 31	340 798 ± 34
966	qazsmartvisionai-003	1365926	102265	913 2894	187 544 ± 0	1131 1056 ± 5	833 1078 ± 12	804 1079 ± 10	724 1078 ± 10	630 1088 ± 18	349 822 ± 28	347 803 ± 32
967	qazsmartvisionai-004	1564226	102265	942 3281	189 544 ± 0	1296 1280 ± 13	1021 1310 ± 14	1013 1338 ± 10	928 1323 ± 13	649 1115 ± 2	315 758 ± 38	324 758 ± 31
968	qluevision-001	173605	205230	123 376	1457 8192 ± 0	160 229 ± 1	101 230 ± 1	80 231 ± 1	69 233 ± 1	50 239 ± 1	939 3374 ± 38	944 3365 ± 41

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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 44: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480X720	960X1440	1600X2400	3000X4500	GENUINE	IMPOSTOR
969	qnap-000	186731	15598	-	363 2048 ± 0	790 726 ± 9	280 457 ± 1	240 458 ± 0	214 464 ± 1	173 482 ± 2	259 660 ± 25	270 654 ± 29
970	qnap-001	196210	13399	88 287	902 2048 ± 0	625 614 ± 1	419 615 ± 1	397 627 ± 1	347 623 ± 1	283 634 ± 2	251 649 ± 11	265 648 ± 14
971	qnap-002	346963	33284	383 882	776 2048 ± 0	895 821 ± 1	621 824 ± 1	573 824 ± 1	531 826 ± 1	450 832 ± 1	75 293 ± 13	80 287 ± 17
972	qnap-003	245476	61427	335 807	962 2048 ± 0	343 387 ± 0	221 393 ± 0	182 393 ± 0	158 393 ± 1	126 400 ± 2	267 683 ± 20	267 651 ± 17
973	qnap-004	371274	61422	431 1002	360 2048 ± 0	315 365 ± 0	195 365 ± 1	159 367 ± 0	138 368 ± 0	106 375 ± 3	265 681 ± 7	281 682 ± 13
974	qnap-005	977818	61429	907 2828	831 2048 ± 0	1057 969 ± 5	773 988 ± 8	722 992 ± 6	664 1000 ± 7	572 1008 ± 6	238 633 ± 1	250 632 ± 13
975	qnap-006	1244083	61533	948 3351	1024 2052 ± 0	1106 1025 ± 5	818 1051 ± 6	770 1041 ± 6	704 1055 ± 8	605 1057 ± 10	275 688 ± 29	274 664 ± 10
976	quantasoft-003	370518	211354	465 1063	472 2048 ± 0	655 632 ± 2	454 634 ± 0	404 632 ± 0	351 631 ± 1	284 634 ± 0	43 201 ± 7	45 203 ± 8
977	rankone-000	0	96896	-	14 144 ± 0	31 82 ± 9	-	-	-	-	1387 39932 ± 468	1176 8722 ± 171
978	rankone-001	0	97092	-	21 208 ± 0	18 56 ± 4	-	-	-	-	1443 72754 ± 658	108 345 ± 29
979	rankone-002	0	98585	-	22 224 ± 0	25 75 ± 1	-	-	-	-	1438 69113 ± 3802	113 369 ± 26
980	rankone-003	0	93554	-	23 228 ± 0	10 37 ± 1	-	-	-	-	88 322 ± 20	94 324 ± 19
981	rankone-004	0	135727	-	12 133 ± 0	27 78 ± 1	-	-	-	-	95 340 ± 19	84 298 ± 17
982	rankone-005	0	104048	-	13 133 ± 0	24 71 ± 2	-	-	-	-	117 403 ± 10	103 337 ± 19
983	rankone-006	0	133045	-	16 165 ± 0	146 210 ± 1	-	-	-	-	127 443 ± 26	122 395 ± 22
984	rankone-007	0	136942	-	18 165 ± 0	176 245 ± 5	-	-	-	-	274 688 ± 20	215 601 ± 16
985	rankone-008	2	141234	-	17 165 ± 0	199 272 ± 3	-	-	-	-	312 750 ± 20	230 613 ± 23
986	rankone-009	0	107688	-	51 260 ± 0	108 179 ± 4	-	-	-	-	292 710 ± 32	177 552 ± 25
987	rankone-010	0	138435	-	53 261 ± 0	127 193 ± 1	-	-	-	-	72 282 ± 13	60 234 ± 16
988	rankone-011	0	179209	-	52 261 ± 0	549 567 ± 1	365 557 ± 1	335 567 ± 1	308 586 ± 1	322 682 ± 3	73 283 ± 14	51 220 ± 19
989	rankone-012	0	264182	-	54 261 ± 0	543 564 ± 3	362 554 ± 1	332 564 ± 1	310 586 ± 1	337 695 ± 1	69 273 ± 17	57 231 ± 14
990	rankone-013	0	228729	49 150	57 261 ± 0	746 690 ± 5	488 672 ± 1	489 712 ± 1	495 780 ± 1	652 1118 ± 3	102 356 ± 23	87 304 ± 23
991	rankone-014	0	243130	52 164	56 261 ± 0	763 701 ± 1	524 705 ± 0	508 732 ± 1	507 800 ± 1	647 1113 ± 1	83 306 ± 16	65 251 ± 13
992	rankone-015	0	289078	68 190	55 261 ± 0	526 547 ± 5	353 543 ± 6	347 578 ± 2	349 629 ± 7	494 894 ± 9	44 201 ± 19	32 172 ± 16
993	realnetworks-000	99044	107893	-	1346 4100 ± 0	77 144 ± 2	-	-	-	-	809 2458 ± 26	816 2467 ± 35
994	realnetworks-001	99044	107893	-	1343 4100 ± 0	78 144 ± 2	-	-	-	-	819 2500 ± 47	802 2420 ± 34
995	realnetworks-002	95328	107088	-	332 1848 ± 0	182 250 ± 2	105 242 ± 4	99 282 ± 5	147 381 ± 10	408 774 ± 15	515 1285 ± 17	509 1247 ± 42
996	realnetworks-003	95334	104498	-	333 1848 ± 0	106 177 ± 10	-	-	-	-	595 1516 ± 29	600 1522 ± 60
997	realnetworks-004	172335	913988	-	1078 2056 ± 0	270 330 ± 4	166 333 ± 3	189 402 ± 7	307 585 ± 15	852 1402 ± 51	490 1210 ± 29	499 1202 ± 17
998	realnetworks-005	172253	56755	-	1110 2056 ± 0	147 211 ± 4	84 205 ± 3	103 290 ± 6	254 515 ± 17	791 1312 ± 78	491 1213 ± 17	501 1207 ± 16
999	realnetworks-006	466225	56771	671 1596	1069 2056 ± 0	665 638 ± 4	443 630 ± 3	448 672 ± 5	431 706 ± 5	407 774 ± 5	137 469 ± 19	151 478 ± 25
1000	realnetworks-007	570797	101527	936 3145	1079 2056 ± 0	1333 1348 ± 2	1048 1358 ± 11	1029 1363 ± 10	969 1386 ± 9	928 1517 ± 6	172 559 ± 31	172 539 ± 35
1001	realnetworks-008	73346	75421	133 402	1097 2056 ± 0	228 296 ± 3	140 294 ± 3	146 353 ± 4	131 361 ± 5	175 485 ± 5	166 539 ± 31	173 543 ± 29
1002	realsense-001	363245	116558	768 1995	1046 2052 ± 0	1463 1867 ± 0	1171 1845 ± 0	1160 1854 ± 0	1120 1875 ± 1	1020 1952 ± 2	1270 16252 ± 121	1273 16305 ± 100
1003	realsense-002	406105	116558	739 1852	822 2048 ± 0	1472 1939 ± 0	1179 1923 ± 0	1169 1939 ± 0	1133 1940 ± 1	1033 2042 ± 2	1271 16369 ± 206	1274 16347 ± 167
1004	rebs-000	1375350	62664	917 2934	1289 4096 ± 0	641 623 ± 0	451 633 ± 10	393 626 ± 10	335 616 ± 1	272 617 ± 0	470 1150 ± 1	480 1150 ± 2
1005	rebs-001	2038561	62673	870 2582	1340 4100 ± 0	870 802 ± 1	614 812 ± 8	581 829 ± 1	512 807 ± 2	433 811 ± 4	472 1151 ± 8	475 1145 ± 4
1006	recognito-000	549308	247223	1044 5903	1185 2564 ± 0	1147 1086 ± 2	854 1096 ± 1	823 1102 ± 2	777 1142 ± 3	719 1219 ± 7	37 193 ± 8	40 190 ± 16
1007	recognito-001	517547	62334	979 3833	1184 2564 ± 0	1141 1072 ± 1	842 1089 ± 1	818 1095 ± 4	745 1112 ± 2	713 1205 ± 4	33 183 ± 10	35 183 ± 7
1008	regula-000	262444	29384	234 610	752 2048 ± 0	1227 1187 ± 1	876 1126 ± 1	845 1129 ± 0	767 1132 ± 1	686 1159 ± 1	150 491 ± 16	158 500 ± 22
1009	regula-001	256075	25980	426 987	936 2048 ± 0	1298 1284 ± 1	954 1220 ± 1	919 1222 ± 1	853 1226 ± 1	750 1255 ± 1	104 361 ± 10	106 342 ± 25
1010	regula-002	355812	42119	467 1066	1033 2052 ± 0	1046 962 ± 0	702 918 ± 0	650 917 ± 0	605 932 ± 0	564 1000 ± 1	486 1194 ± 14	504 1233 ± 53
1011	regula-003	495453	49195	617 1428	1109 2056 ± 0	1165 1110 ± 0	817 1048 ± 0	784 1055 ± 0	721 1077 ± 1	671 1140 ± 1	1451 77846 ± 428	1449 77865 ± 437
1012	remarkai-000	240152	868276	-	508 2048 ± 0	902 829 ± 7	-	-	-	-	364 873 ± 4	361 835 ± 35

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	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1013	remarkai-001	241857	868314	297 730	1016 2052 ± 0	905 831 ± 6	644 849 ± 18	783 1055 ± 25	823 1198 ± 34	929 1519 ± 38	497 1229 ± 20	352 805 ± 56
1014	remarkai-002	224157	808777	-	482 2048 ± 0	1035 950 ± 6	722 935 ± 4	686 953 ± 3	634 964 ± 4	624 1073 ± 4	456 1115 ± 25	441 1068 ± 54
1015	remarkai-003	280516	58559	985 3909	1351 4100 ± 0	1074 986 ± 1	778 993 ± 1	724 992 ± 1	662 999 ± 3	578 1019 ± 2	328 787 ± 20	338 793 ± 22
1016	rendip-000	0	437653	304 735	663 2048 ± 0	431 464 ± 2	283 458 ± 0	256 473 ± 0	230 483 ± 1	233 556 ± 4	181 576 ± 13	187 573 ± 11
1017	revealmedia-005	293933	202465	317 767	1336 4100 ± 0	393 428 ± 0	248 428 ± 0	214 430 ± 0	185 433 ± 0	149 442 ± 0	716 2023 ± 38	716 2009 ± 26
1018	revealmedia-006	293933	200912	307 745	1020 2052 ± 0	335 381 ± 0	210 381 ± 0	174 382 ± 0	149 384 ± 0	122 394 ± 0	229 626 ± 35	214 600 ± 2
1019	roc-016	0	674869	69 191	81 392 ± 0	541 562 ± 1	352 543 ± 2	327 553 ± 2	292 564 ± 2	278 626 ± 2	38 194 ± 8	28 161 ± 26
1020	roc-017	0	565239	91 291	82 396 ± 0	676 651 ± 2	470 660 ± 1	450 675 ± 1	392 669 ± 2	367 736 ± 2	40 197 ± 4	34 178 ± 7
1021	roc-018	0	932665	156 459	209 792 ± 0	1261 1231 ± 3	961 1227 ± 4	949 1253 ± 2	867 1243 ± 4	789 1308 ± 3	50 217 ± 10	90 316 ± 14
1022	roc-019	0	944816	116 360	86 444 ± 0	1232 1196 ± 4	931 1191 ± 5	916 1218 ± 6	855 1232 ± 4	760 1272 ± 5	82 306 ± 7	71 259 ± 8
1023	roc-020	0	1095173	121 375	87 444 ± 0	1466 1906 ± 0	1181 1936 ± 0	1174 1985 ± 0	1140 2030 ± 1	1043 2112 ± 1	206 602 ± 53	183 564 ± 22
1024	roc-021	0	1004172	122 375	88 448 ± 0	1450 1738 ± 0	1160 1781 ± 0	1159 1853 ± 0	1116 1863 ± 0	1023 1960 ± 23	175 561 ± 11	175 545 ± 11
1025	rokid-000	258612	396624	541 1218	1071 2056 ± 0	525 546 ± 3	351 542 ± 2	322 545 ± 1	259 522 ± 3	238 563 ± 4	949 3457 ± 62	958 3463 ± 77
1026	rokid-001	641223	413733	471 1071	1126 2060 ± 0	988 911 ± 2	687 901 ± 5	636 899 ± 2	585 900 ± 3	497 901 ± 3	936 3345 ± 50	942 3346 ± 149
1027	s1-001	435491	844340	-	1156 2092 ± 0	609 605 ± 24	433 623 ± 20	655 920 ± 39	1069 1567 ± 92	-	554 1428 ± 34	553 1415 ± 85
1028	s1-002	520163	95479	-	1425 6144 ± 0	1282 1257 ± 1	986 1260 ± 1	951 1261 ± 1	883 1262 ± 1	762 1273 ± 1	1036 4513 ± 25	1032 4479 ± 25
1029	s1-003	145509	95446	-	1242 4096 ± 0	1029 947 ± 0	743 959 ± 0	684 952 ± 0	621 952 ± 1	533 955 ± 1	972 3657 ± 19	977 3652 ± 16
1030	s1-004	246514	202623	-	649 2048 ± 0	885 815 ± 0	616 818 ± 1	571 818 ± 1	524 820 ± 1	441 828 ± 1	928 3245 ± 100	923 3161 ± 88
1031	s1-005	482369	95685	507 1146	934 2048 ± 0	1085 1001 ± 0	786 1002 ± 0	740 1004 ± 0	675 1008 ± 0	584 1029 ± 2	230 626 ± 74	199 589 ± 14
1032	s1-006	482372	95681	506 1146	549 2048 ± 0	1039 951 ± 0	741 956 ± 0	689 957 ± 0	631 962 ± 0	550 983 ± 0	278 696 ± 23	292 696 ± 29
1033	s1-007	482385	59657	505 1144	530 2048 ± 0	1076 988 ± 1	747 965 ± 0	699 966 ± 0	638 972 ± 0	555 991 ± 0	249 648 ± 26	271 658 ± 25
1034	s1-008	786695	102436	866 2573	574 2048 ± 0	869 800 ± 5	615 814 ± 3	570 818 ± 10	515 809 ± 8	459 841 ± 3	252 651 ± 32	264 647 ± 34
1035	s1-009	1457134	102436	1010 4435	921 2048 ± 0	1245 1213 ± 9	959 1224 ± 13	926 1230 ± 4	871 1251 ± 3	752 1258 ± 11	235 632 ± 13	239 622 ± 2
1036	saffe-001	85973	62488	55 168	301 1280 ± 0	210 281 ± 1	-	-	-	-	508 1274 ± 19	519 1277 ± 26
1037	saffe-002	260622	28285	371 855	946 2048 ± 0	888 817 ± 11	610 805 ± 15	559 809 ± 19	519 815 ± 29	434 813 ± 23	297 717 ± 7	308 714 ± 29
1038	samsungsds-000	0	307431	-	349 2048 ± 0	256 316 ± 0	161 326 ± 5	126 328 ± 4	113 327 ± 1	89 343 ± 0	1321 23722 ± 295	1323 23874 ± 305
1039	samsungsds-001	1189592	147444	987 3922	1269 4096 ± 0	1184 1140 ± 3	894 1145 ± 4	1016 1344 ± 5	957 1366 ± 5	925 1514 ± 7	1409 51559 ± 773	1409 51721 ± 1003
1040	samsungsds-002	1040732	147475	945 3321	1080 2056 ± 0	1174 1118 ± 1	916 1175 ± 12	1033 1372 ± 6	929 1324 ± 2	913 1489 ± 4	1376 35803 ± 266	1379 36181 ± 674
1041	samtech-000	109774	12807	-	1102 2056 ± 0	189 262 ± 2	-	-	-	-	1040 4550 ± 26	1038 4541 ± 28
1042	samtech-001	288082	219883	229 605	1072 2056 ± 0	227 294 ± 3	-	-	-	-	1157 7694 ± 59	1158 7678 ± 91
1043	samtech-002	192220	474779	289 714	1113 2056 ± 0	150 212 ± 0	89 212 ± 0	73 215 ± 0	65 225 ± 0	64 285 ± 1	1262 15231 ± 50	1266 15215 ± 59
1044	scanalitix-001	270730	152135	222 591	1008 2052 ± 0	1383 1418 ± 1	1091 1421 ± 1	1056 1401 ± 1	1018 1442 ± 1	922 1507 ± 3	1346 27691 ± 59	1349 27723 ± 65
1045	scanovate-001	257083	328532	-	836 2048 ± 0	558 577 ± 24	382 577 ± 21	407 632 ± 27	488 770 ± 28	855 1404 ± 32	1225 12054 ± 699	1245 13795 ± 705
1046	scanovate-002	256986	457227	369 850	696 2048 ± 0	755 696 ± 32	534 713 ± 33	512 738 ± 28	493 779 ± 32	696 1172 ± 53	902 3021 ± 38	915 3120 ± 163
1047	scanovate-003	135585	89469	334 805	899 2048 ± 0	574 585 ± 1	415 613 ± 12	356 591 ± 1	326 610 ± 2	323 684 ± 1	889 2926 ± 22	890 2925 ± 20
1048	sdc-000	256814	481583	323 784	418 2048 ± 0	990 913 ± 14	692 906 ± 9	859 1142 ± 19	1100 1774 ± 45	1137 4719 ± 222	1364 32645 ± 93	1367 32653 ± 112
1049	seamfix-001	429267	86391	101 320	373 2048 ± 0	780 718 ± 8	527 707 ± 25	680 947 ± 18	1056 1521 ± 55	1131 4179 ± 226	1101 5897 ± 118	1101 5867 ± 123
1050	securifai-001	120291	12456	-	1357 4104 ± 0	148 211 ± 1	88 211 ± 1	71 211 ± 1	57 211 ± 1	41 211 ± 1	631 1681 ± 29	638 1701 ± 25
1051	securifai-002	198824	13496	-	278 1032 ± 0	64 123 ± 0	37 123 ± 0	30 123 ± 0	26 123 ± 0	18 123 ± 0	220 615 ± 22	210 597 ± 19
1052	securifai-003	303794	13512	-	1355 4104 ± 0	529 549 ± 7	360 550 ± 7	324 549 ± 7	279 546 ± 6	225 546 ± 6	640 1714 ± 26	642 1713 ± 37
1053	securifai-004	282177	12027	250 636	924 2048 ± 0	941 869 ± 1	657 867 ± 1	612 867 ± 1	559 867 ± 1	474 865 ± 1	636 1711 ± 19	640 1705 ± 29
1054	securifai-005	252532	81777	191 526	789 2048 ± 0	1349 1377 ± 2	1046 1355 ± 1	1024 1353 ± 0	946 1357 ± 0	824 1356 ± 0	683 1873 ± 25	681 1847 ± 35
1055	securifai-006	452474	81856	320 771	1218 4096 ± 0	1151 1090 ± 2	839 1086 ± 3	813 1093 ± 1	738 1104 ± 10	634 1090 ± 2	941 3376 ± 42	949 3399 ± 40
1056	securifai-007	94589	88169	216 583	713 2048 ± 0	473 500 ± 0	323 506 ± 0	279 499 ± 1	242 501 ± 1	194 505 ± 1	581 1484 ± 11	585 1483 ± 13

Notes	
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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 46: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1057	securifai-008	111759	84658	287 711	769 2048 ± 0	666 641 ± 4	461 647 ± 3	442 663 ± 3	365 645 ± 3	301 649 ± 1	589 1498 ± 11	586 1487 ± 13
1058	securitypitch-000	271112	45153	207 559	1137 2066 ± 0	811 750 ± 0	565 749 ± 0	522 750 ± 0	469 751 ± 0	387 758 ± 0	624 1644 ± 14	630 1669 ± 20
1059	sensetime-001	531783	7203	-	1015 2052 ± 0	788 725 ± 3	-	-	-	-	824 2546 ± 102	790 2371 ± 45
1060	sensetime-002	531783	7203	-	996 2052 ± 0	865 797 ± 3	-	-	-	-	849 2713 ± 90	776 2301 ± 25
1061	sensetime-003	787853	74950	-	1009 2052 ± 0	985 908 ± 4	-	-	-	-	821 2527 ± 65	904 3004 ± 174
1062	sensetime-004	954663	30733	-	269 1028 ± 0	1392 1437 ± 15	-	-	-	-	499 1239 ± 31	490 1171 ± 22
1063	sensetime-005	765353	37673	-	272 1028 ± 0	1340 1361 ± 27	1019 1304 ± 1	998 1319 ± 1	949 1360 ± 1	926 1514 ± 1	494 1223 ± 28	493 1184 ± 29
1064	sensetime-006	765353	37673	8 33	275 1028 ± 0	1337 1352 ± 17	1022 1311 ± 1	999 1323 ± 1	947 1357 ± 1	932 1523 ± 2	482 1179 ± 28	488 1157 ± 29
1065	sensetime-007	765353	37533	1046 5950	273 1028 ± 0	1360 1386 ± 41	1034 1323 ± 2	1019 1347 ± 2	956 1366 ± 2	959 1593 ± 8	565 1460 ± 29	559 1425 ± 26
1066	sensetime-008	1176483	60067	1041 5882	271 1028 ± 0	1418 1479 ± 31	1101 1436 ± 4	1106 1482 ± 4	1058 1525 ± 5	978 1669 ± 2	514 1283 ± 51	505 1240 ± 47
1067	sensetime-010	1175952	62384	1054 6671	270 1028 ± 0	1215 1176 ± 36	893 1144 ± 2	865 1153 ± 9	802 1172 ± 6	727 1229 ± 7	98 351 ± 1	93 324 ± 5
1068	sensetime-011	1176053	63554	1045 5909	285 1060 ± 0	1428 1531 ± 1	1128 1502 ± 0	1122 1547 ± 0	1054 1508 ± 0	958 1586 ± 0	154 501 ± 1	160 503 ± 2
1069	sensetime-012	1252842	63563	1047 6047	288 1064 ± 0	1447 1711 ± 0	1149 1651 ± 0	1137 1656 ± 0	1083 1651 ± 0	990 1753 ± 0	207 603 ± 77	268 652 ± 4
1070	serendipity-000	0	346443	157 462	524 2048 ± 0	977 900 ± 8	688 901 ± 3	659 922 ± 2	610 936 ± 1	552 987 ± 2	205 602 ± 1	224 607 ± 10
1071	sertis-000	265572	68770	142 427	547 2048 ± 0	817 754 ± 0	574 759 ± 0	533 764 ± 0	477 760 ± 0	392 763 ± 0	586 1497 ± 29	613 1582 ± 38
1072	sertis-001	292591	68831	-	501 2048 ± 0	660 634 ± 1	-	-	-	-	807 2449 ± 23	849 2685 ± 41
1073	sertis-002	460790	68929	265 660	657 2048 ± 0	1224 1181 ± 1	918 1178 ± 0	890 1183 ± 0	820 1187 ± 0	721 1221 ± 0	445 1086 ± 32	446 1076 ± 31
1074	sertis-003	0	451507	763 1965	498 2048 ± 0	410 441 ± 0	260 436 ± 2	244 461 ± 2	212 463 ± 1	170 473 ± 4	381 915 ± 10	386 917 ± 8
1075	seventhsense-000	369850	1561668	-	1042 2052 ± 0	1278 1250 ± 3	985 1257 ± 1	953 1261 ± 1	880 1259 ± 1	761 1272 ± 2	667 1800 ± 35	665 1787 ± 32
1076	seventhsense-001	369850	3183365	339 812	1034 2052 ± 0	1281 1255 ± 2	1011 1294 ± 15	967 1277 ± 3	894 1275 ± 2	772 1288 ± 3	701 1936 ± 26	705 1943 ± 34
1077	seventhsense-002	452197	1567903	411 944	1002 2052 ± 0	1279 1252 ± 1	995 1271 ± 1	959 1269 ± 1	890 1272 ± 1	774 1290 ± 1	739 2131 ± 45	737 2123 ± 45
1078	seventhsense-003	603091	453847	484 1090	995 2052 ± 0	1087 1003 ± 2	760 977 ± 0	579 828 ± 1	521 817 ± 1	440 826 ± 2	597 1523 ± 25	602 1530 ± 22
1079	seventhsense-005	4000865	453775	869 2581	1345 4100 ± 0	1175 1119 ± 1	870 1120 ± 1	851 1134 ± 4	766 1131 ± 2	675 1144 ± 1	863 2787 ± 28	864 2777 ± 22
1080	seventhsense-006	1444389	451757	867 2573	1338 4100 ± 0	1203 1162 ± 1	907 1161 ± 2	868 1158 ± 2	787 1156 ± 0	700 1182 ± 2	853 2724 ± 16	854 2728 ± 16
1081	shaman-000	0	120033	179 507	1240 4096 ± 0	682 653 ± 16	-	-	-	-	112 380 ± 25	118 379 ± 31
1082	shaman-001	0	174446	185 511	1245 4096 ± 0	226 294 ± 2	-	-	-	-	240 635 ± 19	133 441 ± 25
1083	shu-001	329513	62157	-	631 2048 ± 0	621 612 ± 5	-	-	-	-	835 2619 ± 19	901 2987 ± 143
1084	shu-002	731250	148309	391 890	1237 4096 ± 0	814 751 ± 2	584 769 ± 4	658 922 ± 4	1008 1431 ± 9	1119 3489 ± 47	1482 2930763 ± 47355	1482 2929759 ± 39149
1085	shu-003	428774	146940	184 511	916 2048 ± 0	892 820 ± 6	628 828 ± 3	675 941 ± 9	917 1308 ± 15	1108 3045 ± 44	820 2506 ± 26	824 2512 ± 38
1086	shufti-001	442529	2108442	757 1941	1017 2052 ± 0	737 687 ± 0	510 694 ± 0	574 824 ± 1	842 1216 ± 4	1099 2875 ± 1	913 3133 ± 19	920 3153 ± 54
1087	siat-002	486842	7738	838 2434	1019 2052 ± 0	564 579 ± 0	-	-	-	-	322 769 ± 13	323 750 ± 13
1088	siat-004	940063	6984	-	1352 4100 ± 0	704 670 ± 0	482 671 ± 7	470 693 ± 10	464 742 ± 10	519 935 ± 17	1009 4013 ± 45	992 3782 ± 173
1089	siat-005	380936	16935	571 1299	528 2048 ± 0	361 403 ± 0	229 400 ± 0	185 401 ± 0	168 403 ± 1	139 422 ± 7	182 577 ± 13	191 580 ± 17
1090	sightropic-000	355144	205017	365 847	893 2048 ± 0	488 515 ± 0	337 524 ± 0	298 519 ± 0	255 516 ± 0	224 544 ± 0	1196 10207 ± 35	1197 10214 ± 59
1091	sightropic-001	765429	490992	1028 4914	1085 2056 ± 0	1179 1134 ± 0	1030 1321 ± 6	856 1140 ± 1	771 1136 ± 2	687 1162 ± 6	1305 21280 ± 192	1308 21306 ± 176
1092	sightropic-002	611800	467645	761 1954	1018 2052 ± 0	1028 946 ± 0	731 947 ± 0	678 946 ± 0	625 955 ± 0	546 978 ± 0	1458 89034 ± 130	1457 89056 ± 133
1093	sjtu-001	347115	148322	-	788 2048 ± 0	675 650 ± 2	-	-	-	-	1481 2243804 ± 12751	1481 2249915 ± 19380
1094	sjtu-002	446215	148306	-	909 2048 ± 0	911 835 ± 2	-	-	-	-	1480 2211198 ± 26052	1480 2210428 ± 21877
1095	sjtu-003	480795	148243	199 539	806 2048 ± 0	894 821 ± 2	617 820 ± 2	660 923 ± 3	826 1201 ± 3	1071 2373 ± 9	605 1560 ± 20	608 1560 ± 14
1096	sjtu-004	1953267	241108	894 2741	1406 4608 ± 0	1266 1236 ± 2	947 1209 ± 2	983 1294 ± 4	1062 1554 ± 5	1091 2738 ± 8	905 3057 ± 14	912 3070 ± 20
1097	sjtu-005	1953267	241151	892 2734	1410 4608 ± 0	600 601 ± 1	423 617 ± 1	499 724 ± 2	762 1127 ± 3	1098 2855 ± 11	883 2898 ± 4	886 2900 ± 4
1098	sjtu-006	2685387	241394	927 3071	1405 4608 ± 0	678 651 ± 1	486 672 ± 4	546 788 ± 9	640 979 ± 2	625 1074 ± 7	882 2897 ± 4	884 2896 ± 4
1099	sjtu-007	1126310	244296	900 2759	1409 4608 ± 0	992 914 ± 12	739 956 ± 12	886 1181 ± 37	1089 1707 ± 169	1045 2118 ± 191	887 2903 ± 14	888 2902 ± 11
1100	sjtu-008	1451928	245324	923 2997	1408 4608 ± 0	1458 1832 ± 1	1167 1836 ± 2	1155 1835 ± 0	1112 1839 ± 1	1011 1877 ± 1	1042 4621 ± 10	1041 4619 ± 25

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5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 47: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480X720	960X1440	1600X2400	3000X4500	GENUINE	IMPOSTOR
1101	sjtu-009	894841	245301	922 2997	1407 4608 ± 0	1459 1832 ± 0	1172 1845 ± 1	1158 1846 ± 1	1113 1846 ± 1	1008 1874 ± 2	978 3688 ± 17	982 3694 ± 11
1102	sktelecom-000	527132	298496	577 1321	307 1536 ± 0	1167 1110 ± 1	864 1113 ± 1	833 1114 ± 1	752 1120 ± 1	682 1155 ± 1	1341 26583 ± 128	1340 26508 ± 126
1103	smargtech-001	164944	16729	276 683	754 2048 ± 0	1443 1664 ± 6	1150 1659 ± 3	1141 1717 ± 11	1114 1847 ± 31	1072 2375 ± 105	948 3448 ± 10	961 3492 ± 17
1104	smartbiometrik-001	30875	92620	16 72	97 512 ± 0	638 620 ± 7	436 625 ± 7	419 640 ± 4	449 728 ± 6	595 1047 ± 8	282 703 ± 31	303 710 ± 40
1105	smartengines-000	1711	3025	10 50	70 288 ± 0	97 168 ± 7	63 180 ± 1	55 188 ± 3	60 217 ± 3	63 275 ± 1	39 197 ± 5	29 167 ± 11
1106	smartengines-001	7095	4601	9 47	69 288 ± 0	278 333 ± 89	234 408 ± 1	207 423 ± 1	205 460 ± 2	230 553 ± 5	26 153 ± 11	23 143 ± 13
1107	smartvist-000	5959	134084	57 169	135 512 ± 0	20 59 ± 0	14 56 ± 0	9 56 ± 0	8 58 ± 0	12 90 ± 1	558 1435 ± 31	557 1422 ± 48
1108	smartvist-001	349219	25644	393 904	913 2048 ± 0	322 370 ± 0	204 374 ± 0	168 377 ± 1	179 422 ± 0	153 445 ± 0	868 2835 ± 99	873 2836 ± 80
1109	smilart-001	67339	45737	-	107 512 ± 0	161 230 ± 11	-	-	-	-	533 1371 ± 29	539 1356 ± 57
1110	smilart-002	111826	87805	82 263	249 1024 ± 0	105 176 ± 16	-	-	-	-	1290 18784 ± 136	1294 18795 ± 151
1111	smilart-003	67339	91670	70 192	146 512 ± 0	110 180 ± 12	65 181 ± 10	117 313 ± 22	387 665 ± 49	1065 2299 ± 196	540 1395 ± 74	422 1027 ± 66
1112	sodec-000	836592	13142	915 2902	1278 4096 ± 0	1122 1041 ± 2	809 1032 ± 1	766 1035 ± 1	691 1037 ± 2	613 1061 ± 2	661 1794 ± 37	662 1775 ± 23
1113	softtech-000	656559	88676	500 1127	455 2048 ± 0	1124 1051 ± 0	819 1055 ± 0	782 1055 ± 0	701 1053 ± 0	616 1064 ± 0	609 1585 ± 29	615 1600 ± 40
1114	sparktek-000	2050067	115114	1077 9564	1322 4096 ± 0	1442 1656 ± 0	1151 1661 ± 0	1139 1671 ± 0	1084 1657 ± 0	975 1662 ± 0	152 495 ± 11	190 577 ± 32
1115	sparktek-001	2095581	116978	1078 10113	1390 4180 ± 0	1457 1806 ± 10	1164 1799 ± 0	1154 1818 ± 0	1108 1820 ± 0	1002 1818 ± 0	347 816 ± 13	351 804 ± 10
1116	sparsh-001	0	366290	1012 4525	741 2048 ± 0	725 680 ± 2	462 648 ± 1	416 638 ± 0	363 642 ± 0	351 717 ± 0	1187 9600 ± 275	1186 9601 ± 290
1117	sqisoft-001	278968	386291	280 689	1091 2056 ± 0	448 477 ± 5	1044 1348 ± 18	1023 1353 ± 26	936 1340 ± 14	849 1393 ± 28	337 797 ± 22	334 788 ± 22
1118	sqisoft-002	278039	386291	272 675	1054 2056 ± 0	436 466 ± 8	288 466 ± 2	252 468 ± 11	208 461 ± 6	169 472 ± 4	316 758 ± 11	325 760 ± 23
1119	sqisoft-003	362737	607964	338 812	1066 2056 ± 0	664 638 ± 2	490 674 ± 7	494 718 ± 17	388 665 ± 6	354 720 ± 6	356 844 ± 11	364 844 ± 23
1120	staqu-000	879661	624676	466 1064	1280 4096 ± 0	882 813 ± 25	-	-	-	-	895 2979 ± 31	906 3007 ± 75
1121	starhybrid-001	100509	289356	362 845	429 2048 ± 0	305 358 ± 82	182 355 ± 49	171 379 ± 58	165 401 ± 79	119 393 ± 67	439 1075 ± 51	447 1078 ± 53
1122	stcon-000	408095	49619	502 1137	245 1024 ± 0	633 617 ± 1	449 632 ± 4	412 634 ± 1	366 645 ± 2	319 676 ± 6	125 437 ± 10	129 434 ± 11
1123	stcon-001	322871	64463	374 860	236 1024 ± 0	1070 980 ± 1	775 991 ± 1	730 995 ± 5	668 1003 ± 6	591 1042 ± 4	91 336 ± 14	101 334 ± 14
1124	stcon-002	1255490	62667	787 2095	1252 4096 ± 0	570 583 ± 0	378 570 ± 2	342 574 ± 2	303 582 ± 1	280 630 ± 0	465 1143 ± 2	472 1142 ± 4
1125	stcon-003	2119710	62673	981 3862	1347 4100 ± 0	899 823 ± 0	619 821 ± 1	584 831 ± 3	527 823 ± 1	458 838 ± 2	473 1151 ± 5	487 1157 ± 5
1126	stcon-004	2831397	64930	977 3781	1037 2052 ± 0	1347 1375 ± 1	1057 1384 ± 4	1020 1348 ± 1	954 1365 ± 10	835 1375 ± 8	187 591 ± 4	198 589 ± 1
1127	stcon-005	3126319	50632	953 3424	1334 4100 ± 0	1372 1401 ± 2	1085 1415 ± 8	1054 1400 ± 10	1012 1435 ± 1	842 1382 ± 1	483 1181 ± 31	489 1162 ± 5
1128	stcon-006	3126319	50636	954 3426	1342 4100 ± 0	1356 1384 ± 1	1093 1426 ± 3	1068 1410 ± 1	999 1419 ± 21	860 1412 ± 7	467 1148 ± 1	479 1148 ± 1
1129	stcon-007	3126319	50636	952 3413	1349 4100 ± 0	1390 1431 ± 1	1094 1427 ± 9	1114 1503 ± 8	991 1406 ± 7	903 1473 ± 1	478 1163 ± 10	494 1185 ± 32
1130	stcon-008	2124613	56551	863 2552	1335 4100 ± 0	1431 1558 ± 0	1136 1564 ± 0	1127 1560 ± 0	1067 1565 ± 0	948 1561 ± 0	587 1497 ± 14	593 1505 ± 11
1131	stcon-009	2186212	56547	872 2599	1365 4108 ± 0	1433 1560 ± 0	1135 1561 ± 0	1128 1568 ± 2	1073 1580 ± 0	947 1555 ± 0	582 1485 ± 2	596 1515 ± 5
1132	stcon-010	2227081	56547	875 2644	1366 4108 ± 0	1430 1558 ± 0	1141 1573 ± 0	1131 1574 ± 0	1070 1568 ± 0	952 1572 ± 0	591 1504 ± 2	601 1523 ± 5
1133	stellartecs-001	1431086	33554	770 1997	1290 4096 ± 0	1336 1351 ± 0	1084 1414 ± 0	1065 1409 ± 2	1017 1441 ± 0	963 1605 ± 1	710 1987 ± 8	714 1986 ± 31
1134	stengg-000	361104	34674	295 724	540 2048 ± 0	875 809 ± 1	707 921 ± 2	737 1002 ± 1	504 794 ± 1	1048 2130 ± 5	712 2005 ± 22	717 2011 ± 34
1135	sukshi-000	94035	688738	124 382	1481 32768 ± 0	369 407 ± 11	236 413 ± 8	285 504 ± 8	412 689 ± 11	953 1574 ± 28	1192 9817 ± 50	1191 9787 ± 62
1136	sumsub-000	1120840	252	472 1074	401 2048 ± 0	1475 1947 ± 1	1182 1947 ± 0	1172 1947 ± 1	1134 1950 ± 1	1022 1958 ± 0	1439 69114 ± 201	1437 69212 ± 246
1137	sumsub-001	848331	252	523 1183	652 2048 ± 0	1470 1916 ± 0	1180 1934 ± 0	1167 1928 ± 0	-	1025 1978 ± 0	384 939 ± 5	389 926 ± 13
1138	suprema-000	246761	38507	-	725 2048 ± 0	835 771 ± 2	592 778 ± 1	608 864 ± 2	742 1109 ± 2	1051 2150 ± 4	633 1690 ± 17	633 1688 ± 13
1139	suprema-001	373423	41460	-	410 2048 ± 0	853 788 ± 1	622 826 ± 2	646 914 ± 2	780 1146 ± 7	1080 2443 ± 4	922 3212 ± 16	930 3220 ± 22
1140	suprema-002	373808	41473	706 1733	707 2048 ± 0	852 787 ± 3	633 833 ± 3	661 924 ± 4	815 1185 ± 6	1082 2479 ± 3	931 3255 ± 17	936 3253 ± 14
1141	suprema-003	498231	116054	546 1244	971 2048 ± 0	1399 1448 ± 1	1088 1417 ± 4	1072 1418 ± 3	1002 1421 ± 4	890 1451 ± 5	752 2201 ± 10	752 2198 ± 13
1142	suprema-004	1430475	116085	930 3096	1319 4096 ± 0	1417 1478 ± 2	1120 1472 ± 2	1102 1469 ± 1	1040 1476 ± 1	917 1496 ± 1	921 3201 ± 14	926 3202 ± 22
1143	suprema-005	1430177	116089	929 3093	1310 4096 ± 0	859 794 ± 1	656 863 ± 1	558 808 ± 5	572 883 ± 1	445 830 ± 5	1004 3950 ± 7	1009 3948 ± 4
1144	supremaid-001	258193	23479	200 541	841 2048 ± 0	449 479 ± 1	301 481 ± 0	261 481 ± 0	233 490 ± 0	212 522 ± 0	285 704 ± 19	269 652 ± 19

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	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1145	supremaid-002	256273	23899	106 334	685 2048 ± 0	457 483 ± 0	319 501 ± 0	268 488 ± 0	244 503 ± 0	240 565 ± 0	711 1990 ± 19	697 1923 ± 29
1146	surrey-cvssp-000	158030	70795	381 879	883 2048 ± 0	1185 1141 ± 3	903 1157 ± 3	869 1158 ± 4	793 1163 ± 3	737 1245 ± 3	1184 9557 ± 143	1187 9602 ± 186
1147	surrey-cvssp-001	900280	76392	696 1705	706 2048 ± 0	1253 1221 ± 1	970 1238 ± 2	933 1240 ± 0	865 1243 ± 0	751 1257 ± 0	1293 18970 ± 161	1296 18999 ± 176
1148	surrey-cvssp-002	903493	87776	700 1720	358 2048 ± 0	1246 1213 ± 1	948 1211 ± 1	968 1279 ± 0	901 1282 ± 0	783 1300 ± 0	1194 9906 ± 120	1194 9911 ± 167
1149	surrey-cvssp-003	1013099	109557	811 2208	457 2048 ± 0	854 789 ± 1	641 846 ± 1	604 854 ± 1	857 1234 ± 2	1114 3283 ± 3	1149 7358 ± 130	1147 7336 ± 93
1150	surrey-cvssp-004	1142444	108961	810 2203	669 2048 ± 0	808 747 ± 1	1139 1568 ± 2	1196 4350 ± 12	1193 11245 ± 7	-	1159 7794 ± 109	1159 7738 ± 118
1151	swsam-001	265887	95072	254 642	556 2048 ± 0	1132 1058 ± 34	834 1079 ± 31	812 1091 ± 35	719 1076 ± 9	643 1109 ± 45	885 2902 ± 31	883 2892 ± 32
1152	synesis-000	340264	76158	-	103 512 ± 0	58 117 ± 0	-	-	-	-	324 775 ± 206	186 568 ± 48
1153	synesis-001	340264	78195	-	127 512 ± 0	63 122 ± 0	-	-	-	-	354 828 ± 182	293 698 ± 44
1154	synesis-003	256748	155901	-	1247 4096 ± 0	36 90 ± 0	-	-	-	-	1144 7118 ± 121	1143 7162 ± 100
1155	synesis-004	270628	16252	-	538 2048 ± 0	798 735 ± 15	-	-	-	-	119 424 ± 14	128 430 ± 22
1156	synesis-005	146509	17726	-	592 2048 ± 0	149 211 ± 9	-	-	-	-	199 599 ± 23	192 581 ± 32
1157	synesis-006	731941	21817	628 1472	1360 4104 ± 0	528 549 ± 1	356 546 ± 1	326 552 ± 1	288 558 ± 2	290 639 ± 28	279 697 ± 32	289 688 ± 31
1158	synesis-007	1442961	24145	841 2443	1202 3080 ± 0	1247 1215 ± 5	991 1268 ± 30	989 1306 ± 67	919 1311 ± 58	869 1423 ± 52	270 684 ± 32	286 686 ± 25
1159	synology-000	221021	25809	152 453	955 2048 ± 0	366 407 ± 14	237 415 ± 14	471 694 ± 31	979 1396 ± 58	1133 4568 ± 211	1298 19720 ± 203	1299 19767 ± 379
1160	synology-001	256895	25804	-	927 2048 ± 0	1018 942 ± 21	-	-	-	-	546 1402 ± 29	651 1730 ± 220
1161	synology-002	256713	25943	171 488	708 2048 ± 0	969 886 ± 4	682 892 ± 3	654 920 ± 2	663 1000 ± 5	796 1317 ± 12	569 1466 ± 32	591 1496 ± 45
1162	syscake-000	50022	12609	63 184	8 128 ± 0	172 243 ± 1	106 245 ± 0	84 241 ± 1	81 269 ± 0	97 357 ± 1	34 185 ± 1	37 185 ± 1
1163	szpu-000	283522	15225	880 2660	430 2048 ± 0	122 190 ± 1	77 192 ± 1	90 259 ± 6	190 438 ± 11	712 1204 ± 32	1010 4026 ± 57	1015 4021 ± 56
1164	sztu-000	338637	15871	570 1298	668 2048 ± 0	503 531 ± 0	341 532 ± 0	309 533 ± 0	270 537 ± 0	228 548 ± 0	185 585 ± 11	202 592 ± 13
1165	sztu-001	338650	15871	572 1299	723 2048 ± 0	508 535 ± 0	346 537 ± 0	314 538 ± 0	271 540 ± 0	231 553 ± 0	200 599 ± 10	212 598 ± 10
1166	t4isb-000	234227	115237	111 345	413 2048 ± 0	1091 1006 ± 5	784 1001 ± 1	742 1006 ± 1	676 1009 ± 1	580 1022 ± 2	969 3586 ± 34	970 3534 ± 34
1167	tahakom-001	198869	724103	214 581	842 2048 ± 0	236 302 ± 2	129 277 ± 3	156 364 ± 7	139 368 ± 4	117 389 ± 4	794 2387 ± 28	793 2381 ± 22
1168	tahakom-002	609724	37452	525 1184	1250 4096 ± 0	1262 1231 ± 1	977 1250 ± 1	930 1234 ± 0	843 1218 ± 0	756 1265 ± 0	592 1513 ± 2	595 1515 ± 2
1169	taiwanmobile-001	170523	1947147	211 570	1181 2560 ± 0	562 579 ± 8	590 775 ± 16	458 683 ± 4	443 718 ± 8	413 781 ± 6	1382 38086 ± 62	1384 38109 ± 66
1170	taiwanmobile-002	170523	1969728	92 295	505 2048 ± 0	88 156 ± 0	50 162 ± 0	98 281 ± 0	282 549 ± 3	1026 1978 ± 21	1414 55746 ± 469	1412 55598 ± 410
1171	tcc-000	54057	1188952	98 318	247 1024 ± 0	326 375 ± 93	192 363 ± 76	662 924 ± 279	1163 2339 ± 806	1166 9265 ± 3411	422 1053 ± 38	440 1064 ± 23
1172	tcc-001	989049	1238700	256 644	801 2048 ± 0	371 408 ± 0	238 417 ± 0	200 415 ± 0	175 413 ± 0	155 450 ± 0	447 1090 ± 5	458 1105 ± 29
1173	tcv-001	171624	816225	117 363	615 2048 ± 0	44 98 ± 0	27 98 ± 0	22 99 ± 0	18 100 ± 0	17 122 ± 0	435 1070 ± 31	430 1045 ± 17
1174	tech5-000	1451185	470097	-	262 1024 ± 0	469 498 ± 4	-	-	-	-	320 763 ± 23	328 766 ± 28
1175	tech5-001	636346	470101	-	1311 4096 ± 0	493 522 ± 5	-	-	-	-	446 1087 ± 34	342 799 ± 44
1176	tech5-002	1150887	111744	-	299 1280 ± 0	841 780 ± 10	-	-	-	-	549 1406 ± 120	433 1048 ± 57
1177	tech5-003	1427464	118901	-	308 1536 ± 0	968 885 ± 20	-	-	-	-	471 1151 ± 37	467 1128 ± 32
1178	tech5-004	2410272	118858	-	75 321 ± 0	945 872 ± 2	869 1117 ± 164	834 1114 ± 182	770 1134 ± 179	562 999 ± 44	192 597 ± 13	201 592 ± 16
1179	tech5-005	1178769	120517	613 1426	137 512 ± 0	1291 1272 ± 109	812 1038 ± 63	776 1046 ± 39	756 1124 ± 38	821 1351 ± 44	830 2573 ± 37	830 2545 ± 32
1180	tech5-007	0	340324	877 2647	122 512 ± 0	1339 1360 ± 0	1050 1366 ± 0	1036 1376 ± 0	963 1373 ± 0	882 1438 ± 6	165 538 ± 19	162 516 ± 22
1181	tech5-008	0	365877	1039 5567	765 2048 ± 0	687 655 ± 3	435 624 ± 11	392 625 ± 11	350 631 ± 11	346 710 ± 12	1151 7371 ± 81	1148 7367 ± 99
1182	tech5-009	0	361983	847 2453	855 2048 ± 0	1114 1029 ± 25	774 988 ± 21	759 1028 ± 23	672 1006 ± 20	644 1111 ± 22	1155 7478 ± 131	1154 7464 ± 120
1183	techainer-001	198738	703687	767 1977	866 2048 ± 0	334 381 ± 9	219 391 ± 5	187 402 ± 5	156 391 ± 1	162 461 ± 5	1206 10726 ± 189	1206 10714 ± 243
1184	techainer-002	286340	1013105	567 1293	1281 4096 ± 0	391 427 ± 2	249 429 ± 4	209 424 ± 1	189 438 ± 11	195 506 ± 6	571 1472 ± 10	574 1462 ± 16
1185	techainer-003	557482	1098186	653 1550	1306 4096 ± 0	825 763 ± 0	579 765 ± 0	547 789 ± 0	510 803 ± 0	583 1025 ± 2	1002 3940 ± 17	1011 3978 ± 37
1186	techbiometricid-001	172553	886937	141 427	966 2048 ± 0	154 218 ± 8	99 226 ± 6	238 455 ± 19	685 1023 ± 36	1121 3669 ± 176	1423 59682 ± 512	1423 61123 ± 662
1187	techbiometricid-002	601291	465337	147 439	391 2048 ± 0	523 543 ± 5	349 541 ± 4	325 552 ± 5	309 586 ± 5	396 764 ± 20	1272 16395 ± 120	1275 16374 ± 124
1188	techsign-000	0	1101622	765 1969	644 2048 ± 0	316 366 ± 1	227 398 ± 1	880 1172 ± 3	1181 3065 ± 18	1176 10460 ± 65	1048 4758 ± 112	1052 4789 ± 93

Notes	
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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 49: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1189	techsign-001	0	586983	748 1881	620 2048 ± 0	837 772 ± 35	596 788 ± 23	556 802 ± 42	620 949 ± 10	857 1409 ± 26	189 592 ± 11	204 592 ± 13
1190	telexper-000	266638	74277	244 625	352 2048 ± 0	578 586 ± 1	381 576 ± 0	358 592 ± 1	361 639 ± 0	488 886 ± 0	1088 5559 ± 54	1086 5553 ± 47
1191	tevian-000	455746	16296	-	798 2048 ± 0	170 242 ± 14	-	-	-	-	122 430 ± 38	127 427 ± 23
1192	tevian-001	682319	16344	-	898 2048 ± 0	175 245 ± 13	-	-	-	-	96 350 ± 22	97 327 ± 13
1193	tevian-002	724437	16352	-	982 2049 ± 0	232 298 ± 15	-	-	-	-	108 369 ± 16	121 386 ± 25
1194	tevian-003	791725	16348	-	990 2049 ± 0	362 404 ± 15	-	-	-	-	97 350 ± 11	104 338 ± 25
1195	tevian-004	863474	16544	-	746 2048 ± 0	479 506 ± 30	-	-	-	-	139 474 ± 31	96 326 ± 20
1196	tevian-005	921043	16556	-	377 2048 ± 0	656 633 ± 21	485 672 ± 25	569 818 ± 37	747 1117 ± 64	1069 2364 ± 121	179 568 ± 22	223 607 ± 35
1197	tevian-006	692492	19339	-	823 2048 ± 0	616 611 ± 1	477 666 ± 41	437 661 ± 32	395 672 ± 37	359 723 ± 31	188 591 ± 19	188 573 ± 28
1198	tevian-007	779934	19523	698 1712	276 1032 ± 0	571 583 ± 1	385 579 ± 0	349 580 ± 0	313 588 ± 1	288 636 ± 0	1060 4894 ± 65	1058 4841 ± 83
1199	tevian-008	847177	19519	966 3610	277 1032 ± 0	966 884 ± 2	690 903 ± 1	639 903 ± 1	590 911 ± 1	526 946 ± 1	1054 4828 ± 40	1055 4811 ± 41
1200	tiger-001	171041	102515	-	1466 8632 ± 0	249 312 ± 12	-	-	-	-	1459 98339 ± 34	1459 98327 ± 32
1201	tiger-002	341638	178194	-	1056 2056 ± 0	351 393 ± 20	-	-	-	-	741 2135 ± 29	742 2137 ± 38
1202	tiger-003	426164	560292	-	1090 2056 ± 0	425 458 ± 21	-	-	-	-	717 2031 ± 35	719 2029 ± 38
1203	tiger-004	342866	253683	-	1029 2052 ± 0	866 798 ± 2	-	-	-	-	191 595 ± 5	213 598 ± 7
1204	tiger-005	342866	253734	643 1532	1012 2052 ± 0	1153 1097 ± 2	828 1065 ± 2	798 1066 ± 2	711 1067 ± 3	629 1088 ± 3	223 620 ± 19	234 615 ± 16
1205	tiger-006	421186	394688	288 714	997 2052 ± 0	1363 1392 ± 16	1082 1411 ± 10	1084 1444 ± 10	1060 1531 ± 11	1004 1848 ± 10	672 1810 ± 20	673 1801 ± 13
1206	tiger-007	516470	192057	266 661	1025 2052 ± 0	422 455 ± 1	287 464 ± 2	247 464 ± 3	221 472 ± 4	221 537 ± 1	417 1033 ± 8	427 1037 ± 19
1207	tinkoff-001	274660	389272	238 616	584 2048 ± 0	1214 1176 ± 3	919 1179 ± 3	884 1178 ± 3	800 1169 ± 2	709 1203 ± 3	1026 4361 ± 74	1025 4364 ± 75
1208	tngd-001	73167	519258	208 560	1035 2052 ± 0	103 175 ± 0	72 190 ± 0	190 404 ± 0	206 460 ± 1	141 432 ± 0	769 2274 ± 29	768 2265 ± 20
1209	tngd-002	134027	35925	358 843	953 2048 ± 0	751 694 ± 0	545 728 ± 0	1021 1348 ± 0	1177 2974 ± 1	1178 10990 ± 7	410 1016 ± 4	417 1019 ± 4
1210	tnitech-000	234032	13976	1037 5471	804 2048 ± 0	376 415 ± 1	247 427 ± 1	204 419 ± 4	181 427 ± 3	143 436 ± 2	803 2438 ± 14	804 2436 ± 19
1211	tongyi-004	947322	138901	-	1166 2167 ± 0	85 153 ± 0	-	-	-	-	1291 18916 ± 120	1295 18909 ± 126
1212	tongyi-005	1140701	138919	791 2121	1155 2089 ± 0	93 165 ± 1	-	-	-	-	1292 18924 ± 65	1302 20158 ± 103
1213	tongyitrans-001	625339	75069	-	1122 2058 ± 0	247 310 ± 20	-	-	-	-	1280 17769 ± 74	1283 17750 ± 63
1214	tongyitrans-002	625336	75081	-	1121 2058 ± 0	301 356 ± 35	-	-	-	-	1351 29816 ± 281	1284 17799 ± 127
1215	tongyitrans-003	1769974	138883	-	1404 4386 ± 0	440 468 ± 0	-	-	-	-	1427 61793 ± 154	1381 37122 ± 155
1216	toppanidgate-000	671181	711850	726 1798	1274 4096 ± 0	996 915 ± 1	699 916 ± 1	649 916 ± 1	593 917 ± 1	507 917 ± 1	1336 25262 ± 84	1334 25264 ± 97
1217	toppansecurity-001	1480690	130296	668 1595	1368 4113 ± 0	1268 1238 ± 3	950 1217 ± 0	977 1292 ± 0	849 1223 ± 0	738 1247 ± 1	908 3098 ± 28	928 3207 ± 38
1218	toshiba-000	3893310	235400	-	331 1812 ± 0	497 528 ± 3	-	-	-	-	763 2255 ± 60	761 2251 ± 53
1219	toshiba-001	3893310	235400	-	1186 2580 ± 0	629 615 ± 3	-	-	-	-	884 2900 ± 80	882 2881 ± 60
1220	toshiba-002	813606	114260	-	322 1560 ± 0	519 541 ± 0	-	-	-	-	960 3521 ± 369	809 2449 ± 124
1221	toshiba-003	984125	114264	-	323 1560 ± 0	517 540 ± 0	-	-	-	-	796 2390 ± 41	799 2407 ± 81
1222	toshiba-004	599297	27880	670 1596	1107 2056 ± 0	1398 1447 ± 3	1111 1453 ± 2	1097 1457 ± 9	1029 1457 ± 3	908 1479 ± 4	411 1020 ± 25	408 998 ± 32
1223	toshiba-005	599298	61113	-	1099 2056 ± 0	1406 1456 ± 4	1113 1454 ± 2	1099 1461 ± 2	1025 1455 ± 2	893 1459 ± 2	617 1613 ± 34	616 1607 ± 28
1224	toshiba-006	599566	44078	667 1588	1070 2056 ± 0	1419 1481 ± 16	1131 1515 ± 7	1115 1506 ± 6	1057 1521 ± 2	945 1546 ± 30	412 1022 ± 17	419 1022 ± 23
1225	toshiba-007	547545	76675	569 1297	1052 2056 ± 0	785 722 ± 1	595 784 ± 0	543 777 ± 4	483 766 ± 2	451 832 ± 15	291 709 ± 22	305 712 ± 26
1226	toshiba-008	707290	76687	724 1791	1061 2056 ± 0	1068 979 ± 1	759 976 ± 0	716 986 ± 3	670 1005 ± 2	603 1056 ± 14	260 674 ± 68	283 683 ± 71
1227	toshiba-009	651747	76688	678 1639	1093 2056 ± 0	1001 921 ± 1	721 933 ± 1	652 919 ± 2	608 935 ± 3	560 996 ± 21	298 719 ± 48	295 700 ± 31
1228	toshiba-010	1091350	76688	889 2710	1112 2056 ± 0	1364 1393 ± 2	1147 1638 ± 15	1096 1457 ± 1	1006 1426 ± 14	901 1473 ± 19	204 601 ± 25	222 604 ± 23
1229	toshiba-011	1106463	76900	891 2733	1073 2056 ± 0	1351 1379 ± 1	1080 1406 ± 1	1061 1407 ± 1	961 1370 ± 1	889 1448 ± 18	290 708 ± 56	261 641 ± 37
1230	toshiba-012	1151623	85416	816 2253	1101 2056 ± 0	1477 2020 ± 0	1189 2048 ± 0	1177 2049 ± 0	1146 2073 ± 4	1047 2125 ± 5	513 1281 ± 17	529 1315 ± 40
1231	toshiba-013	993674	205625	937 3146	1083 2056 ± 0	1453 1754 ± 0	1158 1752 ± 0	1146 1760 ± 0	1106 1797 ± 0	1018 1947 ± 5	500 1248 ± 23	532 1336 ± 19
1232	toss-000	993640	94527	998 4182	889 2048 ± 0	1111 1028 ± 0	938 1200 ± 3	939 1246 ± 7	746 1116 ± 2	623 1072 ± 0	378 909 ± 78	384 914 ± 60

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	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1233	toss-001	1006455	93901	1021 4801	468 2048 ± 0	654 631 ± 0	441 629 ± 0	391 625 ± 0	356 636 ± 0	307 661 ± 0	360 855 ± 10	387 918 ± 26
1234	touchlessid-000	92561	64467	107 341	489 2048 ± 0	246 309 ± 5	148 305 ± 2	116 312 ± 5	84 277 ± 4	93 349 ± 17	1362 31935 ± 292	1363 31958 ± 243
1235	touchlessid-001	255274	14355	198 538	643 2048 ± 0	289 344 ± 1	177 347 ± 1	199 414 ± 3	318 595 ± 10	988 1732 ± 61	671 1806 ± 35	672 1800 ± 35
1236	touchlessid-002	255586	14284	119 367	359 2048 ± 0	324 371 ± 1	205 375 ± 1	221 438 ± 3	362 640 ± 10	946 1548 ± 57	695 1915 ± 41	685 1871 ± 38
1237	touchlessid-003	271397	14226	186 512	610 2048 ± 0	372 410 ± 0	243 419 ± 0	213 429 ± 0	229 479 ± 1	324 685 ± 6	208 608 ± 17	209 596 ± 1
1238	touchlessid-004	386383	14727	253 642	874 2048 ± 0	931 854 ± 0	719 932 ± 2	938 1244 ± 6	1139 2001 ± 2	1140 5013 ± 7	598 1527 ± 28	592 1501 ± 14
1239	transmitsecurity-001	1786905	2031398	594 1363	1429 6144 ± 0	286 342 ± 13	172 344 ± 15	130 332 ± 22	102 312 ± 15	102 370 ± 19	1191 9692 ± 23	1192 9847 ± 34
1240	trueface-000	255123	796861	-	886 2048 ± 0	318 367 ± 8	-	-	-	-	144 482 ± 13	167 528 ± 20
1241	trueface-001	255123	186754	-	93 500 ± 0	347 390 ± 1	-	-	-	-	262 676 ± 26	178 558 ± 50
1242	trueface-002	253947	123116	170 486	339 2000 ± 0	308 360 ± 0	188 361 ± 0	208 423 ± 0	315 590 ± 1	-	36 192 ± 14	38 186 ± 19
1243	trueface-003	346530	24308	986 3916	621 2048 ± 0	1161 1107 ± 22	495 677 ± 3	507 732 ± 7	587 905 ± 5	-	12 103 ± 11	14 112 ± 29
1244	trueidvng-001	766071	37721	722 1781	1428 6144 ± 0	1063 975 ± 1	768 985 ± 1	719 989 ± 1	681 1016 ± 1	664 1128 ± 2	1378 37129 ± 216	1443 72067 ± 305
1245	truststamp-001	1795742	275073	641 1524	207 768 ± 0	128 194 ± 0	76 191 ± 0	53 187 ± 0	56 210 ± 1	83 338 ± 8	1170 8415 ± 66	1172 8392 ± 71
1246	truststamp-002	1060439	10841	780 2066	398 2048 ± 0	235 300 ± 0	163 330 ± 1	124 325 ± 0	117 335 ± 2	156 452 ± 0	1212 11281 ± 179	1214 11273 ± 203
1247	truststamp-003	1206119	10474	747 1875	908 2048 ± 0	412 445 ± 0	271 445 ± 0	230 449 ± 0	210 462 ± 1	222 539 ± 2	1255 14901 ± 619	1257 14813 ± 472
1248	tsight-000	1218024	107998	868 2579	232 1024 ± 0	1462 1861 ± 0	1173 1855 ± 1	1162 1874 ± 1	1117 1864 ± 1	1010 1877 ± 1	791 2373 ± 4	806 2438 ± 13
1249	ttigroup-000	466606	454674	825 2320	348 2048 ± 0	986 909 ± 2	712 925 ± 8	634 897 ± 1	588 909 ± 3	512 923 ± 2	792 2382 ± 17	796 2387 ± 23
1250	tuputech-000	11476	17185	7 33	872 2048 ± 0	62 122 ± 4	35 120 ± 1	33 142 ± 2	49 196 ± 5	131 411 ± 14	1323 23893 ± 406	1335 25279 ± 406
1251	turingtechvip-001	399874	54535	237 615	456 2048 ± 0	1357 1384 ± 4	1063 1391 ± 1	1048 1393 ± 1	994 1411 ± 1	907 1476 ± 2	646 1733 ± 19	653 1734 ± 20
1252	turingtechvip-002	167556	140995	379 877	421 2048 ± 0	1424 1493 ± 2	1020 1306 ± 1	1040 1382 ± 1	934 1337 ± 1	871 1426 ± 3	1243 13819 ± 103	1246 13807 ± 137
1253	turkcell-000	271083	133553	251 636	417 2048 ± 0	1166 1110 ± 1	849 1094 ± 0	824 1103 ± 0	760 1126 ± 1	705 1201 ± 1	877 2866 ± 23	881 2873 ± 40
1254	turkcell-001	287616	133870	263 657	355 2048 ± 0	730 682 ± 0	511 695 ± 0	463 688 ± 1	415 692 ± 0	341 702 ± 0	1024 4342 ± 71	1022 4310 ± 72
1255	turkcell-002	542178	139150	248 633	596 2048 ± 0	856 791 ± 1	367 559 ± 1	550 793 ± 3	503 793 ± 2	243 574 ± 0	929 3249 ± 65	932 3231 ± 53
1256	twface-000	661735	11782	905 2796	582 2048 ± 0	943 871 ± 1	667 873 ± 1	618 873 ± 2	564 876 ± 2	496 898 ± 1	590 1504 ± 29	594 1510 ± 34
1257	twface-001	671511	11782	908 2851	464 2048 ± 0	1006 923 ± 1	713 925 ± 2	665 926 ± 1	602 929 ± 2	522 940 ± 2	544 1400 ± 32	550 1402 ± 37
1258	ulsee-001	370519	57261	-	378 2048 ± 0	683 654 ± 2	-	-	-	-	1109 6065 ± 94	1114 6228 ± 77
1259	ultinous-000	90803	503529	-	1327 4100 ± 0	17 53 ± 0	-	-	-	-	1022 4263 ± 50	1021 4262 ± 42
1260	ultinous-001	403572	501651	-	1470 12304 ± 0	138 200 ± 0	-	-	-	-	1377 36357 ± 223	1380 36332 ± 210
1261	uluface-002	0	480761	482 1088	728 2048 ± 0	950 873 ± 42	647 855 ± 9	707 978 ± 24	888 1271 ± 40	1067 2333 ± 68	1295 19207 ± 1114	1291 18501 ± 274
1262	uluface-003	97357	529422	553 1264	1198 3072 ± 0	1050 965 ± 11	748 968 ± 10	808 1087 ± 20	972 1387 ± 36	1081 2469 ± 86	1337 26057 ± 195	1343 26865 ± 566
1263	unicc-001	24840	5123261	738 1849	569 2048 ± 0	254 316 ± 4	179 349 ± 4	590 838 ± 14	736 1102 ± 11	792 1312 ± 15	893 2950 ± 29	891 2929 ± 26
1264	unicc-002	277015	1705310	688 1673	511 2048 ± 0	612 609 ± 1	427 620 ± 1	376 612 ± 0	382 660 ± 2	393 763 ± 3	772 2289 ± 13	773 2295 ± 22
1265	unicc-003	361274	1737967	728 1806	839 2048 ± 0	1329 1343 ± 3	1036 1328 ± 1	1018 1345 ± 1	955 1366 ± 1	911 1484 ± 6	784 2339 ± 74	784 2327 ± 29
1266	unicc-004	361274	1737981	725 1793	894 2048 ± 0	1331 1348 ± 2	1040 1334 ± 1	1025 1355 ± 1	980 1397 ± 1	966 1613 ± 4	778 2306 ± 13	779 2304 ± 16
1267	unissey-001	0	1956593	406 934	1215 4096 ± 0	961 880 ± 3	683 892 ± 3	1093 1452 ± 8	1180 3048 ± 12	1171 10017 ± 387	567 1463 ± 35	577 1471 ± 34
1268	unissey-002	0	1443765	316 765	1283 4096 ± 0	799 736 ± 1	569 752 ± 1	726 994 ± 1	1005 1426 ± 1	1117 3331 ± 2	1229 12308 ± 91	1231 12302 ± 137
1269	unissey-003	0	814526	258 650	1314 4096 ± 0	779 718 ± 1	560 744 ± 0	687 956 ± 1	989 1403 ± 1	1109 3055 ± 2	612 1594 ± 20	611 1570 ± 44
1270	unissey-004	1	1065839	446 1030	1317 4096 ± 0	296 354 ± 2	196 366 ± 0	201 416 ± 0	176 417 ± 0	129 409 ± 0	1021 4261 ± 124	1039 4557 ± 480
1271	unissey-005	0	1125043	457 1050	1303 4096 ± 0	295 352 ± 0	200 369 ± 0	188 402 ± 0	162 397 ± 0	127 408 ± 0	1043 4630 ± 97	1062 4964 ± 161
1272	unissey-006	0	1315567	529 1188	1271 4096 ± 0	993 914 ± 0	718 930 ± 0	773 1043 ± 0	698 1044 ± 1	597 1048 ± 0	1115 6189 ± 111	1123 6605 ± 186
1273	unissey-007	1	1763338	516 1173	1243 4096 ± 0	1258 1224 ± 22	1000 1283 ± 17	1043 1390 ± 23	982 1401 ± 23	858 1410 ± 23	1329 24488 ± 173	1330 24524 ± 185
1274	uniubi-000	402734	49121	399 915	1077 2056 ± 0	1096 1011 ± 1	791 1009 ± 1	752 1022 ± 1	686 1025 ± 1	598 1051 ± 1	785 2342 ± 8	789 2369 ± 17
1275	universeai-000	341437	1395338	100 320	178 521 ± 0	264 324 ± 0	159 325 ± 0	149 357 ± 0	186 434 ± 0	444 829 ± 0	1245 14062 ± 60	1248 14142 ± 74
1276	upc-001	0	89914	474 1077	284 1052 ± 0	531 551 ± 15	522 703 ± 56	497 724 ± 51	470 751 ± 49	473 863 ± 33	909 3114 ± 44	924 3165 ± 97

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 51: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1277	urielsoft-000	256261	67932	1018 4700	683 2048 ± 0	133 197 ± 0	98 226 ± 0	96 281 ± 2	91 289 ± 2	214 528 ± 2	234 631 ± 10	246 630 ± 2
1278	useb-000	1002886	493465	624 1437	690 2048 ± 0	624 613 ± 2	407 605 ± 1	384 621 ± 2	373 653 ± 2	464 848 ± 3	743 2153 ± 45	756 2222 ± 115
1279	useb-001	1000960	493816	619 1429	625 2048 ± 0	709 673 ± 4	383 577 ± 4	348 579 ± 8	320 596 ± 9	401 768 ± 2	757 2230 ± 65	775 2299 ± 90
1280	useb-002	1000967	493456	612 1423	437 2048 ± 0	506 532 ± 2	593 778 ± 2	431 657 ± 4	377 657 ± 2	460 843 ± 3	783 2336 ± 47	782 2323 ± 60
1281	useb-003	1001019	493456	633 1484	642 2048 ± 0	485 512 ± 2	333 521 ± 3	307 528 ± 12	284 550 ± 2	358 723 ± 2	746 2172 ± 38	748 2175 ± 75
1282	useb-004	2086479	454237	902 2765	681 2048 ± 0	1343 1369 ± 2	1095 1428 ± 12	1076 1428 ± 5	1063 1557 ± 10	1017 1945 ± 4	777 2303 ± 54	786 2350 ± 94
1283	uxlabs-001	291127	39378	282 700	1261 4096 ± 0	353 395 ± 0	214 387 ± 0	178 388 ± 0	154 390 ± 0	123 396 ± 0	681 1863 ± 31	696 1921 ± 45
1284	uxlabs-003	1051837	37767	677 1638	1263 4096 ± 0	1362 1390 ± 1	1081 1410 ± 5	1045 1391 ± 3	987 1402 ± 6	878 1435 ± 1	625 1645 ± 10	622 1641 ± 37
1285	uzinfocom-000	271124	508385	386 885	1145 2078 ± 0	107 178 ± 0	61 180 ± 0	60 191 ± 0	54 208 ± 12	76 325 ± 0	430 1064 ± 41	439 1055 ± 26
1286	uzinfocom-001	377667	396457	485 1091	1146 2078 ± 0	399 435 ± 0	254 433 ± 0	231 450 ± 0	251 511 ± 0	249 589 ± 1	1422 59486 ± 145	1420 59551 ± 123
1287	vaidio-001	461288	112836	469 1068	848 2048 ± 0	632 616 ± 0	429 621 ± 0	386 622 ± 0	346 623 ± 0	330 690 ± 0	1415 56447 ± 44	1416 56451 ± 63
1288	vcog-002	3229434	118946	972 3666	1483 61504 ± 5	303 357 ± 25	-	-	-	-	1472 296154 ± 3077	1472 296436 ± 4183
1289	vcortex-001	105305	55240	177 502	214 896 ± 0	139 201 ± 2	81 197 ± 0	70 209 ± 3	73 244 ± 6	105 371 ± 19	70 279 ± 4	77 277 ± 4
1290	vd-000	213686	12962	-	274 1028 ± 0	74 134 ± 3	-	-	-	-	570 1467 ± 19	569 1456 ± 16
1291	vd-001	170262	44058	-	1007 2052 ± 0	257 316 ± 6	-	-	-	-	502 1258 ± 38	476 1148 ± 109
1292	vd-002	254498	34389	6 31	169 516 ± 0	735 684 ± 5	496 679 ± 4	451 676 ± 5	416 693 ± 5	383 754 ± 5	80 300 ± 14	91 319 ± 32
1293	vd-003	254505	44051	3 29	1043 2052 ± 0	748 691 ± 5	508 690 ± 5	459 683 ± 4	414 691 ± 5	357 722 ± 5	406 1003 ± 11	413 1001 ± 7
1294	veertec-001	106668	28071	271 674	1129 2060 ± 0	1452 1751 ± 0	1157 1751 ± 0	1143 1749 ± 0	1094 1747 ± 0	998 1792 ± 15	1033 4453 ± 20	1029 4449 ± 25
1295	veridas-001	196540	190018	-	647 2048 ± 0	706 671 ± 21	-	-	-	-	1095 5748 ± 20	1140 7111 ± 148
1296	veridas-002	193466	190018	-	149 512 ± 0	702 669 ± 20	-	-	-	-	647 1733 ± 81	703 1934 ± 44
1297	veridas-003	293109	141587	-	717 2048 ± 0	973 890 ± 33	-	-	-	-	1080 5484 ± 42	1146 7306 ± 410
1298	veridas-004	196585	160684	-	743 2048 ± 0	723 678 ± 22	564 749 ± 27	1103 1470 ± 52	1183 3228 ± 95	1173 10349 ± 273	1084 5516 ± 42	1151 7425 ± 130
1299	veridas-005	196585	269478	-	796 2048 ± 0	611 606 ± 12	-	-	-	-	1419 56976 ± 200	1430 66298 ± 216
1300	veridas-006	355669	896424	-	369 2048 ± 0	960 880 ± 8	678 885 ± 8	961 1271 ± 18	1159 2242 ± 38	1153 6414 ± 156	1418 56940 ± 149	1429 66077 ± 194
1301	veridas-007	355105	891492	857 2522	533 2048 ± 0	946 872 ± 9	668 875 ± 8	952 1261 ± 18	1158 2238 ± 38	1151 6374 ± 147	255 655 ± 16	273 660 ± 19
1302	veridas-008	1100495	1190915	1071 8633	422 2048 ± 0	1025 944 ± 12	728 945 ± 11	1009 1334 ± 27	1164 2382 ± 48	1155 6959 ± 172	299 723 ± 14	317 731 ± 16
1303	veridas-009	1100503	625120	1075 9163	871 2048 ± 0	584 589 ± 6	373 565 ± 5	345 576 ± 5	305 583 ± 5	314 666 ± 6	245 638 ± 5	260 638 ± 16
1304	veridas-010	1103722	623326	1073 8991	732 2048 ± 0	777 714 ± 4	502 686 ± 4	482 705 ± 6	411 687 ± 4	335 693 ± 5	247 644 ± 2	263 644 ± 4
1305	veridium-000	0	47198	26 98	1478 29399 ± 2045	29 79 ± 0	18 80 ± 0	18 89 ± 0	14 90 ± 0	14 111 ± 0	1430 64880 ± 171	1428 64697 ± 247
1306	veridium-001	0	40561	45 142	1174 2489 ± 0	13 44 ± 0	9 45 ± 0	6 48 ± 0	6 50 ± 0	7 72 ± 0	1429 63417 ± 1061	1426 63225 ± 2133
1307	veridium-002	0	52482	51 161	1172 2486 ± 0	4 25 ± 0	2 25 ± 0	3 29 ± 0	2 30 ± 0	3 51 ± 0	1389 43891 ± 177	1390 43918 ± 167
1308	veridium-003	0	25975	39 136	1175 2490 ± 0	5 27 ± 0	3 25 ± 0	2 28 ± 0	3 30 ± 0	4 53 ± 0	1372 34755 ± 140	1374 34828 ± 177
1309	veridium-004	0	30232	62 181	1173 2486 ± 0	14 46 ± 0	10 46 ± 0	7 51 ± 0	7 53 ± 0	10 87 ± 0	1347 28700 ± 167	1351 28723 ± 149
1310	verigram-000	256209	7798	794 2137	459 2048 ± 0	872 807 ± 1	618 821 ± 1	703 972 ± 2	948 1358 ± 3	1097 2848 ± 13	493 1222 ± 17	502 1219 ± 17
1311	verigram-001	282155	11773	874 2638	475 2048 ± 0	699 664 ± 2	491 675 ± 2	587 833 ± 4	829 1202 ± 7	1090 2733 ± 32	628 1664 ± 60	623 1648 ± 56
1312	verigram-002	1588249	20691	777 2057	1254 4096 ± 0	1409 1464 ± 3	1079 1406 ± 0	1108 1486 ± 1	1009 1431 ± 0	957 1586 ± 5	1473 453119 ± 13595	1473 452140 ± 9654
1313	verigram-003	1590775	20692	776 2052	1287 4096 ± 0	749 691 ± 4	493 677 ± 1	452 677 ± 1	419 694 ± 1	381 746 ± 1	1383 38843 ± 1140	1386 38665 ± 1071
1314	verigram-004	1521843	23835	815 2247	1460 8216 ± 0	1437 1575 ± 2	1138 1565 ± 1	1125 1558 ± 1	1064 1558 ± 1	964 1606 ± 1	890 2928 ± 17	889 2915 ± 11
1315	verihubs-inteligensia-000	209562	51877	143 430	387 2048 ± 0	548 567 ± 0	1134 1558 ± 8	1126 1560 ± 8	1071 1568 ± 8	968 1621 ± 8	1314 22351 ± 91	1318 22371 ± 81
1316	verihubs-inteligensia-001	216524	51916	145 437	688 2048 ± 0	544 564 ± 0	369 562 ± 0	333 566 ± 1	293 566 ± 0	260 600 ± 0	1306 21770 ± 84	1310 21735 ± 102
1317	verihubs-inteligensia-002	361105	17045	296 727	747 2048 ± 0	443 471 ± 0	282 458 ± 0	242 459 ± 0	211 463 ± 1	168 471 ± 0	1284 18131 ± 558	1286 17851 ± 197
1318	verijelas-000	254540	10322	708 1736	565 2048 ± 0	260 321 ± 0	160 325 ± 1	129 329 ± 0	119 335 ± 5	99 360 ± 0	1198 10267 ± 143	1198 10218 ± 109
1319	verismart-001	271195	15029	262 657	1139 2068 ± 0	288 343 ± 0	176 346 ± 0	138 346 ± 0	122 345 ± 0	91 346 ± 0	368 885 ± 5	390 926 ± 14
1320	via-000	124422	11151	-	919 2048 ± 0	768 707 ± 8	558 740 ± 5	640 906 ± 41	614 941 ± 40	589 1040 ± 5	394 966 ± 28	418 1021 ± 44

Notes	
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4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 52: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1321	via-001	370255	11151	695 1697	890 2048 ± 0	1049 964 ± 3	793 1011 ± 3	755 1026 ± 4	699 1045 ± 3	669 1137 ± 28	397 983 ± 31	406 989 ± 40
1322	via-004	530941	93634	593 1361	775 2048 ± 0	320 368 ± 0	223 397 ± 1	144 352 ± 1	134 364 ± 2	180 490 ± 1	62 255 ± 7	69 255 ± 8
1323	via-005	510729	93629	817 2257	698 2048 ± 0	514 538 ± 0	338 528 ± 0	304 525 ± 1	268 535 ± 2	262 601 ± 1	52 223 ± 2	53 222 ± 7
1324	via-006	1087232	93594	993 4074	345 2048 ± 0	948 872 ± 1	686 898 ± 1	673 941 ± 1	609 936 ± 2	558 993 ± 1	53 224 ± 2	55 230 ± 11
1325	viante-000	1244225	19718	644 1532	1094 2056 ± 0	1127 1054 ± 0	824 1059 ± 1	801 1075 ± 1	716 1070 ± 4	627 1087 ± 1	1124 6461 ± 44	1120 6424 ± 32
1326	viante-001	1401078	19722	733 1812	1088 2056 ± 0	1152 1094 ± 1	868 1117 ± 5	836 1119 ± 6	753 1120 ± 3	641 1103 ± 1	917 3158 ± 45	918 3143 ± 53
1327	viante-002	1247402	19718	647 1538	1104 2056 ± 0	1196 1152 ± 1	865 1113 ± 3	829 1109 ± 1	741 1107 ± 2	656 1121 ± 2	923 3213 ± 90	929 3207 ± 90
1328	viante-003	1247402	19705	645 1534	1087 2056 ± 0	1455 1768 ± 2	1162 1787 ± 0	1150 1775 ± 0	1103 1786 ± 8	997 1791 ± 0	1046 4666 ± 111	1046 4690 ± 83
1329	viante-004	1247402	19705	649 1542	1106 2056 ± 0	1454 1760 ± 0	1169 1840 ± 0	1153 1809 ± 0	1104 1786 ± 0	-	1025 4346 ± 41	1023 4357 ± 44
1330	videmo-000	139643	39470	-	843 2048 ± 0	76 142 ± 5	46 150 ± 4	38 150 ± 6	32 151 ± 4	23 155 ± 8	159 513 ± 16	165 523 ± 38
1331	videmo-001	212051	95063	97 309	784 2048 ± 0	137 199 ± 0	52 164 ± 0	43 164 ± 0	35 164 ± 0	24 165 ± 0	76 296 ± 17	81 288 ± 16
1332	videmo-002	212053	32963	105 333	930 2048 ± 0	135 199 ± 0	55 169 ± 0	46 169 ± 0	36 170 ± 0	26 170 ± 0	47 209 ± 7	47 208 ± 8
1333	videmo-003	173512	43964	150 447	131 512 ± 0	73 133 ± 0	32 111 ± 0	26 115 ± 0	23 114 ± 0	16 115 ± 0	3 75 ± 10	3 75 ± 16
1334	videonetics-001	30875	5963	11 61	129 512 ± 0	190 262 ± 3	124 273 ± 1	222 439 ± 3	523 820 ± 3	1075 2393 ± 43	475 1153 ± 38	473 1142 ± 65
1335	videonetics-002	121981	6289	30 115	1041 2052 ± 0	212 282 ± 5	141 295 ± 1	293 513 ± 4	687 1029 ± 3	1111 3151 ± 46	492 1219 ± 57	513 1262 ± 56
1336	viettelai-000	284612	132031	240 619	521 2048 ± 0	252 315 ± 0	155 317 ± 0	122 321 ± 0	111 322 ± 0	80 334 ± 1	1311 22074 ± 112	1317 22173 ± 163
1337	viettelai-001	528186	132058	530 1189	384 2048 ± 0	778 715 ± 1	-	-	-	-	1316 22740 ± 167	1456 86287 ± 1073
1338	viettelai-002	528186	132049	488 1094	557 2048 ± 0	772 712 ± 1	547 729 ± 0	498 724 ± 0	458 735 ± 0	370 738 ± 5	1339 26231 ± 154	1458 89838 ± 1111
1339	viettelai-003	528138	130904	487 1093	382 2048 ± 0	771 710 ± 1	540 720 ± 0	518 743 ± 3	496 782 ± 0	716 1207 ± 1	648 1743 ± 11	650 1728 ± 7
1340	viettelai-004	1506350	203627	932 3115	1010 2052 ± 0	1483 2910 ± 0	1196 2922 ± 1	1191 2915 ± 0	1175 2917 ± 0	1105 2944 ± 0	507 1274 ± 17	524 1290 ± 32
1341	viettelai-005	1865780	365644	928 3089	994 2052 ± 0	1467 1908 ± 0	1190 2054 ± 2	1179 2053 ± 2	1143 2051 ± 2	1038 2095 ± 4	551 1413 ± 14	558 1425 ± 20
1342	viettelai-006	1531507	365644	931 3101	1032 2052 ± 0	1468 1910 ± 0	1178 1910 ± 0	1163 1902 ± 0	1132 1925 ± 0	1021 1953 ± 1	541 1398 ± 10	552 1405 ± 13
1343	viettelhightech-000	259471	215557	139 421	602 2048 ± 0	429 461 ± 1	285 461 ± 2	246 461 ± 1	218 467 ± 2	181 494 ± 0	198 599 ± 11	200 591 ± 13
1344	viettelhightech-001	461523	147432	483 1088	712 2048 ± 0	1138 1068 ± 1	837 1084 ± 1	860 1145 ± 4	887 1270 ± 4	1032 2041 ± 2	197 598 ± 14	197 589 ± 5
1345	viettelhightech-002	461606	115515	443 1022	603 2048 ± 0	1446 1700 ± 1	1152 1665 ± 0	1142 1723 ± 1	1125 1893 ± 1	1092 2742 ± 1	326 778 ± 4	333 783 ± 8
1346	vigilantsolutions-000	352218	389974	-	1480 31540 ± 0	965 884 ± 23	-	-	-	-	1285 18201 ± 94	1240 13030 ± 83
1347	vigilantsolutions-001	344685	89145	-	312 1544 ± 0	1002 921 ± 2	-	-	-	-	246 644 ± 13	266 649 ± 16
1348	vigilantsolutions-002	344137	89162	-	1124 2060 ± 0	922 844 ± 2	-	-	-	-	263 679 ± 10	279 680 ± 7
1349	vigilantsolutions-003	343048	94301	-	316 1548 ± 0	900 824 ± 3	-	-	-	-	239 634 ± 19	259 638 ± 17
1350	vigilantsolutions-004	343048	125719	-	313 1548 ± 0	915 837 ± 6	-	-	-	-	273 686 ± 14	288 688 ± 17
1351	vigilantsolutions-005	343048	125719	-	314 1548 ± 0	914 837 ± 13	-	-	-	-	365 874 ± 22	257 637 ± 16
1352	vigilantsolutions-006	343048	125719	-	320 1548 ± 0	919 841 ± 8	-	-	-	-	385 939 ± 32	304 711 ± 37
1353	vigilantsolutions-007	255600	125715	-	318 1548 ± 0	465 493 ± 6	-	-	-	-	340 803 ± 35	345 800 ± 40
1354	vigilantsolutions-008	441835	47824	-	321 1548 ± 0	672 647 ± 2	-	-	-	-	370 889 ± 23	378 903 ± 44
1355	vigilantsolutions-009	348798	49973	-	317 1548 ± 0	628 615 ± 0	445 631 ± 0	406 632 ± 1	357 636 ± 0	304 659 ± 0	129 452 ± 11	136 451 ± 11
1356	vigilantsolutions-010	348798	49973	355 841	315 1548 ± 0	631 615 ± 0	444 631 ± 0	405 632 ± 0	355 636 ± 0	305 659 ± 0	148 490 ± 13	154 488 ± 11
1357	vigilantsolutions-011	255661	49973	221 591	319 1548 ± 0	360 402 ± 0	241 418 ± 0	202 418 ± 0	177 422 ± 0	154 445 ± 0	94 339 ± 20	111 366 ± 37
1358	vinai-000	402391	866522	450 1034	479 2048 ± 0	1157 1099 ± 1	851 1095 ± 1	815 1093 ± 1	735 1099 ± 1	661 1126 ± 1	896 2996 ± 20	903 2993 ± 26
1359	vinbigdata-001	271405	44746	223 591	539 2048 ± 0	1371 1400 ± 5	1065 1393 ± 2	1044 1391 ± 2	976 1393 ± 1	854 1404 ± 1	528 1351 ± 50	528 1310 ± 38
1360	vinbigdata-002	256322	138864	232 607	810 2048 ± 0	551 569 ± 2	379 572 ± 1	338 571 ± 1	296 572 ± 1	256 596 ± 1	747 2175 ± 44	744 2160 ± 53
1361	vinbigdata-003	259968	183557	109 342	525 2048 ± 0	926 849 ± 6	1023 1312 ± 1	603 853 ± 2	927 1321 ± 2	449 832 ± 2	438 1074 ± 37	453 1086 ± 40
1362	vinbigdata-004	1570839	333050	941 3247	1238 4096 ± 0	1204 1162 ± 3	900 1155 ± 14	858 1141 ± 2	801 1171 ± 5	672 1140 ± 2	519 1299 ± 17	511 1252 ± 28
1363	vinbigdata-005	1106052	333050	820 2294	833 2048 ± 0	1242 1210 ± 1	969 1237 ± 1	944 1251 ± 1	858 1234 ± 1	767 1282 ± 7	61 251 ± 17	61 234 ± 16
1364	vinbigdata-006	1106052	333624	821 2294	404 2048 ± 0	1244 1210 ± 2	944 1205 ± 5	881 1175 ± 2	807 1178 ± 3	707 1202 ± 16	57 246 ± 16	59 232 ± 16

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3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 53: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1365	vinbigdata-007	1106071	335755	822 2295	655 2048 ± 0	1237 1201 ± 10	911 1166 ± 3	946 1252 ± 6	876 1256 ± 15	748 1254 ± 20	59 249 ± 10	63 238 ± 16
1366	vion-000	228219	7533	176 498	1011 2052 ± 0	277 333 ± 1	-	-	-	-	1385 39839 ± 3561	1342 26830 ± 2241
1367	visage-000	49218	70150	17 72	98 512 ± 0	6 27 ± 0	4 27 ± 0	4 31 ± 0	5 38 ± 0	6 63 ± 0	755 2220 ± 14	754 2218 ± 14
1368	visica-001	1	59680	14 68	240 1024 ± 0	79 144 ± 1	44 143 ± 1	40 151 ± 1	45 188 ± 2	111 383 ± 11	418 1035 ± 41	416 1017 ± 56
1369	visica-002	1	63831	34 126	438 2048 ± 0	67 124 ± 1	38 125 ± 1	41 156 ± 2	82 271 ± 4	492 892 ± 29	1049 4759 ± 31	1043 4630 ± 26
1370	visica-003	1	252112	161 470	599 2048 ± 0	231 297 ± 2	145 297 ± 1	135 341 ± 2	220 471 ± 5	779 1297 ± 32	1103 5939 ± 47	1105 5987 ± 81
1371	visionbox-000	176501	190645	-	755 2048 ± 0	239 304 ± 7	-	-	-	-	626 1648 ± 57	496 1192 ± 42
1372	visionbox-001	256869	190645	-	954 2048 ± 0	1071 983 ± 7	848 1093 ± 46	1027 1360 ± 68	1155 2181 ± 105	1148 5955 ± 281	477 1161 ± 22	483 1154 ± 20
1373	visionbox-002	259063	135281	235 613	1123 2059 ± 0	456 482 ± 1	302 482 ± 0	263 484 ± 1	237 492 ± 1	206 517 ± 3	709 1969 ± 44	701 1931 ± 42
1374	visionbox-003	259542	156891	552 1263	1140 2068 ± 0	722 678 ± 2	499 682 ± 1	456 682 ± 2	421 695 ± 2	373 739 ± 5	623 1643 ± 56	625 1649 ± 66
1375	visionbox-004	309730	154797	611 1422	1369 4116 ± 0	330 378 ± 1	199 369 ± 0	179 389 ± 5	166 402 ± 2	146 437 ± 4	727 2071 ± 31	732 2076 ± 41
1376	visionbox-005	309402	197528	718 1771	1371 4116 ± 0	293 350 ± 1	171 343 ± 1	142 351 ± 2	136 367 ± 5	158 454 ± 19	729 2074 ± 17	730 2067 ± 29
1377	visionbox-006	479723	170096	921 2995	1362 4107 ± 0	1011 932 ± 2	755 973 ± 4	729 994 ± 6	667 1003 ± 11	655 1120 ± 20	719 2038 ± 74	718 2022 ± 31
1378	visionbox-007	479723	364416	848 2456	1363 4107 ± 0	635 619 ± 1	421 616 ± 0	400 629 ± 2	390 666 ± 7	421 794 ± 32	961 3524 ± 29	968 3527 ± 40
1379	visionbox-008	479723	449470	851 2476	1364 4107 ± 0	636 619 ± 1	425 619 ± 1	399 628 ± 2	384 663 ± 7	425 799 ± 36	980 3704 ± 35	980 3692 ± 40
1380	visionlabs-001	343661	127922	-	20 204 ± 0	1021 943 ± 8	-	-	-	-	539 1395 ± 45	478 1148 ± 53
1381	visionlabs-002	591936	46855	-	62 264 ± 0	193 265 ± 8	-	-	-	-	336 796 ± 35	341 799 ± 31
1382	visionlabs-003	639849	15246	-	50 260 ± 0	162 231 ± 5	-	-	-	-	126 439 ± 23	138 454 ± 20
1383	visionlabs-004	313016	90858	-	28 256 ± 0	240 304 ± 0	-	-	-	-	352 827 ± 26	368 865 ± 32
1384	visionlabs-005	369602	16523	-	126 512 ± 0	250 313 ± 0	-	-	-	-	358 848 ± 26	375 889 ± 37
1385	visionlabs-006	353044	16655	-	104 512 ± 0	195 270 ± 0	-	-	-	-	280 698 ± 19	320 734 ± 28
1386	visionlabs-007	357204	16643	-	115 512 ± 0	198 272 ± 0	-	-	-	-	393 965 ± 41	401 972 ± 31
1387	visionlabs-008	706099	19705	-	105 512 ± 0	438 467 ± 1	-	-	-	-	389 955 ± 23	395 962 ± 25
1388	visionlabs-009	706099	19862	-	160 513 ± 0	489 515 ± 41	292 472 ± 1	257 474 ± 1	225 476 ± 1	211 521 ± 1	391 957 ± 28	399 965 ± 32
1389	visionlabs-010	1067280	19357	385 883	159 513 ± 0	793 730 ± 0	539 717 ± 1	486 709 ± 0	438 713 ± 1	374 739 ± 0	202 600 ± 41	242 626 ± 35
1390	visionlabs-011	1067280	19353	403 923	157 513 ± 0	795 731 ± 1	538 717 ± 1	487 710 ± 1	439 714 ± 1	377 741 ± 1	170 556 ± 26	180 559 ± 25
1391	visteam-000	32729	17740	-	311 1536 ± 0	41 96 ± 7	29 102 ± 8	151 358 ± 19	684 1022 ± 50	1125 3987 ± 211	1121 6361 ± 87	1125 6668 ± 277
1392	visteam-001	186440	30878	-	1318 4096 ± 0	940 869 ± 7	665 872 ± 6	840 1121 ± 15	1090 1719 ± 38	1132 4375 ± 157	1141 7054 ± 108	1138 7025 ± 109
1393	visteam-002	186440	30888	-	1270 4096 ± 0	903 829 ± 5	632 832 ± 6	592 839 ± 7	550 853 ± 6	576 1013 ± 14	1137 6952 ± 118	1134 6970 ± 120
1394	visteam-003	215359	33730	173 493	1222 4096 ± 0	1277 1249 ± 4	978 1251 ± 4	957 1266 ± 5	889 1272 ± 5	830 1370 ± 9	1132 6816 ± 111	1130 6816 ± 105
1395	visteam-004	61594	35369	54 168	351 2048 ± 0	238 303 ± 5	149 313 ± 6	94 278 ± 4	89 288 ± 4	108 377 ± 7	1001 3936 ± 72	1008 3938 ± 79
1396	visteam-005	288140	35427	112 347	446 2048 ± 0	1173 1117 ± 6	858 1106 ± 6	790 1060 ± 4	717 1071 ± 4	683 1156 ± 8	999 3932 ± 97	1004 3932 ± 71
1397	visteam-006	339353	47921	174 493	716 2048 ± 0	1094 1009 ± 1	794 1015 ± 1	765 1035 ± 2	680 1014 ± 1	596 1047 ± 2	848 2711 ± 32	851 2704 ± 41
1398	visteam-007	331673	264832	169 485	881 2048 ± 0	613 611 ± 1	438 625 ± 1	396 626 ± 1	334 615 ± 1	297 647 ± 0	847 2704 ± 38	856 2732 ± 54
1399	visteam-008	582800	264832	298 731	827 2048 ± 0	1219 1179 ± 1	914 1173 ± 1	912 1210 ± 1	821 1190 ± 3	717 1208 ± 1	838 2632 ± 28	839 2647 ± 42
1400	visteam-009	582793	48090	300 731	527 2048 ± 0	1212 1174 ± 1	926 1184 ± 1	915 1218 ± 2	844 1219 ± 2	747 1253 ± 2	840 2668 ± 34	844 2671 ± 38
1401	visteam-010	582793	48090	302 731	375 2048 ± 0	1213 1174 ± 1	942 1203 ± 2	914 1215 ± 1	845 1221 ± 1	745 1252 ± 1	842 2675 ± 51	848 2681 ± 57
1402	vivotek-001	15240	247926	36 128	543 2048 ± 0	46 100 ± 0	28 101 ± 0	25 106 ± 0	29 134 ± 0	49 238 ± 1	1340 26504 ± 57	1341 26520 ± 69
1403	vivotek-002	12769	247949	33 124	850 2048 ± 0	38 93 ± 0	23 93 ± 0	21 98 ± 0	16 95 ± 0	22 150 ± 0	793 2384 ± 19	795 2385 ± 20
1404	vixvizion-000	183453	599571	-	494 2048 ± 0	388 425 ± 24	-	-	-	-	466 1145 ± 25	645 1718 ± 63
1405	vixvizion-001	282680	256875	-	627 2048 ± 0	871 807 ± 29	-	-	-	-	420 1045 ± 22	391 934 ± 45
1406	vixvizion-002	227766	318409	-	442 2048 ± 0	851 786 ± 1	580 766 ± 2	627 885 ± 3	1007 1430 ± 3	1126 4080 ± 10	261 676 ± 16	243 630 ± 20
1407	vixvizion-003	254860	378019	-	634 2048 ± 0	938 865 ± 2	723 938 ± 1	1132 1577 ± 1	1179 3002 ± 3	1169 9521 ± 16	310 738 ± 26	284 683 ± 25
1408	vixvizion-004	254405	380049	-	491 2048 ± 0	642 624 ± 1	393 587 ± 10	394 626 ± 3	317 592 ± 3	350 717 ± 6	317 760 ± 22	297 703 ± 28

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Table 54: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1409	vixvizion-005	38886	534579	322 776	650 2048 ± 0	398 433 ± 4	209 381 ± 3	175 383 ± 3	141 373 ± 1	130 411 ± 1	305 731 ± 63	249 632 ± 32
1410	vixvizion-006	594053	396294	591 1359	875 2048 ± 0	956 876 ± 9	627 828 ± 3	568 817 ± 1	529 825 ± 2	479 871 ± 1	203 600 ± 23	227 611 ± 25
1411	vixvizion-007	594053	470119	623 1435	571 2048 ± 0	967 885 ± 35	626 828 ± 1	567 816 ± 1	530 825 ± 1	478 870 ± 1	201 600 ± 28	216 602 ± 34
1412	vnis-000	285395	1404886	203 546	518 2048 ± 0	405 437 ± 1	268 443 ± 1	239 456 ± 2	202 458 ± 2	172 477 ± 2	536 1386 ± 10	548 1382 ± 16
1413	vnpay-000	258919	1404844	178 504	598 2048 ± 0	168 239 ± 1	108 248 ± 0	89 252 ± 0	74 249 ± 0	58 260 ± 0	529 1353 ± 11	537 1353 ± 23
1414	vnpay-001	1044099	3060240	407 937	545 2048 ± 0	1200 1156 ± 1	981 1255 ± 1	945 1252 ± 2	882 1261 ± 2	771 1288 ± 10	552 1421 ± 32	556 1421 ± 26
1415	vnpt-001	266499	535529	-	380 2048 ± 0	472 499 ± 2	314 499 ± 2	271 494 ± 3	243 502 ± 3	199 512 ± 2	720 2049 ± 29	102 337 ± 121
1416	vnpt-002	271649	3203296	-	523 2048 ± 0	802 739 ± 2	548 731 ± 2	516 740 ± 1	463 742 ± 2	394 763 ± 2	321 766 ± 13	326 762 ± 13
1417	vnpt-003	369956	297799	-	1226 4096 ± 0	1319 1315 ± 4	1025 1315 ± 4	997 1318 ± 2	943 1350 ± 3	873 1428 ± 3	1152 7397 ± 31	1149 7384 ± 29
1418	vnpt-004	370110	240841	428 992	914 2048 ± 0	1269 1238 ± 1	972 1241 ± 1	937 1242 ± 2	914 1307 ± 2	920 1505 ± 2	1013 4047 ± 48	1013 4008 ± 108
1419	vnpt-005	560630	240888	510 1157	550 2048 ± 0	1375 1403 ± 0	1075 1404 ± 6	1057 1403 ± 6	1027 1456 ± 0	969 1630 ± 10	966 3562 ± 23	972 3554 ± 29
1420	vnpt-006	1114807	360198	814 2245	733 2048 ± 0	1293 1277 ± 2	1061 1390 ± 19	1067 1410 ± 14	1045 1481 ± 30	983 1703 ± 69	942 3400 ± 20	948 3393 ± 22
1421	vnpt-007	2113252	360190	968 3622	400 2048 ± 0	1189 1143 ± 1	902 1157 ± 4	877 1170 ± 2	811 1183 ± 3	720 1219 ± 3	953 3486 ± 41	959 3485 ± 35
1422	vnpt-008	1114270	360190	970 3641	436 2048 ± 0	1207 1165 ± 1	927 1186 ± 2	883 1177 ± 9	825 1201 ± 2	736 1244 ± 2	956 3506 ± 16	965 3510 ± 23
1423	vnpt-009	1114270	360190	969 3641	869 2048 ± 0	1208 1166 ± 1	935 1197 ± 6	897 1193 ± 6	835 1209 ± 3	742 1251 ± 4	963 3539 ± 17	971 3551 ± 31
1424	vocalize-001	0	38841	25 96	96 512 ± 0	19 57 ± 0	13 55 ± 0	8 56 ± 0	9 60 ± 0	5 62 ± 0	542 1400 ± 59	542 1365 ± 26
1425	vocord-001	616989	109135	-	1434 6194 ± 0	981 908 ± 16	-	-	-	-	1479 1094730 ± 64282	1479 1107193 ± 66523
1426	vocord-002	918292	648259	-	304 1330 ± 0	843 782 ± 36	-	-	-	-	1456 83063 ± 517	1454 83072 ± 714
1427	vocord-003	900909	1140643	-	210 817 ± 0	229 296 ± 3	-	-	-	-	1397 46683 ± 378	1378 36007 ± 292
1428	vocord-004	1754886	129419	-	213 896 ± 0	637 619 ± 1	-	-	-	-	686 1885 ± 66	690 1889 ± 71
1429	vocord-005	1060445	189974	-	205 768 ± 0	836 771 ± 1	-	-	-	-	654 1768 ± 48	641 1713 ± 77
1430	vocord-006	559457	189913	-	206 768 ± 0	971 886 ± 1	-	-	-	-	715 2020 ± 72	711 1969 ± 62
1431	vocord-007	587489	344995	-	328 1664 ± 0	842 780 ± 2	-	-	-	-	832 2593 ± 83	825 2526 ± 59
1432	vocord-008	603867	345047	-	1188 2688 ± 0	1047 962 ± 2	757 976 ± 2	793 1061 ± 3	860 1236 ± 23	1005 1851 ± 9	899 3015 ± 50	902 2988 ± 62
1433	vocord-009	1380132	201560	1007 4372	334 1920 ± 0	1413 1472 ± 2	1118 1472 ± 1	1123 1549 ± 1	1085 1667 ± 2	1036 2064 ± 2	723 2052 ± 50	727 2056 ± 39
1434	vocord-010	902552	206873	982 3866	292 1088 ± 0	1407 1459 ± 2	1116 1459 ± 1	1100 1463 ± 2	1046 1484 ± 1	940 1535 ± 3	854 2724 ± 31	842 2653 ± 45
1435	vogic-000	237025	23230	164 477	1361 4104 ± 0	468 497 ± 0	315 499 ± 0	280 499 ± 0	241 499 ± 1	185 499 ± 0	584 1490 ± 1	590 1495 ± 2
1436	vsoft-001	10202	61965	59 175	341 2048 ± 0	66 123 ± 8	40 131 ± 7	251 468 ± 19	1003 1422 ± 48	1146 5622 ± 229	827 2553 ± 48	831 2559 ± 44
1437	vsoft-002	10202	62518	56 169	411 2048 ± 0	91 160 ± 10	51 164 ± 11	313 537 ± 21	1074 1581 ± 53	1149 6279 ± 228	1011 4030 ± 260	999 3902 ± 137
1438	vsoft-003	10202	62894	78 258	1118 2057 ± 0	124 191 ± 14	74 191 ± 14	398 628 ± 27	1126 1895 ± 66	1159 7545 ± 376	1274 16609 ± 97	1278 16649 ± 154
1439	vsoft-004	488414	110735	520 1180	1117 2057 ± 0	384 420 ± 13	240 418 ± 15	605 857 ± 28	1152 2135 ± 73	1161 8068 ± 378	1297 19601 ± 136	1300 19776 ± 152
1440	vts-000	256589	169760	694 1697	868 2048 ± 0	460 486 ± 1	300 481 ± 0	264 484 ± 0	231 485 ± 1	205 517 ± 0	1463 124209 ± 352	1463 123652 ± 358
1441	vts-001	293000	475743	241 622	903 2048 ± 0	717 676 ± 1	501 683 ± 6	462 687 ± 3	420 695 ± 2	343 709 ± 2	1190 9620 ± 44	1188 9618 ± 54
1442	vts-002	361977	904603	354 840	828 2048 ± 0	379 416 ± 5	257 434 ± 3	215 432 ± 3	195 449 ± 3	235 556 ± 4	1154 7447 ± 34	1152 7450 ± 54
1443	wicket-000	826392	641802	782 2071	730 2048 ± 0	1384 1419 ± 2	1096 1429 ± 3	1085 1444 ± 4	1032 1460 ± 3	941 1537 ± 6	1425 60976 ± 232	1422 61096 ± 323
1444	winsense-000	270819	32034	-	302 1280 ± 0	214 283 ± 1	-	-	-	-	604 1551 ± 31	604 1532 ± 42
1445	winsense-001	264428	32035	402 922	300 1280 ± 0	830 766 ± 7	823 1058 ± 47	713 983 ± 97	703 1053 ± 119	797 1320 ± 84	621 1631 ± 28	709 1964 ± 171
1446	winsense-002	281379	25780	742 1855	674 2048 ± 0	466 494 ± 2	313 498 ± 1	297 519 ± 1	269 537 ± 1	285 634 ± 1	632 1683 ± 8	632 1685 ± 7
1447	wiseai-001	189467	60781	75 245	761 2048 ± 0	169 240 ± 0	111 251 ± 0	127 328 ± 1	114 327 ± 0	79 332 ± 0	873 2850 ± 29	877 2852 ± 31
1448	wuhantianyu-001	465118	66457	480 1086	440 2048 ± 0	667 642 ± 1	457 642 ± 1	424 644 ± 0	372 652 ± 0	338 697 ± 0	1180 9502 ± 151	1195 9920 ± 253
1449	x-laboratory-000	520020	197310	642 1524	1049 2056 ± 0	874 808 ± 7	685 897 ± 113	641 907 ± 103	573 886 ± 103	317 673 ± 39	300 725 ± 19	322 749 ± 34
1450	x-laboratory-001	625140	398792	737 1844	1062 2056 ± 0	577 586 ± 2	401 596 ± 5	367 603 ± 6	342 620 ± 7	420 793 ± 14	346 813 ± 28	369 872 ± 32
1451	xforwardai-000	242457	175556	-	672 2048 ± 0	820 757 ± 6	-	-	-	-	485 1185 ± 44	486 1157 ± 44
1452	xforwardai-001	340100	51163	803 2176	969 2048 ± 0	1222 1180 ± 2	925 1182 ± 1	898 1194 ± 1	819 1186 ± 2	708 1203 ± 1	327 779 ± 17	339 797 ± 13

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5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 55: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1453	xforwardai-002	707715	51163	772 2003	1219 4096 ± 0	1024 944 ± 1	727 942 ± 1	676 943 ± 4	607 935 ± 1	538 967 ± 1	548 1406 ± 8	551 1405 ± 13
1454	xm-000	578041	148920	277 684	1013 2052 ± 0	959 878 ± 2	675 882 ± 1	718 988 ± 2	878 1258 ± 3	1079 2434 ± 7	622 1634 ± 17	619 1632 ± 20
1455	yisheng-000	122704	41760	-	660 2048 ± 0	49 103 ± 1	-	-	-	-	330 790 ± 23	335 789 ± 20
1456	yisheng-001	120112	41760	-	948 2048 ± 0	48 103 ± 1	-	-	-	-	375 906 ± 31	381 905 ± 25
1457	yisheng-002	604097	42014	-	305 1348 ± 0	274 332 ± 1	-	-	-	-	217 613 ± 40	195 588 ± 26
1458	yisheng-004	486351	38653	560 1279	1213 3704 ± 0	331 378 ± 12	-	-	-	-	277 693 ± 137	166 526 ± 34
1459	yitu-000	2211068	76146	-	1374 4130 ± 0	707 672 ± 2	-	-	-	-	1374 35352 ± 114	1383 37848 ± 1773
1460	yitu-001	2069440	138901	-	1382 4146 ± 0	813 751 ± 0	-	-	-	-	1380 37810 ± 127	1377 35486 ± 203
1461	yitu-002	1726673	138901	-	1383 4147 ± 0	750 691 ± 0	-	-	-	-	1375 35435 ± 111	1375 35325 ± 131
1462	yitu-003	1525719	138919	975 3737	1153 2082 ± 0	935 860 ± 0	-	-	-	-	1286 18305 ± 71	1289 18286 ± 62
1463	yoonik-000	290414	206059	-	677 2048 ± 0	1016 941 ± 3	746 965 ± 13	698 964 ± 10	633 964 ± 9	543 974 ± 23	457 1116 ± 34	464 1113 ± 54
1464	yoonik-001	346629	265353	-	926 2048 ± 0	1254 1223 ± 3	971 1238 ± 1	932 1238 ± 1	864 1240 ± 1	731 1240 ± 1	287 706 ± 29	291 690 ± 26
1465	yoonik-002	453720	265415	893 2737	783 2048 ± 0	1190 1145 ± 4	872 1123 ± 2	844 1124 ± 2	758 1125 ± 2	662 1126 ± 3	319 761 ± 32	321 736 ± 32
1466	yoonik-003	346691	265415	801 2161	920 2048 ± 0	1077 991 ± 3	764 980 ± 1	714 984 ± 4	642 982 ± 1	549 983 ± 1	268 684 ± 45	278 678 ± 41
1467	yoonik-004	469597	88673	328 792	795 2048 ± 0	1250 1219 ± 1	1001 1284 ± 1	950 1255 ± 1	885 1266 ± 0	743 1251 ± 1	419 1039 ± 42	424 1030 ± 40
1468	yoti-001	267246	470114	499 1125	495 2048 ± 0	1270 1238 ± 4	1027 1316 ± 17	1195 4099 ± 13	1186 4160 ± 45	1138 4774 ± 95	1200 10352 ± 247	1202 10364 ± 246
1469	yoti-002	164043	470114	479 1086	639 2048 ± 0	607 604 ± 1	608 802 ± 3	561 812 ± 4	611 938 ± 15	831 1373 ± 4	1203 10662 ± 805	1201 10320 ± 265
1470	ytu-000	1477360	44032	852 2484	963 2048 ± 0	499 530 ± 0	343 533 ± 0	421 640 ± 0	557 861 ± 2	1019 1949 ± 8	1360 31797 ± 131	1361 31794 ± 133
1471	yuan-000	370472	331437	-	742 2048 ± 0	1169 1112 ± 6	-	-	-	-	862 2777 ± 44	893 2937 ± 87
1472	yuan-001	370473	189558	-	635 2048 ± 0	1354 1383 ± 3	1068 1394 ± 3	1063 1408 ± 3	1033 1461 ± 4	967 1615 ± 4	787 2344 ± 25	783 2325 ± 42
1473	yuan-002	370472	165662	-	484 2048 ± 0	1386 1420 ± 3	1097 1429 ± 4	1117 1511 ± 4	1086 1695 ± 4	1076 2408 ± 5	776 2297 ± 23	780 2310 ± 31
1474	yuan-003	370419	147783	-	611 2048 ± 0	1376 1405 ± 2	1083 1413 ± 3	1088 1446 ± 3	1061 1547 ± 5	1012 1878 ± 5	781 2320 ± 32	772 2287 ± 34
1475	yuan-004	428665	50011	590 1356	1285 4096 ± 0	547 567 ± 0	377 569 ± 0	340 573 ± 0	300 579 ± 0	266 607 ± 0	1097 5816 ± 35	1098 5800 ± 31
1476	yuan-005	258312	145564	357 843	454 2048 ± 0	336 381 ± 0	212 386 ± 0	177 387 ± 2	153 390 ± 4	138 421 ± 3	476 1156 ± 8	497 1196 ± 26
1477	yuan-006	1622733	145572	1006 4371	1315 4096 ± 0	1294 1280 ± 2	992 1268 ± 1	964 1273 ± 0	891 1272 ± 2	788 1306 ± 3	753 2202 ± 19	750 2190 ± 20
1478	yuan-007	258312	145568	674 1614	613 2048 ± 0	307 359 ± 0	190 361 ± 1	172 379 ± 4	146 380 ± 0	134 417 ± 0	295 715 ± 4	315 726 ± 16
1479	yuan-008	620369	167833	675 1626	1256 4096 ± 0	1209 1169 ± 0	571 752 ± 0	524 752 ± 3	490 775 ± 4	414 782 ± 0	531 1362 ± 16	545 1377 ± 19
1480	yuan-009	1161457	167833	855 2507	1296 4096 ± 0	1377 1408 ± 2	1089 1418 ± 2	1104 1471 ± 1	971 1387 ± 8	863 1414 ± 12	522 1326 ± 11	533 1336 ± 29
1481	yuan-010	1054332	167833	854 2499	1295 4096 ± 0	1292 1276 ± 1	1078 1406 ± 8	980 1294 ± 1	916 1308 ± 2	837 1375 ± 2	526 1331 ± 22	535 1349 ± 47
1482	yuanli-001	136632	47927	67 190	260 1024 ± 0	75 142 ± 0	43 143 ± 0	36 145 ± 0	31 150 ± 0	40 210 ± 1	730 2080 ± 16	733 2083 ± 22

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Table 56: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
1	2060io-001	-		-		0.9149	1351	0.9720	1347	0.9773	1306	0.9768	1153	0.8672	1184	0.9993	798		
2	2060io-002	-		-		0.8904	1349	0.9587	1345	0.9020	1292	1.0000	1304	1.0000	1324	1.0000	895		
3	20face-000	0.1268	936	0.1828	927	0.1748	1246	0.2768	1247	0.1765	1169	0.1864	852	0.0927	934	0.5943	588		
4	20face-001	0.0521	859	0.0732	852	0.1414	1239	0.2549	1242	0.0769	1114	0.1354	825	0.0419	841	0.8953	690		
5	3divi-000	-		0.1335	901	-		-		-		-		-		-			
6	3divi-001	0.0777	898	0.1540	914	-		-		-		-		-		-			
7	3divi-002	0.0788	900	0.1539	912	-		-		-		-		-		-			
8	3divi-003	0.0318	821	0.0588	831	0.0389	1147	0.0639	1142	0.0619	1097	0.0928	779	0.0555	881	-			
9	3divi-004	0.0095	550	0.0153	563	0.0097	900	0.0145	888	0.0175	881	0.0330	615	0.0242	741	-			
10	3divi-005	0.0094	546	0.0151	560	0.0078	821	0.0121	820	0.0135	803	0.0231	537	0.0156	616	-			
11	3divi-006	0.0064	430	0.0094	430	0.0047	635	0.0066	639	0.0091	675	0.0191	489	0.0113	508	0.2854	502		
12	3divi-007	0.0024	141	0.0038	148	0.0028	322	0.0034	322	0.0046	439	0.0101	319	0.0082	392	0.2515	481		
13	accurascan-001	0.3670	1002	0.4777	1008	0.5313	1301	0.6155	1293	0.5583	1240	0.6120	1017	0.2569	1038	0.8833	681		
14	accurascan-002	0.4209	1012	0.5213	1018	0.5826	1309	0.6642	1298	0.6295	1244	0.6937	1035	0.3116	1061	0.9162	697		
15	accurascan-003	0.0017	81	0.0028	92	0.0024	155	0.0024	124	0.0029	222	0.0057	130	0.0048	168	0.0629	133		
16	acer-000	0.1393	941	0.9075	1068	0.9981	1366	1.0000	1374	1.0000	1331	1.0000	1275	0.9998	1294	-			
17	acer-001	0.0294	806	0.0504	811	0.0240	1097	0.0463	1104	0.0436	1056	0.0622	735	0.0360	820	-			
18	acer-002	0.0169	708	0.0262	708	0.0103	917	0.0167	937	0.0182	891	0.0281	575	0.0159	619	0.1555	405		
19	acisw-003	0.9682	1067	0.9971	1078	0.7892	1342	0.8738	1336	0.8752	1286	0.8275	1084	0.6698	1142	-			
20	acisw-006	0.2945	990	0.9788	1074	0.6044	1311	0.7588	1309	0.9900	1310	1.0000	1280	0.9999	1301	-			
21	acisw-007	0.4276	1014	0.5493	1025	0.8425	1345	0.9185	1343	0.8424	1277	0.9976	1213	0.9930	1255	0.9999	831		
22	acisw-008	0.0100	566	0.0147	551	0.0094	886	0.0126	830	0.1740	1168	0.6651	1030	0.4545	1100	0.9302	703		
23	adera-001	0.1021	918	0.1757	921	0.1823	1247	0.2967	1250	0.1714	1167	0.6357	1021	0.1127	951	-			
24	adera-002	0.0052	352	0.0071	340	0.0047	628	0.0064	629	0.0087	656	0.0159	436	0.0136	570	-			
25	adera-003	0.0043	292	0.0059	286	0.0036	499	0.0043	452	0.0076	612	0.0151	420	0.0128	549	0.1526	404		
26	adera-004	0.0014	52	0.0022	55	0.0035	488	0.0050	515	0.0023	102	0.0212	510	0.0058	242	0.1469	395		
27	adera-005	0.0017	74	0.0026	76	0.0027	293	0.0030	250	0.0024	114	0.0134	388	0.0049	172	0.1246	354		
28	advance-002	0.0089	527	0.0137	533	0.0073	800	0.0115	808	0.0400	1041	0.0722	758	0.0593	890	-			
29	advance-003	0.0060	411	0.0087	412	0.0052	672	0.0067	642	0.0389	1037	0.4914	980	0.1291	964	0.7998	645		
30	advance-004	0.0083	513	0.0101	463	0.0037	526	0.0054	552	0.0051	477	0.3555	940	0.1088	947	0.8285	653		
31	advance-005	0.0059	403	0.0084	397	0.0043	593	0.0060	596	0.0065	566	0.9998	1261	0.9993	1281	1.0000	921		
32	advance-006	-		-		0.0032	434	0.0042	440	0.0034	310	0.9982	1220	0.5024	1109	1.0000	910		
33	advance-007	-		-		0.0025	193	0.0028	208	0.0025	136	0.0801	767	0.0258	755	0.3283	516		
34	affinsys-ai-000	-		-		0.1154	1228	0.1788	1221	0.1886	1176	0.2159	874	0.1383	973	0.6126	593		
35	afisbiometrics-000	0.0051	349	0.0073	349	0.0030	381	0.0050	520	0.0044	413	0.0077	216	0.0057	238	0.0796	222		
36	afrengine-000	0.6244	1045	0.7336	1052	0.8318	1344	0.9083	1340	0.8122	1269	0.9980	1218	0.9895	1250	1.0000	836		
37	afrengine-001	0.0071	462	0.0129	521	0.0094	891	0.0201	977	0.0097	707	0.9737	1151	0.8179	1175	0.9999	815		
38	afrengine-002	0.0052	358	0.0071	338	0.0032	423	0.0043	457	0.0194	914	0.2125	872	0.0450	852	0.2330	473		
39	afrengine-003	-		-		0.0028	335	0.0031	283	0.0116	763	0.0224	527	0.0208	707	0.0860	244		
40	afrengine-004	-		-		0.0027	310	0.0028	220	0.1475	1157	0.2477	894	0.2470	1032	0.2062	457		
41	afrengine-005	-		-		0.0028	339	0.0031	268	0.0035	319	0.0078	221	0.0059	256	0.0768	209		
42	aifirst-001	0.0119	620	0.0170	595	0.0084	852	0.0127	840	0.0131	793	0.0212	509	0.0138	575	-			
43	aigen-001	0.0124	631	0.0219	662	0.0143	1014	0.0217	989	0.0236	957	0.8960	1105	0.3255	1067	-			
44	aigen-002	0.0192	736	0.0343	752	0.0256	1101	0.0402	1081	0.0389	1036	0.9196	1114	0.3876	1084	0.9994	803		

Table 57: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
45	aigroup-000	-		-		0.0683	1198	0.0774	1162	0.0612	1093	0.9816	1160	0.9069	1201	1.0000	1073
46	aigroup-001	-		-		0.0114	959	0.0192	968	0.0179	884	0.9998	1263	0.9986	1277	1.0000	1461
47	aigroup-002	-		-		0.0034	467	0.0049	512	0.0041	382	0.9104	1110	0.7156	1156	0.9949	768
48	ailabs-001	0.0158	695	0.0276	716	0.0192	1056	0.0317	1054	0.0352	1021	0.0608	731	0.0434	846	-	
49	aimall-001	0.0176	719	0.0324	741	0.0225	1086	0.0407	1085	-		-		-		-	
50	aimall-002	0.0119	619	0.0167	589	0.0224	1083	0.0411	1086	0.0233	949	0.0373	656	0.0235	734	-	
51	aimall-003	0.0033	218	0.0041	170	0.0033	457	0.0035	353	0.0056	514	0.0109	344	0.0087	413	-	
52	aipractix-000	-		-		0.0649	1192	0.0749	1158	0.0065	567	0.9940	1193	0.9055	1200	0.9999	827
53	aiseemu-001	0.0021	109	0.0029	98	0.0027	290	0.0033	307	0.0038	351	0.0339	628	0.0057	240	0.4819	557
54	aiseemu-002	0.0023	131	0.0032	113	0.0026	247	0.0027	186	0.0036	326	0.0439	674	0.0057	234	0.4259	544
55	aiunionface-000	0.0104	577	0.0154	564	0.0082	844	0.0122	823	0.0141	814	0.0243	544	0.0169	638	-	
56	aize-001	0.0223	762	0.0344	754	0.0199	1063	0.0313	1050	0.0367	1028	0.0522	707	0.0359	818	0.2506	480
57	aize-002	0.0210	752	0.0327	742	0.0280	1111	0.0489	1115	0.0504	1073	0.0692	753	0.0434	845	0.2657	493
58	aize-003	0.0086	520	0.0125	517	0.0072	791	0.0126	834	0.0516	1077	0.6137	1018	0.1038	943	0.9911	757
59	aize-006	-		-		0.0081	838	0.0160	920	0.0133	798	0.9925	1187	0.9473	1215	1.0000	842
60	ajou-001	0.0093	542	0.0147	552	0.0071	790	0.0126	831	0.0173	878	0.0274	569	0.0186	672	-	
61	akiyama-000	0.2564	982	0.3776	990	0.7240	1329	0.8699	1333	0.8899	1289	0.9872	1169	0.9483	1216	-	
62	akiyama-001	0.1311	937	0.2306	948	0.5113	1299	0.6797	1300	0.8748	1285	0.9908	1179	0.9649	1229	-	
63	akiyama-002	-		-		0.0119	969	0.0157	910	0.0104	730	0.9974	1211	0.9791	1239	1.0000	868
64	akiyama-003	-		-		0.0431	1156	0.0516	1121	0.0400	1042	0.2055	868	0.0835	921	0.5380	570
65	akor-000	-		-		0.0022	65	0.0024	107	0.0017	29	0.0035	34	0.0028	20	0.0506	62
66	alchera-000	0.0165	704	0.0243	687	0.0125	982	0.0186	960	0.0204	923	0.0349	635	0.0243	743	-	
67	alchera-001	0.0183	728	0.0299	727	0.0142	1009	0.0234	1006	0.0239	964	0.0388	663	0.0267	766	-	
68	alchera-002	0.0107	588	0.0157	574	0.0104	921	0.0229	1001	0.0144	822	0.0246	548	0.0198	694	-	
69	alchera-003	0.0044	298	0.0055	269	0.0031	398	0.0039	402	0.0042	391	0.0077	218	0.0065	278	0.0777	213
70	alchera-004	0.0035	242	0.0052	255	0.0028	343	0.0039	404	0.0029	226	0.0075	206	0.0044	145	0.0742	189
71	alchera-005	0.0027	166	0.0040	166	0.0026	237	0.0030	244	0.0025	149	0.0055	126	0.0040	115	0.0642	146
72	alchera-006	0.0028	178	0.0039	157	0.0026	246	0.0031	274	0.0025	143	0.0090	275	0.0042	124	0.0867	246
73	alchera-007	0.0034	236	0.0049	234	0.0026	245	0.0031	276	0.0026	172	0.0096	302	0.0044	140	0.0834	240
74	alchera-008	-		-		0.0026	233	0.0030	252	0.0024	134	0.0099	310	0.0043	133	0.0951	290
75	alchera-009	-		-		0.0026	268	0.0029	234	0.0022	84	0.5197	990	0.2232	1027	0.8470	657
76	alfabeta-001	0.4867	1030	0.5831	1033	0.6855	1321	0.8156	1322	0.8253	1275	0.7765	1060	0.6416	1140	0.9627	717
77	alice-000	0.0119	621	0.0192	631	0.0106	929	0.0170	940	0.0167	864	0.0265	563	0.0150	606	0.1824	435
78	alice-001	0.0034	231	0.0046	219	0.0032	421	0.0037	381	0.0030	251	0.3164	925	0.1088	948	0.8336	654
79	alleyes-000	0.0058	391	0.0090	419	0.0055	695	0.0087	739	0.0068	575	0.0105	332	0.0076	360	-	
80	allgovision-000	0.0346	830	0.0527	818	0.0232	1091	0.0339	1057	0.0372	1032	0.0620	734	0.0443	851	-	
81	alphaface-001	0.0065	436	0.0097	445	0.0039	548	0.0063	625	0.0083	641	-		-		-	
82	alphaface-002	0.0052	357	0.0075	360	0.0030	367	0.0044	466	1.0000	1341	0.0115	362	0.0084	403	-	
83	alttekglobal-000	-		-		0.0031	388	0.0046	476	0.0051	476	0.0097	308	0.0075	348	0.0927	282
84	amplifiedgroup-001	0.5034	1032	0.5848	1034	0.6973	1325	0.8316	1323	0.7807	1264	0.7724	1057	0.6354	1135	-	
85	androvideo-000	0.0243	774	0.0438	789	0.0239	1095	0.0365	1072	0.0483	1067	0.1870	853	0.0635	895	-	
86	anke-000	0.0194	739	0.0260	705	-		-		-		-		-		-	
87	anke-001	0.0186	732	0.0238	684	-		-		-		-		-		-	
88	anke-002	0.0402	840	0.0759	858	0.0271	1109	0.0484	1113	-		-		-		-	

Table 58: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
89	anke-003	0.0131	649	0.0213	655	0.0094	888	0.0175	943	0.0134	800	0.9951	1198	0.0150	605	-	
90	anke-004	0.0080	502	0.0154	565	0.0073	799	0.0112	802	0.0102	725	0.0178	470	0.0118	523	-	
91	anke-005	0.0070	455	0.0109	481	0.0059	726	0.0094	753	0.0105	732	0.0142	399	0.0102	471	-	
92	anyvision-001	0.0716	891	0.0859	869	-		-		-		-		-		-	
93	anyvision-002	0.0660	886	0.0898	875	0.0928	1215	0.1512	1210	0.0899	1129	0.1191	811	0.0801	915	-	
94	anyvision-004	0.0267	792	0.0385	772	0.0258	1103	0.0487	1114	0.0234	954	0.0301	588	0.0191	679	-	
95	anyvision-005	0.0023	132	0.0037	146	0.0027	317	0.0035	343	0.0049	464	0.0084	255	0.0069	304	-	
96	aratek-001	0.0033	222	0.0045	212	0.0028	332	0.0031	284	0.0043	398	0.0092	285	0.0069	307	0.0897	261
97	armatura-001	0.0033	214	0.0042	184	0.0031	397	0.0037	374	0.0056	513	0.0110	345	0.0092	435	0.4625	553
98	armatura-002	0.0041	280	0.0052	253	0.0034	468	0.0044	459	0.0040	368	0.8502	1091	0.0275	771	0.9970	782
99	armatura-003	0.0020	106	0.0029	96	0.0026	261	0.0028	197	0.0025	154	0.0049	99	0.0043	134	0.0608	119
100	arvis-000	-		-		0.3652	1280	0.4777	1275	1.0000	1345	1.0000	1405	1.0000	1340	1.0000	1086
101	asr-000	-		-		0.5523	1303	0.7662	1310	0.8807	1287	1.0000	1310	1.0000	1309	1.0000	897
102	asr-001	-		-		0.0030	365	0.0039	400	0.0027	192	0.3390	931	0.1327	970	0.7167	620
103	asusaics-000	0.0125	638	0.0209	650	0.0085	854	0.0134	856	0.0143	820	0.7189	1041	0.0285	778	-	
104	asusaics-001	0.0125	636	0.0210	652	0.0085	856	0.0134	857	0.0143	821	0.7437	1047	0.0289	783	-	
105	autentika-000	0.1415	943	0.1916	932	0.4130	1288	0.5521	1284	0.4217	1228	0.9998	1259	0.9954	1262	1.0000	922
106	autentika-001	0.0444	850	0.0688	847	0.0987	1217	0.1531	1211	0.1547	1160	1.0000	1276	0.9997	1293	1.0000	887
107	autentika-002	0.0536	863	0.0866	871	0.0657	1195	0.1084	1189	0.1548	1161	0.9993	1237	0.9939	1258	1.0000	899
108	autentika-003	-		-		0.0203	1066	0.0314	1051	0.0404	1045	0.9791	1156	0.9294	1209	0.9994	802
109	authenmetric-002	0.0092	539	0.0134	527	0.0095	892	0.0177	945	0.0192	907	0.0463	687	0.0236	736	-	
110	authenmetric-003	0.0036	250	0.0053	264	0.0039	556	0.0051	526	0.0095	698	0.9930	1188	0.5932	1126	1.0000	861
111	authenmetric-004	0.0027	168	0.0042	185	0.0033	447	0.0036	370	0.0083	644	0.9879	1172	0.4058	1091	1.0000	848
112	authme-001	0.0036	245	0.0049	235	0.0037	528	0.0040	412	0.0030	248	0.3075	918	0.1310	967	0.7046	614
113	authme-002	-		-		0.0032	428	0.0035	347	0.0026	180	0.4284	966	0.2115	1022	0.7861	640
114	aware-000	0.1965	965	0.3090	972	-		-		-		-		-		-	
115	aware-001	0.1965	964	0.3092	973	-		-		-		-		-		-	
116	aware-002	0.1951	963	0.3202	976	-		-		-		-		-		-	
117	aware-003	0.0793	901	0.1161	895	0.1028	1219	0.1708	1217	0.1698	1166	0.2333	885	0.1174	957	-	
118	aware-004	0.0690	889	0.0949	881	0.0837	1210	0.1436	1208	0.1171	1144	0.8137	1074	0.1056	944	-	
119	aware-005	0.0457	853	0.0643	837	0.0603	1187	0.1094	1193	0.0613	1094	0.1075	799	0.0491	864	-	
120	aware-006	0.0487	856	0.0819	865	0.0529	1177	0.1090	1191	0.1011	1139	0.1058	794	0.0502	868	0.5309	567
121	aware-007	0.0101	570	0.0198	642	0.0060	737	0.0133	853	0.0224	940	0.5813	1008	0.1673	1002	0.8871	685
122	aware-008	-		-		0.0062	746	0.0143	885	0.0135	802	0.0332	618	0.0133	558	0.3622	522
123	awiros-001	0.4044	1008	0.4622	1005	0.5530	1304	0.6518	1296	0.2008	1179	0.1994	861	0.1386	975	-	
124	awiros-002	0.1990	968	0.2561	955	0.3319	1276	0.4411	1271	0.3821	1221	0.9938	1191	0.2634	1044	-	
125	awiros-003	-		-		0.0128	989	0.0158	915	0.9978	1314	1.0000	1301	1.0000	1316	1.0000	914
126	awiros-004	-		-		0.0158	1039	0.0255	1017	0.9981	1317	1.0000	1303	1.0000	1321	1.0000	866
127	aximetria-001	0.0111	597	0.0186	621	0.0110	939	0.0148	896	0.0170	871	0.3928	956	0.2090	1020	0.7924	642
128	ayftech-001	0.0946	913	0.1941	934	0.2438	1262	0.3625	1260	0.1558	1162	0.1589	839	0.0936	936	-	
129	ayftech-003	0.0023	135	0.0037	144	0.0026	269	0.0030	265	0.0072	597	0.5411	998	0.2515	1034	0.8902	686
130	ayonix-000	0.4351	1017	0.4872	1011	0.6150	1313	0.7510	1308	0.6557	1250	0.6361	1022	0.4981	1107	-	
131	beethedata-000	0.0127	642	0.0195	635	0.0092	876	0.0157	911	0.0171	874	0.0306	593	0.0204	700	0.1617	417
132	betaface-000	0.8804	1060	0.8998	1064	-		-		-		-		-		-	

Table 59: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
133	betaface-002	0.8813	1061	0.8998	1065	-		-		-		-		-		-			
134	beyneai-000	0.0071	468	0.0107	479	0.0104	922	0.0131	849	0.0170	872	0.9837	1163	0.6171	1131	1.0000	834		
135	biocube-001	0.5596	1040	0.6834	1046	0.7700	1340	0.8712	1334	0.8446	1278	0.9661	1146	0.7922	1171	0.9846	746		
136	biocube-002	0.0612	875	0.0856	868	0.2330	1260	0.2972	1251	0.2365	1196	0.9327	1128	0.6947	1146	0.9989	793		
137	bioid-001	-		-		0.0121	974	0.0197	975	0.0148	828	0.9883	1173	0.7807	1168	0.9997	809		
138	bioid-002	-		-		0.0127	987	0.0205	980	0.0158	844	0.7684	1056	0.0224	723	0.9859	748		
139	bioid-003	-		-		0.0083	848	0.0137	865	0.0112	754	0.1141	805	0.0135	563	0.3783	530		
140	bioid-004	-		-		0.0083	851	0.0136	862	0.0109	746	0.1377	829	0.0139	579	0.5446	573		
141	bioidtechswiss-000	0.0066	442	0.0082	393	0.0113	955	0.0225	996	0.0078	621	0.0139	395	0.0092	430	-			
142	bioidtechswiss-001	0.0054	367	0.0072	344	0.0069	780	0.0124	826	0.0060	536	0.0094	295	0.0065	286	-			
143	bioidtechswiss-002	0.0049	329	0.0067	323	0.0064	754	0.0116	810	0.0067	572	0.0117	368	0.0086	409	-			
144	bioidtechswiss-003	-		-		0.0045	613	0.0050	518	0.0193	911	0.0360	643	0.0356	814	0.1320	363		
145	biometric-vision-000	0.0023	128	0.0036	137	0.0028	336	0.0034	336	0.0028	217	0.3109	921	0.1504	984	0.6849	610		
146	biometrik-001	-		-		0.7415	1333	0.8509	1326	0.9980	1316	1.0000	1329	0.9877	1247	1.0000	1411		
147	biometrik-002	-		-		0.0641	1191	0.0984	1182	0.0419	1050	0.3401	932	0.0292	787	0.9219	702		
148	bm-001	0.7431	1052	0.9494	1072	0.9586	1356	0.9843	1352	0.9049	1293	0.9021	1109	0.8395	1179	-			
149	boetech-001	0.0662	887	0.0802	863	0.0493	1170	0.0791	1164	0.0682	1106	0.1074	798	0.0758	909	0.9951	769		
150	boetech-002	0.0535	862	0.0565	827	0.0114	957	0.0136	861	0.0403	1043	0.0650	738	0.0606	891	0.3470	520		
151	bresee-000	0.8467	1058	0.9472	1070	0.9819	1357	0.9903	1353	0.9940	1312	0.9919	1182	0.9760	1236	-			
152	bresee-001	0.0085	517	0.0143	542	0.0086	862	0.0153	904	0.0108	741	0.0168	455	0.0115	515	-			
153	bresee-002	0.0079	497	0.0101	460	0.0065	763	0.0079	707	0.0129	785	0.0263	561	0.0224	724	0.2137	464		
154	btsdigital-001	-		-		0.0046	622	0.0072	672	0.0077	615	0.0141	397	0.0092	433	0.1256	357		
155	bynn-000	-		-		0.0193	1059	0.0234	1005	0.1302	1148	0.9999	1267	0.9996	1289	1.0000	920		
156	cakevn-001	-		-		0.0311	1128	0.0677	1147	0.0212	930	0.1427	831	0.0185	669	0.9170	699		
157	cakevn-002	-		-		0.0027	298	0.0039	403	0.0026	184	0.0049	100	0.0037	77	0.0569	96		
158	camvi-001	0.3822	1006	0.5375	1022	-		-		-		-		-		-			
159	camvi-002	0.0125	637	0.0221	665	0.0089	869	0.0145	890	0.0142	817	0.2650	902	0.0166	636	-			
160	camvi-003	0.0184	730	0.0320	739	0.0134	998	0.0255	1018	-		-		-		-			
161	camvi-004	0.0171	712	0.0316	738	0.0042	585	0.0049	508	0.0097	709	0.6636	1029	0.0141	583	-			
162	candour-001	0.1048	921	0.3189	975	0.0130	992	0.0182	952	0.3879	1223	0.9216	1116	0.7071	1151	0.9836	740		
163	candour-002	-		-		0.0036	506	0.0061	612	0.1489	1159	0.4137	963	0.2751	1048	0.8795	677		
164	candour-003	-		-		0.0040	567	0.0068	649	0.0041	383	0.1321	821	0.0225	726	0.6090	592		
165	candour-004	-		-		0.0027	306	0.0039	409	0.0030	241	0.1049	793	0.0121	534	0.5691	580		
166	canon-002	0.0034	230	0.0050	245	0.0026	270	0.0033	301	0.0043	407	0.0182	474	0.0065	284	-			
167	canon-003	0.0041	287	0.0059	282	0.0030	366	0.0040	414	0.0040	363	0.0073	199	0.0059	250	0.0735	186		
168	canon-004	0.0052	356	0.0091	422	0.0033	455	0.0058	580	0.0037	335	0.0770	765	0.0494	865	0.3192	514		
169	canon-005	0.0024	140	0.0035	131	0.0025	212	0.0033	319	0.0052	487	0.0096	305	0.0090	422	0.0542	80		
170	cchonolulu-000	0.7910	1056	0.8611	1060	0.9824	1358	0.9926	1355	0.9940	1311	0.9861	1164	0.9534	1218	1.0000	837		
171	cchonolulu-001	0.4202	1011	0.4853	1010	0.6298	1314	0.7694	1311	0.7402	1257	0.9720	1150	0.8870	1194	0.9951	770		
172	cdachyd-001	-		-		0.7397	1332	0.8335	1324	0.7630	1259	0.7410	1045	0.5756	1124	0.9588	714		
173	ceiaufgbr-000	-		-		0.9990	1371	1.0000	1373	0.9998	1325	1.0000	1277	0.9999	1304	1.0000	896		
174	ceiec-001	0.0328	825	0.0475	802	0.0295	1125	0.0478	1111	0.0621	1099	0.1059	795	0.0932	935	-			
175	ceiec-002	0.0161	701	0.0193	632	0.0122	977	0.0164	933	0.0270	981	0.0555	714	0.0472	861	-			
176	ceiec-003	0.0071	466	0.0107	476	0.0061	742	0.0079	712	0.0160	849	0.0316	601	0.0260	759	-			

Table 60: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
177	ceiec-004	0.0038	262	0.0051	249	0.0045	619	0.0053	540	0.0062	554	0.3939	958	0.0104	481	-	
178	celloscope-001	-		-		0.1245	1232	0.1840	1222	0.1886	1175	0.2636	901	0.1485	983	0.7055	617
179	chosun-000	0.8481	1059	1.0000	1296	1.0000	1382	0.9999	1371	1.0000	1379	1.0000	1454	1.0000	1384	-	
180	chosun-001	0.0525	860	0.0936	879	0.0742	1202	0.1263	1202	0.0978	1136	1.0000	1305	0.9354	1210	-	
181	chosun-002	0.0390	839	0.0646	838	0.0339	1135	0.0576	1133	0.0455	1063	0.6904	1033	0.1746	1003	-	
182	chtface-001	0.9993	1071	0.9994	1079	0.9999	1374	0.9996	1363	1.0000	1329	1.0000	1281	0.9999	1303	-	
183	chtface-002	0.0150	683	0.0268	710	0.0096	899	0.0140	876	0.0186	896	0.0320	607	0.0194	687	-	
184	chtface-003	0.0091	534	0.0146	548	0.0083	849	0.0128	842	0.0132	794	0.0220	522	0.0149	603	-	
185	chtface-004	0.0046	316	0.0062	306	0.0052	671	0.0080	714	0.0088	667	0.0152	421	0.0106	486	0.1152	334
186	chtface-005	0.0033	216	0.0049	231	0.0029	350	0.0041	428	0.0044	412	0.0317	604	0.0066	289	0.1952	450
187	chtface-006	0.0029	185	0.0043	187	0.0026	276	0.0034	337	0.0040	369	0.2701	906	0.0065	282	0.4464	548
188	chtface-007	-		-		0.0025	210	0.0032	291	0.0040	365	0.2018	863	0.0064	277	0.4751	556
189	chtface-008	-		-		0.0022	71	0.0026	155	0.0025	150	0.1246	815	0.0400	833	0.4384	547
190	chtface-009	-		-		0.0023	97	0.0027	176	0.0024	127	0.3659	946	0.1226	962	0.7714	632
191	canon-000	0.0084	514	0.0156	572	0.0049	650	0.0051	528	0.0275	989	0.1371	827	0.0138	577	-	
192	canon-001	0.0041	279	0.0061	297	0.0030	386	0.0041	426	0.0048	456	0.0578	720	0.0069	306	-	
193	cist-001	0.0046	311	0.0065	314	0.0042	589	0.0063	623	0.9675	1305	0.9997	1253	0.9994	1283	1.0000	852
194	cist-002	0.0073	475	0.0096	443	0.0085	857	0.0090	746	0.9990	1319	0.9997	1252	0.9996	1291	0.9998	811
195	cist-003	0.0065	437	0.0087	409	0.0046	626	0.0068	648	0.9993	1322	0.9994	1241	0.9993	1282	0.9995	805
196	cist-004	0.0043	297	0.0062	298	0.0031	395	0.0041	433	0.0031	257	0.0363	649	0.0069	309	0.2732	495
197	clearviewai-000	0.0010	29	0.0019	44	0.0024	138	0.0028	204	0.0030	236	0.0058	133	0.0050	177	0.0604	116
198	clearviewai-001	0.0015	62	0.0021	50	0.0024	143	0.0030	254	0.0022	94	0.0745	760	0.0194	686	0.2740	496
199	clearviewai-002	-		-		0.0024	141	0.0031	282	0.0022	92	0.0042	62	0.0037	80	0.0531	69
200	clearviewai-003	-		-		0.0023	80	0.0025	132	0.0022	87	0.0042	64	0.0038	83	0.0533	71
201	closeli-001	0.0136	656	0.0163	580	0.0039	553	0.0054	550	0.0072	595	1.0000	1282	0.0094	443	1.0000	913
202	cloudmatrix-000	0.0192	737	0.0340	749	0.0133	997	0.0220	992	0.9837	1308	1.0000	1288	0.0281	772	-	
203	cloudmatrix-001	0.0668	888	0.1141	894	0.0539	1178	0.0905	1178	0.3509	1217	0.9819	1161	0.9010	1198	0.9999	816
204	cloudmatrix-002	0.0075	482	0.0113	495	0.0084	853	0.0120	814	0.9248	1298	0.9997	1251	0.9985	1276	1.0000	923
205	cloudwalk-hr-000	0.0265	791	0.0434	787	0.0340	1136	0.0646	1144	-		-		-		-	
206	cloudwalk-hr-001	0.0044	299	0.0072	346	0.0073	802	0.0108	796	0.0125	778	0.0228	534	0.0145	594	-	
207	cloudwalk-hr-002	0.0043	294	0.0059	284	0.0054	687	0.0076	690	0.0076	613	0.5932	1013	0.0093	438	-	
208	cloudwalk-hr-003	0.0026	162	0.0041	172	0.0040	563	0.0058	579	0.0060	546	0.9992	1234	0.0094	439	-	
209	cloudwalk-hr-004	0.0009	17	0.0018	38	0.0034	461	0.0028	216	0.0052	482	0.9992	1235	0.0093	437	-	
210	cloudwalk-mt-000	0.0100	564	0.0170	594	0.0074	807	0.0118	812	0.0127	781	0.0197	496	0.0129	551	-	
211	cloudwalk-mt-002	0.0064	433	0.0085	402	0.0054	690	0.0098	772	0.0070	587	0.0108	341	0.0076	359	-	
212	cloudwalk-mt-003	0.0013	49	0.0022	52	0.0026	235	0.0027	185	0.0039	355	0.0076	207	0.0067	292	-	
213	cloudwalk-mt-004	0.0009	25	0.0013	14	0.0024	153	0.0021	33	0.0028	206	0.0054	122	0.0050	179	0.0572	98
214	cloudwalk-mt-005	0.0006	8	0.0009	10	0.0025	206	0.0022	59	0.0017	35	0.9286	1122	0.5956	1127	0.9948	767
215	cloudwalk-mt-006	0.0006	10	0.0006	1	0.0023	94	0.0019	1	0.0016	20	0.0032	19	0.0030	37	0.0427	21
216	cloudwalk-mt-007	0.0006	7	0.0007	3	0.0023	101	0.0019	2	0.0016	18	0.0032	16	0.0030	35	0.0419	18
217	clova-000	0.0099	560	0.0150	554	0.0094	889	0.0147	894	0.0136	805	0.0213	513	0.0152	610	-	
218	cmcuni-001	0.0025	155	0.0038	150	0.0025	199	0.0025	149	0.0022	89	0.0052	113	0.0034	56	0.0660	160
219	codeline-000	0.0057	381	0.0079	379	0.0037	519	0.0053	548	0.2721	1202	1.0000	1285	0.9763	1237	1.0000	917
220	cogent-000	0.0435	848	0.0835	867	-		-		-		-		-		-	

Table 61: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
221	cogent-001	0.0176	721	0.0385	773	-		-		-		-		-		-	
222	cogent-002	0.0148	678	0.0321	740	0.0095	894	0.0127	837	-		-		-		-	
223	cogent-003	0.0091	533	0.0188	624	0.0098	902	0.0132	850	0.0187	897	0.0319	605	0.0224	725	-	
224	cogent-004	0.0064	435	0.0116	497	0.0096	897	0.0134	858	0.0157	841	0.0325	611	0.0204	699	-	
225	cogent-005	0.0060	408	0.0112	490	0.0064	758	0.0070	660	0.0095	696	0.0184	478	0.0135	564	-	
226	cogent-006	0.0046	317	0.0059	290	0.0036	491	0.0047	482	0.0058	532	0.0113	357	0.0091	425	0.1522	403
227	cogent-007	0.0022	124	0.0038	149	0.0028	340	0.0031	266	0.0040	371	0.0082	241	0.0067	291	0.1395	377
228	cogent-008	0.0015	60	0.0027	84	0.0023	103	0.0025	140	0.0033	294	0.0063	162	0.0055	221	0.0884	254
229	cognitec-000	0.0116	610	0.0177	608	0.0118	968	0.0167	936	0.0285	993	0.9924	1186	0.0435	847	-	
230	cognitec-001	0.0126	641	0.0185	619	0.0120	973	0.0168	938	0.0270	979	0.0554	712	0.0357	815	-	
231	cognitec-002	0.0066	441	0.0101	458	0.0079	824	0.0108	794	0.0181	888	0.0317	603	0.0237	737	-	
232	cognitec-003	0.0038	259	0.0052	254	0.0054	692	0.0057	574	0.0225	941	0.0416	672	0.0388	827	0.1397	378
233	cognitec-004	0.0036	247	0.0053	258	0.0053	678	0.0056	565	0.0098	711	0.0202	503	0.0154	611	0.3436	519
234	cognitec-005	0.0022	119	0.0030	104	0.0035	485	0.0041	420	0.0079	623	0.0480	693	0.0233	732	0.2134	462
235	cognitec-006	-		-		0.0034	473	0.0039	408	0.0069	579	0.0160	437	0.0108	492	0.1197	345
236	cognitec-007	-		-		0.0035	490	0.0042	444	0.0069	583	0.0114	360	0.0091	424	0.1112	327
237	cognitec-008	-		-		0.0030	385	0.0036	366	0.0058	525	0.0106	333	0.0091	426	0.0884	253
238	compal-000	-		-		0.0028	347	0.0033	320	0.0039	359	0.0064	167	0.0051	191	0.0736	187
239	compal-001	-		-		0.0073	798	0.0053	543	0.0092	682	0.1965	860	0.0165	633	0.2610	491
240	compal-002	-		-		0.0073	797	0.0053	542	0.0092	681	0.1923	858	0.0165	632	0.2592	487
241	cor-001	0.0075	483	0.0113	492	0.0055	700	0.0084	727	0.0091	677	0.0148	412	0.0092	434	-	
242	coretech-000	0.7699	1055	1.0000	1246	1.0000	1477	0.9999	1372	1.0000	1441	1.0000	1356	1.0000	1451	1.0000	1424
243	coretech-001	0.0052	351	0.0067	325	0.0083	850	0.0092	750	0.0346	1018	0.1363	826	0.0252	750	0.6210	598
244	coretech-002	0.0020	100	0.0036	136	0.0064	760	0.0064	628	0.0118	765	0.0157	434	0.0110	497	0.3091	509
245	coretech-003	-		-		0.0045	617	0.0051	525	0.1155	1142	0.1539	836	0.1515	985	0.1180	339
246	coretech-004	-		-		0.0044	605	0.0048	493	0.0051	480	0.0093	290	0.0070	314	0.0893	258
247	corsight-001	0.0040	274	0.0057	278	0.0033	456	0.0047	481	0.0045	429	0.0095	298	0.0063	269	-	
248	corsight-002	0.0053	361	0.0068	329	0.0030	375	0.0041	432	0.0039	360	0.0079	225	0.0054	214	0.0747	194
249	corsight-003	0.0026	161	0.0040	168	0.0028	323	0.0045	472	0.0035	323	0.0059	145	0.0046	153	0.0637	142
250	csc-001	0.9158	1063	0.9346	1069	0.9899	1359	0.9993	1361	0.9999	1326	0.9959	1201	0.9906	1252	-	
251	csc-002	0.0099	562	0.0132	524	0.0077	815	0.0142	884	0.0126	779	0.0195	495	0.0146	597	-	
252	csc-003	0.0053	362	0.0065	313	0.0037	521	0.0047	486	0.0074	606	0.0124	378	0.0112	506	0.0928	283
253	ctbcbank-000	0.0168	706	0.0250	697	0.0146	1019	0.0224	994	0.0211	929	0.8964	1107	0.3779	1081	-	
254	ctbcbank-001	0.0155	688	0.0235	682	0.0148	1026	0.0243	1011	0.0207	924	0.9279	1121	0.3469	1072	-	
255	ctbcbank-002	-		-		0.0031	387	0.0049	504	0.0037	338	0.7785	1061	0.4878	1105	0.9727	725
256	cu-face-002	0.0105	582	0.0116	498	0.0650	1193	0.0568	1130	0.0271	983	0.0139	394	0.0076	355	0.1122	328
257	cu-face-003	0.0036	244	0.0049	237	0.0031	408	0.0037	380	0.0030	244	0.3075	917	0.1310	968	0.7046	615
258	cu-face-004	-		-		0.0032	429	0.0035	346	0.0026	181	0.4284	967	0.2115	1021	0.7861	641
259	cubox-001	0.0064	431	0.0080	380	0.0037	514	0.0055	558	0.0060	537	0.0111	350	0.0077	366	-	
260	cubox-002	0.0034	232	0.0041	173	0.0025	190	0.0025	151	0.0033	293	0.0064	165	0.0058	244	0.0519	64
261	cubox-003	0.0021	112	0.0028	91	0.0024	176	0.0026	166	0.0025	152	0.0046	79	0.0037	81	0.0495	52
262	cubox-004	-		-		0.0025	217	0.0027	190	0.0023	109	0.0044	75	0.0038	84	0.0490	48
263	cubox-005	-		-		0.0026	252	0.0030	261	0.0025	144	0.0047	89	0.0039	99	0.0515	63
264	cudocommunication-001	0.4777	1026	1.0000	1363	0.4373	1290	0.5360	1281	1.0000	1399	1.0000	1480	1.0000	1406	1.0000	1017

Table 62: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.			
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
265	cuhkee-001	0.0036	248	0.0045	209	0.0031	410	0.0046	477	0.0051	481	0.0095	299	0.0079	372	-	
266	cybercore-000	0.0728	893	0.1110	890	0.1521	1242	0.2375	1236	0.1874	1172	0.1907	856	0.1178	959	-	
267	cybercore-001	0.3759	1004	0.5677	1029	0.6928	1324	0.7926	1315	0.8118	1268	0.9291	1125	0.7080	1152	0.9782	732
268	cybercore-002	0.0092	541	0.0119	504	0.0049	648	0.0072	669	0.9105	1296	1.0000	1302	1.0000	1311	1.0000	918
269	cybercore-003	0.0155	687	0.0164	582	0.0032	433	0.0033	316	0.0024	118	0.9719	1149	0.8213	1177	0.9993	799
270	cyberextruder-001	0.1972	966	0.2547	954	0.4686	1297	0.6387	1295	0.3807	1220	0.3806	953	0.2582	1039	-	
271	cyberextruder-002	0.0811	906	0.1336	902	0.1465	1240	0.2266	1235	0.2086	1185	1.0000	1324	1.0000	1414	-	
272	cyberextruder-003	0.0109	593	0.0169	593	0.0071	788	0.0112	804	0.0165	860	0.0410	670	0.0272	770	0.2308	470
273	cyberextruder-004	0.0118	617	0.0181	613	0.0081	835	0.0133	852	0.0191	906	0.0329	614	0.0268	768	0.1904	442
274	cyberextruder-005	-		-		0.0049	647	0.0063	621	0.0055	500	0.0105	330	0.0079	377	0.0806	227
275	cyberextruder-006	-		-		0.0083	845	0.0138	868	0.0100	719	0.0180	472	0.0114	513	0.1276	359
276	cyberlink-000	0.0181	726	0.0274	714	0.0125	984	0.0180	950	-		-		-		-	
277	cyberlink-001	0.0131	647	0.0210	651	0.0439	1161	0.0517	1122	0.0173	879	0.0333	622	0.0209	708	-	
278	cyberlink-002	0.0114	604	0.0195	633	0.0101	912	0.0163	929	0.0160	850	0.0302	589	0.0192	682	-	
279	cyberlink-003	0.0118	615	0.0192	630	0.0098	905	0.0161	923	0.0153	834	0.0288	582	0.0159	621	-	
280	cyberlink-004	0.0074	478	0.0105	472	0.0068	776	0.0089	742	0.0094	692	0.0176	468	0.0117	520	-	
281	cyberlink-005	0.0060	409	0.0092	427	0.0058	721	0.0067	645	0.0074	602	0.0146	409	0.0105	485	-	
282	cyberlink-006	0.0042	289	0.0054	267	0.0043	592	0.0049	501	0.0052	486	0.0097	307	0.0077	367	-	
283	cyberlink-007	0.0032	211	0.0053	260	0.0041	573	0.0043	450	0.0052	485	0.0243	543	0.0084	404	0.3366	517
284	cyberlink-008	0.0042	288	0.0056	277	0.0038	545	0.0048	494	0.0053	491	0.0099	313	0.0074	342	0.1354	374
285	cyberlink-009	0.0018	89	0.0027	81	0.0047	627	0.0046	474	0.0040	375	0.0086	269	0.0062	266	0.1104	321
286	cyberlink-010	0.0011	39	0.0019	43	0.0041	570	0.0041	421	0.0039	354	0.1829	850	0.0054	215	0.8629	667
287	cyberlink-011	0.0010	28	0.0017	31	0.0035	484	0.0034	325	0.0045	421	0.0080	228	0.0070	317	0.0877	250
288	cyberlink-012	0.0010	27	0.0016	29	0.0036	495	0.0035	351	0.0047	452	0.0099	312	0.0074	343	0.1350	372
289	cyberlink-013	0.0011	35	0.0017	36	0.0037	516	0.0036	362	0.0045	423	0.0090	277	0.0080	381	0.0924	280
290	cyberlink-014	-		-		0.0026	271	0.0027	174	0.0030	234	0.0065	171	0.0052	198	0.0869	247
291	cyberlink-015	-		-		0.0025	200	0.0026	160	0.0028	208	0.0063	161	0.0050	182	0.0835	242
292	cyberlink-016	-		-		0.0025	188	0.0025	137	0.0028	202	0.0063	160	0.0050	178	0.0827	237
293	dahua-000	0.0249	777	0.0477	804	-		-		-		-		-		-	
294	dahua-001	0.0250	782	0.0466	799	0.0228	1089	0.0379	1075	-		-		-		-	
295	dahua-002	0.0129	643	0.0157	573	0.0116	963	0.0153	905	0.0134	799	0.0222	525	0.0170	641	-	
296	dahua-003	0.0052	355	0.0068	330	0.0056	704	0.0062	616	0.0113	757	0.0207	507	0.0185	671	-	
297	dahua-004	0.0045	308	0.0058	280	0.0036	501	0.0048	492	0.0051	478	0.0086	265	0.0070	318	-	
298	dahua-005	0.0031	201	0.0046	215	0.0035	487	0.0049	509	0.0046	433	0.0076	209	0.0062	265	-	
299	dahua-006	0.0027	165	0.0039	160	0.0031	403	0.0039	406	0.0039	358	0.0067	182	0.0058	241	-	
300	dahua-007	0.0017	76	0.0023	58	0.0026	258	0.0032	288	0.0033	286	0.0060	149	0.0054	213	0.0678	166
301	daon-000	0.0095	552	0.0117	500	0.0068	772	0.0077	699	0.0092	686	0.0174	464	0.0137	573	0.1430	387
302	daon-001	-		-		0.0036	496	0.0058	578	0.0050	470	0.0084	250	0.0057	235	0.0805	225
303	daon-002	-		-		0.0030	378	0.0042	439	0.0039	353	0.0146	410	0.0055	223	0.7444	626
304	daon-003	-		-		0.0024	174	0.0029	228	0.0030	242	0.0059	147	0.0048	169	0.0639	144
305	datech-000	0.0020	107	0.0031	111	0.0031	400	0.0041	429	0.0033	292	0.1646	841	0.0667	899	0.4898	560
306	datech-001	-		-		0.0028	327	0.0034	331	0.0026	187	0.2308	883	0.0818	920	0.5505	574
307	datech-002	-		-		0.0028	326	0.0034	333	0.0026	166	0.3779	951	0.1654	1001	0.7940	643
308	decatur-000	0.0714	890	0.1115	892	0.0608	1188	0.1106	1194	0.0866	1125	1.0000	1295	0.0714	906	-	

Table 63: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)													
	Algorithm	CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
309	decatur-001	0.0424	844	0.0711	849	0.0237	1092	0.0458	1100	0.0447	1061	1.0000	1272	0.9969	1268	1.0000	933
310	decloakface-001	0.0795	902	0.1131	893	0.3127	1272	0.4259	1267	0.2118	1188	0.7926	1065	0.6088	1130	0.9589	715
311	decloakface-002	-		-		0.0048	637	0.0077	698	1.0000	1339	0.9998	1260	0.9997	1292	1.0000	911
312	deepglint-001	0.0040	278	0.0062	299	0.0047	633	0.0067	643	0.0069	580	1.0000	1299	1.0000	1310	-	
313	deepglint-002	0.0016	65	0.0027	86	0.0032	419	0.0033	318	0.0043	406	0.0084	256	0.0077	363	-	
314	deepglint-003	0.0027	169	0.0038	151	0.0030	374	0.0032	286	0.0043	396	0.0082	243	0.0076	356	-	
315	deepglint-004	0.0025	150	0.0034	123	0.0039	554	0.0061	611	0.0050	472	0.0091	281	0.0082	391	0.1003	297
316	deepglint-005	0.0052	353	0.0059	288	0.0030	368	0.0031	267	0.0033	298	0.7620	1054	0.1535	986	0.9912	758
317	deepsea-001	0.0136	657	0.0215	657	0.0142	1011	0.0214	986	0.0163	854	0.0250	550	0.0192	683	-	
318	deepsense-000	0.0145	672	0.0265	709	0.0113	952	0.0196	973	0.0151	831	0.0215	516	0.0129	552	-	
319	deepsense-001	0.0013	48	0.0019	41	0.0024	164	0.0025	142	0.0027	197	0.0115	363	0.0053	204	0.1394	376
320	deepsense-002	0.0010	26	0.0016	30	0.0024	127	0.0023	90	0.0026	171	0.0052	116	0.0043	135	0.0781	214
321	deepsense-003	0.0009	20	0.0015	25	0.0025	213	0.0024	101	0.0023	107	0.0457	684	0.0097	453	0.2547	484
322	deepsense-004	-		-		0.0023	77	0.0021	39	0.0020	63	0.0111	349	0.0040	118	0.1195	343
323	deepsense-005	-		-		0.0023	75	0.0021	45	0.0019	50	0.0081	236	0.0035	67	0.0922	279
324	deepsense-006	-		-		0.0022	54	0.0022	65	0.0018	42	0.0336	625	0.0059	249	0.2813	501
325	deepsense-007	-		-		0.0022	56	0.0022	64	0.0018	41	0.0035	35	0.0030	38	0.0440	28
326	deepsense-008	-		-		0.0022	59	0.0022	67	0.0018	39	0.0034	33	0.0030	36	0.0434	23
327	dermalog-001	-		0.3049	971	-		-		-		-		-		-	
328	dermalog-002	-		0.3154	974	-		-		-		-		-		-	
329	dermalog-003	0.2355	976	0.2795	964	-		-		-		-		-		-	
330	dermalog-004	0.1810	959	0.2399	951	-		-		-		-		-		-	
331	dermalog-005	0.1526	947	0.1823	926	0.2580	1263	0.4018	1264	-		0.2651	903	0.1585	992	-	
332	dermalog-006	0.0253	784	0.0369	768	0.0171	1047	0.0283	1037	0.0217	936	0.0358	640	0.0230	728	-	
333	dermalog-007	0.0564	872	0.0798	862	0.1351	1236	0.2471	1238	0.2343	1195	0.2223	876	0.1397	976	-	
334	dermalog-008	0.0096	556	0.0166	588	0.0086	859	0.0133	851	0.0165	859	0.0586	725	0.0226	727	-	
335	dermalog-009	0.0067	444	0.0094	432	0.0051	666	0.0069	653	0.0116	764	0.0312	595	0.0177	657	0.3747	529
336	dermalog-010	0.0030	190	0.0041	174	0.0034	471	0.0037	384	0.0075	607	0.5181	988	0.2530	1035	0.8853	683
337	dermalog-011	0.0045	309	0.0062	304	0.0035	486	0.0059	589	0.0057	524	0.2242	878	0.0407	836	0.6329	599
338	dermalog-012	0.0080	503	0.0101	464	0.0037	525	0.0057	575	0.0055	506	0.0095	300	0.0068	300	0.0834	241
339	dermalog-013	-		-		0.0036	503	0.0057	572	0.0058	528	0.0103	322	0.0073	332	0.0894	259
340	dermalog-014	-		-		0.0029	357	0.0039	407	0.0037	340	0.0069	189	0.0057	233	0.0646	149
341	dermalog-015	-		-		0.0025	205	0.0025	128	0.0033	284	0.0064	168	0.0054	216	0.0587	108
342	dermalog-016	-		-		0.0023	105	0.0023	85	0.0025	162	0.0050	108	0.0040	109	0.0550	84
343	dicio-001	0.5486	1038	0.6442	1039	0.7516	1335	0.8607	1329	0.8678	1284	0.8268	1082	0.7034	1149	0.9752	730
344	dicio-002	-		-		0.0057	713	0.0095	760	0.0097	710	0.9998	1256	0.9989	1279	1.0000	870
345	didiglobalface-001	0.0055	373	0.0092	424	0.0030	372	0.0045	471	0.0088	660	0.0119	372	0.0085	407	-	
346	didiglobalface-002	0.0033	226	0.0051	247	0.0026	259	0.0034	339	0.0033	291	0.0085	258	0.0047	161	0.1135	331
347	didiglobalface-003	-		-		0.0023	85	0.0025	150	0.0027	194	0.1115	803	0.0052	201	0.5374	569
348	digidata-000	0.0967	914	0.1410	909	0.2596	1264	0.3462	1258	0.0293	999	0.0363	648	0.0212	710	0.1849	440
349	digidata-001	0.0224	763	0.0352	757	0.0330	1131	0.0570	1132	0.0109	747	0.0481	694	0.0123	539	0.3055	508
350	digitalbarriers-000	0.3683	1003	0.4630	1006	-		-		-		-		-		-	
351	digitalbarriers-001	0.4683	1024	0.5024	1015	-		-		-		-		-		-	
352	digitalbarriers-002	0.3360	998	0.3690	988	0.0877	1212	0.1557	1212	0.0971	1135	0.0951	785	0.0497	866	-	

Table 64: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
353	dps-000	0.0115	607	0.0176	606	0.0149	1029	0.0185	959	0.0173	877	0.0275	571	0.0180	661	0.1757	430
354	dralmey-001	-		-		0.0036	500	0.0052	538	0.0057	519	0.9985	1226	0.9872	1246	1.0000	869
355	dsk-000	0.1526	946	0.2169	942	0.3787	1282	0.5426	1283	0.3115	1209	0.3089	919	0.1994	1016	-	
356	einetworks-000	0.0099	561	0.0180	611	0.0088	867	0.0140	878	0.0130	788	0.0225	530	0.0147	599	-	
357	einetworksindia-000	0.0058	393	0.0074	355	0.0051	665	0.0061	613	0.0074	603	0.0193	493	0.0102	472	0.1819	434
358	einetworksindia-001	0.0042	290	0.0062	300	0.0048	636	0.0064	626	0.0057	522	0.0473	691	0.0092	432	0.3984	540
359	einetworksindia-002	-		-		0.0024	152	0.0027	175	0.0023	99	0.1534	835	0.0513	870	0.4614	551
360	ekin-002	0.1168	929	0.2042	938	0.1530	1243	0.2524	1241	0.1777	1171	0.2773	908	0.1347	971	0.8954	691
361	element-000	0.0023	133	0.0035	133	0.0024	163	0.0026	157	0.0029	225	0.0060	152	0.0048	165	0.0631	137
362	enface-000	0.0028	177	0.0049	238	0.0043	598	0.0072	667	0.0058	533	0.0150	417	0.0090	423	0.7851	639
363	enface-001	0.0072	471	0.0107	477	0.0071	784	0.0138	869	0.0068	576	0.0515	704	0.0094	444	0.9942	765
364	enface-002	0.0033	221	0.0052	257	0.0038	540	0.0073	675	0.0026	186	0.9998	1262	0.9995	1287	1.0000	909
365	enqura-001	-		-		0.0155	1035	0.0144	887	0.0312	1009	0.0597	727	0.0485	863	0.2970	506
366	enqura-002	-		-		0.0082	841	0.0150	899	0.0105	731	0.0193	490	0.0111	501	0.1302	361
367	eocortex-000	0.3485	1000	0.6943	1048	0.1122	1225	0.1574	1213	0.2155	1189	0.2257	882	0.1606	997	-	
368	ercacat-001	0.0036	252	0.0044	198	0.0033	446	0.0047	487	0.0106	735	0.0202	502	0.0184	667	-	
369	ethiopianlmis-000	-		-		0.2967	1268	0.4098	1265	0.9992	1321	1.0000	1278	0.9999	1299	1.0000	924
370	euronovate-001	0.2786	985	0.3608	987	0.4489	1293	0.6105	1292	0.5010	1234	0.5392	997	0.3769	1079	0.8802	679
371	euronovate-002	0.0256	787	0.0436	788	0.0226	1087	0.0425	1092	0.0301	1004	0.0691	752	0.0319	796	0.3687	525
372	euronovate-003	0.0061	415	0.0088	414	0.0040	568	0.0060	598	0.0065	568	0.7815	1063	0.5034	1110	0.9823	736
373	euronovate-004	0.0082	508	0.0110	484	0.0050	660	0.0077	697	0.0074	604	0.6949	1037	0.3354	1070	0.9791	734
374	everai-000	0.0122	626	0.0225	670	-		-		-		-		-		-	
375	everai-001	0.0085	518	0.0156	570	0.0072	794	0.0114	807	-		-		-		-	
376	everai-002	0.0104	580	0.0159	575	0.0063	752	0.0112	803	0.0182	890	0.0353	637	0.0264	765	-	
377	paravision-003	0.0034	237	0.0050	242	0.0036	508	0.0052	537	0.0092	685	0.0193	492	0.0156	615	-	
378	expasoft-000	0.0427	845	0.0655	842	0.0239	1094	0.0393	1078	0.0673	1105	0.8963	1106	0.3832	1083	-	
379	expasoft-001	0.0328	826	0.0488	808	0.0211	1075	0.0342	1059	0.0629	1100	0.6483	1025	0.2816	1051	-	
380	expasoft-002	0.0170	709	0.0274	713	0.0787	1206	0.0768	1161	0.1629	1163	0.9996	1248	0.9631	1228	1.0000	876
381	f8-001	0.0249	780	0.0336	747	0.0178	1050	0.0232	1003	0.0303	1005	0.0615	733	0.0408	837	-	
382	f8-002	0.0340	828	0.0591	833	0.0213	1077	0.0374	1073	0.0452	1062	0.0760	761	0.0502	869	0.2708	494
383	facehawk-000	1.0000	1476	1.0000	1098	1.0000	1440	1.0000	1419	1.0000	1475	1.0000	1366	1.0000	1454	1.0000	1276
384	facehawk-001	-		-		0.0545	1181	0.0748	1157	0.0751	1112	0.9919	1183	0.9566	1222	1.0000	851
385	faceint-001	-		-		0.0105	926	0.0124	827	0.0313	1010	0.3318	928	0.0557	883	0.9851	747
386	facelocate-001	-		-		0.0664	1196	0.1171	1197	0.0950	1132	0.9219	1117	0.7041	1150	0.9985	788
387	facelocate-002	-		-		0.0335	1132	0.0634	1139	0.0393	1038	0.9988	1232	0.8977	1197	1.0000	853
388	faceonlive-001	0.0269	794	0.0359	763	0.0387	1146	0.0721	1154	0.0246	969	0.0349	634	0.0220	717	0.2408	475
389	faceonlive-002	0.0121	622	0.0135	530	0.0033	451	0.0041	424	0.0037	342	0.9427	1131	0.7927	1172	0.9993	797
390	faceonlive-003	-		-		0.0045	612	0.0050	519	0.0193	912	0.0360	644	0.0356	813	0.1320	362
391	facephi-000	0.0044	300	0.0059	287	0.0047	629	0.0057	576	0.0088	666	1.0000	1316	1.0000	1318	1.0000	925
392	facephi-001	-		-		0.0031	405	0.0037	379	0.0058	531	0.0111	347	0.0098	458	0.1048	309
393	facephi-002	-		-		0.0024	128	0.0022	61	0.0036	329	0.0093	289	0.0070	311	0.1136	332
394	facephi-003	-		-		0.0023	107	0.0023	82	0.0031	263	0.0073	200	0.0057	239	0.1060	311
395	facesoft-000	0.0085	516	0.0112	491	0.0064	756	0.0107	791	0.0091	676	0.0171	460	0.0107	488	-	
396	facetag-000	0.2836	986	0.4081	997	0.2933	1267	0.4303	1268	0.3448	1215	0.6312	1020	0.3530	1073	0.9753	731

Table 65: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)													
	Algorithm	CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
397	facetag-001	0.2920	988	0.4538	1003	0.3220	1274	0.4371	1269	1.0000	1355	1.0000	1402	0.3254	1066	1.0000	1130
398	facetag-002	0.0098	559	0.0147	550	0.0064	761	0.0110	799	0.0116	762	0.0190	488	0.0119	529	0.1334	368
399	facex-001	1.0000	1301	1.0000	1255	1.0000	1375	1.0000	1376	1.0000	1444	1.0000	1351	1.0000	1328	-	
400	facex-002	0.0803	904	0.1404	907	0.1283	1233	0.1979	1229	0.1440	1156	0.1952	859	0.1299	966	0.7427	624
401	facia-001	0.0083	512	0.0124	509	0.0064	759	0.0097	765	0.0098	712	0.9983	1222	0.9849	1243	1.0000	903
402	facia-002	-		-		0.0072	793	0.0109	797	0.0122	772	0.0460	685	0.0135	565	0.4464	549
403	faicetech-000	-		-		0.0026	274	0.0031	272	0.0023	110	0.0116	366	0.0037	79	0.0904	266
404	farfaces-001	0.4890	1031	0.5860	1035	0.5650	1305	0.7268	1305	0.8015	1267	0.7511	1050	0.5892	1125	0.9536	710
405	fastenterprises-000	0.0093	544	0.0151	559	0.0098	903	0.0140	879	0.0209	926	0.2991	913	0.1608	998	0.6727	607
406	fci-000	-		-		0.0027	288	0.0027	183	0.0028	209	0.1269	818	0.0563	884	0.3886	535
407	fdu-000	0.0386	838	0.0576	829	-		-		-		-		-		-	
408	fdu-001	0.0347	831	0.0439	791	-		-		-		-		-		-	
409	fedu-001	-		-		0.0033	440	0.0047	480	0.0128	782	0.9816	1159	0.9216	1206	0.9999	821
410	fiberhome-nanjing-002	0.0217	758	0.0381	771	0.0874	1211	0.1770	1220	0.0271	982	0.0351	636	0.0188	675	-	
411	fiberhome-nanjing-003	0.0090	528	0.0139	537	0.0082	842	0.0144	886	0.0110	749	0.0174	462	0.0107	490	-	
412	fiberhome-nanjing-004	0.0037	258	0.0056	276	0.0031	399	0.0043	451	0.0043	408	0.0083	249	0.0061	262	-	
413	fincore-000	0.0309	815	0.0502	810	0.0281	1112	0.0510	1119	0.0521	1079	0.0815	769	0.0522	872	0.3100	510
414	firstcreditkz-001	0.0024	144	0.0034	120	0.0024	171	0.0024	125	0.0034	299	1.0000	1371	1.0000	1467	1.0000	1233
415	firstcreditkz-002	0.0018	87	0.0026	75	0.0024	154	0.0024	112	0.0029	232	0.0056	128	0.0049	173	0.0624	130
416	firstcreditkz-003	-		-		0.0023	113	0.0024	117	0.0026	176	0.0104	328	0.0047	158	0.0910	271
417	firstcreditkz-004	-		-		0.0023	93	0.0024	103	0.0024	120	0.0360	645	0.0061	261	0.2315	471
418	foomobi-001	0.4827	1029	0.5795	1030	0.6823	1320	0.8132	1318	0.8217	1273	1.0000	1379	1.0000	1464	1.0000	1241
419	foomobi-002	0.4827	1028	0.5795	1031	0.6823	1319	0.8132	1319	0.8217	1272	1.0000	1427	1.0000	1349	1.0000	1205
420	fourthlinecom-001	-		-		0.0442	1162	0.0696	1150	0.0609	1092	0.8783	1098	0.7523	1163	0.9917	759
421	fpt-000	-		-		0.0213	1078	0.0375	1074	0.0364	1027	0.9528	1135	0.8729	1187	0.9993	800
422	fpt-001	-		-		0.0099	906	0.0148	895	0.8158	1270	0.9986	1227	0.9934	1256	1.0000	926
423	fpttelecom-000	-		-		0.0586	1185	0.0761	1160	0.2308	1194	0.9988	1231	0.9937	1257	1.0000	885
424	fraudcom-000	0.0110	595	0.0173	597	0.0083	846	0.0139	874	0.0166	861	0.0258	560	0.0177	654	0.1488	399
425	frpkauai-001	0.0023	130	0.0035	132	0.0026	232	0.0030	260	0.0040	373	0.0080	232	0.0072	327	0.0706	175
426	frpkauai-002	0.0024	146	0.0035	124	0.0024	168	0.0024	120	0.0033	296	0.0065	174	0.0058	245	0.0599	112
427	frpkauai-003	-		-		0.0023	102	0.0023	95	0.0024	113	0.0075	205	0.0039	100	0.0762	204
428	frpkauai-004	-		-		0.0037	513	0.0036	371	0.0069	582	0.0226	531	0.0138	574	0.1354	373
429	frpkauai-005	-		-		0.0023	73	0.0023	77	0.0021	79	0.0082	239	0.0042	125	0.0760	203
430	fujitsulab-000	0.0123	627	0.0212	654	0.0091	874	0.0133	854	0.0251	971	0.4200	965	0.0360	819	-	
431	fujitsulab-001	0.0059	404	0.0082	388	0.0132	995	0.0258	1020	0.0140	813	0.0246	547	0.0142	586	-	
432	fujitsulab-002	0.0091	535	0.0124	512	0.0105	923	0.0156	908	0.0169	870	0.0345	632	0.0146	598	-	
433	fujitsulab-003	0.0045	307	0.0065	315	0.0057	716	0.0083	723	0.0080	631	0.0154	427	0.0101	464	0.5564	576
434	g42-intellibrain-001	0.0006	9	0.0009	9	0.0037	518	0.0044	460	0.0030	252	0.0059	142	0.0053	210	0.0750	200
435	g42-intellibrain-002	0.0006	11	0.0011	11	0.0029	348	0.0037	382	0.0028	205	0.0212	511	0.0091	427	0.1398	380
436	geo-000	0.0543	866	0.0814	864	0.1042	1221	0.1592	1214	0.1637	1164	0.8533	1092	0.1800	1008	-	
437	geo-001	0.0180	725	0.0198	641	0.0037	517	0.0055	556	0.0070	585	0.0129	385	0.0094	440	-	
438	geo-002	0.0171	713	0.0187	623	0.0035	479	0.0051	530	0.0064	558	0.0117	369	0.0083	400	0.0938	286
439	geo-003	0.0180	724	0.0313	735	0.0239	1096	0.0552	1124	0.0319	1013	0.0487	697	0.0222	720	0.1962	452
440	geo-004	0.0030	186	0.0041	175	0.0025	214	0.0030	241	0.0035	322	0.0065	172	0.0053	209	0.0648	151

Table 66: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
441	gistouch-000	0.0296	808	0.0518	815	0.0342	1137	0.0609	1136	0.0493	1071	1.0000	1315	1.0000	1327	1.0000	934
442	gistouch-001	0.0176	720	0.0312	734	0.8931	1350	0.9735	1348	0.8625	1280	1.0000	1320	1.0000	1420	1.0000	1381
443	glory-000	0.1094	925	0.1286	898	0.2179	1257	0.2656	1245	-		-		-		-	
444	glory-001	0.0902	910	0.1082	888	0.1642	1245	0.2065	1234	0.2186	1190	0.2669	904	0.2089	1019	-	
445	glory-002	0.0241	769	0.0311	733	0.0116	964	0.0151	903	0.0157	843	0.0264	562	0.0188	677	-	
446	glory-003	0.0076	485	0.0125	514	0.0077	818	0.0103	785	0.0130	787	0.0205	504	0.0143	589	-	
447	glory-004	0.0077	489	0.0123	508	0.0074	808	0.0098	773	0.0122	775	0.0193	491	0.0134	562	0.1520	402
448	glory-005	0.0056	376	0.0076	363	0.0054	693	0.0072	670	0.0075	608	0.9237	1119	0.7853	1169	0.9974	784
449	glory-006	0.0050	335	0.0068	326	0.0051	668	0.0070	659	0.0069	584	0.7869	1064	0.5076	1111	0.9719	724
450	glory-007	0.0050	342	0.0068	327	0.0050	658	0.0069	655	0.0068	573	0.0126	383	0.0083	399	0.1066	312
451	gorilla-000	0.0802	903	0.1409	908	-		-		-		-		-		-	
452	gorilla-001	0.0488	858	0.0756	855	0.1218	1231	0.2034	1231	-		-		-		-	
453	gorilla-002	0.0256	788	0.0413	779	0.0478	1167	0.0912	1179	-		-		-		-	
454	gorilla-003	0.0165	705	0.0291	719	0.0205	1072	0.0437	1093	0.0297	1002	0.9658	1145	0.0262	760	-	
455	gorilla-004	0.0138	660	0.0239	686	0.0243	1098	0.0444	1095	0.0275	988	0.0369	651	0.0204	702	-	
456	gorilla-005	-		-		0.0142	1012	0.0267	1030	0.0228	945	0.0358	639	0.0195	688	-	
457	gorilla-006	0.0105	587	0.0152	561	0.0106	927	0.0203	979	0.0155	836	0.0218	520	0.0136	567	-	
458	gorilla-007	0.0074	477	0.0111	487	0.0065	764	0.0126	832	0.0100	722	0.0151	419	0.0102	468	0.1450	391
459	gorilla-008	0.0058	395	0.0091	420	0.0049	646	0.0079	711	0.0079	628	0.0126	382	0.0091	428	0.1217	350
460	gorilla-009	0.0049	328	0.0072	343	0.0038	534	0.0056	566	0.0065	563	0.0104	324	0.0070	312	0.1103	320
461	gpstechvn-000	0.0202	747	0.0355	760	0.0142	1010	0.0248	1013	0.0208	925	0.0495	699	0.0207	705	0.5098	564
462	graymatics-001	0.1039	919	0.1620	918	0.1344	1235	0.1917	1225	0.1648	1165	0.5160	986	0.2689	1045	0.9194	700
463	griaule-000	0.0071	467	0.0099	450	0.0050	653	0.0072	665	0.0160	847	0.0304	590	0.0267	767	0.1480	397
464	griaule-001	0.0057	383	0.0078	372	0.0045	618	0.0065	633	0.0070	588	0.7515	1051	0.5106	1113	0.9747	728
465	griaule-002	0.0021	117	0.0032	112	0.0025	227	0.0027	191	0.0034	311	0.0996	791	0.0201	696	0.3978	538
466	gtelots-000	-		-		0.0059	723	0.0098	769	0.0090	672	0.9995	1245	0.9975	1269	1.0000	881
467	gtelots-001	-		-		0.0053	680	0.0102	783	0.0071	589	0.0111	351	0.0070	310	0.1173	337
468	hertasecurity-000	0.0630	880	0.0780	861	0.0503	1173	0.0898	1174	0.0738	1109	0.0693	755	0.0420	842	-	
469	hertasecurity-001	0.0249	781	0.0309	732	0.0105	924	0.0161	922	0.0245	967	0.0447	679	0.0359	817	0.1850	441
470	hertasecurity-002	0.0206	750	0.0315	736	0.0060	730	0.0078	705	0.0253	974	0.0696	756	0.0457	853	0.4565	550
471	hertasecurity-003	0.0079	501	0.0104	470	0.0060	731	0.0078	701	0.0255	975	0.0732	759	0.0460	854	0.4731	554
472	hik-001	0.0096	555	0.0125	513	0.0093	883	0.0164	932	0.0108	742	0.0937	781	0.0127	546	-	
473	hisign-001	0.0036	249	0.0050	241	0.0034	459	0.0046	475	0.0079	627	0.0153	426	0.0133	559	0.1080	314
474	hisign-002	0.0029	183	0.0044	200	0.0027	313	0.0032	300	0.0028	215	0.0409	669	0.0132	557	0.2016	456
475	hisign-003	-		-		0.0030	370	0.0036	360	0.0036	332	0.0582	723	0.0201	697	0.2584	486
476	hyperverge-001	1.0000	1297	1.0000	1251	1.0000	1476	1.0000	1387	1.0000	1445	1.0000	1352	1.0000	1450	-	
477	hyperverge-002	0.0050	333	0.0066	320	0.0035	478	0.0051	524	0.0062	552	0.0107	337	0.0074	344	0.0921	278
478	hyperverge-003	0.0019	98	0.0030	100	0.0025	195	0.0029	230	0.0027	189	0.0049	106	0.0042	129	0.0576	100
479	hyperverge-004	0.0072	472	0.0116	499	0.0040	561	0.0071	664	0.0058	530	0.0080	230	0.0057	236	0.0621	127
480	hyperverge-005	0.0026	164	0.0039	162	0.0027	284	0.0031	281	0.0036	334	0.0066	176	0.0055	220	0.0658	158
481	hyperverge-006	-		-		0.0025	223	0.0030	246	0.0033	290	0.0060	151	0.0051	193	0.0601	114
482	hyperverge-007	-		-		0.0026	241	0.0030	256	0.0034	308	0.0061	154	0.0052	196	0.0605	117
483	hyundai-roboticslab-000	-		-		0.0037	515	0.0052	534	0.0063	556	1.0000	1404	0.0872	928	1.0000	1084
484	hyundai-roboticslab-001	-		-		0.0048	644	0.0066	640	0.0193	910	0.8620	1093	0.6861	1144	0.9861	749

Table 67: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
485	hyundai-roboticslab-002	-		-		0.0039	559	0.0054	551	0.0086	655	0.9589	1138	0.8817	1192	0.9986	790
486	hyundai-roboticslab-003	-		-		0.0024	132	0.0028	200	0.0027	198	0.6384	1023	0.2934	1055	0.8692	670
487	hyundai-roboticslab-004	-		-		0.0023	121	0.0028	198	0.0028	218	0.2635	900	0.0802	916	0.5392	571
488	hzailu-001	0.0122	624	0.0164	584	0.0095	893	0.0196	971	0.0079	625	0.0118	370	0.0090	421	0.1173	338
489	hzailu-002	0.0051	344	0.0072	342	0.0038	546	0.0055	561	0.0040	370	0.0077	210	0.0062	267	0.0928	285
490	hzailu-003	0.0178	723	0.0291	720	0.0031	416	0.0042	447	0.0035	313	0.0061	157	0.0052	195	0.0799	223
491	hzailu-004	0.0175	718	0.0197	639	0.0031	417	0.0042	442	0.0034	307	0.0060	153	0.0051	185	0.0775	212
492	hzailu-005	0.0172	715	0.0197	638	0.0028	338	0.0036	365	0.0032	280	0.0058	134	0.0048	166	0.0751	201
493	hzailu-006	0.0161	700	0.0174	599	0.0024	160	0.0027	170	0.0026	177	0.0049	105	0.0044	146	0.0654	155
494	hzailu-007	-		-		0.0025	186	0.0027	187	0.0026	167	0.0048	95	0.0044	143	0.0653	154
495	hzailu-008	-		-		0.0024	179	0.0026	169	0.0025	159	0.0048	93	0.0044	141	0.0652	153
496	i2v-001	-		-		0.0946	1216	0.1369	1204	0.3989	1225	0.9998	1258	0.9966	1265	1.0000	877
497	i2v-002	-		-		0.0035	482	0.0044	464	0.0037	337	0.9355	1129	0.4823	1103	0.9960	773
498	icehouse-000	-		-		0.0027	285	0.0036	369	0.0047	445	0.0088	274	0.0072	323	0.0726	181
499	icm-002	0.0143	670	0.0249	695	0.0144	1017	0.0256	1019	0.0236	959	0.0386	662	0.0263	764	-	
500	icm-003	0.0138	661	0.0222	667	0.0149	1027	0.0282	1036	0.0227	944	0.0384	660	0.0257	754	0.1938	447
501	icm-004	0.0079	498	0.0120	505	0.0074	805	0.0107	790	0.0091	678	0.0281	576	0.0128	550	0.2778	498
502	icm-005	0.0100	565	0.0140	539	0.0091	870	0.0108	792	0.0169	869	0.0319	606	0.0290	784	0.1775	433
503	icomm-000	0.0019	99	0.0033	119	0.0033	450	0.0044	461	0.0031	270	0.1350	824	0.0552	880	0.4196	543
504	icthtc-000	0.0260	790	0.0396	777	0.0207	1073	0.0339	1058	0.0291	998	0.0474	692	0.0346	808	-	
505	id3-001	0.1698	953	0.2499	953	-		-		-		-		-		-	
506	id3-002	0.1615	952	0.2393	950	-		-		-		-		-		-	
507	id3-003	0.0361	835	0.0757	857	0.0292	1119	0.0476	1108	0.0447	1060	0.0674	747	0.0396	830	-	
508	id3-004	0.0198	744	0.0344	753	0.0238	1093	0.0423	1091	0.0289	997	0.0416	671	0.0257	753	-	
509	id3-005	0.0104	579	0.0169	592	0.0080	833	0.0133	855	0.0194	913	0.0334	623	0.0262	761	-	
510	id3-006	0.0072	470	0.0103	466	0.0049	649	0.0074	680	0.0095	694	0.0165	449	0.0119	528	-	
511	id3-007	0.0056	377	0.0078	369	0.0060	735	0.0072	674	0.0275	986	0.1374	828	0.0519	871	-	
512	id3-008	0.0039	265	0.0055	271	0.0032	431	0.0042	438	0.0081	636	0.0155	429	0.0134	560	0.1186	341
513	id3-009	-		-		0.0035	483	0.0049	510	0.0046	432	0.4689	975	0.0214	714	0.9377	705
514	id3-010	-		-		0.0032	437	0.0045	468	0.0042	394	0.8447	1090	0.2140	1023	0.9932	763
515	idbio-000	-		-		0.0044	602	0.0067	644	0.0068	577	0.0557	715	0.0222	721	0.2871	505
516	idemia-003	0.0222	760	0.0316	737	0.0188	1053	0.0325	1055	0.0548	1084	-		-		-	
517	idemia-004	0.0160	698	0.0244	691	0.0199	1064	0.0354	1064	0.0268	977	0.1489	834	0.0263	763	-	
518	idemia-005	0.0132	650	0.0216	660	0.0121	975	0.0218	991	0.0215	932	0.0323	610	0.0188	676	-	
519	idemia-006	0.0046	318	0.0062	305	0.0047	631	0.0066	635	0.0073	599	0.2882	910	0.0094	445	-	
520	idemia-007	0.0024	138	0.0039	163	0.0032	438	0.0038	394	0.0046	436	0.0092	286	0.0070	319	-	
521	idemia-008	0.0023	129	0.0032	115	0.0023	115	0.0028	194	0.0034	312	0.0067	181	0.0056	231	0.0765	206
522	idemia-009	0.0022	127	0.0030	102	0.0022	63	0.0023	97	0.0023	98	0.0046	80	0.0039	95	0.0533	72
523	idemia-010	0.0015	61	0.0026	71	0.0022	69	0.0020	8	0.0020	61	0.0040	53	0.0033	49	0.0465	37
524	idemia-011	-		-		0.0021	30	0.0021	40	0.0018	45	0.0077	215	0.0038	88	0.0782	217
525	identy-000	0.0073	474	0.0095	438	0.0050	654	0.0067	641	0.0071	592	0.8257	1081	0.4310	1095	0.9839	742
526	identy-001	-		-		0.0029	356	0.0034	321	0.0037	343	0.0066	177	0.0055	222	0.0748	199
527	identy-002	-		-		0.0027	294	0.0029	240	0.0030	235	0.0058	135	0.0051	188	0.0691	170
528	identy-003	-		-		0.0025	220	0.0030	248	0.0026	164	0.0050	111	0.0043	130	0.0547	81

Table 68: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
529	idrnd-000	-		-		0.0041	581	0.0053	547	0.0050	467	0.0094	292	0.0075	346	0.1026	303
530	idrnd-001	-		-		0.0025	226	0.0033	311	0.0032	276	0.0528	709	0.0173	649	0.2243	466
531	igearx-face-000	0.0091	532	0.0146	549	0.0163	1042	0.0362	1069	0.0399	1040	0.6436	1024	0.3305	1068	0.7832	638
532	iit-000	0.1516	945	0.1981	935	0.0828	1209	0.1442	1209	0.2214	1191	0.2943	911	0.1877	1011	-	
533	iit-001	0.0104	576	0.0179	610	0.0099	909	0.0142	883	0.1222	1145	0.2677	905	0.2171	1025	-	
534	iit-002	0.0111	601	0.0177	609	0.0085	855	0.0140	877	0.0193	909	0.0332	620	0.0260	757	-	
535	iit-003	0.0082	511	0.0151	558	0.0053	681	0.0084	728	0.0122	774	0.0199	500	0.0137	572	-	
536	imds-software-001	0.0126	640	0.0228	673	0.0130	991	0.0221	993	0.0231	947	0.0469	690	0.0199	695	0.4260	545
537	imds-software-002	0.0048	323	0.0072	348	0.0036	498	0.0052	539	0.0047	450	0.9981	1219	0.0078	370	1.0000	883
538	imds-software-003	0.0032	204	0.0047	224	0.0028	320	0.0036	368	0.0036	333	0.9987	1228	0.0063	270	1.0000	872
539	imperial-000	0.0067	446	0.0108	480	0.0080	831	0.0134	859	0.0087	657	0.0581	722	0.0102	473	-	
540	imperial-001	0.0094	547	0.0154	566	0.0072	792	0.0120	816	0.0103	726	0.0190	486	0.0120	532	-	
541	imperial-002	0.0058	389	0.0081	387	0.0055	698	0.0085	731	0.0083	642	0.0157	431	0.0103	474	-	
542	incode-000	0.0766	896	0.1029	887	-		-		-		-		-		-	
543	incode-001	0.0550	869	0.0773	859	-		-		-		-		-		-	
544	incode-002	0.0293	805	0.0548	825	0.0436	1160	0.0785	1163	-		-		-		-	
545	incode-003	0.0142	668	0.0249	696	0.0448	1163	0.0869	1170	0.0354	1022	0.0661	742	0.0320	797	-	
546	incode-004	0.0077	490	0.0132	523	0.0096	898	0.0160	918	0.0171	873	0.0281	574	0.0179	659	-	
547	incode-005	0.0111	599	0.0189	625	0.0156	1037	0.0304	1046	0.0195	916	0.0307	594	0.0172	645	-	
548	incode-006	0.0156	689	0.0273	712	0.0117	965	0.0238	1009	0.0202	921	0.0339	629	0.0164	630	-	
549	incode-007	0.0109	592	0.0155	569	0.0056	708	0.0099	774	0.0107	737	0.0168	454	0.0102	469	-	
550	incode-008	0.0063	425	0.0101	461	0.0046	623	0.0086	737	0.0057	520	0.0104	325	0.0074	337	-	
551	incode-009	0.0044	301	0.0067	324	0.0034	465	0.0051	523	0.0049	465	0.0091	280	0.0067	294	0.0892	257
552	incode-010	0.0041	281	0.0063	309	0.0028	341	0.0043	449	0.0047	454	0.0077	217	0.0061	260	0.0770	210
553	incode-011	0.0032	205	0.0044	199	0.0026	266	0.0034	334	0.0032	271	0.0359	641	0.0140	581	0.1840	439
554	incode-012	0.0027	167	0.0043	186	0.0025	198	0.0035	348	0.0034	304	0.0966	789	0.0385	825	0.3387	518
555	incode-013	0.0035	241	0.0049	230	0.0032	422	0.0039	405	0.0051	474	0.0094	293	0.0081	389	0.0768	208
556	incode-014	-		-		0.0022	66	0.0023	91	0.0034	303	0.0066	179	0.0062	268	0.0569	95
557	incode-015	-		-		0.0021	16	0.0022	66	0.0020	65	0.0040	55	0.0037	76	0.0537	76
558	infocert-001	0.0105	585	0.0172	596	0.0078	820	0.0125	829	0.0159	845	0.1573	838	0.0565	885	0.5716	581
559	infocert-002	-		-		1.0000	1457	0.9993	1360	1.0000	1426	1.0000	1339	1.0000	1427	1.0000	1366
560	infocert-003	-		-		0.0202	1065	0.0284	1038	0.0620	1098	0.2624	899	0.1316	969	0.6450	601
561	innefulabs-000	0.0122	625	0.0199	643	0.0112	949	0.0197	974	0.0222	939	0.0372	655	0.0271	769	-	
562	innominds-001	0.5569	1039	0.6541	1042	0.7642	1339	0.8665	1332	0.7629	1258	0.8200	1076	0.6359	1136	0.9841	744
563	innov8tif-000	-		-		0.1842	1248	0.3028	1255	0.2056	1183	0.2348	887	0.1449	979	0.6142	594
564	innov8tif-001	-		-		0.1865	1250	0.2951	1249	0.1976	1178	0.2564	897	0.1593	993	0.6665	604
565	innov8tif-002	-		-		0.2074	1255	0.2575	1244	0.1162	1143	0.1822	849	0.1782	1005	0.5224	566
566	innovativetechnologyltd-001	0.0578	873	0.0938	880	0.0501	1172	0.0981	1181	0.0592	1091	0.0779	766	0.0422	843	-	
567	innovativetechnologyltd-002	0.0451	851	0.0716	851	0.0541	1179	0.1009	1187	0.0506	1074	0.0682	749	0.0371	822	-	
568	innovatrics-000	0.1105	926	0.1905	931	-		-		-		-		-		-	
569	innovatrics-001	0.1077	924	0.1828	928	-		-		-		-		-		-	
570	innovatrics-002	0.4049	1009	0.6142	1036	-		-		-		-		-		-	
571	innovatrics-003	0.0349	832	0.0483	806	-		-		-		-		-		-	
572	innovatrics-004	0.0194	740	0.0292	721	0.0344	1138	0.0617	1137	0.0562	1086	0.0762	762	0.0412	838	-	

Table 69: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
573	innovatrics-005	0.0230	766	0.0353	758	0.0398	1150	0.0721	1155	-		-		-		-	
574	innovatrics-006	0.0058	390	0.0089	418	0.0061	743	0.0096	763	0.0096	704	0.0165	450	0.0103	475	-	
575	innovatrics-007	0.0040	272	0.0054	266	0.0057	711	0.0078	700	0.0079	626	0.0123	375	0.0088	416	-	
576	innovatrics-008	0.0047	321	0.0064	312	0.0038	543	0.0052	536	0.0053	492	0.0088	271	0.0069	308	0.0899	263
577	innovatrics-009	0.0022	118	0.0031	110	0.0028	319	0.0032	297	0.0034	302	0.1165	808	0.0326	798	0.3866	534
578	innovatrics-010	0.0017	82	0.0031	109	0.0027	309	0.0028	199	0.0041	378	0.0078	219	0.0066	290	0.0744	192
579	innovatrics-011	-		-		0.0026	267	0.0027	173	0.0032	278	0.0060	148	0.0044	142	0.0615	124
580	innovatrics-012	-		-		0.0025	202	0.0026	154	0.0034	306	0.9309	1126	0.4977	1106	0.9960	772
581	innovatrics-013	-		-		0.0022	45	0.0021	42	0.0026	168	0.9984	1223	0.3780	1082	1.0000	873
582	innovatrics-014	-		-		0.0021	25	0.0020	6	0.0017	32	0.0033	25	0.0029	22	0.0417	17
583	innovatrics-015	-		-		0.0021	15	0.0020	23	0.0019	53	0.0036	39	0.0031	41	0.0468	38
584	innovatrics-016	-		-		0.0021	19	0.0020	24	0.0018	38	0.0034	31	0.0029	29	0.0451	32
585	innovatrics-017	-		-		0.0020	9	0.0021	37	0.0017	30	0.0033	21	0.0029	24	0.0437	25
586	insightface-000	0.0018	93	0.0027	83	0.0029	349	0.0030	263	0.0038	349	0.0077	214	0.0068	298	0.0765	205
587	insightface-001	0.0009	18	0.0014	18	0.0027	289	0.0024	123	0.0035	321	0.0070	191	0.0065	280	0.0692	171
588	insightface-002	0.0011	36	0.0019	40	0.0027	307	0.0026	159	0.0036	331	0.0069	187	0.0065	279	0.0694	172
589	insightface-003	0.0015	59	0.0021	51	0.0045	616	0.0034	338	0.1298	1147	1.0000	1418	0.9407	1212	1.0000	1177
590	insightface-004	0.0015	58	0.0023	59	0.0052	674	0.0036	367	0.1403	1154	0.9923	1185	0.9128	1202	0.9989	792
591	inspur-000	0.0060	410	0.0078	371	0.7415	1334	0.9093	1341	0.2838	1205	0.9996	1246	0.9976	1270	1.0000	916
592	inspur-001	0.0035	238	0.0049	232	0.0025	201	0.0030	251	0.0029	230	0.9951	1197	0.9592	1225	1.0000	857
593	inspur-002	0.0025	147	0.0039	161	0.0024	130	0.0027	172	0.0031	256	0.0062	159	0.0047	159	0.1112	324
594	inspur-003	-		-		0.0023	98	0.0026	163	0.0031	258	0.0060	150	0.0046	154	0.0812	232
595	intellicloudai-001	0.0142	669	0.0234	680	0.0092	878	0.0145	889	0.0162	852	0.0371	654	0.0171	644	-	
596	intellicloudai-002	0.0059	402	0.0085	400	0.0060	733	0.0069	656	0.0108	740	0.2477	895	0.0171	643	-	
597	intellifusion-001	0.0072	469	0.0094	436	0.0056	710	0.0085	732	0.0111	753	0.0212	512	0.0143	588	-	
598	intellifusion-002	0.0059	398	0.0077	367	0.0040	562	0.0074	679	0.0085	651	0.5352	995	0.0104	482	-	
599	intellivision-001	0.1335	938	0.2205	945	0.1090	1223	0.1670	1216	0.1385	1151	0.1676	843	0.1170	956	-	
600	intellivision-002	0.1000	915	0.1775	922	0.0610	1189	0.1009	1186	0.0805	1119	0.1074	797	0.0682	900	-	
601	intellivision-003	0.1177	931	0.2006	937	0.0760	1203	0.1244	1201	0.1069	1140	0.1431	832	0.0839	922	0.4940	562
602	intellivision-004	0.0271	796	0.0559	826	0.0294	1123	0.0503	1118	0.0327	1014	0.0461	686	0.0293	789	0.2860	503
603	intellivision-005	0.0105	583	0.0182	616	0.0093	884	0.0141	880	0.0146	825	0.0252	552	0.0172	647	0.1763	432
604	intellivision-006	0.0111	600	0.0174	600	0.0110	941	0.0162	924	0.0181	886	0.0313	596	0.0191	680	0.2416	477
605	intellivision-007	0.0086	519	0.0128	520	0.0098	904	0.0159	916	0.0133	795	0.0218	521	0.0154	612	0.1751	429
606	intellivix-001	0.0064	434	0.0087	408	0.0046	620	0.0063	624	0.0072	593	0.9233	1118	0.7856	1170	0.9977	786
607	intellivix-002	0.0062	416	0.0085	401	0.0039	551	0.0056	564	0.0060	543	0.3464	935	0.0857	926	0.8759	673
608	intellivix-003	0.0075	481	0.0125	516	0.0052	673	0.0091	749	0.0066	570	0.0297	585	0.0096	452	0.1907	444
609	intellivix-004	0.0032	213	0.0045	210	0.0025	219	0.0031	280	0.0040	367	0.0074	202	0.0063	271	0.0662	161
610	intellivix-005	0.0034	234	0.0048	228	0.0024	139	0.0027	189	0.0035	320	0.0067	180	0.0059	258	0.0609	120
611	intellivix-006	-		-		0.0024	178	0.0030	258	0.0026	169	0.0085	257	0.0040	108	0.0918	275
612	intelresearch-000	0.0307	814	0.0578	830	0.0385	1144	0.0751	1159	0.0409	1047	0.9870	1168	0.5380	1118	-	
613	intelresearch-001	0.0242	771	0.0595	834	0.0129	990	0.0292	1042	0.0351	1020	0.9993	1236	0.1854	1010	-	
614	intelresearch-002	0.0058	396	0.0082	389	0.0050	662	0.0086	734	0.0136	804	0.0434	673	0.0216	715	-	
615	intelresearch-003	0.0046	314	0.0062	303	0.0038	539	0.0060	599	0.0088	662	0.0168	456	0.0136	568	-	
616	intelresearch-004	0.0025	148	0.0035	130	0.0032	425	0.0038	389	0.0049	466	0.0094	291	0.0072	325	-	

Table 70: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
617	intelresearch-005	0.0016	67	0.0023	57	0.0028	318	0.0034	328	0.0042	393	0.0084	252	0.0066	288	0.1433	388		
618	intelresearch-006	0.0010	33	0.0015	24	0.0026	282	0.0028	210	0.0032	277	0.8123	1073	0.4742	1101	0.9907	755		
619	intelresearch-007	0.0009	23	0.0013	16	0.0025	196	0.0025	130	0.0035	317	0.0069	188	0.0057	232	0.0746	193		
620	intelresearch-008	0.0009	19	0.0013	17	0.0025	197	0.0025	129	0.0028	212	0.0304	591	0.0048	167	0.2375	474		
621	intelresearch-009	-		-		0.0024	180	0.0025	135	0.0028	214	0.0058	138	0.0046	155	0.0665	162		
622	intelresearch-010	-		-		0.0023	108	0.0024	105	0.0029	231	0.0058	139	0.0050	180	0.0683	169		
623	intelresearch-011	-		-		0.0023	78	0.0023	69	0.0025	148	0.0049	103	0.0040	116	0.0549	83		
624	intelresearch-012	-		-		0.0023	92	0.0023	75	0.0025	140	0.0048	96	0.0040	114	0.0549	82		
625	intema-000	0.0012	44	0.0017	35	0.0023	72	0.0022	62	0.0022	91	0.0172	461	0.0061	259	0.0996	296		
626	intema-001	0.0010	30	0.0014	22	0.0021	23	0.0020	25	0.0019	56	0.0037	41	0.0030	40	0.0457	35		
627	interlinklabs-001	-		-		0.0102	915	0.0205	981	0.0195	915	0.0237	539	0.0115	517	0.1579	407		
628	interlinklabs-002	-		-		0.0033	452	0.0049	505	0.0040	364	0.9968	1208	0.9807	1240	1.0000	840		
629	interlinklabs-003	-		-		0.0029	360	0.0034	341	0.0046	437	0.0083	247	0.0074	336	0.0620	126		
630	intozi-001	0.9982	1070	0.9998	1081	0.9998	1372	1.0000	1377	1.0000	1335	1.0000	1307	1.0000	1320	1.0000	1350		
631	intozi-002	-		-		0.0038	537	0.0050	521	0.0054	494	0.9998	1257	0.9989	1280	1.0000	900		
632	intsysmsu-000	0.0135	653	0.0204	644	0.0112	948	0.0161	921	0.0148	829	0.0244	545	0.0172	646	-			
633	intsysmsu-001	0.9543	1066	0.9888	1075	0.9923	1360	0.9948	1356	0.9977	1313	0.9955	1199	0.9892	1249	-			
634	intsysmsu-002	0.0130	644	0.0254	699	0.0137	1003	0.0267	1031	0.0160	848	0.0267	565	0.0145	595	-			
635	ionetworks-000	0.0060	412	0.0087	411	0.0044	601	0.0058	584	0.0080	635	0.0144	405	0.0112	504	0.1149	333		
636	ionetworks-001	0.0254	785	0.0388	774	0.0435	1157	0.0662	1145	0.0426	1053	0.8936	1104	0.7391	1160	0.9964	776		
637	ionetworks-002	0.0045	302	0.0066	319	0.0033	445	0.0047	488	0.0544	1082	0.9648	1143	0.8765	1189	0.9994	801		
638	ionetworks-003	-		-		0.0034	472	0.0049	513	0.0043	403	0.0079	224	0.0056	228	0.0812	231		
639	ionetworks-004	-		-		0.0034	476	0.0050	517	0.0043	399	0.0079	223	0.0056	226	0.0808	229		
640	iqface-000	0.0091	538	0.0143	540	0.0075	809	0.0110	800	0.0171	875	0.2234	877	0.0359	816	-			
641	iqface-001	0.0088	525	0.0117	501	0.0177	1048	0.0230	1002	0.0165	858	0.5514	1000	0.0286	779	-			
642	iqface-003	0.0058	397	0.0079	378	0.0051	667	0.0058	585	0.0104	729	0.0200	501	0.0193	684	-			
643	iqsec-001	-		-		0.7050	1327	0.8569	1328	0.7728	1260	0.9962	1206	0.9840	1241	1.0000	858		
644	iqsec-002	-		-		0.7050	1326	0.8569	1327	0.7728	1261	0.9962	1205	0.9841	1242	1.0000	859		
645	iqsec-003	-		-		0.9924	1362	0.9984	1359	0.9998	1324	0.9996	1249	0.9981	1272	1.0000	892		
646	irex-000	0.0052	350	0.0099	454	0.0056	707	0.0083	725	0.0137	809	0.0163	446	0.0078	369	-			
647	irisid-001	-		-		0.0034	469	0.0048	491	0.0058	529	0.0108	342	0.0079	373	0.1024	301		
648	irisid-002	-		-		0.0031	413	0.0042	443	0.0055	504	0.0102	321	0.0077	365	0.1024	302		
649	isap-001	0.5092	1033	0.6588	1043	0.6899	1323	0.7978	1316	0.7200	1254	0.7253	1042	0.5373	1117	-			
650	isap-002	0.0114	606	0.0186	622	0.0087	864	0.0151	902	0.0156	840	0.5134	985	0.0333	802	-			
651	isityou-000	0.5682	1041	0.7033	1050	1.0000	1456	0.9999	1370	1.0000	1423	1.0000	1340	1.0000	1426	-			
652	istl-001	-		-		0.0041	580	0.0059	595	0.0053	490	0.5416	999	0.2534	1036	0.8765	674		
653	isystems-000	0.1267	935	0.1795	924	-		-		-		-		-		-			
654	isystems-001	0.0149	682	0.0245	692	0.0138	1006	0.0210	984	0.0209	928	0.0332	619	0.0223	722	-			
655	isystems-002	0.0118	614	0.0182	615	0.0111	944	0.0162	927	0.0166	863	0.0284	578	0.0195	689	-			
656	itmo-001	-		0.4408	1001	-		-		-		-		-		-			
657	itmo-002	0.1607	950	0.2866	967	-		-		-		-		-		-			
658	itmo-003	0.0749	895	0.1398	906	-		-		-		-		-		-			
659	itmo-004	0.0255	786	0.0543	823	-		-		-		-		-		-			
660	itmo-005	0.0182	727	0.0345	755	0.0181	1051	0.0348	1063	0.0225	942	0.0378	658	0.0230	729	-			

Table 71: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

		FALSE NON-MATCH RATE (FNMR)																	
	Algorithm	CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
661	itmo-006	0.0125	634	0.0220	664	0.0149	1028	0.0266	1027	0.0233	950	0.0383	659	0.0285	777	-	
662	itmo-007	0.0080	504	0.0125	515	0.0107	930	0.0185	957	0.0167	865	0.0222	526	0.0144	592	-	
663	itmo-008	0.0090	530	0.0150	556	0.0058	719	0.0059	592	0.0187	898	0.0355	638	0.0339	803	0.1758	431
664	ivacognitive-001	0.0189	734	0.0351	756	0.0123	978	0.0235	1007	0.0198	918	0.0274	570	0.0155	613	-	
665	iws-000	0.4824	1027	0.5801	1032	0.6859	1322	0.8155	1321	0.8251	1274	0.7756	1059	0.6400	1139	-	
666	jaakit-001	0.5830	1042	0.7146	1051	0.8173	1343	0.8893	1338	0.8950	1290	0.8387	1088	0.7091	1153	0.9733	727
667	jaakit-002	-		-		0.5729	1307	0.6951	1302	0.6495	1248	0.9894	1177	0.9721	1233	0.9999	823
668	kaizenvoiz-001	-		-		0.4082	1286	0.5607	1285	0.4678	1231	0.4894	979	0.3770	1080	0.8558	663
669	kakao-001	0.4553	1022	0.5532	1027	0.6580	1317	0.8150	1320	0.4831	1233	1.0000	1420	1.0000	1348	-	
670	kakao-002	0.0625	879	0.1779	923	0.0791	1207	0.1381	1206	0.0636	1101	1.0000	1392	1.0000	1483	-	
671	kakao-003	0.0130	646	0.0185	620	0.0261	1105	0.0464	1105	0.0252	973	0.2380	889	0.0284	776	-	
672	kakao-004	0.0078	493	0.0103	467	0.0059	728	0.0102	780	0.0155	837	0.1182	810	0.0230	730	-	
673	kakao-005	0.0040	269	0.0059	285	0.0036	507	0.0057	573	0.0085	649	0.0239	541	0.0125	542	-	
674	kakao-007	0.0019	97	0.0028	89	0.0024	137	0.0026	161	0.0033	289	0.0061	155	0.0053	205	0.0580	101
675	kakao-008	0.0011	38	0.0018	39	0.0023	83	0.0023	81	0.0021	78	0.0041	60	0.0035	64	0.0475	43
676	kakao-009	0.0009	24	0.0013	15	0.0023	90	0.0023	86	0.0020	66	0.0039	50	0.0034	54	0.0468	39
677	kakaobank-000	0.0041	282	0.0058	281	0.0041	572	0.0058	587	0.0060	545	0.9988	1230	0.3243	1065	1.0000	871
678	kakaobank-001	0.0030	189	0.0046	216	0.0032	435	0.0039	395	0.0049	463	0.9998	1264	0.9994	1284	1.0000	931
679	kakaobank-002	-		-		0.0026	255	0.0036	358	0.0038	346	0.3184	926	0.1174	958	0.6931	612
680	kakaobank-003	-		-		0.0023	112	0.0025	148	0.0029	227	0.0065	169	0.0052	200	0.0733	184
681	kakaopay-001	0.0152	686	0.0252	698	0.0145	1018	0.0270	1032	0.0232	948	0.0344	631	0.0194	685	0.2005	455
682	kasikornlabs-000	0.0112	602	0.0184	618	0.0086	860	0.0137	867	0.0130	790	0.0225	529	0.0148	602	0.1660	422
683	kasikornlabs-001	0.0138	662	0.0206	646	0.0087	863	0.0139	875	0.0142	815	0.0236	538	0.0171	642	0.1684	425
684	kasikornlabs-002	0.0069	453	0.0091	421	0.0048	639	0.0063	619	0.0076	611	0.0144	404	0.0110	499	0.1249	355
685	kasikornlabs-003	0.0057	379	0.0076	362	0.0042	588	0.0056	567	0.0059	534	0.0123	374	0.0098	454	0.1212	349
686	kasikornlabs-004	-		-		0.0041	578	0.0054	553	0.0057	516	0.0112	354	0.0095	446	0.1101	318
687	kasikornlabs-005	-		-		0.0037	510	0.0043	454	0.0054	495	0.0107	338	0.0093	436	0.1069	313
688	kbyai-000	-		-		0.0027	303	0.0032	295	0.0031	267	0.0079	226	0.0049	174	0.0655	156
689	kedacom-000	0.0055	370	0.0081	386	0.0111	947	0.0120	815	0.0415	1048	0.0966	788	0.0686	901	-	
690	keyless-000	-		-		0.0025	215	0.0034	326	0.0031	265	0.1874	854	0.0568	886	0.5543	575
691	kiwitech-000	0.0076	487	0.0105	471	0.0081	836	0.0128	844	0.0096	700	0.0163	445	0.0101	466	-	
692	kloudspot-001	-		-		0.0140	1007	0.0259	1021	0.9999	1328	1.0000	1313	1.0000	1325	1.0000	882
693	kloudspot-002	-		-		0.0255	1100	0.0423	1090	0.0192	908	0.8748	1096	0.7143	1155	0.9929	760
694	kneron-003	0.0542	865	0.0902	876	0.0346	1139	0.0562	1128	0.0919	1130	0.1251	816	0.0973	938	-	
695	kneron-005	0.0157	691	0.0259	703	0.0126	986	0.0212	985	0.0406	1046	0.0693	754	0.0542	877	-	
696	kneron-006	-		-		0.0094	887	0.0125	828	0.1305	1149	0.9751	1152	0.9033	1199	0.9987	791
697	knowutech-000	0.0039	267	0.0055	272	0.0028	346	0.0042	436	0.0042	386	0.0077	212	0.0059	252	0.0820	234
698	kogniza-001	-		-		0.0028	345	0.0031	285	0.0025	147	0.0048	92	0.0042	122	0.0557	86
699	kookmin-001	0.0462	855	0.0750	853	0.0489	1169	0.0842	1167	0.0659	1103	0.8380	1087	0.3212	1064	-	
700	kookmin-002	0.0054	366	0.0077	368	0.0043	595	0.0065	632	0.0123	776	0.7591	1053	0.0198	693	-	
701	koreaid-001	0.0031	202	0.0045	207	0.0026	257	0.0032	292	0.0043	402	0.0083	244	0.0068	302	0.0743	190
702	krungthai-002	0.0105	586	0.0161	578	0.0091	875	0.0141	881	0.7350	1256	0.9889	1175	0.9605	1226	0.9999	820
703	kuke3d-001	0.0058	388	0.0104	468	0.0083	847	0.0093	752	0.0270	980	0.9901	1178	0.8341	1178	1.0000	860
704	kuke3d-002	0.0077	488	0.0135	529	0.0069	779	0.0098	770	0.0111	752	1.0000	1312	1.0000	1319	1.0000	888

Table 72: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
705	lebentech-000	0.5940	1043	0.7032	1049	0.8854	1348	0.9511	1344	0.9089	1295	0.9970	1209	0.9861	1245	0.9999	828
706	lebentech-001	0.0304	812	0.9032	1066	0.9998	1373	1.0000	1375	1.0000	1338	1.0000	1293	0.9999	1308	1.0000	930
707	lemalabs-001	0.0111	598	0.0175	603	0.0088	865	0.0142	882	0.0143	818	0.0228	533	0.0140	580	0.1646	419
708	lightvision-000	-		-		0.0146	1022	0.0232	1004	0.0237	960	0.9638	1142	0.7768	1167	0.9986	789
709	lightvision-001	-		-		0.0068	774	0.0096	762	0.0084	647	0.9948	1196	0.9288	1208	1.0000	844
710	lineclova-000	0.0172	714	0.0236	683	0.0109	936	0.0194	970	0.0183	892	0.0291	584	0.0204	701	0.1936	446
711	lineclova-001	0.0025	149	0.0040	165	0.0026	281	0.0034	340	0.0045	431	0.4127	962	0.0080	379	0.9824	737
712	lineclova-002	0.0021	113	0.0035	125	0.0025	185	0.0027	180	0.0041	376	0.0086	263	0.0072	326	0.1595	411
713	lineclova-003	0.0018	91	0.0030	101	0.0028	342	0.0031	269	0.0041	377	0.0083	248	0.0075	353	0.1014	300
714	lookman-002	0.0297	809	0.0547	824	0.0339	1134	0.0562	1127	0.0614	1095	0.0960	787	0.0790	912	-	
715	lookman-004	0.0074	480	0.0099	449	0.0124	981	0.0149	897	0.0430	1055	0.0866	775	0.0694	903	-	
716	lssytech-000	-		-		0.0023	84	0.0023	92	0.0021	80	0.0040	57	0.0036	73	0.0574	99
717	lssytech-001	-		-		0.0021	39	0.0020	18	0.0020	64	0.0039	49	0.0034	55	0.0496	53
718	luxand-000	0.2056	970	0.2814	965	0.4053	1284	0.5365	1282	0.3497	1216	0.3743	949	0.2605	1042	-	
719	luxand-001	0.0116	608	0.0925	877	0.0144	1016	0.0267	1028	1.0000	1438	1.0000	1328	1.0000	1410	1.0000	1410
720	luxand-003	-		-		0.9938	1363	0.9999	1367	0.2905	1206	0.2256	881	0.0849	925	0.4876	558
721	luxand-004	-		-		0.0031	411	0.0042	437	0.0110	748	0.9994	1242	0.9969	1267	1.0000	1154
722	mantra-000	0.0037	256	0.0052	256	0.0054	689	0.0056	569	0.0097	708	0.0181	473	0.0151	607	-	
723	matrixcomsec-000	-		-		0.1032	1220	0.0895	1173	1.0000	1392	1.0000	1448	1.0000	1368	1.0000	989
724	matrixcomsec-001	-		-		0.0070	781	0.0136	860	0.8826	1288	1.0000	1373	1.0000	1466	1.0000	1236
725	matrixcomsec-002	-		-		0.0091	872	0.0172	941	0.0168	867	0.3770	950	0.1584	991	0.8585	664
726	maverickresearch-000	-		-		0.0143	1013	0.0192	966	0.0216	933	0.0362	647	0.0250	748	0.2293	467
727	maxvision-000	0.0078	494	0.0106	474	0.0110	940	0.0147	892	0.0368	1030	1.0000	1409	0.1545	987	1.0000	1077
728	maxvision-001	0.0305	813	0.0528	819	0.1028	1218	0.1921	1226	0.0650	1102	0.3001	914	0.1553	989	0.6188	597
729	maxvision-002	0.0070	461	0.0107	475	0.0061	741	0.0093	751	0.0080	632	0.5726	1005	0.2943	1056	0.9160	696
730	maxvision-003	0.0056	374	0.0083	396	0.0038	544	0.0060	602	0.0061	548	0.2614	898	0.0650	896	0.5812	583
731	maxvision-004	0.0024	145	0.0038	155	0.0025	231	0.0029	233	0.0029	223	0.4732	976	0.1965	1013	0.8141	649
732	maxvision-005	0.0014	55	0.0026	73	0.0024	146	0.0024	104	0.0025	137	0.1216	812	0.0332	800	0.3922	537
733	maxvision-006	0.0016	69	0.0026	78	0.0024	151	0.0024	109	0.0024	132	0.4012	960	0.1799	1007	0.7634	627
734	mbbankvn-000	-		-		0.0030	369	0.0035	342	0.0026	165	0.3822	954	0.1130	952	0.7052	616
735	megvii-000	0.0159	697	0.0246	694	-		-		-		-		-		-	
736	megvii-001	0.0157	692	0.0244	689	0.0392	1149	0.0671	1146	0.0168	868	0.0455	683	0.0164	629	-	
737	megvii-002	0.0104	578	0.0145	545	0.0225	1084	0.0345	1061	0.0099	718	0.0286	579	0.0240	740	-	
738	megvii-003	0.0064	429	0.0094	431	0.0136	1002	0.0260	1023	0.0050	468	0.0080	231	0.0059	255	-	
739	megvii-004	0.0020	103	0.0033	118	0.0028	334	0.0035	354	0.0037	345	0.0074	203	0.0068	303	0.0736	188
740	megvii-005	0.0010	31	0.0015	23	0.0026	248	0.0031	275	0.0019	51	0.0500	700	0.0057	237	0.1991	453
741	megvii-006	0.0011	34	0.0016	26	0.0026	275	0.0033	312	0.0025	145	0.0050	107	0.0048	163	0.0588	110
742	megvii-007	0.0006	12	0.0016	27	0.0025	216	0.0028	201	0.0024	128	0.0048	97	0.0047	160	0.0586	104
743	megvii-008	0.0003	3	0.0008	6	0.0023	116	0.0021	48	0.0024	133	0.0049	101	0.0047	162	0.0541	78
744	megvii-009	0.0003	4	0.0008	7	0.0023	117	0.0021	47	0.0018	36	0.0149	415	0.0036	69	0.1096	317
745	meituan-000	0.0197	741	0.0424	785	0.0078	822	0.0074	678	0.0103	727	0.0193	494	0.0164	628	0.1133	330
746	meituan-001	0.0164	703	0.1886	930	0.0025	204	0.0026	156	0.0030	240	0.0074	201	0.0051	189	0.0791	219
747	meituan-002	0.0017	78	0.0025	64	0.0024	150	0.0023	76	0.0024	130	0.0067	183	0.0044	148	0.0773	211
748	meituan-003	0.0017	79	0.0021	48	0.0024	142	0.0023	98	0.0024	125	0.0084	253	0.0046	151	0.0883	252

Table 73: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)													
		CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
749	meituan-004	0.0014	51	0.0021	46	0.0024	144	0.0024	118	0.0023	111	0.0070	190	0.0044	144	0.0813	233
750	meiya-001	0.0171	711	0.0275	715	0.0159	1041	0.0261	1026	0.0311	1007	0.2250	880	0.0245	745	-	
751	mendaxiatech-000	0.0027	170	0.0036	134	0.0029	351	0.0036	372	0.0031	261	0.0057	132	0.0051	190	-	
752	metsakuurcompany-001	0.0068	451	0.0087	407	0.0068	777	0.0078	703	0.0095	695	0.8972	1108	0.5635	1121	0.9970	781
753	metsakuurcompany-002	0.0048	324	0.0071	339	0.0030	379	0.0043	456	0.0032	279	0.2059	869	0.0665	898	0.5374	568
754	metsakuurcompany-003	0.0034	228	0.0051	246	0.0026	253	0.0036	356	0.0028	207	0.2418	890	0.1001	940	0.5866	587
755	metsakuurcompany-004	-		-		0.0026	278	0.0032	287	0.0032	272	0.4488	972	0.2015	1017	0.8129	648
756	metsakuurcompany-005	-		-		0.0030	380	0.0041	425	0.0039	356	0.1725	845	0.0054	218	0.8274	652
757	metsakuurcompany-006	-		-		0.0034	463	0.0061	608	0.0042	384	0.0896	778	0.0399	832	0.2791	499
758	miaxis-001	0.0068	449	0.0099	452	0.0059	727	0.0097	767	0.0096	703	0.0231	536	0.0109	496	0.1592	408
759	miaxis-002	0.0745	894	0.0774	860	0.3215	1273	0.4000	1263	0.1485	1158	0.2087	870	0.2058	1018	0.3846	533
760	miaxis-003	0.0045	304	0.0067	321	0.0037	527	0.0050	516	0.0061	550	0.0139	393	0.0088	414	0.1322	364
761	miaxis-004	-		-		0.0037	509	0.0046	478	0.0057	521	0.0107	339	0.0084	405	0.1261	358
762	miaxis-005	-		-		0.0033	448	0.0041	435	0.0053	489	0.0106	335	0.0079	375	0.1236	353
763	microfocus-000	0.4656	1023	0.5624	1028	-		-		-		-		-		-	
764	microfocus-001	0.4482	1020	0.5524	1026	0.7256	1330	0.8416	1325	0.7301	1255	0.6926	1034	0.5180	1114	-	
765	microfocus-002	0.3605	1001	0.5057	1016	0.5783	1308	0.7223	1304	0.5909	1241	0.5963	1015	0.4160	1092	-	
766	microfocus-003	0.0416	842	0.0607	836	0.0652	1194	0.1025	1188	0.0793	1116	0.2248	879	0.0923	933	0.6153	595
767	minivision-000	0.0033	219	0.0048	225	0.0038	541	0.0049	500	0.0055	502	0.0094	297	0.0079	376	-	
768	mitek-000	0.6882	1050	0.8192	1058	0.9568	1355	0.9788	1351	0.9545	1303	0.9437	1132	0.8494	1181	0.9931	762
769	mitek-001	-		-		0.0050	657	0.0068	647	0.0053	488	0.0096	304	0.0077	364	0.1130	329
770	mitek-002	-		-		0.0039	557	0.0054	549	0.0044	415	0.0085	260	0.0065	283	0.1112	325
771	mobai-000	0.0360	834	0.0439	792	0.0372	1142	0.0700	1151	0.0367	1029	0.0939	782	0.0795	914	-	
772	mobai-001	0.0199	746	0.0219	661	0.0047	630	0.0061	607	0.0093	691	0.0174	463	0.0138	576	-	
773	mobai-002	-		-		0.0048	640	0.0070	658	0.0088	668	0.5904	1011	0.3167	1063	0.8952	689
774	mobai-003	-		-		0.0053	684	0.0090	745	0.0689	1107	0.9994	1240	0.9982	1273	1.0000	901
775	mobbl-000	0.2938	989	0.3861	993	0.5391	1302	0.6888	1301	0.6545	1249	0.8027	1068	0.6207	1132	-	
776	mobbl-001	0.3208	993	0.4375	1000	0.5680	1306	0.7193	1303	0.6282	1243	0.5783	1006	0.3984	1087	0.8864	684
777	mobbl-002	0.9914	1069	0.9970	1077	0.9355	1352	0.9923	1354	1.0000	1334	1.0000	1283	0.9999	1300	1.0000	894
778	mobbl-003	0.0087	521	0.0134	528	0.0062	745	0.0087	738	0.0099	714	0.0197	497	0.0122	538	0.1618	418
779	mobbl-004	-		-		0.0031	409	0.0043	453	0.0088	663	1.0000	1290	1.0000	1312	1.0000	902
780	mobbl-005	-		-		0.0031	401	0.0044	467	0.0045	419	0.4420	970	0.1465	980	0.7973	644
781	mobipintech-000	0.0090	531	0.0149	553	0.0039	560	0.0057	570	0.0115	761	0.0465	688	0.0182	664	0.6443	600
782	momovn-001	0.1044	920	0.1694	920	0.3253	1275	0.5244	1279	0.3861	1222	0.9995	1244	0.9985	1274	1.0000	904
783	momovn-002	-		-		0.0042	586	0.0060	604	0.0062	553	0.7672	1055	0.5456	1119	0.9731	726
784	momovn-003	-		-		0.0029	353	0.0033	305	0.0056	511	0.0093	288	0.0076	357	0.0612	122
785	moredian-000	0.3874	1007	0.4912	1012	0.9988	1369	0.9999	1369	0.9990	1320	0.9999	1269	0.9998	1298	-	
786	morpho-000	0.1020	917	0.1343	903	-		-		-		-		-		-	
787	morpho-001	-		0.2772	962	-		-		-		-		-		-	
788	morpho-002	0.0377	836	0.0680	846	0.0368	1141	0.0716	1153	-		-		-		-	
789	mukh-001	0.0170	710	0.0285	717	0.0225	1085	0.0405	1084	0.0272	984	0.0950	784	0.0291	786	0.5846	586
790	mukh-002	0.0269	795	0.0357	761	0.0435	1159	0.0799	1165	0.0143	819	0.0213	514	0.0122	537	0.1908	445
791	mukh-003	0.0077	491	0.0113	493	0.0111	945	0.0224	995	0.0061	549	0.7813	1062	0.4504	1098	0.9899	754
792	mukh-004	0.0050	338	0.0069	335	0.0051	664	0.0099	776	0.0045	422	0.0083	246	0.0059	254	0.1030	305

Table 74: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
793	mukh-005	-		-		0.0023	81	0.0024	126	0.0056	510	0.0161	440	0.0067	293	0.1398	379
794	multimodality-000	0.0034	227	0.0047	221	0.0036	504	0.0044	465	0.0077	617	0.9976	1214	0.4456	1097	1.0000	884
795	multimodality-001	0.0029	182	0.0042	179	0.0031	393	0.0035	345	0.0038	347	0.0071	194	0.0059	251	0.0732	182
796	mvision-001	0.0191	735	0.0233	677	0.0204	1069	0.0356	1065	0.0198	919	0.0337	626	0.0242	742	-	
797	nazhiai-000	0.0040	277	0.0059	289	0.0036	492	0.0048	497	0.0057	523	0.0125	379	0.0083	398	-	
798	ncsgroup-000	-		-		0.0126	985	0.0226	997	0.0179	883	0.1715	844	0.0287	781	0.6160	596
799	ncssg-001	0.0207	751	0.1540	913	0.0045	610	0.0067	646	0.0056	515	0.1180	809	0.0301	790	0.3908	536
800	ncssg-002	-		-		0.0032	430	0.0038	391	0.0033	285	0.3144	923	0.1061	945	0.7441	625
801	ncssg-003	-		-		0.0033	449	0.0040	416	0.0033	297	0.2511	896	0.0959	937	0.6553	603
802	ncssg-004	-		-		0.0033	443	0.0041	434	0.0029	233	0.2454	892	0.0860	927	0.6010	589
803	neosystems-001	1.0000	1460	1.0000	1082	0.2987	1270	0.4382	1270	0.5173	1236	0.6570	1027	0.4043	1090	-	
804	neosystems-002	0.2905	987	0.4077	996	0.2028	1254	0.3252	1256	0.4088	1227	0.5519	1001	0.3331	1069	0.9561	712
805	neosystems-003	0.2429	978	0.3349	981	0.1844	1249	0.2999	1253	0.5942	1242	0.3936	957	0.2292	1029	0.8482	659
806	neosystems-004	0.0279	799	0.0495	809	0.0289	1117	0.0585	1135	0.0439	1058	0.9621	1140	0.1296	965	0.9996	807
807	netbridgetech-001	0.4749	1025	0.6599	1044	0.4438	1292	0.5676	1286	0.4491	1230	1.0000	1274	0.9541	1219	-	
808	netbridgetech-002	0.0101	569	0.0166	587	0.0077	817	0.0127	839	0.0133	797	0.8215	1079	0.0523	873	-	
809	neurotechnology-000	-		0.2371	949	-		-		-		-		-		-	
810	neurotechnology-001	0.1393	940	0.2218	946	-		-		-		-		-		-	
811	neurotechnology-002	0.1115	927	0.1663	919	-		-		-		-		-		-	
812	neurotechnology-003	0.0617	876	0.0880	872	-		-		-		-		-		-	
813	neurotechnology-004	0.0146	674	0.0297	725	0.0107	932	0.0164	931	-		-		-		-	
814	neurotechnology-005	0.0141	667	0.0300	728	0.0108	935	0.0163	930	0.0204	922	0.0875	776	0.0246	747	-	
815	neurotechnology-006	0.0098	558	0.0136	532	0.0105	925	0.0182	951	0.0164	856	0.0843	772	0.0214	711	-	
816	neurotechnology-008	0.0091	536	0.0184	617	0.0076	813	0.0116	809	0.0157	842	0.9973	1210	0.1803	1009	-	
817	neurotechnology-009	0.0049	331	0.0087	406	0.0057	715	0.0094	754	0.0079	630	1.0000	1279	0.0101	465	-	
818	neurotechnology-010	0.0048	325	0.0085	403	0.0056	706	0.0090	744	0.0077	620	1.0000	1284	0.0113	507	-	
819	neurotechnology-011	0.0050	339	0.0087	405	0.0061	744	0.0097	766	0.0077	619	0.0404	668	0.0092	431	-	
820	neurotechnology-012	0.0051	347	0.0070	337	0.0038	531	0.0056	568	0.0066	569	0.0112	355	0.0075	350	0.1009	299
821	neurotechnology-013	0.0032	212	0.0045	206	0.0026	279	0.0036	357	0.0037	336	0.0068	184	0.0052	203	0.0747	195
822	neurotechnology-015	0.0022	123	0.0036	141	0.0024	136	0.0028	207	0.0030	237	0.0052	117	0.0041	121	0.0646	150
823	neurotechnology-016	0.0022	122	0.0032	114	0.0023	100	0.0027	181	0.0027	188	0.0049	102	0.0039	106	0.0613	123
824	neurotechnology-017	0.0017	83	0.0031	107	0.0023	79	0.0025	143	0.0025	153	0.0047	90	0.0038	92	0.0612	121
825	neurotechnology-018	0.0020	104	0.0029	93	0.0022	58	0.0024	113	0.0024	129	0.0046	81	0.0037	78	0.0586	105
826	neurotechnology-019	-		-		0.0022	55	0.0024	99	0.0022	86	0.0042	63	0.0034	51	0.0526	66
827	neurotechnology-020	-		-		0.0023	118	0.0028	226	0.0024	115	0.0043	70	0.0035	62	0.0562	90
828	neurotechnology-021	-		-		0.0022	43	0.0023	87	0.0019	55	0.0035	37	0.0029	27	0.0441	30
829	nextmetalabs-001	-		-		0.1126	1226	0.1962	1228	0.1137	1141	1.0000	1289	0.9998	1295	1.0000	905
830	nextmetalabs-002	-		-		0.0066	768	0.0081	721	0.0097	706	0.9946	1195	0.9565	1221	1.0000	854
831	nextmetalabs-003	-		-		0.0023	125	0.0028	219	0.0028	200	0.9681	1147	0.8769	1190	0.9995	806
832	nextmetalabs-004	-		-		0.0023	126	0.0028	218	0.0029	221	0.0055	125	0.0048	164	0.0716	178
833	nhn-001	0.0066	443	0.0098	446	0.0053	682	0.0079	713	0.0093	687	0.0156	430	0.0109	494	-	
834	nhn-002	0.0068	452	0.0096	439	0.0057	717	0.0087	740	0.0136	808	0.0253	554	0.0186	674	0.1464	393
835	nhn-003	0.0033	217	0.0048	227	0.0027	302	0.0038	388	0.0036	328	0.0198	498	0.0071	321	0.1280	360
836	nhn-004	0.0030	187	0.0044	204	0.0025	218	0.0029	237	0.0030	253	0.0331	616	0.0113	510	0.1598	413

Table 75: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
837	nhn-005	0.0028	172	0.0043	192	0.0023	106	0.0027	171	0.0025	138	0.0047	84	0.0038	91	0.0554	85
838	nmetics-001	-		-		0.0026	280	0.0033	315	0.0027	196	0.0052	115	0.0043	131	0.0594	111
839	noblis-000	0.4424	1019	0.5415	1024	-		-		-		-		-		-	
840	nodeflux-000	1.0000	1111	1.0000	1439	1.0000	1429	-		0.9079	1294	-		-		-	
841	nodeflux-001	1.0000	1347	1.0000	1202	1.0000	1462	0.9998	1366	0.5169	1235	1.0000	1332	0.5654	1122	-	
842	nodeflux-002	0.0186	731	0.0340	748	0.0261	1104	0.0451	1097	0.0548	1083	1.0000	1300	1.0000	1317	-	
843	nominder-000	0.0025	157	0.0036	138	0.0029	359	0.0033	310	0.0026	179	0.9919	1184	0.8877	1195	1.0000	864
844	nominder-001	0.0029	184	0.0042	183	0.0031	414	0.0038	390	0.0031	260	0.9975	1212	0.9180	1204	0.9999	830
845	nominder-002	0.0025	156	0.0037	143	0.0033	442	0.0037	378	0.0026	185	0.5785	1007	0.1435	978	0.8381	655
846	nominder-003	-		-		0.0031	390	0.0034	330	0.0026	170	0.7010	1038	0.2159	1024	0.8909	687
847	nominder-004	-		-		0.0028	337	0.0033	306	0.0024	116	0.0316	602	0.0045	149	0.1348	370
848	nominder-005	-		-		0.0025	203	0.0029	232	0.0024	124	0.0046	82	0.0037	82	0.0537	77
849	nominder-006	-		-		0.0027	304	0.0032	290	0.0030	250	0.0059	141	0.0052	197	0.0644	148
850	nominder-007	-		-		0.0027	287	0.0031	271	0.0023	108	0.0072	197	0.0036	68	0.0946	288
851	nominder-008	-		-		0.0026	260	0.0028	224	0.0022	90	0.0069	186	0.0035	65	0.0928	284
852	notiontag-000	0.6669	1048	0.7885	1056	0.3715	1281	0.4978	1278	0.8571	1279	0.8102	1072	0.6460	1141	-	
853	notiontag-001	0.6846	1049	0.8006	1057	0.3955	1283	0.5247	1280	0.8669	1282	0.8313	1085	0.6362	1138	-	
854	notiontag-002	0.0066	439	0.0089	416	0.0045	609	0.0061	609	0.0077	618	0.0137	392	0.0104	479	0.1106	322
855	npt-000	-		-		0.0035	489	0.0050	514	0.0042	395	0.8079	1070	0.5102	1112	0.9695	722
856	nsensecorp-001	0.9909	1068	0.9994	1080	0.9987	1368	0.9999	1368	1.0000	1336	1.0000	1309	1.0000	1322	-	
857	nsensecorp-002	0.4277	1015	0.5375	1021	0.6734	1318	0.7924	1314	0.7194	1253	0.6937	1036	0.5617	1120	0.9392	706
858	nsensecorp-003	0.0251	783	0.0295	724	0.0212	1076	0.0305	1048	0.0131	792	0.2139	873	0.0141	584	0.9968	779
859	nsensecorp-004	0.1370	939	0.1397	905	0.0066	767	0.0094	755	1.0000	1395	1.0000	1444	1.0000	1365	1.0000	1004
860	nsensecorp-005	0.0055	371	0.0080	384	0.0039	552	0.0058	586	0.2723	1203	0.9999	1268	0.9949	1259	1.0000	915
861	ntechlab-000	-		0.0862	870	-		-		-		-		-		-	
862	ntechlab-001	-		0.0825	866	-		-		-		-		-		-	
863	ntechlab-002	0.0558	871	0.0649	839	-		-		-		-		-		-	
864	ntechlab-003	0.0286	803	0.0390	775	-		-		-		-		-		-	
865	ntechlab-004	0.0090	529	0.0132	525	-		-		-		-		-		-	
866	ntechlab-005	0.0093	543	0.0145	546	0.0131	994	0.0275	1034	-		-		-		-	
867	ntechlab-006	0.0078	495	0.0111	485	0.0112	951	0.0227	1000	0.0108	739	0.0149	416	0.0098	456	-	
868	ntechlab-007	0.0056	375	0.0076	361	0.0073	803	0.0128	843	0.0079	629	0.0144	403	0.0102	467	-	
869	ntechlab-008	0.0041	283	0.0061	295	0.0056	702	0.0108	793	0.0042	392	0.0080	229	0.0063	274	-	
870	ntechlab-009	0.0039	266	0.0054	268	0.0042	590	0.0063	620	0.0275	987	0.0674	746	0.0532	874	-	
871	ntechlab-010	0.0013	50	0.0017	32	0.0024	170	0.0029	235	0.0031	262	0.0058	140	0.0050	181	0.0656	157
872	ntechlab-011	0.0012	40	0.0019	42	0.0024	161	0.0028	222	0.0029	229	0.0055	123	0.0047	157	0.0630	135
873	ntechlab-012	0.0011	37	0.0016	28	0.0023	122	0.0030	249	0.0026	173	0.0050	110	0.0043	132	0.0587	107
874	ntnu-000	-		-		0.0040	566	0.0066	637	0.0069	581	0.9942	1194	0.9744	1235	0.9999	832
875	ocrstudio-001	-		-		0.4396	1291	0.6256	1294	0.3190	1210	0.3128	922	0.1754	1004	0.7785	635
876	omface-000	0.2573	983	0.3835	992	0.3590	1279	0.4903	1276	0.3956	1224	0.5003	981	0.2595	1041	0.9137	695
877	omface-001	0.0137	659	0.0212	653	0.0114	961	0.0187	961	0.0174	880	1.0000	1442	0.0214	713	1.0000	971
878	omnigarde-000	0.0633	881	0.1002	885	0.1109	1224	0.2042	1232	0.1288	1146	0.5113	984	0.1227	963	0.9604	716
879	omnigarde-001	0.0168	707	0.0260	704	0.0203	1067	0.0402	1080	0.0243	965	0.0327	612	0.0177	655	0.2164	465
880	omnigarde-002	0.0033	223	0.0046	217	0.0027	316	0.0039	397	0.0041	379	0.0076	208	0.0059	257	0.0806	228

Table 76: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
881	omnigarde-003	0.0024	137	0.0038	152	0.0024	177	0.0029	227	0.0032	281	0.0059	144	0.0049	176	0.0706	176		
882	omnigarde-004	-		-		0.0022	52	0.0024	102	0.0023	101	0.0044	74	0.0038	93	0.0559	88		
883	omnigarde-005	-		-		0.0022	64	0.0024	111	0.0024	121	0.0045	78	0.0039	105	0.0582	102		
884	onfido-000	0.1472	944	0.2881	968	0.0335	1133	0.0731	1156	0.0515	1076	0.9915	1181	0.9579	1223	1.0000	835		
885	onfido-001	-		-		0.0146	1020	0.0294	1043	0.0190	903	0.0702	757	0.0352	811	0.3117	511		
886	onfido-002	-		-		0.0067	769	0.0098	771	0.0062	555	0.0167	453	0.0091	429	0.1228	352		
887	onfido-003	-		-		0.0045	614	0.0070	657	0.0043	400	0.0164	448	0.0076	358	0.1153	335		
888	onfido-004	-		-		0.0033	458	0.0047	485	0.0032	283	0.0305	592	0.0086	412	0.1450	392		
889	oorudigital-000	-		-		0.0106	928	0.0190	964	0.0146	826	0.9930	1189	0.9585	1224	1.0000	849		
890	ooto-001	-		-		0.0193	1058	0.0290	1041	0.0529	1081	0.0272	567	0.0255	751	0.1956	451		
891	openedge-000	0.0045	305	0.0062	307	0.0049	645	0.0069	654	0.0049	462	0.9889	1176	0.7991	1174	0.9999	825		
892	openface-001	0.1804	958	0.2921	970	0.2878	1266	0.3906	1262	0.2054	1182	0.2338	886	0.1549	988	0.6531	602		
893	ovisionllc-001	-		-		0.0058	722	0.0081	715	0.8171	1271	1.0000	1311	1.0000	1314	1.0000	880		
894	ovisionllc-002	-		-		0.1963	1251	0.2522	1240	1.0000	1340	1.0000	1314	1.0000	1326	1.0000	891		
895	oz-001	0.0133	651	0.0215	659	0.0109	937	0.0160	919	0.0235	956	1.0000	1441	1.0000	1374	-			
896	oz-002	0.0071	464	0.0099	457	0.0099	910	0.0100	777	0.0139	811	0.0502	702	0.0202	698	-			
897	oz-003	0.0095	554	0.0143	541	0.0054	691	0.0077	696	0.0096	702	0.0175	467	0.0118	524	0.1404	381		
898	oz-004	0.0033	224	0.0049	236	0.0038	547	0.0055	557	0.0081	638	0.0163	447	0.0142	587	-			
899	pa-002	0.1847	960	0.2859	966	-		-		-		-		-		-			
900	pa-003	0.0453	852	0.0679	845	-		-		-		-		-		-			
901	palit-000	0.0062	419	0.0084	398	0.0039	550	0.0055	554	0.0055	505	0.4610	973	0.2468	1031	0.8695	671		
902	palit-001	0.0050	332	0.0068	331	0.0032	436	0.0047	489	0.0045	426	0.0110	346	0.0058	248	0.1446	389		
903	pangiam-000	0.0031	200	0.0043	197	0.0026	238	0.0030	264	0.0038	348	0.0071	196	0.0061	264	0.0667	164		
904	pangiam-001	0.0031	198	0.0044	203	0.0029	352	0.0040	419	0.0028	213	0.0362	646	0.0056	230	0.1711	427		
905	pangiam-002	0.0030	193	0.0042	182	0.0026	272	0.0031	277	0.0030	247	0.1772	846	0.0385	824	0.6076	591		
906	papago-001	0.0067	447	0.0096	442	0.0051	670	0.0077	694	0.0071	590	0.0126	380	0.0086	410	0.1033	307		
907	papil11-000	0.0021	114	0.0030	103	0.0024	129	0.0024	121	0.0025	157	0.0047	87	0.0040	112	0.0536	74		
908	papil11-001	-		-		0.0023	123	0.0024	119	0.0023	97	0.0185	480	0.0040	119	0.1449	390		
909	papil11-002	-		-		0.0023	119	0.0024	114	0.0023	96	0.0130	386	0.0039	101	0.1112	326		
910	papil11-003	-		-		0.0023	110	0.0024	115	0.0023	106	0.0042	65	0.0035	63	0.0529	68		
911	papil11-004	-		-		0.0027	297	0.0029	238	0.0025	158	0.0153	424	0.0058	247	0.0972	292		
912	papsav1923-001	0.0078	496	0.0130	522	0.0068	775	0.0105	787	0.0119	767	0.0221	524	0.0136	569	-			
913	papsav1923-002	0.0021	115	0.0034	121	0.0026	240	0.0030	262	0.0048	458	0.0093	287	0.0086	408	0.0630	134		
914	papsav1923-003	0.0025	154	0.0035	129	0.0024	173	0.0025	131	0.0034	300	0.0066	175	0.0058	246	0.0603	115		
915	paravision-004	0.0030	191	0.0046	213	0.0030	373	0.0036	361	0.0091	680	0.0188	484	0.0173	648	-			
916	paravision-006	0.0016	70	0.0025	68	0.0026	250	0.0026	168	0.0075	609	0.9933	1190	0.0142	585	-			
917	paravision-008	0.0018	86	0.0025	66	0.0024	149	0.0025	141	0.0036	327	0.0070	192	0.0063	273	0.0747	196		
918	paravision-010	0.0012	41	0.0021	47	0.0022	51	0.0021	41	0.0027	195	0.0055	124	0.0050	183	0.0534	73		
919	paravision-011	0.0008	15	0.0020	45	0.0021	33	0.0020	22	0.0026	174	0.0053	118	0.0049	175	0.0498	54		
920	paravision-013	0.0010	32	0.0018	37	0.0021	24	0.0019	4	0.0019	58	0.0041	61	0.0036	72	0.0442	31		
921	paravision-015	-		-		0.0021	31	0.0021	46	0.0021	69	0.0043	67	0.0039	97	0.0528	67		
922	paravision-017	-		-		0.0021	21	0.0021	30	0.0019	47	0.0038	45	0.0035	58	0.0458	36		
923	paravision-018	-		-		0.0020	7	0.0020	20	0.0016	19	0.0034	29	0.0029	30	0.0414	16		
924	paravision-020	-		-		0.0020	2	0.0020	13	0.0015	15	0.0033	22	0.0028	19	0.0410	14		

Table 77: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
925	paravision-022	-		-		0.0020	4	0.0019	3	0.0015	10	0.0032	18	0.0028	17	0.0414	15		
926	pensees-001	0.0087	523	0.0133	526	0.0071	786	0.0122	825	0.0145	823	0.0252	553	0.0195	690	-			
927	pentacomp-000	-		-		0.0041	579	0.0061	606	0.0047	447	0.3501	937	0.1211	961	0.7162	619		
928	pixelall-002	0.0193	738	0.0340	750	0.0127	988	0.0209	982	0.0234	955	0.9182	1113	0.2728	1047	-			
929	pixelall-003	0.0074	479	0.0118	502	0.0057	712	0.0079	708	0.0121	771	0.0390	664	0.0170	640	-			
930	pixelall-004	0.0040	275	0.0061	293	0.0055	697	0.0069	652	0.0100	720	0.5663	1004	0.0371	821	-			
931	pixelall-005	0.0038	260	0.0052	252	0.0043	596	0.0051	527	0.0077	616	0.0839	771	0.0136	571	-			
932	pixelall-006	0.0032	208	0.0042	180	0.0032	424	0.0039	399	0.0063	557	0.9960	1203	0.0723	907	1.0000	867		
933	pixelall-007	0.0036	253	0.0049	233	0.0039	549	0.0044	458	0.0068	574	0.9873	1170	0.0217	716	1.0000	843		
934	pixelall-008	0.0015	57	0.0023	60	0.0034	474	0.0049	499	0.0031	259	0.0057	131	0.0052	194	0.0682	168		
935	pixelall-009	0.0018	85	0.0025	69	0.0024	165	0.0026	162	0.0031	269	0.3475	936	0.0053	208	0.9006	692		
936	pixelall-010	0.0020	101	0.0030	105	0.0025	209	0.0028	225	0.0031	264	0.3512	938	0.0054	212	0.8813	680		
937	pnpsecure-000	-		-		0.0190	1055	0.0274	1033	0.9980	1315	0.9997	1254	0.9996	1290	0.9997	810		
938	pnpsecure-001	-		-		0.0050	659	0.0040	415	0.0037	344	0.0166	452	0.0075	349	0.1059	310		
939	pnpsecure-002	-		-		0.0025	211	0.0028	214	0.0031	266	0.5367	996	0.2784	1049	0.8498	661		
940	privid-001	0.3350	997	0.5013	1014	0.4327	1289	0.5880	1287	0.9790	1307	1.0000	1291	0.9998	1296	1.0000	929		
941	privid-002	0.0460	854	0.0652	841	0.0524	1176	0.0882	1172	0.0671	1104	0.9769	1154	0.8960	1196	0.9999	819		
942	privid-003	-		-		0.0030	376	0.0039	396	0.0045	428	0.9996	1250	0.9985	1275	1.0000	878		
943	privid-004	-		-		0.0026	273	0.0028	205	0.0034	301	0.9939	1192	0.9657	1230	1.0000	855		
944	psl-000	0.0216	754	0.0333	745	-		-		-		-		-		-			
945	psl-001	0.0549	868	0.0927	878	0.0096	896	0.0149	898	-		-		-		-			
946	psl-002	0.0107	589	0.0180	612	0.0089	868	0.0120	818	0.0790	1115	0.0270	566	0.0192	681	-			
947	psl-003	0.0065	438	0.0099	451	0.0055	694	0.0075	686	1.0000	1337	0.0188	485	0.0129	553	-			
948	psl-004	0.0064	428	0.0094	437	0.0035	480	0.0050	522	0.0084	645	0.0145	408	0.0106	487	-			
949	psl-005	0.0060	407	0.0094	429	0.0034	475	0.0048	495	0.0081	637	0.0140	396	0.0103	476	-			
950	psl-006	0.0025	153	0.0039	164	0.0028	333	0.0030	257	0.0160	846	0.0482	695	0.0310	792	-			
951	psl-007	0.0026	163	0.0040	167	0.0027	291	0.0030	253	0.0054	493	0.0101	320	0.0081	387	-			
952	psl-008	0.0026	159	0.0040	169	0.0024	159	0.0028	217	0.0041	381	0.0077	211	0.0055	219	0.0719	179		
953	psl-009	0.0161	699	0.0294	723	0.0023	95	0.0025	133	0.0036	330	0.0065	173	0.0048	170	0.0659	159		
954	psl-010	0.0017	84	0.0035	128	0.0023	87	0.0025	127	0.0035	318	0.0104	326	0.0052	202	0.0994	295		
955	psl-011	0.0013	47	0.0026	77	0.0021	10	0.0021	36	0.0024	119	0.0047	86	0.0035	66	0.0565	93		
956	psl-012	0.0017	75	0.0027	85	0.0021	36	0.0023	71	0.0021	70	0.0039	48	0.0032	45	0.0494	50		
957	psl-013	-		-		0.0022	44	0.0023	73	0.0021	77	0.0039	52	0.0033	47	0.0499	58		
958	psl-014	-		-		0.0021	37	0.0023	70	0.0021	75	0.0039	51	0.0033	46	0.0498	55		
959	ptakuratsatu-000	0.0060	405	0.0089	417	0.0070	783	0.0104	786	0.0096	705	0.0152	423	0.0100	462	-			
960	pxl-001	0.0488	857	0.0752	854	0.0586	1186	0.1087	1190	0.0946	1131	0.1065	796	0.0625	894	-			
961	pyramid-000	0.0136	655	0.0233	679	0.0117	966	0.0192	967	0.0185	895	0.0322	609	0.0206	704	-			
962	pyramid-001	-		-		0.0025	187	0.0030	259	0.0025	146	0.0099	311	0.0044	139	0.1031	306		
963	qazbs-000	0.0058	386	0.0083	394	0.0046	625	0.0072	668	0.0130	789	0.0244	546	0.0196	691	0.7784	634		
964	qazsmartvisionai-000	0.0008	16	0.0011	12	0.0021	14	0.0022	51	0.0018	43	0.0035	36	0.0030	34	0.0441	29		
965	qazsmartvisionai-001	0.0003	1	0.0008	8	0.0020	8	0.0020	27	0.0015	14	0.0031	13	0.0027	13	0.0394	11		
966	qazsmartvisionai-002	-		-		0.0020	3	0.0020	11	0.0014	3	0.0028	3	0.0025	4	0.0380	5		
967	qazsmartvisionai-003	-		-		0.0020	5	0.0020	12	0.0015	5	0.0029	7	0.0025	6	0.0381	7		
968	qazsmartvisionai-004	-		-		0.0020	6	0.0020	17	0.0014	2	0.0028	2	0.0025	2	0.0377	4		

Table 78: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
969	qluevision-001	0.0223	761	0.0419	781	0.0205	1071	0.0343	1060	0.0327	1015	0.8762	1097	0.7413	1161	0.9908	756		
970	qnap-000	0.0149	680	0.0228	672	0.0155	1034	0.0267	1029	0.0238	963	0.8329	1086	0.0396	829	0.9992	796		
971	qnap-001	0.0148	677	0.0215	658	0.0103	918	0.0162	926	0.0183	894	0.0301	587	0.0186	673	0.1995	454		
972	qnap-002	0.0122	623	0.0191	628	0.0075	812	0.0095	759	0.0146	824	0.0281	577	0.0184	666	0.1829	437		
973	qnap-003	0.0637	882	0.0657	843	0.0058	720	0.0078	702	0.0082	640	0.9985	1225	0.9658	1231	1.0000	886		
974	qnap-004	0.0046	319	0.0061	294	0.0038	535	0.0059	593	0.0042	390	0.9960	1202	0.9442	1213	1.0000	850		
975	qnap-005	0.0033	225	0.0047	222	0.0068	773	0.0071	663	0.0096	701	0.0184	479	0.0163	626	0.0948	289		
976	qnap-006	-		-		0.0030	382	0.0041	430	0.0044	414	0.0088	273	0.0065	285	0.0799	224		
977	quantasoft-003	0.0081	507	0.0113	494	0.0056	705	0.0076	689	0.0091	679	0.0161	442	0.0107	491	0.1193	342		
978	rankone-000	-		0.2760	960	-		-		-		-		-		-			
979	rankone-001	-		0.4198	998	-		-		-		-		-		-			
980	rankone-002	0.1547	948	0.2170	943	-		-		-		-		-		-			
981	rankone-003	0.1175	930	0.1838	929	-		-		-		-		-		-			
982	rankone-004	0.0548	867	0.1015	886	-		-		-		-		-		-			
983	rankone-005	0.0405	841	0.0757	856	0.0170	1046	0.0299	1045	-		-		-		-			
984	rankone-006	0.0242	770	0.0460	797	0.0119	970	0.0188	963	0.0315	1011	0.0539	711	0.0385	826	-			
985	rankone-007	0.0197	742	0.0366	766	0.0113	954	0.0177	946	0.0286	994	0.0454	682	0.0333	801	-			
986	rankone-008	0.0124	629	0.0232	676	0.0082	840	0.0107	789	0.0188	900	0.0320	608	0.0244	744	-			
987	rankone-009	0.0087	522	0.0119	503	0.0065	762	0.0086	736	0.0088	664	0.0161	439	0.0121	535	-			
988	rankone-010	0.0079	499	0.0112	489	0.0061	738	0.0081	719	0.0088	661	0.0149	413	0.0117	522	-			
989	rankone-011	0.0049	326	0.0075	359	0.0038	530	0.0048	496	0.0060	542	0.0143	402	0.0080	382	0.1839	438		
990	rankone-012	0.0043	293	0.0058	279	0.0031	418	0.0038	386	0.0047	446	0.0081	235	0.0065	281	0.0842	243		
991	rankone-013	0.0028	174	0.0041	171	0.0026	242	0.0033	308	0.0028	219	0.0055	127	0.0040	111	0.0725	180		
992	rankone-014	0.0016	72	0.0021	49	0.0024	133	0.0027	179	0.0022	93	0.0047	88	0.0035	60	0.0666	163		
993	rankone-015	0.0009	22	0.0014	20	0.0024	140	0.0026	158	0.0024	131	0.0044	76	0.0037	74	0.0635	141		
994	realnetworks-000	0.0325	823	0.0463	798	-		-		-		-		-		-			
995	realnetworks-001	0.0315	820	0.0455	795	0.0920	1214	0.1938	1227	-		-		-		-			
996	realnetworks-002	0.0248	776	0.0358	762	0.0513	1174	0.1127	1196	0.0371	1031	0.0614	732	0.0316	795	-			
997	realnetworks-003	0.0259	789	0.0372	769	0.0541	1180	0.1208	1200	0.0378	1034	0.0578	721	0.0306	791	-			
998	realnetworks-004	0.0075	484	0.0101	459	0.0066	765	0.0097	768	0.0108	745	0.0187	482	0.0131	555	0.1430	386		
999	realnetworks-005	0.0070	454	0.0093	428	0.0063	750	0.0089	743	0.0092	684	0.0161	443	0.0104	480	0.1249	356		
1000	realnetworks-006	0.0040	273	0.0056	274	0.8657	1347	0.9705	1346	0.0059	535	0.0112	352	0.0085	406	0.0911	272		
1001	realnetworks-007	0.0031	203	0.0051	248	0.0028	330	0.0035	349	0.0048	459	0.0091	279	0.0074	339	0.0945	287		
1002	realnetworks-008	0.0022	125	0.0039	158	0.0038	532	0.0045	469	0.0055	499	0.0100	318	0.0080	386	0.0889	255		
1003	realsense-001	-		-		0.0023	86	0.0025	145	0.0023	112	0.0046	83	0.0039	96	0.0537	75		
1004	realsense-002	-		-		0.0022	57	0.0023	88	0.0023	100	0.0045	77	0.0038	87	0.0531	70		
1005	rebs-000	0.0030	194	0.0043	194	0.0024	147	0.0032	289	0.0024	117	0.3069	916	0.1170	955	0.7667	629		
1006	rebs-001	0.0037	254	0.0049	239	0.0024	172	0.0033	304	0.0025	139	0.0257	559	0.0096	451	0.1487	398		
1007	recognito-000	0.0004	5	0.0006	2	0.0021	32	0.0022	63	0.0016	22	0.0555	713	0.0207	706	0.1943	448		
1008	recognito-001	0.0003	2	0.0007	5	0.0021	22	0.0022	56	0.0016	16	0.0663	743	0.0257	752	0.1825	436		
1009	regula-000	0.0184	729	0.0376	770	0.0103	919	0.0185	956	0.0120	769	0.9983	1221	0.0231	731	1.0000	889		
1010	regula-001	0.0072	473	0.0107	478	0.0102	916	0.0179	948	0.0123	777	0.0333	621	0.0174	650	0.3711	528		
1011	regula-002	-		-		0.0053	677	0.0058	583	0.0044	418	0.9999	1266	0.9995	1285	1.0000	890		
1012	regula-003	-		-		0.0056	701	0.0059	594	0.0060	544	0.5284	993	0.4298	1094	0.8213	651		

Table 79: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)													
		CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1013	remarkai-000	0.0147	676	0.0257	702	0.0102	913	0.0158	914	0.0163	855	0.0577	719	0.0185	668	-	
1014	remarkai-001	0.0144	671	0.0256	700	0.0102	914	0.0159	917	0.0162	853	0.0582	724	0.0185	670	-	
1015	remarkai-002	0.0151	684	0.0197	640	0.0075	810	0.0108	795	0.0119	768	0.0187	483	0.0127	545	-	
1016	remarkai-003	0.0047	322	0.0063	310	0.0033	454	0.0049	503	0.0054	496	0.0100	317	0.0072	328	0.0908	269
1017	rendip-000	0.0055	372	0.0077	365	0.0048	642	0.0060	603	0.0080	633	0.0142	401	0.0110	500	0.1027	304
1018	revealmedia-005	0.0050	340	0.0074	354	0.0050	655	0.0068	650	0.0075	610	0.0124	376	0.0104	483	0.0895	260
1019	revealmedia-006	0.0040	276	0.0067	322	0.0041	576	0.0056	563	0.0056	508	0.0085	262	0.0068	297	0.0872	249
1020	roc-016	0.0009	21	0.0014	21	0.0023	91	0.0024	106	0.0024	122	0.0042	66	0.0035	59	0.0558	87
1021	roc-017	-		-		0.0023	82	0.0025	147	0.0021	76	0.0039	47	0.0032	42	0.0542	79
1022	roc-018	0.0007	13	0.0012	13	0.0022	53	0.0022	58	0.0018	44	0.0034	32	0.0029	25	0.0493	49
1023	roc-019	-		-		0.0021	40	0.0021	35	0.0016	23	0.0032	15	0.0027	15	0.0439	27
1024	roc-020	-		-		0.0021	17	0.0021	28	0.0016	21	0.0032	17	0.0027	14	0.0452	33
1025	roc-021	-		-		0.0021	11	0.0020	19	0.0015	11	0.0030	12	0.0026	9	0.0409	13
1026	rokid-000	0.0093	545	0.0145	544	0.0073	801	0.0102	782	0.0164	857	0.0280	573	0.0214	712	-	
1027	rokid-001	0.0105	584	0.0162	579	0.0094	890	0.0163	928	0.0181	887	0.0276	572	0.0165	634	-	
1028	s1-001	0.0314	819	0.0651	840	0.0252	1099	0.0357	1066	0.0444	1059	0.0653	739	0.0429	844	-	
1029	s1-002	0.0095	553	0.0144	543	0.0112	950	0.0196	972	0.0234	953	0.0371	653	0.0282	775	-	
1030	s1-003	0.0051	346	0.0073	350	0.0044	604	0.0063	622	0.0052	484	0.0096	303	0.0070	313	0.0870	248
1031	s1-004	0.0053	360	0.0080	383	0.0038	533	0.0059	591	0.0057	518	0.0103	323	0.0073	334	0.0899	264
1032	s1-005	0.0024	143	0.0036	139	0.0025	225	0.0029	236	0.0026	182	0.0048	98	0.0038	89	0.0638	143
1033	s1-006	0.0029	181	0.0044	201	0.0028	321	0.0033	302	0.0035	315	0.0073	198	0.0044	147	0.7160	618
1034	s1-007	0.0023	136	0.0056	273	0.0024	134	0.0028	195	0.0025	151	0.0048	94	0.0039	98	0.0608	118
1035	s1-008	-		-		0.0093	882	0.0137	864	0.0179	885	0.9118	1111	0.0286	780	0.9999	822
1036	s1-009	-		-		0.0023	124	0.0023	94	0.0022	85	0.0043	68	0.0037	75	0.0504	61
1037	saffe-001	0.4339	1016	0.5261	1019	0.7539	1337	0.8736	1335	0.7977	1266	0.9810	1158	0.7435	1162	-	
1038	saffe-002	0.0119	618	0.0206	645	0.0107	934	0.0177	944	0.0244	966	0.9998	1255	0.2785	1050	-	
1039	samsungsds-000	0.0046	315	0.0069	332	0.0132	996	0.0081	716	0.0099	715	0.0179	471	0.0162	625	0.1101	319
1040	samsungsds-001	0.0015	63	0.0026	74	0.0023	96	0.0023	93	0.0024	123	0.1660	842	0.0536	875	0.4995	563
1041	samsungsds-002	0.0017	80	0.0027	80	0.0023	89	0.0022	52	0.0021	73	0.0043	69	0.0036	71	0.0586	106
1042	samtech-000	0.3418	999	0.4425	1002	-		-		-		-		-		-	
1043	samtech-001	0.0197	743	0.0365	764	0.0146	1024	0.0241	1010	0.0238	962	0.0394	665	0.0251	749	-	
1044	samtech-002	0.0084	515	0.0127	519	0.0093	880	0.0158	912	0.0113	758	0.0442	676	0.0132	556	0.4110	542
1045	scanalitix-001	-		-		0.9968	1365	0.9997	1364	1.0000	1330	1.0000	1297	1.0000	1313	1.0000	912
1046	scanovate-001	0.0175	716	0.0331	744	0.0163	1044	0.0248	1012	0.2476	1197	0.3801	952	0.3740	1078	-	
1047	scanovate-002	0.0175	717	0.0355	759	0.0146	1021	0.0286	1039	0.0269	978	0.0301	586	0.0178	658	-	
1048	scanovate-003	0.0054	365	0.0080	382	0.0054	685	0.0072	673	0.0312	1008	0.0599	728	0.0568	887	-	
1049	sdc-000	0.0303	811	0.0526	817	0.0572	1184	0.1094	1192	0.0867	1126	0.6230	1019	0.3682	1077	0.8797	678
1050	seamfix-001	-		-		0.1973	1252	0.2878	1248	0.2613	1200	0.3156	924	0.1976	1015	0.8176	650
1051	securifai-001	0.4538	1021	0.6142	1037	0.5844	1310	0.7428	1306	0.7051	1252	0.9961	1204	0.9558	1220	-	
1052	securifai-002	0.7557	1054	0.8574	1059	0.4550	1294	0.5953	1288	0.9860	1309	0.9796	1157	0.9158	1203	-	
1053	securifai-003	0.4086	1010	0.7577	1054	0.7233	1328	0.8070	1317	0.7787	1263	1.0000	1306	0.9988	1278	1.0000	907
1054	securifai-004	0.0136	654	0.0192	629	0.0064	755	0.0099	775	0.0115	760	0.0272	568	0.0127	547	0.4881	559
1055	securifai-005	0.0125	635	0.0190	626	0.0080	832	0.0126	833	0.0134	801	0.9861	1165	0.9205	1205	1.0000	841
1056	securifai-006	0.0140	666	0.0196	636	0.0067	771	0.0102	779	0.0113	756	0.9888	1174	0.9239	1207	1.0000	846

Table 80: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)													
	Algorithm	CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1057	securifai-007	0.0124	630	0.0175	604	0.0064	757	0.0105	788	0.0108	738	0.9999	1265	0.9966	1266	1.0000	908
1058	securifai-008	-		-		0.0060	736	0.0102	781	0.2096	1186	0.9649	1144	0.8757	1188	0.9990	794
1059	securitypitch-000	-		-		0.0286	1115	0.0382	1076	0.0393	1039	0.9979	1217	0.9881	1248	1.0000	879
1060	sensetime-001	0.0063	427	0.0092	426	0.0130	993	-		0.0252	972	0.0886	777	0.0886	931	-	
1061	sensetime-002	0.0068	450	0.0098	447	0.0143	1015	1.0000	1379	0.0278	991	0.0502	701	0.0502	867	-	
1062	sensetime-003	0.0021	116	0.0027	82	0.0027	283	0.0027	188	0.0051	479	0.0100	316	0.0089	418	-	
1063	sensetime-004	0.0026	158	0.0038	154	0.0022	60	0.0023	78	0.0042	387	0.0082	240	0.0078	368	-	
1064	sensetime-005	0.0019	95	0.0029	95	0.0022	49	0.0021	38	0.0023	104	0.0044	72	0.0039	94	0.0564	91
1065	sensetime-006	0.0014	54	0.0024	62	0.0021	35	0.0020	15	0.0021	67	0.0040	56	0.0036	70	0.0525	65
1066	sensetime-007	0.0012	43	0.0022	53	0.0021	38	0.0020	21	0.0018	40	0.0034	27	0.0029	32	0.0479	46
1067	sensetime-008	0.0008	14	0.0014	19	0.0021	18	0.0020	7	0.0018	46	0.0036	38	0.0033	48	0.0504	60
1068	sensetime-010	-		-		0.0022	42	0.0021	34	0.0018	37	0.0034	30	0.0029	31	0.0437	24
1069	sensetime-011	-		-		0.0022	41	0.0020	26	0.0017	28	0.0033	24	0.0029	23	0.0438	26
1070	sensetime-012	-		-		0.0022	50	0.0023	80	0.0015	9	0.0029	9	0.0026	8	0.0372	1
1071	serendipity-000	0.0046	312	0.0069	334	0.0042	591	0.0056	562	0.0073	600	0.9129	1112	0.7189	1158	0.9952	771
1072	sertis-000	0.0118	616	0.0208	648	0.0080	828	0.0127	836	0.0110	751	0.0176	469	0.0114	511	-	
1073	sertis-001	0.0113	603	0.0182	614	0.0204	1070	0.0403	1083	0.0216	934	0.0763	764	0.0282	774	-	
1074	sertis-002	0.0049	327	0.0061	292	0.0039	558	0.0061	614	0.0055	501	0.0099	314	0.0070	315	0.0921	277
1075	sertis-003	0.0049	330	0.0059	291	0.0037	520	0.0058	582	0.0055	507	0.0100	315	0.0072	329	0.0881	251
1076	seventhsense-000	0.0067	448	0.0099	455	0.0045	607	0.0065	634	0.0093	688	0.0169	459	0.0124	541	0.1325	366
1077	seventhsense-001	0.0034	235	0.0047	223	0.0025	224	0.0031	273	0.0029	228	0.0338	627	0.0109	495	0.2135	463
1078	seventhsense-002	0.0036	251	0.0050	243	0.0028	329	0.0036	359	0.0035	316	0.0811	768	0.0183	665	0.5444	572
1079	seventhsense-003	0.0035	240	0.0046	214	0.0026	265	0.0033	303	0.0062	551	0.0448	680	0.0175	652	0.2594	488
1080	seventhsense-005	0.0031	199	0.0045	208	0.0028	324	0.0030	242	0.0028	211	0.0466	689	0.0165	635	0.1949	449
1081	seventhsense-006	-		-		0.0029	355	0.0034	335	0.0035	324	0.2052	867	0.0802	917	0.4742	555
1082	shaman-000	0.9297	1065	0.9774	1073	0.9990	1370	0.9997	1365	0.9999	1327	1.0000	1286	0.9999	1306	-	
1083	shaman-001	0.3346	996	0.4616	1004	0.2368	1261	0.3723	1261	0.3574	1218	0.3527	939	0.2304	1030	-	
1084	shu-001	0.0103	574	0.0140	538	0.0293	1122	0.0688	1149	0.0172	876	0.0182	475	0.0120	531	-	
1085	shu-002	-		0.0079	377	0.0146	1023	0.0308	1049	1.0000	1333	0.0183	476	0.0115	514	-	
1086	shu-003	0.0028	173	0.0041	177	0.0050	652	0.0088	741	0.0081	639	0.0133	387	0.0094	441	-	
1087	shufti-001	-		-		0.0053	676	0.0083	724	0.0085	650	1.0000	1395	0.0098	457	1.0000	1121
1088	siat-002	0.0091	537	0.0126	518	0.0109	938	0.0190	965	0.0276	990	0.0516	705	0.0464	859	-	
1089	siat-004	0.0067	445	0.0099	453	0.0152	1032	1.0000	1378	0.0275	985	0.4823	978	0.4823	1102	-	
1090	siat-005	0.0021	110	0.0038	153	0.0059	725	0.0049	506	0.0742	1110	0.9623	1141	0.6801	1143	0.9833	739
1091	sightropic-000	-		-		0.0028	344	0.0038	393	0.0029	224	0.2207	875	0.0810	919	0.5137	565
1092	sightropic-001	-		-		0.0052	675	0.0049	507	0.0050	471	0.0094	294	0.0081	390	0.0672	165
1093	sightropic-002	-		-		0.0027	308	0.0037	377	0.0028	201	0.0051	112	0.0038	85	0.0599	113
1094	sjtu-001	0.0051	345	0.0080	381	0.0211	1074	0.0446	1096	0.0131	791	0.0175	465	0.0110	498	-	
1095	sjtu-002	0.0053	363	0.0078	374	0.0138	1004	0.0296	1044	0.0084	646	0.0136	390	0.0095	447	-	
1096	sjtu-003	0.0017	73	0.0033	116	0.0030	377	0.0037	375	0.0058	526	0.0104	327	0.0081	388	-	
1097	sjtu-004	0.0014	53	0.0025	65	0.0027	292	0.0028	223	0.0046	435	0.0086	266	0.0073	333	0.0906	267
1098	sjtu-005	0.0029	179	0.0036	140	0.0025	229	0.0027	182	0.0044	410	0.0120	373	0.0072	324	0.1325	365
1099	sjtu-006	-		-		0.0027	300	0.0028	202	0.0044	417	0.0081	238	0.0071	322	0.0715	177
1100	sjtu-007	-		-		0.0023	120	0.0023	68	0.0040	366	0.0251	551	0.0126	544	0.1694	426

Table 81: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)																	
	Algorithm	CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1101	sjtu-008	-		-		0.0022	47	0.0023	72	0.0029	220	0.0078	222	0.0055	224	0.0748	197
1102	sjtu-009	-		-		0.0022	48	0.0021	49	0.0028	204	0.5621	1002	0.0397	831	0.6795	609
1103	sktelecom-000	0.0038	264	0.0054	265	0.0031	389	0.0051	529	0.0042	385	0.3418	933	0.0061	263	0.9499	709
1104	smargtech-001	-		-		0.4604	1296	0.6005	1290	0.5247	1237	0.5858	1009	0.3098	1060	0.9826	738
1105	smartbiometrik-001	0.5485	1037	0.6442	1040	0.7550	1338	0.8611	1331	0.8677	1283	0.8270	1083	0.7030	1148	0.9752	729
1106	smartengines-000	0.6240	1044	0.7562	1053	0.9552	1354	0.9784	1350	0.9515	1302	0.9288	1124	0.8200	1176	0.9930	761
1107	smartengines-001	0.6434	1046	0.7666	1055	0.9446	1353	0.9750	1349	0.9387	1300	0.9556	1136	0.8647	1183	0.9972	783
1108	smartvist-000	0.0912	911	0.1587	915	0.1163	1229	0.1841	1223	0.1397	1153	0.9372	1130	0.7107	1154	0.9990	795
1109	smartvist-001	0.0137	658	0.0244	688	0.0099	908	0.0158	913	0.0151	832	0.1349	823	0.0419	840	0.5589	578
1110	smilart-001	0.6476	1047	0.8779	1061	-		-		-		-		-		-	
1111	smilart-002	0.2440	979	0.3532	986	-		-		0.3785	1219	0.4145	964	0.2611	1043	-	
1112	smilart-003	0.6944	1051	0.8836	1062	0.0695	1199	0.1193	1198	0.0894	1128	0.1221	813	0.0737	908	-	
1113	sodec-000	0.0033	220	0.0044	205	0.0040	564	0.0053	545	0.0054	497	0.0096	301	0.0080	378	0.0915	273
1114	softtech-000	-		-		0.0024	166	0.0023	89	0.0022	88	0.0043	71	0.0038	86	0.0641	145
1115	sparktek-000	-		-		0.0025	182	0.0025	139	0.0031	268	0.0058	137	0.0053	206	0.0643	147
1116	sparktek-001	-		-		0.0021	26	0.0020	14	0.0017	31	0.0034	26	0.0029	26	0.0470	40
1117	sparsh-001	0.0018	92	0.0029	97	0.0026	251	0.0028	213	0.0027	190	0.8806	1099	0.6313	1134	0.9963	774
1118	sqisoft-001	0.1220	933	0.2088	939	0.1978	1253	0.3386	1257	0.2111	1187	0.2798	909	0.1474	982	0.7820	637
1119	sqisoft-002	0.0082	509	0.0124	511	0.0051	663	0.0086	735	0.0102	724	0.0183	477	0.0122	536	0.1498	400
1120	sqisoft-003	0.0041	285	0.0055	270	0.0026	236	0.0032	299	0.0039	361	1.0000	1319	1.0000	1415	1.0000	1393
1121	staqu-000	0.0139	663	0.0208	649	0.0104	920	0.0145	891	0.0156	838	0.8063	1069	0.1408	977	-	
1122	starhybrid-001	0.0108	590	0.0138	534	0.0081	834	0.0113	805	0.0152	833	0.0265	564	0.0189	678	-	
1123	stcon-000	0.0040	271	0.0056	275	0.0031	404	0.0047	483	0.0048	460	0.9863	1166	0.3562	1075	0.9999	824
1124	stcon-001	0.0040	270	0.0053	263	0.0030	383	0.0045	470	0.0047	444	0.8632	1095	0.1788	1006	0.9892	752
1125	stcon-002	0.0031	195	0.0043	189	0.0025	192	0.0034	329	0.0030	246	0.9988	1229	0.9855	1244	1.0000	862
1126	stcon-003	0.0038	261	0.0050	244	0.0024	135	0.0032	296	0.0023	103	0.1082	801	0.0350	810	0.3982	539
1127	stcon-004	-		-		0.0022	62	0.0025	146	0.0020	62	0.0519	706	0.0145	593	0.2564	485
1128	stcon-005	-		-		0.0021	27	0.0021	31	0.0017	26	0.0336	624	0.0115	516	0.1725	428
1129	stcon-006	-		-		0.0021	29	0.0021	32	0.0017	25	0.0034	28	0.0028	18	0.0425	20
1130	stcon-007	-		-		0.0021	28	0.0021	29	0.0017	24	0.0033	23	0.0028	16	0.0424	19
1131	stcon-008	-		-		0.0021	20	0.0020	5	0.0015	7	0.0029	11	0.0026	7	0.0390	10
1132	stcon-009	-		-		0.0021	13	0.0020	9	0.0015	4	0.0029	5	0.0025	5	0.0380	6
1133	stcon-010	-		-		0.0021	12	0.0020	10	0.0015	6	0.0028	4	0.0025	3	0.0376	3
1134	stellartecs-001	-		-		0.0046	624	0.0060	601	0.0046	434	0.0658	741	0.0181	663	0.3818	532
1135	stengg-000	0.0059	401	0.0082	391	0.0037	524	0.0060	600	0.0050	469	0.5065	983	0.0792	913	0.9348	704
1136	sukshi-000	0.5409	1035	0.6612	1045	0.4556	1295	0.6567	1297	0.9296	1299	0.8898	1103	0.7384	1159	0.9899	753
1137	sumsub-000	-		-		0.0041	575	0.0043	455	0.0038	350	0.5301	994	0.2536	1037	0.9068	693
1138	sumsub-001	-		-		0.0023	88	0.0025	152	0.0028	216	0.0052	114	0.0044	136	0.0633	138
1139	suprema-000	0.0064	432	0.0092	423	0.0081	837	0.0096	764	0.0139	812	0.0254	557	0.0220	718	-	
1140	suprema-001	0.0041	284	0.0053	259	0.0038	542	0.0047	484	0.0060	539	0.0111	348	0.0095	450	0.0907	268
1141	suprema-002	0.0030	192	0.0041	176	0.0034	462	0.0040	411	0.0045	420	0.0085	259	0.0072	330	0.0781	216
1142	suprema-003	0.0028	175	0.0041	178	0.0034	460	0.0039	398	0.0030	249	0.3095	920	0.0580	889	0.6672	605
1143	suprema-004	0.0024	139	0.0035	126	0.0032	427	0.0036	363	0.0028	199	0.0053	119	0.0045	150	0.0625	131
1144	suprema-005	0.0022	120	0.0036	135	0.0034	477	0.0031	279	0.0026	183	0.0050	109	0.0044	138	0.0588	109

Table 82: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1145	supremaid-001	0.0053	364	0.0073	351	0.0045	608	0.0066	638	0.0099	717	0.0186	481	0.0148	601	0.1040	308
1146	supremaid-002	0.0063	426	0.0094	435	0.0044	599	0.0062	617	0.0072	598	0.0229	535	0.0095	449	0.2986	507
1147	surrey-cvssp-000	0.9084	1062	0.9909	1076	0.9923	1361	0.9950	1357	0.9981	1318	0.9994	1239	0.9979	1271	0.9998	814
1148	surrey-cvssp-001	1.0000	1402	1.0000	1169	0.0077	819	0.0079	710	0.0266	976	0.3822	955	0.0551	879	0.7285	621
1149	surrey-cvssp-002	0.0019	94	0.0028	87	0.0027	312	0.0028	203	0.0024	126	0.5978	1016	0.1206	960	0.8488	660
1150	surrey-cvssp-003	0.0016	71	0.0028	88	0.0022	61	0.0024	108	0.0022	95	0.3048	915	0.0539	876	0.5830	585
1151	surrey-cvssp-004	-		-		0.0029	362	0.0025	153	0.0025	156	0.0690	751	0.0076	361	0.2741	497
1152	swsam-001	0.0268	793	0.0476	803	0.0271	1108	0.0460	1101	0.0584	1090	0.7745	1058	0.5013	1108	0.9642	718
1153	synesis-000	0.1752	955	0.3241	979	-		-		-		-		-		-	
1154	synesis-001	0.1777	956	1.0000	1090	1.0000	1443	-		-		-		-		-	
1155	synesis-003	0.8375	1057	0.8857	1063	0.8548	1346	0.9178	1342	-		-		-		-	
1156	synesis-004	0.0310	816	0.0480	805	0.0476	1166	0.0443	1094	0.0764	1113	-		-		-	
1157	synesis-005	0.0147	675	0.0226	671	0.0153	1033	0.0226	998	0.0213	931	0.0375	657	0.0262	762	-	
1158	synesis-006	0.0070	460	0.0096	440	0.0107	931	0.0166	935	-		0.0128	384	0.0089	417	-	
1159	synesis-007	0.0050	336	0.0073	353	0.0062	749	0.0076	688	-		0.0105	329	0.0080	383	-	
1160	synology-000	0.0149	681	0.0238	685	0.0148	1025	0.0261	1024	0.0221	938	0.0331	617	0.0209	709	-	
1161	synology-001	0.0214	753	0.0307	730	0.0266	1107	0.0513	1120	0.0283	992	0.8884	1101	0.2589	1040	-	
1162	synology-002	0.0104	581	0.0153	562	0.0107	933	0.0184	953	0.0189	902	0.2032	865	0.0180	660	-	
1163	syscake-000	-		-		0.0091	871	0.0155	907	0.0142	816	0.0287	580	0.0145	596	0.1682	424
1164	szpu-000	-		-		0.0034	464	0.0037	376	0.0039	357	0.8444	1089	0.3886	1085	0.8849	682
1165	sztu-000	0.0092	540	0.0139	536	0.0091	873	0.0201	978	0.0136	806	0.0685	750	0.0118	526	-	
1166	sztu-001	0.0031	197	0.0043	195	0.0025	208	0.0028	215	0.0051	473	0.0113	358	0.0089	419	0.9963	775
1167	t4isb-000	0.0058	387	0.0087	410	0.0041	584	0.0064	631	0.0083	643	0.0157	432	0.0103	477	0.1196	344
1168	tahakom-001	-		-		0.0223	1082	0.0364	1070	0.0361	1026	1.0000	1287	0.9998	1297	1.0000	893
1169	tahakom-002	-		-		0.0380	1143	0.0456	1098	0.0403	1044	0.4790	977	0.2494	1033	0.8529	662
1170	taiwanmobile-001	-		-		0.2136	1256	0.3014	1254	0.3408	1214	0.9439	1133	0.7559	1164	0.9966	778
1171	taiwanmobile-002	-		-		0.0088	866	0.0114	806	0.0108	743	0.9776	1155	0.8683	1185	0.9998	813
1172	tcc-000	-		-		0.1374	1237	0.2052	1233	0.6334	1245	0.9985	1224	0.9897	1251	0.9999	818
1173	tcc-001	-		-		0.0120	972	0.0217	990	0.2620	1201	0.9958	1200	0.9791	1238	1.0000	845
1174	tcv-001	-		-		0.0301	1126	0.0556	1125	0.0570	1088	0.1222	814	0.0786	910	0.3172	513
1175	tech5-000	0.0114	605	0.0160	577	-		-		-		-		-		-	
1176	tech5-001	0.0130	645	0.0176	607	0.0218	1080	0.0419	1088	0.0156	839	-		-		-	
1177	tech5-002	0.0046	313	0.0063	311	0.0113	956	0.0216	988	0.0233	951	0.9609	1139	0.7704	1166	-	
1178	tech5-003	0.0053	359	0.0070	336	0.0099	907	0.0185	955	0.0095	697	0.6670	1031	0.1024	942	-	
1179	tech5-004	0.0123	628	0.0234	681	0.0086	861	0.0162	925	0.0065	565	0.0112	353	0.0082	394	-	
1180	tech5-005	0.0054	369	0.0072	345	0.0069	778	0.0122	824	0.0060	538	0.0094	296	0.0066	287	-	
1181	tech5-007	0.0020	105	0.0029	94	0.0024	148	0.0028	196	0.0034	309	0.8622	1094	0.5335	1116	0.9867	750
1182	tech5-008	0.0020	102	0.0031	106	0.0026	249	0.0028	209	0.0026	175	0.9238	1120	0.7170	1157	0.9984	787
1183	tech5-009	-		-		0.0041	577	0.0040	410	0.0045	424	0.0087	270	0.0080	380	0.0821	235
1184	techainer-001	0.0061	413	0.0092	425	0.0046	621	0.0073	676	0.0060	540	0.9575	1137	0.8567	1182	0.9995	804
1185	techainer-002	-		-		0.0058	718	0.0121	821	0.0136	807	0.0853	774	0.0313	793	0.1610	415
1186	techainer-003	-		-		0.0029	354	0.0033	317	0.0046	443	0.0084	251	0.0074	340	0.0623	129
1187	techbiometricid-001	-		-		0.0877	1213	0.1663	1215	0.1000	1138	0.5946	1014	0.3065	1058	0.9792	735
1188	techbiometricid-002	-		-		0.0060	732	0.0091	747	0.8984	1291	1.0000	1308	1.0000	1323	1.0000	928

Table 83: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1189	techsign-000	0.0325	822	0.0511	813	0.0435	1158	0.0710	1152	0.0746	1111	0.1104	802	0.0841	923	0.4288	546
1190	techsign-001	0.0110	594	0.0196	637	0.0067	770	0.0120	817	0.0087	659	0.2475	893	0.0883	930	0.6767	608
1191	telexper-000	-		-		0.0056	703	0.0077	693	0.0104	728	0.9976	1215	0.8691	1186	0.9999	826
1192	tevian-000	0.0871	908	0.1289	900	-		-		-		-		-		-	
1193	tevian-001	0.0813	907	0.1266	897	-		-		-		-		-		-	
1194	tevian-002	0.0279	800	0.0392	776	-		-		-		-		-		-	
1195	tevian-003	0.0217	756	0.0298	726	0.0230	1090	0.0484	1112	-		-		-		-	
1196	tevian-004	0.0228	765	0.0304	729	0.0226	1088	0.0478	1109	0.0128	784	0.0228	532	0.0138	578	-	
1197	tevian-005	0.0043	296	0.0062	301	0.0057	714	0.0085	733	0.0070	586	0.0135	389	0.0119	527	-	
1198	tevian-006	0.0045	306	0.0061	296	0.0045	615	0.0066	636	0.0046	441	0.0091	278	0.0075	352	-	
1199	tevian-007	0.0019	96	0.0027	79	0.0032	432	0.0041	423	0.0045	425	0.0086	264	0.0078	371	0.0925	281
1200	tevian-008	0.0012	46	0.0017	33	0.0033	439	0.0042	446	0.0042	388	0.0081	234	0.0068	299	0.0909	270
1201	tiger-001	0.3120	992	0.3974	994	-		-		-		-		-		-	
1202	tiger-002	0.0658	884	0.0889	874	0.1083	1222	0.1766	1219	0.0952	1133	0.1568	837	0.0661	897	-	
1203	tiger-003	0.0313	818	0.0602	835	0.0188	1054	0.0359	1068	0.0344	1017	-		-		-	
1204	tiger-004	0.0779	899	0.1393	904	0.0488	1168	0.0905	1177	0.0800	1117	0.1640	840	0.0617	892	-	
1205	tiger-005	0.0624	878	0.2450	952	0.0292	1120	0.0556	1126	0.0430	1054	1.0000	1271	0.9964	1263	1.0000	863
1206	tiger-006	0.0066	440	0.0101	462	0.0050	661	0.0075	687	0.0089	670	0.0158	435	0.0117	521	0.1466	394
1207	tiger-007	-		-		0.0041	583	0.0063	618	0.0088	665	0.7340	1044	0.3935	1086	0.9464	708
1208	tinkoff-001	0.0145	673	0.0244	690	0.0318	1129	0.0636	1140	0.0236	958	1.0000	1445	0.0339	804	1.0000	984
1209	tngd-001	-		-		0.0827	1208	0.1404	1207	0.0981	1137	0.1409	830	0.0808	918	0.7653	628
1210	tngd-002	-		-		0.1471	1241	0.2504	1239	0.1879	1174	0.7463	1048	0.6359	1137	0.9555	711
1211	tnitech-000	0.0025	151	0.0037	145	0.0033	444	0.0035	350	0.0025	142	0.8893	1102	0.6283	1133	0.9883	751
1212	tongyi-004	0.0428	846	0.0540	820	-		-		-		-		-		-	
1213	tongyi-005	0.0073	476	0.0146	547	0.0187	1052	0.0421	1089	0.0161	851	0.0215	515	0.0149	604	-	
1214	tongyitrans-001	0.0617	877	0.0715	850	-		-		-		-		-		-	
1215	tongyitrans-002	0.0437	849	0.0657	844	-		-		-		-		-		-	
1216	tongyitrans-003	0.0249	779	0.0474	801	-		-		-		-		-		-	
1217	toppanidgate-000	0.0021	108	0.0033	117	0.0026	244	0.0028	206	0.0039	362	0.0075	204	0.0068	295	0.0631	136
1218	toppansecurity-001	-		-		0.0025	189	0.0027	193	0.0032	273	0.0059	146	0.0051	187	0.0499	57
1219	toshiba-000	0.1906	962	0.2896	969	-		-		-		-		-		-	
1220	toshiba-001	0.1063	923	0.1926	933	-		-		-		-		-		-	
1221	toshiba-002	0.0134	652	0.0222	668	0.0097	901	0.0154	906	-		0.0327	613	0.0158	618	-	
1222	toshiba-003	0.0125	633	0.0214	656	0.0085	858	0.0131	848	-		0.0241	542	0.0151	609	-	
1223	toshiba-004	0.0030	188	0.0042	181	0.0025	207	0.0027	192	0.0034	305	0.0063	164	0.0053	211	0.0700	174
1224	toshiba-005	0.0023	134	0.0037	147	0.0024	156	0.0026	164	0.0072	596	0.0141	398	0.0130	554	0.0918	276
1225	toshiba-006	0.0022	126	0.0035	127	0.0024	167	0.0025	144	0.0027	191	0.7425	1046	0.3070	1059	0.9691	721
1226	toshiba-007	0.0016	66	0.0023	61	0.0024	175	0.0023	74	0.0021	71	0.0629	736	0.0056	227	0.2301	468
1227	toshiba-008	0.0012	45	0.0017	34	0.0024	158	0.0022	60	0.0019	54	0.0108	340	0.0039	104	0.0899	262
1228	toshiba-009	-		-		0.0024	145	0.0021	50	0.0019	52	0.0256	558	0.0055	225	0.1500	401
1229	toshiba-010	-		-		0.0024	131	0.0021	43	0.0017	33	0.0316	600	0.0053	207	0.1331	367
1230	toshiba-011	-		-		0.0024	157	0.0022	55	0.0019	49	0.0037	44	0.0033	50	0.0453	34
1231	toshiba-012	-		-		0.0022	68	0.0022	54	0.0019	48	0.0040	54	0.0034	52	0.0503	59
1232	toshiba-013	-		-		0.0021	34	0.0022	53	0.0017	34	0.0288	581	0.0046	152	0.0917	274

Table 84: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)													
		CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1233	toss-000	-		-		0.0023	74	0.0022	57	0.0021	74	0.0041	59	0.0035	57	0.0474	41
1234	toss-001	-		-		0.0023	104	0.0024	100	0.0021	81	0.0040	58	0.0035	61	0.0494	51
1235	touchlessid-000	0.3296	995	0.4804	1009	0.4111	1287	0.6026	1291	0.5324	1239	0.9996	1247	0.9964	1264	1.0000	875
1236	touchlessid-001	0.0076	486	0.0104	469	0.0680	1197	0.0842	1168	0.1358	1150	1.0000	1273	0.9995	1286	1.0000	932
1237	touchlessid-002	0.0070	458	0.0096	444	0.0407	1152	0.0469	1106	0.4234	1229	0.9991	1233	0.9953	1261	1.0000	865
1238	touchlessid-003	0.0061	414	0.0081	385	0.0040	569	0.0053	546	0.0056	512	0.0155	428	0.0082	393	0.1370	375
1239	touchlessid-004	-		-		0.0035	481	0.0049	498	0.0043	404	0.0077	213	0.0058	243	0.0793	221
1240	transmitsecurity-001	-		-		0.0051	669	0.0078	704	0.0089	671	0.0151	418	0.0111	503	0.1181	340
1241	trueface-000	0.0249	778	0.4321	999	0.0119	971	0.0180	949	0.0297	1003	0.7467	1049	0.1602	996	-	
1242	trueface-001	0.0204	749	0.0438	790	0.0095	895	0.0138	872	0.0154	835	0.0253	555	0.0169	639	-	
1243	trueface-002	0.0060	406	0.0096	441	0.0048	641	0.0061	610	0.0112	755	0.0198	499	0.0155	614	-	
1244	trueface-003	0.0070	459	0.0094	434	0.0053	679	0.0081	720	0.0122	773	0.0217	519	0.0159	620	0.1204	347
1245	trueidvng-001	0.0063	422	0.0077	366	0.0033	453	0.0044	462	0.0046	438	0.0086	267	0.0069	305	0.0809	230
1246	truststamp-001	0.0242	772	0.0484	807	0.0194	1060	0.0359	1067	0.0381	1035	0.8808	1100	0.6898	1145	0.9948	766
1247	truststamp-002	-		-		0.0080	827	0.0138	873	0.0128	783	0.9718	1148	0.8779	1191	0.9998	812
1248	truststamp-003	-		-		0.0047	634	0.0070	661	0.0073	601	0.7132	1040	0.4538	1099	0.9578	713
1249	tsight-000	-		-		0.0020	1	0.0020	16	0.0014	1	0.0028	1	0.0025	1	0.0373	2
1250	ttigroup-000	-		-		0.0028	331	0.0034	324	0.0043	409	0.9208	1115	0.7608	1165	0.9975	785
1251	tuputech-000	0.3218	994	0.3696	989	-		-		0.3237	1211	0.4304	968	0.2973	1057	-	
1252	turingtechvip-001	0.0330	827	0.0540	821	0.0458	1165	0.1007	1184	0.4715	1232	0.9286	1123	0.8448	1180	0.9964	777
1253	turingtechvip-002	0.0126	639	0.0163	581	0.0092	879	0.0118	811	0.2264	1193	1.0000	1407	0.9925	1254	0.9999	829
1254	turkcell-000	0.1134	928	0.1288	899	0.0770	1205	0.1112	1195	0.2570	1198	1.0000	1270	0.9999	1305	1.0000	898
1255	turkcell-001	0.0043	291	0.0066	318	0.0037	529	0.0055	560	0.0043	397	0.0440	675	0.0082	395	0.3148	512
1256	turkcell-002	-		-		0.0036	497	0.0053	544	0.0044	411	0.0371	652	0.0075	347	0.2871	504
1257	twface-000	0.0051	343	0.0072	347	0.0041	574	0.0058	577	0.0071	591	0.0153	425	0.0100	459	0.1595	410
1258	twface-001	0.0036	243	0.0051	250	0.0031	406	0.0038	385	0.0049	461	0.0091	283	0.0075	351	0.0831	239
1259	ulsee-001	0.0151	685	0.0246	693	0.0113	953	0.0185	958	0.0187	899	0.6766	1032	0.0181	662	-	
1260	ultinous-000	0.2343	975	0.3484	984	-		-		-		-		-		-	
1261	ultinous-001	0.2485	980	0.4003	995	-		-		-		-		-		-	
1262	uluface-002	0.0081	506	0.0123	507	0.0071	785	0.0095	761	0.0107	736	1.0000	1419	0.0140	582	-	
1263	uluface-003	0.0100	567	0.0150	555	0.0079	823	0.0128	841	-		-		-		-	
1264	unicc-001	0.5101	1034	0.6167	1038	0.6577	1316	0.7894	1313	0.9208	1297	0.9866	1167	0.9386	1211	0.9999	833
1265	unicc-002	0.2153	972	0.2190	944	0.0417	1154	0.0526	1123	0.0891	1127	0.1321	822	0.1125	950	0.3813	531
1266	unicc-003	0.0217	757	0.0260	706	0.0093	885	0.0136	863	0.0129	786	0.0216	517	0.0143	590	0.2091	458
1267	unicc-004	-		-		0.0055	696	0.0077	695	0.0076	614	0.0147	411	0.0102	470	0.1421	385
1268	unissey-001	0.0095	551	0.0160	576	0.0134	999	0.0150	900	0.0147	827	0.0253	556	0.0163	627	0.1651	421
1269	unissey-002	0.0094	548	0.0151	557	0.0079	825	0.0110	798	0.0114	759	0.4424	971	0.1914	1012	0.8051	646
1270	unissey-003	0.0057	380	0.0082	390	0.0047	632	0.0082	722	0.0067	571	0.5179	987	0.2863	1053	0.8684	669
1271	unissey-004	0.0043	295	0.0066	316	0.0037	511	0.0052	533	0.0048	457	0.5272	992	0.2903	1054	0.8777	676
1272	unissey-005	-		-		0.0037	512	0.0052	532	0.0086	654	0.0161	441	0.0134	561	0.1006	298
1273	unissey-006	-		-		0.0073	795	0.0121	822	0.0288	995	0.0529	710	0.0464	858	0.2808	500
1274	unissey-007	-		-		0.3124	1271	0.4571	1272	0.6912	1251	0.9823	1162	0.8827	1193	1.0000	847
1275	uniubi-000	-		-		0.0037	523	0.0064	630	0.0047	448	0.9321	1127	0.0147	600	0.9969	780
1276	universeai-000	-		-		0.0023	114	0.0025	138	0.0043	405	0.0082	242	0.0075	345	0.0616	125

Table 85: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1277	upc-001	0.0234	767	0.0519	816	0.0291	1118	0.0490	1116	0.0294	1000	0.2316	884	0.0389	828	-	
1278	urielsoft-000	-		-		0.0026	256	0.0032	293	0.0043	401	0.0083	245	0.0068	301	0.0744	191
1279	useb-000	0.0158	693	0.0222	669	0.0123	980	0.0168	939	0.0182	889	0.0289	583	0.0167	637	0.1614	416
1280	useb-001	0.0035	239	0.0053	261	0.0027	301	0.0033	314	0.0027	193	0.0085	261	0.0044	137	0.0806	226
1281	useb-002	0.0028	176	0.0039	159	0.0027	299	0.0031	278	0.0025	141	0.0090	276	0.0042	123	0.0867	245
1282	useb-003	-		-		0.0026	234	0.0030	243	0.0025	135	0.0091	282	0.0042	126	0.0901	265
1283	useb-004	-		-		0.0025	222	0.0028	221	0.0022	83	0.5197	991	0.2232	1028	0.8470	658
1284	uxlabs-001	0.0534	861	0.0570	828	0.0118	967	0.0131	847	0.0237	961	0.0399	667	0.0288	782	0.2097	460
1285	uxlabs-003	0.2133	971	0.2143	940	0.0386	1145	0.0457	1099	0.0374	1033	0.3343	929	0.1554	990	0.6941	613
1286	uzinfocom-000	-		-		0.0070	782	0.0120	819	0.0086	653	0.9874	1171	0.9469	1214	0.9999	817
1287	uzinfocom-001	-		-		0.0029	361	0.0038	387	0.0041	380	0.9963	1207	0.9725	1234	1.0000	856
1288	vaidio-001	-		-		0.0135	1000	0.0166	934	0.0079	624	0.6587	1028	0.4279	1093	0.9123	694
1289	vcog-002	0.7522	1053	0.9033	1067	-		-		-		-		-		-	
1290	vcortex-001	0.0131	648	0.0207	647	0.0192	1057	0.0315	1052	0.2238	1192	0.3586	942	0.3536	1074	0.3569	521
1291	vd-000	0.3790	1005	0.4973	1013	-		-		-		-		-		-	
1292	vd-001	0.0243	773	0.0452	794	0.0271	1110	0.0402	1082	0.0424	1052	-		-		-	
1293	vd-002	0.0429	847	0.0704	848	0.0569	1183	0.0844	1169	0.0801	1118	0.0937	780	0.0577	888	-	
1294	vd-003	0.0199	745	0.0222	666	0.0115	962	0.0130	846	0.0138	810	0.0239	540	0.0177	656	0.1647	420
1295	veertec-001	-		-		0.0564	1182	0.0984	1183	0.0845	1123	0.1076	800	0.0691	902	0.4622	552
1296	veridas-001	0.1998	969	0.2724	959	0.2987	1269	0.4587	1273	0.2599	1199	0.2454	891	0.1357	972	-	
1297	veridas-002	0.1733	954	0.2257	947	0.2617	1265	0.4147	1266	0.2073	1184	0.2028	864	0.1112	949	-	
1298	veridas-003	0.0557	870	0.0983	883	0.0734	1201	0.1267	1203	0.0694	1108	0.0951	786	0.0480	862	-	
1299	veridas-004	0.0281	801	0.0467	800	0.0353	1140	0.0643	1143	0.0424	1051	0.0644	737	0.0342	805	-	
1300	veridas-005	0.0599	874	0.1110	891	0.0294	1124	0.0639	1141	0.0811	1121	0.1130	804	0.0548	878	-	
1301	veridas-006	0.0098	557	0.0167	590	0.0079	826	0.0127	835	0.0127	780	0.0217	518	0.0151	608	0.1573	406
1302	veridas-007	0.0063	424	0.0083	395	0.0044	603	0.0058	581	0.0080	634	0.0152	422	0.0120	533	0.1199	346
1303	veridas-008	0.0032	210	0.0045	211	0.0030	371	0.0033	309	0.0085	652	0.0206	505	0.0143	591	0.1228	351
1304	veridas-009	0.0031	196	0.0043	190	0.0030	384	0.0035	344	0.0055	498	0.0106	334	0.0100	460	0.0959	291
1305	veridas-010	-		-		0.0027	311	0.0033	313	0.0042	389	0.0080	233	0.0073	331	0.0649	152
1306	veridium-000	0.0726	892	0.1248	896	0.5226	1300	0.6652	1299	0.6425	1247	0.8150	1075	0.7989	1173	0.8772	675
1307	veridium-001	0.0274	797	0.0368	767	0.0292	1121	0.0475	1107	0.0488	1068	0.0673	745	0.0463	856	0.3691	527
1308	veridium-002	0.0343	829	0.0427	786	0.0283	1114	0.0460	1102	0.0489	1069	0.0673	744	0.0464	857	0.3658	523
1309	veridium-003	-		-		0.0286	1116	0.0462	1103	0.0492	1070	0.0679	748	0.0470	860	0.3672	524
1310	veridium-004	-		-		0.0141	1008	0.0254	1015	0.0217	935	0.0315	597	0.0197	692	0.2126	461
1311	verigram-000	0.0032	206	0.0043	188	0.0031	392	0.0034	327	0.0093	690	0.0175	466	0.0164	631	0.1158	336
1312	verigram-001	0.0032	209	0.0044	202	0.0027	295	0.0032	294	0.0030	238	0.9995	1243	0.9953	1260	1.0000	927
1313	verigram-002	0.0079	500	0.0089	415	0.0038	536	0.0040	413	0.0060	541	0.0113	359	0.0107	489	0.0634	139
1314	verigram-003	0.0101	568	0.0109	482	0.0034	466	0.0039	401	0.0037	339	0.0070	193	0.0063	272	0.0559	89
1315	verigram-004	-		-		0.0023	111	0.0023	79	0.0021	68	1.0000	1296	0.9999	1302	1.0000	906
1316	verihubs-inteligensia-000	0.0070	457	0.0098	448	0.0048	643	0.0076	692	0.0092	683	0.0160	438	0.0117	519	0.1348	371
1317	verihubs-inteligensia-001	0.0071	463	0.0114	496	0.0050	656	0.0076	691	0.0096	699	0.0165	451	0.0114	512	0.1409	383
1318	verihubs-inteligensia-002	0.0028	171	0.0043	193	0.0025	230	0.0030	245	0.0047	449	0.0092	284	0.0083	401	0.0983	294
1319	verijelas-000	0.2488	981	0.3431	982	0.4861	1298	0.6004	1289	0.0811	1120	0.1148	806	0.0440	848	0.4077	541
1320	verismart-001	-		-		0.0044	600	0.0068	651	0.0064	561	0.1877	855	0.0080	384	0.8127	647

Table 86: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1321	via-000	0.0216	755	0.0365	765	0.0177	1049	0.0287	1040	0.0296	1001	0.0572	717	0.0290	785	-	
1322	via-001	0.0149	679	0.0229	675	0.0114	960	0.0177	947	0.0183	893	0.4056	961	0.0176	653	-	
1323	via-004	0.0099	563	0.0135	531	0.0031	412	0.0044	463	0.0055	503	0.4009	959	0.0080	385	0.9840	743
1324	via-005	0.0063	421	0.0082	392	0.0031	394	0.0037	383	0.0032	274	0.3659	945	0.0068	296	0.8709	672
1325	via-006	-		-		0.0025	181	0.0027	177	0.0025	160	0.5904	1012	0.0315	794	0.9455	707
1326	viante-000	0.0004	6	0.0007	4	0.0026	263	0.0026	167	0.0017	27	0.0033	20	0.0029	28	0.0475	44
1327	viante-001	-		-		0.0025	228	0.0025	134	0.0016	17	0.0031	14	0.0028	21	0.0432	22
1328	viante-002	-		-		0.0025	191	0.0024	110	0.0015	12	0.0029	10	0.0027	12	0.0399	12
1329	viante-003	-		-		0.0025	184	0.0023	83	0.0015	8	0.0029	6	0.0026	10	0.0388	8
1330	viante-004	-		-		0.0025	194	0.0023	84	0.0015	13	0.0029	8	0.0027	11	0.0390	9
1331	videmo-000	0.0298	810	0.0423	783	0.0155	1036	0.0260	1022	0.0246	968	0.0397	666	0.0239	739	-	
1332	videmo-001	0.0295	807	0.0417	780	0.0164	1045	0.0261	1025	0.0355	1023	0.0603	730	0.0442	850	0.2428	478
1333	videmo-002	0.0158	694	0.0288	718	0.0149	1030	0.0249	1014	0.0230	946	0.3429	934	0.1468	981	0.8393	656
1334	videmo-003	-		-		0.0082	843	0.0138	871	0.0133	796	0.0220	523	0.0119	530	0.1596	412
1335	videonetics-001	0.5483	1036	0.6446	1041	0.7517	1336	0.8607	1330	0.8664	1281	0.8255	1080	0.6956	1147	-	
1336	videonetics-002	0.4274	1013	0.5329	1020	0.6081	1312	0.7438	1307	0.7775	1262	0.7297	1043	0.5756	1123	-	
1337	viettelai-000	0.0051	348	0.0068	328	0.0054	686	0.0064	627	0.9451	1301	1.0000	1383	1.0000	1460	1.0000	1246
1338	viettelai-001	0.0054	368	0.0076	364	0.0049	651	0.0079	709	0.0064	560	0.0225	528	0.0083	397	1.0000	1134
1339	viettelai-002	-		-		0.0029	358	0.0041	427	0.0069	578	0.0126	381	0.0074	341	0.1095	316
1340	viettelai-003	-		-		0.0034	470	0.0041	431	0.2038	1180	0.3193	927	0.1067	946	0.8681	668
1341	viettelai-004	-		-		0.0023	109	0.0028	212	0.0023	105	0.0044	73	0.0034	53	0.0566	94
1342	viettelai-005	-		-		0.0023	99	0.0027	184	0.0021	72	0.0038	46	0.0030	39	0.0499	56
1343	viettelai-006	-		-		0.0022	70	0.0024	122	0.0020	60	0.0037	42	0.0029	33	0.0481	47
1344	viettelhightech-000	0.0117	612	0.0166	586	0.0110	942	0.0198	976	0.0167	866	0.0249	549	0.0158	617	0.1604	414
1345	viettelhightech-001	-		-		0.0031	407	0.0040	418	0.0032	282	0.0063	163	0.0051	186	0.0781	215
1346	viettelhightech-002	-		-		0.0031	391	0.0042	448	0.0033	287	0.0061	158	0.0051	184	0.0758	202
1347	vigilantsolutions-000	-		0.6881	1047	-		-		-		-		-		-	
1348	vigilantsolutions-001	-		0.3475	983	-		-		-		-		-		-	
1349	vigilantsolutions-002	0.2290	974	0.3214	977	-		-		-		-		-		-	
1350	vigilantsolutions-003	0.1878	961	0.2673	958	-		-		-		-		-		-	
1351	vigilantsolutions-004	0.1780	957	0.2598	956	-		-		-		-		-		-	
1352	vigilantsolutions-005	0.1613	951	0.4702	1007	0.0163	1043	0.0279	1035	-		-		-		-	
1353	vigilantsolutions-006	0.1264	934	0.3221	978	0.0150	1031	0.0254	1016	0.0493	1072	0.2991	912	0.0403	835	-	
1354	vigilantsolutions-007	0.0202	748	0.0307	731	0.0136	1001	0.0227	999	0.0356	1025	1.0000	1440	1.0000	1371	-	
1355	vigilantsolutions-008	0.0156	690	0.0233	678	0.0101	911	0.0147	893	0.0355	1024	0.0492	698	0.0293	788	-	
1356	vigilantsolutions-009	0.0117	613	0.0165	585	0.0075	811	0.0101	778	0.0219	937	0.0385	661	0.0238	738	-	
1357	vigilantsolutions-010	0.0109	591	0.0164	583	0.0074	806	0.0095	758	0.0209	927	0.0365	650	0.0233	733	0.2465	479
1358	vigilantsolutions-011	0.0124	632	0.0176	605	0.0073	796	0.0095	757	0.0196	917	0.0360	642	0.0221	719	0.2633	492
1359	vinai-000	0.0081	505	0.0124	510	0.0045	606	0.0072	671	0.0089	669	0.1814	848	0.0112	505	-	
1360	vinbigdata-001	0.2576	984	0.2763	961	0.1404	1238	0.1988	1230	0.1407	1155	0.1150	807	0.0703	905	0.3688	526
1361	vinbigdata-002	0.0102	573	0.0175	602	0.0071	789	0.0084	729	0.0090	674	0.8017	1067	0.3134	1062	0.9936	764
1362	vinbigdata-003	0.0771	897	0.0992	884	0.1202	1230	0.1756	1218	0.2762	1204	0.2369	888	0.0100	461	0.8914	688
1363	vinbigdata-004	-		-		0.0031	396	0.0046	473	0.0037	341	0.1476	833	0.0063	275	0.4917	561
1364	vinbigdata-005	-		-		0.0023	76	0.0024	116	0.0022	82	0.0068	185	0.0038	90	0.0699	173

Table 87: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)													
		CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1365	vinbigdata-006	-		-		0.0022	67	0.0023	96	0.0019	59	0.0037	43	0.0032	43	0.0479	45
1366	vinbigdata-007	-		-		0.0022	46	0.0021	44	0.0019	57	0.0036	40	0.0032	44	0.0475	42
1367	vion-000	0.0419	843	0.0590	832	0.0422	1155	0.0478	1110	0.0581	1089	0.0968	790	0.0847	924	-	
1368	visage-000	0.0933	912	0.1441	911	0.1316	1234	0.2416	1237	0.1395	1152	0.1920	857	0.1001	939	-	
1369	visica-001	-		-		0.2293	1259	0.3588	1259	0.2936	1207	0.3564	941	0.1967	1014	0.7795	636
1370	visica-002	-		-		0.0520	1175	0.0898	1175	0.2960	1208	0.8202	1077	0.5956	1128	0.9838	741
1371	visica-003	-		-		0.3326	1277	0.4772	1274	0.3237	1212	0.3591	943	0.2203	1026	0.7317	622
1372	visionbox-000	0.0293	804	0.0541	822	0.0197	1062	0.0339	1056	0.0349	1019	0.0593	726	0.0329	799	-	
1373	visionbox-001	0.0159	696	0.0270	711	0.0111	946	0.0173	942	0.0190	904	0.0315	598	0.0205	703	-	
1374	visionbox-002	0.0058	385	0.0079	376	0.0060	729	0.0074	677	0.0084	648	0.0149	414	0.0113	509	0.0979	293
1375	visionbox-003	0.0057	384	0.0075	358	0.0062	747	0.0083	726	0.0100	721	0.9915	1180	0.9625	1227	1.0000	839
1376	visionbox-004	0.0046	320	0.0075	357	0.0060	734	0.0075	685	0.0056	509	0.0316	599	0.0070	316	0.2528	482
1377	visionbox-005	-		-		0.0041	571	0.0057	571	0.0046	442	0.0109	343	0.0088	415	0.0767	207
1378	visionbox-006	-		-		0.0063	751	0.0075	682	0.0048	455	0.0116	365	0.0094	442	0.0789	218
1379	visionbox-007	-		-		0.0063	753	0.0075	681	0.0047	453	0.0116	367	0.0095	448	0.0793	220
1380	visionbox-008	-		-		0.0056	709	0.0072	666	0.0044	416	0.0115	364	0.0089	420	0.0828	238
1381	visionlabs-001	0.1049	922	0.1803	925	-		-		-		-		-		-	
1382	visionlabs-002	0.0326	824	0.0507	812	-		-		-		-		-		-	
1383	visionlabs-003	0.0178	722	0.0293	722	-		-		-		-		-		-	
1384	visionlabs-004	0.0219	759	0.0457	796	-		-		-		-		-		-	
1385	visionlabs-005	0.0088	524	0.0156	571	0.0048	638	0.0070	662	-		-		-		-	
1386	visionlabs-006	0.0037	255	0.0066	317	0.0041	582	0.0060	597	0.0061	547	0.0106	336	0.0076	362	-	
1387	visionlabs-007	0.0038	263	0.0048	226	0.0036	502	0.0048	490	0.0057	517	0.0097	306	0.0071	320	-	
1388	visionlabs-008	0.0026	160	0.0036	142	0.0031	415	0.0040	417	0.0045	427	0.0079	227	0.0064	276	-	
1389	visionlabs-009	0.0018	88	0.0025	67	0.0026	243	0.0029	239	0.0035	314	0.0064	166	0.0054	217	-	
1390	visionlabs-010	0.0017	77	0.0024	63	0.0026	262	0.0030	255	0.0033	295	0.0061	156	0.0052	199	-	
1391	visionlabs-011	0.0012	42	0.0022	54	0.0024	169	0.0026	165	0.0028	203	0.0053	120	0.0046	156	0.0635	140
1392	visteam-000	0.9200	1064	0.9489	1071	0.9959	1364	0.9984	1358	0.9994	1323	0.9978	1216	0.9914	1253	-	
1393	visteam-001	0.4417	1018	0.5385	1023	0.6410	1315	0.7788	1312	0.6386	1246	0.5904	1010	0.4023	1089	-	
1394	visteam-002	0.1564	949	0.2789	963	0.1581	1244	0.2567	1243	0.1776	1170	0.2090	871	0.1021	941	0.5820	584
1395	visteam-003	0.0804	905	0.2166	941	0.0613	1190	0.1204	1199	0.0963	1134	0.1269	819	0.0441	849	0.6049	590
1396	visteam-004	0.0541	864	0.5202	1017	0.0406	1151	0.0827	1166	0.1879	1173	0.1795	847	0.0347	809	0.9673	720
1397	visteam-005	0.0235	768	0.0333	746	0.0265	1106	0.0583	1134	0.0341	1016	0.0524	708	0.0259	756	0.2413	476
1398	visteam-006	0.0188	733	0.0256	701	0.0158	1040	0.0365	1071	0.0198	920	0.0340	630	0.0174	651	0.1904	443
1399	visteam-007	0.0312	817	0.0448	793	0.0416	1153	0.1008	1185	0.0466	1064	0.0602	729	0.0281	773	0.2601	489
1400	visteam-008	0.0378	837	0.0515	814	0.0391	1148	0.0937	1180	0.0437	1057	0.0567	716	0.0260	758	0.2533	483
1401	visteam-009	-		-		0.0138	1005	0.0304	1047	0.0166	862	0.9993	1238	0.9719	1232	0.9997	808
1402	visteam-010	-		-		0.0156	1038	0.0347	1062	0.0189	901	1.0000	1292	0.9996	1288	1.0000	838
1403	vivotek-001	-		-		0.9982	1367	0.9994	1362	1.0000	1332	1.0000	1294	0.9999	1307	1.0000	874
1404	vivotek-002	-		-		0.0761	1204	0.1380	1205	0.9587	1304	1.0000	1298	1.0000	1315	1.0000	919
1405	vixvizion-000	0.0642	883	0.0882	873	0.0497	1171	0.0905	1176	0.0848	1124	0.1026	792	0.0618	893	-	
1406	vixvizion-001	0.0245	775	0.0407	778	0.0257	1102	0.0497	1117	0.0514	1075	0.0656	740	0.0400	834	-	
1407	vixvizion-002	0.0062	420	0.0086	404	0.0053	683	0.0075	683	0.0121	770	0.0207	506	0.0161	623	-	
1408	vixvizion-003	0.0059	400	0.0084	399	0.0059	724	0.0081	717	0.0119	766	0.0209	508	0.0162	624	-	

Table 88: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
1409	vixvizion-004	0.0063	423	0.0094	433	0.0055	699	0.0081	718	0.0098	713	0.0157	433	0.0111	502	0.1417	384		
1410	vixvizion-005	0.0276	798	0.0420	782	0.0302	1127	0.0629	1138	0.0288	996	0.0447	678	0.0235	735	0.2304	469		
1411	vixvizion-006	0.0082	510	0.0122	506	0.0093	881	0.0194	969	0.0099	716	0.0169	458	0.0108	493	0.1338	369		
1412	vixvizion-007	0.0110	596	0.0191	627	0.0080	830	0.0157	909	0.0101	723	0.0190	487	0.0118	525	0.1407	382		
1413	vnis-000	0.1397	942	0.1608	917	0.2184	1258	0.2976	1252	0.5277	1238	0.3603	944	0.2714	1046	0.6929	611		
1414	vnpay-000	0.2380	977	0.2614	957	0.4054	1285	0.4968	1277	0.3993	1226	0.4389	969	0.4006	1088	0.7693	631		
1415	vnpay-001	-		-		0.0026	264	0.0030	247	0.0065	564	0.0124	377	0.0116	518	0.0826	236		
1416	vnpt-001	0.3117	991	0.3523	985	0.3474	1278	0.2747	1246	0.3405	1213	0.5015	982	0.4827	1104	-			
1417	vnpt-002	0.0351	833	0.0424	784	0.0220	1081	0.0316	1053	0.0471	1065	0.0817	770	0.0698	904	0.2610	490		
1418	vnpt-003	0.0117	611	0.0138	535	0.0040	565	0.0058	588	0.0087	658	0.0161	444	0.0126	543	0.1110	323		
1419	vnpt-004	0.0058	394	0.0078	373	0.0037	522	0.0053	541	0.0051	475	0.4640	974	0.1384	974	0.7753	633		
1420	vnpt-005	0.0036	246	0.0052	251	0.0027	296	0.0031	270	0.0036	325	0.0066	178	0.0056	229	0.0621	128		
1421	vnpt-006	0.0034	233	0.0050	240	0.0027	286	0.0035	355	0.0026	178	0.0049	104	0.0040	117	0.0585	103		
1422	vnpt-007	-		-		0.0025	221	0.0029	231	0.0025	155	0.0047	85	0.0039	103	0.0564	92		
1423	vnpt-008	-		-		0.0026	239	0.0029	229	0.0025	161	0.0047	91	0.0040	107	0.0570	97		
1424	vnpt-009	-		-		0.0030	364	0.0042	441	0.0030	245	0.0053	121	0.0042	127	0.0628	132		
1425	vocalize-001	-		-		0.7356	1331	0.8750	1337	0.7830	1265	0.7067	1039	0.5268	1115	0.9164	698		
1426	vocord-001	-		0.1415	910	-		-		-		-		-		-			
1427	vocord-002	0.0228	764	0.0341	751	-		-		-		-		-		-			
1428	vocord-003	0.1003	916	0.1092	889	-		-		-		-		-		-			
1429	vocord-004	0.0101	571	0.0174	601	0.0196	1061	0.0384	1077	-		-		-		-			
1430	vocord-005	0.0089	526	0.0155	568	0.0216	1079	0.0414	1087	-		-		-		-			
1431	vocord-006	0.0062	417	0.0102	465	0.0082	839	0.0151	901	0.0475	1066	0.0852	773	0.0788	911	-			
1432	vocord-007	0.0039	268	0.0053	262	0.0061	740	0.0094	756	0.0520	1078	0.0939	783	0.0910	932	-			
1433	vocord-008	0.0029	180	0.0038	156	0.0042	587	0.0055	555	0.0045	430	0.0086	268	0.0073	335	-			
1434	vocord-009	0.0022	121	0.0029	99	0.0036	493	0.0046	479	0.0052	483	0.0098	309	0.0086	411	-			
1435	vocord-010	0.0024	142	0.0031	108	0.0036	494	0.0049	511	0.0025	163	0.0065	170	0.0040	110	0.0679	167		
1436	vogic-000	-		-		0.0043	594	0.0061	615	0.0078	622	0.8206	1078	0.6006	1129	0.9843	745		
1437	vsoft-001	-		-		0.0707	1200	0.0877	1171	0.8390	1276	0.9519	1134	0.9506	1217	0.9655	719		
1438	vsoft-002	-		-		0.0451	1164	0.0679	1148	0.0550	1085	0.3671	947	0.1633	999	0.8586	665		
1439	vsoft-003	-		-		0.0326	1130	0.0563	1129	0.0526	1080	0.3671	948	0.1633	1000	0.8586	666		
1440	vsoft-004	-		-		0.0061	739	0.0103	784	0.0090	673	0.1830	851	0.0344	807	0.5669	579		
1441	vts-000	0.0103	575	0.0174	598	0.0080	829	0.0129	845	0.0250	970	0.0450	681	0.0372	823	-			
1442	vts-001	0.0033	215	0.0048	229	0.0027	305	0.0036	364	0.0032	275	0.6519	1026	0.3563	1076	0.9218	701		
1443	vts-002	-		-		0.0025	183	0.0032	298	0.0030	239	0.0113	356	0.0042	128	0.1593	409		
1444	wicket-000	0.0018	90	0.0028	90	0.0024	162	0.0027	178	0.0031	255	0.7968	1066	0.4340	1096	0.9788	733		
1445	winsense-000	0.0140	665	0.0228	674	0.0125	983	0.0215	987	0.0226	943	0.8091	1071	0.0461	855	-			
1446	winsense-001	0.0062	418	0.0099	456	0.0092	877	0.0210	983	0.0093	689	0.0144	407	0.0098	455	-			
1447	winsense-002	0.0050	337	0.0073	352	0.0038	538	0.0059	590	0.0064	559	0.0118	371	0.0084	402	-			
1448	wiseai-001	0.0658	885	0.0964	882	0.7743	1341	0.8956	1339	0.1967	1177	0.7526	1052	0.3419	1071	0.9695	723		
1449	wuhantianyu-001	0.0163	702	0.0262	707	0.0281	1113	0.0569	1131	0.0316	1012	0.0486	696	0.0344	806	0.2097	459		
1450	x-laboratory-000	0.0071	465	0.0106	473	0.0123	979	0.0138	870	0.0419	1049	0.5629	1003	0.2852	1052	-			
1451	x-laboratory-001	0.0059	399	0.0110	483	0.0054	688	0.0078	706	0.0094	693	0.0142	400	0.0100	463	-			
1452	xforwardai-000	0.0050	341	0.0072	341	0.0036	505	0.0051	531	0.0074	605	0.0136	391	0.0105	484	-			

Table 89: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1453	xforwardai-001	0.0021	111	0.0034	122	0.0027	314	0.0028	211	0.0046	440	0.0088	272	0.0079	374	-	
1454	xforwardai-002	0.0016	68	0.0023	56	0.0026	277	0.0025	136	0.0040	372	0.0081	237	0.0074	338	-	
1455	xm-000	0.0015	56	0.0026	72	0.0031	402	0.0038	392	0.0058	527	0.0105	331	0.0082	396	-	
1456	yisheng-000	0.1215	932	0.1990	936	-		-		-		-		-		-	
1457	yisheng-001	0.0892	909	0.1600	916	-		-		-		-		-		-	
1458	yisheng-002	0.2241	973	0.3795	991	-		-		-		-		-		-	
1459	yisheng-004	0.1988	967	0.3329	980	0.1147	1227	0.1849	1224	0.2044	1181	-		-		-	
1460	yitu-000	0.0284	802	0.0330	743	-		-		-		-		-		-	
1461	yitu-001	0.0050	334	0.0074	356	-		-		-		-		-		-	
1462	yitu-002	0.0025	152	0.0043	196	-		-		-		-		-		-	
1463	yitu-003	0.0015	64	0.0026	70	0.0066	766	0.0085	730	0.0064	562	0.0114	361	0.0103	478	-	
1464	yoonik-000	0.0070	456	0.0112	488	0.0074	804	0.0118	813	0.0564	1087	0.2013	862	0.1160	953	-	
1465	yoonik-001	0.0057	378	0.0079	375	0.0043	597	0.0061	605	0.0307	1006	0.0762	763	0.0556	882	-	
1466	yoonik-002	0.0052	354	0.0062	308	0.0029	363	0.0034	332	0.0615	1096	0.1279	820	0.1166	954	0.2329	472
1467	yoonik-003	0.0034	229	0.0047	220	0.0032	426	0.0037	373	0.0816	1122	0.2033	866	0.1601	995	0.3210	515
1468	yoonik-004	0.0041	286	0.0059	283	0.0033	441	0.0052	535	0.0040	374	0.0071	195	0.0051	192	0.0732	183
1469	yoti-001	0.0139	664	0.0195	634	0.0110	943	0.0187	962	0.0191	905	0.1255	817	0.0415	839	0.5578	577
1470	yoti-002	-		-		0.0076	814	0.0137	866	0.0106	733	0.3375	930	0.1599	994	0.7672	630
1471	ytu-000	0.0057	382	0.0087	413	0.0121	976	0.0238	1008	0.0047	451	0.0078	220	0.0059	253	-	
1472	yuan-000	0.0101	572	0.0168	591	0.0077	816	0.0127	838	0.0178	882	0.5189	989	0.0245	746	-	
1473	yuan-001	0.0116	609	0.0220	663	0.0114	958	0.0184	954	0.0149	830	0.0574	718	0.0160	622	-	
1474	yuan-002	0.0094	549	0.0154	567	0.0071	787	0.0110	801	0.0108	744	0.0348	633	0.0127	548	0.5738	582
1475	yuan-003	0.0078	492	0.0111	486	0.0062	748	0.0091	748	0.0106	734	0.0511	703	0.0123	540	0.7337	623
1476	yuan-004	0.0058	392	0.0078	370	0.0039	555	0.0055	559	0.0234	952	0.0442	677	0.0353	812	0.1673	423
1477	yuan-005	0.0037	257	0.0046	218	0.0027	315	0.0035	352	0.0033	288	0.2706	907	0.0876	929	0.6695	606
1478	yuan-006	0.0045	303	0.0062	302	0.0032	420	0.0049	502	0.0038	352	0.0084	254	0.0049	171	0.1211	348
1479	yuan-007	0.0046	310	0.0069	333	0.0045	611	0.0075	684	0.0072	594	0.0144	406	0.0076	354	0.1081	315
1480	yuan-008	0.0032	207	0.0043	191	0.0026	254	0.0034	323	0.0028	210	0.0059	143	0.0039	102	0.0891	256
1481	yuan-009	-		-		0.0028	328	0.0042	445	0.0030	243	0.0057	129	0.0040	113	0.0733	185
1482	yuan-010	-		-		0.0028	325	0.0041	422	0.0030	254	0.0058	136	0.0041	120	0.0748	198
1483	yuanli-001	-		-		0.0203	1068	0.0399	1079	0.0110	750	0.0168	457	0.0135	566	0.1476	396

Table 90: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

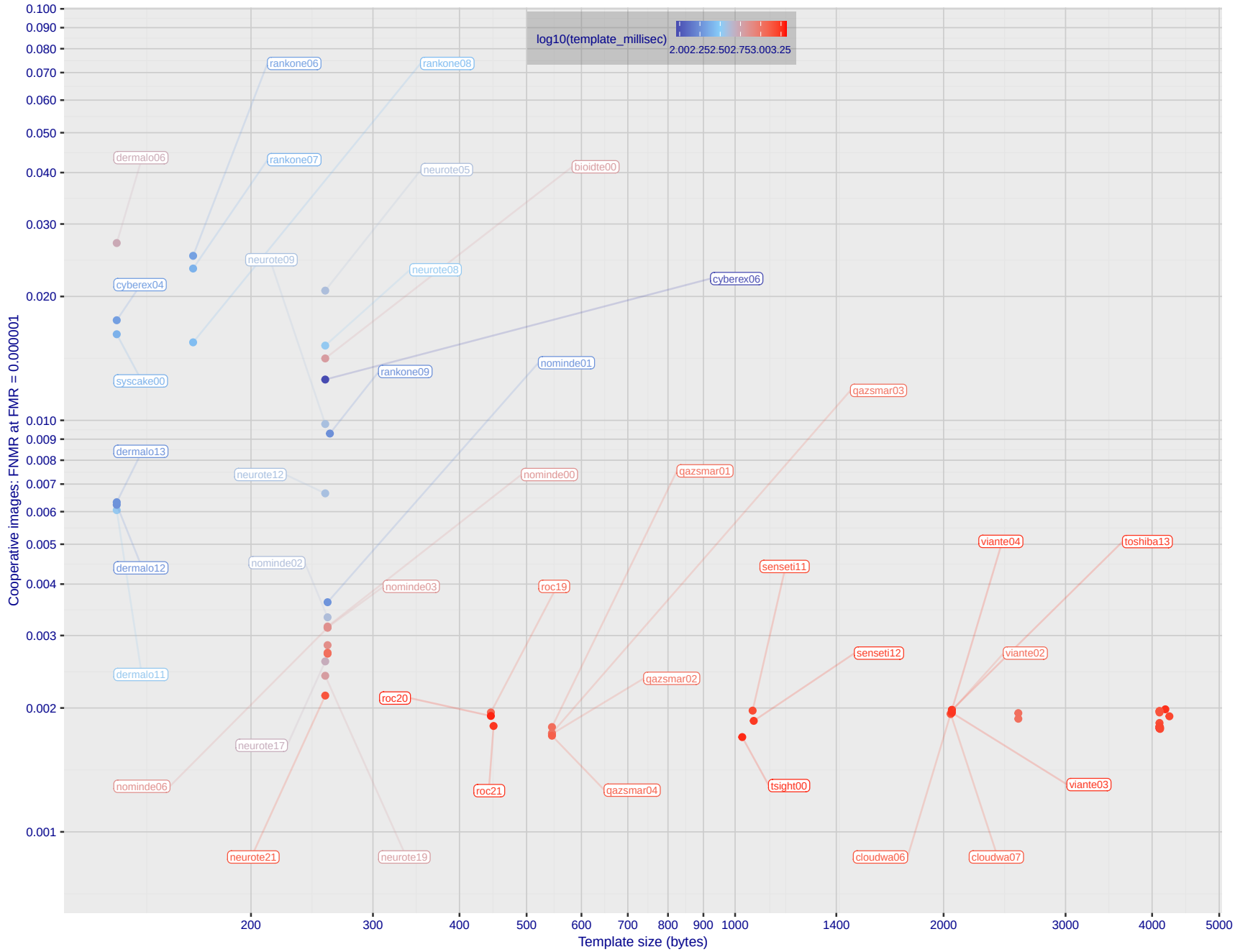


Figure 1: The points show false non-match rates (FNMR) versus the size of the encoded template. FNMR is the geometric mean of FNMR values for visa-border and mugshot images (from Figs. 224 and 294) at the false match rate (FMR) given in the y-axis label. The color of the points encodes template generation time - which spans at least one order of magnitude. Durations are measured on a single core of a c. 2016 Intel Xeon CPU E5-2630 v4 running at 2.20GHz. Algorithms with poor FNMR are omitted.

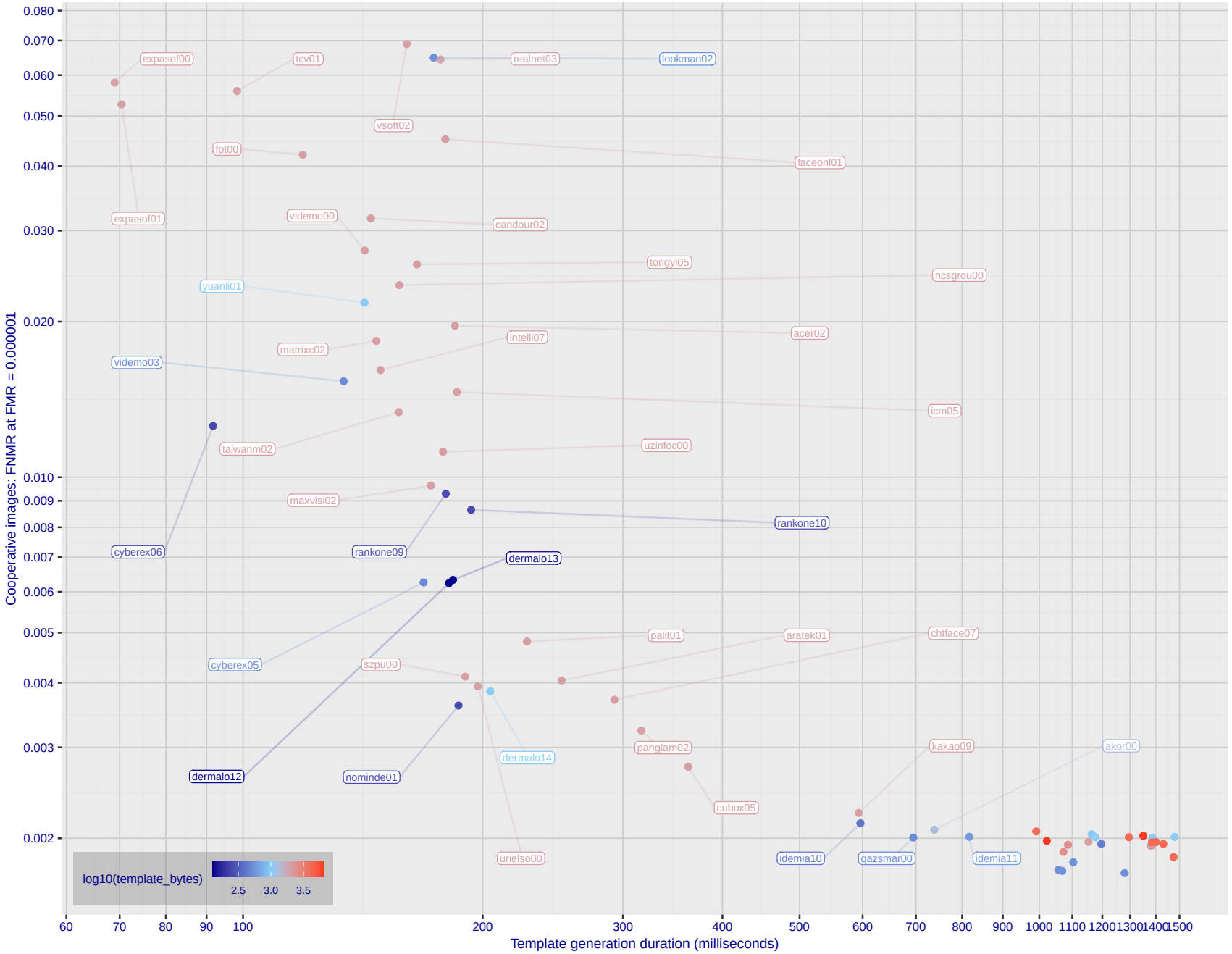


Figure 2: The points show false non-match rates (FNMR) versus the duration of the template generation operation. FNMR is the geometric mean of FNMR values for visa-border and mugshot images (from Figs. 224 and 294) at a false match rate (FMR) given in the y-axis label. Template generation time is a median estimated over 640 x 480 pixel portraits. It is measured on a single core of a c. 2016 Intel Xeon CPU E5-2630 v4 running at 2.20GHz. The color of the points encodes template size - which spans two orders of magnitude. Algorithms with poor FNMR are omitted.

1 Metrics

1.1 Core accuracy

Given a vector of N genuine scores, u, the false non-match rate (FNMR) is computed as the proportion below some threshold, T:

$$\text{FNMR}(T) = 1 - \frac{1}{N}\sum_{i=1}^{N} H(u_i - T) \tag{1}$$

where $H(x)$ is the unit step function, and $H(0)$ taken to be 1.

Similarly, given a vector of N impostor scores, v, the false match rate (FMR) is computed as the proportion above T:

$$\text{FMR}(T) = \frac{1}{N}\sum_{i=1}^{N} H(v_i - T) \tag{2}$$

The threshold, T, can take on any value. We typically generate a set of thresholds from quantiles of the observed impostor scores, v, as follows. Given some interesting false match rate range, $[\text{FMR}_L, \text{FMR}_U]$, we form a vector of K thresholds corresponding to FMR measurements evenly spaced on a logarithmic scale

$$T_k = Q_v(1 - \text{FMR}_k) \tag{3}$$

where Q is the quantile function, and FMR_k comes from

$$\log_{10}\text{FMR}_k = \log_{10}\text{FMR}_L + \frac{k}{K}\ [\log_{10}\text{FMR}_U - \log_{10}\text{FMR}_L] \tag{4}$$

Error tradeoff characteristics are plots of FNMR(T) vs. FMR(T). These are plotted with $\text{FMR}_U \to 1$ and FMR_L as low as is sustained by the number of impostor comparisons, N. This is somewhat higher than the "rule of three" limit $3/N$ because samples are not independent, due to re-use of images.

1.2 Multi-template scoring methodology

There are some scenarios when one or more people exist and are detected in an image, and some of the proposed test images include K $>$ 1 persons for some images and situations where the subject of interest may or may not be the foreground face (largest face in the image). The NIST FRTE 1:1 API supports this by allowing generation of multiple templates representing each person detected in an image. When this occurs, NIST will match all templates generated from the enrollment image with all templates generated from the verification image and use the **maximum** similarity score across all template comparisons. This scoring approach will be used in our calculation of FMR and FNMR (this applies to both genuine and imposter comparisons).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

2 Datasets

2.1 Visa images

- The number of images is on the order of 10^5.
- The number of subjects is on the order of 10^5.
- The number of subjects with two images is on the order of 10^4.
- The images have geometry in reasonable conformance with the ISO/IEC 19794-5 Full Frontal image type. Pose is generally excellent.
- The images are of size 252x300 pixels. The mean interocular distance (IOD) is 69 pixels.
- The images are of subjects from greater than 100 countries, with significant imbalance due to visa issuance patterns.
- The images are of subjects of all ages, including children, again with imbalance due to visa issuance demand.
- Many of the images are live capture. A substantial number of the images are photographs of paper photographs.
- When these images are input to the algorithm, they are labelled as being of type "ISO" - see Table 4 of the FRTE API.

2.2 Application images

- The number of images is on the order of 10^6.
- The number of subjects is on the order of 10^6.
- The number of subjects with two images is on the order of 10^6.
- The images have geometry in good conformance with the ISO/IEC 19794-5 Full Frontal image type. Pose is generally excellent.
- The images are of size 300x300 pixels. The mean interocular distance (IOD) is 61 pixels.
- The images are of subjects from greater than 100 countries, with significant imbalance due to population and immigration patterns.
- The images are of subjects of adults.
- All of the images are live capture.
- When these images are input to the algorithm, they are labelled as being of type "ISO" - see Table 4 of the FRTE API.

2.3 Application images with head yaw

- The number of images is on the order of 10^5.
- The number of subjects is on the order of 10^5.
- The number of subjects with two images is on the order of 10^5.
- The images have geometry in good conformance with the ISO/IEC 19794-5 Full Frontal image type *except* the yaw angle is between 25 and 85 degrees. Our pose estimates are approximate, with an angular error that increases with yaw. The angular estimates will be improved over time.
- The images are of size 300x300 pixels. The mean interocular distance (IOD), if frontal, would be about pixels, but reduces with cosine of yaw.
- The images are of subjects from greater than 100 countries, with significant imbalance due to population and immigration patterns.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ The images are of subjects of adults.
- ▷ All of the images are live capture.
- ▷ When these images are input to the algorithm, they are labelled as being of type ”WILD” - see Table 4 of the FRTE API.

2.4 Border crossing images

- ▷ The number of images is on the order of 10^6.
- ▷ The number of subjects is on the order of 10^6.
- ▷ The number of subjects with two images is on the order of 10^6.
- ▷ The images are taken with at camera oriented by an attendant toward a cooperating subject. This is done under time constraints so there are role, pitch and yaw angle variations. Also background illumination is sometimes strong, so the face is under-exposed. There is some perspective distortion due to close range images. Some faces are partially cropped.
- ▷ The images have mean IOD of 38 pixels.
- ▷ The images are of subjects of adults and childen aged 12 or above.
- ▷ The images are of subjects from greater than 100 countries, with significant imbalance due to population and immigration patterns.
- ▷ The images are all live capture.
- ▷ When these images are input to the algorithm, they are labelled as being of type ”WILD” - see Table 4 of the FRTE API.

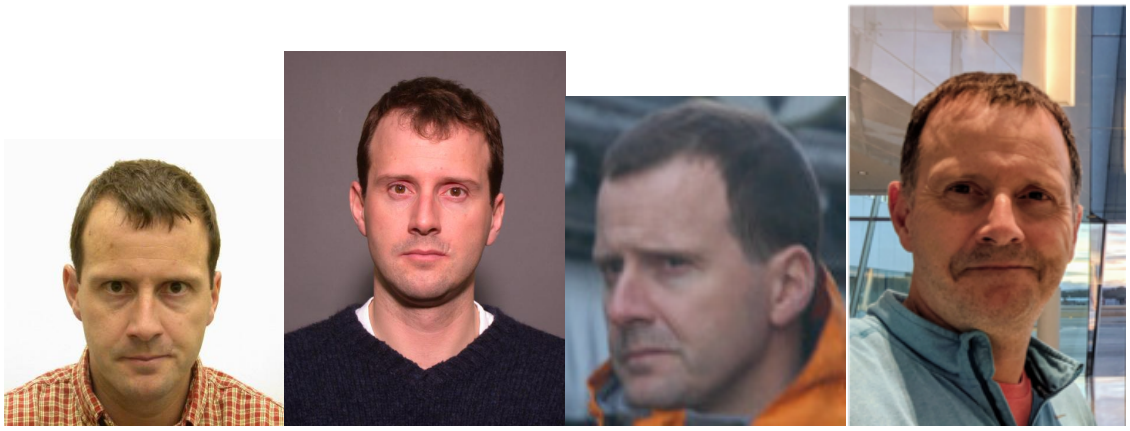
2.5 Mugshot images

- ▷ The number of images is on the order of 10^6.
- ▷ The number of subjects is on the order of 10^6.
- ▷ The number of subjects with two images is on the order of 10^6.
- ▷ The images have geometry in reasonable conformance with the ISO/IEC 19794-5 Full Frontal image type.
- ▷ The images are of variable sizes. The median IOD is 105 pixels. The mean IOD is 113 pixels. The 1-st, 5-th, 10-th, 25-th, 75-th, 90-th and 99-th percentiles are 34, 58, 70, 87, 121, 161 and 297 pixels.
- ▷ The images are of subjects from the United States.
- ▷ The images are of adults.
- ▷ The images are all live capture.
- ▷ When these images are input to the algorithm, they are labelled as being of type ”mugshot” - see Table 4 of the FRTE API.

2.6 Kiosk images

- ▷ The number of images is on the order of 10^6.
- ▷ The number of subjects is on the order of 10^5.
- ▷ The number of subjects with multiple images is the order of 10^5.

FNMR(T) “False non-match rate”
FMR(T) “False match rate”



(a) Visa (b) Mugshot (c) Wild (d) Border

Figure 3: The figure gives simulated samples of image types used in this report.

- The images are taken at kiosk equipped with a camera intended to capture a centered face. However the images have specific quality defects arising from the camera triggering before the subject looks at it. These are downward pitch of the face relative to the optical axis; cropping of the forehead; and cropping of left or right part of the face. Partial cropping affects perhaps 10% of the images. Resolution does not vary widely.
- The images are of adults.
- The images have mean IOD of 44 pixels, with maximum below 75, and minimum when both eyes are present above 25 pixels.
- All of the images are live capture, none are scanned.
- When these images are input to the algorithm, they are labelled as being of type ”WILD” - see Table 4 of the FRTE API.

2.7 Wild images

- The number of images is on the order of 10^5.
- The number of subjects is on the order of 10^4.
- The number of subjects with two images on the order of 10^4.
- The images include many photojournalism-style images. Images are given to the algorithm using a variable but generally tight crop of the head. Resolution varies very widely. The images are very unconstrained, with wide yaw and pitch pose variation. Faces can be occluded, including hair and hands.
- The images are of adults.
- All of the images are live capture, none are scanned.
- When these images are input to the algorithm, they are labelled as being of type ”WILD” - see Table 4 of the FRTE API.

3 Results

3.1 Test goals

- To state absolute accuracy for different kinds of images, including those with and without subject cooperation.

FNMR(T) “False non-match rate”
FMR(T) “False match rate”

▷ To state comparative accuracy, across algorithms.

3.2 Test design

Method: For visa images:

▷ The comparisons are of visa photos against visa photos.
▷ The number of genuine comparisons is on the order of 10^4.
▷ The number of impostor comparisons is on the order of 10^{10}.
▷ The comparisons are fully zero-effort, meaning impostors are paired without attention to sex, age or other covariates. However, later analysis is conducted on subsets.
▷ The number of persons is on the order of 10^5.
▷ The number of images used to make a template is one.
▷ The number of templates used to make each comparison score is two corresponding to simple one-to-one verification.

Method: For mugshot images:

▷ The comparisons are of mugshot photos against mugshot photos.
▷ The number of genuine comparisons is on the order of 10^6.
▷ The number of impostor comparisons is on the order of 10^8.
▷ The impostors are paired by sex, but not by age or other covariates.
▷ The number of persons is on the order of 10^6.
▷ The number of images used to make a template is one.
▷ The number of templates used to make each comparison score is two, corresponding to simple one-to-one verification.

Method: For visa-border comparisons:

▷ The comparisons are of visa-like frontals against border crossing webcam photos.
▷ The number of genuine comparisons is on the order of 10^6.
▷ The number of impostor comparisons is on the order of 10^8.
▷ The impostors are paired by sex, but not by age or other covariates.
▷ The number of persons is on the order of 10^6.
▷ The number of images used to make a template is one.
▷ The number of templates used to make each comparison score is two, corresponding to simple one-to-one verification.

Method: For visa-border non-frontal yaw comparisons:

▷ The comparisons are of visa-like images with yaw 25 to 85 degrees against border crossing webcam photos.
▷ The number of genuine comparisons is on the order of 10^5.
▷ The number of impostor comparisons is on the order of 10^8.
▷ The impostors are paired by sex, but not by age or other covariates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ The number of persons is on the order of 10^5.
- ▷ The number of images used to make a template is one.
- ▷ The number of templates used to make each comparison score is two, corresponding to simple one-to-one verification.

Method: For kiosk-border comparisons:

- ▷ The comparisons are of visa-like frontals against kiosk-style photos.
- ▷ The number of genuine comparisons is on the order of 10^6.
- ▷ The number of impostor comparisons is on the order of 10^8.
- ▷ The impostors are paired by sex, but not by age or other covariates.
- ▷ The number of persons is on the order of 10^5.
- ▷ The number of images used to make a template is one.
- ▷ The number of templates used to make each comparison score is two, corresponding to simple one-to-one verification.

Method: For border-border comparisons:

- ▷ The comparisons are of border crossing webcam photos.
- ▷ The number of genuine comparisons is on the order of 10^6.
- ▷ The number of impostor comparisons is on the order of 10^8.
- ▷ The impostors are paired by sex, but not by age or other covariates.
- ▷ The number of persons is on the order of 10^6.
- ▷ The number of images used to make a template is one.
- ▷ The number of templates used to make each comparison score is two corresponding to simple one-to-one verification.

Method: For wild images:

- ▷ The comparisons are of wild photos against wild photos.
- ▷ The number of genuine comparisons is on the order of 10^6.
- ▷ The number of impostor comparisons is on the order of 10^8.
- ▷ The comparisons are fully zero-effort, meaning impostors are paired without attention to sex, age or other covariates.
- ▷ The number of persons is on the order of 10^4.
- ▷ The number of images used to make a template is one.
- ▷ The number of templates used to make each comparison score is two corresponding to simple one-to-one verification.

Method: For child exploitation images:

- ▷ The comparisons are of unconstrained child exploitation photos against others of the same type.
- ▷ The number of genuine comparisons is on the order of 10^4.
- ▷ The number of impostor comparisons is on the order of 10^7.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ The comparisons are fully zero-effort, meaning impostors are paired without attention to sex, age or other covariates.
- ▷ The number of persons is on the order of 10^3.
- ▷ The number of images used to make 1 template is 1.
- ▷ The number of templates used to make each comparison score is two corresponding to simple one-to-one verification.
- ▷ We produce two performance statements. First, is a DET as used for visa and mugshot images. The second is a cumulative match characteristic (CMC) summarizing a simulated one-to-many search process. This is done as follows.
  - We regard M enrollment templates as items in a gallery.
  - These M templates come from $M > N$ individuals, because multiple images of a subject are present in the gallery under separate identifiers.
  - We regard the verification templates as search templates.
  - For each search we compute the rank of the highest scoring mate.
  - This process should properly be conducted with a 1:N algorithm, such as those tested in NIST IR 8009. We use the 1:1 algorithms in a simulated 1:N mode here to a) better reflect what a child exploitation analyst does, and b) to show algorithm efficacy is better than that revealed in the verification DETs.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.3 Failure to enroll

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1	2060io-001	0.0160	1315	0.0998	1320	0.4698	963	0.0193	1375	-	1234
2	2060io-002	0.0000	1030	0.0015	947	0.0238	646	0.0028	1328	-	1323
3	20face-000	0.0000	599	0.0008	696	0.0217	594	0.0000	352	0.0004	613
4	20face-001	0.0000	774	0.0008	695	0.0000	23	0.0000	368	0.0004	608
5	3divi-000	-	1372	-	1409	-	1353	-	1412	0.0008	933
6	3divi-001	-	1410	-	1377	-	993	-	1443	0.0007	923
7	3divi-002	-	1465	-	1347	-	1138	-	1469	0.0010	952
8	3divi-003	0.0001	1225	0.0059	1214	-	1450	0.0007	1209	0.0006	892
9	3divi-004	0.0001	1224	0.0059	1215	-	1318	0.0008	1224	0.0006	899
10	3divi-005	0.0000	900	0.0008	667	-	1301	0.0000	497	0.0002	272
11	3divi-006	0.0000	811	0.0007	612	0.0214	589	0.0001	717	0.0002	278
12	3divi-007	0.0000	835	0.0007	611	0.0214	590	0.0001	719	0.0002	277
13	accurascan-001	0.0001	1215	0.0060	1219	0.0838	922	0.0006	1198	0.0022	1030
14	accurascan-002	0.0001	1214	0.0060	1220	0.0838	923	0.0006	1197	0.0022	1031
15	accurascan-003	0.0000	600	0.0009	783	0.0226	612	0.0000	499	0.0003	403
16	acer-000	0.0000	1068	0.0024	1063	-	1194	0.0002	913	0.0004	731
17	acer-001	0.0000	634	0.0011	855	-	1356	0.0001	633	0.0004	649
18	acer-002	0.0000	1121	0.0008	678	0.0191	527	0.0003	1004	0.0004	639
19	acisw-003	0.0000	320	0.0000	98	-	1059	0.0000	190	0.0000	78
20	acisw-006	0.0000	191	0.0000	168	-	1439	0.0000	124	0.0000	128
21	acisw-007	0.0000	30	0.0000	207	0.0000	94	0.0000	36	0.0000	204
22	acisw-008	0.0000	532	0.0009	762	0.0173	466	0.0004	1103	0.0004	445
23	adera-001	0.0000	1067	0.0034	1139	-	1176	0.0003	1025	0.0005	858
24	adera-002	0.0000	1066	0.0034	1140	-	1073	0.0003	1024	0.0005	859
25	adera-003	0.0000	1064	0.0034	1141	0.0403	823	0.0003	1023	0.0005	862
26	adera-004	0.0000	662	0.0008	675	0.0202	547	0.0003	1057	0.0004	569
27	adera-005	0.0000	622	0.0008	674	0.0202	546	0.0003	1055	0.0004	570
28	advance-002	0.0000	758	0.0013	921	-	1023	0.0000	587	0.0004	642
29	advance-003	0.0000	1021	0.0012	881	0.0247	658	0.0001	782	0.0004	745
30	advance-004	0.0001	1221	0.0010	827	0.0157	432	0.0008	1235	0.0006	900
31	advance-005	0.0000	579	0.0003	320	0.0074	238	0.0002	902	0.0003	290
32	advance-006	0.0001	1217	0.0012	894	0.0202	545	0.0008	1231	-	1219
33	advance-007	0.0000	719	0.0007	616	0.0212	577	0.0002	852	-	1311
34	affinsys-ai-000	0.0000	1132	0.0068	1223	0.0641	882	0.0016	1292	-	1221
35	afisbiometrics-000	0.0000	699	0.0008	652	0.0213	587	0.0000	361	0.0004	645
36	afrengine-000	0.0000	620	0.0015	951	0.0254	681	0.0008	1232	0.0004	518
37	afrengine-001	0.0000	750	0.0007	630	0.0183	491	0.0005	1141	0.0004	739
38	afrengine-002	0.0000	717	0.0007	631	0.0183	492	0.0005	1139	0.0004	741
39	afrengine-003	0.0000	829	0.0099	1252	0.0384	807	0.0005	1142	-	1183
40	afrengine-004	0.0000	735	0.1602	1324	0.0528	853	0.0005	1140	-	1282
41	afrengine-005	0.0000	743	0.0012	891	0.0232	636	0.0006	1182	-	1288
42	aifirst-001	0.0000	62	0.0000	225	-	1300	0.0000	50	0.0000	189
43	aigen-001	0.0000	133	0.0000	136	-	1359	0.0000	79	0.0000	166
44	aigen-002	0.0000	61	0.0000	205	0.0000	89	0.0000	22	0.0000	216
45	aigroup-000	0.0000	370	0.0005	429	0.0103	300	0.0000	200	-	1203
46	aigroup-001	0.0000	972	0.0005	445	0.0104	303	0.0001	845	-	1094
47	aigroup-002	0.0000	69	0.0003	350	0.0083	269	0.0001	720	-	1422
48	ailabs-001	0.0000	795	0.0090	1248	-	1086	0.0007	1222	0.0005	797
49	aimall-001	0.0000	499	0.0000	63	-	1200	0.0000	280	0.0000	5
50	aimall-002	0.0000	1078	0.0043	1177	-	1322	0.0012	1268	0.0005	851
51	aimall-003	0.0000	1019	0.0012	895	-	1195	0.0004	1080	0.0005	777
52	aipractix-000	0.0021	1287	0.0021	1038	0.0175	472	0.0390	1382	-	1463
53	aiseemu-001	0.0000	400	0.0000	17	0.0000	45	0.0000	228	0.0000	51
54	aiseemu-002	0.0000	35	0.0000	212	0.0000	95	0.0000	30	0.0000	208
55	aiunionface-000	0.0000	114	0.0000	231	-	1326	0.0000	54	0.0000	185
56	aize-001	0.0001	1223	0.0040	1159	0.0652	884	0.0026	1324	0.0022	1029
57	aize-002	0.0000	466	0.0014	939	0.0230	630	0.0005	1180	0.0004	605
58	aize-003	0.0000	300	0.0000	255	0.0001	157	0.0000	168	0.0000	241

Table 91: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
59	aize-006	0.0000	636	0.0004	369	0.0090	286	0.0001	842	-	1389
60	ajou-001	0.0000	720	0.0020	1027	-	1449	0.0001	724	0.0004	752
61	akiyama-000	0.0000	397	0.0000	13	-	1104	0.0000	223	0.0000	47
62	akiyama-001	0.0000	262	0.0000	72	-	998	0.0000	151	0.0000	103
63	akiyama-002	0.0000	1167	0.0015	949	0.0213	581	0.0044	1346	-	1351
64	akiyama-003	0.0325	1327	0.0079	1236	0.0216	592	0.0270	1379	-	1182
65	akor-000	0.0000	849	0.0004	422	0.0112	334	0.0000	526	-	1119
66	alchera-000	0.0000	1180	0.0041	1172	-	1008	0.0004	1101	0.0014	1007
67	alchera-001	0.0000	1181	0.0041	1171	-	1132	0.0004	1102	0.0014	1006
68	alchera-002	0.0000	544	0.0008	705	-	1223	0.0001	803	0.0004	499
69	alchera-003	0.0001	1244	0.0013	917	0.0317	739	0.0002	940	0.0004	662
70	alchera-004	0.0000	640	0.0009	757	0.0228	624	0.0001	808	0.0004	520
71	alchera-005	0.0000	803	0.0009	755	0.0228	617	0.0001	810	0.0004	508
72	alchera-006	0.0000	612	0.0009	760	0.0228	623	0.0001	807	0.0004	523
73	alchera-007	0.0000	856	0.0009	753	0.0228	620	0.0001	814	0.0004	501
74	alchera-008	0.0000	575	0.0009	759	0.0228	622	0.0001	806	-	1415
75	alchera-009	0.0000	794	0.0004	415	0.0112	332	0.0002	921	-	1196
76	alfabeta-001	0.0005	1269	0.0650	1308	0.2142	953	0.0024	1317	0.0018	1020
77	alice-000	0.0000	452	0.0006	500	0.0133	370	0.0000	444	0.0004	484
78	alice-001	0.0000	43	0.0006	502	0.0133	371	0.0000	428	0.0004	498
79	alleyes-000	0.0000	663	0.0010	804	-	1377	0.0002	846	0.0004	698
80	allgovision-000	0.0007	1277	0.0062	1221	-	1327	0.0026	1322	0.0052	1069
81	alphaface-001	0.0000	797	0.0012	869	-	1077	0.0000	602	0.0004	690
82	alphaface-002	0.0000	659	0.0012	870	-	1393	0.0000	596	0.0004	697
83	alttekglobal-000	0.0000	705	0.0020	1019	0.0471	839	0.0000	608	-	1301
84	amplifiedgroup-001	0.0114	1310	0.1023	1321	-	1220	0.0189	1374	0.0279	1084
85	androvideo-000	0.0000	310	0.0000	82	-	1017	0.0000	164	0.0000	93
86	anke-000	-	1374	-	1411	-	1340	-	1410	0.0021	1025
87	anke-001	-	1482	-	1363	-	1185	-	1480	0.0047	1068
88	anke-002	-	1391	-	1431	-	1382	0.0004	1087	0.0009	939
89	anke-003	-	1359	-	1471	-	1282	0.0001	748	0.0004	580
90	anke-004	0.0000	832	0.0011	850	-	1089	0.0001	753	0.0004	708
91	anke-005	0.0000	711	0.0012	877	-	1457	0.0001	790	0.0004	736
92	anyvision-001	-	1425	-	1386	-	1009	-	1446	0.0091	1077
93	anyvision-002	0.0060	1299	0.0230	1280	-	1470	0.0070	1359	0.0090	1076
94	anyvision-004	0.0000	1032	0.0017	986	-	984	0.0001	791	0.0004	623
95	anyvision-005	0.0000	726	0.0013	904	-	1000	0.0000	509	0.0004	489
96	aratek-001	0.0000	358	0.0000	115	0.0000	36	0.0000	208	0.0000	65
97	armatura-001	0.0000	1096	0.0021	1039	0.0257	687	0.0005	1163	0.0005	807
98	armatura-002	0.0000	1031	0.0018	992	0.0206	557	0.0003	1059	0.0004	740
99	armatura-003	0.0000	574	0.0012	882	0.0333	770	0.0004	1095	0.0004	602
100	arvis-000	0.0008	1281	0.0014	931	0.0195	535	0.0083	1361	-	1409
101	asr-000	0.0000	969	0.0007	615	0.0152	413	0.0002	967	-	1136
102	asr-001	0.0000	961	0.0006	546	0.0138	378	0.0002	985	-	1177
103	asusaics-000	0.0000	509	0.0000	49	-	1197	0.0000	276	0.0000	9
104	asusaics-001	0.0000	73	0.0000	219	-	1290	0.0000	44	0.0000	198
105	autentika-000	0.0000	470	0.0000	48	0.0000	67	0.0000	260	0.0000	22
106	autentika-001	0.0000	106	0.0000	238	0.0000	108	0.0000	57	0.0000	183
107	autentika-002	0.0000	442	0.0000	25	0.0000	58	0.0000	248	0.0000	36
108	autentika-003	0.0000	223	0.0000	161	0.0000	131	0.0000	103	-	1338
109	authenmetric-002	0.0000	260	0.0000	71	-	992	0.0000	154	0.0000	100
110	authenmetric-003	0.0000	362	0.0000	116	0.0000	30	0.0000	202	0.0000	66
111	authenmetric-004	0.0000	367	0.0000	120	0.0000	32	0.0000	198	0.0000	71
112	authme-001	0.0000	1130	0.0009	774	0.0245	656	0.0003	1047	0.0005	786
113	authme-002	0.0000	938	0.0007	639	0.0179	479	0.0001	785	-	1278
114	aware-000	-	1421	-	1389	-	1025	-	1447	0.0005	804
115	aware-001	-	1387	-	1426	-	1374	-	1420	0.0002	261
116	aware-002	-	1336	-	1458	-	1243	-	1388	0.0006	872

Table 92: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
117	aware-003	-	1428	-	1397	-	1061	0.0016	1290	0.0013	996
118	aware-004	0.0000	967	0.0023	1053	-	1172	0.0002	879	0.0005	778
119	aware-005	0.0000	979	0.0020	1023	-	1180	0.0001	844	0.0004	705
120	aware-006	0.0000	669	0.0009	741	0.0249	664	0.0000	524	0.0004	628
121	aware-007	0.0000	555	0.0012	883	0.0325	756	0.0000	534	0.0004	672
122	aware-008	0.0000	541	0.0012	889	0.0327	758	0.0000	532	-	1476
123	awiros-001	0.0039	1291	0.0369	1297	-	1063	0.0386	1381	0.0872	1089
124	awiros-002	0.0000	1126	0.0038	1155	-	1476	0.0007	1218	0.0012	995
125	awiros-003	0.0000	360	0.0000	114	0.0000	37	0.0000	209	-	1199
126	awiros-004	0.0000	390	0.0000	9	0.0000	50	0.0000	235	-	1167
127	aximetria-001	0.0000	886	0.0010	833	0.0217	595	0.0001	840	0.0004	618
128	ayftech-001	0.0002	1252	0.0046	1184	-	1196	0.0043	1345	0.0011	960
129	ayftech-003	0.0000	350	0.0006	559	0.0123	352	0.0000	211	0.0000	60
130	ayonix-000	0.0053	1296	0.0341	1293	-	1141	0.0113	1368	0.0137	1079
131	beethedata-000	0.0005	1265	0.0042	1174	0.0366	792	0.0002	887	0.0010	948
132	betaface-000	-	1477	-	1366	-	1203	-	1483	0.0011	965
133	betaface-002	-	1366	-	1415	-	1372	-	1417	0.0010	950
134	beyneai-000	0.0000	205	0.0000	155	0.0000	135	0.0000	115	0.0000	138
135	biocube-001	0.0006	1272	0.0391	1298	0.1207	934	0.0015	1284	0.0020	1024
136	biocube-002	0.0000	734	0.0046	1187	0.0721	910	0.0008	1230	0.0012	985
137	bioid-001	0.0005	1267	0.0019	1015	0.0324	755	0.0005	1152	-	1233
138	bioid-002	0.0000	496	0.0000	60	0.0000	79	0.0000	284	-	1098
139	bioid-003	0.0000	173	0.0000	150	0.0000	126	0.0000	94	-	1355
140	bioid-004	0.0000	44	0.0000	214	0.0000	90	0.0000	26	-	1447
141	bioidtechswiss-000	0.0000	718	0.0008	658	-	1446	0.0003	1017	0.0004	753
142	bioidtechswiss-001	0.0000	656	0.0007	603	-	1397	0.0000	458	0.0004	684
143	bioidtechswiss-002	0.0000	670	0.0007	610	-	1428	0.0000	480	0.0004	666
144	bioidtechswiss-003	0.0000	992	0.0202	1276	0.0980	929	0.0005	1154	-	1404
145	biometric-vision-000	0.0000	850	0.0005	441	0.0107	320	0.0002	970	0.0004	533
146	biometrik-001	0.0000	96	0.0000	236	0.0000	112	0.0000	63	-	1395
147	biometrik-002	0.0000	451	0.0000	27	0.0000	55	0.0000	241	-	1158
148	bm-001	0.0000	213	0.0000	159	-	1408	0.0000	299	0.0000	143
149	boetech-001	0.0087	1306	0.0272	1284	0.2117	950	0.0032	1333	0.0160	1080
150	boetech-002	0.0087	1305	0.0272	1285	0.2117	951	0.0032	1332	0.0160	1081
151	bresee-000	0.0000	571	0.0010	820	-	1257	0.0002	885	0.0003	369
152	bresee-001	0.0000	736	0.0010	819	-	973	0.0002	888	0.0003	363
153	bresee-002	0.0000	1045	0.0020	1031	0.0219	602	0.0008	1223	0.0004	579
154	btsdigital-001	0.0000	410	0.0009	738	0.0265	697	0.0000	421	-	1181
155	bynn-000	0.0130	1312	0.0048	1191	0.0089	281	0.0112	1367	-	1212
156	cakevn-001	0.0000	517	0.0007	620	0.0130	365	0.0000	272	-	1115
157	cakevn-002	0.0000	94	0.0008	654	0.0163	449	0.0000	62	-	1393
158	camvi-001	-	1360	-	1482	-	1336	-	1406	0.0010	956
159	camvi-002	0.0000	206	0.0000	154	-	1421	0.0000	117	0.0000	137
160	camvi-003	-	1457	-	1335	-	1090	0.0000	216	0.0000	58
161	camvi-004	0.0000	288	0.0000	252	-	968	0.0000	144	0.0000	110
162	candour-001	0.0000	196	0.0000	169	0.0000	137	0.0000	122	0.0000	132
163	candour-002	0.0000	354	0.0000	113	0.0000	35	0.0000	206	-	1198
164	candour-003	0.0000	79	0.0000	222	0.0000	96	0.0000	37	-	1430
165	candour-004	0.0000	493	0.0000	62	0.0000	78	0.0000	282	-	1100
166	canon-002	0.0000	495	0.0000	61	-	1216	0.0000	283	0.0000	4
167	canon-003	0.0000	655	0.0008	650	0.0234	639	0.0000	579	0.0004	647
168	canon-004	0.0000	560	0.0008	653	0.0234	638	0.0000	577	0.0004	655
169	canon-005	0.0000	16	0.0044	1179	0.0269	702	0.0000	451	0.0004	551
170	cchonolulu-000	0.0054	1297	0.0395	1300	0.2802	958	0.0036	1336	0.0012	984
171	cchonolulu-001	0.0056	1298	0.0414	1301	0.2894	960	0.0038	1342	0.0012	987
172	cdachyd-001	0.0005	1266	0.0639	1305	0.2112	949	0.0023	1312	-	1147
173	ceiaufgbr-000	0.0000	333	0.0000	103	0.0000	21	0.0000	188	-	1225
174	ceiec-001	-	1415	-	1371	-	975	0.0029	1330	0.0023	1035

Table 93: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
175	ceiec-002	0.0002	1251	0.0056	1211	-	1145	0.0036	1337	0.0031	1051
176	ceiec-003	0.0000	242	0.0013	926	-	1466	0.0001	659	0.0004	681
177	ceiec-004	0.0000	107	0.0008	693	-	1330	0.0000	498	0.0004	522
178	celloscope-001	0.0000	1115	0.0056	1210	0.0540	859	0.0022	1309	-	1101
179	chosun-000	0.0000	498	0.0000	64	-	1199	0.0000	279	0.0000	6
180	chosun-001	0.0000	189	0.0000	167	-	1438	0.0000	125	0.0000	129
181	chosun-002	0.0000	343	0.0000	96	-	1036	0.0000	178	0.0000	87
182	chtface-001	0.0000	20	0.0000	192	-	1219	0.0000	4	0.0000	228
183	chtface-002	0.0000	1119	0.0021	1041	-	1139	0.0002	996	0.0007	915
184	chtface-003	0.0000	1046	0.0018	997	-	1364	0.0001	677	0.0006	874
185	chtface-004	0.0000	309	0.0017	978	0.0320	742	0.0000	549	0.0004	691
186	chtface-005	0.0000	215	0.0017	979	0.0320	745	0.0000	545	0.0004	696
187	chtface-006	0.0000	486	0.0017	977	0.0320	743	0.0000	552	0.0004	687
188	chtface-007	0.0000	97	0.0017	980	0.0320	744	0.0000	544	-	1394
189	chtface-008	0.0000	871	0.0009	722	0.0221	604	0.0000	494	-	1102
190	chtface-009	0.0000	672	0.0009	723	0.0221	605	0.0000	481	-	1320
191	cib-000	0.0000	170	0.0000	151	-	1399	0.0001	815	0.0000	154
192	cib-001	0.0000	253	0.0000	179	-	1448	0.0000	127	0.0000	125
193	cist-001	0.0000	192	0.0005	483	0.0087	279	0.0000	126	0.0000	127
194	cist-002	0.0000	396	0.0048	1189	0.0128	359	0.0000	233	0.0000	45
195	cist-003	0.0000	353	0.0011	846	0.0343	781	0.0000	213	0.0000	63
196	cist-004	0.0000	159	0.0005	446	0.0108	323	0.0000	99	0.0000	147
197	clearviewai-000	0.0000	773	0.0003	327	0.0081	258	0.0000	563	0.0003	314
198	clearviewai-001	0.0000	584	0.0006	524	0.0174	468	0.0001	612	0.0004	469
199	clearviewai-002	0.0000	889	0.0009	782	0.0244	650	0.0001	638	-	1483
200	clearviewai-003	0.0000	918	0.0009	781	0.0244	651	0.0001	643	-	1376
201	closeli-001	0.0000	311	0.0000	81	0.0000	9	0.0000	165	0.0000	92
202	cloudmatrix-000	0.0000	962	0.0012	884	-	1095	0.0001	650	0.0004	454
203	cloudmatrix-001	0.0000	920	0.0028	1082	0.0225	608	0.0001	642	0.0004	466
204	cloudmatrix-002	0.0000	928	0.0028	1083	0.0225	609	0.0001	646	0.0004	463
205	cloudwalk-hr-000	-	1426	-	1385	-	1010	0.0003	1040	0.0008	930
206	cloudwalk-hr-001	0.0000	716	0.0007	629	-	1445	0.0001	620	0.0004	566
207	cloudwalk-hr-002	0.0000	724	0.0007	625	-	999	0.0002	927	0.0004	709
208	cloudwalk-hr-003	0.0000	802	0.0008	697	-	1082	0.0001	676	0.0004	487
209	cloudwalk-hr-004	0.0000	800	0.0011	865	-	1078	0.0004	1089	0.0003	406
210	cloudwalk-mt-000	0.0000	561	0.0008	666	-	1265	0.0002	928	0.0004	716
211	cloudwalk-mt-002	0.0000	539	0.0003	319	-	1239	0.0001	618	0.0004	475
212	cloudwalk-mt-003	0.0000	617	0.0007	588	-	1316	0.0002	915	0.0004	714
213	cloudwalk-mt-004	0.0000	613	0.0009	712	0.0191	525	0.0002	942	0.0004	743
214	cloudwalk-mt-005	0.0000	864	0.0005	461	0.0130	366	0.0003	998	0.0004	706
215	cloudwalk-mt-006	0.0000	609	0.0006	511	0.0158	434	0.0002	917	0.0004	713
216	cloudwalk-mt-007	0.0000	533	0.0006	509	0.0158	433	0.0002	914	0.0004	718
217	clova-000	0.0000	1100	0.0022	1049	-	1022	0.0006	1195	0.0005	788
218	cmcuni-001	0.0000	7	0.0005	469	0.0119	348	0.0000	12	0.0000	222
219	codeline-000	0.0000	6	0.0000	195	0.0000	84	0.0000	15	0.0000	220
220	cogent-000	-	1389	-	1427	-	1378	-	1419	0.0005	836
221	cogent-001	-	1456	-	1333	-	1101	-	1458	0.0004	757
222	cogent-002	-	1335	-	1457	-	1241	0.0002	854	0.0004	765
223	cogent-003	0.0000	1058	0.0018	998	-	1103	0.0001	678	0.0004	607
224	cogent-004	0.0000	368	0.0000	119	-	1080	0.0000	199	0.0000	70
225	cogent-005	0.0000	282	0.0000	66	-	978	0.0000	146	0.0000	109
226	cogent-006	0.0000	401	0.0000	18	0.0000	44	0.0000	227	0.0000	50
227	cogent-007	0.0000	1052	0.0000	249	0.0000	152	0.0000	511	0.0000	246
228	cogent-008	0.0000	199	0.0010	834	0.0304	730	0.0000	571	0.0004	437
229	cognitec-000	0.0005	1262	0.0112	1259	-	1412	0.0007	1214	0.0007	922
230	cognitec-001	0.0003	1258	0.0110	1258	-	1067	0.0008	1233	0.0010	944
231	cognitec-002	0.0001	1200	0.0069	1225	-	1136	0.0003	1054	0.0005	799
232	cognitec-003	0.0001	1201	0.0194	1274	0.0820	919	0.0003	1043	0.0005	815

Table 94: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
233	cognitec-004	0.0001	1202	0.0037	1153	0.0580	866	0.0003	1044	0.0005	813
234	cognitec-005	0.0000	1104	0.0026	1068	0.0550	861	0.0002	868	0.0005	796
235	cognitec-006	0.0000	1112	0.0016	969	0.0511	845	0.0002	847	-	1204
236	cognitec-007	0.0000	1193	0.0025	1067	0.0736	912	0.0002	898	-	1118
237	cognitec-008	0.0000	1050	0.0030	1112	0.0523	852	0.0001	816	-	1293
238	compal-000	0.0000	28	0.0000	208	0.0000	93	0.0000	34	-	1435
239	compal-001	0.0000	1002	0.0060	1217	0.0320	746	0.0012	1266	-	1274
240	compal-002	0.0000	994	0.0060	1218	0.0320	747	0.0012	1265	-	1381
241	cor-001	0.0000	548	0.0006	522	-	1269	0.0002	971	0.0004	633
242	coretech-000	0.0000	95	0.0000	235	0.0000	111	0.0000	64	0.0000	180
243	coretech-001	0.0000	1176	0.0033	1133	0.0677	893	0.0005	1176	0.0011	974
244	coretech-002	0.0000	269	0.0013	927	0.0301	724	0.0000	363	0.0004	644
245	coretech-003	0.0000	53	0.0012	896	0.0282	707	0.0000	476	-	1458
246	coretech-004	0.0000	99	0.0013	928	0.0301	725	0.0000	351	-	1398
247	corsight-001	0.0000	860	0.0006	547	-	1152	0.0001	825	0.0004	574
248	corsight-002	0.0000	744	0.0005	489	0.0152	414	0.0001	757	0.0004	594
249	corsight-003	0.0000	559	0.0006	549	0.0175	471	0.0001	727	0.0004	617
250	csc-001	0.0000	1151	0.0030	1109	-	1183	0.0002	874	0.0005	787
251	csc-002	0.0015	1286	0.0033	1126	-	1019	0.0006	1203	0.0006	904
252	csc-003	0.0015	1285	0.0033	1127	0.0445	837	0.0006	1202	0.0006	905
253	ctbcbank-000	0.0001	1207	0.0051	1198	-	1409	0.0011	1255	0.0019	1021
254	ctbcbank-001	0.0000	1128	0.0036	1151	-	1068	0.0005	1170	0.0010	945
255	ctbcbank-002	0.0000	445	0.0006	492	0.0150	408	0.0000	405	-	1156
256	cu-face-002	0.0000	219	0.0000	162	0.0000	129	0.0000	102	0.0000	145
257	cu-face-003	0.0000	1063	0.0009	771	0.0245	655	0.0003	1007	0.0004	704
258	cu-face-004	0.0000	909	0.0007	641	0.0179	480	0.0001	783	-	1433
259	cubox-001	0.0000	88	0.0000	233	-	1335	0.0000	65	0.0000	176
260	cubox-002	0.0000	930	0.0006	536	0.0159	437	0.0002	972	0.0005	854
261	cubox-003	0.0000	423	0.0010	826	0.0210	573	0.0002	870	0.0004	675
262	cubox-004	0.0000	1033	0.0009	772	0.0193	530	0.0003	1064	-	1289
263	cubox-005	0.0000	1034	0.0009	773	0.0193	531	0.0003	1065	-	1249
264	cudocommunication-001	0.0000	326	0.0000	100	0.0000	24	0.0000	189	0.0000	80
265	cuhkee-001	0.0000	681	0.0011	861	-	1420	0.0000	358	0.0004	596
266	cybercore-000	0.0000	826	0.0073	1229	-	1114	0.0001	710	0.0005	801
267	cybercore-001	0.0000	1073	0.0001	272	0.0014	167	0.0002	860	0.0002	270
268	cybercore-002	0.0000	1088	0.0001	271	0.0014	166	0.0002	865	0.0002	264
269	cybercore-003	0.0000	641	0.0003	331	0.0060	206	0.0005	1177	0.0003	326
270	cyberextruder-001	0.0029	1290	0.0293	1286	-	1168	0.0024	1315	0.0029	1048
271	cyberextruder-002	0.0013	1282	0.0840	1317	-	1047	0.0027	1325	0.0028	1044
272	cyberextruder-003	0.0000	1075	0.0077	1233	0.0887	925	0.0001	826	0.0006	894
273	cyberextruder-004	0.0000	1069	0.0097	1250	0.1025	931	0.0001	800	0.0007	907
274	cyberextruder-005	0.0000	71	0.0000	229	0.0000	103	0.0000	46	-	1424
275	cyberextruder-006	0.0000	153	0.0000	128	0.0000	114	0.0000	71	-	1387
276	cyberlink-000	-	1446	-	1338	-	1119	0.0006	1200	0.0008	927
277	cyberlink-001	-	1341	-	1464	-	1277	0.0073	1360	0.0005	792
278	cyberlink-002	-	1481	-	1364	-	1179	0.0004	1077	0.0004	766
279	cyberlink-003	0.0000	145	0.0007	618	-	1352	0.0001	632	0.0004	519
280	cyberlink-004	0.0000	538	0.0010	842	-	1238	0.0004	1078	0.0003	415
281	cyberlink-005	0.0000	547	0.0009	740	-	1226	0.0003	1060	0.0003	416
282	cyberlink-006	0.0000	511	0.0005	431	-	1177	0.0000	347	0.0003	354
283	cyberlink-007	0.0000	36	0.0003	298	0.0067	225	0.0000	318	0.0003	368
284	cyberlink-008	0.0000	411	0.0004	400	0.0106	309	0.0000	336	0.0003	357
285	cyberlink-009	0.0000	480	0.0004	404	0.0106	313	0.0000	344	0.0003	355
286	cyberlink-010	0.0000	26	0.0004	408	0.0106	314	0.0000	319	0.0003	367
287	cyberlink-011	0.0000	387	0.0004	401	0.0106	310	0.0000	339	0.0003	356
288	cyberlink-012	0.0000	87	0.0004	410	0.0106	316	0.0000	321	0.0003	366
289	cyberlink-013	0.0000	304	0.0004	406	0.0106	308	0.0000	330	0.0003	362
290	cyberlink-014	0.0000	443	0.0004	402	0.0106	312	0.0000	342	-	1151

Table 95: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
291	cyberlink-015	0.0000	453	0.0004	403	0.0106	311	0.0000	341	-	1160
292	cyberlink-016	0.0000	78	0.0004	409	0.0106	315	0.0000	320	-	1431
293	dahua-000	-	1460	-	1353	-	1143	-	1472	0.0000	30
294	dahua-001	-	1344	-	1460	-	1258	0.0000	24	0.0000	214
295	dahua-002	-	1369	-	1417	-	1362	0.0024	1318	0.0022	1032
296	dahua-003	0.0000	461	0.0019	1004	-	1174	0.0002	930	0.0003	309
297	dahua-004	0.0000	327	0.0009	779	-	1062	0.0001	787	0.0003	287
298	dahua-005	0.0000	382	0.0000	244	-	1064	0.0000	514	0.0003	360
299	dahua-006	0.0000	64	0.0000	247	-	1310	0.0000	568	0.0003	413
300	dahua-007	0.0000	162	0.0000	245	0.0000	154	0.0000	570	0.0003	410
301	daon-000	0.0000	1135	0.0028	1091	0.0577	865	0.0014	1278	0.0015	1010
302	daon-001	0.0000	323	0.0009	720	0.0206	556	0.0000	488	-	1218
303	daon-002	0.0000	1	0.0004	421	0.0094	291	0.0000	16	-	1464
304	daon-003	0.0000	232	0.0008	671	0.0174	469	0.0000	386	-	1297
305	datech-000	0.0000	589	0.0004	356	0.0089	284	0.0000	506	0.0003	295
306	datech-001	0.0000	809	0.0004	354	0.0089	283	0.0000	513	-	1211
307	datech-002	0.0000	696	0.0004	355	0.0089	285	0.0000	508	-	1294
308	decatur-000	0.0000	945	0.0020	1021	-	1021	0.0004	1120	0.0005	781
309	decatur-001	0.0000	841	0.0009	766	0.0194	533	0.0001	687	0.0004	557
310	decloakface-001	0.0000	851	0.0016	968	0.0392	813	0.0001	740	0.0005	863
311	decloakface-002	0.0000	420	0.0011	851	0.0218	599	0.0001	623	-	1193
312	deepglint-001	0.0000	29	0.0000	209	-	1270	0.0000	33	0.0000	205
313	deepglint-002	0.0000	882	0.0004	405	-	1188	0.0002	944	0.0004	531
314	deepglint-003	0.0000	630	0.0004	407	-	1347	0.0002	943	0.0004	547
315	deepglint-004	0.0000	535	0.0005	434	0.0130	361	0.0002	962	0.0004	528
316	deepglint-005	0.0000	984	0.0019	1007	0.0438	834	0.0006	1193	0.0006	902
317	deepsea-001	0.0000	407	0.0000	21	-	1113	0.0000	222	0.0000	53
318	deepsense-000	0.0000	13	0.0006	551	-	1246	0.0000	409	0.0004	431
319	deepsense-001	0.0000	433	0.0006	558	0.0191	519	0.0000	446	0.0004	433
320	deepsense-002	0.0000	109	0.0006	560	0.0191	520	0.0000	429	0.0004	441
321	deepsense-003	0.0000	292	0.0004	390	0.0110	325	0.0000	362	0.0003	325
322	deepsense-004	0.0000	834	0.0007	604	0.0179	481	0.0001	697	-	1190
323	deepsense-005	0.0000	412	0.0005	487	0.0144	394	0.0001	661	-	1185
324	deepsense-006	0.0000	112	0.0005	436	0.0113	337	0.0000	431	-	1407
325	deepsense-007	0.0000	436	0.0005	433	0.0113	335	0.0000	447	-	1150
326	deepsense-008	0.0000	18	0.0005	435	0.0113	336	0.0000	426	-	1478
327	dermalog-001	-	1348	-	1476	-	1308	-	1399	0.0013	998
328	dermalog-002	-	1370	-	1419	-	1363	-	1416	0.0013	997
329	dermalog-003	-	1466	-	1350	-	1123	-	1467	0.0025	1037
330	dermalog-004	-	1478	-	1368	-	1205	-	1481	0.0090	1075
331	dermalog-005	-	1461	0.0031	1115	-	1131	0.0013	1274	0.0041	1062
332	dermalog-006	0.0005	1264	0.0031	1119	-	1292	0.0013	1273	0.0041	1063
333	dermalog-007	0.0000	714	0.0006	505	-	1443	0.0000	547	0.0002	274
334	dermalog-008	0.0000	1114	0.0031	1116	-	1148	0.0006	1185	0.0003	297
335	dermalog-009	0.0000	1105	0.0031	1118	0.0148	402	0.0006	1183	0.0003	299
336	dermalog-010	0.0000	1111	0.0031	1117	0.0148	401	0.0006	1184	0.0003	298
337	dermalog-011	0.0000	1164	0.0005	438	0.0116	342	0.0001	634	0.0003	315
338	dermalog-012	0.0000	522	0.0003	342	0.0068	230	0.0003	1011	0.0005	870
339	dermalog-013	0.0000	761	0.0002	294	0.0051	194	0.0002	857	-	1260
340	dermalog-014	0.0000	793	0.0003	340	0.0068	229	0.0003	1013	-	1194
341	dermalog-015	0.0000	769	0.0003	339	0.0068	228	0.0003	1012	-	1267
342	dermalog-016	0.0000	216	0.0003	341	0.0068	231	0.0001	613	-	1332
343	dicio-001	0.0005	1270	0.0649	1306	0.2136	952	0.0024	1314	0.0012	983
344	dicio-002	0.0000	426	0.0007	594	0.0185	501	0.0000	423	-	1144
345	didiglobalface-001	0.0000	536	0.0012	871	-	1233	0.0000	594	0.0004	703
346	didiglobalface-002	0.0000	611	0.0012	872	0.0247	661	0.0000	595	0.0004	699
347	didiglobalface-003	0.0000	677	0.0012	874	0.0247	662	0.0000	598	-	1326
348	digidata-000	0.0000	530	0.0023	1056	0.0375	803	0.0004	1130	0.0006	888

Table 96: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
349	digidata-001	0.0000	843	0.0023	1054	0.0375	801	0.0004	1132	0.0006	887
350	digitalbarriers-000	-	1403	-	1441	-	1455	-	1432	0.0019	1022
351	digitalbarriers-001	-	1448	-	1341	-	1106	-	1462	0.0018	1019
352	digitalbarriers-002	0.0001	1233	0.0045	1181	-	1011	0.0028	1327	0.0027	1041
353	dps-000	0.0000	11	0.0000	197	0.0000	82	0.0000	11	0.0000	223
354	dralmey-001	0.0000	833	0.0007	590	0.0185	500	0.0000	420	-	1188
355	dsk-000	0.0000	342	0.0000	94	-	1053	0.0000	180	0.0000	86
356	einetworks-000	0.0000	1127	0.0017	984	-	971	0.0002	926	0.0005	841
357	einetworksindia-000	0.0000	1094	0.0018	994	0.0516	847	0.0002	993	0.0005	845
358	einetworksindia-001	0.0000	1089	0.0018	995	0.0516	848	0.0002	989	0.0005	847
359	einetworksindia-002	0.0000	621	0.0006	506	0.0159	439	0.0000	607	-	1366
360	ekin-002	0.0000	437	0.0000	251	0.0004	161	0.0000	343	0.0000	239
361	element-000	0.0000	821	0.0008	703	0.0192	528	0.0000	504	0.0003	370
362	enface-000	0.0000	221	0.0012	893	0.0305	732	0.0000	537	0.0004	612
363	enface-001	0.0000	508	0.0012	890	0.0304	731	0.0000	473	0.0004	583
364	enface-002	0.0000	604	0.0004	371	0.0084	272	0.0000	398	0.0004	440
365	enqura-001	0.0003	1253	0.0203	1277	0.1737	945	0.0023	1311	-	1305
366	enqura-002	0.0000	371	0.0000	121	0.0000	33	0.0000	201	-	1202
367	eocortex-000	0.0095	1308	0.0602	1304	-	1441	0.0094	1364	0.0059	1070
368	ercacat-001	0.0000	271	0.0005	471	-	983	0.0000	538	0.0003	372
369	ethiopianlmis-000	0.0001	1204	0.0019	1013	0.0159	441	0.0009	1240	-	1312
370	euronovate-001	0.0255	1323	0.0102	1255	0.0517	850	0.0021	1305	0.0004	764
371	euronovate-002	0.0000	874	0.0010	791	0.0265	695	0.0000	450	0.0004	686
372	euronovate-003	0.0000	839	0.0006	515	0.0159	436	0.0002	922	0.0004	688
373	euronovate-004	0.0000	1097	0.0016	961	0.0236	640	0.0015	1285	0.0006	891
374	everai-000	-	1468	-	1358	-	1170	-	1478	0.0004	448
375	everai-001	-	1400	-	1448	-	1463	0.0004	1106	0.0004	762
376	everai-002	-	1442	-	1403	-	1075	0.0002	933	0.0004	455
377	everai-paravision-003	0.0000	915	0.0008	706	-	1317	0.0002	866	0.0004	524
378	expasoft-000	0.0000	24	0.0000	194	-	1225	0.0000	1	0.0000	230
379	expasoft-001	0.0000	450	0.0000	28	-	1121	0.0000	240	0.0000	39
380	expasoft-002	0.0000	373	0.0000	105	0.0000	27	0.0000	196	0.0000	72
381	f8-001	0.0003	1256	0.0059	1213	-	1046	0.0035	1335	0.0030	1050
382	f8-002	0.0000	1183	0.0150	1268	0.0685	897	0.0005	1136	0.0013	1002
383	facehawk-000	0.0000	1054	0.0357	1294	0.0720	909	0.0021	1306	0.0009	938
384	facehawk-001	0.0299	1325	0.0098	1251	0.0236	641	0.0253	1378	-	1319
385	faceint-001	0.0001	1222	0.0190	1272	0.0898	926	0.0014	1283	-	1300
386	facelocate-001	0.0000	154	0.0019	1010	0.0329	762	0.0000	560	-	1390
387	facelocate-002	0.0000	772	0.0025	1065	0.0387	811	0.0001	622	-	1214
388	faceonlive-001	0.0000	1155	0.0029	1098	0.0481	842	0.0013	1276	0.0011	961
389	faceonlive-002	0.0002	1249	0.0009	769	0.0075	244	0.0008	1227	0.0008	935
390	faceonlive-003	0.0000	1003	0.0202	1275	0.0980	928	0.0005	1155	-	1280
391	facephi-000	0.0000	283	0.0004	358	0.0090	287	0.0001	758	0.0004	460
392	facephi-001	0.0000	963	0.0034	1138	0.0499	844	0.0003	1027	-	1189
393	facephi-002	0.0000	566	0.0016	966	0.0286	714	0.0002	938	-	1453
394	facephi-003	0.0000	274	0.0006	535	0.0153	416	0.0001	731	-	1276
395	facesoft-000	0.0000	184	0.0000	144	-	1375	0.0000	87	0.0000	159
396	facetag-000	0.0000	393	0.0000	10	0.0000	53	0.0000	231	0.0000	44
397	facetag-001	0.0000	267	0.0000	75	0.0000	2	0.0000	148	0.0000	105
398	facetag-002	0.0000	166	0.0000	148	0.0000	128	0.0000	97	0.0000	151
399	facex-001	0.0001	1243	0.0360	1295	-	1351	0.0047	1349	0.0027	1042
400	facex-002	0.0001	1242	0.0360	1296	0.2663	956	0.0047	1348	0.0027	1043
401	facia-001	0.0000	488	0.0005	432	0.0122	351	0.0001	617	0.0004	477
402	facia-002	0.0000	1010	0.0017	982	0.0185	499	0.0002	978	-	1237
403	faicetech-000	0.0000	869	0.0007	608	0.0130	364	0.0002	983	-	1097
404	farfaces-001	0.0000	1123	0.0007	607	0.0061	208	0.0003	1034	0.0003	335
405	fastenterprises-000	0.0000	650	0.0082	1242	0.0169	462	0.0001	821	0.0003	365
406	fci-000	0.0000	182	0.0000	143	0.0000	125	0.0000	90	-	1359

Table 97: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
407	fdu-000	-	1377	-	1413	-	1344	-	1409	0.0011	978
408	fdu-001	-	1396	-	1439	-	1433	-	1427	0.0009	943
409	fedu-001	0.0000	12	0.0000	198	0.0000	83	0.0000	10	-	1470
410	fiberhome-nanjing-002	0.0000	974	0.0006	554	-	1215	0.0001	701	0.0004	500
411	fiberhome-nanjing-003	0.0000	2	0.0004	380	-	1247	0.0000	17	0.0003	305
412	fiberhome-nanjing-004	0.0000	177	0.0004	379	-	1385	0.0000	93	0.0003	304
413	fincore-000	0.0000	848	0.0008	699	0.0185	504	0.0001	624	0.0004	674
414	firstcreditkz-001	0.0000	1008	0.0019	1017	0.0321	748	0.0000	588	0.0004	592
415	firstcreditkz-002	0.0000	866	0.0010	836	0.0232	637	0.0000	542	0.0004	504
416	firstcreditkz-003	0.0000	1131	0.0010	828	0.0219	600	0.0000	520	-	1316
417	firstcreditkz-004	0.0000	807	0.0004	396	0.0124	353	0.0000	335	-	1207
418	foomobi-001	0.0007	1273	0.0000	176	0.0000	140	0.0020	1302	0.0011	964
419	foomobi-002	0.0007	1274	0.0000	93	0.0000	18	0.0020	1303	0.0011	963
420	fourthlinecom-001	0.0000	953	0.0004	393	0.0078	251	0.0005	1149	-	1268
421	fpt-000	0.0001	1211	0.0007	614	0.0076	246	0.0006	1205	-	1159
422	fpt-001	0.0001	1213	0.0005	477	0.0067	226	0.0006	1189	-	1104
423	fpttelecom-000	0.0000	395	0.0000	11	0.0000	54	0.0000	232	-	1169
424	fraudcom-000	0.0000	958	0.0055	1207	0.0308	736	0.0002	936	0.0004	676
425	frpkauai-001	0.0000	983	0.0024	1064	0.0360	790	0.0001	639	0.0004	730
426	frpkauai-002	0.0000	998	0.0019	1018	0.0321	750	0.0000	586	0.0004	595
427	frpkauai-003	0.0000	873	0.0006	525	0.0145	398	0.0000	493	-	1105
428	frpkauai-004	0.0001	1232	0.0051	1196	0.0730	911	0.0011	1256	-	1141
429	frpkauai-005	0.0000	881	0.0006	527	0.0145	397	0.0000	492	-	1113
430	fujitsulab-000	0.0000	1170	0.0033	1129	-	1366	0.0005	1153	0.0002	263
431	fujitsulab-001	0.0000	277	0.0014	937	-	974	0.0002	951	0.0003	396
432	fujitsulab-002	0.0000	200	0.0009	730	-	1432	0.0001	784	0.0003	303
433	fujitsulab-003	0.0000	150	0.0008	673	0.0166	455	0.0001	749	0.0001	257
434	g42-intellibrain-001	0.0000	118	0.0000	232	0.0000	105	0.0000	52	0.0000	187
435	g42-intellibrain-002	0.0000	340	0.0000	91	0.0000	17	0.0000	183	0.0000	84
436	geo-000	0.0000	479	0.0000	36	-	1159	0.0000	255	0.0000	26
437	geo-001	0.0000	819	0.0011	848	-	1120	0.0000	590	0.0004	587
438	geo-002	0.0000	778	0.0015	948	0.0332	768	0.0001	615	0.0004	732
439	geo-003	0.0000	651	0.0010	805	0.0244	652	0.0000	383	0.0004	726
440	geo-004	0.0000	766	0.0005	485	0.0138	380	0.0001	716	0.0004	540
441	gistouch-000	0.0000	1182	0.0188	1271	0.1345	937	0.0018	1296	0.0007	908
442	gistouch-001	0.0000	98	0.0002	277	0.0027	178	0.0001	704	0.0003	316
443	glory-000	-	1381	-	1420	-	1384	0.0053	1354	0.0013	1000
444	glory-001	0.0160	1314	0.0314	1290	-	1035	0.0051	1353	0.0010	949
445	glory-002	0.0003	1254	0.0045	1180	-	1093	0.0015	1286	0.0011	976
446	glory-003	0.0000	999	0.0027	1081	-	1410	0.0004	1088	0.0005	795
447	glory-004	0.0000	908	0.0020	1029	0.0345	783	0.0001	762	0.0004	728
448	glory-005	0.0000	911	0.0020	1030	0.0345	784	0.0001	764	0.0004	727
449	glory-006	0.0000	956	0.0020	1028	0.0345	782	0.0001	771	0.0004	722
450	glory-007	0.0000	1015	0.0020	1036	0.0403	824	0.0002	945	0.0005	779
451	gorilla-000	-	1473	-	1356	-	1155	-	1475	0.0004	768
452	gorilla-001	-	1436	-	1395	-	1044	0.0003	1066	0.0004	769
453	gorilla-002	-	1402	-	1440	-	1453	0.0003	1063	0.0004	771
454	gorilla-003	-	1471	-	1361	-	1161	0.0003	1068	0.0004	767
455	gorilla-004	0.0000	111	0.0008	685	-	1320	0.0000	455	0.0003	343
456	gorilla-005	0.0000	421	0.0008	692	-	1091	0.0000	470	0.0003	336
457	gorilla-006	0.0000	152	0.0006	570	-	1342	0.0000	412	0.0003	341
458	gorilla-007	0.0000	746	0.0009	784	0.0252	676	0.0001	684	0.0004	661
459	gorilla-008	0.0000	652	0.0009	785	0.0259	688	0.0001	681	0.0004	682
460	gorilla-009	0.0000	825	0.0010	813	0.0276	705	0.0001	651	0.0004	640
461	gpstechvn-000	0.0000	518	0.0000	54	0.0000	72	0.0000	273	0.0000	12
462	graymatics-001	0.0000	254	0.0010	789	0.0210	569	0.0001	822	0.0004	567
463	griaule-000	0.0000	1146	0.0026	1079	0.0531	857	0.0004	1131	0.0010	946
464	griaule-001	0.0000	432	0.0012	892	0.0366	795	0.0000	422	0.0004	658

Table 98: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
465	griaule-002	0.0000	3	0.0007	619	0.0209	560	0.0000	581	0.0004	582
466	gtelots-000	0.0000	996	0.0003	347	0.0081	265	0.0004	1085	-	1383
467	gtelots-001	0.0000	1022	0.0009	745	0.0177	475	0.0004	1099	-	1479
468	hertasecurity-000	0.0133	1313	0.0077	1234	-	1032	0.0025	1321	0.0243	1082
469	hertasecurity-001	0.0000	357	0.0000	254	0.0000	155	0.0000	392	0.0001	248
470	hertasecurity-002	0.0000	507	0.0000	50	0.0000	74	0.0000	396	0.0000	234
471	hertasecurity-003	0.0000	241	0.0000	188	0.0000	148	0.0000	327	0.0000	238
472	hik-001	0.0000	202	0.0000	257	-	1417	0.0000	113	0.0000	135
473	hisign-001	0.0000	296	0.0000	88	0.0000	12	0.0000	172	0.0000	88
474	hisign-002	0.0000	926	0.0006	539	0.0150	409	0.0001	776	0.0003	401
475	hisign-003	0.0000	951	0.0008	700	0.0237	643	0.0001	777	-	1265
476	hyperverge-001	0.0000	1179	0.0072	1226	-	1230	0.0015	1288	0.0014	1005
477	hyperverge-002	0.0000	471	0.0008	668	0.0210	575	0.0002	982	0.0004	554
478	hyperverge-003	0.0000	418	0.0008	662	0.0210	570	0.0002	980	0.0004	559
479	hyperverge-004	0.0000	595	0.0008	698	0.0218	596	0.0002	931	0.0004	599
480	hyperverge-005	0.0000	9	0.0008	664	0.0210	571	0.0002	975	0.0004	572
481	hyperverge-006	0.0000	128	0.0008	663	0.0210	572	0.0002	977	-	1368
482	hyperverge-007	0.0000	352	0.0008	669	0.0210	574	0.0002	979	-	1195
483	hyundai-roboticslab-000	0.0000	37	0.0011	864	0.0246	657	0.0001	746	-	1440
484	hyundai-roboticslab-001	0.0000	380	0.0105	1257	0.1972	946	0.0007	1217	-	1208
485	hyundai-roboticslab-002	0.0000	193	0.0020	1035	0.0299	723	0.0002	960	-	1317
486	hyundai-roboticslab-003	0.0000	142	0.0005	437	0.0136	374	0.0000	355	-	1378
487	hyundai-roboticslab-004	0.0000	130	0.0007	598	0.0200	537	0.0000	379	-	1367
488	hzailu-001	0.0000	1091	0.0016	967	0.0449	838	0.0003	1056	0.0005	850
489	hzailu-002	0.0000	1093	0.0015	955	0.0424	827	0.0003	1058	0.0005	849
490	hzailu-003	0.0000	639	0.0004	362	0.0081	264	0.0002	864	0.0003	385
491	hzailu-004	0.0000	614	0.0004	365	0.0081	262	0.0002	862	0.0003	387
492	hzailu-005	0.0000	520	0.0004	363	0.0081	260	0.0002	859	0.0003	391
493	hzailu-006	0.0000	637	0.0004	361	0.0081	263	0.0002	863	0.0003	384
494	hzailu-007	0.0000	892	0.0004	364	0.0081	261	0.0003	1018	-	1439
495	hzailu-008	0.0000	968	0.0004	360	0.0081	259	0.0003	1020	-	1134
496	i2v-001	0.0000	1095	0.0018	1001	0.0341	780	0.0001	709	-	1176
497	i2v-002	0.0000	1006	0.0012	867	0.0244	653	0.0000	528	-	1216
498	icehouse-000	0.0000	220	0.0000	163	0.0000	130	0.0000	101	-	1339
499	icm-002	0.0000	416	0.0001	262	-	1099	0.0000	218	0.0000	231
500	icm-003	0.0000	110	0.0001	263	0.0023	173	0.0000	55	0.0000	232
501	icm-004	0.0000	1143	0.0033	1137	0.0698	901	0.0006	1201	0.0010	957
502	icm-005	0.0002	1248	0.0117	1262	0.1223	935	0.0018	1299	0.0014	1009
503	icomm-000	0.0000	572	0.0004	386	0.0098	298	0.0000	556	0.0003	344
504	icthtc-000	0.0001	1240	0.0047	1188	-	1102	0.0028	1329	0.0029	1045
505	id3-001	-	1435	-	1394	-	1041	-	1449	0.0043	1066
506	id3-002	-	1452	-	1330	-	1094	-	1459	0.0032	1052
507	id3-003	0.0001	1203	0.0052	1201	-	1325	0.0016	1291	0.0011	975
508	id3-004	0.0000	1195	0.0052	1199	-	1214	0.0015	1289	0.0011	973
509	id3-005	0.0000	741	0.0074	1231	-	977	0.0002	884	0.0005	794
510	id3-006	0.0000	1016	0.0009	778	-	1163	0.0004	1105	0.0005	835
511	id3-007	0.0000	626	0.0041	1165	-	1358	0.0001	751	0.0004	631
512	id3-008	0.0000	286	0.0006	545	0.0184	496	0.0001	817	0.0004	436
513	id3-009	0.0000	937	0.0003	349	0.0088	280	0.0001	768	-	1272
514	id3-010	0.0000	504	0.0001	274	0.0030	179	0.0000	474	-	1106
515	idbio-000	0.0000	446	0.0008	679	0.0189	508	0.0000	603	-	1155
516	idemia-003	-	1394	-	1436	-	1437	0.0000	434	0.0004	461
517	idemia-004	0.0000	552	0.0006	523	-	1275	0.0000	477	0.0004	497
518	idemia-005	0.0000	83	0.0004	418	-	1283	0.0000	374	0.0003	389
519	idemia-006	0.0000	190	0.0004	417	-	1440	0.0000	384	0.0003	381
520	idemia-007	0.0000	116	0.0004	419	-	1311	0.0000	377	0.0003	388
521	idemia-008	0.0000	394	0.0004	411	0.0078	255	0.0000	394	0.0003	376
522	idemia-009	0.0000	316	0.0004	397	0.0077	249	0.0000	388	0.0003	378

Table 99: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
523	idemia-010	0.0000	92	0.0004	353	0.0074	239	0.0000	67	0.0003	342
524	idemia-011	0.0000	329	0.0003	302	0.0055	199	0.0000	185	-	1224
525	identy-000	0.0000	1184	0.0020	1032	0.0152	412	0.0012	1262	0.0004	720
526	identy-001	0.0000	1189	0.0013	925	0.0398	819	0.0002	968	-	1330
527	identy-002	0.0000	1192	0.0013	924	0.0401	820	0.0002	966	-	1230
528	identy-003	0.0000	1137	0.0011	857	0.0247	660	0.0001	714	-	1322
529	idrnd-000	0.0000	912	0.0020	1022	0.0566	863	0.0003	1062	-	1402
530	idrnd-001	0.0000	513	0.0006	526	0.0182	485	0.0000	425	-	1111
531	igearx-face-000	0.0000	654	0.0006	501	0.0153	418	0.0004	1097	0.0004	668
532	iit-000	-	1438	-	1406	-	1085	0.0007	1208	0.0011	966
533	iit-001	0.0000	1005	0.0806	1316	-	1012	0.0001	838	0.0004	643
534	iit-002	0.0000	1138	0.0021	1037	-	1186	0.0009	1242	0.0005	857
535	iit-003	0.0000	543	0.0008	701	-	1221	0.0000	495	0.0004	476
536	imds-software-001	0.0000	331	0.0000	101	0.0000	20	0.0000	187	0.0000	81
537	imds-software-002	0.0000	483	0.0000	38	0.0000	61	0.0000	254	0.0000	28
538	imds-software-003	0.0000	185	0.0000	166	0.0000	139	0.0000	123	0.0000	126
539	imperial-000	0.0000	429	0.0000	32	-	1147	0.0000	251	0.0000	29
540	imperial-001	-	1333	-	1454	-	1249	0.0000	18	0.0000	218
541	imperial-002	0.0000	197	0.0000	170	-	1430	0.0000	121	0.0000	131
542	incode-000	-	1447	-	1340	-	1105	-	1463	0.0035	1054
543	incode-001	-	1378	-	1432	-	1401	-	1426	0.0035	1055
544	incode-002	-	1423	-	1384	-	1003	0.0004	1126	0.0007	909
545	incode-003	-	1388	-	1429	-	1376	0.0004	1125	0.0007	910
546	incode-004	-	1357	-	1470	-	1279	0.0004	1123	0.0007	911
547	incode-005	0.0000	33	0.0008	661	-	1272	0.0001	690	0.0004	654
548	incode-006	0.0000	664	0.0005	451	-	1381	0.0001	799	0.0004	425
549	incode-007	0.0000	884	0.0009	751	-	1254	0.0002	891	0.0004	550
550	incode-008	0.0000	978	0.0009	746	-	1178	0.0002	899	0.0004	530
551	incode-009	0.0000	954	0.0009	748	0.0255	683	0.0002	897	0.0004	537
552	incode-010	0.0000	890	0.0009	750	0.0255	684	0.0002	889	0.0004	552
553	incode-011	0.0000	949	0.0009	747	0.0255	682	0.0002	896	0.0004	539
554	incode-012	0.0000	925	0.0009	749	0.0255	685	0.0002	894	0.0004	545
555	incode-013	0.0001	1245	0.0012	885	0.0254	680	0.0005	1162	0.0004	749
556	incode-014	0.0000	473	0.0006	555	0.0166	454	0.0001	831	-	1128
557	incode-015	0.0000	447	0.0007	634	0.0195	534	0.0000	445	-	1154
558	infocert-001	0.0000	1026	0.0059	1216	0.0424	828	0.0001	691	0.0006	878
559	infocert-002	0.0084	1304	0.0041	1173	0.0084	273	0.0085	1362	-	1348
560	infocert-003	0.0089	1307	0.0042	1175	0.0077	248	0.0087	1363	-	1186
561	innefulabs-000	0.0000	703	0.0024	1062	-	1461	0.0003	1031	0.0005	829
562	innominds-001	0.0000	1087	0.0030	1108	0.0637	881	0.0010	1253	0.0008	932
563	innov8tif-000	0.0000	1129	0.0029	1104	0.0657	885	0.0004	1118	-	1173
564	innov8tif-001	0.0000	1125	0.0029	1105	0.0657	886	0.0004	1116	-	1446
565	innov8tif-002	0.0258	1324	0.0862	1319	0.4368	962	0.1312	1383	-	1107
566	innovativetechnologyltd-001	0.0001	1237	0.0050	1195	-	1107	0.0024	1320	0.0025	1038
567	innovativetechnologyltd-002	0.0000	1018	0.0046	1183	-	1190	0.0057	1356	0.0005	832
568	innovatrics-000	-	1434	-	1393	-	1040	-	1450	0.0004	733
569	innovatrics-001	-	1358	-	1472	-	1280	-	1394	0.0004	738
570	innovatrics-002	-	1332	-	1455	-	1248	-	1390	0.0004	473
571	innovatrics-003	-	1368	-	1418	-	1361	-	1415	0.0005	785
572	innovatrics-004	0.0000	971	0.0017	983	-	1210	0.0000	593	0.0004	673
573	innovatrics-005	-	1454	-	1332	-	1097	0.0000	589	0.0004	660
574	innovatrics-006	0.0000	537	0.0009	775	-	1237	0.0000	522	0.0004	474
575	innovatrics-007	0.0000	788	0.0007	628	-	1042	0.0001	614	0.0003	337
576	innovatrics-008	0.0000	808	0.0009	765	0.0204	548	0.0000	529	0.0004	420
577	innovatrics-009	0.0000	278	0.0005	428	0.0142	389	0.0000	147	0.0000	240
578	innovatrics-010	0.0000	678	0.0019	1009	0.0393	814	0.0001	682	0.0004	438
579	innovatrics-011	0.0000	855	0.0001	264	0.0017	171	0.0000	346	-	1123
580	innovatrics-012	0.0000	828	0.0004	425	0.0118	343	0.0000	515	-	1178

Table 100: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
581	innovatrics-013	0.0000	745	0.0001	267	0.0027	177	0.0000	303	-	1290
582	innovatrics-014	0.0000	608	0.0007	596	0.0165	450	0.0000	430	-	1406
583	innovatrics-015	0.0000	526	0.0008	690	0.0236	642	0.0000	410	-	1467
584	innovatrics-016	0.0000	818	0.0006	499	0.0177	477	0.0000	310	-	1168
585	innovatrics-017	0.0000	675	0.0006	531	0.0189	511	0.0000	359	-	1325
586	insightface-000	0.0000	467	0.0000	44	0.0000	66	0.0000	262	0.0000	18
587	insightface-001	0.0000	201	0.0000	172	0.0000	138	0.0000	120	0.0000	134
588	insightface-002	0.0000	148	0.0000	126	0.0000	117	0.0000	74	0.0000	172
589	insightface-003	0.0000	19	0.0000	191	0.0000	80	0.0000	5	0.0000	229
590	insightface-004	0.0000	378	0.0000	107	0.0000	29	0.0000	194	0.0000	74
591	inspur-000	0.0000	66	0.0000	228	0.0000	101	0.0000	47	0.0000	193
592	inspur-001	0.0000	349	0.0000	110	0.0000	34	0.0000	212	0.0000	61
593	inspur-002	0.0000	249	0.0000	178	0.0000	142	0.0000	129	0.0000	124
594	inspur-003	0.0000	386	0.0000	109	0.0000	26	0.0000	193	-	1213
595	intellicloudai-001	0.0000	229	0.0000	182	-	1474	0.0000	142	0.0000	113
596	intellicloudai-002	0.0000	129	0.0008	683	-	1371	0.0000	500	0.0004	439
597	intellifusion-001	0.0000	573	0.0005	479	-	1259	0.0001	665	0.0003	405
598	intellifusion-002	0.0000	169	0.0000	253	-	1395	0.0000	296	0.0000	153
599	intellivision-001	0.0042	1294	0.0296	1287	-	1024	0.0048	1350	0.0042	1065
600	intellivision-002	0.0000	1188	0.0046	1182	-	1125	0.0012	1264	0.0005	868
601	intellivision-003	0.0000	645	0.0012	880	0.0308	734	0.0003	1019	0.0004	772
602	intellivision-004	0.0000	739	0.0011	852	0.0266	699	0.0002	986	0.0004	761
603	intellivision-005	0.0001	1230	0.0041	1167	0.0528	856	0.0005	1159	0.0007	919
604	intellivision-006	0.0001	1229	0.0041	1169	0.0528	855	0.0005	1157	0.0007	920
605	intellivision-007	0.0001	1228	0.0041	1168	0.0528	854	0.0005	1156	0.0007	921
606	intellivix-001	0.0000	449	0.0000	26	0.0000	59	0.0000	244	0.0000	38
607	intellivix-002	0.0000	102	0.0009	770	0.0184	497	0.0000	59	0.0000	181
608	intellivix-003	0.0000	60	0.0000	206	0.0000	88	0.0000	21	0.0000	217
609	intellivix-004	0.0000	464	0.0022	1045	0.0333	772	0.0000	267	0.0000	17
610	intellivix-005	0.0000	434	0.0022	1044	0.0333	771	0.0000	249	0.0000	31
611	intellivix-006	0.0000	295	0.0007	597	0.0180	482	0.0000	176	-	1247
612	intelresearch-000	0.0000	629	0.0006	493	-	1346	0.0000	561	0.0003	402
613	intelresearch-001	0.0000	1165	0.0082	1241	-	1424	0.0005	1168	0.0010	947
614	intelresearch-002	0.0000	582	0.0022	1042	-	1307	0.0000	559	0.0004	427
615	intelresearch-003	0.0000	791	0.0006	513	-	1045	0.0000	438	0.0004	511
616	intelresearch-004	0.0000	707	0.0006	518	-	1467	0.0000	435	0.0004	516
617	intelresearch-005	0.0000	787	0.0006	517	0.0144	396	0.0000	439	0.0004	510
618	intelresearch-006	0.0000	408	0.0000	248	0.0004	162	0.0000	338	0.0004	505
619	intelresearch-007	0.0000	472	0.0007	580	0.0168	458	0.0000	345	0.0004	502
620	intelresearch-008	0.0000	203	0.0006	571	0.0169	463	0.0000	324	0.0004	543
621	intelresearch-009	0.0000	42	0.0007	640	0.0232	635	0.0000	317	-	1443
622	intelresearch-010	0.0000	786	0.0010	835	0.0335	774	0.0000	512	-	1236
623	intelresearch-011	0.0000	737	0.0006	562	0.0175	470	0.0000	483	-	1285
624	intelresearch-012	0.0000	812	0.0007	606	0.0191	526	0.0000	490	-	1165
625	intema-000	0.0000	298	0.0005	448	0.0126	355	0.0000	573	0.0004	457
626	intema-001	0.0000	521	0.0004	395	0.0106	317	0.0000	315	0.0003	414
627	interlinklabs-001	0.0000	1142	0.0004	412	0.0085	276	0.0005	1148	-	1120
628	interlinklabs-002	0.0000	1140	0.0004	413	0.0085	275	0.0005	1147	-	1238
629	interlinklabs-003	0.0000	903	0.0031	1123	0.0249	667	0.0003	1036	-	1423
630	intozi-001	0.0000	592	0.0005	486	0.0126	354	0.0001	841	0.0004	429
631	intozi-002	0.0000	456	0.0000	40	0.0000	68	0.0000	268	-	1117
632	intsysmsu-000	-	1437	-	1405	-	1084	0.0004	1098	0.0012	979
633	intsysmsu-001	0.0000	188	0.0010	817	-	1436	0.0001	723	0.0004	627
634	intsysmsu-002	0.0000	291	0.0010	815	-	970	0.0001	725	0.0004	625
635	ionetworks-000	0.0000	186	0.0016	974	0.0387	810	0.0004	1071	0.0005	798
636	ionetworks-001	0.0000	337	0.0081	1238	0.0666	887	0.0000	182	0.0000	85
637	ionetworks-002	0.0000	137	0.0003	312	0.0048	190	0.0000	77	0.0000	168
638	ionetworks-003	0.0000	218	0.0004	370	0.0079	257	0.0000	108	-	1337

Table 101: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
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	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
639	ionetworks-004	0.0000	294	0.0004	359	0.0086	278	0.0000	175	-	1250
640	iqface-000	0.0000	227	0.0000	180	-	1471	0.0000	141	0.0000	112
641	iqface-001	0.0000	468	0.0000	47	-	1164	0.0000	311	0.0000	21
642	iqface-003	0.0000	1133	0.0076	1232	-	1158	0.0006	1186	0.0005	864
643	iqsec-001	0.0000	89	0.0763	1315	0.1617	941	0.0000	66	-	1392
644	iqsec-002	0.0000	209	0.0763	1314	0.1617	942	0.0000	109	-	1331
645	iqsec-003	0.0000	419	0.0000	4	0.0000	38	0.0000	214	-	1191
646	irex-000	0.0000	1038	0.0009	768	-	1015	0.0000	562	0.0005	782
647	irisid-001	0.0000	814	0.0016	963	0.0441	835	0.0000	610	-	1163
648	irisid-002	0.0000	542	0.0016	965	0.0441	836	0.0000	604	-	1480
649	isap-001	0.0000	38	0.0000	211	-	1276	0.0000	31	0.0000	207
650	isap-002	0.0000	347	0.0000	111	-	1083	0.0000	210	0.0000	62
651	isityou-000	0.0068	1302	0.0316	1291	-	1048	0.0023	1310	0.0010	953
652	istl-001	0.0000	1079	0.0005	490	0.0162	448	0.0010	1251	-	1360
653	isystems-000	-	1417	-	1372	-	967	-	1438	0.0010	954
654	isystems-001	0.0000	1154	0.0035	1147	-	1380	0.0010	1249	0.0007	916
655	isystems-002	0.0000	1153	0.0035	1148	-	1324	0.0010	1248	0.0007	917
656	itmo-001	-	1427	-	1396	-	1060	-	1453	0.0047	1067
657	itmo-002	-	1334	-	1456	-	1252	-	1389	0.0029	1049
658	itmo-003	-	1398	-	1447	-	1477	-	1434	0.0008	925
659	itmo-004	-	1419	-	1387	-	1030	-	1448	0.0004	758
660	itmo-005	-	1351	-	1477	-	1297	0.0005	1143	0.0002	268
661	itmo-006	0.0000	339	0.0015	954	-	1052	0.0004	1127	0.0004	621
662	itmo-007	0.0000	180	0.0009	725	-	1391	0.0003	1061	0.0000	158
663	itmo-008	0.0000	115	0.0135	1264	0.1239	936	0.0024	1319	0.0000	186
664	ivacognitive-001	0.0000	957	0.0011	854	-	1043	0.0001	649	0.0004	734
665	iws-000	0.0005	1268	0.0650	1309	-	1222	0.0024	1316	0.0012	986
666	jaakit-001	0.0008	1279	0.0858	1318	0.2713	957	0.0042	1343	0.0021	1028
667	jaakit-002	0.0000	344	0.0000	95	0.0000	14	0.0000	179	-	1239
668	kaizenvoiz-001	0.0000	1122	0.0000	39	0.0000	62	0.0037	1338	-	1138
669	kakao-001	-	1405	-	1443	-	1459	0.0002	950	0.0005	812
670	kakao-002	0.0000	986	0.0057	1212	-	1299	0.0002	949	0.0005	817
671	kakao-003	0.0000	266	0.0000	74	-	979	0.0000	149	0.0000	106
672	kakao-004	0.0000	402	0.0000	19	-	1108	0.0000	226	0.0000	49
673	kakao-005	0.0000	444	0.0000	241	-	1135	0.0000	242	0.0000	242
674	kakao-007	0.0000	100	0.0007	578	0.0165	451	0.0001	705	0.0004	521
675	kakao-008	0.0000	76	0.0009	737	0.0209	566	0.0001	703	0.0004	526
676	kakao-009	0.0000	279	0.0009	735	0.0209	564	0.0001	707	0.0004	514
677	kakaobank-000	0.0000	151	0.0000	127	0.0000	113	0.0000	70	0.0000	173
678	kakaobank-001	0.0000	314	0.0006	516	0.0176	473	0.0000	387	0.0004	458
679	kakaobank-002	0.0000	701	0.0010	843	0.0303	729	0.0000	465	-	1298
680	kakaobank-003	0.0000	820	0.0013	922	0.0397	816	0.0000	443	-	1171
681	kakaopay-001	0.0000	906	0.0013	919	0.0322	752	0.0001	654	0.0004	744
682	kasikornlabs-000	0.0000	1168	0.0035	1145	0.0713	907	0.0004	1122	0.0012	991
683	kasikornlabs-001	0.0001	1239	0.0050	1194	0.0885	924	0.0006	1204	0.0035	1057
684	kasikornlabs-002	0.0000	1163	0.0033	1132	0.0698	905	0.0004	1112	0.0012	980
685	kasikornlabs-003	0.0000	1161	0.0033	1130	0.0698	903	0.0004	1111	0.0012	981
686	kasikornlabs-004	0.0000	1162	0.0033	1131	0.0698	904	0.0004	1110	-	1352
687	kasikornlabs-005	0.0000	1166	0.0033	1128	0.0698	902	0.0004	1113	-	1271
688	kbyai-000	0.0000	861	0.0013	903	0.0265	698	0.0001	636	-	1130
689	kedacom-000	0.0000	514	0.0000	52	-	1181	0.0000	275	0.0000	10
690	keyless-000	0.0000	759	0.0010	811	0.0308	735	0.0000	468	-	1257
691	kiwitech-000	0.0000	752	0.0009	719	-	1033	0.0004	1114	0.0005	790
692	kloudspot-001	0.0000	997	0.0010	809	0.0211	576	0.0001	789	-	1363
693	kloudspot-002	0.0000	356	0.0006	563	0.0159	440	0.0002	937	-	1197
694	kneron-003	0.0239	1321	0.0306	1288	-	1175	0.0044	1347	0.0016	1017
695	kneron-005	0.0000	1158	0.0226	1279	-	1038	0.0006	1181	0.0005	823
696	kneron-006	0.0000	1120	0.1109	1322	0.0686	898	0.0012	1263	-	1122

Table 102: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
697	knowutech-000	0.0000	689	0.0008	655	0.0215	591	0.0000	519	0.0004	646
698	kogniza-001	0.0000	709	0.0012	898	0.0337	779	0.0003	1049	-	1302
699	kookmin-001	0.0000	164	0.0000	146	-	1403	0.0000	96	0.0000	149
700	kookmin-002	0.0000	270	0.0000	76	-	981	0.0000	150	0.0000	107
701	koreaid-001	0.0000	167	0.0023	1057	0.0371	799	0.0000	580	0.0005	784
702	krungthai-002	0.0000	708	0.0005	462	0.0111	328	0.0002	925	0.0003	407
703	kuke3d-001	0.0000	375	0.0000	106	0.0000	28	0.0000	197	0.0000	73
704	kuke3d-002	0.0000	239	0.0000	186	0.0000	146	0.0000	133	0.0000	118
705	lebentech-000	0.0042	1292	0.0029	1106	0.0252	675	0.0051	1352	0.0066	1072
706	lebentech-001	0.0000	1194	0.0012	897	0.0317	740	0.0004	1121	0.0005	842
707	lemalabs-001	0.0000	428	0.0005	482	0.0141	387	0.0002	918	0.0004	450
708	lightvision-000	0.0000	415	0.0000	3	0.0000	40	0.0000	219	-	1187
709	lightvision-001	0.0000	144	0.0000	122	0.0000	115	0.0000	76	-	1375
710	lineclova-000	0.0000	86	0.0000	223	0.0000	98	0.0000	40	0.0000	202
711	lineclova-001	0.0000	516	0.0000	55	0.0000	71	0.0000	271	0.0000	13
712	lineclova-002	0.0000	31	0.0007	579	0.0181	484	0.0000	35	0.0000	203
713	lineclova-003	0.0000	1110	0.0023	1051	0.0700	906	0.0002	988	0.0005	780
714	lookman-002	0.0000	127	0.0000	134	-	1370	0.0000	82	0.0000	164
715	lookman-004	0.0000	5	0.0000	196	-	1251	0.0000	14	0.0000	221
716	lssytech-000	0.0000	557	0.0006	552	0.0177	474	0.0000	533	-	1442
717	lssytech-001	0.0000	822	0.0004	381	0.0092	289	0.0000	530	-	1174
718	luxand-000	0.0000	21	0.0000	193	-	1218	0.0000	3	0.0000	227
719	luxand-001	0.0000	1036	0.0016	971	0.0366	793	0.0006	1191	0.0005	839
720	luxand-003	0.0000	1035	0.0016	972	0.0366	794	0.0006	1192	-	1256
721	luxand-004	0.0000	854	0.0003	334	0.0061	210	0.0003	1000	-	1124
722	mantra-000	0.0001	1205	0.0041	1170	0.0680	896	0.0003	1030	0.0004	756
723	matrixcomsec-000	0.0002	1246	1.0000	1449	1.0000	1462	0.0336	1380	-	1299
724	matrixcomsec-001	0.0000	301	0.0009	729	0.0252	674	0.0000	169	-	1255
725	matrixcomsec-002	0.0000	683	0.0009	731	0.0252	677	0.0001	713	-	1328
726	maverickresearch-000	0.0000	1017	0.0096	1249	0.0574	864	0.0013	1275	-	1132
727	maxvision-000	0.0000	280	0.0000	242	0.0000	1	0.0000	145	0.0000	108
728	maxvision-001	0.0000	178	0.0000	142	0.0000	124	0.0000	88	0.0000	156
729	maxvision-002	0.0000	801	0.0009	713	0.0229	625	0.0002	853	0.0004	590
730	maxvision-003	0.0000	524	0.0009	716	0.0229	626	0.0002	849	0.0004	603
731	maxvision-004	0.0000	624	0.0009	714	0.0229	628	0.0002	856	0.0004	597
732	maxvision-005	0.0000	628	0.0009	715	0.0229	627	0.0002	855	0.0004	598
733	maxvision-006	0.0000	798	0.0007	645	0.0226	611	0.0002	877	0.0004	589
734	mbbankvn-000	0.0000	594	0.0004	387	0.0098	299	0.0000	558	-	1432
735	megvii-000	-	1476	-	1357	-	1160	-	1474	0.0003	310
736	megvii-001	0.0000	252	0.0006	521	-	1447	0.0007	1215	0.0004	492
737	megvii-002	0.0000	361	0.0006	520	-	1088	0.0054	1355	0.0004	486
738	megvii-003	0.0000	698	0.0010	837	-	1475	0.0002	965	0.0004	711
739	megvii-004	0.0000	525	0.0010	812	-	1250	0.0002	890	0.0004	657
740	megvii-005	0.0000	616	0.0010	799	0.0206	554	0.0002	955	0.0004	700
741	megvii-006	0.0000	695	0.0010	797	0.0206	555	0.0002	956	0.0004	693
742	megvii-007	0.0000	591	0.0010	798	0.0206	553	0.0002	954	0.0004	702
743	megvii-008	0.0000	840	0.0010	792	0.0206	552	0.0002	959	0.0004	689
744	megvii-009	0.0000	742	0.0010	793	0.0206	551	0.0002	957	0.0004	692
745	meituan-000	0.0000	75	0.0001	268	0.0013	165	0.0000	375	0.0002	275
746	meituan-001	0.0000	684	0.0014	940	0.0295	719	0.0001	730	0.0004	667
747	meituan-002	0.0000	810	0.0013	914	0.0251	670	0.0001	732	0.0004	641
748	meituan-003	0.0000	577	0.0013	915	0.0251	672	0.0001	728	0.0004	652
749	meituan-004	0.0000	877	0.0013	913	0.0251	671	0.0001	733	0.0004	637
750	meiya-001	0.0000	1145	0.0028	1093	-	1398	0.0004	1128	0.0010	955
751	mendaxiatech-000	0.0000	688	0.0010	795	0.0206	550	0.0002	963	0.0004	695
752	metsakuurcompany-001	0.0000	255	0.0011	845	0.0208	559	0.0002	953	0.0004	512
753	metsakuurcompany-002	0.0000	46	0.0000	216	0.0000	91	0.0000	27	0.0000	211
754	metsakuurcompany-003	0.0000	315	0.0000	83	0.0000	5	0.0000	160	0.0000	94

Table 103: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
755	metsakuurcompany-004	0.0000	108	0.0000	240	0.0000	109	0.0000	58	-	1403
756	metsakuurcompany-005	0.0000	80	0.0000	221	0.0000	97	0.0000	38	-	1429
757	metsakuurcompany-006	0.0000	195	0.0005	442	0.0128	357	0.0000	546	-	1318
758	miaxis-001	0.0000	607	0.0013	909	0.0262	692	0.0001	819	0.0003	328
759	miaxis-002	0.0448	1330	0.1162	1323	0.2565	955	0.2128	1385	0.0347	1086
760	miaxis-003	0.0000	605	0.0013	910	0.0262	693	0.0001	820	0.0003	327
761	miaxis-004	0.0000	134	0.0000	137	0.0000	119	0.0000	81	-	1372
762	miaxis-005	0.0000	243	0.0000	174	0.0000	144	0.0000	131	-	1304
763	microfocus-000	-	1386	-	1425	-	1392	-	1421	0.0016	1015
764	microfocus-001	0.0001	1236	0.0053	1204	-	1034	0.0008	1229	0.0016	1013
765	microfocus-002	0.0001	1235	0.0053	1205	-	1444	0.0008	1228	0.0016	1014
766	microfocus-003	0.0001	1227	0.0049	1193	0.0819	918	0.0007	1219	0.0015	1012
767	minivision-000	0.0000	497	0.0000	65	-	1198	0.0000	278	0.0000	7
768	mitek-000	0.0000	1025	0.0029	1097	0.0521	851	0.0002	920	0.0003	352
769	mitek-001	0.0000	981	0.0020	1020	0.0566	862	0.0003	1069	-	1114
770	mitek-002	0.0000	332	0.0012	888	0.0373	800	0.0000	551	-	1222
771	mobai-000	0.0000	1057	0.0114	1260	-	1070	0.0003	1041	0.0012	989
772	mobai-001	0.0000	933	0.0040	1158	-	1482	0.0001	756	0.0012	988
773	mobai-002	0.0000	905	0.0008	659	0.0149	404	0.0002	861	-	1425
774	mobai-003	0.0000	287	0.0000	243	0.0001	156	0.0000	143	-	1287
775	mobbl-000	0.0116	1311	0.0720	1313	-	1217	0.0119	1369	0.0063	1071
776	mobbl-001	0.0000	1134	0.0052	1200	0.0678	894	0.0002	873	0.0005	843
777	mobbl-002	0.0000	1157	0.0029	1100	0.0633	880	0.0002	939	0.0009	941
778	mobbl-003	0.0000	1159	0.0029	1099	0.0633	879	0.0002	941	0.0009	940
779	mobbl-004	0.0000	273	0.0012	887	0.0312	738	0.0000	466	-	1277
780	mobbl-005	0.0000	762	0.0009	726	0.0217	593	0.0000	437	-	1261
781	mobipintech-000	0.0000	455	0.0000	30	0.0000	56	0.0000	238	0.0000	40
782	momovn-001	0.0015	1284	0.0010	787	0.0103	301	0.0011	1259	0.0008	928
783	momovn-002	0.0015	1283	0.0010	788	0.0103	302	0.0011	1258	-	1369
784	momovn-003	0.0000	936	0.0031	1121	0.0249	665	0.0003	1037	-	1269
785	moredian-000	0.0000	615	0.0009	721	-	1314	0.0004	1109	0.0005	791
786	morpho-000	-	1443	-	1404	-	1066	-	1454	0.0000	76
787	morpho-001	-	1352	-	1467	-	1287	-	1396	0.0000	197
788	morpho-002	-	1472	-	1362	-	1165	0.0000	554	0.0004	483
789	mukh-001	0.0000	39	0.0010	818	0.0154	427	0.0001	747	0.0003	292
790	mukh-002	0.0000	346	0.0022	1050	0.0513	846	0.0002	867	0.0004	576
791	mukh-003	0.0000	816	0.0003	304	0.0060	207	0.0001	843	0.0003	392
792	mukh-004	0.0000	585	0.0013	918	0.0327	757	0.0000	567	0.0004	428
793	mukh-005	0.0000	756	0.0009	780	0.0245	654	0.0000	502	-	1254
794	multimodality-000	0.0000	245	0.0000	173	0.0000	145	0.0000	132	0.0000	120
795	multimodality-001	0.0000	183	0.0009	711	0.0259	689	0.0000	91	0.0000	157
796	mvision-001	0.0000	50	0.0000	201	-	1262	0.0000	25	0.0000	212
797	nazhiai-000	0.0000	228	0.0000	181	-	1472	0.0000	140	0.0000	111
798	ncsgroup-000	0.0000	175	0.0000	140	0.0000	123	0.0000	92	-	1357
799	ncssg-001	0.0000	319	0.0000	86	0.0000	6	0.0000	158	0.0000	97
800	ncssg-002	0.0000	398	0.0000	14	0.0000	42	0.0000	224	-	1172
801	ncssg-003	0.0000	668	0.0006	542	0.0149	406	0.0003	1009	-	1314
802	ncssg-004	0.0000	578	0.0006	544	0.0149	405	0.0003	1008	-	1416
803	neosystems-001	0.0000	84	0.0000	246	-	1284	0.0013	1272	0.9994	1093
804	neosystems-002	0.0000	389	0.0000	6	0.0000	49	0.0000	234	0.0000	42
805	neosystems-003	0.0000	482	0.0000	37	0.0000	65	0.0000	256	0.0000	27
806	neosystems-004	0.0000	47	0.0000	215	0.0000	92	0.0000	28	0.0000	210
807	netbridgetech-001	0.0000	264	0.0000	73	-	1001	0.0000	152	0.0000	104
808	netbridgetech-002	0.0000	414	0.0000	1	-	1098	0.0000	217	0.0000	56
809	neurotechnology-000	-	1451	-	1343	-	1111	-	1460	0.0000	54
810	neurotechnology-001	-	1404	-	1442	-	1458	-	1431	0.0000	121
811	neurotechnology-002	-	1347	-	1474	-	1303	-	1402	0.0000	190
812	neurotechnology-003	-	1408	-	1374	-	988	-	1441	0.0005	793

Table 104: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
813	neurotechnology-004	-	1429	-	1399	-	1054	0.0000	416	0.0001	258
814	neurotechnology-005	-	1439	-	1402	-	1071	0.0004	1083	0.0004	561
815	neurotechnology-006	0.0000	1039	0.0006	538	-	1146	0.0004	1084	0.0004	556
816	neurotechnology-008	0.0000	988	0.0006	537	-	1334	0.0004	1082	0.0004	548
817	neurotechnology-009	0.0000	939	0.0008	660	-	985	0.0002	880	0.0004	490
818	neurotechnology-010	0.0000	863	0.0001	265	-	1150	0.0000	565	0.0002	265
819	neurotechnology-011	0.0000	922	0.0013	900	-	1407	0.0002	876	0.0003	400
820	neurotechnology-012	0.0000	1102	0.0010	844	0.0242	649	0.0001	788	0.0004	634
821	neurotechnology-013	0.0000	341	0.0008	704	0.0185	503	0.0000	366	0.0001	249
822	neurotechnology-015	0.0000	293	0.0004	368	0.0082	266	0.0000	174	0.0000	235
823	neurotechnology-016	0.0000	510	0.0003	318	0.0069	233	0.0000	277	0.0000	233
824	neurotechnology-017	0.0000	318	0.0003	344	0.0065	222	0.0000	157	0.0000	236
825	neurotechnology-018	0.0000	563	0.0003	351	0.0074	240	0.0000	427	0.0004	443
826	neurotechnology-019	0.0000	790	0.0004	391	0.0115	340	0.0000	332	-	1244
827	neurotechnology-020	0.0000	846	0.0004	388	0.0115	341	0.0000	340	-	1161
828	neurotechnology-021	0.0000	187	0.0004	394	0.0115	339	0.0000	325	-	1313
829	nextmetalabs-001	0.0000	302	0.0000	90	0.0000	11	0.0000	170	-	1258
830	nextmetalabs-002	0.0000	204	0.0000	152	0.0000	134	0.0000	114	-	1327
831	nextmetalabs-003	0.0000	661	0.0006	574	0.0184	498	0.0000	381	-	1362
832	nextmetalabs-004	0.0000	665	0.0008	709	0.0255	686	0.0000	380	-	1365
833	nhn-001	0.0000	847	0.0019	1003	-	1124	0.0001	685	0.0004	747
834	nhn-002	0.0000	372	0.0004	424	0.0091	288	0.0000	489	0.0003	313
835	nhn-003	0.0000	995	0.0000	125	0.0000	118	0.0001	833	0.0004	671
836	nhn-004	0.0000	1001	0.0000	69	0.0000	3	0.0001	835	0.0004	663
837	nhn-005	0.0000	875	0.0000	51	0.0000	73	0.0000	527	0.0004	481
838	nmetics-001	0.0000	1160	0.0011	849	0.0230	631	0.0002	991	-	1445
839	noblis-000	-	1480	-	1369	-	1207	-	1482	0.0000	8
840	nodeflux-000	-	1350	-	1478	-	1296	0.0001	755	0.0002	286
841	nodeflux-001	0.0000	500	0.0016	964	-	1201	0.0001	759	0.0002	283
842	nodeflux-002	0.0000	748	0.0261	1283	-	1029	0.0008	1226	0.0005	833
843	nominder-000	0.0000	824	0.0003	296	0.0053	196	0.0000	472	0.0003	377
844	nominder-001	0.0000	558	0.0003	297	0.0053	197	0.0000	454	0.0003	390
845	nominder-002	0.0000	697	0.0006	508	0.0119	345	0.0002	934	0.0003	397
846	nominder-003	0.0000	859	0.0005	467	0.0095	295	0.0002	882	-	1129
847	nominder-004	0.0000	813	0.0007	609	0.0130	363	0.0002	981	-	1164
848	nominder-005	0.0000	484	0.0009	728	0.0206	549	0.0000	491	-	1137
849	nominder-006	0.0000	897	0.0017	975	0.0379	806	0.0003	1051	-	1457
850	nominder-007	0.0000	921	0.0006	565	0.0119	346	0.0003	1048	-	1384
851	nominder-008	0.0000	965	0.0006	569	0.0119	344	0.0003	1050	-	1157
852	notiontag-000	0.0000	435	0.0000	33	-	1144	0.0000	250	0.0000	32
853	notiontag-001	0.0000	14	0.0000	200	-	1245	0.0027	1326	0.0000	224
854	notiontag-002	0.0000	476	0.0000	34	0.0000	64	0.0000	258	0.0000	23
855	npt-000	0.0000	224	0.0002	288	0.0042	185	0.0001	744	-	1342
856	nsensecorp-001	0.0000	1178	0.0024	1060	-	1456	0.0014	1282	0.0101	1078
857	nsensecorp-002	0.0000	757	0.0009	724	0.0219	601	0.0003	1005	0.0011	958
858	nsensecorp-003	0.0000	383	0.0000	259	0.0002	158	0.0000	441	0.0007	924
859	nsensecorp-004	0.0406	1329	0.0035	1144	0.0181	483	0.0016	1293	0.0760	1087
860	nsensecorp-005	0.0000	240	0.0000	187	0.0000	147	0.0000	134	0.0000	119
861	ntechlab-000	-	1401	-	1450	-	1468	-	1433	0.0016	1016
862	ntechlab-001	-	1430	-	1398	-	1055	-	1452	0.0005	802
863	ntechlab-002	-	1444	-	1337	-	1115	-	1466	0.0005	800
864	ntechlab-003	-	1383	-	1421	-	1387	-	1423	0.0005	805
865	ntechlab-004	-	1399	-	1446	-	1478	-	1435	0.0004	515
866	ntechlab-005	-	1392	-	1430	-	1383	0.0000	399	0.0004	493
867	ntechlab-006	-	1432	-	1392	-	1037	0.0000	333	0.0004	434
868	ntechlab-007	0.0000	977	0.0019	1005	-	1192	0.0001	653	0.0004	447
869	ntechlab-008	0.0000	929	0.0005	430	-	1426	0.0001	657	0.0004	464
870	ntechlab-009	0.0000	1041	0.0009	742	-	1193	0.0001	792	0.0004	417

Table 105: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
871	ntechlab-010	0.0000	596	0.0005	478	0.0107	321	0.0001	741	0.0004	467
872	ntechlab-011	0.0000	459	0.0003	300	0.0057	203	0.0000	576	0.0004	418
873	ntechlab-012	0.0000	272	0.0003	301	0.0057	202	0.0000	572	0.0004	422
874	ntnu-000	0.0000	425	0.0004	357	0.0075	245	0.0001	828	-	1145
875	ocrstudio-001	0.0000	1149	0.0034	1142	0.0534	858	0.0001	794	-	1240
876	omface-000	0.0000	63	0.0000	224	0.0000	104	0.0000	51	0.0000	188
877	omface-001	0.0000	365	0.0000	250	0.0000	153	0.0000	204	0.0000	68
878	omnigarde-000	0.0000	731	0.0008	647	0.0213	584	0.0000	467	0.0004	622
879	omnigarde-001	0.0000	647	0.0008	649	0.0213	586	0.0000	461	0.0004	630
880	omnigarde-002	0.0000	805	0.0008	648	0.0213	585	0.0000	469	0.0004	619
881	omnigarde-003	0.0000	22	0.0007	644	0.0212	580	0.0000	312	0.0004	529
882	omnigarde-004	0.0000	149	0.0007	643	0.0213	582	0.0000	353	-	1385
883	omnigarde-005	0.0000	141	0.0007	642	0.0213	583	0.0000	354	-	1379
884	onfido-000	0.0000	1150	0.0040	1157	0.0804	917	0.0004	1076	0.0012	982
885	onfido-001	0.0000	658	0.0006	566	0.0183	489	0.0000	295	-	1358
886	onfido-002	0.0000	754	0.0006	564	0.0183	487	0.0000	306	-	1253
887	onfido-003	0.0000	653	0.0006	567	0.0183	490	0.0000	297	-	1354
888	onfido-004	0.0000	586	0.0006	568	0.0183	488	0.0000	291	-	1421
889	oorudigital-000	0.0000	985	0.0002	282	0.0016	170	0.0002	976	-	1436
890	ooto-001	0.0000	226	0.0000	164	0.0000	133	0.0000	105	-	1341
891	openedge-000	0.0000	749	0.0014	929	0.0212	578	0.0003	1026	0.0004	677
892	openface-001	0.0000	1072	0.0104	1256	0.0668	888	0.0004	1096	0.0006	903
893	ovisionllc-001	0.0000	489	0.0000	57	0.0000	75	0.0000	287	-	1095
894	ovisionllc-002	0.0000	58	0.0000	203	0.0000	87	0.0000	20	-	1461
895	oz-001	0.0000	1042	0.0011	863	-	1184	0.0006	1196	0.0004	553
896	oz-002	0.0000	477	0.0003	303	-	1154	0.0000	424	0.0003	346
897	oz-003	0.0000	406	0.0002	279	0.0042	184	0.0000	337	0.0003	289
898	oz-004	0.0000	1099	0.0003	315	0.0041	183	0.0000	364	0.0002	262
899	pa-002	-	1343	-	1461	-	1256	-	1392	0.0000	215
900	pa-003	-	1361	-	1481	-	1337	-	1407	0.0000	178
901	palit-000	0.0000	587	0.0005	475	0.0134	373	0.0002	901	0.0004	525
902	palit-001	0.0000	867	0.0007	646	0.0201	542	0.0002	904	0.0004	503
903	pangiam-000	0.0000	34	0.0021	1040	0.0364	791	0.0001	611	0.0005	789
904	pangiam-001	0.0000	171	0.0005	481	0.0128	358	0.0002	995	0.0003	411
905	pangiam-002	0.0000	258	0.0003	336	0.0072	235	0.0000	436	0.0003	347
906	papago-001	0.0000	896	0.0008	665	0.0159	438	0.0002	990	0.0004	581
907	papil11-000	0.0000	721	0.0008	702	0.0192	529	0.0000	507	0.0003	408
908	papil11-001	0.0000	610	0.0005	444	0.0133	369	0.0000	411	-	1405
909	papil11-002	0.0000	865	0.0004	367	0.0115	338	0.0000	371	-	1135
910	papil11-003	0.0000	236	0.0003	345	0.0077	250	0.0000	328	-	1295
911	papil11-004	0.0000	792	0.0008	656	0.0227	615	0.0003	1032	-	1243
912	papsav1923-001	0.0000	556	0.0007	613	-	1278	0.0001	712	0.0002	280
913	papsav1923-002	0.0000	876	0.0018	999	0.0268	701	0.0000	543	0.0004	638
914	papsav1923-003	0.0000	1014	0.0019	1016	0.0321	749	0.0000	592	0.0004	585
915	paravision-004	0.0000	1027	0.0007	633	-	1312	0.0002	916	0.0004	601
916	paravision-006	0.0000	155	0.0000	129	-	1345	0.0000	69	0.0000	175
917	paravision-008	0.0000	276	0.0010	803	0.0201	538	0.0001	696	0.0004	435
918	paravision-010	0.0000	55	0.0010	806	0.0201	540	0.0001	688	0.0004	444
919	paravision-011	0.0000	431	0.0010	801	0.0201	539	0.0001	698	0.0004	432
920	paravision-013	0.0000	32	0.0010	807	0.0201	541	0.0001	689	0.0004	442
921	paravision-015	0.0000	441	0.0009	763	0.0266	700	0.0000	591	-	1152
922	paravision-017	0.0000	686	0.0009	717	0.0220	603	0.0001	675	-	1335
923	paravision-018	0.0000	176	0.0005	488	0.0142	393	0.0000	356	-	1356
924	paravision-020	0.0000	494	0.0004	373	0.0107	319	0.0000	373	-	1099
925	paravision-022	0.0000	430	0.0004	372	0.0107	318	0.0000	370	-	1149
926	pensees-001	0.0000	674	0.0000	171	-	1431	0.0000	119	0.0000	133
927	pentacomp-000	0.0001	1216	0.0010	786	0.0206	558	0.0004	1086	-	1448
928	pixelall-002	0.0000	438	0.0000	24	-	1129	0.0000	245	0.0000	35

Table 106: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
929	pixelall-003	0.0000	126	0.0000	131	-	1368	0.0000	86	0.0000	161
930	pixelall-004	0.0000	208	0.0000	153	-	1422	0.0000	118	0.0000	136
931	pixelall-005	0.0000	138	0.0000	139	-	1365	0.0000	78	0.0000	169
932	pixelall-006	0.0000	363	0.0000	118	0.0000	31	0.0000	203	0.0000	69
933	pixelall-007	0.0000	463	0.0000	42	0.0000	70	0.0000	266	0.0000	15
934	pixelall-008	0.0000	237	0.0008	688	0.0247	659	0.0000	139	0.0000	117
935	pixelall-009	0.0000	460	0.0000	43	0.0000	69	0.0000	264	0.0000	16
936	pixelall-010	0.0000	72	0.0000	218	0.0000	99	0.0000	43	0.0000	196
937	pnpsecure-000	0.0000	233	0.0030	1107	0.0390	812	0.0003	999	-	1296
938	pnpsecure-001	0.0000	324	0.0023	1052	0.0328	759	0.0002	881	-	1217
939	pnpsecure-002	0.0000	635	0.0006	504	0.0148	403	0.0001	666	-	1388
940	privid-001	0.0001	1241	0.0176	1270	0.0598	873	0.0021	1307	0.0021	1027
941	privid-002	0.0198	1318	0.0040	1161	0.0386	809	0.0014	1279	0.0011	967
942	privid-003	0.0000	1013	0.0002	284	0.0030	180	0.0001	813	-	1143
943	privid-004	0.0000	776	0.0001	275	0.0025	175	0.0001	668	-	1220
944	psl-000	-	1395	-	1438	-	1429	-	1428	0.0000	130
945	psl-001	-	1376	-	1412	-	1343	0.0000	72	0.0000	174
946	psl-002	-	1356	-	1469	-	1294	0.0000	42	0.0000	200
947	psl-003	0.0000	844	0.0007	581	-	1128	0.0000	564	0.0004	606
948	psl-004	0.0000	551	0.0007	585	-	1274	0.0000	557	0.0004	616
949	psl-005	0.0000	680	0.0007	577	-	1418	0.0000	463	0.0004	568
950	psl-006	0.0000	529	0.0002	280	-	1242	0.0000	8	0.0003	288
951	psl-007	0.0000	783	0.0007	587	-	1057	0.0000	574	0.0003	361
952	psl-008	0.0000	693	0.0003	309	0.0068	227	0.0000	357	0.0003	349
953	psl-009	0.0000	676	0.0004	377	0.0095	294	0.0000	301	0.0004	424
954	psl-010	0.0000	565	0.0004	378	0.0095	293	0.0000	289	0.0004	430
955	psl-011	0.0000	597	0.0003	307	0.0063	212	0.0000	292	0.0003	386
956	psl-012	0.0000	649	0.0003	313	0.0067	224	0.0000	298	0.0003	382
957	psl-013	0.0000	780	0.0003	305	0.0063	211	0.0000	550	-	1227
958	psl-014	0.0000	687	0.0003	306	0.0063	213	0.0000	300	-	1334
959	ptakuratsatu-000	0.0000	868	0.0007	627	-	1212	0.0001	616	0.0003	334
960	pxl-001	0.0000	1198	0.0044	1178	-	1321	0.0005	1161	0.0022	1033
961	pyramid-000	0.0001	1231	0.0041	1166	-	1142	0.0005	1160	0.0007	918
962	pyramid-001	0.0000	747	0.0008	670	0.0231	632	0.0001	793	-	1246
963	qazbs-000	0.0000	348	0.0009	739	0.0265	696	0.0000	419	0.0004	560
964	qazsmartvisionai-000	0.0000	623	0.0003	348	0.0083	270	0.0000	323	0.0003	351
965	qazsmartvisionai-001	0.0000	755	0.0003	323	0.0064	216	0.0000	331	0.0003	324
966	qazsmartvisionai-002	0.0000	523	0.0003	325	0.0064	217	0.0000	314	-	1466
967	qazsmartvisionai-003	0.0000	738	0.0003	322	0.0064	215	0.0000	329	-	1284
968	qazsmartvisionai-004	0.0000	581	0.0003	326	0.0064	218	0.0000	322	-	1419
969	qluevision-001	0.0000	1023	0.0008	677	0.0153	417	0.0008	1225	0.0004	775
970	qnap-000	0.0000	93	0.0007	632	0.0174	467	0.0002	893	0.0002	259
971	qnap-001	0.0000	723	0.0000	256	0.0002	159	0.0000	548	0.0001	252
972	qnap-002	0.0000	1136	0.0033	1125	0.0761	914	0.0004	1081	0.0002	260
973	qnap-003	0.0000	156	0.0016	960	0.0402	822	0.0000	583	0.0001	251
974	qnap-004	0.0000	351	0.0016	959	0.0402	821	0.0000	584	0.0001	250
975	qnap-005	0.0001	1212	0.0078	1235	0.0581	867	0.0010	1250	0.0005	867
976	qnap-006	0.0000	941	0.0018	991	0.0427	830	0.0002	929	-	1281
977	quantasoft-003	0.0000	1060	0.0015	957	0.0355	787	0.0005	1150	0.0006	890
978	rankone-000	-	1458	-	1336	-	1092	-	1456	0.0003	359
979	rankone-001	-	1416	-	1370	-	976	-	1439	0.0000	247
980	rankone-002	-	1390	-	1428	-	1379	-	1418	0.0000	244
981	rankone-003	-	1445	-	1339	-	1117	-	1465	0.0000	243
982	rankone-004	-	1345	-	1462	-	1260	-	1391	0.0000	245
983	rankone-005	-	1475	-	1354	-	1157	0.0000	259	0.0000	25
984	rankone-006	-	1397	-	1435	-	1427	0.0000	112	0.0000	139
985	rankone-007	0.0000	1199	0.0048	1190	-	1081	0.0003	1033	0.0004	721
986	rankone-008	0.0000	1177	0.0049	1192	-	1473	0.0003	997	0.0004	423

Table 107: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
987	rankone-009	0.0000	457	0.0000	41	-	1169	0.0000	269	0.0000	14
988	rankone-010	0.0000	179	0.0000	141	-	1388	0.0000	89	0.0000	155
989	rankone-011	0.0000	56	0.0000	202	0.0000	85	0.0000	23	0.0000	213
990	rankone-012	0.0000	417	0.0000	2	0.0000	41	0.0000	220	0.0000	57
991	rankone-013	0.0000	172	0.0005	449	0.0126	356	0.0000	400	0.0003	300
992	rankone-014	0.0000	281	0.0005	439	0.0129	360	0.0000	402	0.0002	266
993	rankone-015	0.0000	248	0.0000	175	0.0000	141	0.0000	128	0.0000	122
994	realnetworks-000	-	1440	-	1401	-	1072	-	1455	0.0003	318
995	realnetworks-001	-	1384	-	1423	-	1389	0.0004	1072	0.0003	321
996	realnetworks-002	0.0000	1007	0.0003	338	-	1056	0.0004	1075	0.0003	319
997	realnetworks-003	0.0000	1004	0.0003	337	-	1027	0.0004	1074	0.0003	320
998	realnetworks-004	0.0000	753	0.0003	299	0.0052	195	0.0000	307	0.0002	281
999	realnetworks-005	0.0000	619	0.0002	290	0.0045	189	0.0000	293	0.0002	285
1000	realnetworks-006	0.0000	760	0.0002	291	0.0045	187	0.0000	305	0.0002	282
1001	realnetworks-007	0.0000	781	0.0013	920	0.0425	829	0.0000	334	0.0004	591
1002	realnetworks-008	0.0000	775	0.0002	292	0.0045	188	0.0000	308	0.0002	284
1003	realsense-001	0.0000	646	0.0006	561	0.0177	476	0.0000	460	-	1350
1004	realsense-002	0.0000	377	0.0006	530	0.0168	461	0.0000	417	-	1206
1005	rebs-000	0.0000	147	0.0005	440	0.0130	362	0.0000	432	0.0004	495
1006	rebs-001	0.0000	506	0.0005	457	0.0144	395	0.0000	505	0.0004	573
1007	recognito-000	0.0000	638	0.0003	311	0.0056	201	0.0000	518	0.0003	383
1008	recognito-001	0.0000	770	0.0003	310	0.0056	200	0.0000	521	0.0003	379
1009	regula-000	0.0000	328	0.0000	102	0.0000	19	0.0000	186	0.0000	82
1010	regula-001	0.0000	90	0.0000	234	0.0000	110	0.0000	68	0.0000	177
1011	regula-002	0.0000	512	0.0009	776	0.0283	708	0.0000	274	-	1108
1012	regula-003	0.0000	120	0.0009	777	0.0283	709	0.0000	53	-	1412
1013	remarkai-000	-	1464	-	1348	-	1137	0.0000	243	0.0000	37
1014	remarkai-001	0.0000	15	0.0000	189	-	1229	0.0000	6	0.0000	225
1015	remarkai-002	0.0000	214	0.0000	158	-	1411	0.0000	106	0.0000	142
1016	remarkai-003	0.0000	870	0.0007	605	0.0187	506	0.0000	555	0.0004	478
1017	rendip-000	0.0000	993	0.0016	970	0.0293	718	0.0002	892	0.0004	763
1018	revealmedia-005	0.0000	1028	0.0007	621	0.0189	513	0.0009	1241	0.0004	773
1019	revealmedia-006	0.0000	265	0.0009	764	0.0238	644	0.0001	738	0.0004	679
1020	roc-016	0.0000	231	0.0000	258	0.0006	163	0.0000	135	0.0000	237
1021	roc-017	0.0000	427	0.0003	352	0.0073	237	0.0000	406	-	1146
1022	roc-018	0.0000	568	0.0006	553	0.0160	443	0.0000	453	0.0004	471
1023	roc-019	0.0000	553	0.0007	602	0.0187	505	0.0000	496	-	1438
1024	roc-020	0.0000	527	0.0006	533	0.0157	431	0.0000	313	-	1469
1025	roc-021	0.0000	77	0.0000	261	0.0010	164	0.0000	290	-	1427
1026	rokid-000	0.0000	376	0.0072	1227	-	1076	0.0001	708	0.0005	821
1027	rokid-001	0.0000	235	0.0013	912	-	1481	0.0000	137	0.0000	114
1028	s1-001	0.0000	1196	0.0073	1230	-	1315	0.0013	1269	0.0007	913
1029	s1-002	0.0000	907	0.0089	1247	-	1293	0.0001	779	0.0005	831
1030	s1-003	0.0000	8	0.0002	283	0.0040	182	0.0007	1210	0.0003	308
1031	s1-004	0.0000	163	0.0000	260	0.0003	160	0.0000	597	0.0001	256
1032	s1-005	0.0000	68	0.0004	383	0.0120	349	0.0001	641	0.0002	267
1033	s1-006	0.0000	369	0.0003	308	0.0074	242	0.0001	635	0.0002	271
1034	s1-007	0.0000	45	0.0006	510	0.0110	326	0.0001	640	0.0002	269
1035	s1-008	0.0000	666	0.0018	1002	0.0252	678	0.0005	1179	-	1364
1036	s1-009	0.0000	168	0.0006	507	0.0110	327	0.0001	644	-	1353
1037	saffe-001	0.0000	17	0.0000	190	-	1231	0.0000	7	0.0000	226
1038	saffe-002	0.0000	161	0.0000	147	-	1402	0.0000	95	0.0000	150
1039	samsungsds-000	0.0000	1011	0.0055	1208	0.0646	883	0.0038	1340	0.0005	822
1040	samsungsds-001	0.0000	194	0.0005	476	0.0146	399	0.0001	693	0.0003	398
1041	samsungsds-002	0.0000	289	0.0004	389	0.0119	347	0.0001	694	0.0003	373
1042	samtech-000	-	1431	-	1390	-	1049	-	1451	0.0042	1064
1043	samtech-001	0.0001	1226	0.0032	1124	-	1151	0.0004	1115	0.0008	926
1044	samtech-002	0.0000	1090	0.0018	996	0.0516	849	0.0002	992	0.0005	846

Table 108: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1045	scanalitix-001	0.0000	852	0.0005	450	0.0095	296	0.0001	836	-	1126
1046	scanovate-001	0.0208	1319	0.2388	1326	-	1452	0.0024	1313	0.0014	1004
1047	scanovate-002	0.0000	943	0.0018	993	-	1028	0.0000	600	0.0004	748
1048	scanovate-003	0.0000	904	0.0233	1281	0.3371	961	0.0006	1187	0.0004	774
1049	sdc-000	0.0000	1175	0.0035	1143	0.0678	895	0.0005	1175	0.0011	969
1050	seamfix-001	0.0000	57	0.0000	204	0.0000	86	0.0000	19	-	1462
1051	securifai-001	0.0000	230	0.0000	184	-	1480	0.0000	302	0.0000	116
1052	securifai-002	0.0000	317	0.0000	85	-	1007	0.0000	161	0.0000	96
1053	securifai-003	0.0000	74	0.0000	220	0.0000	100	0.0000	41	0.0000	199
1054	securifai-004	0.0000	212	0.0000	157	0.0000	136	0.0000	110	0.0000	141
1055	securifai-005	0.0000	225	0.0000	165	0.0000	132	0.0000	104	0.0000	146
1056	securifai-006	0.0000	313	0.0000	84	0.0000	4	0.0000	159	0.0000	95
1057	securifai-007	0.0000	234	0.0000	183	0.0000	149	0.0000	136	0.0000	115
1058	securifai-008	0.0000	338	0.0000	92	0.0000	16	0.0000	181	-	1235
1059	securitypitch-000	0.0175	1317	0.0055	1209	0.0108	322	0.0137	1371	-	1283
1060	sensetime-001	-	1422	-	1382	-	1014	0.0000	485	0.0004	577
1061	sensetime-002	0.0004	1261	0.0082	1240	-	1464	0.0011	1261	0.0005	860
1062	sensetime-003	0.0000	817	0.0011	860	-	1118	0.0000	309	0.0004	586
1063	sensetime-004	0.0000	728	0.0011	858	-	1002	0.0000	304	0.0004	593
1064	sensetime-005	0.0000	275	0.0004	374	0.0106	305	0.0000	484	0.0003	339
1065	sensetime-006	0.0000	238	0.0004	376	0.0106	307	0.0000	482	0.0003	340
1066	sensetime-007	0.0000	307	0.0004	375	0.0106	306	0.0000	486	0.0003	338
1067	sensetime-008	0.0000	122	0.0007	626	0.0250	668	0.0000	294	0.0003	412
1068	sensetime-010	0.0000	305	0.0005	472	0.0111	329	0.0000	539	-	1259
1069	sensetime-011	0.0000	143	0.0005	474	0.0111	331	0.0000	536	-	1377
1070	sensetime-012	0.0000	335	0.0005	473	0.0111	330	0.0000	540	-	1232
1071	serendipity-000	0.0000	765	0.0008	672	0.0202	544	0.0001	830	0.0004	610
1072	sertis-000	0.0000	48	0.0007	617	-	1261	0.0000	605	0.0004	549
1073	sertis-001	0.0000	285	0.0004	426	-	966	0.0002	935	0.0005	869
1074	sertis-002	0.0000	124	0.0007	595	0.0152	411	0.0000	585	0.0004	546
1075	sertis-003	0.0000	727	0.0008	686	0.0183	486	0.0000	510	0.0004	611
1076	seventhsense-000	0.0000	785	0.0006	556	0.0184	494	0.0001	627	0.0004	620
1077	seventhsense-001	0.0000	633	0.0006	557	0.0184	495	0.0001	625	0.0004	632
1078	seventhsense-002	0.0000	364	0.0003	346	0.0076	247	0.0000	609	0.0004	419
1079	seventhsense-003	0.0000	722	0.0003	332	0.0083	271	0.0000	403	0.0002	279
1080	seventhsense-005	0.0000	642	0.0002	295	0.0065	221	0.0000	413	0.0003	306
1081	seventhsense-006	0.0000	733	0.0002	293	0.0065	220	0.0000	414	-	1279
1082	shaman-000	0.0000	259	0.0000	70	-	991	0.0000	155	0.0000	101
1083	shaman-001	0.0000	41	0.0000	213	-	1263	0.0000	29	0.0000	209
1084	shu-001	0.0000	1144	0.0022	1048	-	1264	0.0010	1247	0.0006	886
1085	shu-002	0.0000	934	0.0010	822	-	1483	0.0005	1135	0.0004	723
1086	shu-003	0.0000	256	0.0007	591	-	987	0.0001	630	0.0003	331
1087	shufti-001	0.0000	540	0.0005	484	0.0131	367	0.0001	837	-	1477
1088	siat-002	0.0000	880	0.0012	886	-	1182	0.0000	531	0.0004	584
1089	siat-004	0.0000	690	0.0011	856	-	1415	0.0000	479	0.0004	578
1090	siat-005	0.0000	440	0.0000	23	0.0000	151	0.0000	246	0.0000	33
1091	sightropic-000	0.0000	913	0.0006	548	0.0138	379	0.0002	984	-	1411
1092	sightropic-001	0.0000	973	0.0037	1152	0.0261	691	0.0007	1221	-	1096
1093	sightropic-002	0.0000	570	0.0002	276	0.0022	172	0.0001	778	-	1460
1094	sjtu-001	0.0000	975	0.0010	839	-	1202	0.0005	1151	0.0004	746
1095	sjtu-002	0.0000	924	0.0010	821	-	1394	0.0005	1134	0.0004	725
1096	sjtu-003	0.0000	65	0.0005	491	-	1295	0.0000	566	0.0003	317
1097	sjtu-004	0.0000	404	0.0000	16	0.0000	47	0.0000	229	0.0003	311
1098	sjtu-005	0.0000	1185	0.0000	185	0.0000	150	0.0000	138	0.0006	885
1099	sjtu-006	0.0000	878	0.0026	1069	0.0298	720	0.0002	871	-	1109
1100	sjtu-007	0.0000	23	0.0002	287	0.0054	198	0.0000	288	-	1481
1101	sjtu-008	0.0000	207	0.0003	316	0.0074	243	0.0000	116	-	1329
1102	sjtu-009	0.0000	25	0.0003	317	0.0074	241	0.0000	2	-	1482

Table 109: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1103	sktelecom-000	0.0000	618	0.0008	687	0.0190	517	0.0000	569	0.0004	648
1104	smargtech-001	0.0000	345	0.0000	97	0.0000	15	0.0000	177	-	1241
1105	smartbiometrik-001	0.0005	1263	0.0649	1307	0.2147	954	0.0017	1295	0.0008	934
1106	smartengines-000	0.0066	1301	0.0150	1267	0.1656	943	0.0022	1308	0.0013	1001
1107	smartengines-001	0.0003	1257	0.0073	1228	0.0714	908	0.0007	1213	0.0005	834
1108	smartvist-000	0.0000	325	0.0026	1078	0.0357	789	0.0002	848	0.0011	971
1109	smartvist-001	0.0000	702	0.0005	447	0.0069	232	0.0001	801	0.0003	364
1110	smilart-001	-	1382	-	1422	-	1386	-	1422	0.0273	1083
1111	smilart-002	0.0000	1169	0.0036	1149	-	969	-	1436	0.0011	968
1112	smilart-003	0.0003	1255	0.0100	1253	-	1273	0.0014	1277	0.0013	1003
1113	sodec-000	0.0000	322	0.0000	99	0.0000	22	0.0000	191	0.0000	79
1114	softtech-000	0.0000	409	0.0006	573	0.0166	453	0.0001	780	-	1179
1115	sparktek-000	0.0000	1076	0.0016	962	0.0356	788	0.0001	680	-	1397
1116	sparktek-001	0.0000	857	0.0004	382	0.0092	290	0.0000	553	-	1127
1117	sparsh-001	0.0000	458	0.0007	575	0.0168	457	0.0000	372	0.0004	482
1118	sqisoft-001	0.0000	384	0.0003	333	0.0078	252	0.0000	390	0.0003	393
1119	sqisoft-002	0.0000	210	0.0003	328	0.0078	254	0.0000	401	0.0003	409
1120	sqisoft-003	0.0000	502	0.0003	335	0.0078	256	0.0000	408	0.0003	345
1121	staqu-000	0.0000	211	0.0000	156	-	1425	0.0000	111	0.0000	140
1122	starhybrid-001	0.0001	1234	0.0033	1136	-	1355	0.0009	1239	0.0023	1036
1123	stcon-000	0.0000	831	0.0017	988	0.0301	726	0.0000	471	0.0003	371
1124	stcon-001	0.0000	531	0.0017	989	0.0301	727	0.0000	452	0.0003	375
1125	stcon-002	0.0000	660	0.0011	847	0.0226	613	0.0000	433	0.0003	322
1126	stcon-003	0.0000	503	0.0005	453	0.0142	392	0.0000	449	0.0004	479
1127	stcon-004	0.0000	374	0.0005	455	0.0142	391	0.0000	442	-	1205
1128	stcon-005	0.0000	334	0.0005	454	0.0142	390	0.0000	440	-	1228
1129	stcon-006	0.0000	113	0.0007	586	0.0191	522	0.0000	456	-	1408
1130	stcon-007	0.0000	139	0.0007	583	0.0191	523	0.0000	457	-	1382
1131	stcon-008	0.0000	157	0.0007	584	0.0191	524	0.0000	462	-	1344
1132	stcon-009	0.0000	501	0.0007	582	0.0191	521	0.0000	475	-	1103
1133	stcon-010	0.0000	250	0.0006	572	0.0184	493	0.0000	464	-	1309
1134	stellartecs-001	0.0000	948	0.0012	876	0.0324	754	0.0003	1010	-	1264
1135	stengg-000	0.0000	330	0.0013	916	0.0350	785	0.0000	503	0.0004	609
1136	sukshi-000	0.0000	121	0.0000	130	0.0000	121	0.0000	84	0.0000	160
1137	sumsub-000	0.0000	1051	0.0018	1000	0.0189	512	0.0011	1260	-	1291
1138	sumsub-001	0.0000	564	0.0007	600	0.0197	536	0.0000	316	-	1455
1139	suprema-000	0.0000	932	0.0017	981	-	1469	0.0002	932	0.0004	665
1140	suprema-001	0.0000	990	0.0027	1080	0.0471	840	0.0003	1014	0.0004	755
1141	suprema-002	0.0000	966	0.0010	830	0.0271	704	0.0002	883	0.0004	532
1142	suprema-003	0.0000	715	0.0008	694	0.0231	633	0.0000	385	0.0004	541
1143	suprema-004	0.0000	771	0.0014	933	0.0299	721	0.0000	389	0.0004	536
1144	suprema-005	0.0000	923	0.0014	934	0.0299	722	0.0005	1145	0.0004	669
1145	supremaid-001	0.0000	740	0.0020	1025	0.0330	763	0.0001	726	0.0004	750
1146	supremaid-002	0.0000	673	0.0020	1026	0.0330	764	0.0001	722	0.0004	754
1147	surrey-cvssp-000	0.0000	405	0.0000	20	0.0000	48	0.0000	221	0.0000	52
1148	surrey-cvssp-001	0.0173	1316	0.0007	599	0.0179	478	0.0011	1257	0.0015	1011
1149	surrey-cvssp-002	0.0000	1113	0.0006	519	0.0156	428	0.0001	686	0.0004	575
1150	surrey-cvssp-003	0.0000	485	0.0009	727	0.0165	452	0.0000	253	0.0001	253
1151	surrey-cvssp-004	0.0000	862	0.0015	953	0.0311	737	0.0004	1079	-	1133
1152	swsam-001	0.0000	381	0.0012	899	0.0263	694	0.0000	195	0.0000	75
1153	synesis-000	-	1375	-	1410	-	1341	-	1411	0.0007	914
1154	synesis-001	-	1338	-	1452	-	1236	0.0035	1334	0.0021	1026
1155	synesis-003	-	1365	-	1414	-	1369	0.0102	1365	0.1184	1091
1156	synesis-004	-	1424	-	1383	-	1004	0.0164	1372	0.0035	1056
1157	synesis-005	0.0000	685	0.0019	1006	-	1423	0.0001	700	0.0005	783
1158	synesis-006	0.0000	474	0.0003	343	-	1153	0.0000	575	0.0003	296
1159	synesis-007	0.0000	603	0.0013	911	-	1329	0.0002	952	0.0004	600
1160	synology-000	0.0000	140	0.0000	123	-	1348	0.0000	75	0.0000	170

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A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]							
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1161	synology-001	0.0000	306	0.0000	79	-	1016	0.0000	162	0.0000	90
1162	synology-002	0.0000	515	0.0000	53	-	1187	0.0000	270	0.0000	11
1163	syscake-000	0.0008	1280	0.0017	987	0.0287	715	0.0007	1207	-	1420
1164	szpu-000	0.0000	448	0.0014	944	0.0330	765	0.0001	652	-	1153
1165	sztu-000	0.0000	487	0.0000	56	-	1209	0.0000	285	0.0000	1
1166	sztu-001	0.0000	125	0.0000	132	0.0000	122	0.0000	85	0.0000	162
1167	t4isb-000	0.0000	422	0.0000	5	0.0000	39	0.0000	215	0.0000	59
1168	tahakom-001	0.0025	1288	0.0020	1034	0.0139	381	0.0020	1301	-	1473
1169	tahakom-002	0.0329	1328	0.0083	1243	0.0185	502	0.0238	1377	-	1140
1170	taiwanmobile-001	0.0000	1029	0.0023	1059	0.0331	767	0.0006	1188	-	1324
1171	taiwanmobile-002	0.0000	1037	0.0015	946	0.0238	645	0.0004	1119	-	1263
1172	tcc-000	0.0000	671	0.0699	1312	0.0478	841	0.0000	360	-	1321
1173	tcc-001	0.0000	1187	0.0004	385	0.0089	282	0.0005	1158	-	1386
1174	tcv-001	0.0000	583	0.0013	901	0.0152	415	0.0001	818	-	1418
1175	tech5-000	-	1364	-	1480	-	1313	-	1403	0.0003	404
1176	tech5-001	-	1479	-	1367	-	1206	0.0004	1100	0.0003	353
1177	tech5-002	-	1433	-	1391	-	1039	0.0001	673	0.0003	291
1178	tech5-003	0.0000	887	0.0005	458	-	1232	0.0001	672	0.0003	307
1179	tech5-004	0.0000	632	0.0008	657	-	1349	0.0003	1015	0.0004	742
1180	tech5-005	0.0000	657	0.0007	638	-	1400	0.0000	459	0.0004	683
1181	tech5-007	0.0000	804	0.0014	941	0.0305	733	0.0000	418	0.0004	507
1182	tech5-008	0.0000	321	0.0007	576	0.0168	456	0.0000	367	0.0004	488
1183	tech5-009	0.0000	982	0.0030	1113	0.0542	860	0.0002	923	-	1468
1184	techainer-001	0.0000	1139	0.0004	416	0.0085	277	0.0005	1144	0.0004	670
1185	techainer-002	0.0000	1141	0.0004	414	0.0085	274	0.0005	1146	-	1242
1186	techainer-003	0.0000	895	0.0031	1122	0.0249	666	0.0003	1035	-	1449
1187	techbiometricid-001	0.0000	391	0.0000	8	0.0000	51	0.0000	236	-	1166
1188	techbiometricid-002	0.0000	308	0.0000	78	0.0000	8	0.0000	163	-	1262
1189	techsign-000	0.0007	1275	0.0334	1292	0.2093	948	0.0020	1304	0.0011	962
1190	techsign-001	0.0000	550	0.0008	708	0.0253	679	0.0002	907	0.0004	615
1191	telexper-000	0.0000	902	0.0006	550	0.0147	400	0.0003	1052	-	1417
1192	tevian-000	-	1467	-	1351	-	1127	-	1468	0.0012	994
1193	tevian-001	-	1373	-	1408	-	1354	-	1413	0.0012	992
1194	tevian-002	-	1414	-	1380	-	986	-	1440	0.0005	865
1195	tevian-003	-	1367	-	1416	-	1360	0.0003	1016	0.0005	866
1196	tevian-004	0.0000	1044	0.0024	1061	-	1332	0.0002	878	0.0005	855
1197	tevian-005	0.0001	1238	0.0041	1164	-	1235	0.0006	1190	0.0006	906
1198	tevian-006	0.0000	132	0.0012	878	-	1357	0.0003	1028	0.0005	830
1199	tevian-007	0.0000	562	0.0015	958	0.0429	831	0.0002	924	0.0004	656
1200	tevian-008	0.0000	858	0.0006	503	0.0109	324	0.0000	448	0.0003	329
1201	tiger-001	-	1349	-	1475	-	1309	-	1400	0.0006	893
1202	tiger-002	0.0000	710	0.0010	810	-	1454	0.0001	765	0.0004	565
1203	tiger-003	0.0000	845	-	1349	-	1122	0.0001	773	0.0004	558
1204	tiger-004	0.0000	1191	0.0022	1046	-	1006	0.0001	770	0.0004	735
1205	tiger-005	0.0000	725	0.0009	767	0.0194	532	0.0001	683	0.0004	564
1206	tiger-006	0.0000	899	0.0011	859	0.0396	815	0.0001	805	0.0004	776
1207	tiger-007	0.0000	931	0.0051	1197	0.0201	543	0.0002	895	-	1333
1208	tinkoff-001	0.0000	946	0.0008	684	0.0171	464	0.0001	786	0.0004	562
1209	tngd-001	0.0000	777	0.0009	710	0.0218	597	0.0000	541	-	1223
1210	tngd-002	0.0000	10	0.0000	199	0.0000	81	0.0000	9	-	1471
1211	tnitech-000	0.0000	883	0.0003	330	0.0060	205	0.0005	1178	0.0003	323
1212	tongyi-004	-	1470	-	1359	-	1173	-	1477	0.0006	895
1213	tongyi-005	0.0000	454	0.0000	29	-	1126	0.0000	239	0.0000	41
1214	tongyitrans-001	-	1380	-	1433	-	1406	-	1425	0.0040	1061
1215	tongyitrans-002	-	1407	-	1445	-	1451	-	1429	0.0040	1060
1216	tongyitrans-003	-	1363	-	1479	-	1319	-	1405	0.0006	898
1217	toppanidgate-000	0.0000	700	0.0008	676	0.0232	634	0.0004	1073	0.0004	626
1218	toppansecurity-001	0.0000	916	0.0020	1024	0.0161	447	0.0001	679	-	1374

Table 111: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1219	toshiba-000	-	1409	-	1378	-	990	-	1442	0.0000	102
1220	toshiba-001	-	1450	-	1344	-	1112	-	1461	0.0000	55
1221	toshiba-002	0.0000	27	0.0000	210	-	1268	0.0000	32	0.0000	206
1222	toshiba-003	0.0000	284	0.0001	266	-	965	0.0001	767	0.0001	255
1223	toshiba-004	0.0000	160	0.0000	145	0.0000	127	0.0000	100	0.0000	148
1224	toshiba-005	0.0000	784	0.0004	398	0.0050	192	0.0001	797	0.0003	330
1225	toshiba-006	0.0000	588	0.0004	399	0.0050	193	0.0001	795	0.0003	332
1226	toshiba-007	0.0000	976	0.0005	452	0.0082	267	0.0002	974	0.0004	604
1227	toshiba-008	0.0000	914	0.0005	456	0.0082	268	0.0002	973	0.0004	614
1228	toshiba-009	0.0000	789	0.0004	423	0.0070	234	0.0002	961	-	1245
1229	toshiba-010	0.0000	893	0.0004	384	0.0064	214	0.0002	948	-	1444
1230	toshiba-011	0.0000	644	0.0006	514	0.0137	377	0.0002	946	-	1346
1231	toshiba-012	0.0000	730	0.0004	427	0.0120	350	0.0000	599	-	1275
1232	toshiba-013	0.0000	894	0.0004	392	0.0078	253	0.0000	517	-	1450
1233	toss-000	0.0000	181	0.0009	736	0.0209	567	0.0001	706	-	1361
1234	toss-001	0.0000	481	0.0009	734	0.0209	565	0.0001	711	-	1131
1235	touchlessid-000	0.0042	1293	0.0133	1263	0.2009	947	0.0018	1298	0.0032	1053
1236	touchlessid-001	0.0000	4	0.0036	1150	0.0923	927	0.0000	13	0.0000	219
1237	touchlessid-002	0.0000	81	0.0043	1176	0.1087	932	0.0000	39	0.0000	201
1238	touchlessid-003	0.0000	682	0.0010	814	0.0287	716	0.0002	851	0.0004	737
1239	touchlessid-004	0.0000	1190	0.0013	923	0.0398	818	0.0002	969	-	1162
1240	transmitsecurity-001	0.0000	534	0.0009	733	0.0226	614	0.0000	523	-	1475
1241	trueface-000	0.0000	580	0.0000	226	-	1306	0.0000	49	0.0000	191
1242	trueface-001	0.0000	970	0.0038	1156	-	1208	0.0007	1220	0.0005	828
1243	trueface-002	0.0000	944	0.0046	1185	-	1020	0.0003	1003	0.0005	852
1244	trueface-003	0.0000	888	0.0046	1186	0.0397	817	0.0003	1001	0.0005	856
1245	trueidvng-001	0.0000	980	0.0020	1033	0.0385	808	0.0002	912	0.0005	820
1246	truststamp-001	0.0000	576	0.0010	808	0.0189	516	0.0000	397	0.0003	374
1247	truststamp-002	0.0000	836	0.0010	800	0.0189	514	0.0000	404	-	1192
1248	truststamp-003	0.0000	853	0.0010	802	0.0189	515	0.0000	407	-	1125
1249	tsight-000	0.0000	712	0.0003	324	0.0064	219	0.0000	326	-	1307
1250	ttigroup-000	0.0000	297	0.0000	87	0.0000	13	0.0000	173	-	1251
1251	tuputech-000	0.0003	1259	0.0116	1261	-	1323	-	1404	0.0081	1074
1252	turingtechvip-001	0.0001	1218	0.0007	624	0.0061	209	0.0007	1211	0.0006	882
1253	turingtechvip-002	0.0001	1219	0.0017	985	0.0097	297	0.0007	1212	0.0006	881
1254	turkcell-000	0.0110	1309	0.0234	1282	0.0350	786	0.0103	1366	0.0306	1085
1255	turkcell-001	0.0000	54	0.0002	286	0.0034	181	0.0001	699	0.0003	293
1256	turkcell-002	0.0000	244	0.0002	278	0.0026	176	0.0001	626	-	1303
1257	twface-000	0.0000	251	0.0000	177	0.0000	143	0.0000	130	0.0000	123
1258	twface-001	0.0000	475	0.0000	35	0.0000	63	0.0000	257	0.0000	24
1259	ulsee-001	0.0000	336	0.0000	104	-	1058	0.0000	184	0.0000	83
1260	ultinous-000	-	1412	-	1375	-	995	-	1445	0.0003	348
1261	ultinous-001	-	1379	-	1434	-	1405	-	1424	0.0003	350
1262	uluface-002	0.0000	217	0.0000	160	-	1413	0.0000	107	0.0000	144
1263	uluface-003	0.0000	491	0.0001	273	-	1213	0.0002	858	0.0002	273
1264	unicc-001	0.0000	101	0.0012	875	0.0230	629	0.0004	1124	0.0004	571
1265	unicc-002	0.0316	1326	0.0082	1239	0.0188	507	0.0226	1376	0.0936	1090
1266	unicc-003	0.0007	1276	0.0022	1043	0.0094	292	0.0026	1323	0.0029	1046
1267	unicc-004	0.0000	1186	0.0012	873	0.0066	223	0.0012	1267	-	1286
1268	unissey-001	0.0000	67	0.0000	227	0.0000	102	0.0000	48	0.0000	192
1269	unissey-002	0.0000	492	0.0000	59	0.0000	77	0.0000	281	0.0000	3
1270	unissey-003	0.0000	222	0.0008	651	0.0191	518	0.0001	692	0.0004	517
1271	unissey-004	0.0000	359	0.0009	743	0.0240	647	0.0001	745	0.0004	588
1272	unissey-005	0.0000	136	0.0009	744	0.0240	648	0.0001	742	-	1371
1273	unissey-006	0.0000	827	0.0220	1278	0.0583	868	0.0003	1039	-	1175
1274	unissey-007	0.0000	567	0.0101	1254	0.0376	804	0.0004	1070	-	1452
1275	uniubi-000	0.0000	299	0.0000	89	0.0000	10	0.0000	171	-	1252
1276	universeai-000	0.0000	1065	0.0030	1110	0.0321	751	0.0000	601	-	1215

Table 112: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1277	upc-001	0.0000	1092	0.0003	321	-	1435	0.0003	1006	0.0003	380
1278	urielsoft-000	0.0000	82	0.0023	1058	0.0371	798	0.0000	578	-	1428
1279	useb-000	0.0000	799	0.0009	756	0.0228	618	0.0001	811	0.0004	506
1280	useb-001	0.0000	528	0.0009	758	0.0228	621	0.0001	804	0.0004	527
1281	useb-002	0.0000	782	0.0009	754	0.0228	616	0.0001	809	0.0004	509
1282	useb-003	0.0000	842	0.0009	752	0.0228	619	0.0001	812	-	1148
1283	useb-004	0.0000	598	0.0004	420	0.0112	333	0.0002	919	-	1396
1284	uxlabs-001	0.0000	135	0.0000	138	0.0000	120	0.0000	80	0.0000	167
1285	uxlabs-003	0.0000	146	0.0000	124	0.0000	116	0.0000	73	0.0000	171
1286	uzinfocom-000	0.0000	51	0.0005	443	0.0132	368	0.0001	750	-	1451
1287	uzinfocom-001	0.0000	413	0.0002	285	0.0044	186	0.0001	669	-	1184
1288	vaidio-001	0.0000	627	0.0006	541	0.0157	430	0.0043	1344	-	1373
1289	vcog-002	-	1474	-	1355	-	1156	-	1476	0.0019	1023
1290	vcortex-001	0.0000	1124	0.2295	1325	0.0802	916	0.0017	1294	0.0006	880
1291	vd-000	-	1353	-	1466	-	1288	-	1397	0.0000	195
1292	vd-001	0.0000	1156	0.0030	1114	-	1396	0.0004	1104	0.0009	937
1293	vd-002	0.0000	131	0.0000	135	1.0000	1373	0.0000	83	0.0000	165
1294	vd-003	0.0001	1220	0.0041	1163	0.0676	892	0.0030	1331	0.0029	1047
1295	veertec-001	0.0006	1271	0.0140	1266	0.1098	933	0.0019	1300	-	1410
1296	veridas-001	-	1469	-	1360	-	1171	0.0001	775	0.0005	806
1297	veridas-002	0.0000	1077	0.0026	1077	-	1328	0.0001	763	0.0005	816
1298	veridas-003	0.0000	1084	0.0026	1072	-	1050	0.0001	772	0.0005	809
1299	veridas-004	0.0000	1080	0.0026	1076	-	1460	0.0001	766	0.0005	811
1300	veridas-005	0.0000	1081	0.0026	1071	-	997	0.0001	769	0.0005	810
1301	veridas-006	0.0000	1070	0.0026	1073	0.0595	870	0.0001	760	0.0005	819
1302	veridas-007	0.0000	1086	0.0026	1070	0.0595	869	0.0001	774	0.0005	808
1303	veridas-008	0.0000	1074	0.0026	1074	0.0595	871	0.0001	761	0.0005	818
1304	veridas-009	0.0000	1082	0.0026	1075	0.0597	872	0.0001	695	0.0005	824
1305	veridas-010	0.0000	910	0.0011	862	0.0212	579	0.0004	1117	-	1391
1306	veridium-000	0.0061	1300	0.5956	1328	0.2889	959	0.0050	1351	0.0009	936
1307	veridium-001	0.0001	1209	0.0087	1245	0.1615	939	0.0014	1281	0.0006	873
1308	veridium-002	0.0000	1197	0.0083	1244	0.1591	938	0.0010	1244	0.0005	848
1309	veridium-003	0.0001	1208	0.0087	1246	0.1615	940	0.0014	1280	-	1306
1310	veridium-004	0.0001	1210	0.0028	1092	0.0832	921	0.0005	1164	-	1347
1311	verigram-000	0.0000	1012	0.0068	1222	0.0822	920	0.0003	1067	0.0005	838
1312	verigram-001	0.0000	927	0.0003	329	0.0060	204	0.0002	947	0.0003	399
1313	verigram-002	0.0000	1118	0.0052	1203	0.0378	805	0.0010	1246	0.0006	901
1314	verigram-003	0.0000	1171	0.0025	1066	0.0288	717	0.0009	1243	0.0008	929
1315	verigram-004	0.0000	959	0.0001	270	0.0016	169	0.0001	802	-	1200
1316	verihubs-inteligensia-000	0.0000	768	0.0029	1094	0.0669	889	0.0001	648	0.0004	678
1317	verihubs-inteligensia-001	0.0000	704	0.0029	1095	0.0669	891	0.0001	647	0.0004	680
1318	verihubs-inteligensia-002	0.0000	545	0.0029	1096	0.0669	890	0.0001	637	0.0004	685
1319	verijelas-000	0.0000	546	0.0023	1055	0.0375	802	0.0004	1129	0.0006	889
1320	verismart-001	0.0000	158	0.0002	289	0.0049	191	0.0001	645	-	1343
1321	via-000	0.0000	355	0.0000	112	-	1087	0.0000	207	0.0000	64
1322	via-001	0.0000	312	0.0000	80	-	1018	0.0000	166	0.0000	91
1323	via-004	0.0000	399	0.0000	12	0.0000	43	0.0000	225	0.0000	46
1324	via-005	0.0000	385	0.0000	108	0.0000	25	0.0000	192	0.0000	77
1325	via-006	0.0000	424	0.0000	31	0.0000	60	0.0000	252	-	1142
1326	viante-000	0.0000	940	0.0008	689	0.0168	459	0.0001	737	0.0004	770
1327	viante-001	0.0000	898	0.0008	691	0.0168	460	0.0001	734	-	1459
1328	viante-002	0.0000	838	0.0007	635	0.0154	419	0.0001	739	-	1139
1329	viante-003	0.0000	554	0.0007	637	0.0154	420	0.0001	735	-	1437
1330	viante-004	0.0000	667	0.0007	636	0.0154	421	0.0001	736	-	1315
1331	videmo-000	0.0000	901	0.0019	1011	-	1304	0.0003	1038	0.0012	993
1332	videmo-001	0.0000	1055	0.0170	1269	0.0332	769	0.0010	1254	0.0011	977
1333	videmo-002	0.0000	246	0.0006	543	0.0189	509	0.0001	715	0.0004	491
1334	videmo-003	0.0000	379	0.0006	540	0.0189	510	0.0001	718	-	1210

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A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1335	videonetics-001	0.0004	1260	0.0309	1289	-	1253	0.0015	1287	0.0010	951
1336	videonetics-002	0.0000	987	0.0459	1302	-	1286	0.0006	1194	0.0005	861
1337	viettelai-000	0.0000	303	0.0000	77	0.0000	7	0.0000	167	0.0000	89
1338	viettelai-001	0.0000	917	0.0010	794	0.0161	446	0.0000	582	0.0004	494
1339	viettelai-002	0.0000	1009	0.0019	1008	0.0334	773	0.0002	958	-	1231
1340	viettelai-003	0.0000	891	0.0011	866	0.0218	598	0.0001	628	-	1434
1341	viettelai-004	0.0000	643	0.0005	468	0.0139	382	0.0000	501	-	1345
1342	viettelai-005	0.0000	648	0.0005	460	0.0137	376	0.0000	478	-	1349
1343	viettelai-006	0.0000	779	0.0005	459	0.0137	375	0.0000	487	-	1229
1344	viettelhightech-000	0.0000	1107	0.0019	1014	0.0368	796	0.0007	1216	0.0005	853
1345	viettelhightech-001	0.0000	601	0.0014	932	0.0434	832	0.0001	832	-	1401
1346	viettelhightech-002	0.0000	691	0.0014	930	0.0434	833	0.0001	834	-	1340
1347	vigilantsolutions-000	-	1418	-	1373	-	972	-	1437	0.0007	912
1348	vigilantsolutions-001	-	1406	-	1444	-	1442	-	1430	0.0004	694
1349	vigilantsolutions-002	-	1355	-	1468	-	1291	-	1395	0.0004	701
1350	vigilantsolutions-003	-	1463	-	1346	-	1134	-	1470	0.0004	451
1351	vigilantsolutions-004	-	1411	-	1376	-	994	-	1444	0.0004	459
1352	vigilantsolutions-005	-	1393	-	1437	-	1434	0.0001	658	0.0004	462
1353	vigilantsolutions-006	0.0000	1043	0.0028	1089	-	1338	0.0001	655	0.0004	468
1354	vigilantsolutions-007	0.0000	1059	0.0028	1084	-	1110	0.0001	662	0.0004	453
1355	vigilantsolutions-008	0.0000	1061	0.0028	1085	-	1149	0.0001	663	0.0004	449
1356	vigilantsolutions-009	0.0000	1062	0.0028	1086	-	1211	0.0001	664	0.0004	446
1357	vigilantsolutions-010	0.0000	1048	0.0028	1088	0.0609	875	0.0001	656	0.0004	465
1358	vigilantsolutions-011	0.0000	1053	0.0028	1087	0.0609	874	0.0001	660	0.0004	456
1359	vinai-000	0.0000	70	0.0000	230	-	1298	0.0000	45	0.0000	194
1360	vinbigdata-001	0.0000	490	0.0000	58	0.0000	76	0.0000	286	0.0000	2
1361	vinbigdata-002	0.0000	59	0.0015	952	0.0250	669	0.0000	516	0.0004	717
1362	vinbigdata-003	0.0000	602	0.0007	593	0.0151	410	0.0000	606	0.0004	496
1363	vinbigdata-004	0.0000	830	0.0005	463	0.0141	384	0.0000	393	-	1180
1364	vinbigdata-005	0.0000	879	0.0005	464	0.0141	385	0.0000	395	-	1112
1365	vinbigdata-006	0.0000	806	0.0005	465	0.0141	383	0.0000	391	-	1209
1366	vinbigdata-007	0.0000	631	0.0005	466	0.0141	386	0.0000	378	-	1380
1367	vion-000	0.0050	1295	0.0392	1299	-	1305	0.0130	1370	0.0078	1073
1368	visage-000	0.0000	1101	0.0054	1206	-	1244	0.0009	1236	0.0006	883
1369	visica-001	0.0000	1152	0.0068	1224	0.0692	899	0.0010	1245	-	1474
1370	visica-002	0.0000	960	0.0080	1237	0.1006	930	0.0004	1108	-	1170
1371	visica-003	0.0000	885	0.0022	1047	0.0419	826	0.0004	1107	-	1472
1372	visionbox-000	0.0000	1174	0.0033	1134	-	1065	0.0005	1174	0.0011	970
1373	visionbox-001	0.0000	1173	0.0033	1135	-	1414	0.0005	1173	0.0011	972
1374	visionbox-002	0.0000	263	0.0017	976	0.0270	703	0.0000	525	0.0004	760
1375	visionbox-003	0.0000	989	0.0010	790	0.0209	568	0.0001	721	0.0004	650
1376	visionbox-004	0.0000	935	0.0013	902	0.0248	663	0.0001	796	0.0004	724
1377	visionbox-005	0.0000	1049	0.0037	1154	0.0301	728	0.0005	1133	-	1292
1378	visionbox-006	0.0000	1106	0.0040	1160	0.0331	766	0.0007	1206	-	1270
1379	visionbox-007	0.0000	1098	0.0041	1162	0.0324	753	0.0006	1199	-	1336
1380	visionbox-008	0.0000	1056	0.0035	1146	0.0280	706	0.0001	839	-	1201
1381	visionlabs-001	-	1340	-	1463	-	1267	-	1393	0.0014	1008
1382	visionlabs-002	-	1483	-	1365	-	1189	-	1479	0.0026	1040
1383	visionlabs-003	-	1459	-	1352	-	1140	-	1473	0.0025	1039
1384	visionlabs-004	-	1354	-	1465	-	1289	-	1398	0.0008	931
1385	visionlabs-005	-	1342	-	1459	-	1255	0.0002	994	0.0023	1034
1386	visionlabs-006	-	1385	-	1424	-	1390	0.0003	1046	0.0005	814
1387	visionlabs-007	0.0000	955	0.0014	938	-	1051	0.0003	1042	0.0005	803
1388	visionlabs-008	0.0000	767	0.0015	950	-	1005	0.0002	987	0.0004	759
1389	visionlabs-009	0.0000	692	0.0010	796	-	1416	0.0001	619	0.0004	629
1390	visionlabs-010	0.0000	1020	0.0009	732	-	1234	0.0001	798	0.0004	635
1391	visionlabs-011	0.0000	388	0.0006	534	0.0156	429	0.0001	670	0.0004	485
1392	visteam-000	0.0000	1172	0.0031	1120	-	1069	0.0005	1166	0.0011	959

Table 114: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1393	visteam-001	0.0000	1040	0.0014	935	-	1191	0.0002	900	0.0004	636
1394	visteam-002	0.0000	1024	0.0014	936	0.0285	713	0.0002	886	0.0004	653
1395	visteam-003	0.0000	796	0.0010	824	0.0225	610	0.0001	631	0.0004	535
1396	visteam-004	0.0000	694	0.0010	825	0.0225	607	0.0001	743	0.0004	544
1397	visteam-005	0.0000	764	0.0010	823	0.0224	606	0.0001	667	0.0004	538
1398	visteam-006	0.0000	569	0.0008	682	0.0209	563	0.0000	349	0.0004	470
1399	visteam-007	0.0000	823	0.0008	680	0.0209	561	0.0000	369	0.0004	452
1400	visteam-008	0.0000	519	0.0008	681	0.0209	562	0.0000	348	0.0004	472
1401	visteam-009	0.0000	751	0.0010	840	0.0328	760	0.0000	365	-	1248
1402	visteam-010	0.0000	590	0.0010	841	0.0328	761	0.0000	350	-	1426
1403	vivotek-001	0.0000	462	0.0006	528	0.0161	444	0.0000	265	-	1121
1404	vivotek-002	0.0000	104	0.0006	529	0.0161	445	0.0000	61	-	1400
1405	vixvizion-000	0.0002	1247	0.0028	1090	-	1228	0.0010	1252	0.0012	990
1406	vixvizion-001	0.0000	964	0.0030	1111	-	1130	0.0001	754	0.0004	659
1407	vixvizion-002	0.0000	991	0.0018	990	-	1331	0.0000	535	0.0004	651
1408	vixvizion-003	0.0000	257	0.0000	67	-	989	0.0000	153	0.0000	98
1409	vixvizion-004	0.0000	392	0.0000	7	0.0000	52	0.0000	237	0.0000	43
1410	vixvizion-005	0.0000	403	0.0000	15	0.0000	46	0.0000	230	0.0000	48
1411	vixvizion-006	0.0000	105	0.0000	239	0.0000	107	0.0000	56	0.0000	184
1412	vixvizion-007	0.0000	439	0.0000	22	0.0000	57	0.0000	247	0.0000	34
1413	vnis-000	0.0007	1278	0.0695	1311	0.0756	913	0.0038	1341	0.0009	942
1414	vnpay-000	0.0000	1047	0.0657	1310	0.0693	900	0.0037	1339	0.0006	877
1415	vnpay-001	0.0000	732	0.0052	1202	0.0497	843	0.0001	752	-	1273
1416	vnpt-001	0.0652	1331	0.2829	1327	-	1167	0.2116	1384	0.1598	1092
1417	vnpt-002	0.0000	625	0.0002	281	0.0024	174	0.0003	1045	0.0003	301
1418	vnpt-003	0.0000	91	0.0004	366	0.0105	304	0.0002	850	0.0004	426
1419	vnpt-004	0.0000	837	0.0006	512	0.0160	442	0.0002	875	0.0004	555
1420	vnpt-005	0.0000	198	0.0006	495	0.0154	425	0.0002	908	0.0004	542
1421	vnpt-006	0.0000	478	0.0006	494	0.0154	422	0.0002	910	0.0004	534
1422	vnpt-007	0.0000	52	0.0006	498	0.0154	424	0.0002	906	-	1456
1423	vnpt-008	0.0000	247	0.0006	496	0.0154	426	0.0002	909	-	1308
1424	vnpt-009	0.0000	49	0.0006	497	0.0154	423	0.0002	905	-	1454
1425	vocalize-001	0.0216	1320	0.0193	1273	0.1691	944	0.0068	1358	-	1399
1426	vocord-001	-	1449	-	1342	-	1109	-	1464	0.0038	1059
1427	vocord-002	-	1337	-	1451	-	1227	-	1387	0.0037	1058
1428	vocord-003	-	1453	-	1331	-	1096	-	1457	0.0776	1088
1429	vocord-004	-	1441	-	1400	-	1074	0.0005	1165	0.0003	312
1430	vocord-005	-	1413	-	1379	-	982	0.0013	1271	0.0003	395
1431	vocord-006	-	1455	-	1334	-	1100	0.0003	1029	0.0003	358
1432	vocord-007	0.0000	947	0.0535	1303	-	1026	0.0001	824	0.0004	421
1433	vocord-008	0.0000	679	0.0015	956	-	1419	0.0003	1053	0.0001	254
1434	vocord-009	0.0000	606	0.0006	532	-	1333	0.0001	827	0.0003	302
1435	vocord-010	0.0000	950	0.0005	480	0.0141	388	0.0002	911	0.0003	394
1436	vogic-000	0.0000	713	0.0001	269	0.0015	168	0.0002	872	-	1310
1437	vsoft-001	0.0069	1303	0.8292	1329	0.8200	964	0.0173	1373	-	1110
1438	vsoft-002	0.0026	1289	0.0029	1101	0.0613	876	0.0059	1357	-	1226
1439	vsoft-003	0.0000	1147	0.0029	1103	0.0613	877	0.0005	1137	-	1465
1440	vsoft-004	0.0000	1148	0.0029	1102	0.0613	878	0.0005	1138	-	1370
1441	vts-000	0.0000	1000	0.0011	853	-	1479	0.0001	829	0.0004	751
1442	vts-001	0.0000	174	0.0003	314	0.0073	236	0.0000	382	0.0003	294
1443	vts-002	0.0000	119	0.0008	707	0.0150	407	0.0000	376	-	1413
1444	wicket-000	0.0000	763	0.0009	718	0.0260	690	0.0000	415	0.0004	563
1445	winsense-000	0.0000	165	0.0000	149	-	1404	0.0000	98	0.0000	152
1446	winsense-001	0.0000	261	0.0000	68	-	996	0.0000	156	0.0000	99
1447	winsense-002	0.0000	469	0.0000	46	-	1166	0.0000	263	0.0000	20
1448	wiseai-001	0.0001	1206	0.0137	1265	0.0768	915	0.0018	1297	0.0018	1018
1449	wuhantianyu-001	0.0000	290	0.0007	601	0.0159	435	0.0001	621	0.0004	624
1450	x-laboratory-000	0.0247	1322	0.0000	217	-	1266	0.0005	1172	0.0002	276

Table 115: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1451	x-laboratory-001	0.0000	872	0.0012	879	-	1204	0.0001	781	0.0004	719
1452	xforwardai-000	0.0000	465	0.0000	45	-	1162	0.0000	261	0.0000	19
1453	xforwardai-001	0.0000	593	0.0007	623	-	1281	0.0003	1021	0.0004	715
1454	xforwardai-002	0.0000	815	0.0007	622	-	1116	0.0003	1022	0.0004	707
1455	xm-000	0.0000	85	0.0007	592	-	1285	0.0001	629	0.0003	333
1456	yisheng-000	-	1371	-	1407	-	1350	-	1414	0.0006	876
1457	yisheng-001	-	1339	-	1453	-	1224	-	1386	0.0006	879
1458	yisheng-002	-	1462	-	1345	-	1133	-	1471	0.0006	875
1459	yisheng-004	0.0002	1250	-	1381	-	1013	0.0013	1270	0.0006	897
1460	yitu-000	-	1346	-	1473	-	1302	-	1401	0.0013	999
1461	yitu-001	-	1420	-	1388	-	1031	0.0009	1238	0.0006	896
1462	yitu-002	-	1362	-	1483	-	1339	-	1408	0.0000	179
1463	yitu-003	0.0000	123	0.0000	133	-	1367	0.0009	1237	0.0000	163
1464	yoonik-000	0.0000	549	0.0019	1012	-	1271	0.0001	702	0.0004	729
1465	yoonik-001	0.0000	268	0.0014	942	-	980	0.0001	823	0.0004	710
1466	yoonik-002	0.0000	919	0.0010	816	0.0284	712	0.0003	1002	0.0006	871
1467	yoonik-003	0.0000	942	0.0009	761	0.0214	588	0.0002	869	0.0004	664
1468	yoonik-004	0.0000	505	0.0012	868	0.0251	673	0.0001	671	0.0004	480
1469	yoti-001	0.0000	952	0.0014	943	0.0418	825	0.0008	1234	0.0006	884
1470	yoti-002	0.0000	117	0.0007	589	0.0171	465	0.0001	729	-	1414
1471	ytu-000	0.0000	706	0.0010	838	-	1465	0.0002	964	0.0004	712
1472	yuan-000	0.0000	366	0.0000	117	-	1079	0.0000	205	0.0000	67
1473	yuan-001	0.0000	1071	0.0010	832	-	1240	0.0005	1167	0.0005	827
1474	yuan-002	0.0000	1083	0.0010	831	0.0284	710	0.0005	1169	0.0005	826
1475	yuan-003	0.0000	1085	0.0010	829	0.0284	711	0.0005	1171	0.0005	825
1476	yuan-004	0.0000	103	0.0000	237	0.0000	106	0.0000	60	0.0000	182
1477	yuan-005	0.0000	729	0.0005	470	0.0134	372	0.0002	903	0.0004	513
1478	yuan-006	0.0000	1103	0.0014	945	0.0369	797	0.0004	1090	0.0005	844
1479	yuan-007	0.0000	1116	0.0013	905	0.0336	777	0.0004	1093	0.0005	837
1480	yuan-008	0.0000	1108	0.0013	907	0.0336	776	0.0004	1092	0.0005	840
1481	yuan-009	0.0000	1117	0.0013	906	0.0336	778	0.0004	1094	-	1116
1482	yuan-010	0.0000	1109	0.0013	908	0.0336	775	0.0004	1091	-	1266
1483	yuanli-001	0.0000	40	0.0016	973	0.0318	741	0.0001	674	-	1441

Table 116: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.4 Recognition accuracy

Core algorithm accuracy is stated via:

- ▷ **Cooperative subjects**
  - The summary table of Figure 90;
  - The visa image DETs of Figure 224;
  - The mugshot DETs of Figure 294;
  - The mugshot ageing profiles of Figure 807;
  - The human-difficult pairs of Figure 114
- ▷ **Non-cooperative subjects**
  - The photojournalism DET of Figure 330

Figure 630 shows dependence of false match rate on algorithm score threshold. This allows a deployer to set a threshold to target a particular false match rate appropriate to the security objectives of the application.

Figure 629 likewise shows FMR(T) but for mugshots, and specially four subsets of the population.

Note that in both the mugshot and visa sets false match rates vary with the ethnicity, age, and sex, of the enrollee and impostor. For example figure 399 summarizes FMR for impostors paired from four groups black females, black males, white females, white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

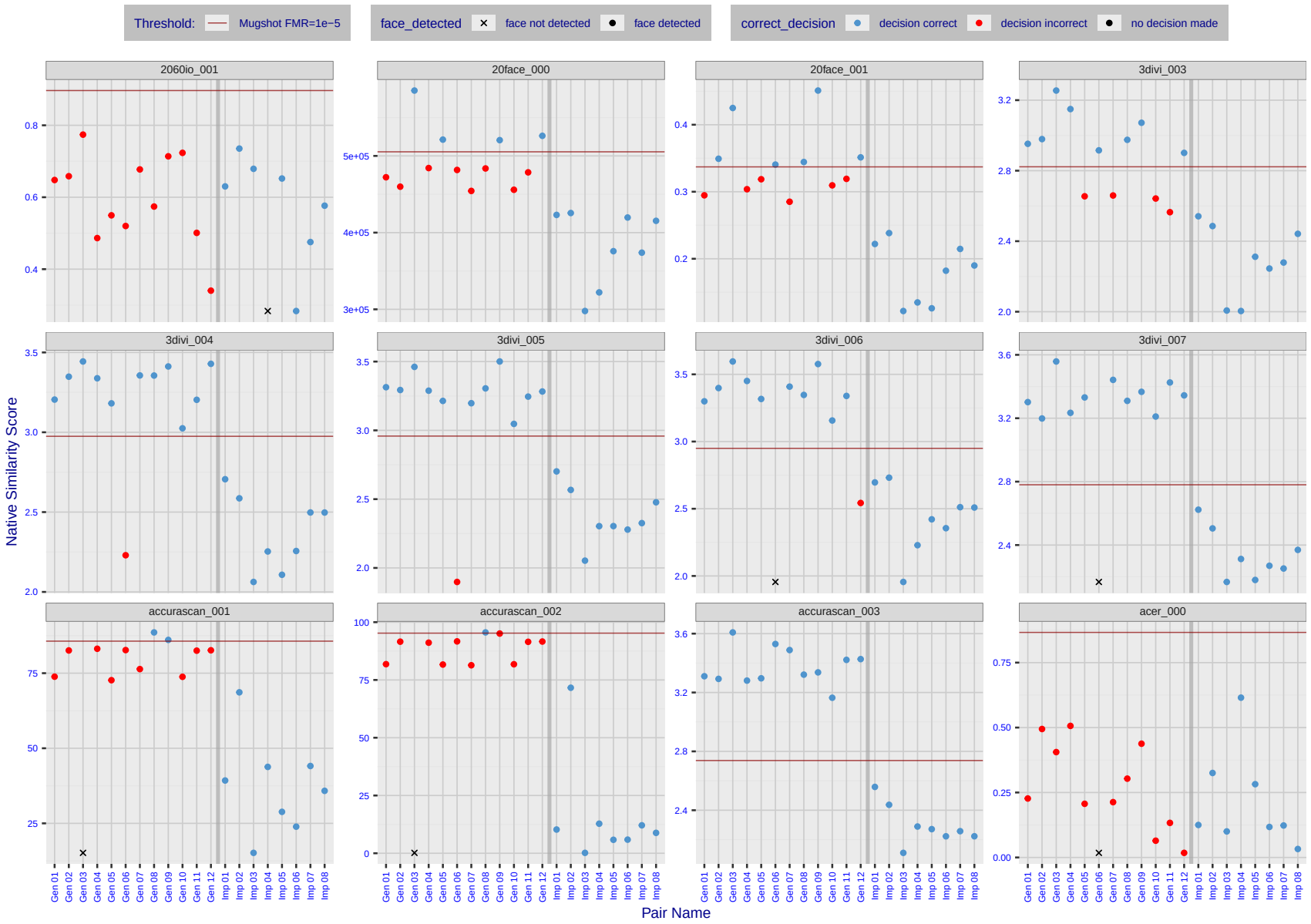

Figure 4: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

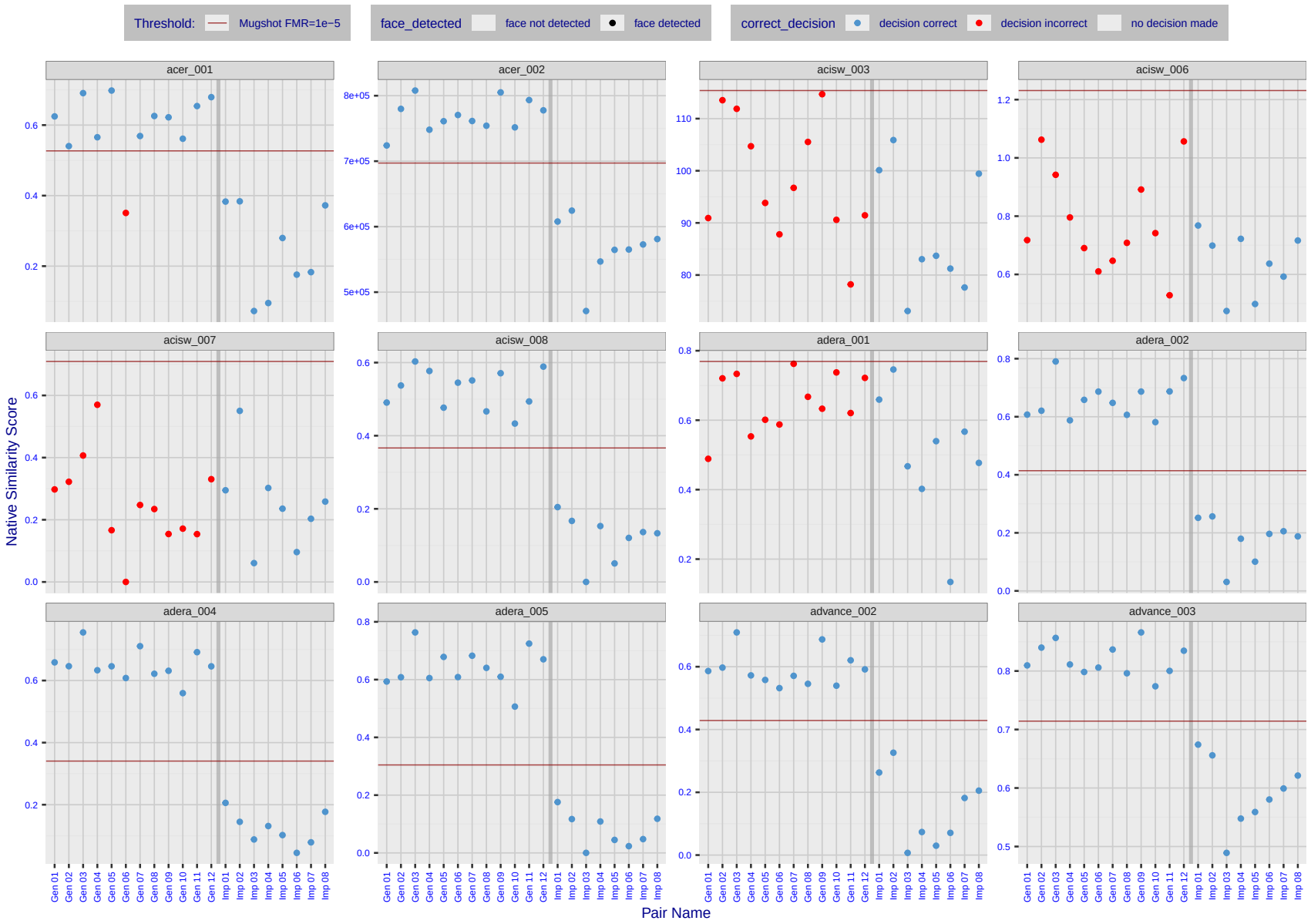

Figure 5: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

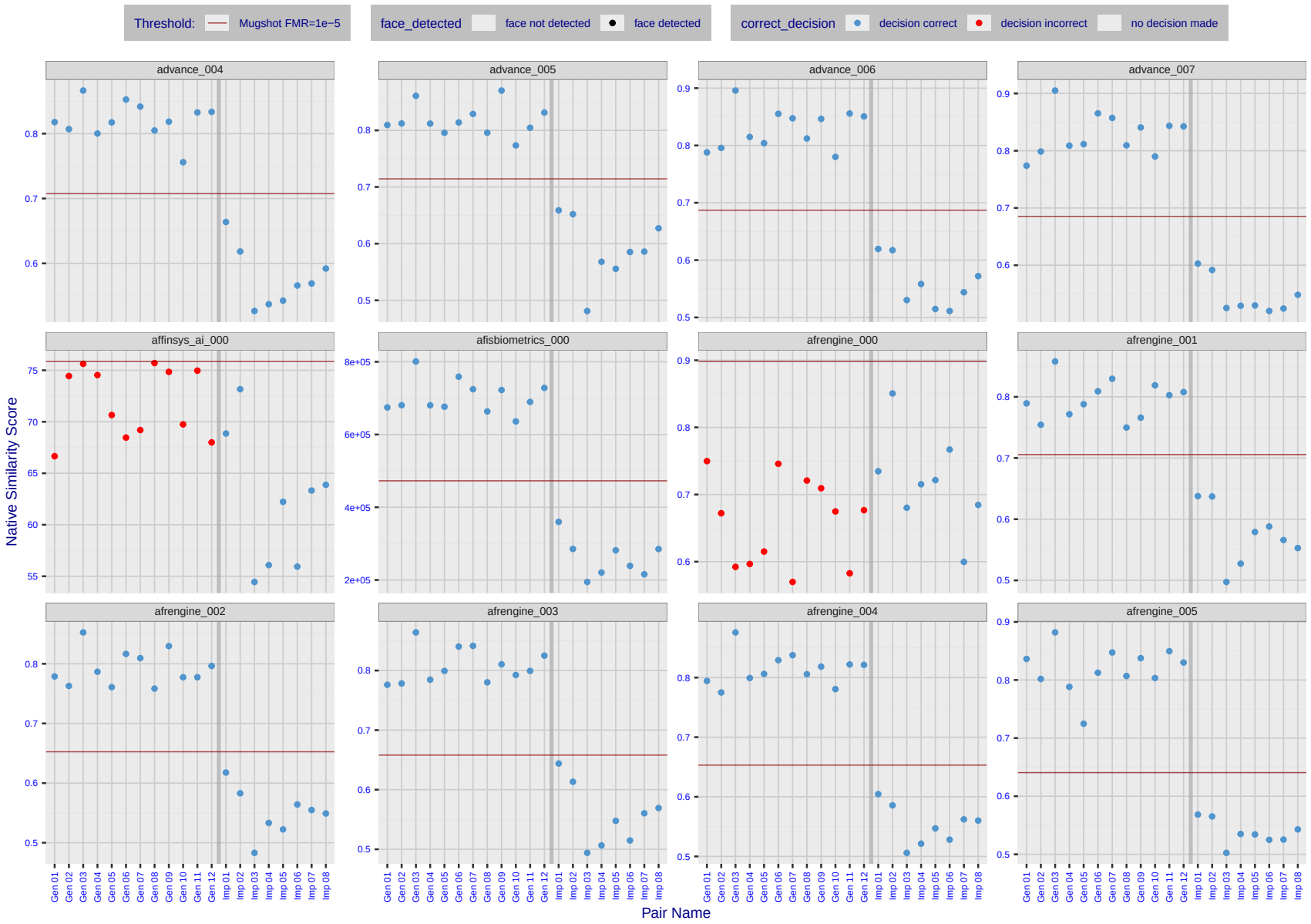

Figure 6: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

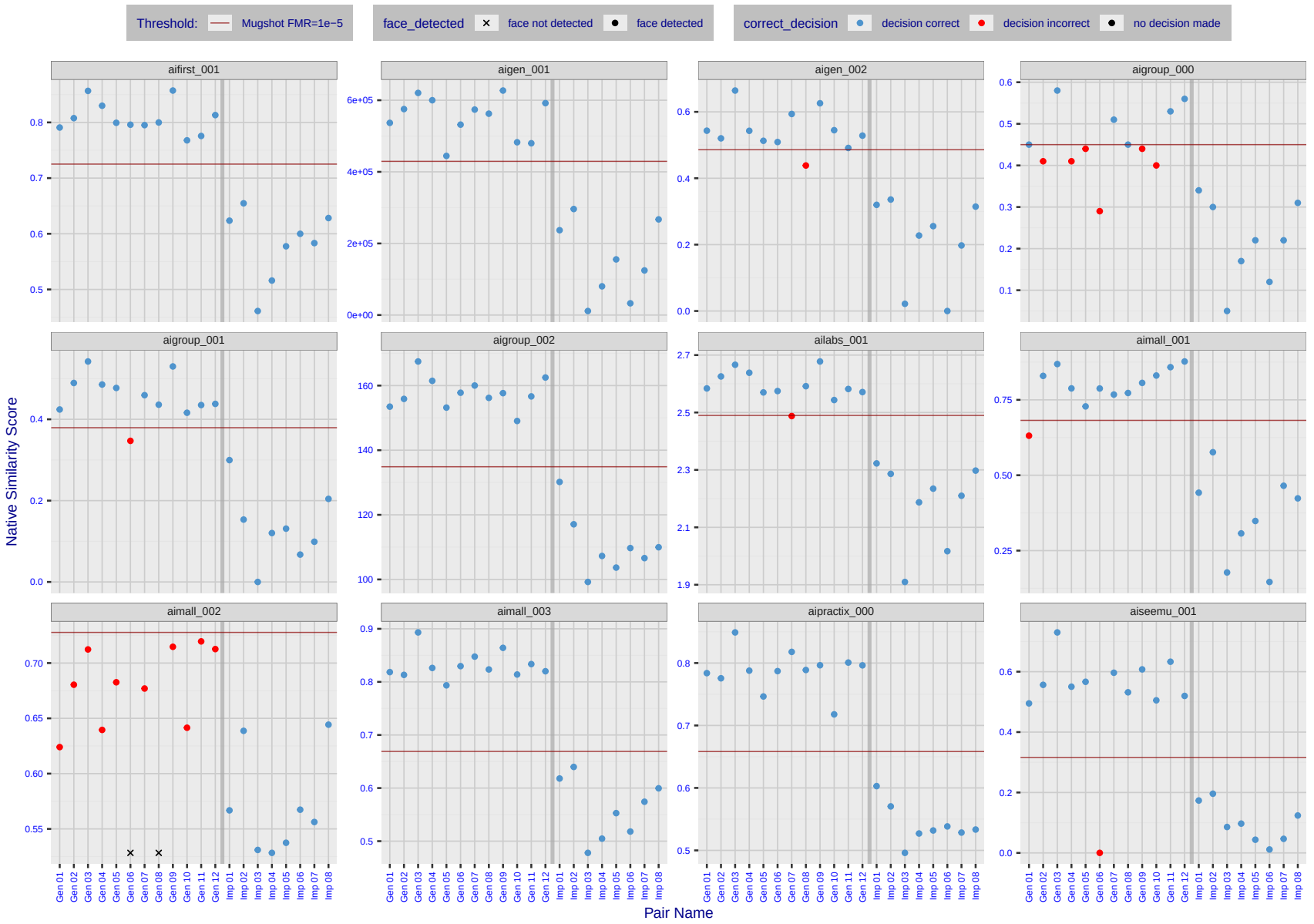

Figure 7: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

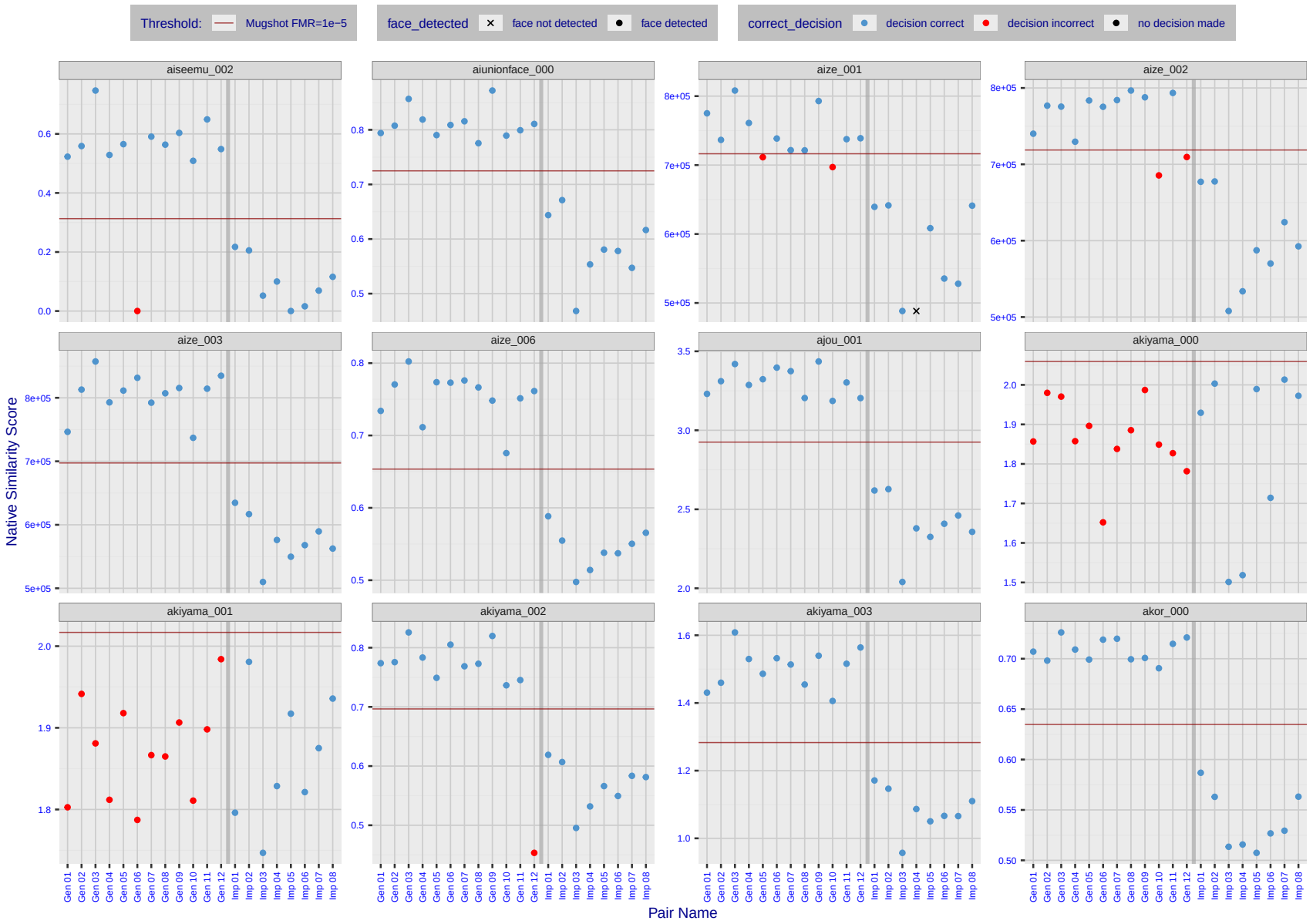

Figure 8: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

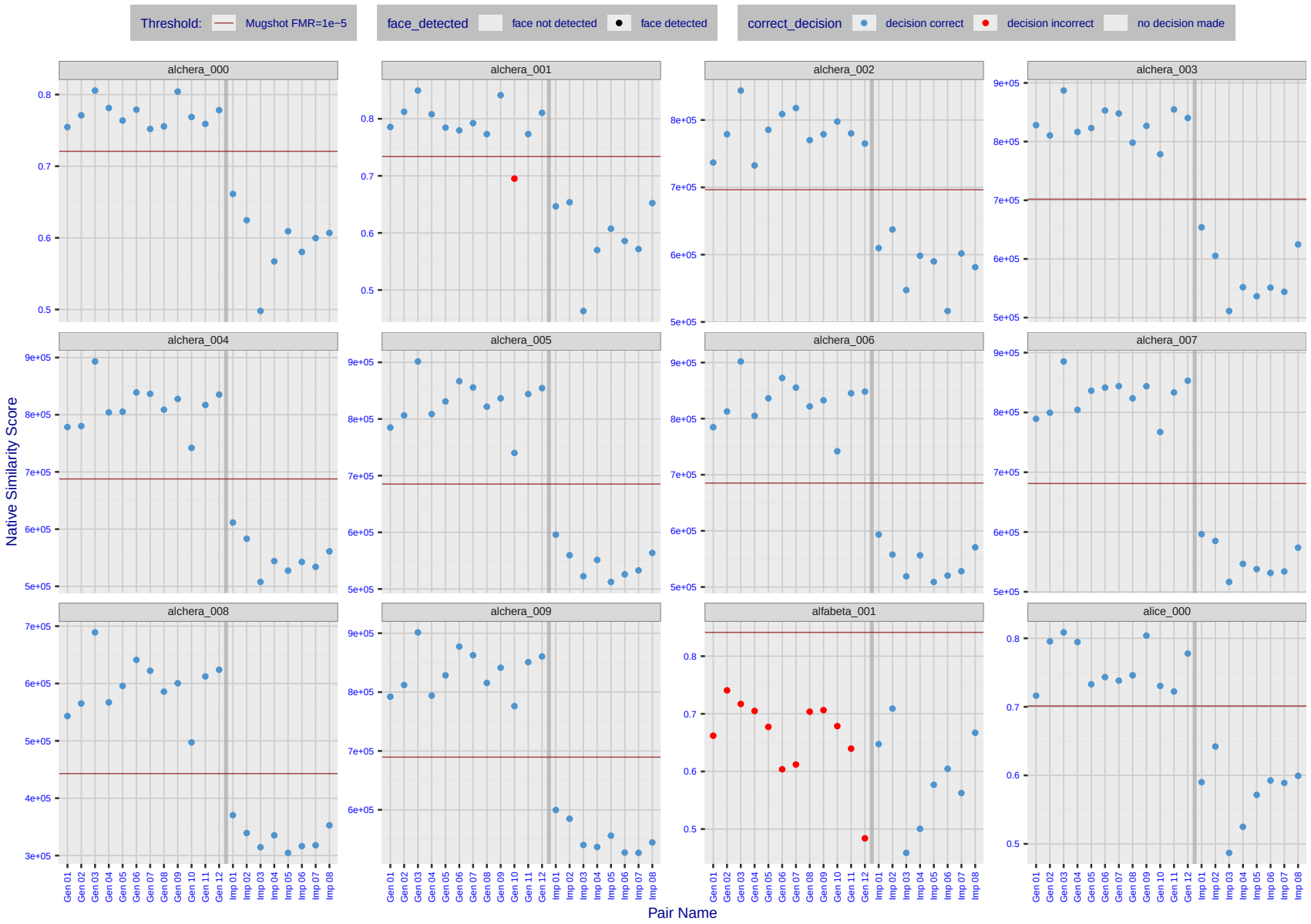

Figure 9: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

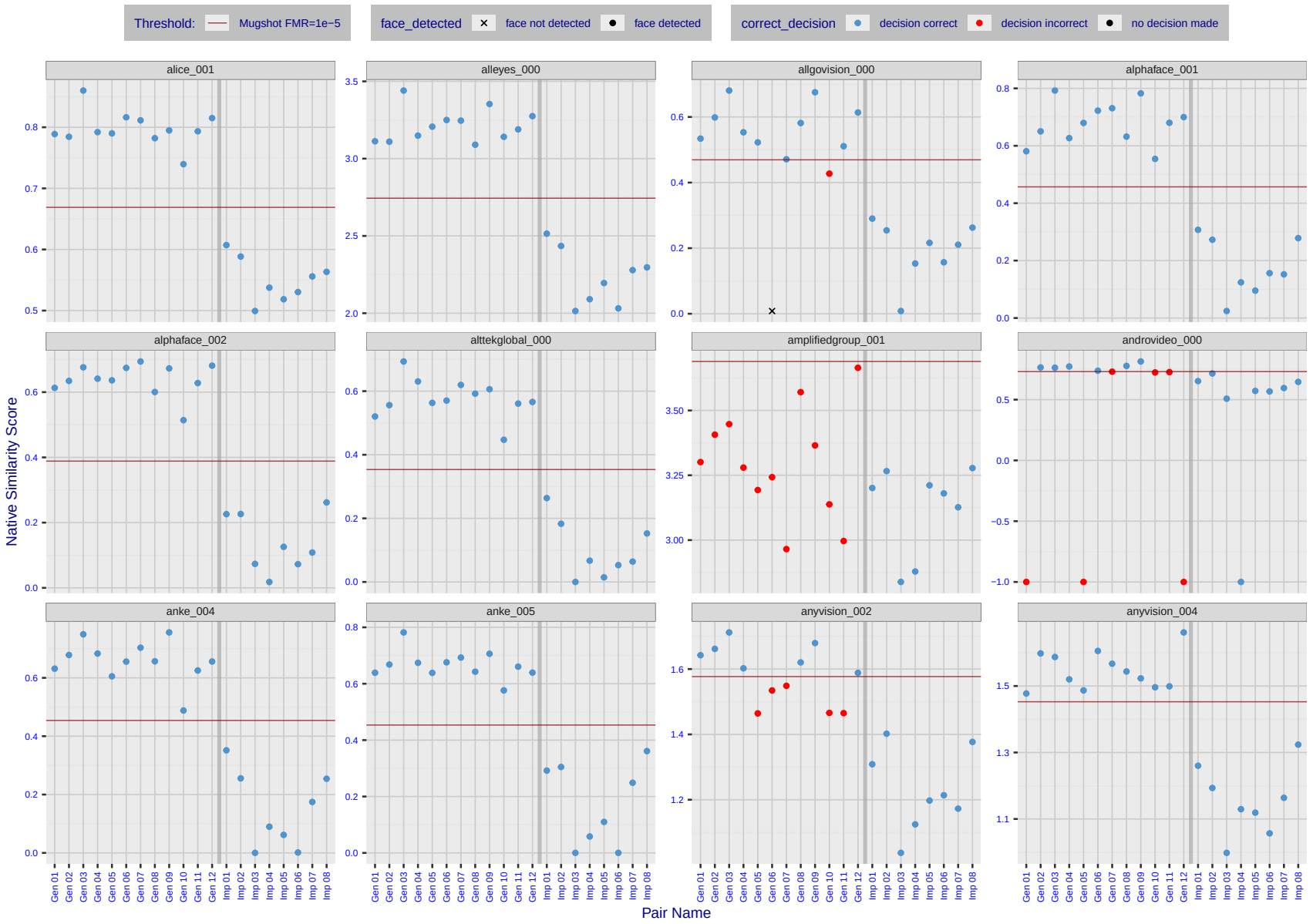

Figure 10: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

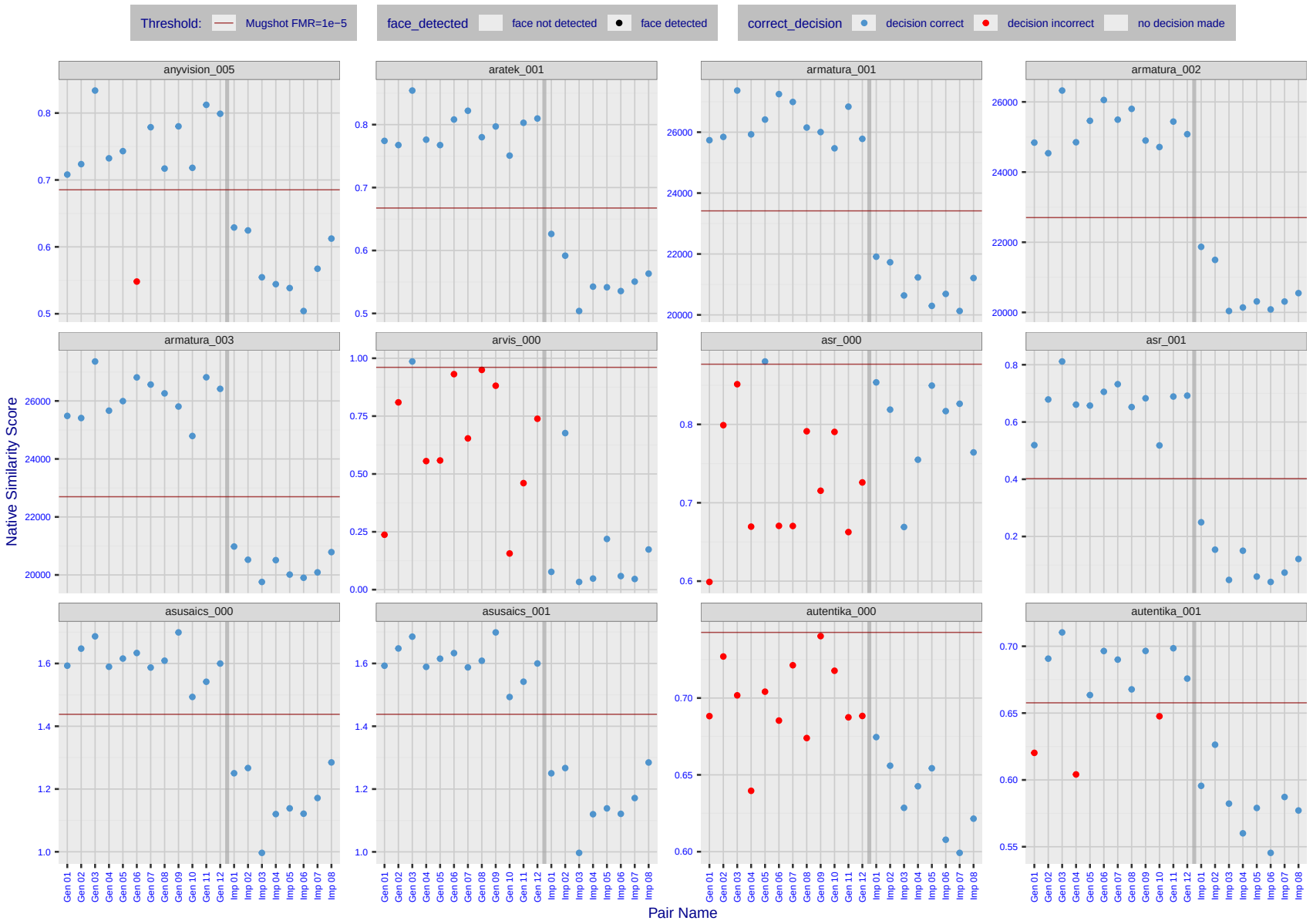

Figure 11: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

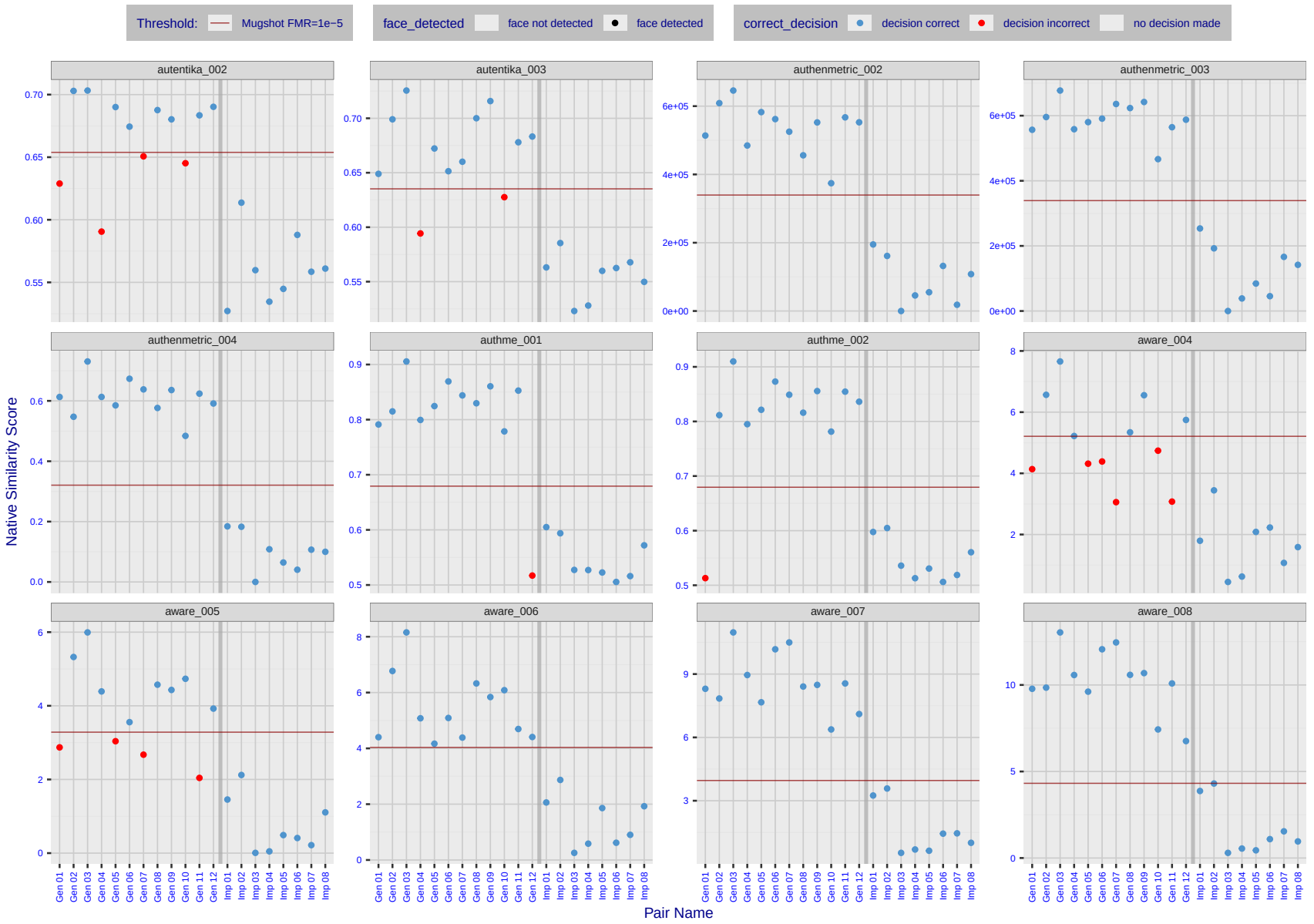

Figure 12: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

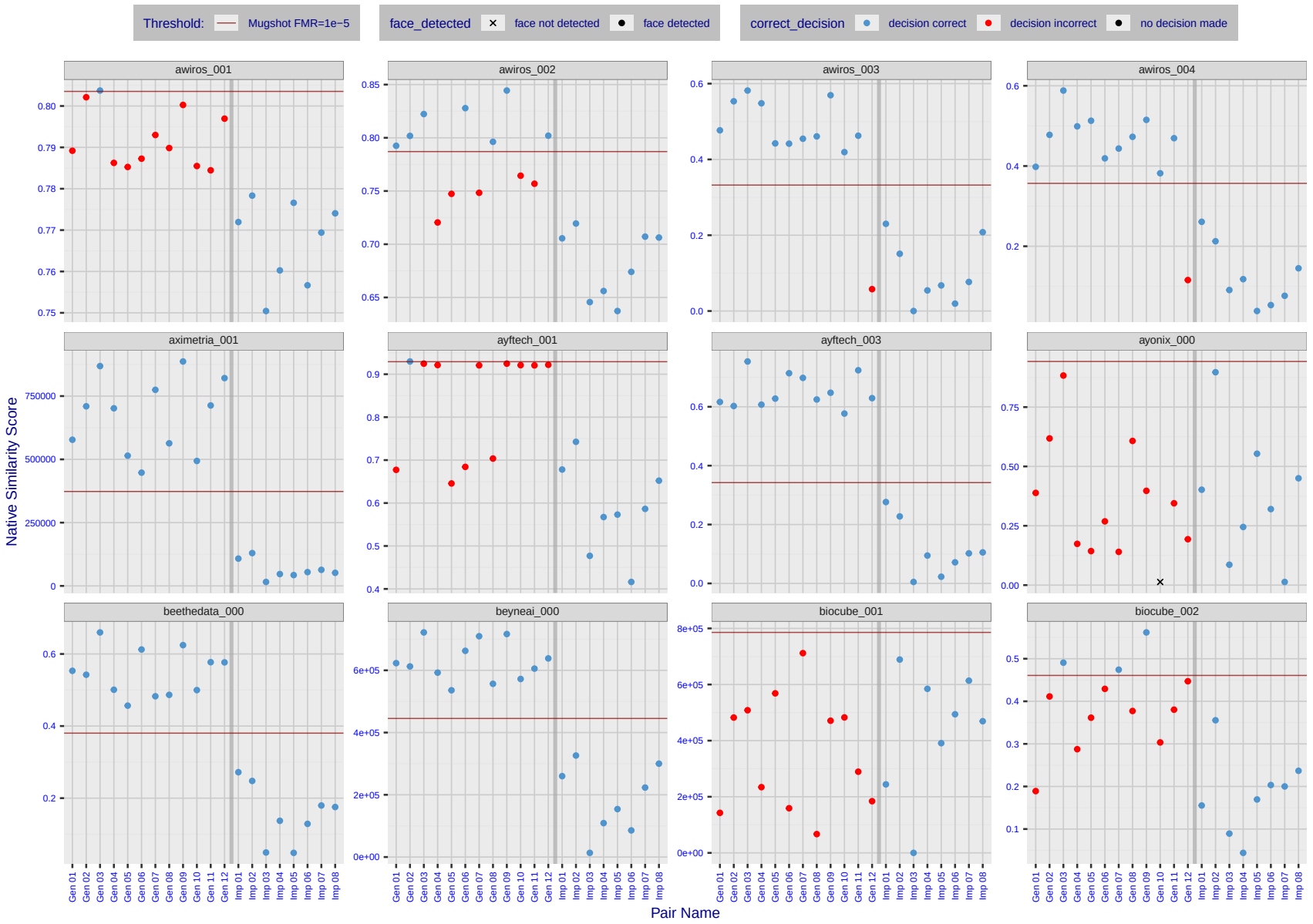

Figure 13: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

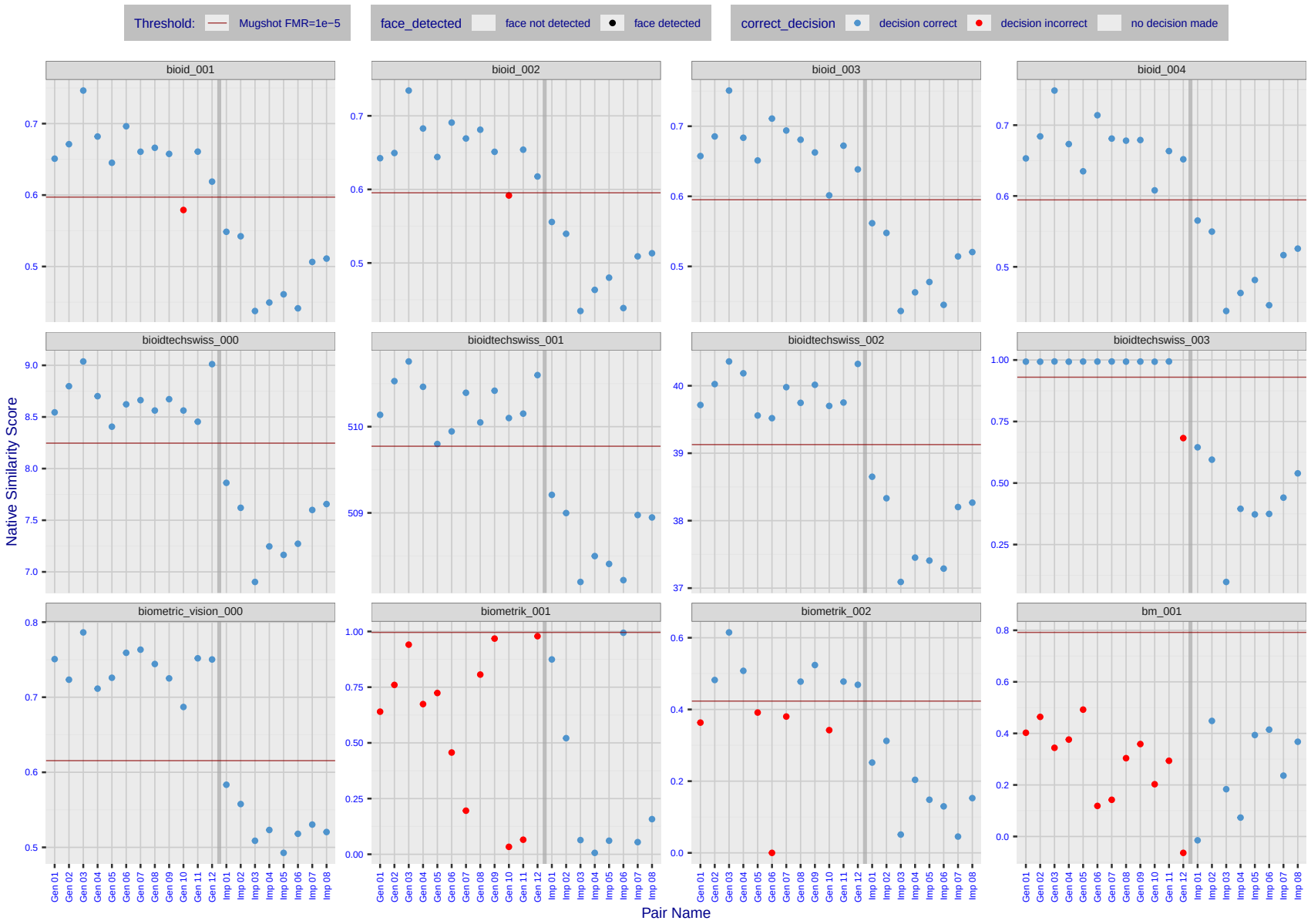

Figure 14: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

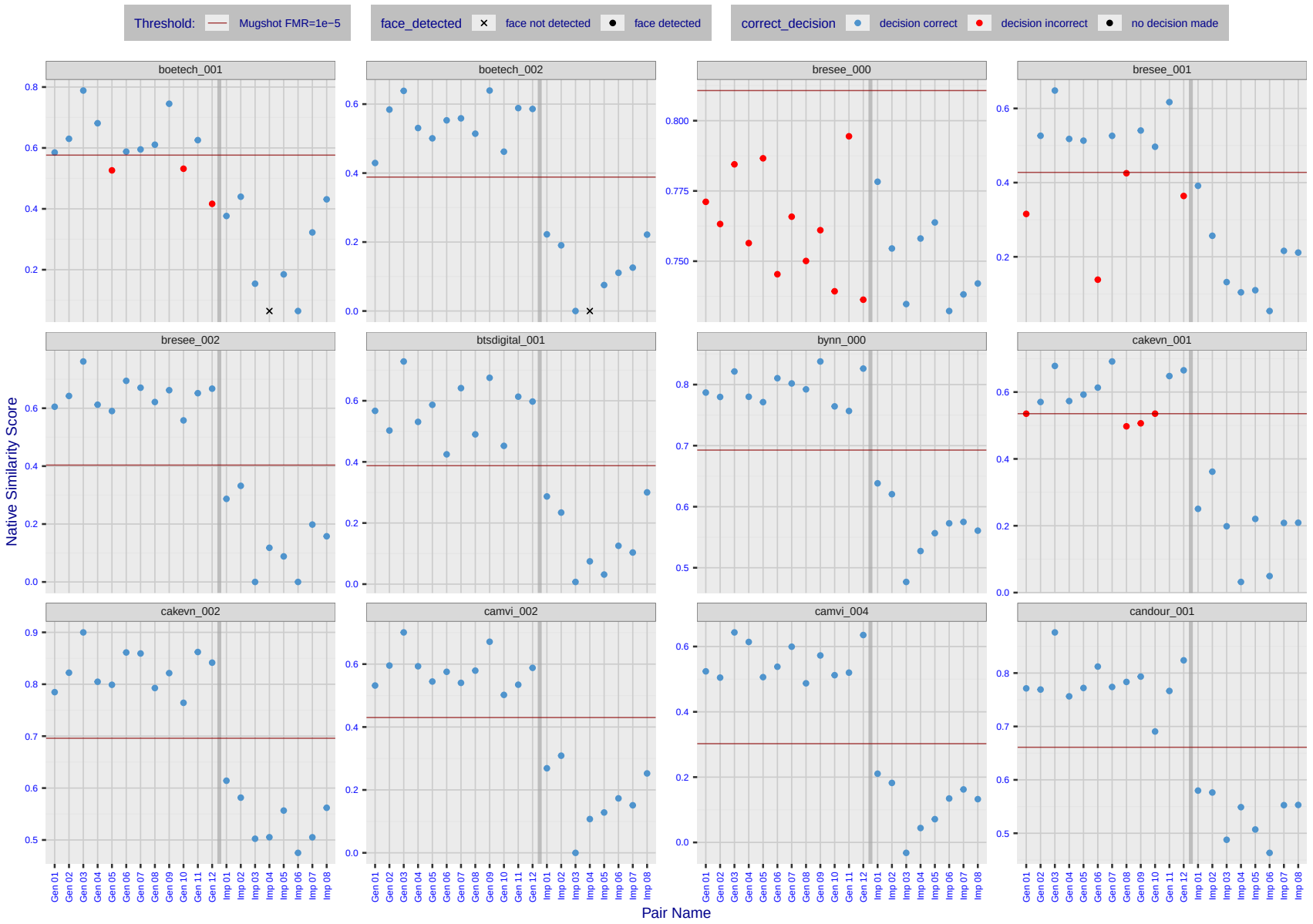

Figure 15: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

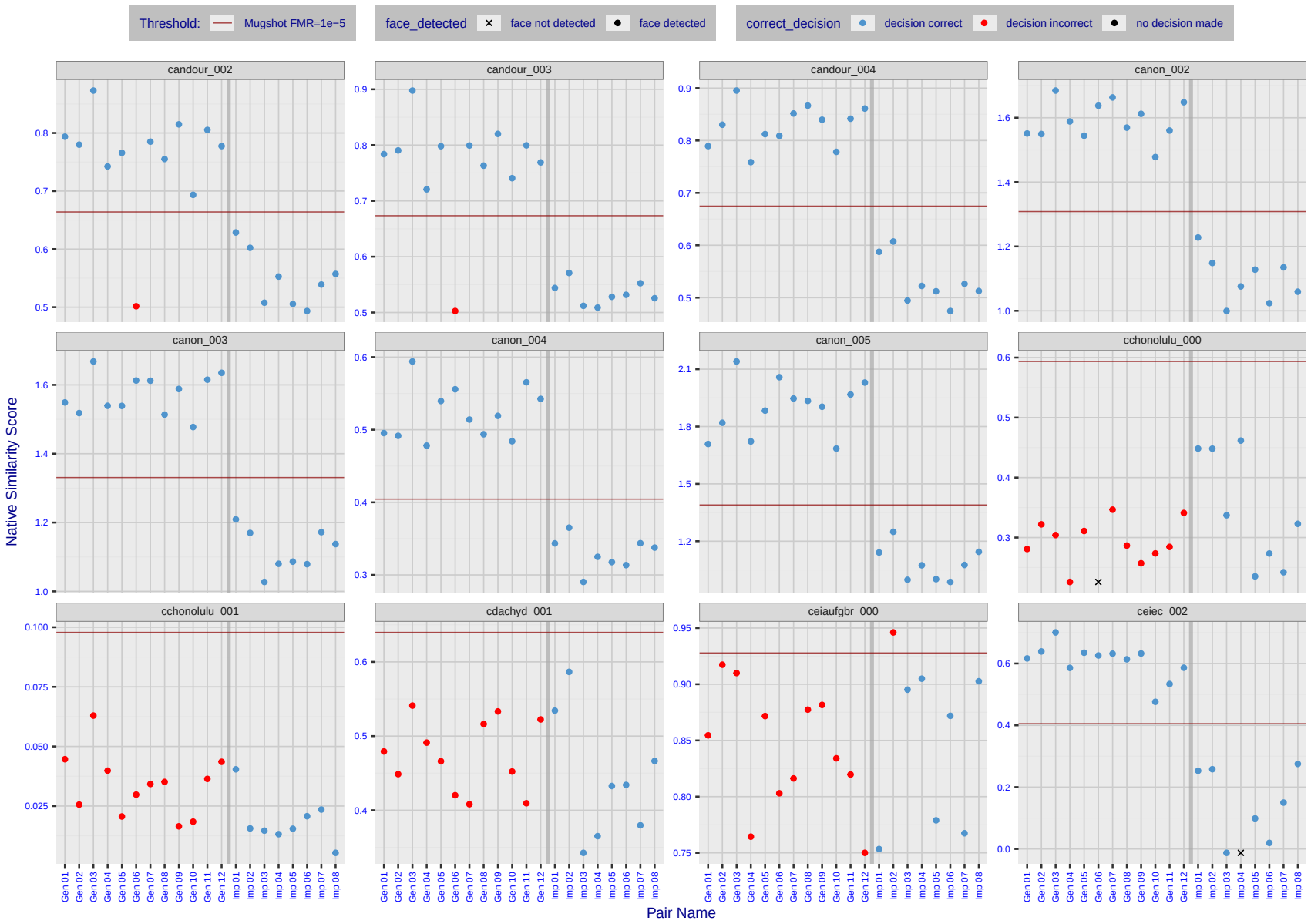

Figure 16: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

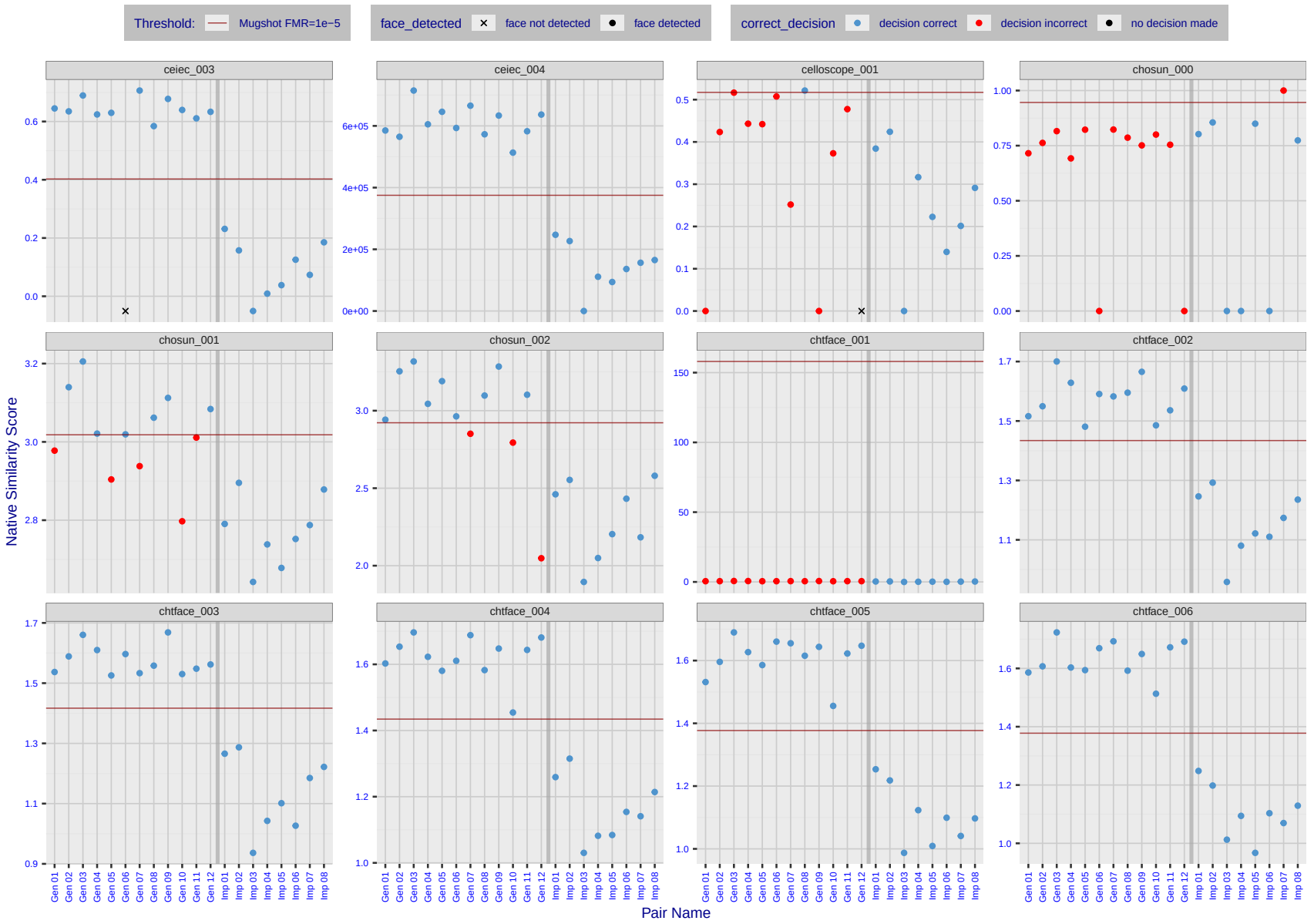

Figure 17: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

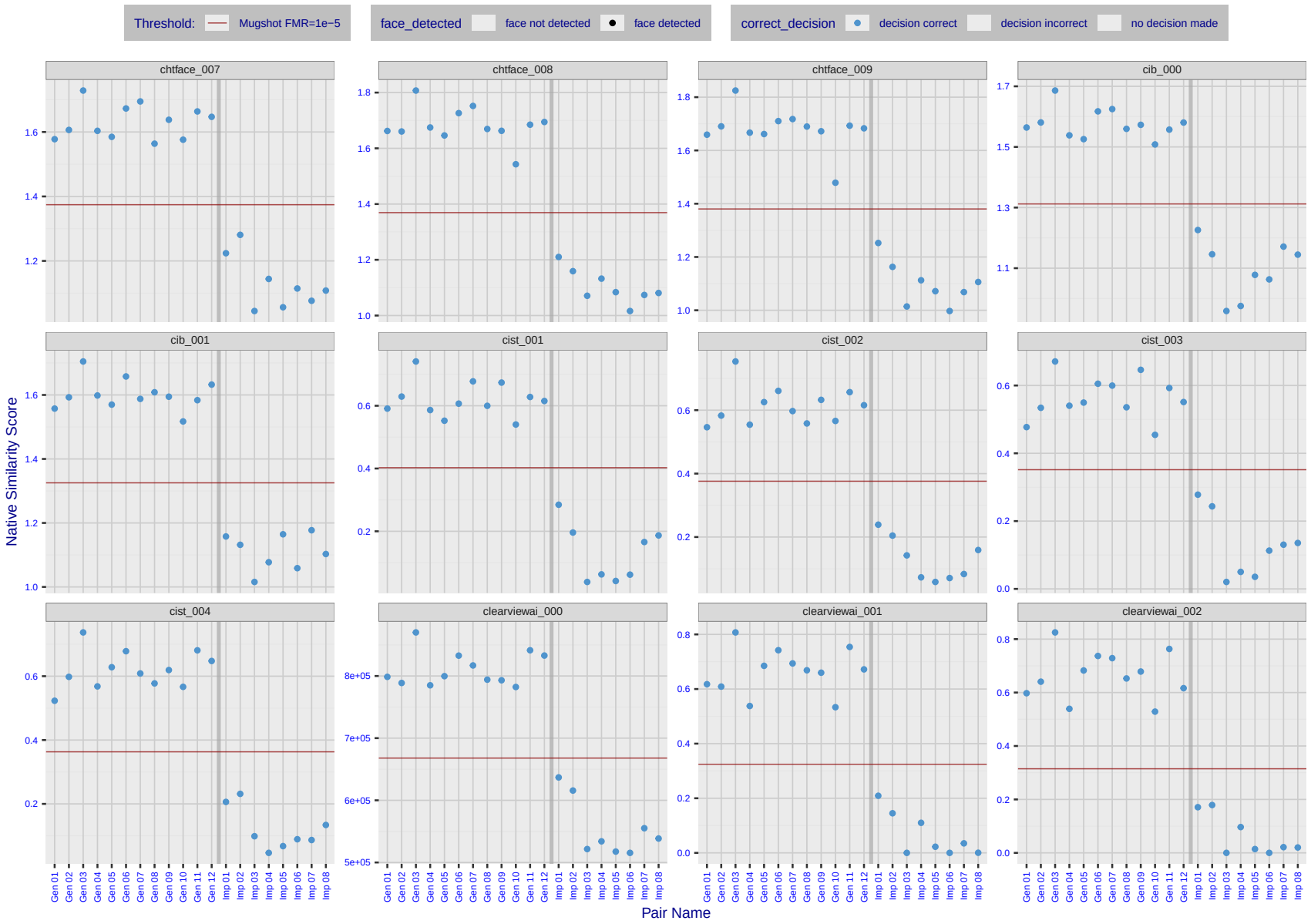

Figure 18: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

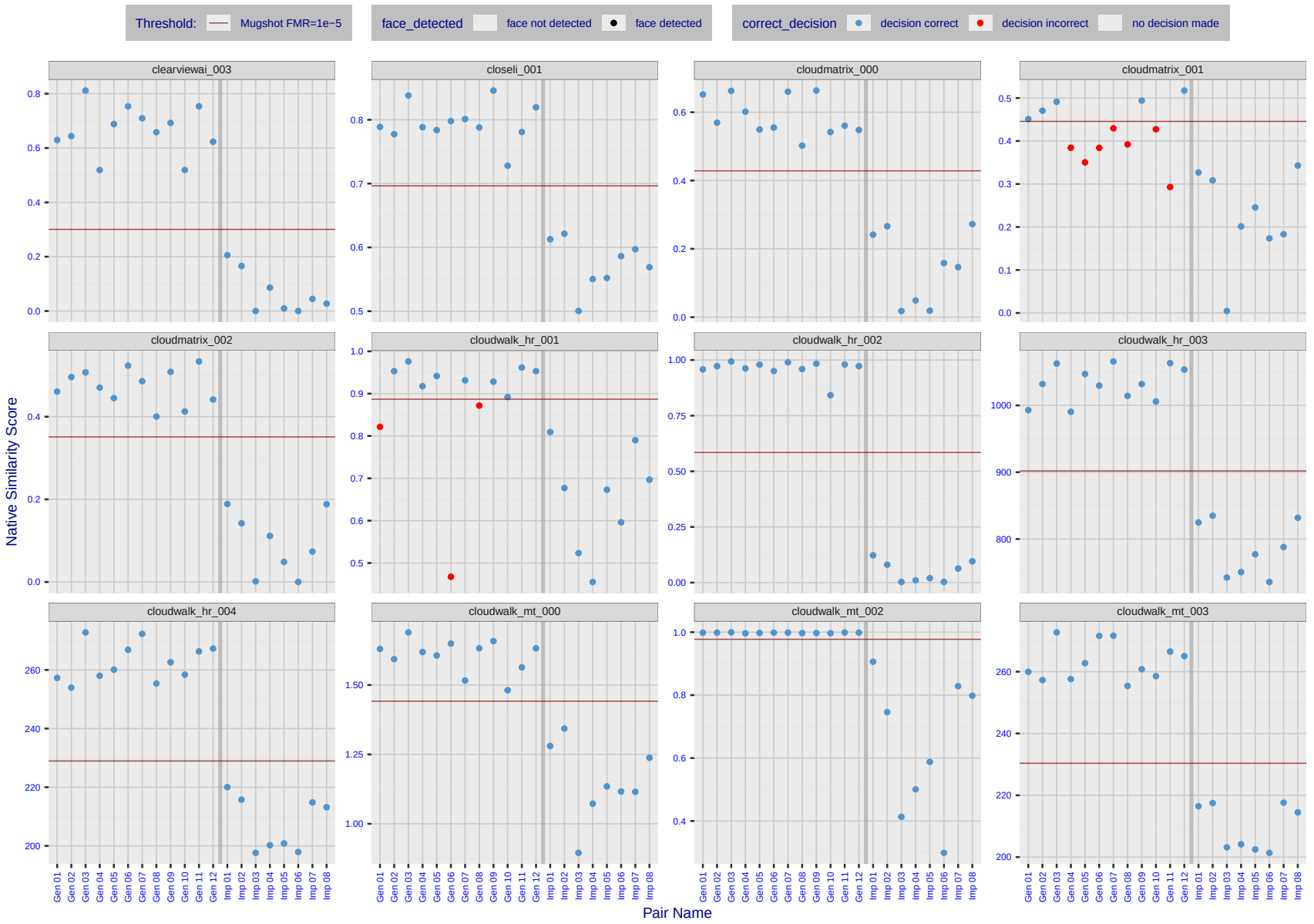

Figure 19: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

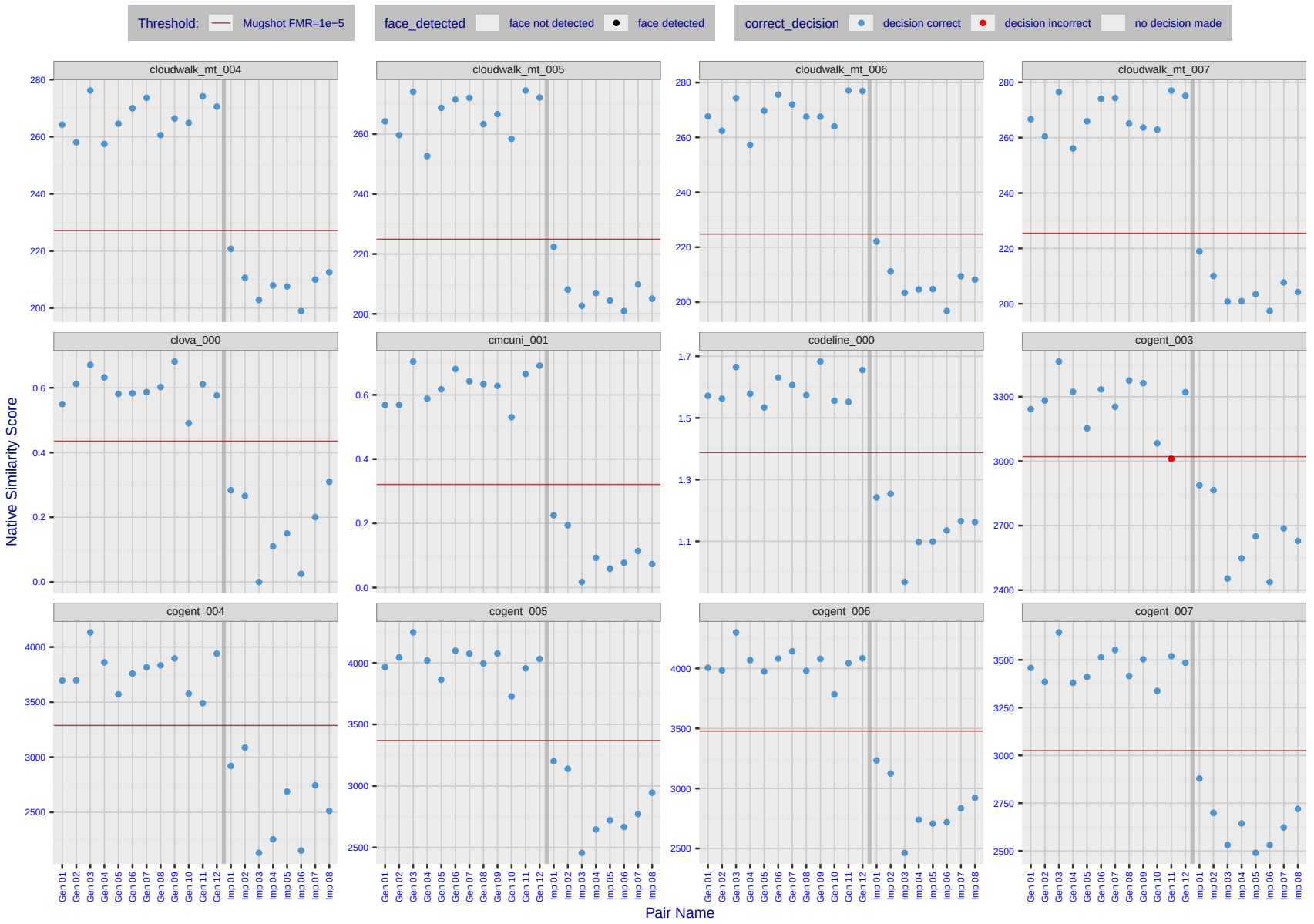

Figure 20: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

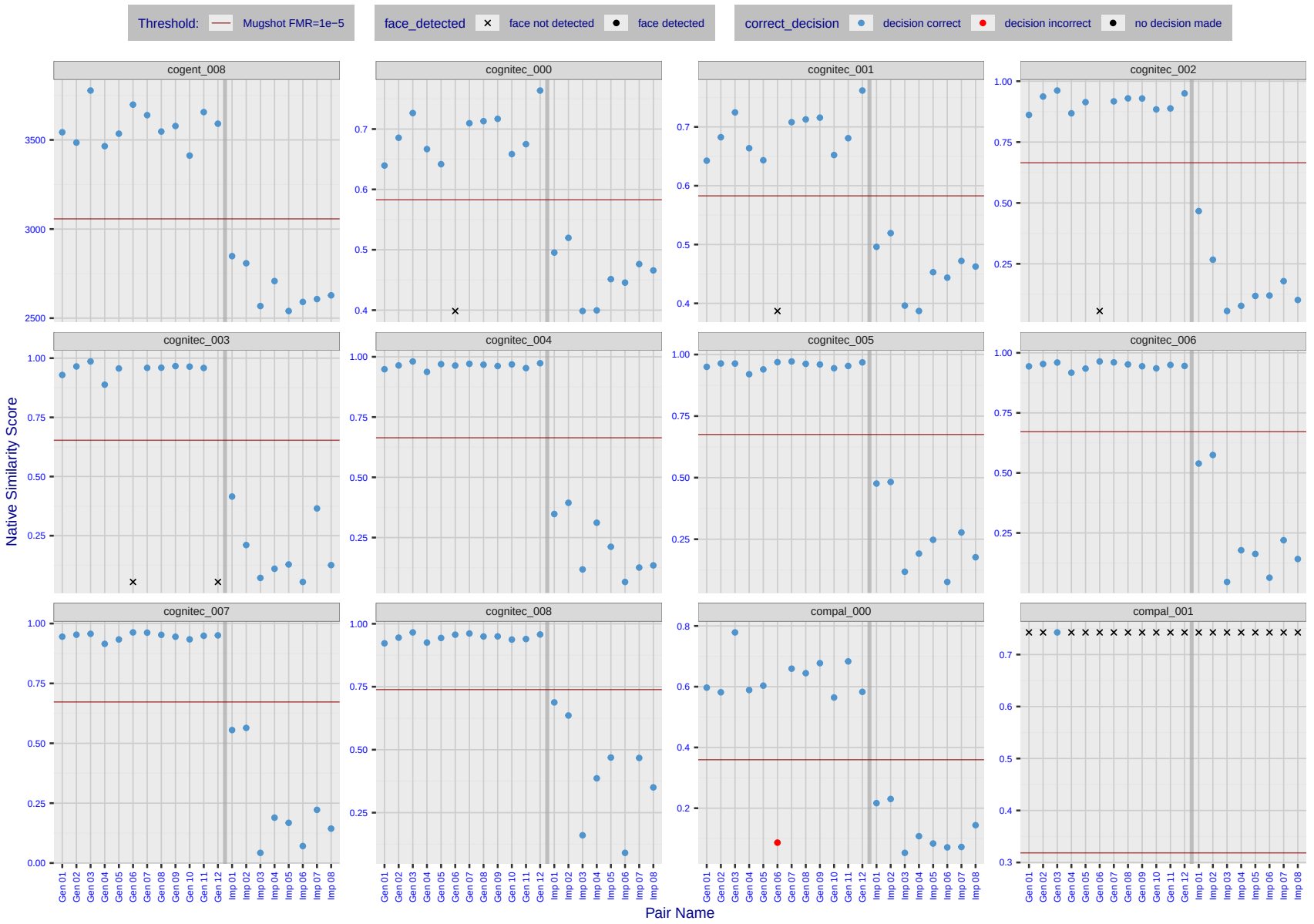

Figure 21: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

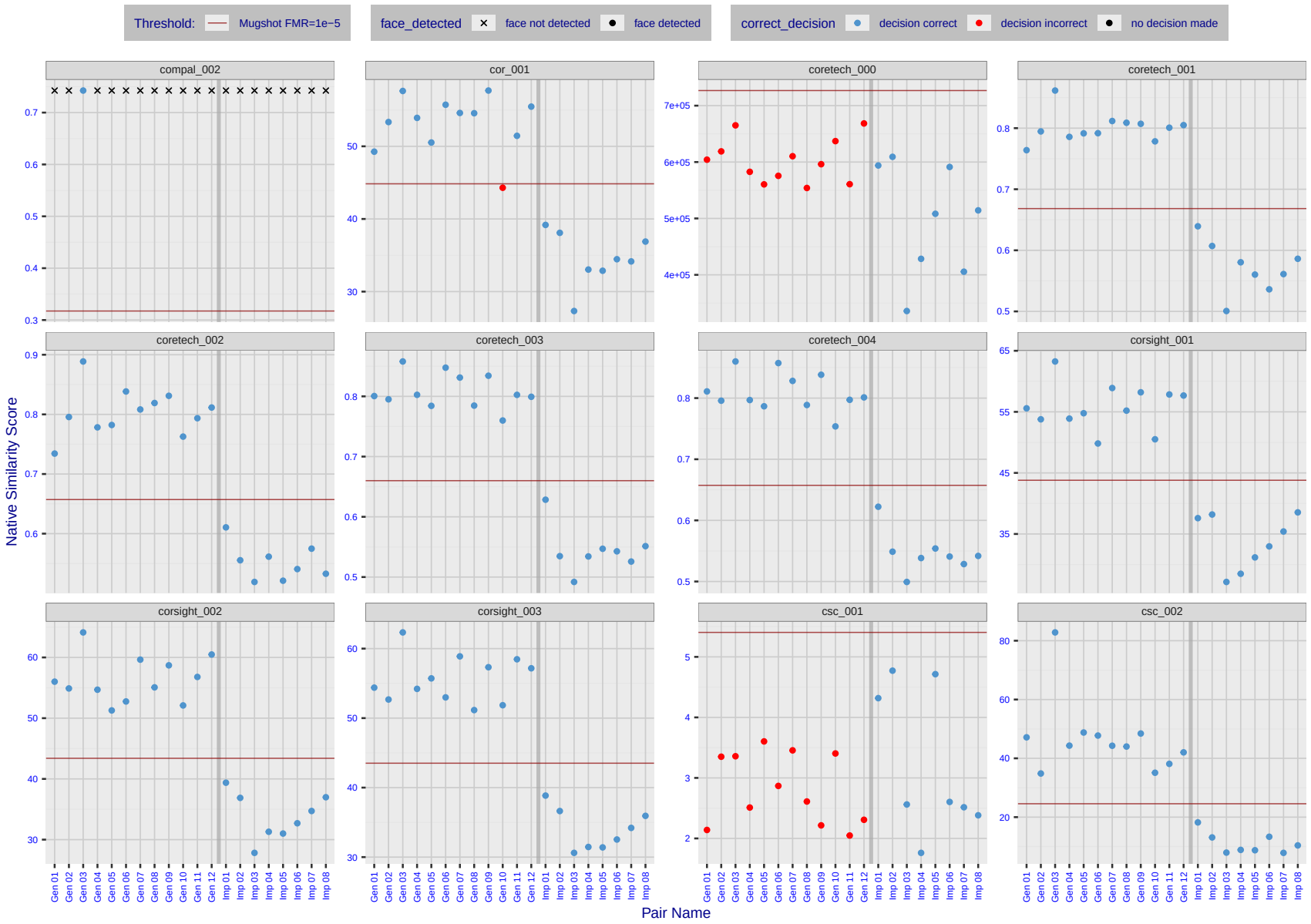

Figure 22: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

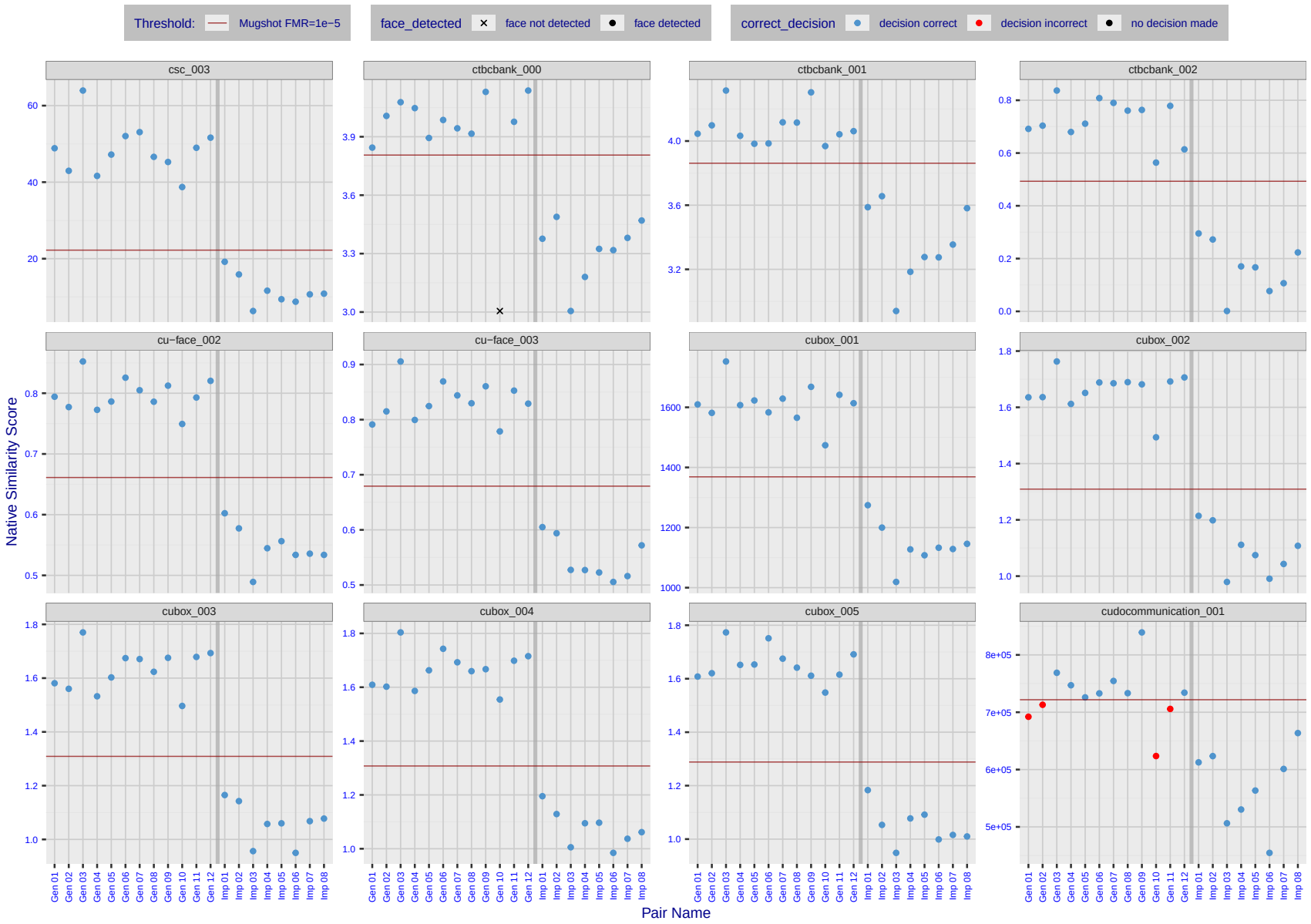

Figure 23: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

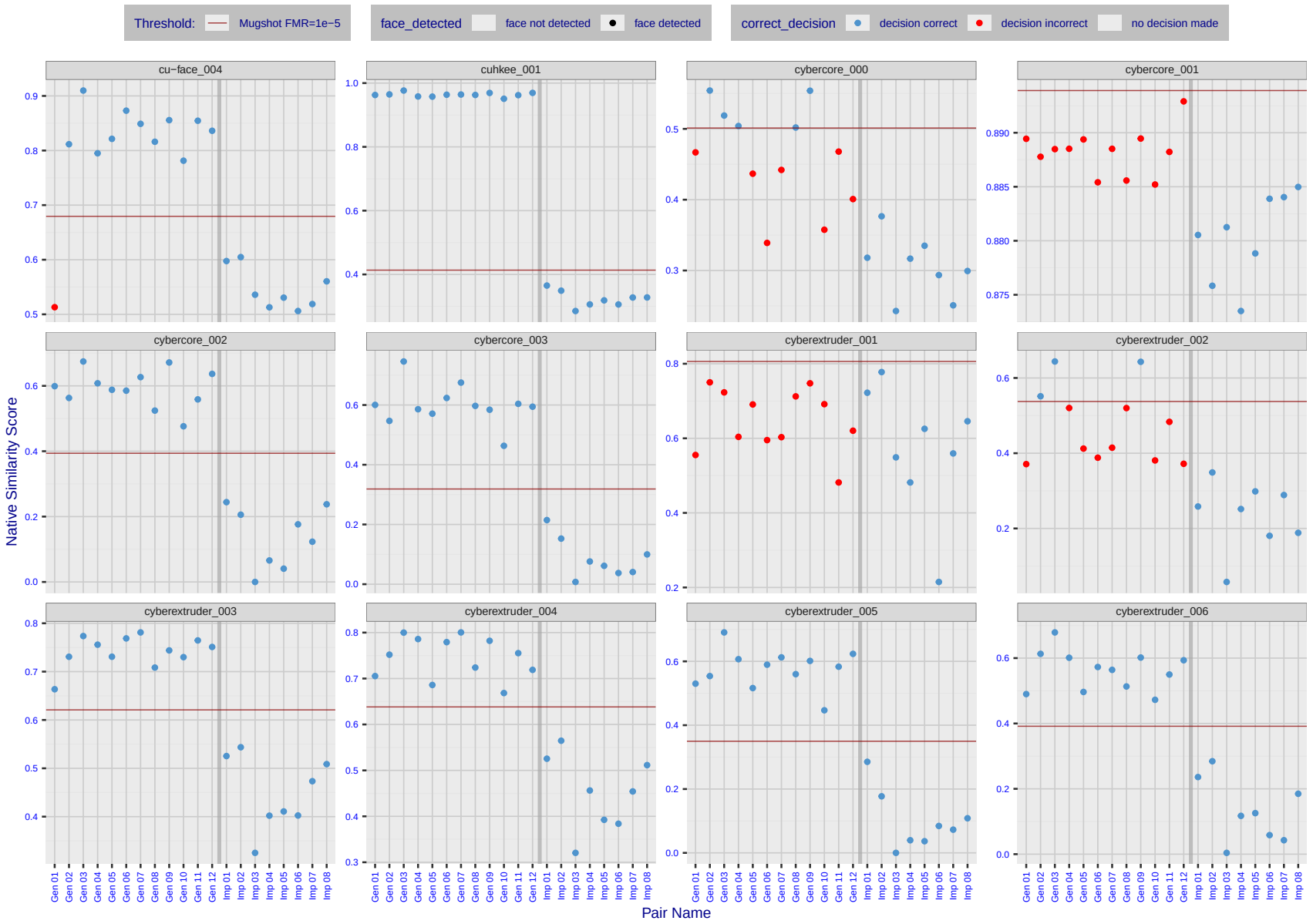

Figure 24: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

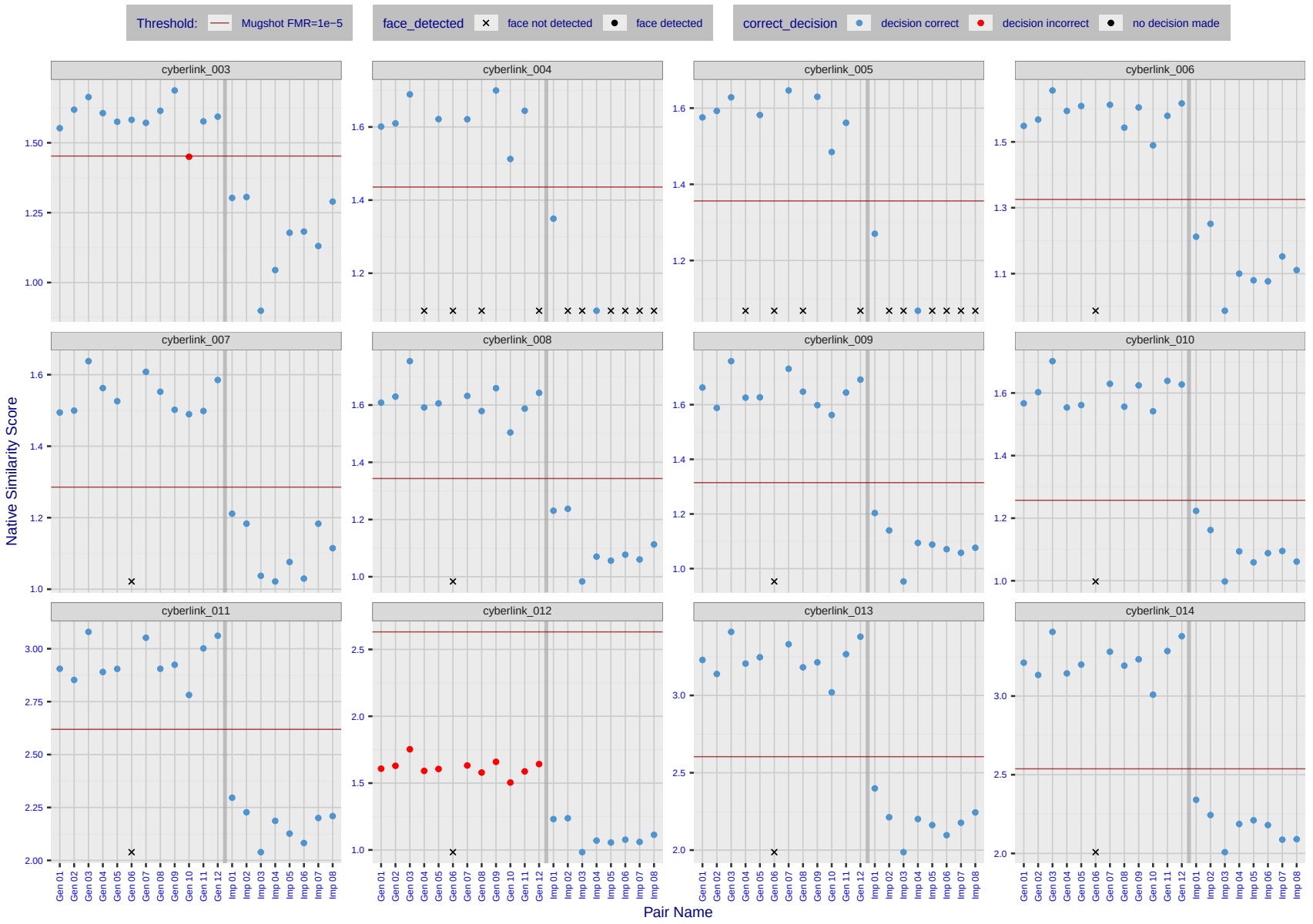

Figure 25: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

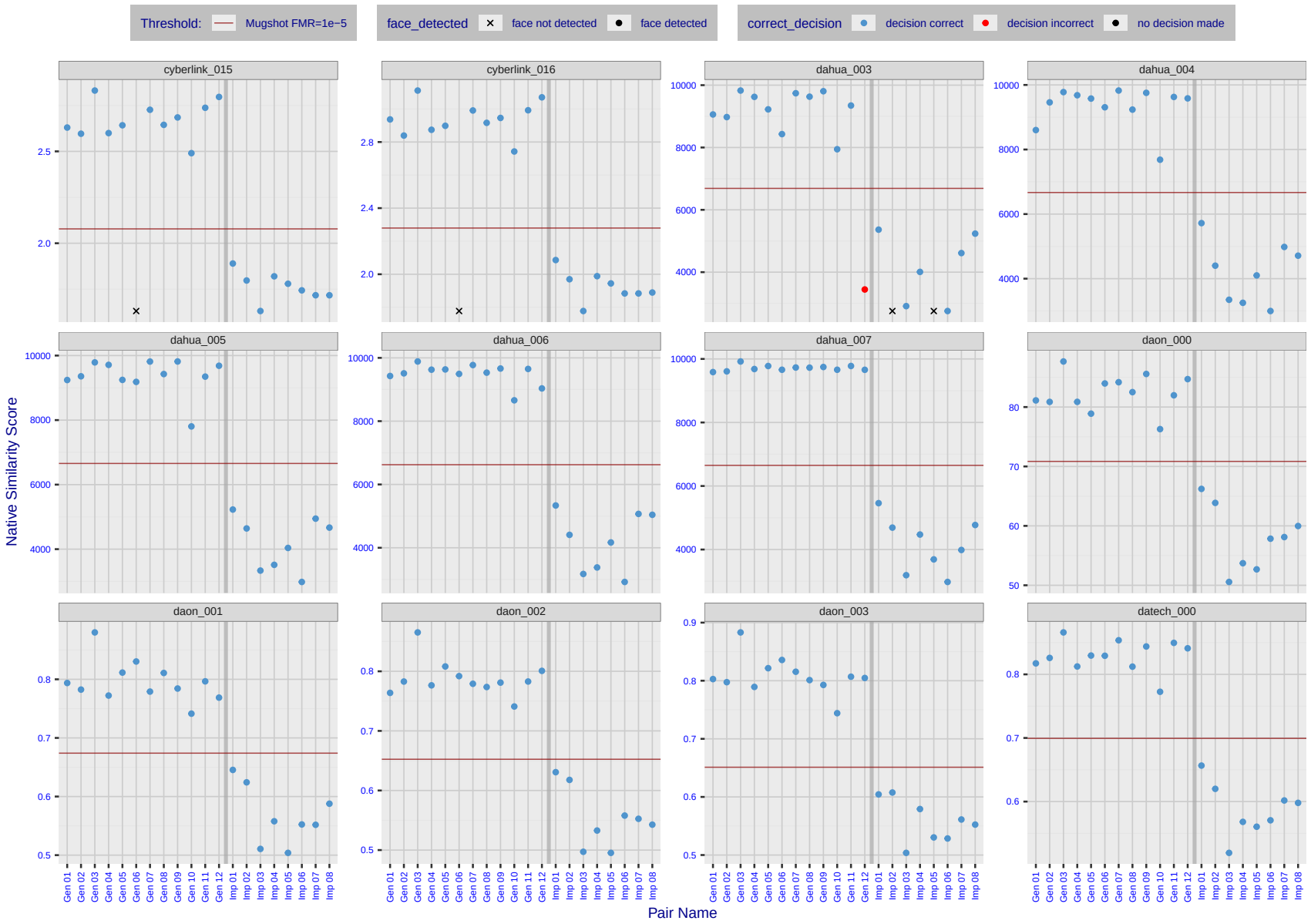

Figure 26: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

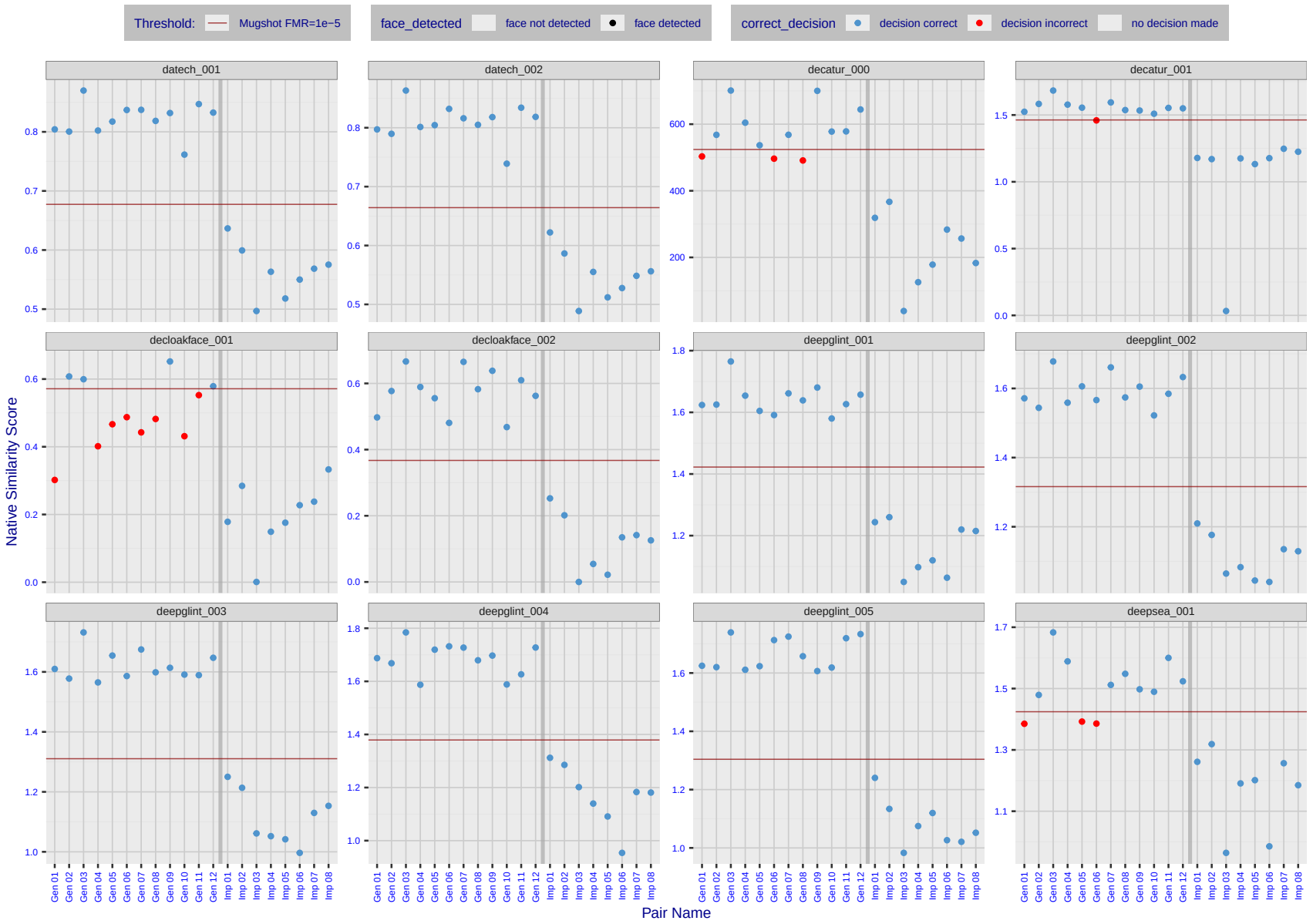

Figure 27: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

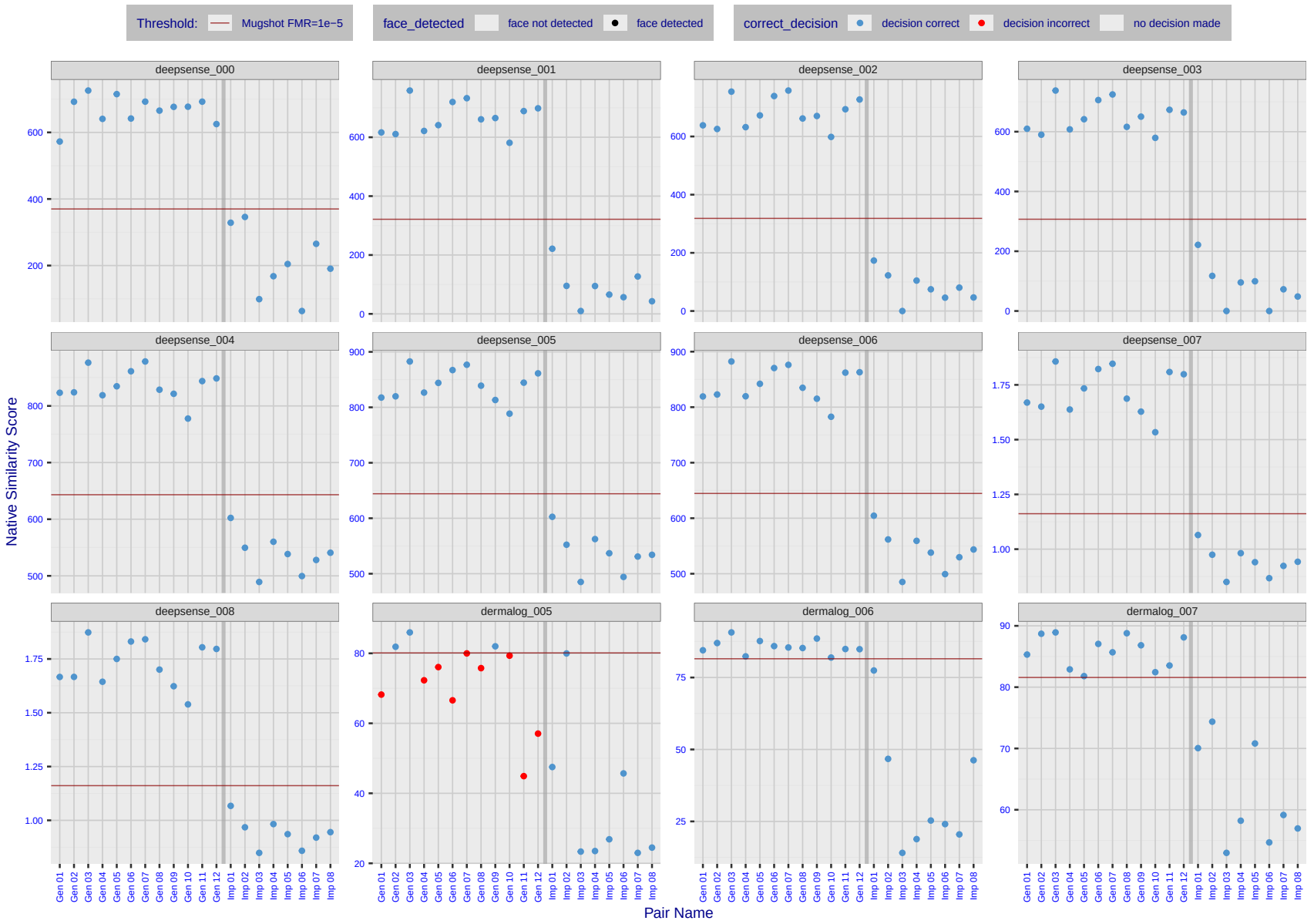

Figure 28: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

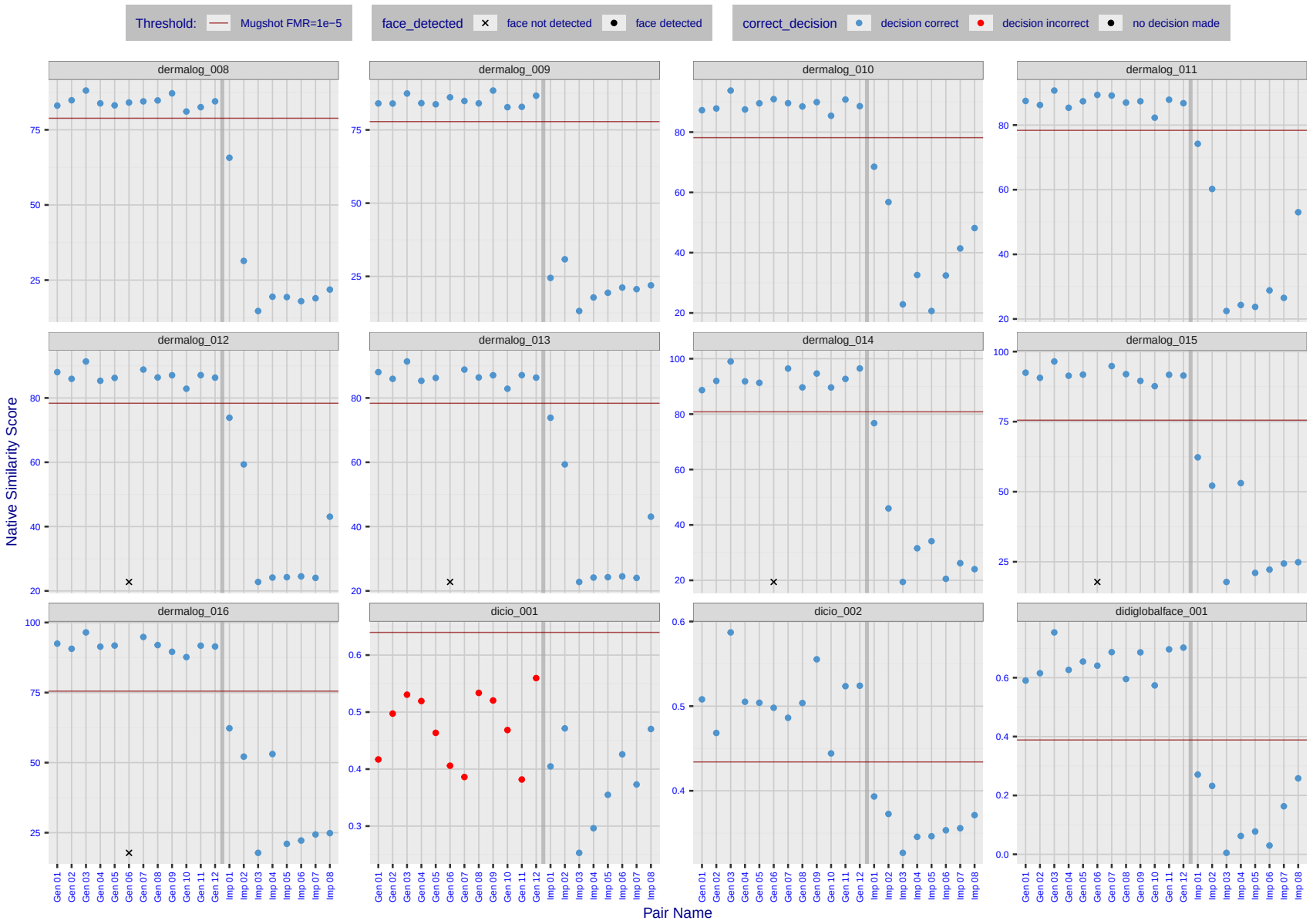

Figure 29: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

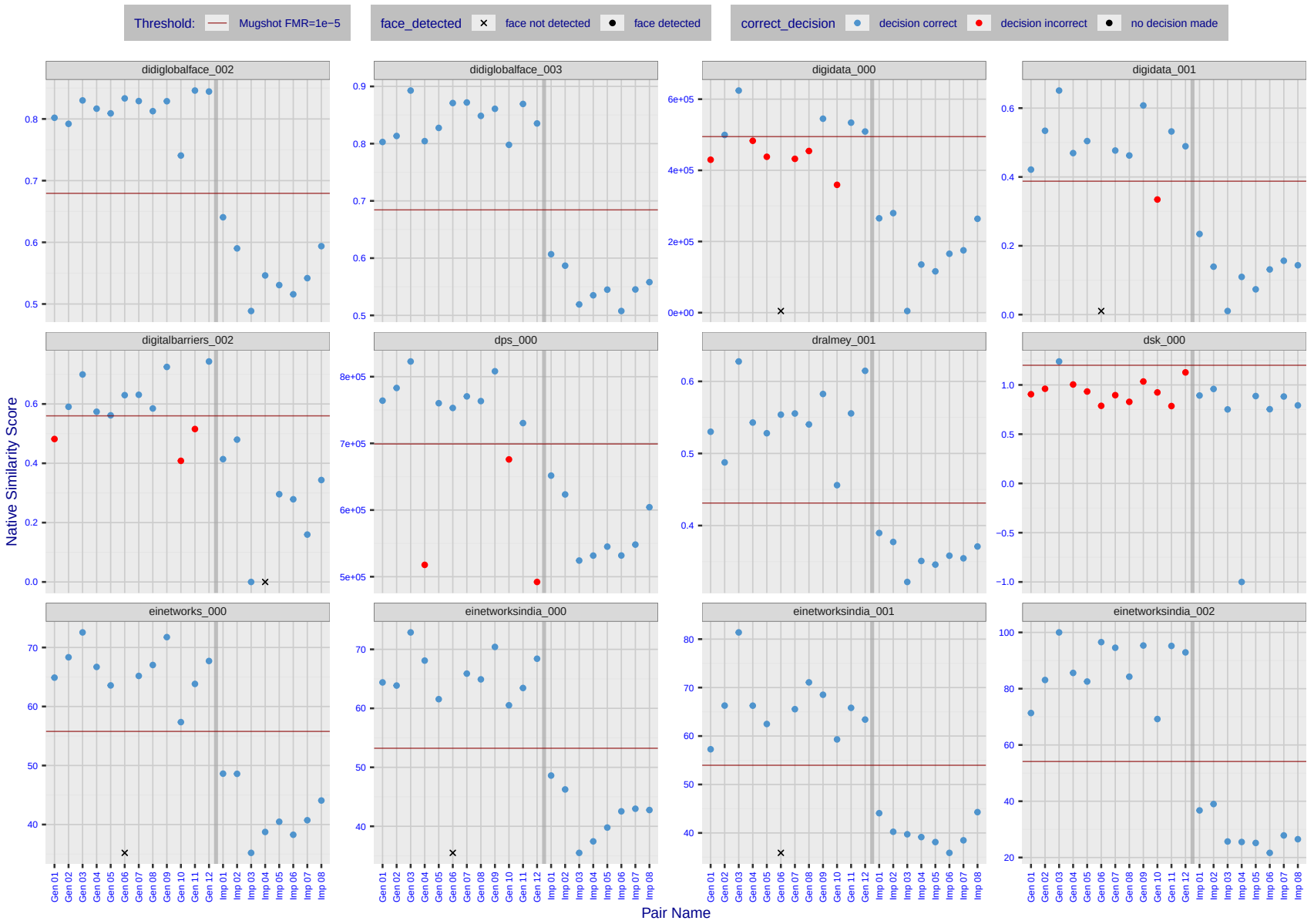

Figure 30: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

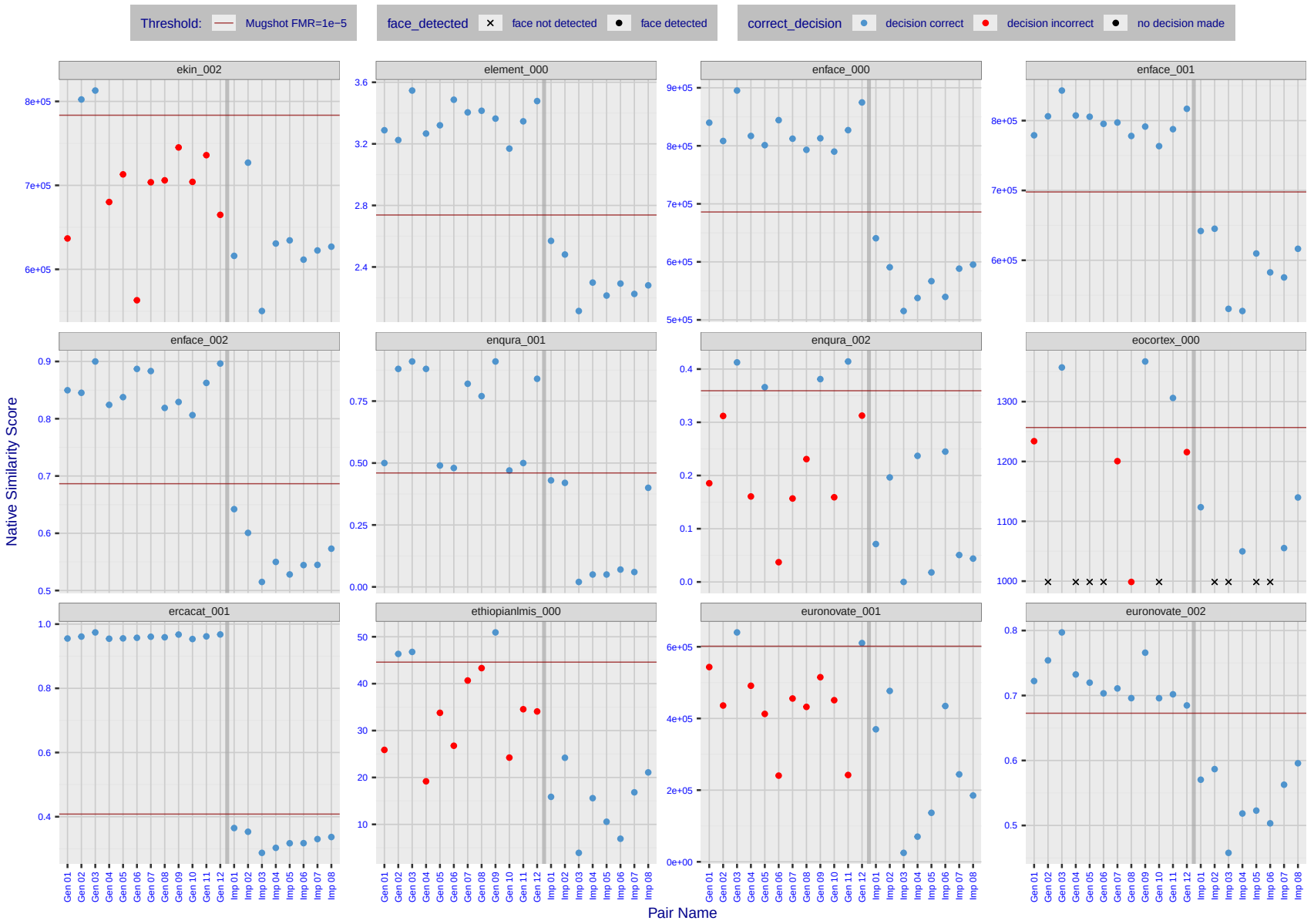

Figure 31: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

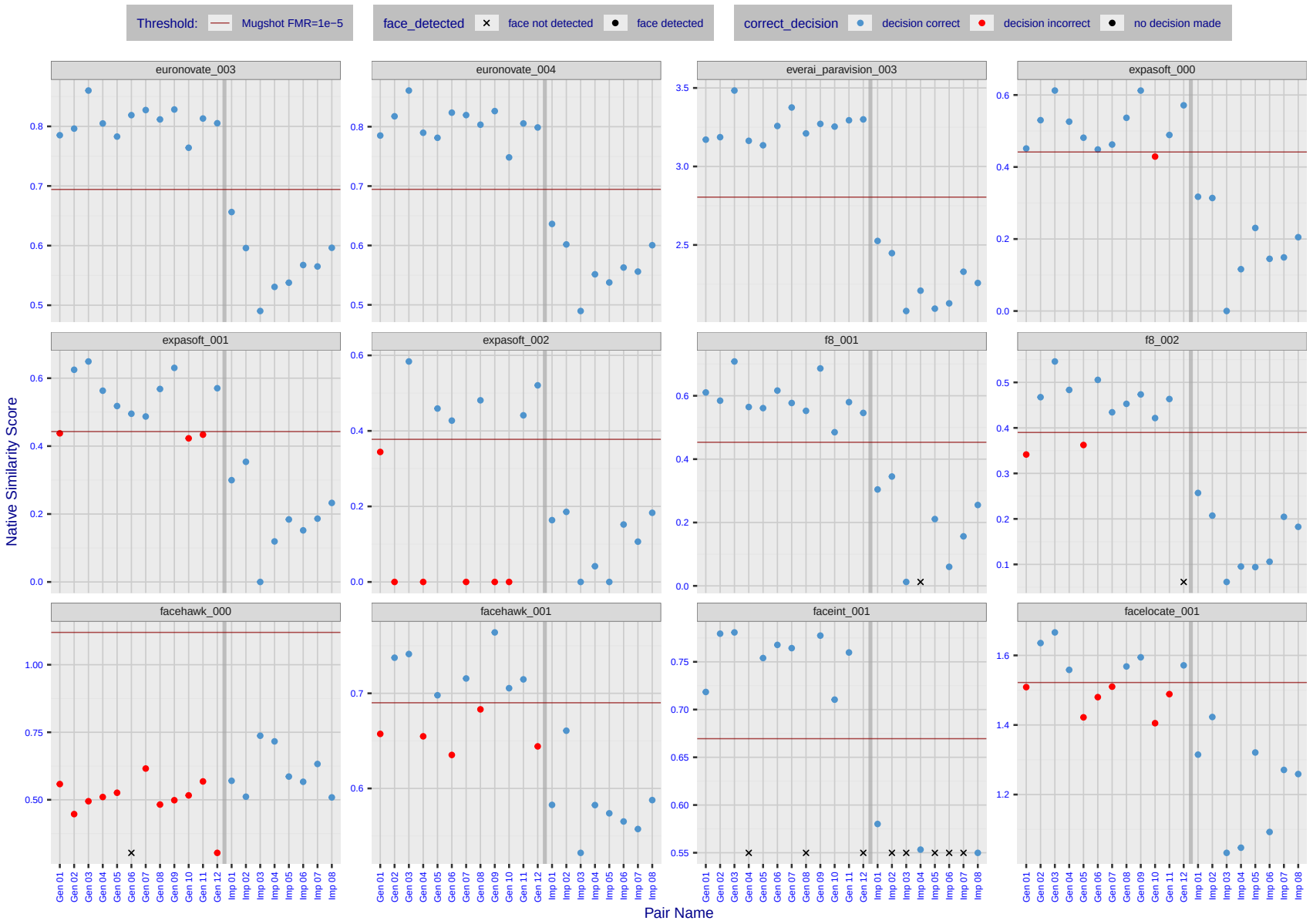

Figure 32: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

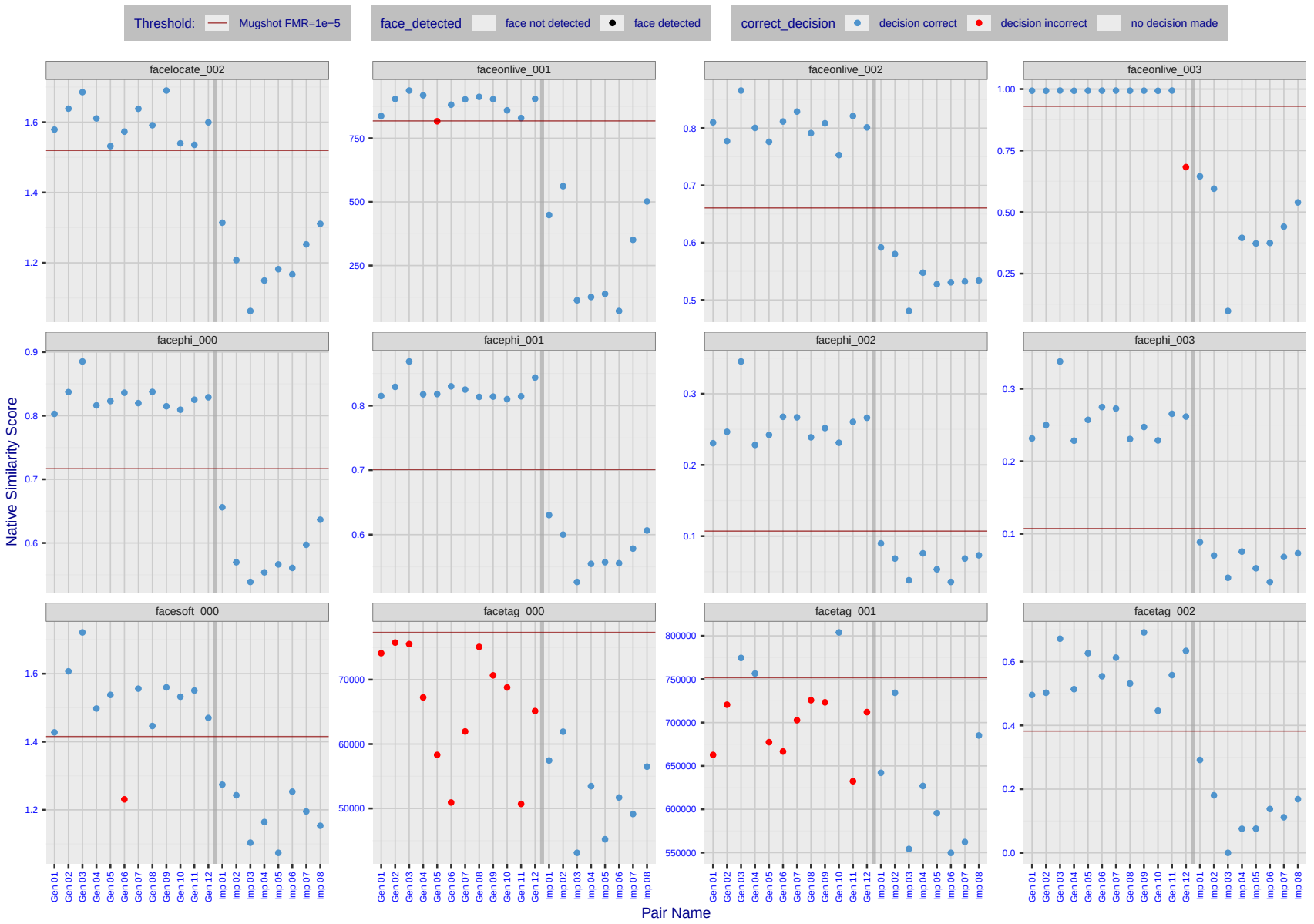

Figure 33: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

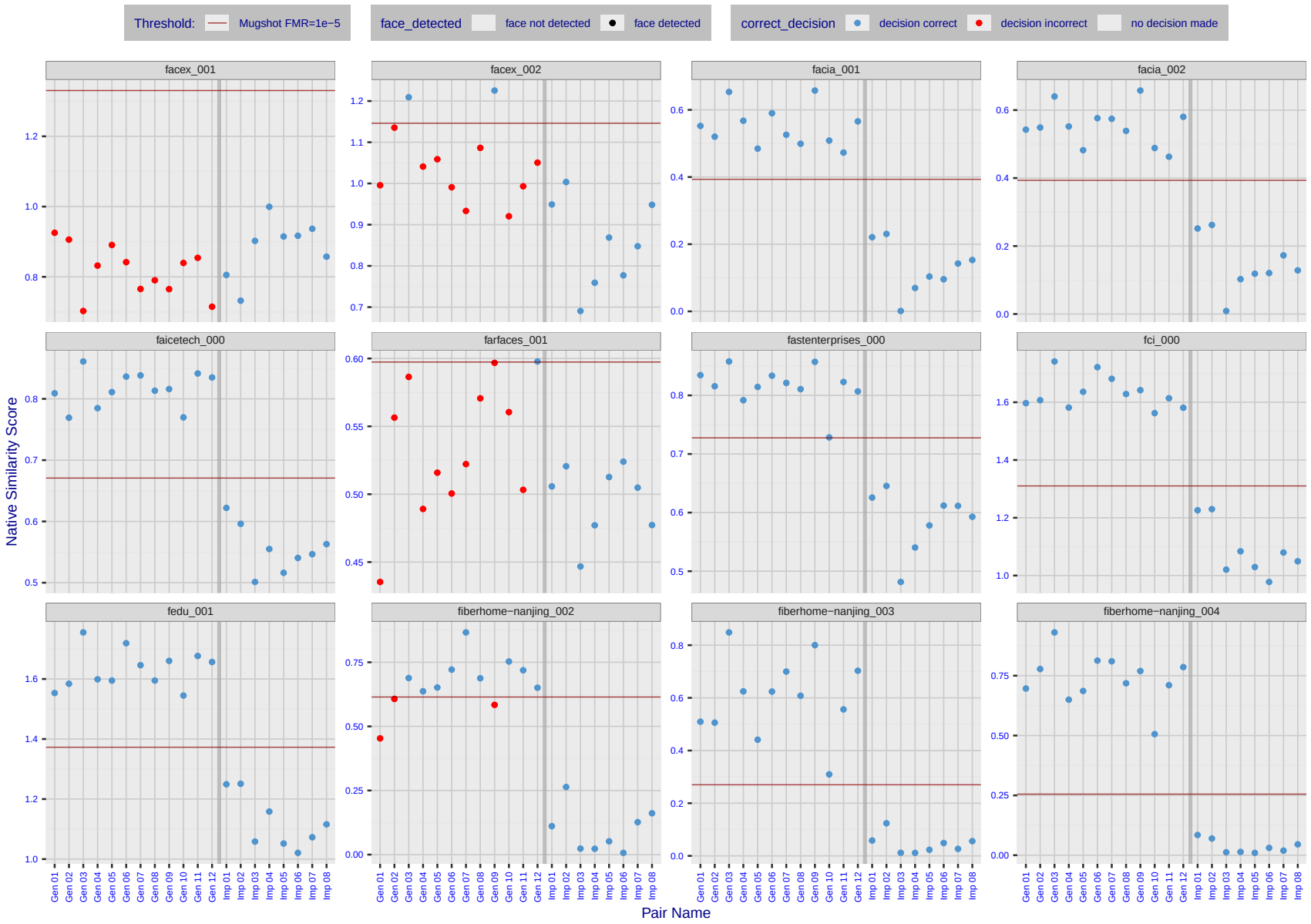

Figure 34: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

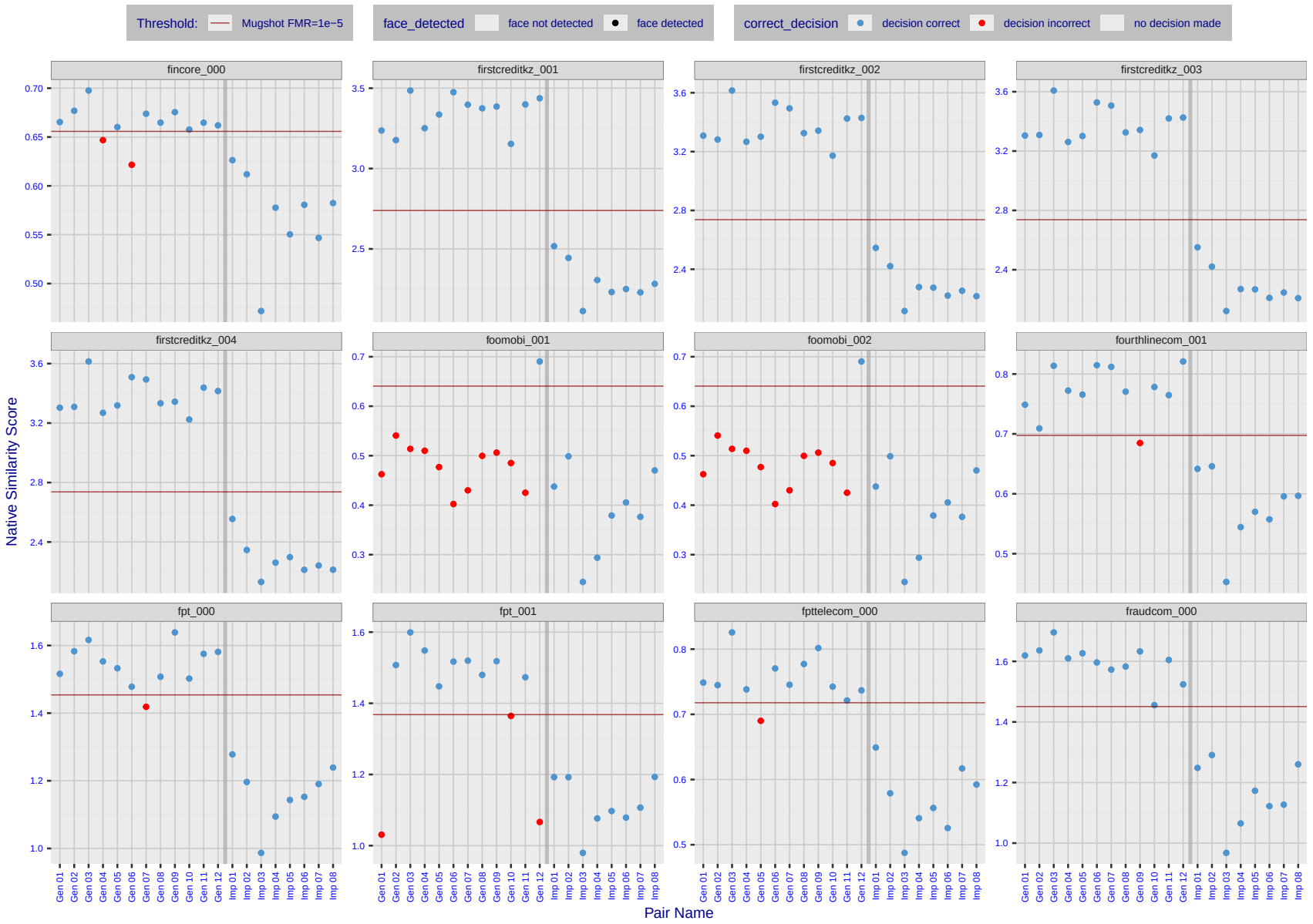

Figure 35: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

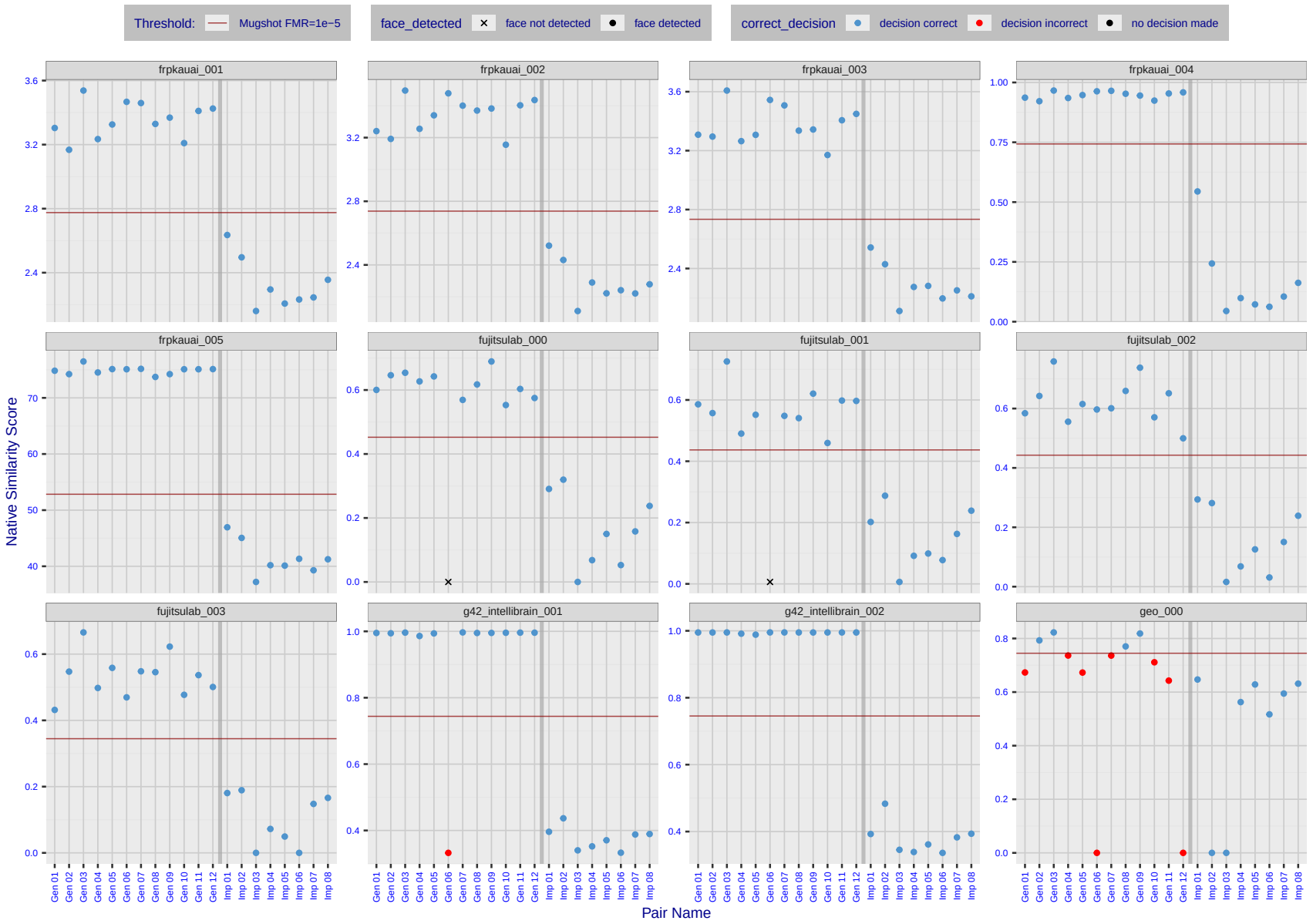

Figure 36: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

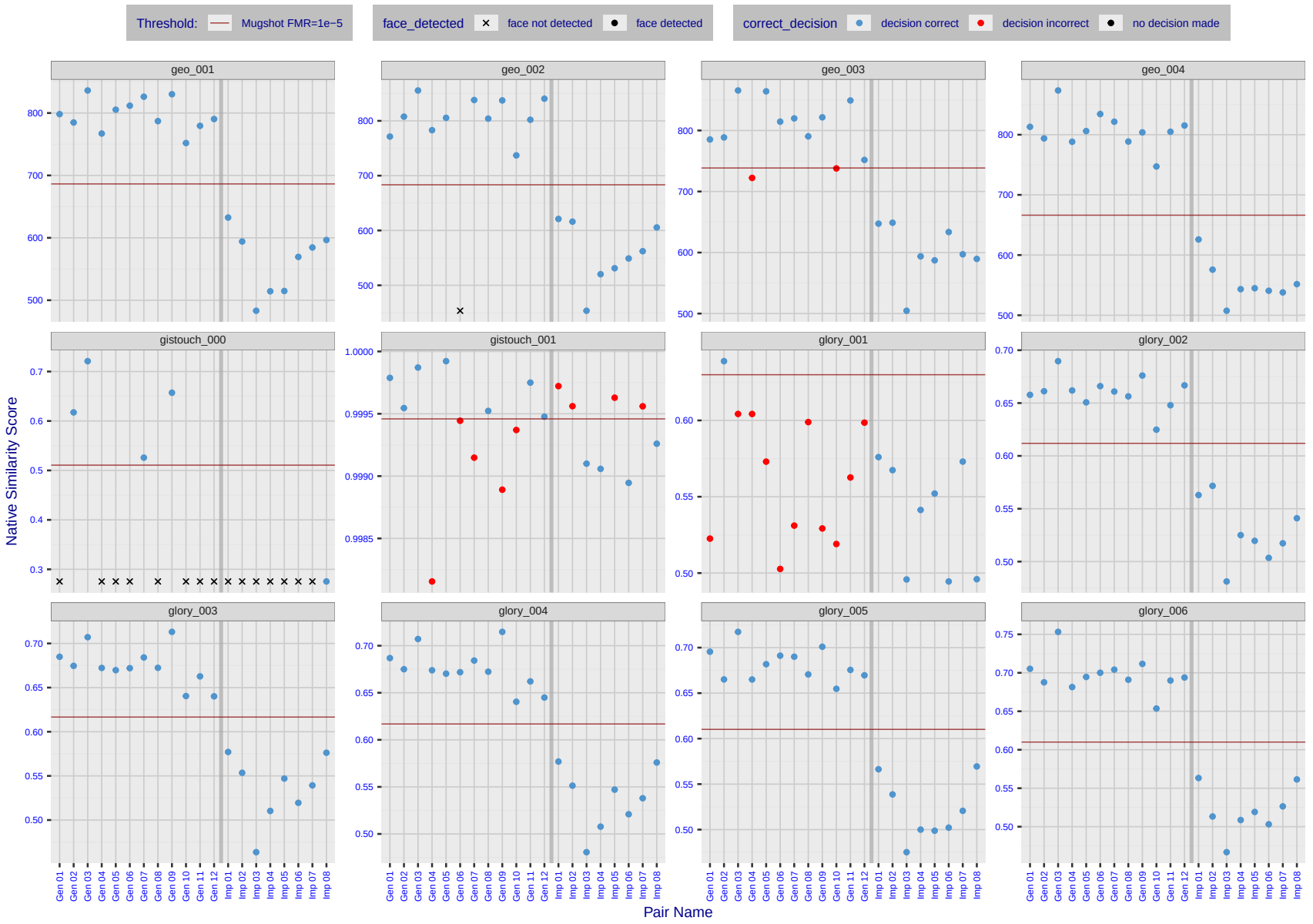

Figure 37: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

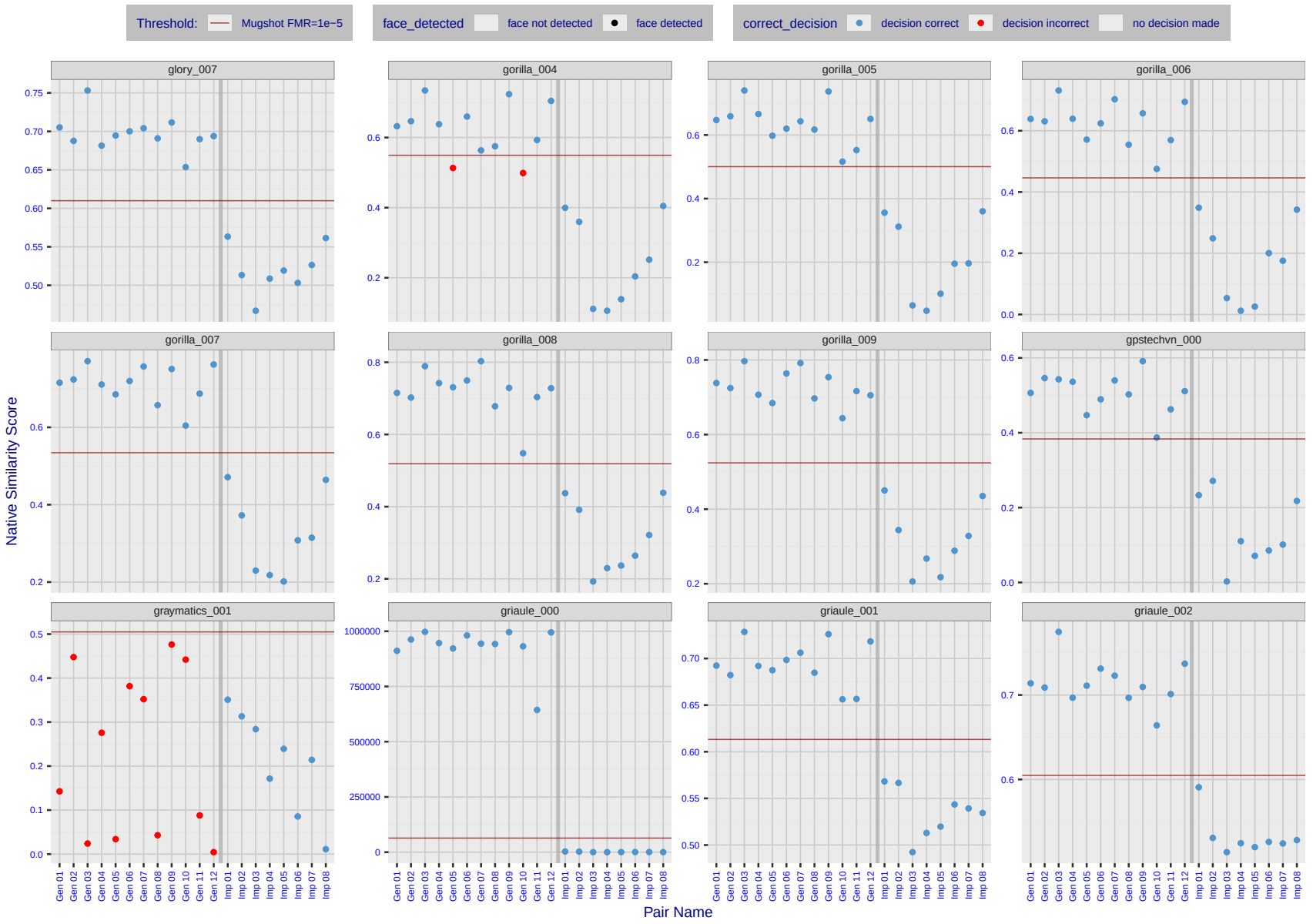

Figure 38: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

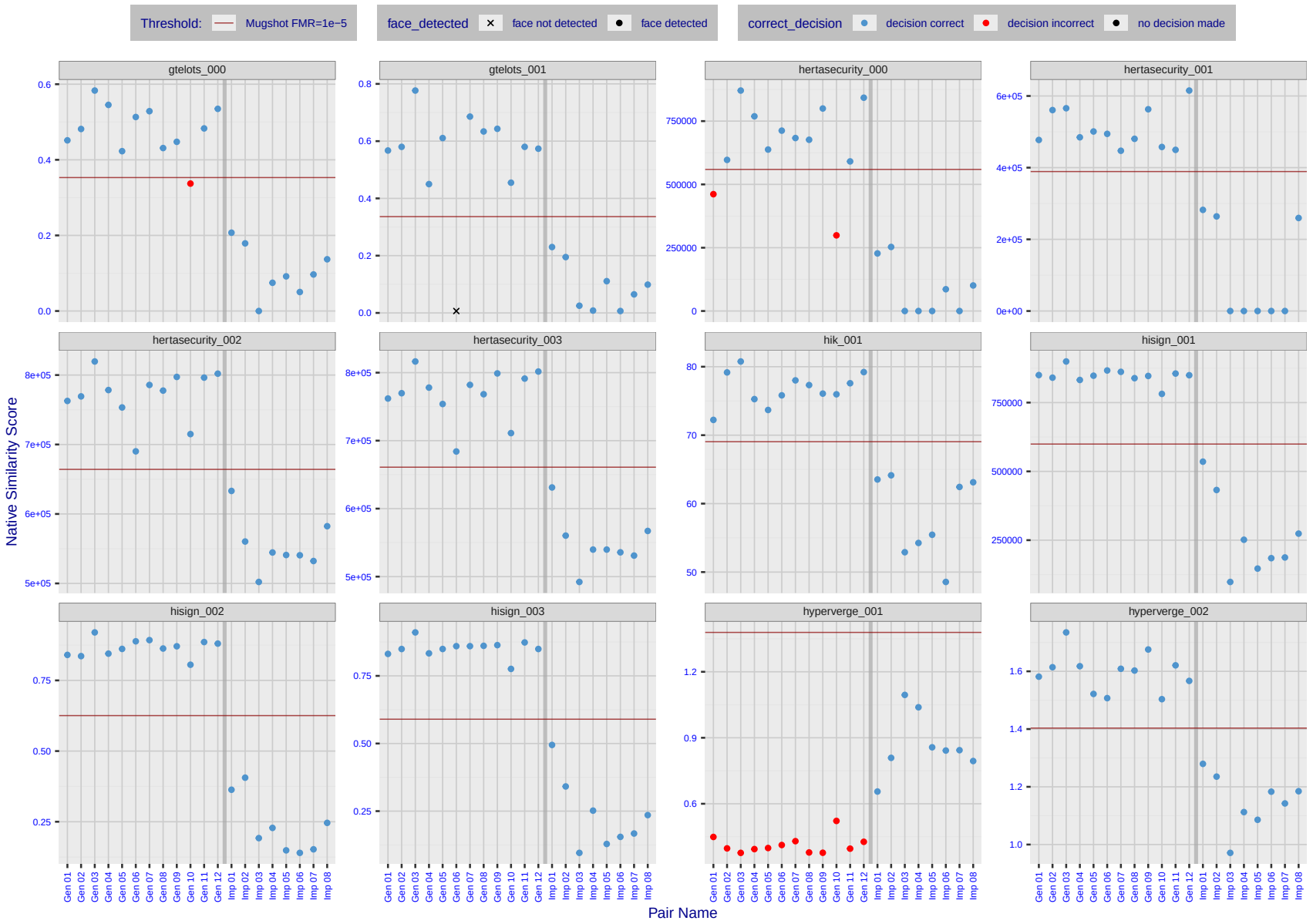

Figure 39: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

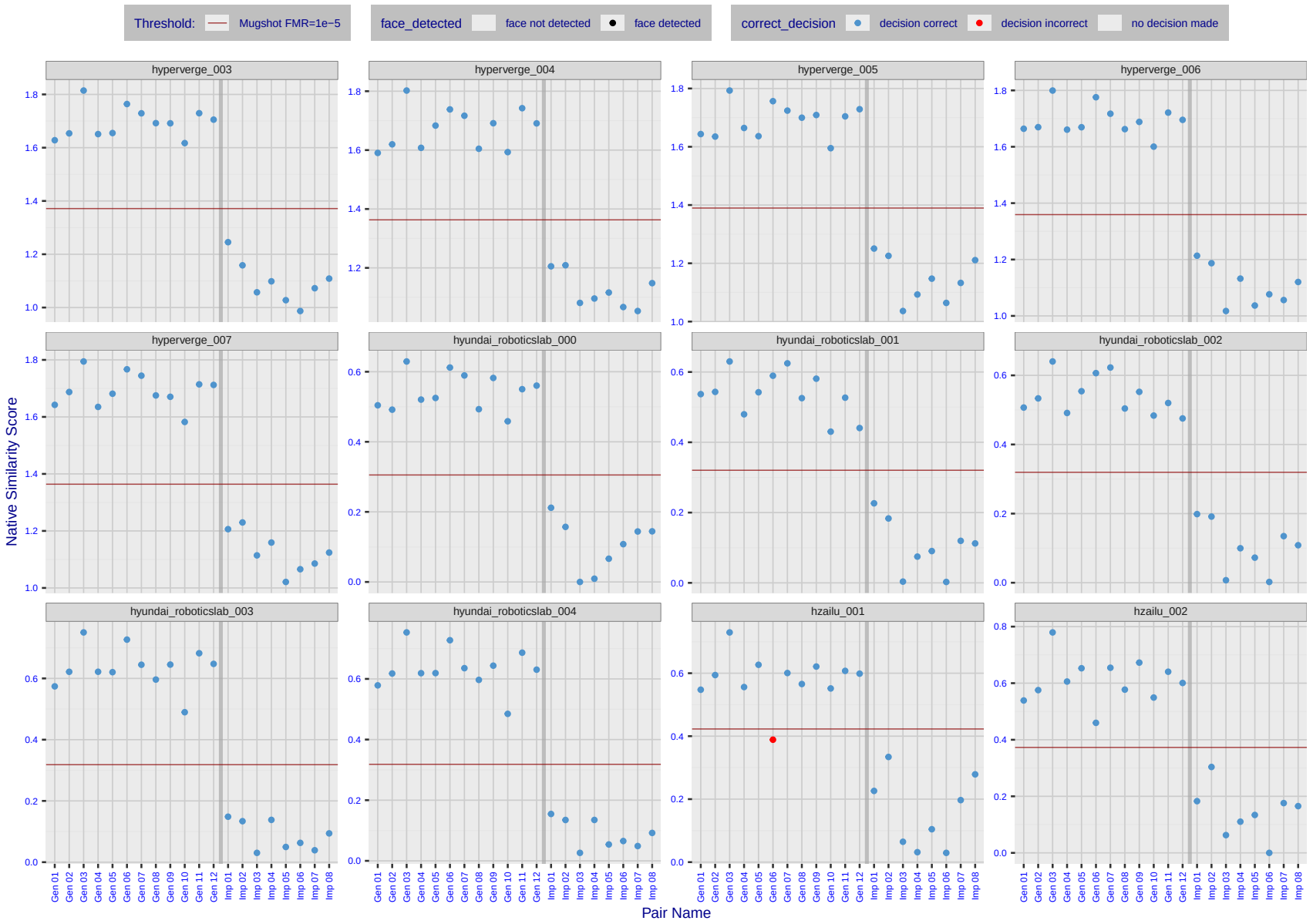

Figure 40: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

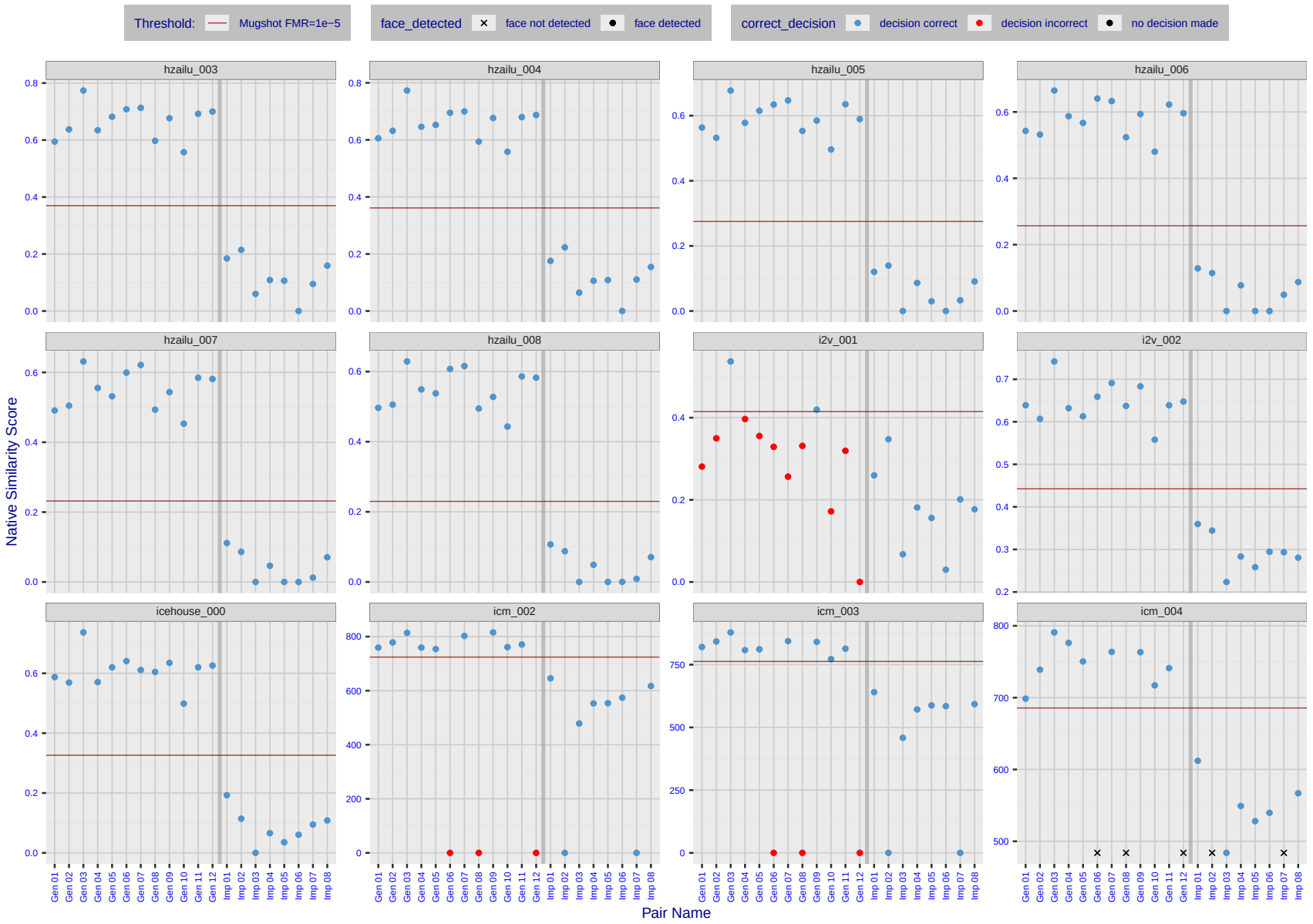

Figure 41: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

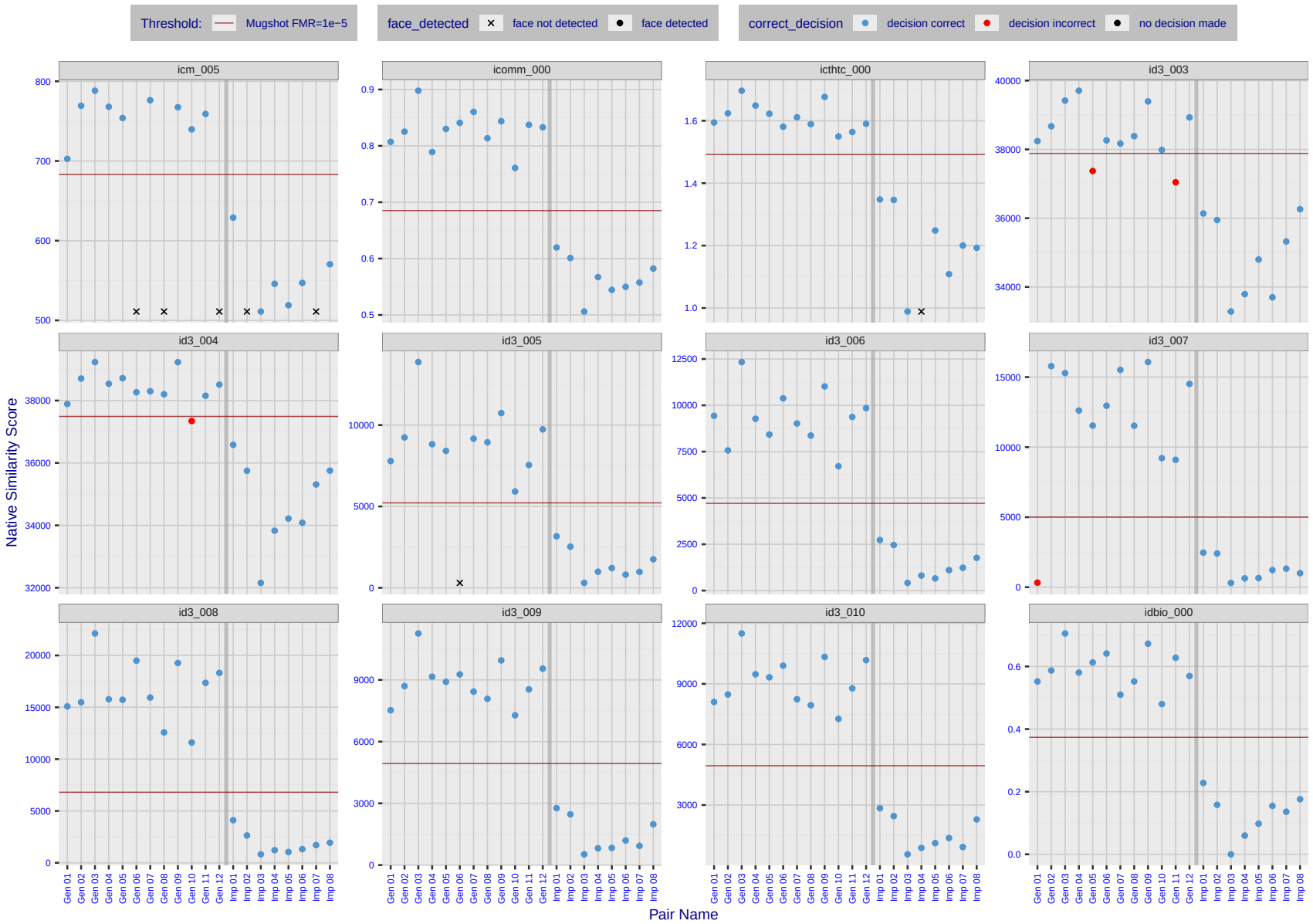

Figure 42: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

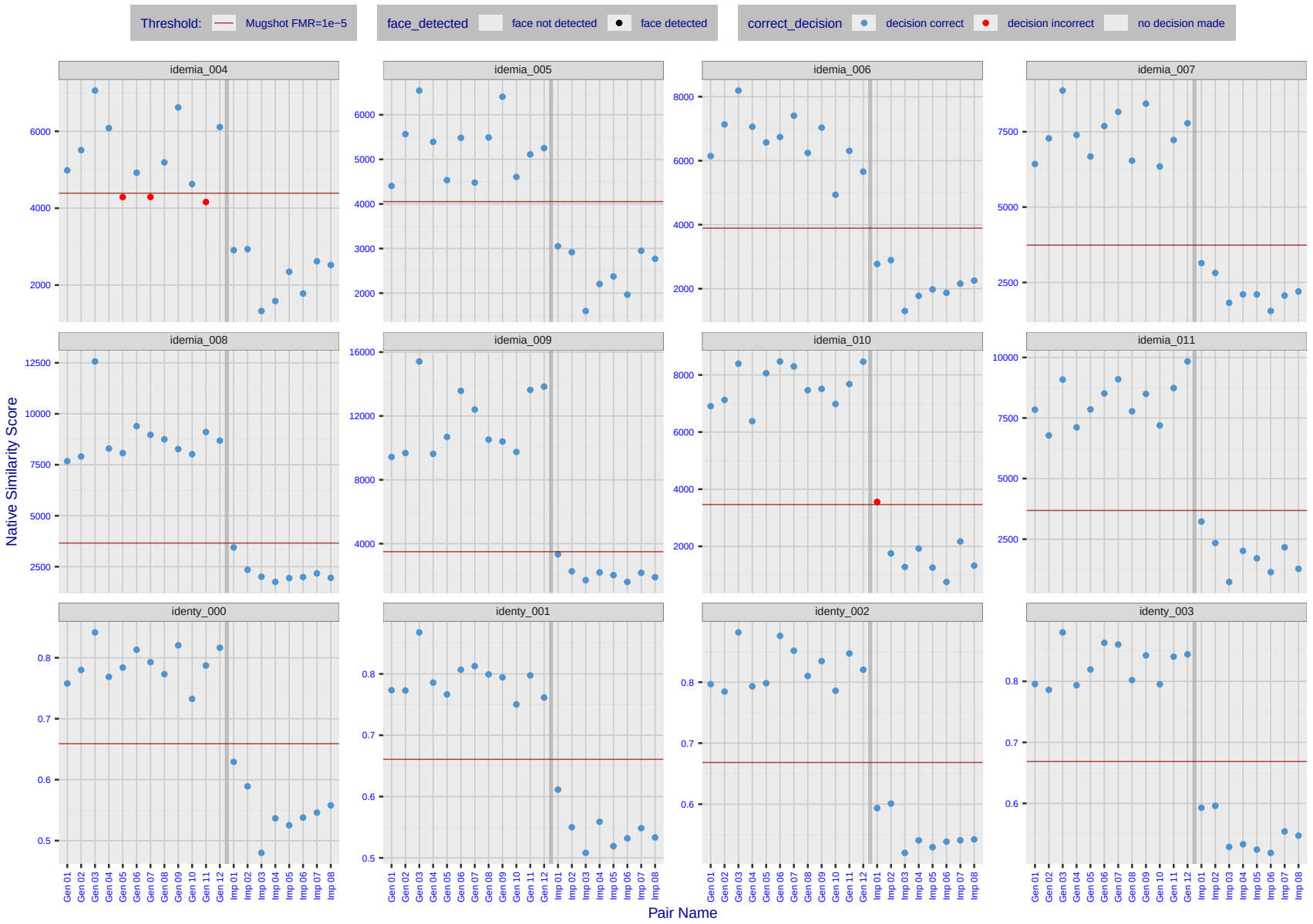

Figure 43: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

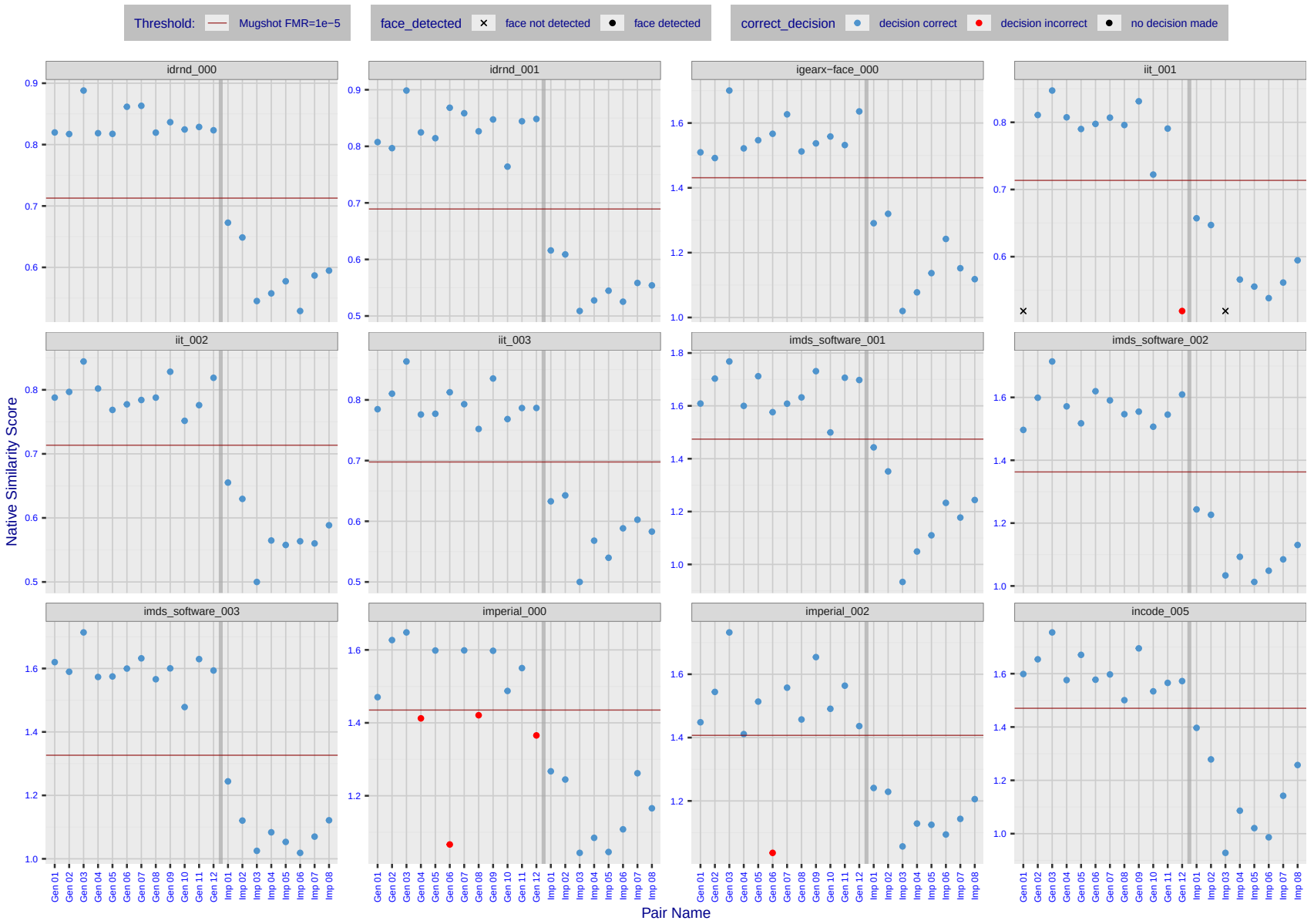

Figure 44: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

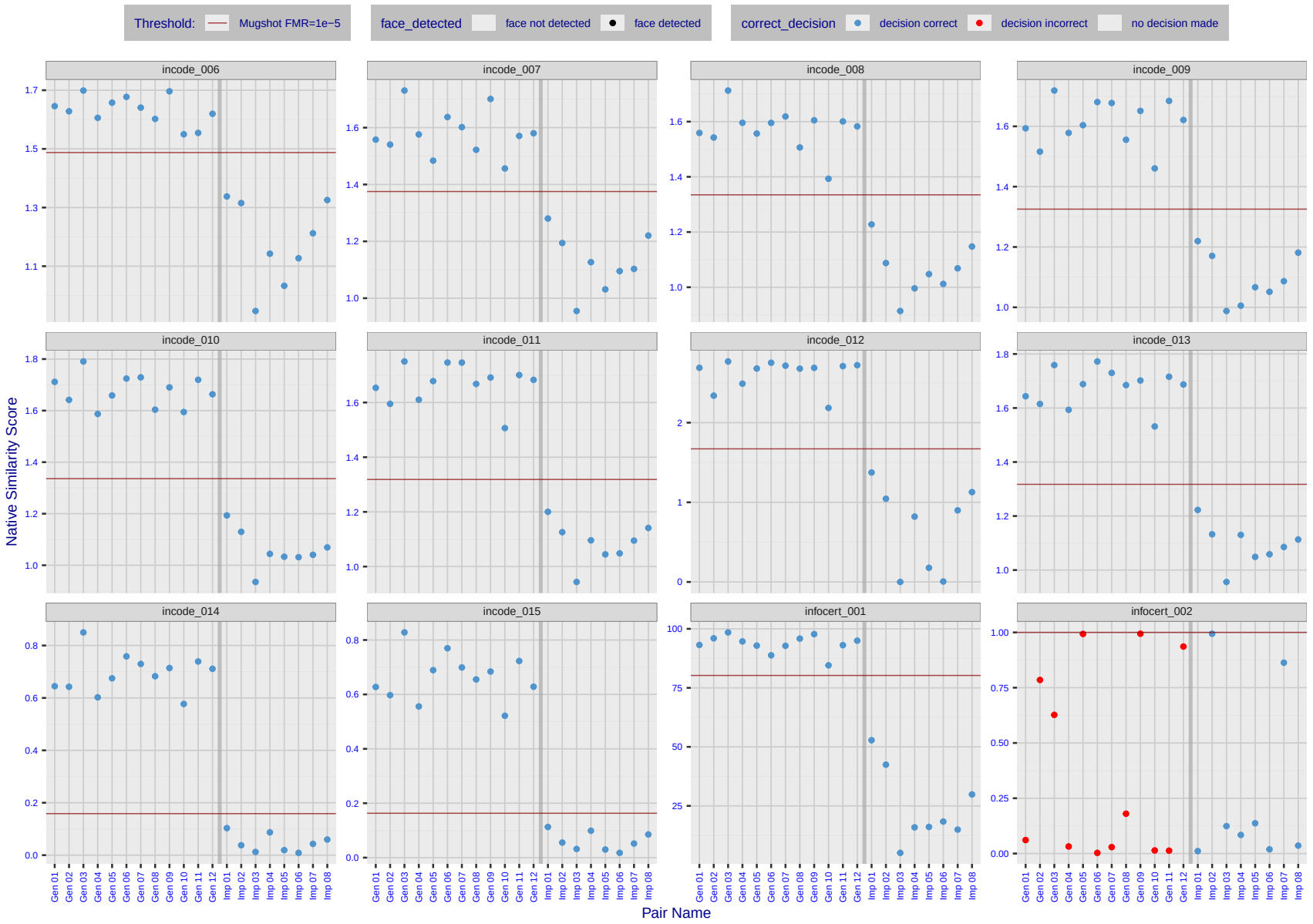

Figure 45: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

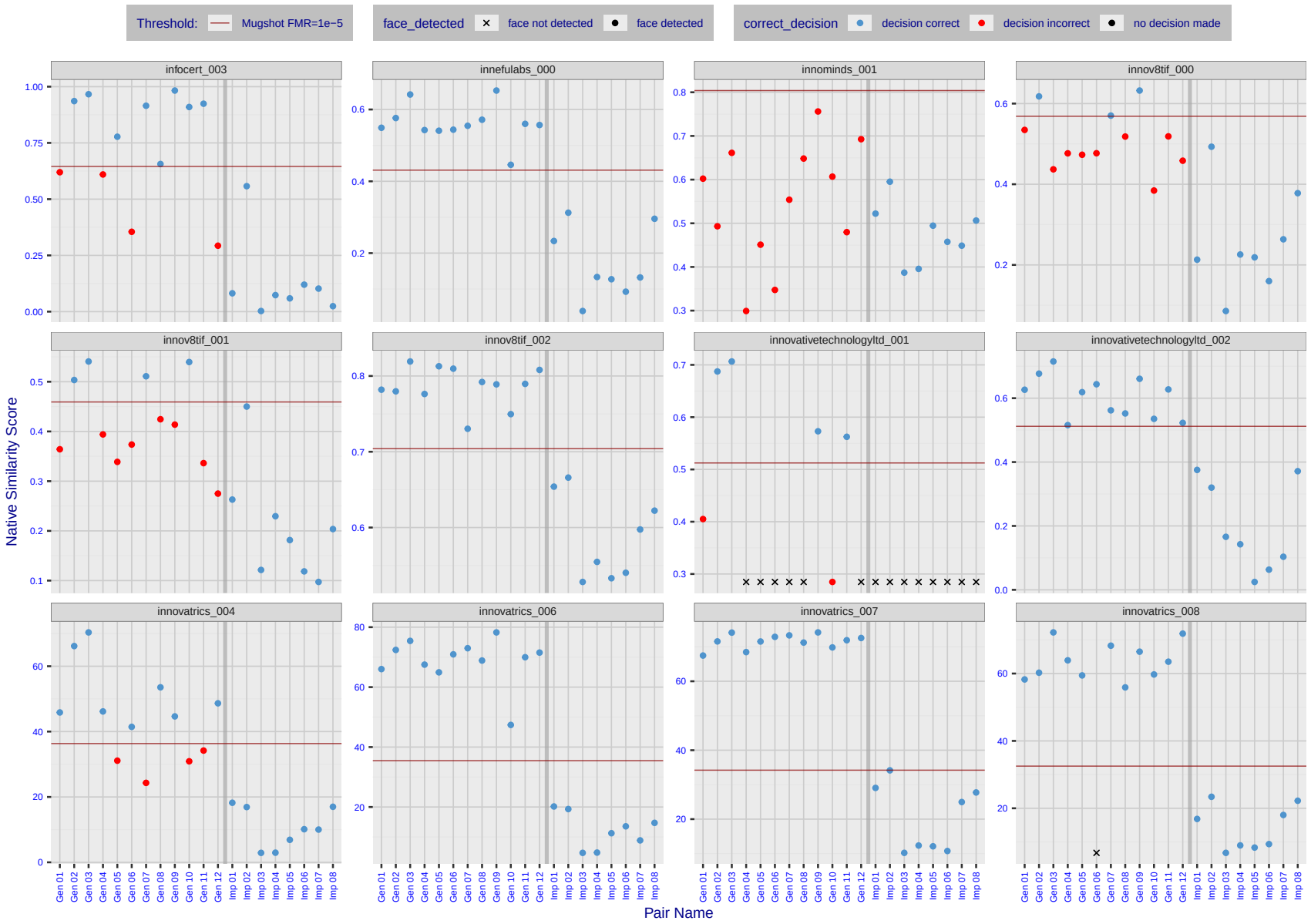

Figure 46: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

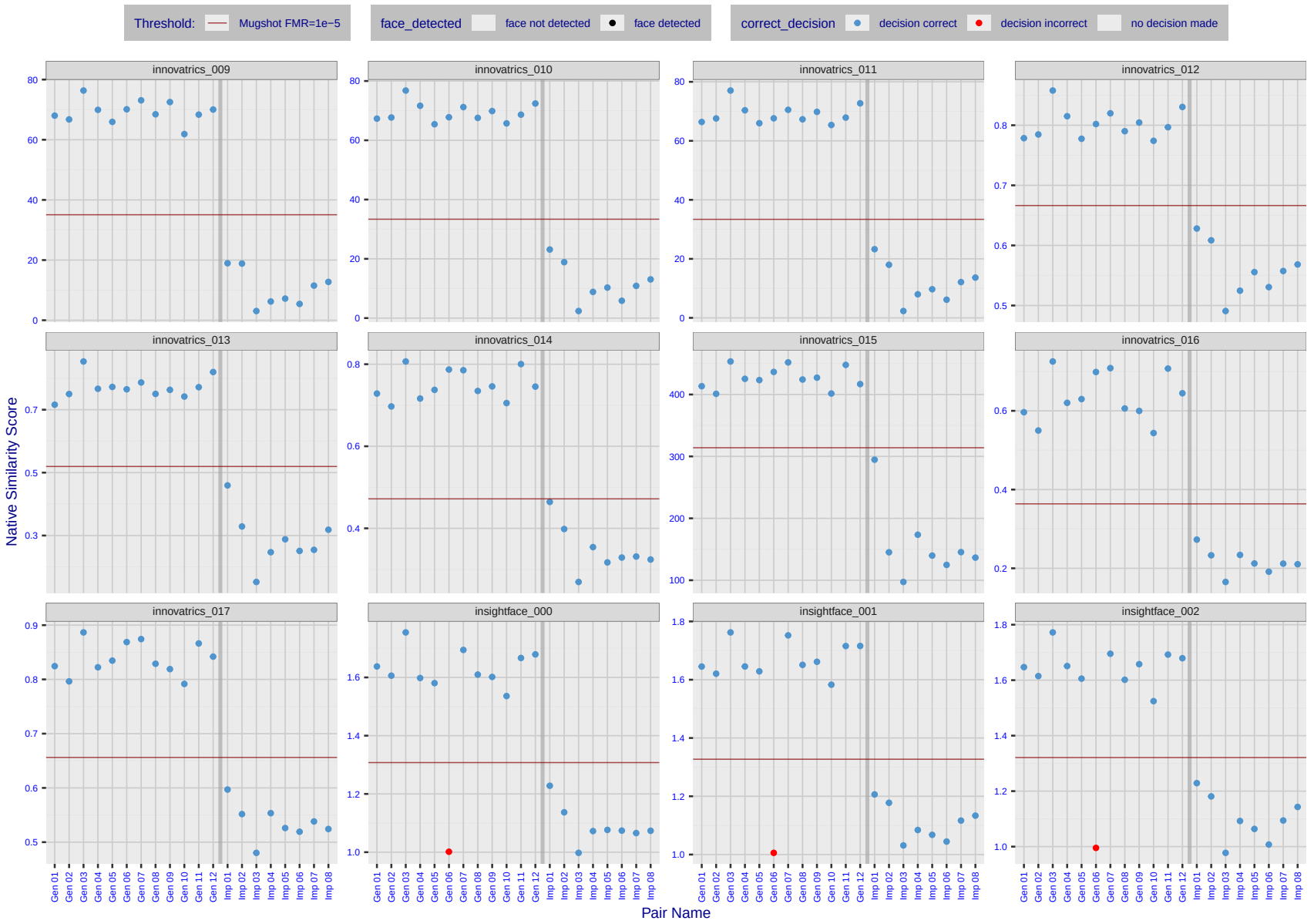

Figure 47: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

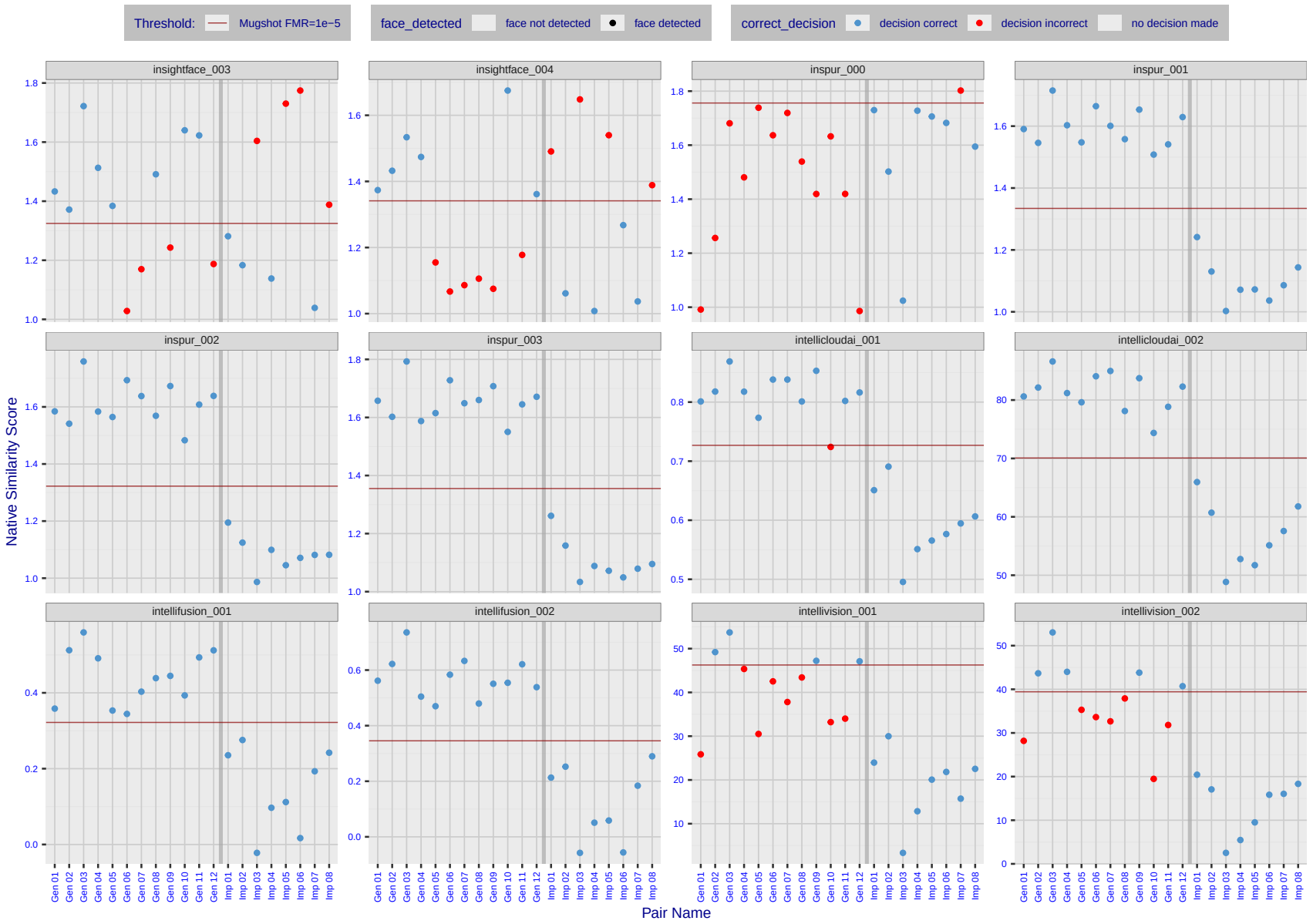

Figure 48: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

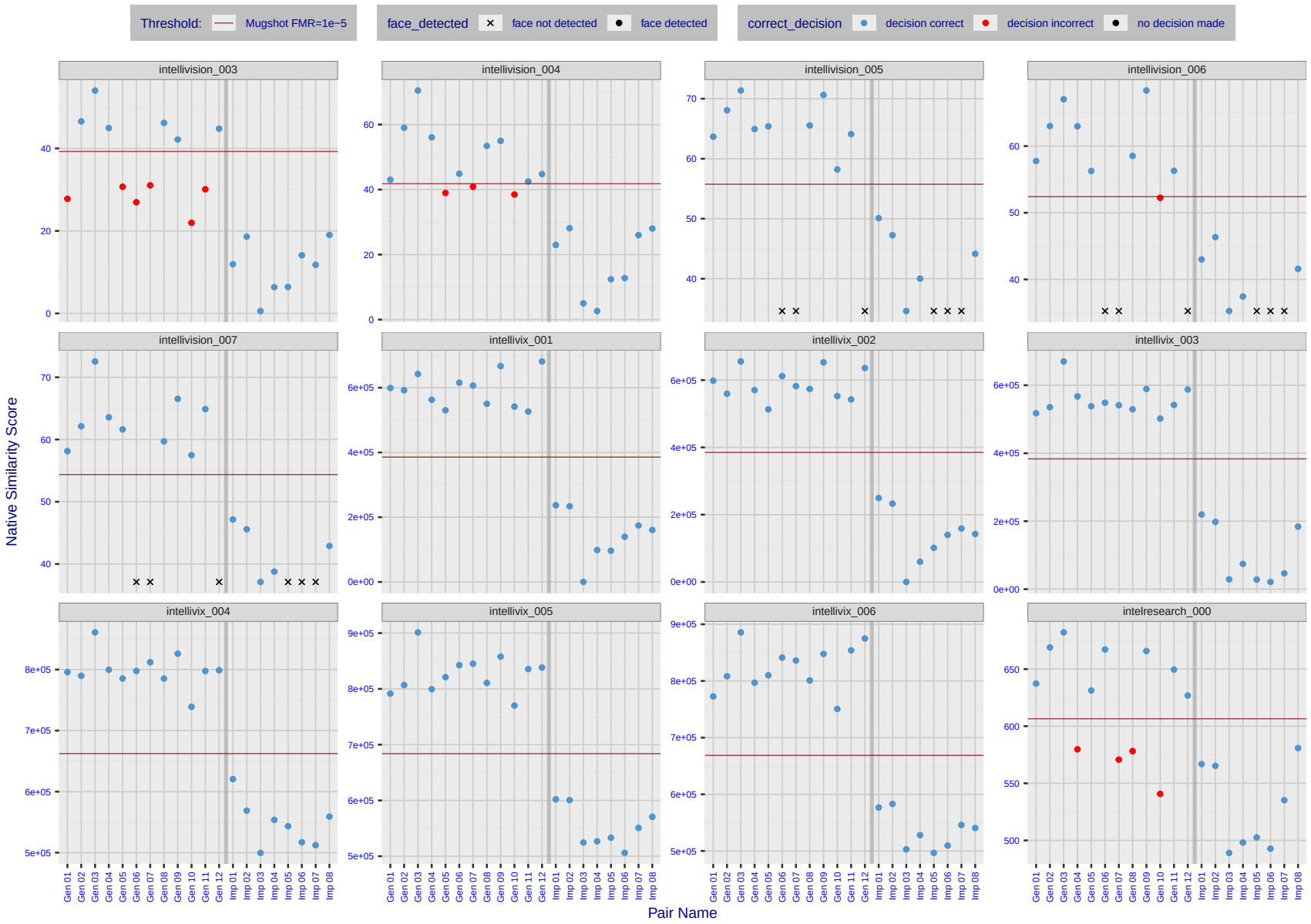

Figure 49: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

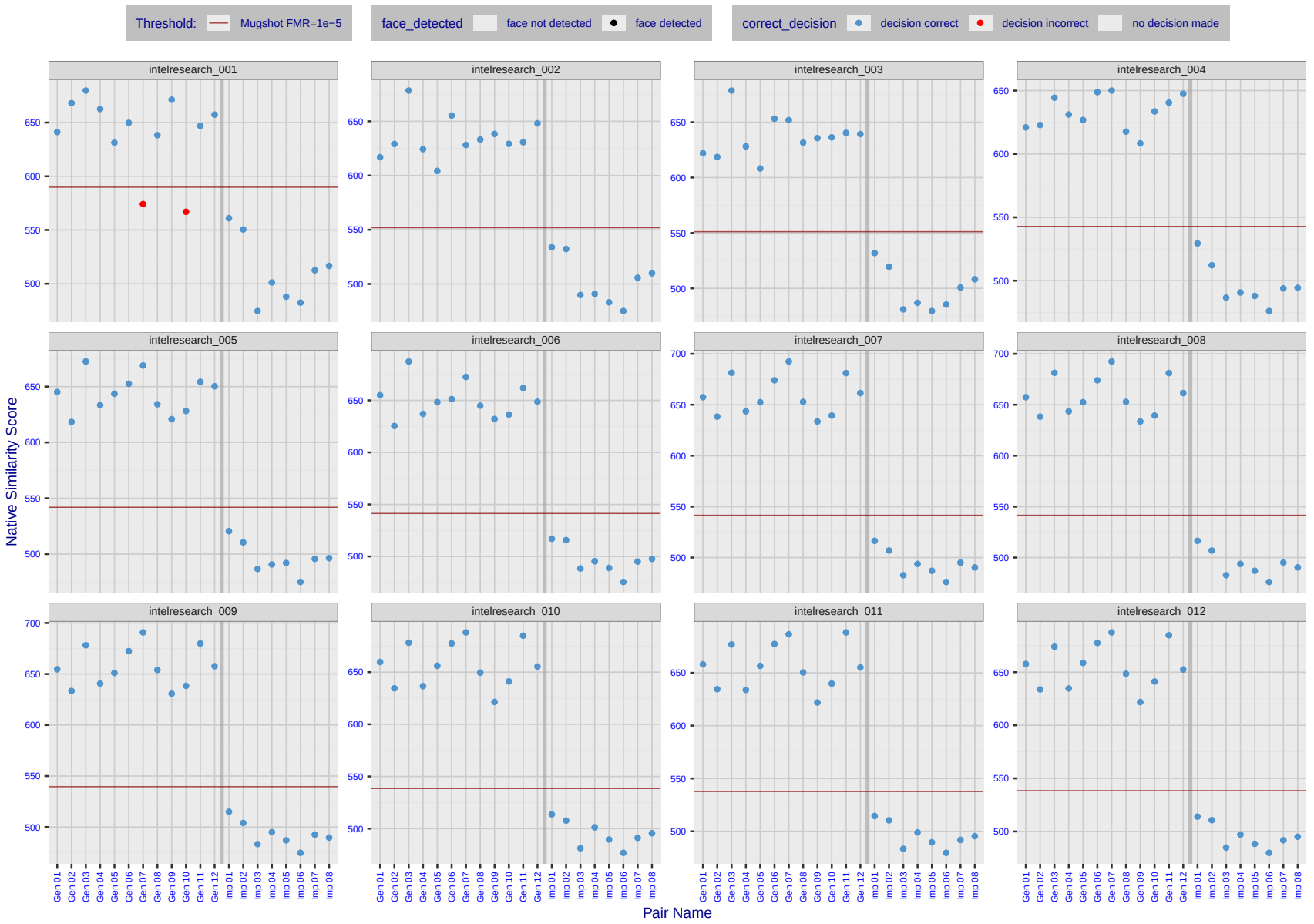

Figure 50: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

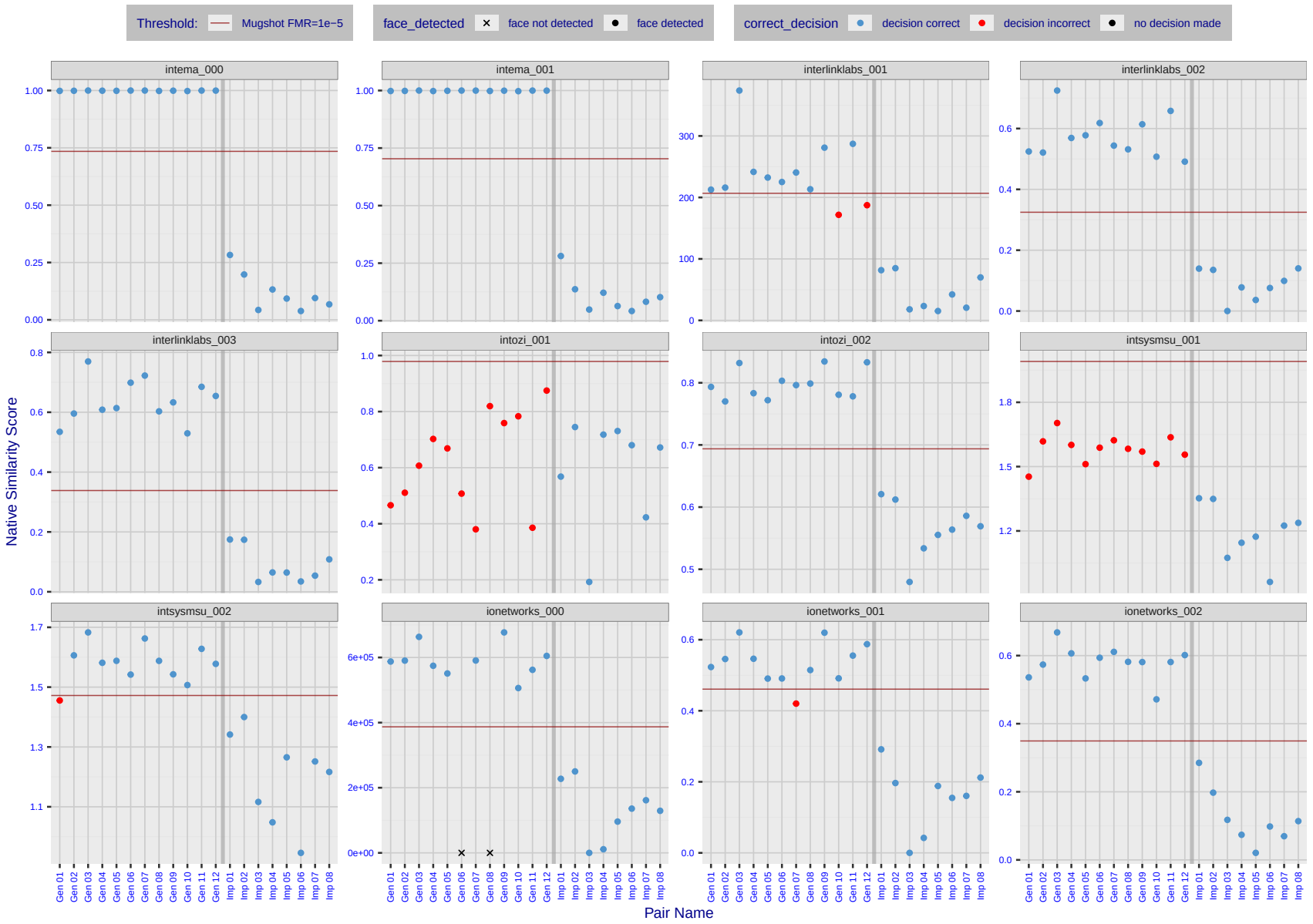

Figure 51: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

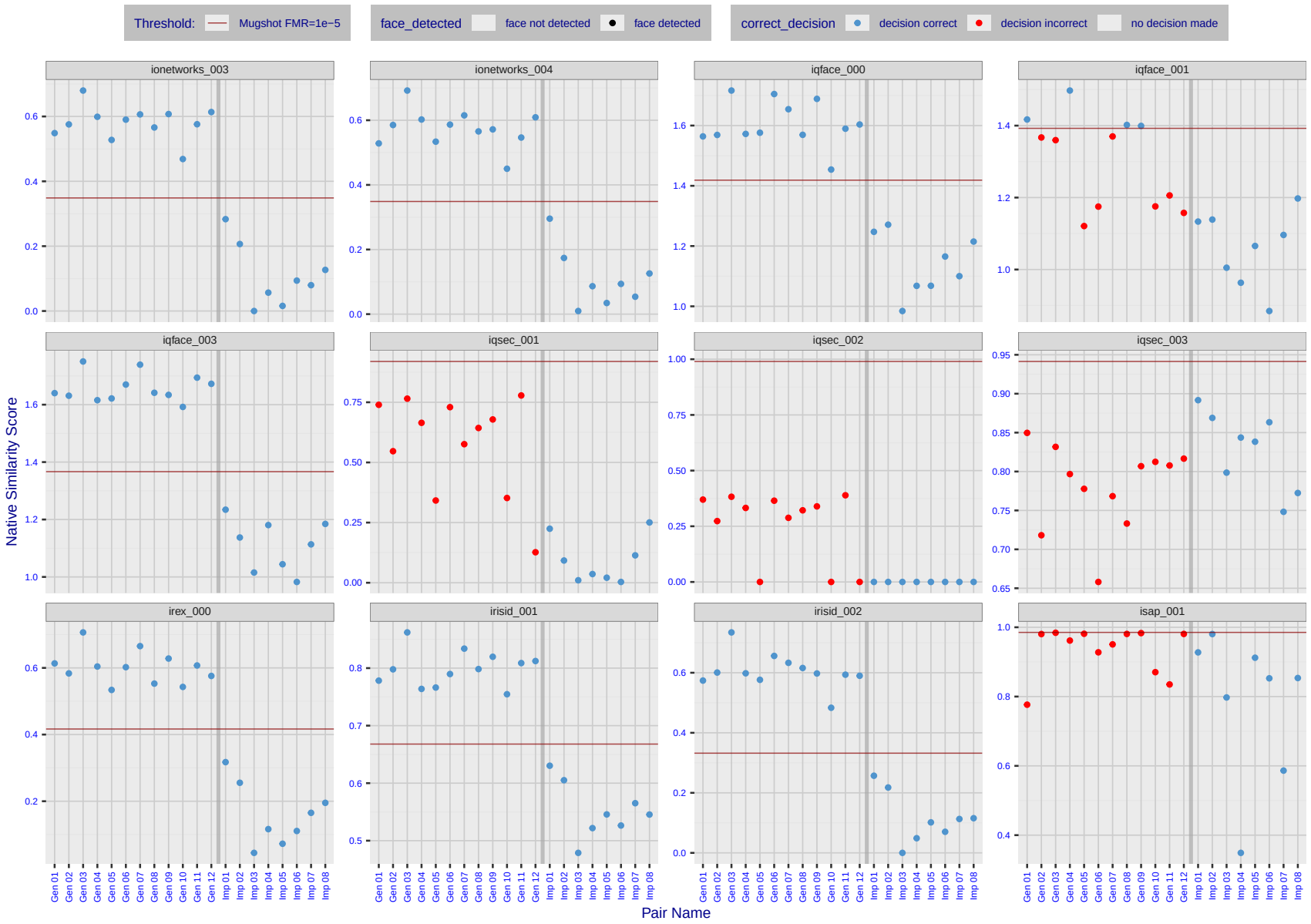

Figure 52: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

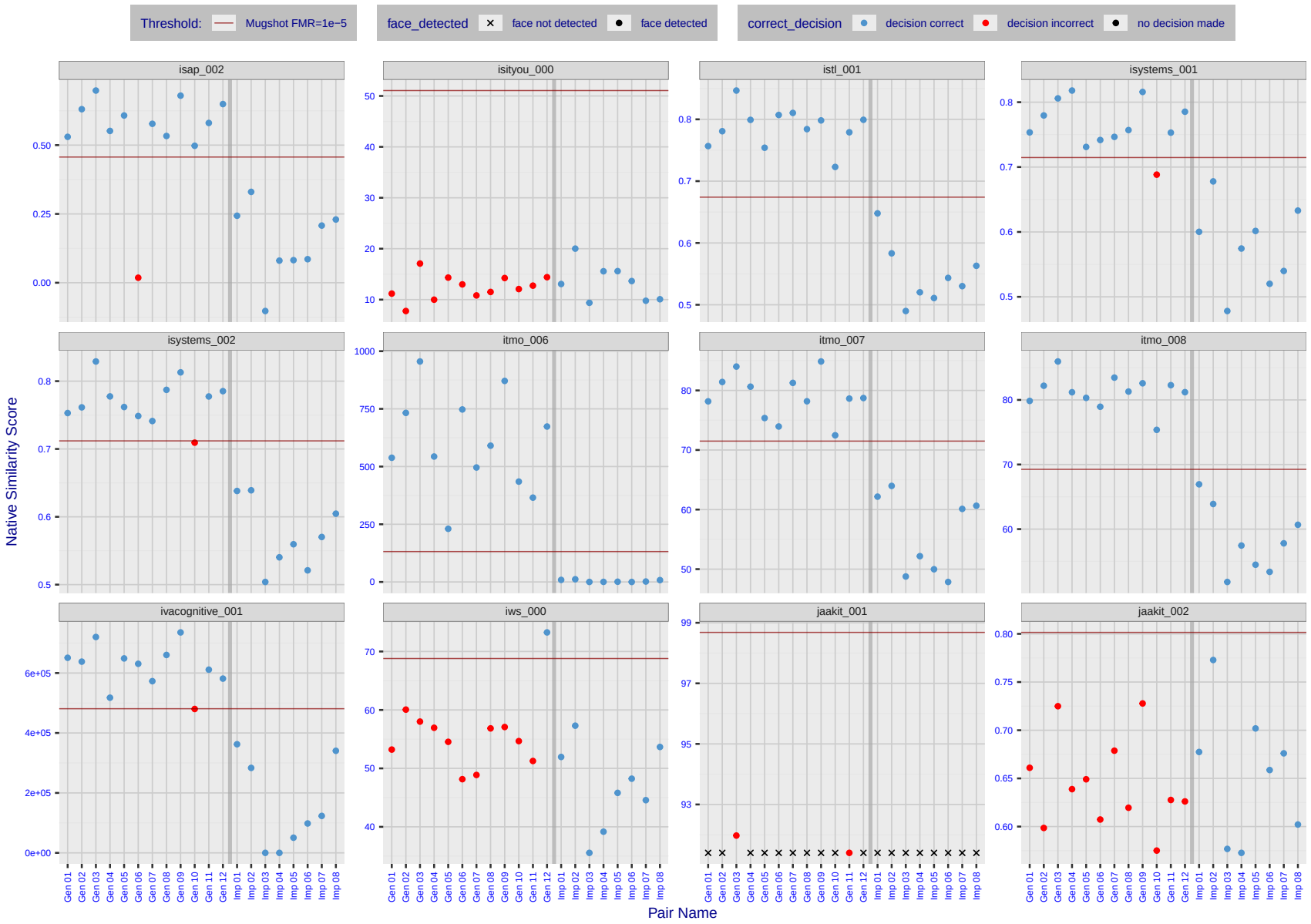

Figure 53: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

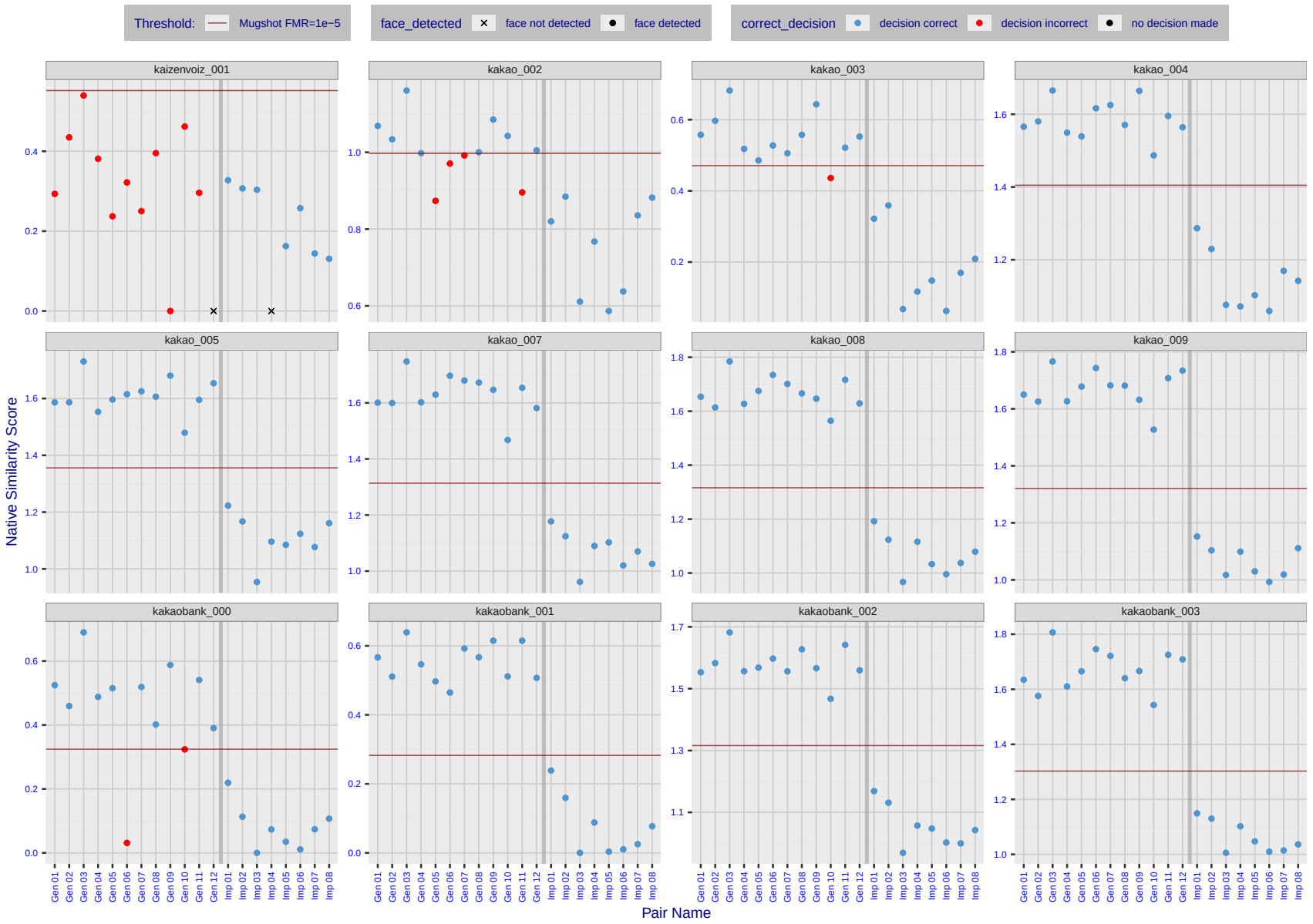

Figure 54: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

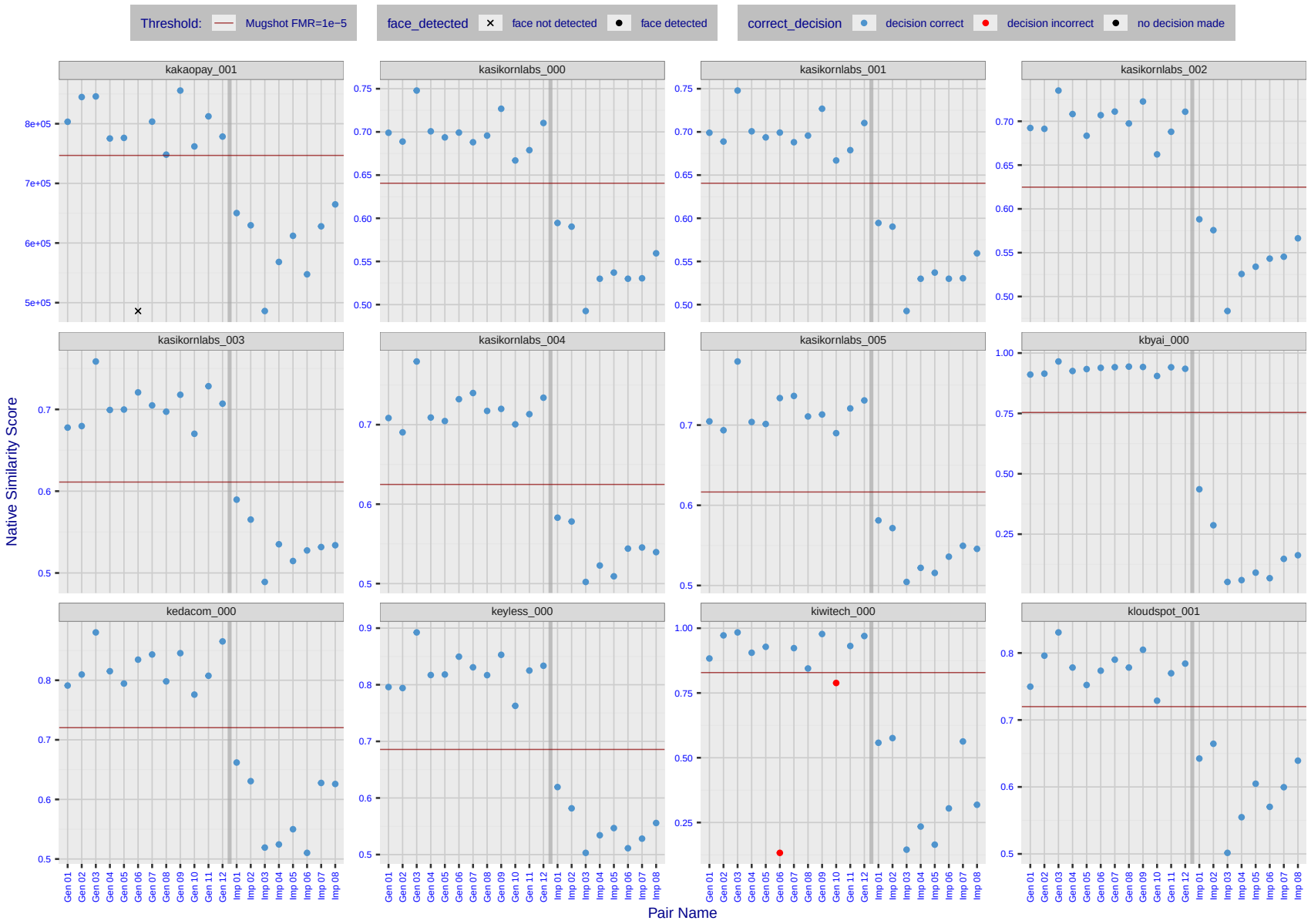

Figure 55: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

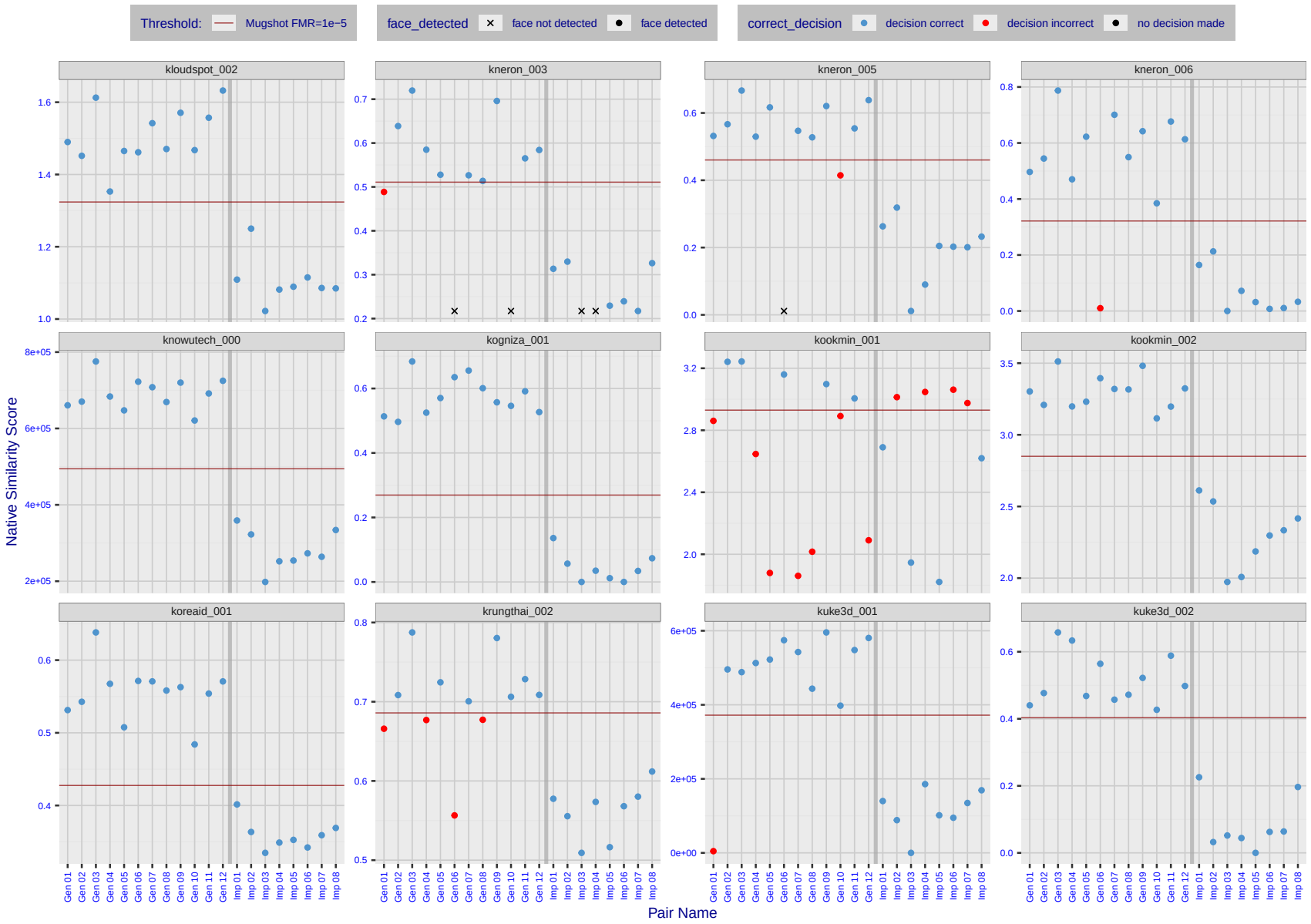

Figure 56: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

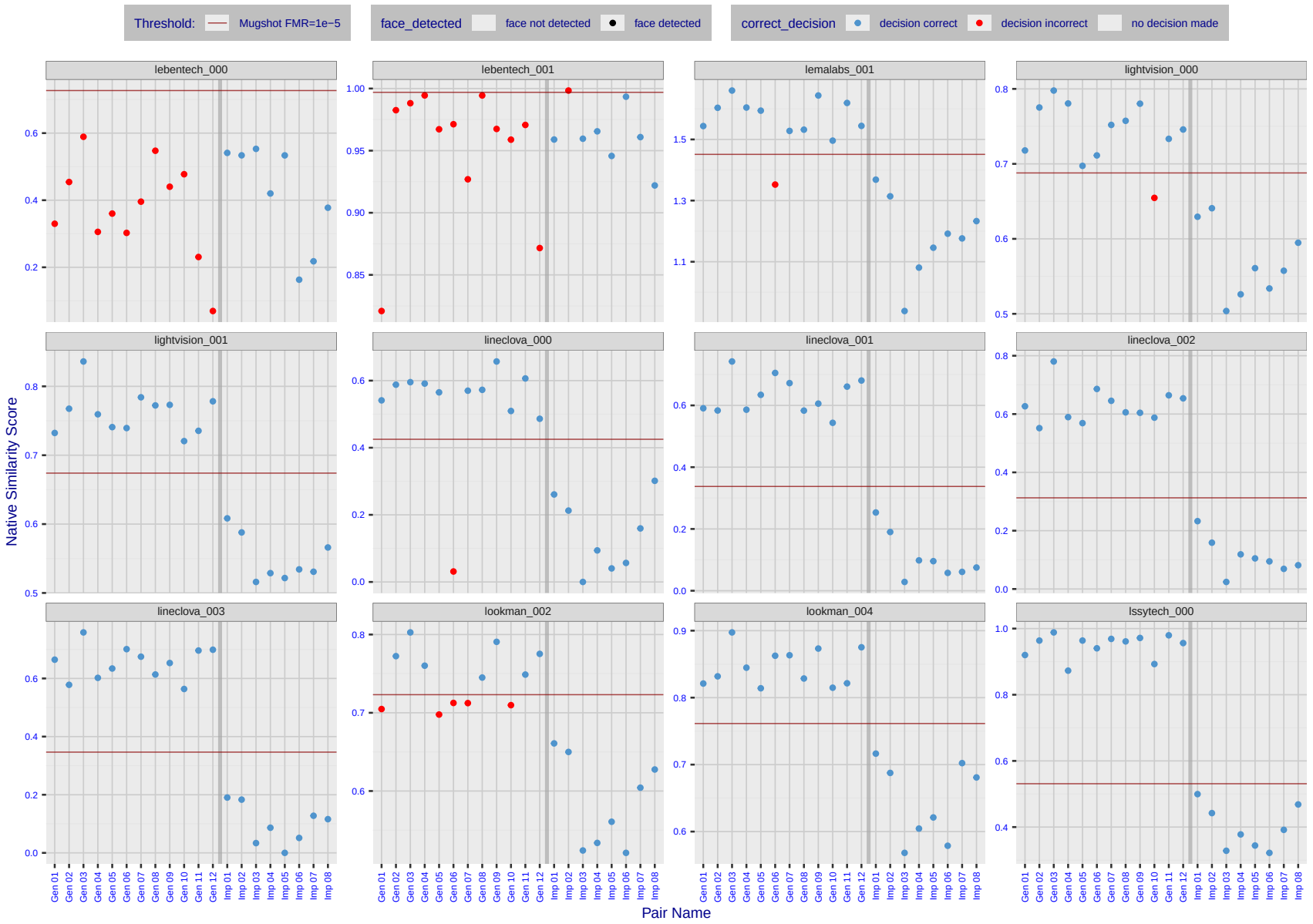

Figure 57: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

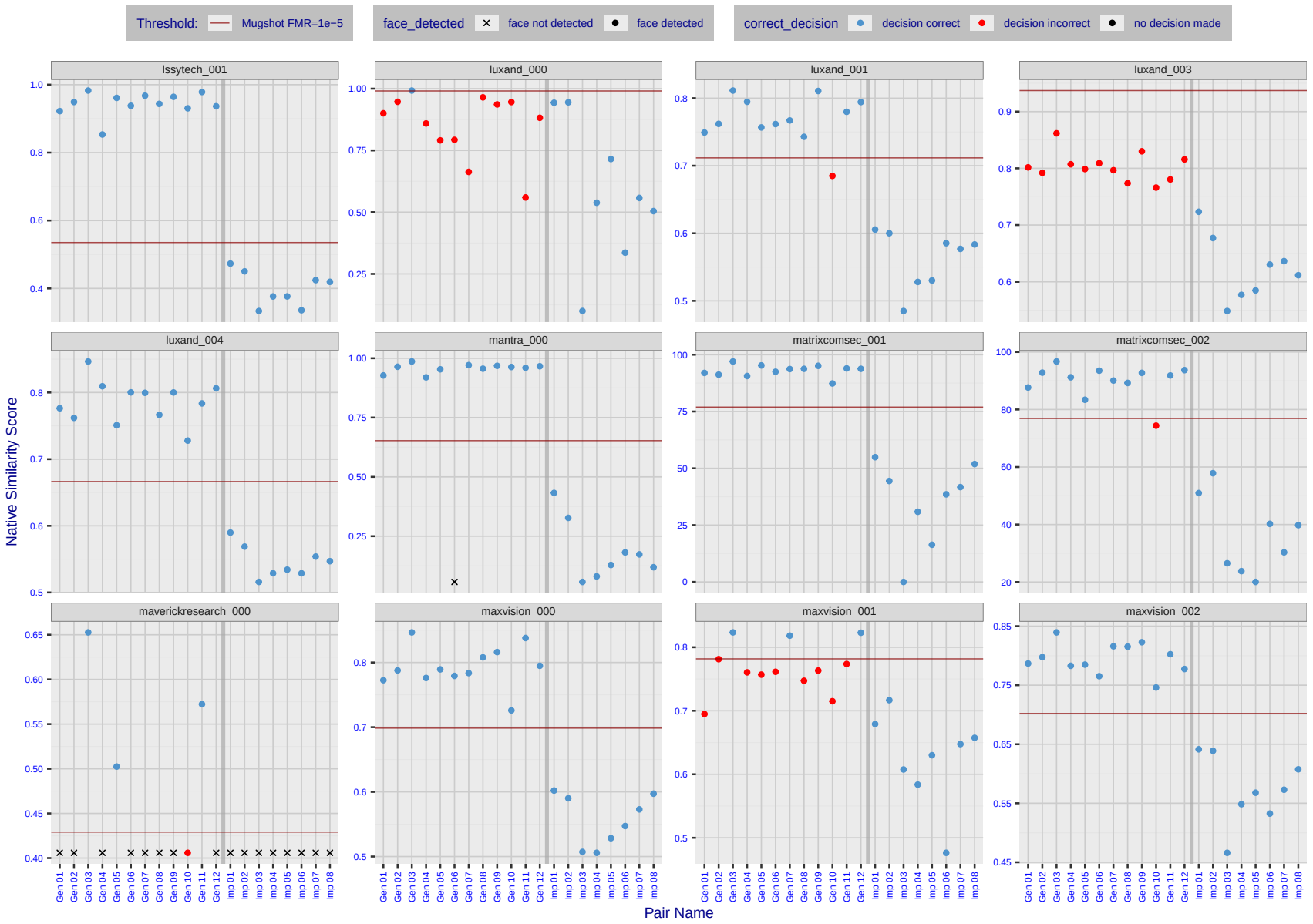

Figure 58: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

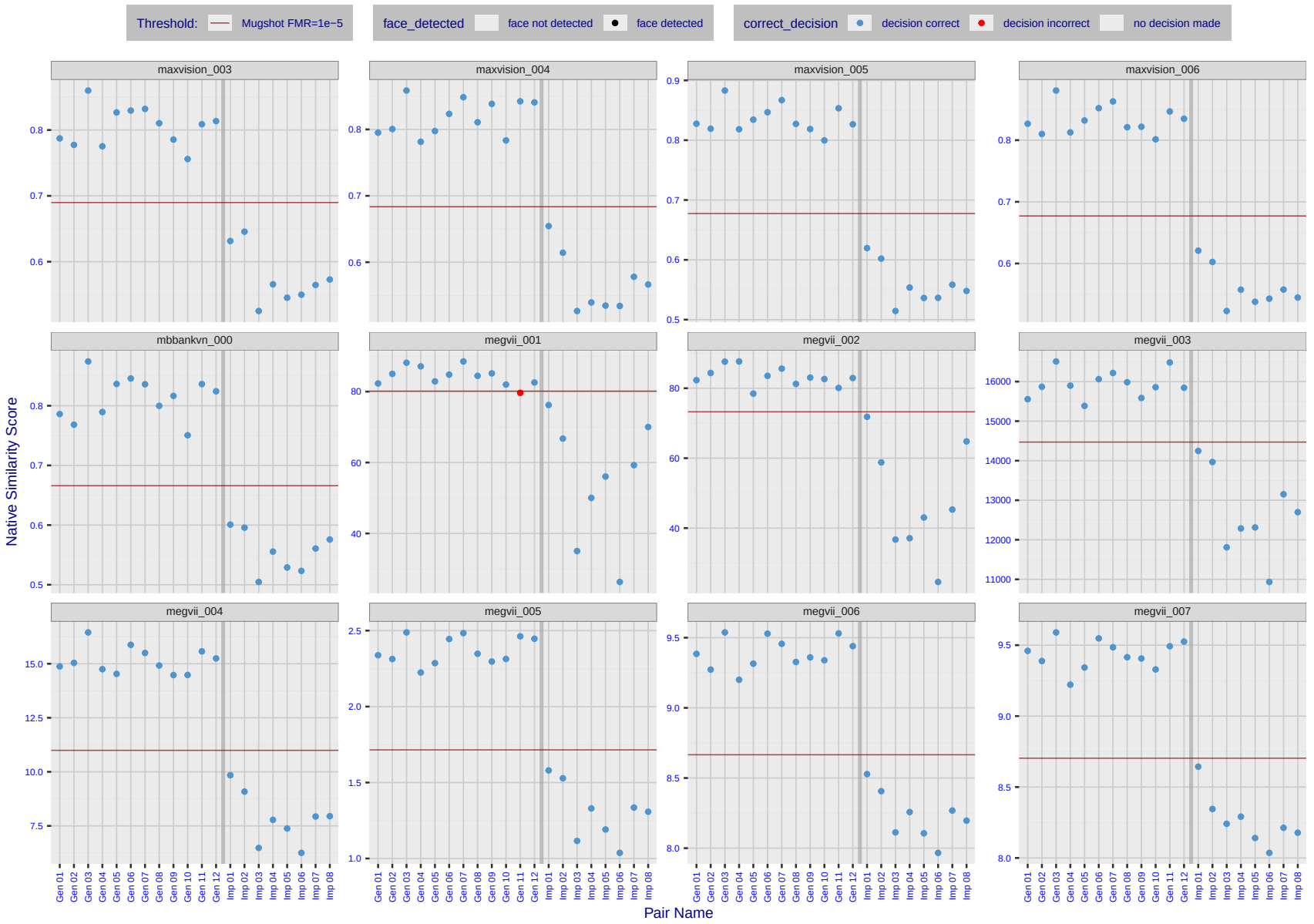

Figure 59: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

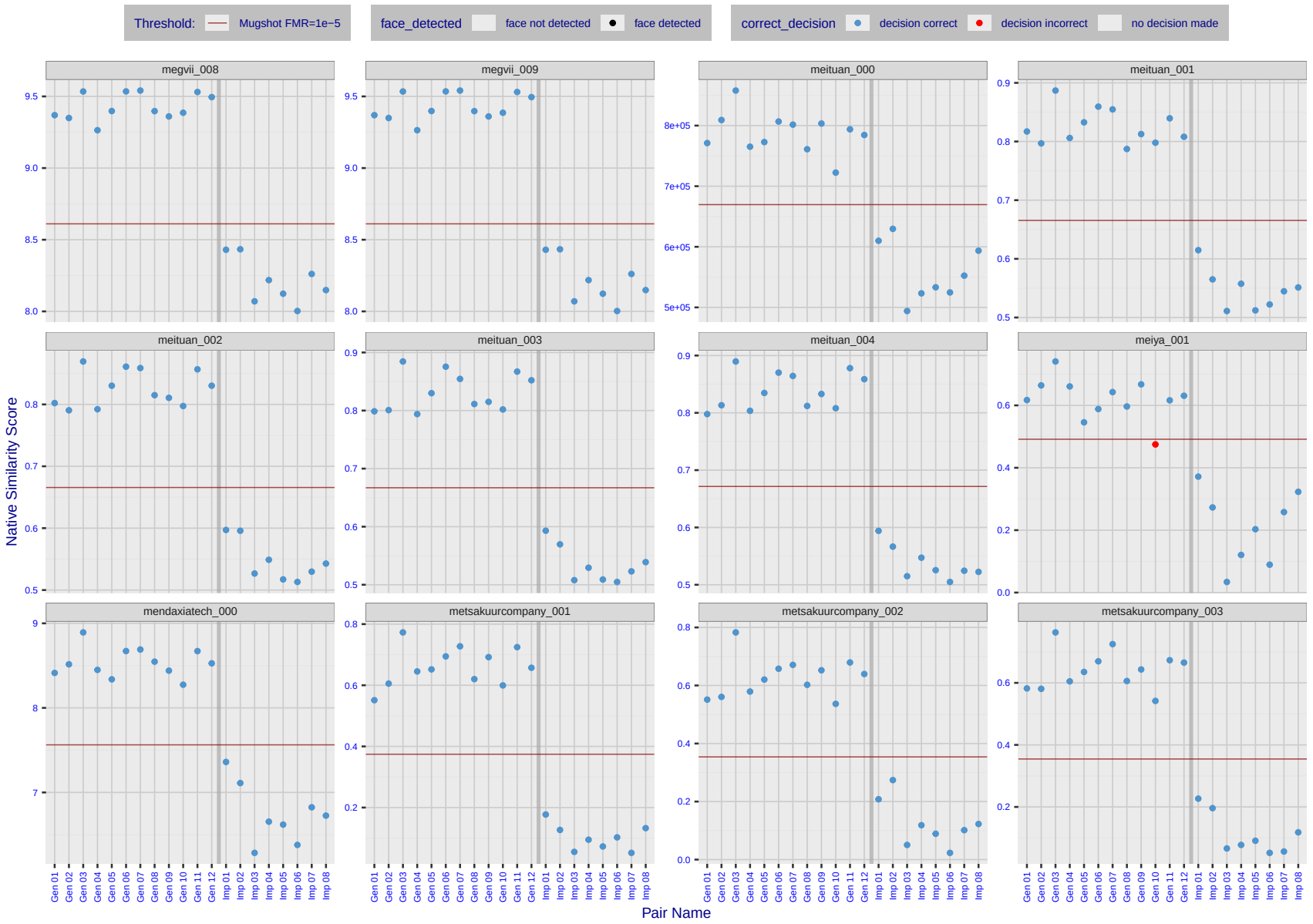

Figure 60: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

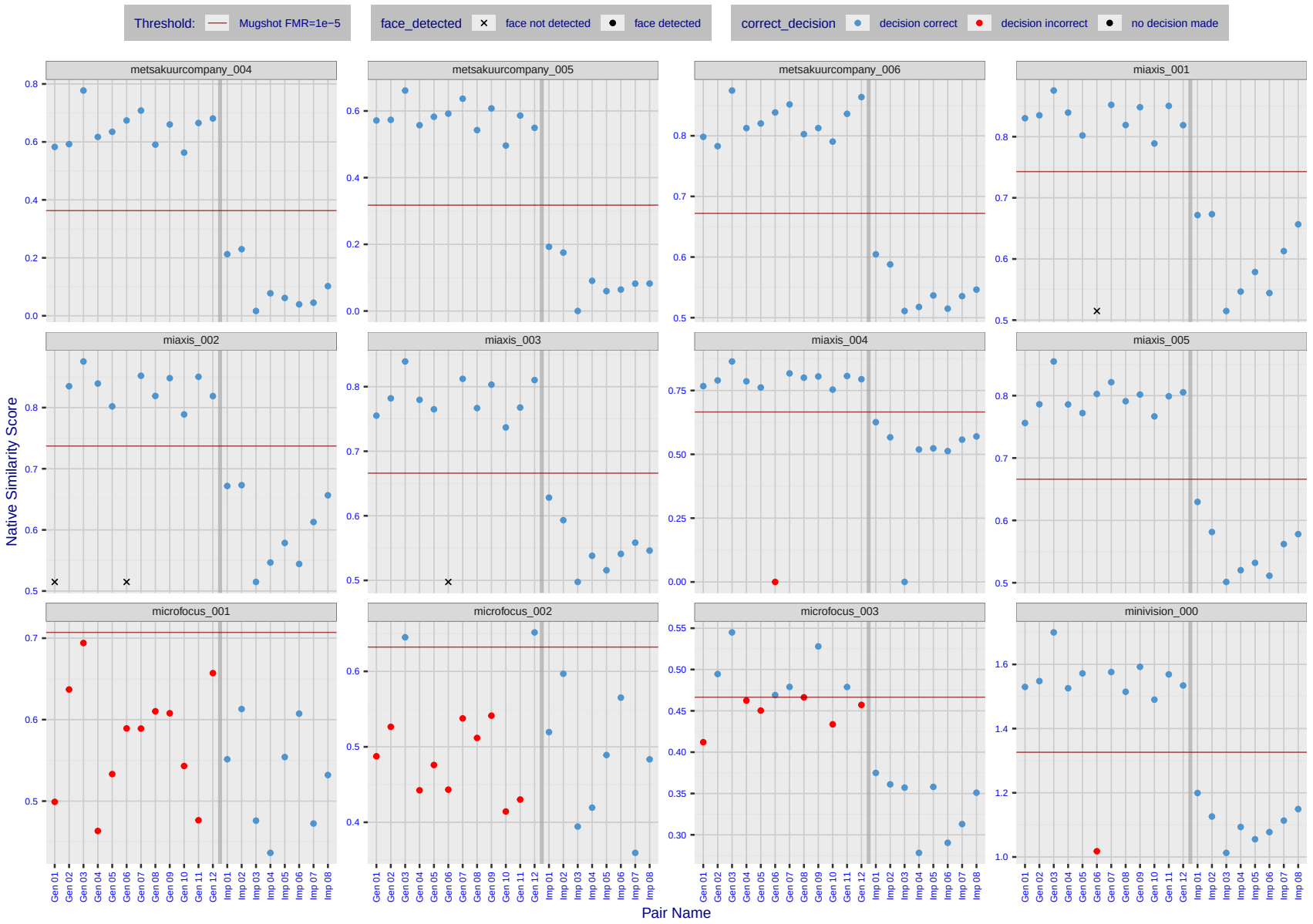

Figure 61: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

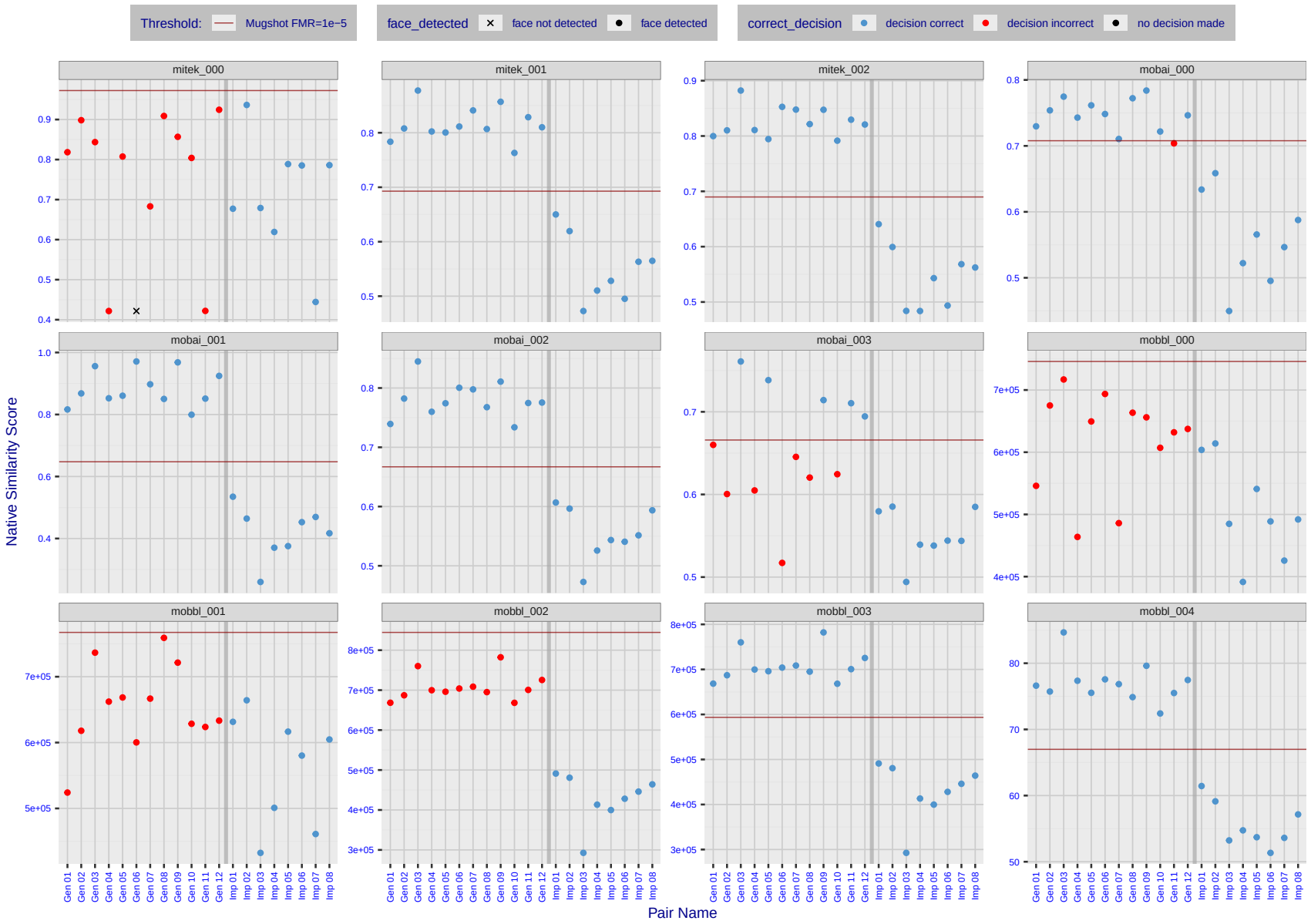

Figure 62: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

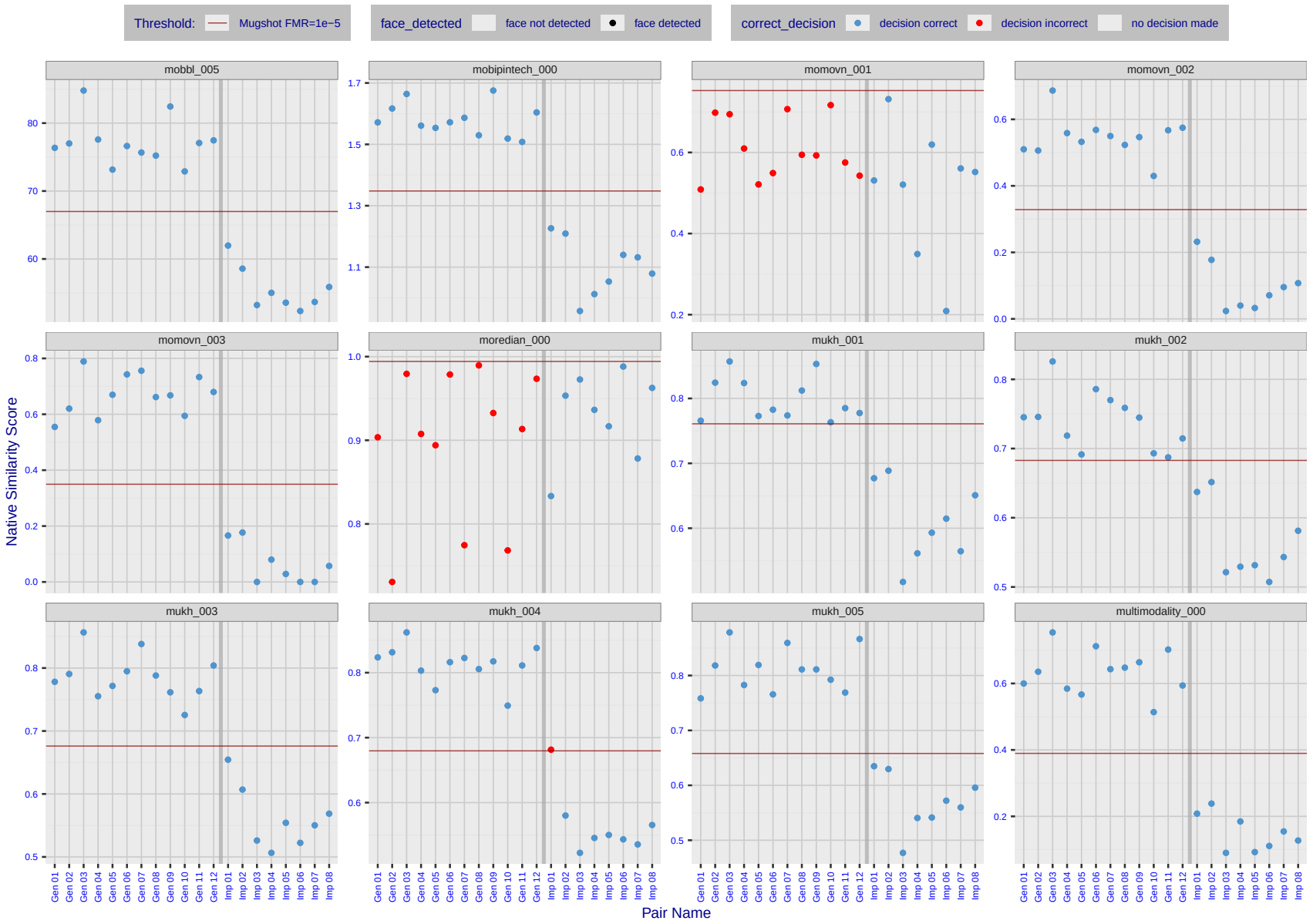

Figure 63: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

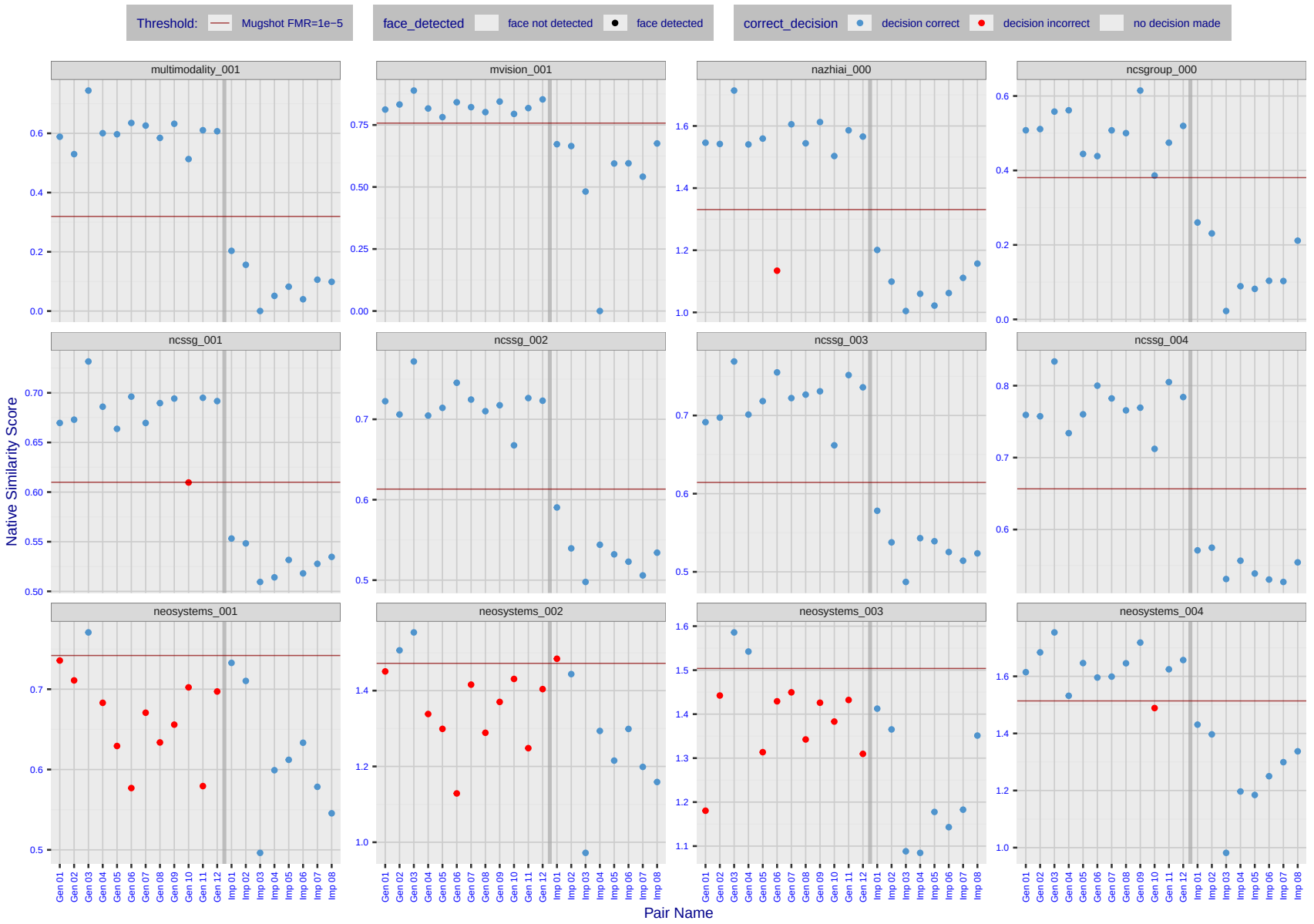

Figure 64: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

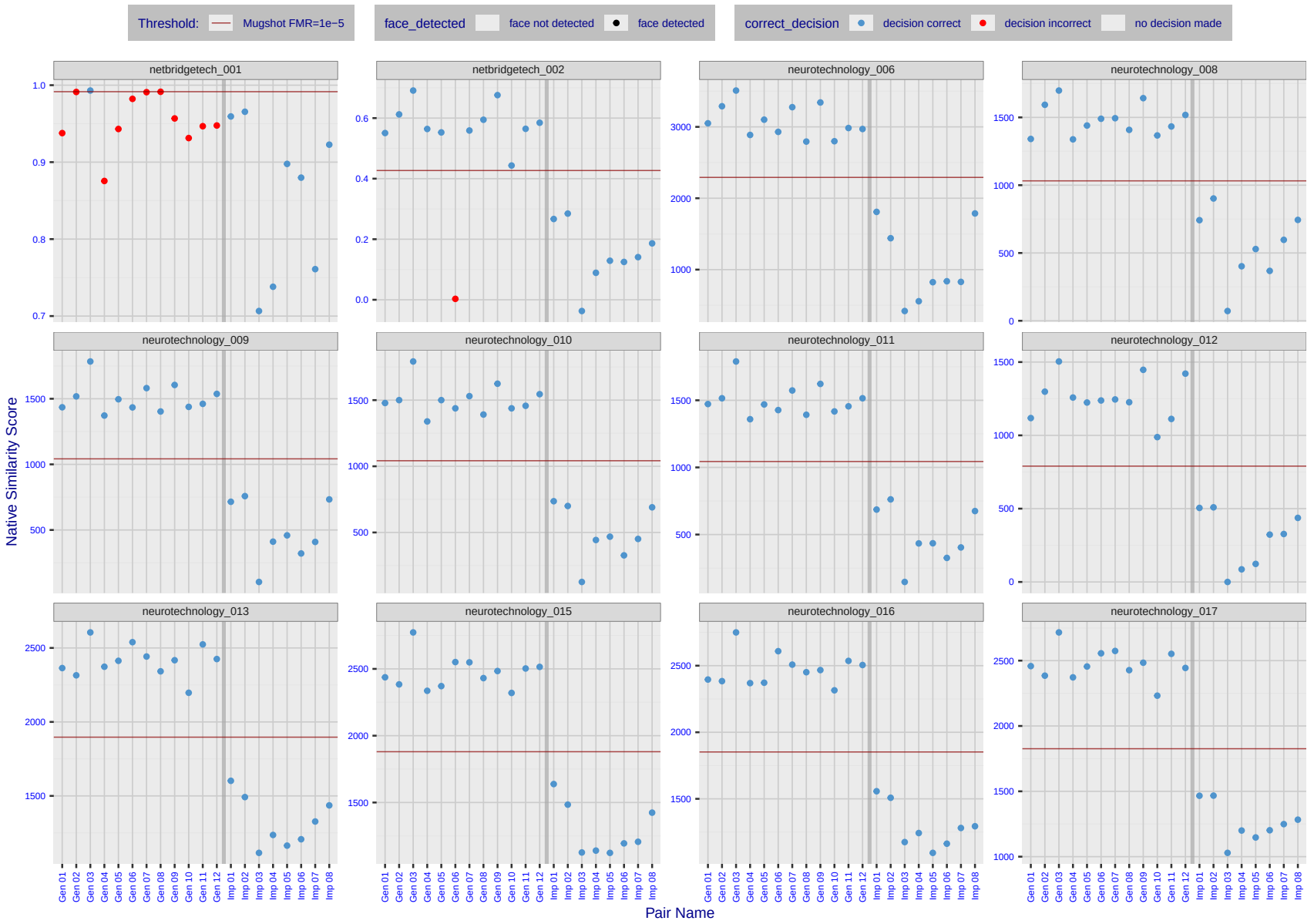

Figure 65: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

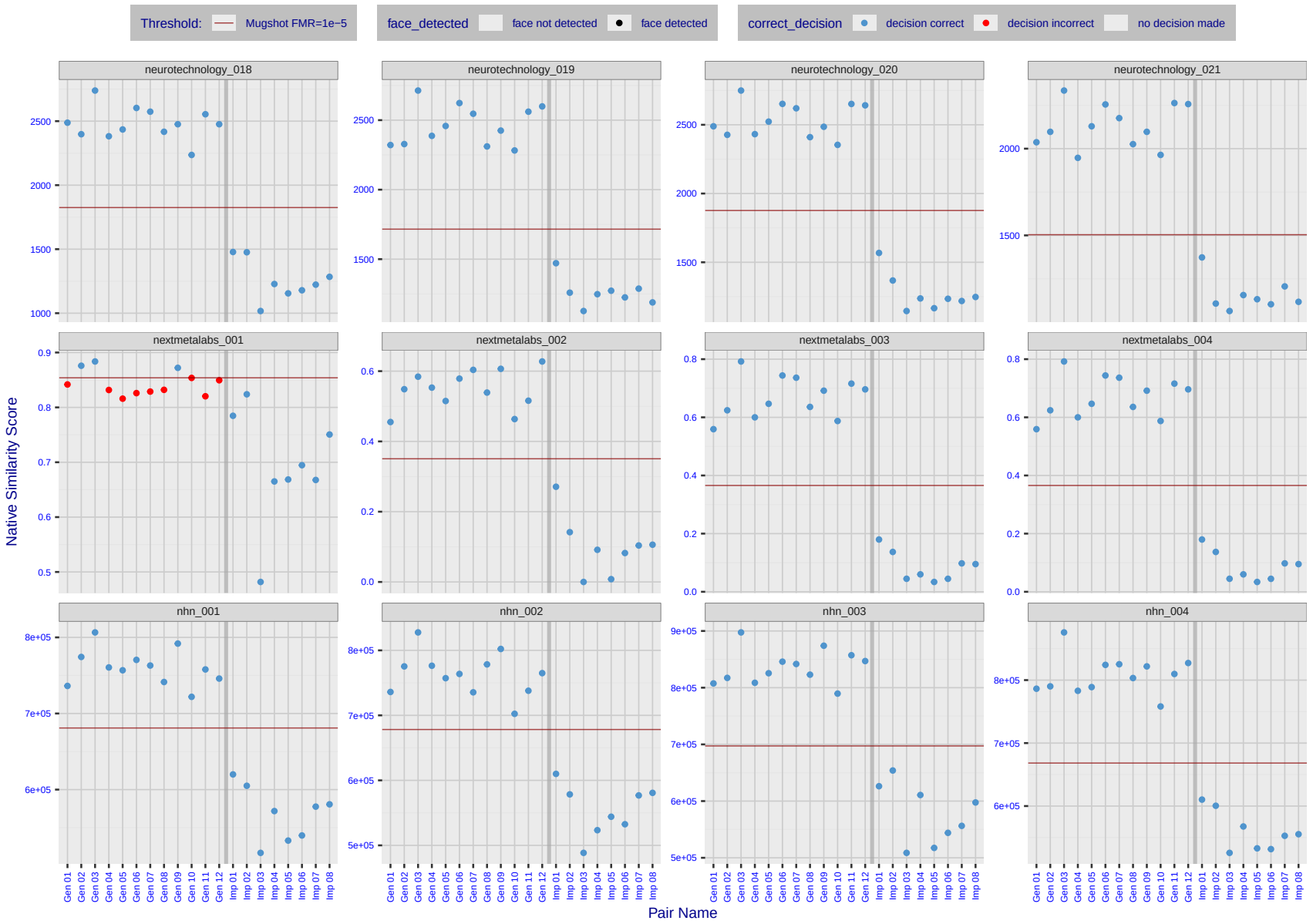

Figure 66: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

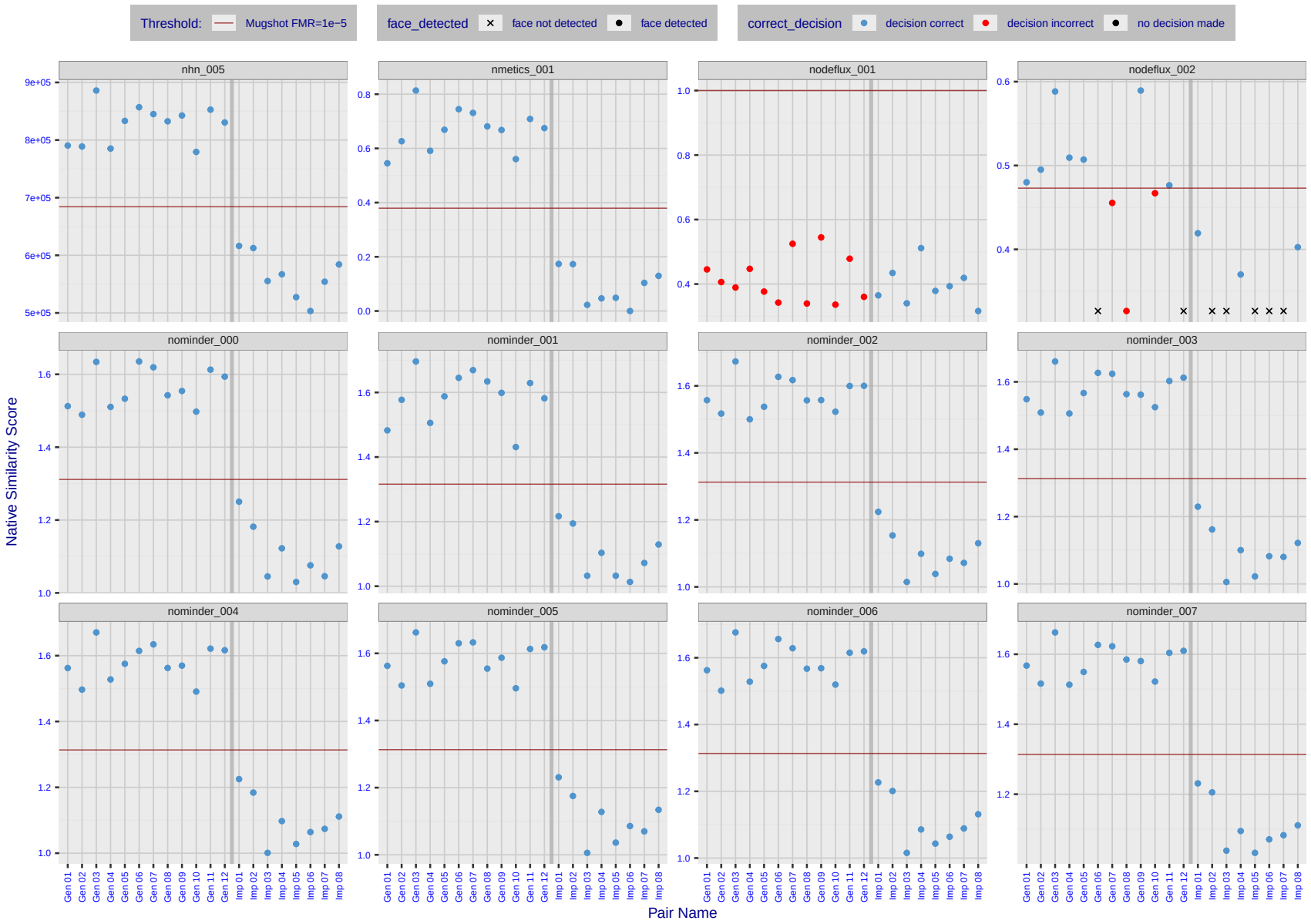

Figure 67: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

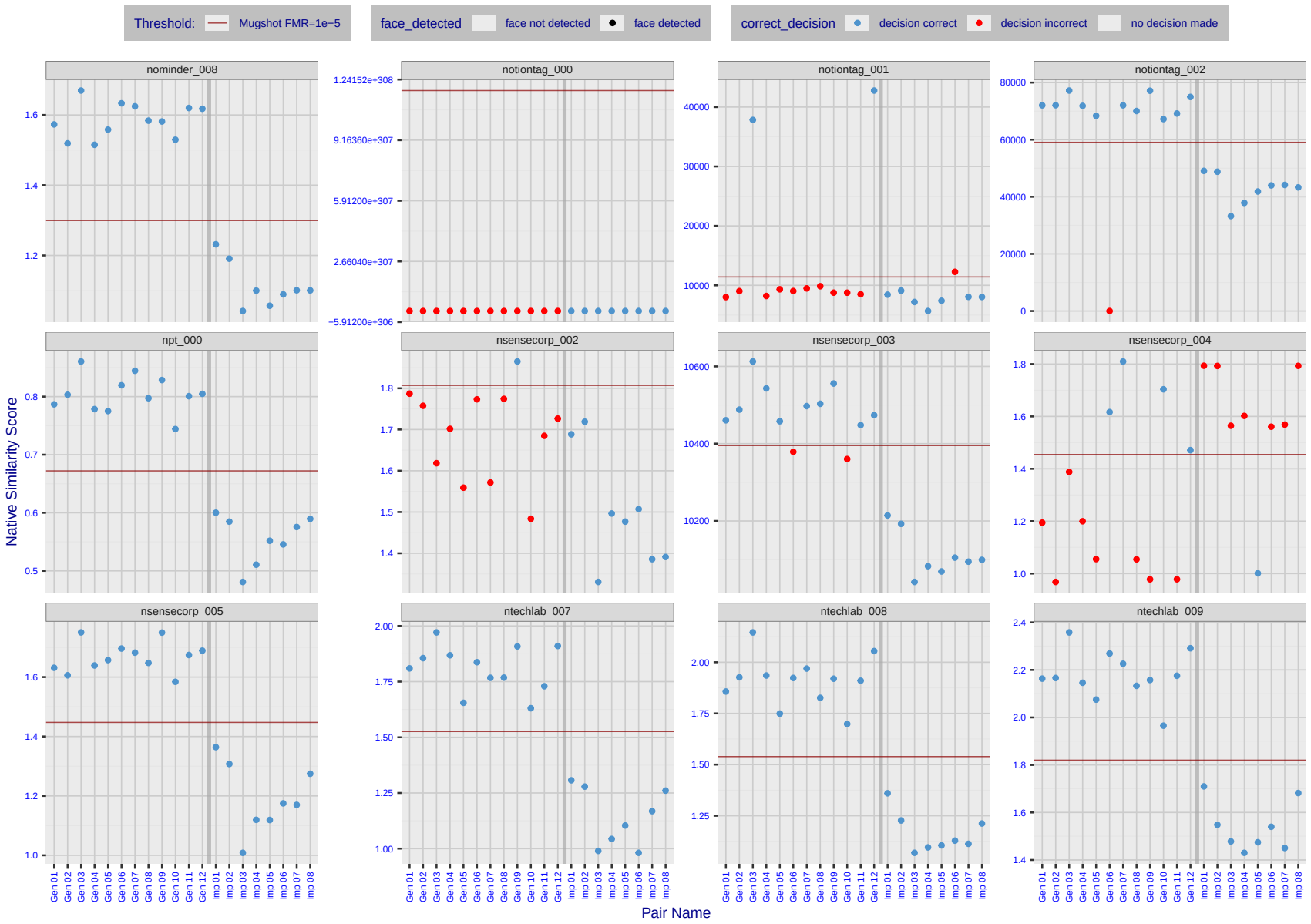

Figure 68: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

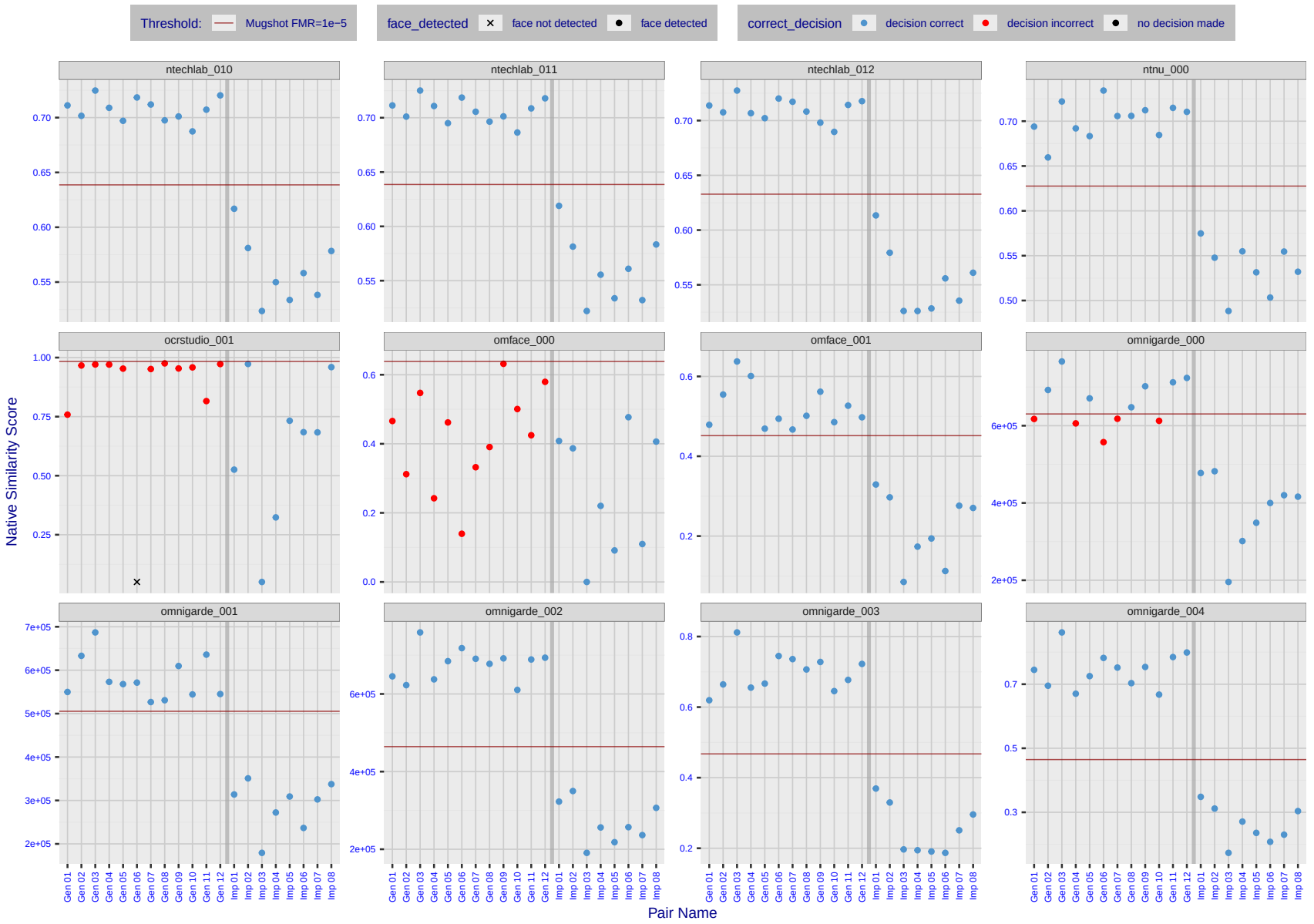

Figure 69: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

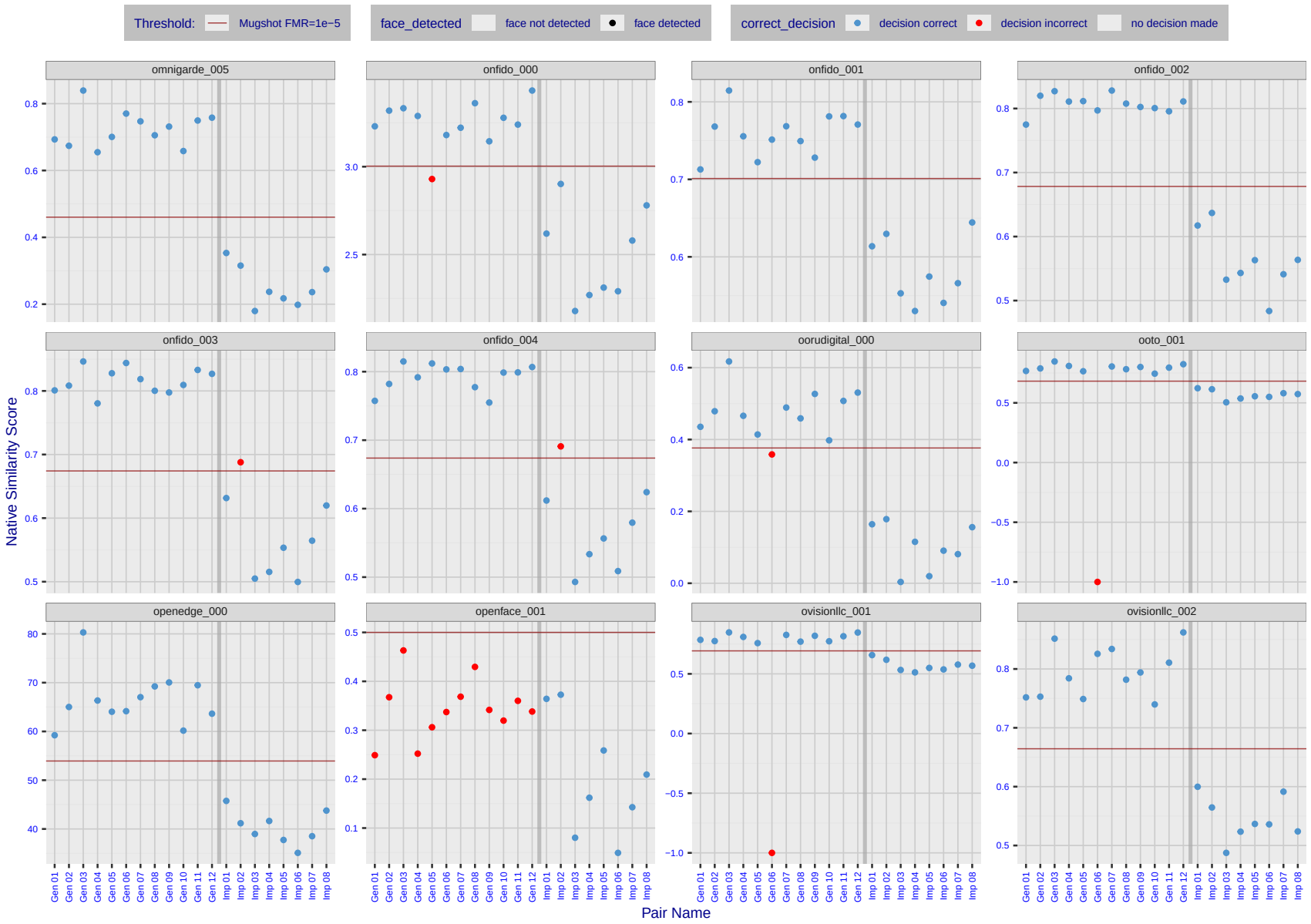

Figure 70: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

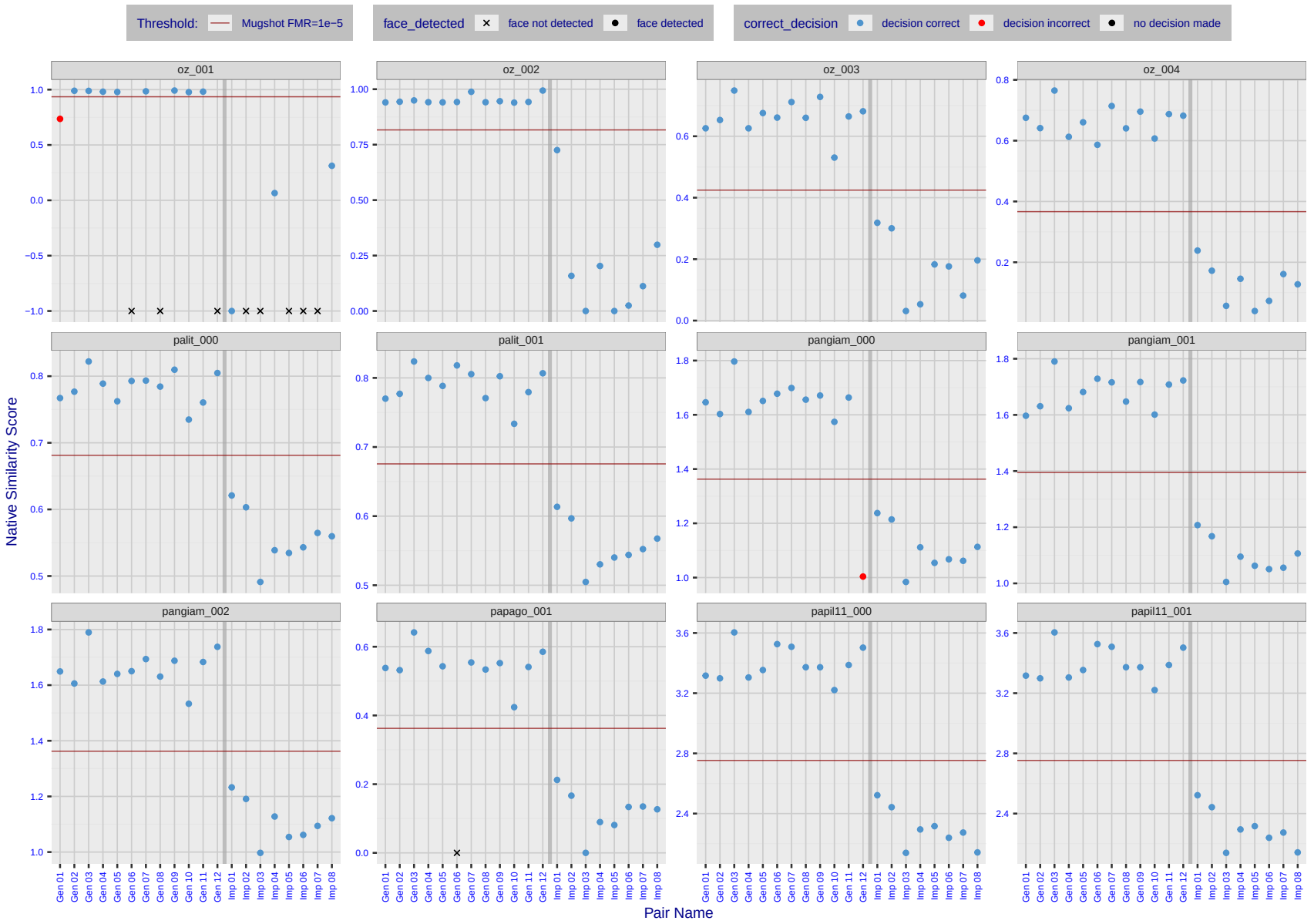

Figure 71: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

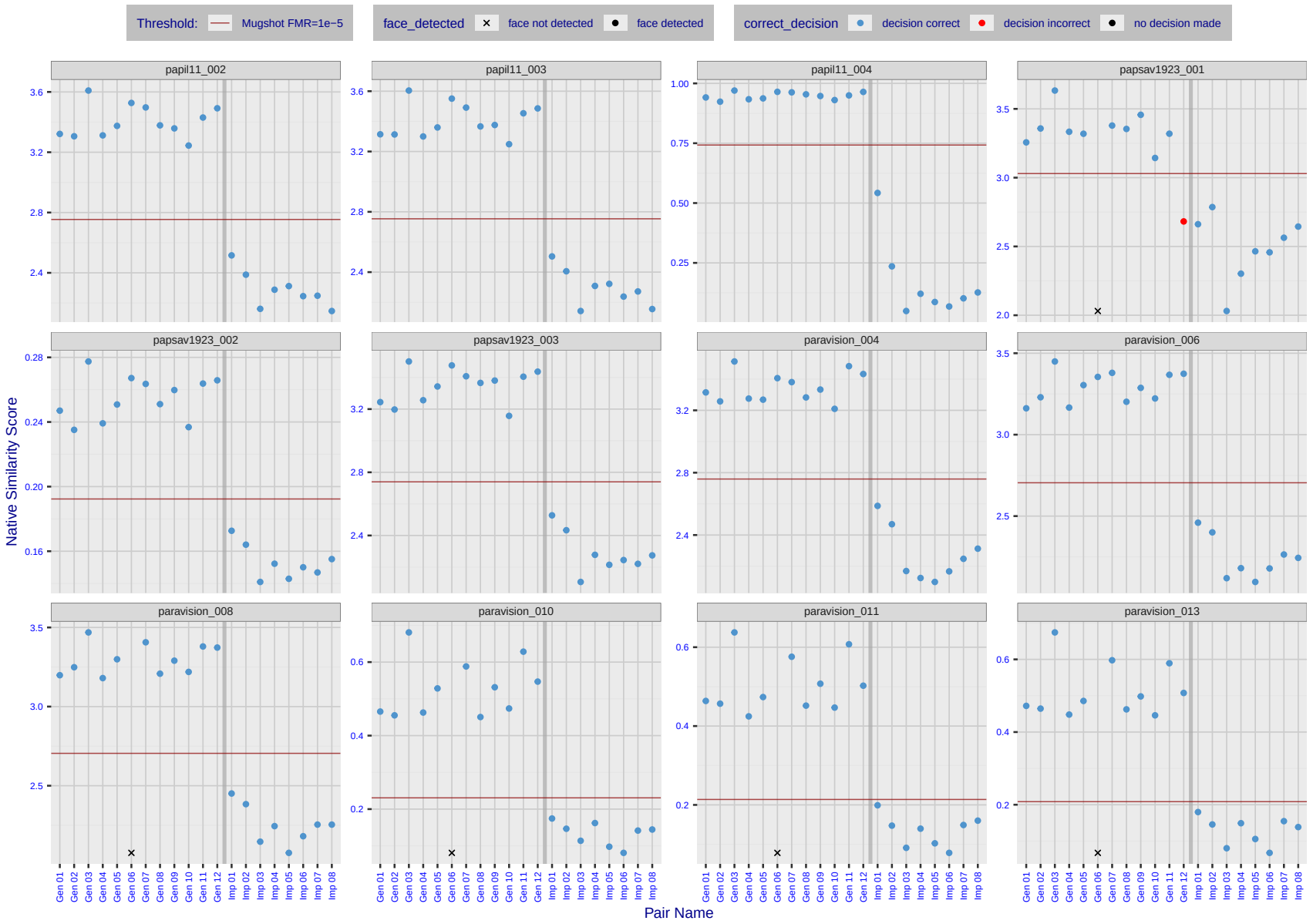

Figure 72: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

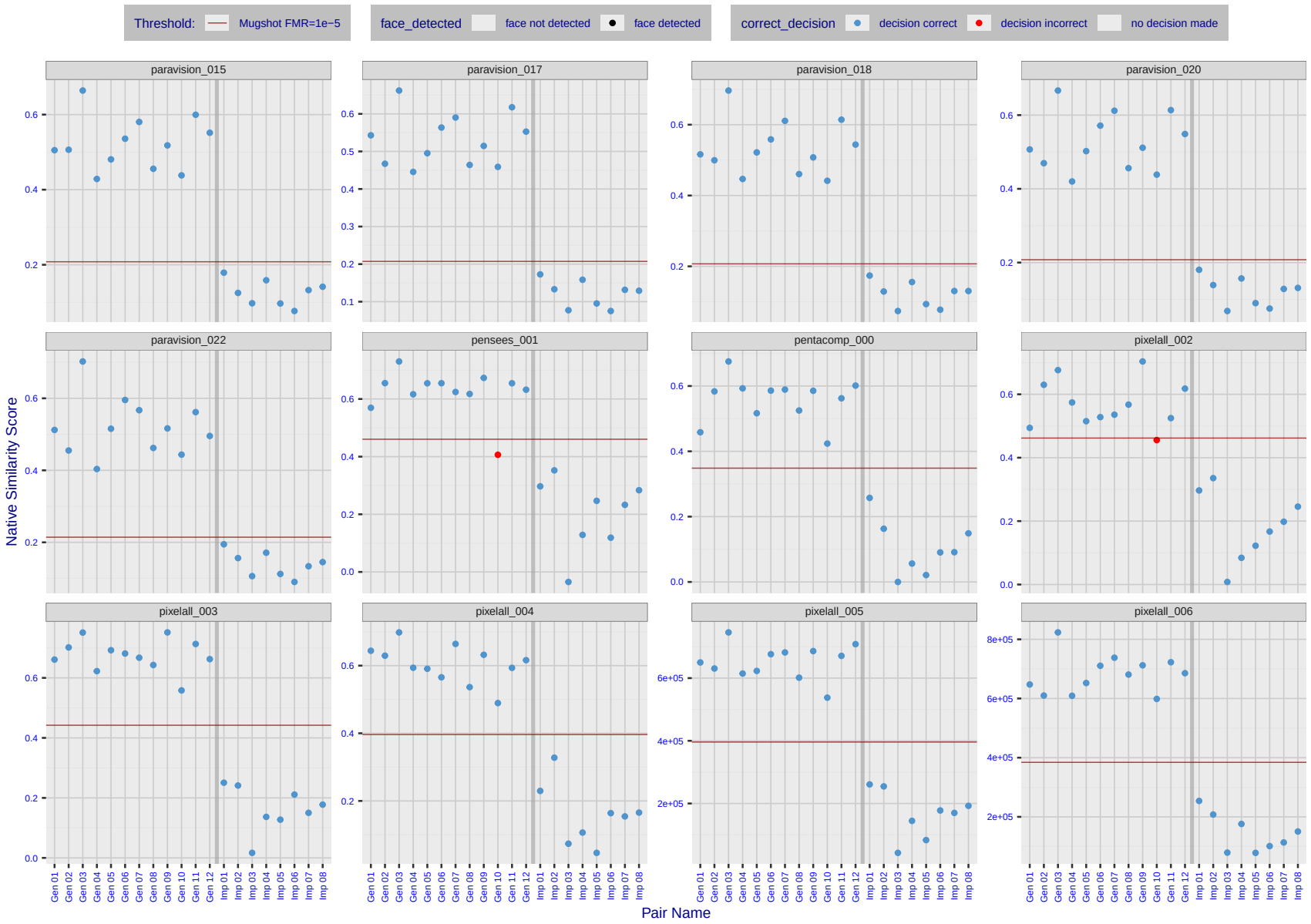

Figure 73: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

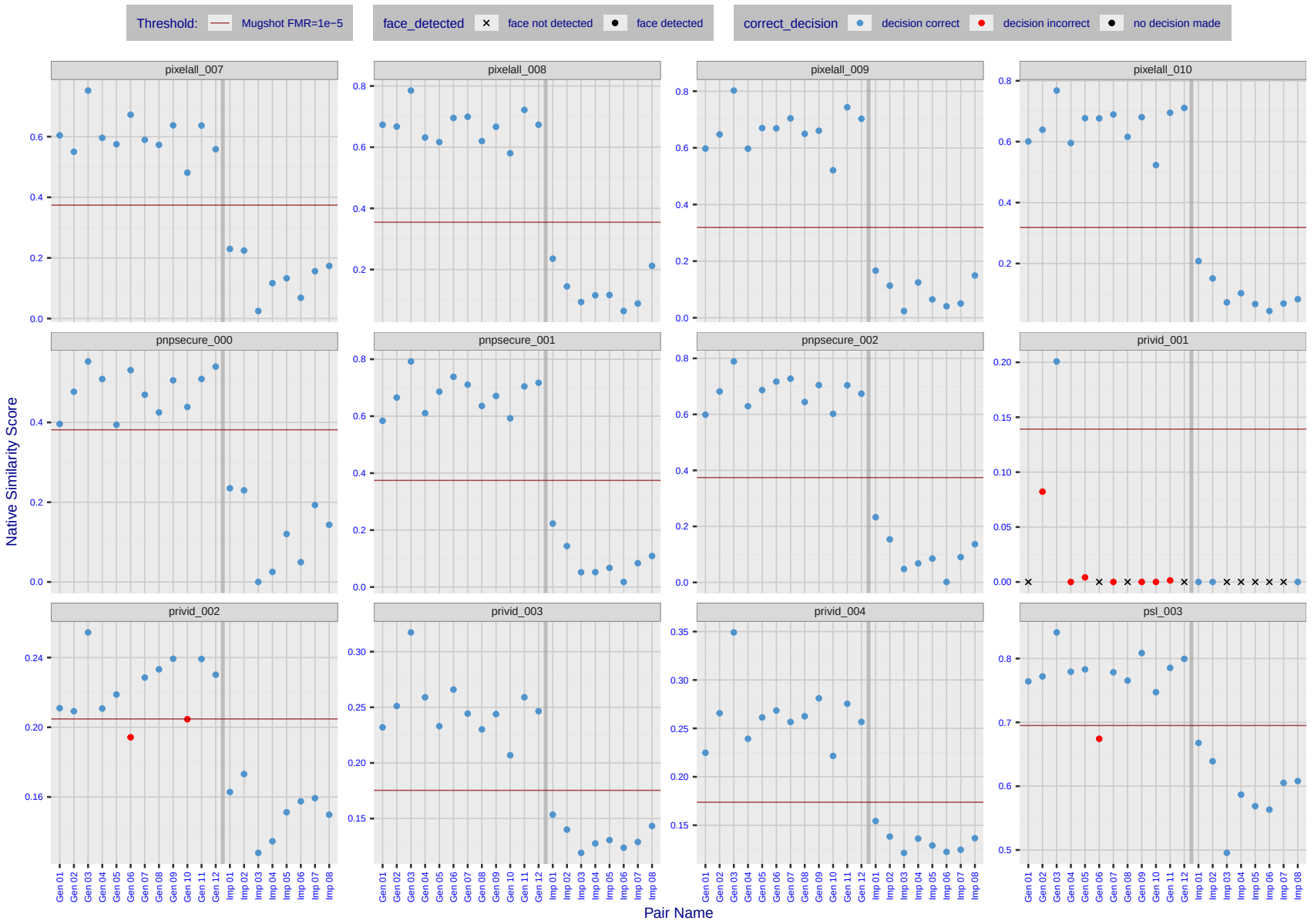

Figure 74: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

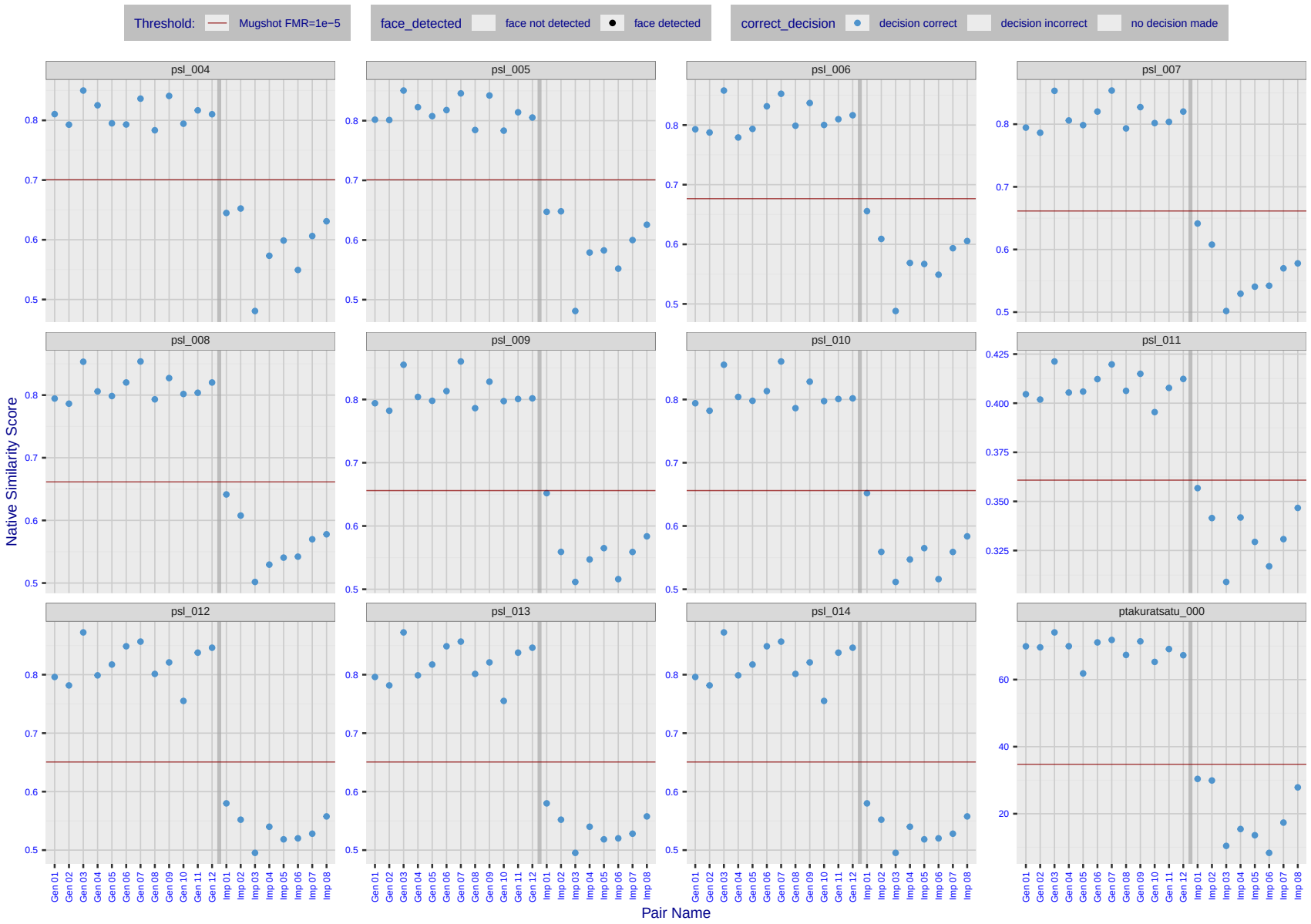

Figure 75: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

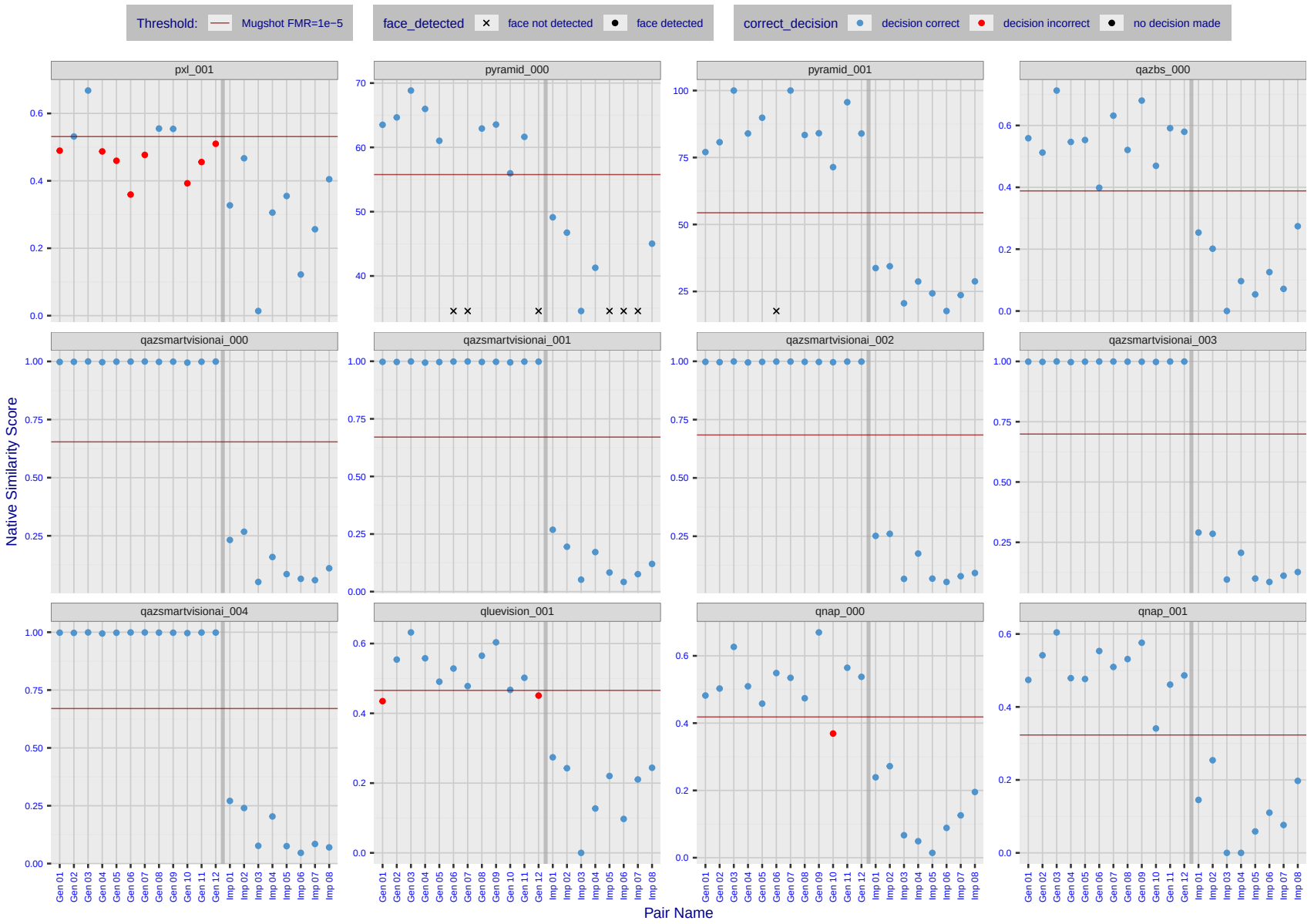

Figure 76: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

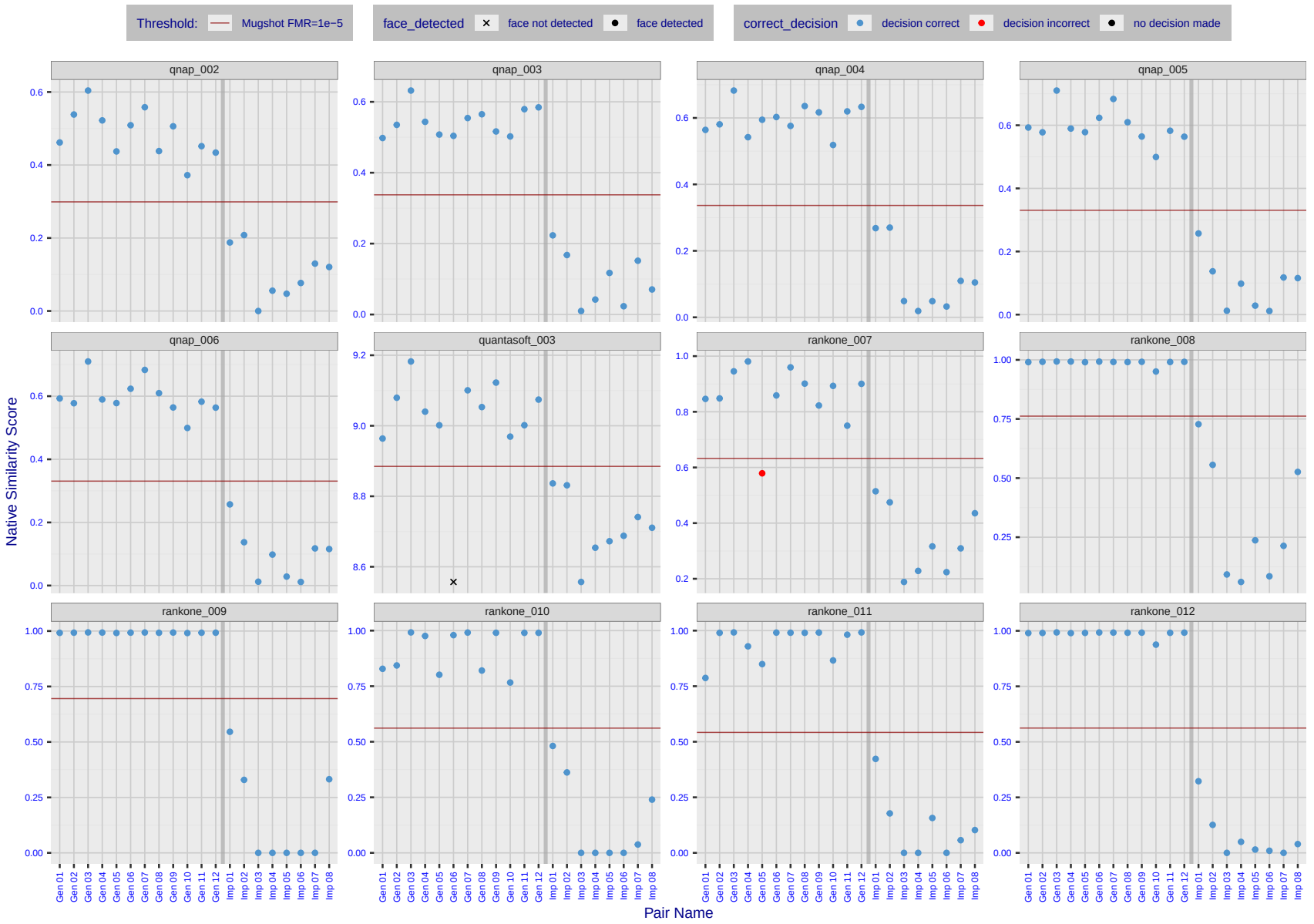

Figure 77: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

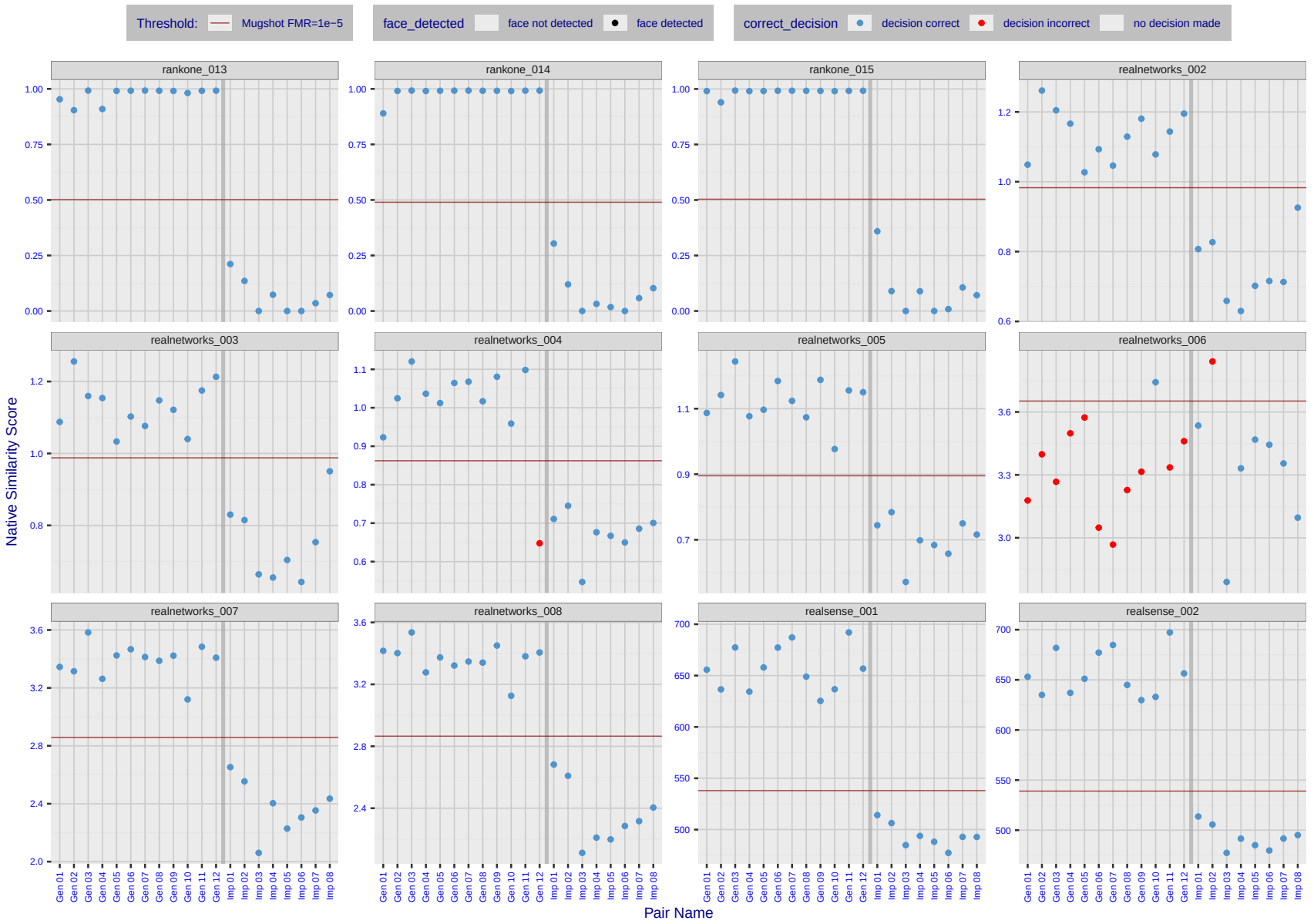

Figure 78: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

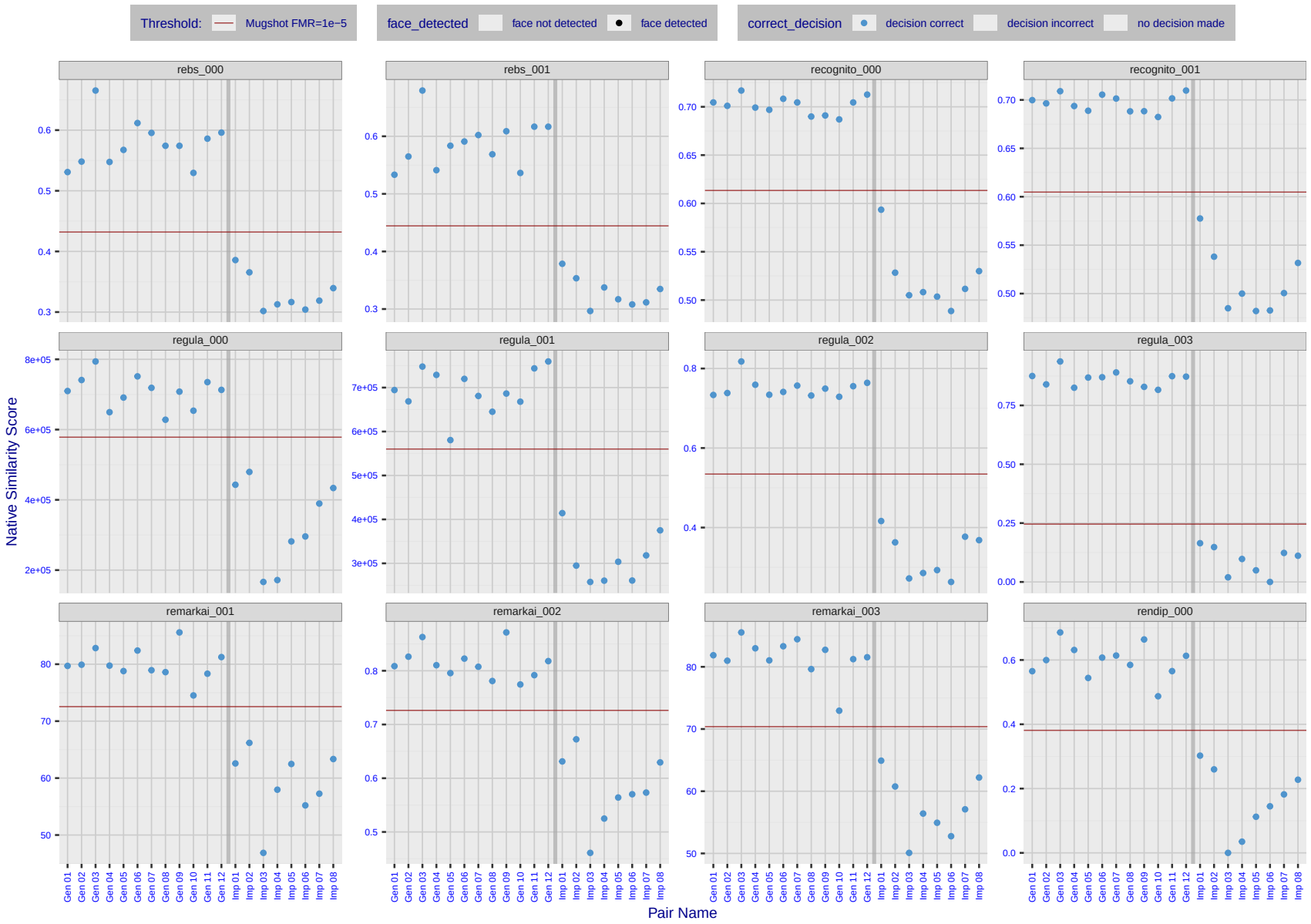

Figure 79: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

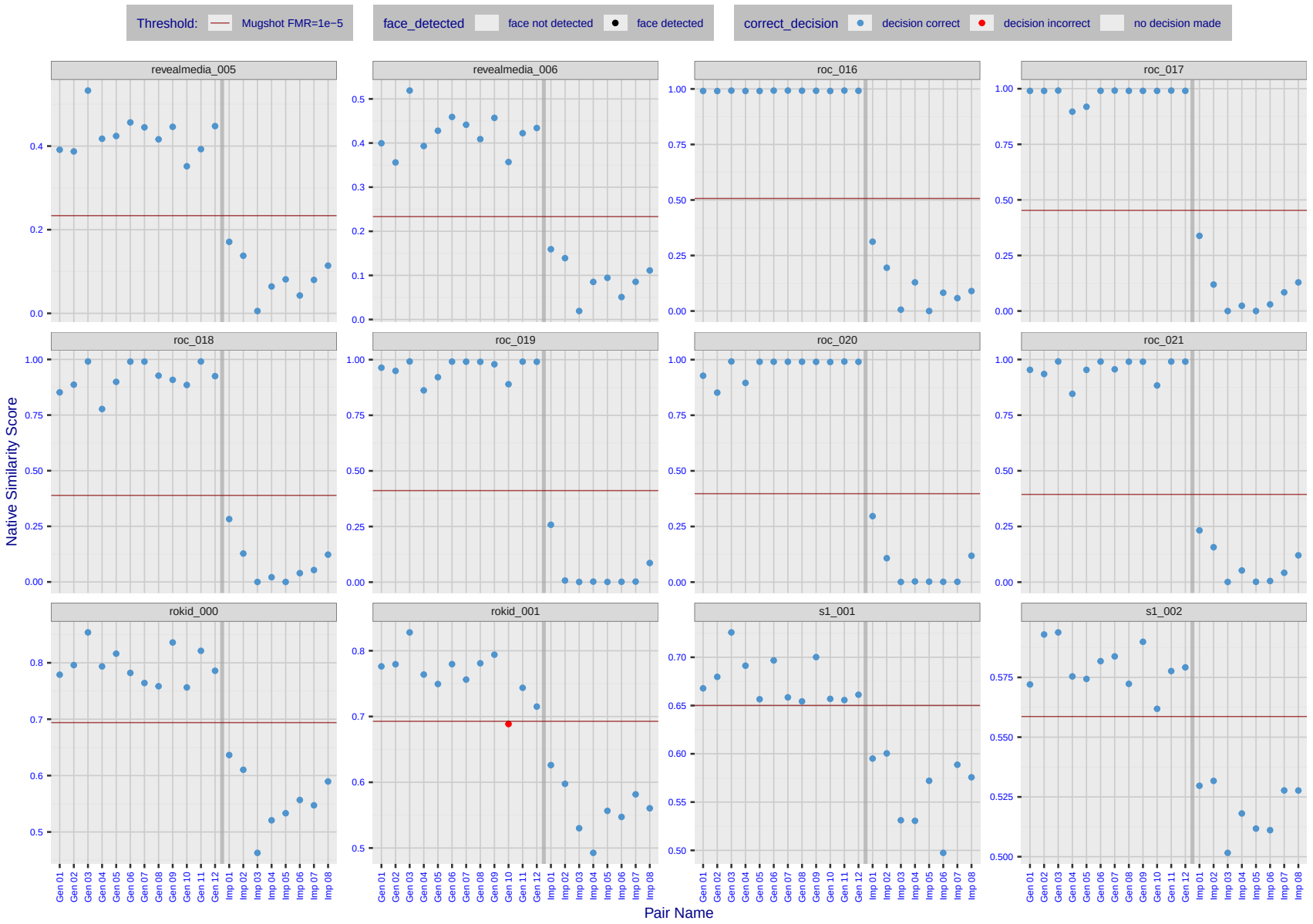

Figure 80: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

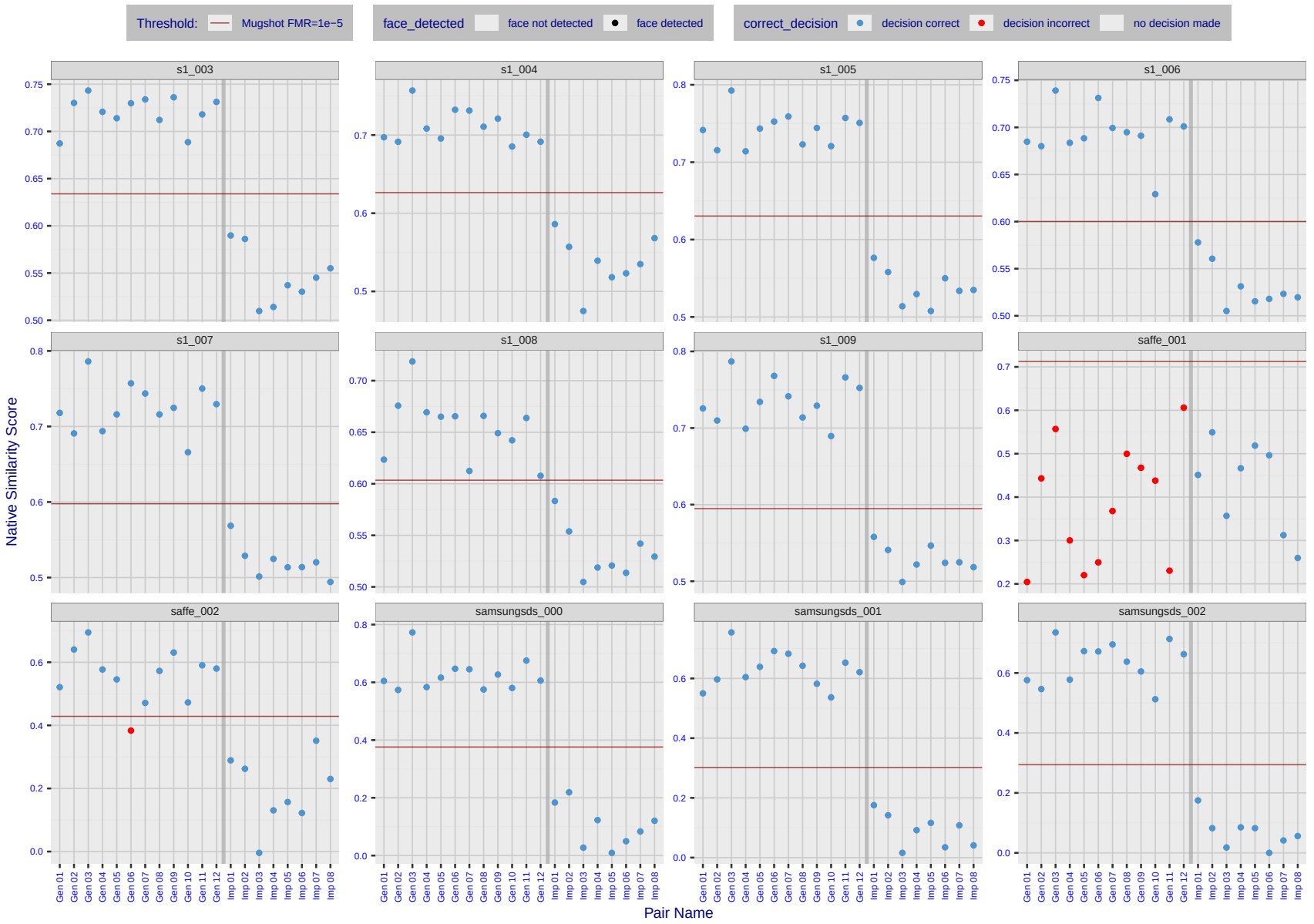

Figure 81: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

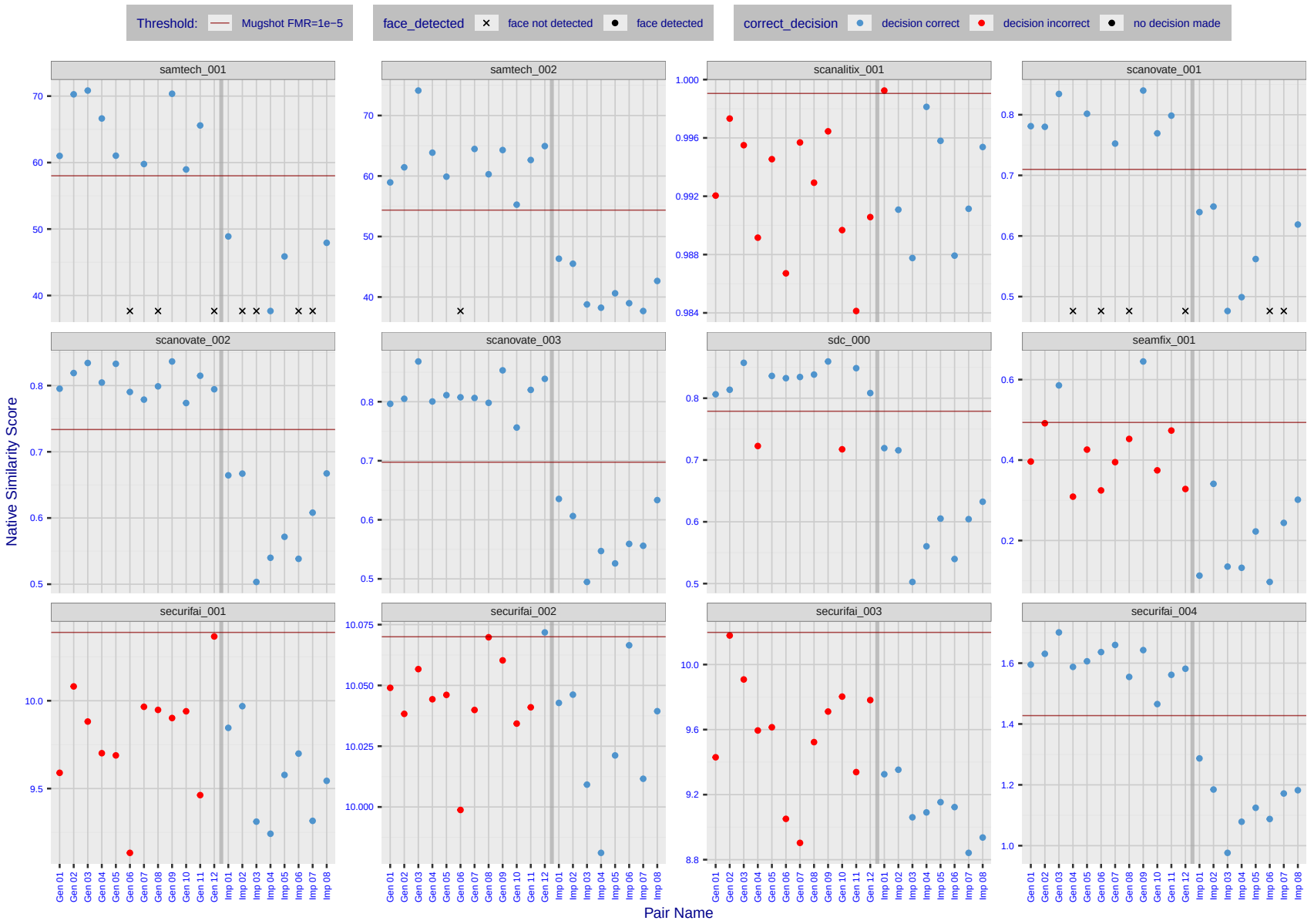

Figure 82: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

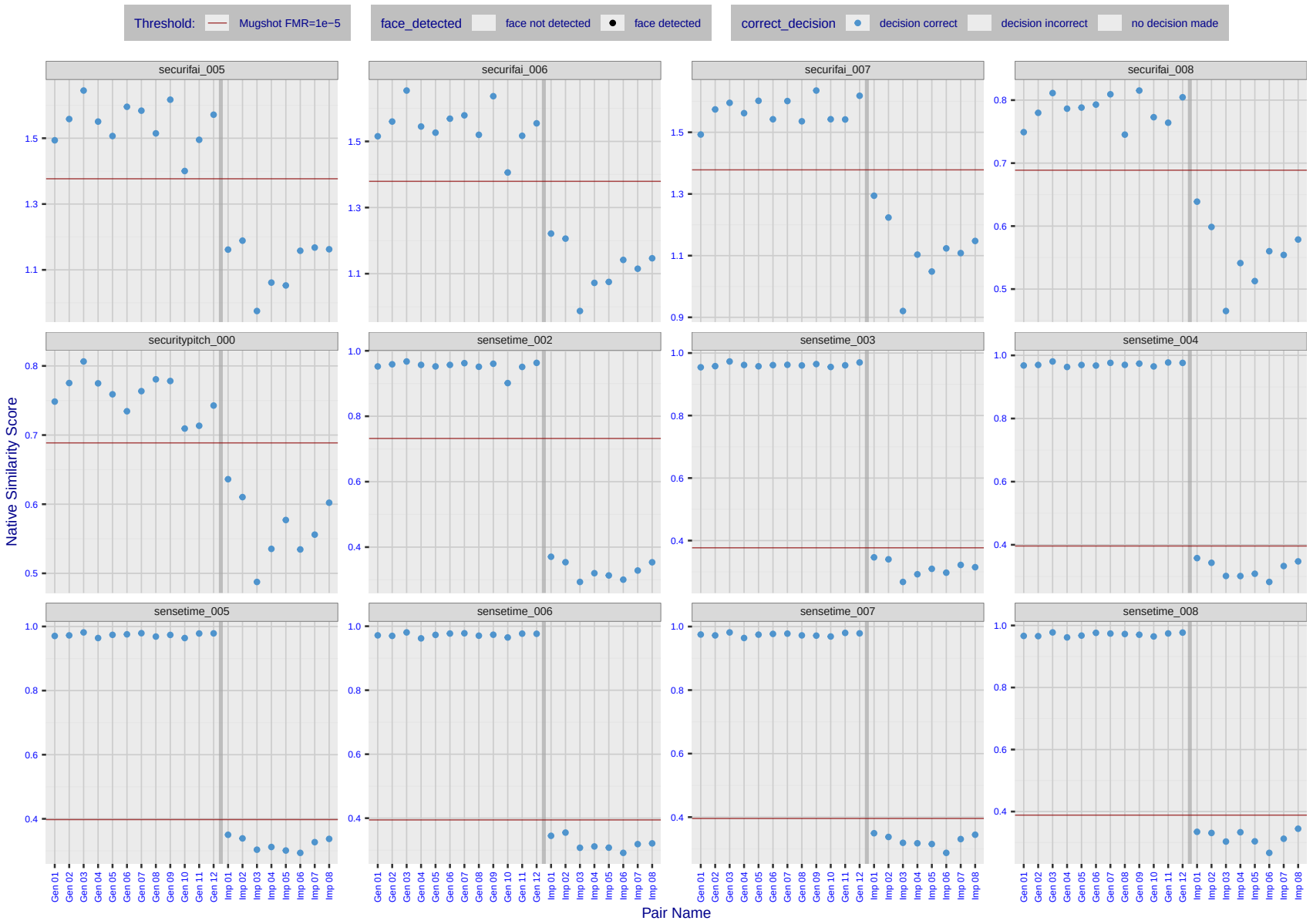

Figure 83: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

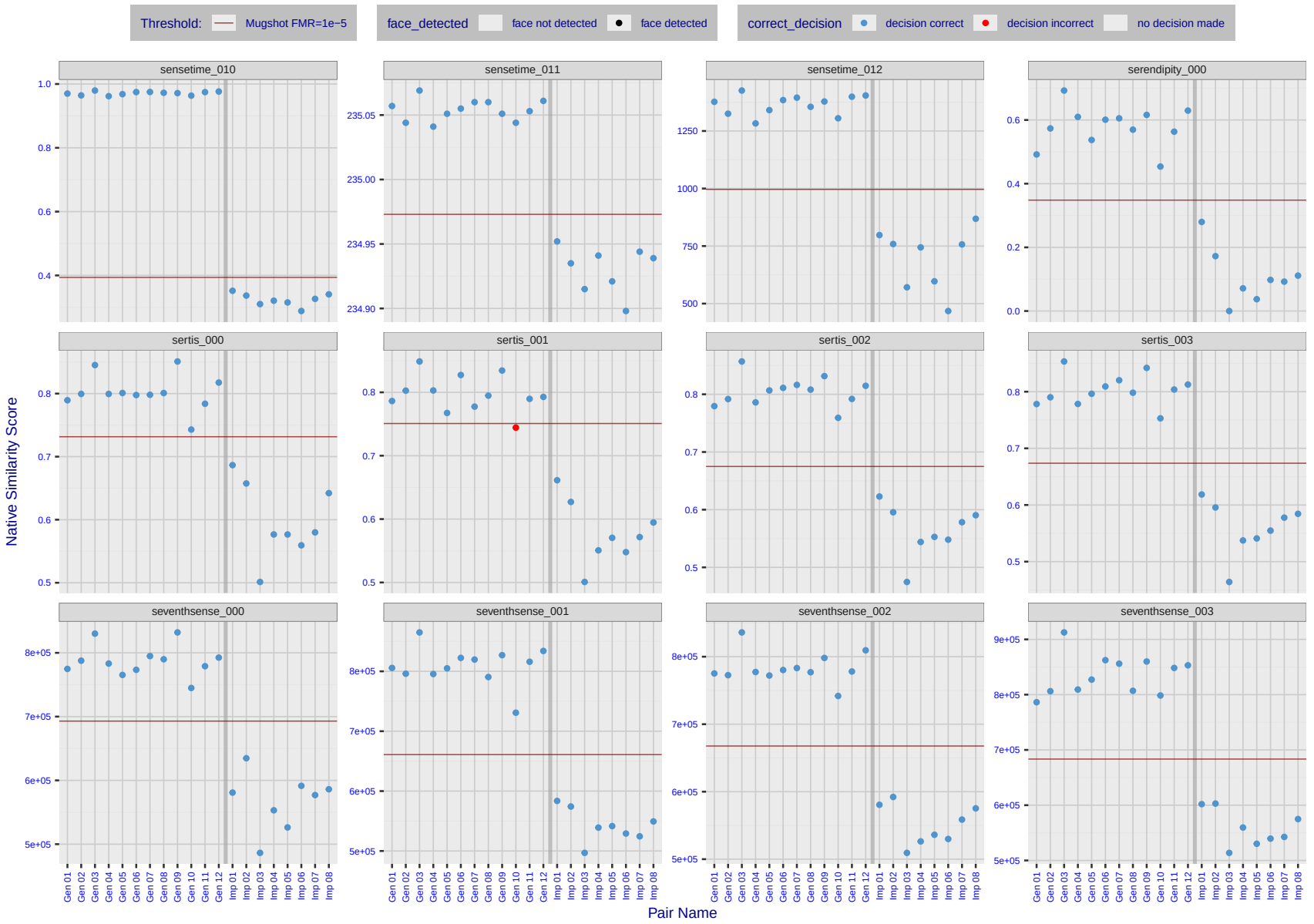

Figure 84: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

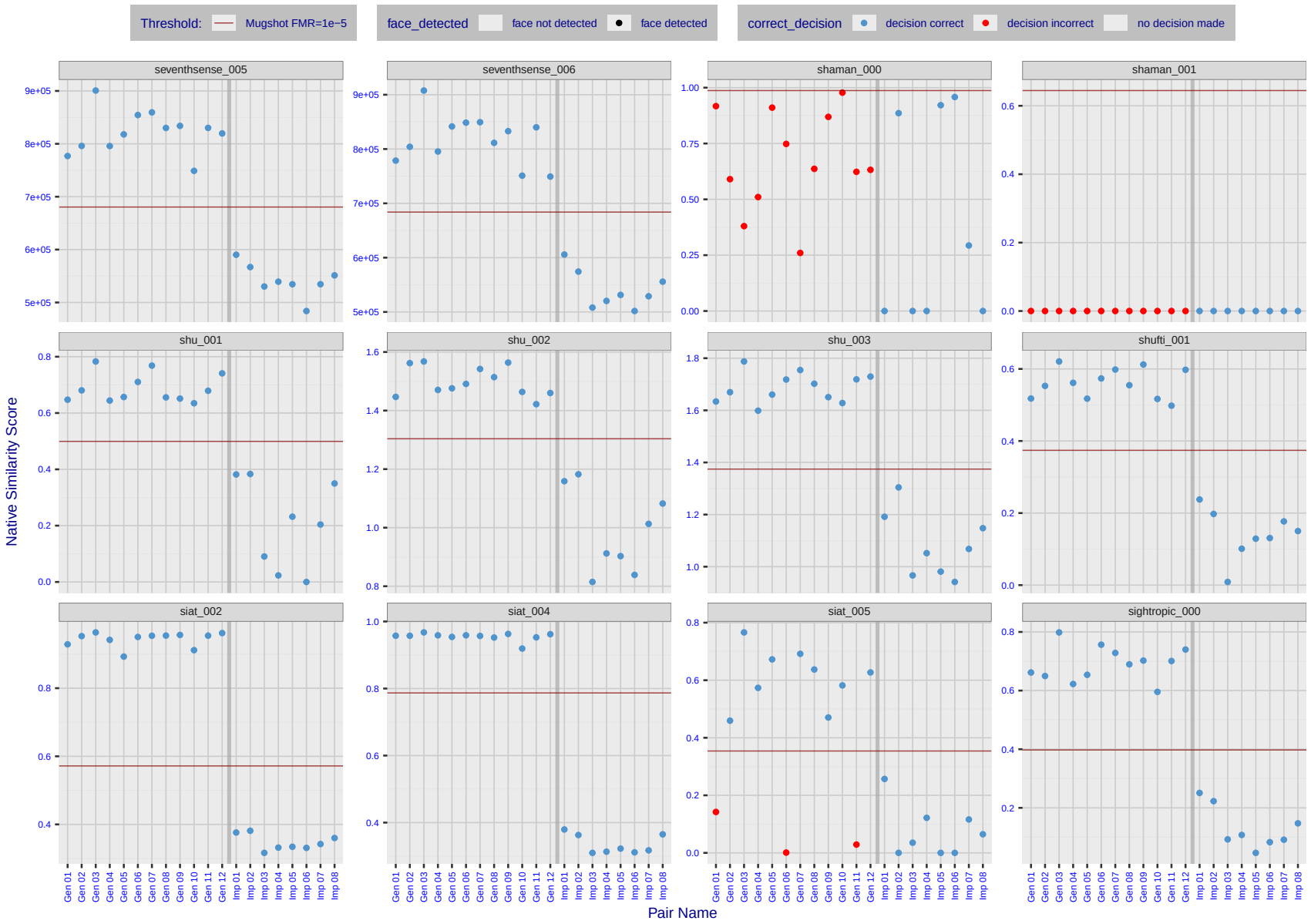

Figure 85: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

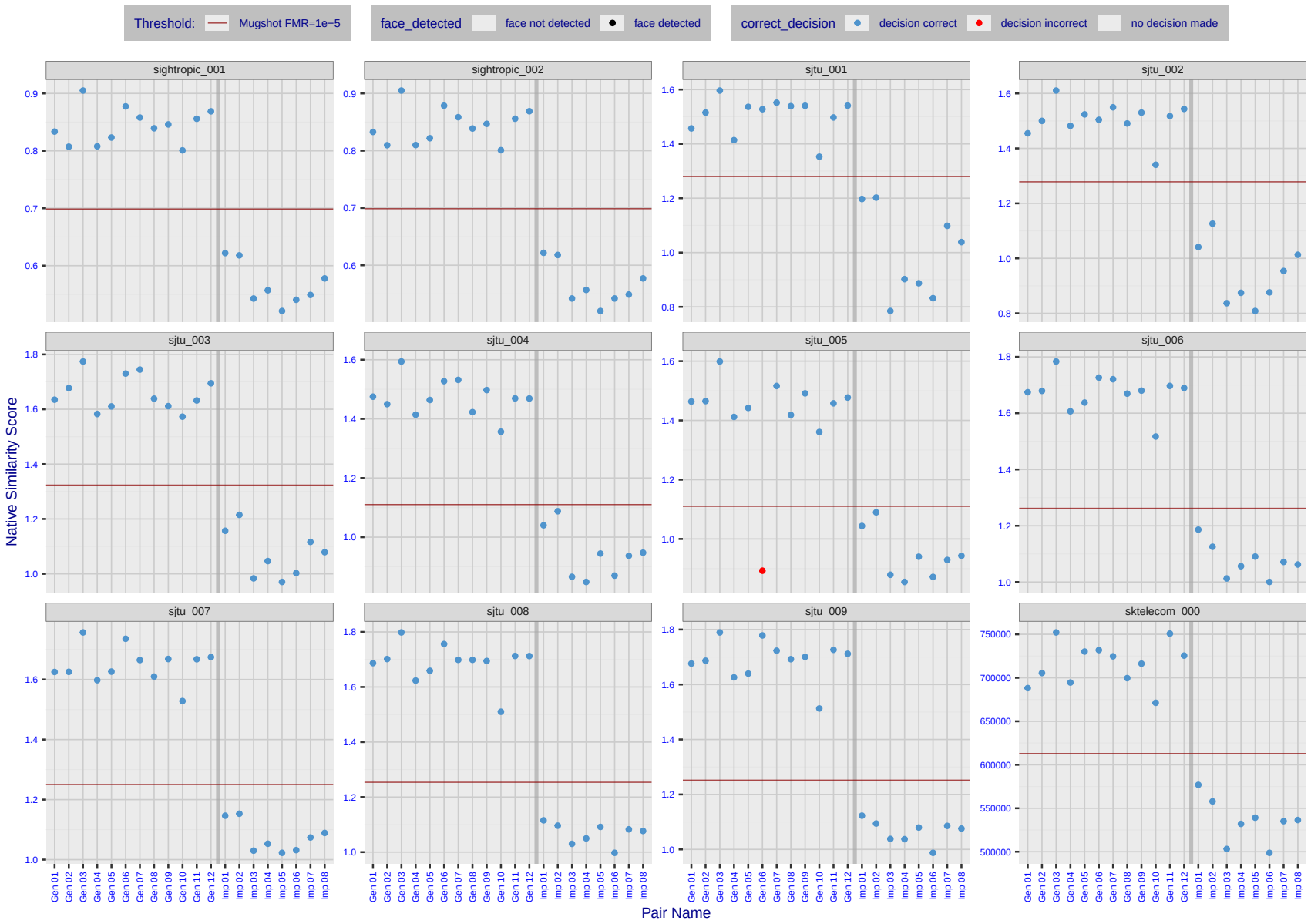

Figure 86: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

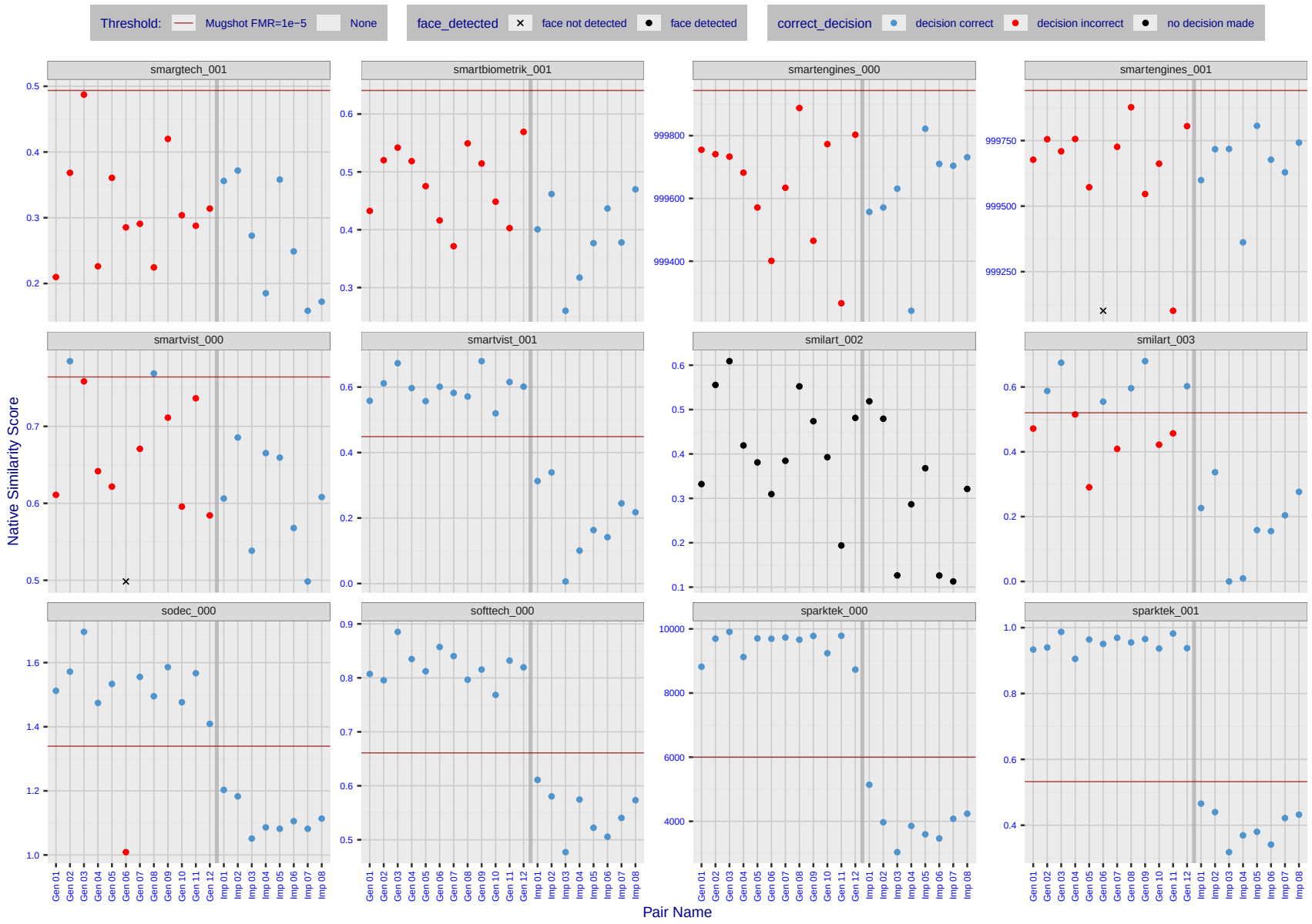

Figure 87: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

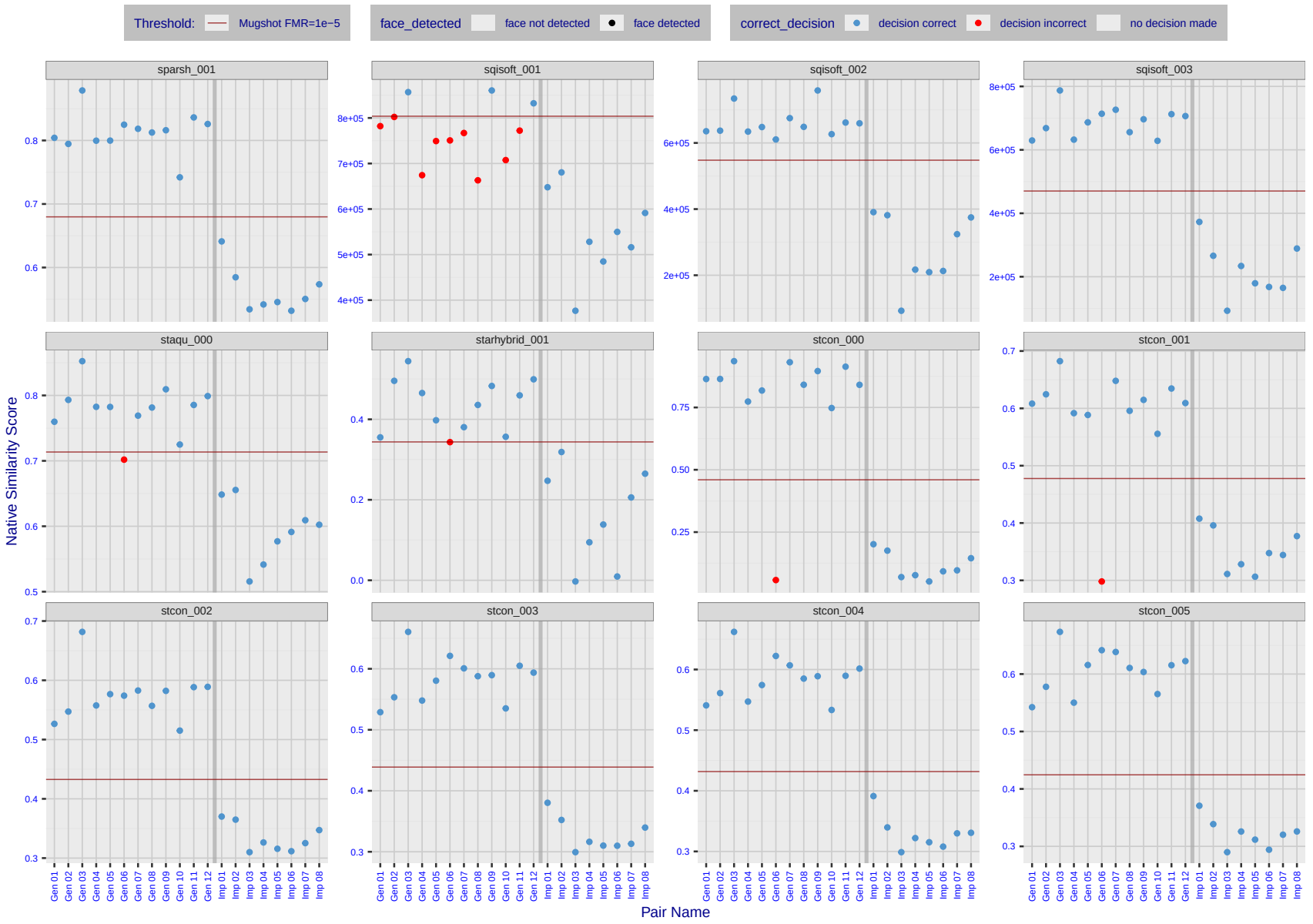

Figure 88: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

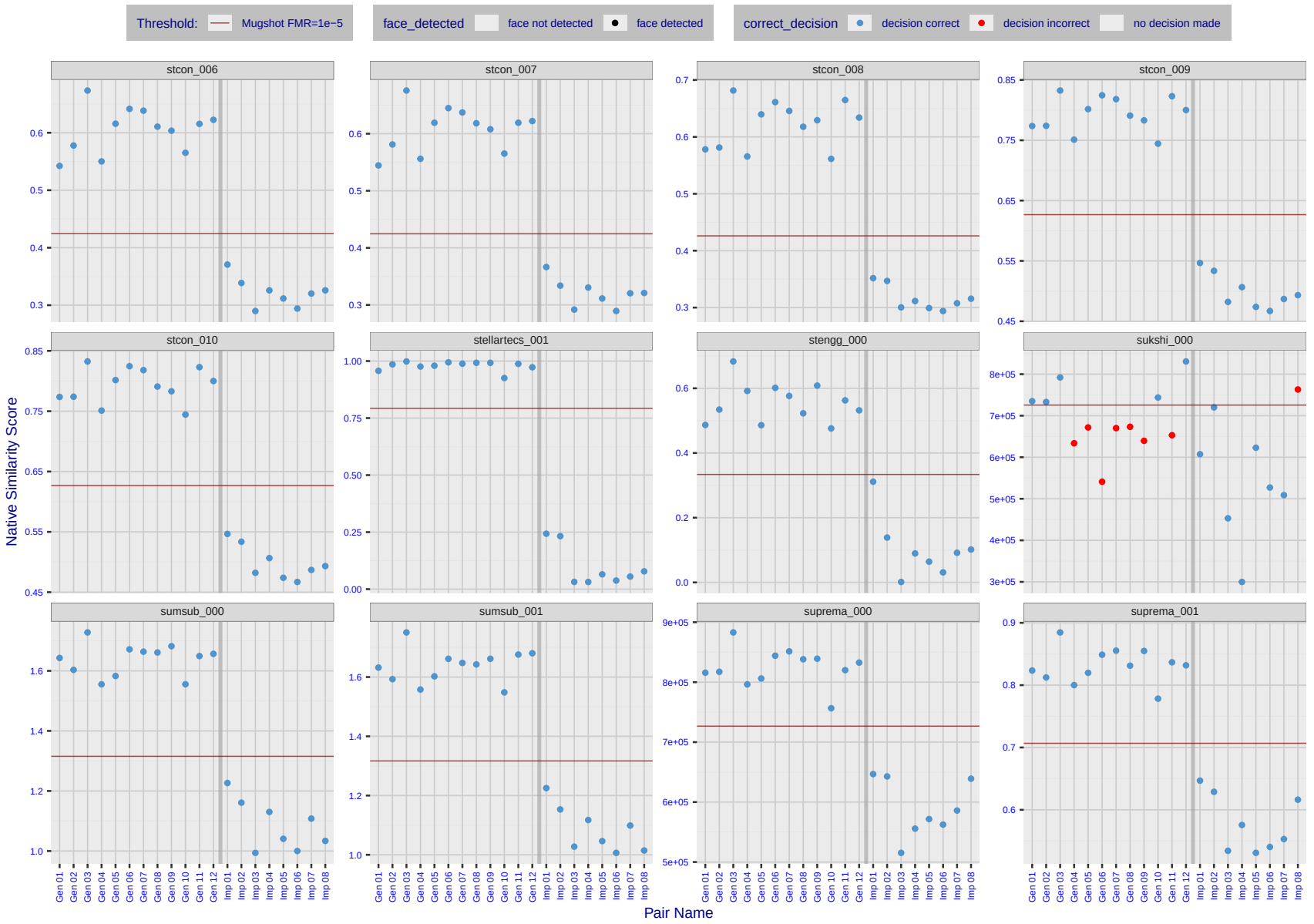

Figure 89: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

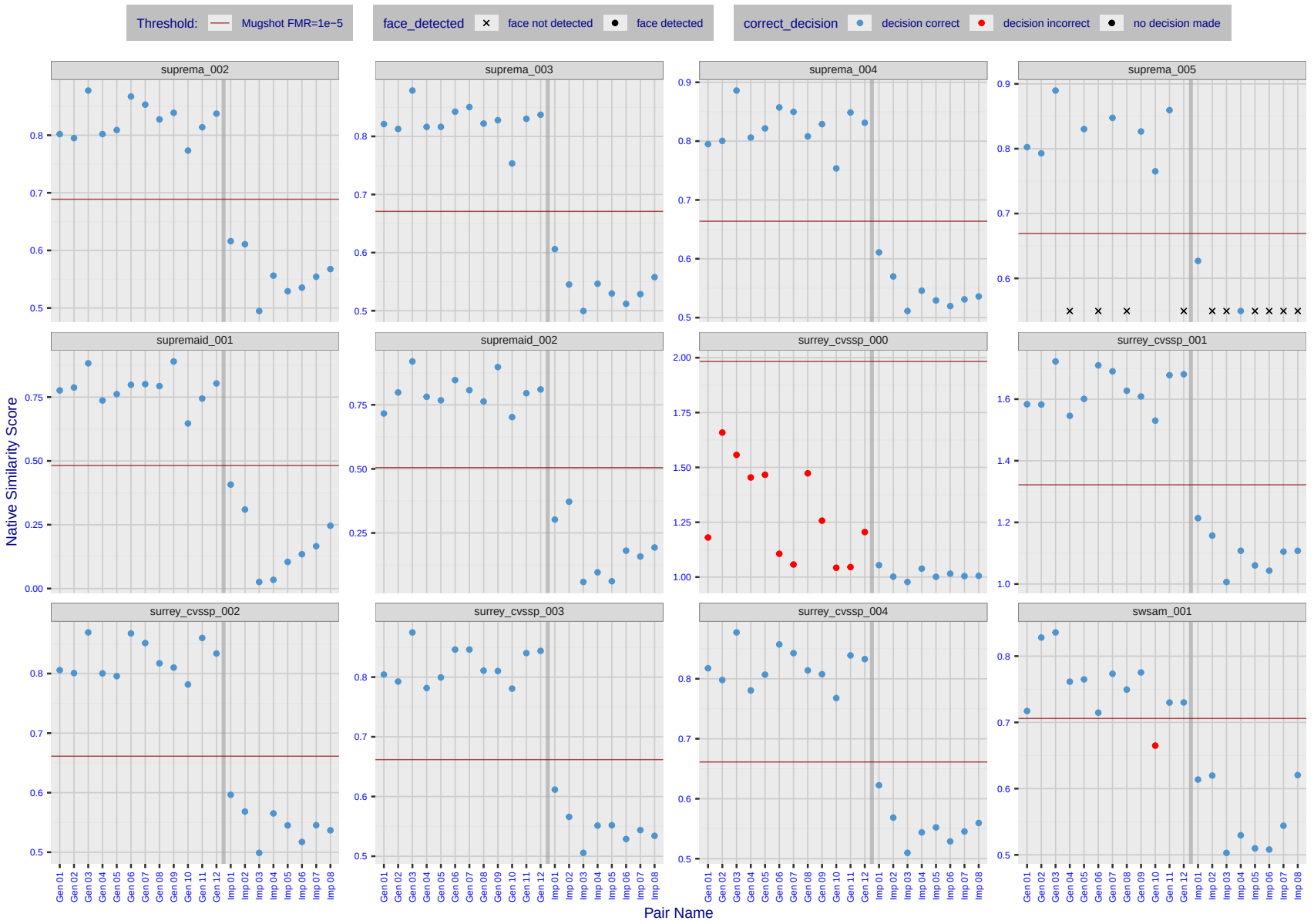

Figure 90: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

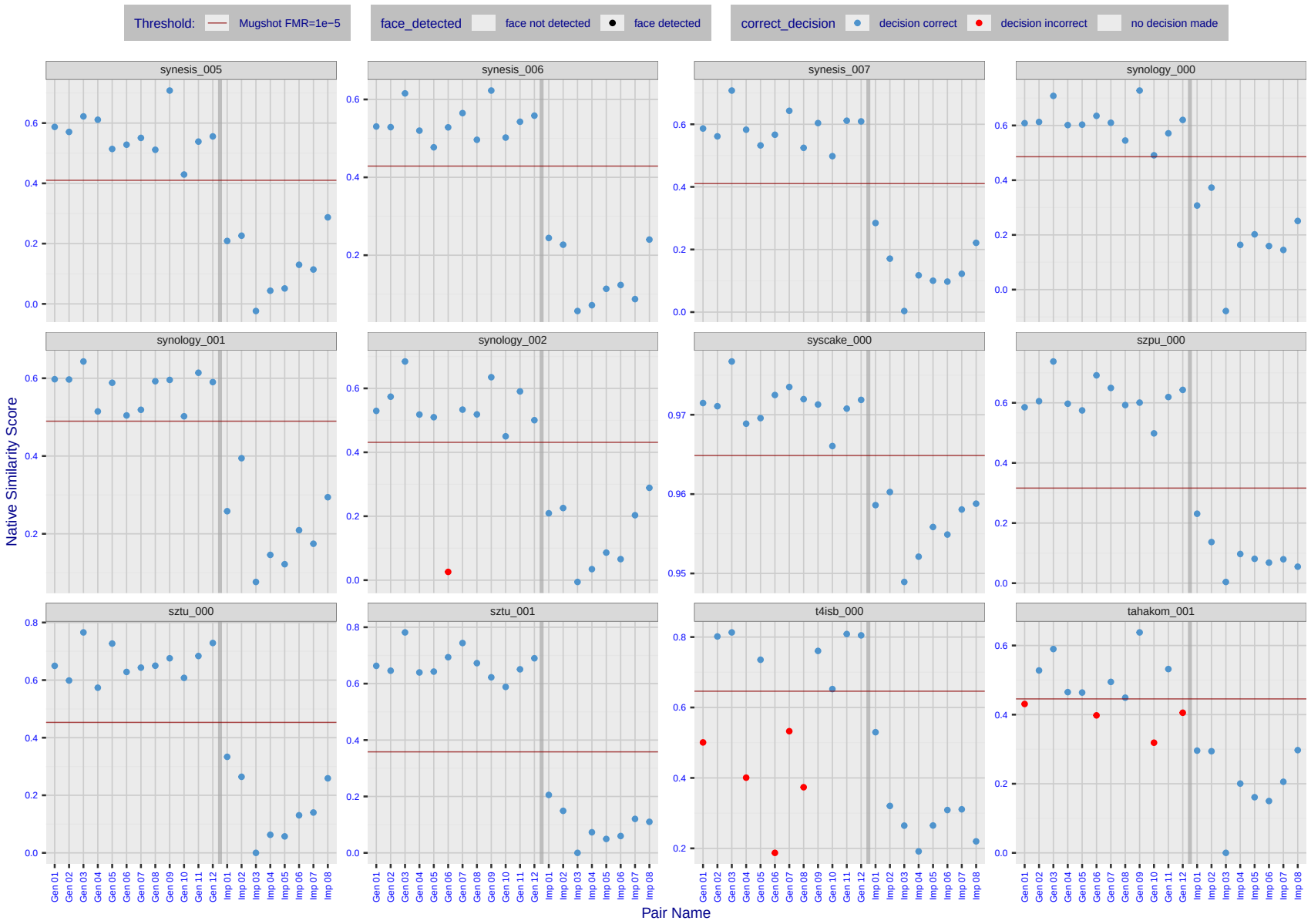

Figure 91: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

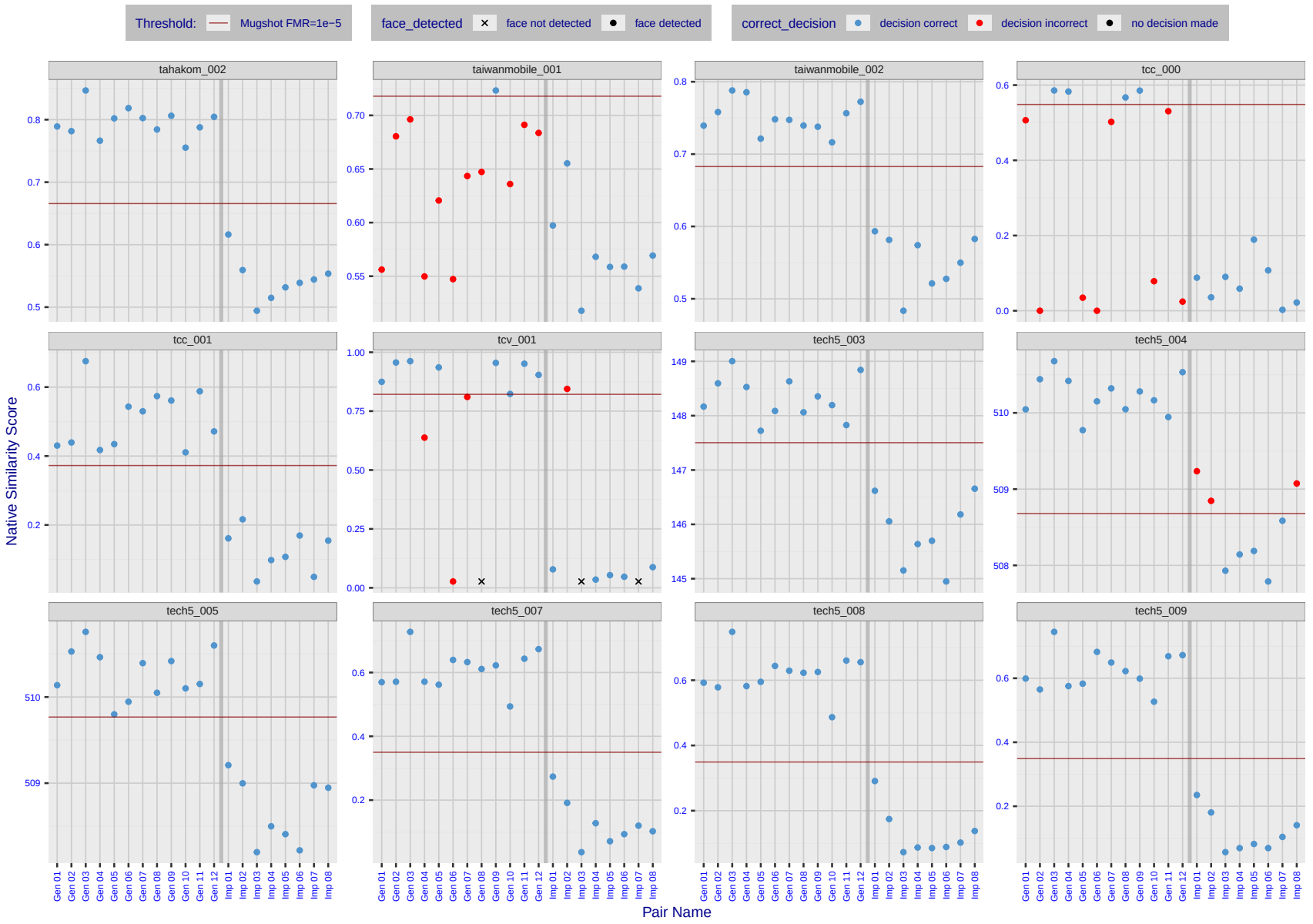

Figure 92: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

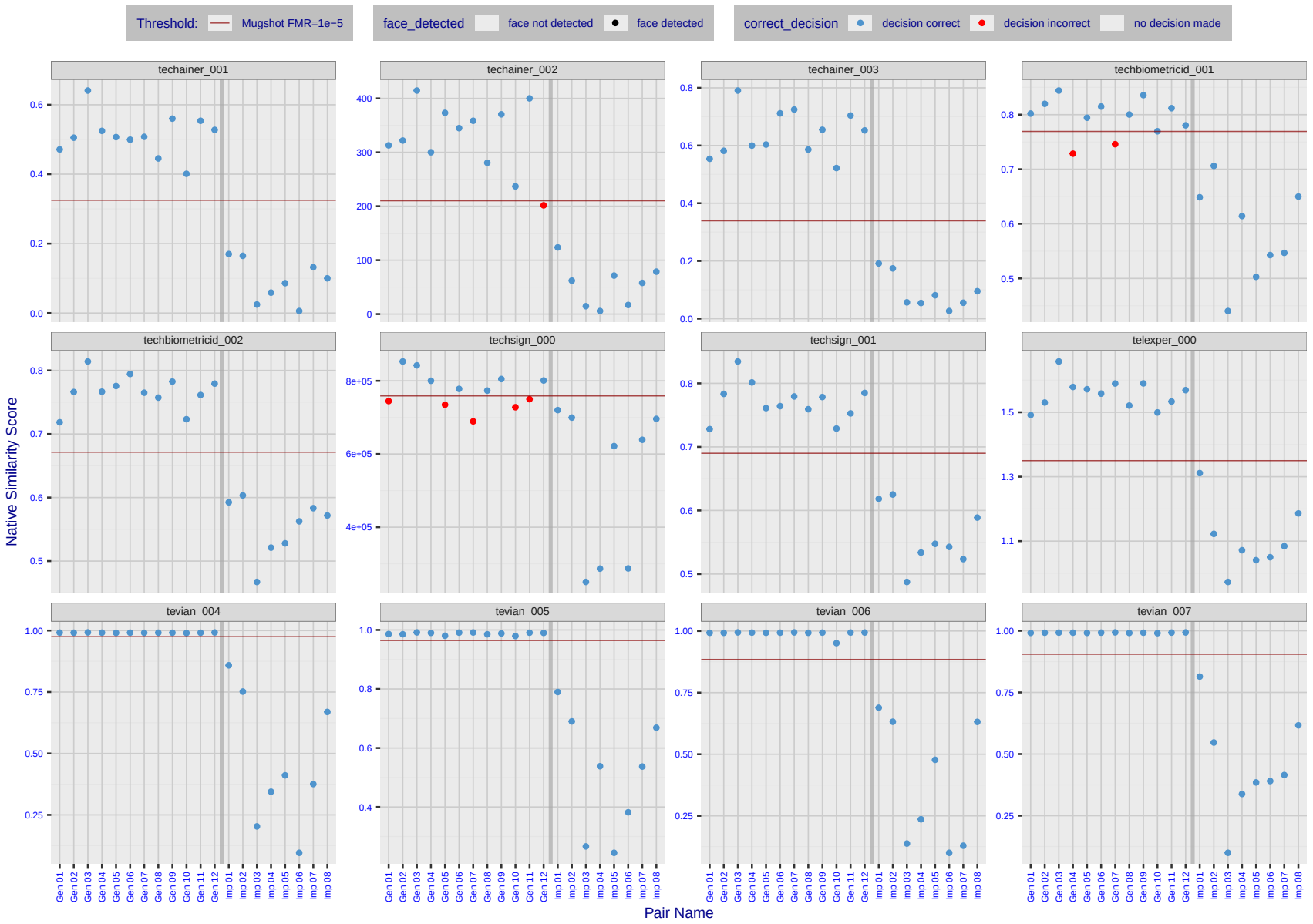

Figure 93: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

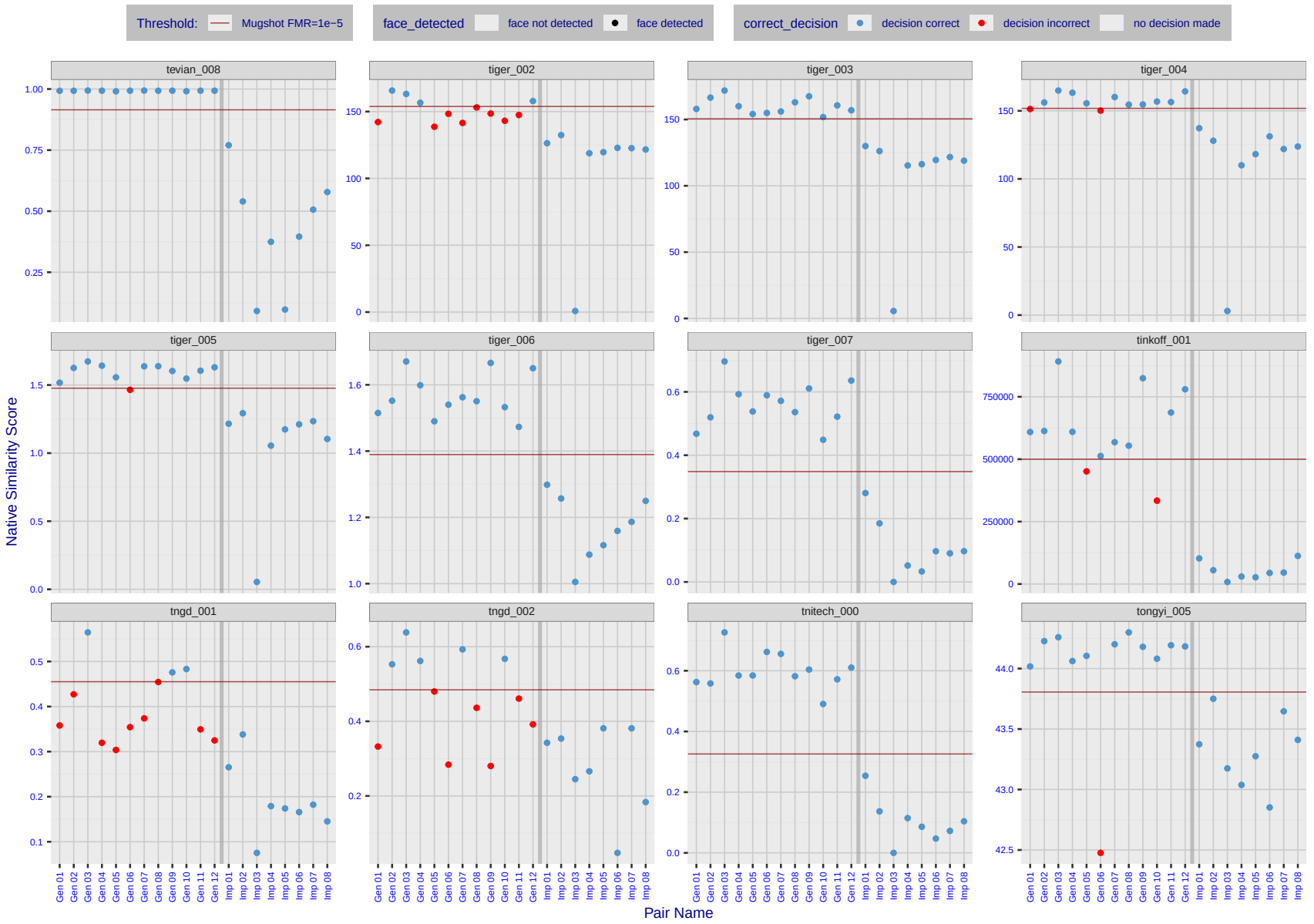

Figure 94: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

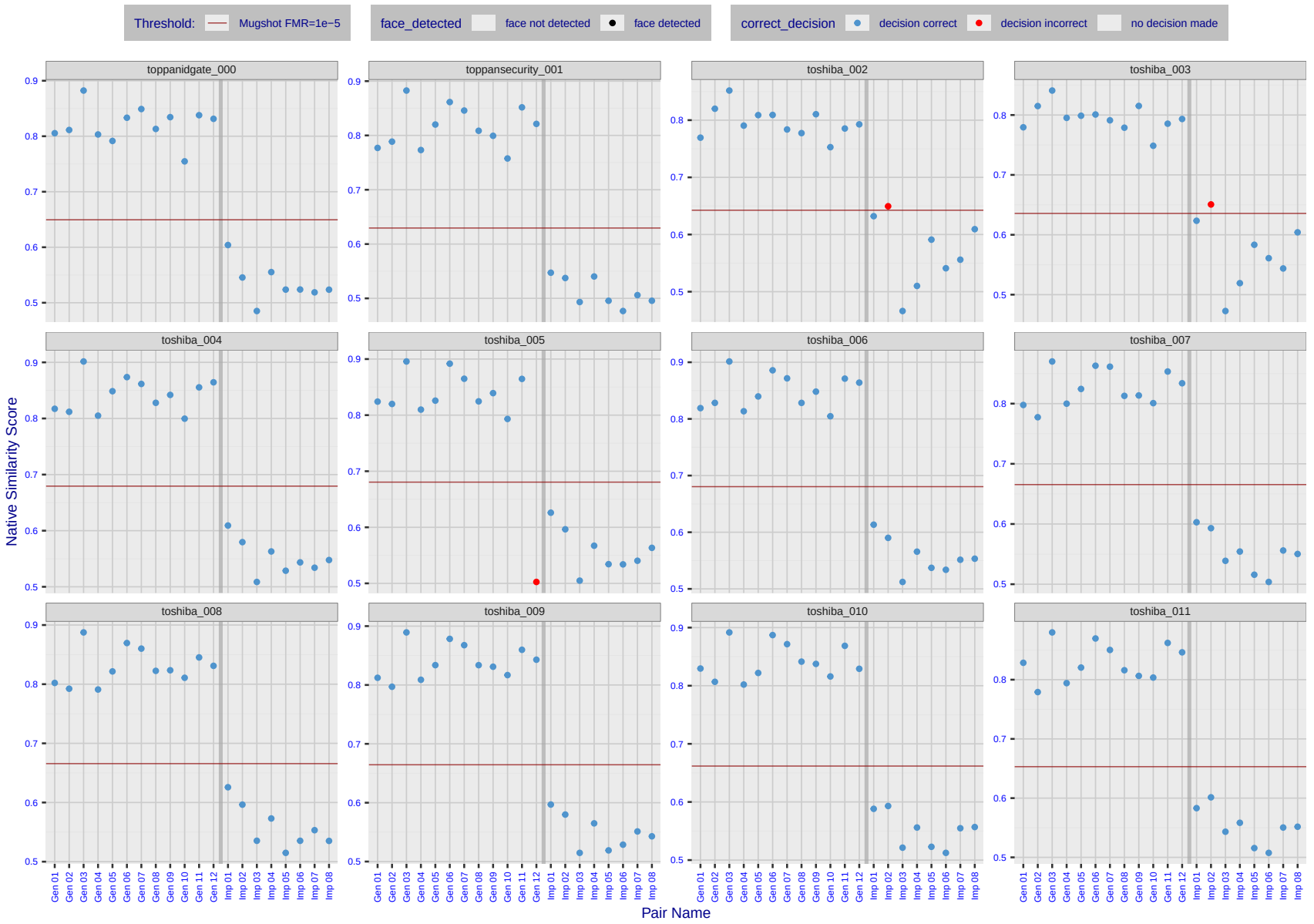

Figure 95: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

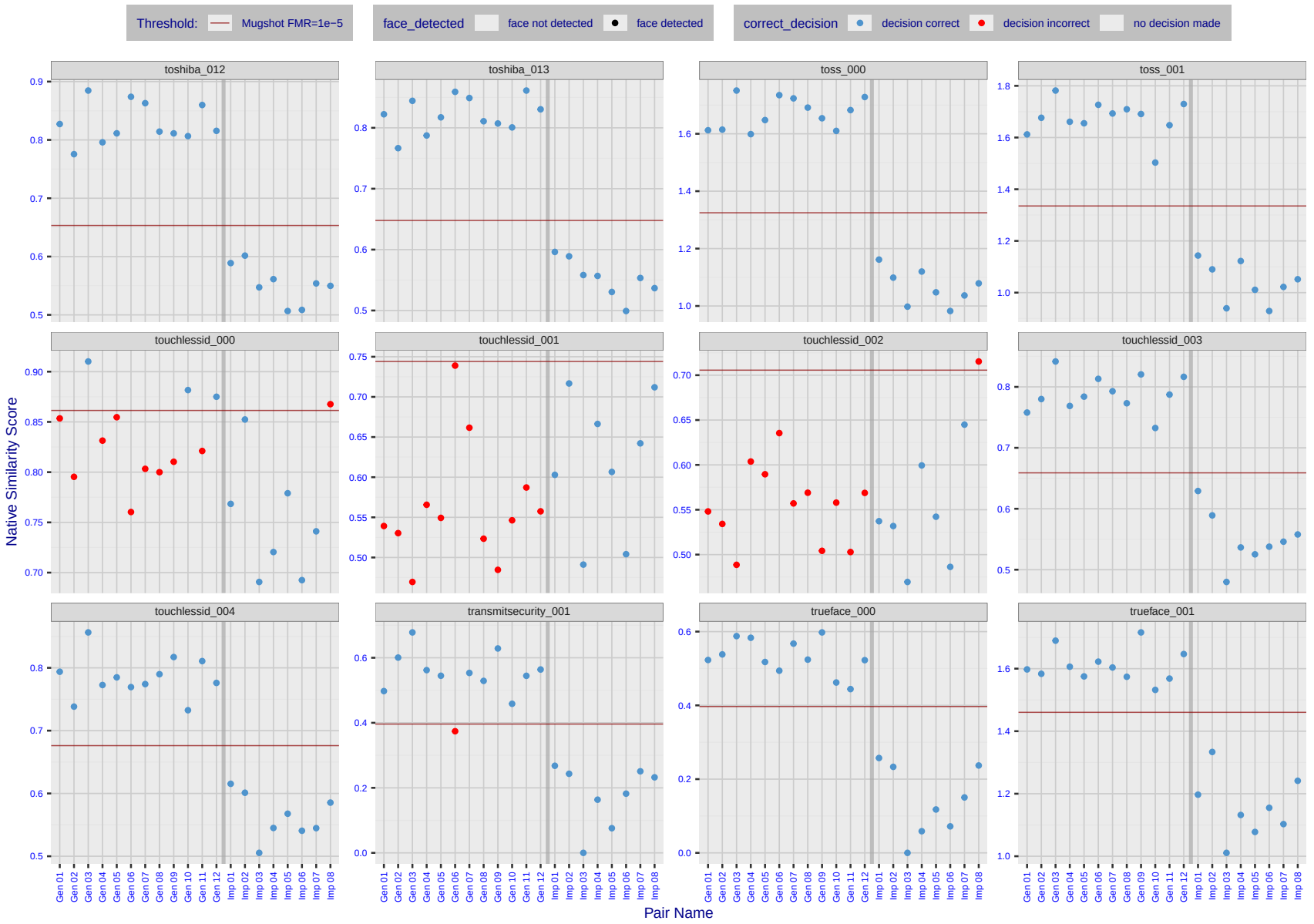

Figure 96: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

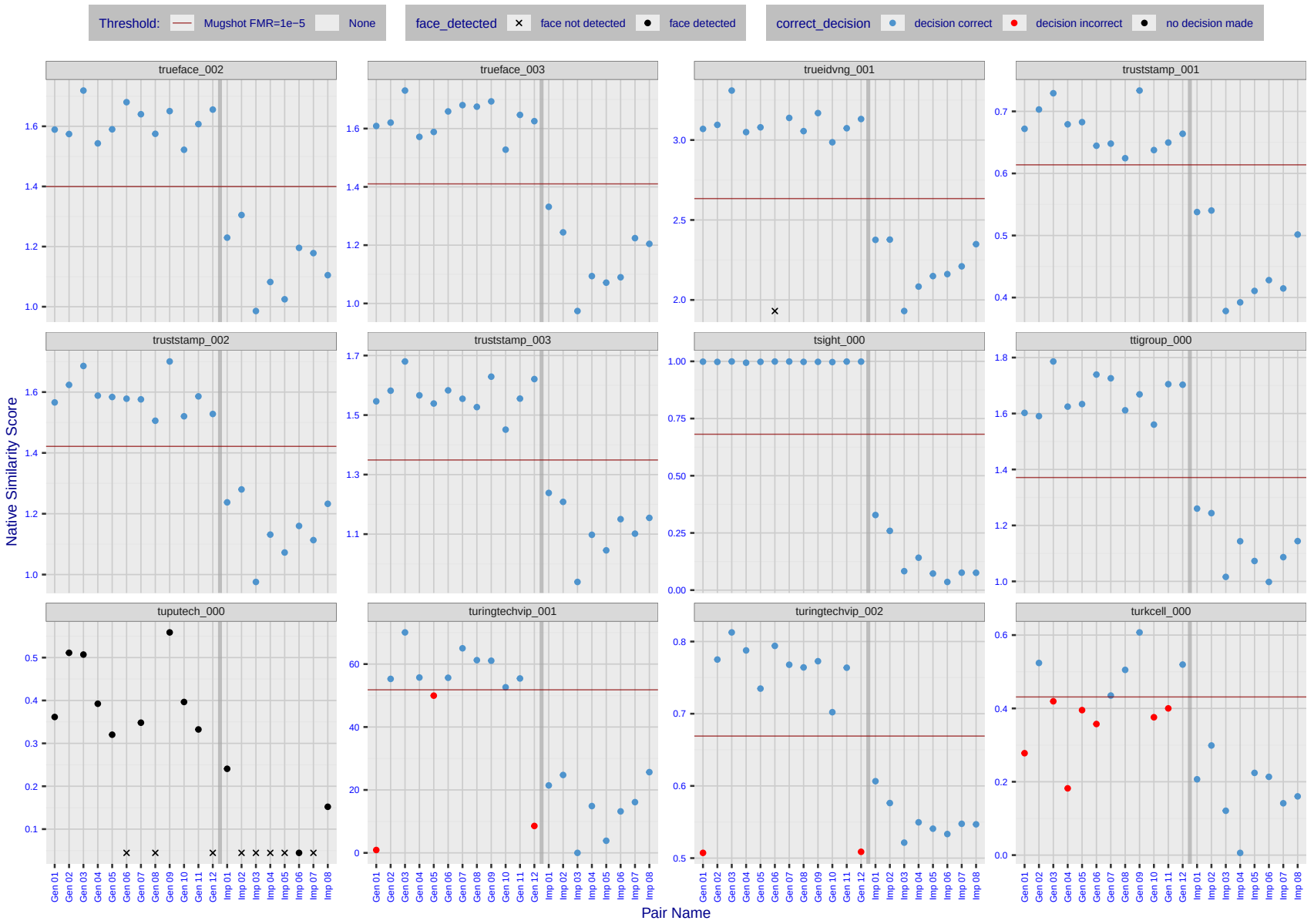

Figure 97: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

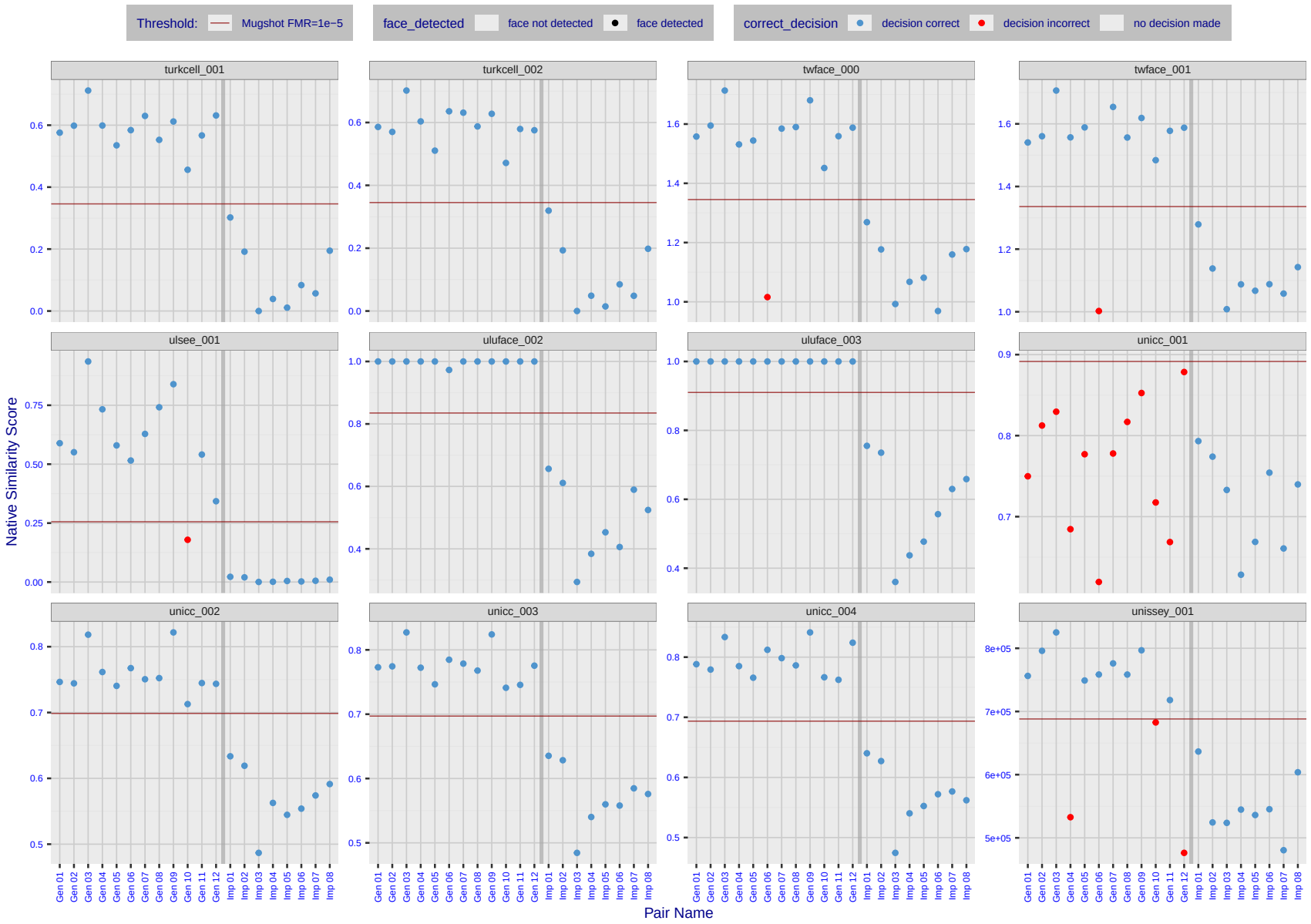

Figure 98: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

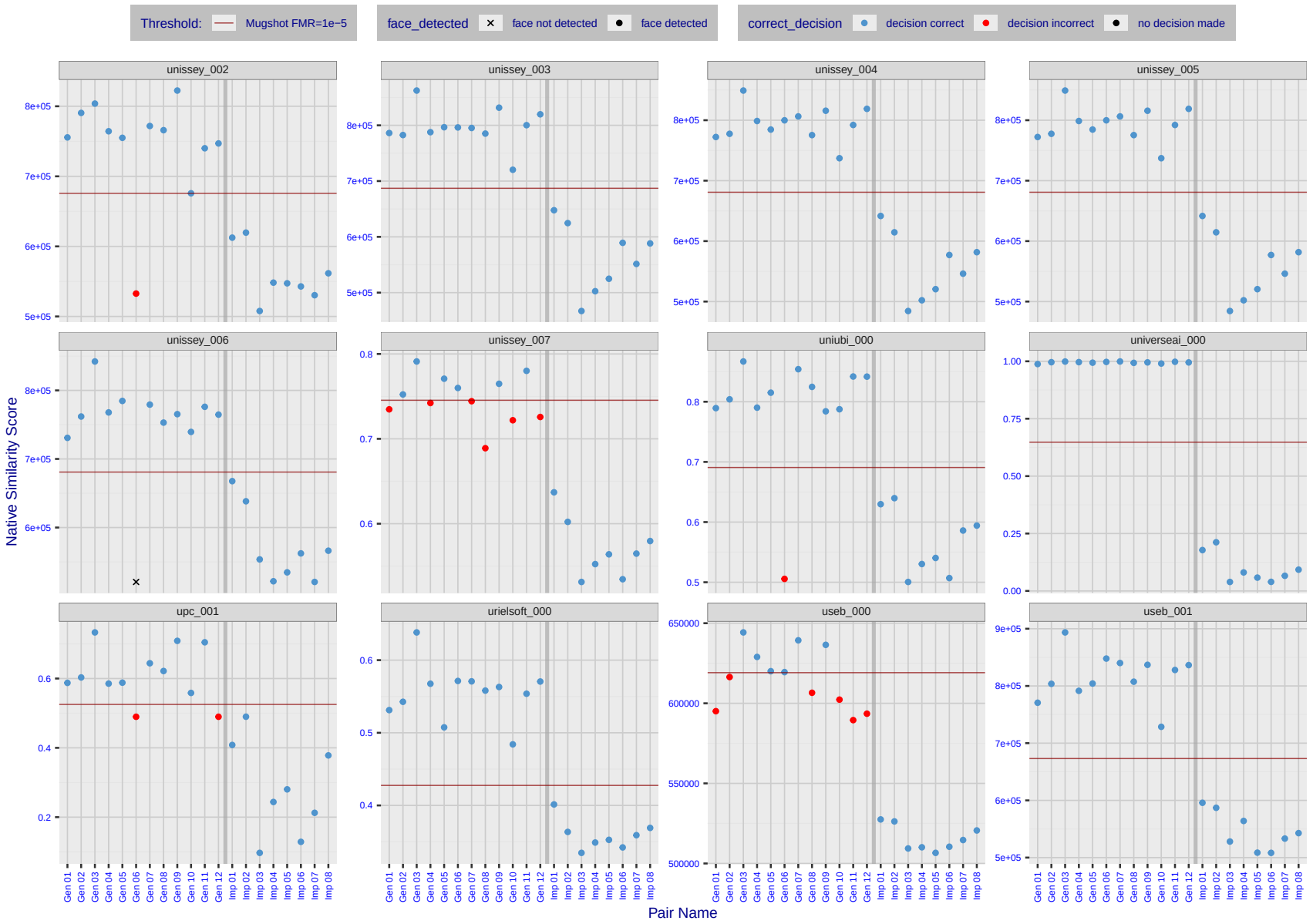

Figure 99: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

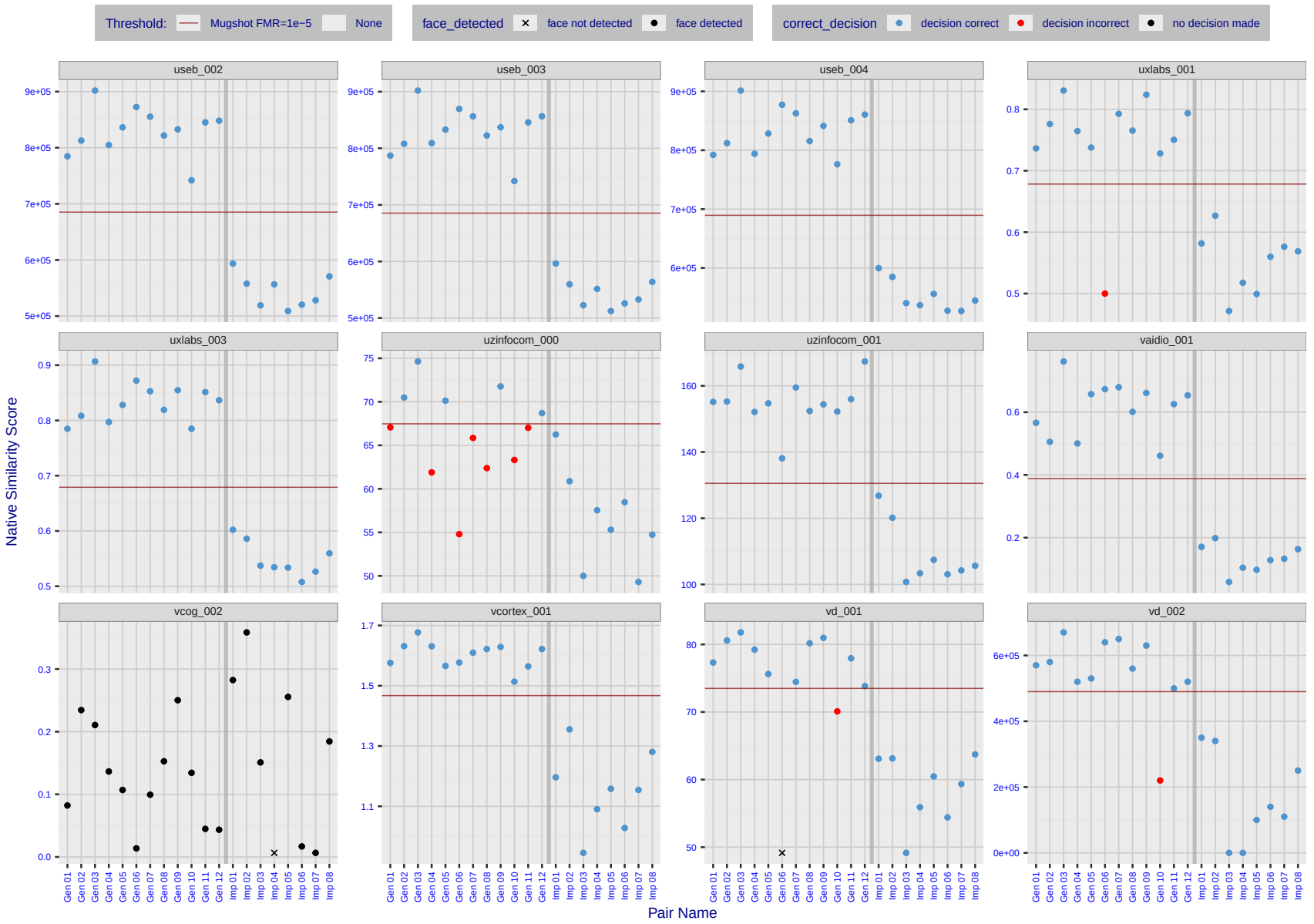

Figure 100: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

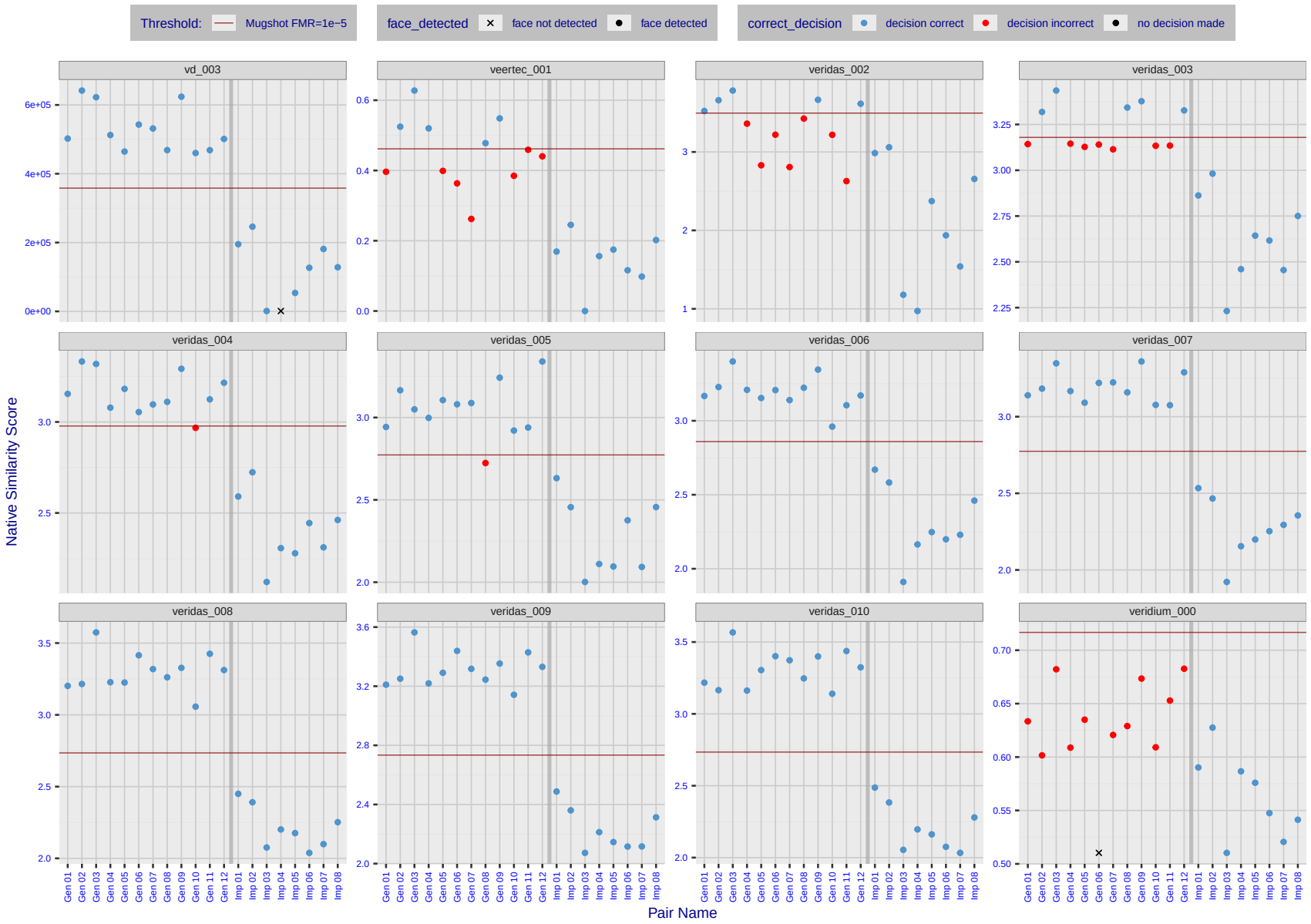

Figure 101: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

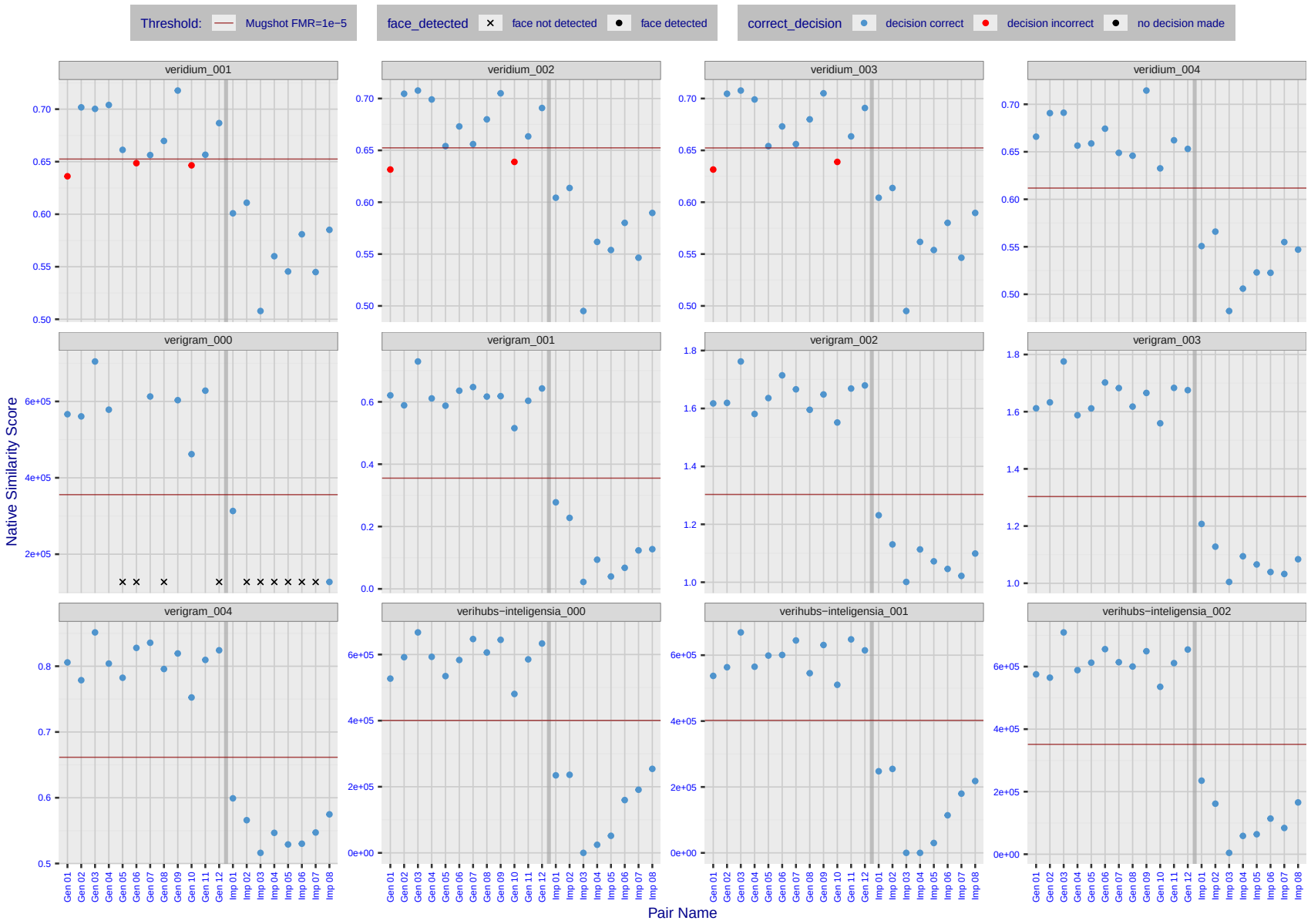

Figure 102: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

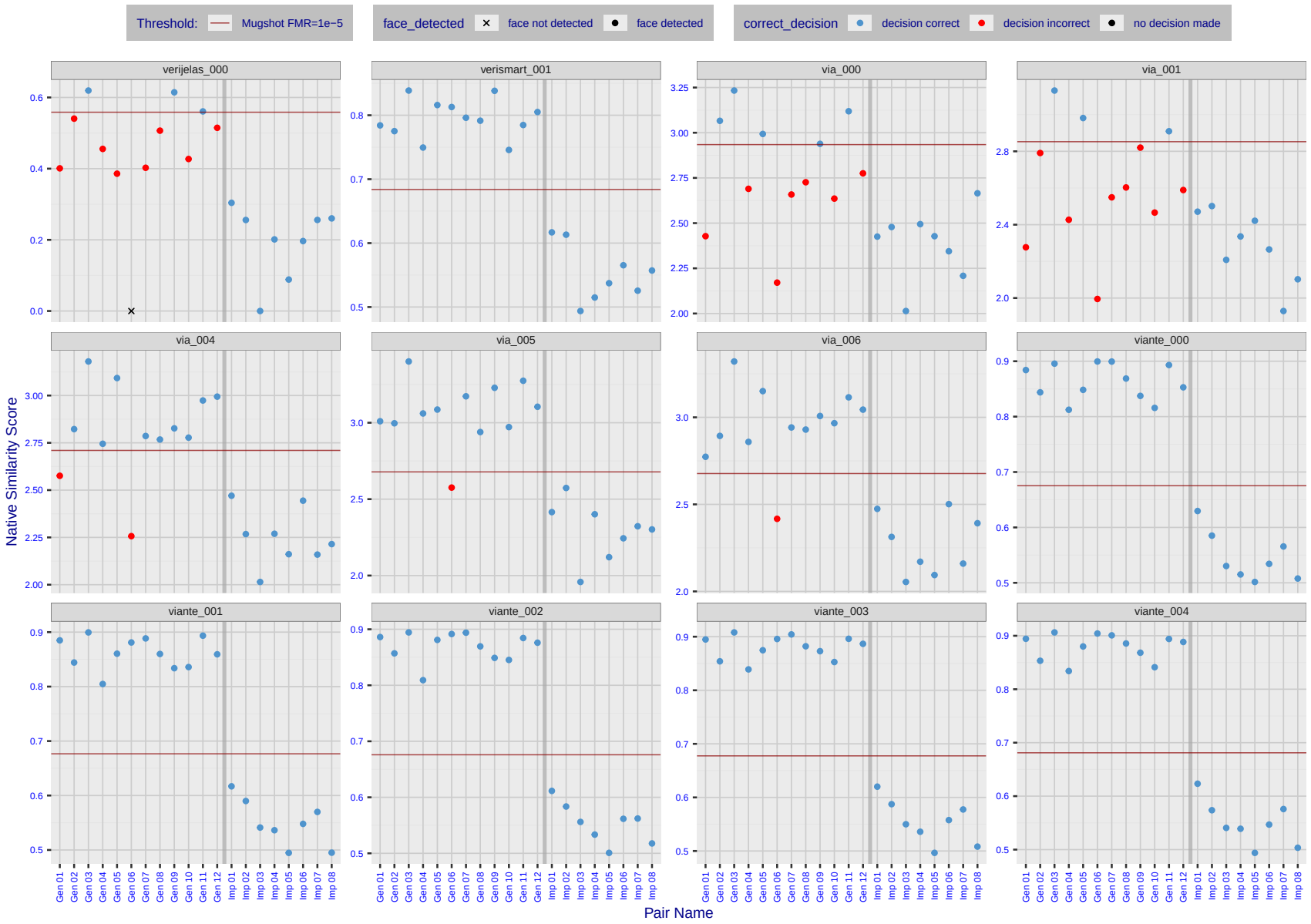

Figure 103: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

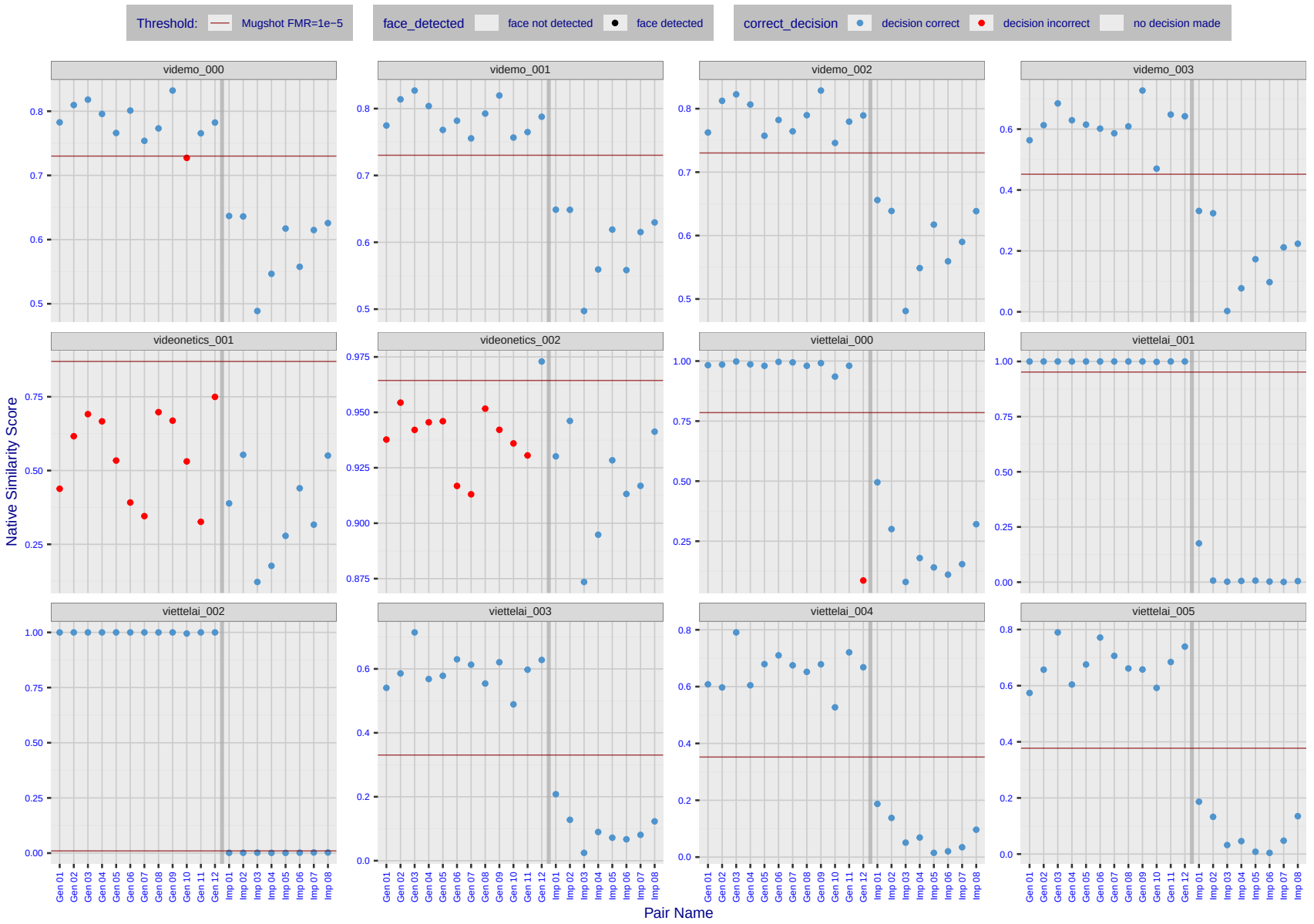

Figure 104: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

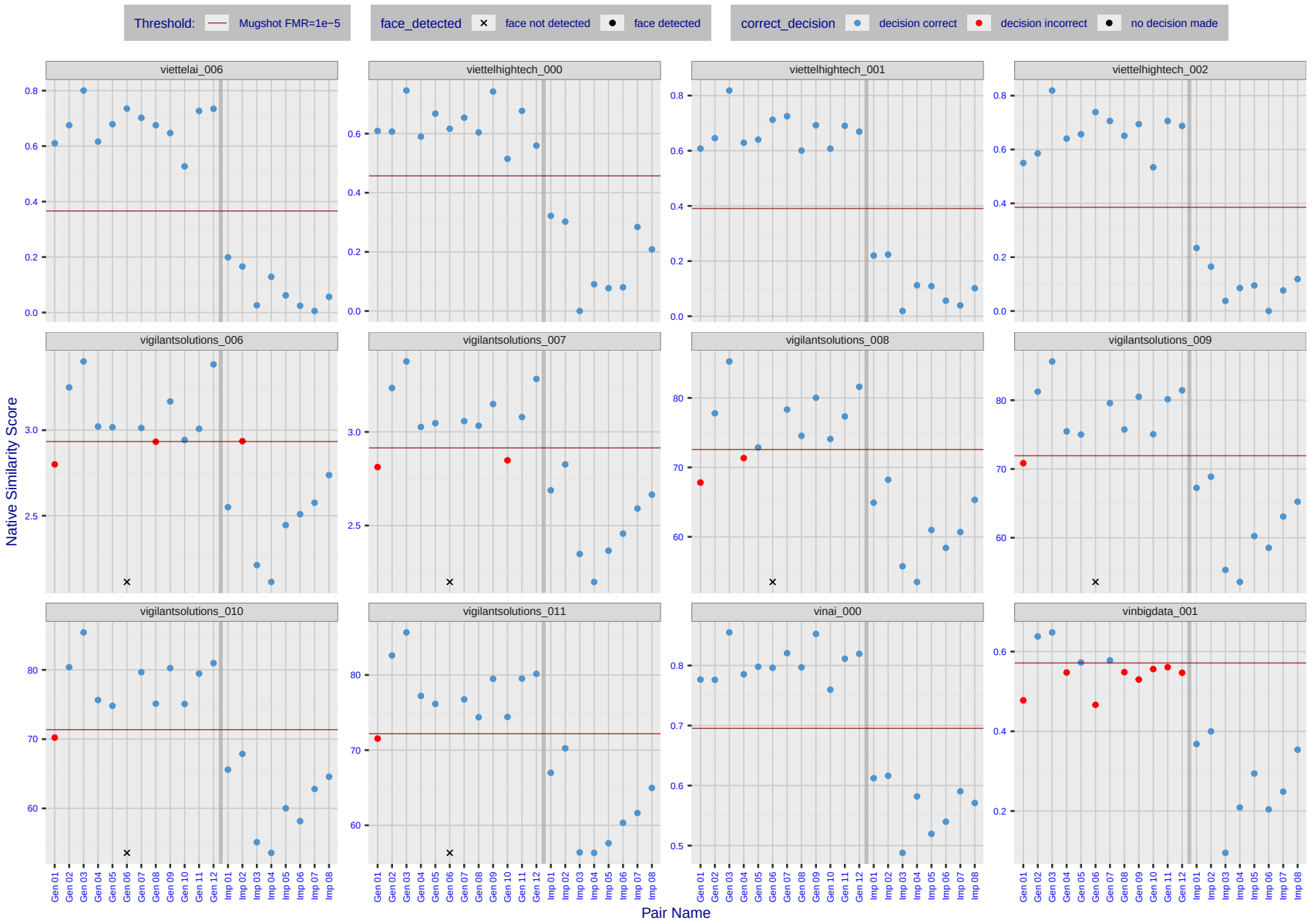

Figure 105: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

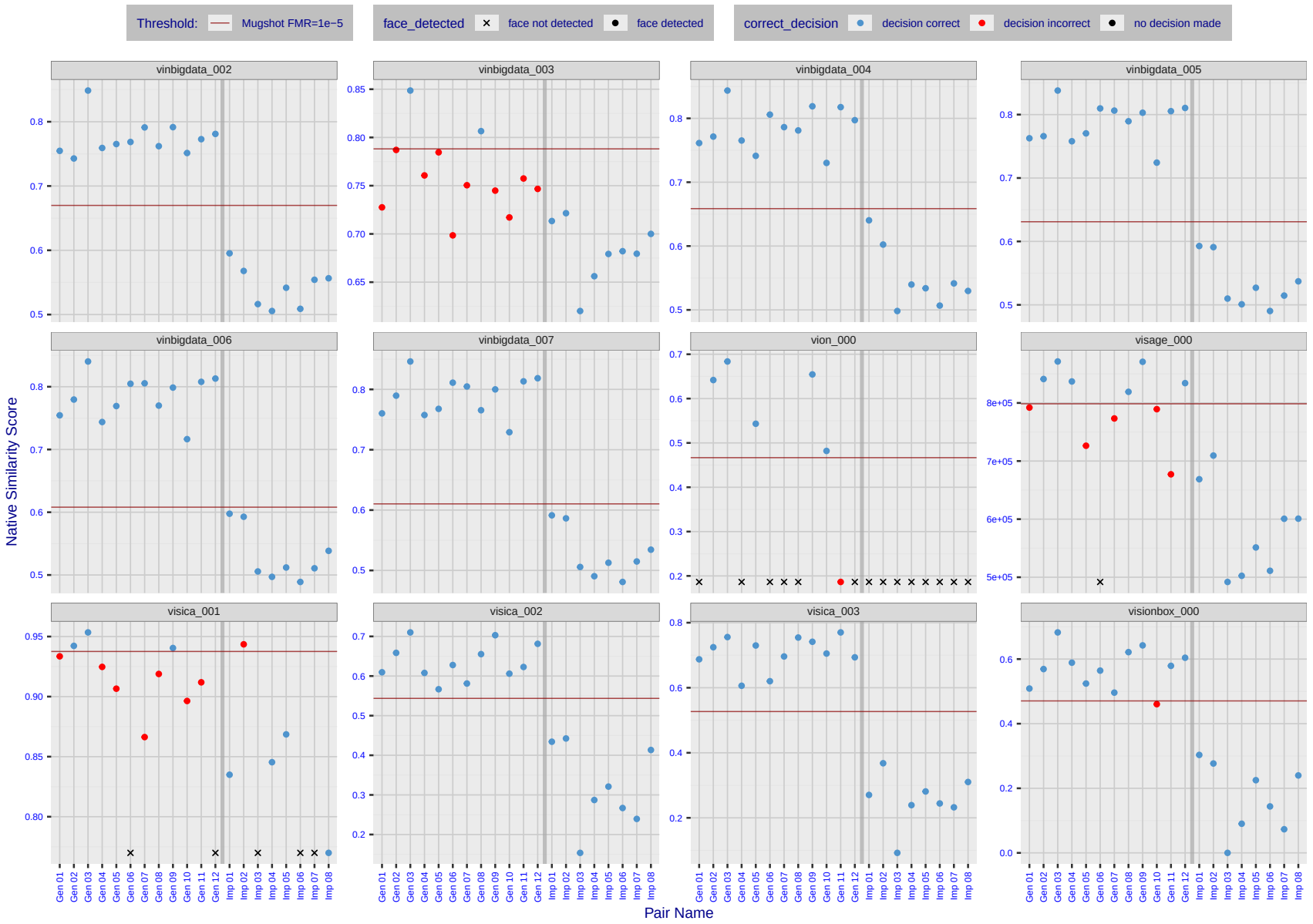

Figure 106: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

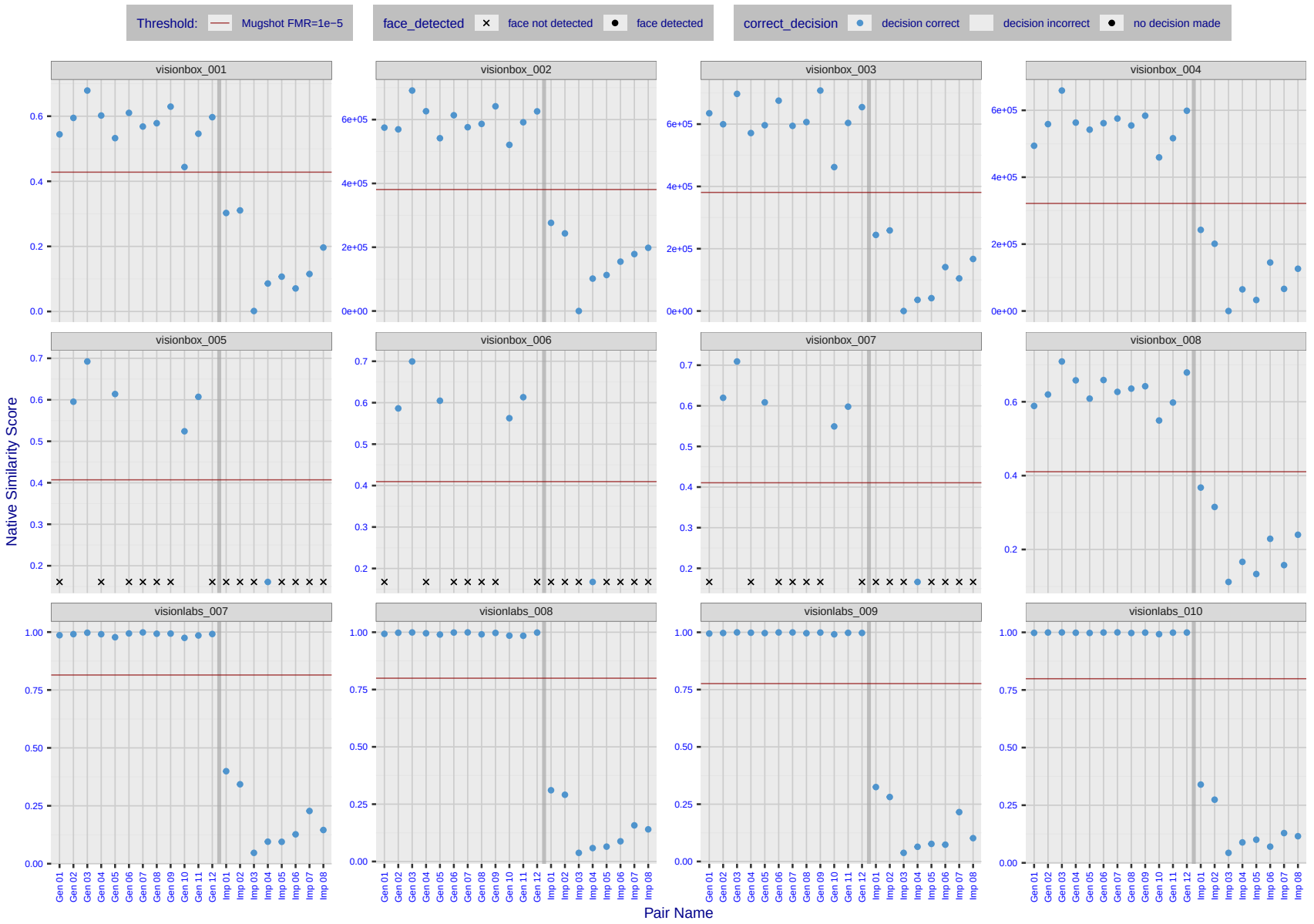

Figure 107: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

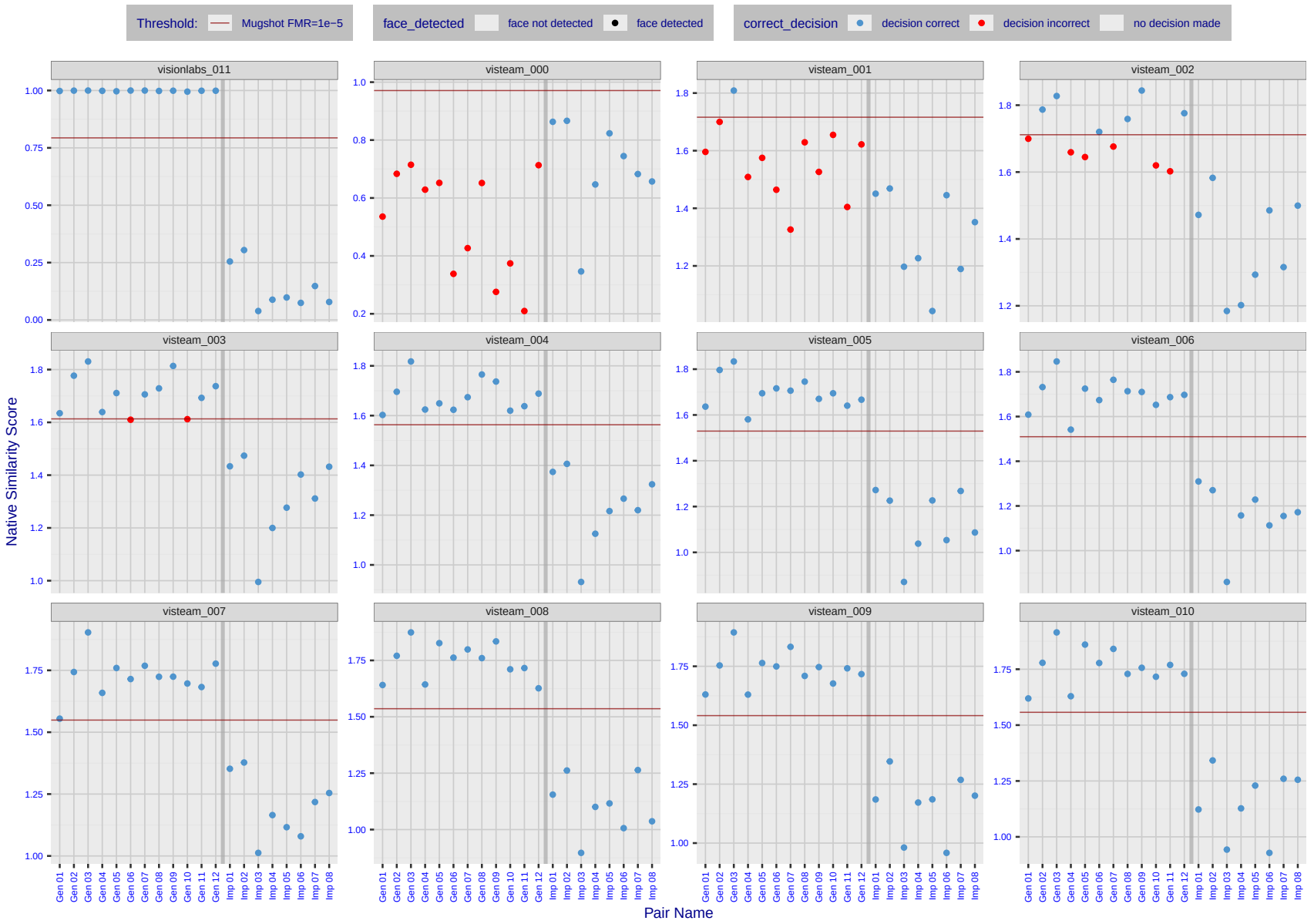

Figure 108: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

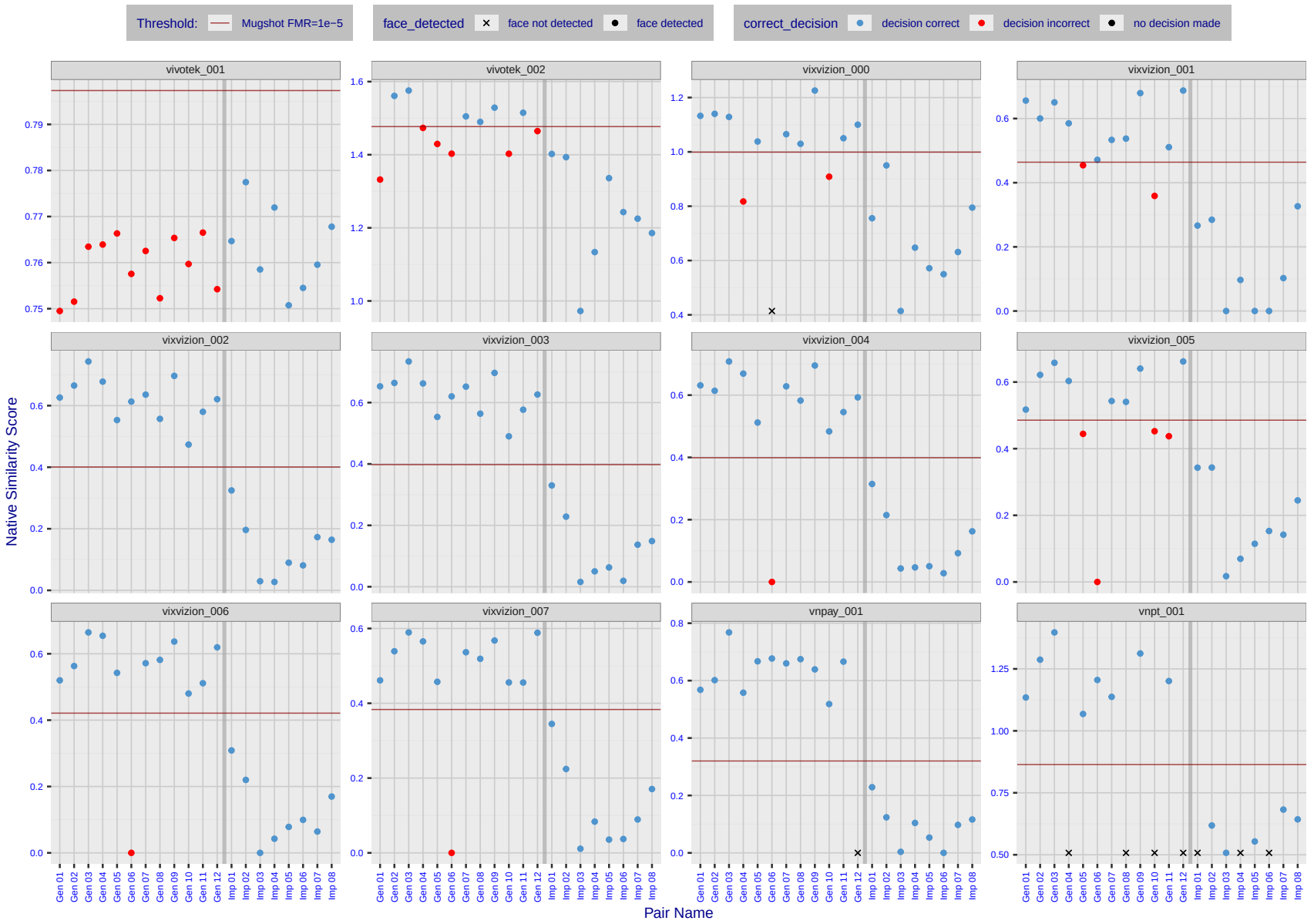

Figure 109: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

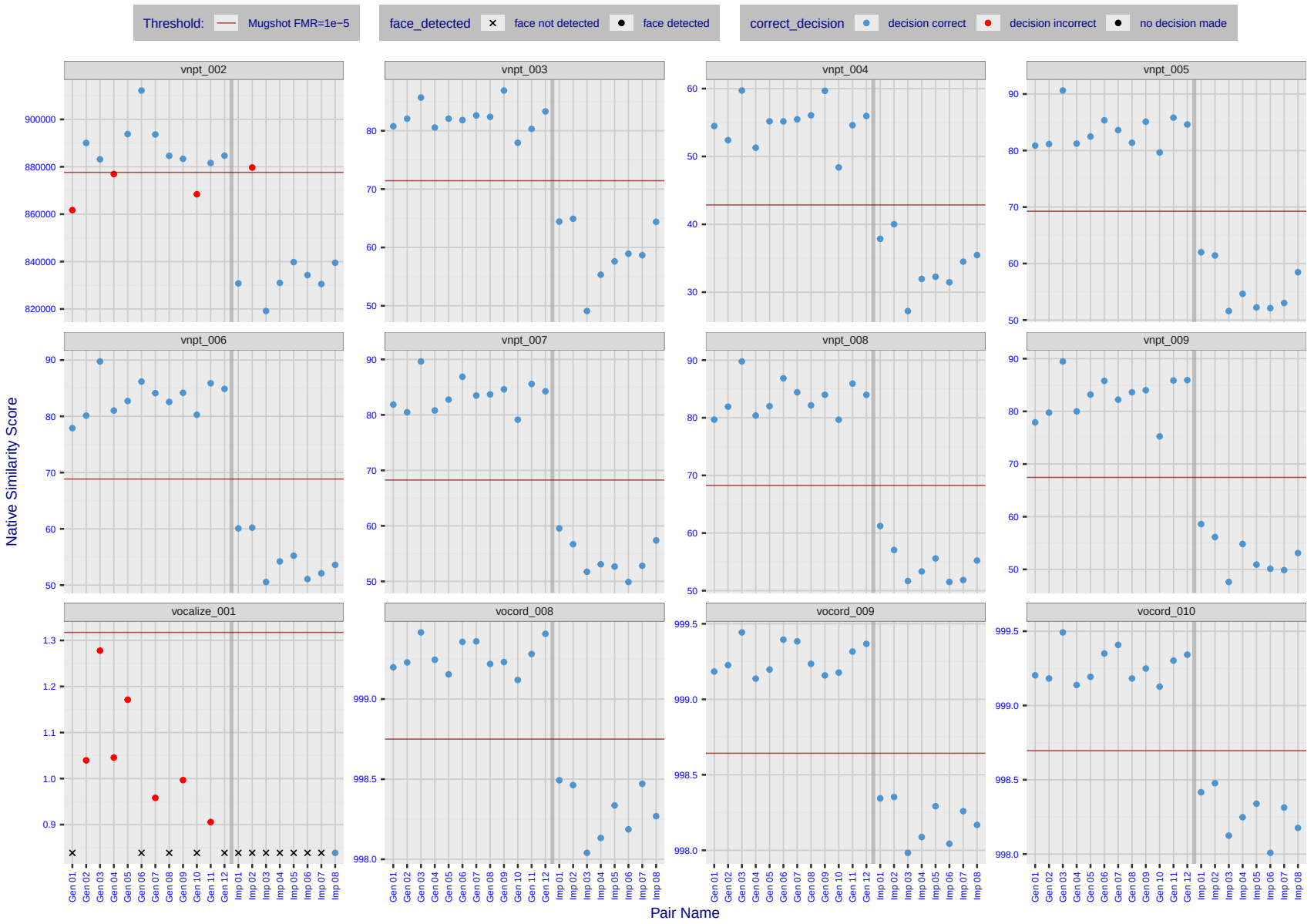

Figure 110: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

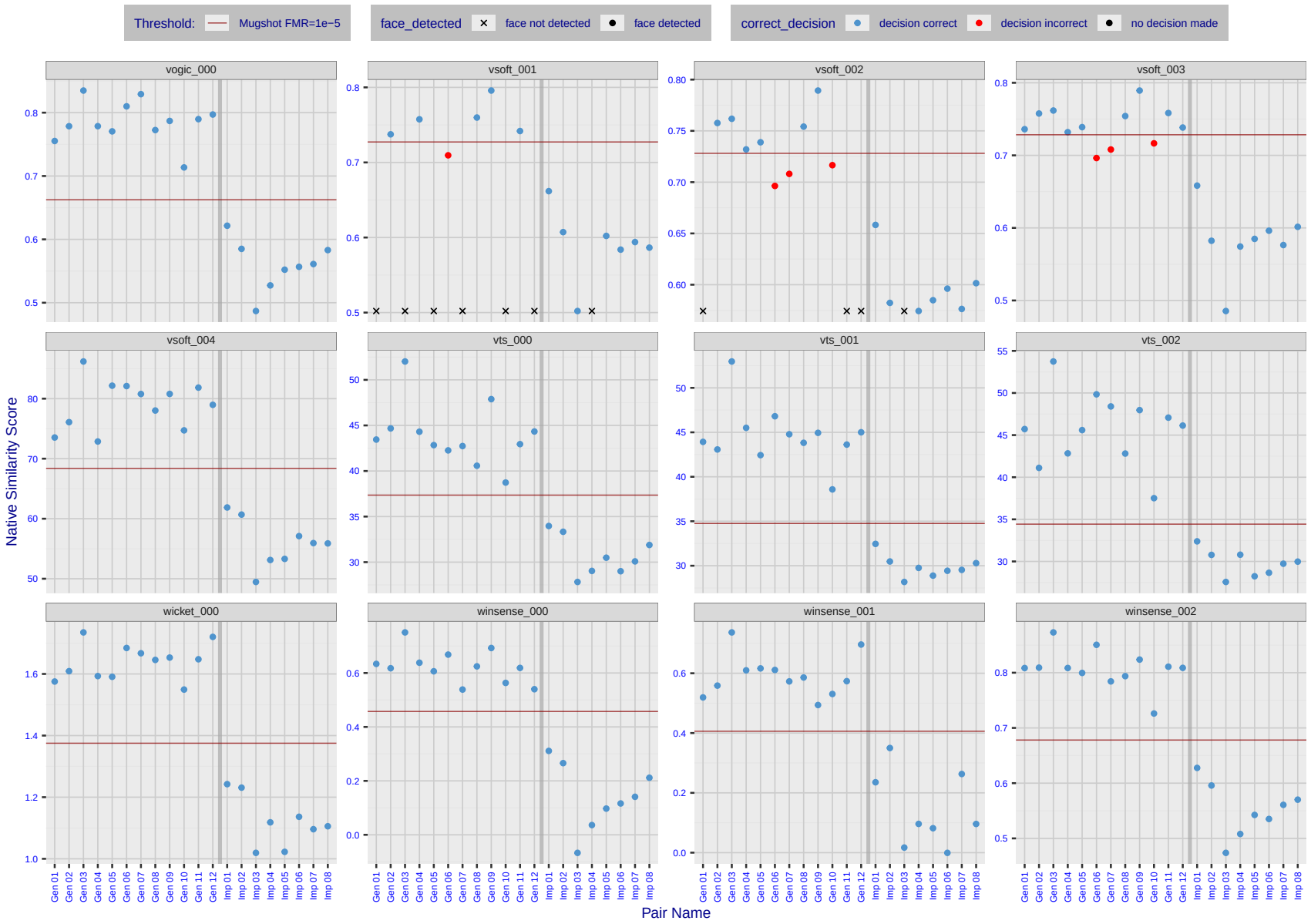

Figure 111: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

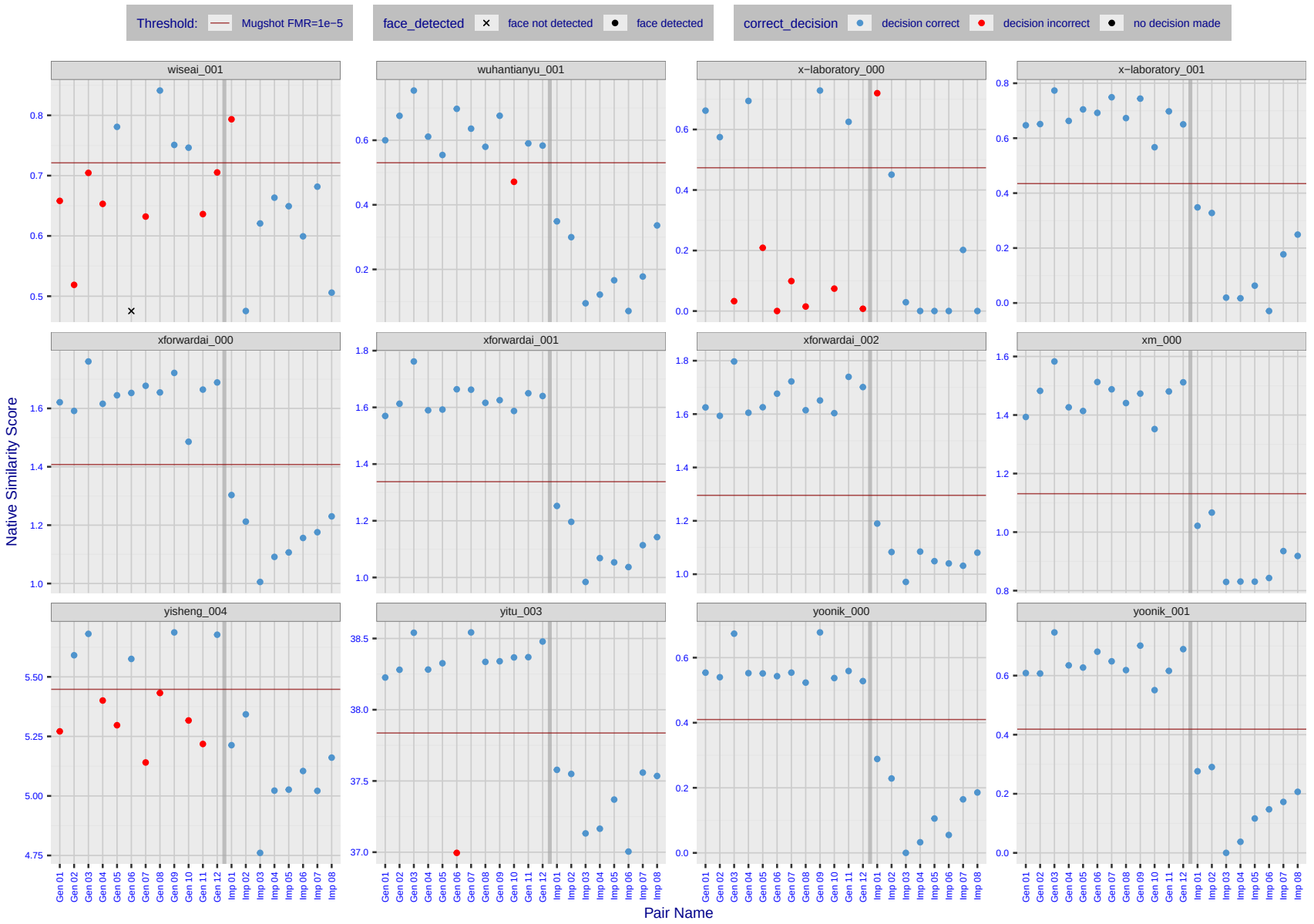

Figure 112: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

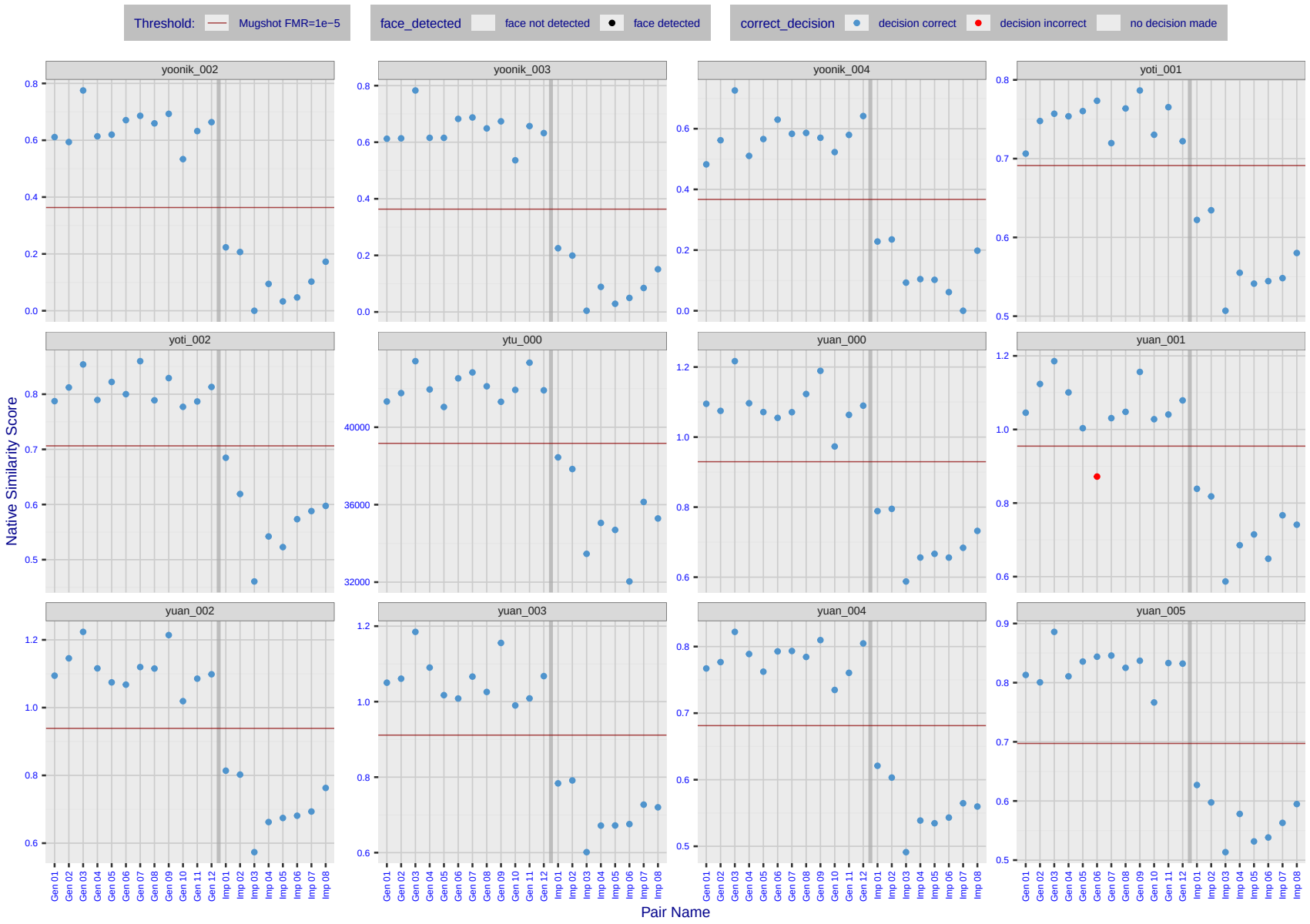

Figure 113: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

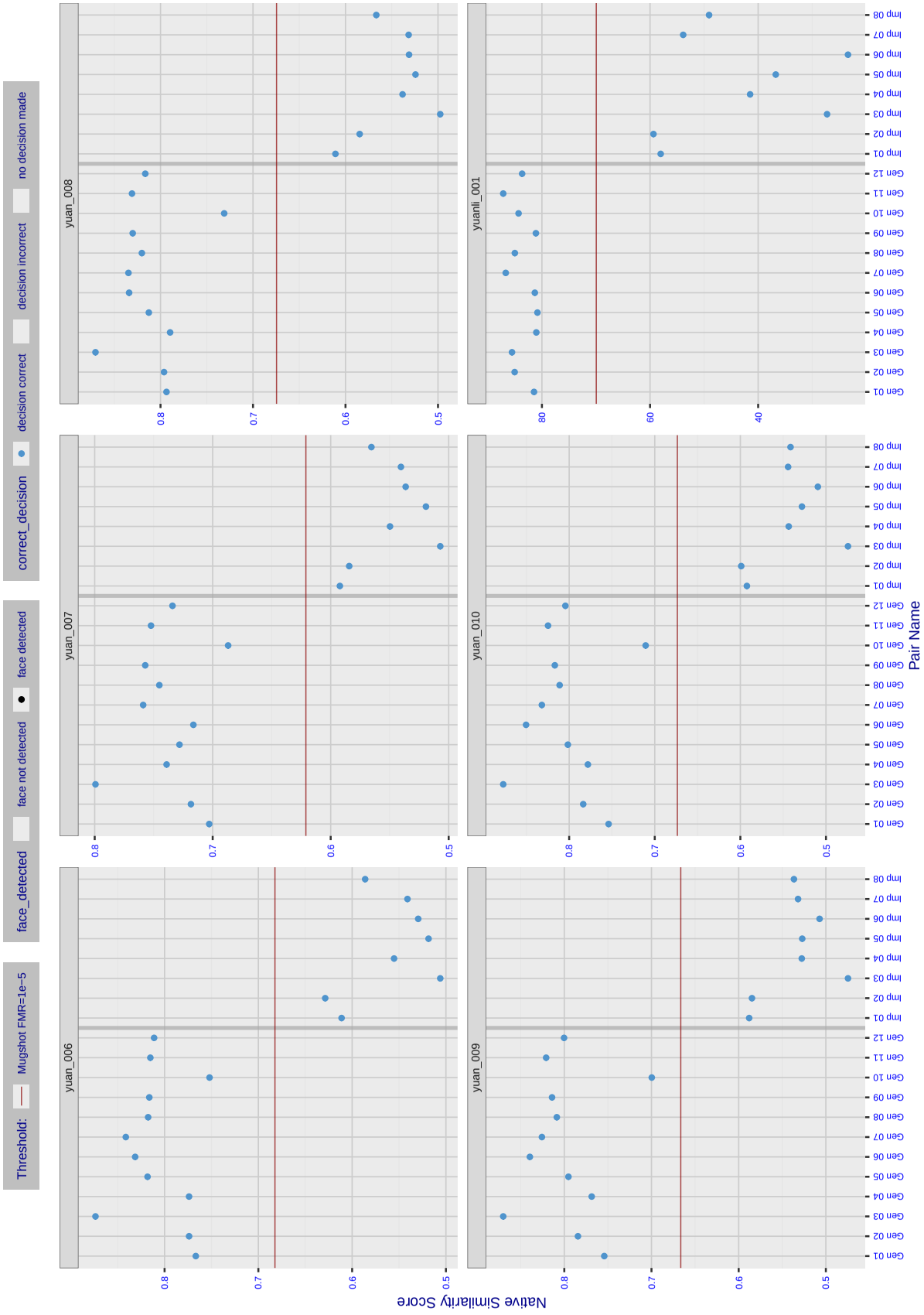

Figure 114: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

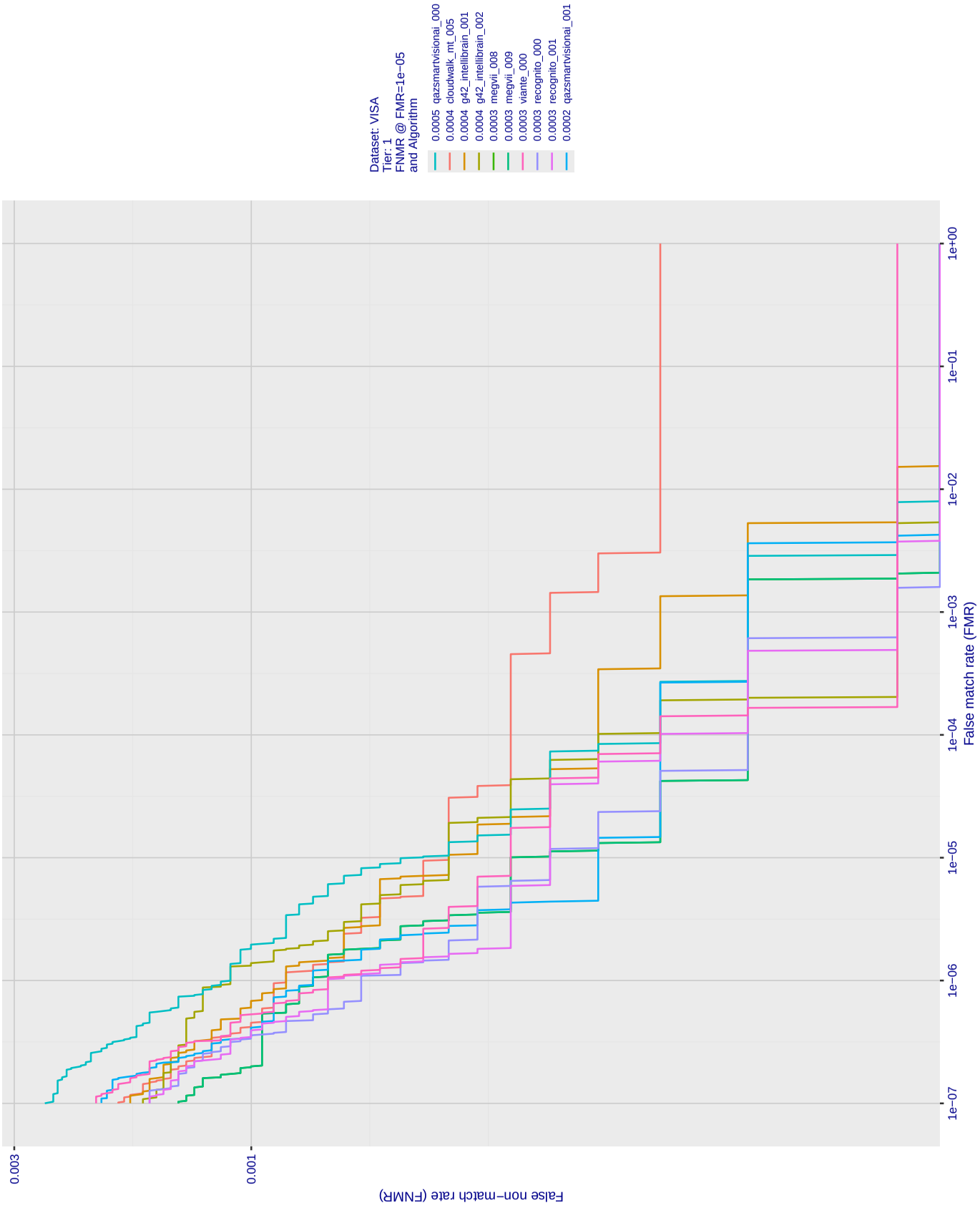

Figure 115: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

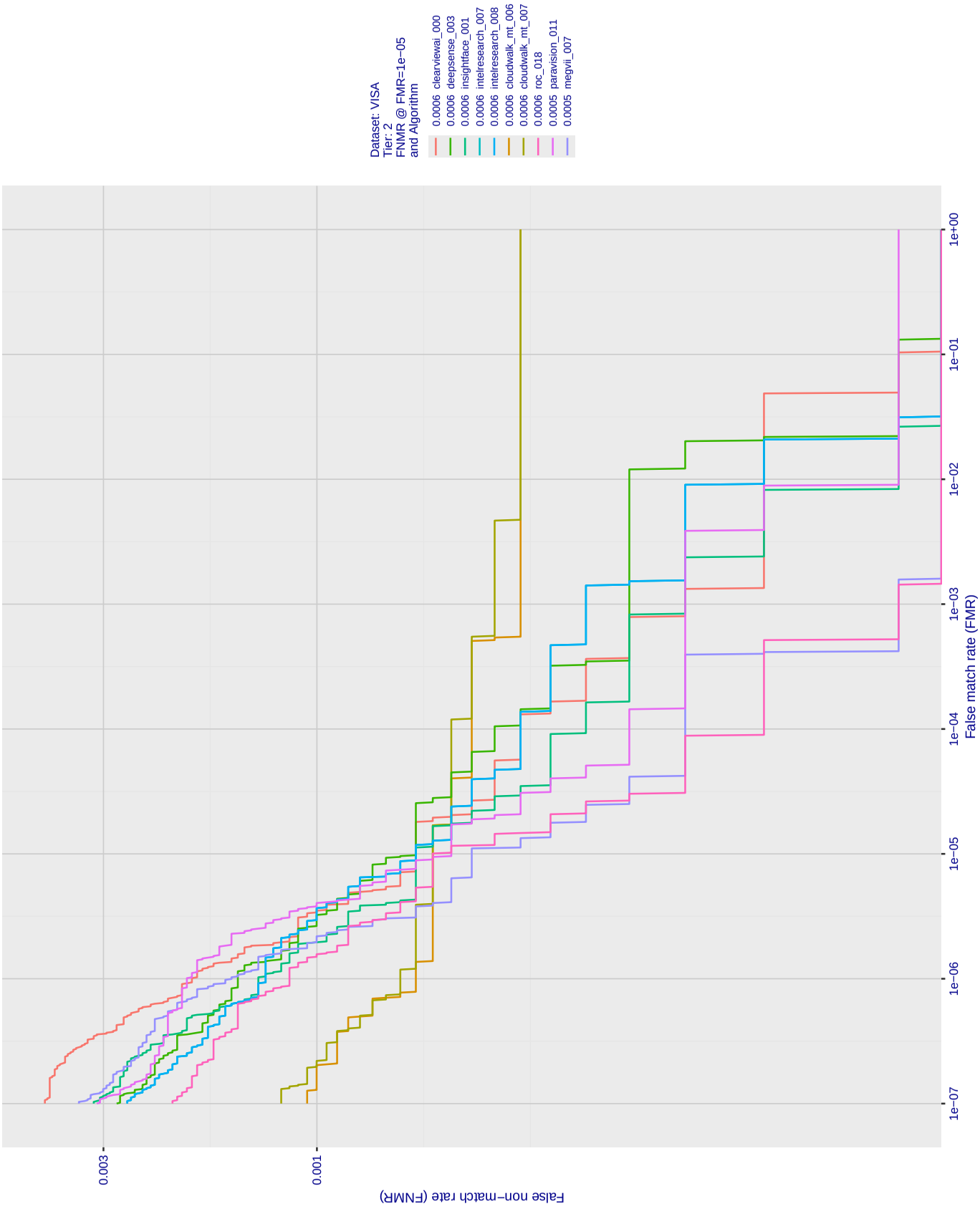

Figure 116: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

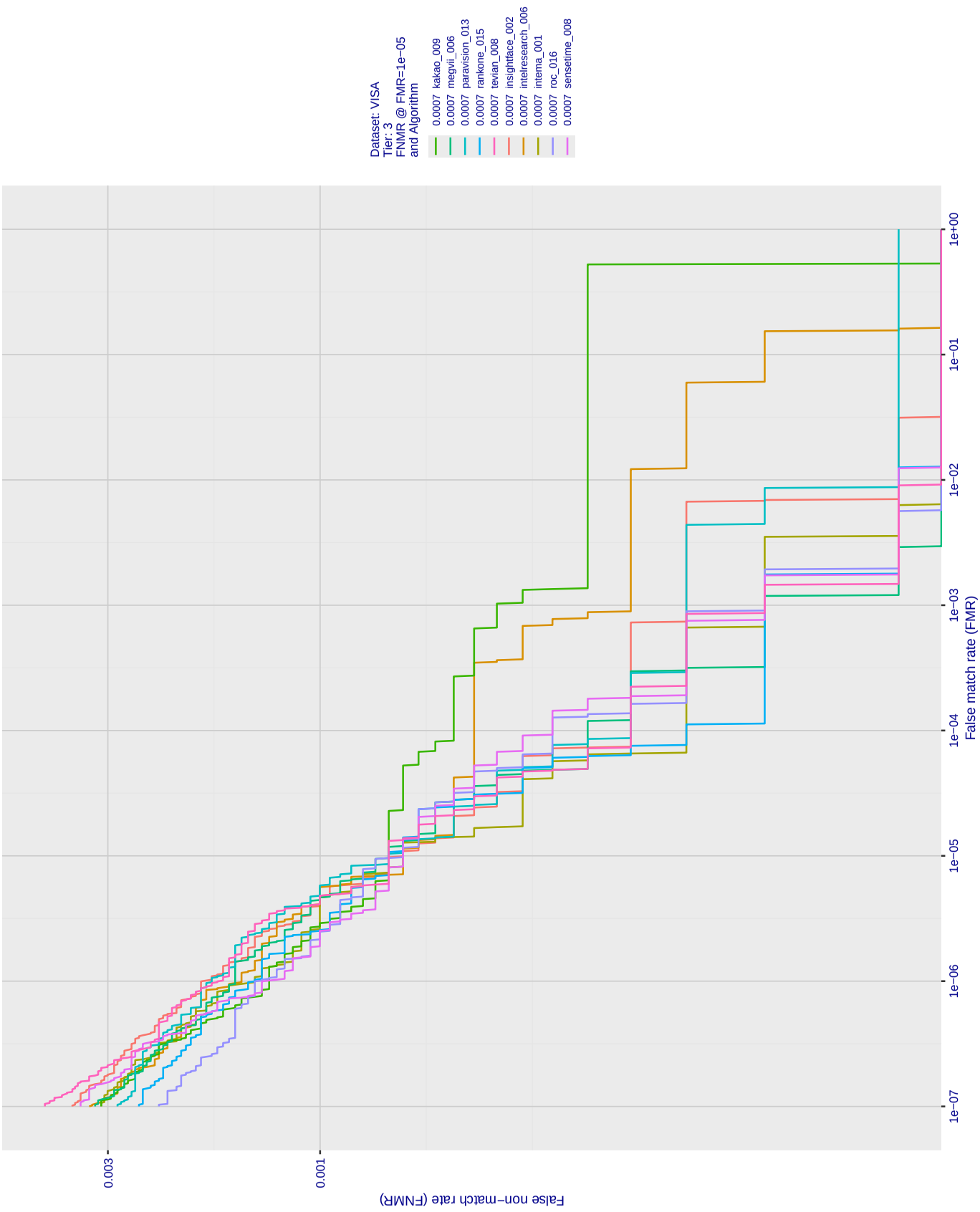

Figure 117: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

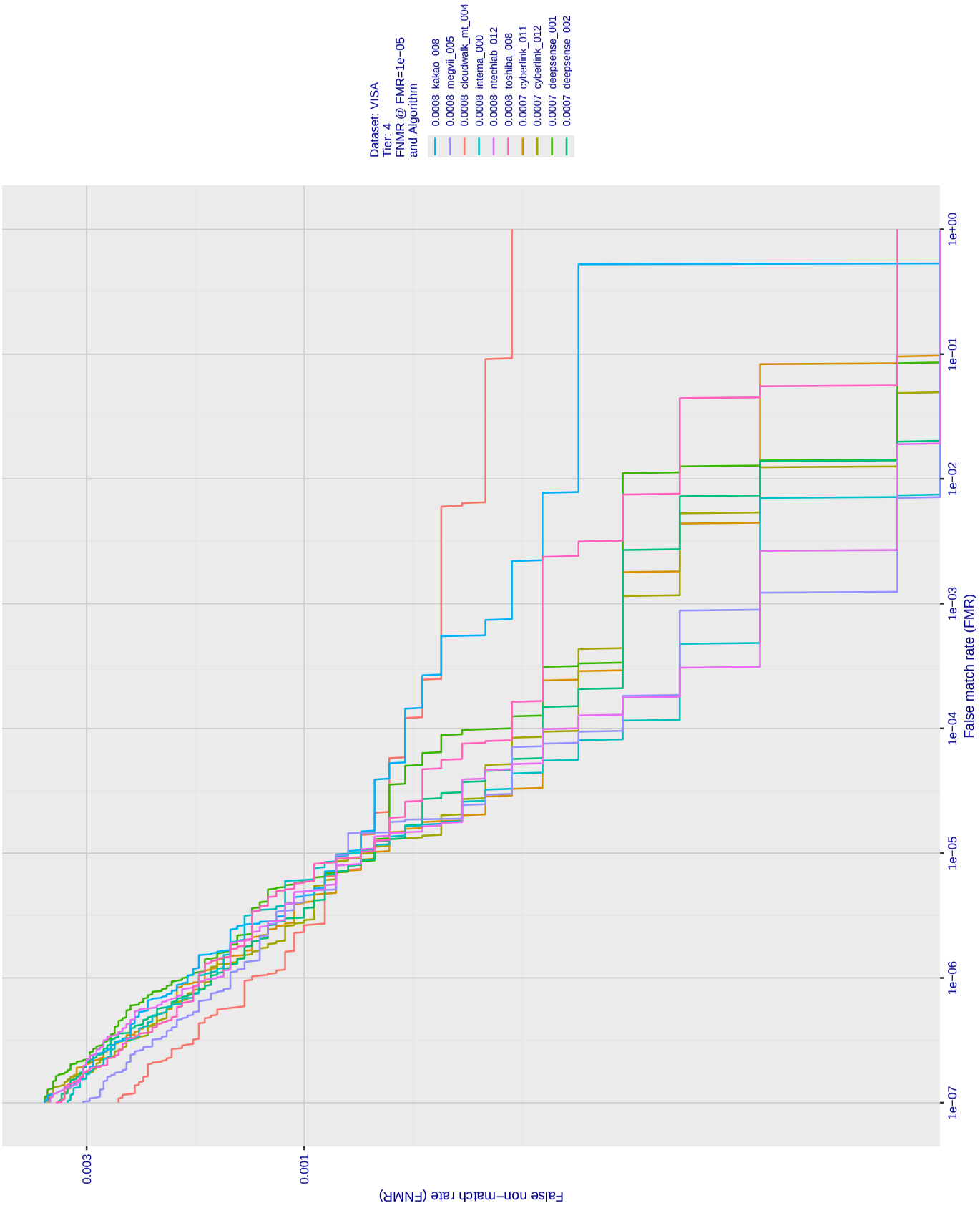

Figure 118: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

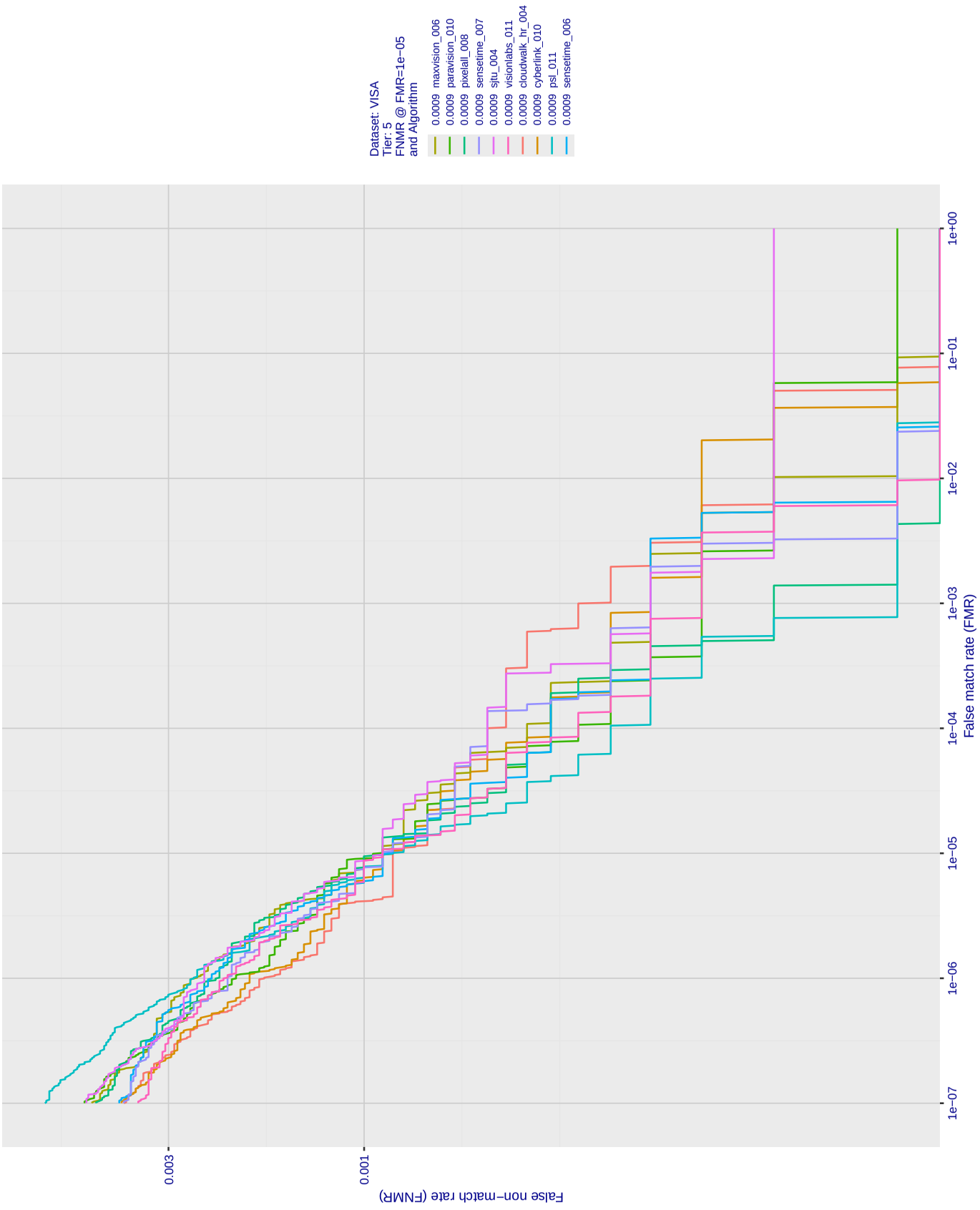

Figure 119: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

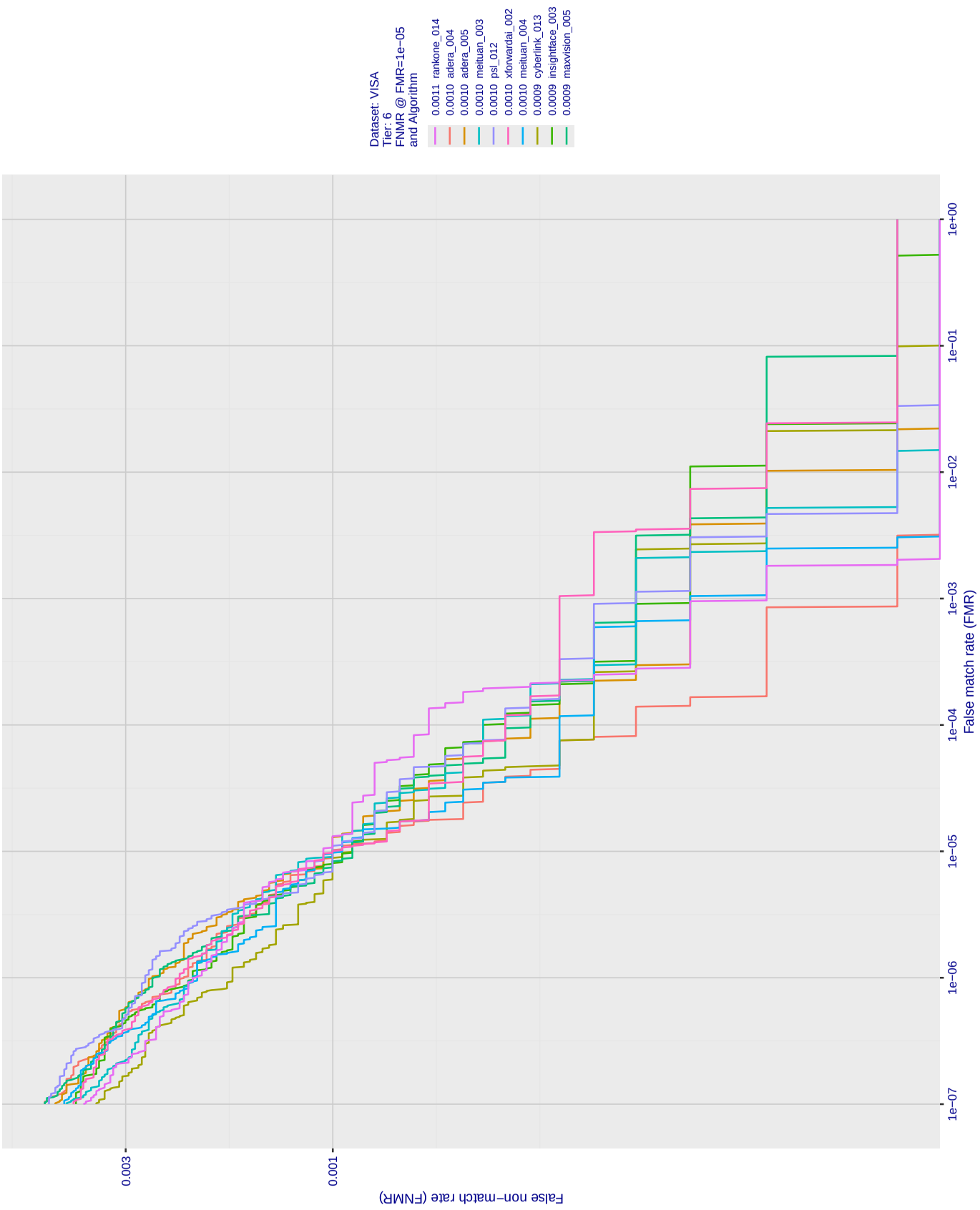

Figure 120: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

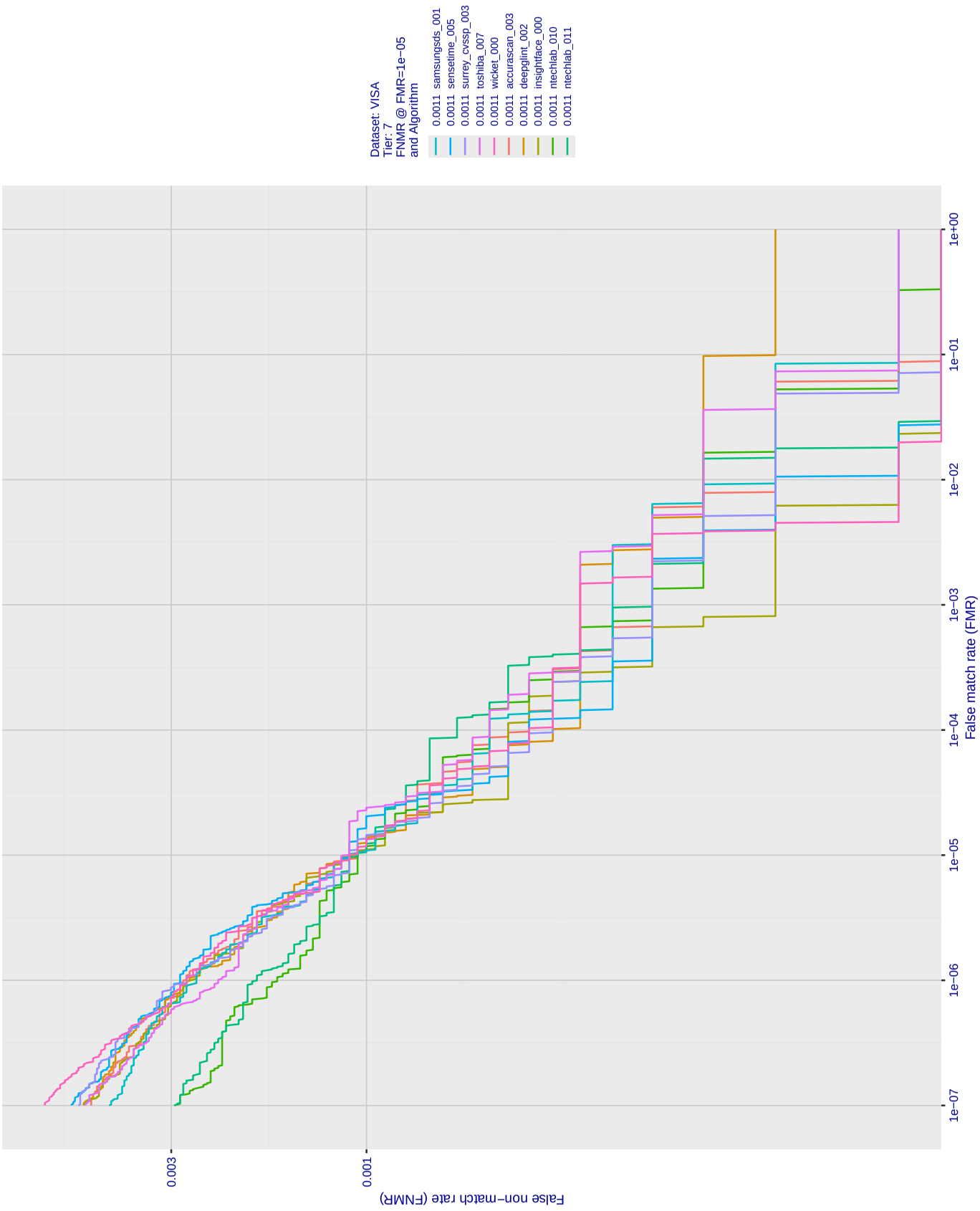

Figure 121: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

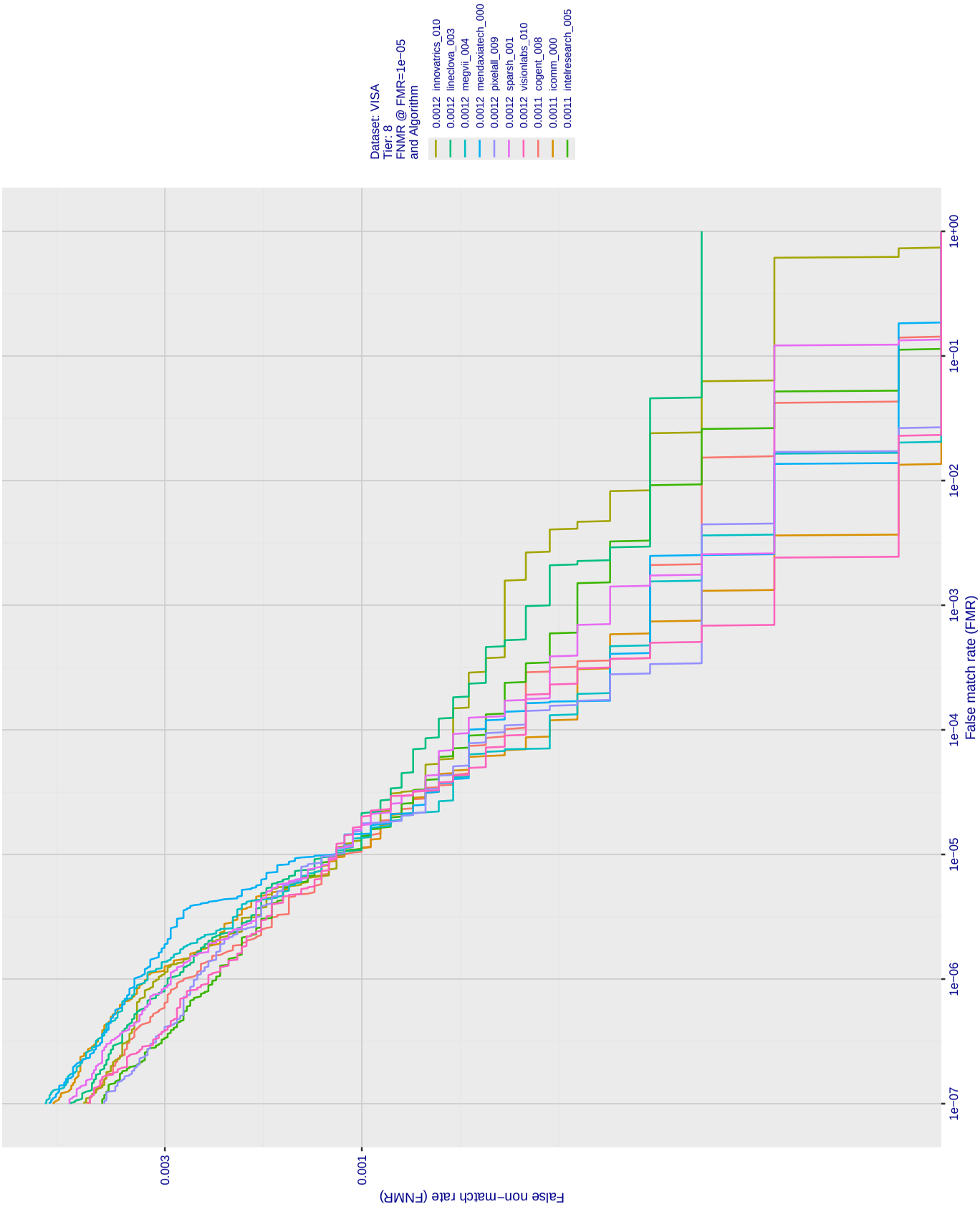

Figure 122: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

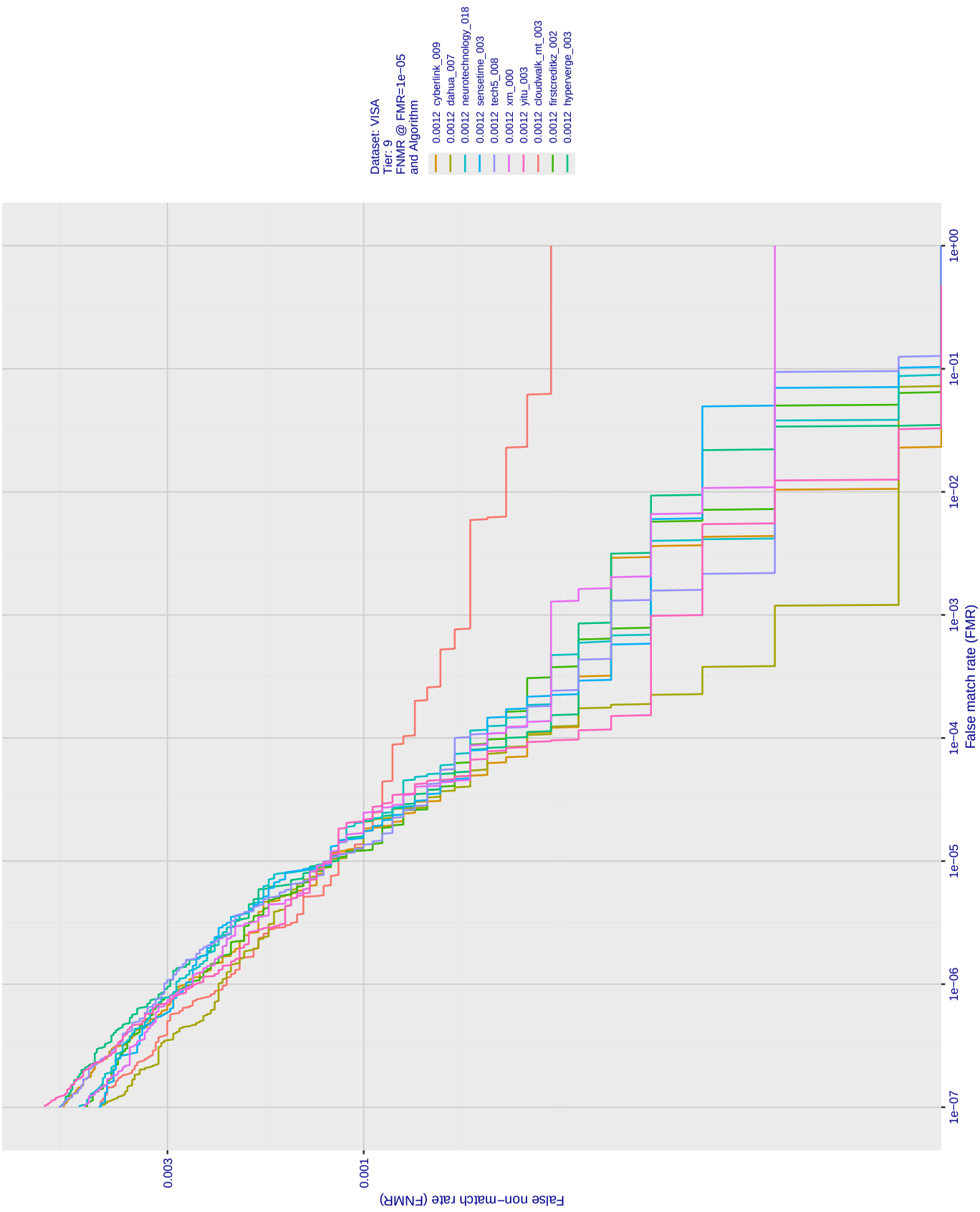

Figure 123: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

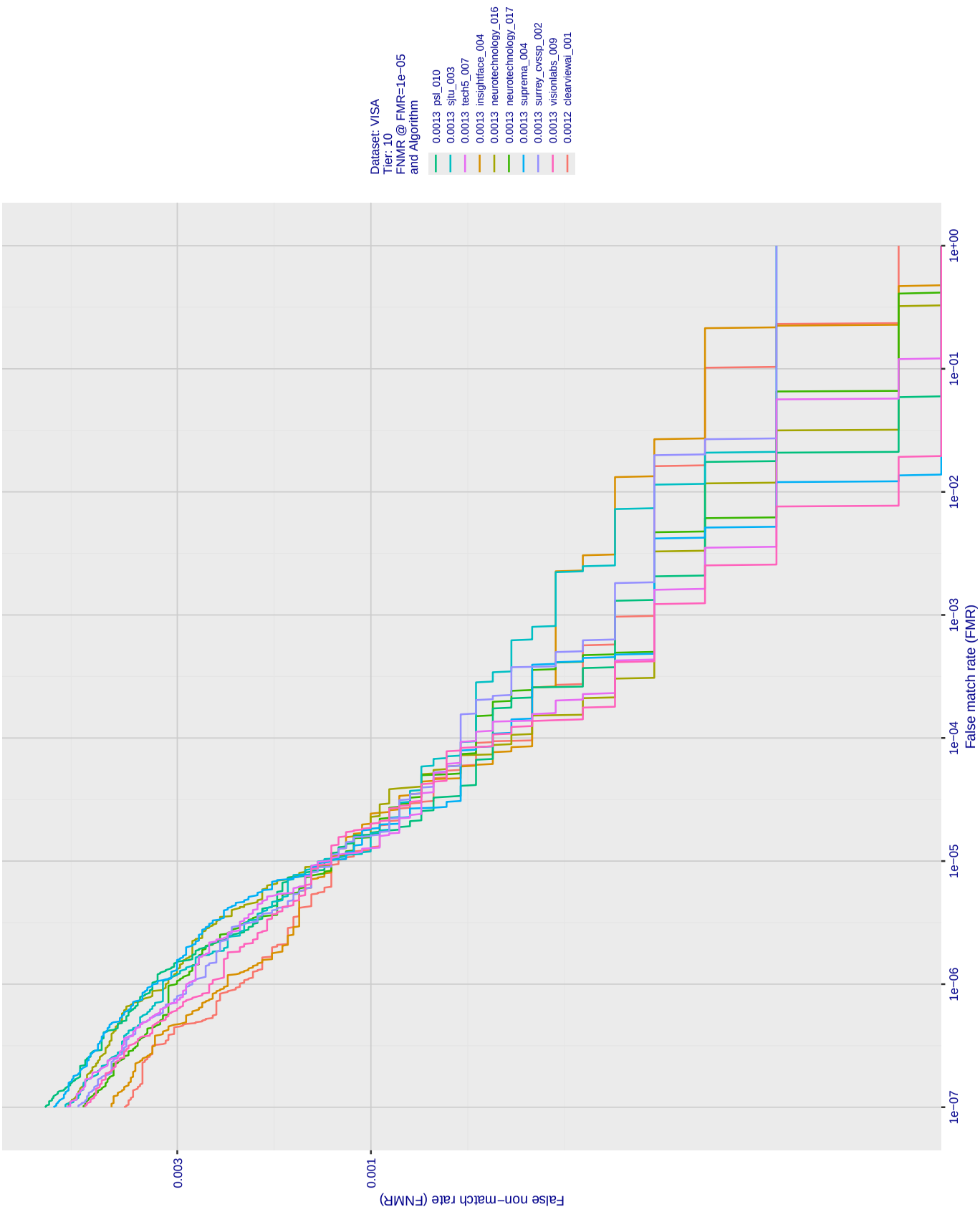

Figure 124: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

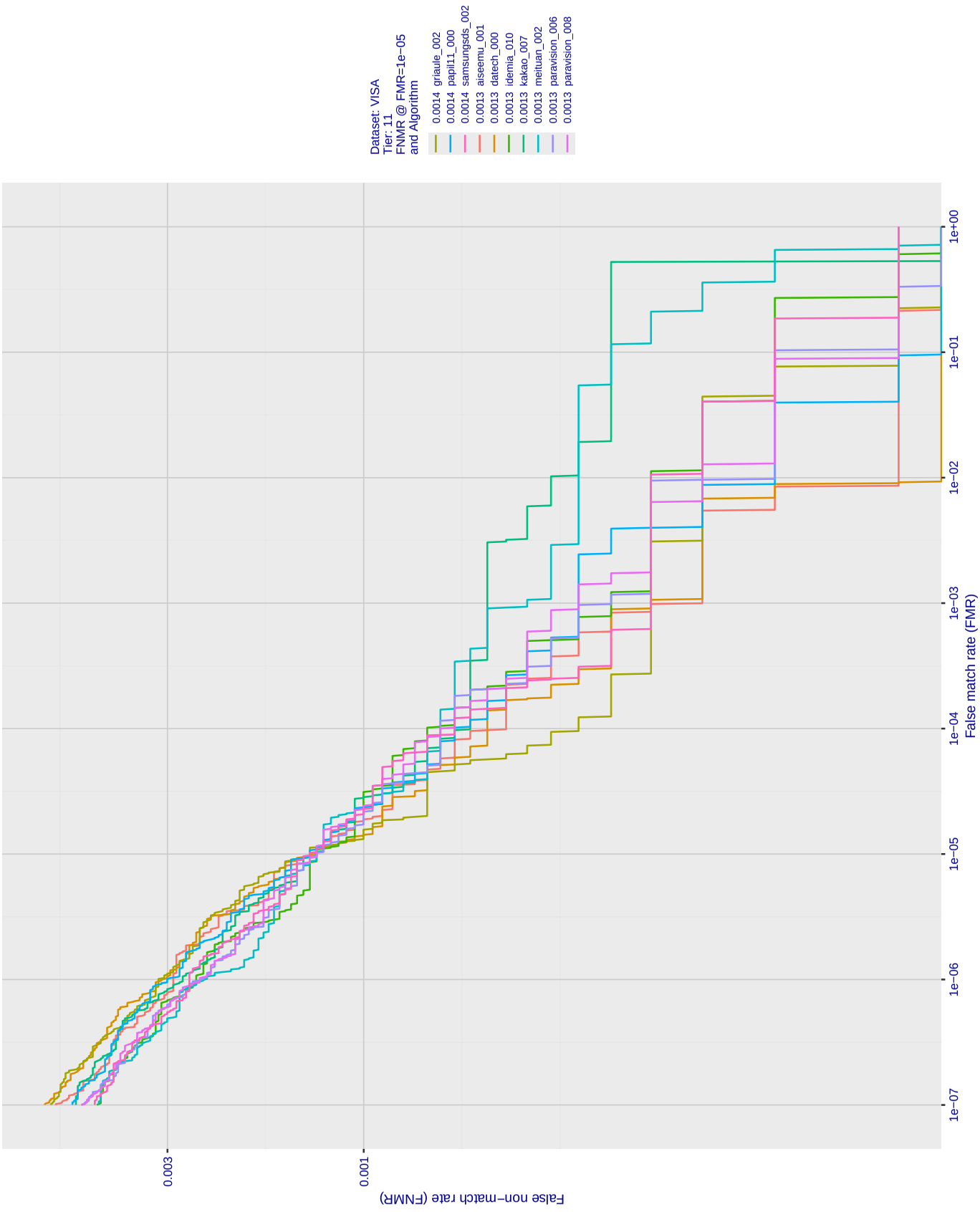

Figure 125: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

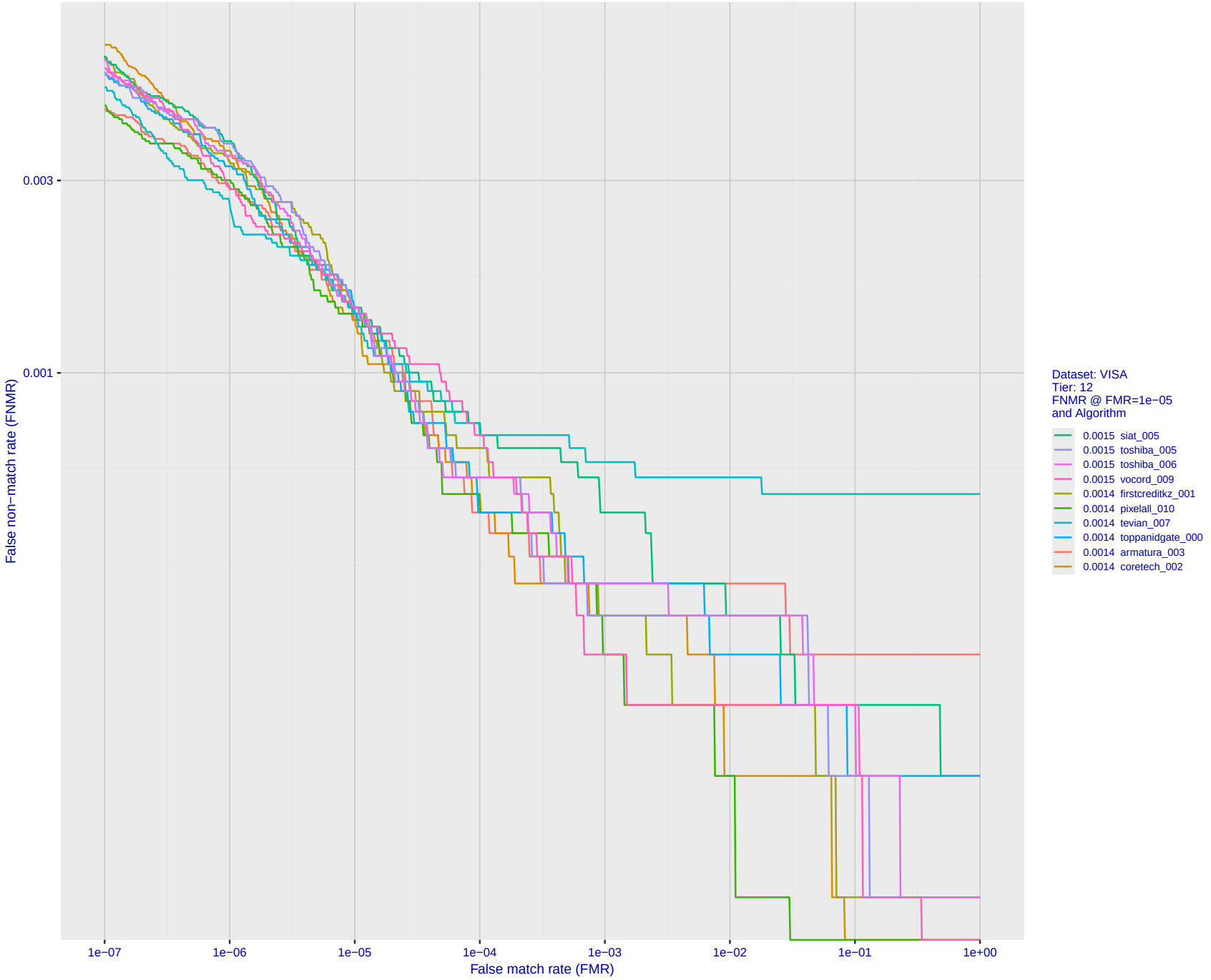

Figure 126: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

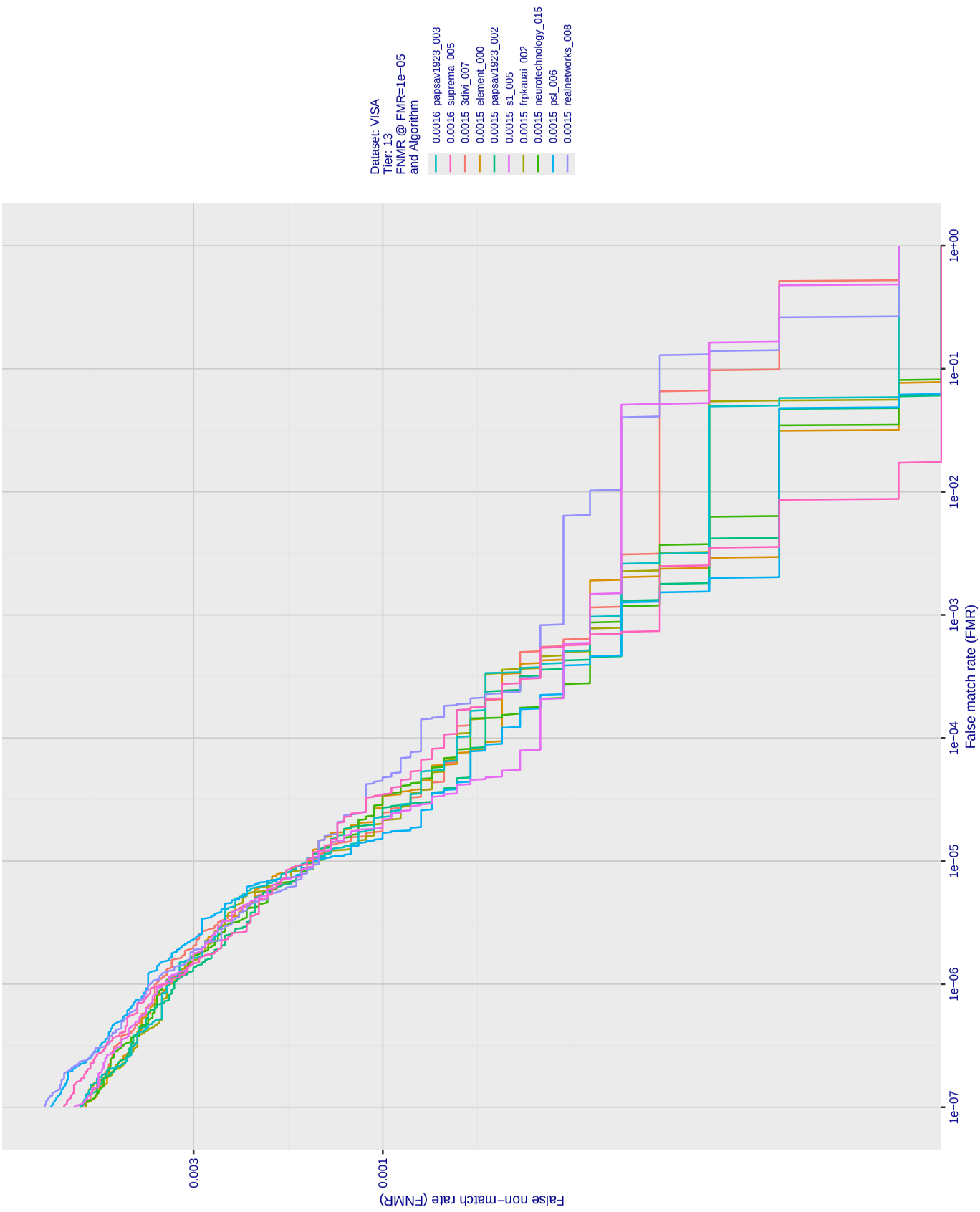

Figure 127: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

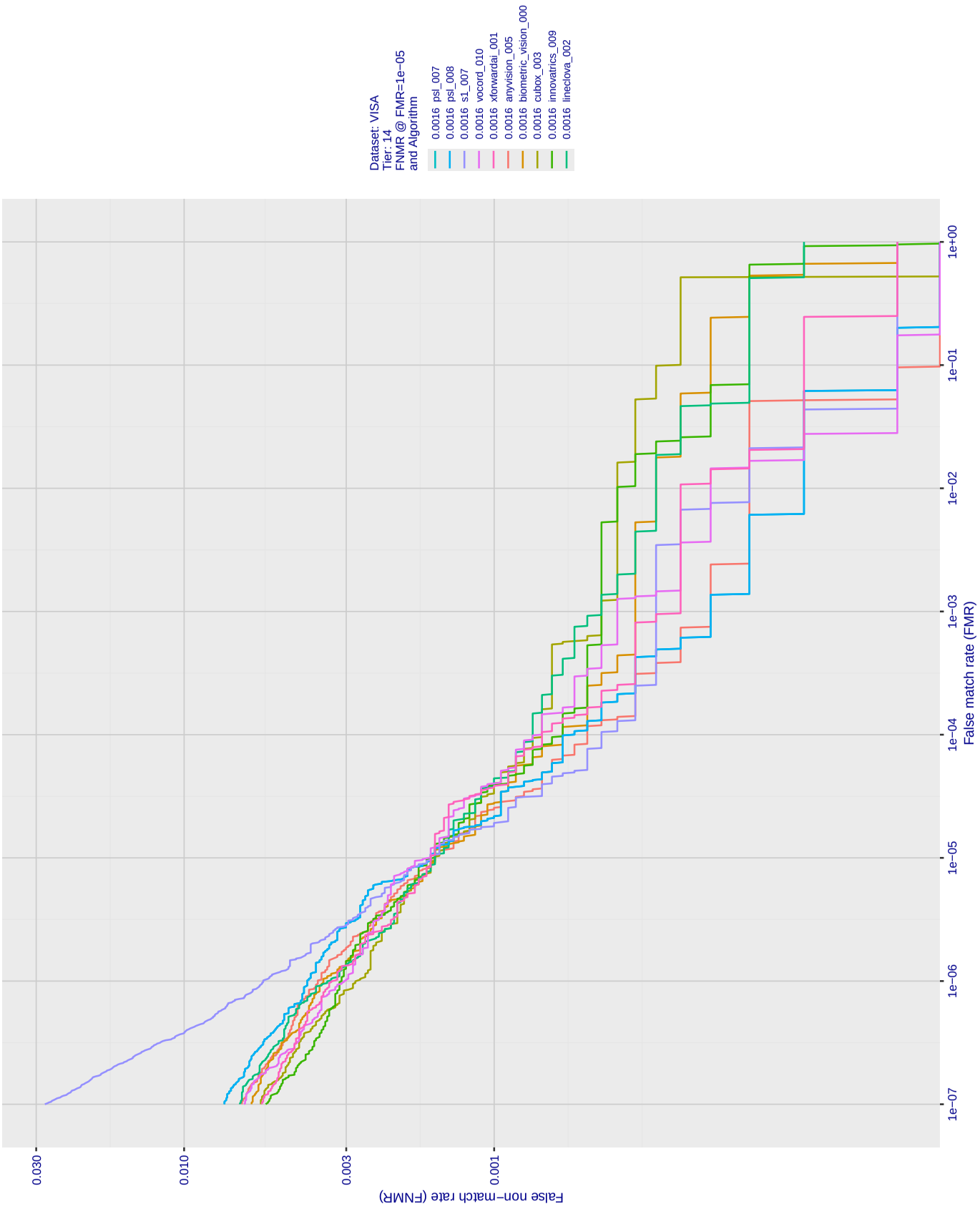

Figure 128: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

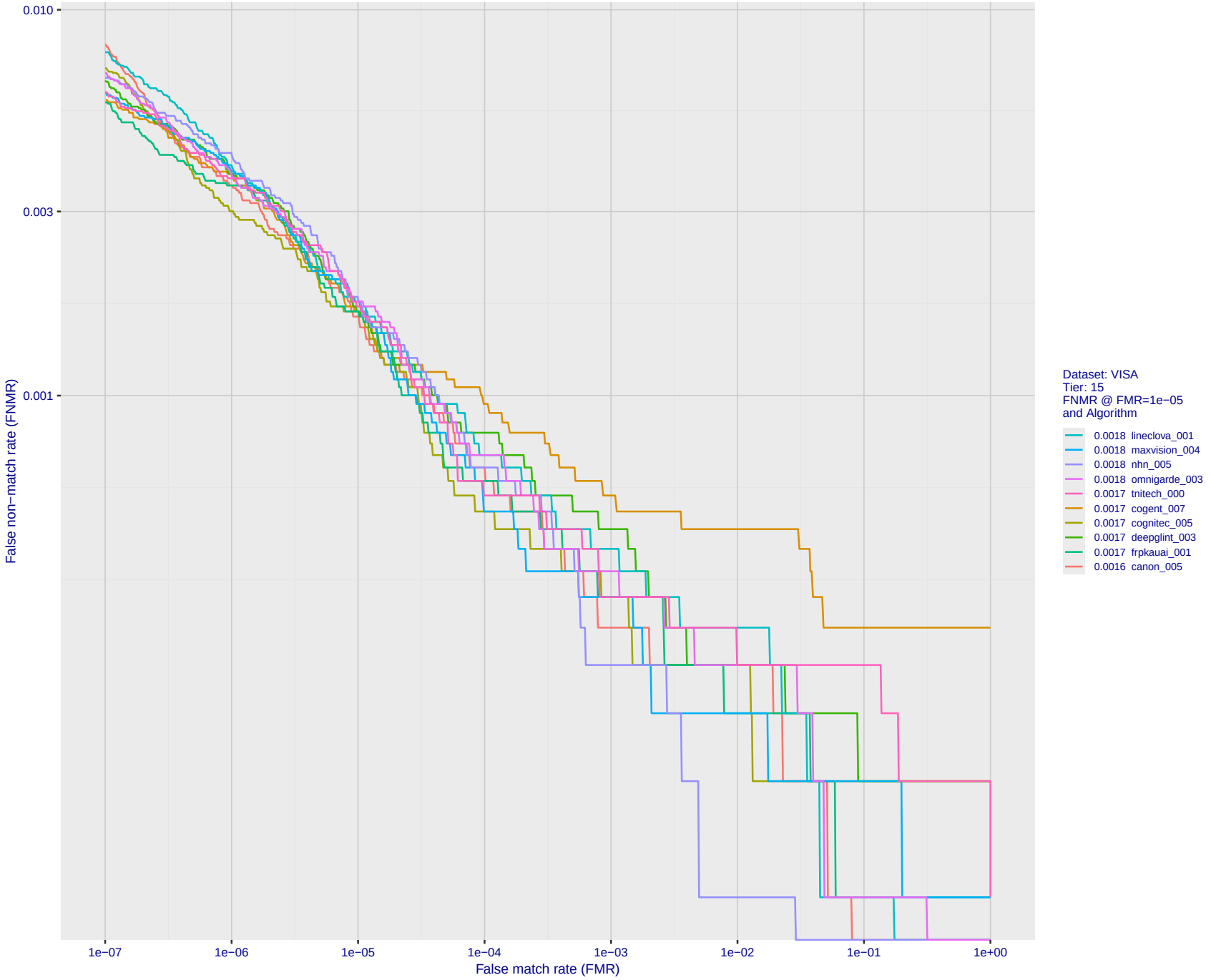

Figure 129: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

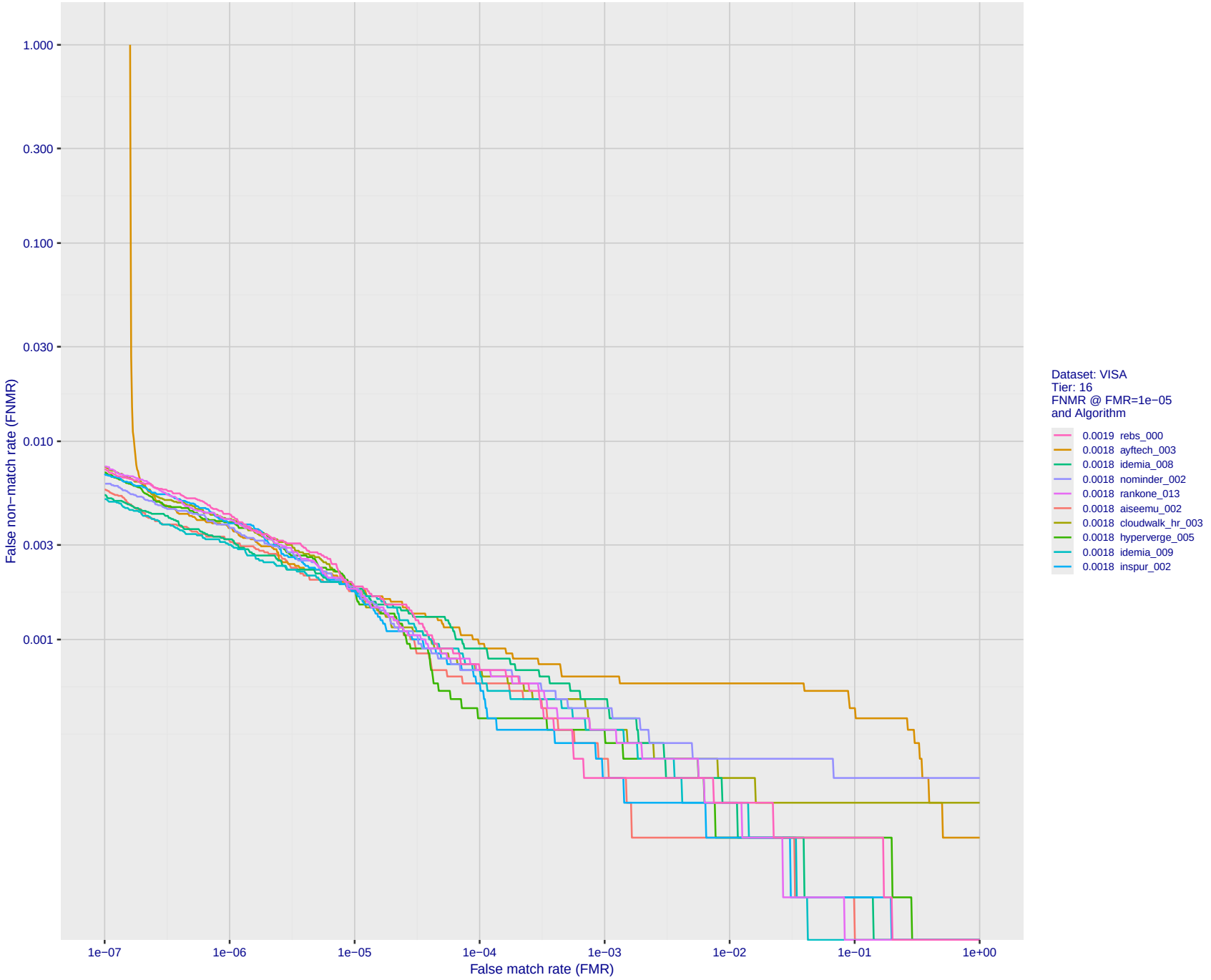

Figure 130: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

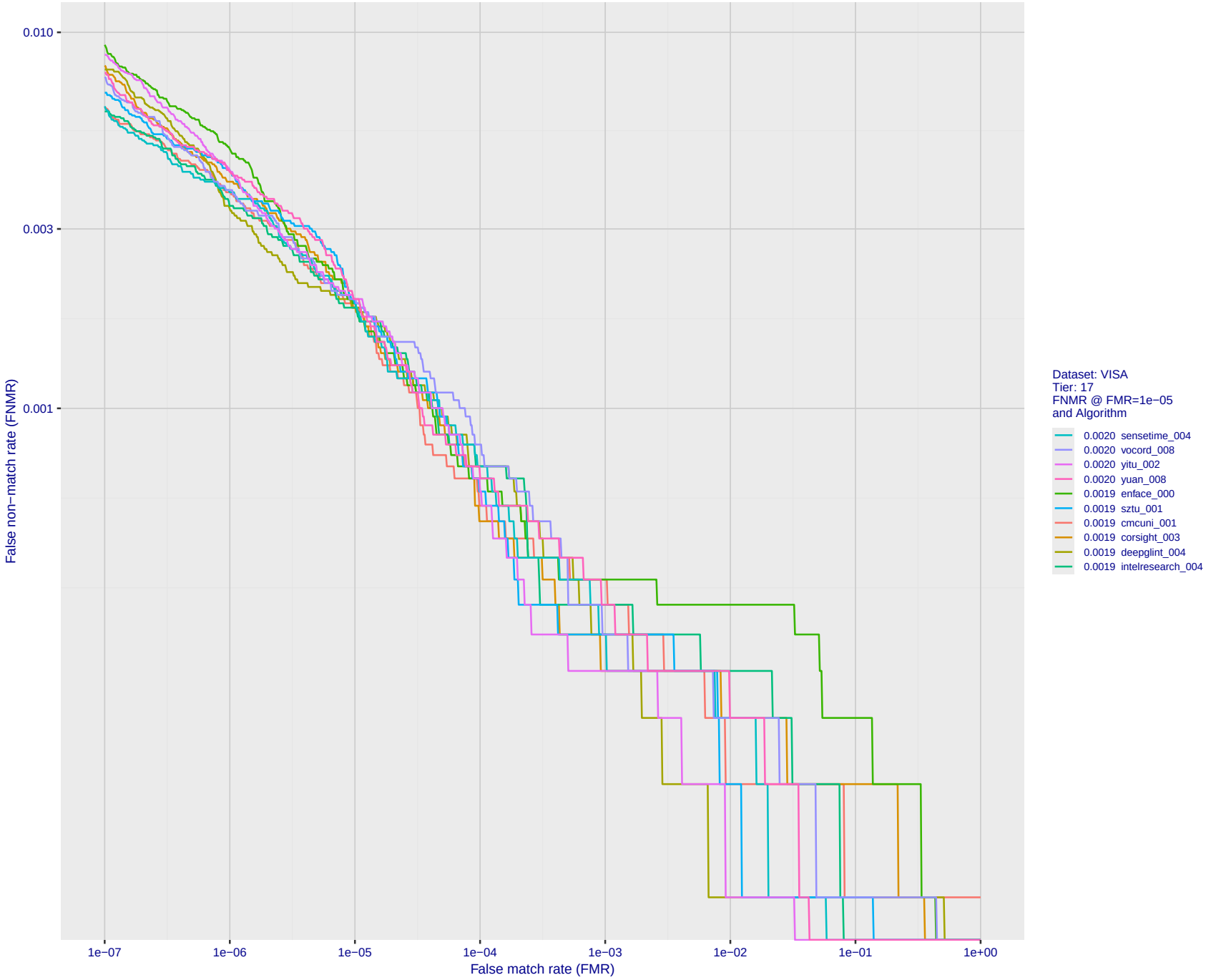

Figure 131: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

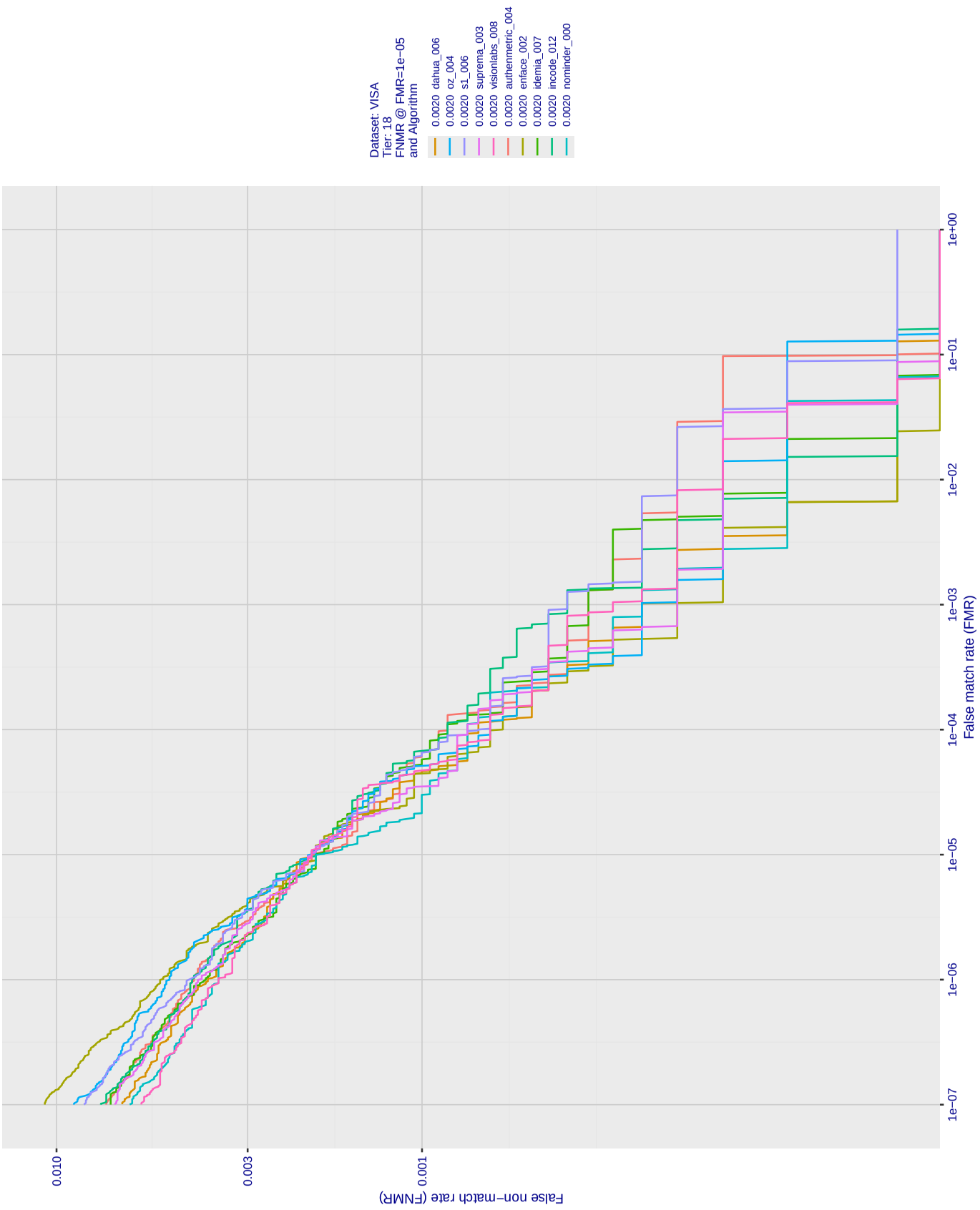

Figure 132: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

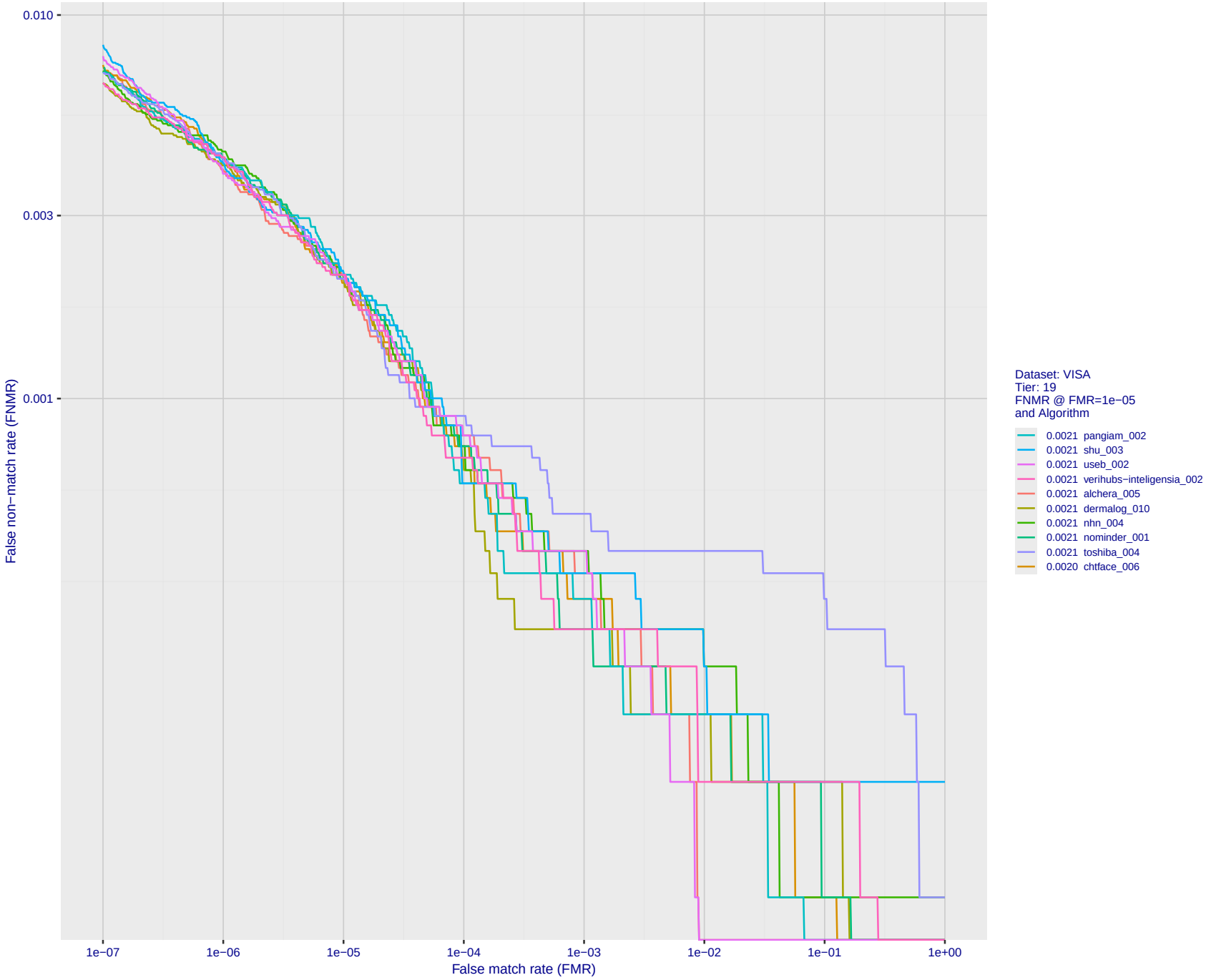

Figure 133: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

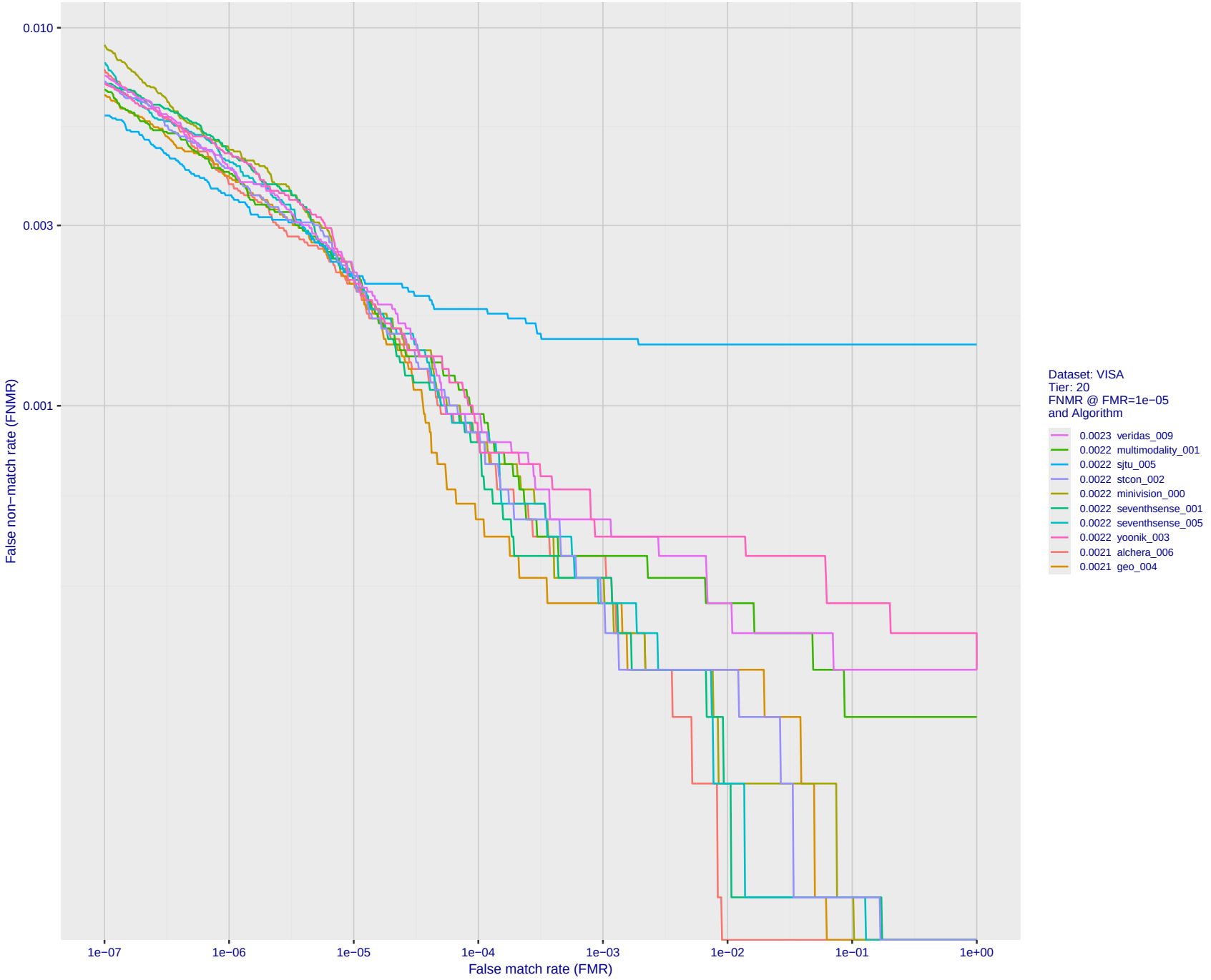

Figure 134: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

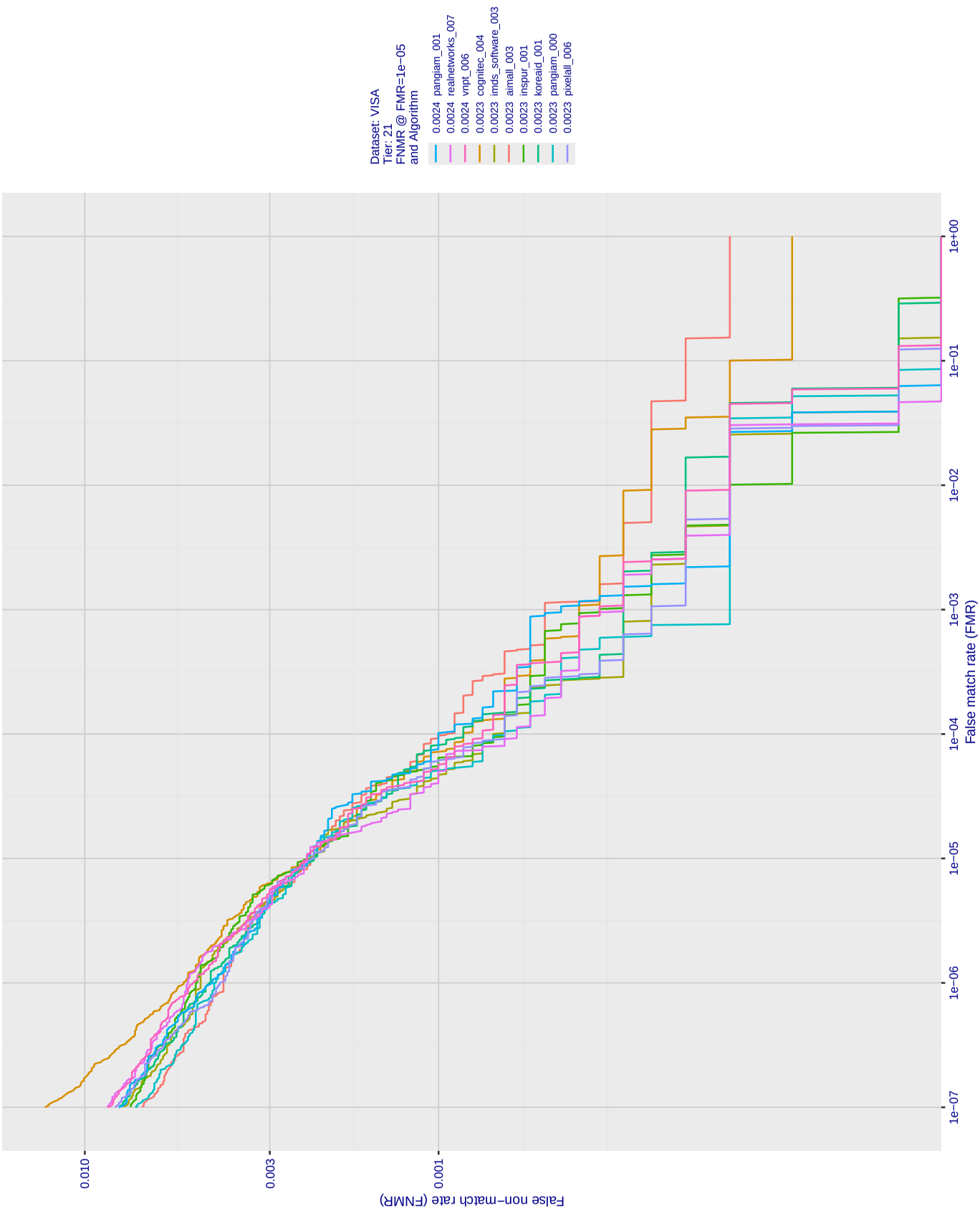

Figure 135: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

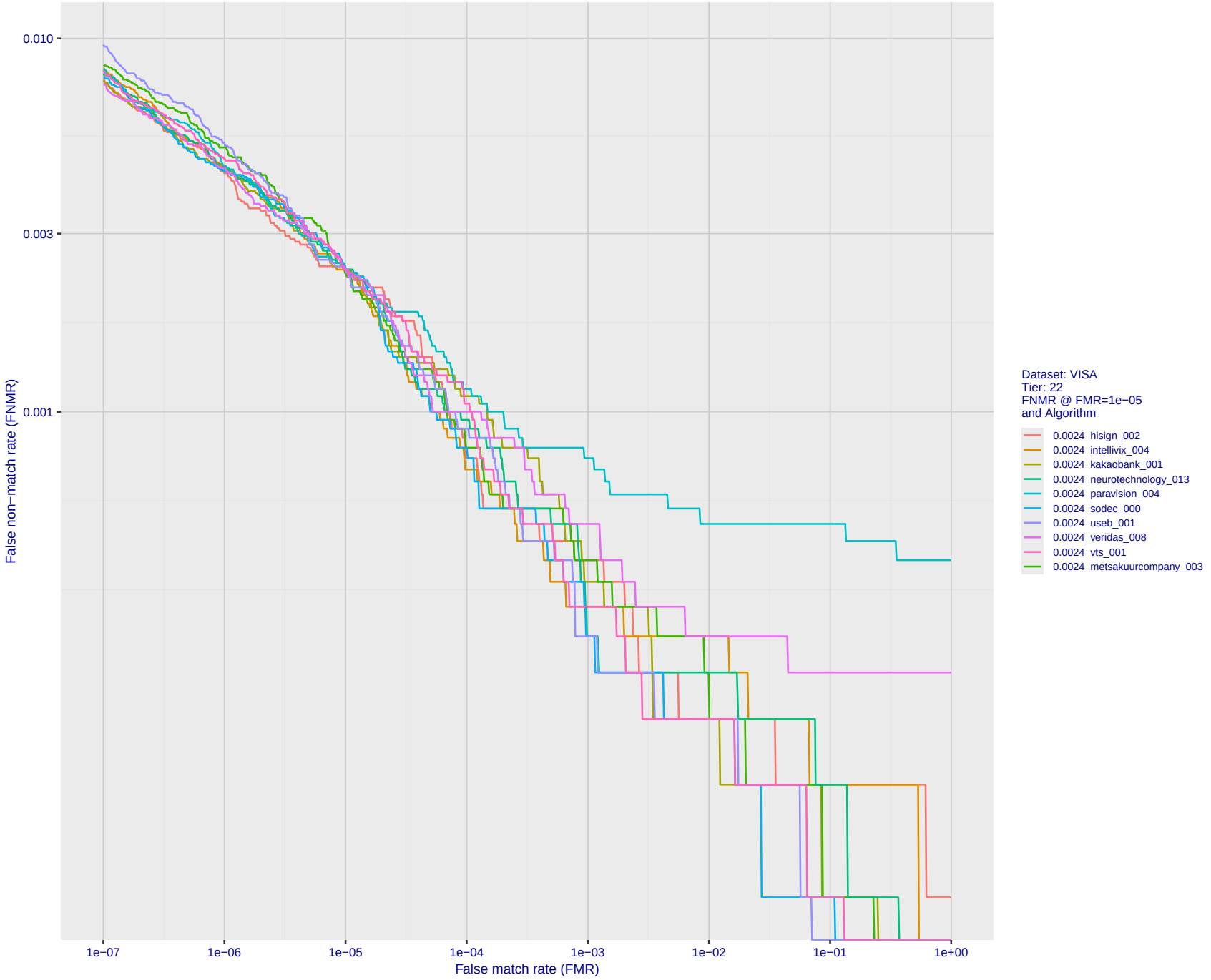

Figure 136: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

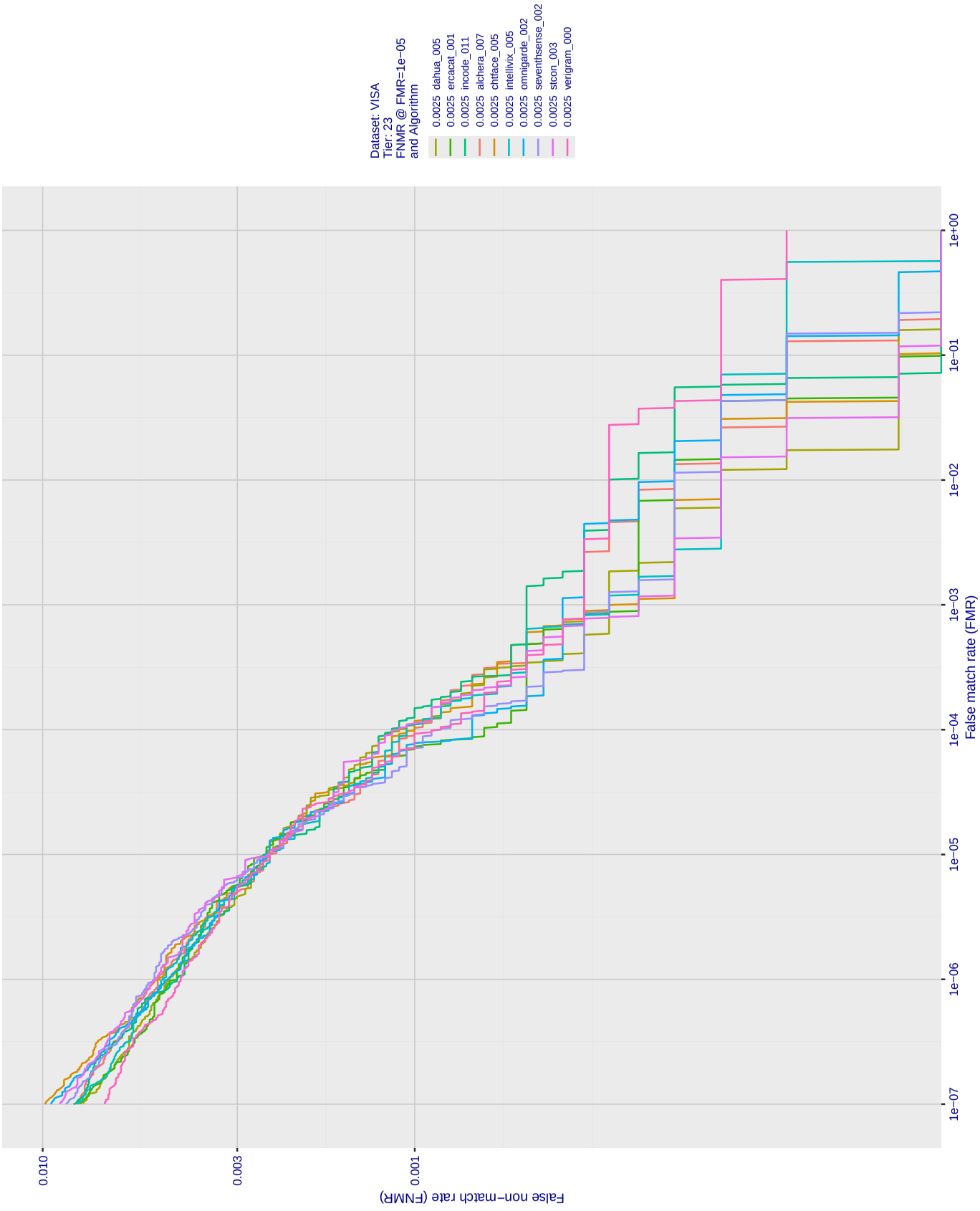

Figure 137: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

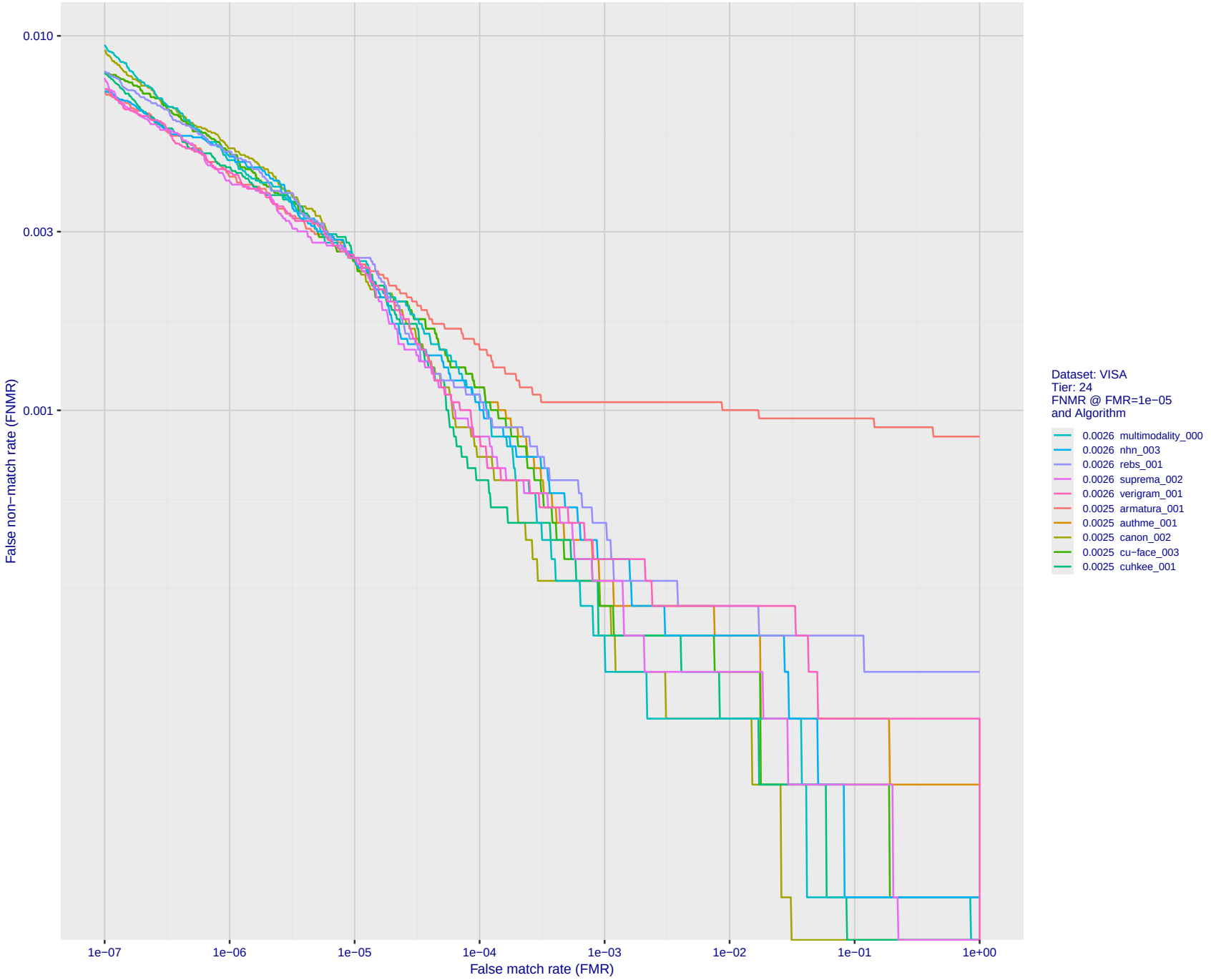

Figure 138: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

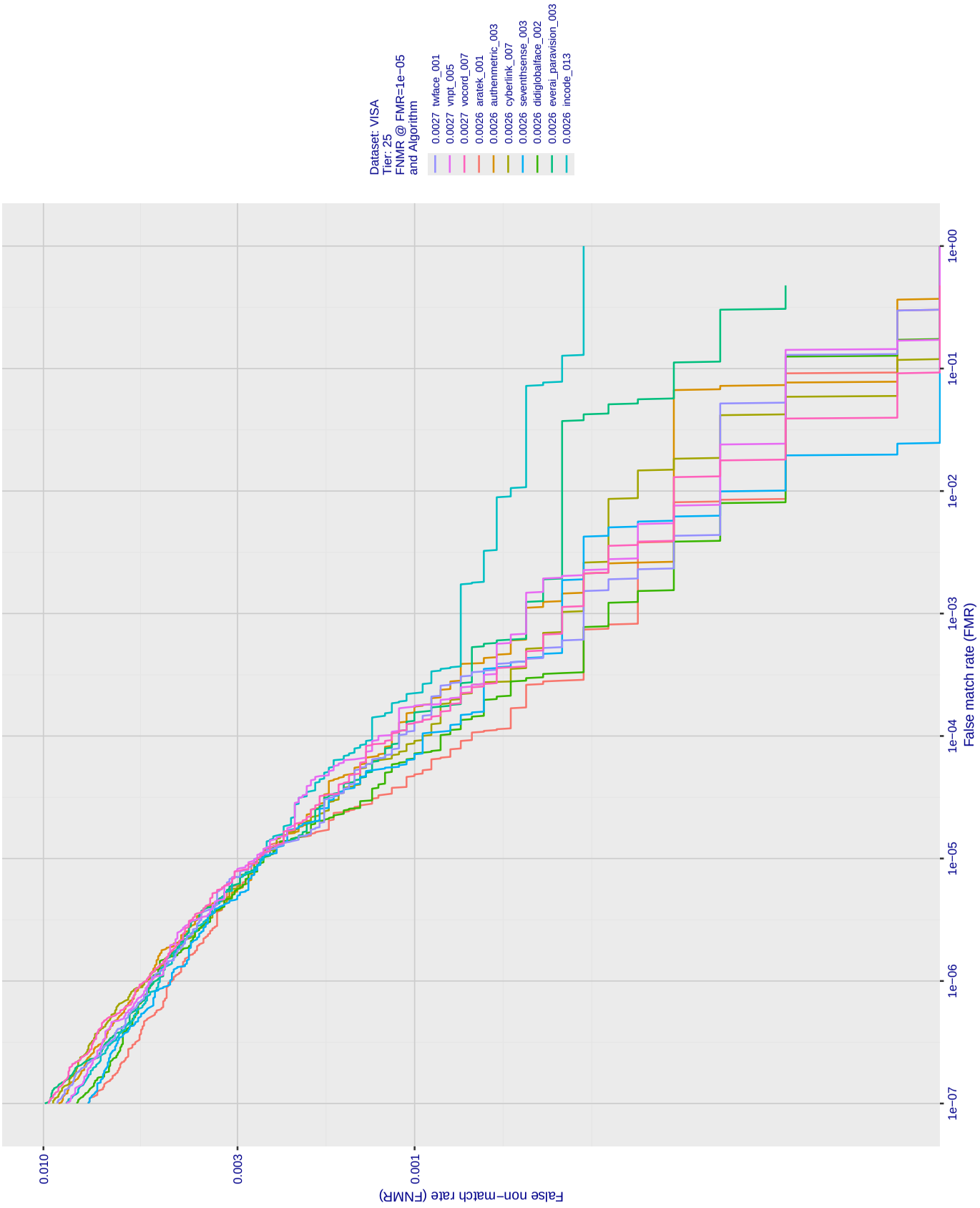

Figure 139: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

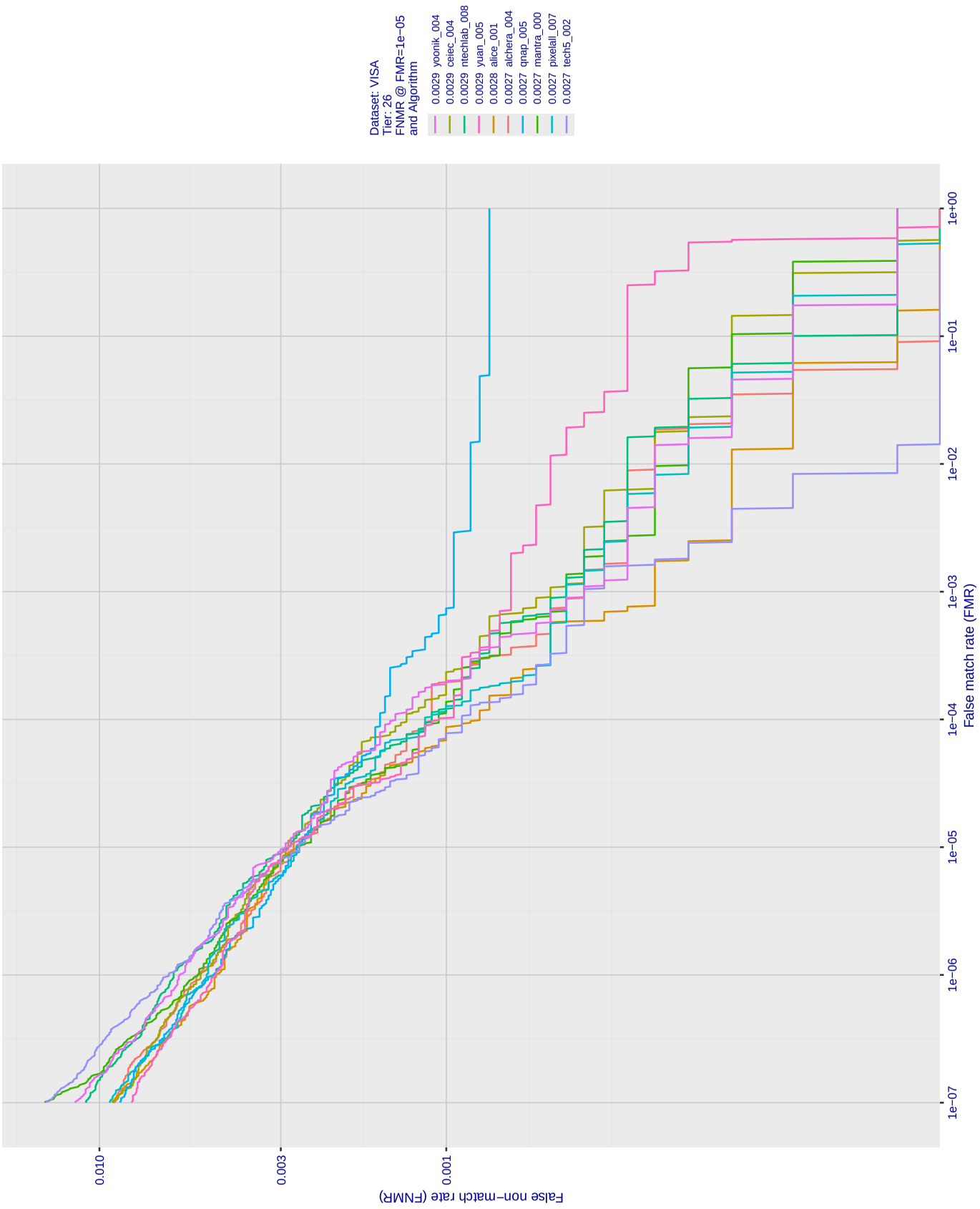

Figure 140: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

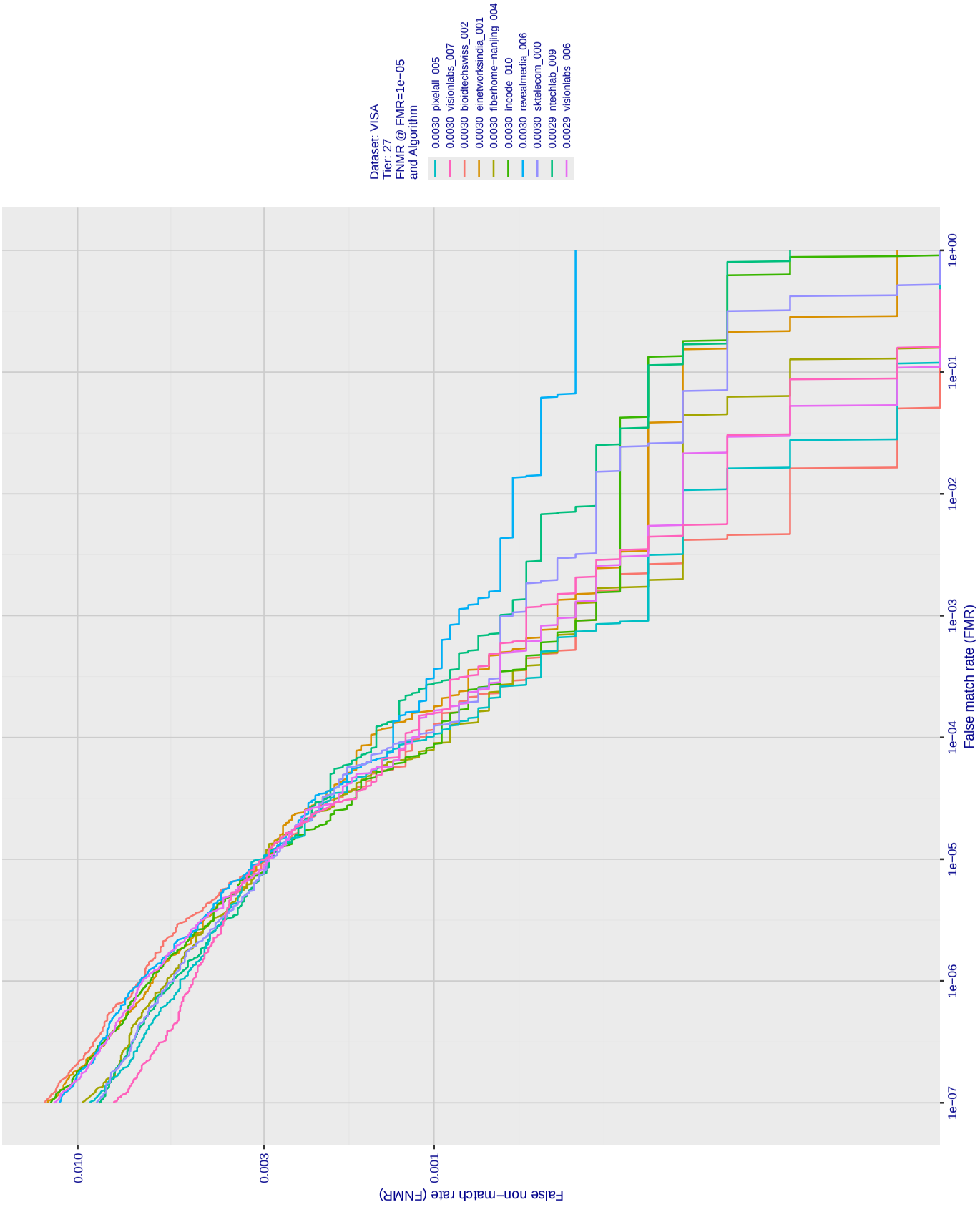

Figure 141: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

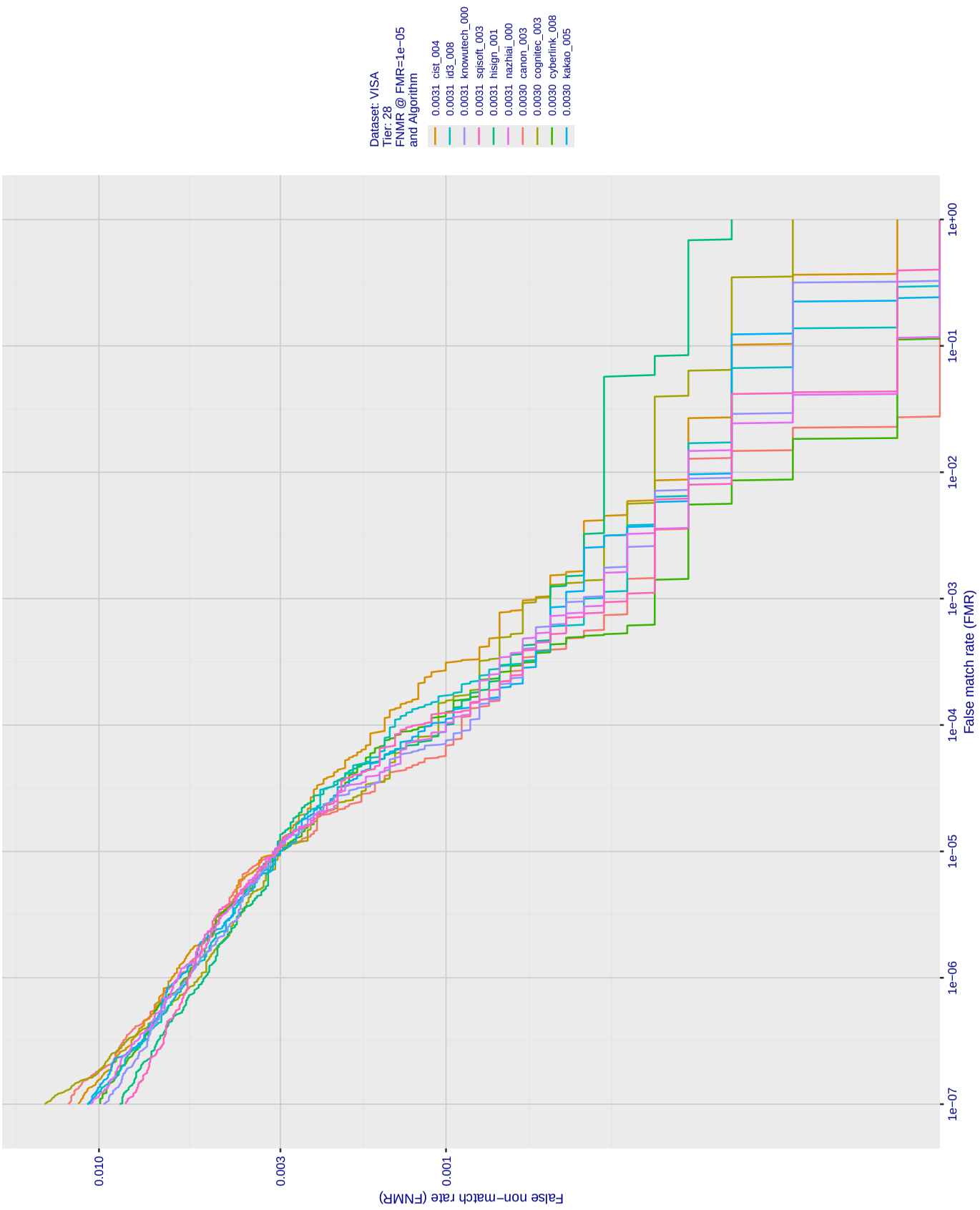

Figure 142: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

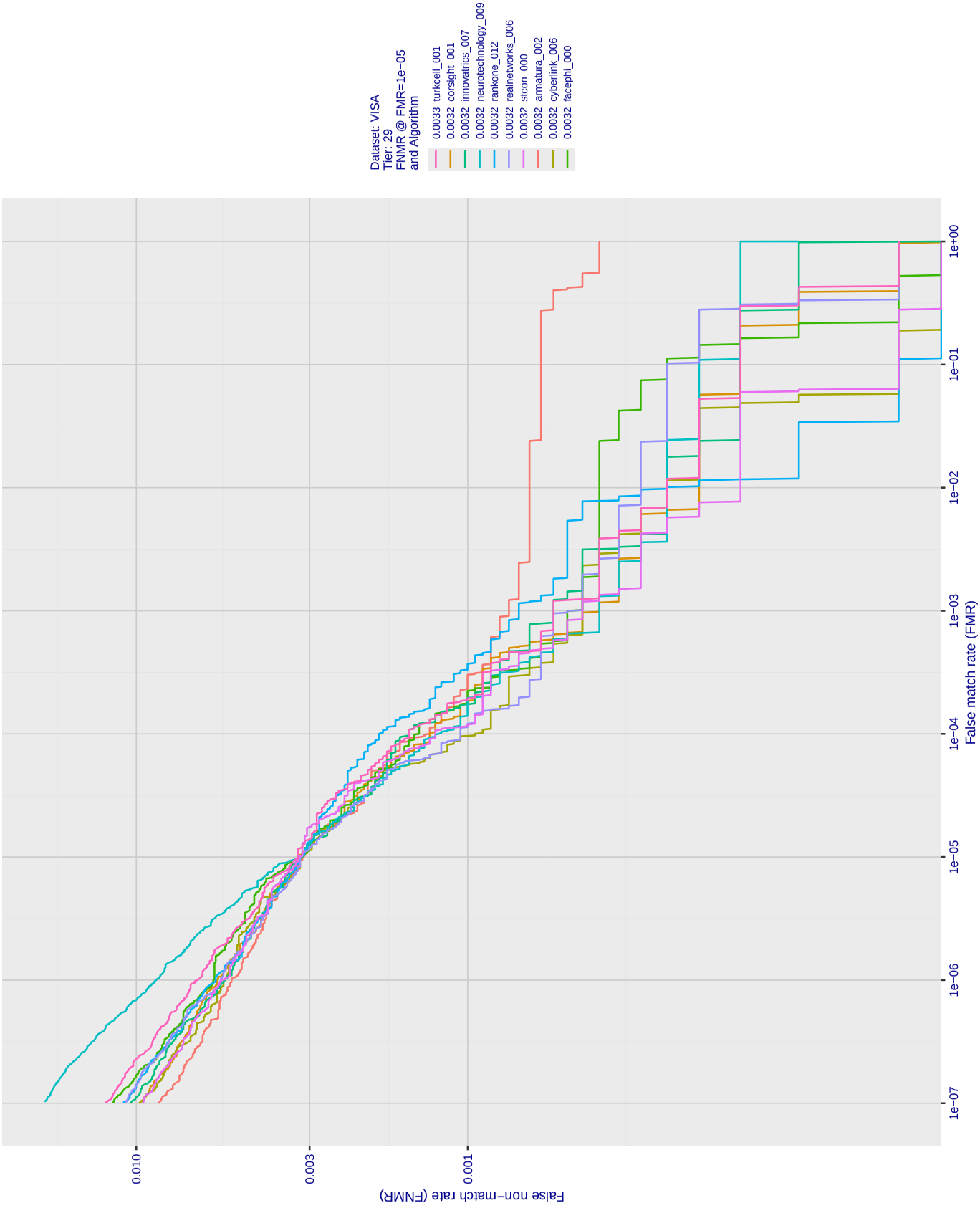

Figure 143: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

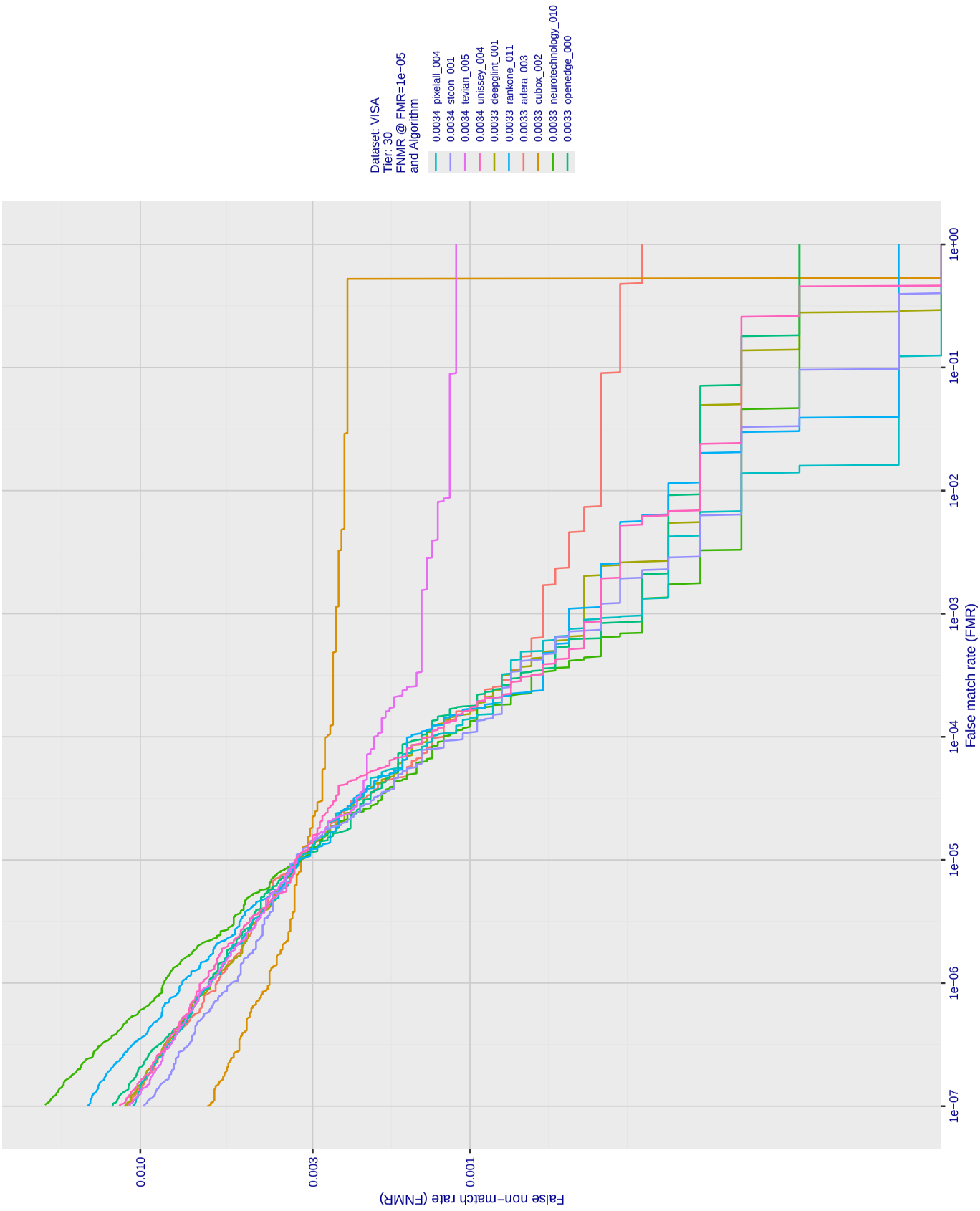

Figure 144: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

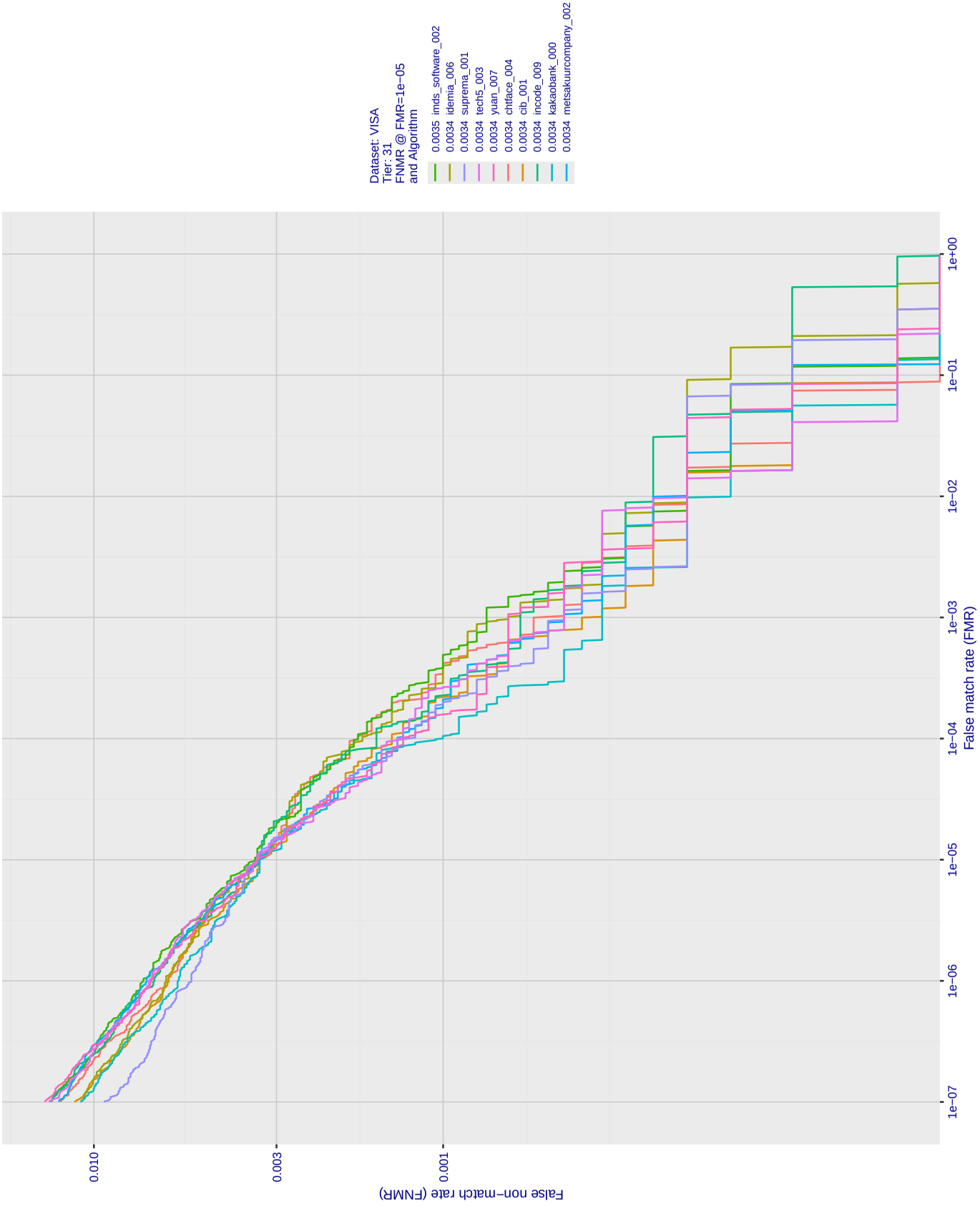

Figure 145: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

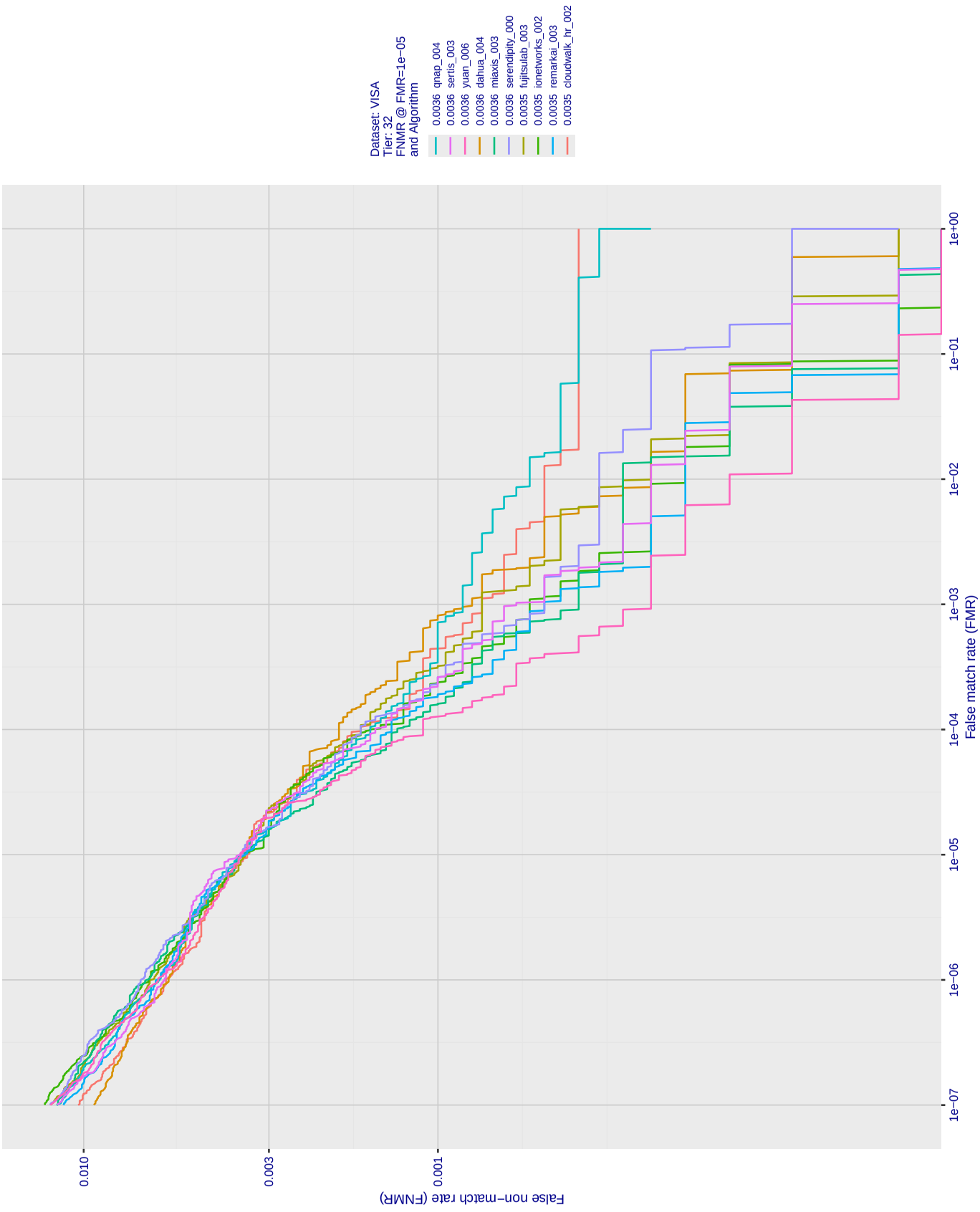

Figure 146: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

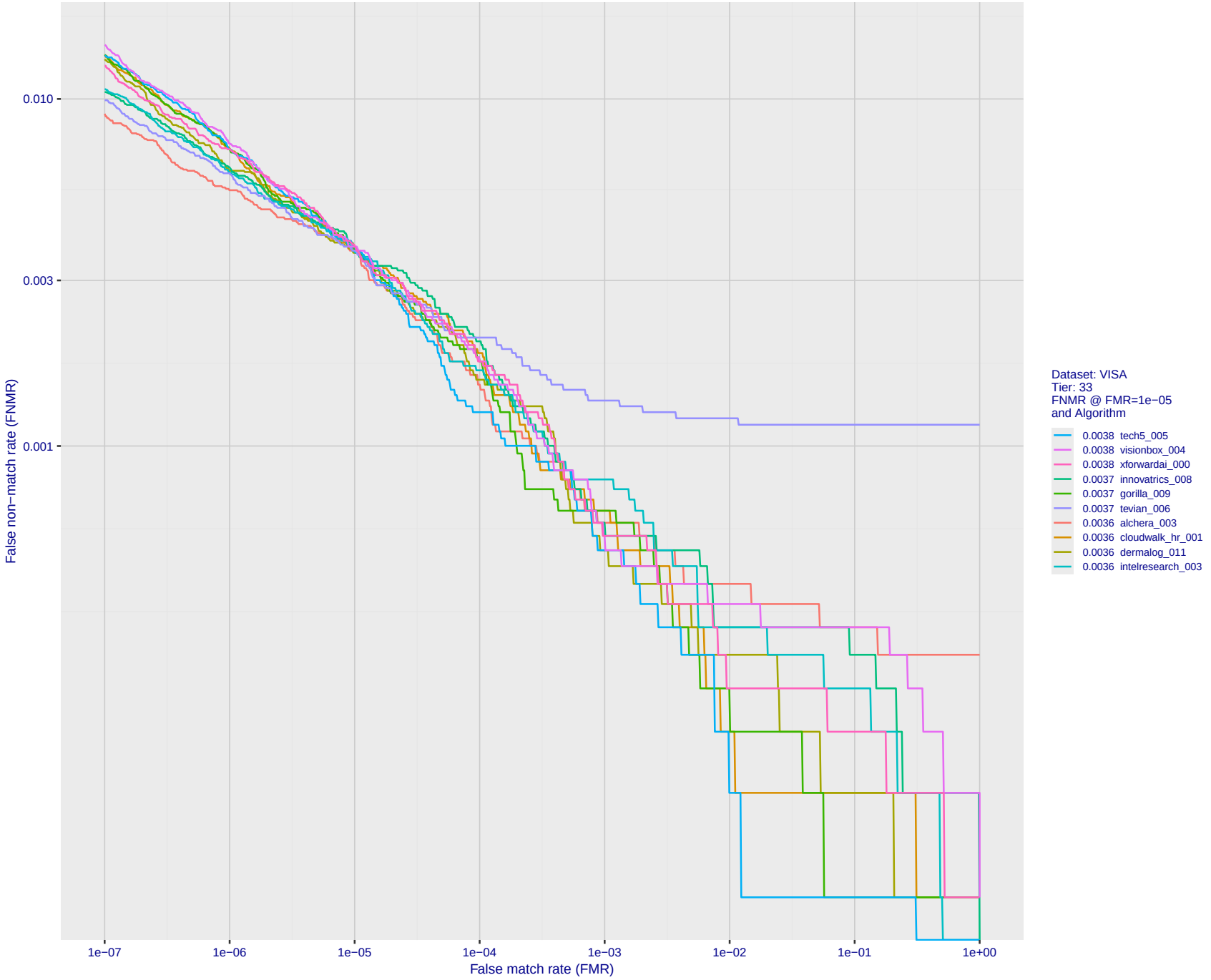

Figure 147: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

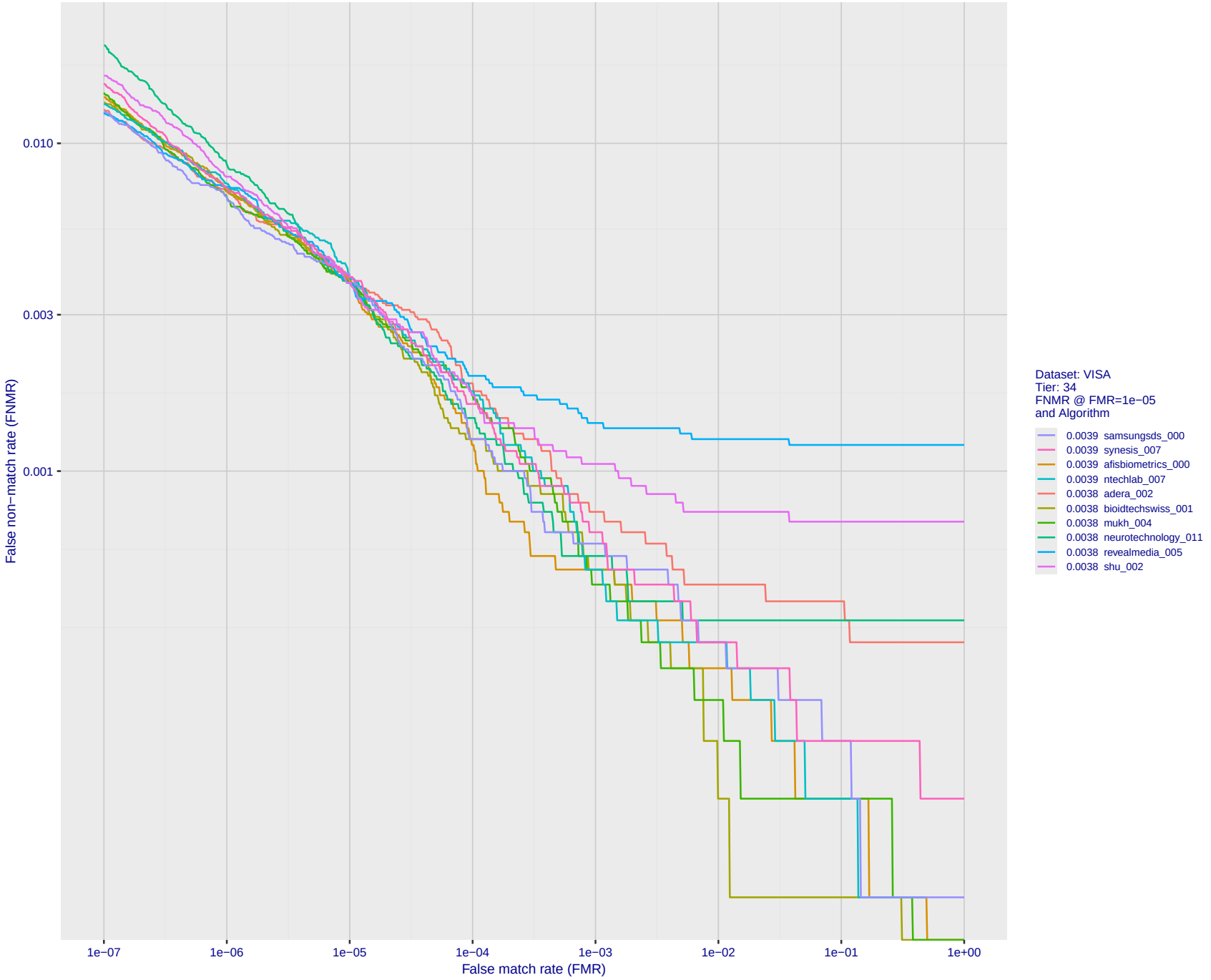

Figure 148: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

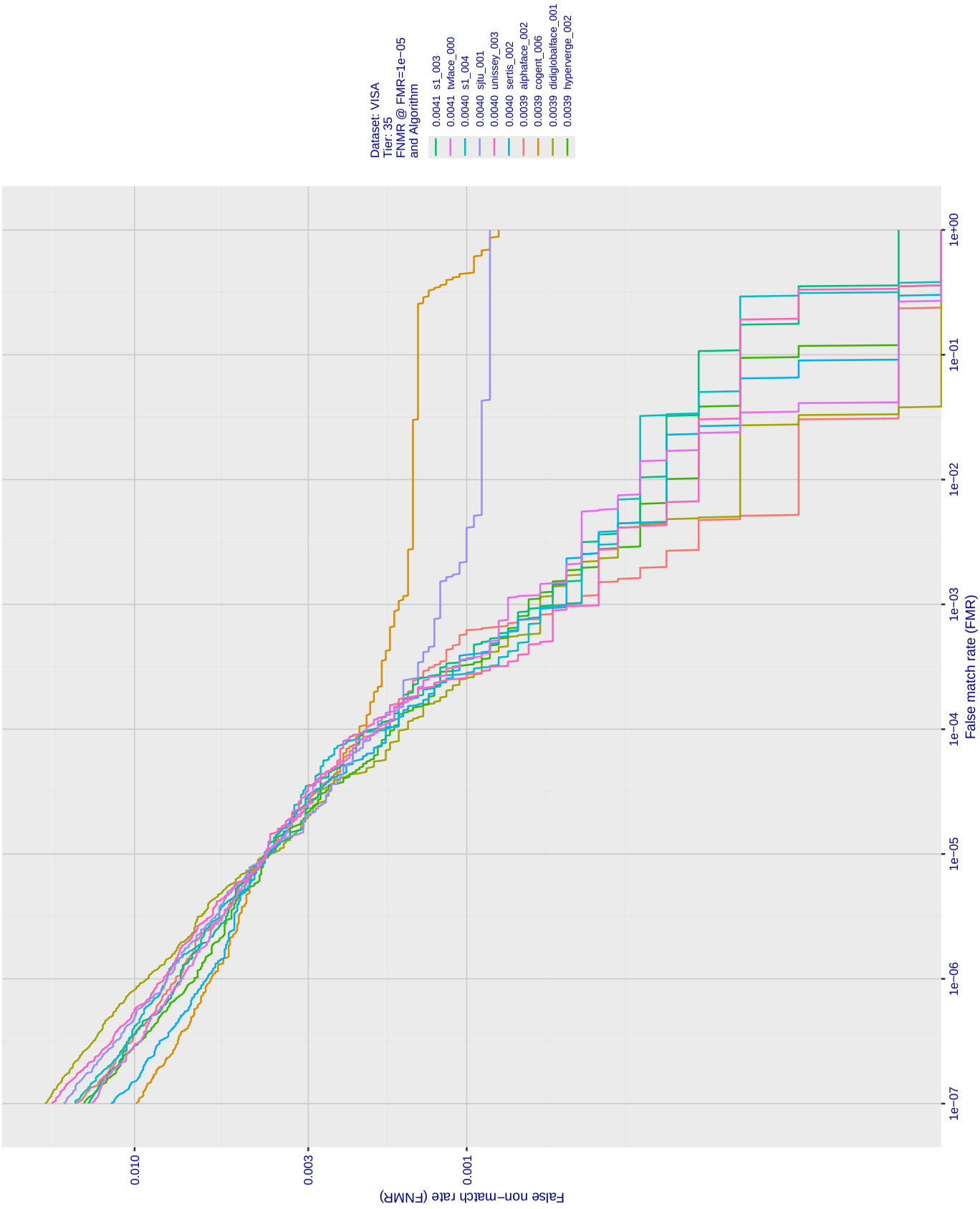

Figure 149: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

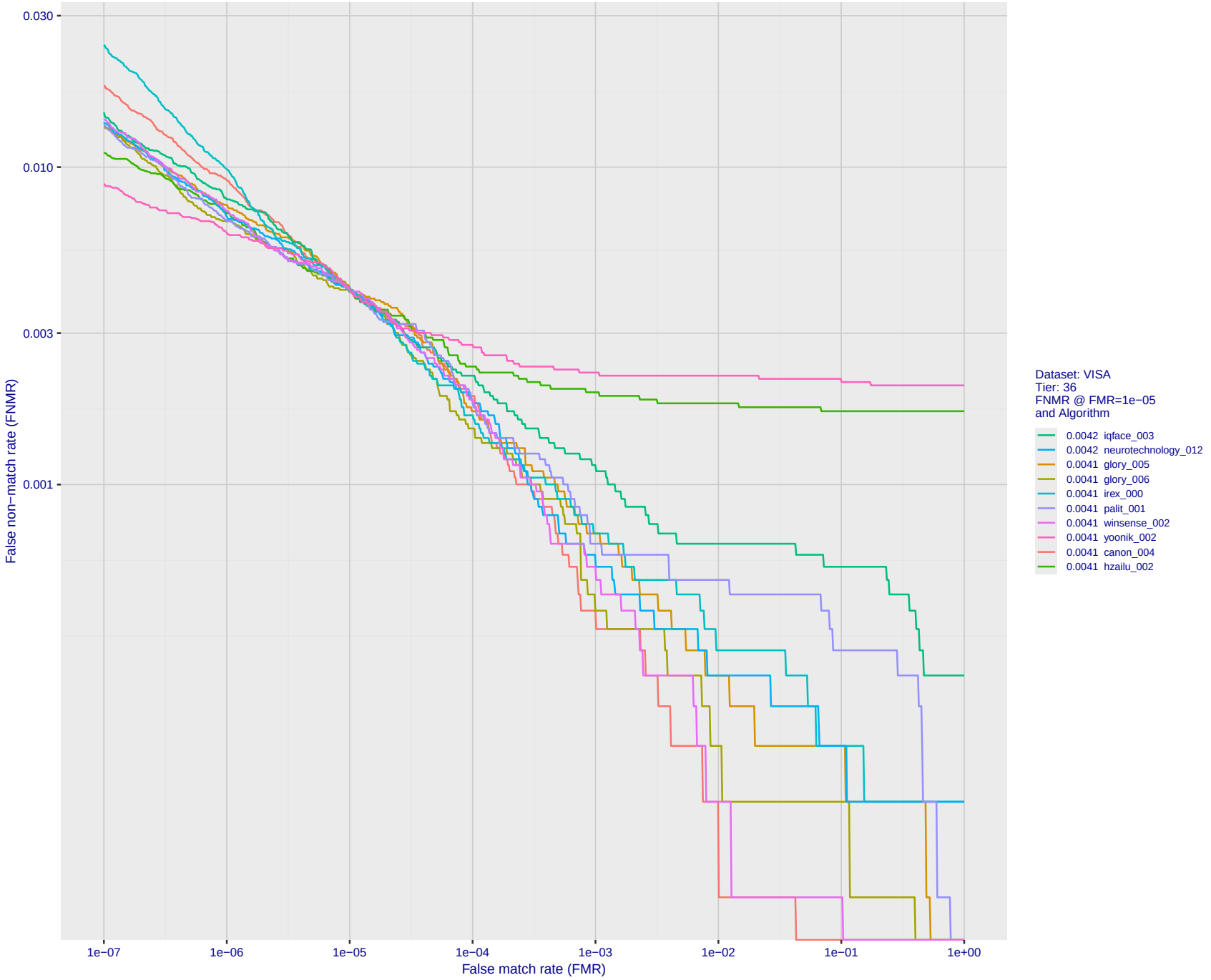

Figure 150: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

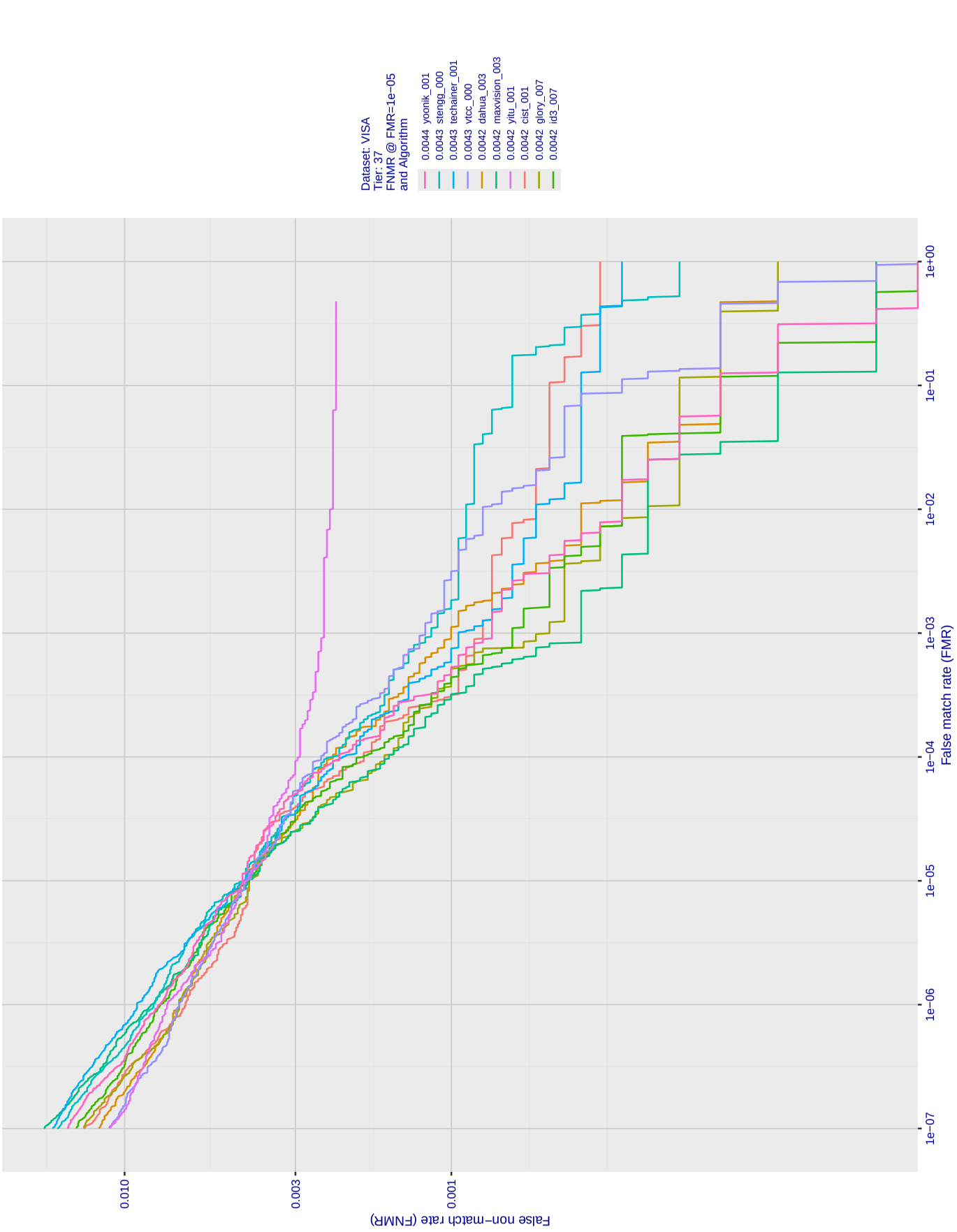

Figure 151: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

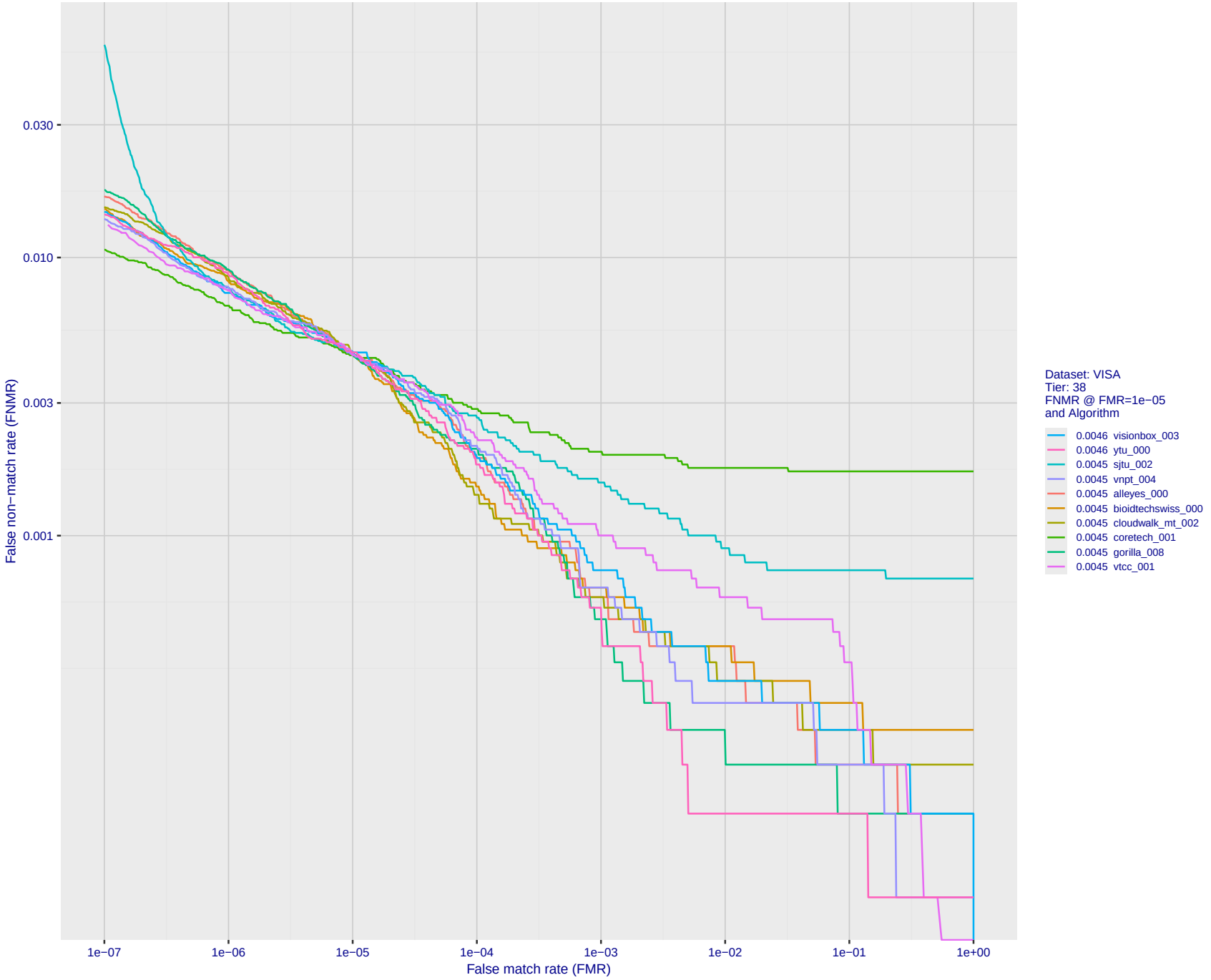

Figure 152: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

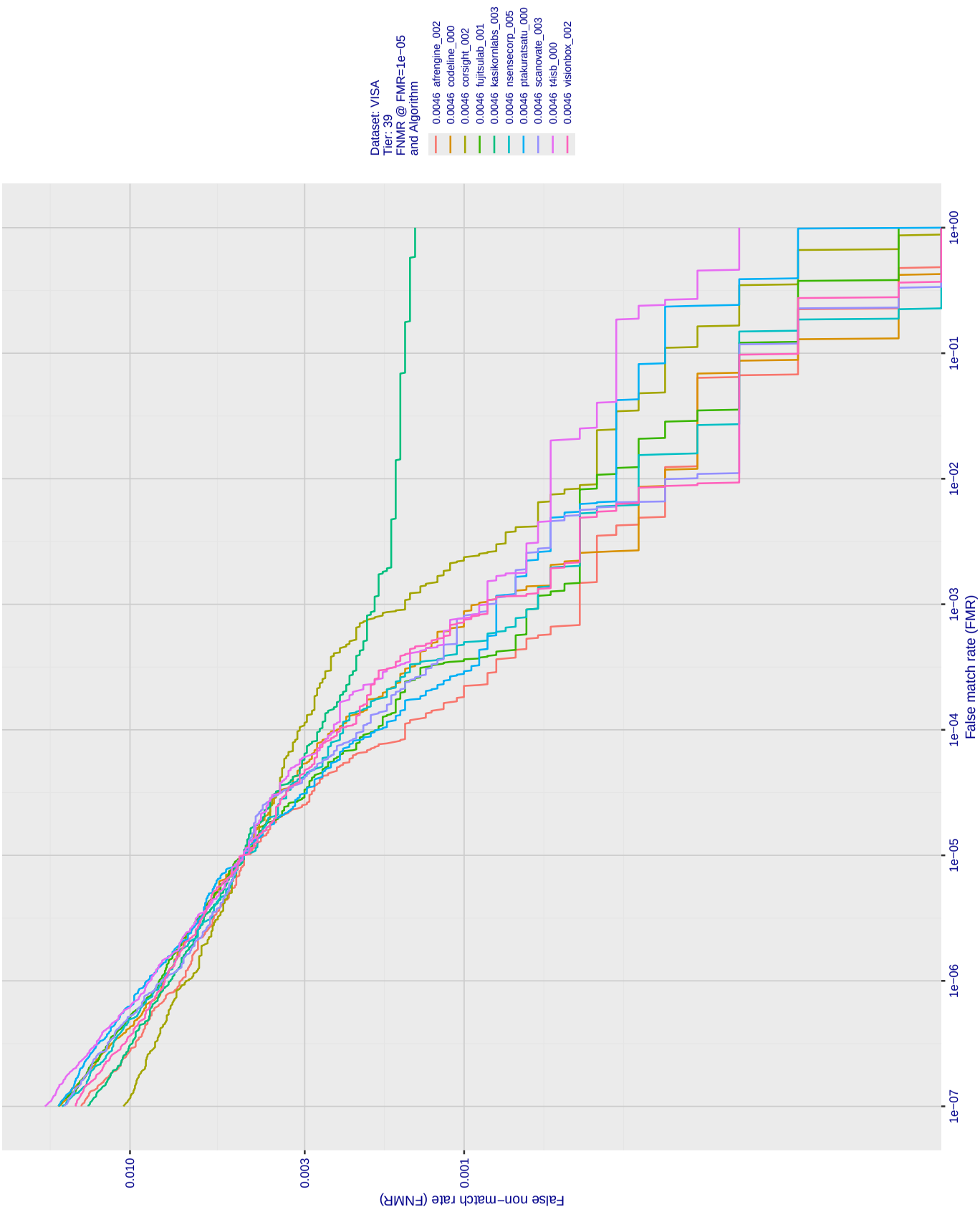

Figure 153: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

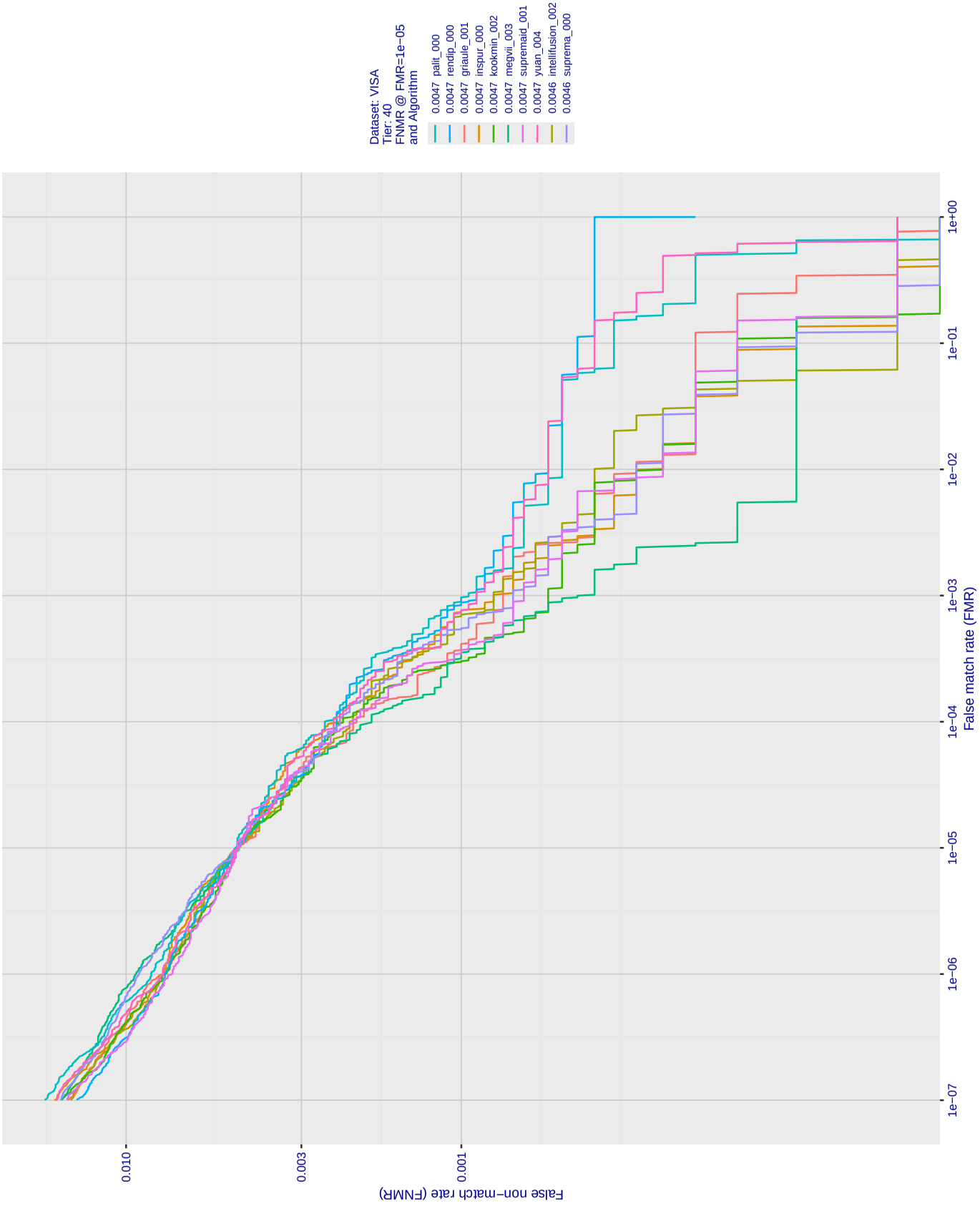

Figure 154: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

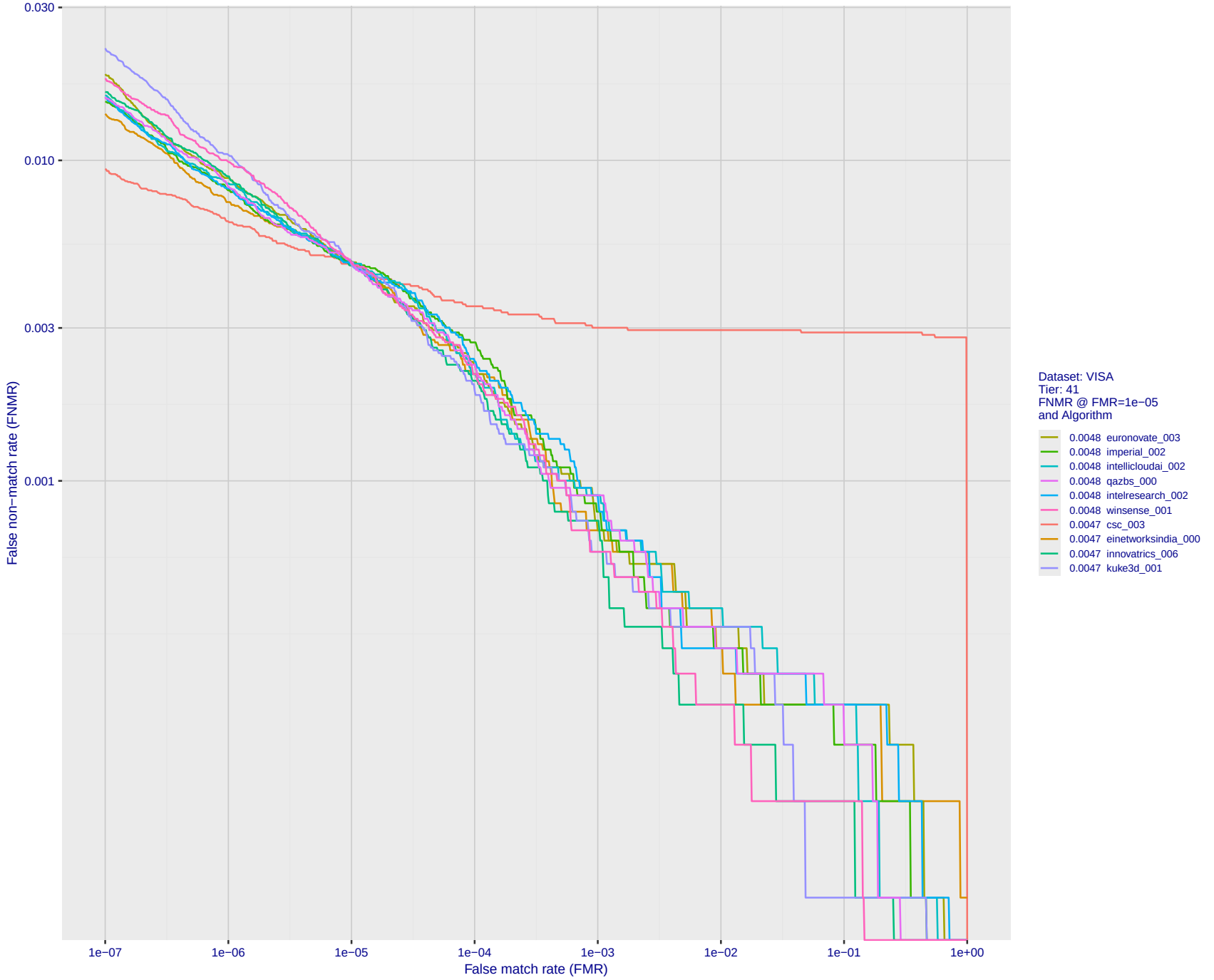

Figure 155: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

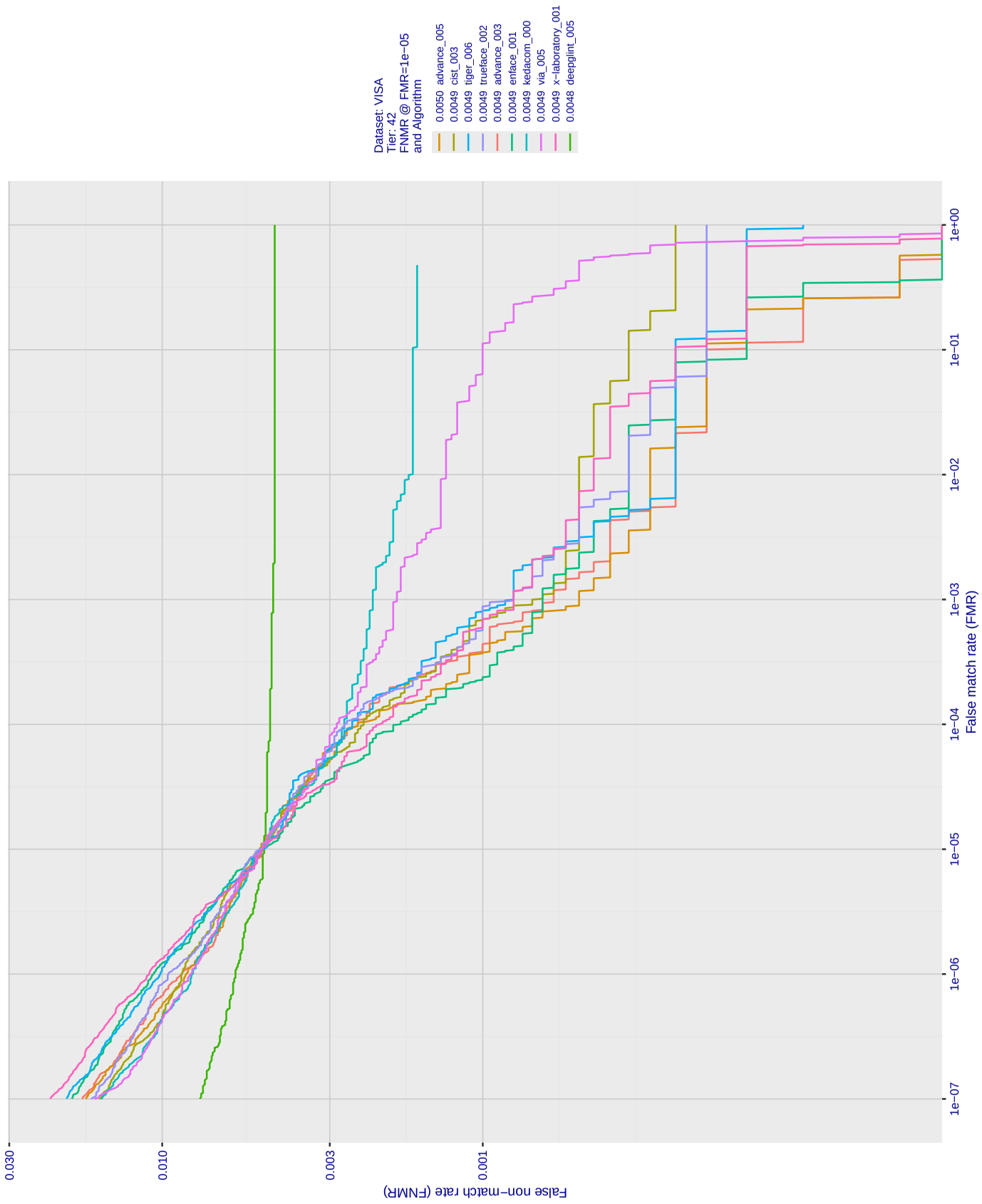

Figure 156: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

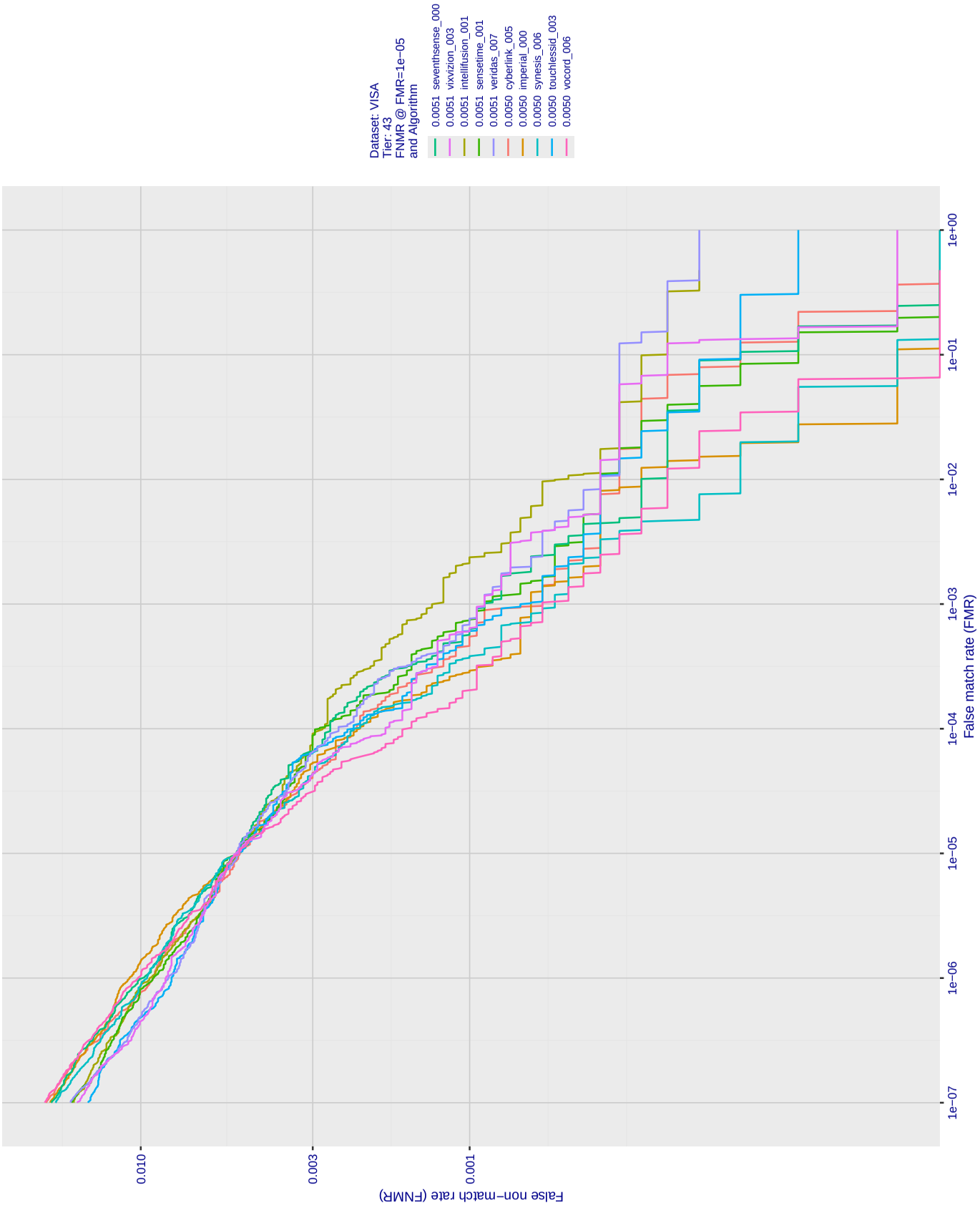

Figure 157: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

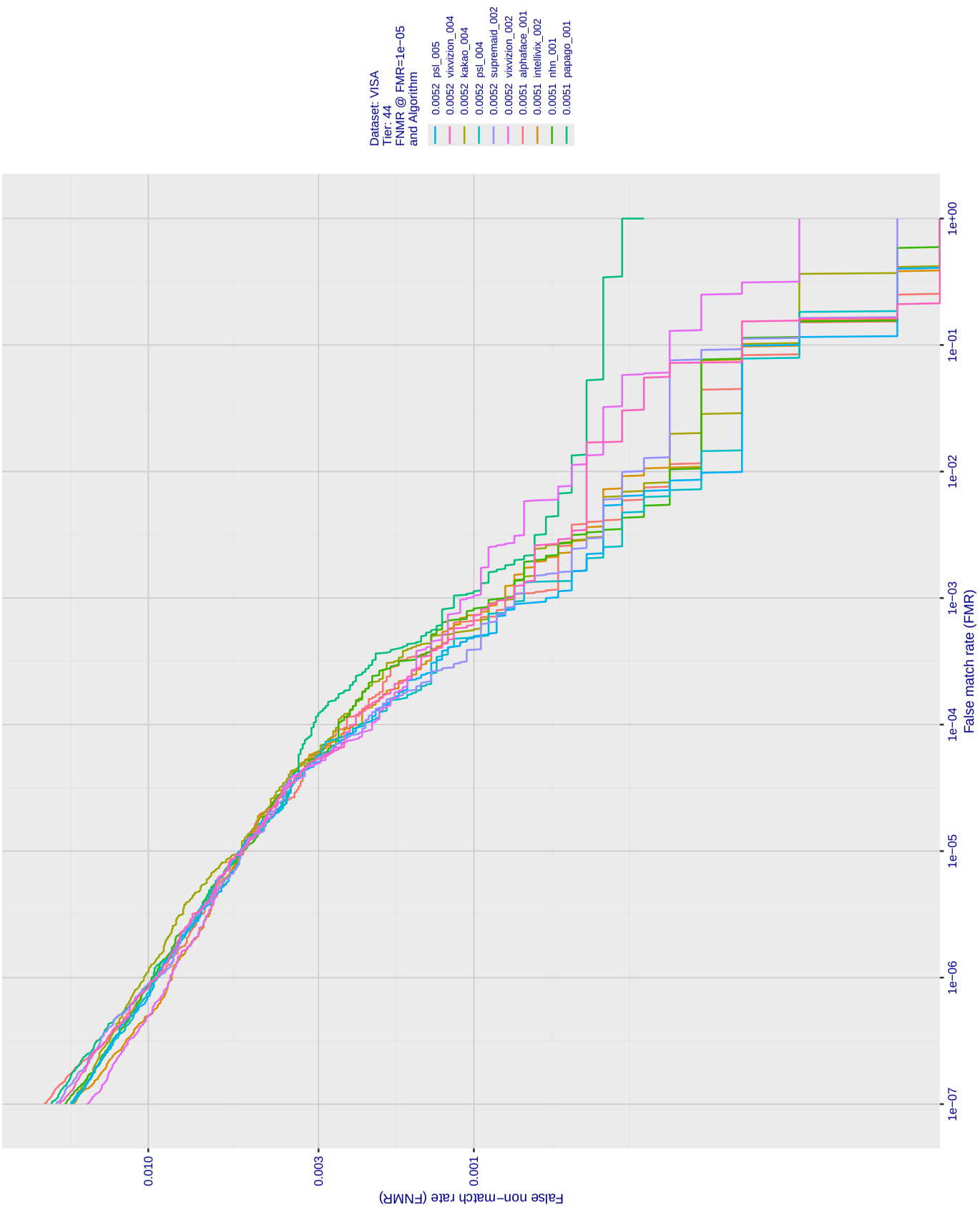

Figure 158: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

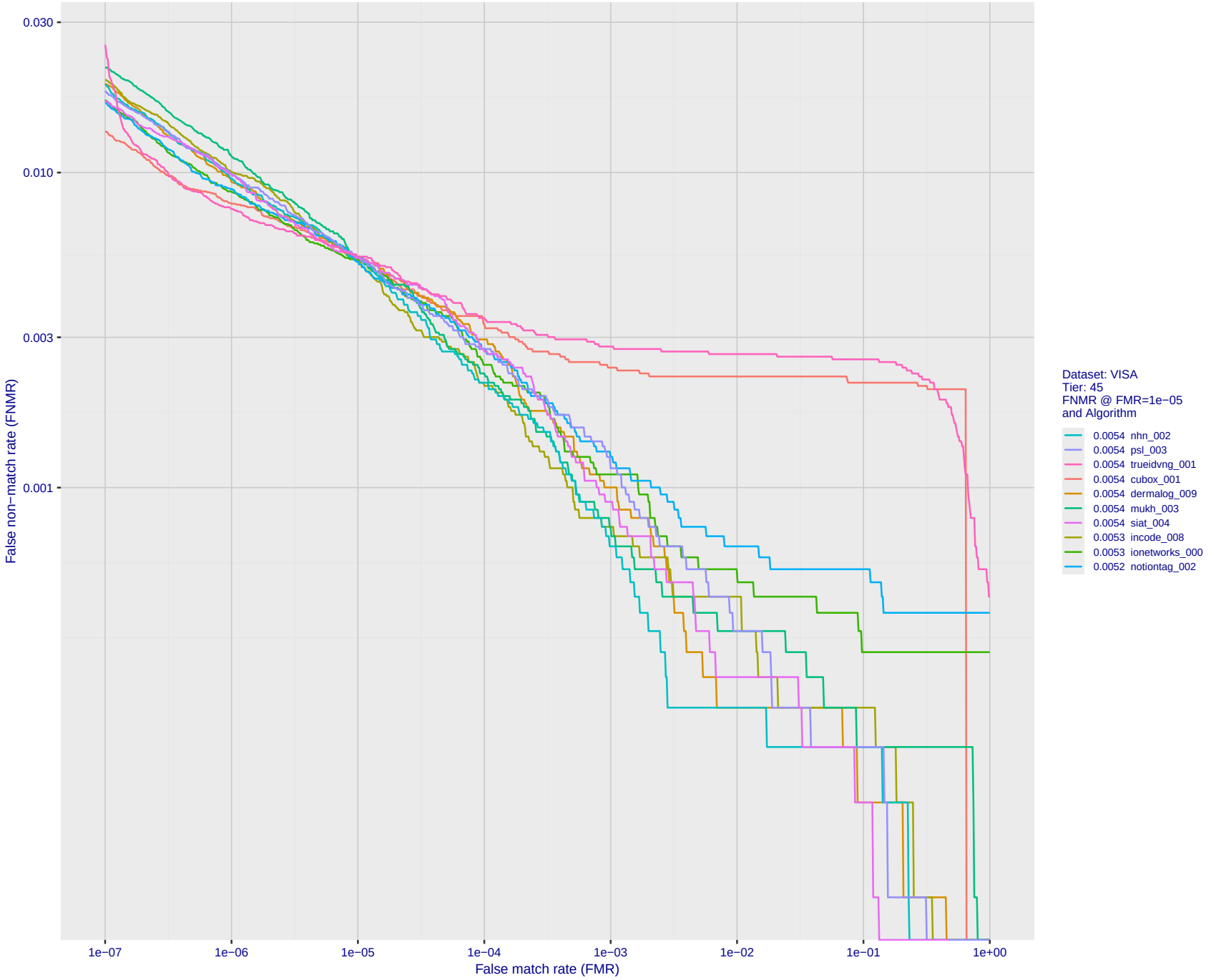

Figure 159: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

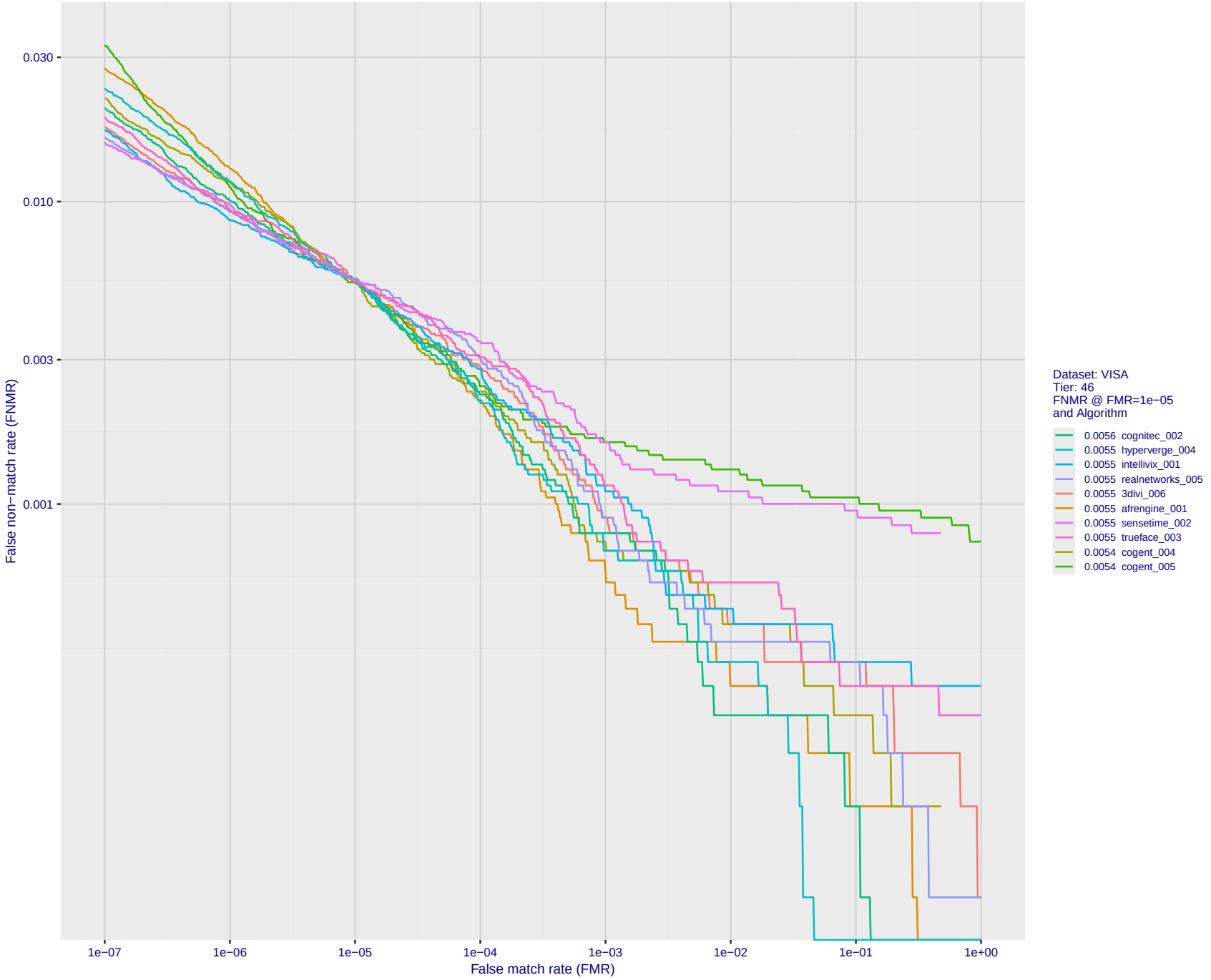

Figure 160: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

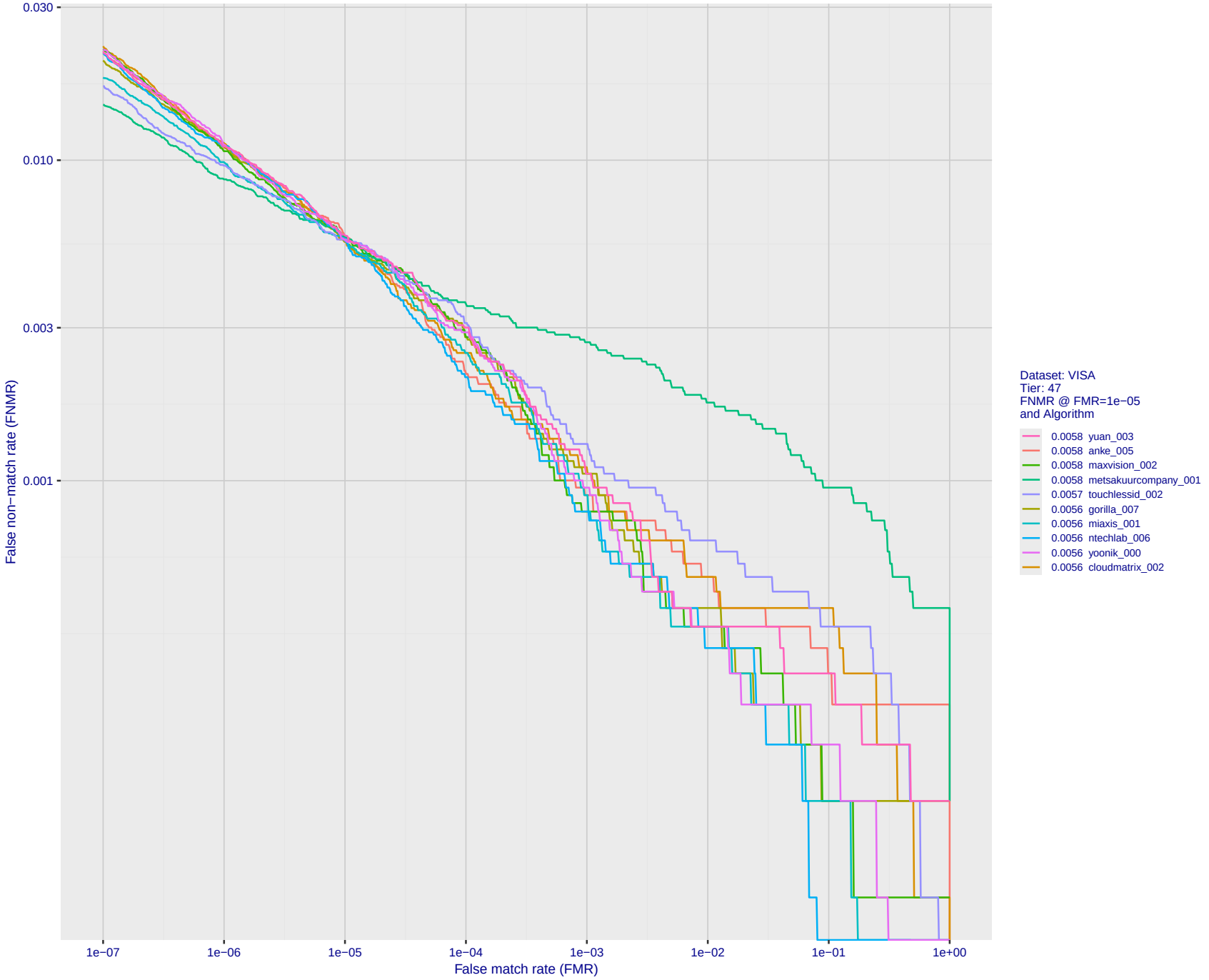

Figure 161: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

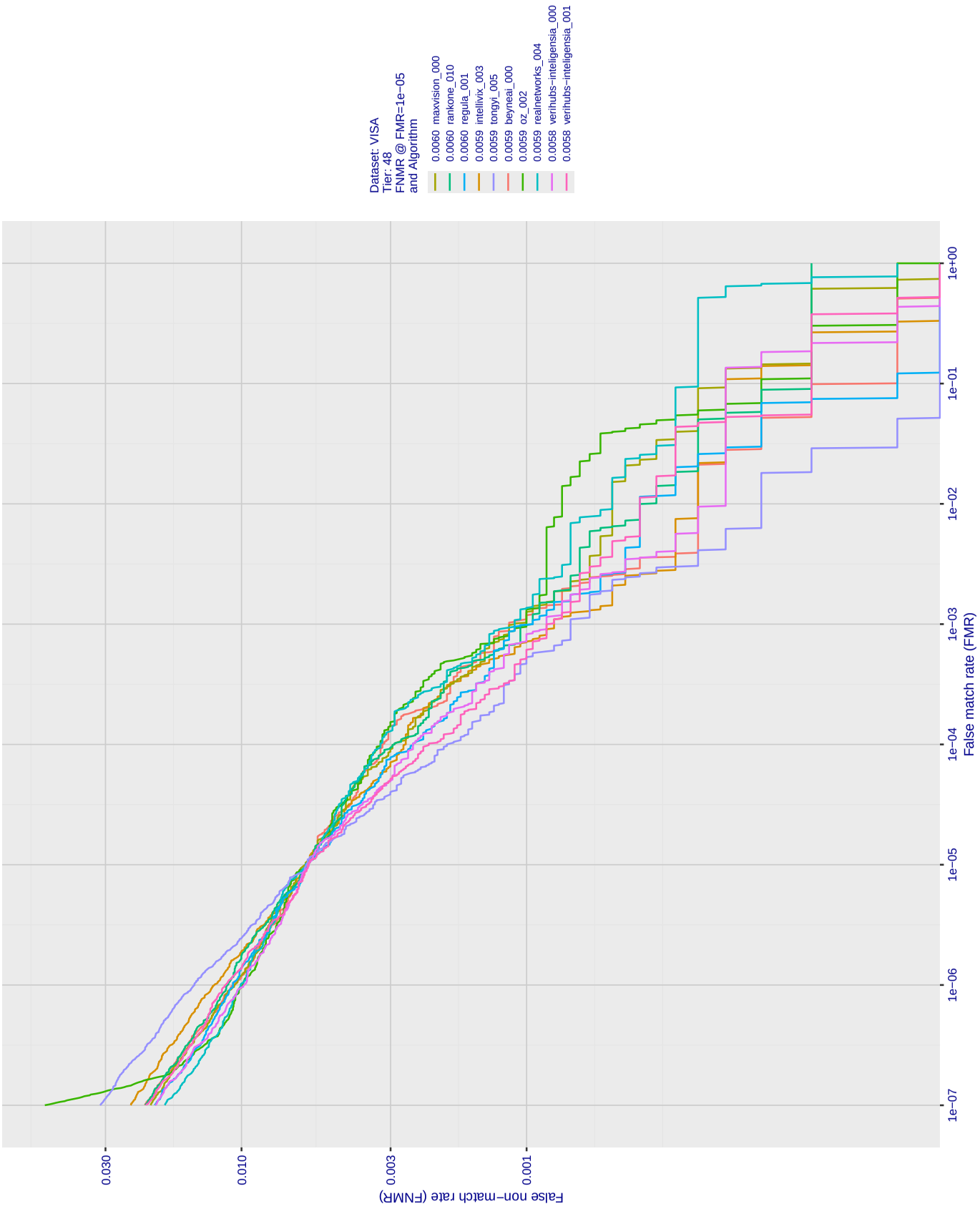

Figure 162: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

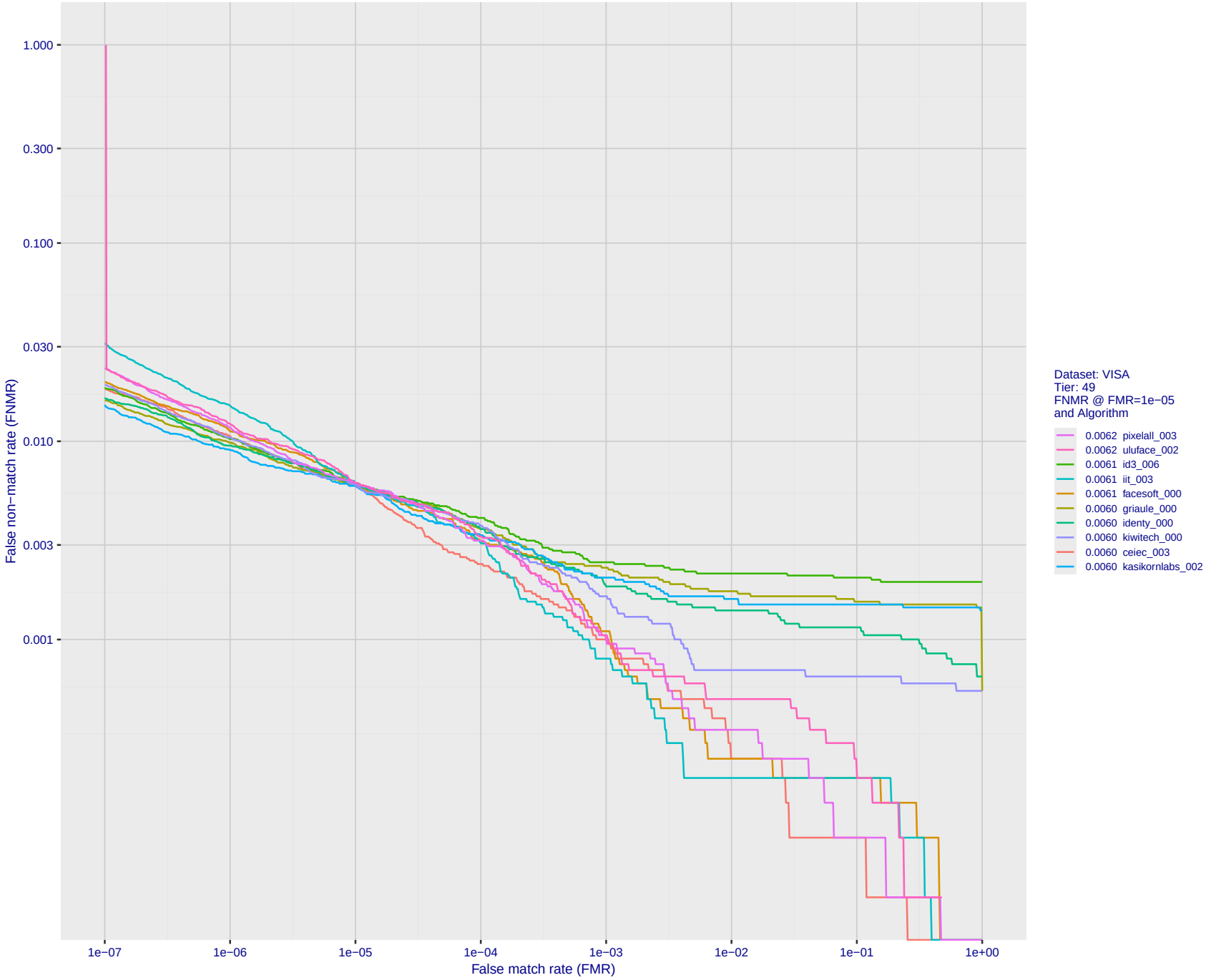

Figure 163: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

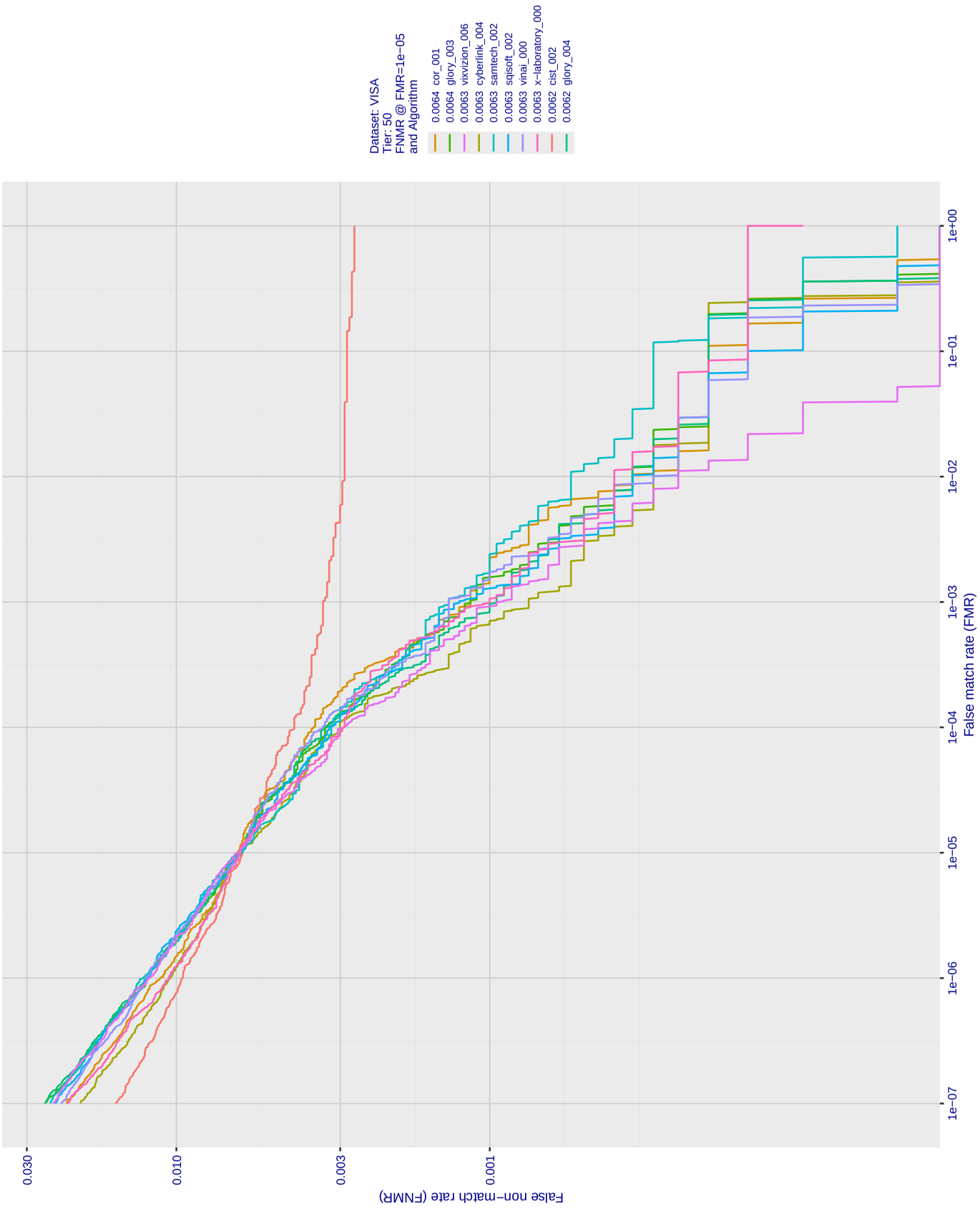

Figure 164: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

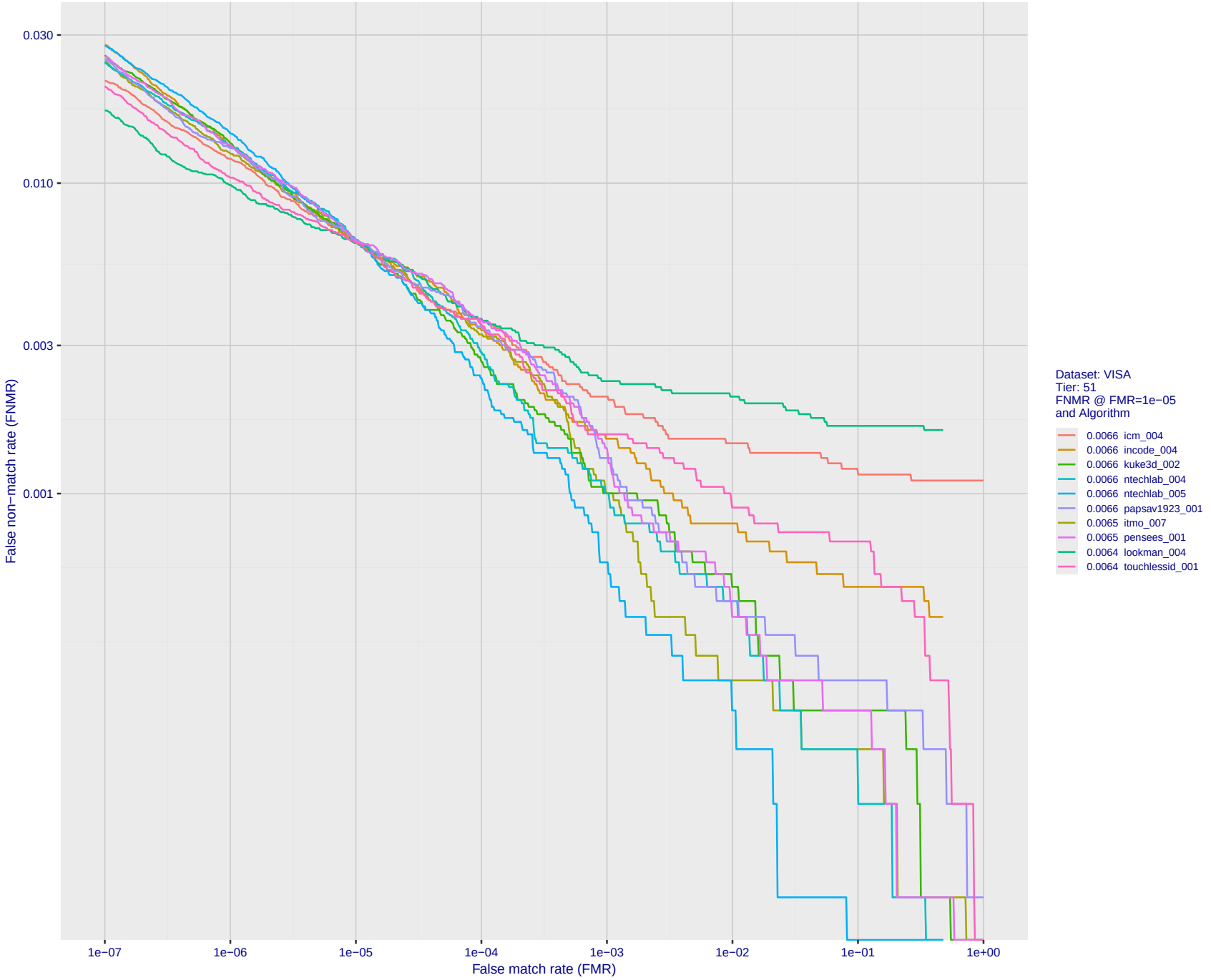

Figure 165: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

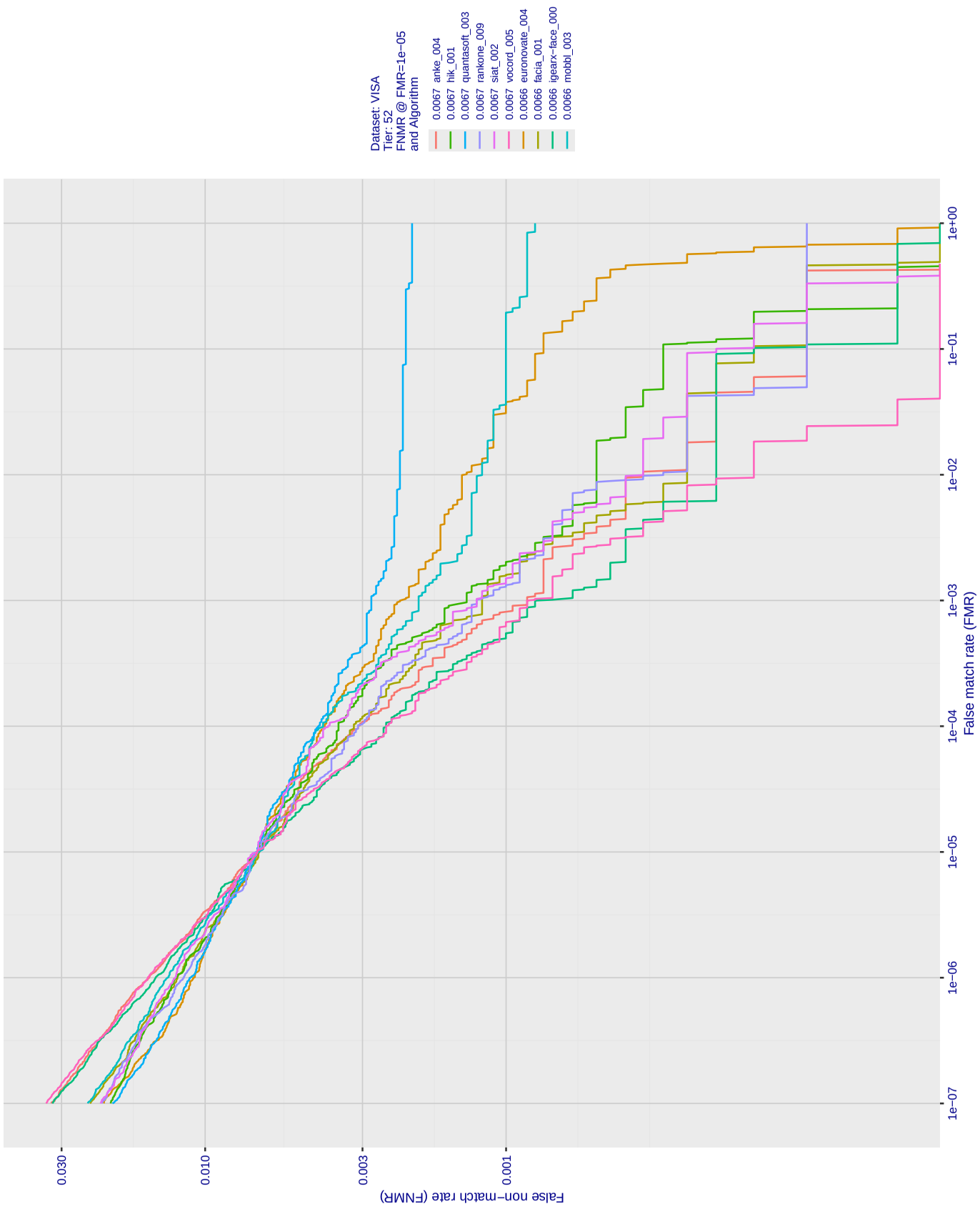

Figure 166: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

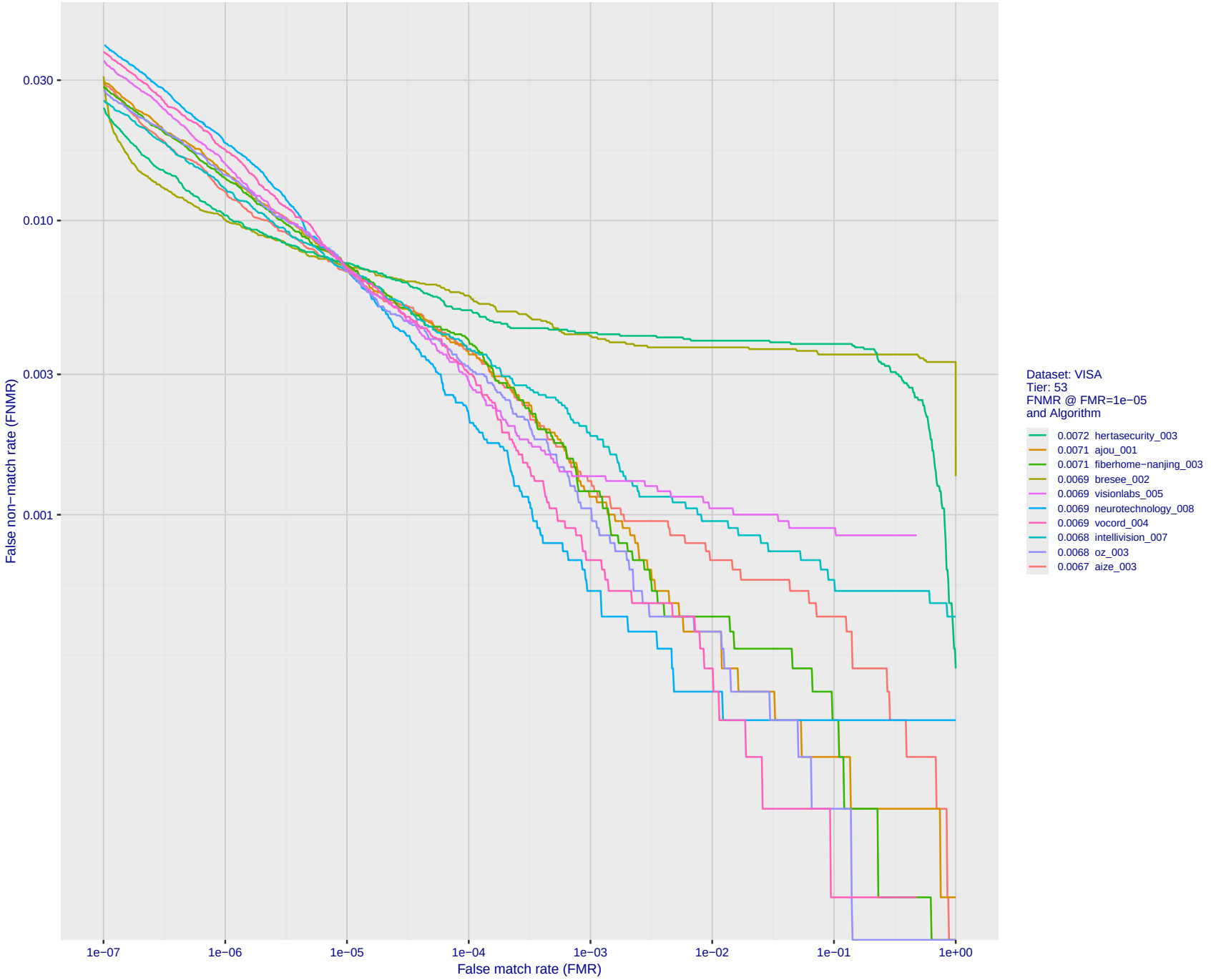

Figure 167: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

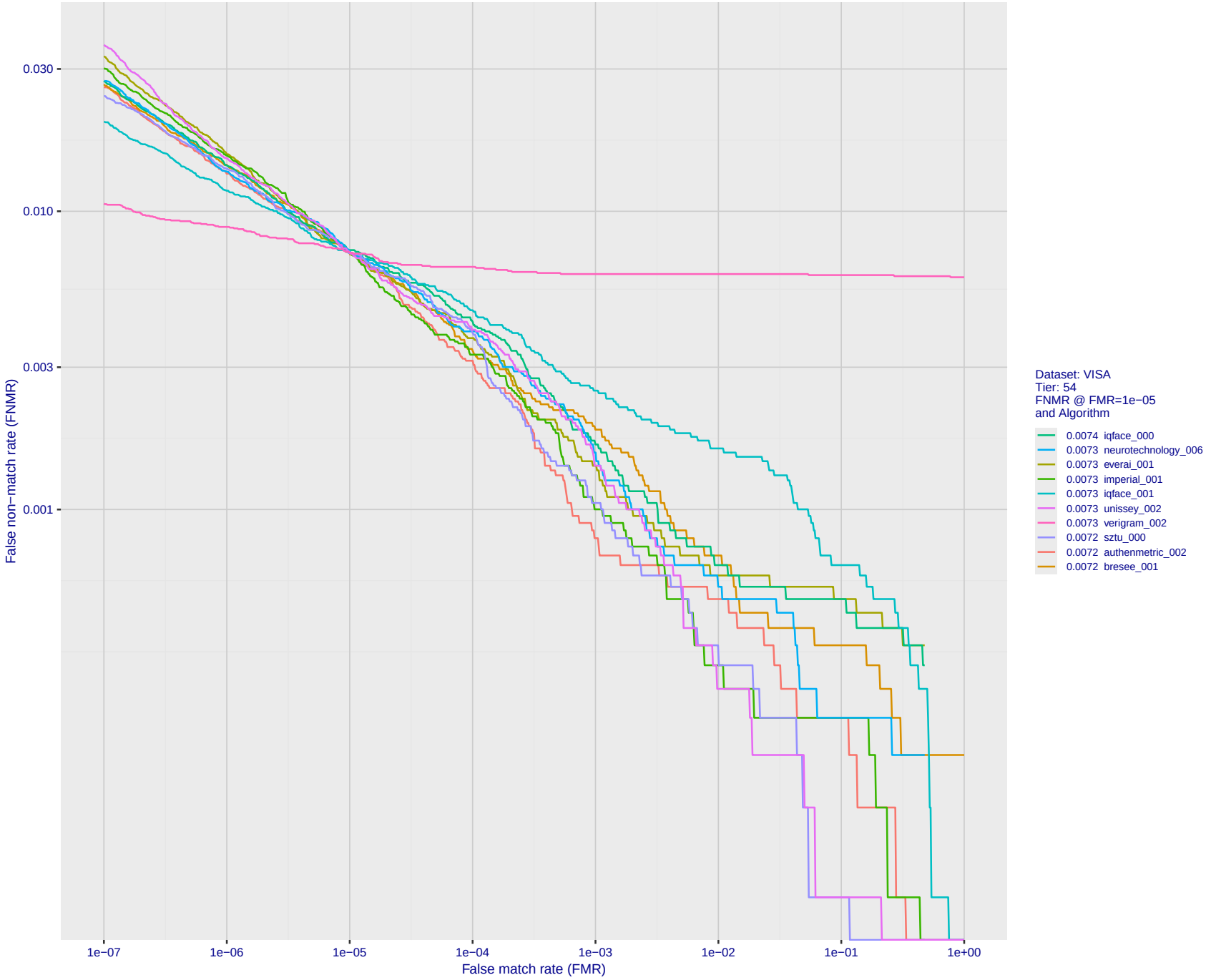

Figure 168: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

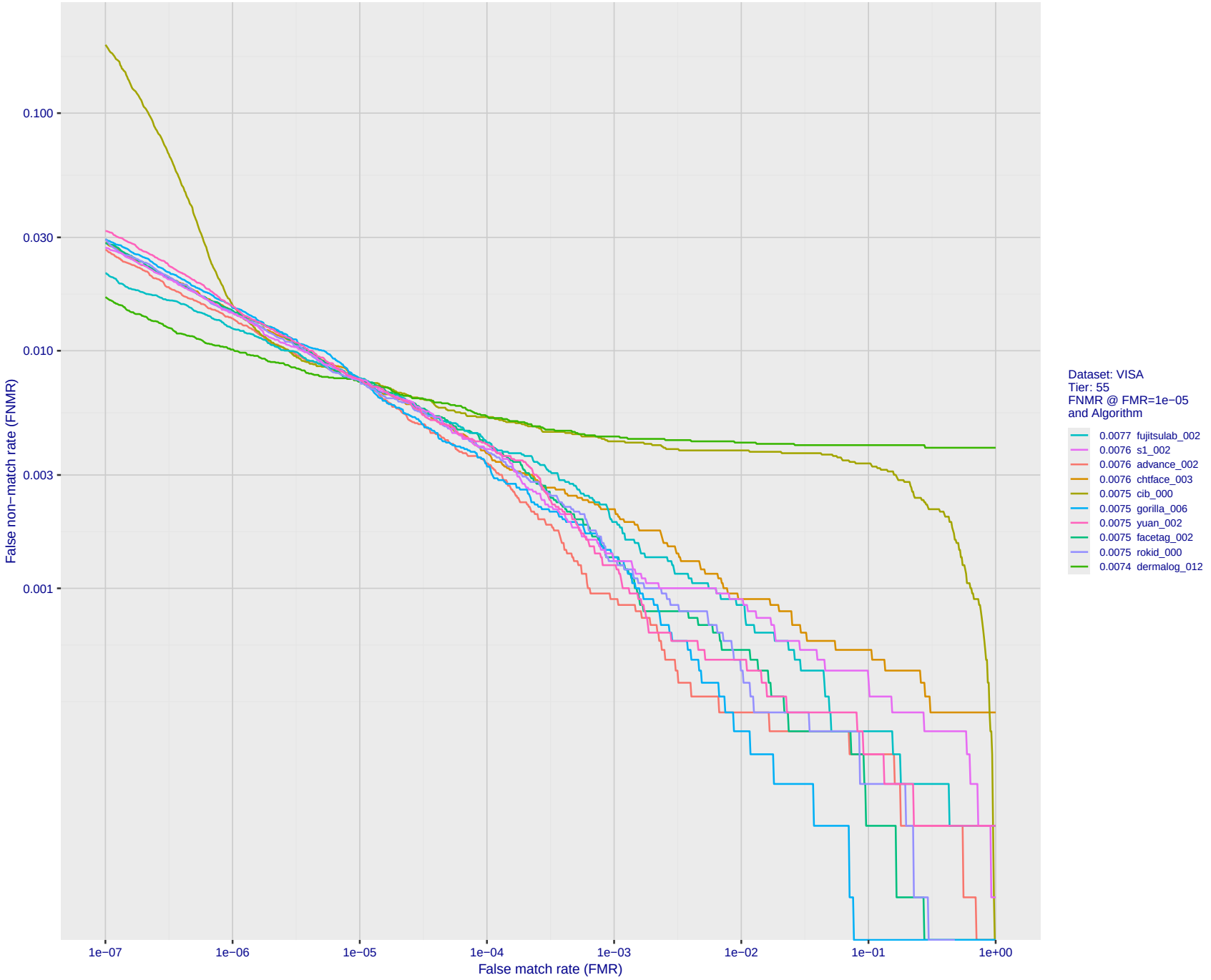

Figure 169: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

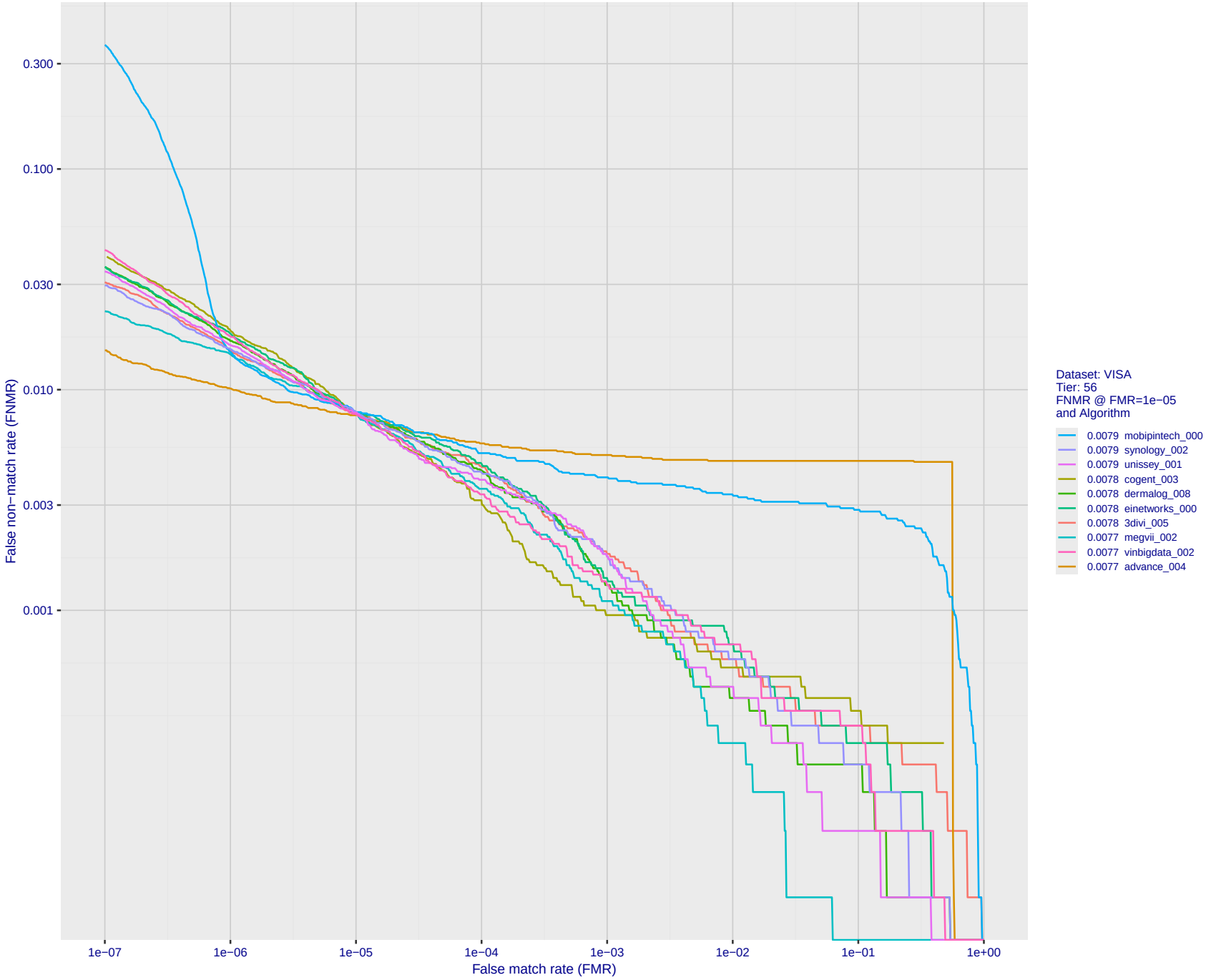

Figure 170: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

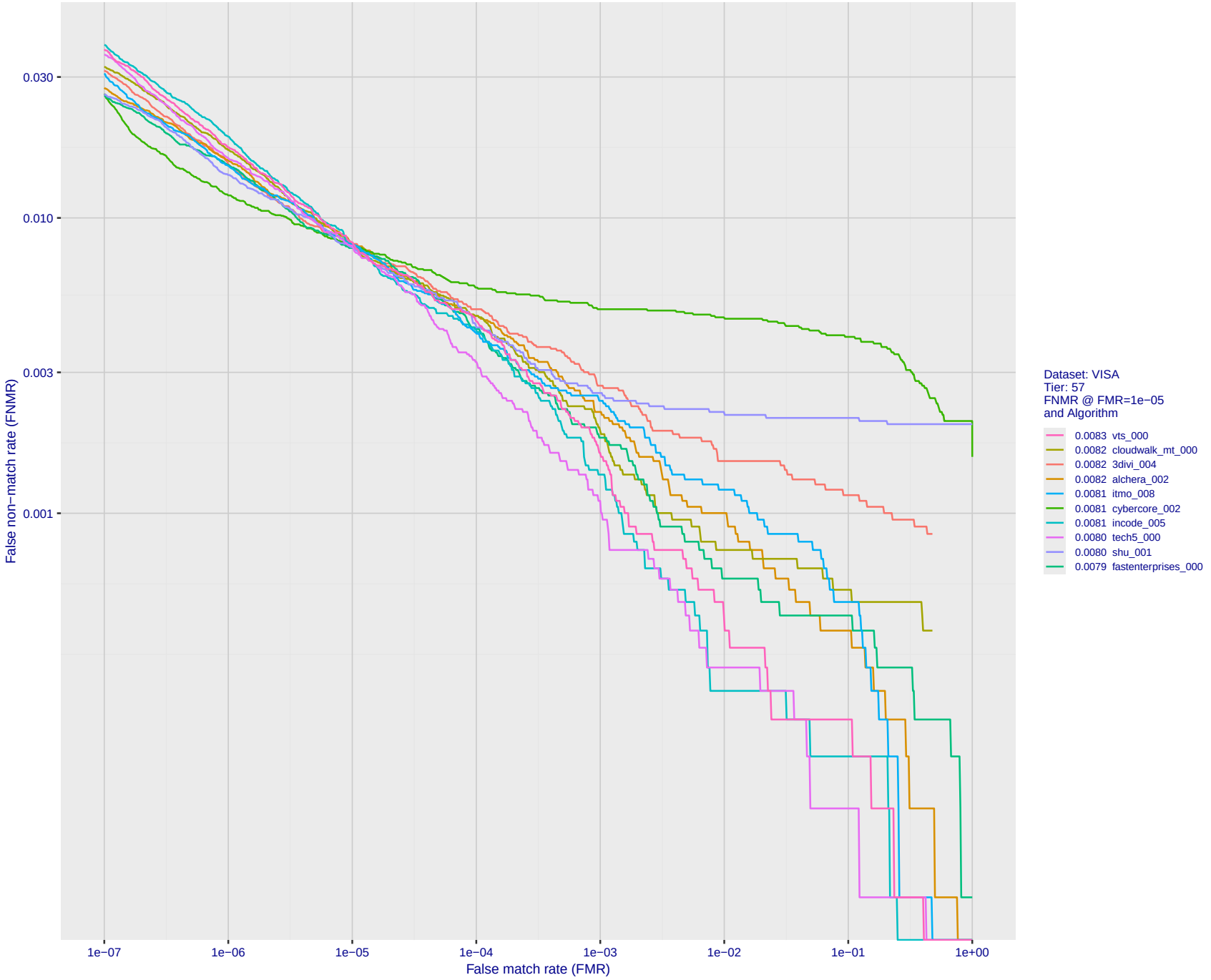

Figure 171: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

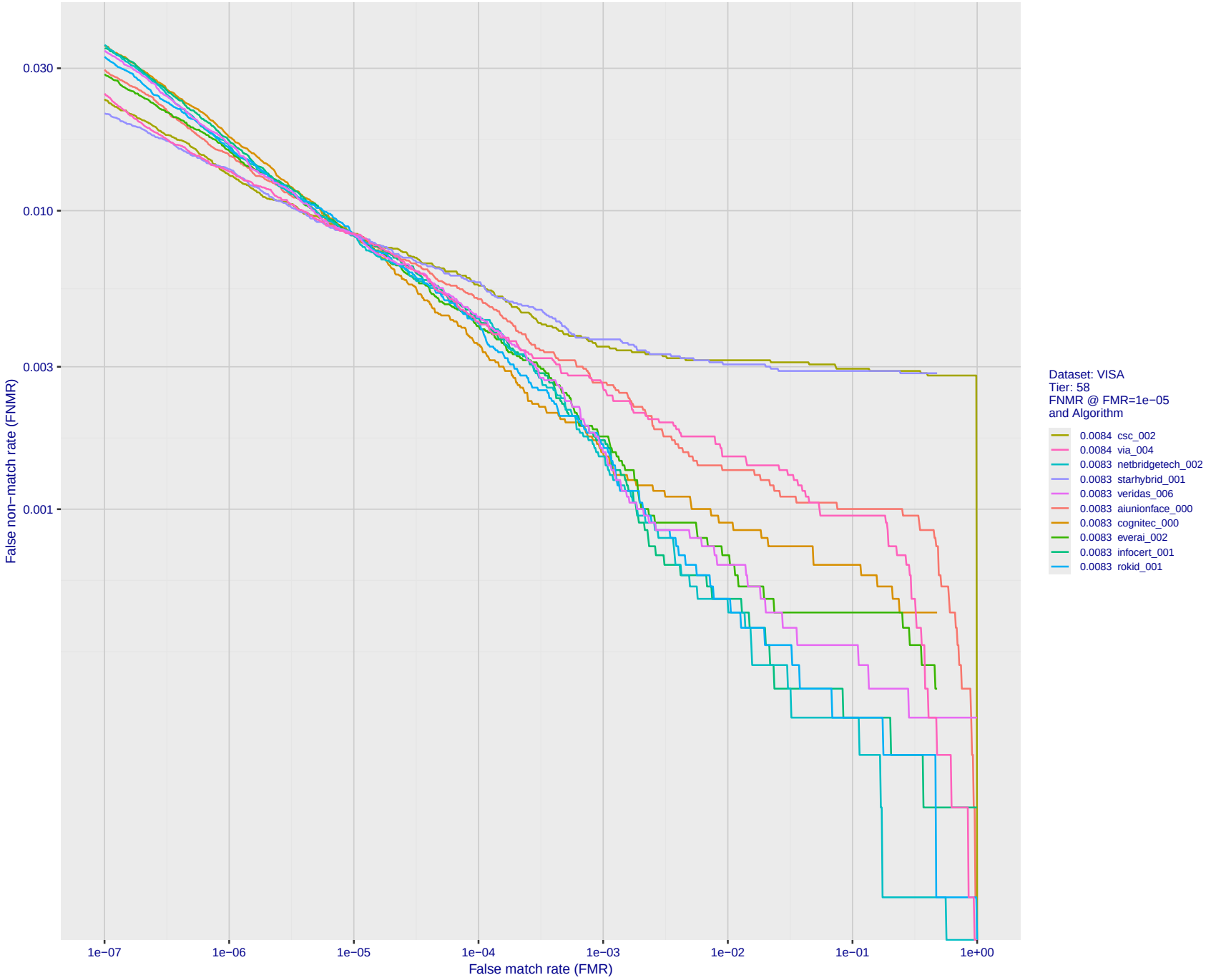

Figure 172: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

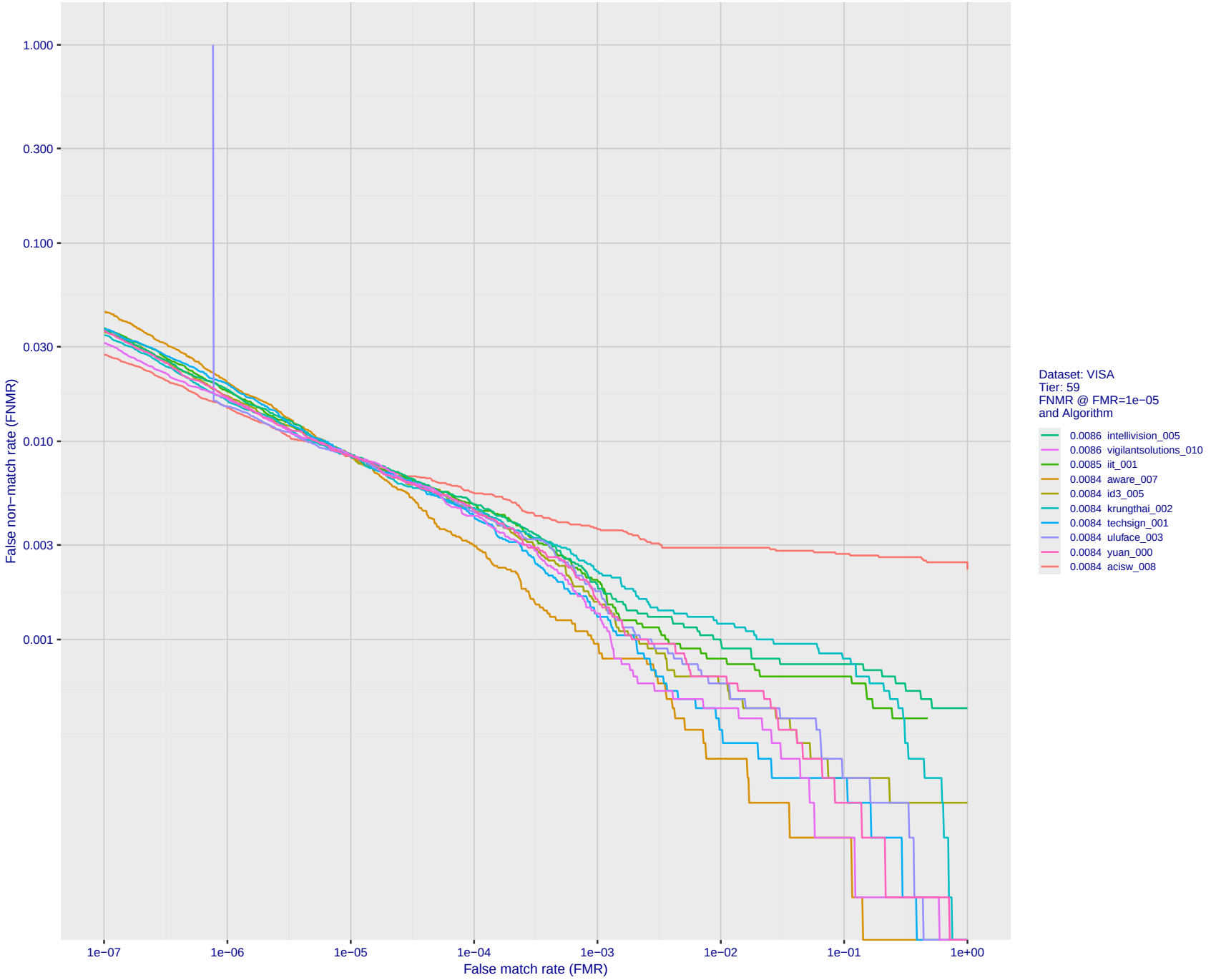

Figure 173: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

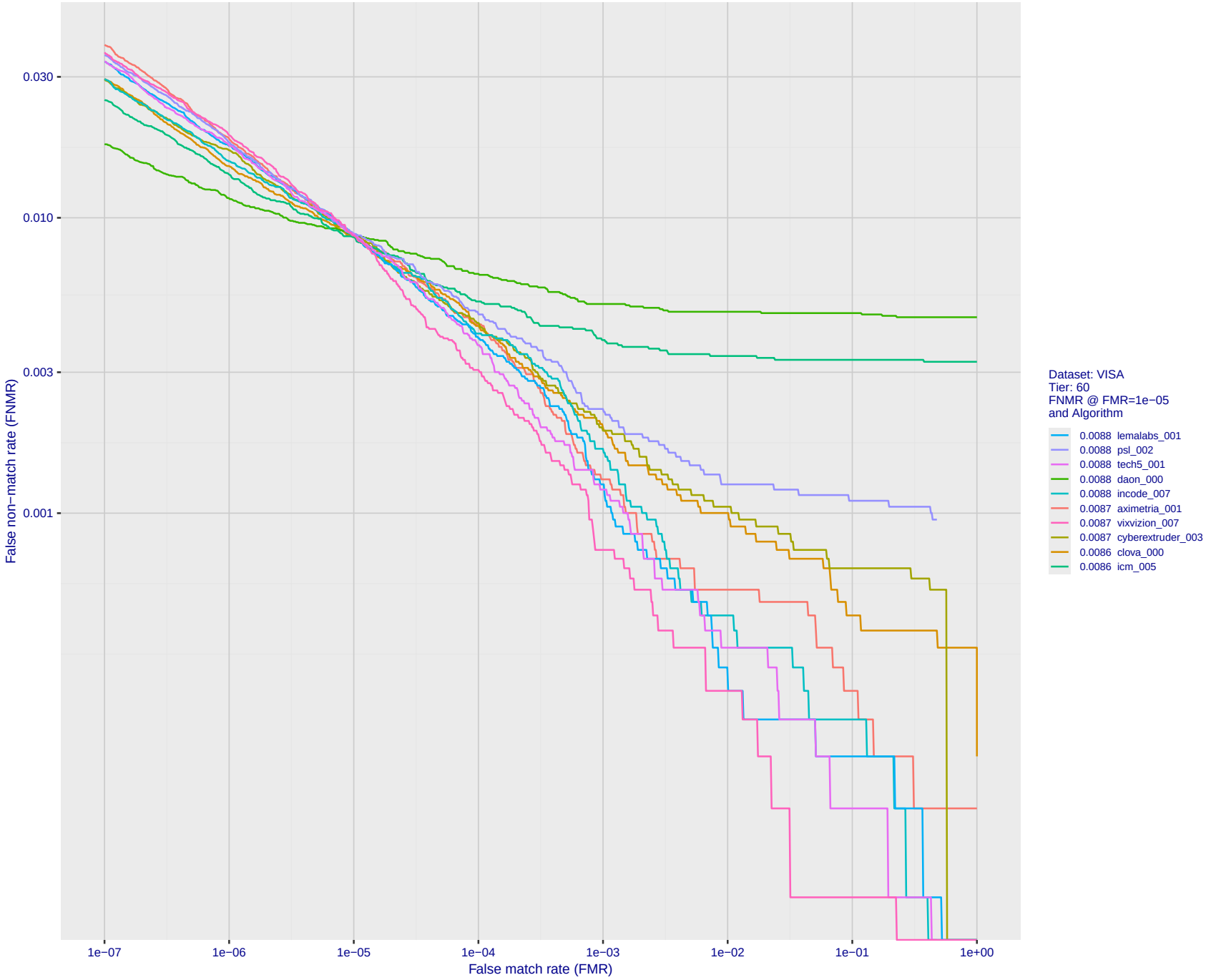

Figure 174: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

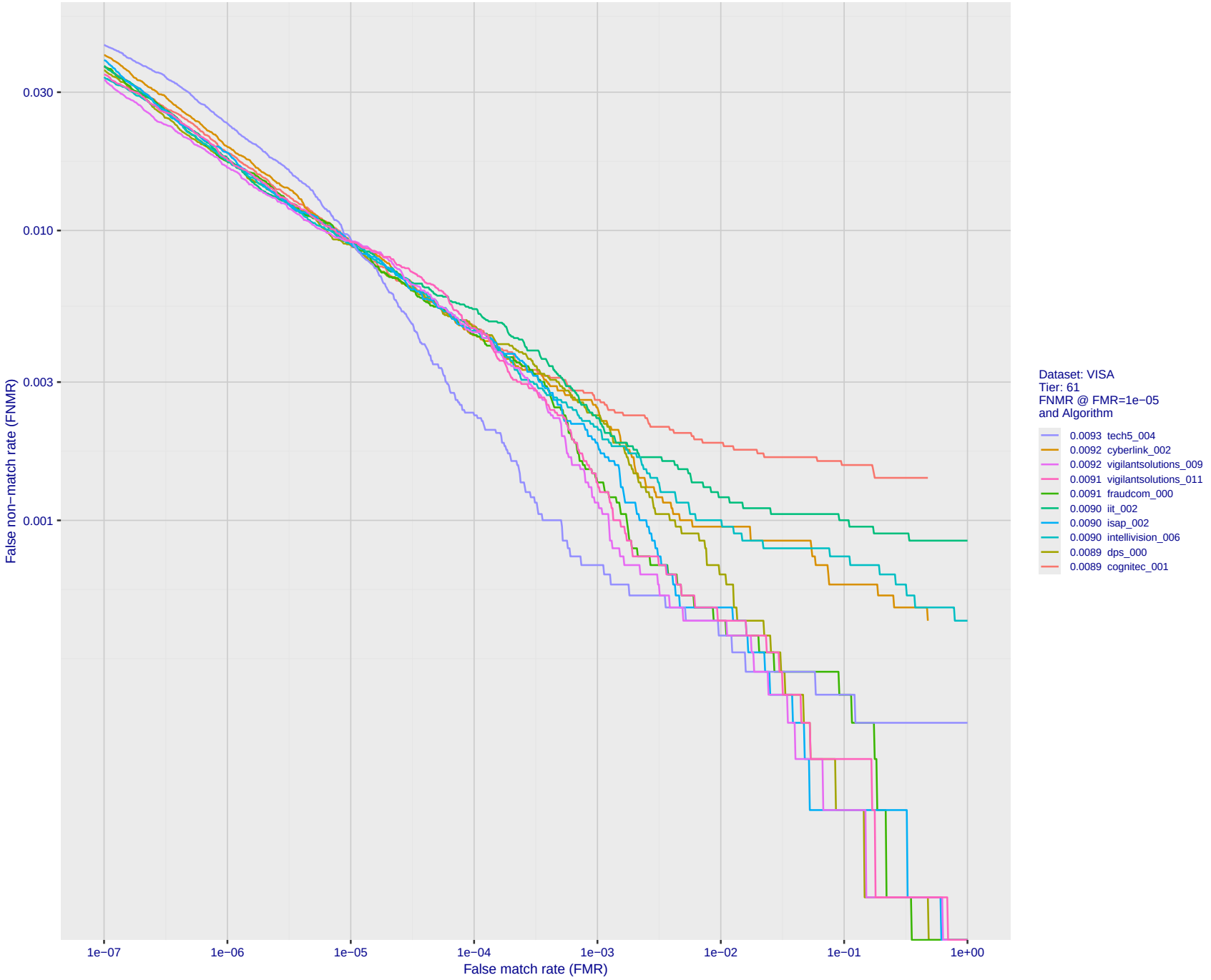

Figure 175: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

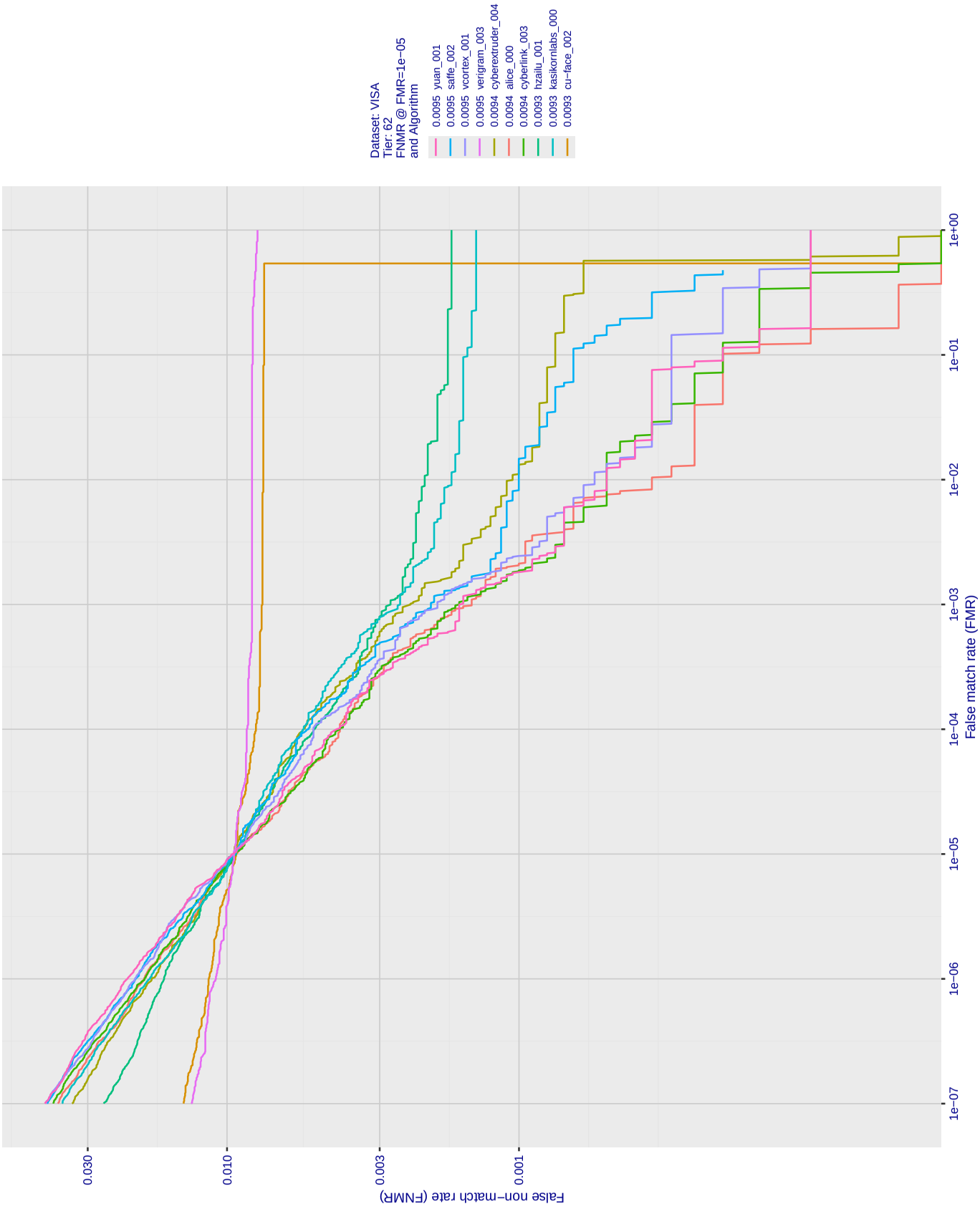

Figure 176: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

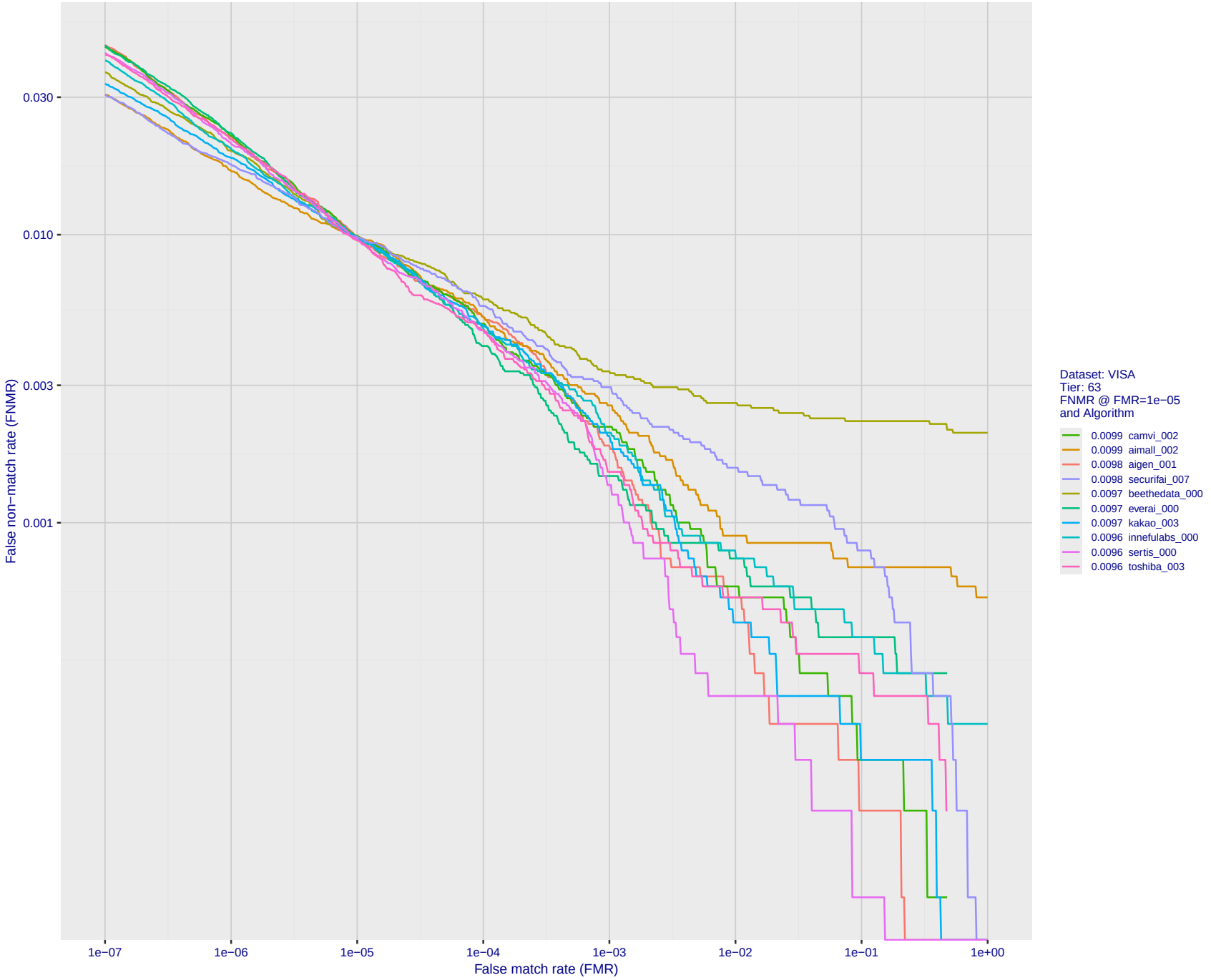

Figure 177: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

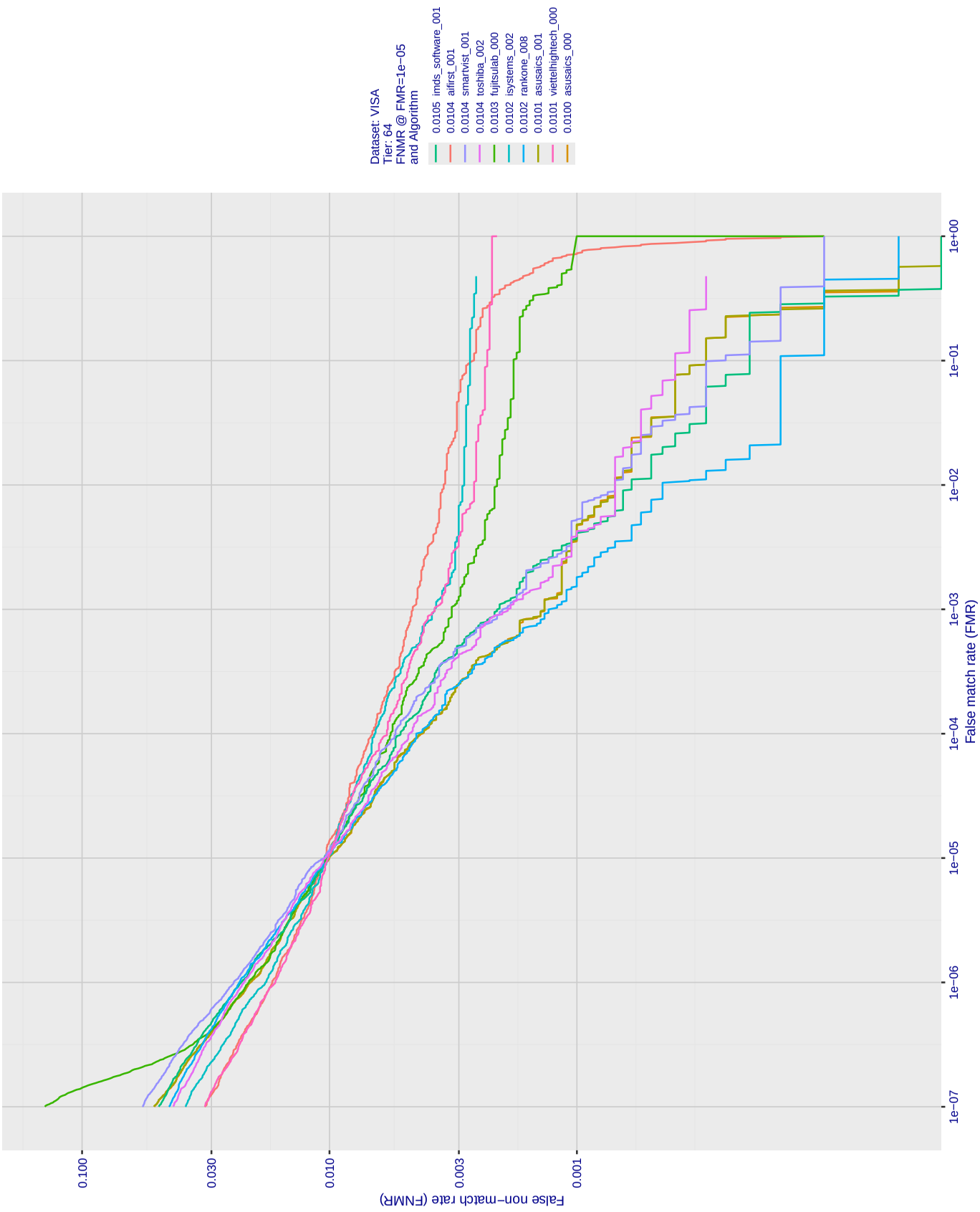

Figure 178: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

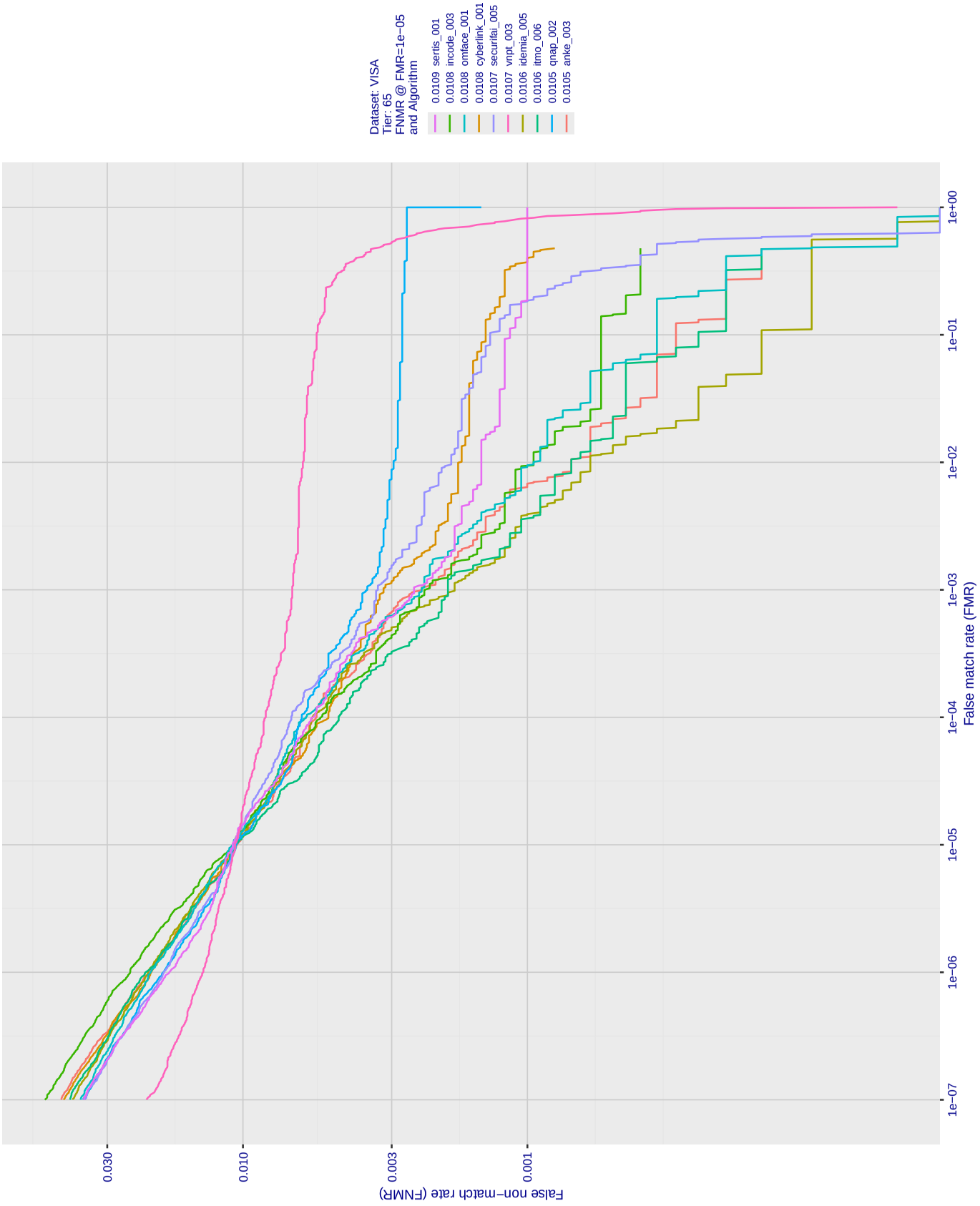

Figure 179: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

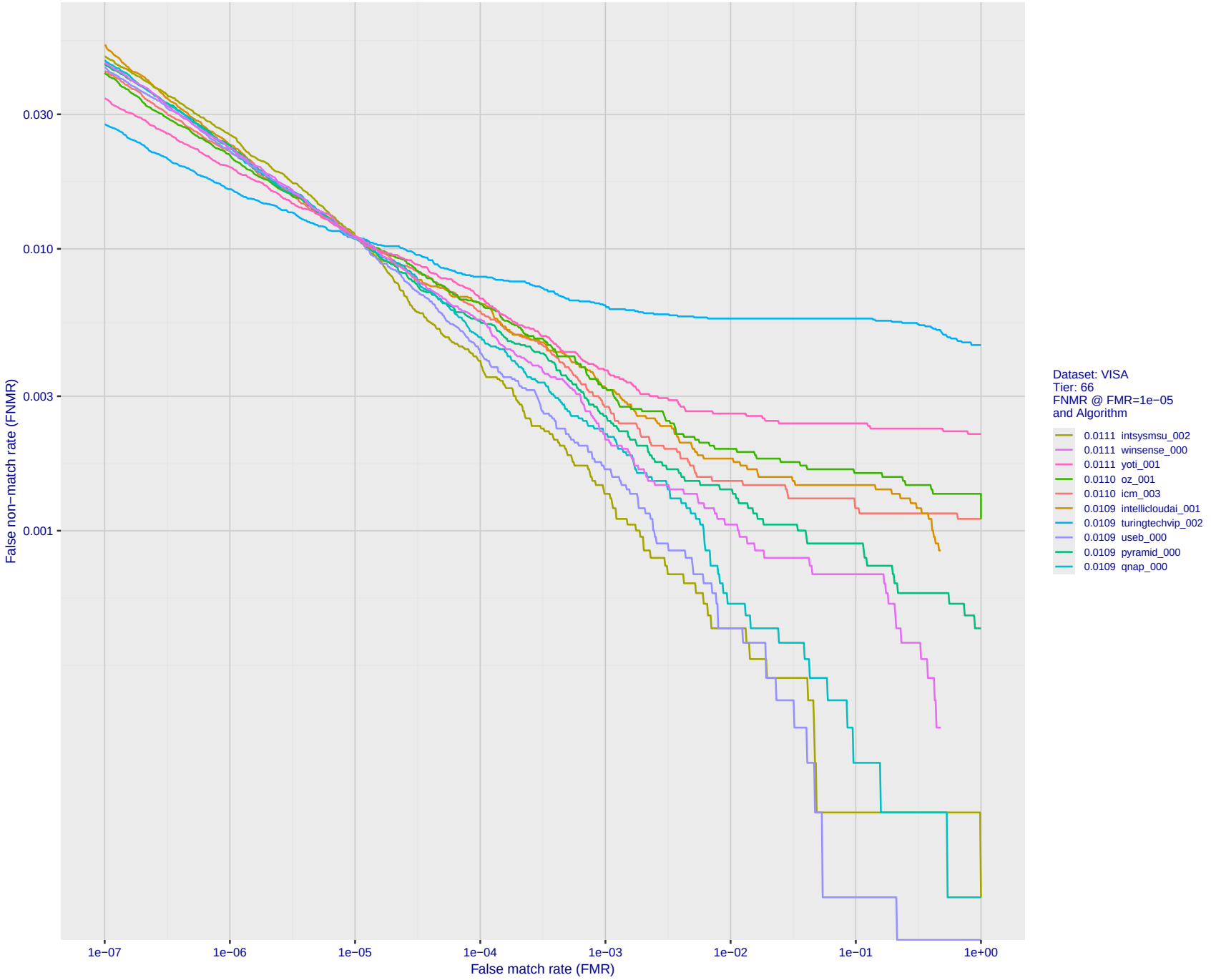

Figure 180: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

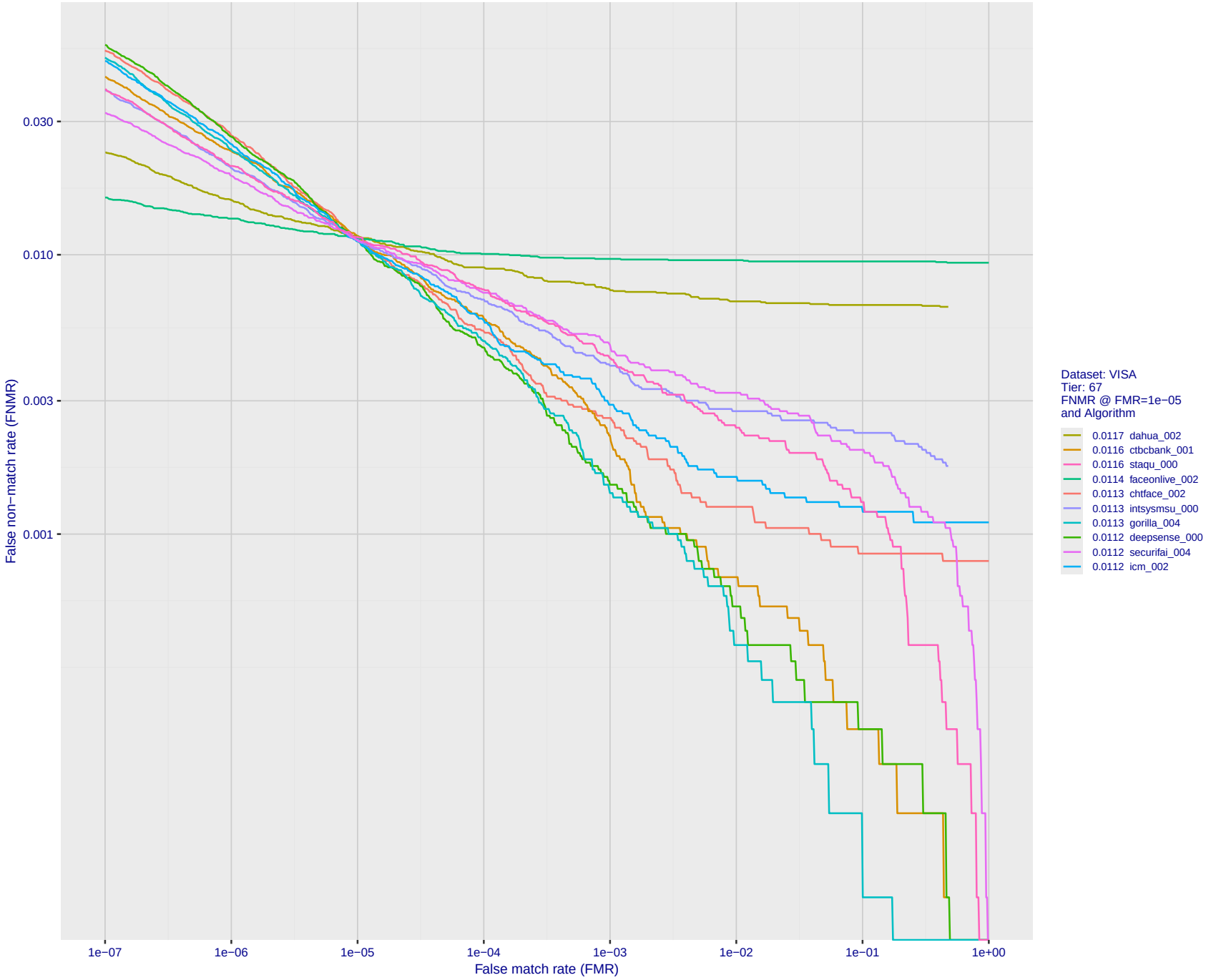

Figure 181: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

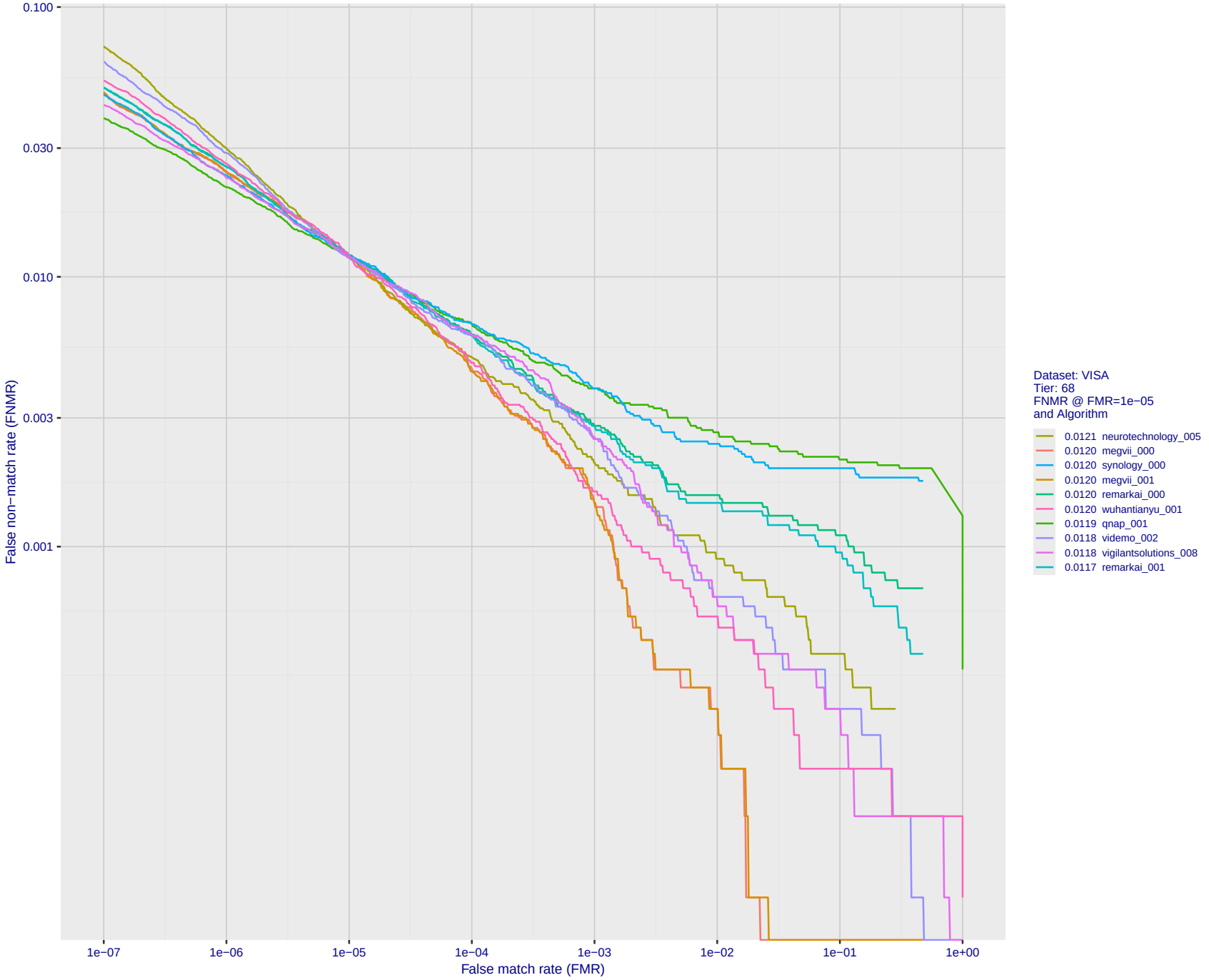

Figure 182: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

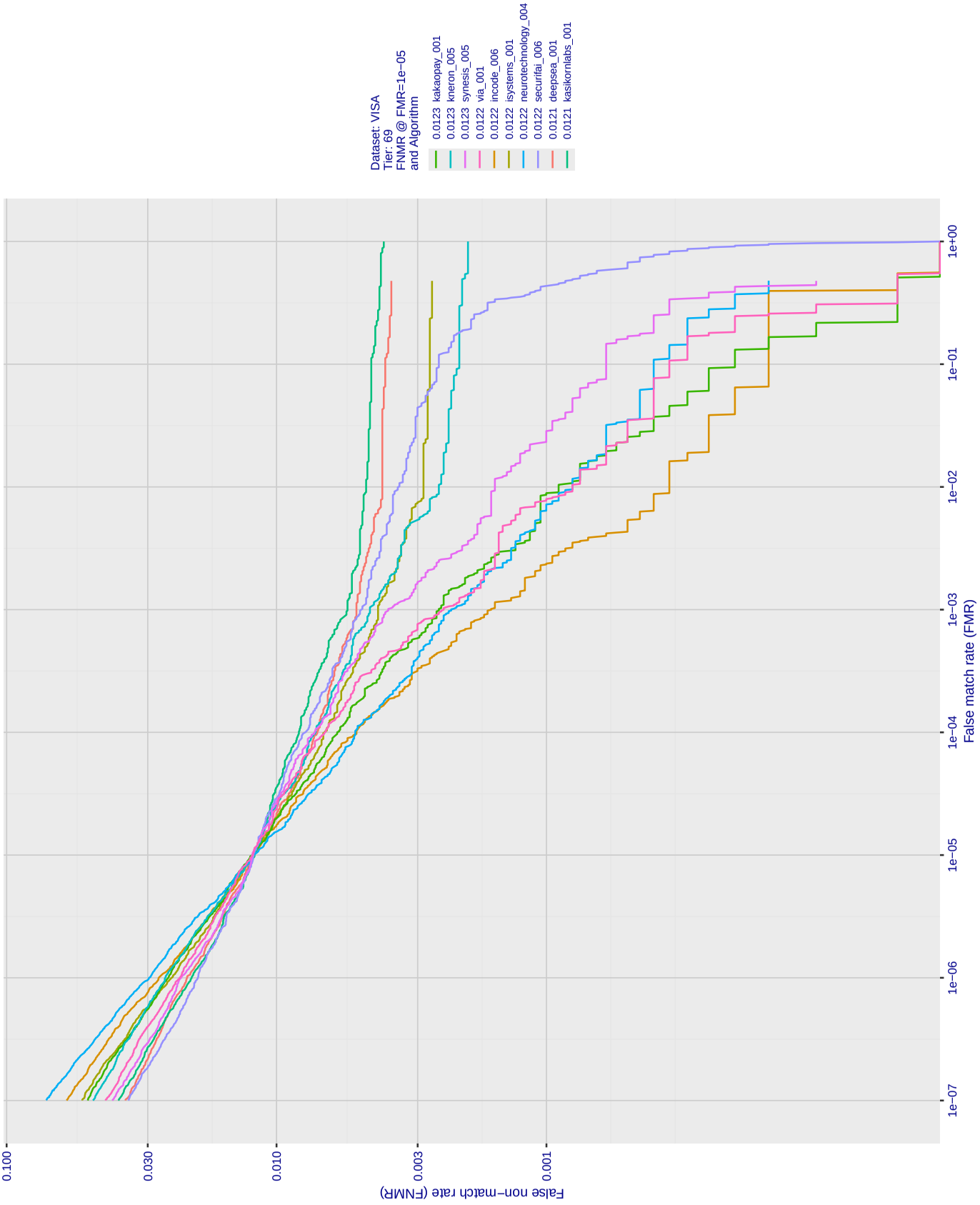

Figure 183: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

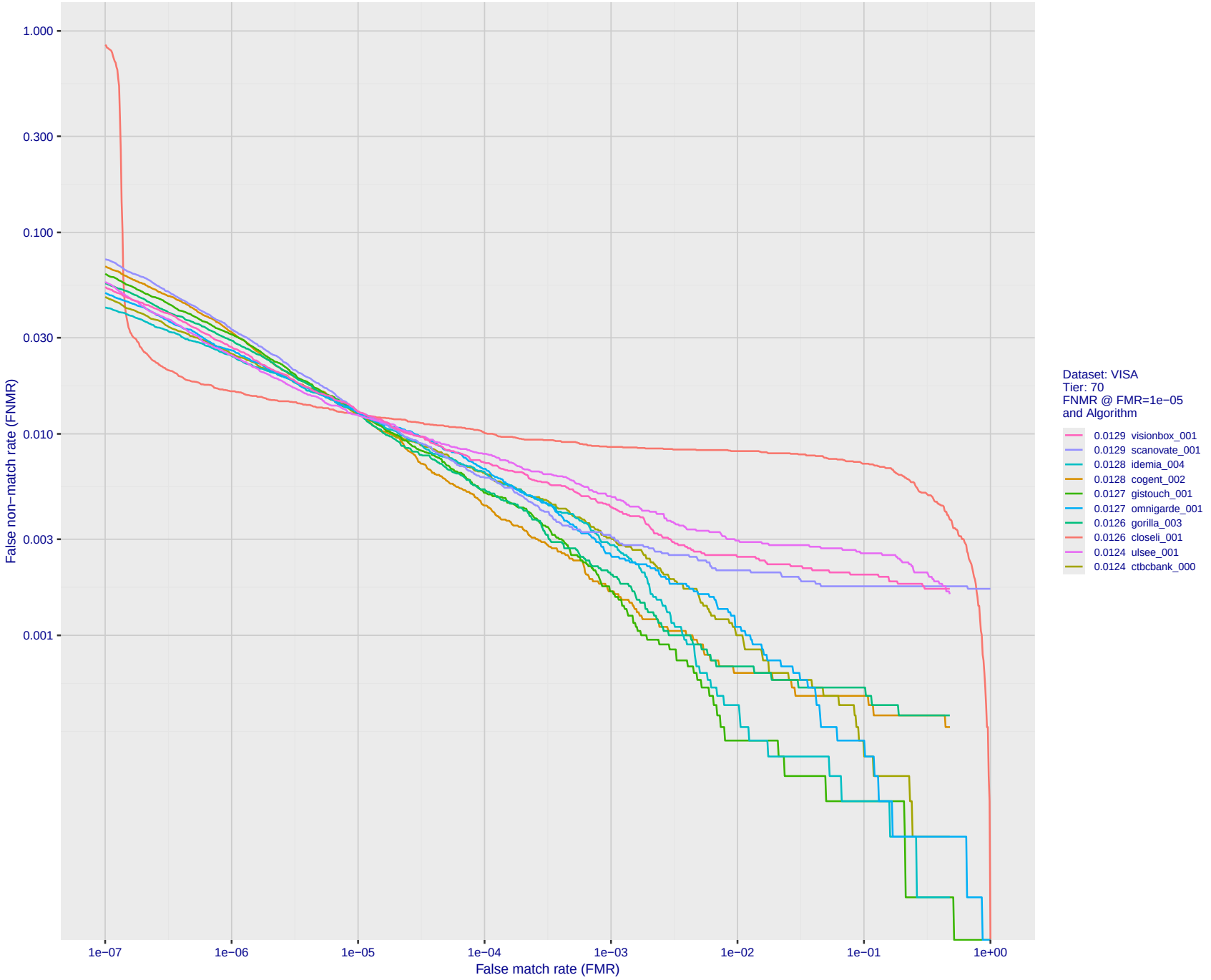

Figure 184: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

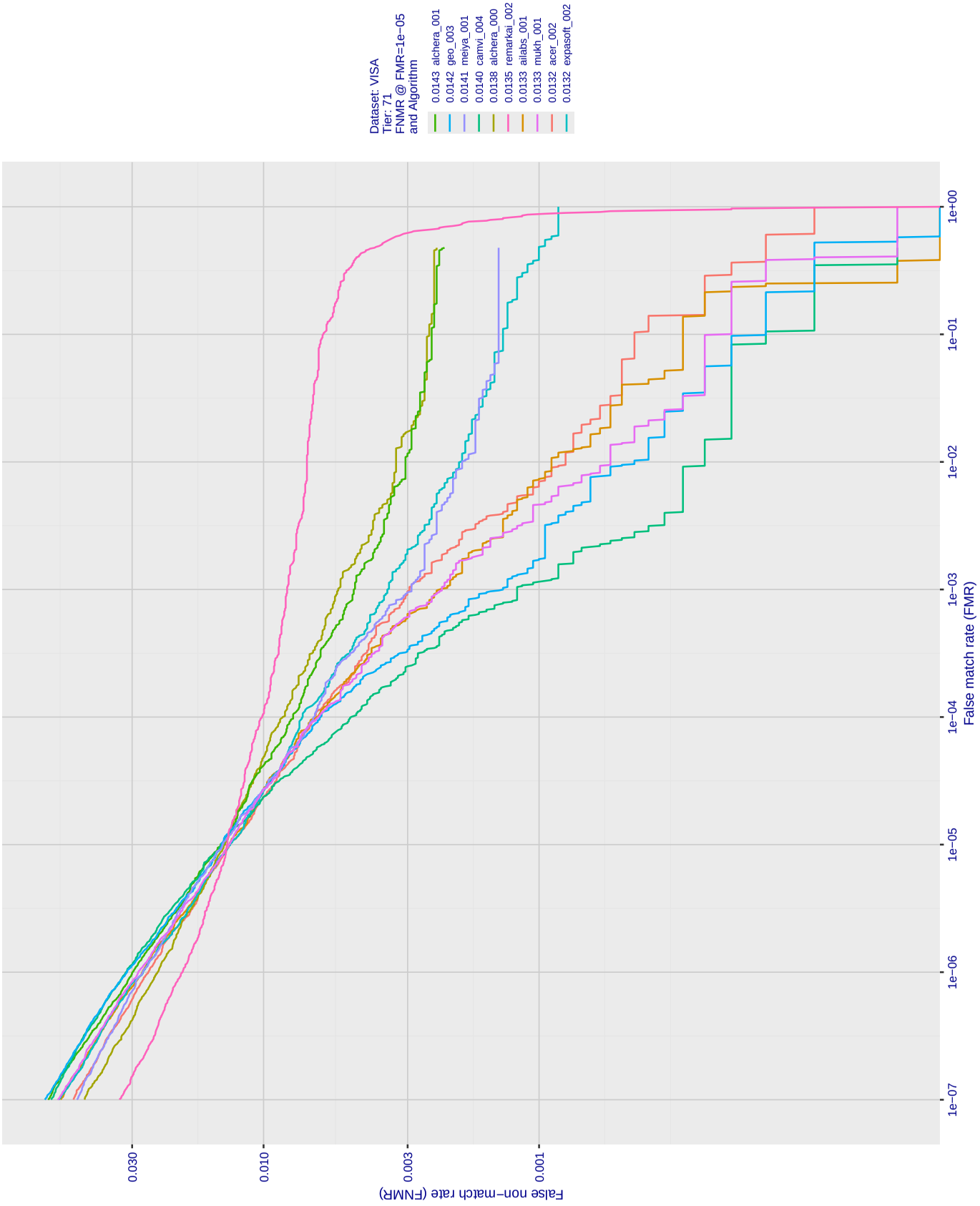

Figure 185: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

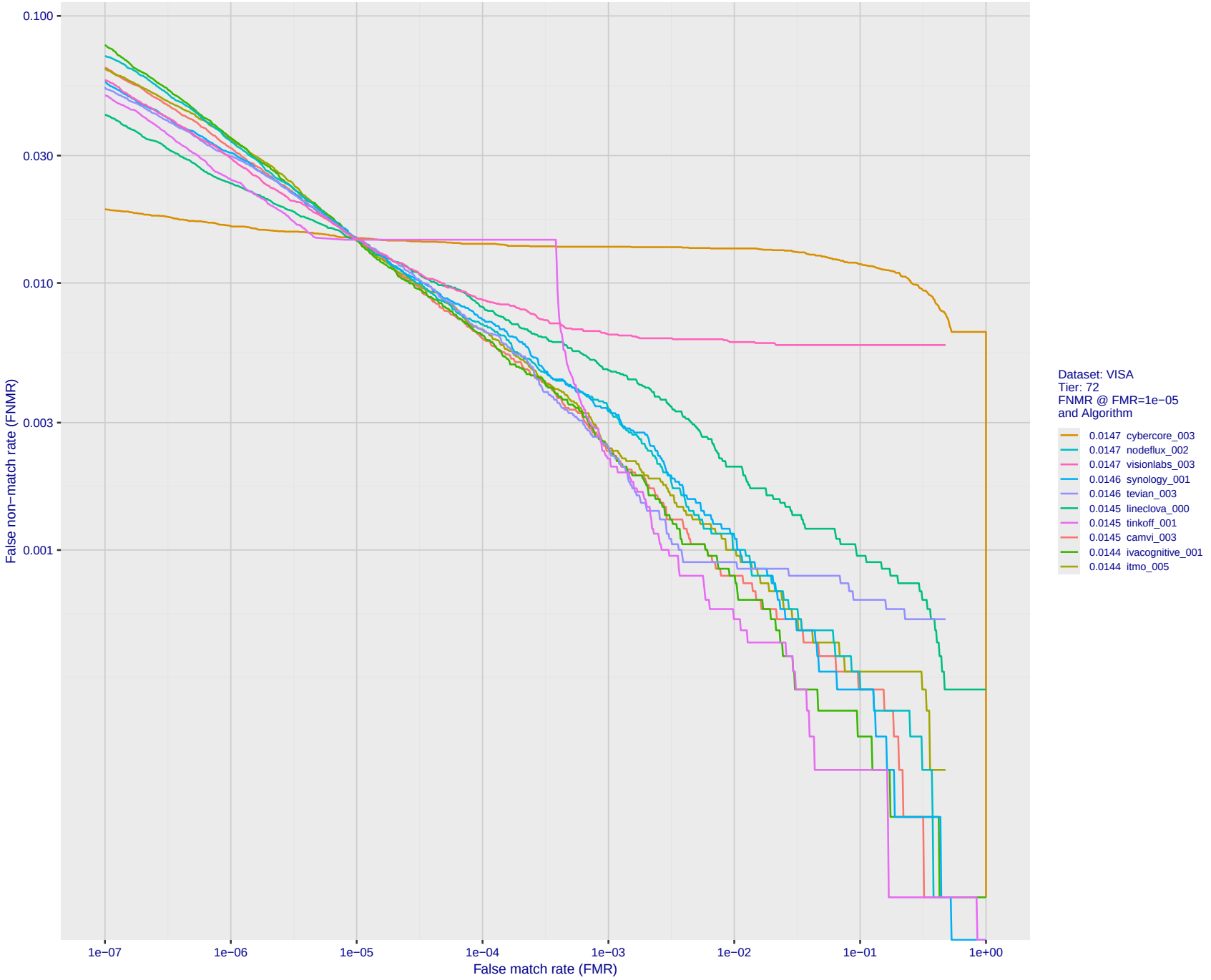

Figure 186: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

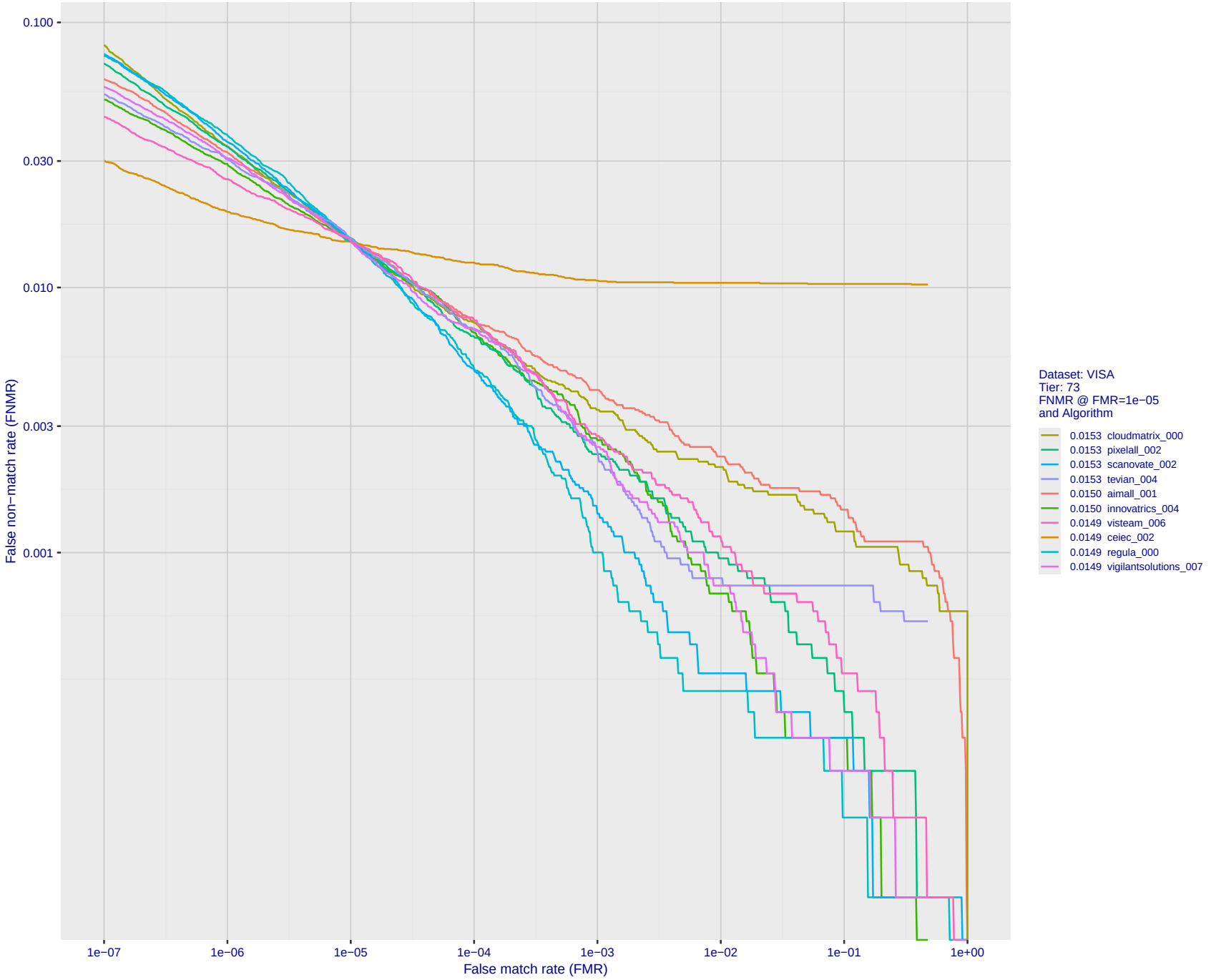

Figure 187: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

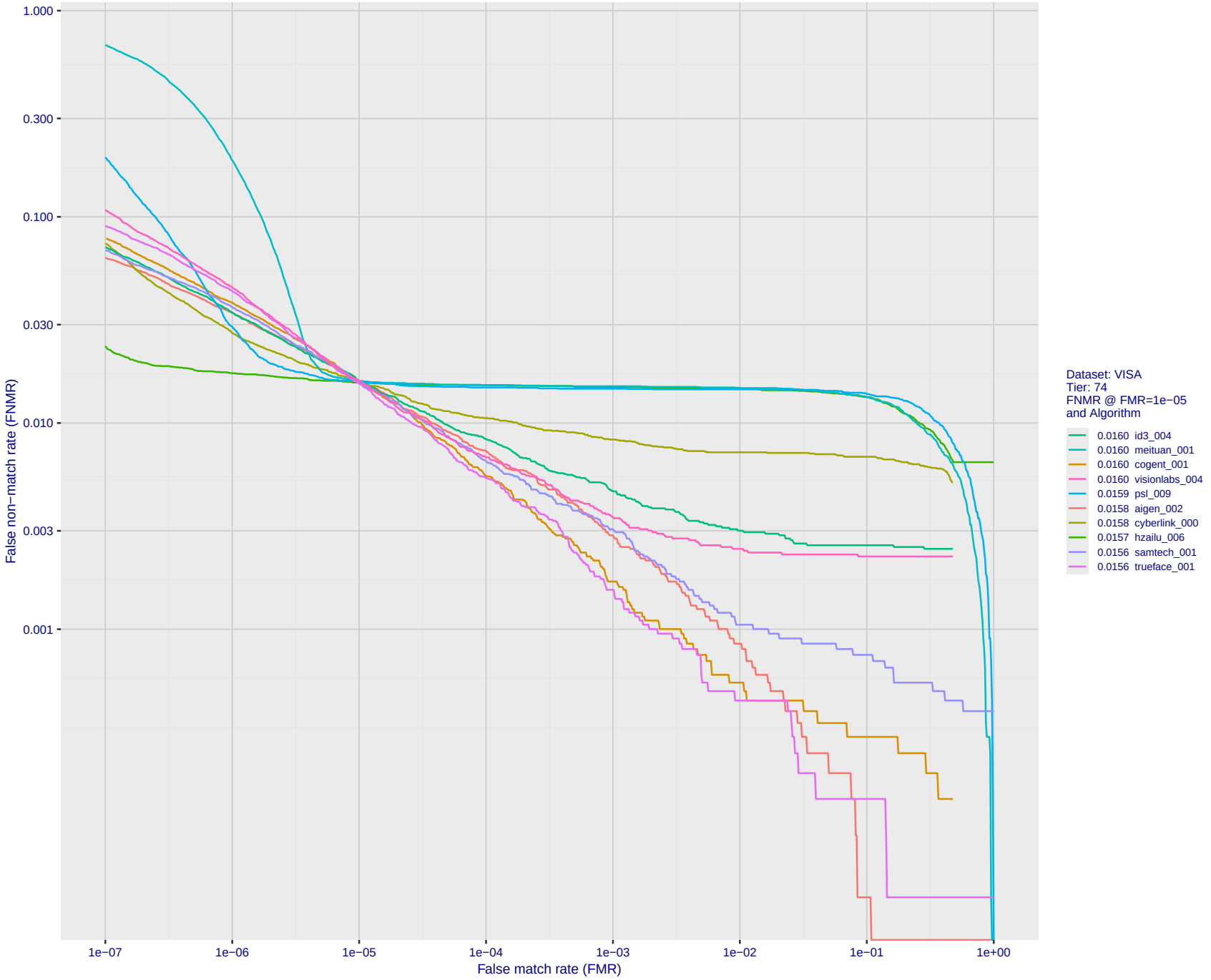

Figure 188: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

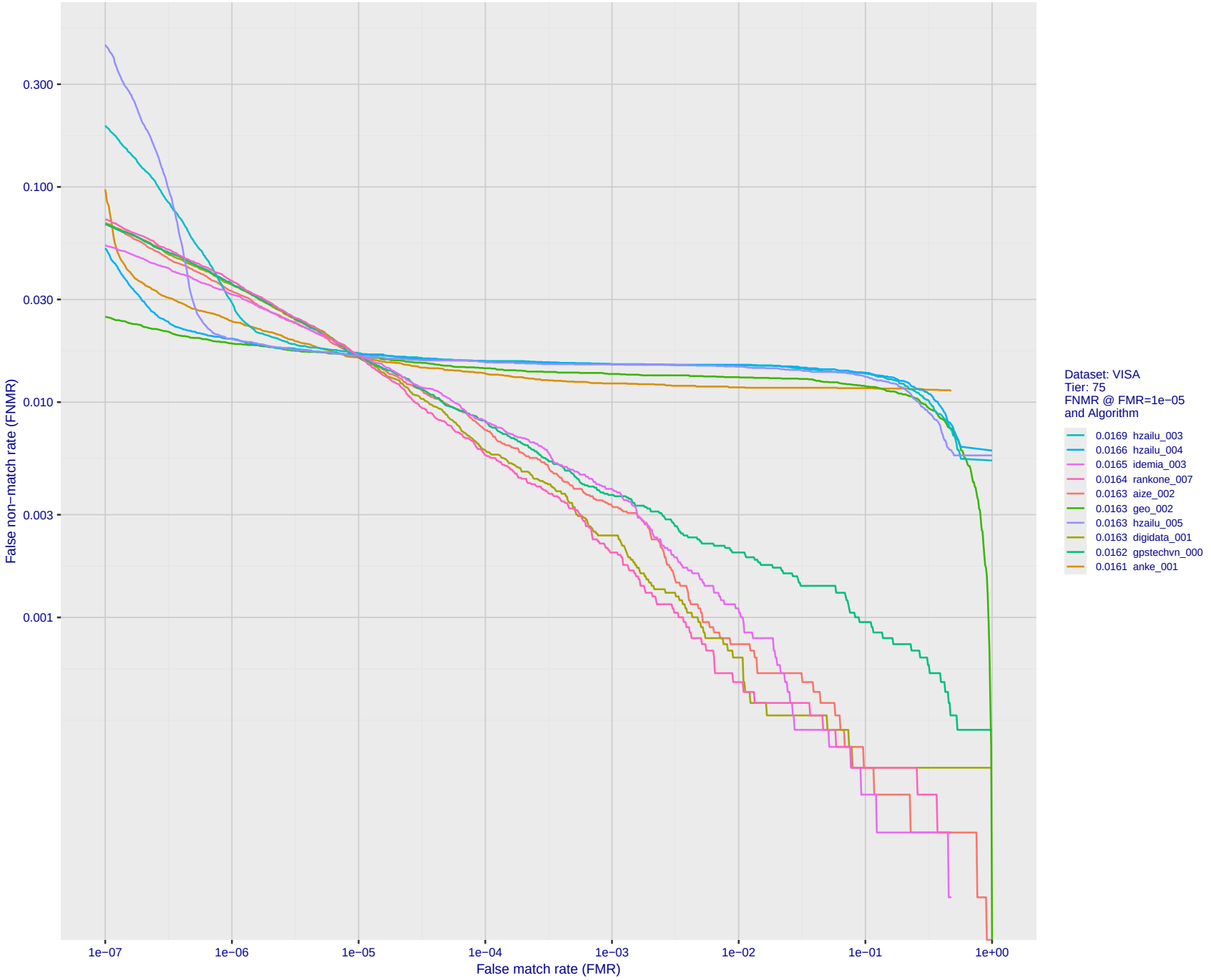

Figure 189: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

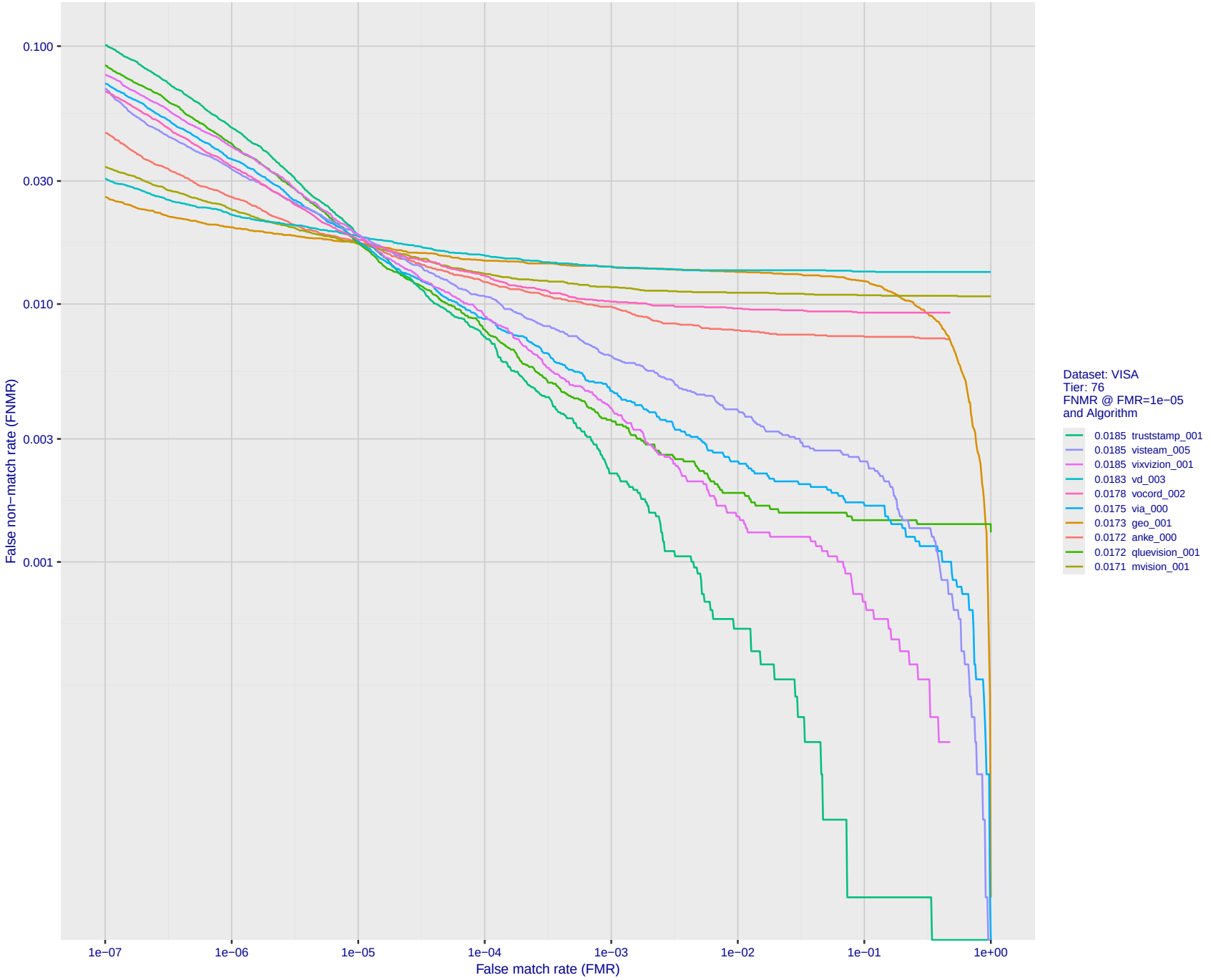

Figure 190: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

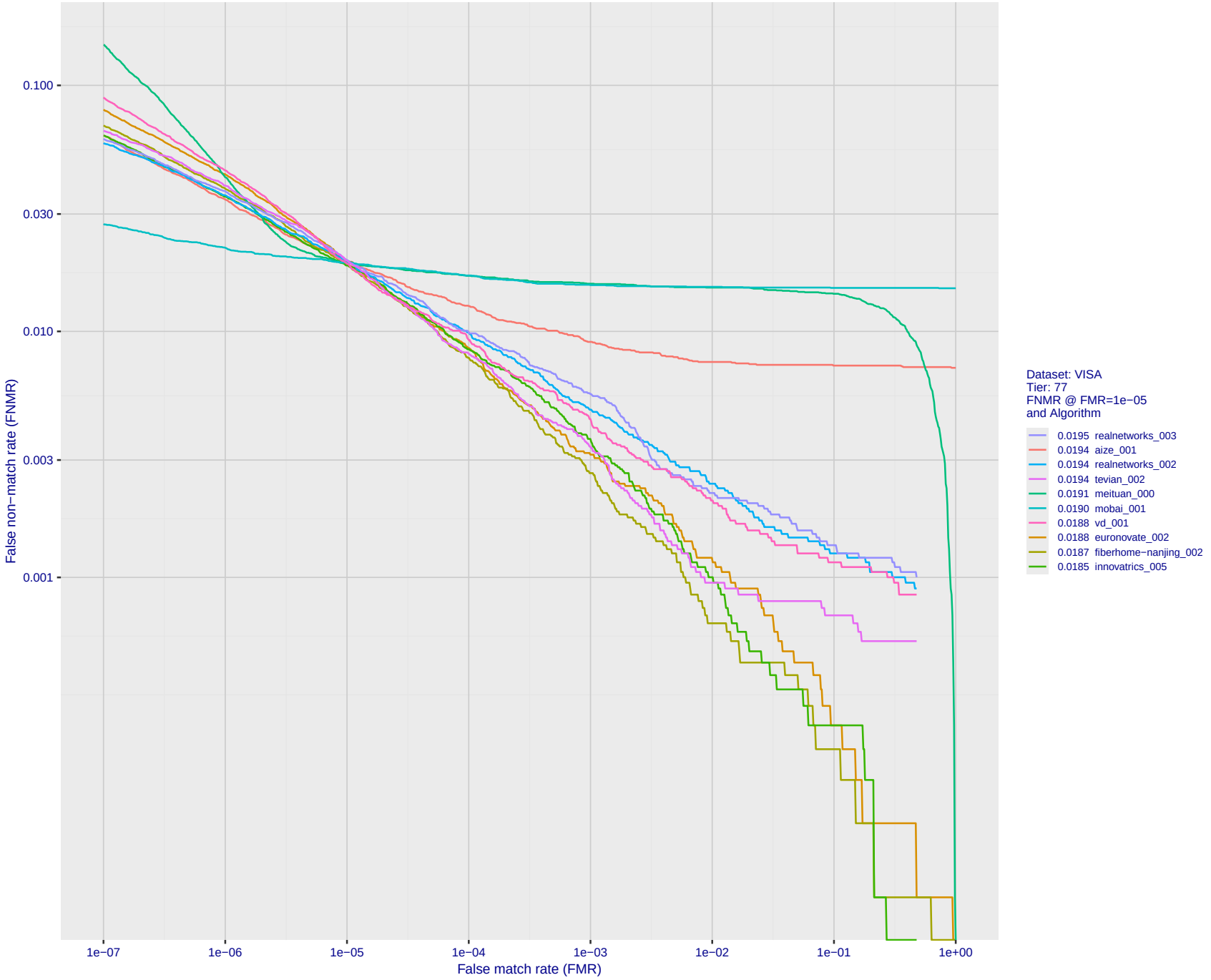

Figure 191: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

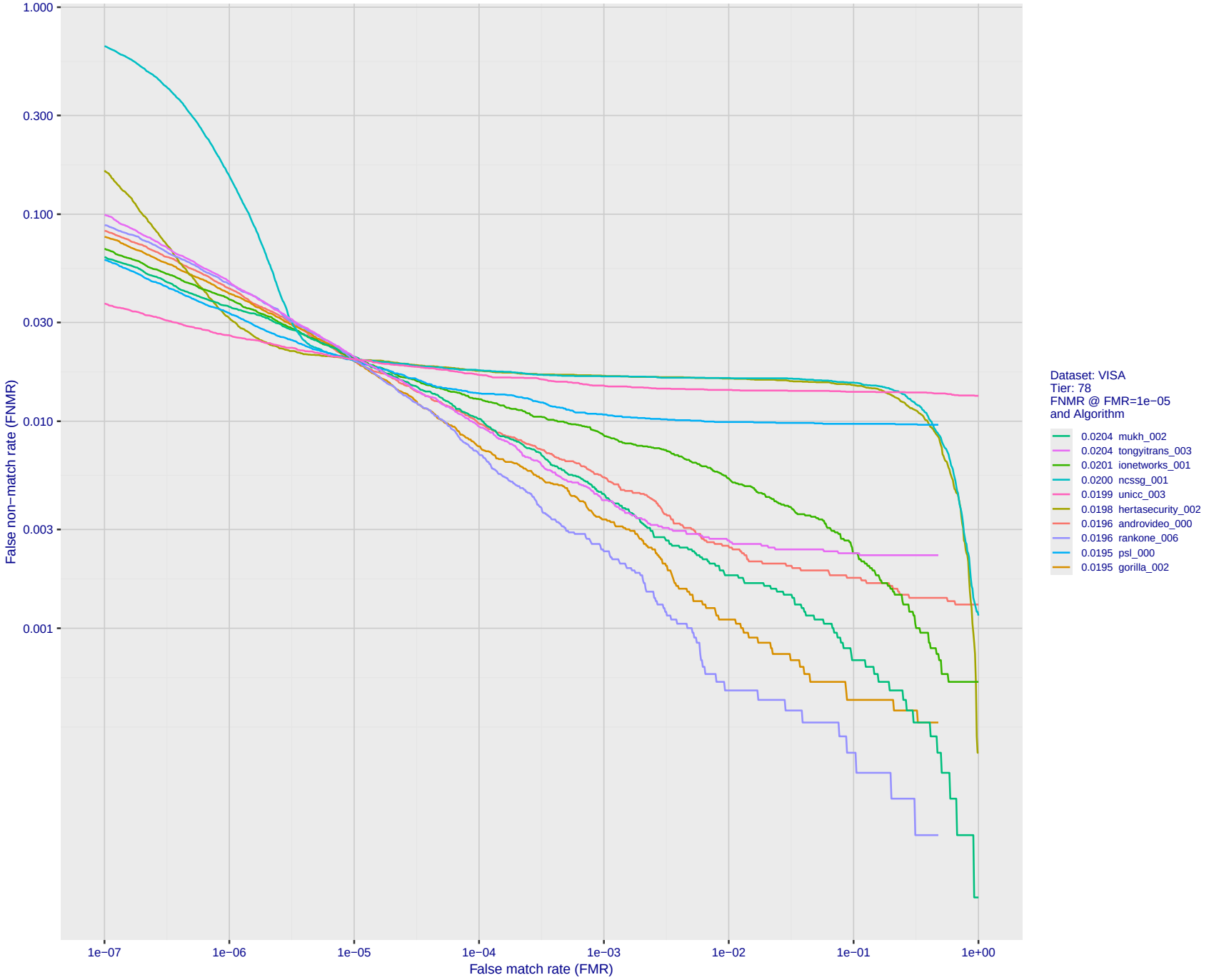

Figure 192: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

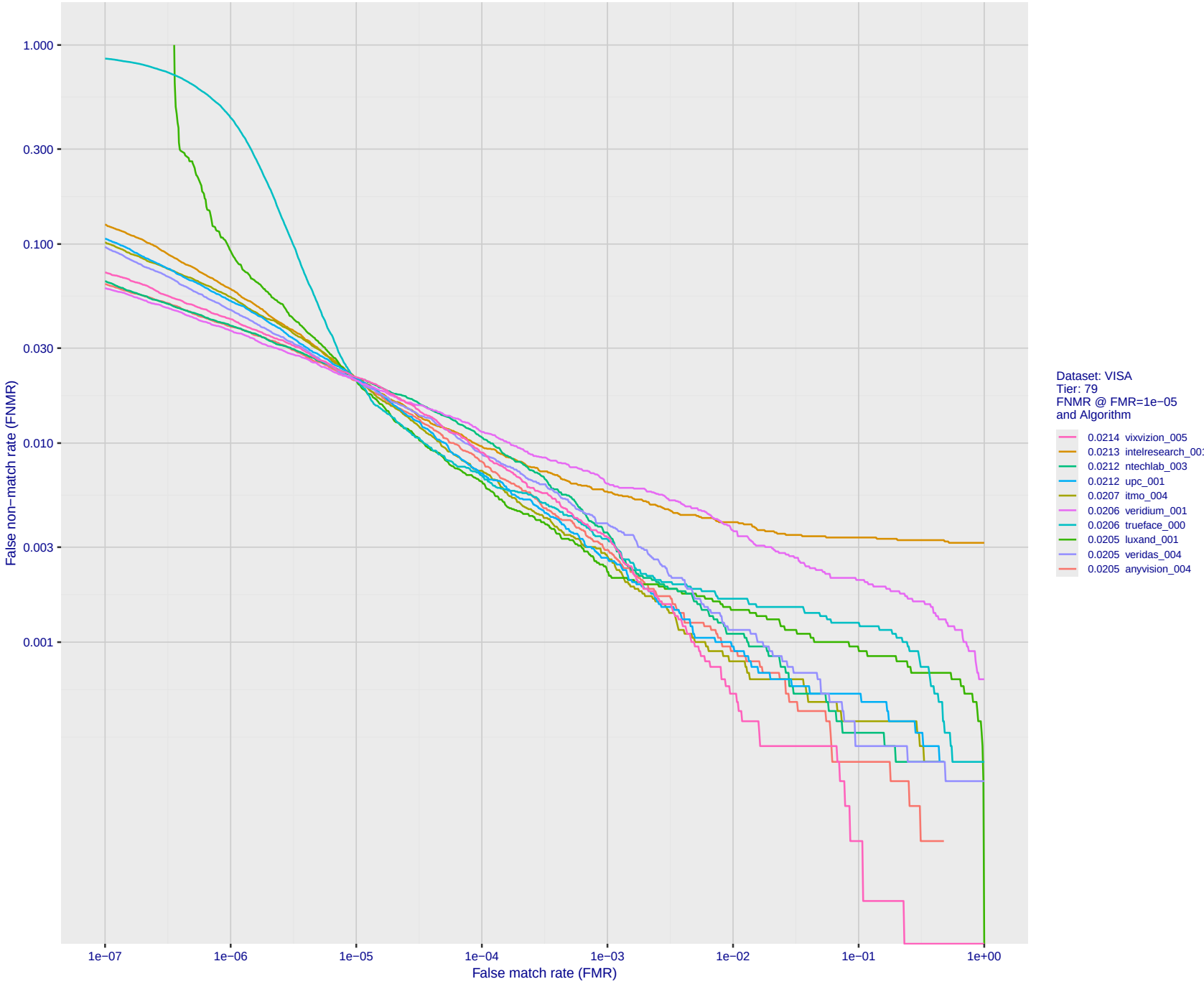

Figure 193: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

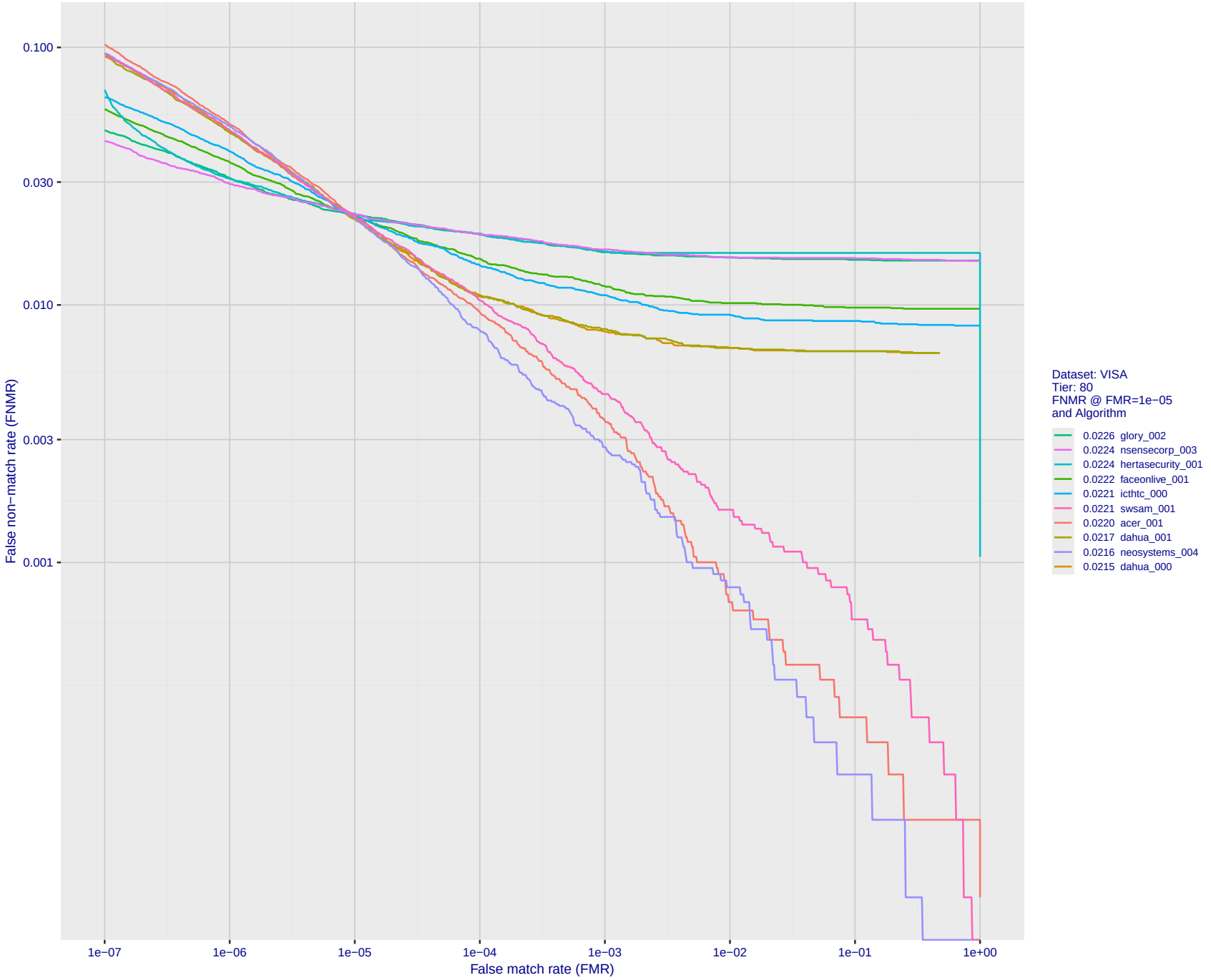

Figure 194: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

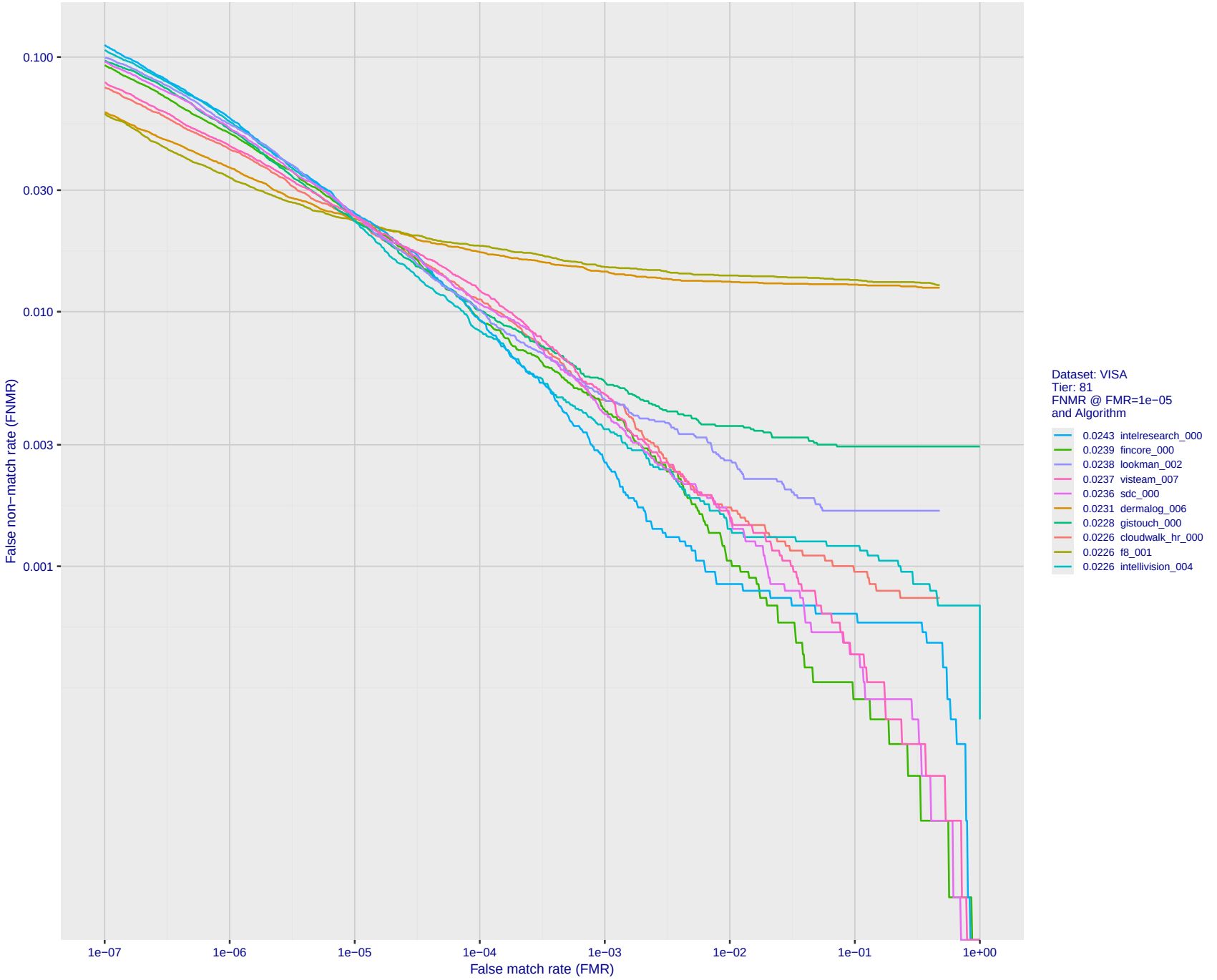

Figure 195: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

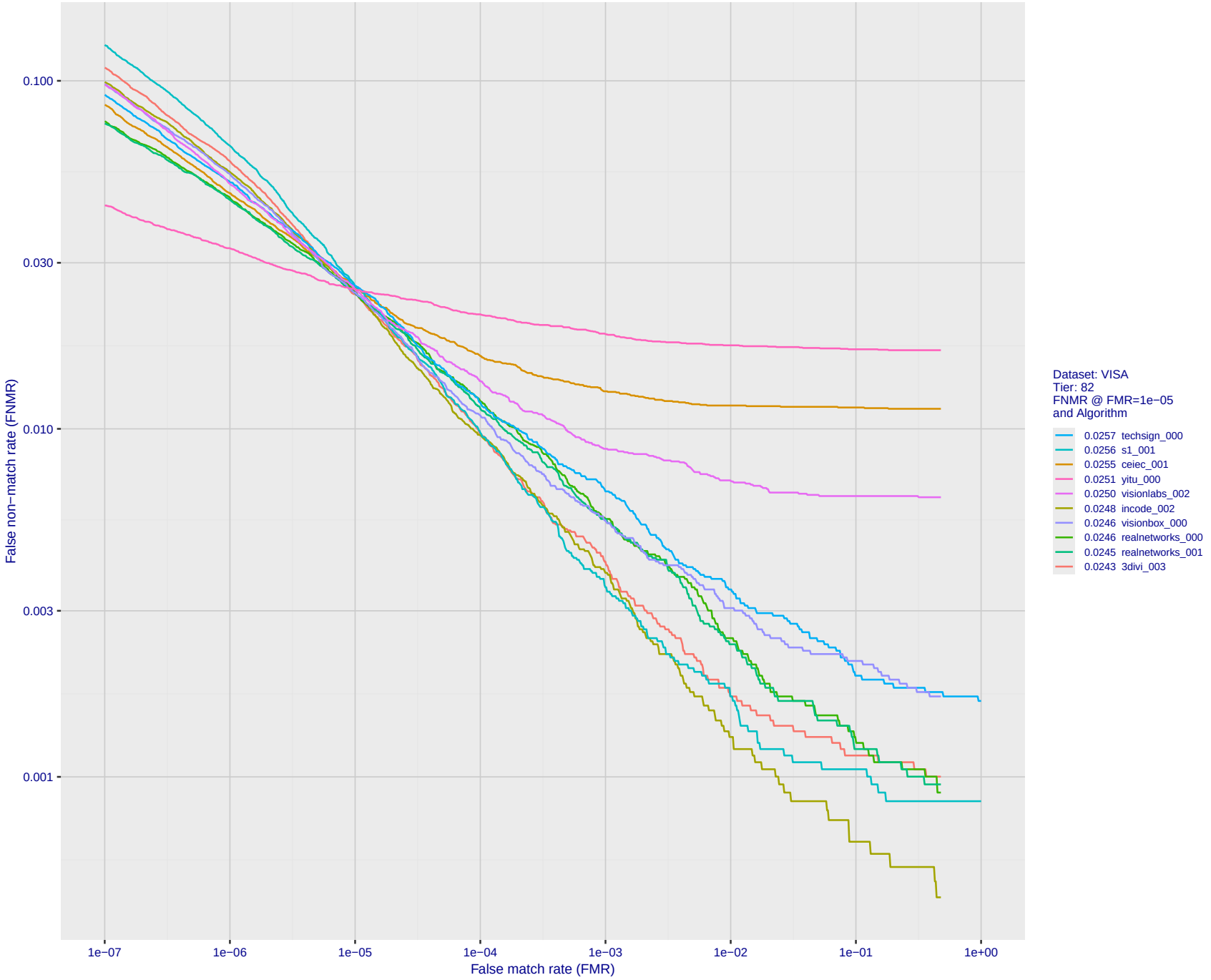

Figure 196: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

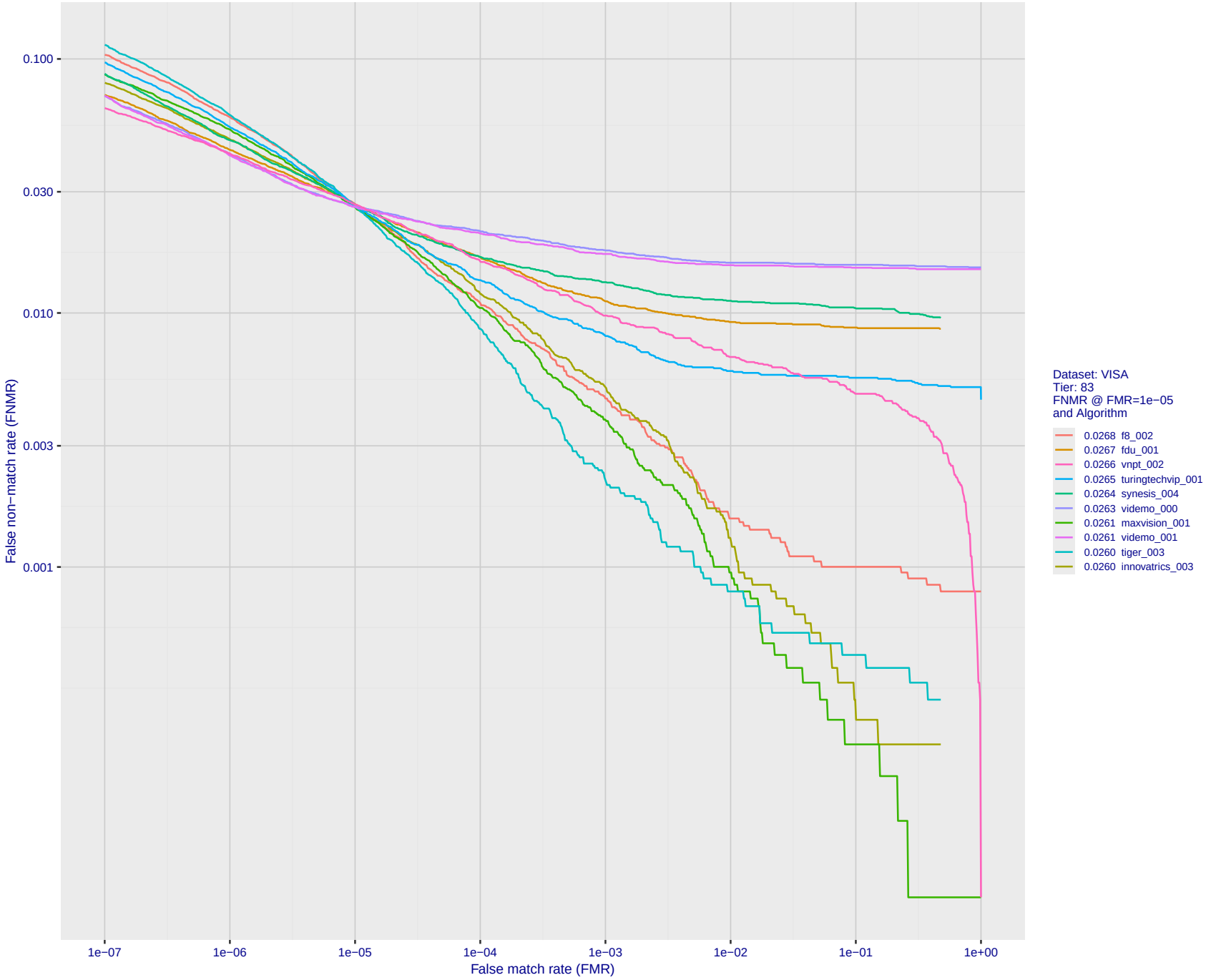

Figure 197: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

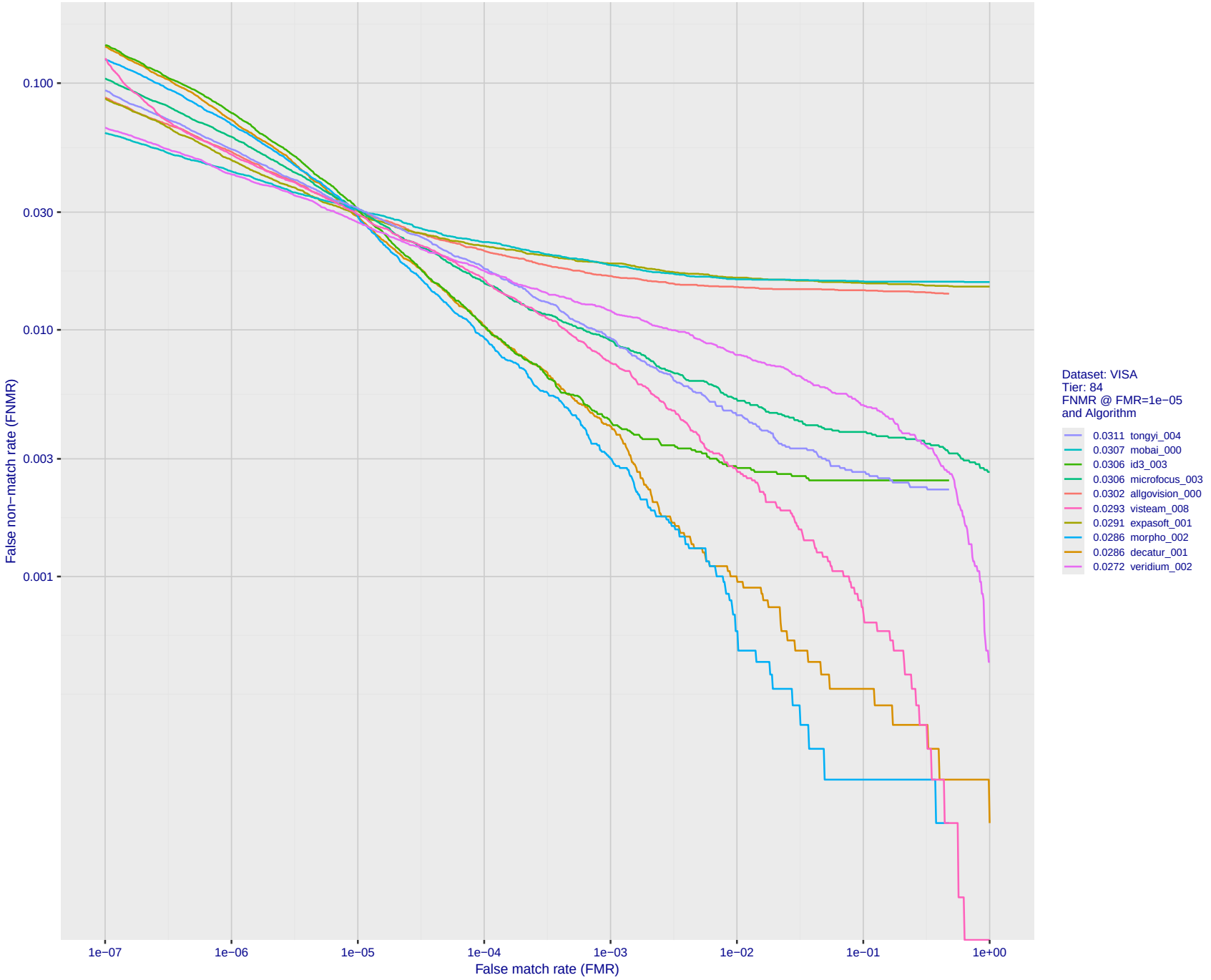

Figure 198: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

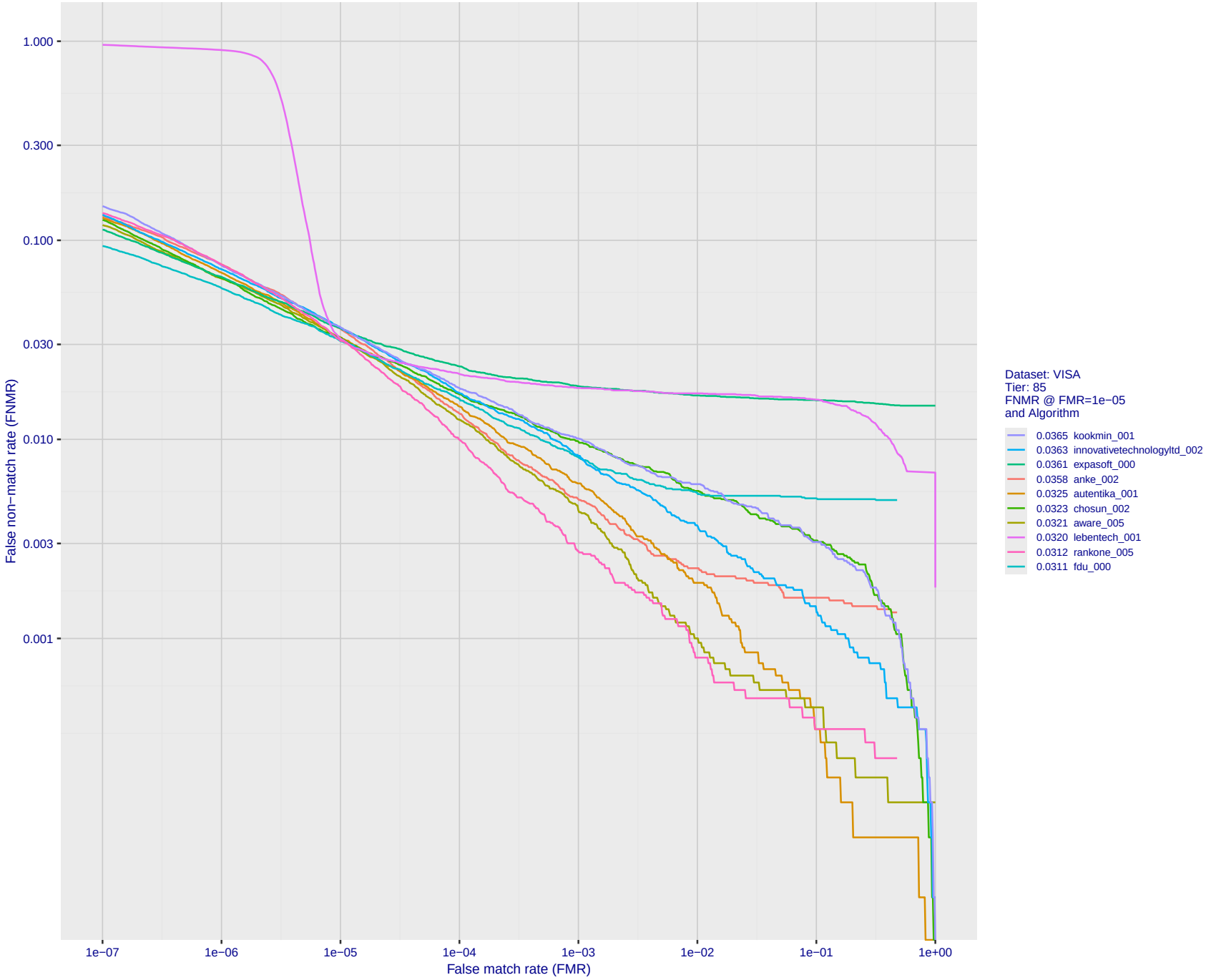

Figure 199: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

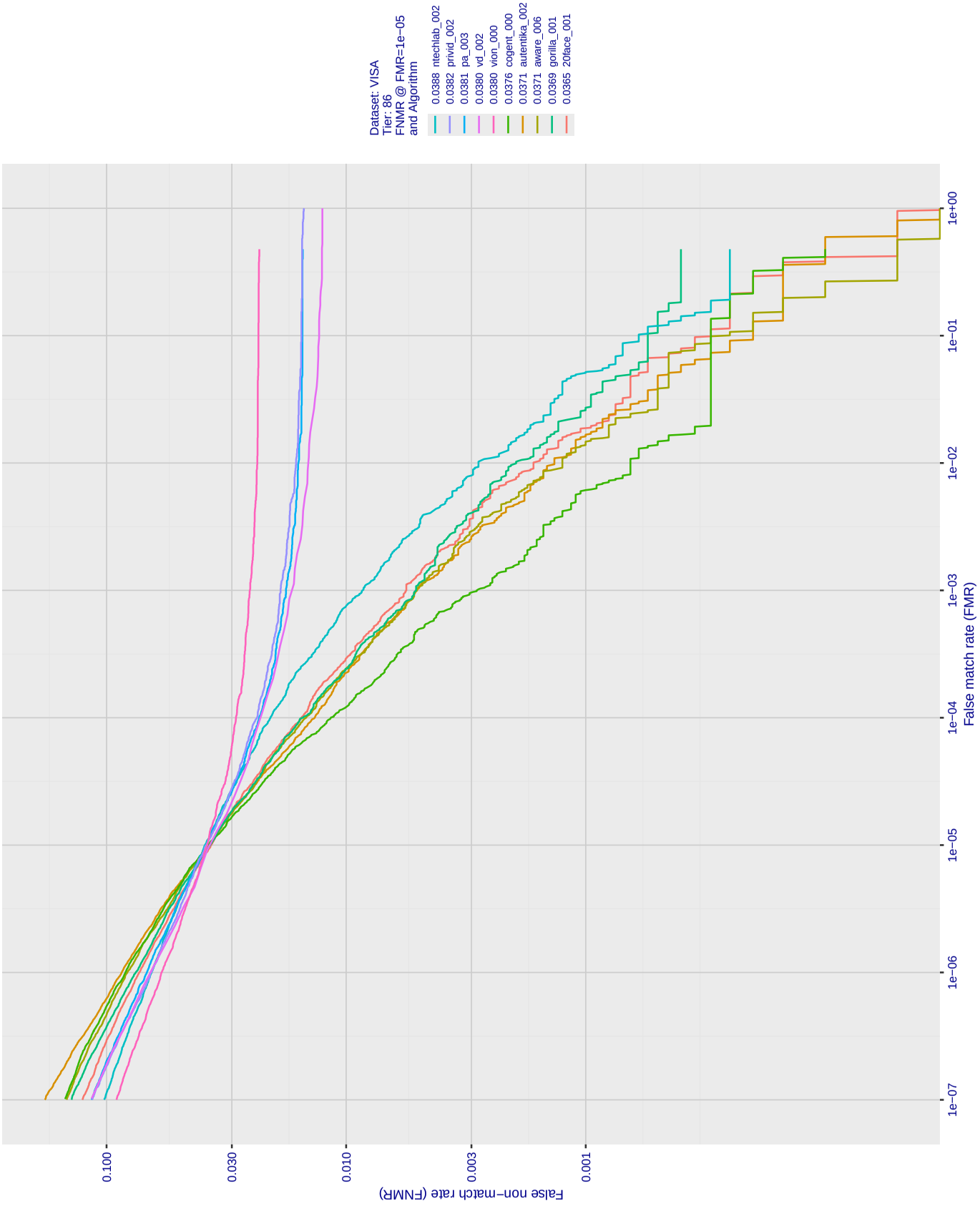

Figure 200: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

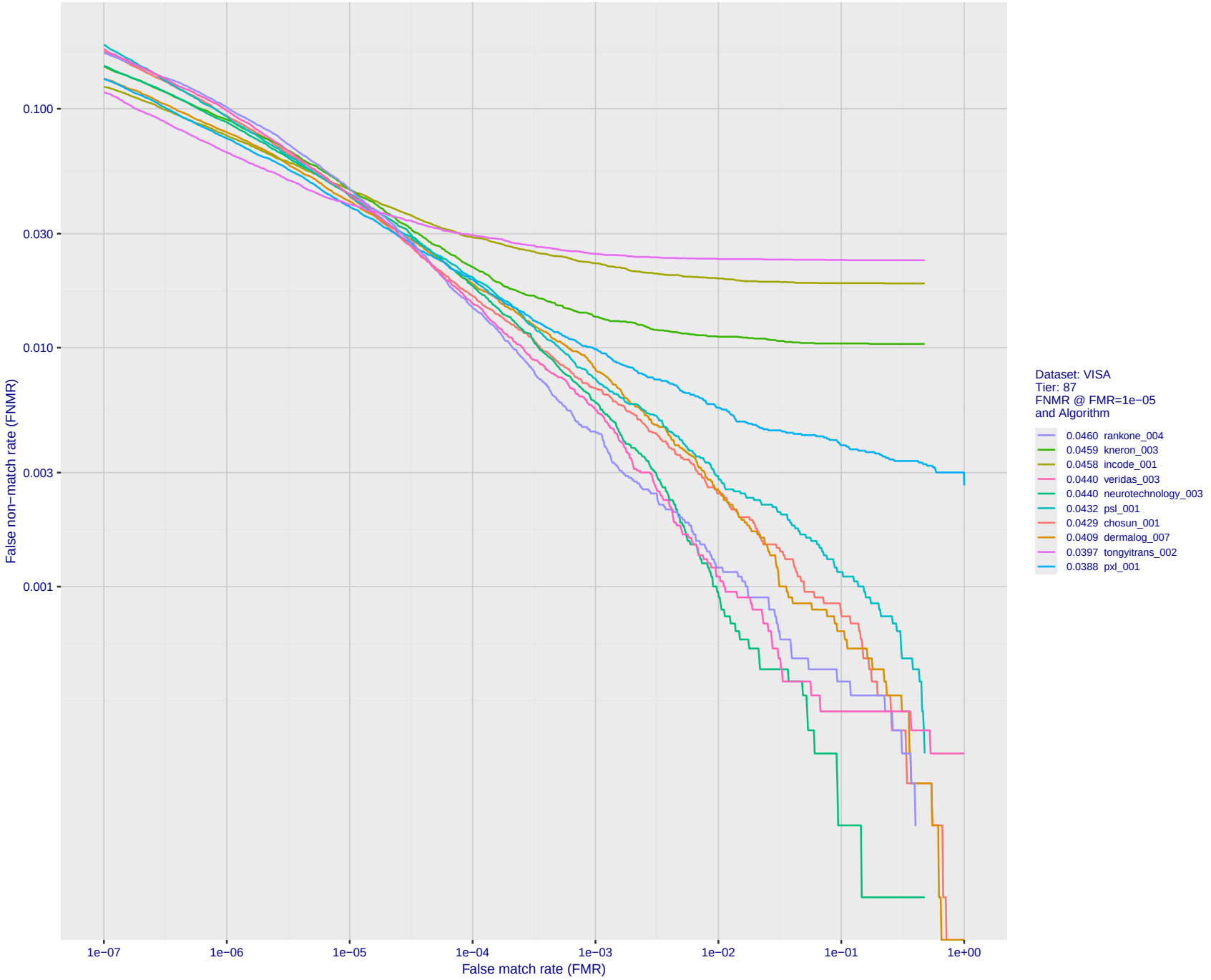

Figure 201: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

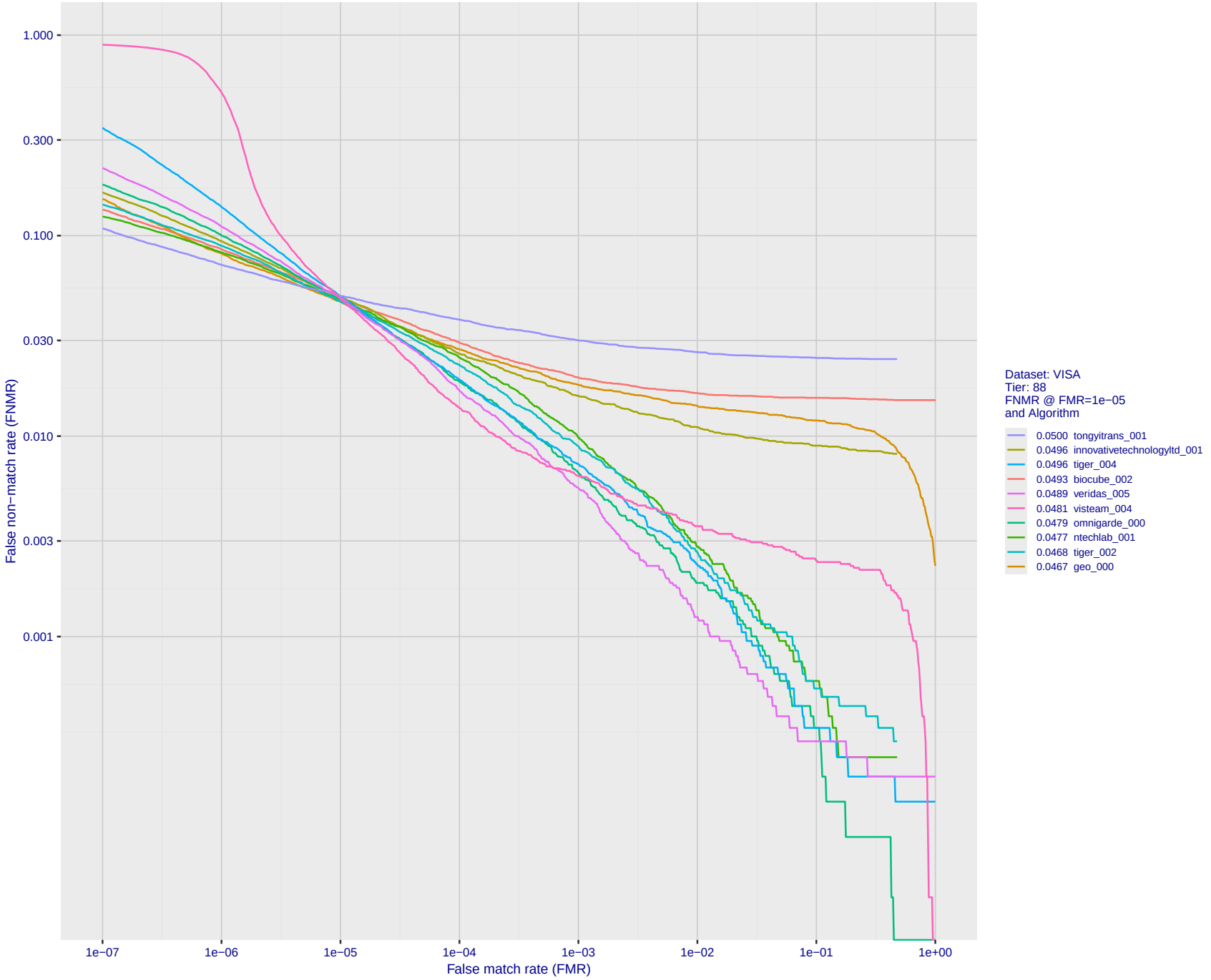

Figure 202: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

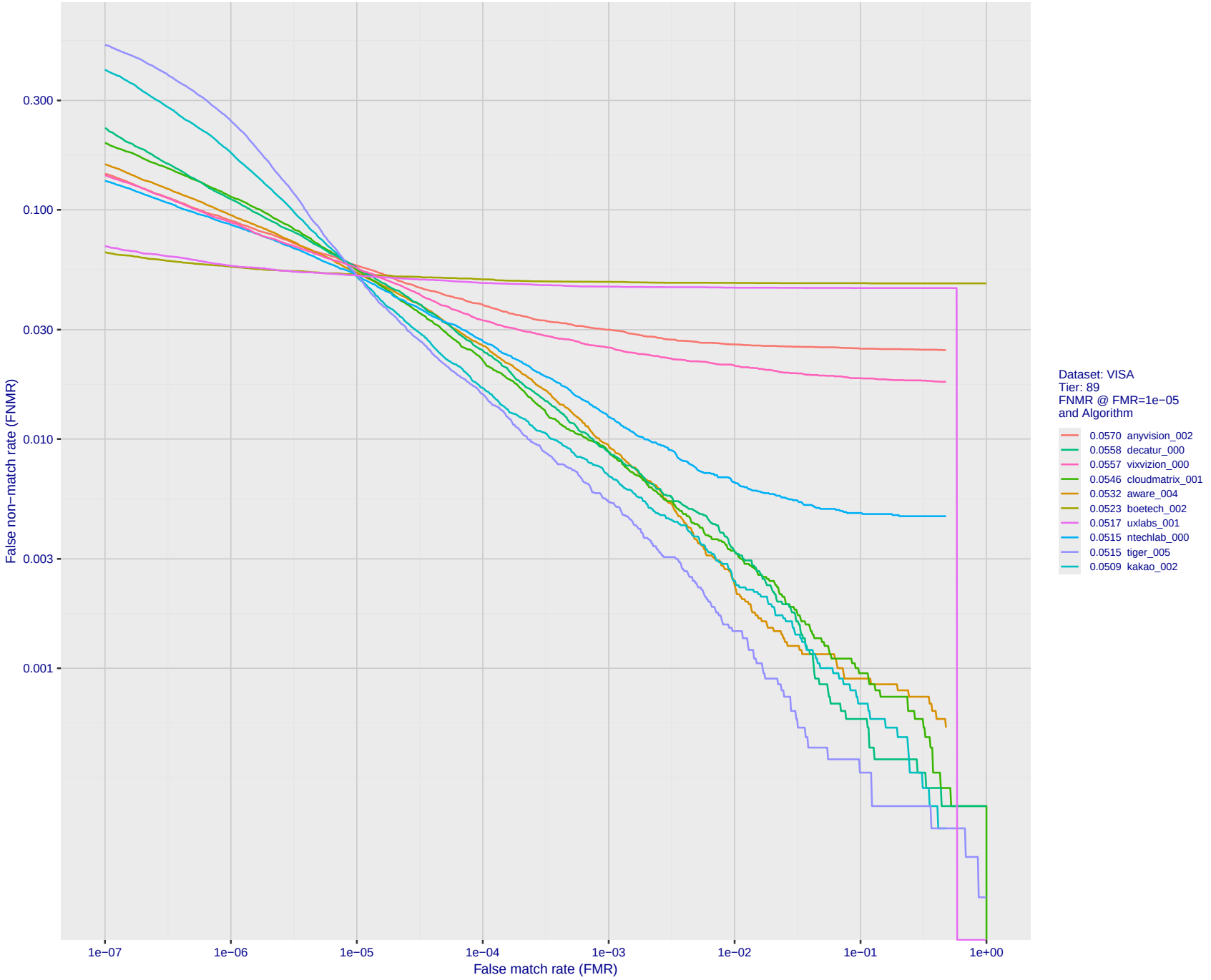

Figure 203: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

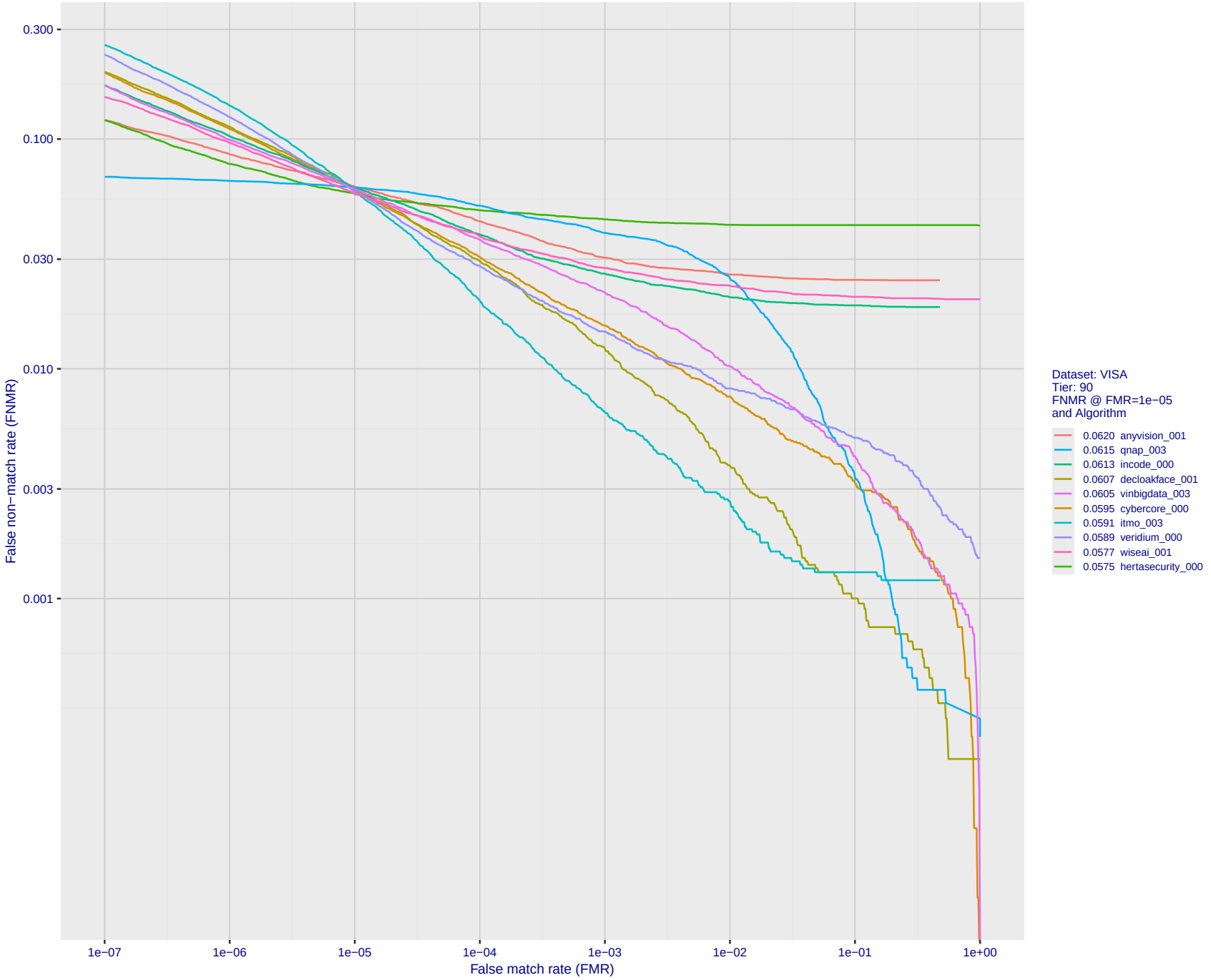

Figure 204: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

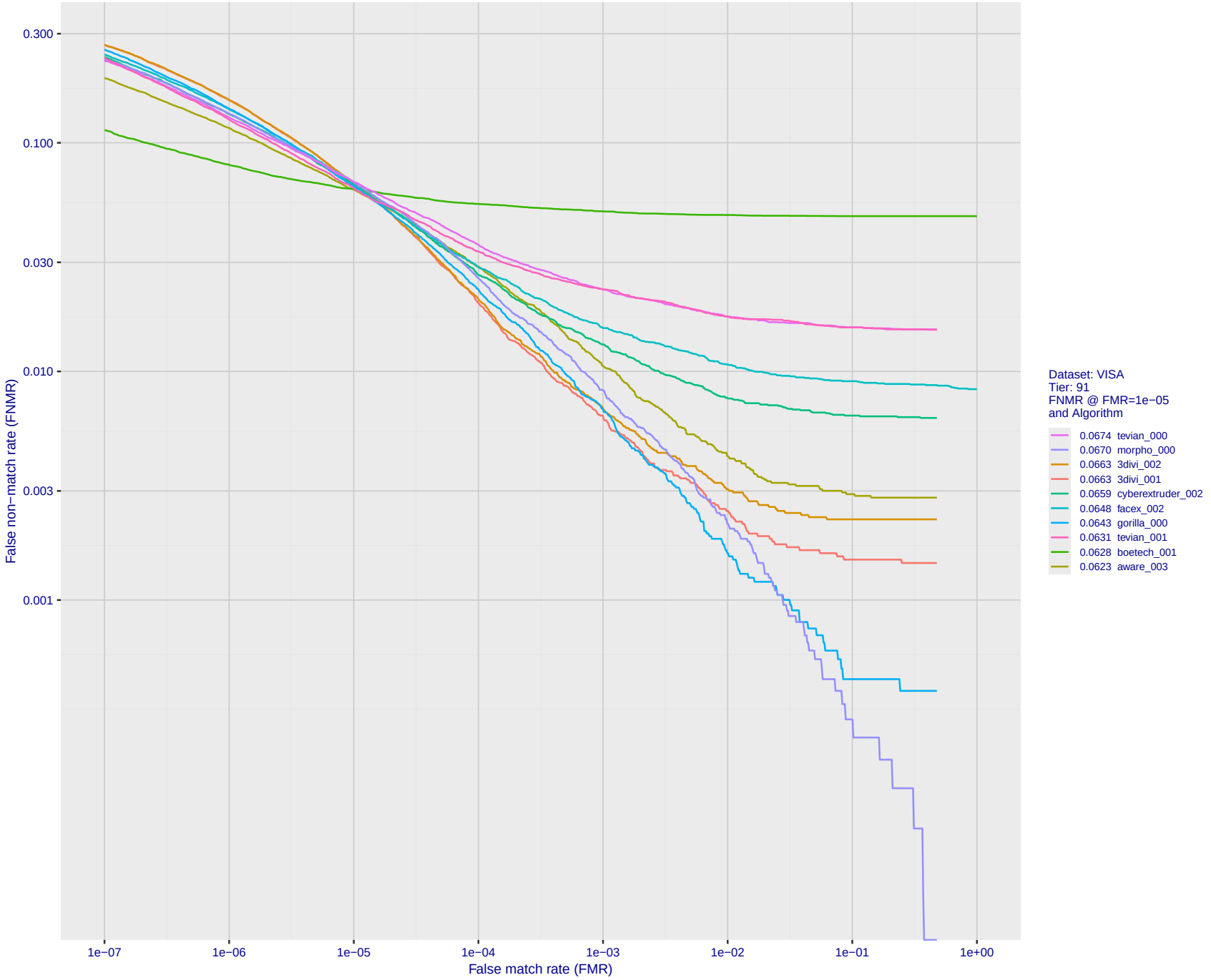

Figure 205: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

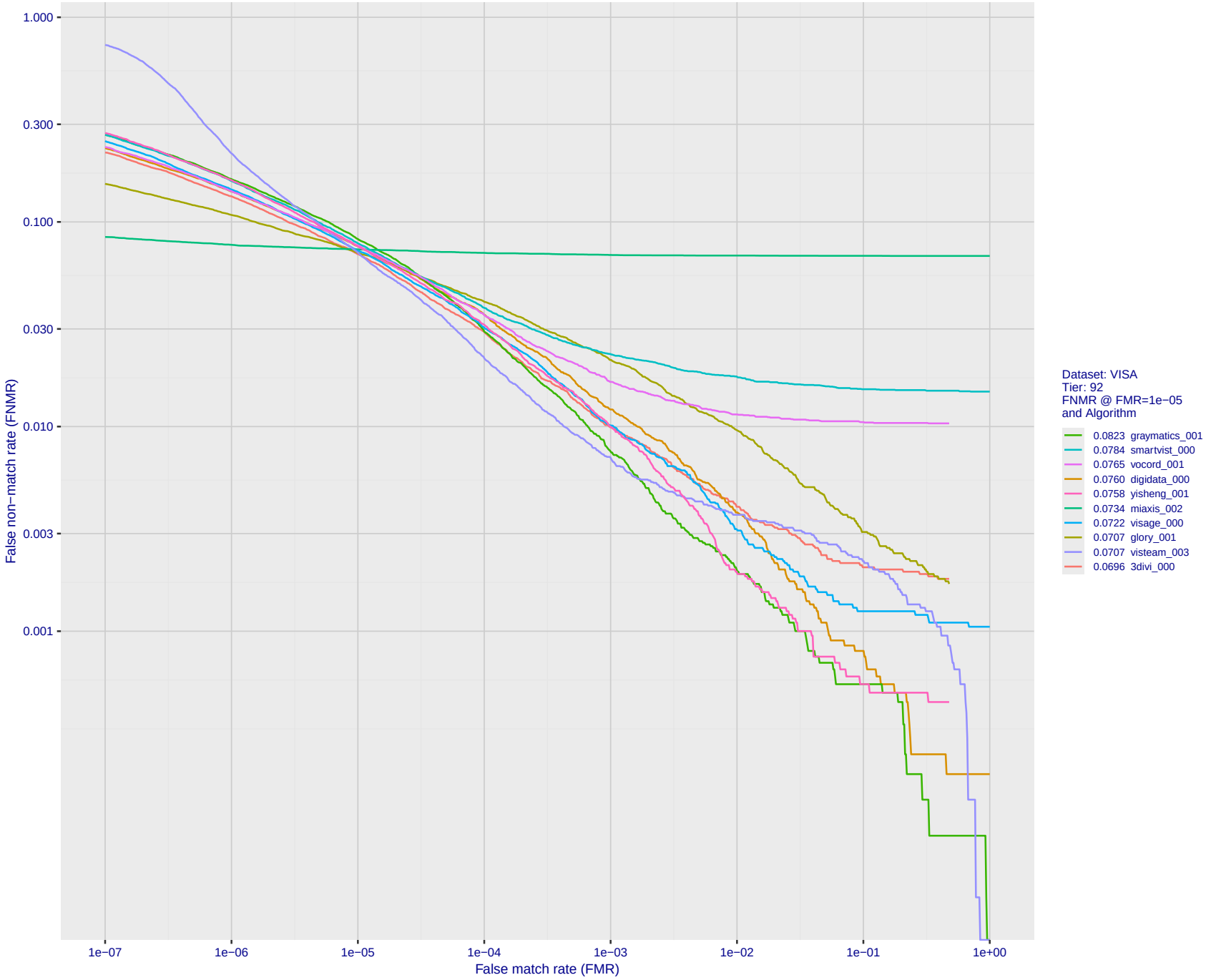

Figure 206: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

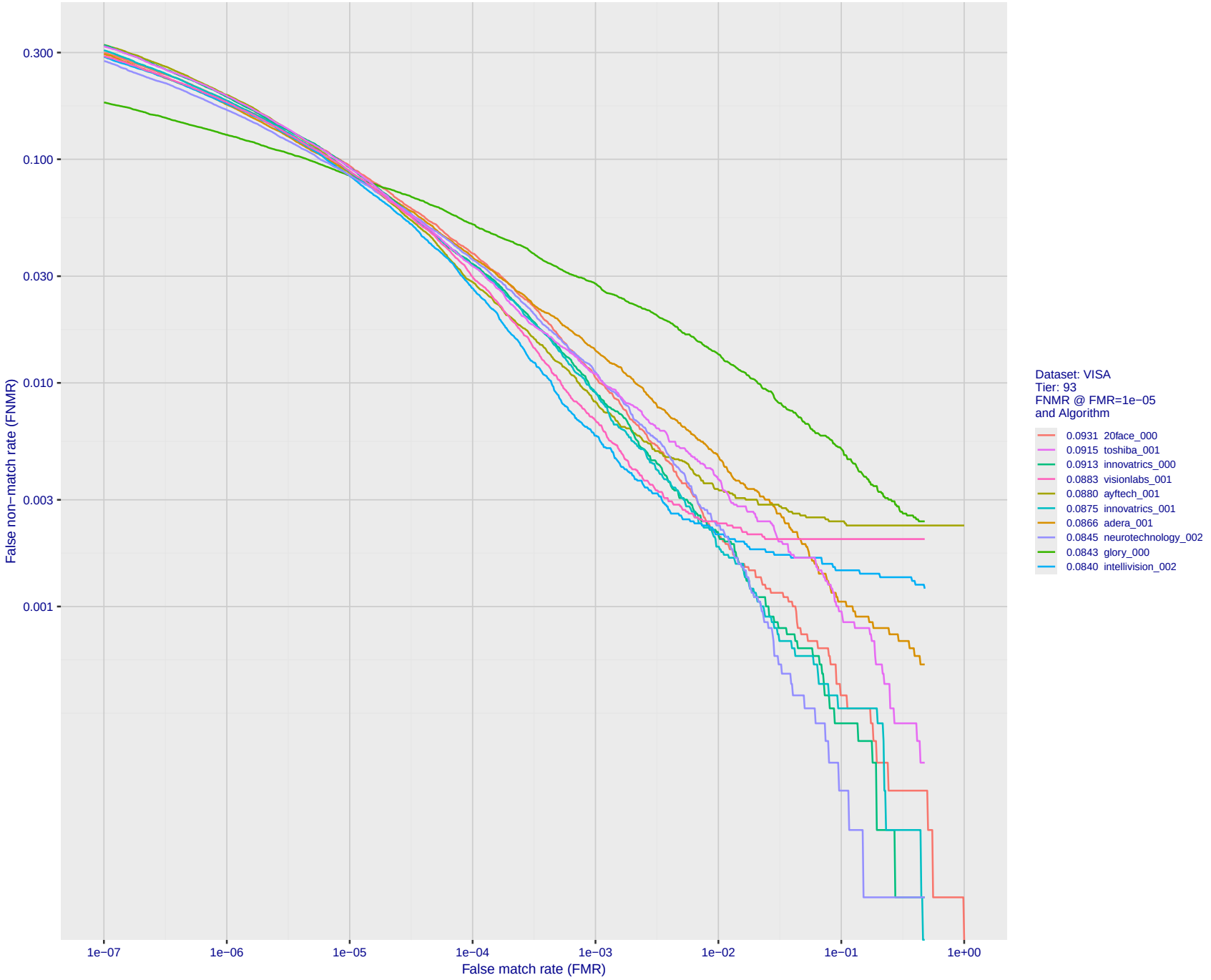

Figure 207: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

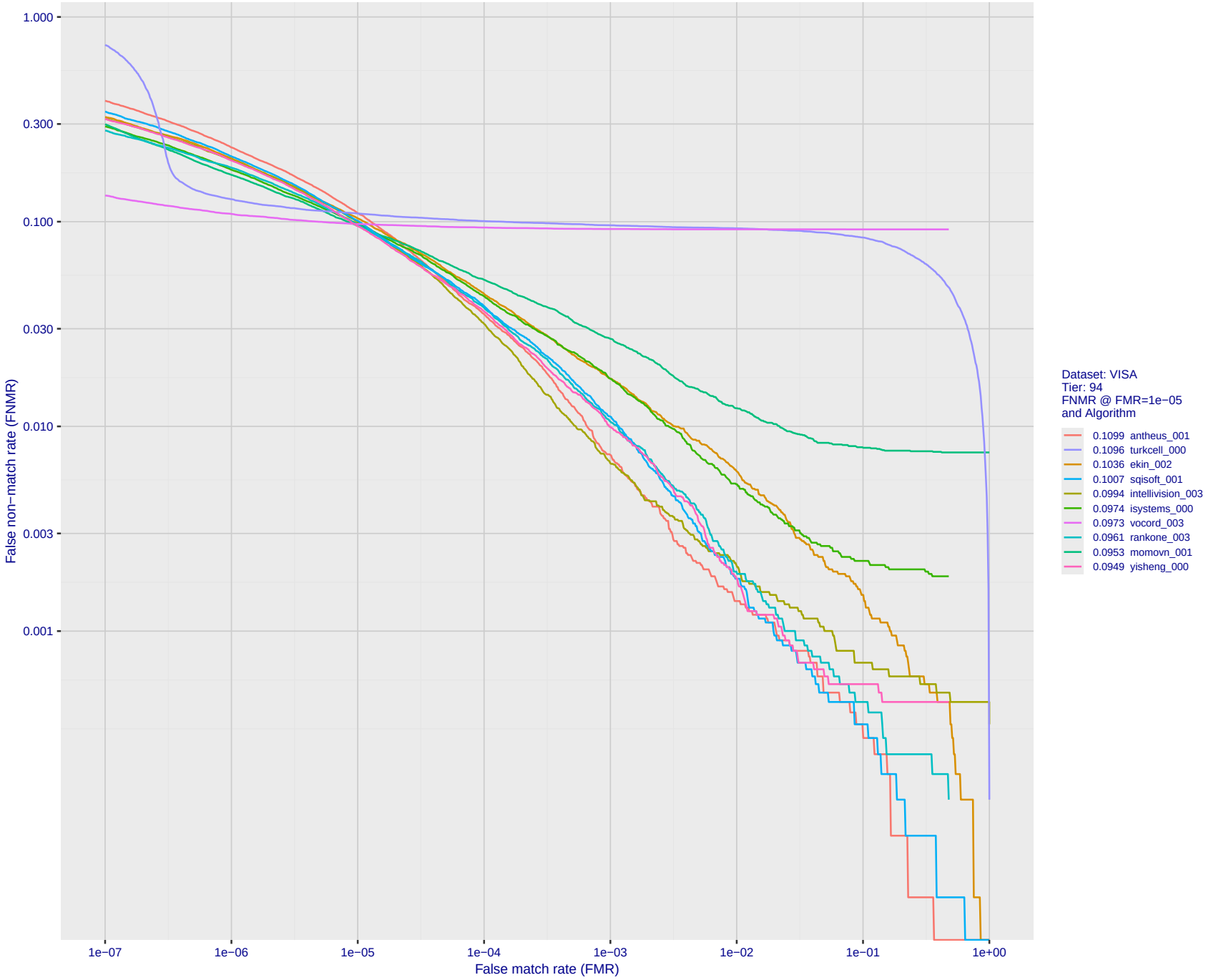

Figure 208: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

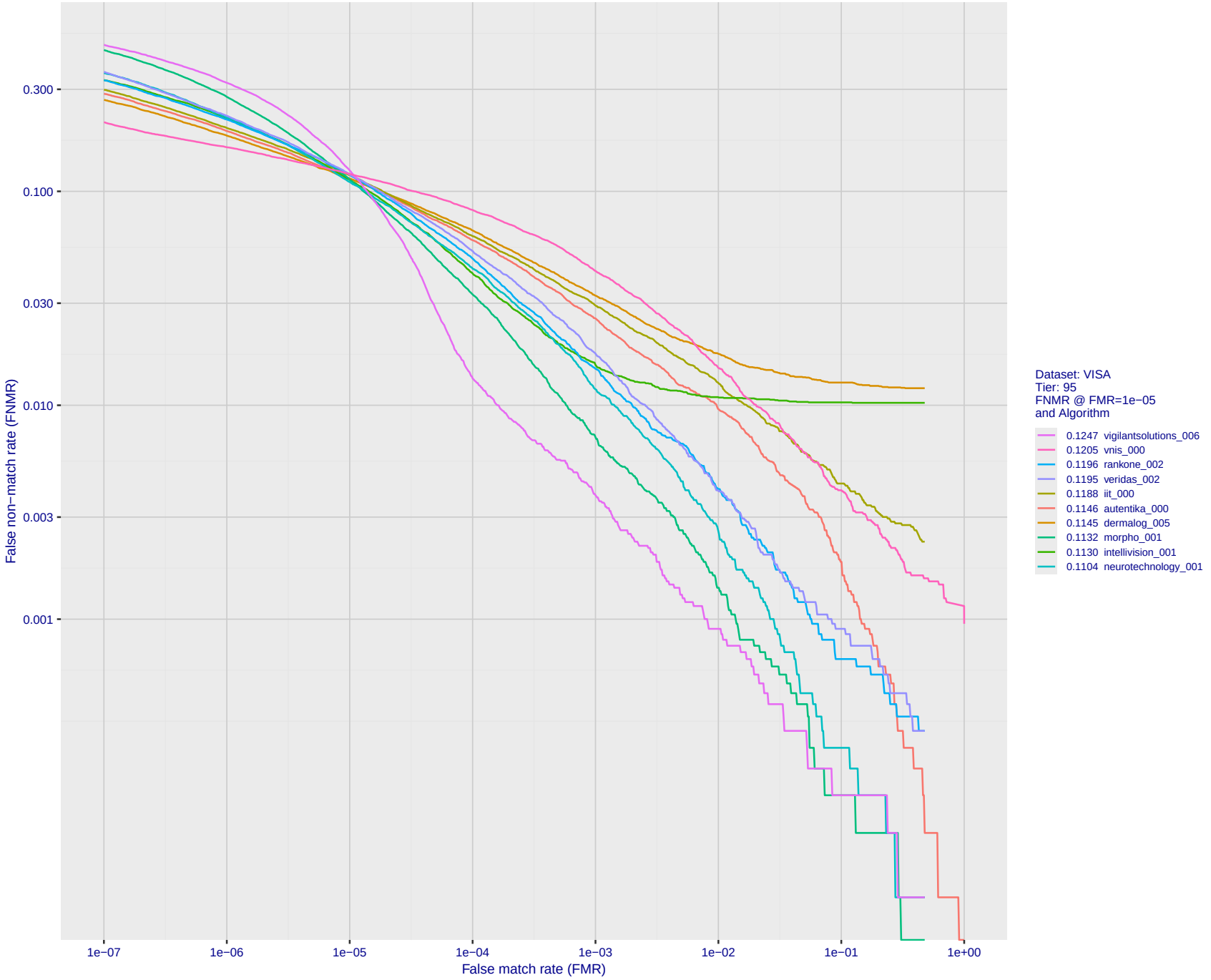

Figure 209: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

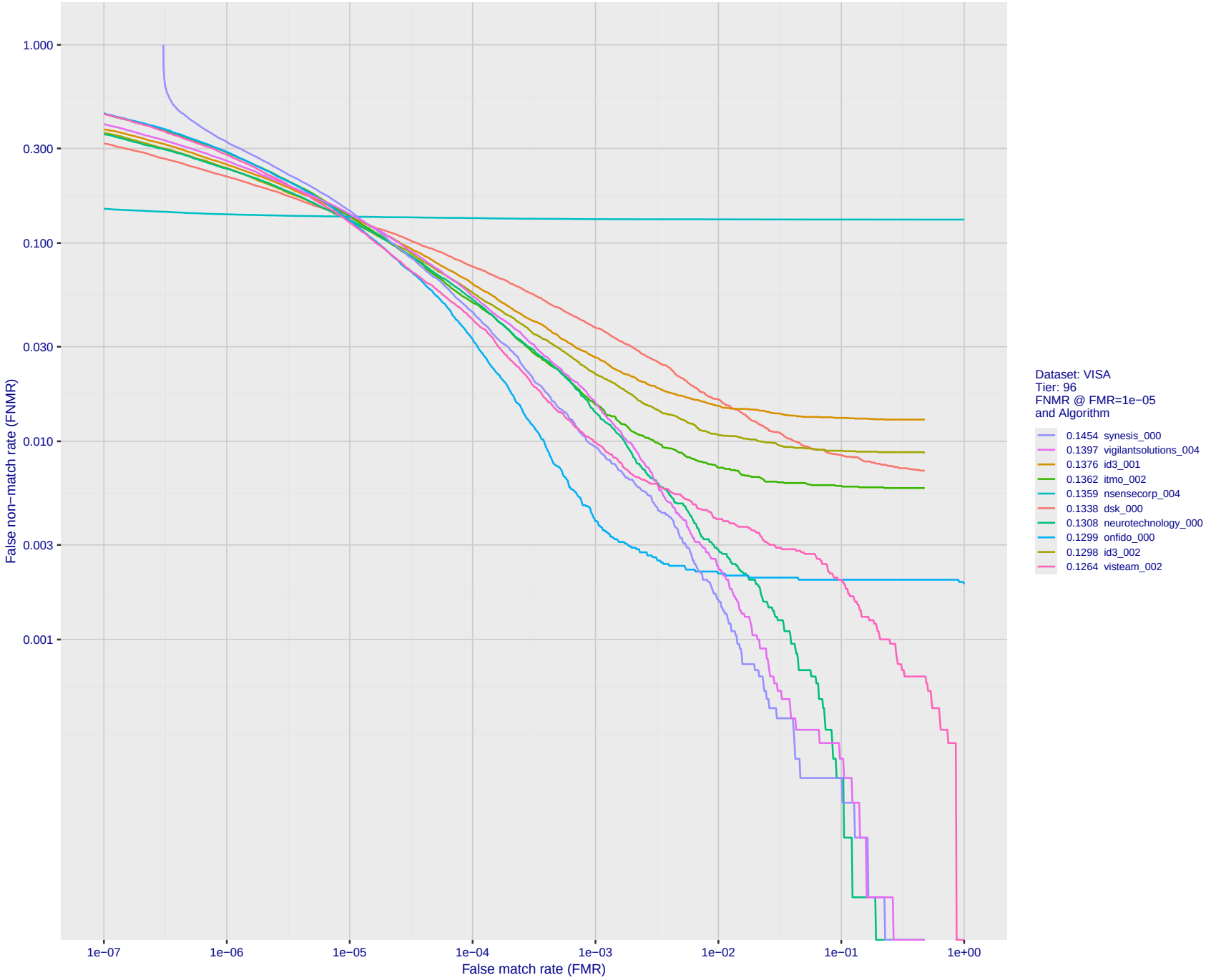

Figure 210: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

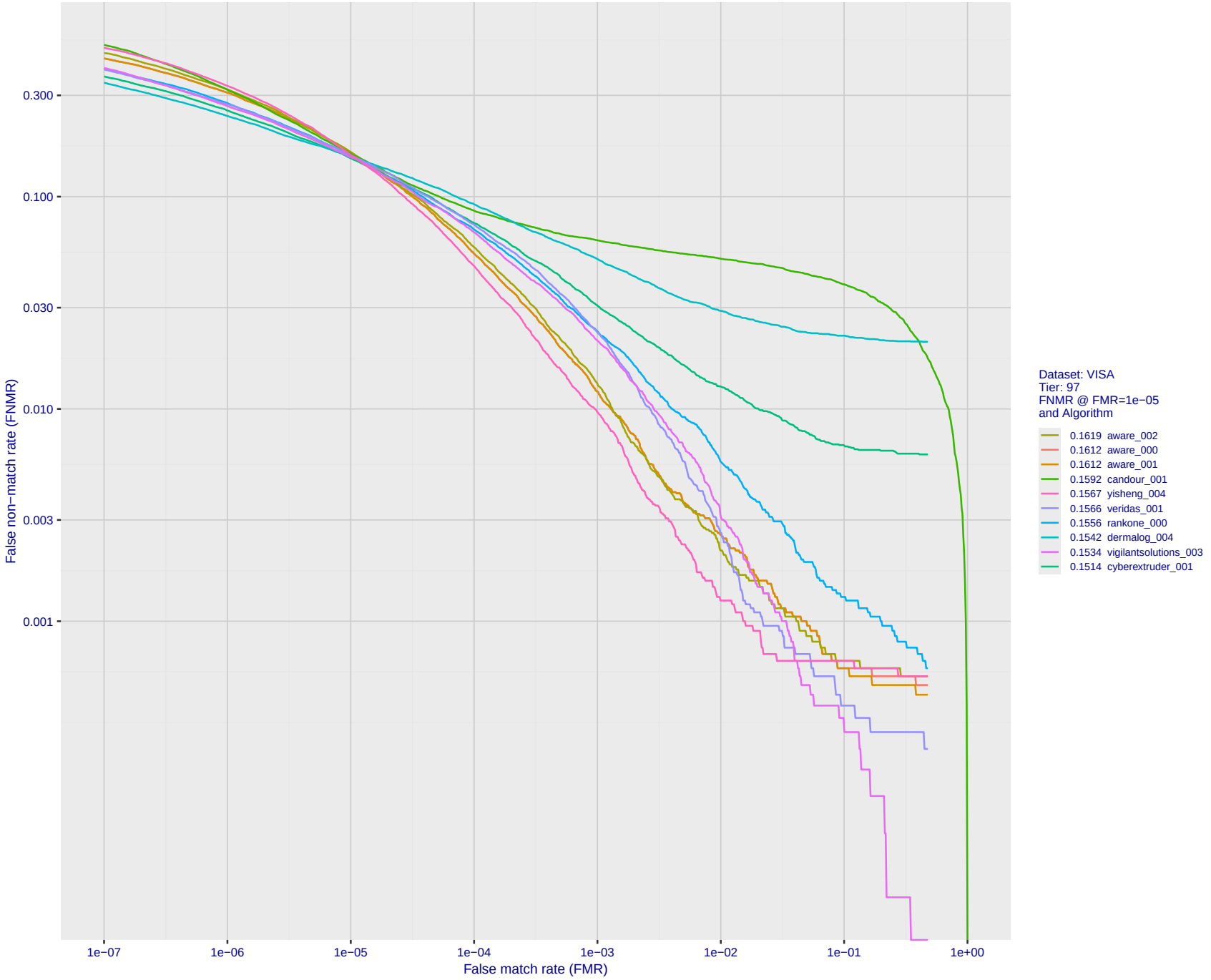

Figure 211: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

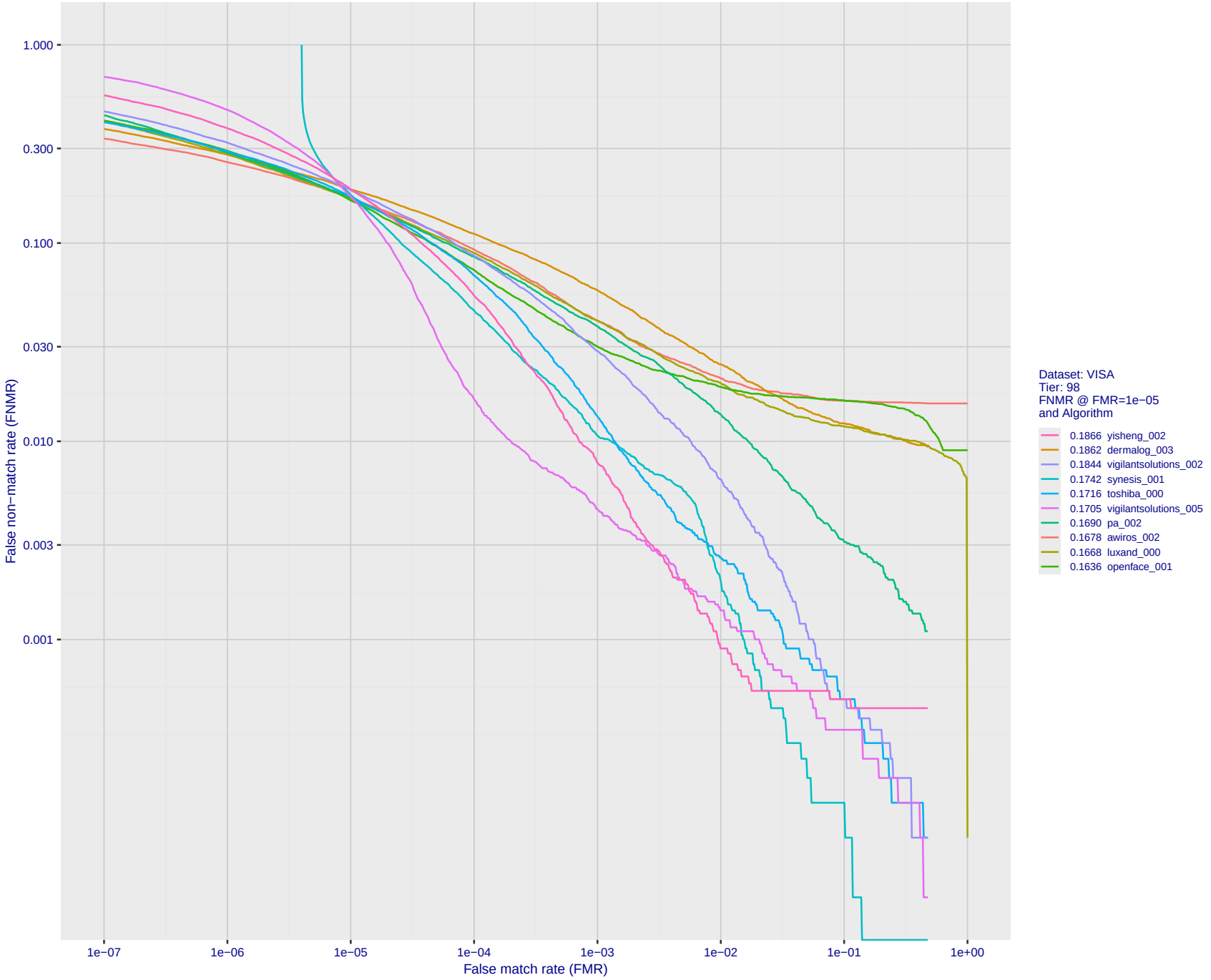

Figure 212: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

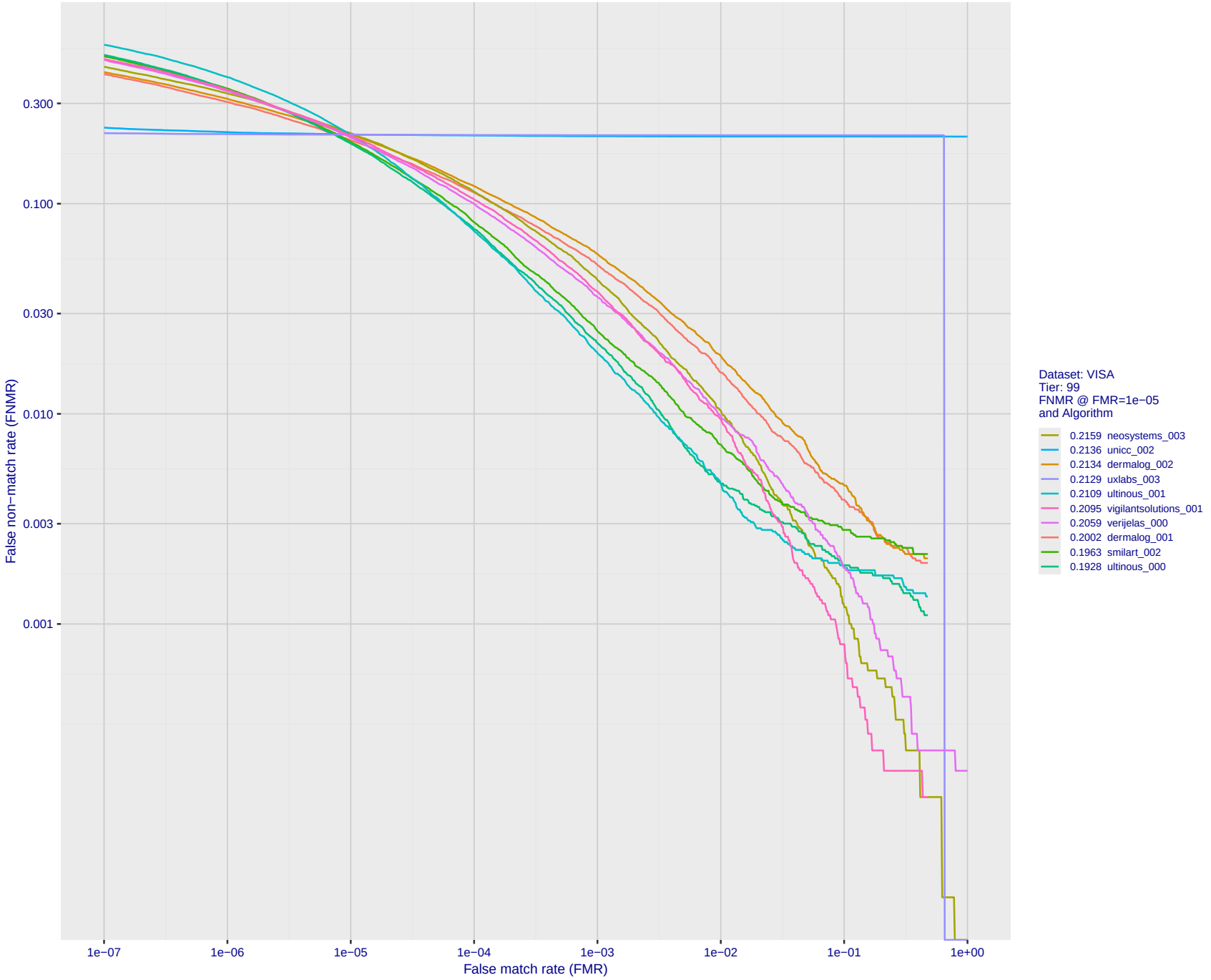

Figure 213: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

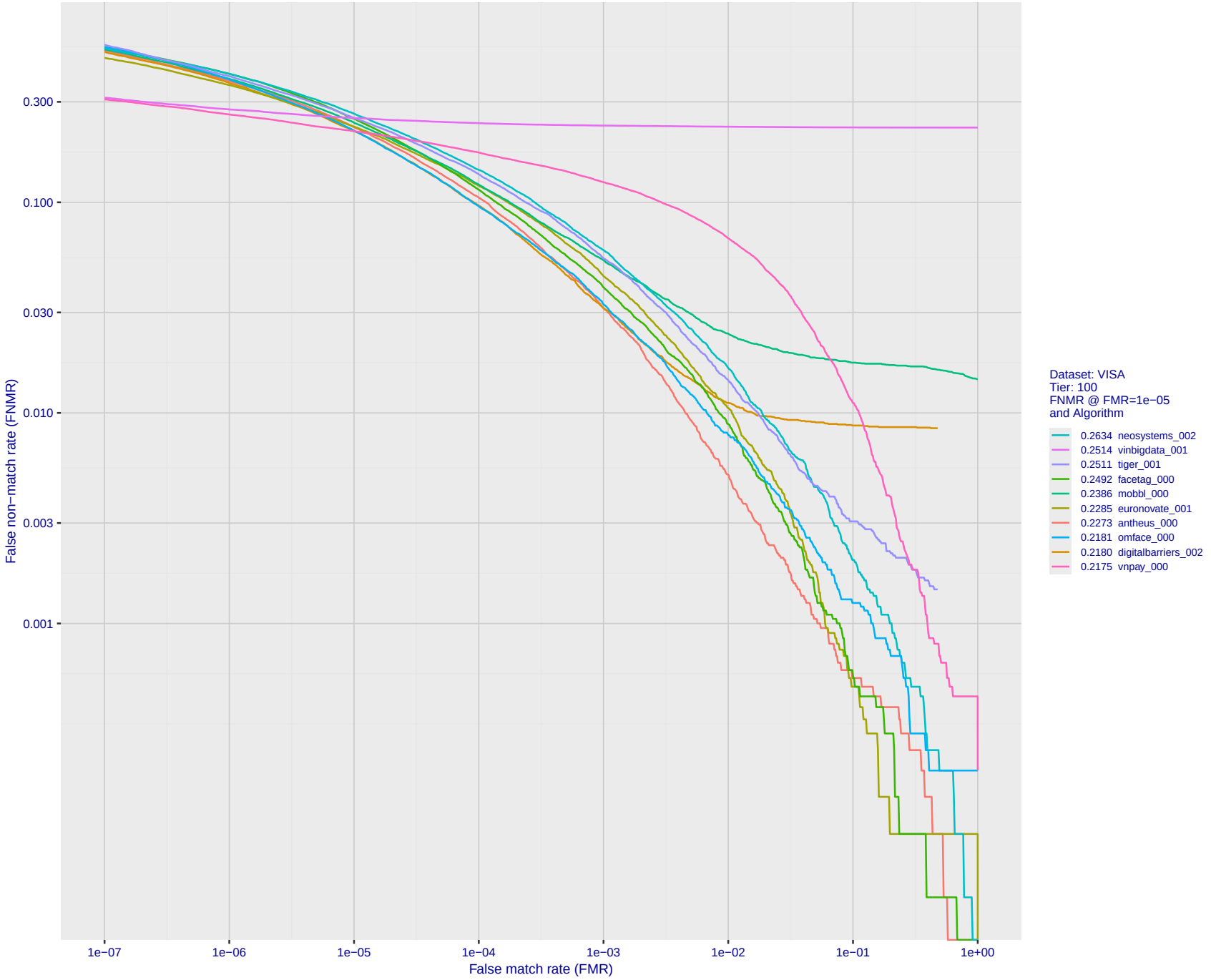

Figure 214: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

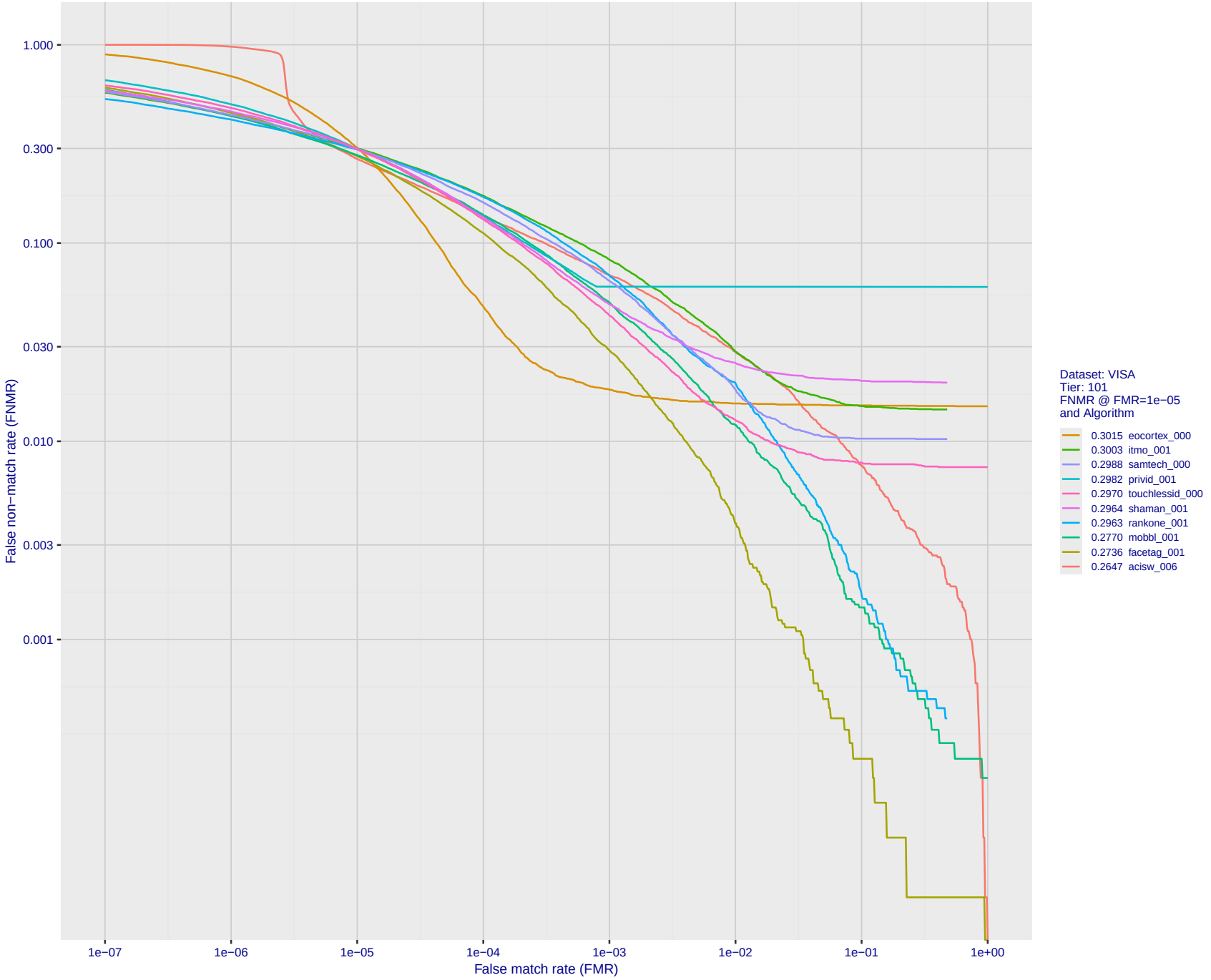

Figure 215: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

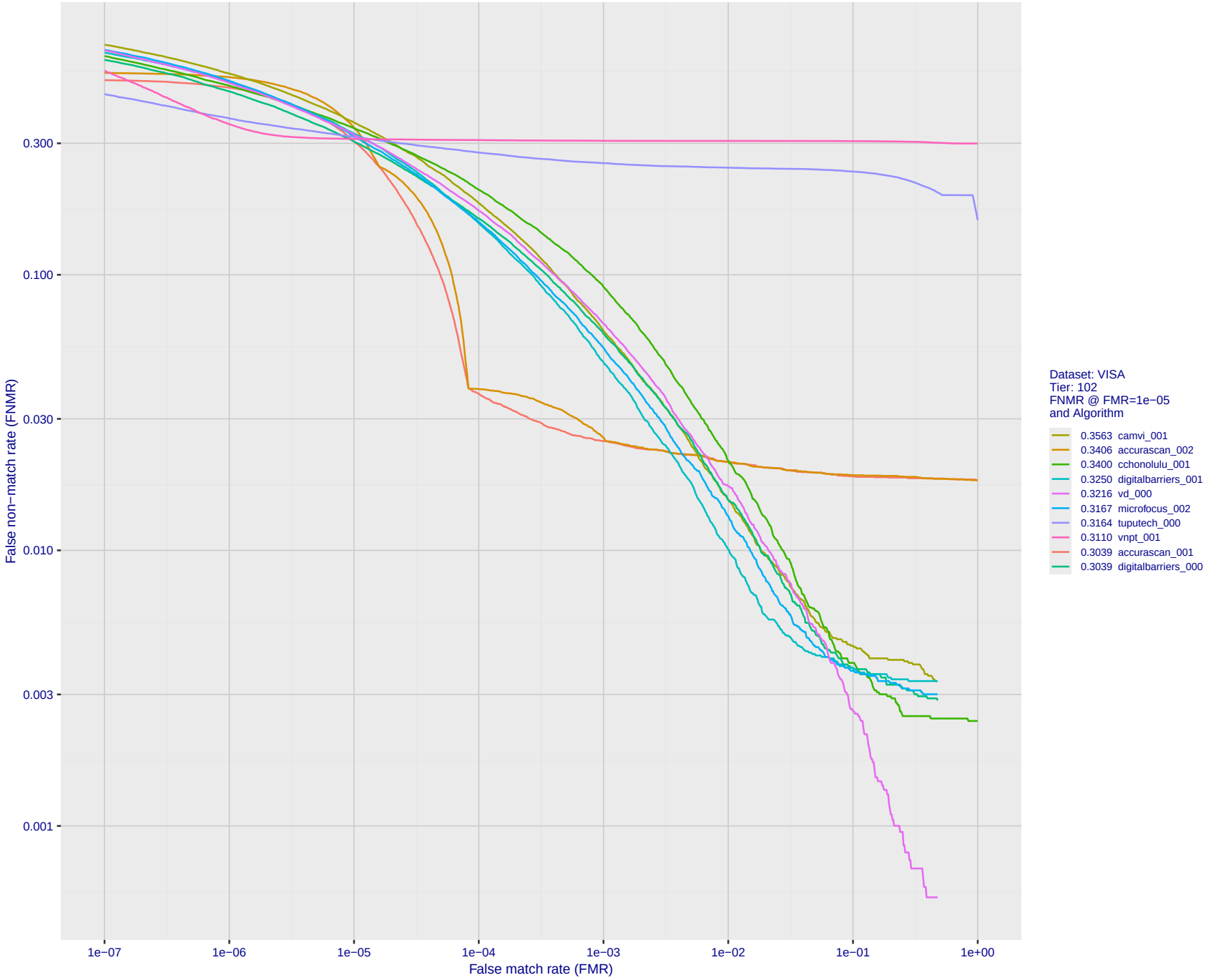

Figure 216: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

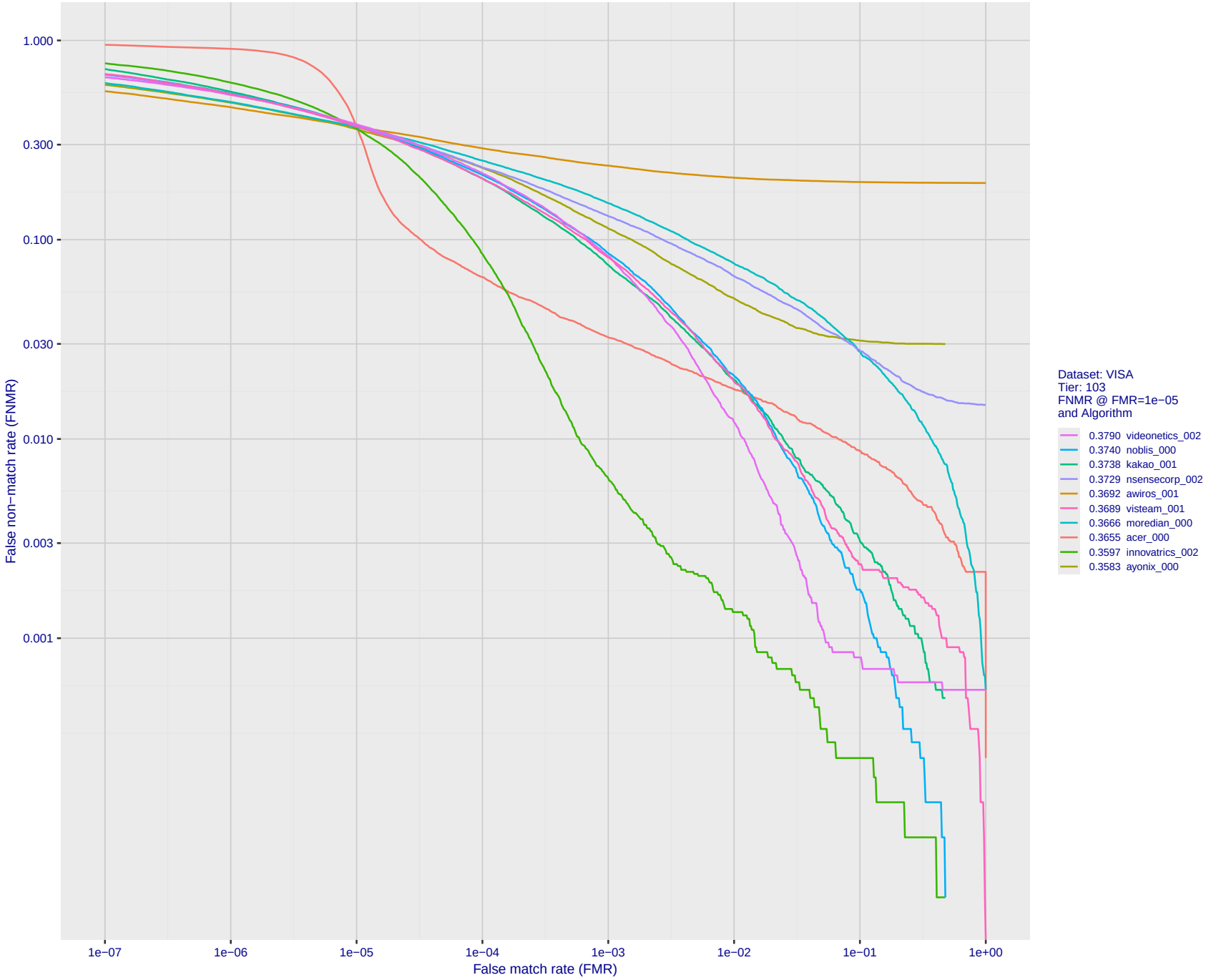

Figure 217: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

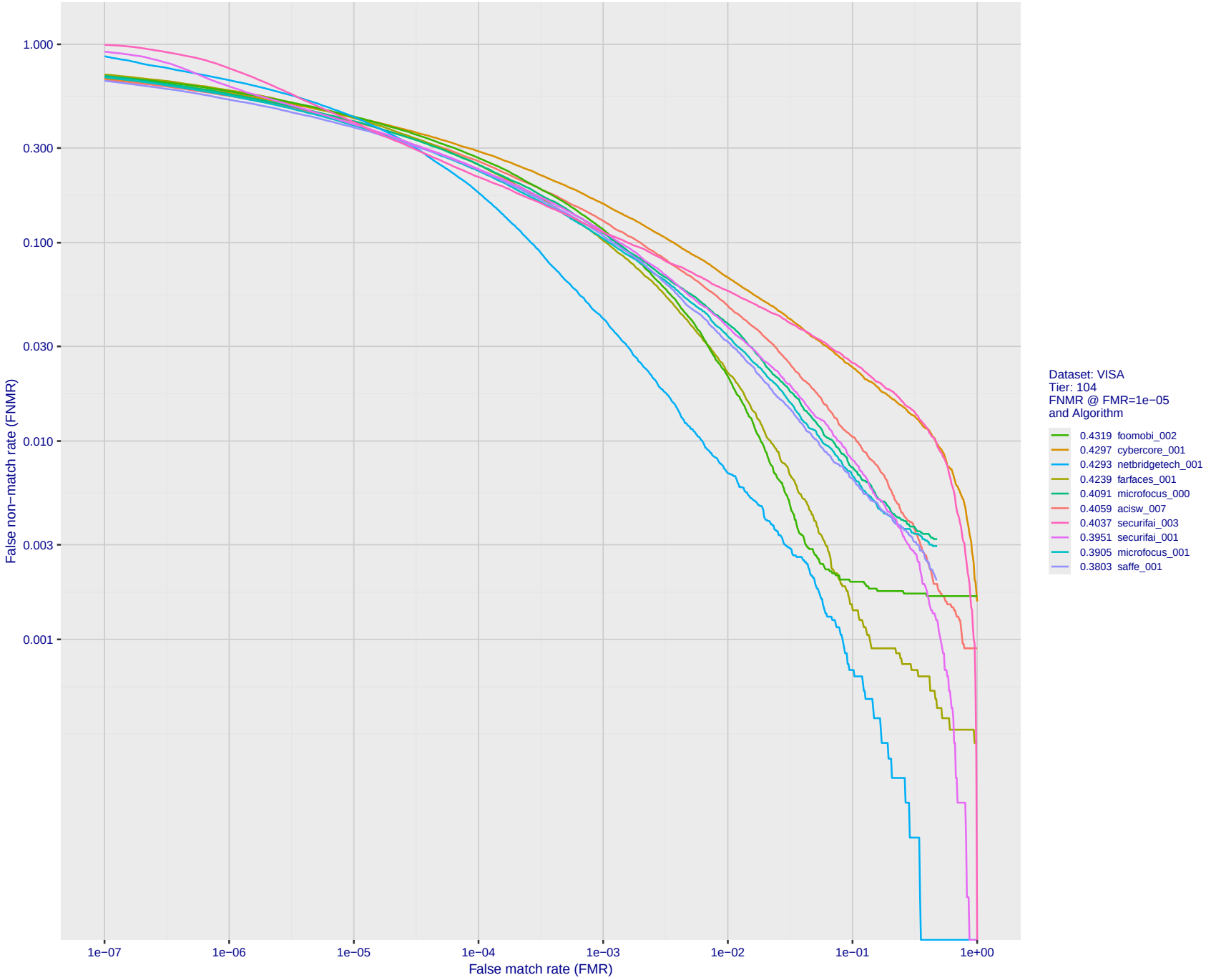

Figure 218: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

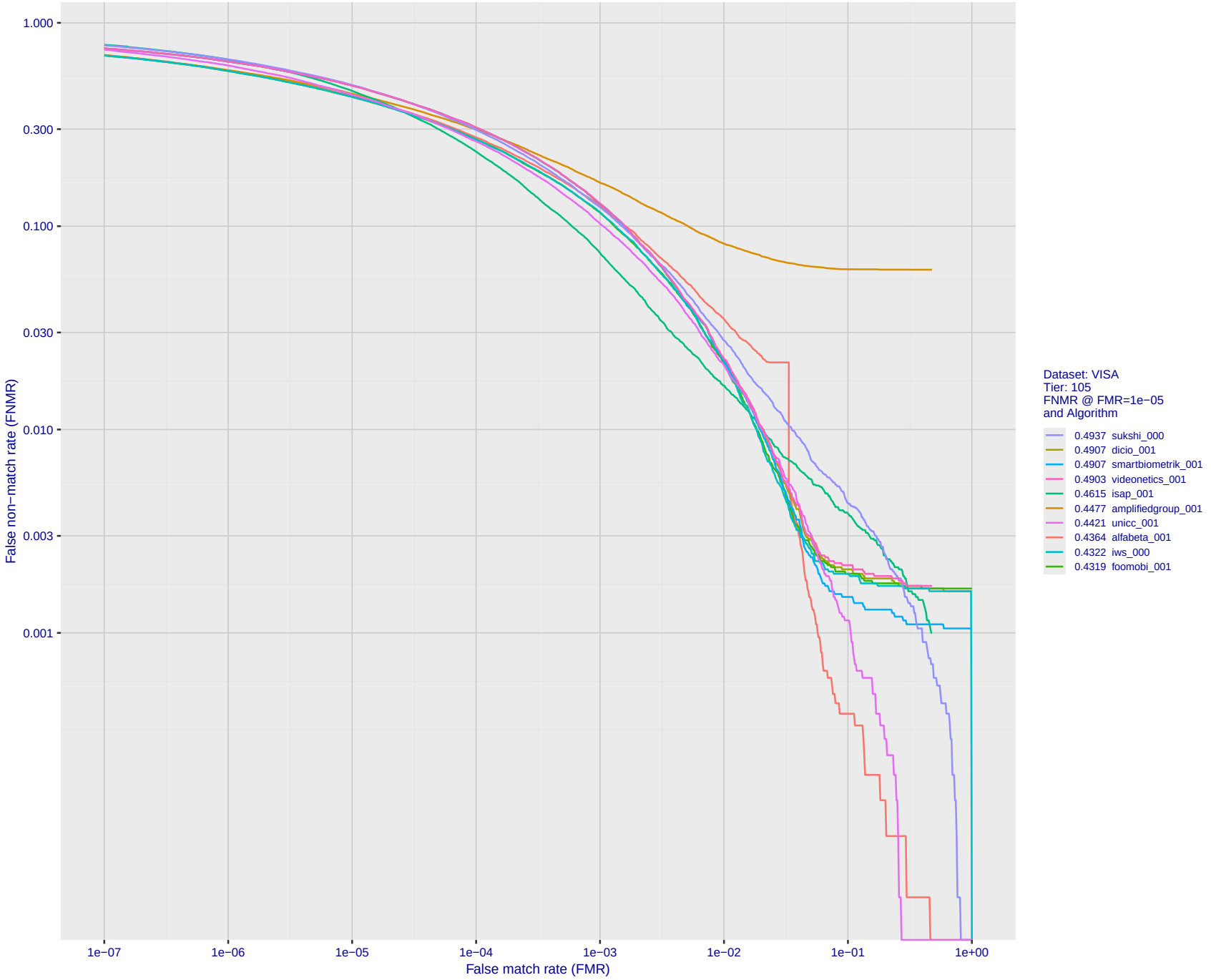

Figure 219: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

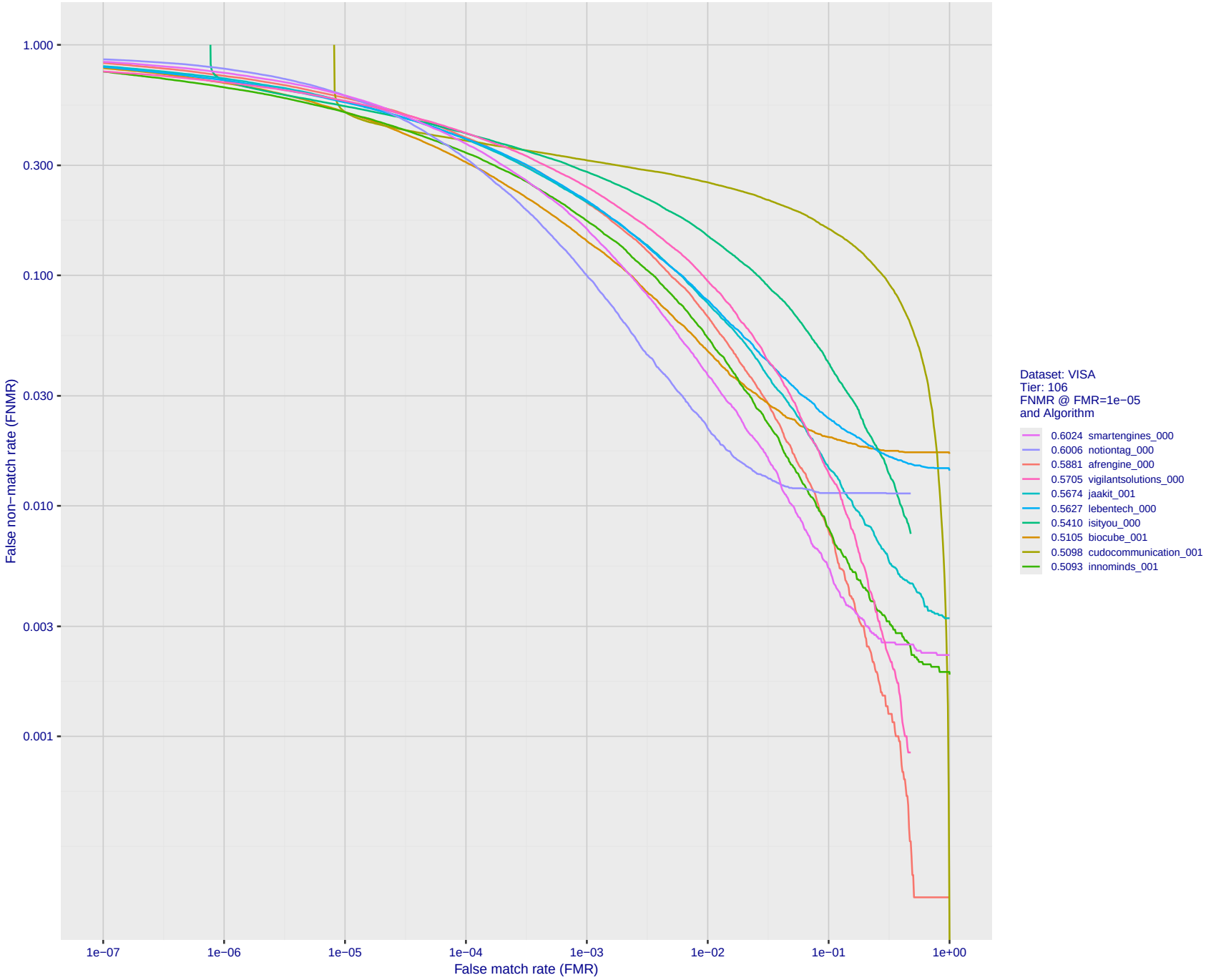

Figure 220: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

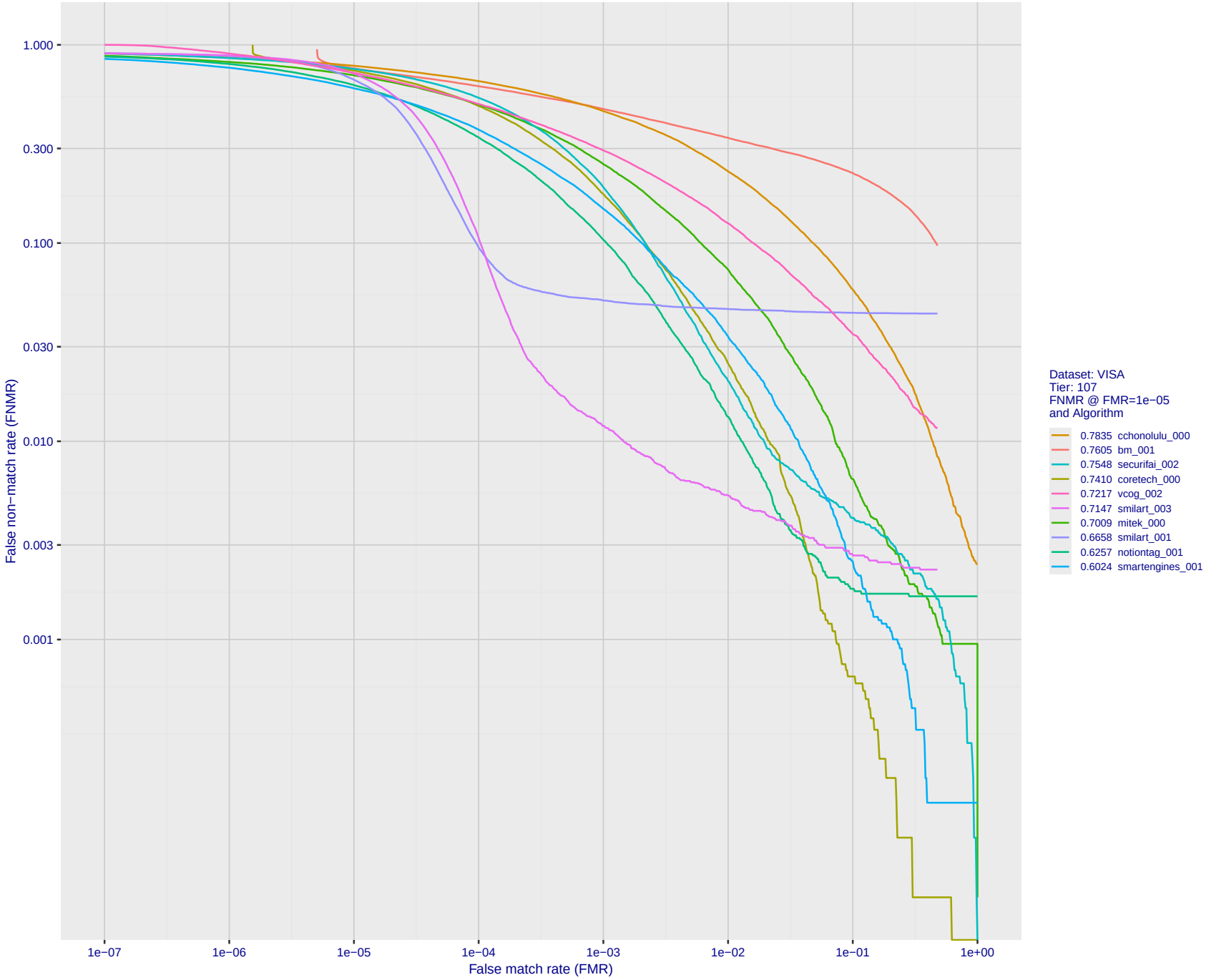

Figure 221: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

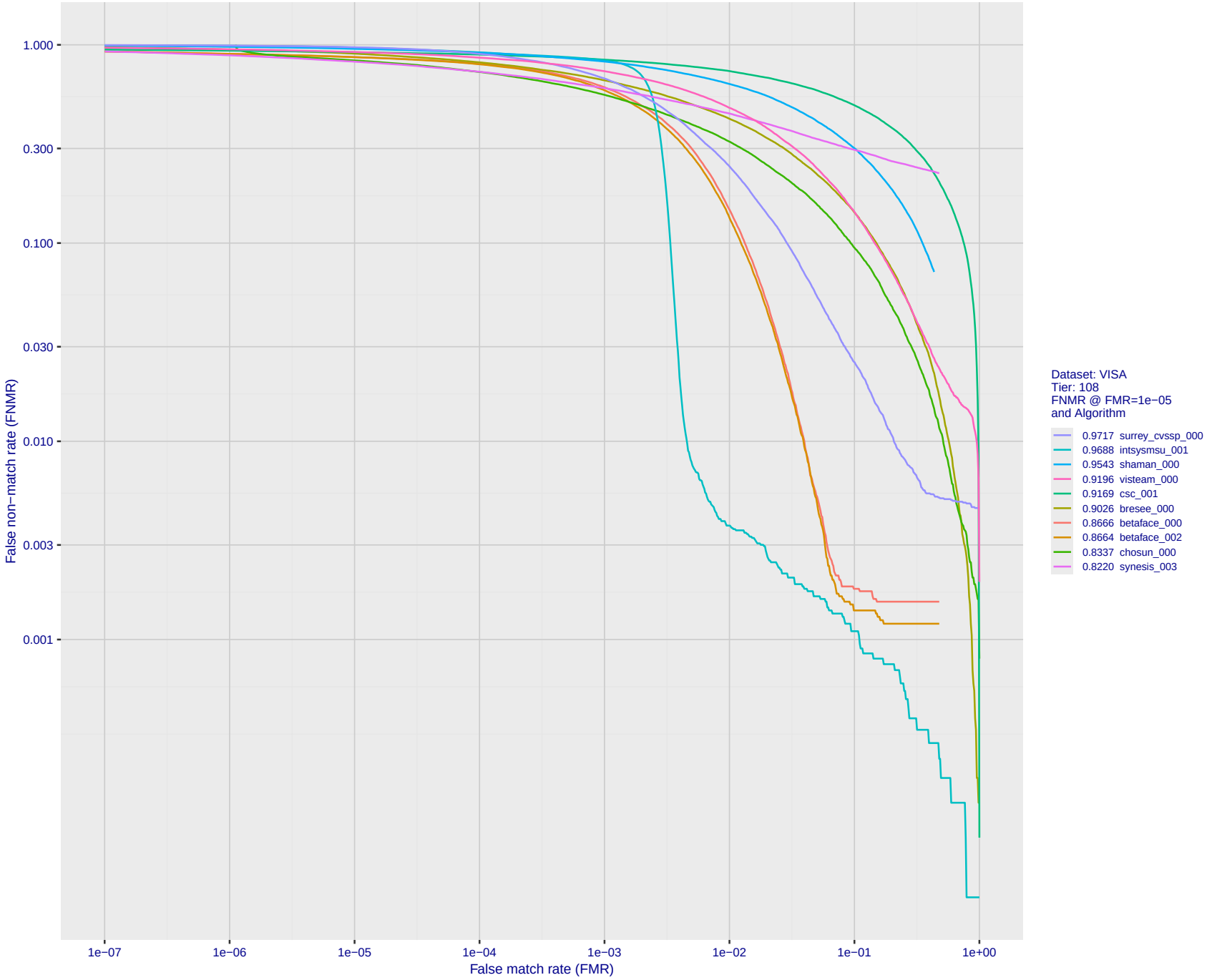

Figure 222: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

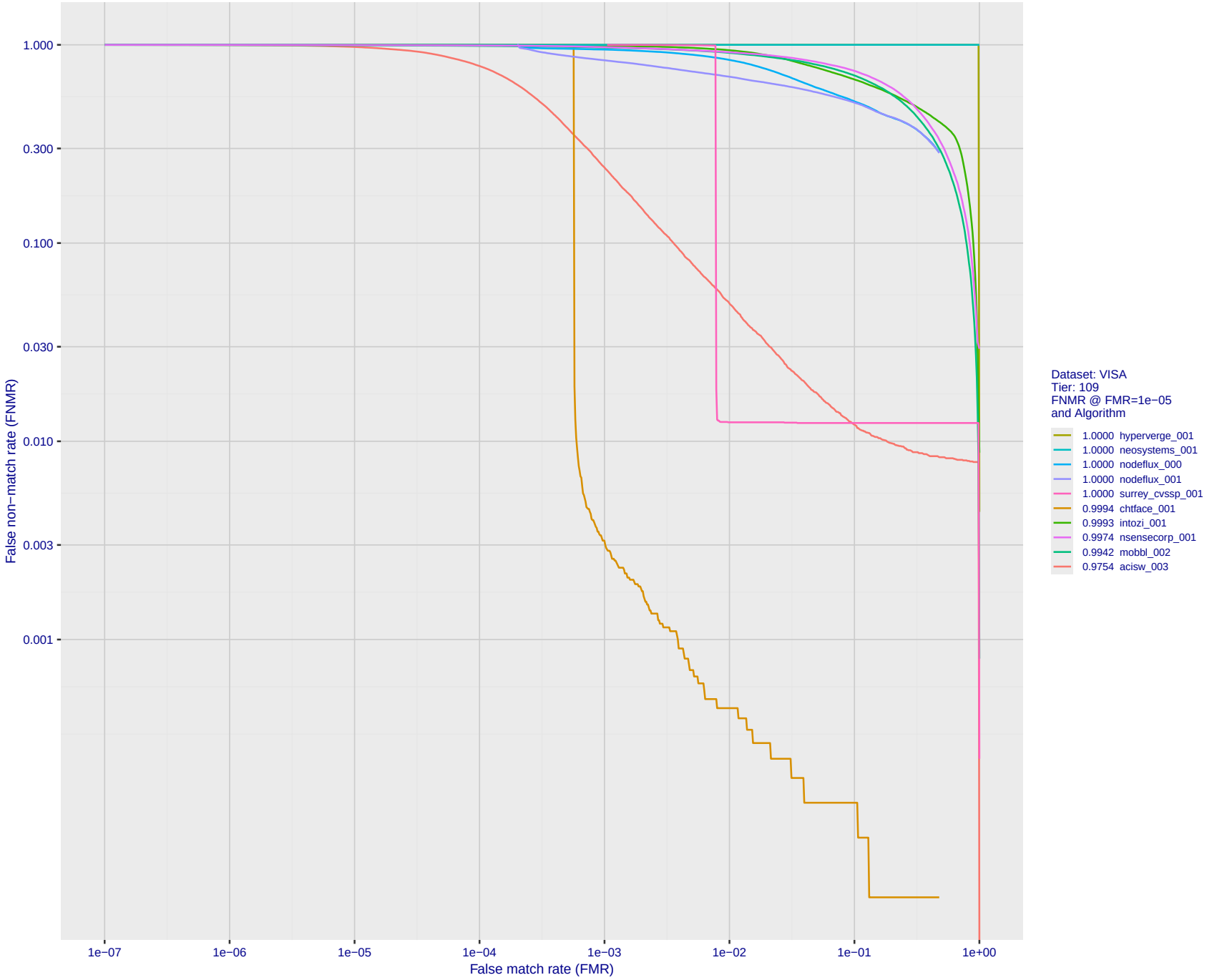

Figure 223: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

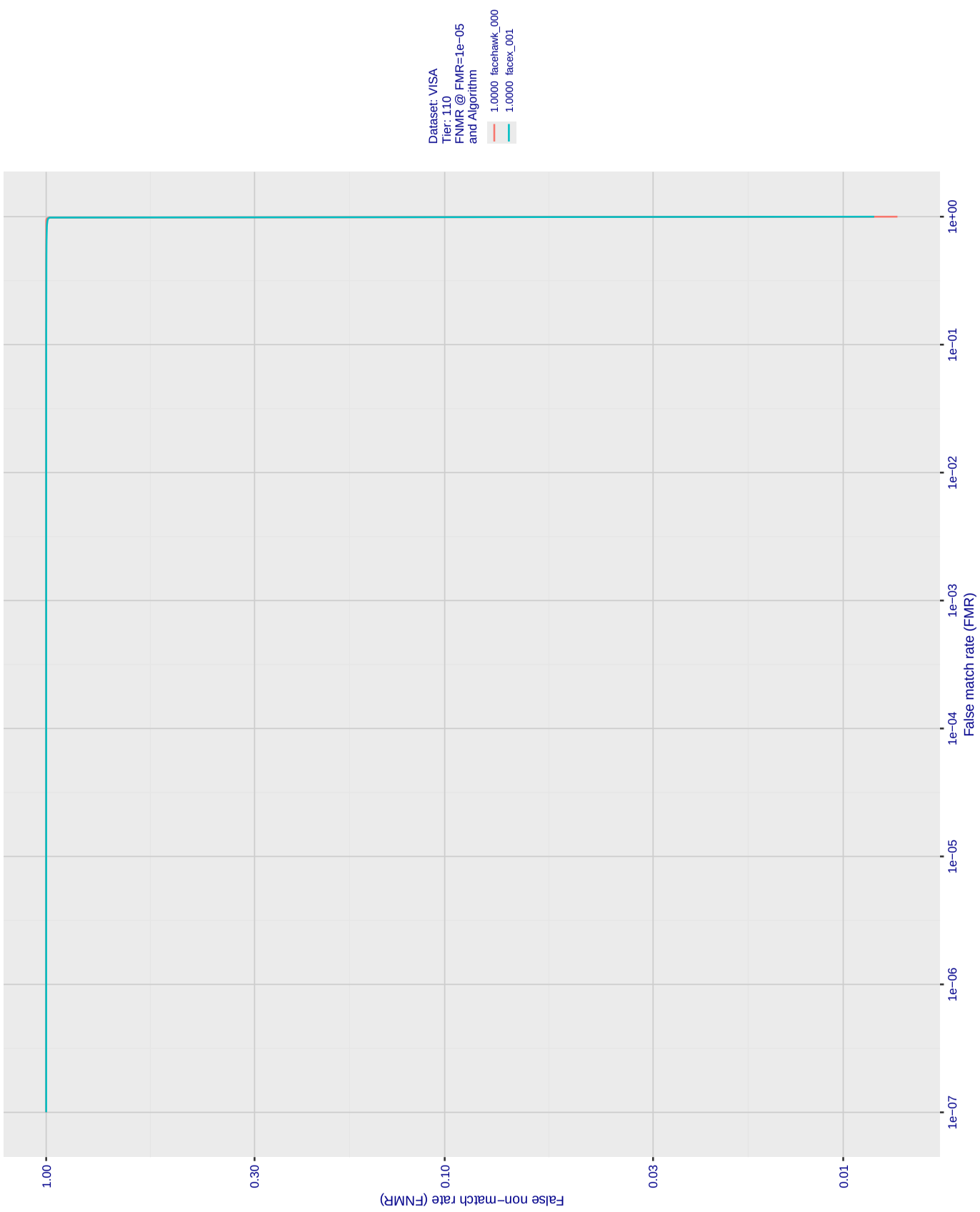

Figure 224: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

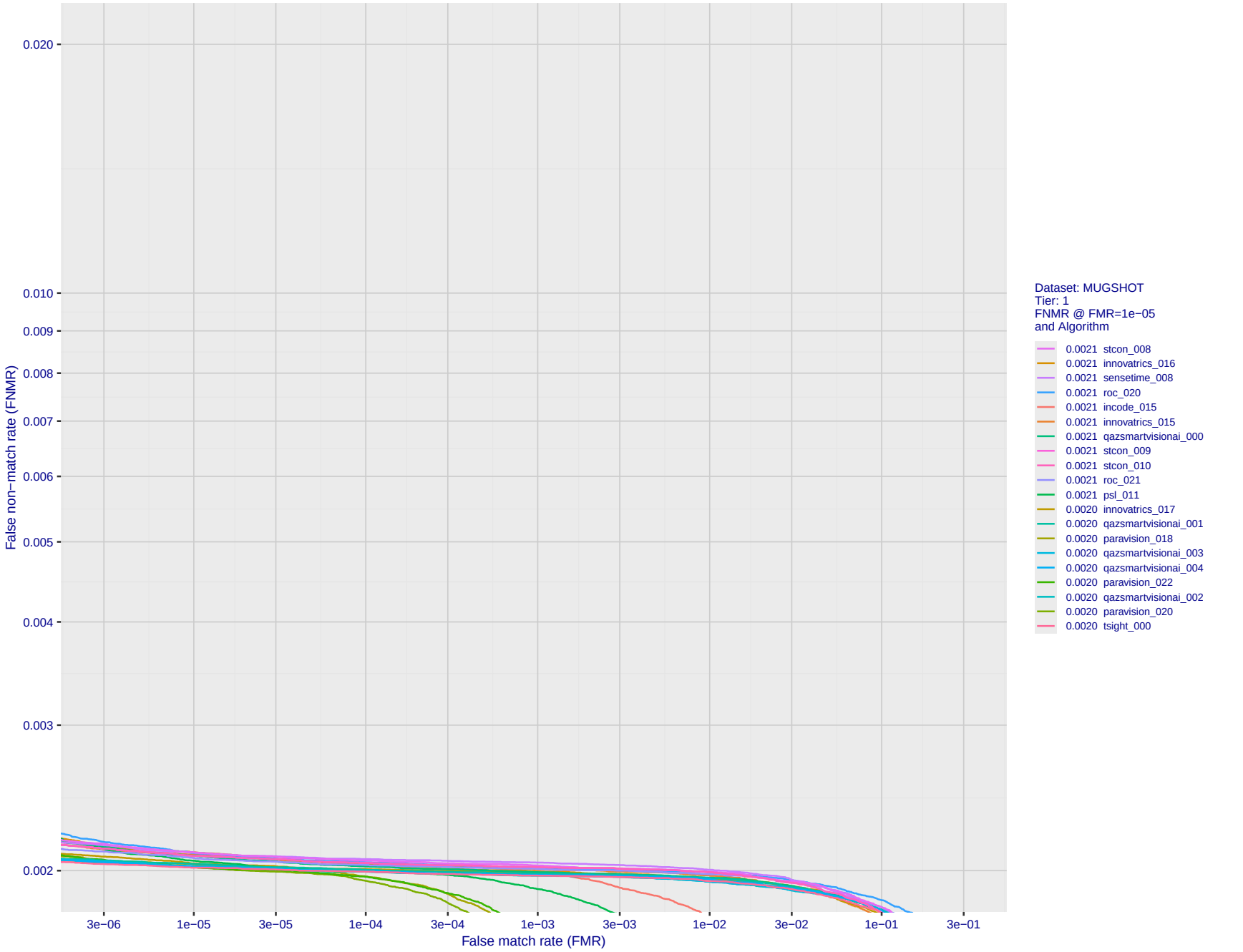

Figure 225: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

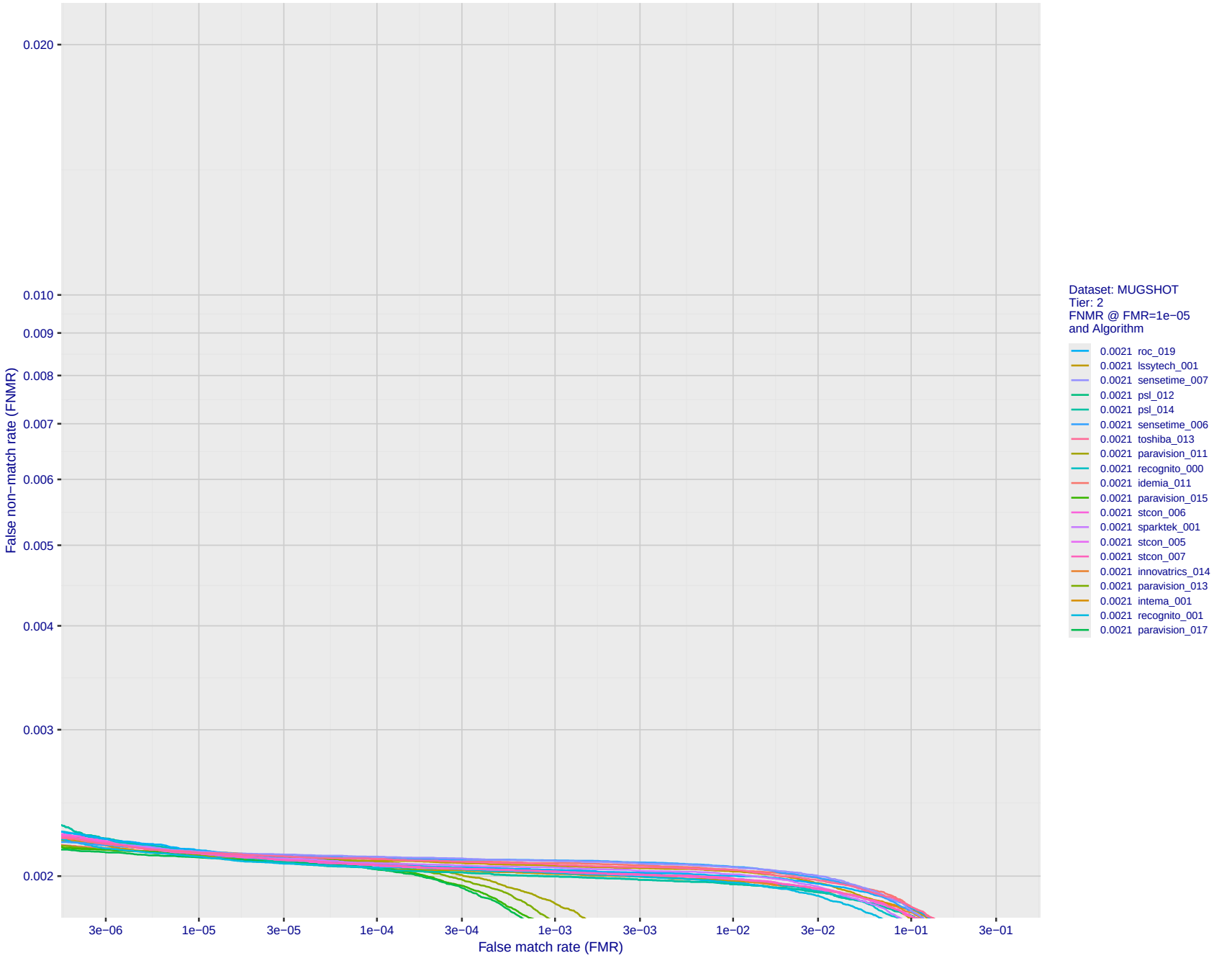

Figure 226: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

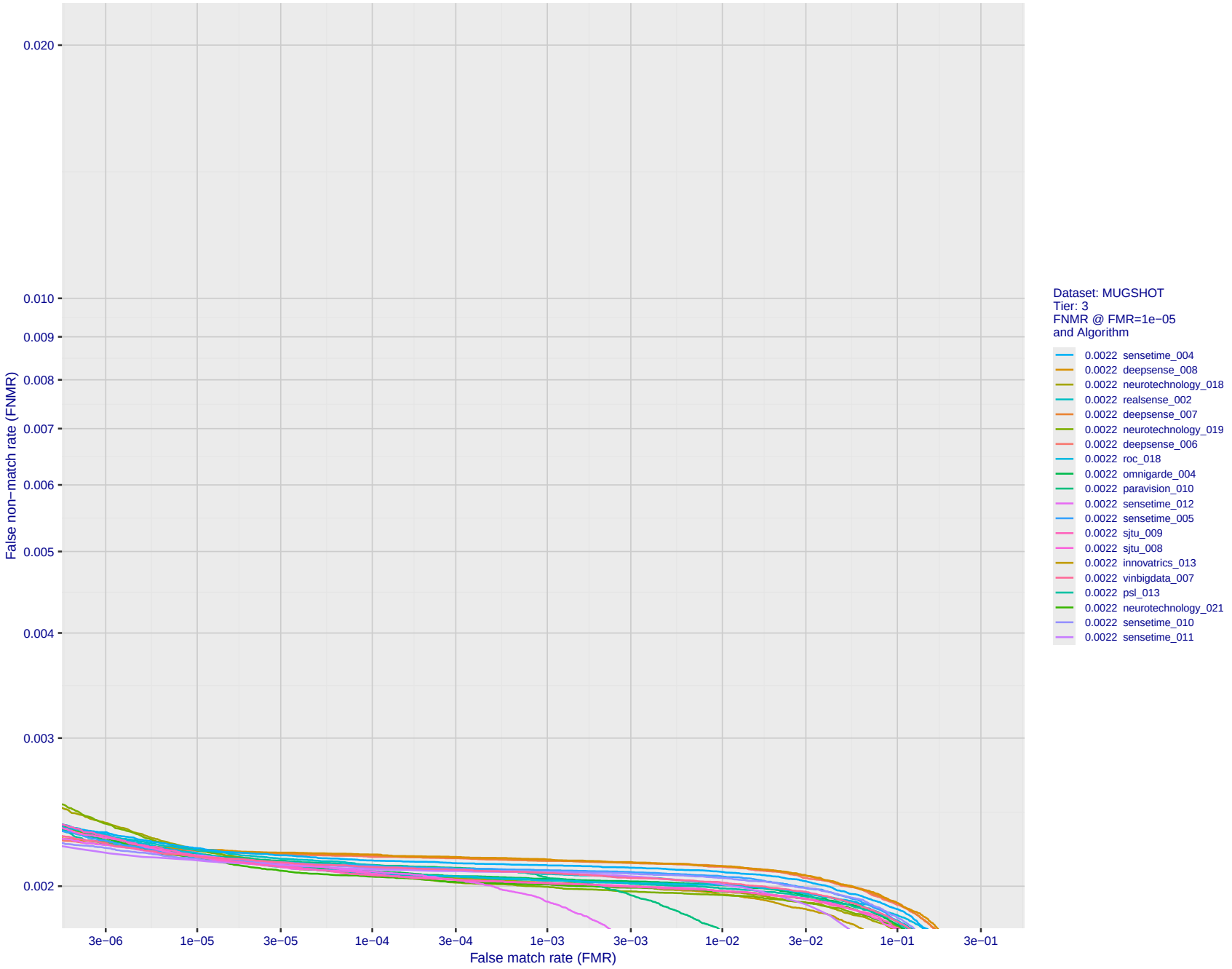

Figure 227: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

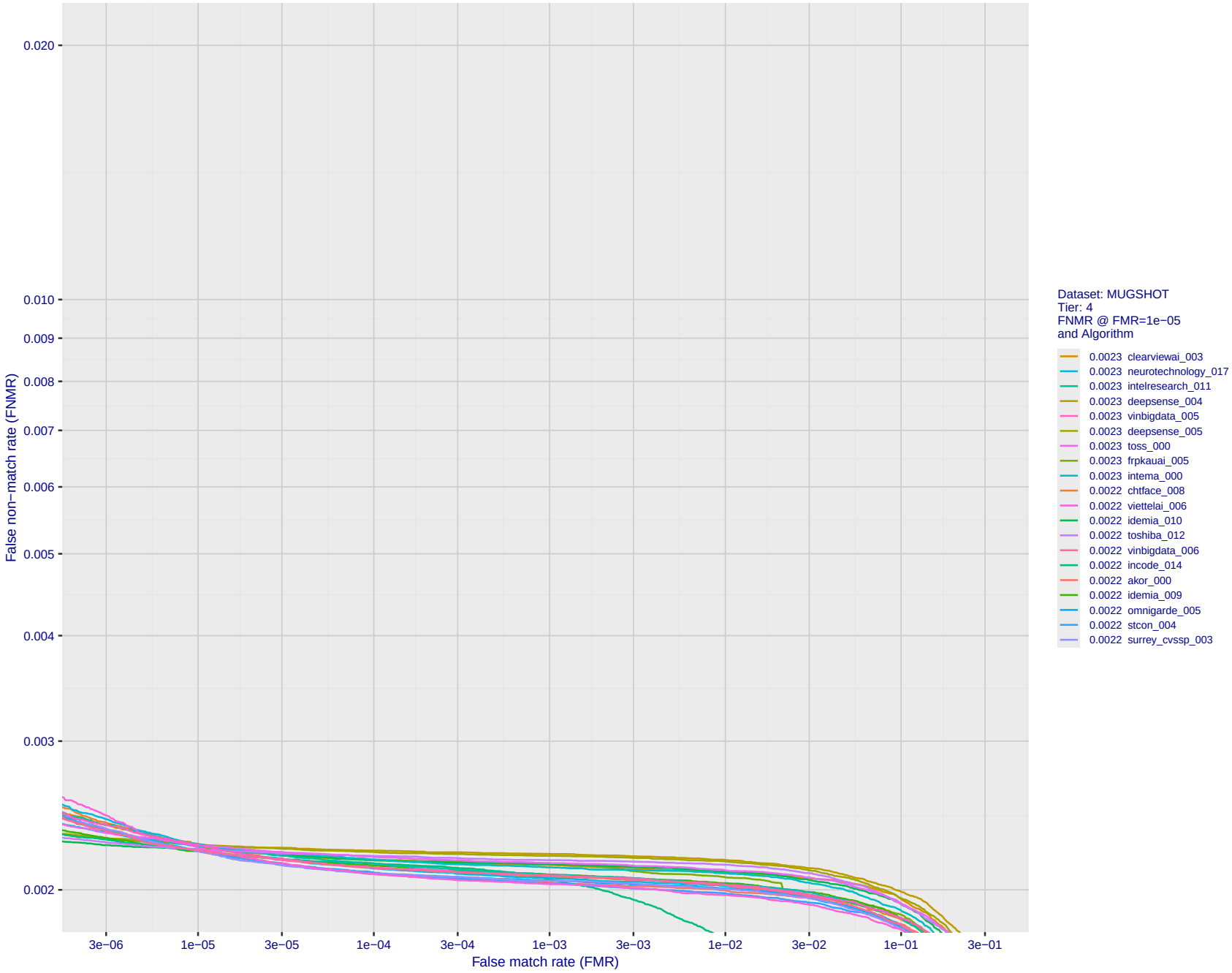

*Figure 228: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR*.*

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

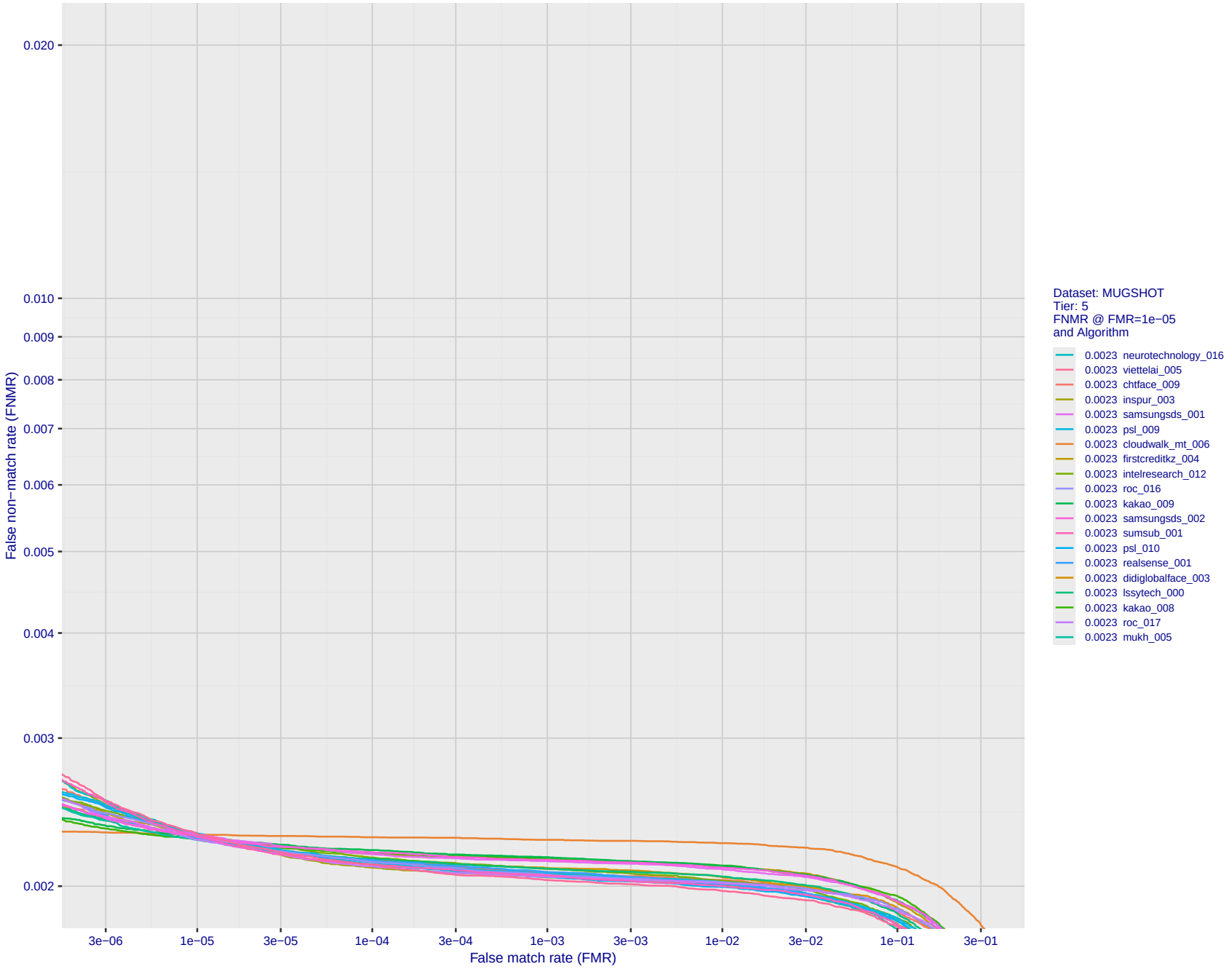

*Figure 229: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR*.*

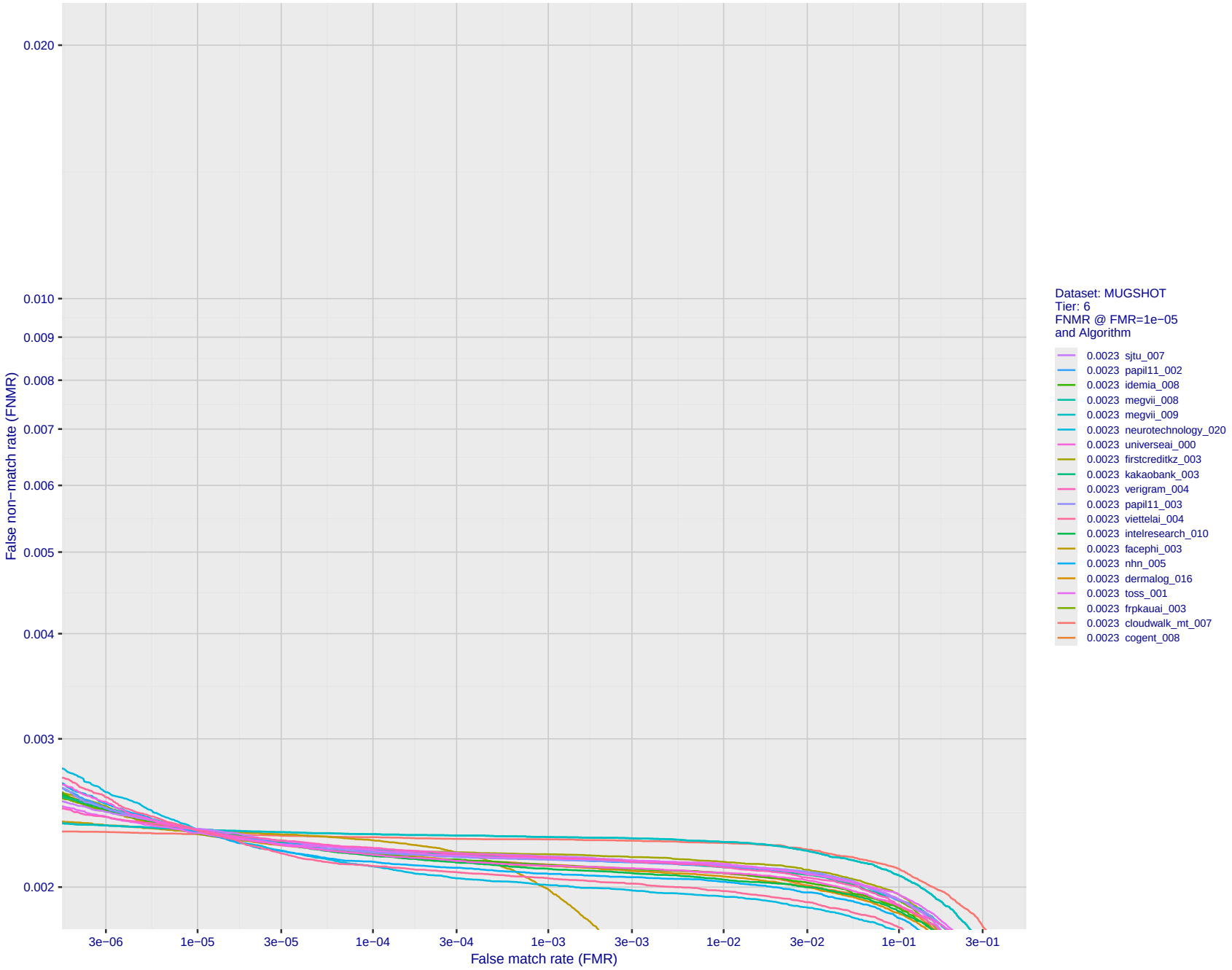

Figure 230: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

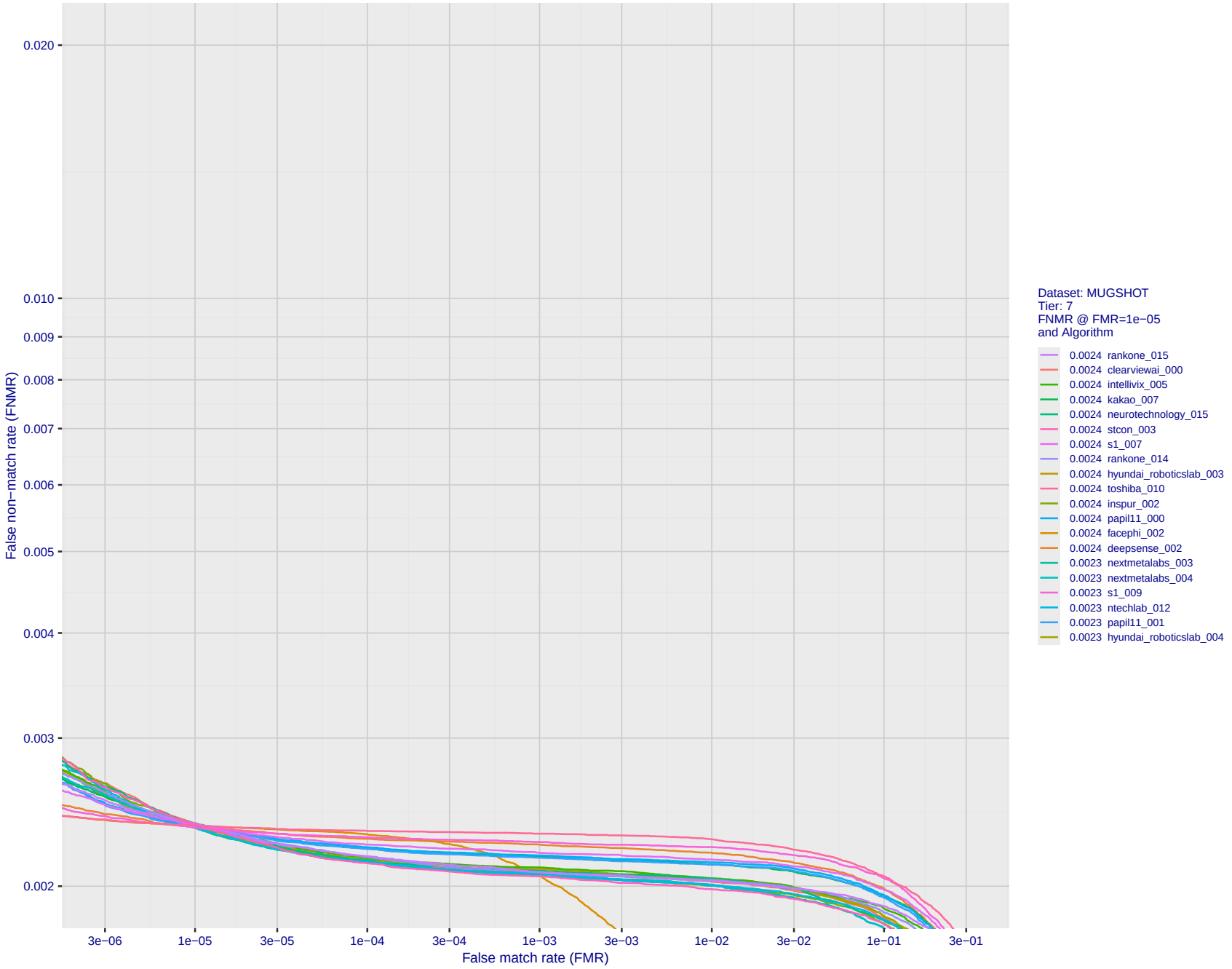

*Figure 231: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

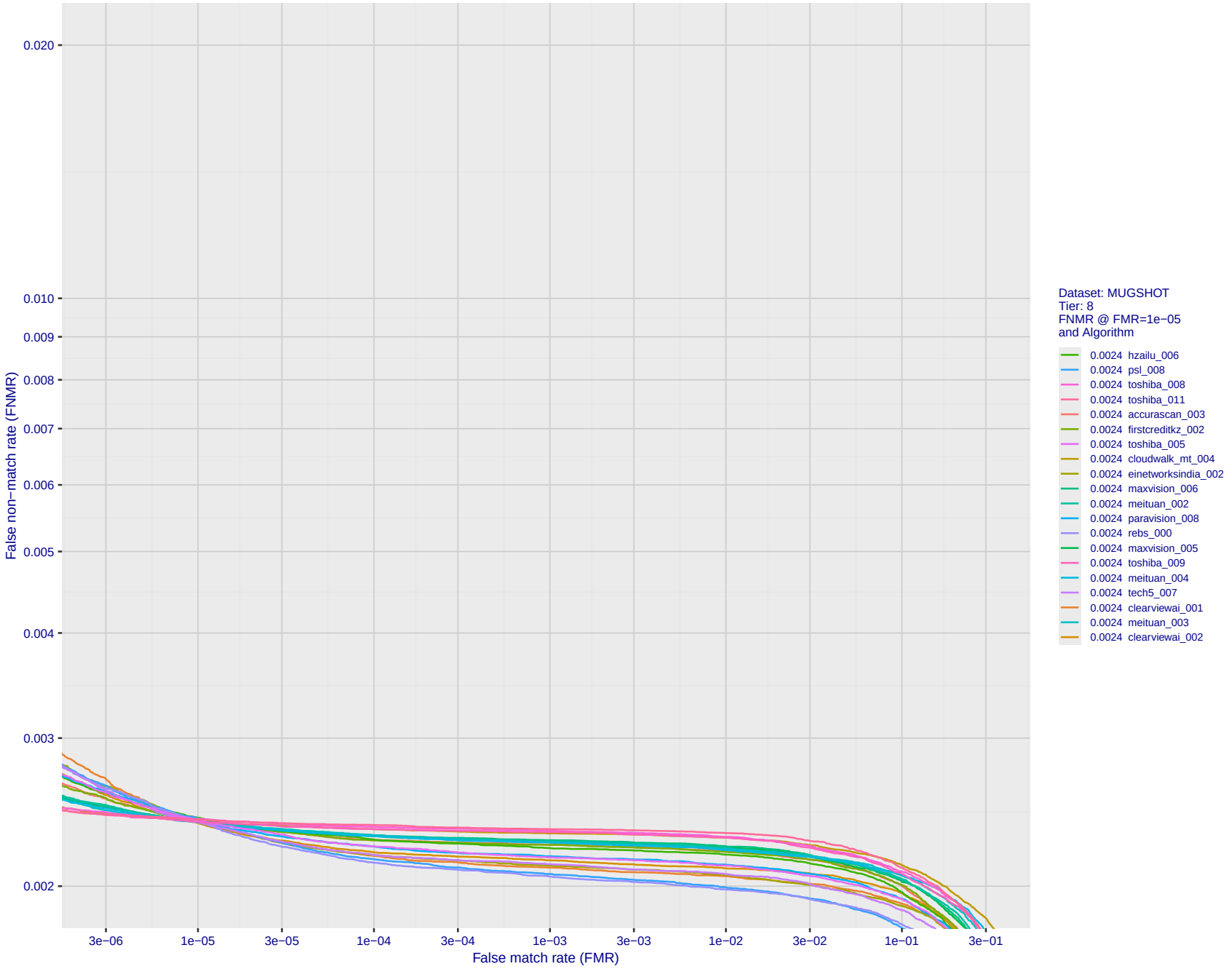

Figure 232: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

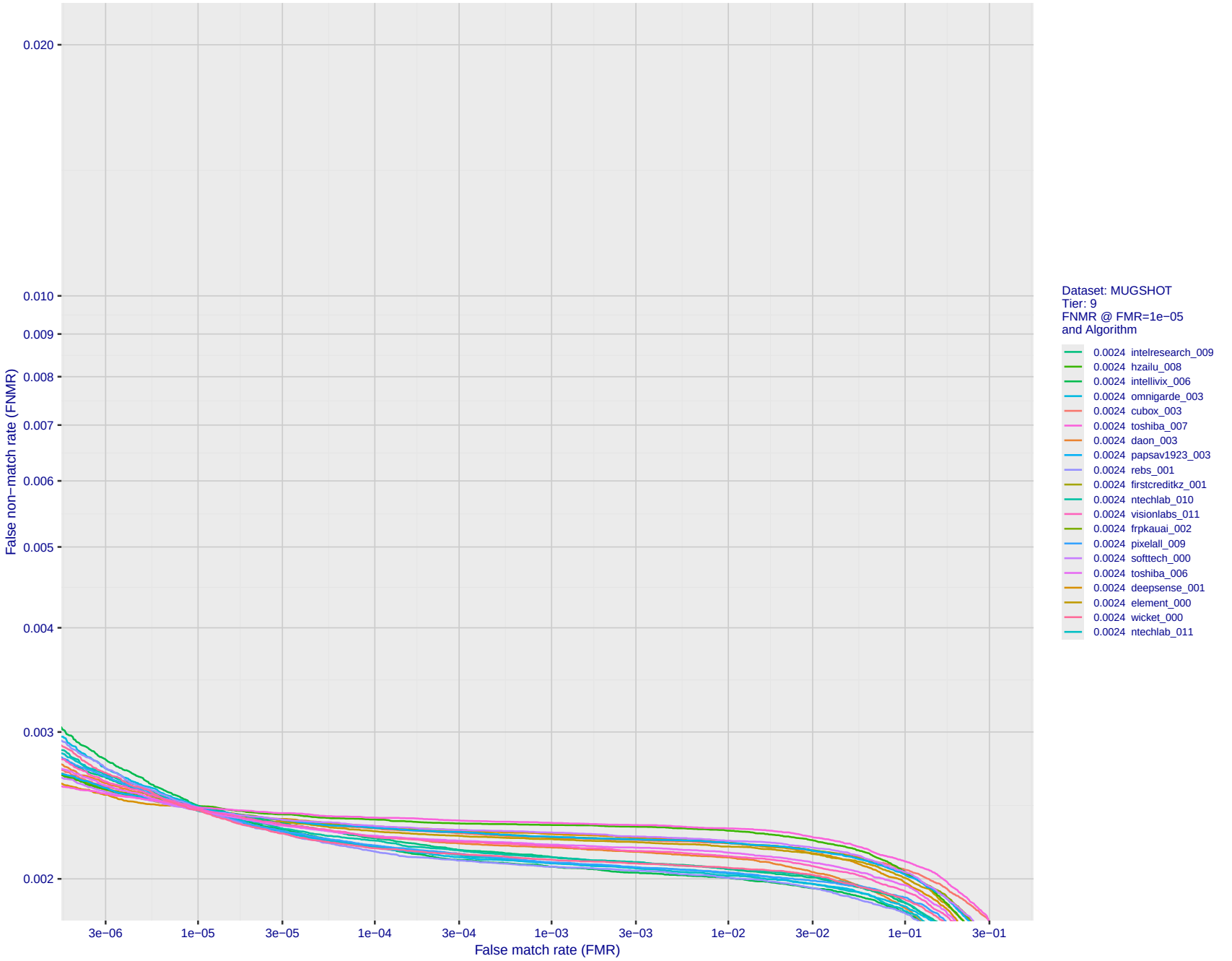

Figure 233: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

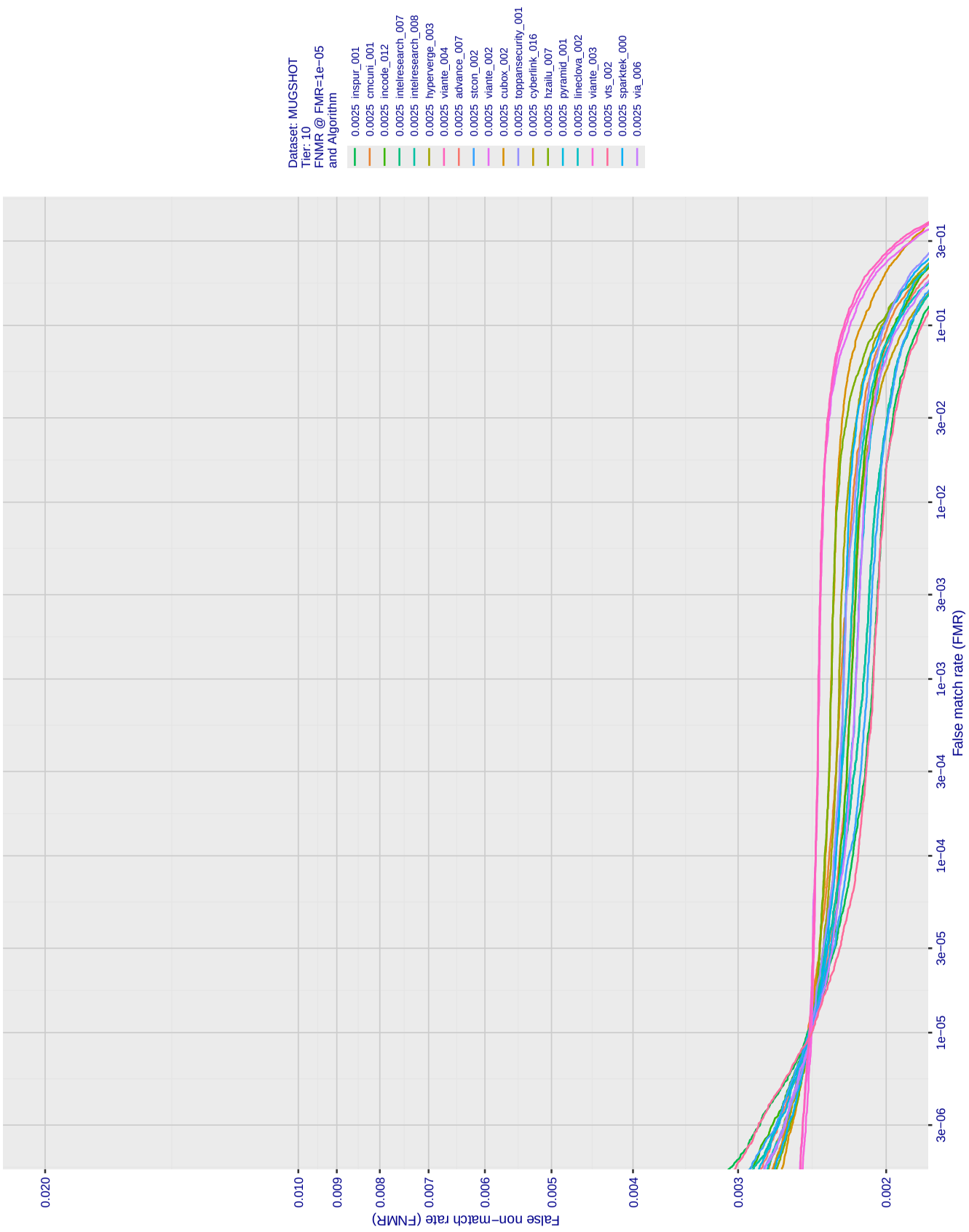

Figure 234: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

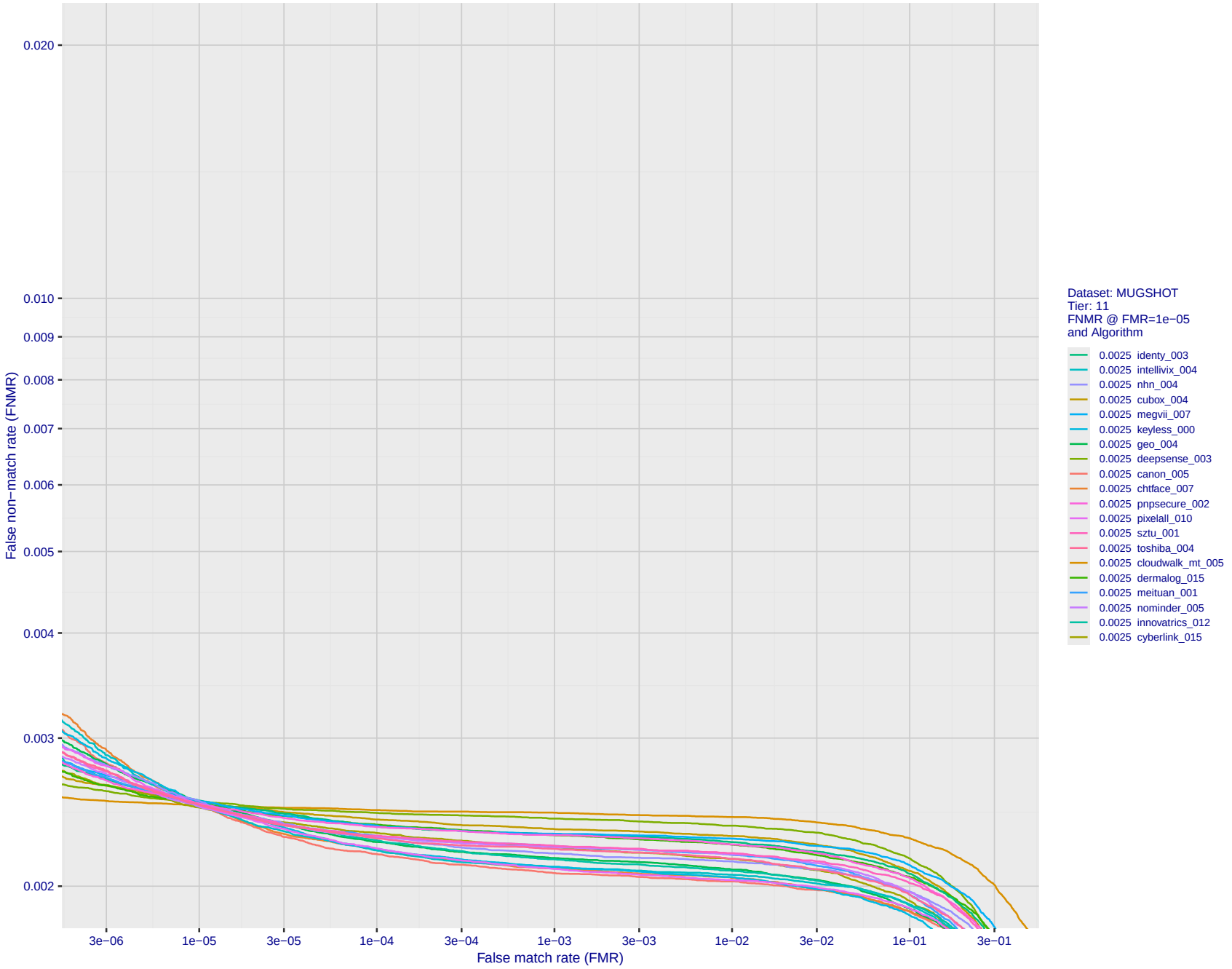

Figure 235: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

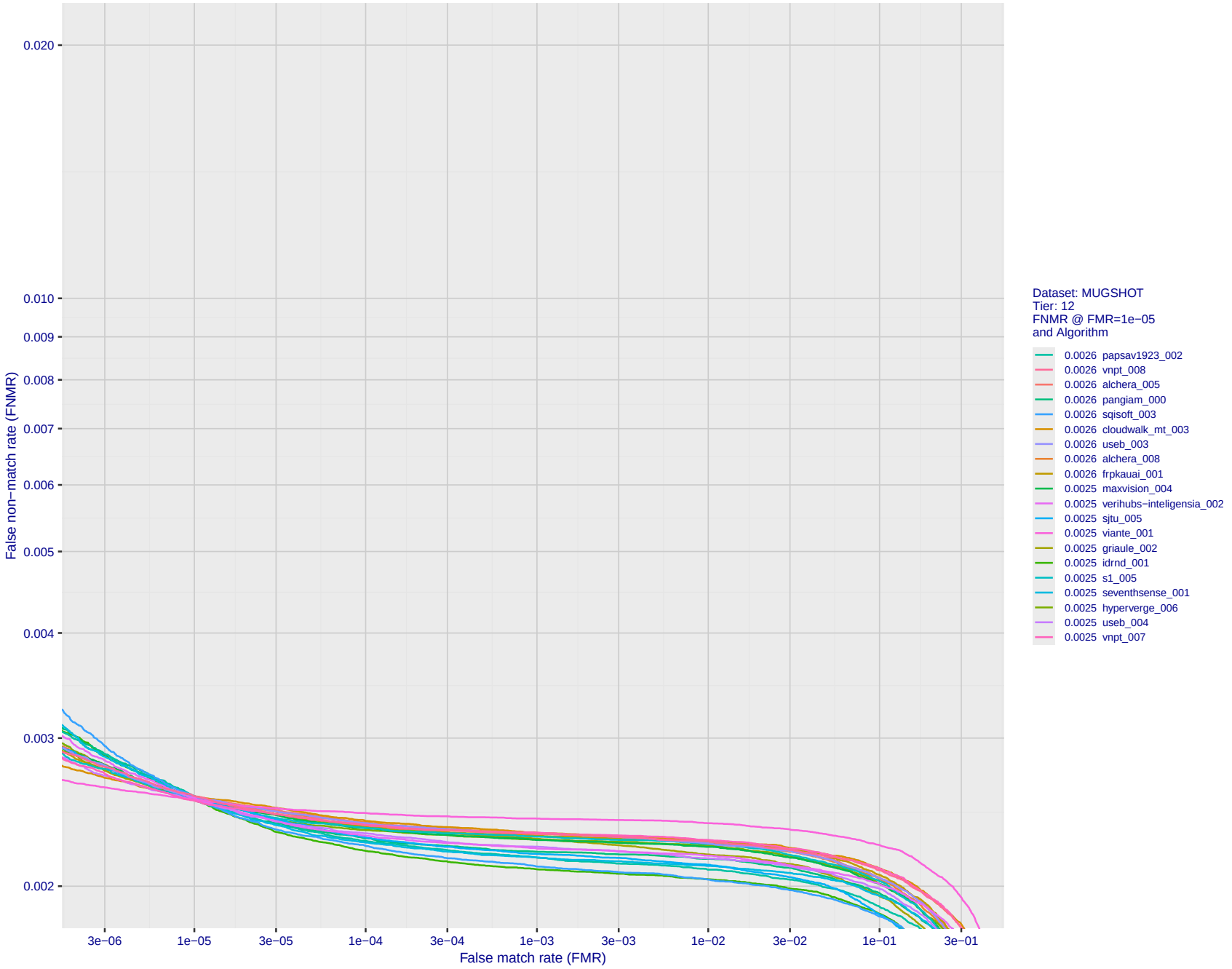

Figure 236: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

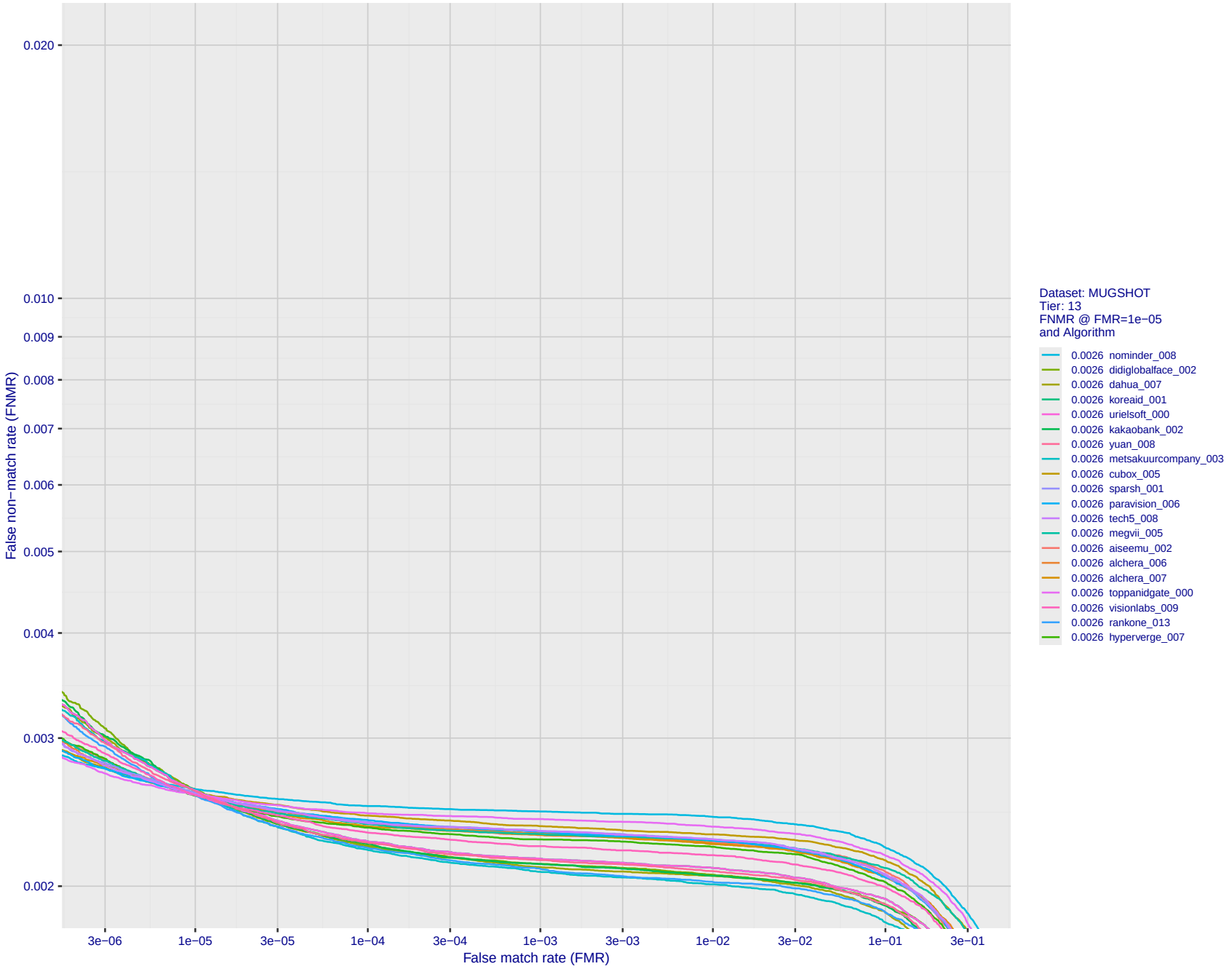

Figure 237: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

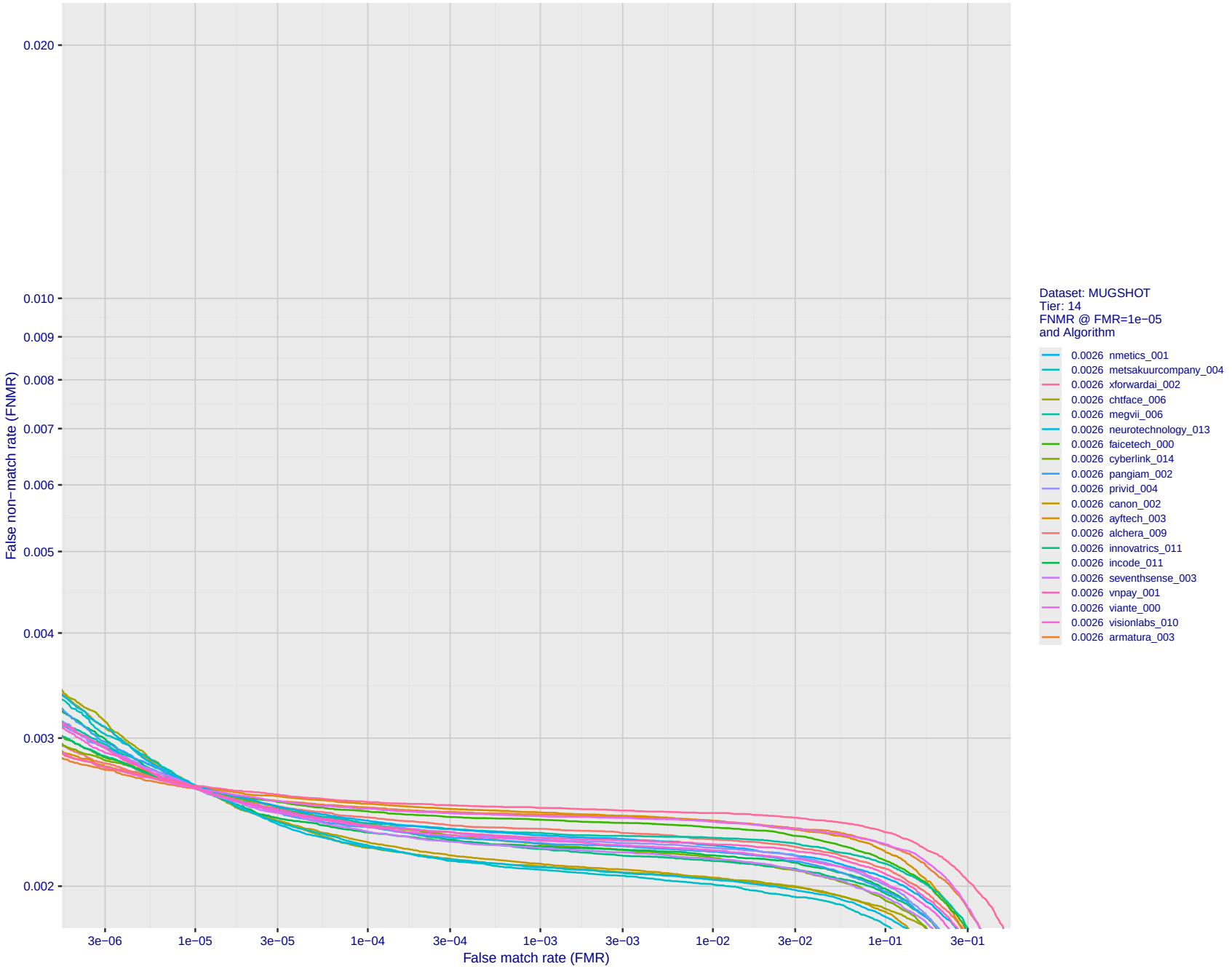

Figure 238: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

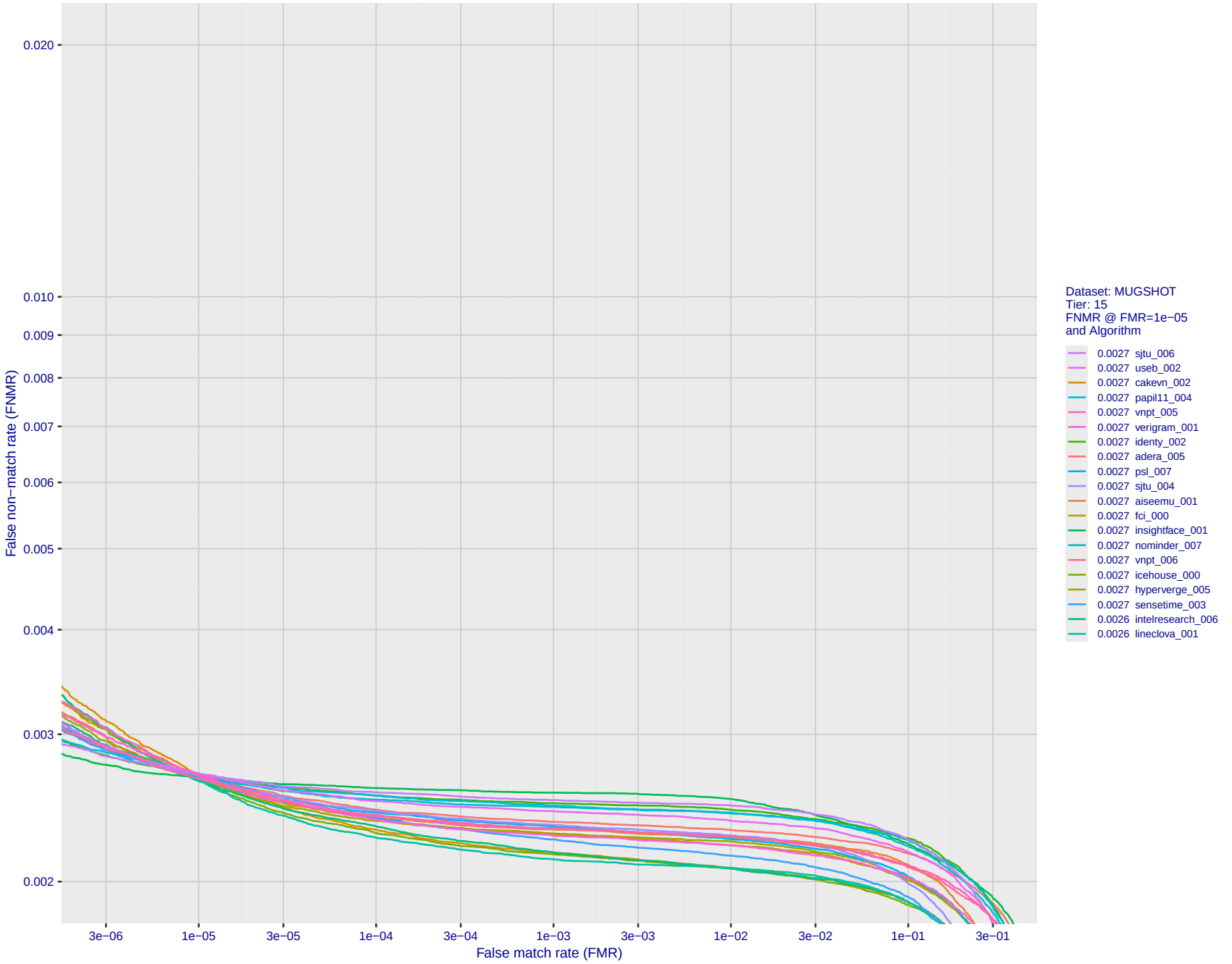

*Figure 239: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR*.*

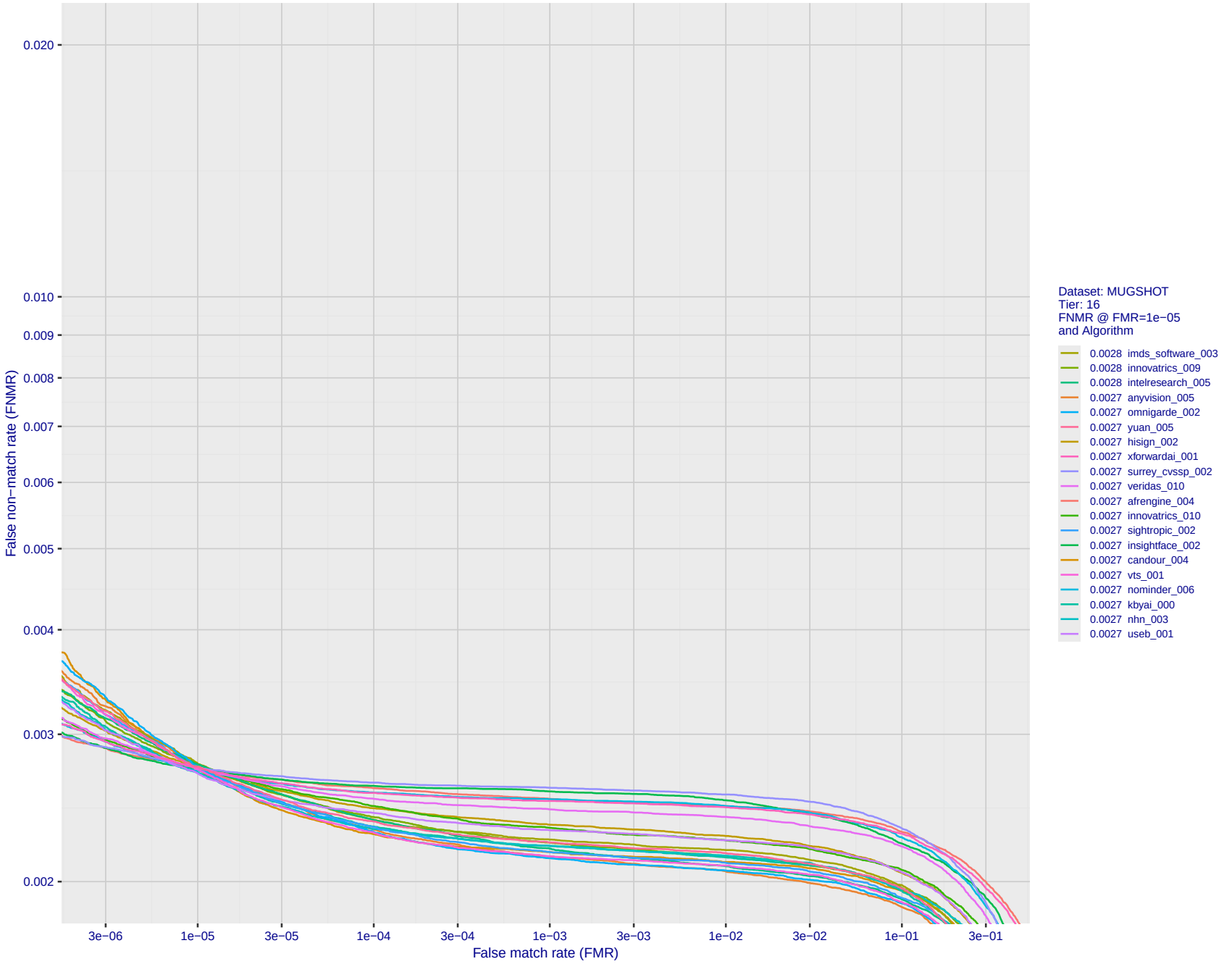

Figure 240: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

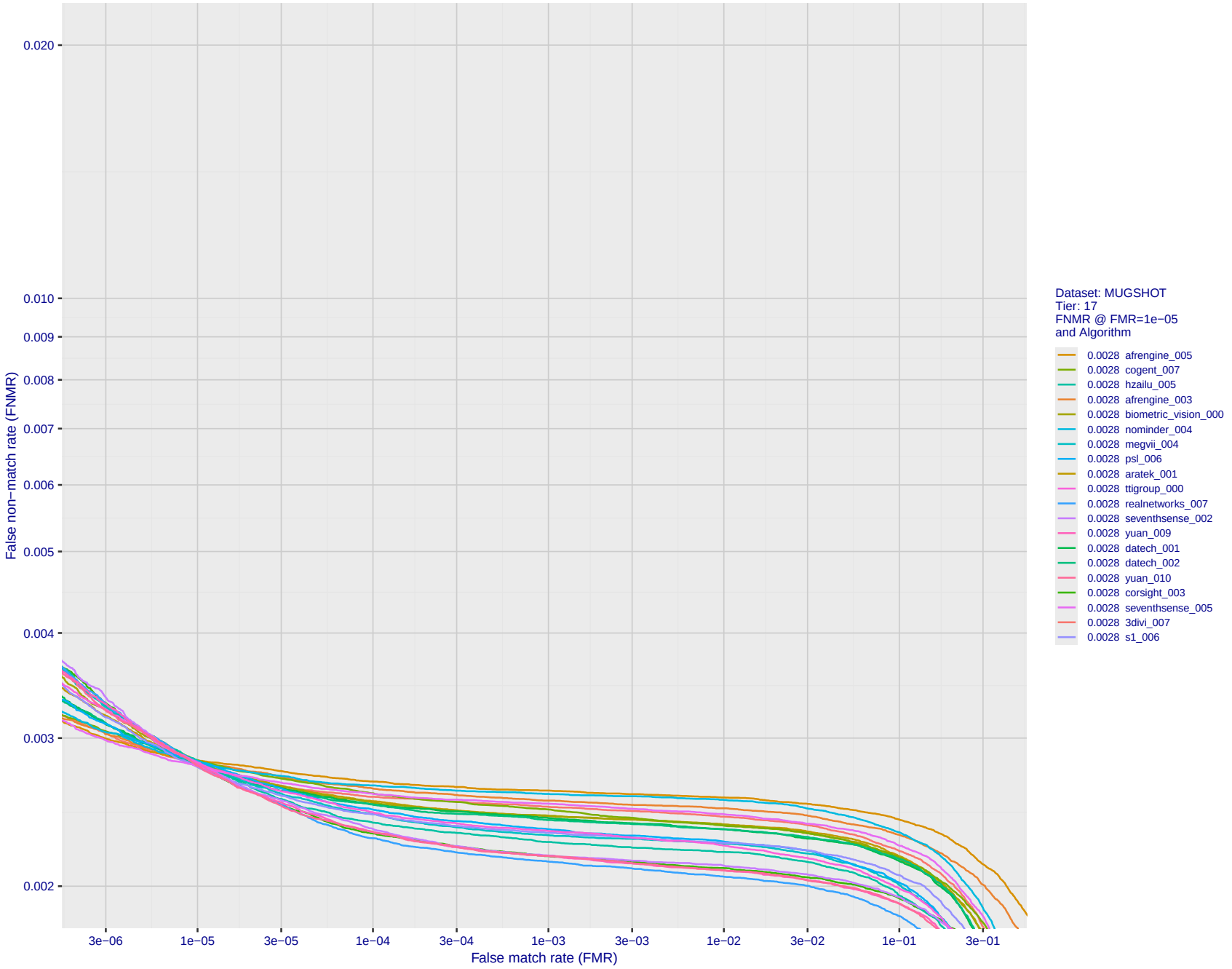

Figure 241: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

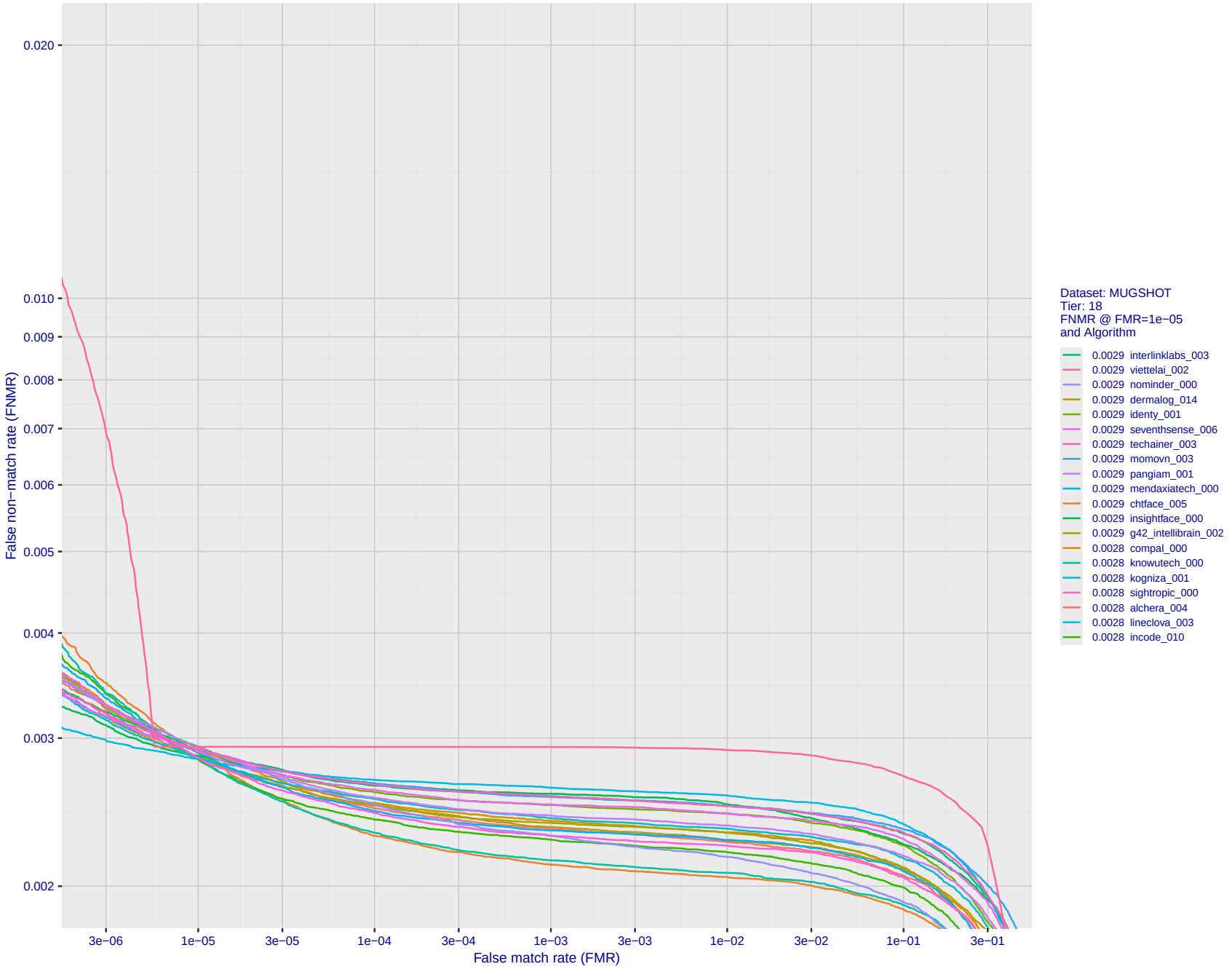

Figure 242: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

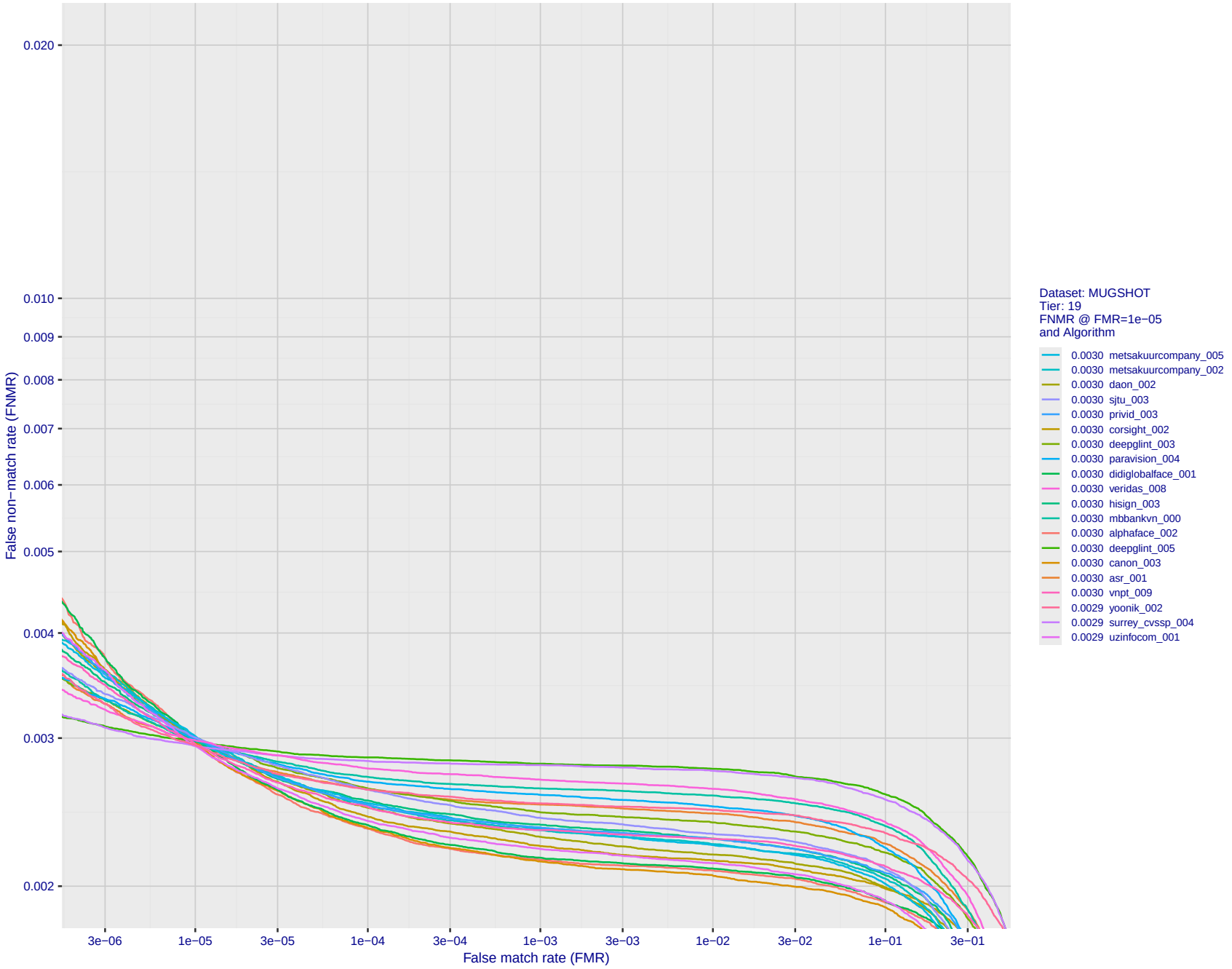

Figure 243: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

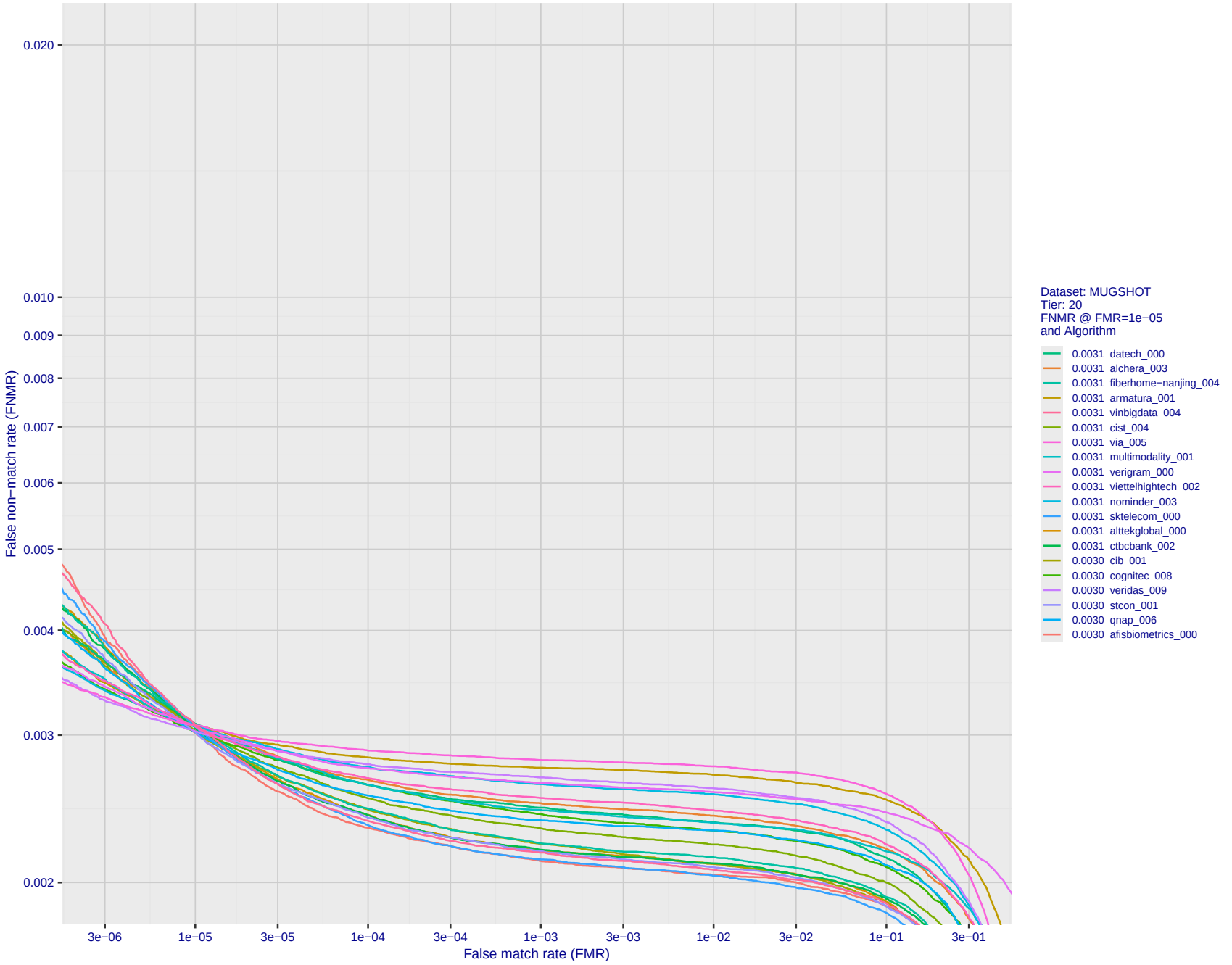

*Figure 244: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR*.*

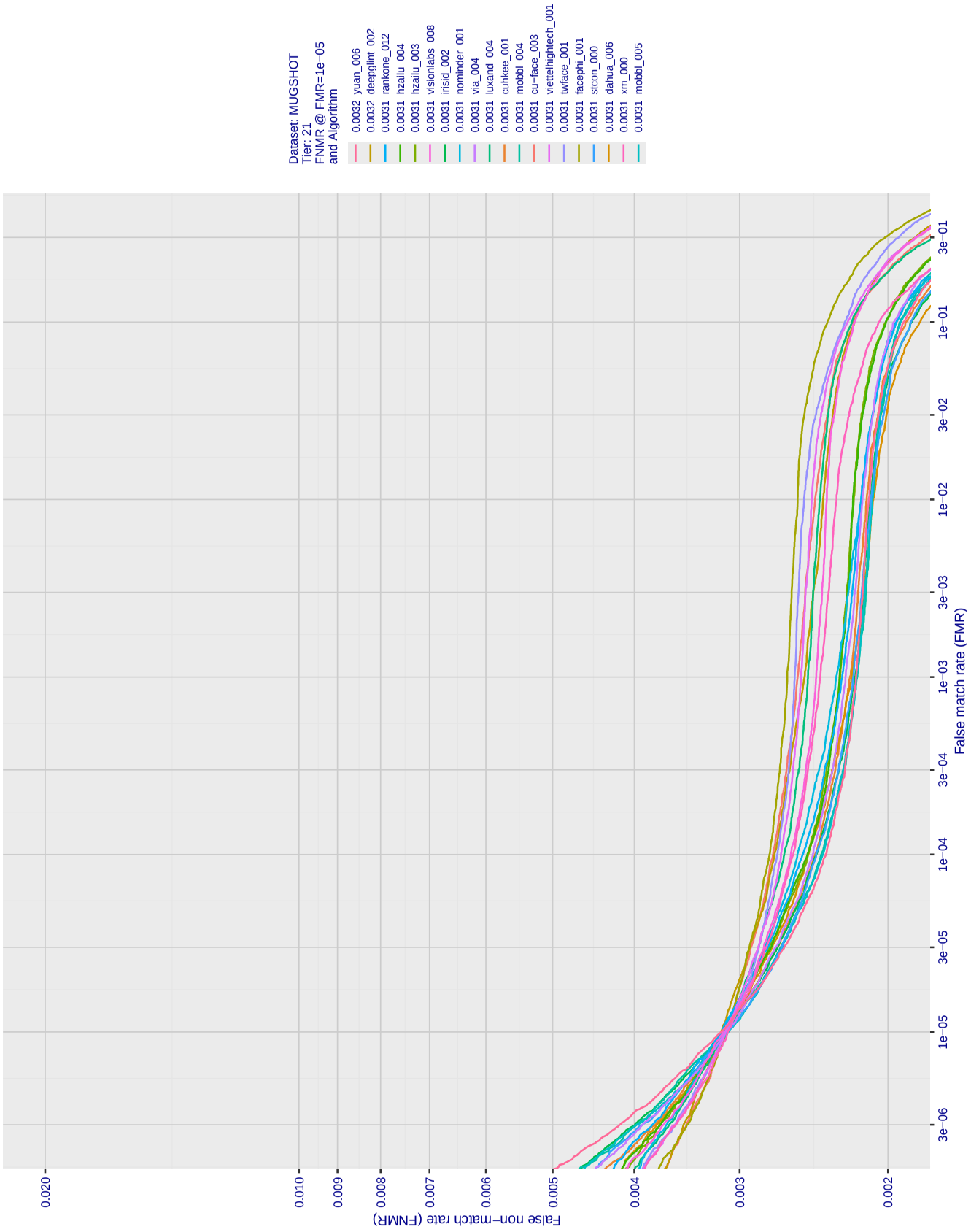

Figure 245: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

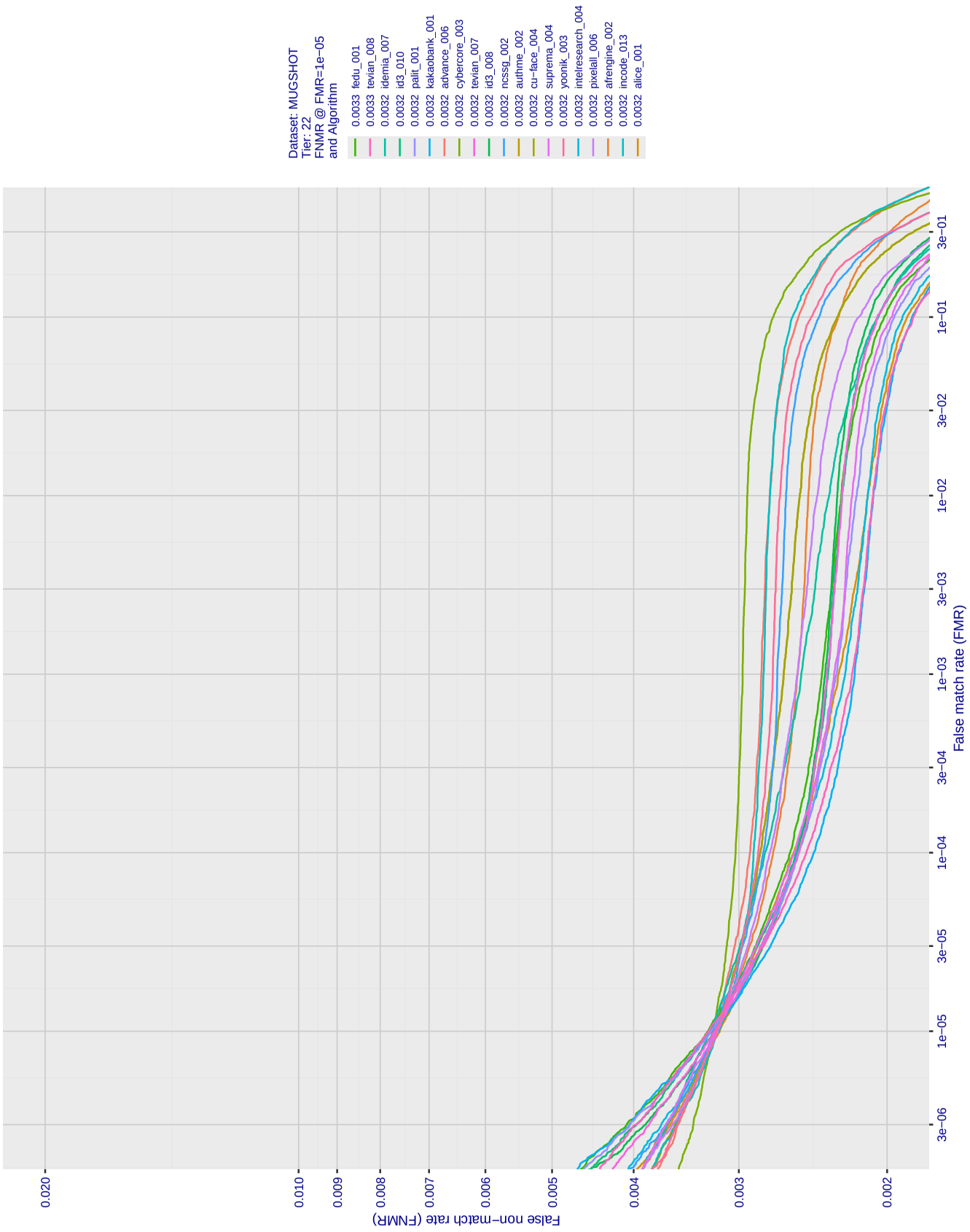

Figure 246: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

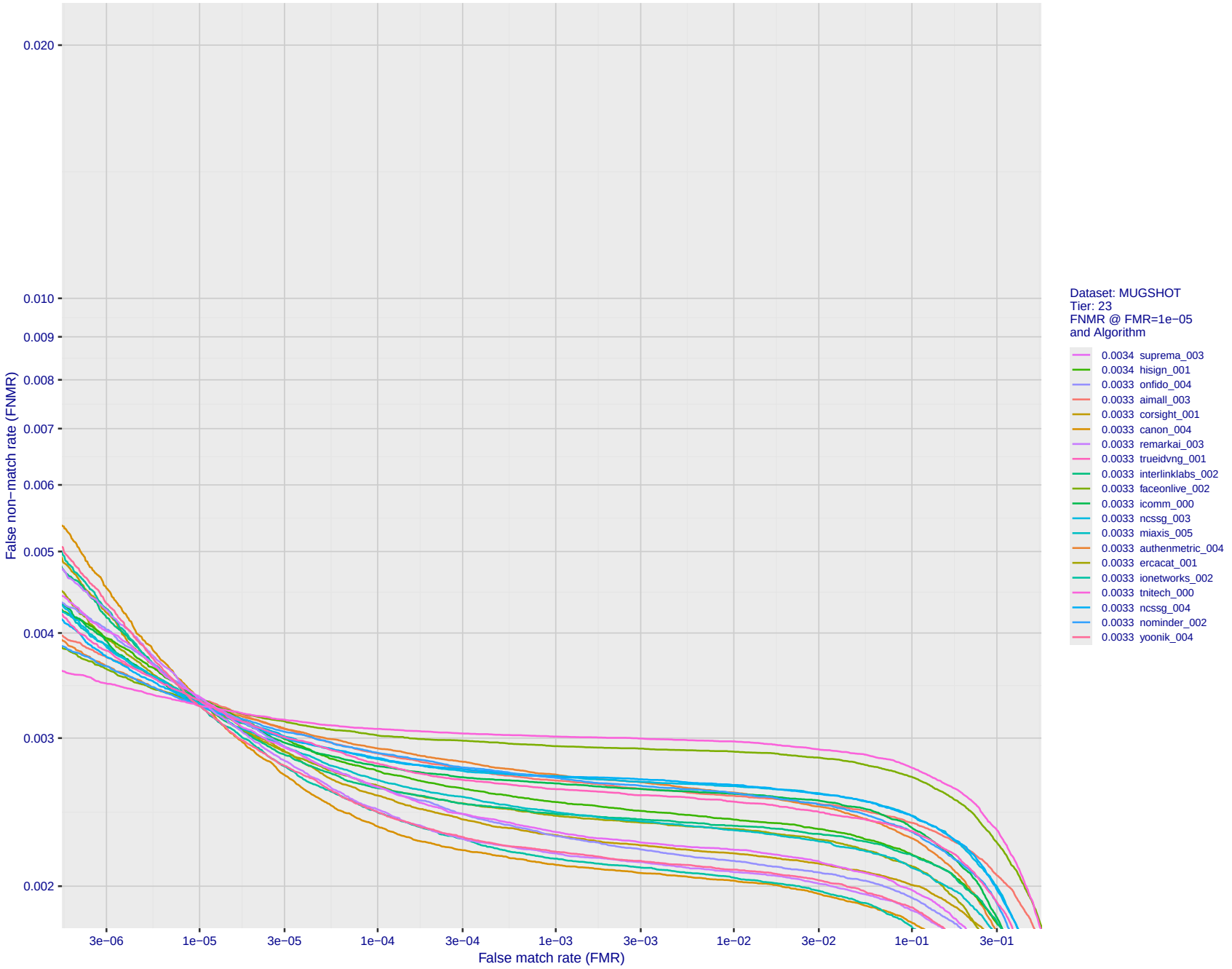

Figure 247: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

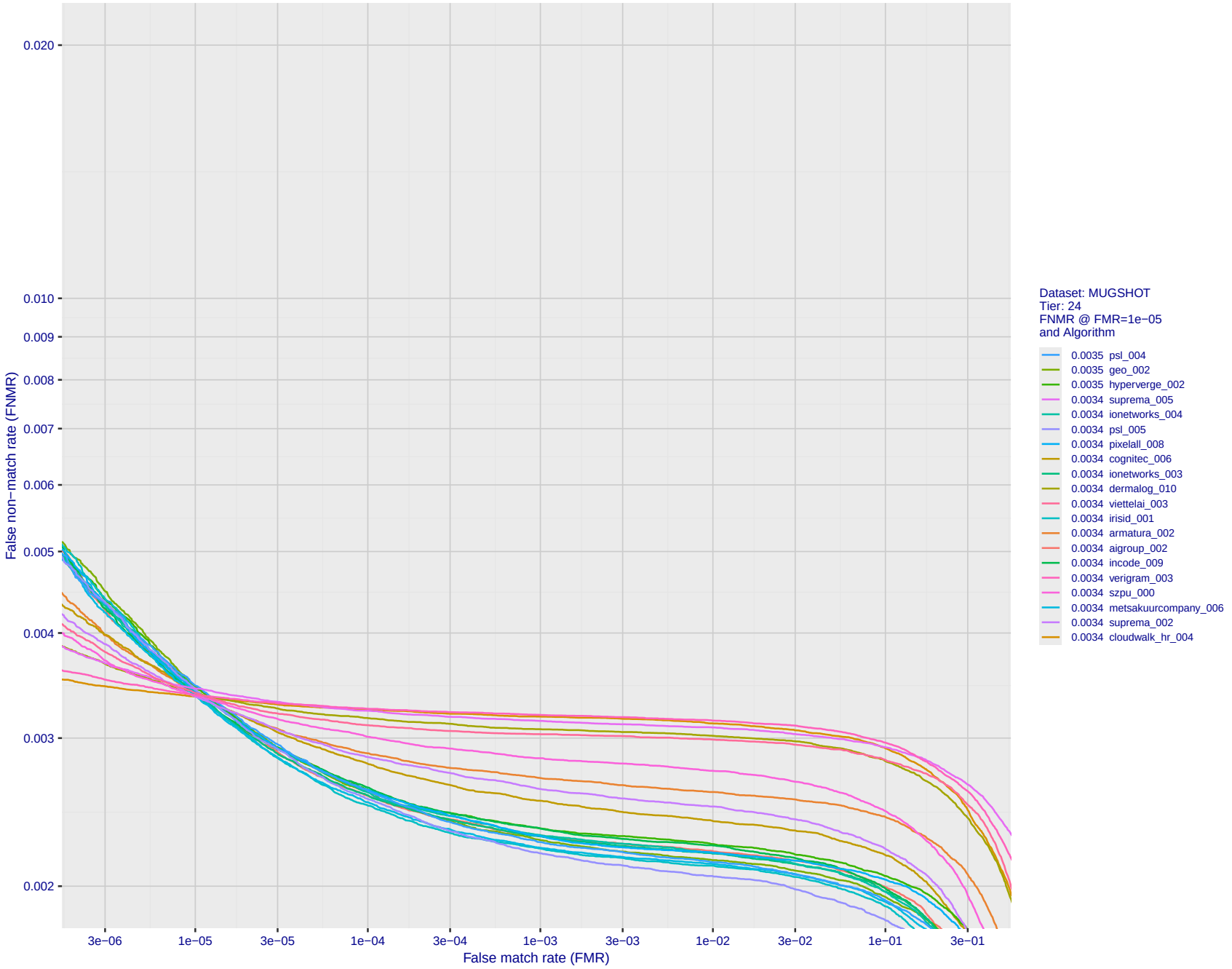

Figure 248: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

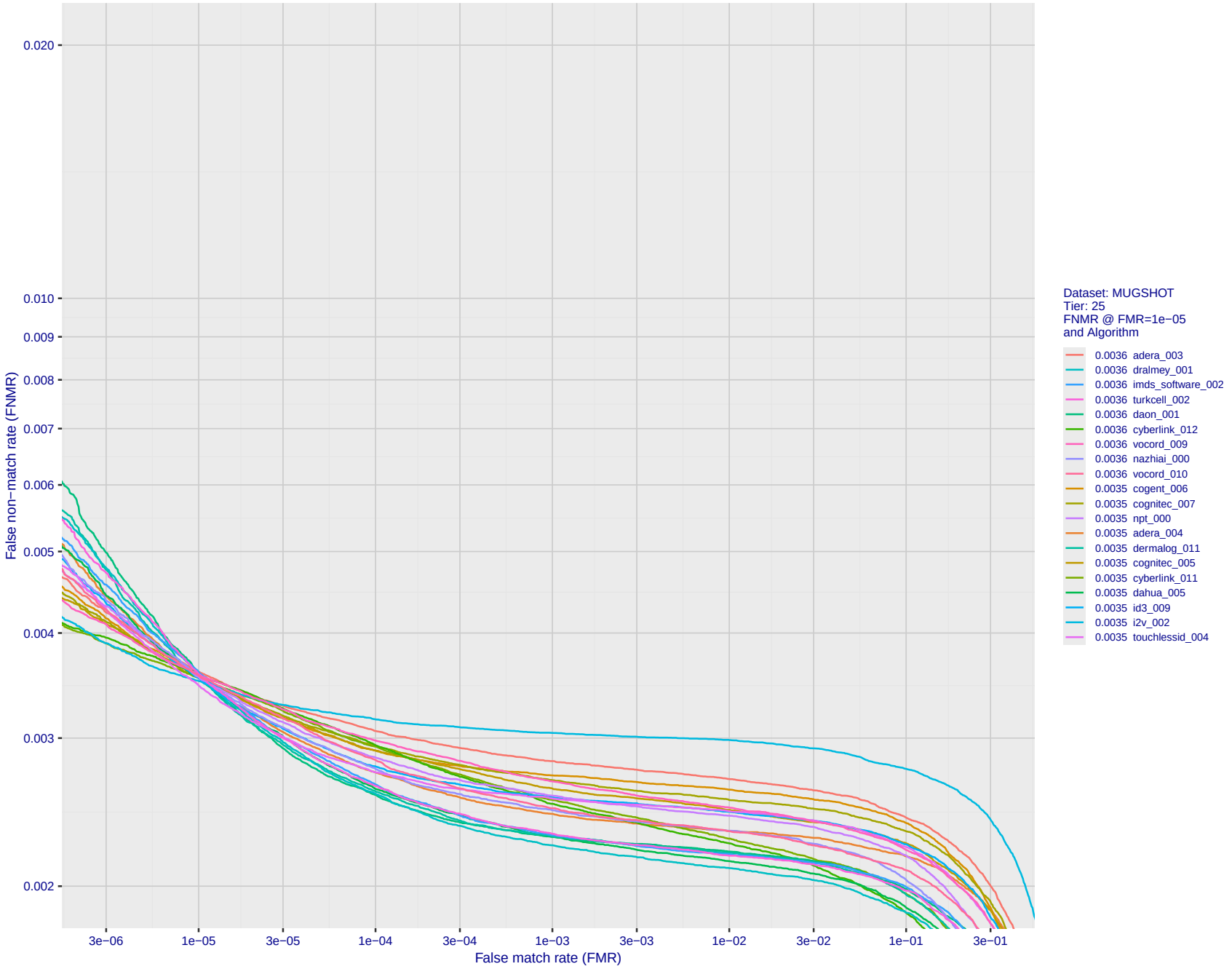

Figure 249: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

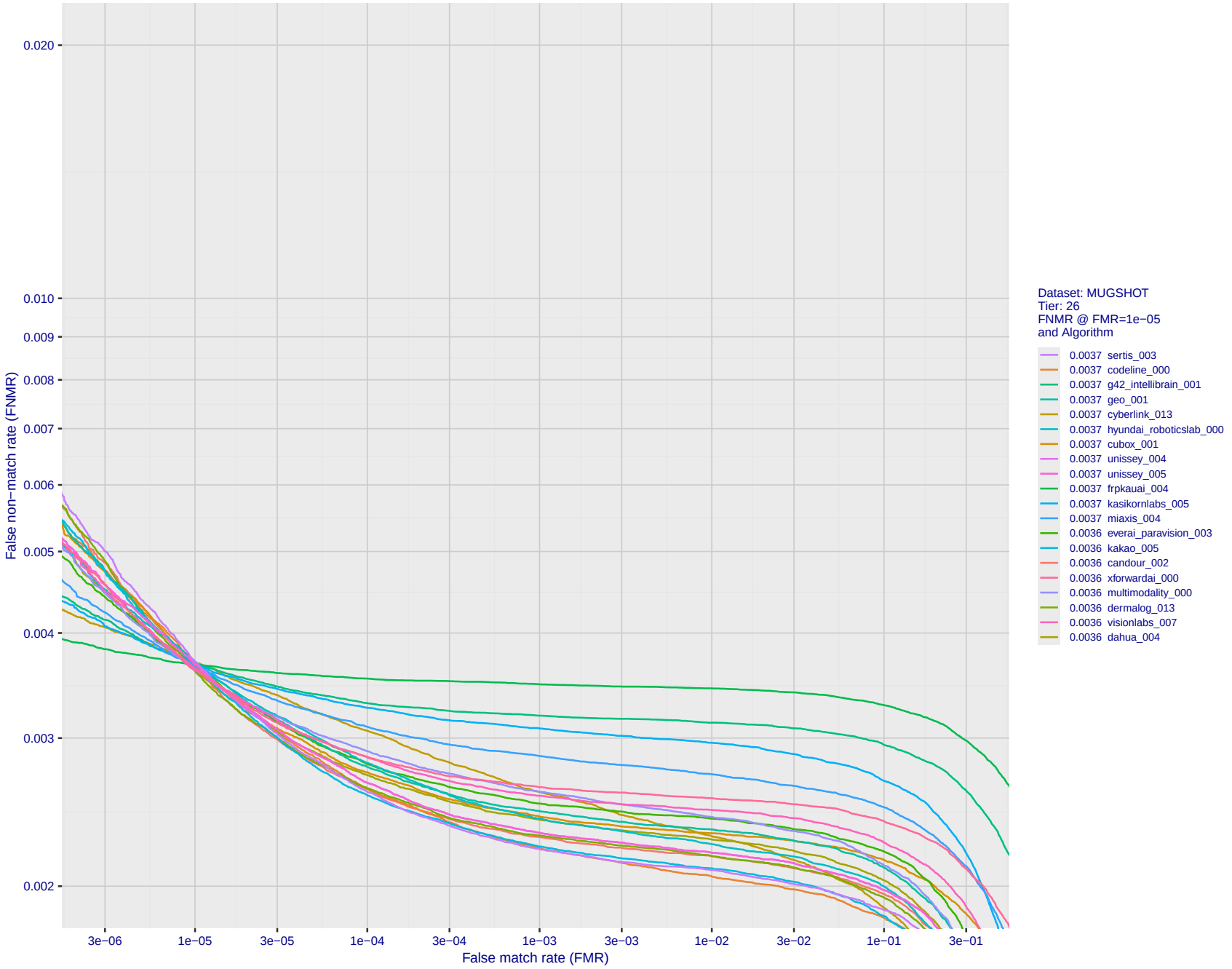

Figure 250: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

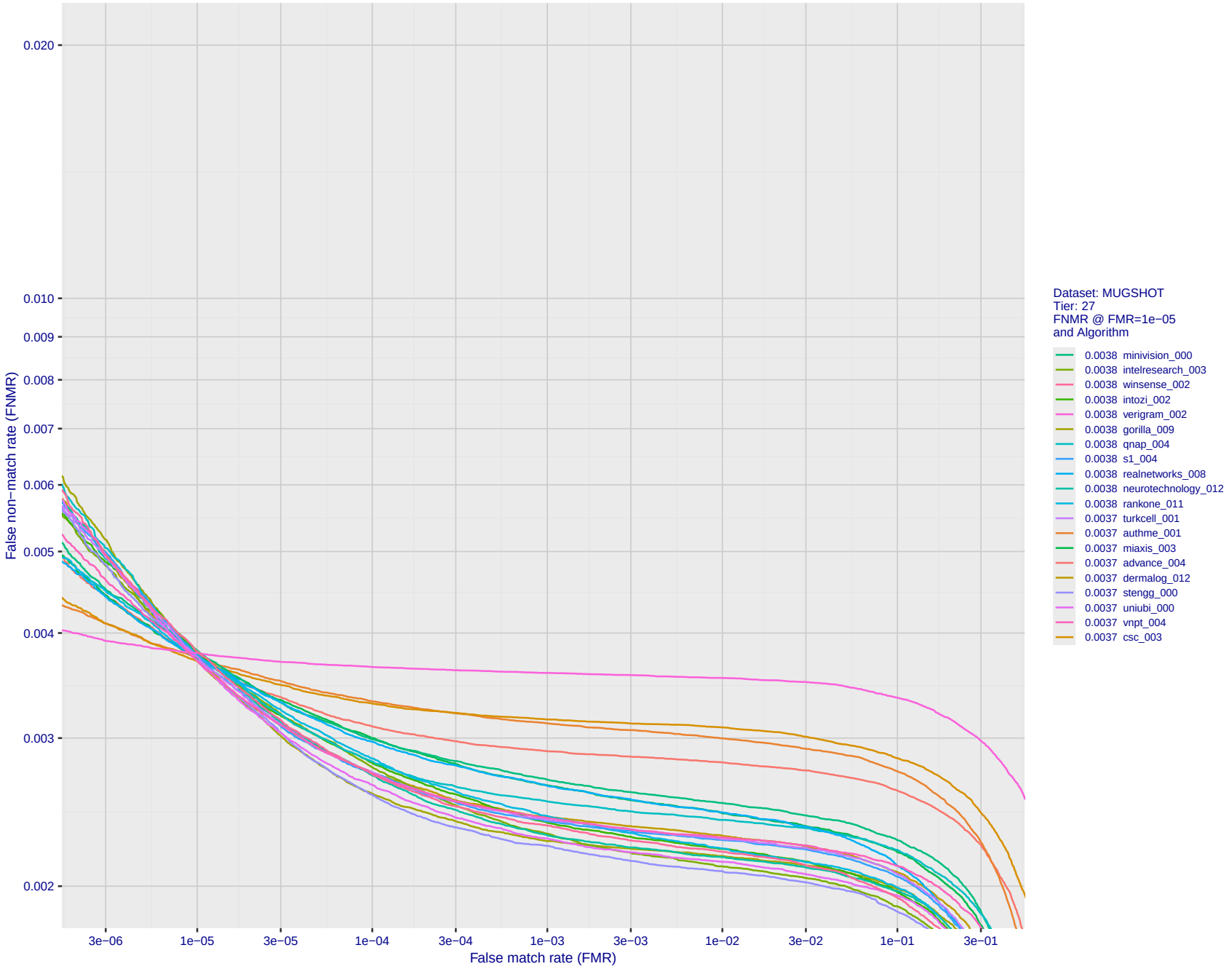

Figure 251: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

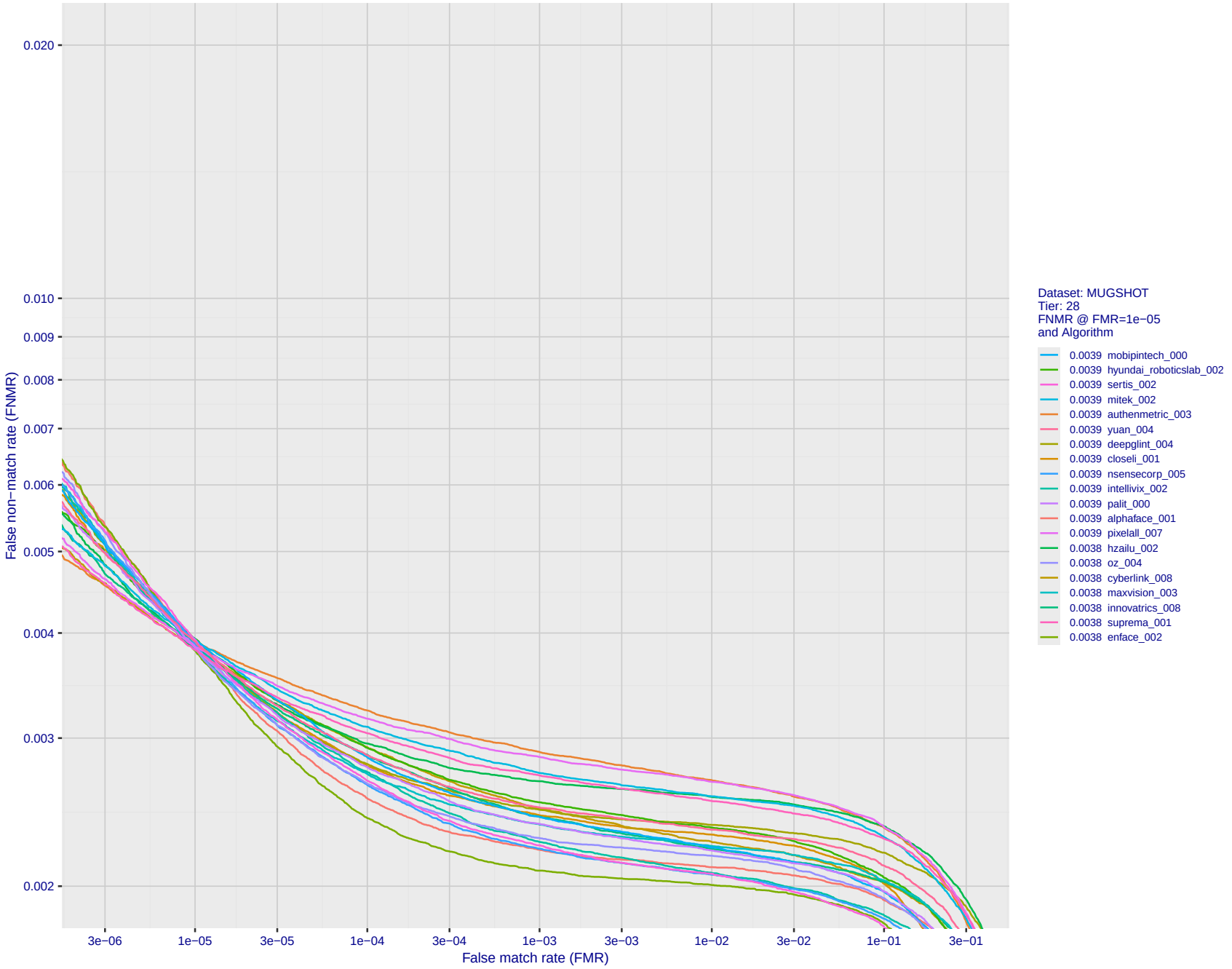

Figure 252: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

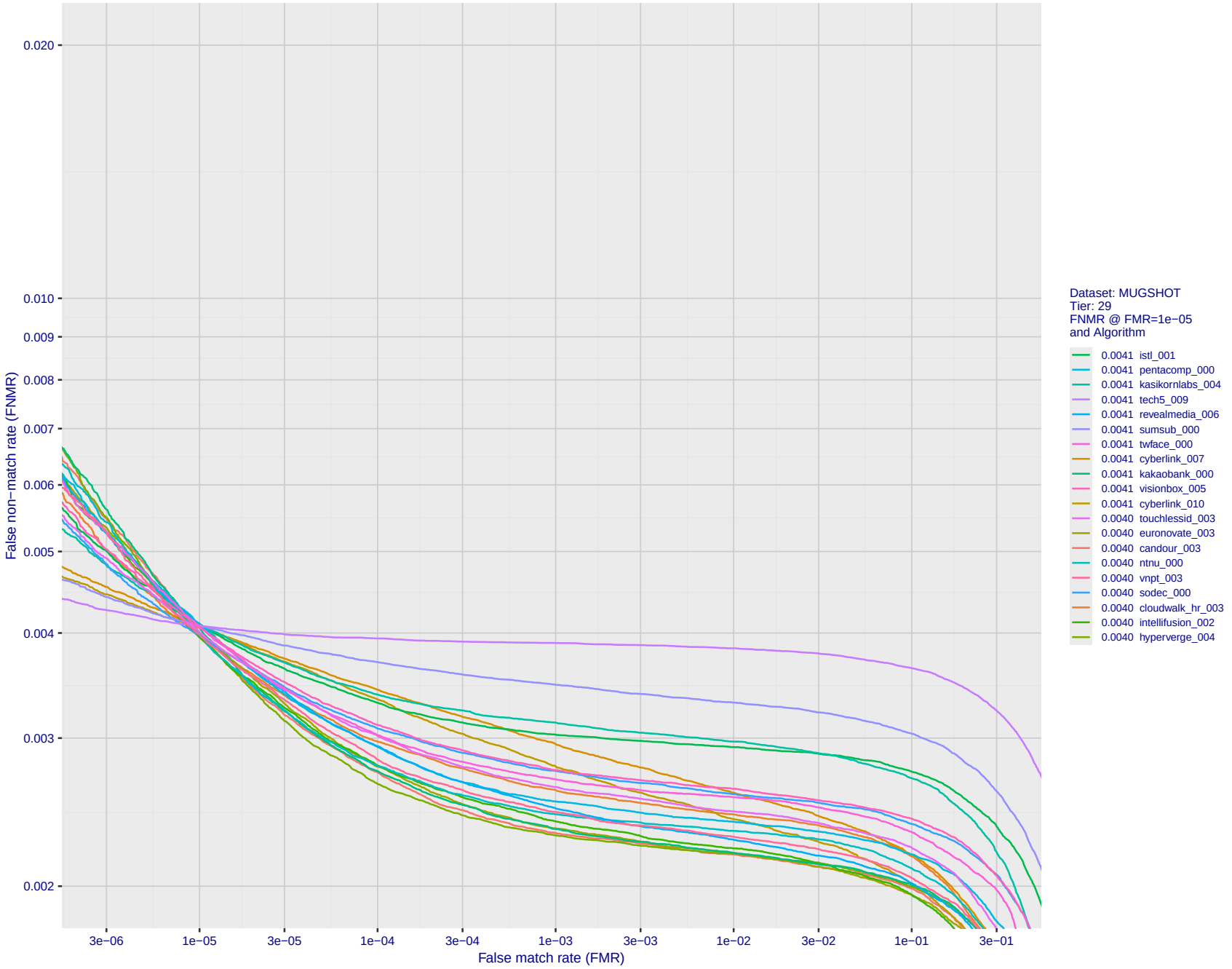

Figure 253: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

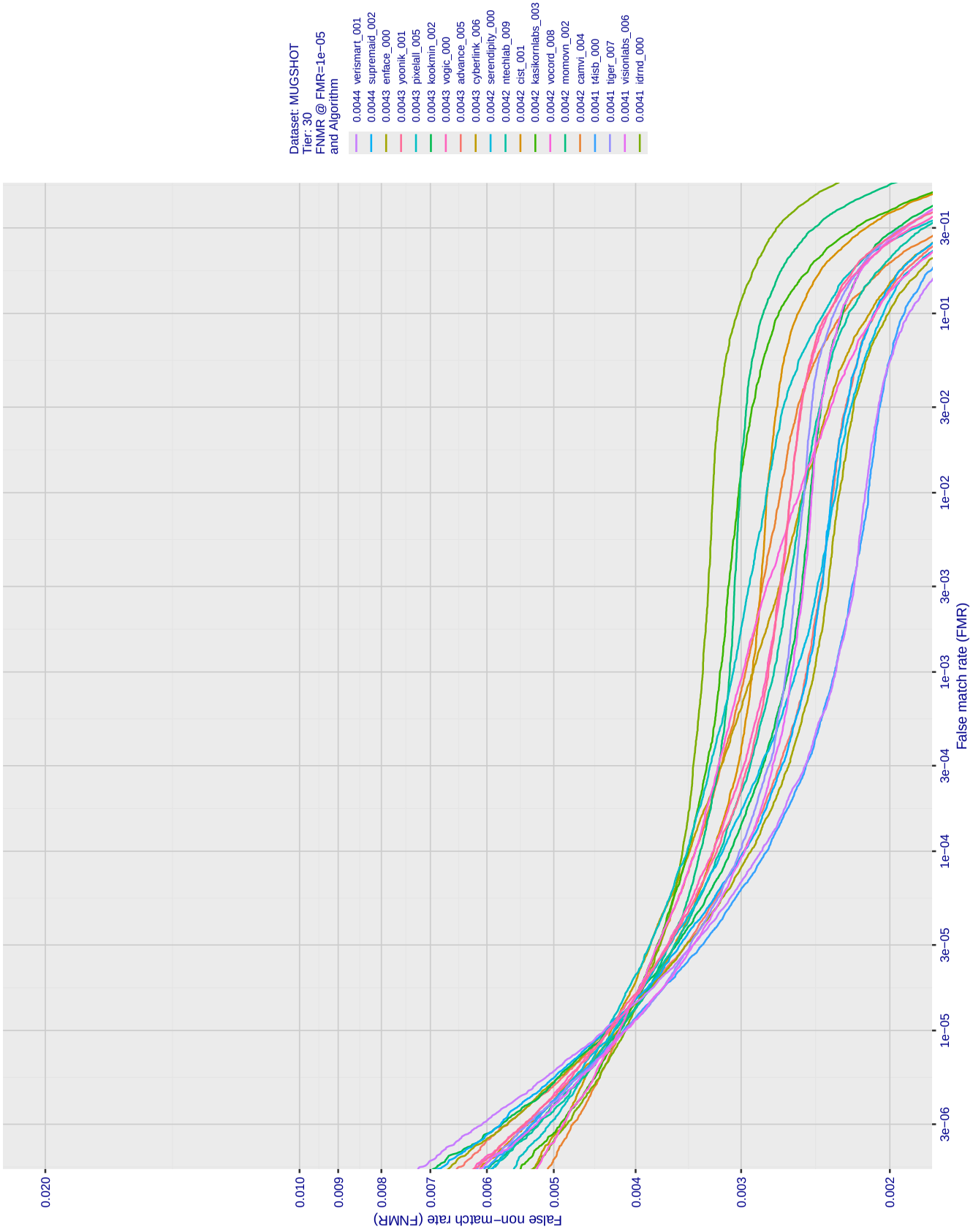

Figure 254: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

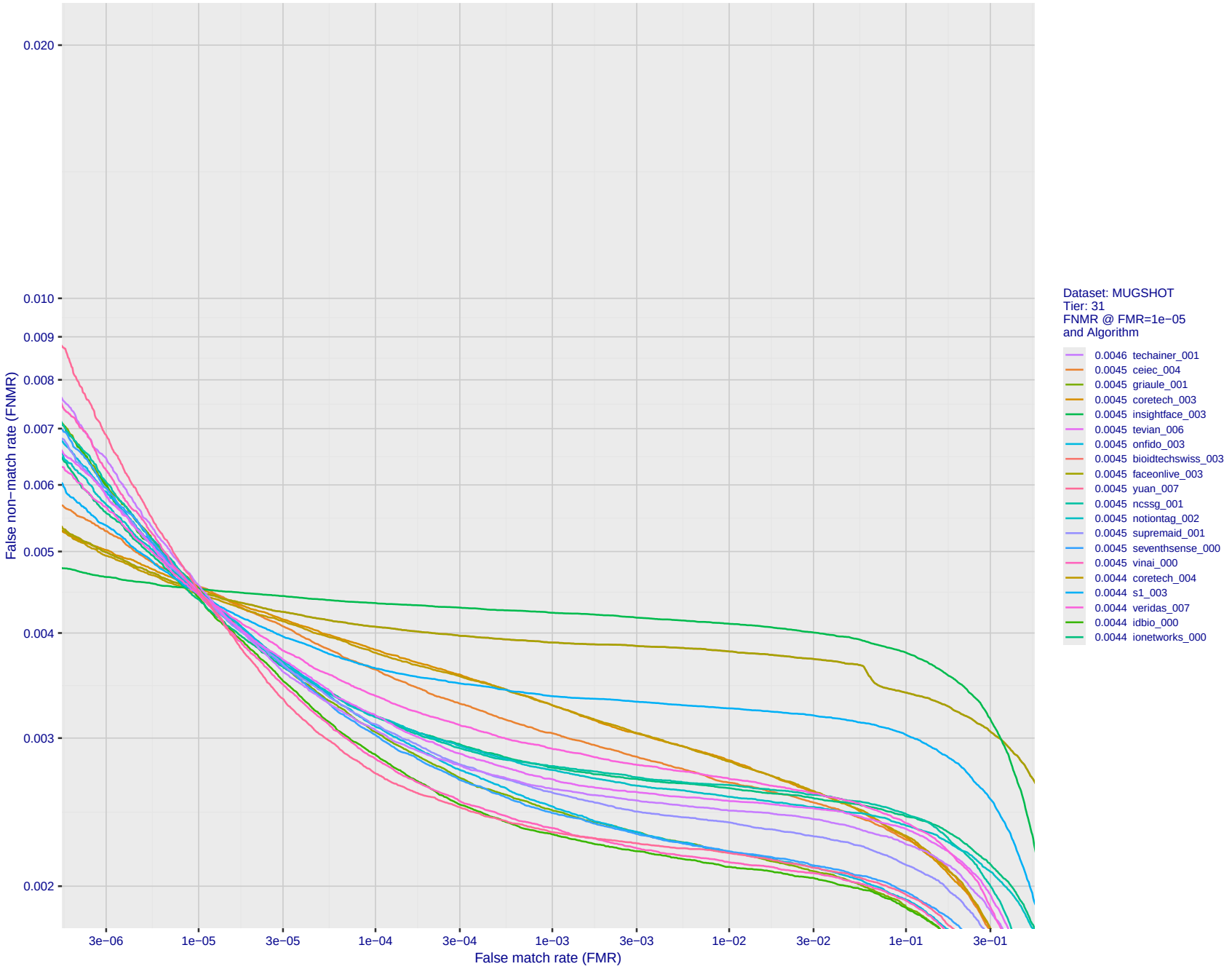

Figure 255: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

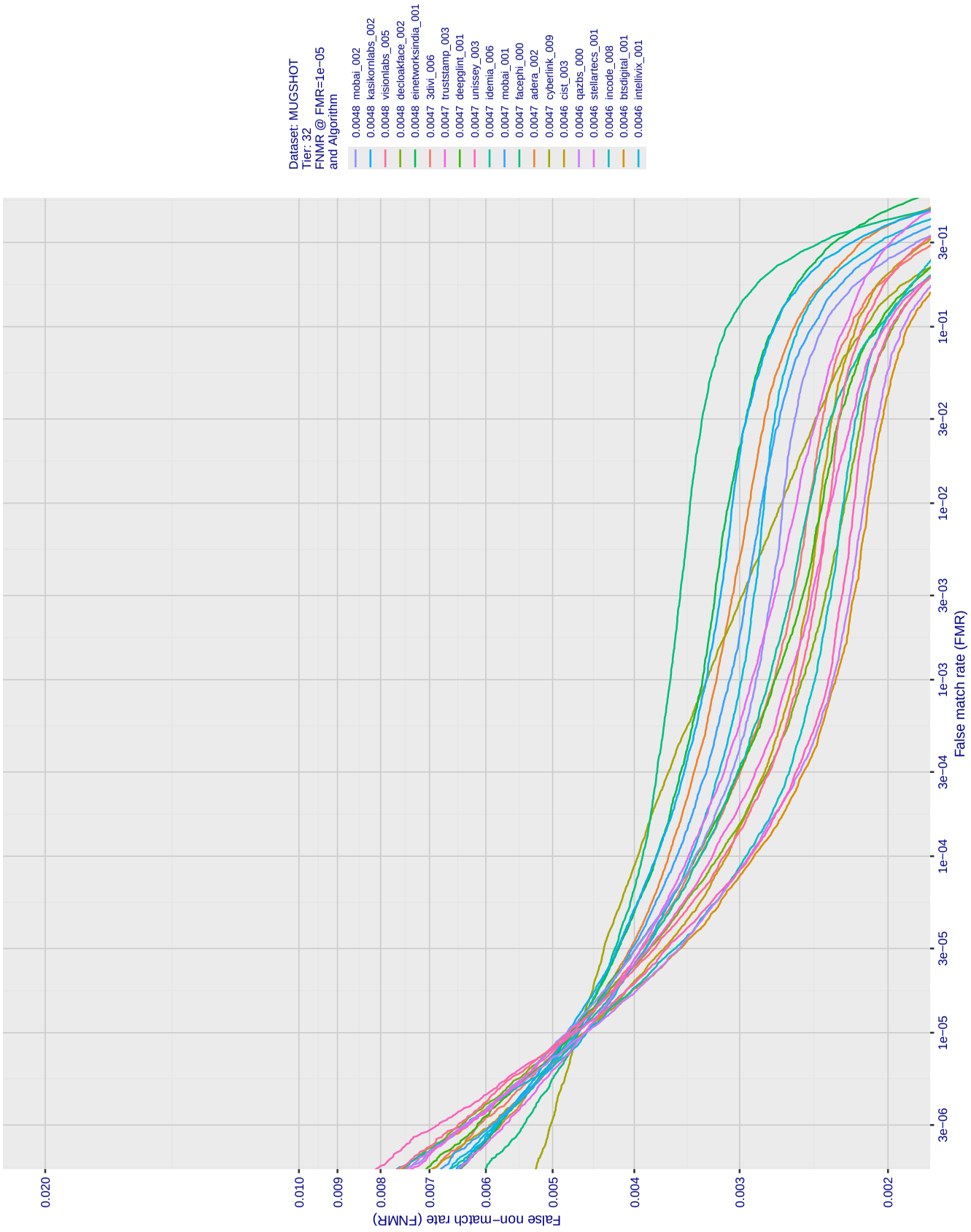

Figure 256: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

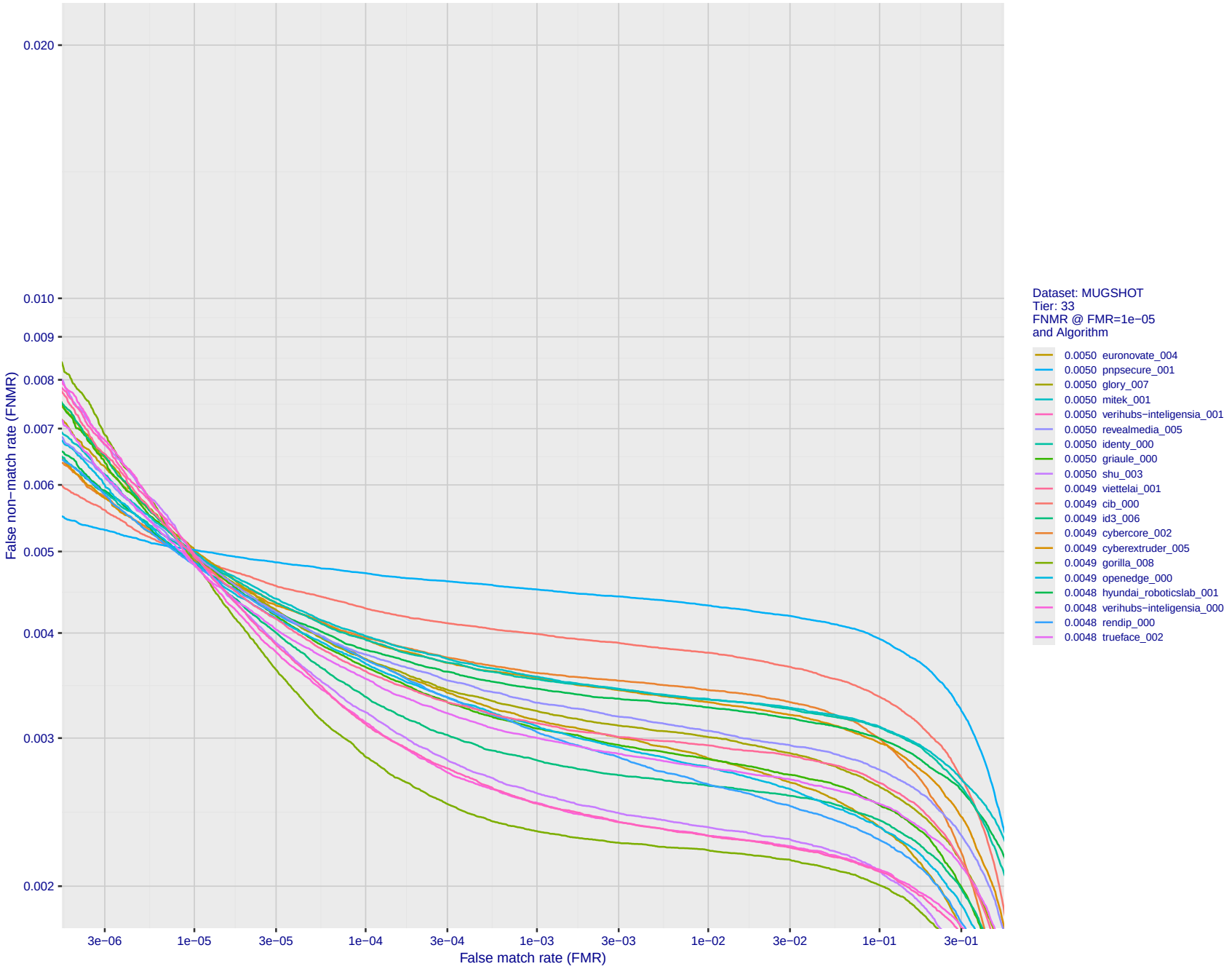

*Figure 257: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR.

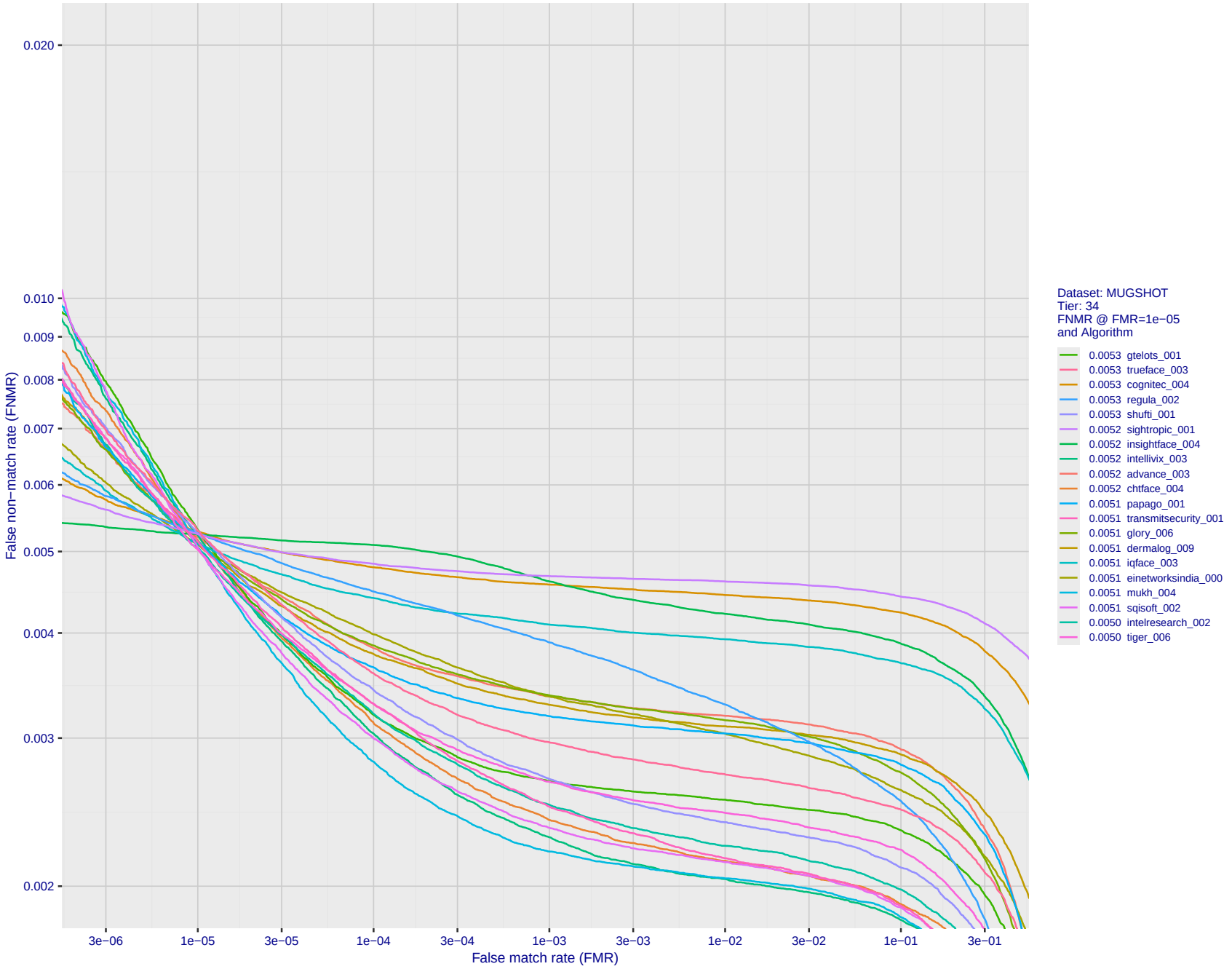

Figure 258: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

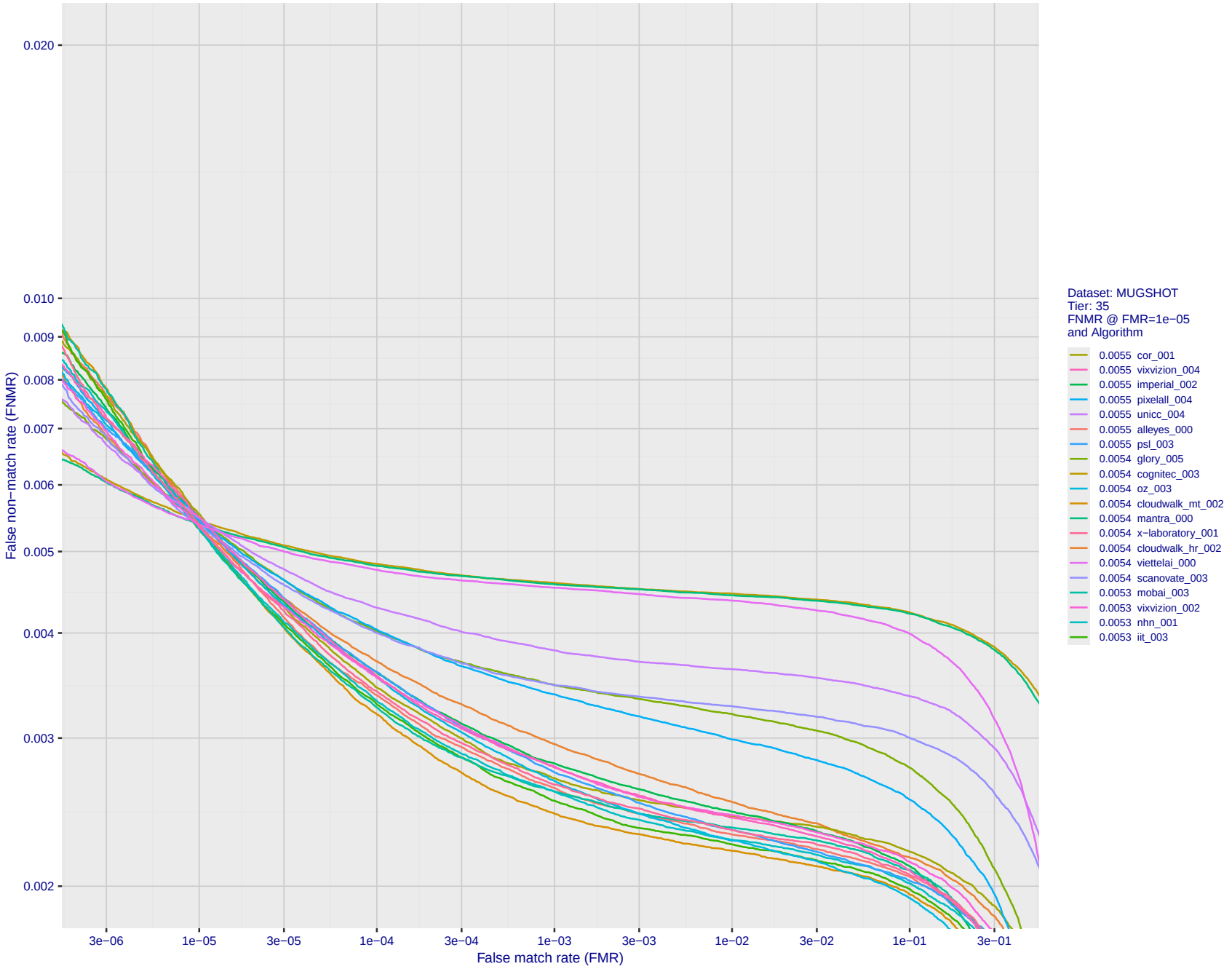

Figure 259: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

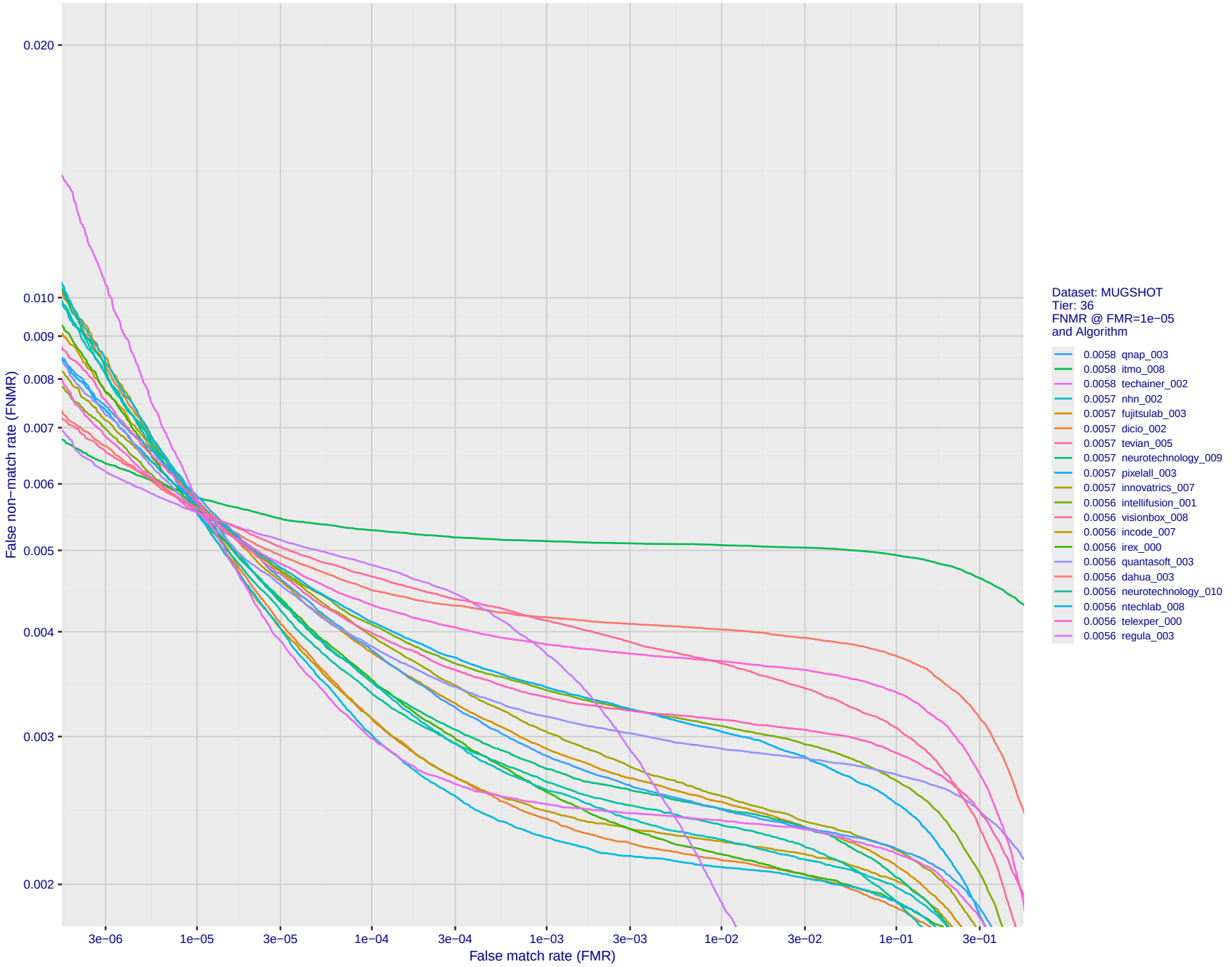

Figure 260: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

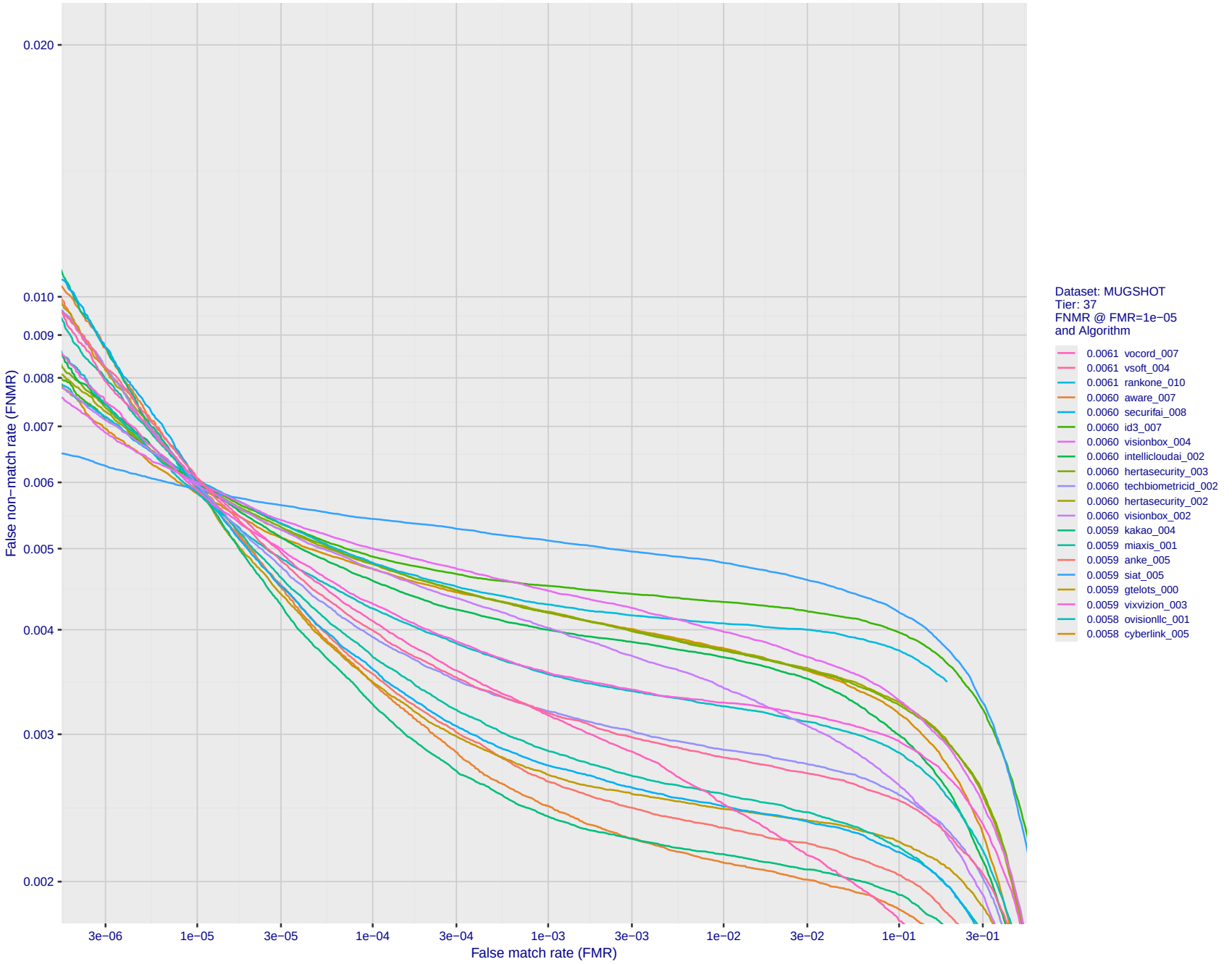

Figure 261: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

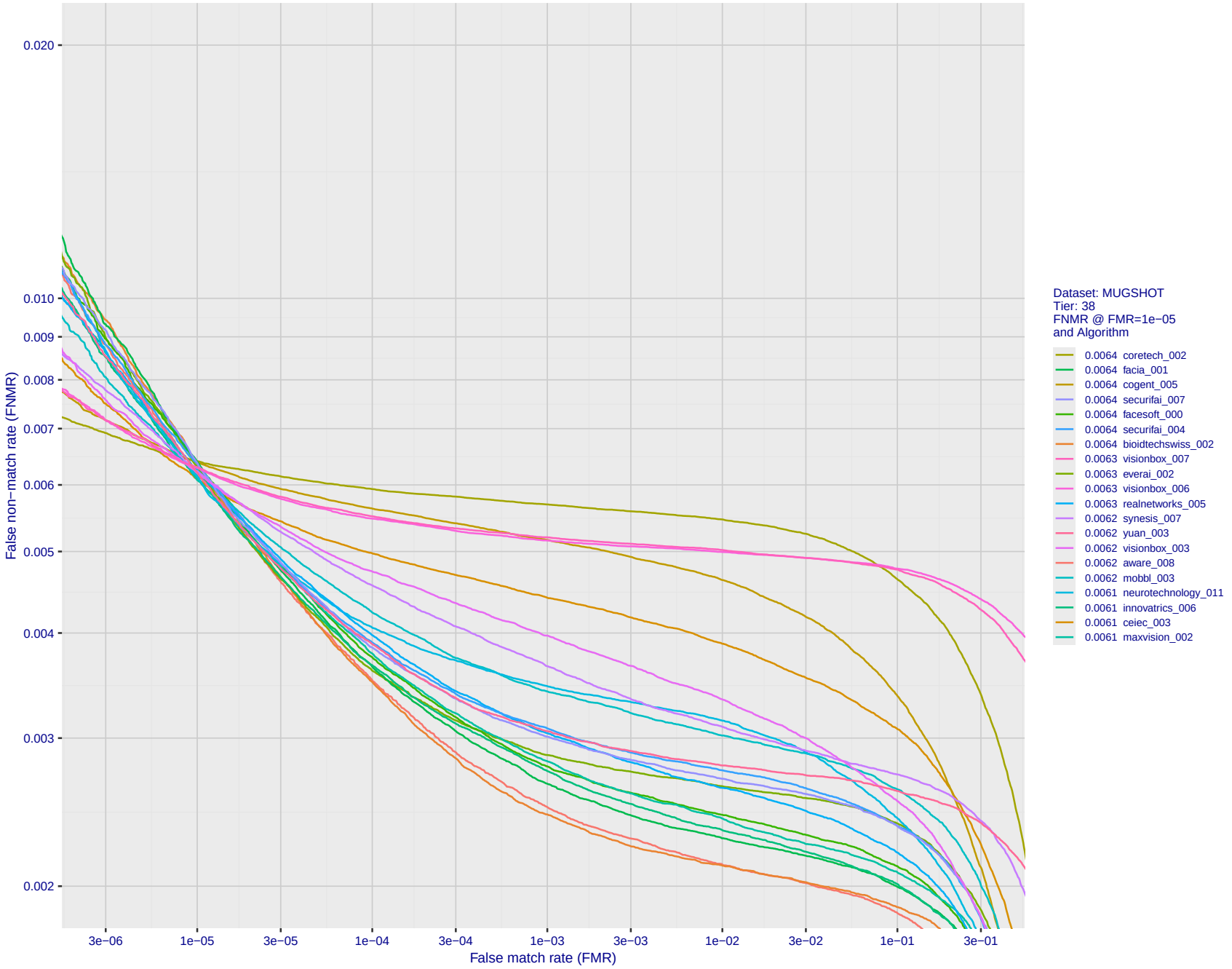

Figure 262: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

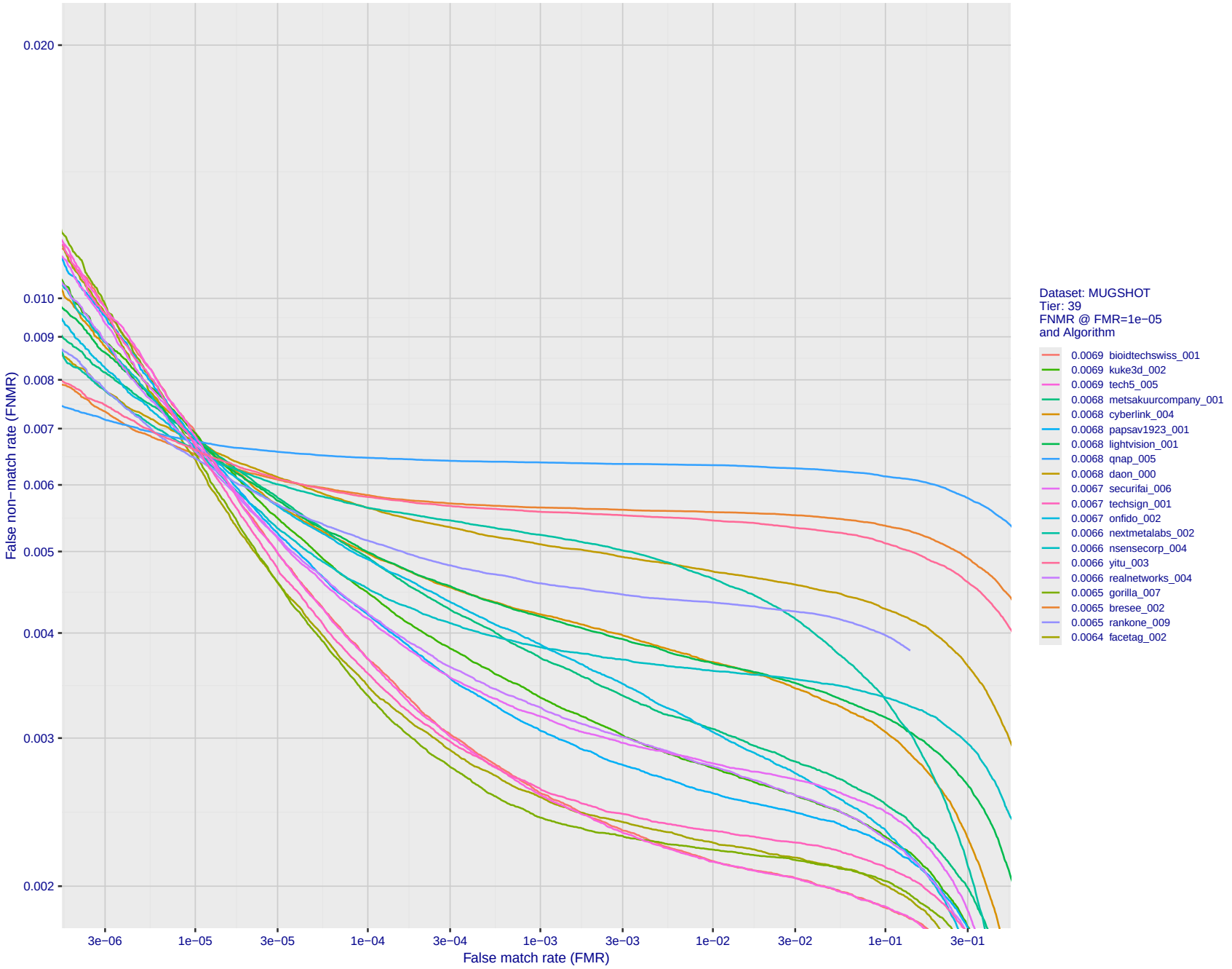

Figure 263: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

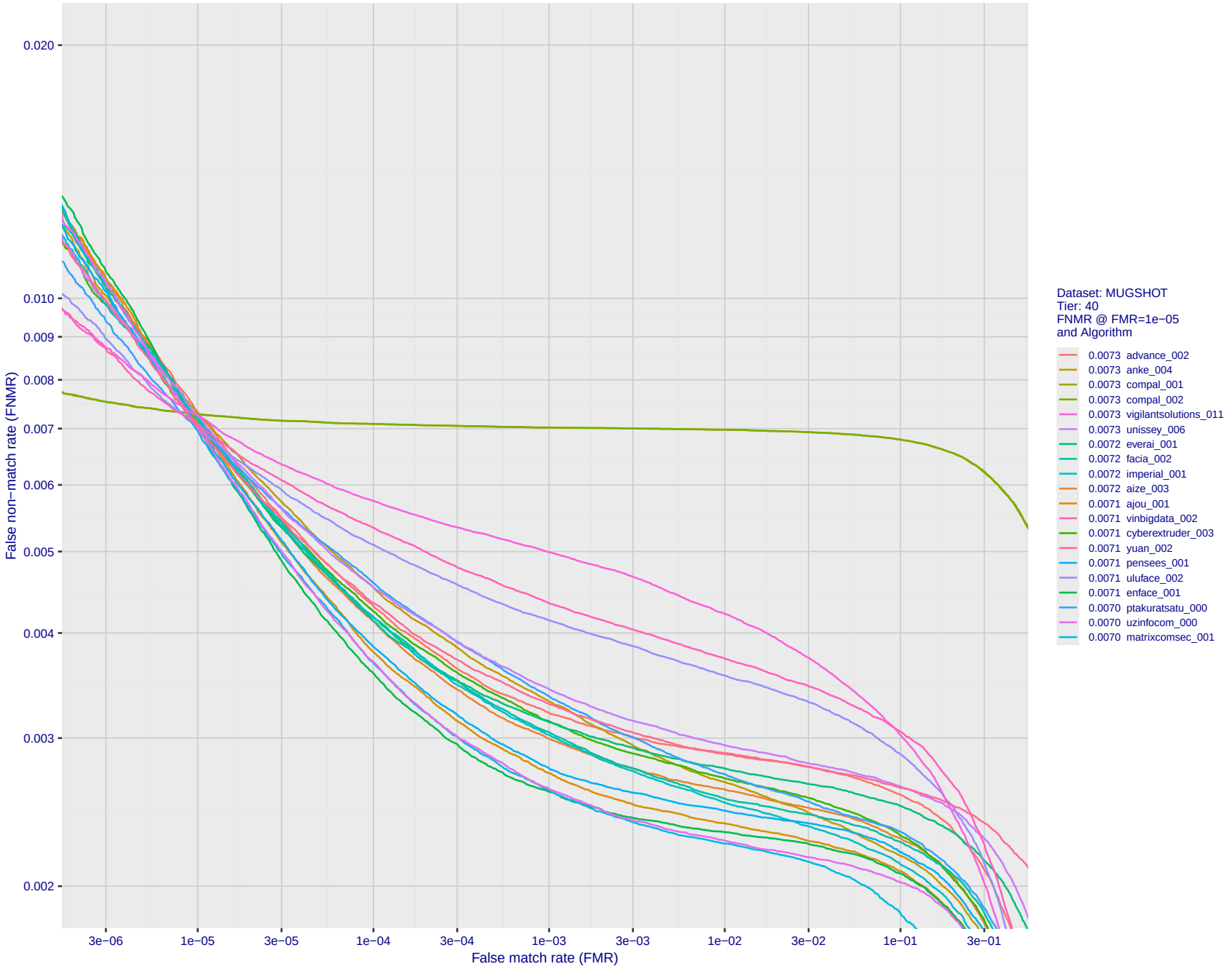

Figure 264: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

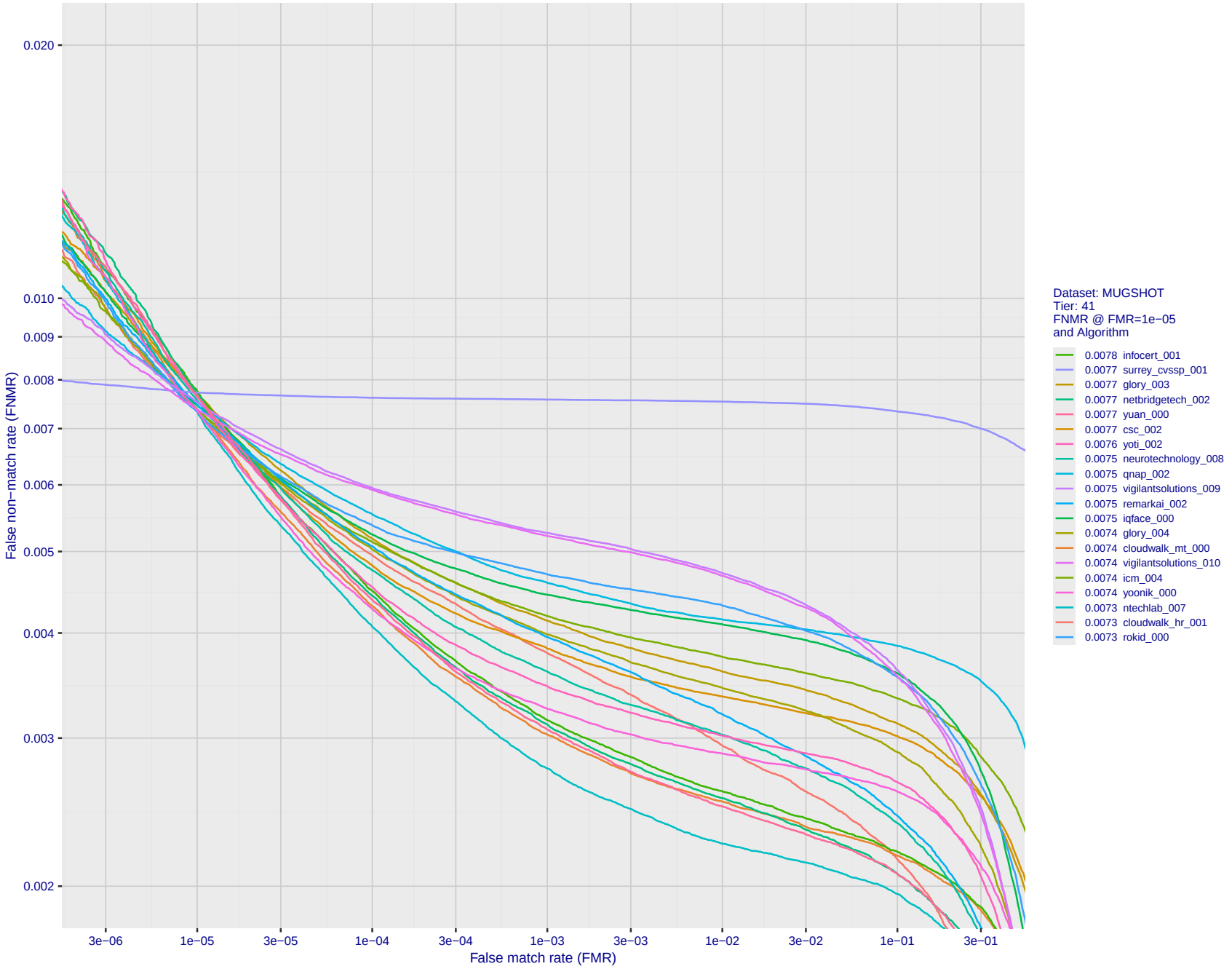

Figure 265: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

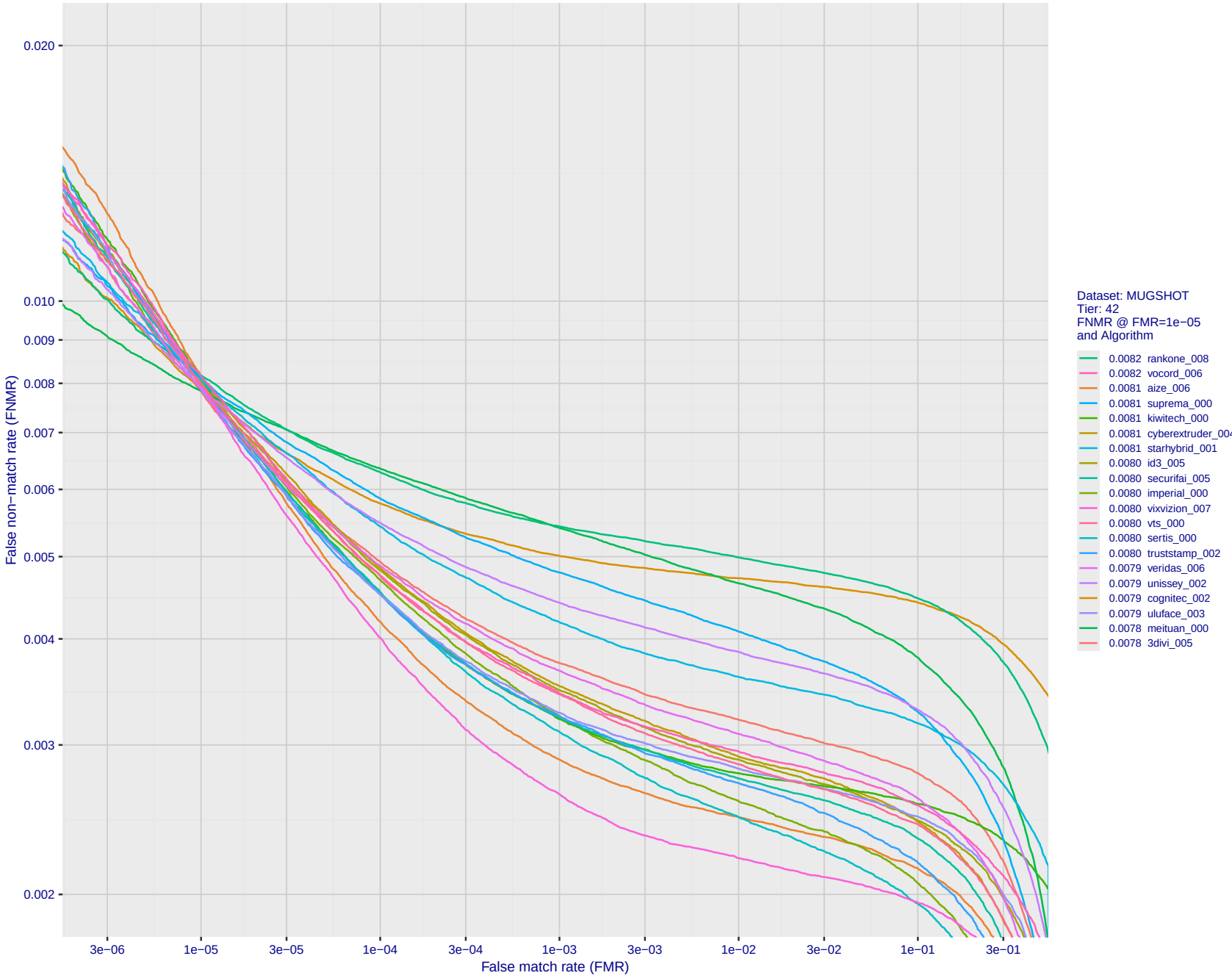

Figure 266: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

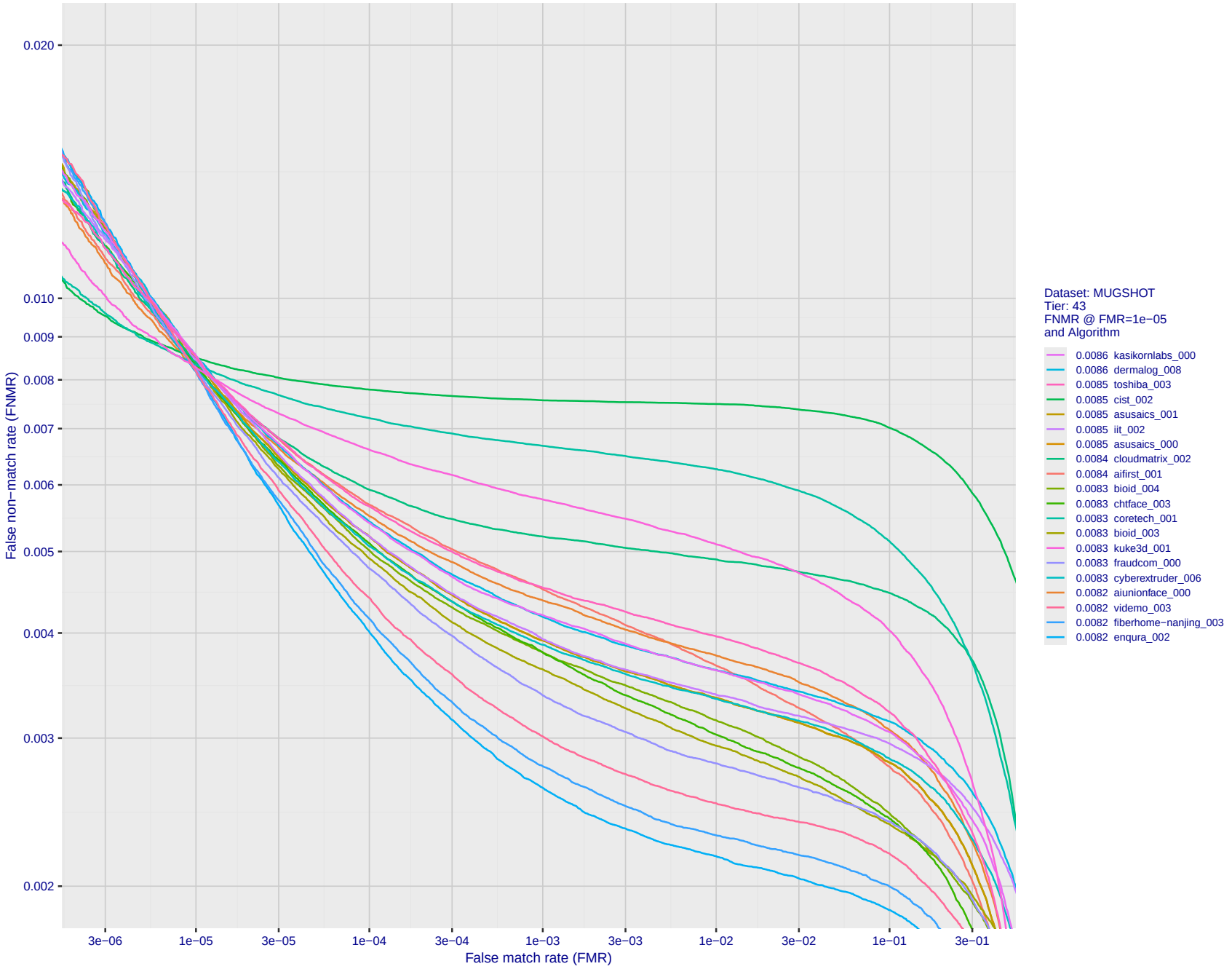

Figure 267: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

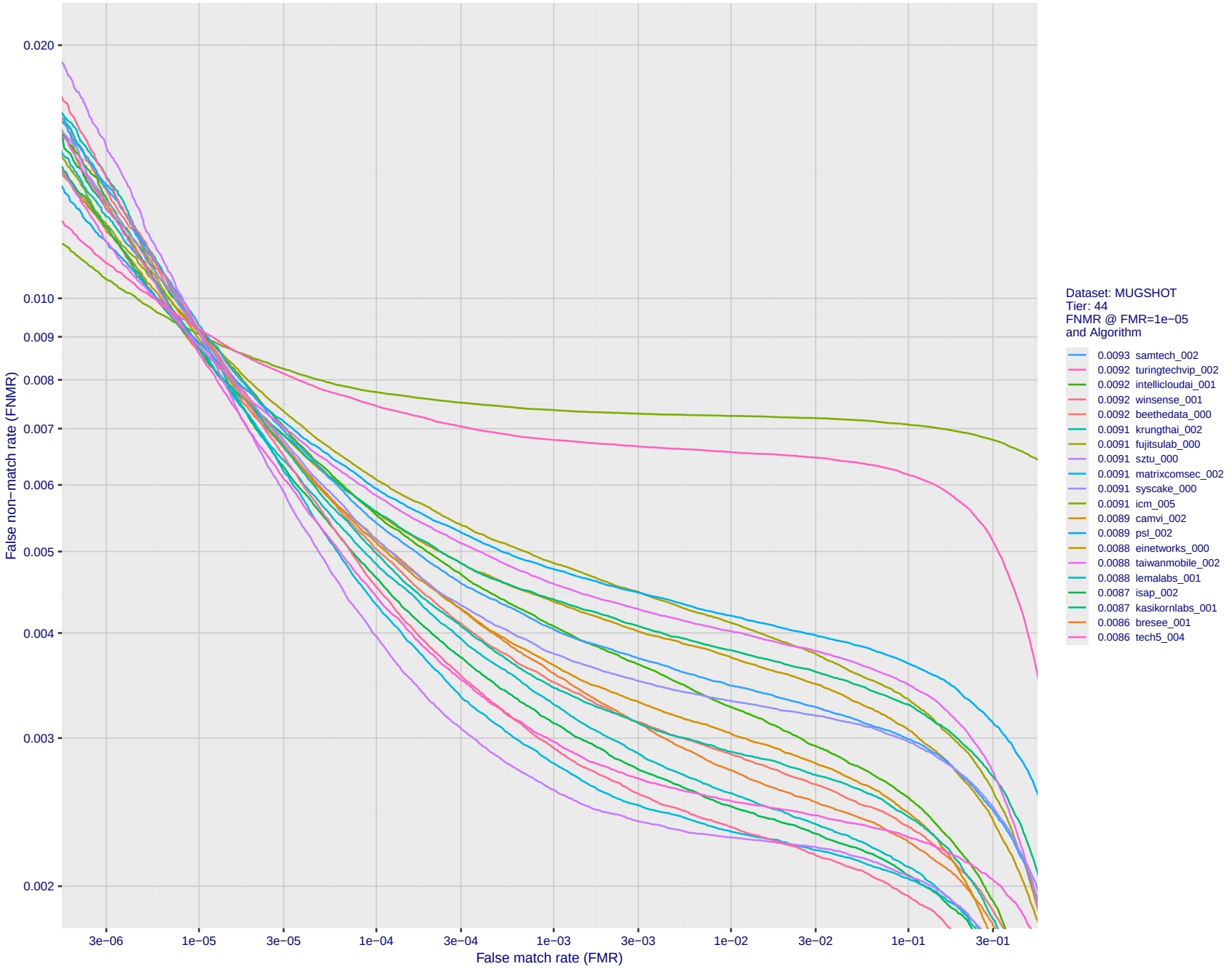

Figure 268: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

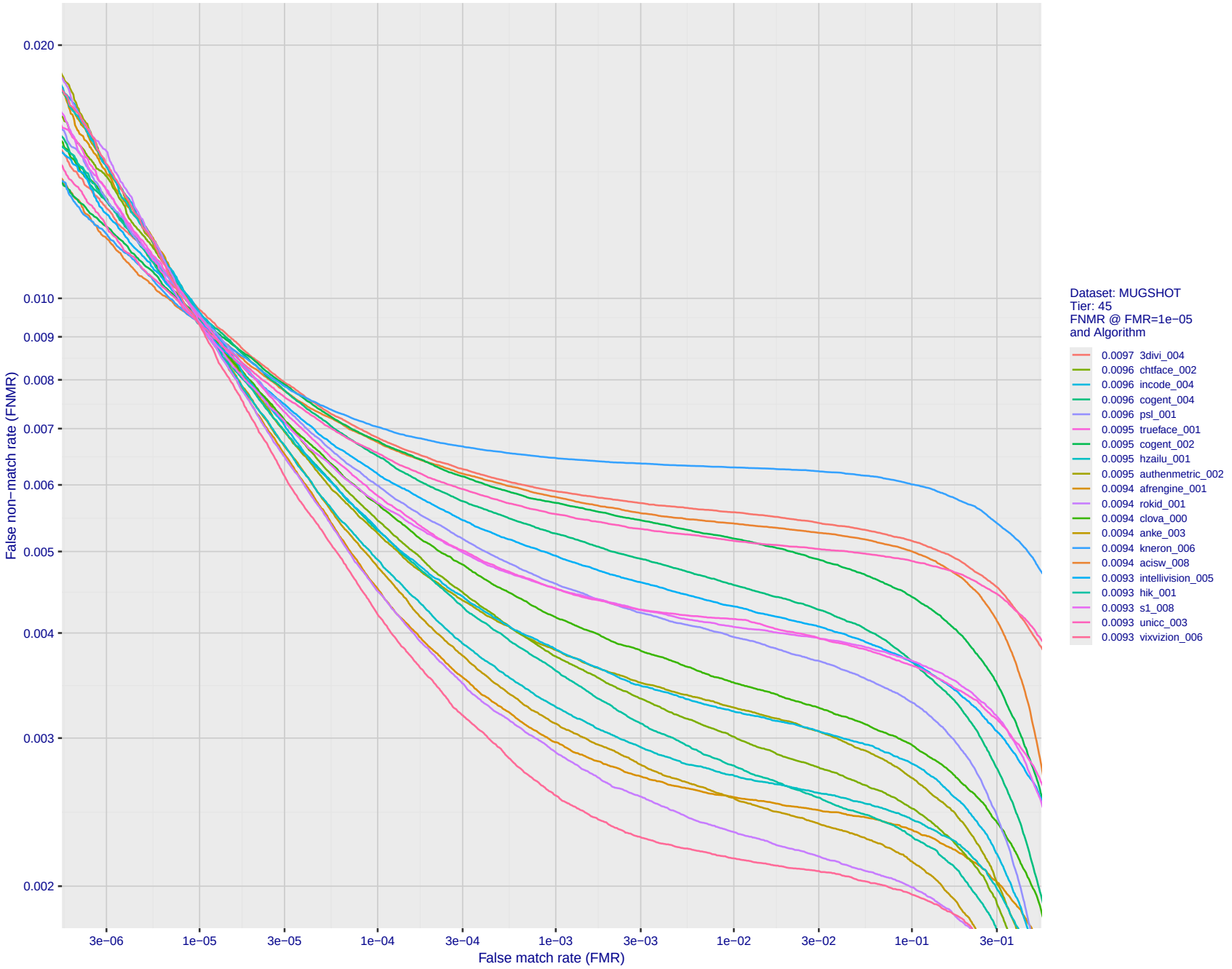

Figure 269: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

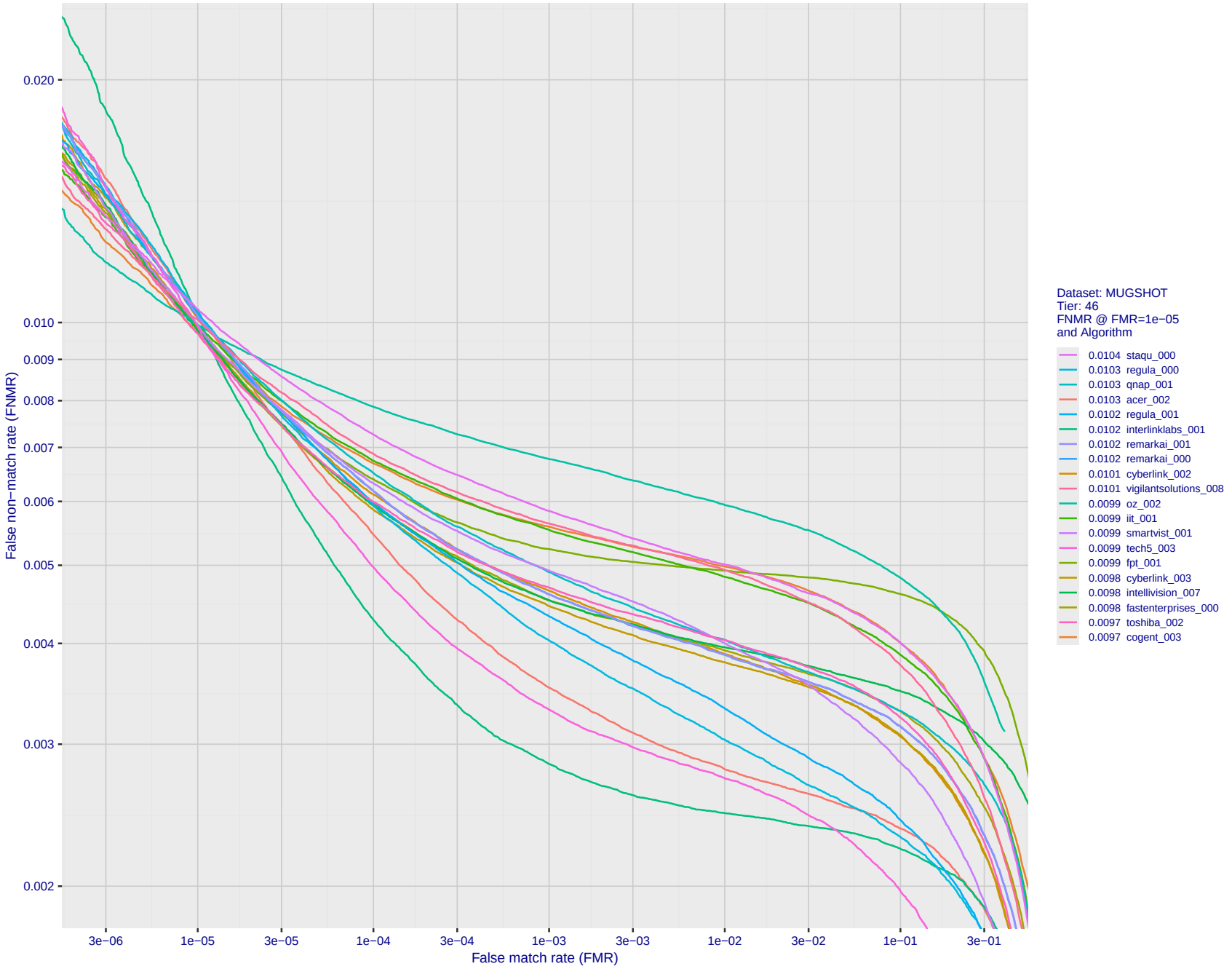

Figure 270: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

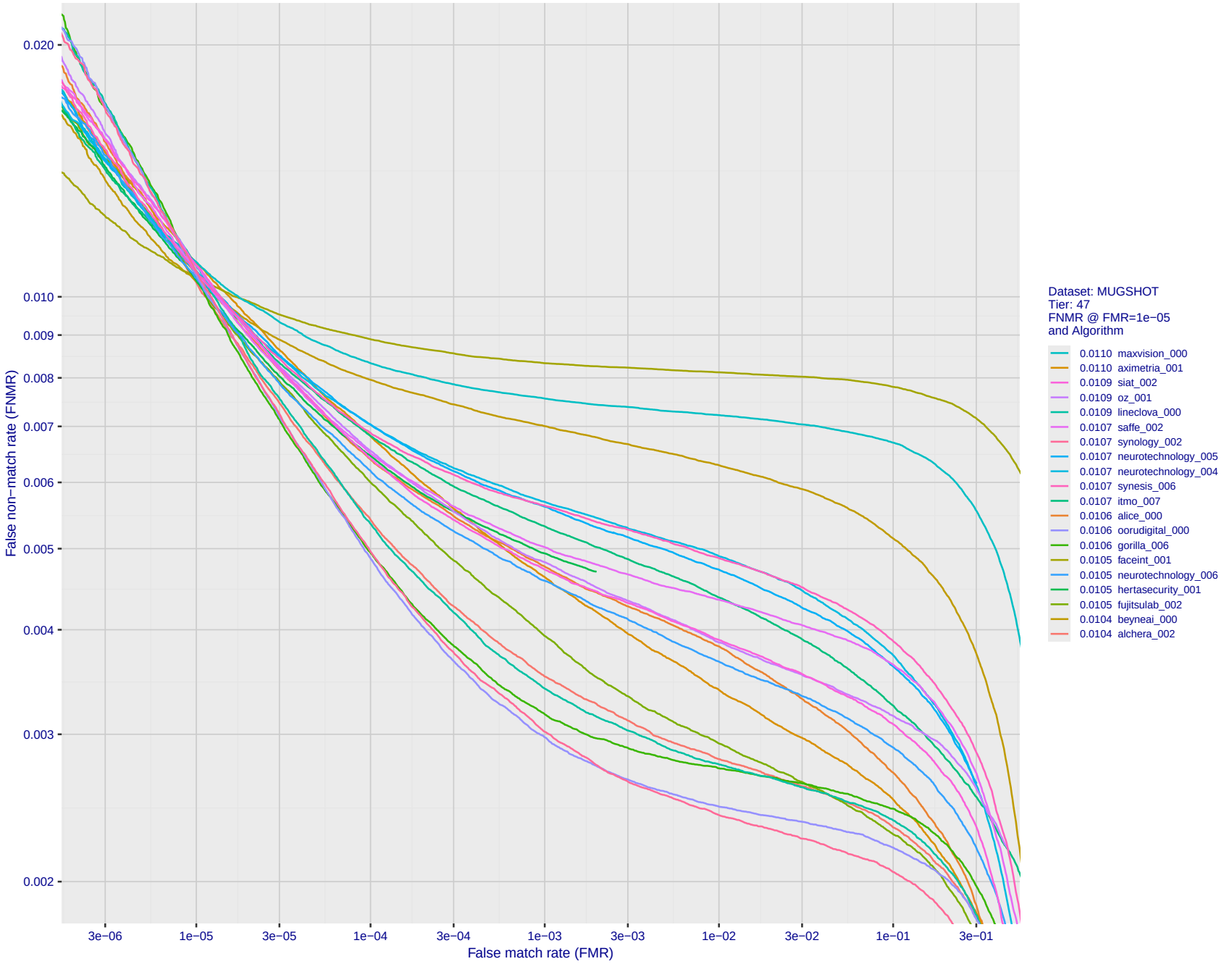

Figure 271: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

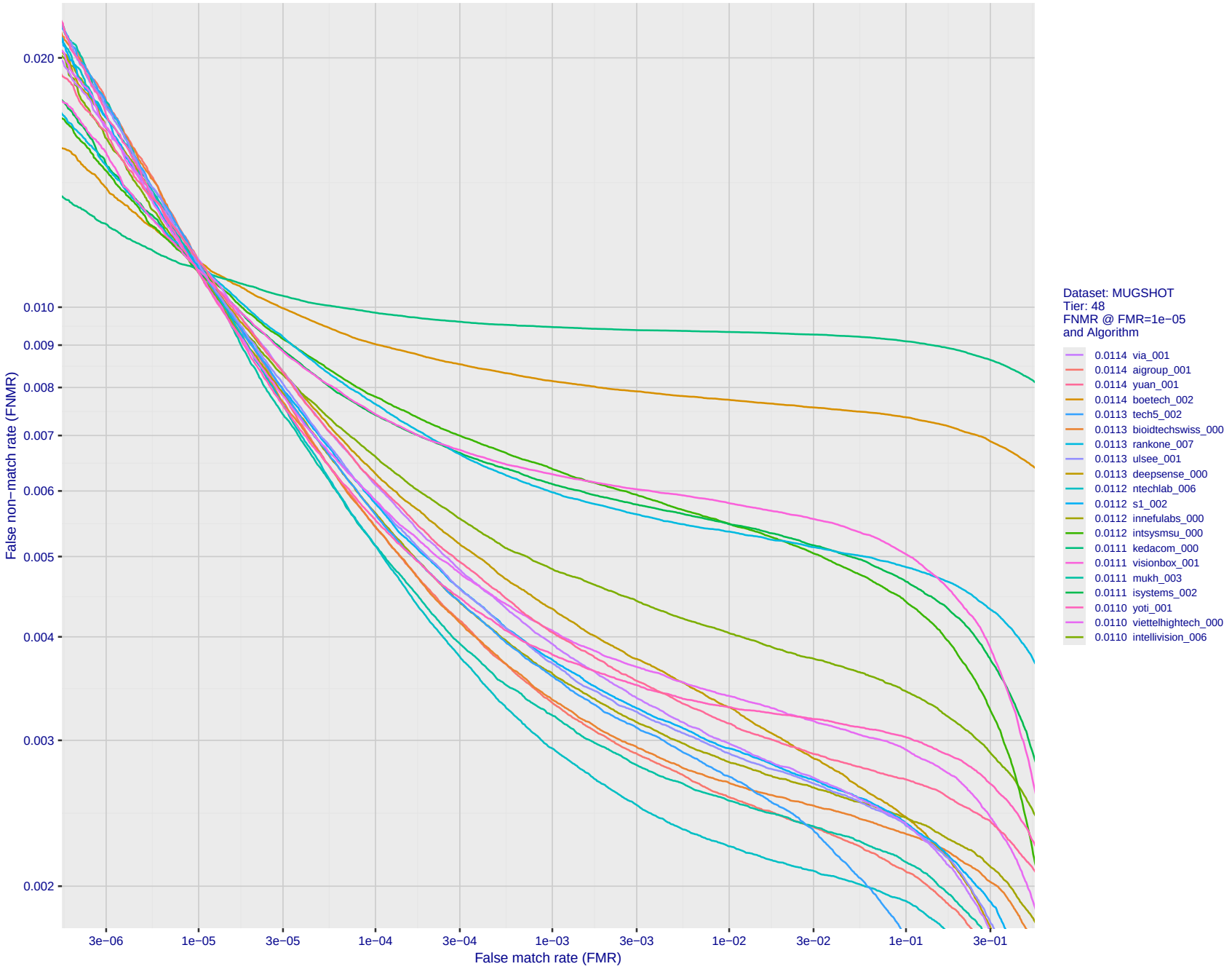

Figure 272: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

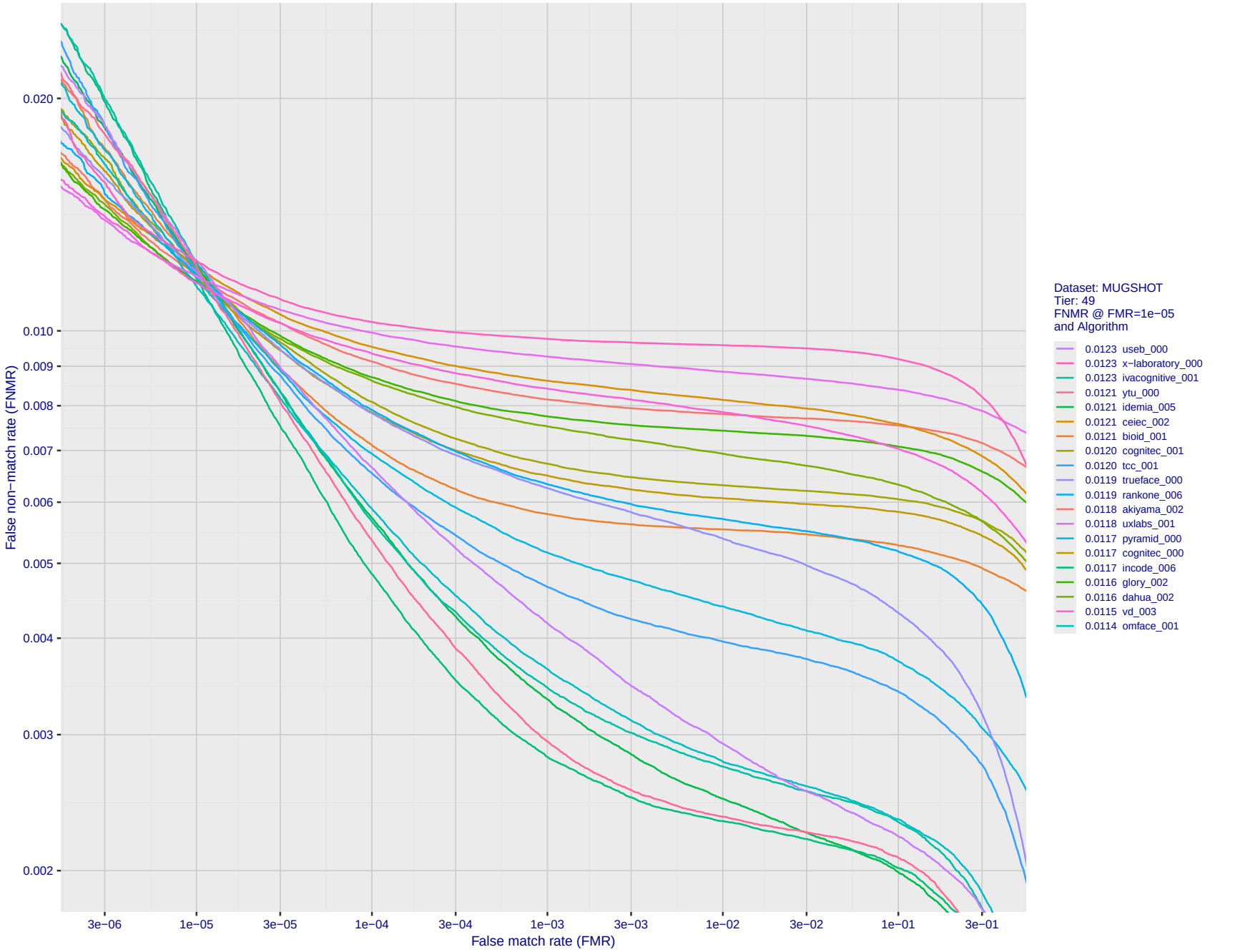

Figure 273: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

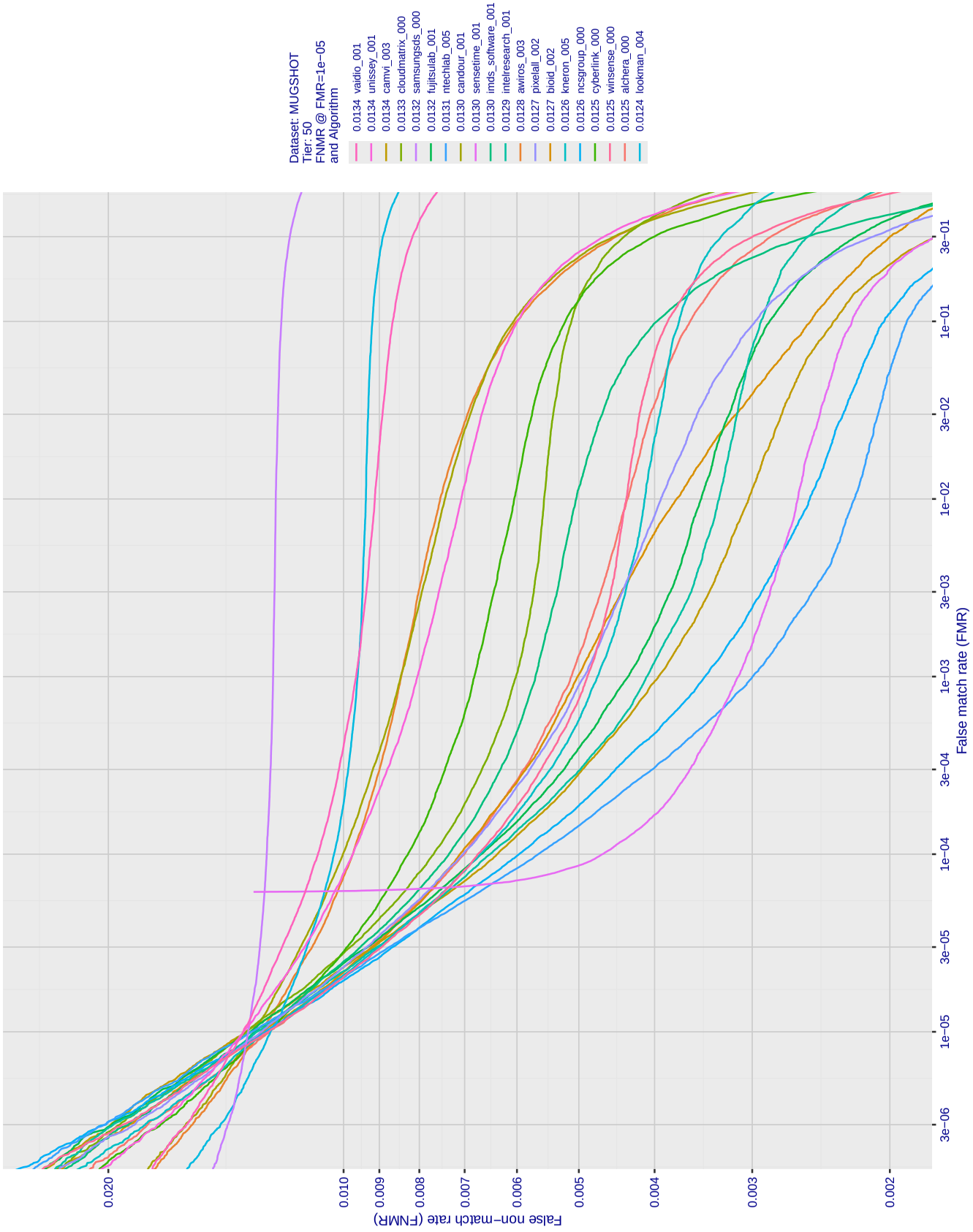

Figure 274: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

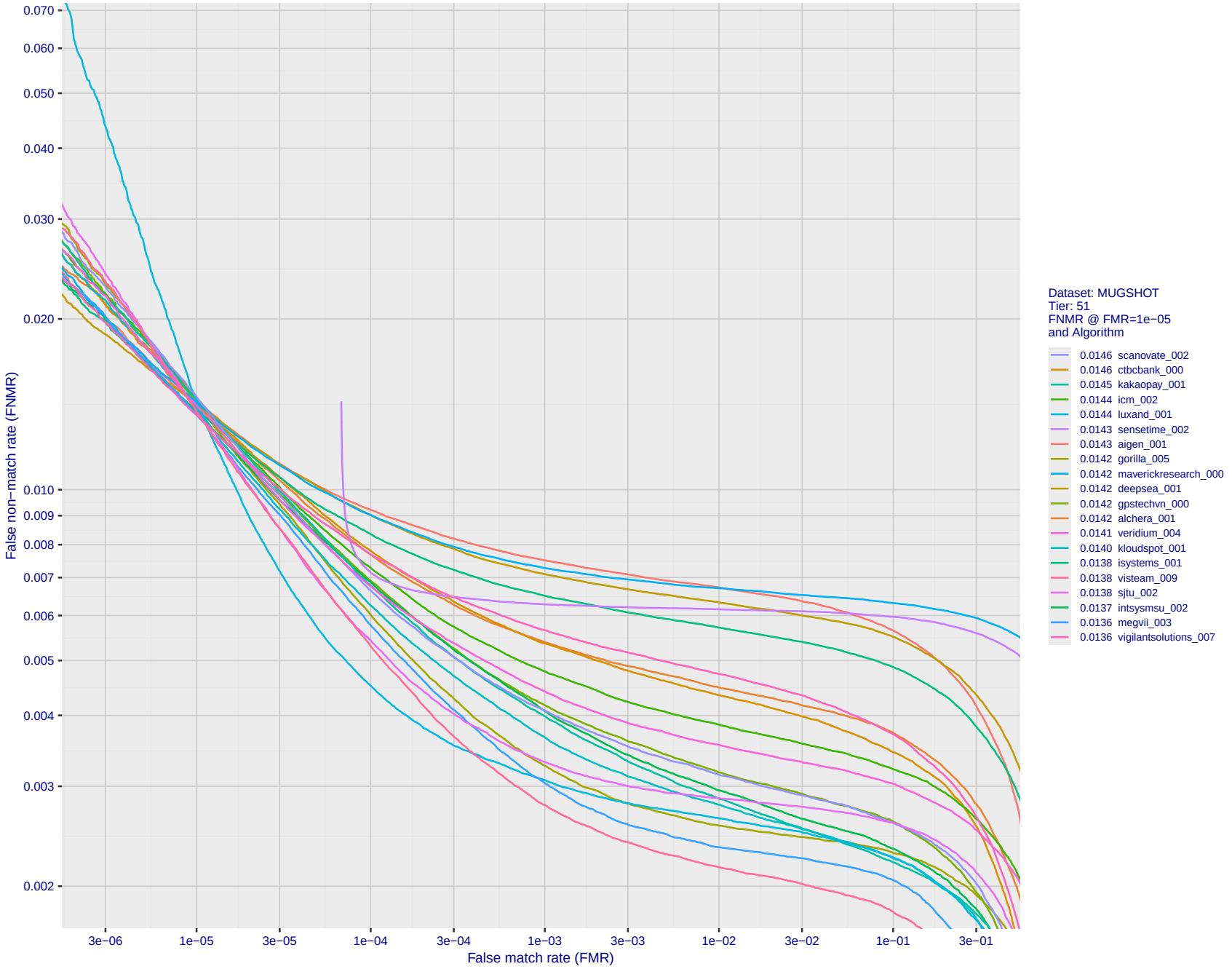

*Figure 275: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR*.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

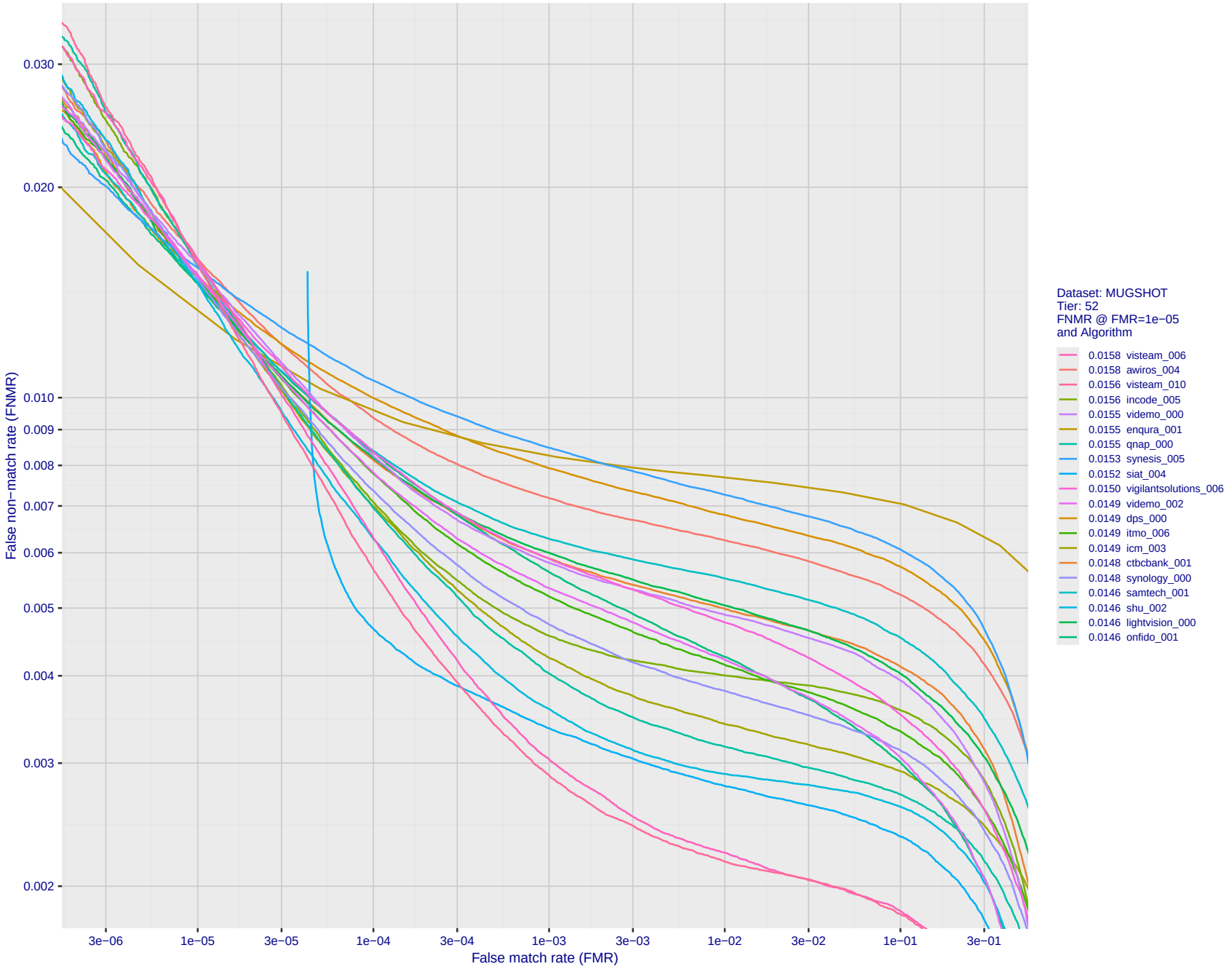

Figure 276: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

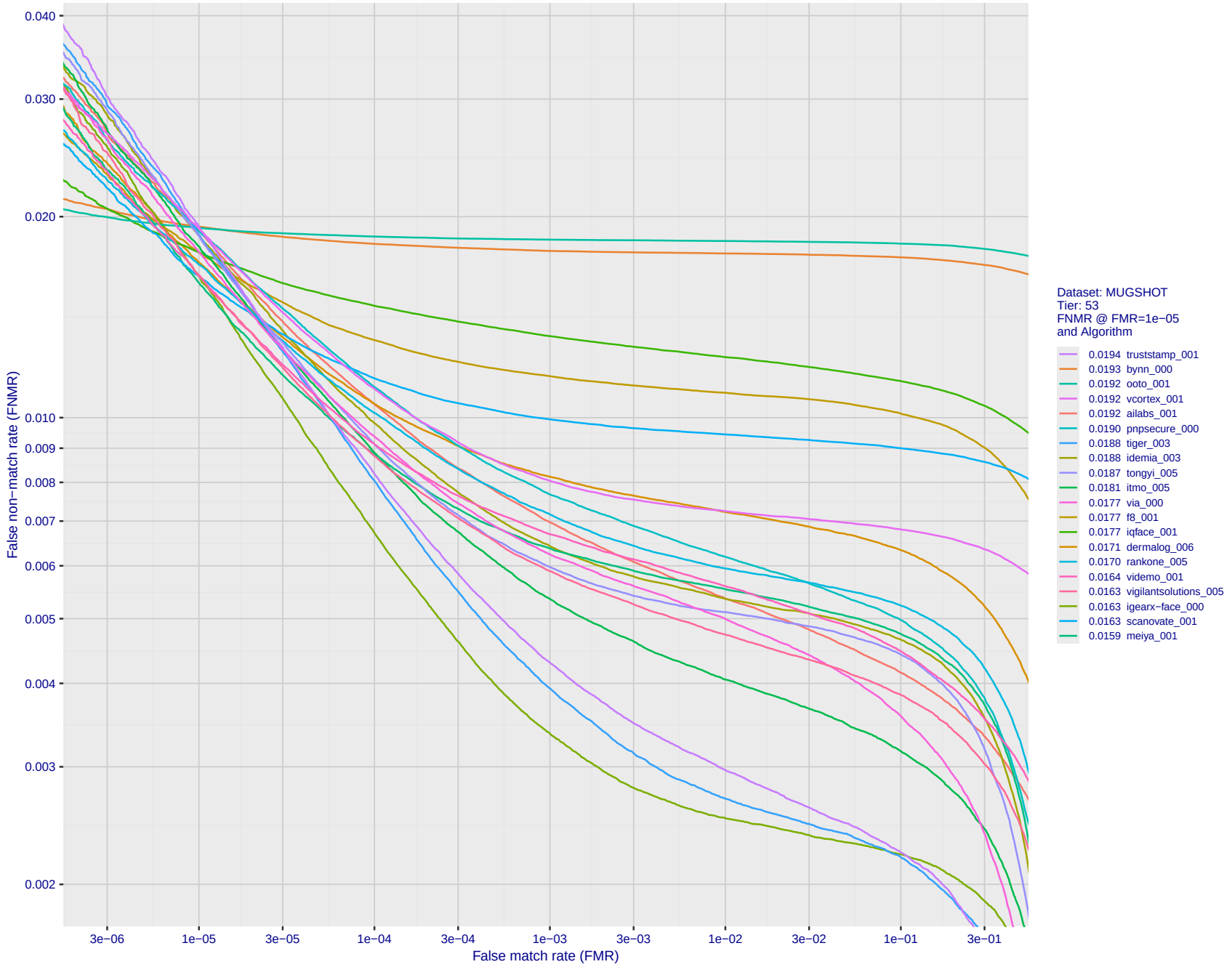

Figure 277: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

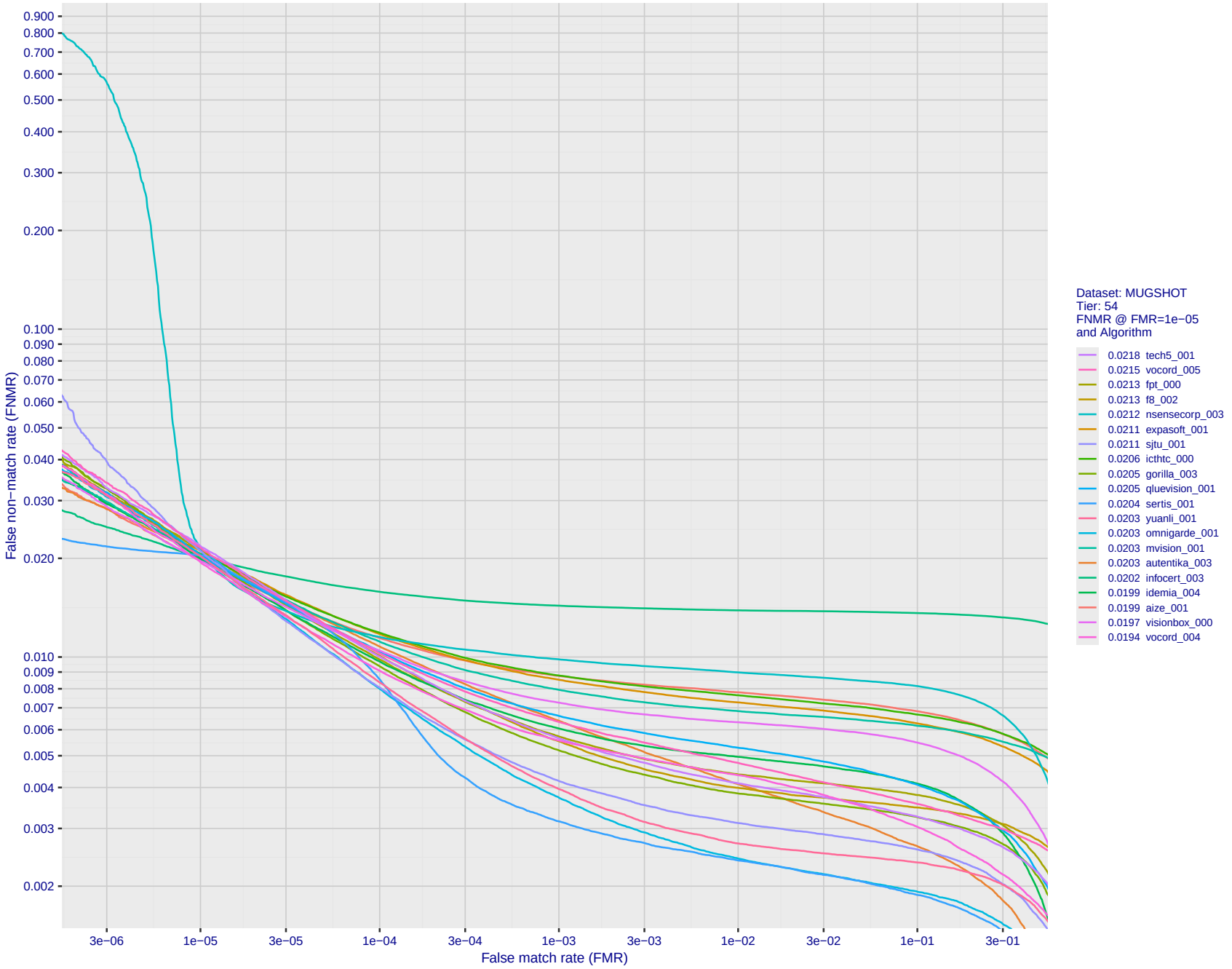

Figure 278: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

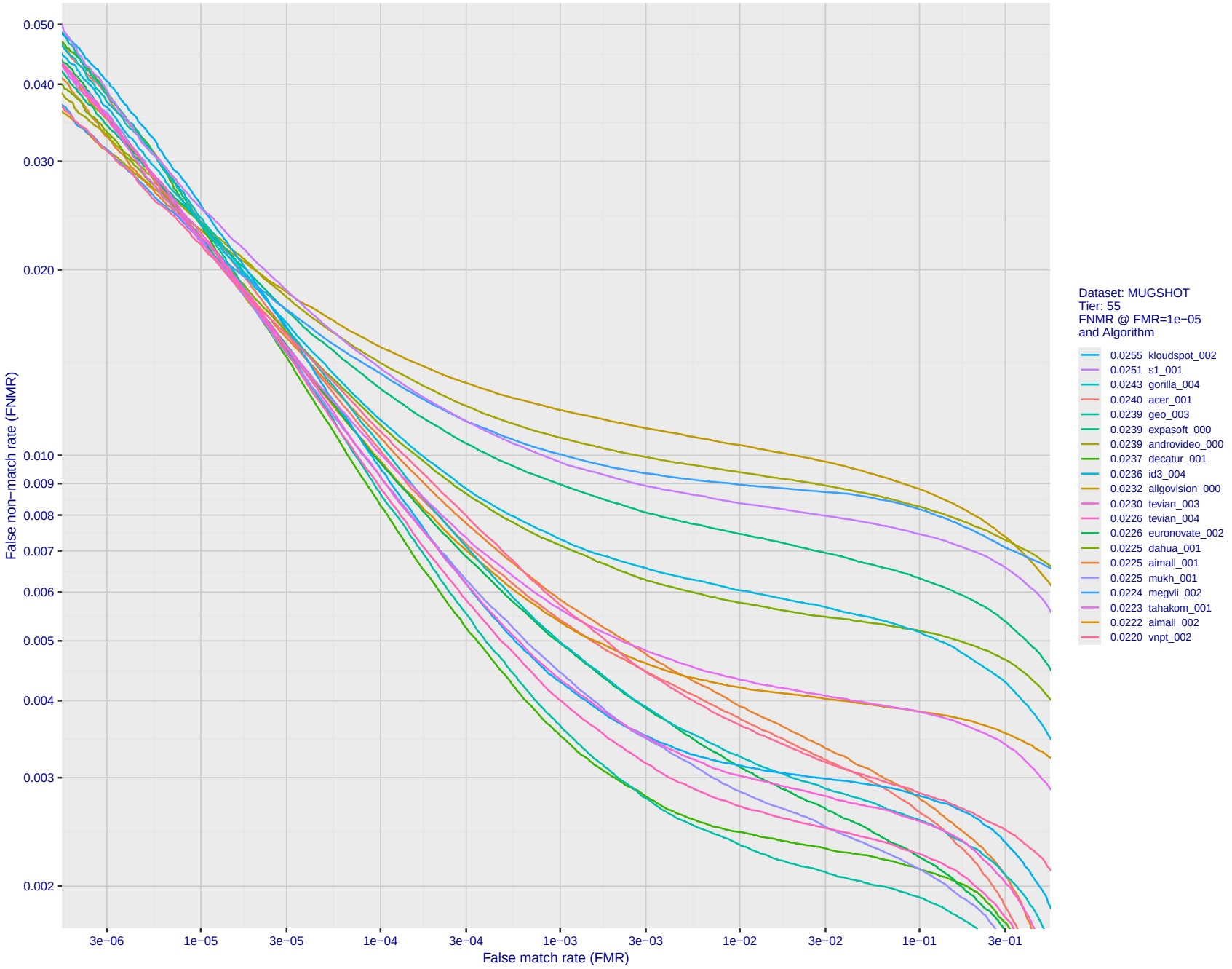

Figure 279: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

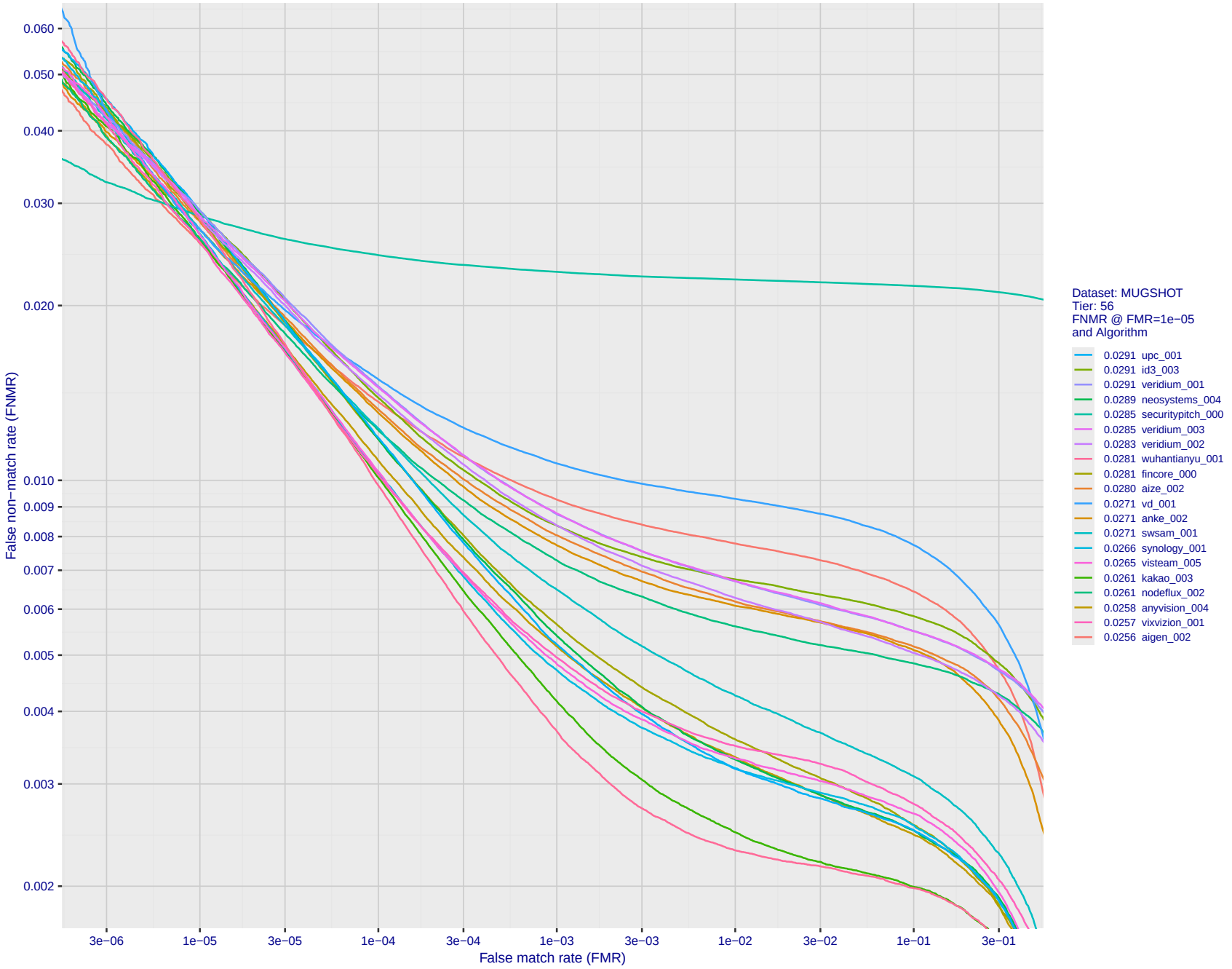

Figure 280: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

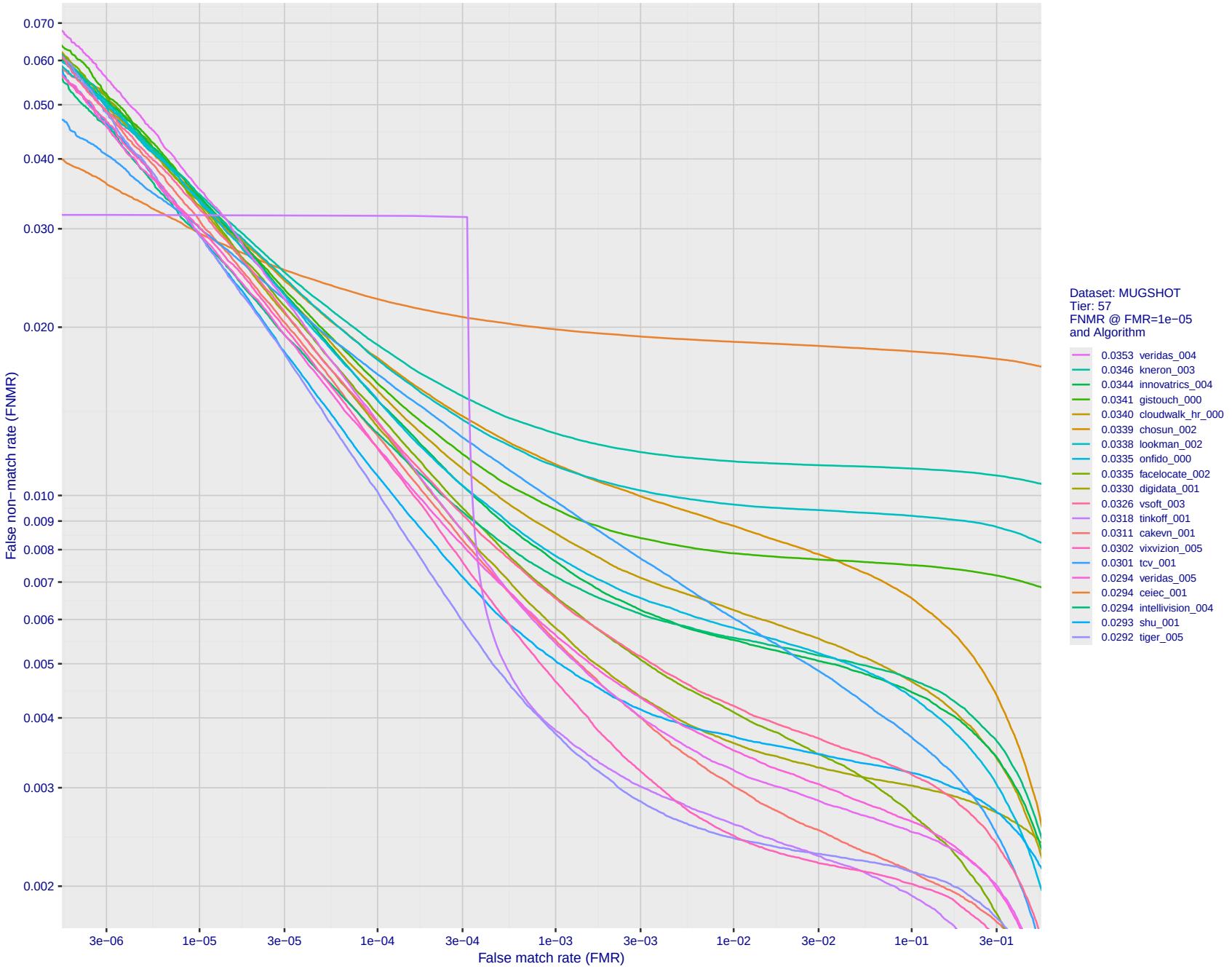

Figure 281: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

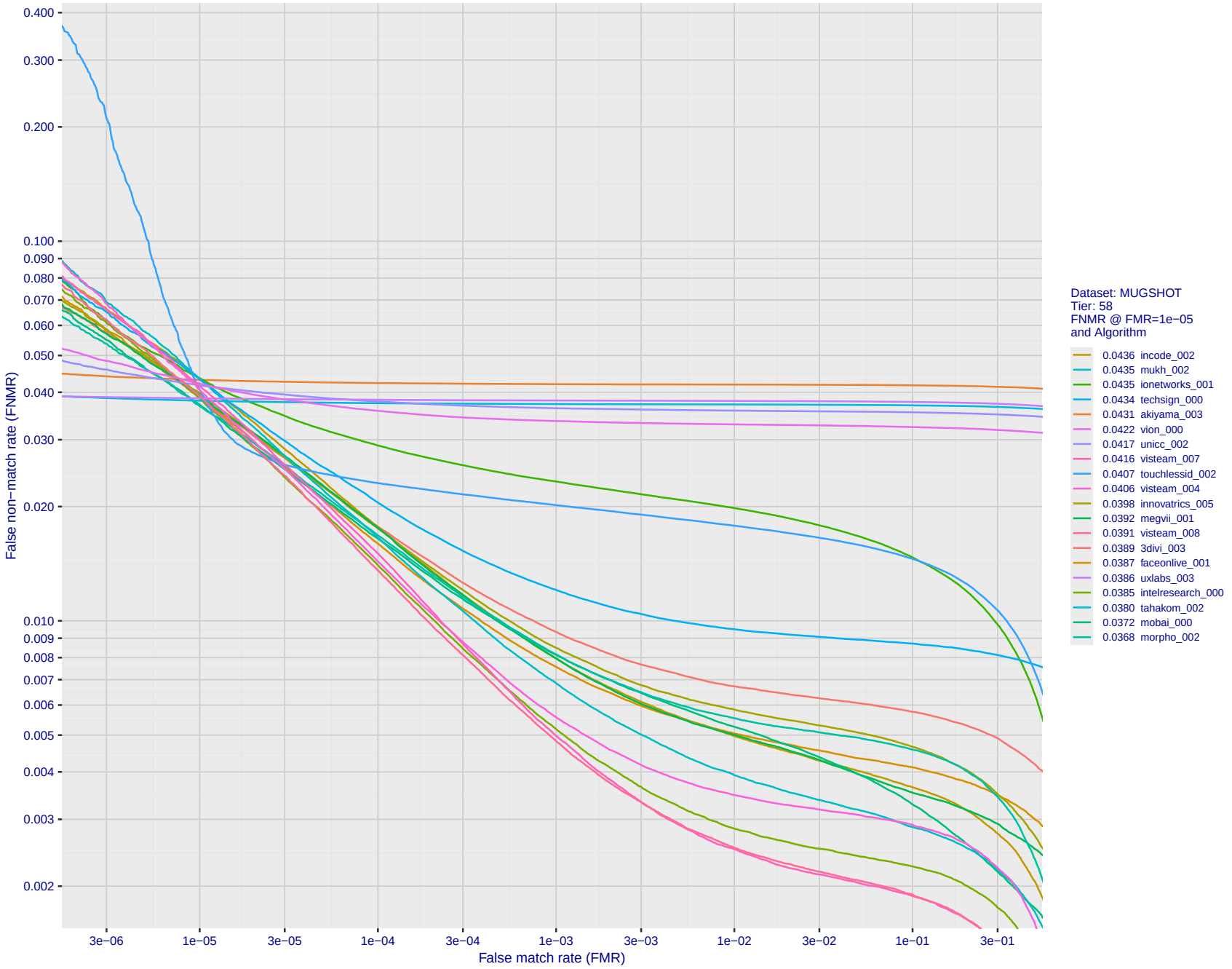

Figure 282: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

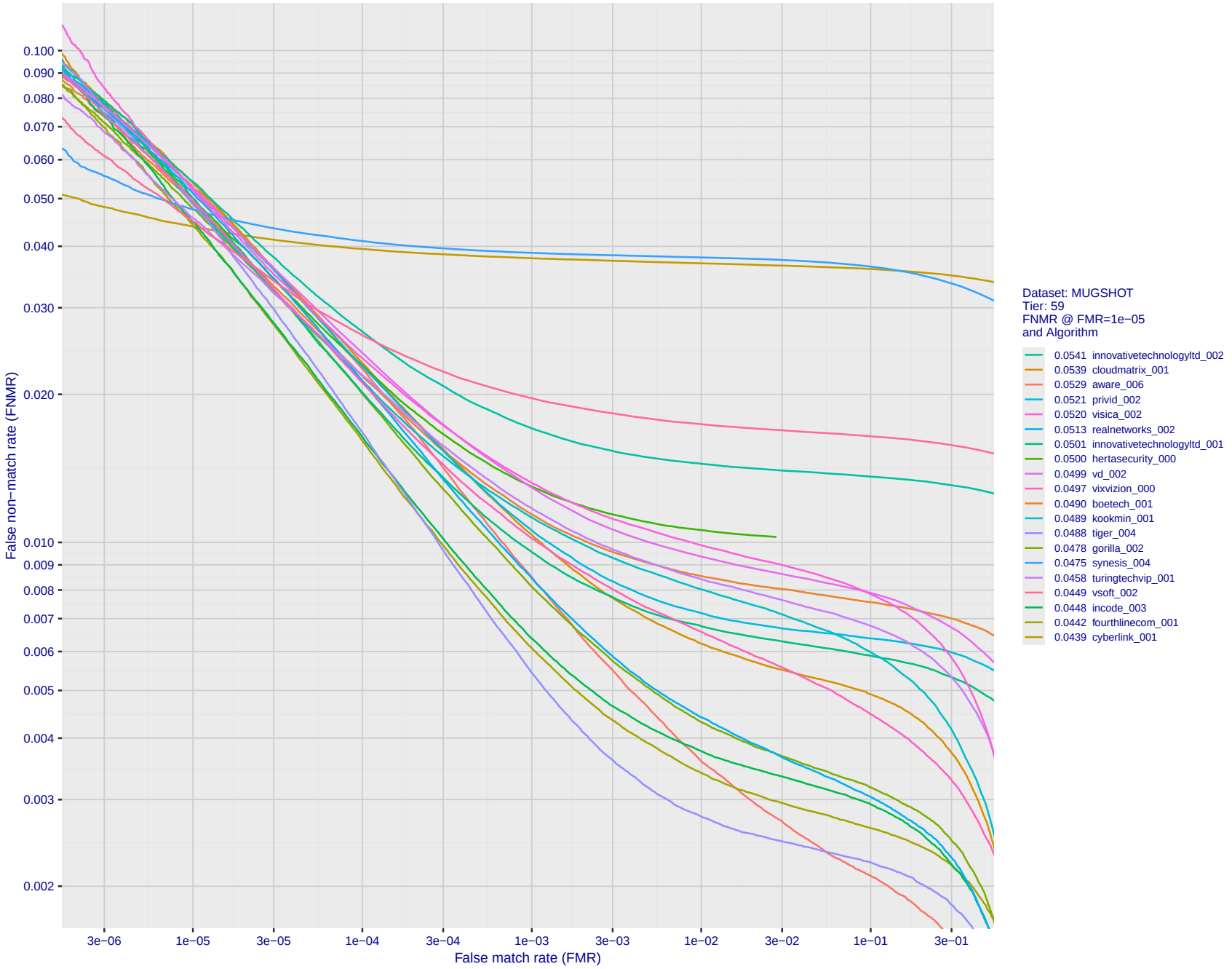

Figure 283: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

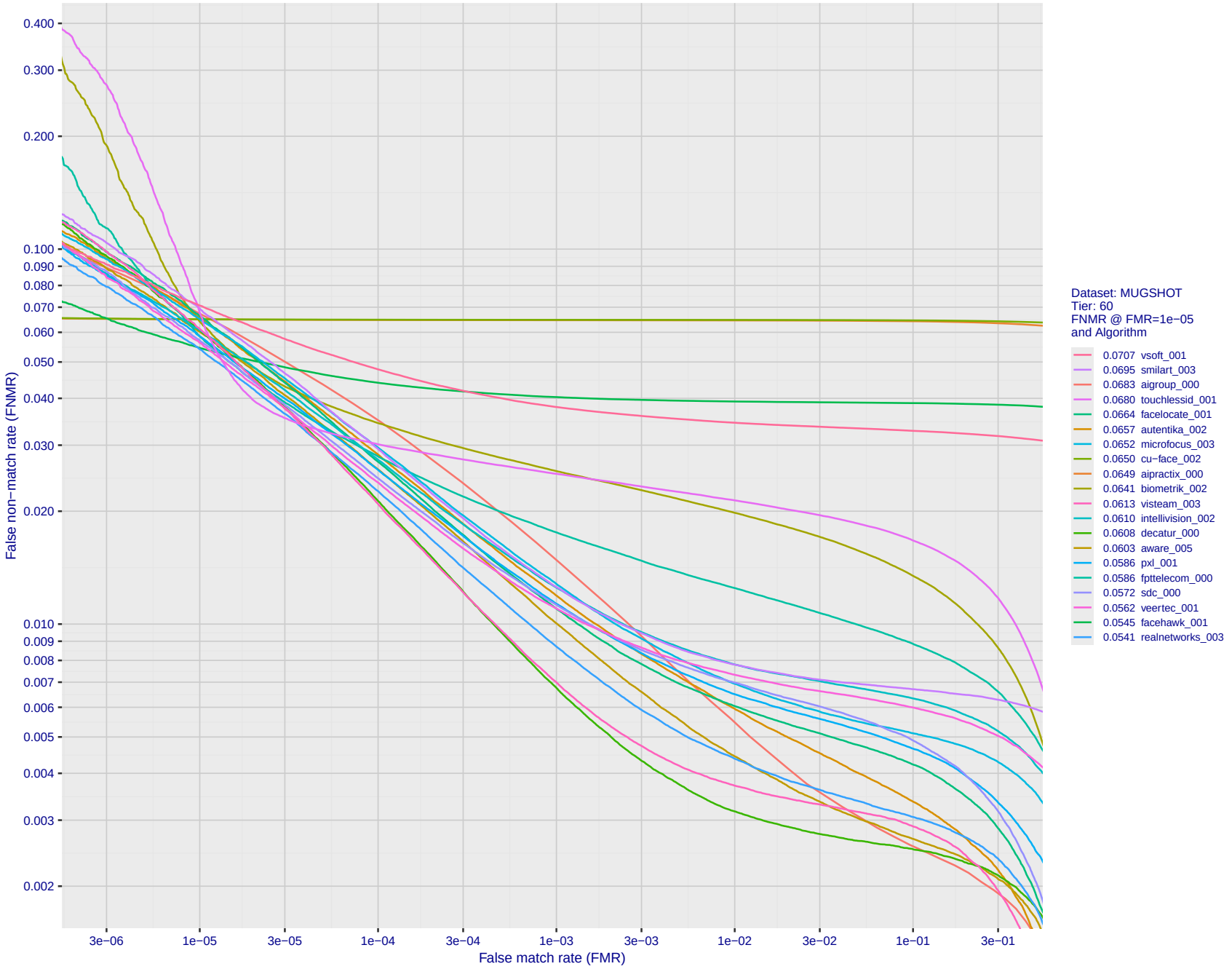

Figure 284: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

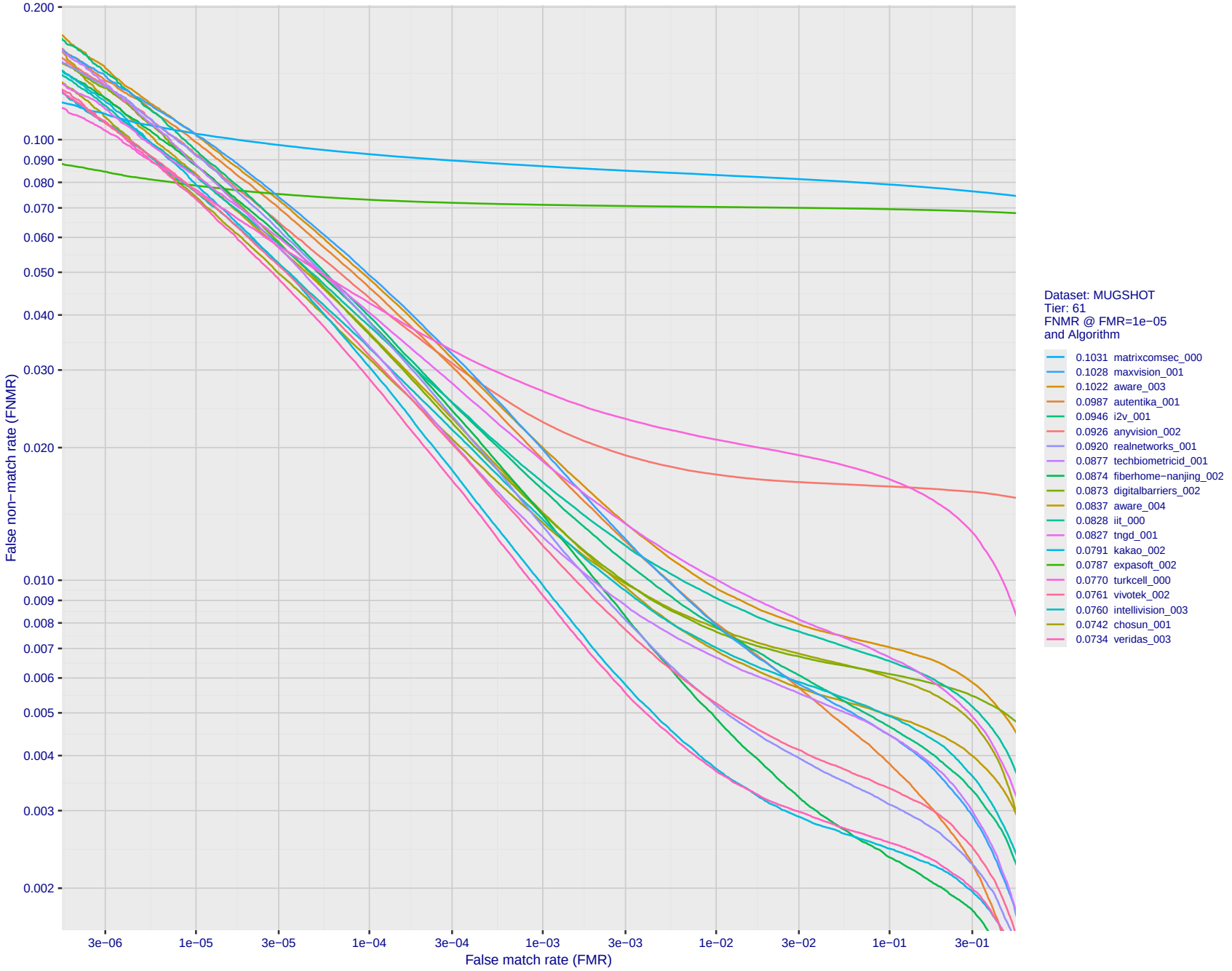

Figure 285: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

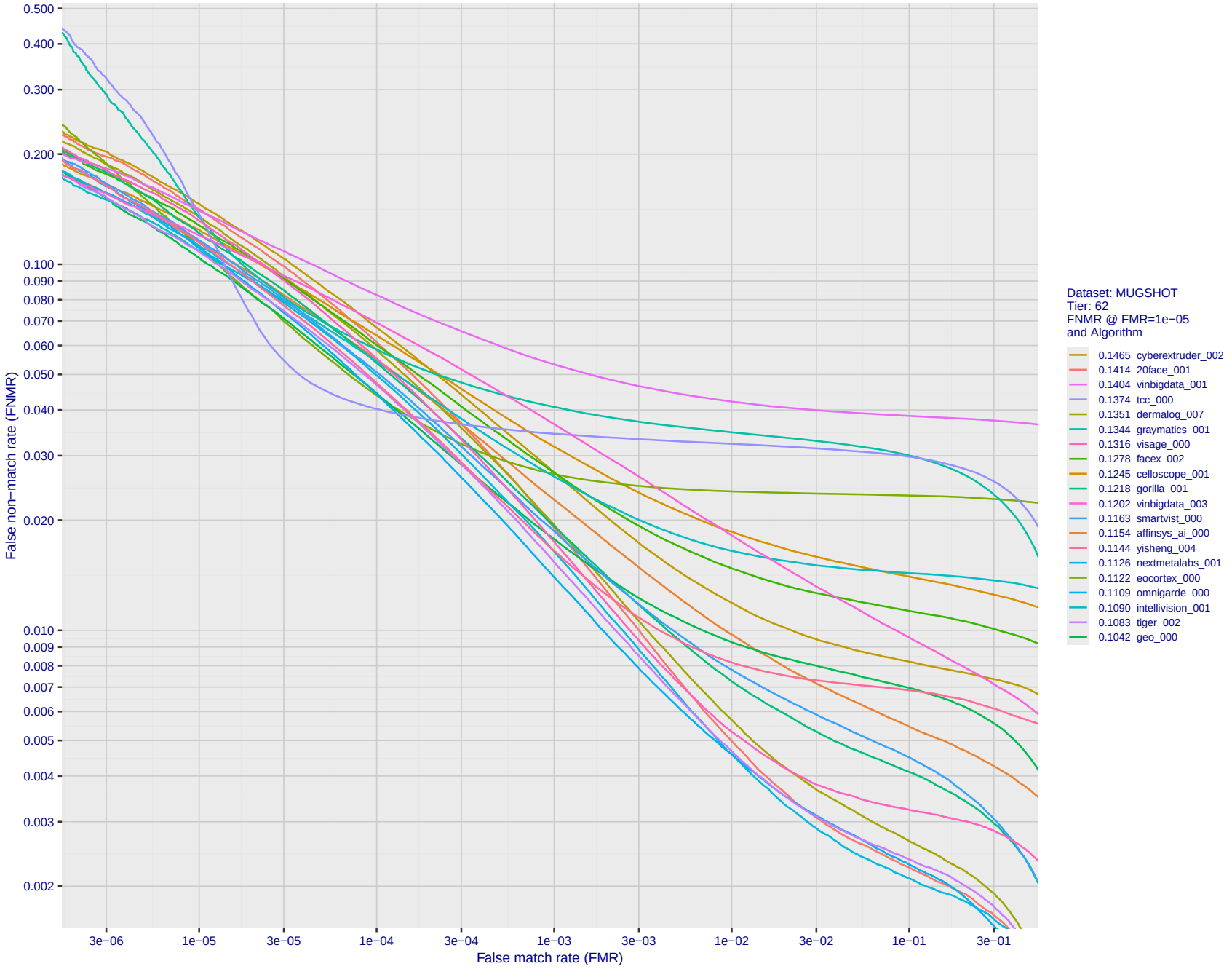

Figure 286: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

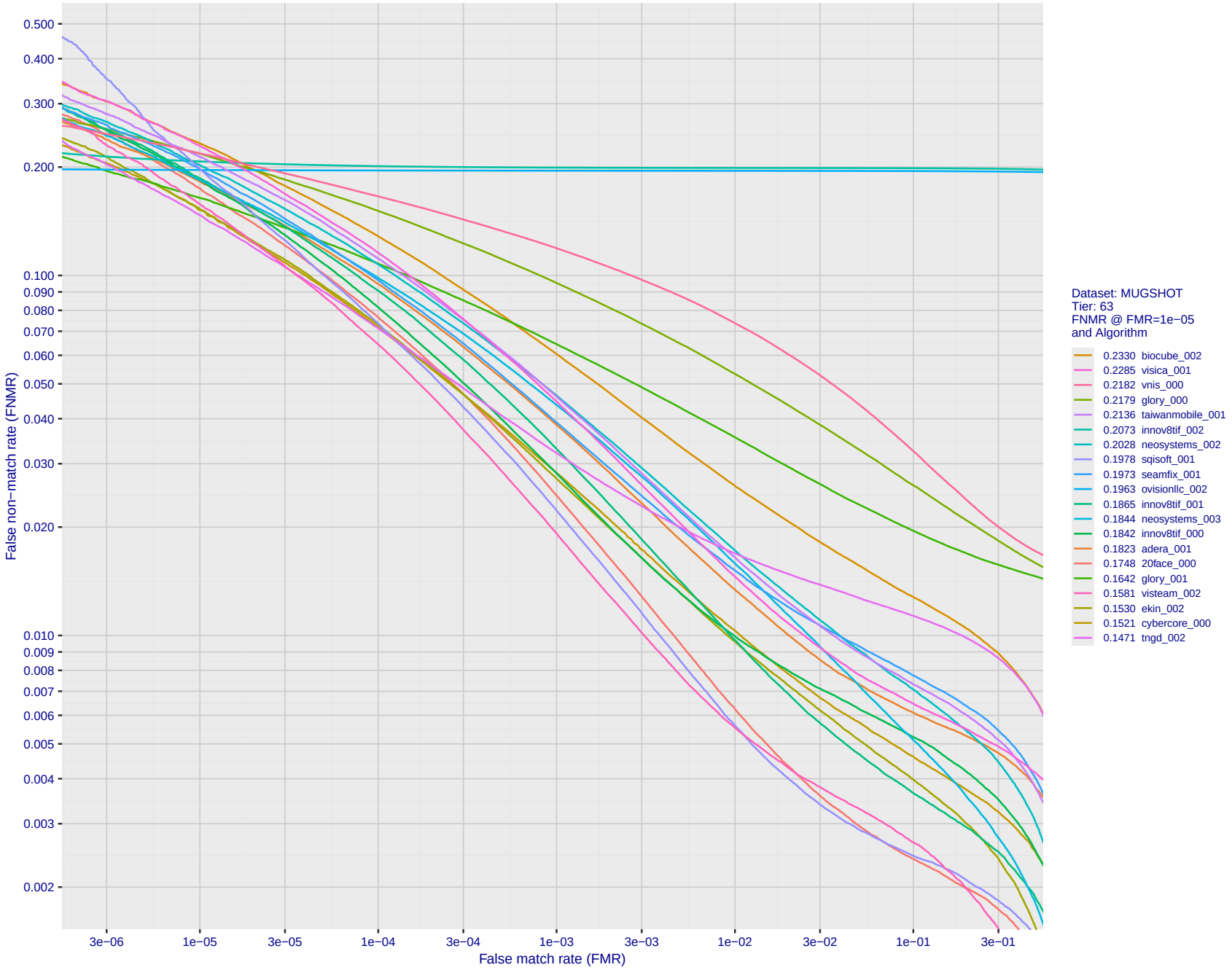

Figure 287: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

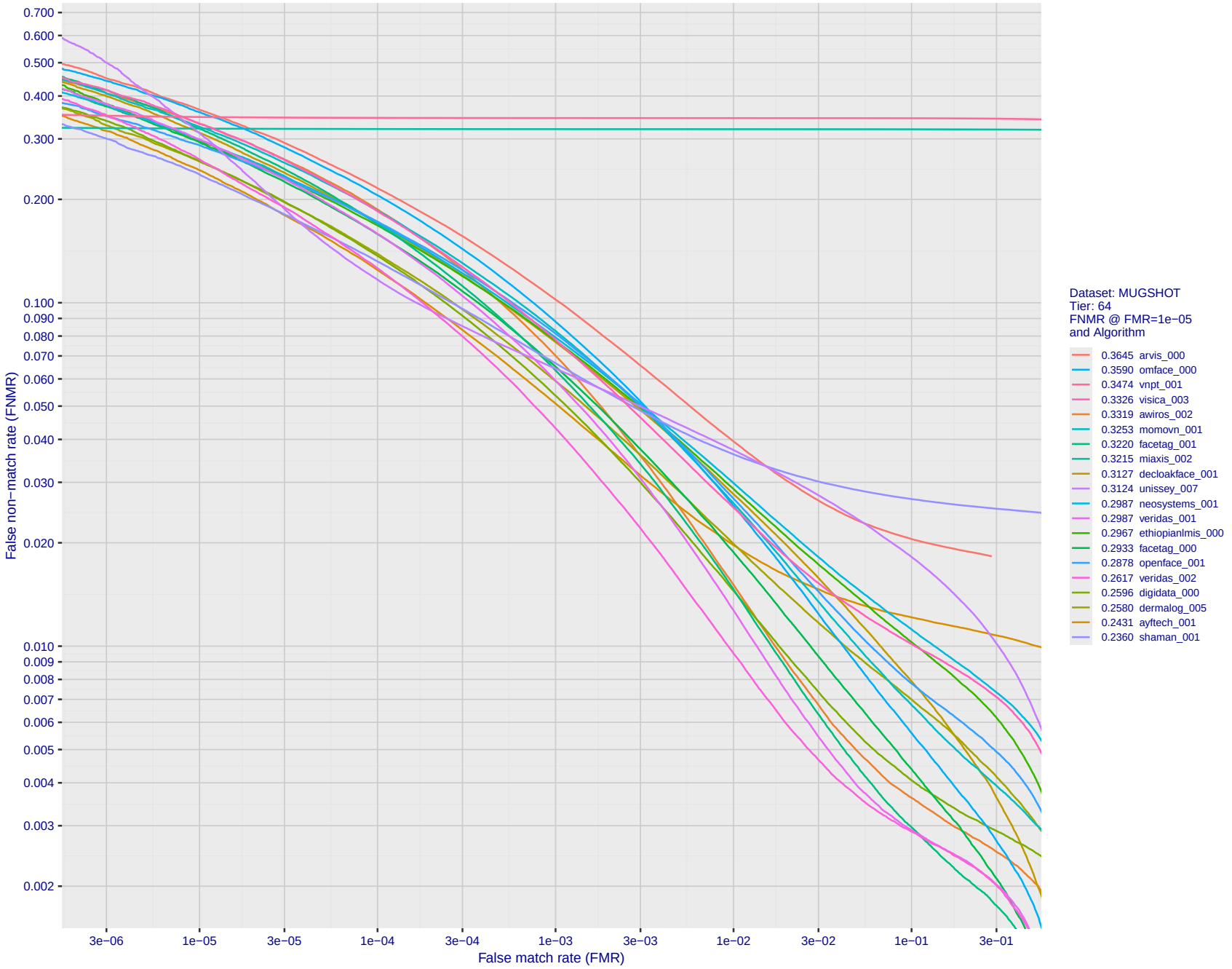

Figure 288: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

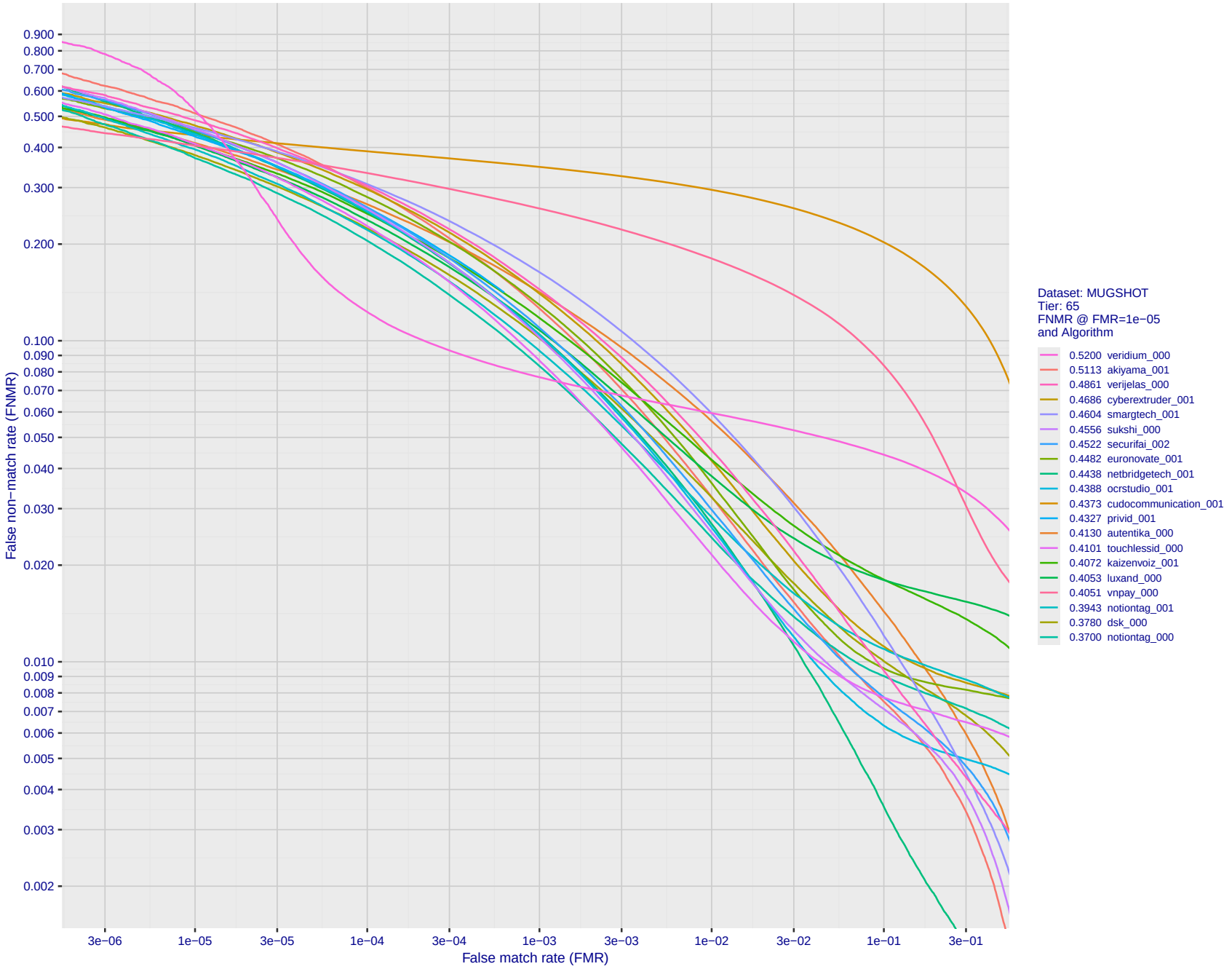

Figure 289: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

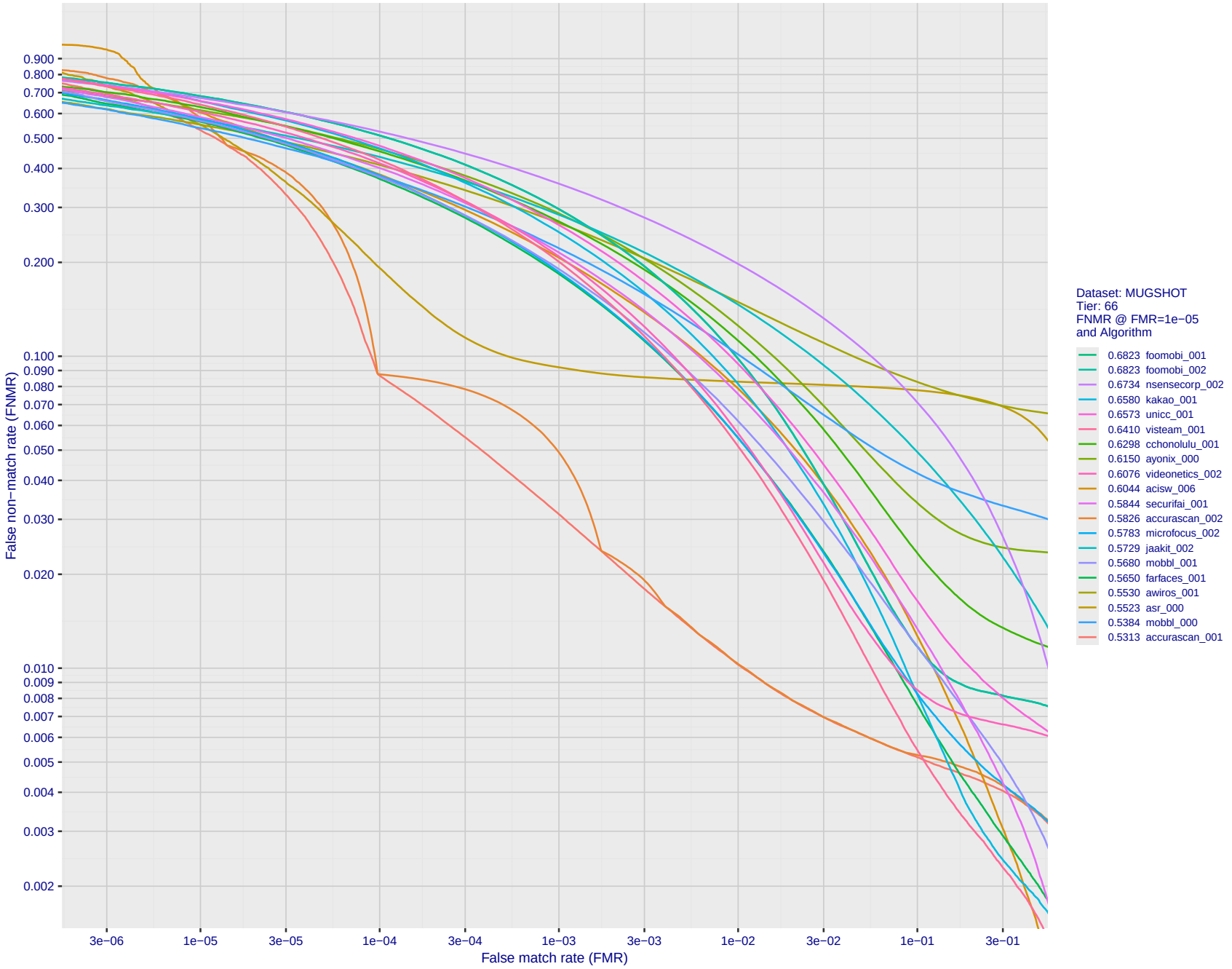

Figure 290: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

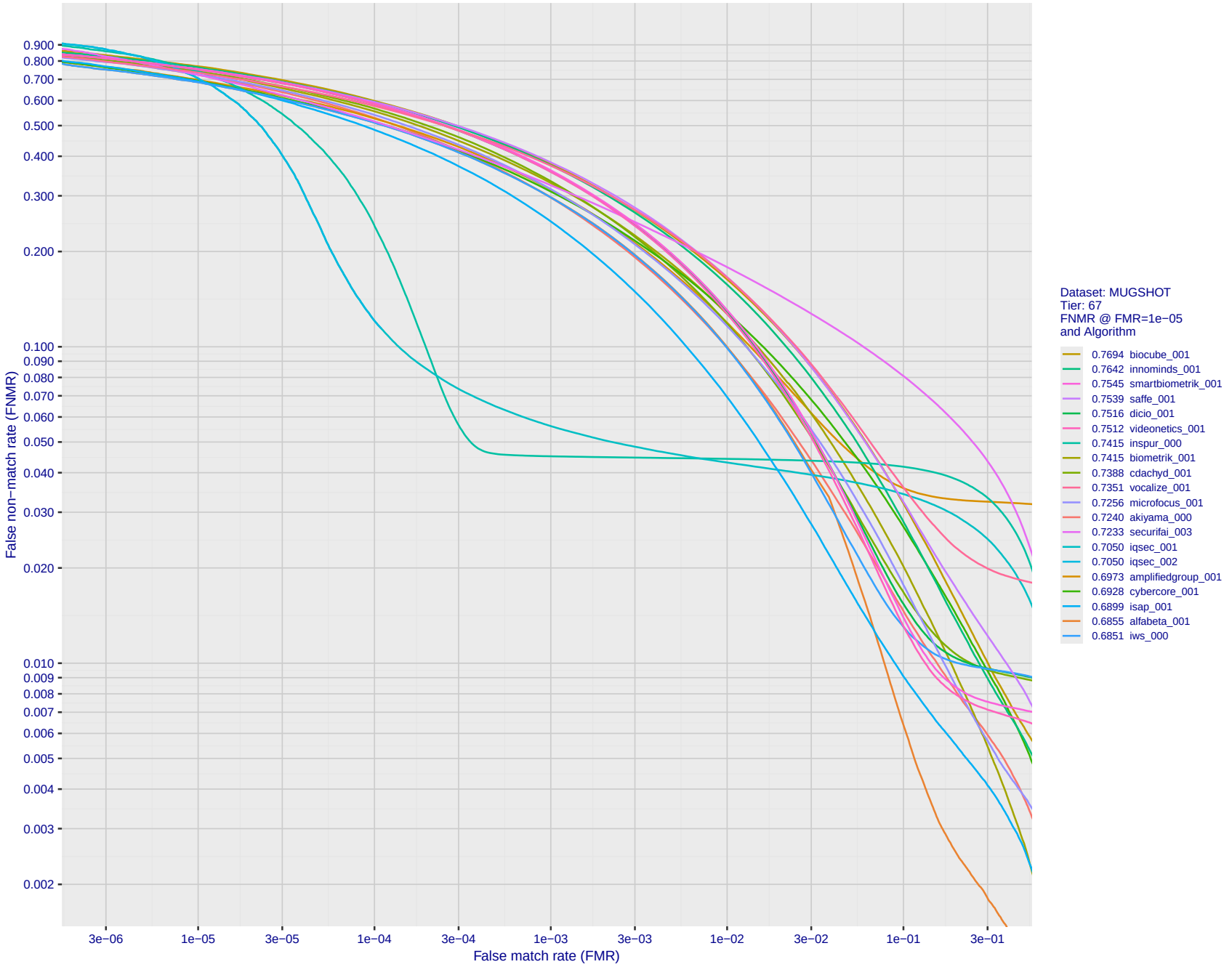

Figure 291: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

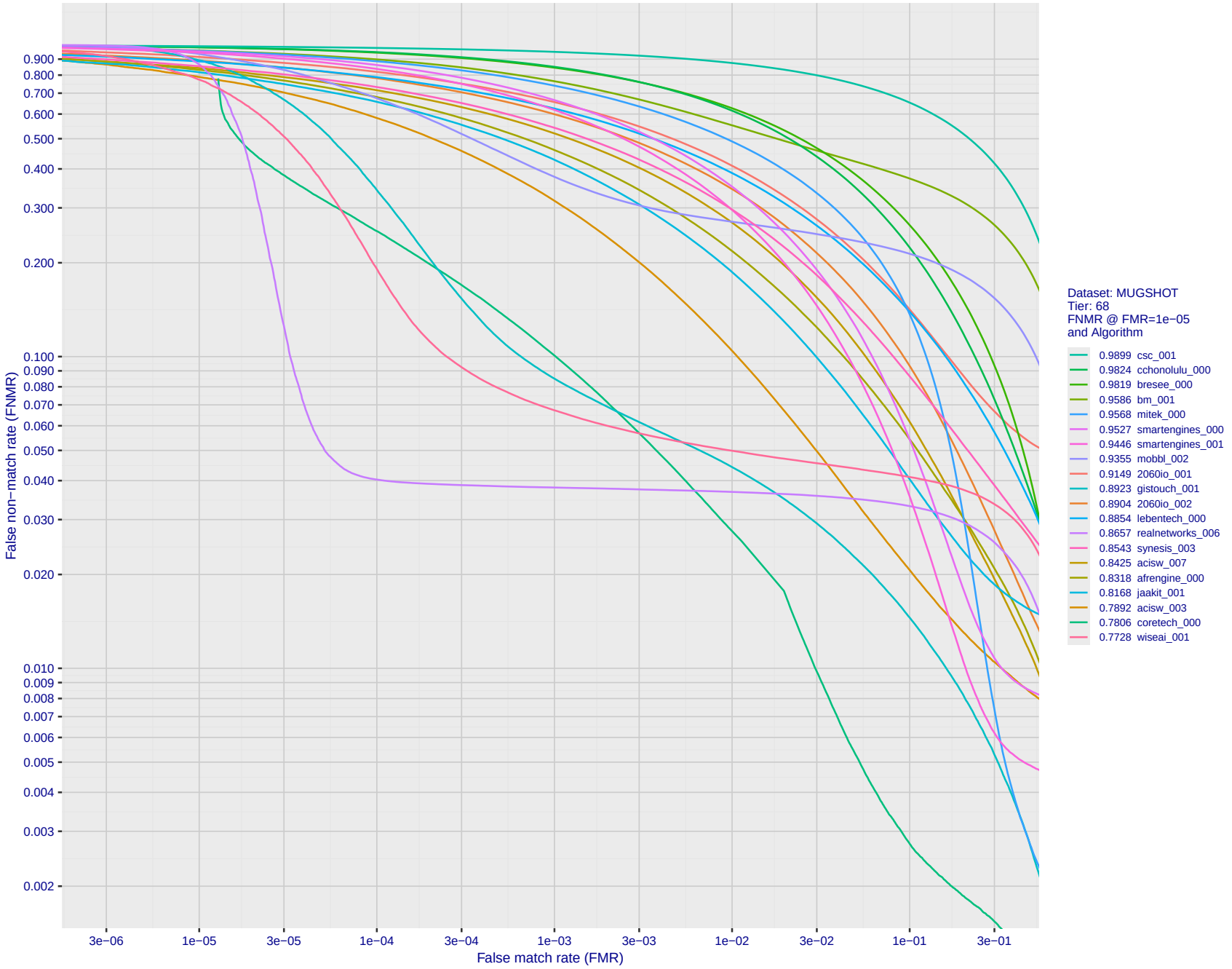

Figure 292: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

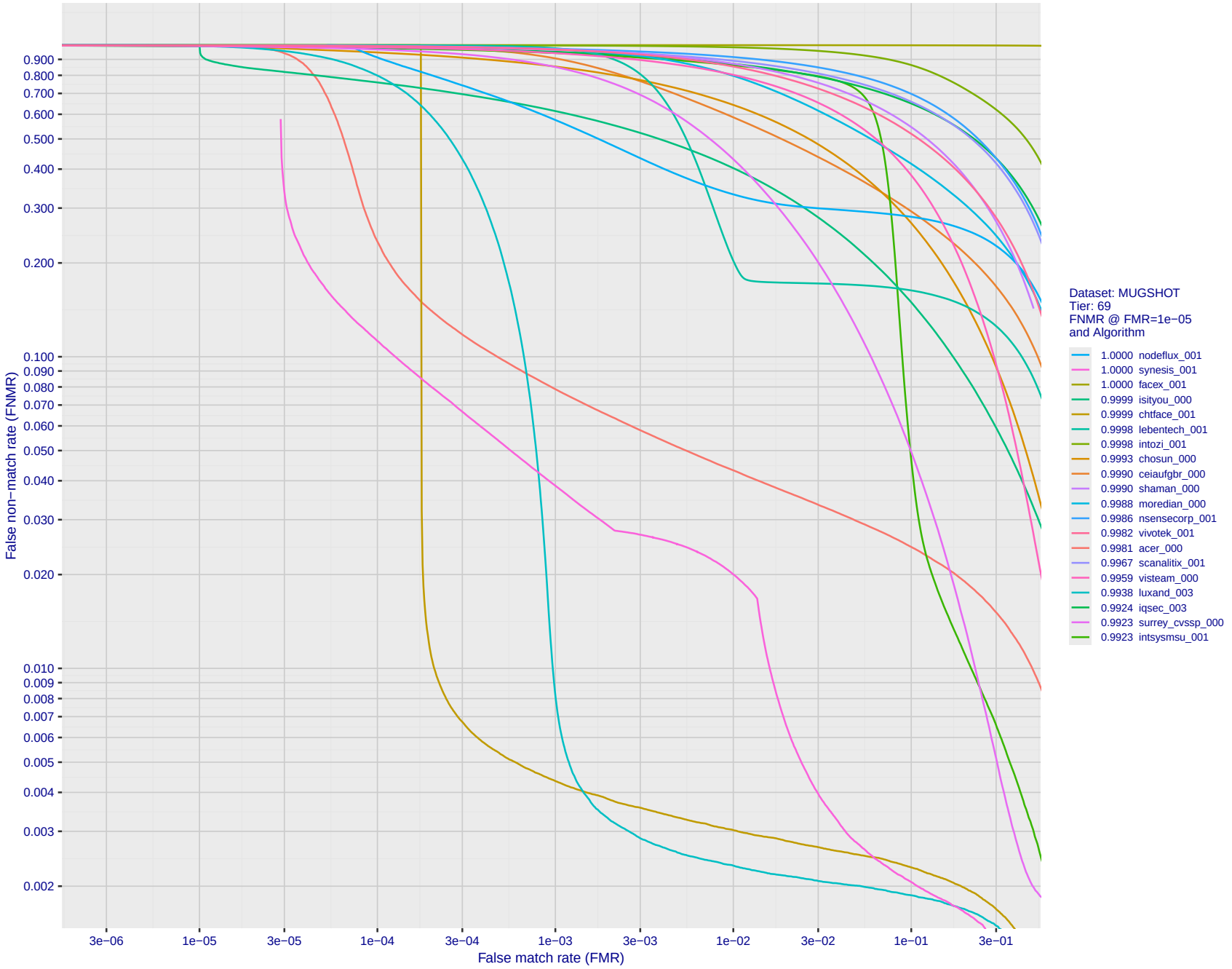

Figure 293: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

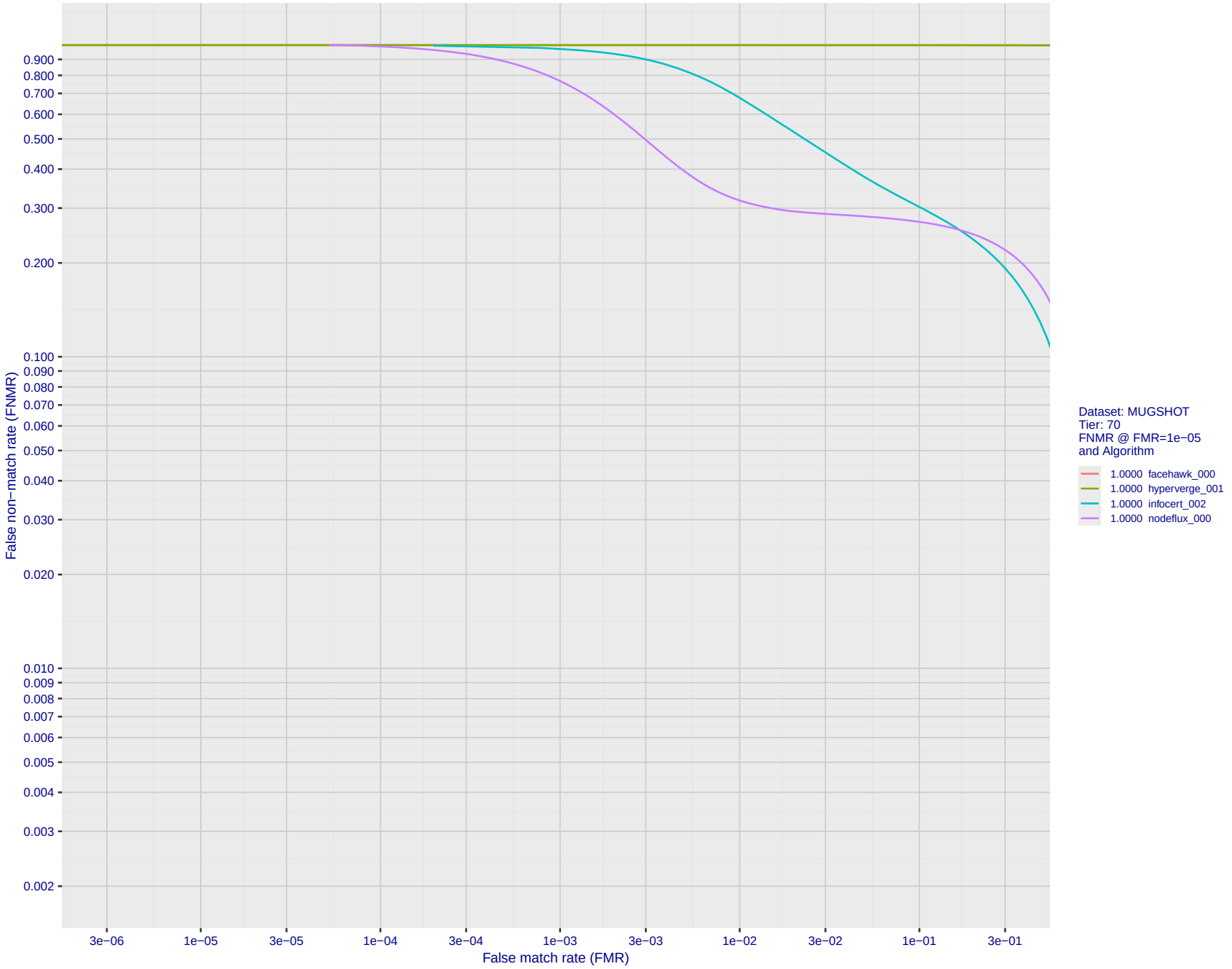

Figure 294: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

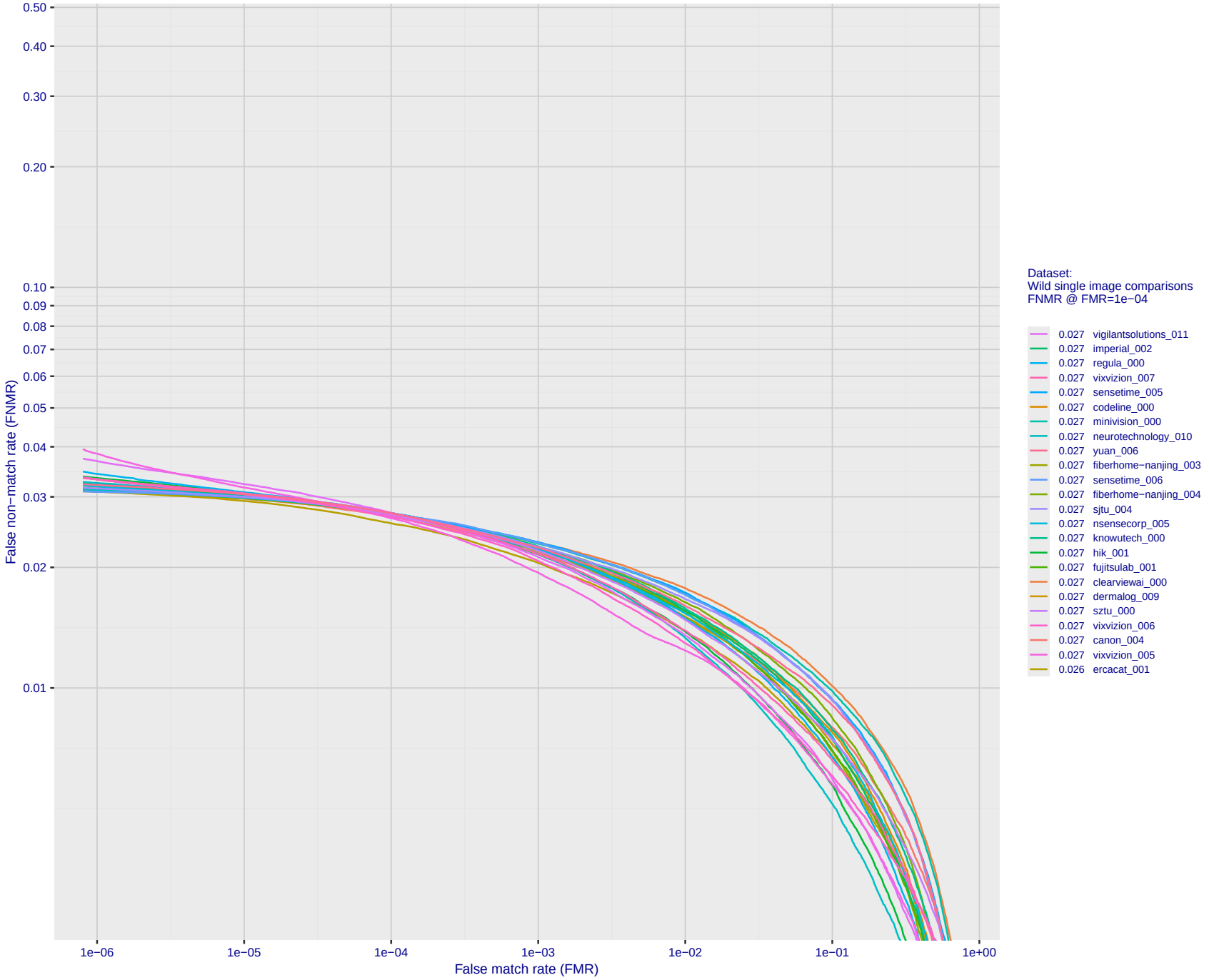

Figure 295: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

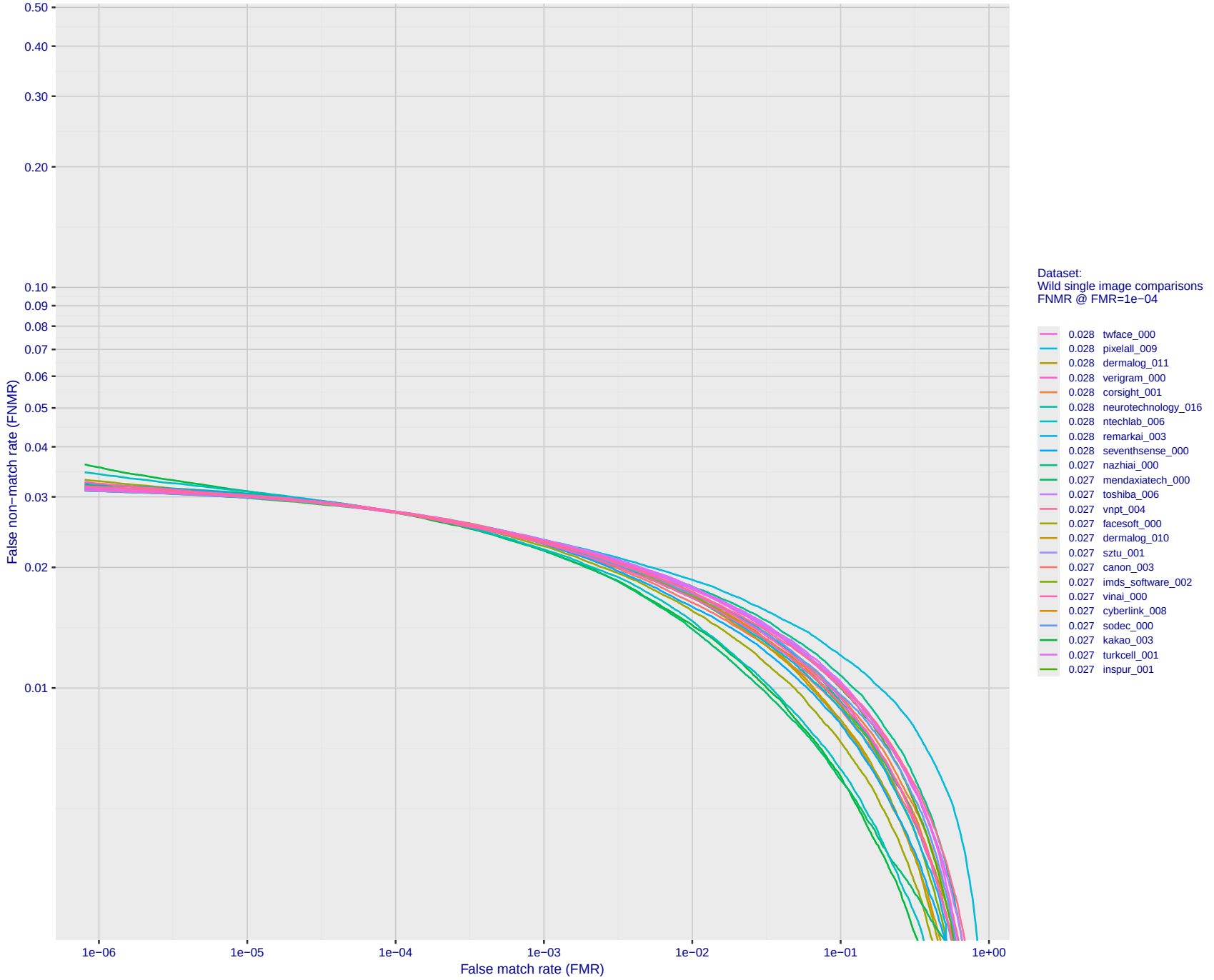

Figure 296: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

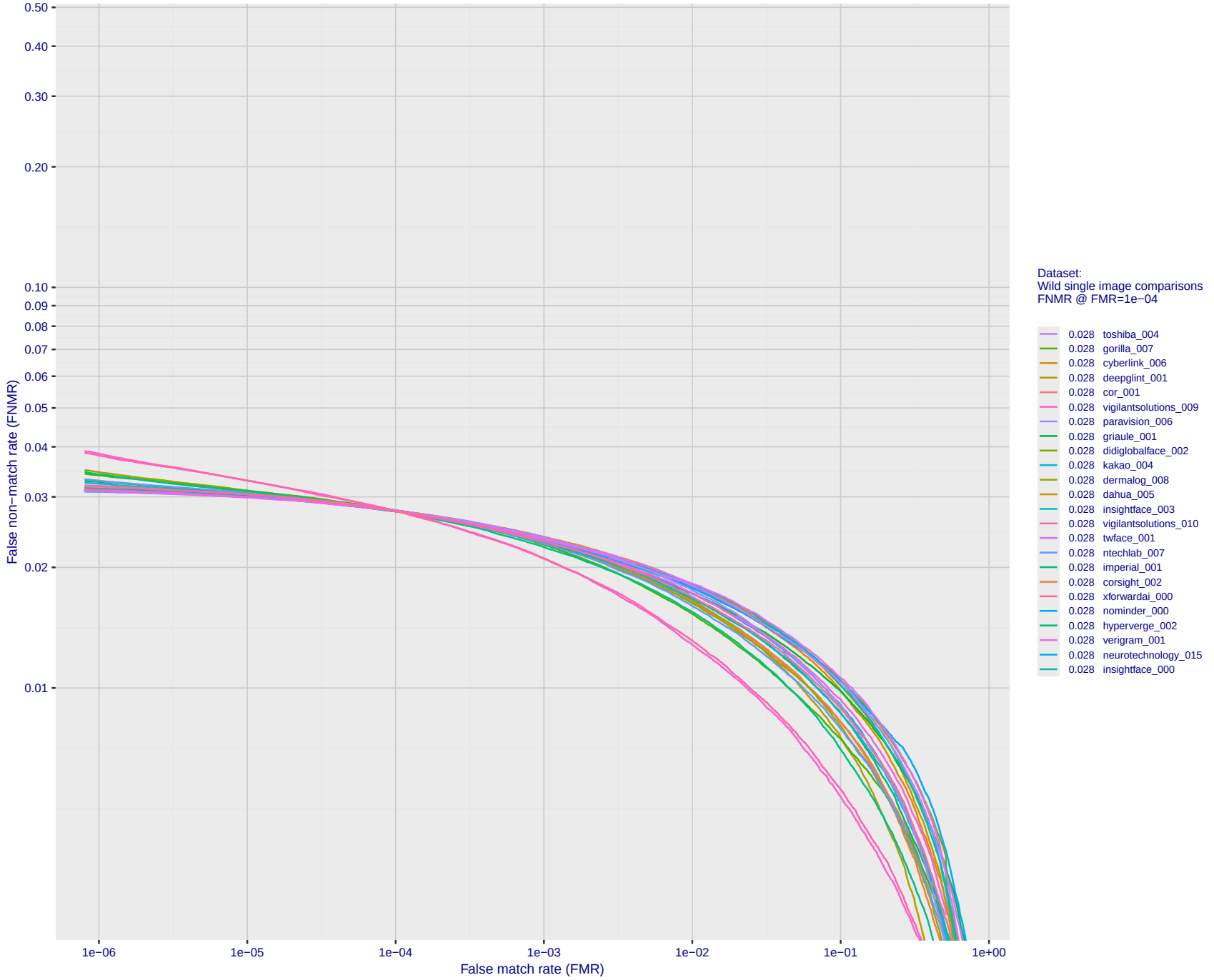

Figure 297: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

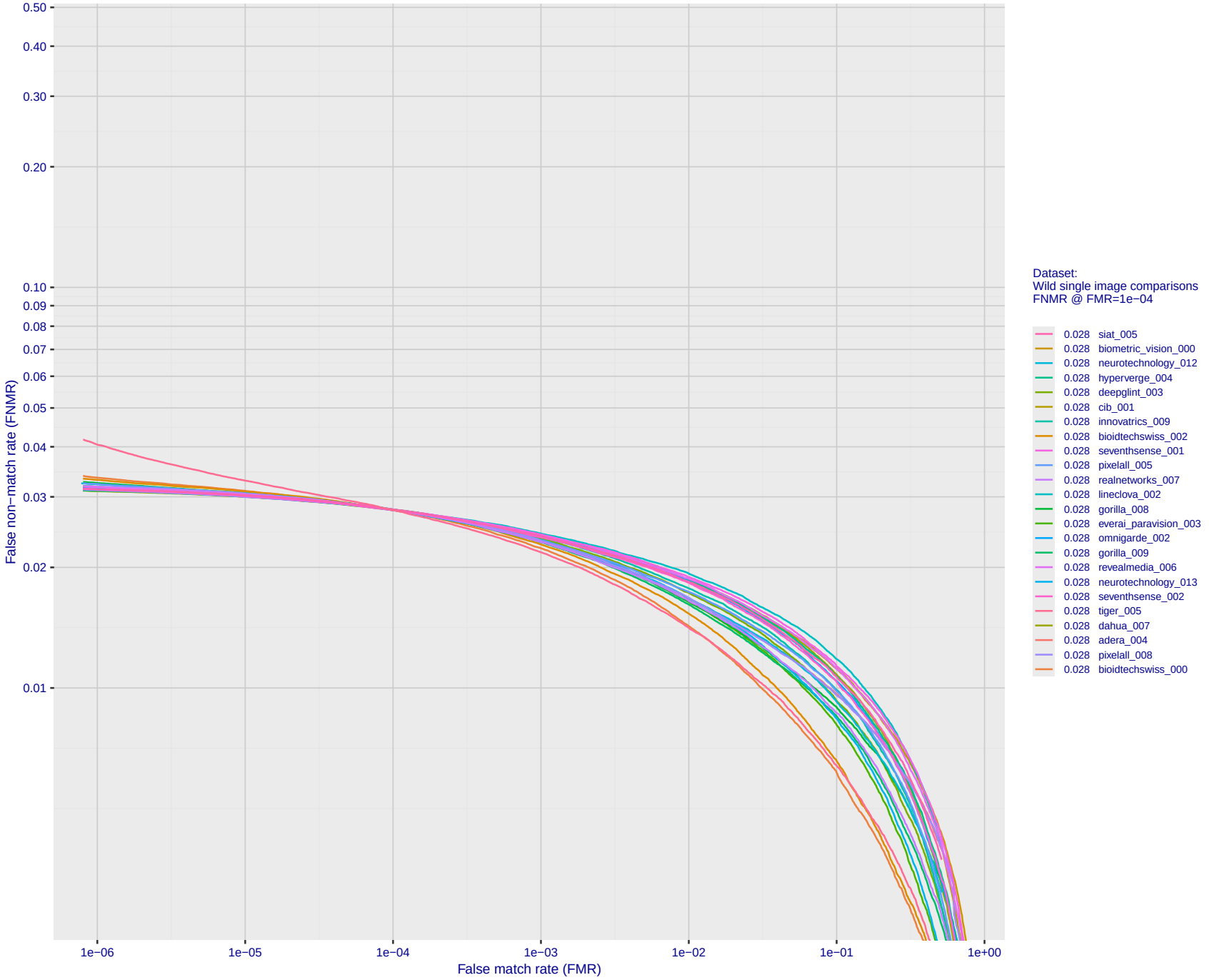

Figure 298: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

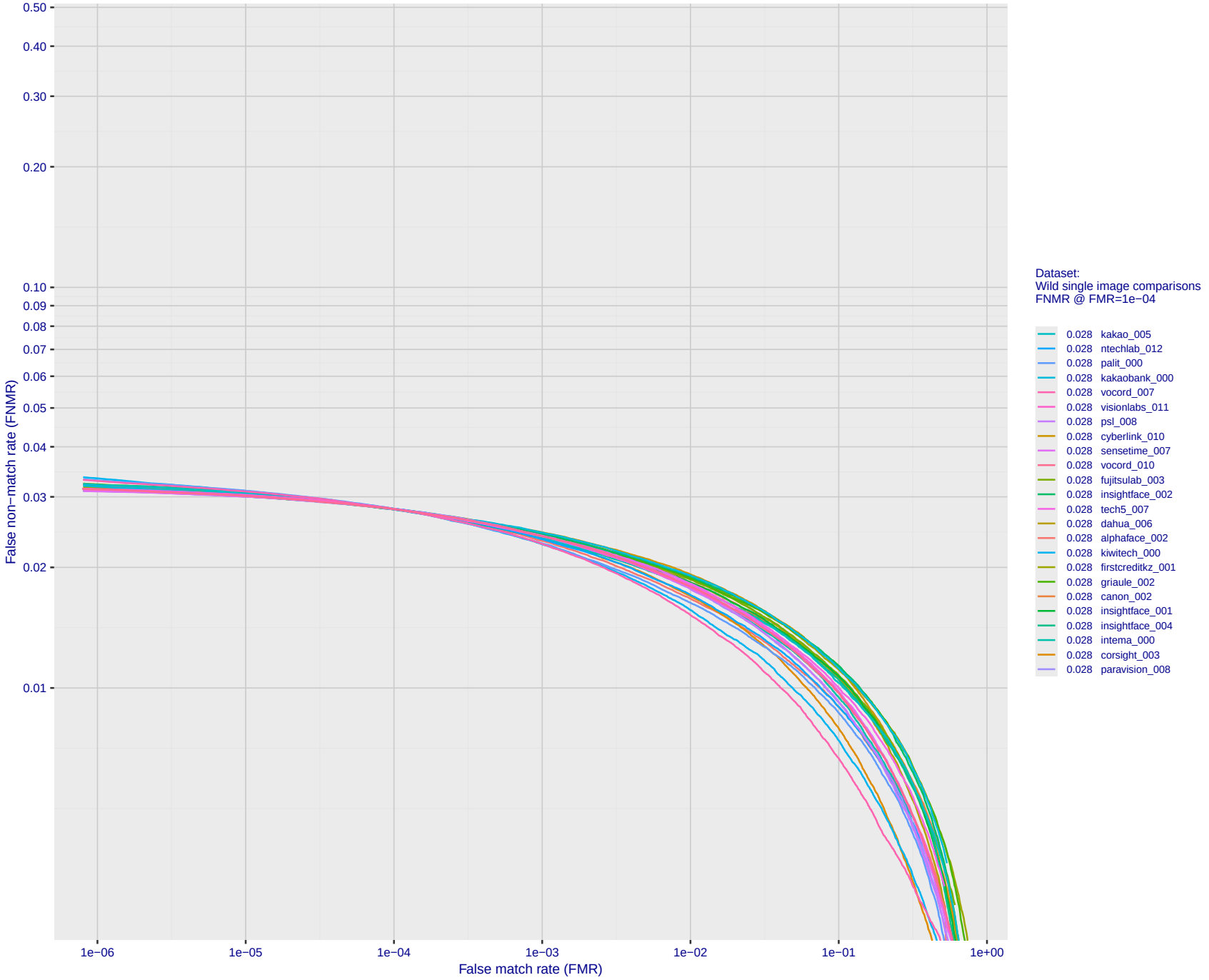

Figure 299: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

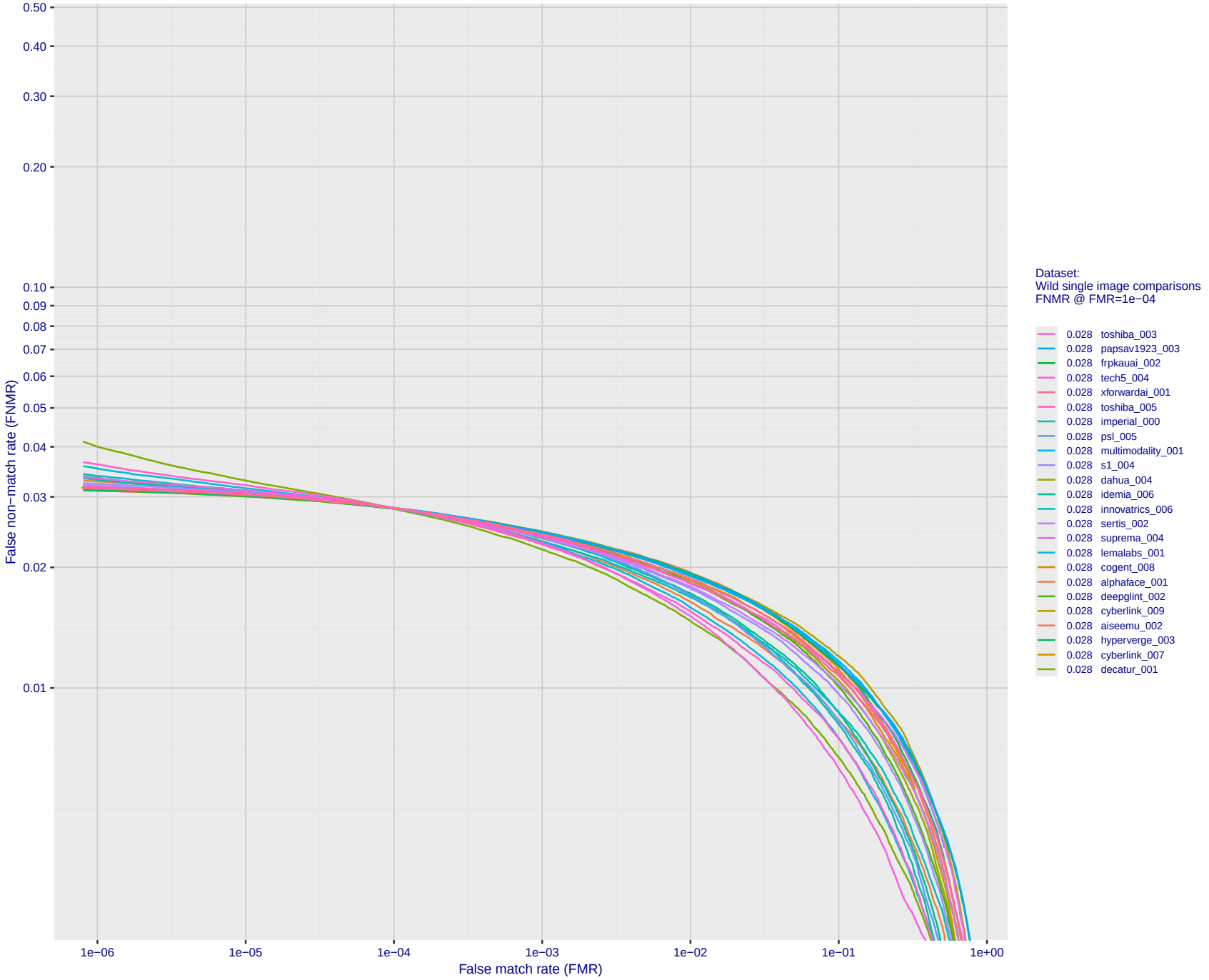

Figure 300: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

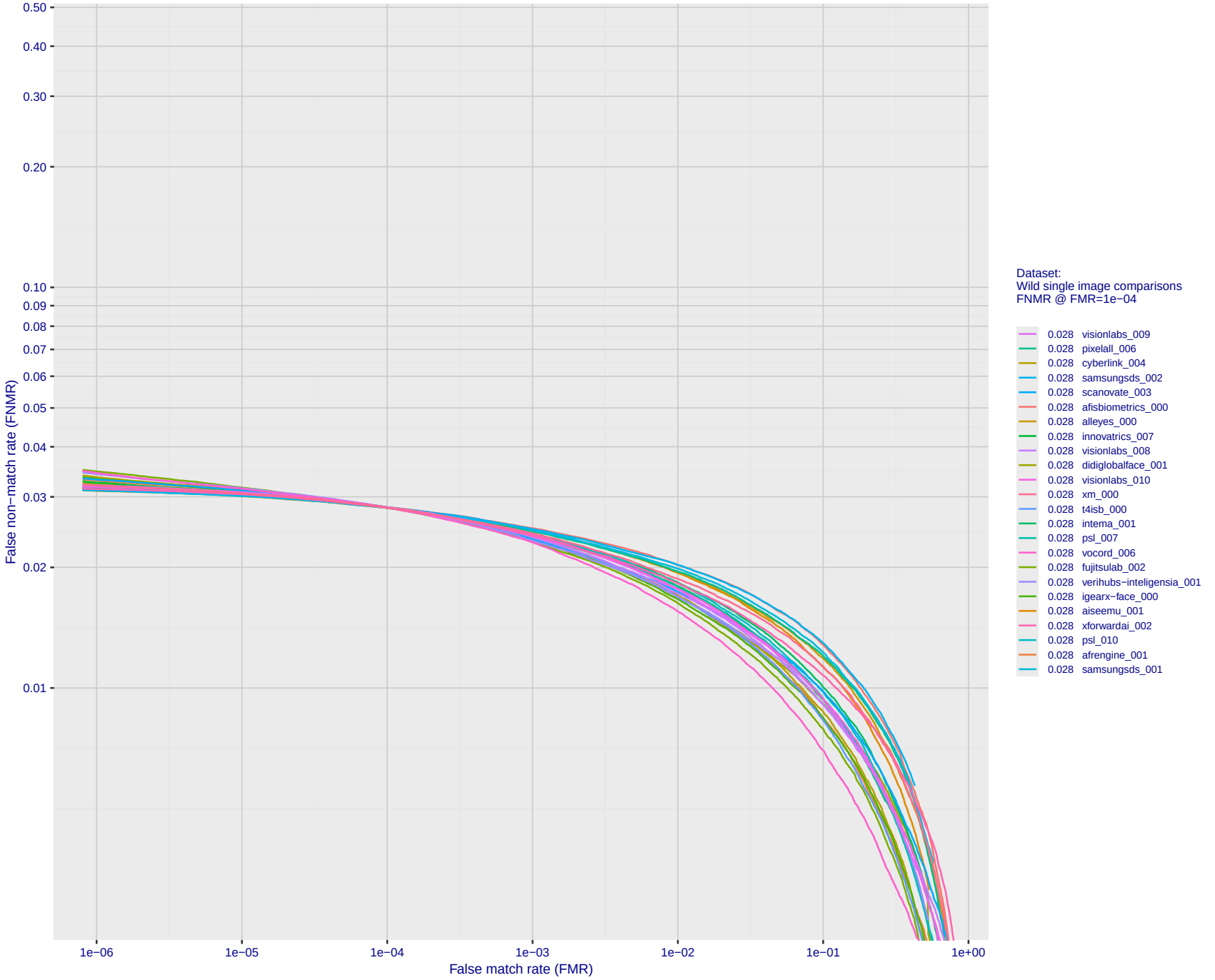

Figure 301: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

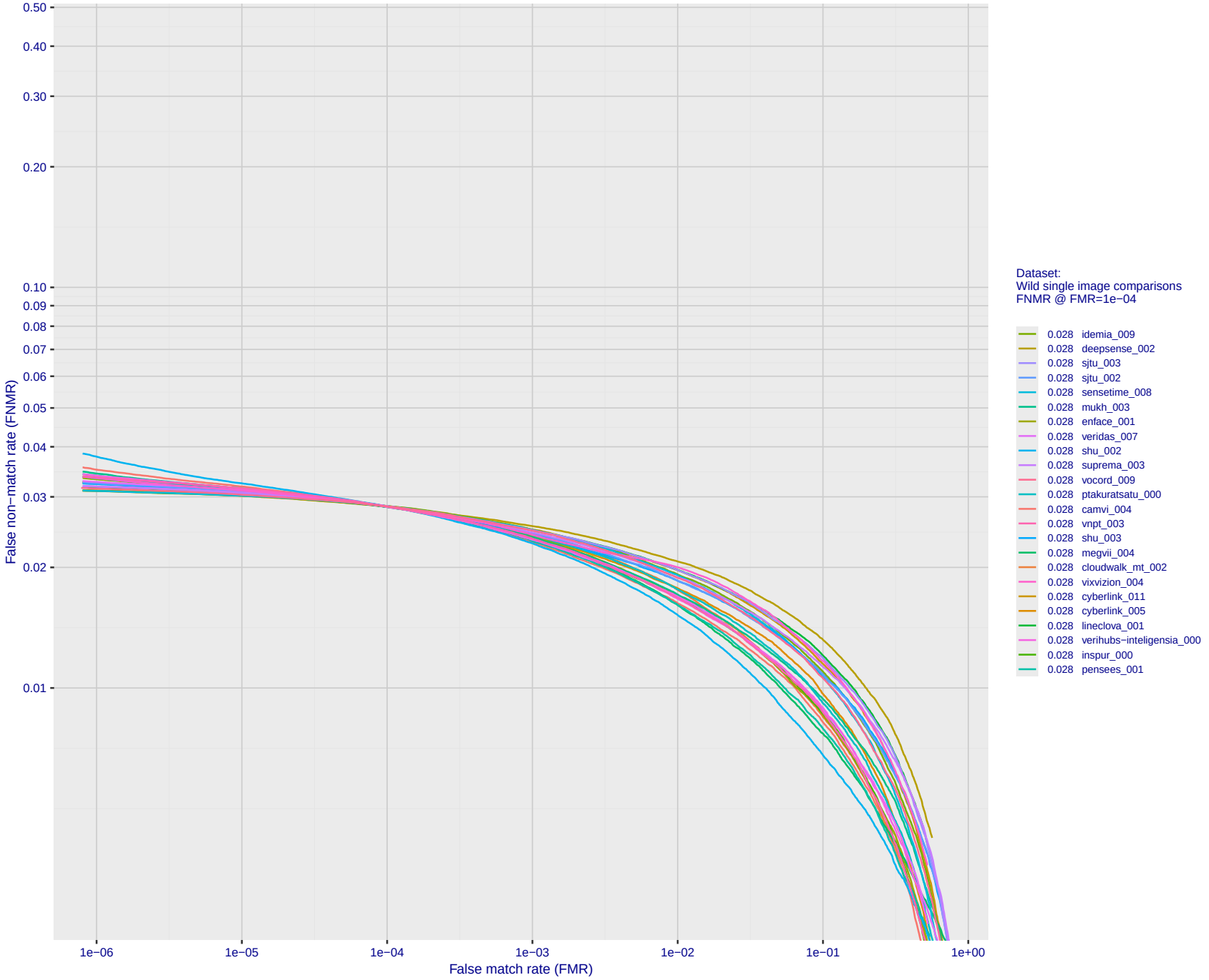

Figure 302: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

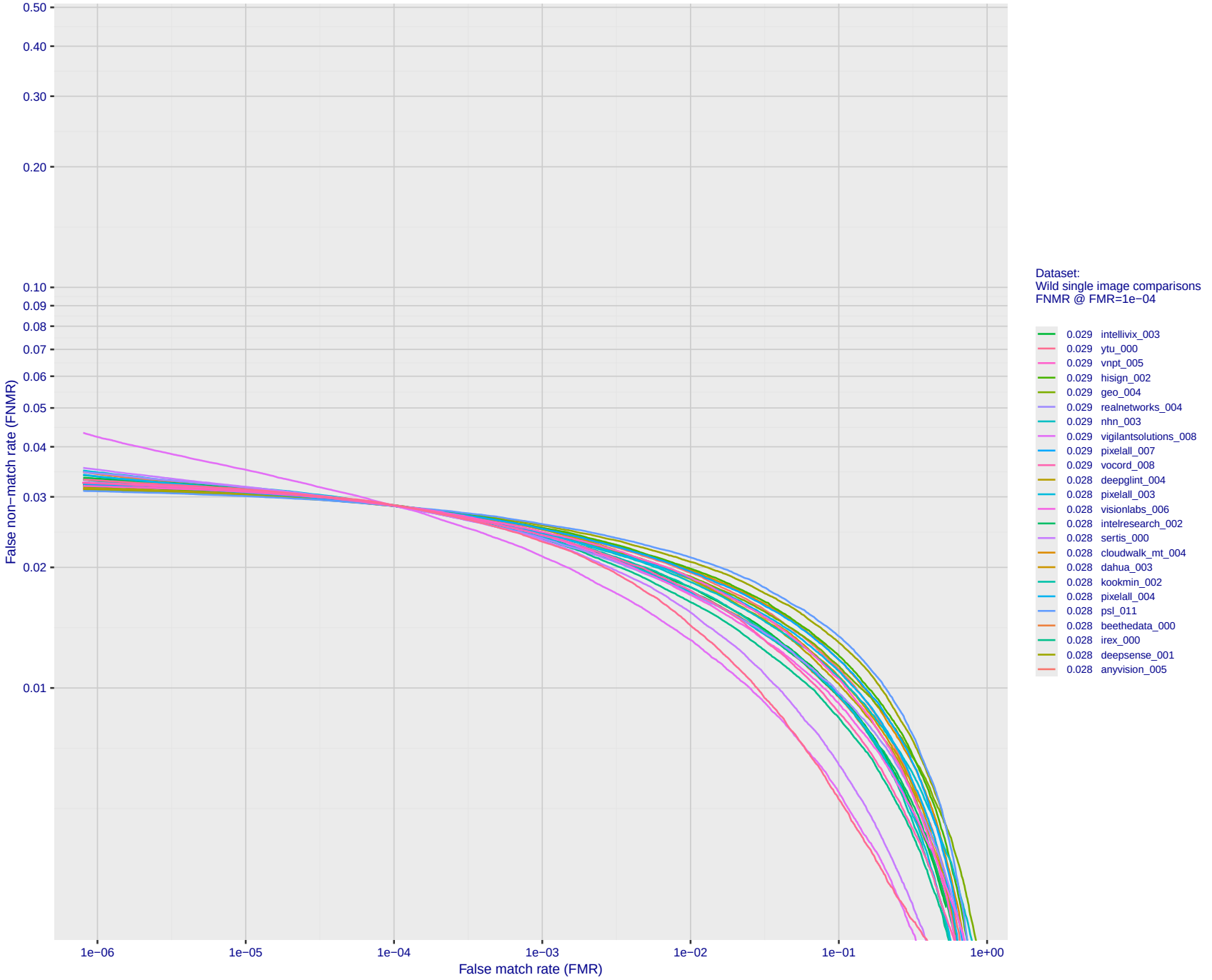

Figure 303: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27 FNMR(T) "False non-match rate" FMR(T) "False match rate"

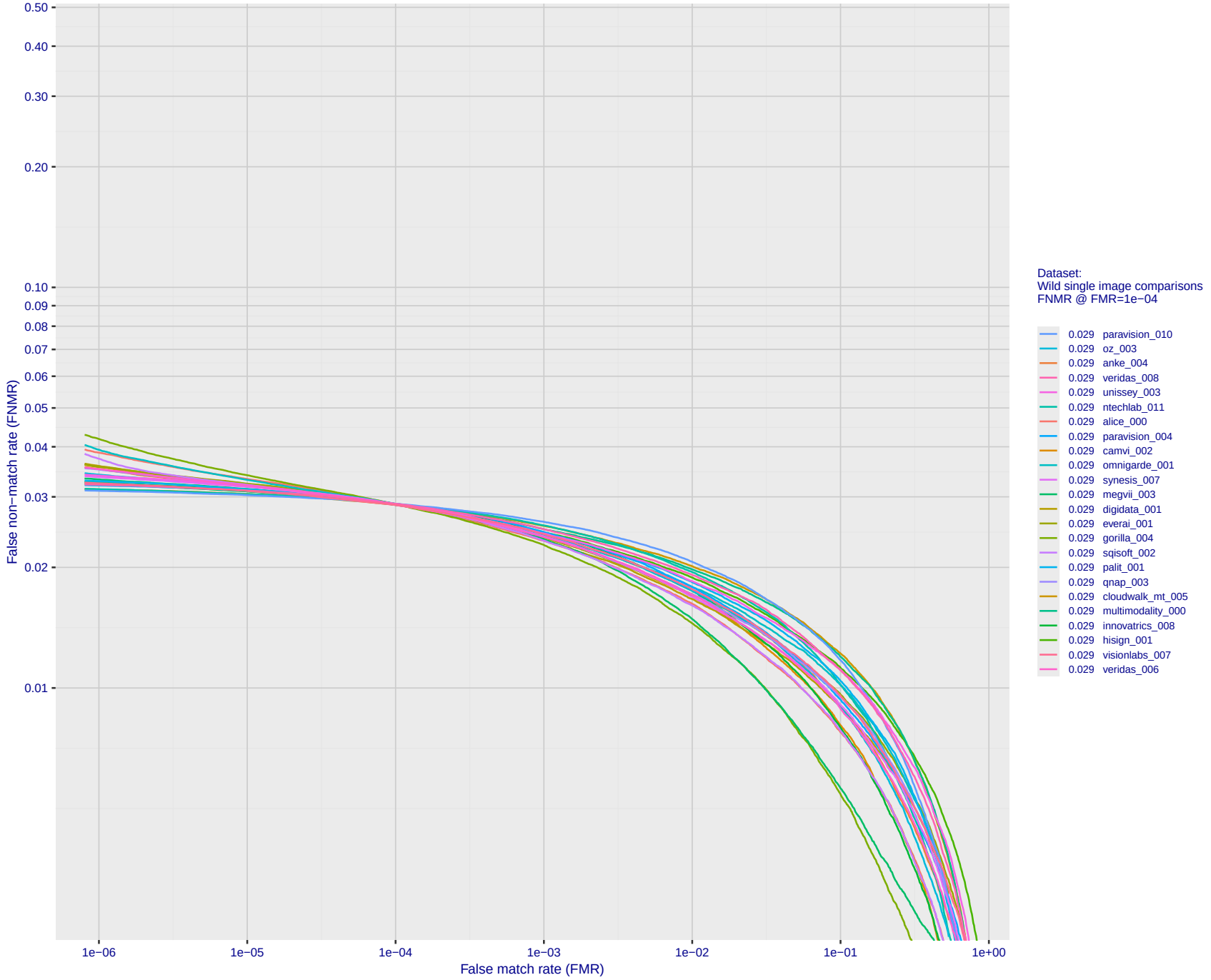

Figure 304: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27 FNMR(T) "False non-match rate" FMR(T) "False match rate"

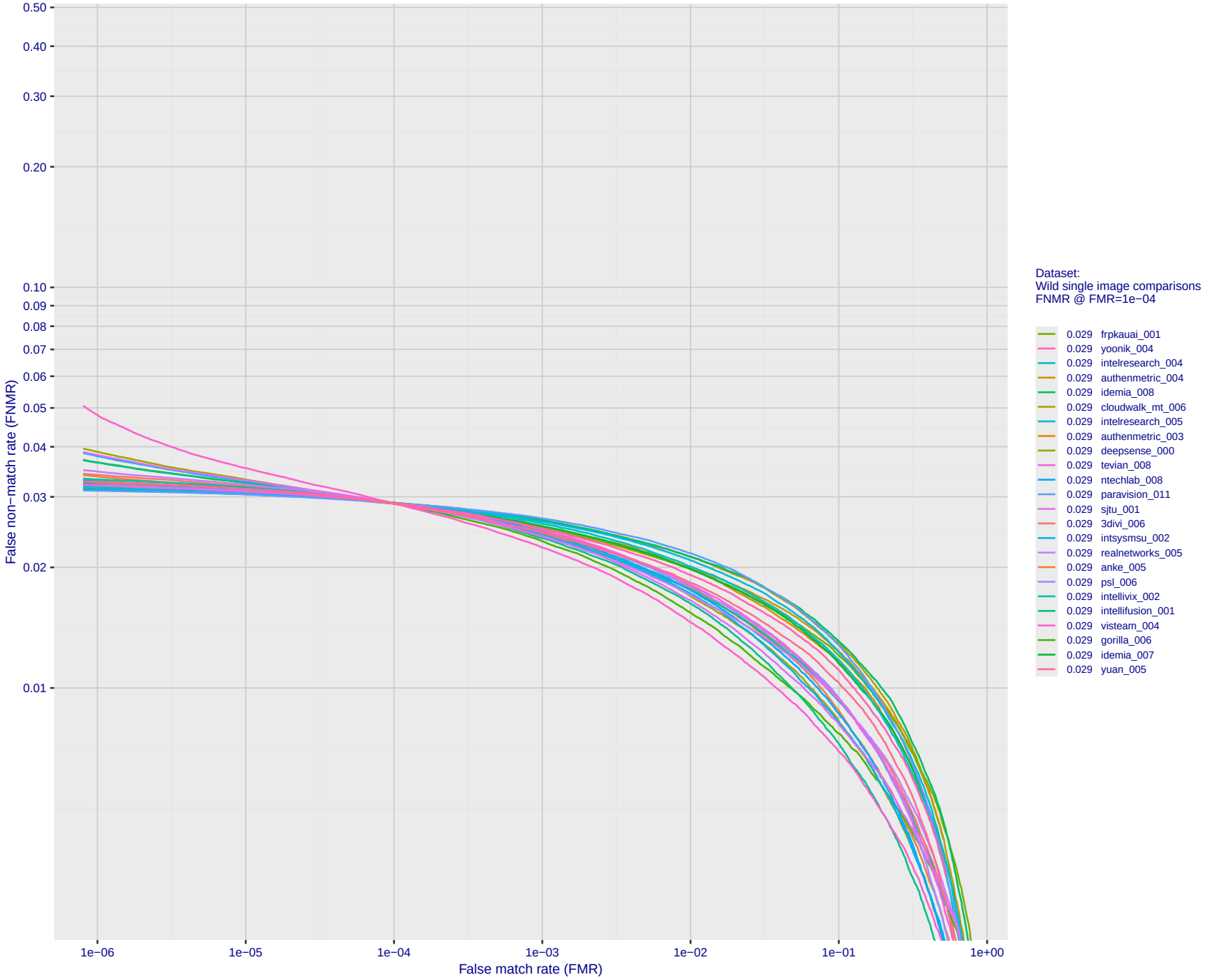

Figure 305: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27 FNMR(T) "False non-match rate" FMR(T) "False match rate"

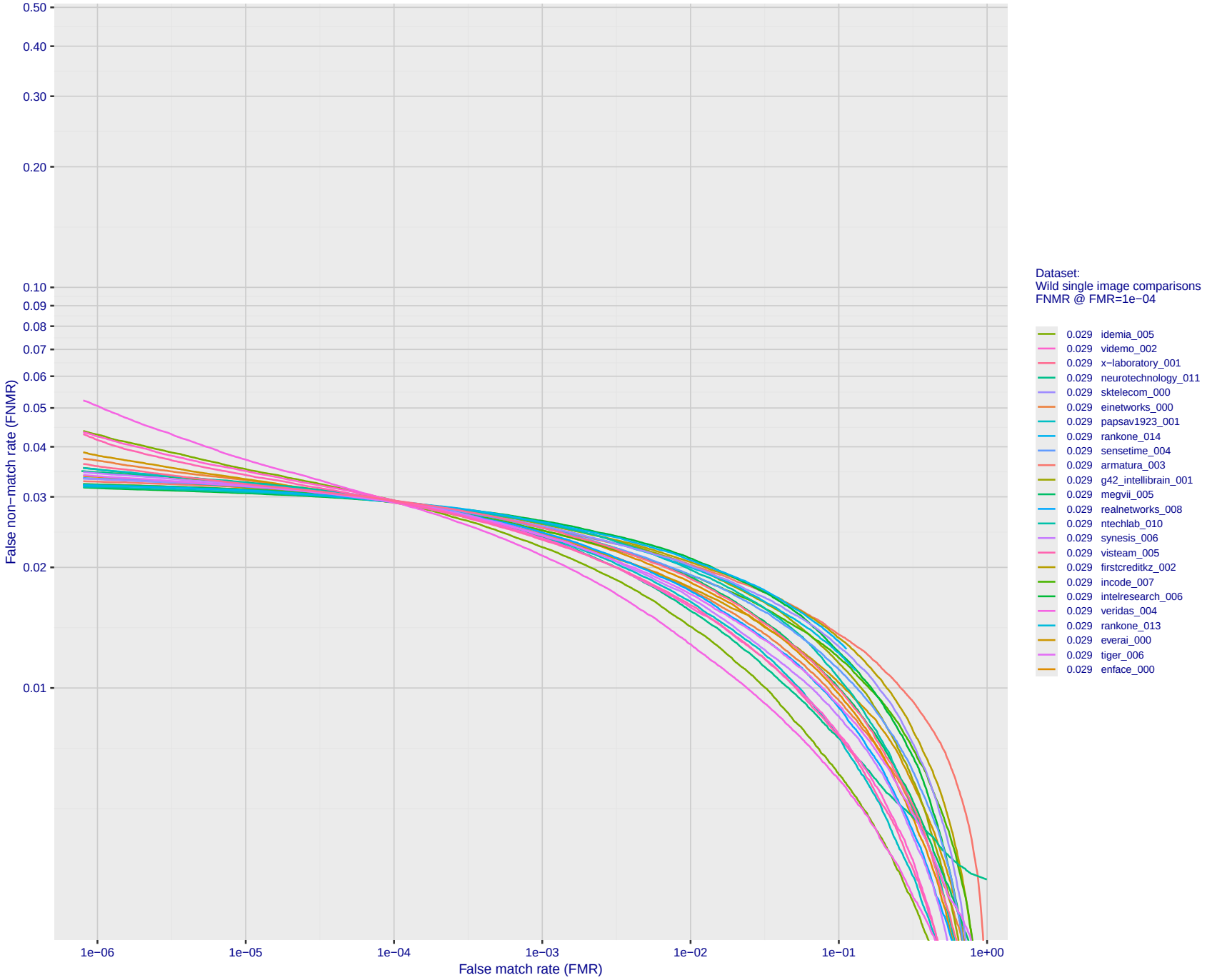

Figure 306: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

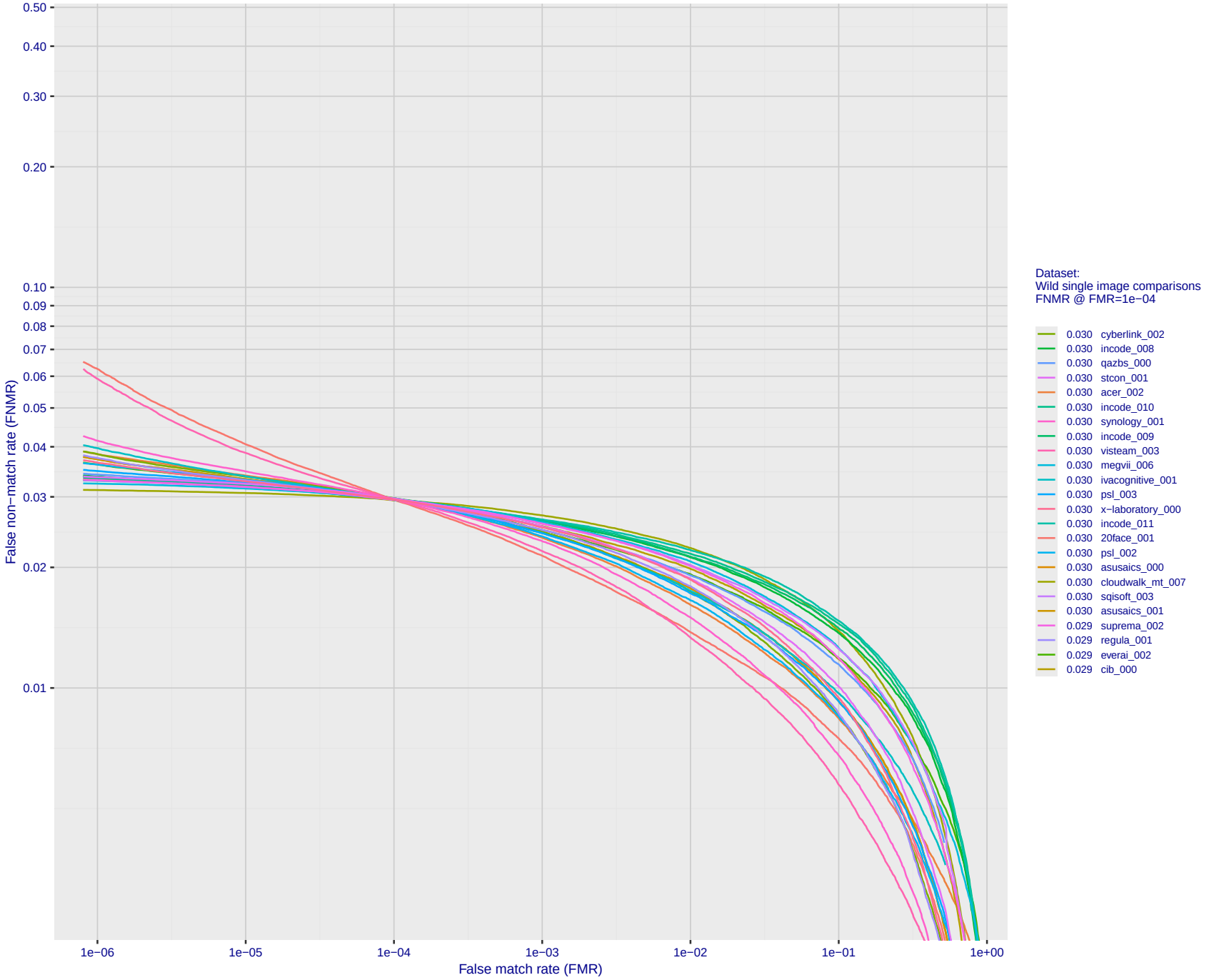

Figure 307: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

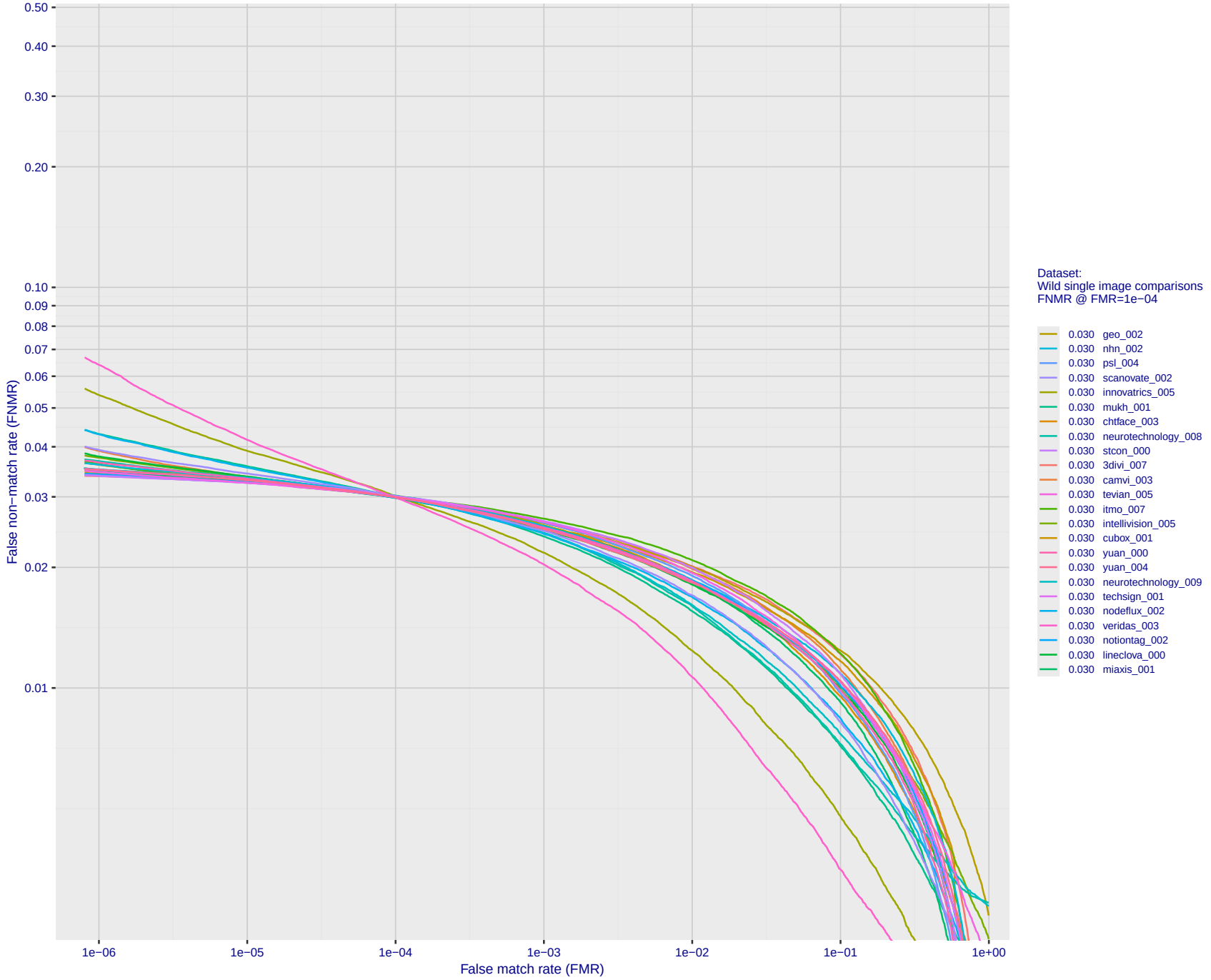

Figure 308: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

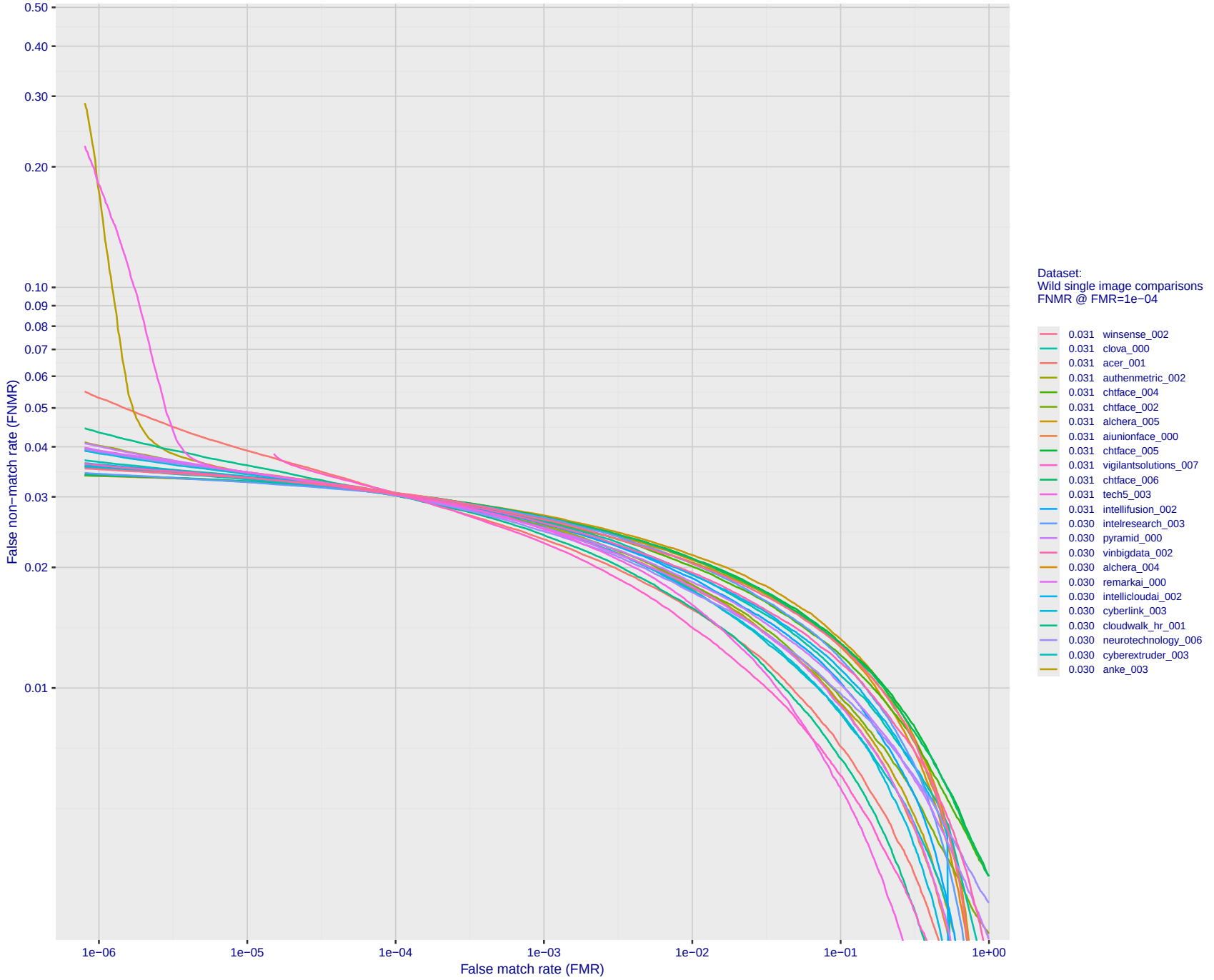

Figure 309: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

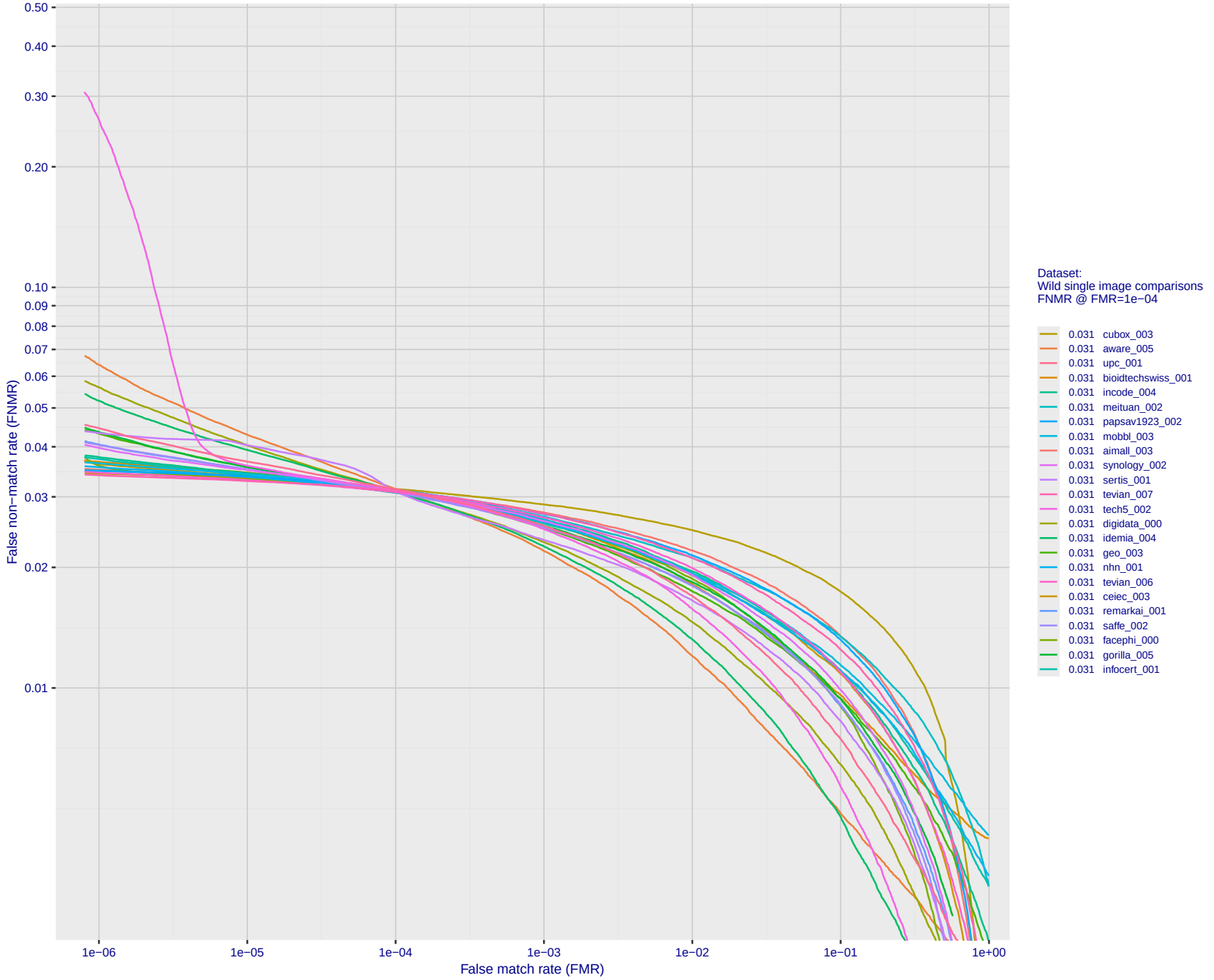

Figure 310: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

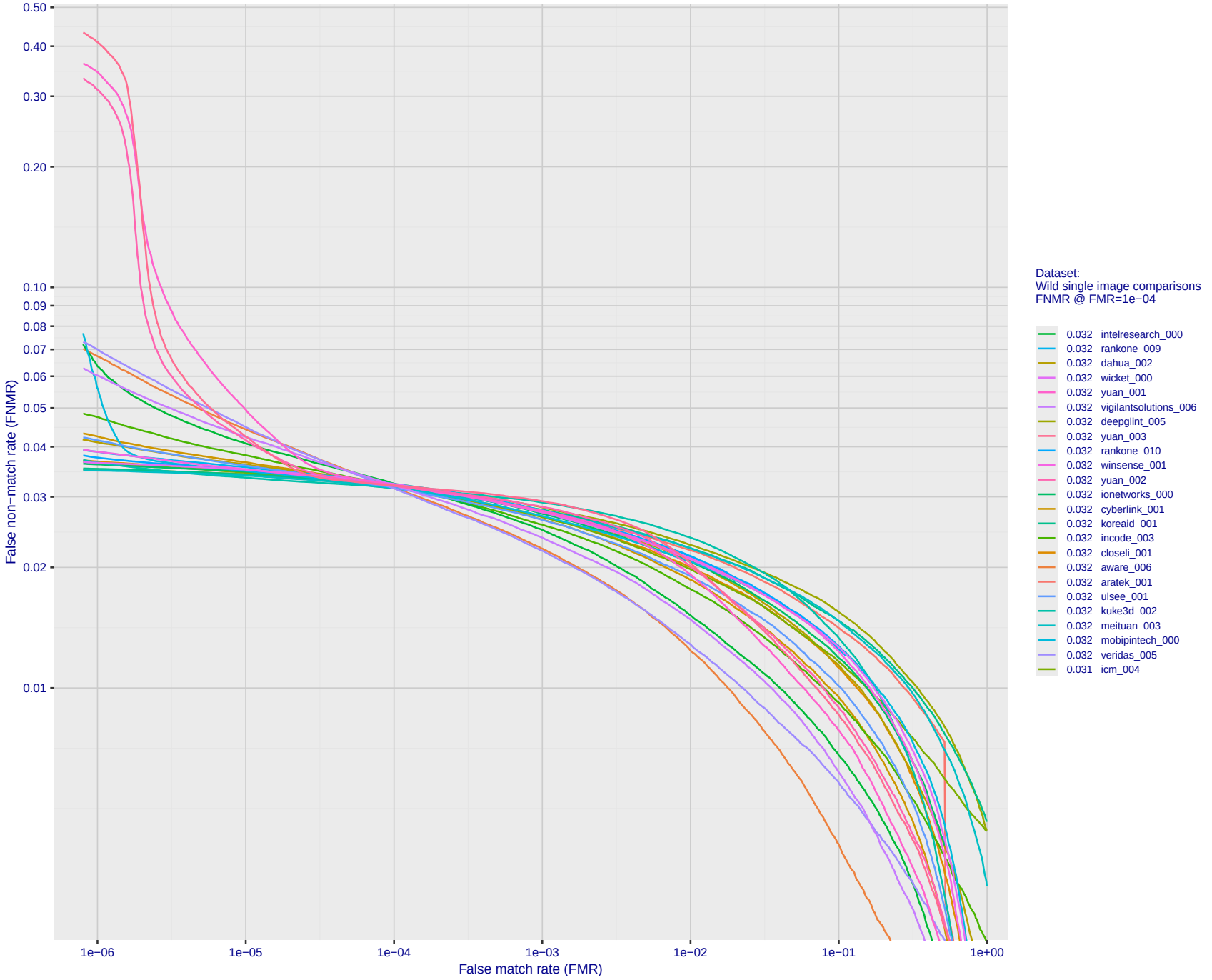

Figure 311: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

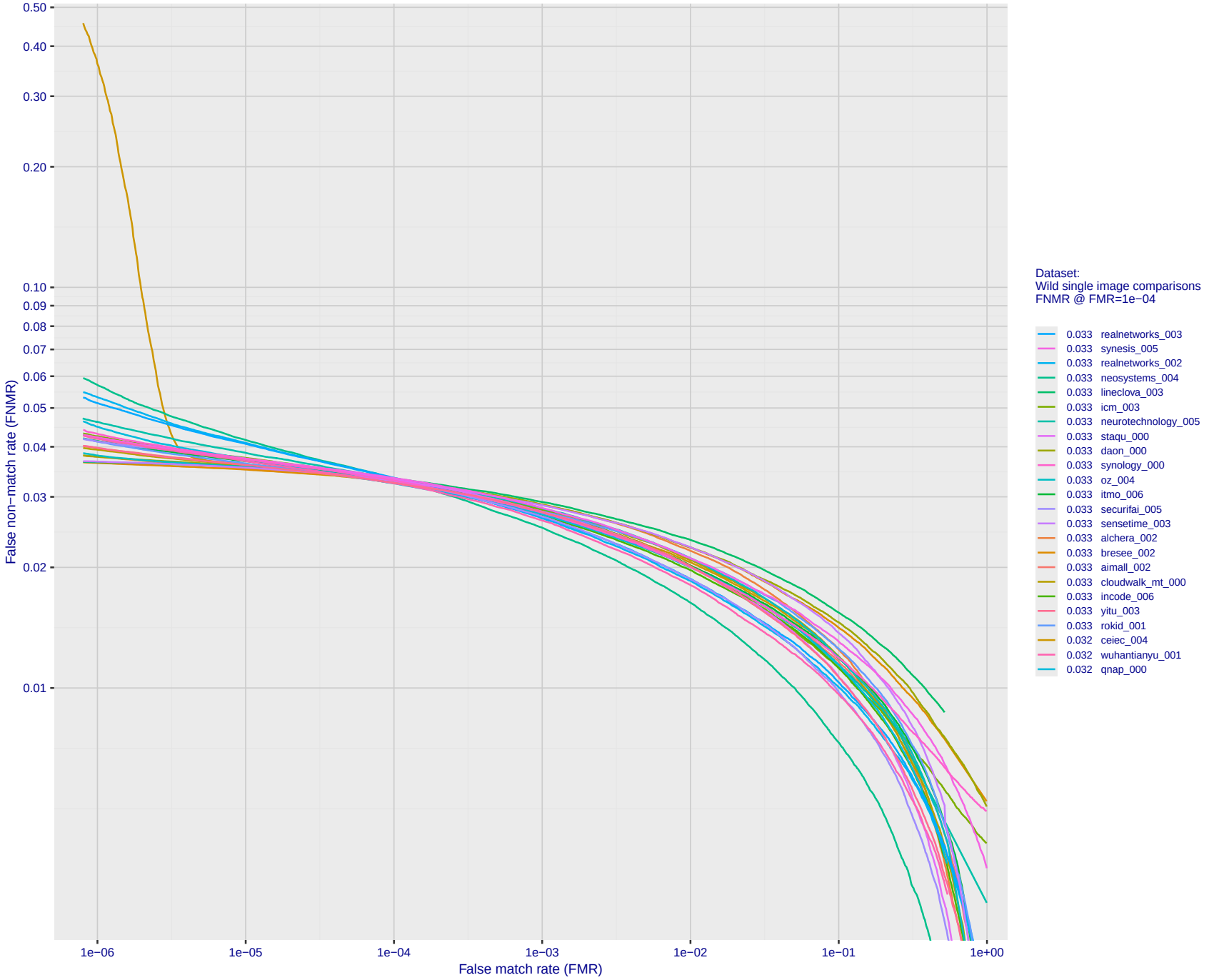

Figure 312: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

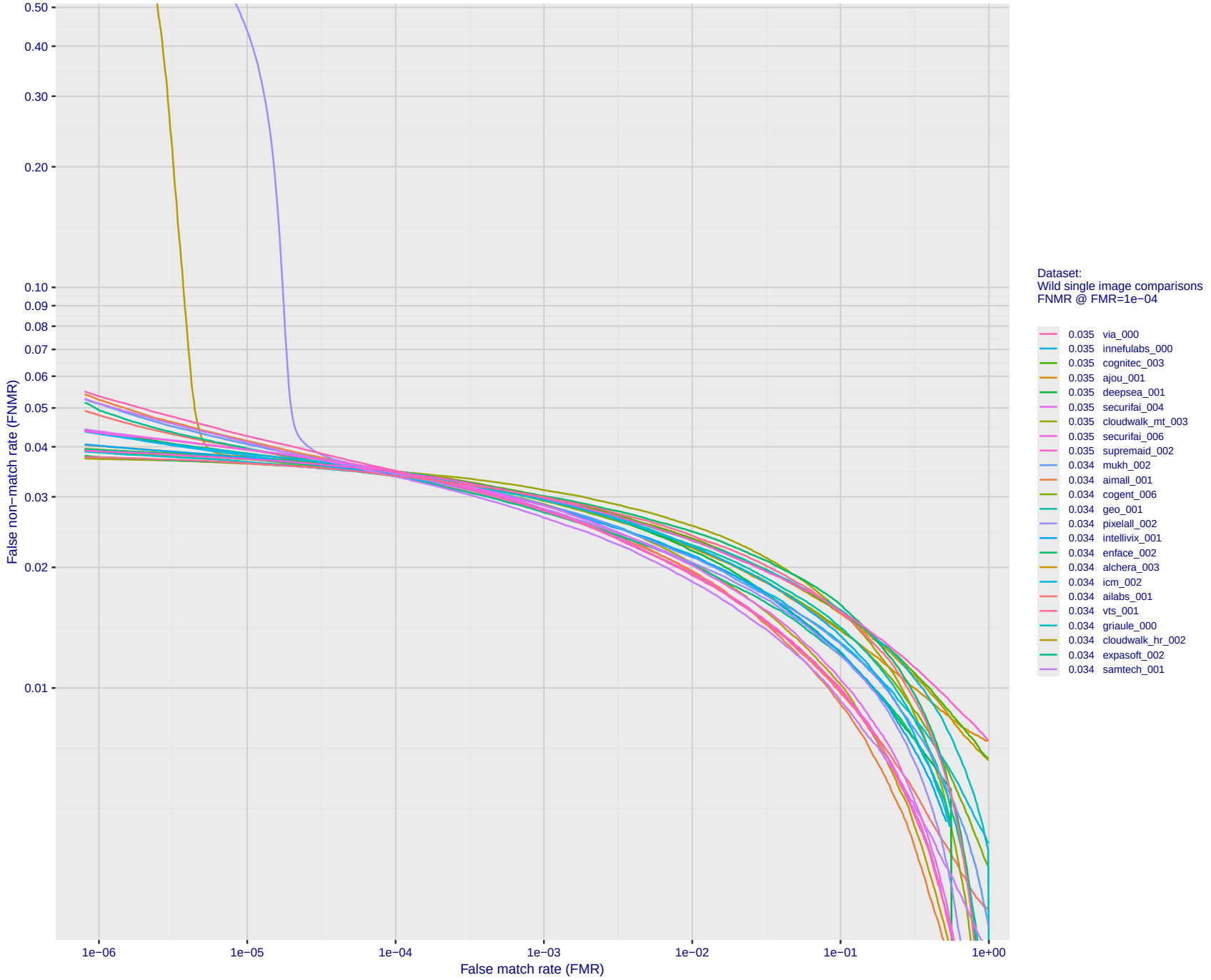

Figure 313: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

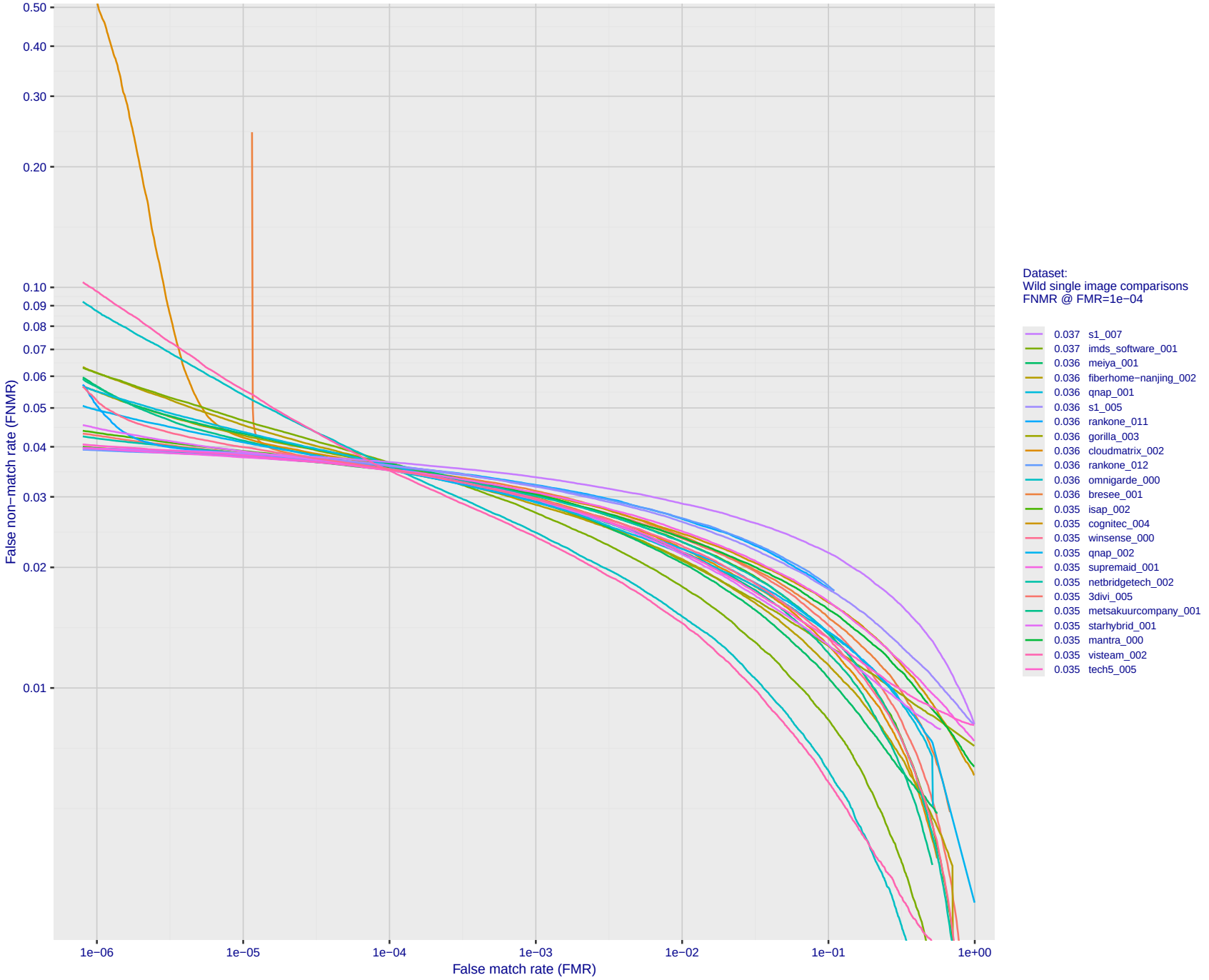

Figure 314: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

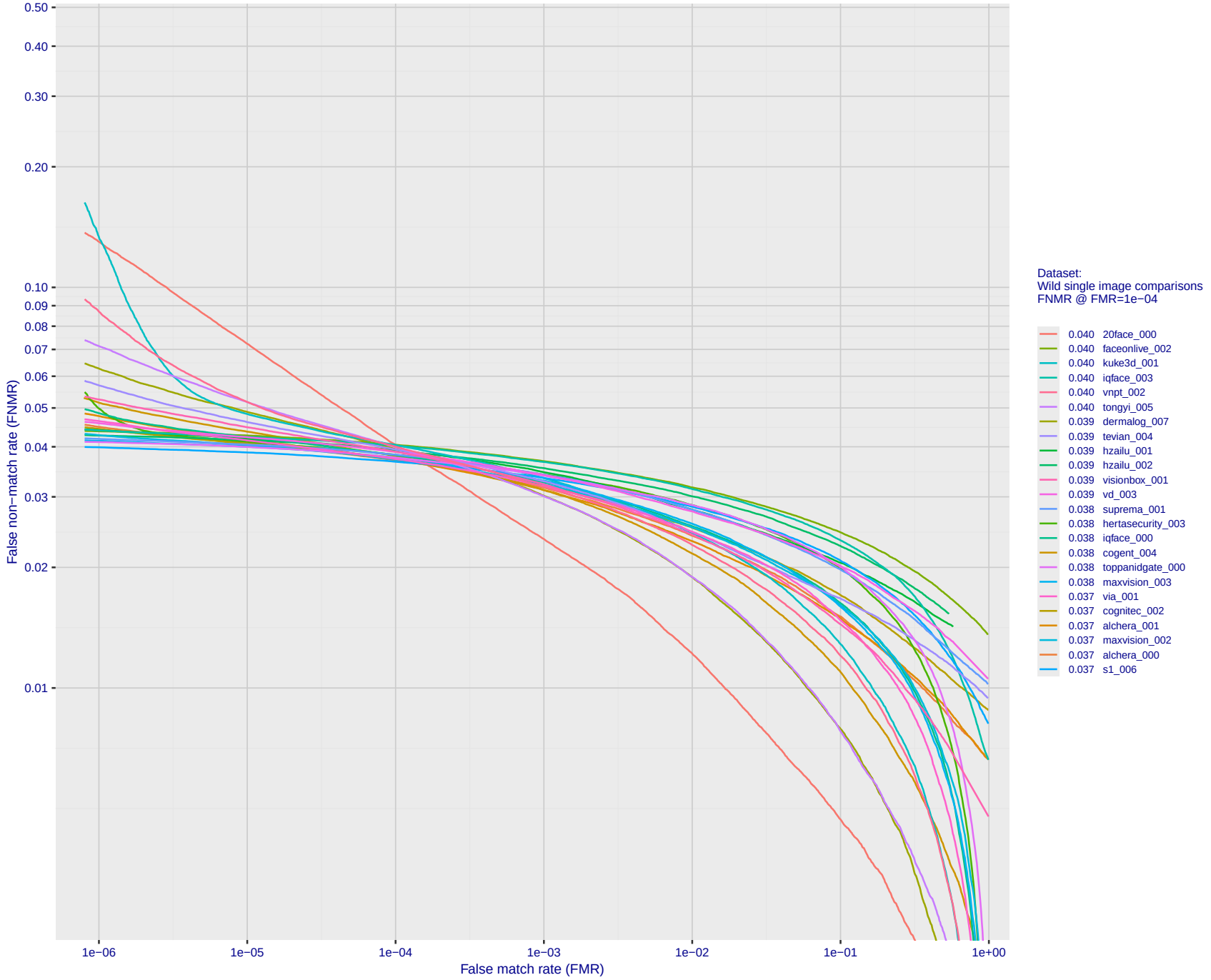

Figure 315: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

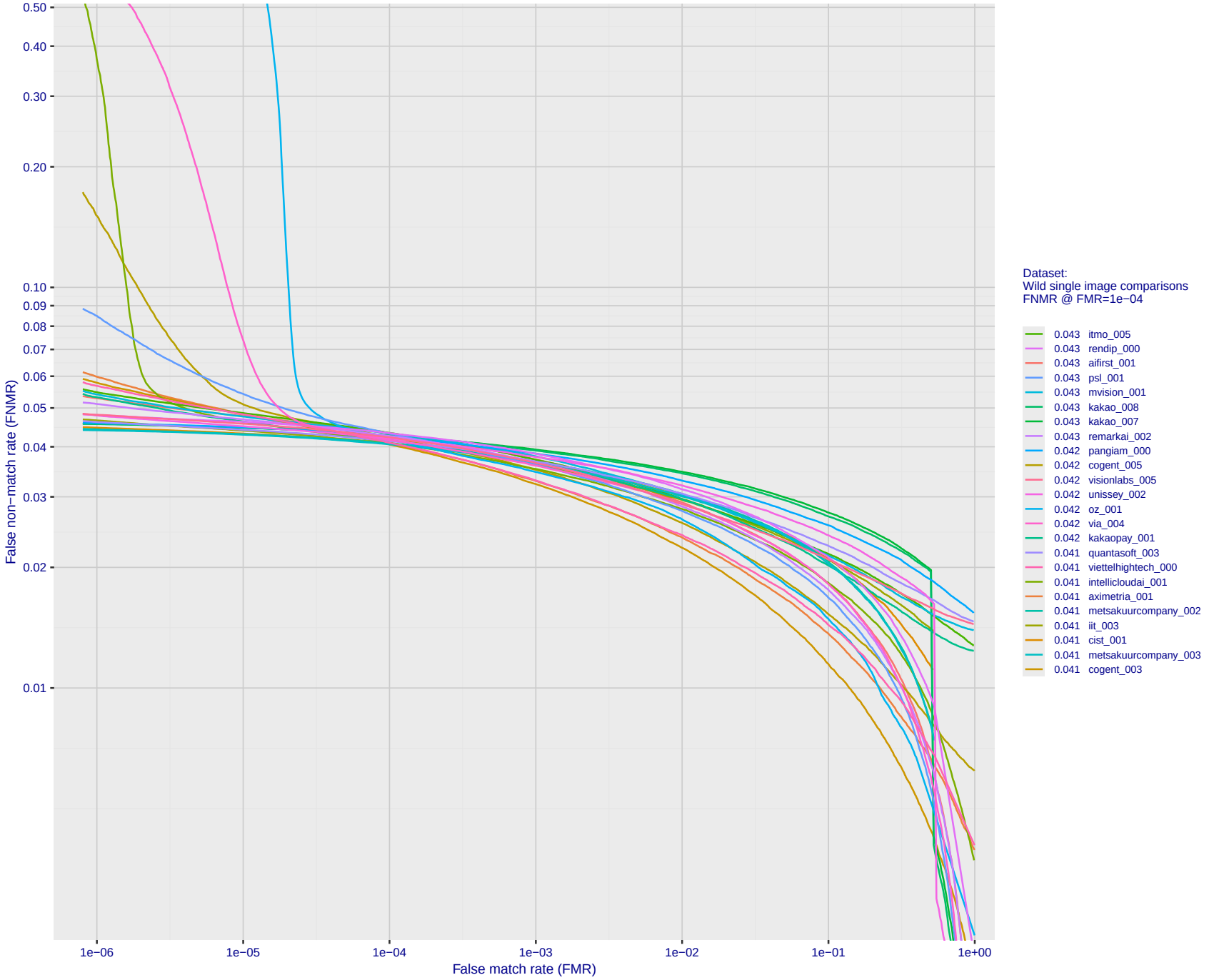

Figure 316: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27 FNMR(T) "False non-match rate" FMR(T) "False match rate"

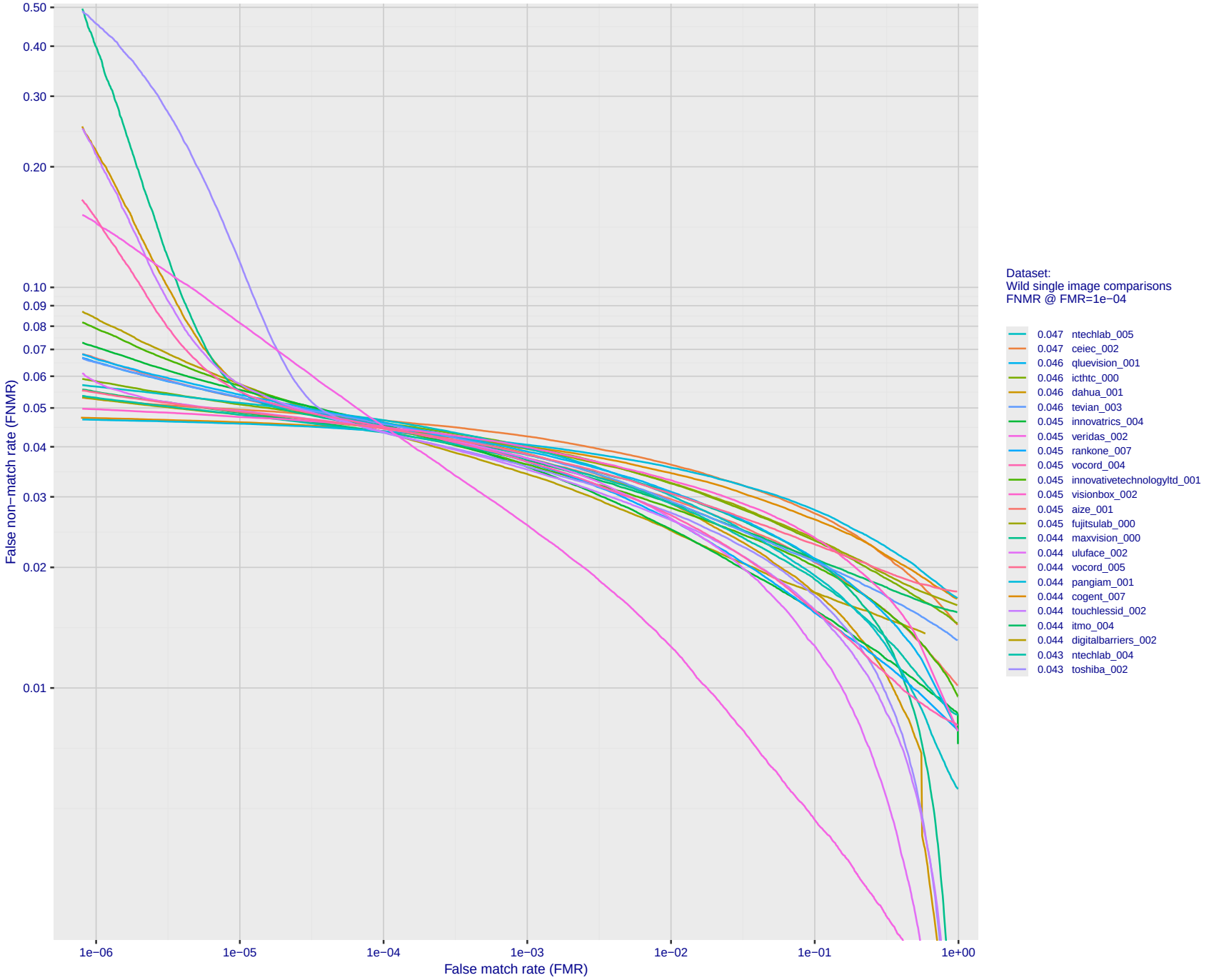

Figure 317: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

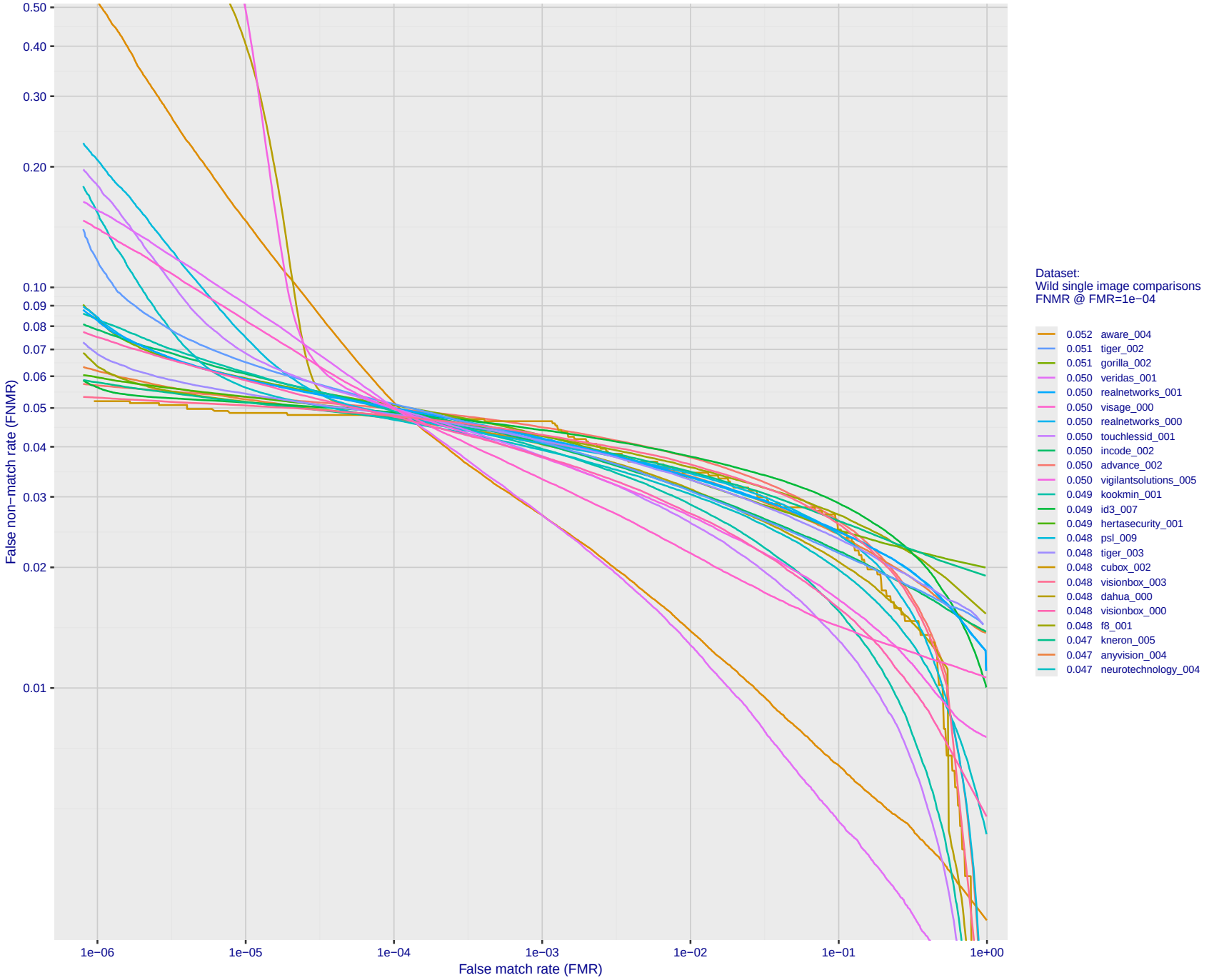

Figure 318: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

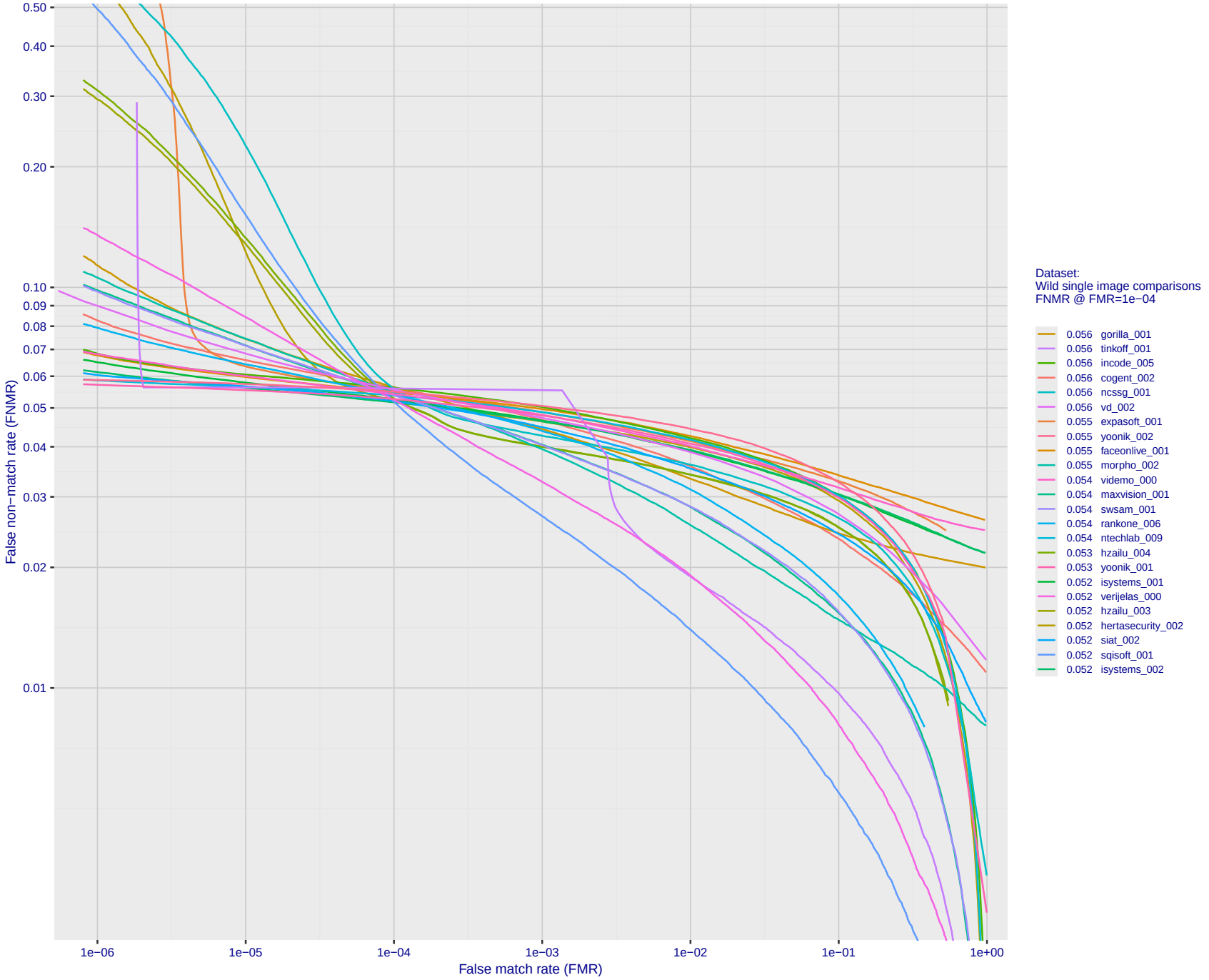

Figure 319: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

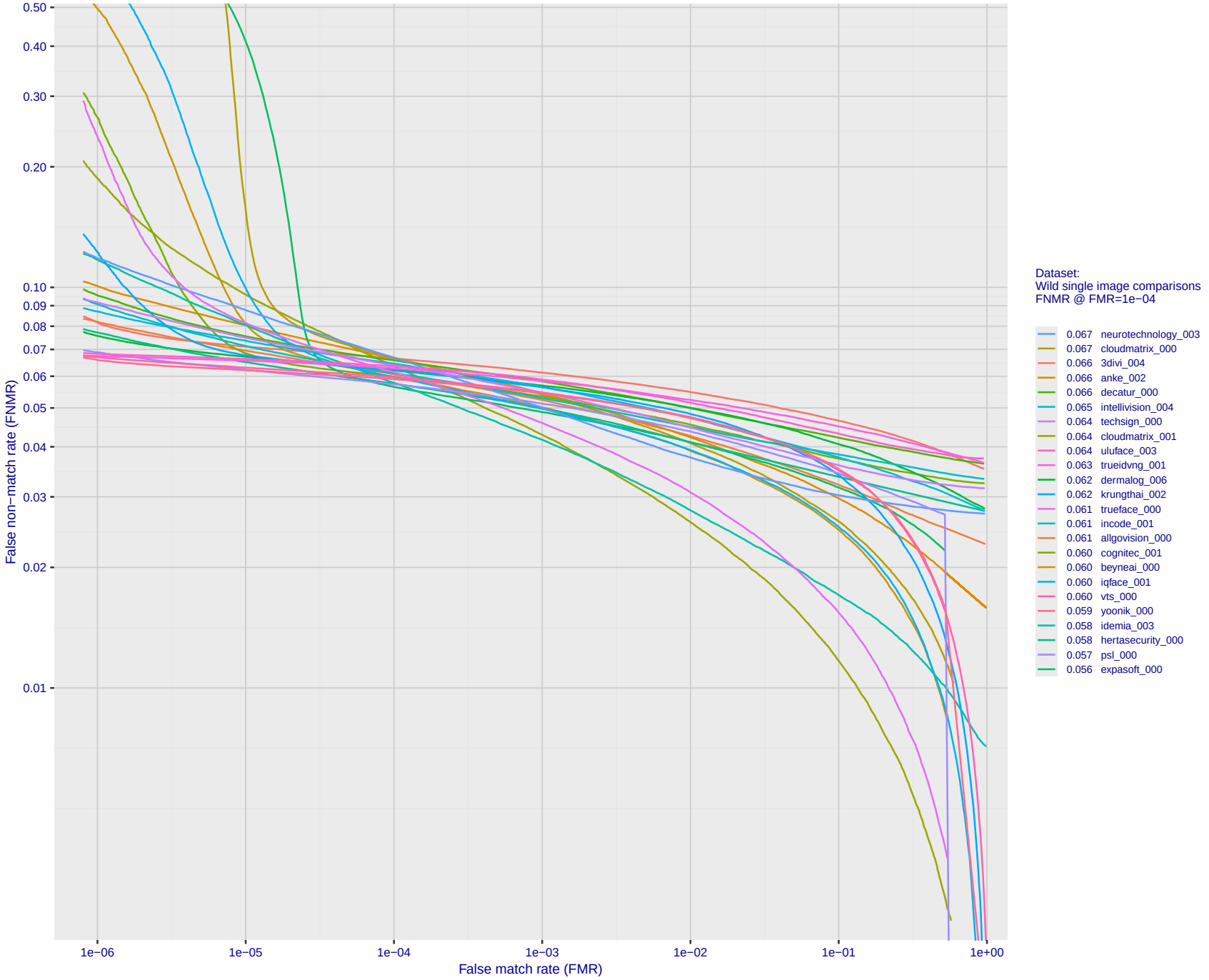

Figure 320: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

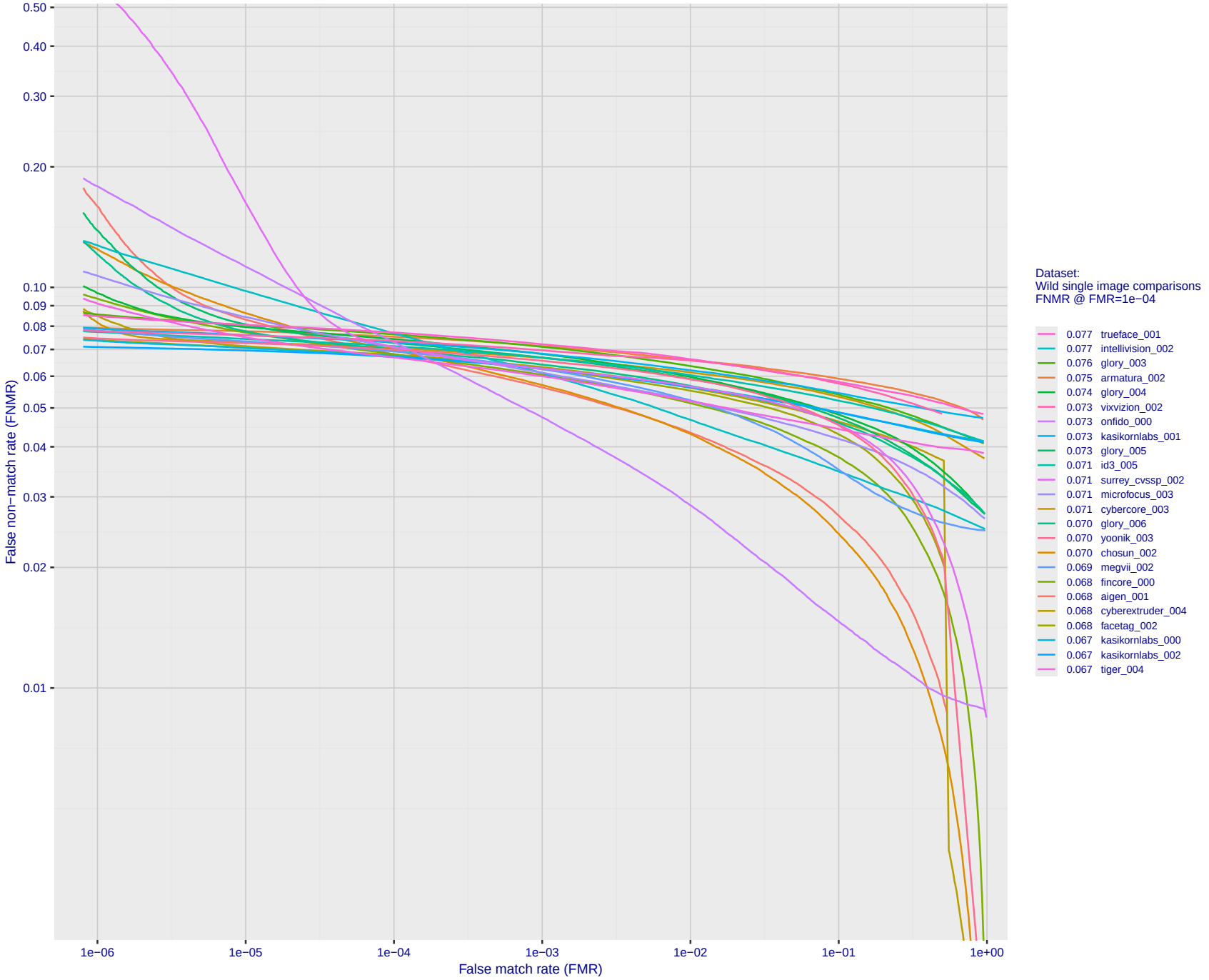

Figure 321: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

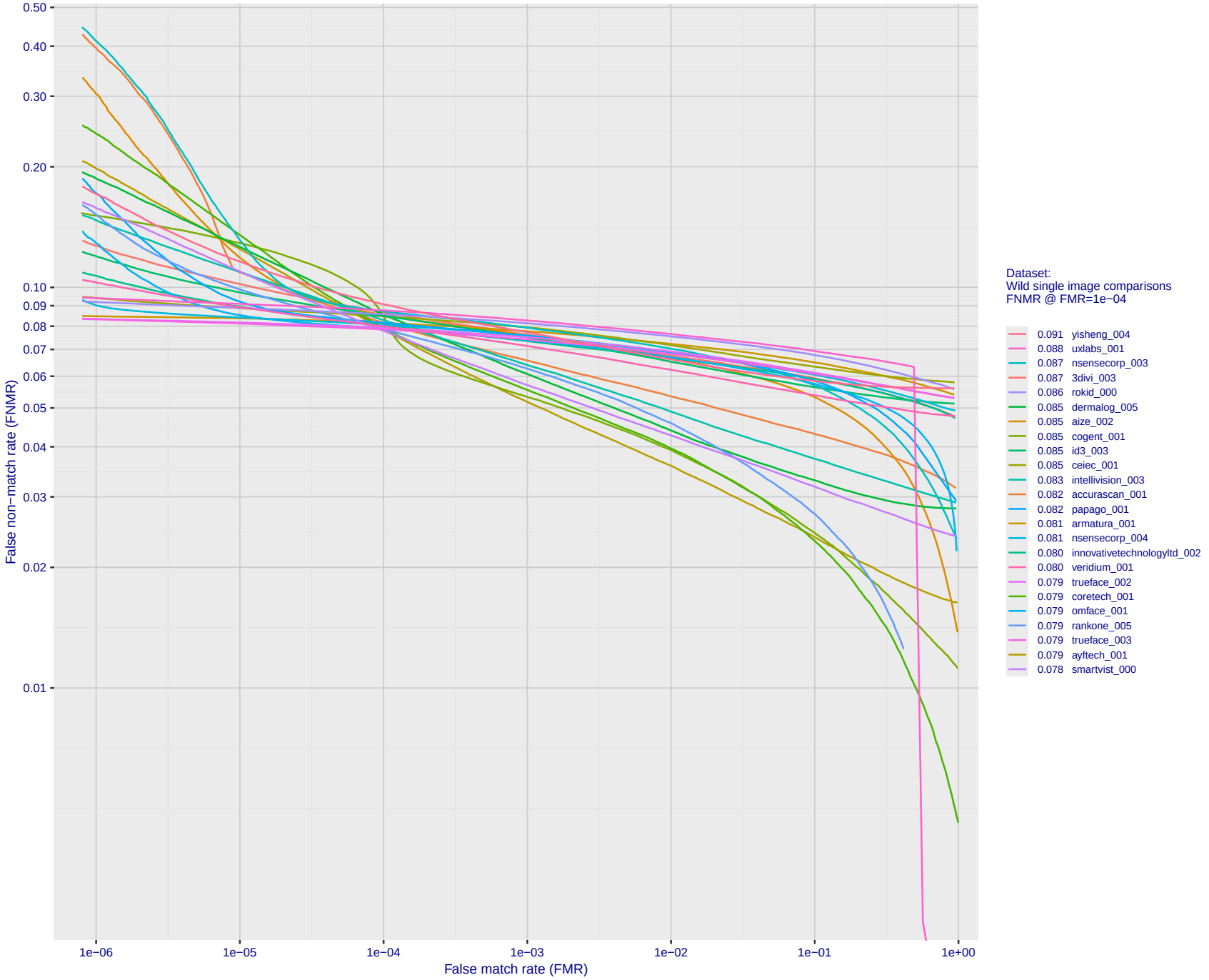

Figure 322: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

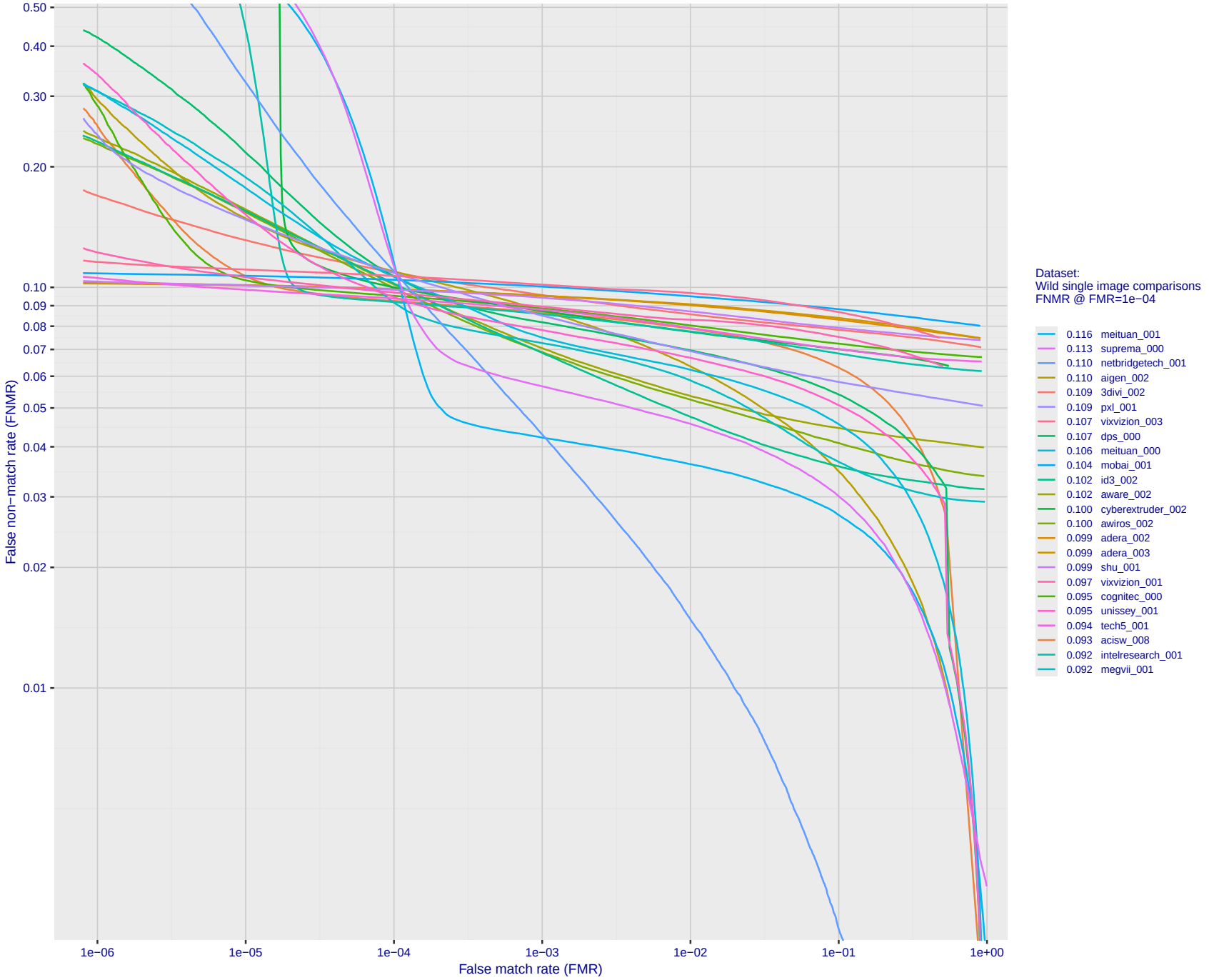

Figure 323: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

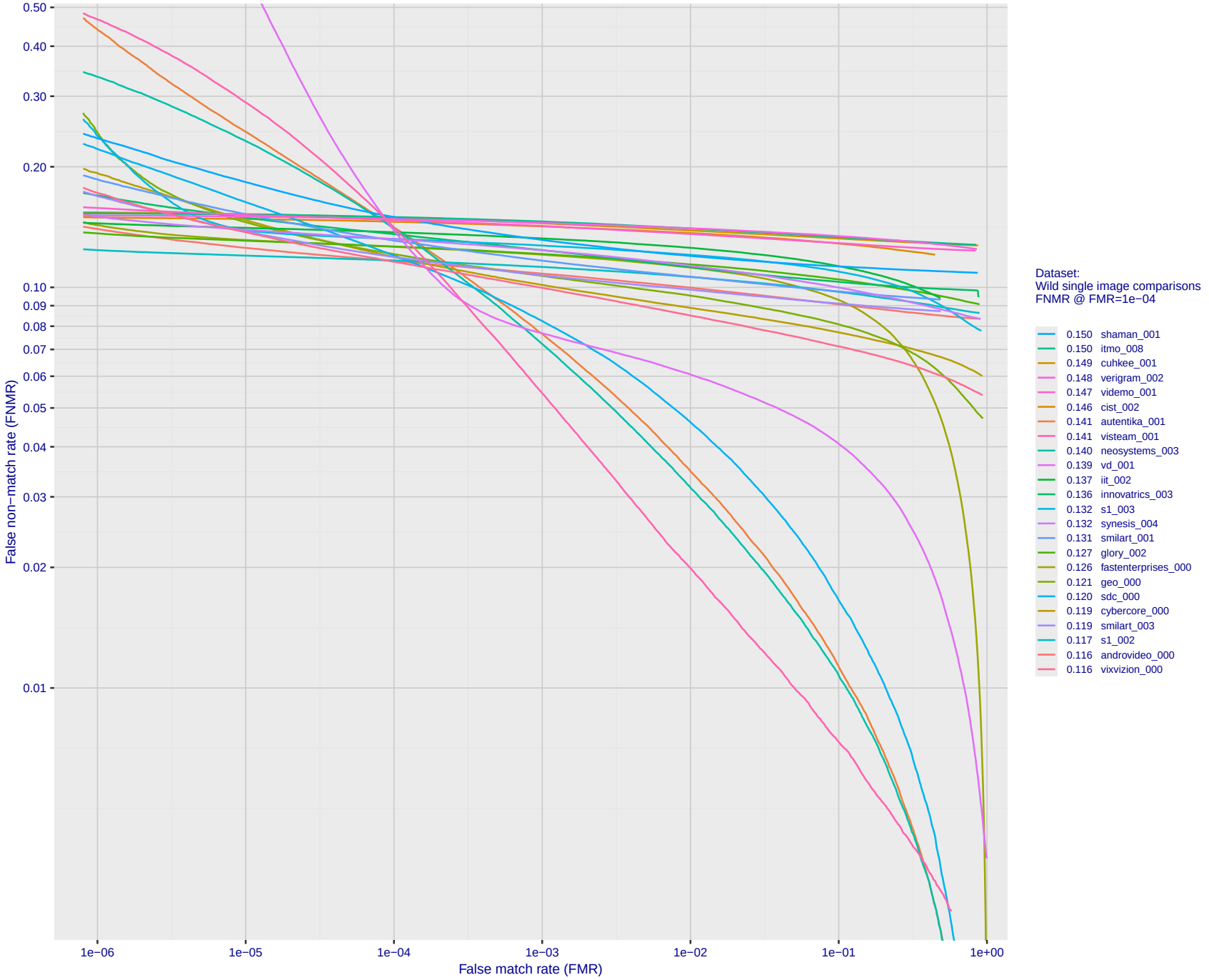

Figure 324: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27 FNMR(T) "False non-match rate" FMR(T) "False match rate"

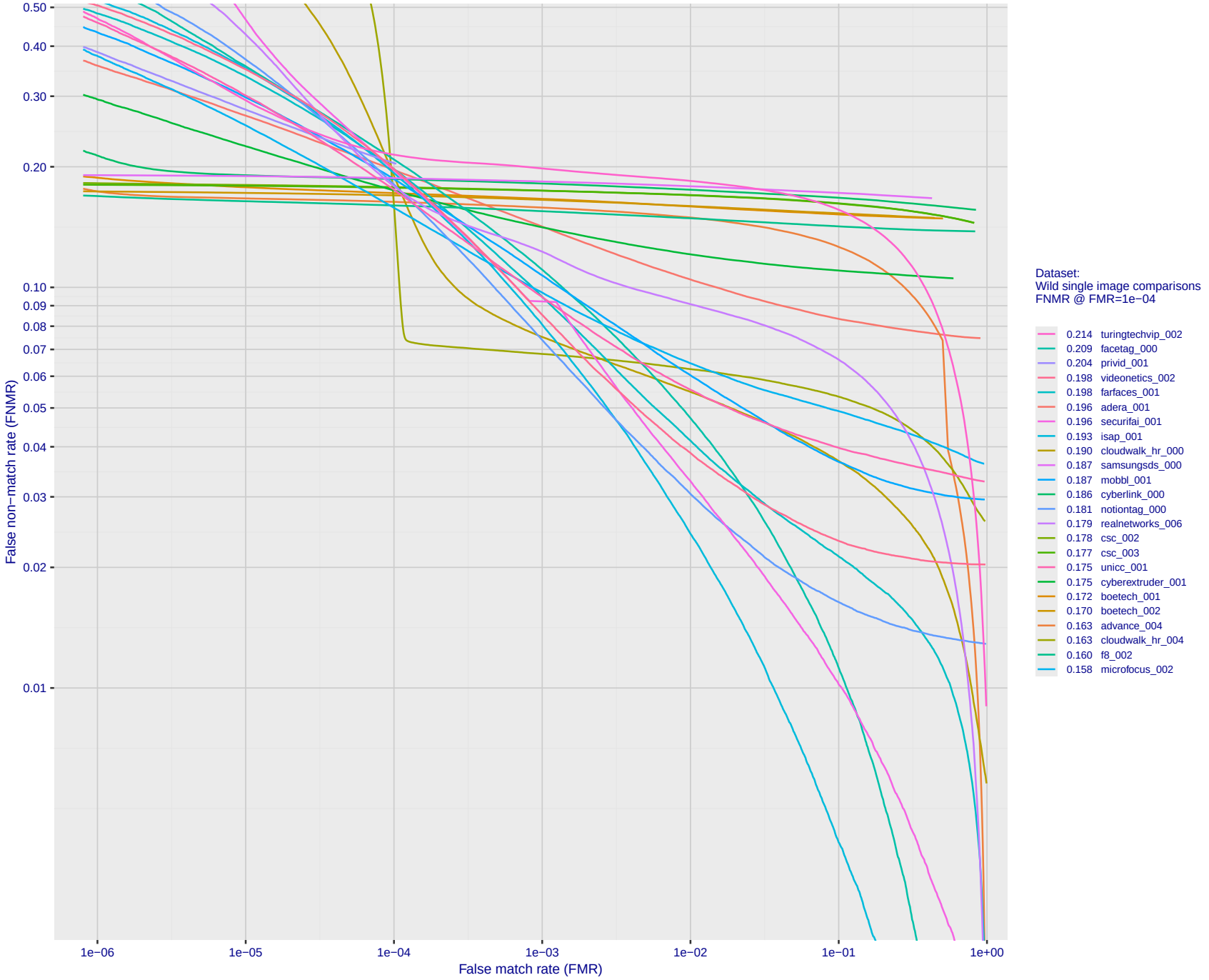

Figure 325: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27 FNMR(T) "False non-match rate" FMR(T) "False match rate"

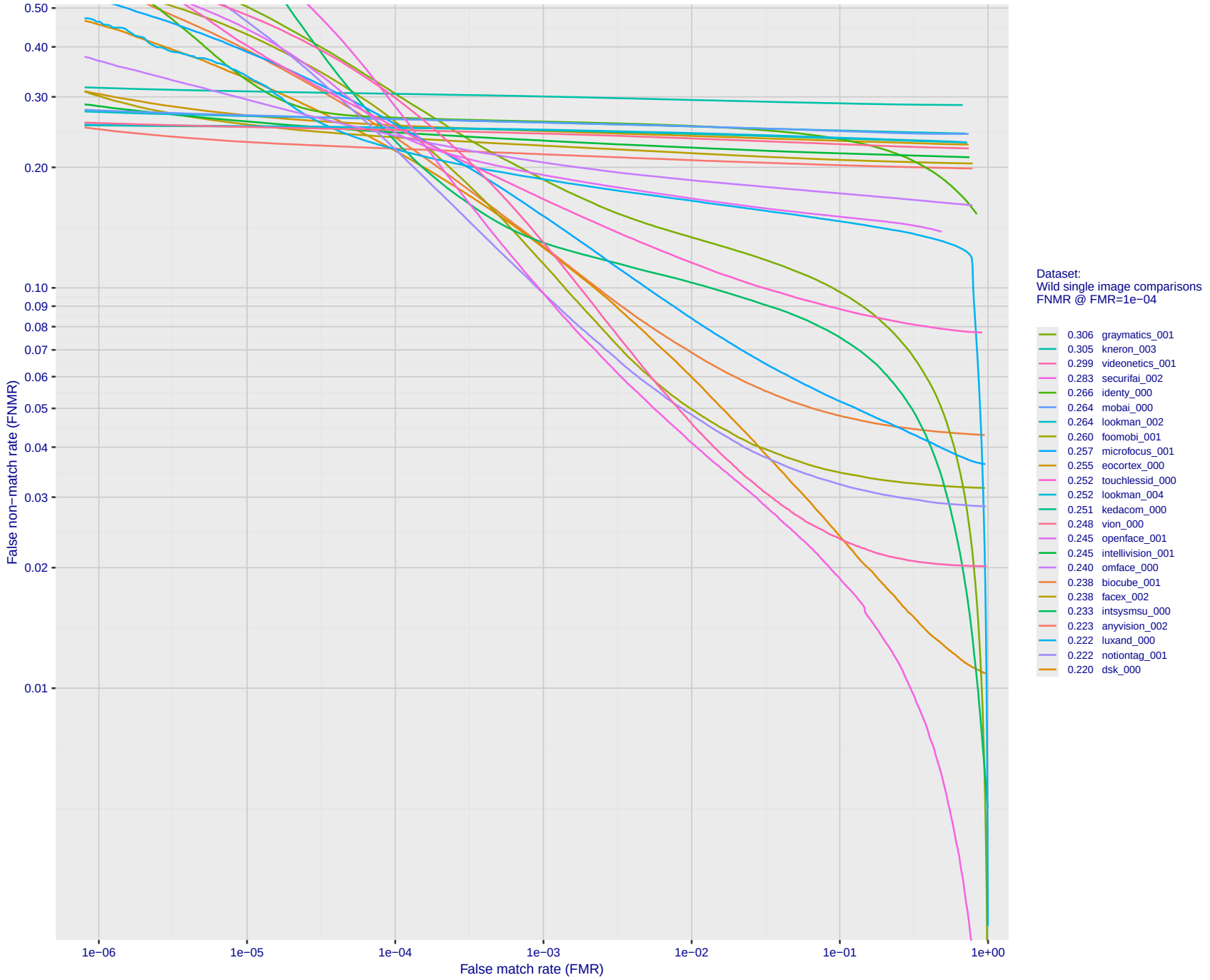

Figure 326: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

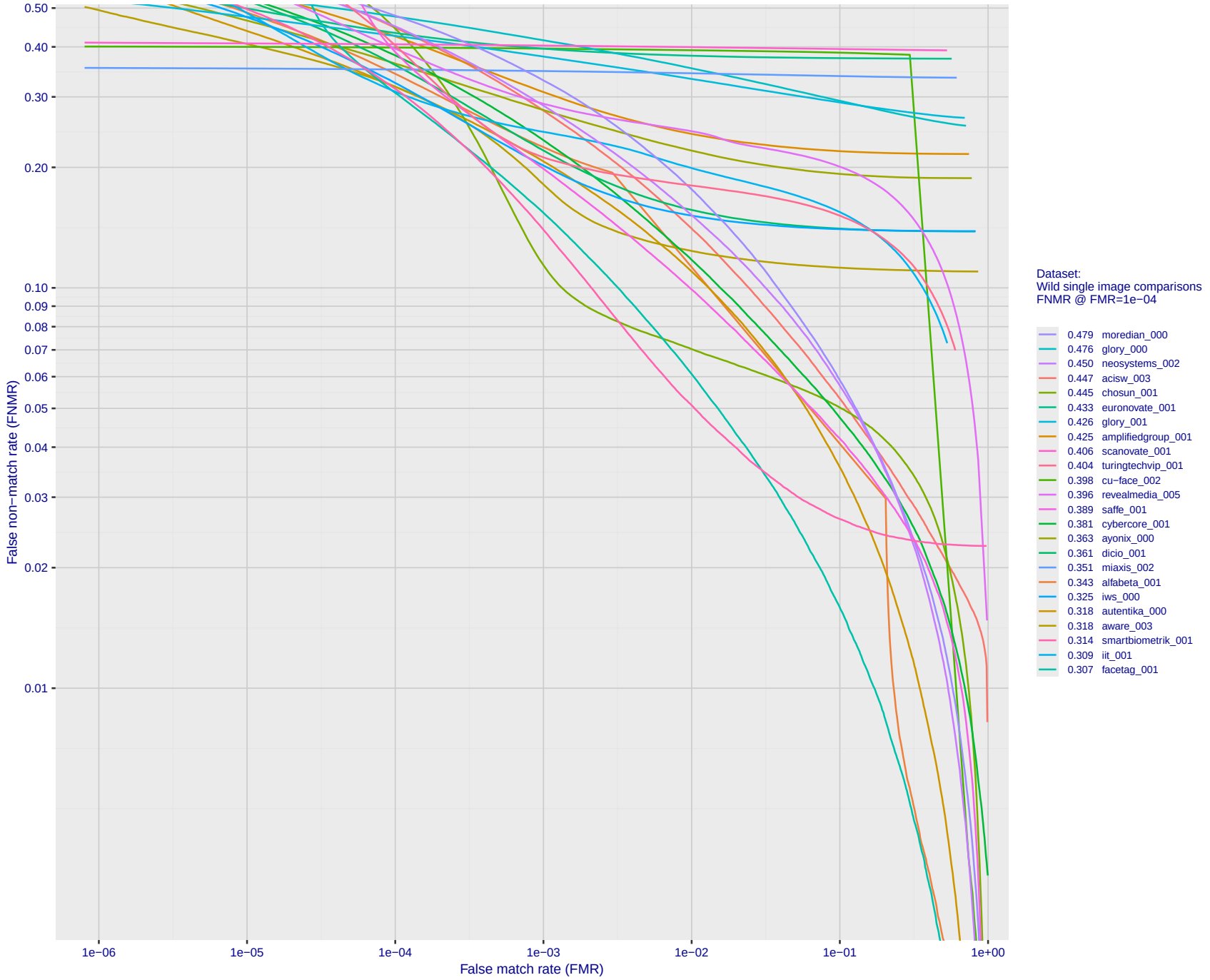

Figure 327: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

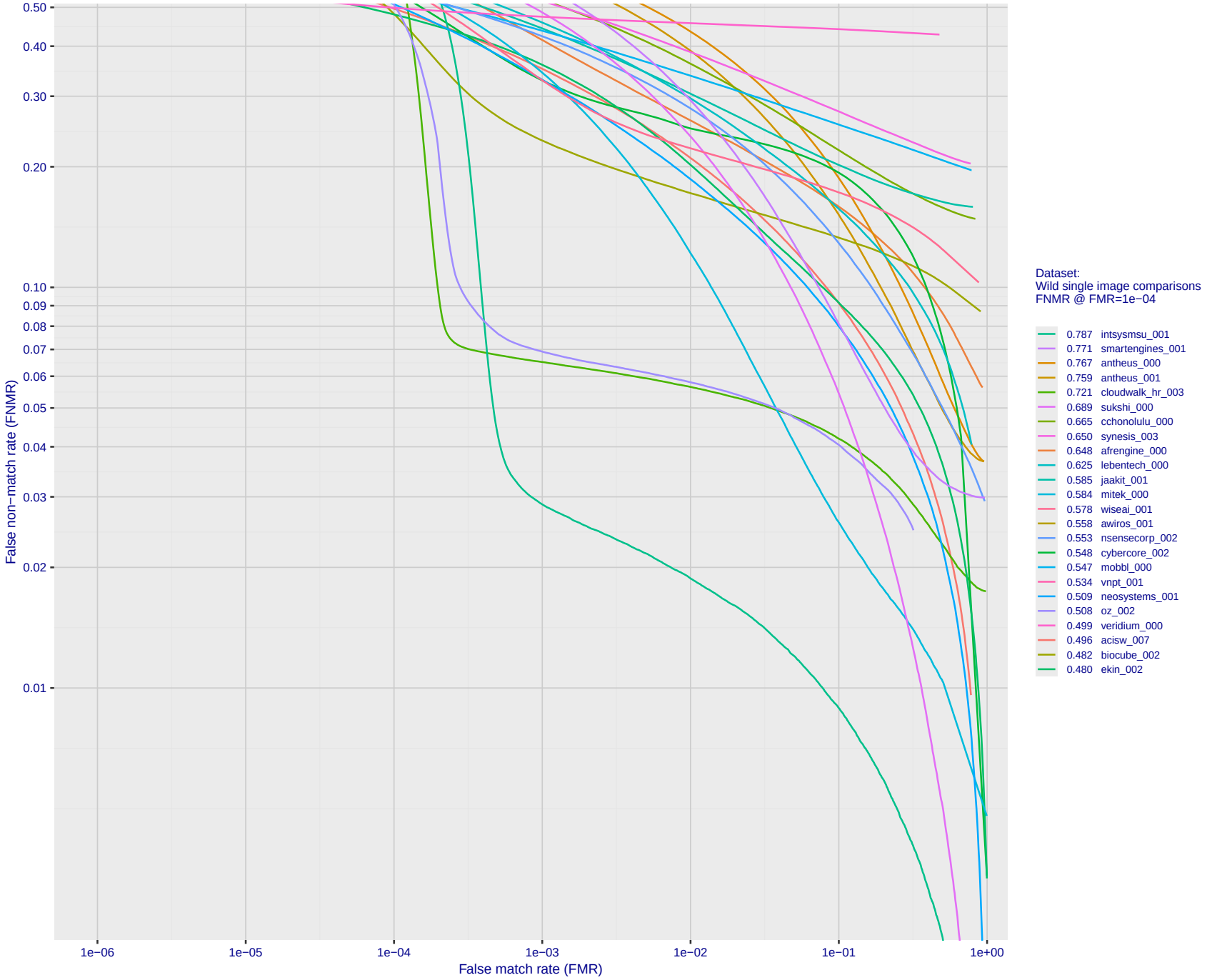

Figure 328: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

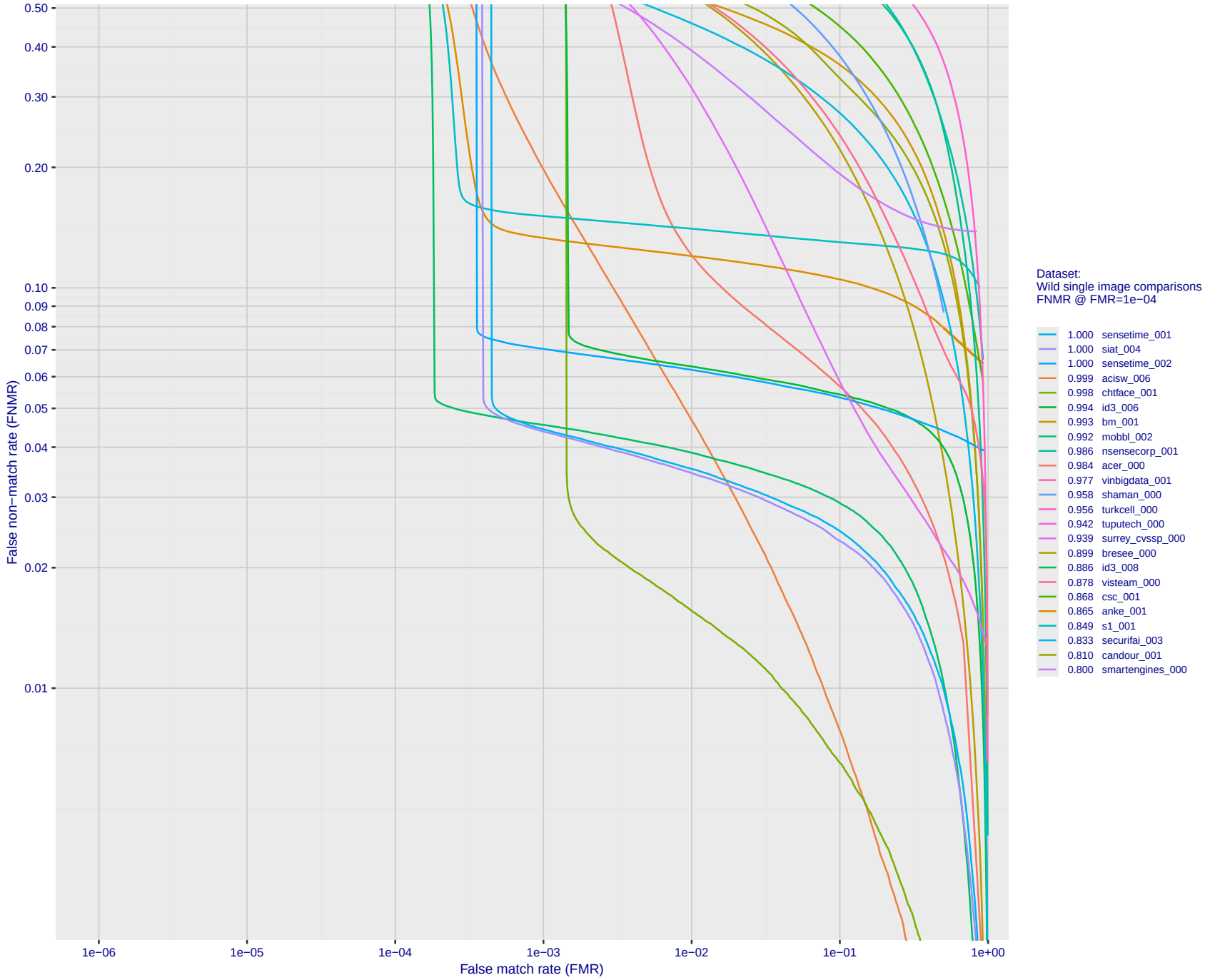

Figure 329: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

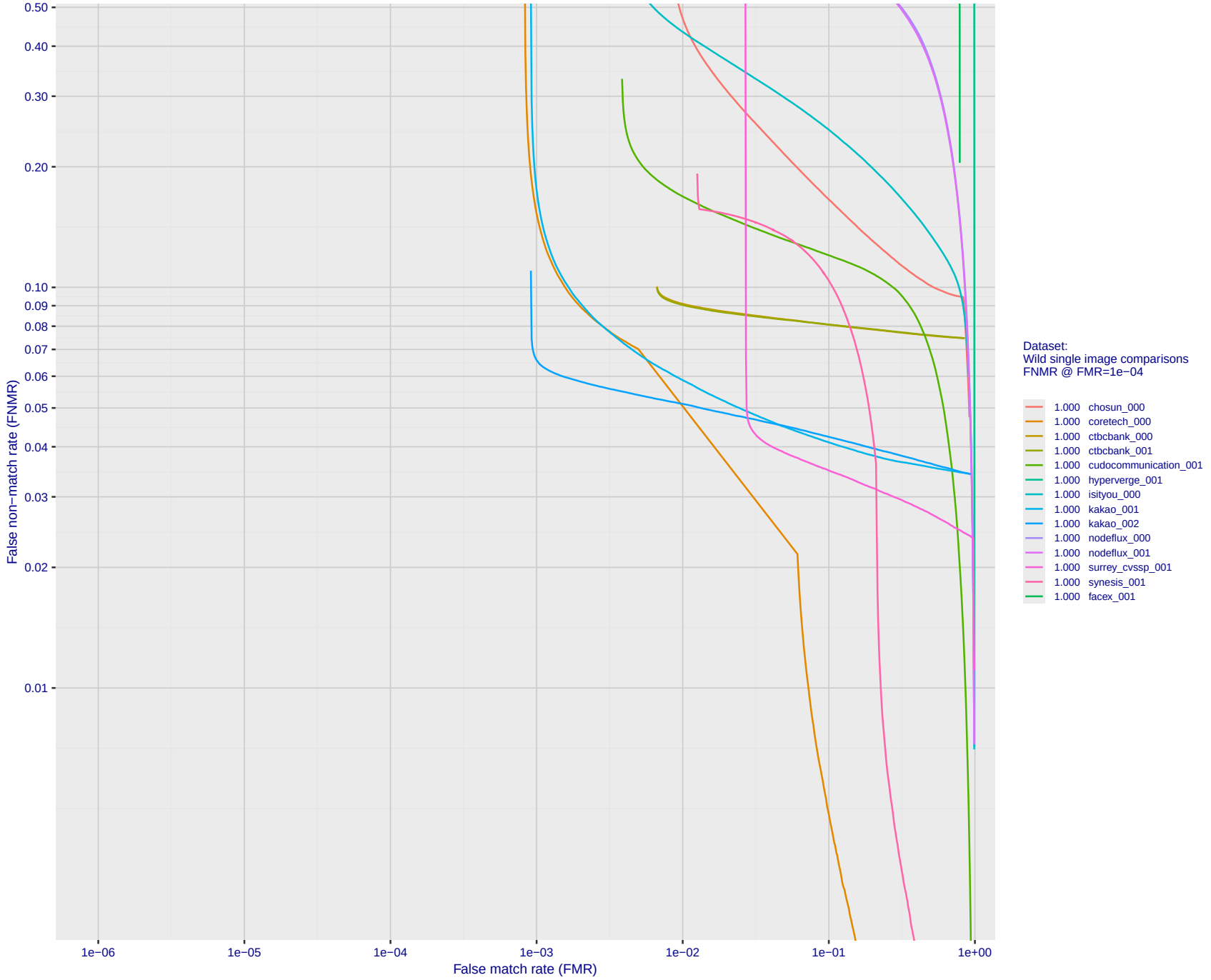

Figure 330: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

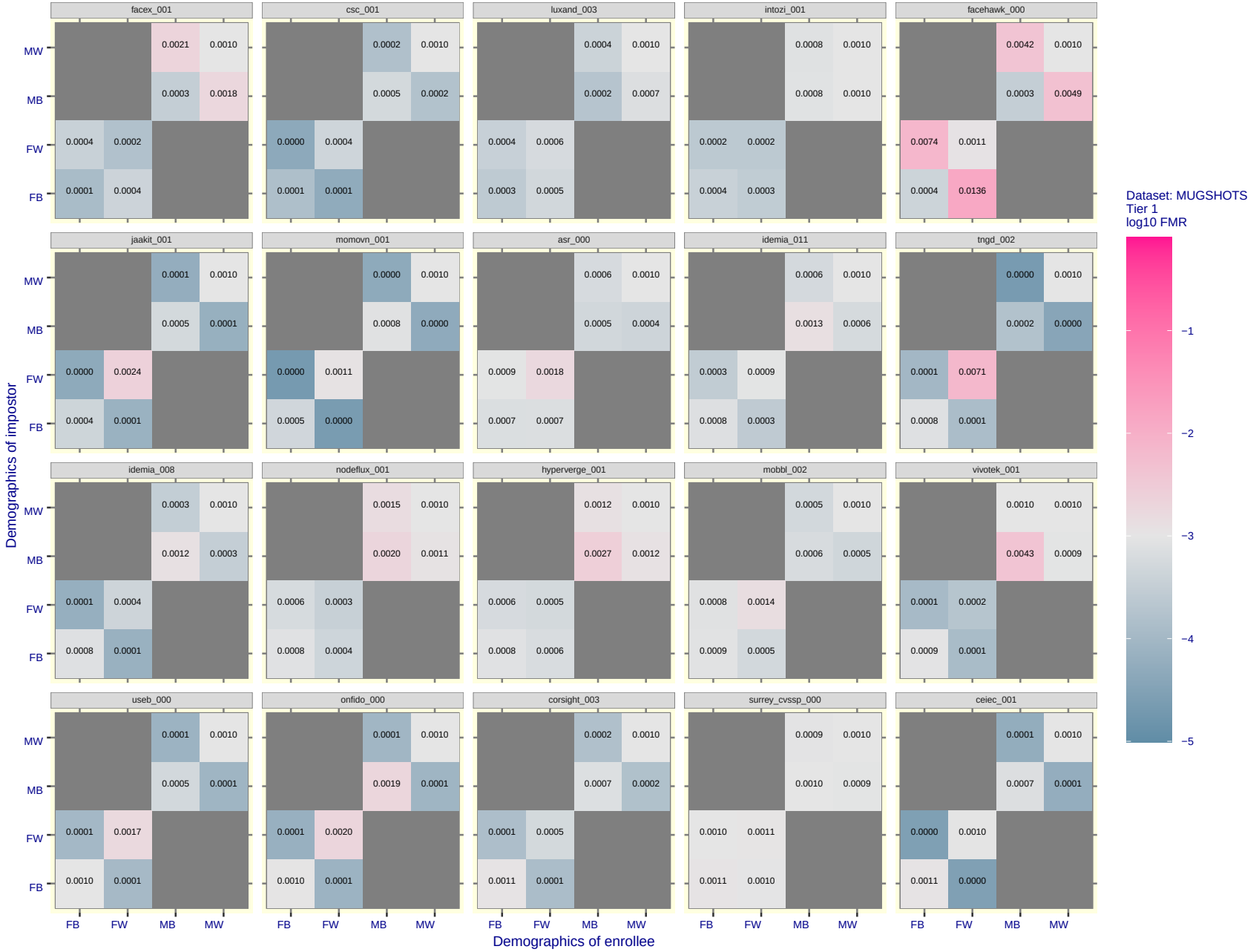

Figure 331: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

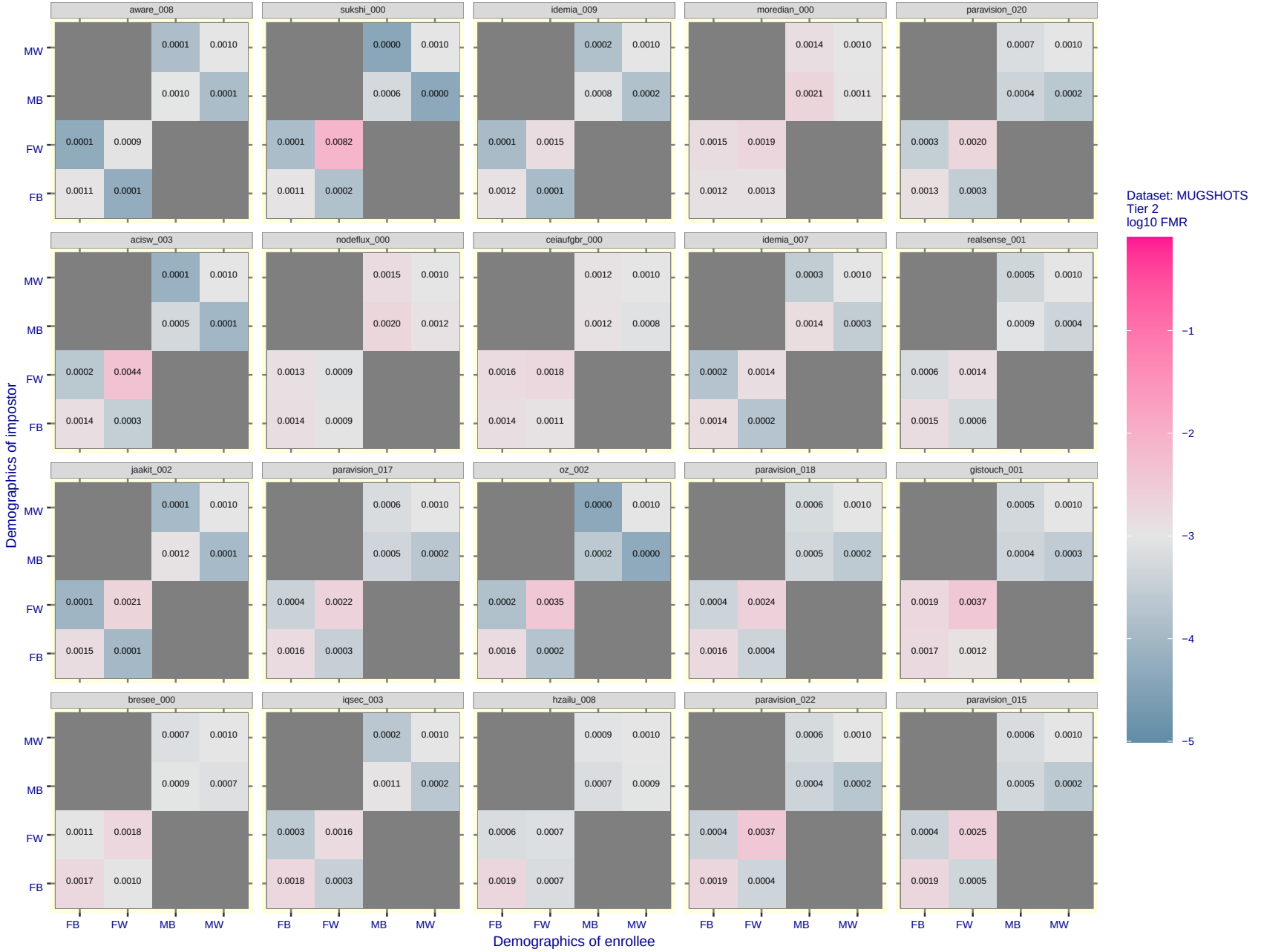

Figure 332: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

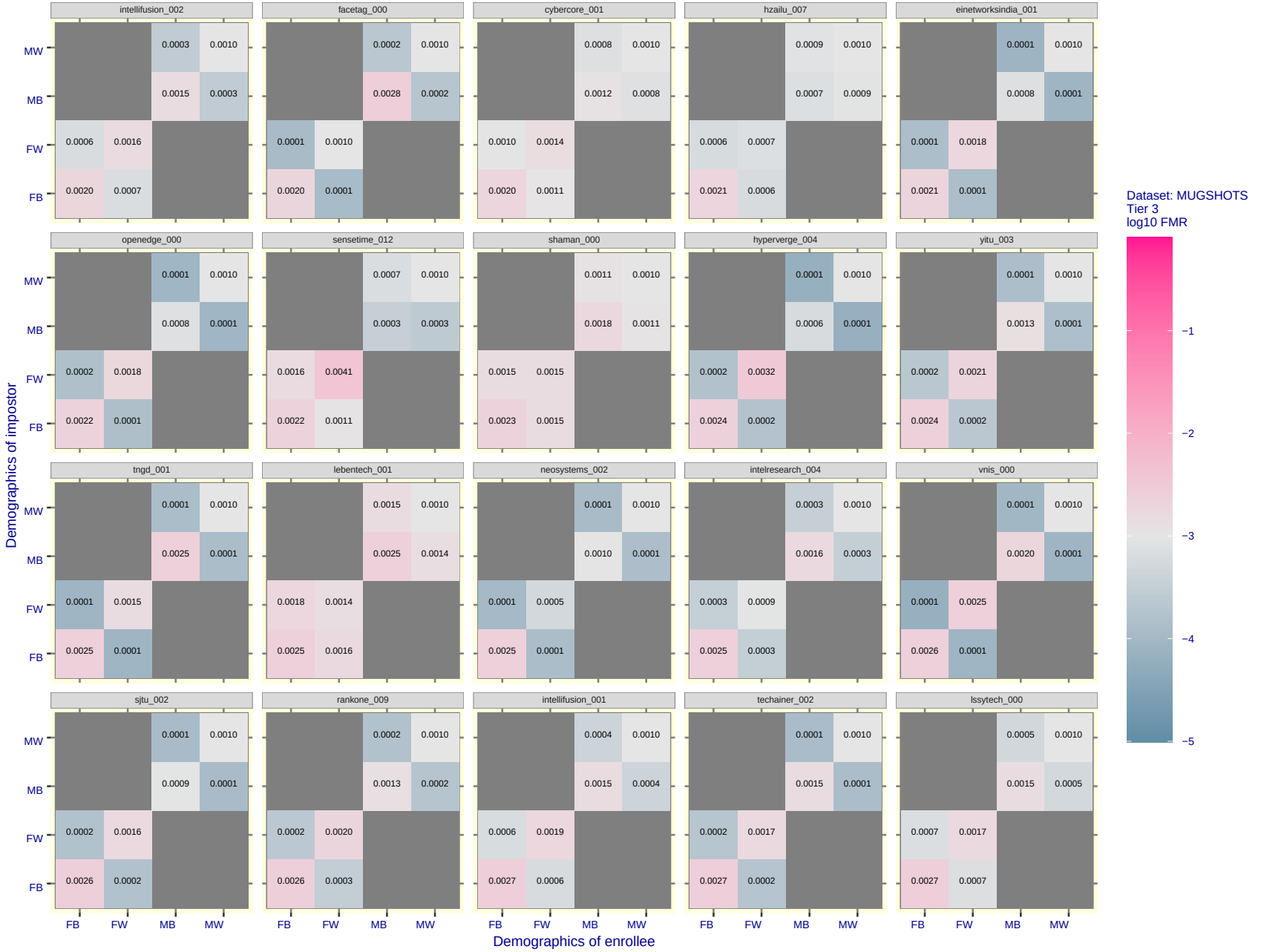

Figure 333: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

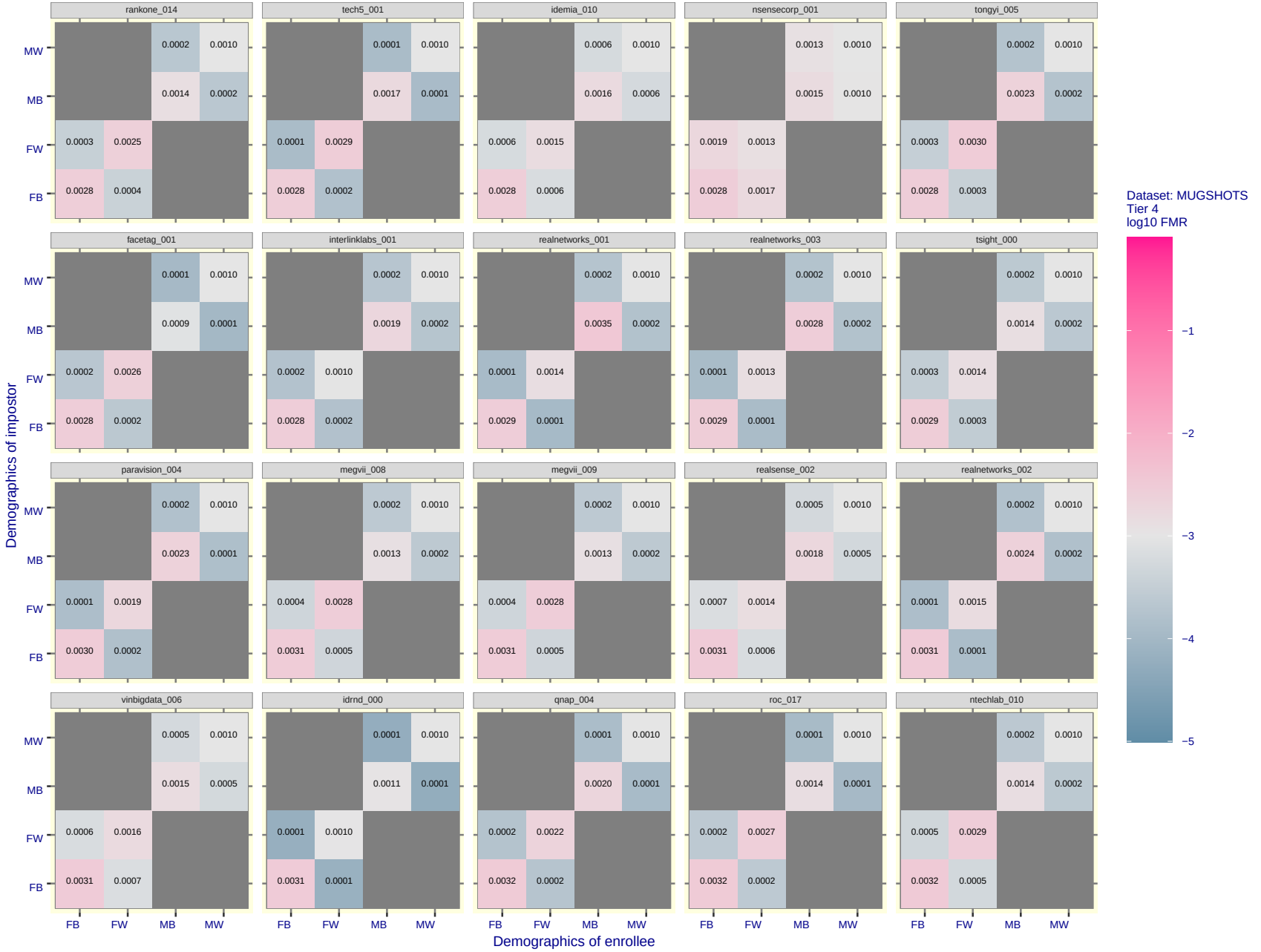

Figure 334: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

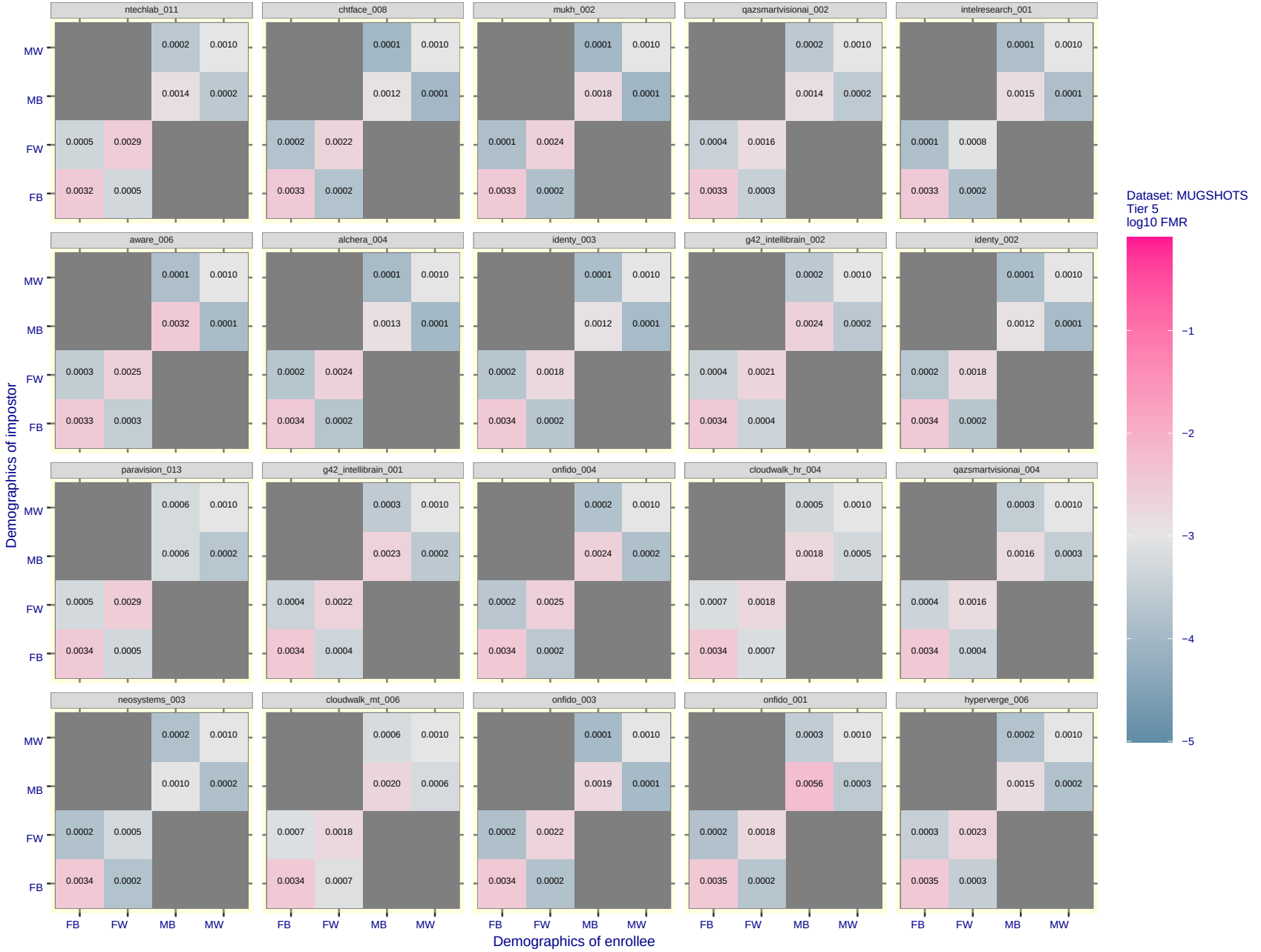

Figure 335: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

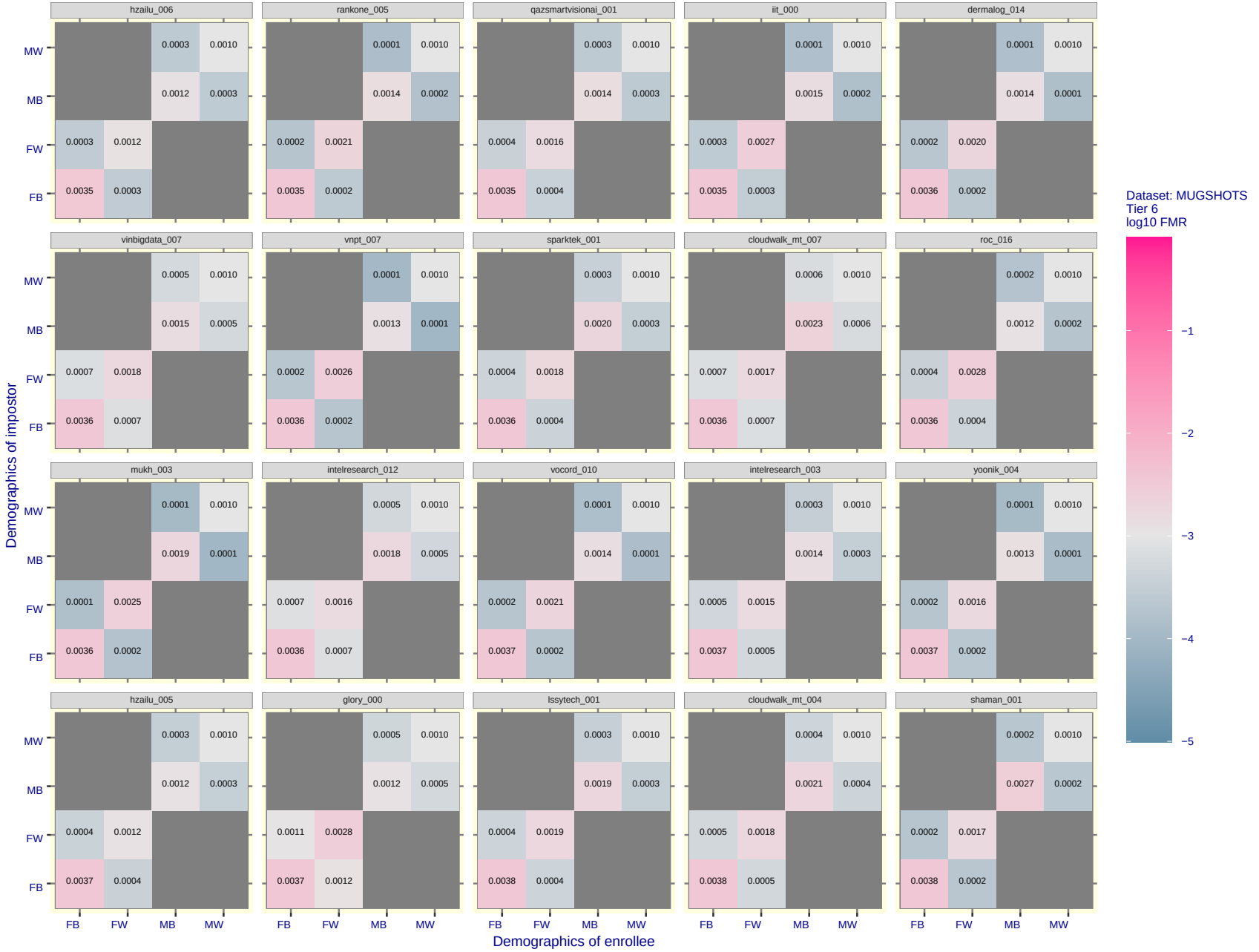

Figure 336: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

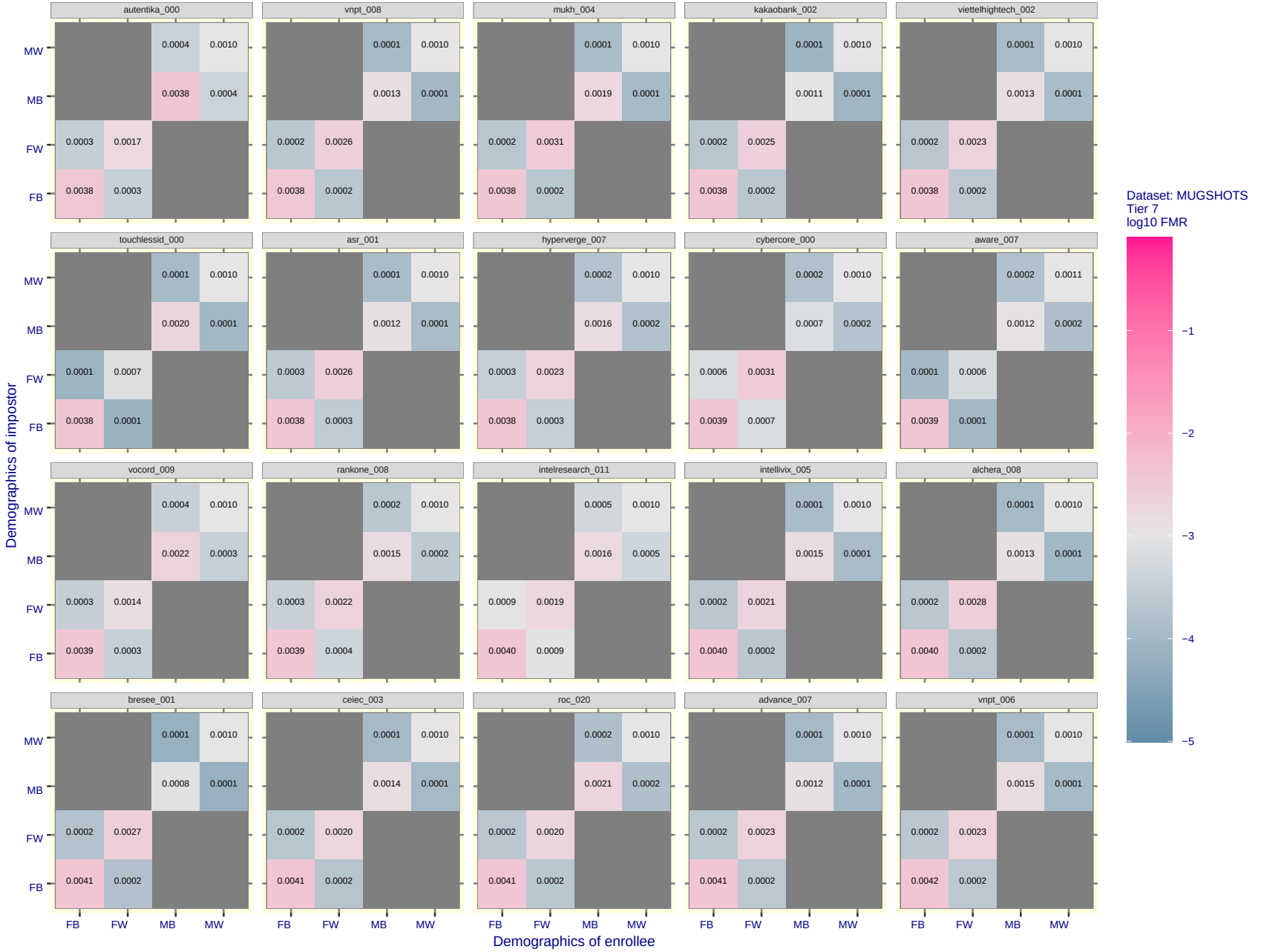

Figure 337: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

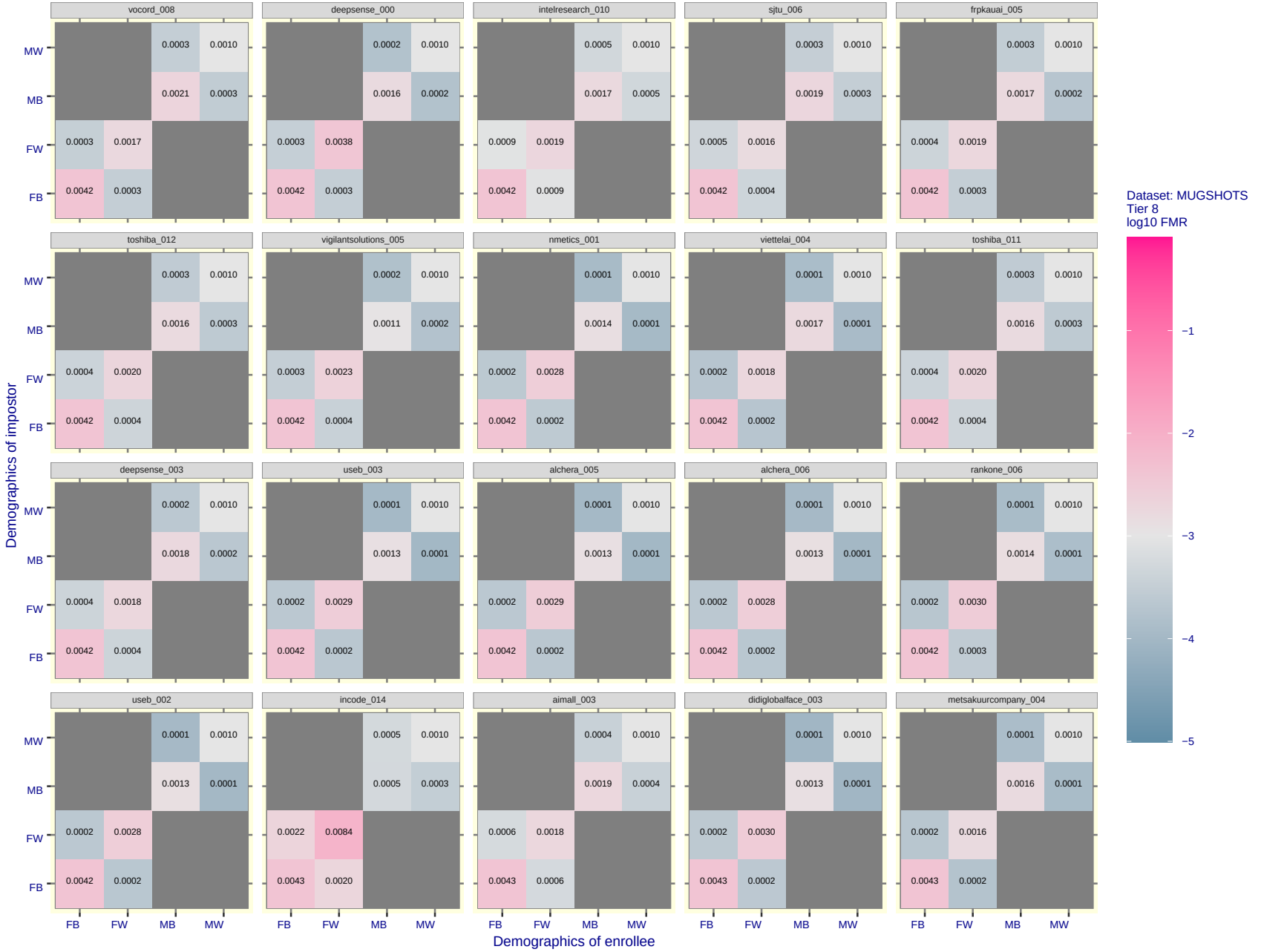

Figure 338: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

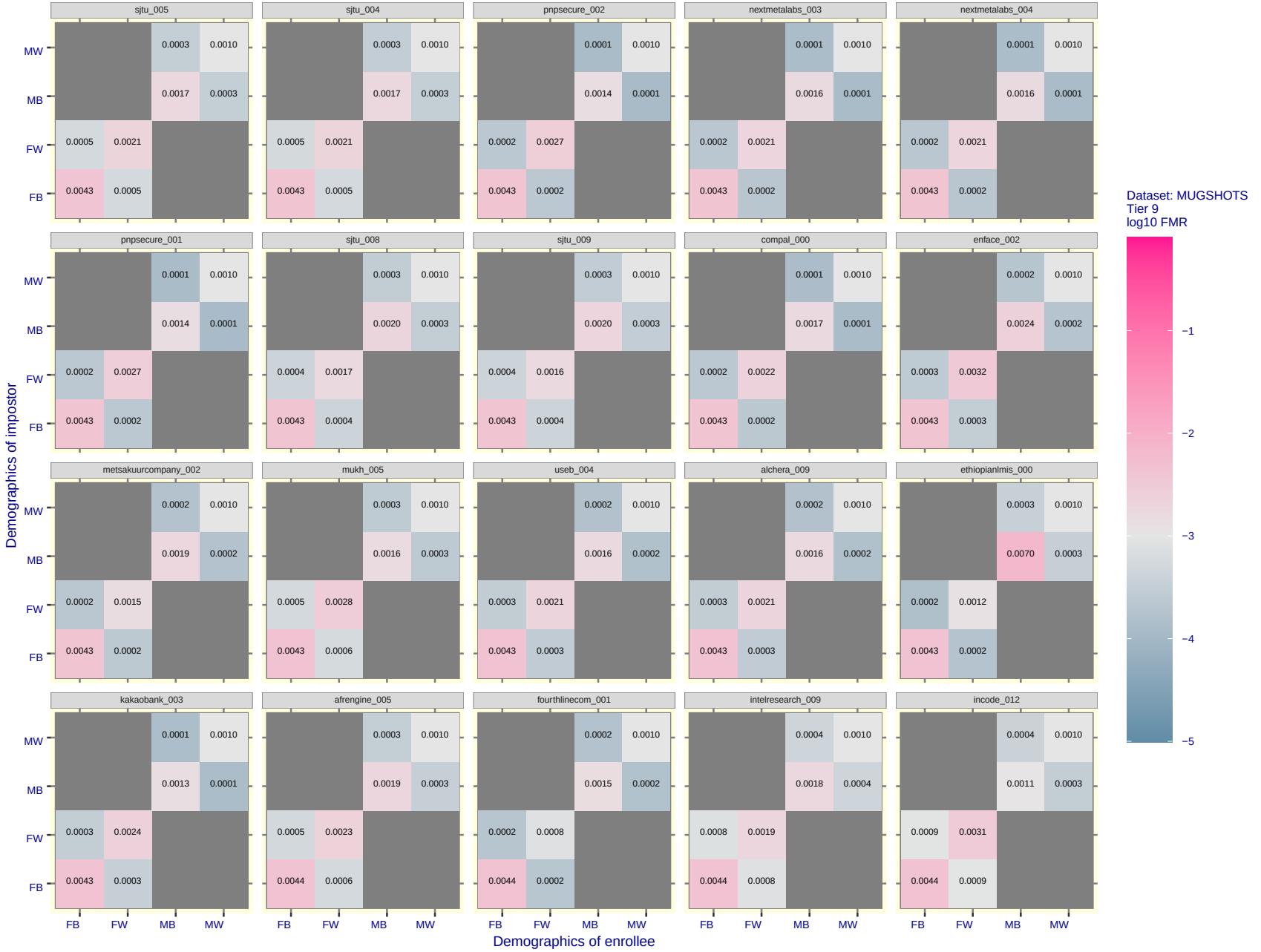

Figure 339: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

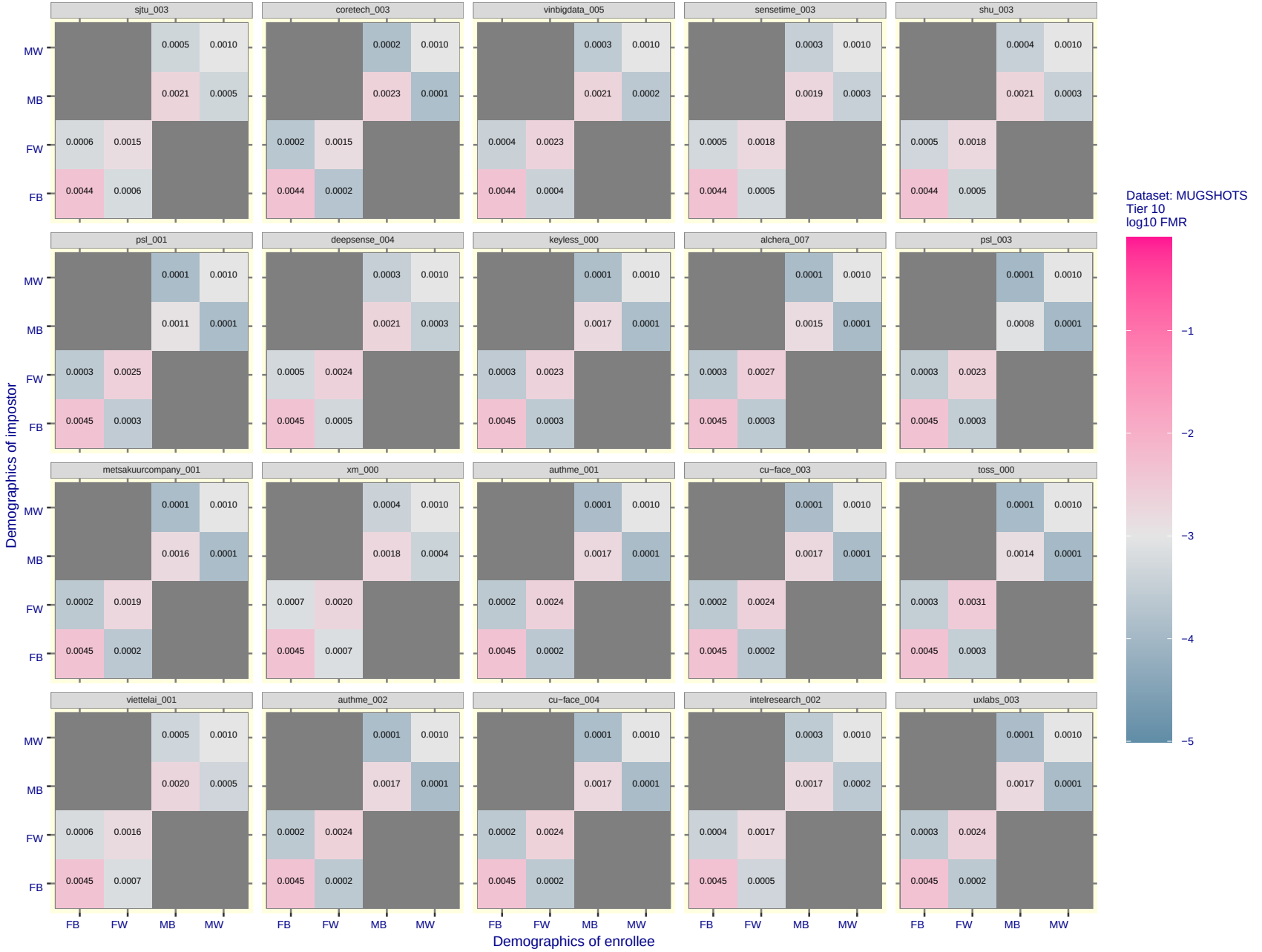

Figure 340: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

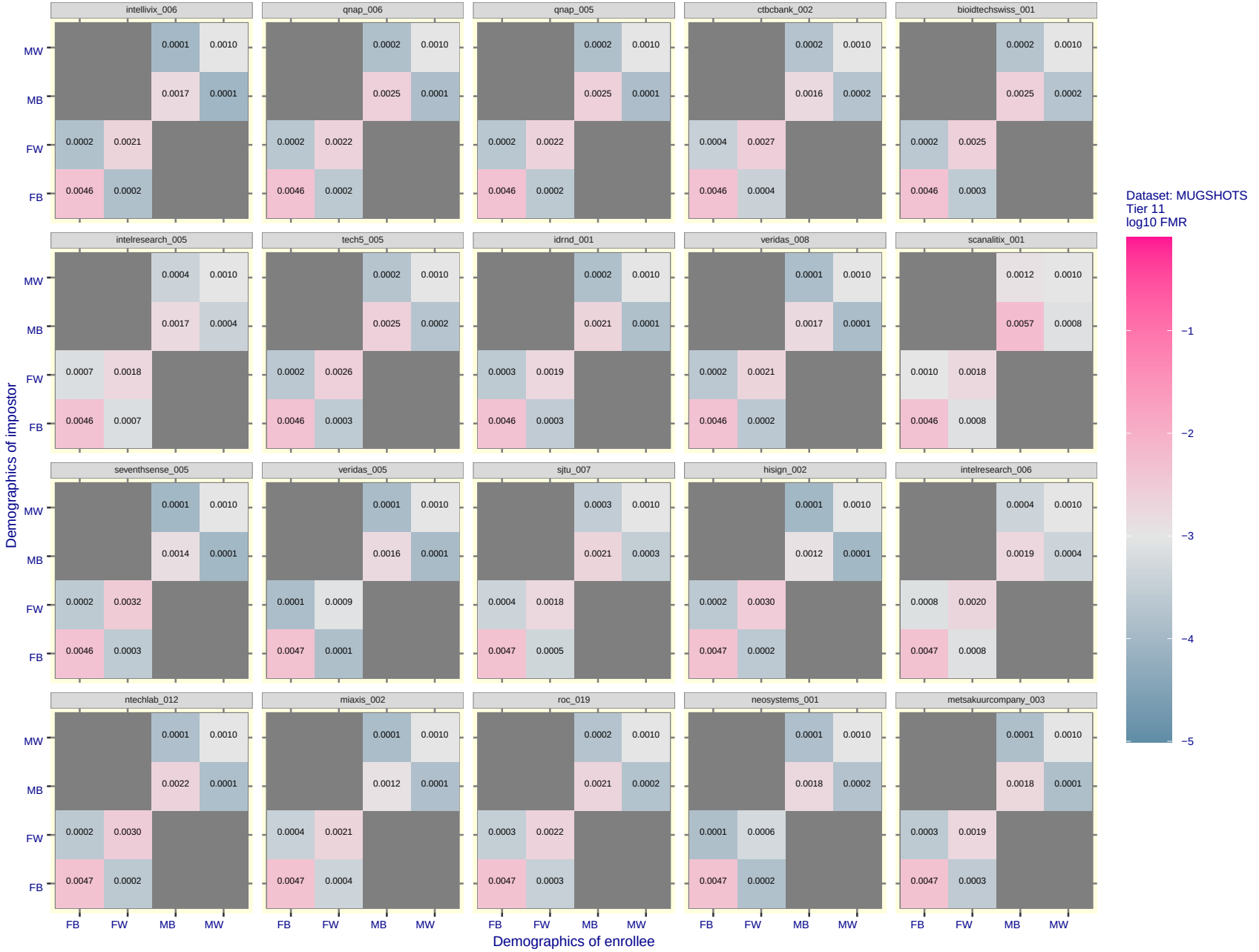

Figure 341: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

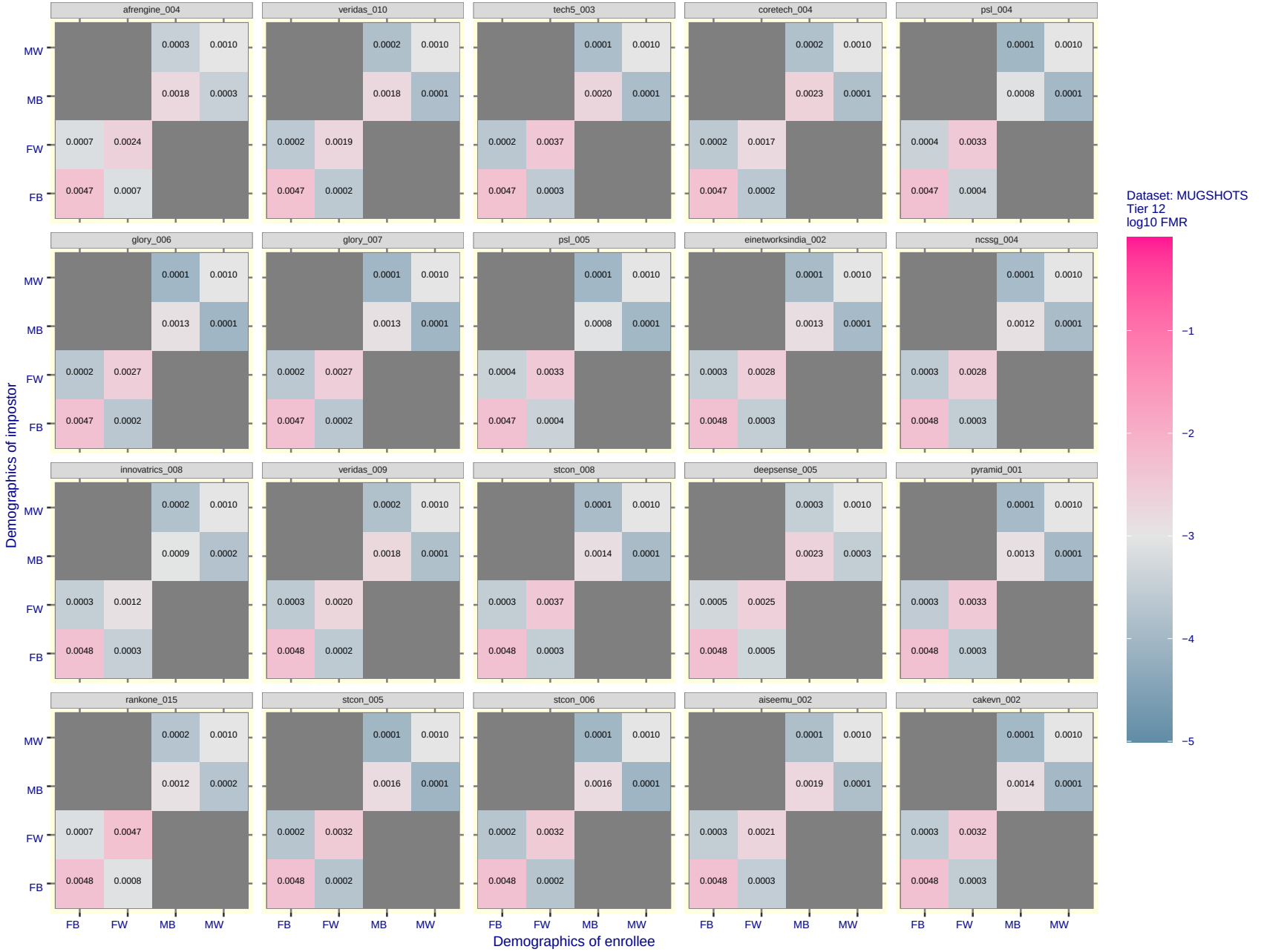

Figure 342: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

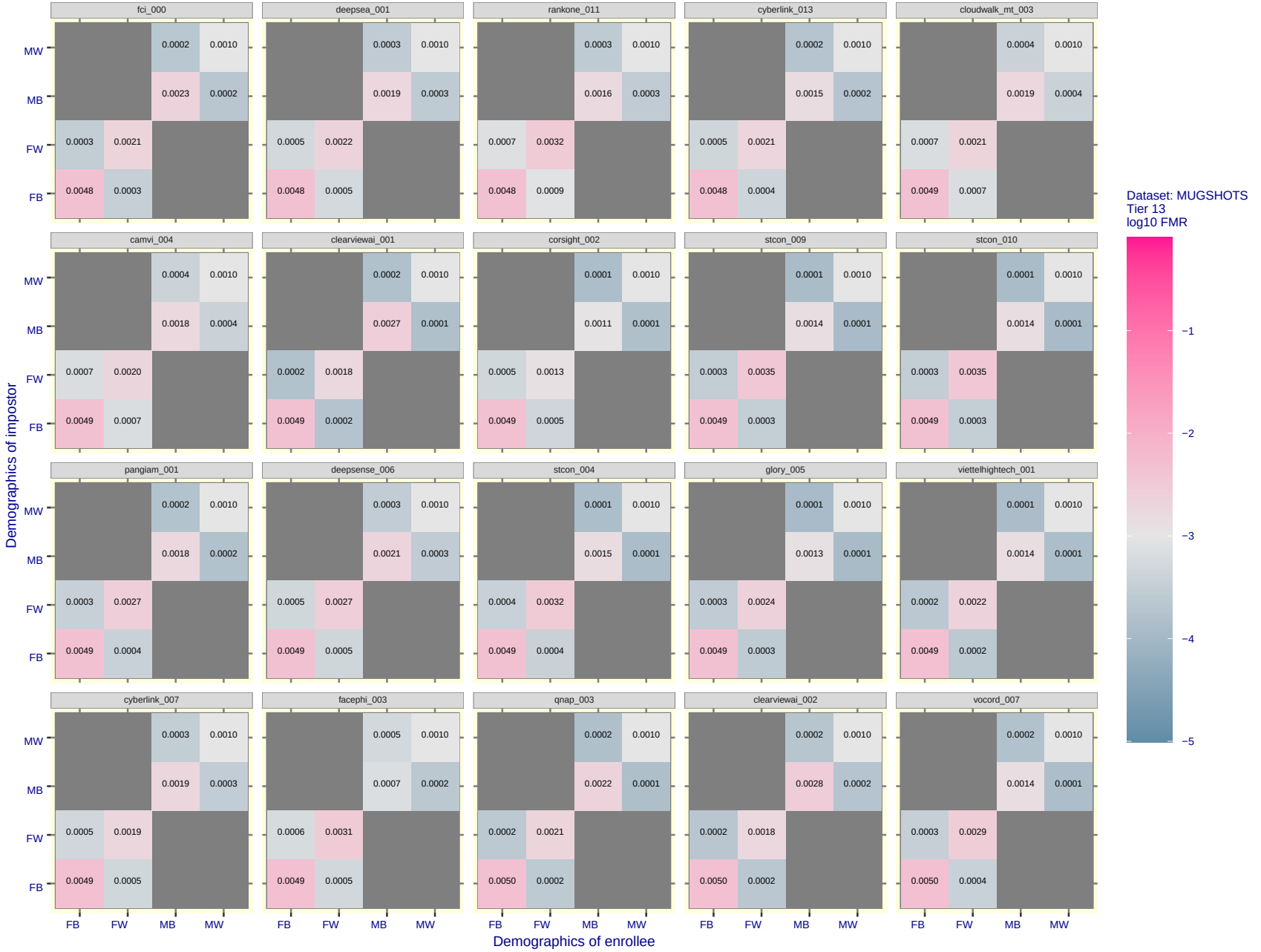

Figure 343: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

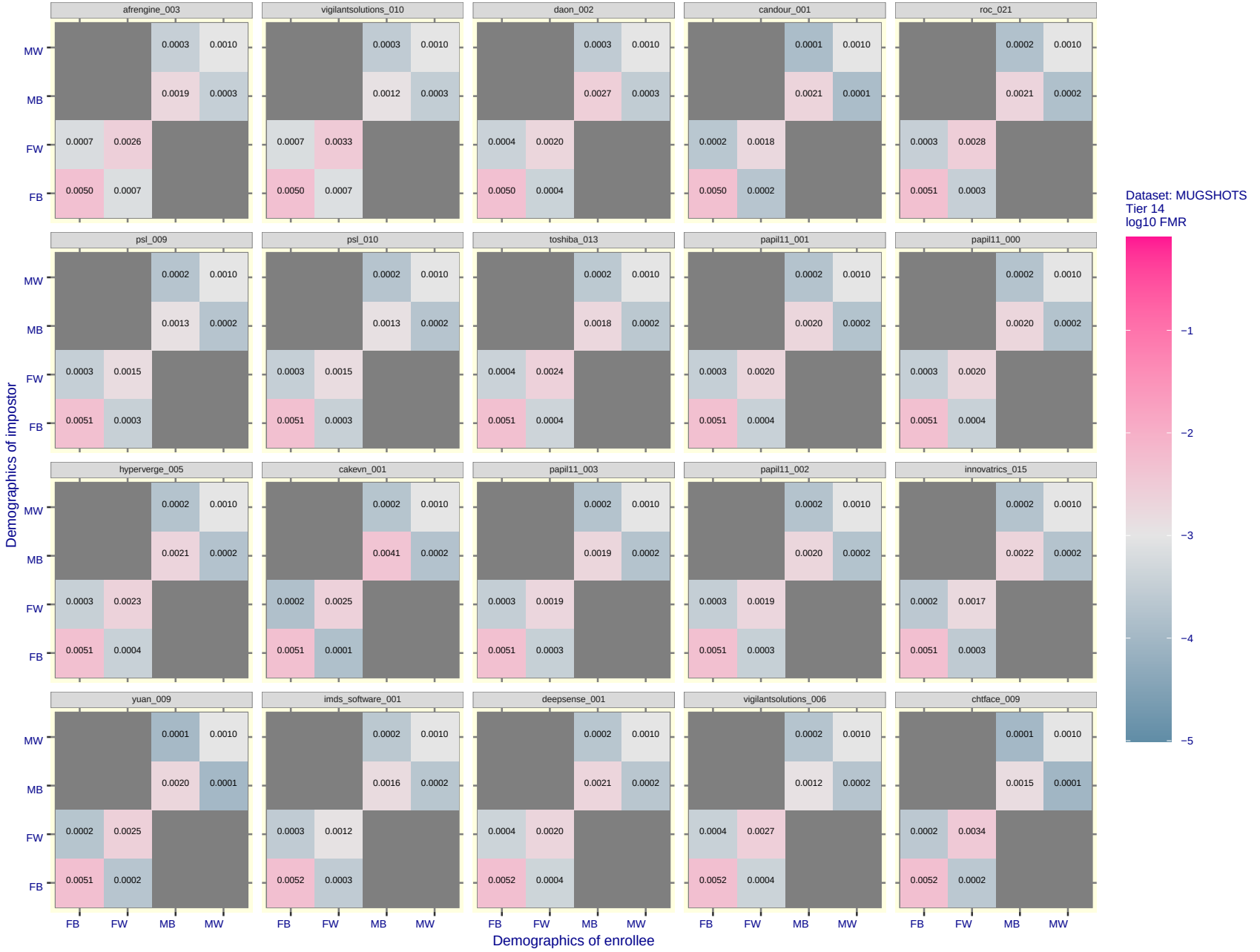

Figure 344: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

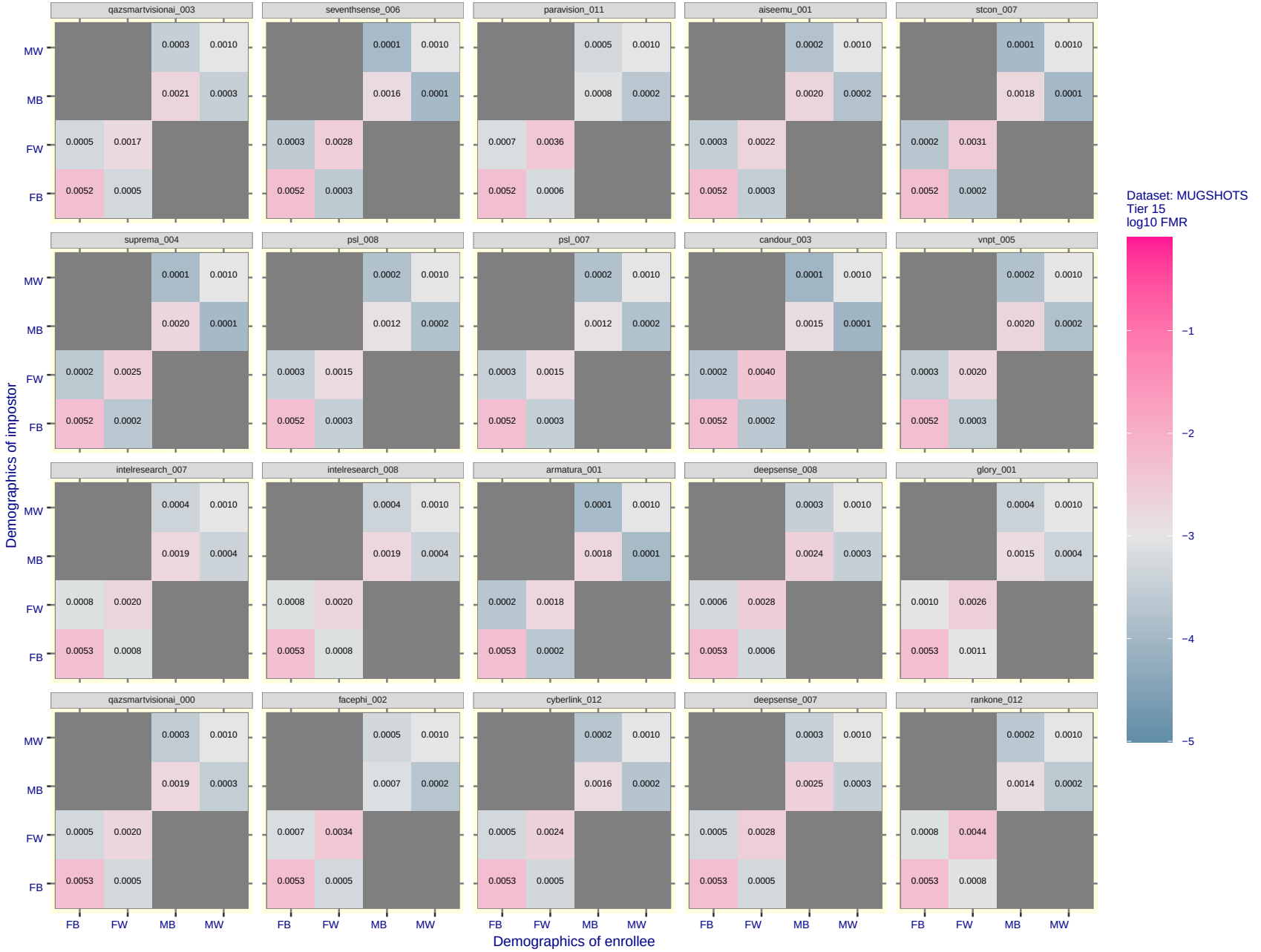

Figure 345: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

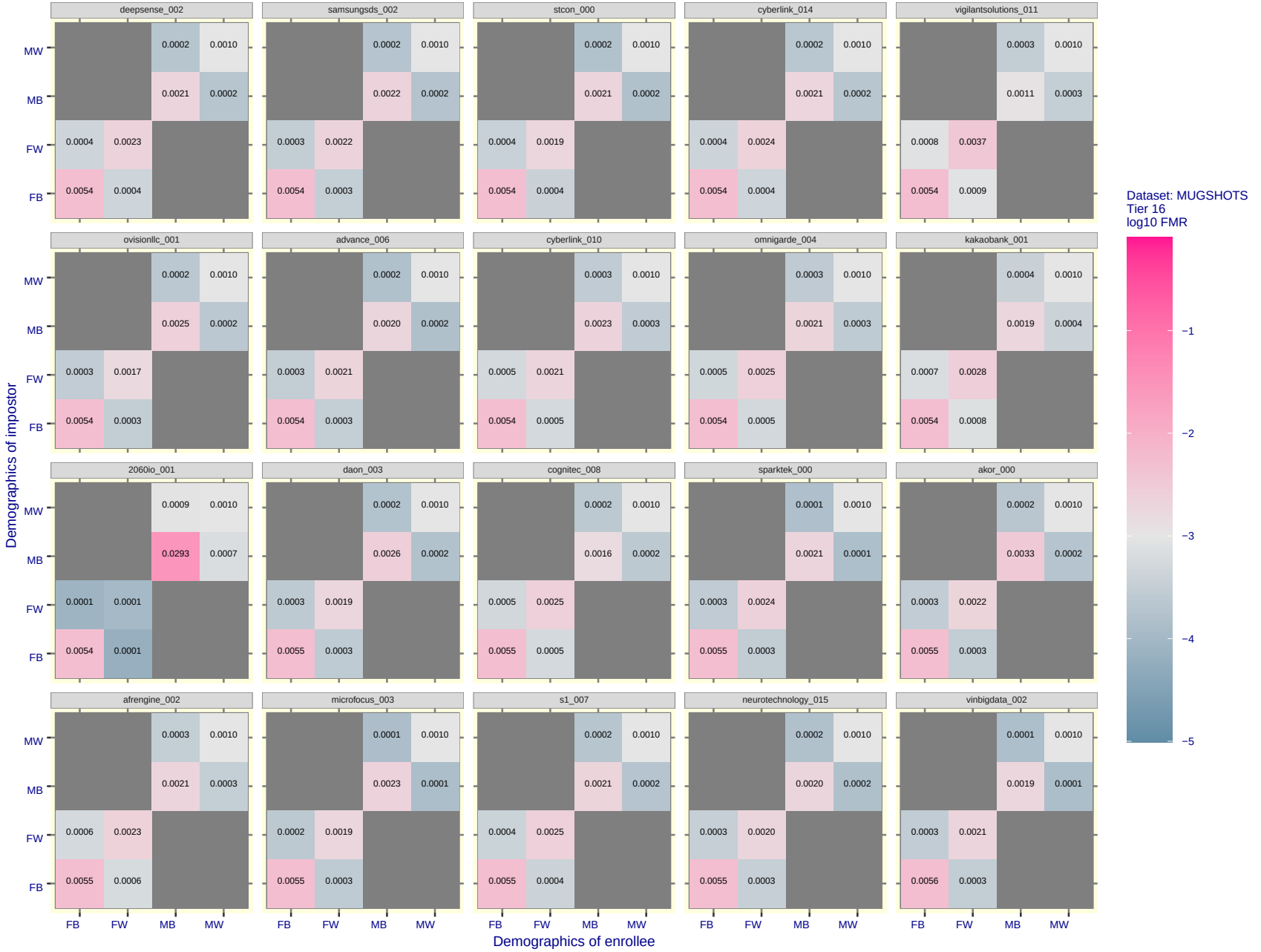

Figure 346: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

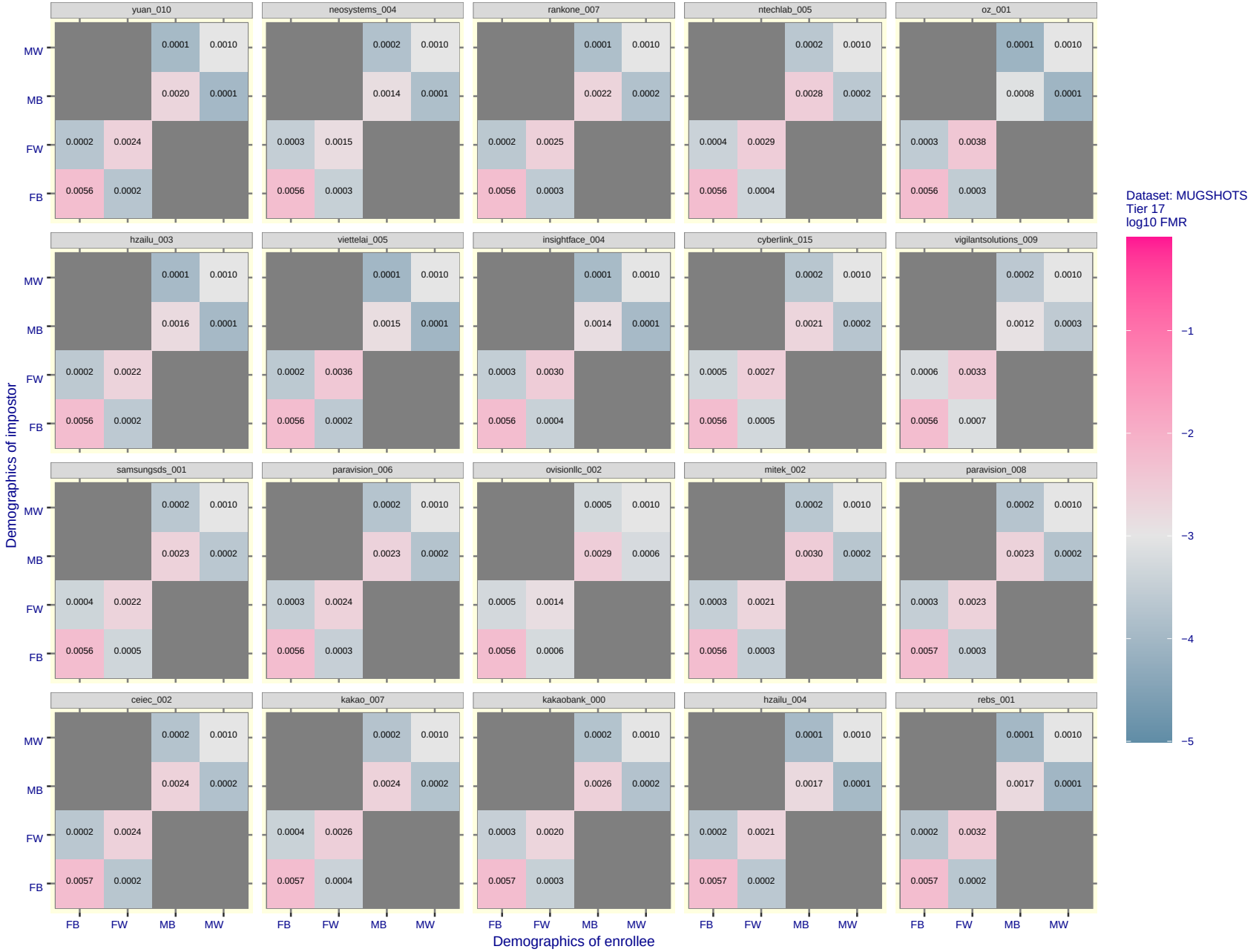

Figure 347: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

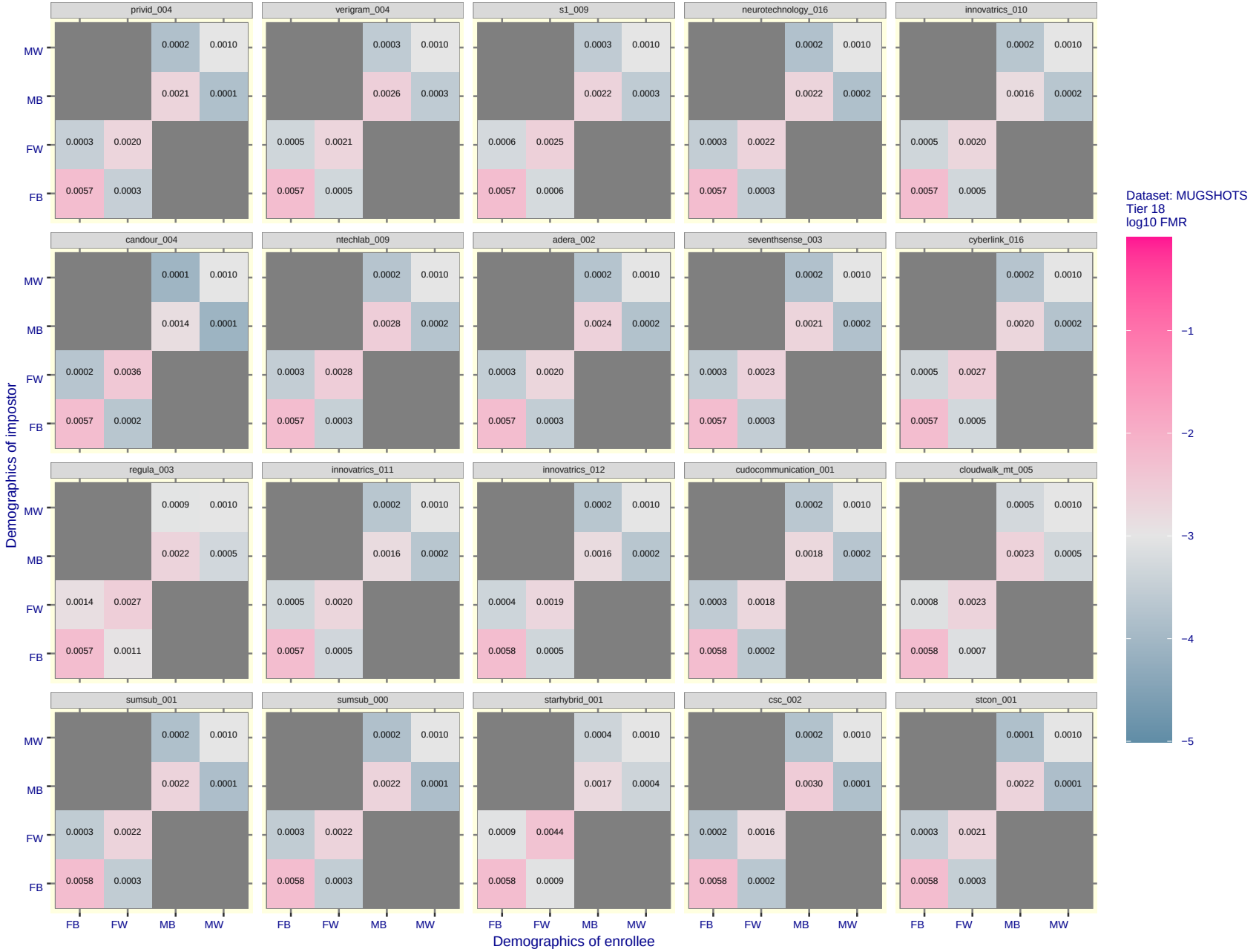

Figure 348: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

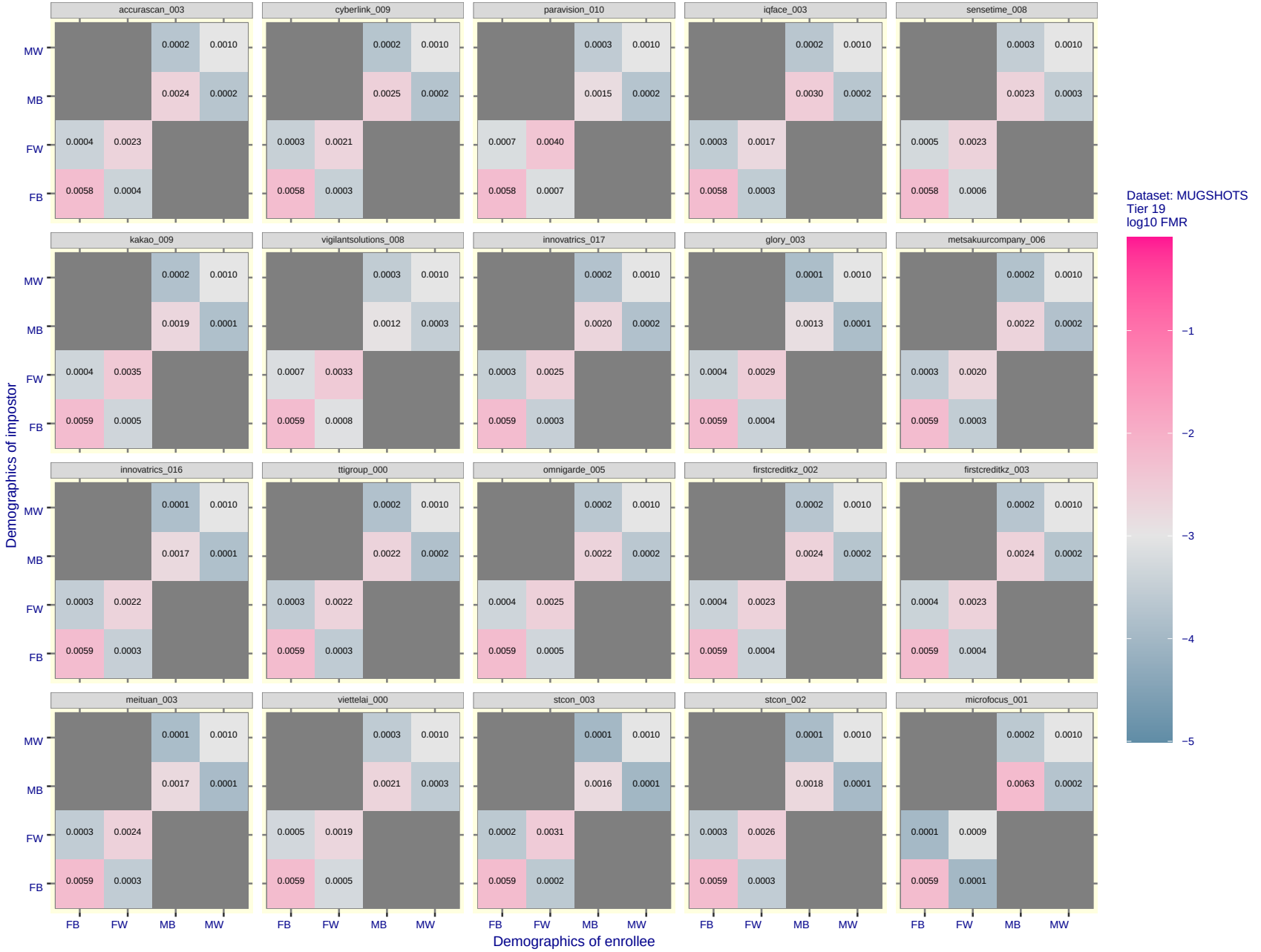

Figure 349: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

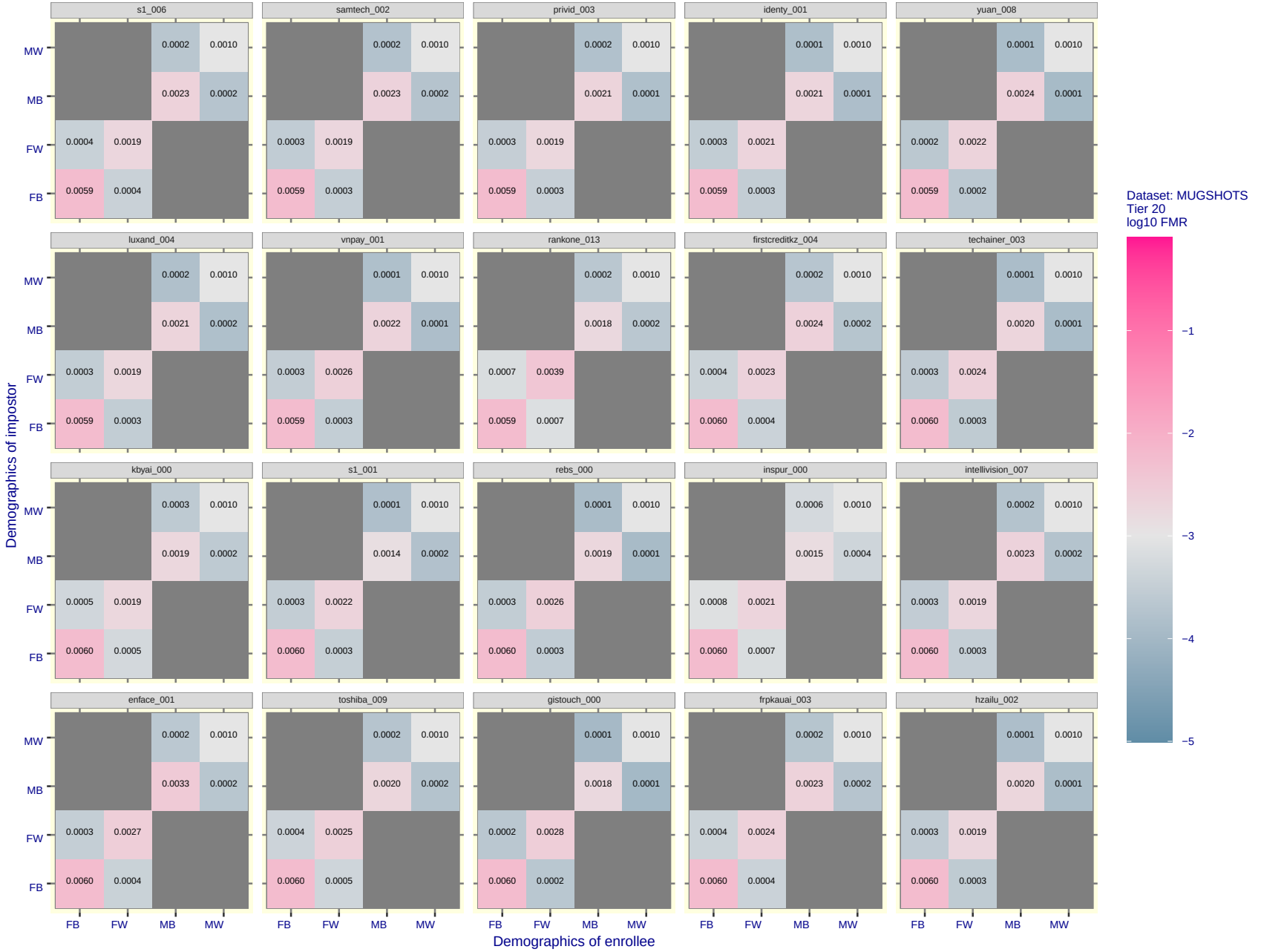

Figure 350: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

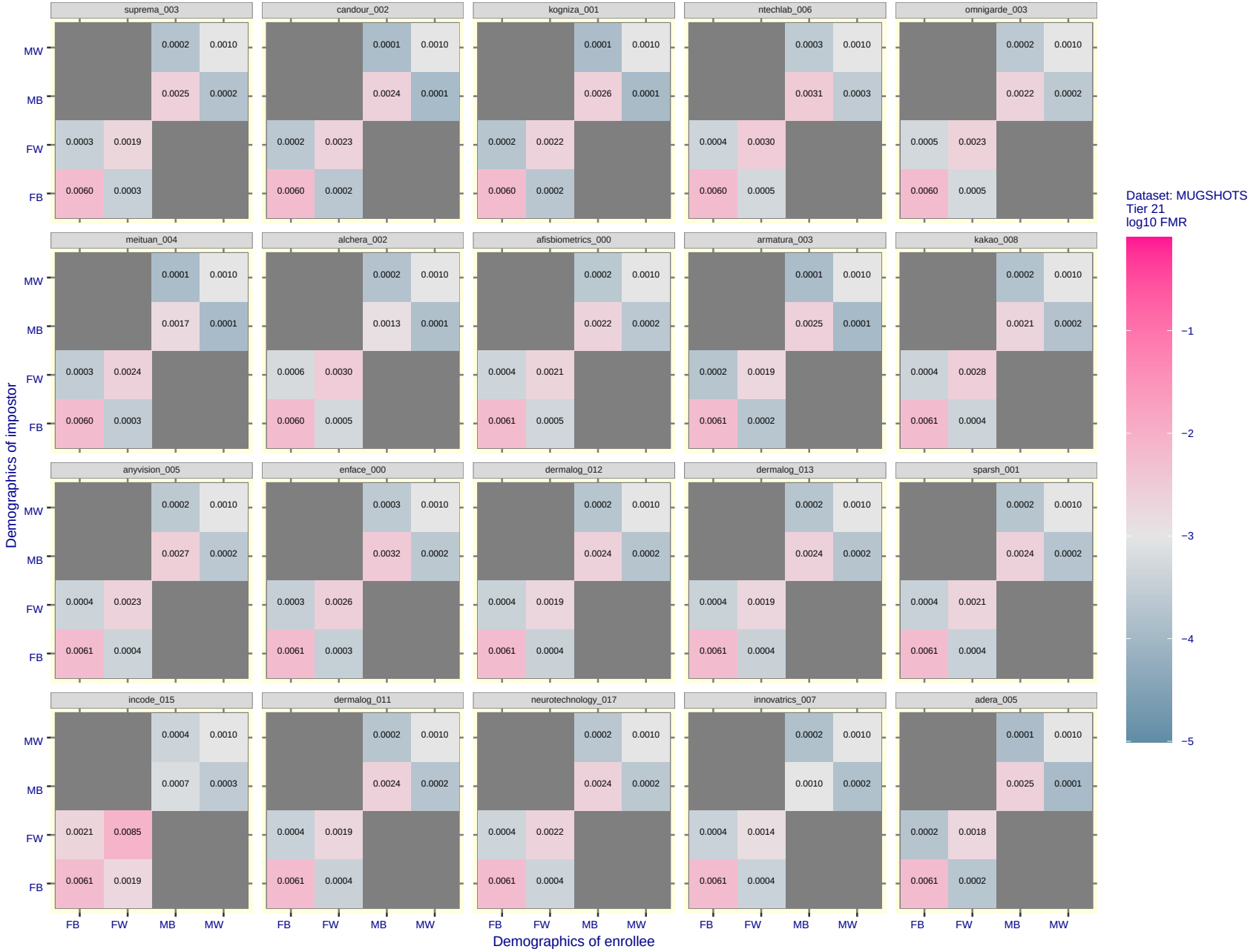

Figure 351: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

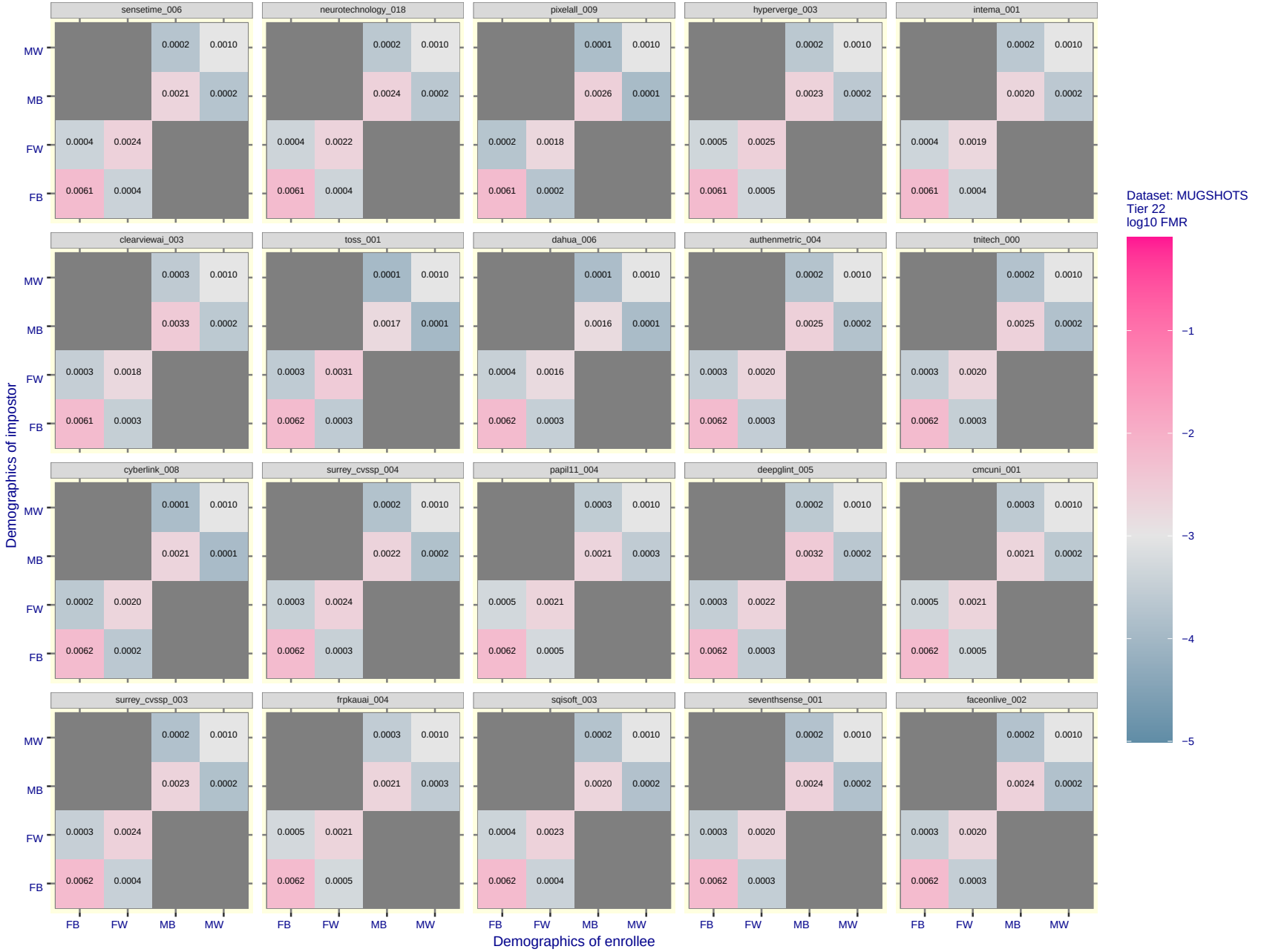

Figure 352: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

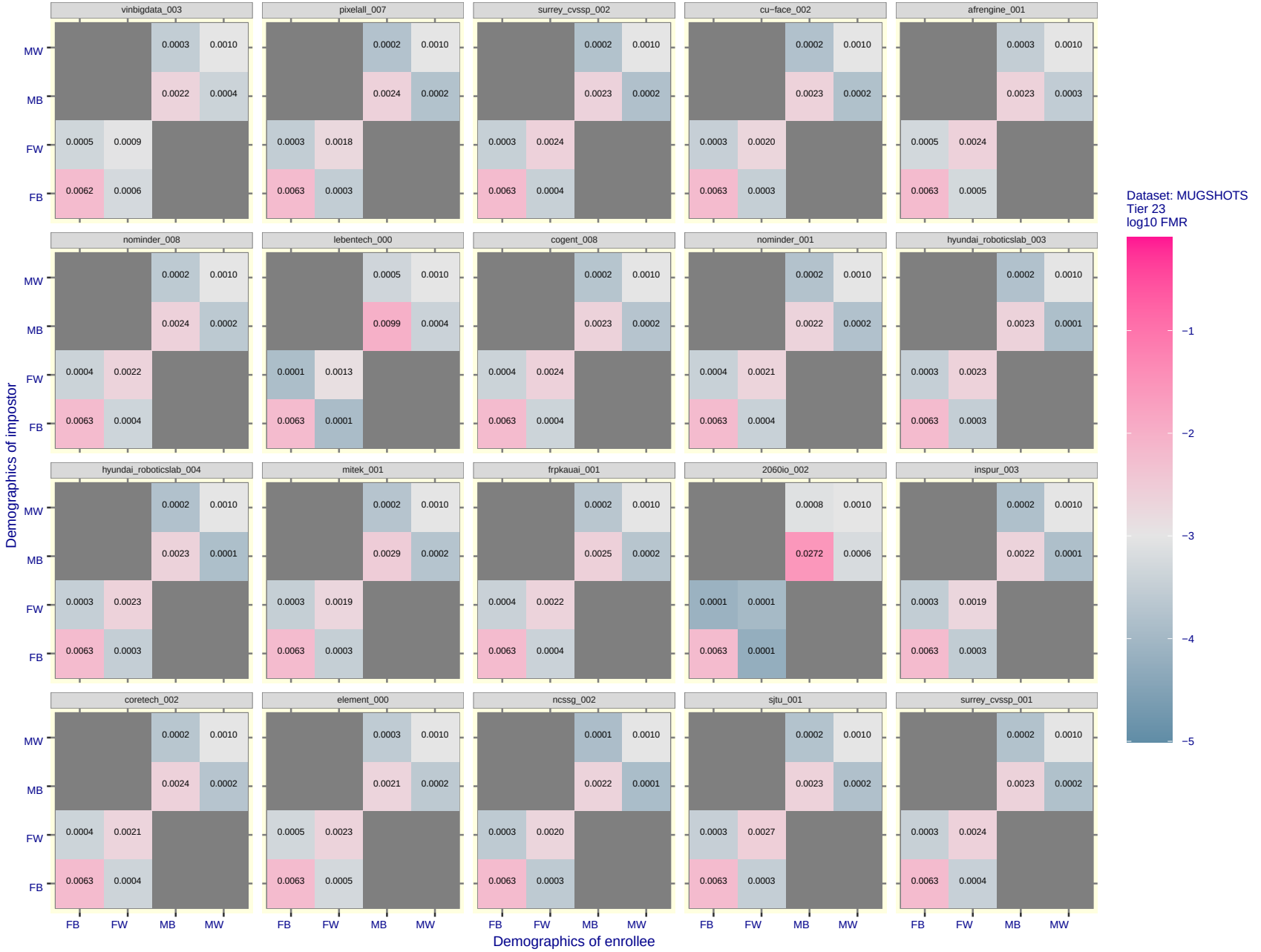

Figure 353: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

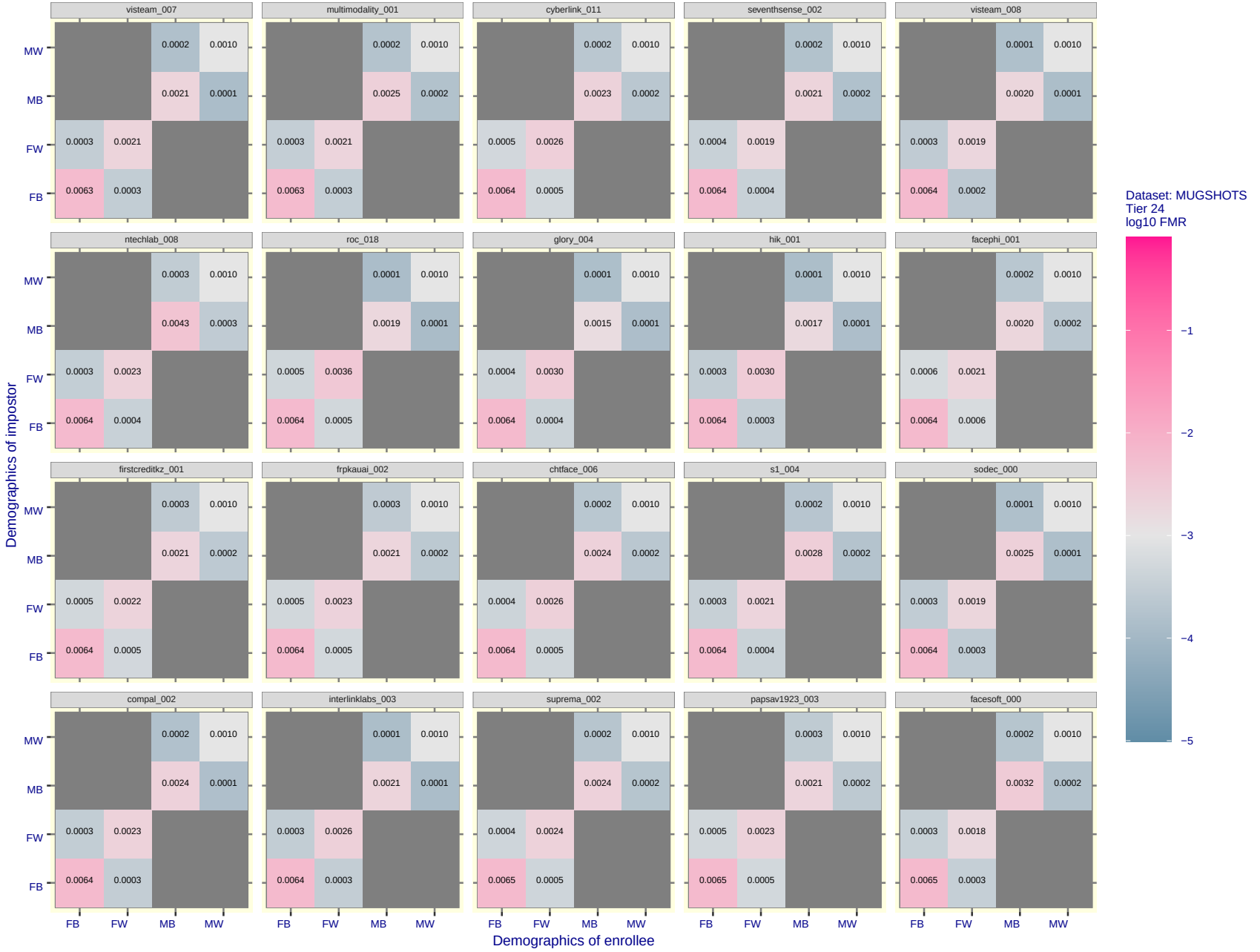

Figure 354: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

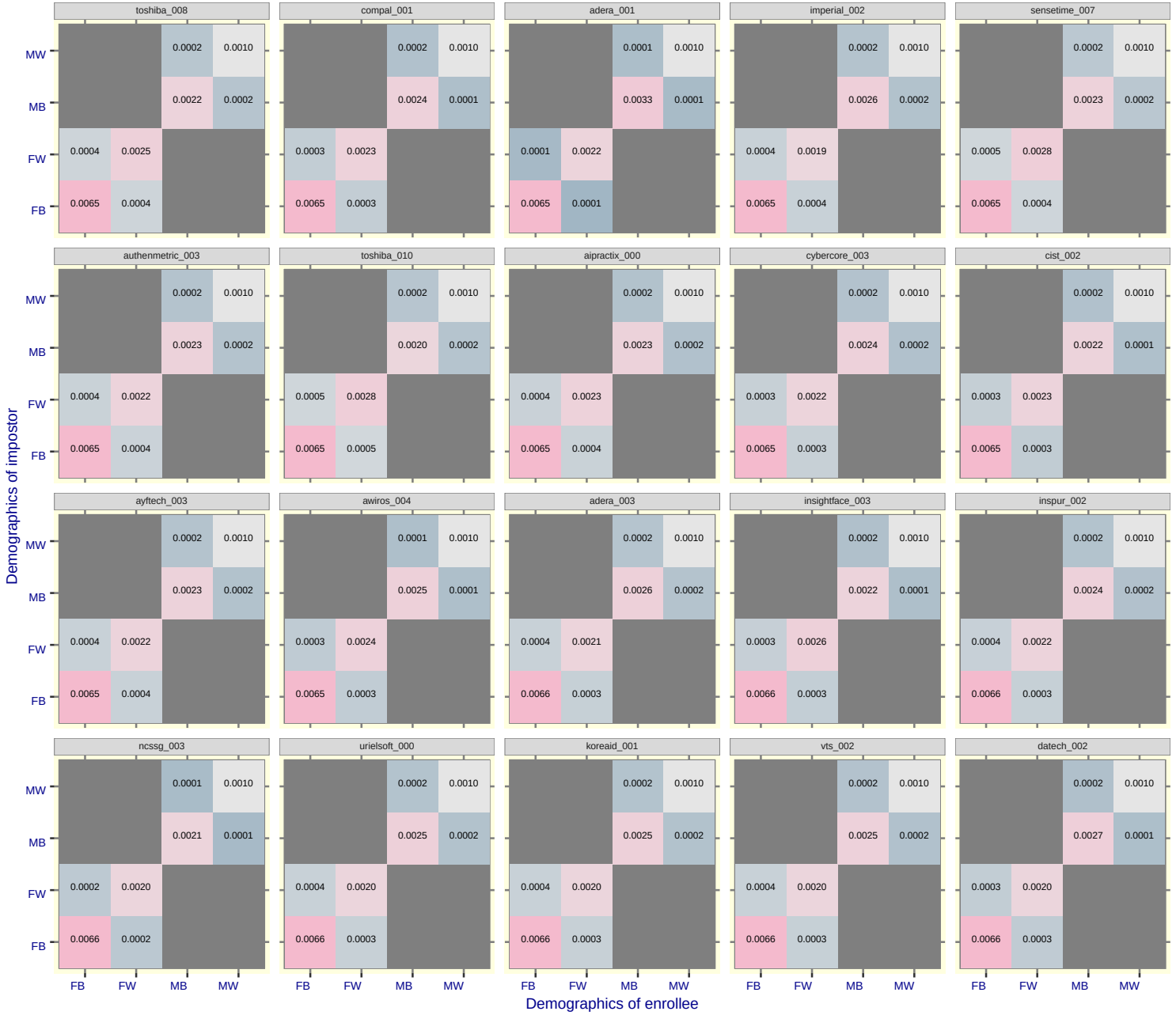

Figure 355: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

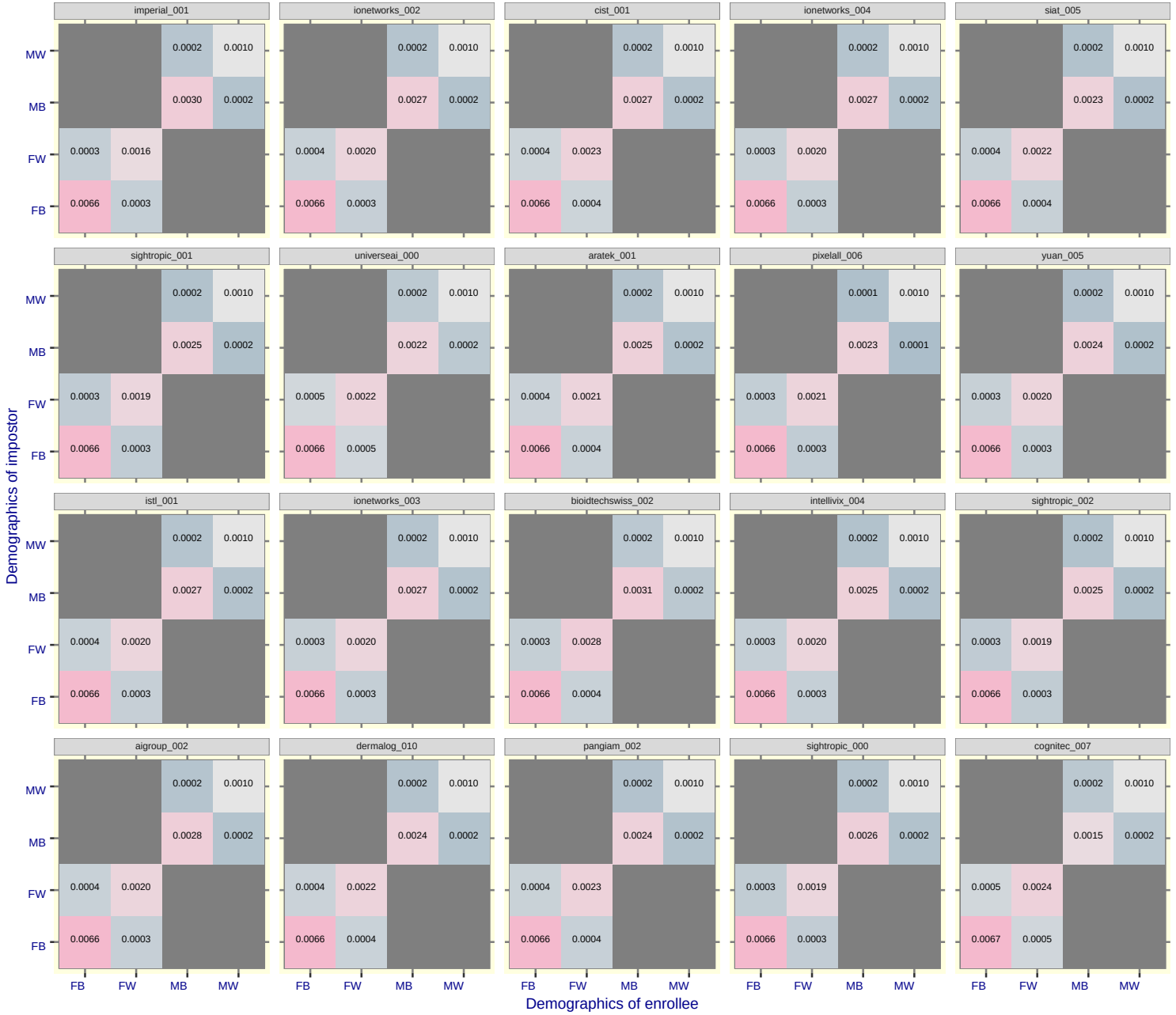

Figure 356: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

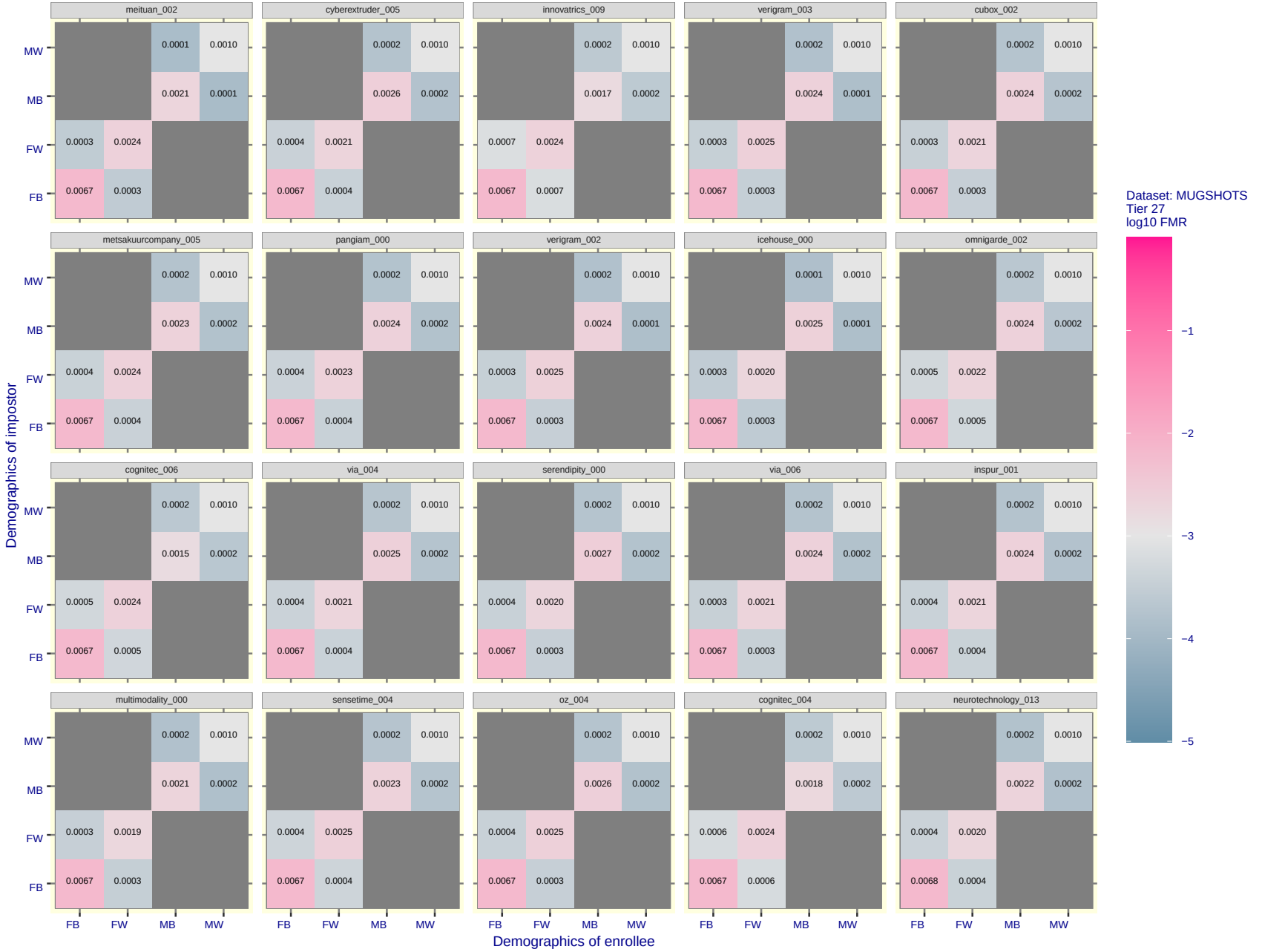

Figure 357: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

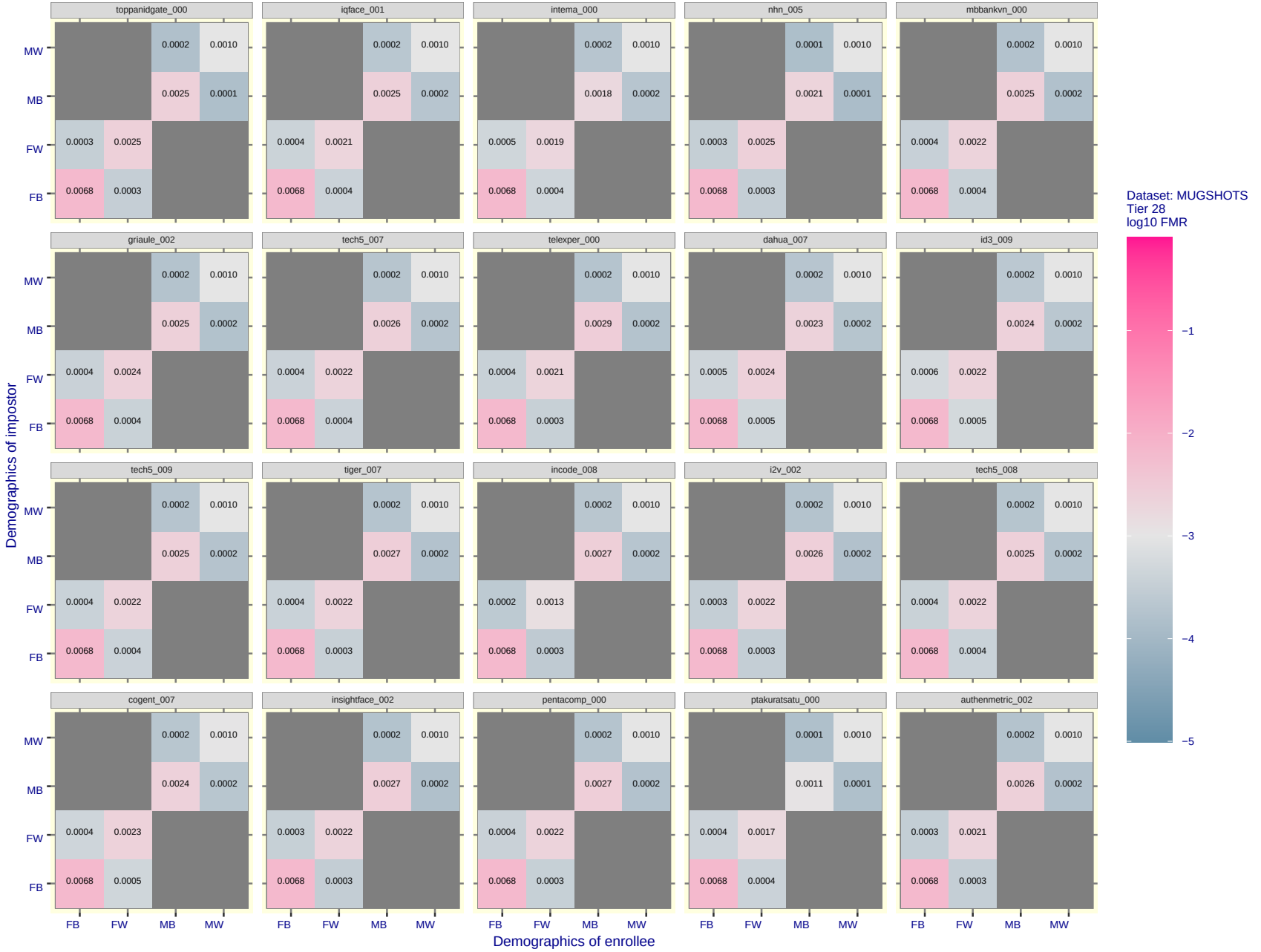

Figure 358: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

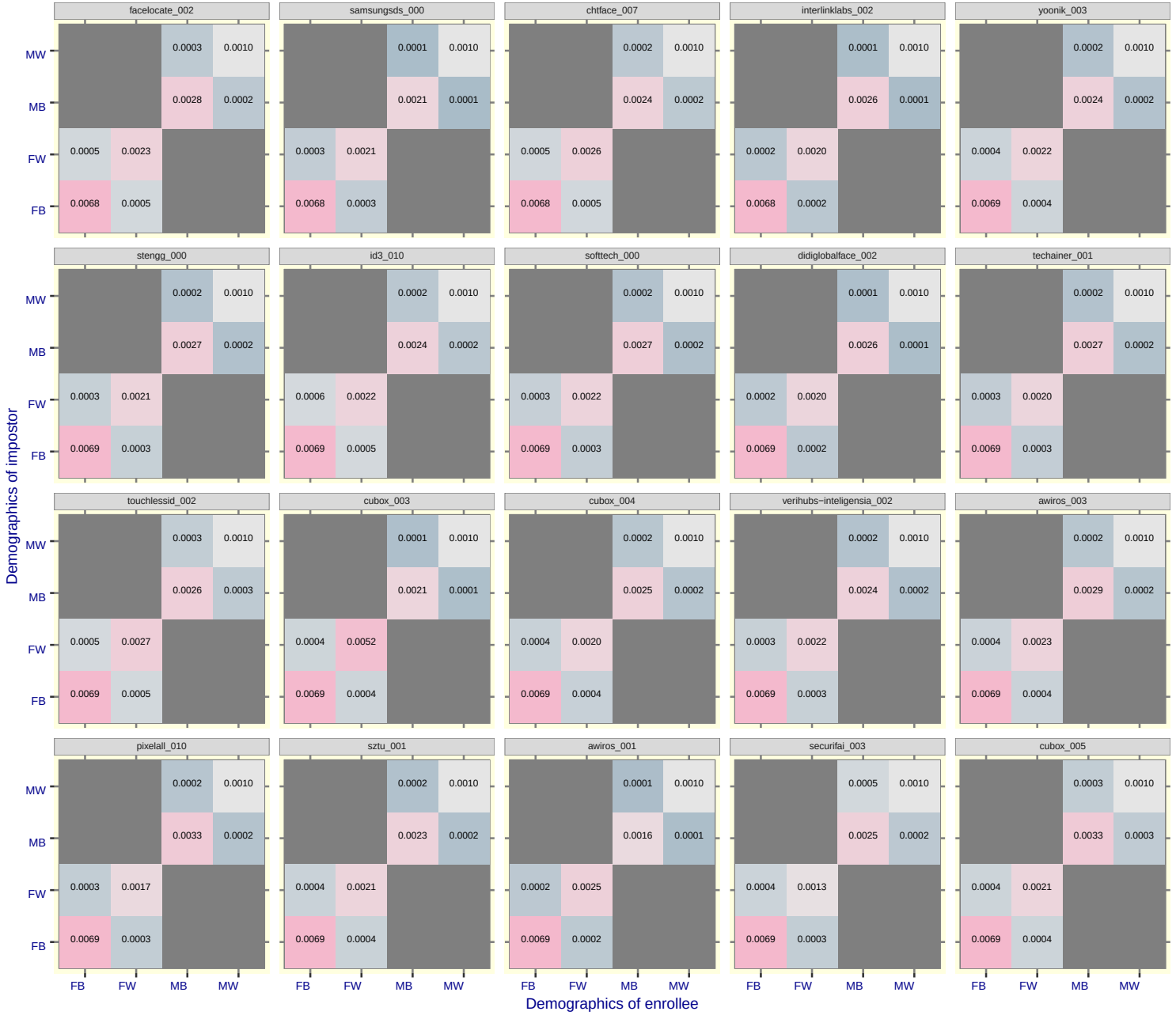

Figure 359: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

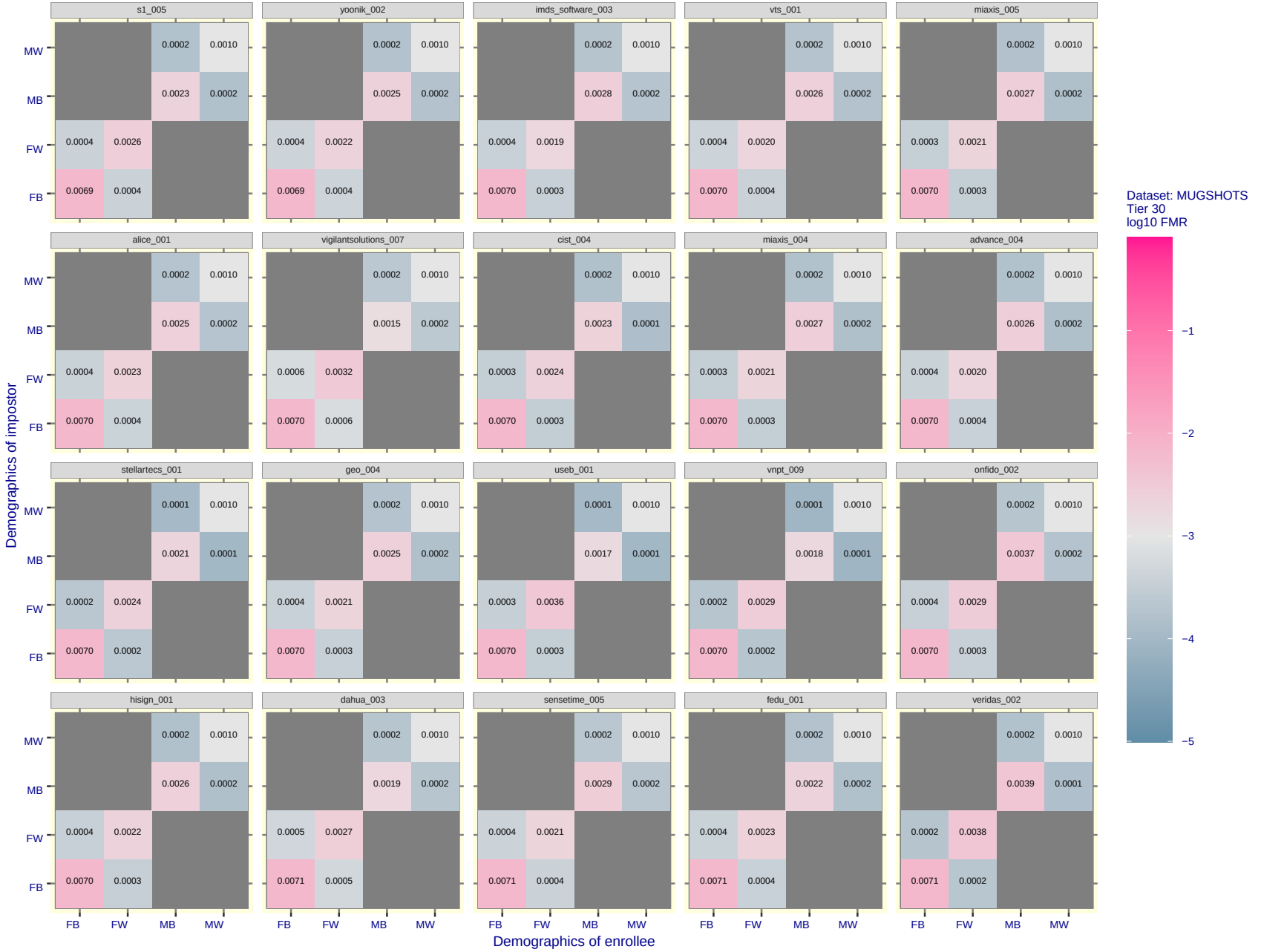

Figure 360: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

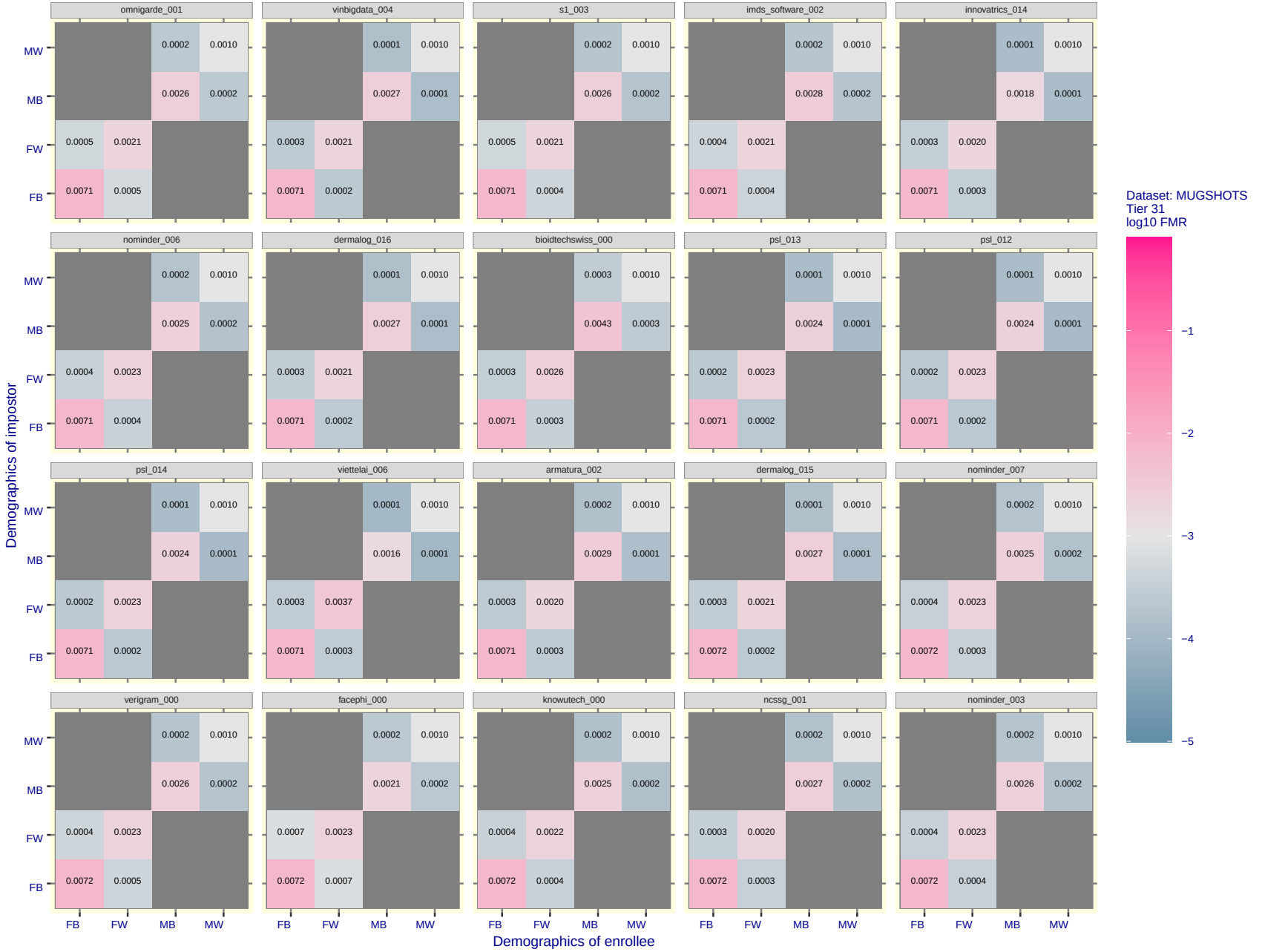

Figure 361: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

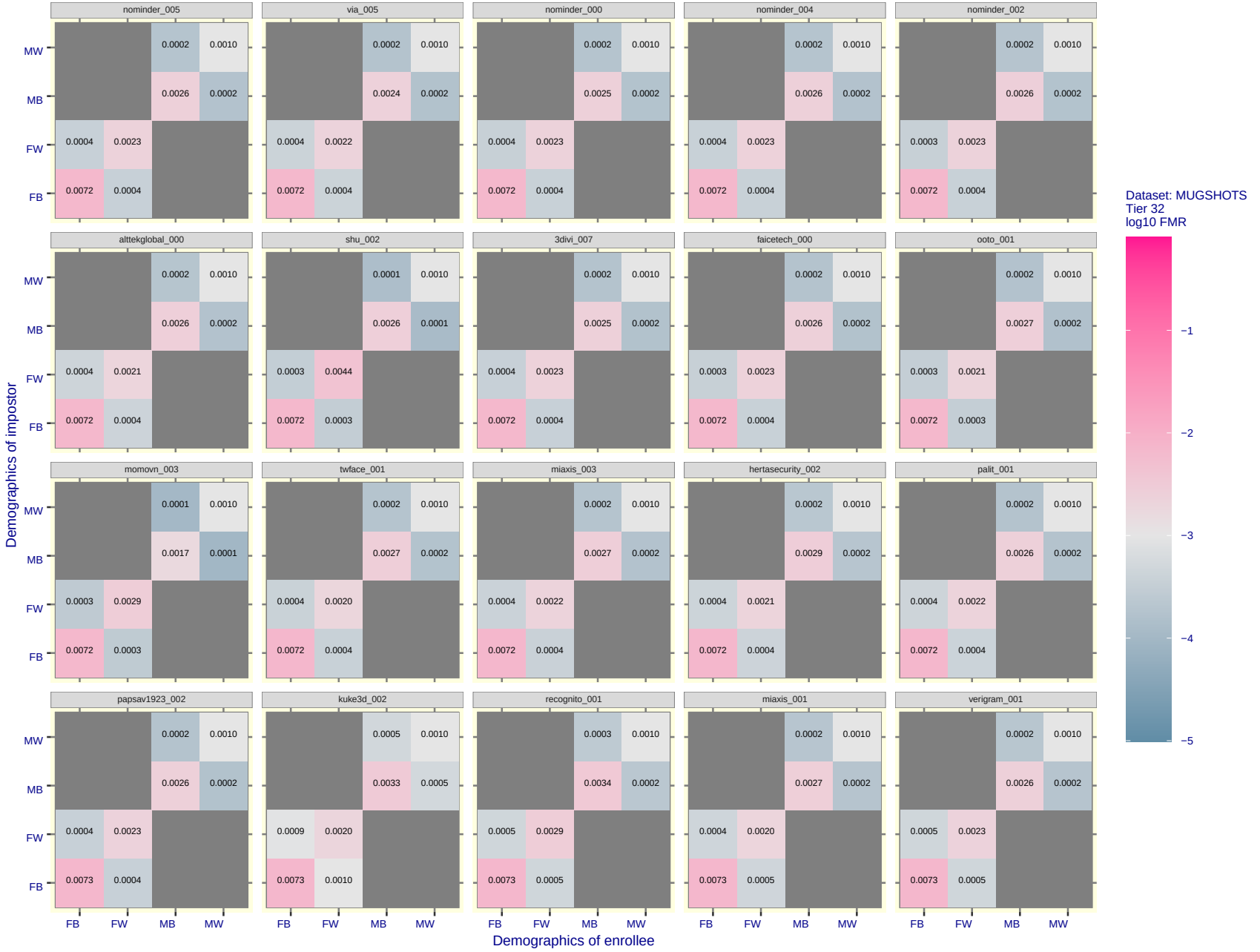

Figure 362: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

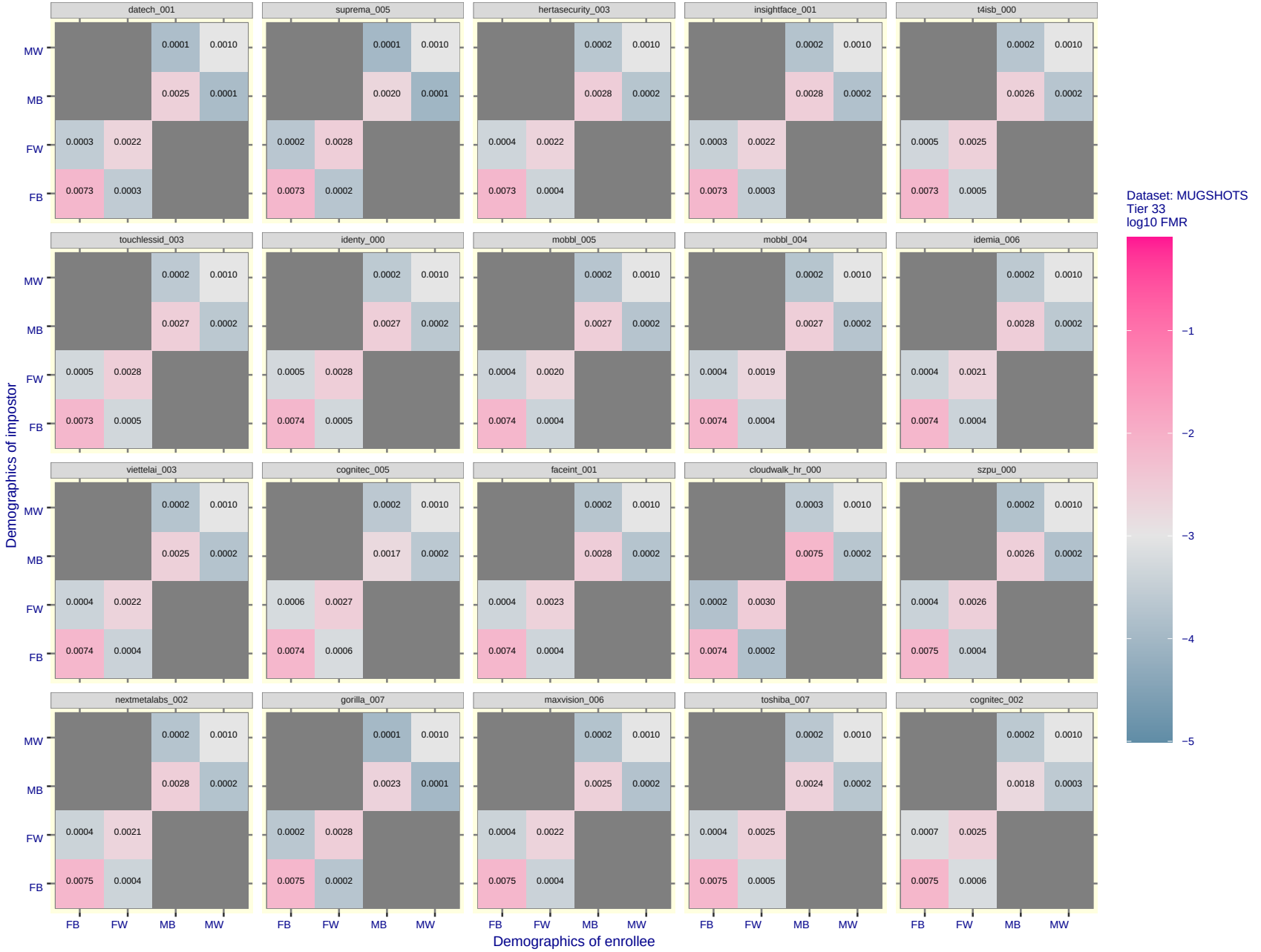

Figure 363: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

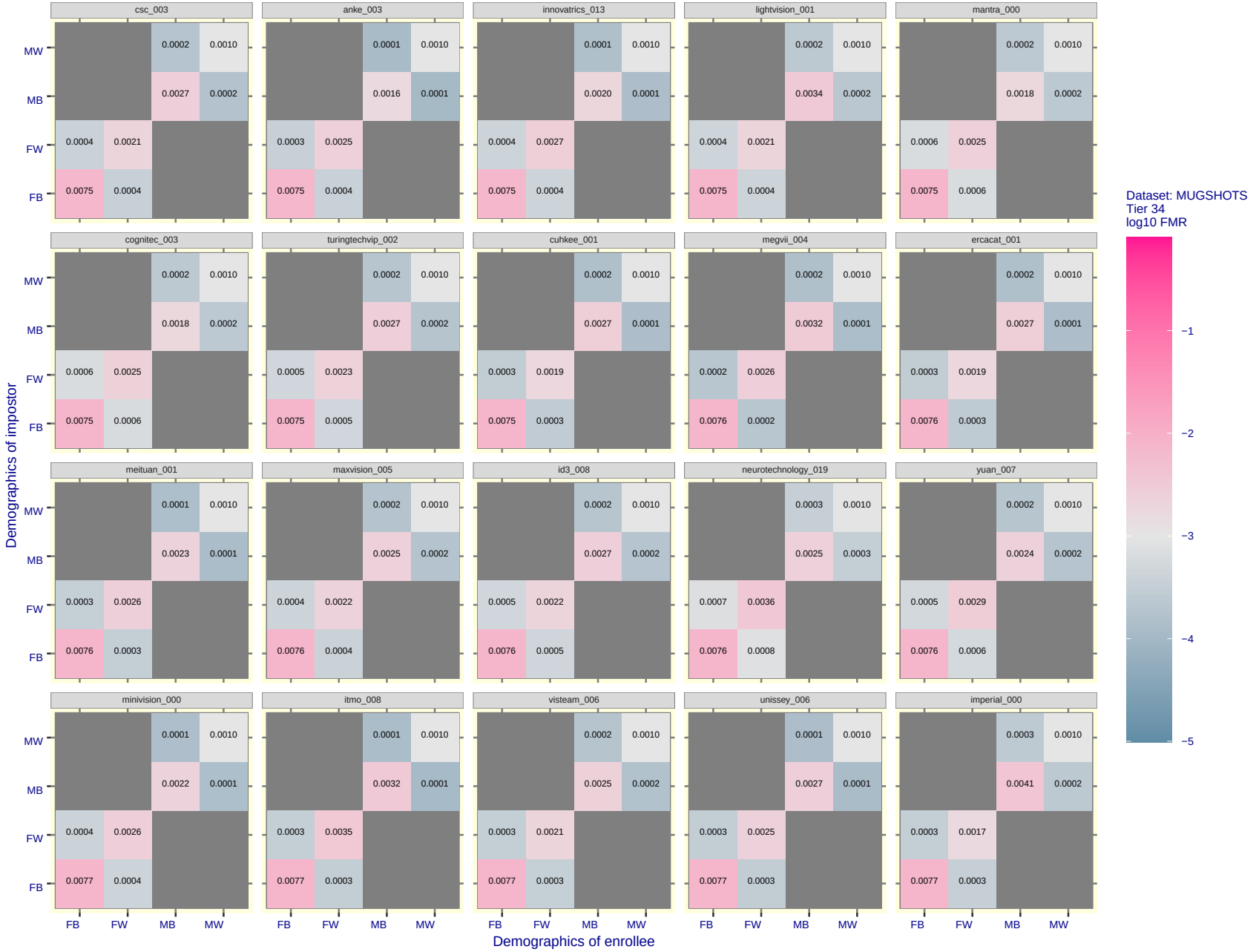

Figure 364: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

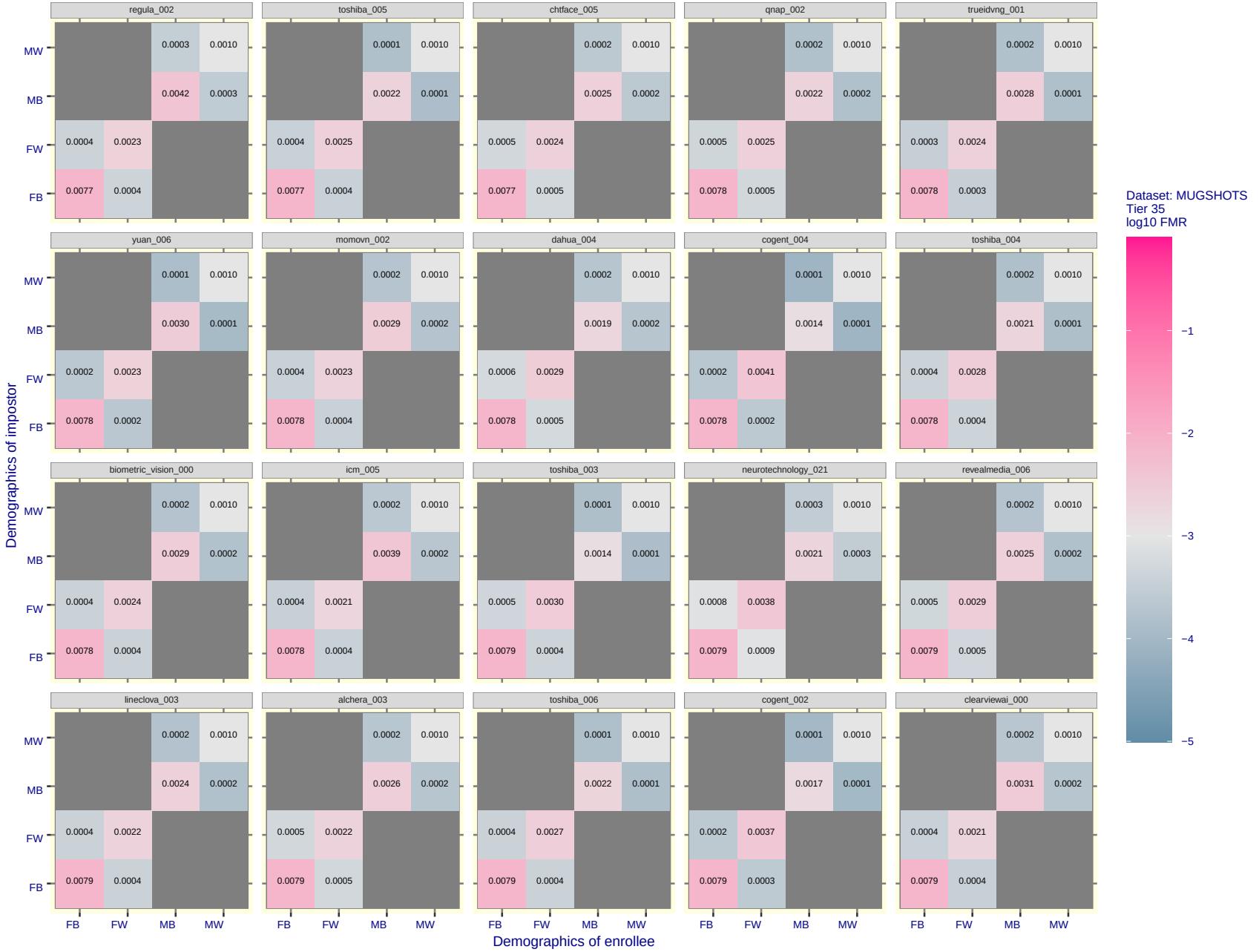

Figure 365: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

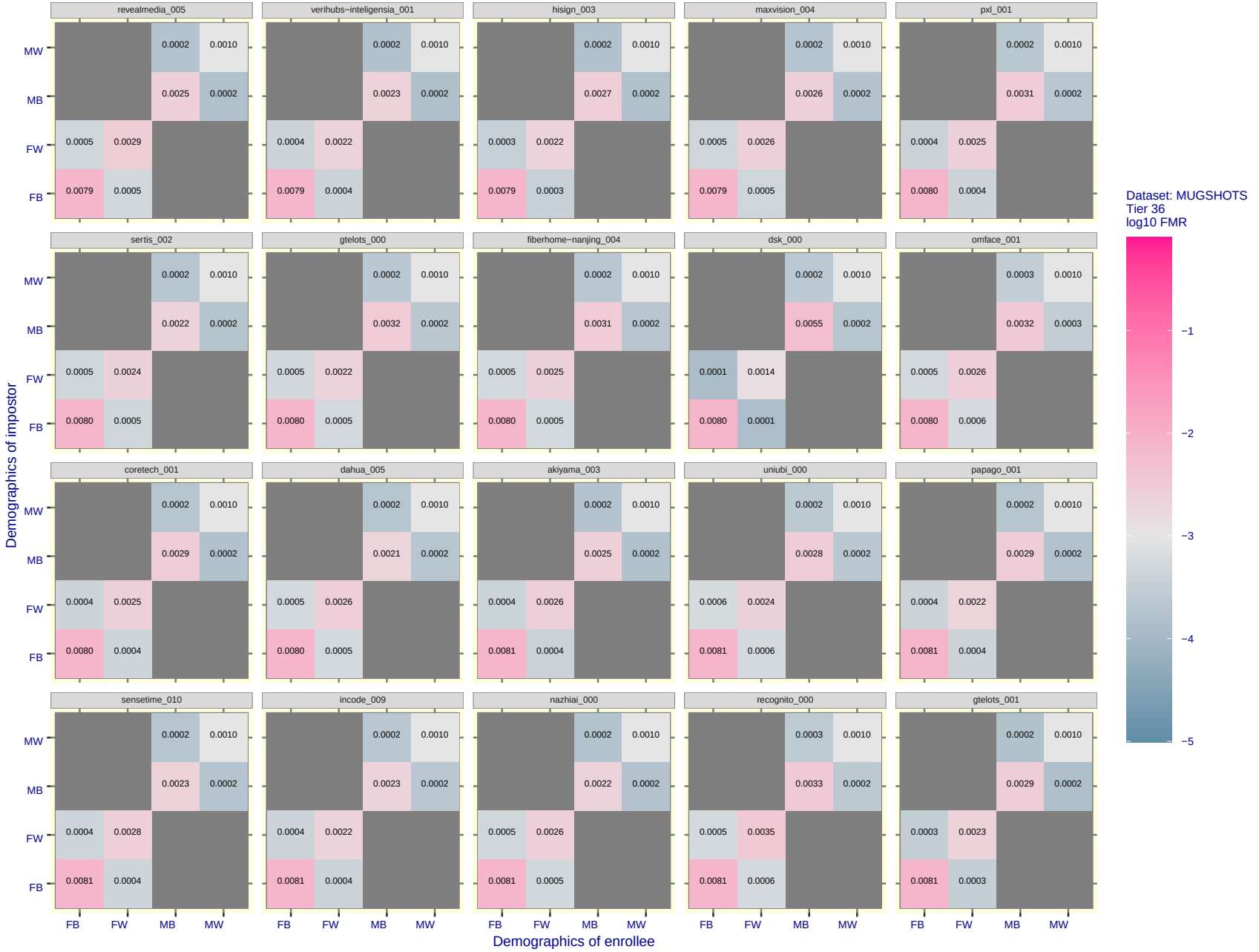

Figure 366: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

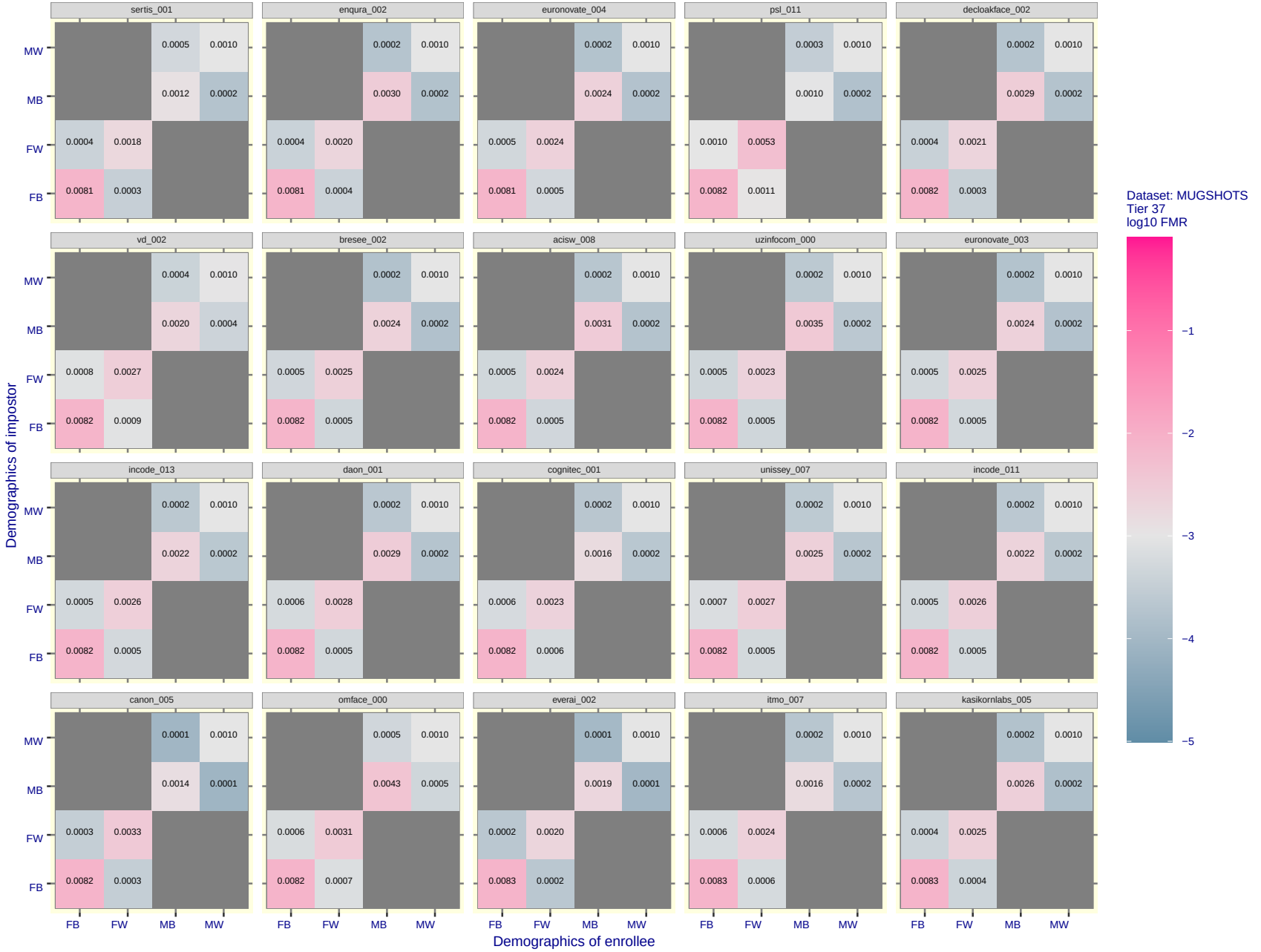

Figure 367: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

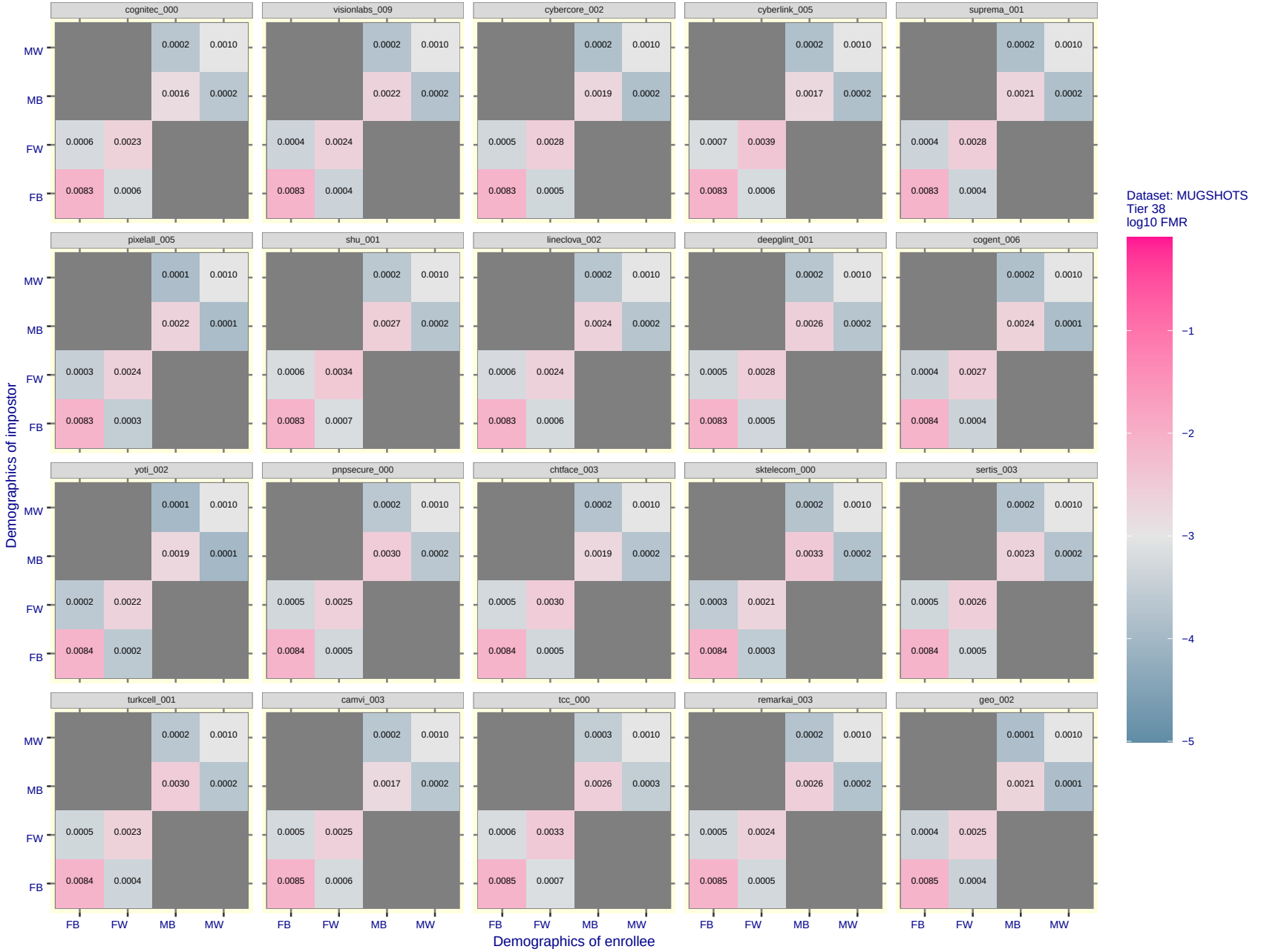

Figure 368: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

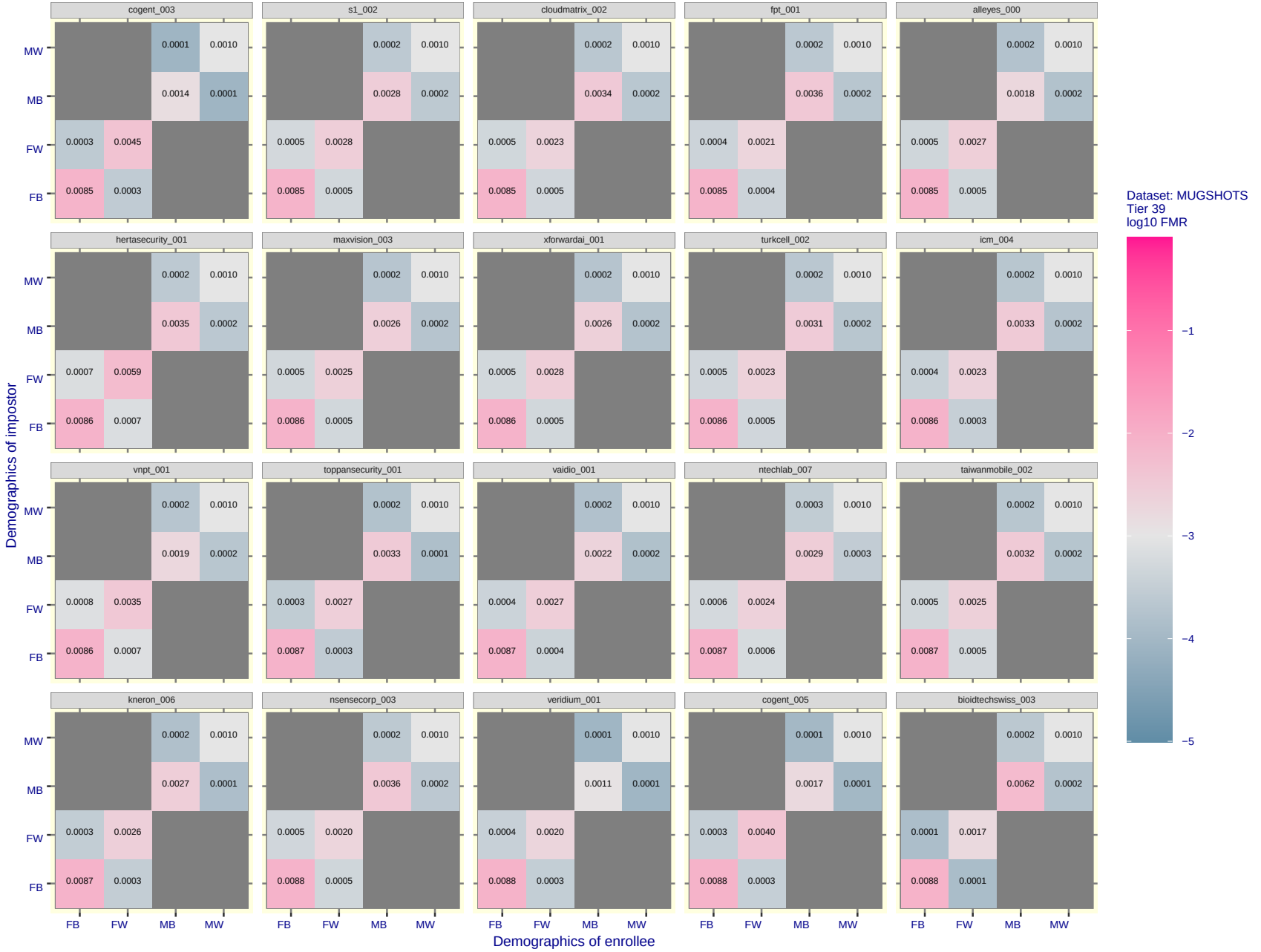

Figure 369: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

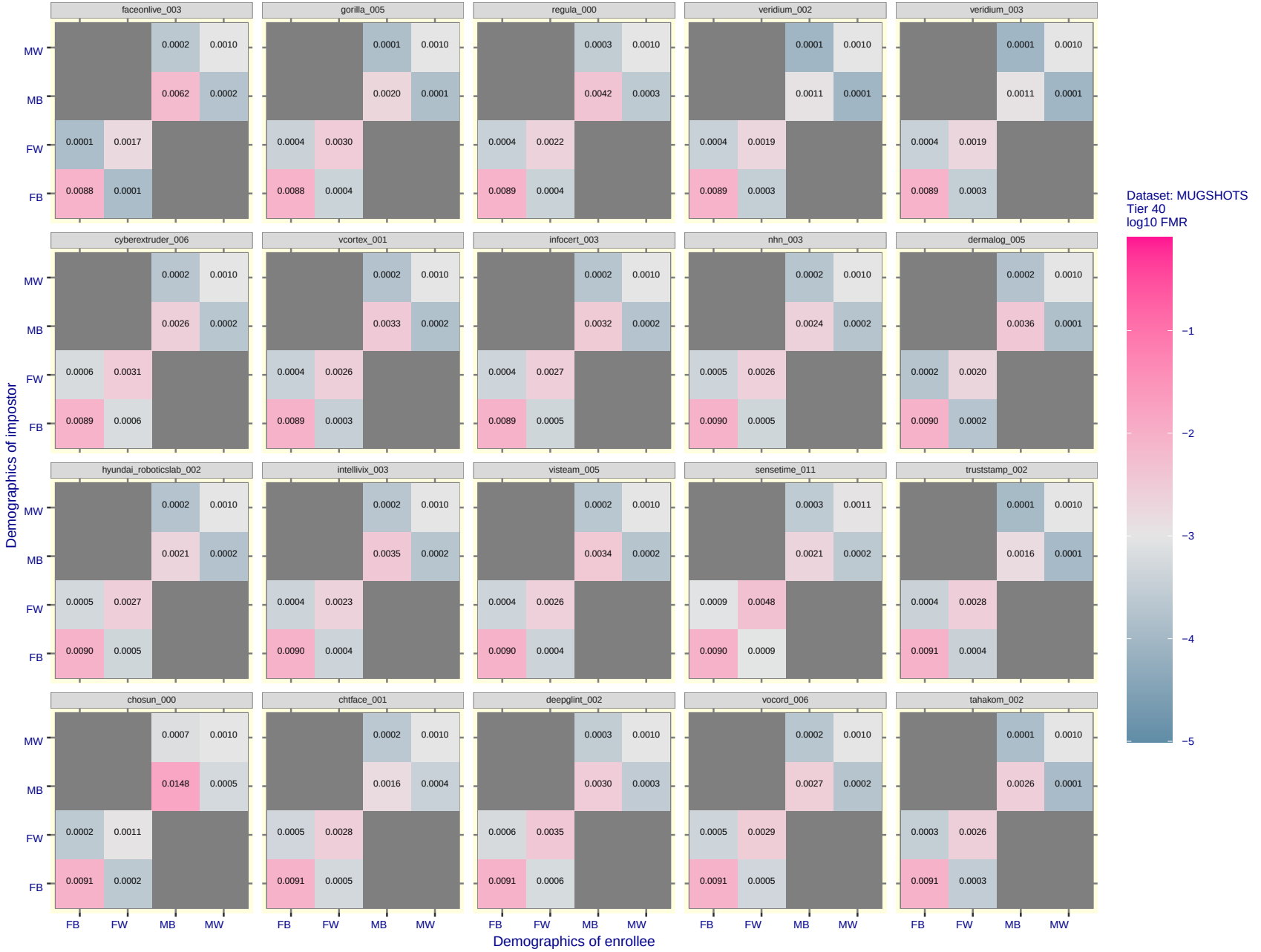

Figure 370: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

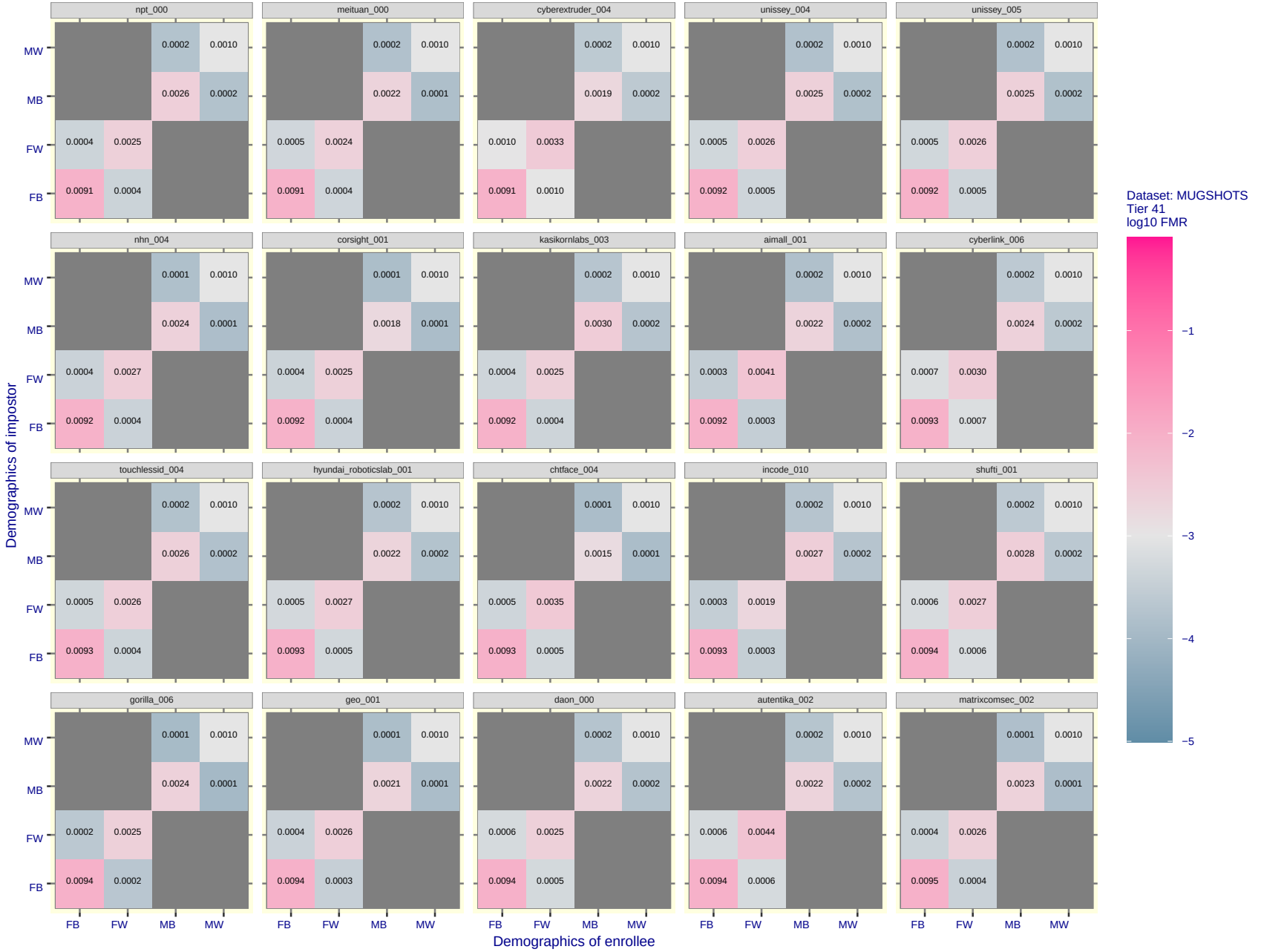

Figure 371: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

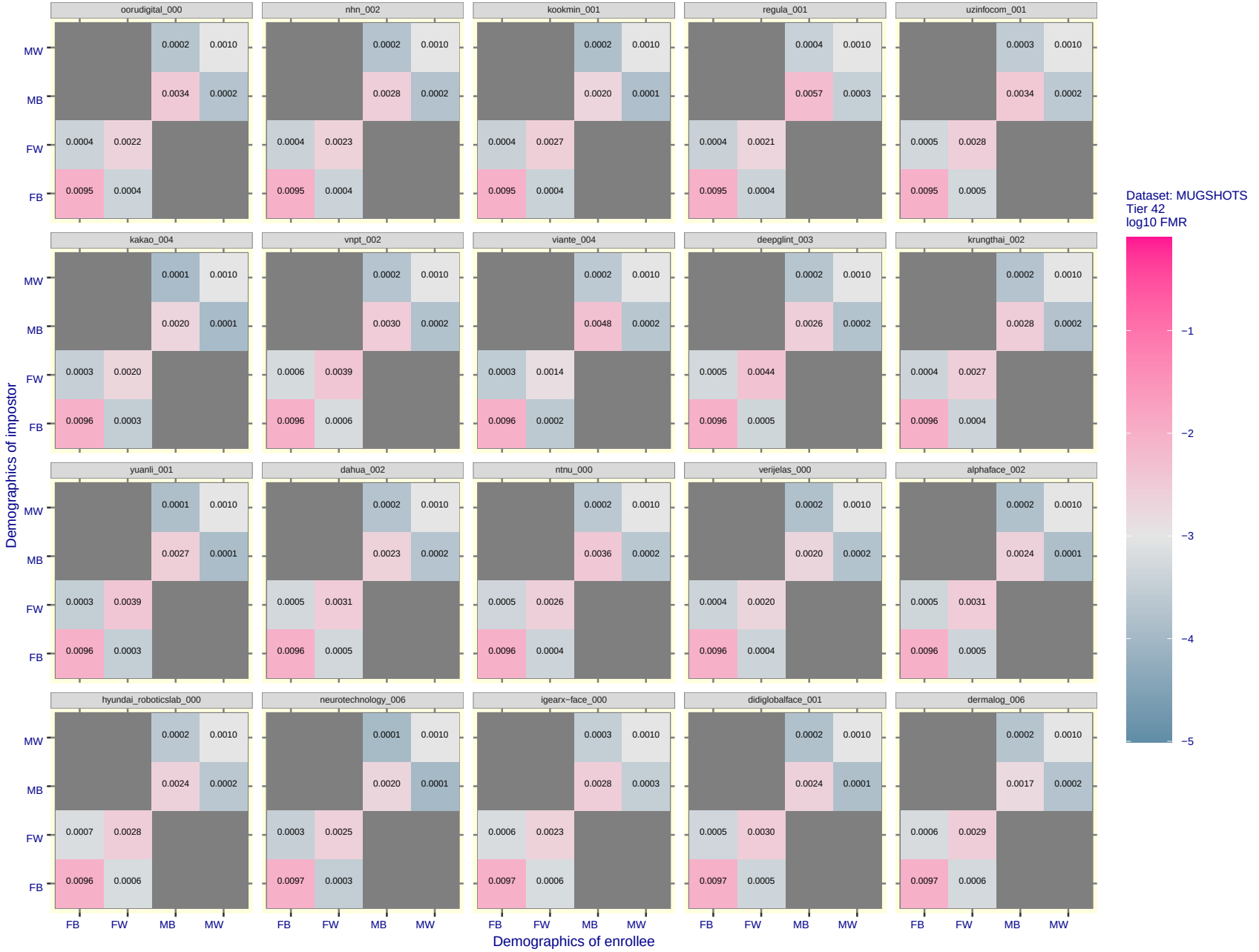

Figure 372: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

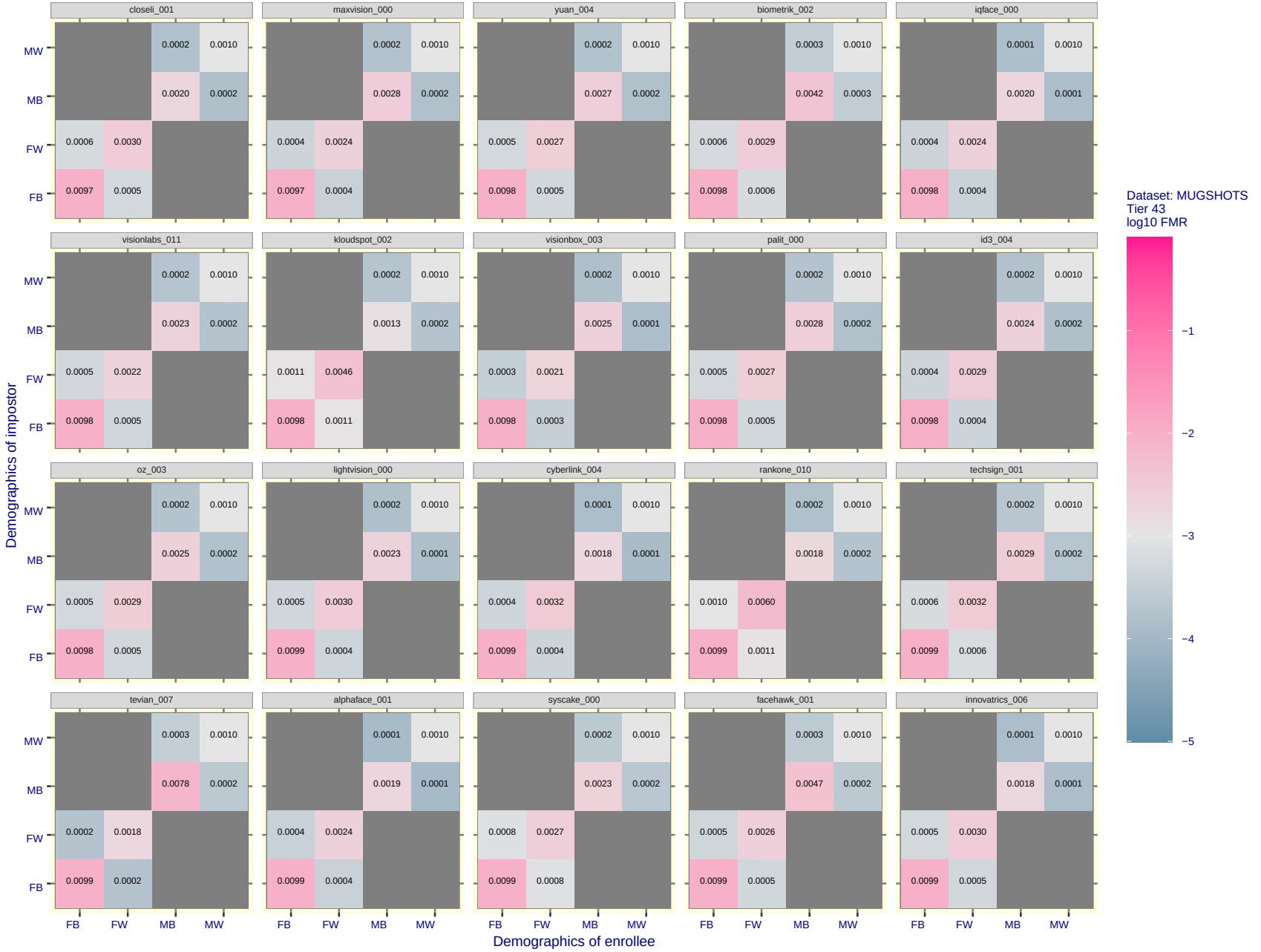

Figure 373: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

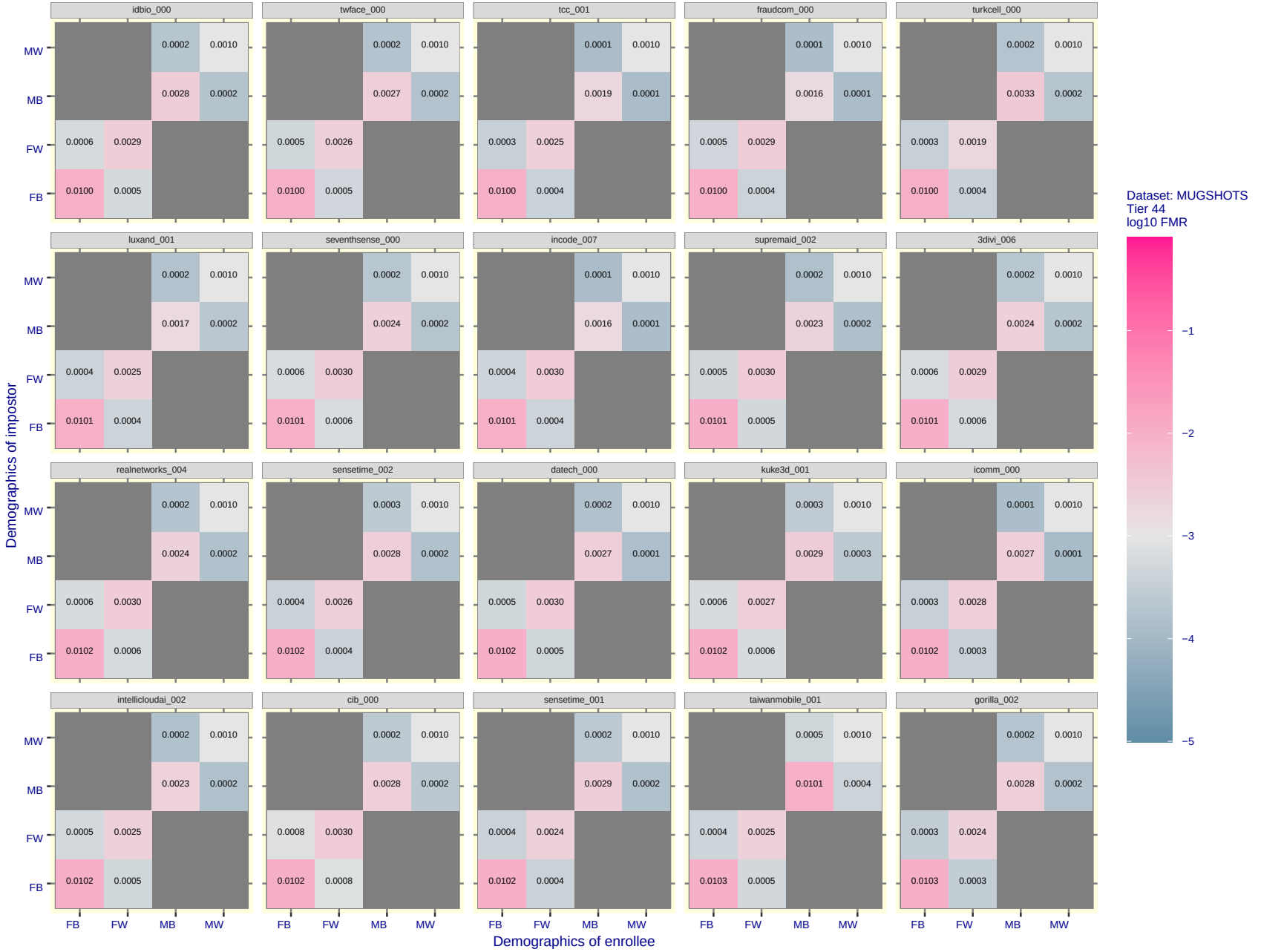

Figure 374: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

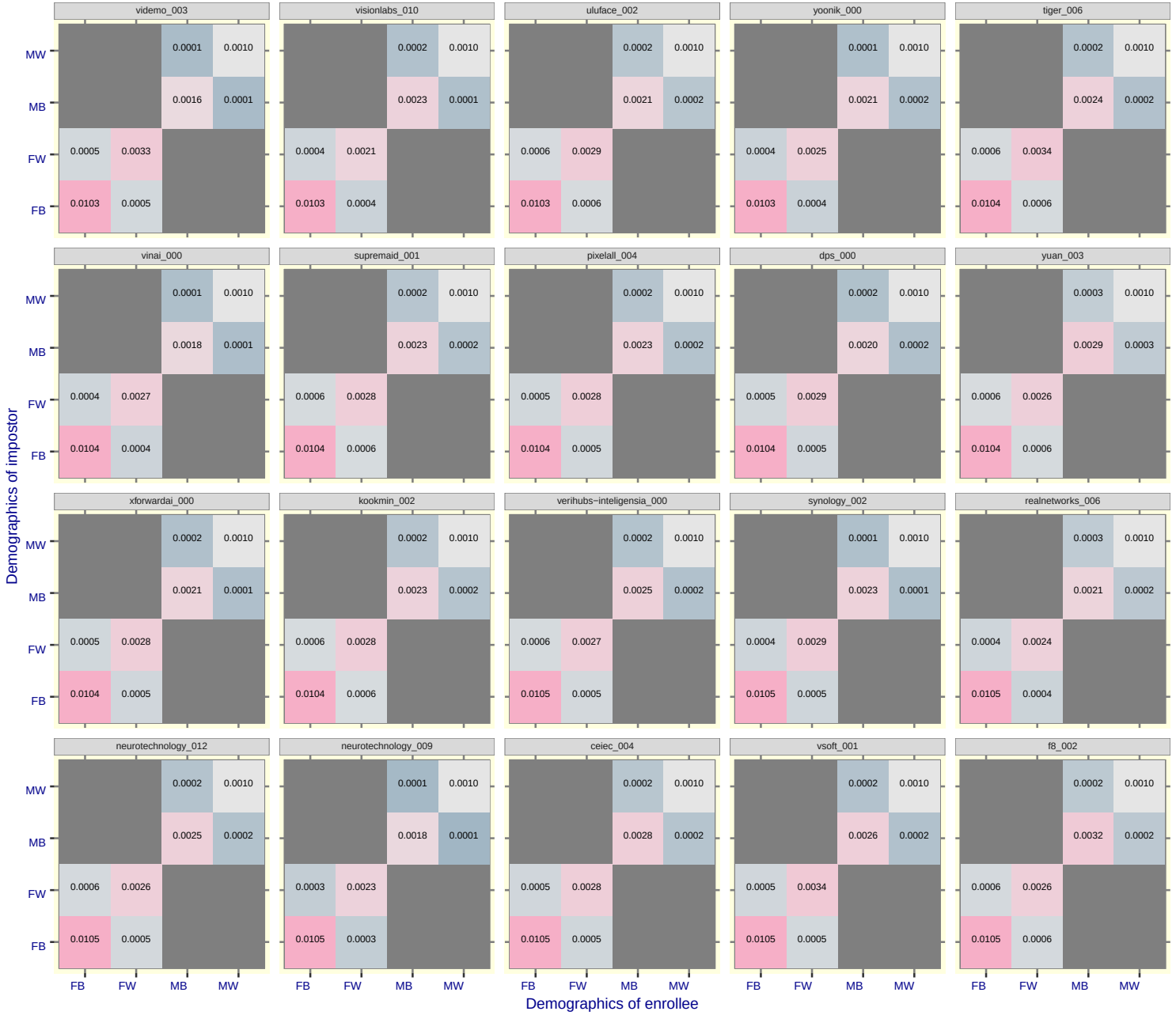

Figure 375: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

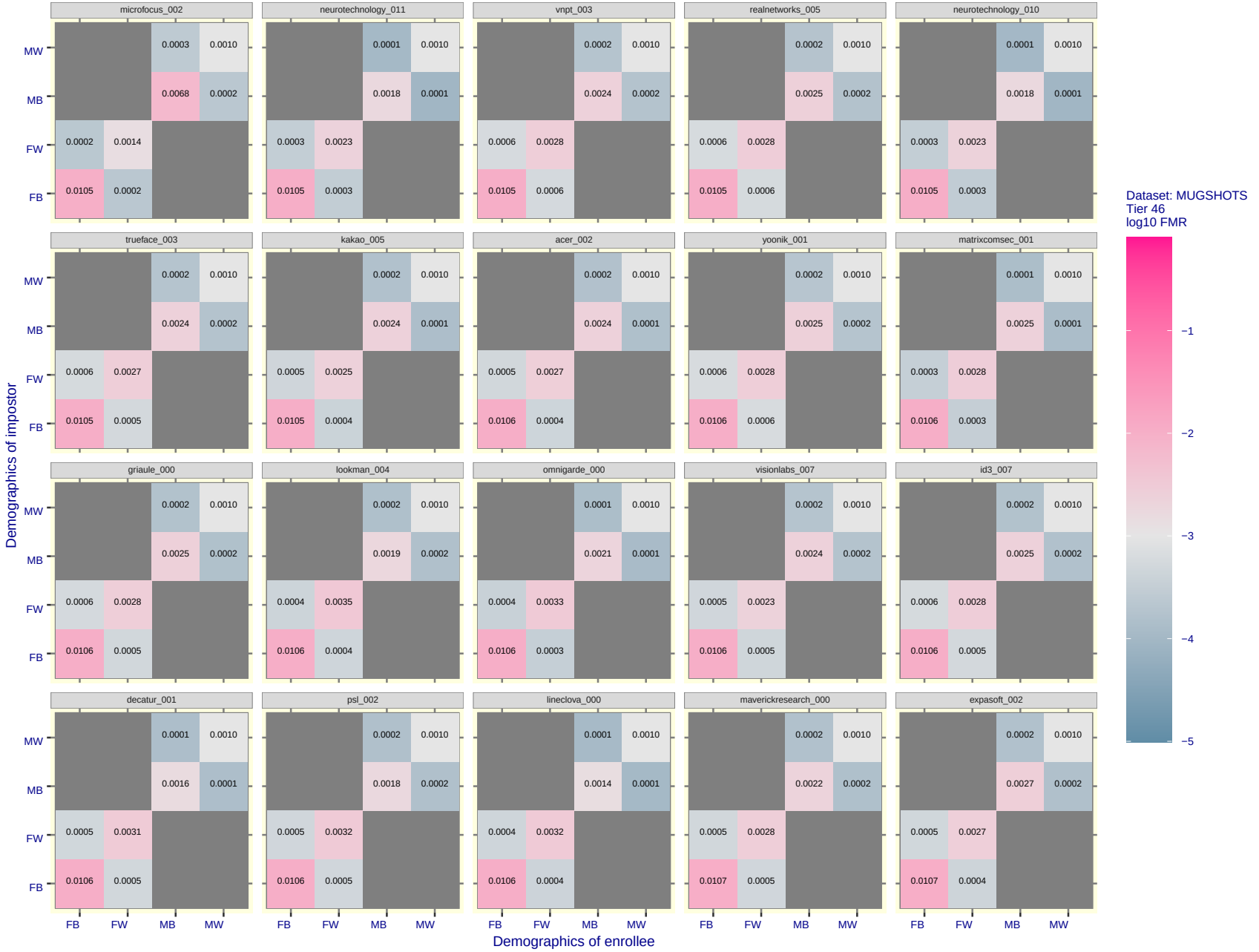

Figure 376: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

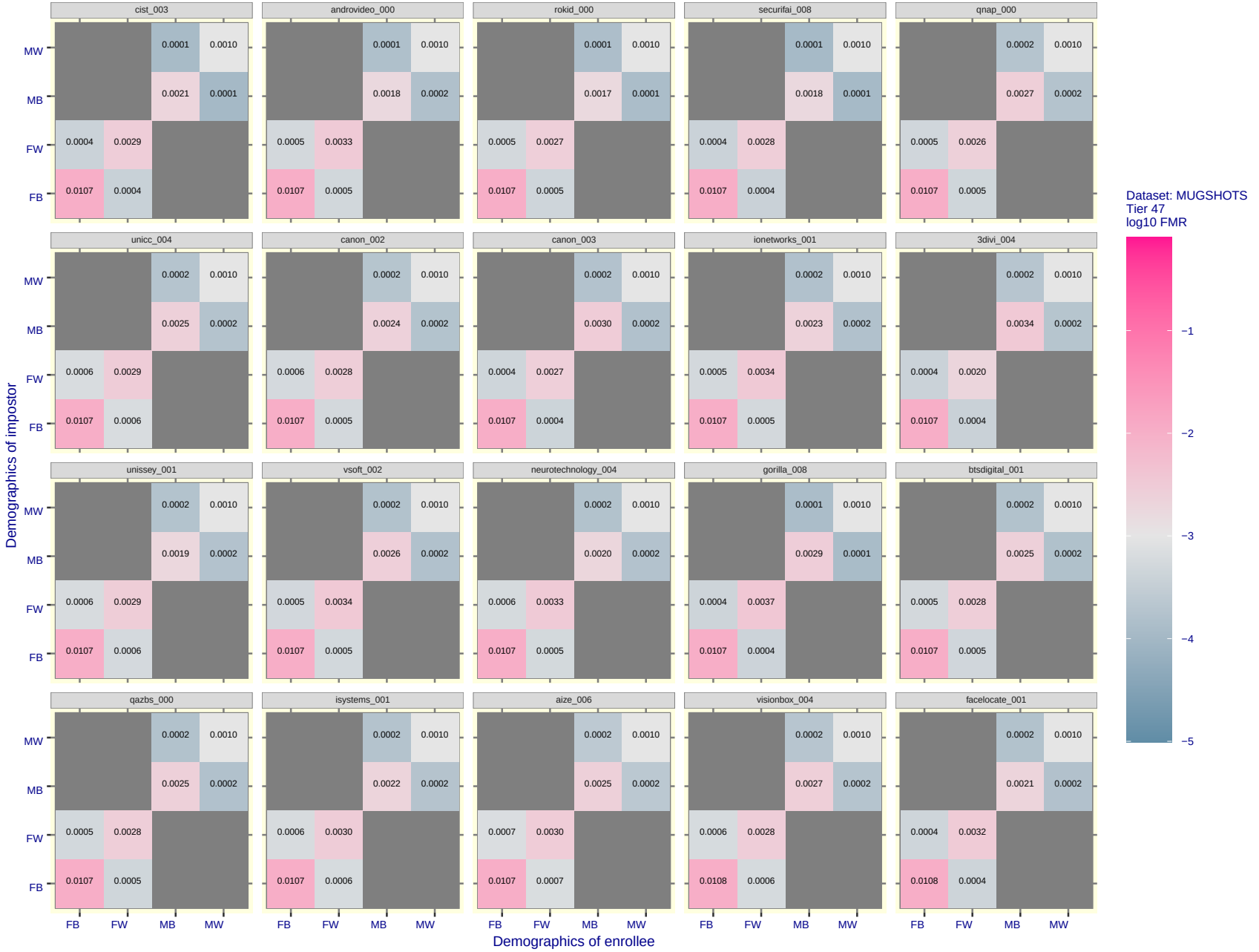

Figure 377: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

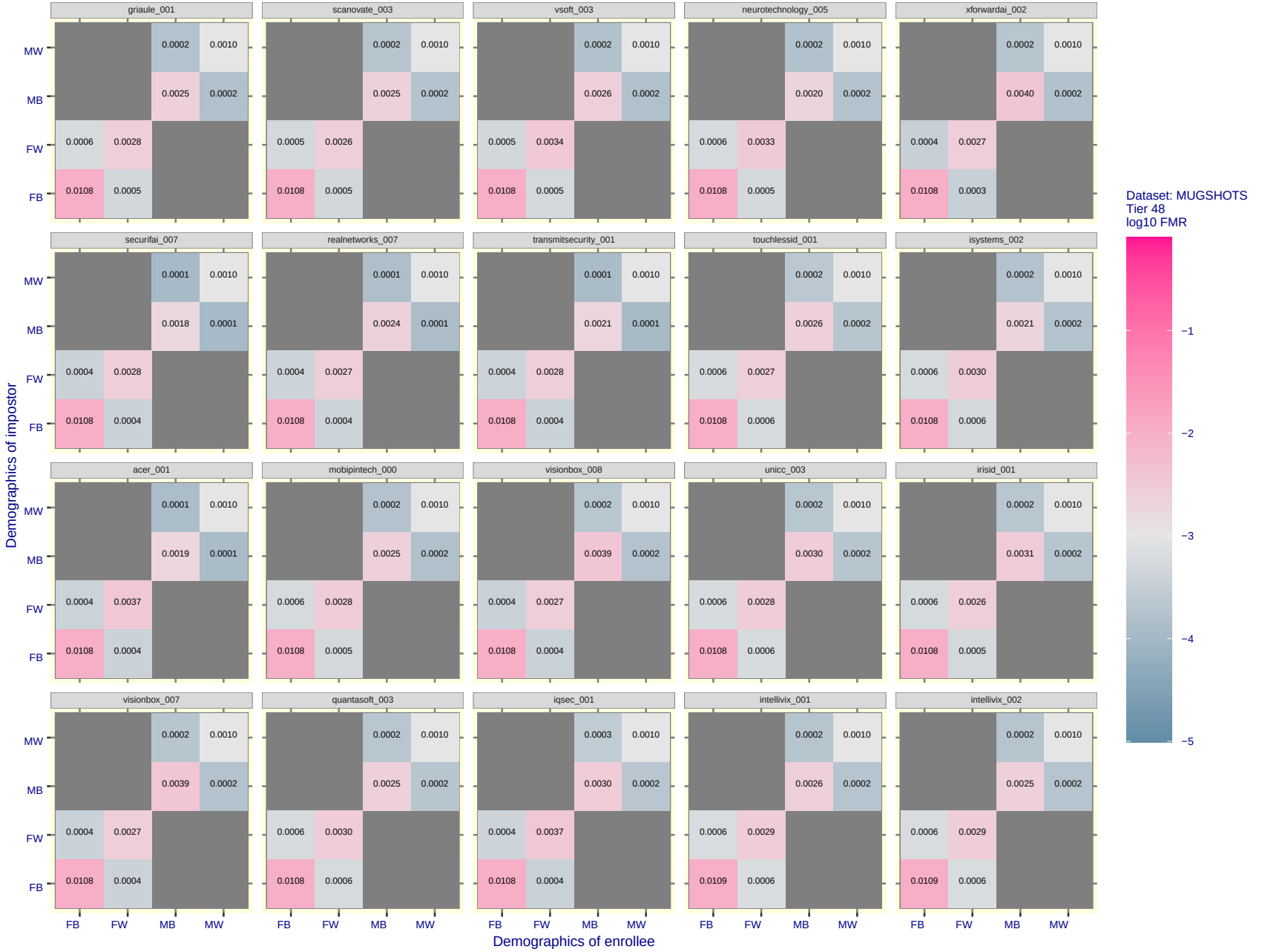

Figure 378: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

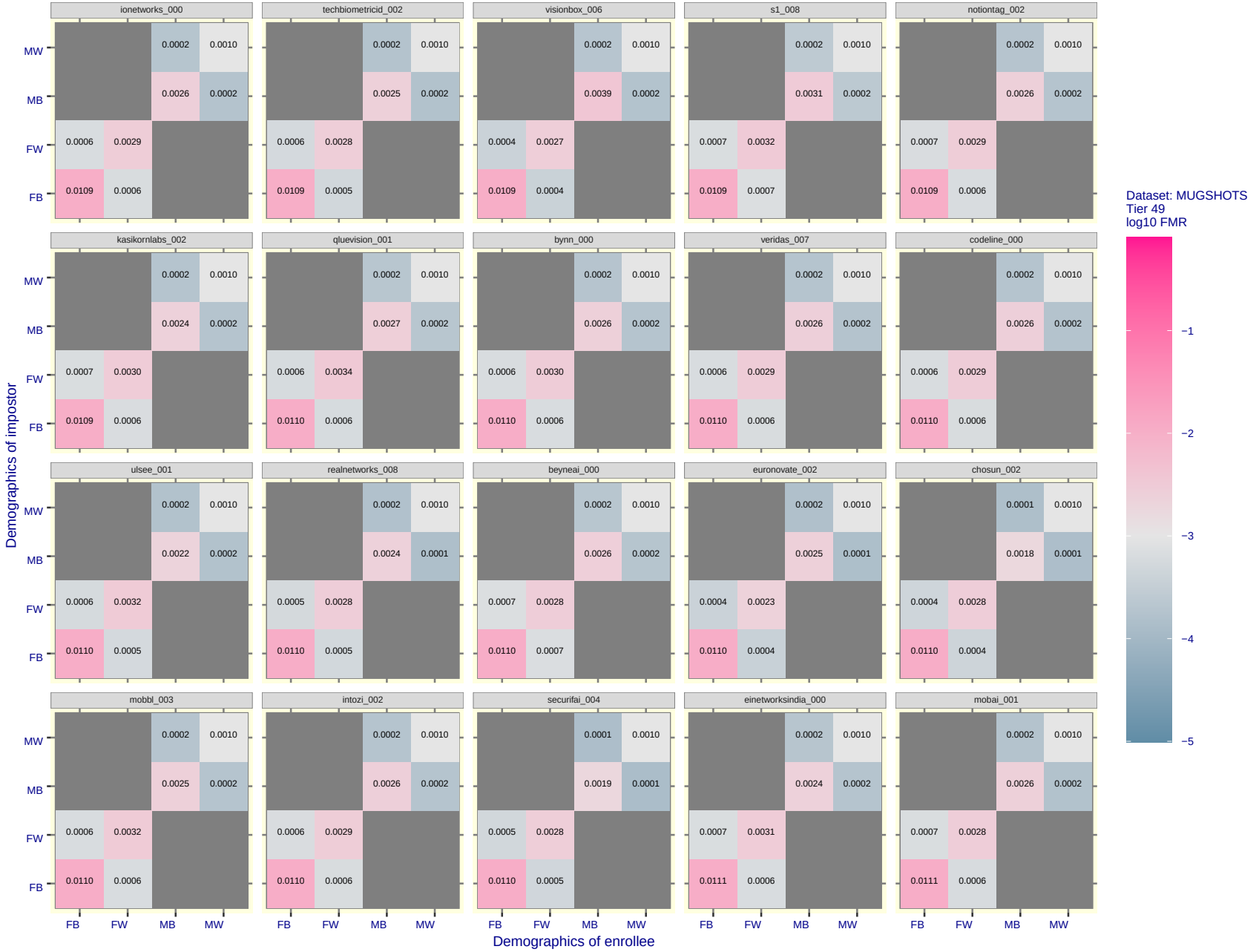

Figure 379: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

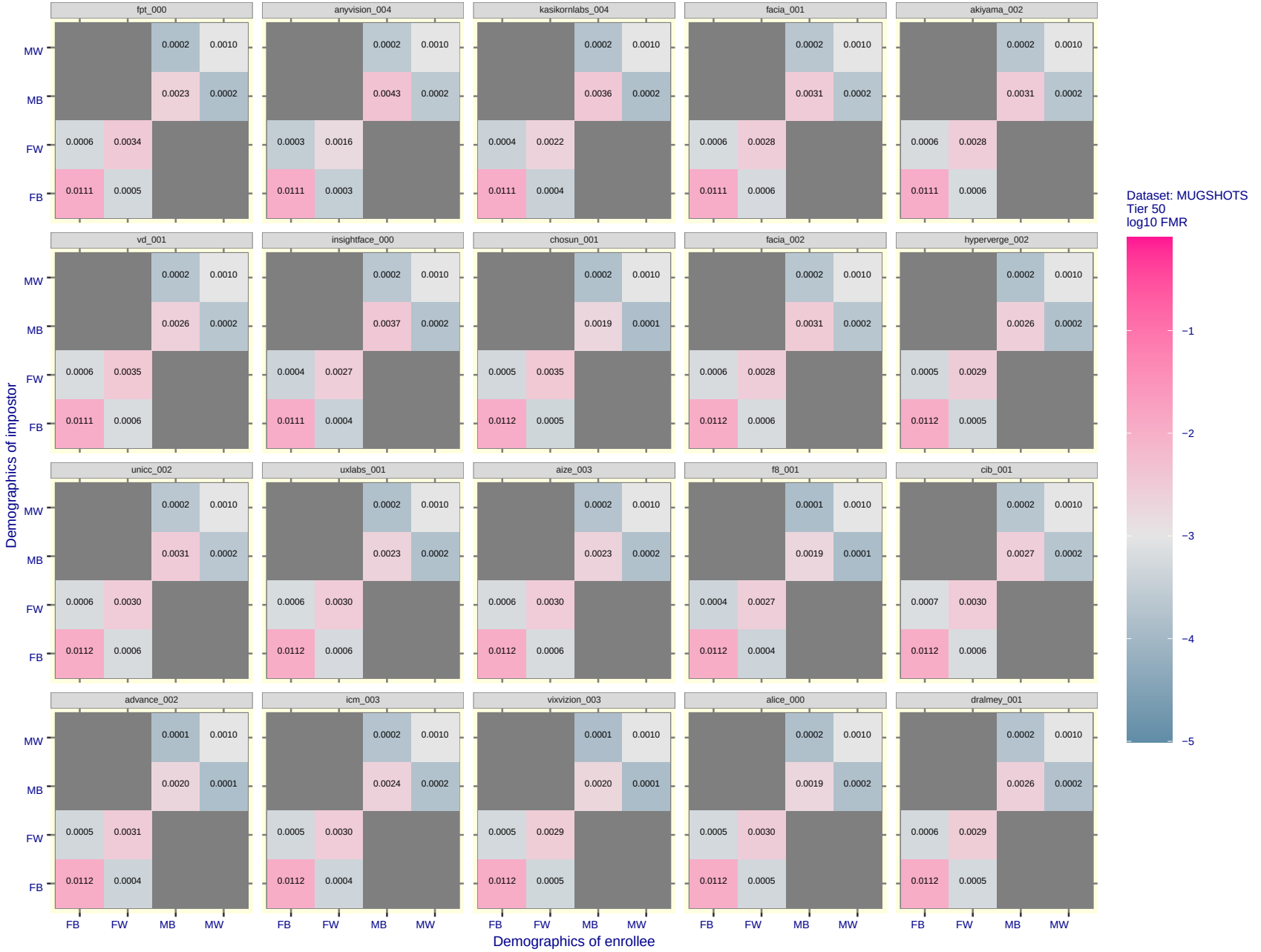

Figure 380: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

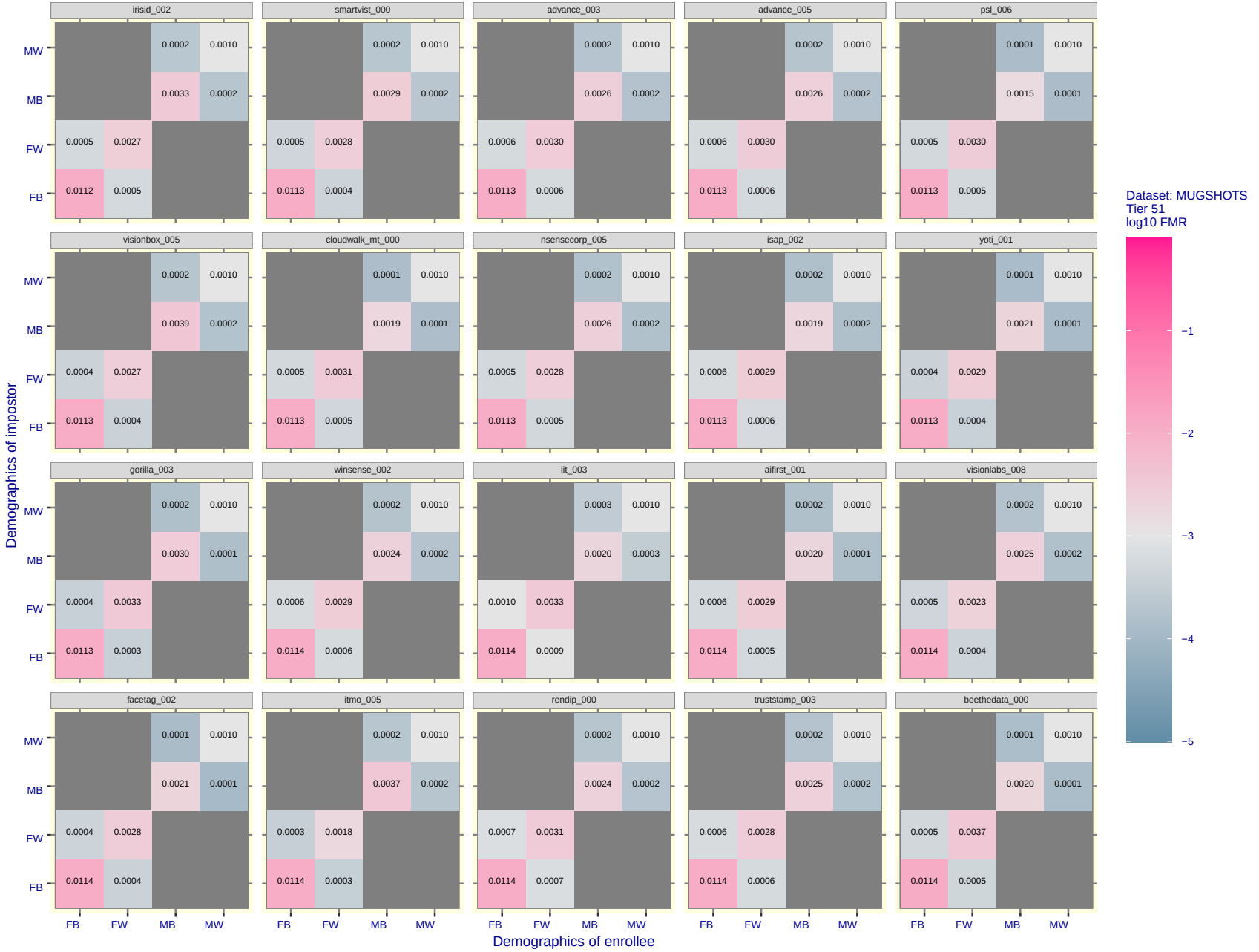

Figure 381: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

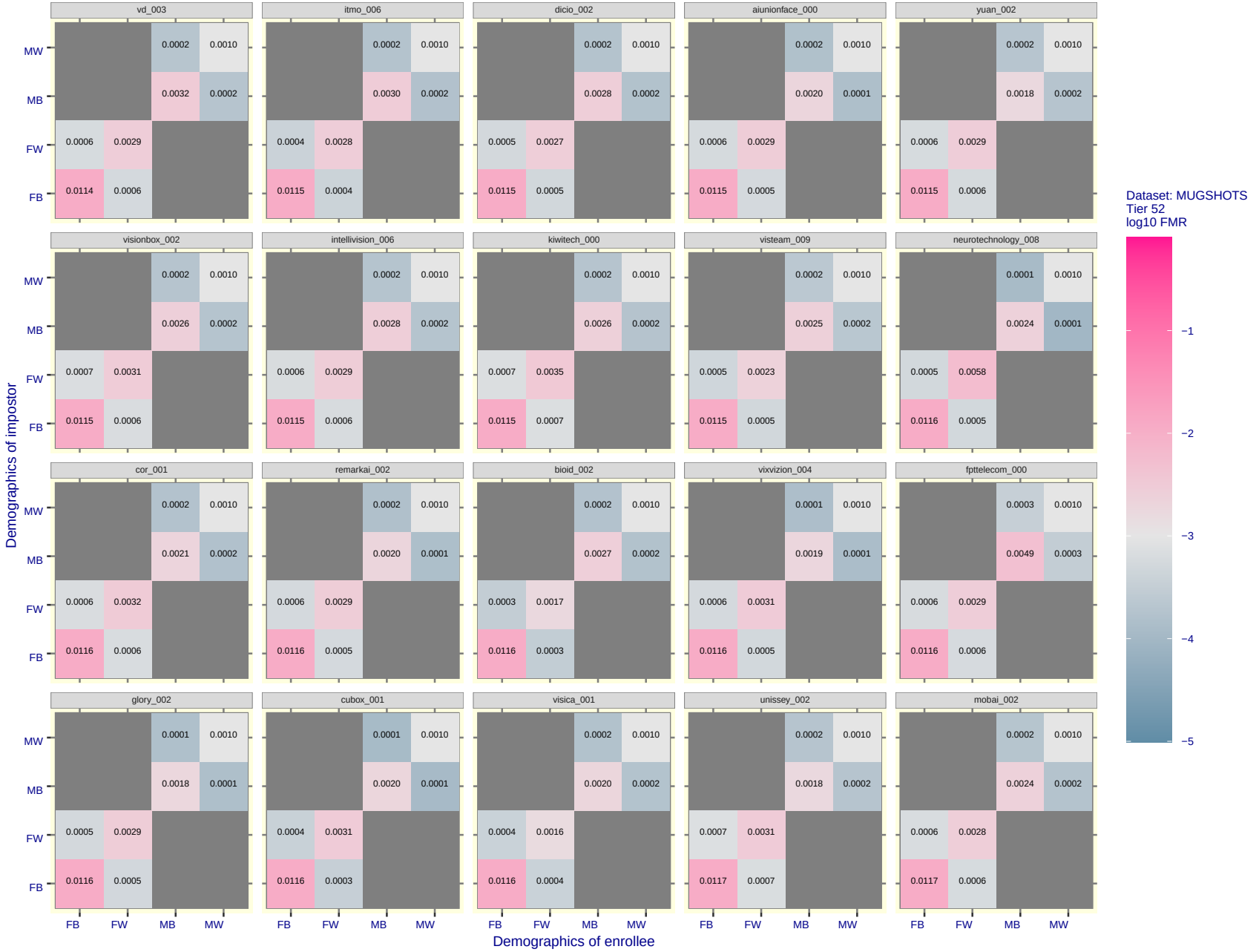

Figure 382: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

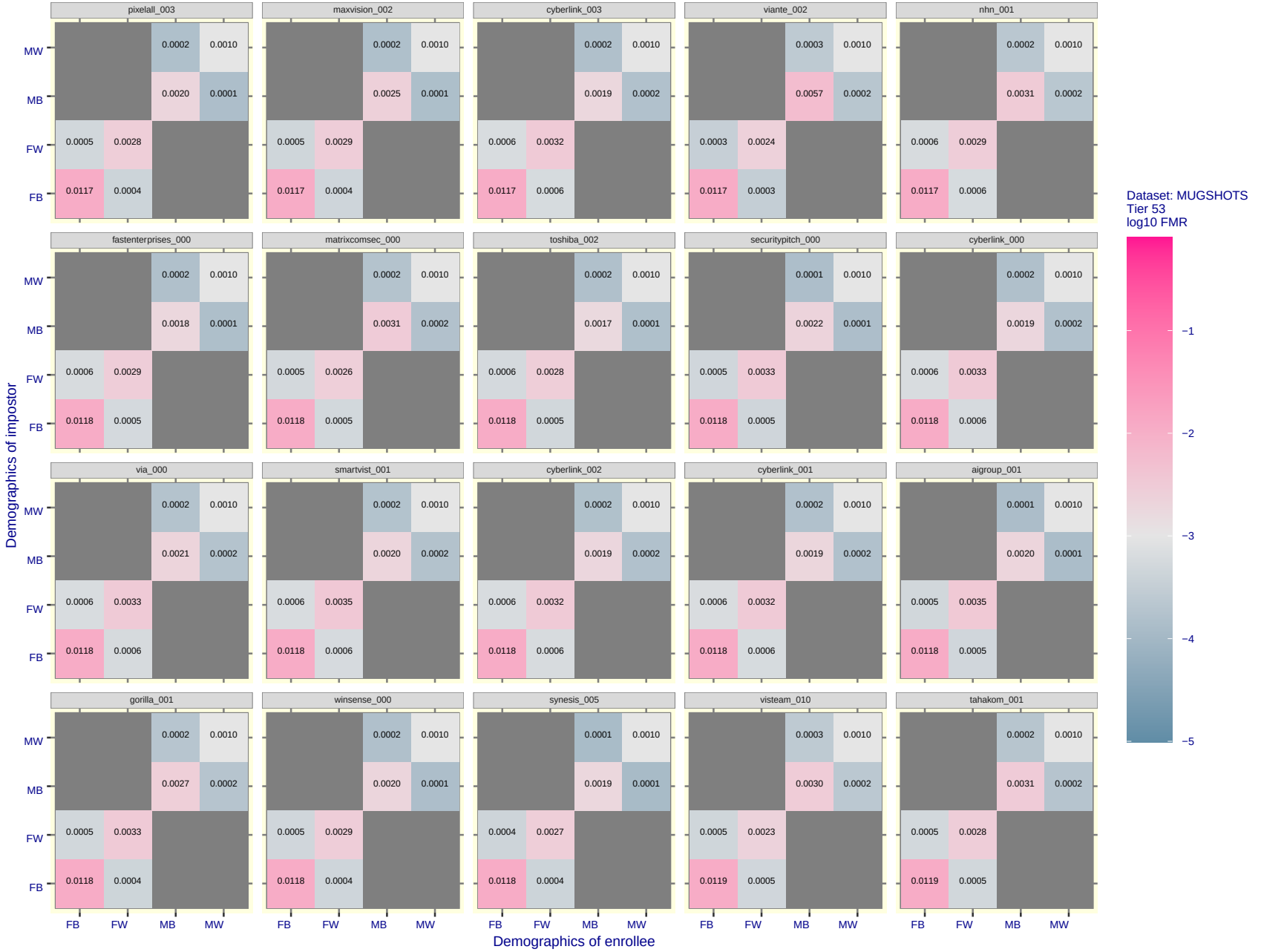

Figure 383: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

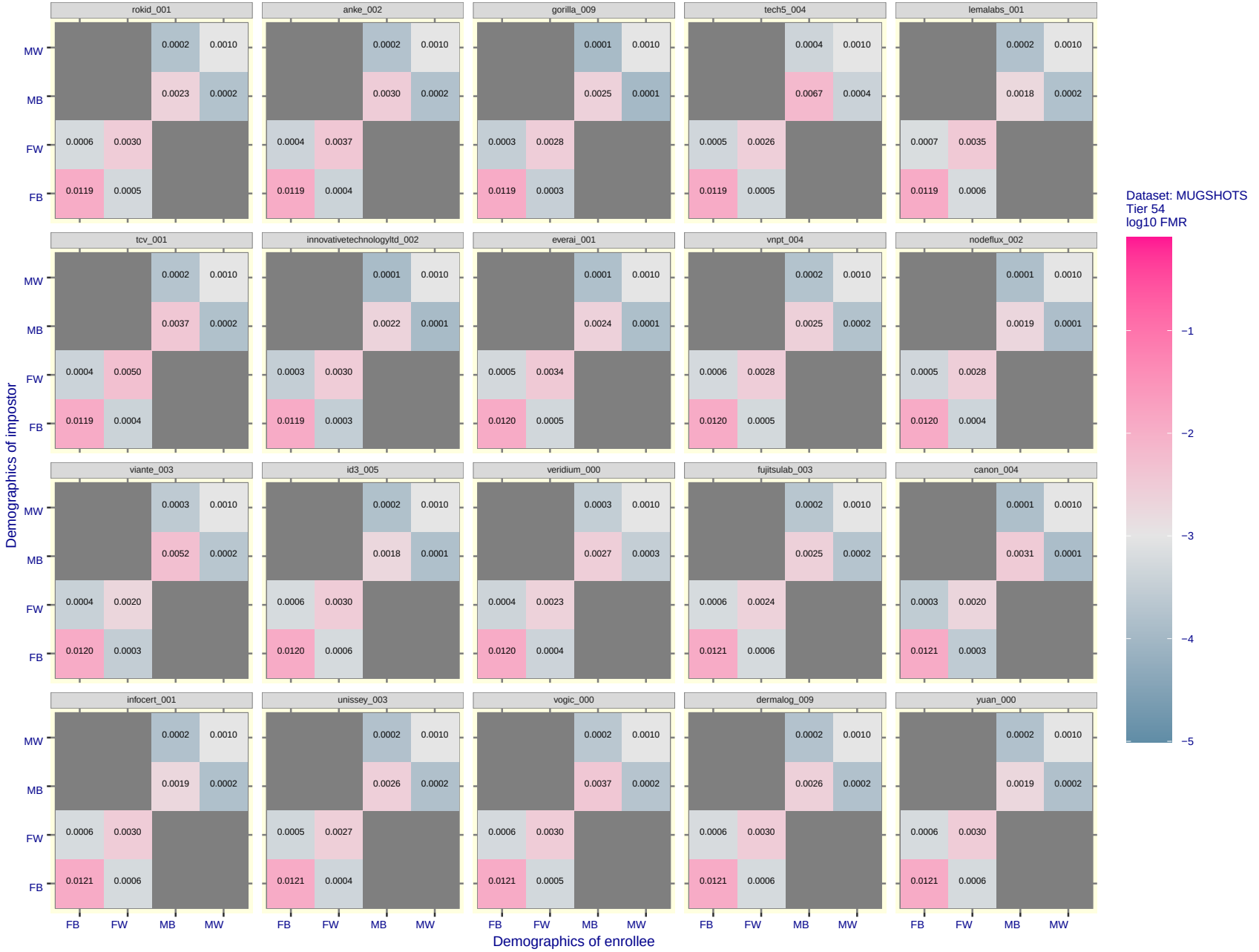

Figure 384: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

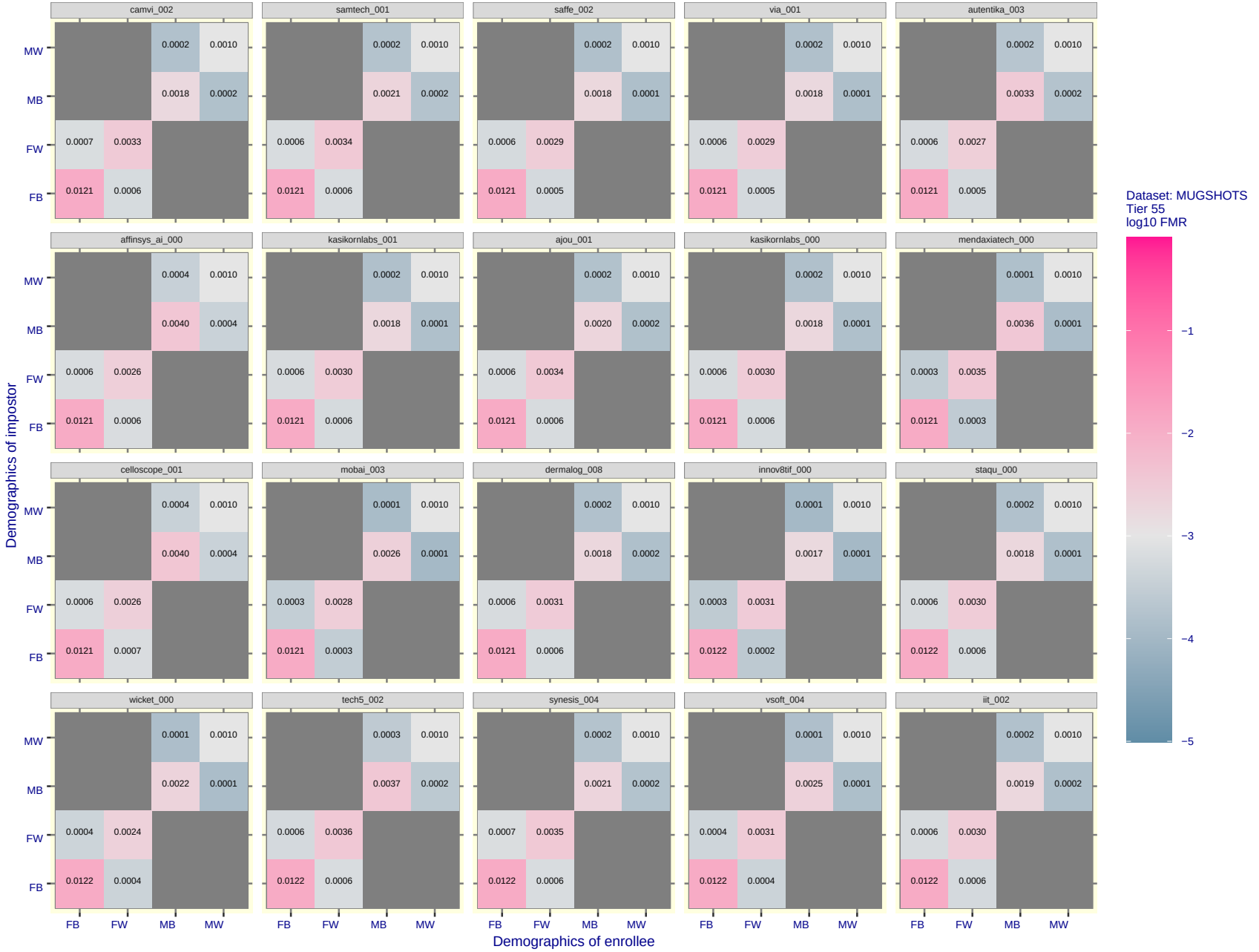

Figure 385: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

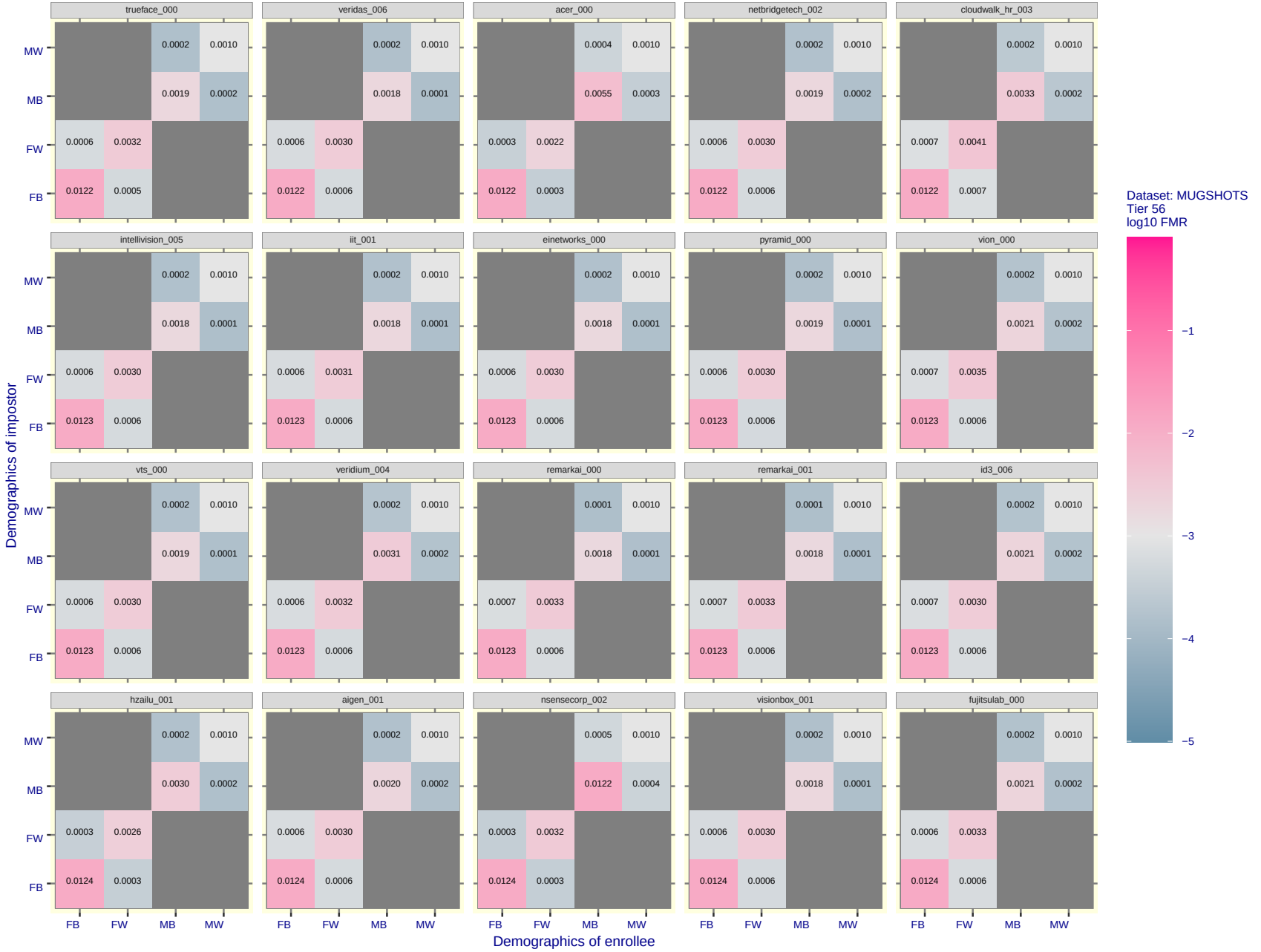

Figure 386: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

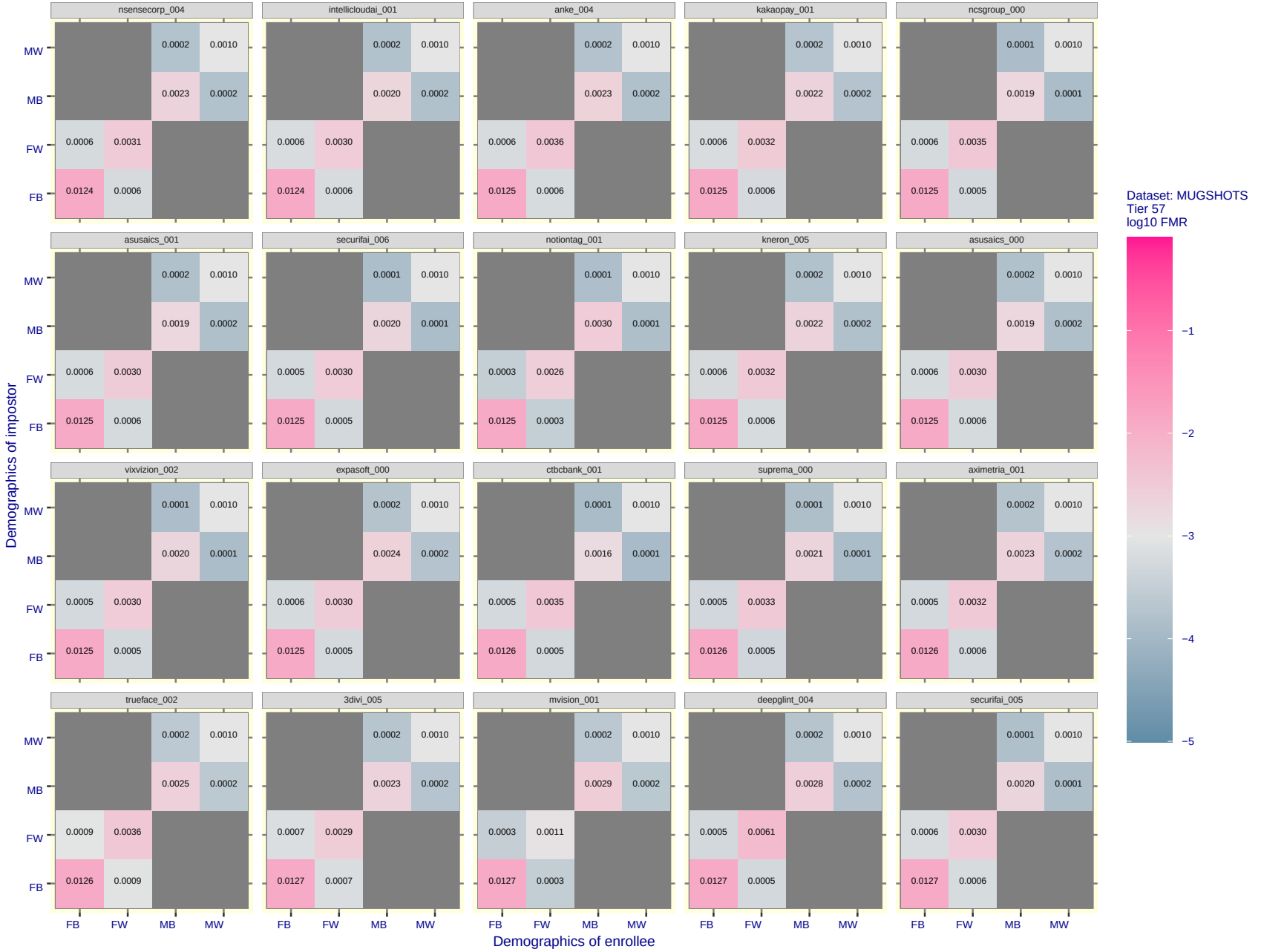

Figure 387: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

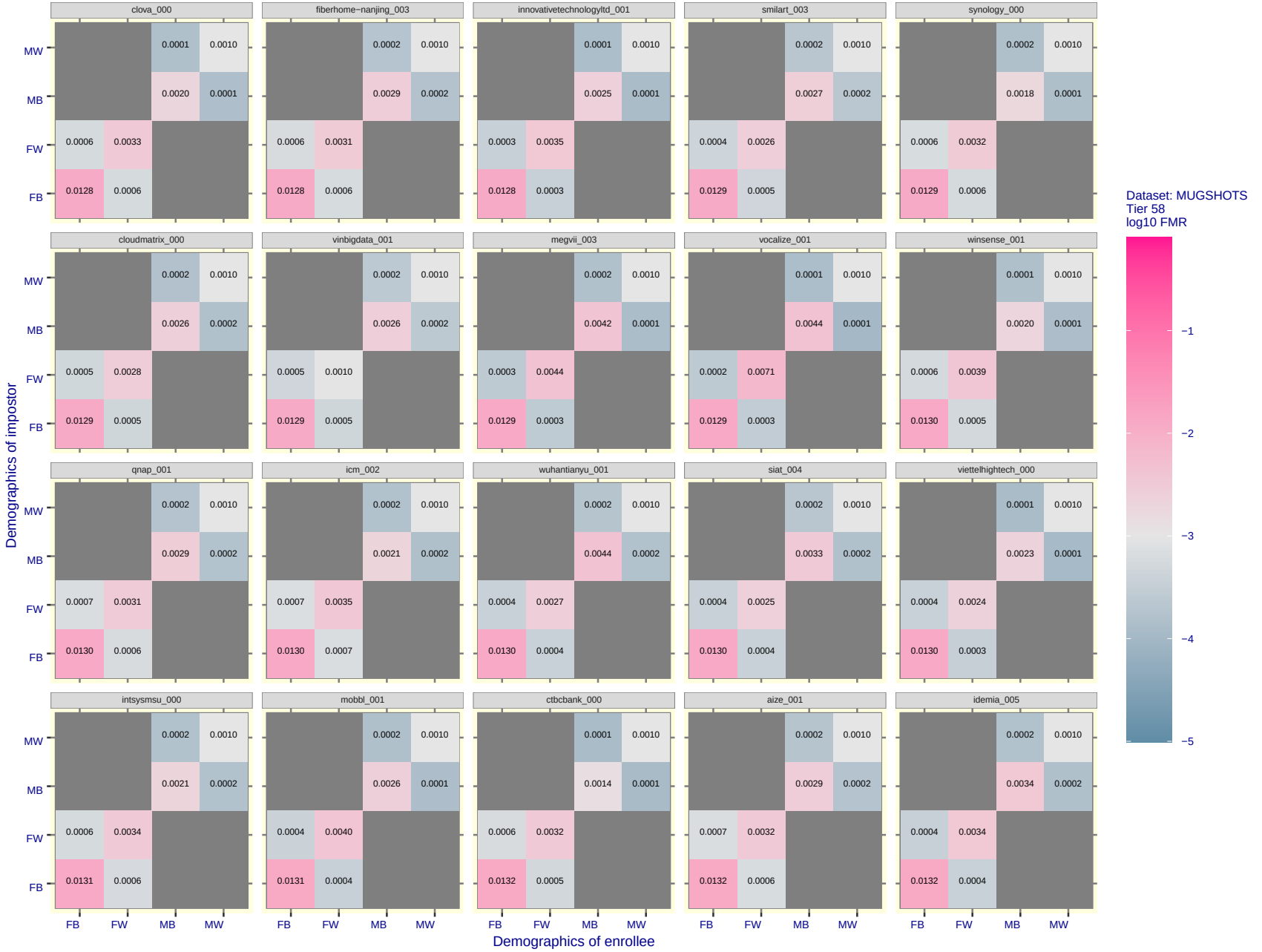

Figure 388: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

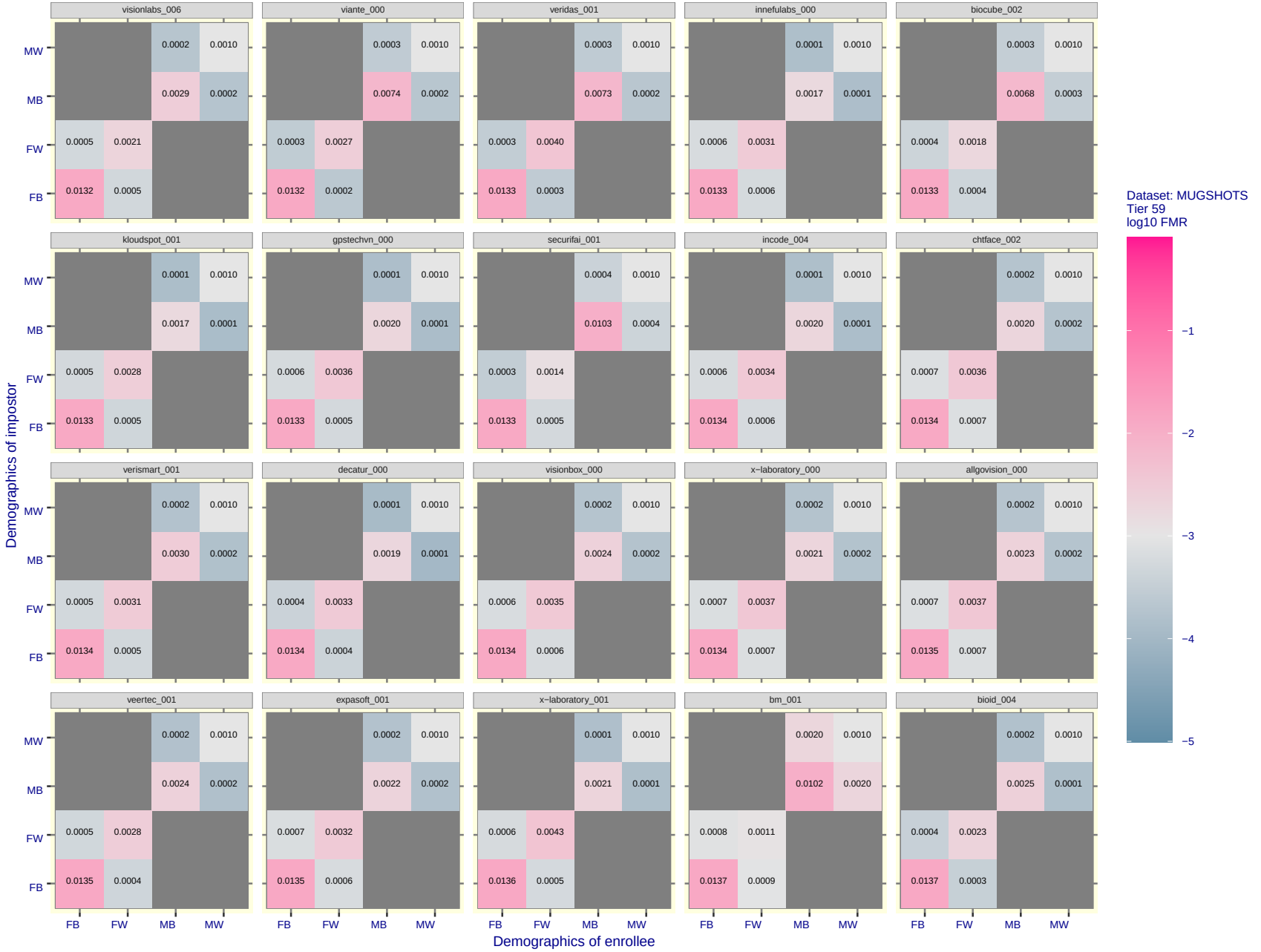

Figure 389: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

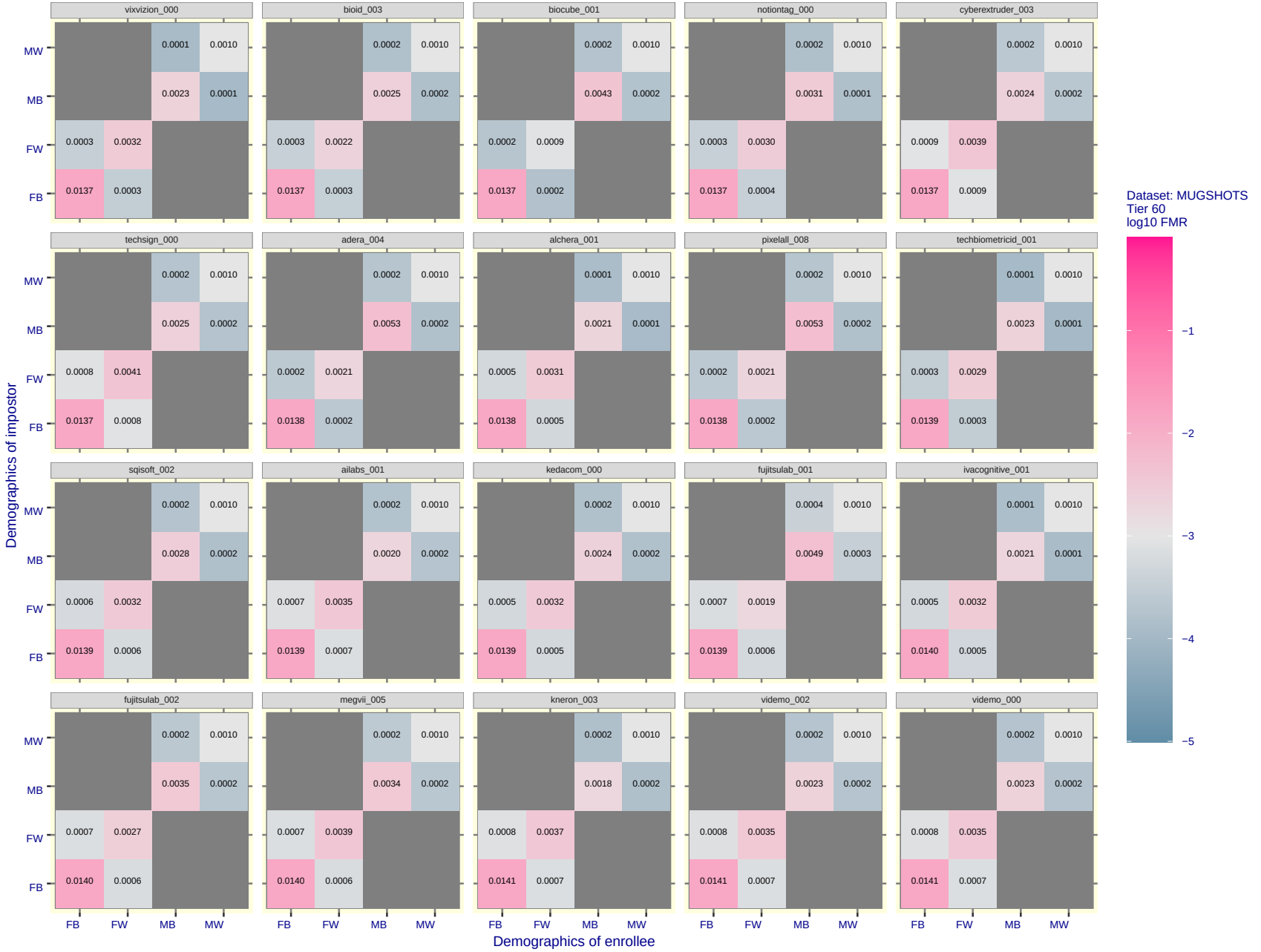

Figure 390: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

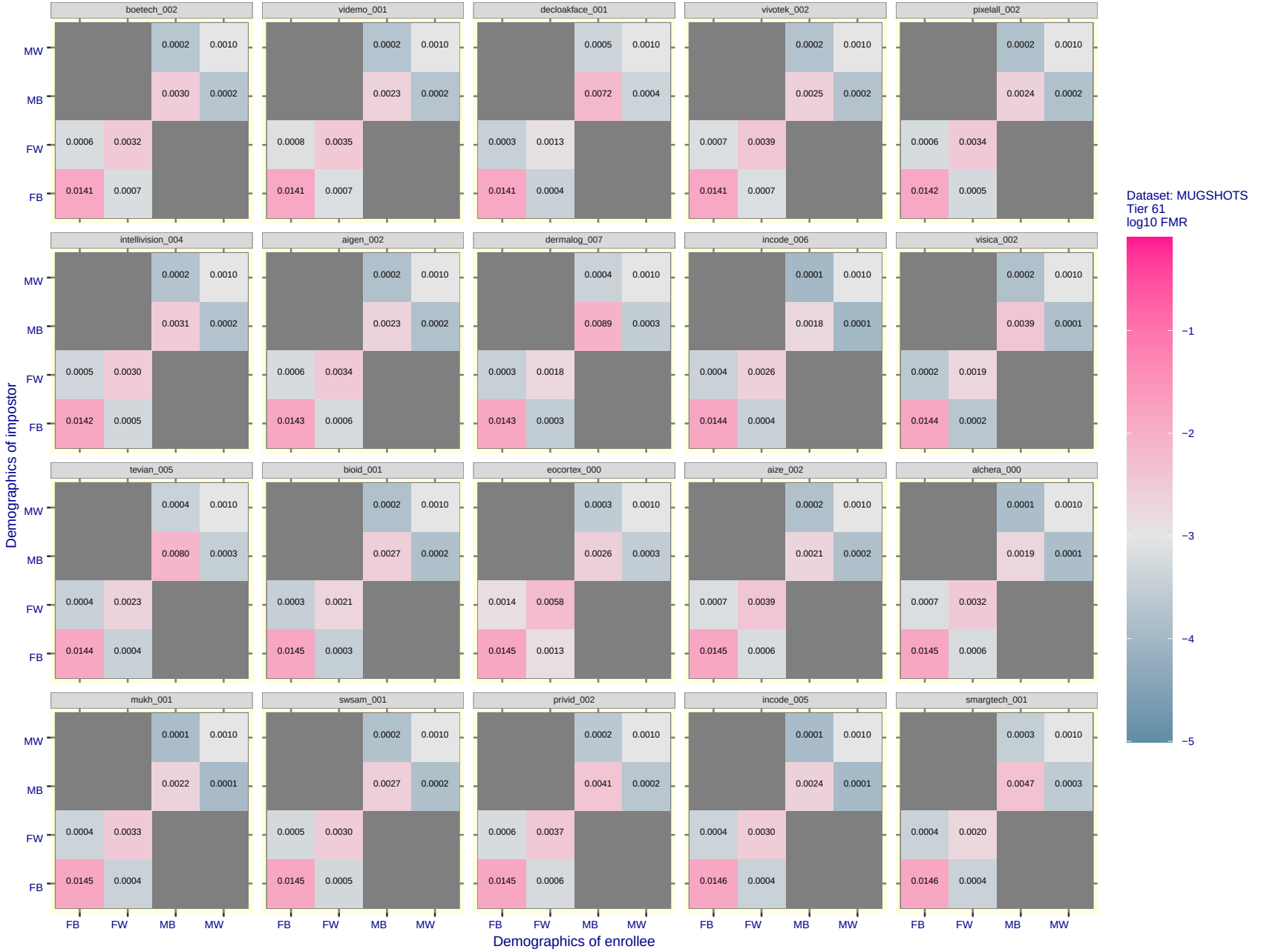

Figure 391: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

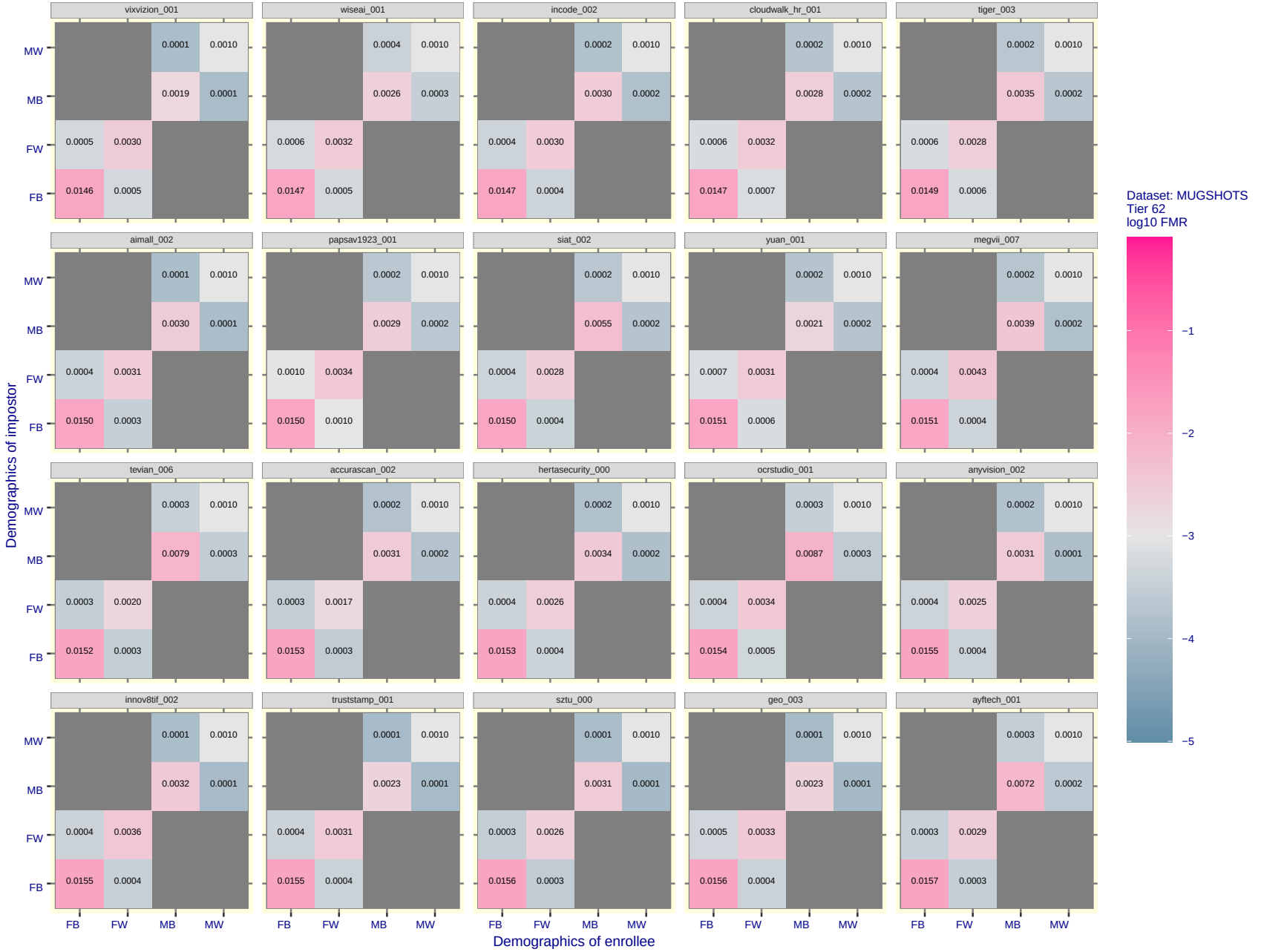

Figure 392: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

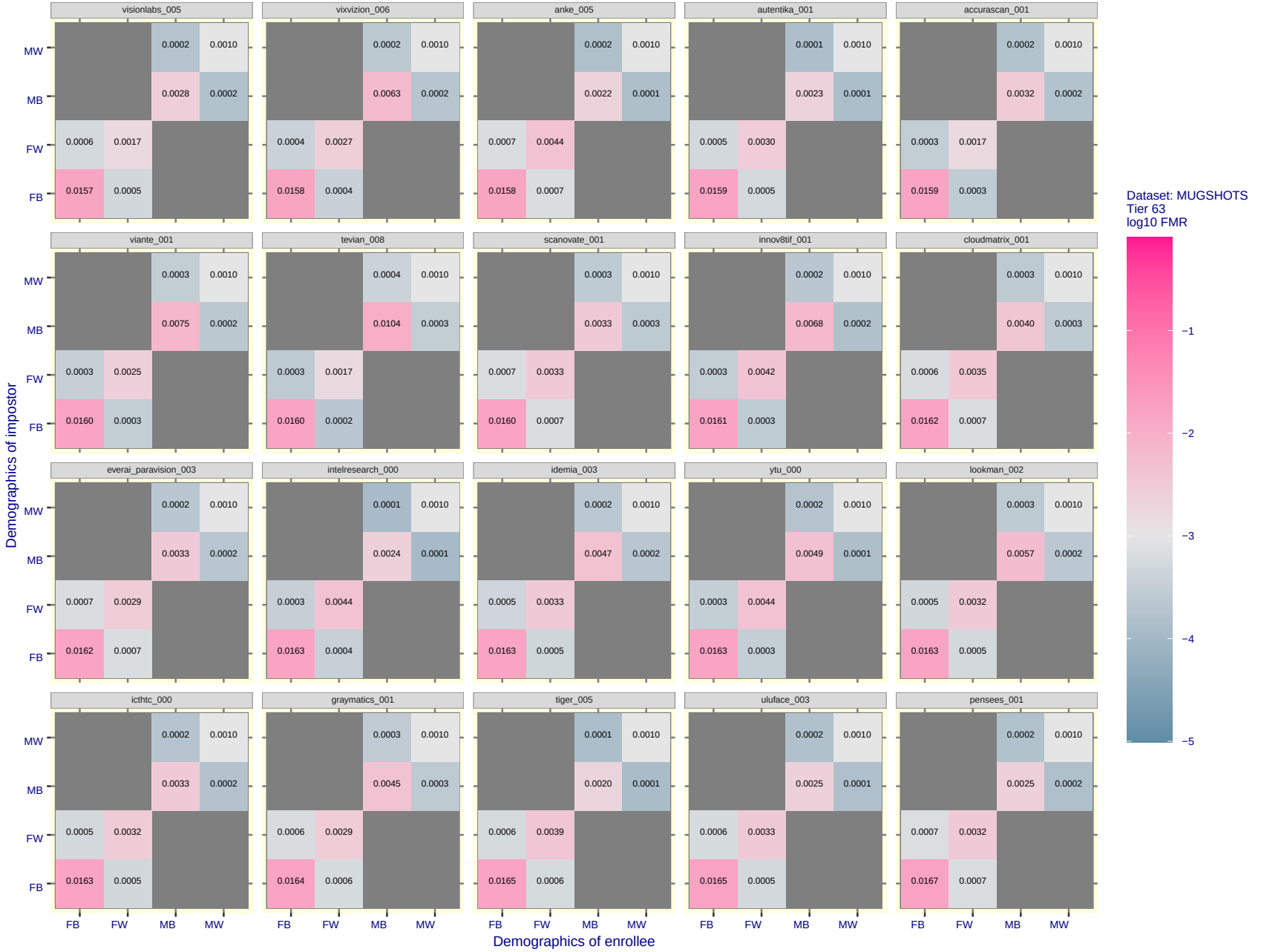

Figure 393: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/10/06 08:01:27
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

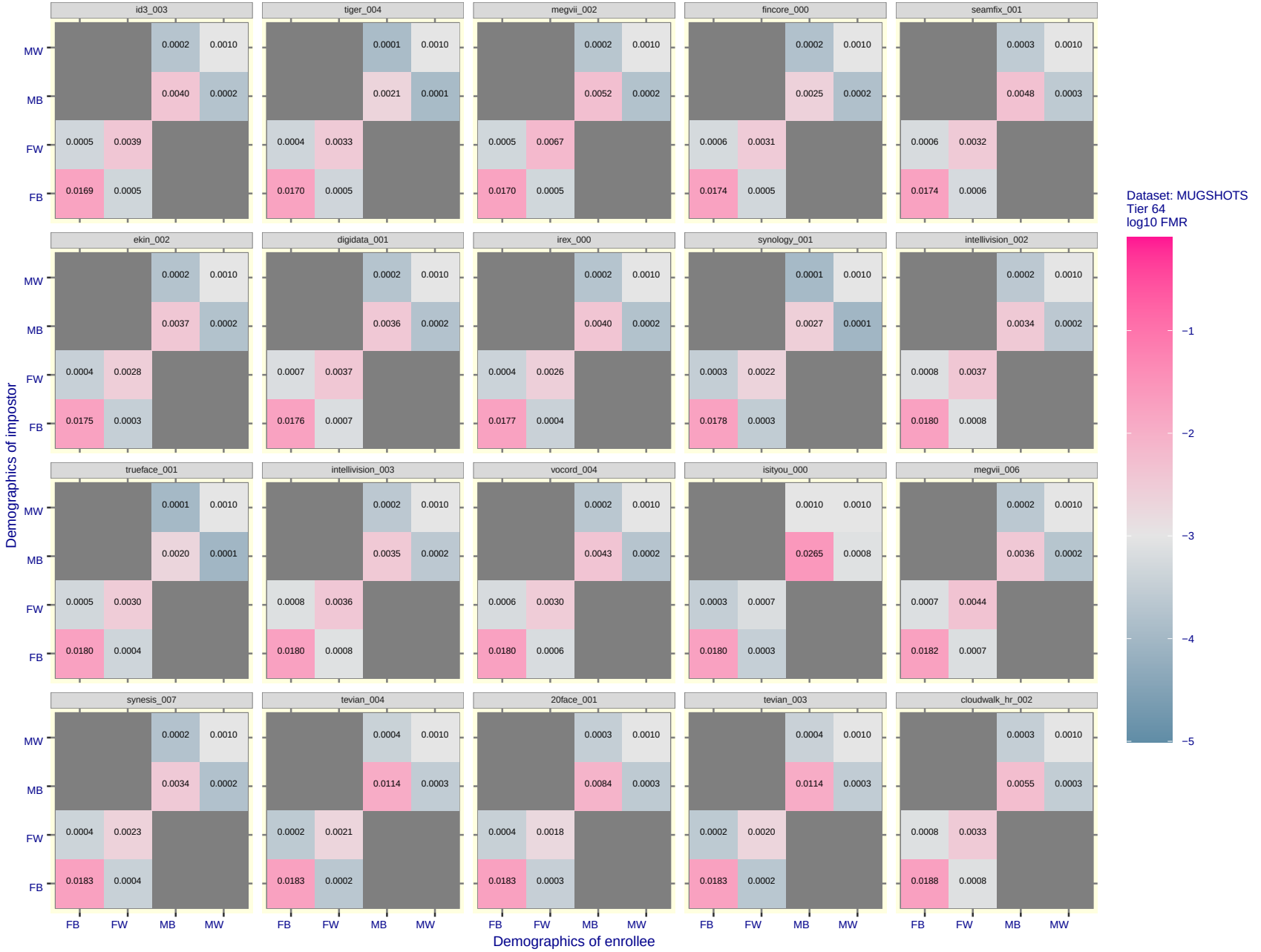

Figure 394: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

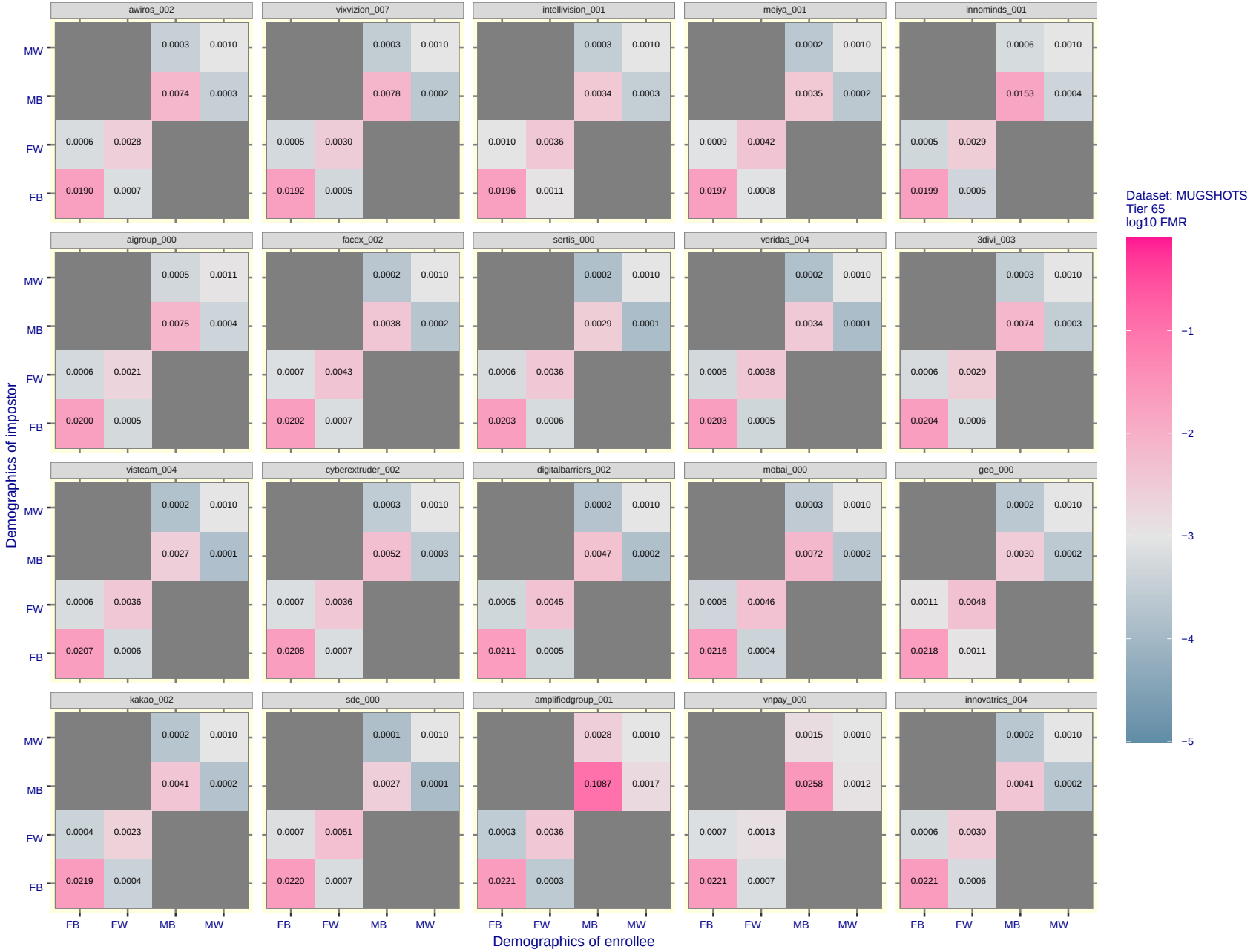

Figure 395: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

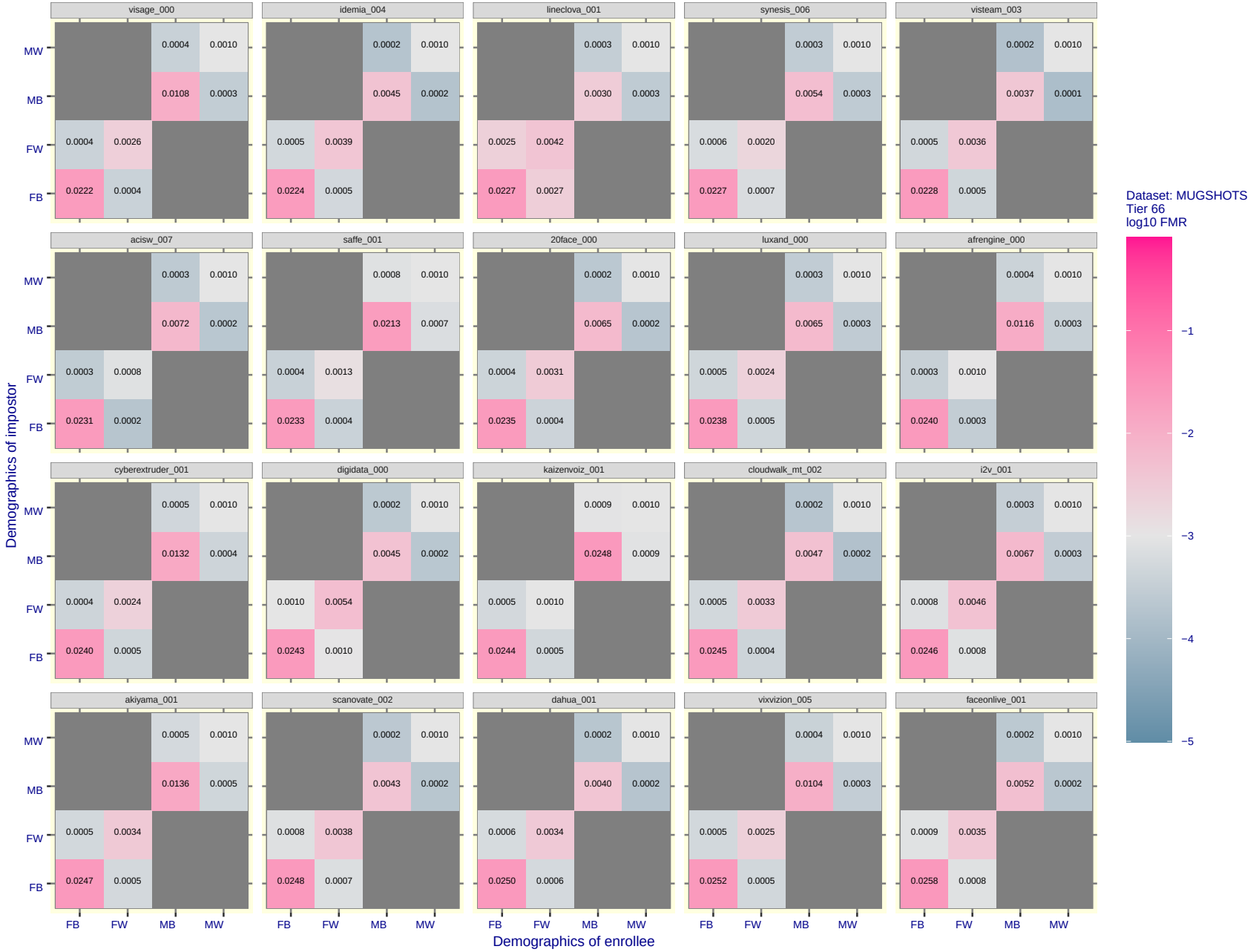

Figure 396: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

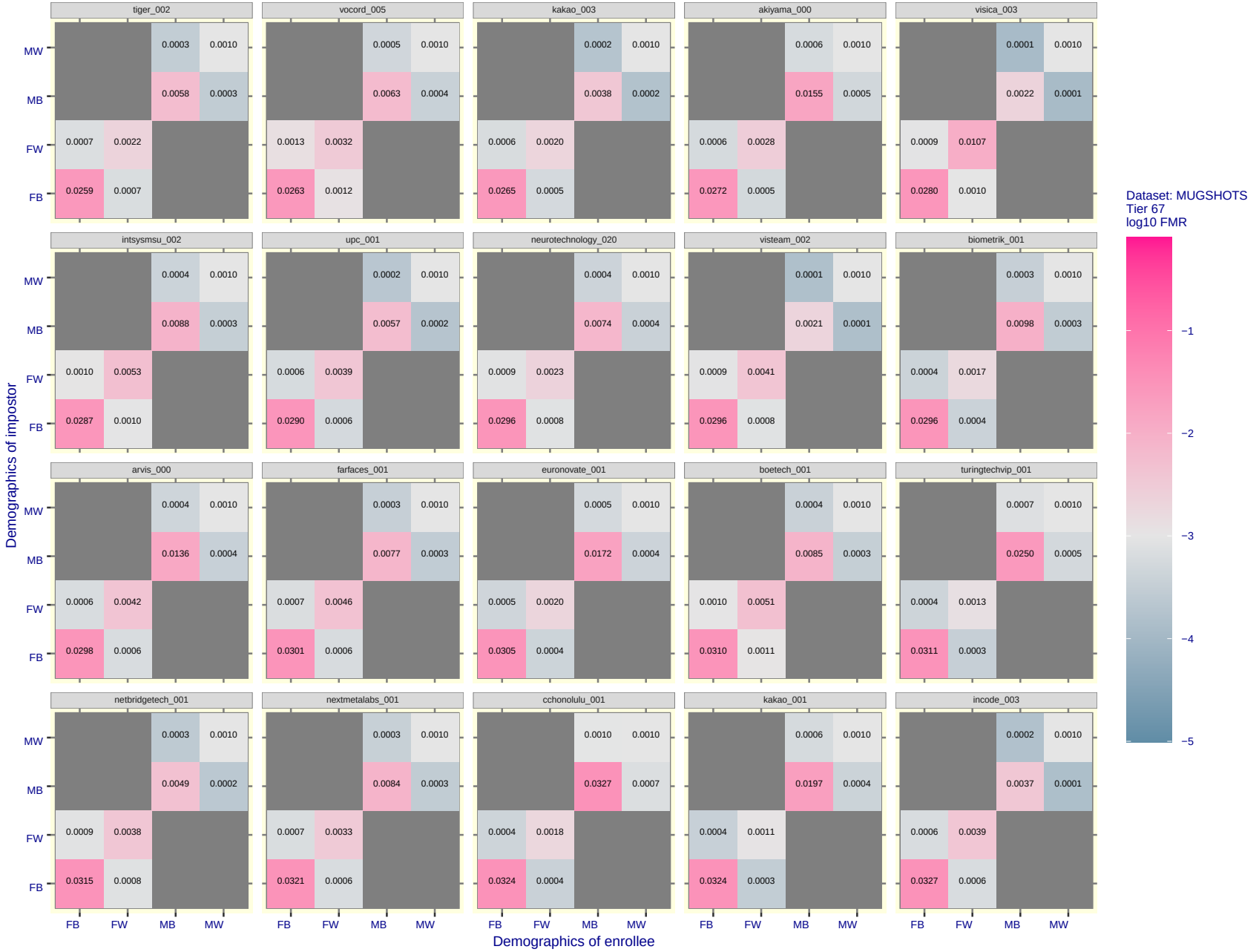

Figure 397: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

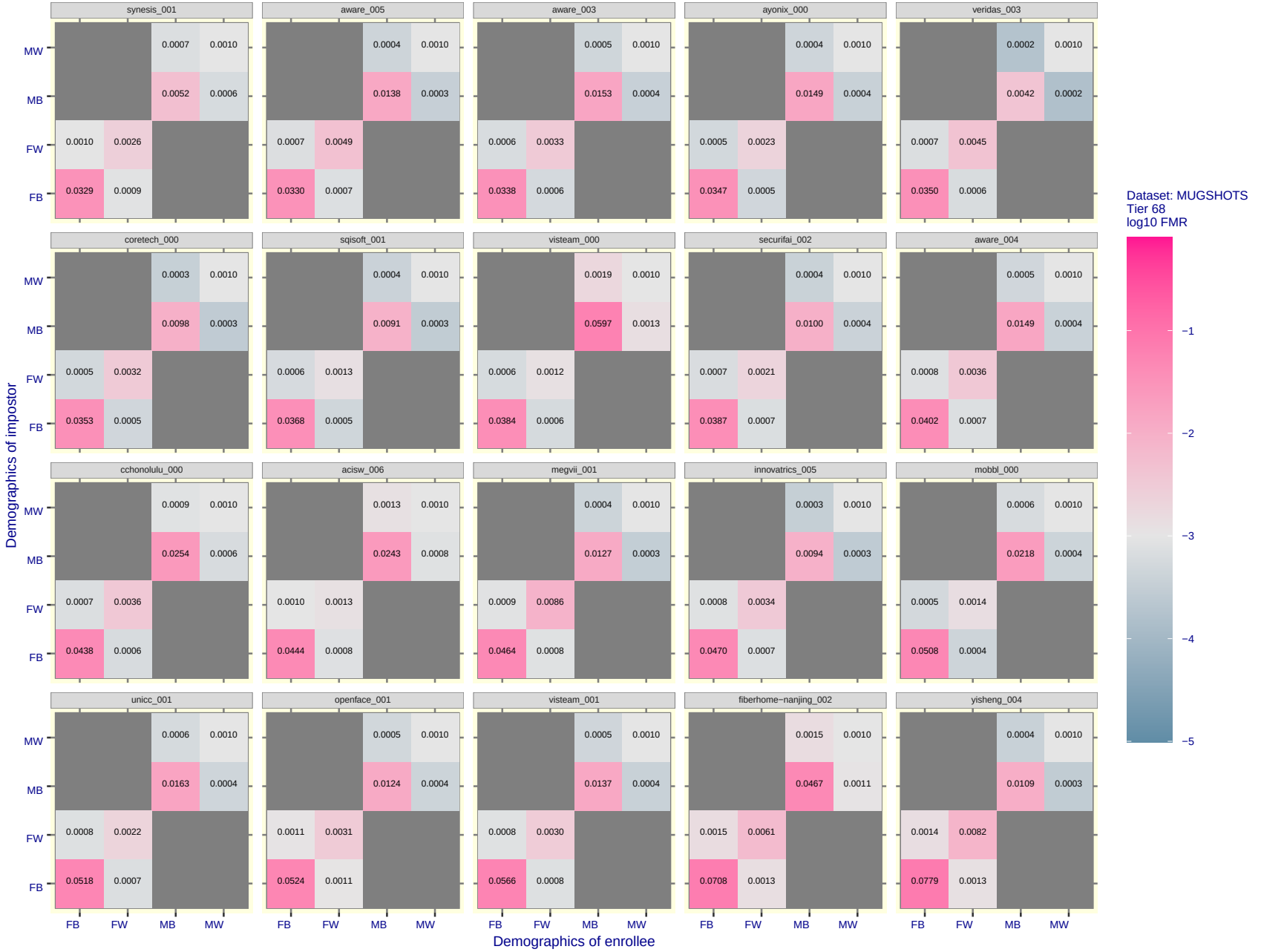

Figure 398: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

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FNMR(T) "False non-match rate"
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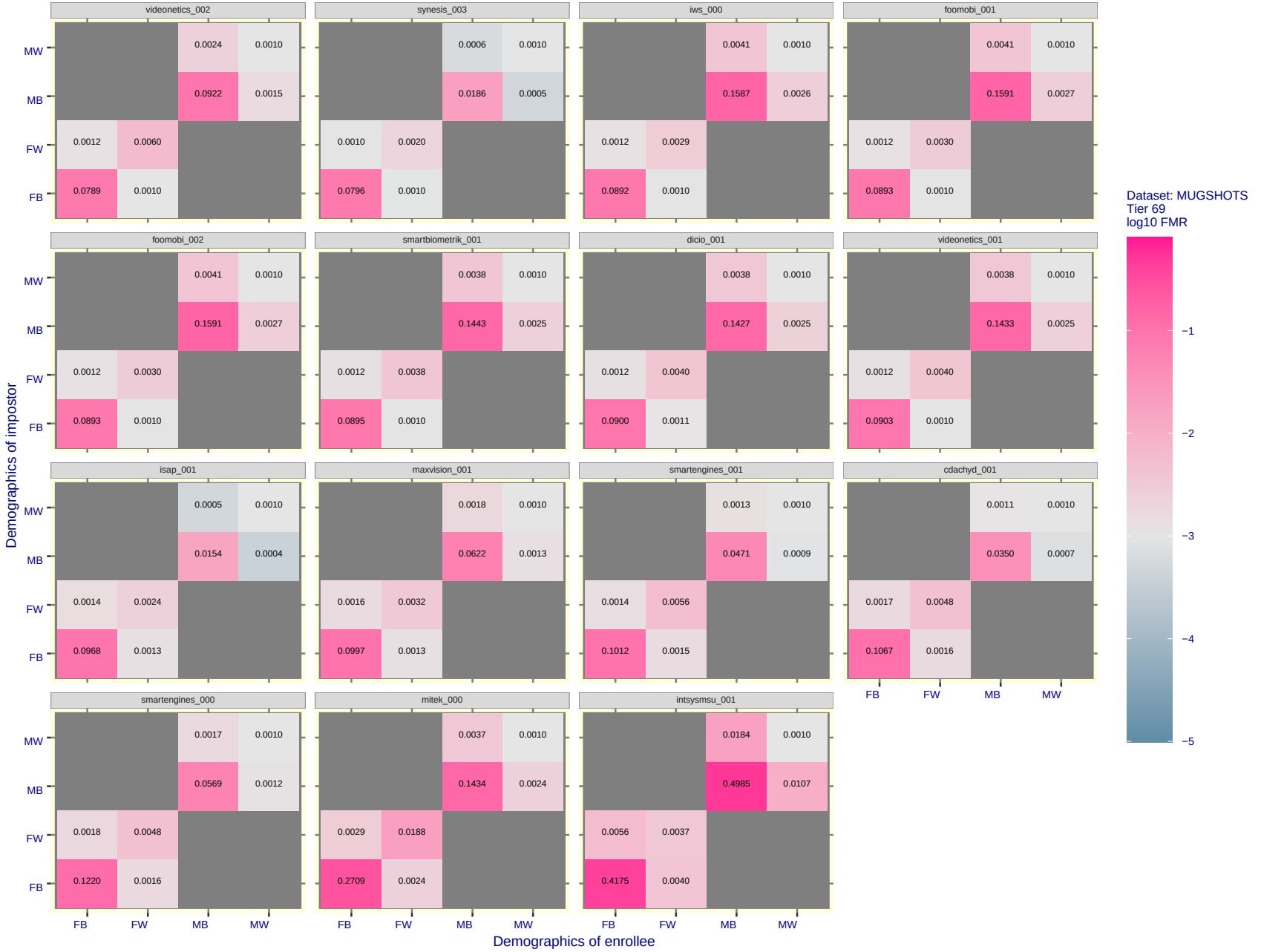

Figure 399: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

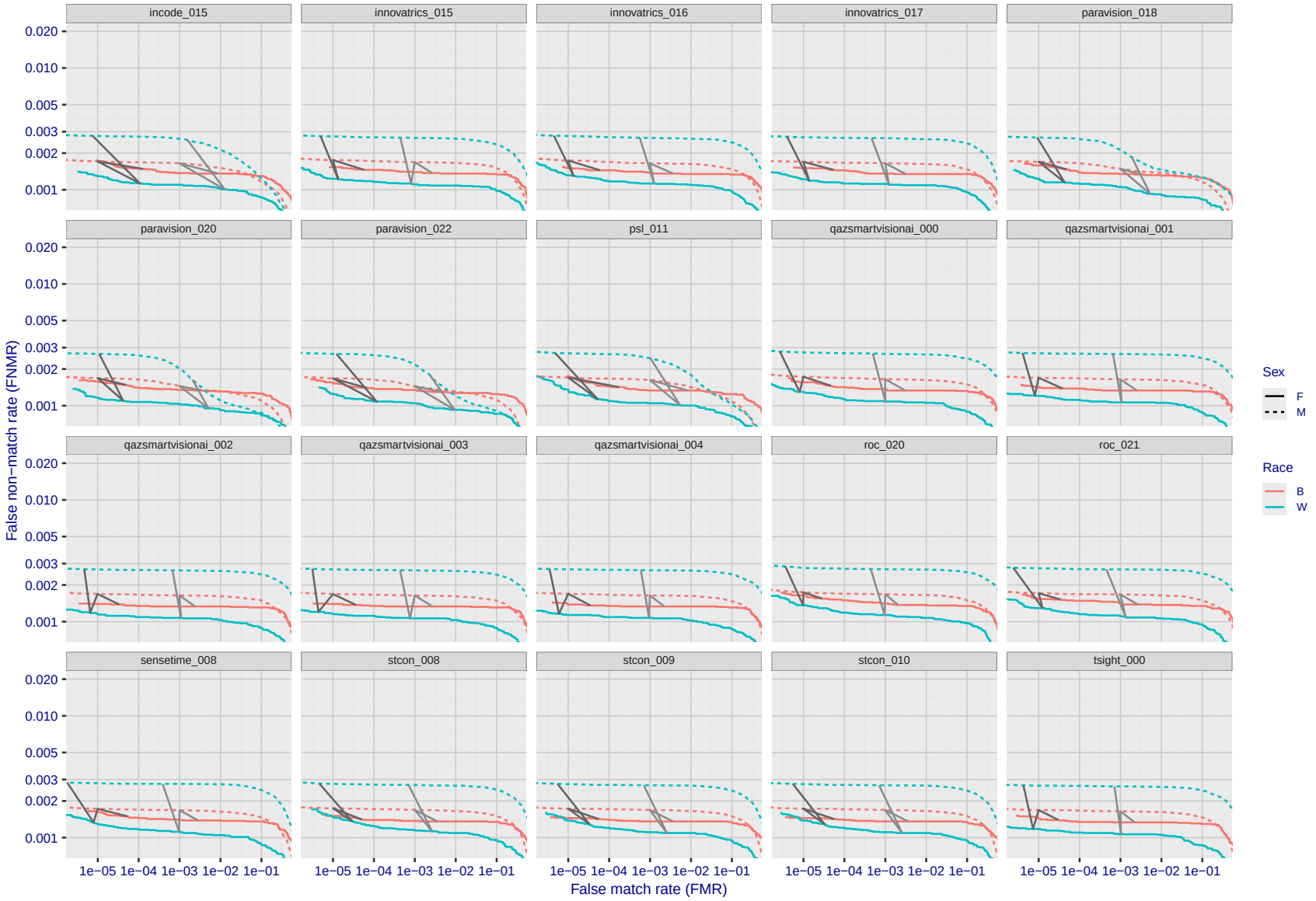

Figure 400: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

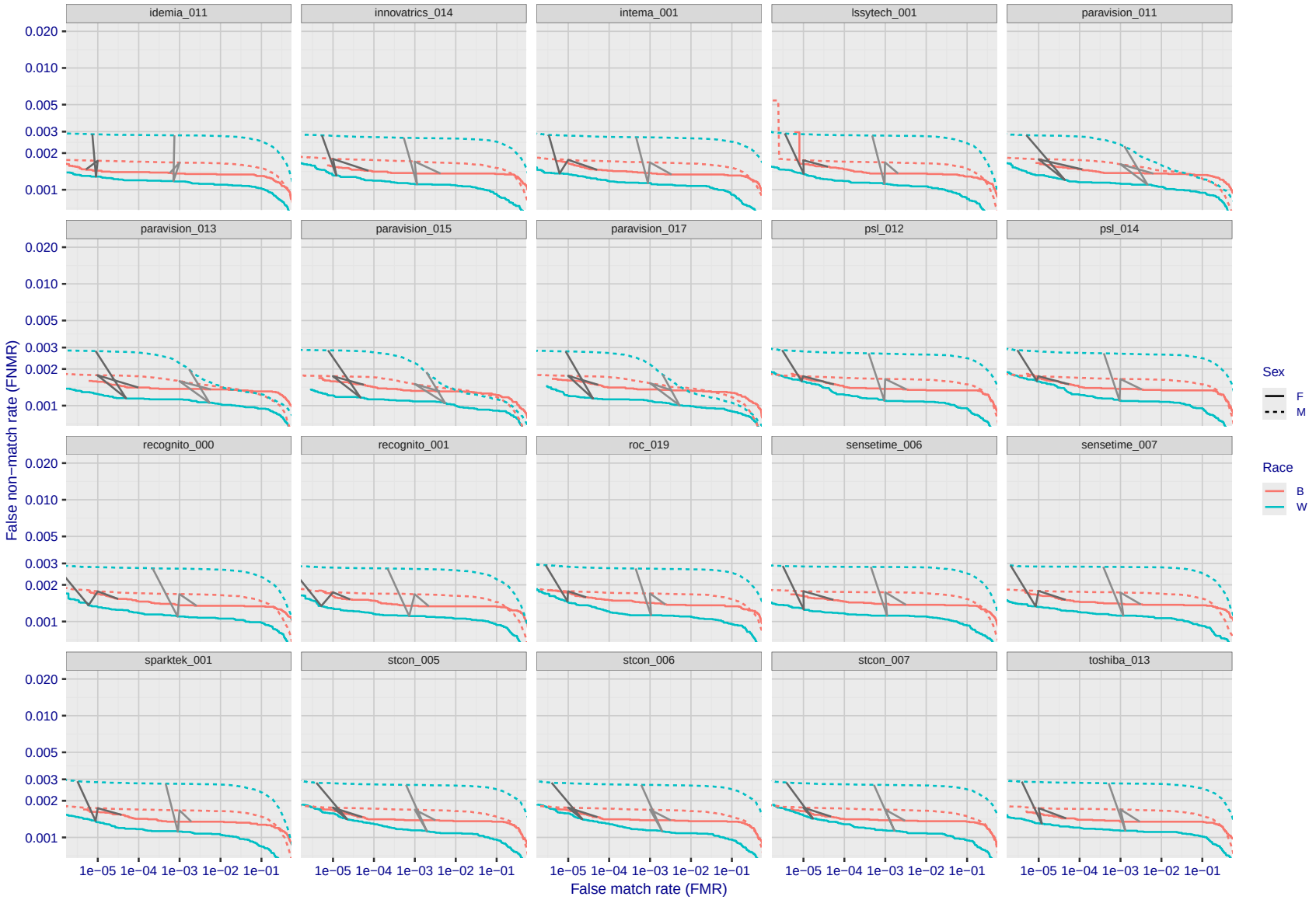

Figure 401: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

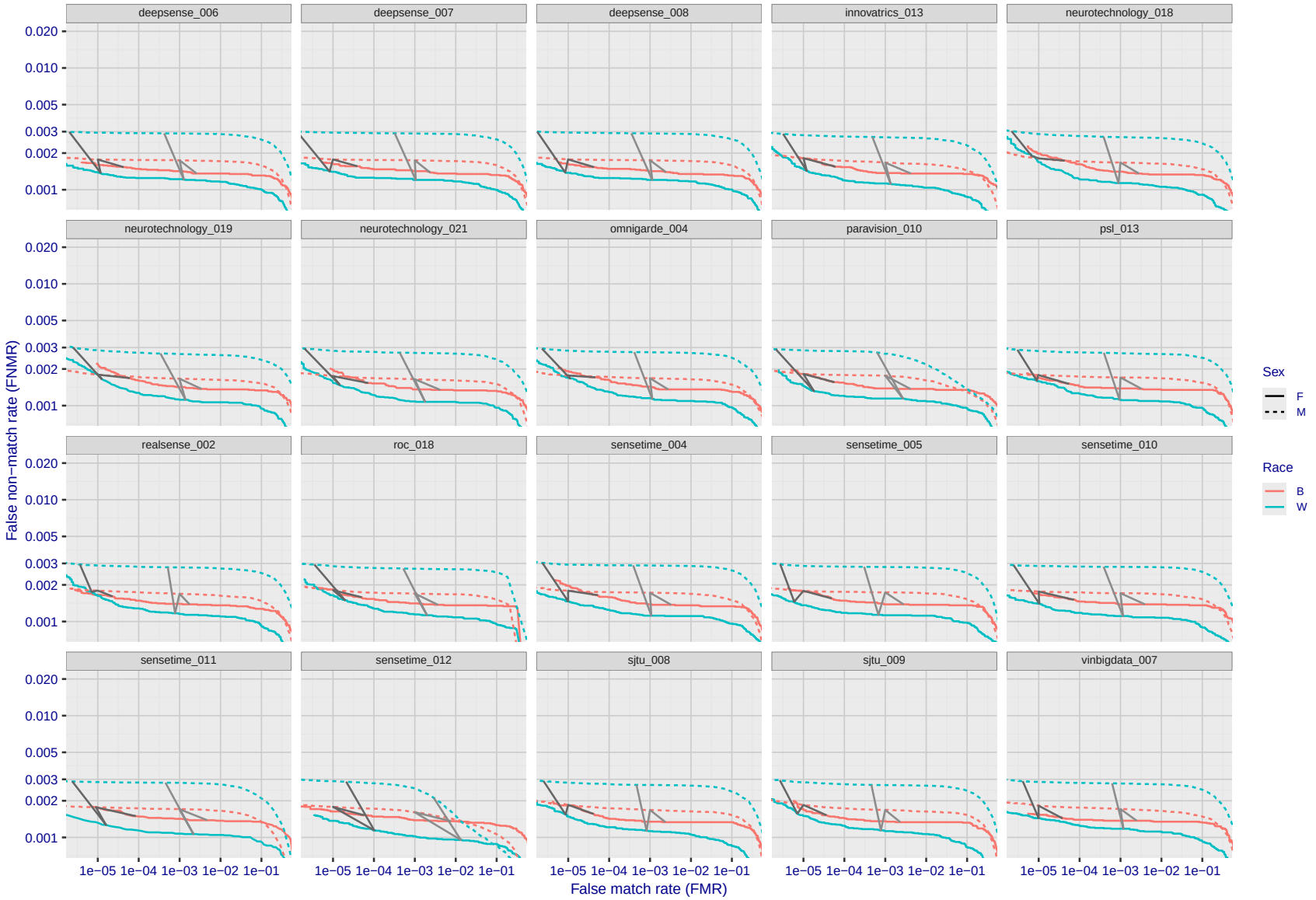

Figure 402: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

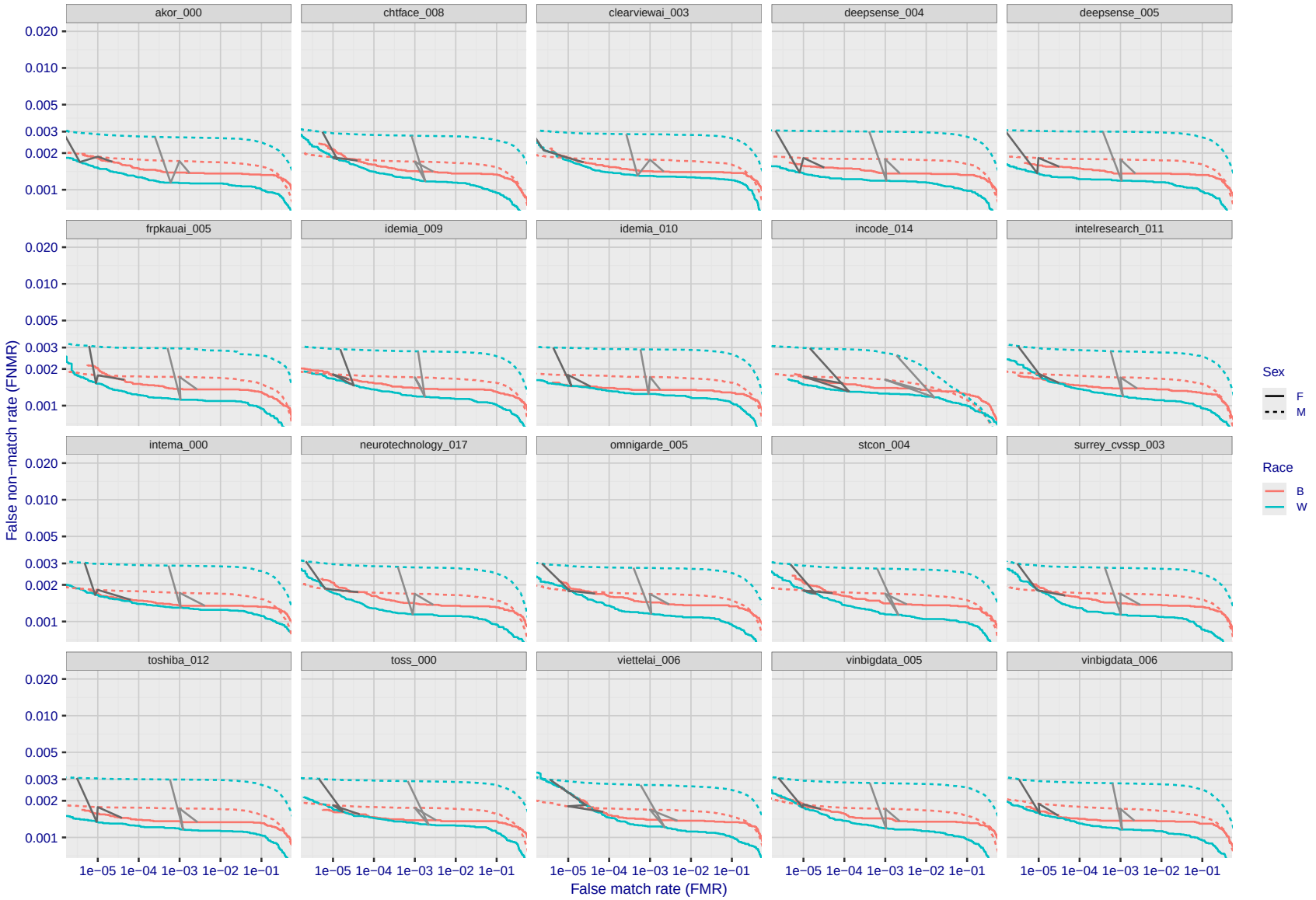


Figure 403: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

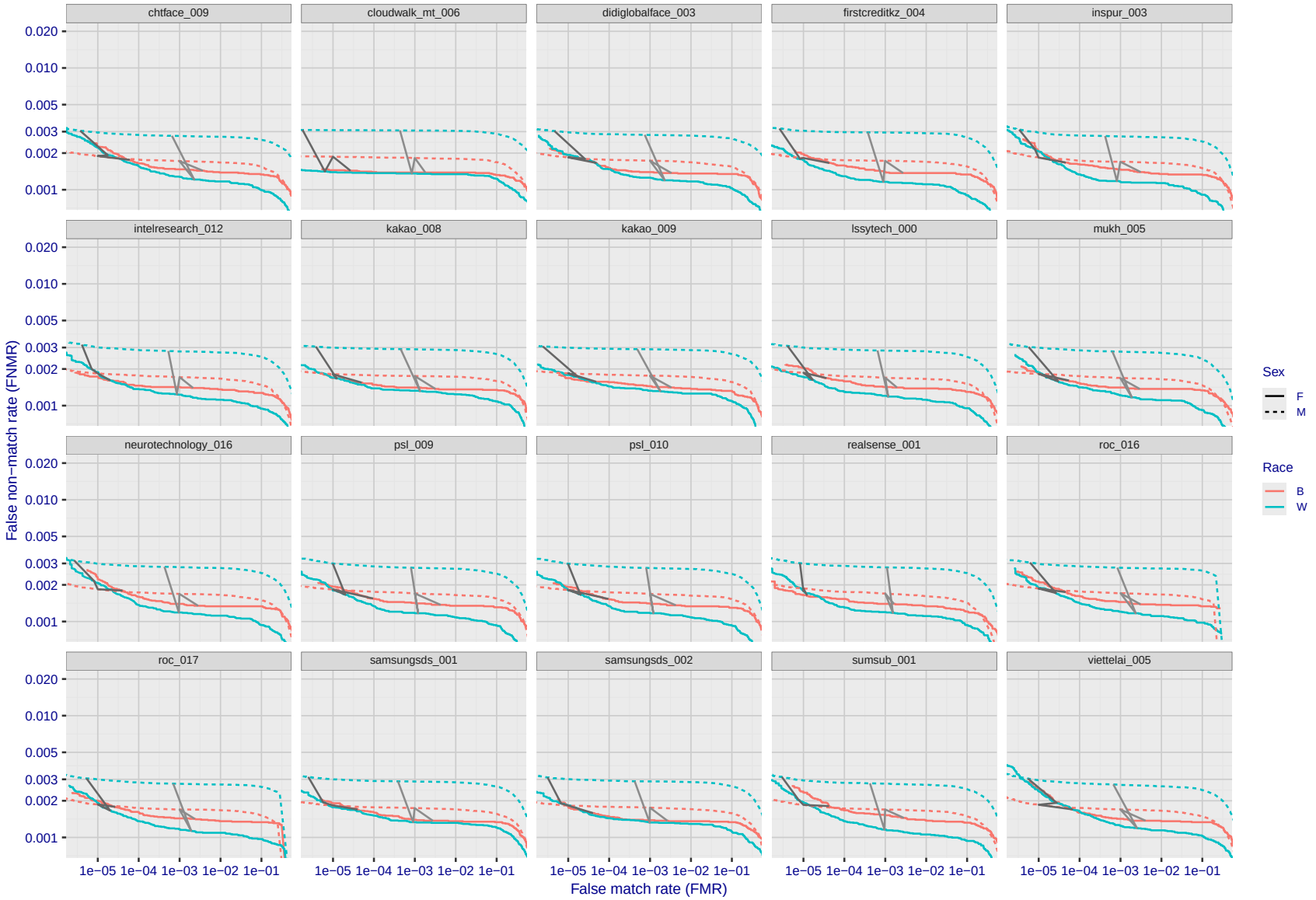


Figure 404: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

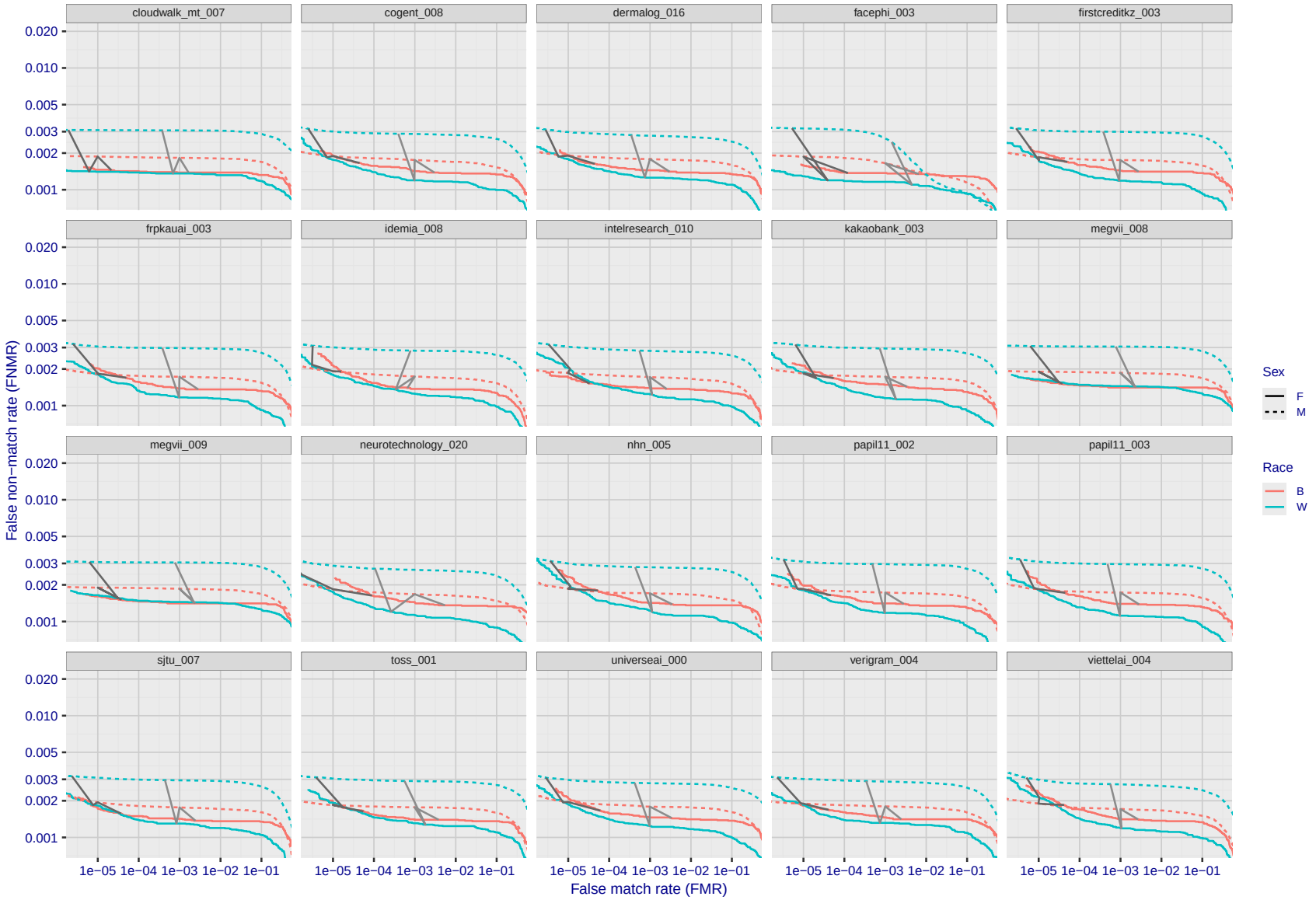


Figure 405: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

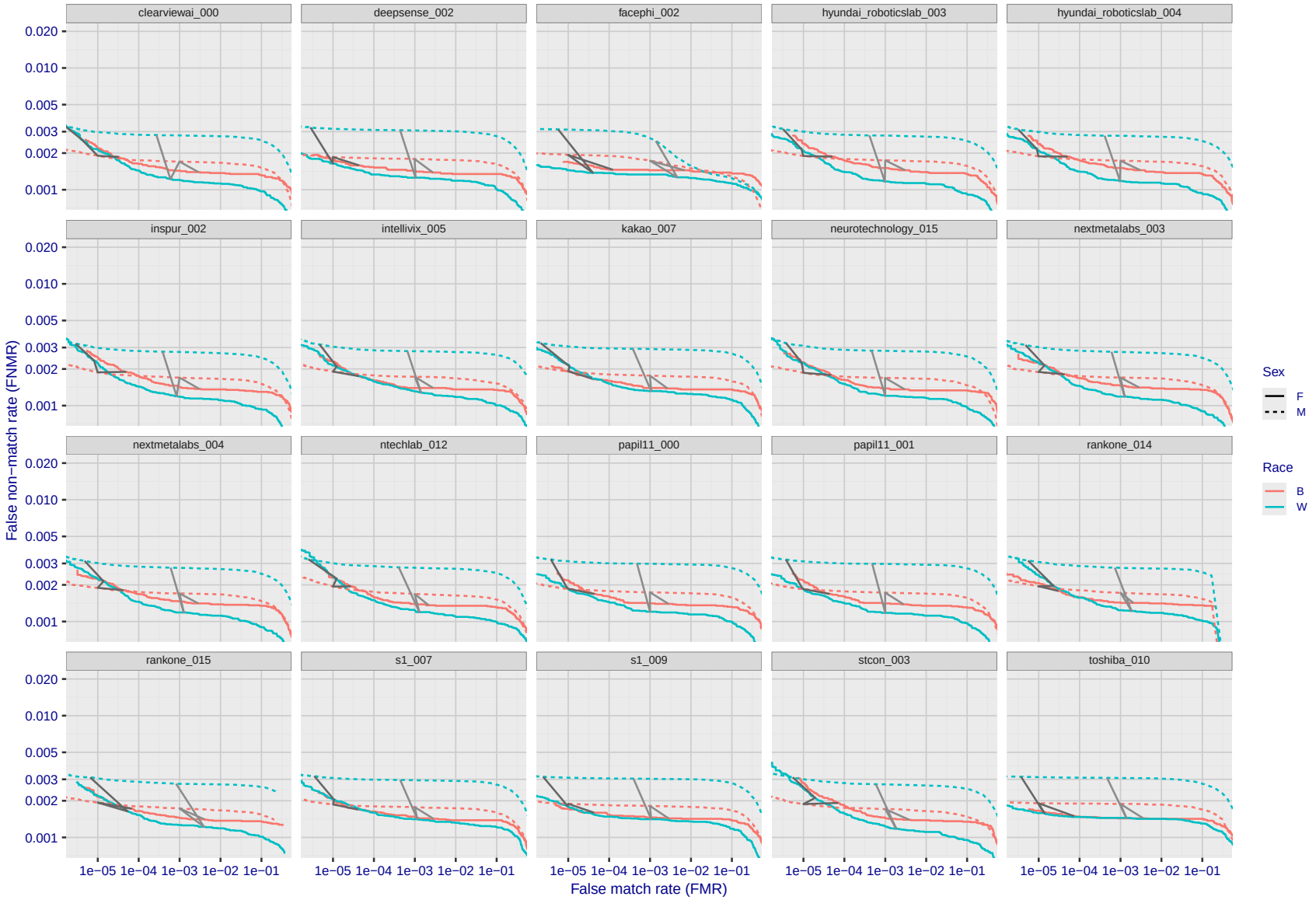


Figure 406: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

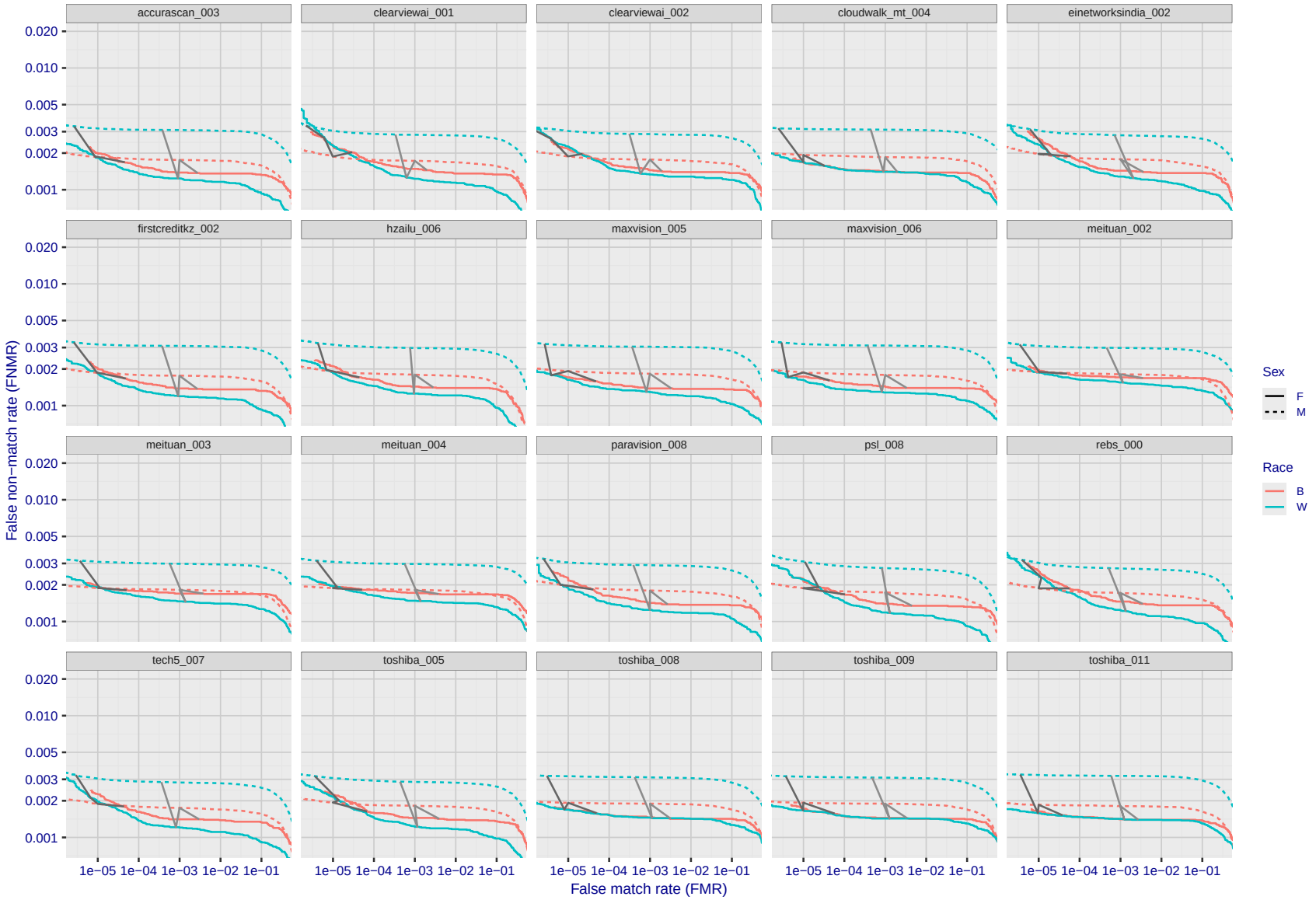


Figure 407: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

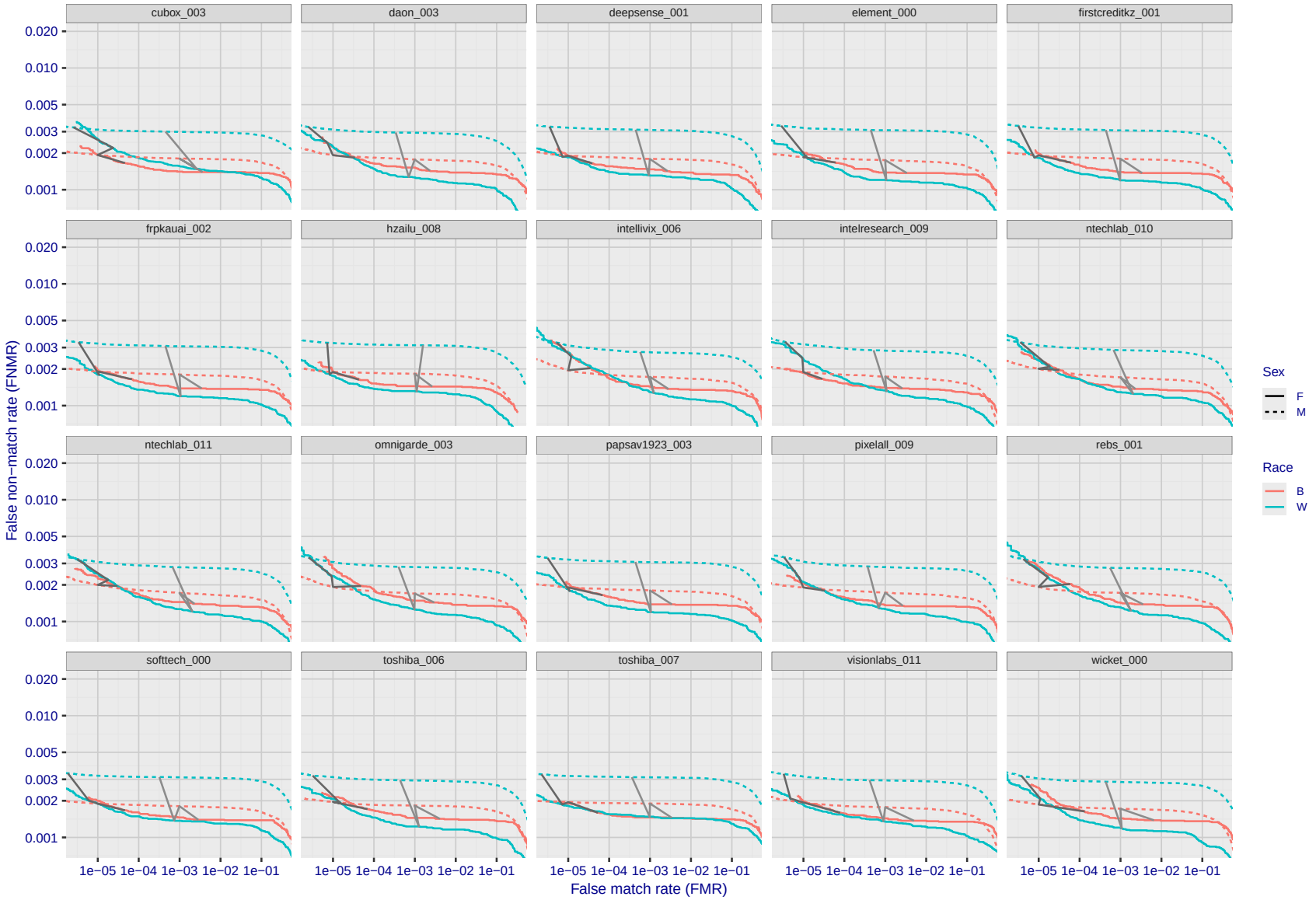


Figure 408: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

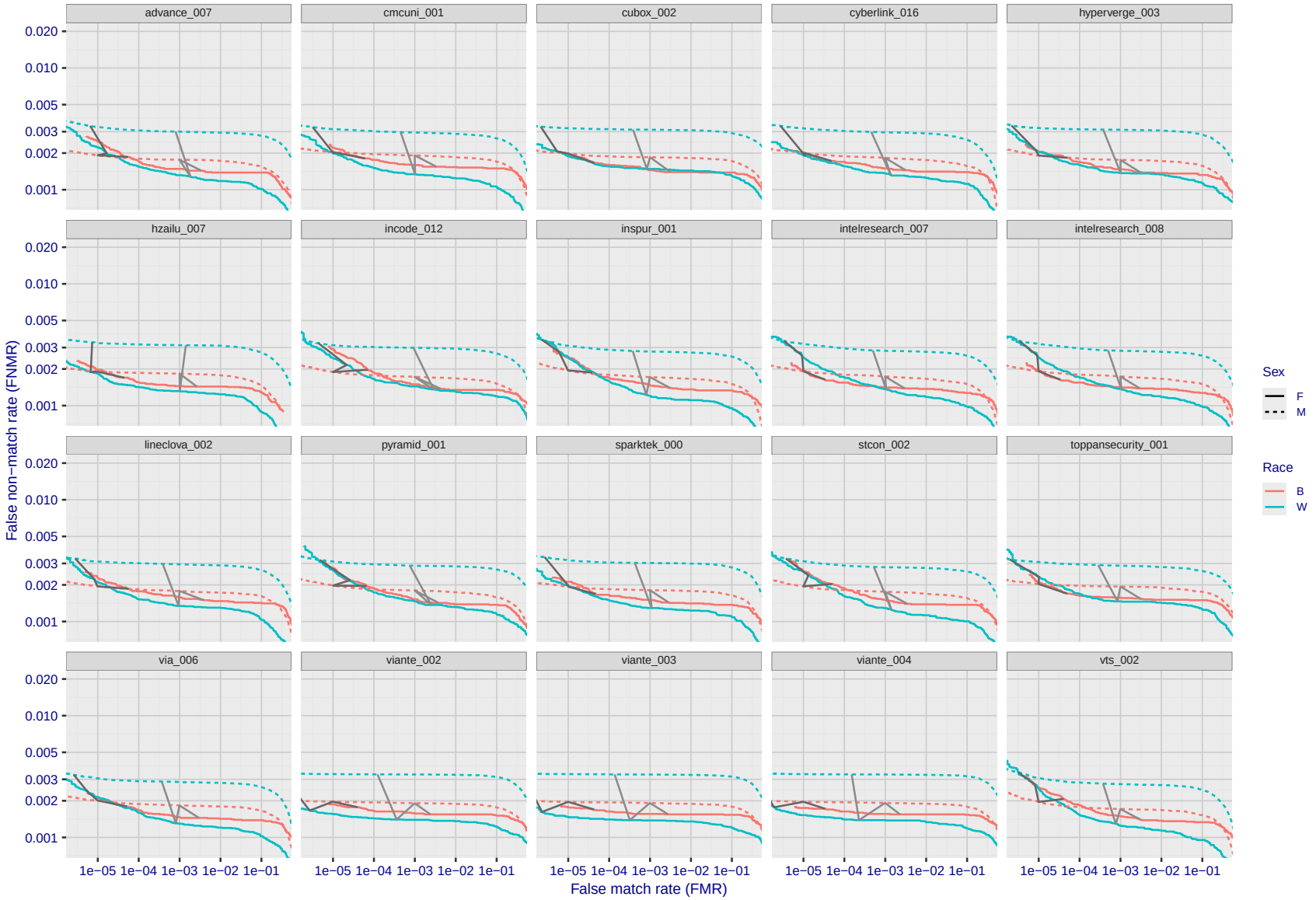


Figure 409: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

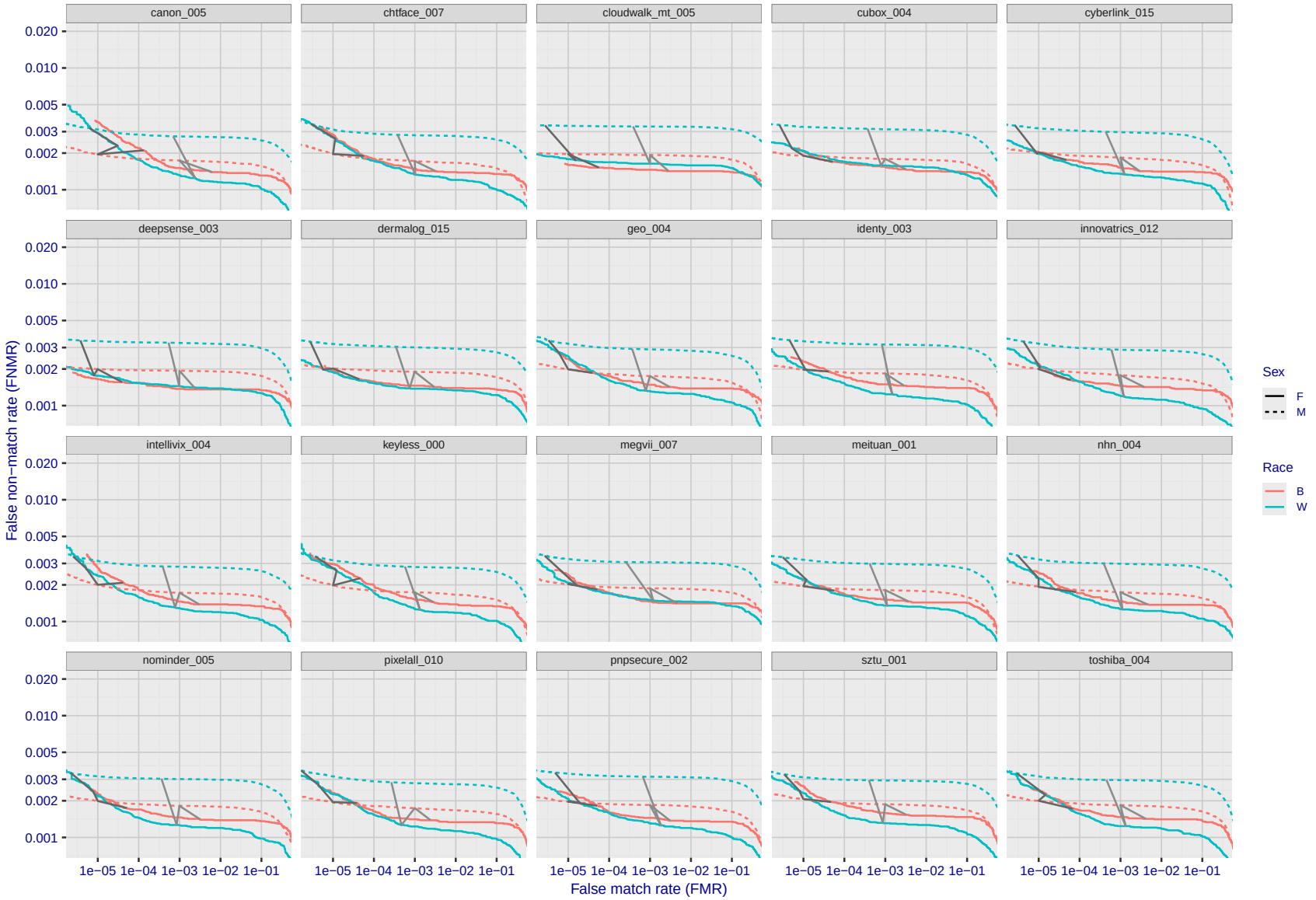


Figure 410: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

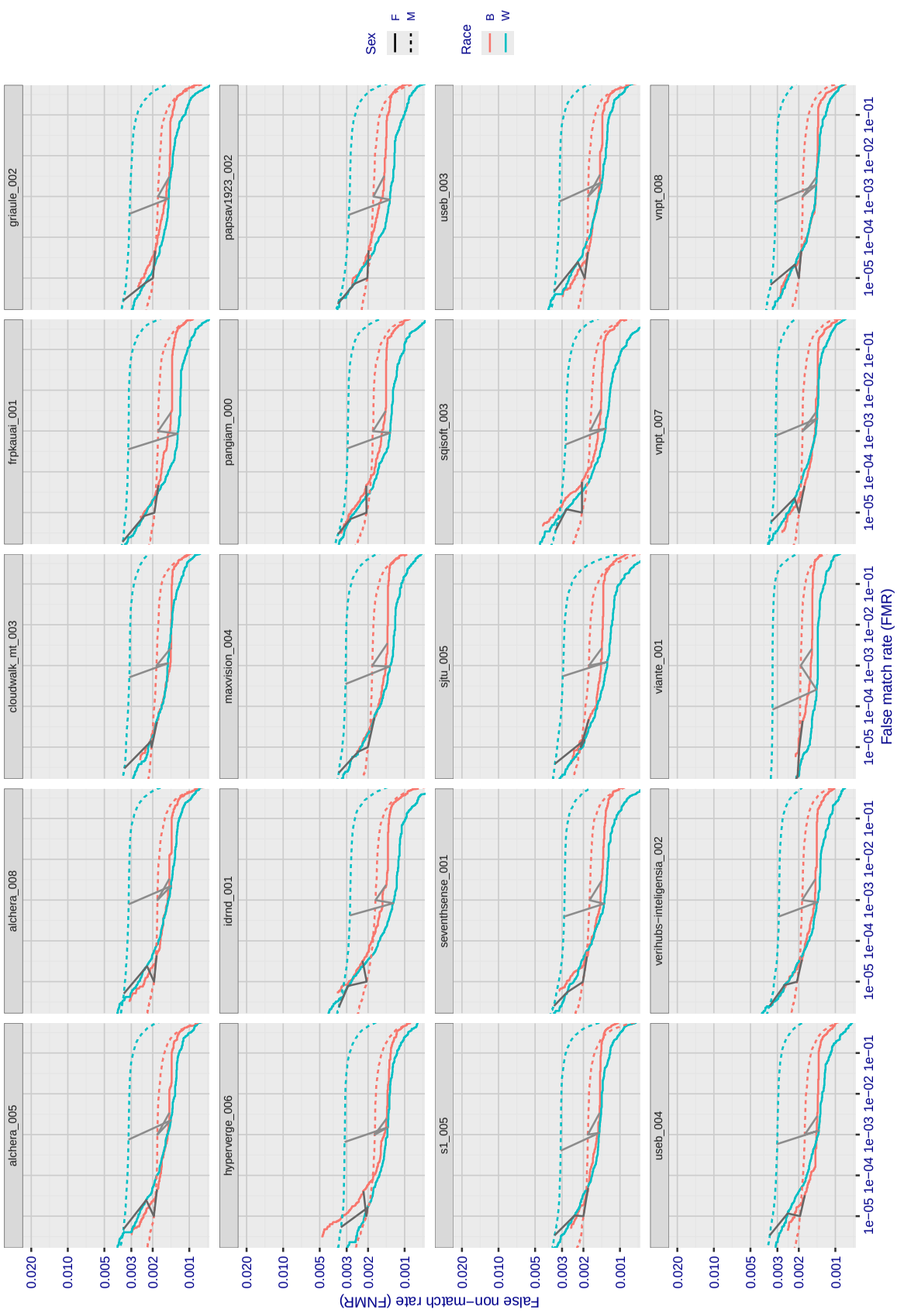


Figure 411: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

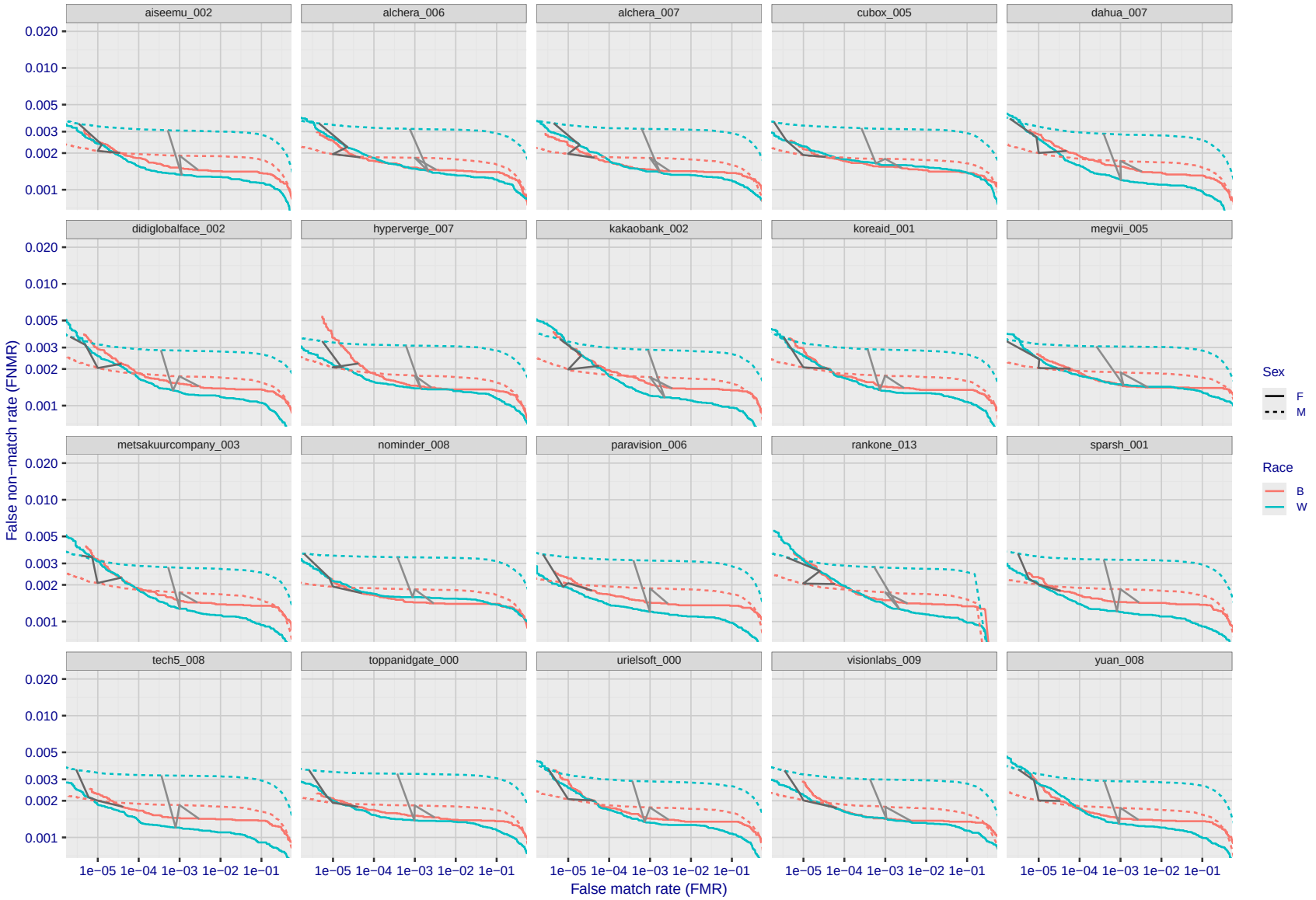


Figure 412: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

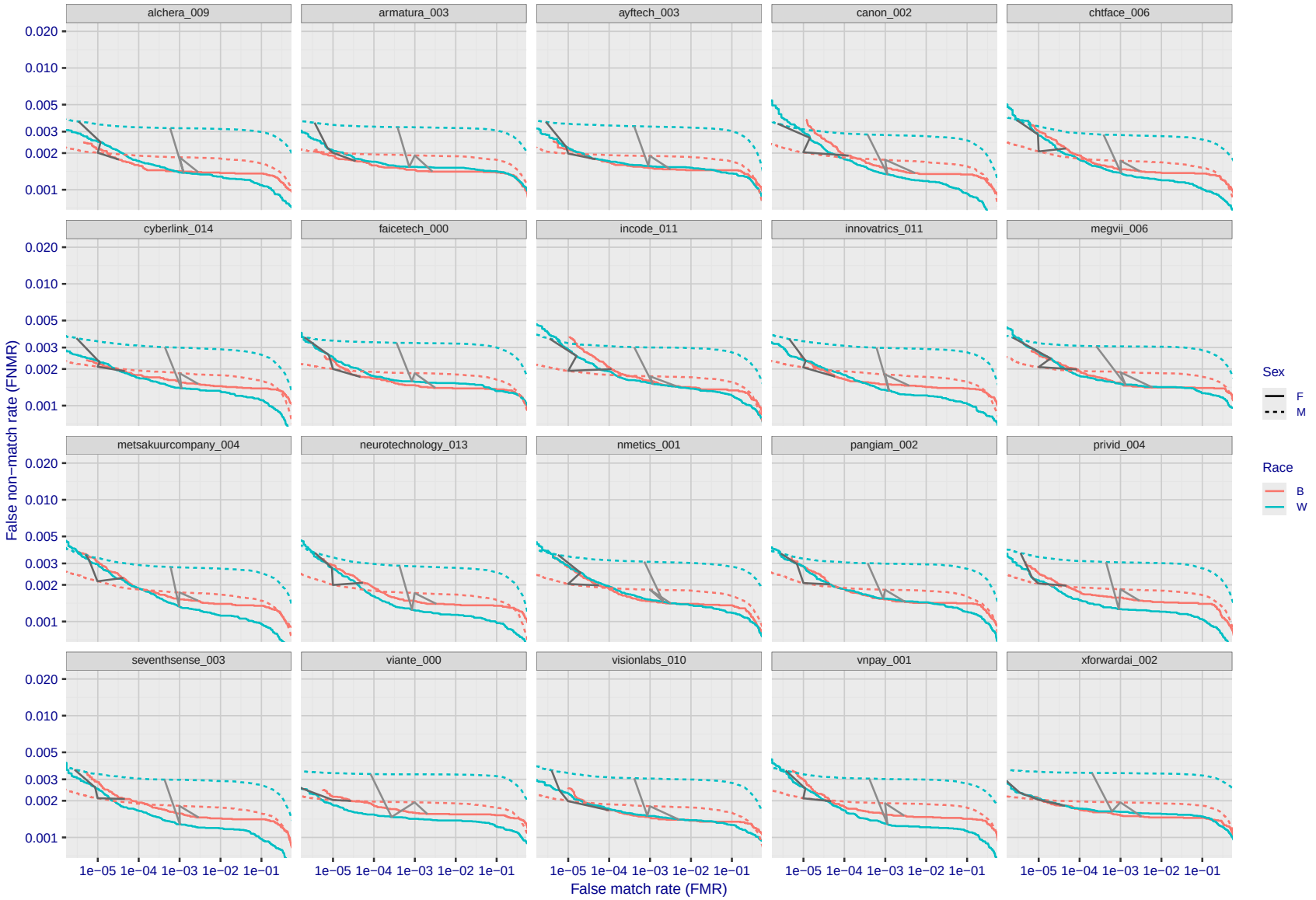


Figure 413: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

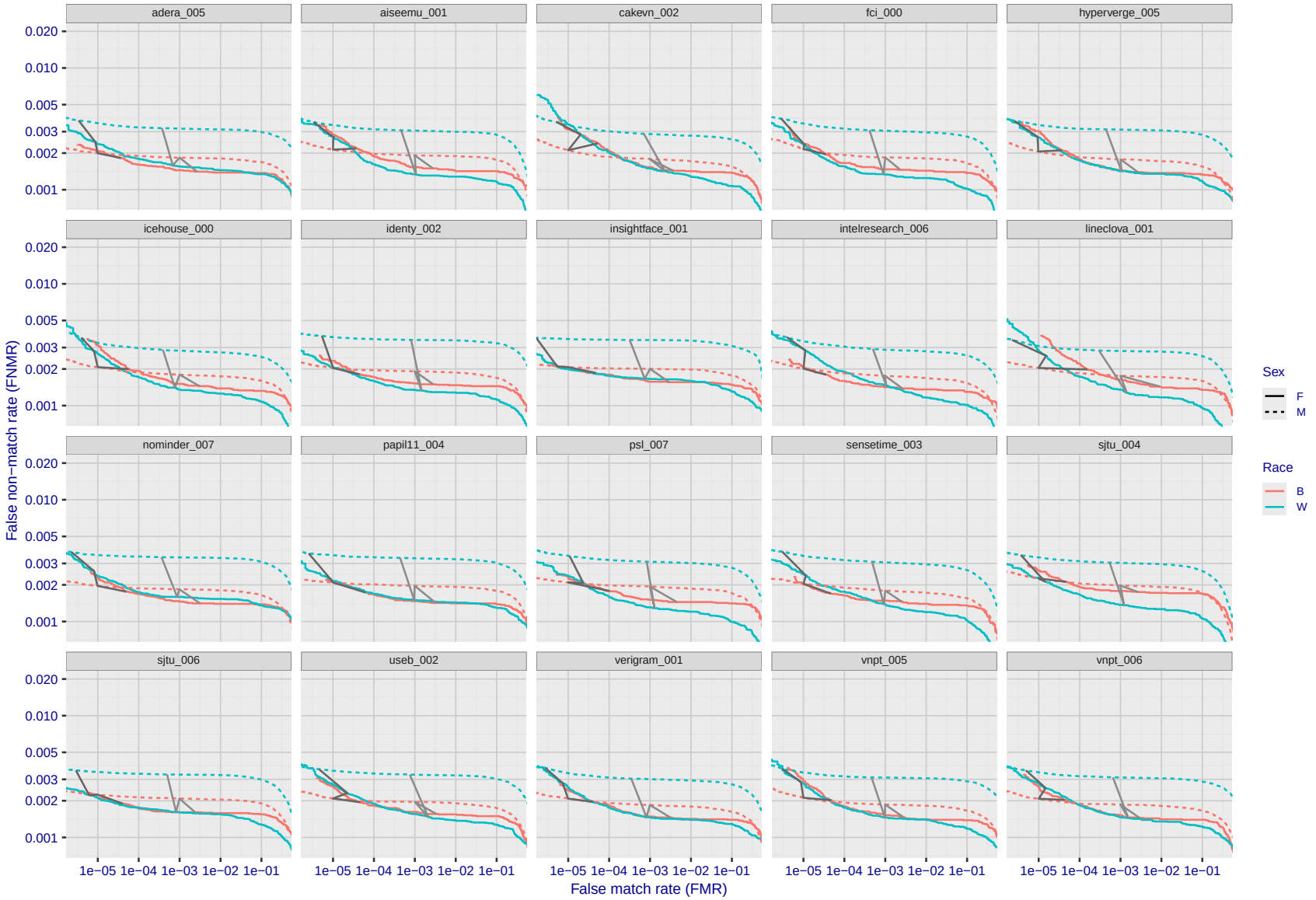


Figure 414: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

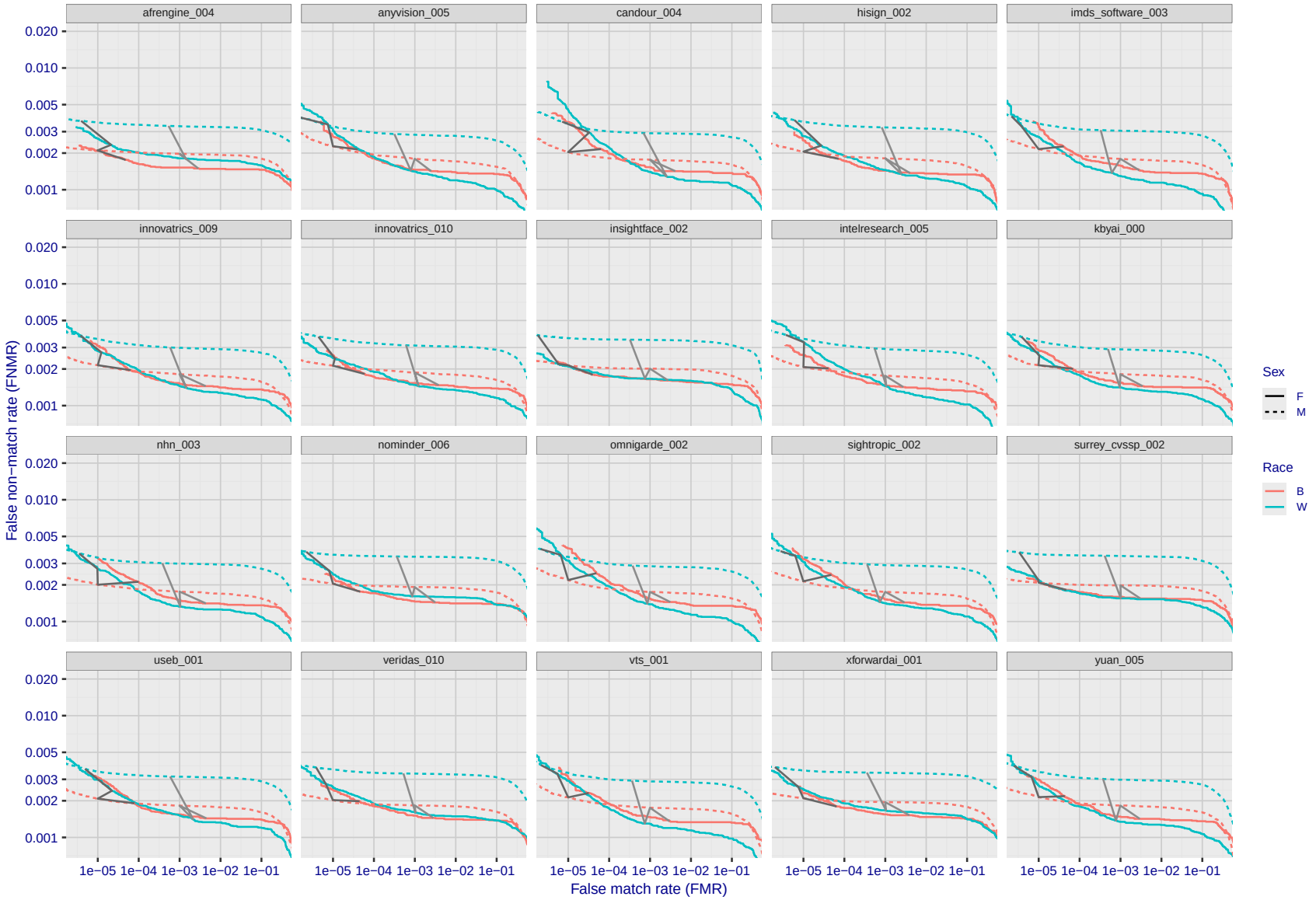


Figure 415: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

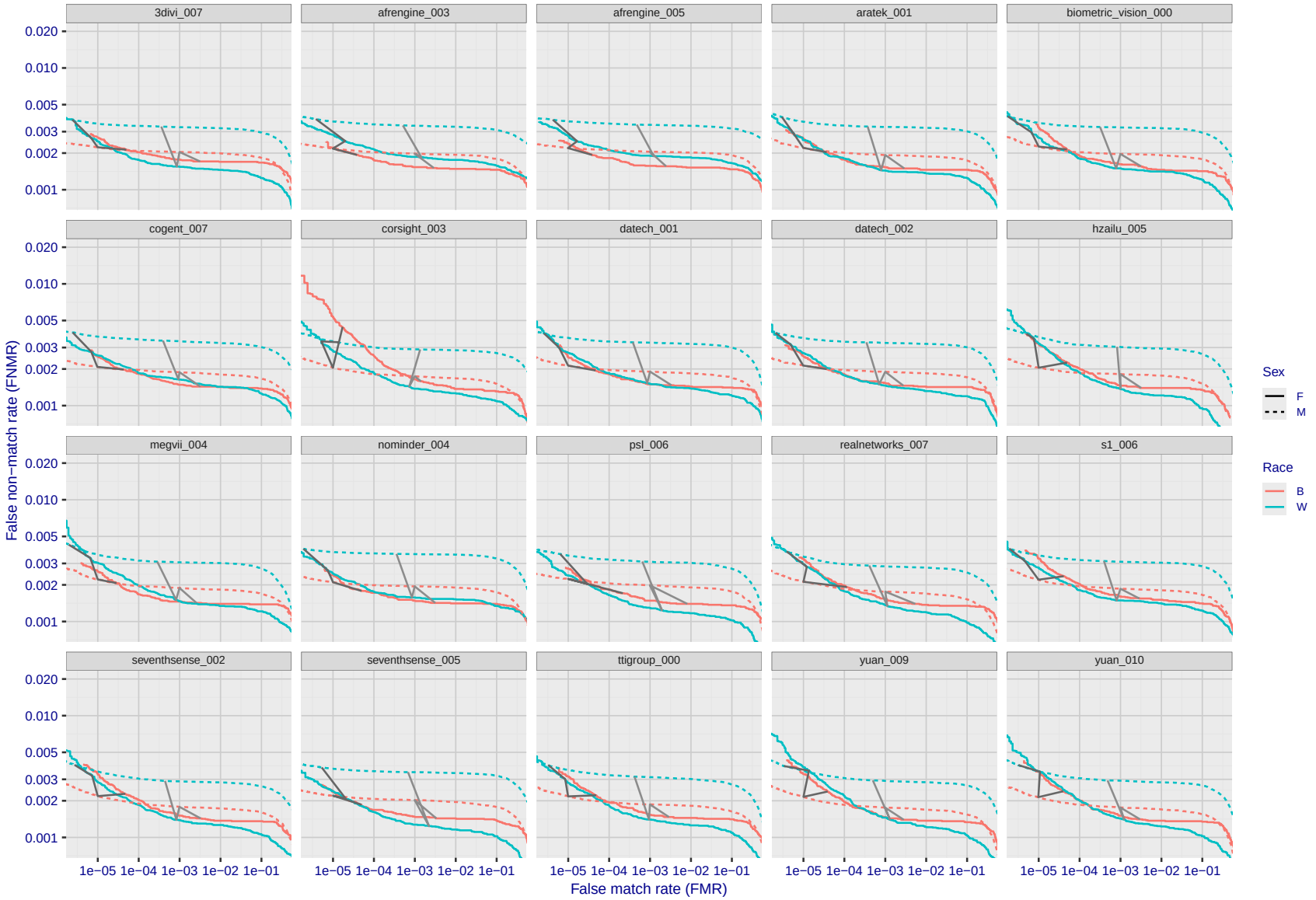


Figure 416: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

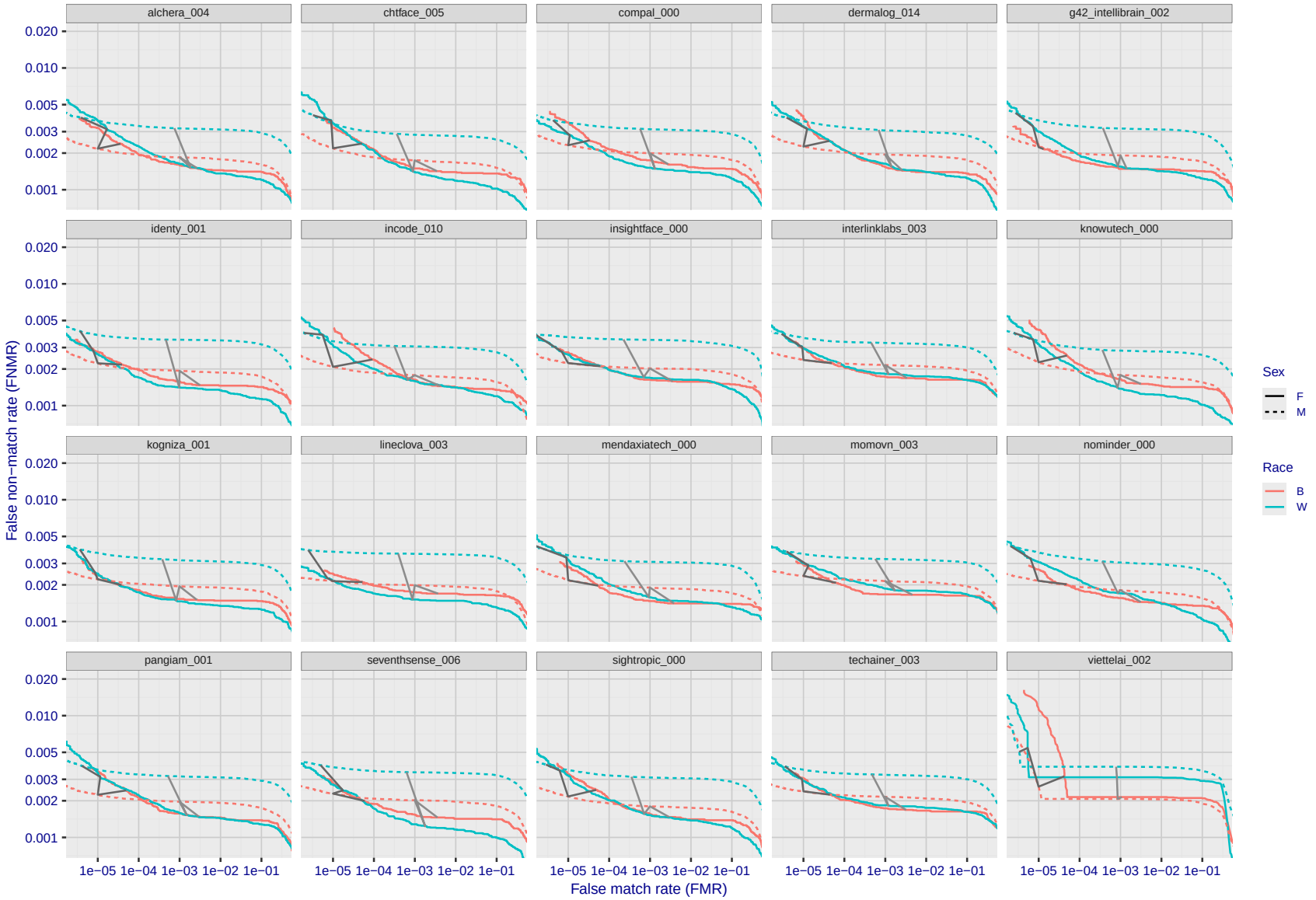


Figure 417: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

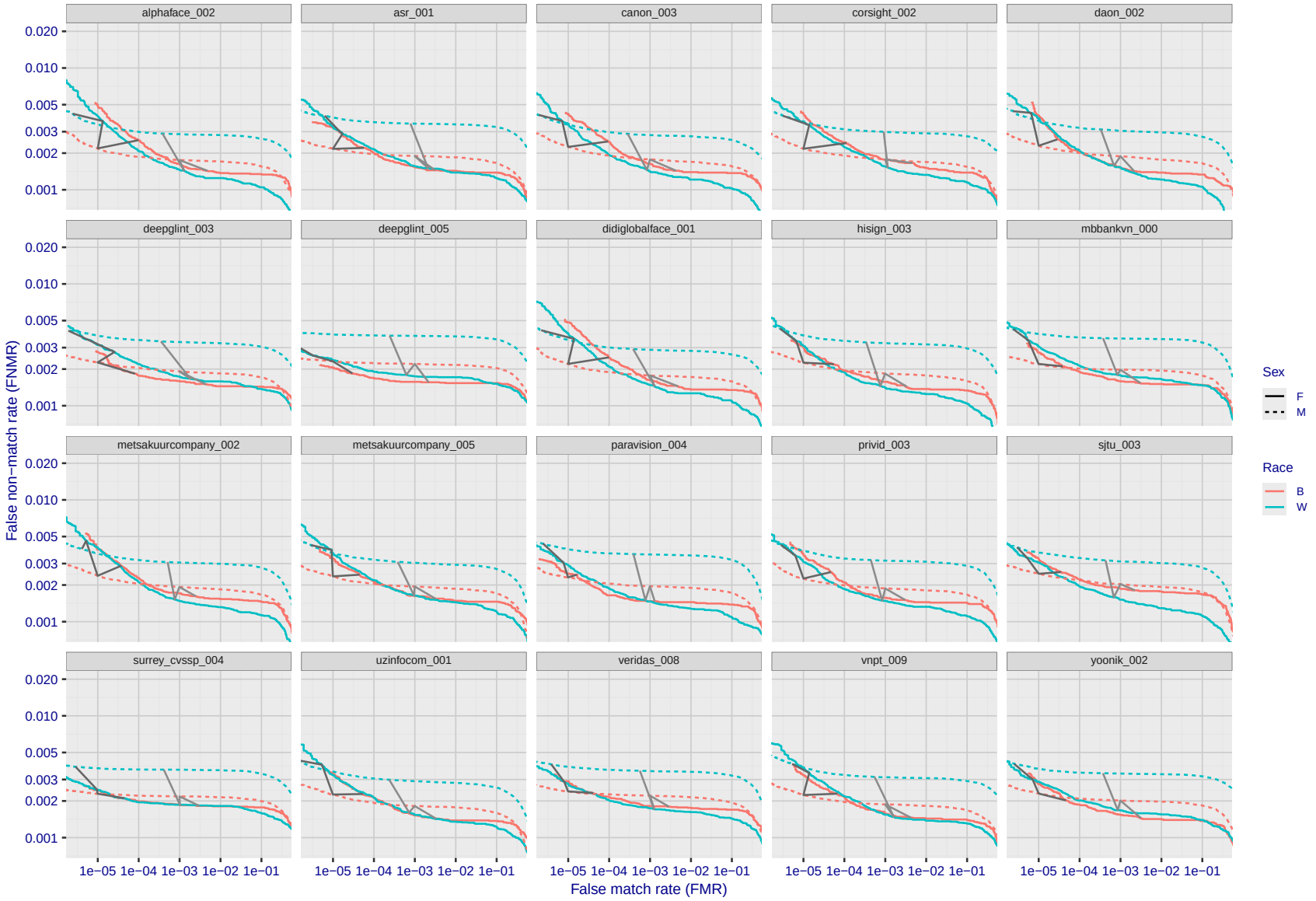


Figure 418: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

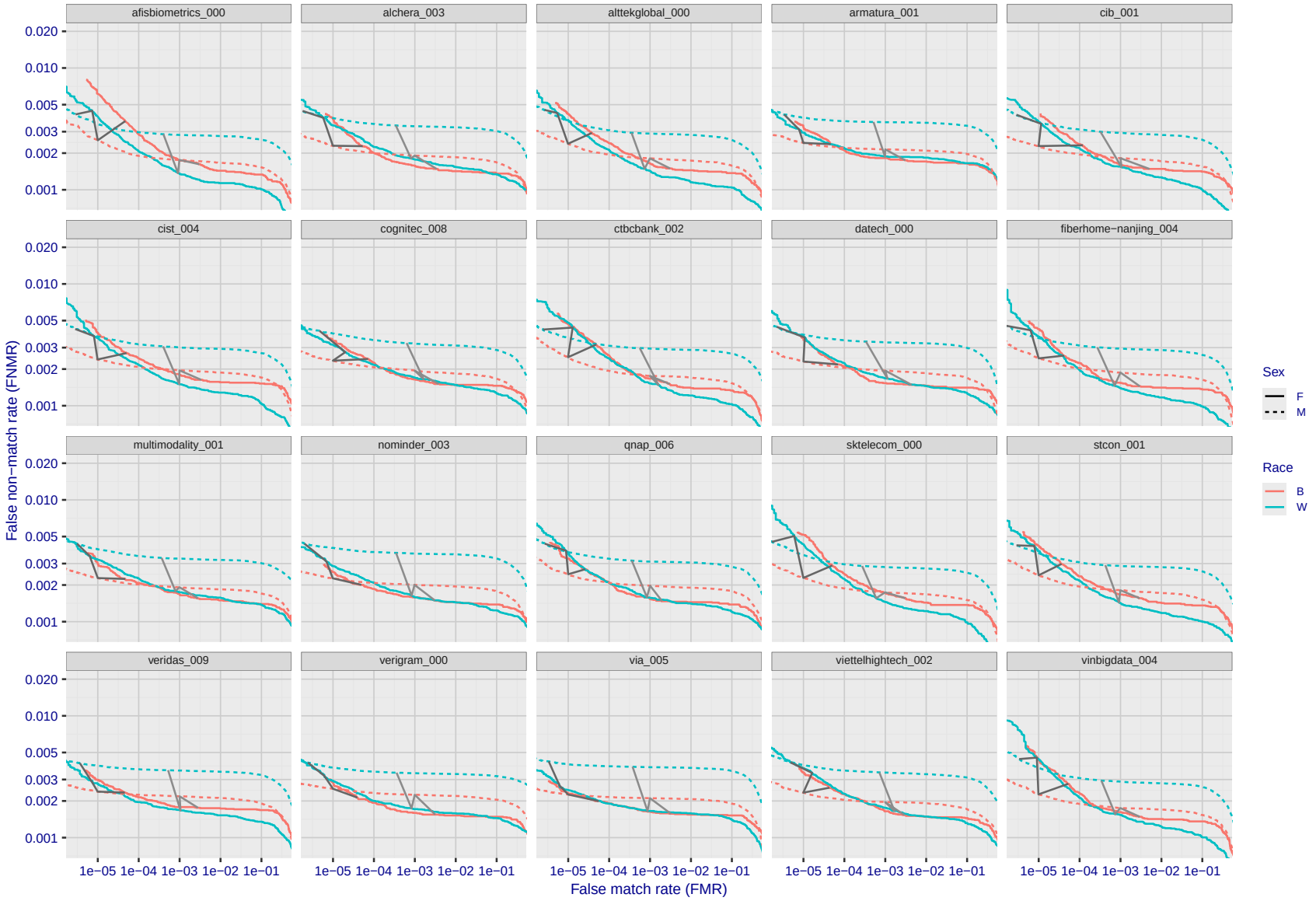


Figure 419: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

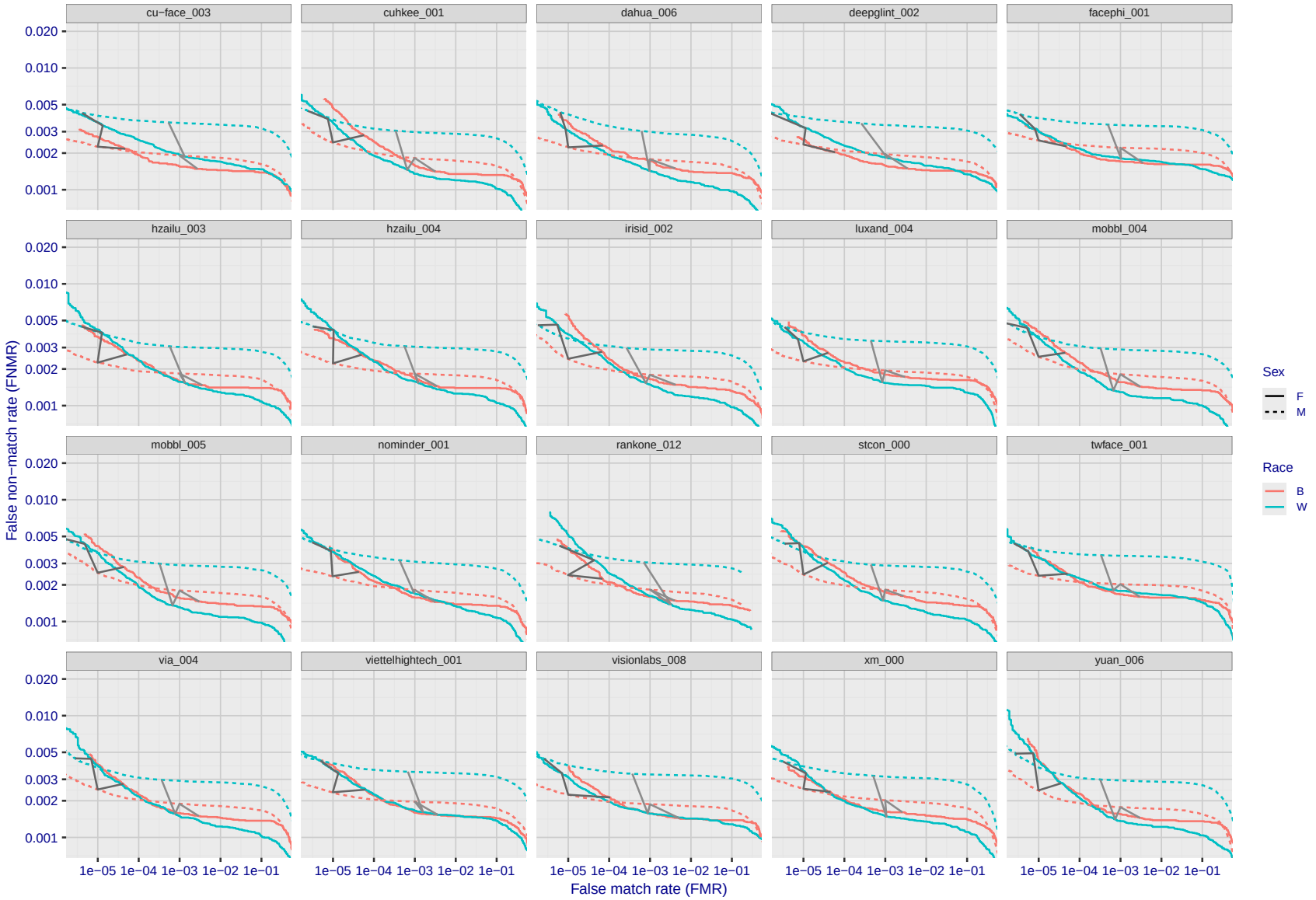


Figure 420: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

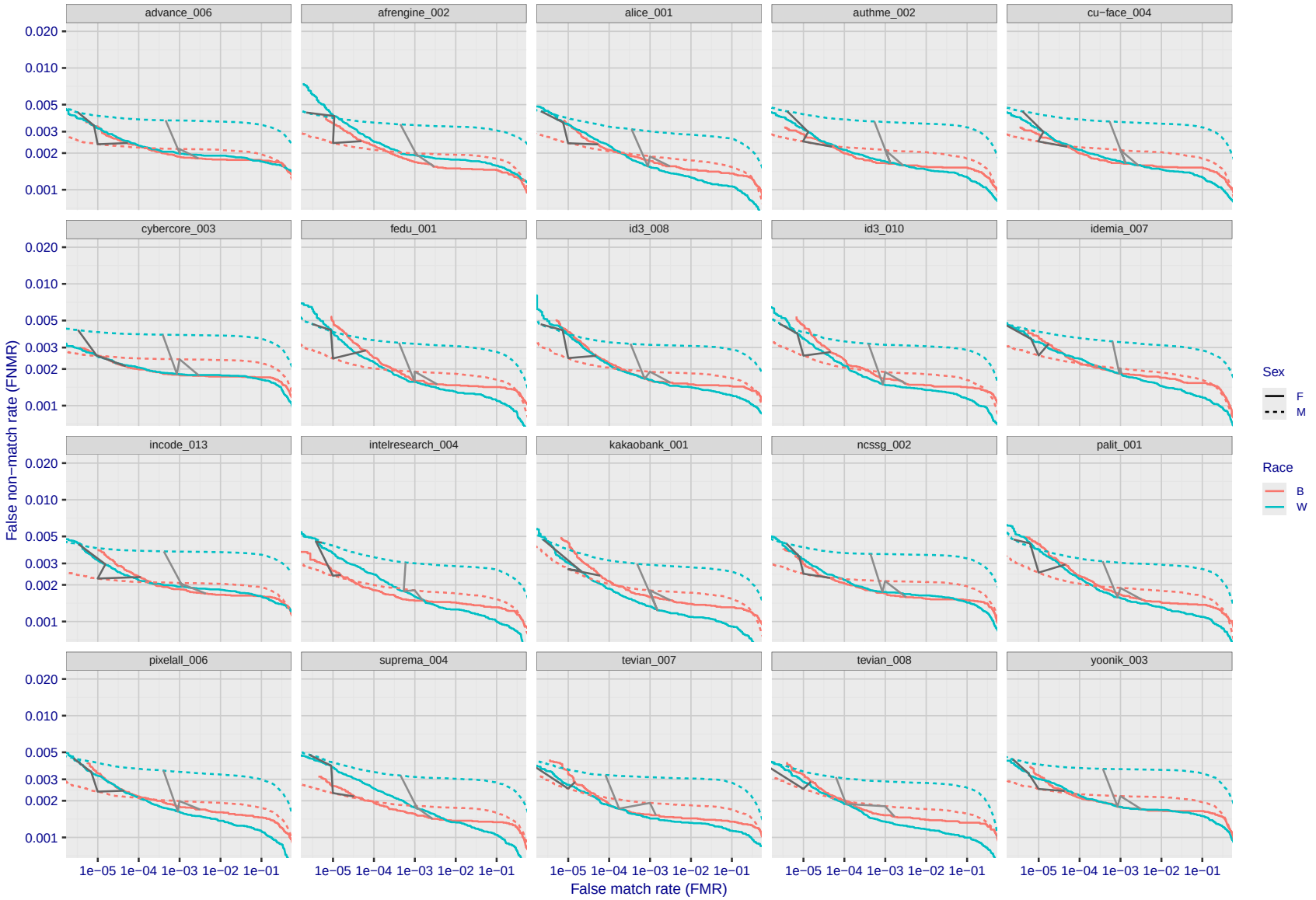


Figure 421: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

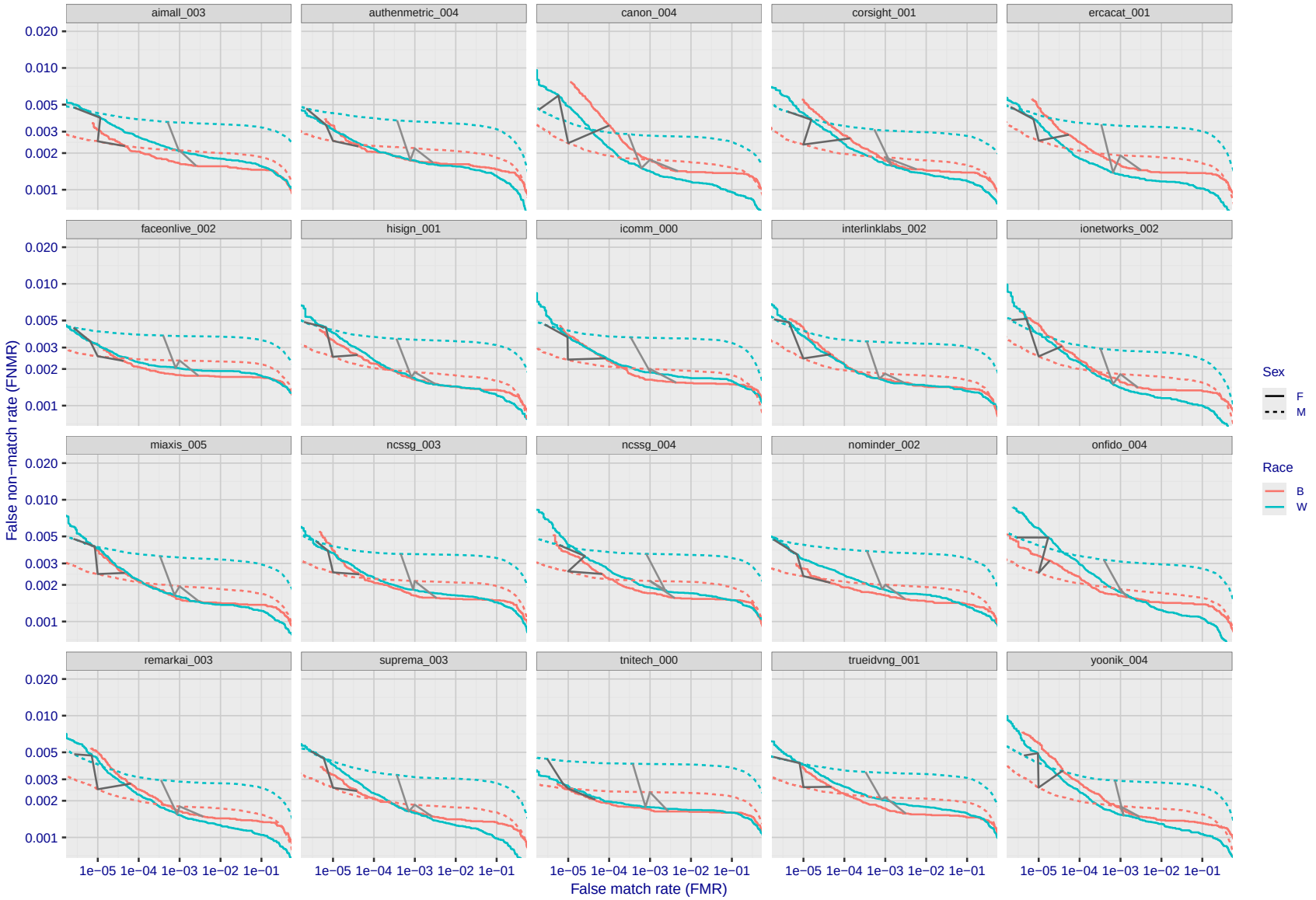


Figure 422: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

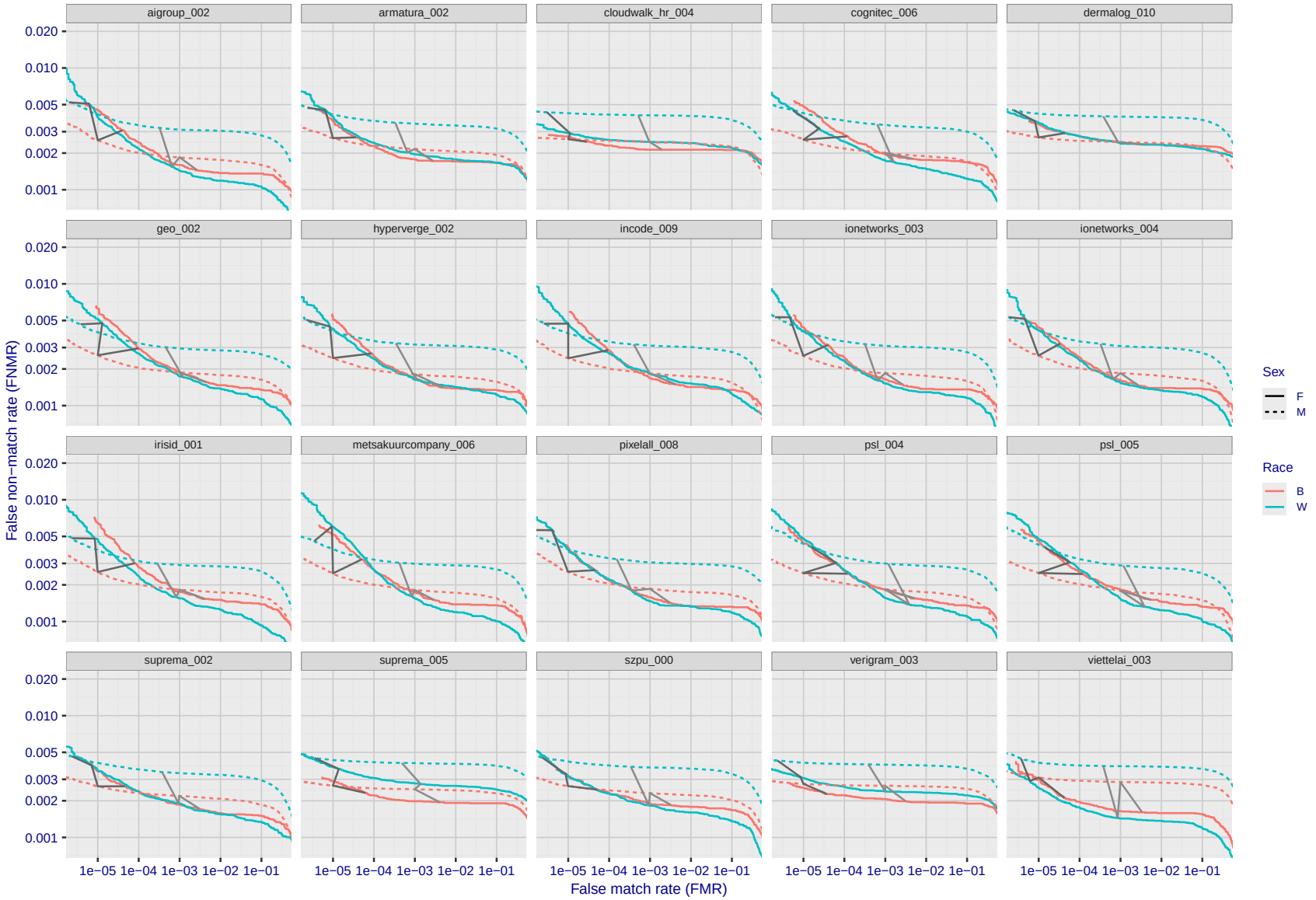


Figure 423: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

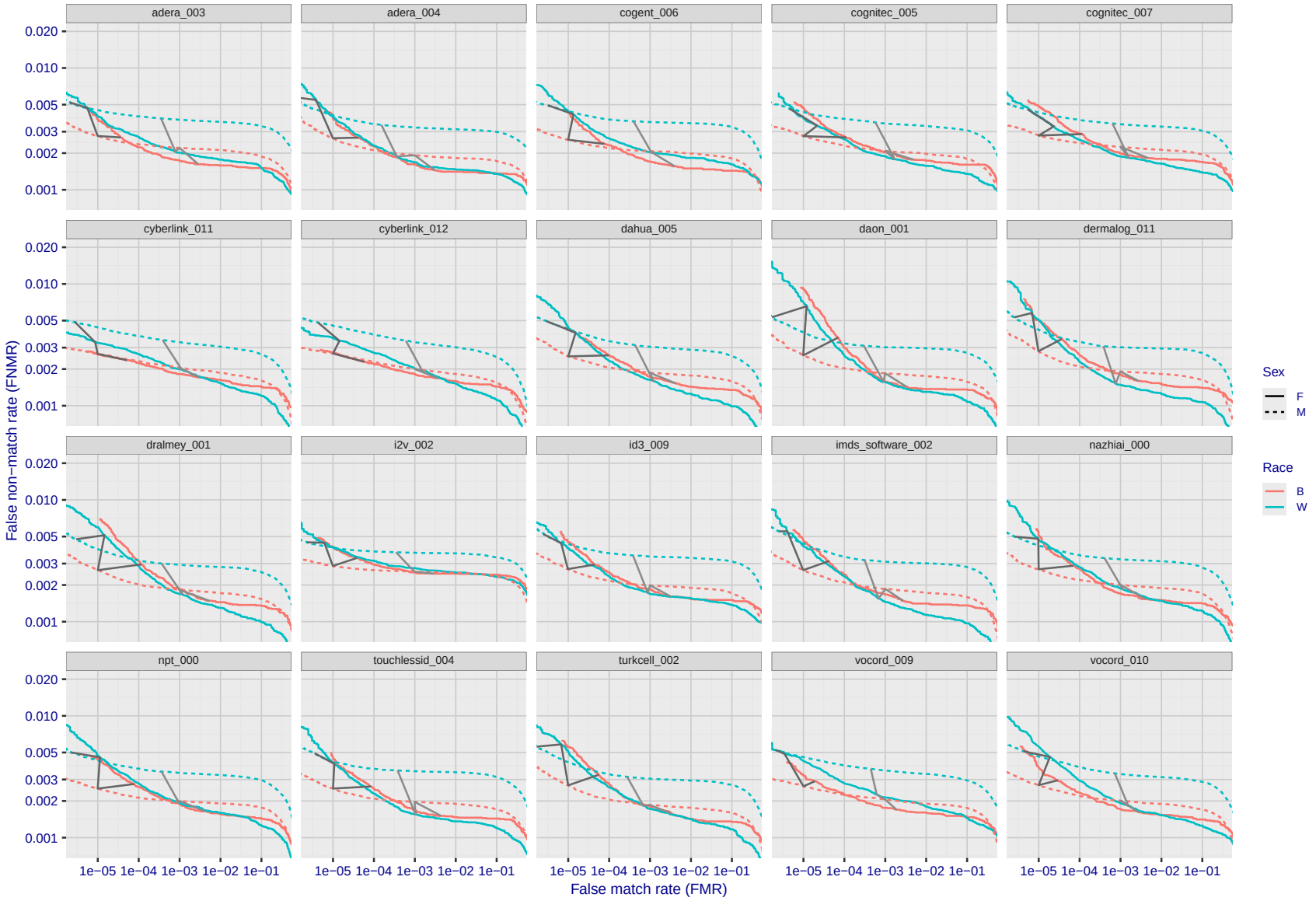


Figure 424: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

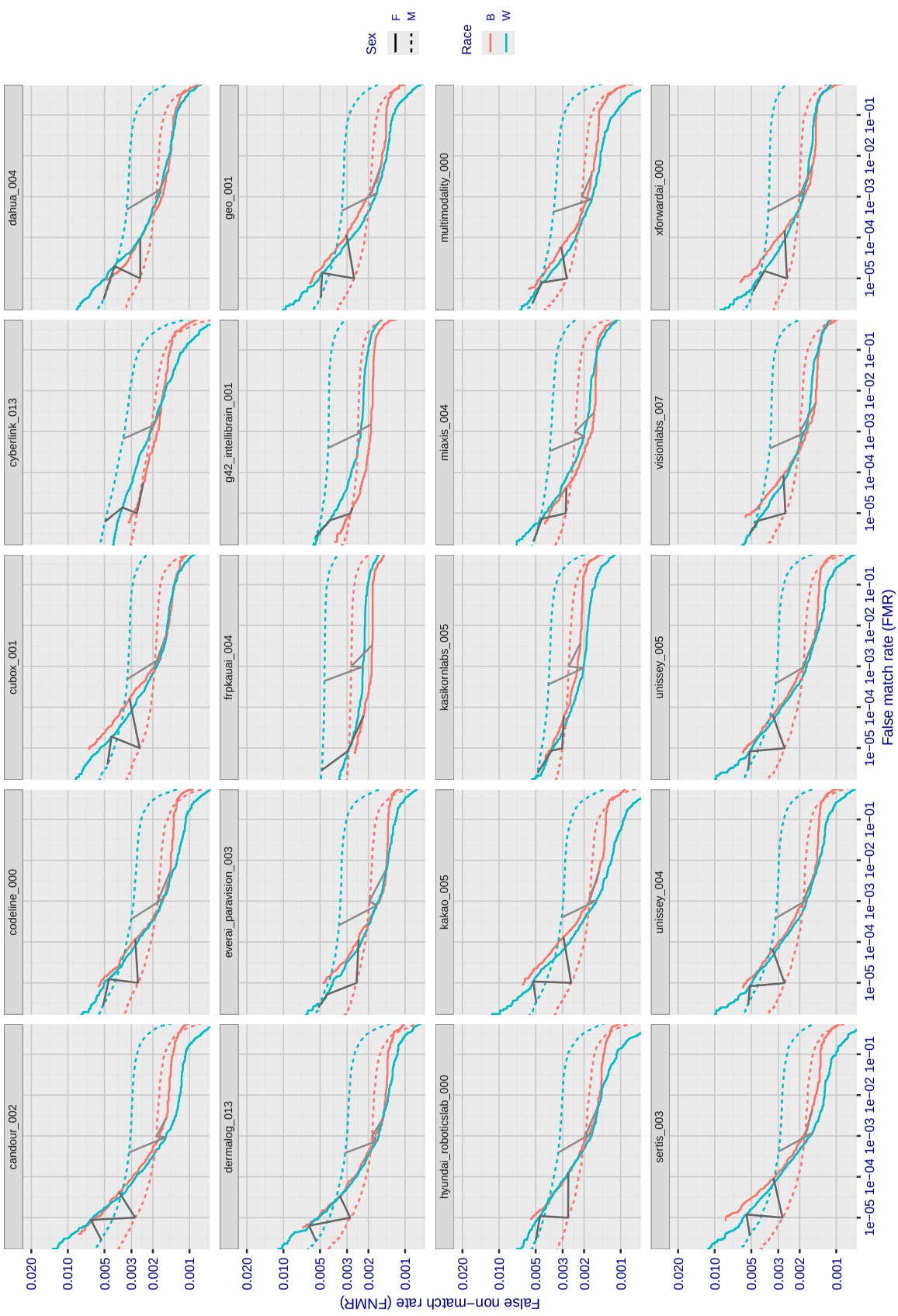


Figure 425: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

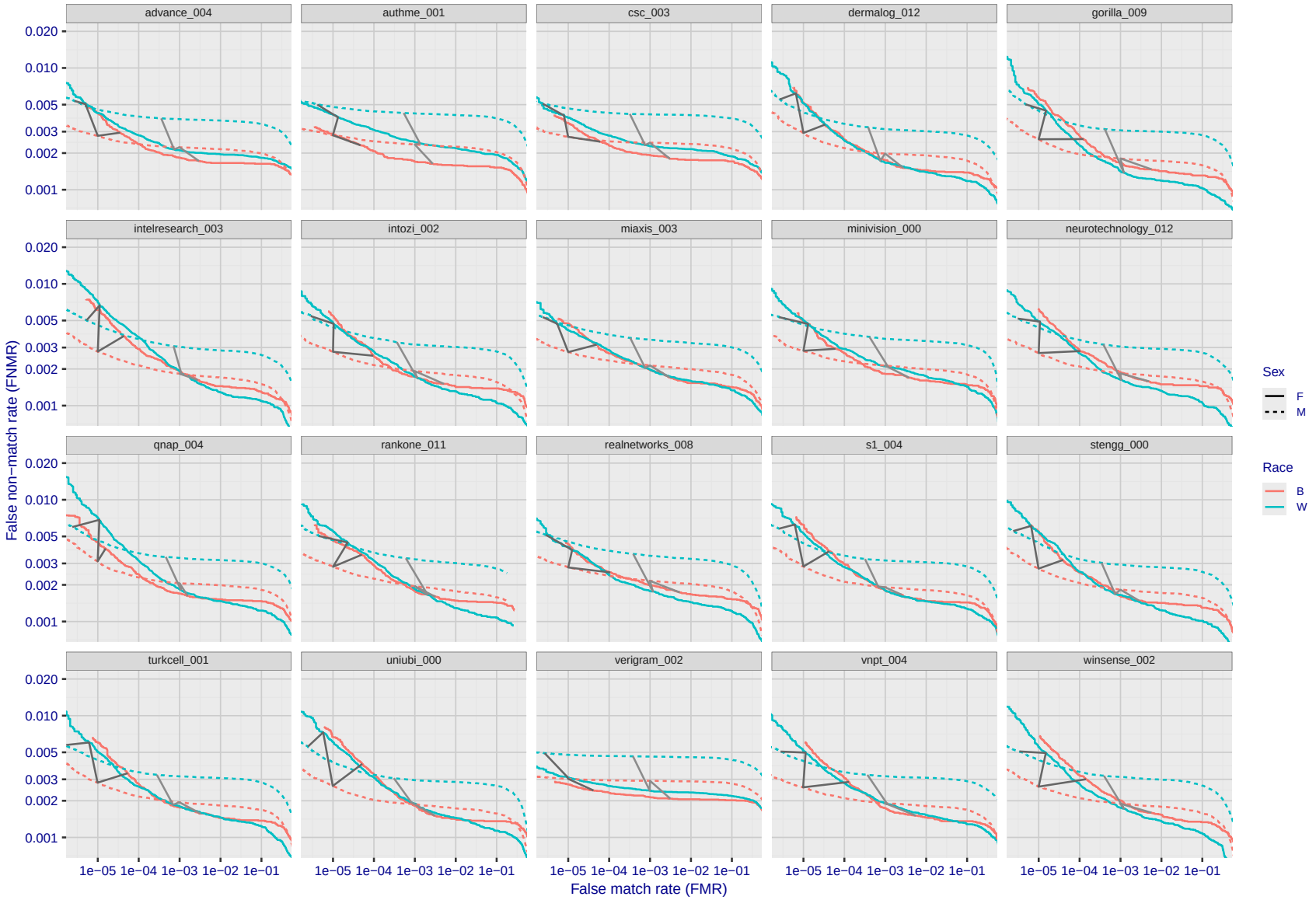


Figure 426: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

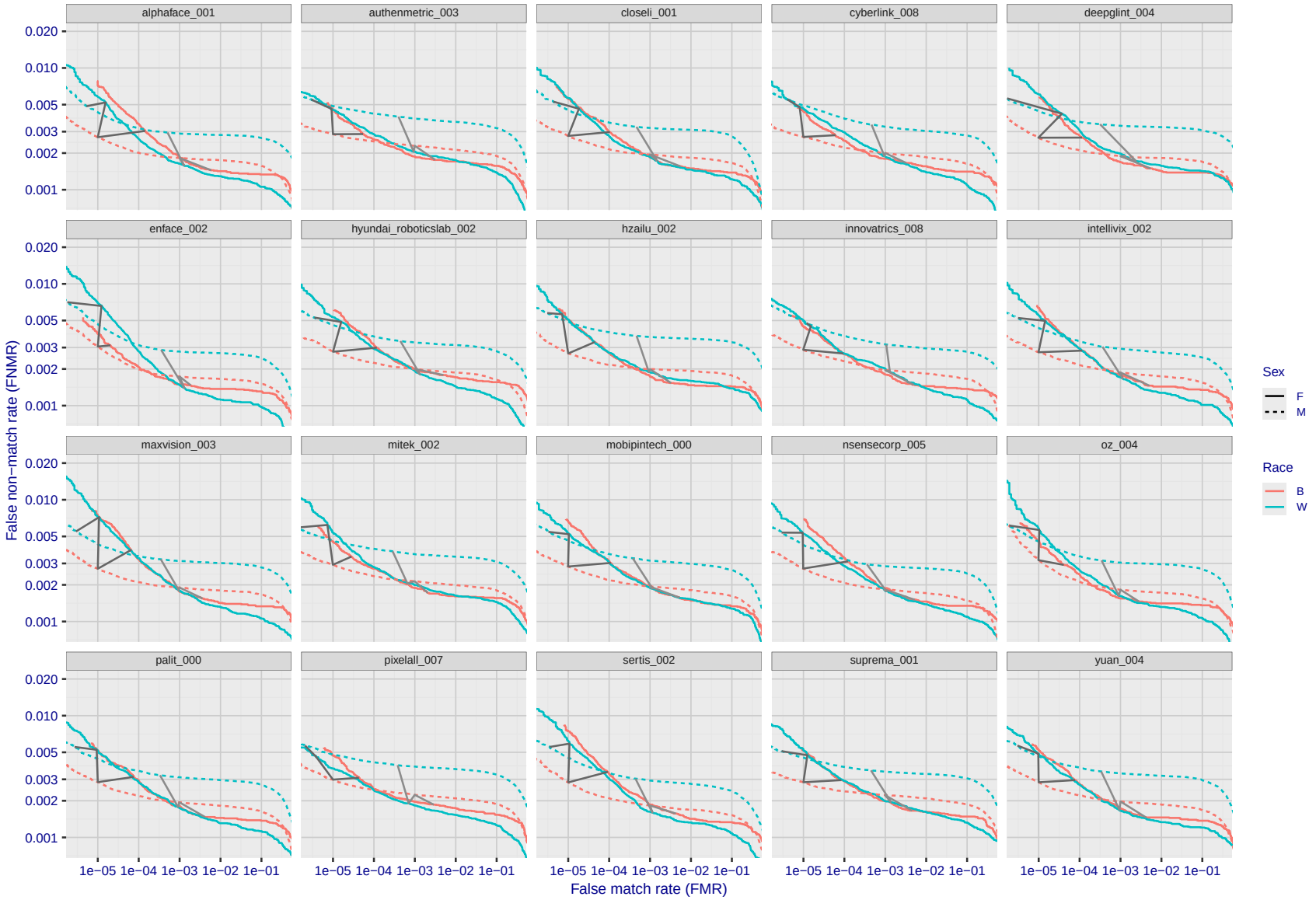


Figure 427: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

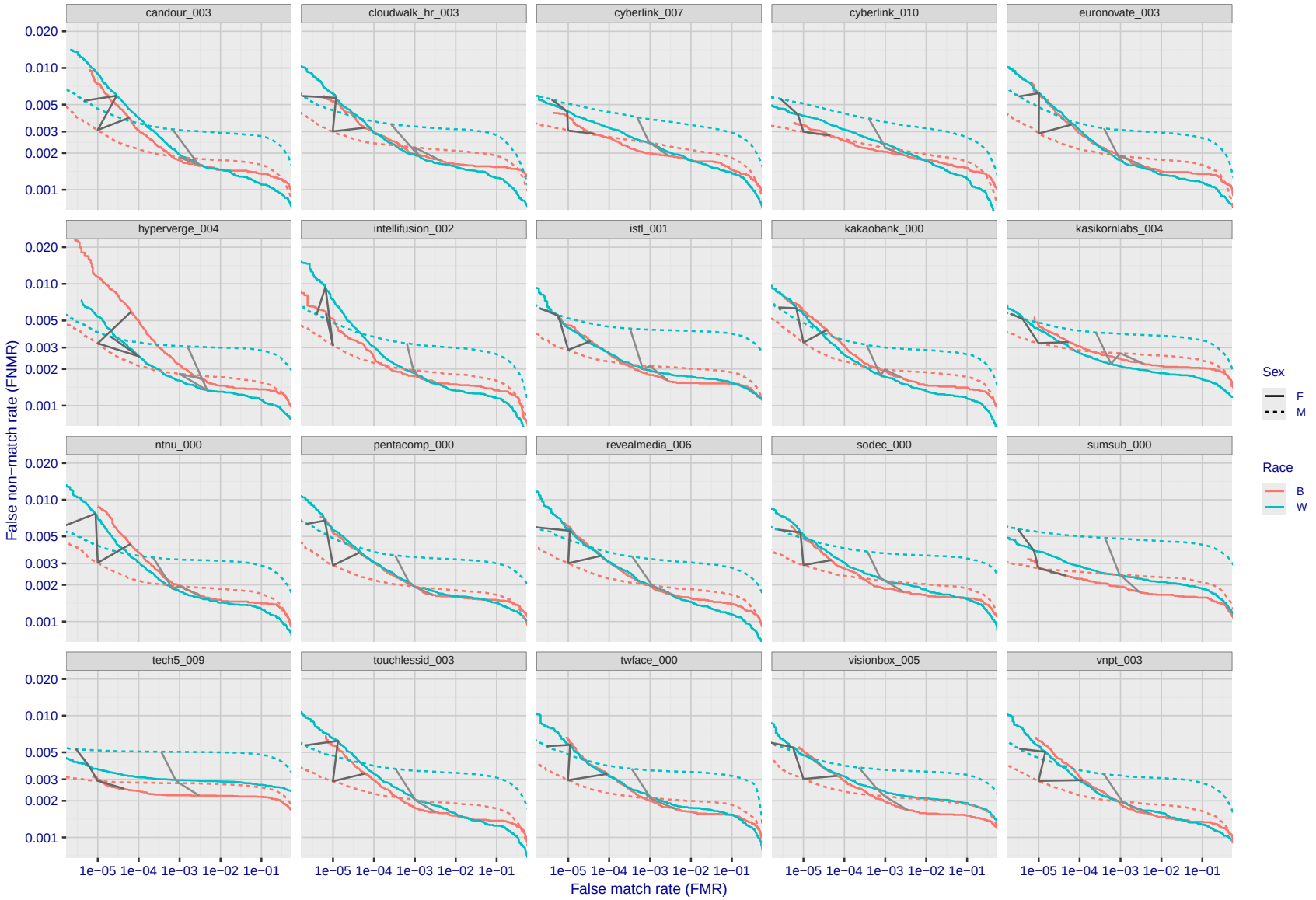


Figure 428: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

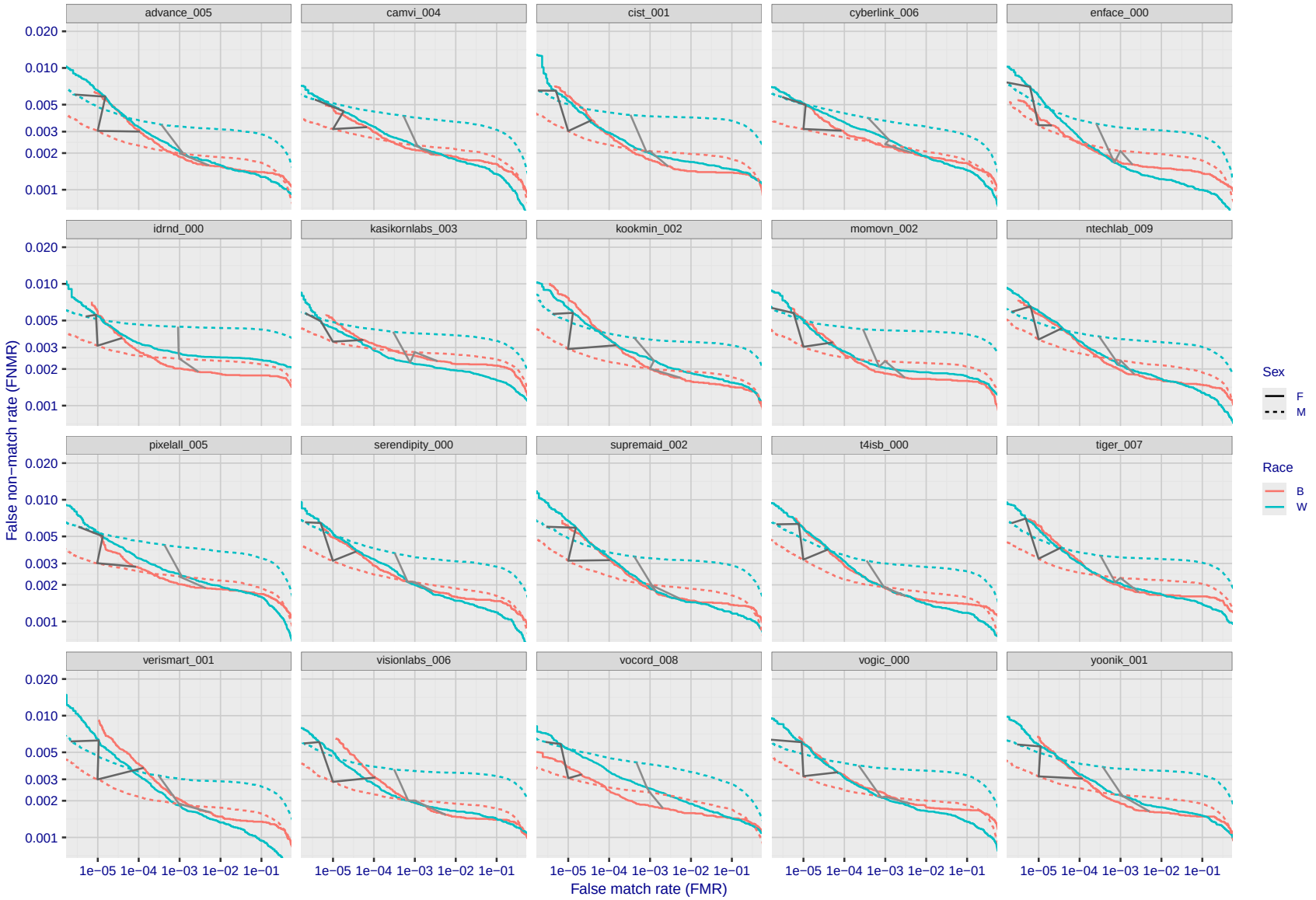


Figure 429: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

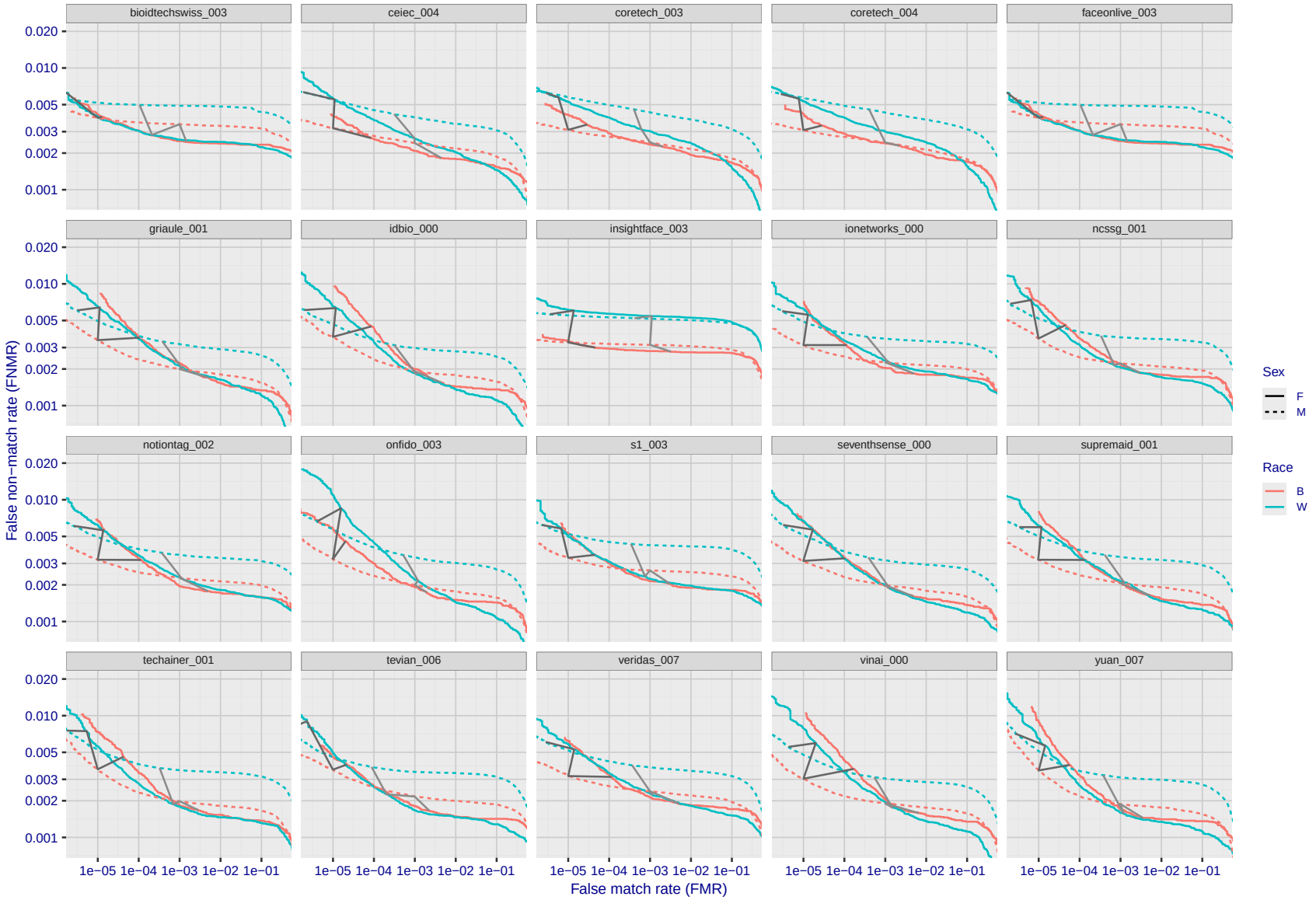


Figure 430: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

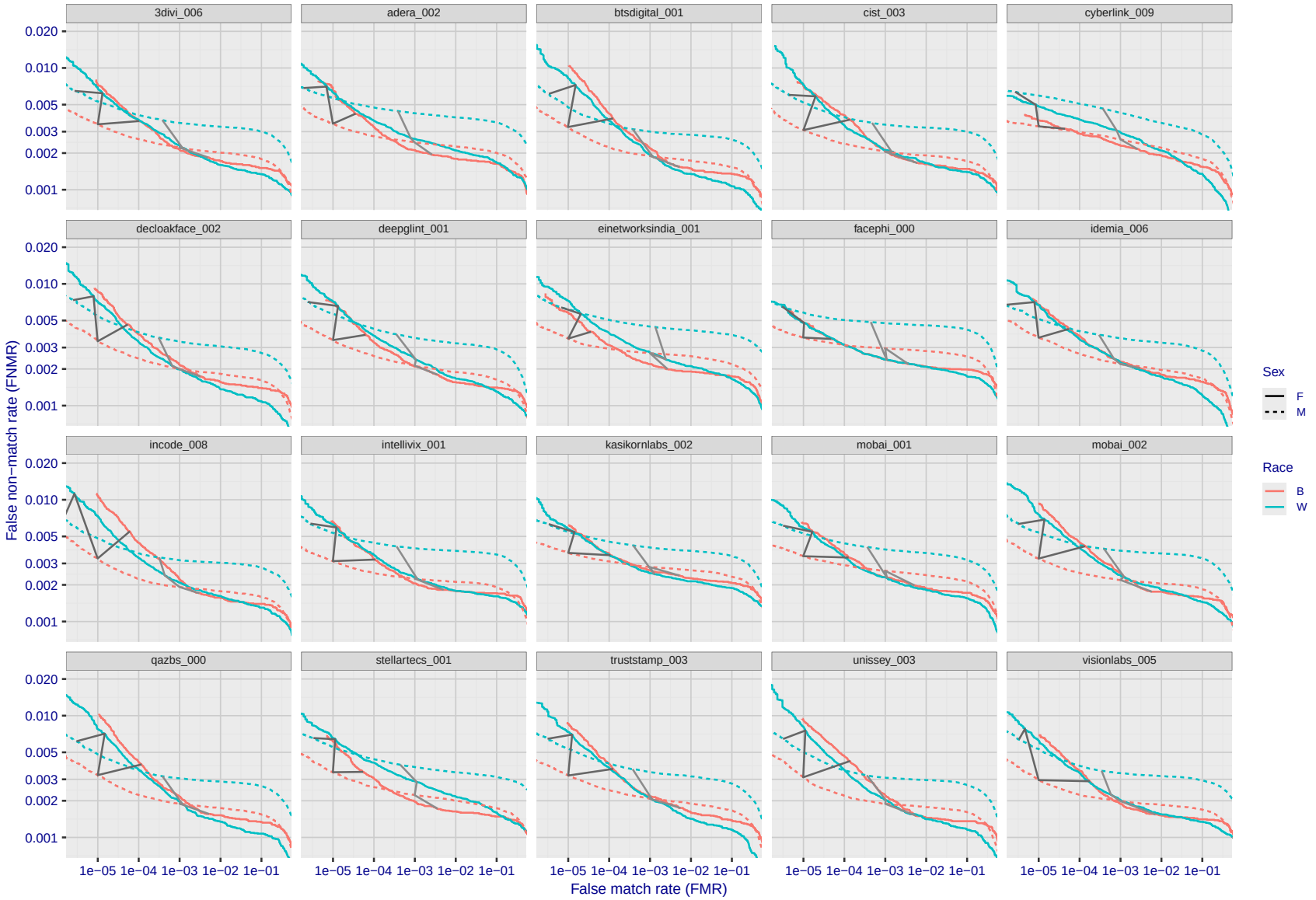


Figure 431: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

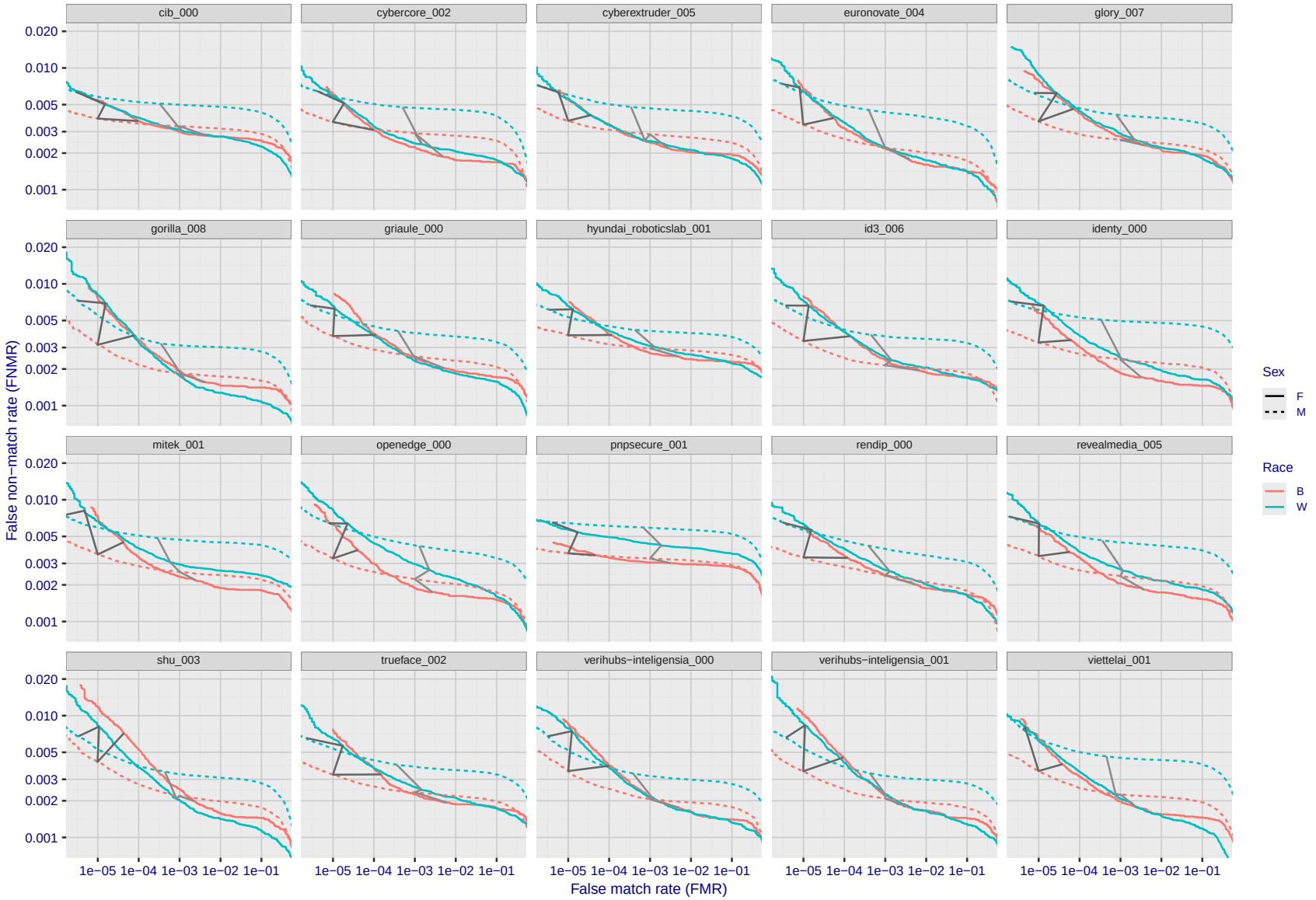


Figure 432: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

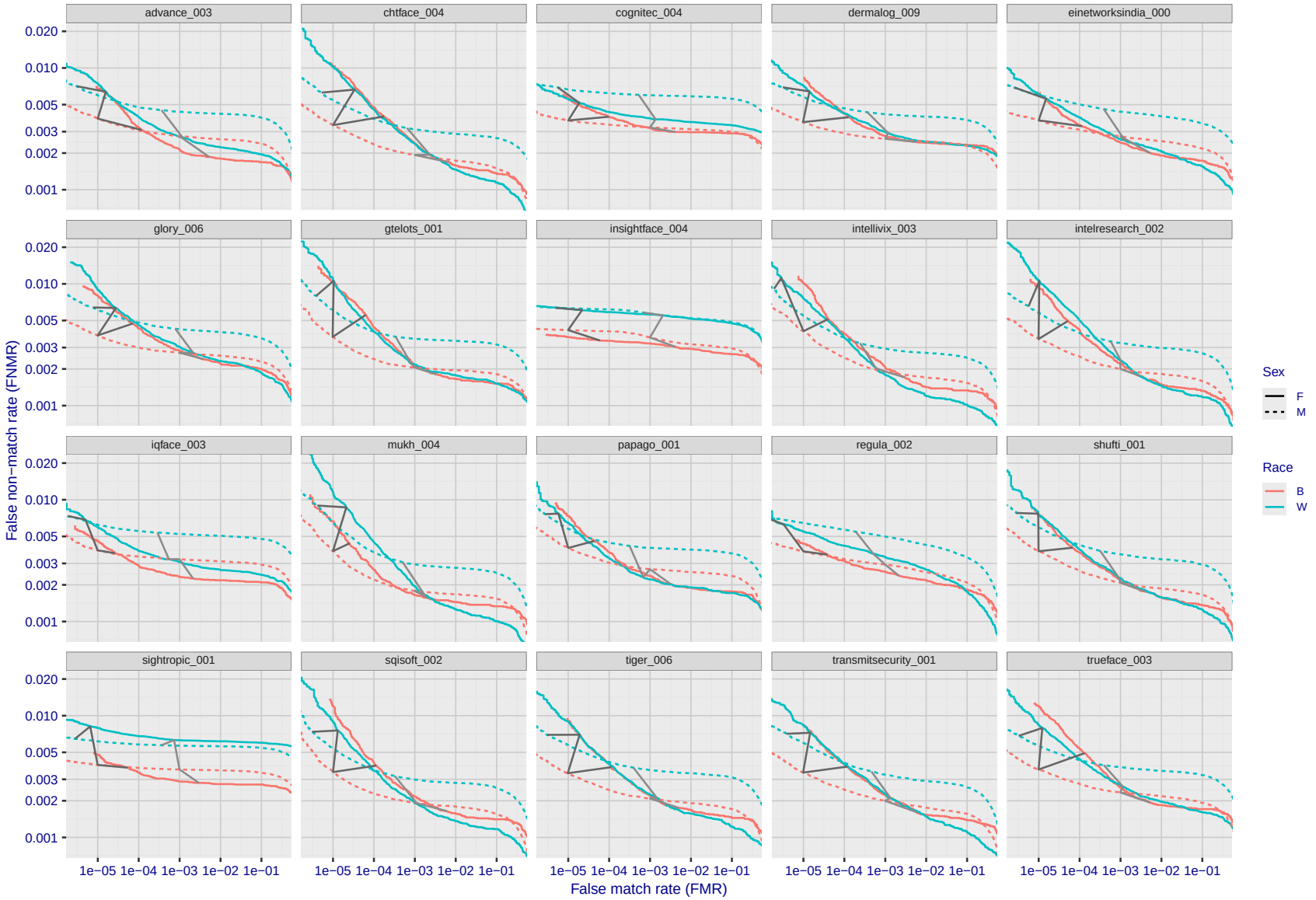


Figure 433: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

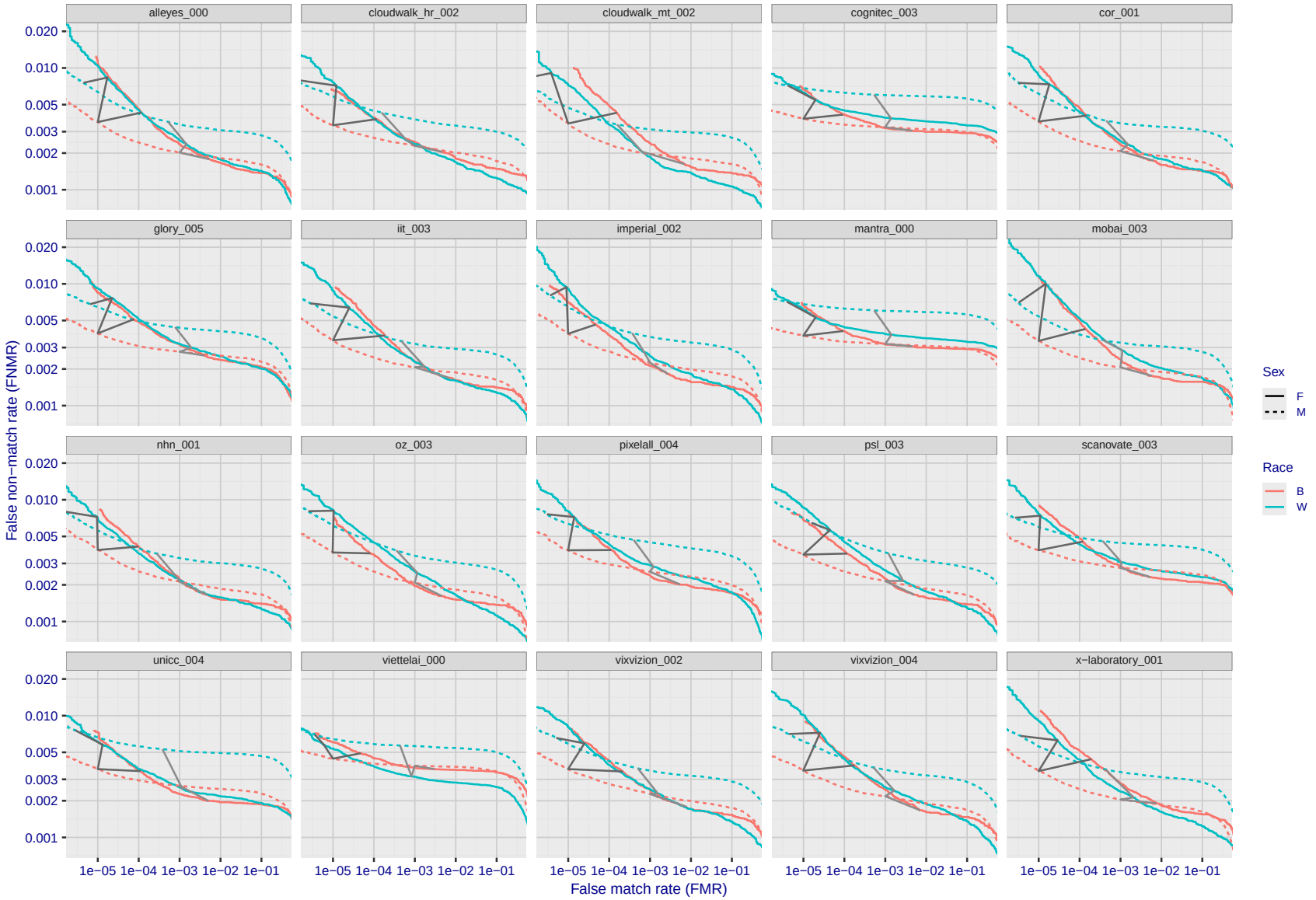


Figure 434: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

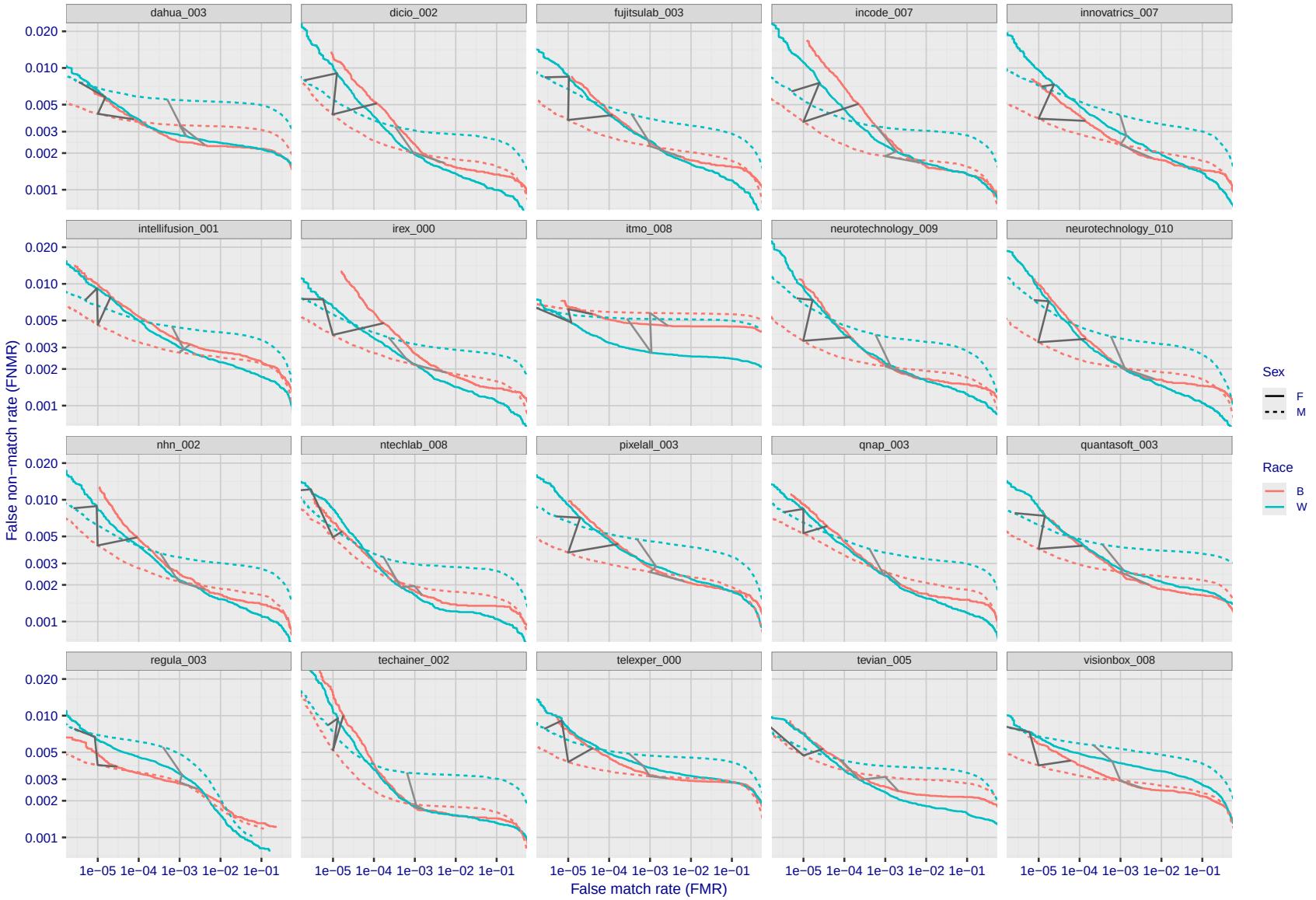


Figure 435: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

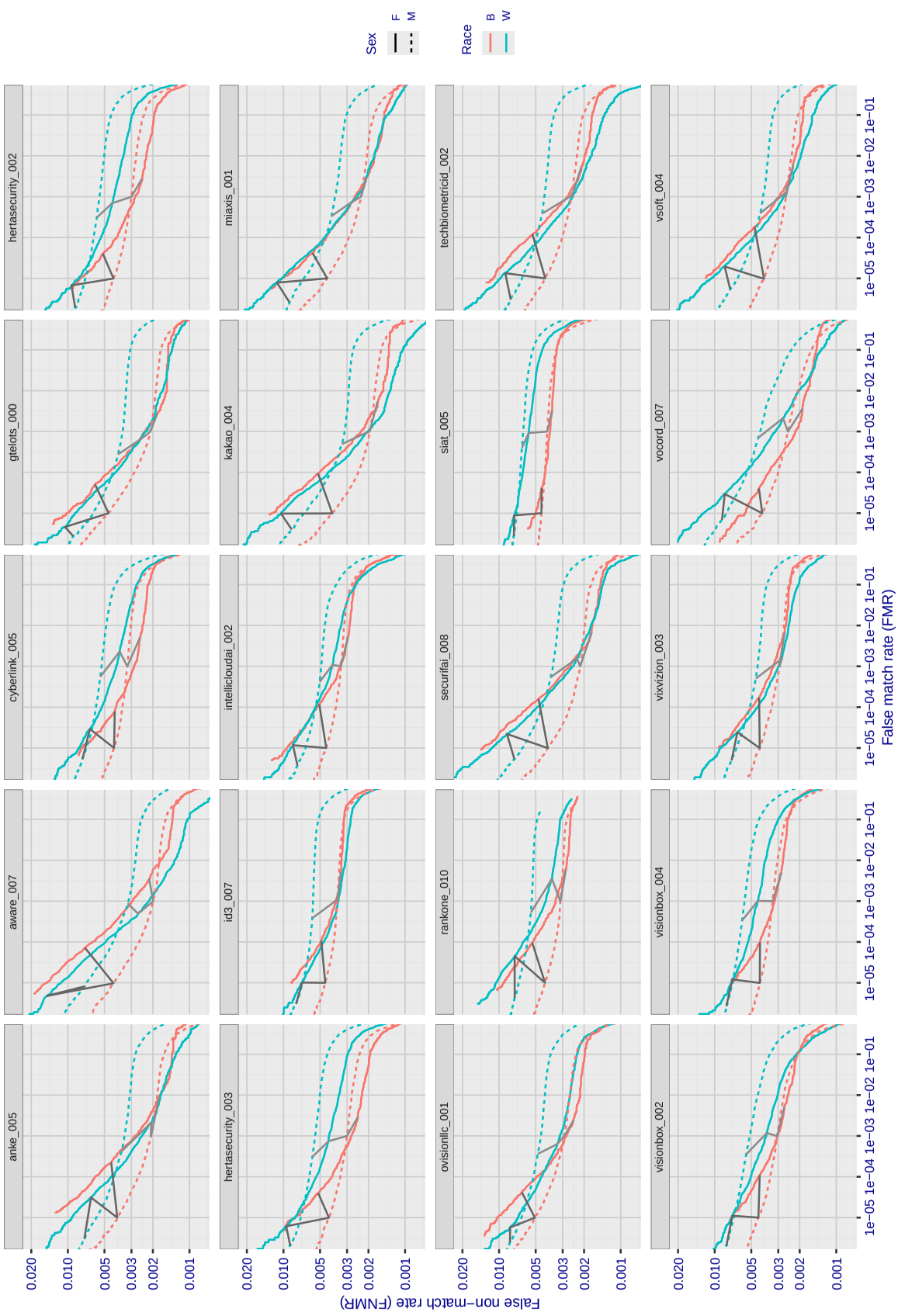
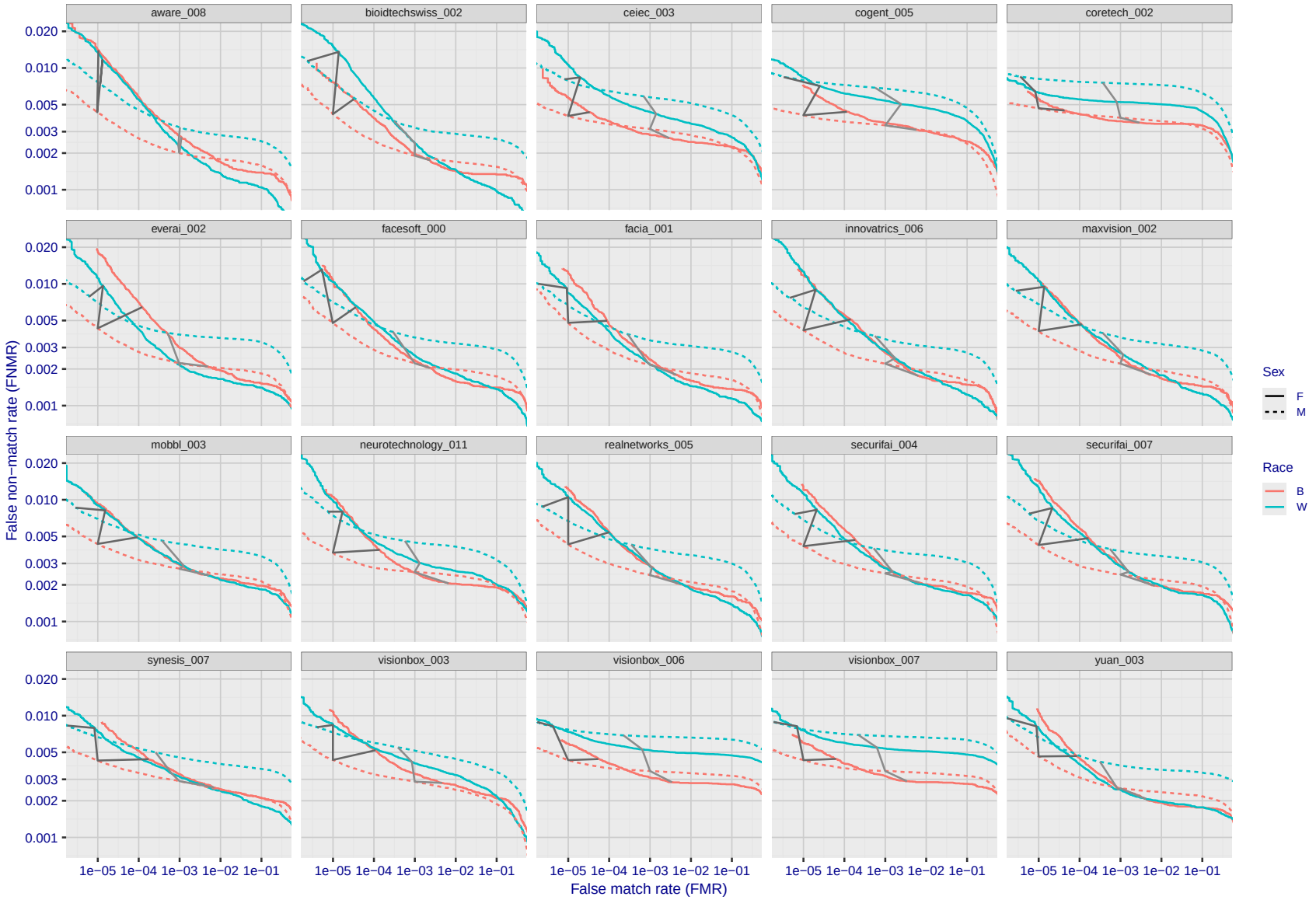


Figure 436: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women at a fixed threshold* but, *at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"



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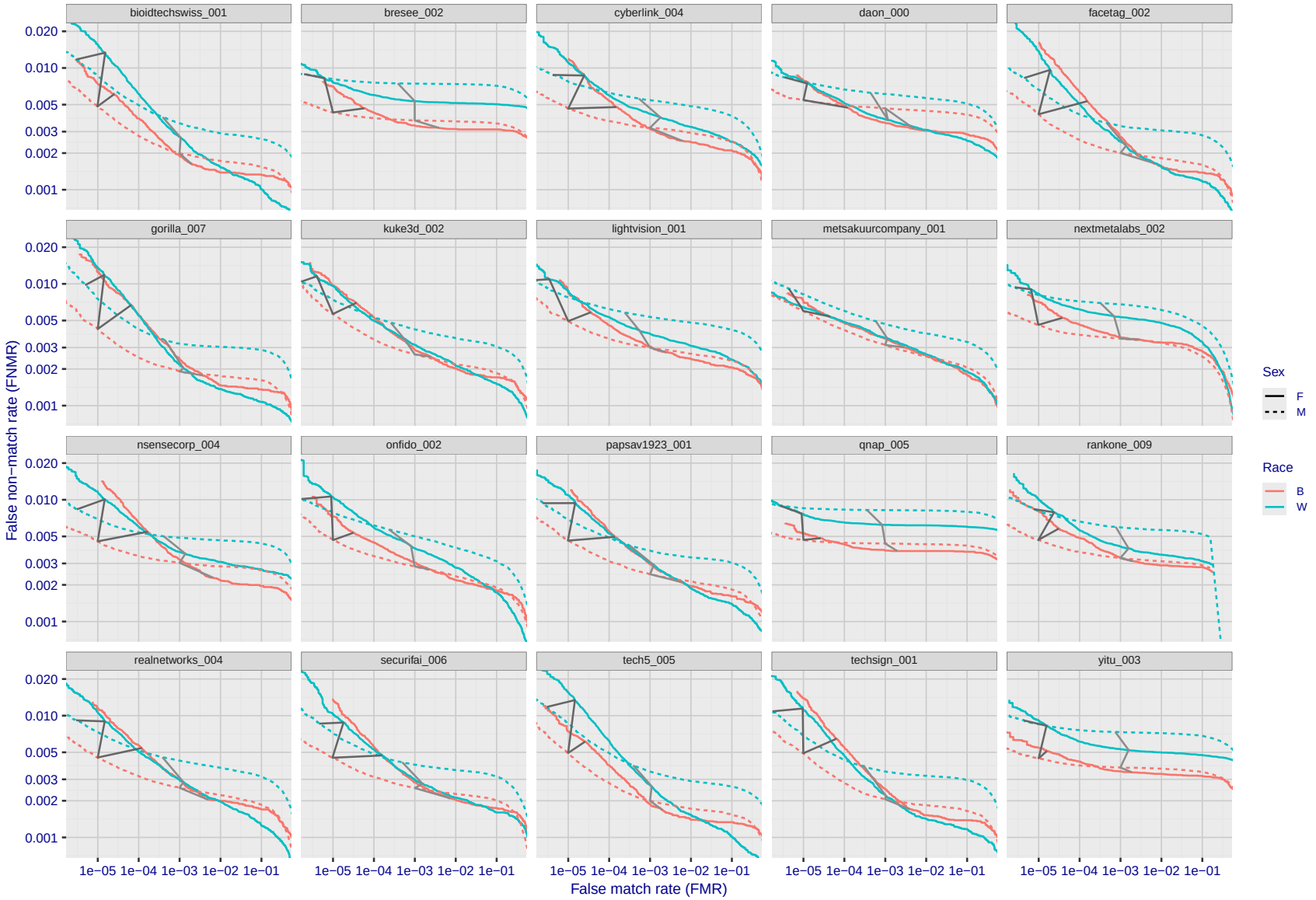

Figure 438: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

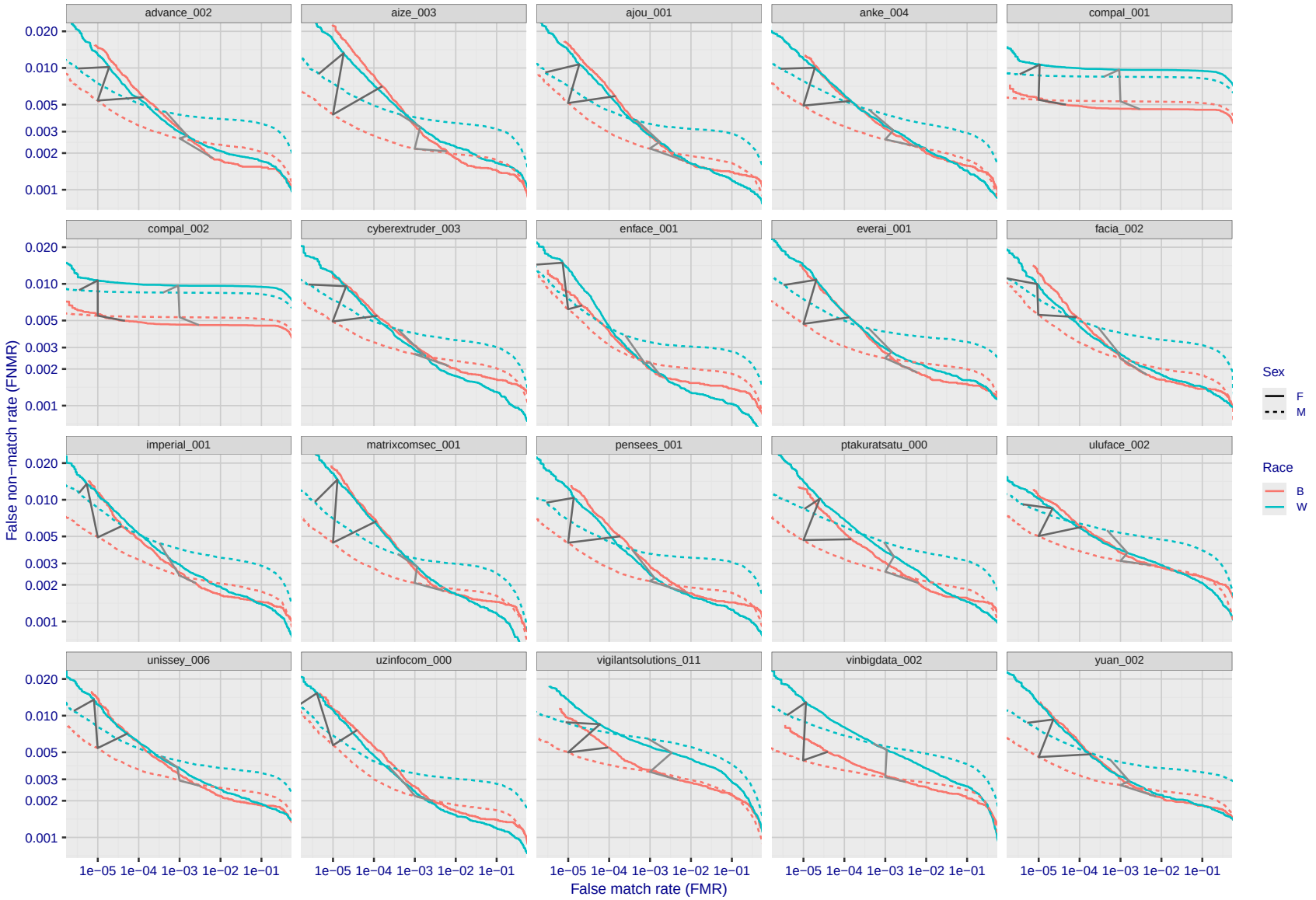

Figure 439: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

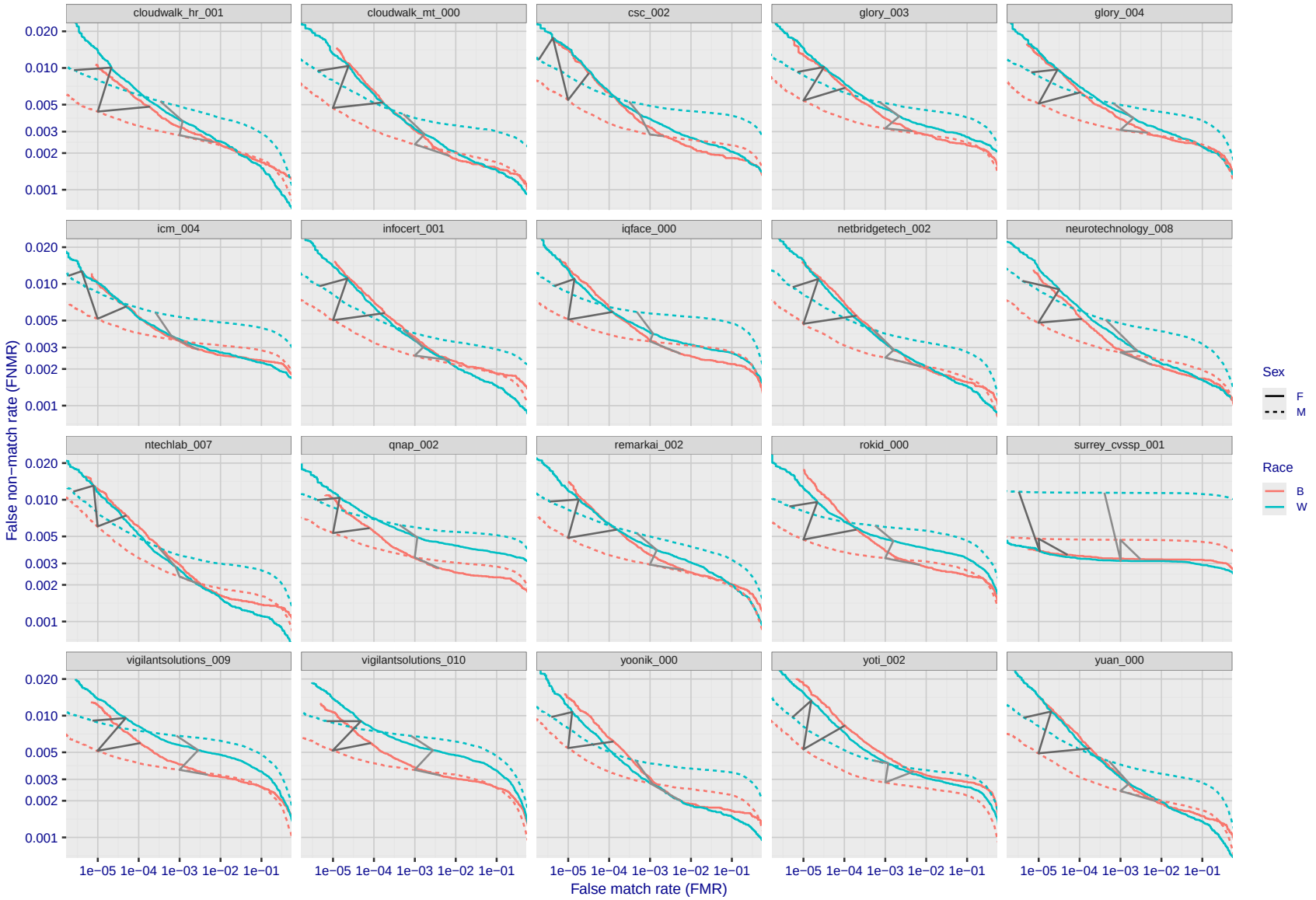

Figure 440: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

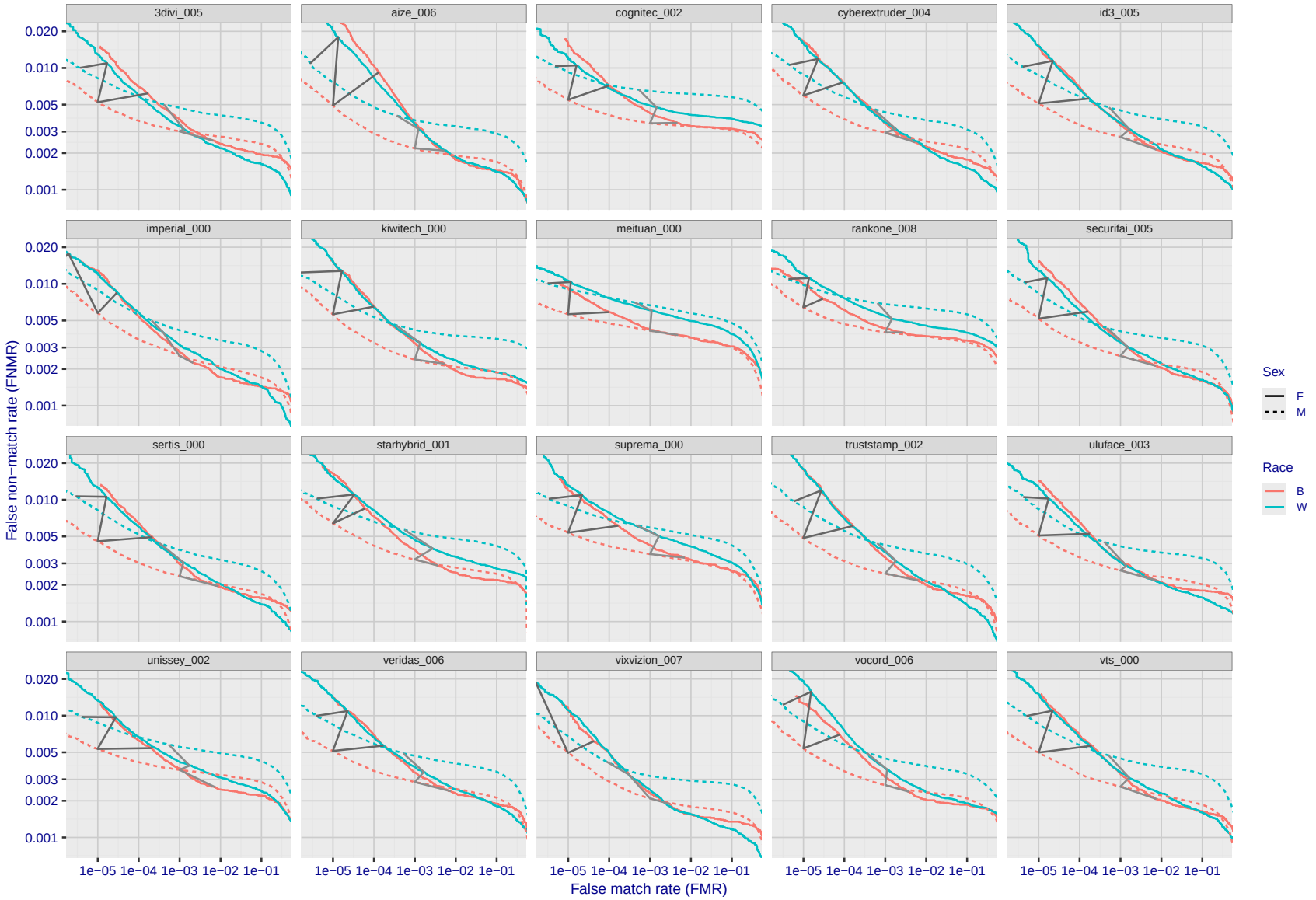

Figure 441: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

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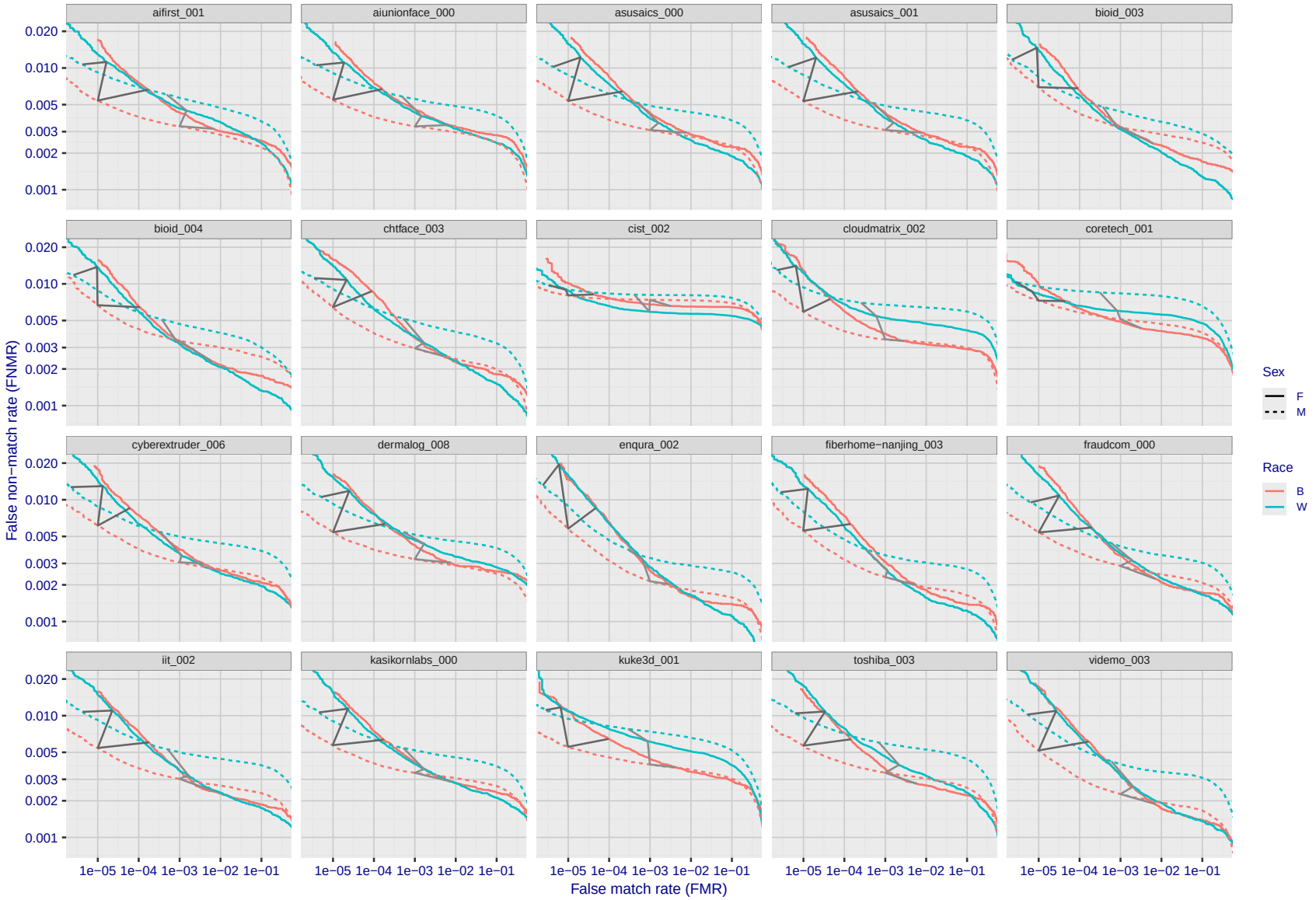

Figure 442: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

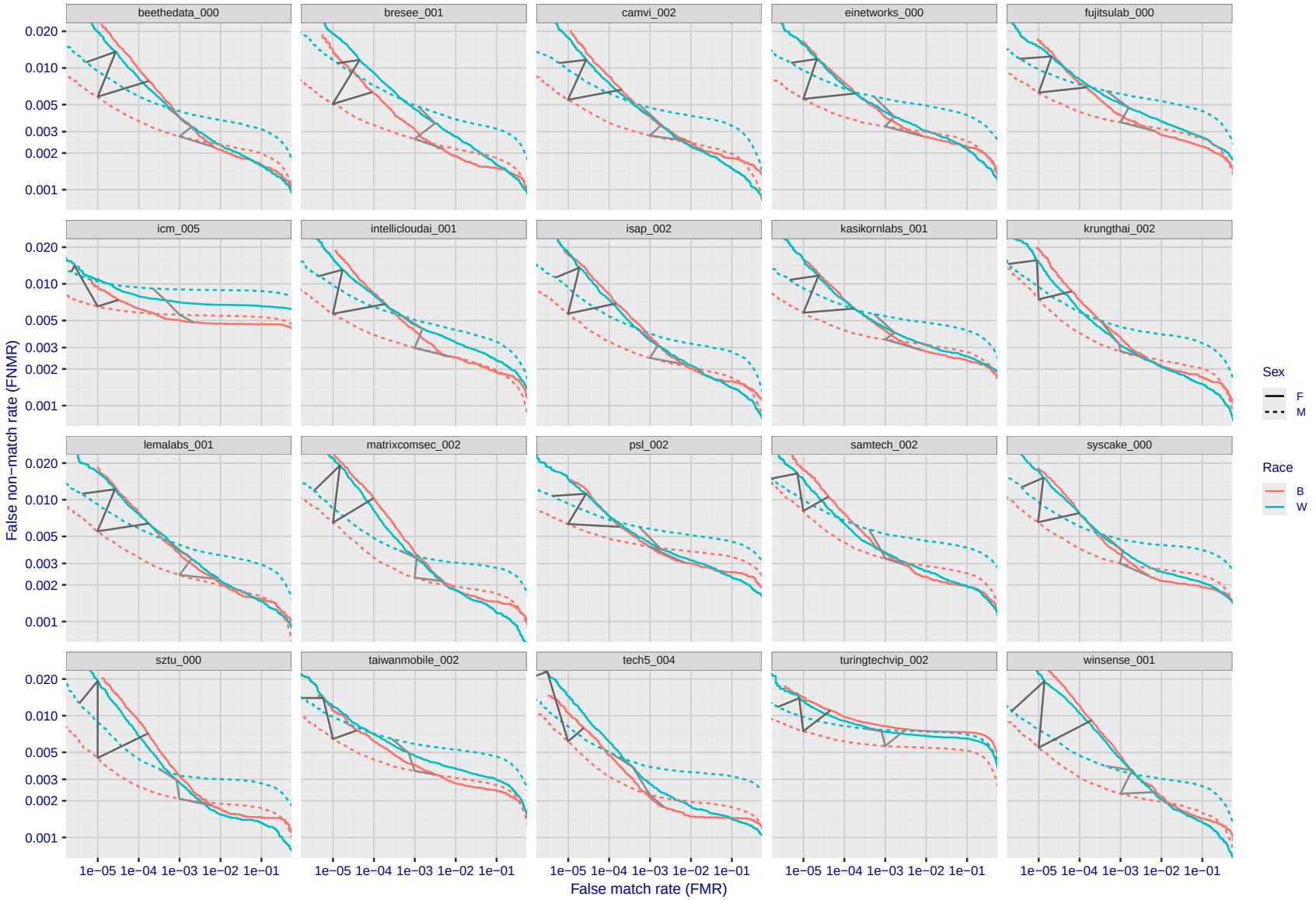

Figure 443: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

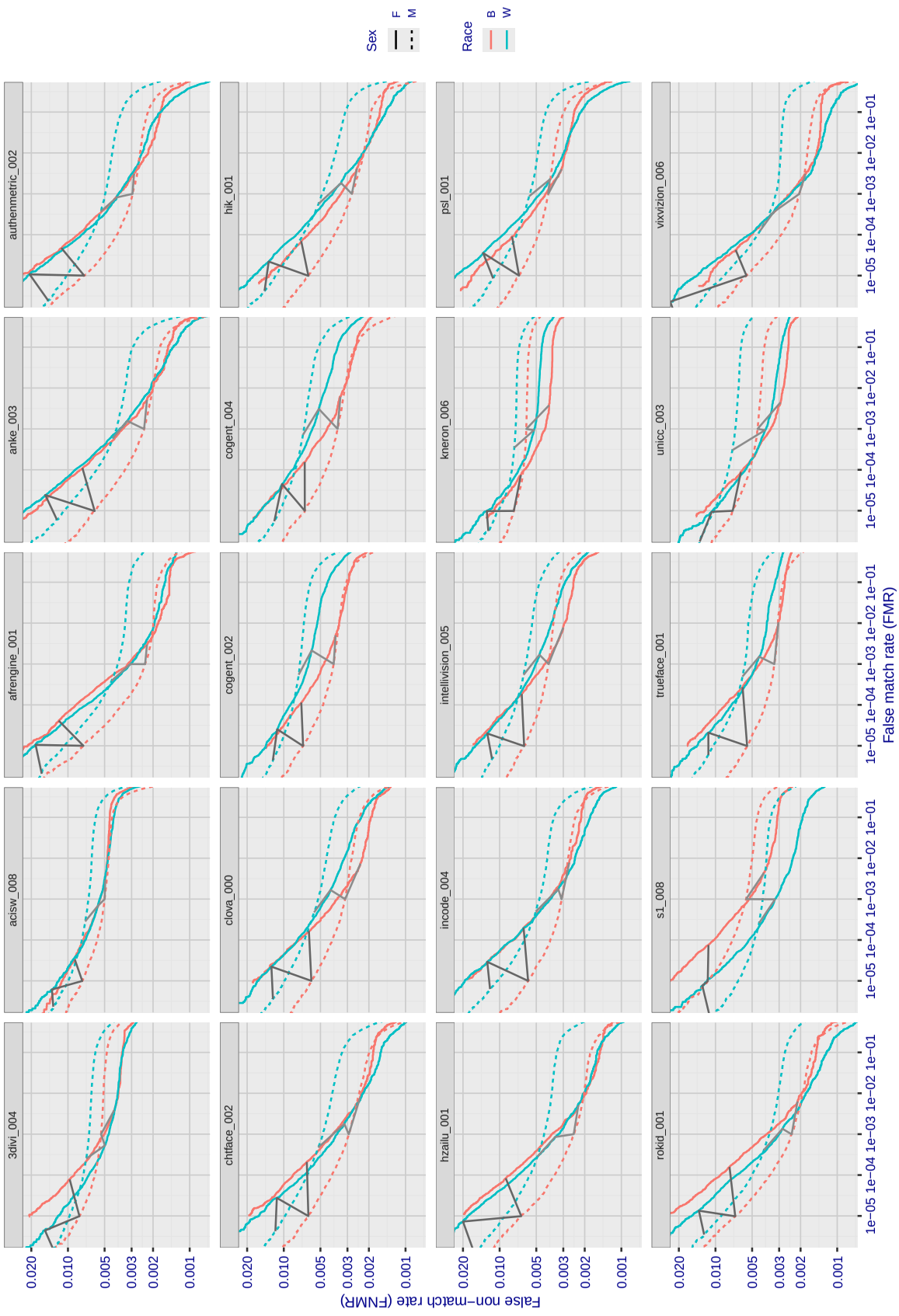

Figure 444: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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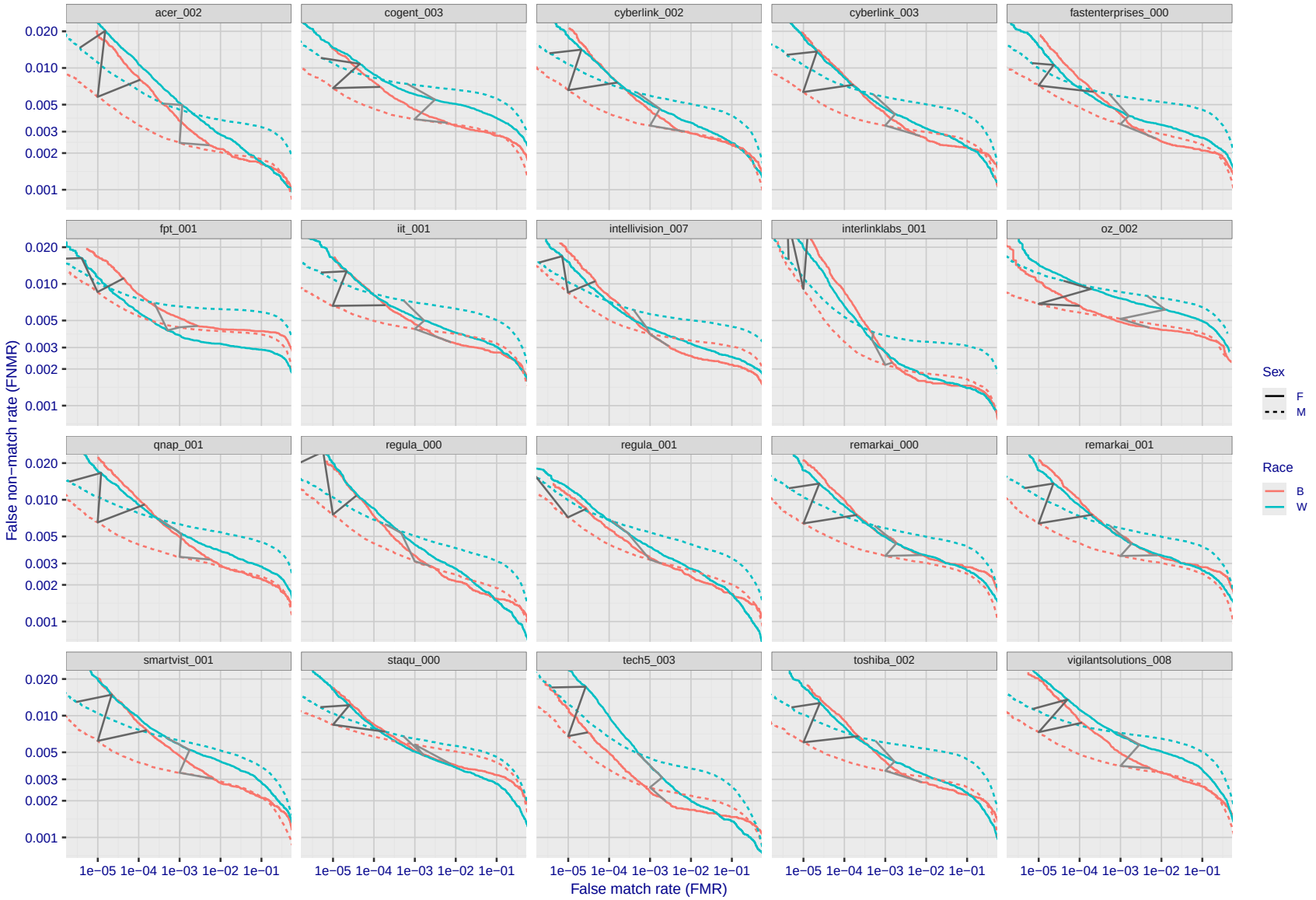

Figure 445: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

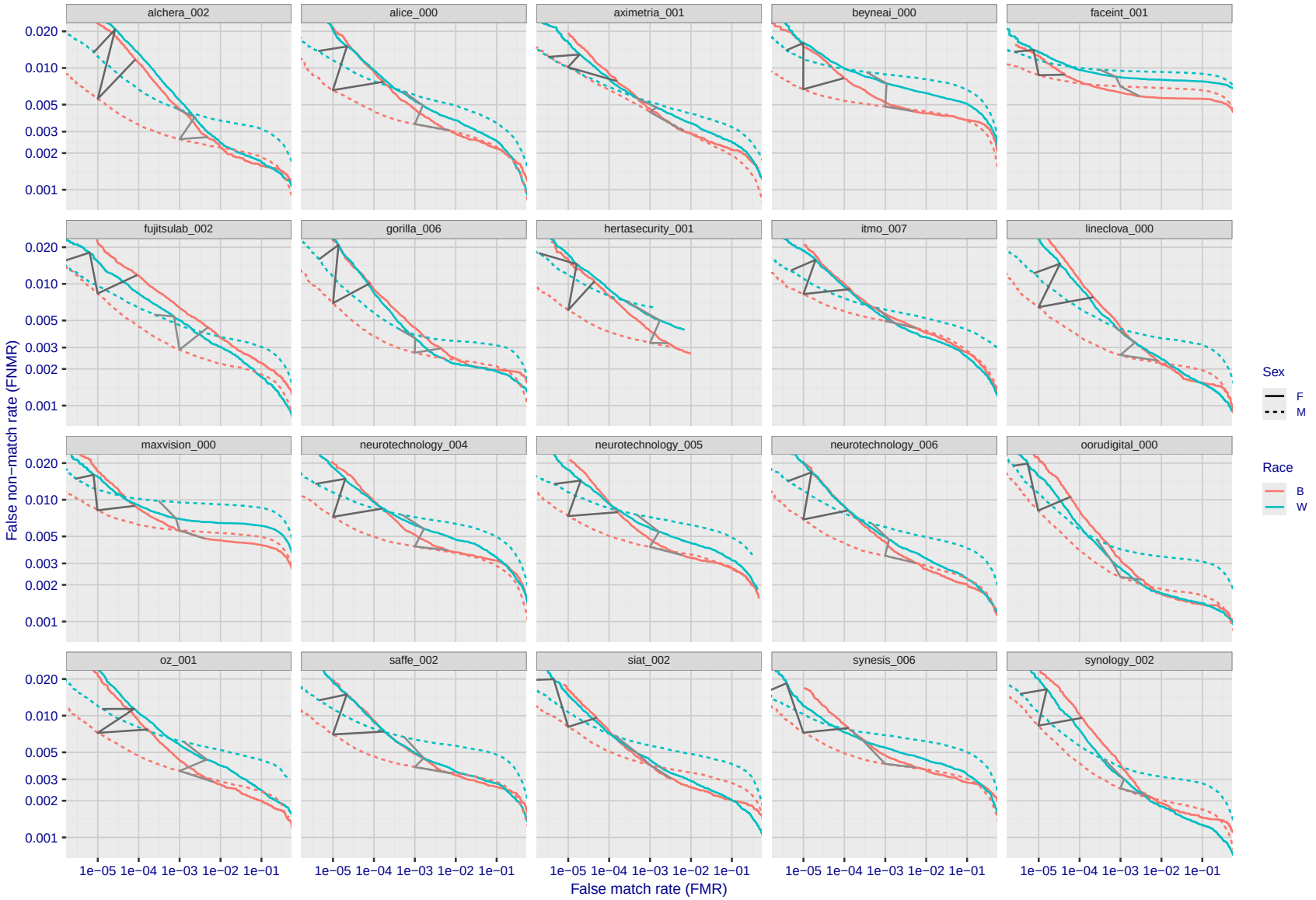

Figure 446: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

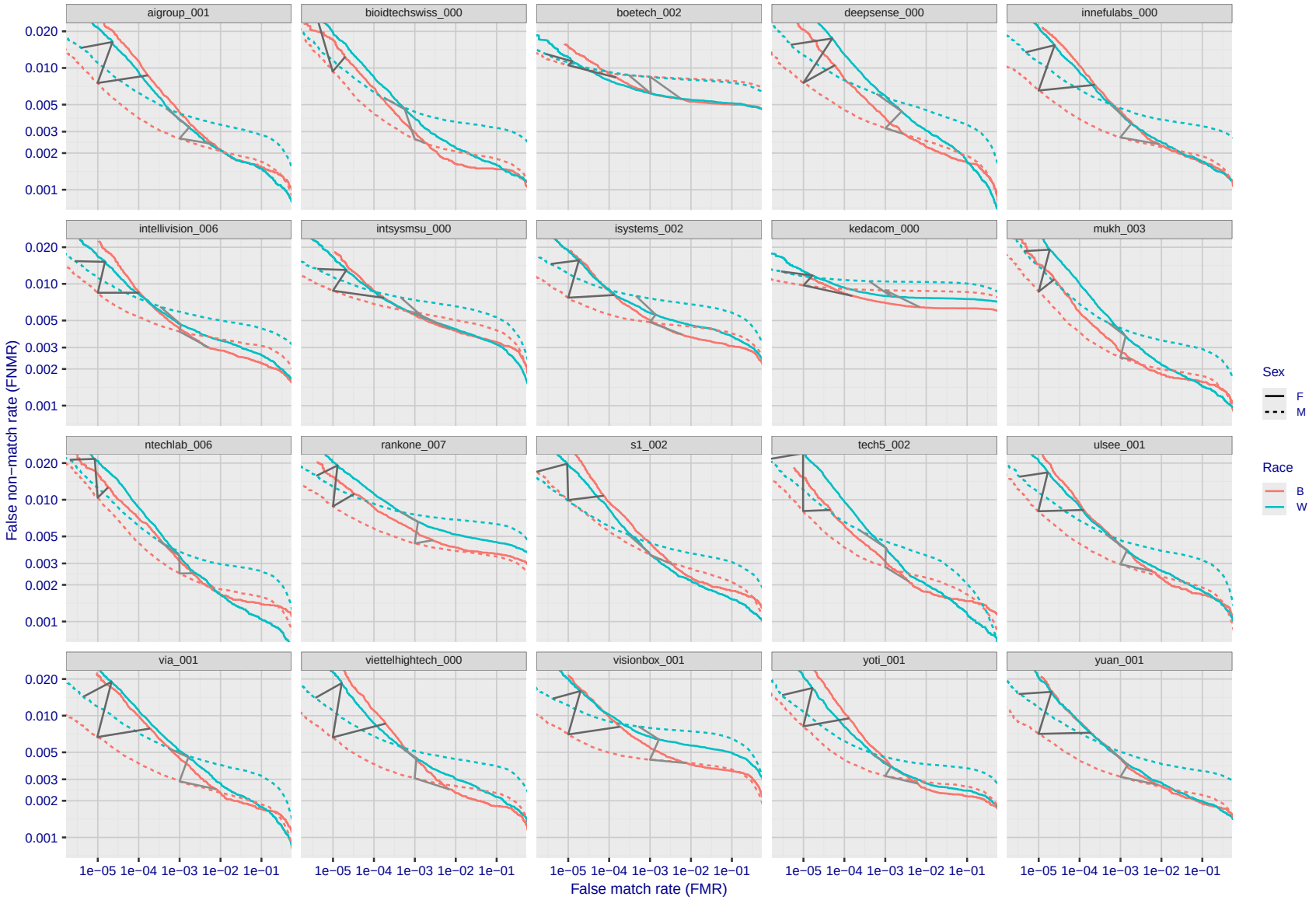

Figure 447: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

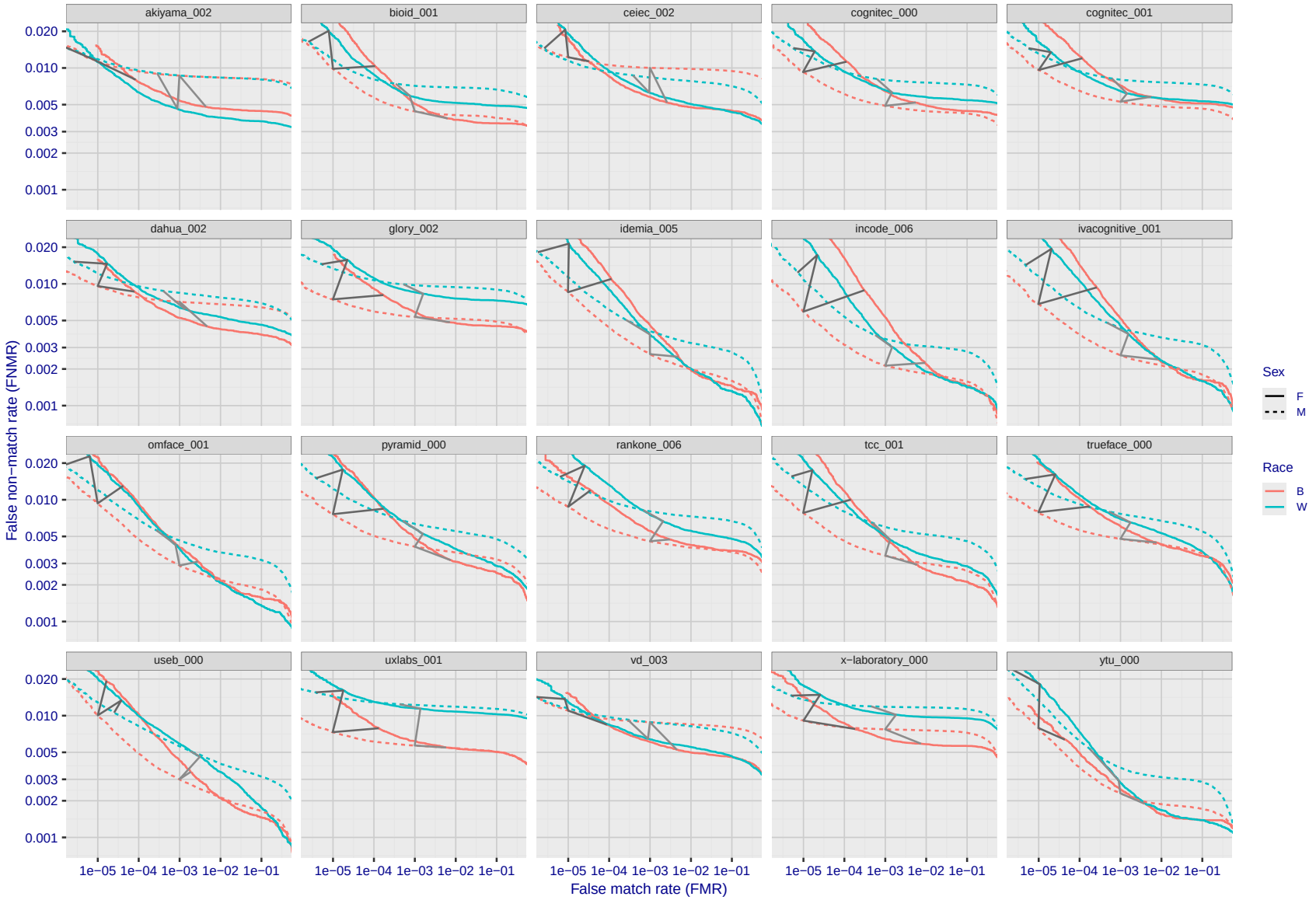

Figure 448: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

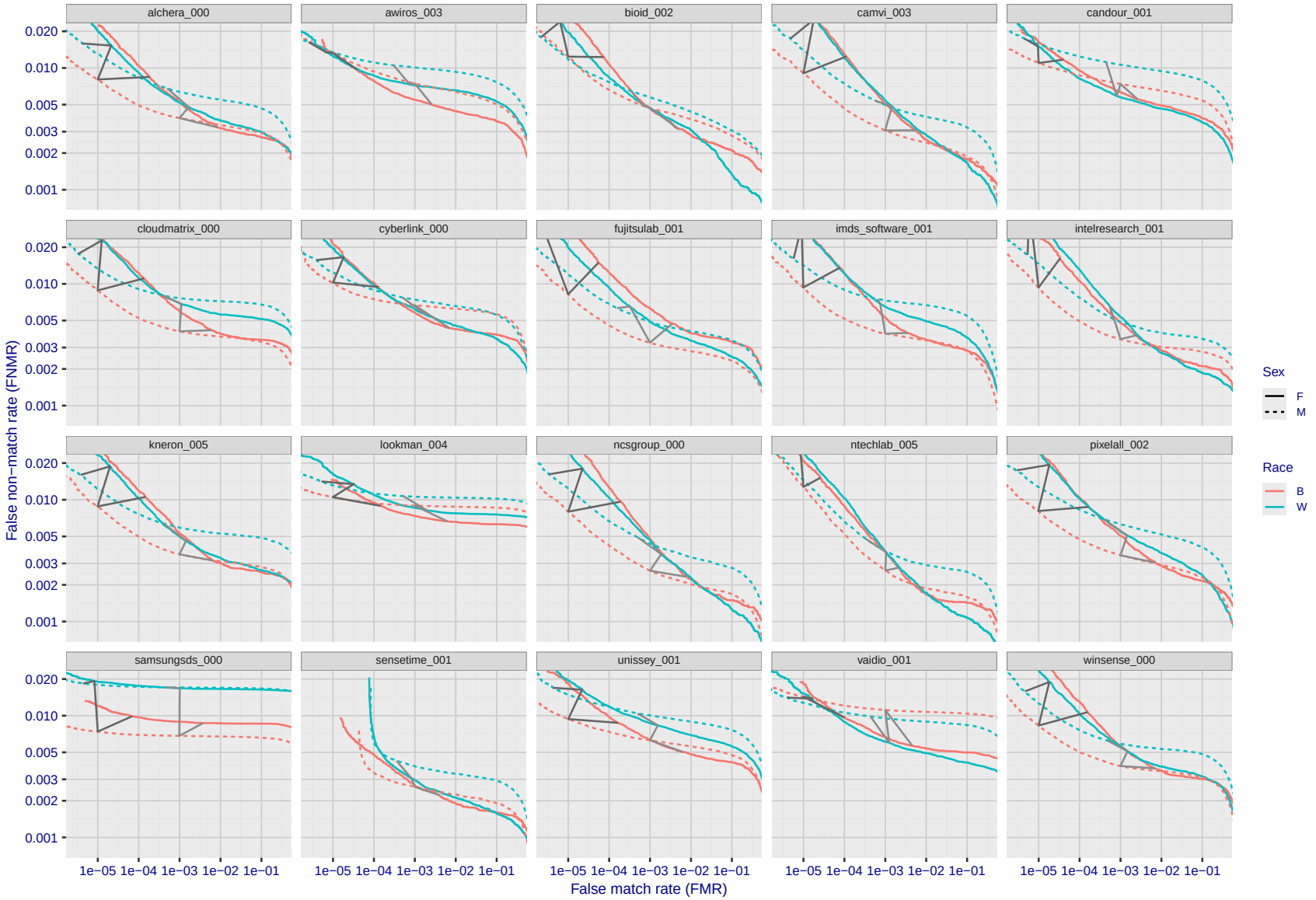

Figure 449: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

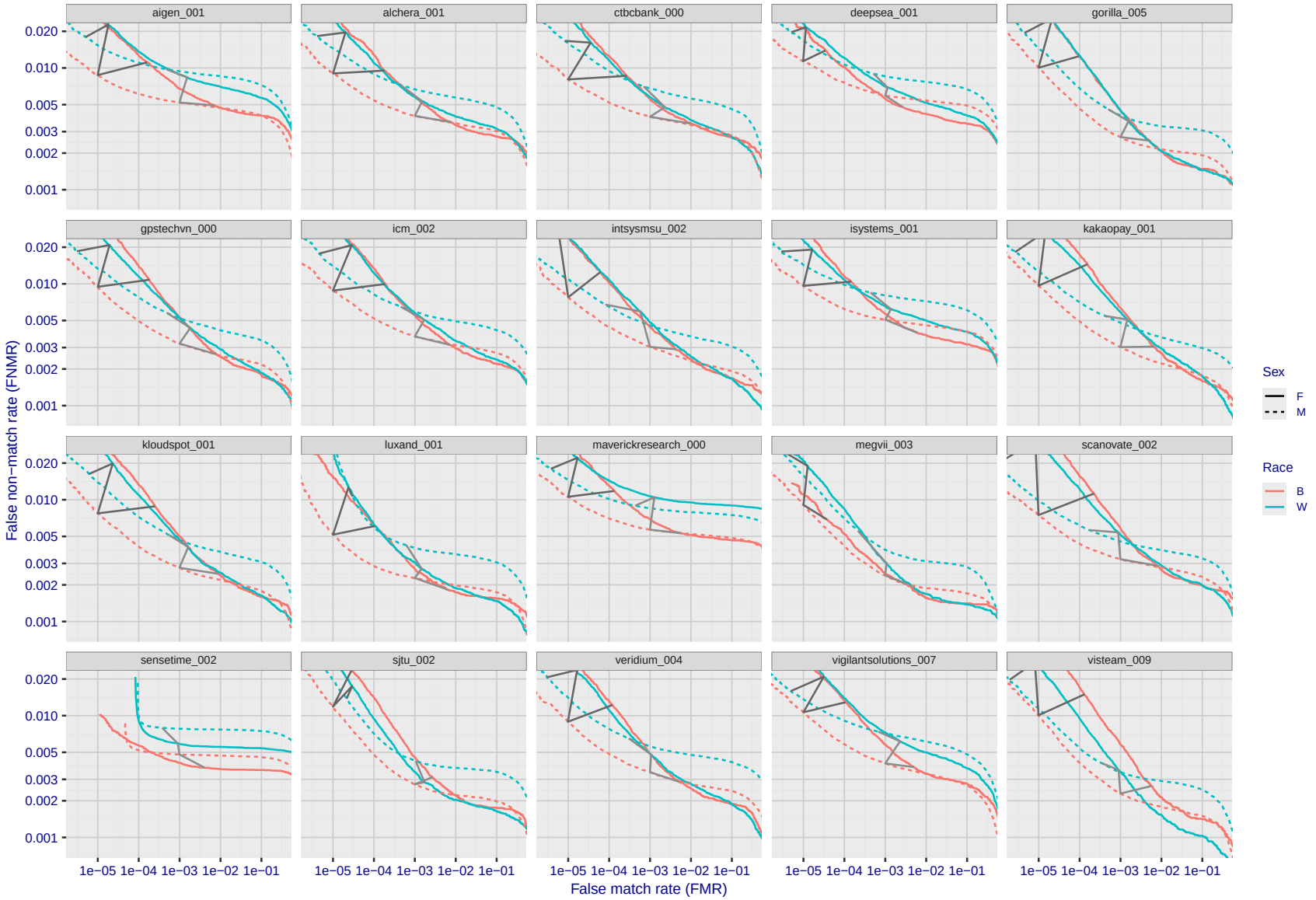

Figure 450: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

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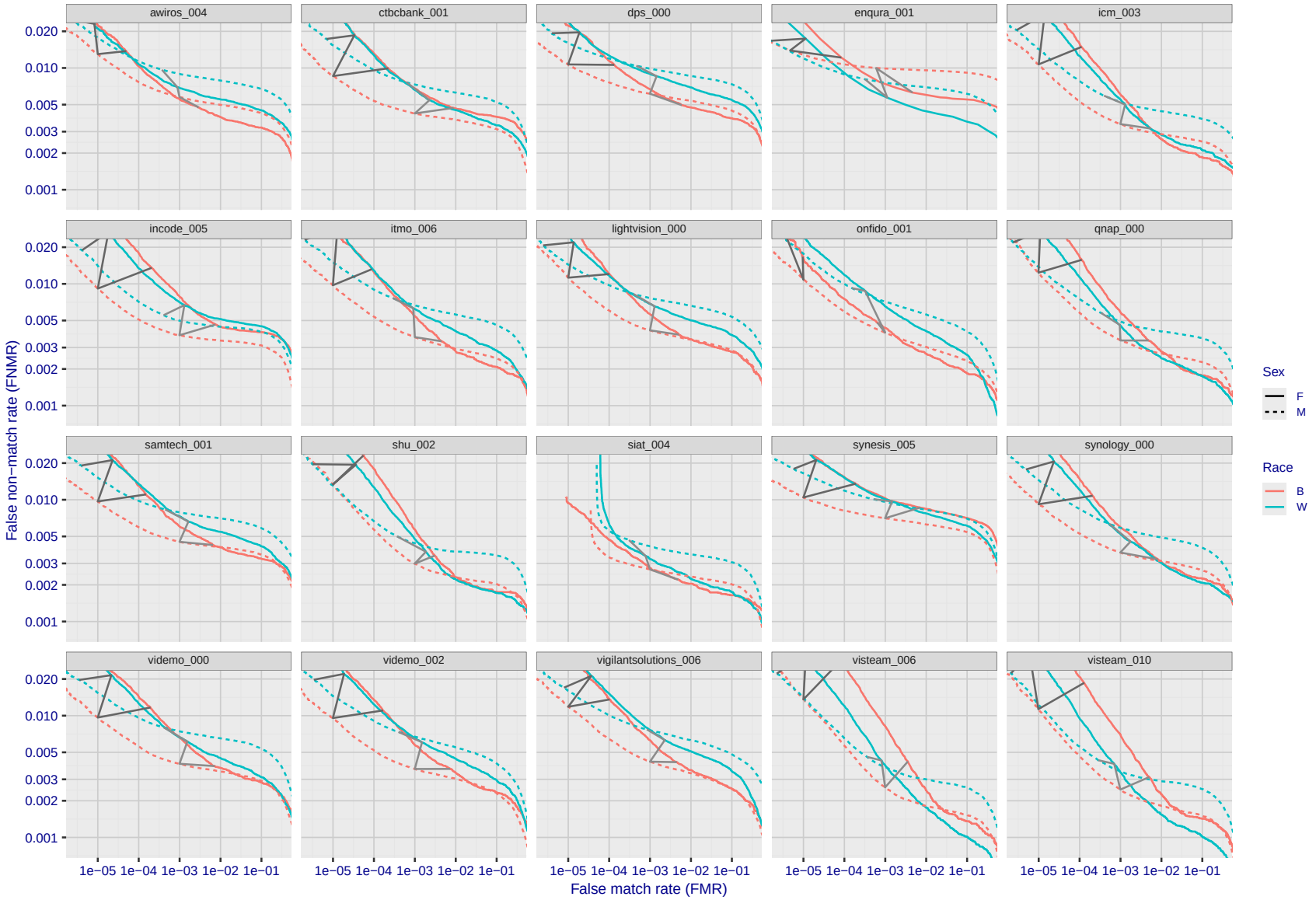

Figure 451: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

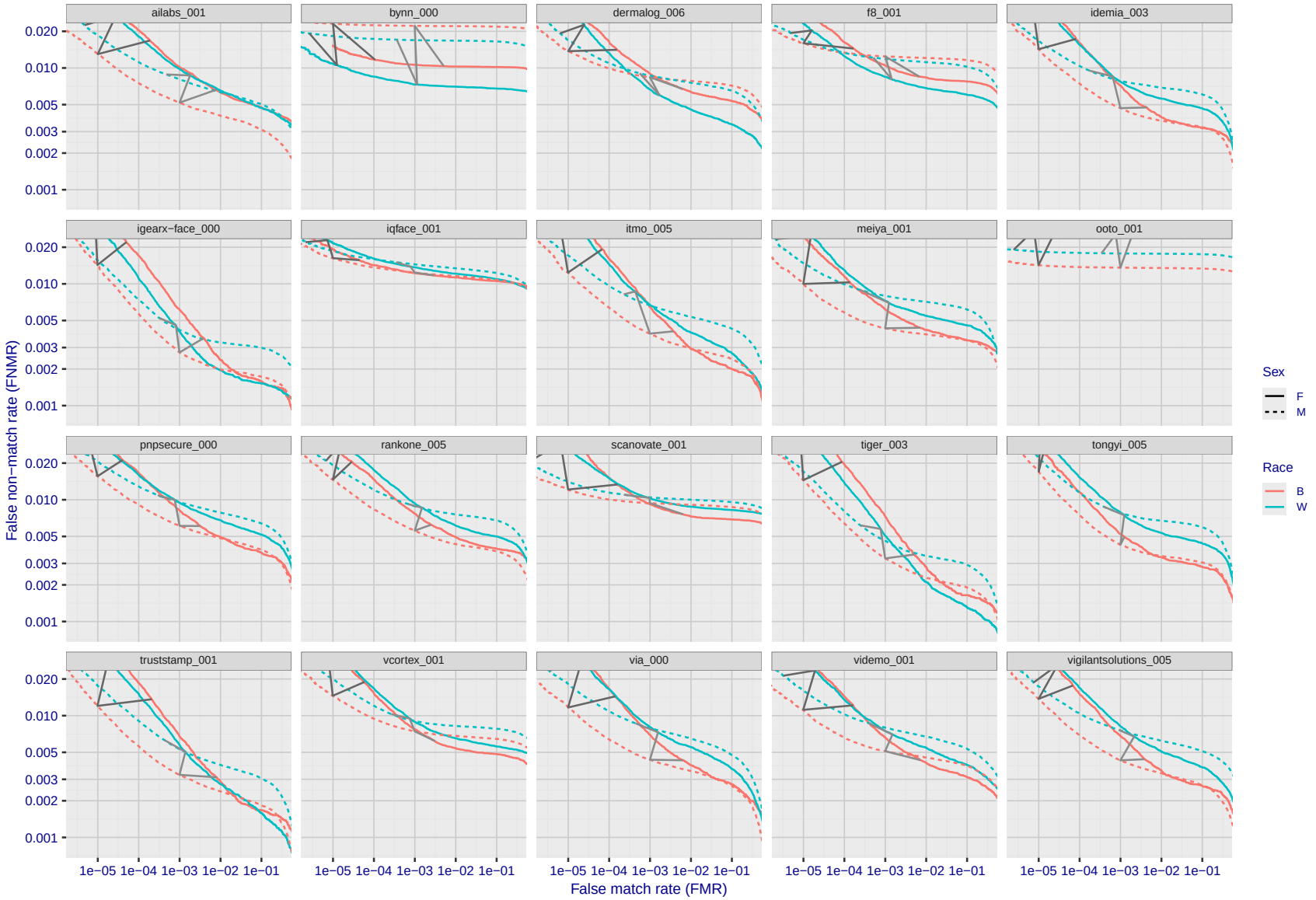

Figure 452: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

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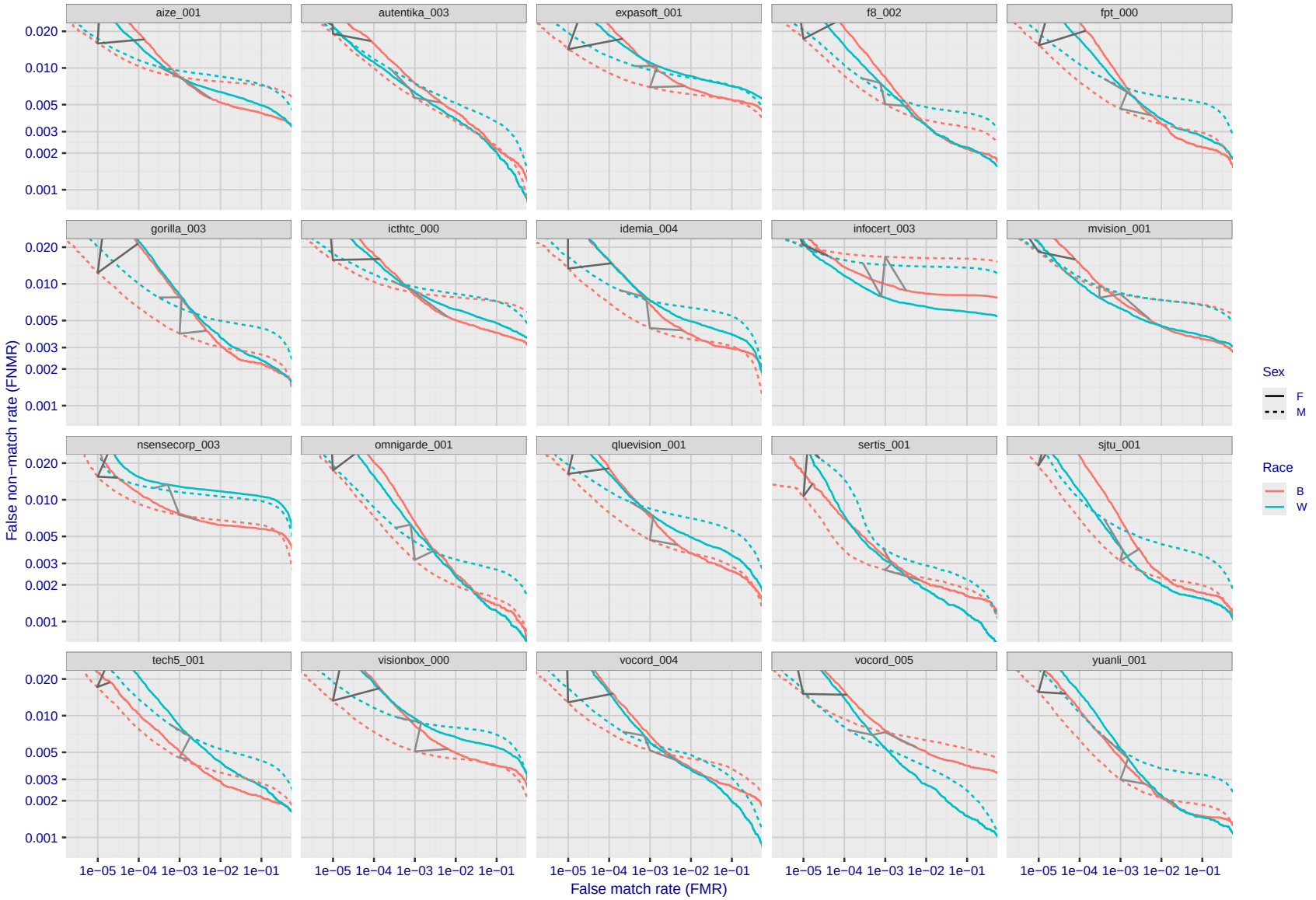

Figure 453: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

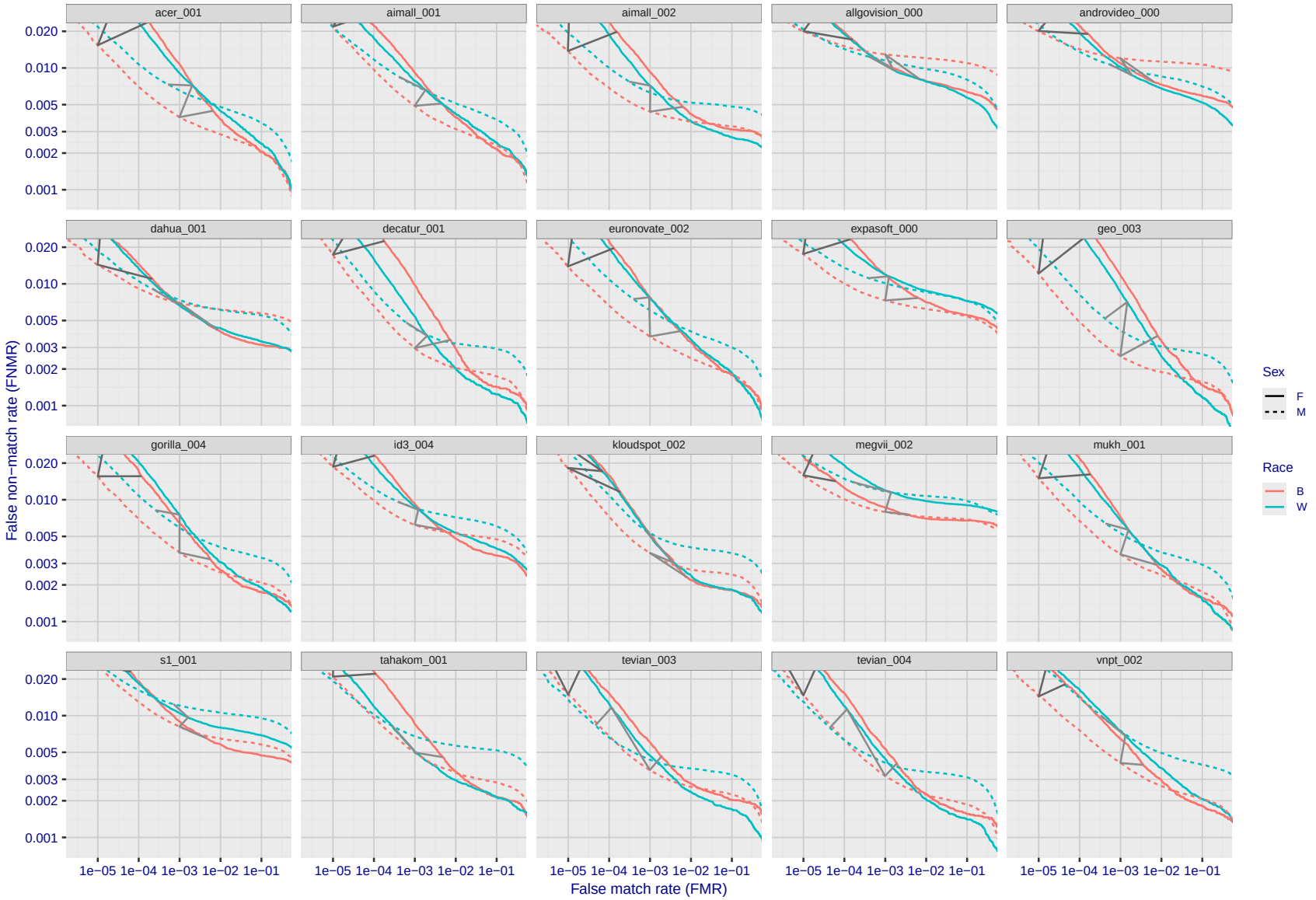

Figure 454: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

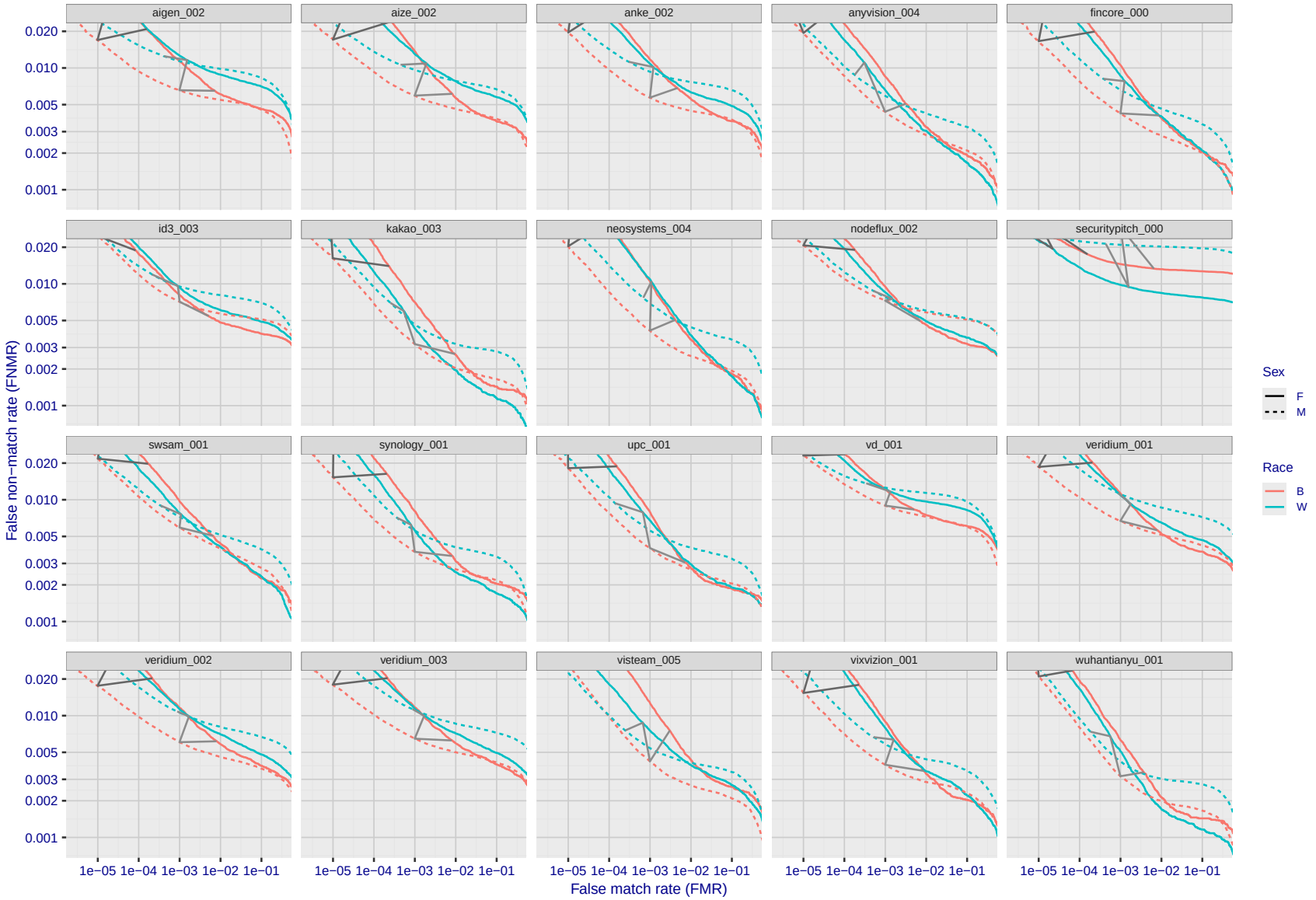

Figure 455: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

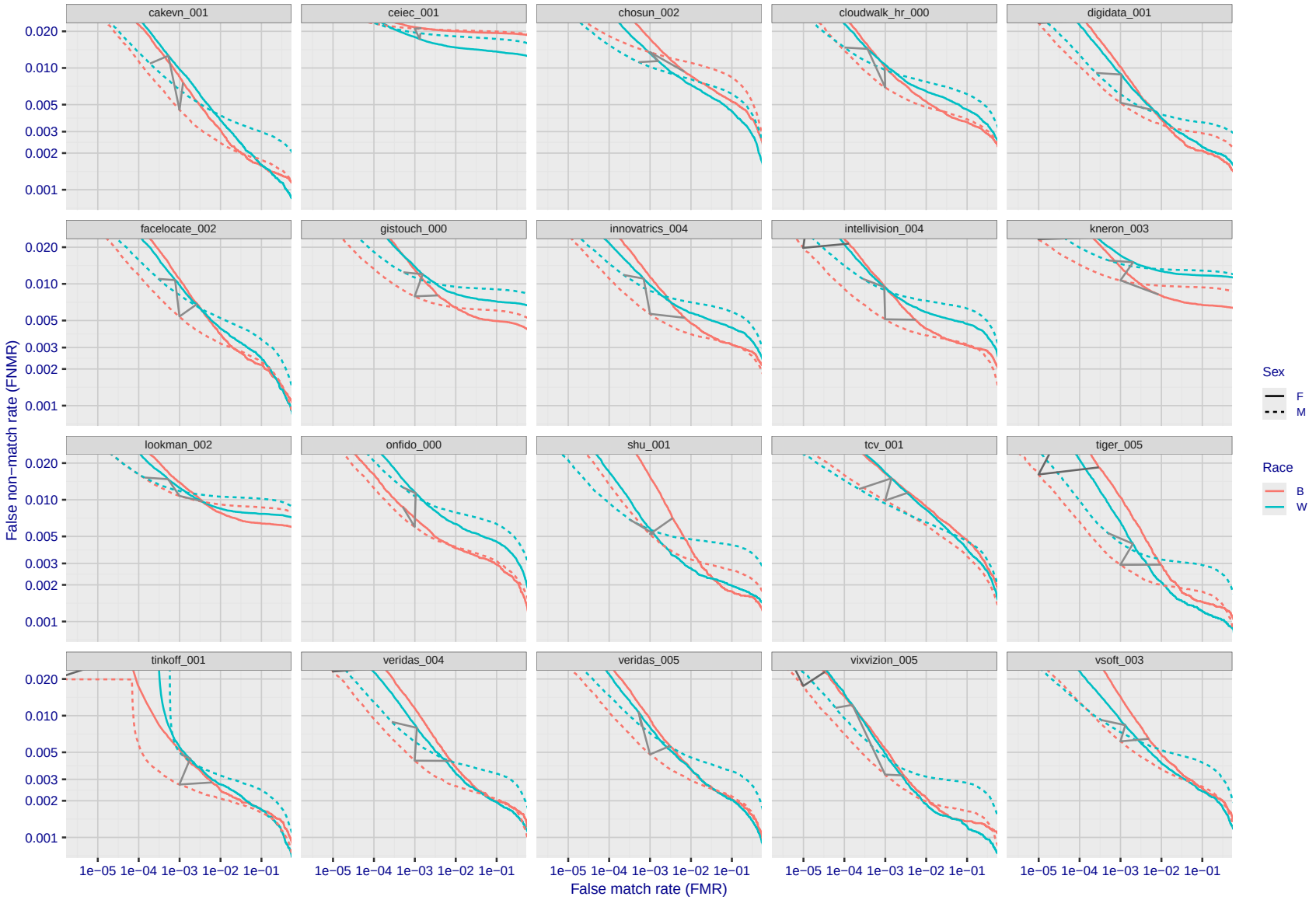

Figure 456: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

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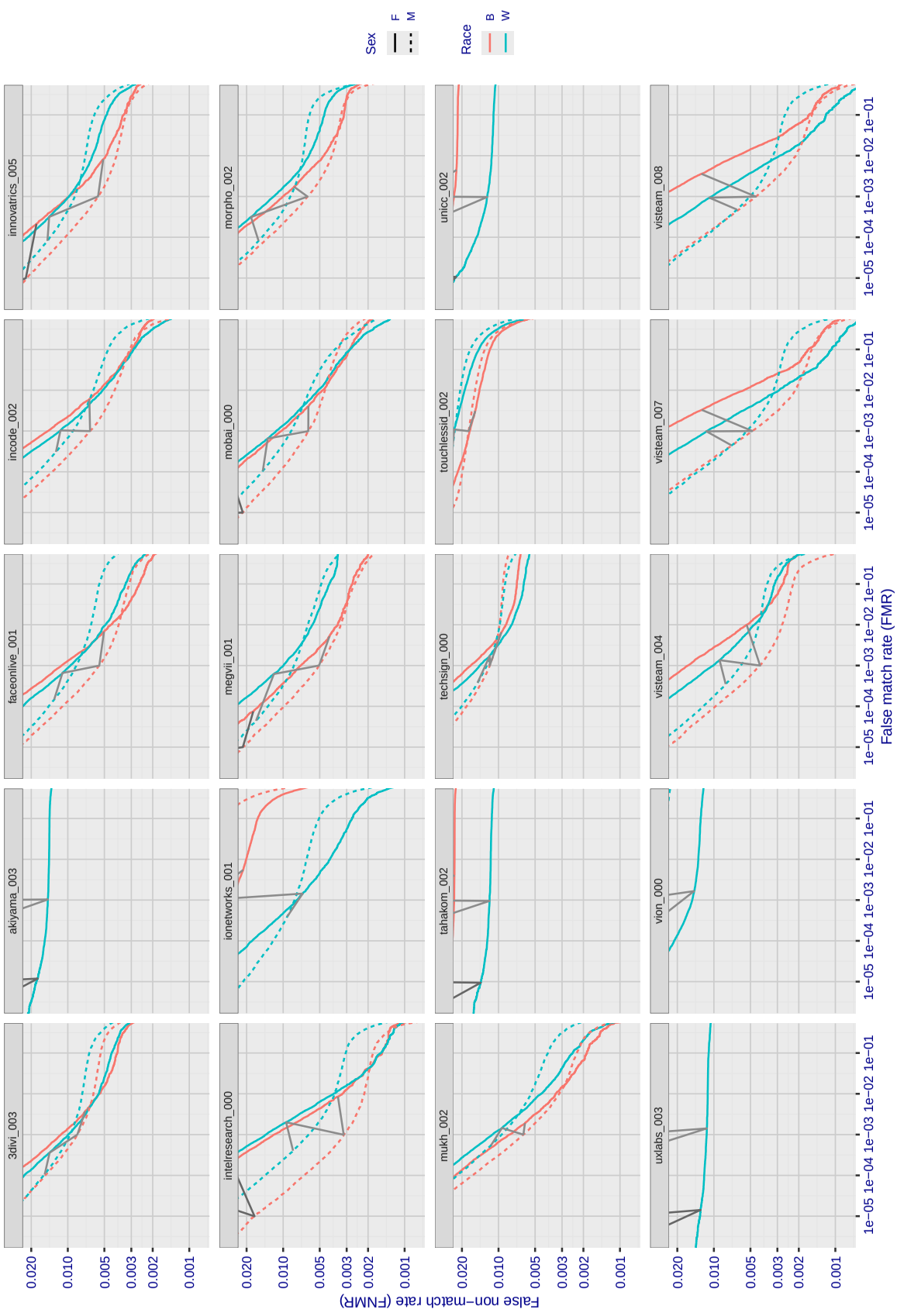

Figure 457: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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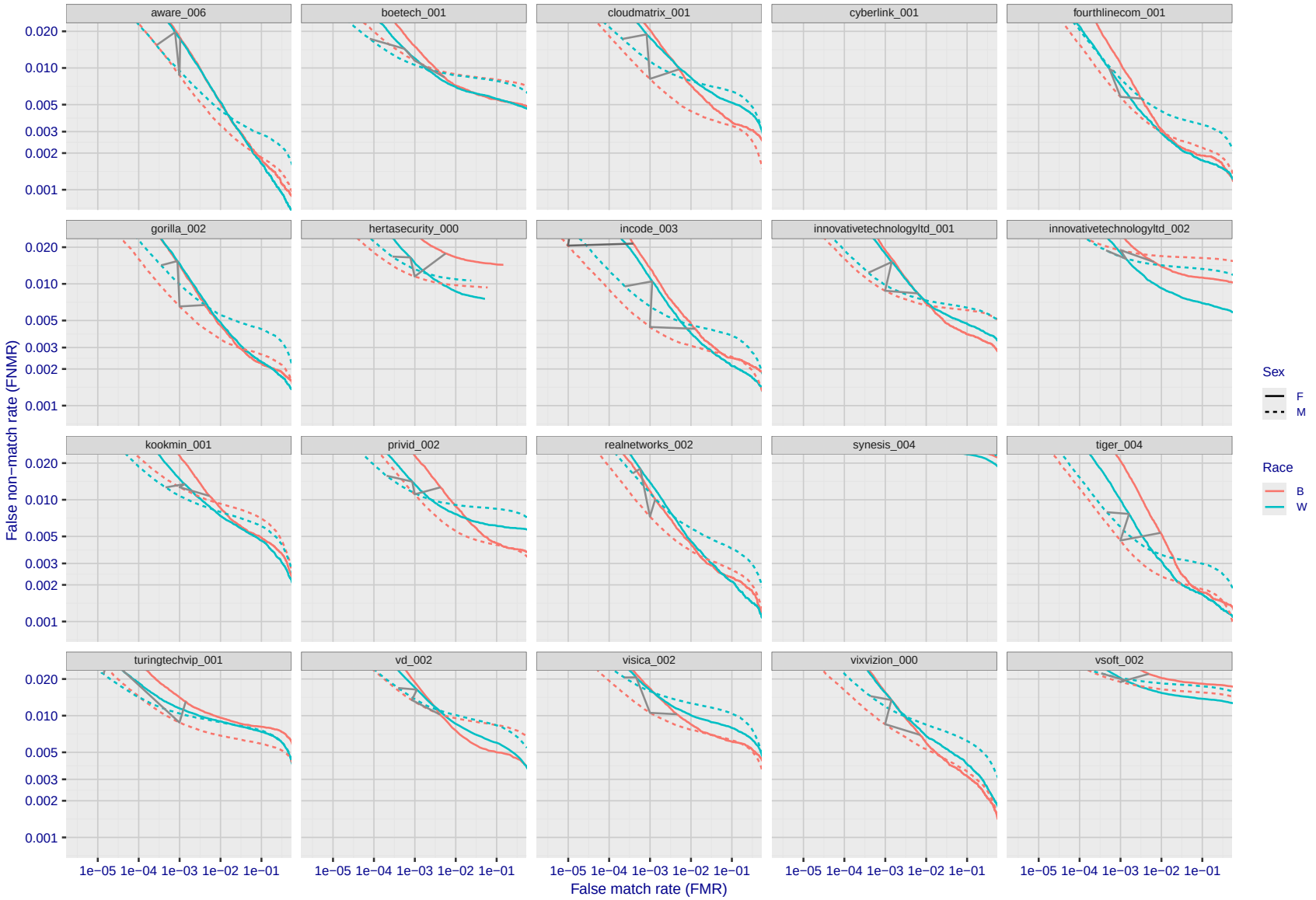

Figure 458: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

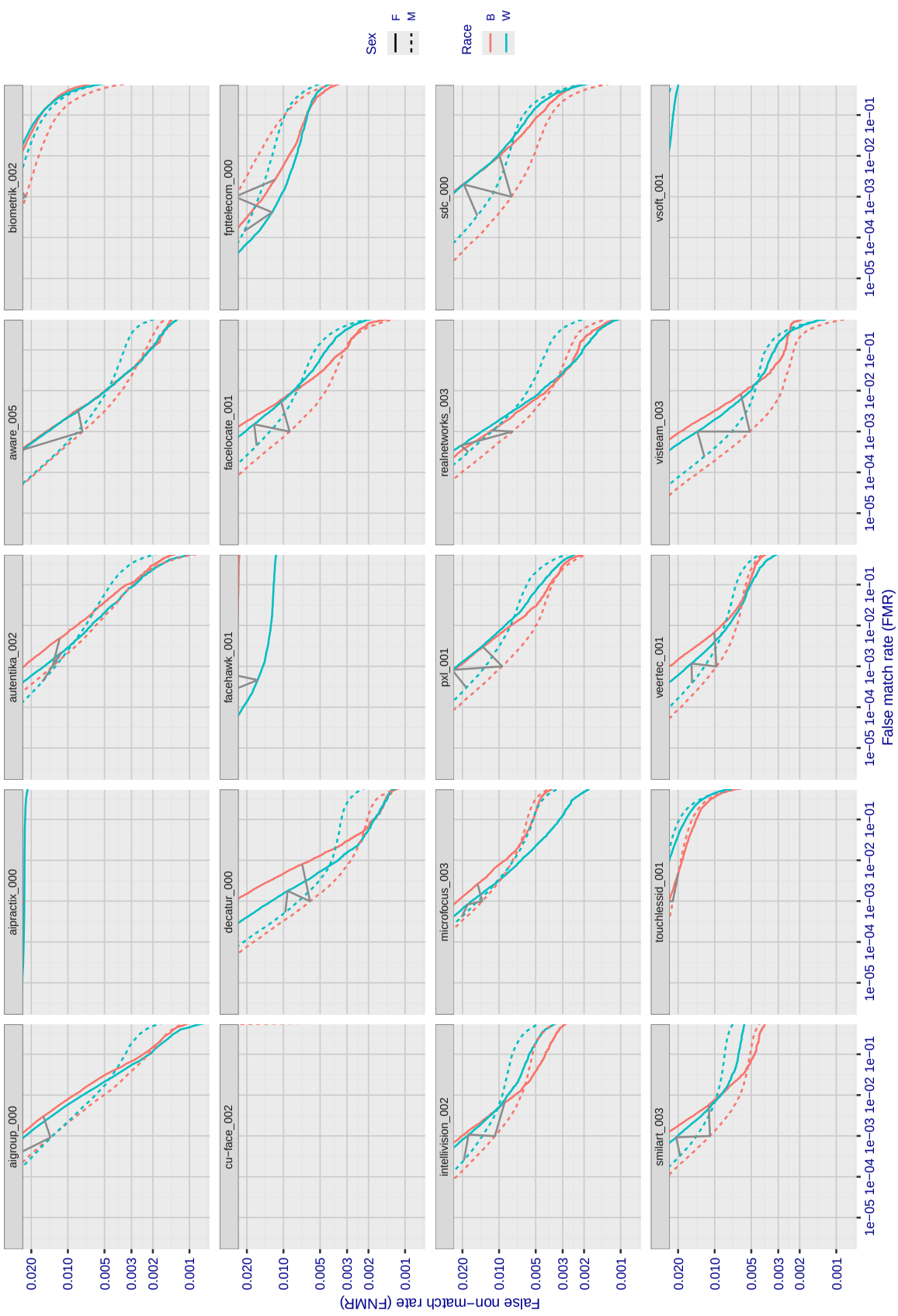

Figure 459: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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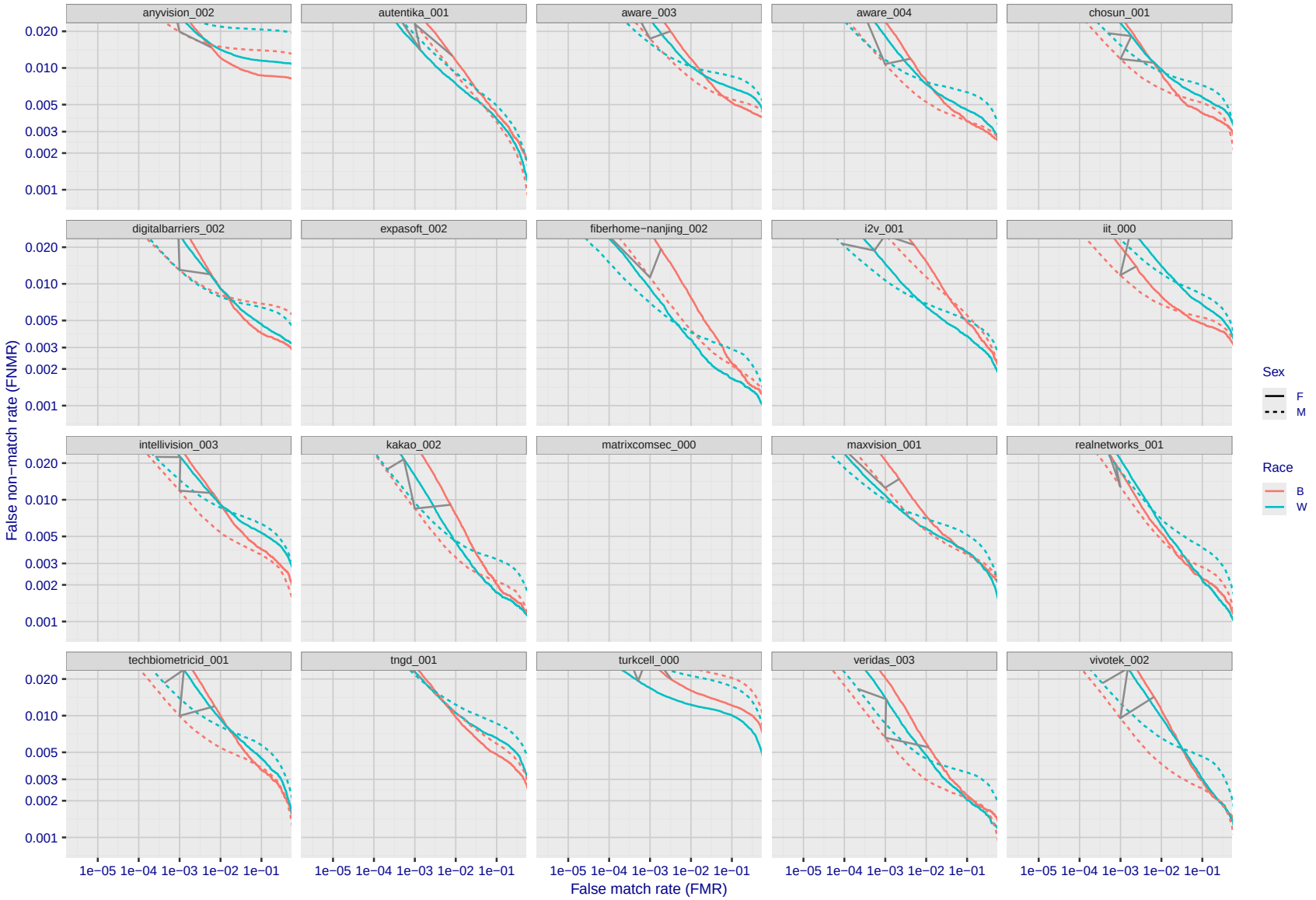

Figure 460: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

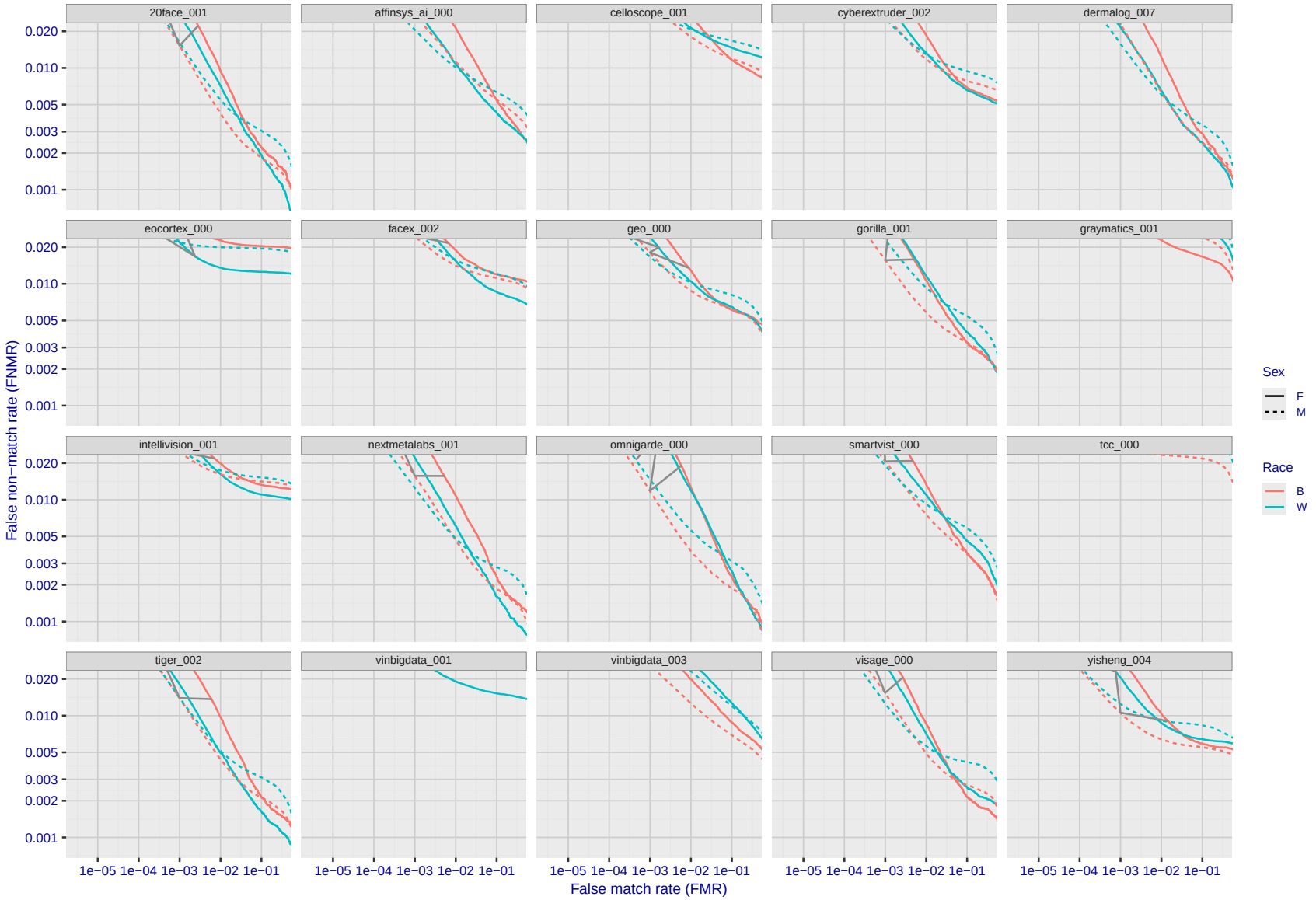

Figure 461: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

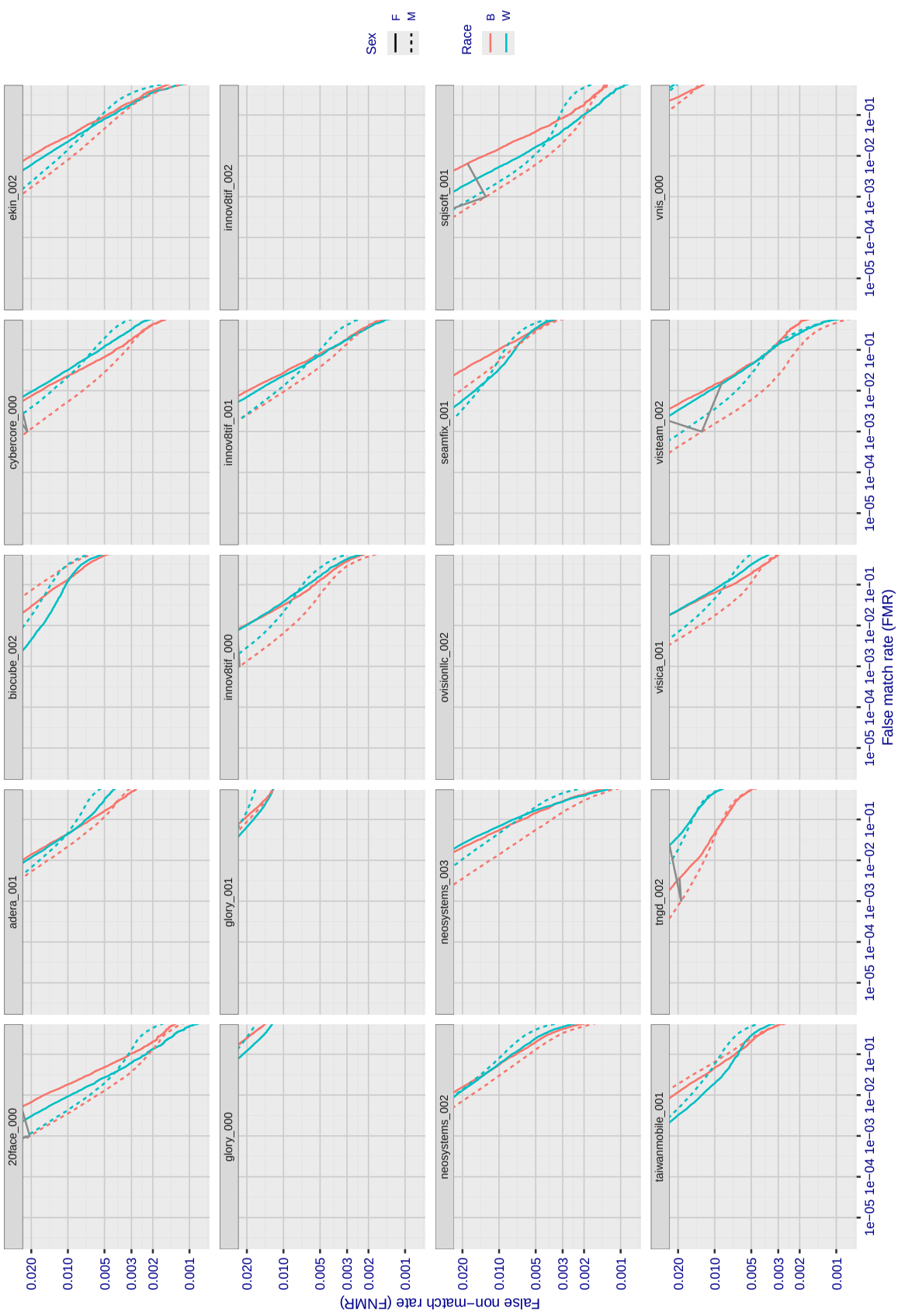

Figure 462: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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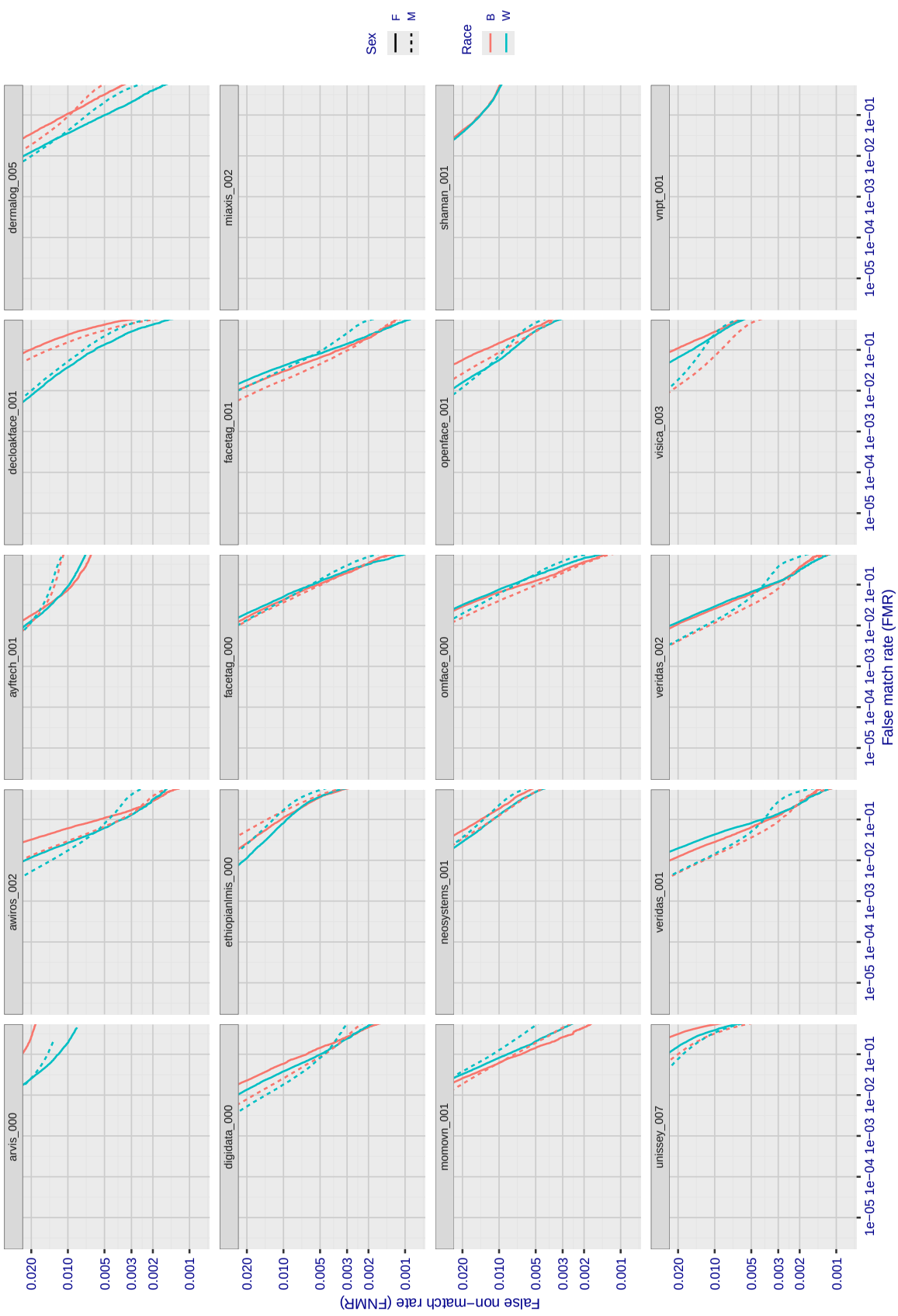

Figure 463: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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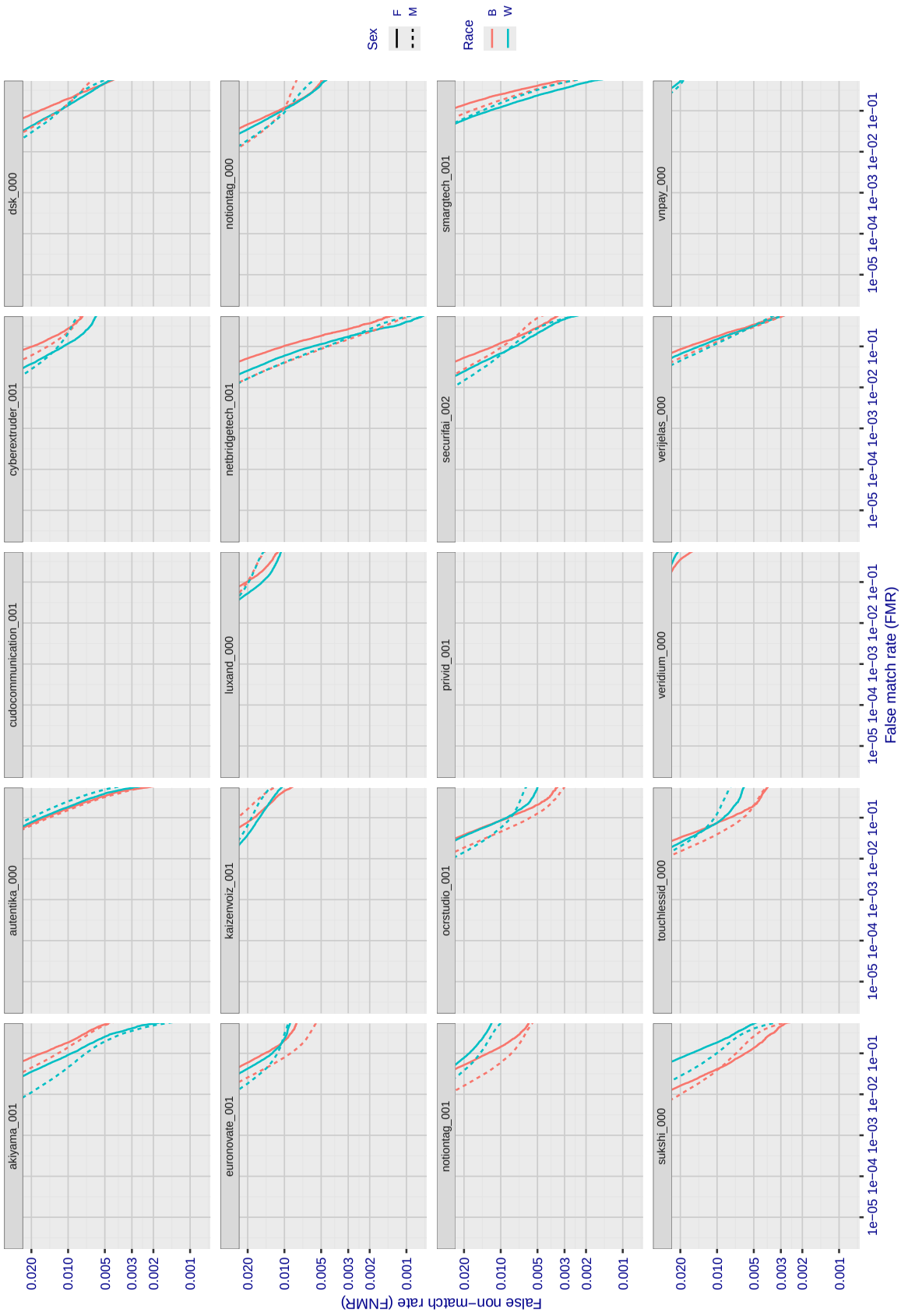

Figure 464: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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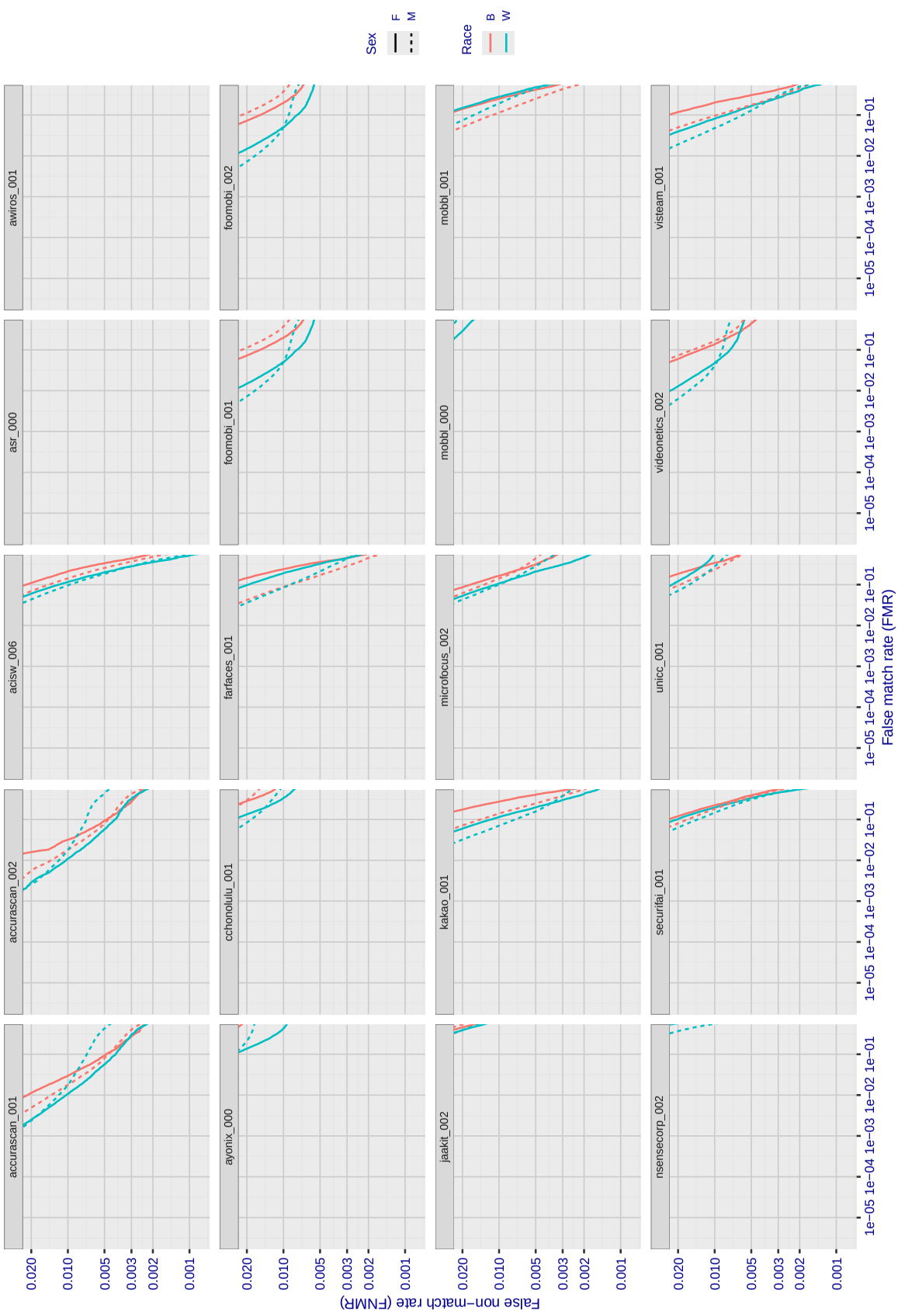

Figure 465: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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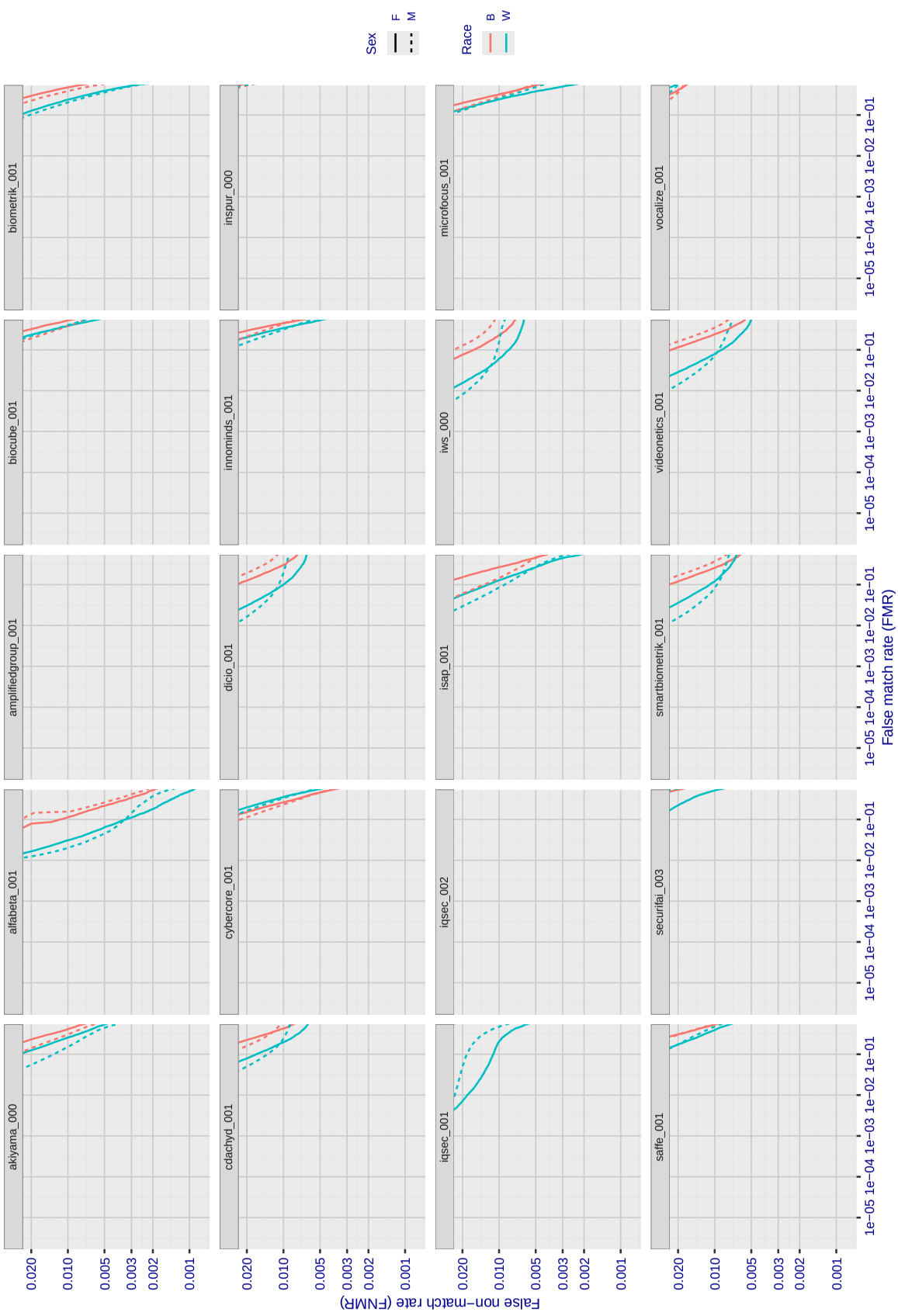

Figure 466: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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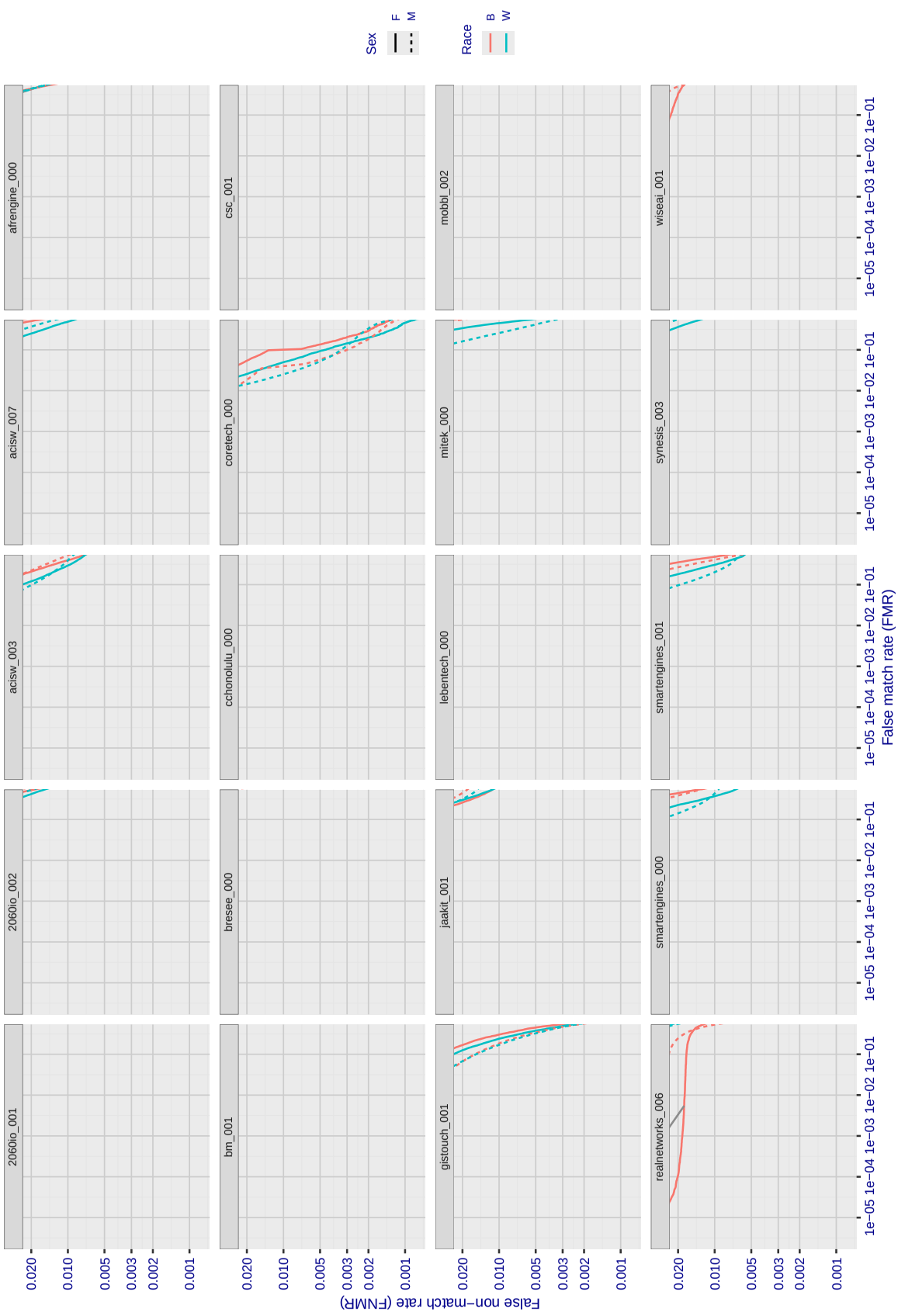

Figure 467: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

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FMR(T) "False match rate"

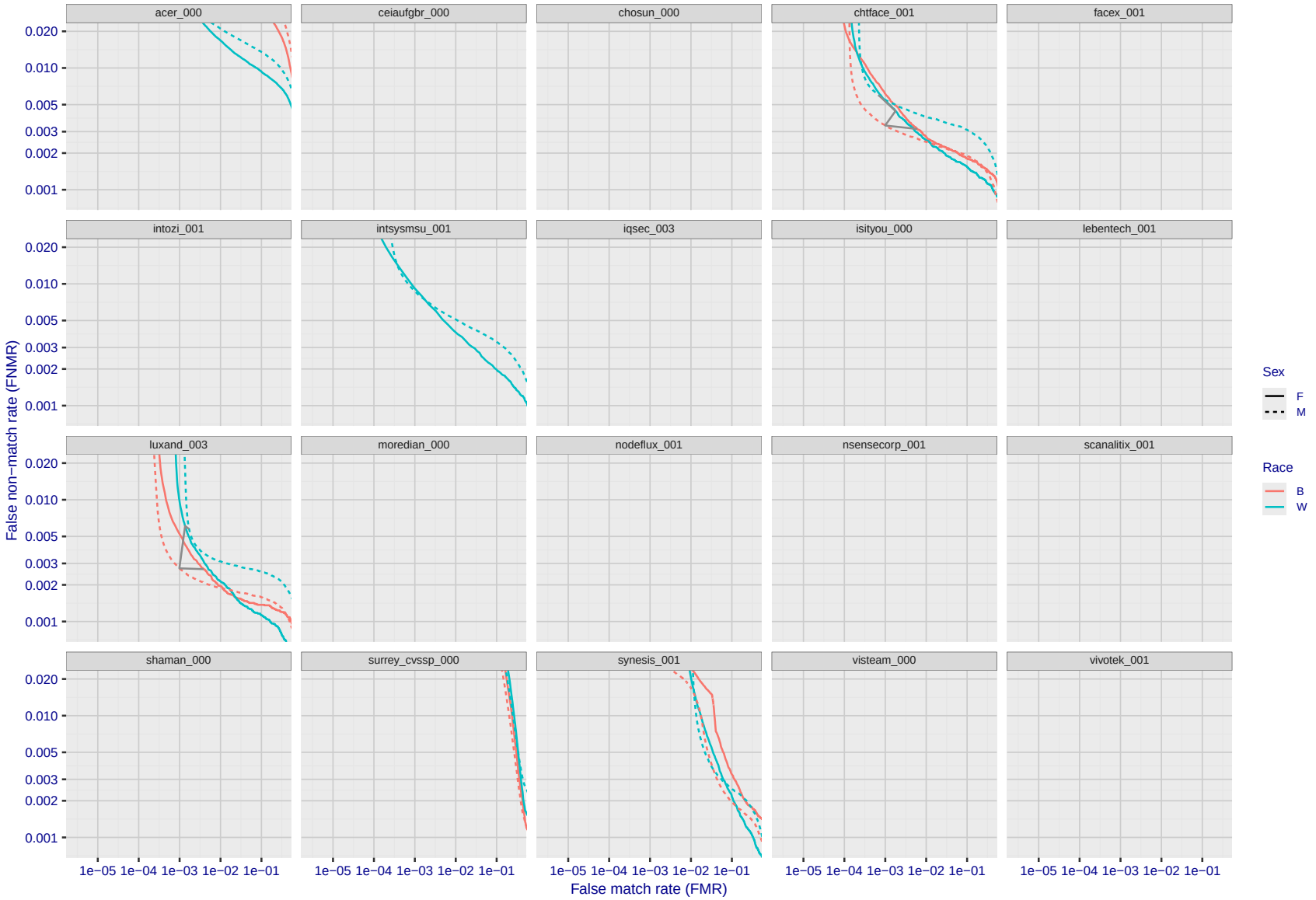

Figure 468: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

Figure 469: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 629. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

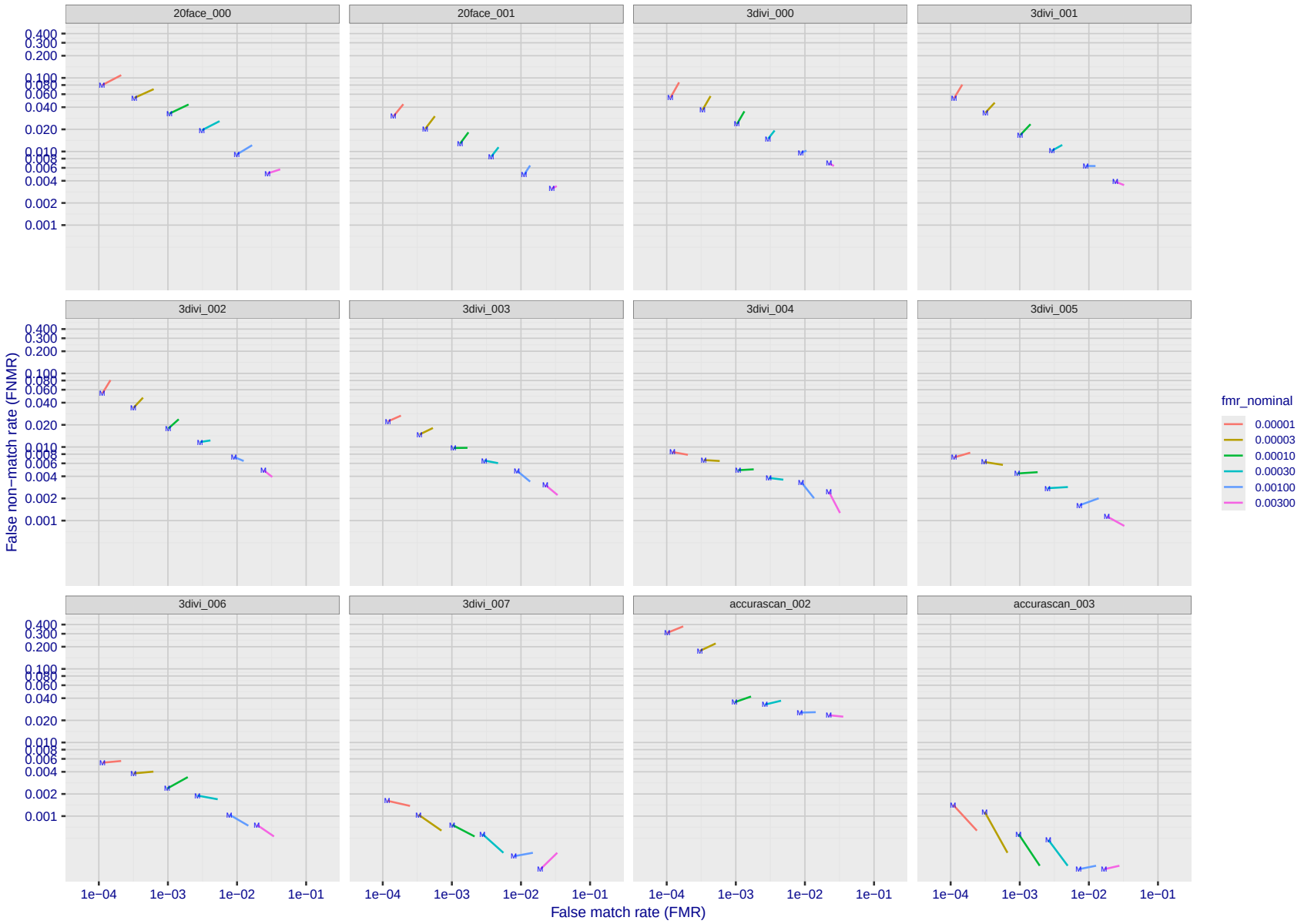

Figure 470: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

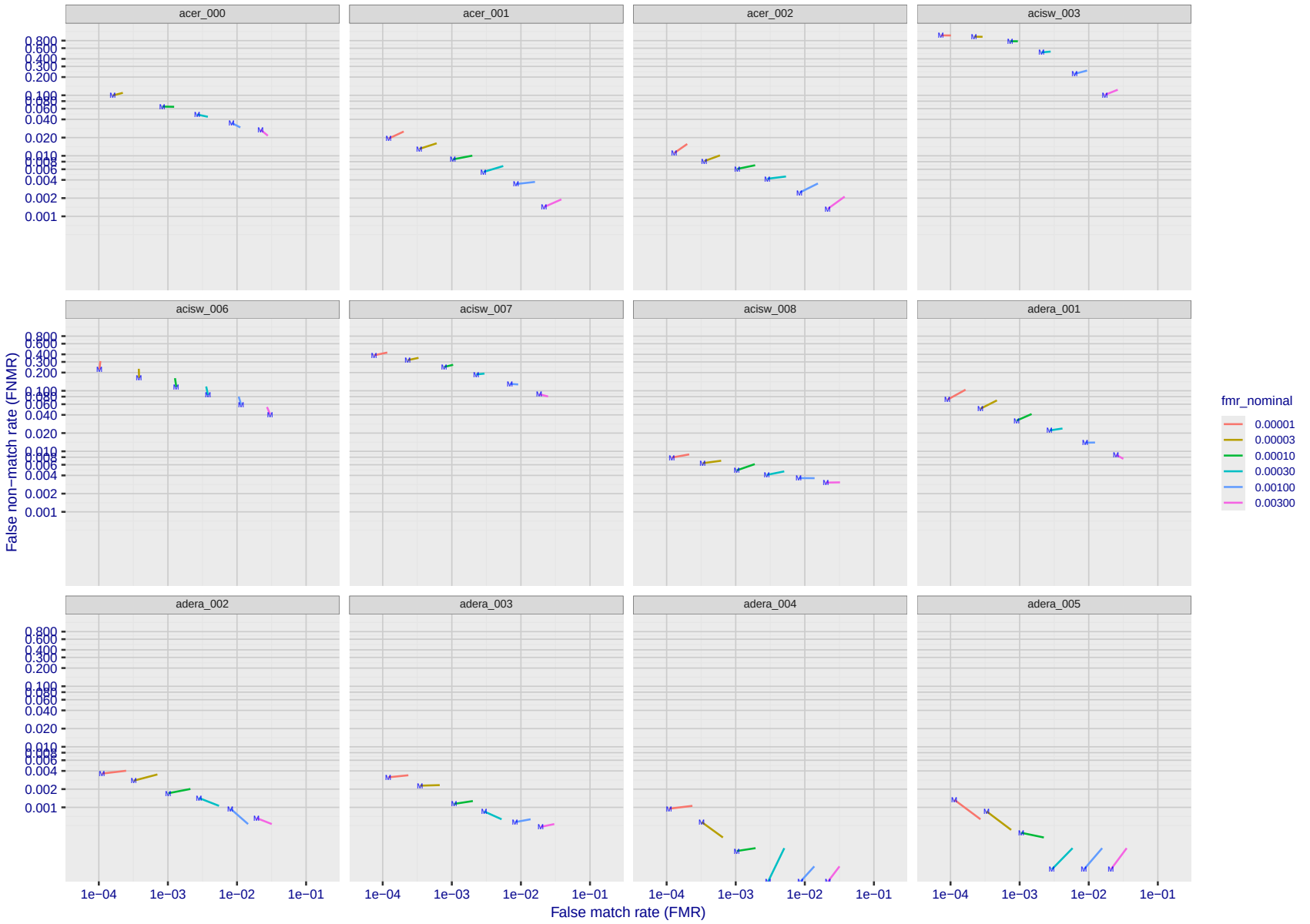

Figure 471: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

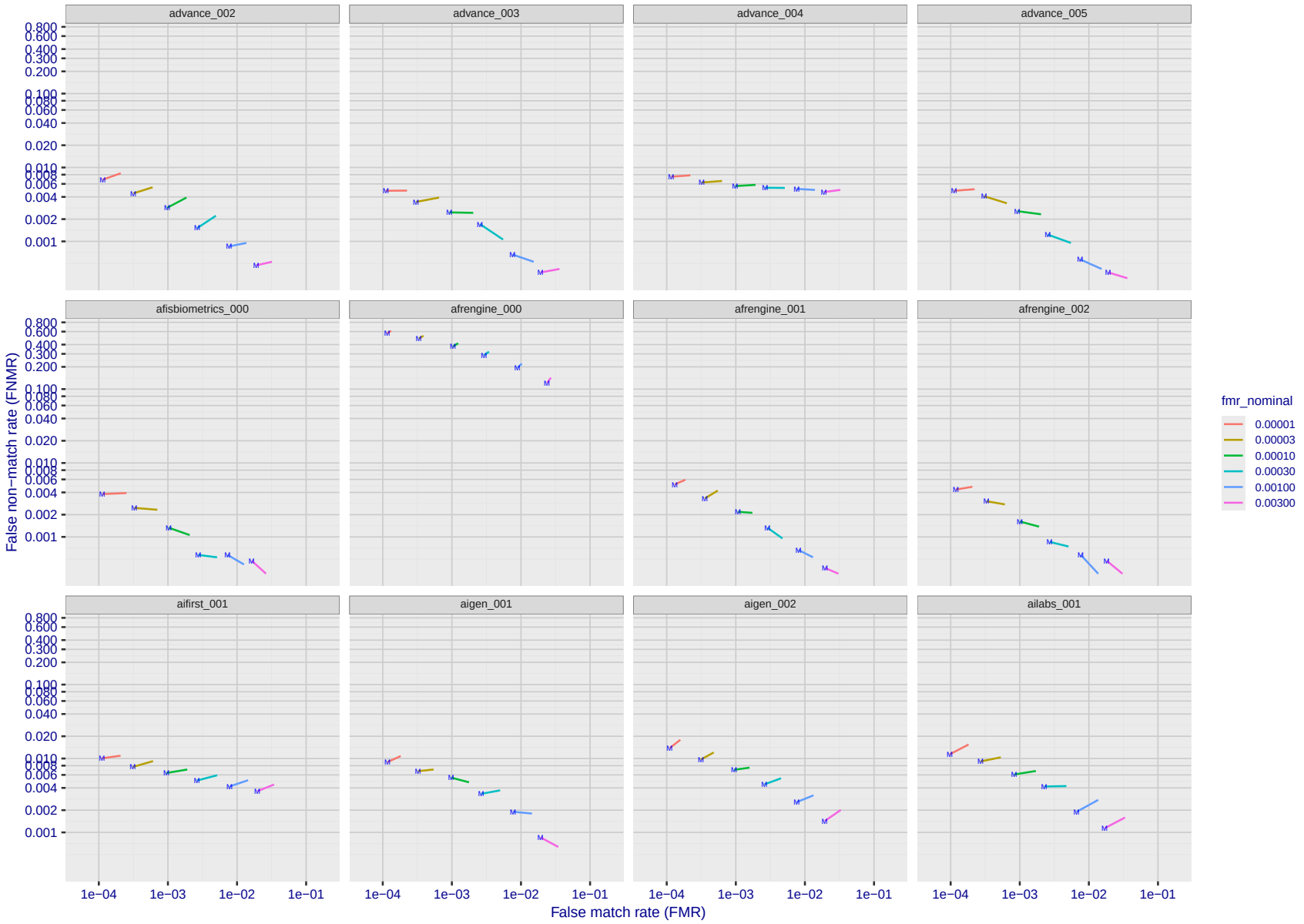

Figure 472: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

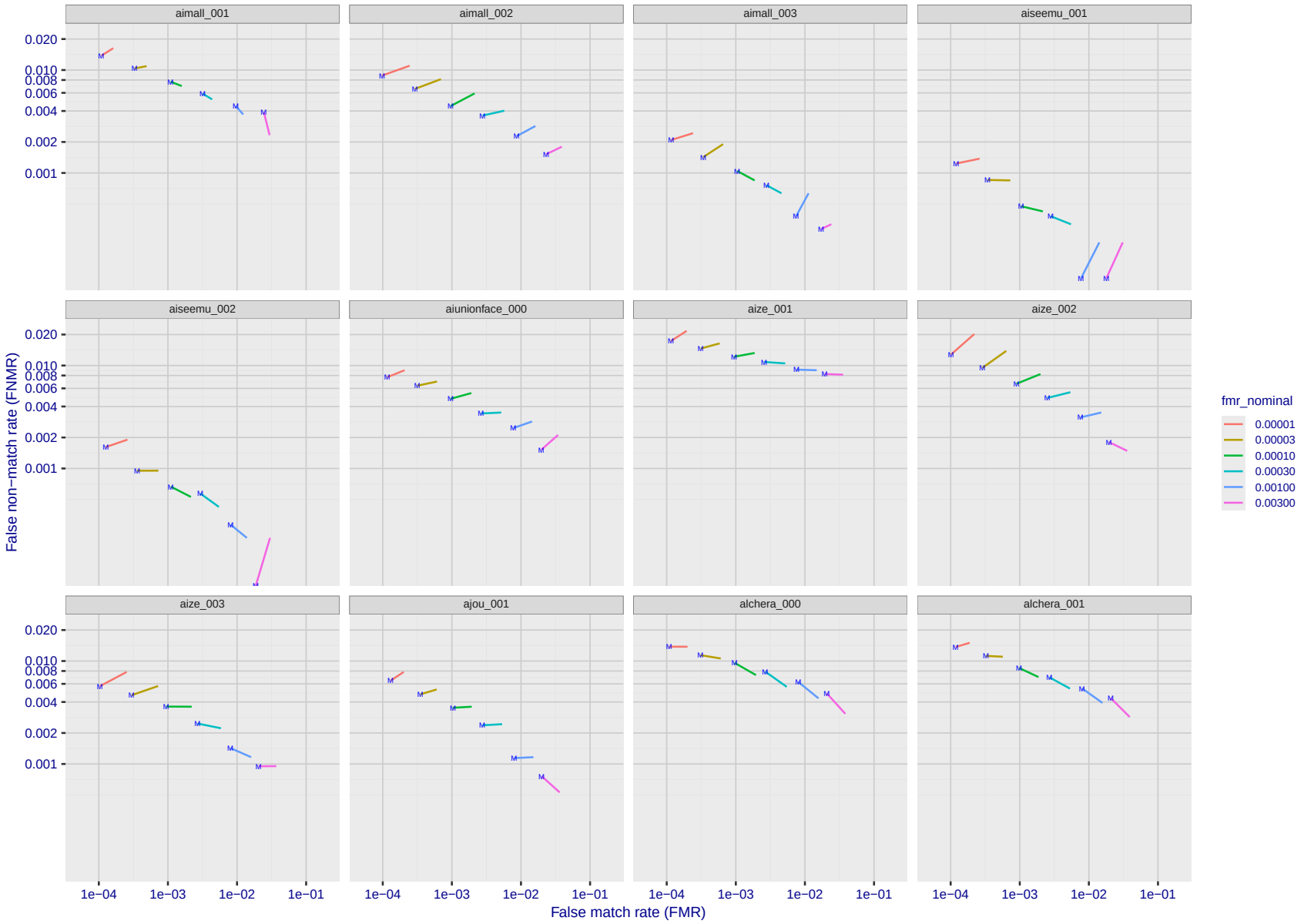

Figure 473: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

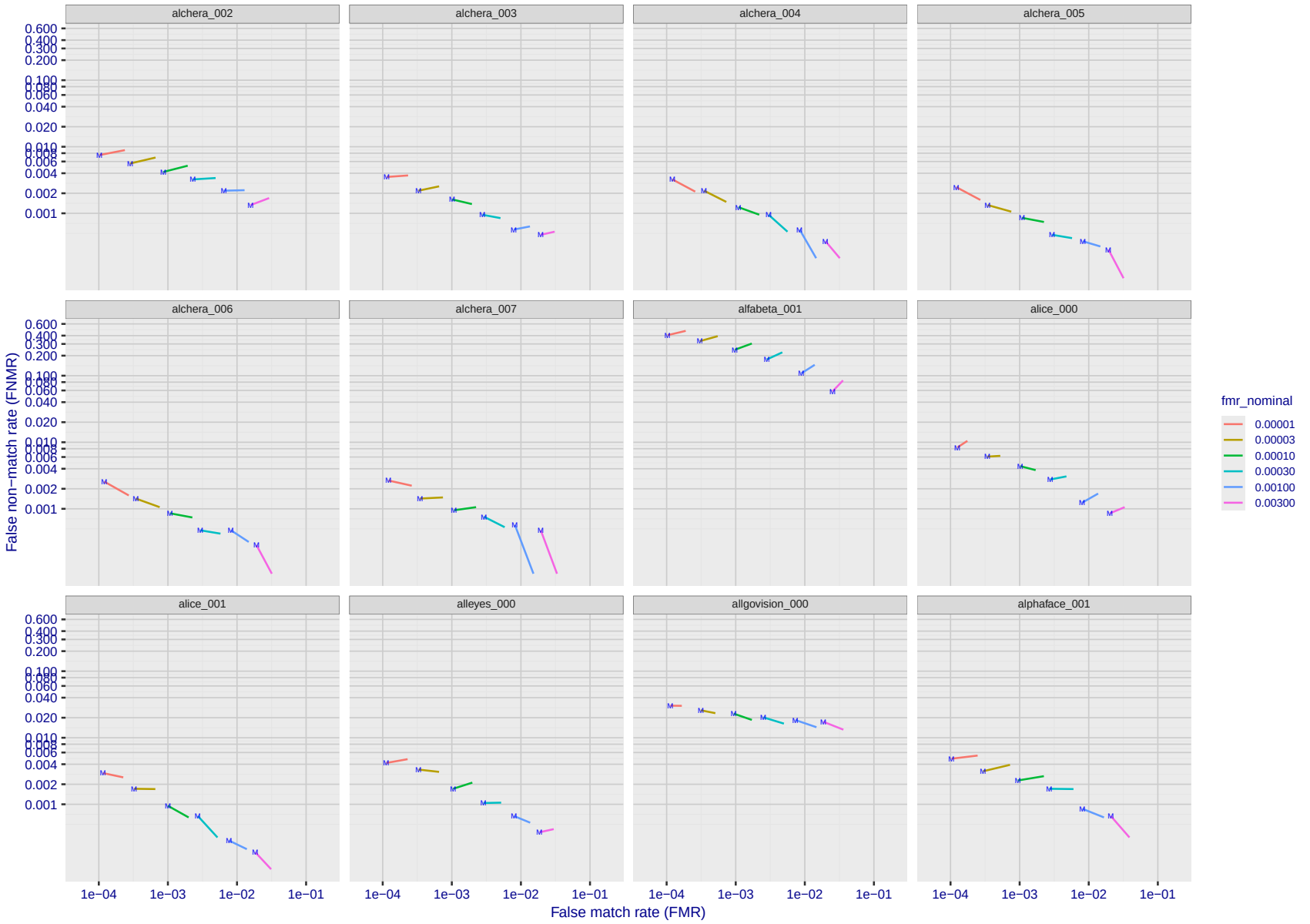

Figure 474: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

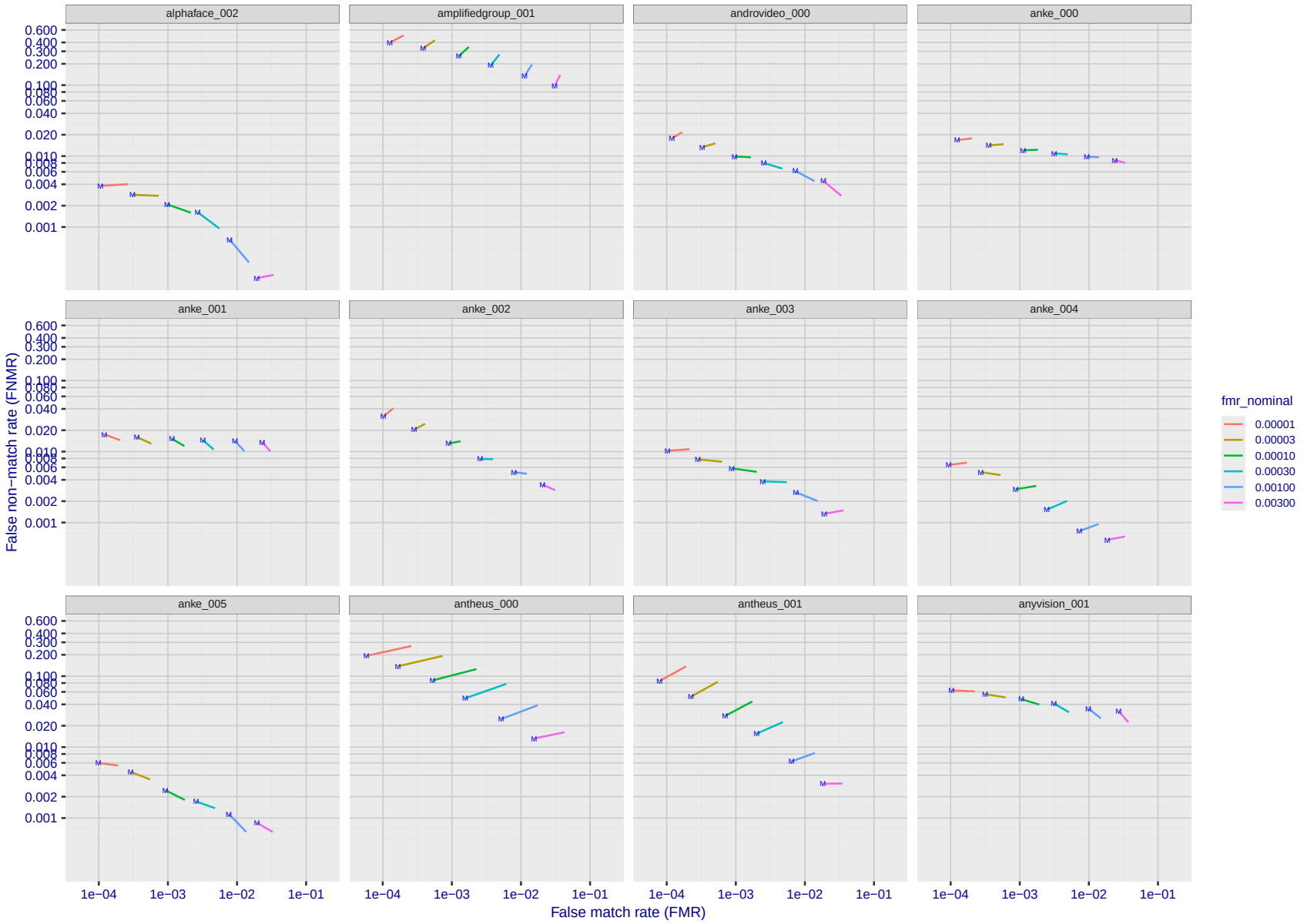

Figure 475: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

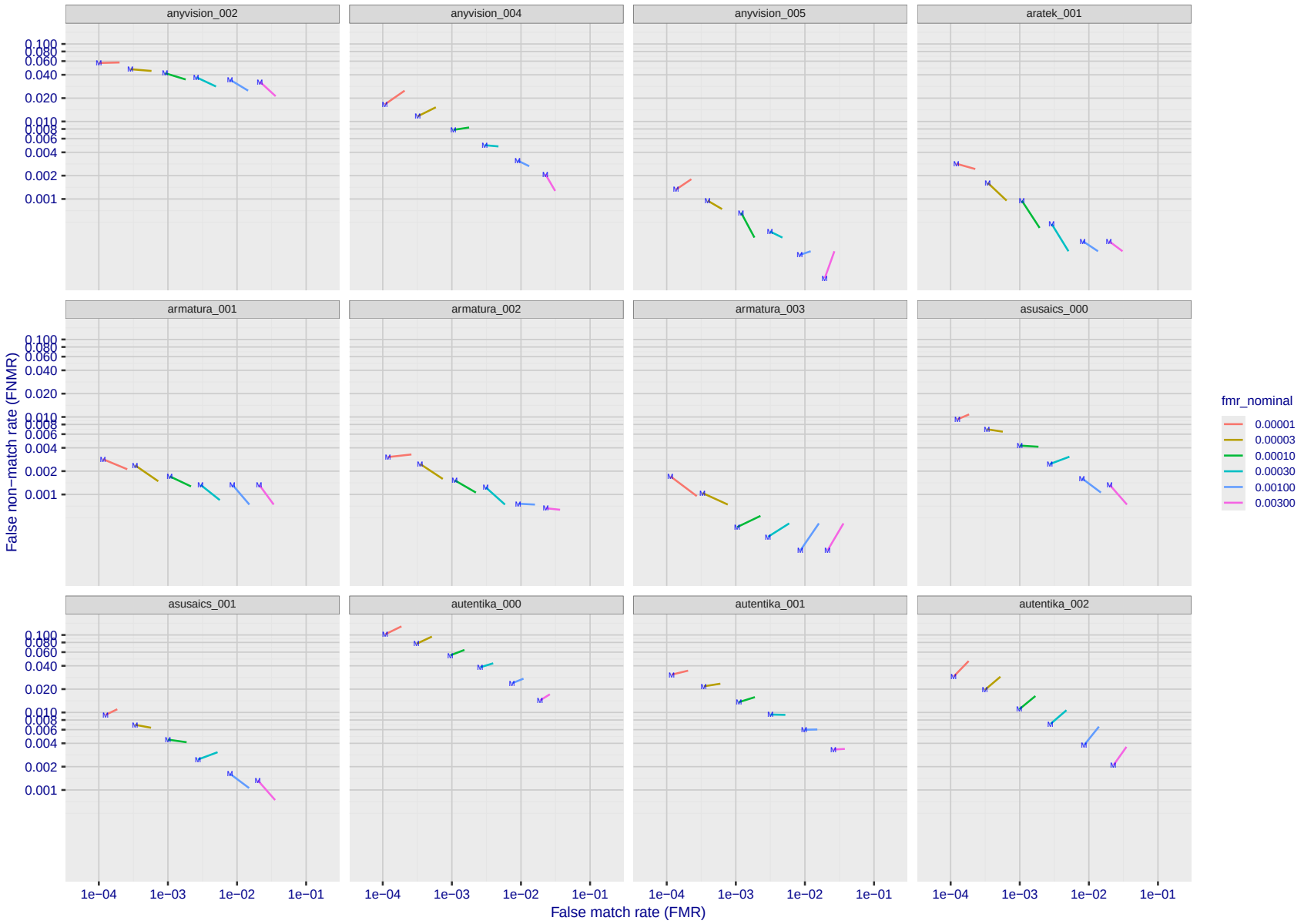

Figure 476: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

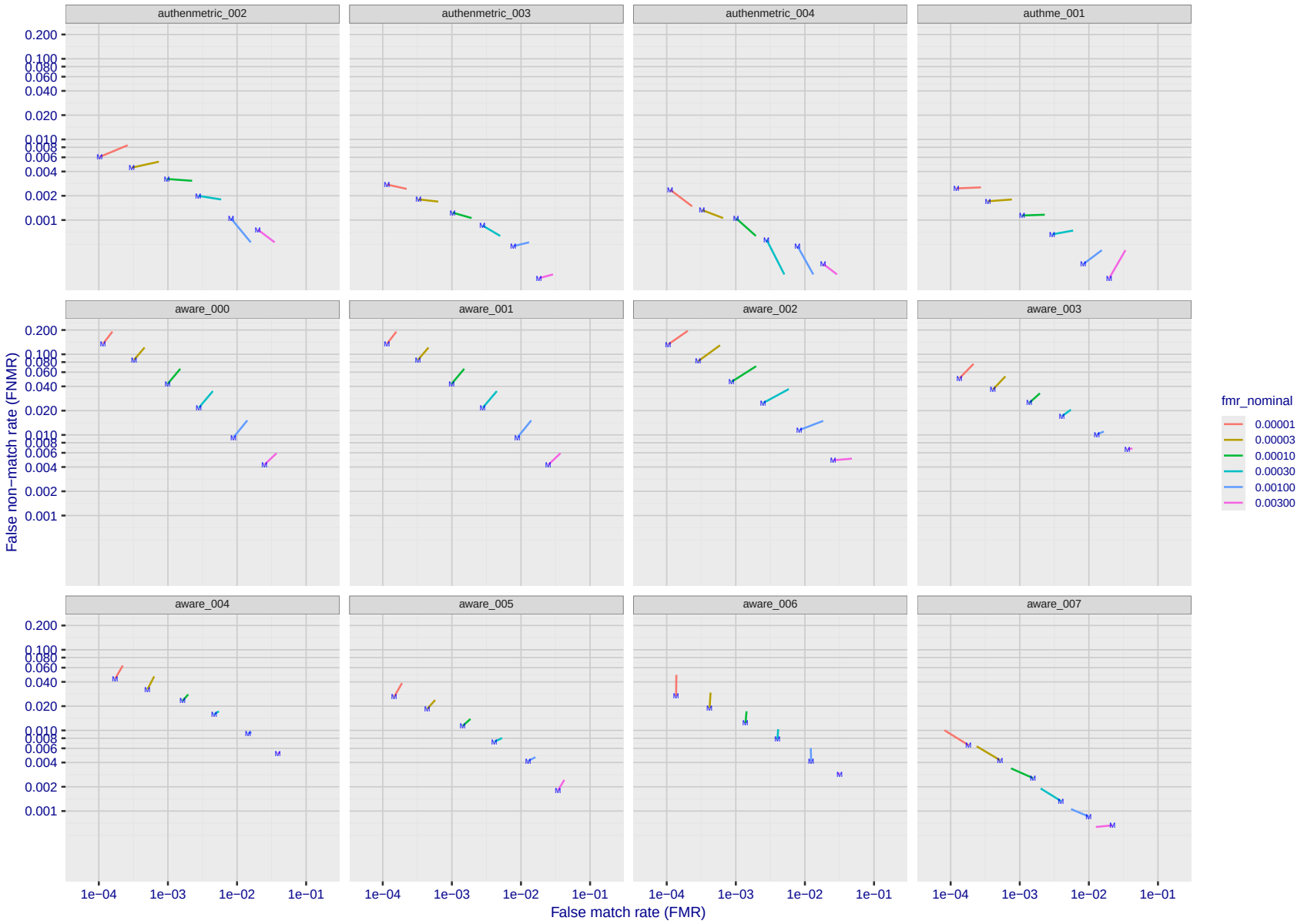

Figure 477: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

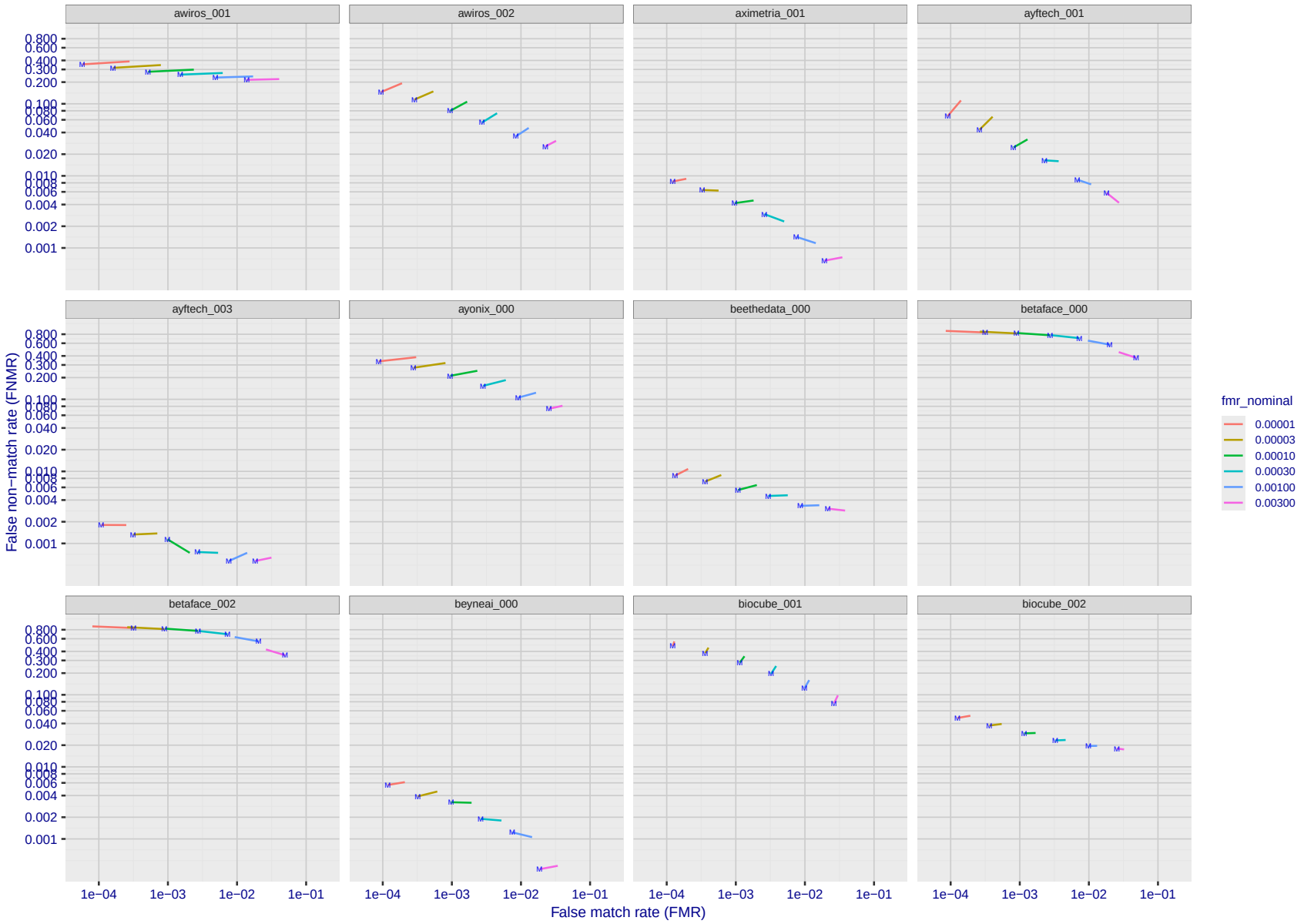

Figure 478: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

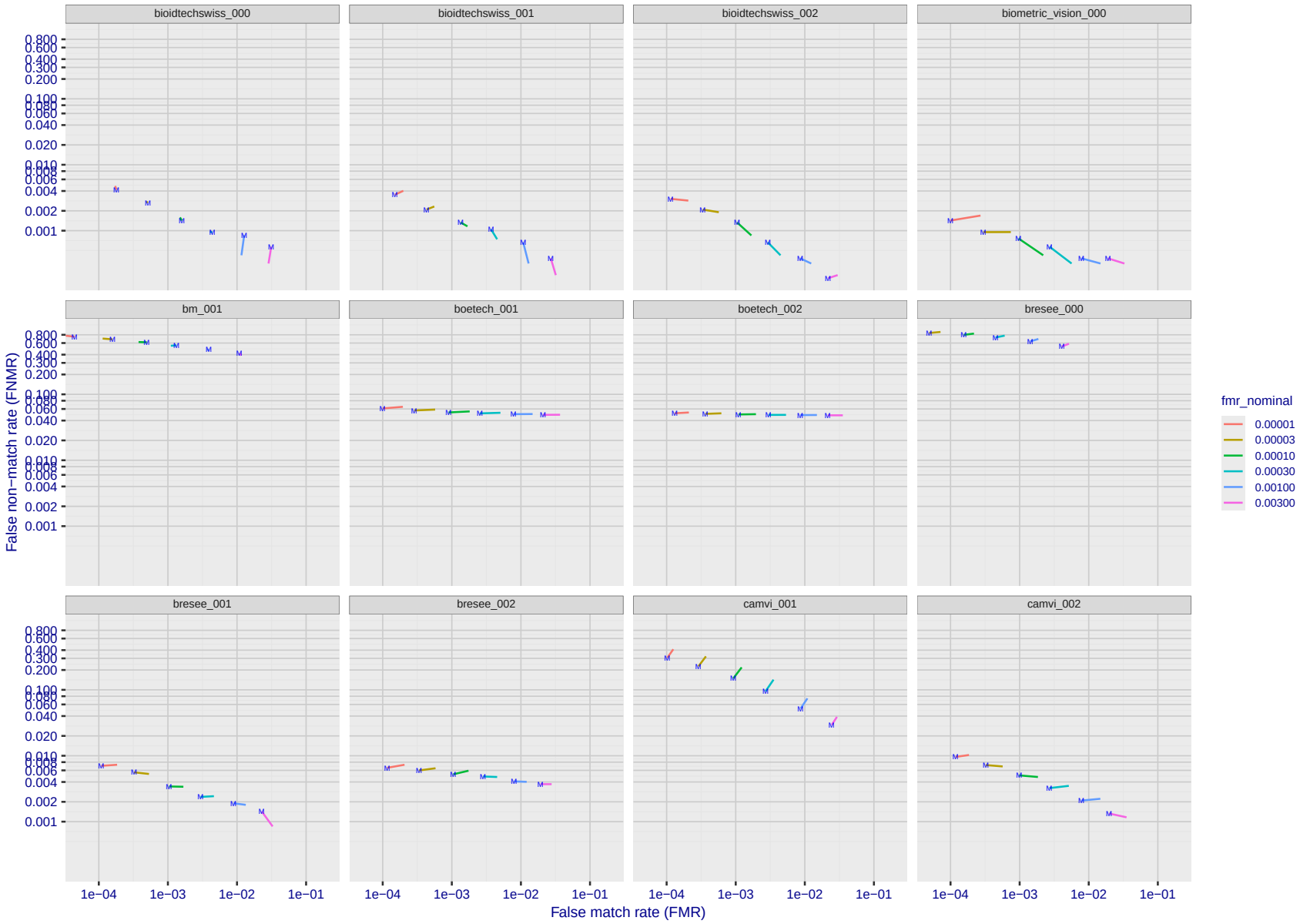

Figure 479: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

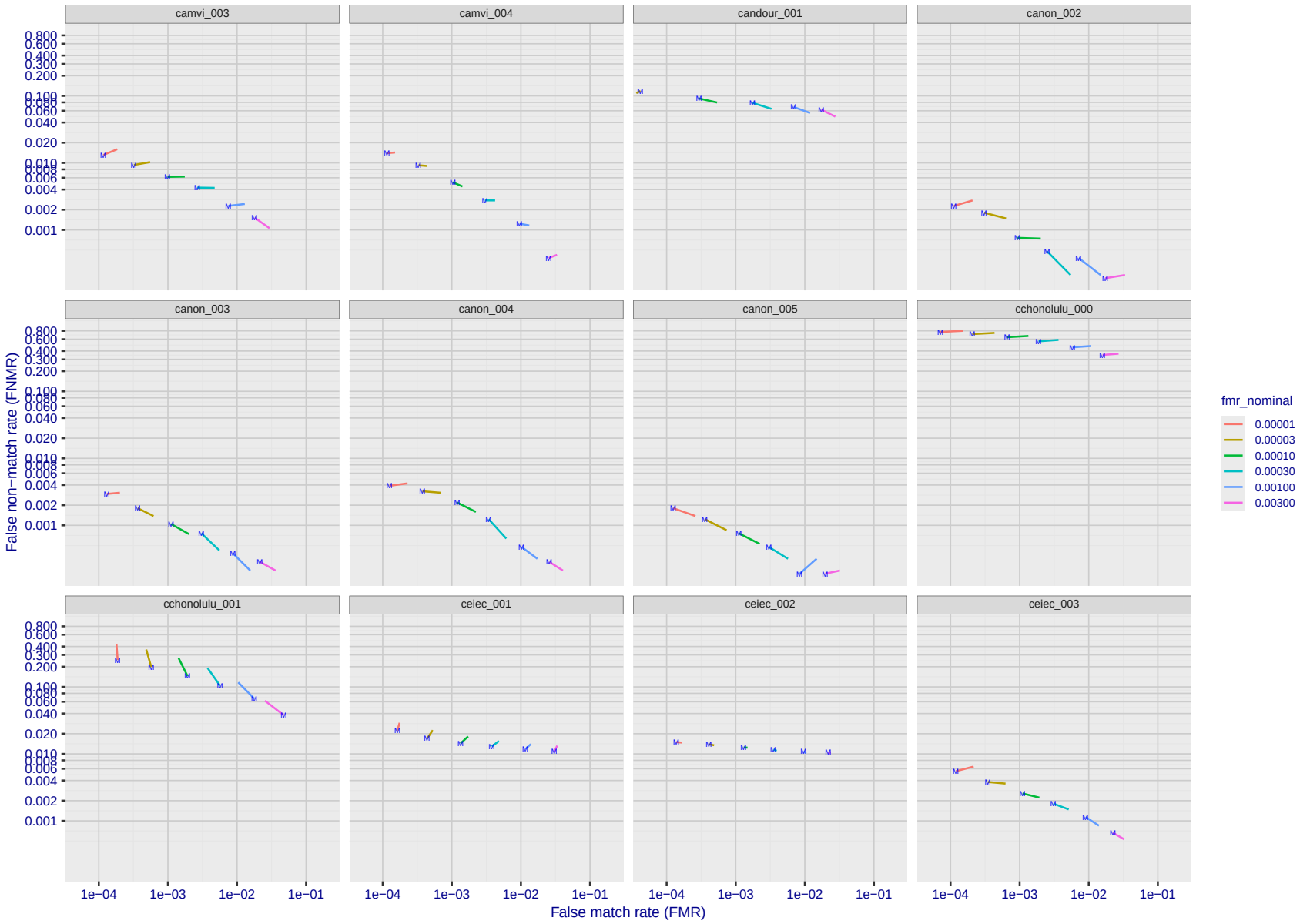

Figure 480: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

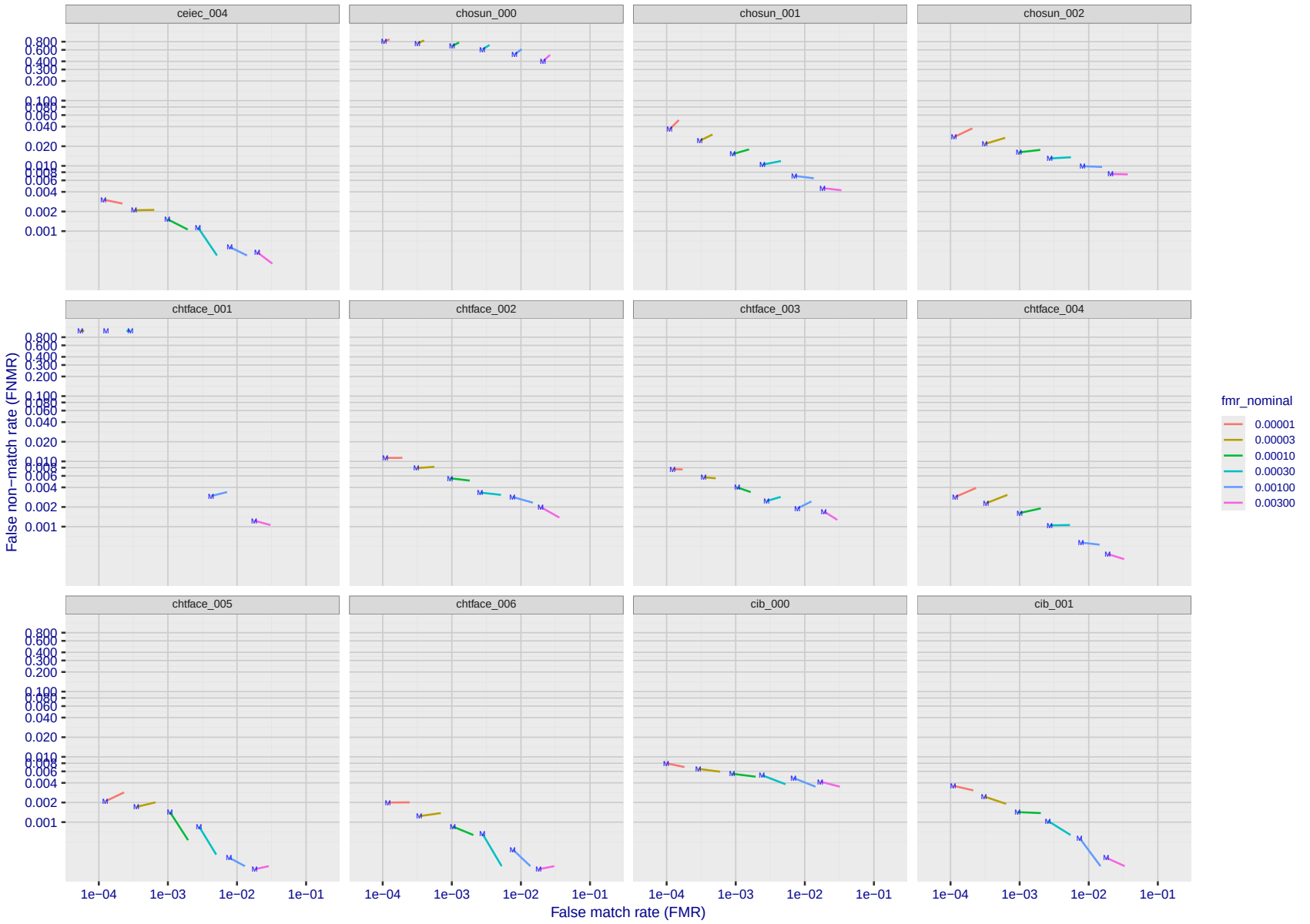

Figure 481: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

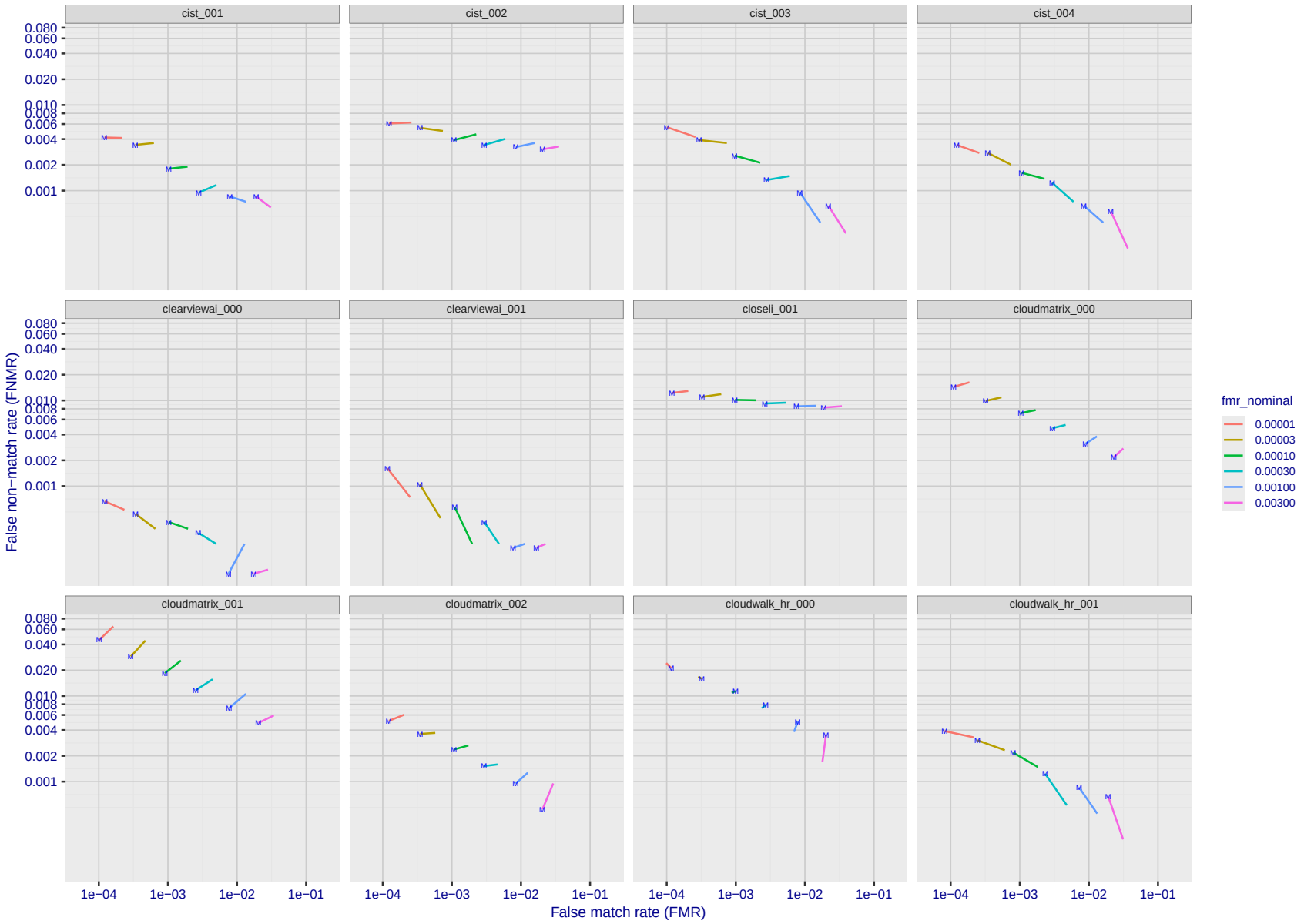

Figure 482: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

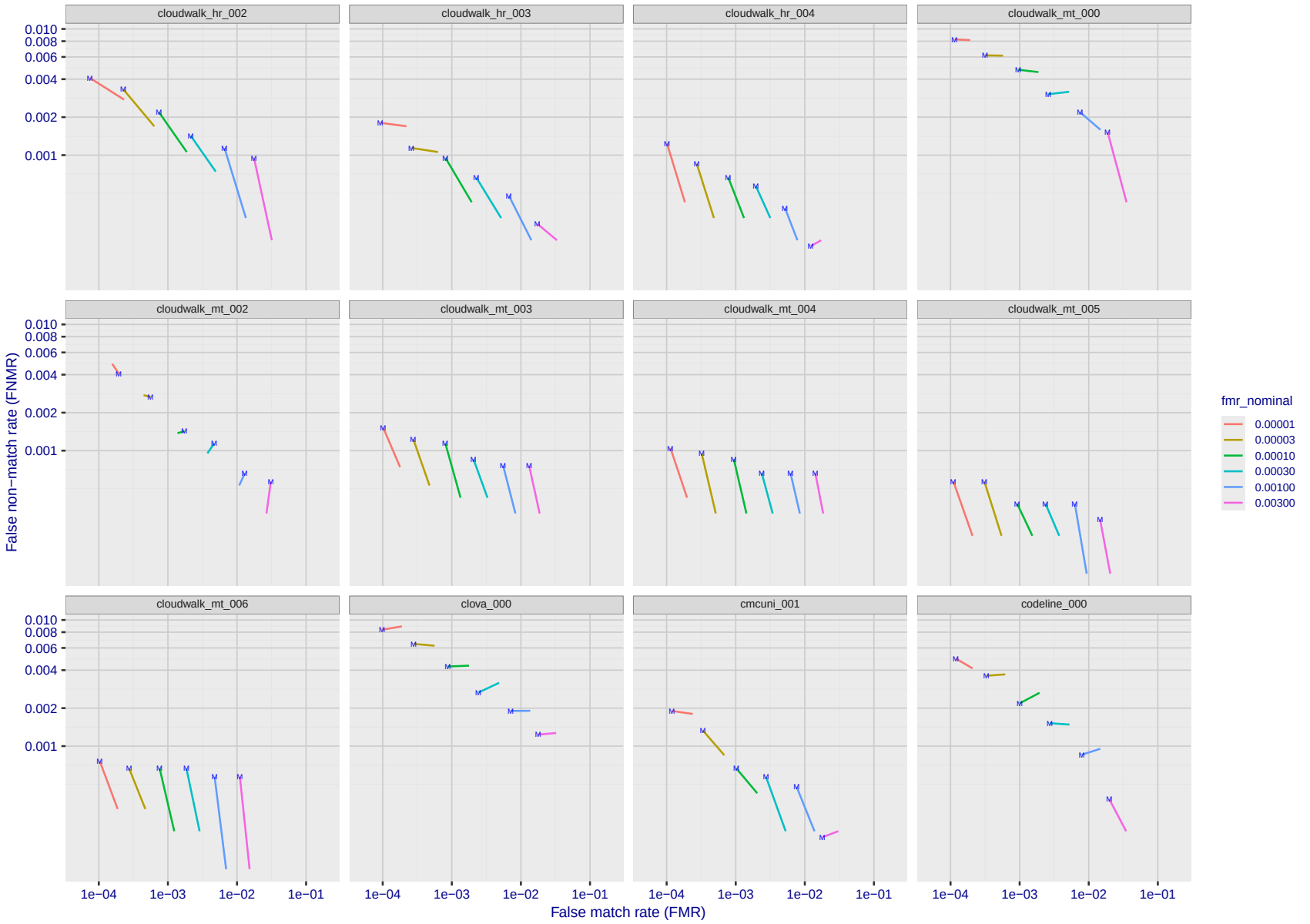

Figure 483: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

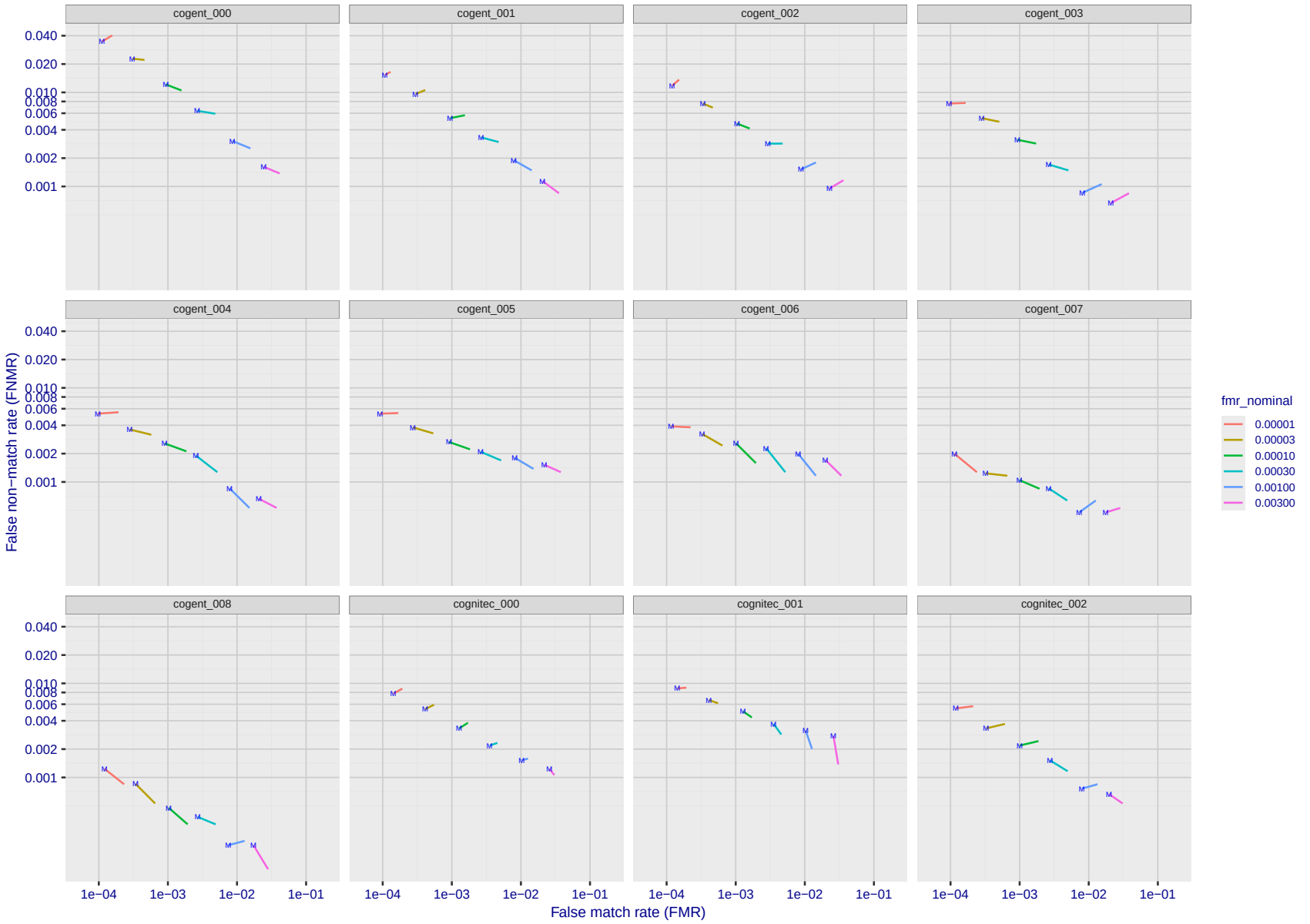

Figure 484: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

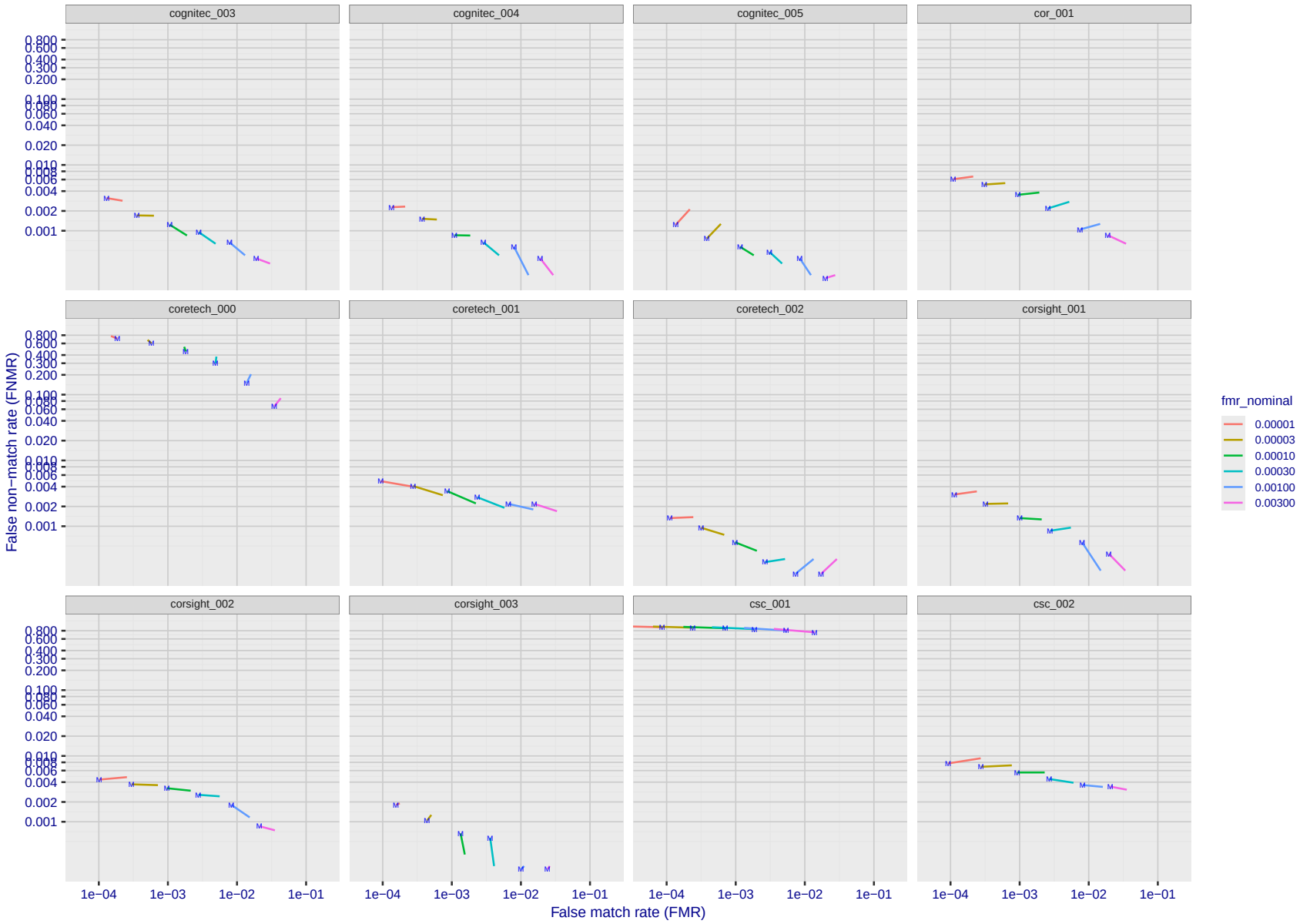

Figure 485: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

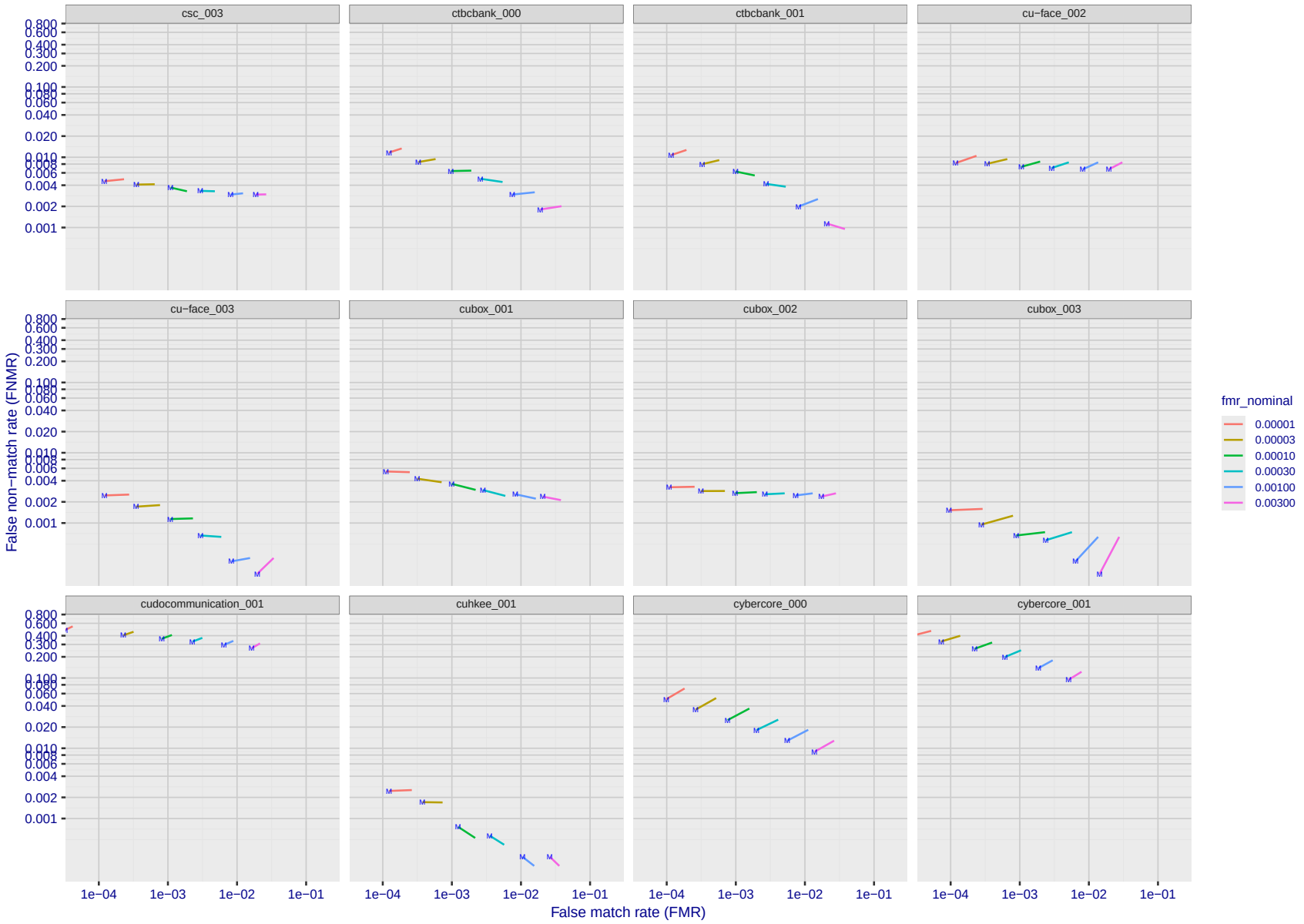

Figure 486: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

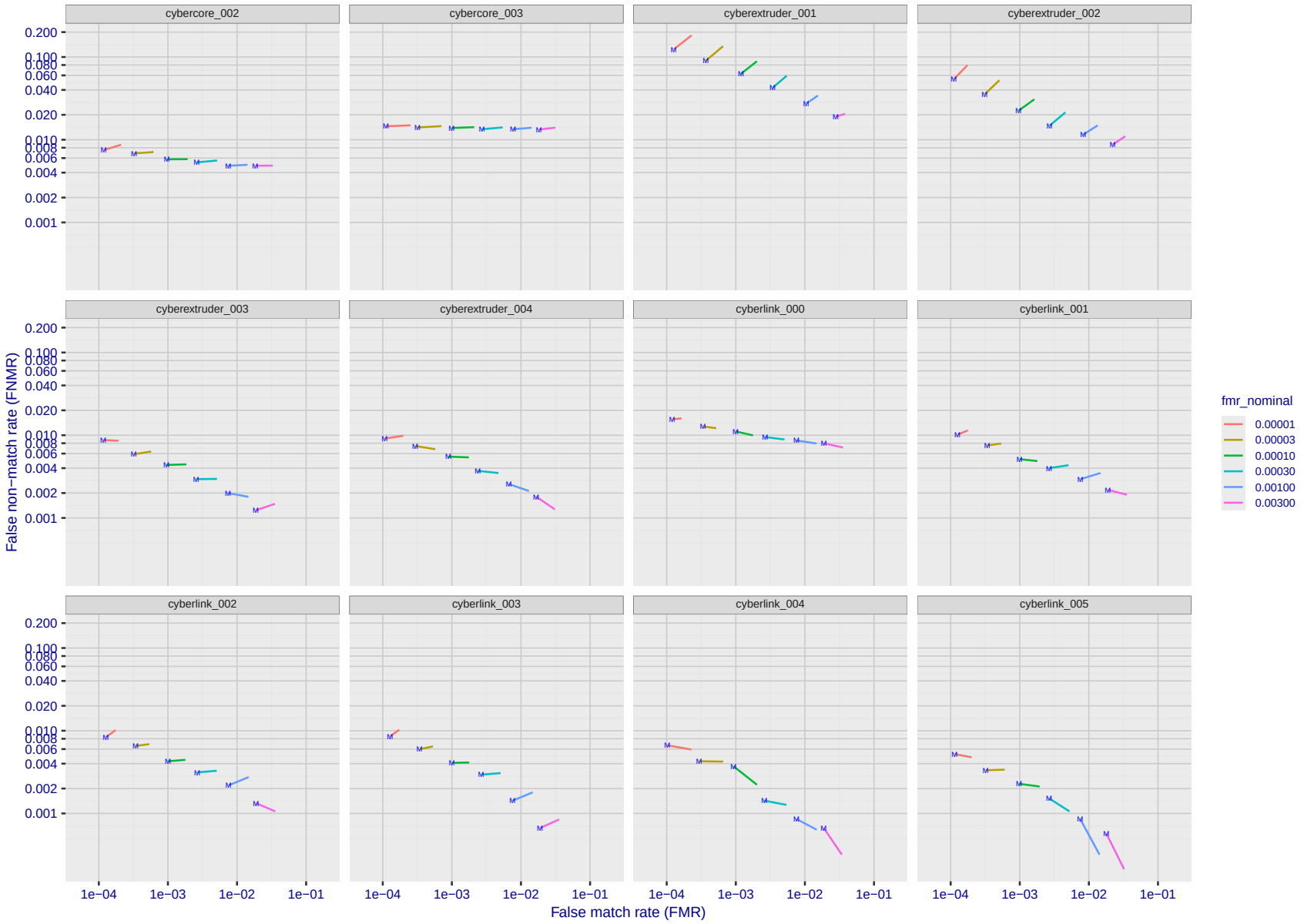

Figure 487: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

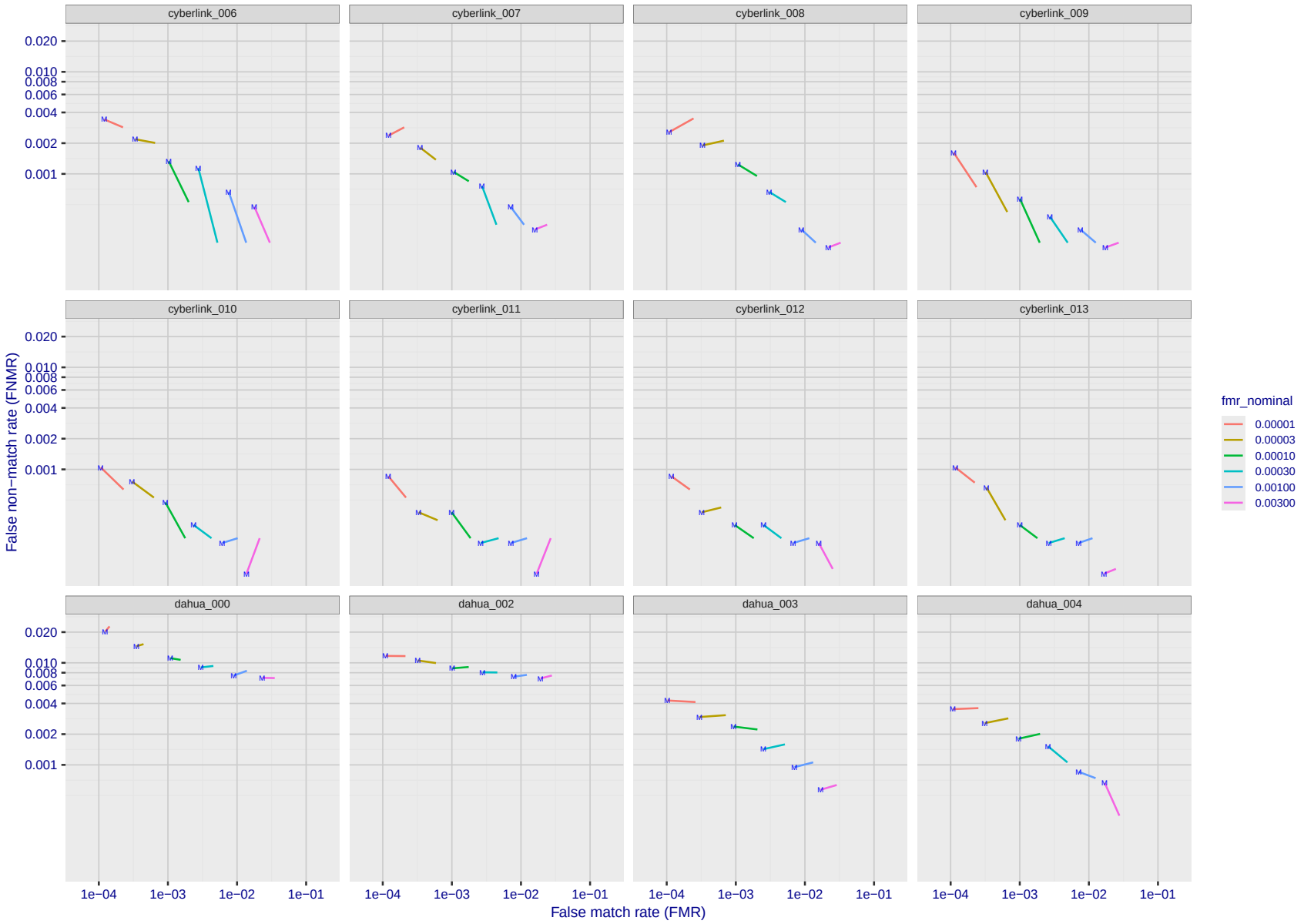

Figure 488: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

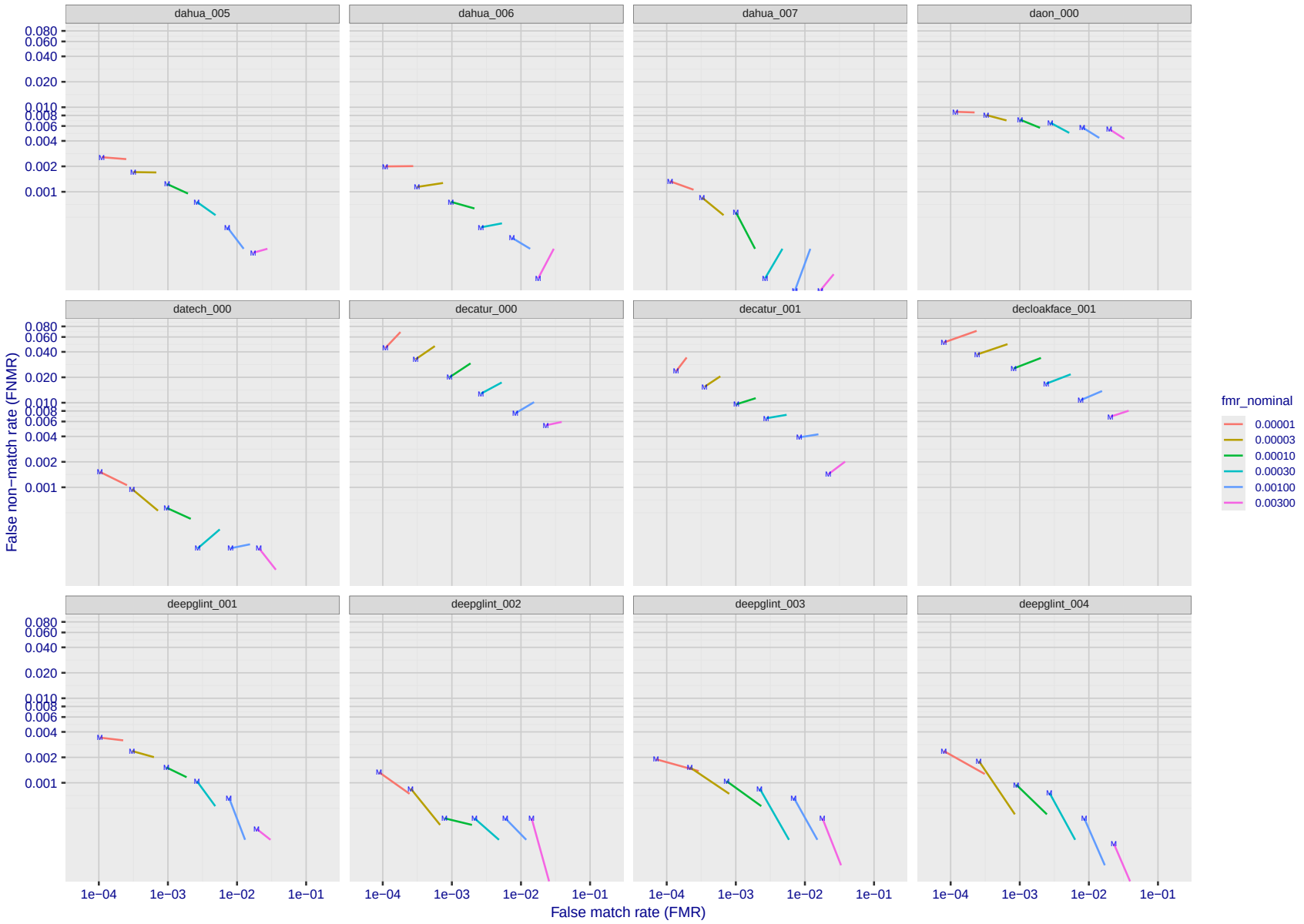

Figure 489: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

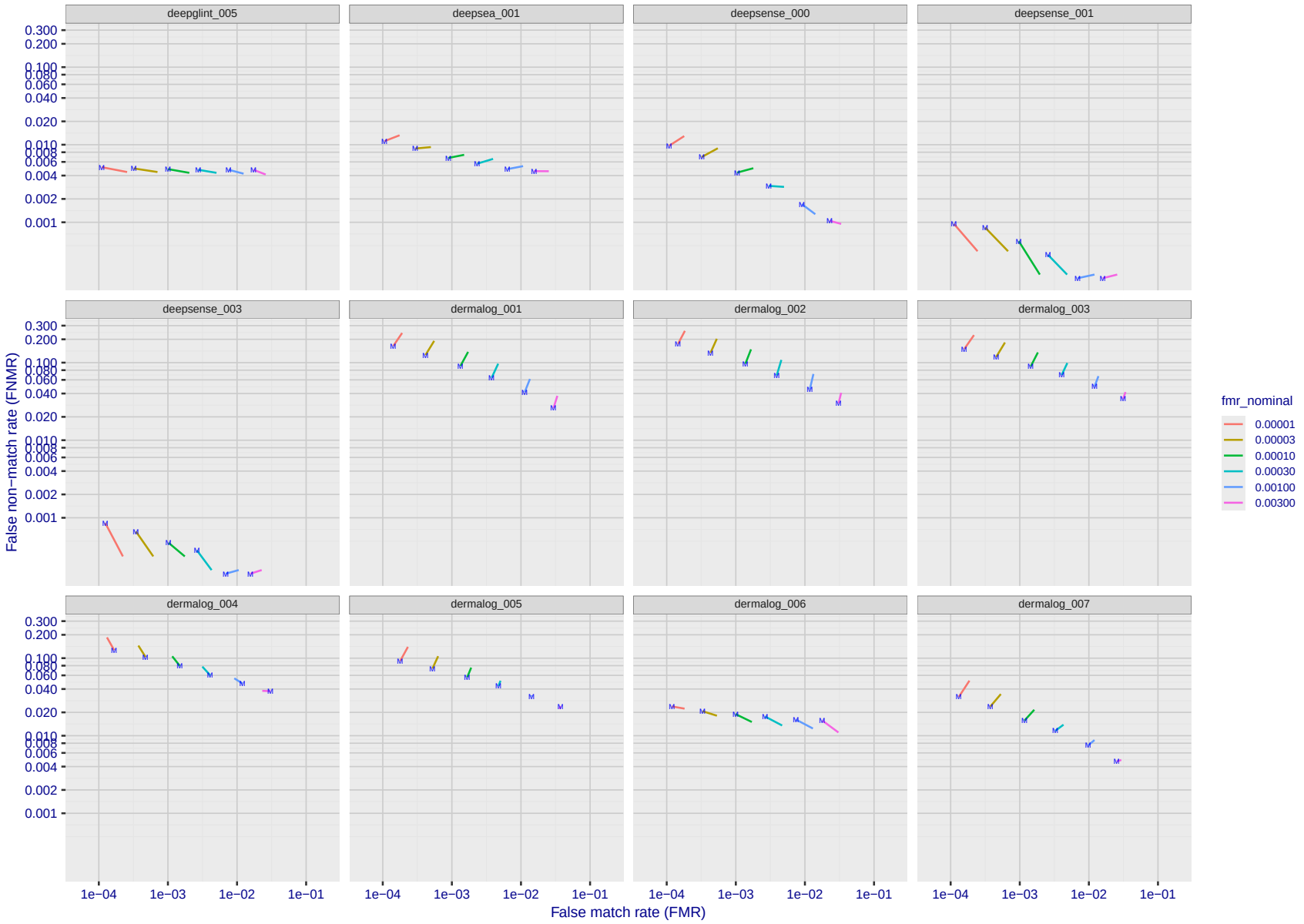

Figure 490: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

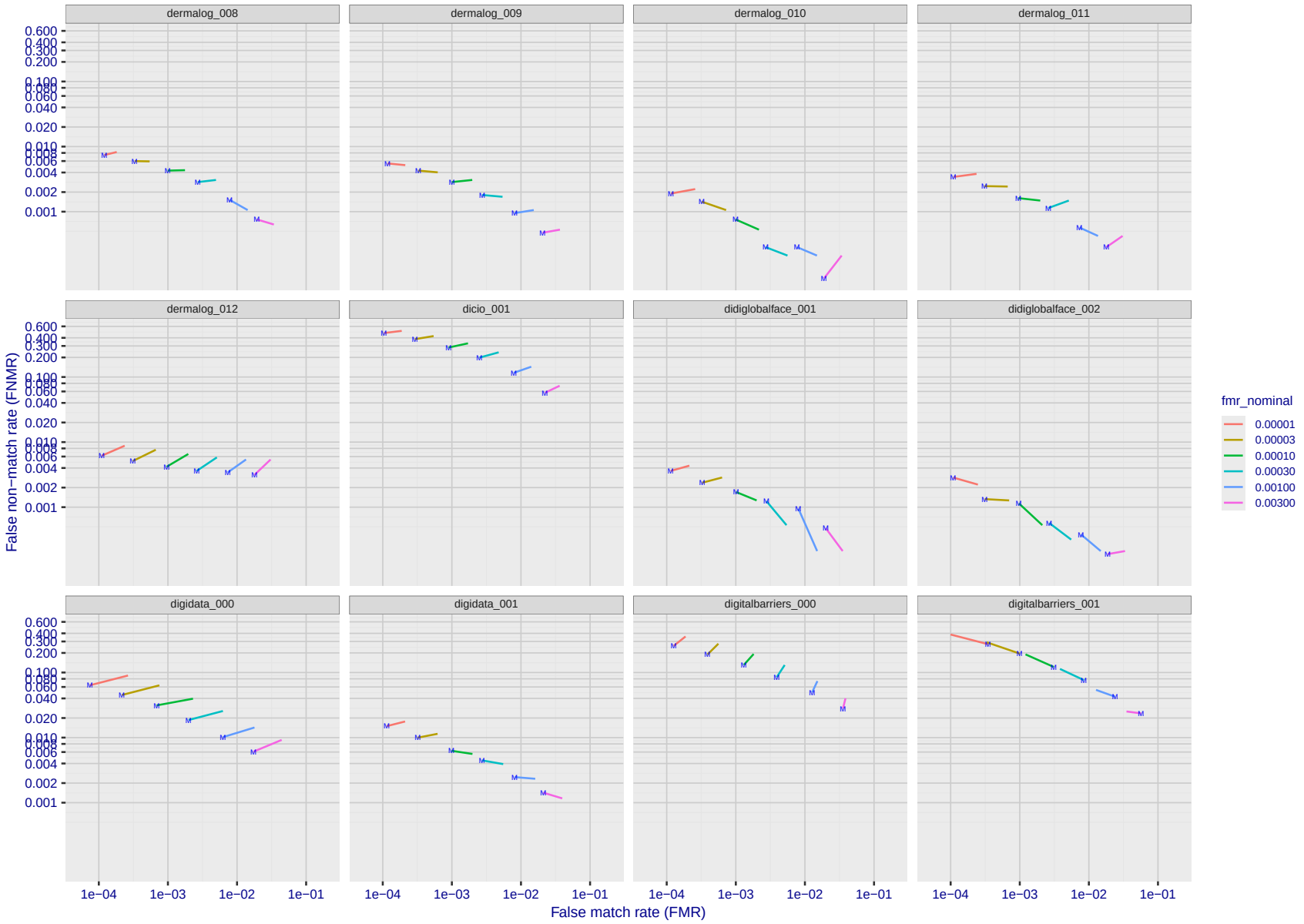

Figure 491: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

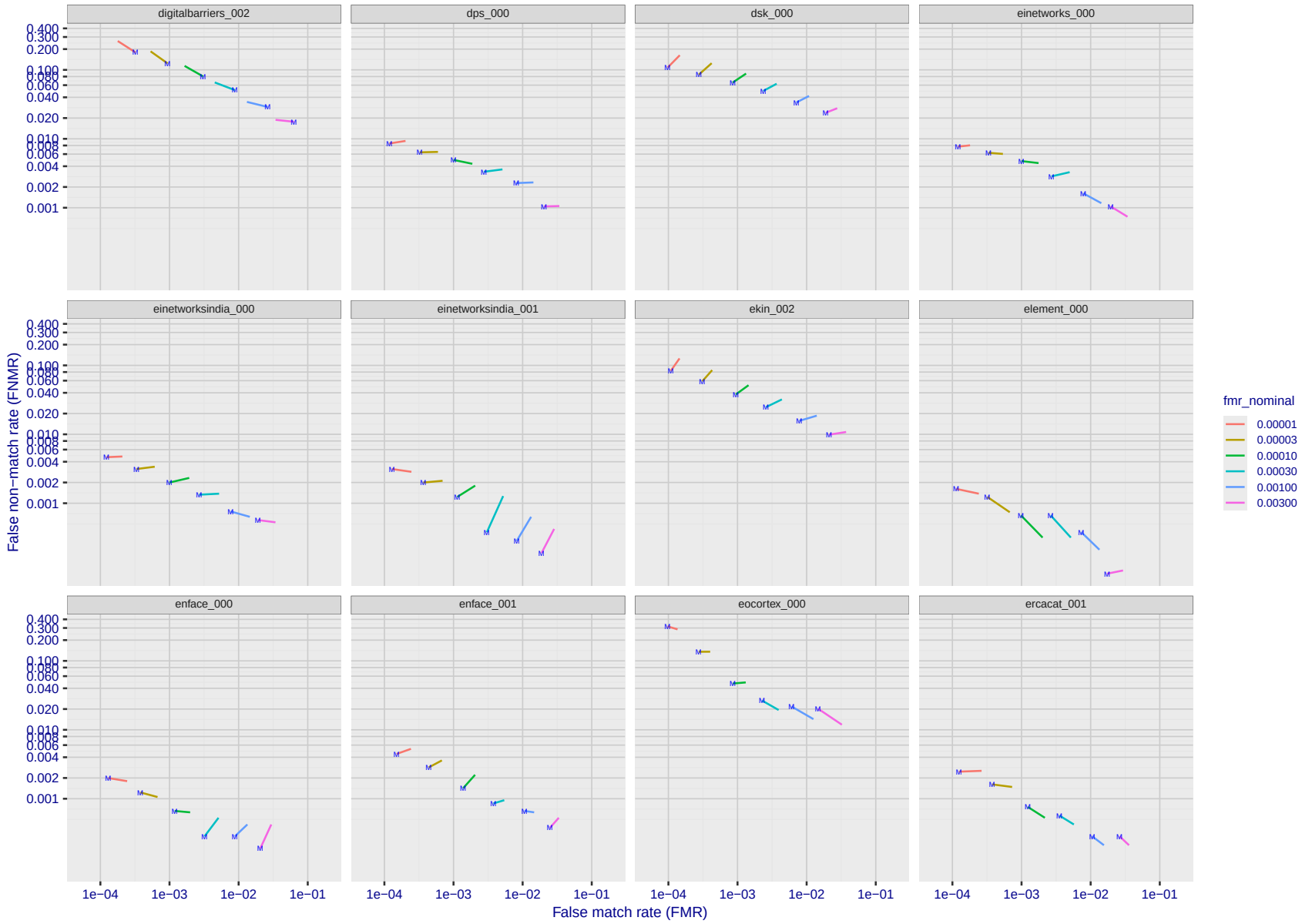

Figure 492: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

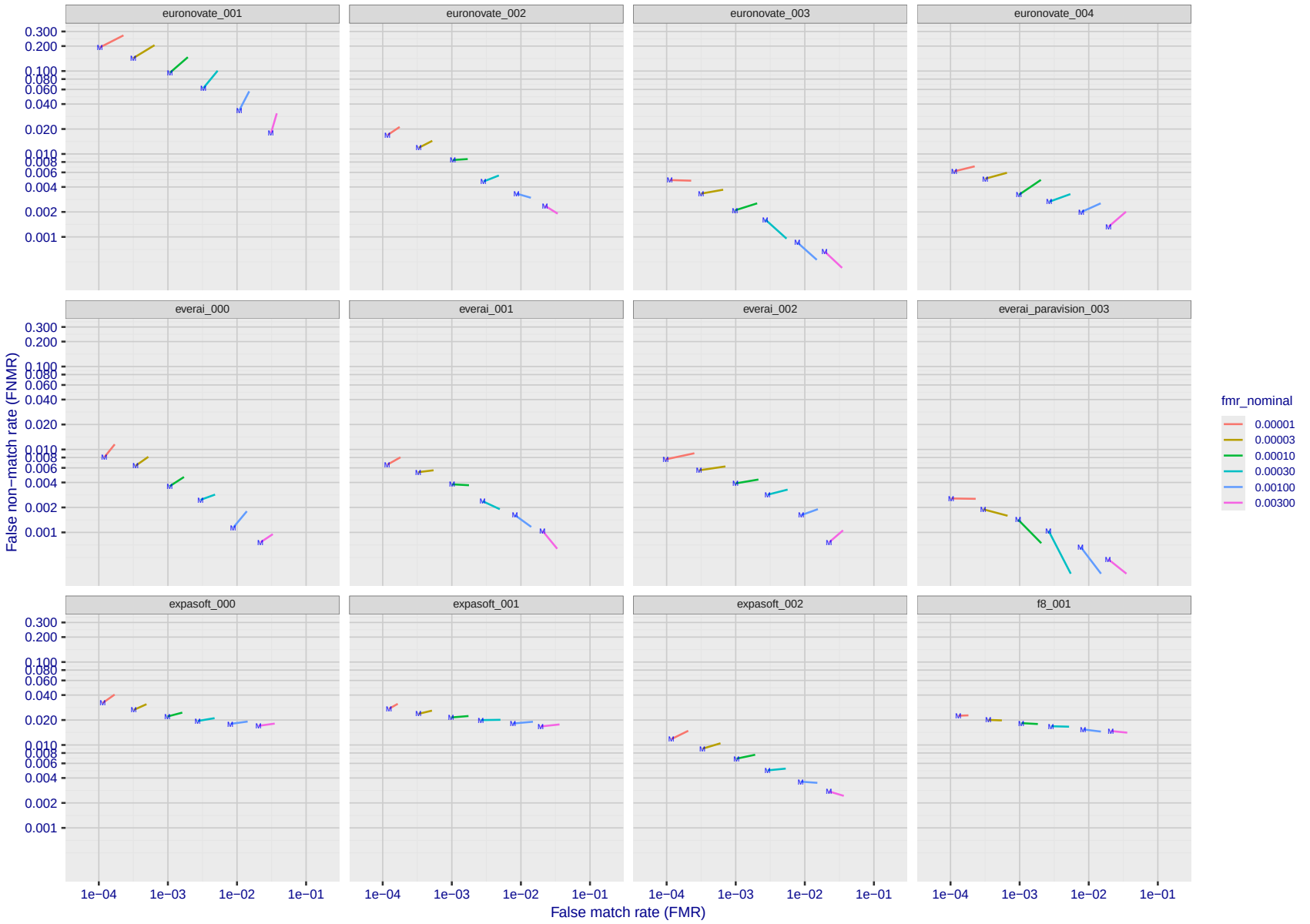

Figure 493: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

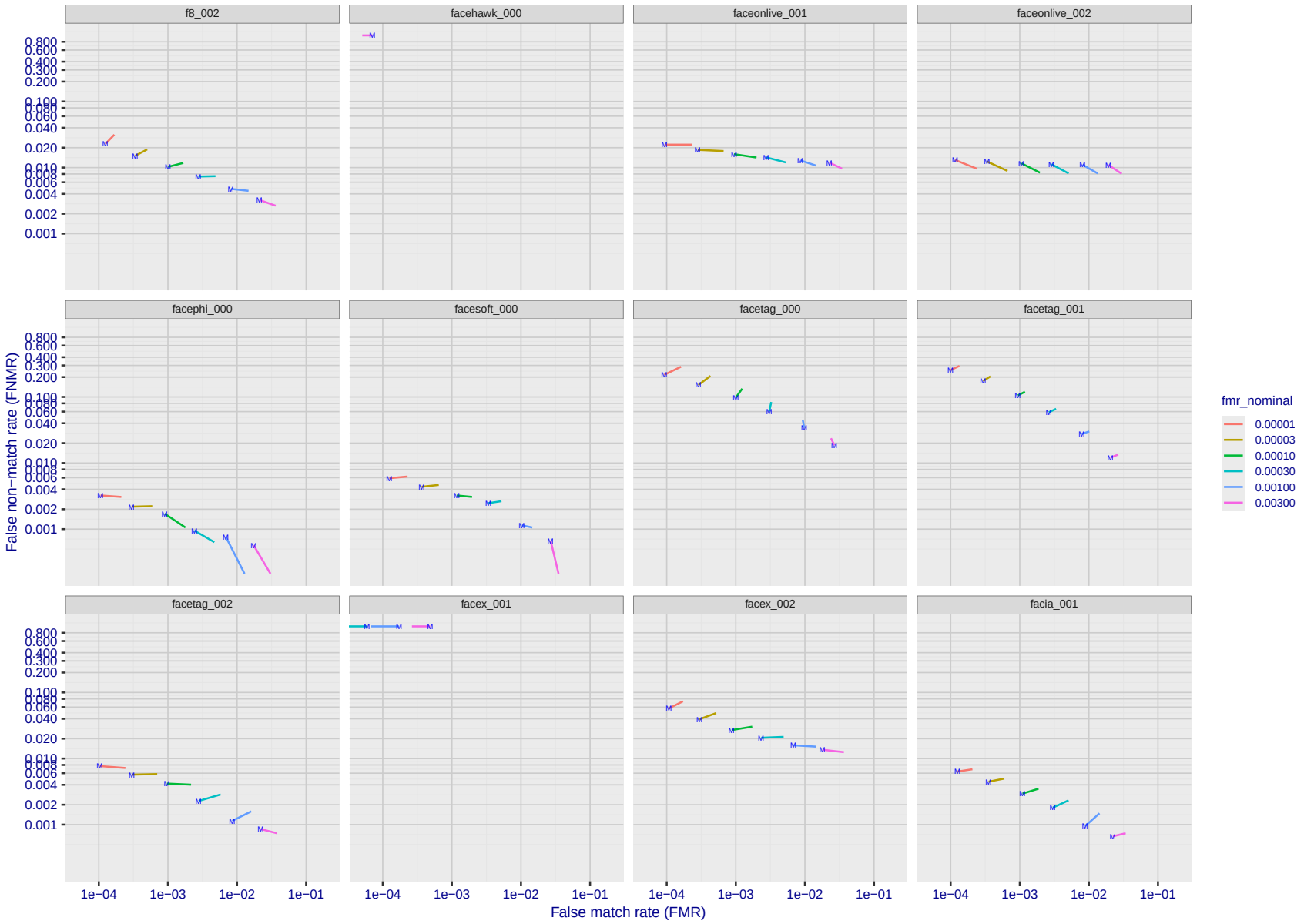

Figure 494: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

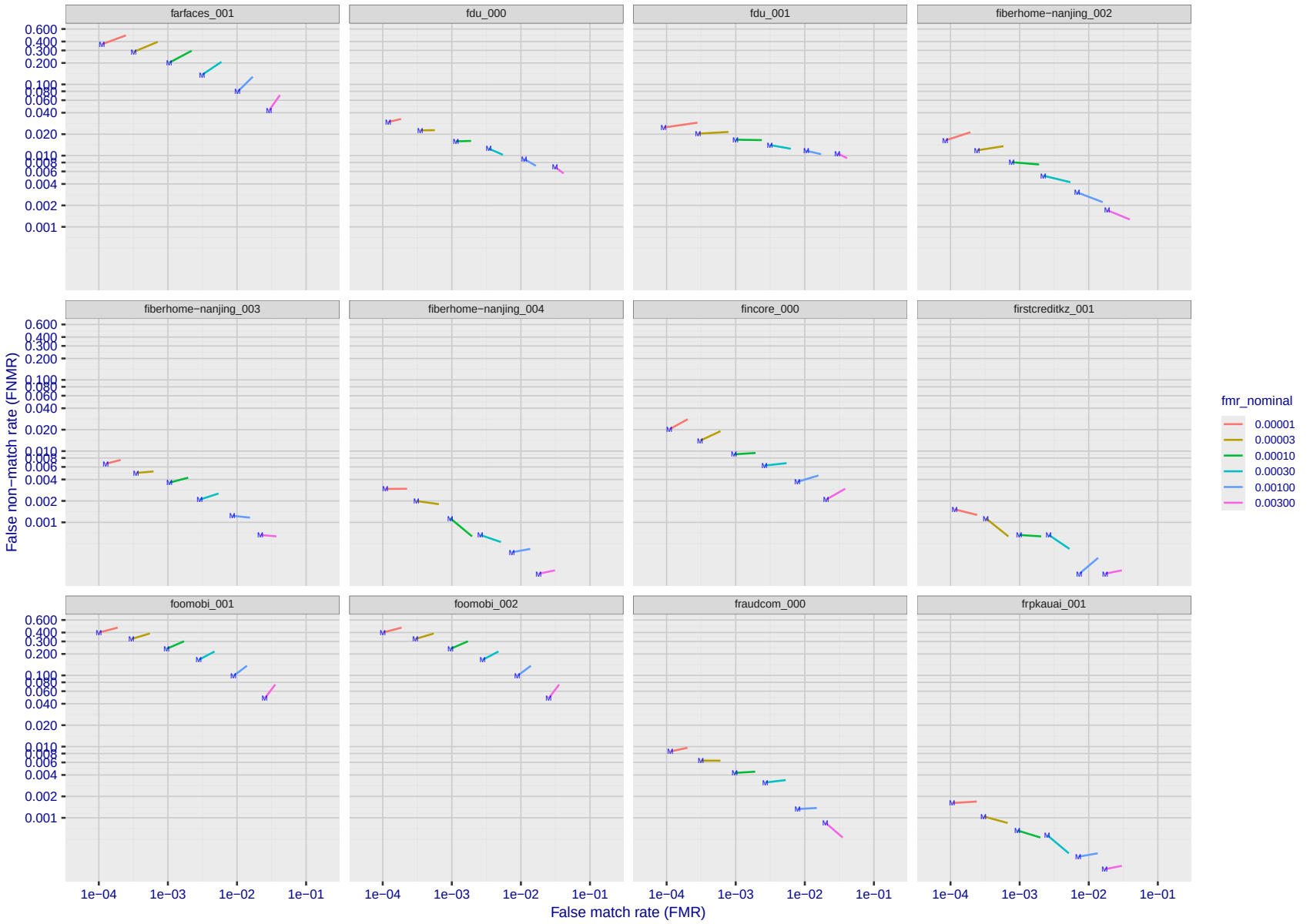

Figure 495: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

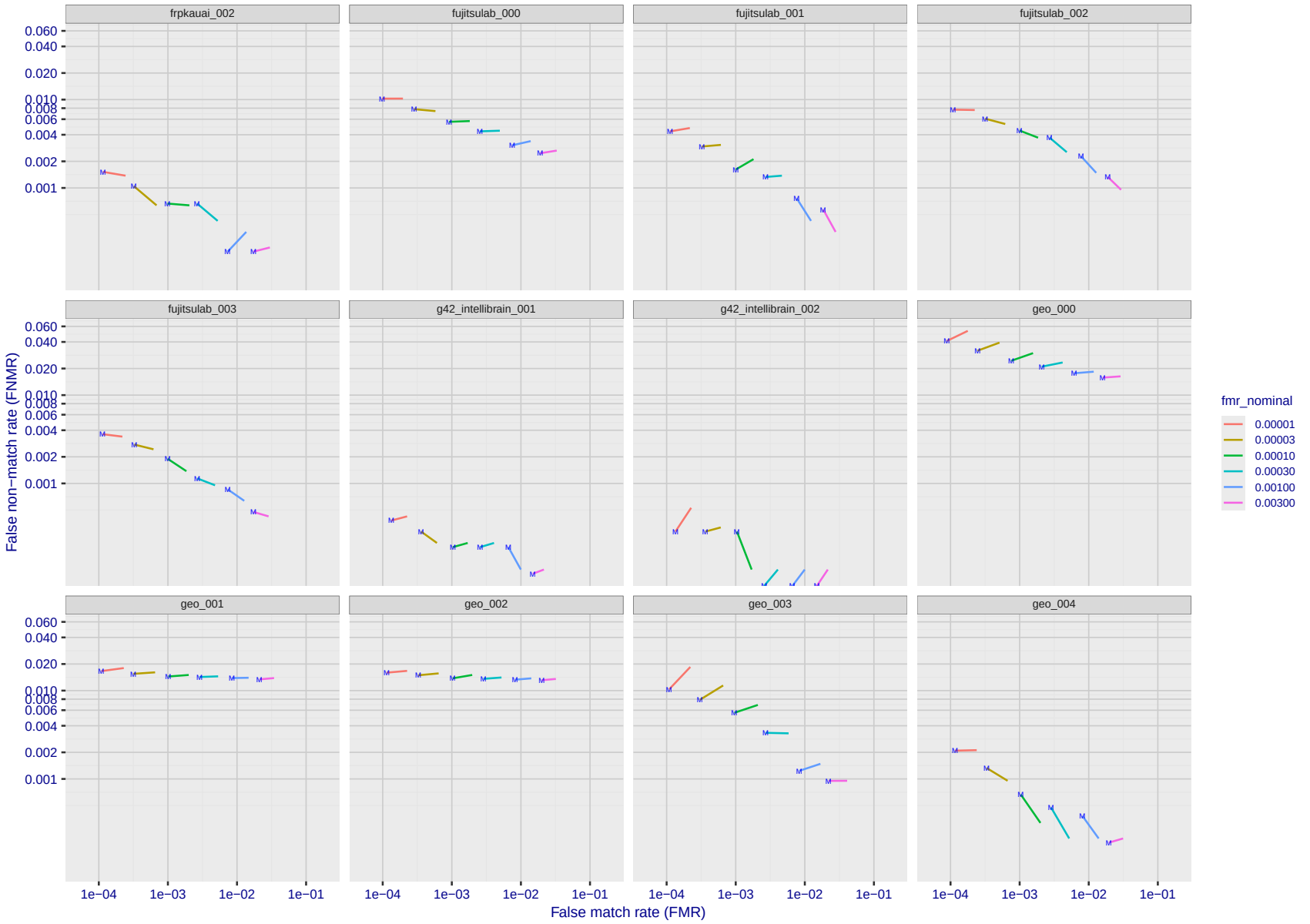

Figure 496: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

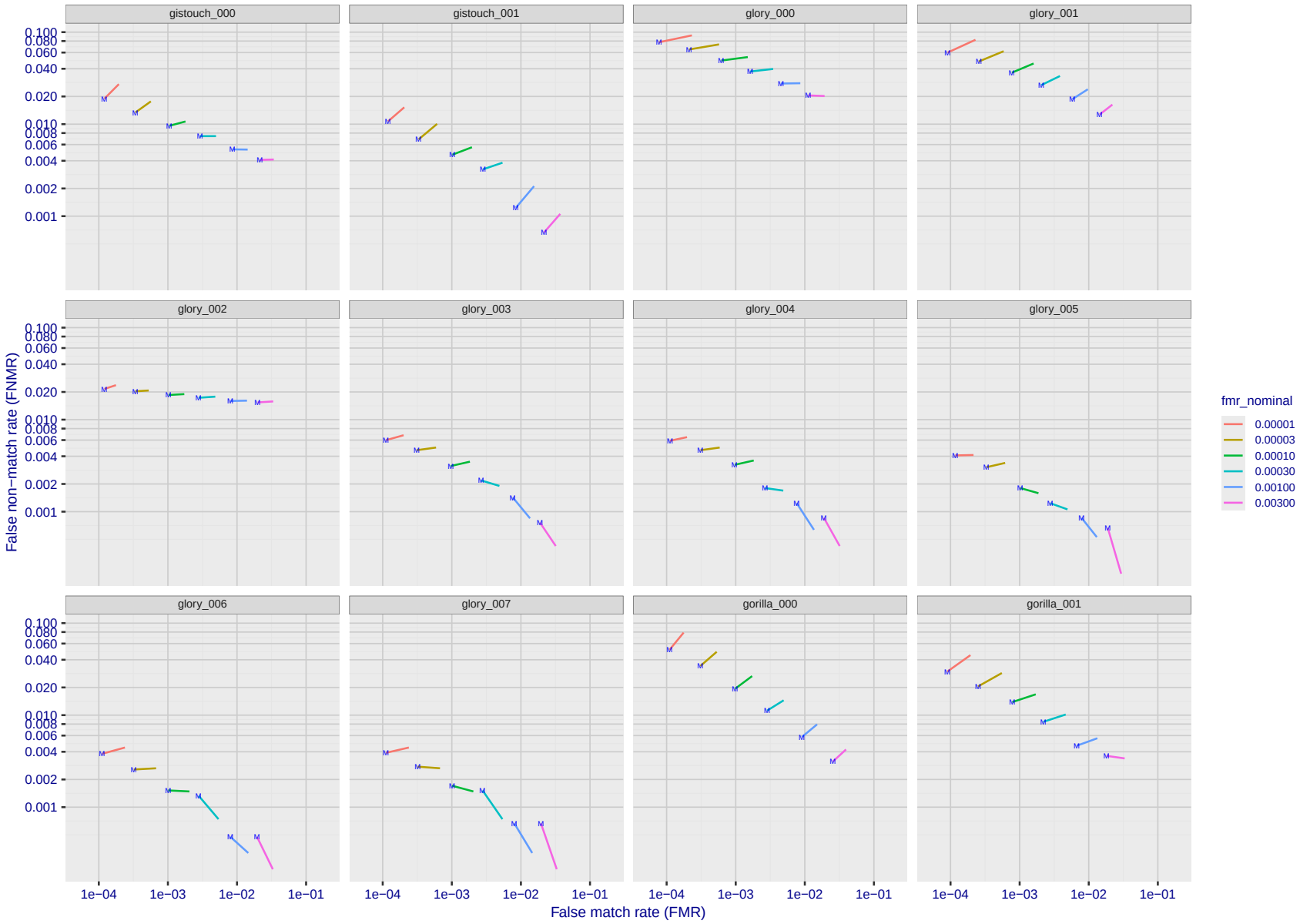

Figure 497: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

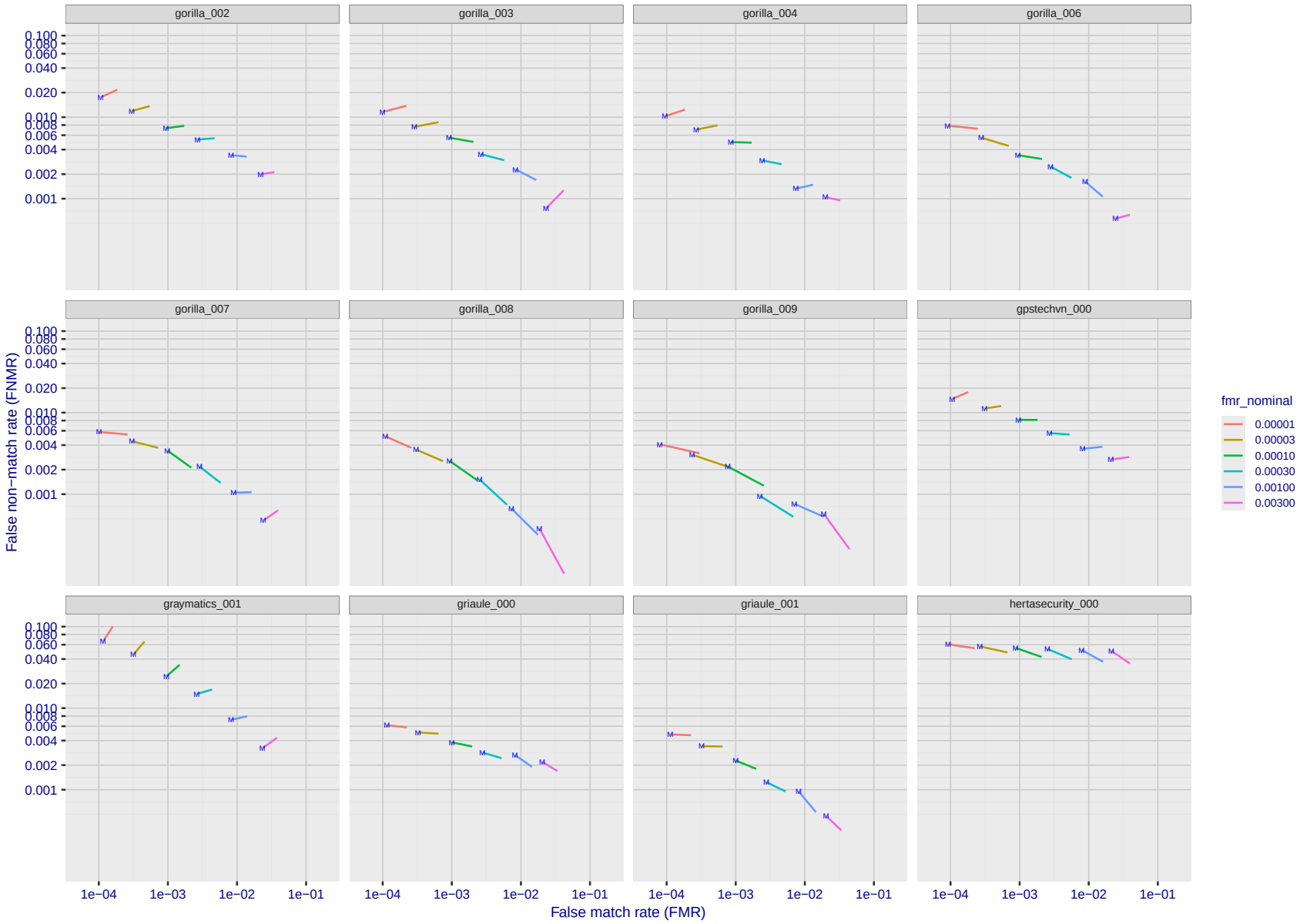

Figure 498: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

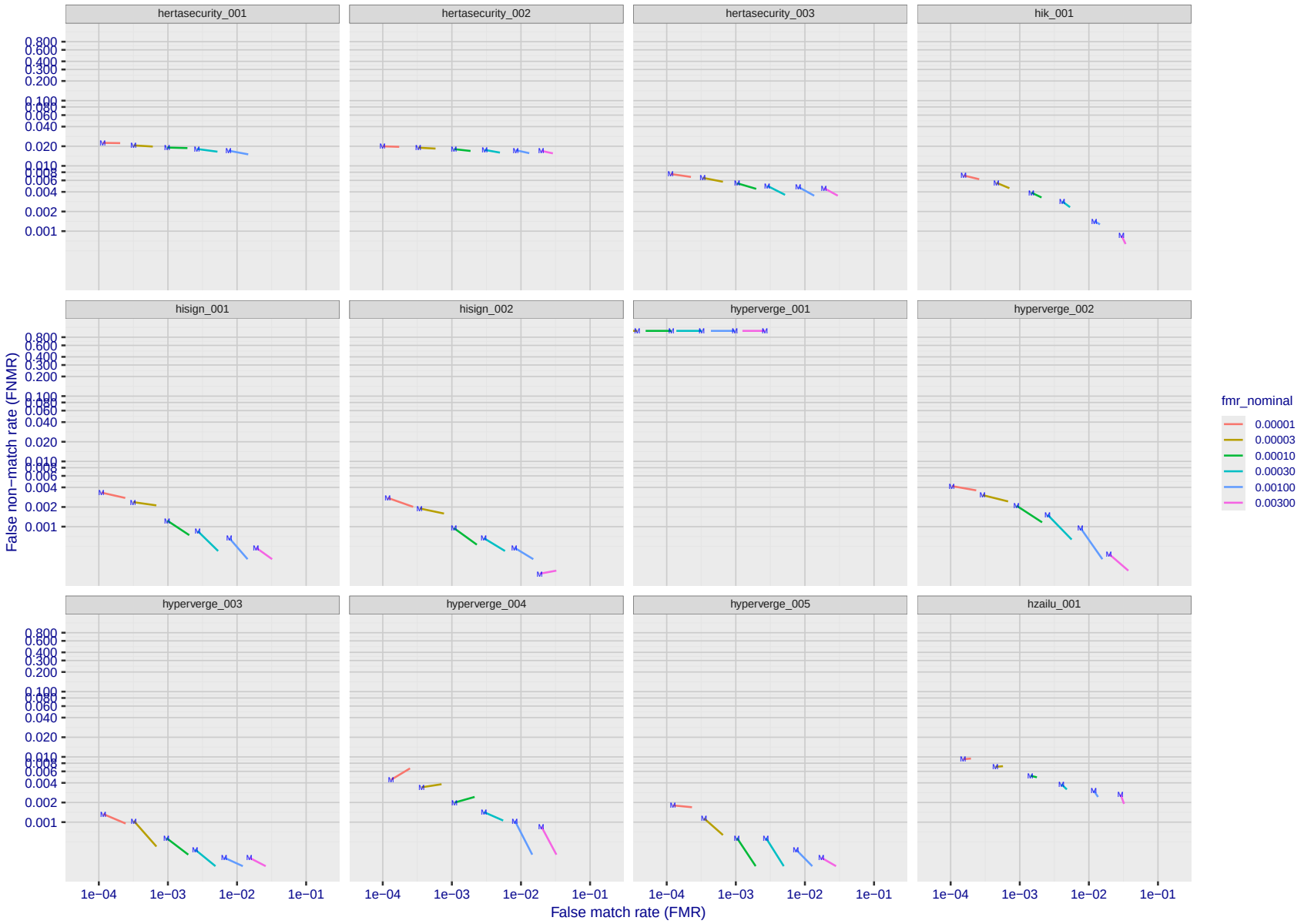

Figure 499: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

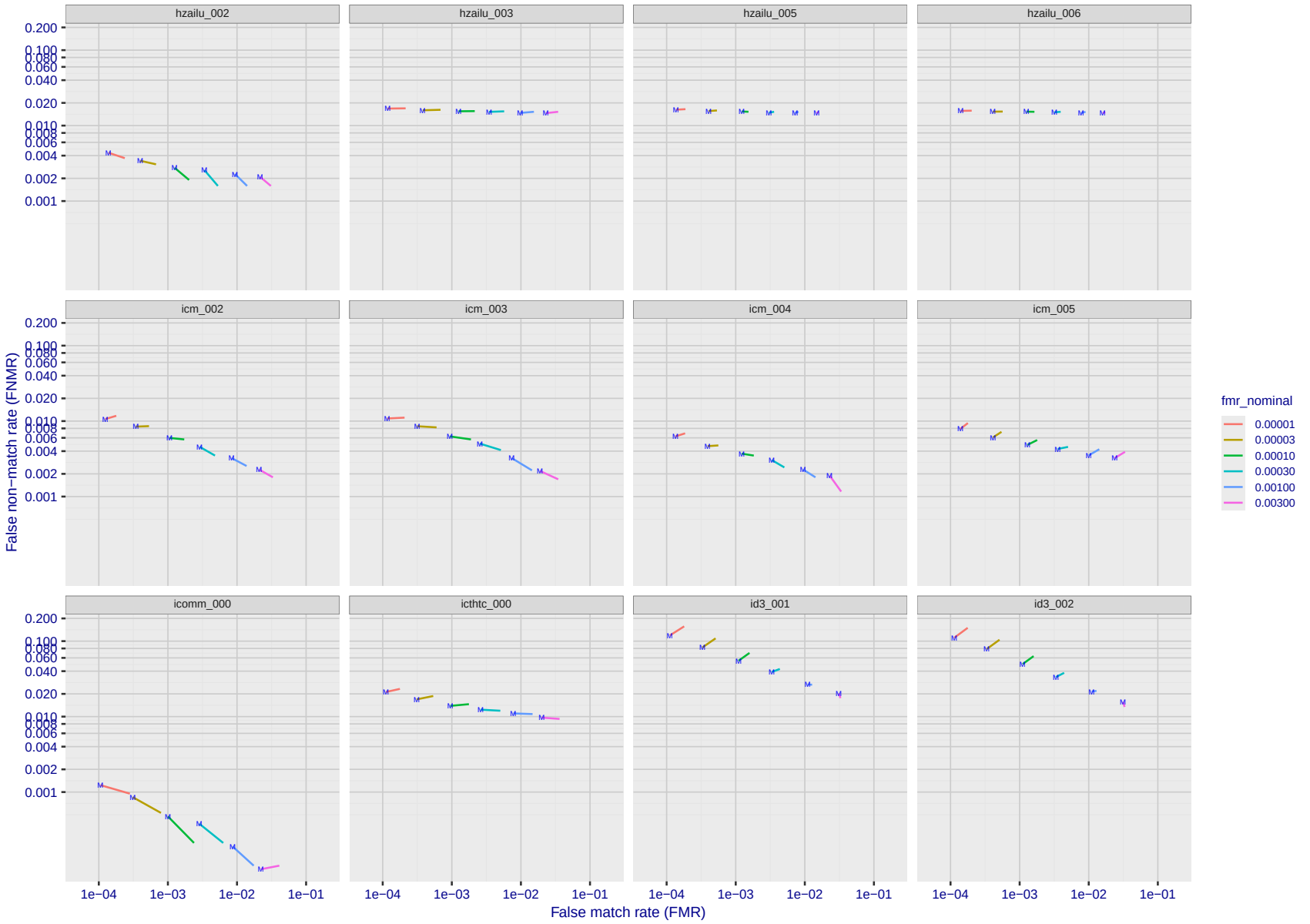

Figure 500: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

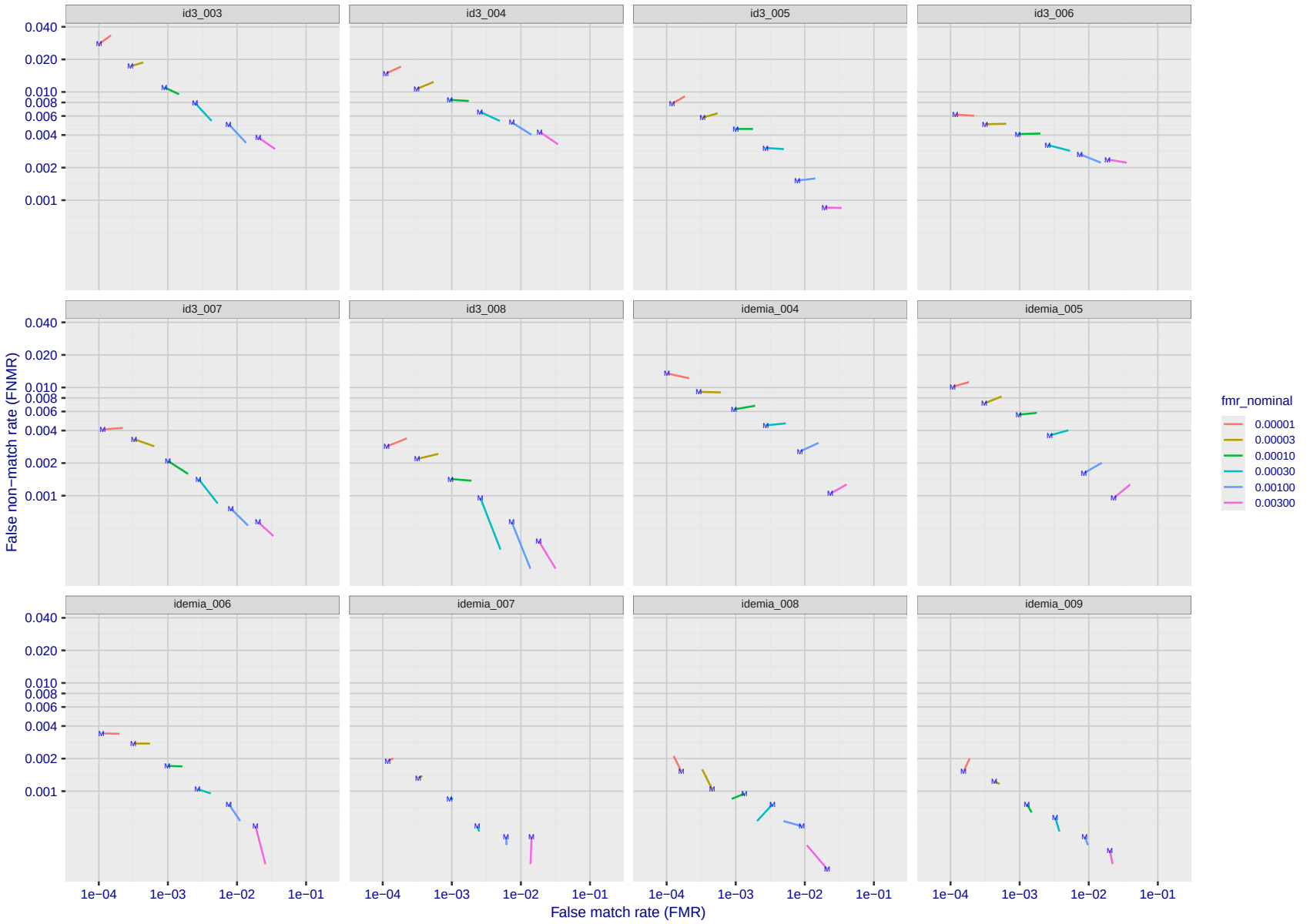

Figure 501: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

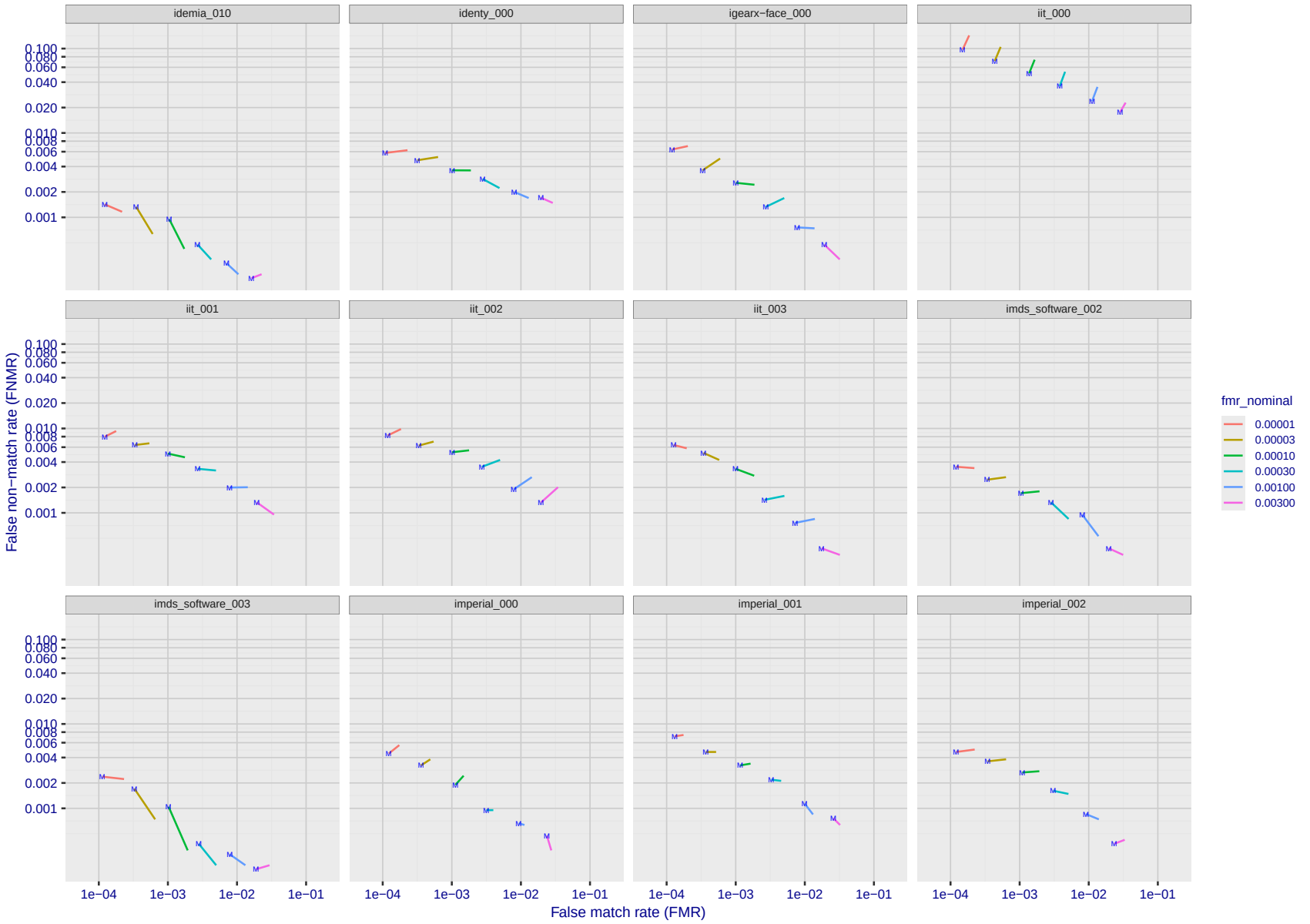

Figure 502: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

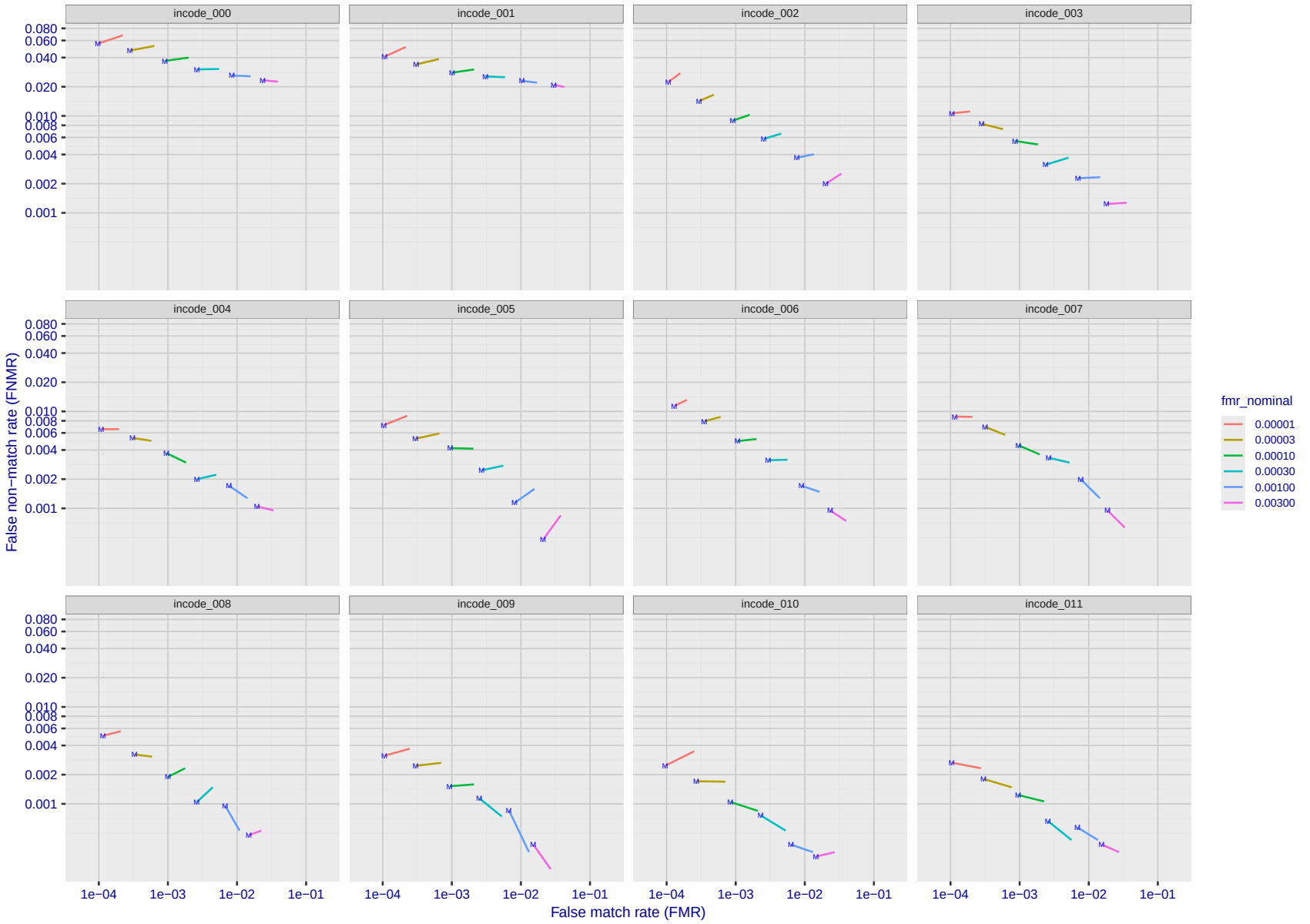

Figure 503: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

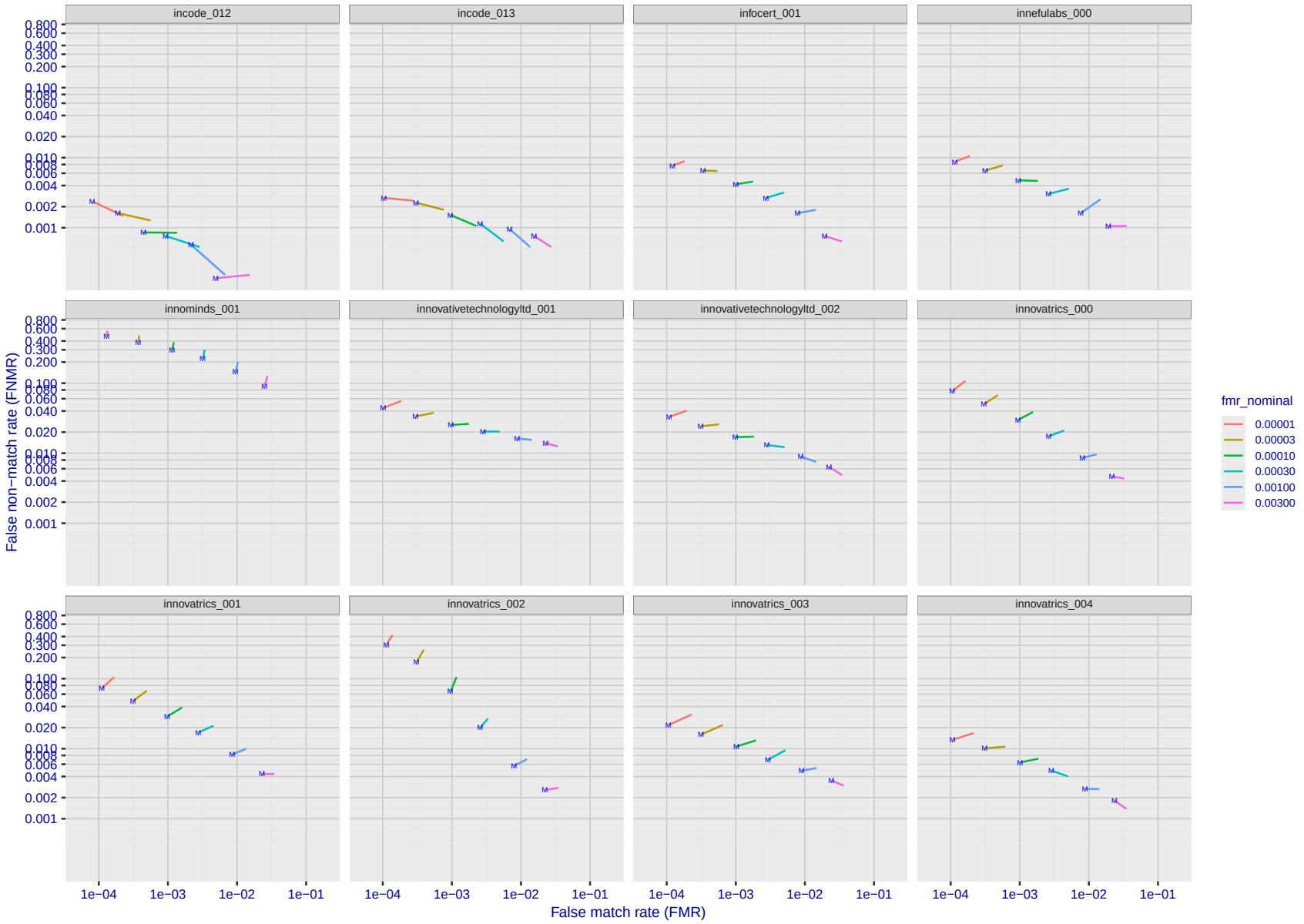

Figure 504: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

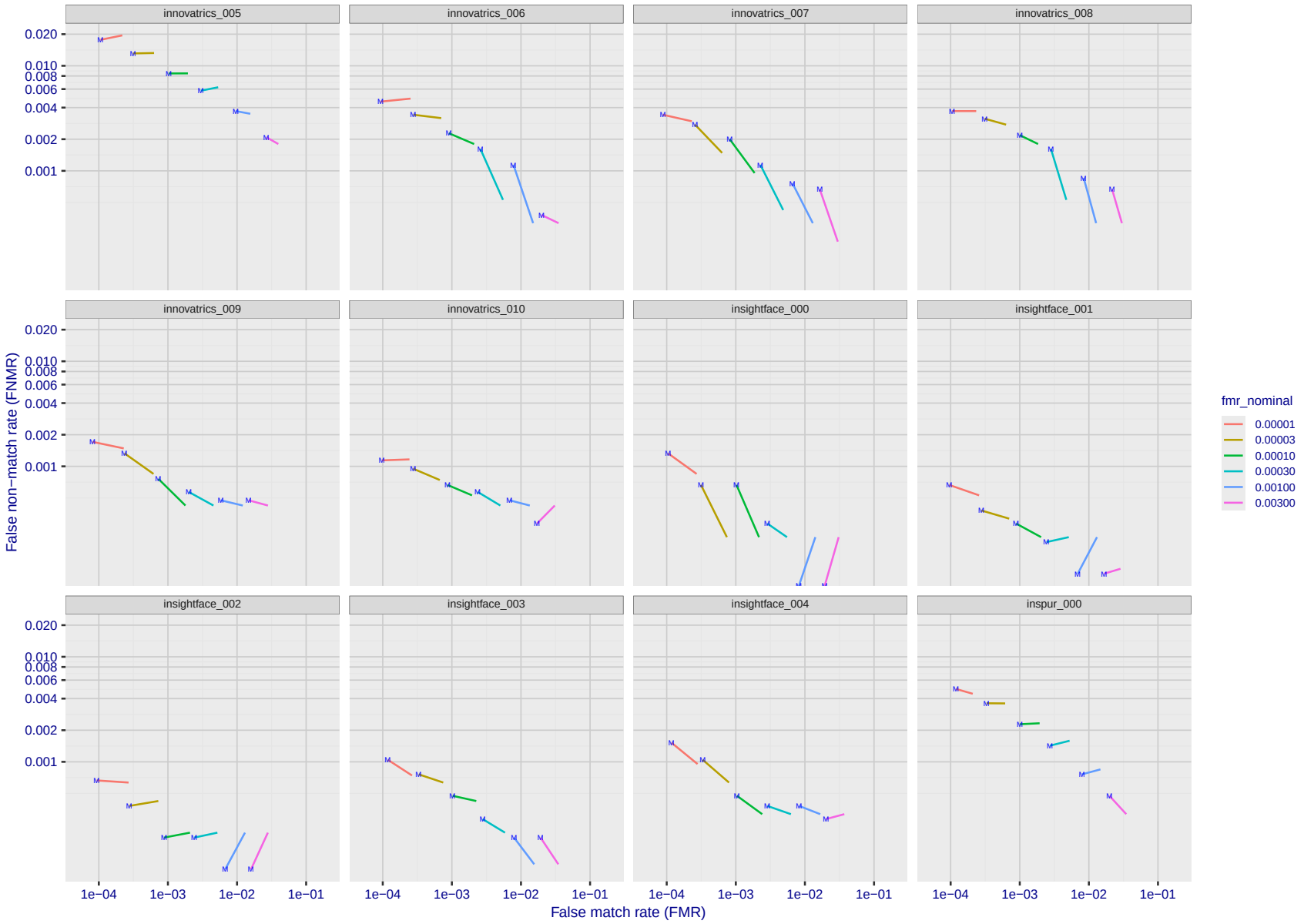

Figure 505: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

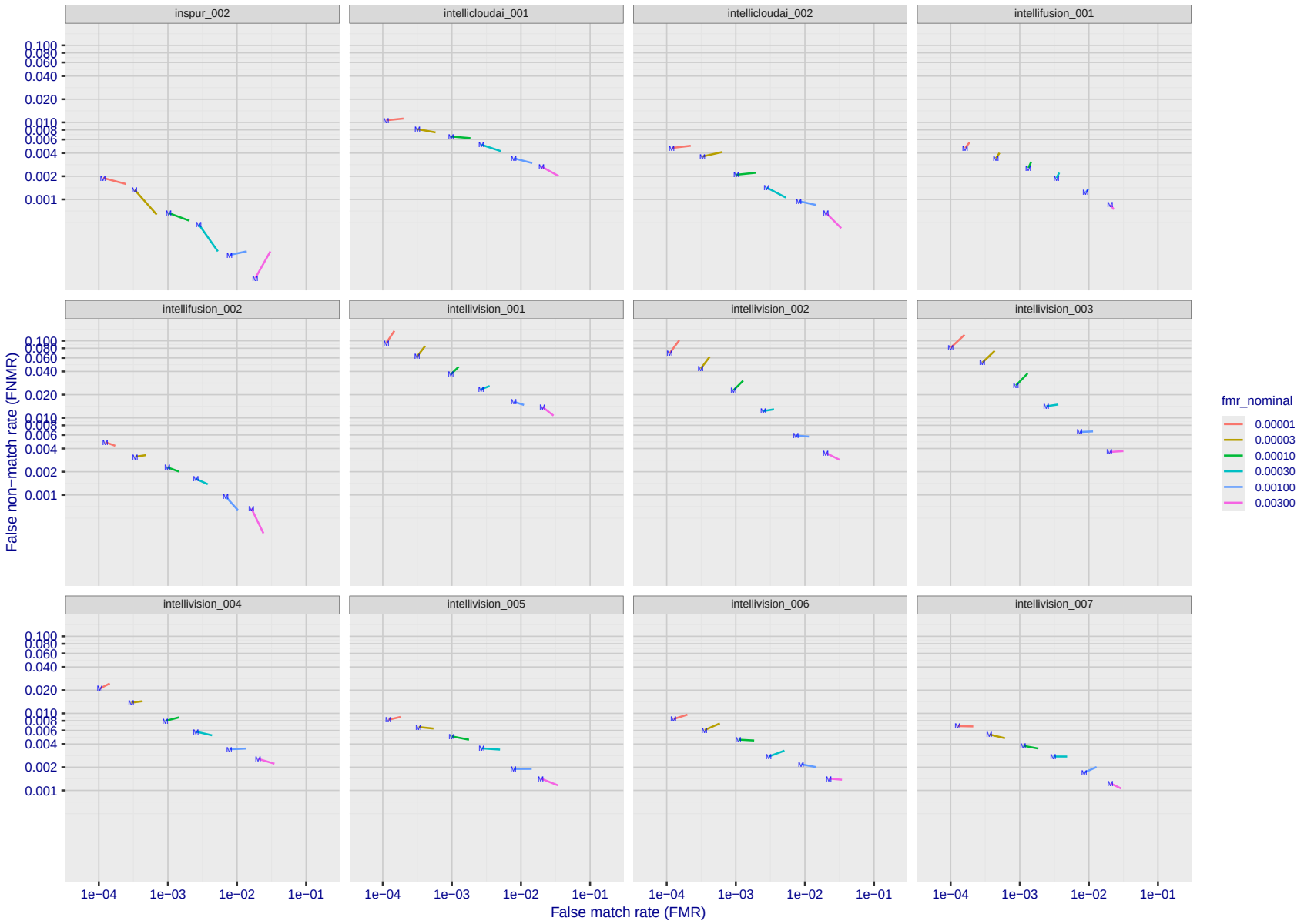

Figure 506: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

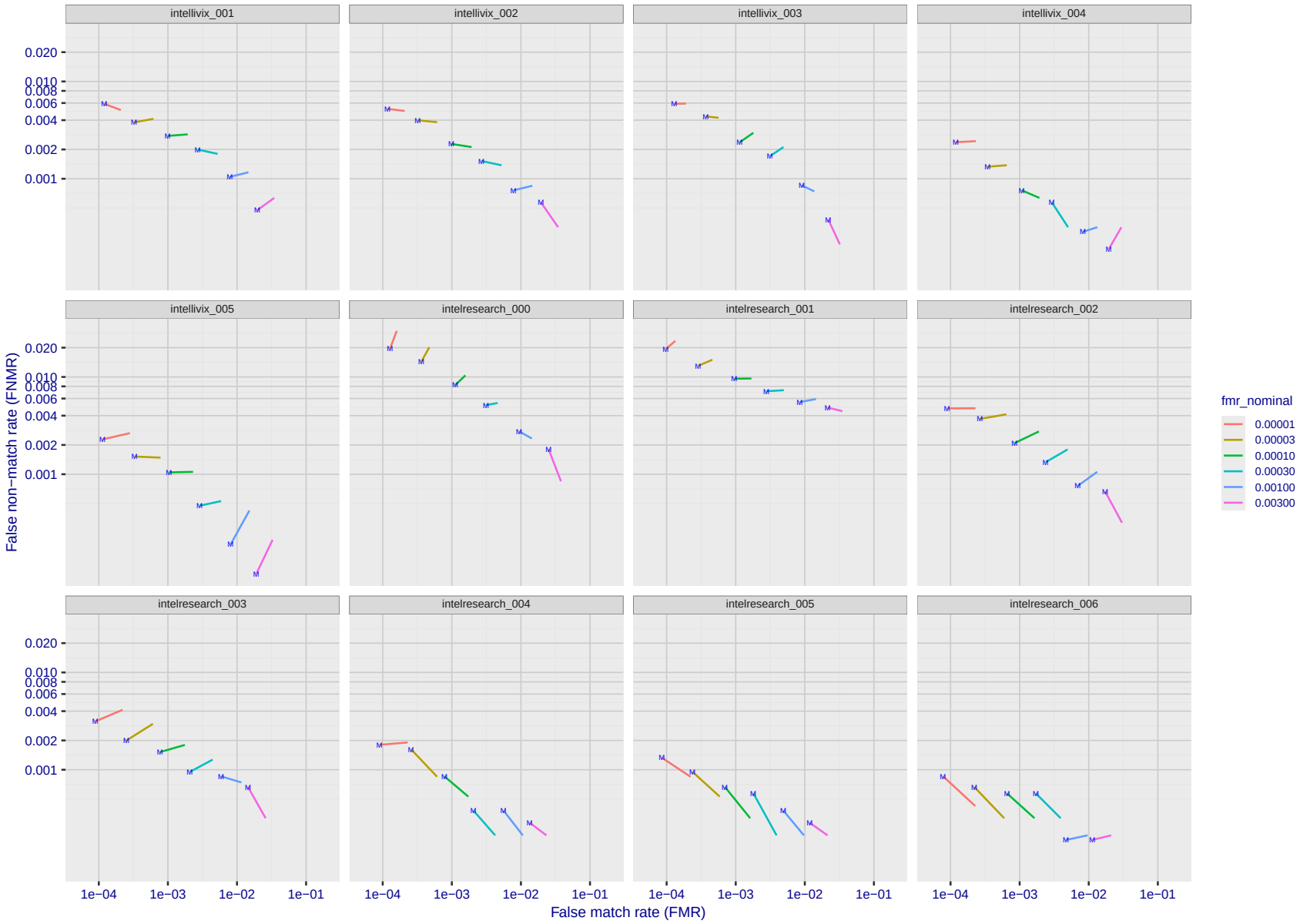

Figure 507: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

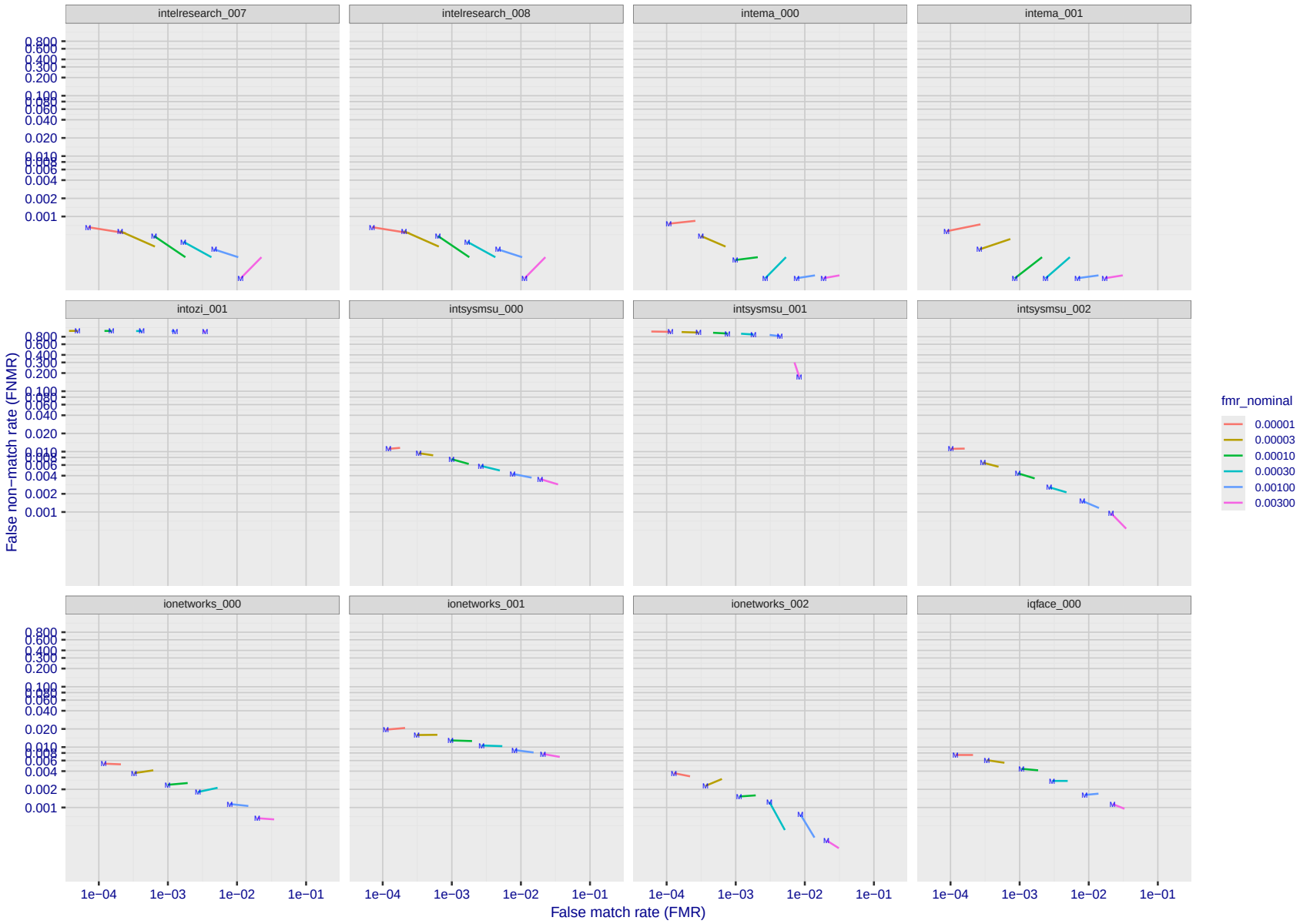

Figure 508: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

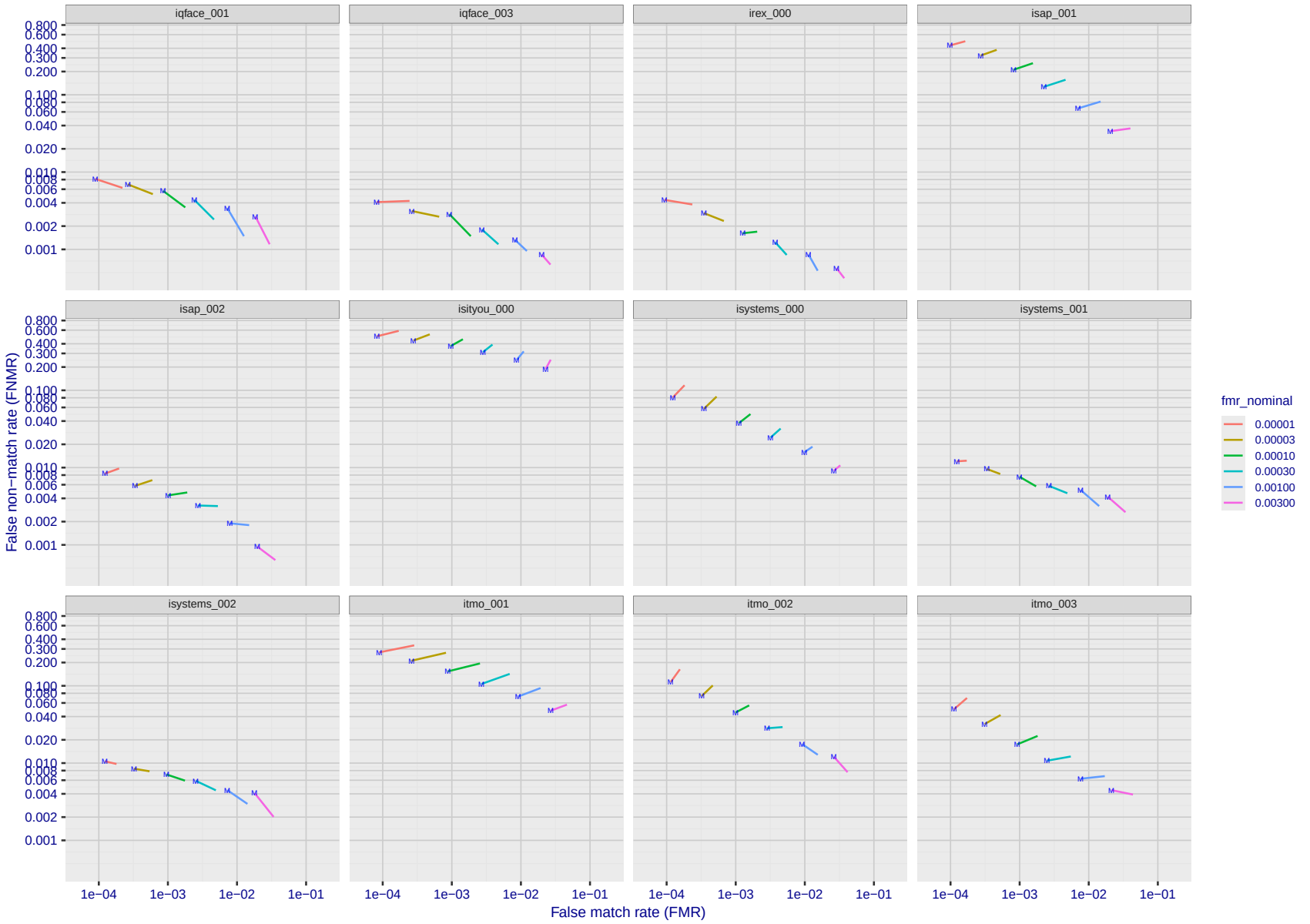

Figure 509: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

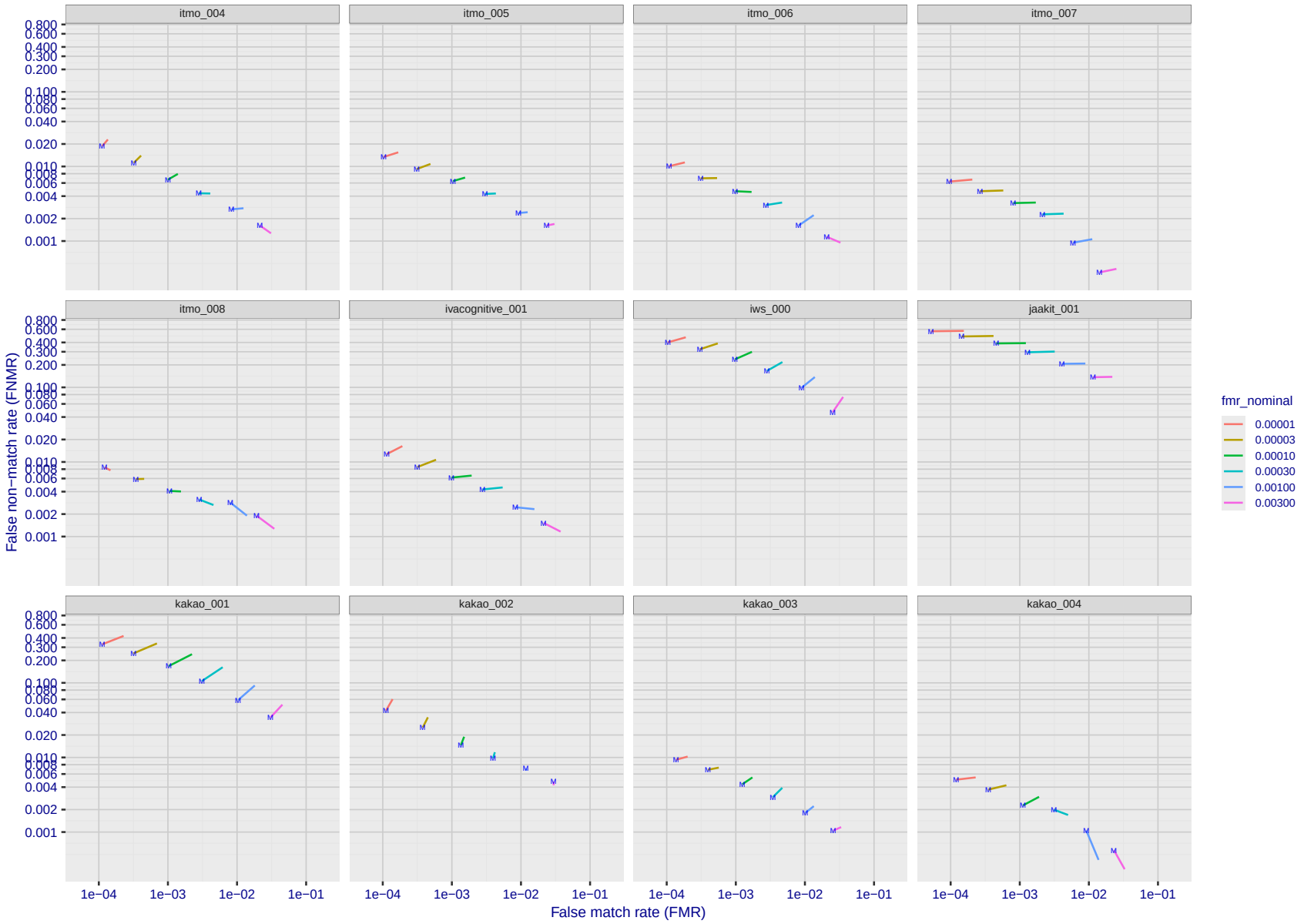

Figure 510: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

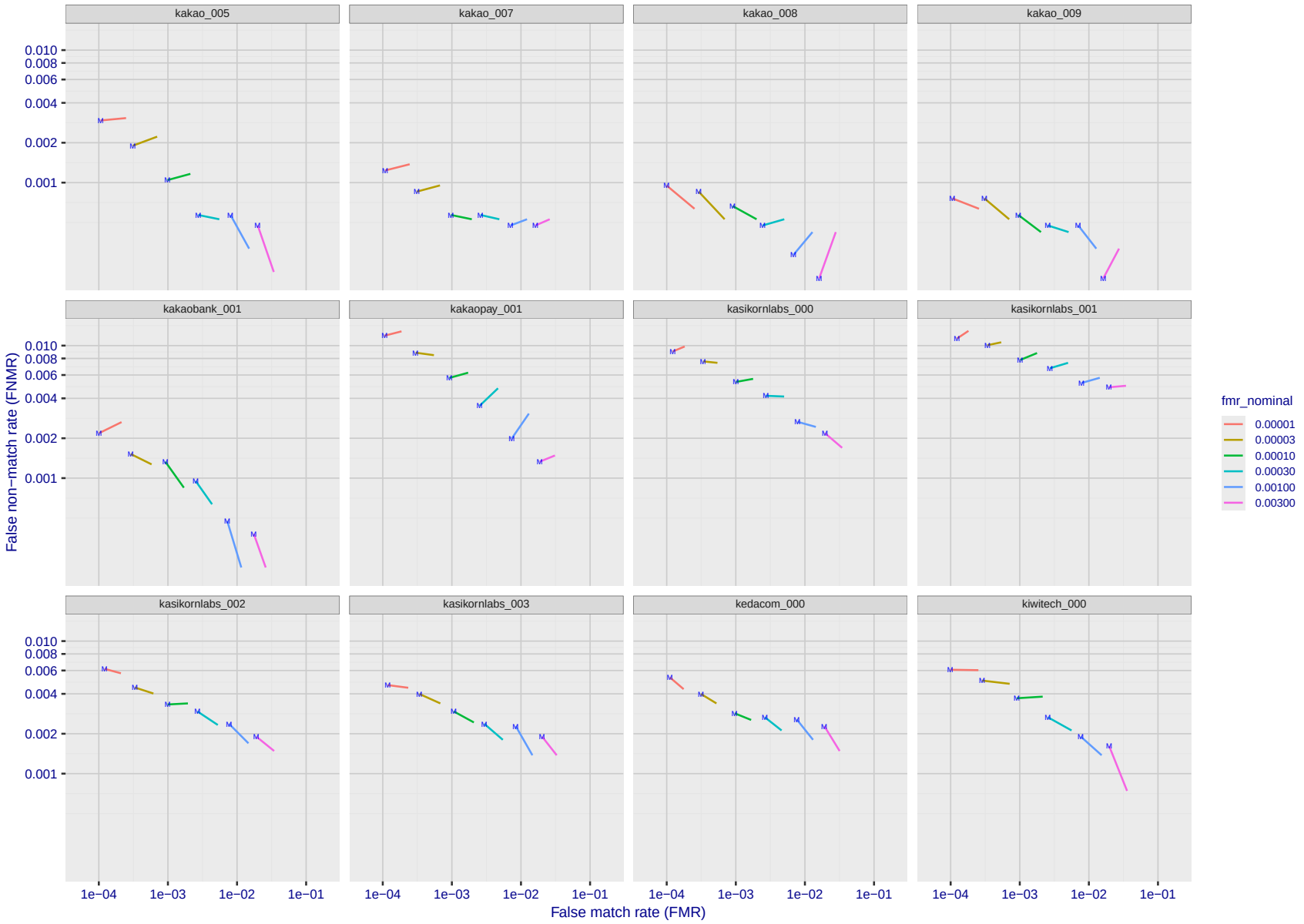

Figure 511: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

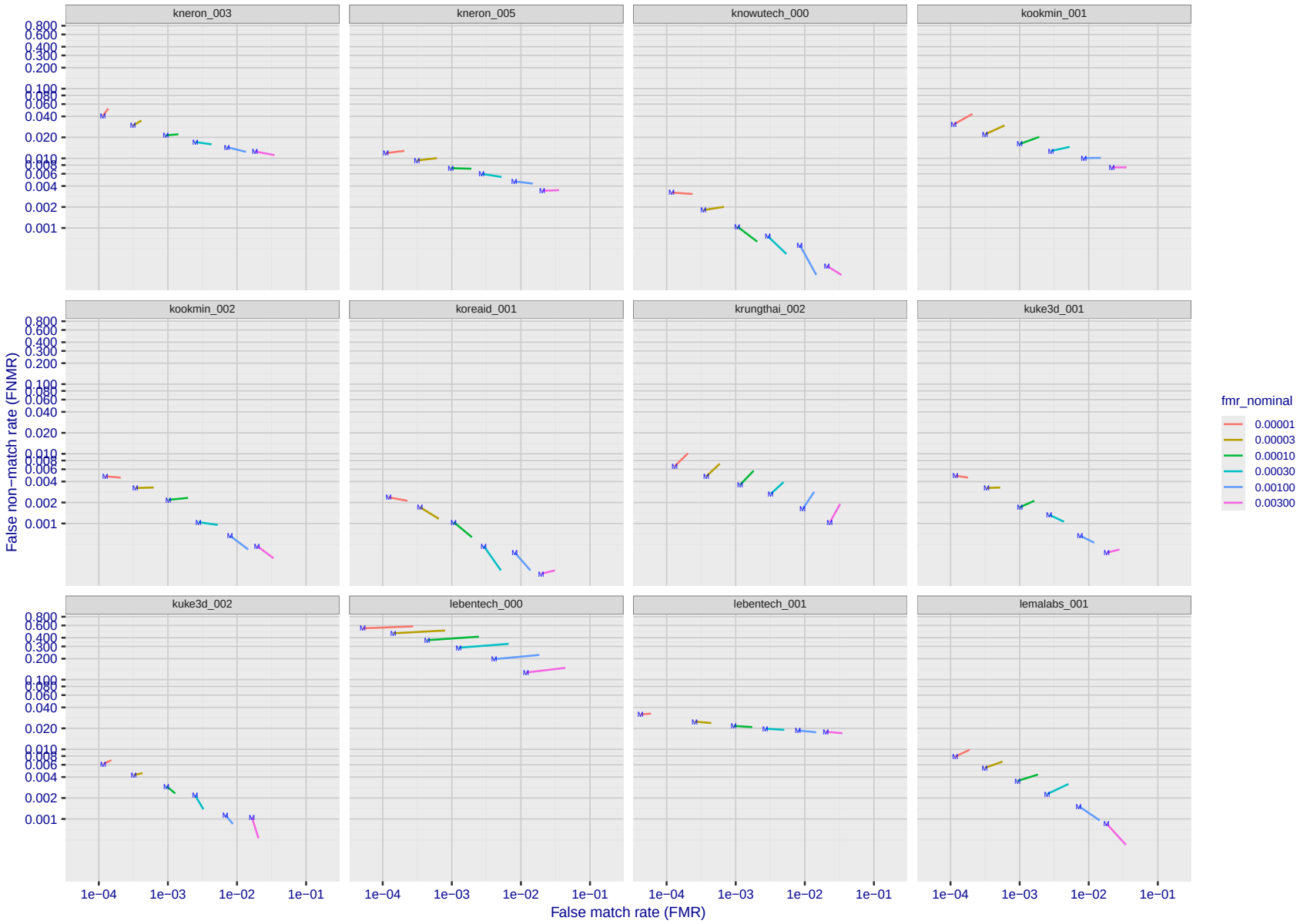

Figure 512: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

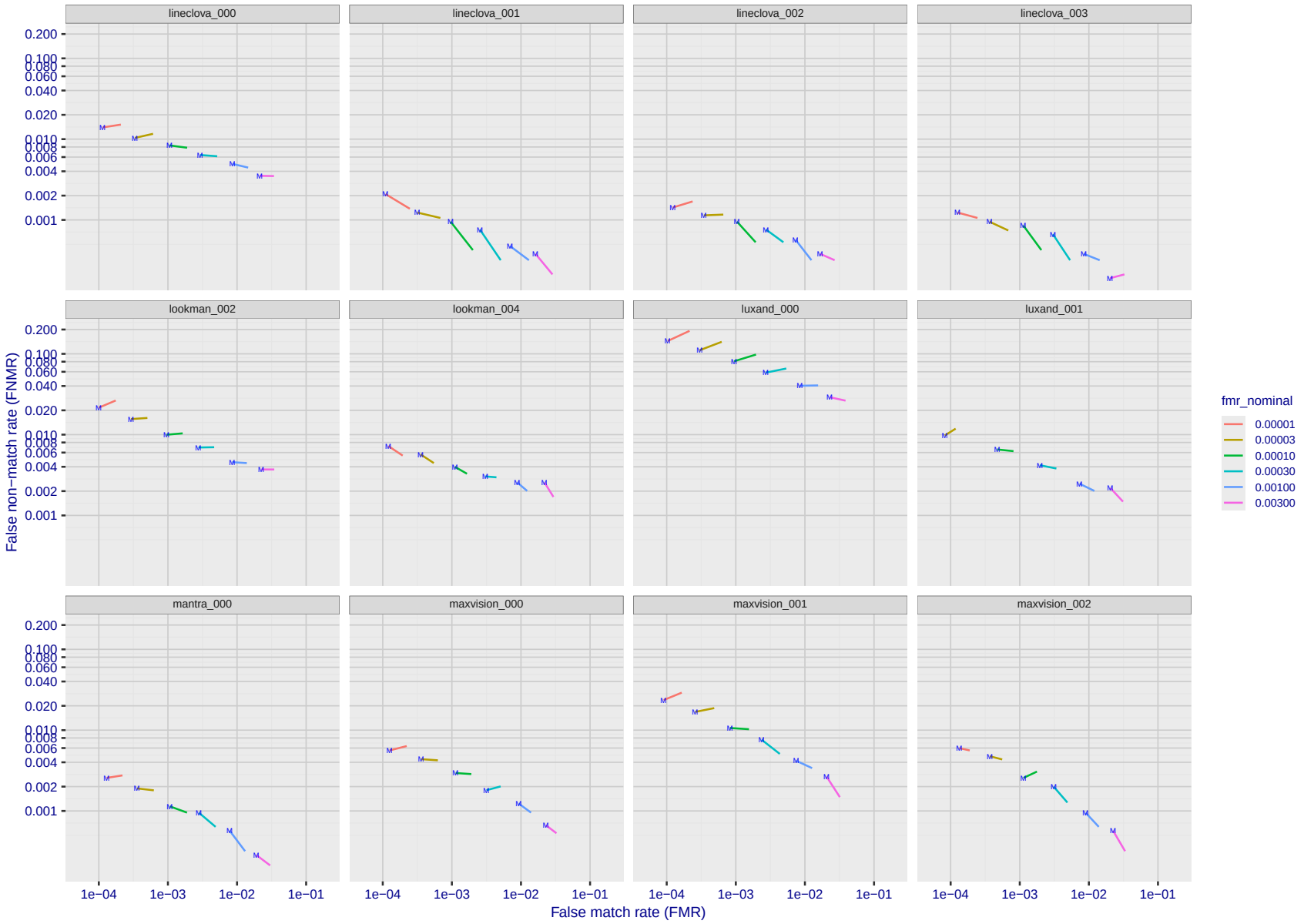

Figure 513: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

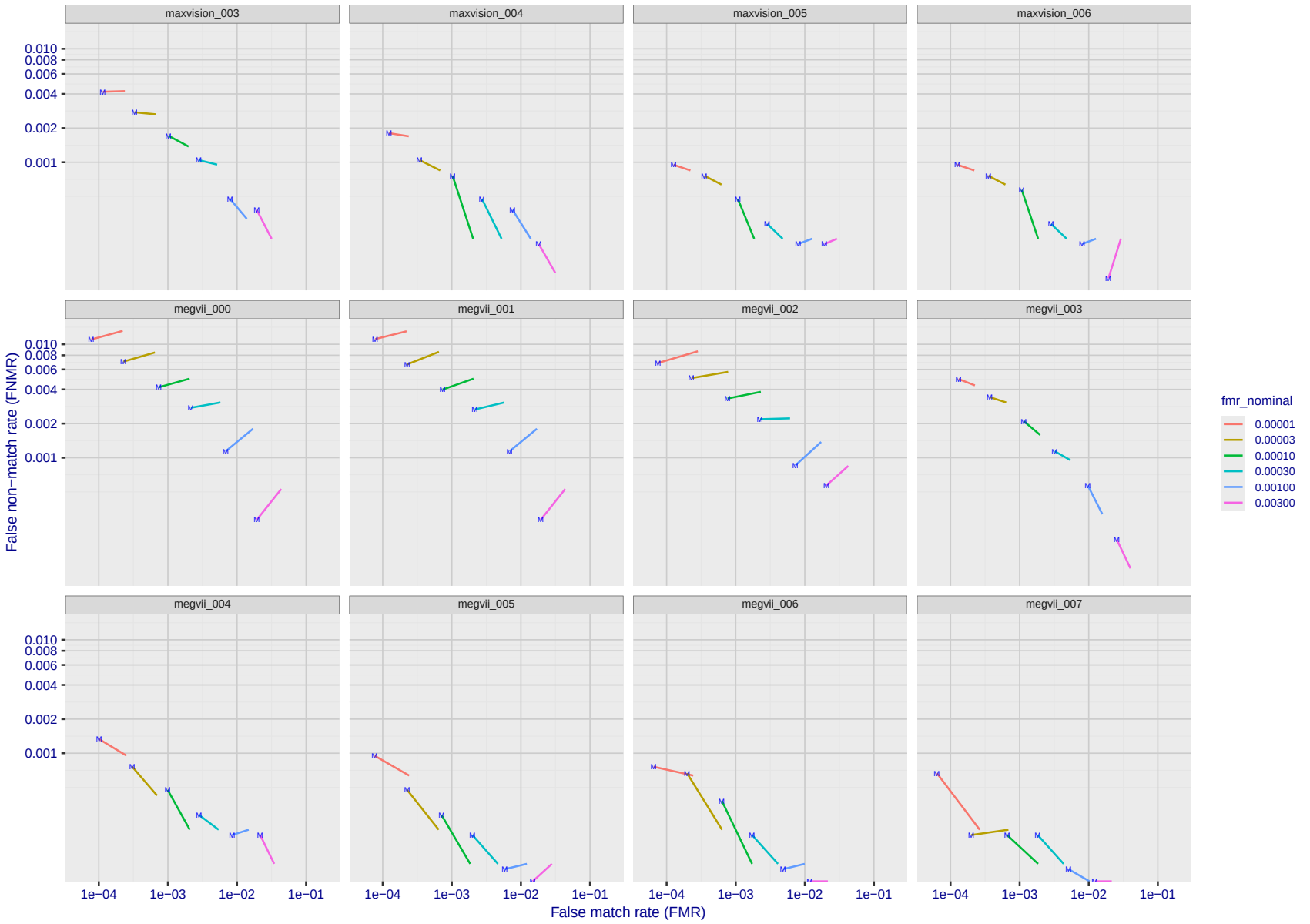

Figure 514: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

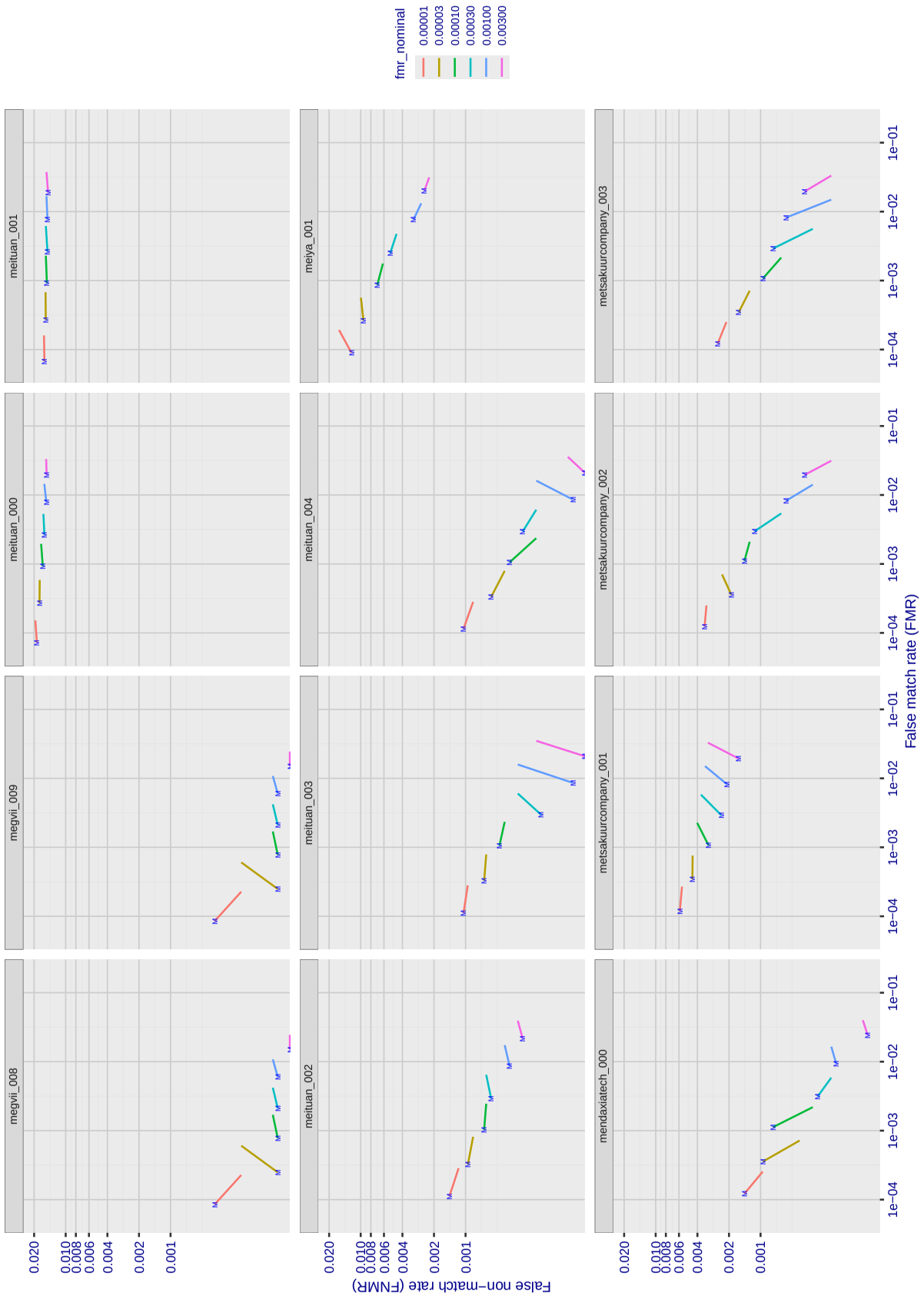

Figure 515: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

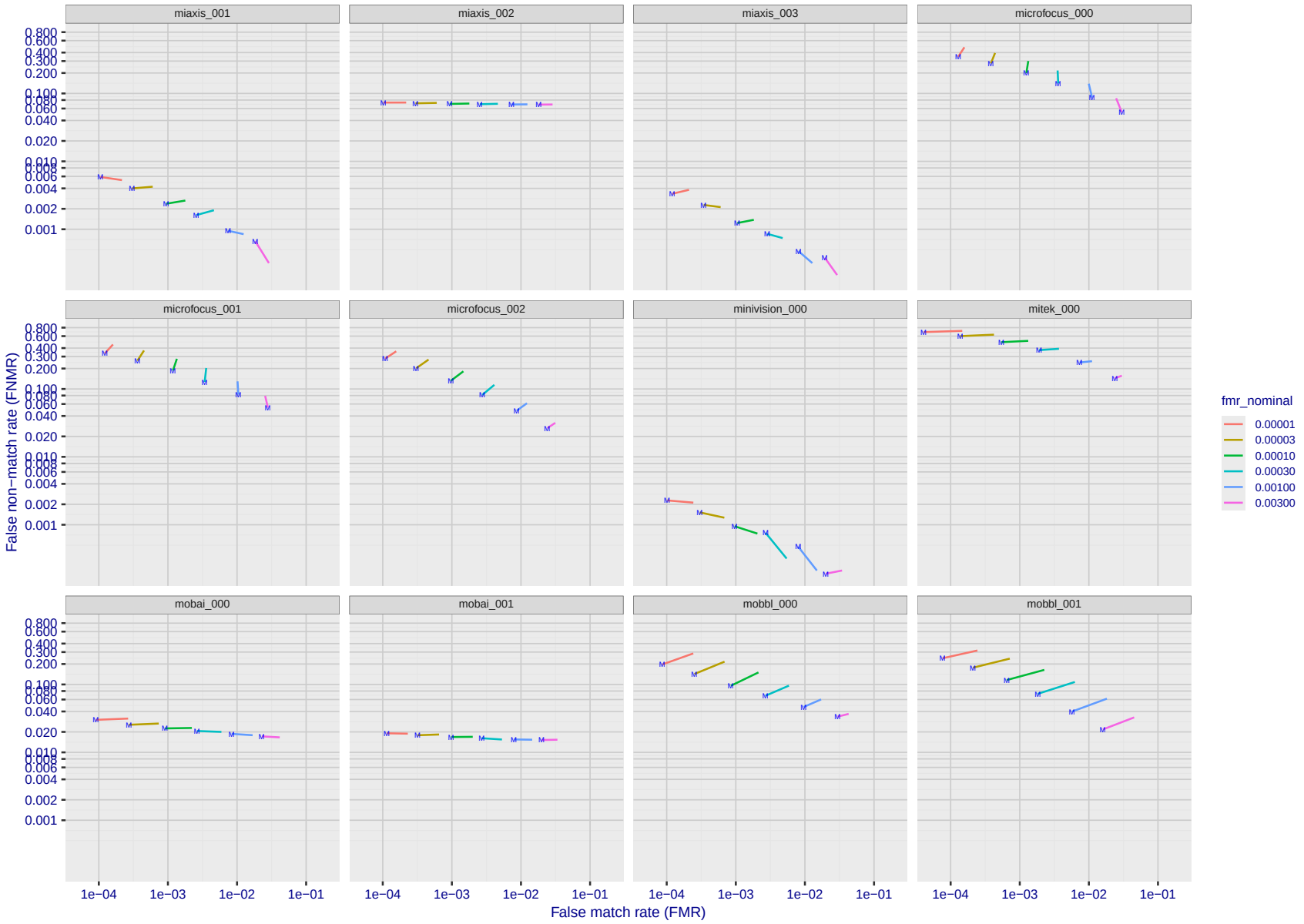

Figure 516: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

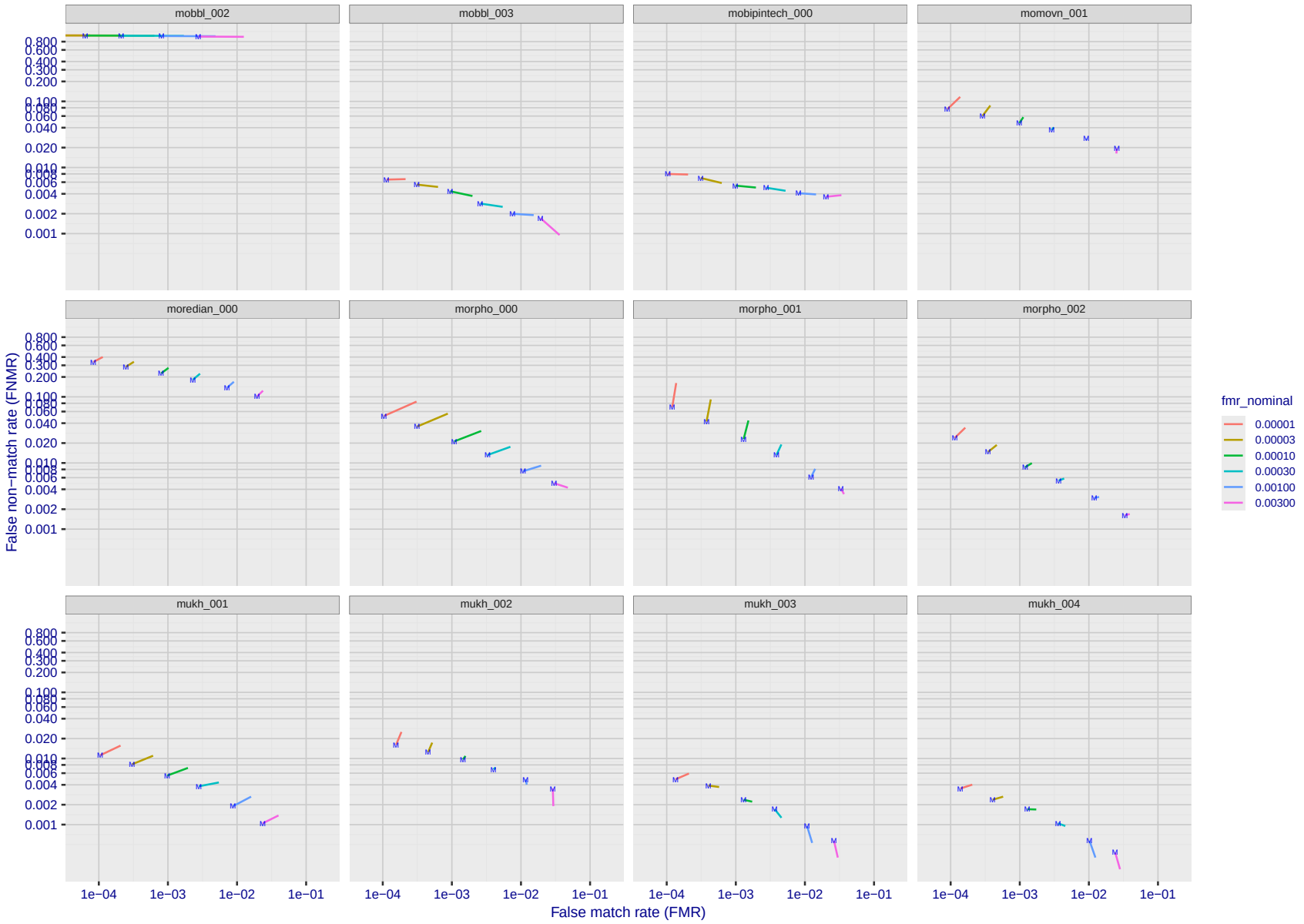

Figure 517: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

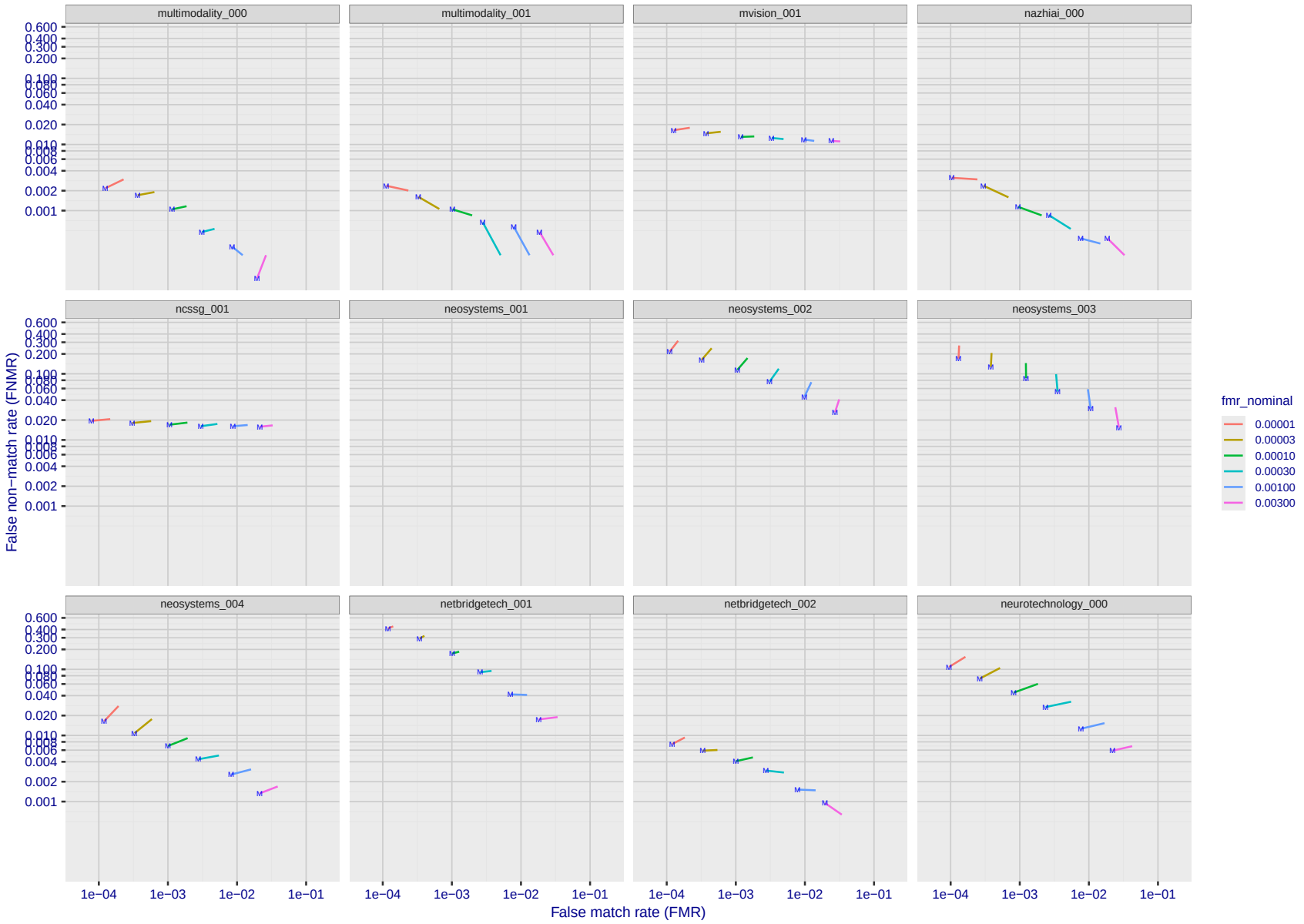

Figure 518: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

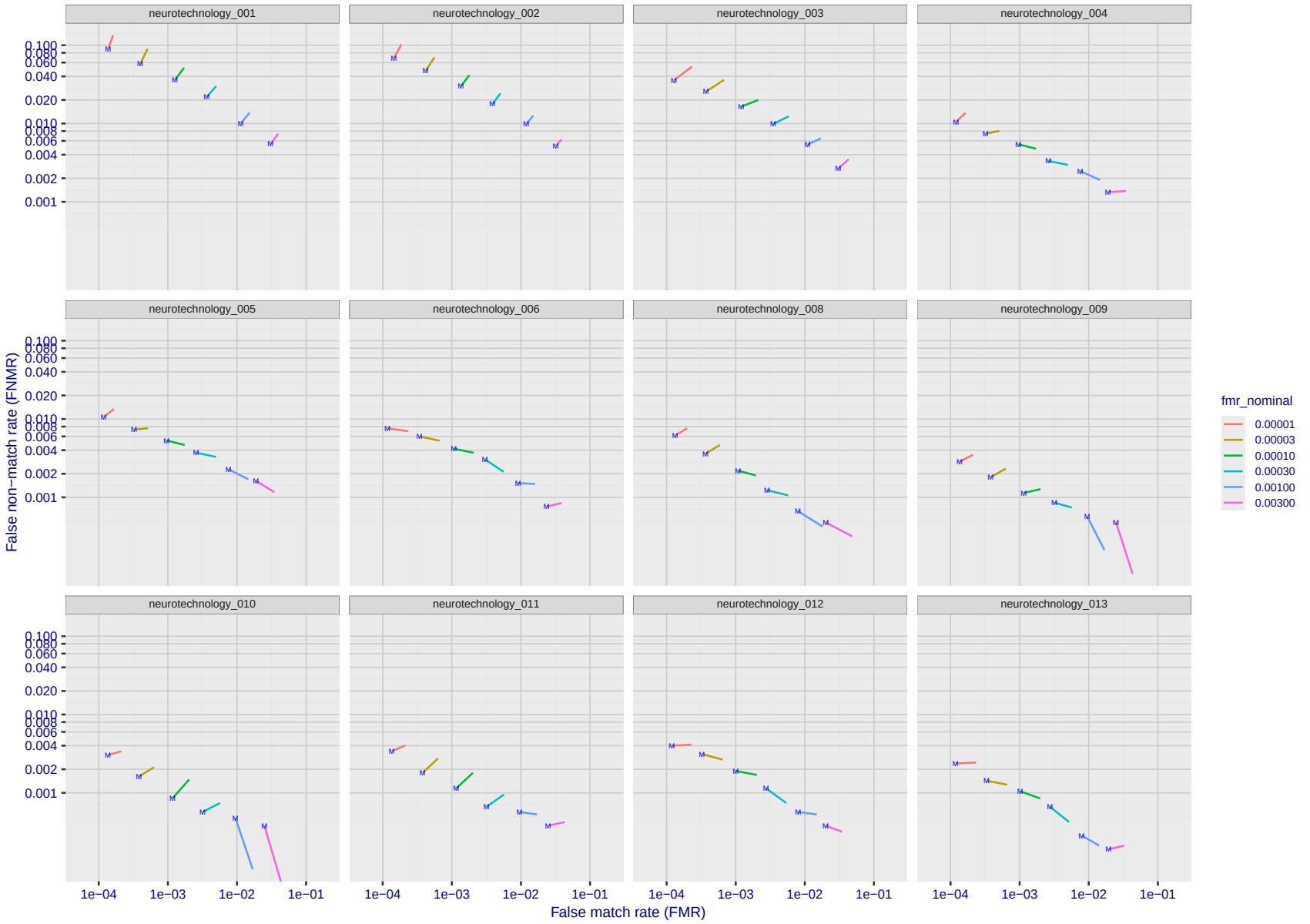

Figure 519: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

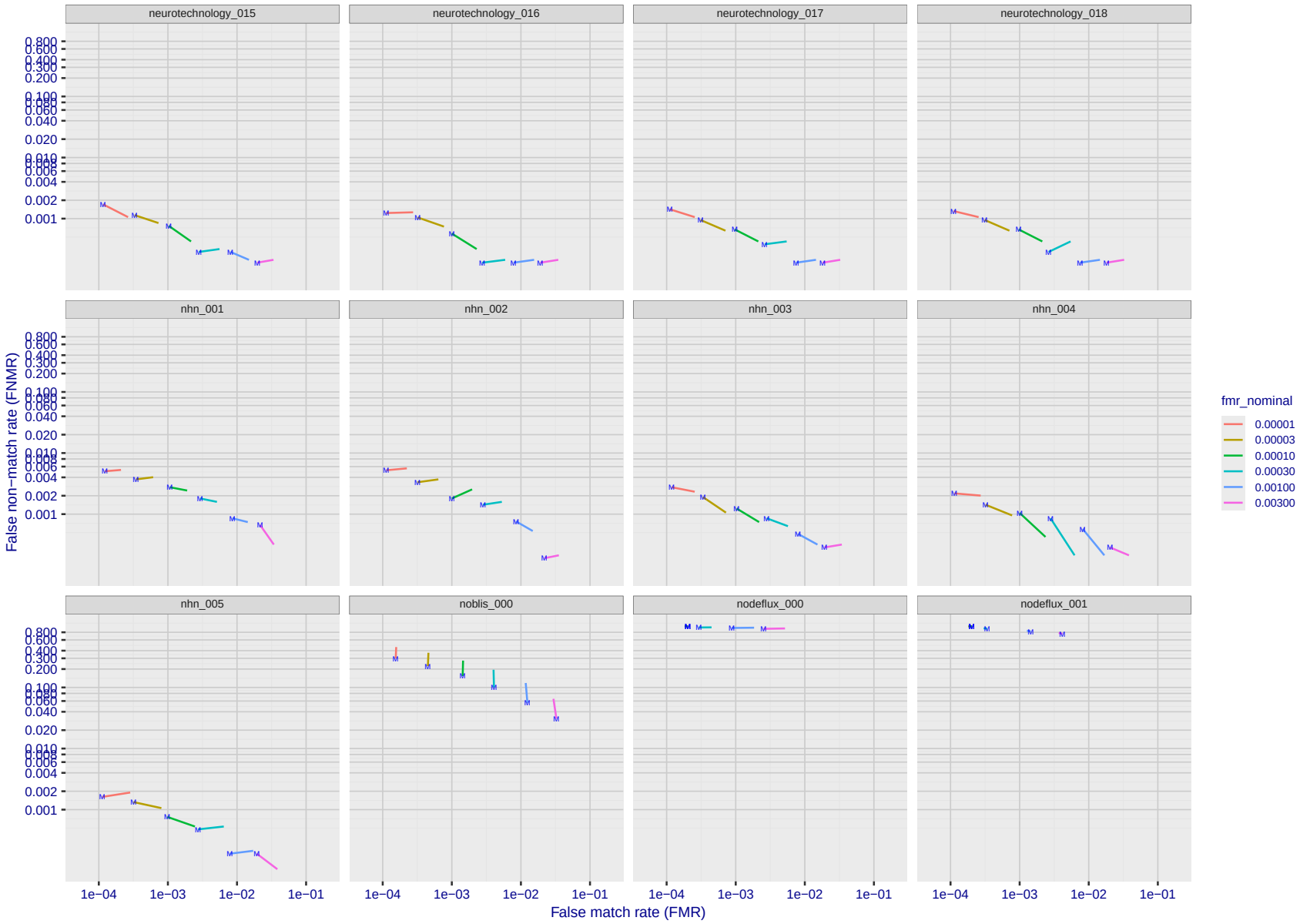

Figure 520: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

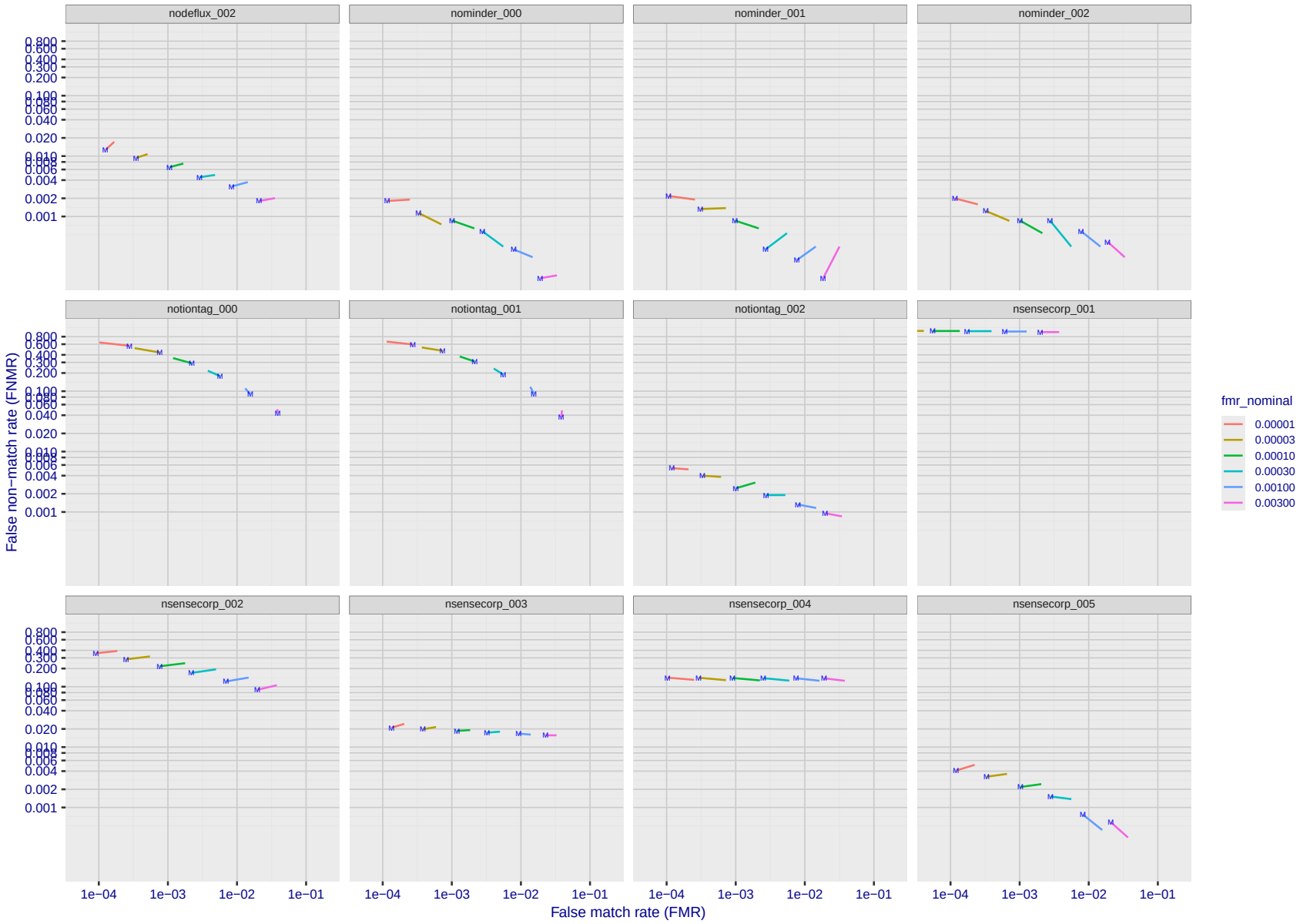

Figure 521: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

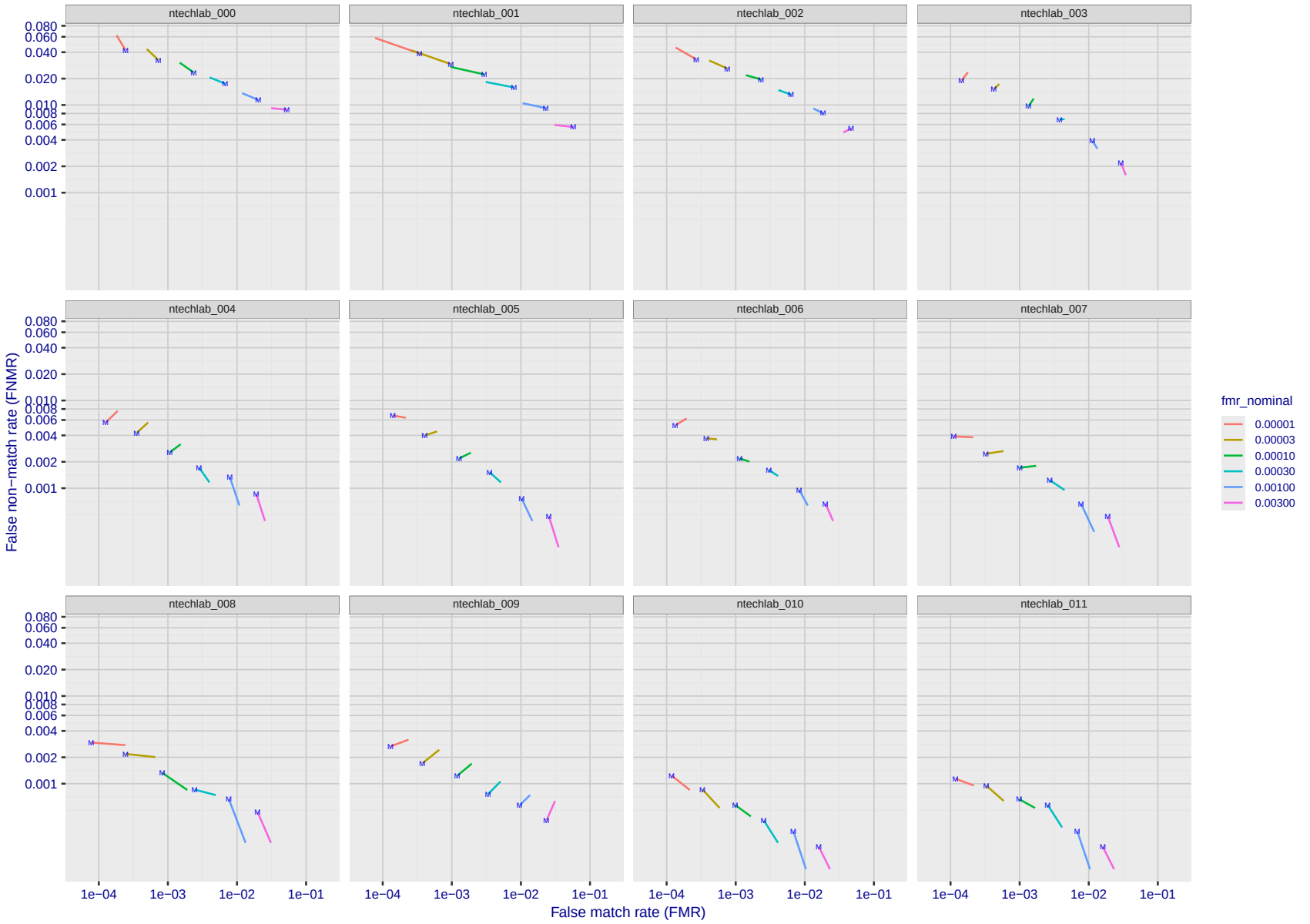

Figure 522: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

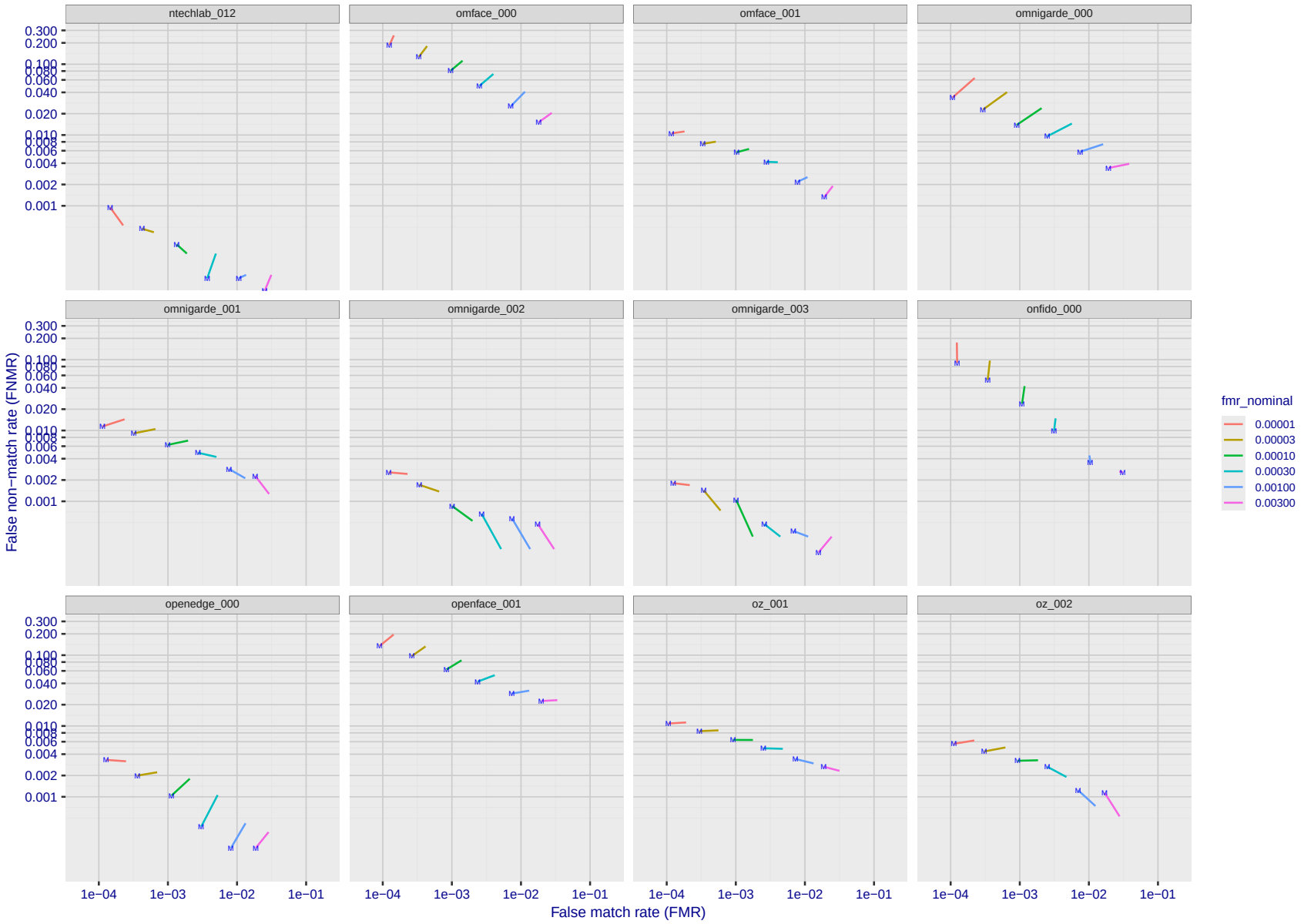

Figure 523: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

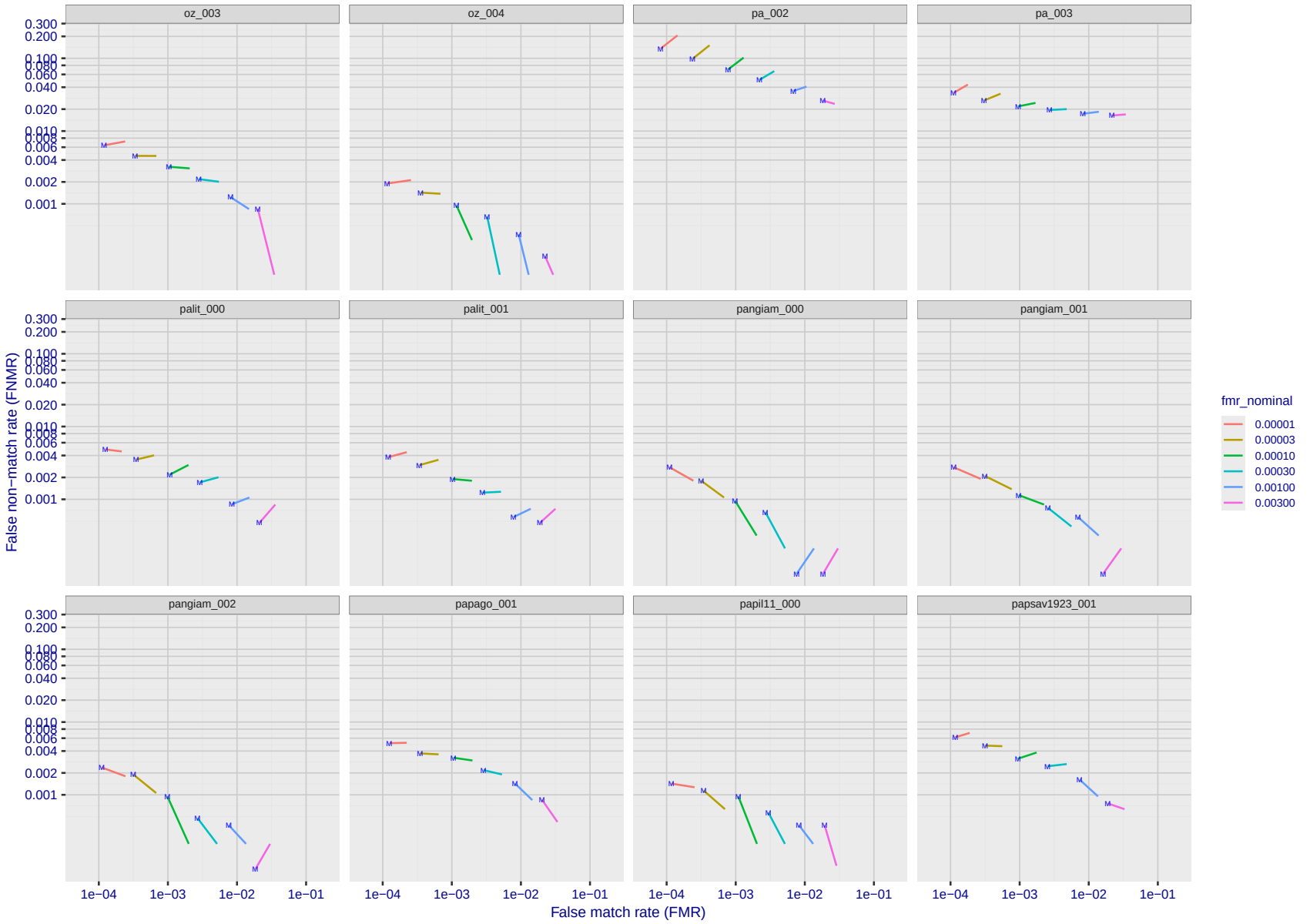

Figure 524: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

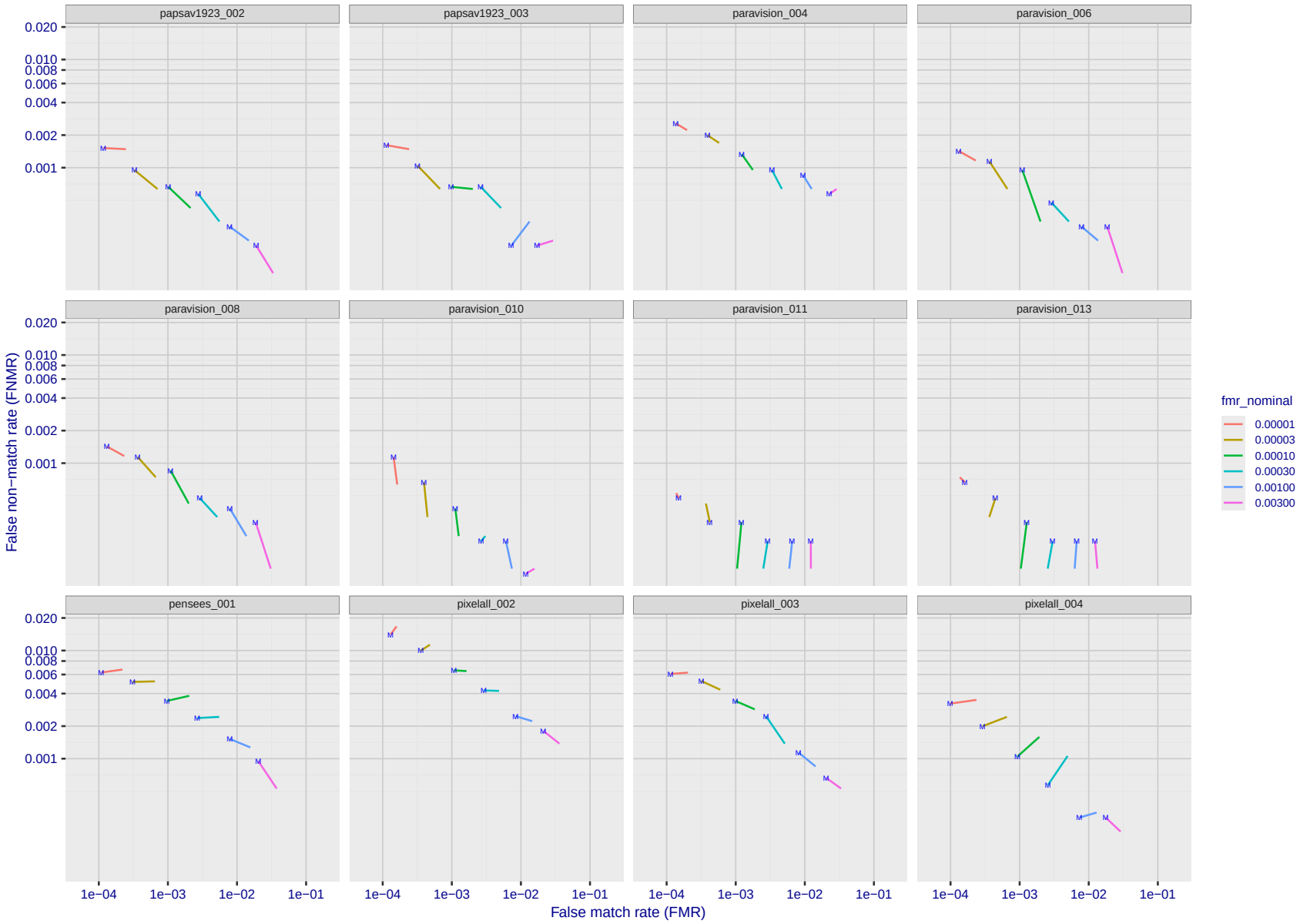

Figure 525: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

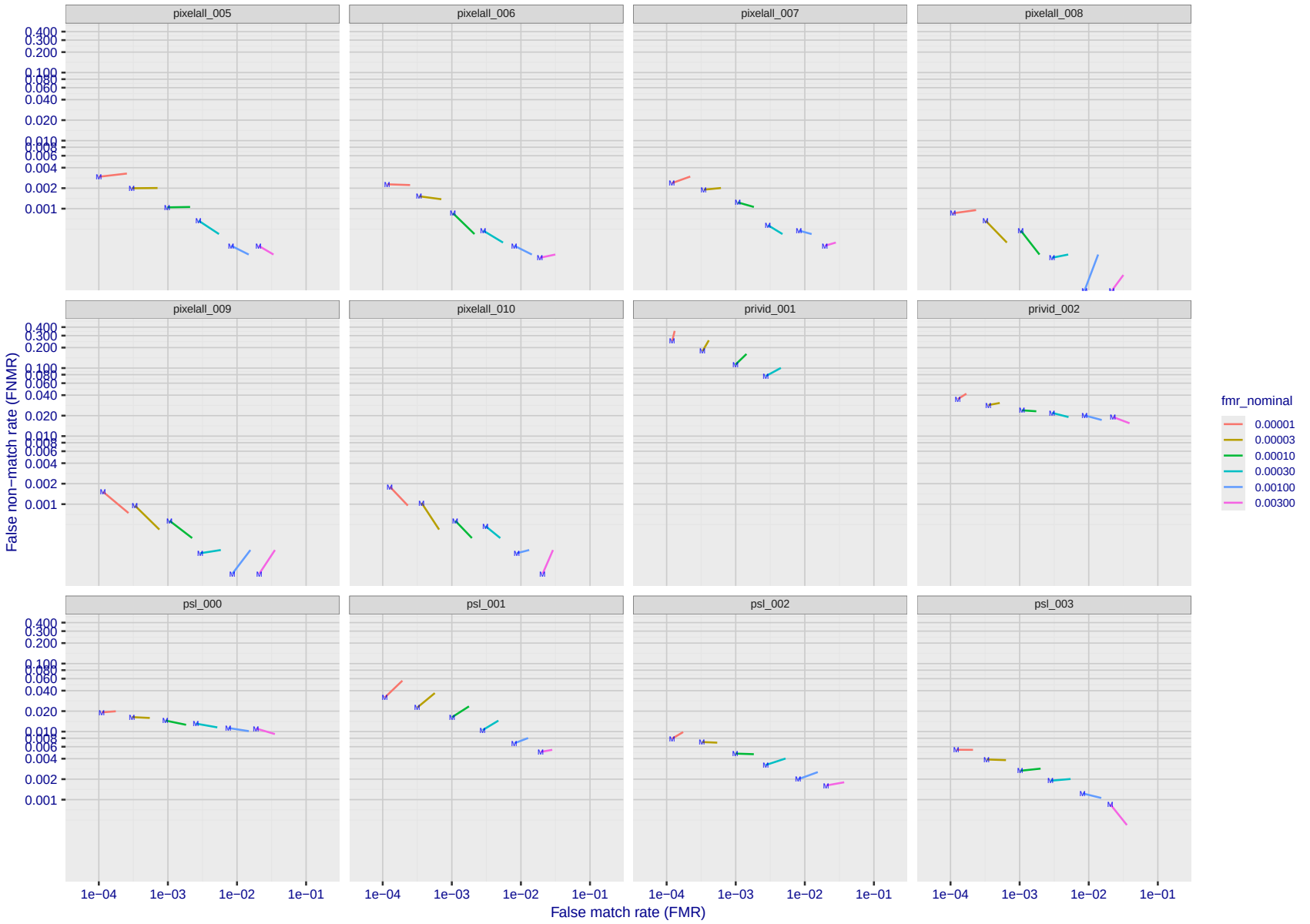

Figure 526: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

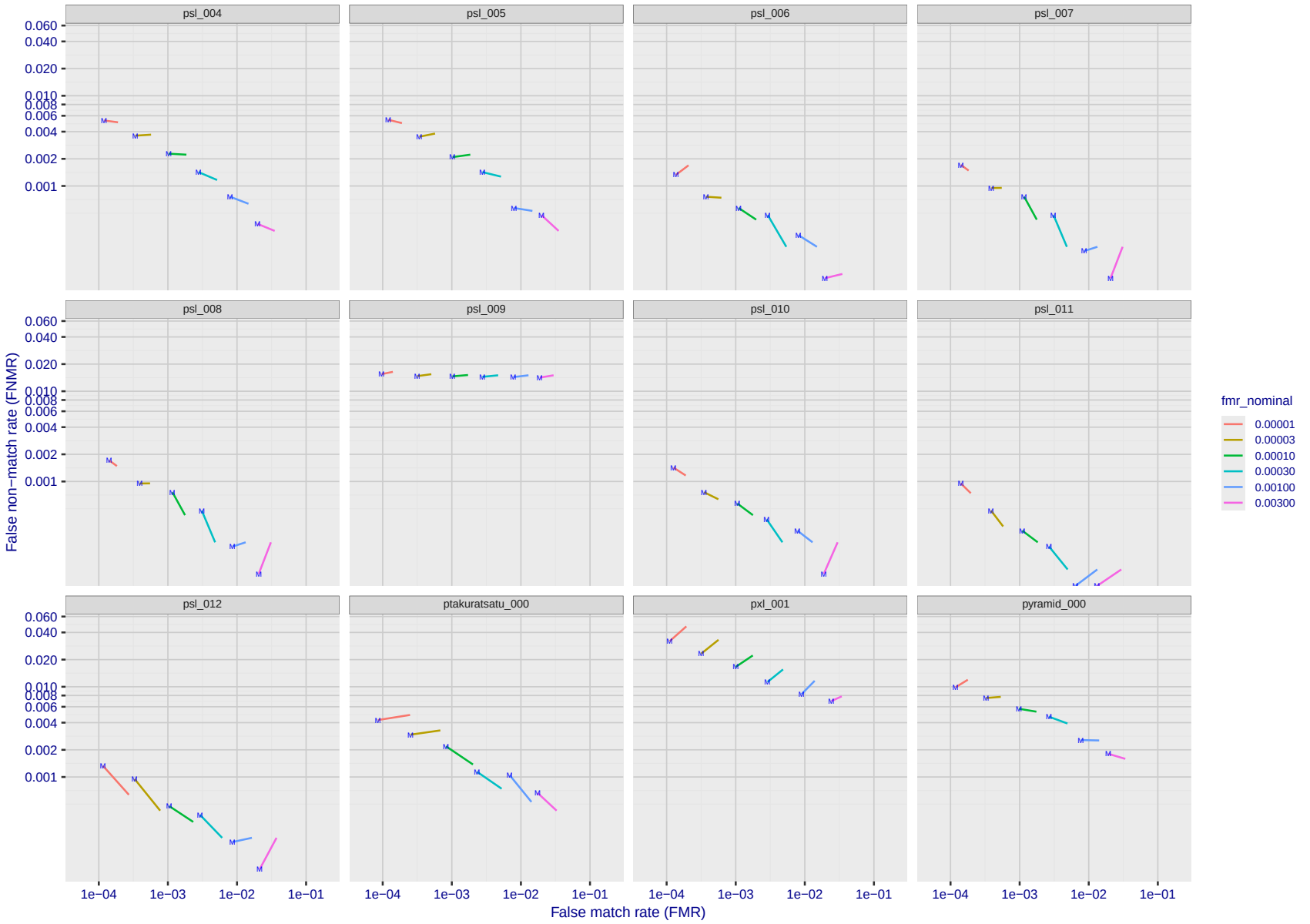

Figure 527: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

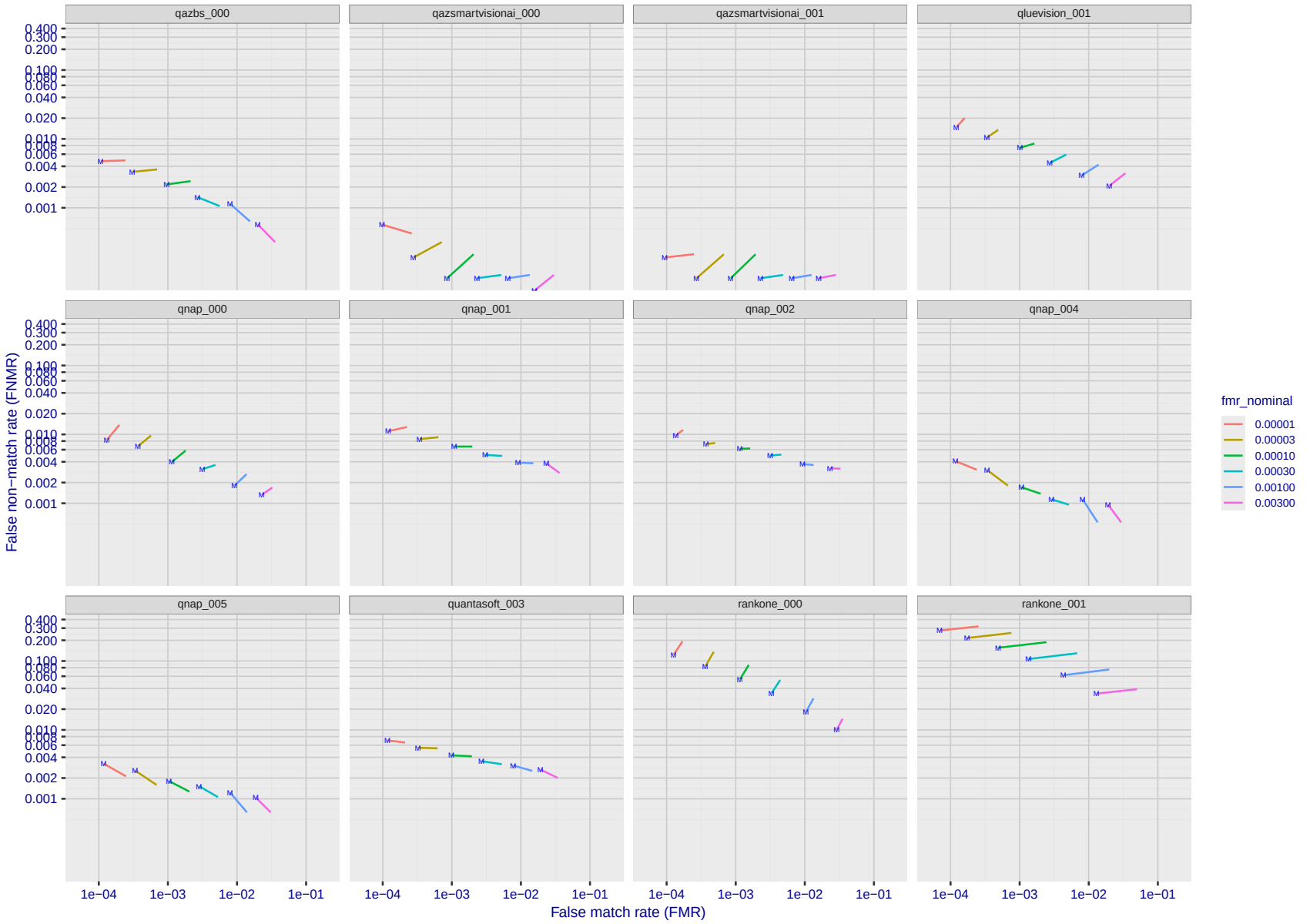

Figure 528: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

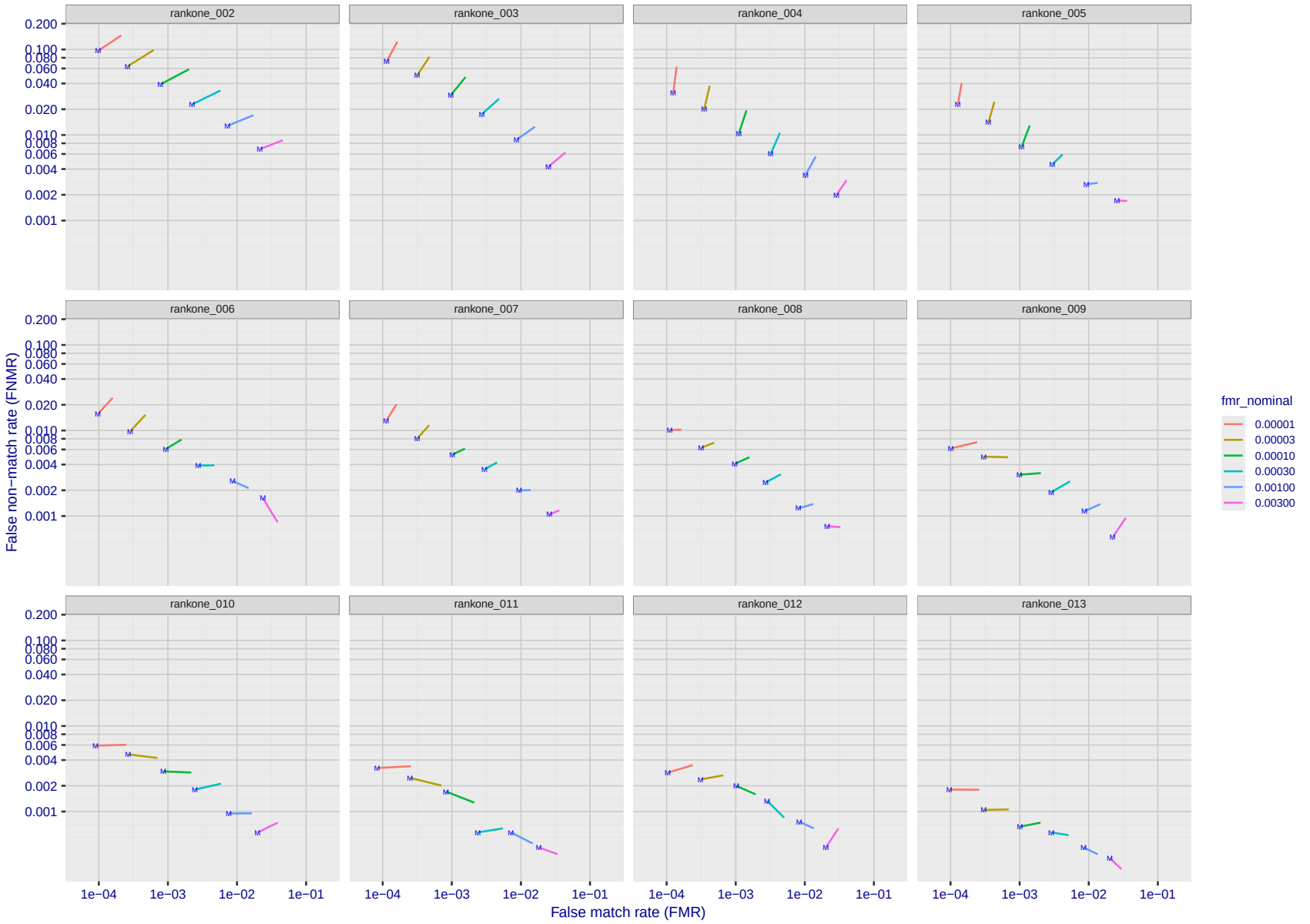

Figure 529: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

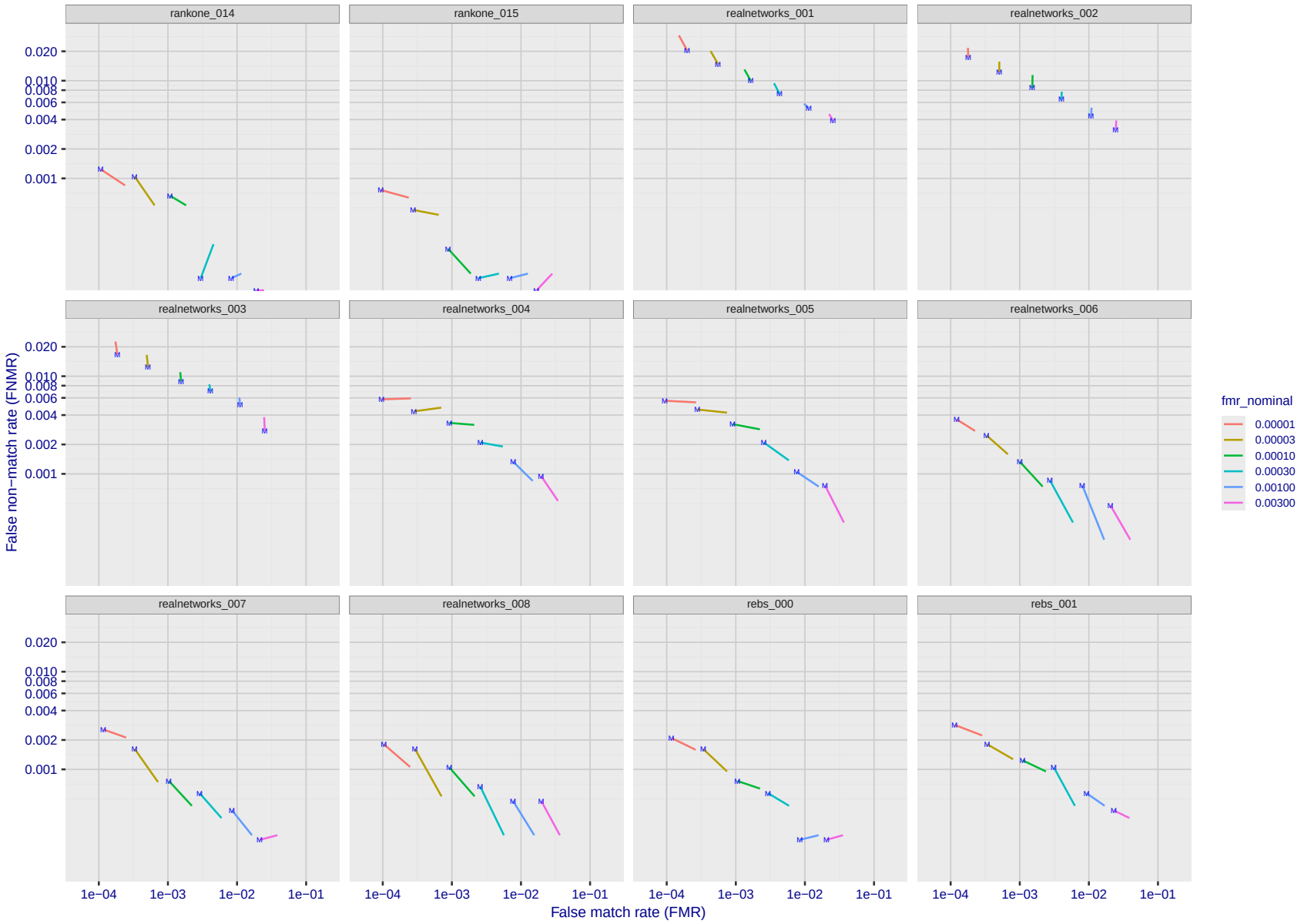

Figure 530: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

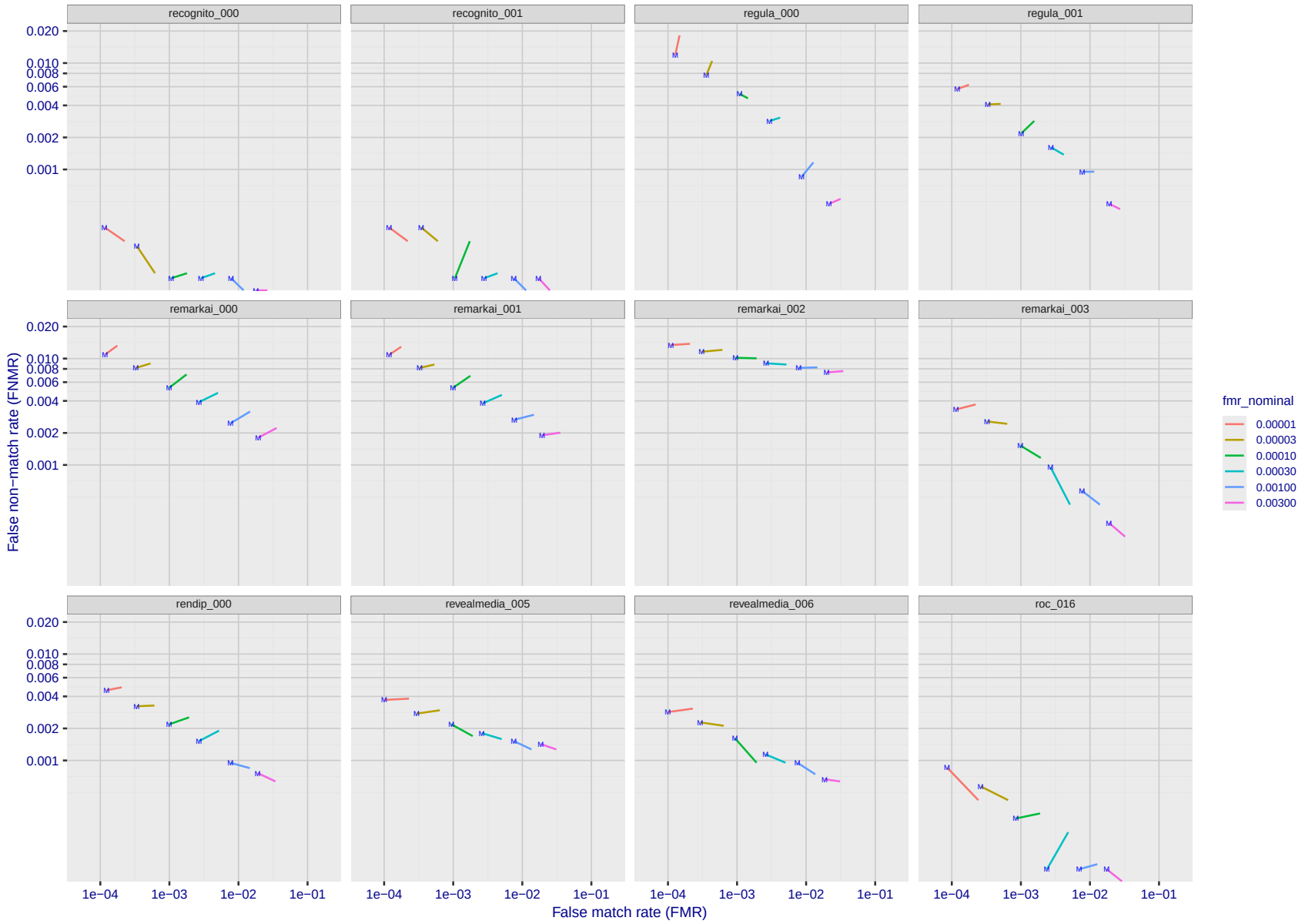

Figure 531: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

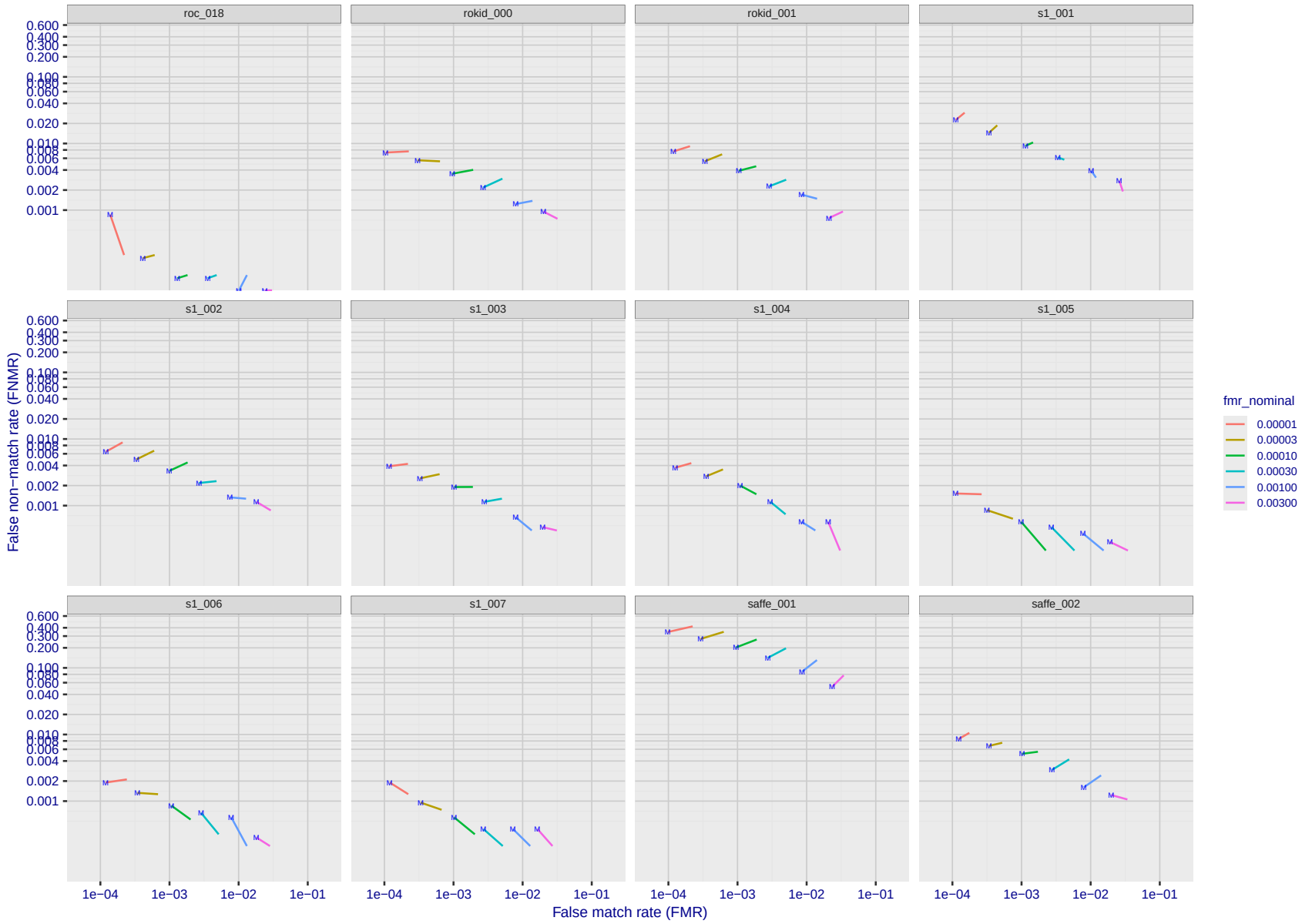

Figure 532: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

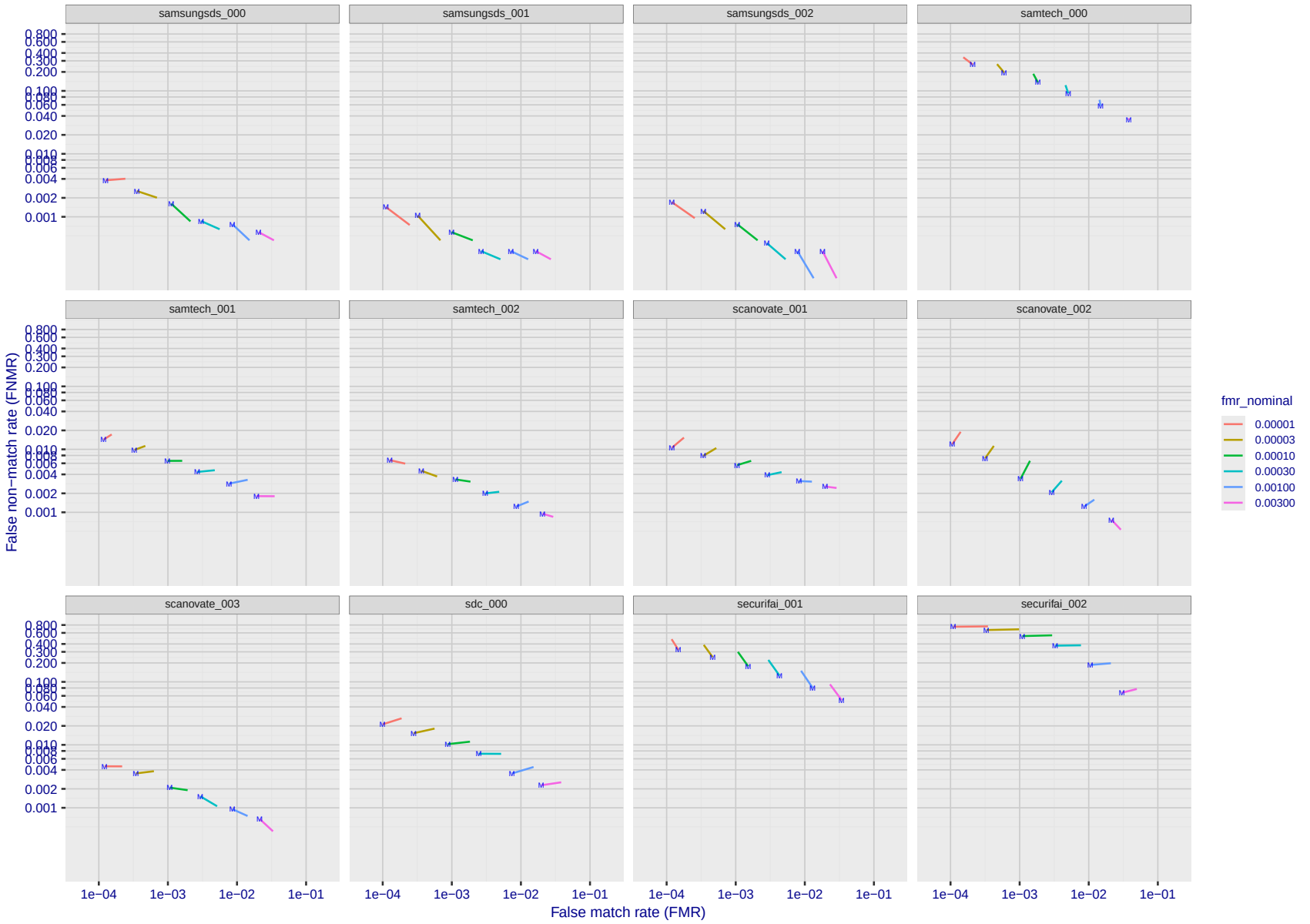

Figure 533: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

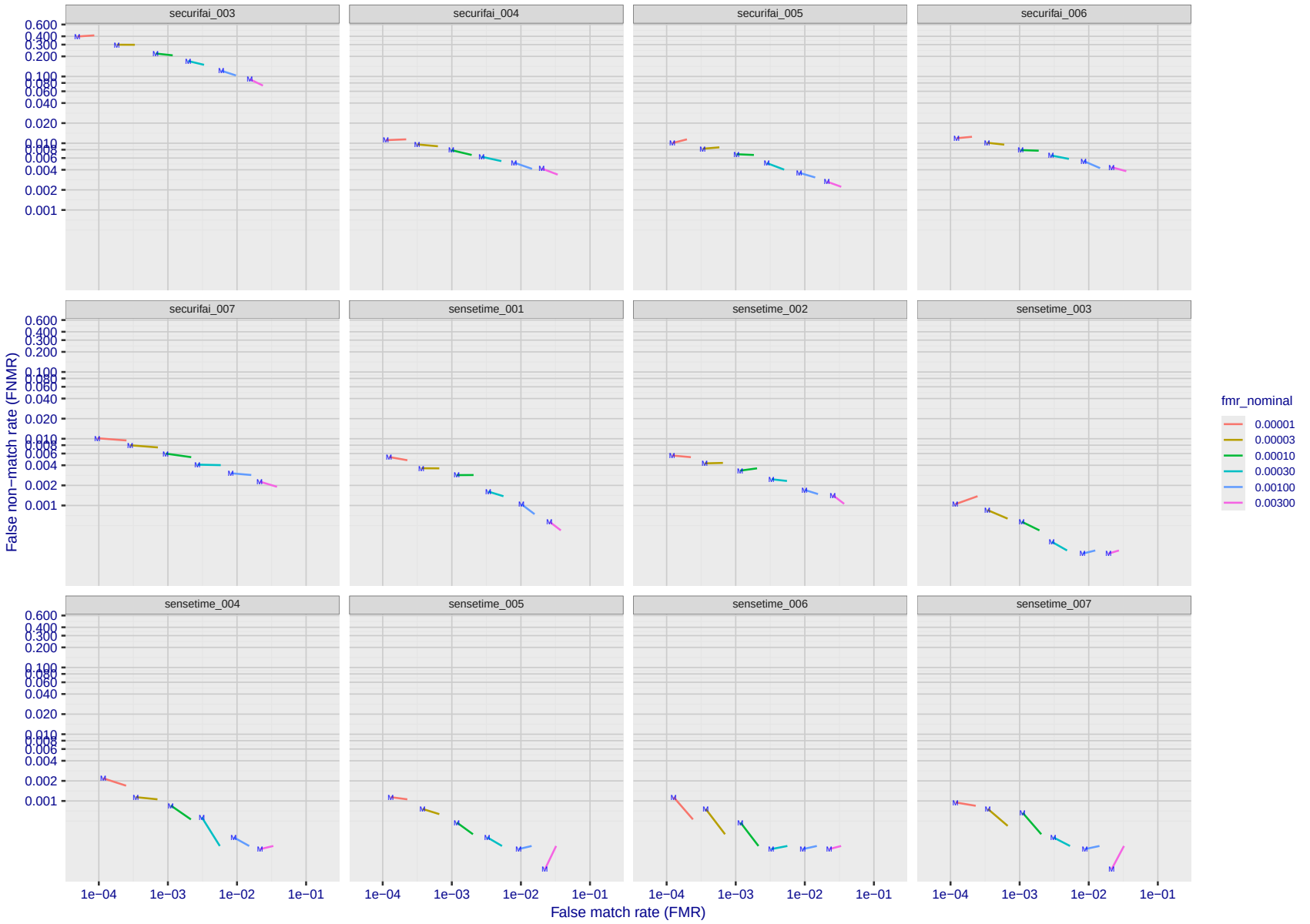

Figure 534: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

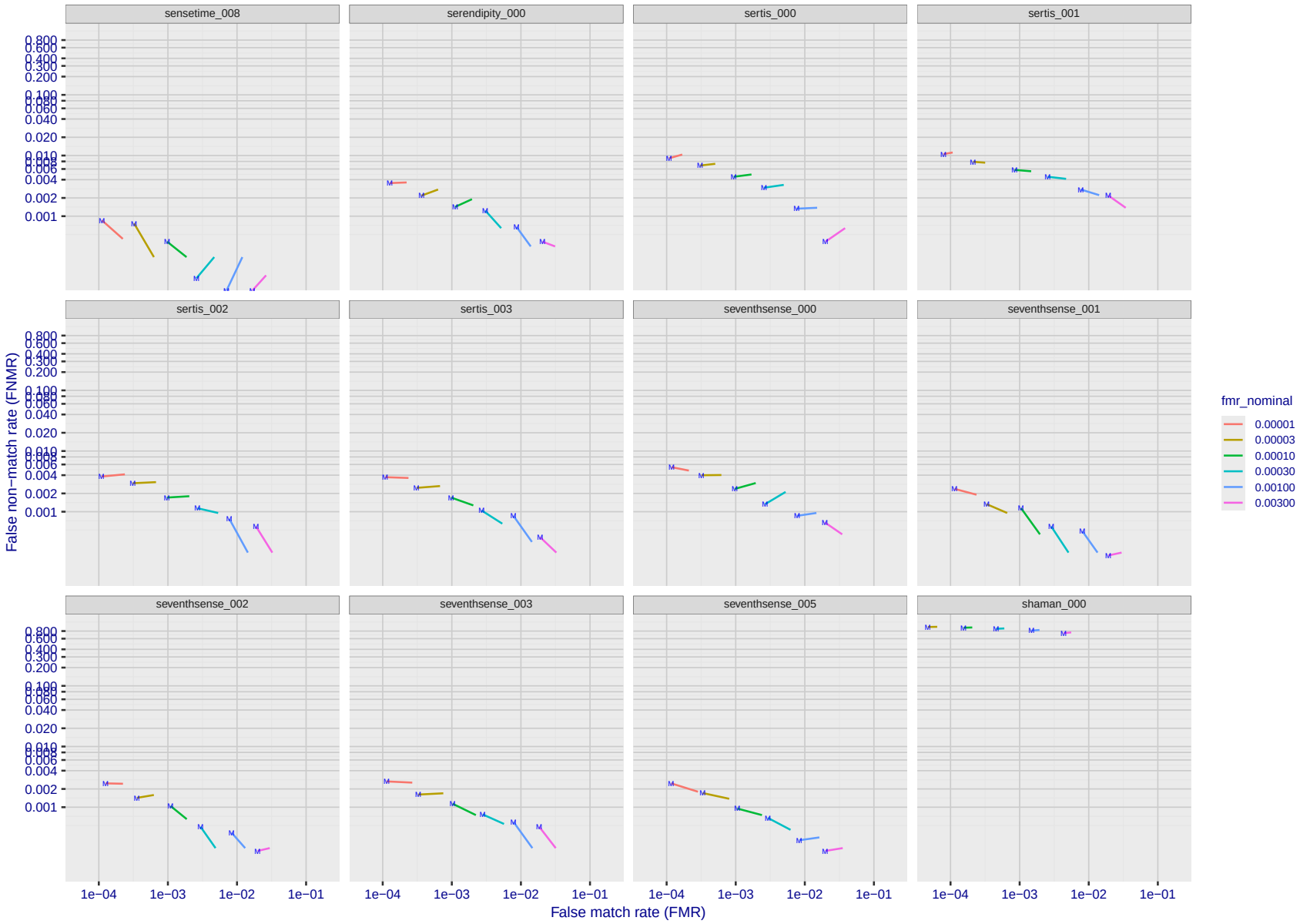

Figure 535: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

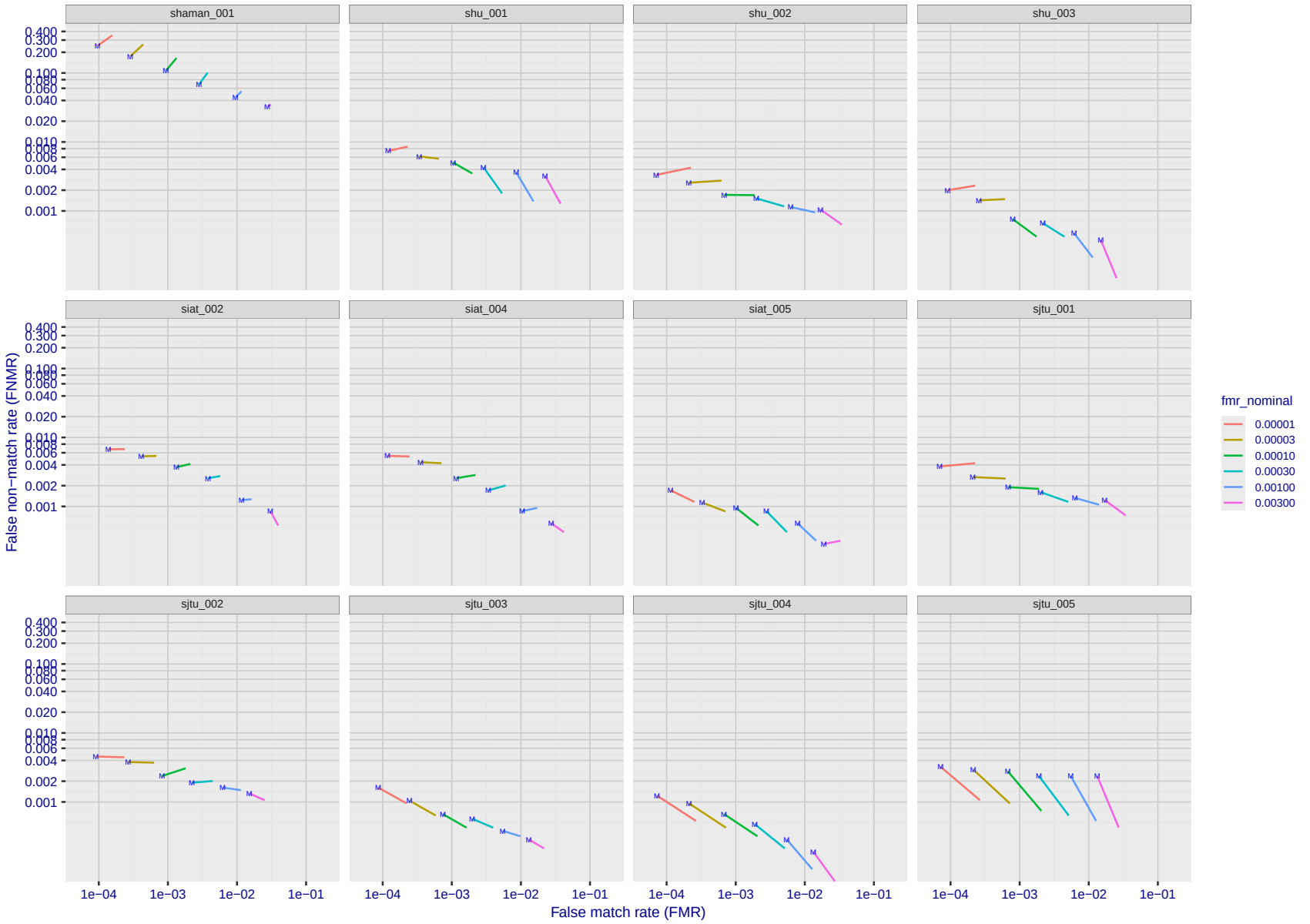

Figure 536: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

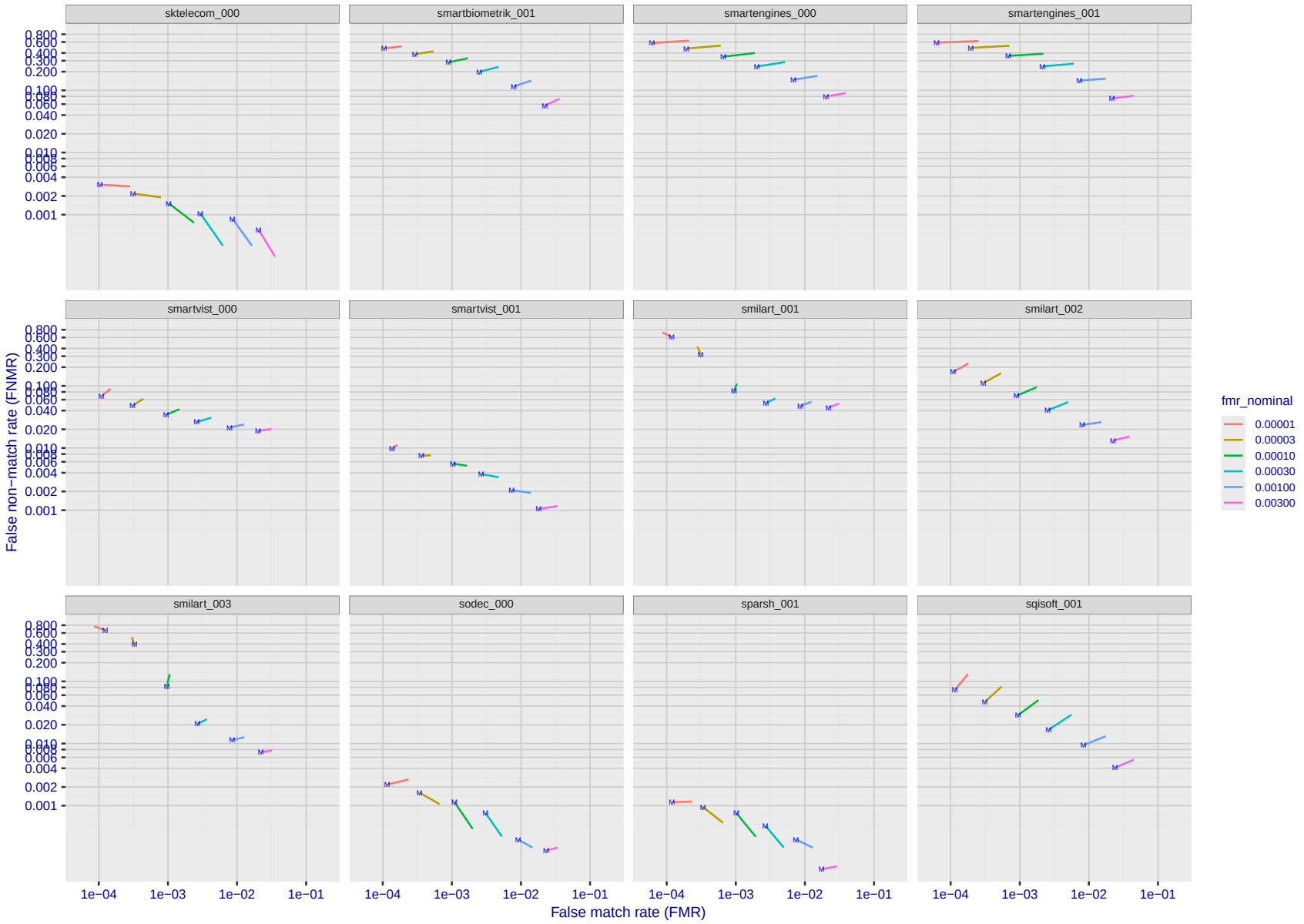

Figure 537: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

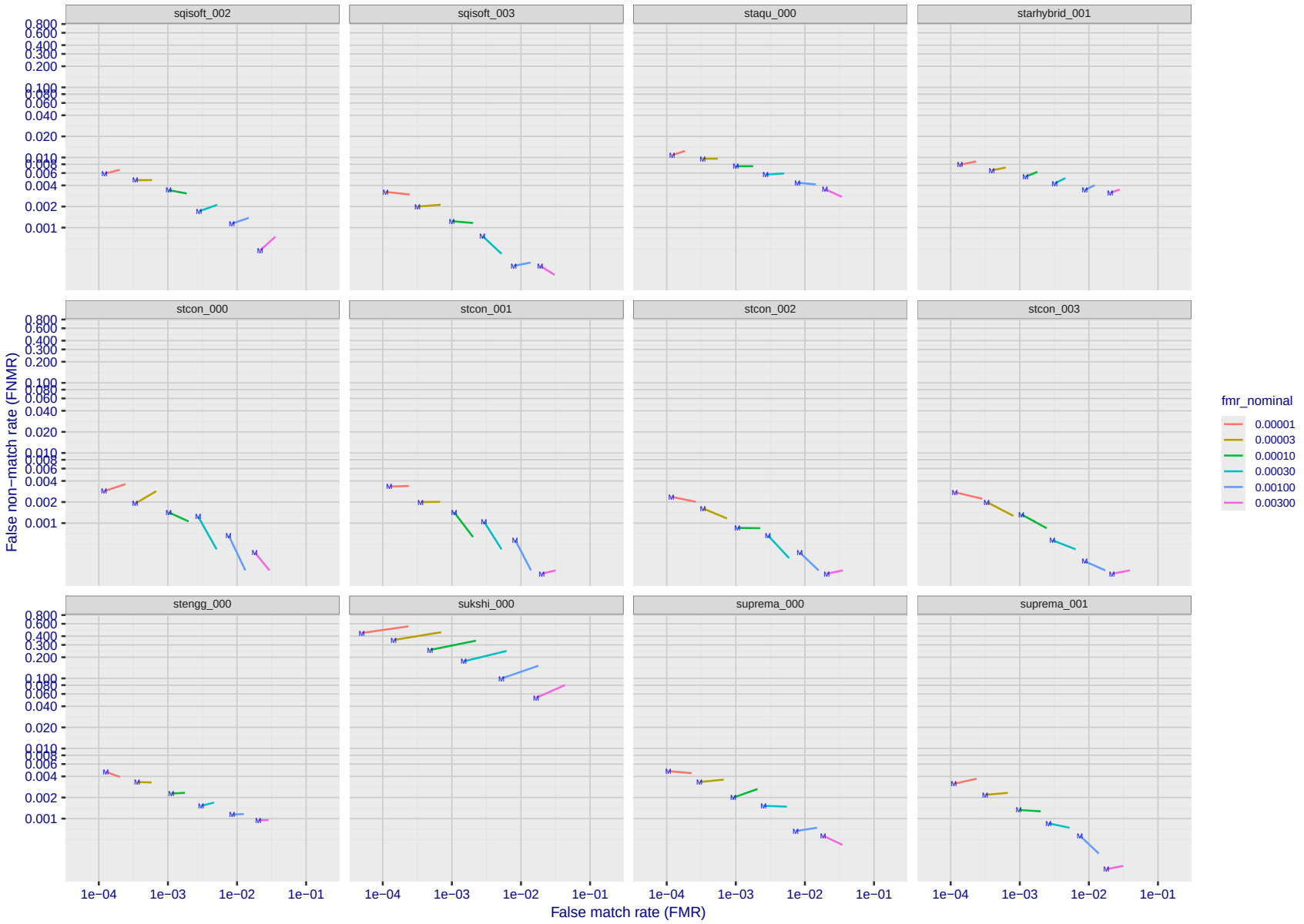

Figure 538: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

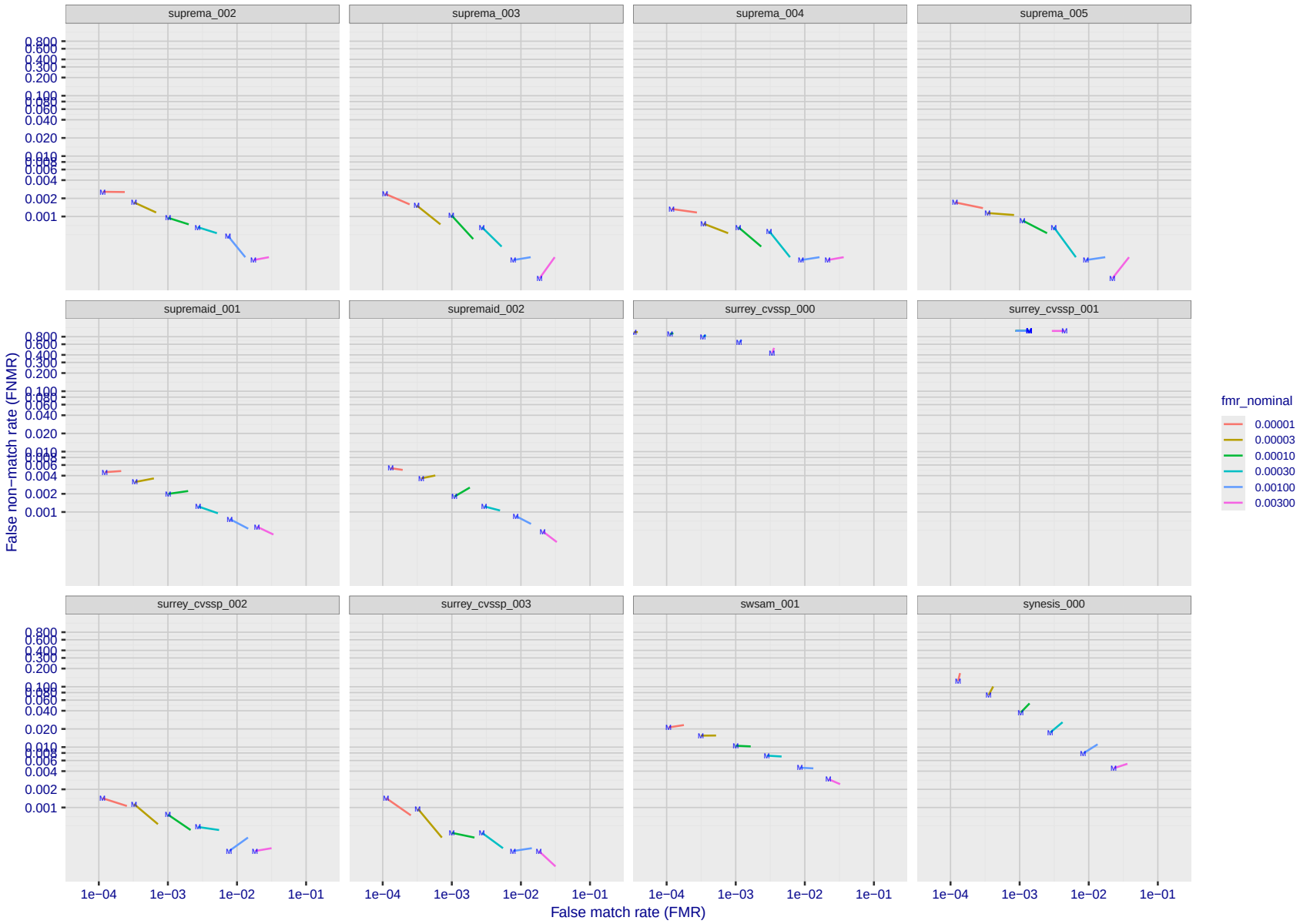

Figure 539: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

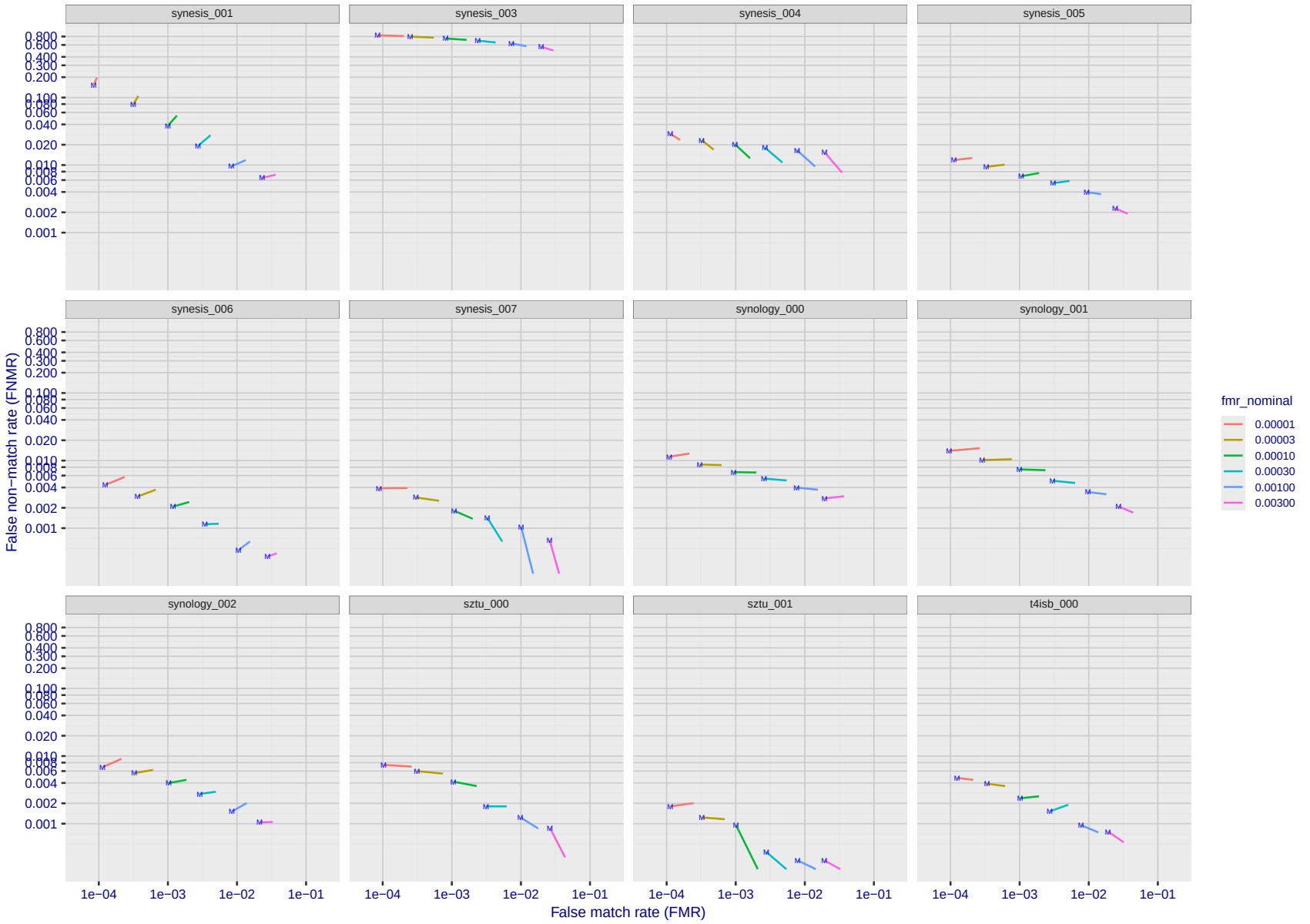

Figure 540: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

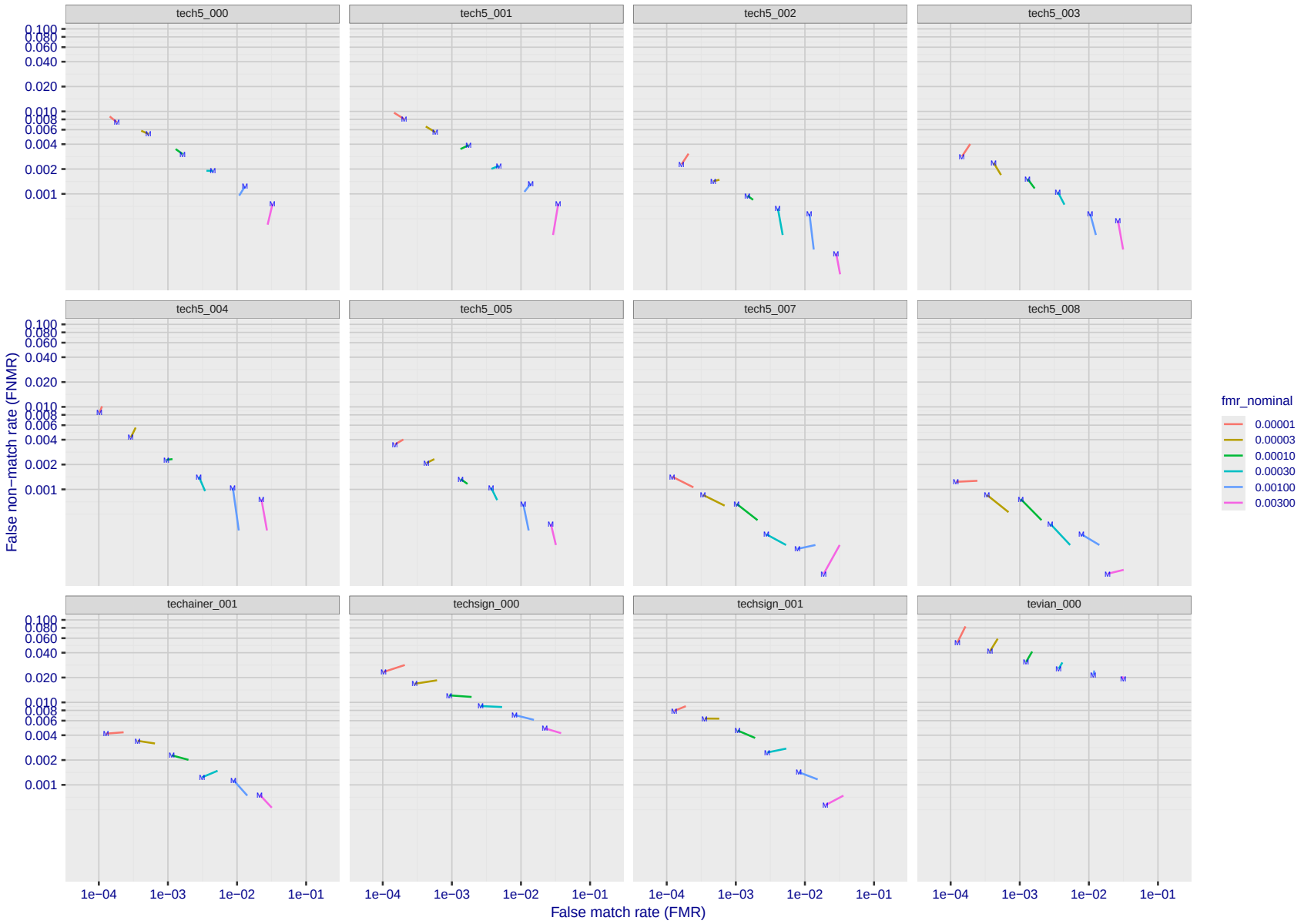

Figure 541: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

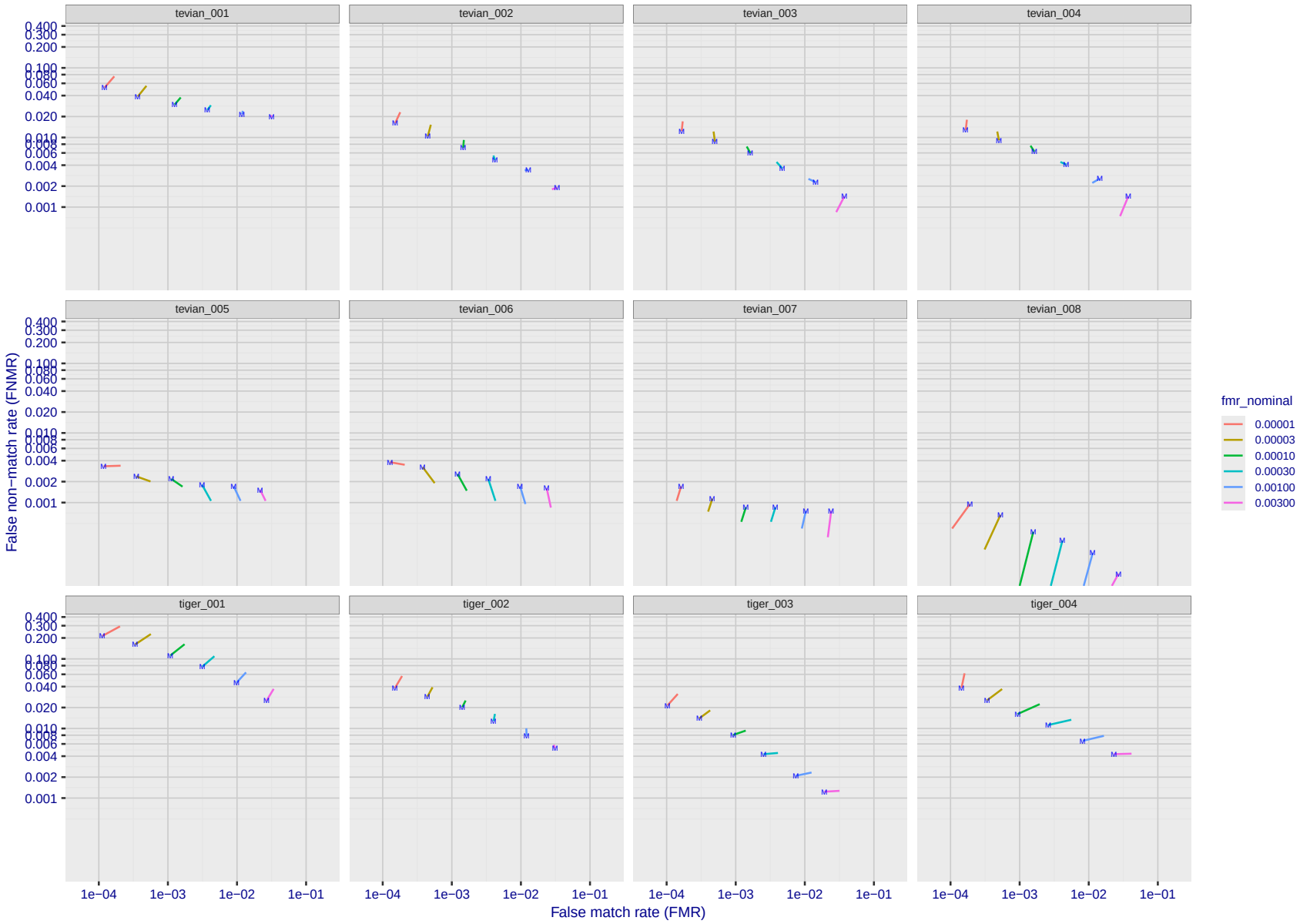

Figure 542: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

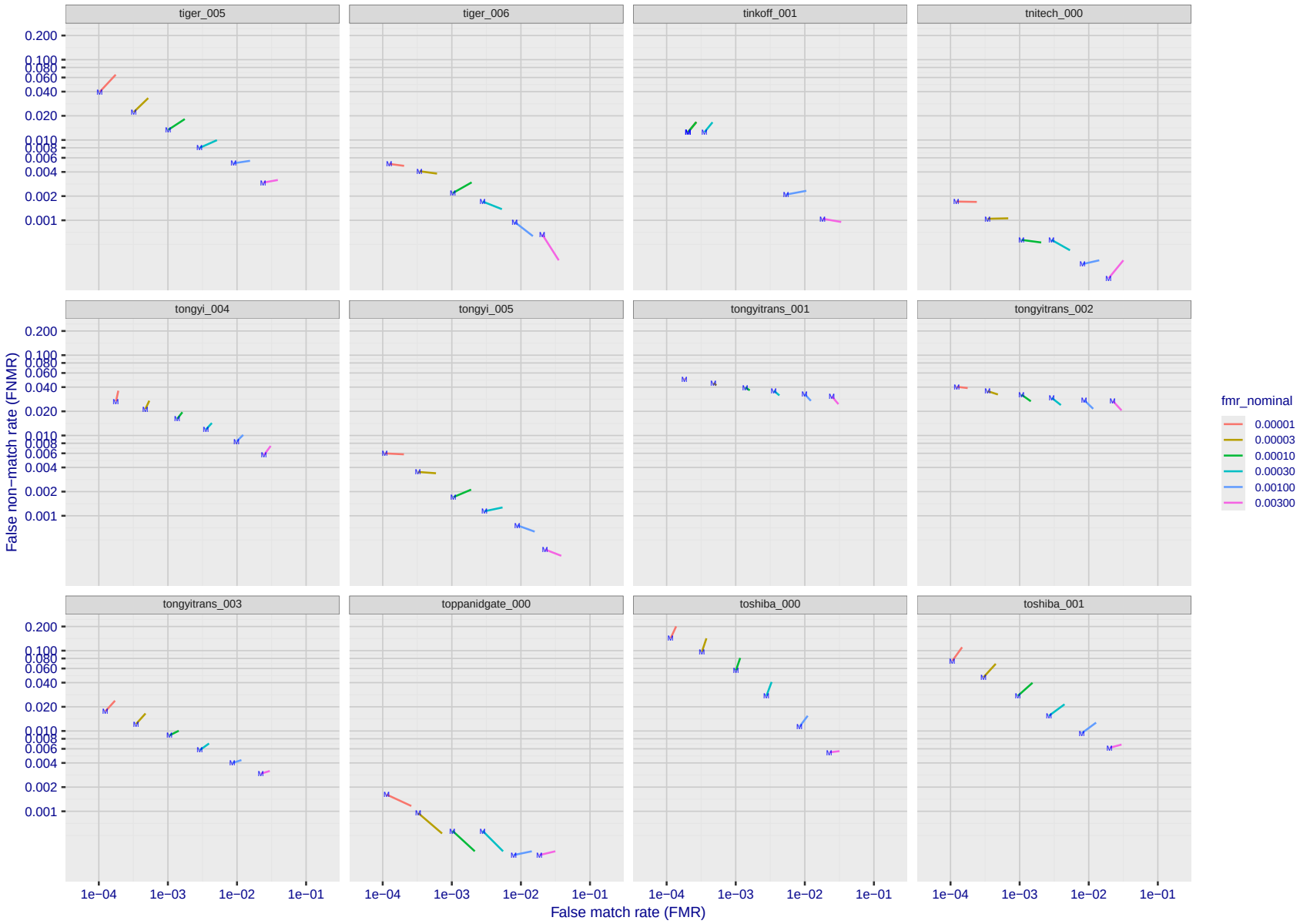

Figure 543: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

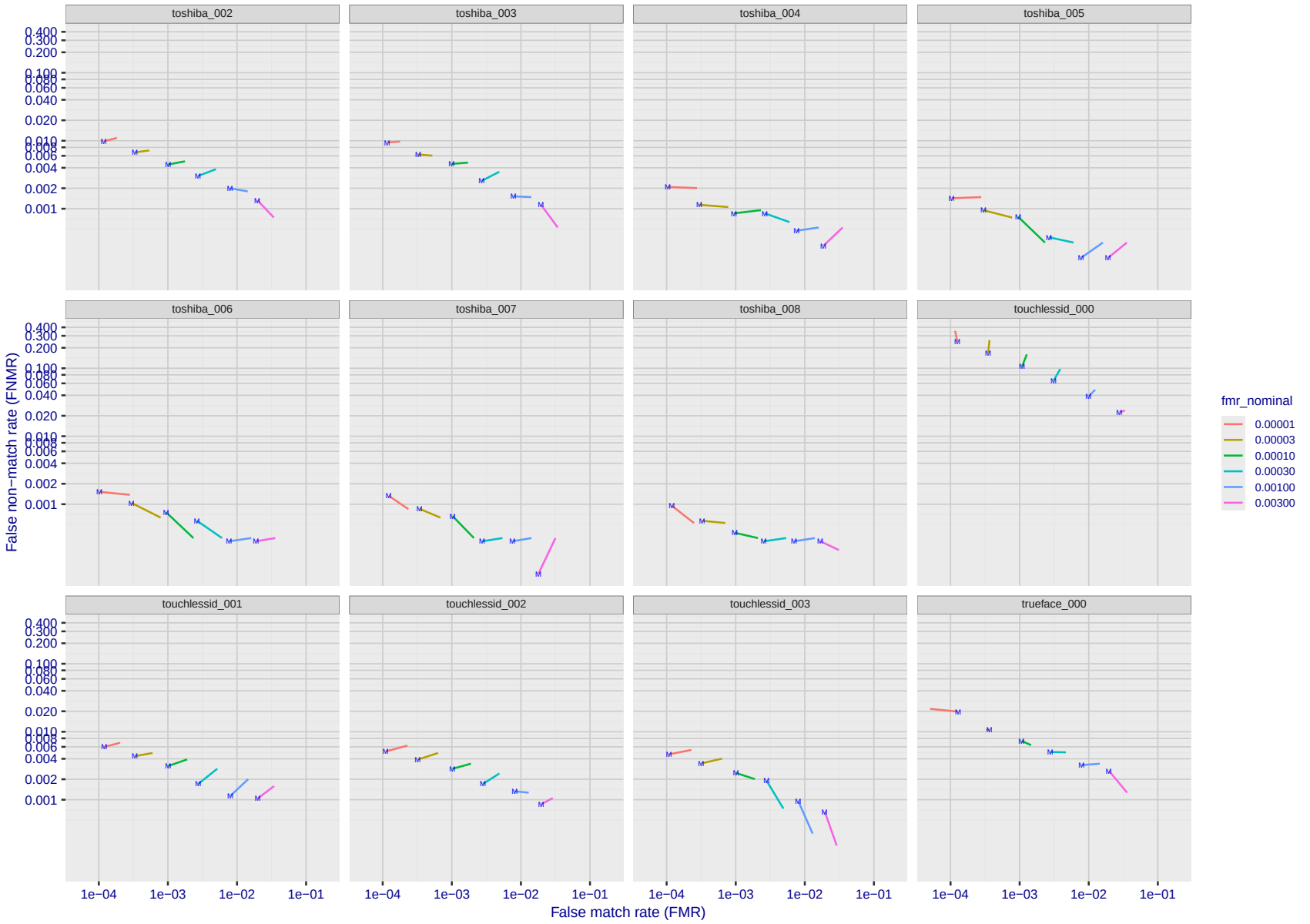

Figure 544: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

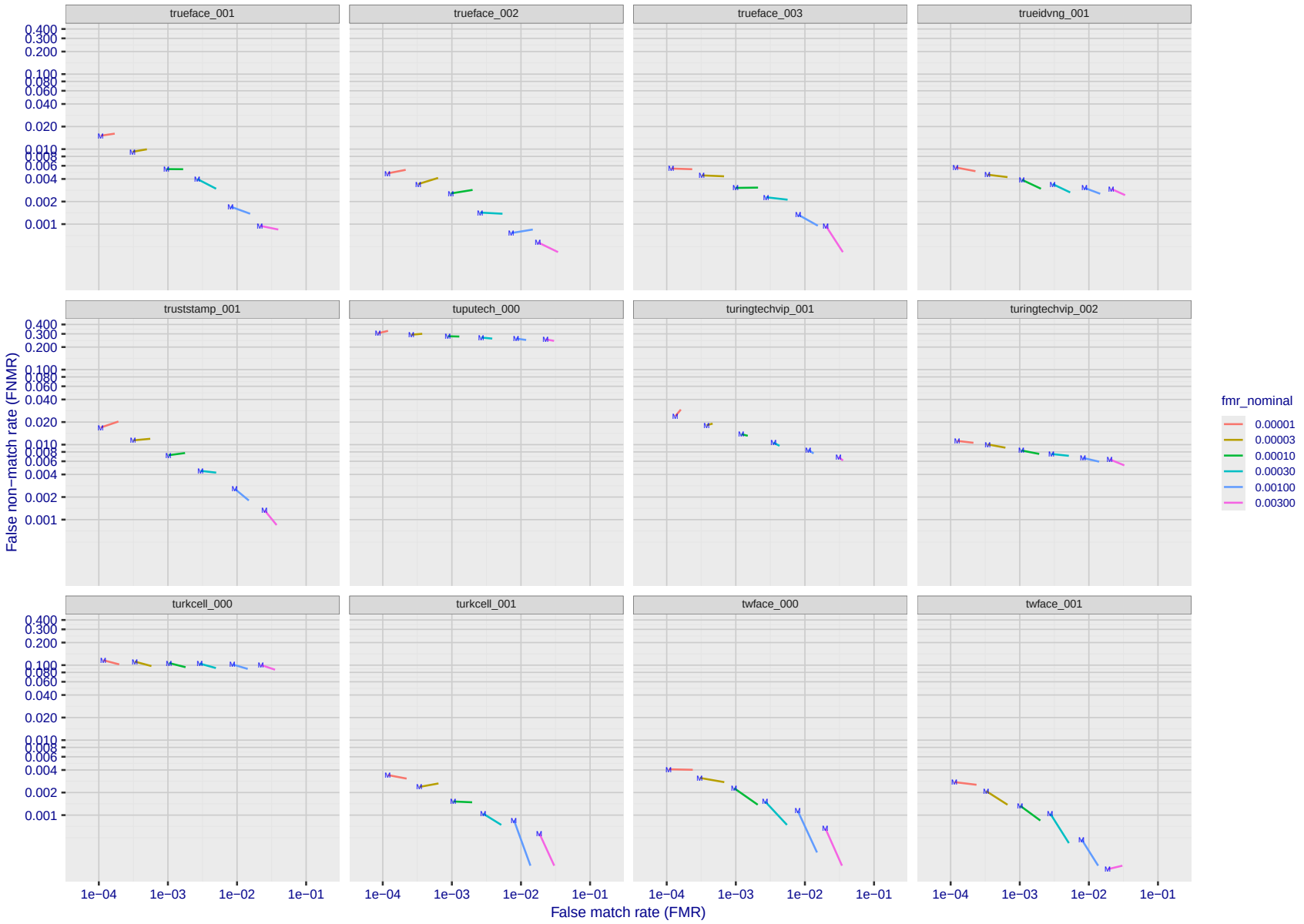

Figure 545: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

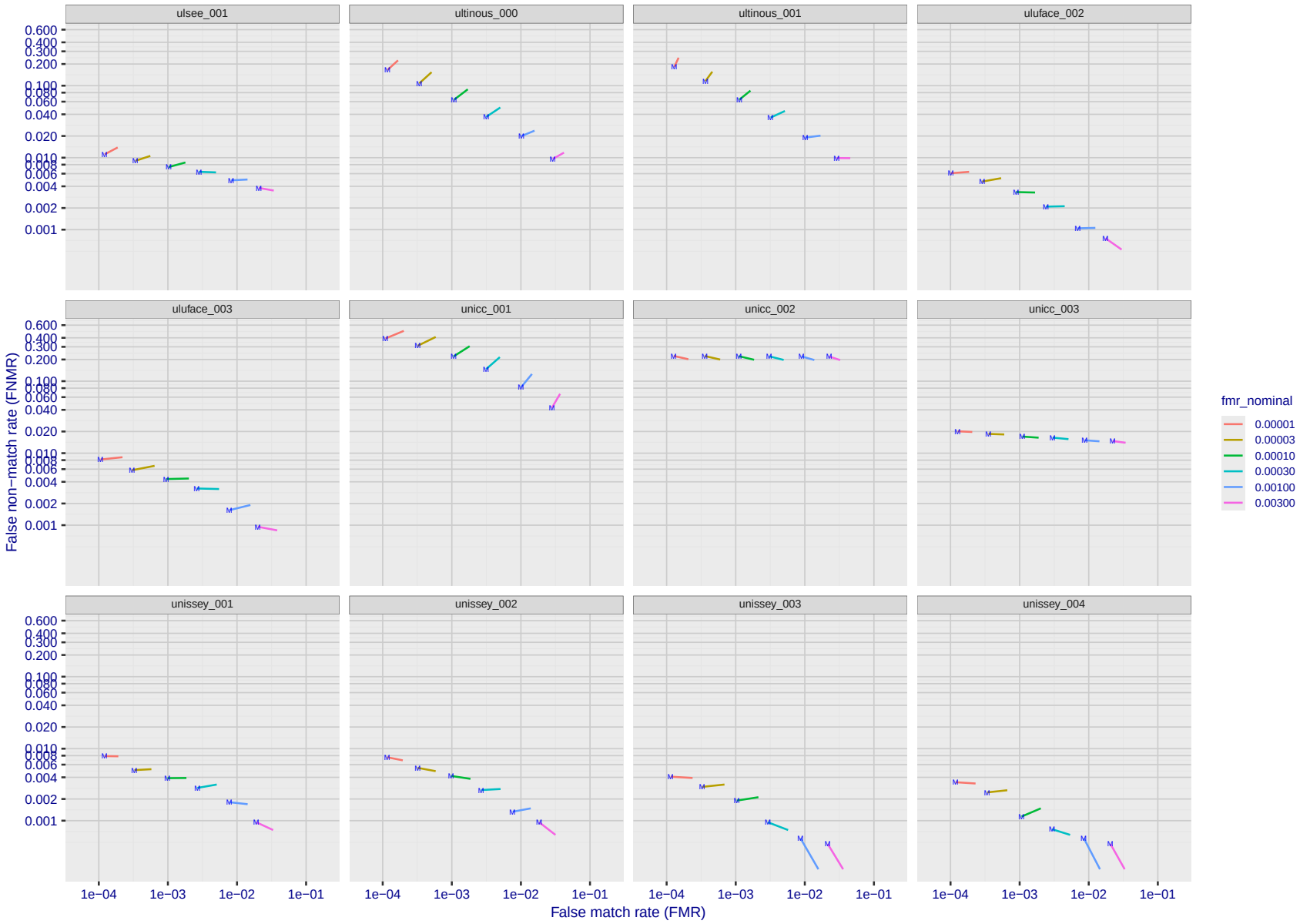

Figure 546: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

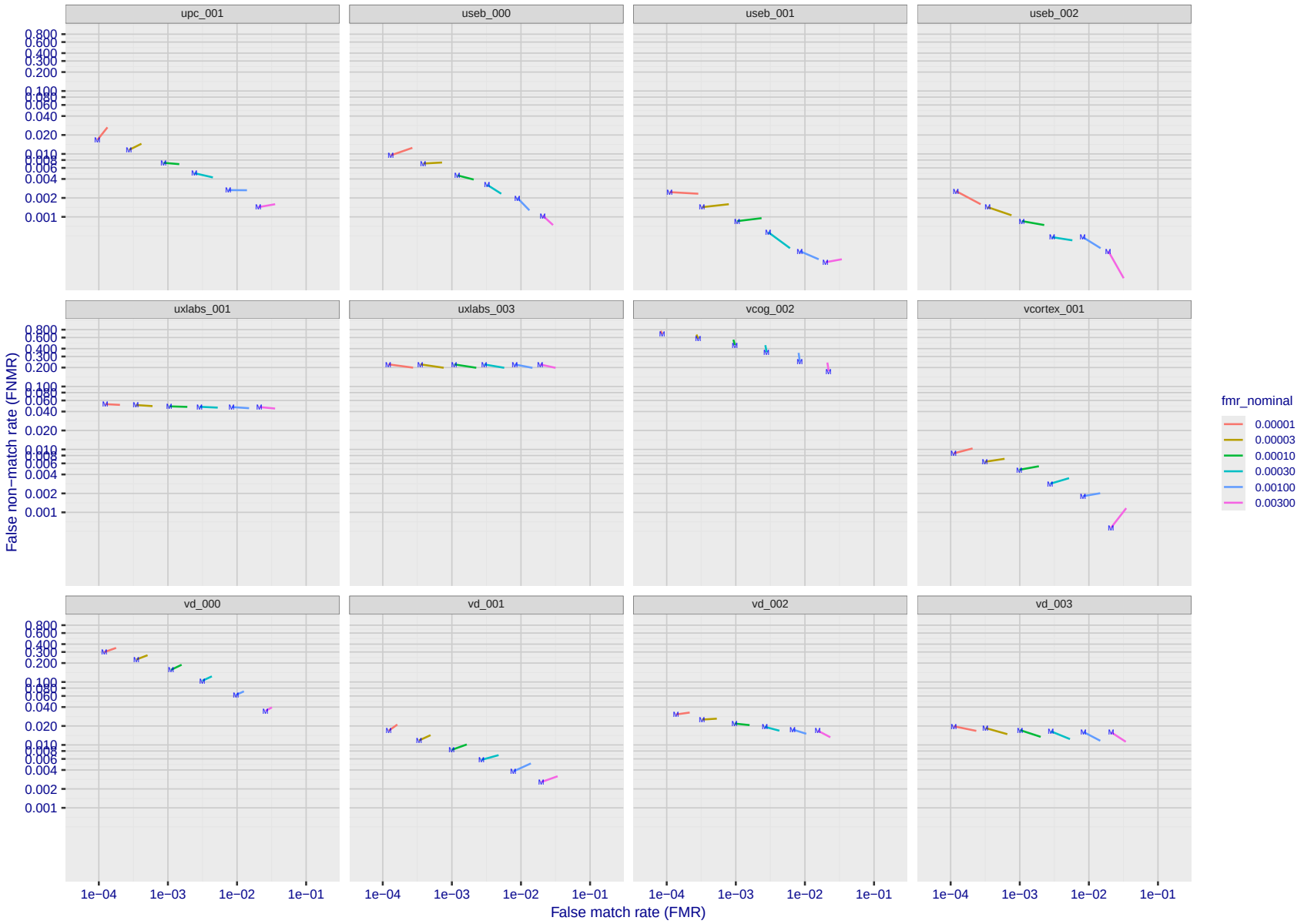

Figure 547: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

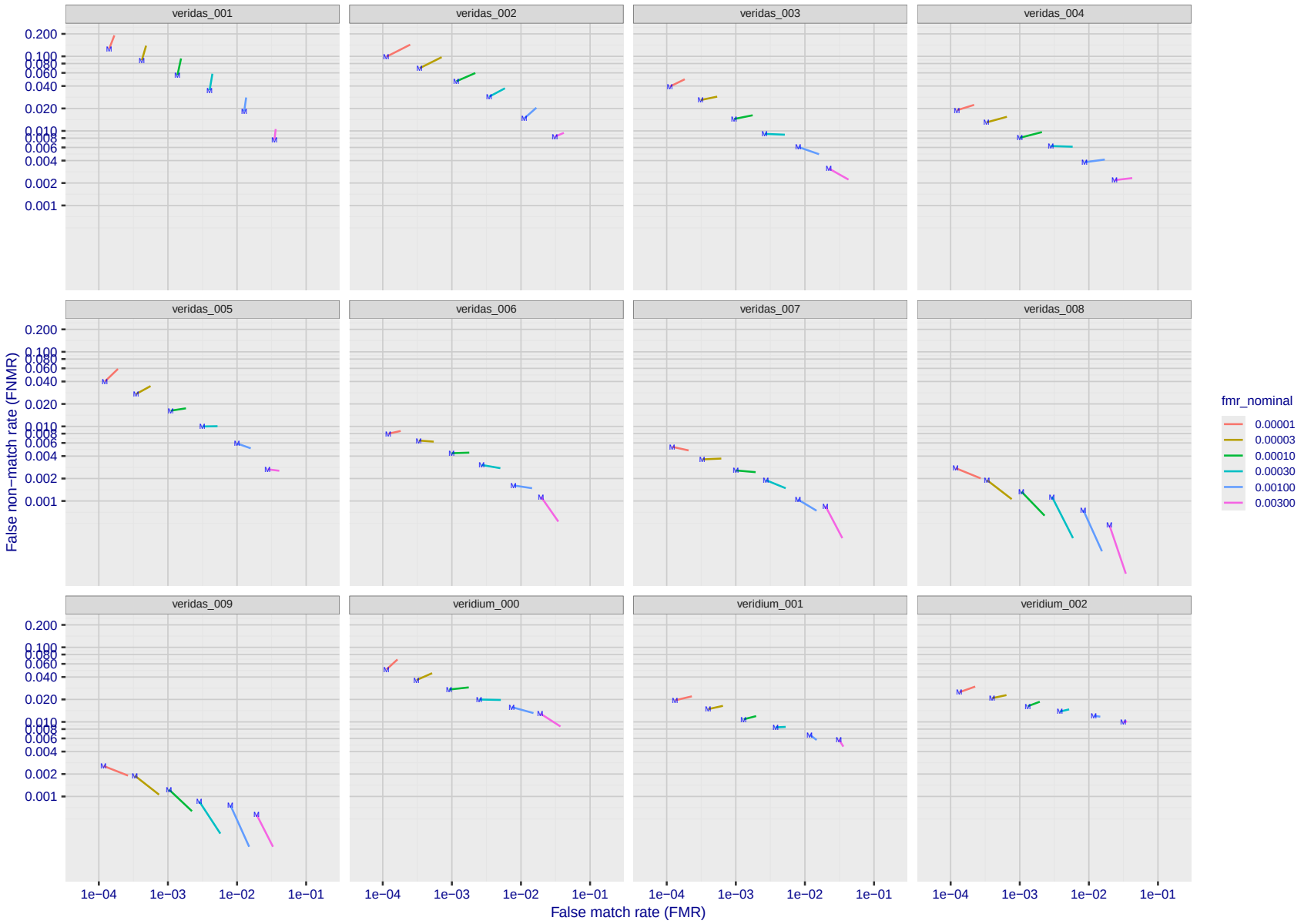

Figure 548: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

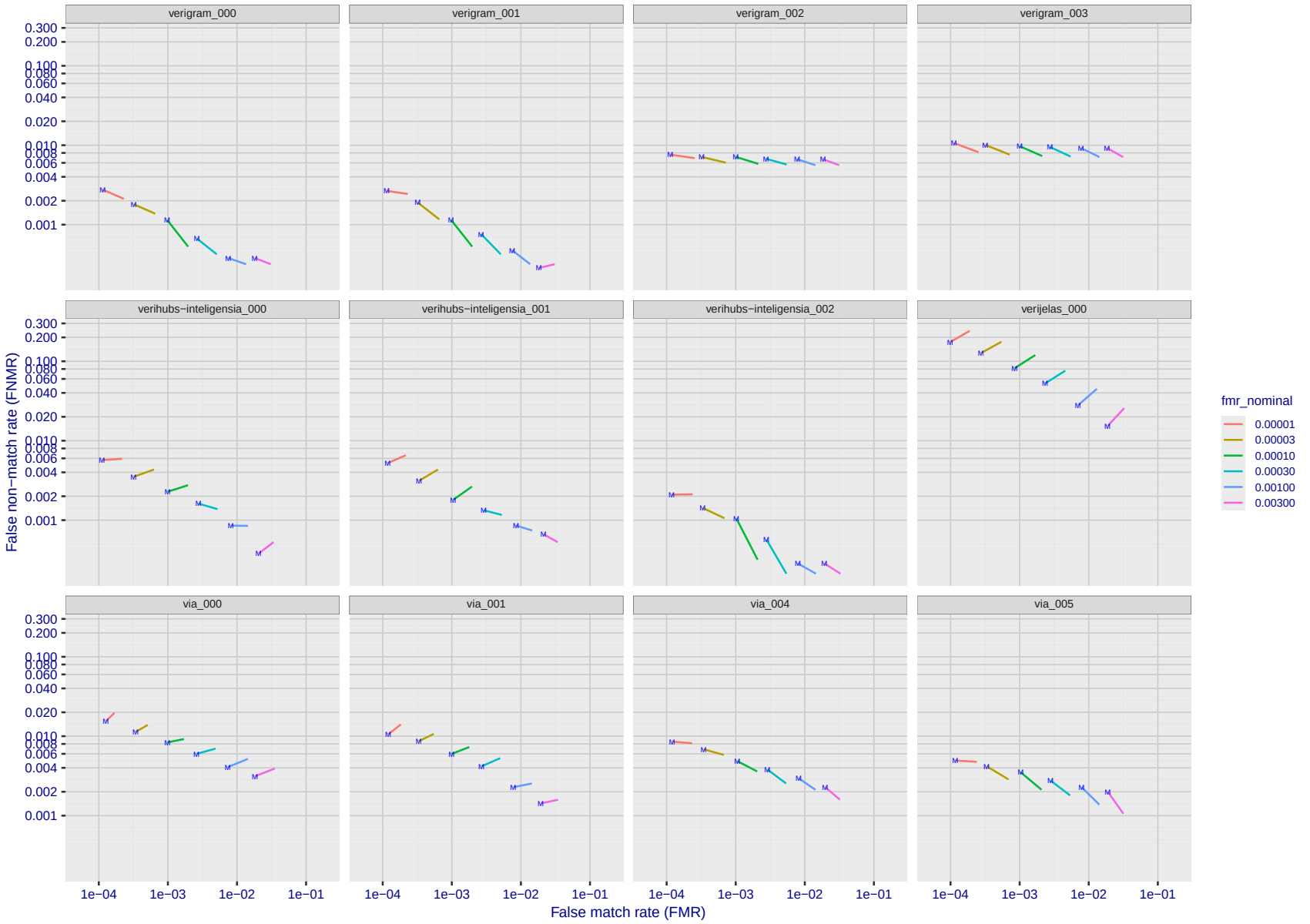

Figure 549: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

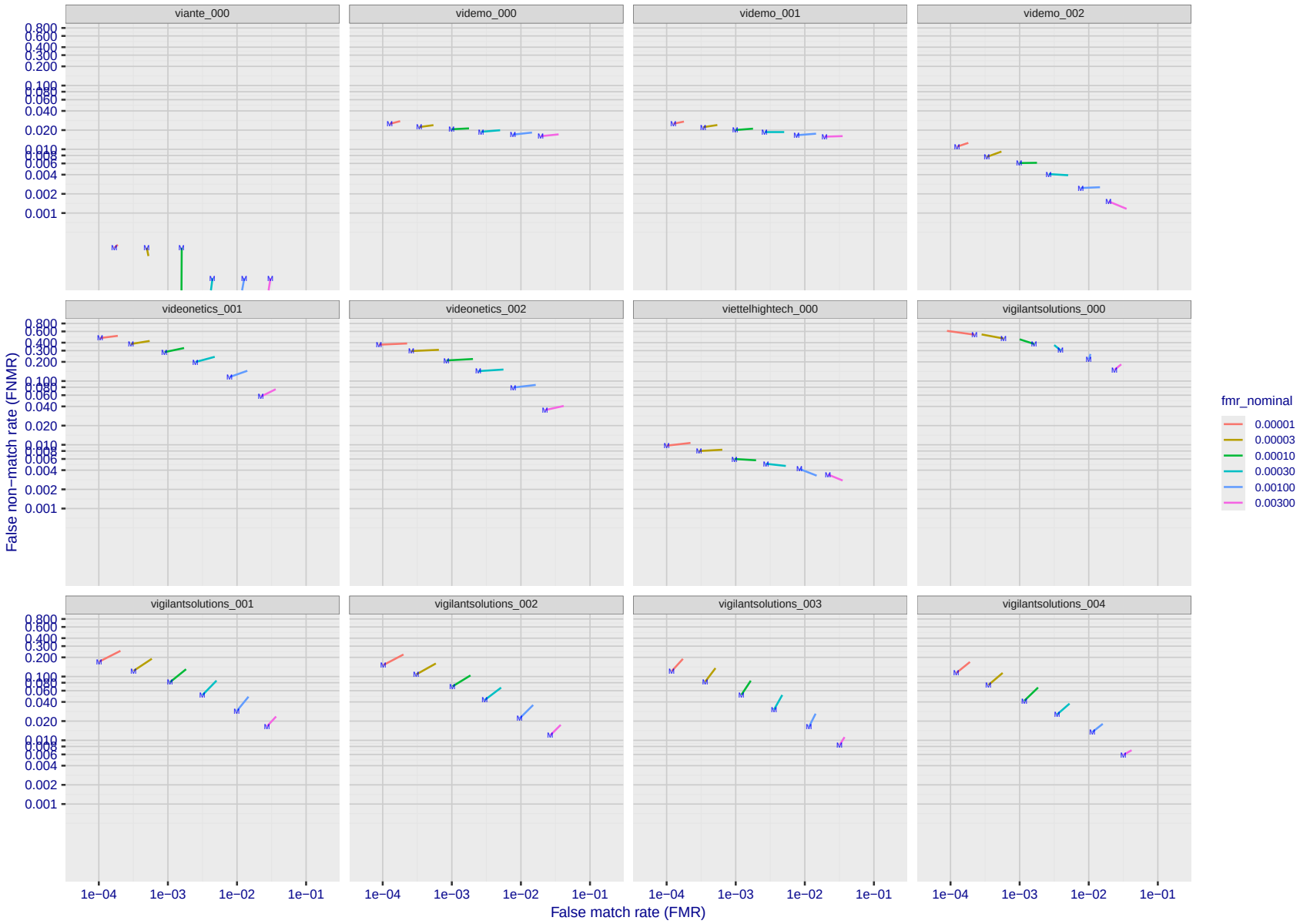

Figure 550: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

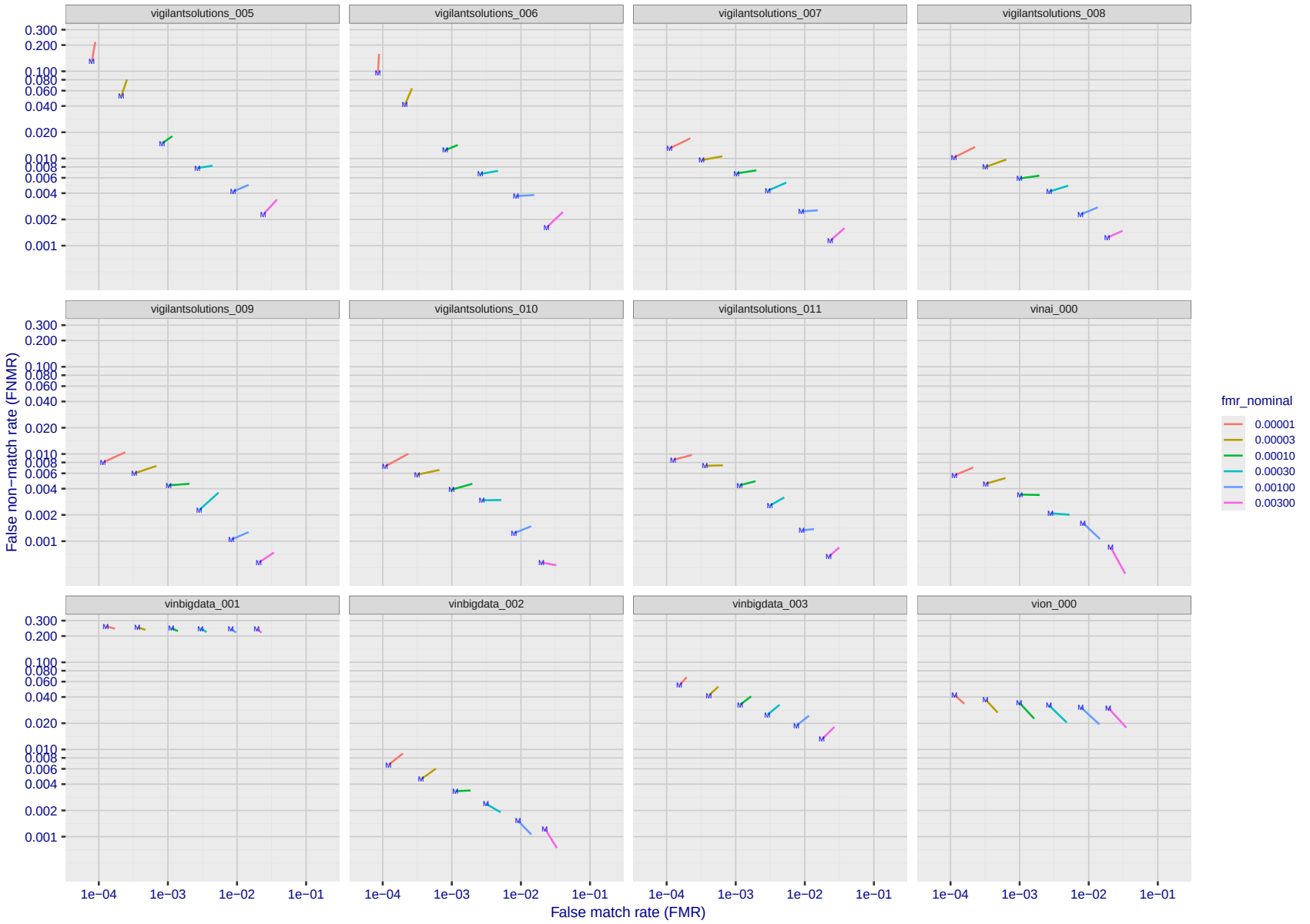

Figure 551: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

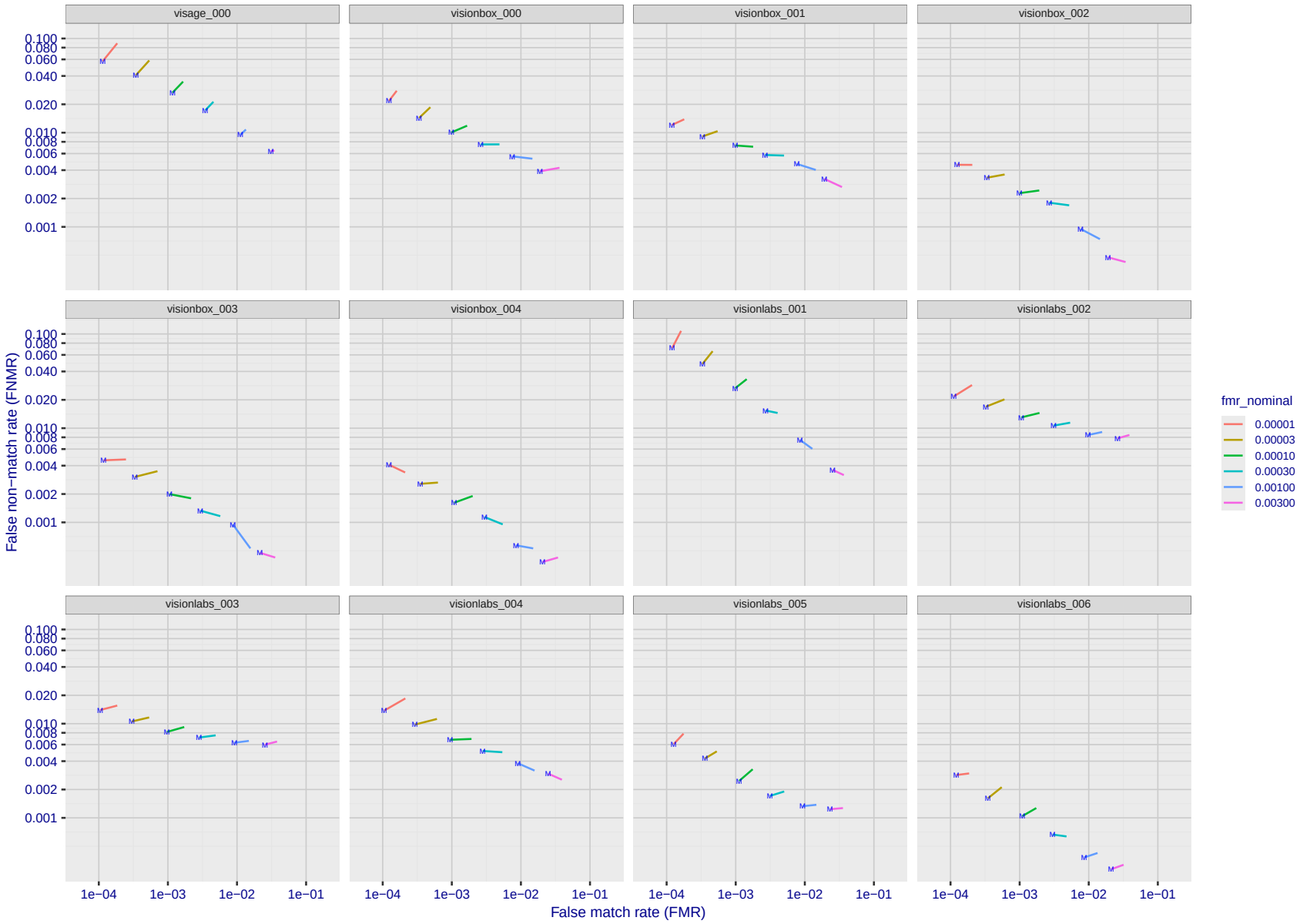

Figure 552: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

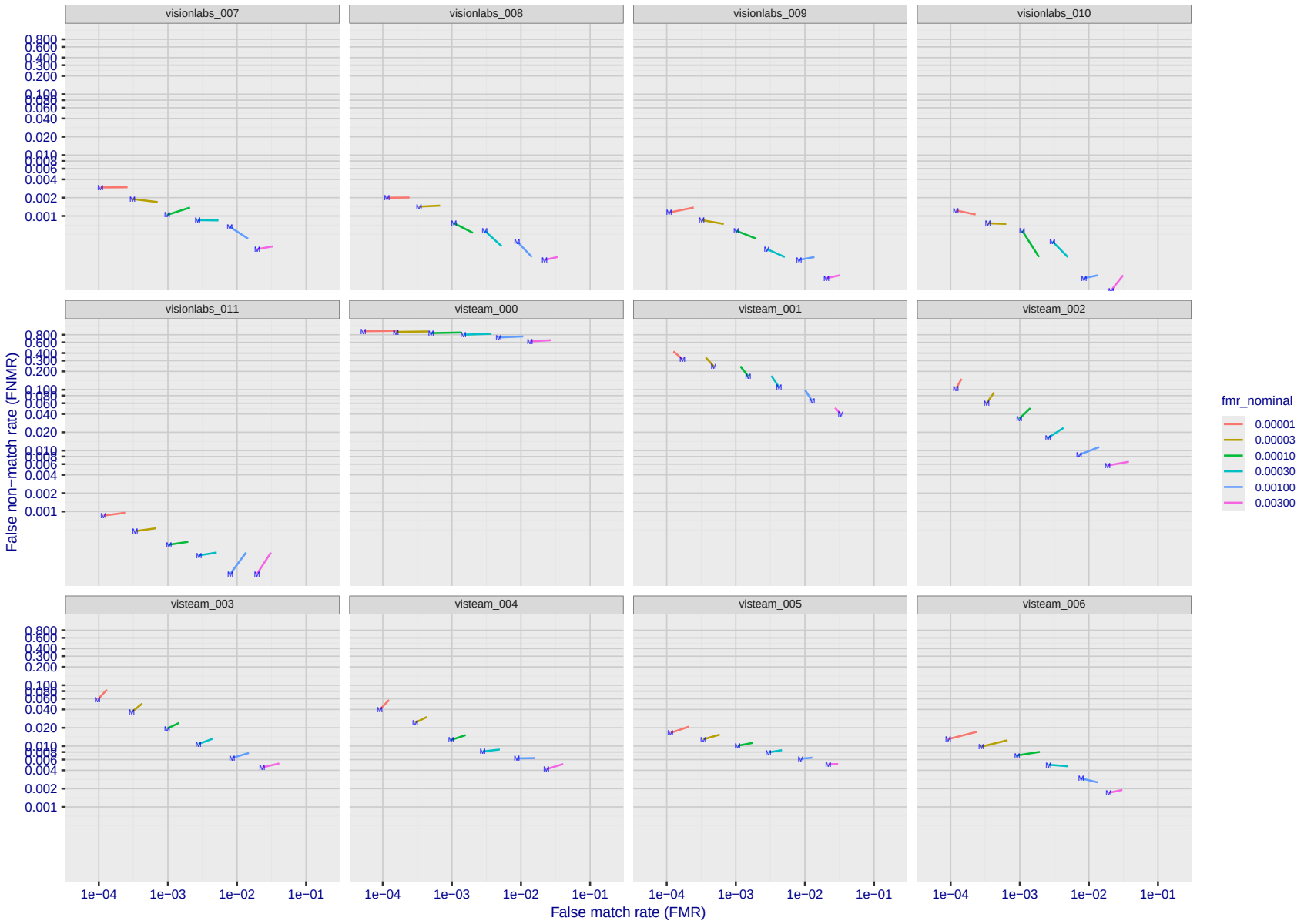

Figure 553: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

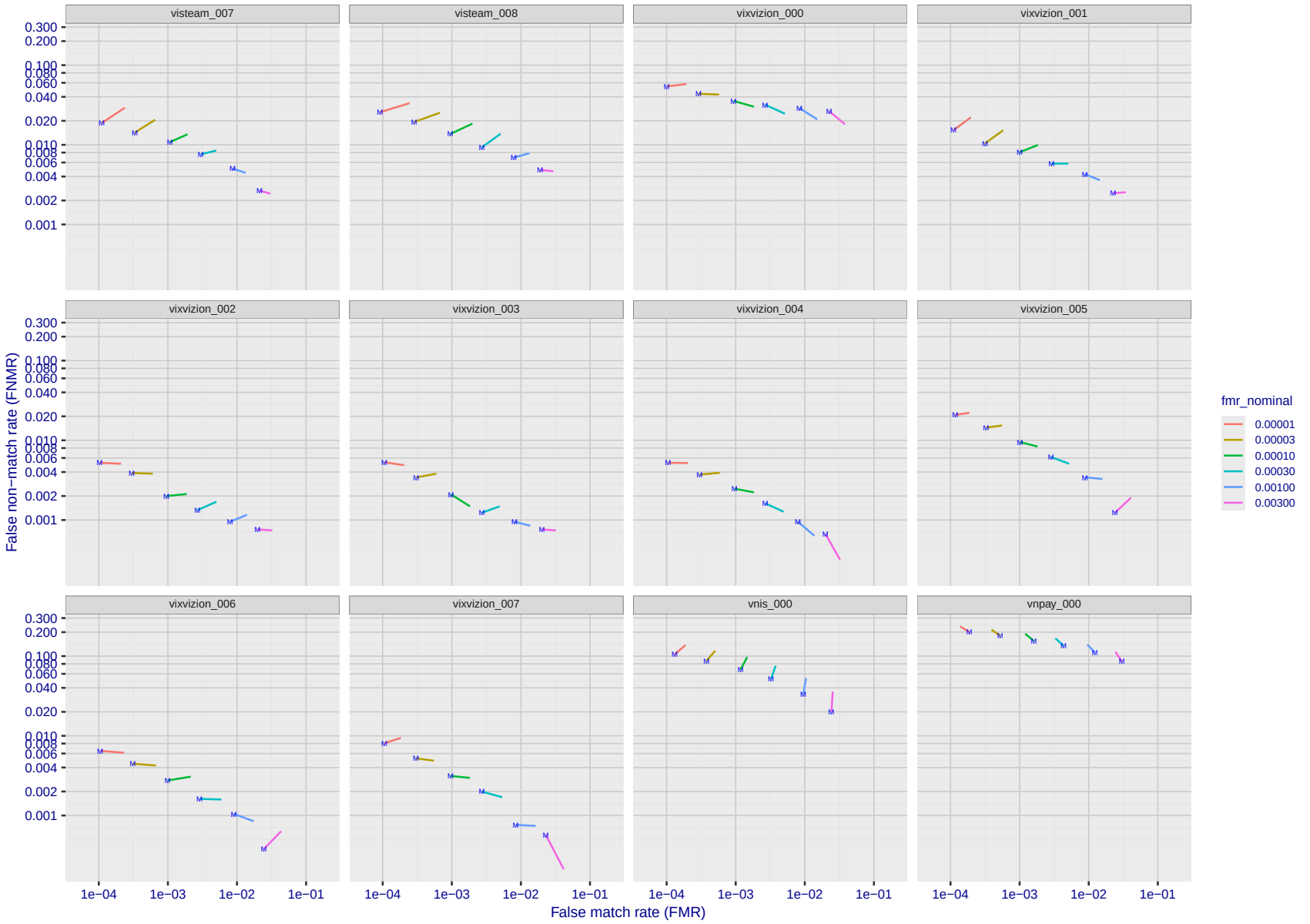

Figure 554: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

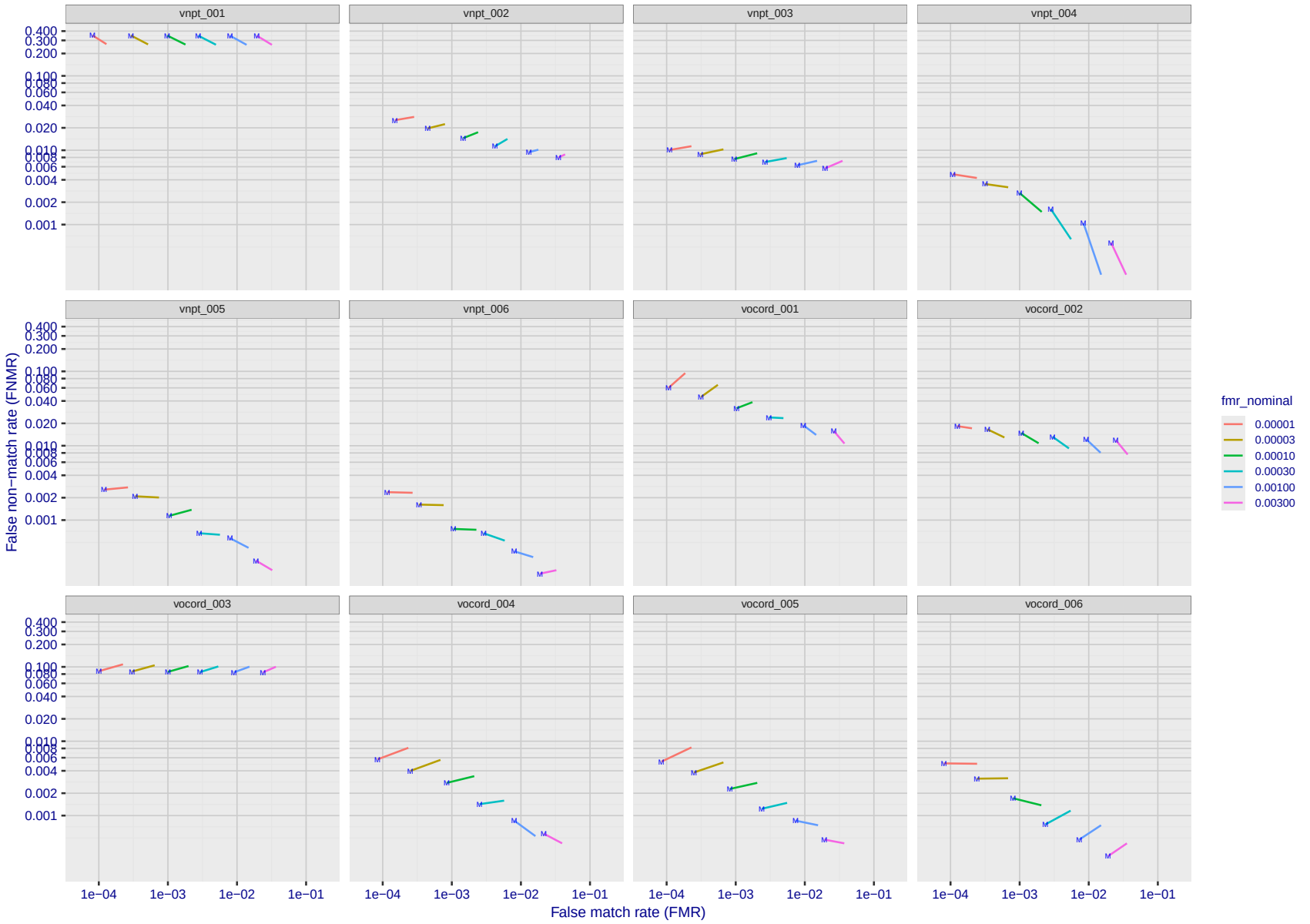

Figure 555: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

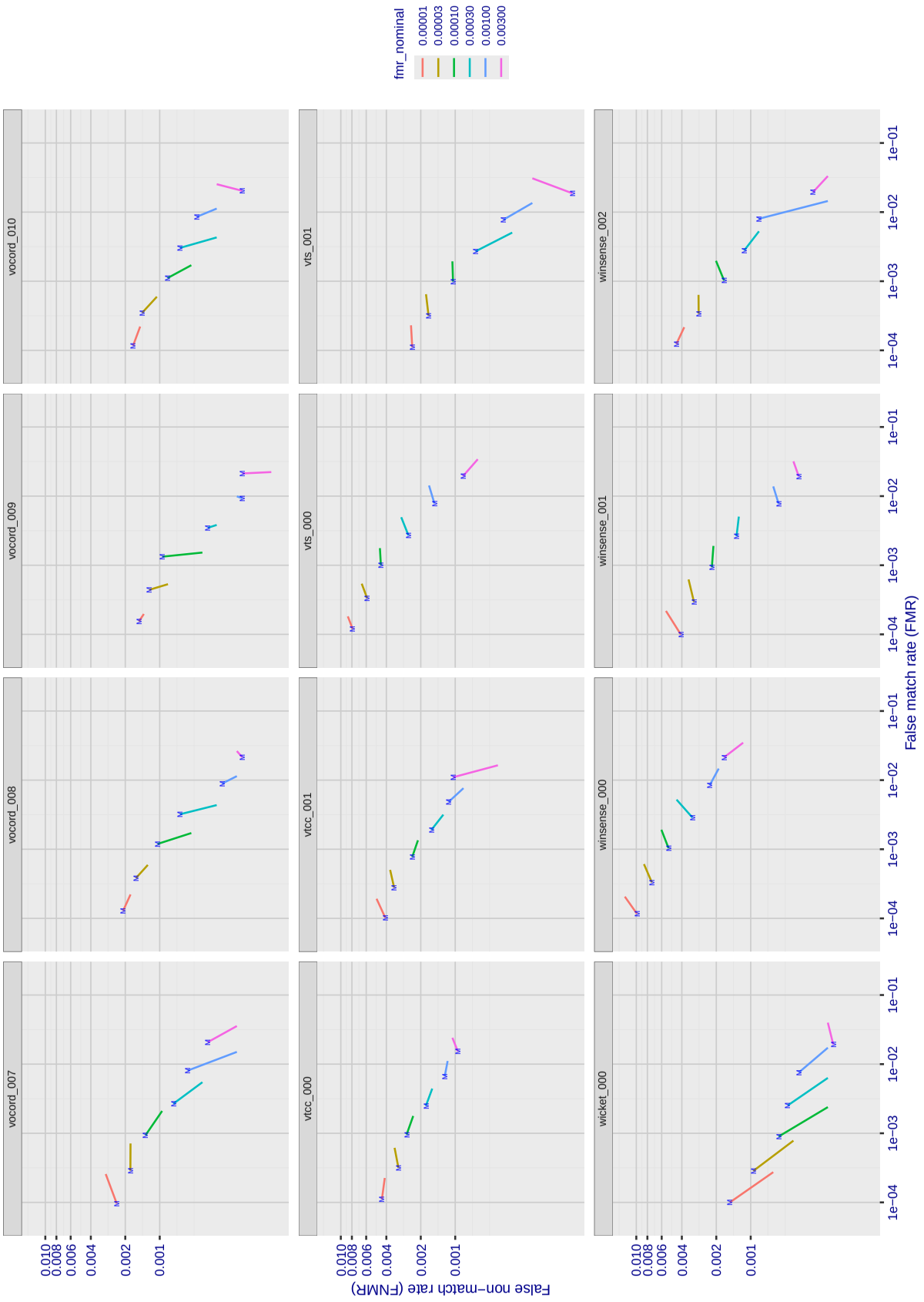

Figure 556: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

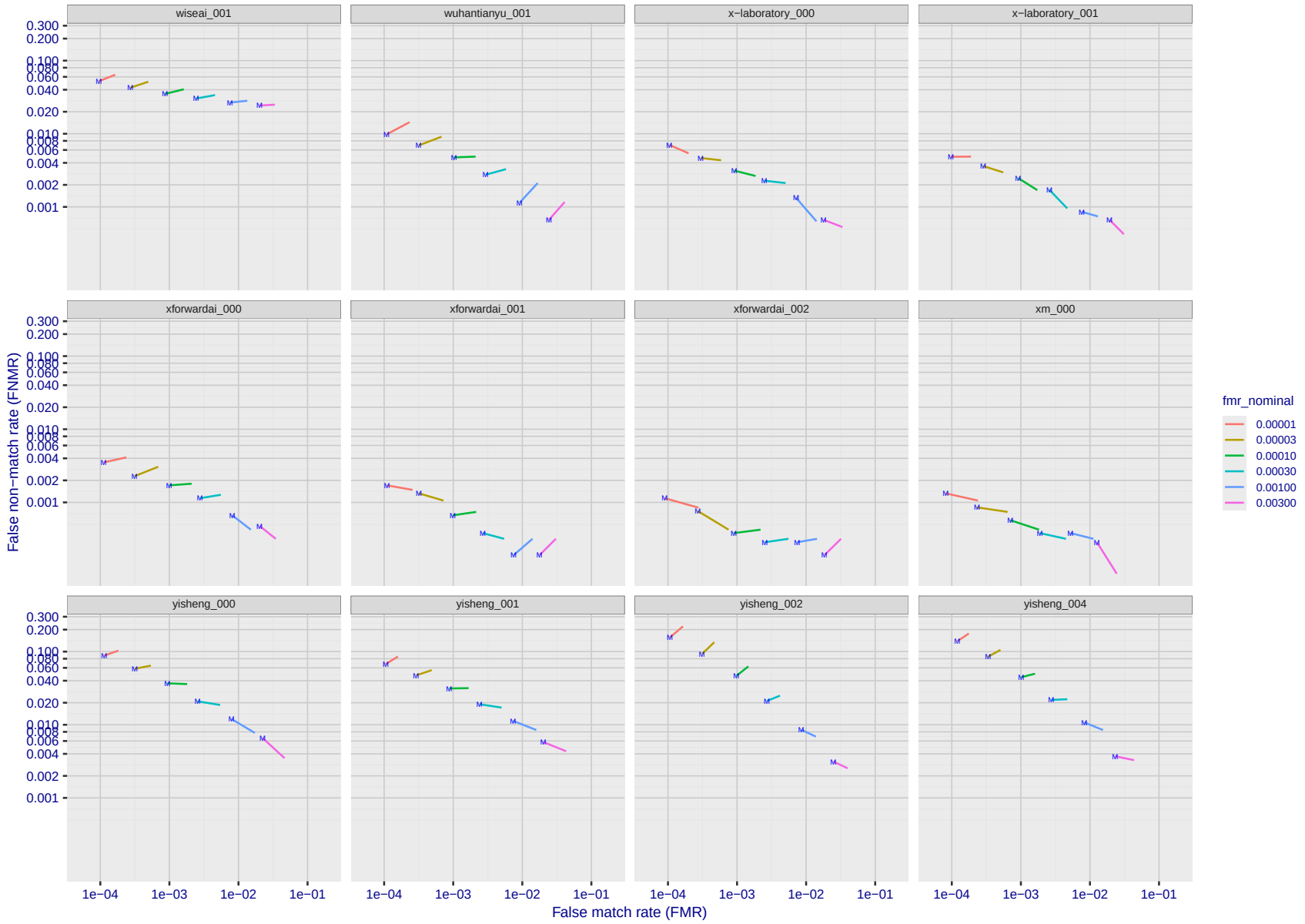

Figure 557: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

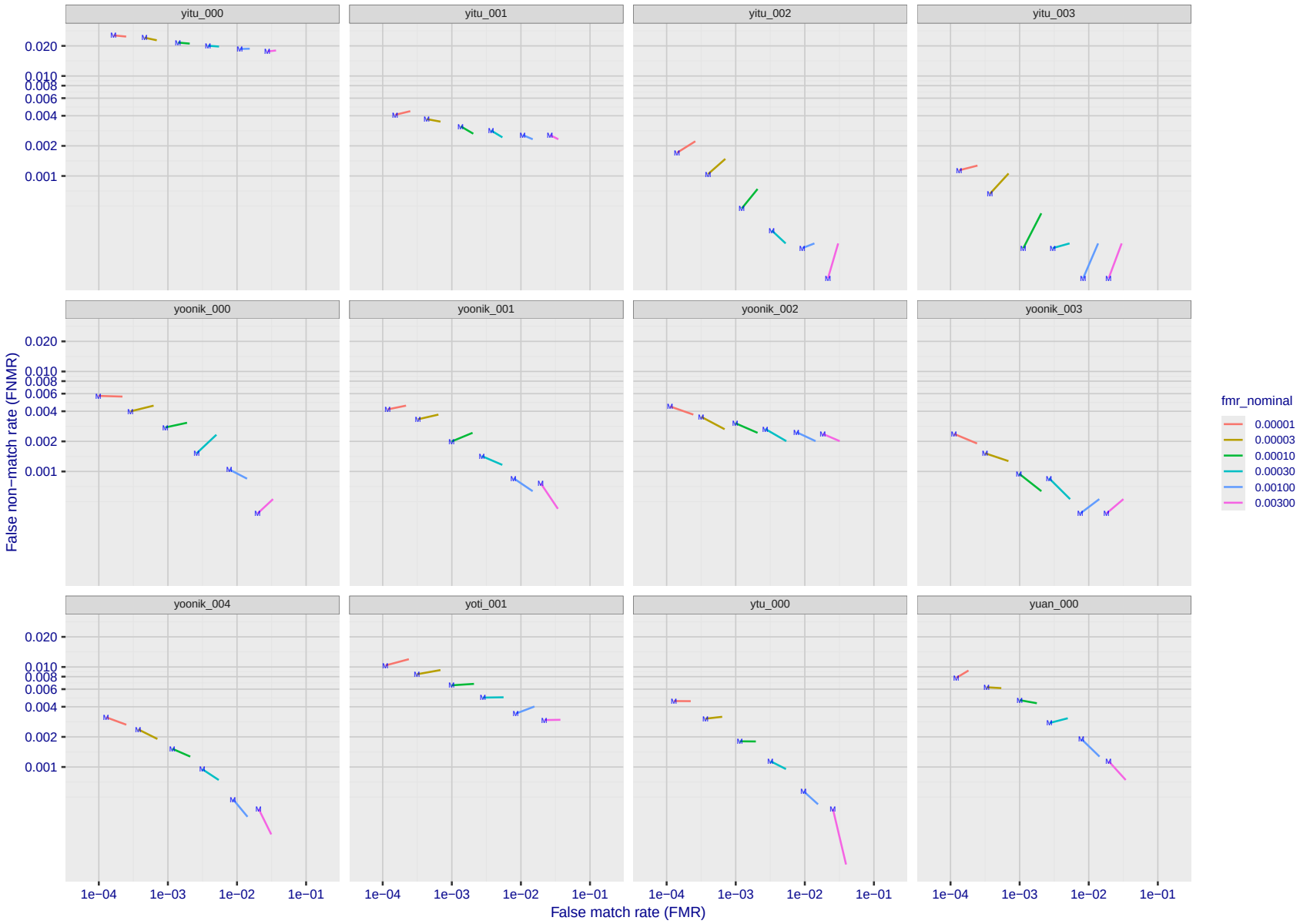

Figure 558: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

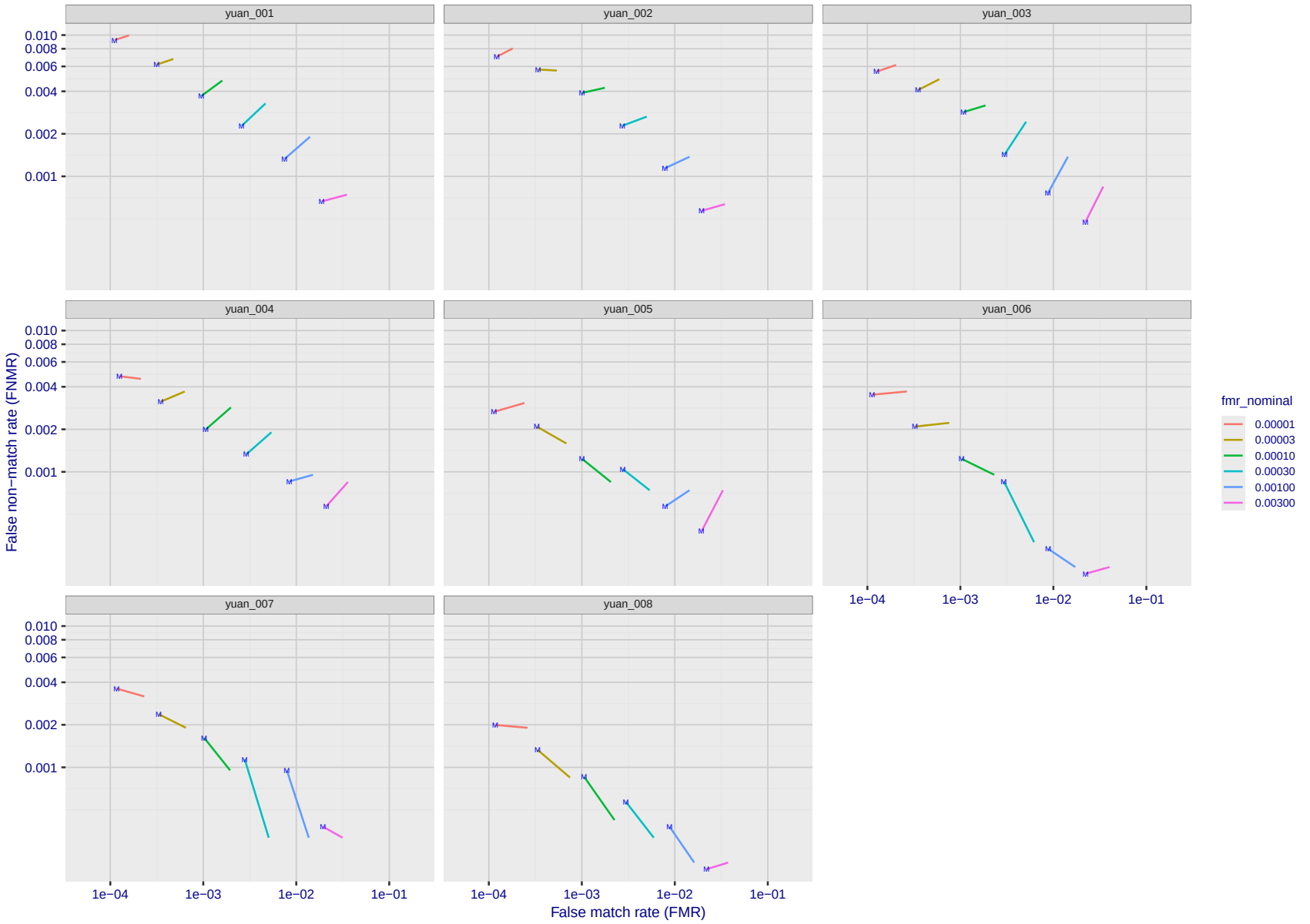

Figure 559: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

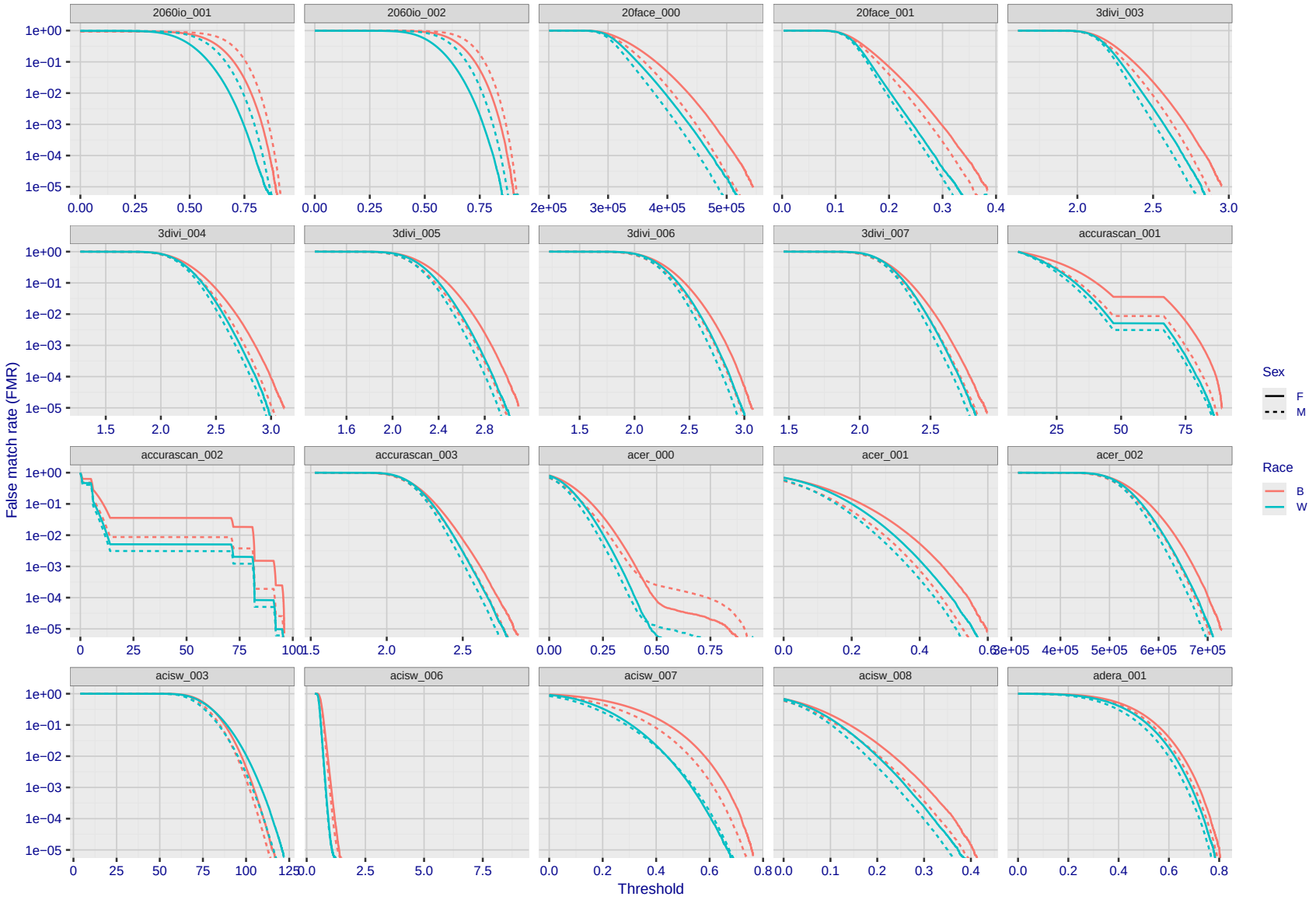

Figure 560: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

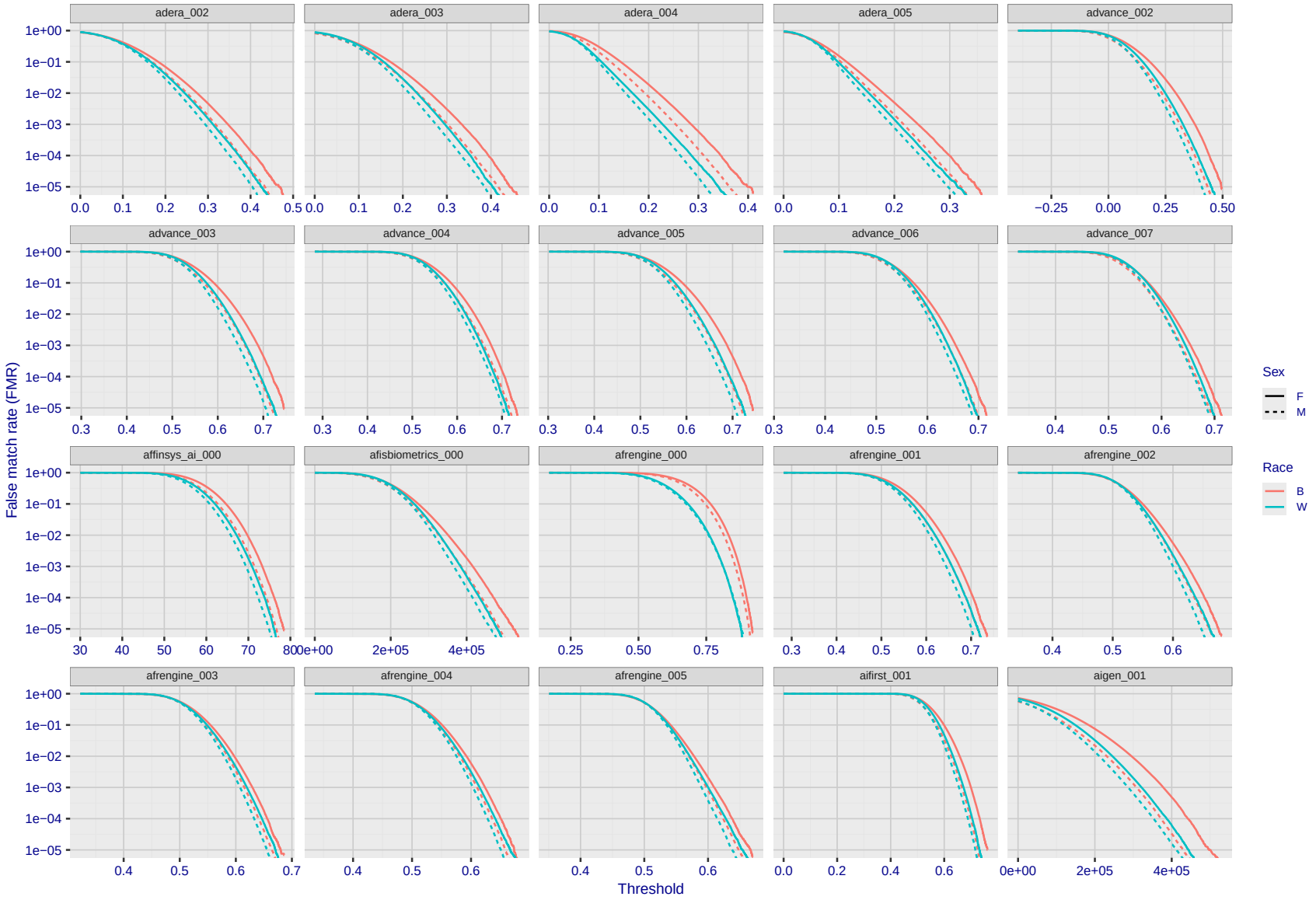

Figure 561: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

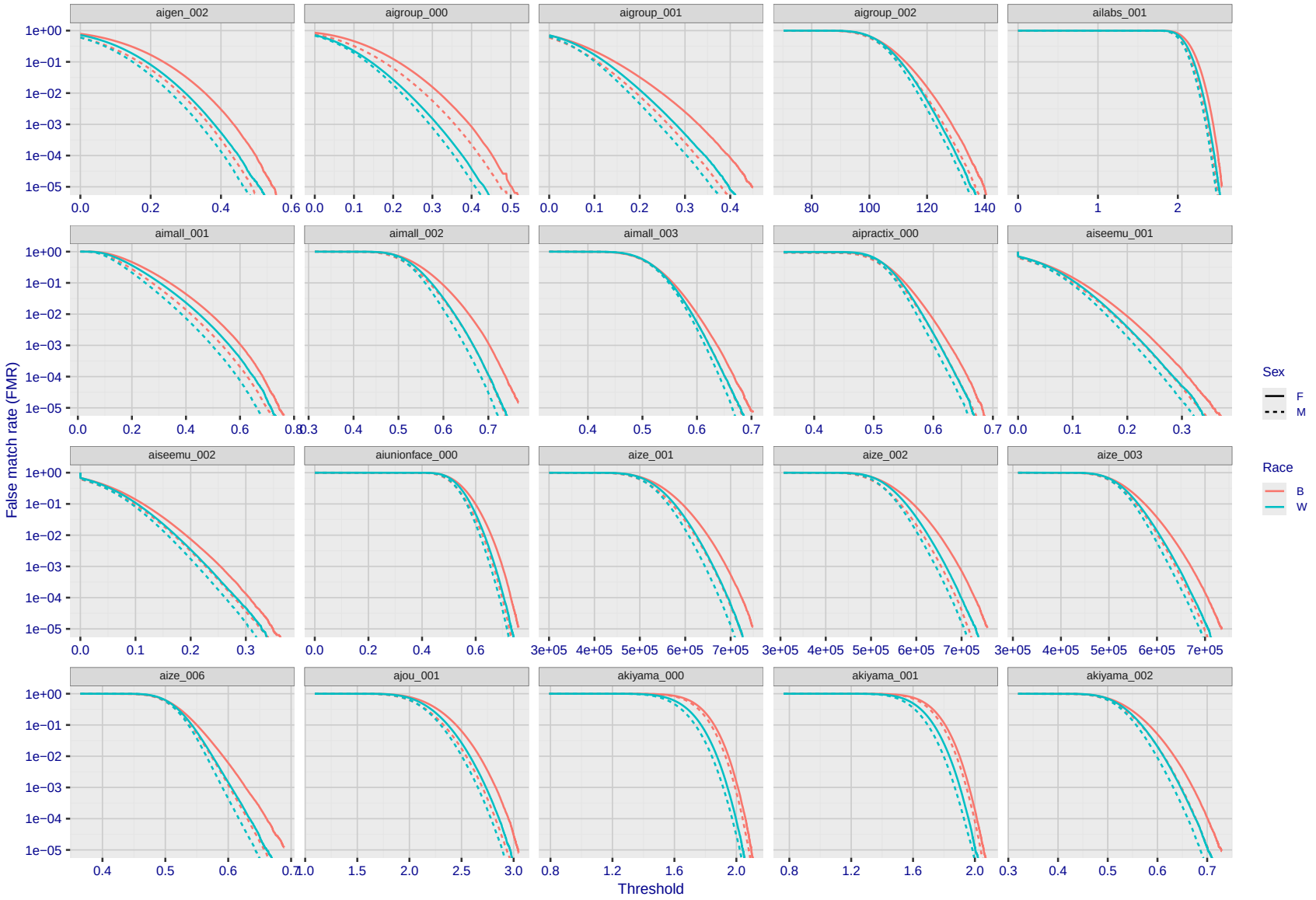

Figure 562: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

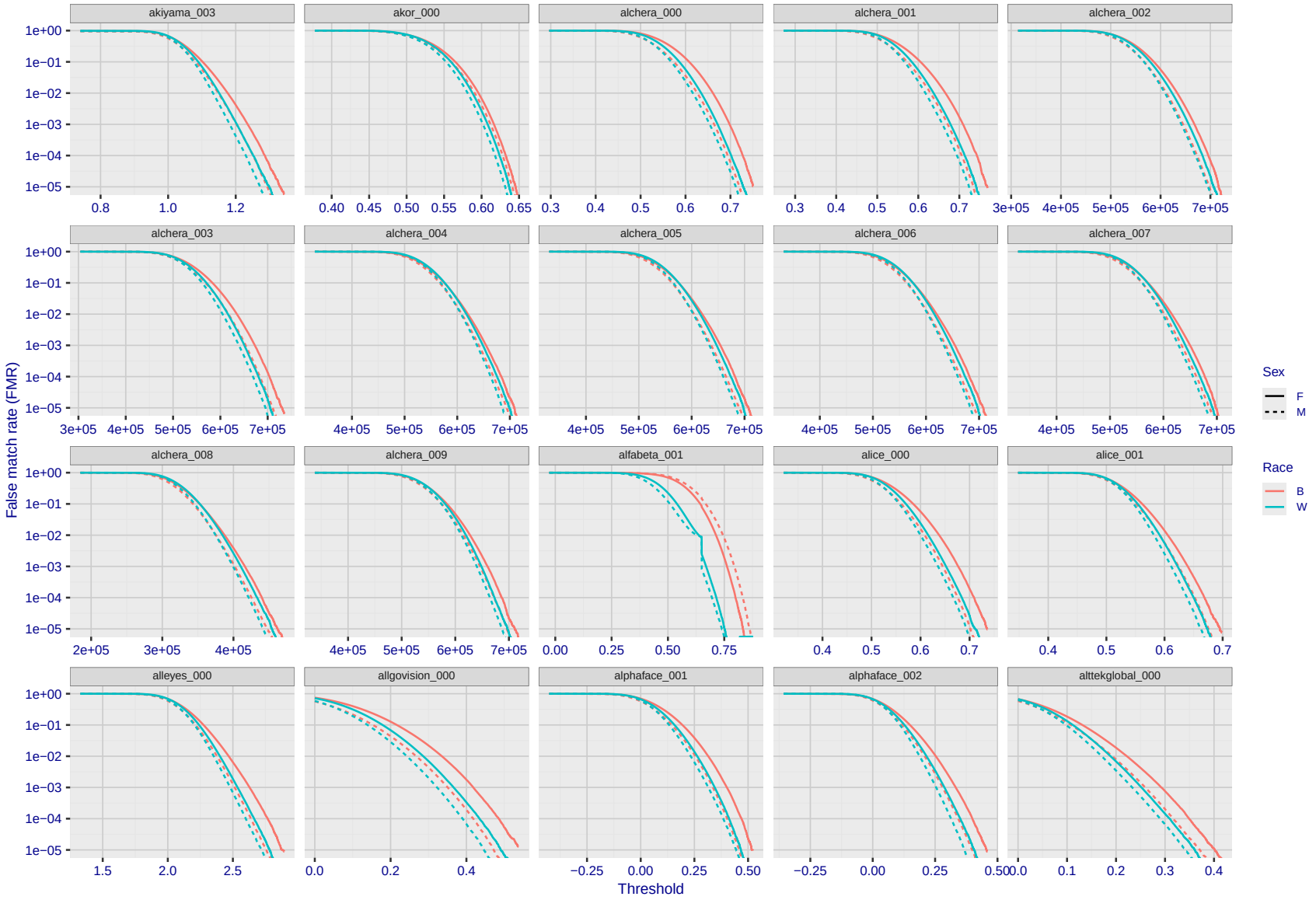

Figure 563: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

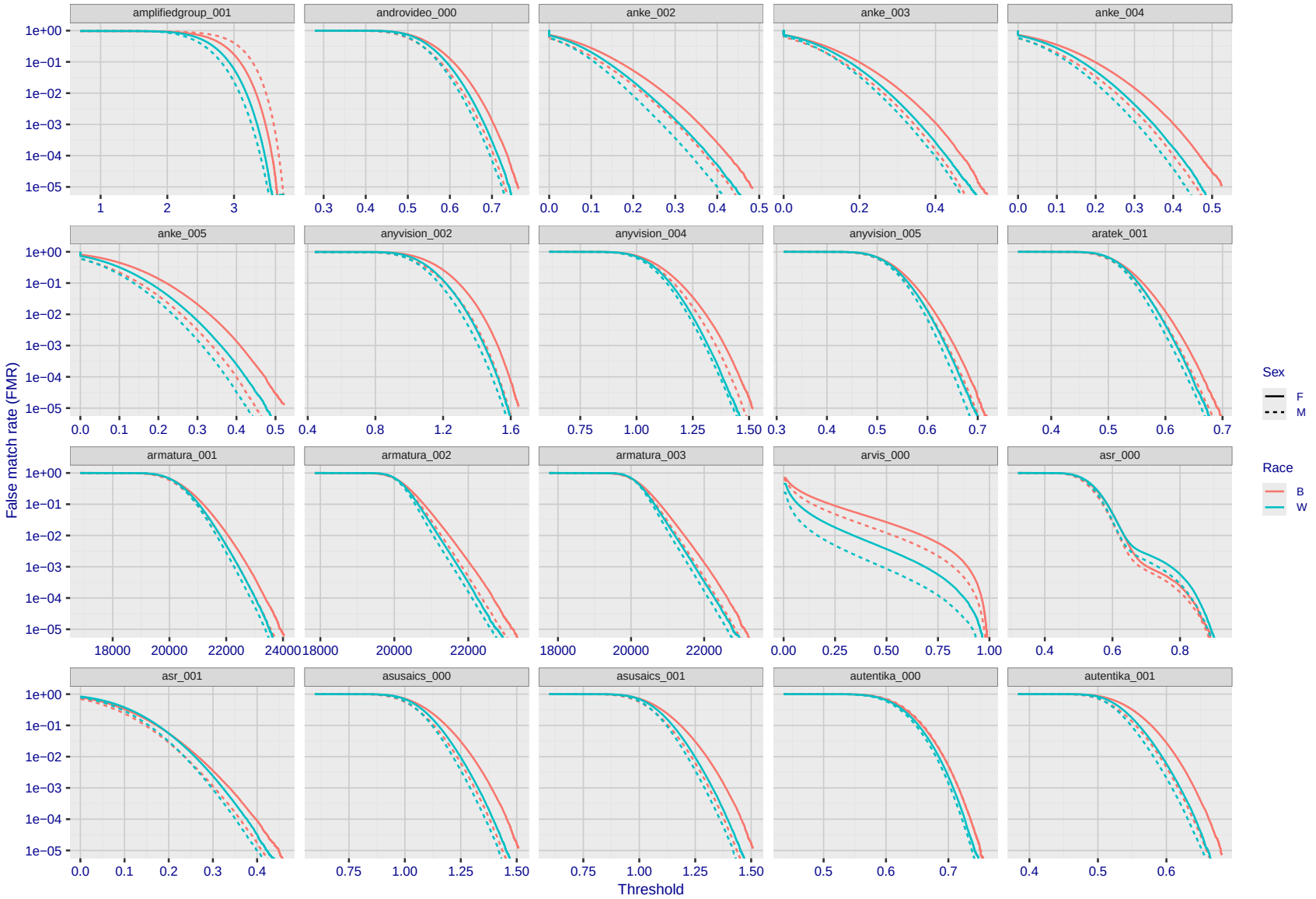

Figure 564: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

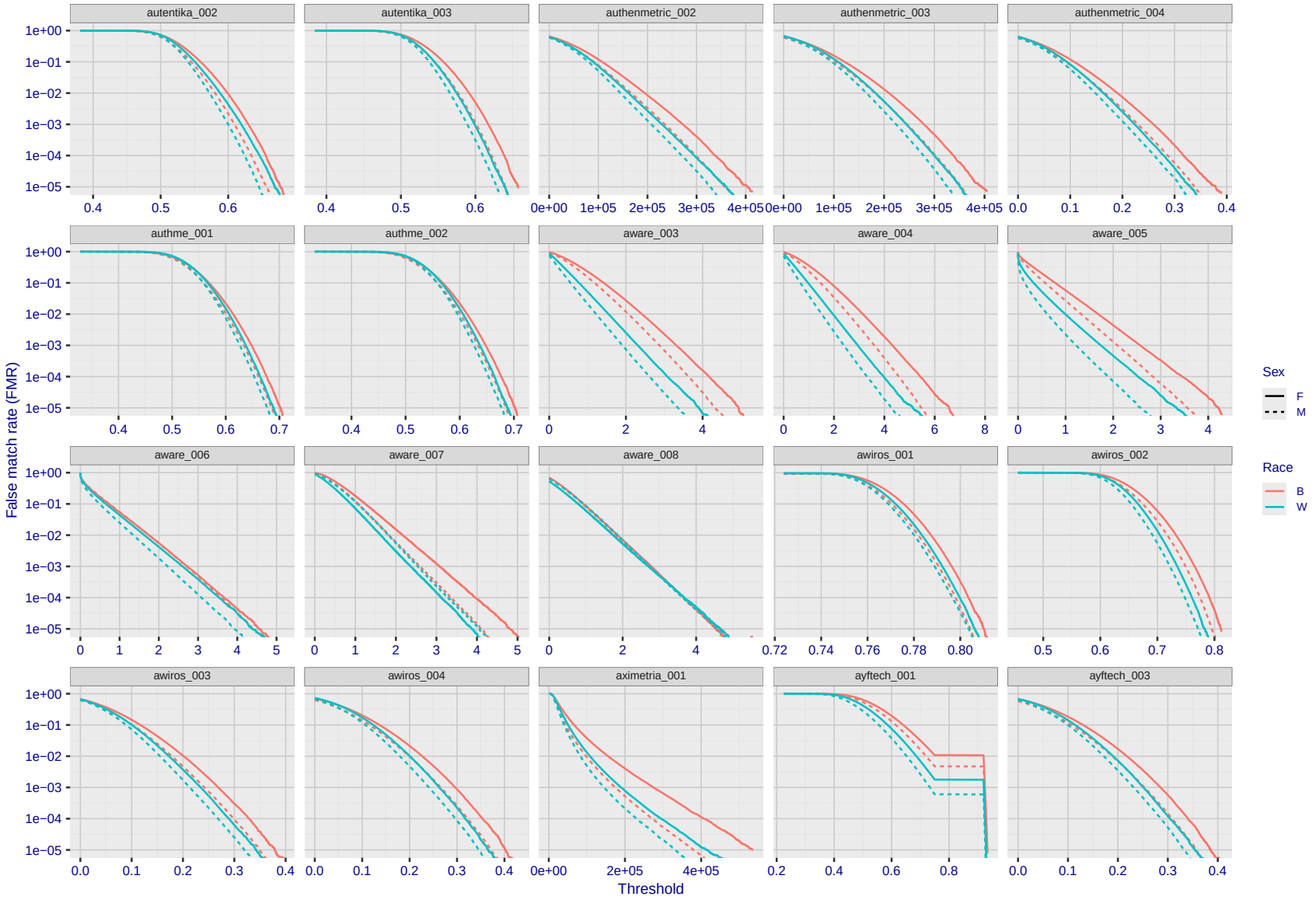

Figure 565: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

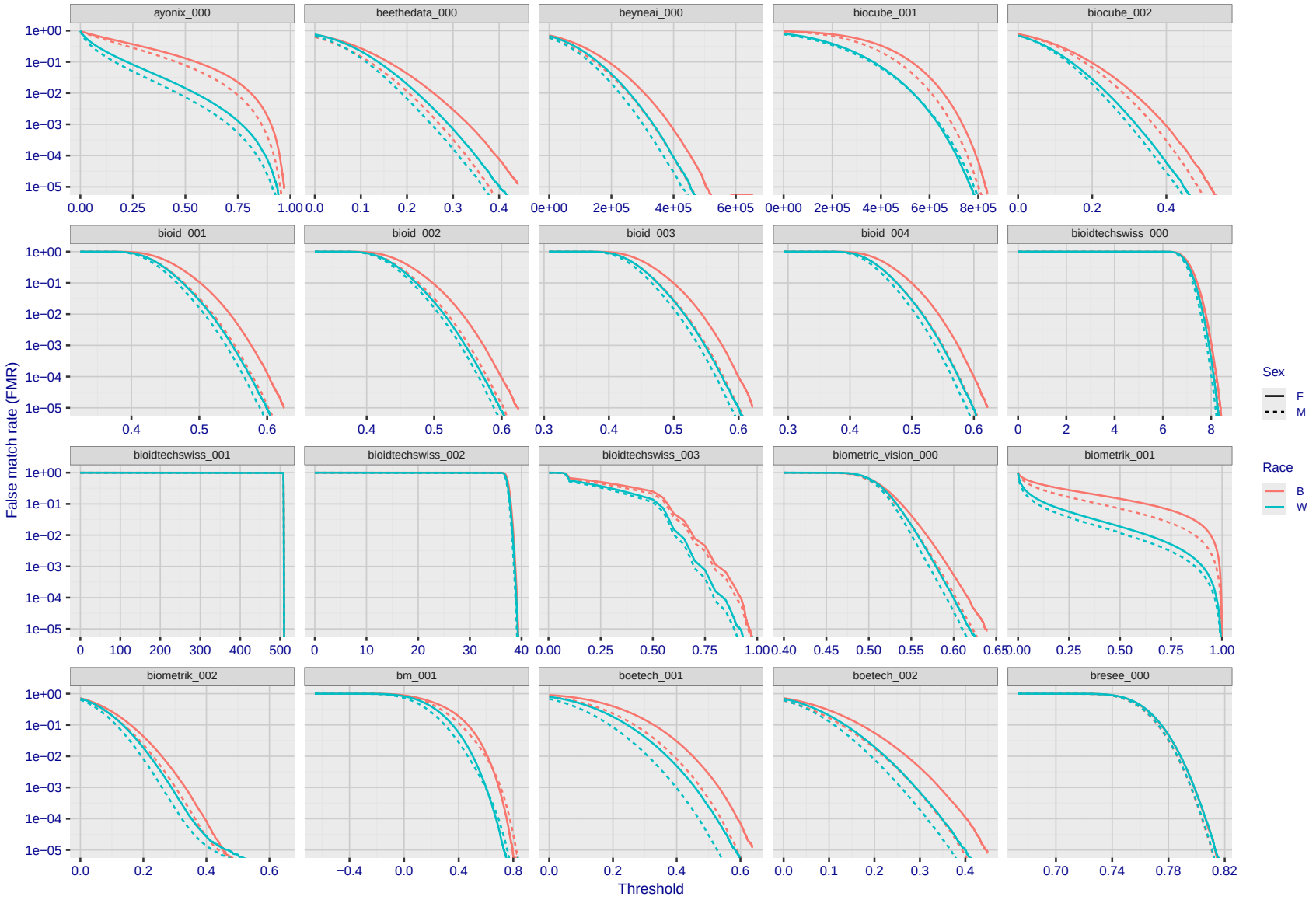

Figure 566: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

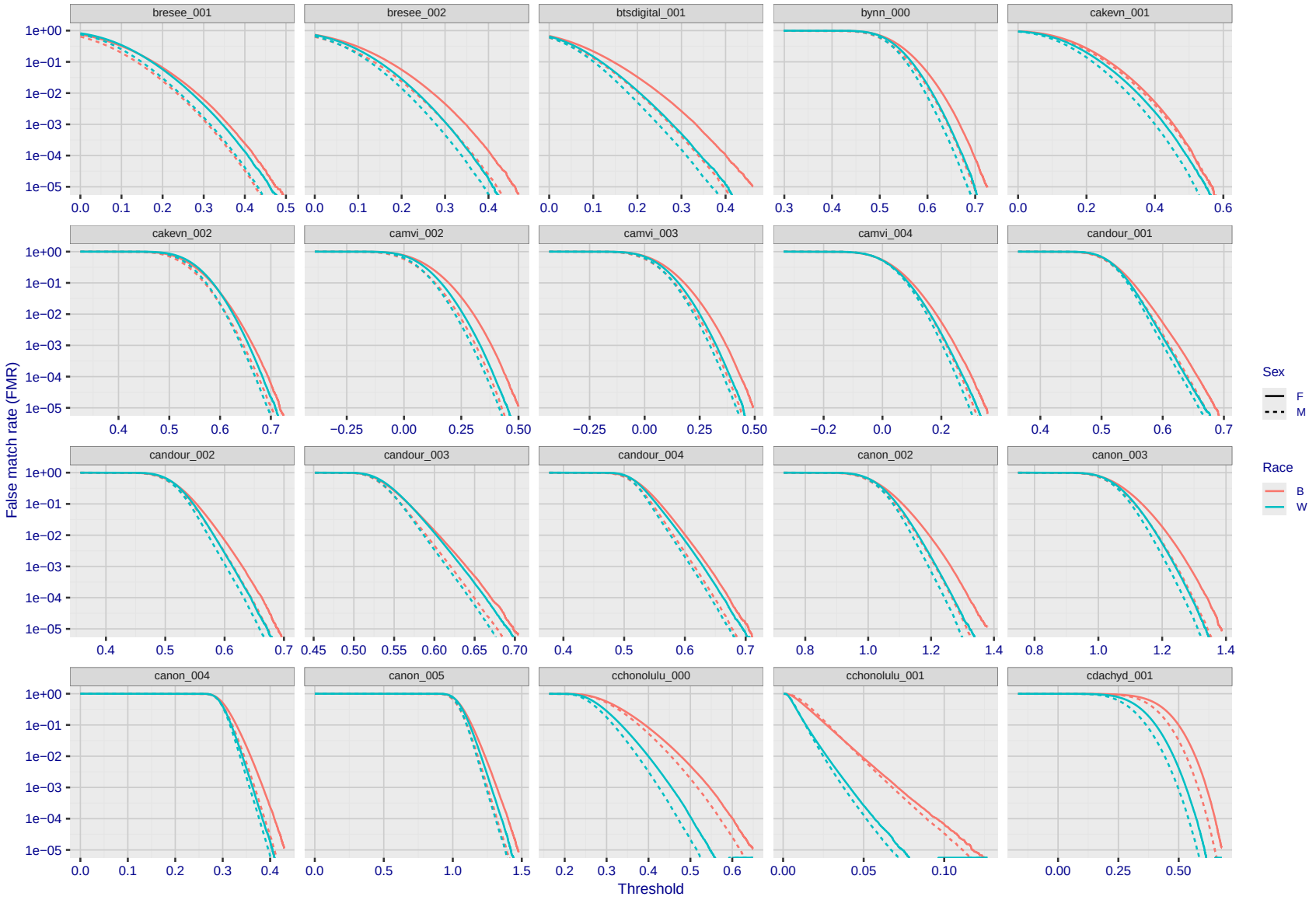

Figure 567: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

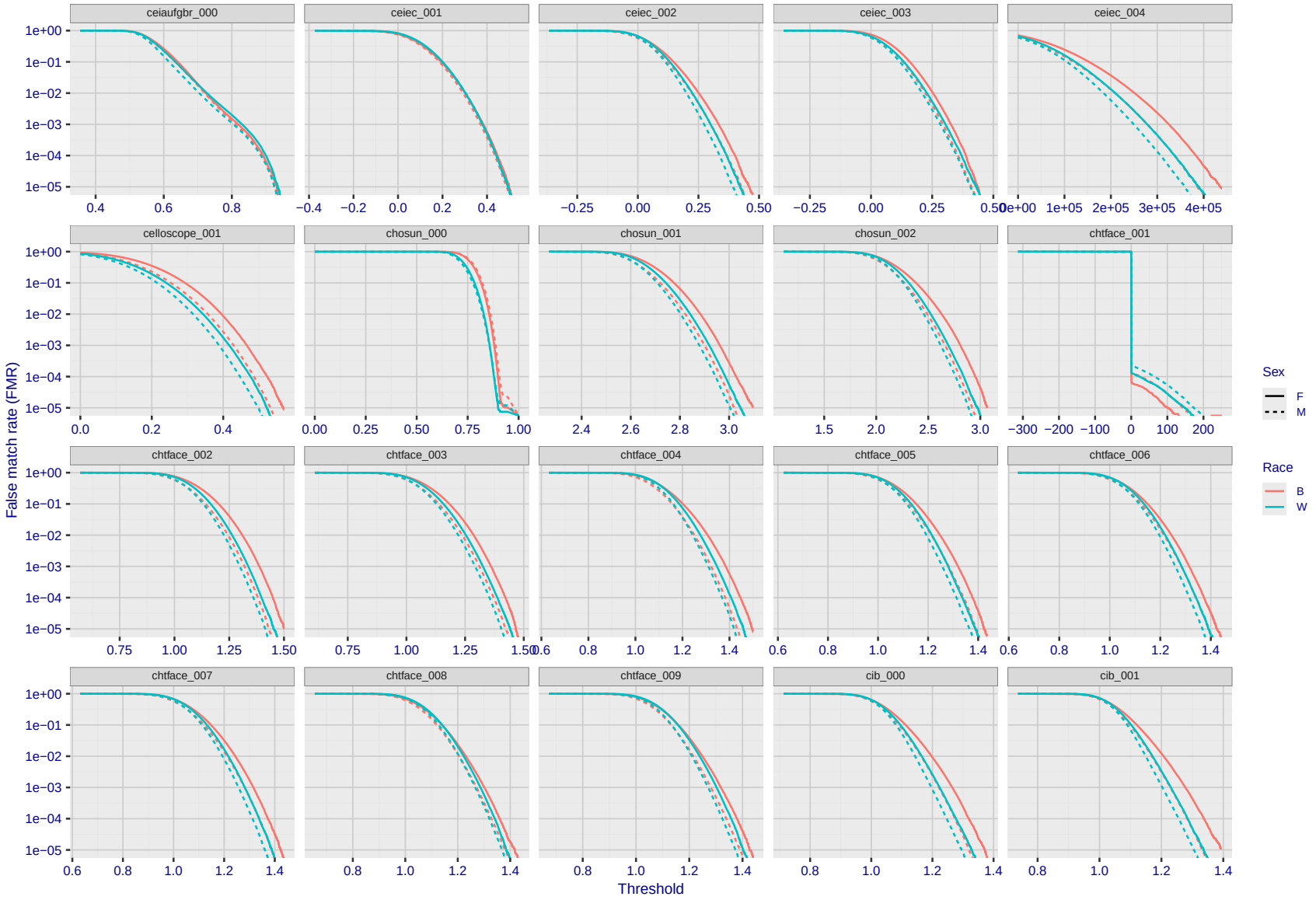

Figure 568: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

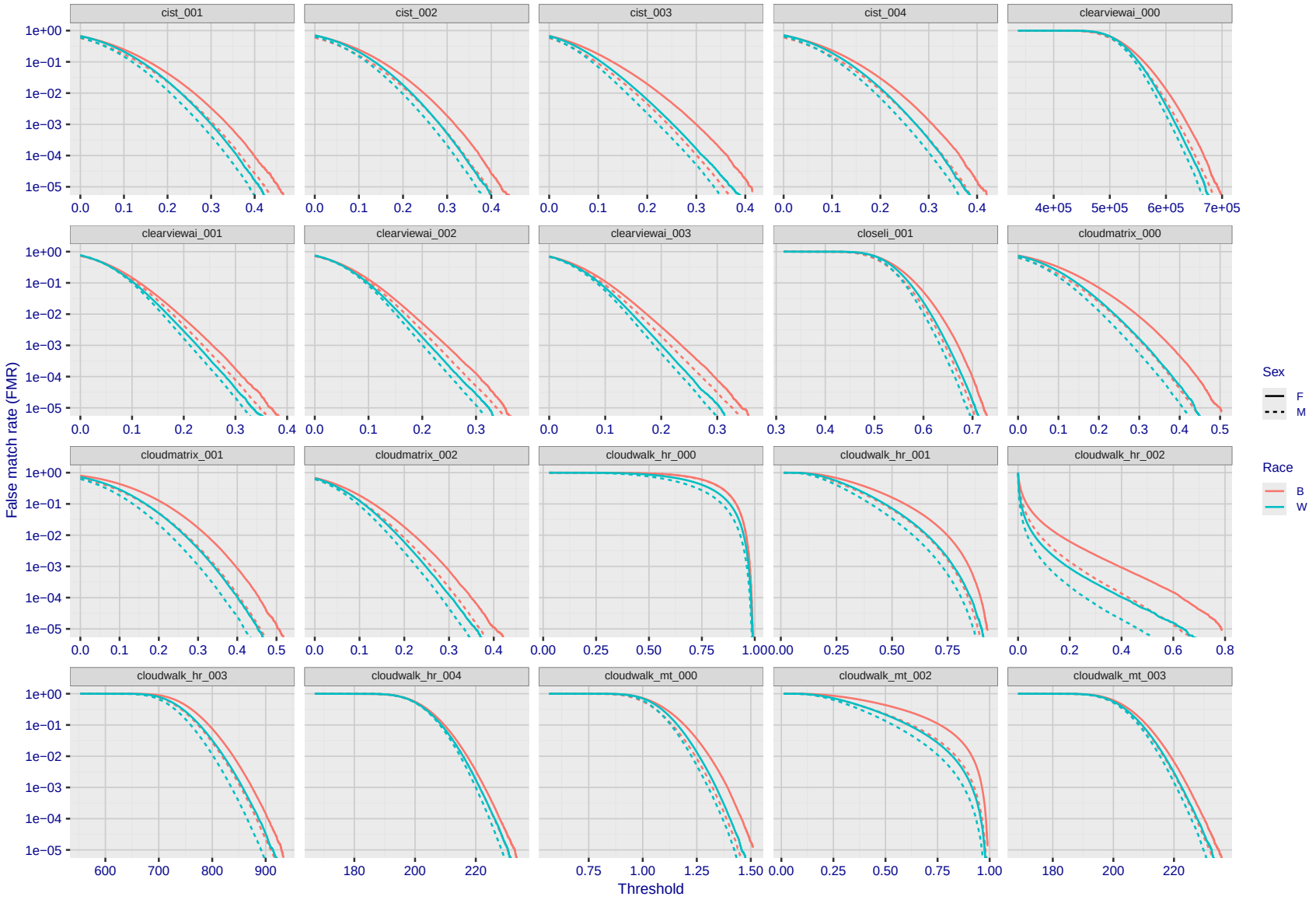

Figure 569: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

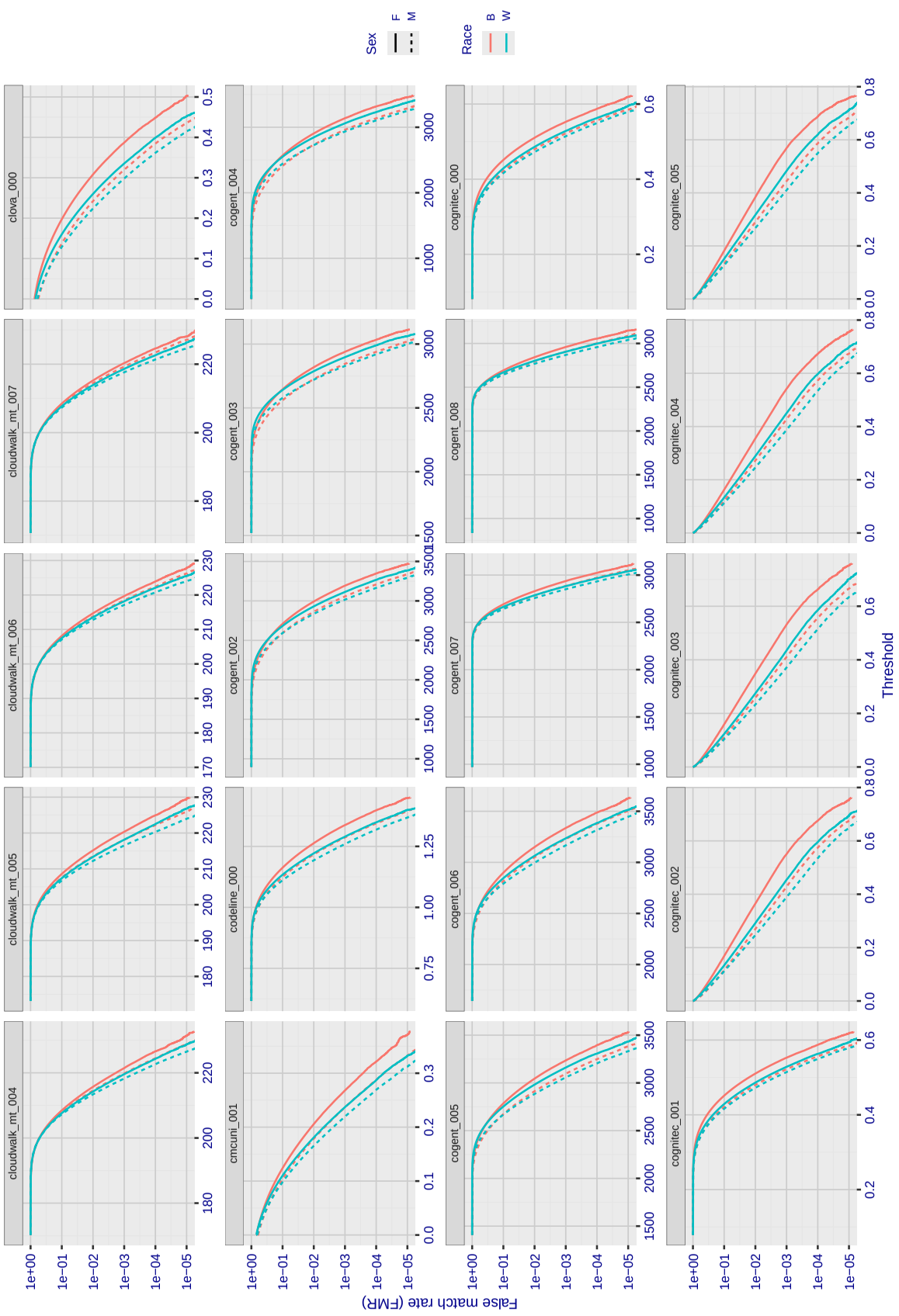

Figure 570: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

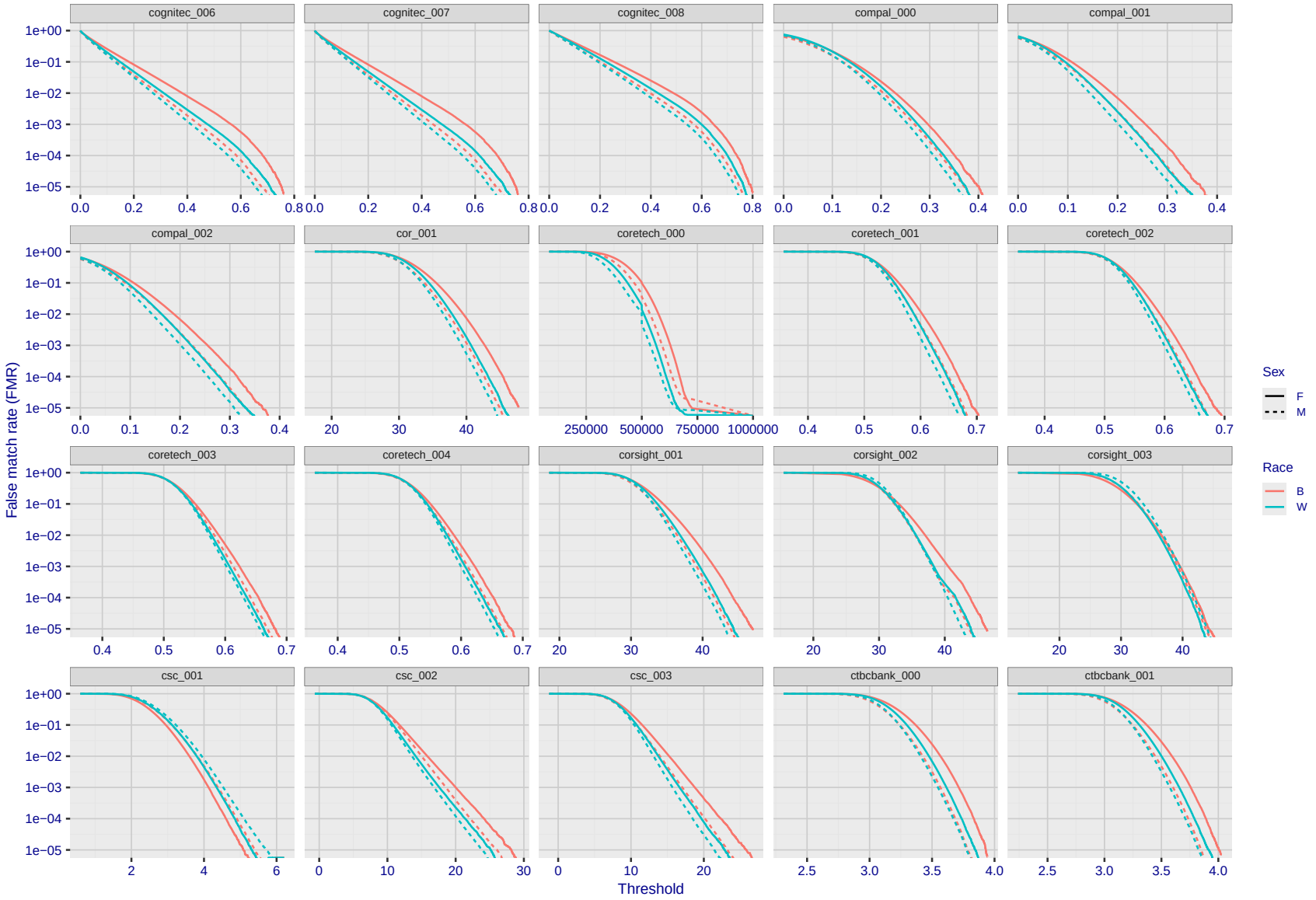

Figure 571: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

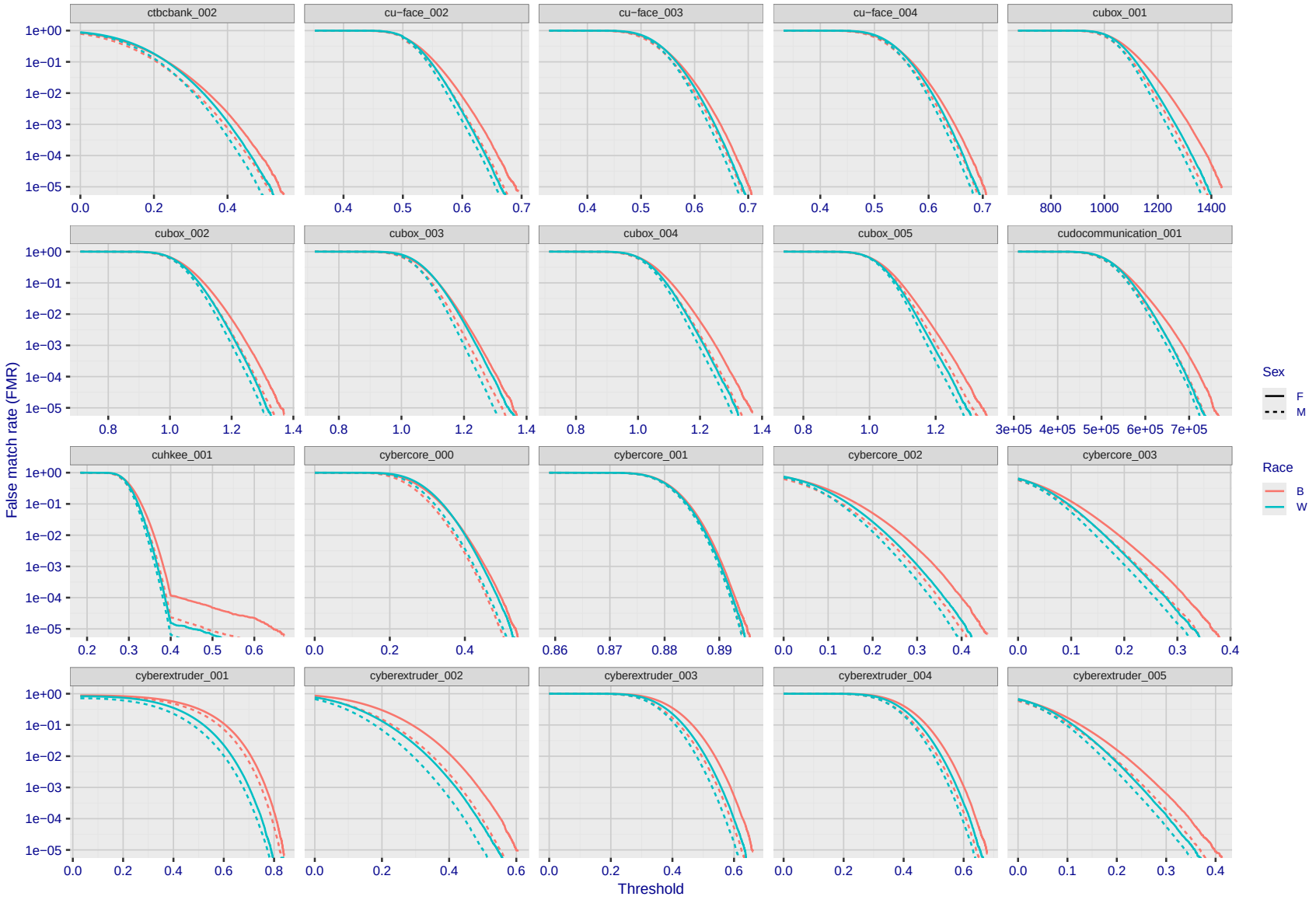

Figure 572: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

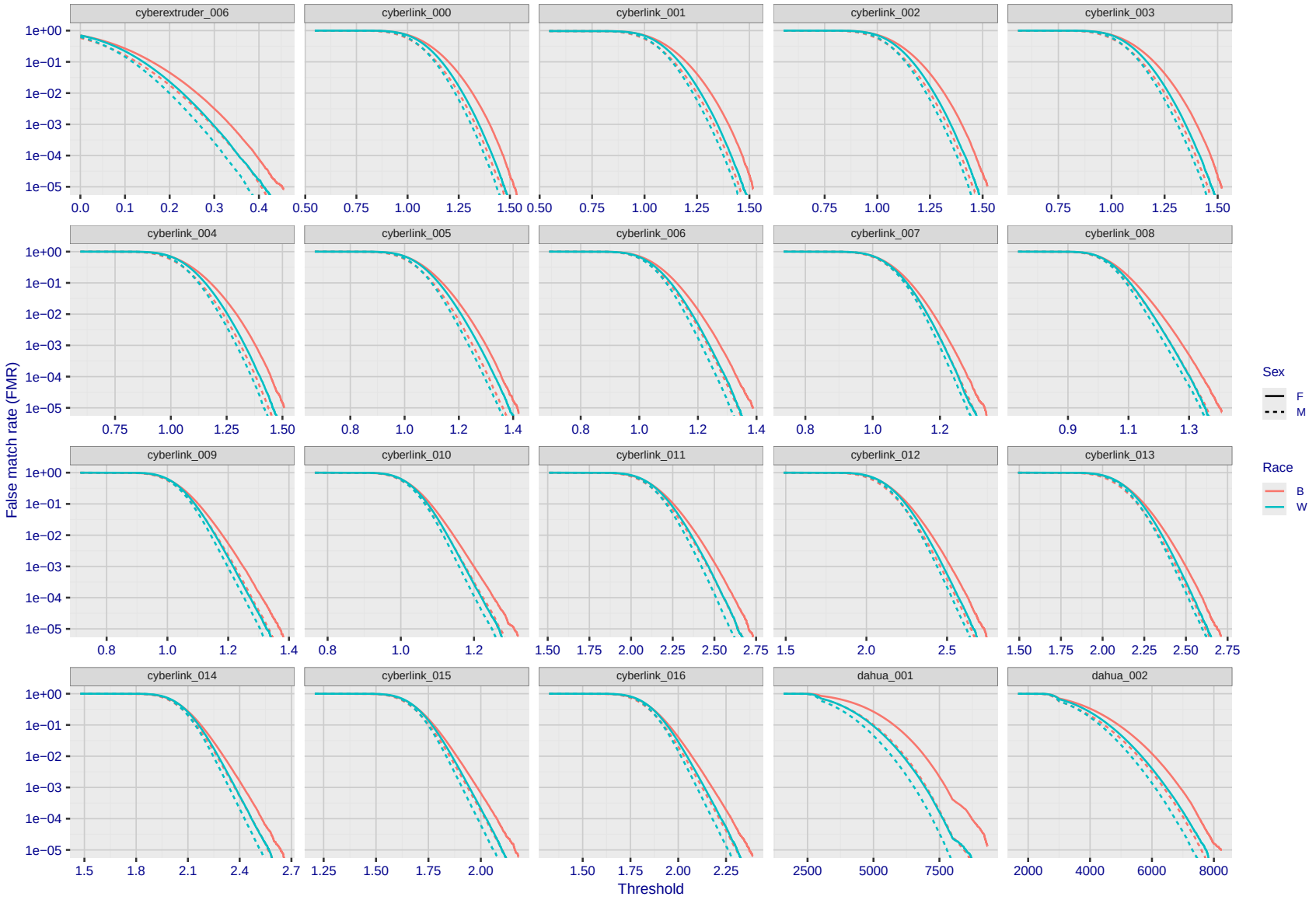

Figure 573: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

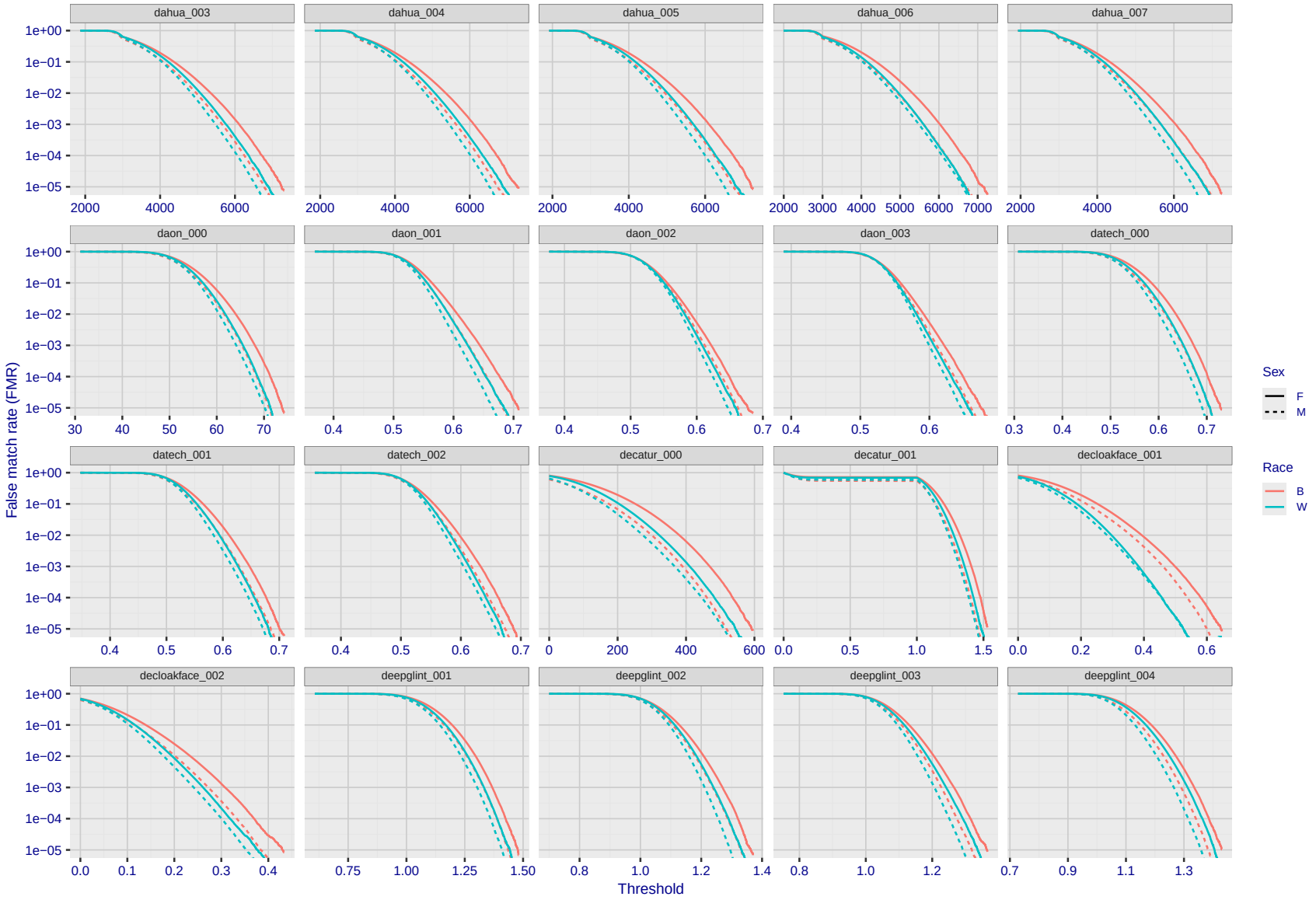

Figure 574: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

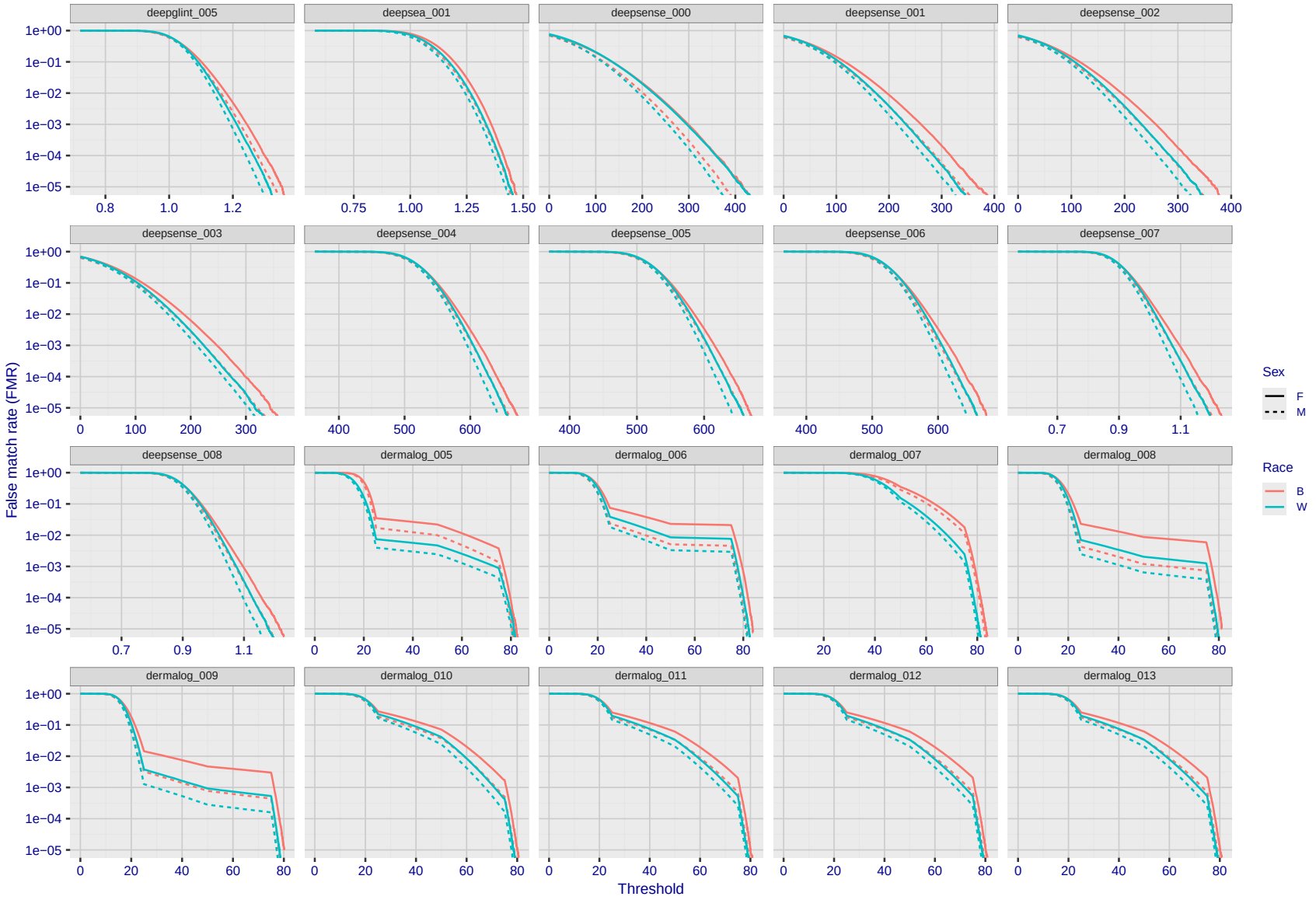

Figure 575: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

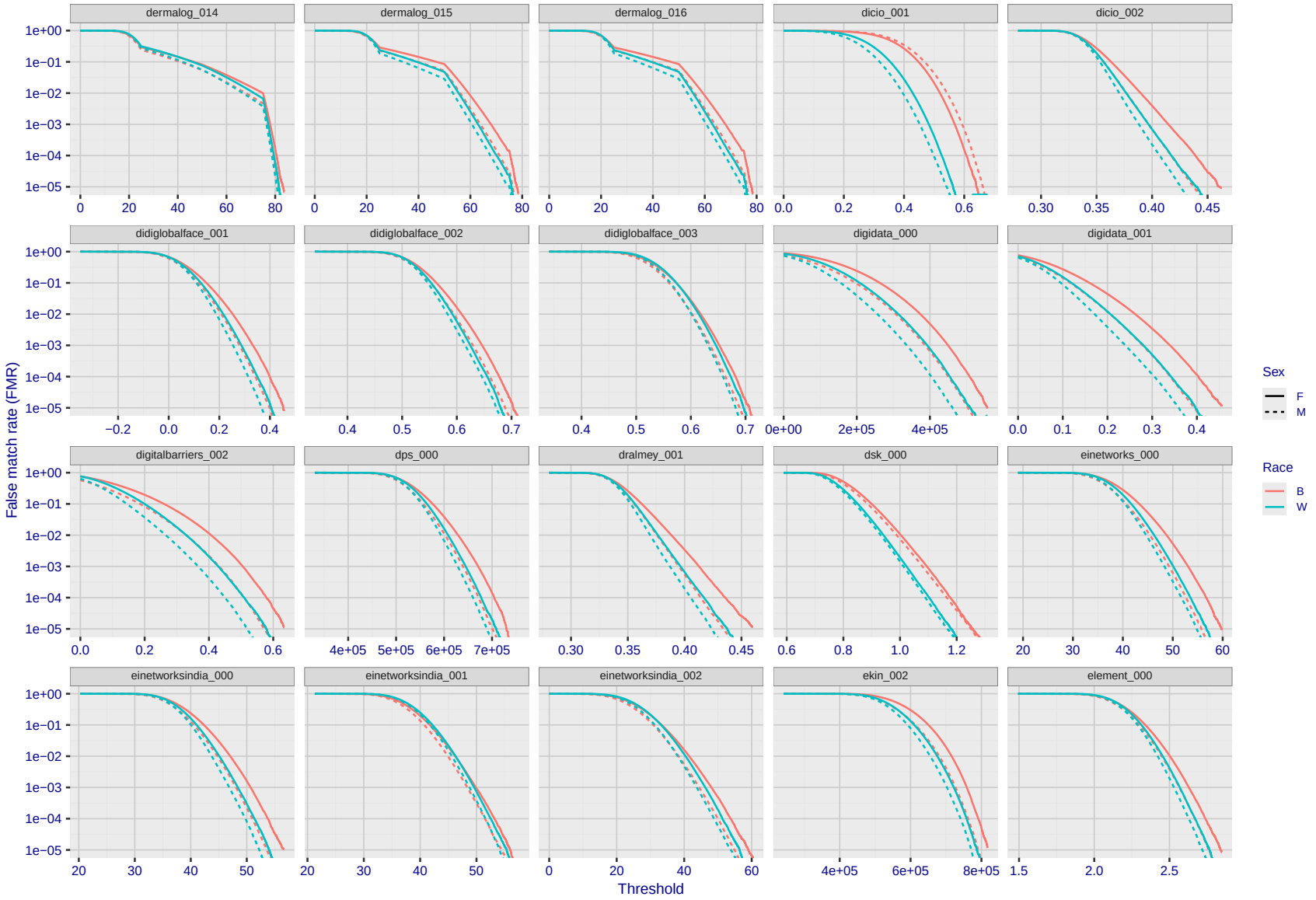

Figure 576: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

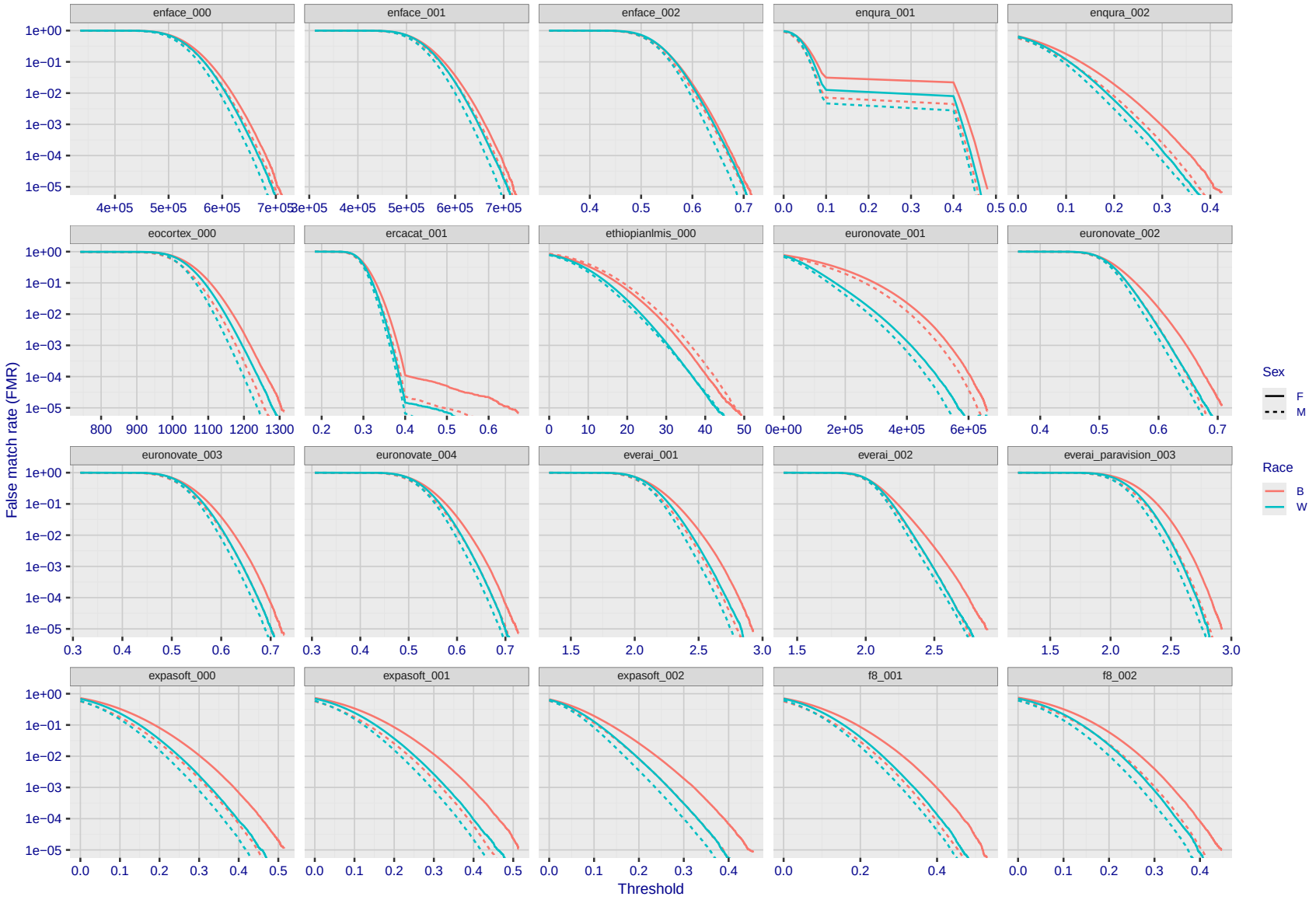

Figure 577: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

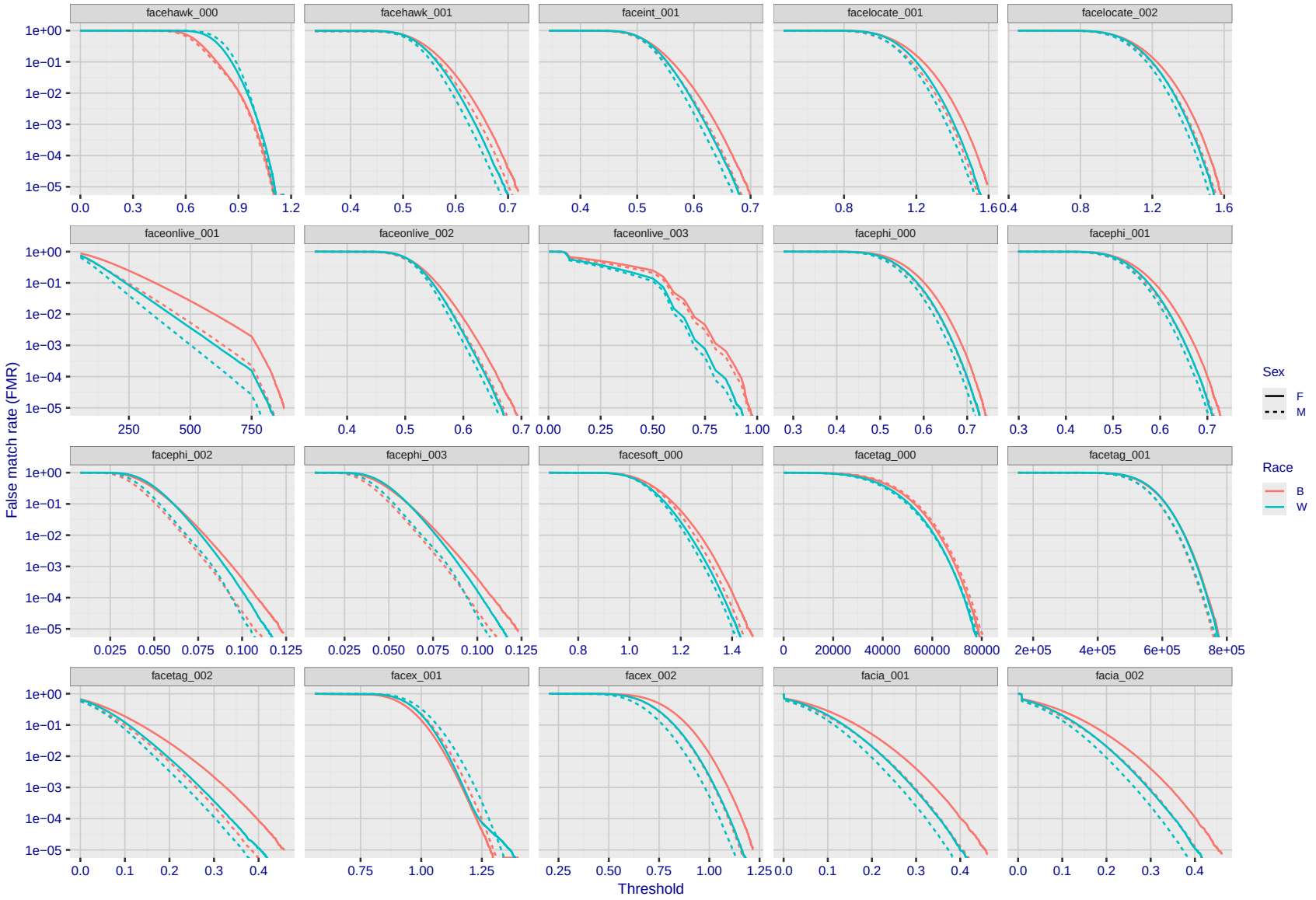

Figure 578: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

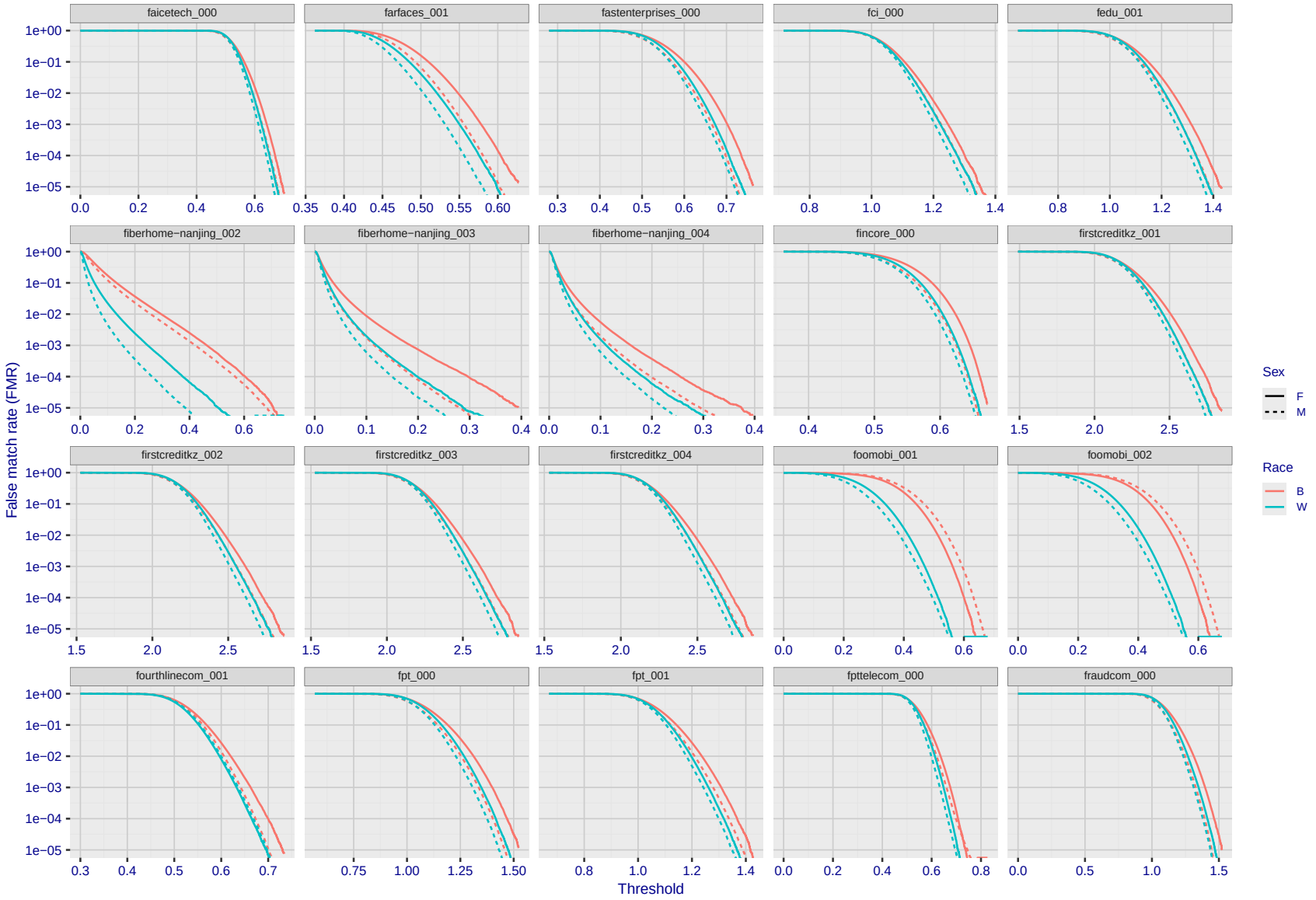

Figure 579: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

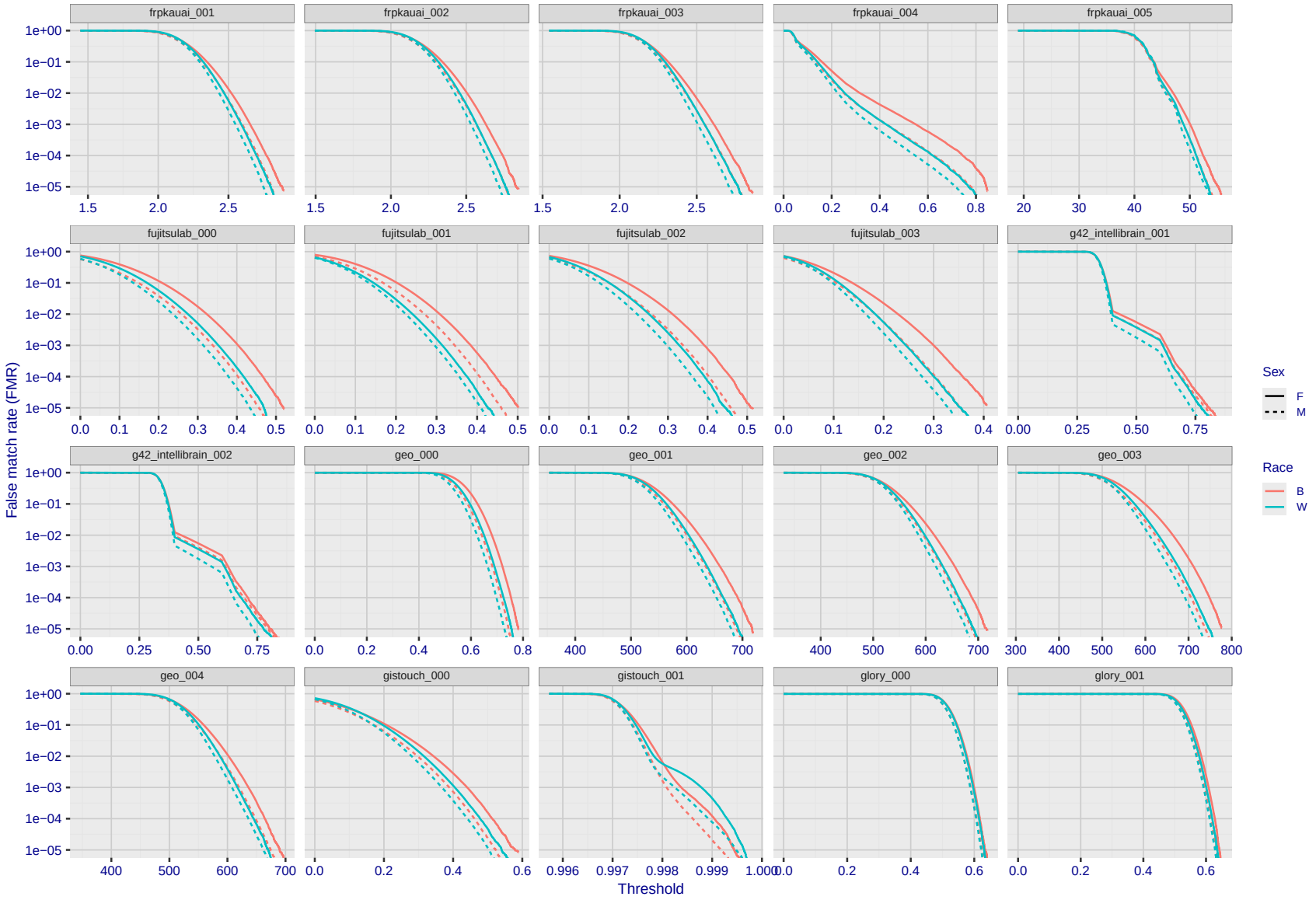

Figure 580: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

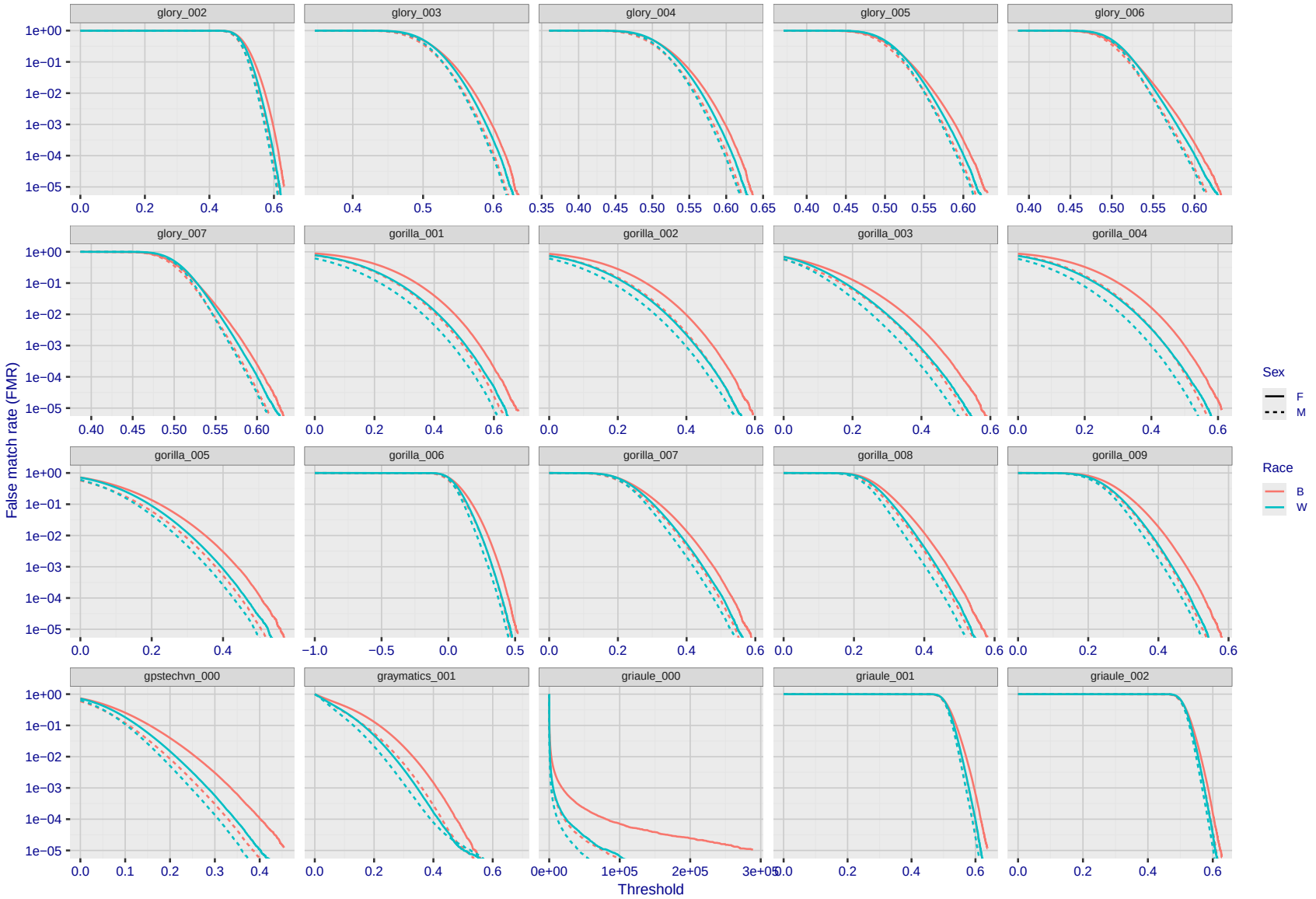

Figure 581: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

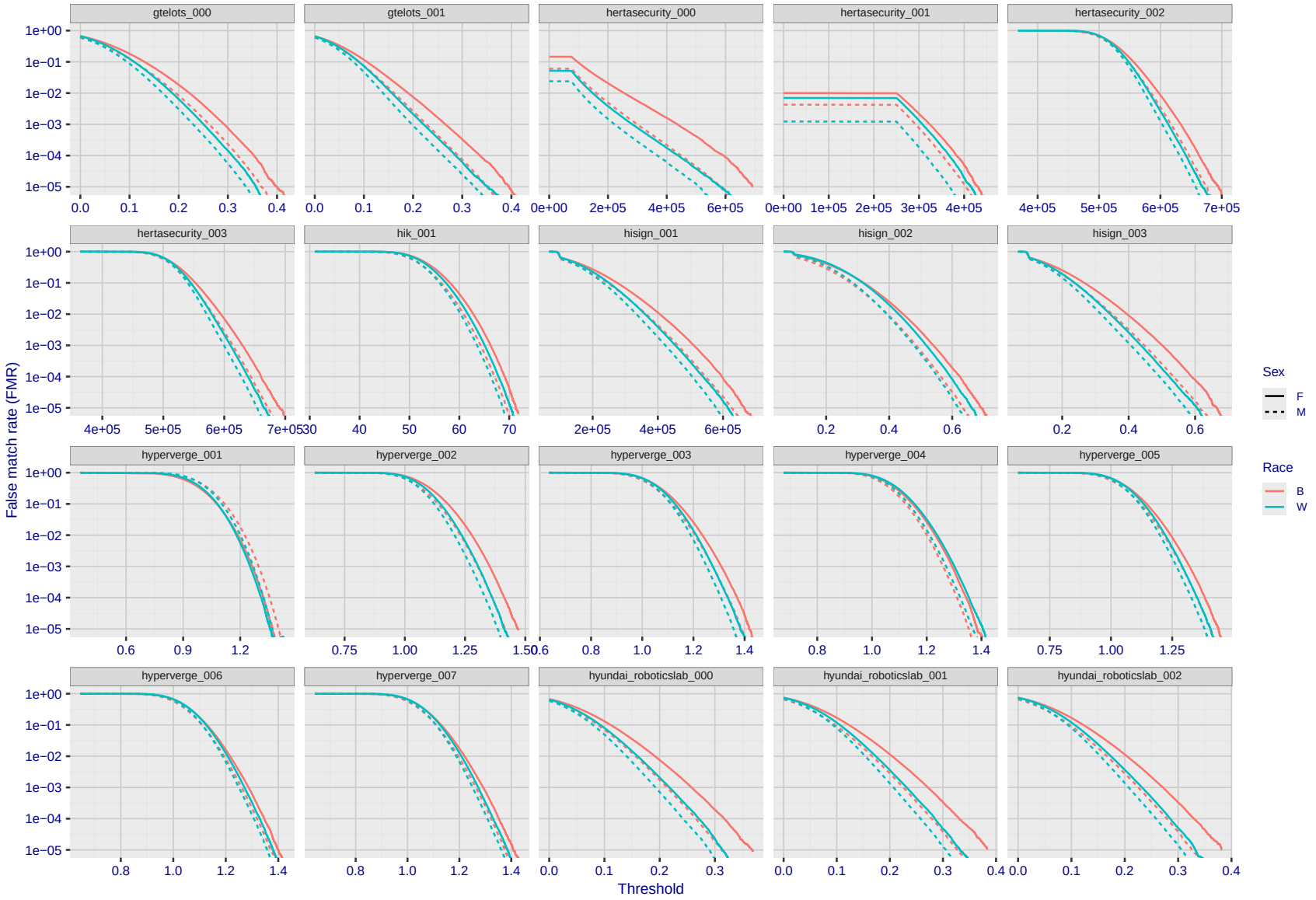

Figure 582: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

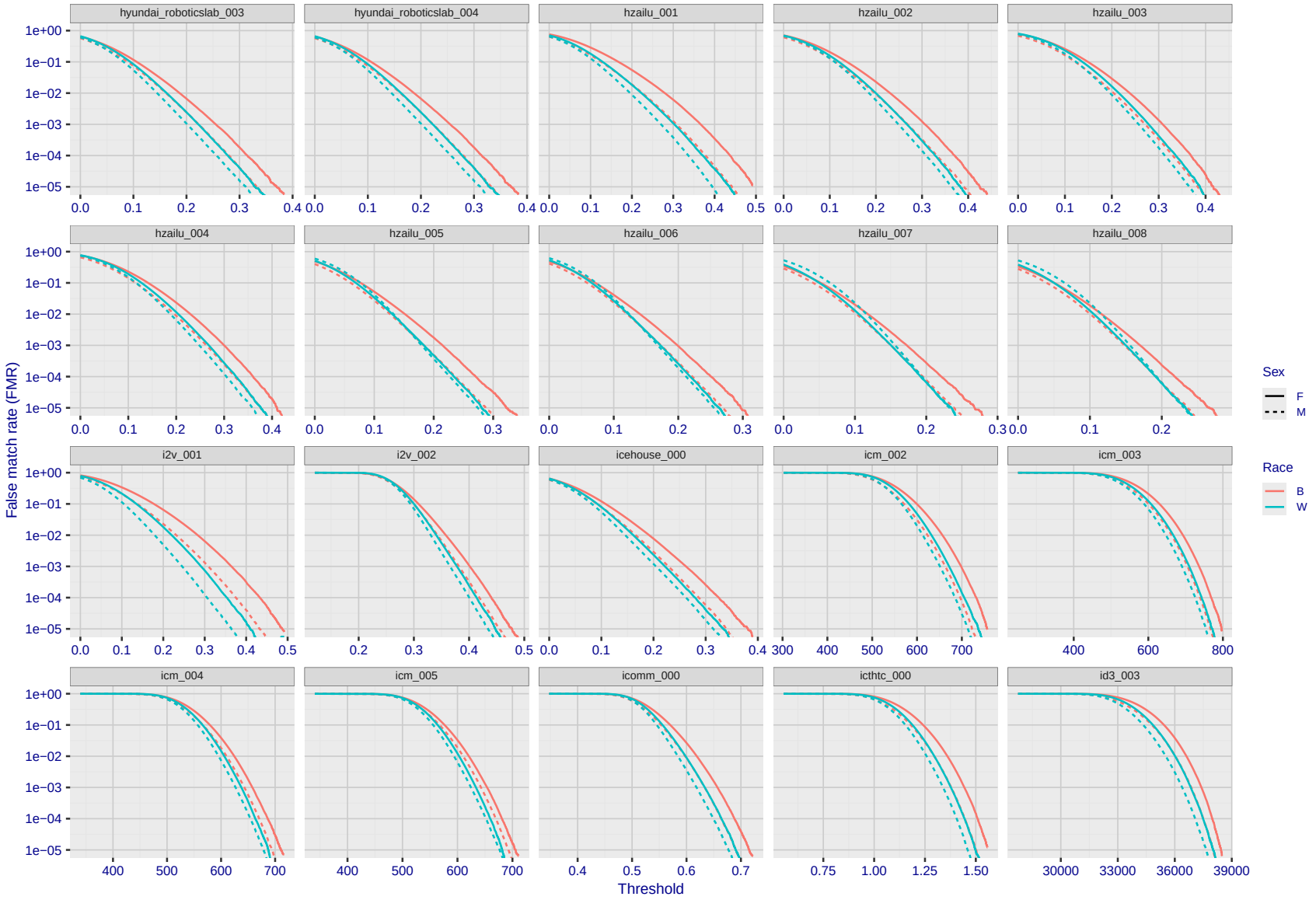

Figure 583: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

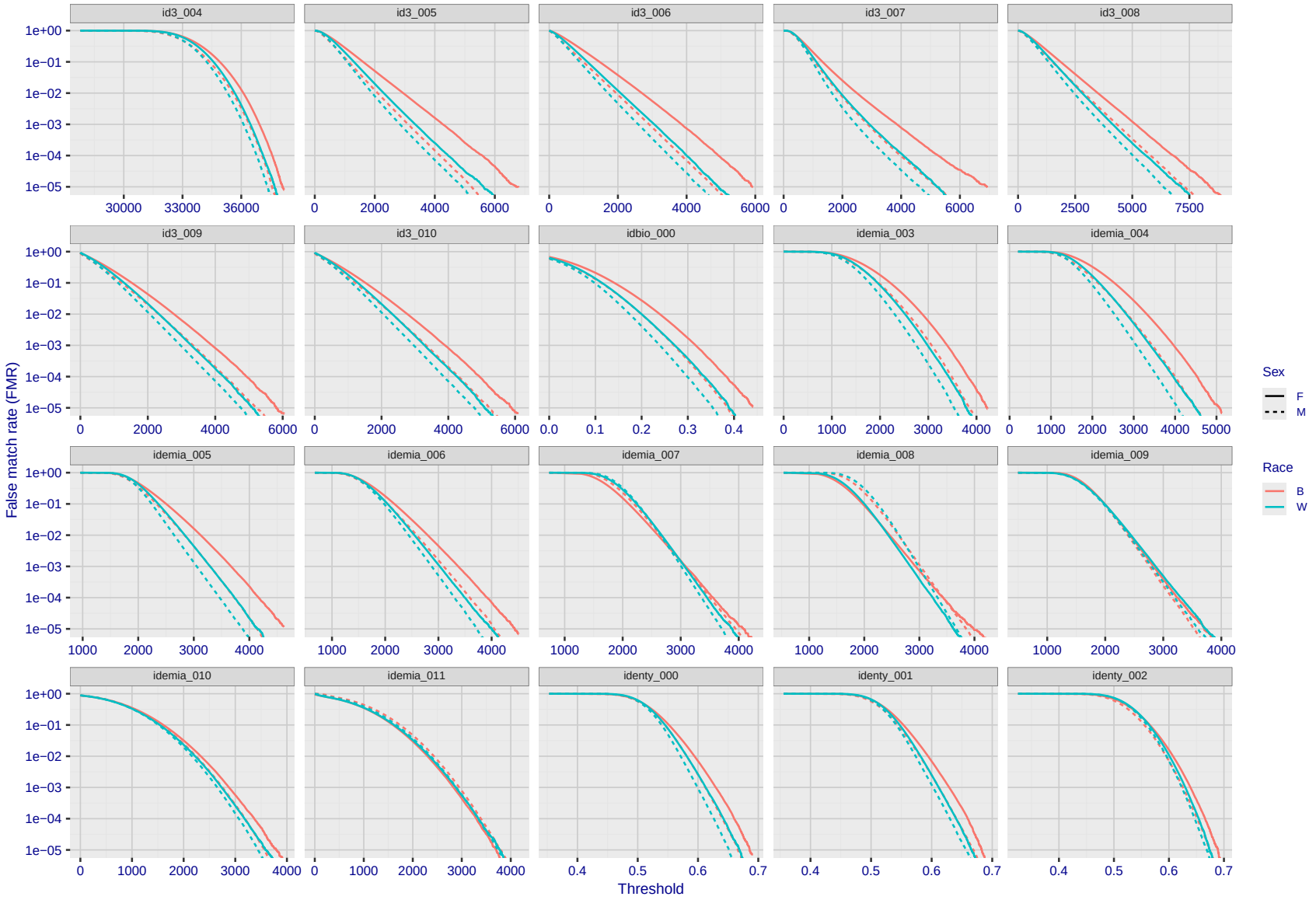

Figure 584: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

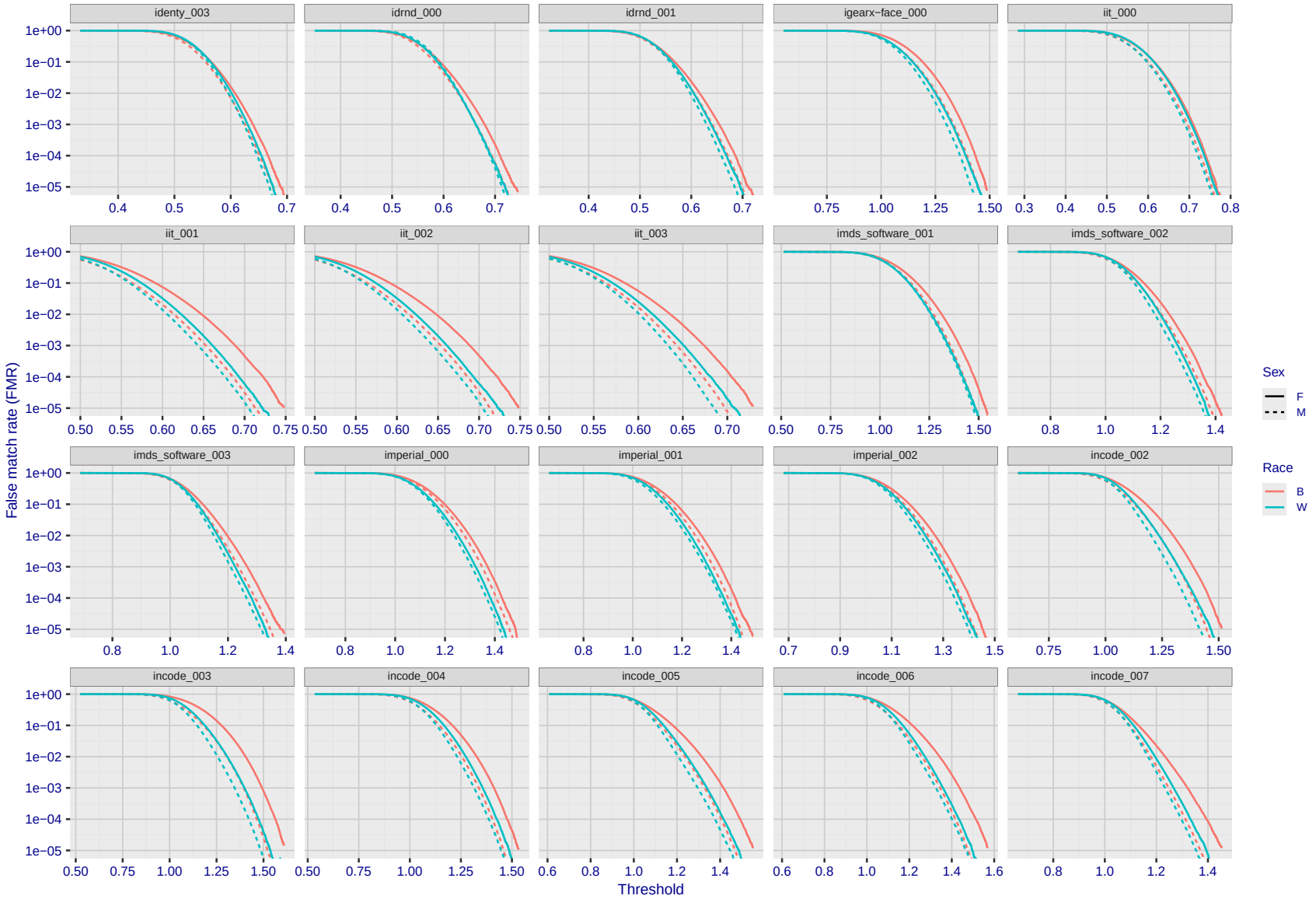

Figure 585: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

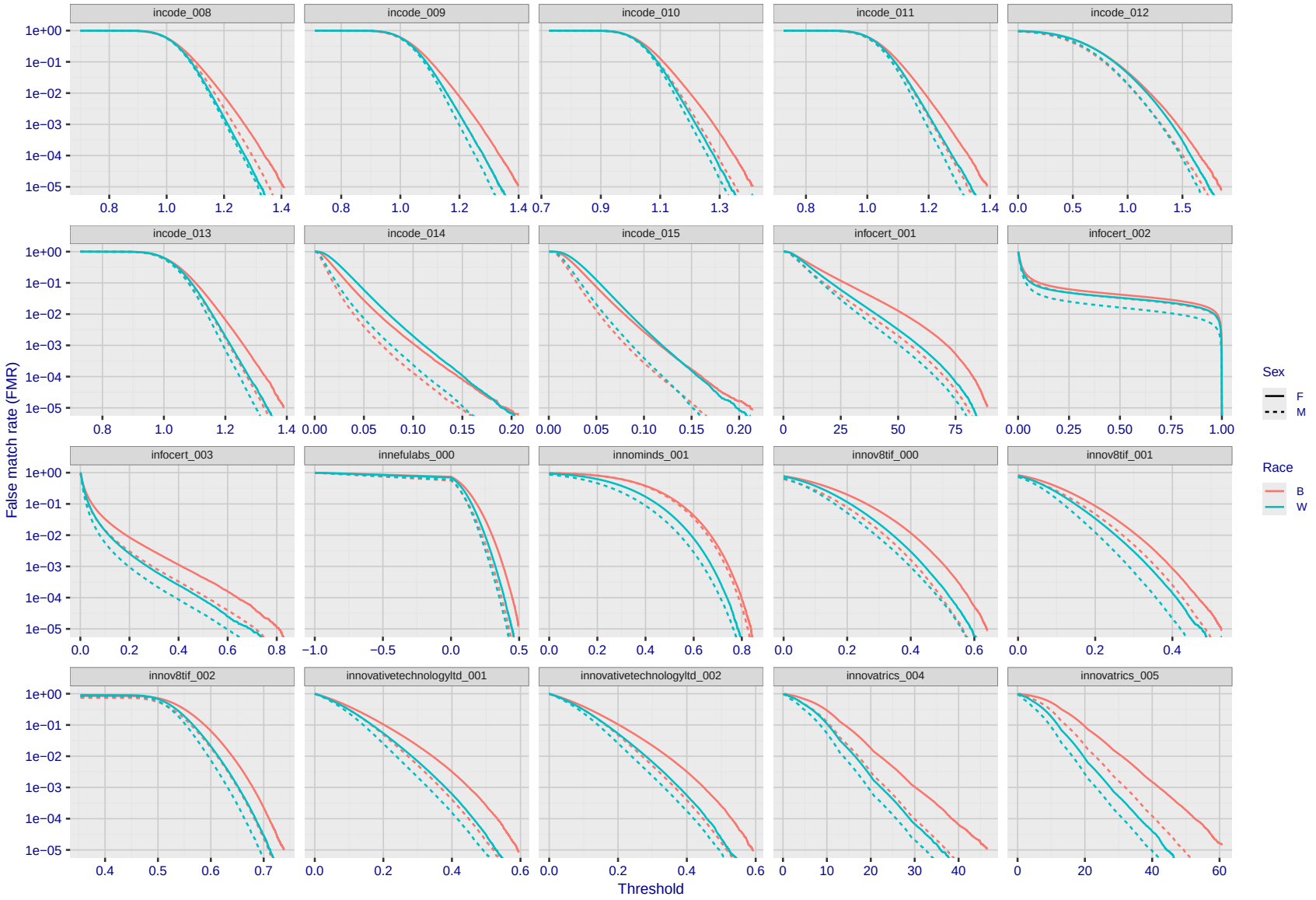

Figure 586: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

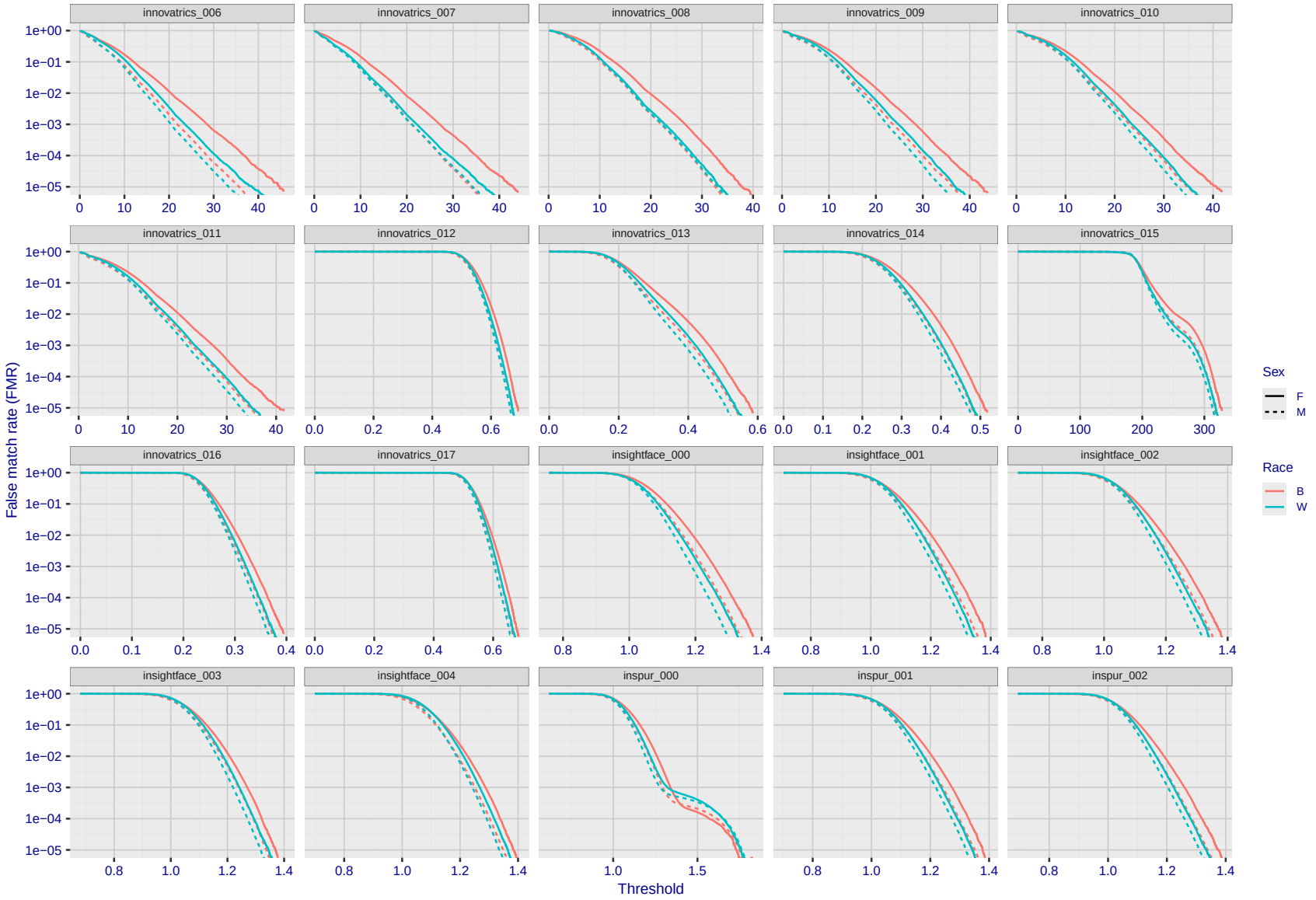

Figure 587: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

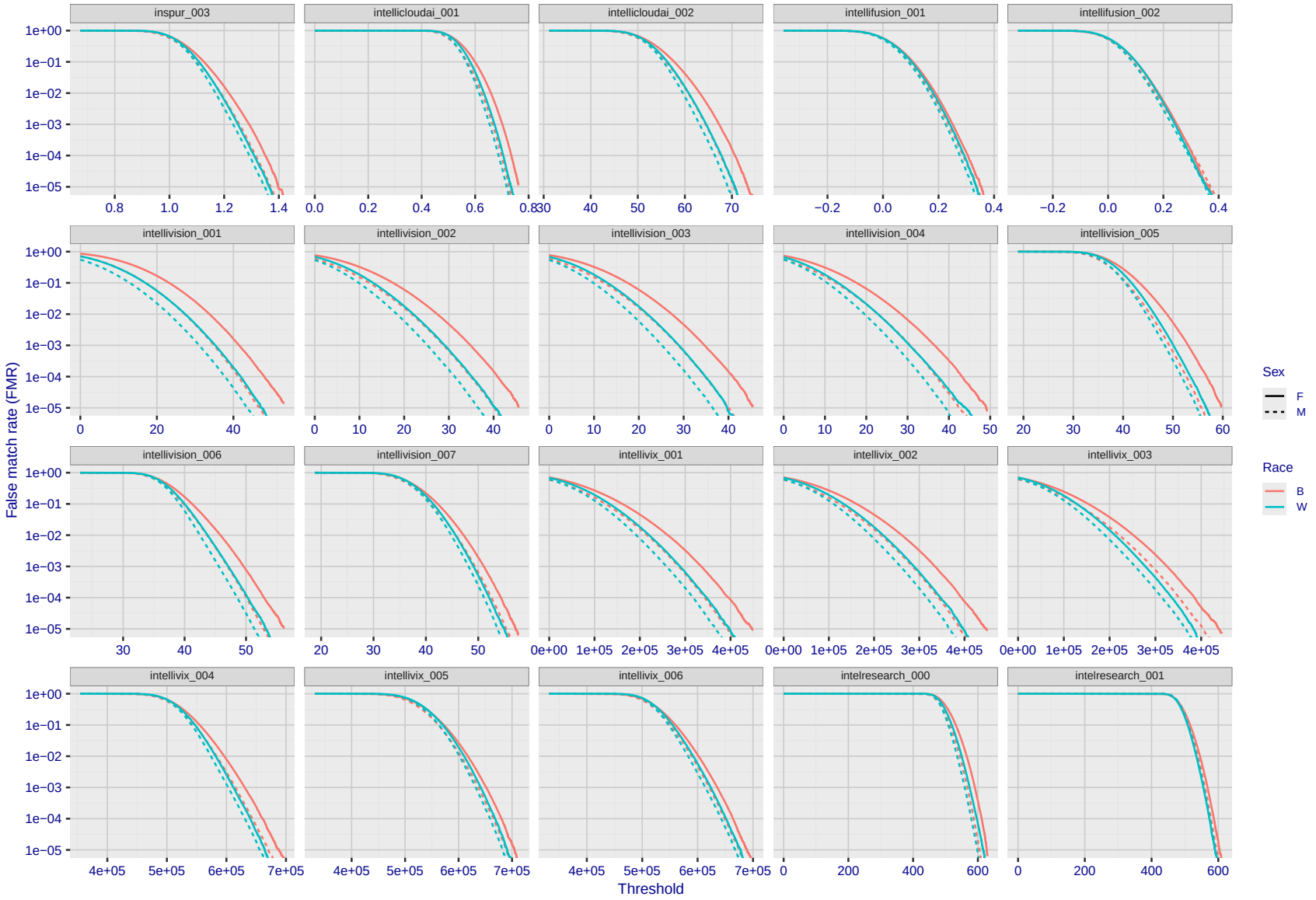

Figure 588: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

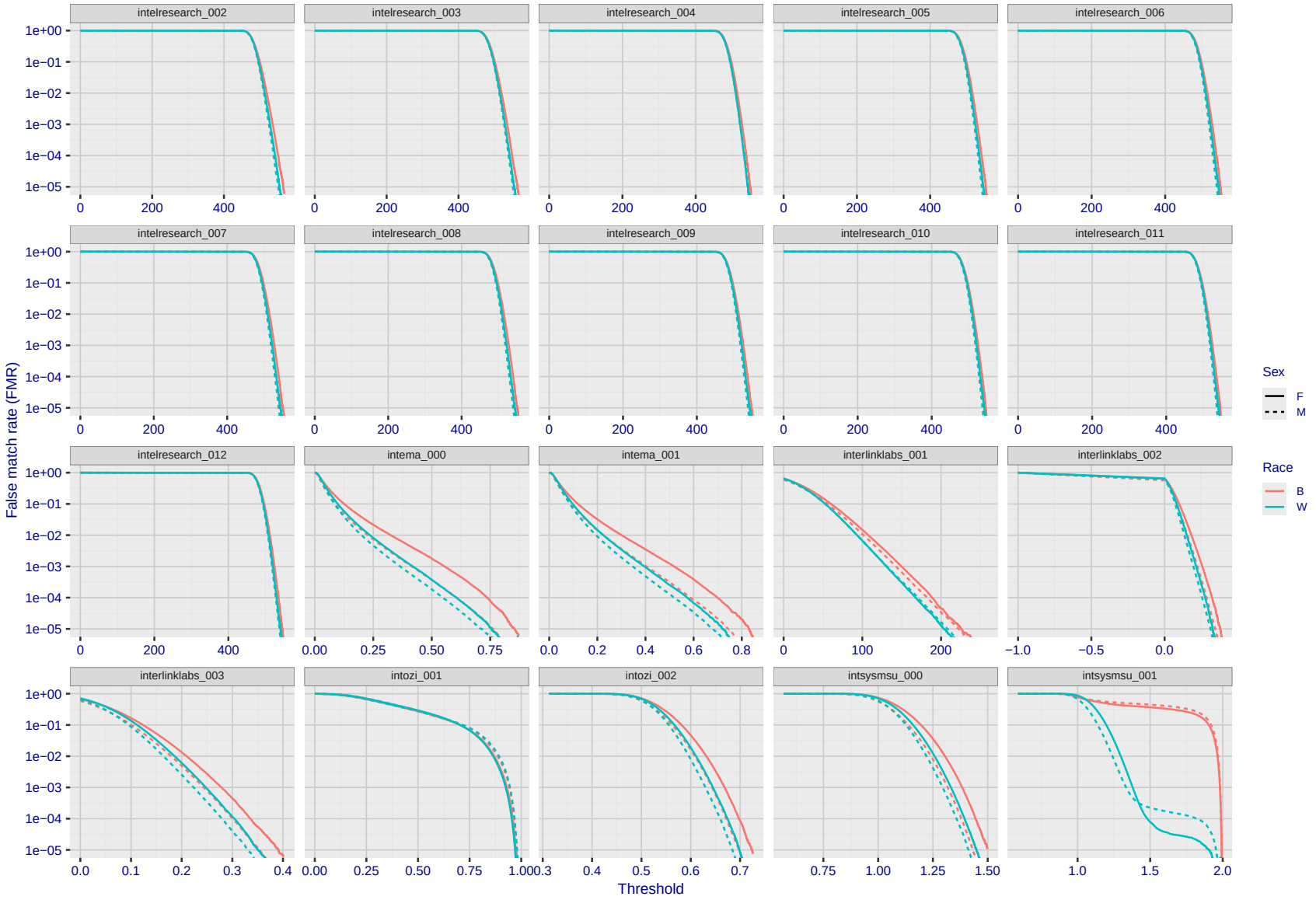

Figure 589: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

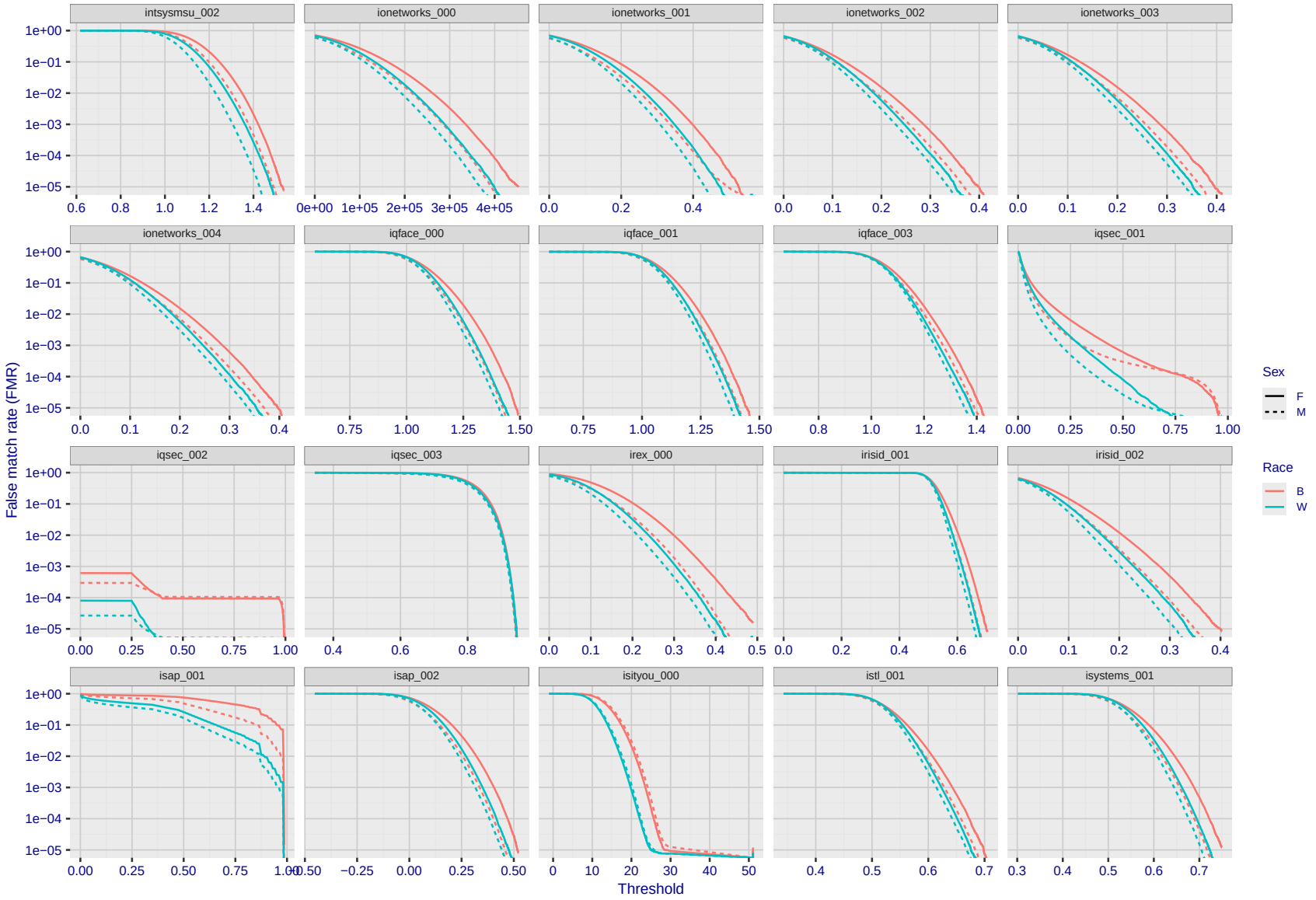

Figure 590: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

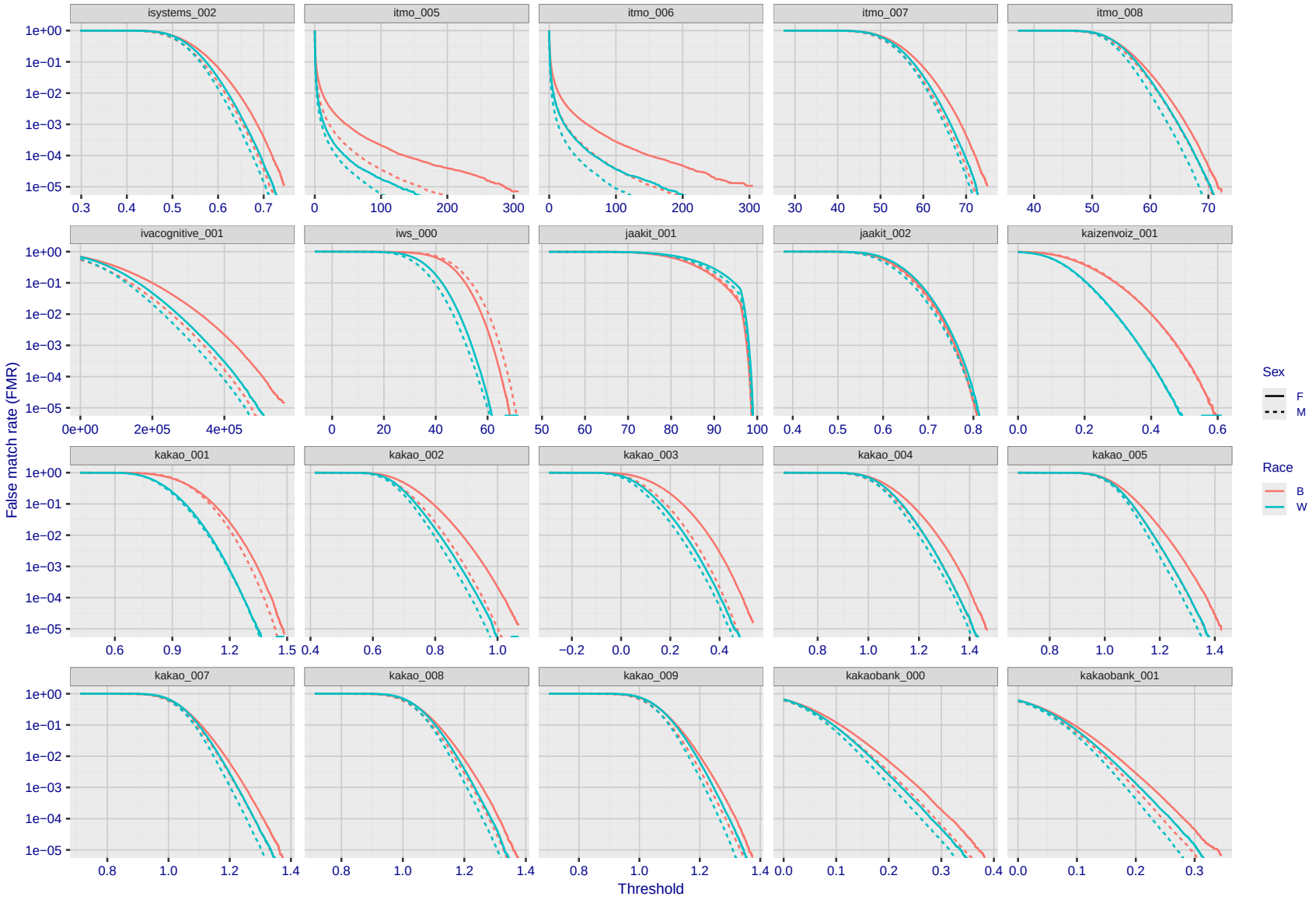

Figure 591: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

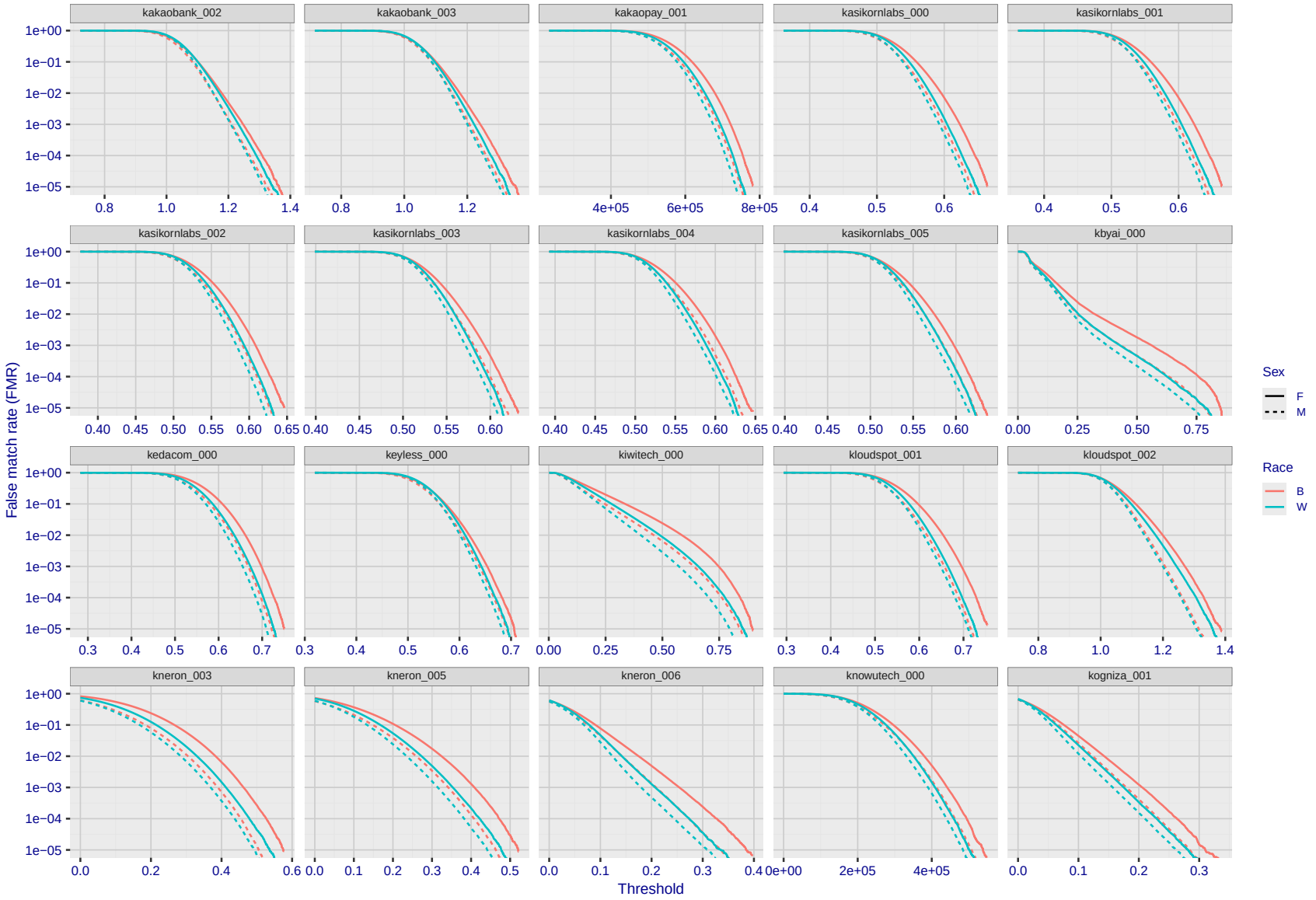

Figure 592: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

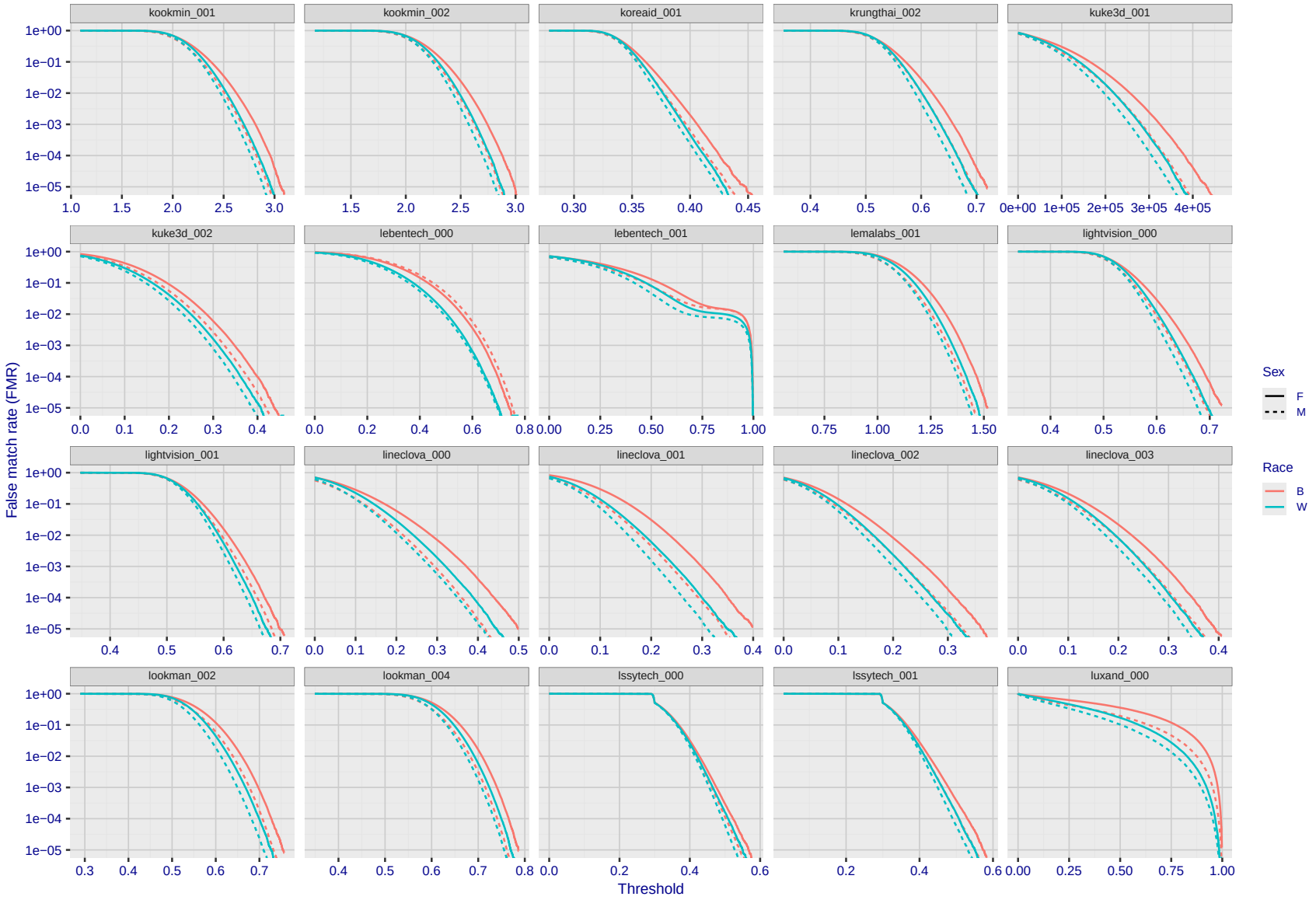

Figure 593: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

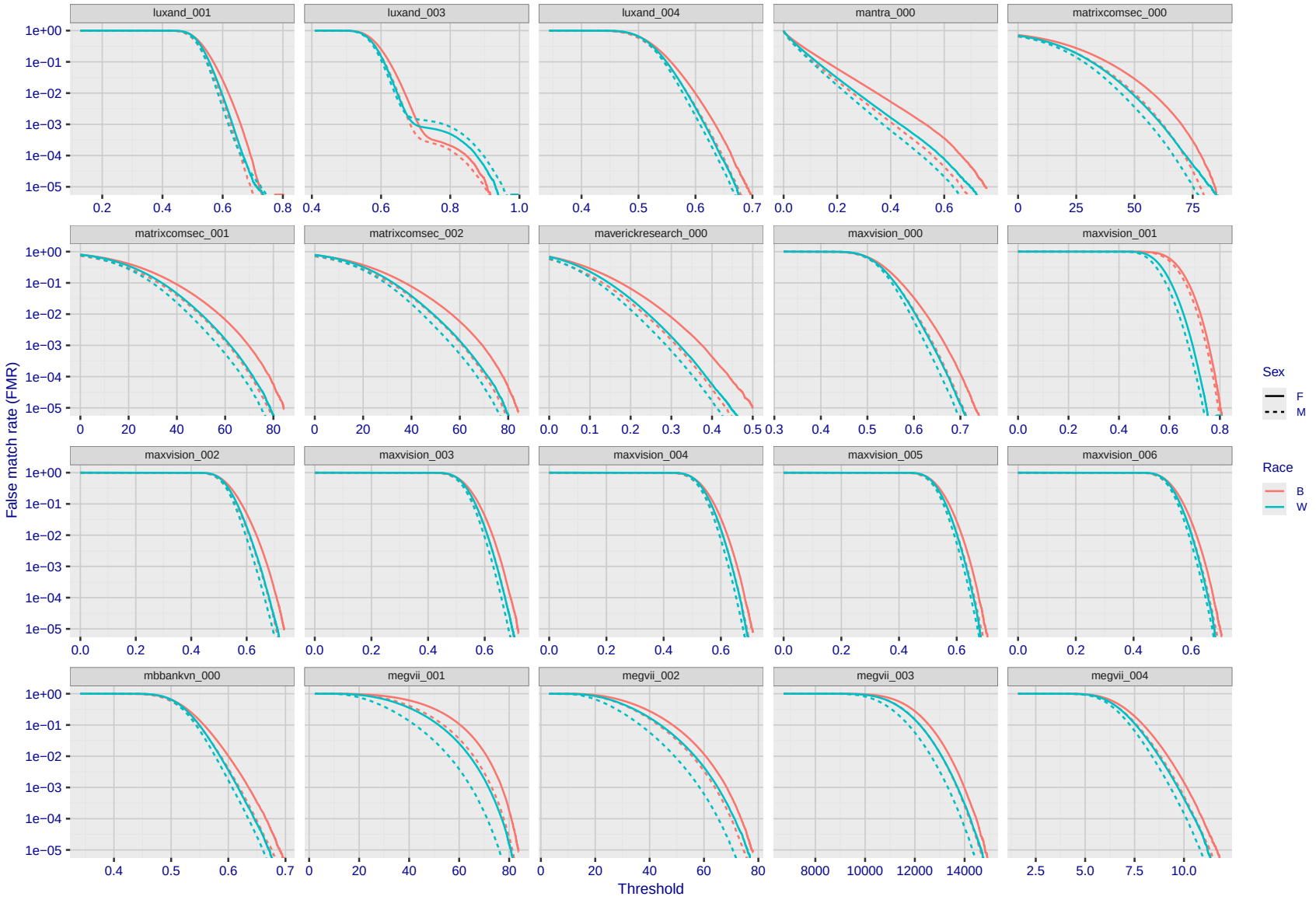

Figure 594: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

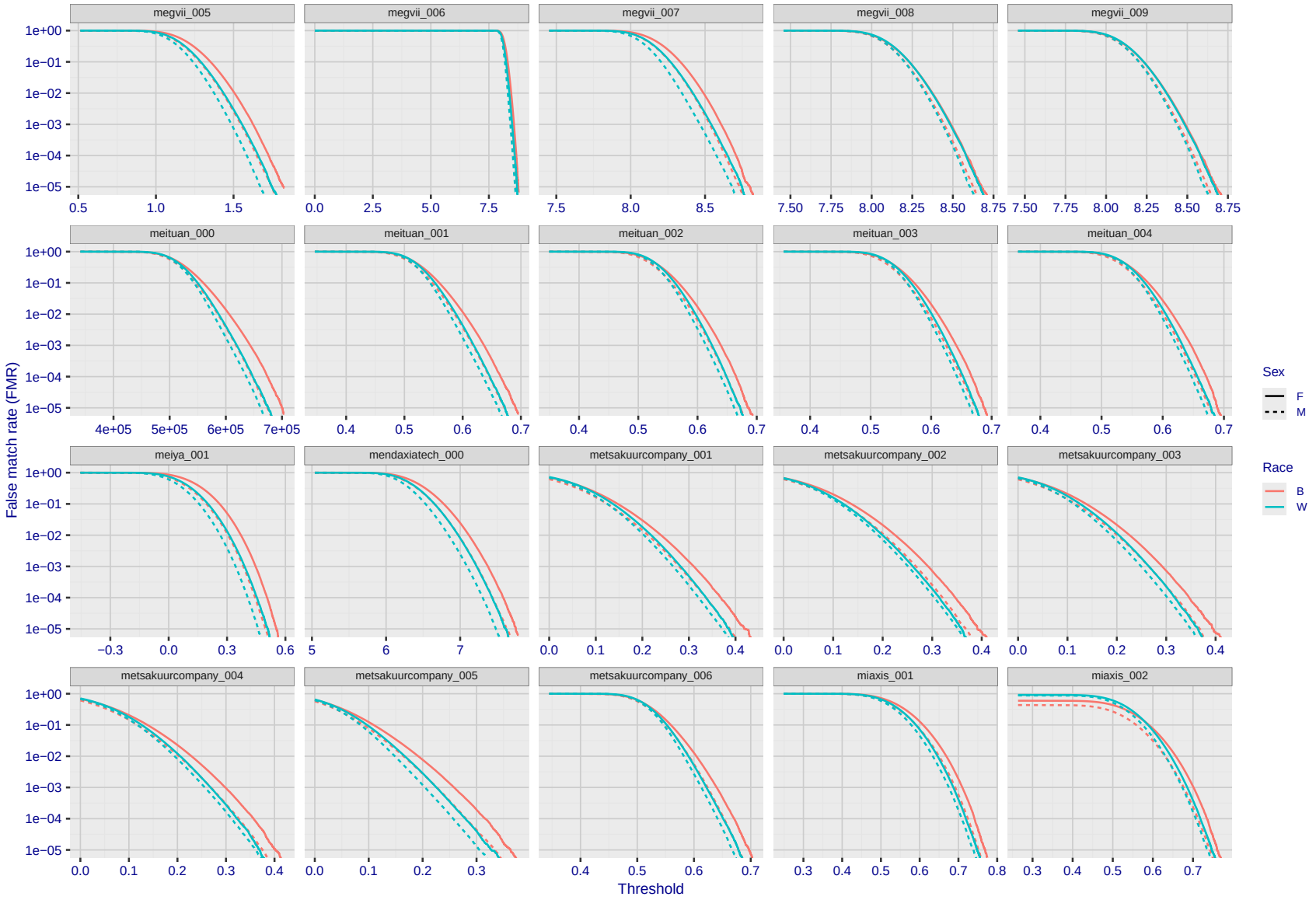

Figure 595: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

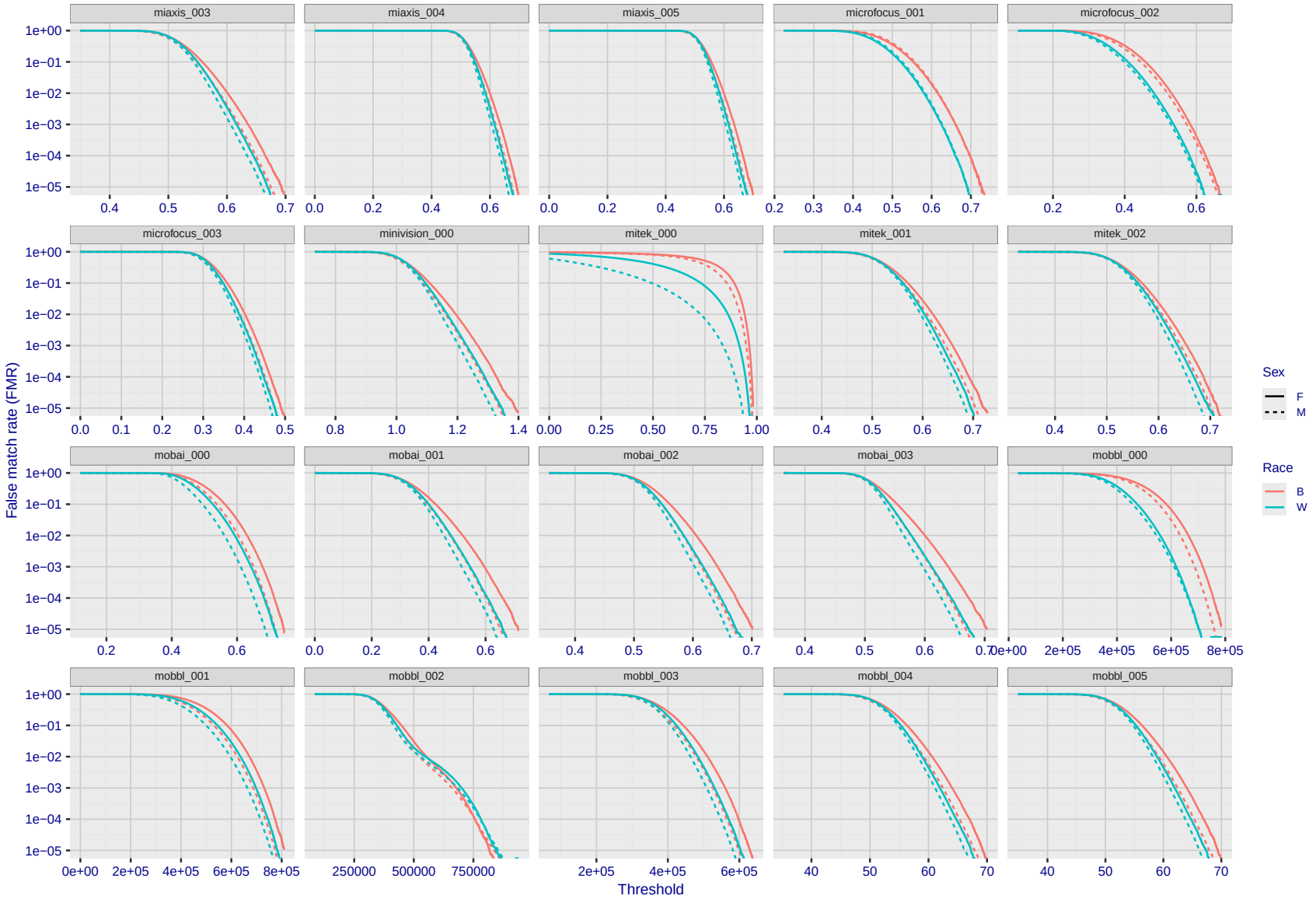

Figure 596: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

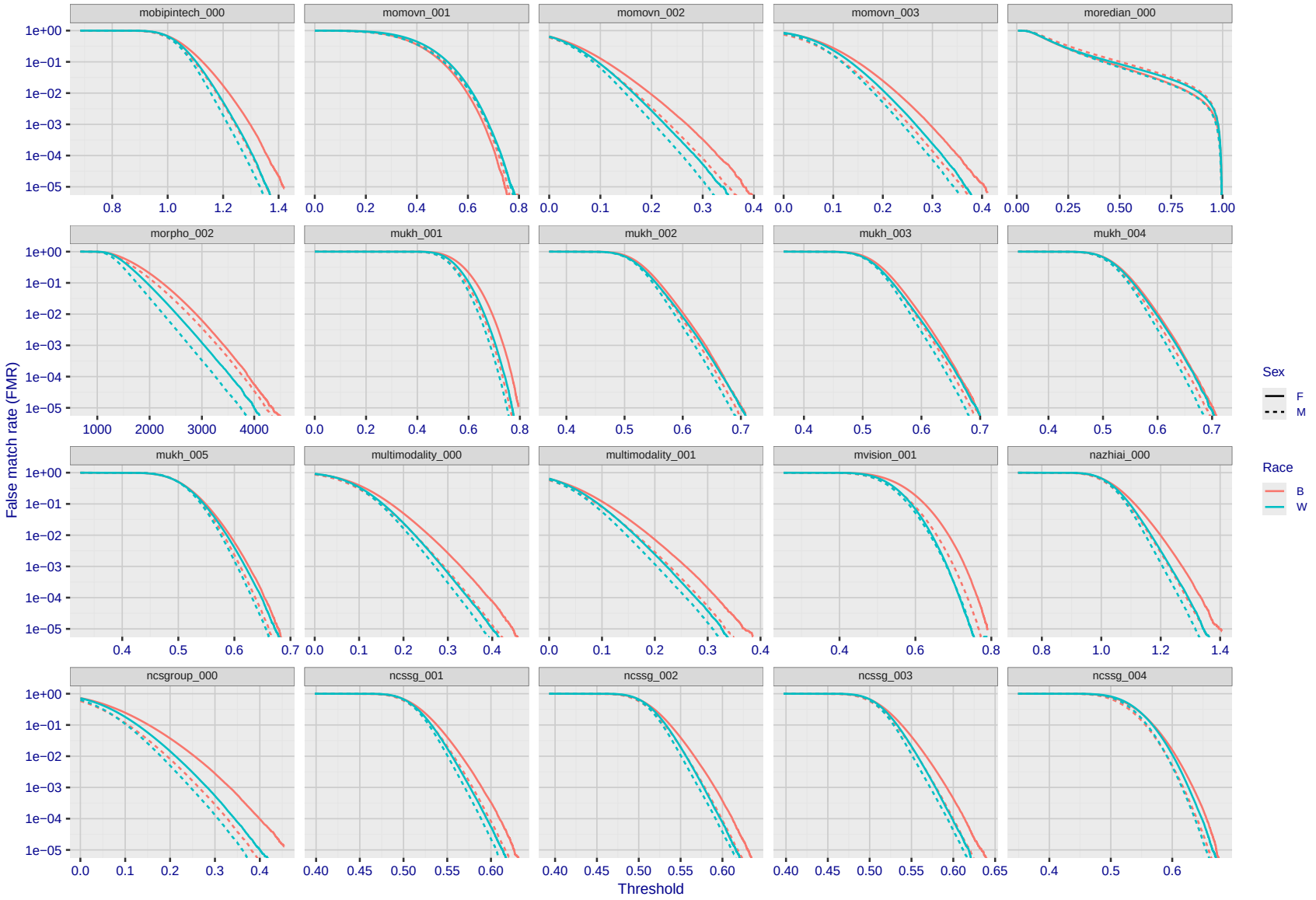

Figure 597: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

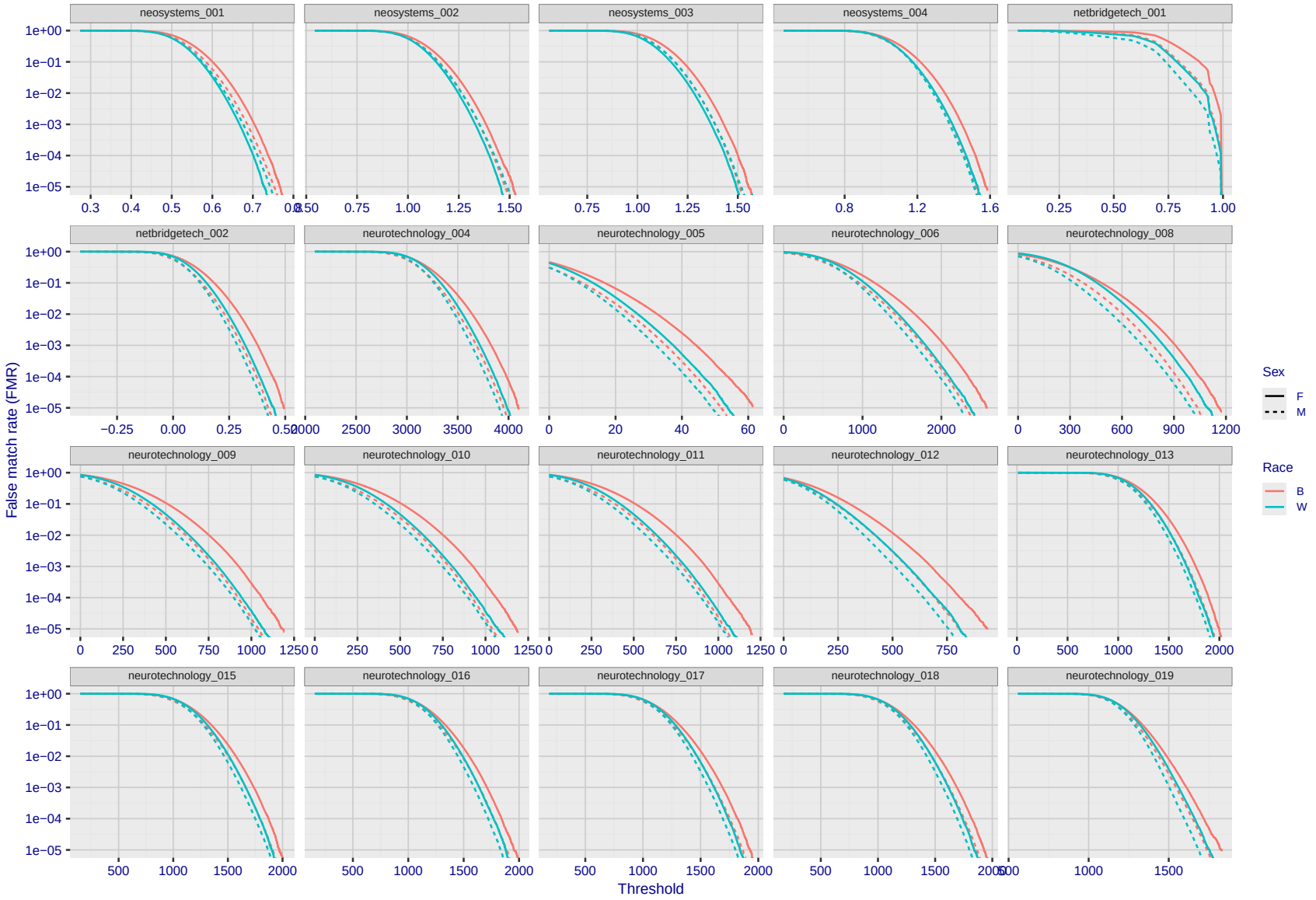

Figure 598: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

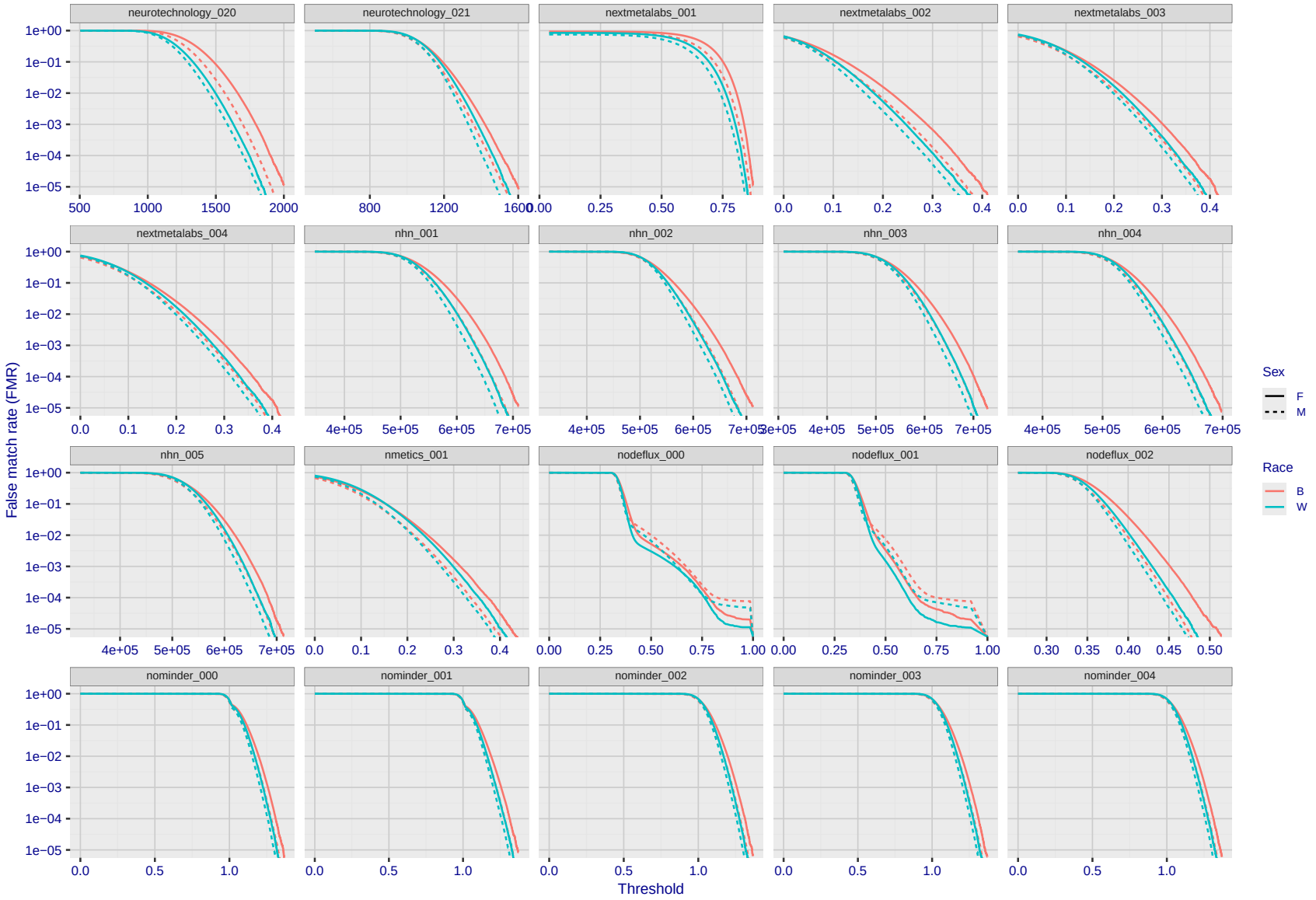

Figure 599: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

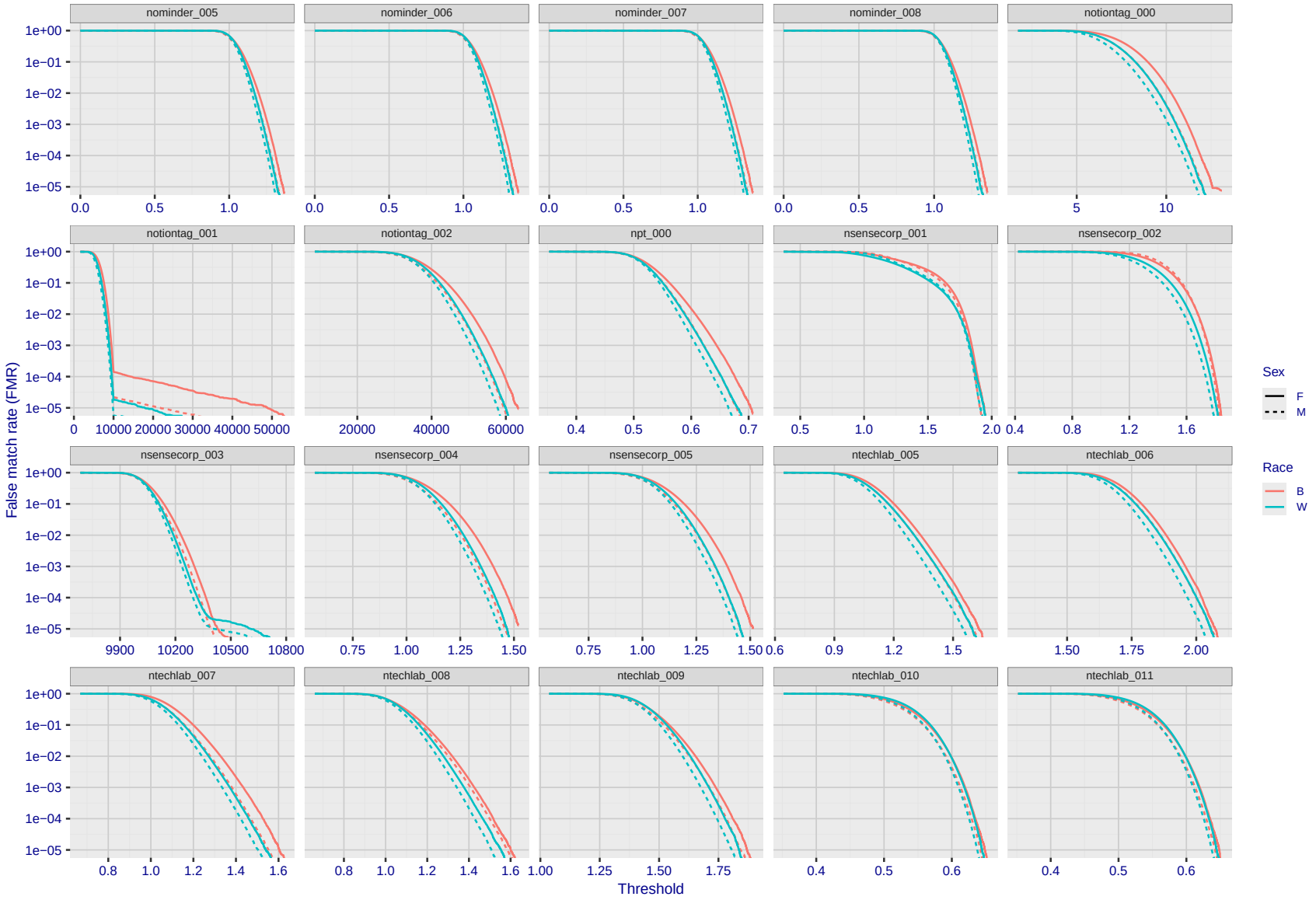

Figure 600: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

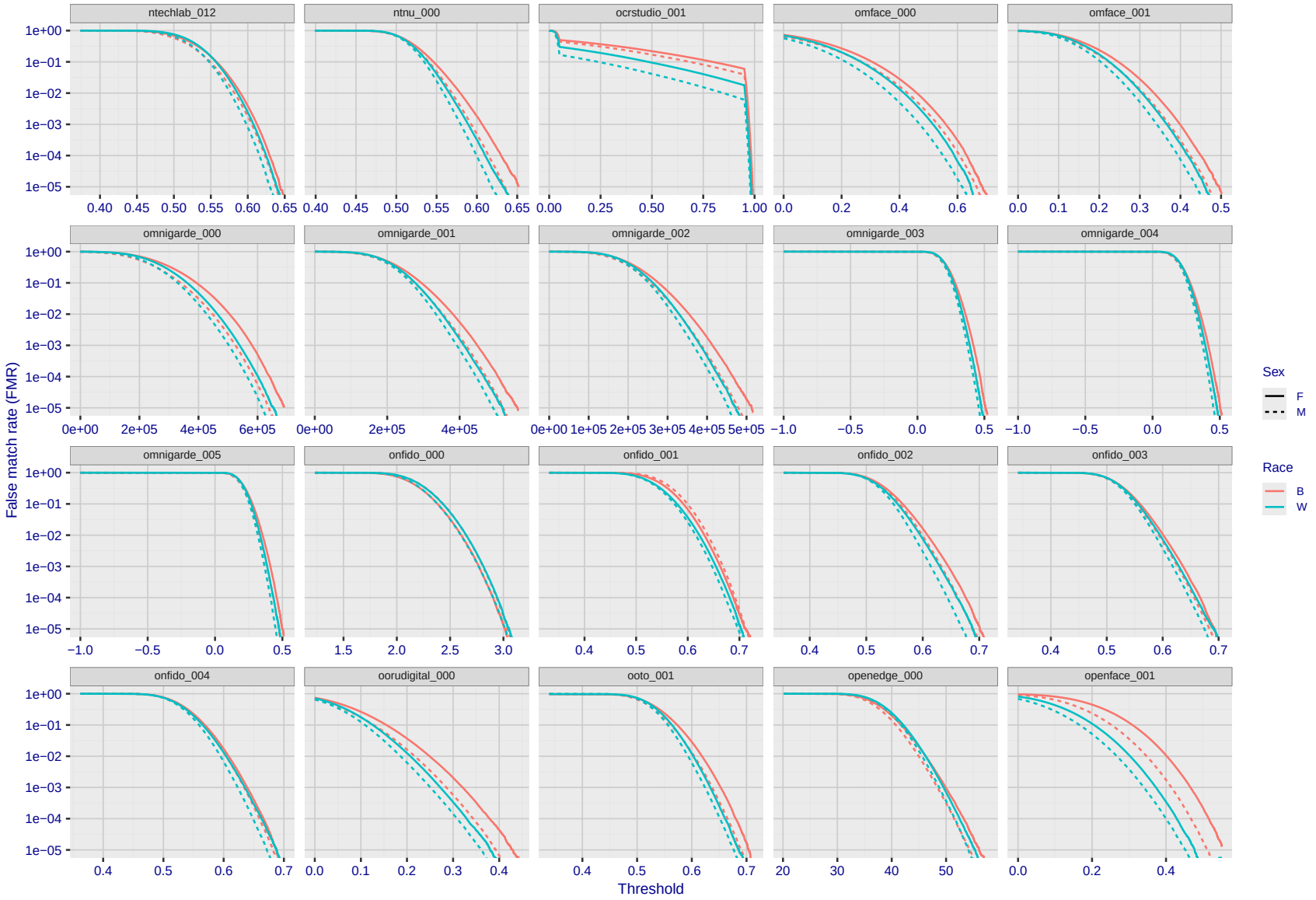

Figure 601: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

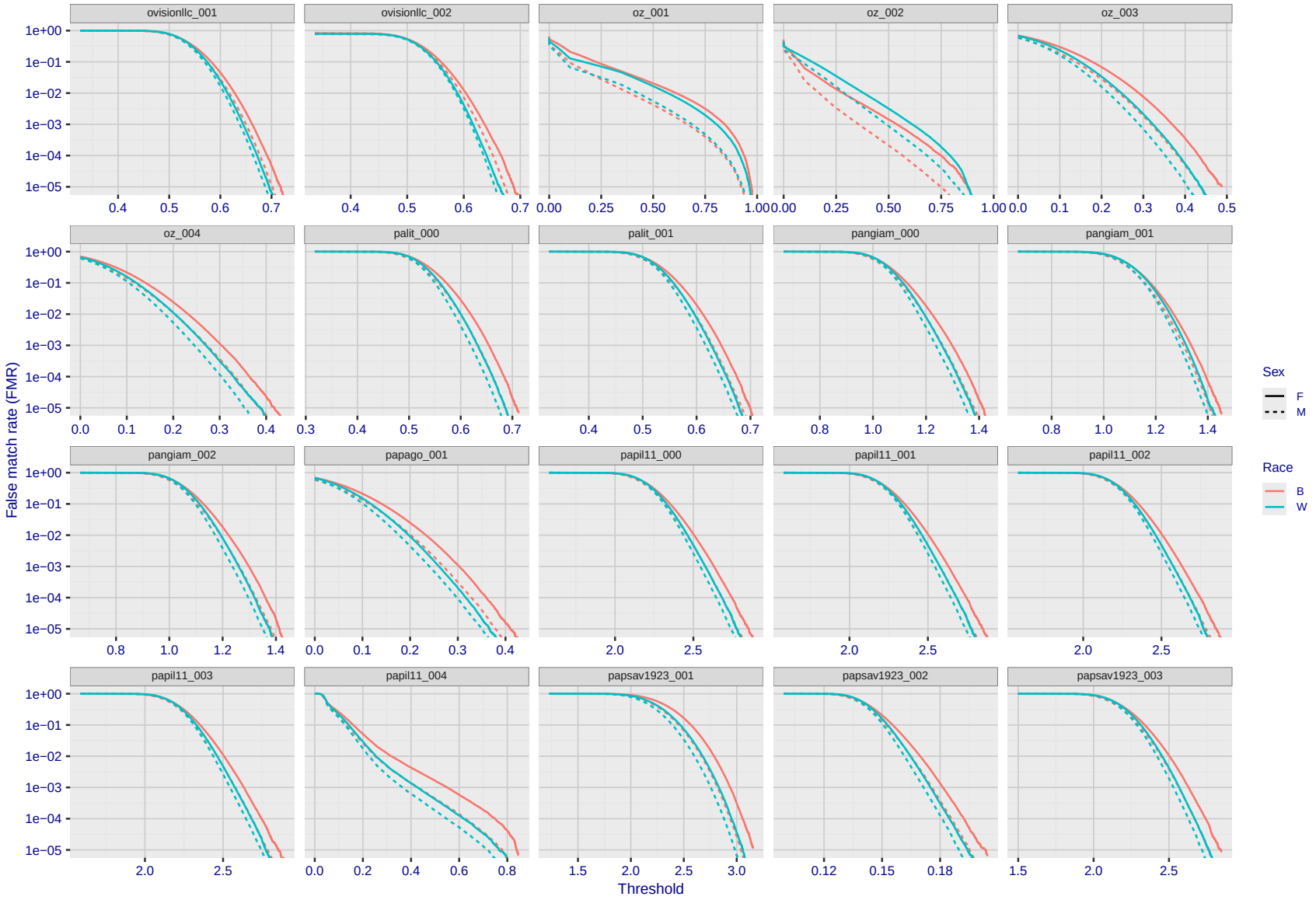

Figure 602: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

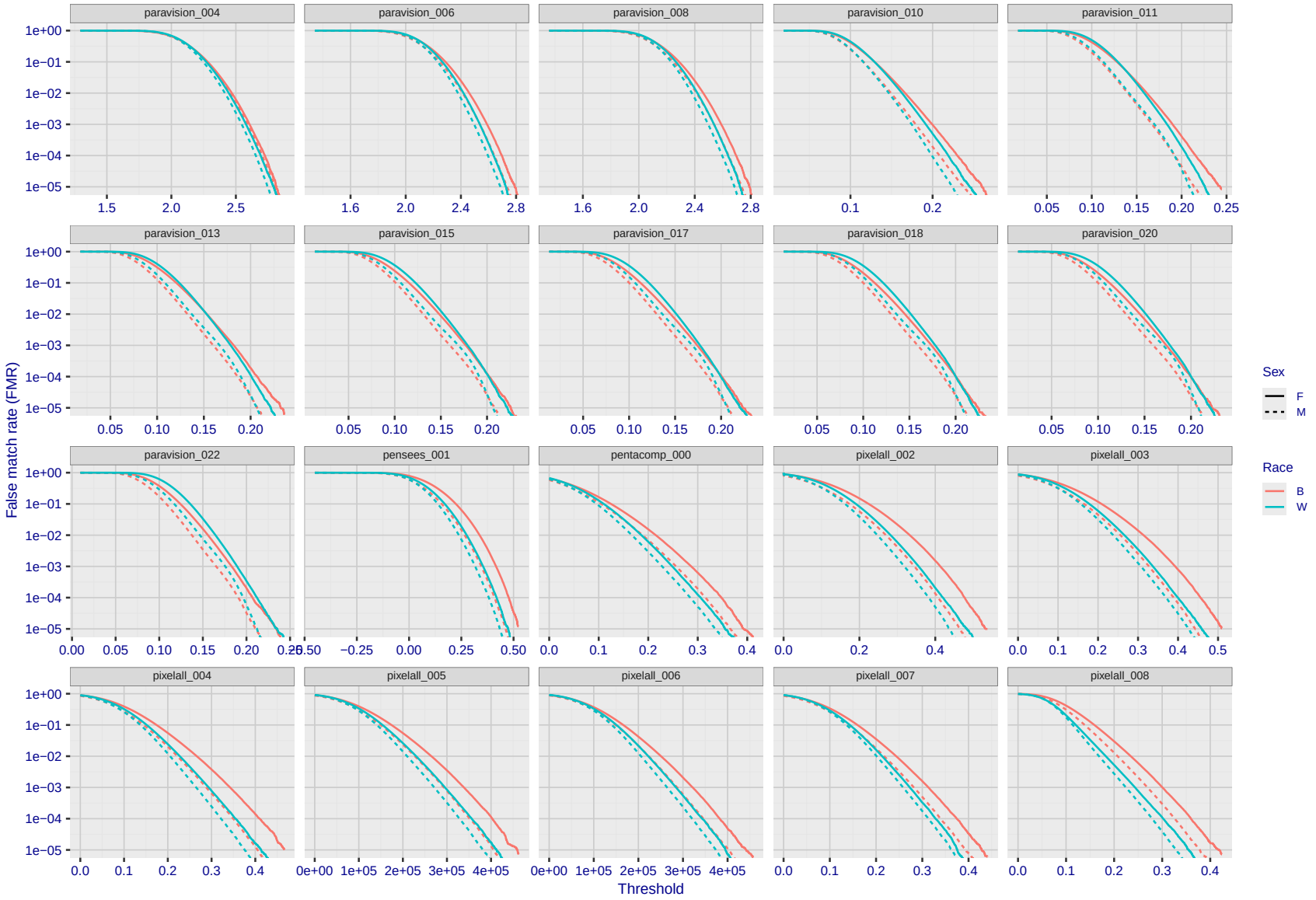

Figure 603: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

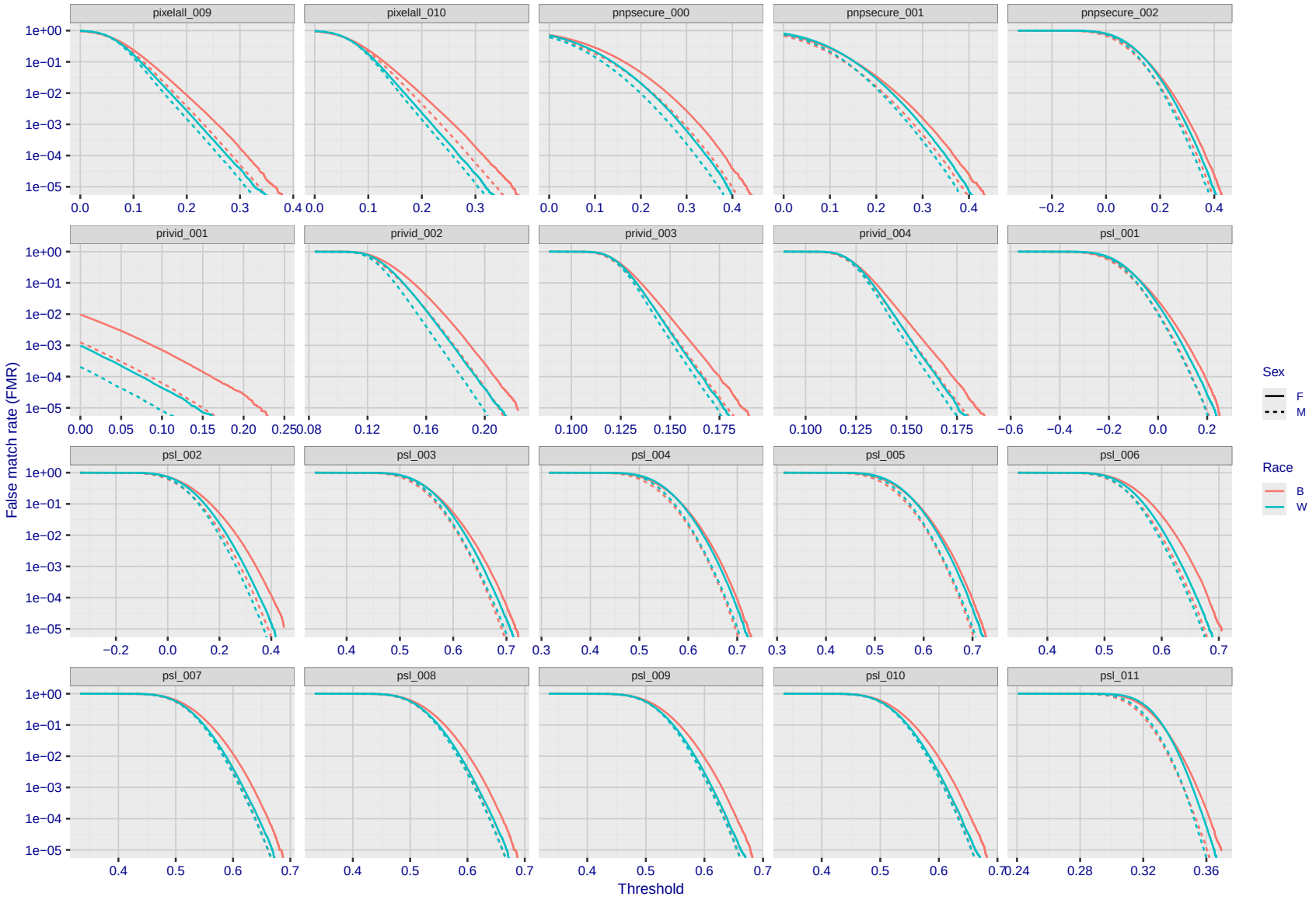

Figure 604: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

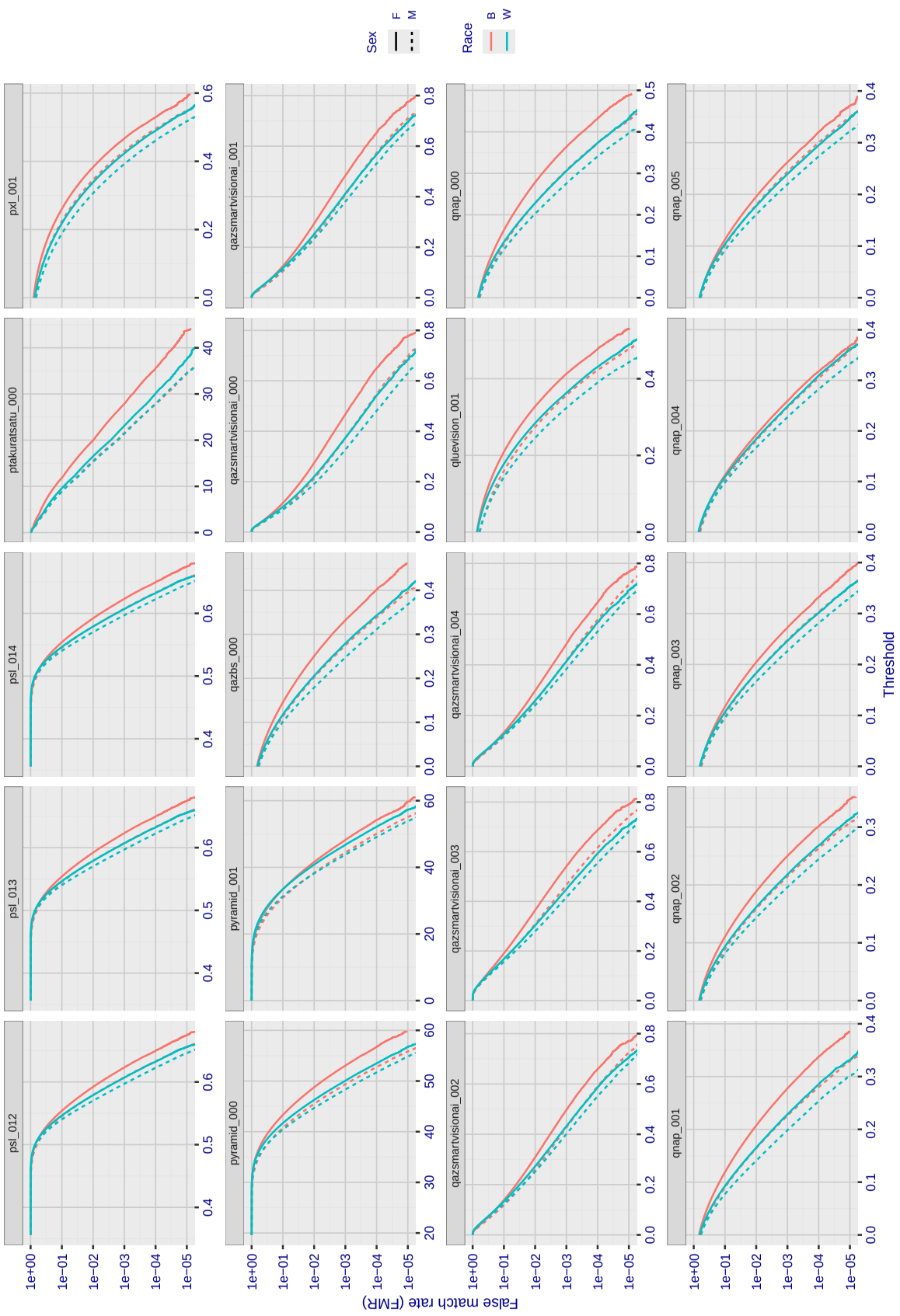

Figure 605: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

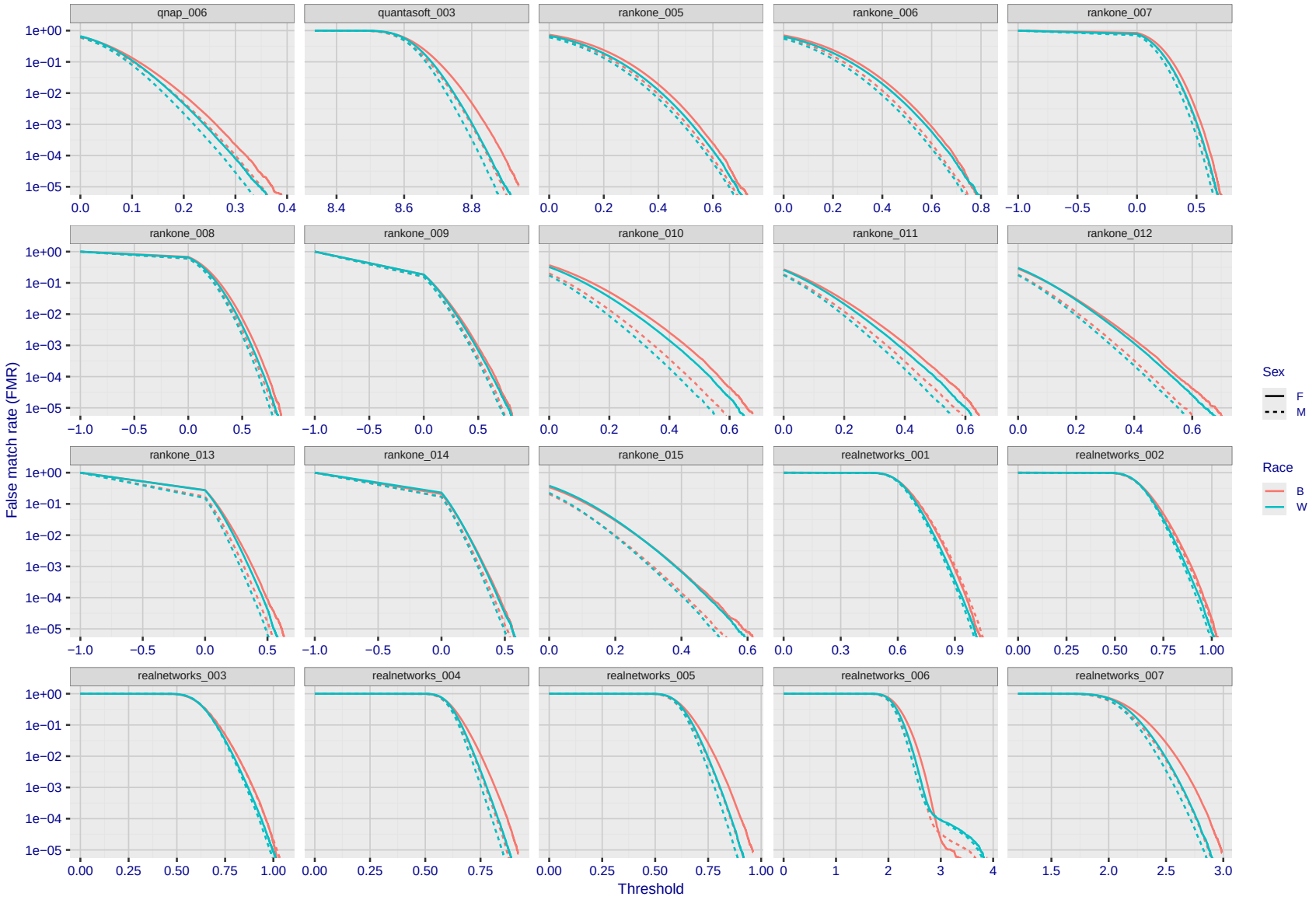

Figure 606: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

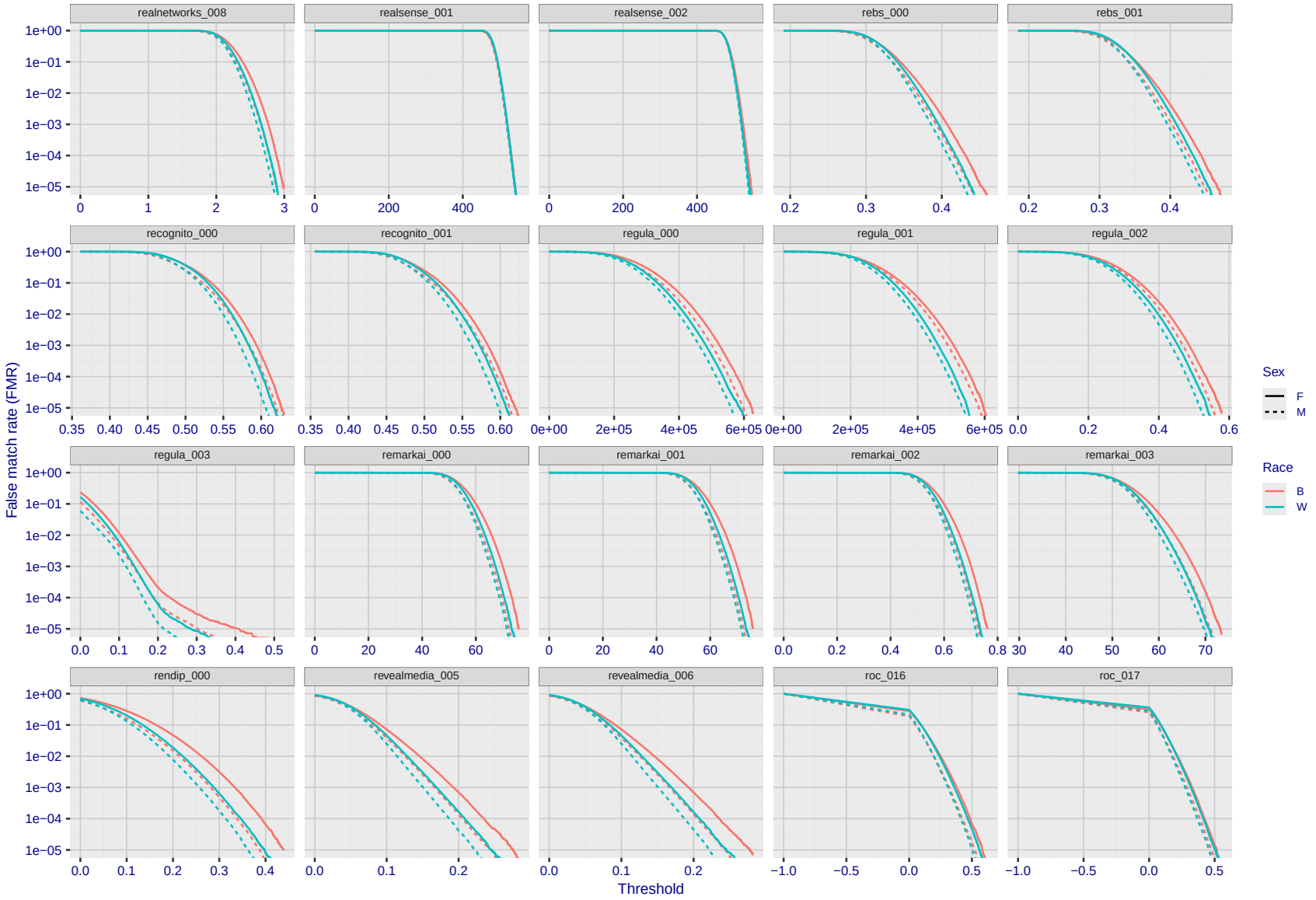

Figure 607: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

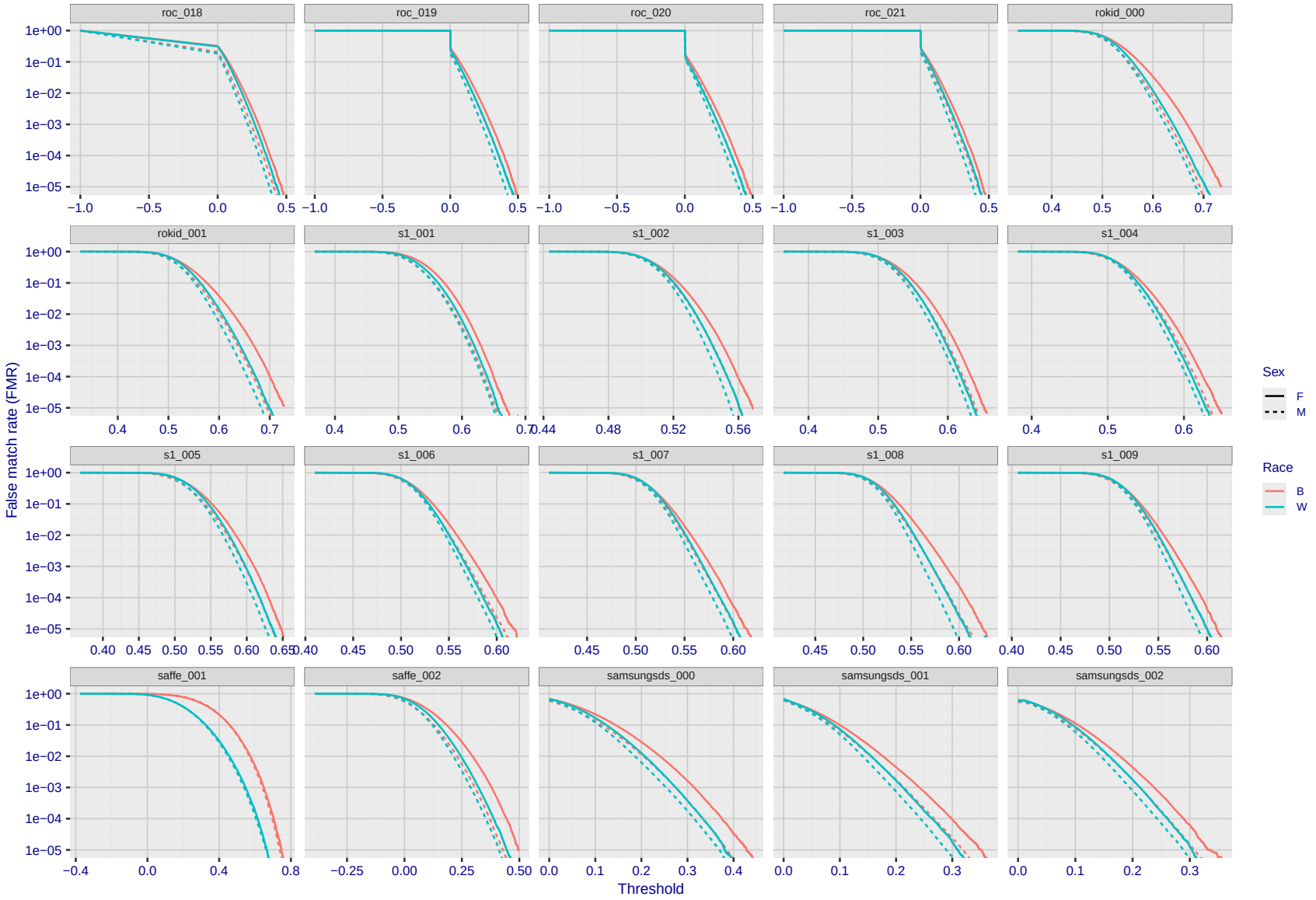

Figure 608: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

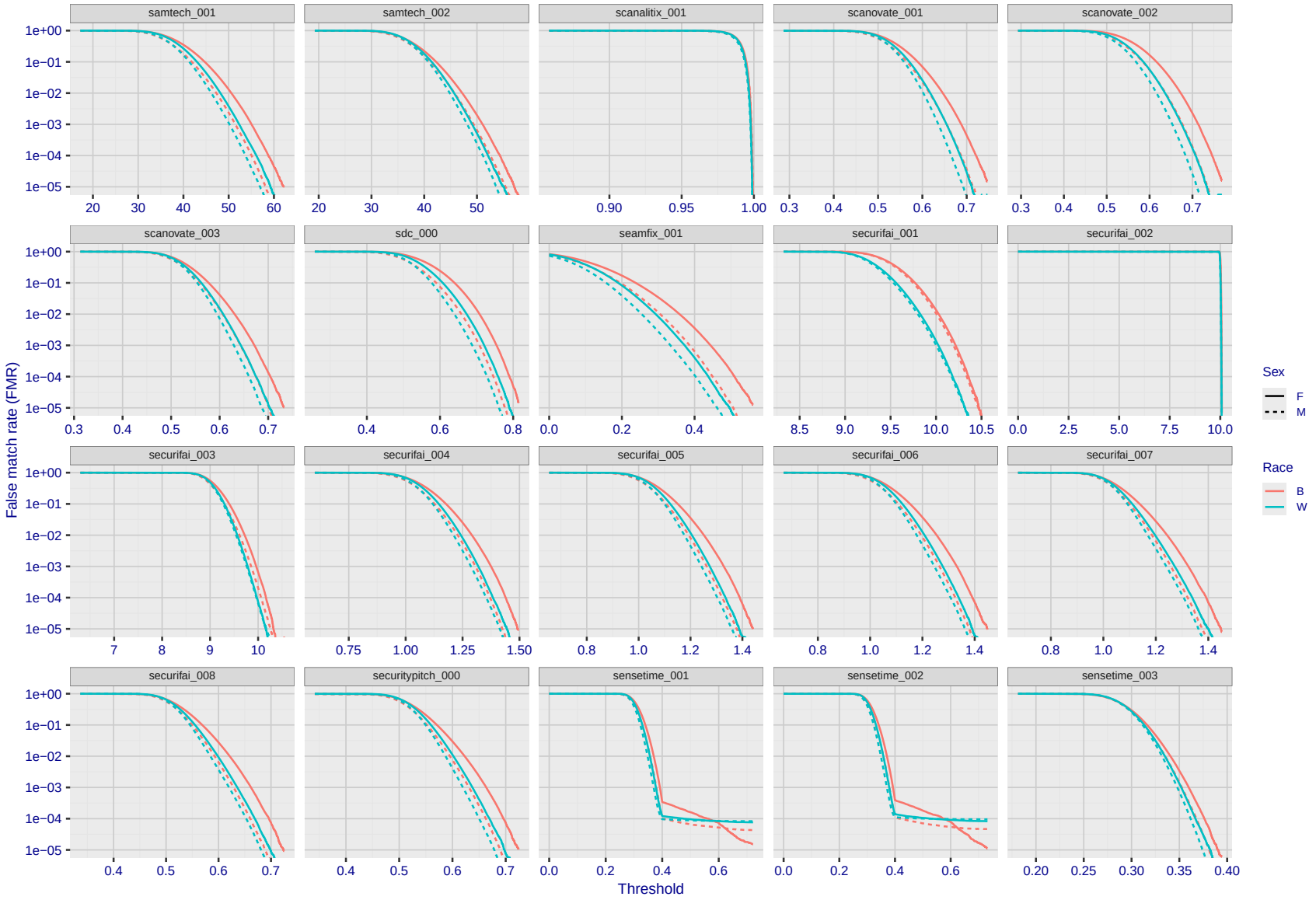

Figure 609: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

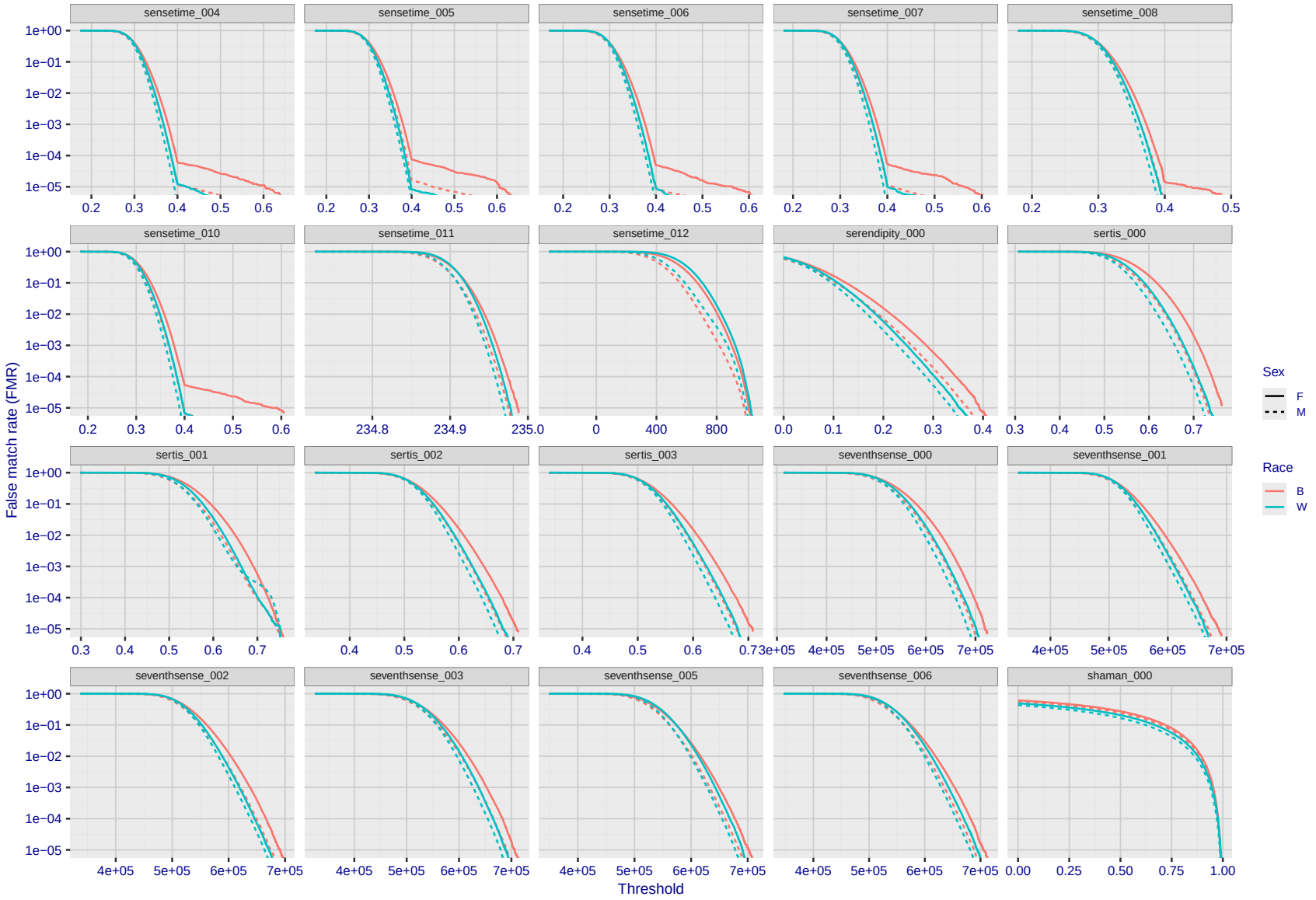

Figure 610: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

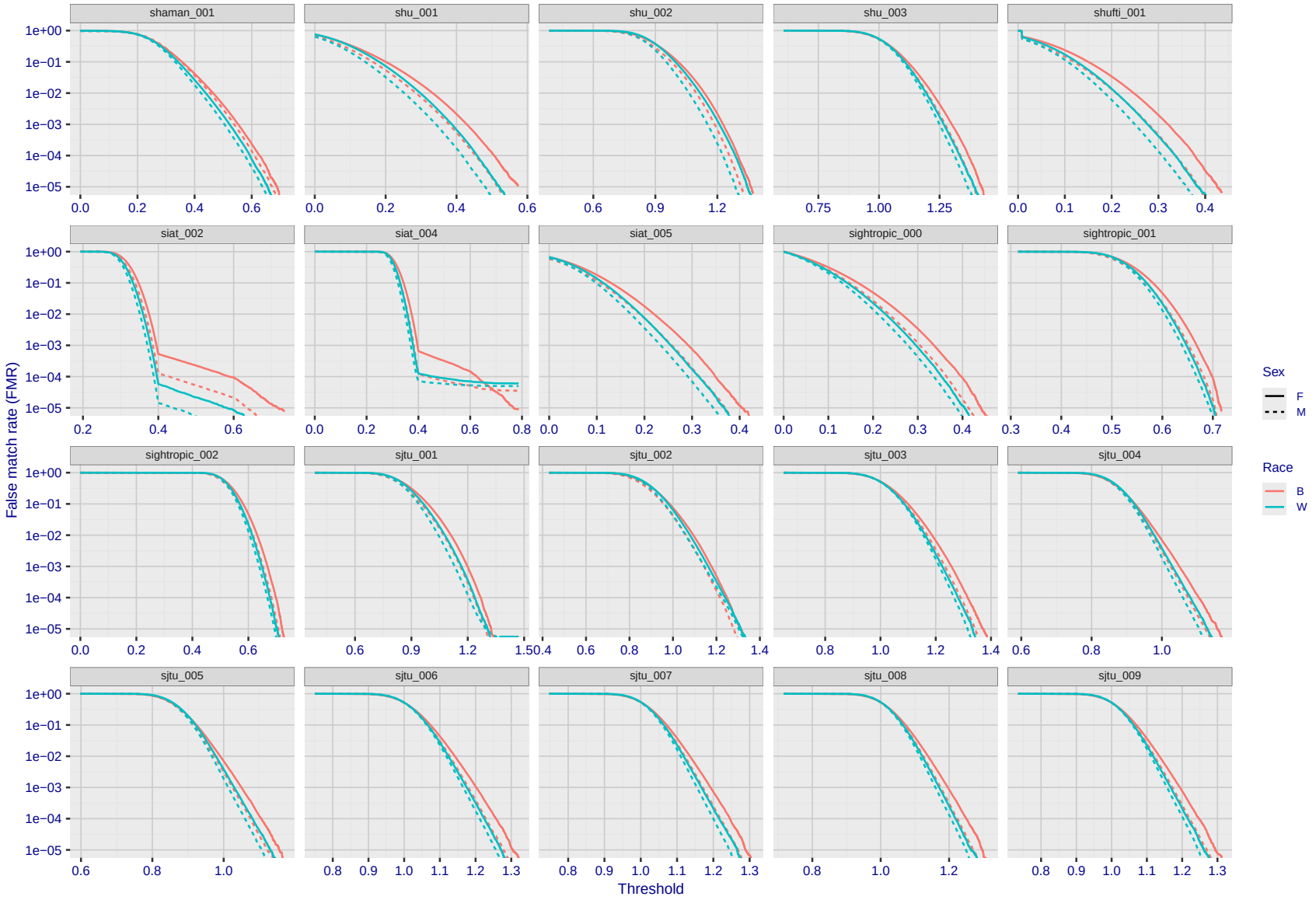

Figure 611: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

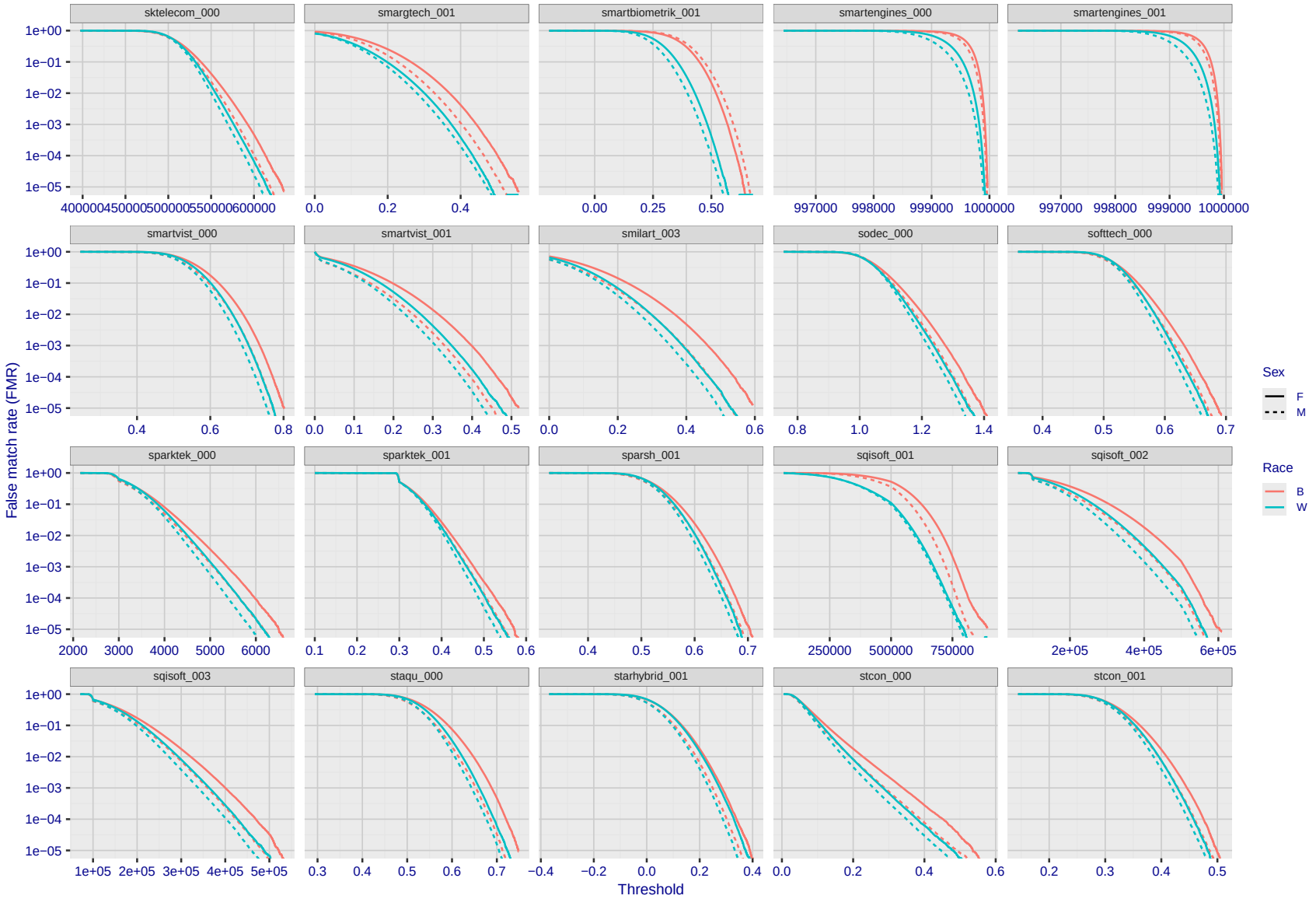

Figure 612: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

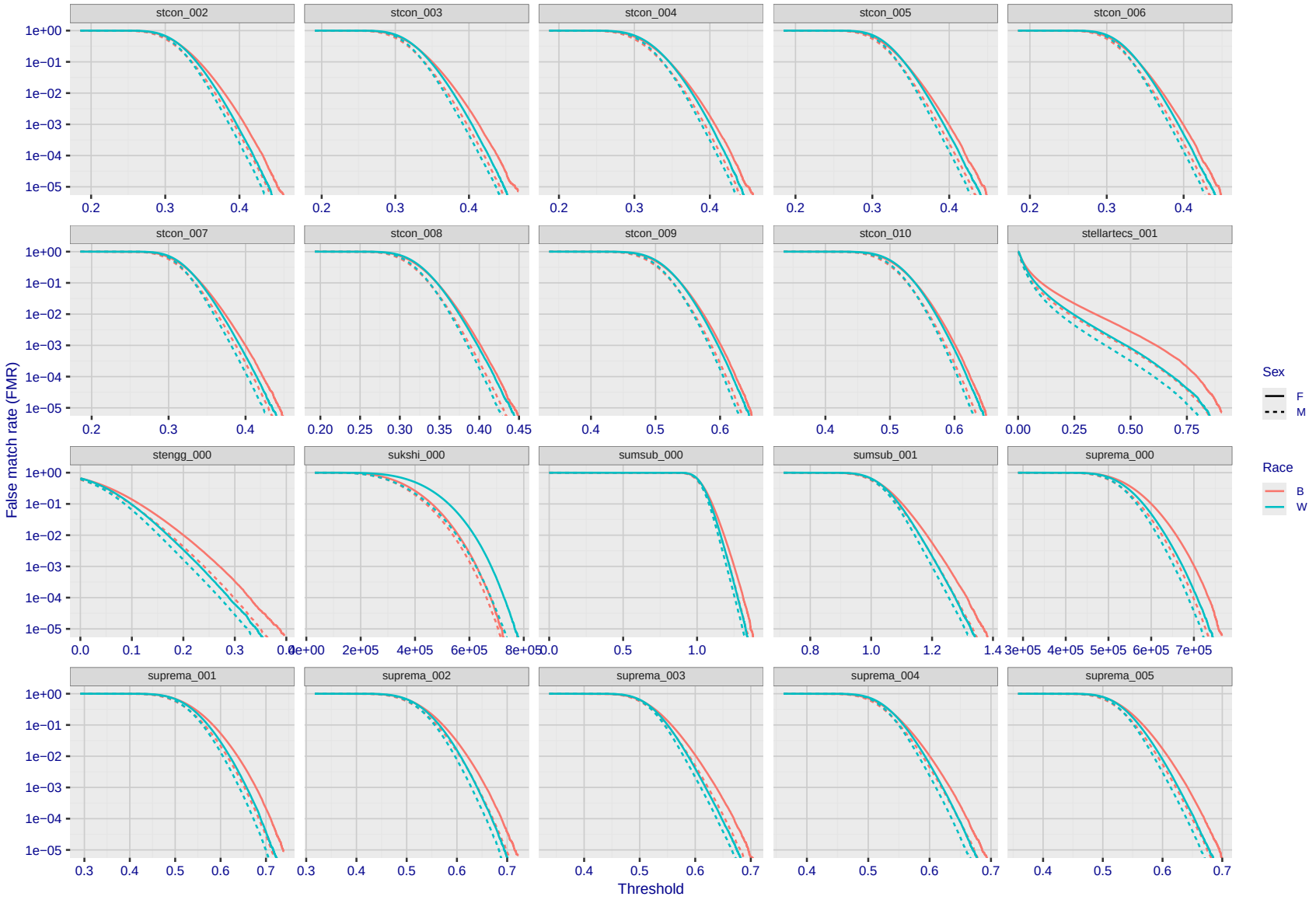

Figure 613: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

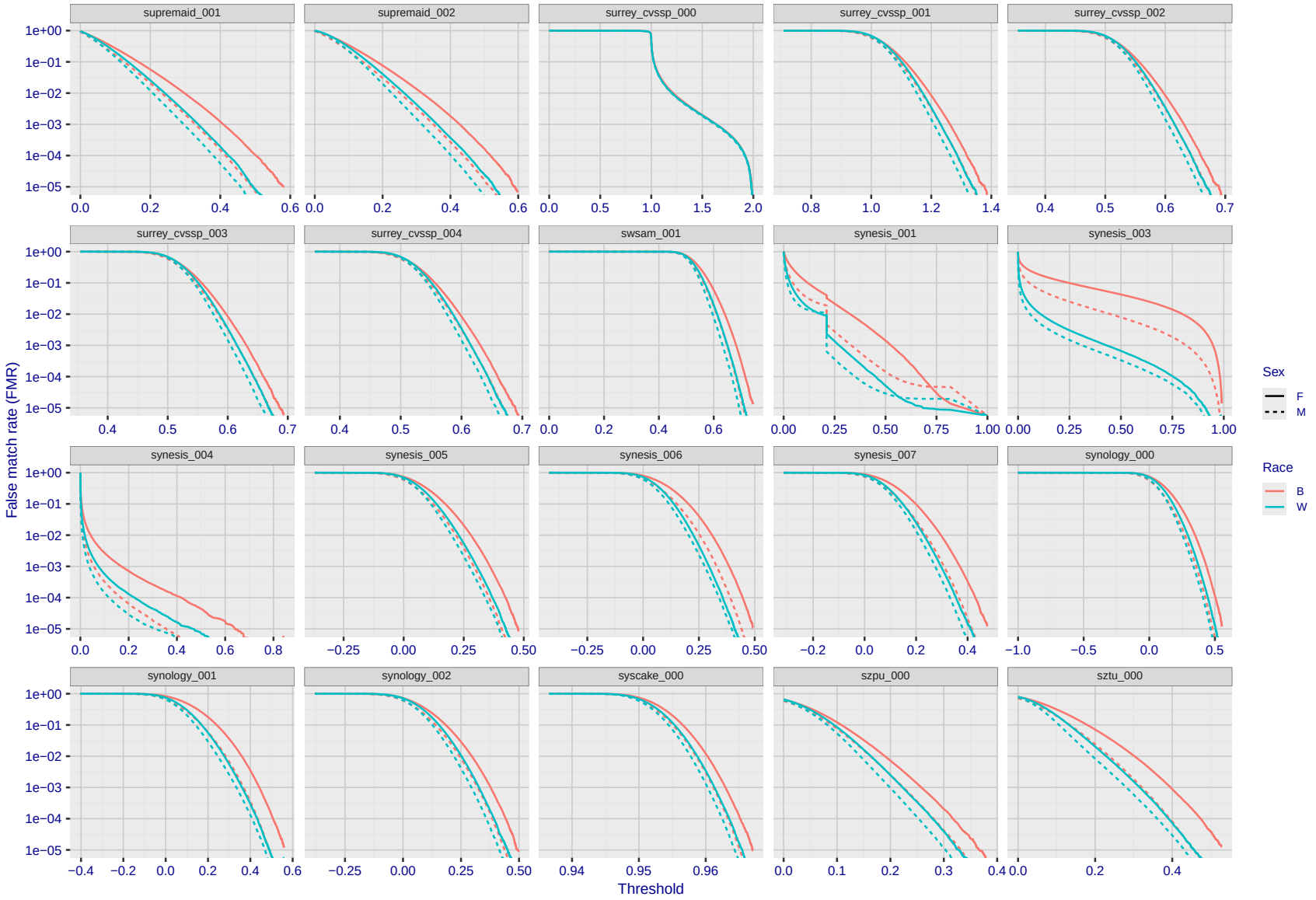

Figure 614: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

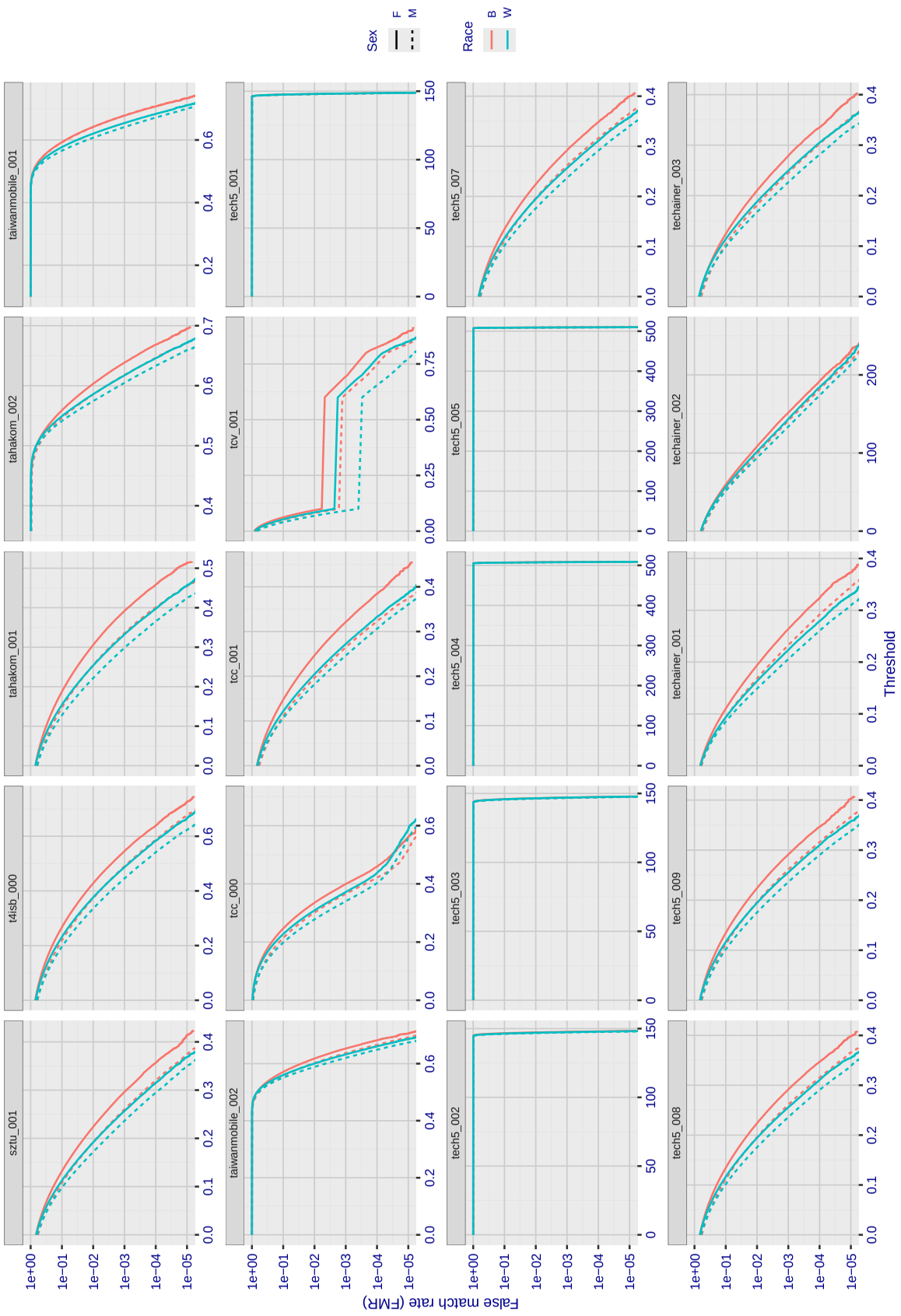

Figure 615: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

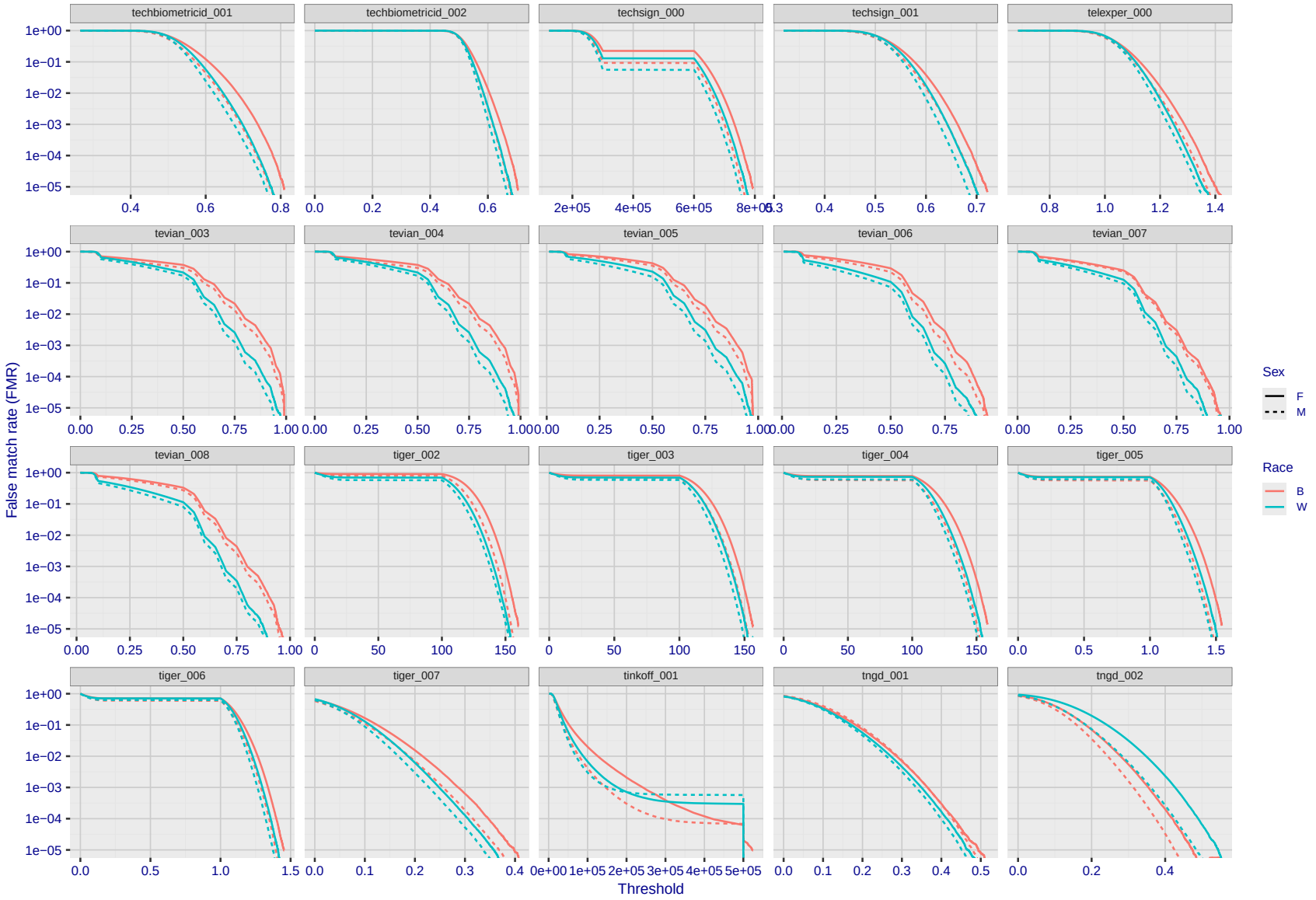

Figure 616: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

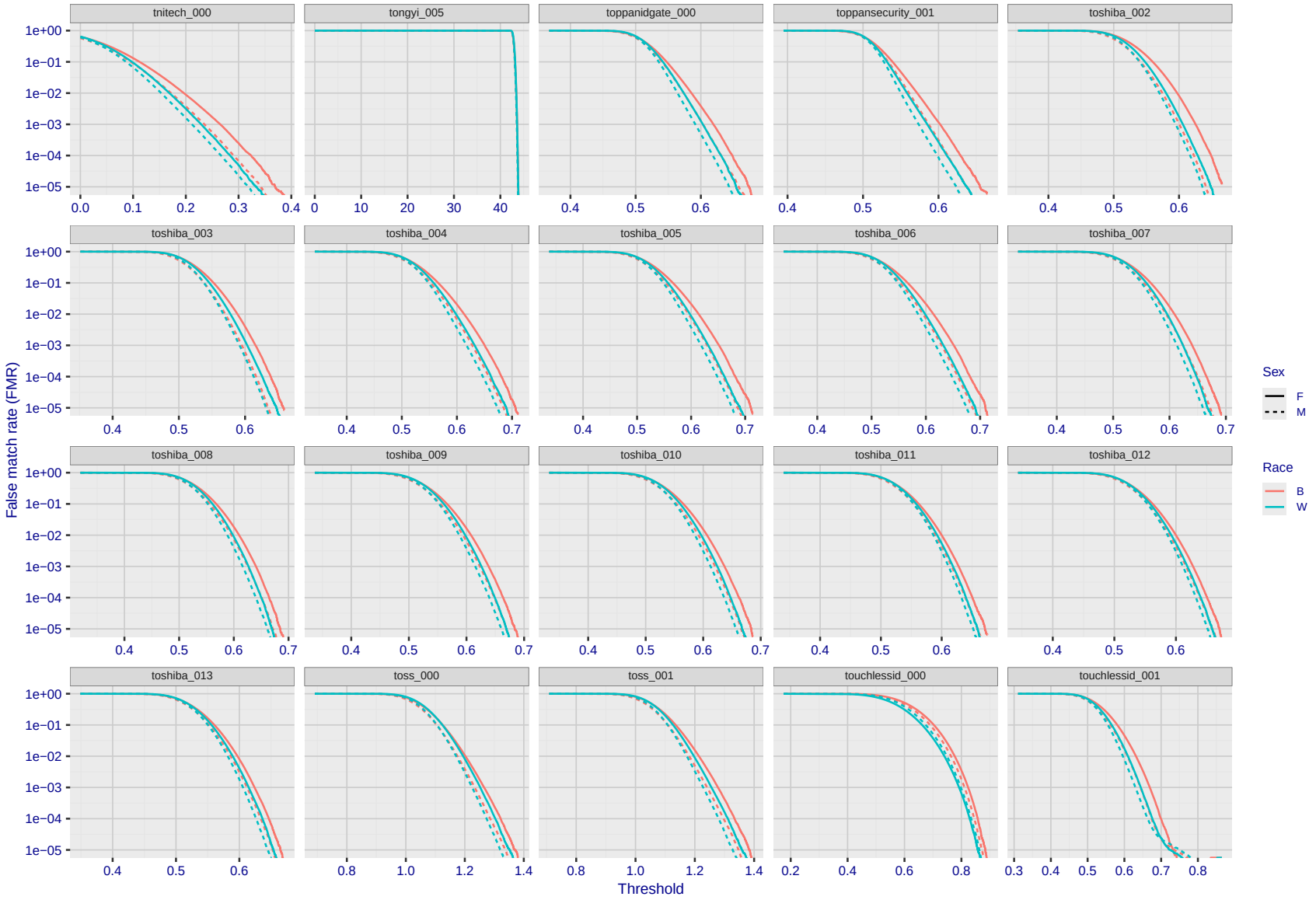

Figure 617: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

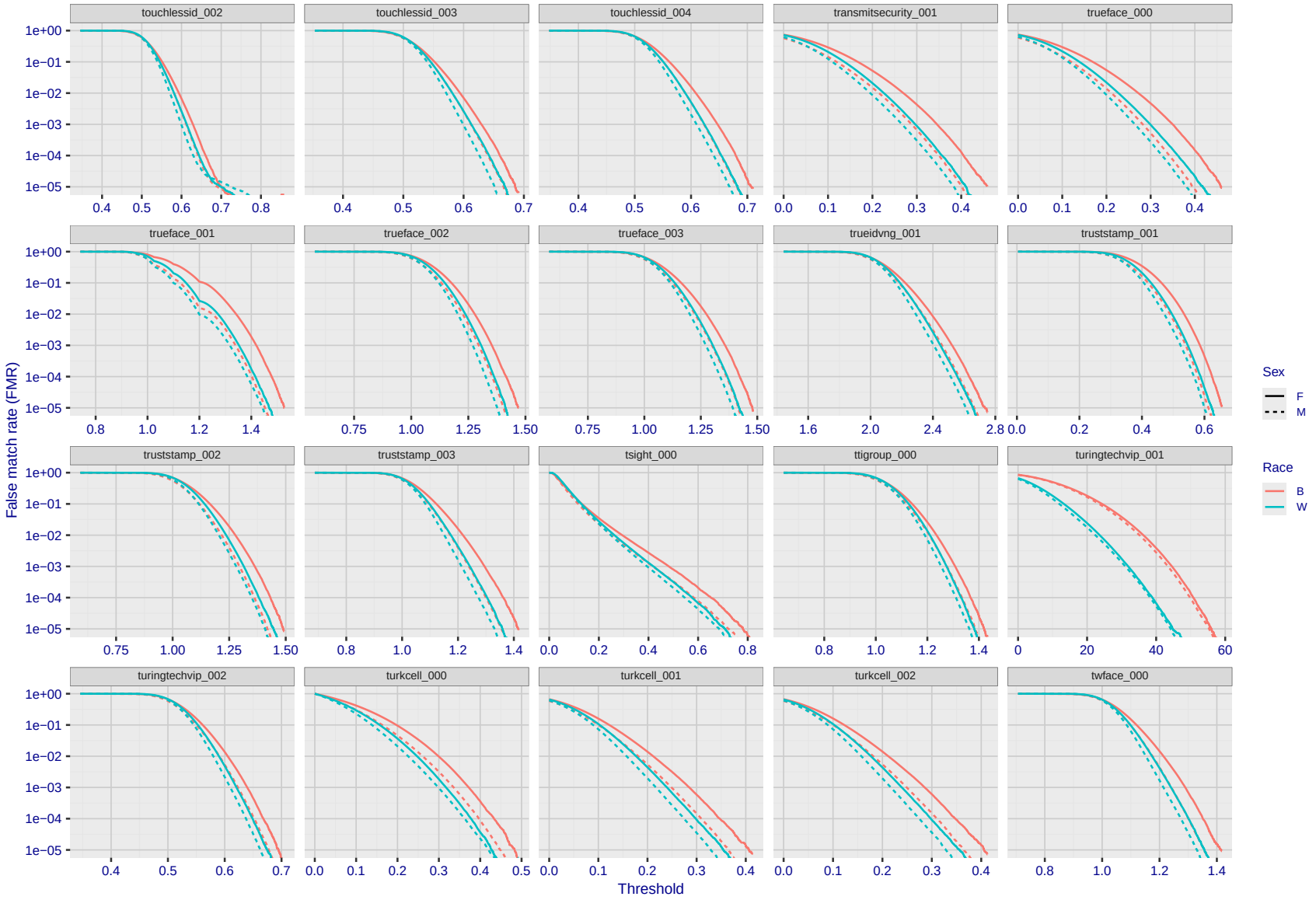

Figure 618: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

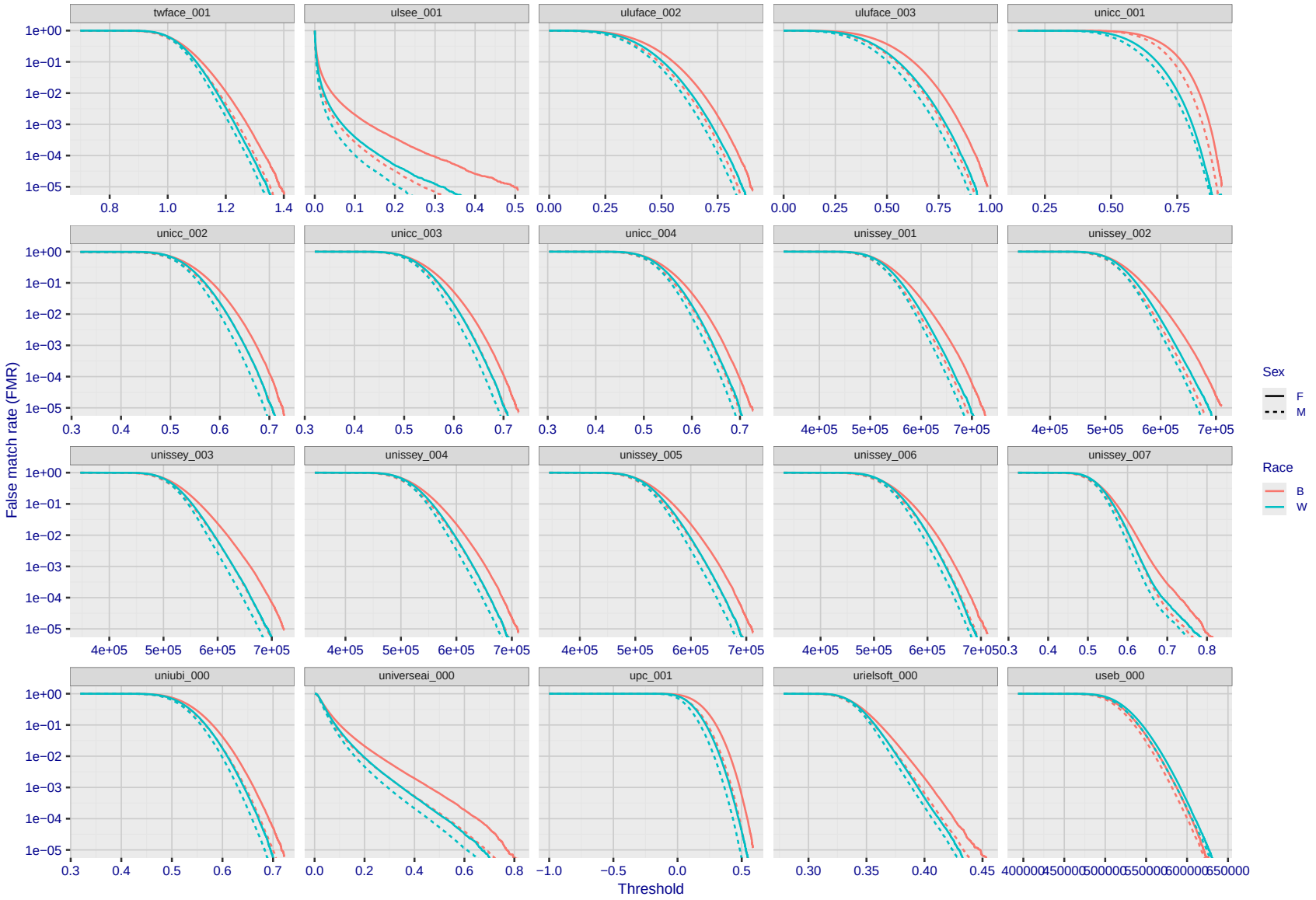

Figure 619: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

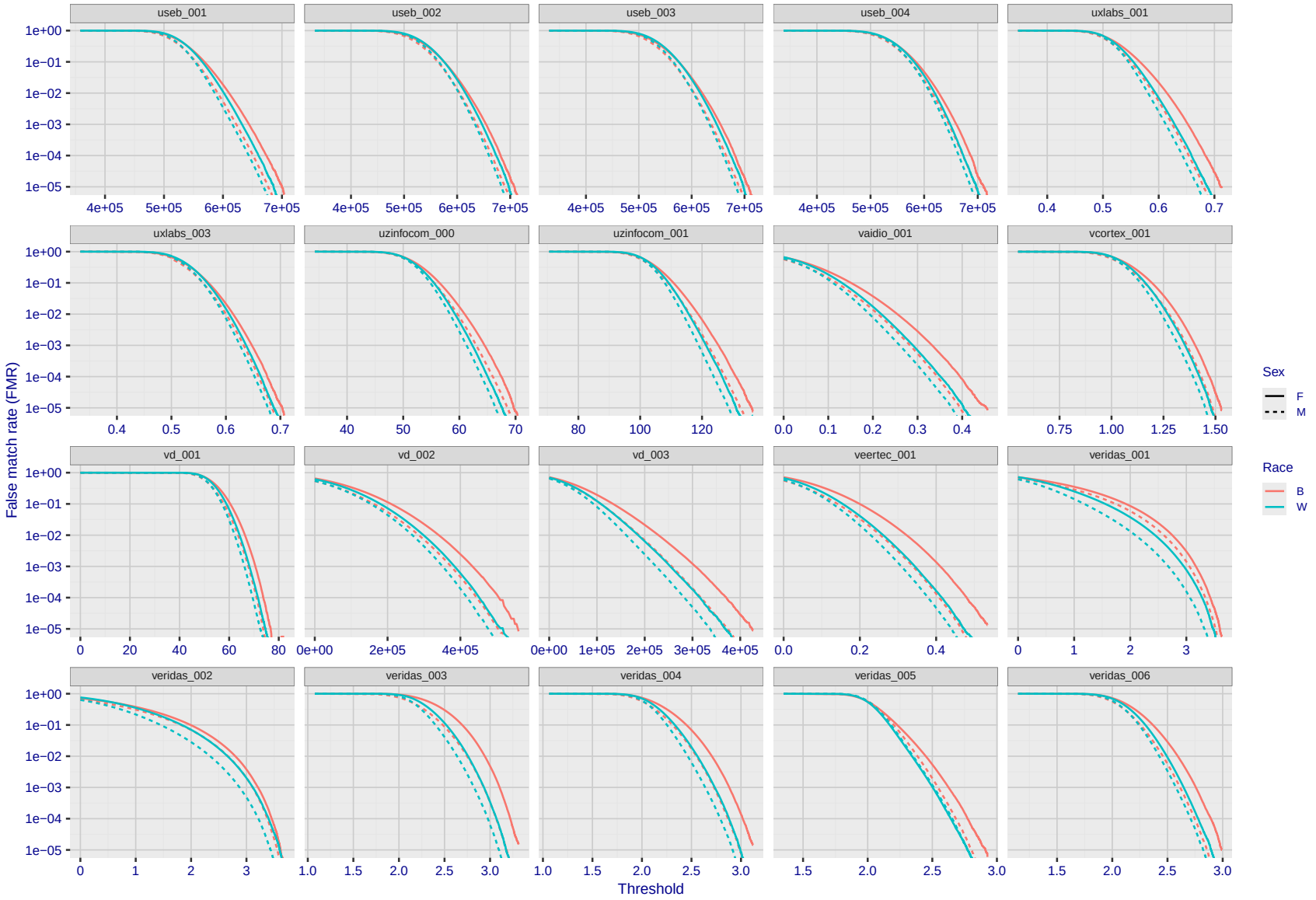

Figure 620: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

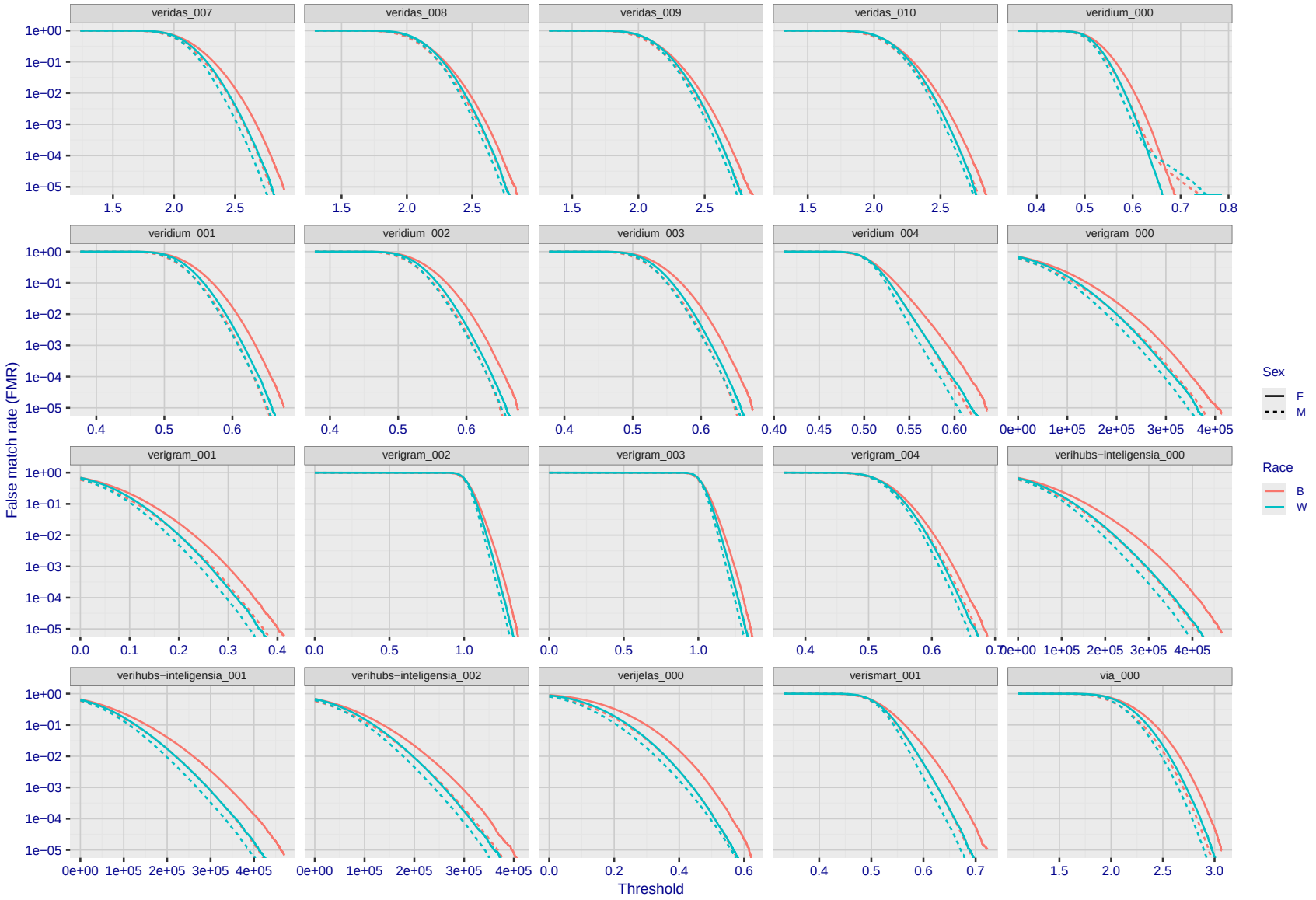

Figure 621: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

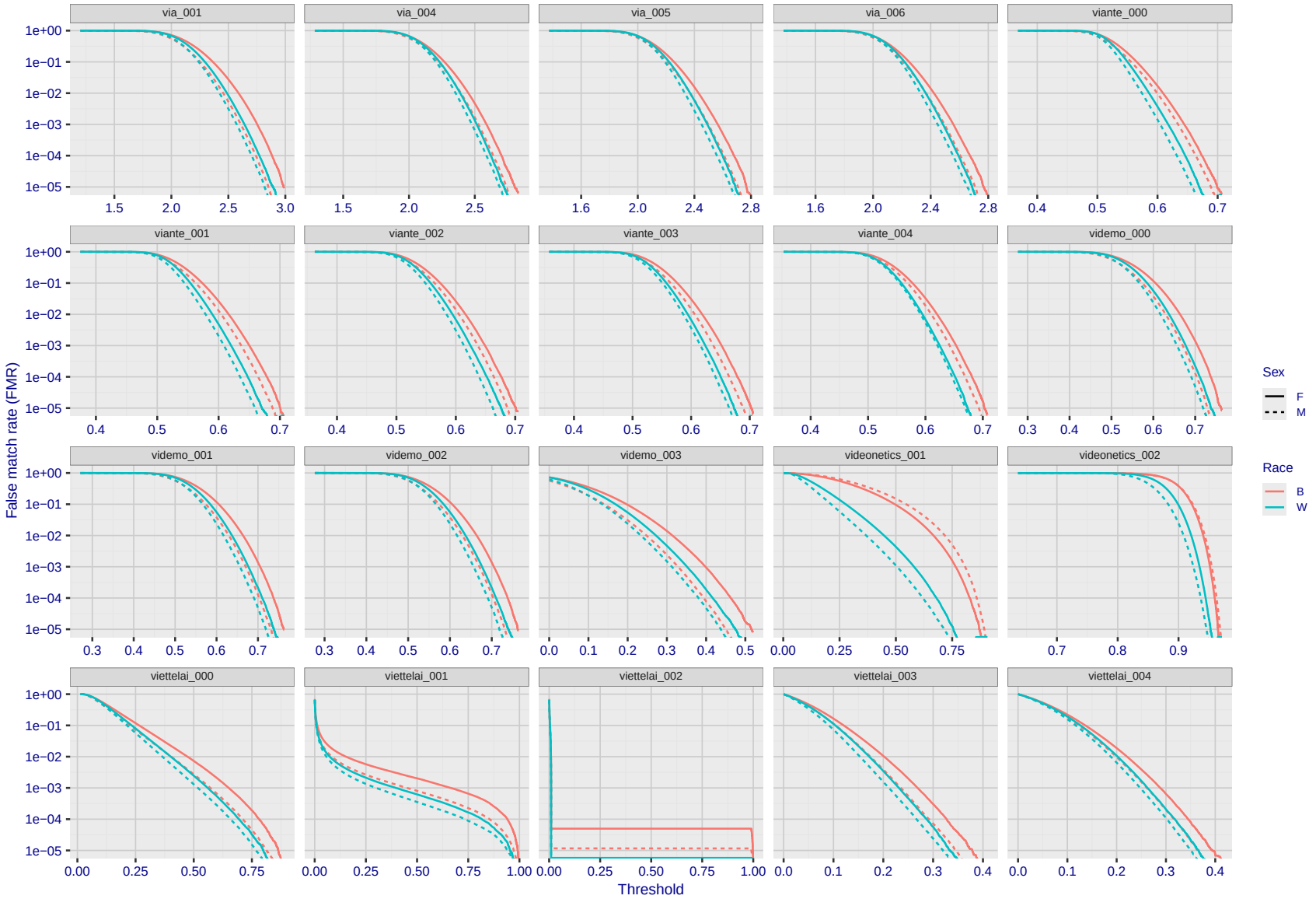

Figure 622: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

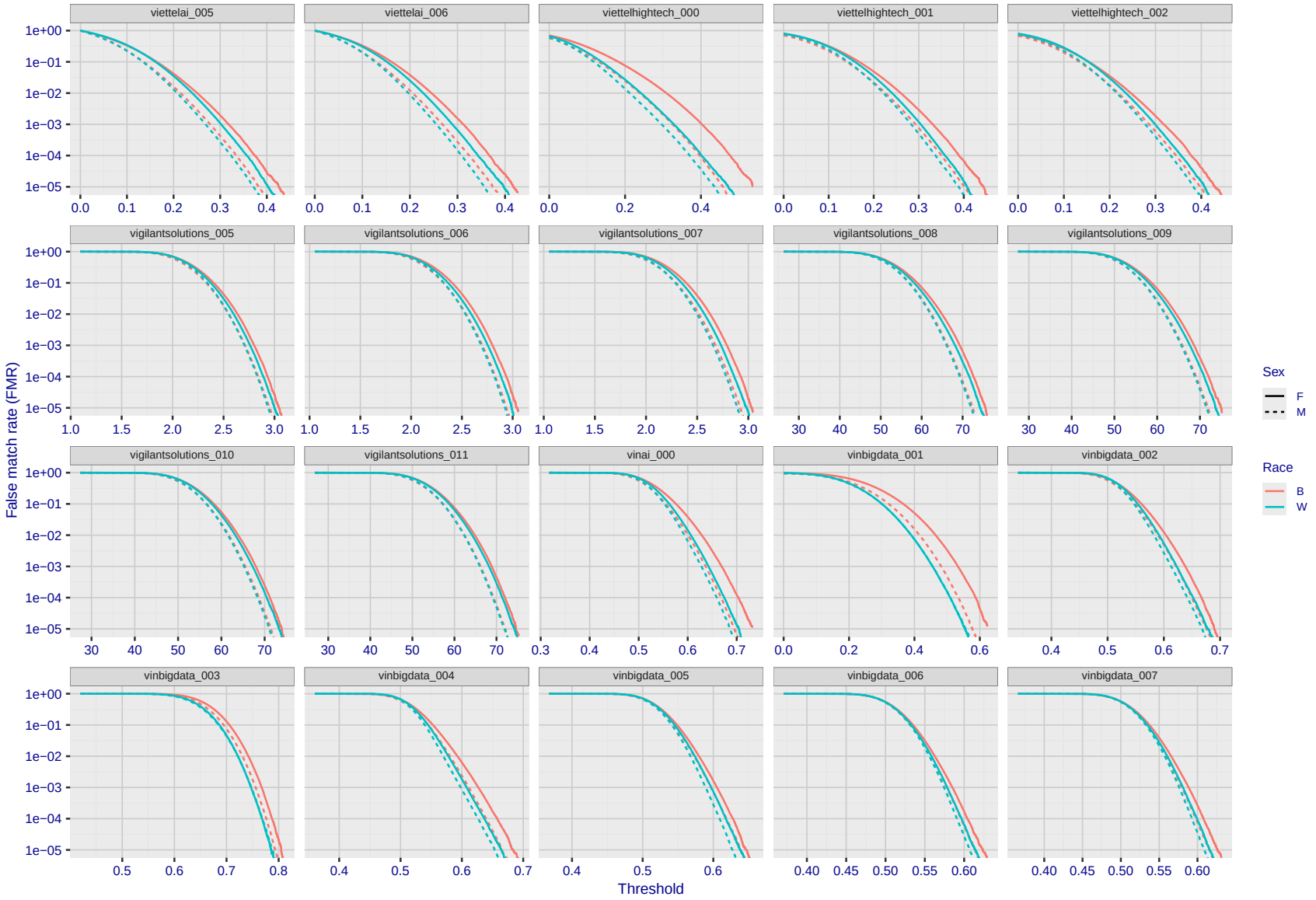

Figure 623: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

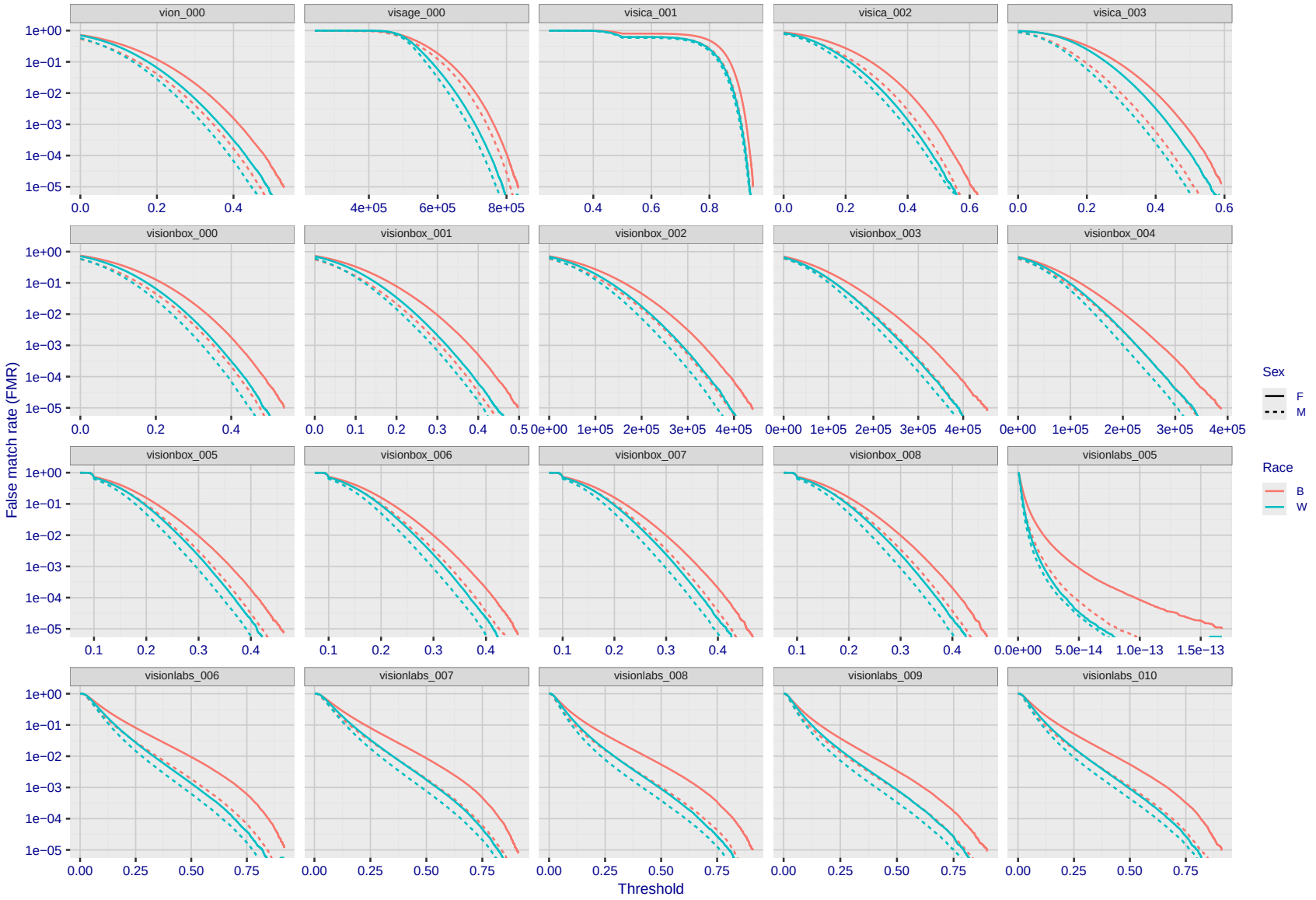

Figure 624: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

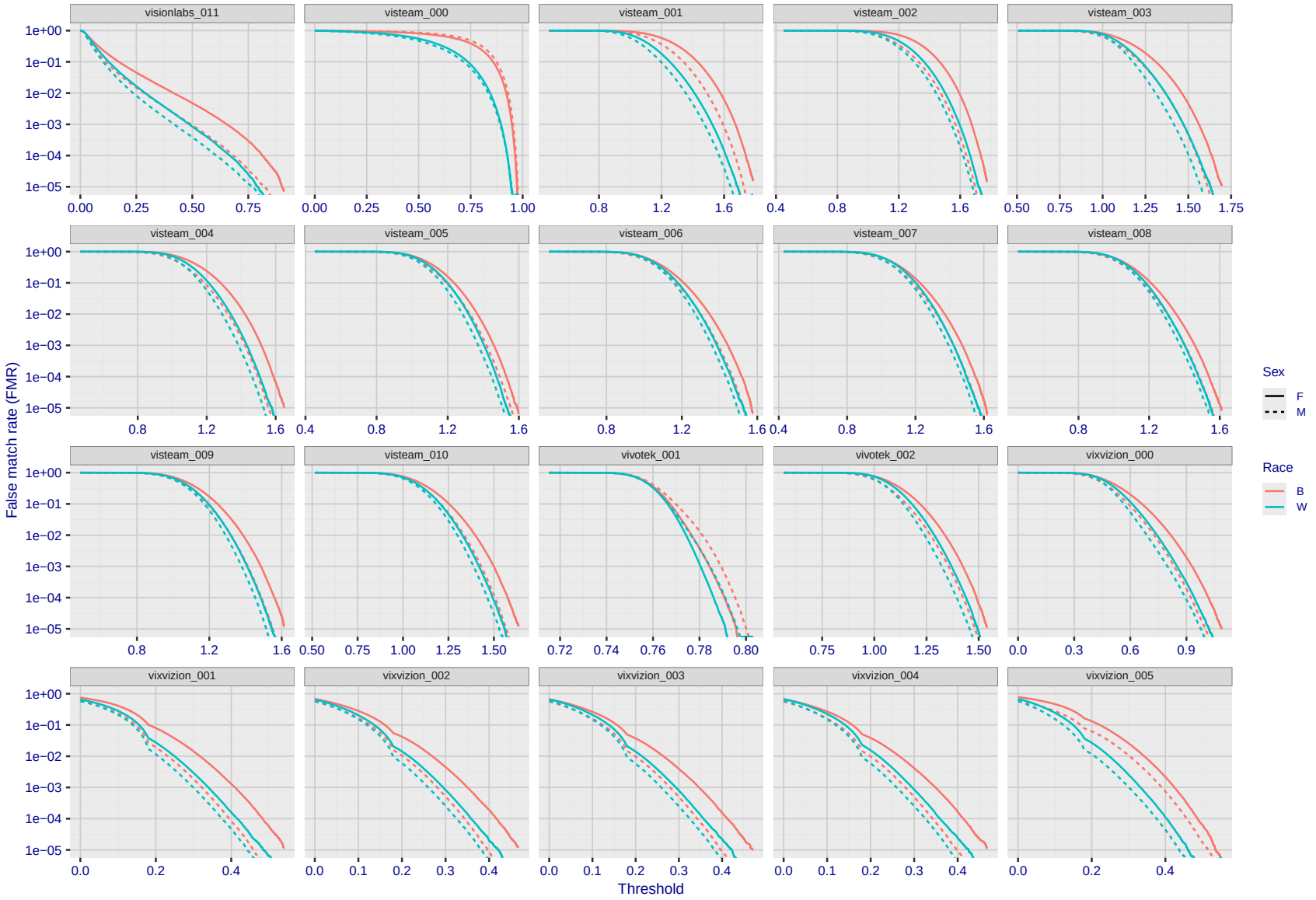

Figure 625: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

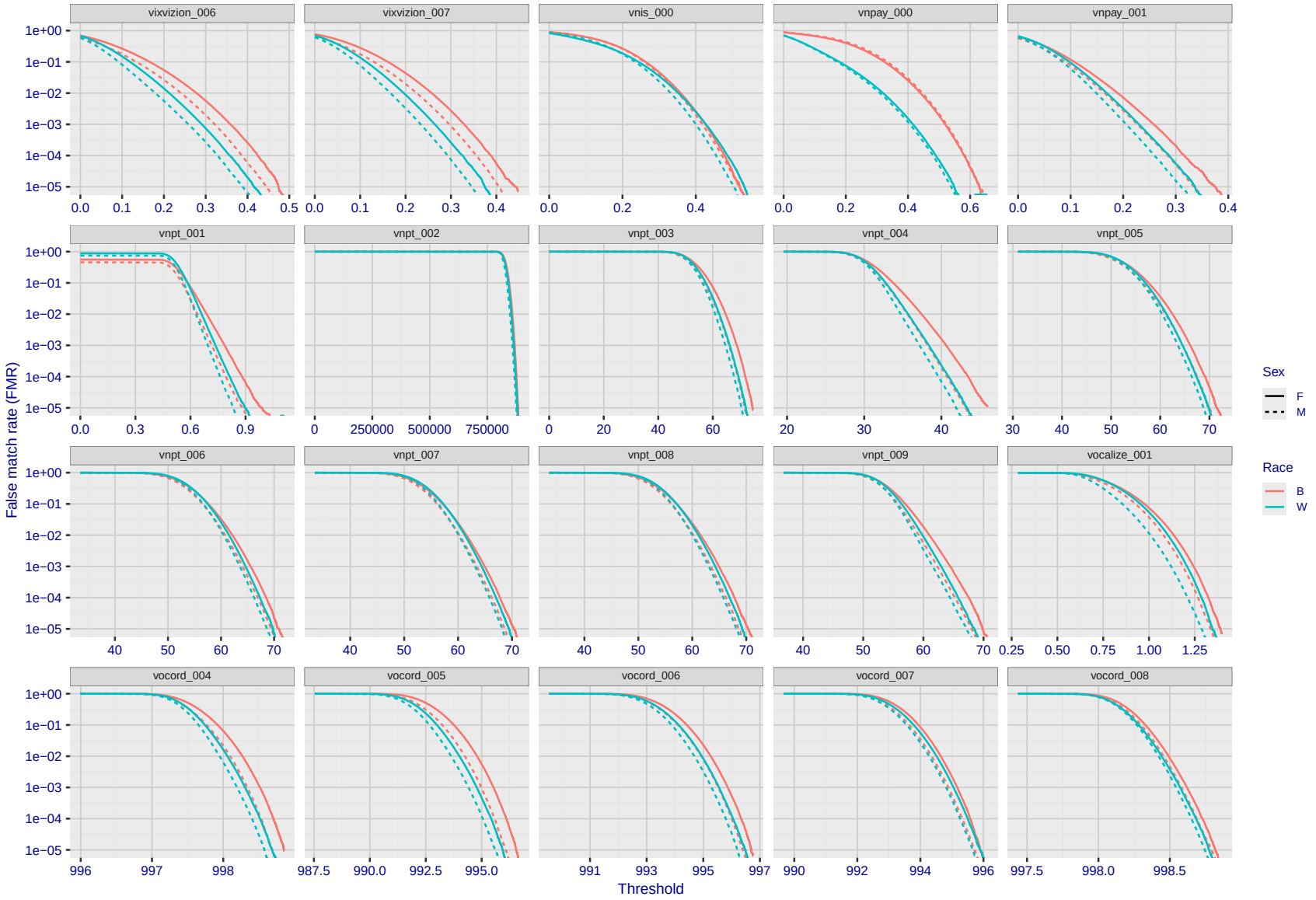

Figure 626: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

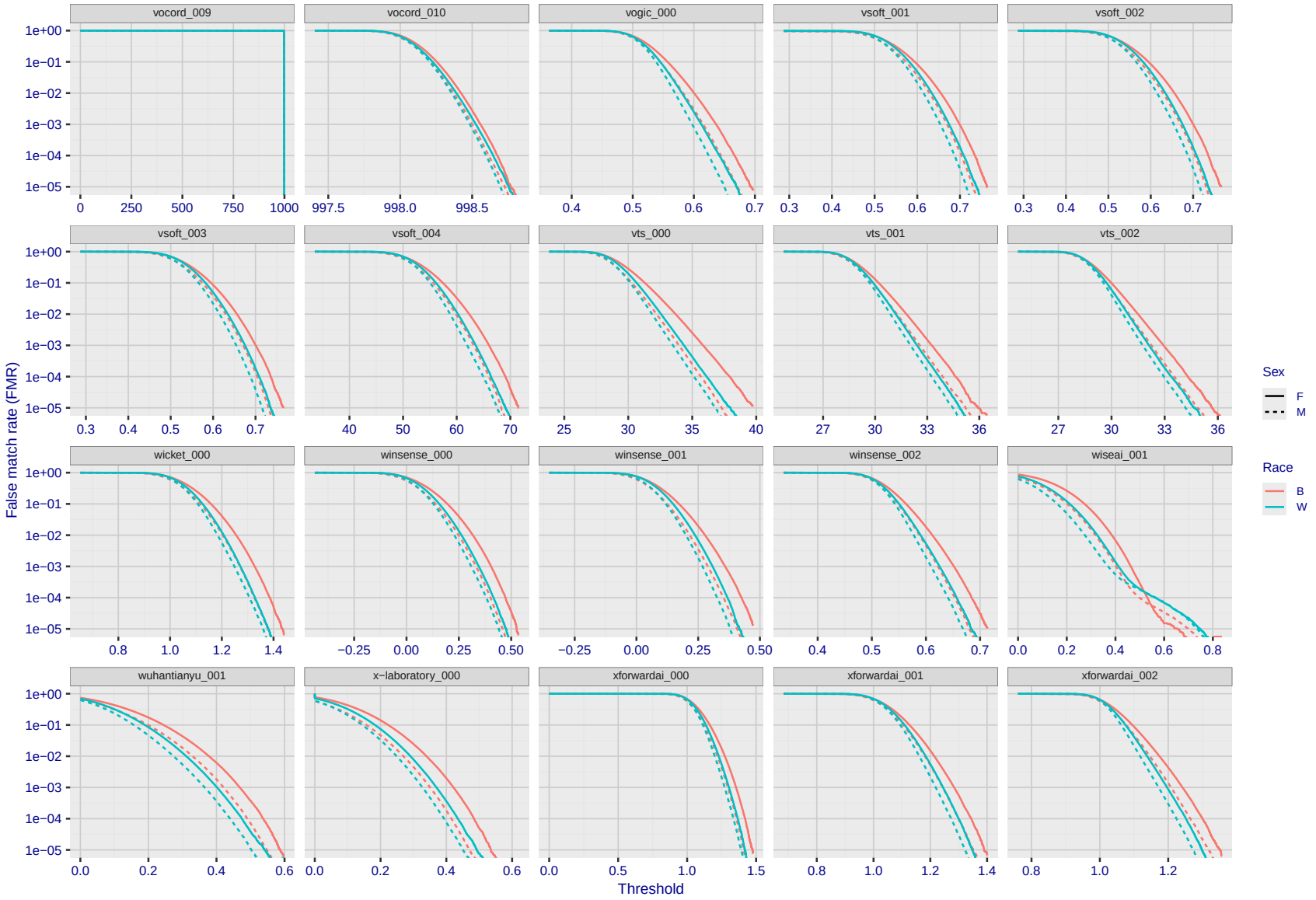

Figure 627: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

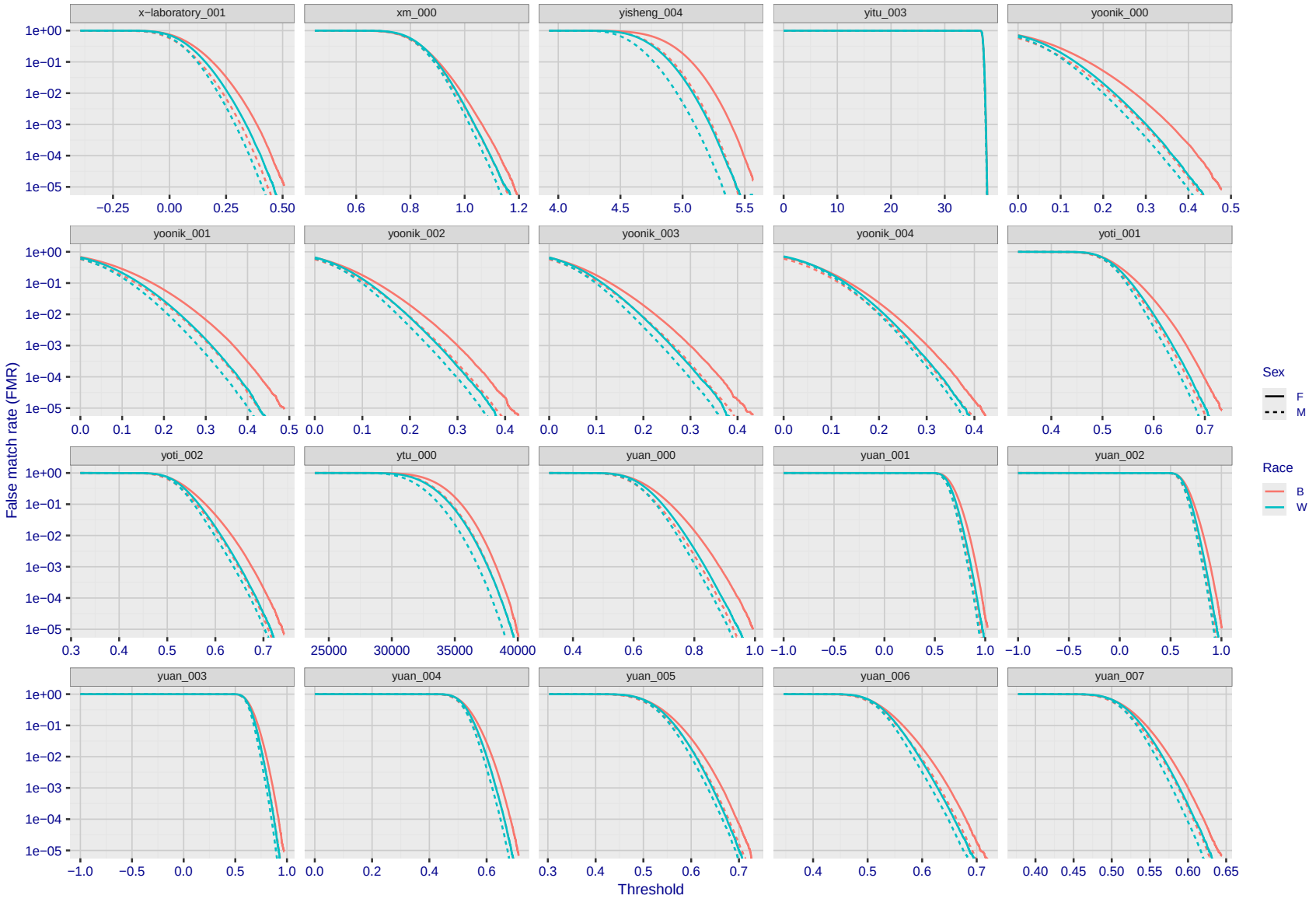

Figure 628: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

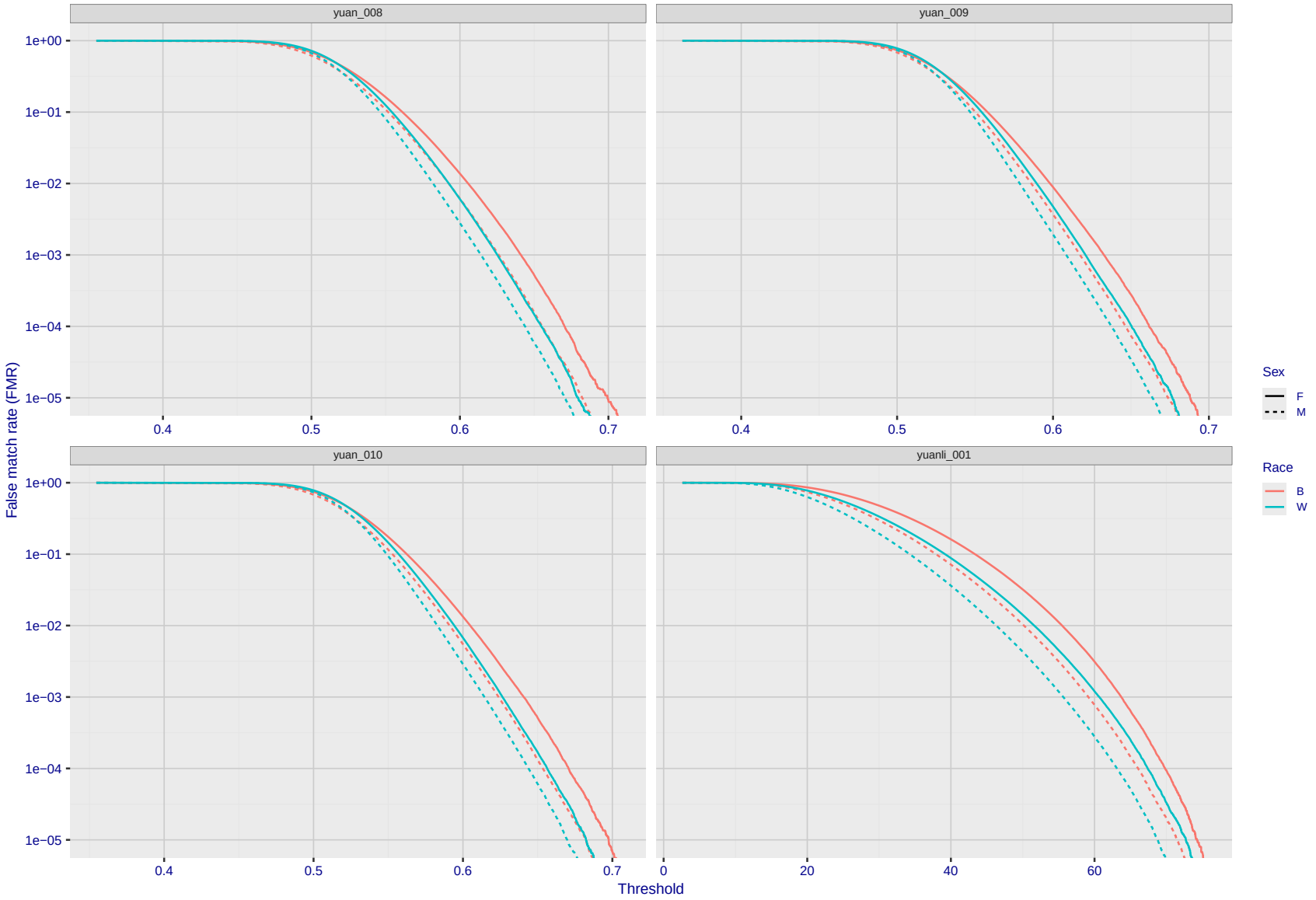

Figure 629: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

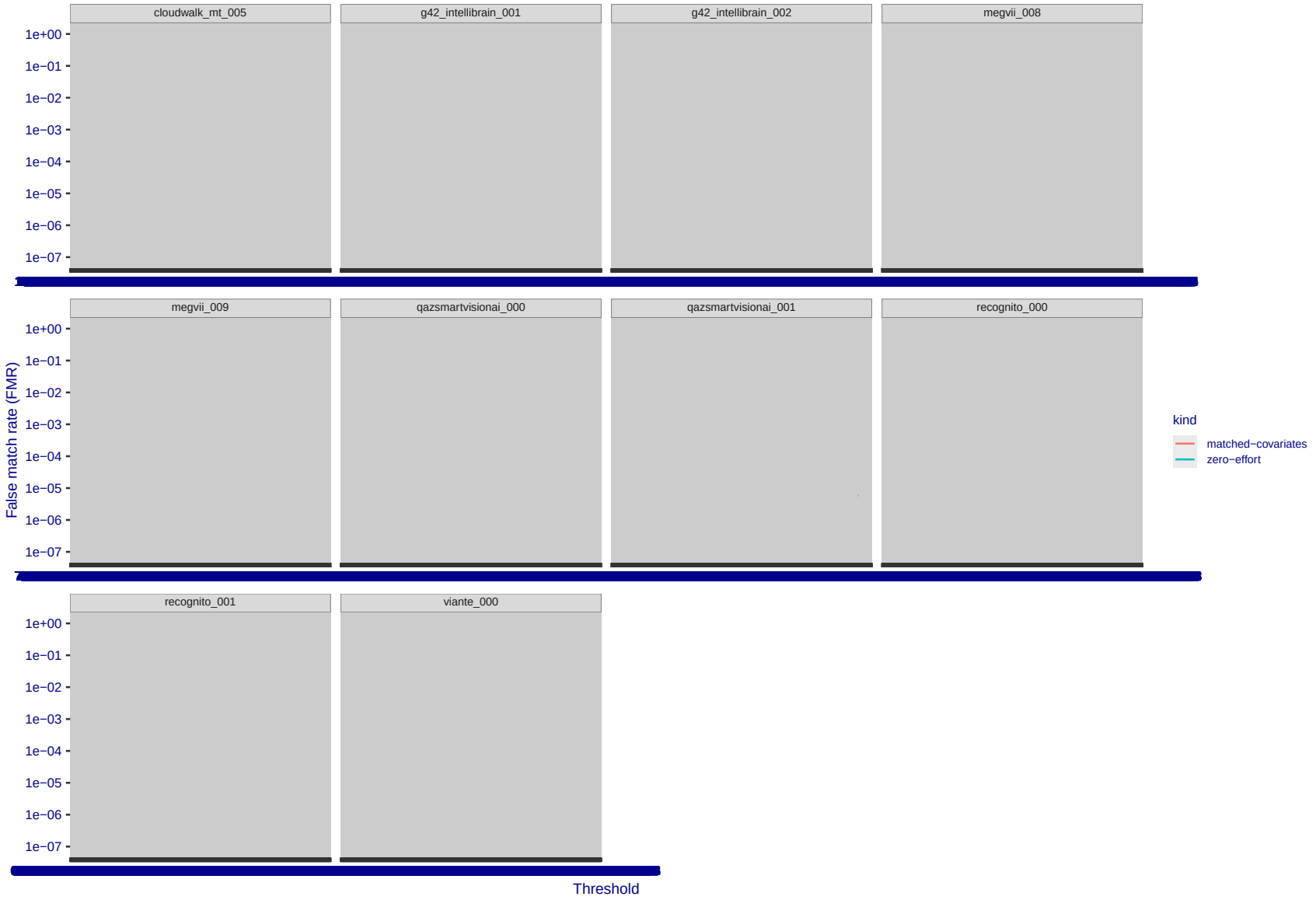

Figure 630: For the visa images, the false match calibration curves show FMR vs. threshold, T. The blue (lower) curves are for zero-effort impostors (i.e. comparing all images against all). The red (upper) curves are for persons of the same-sex, same-age, and same national-origin. This shows that FMR is underestimated (by a factor of 10 or more) by using a zero-effort impostor calculation to calibrate T. As shown later (sec. 3.6), FMR is higher for demographic-matched impostors.

3.5 Genuine distribution stability

3.5.1 Effect of birth place on the genuine distribution

Background: Both skin tone and bone structure vary geographically. Prior studies have reported variations in FNMR and FMR.

Goal: To measure false non-match rate (FNMR) variation with country of birth.

Methods: Thresholds are determined that give FMR = $\{0.001, 0.0001\}$ over the entire impostor set. Then FNMR is measured over 1000 bootstrap replications of the genuine scores. Only those countries with at least 140 individuals are included in the analysis.

Results: Figure 720 shows FNMR by country of birth for the two thresholds.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

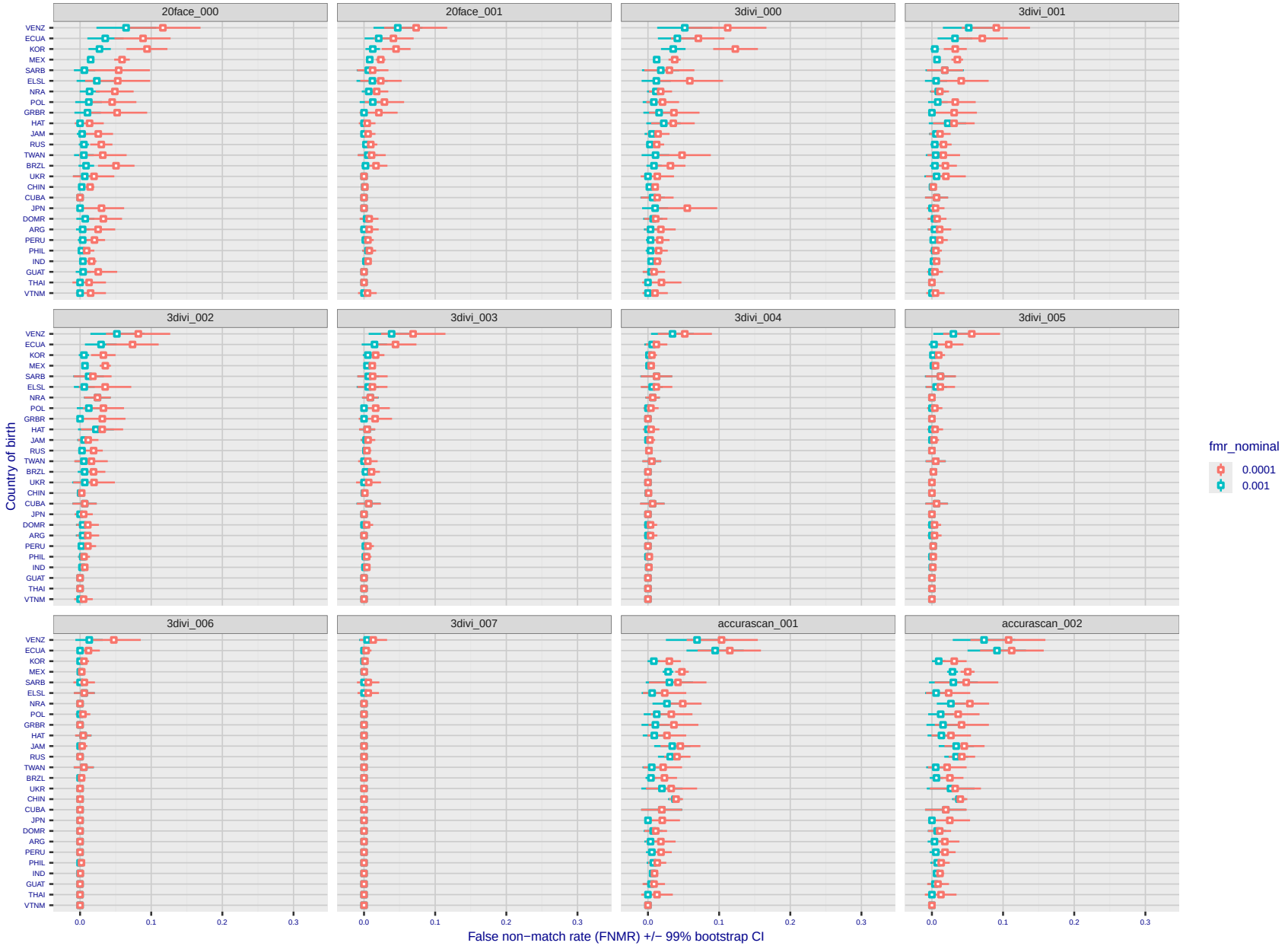

Figure 631: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 632: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

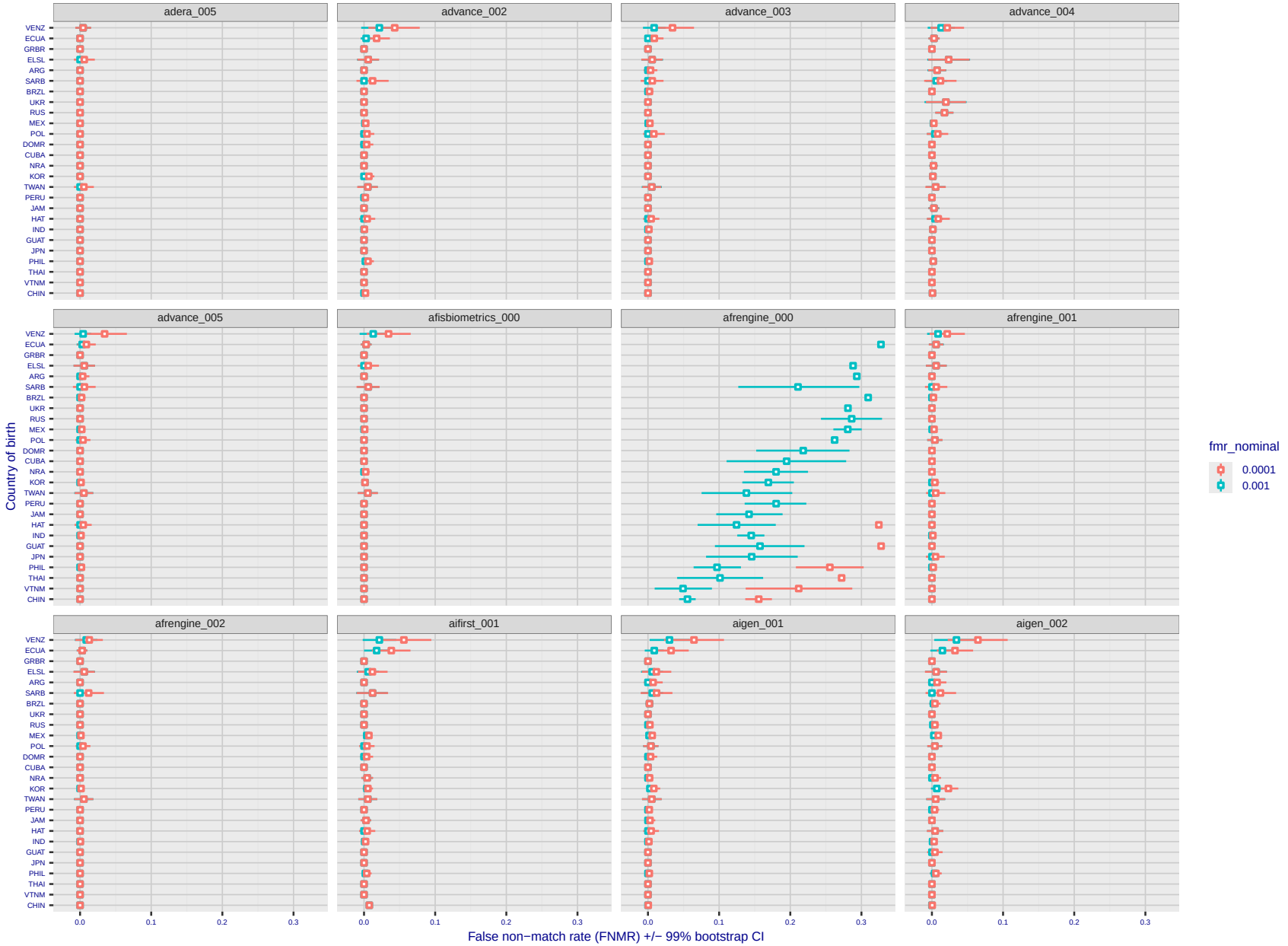

Figure 633: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

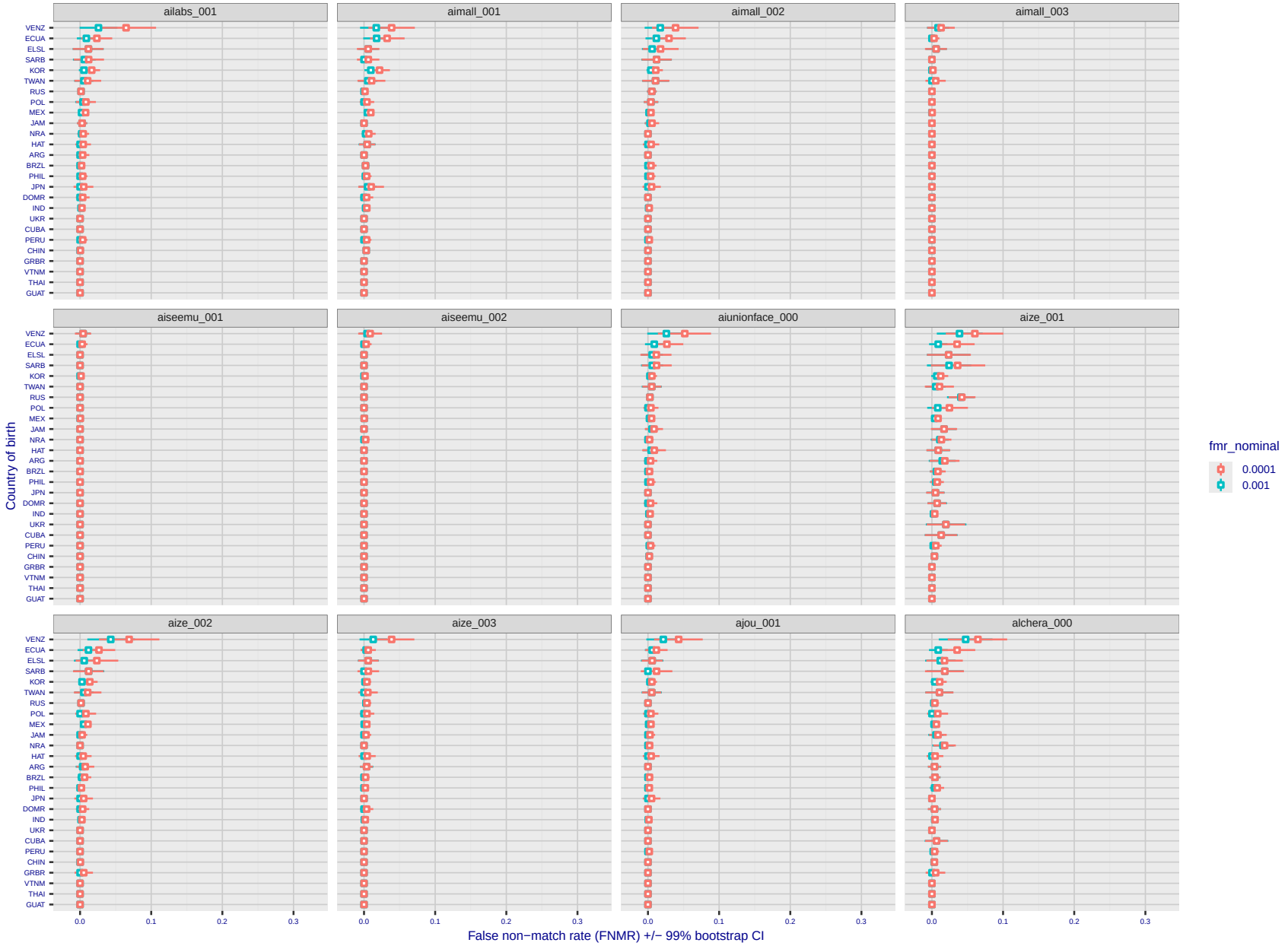

Figure 634: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 635: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

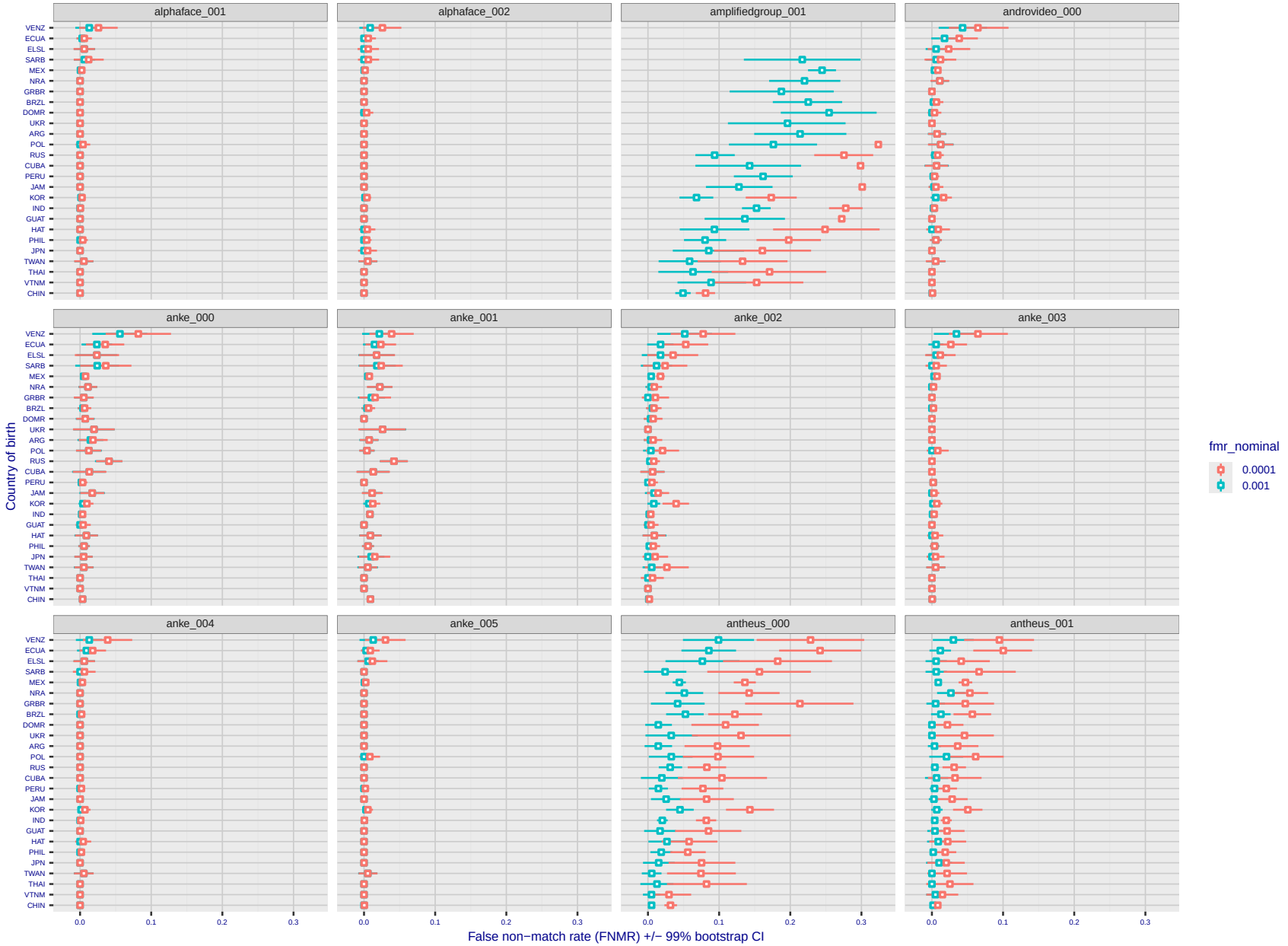

Figure 636: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 637: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 638: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

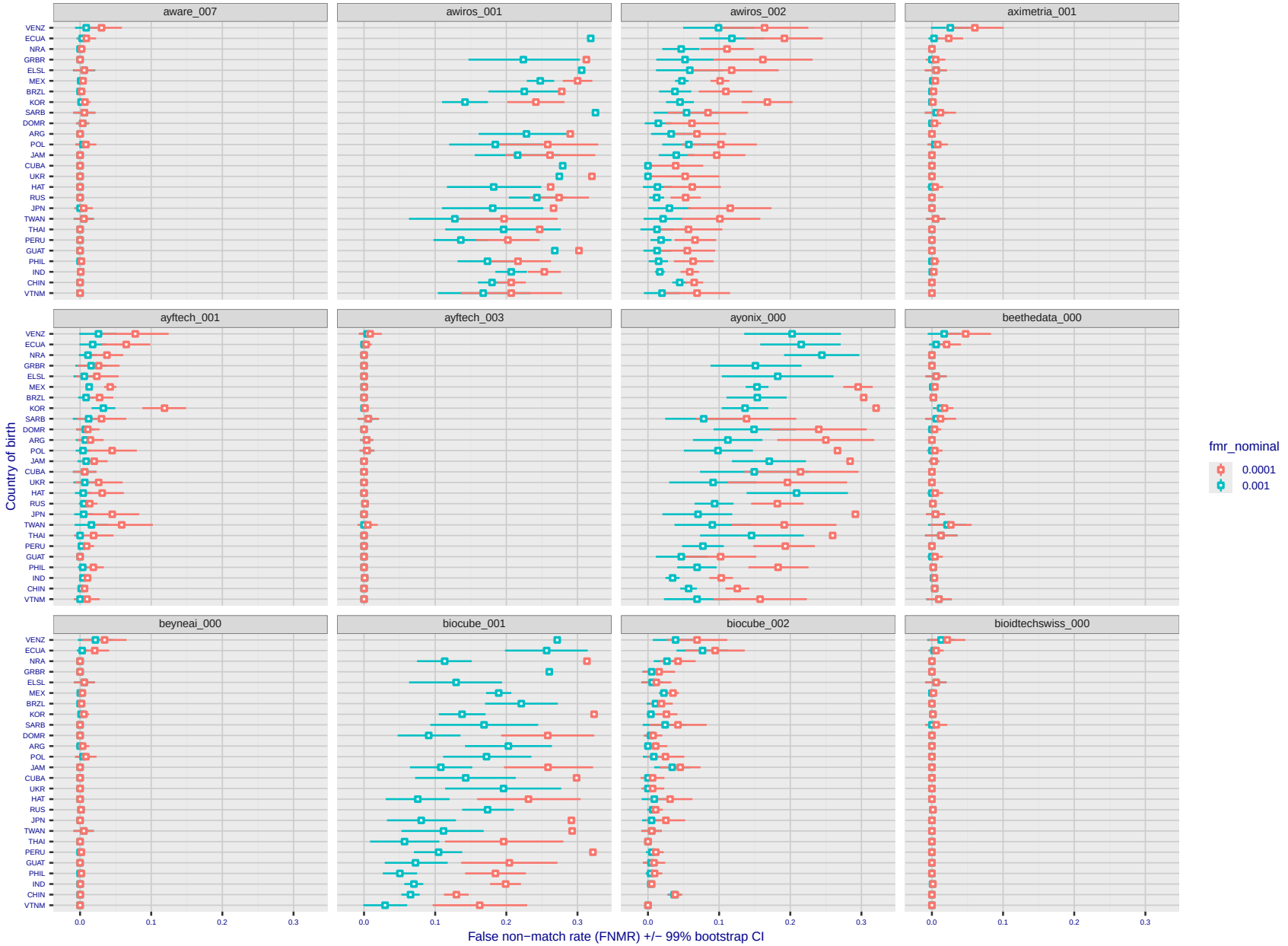

Figure 639: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 640: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 641: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

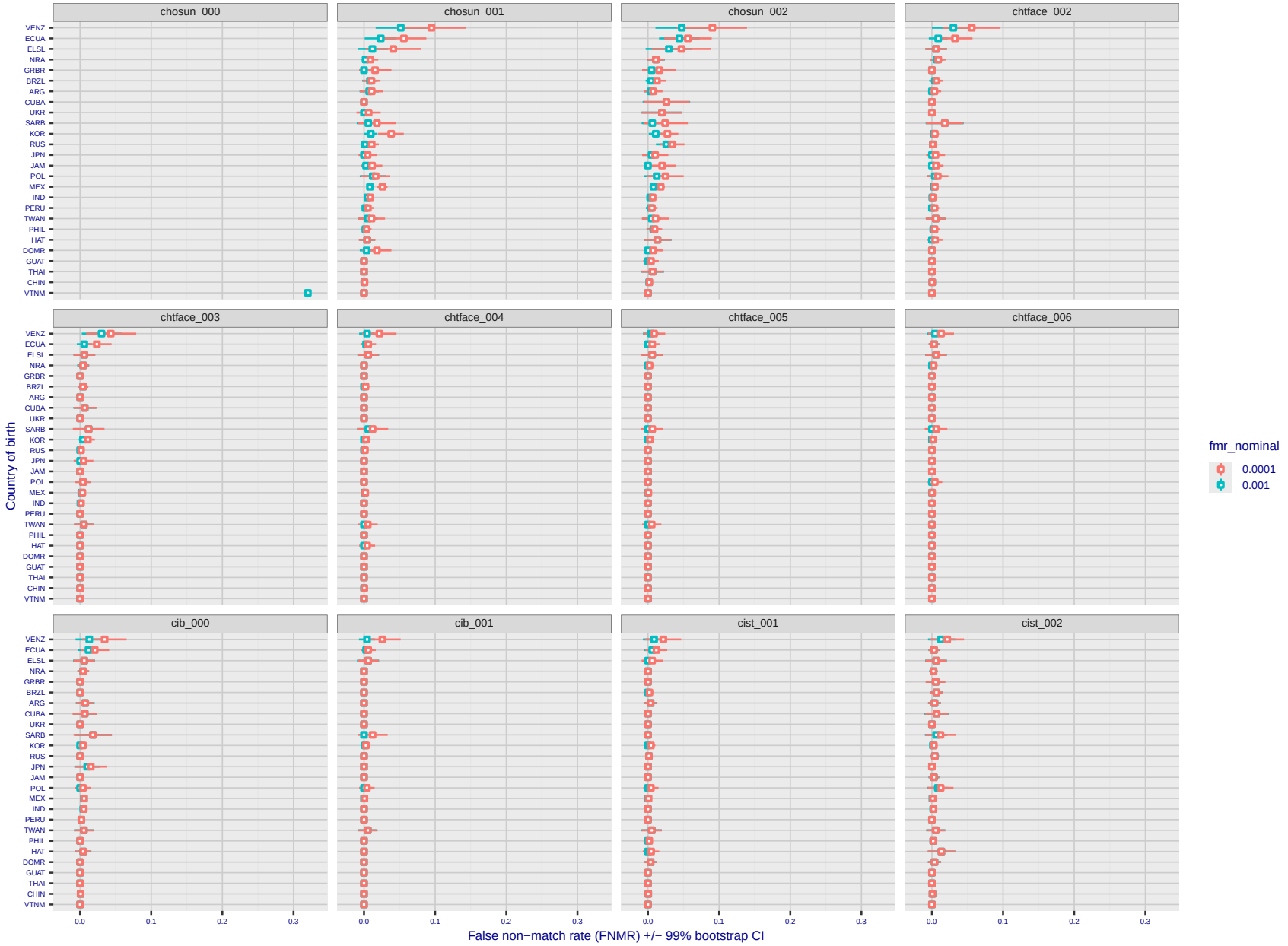

Figure 642: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

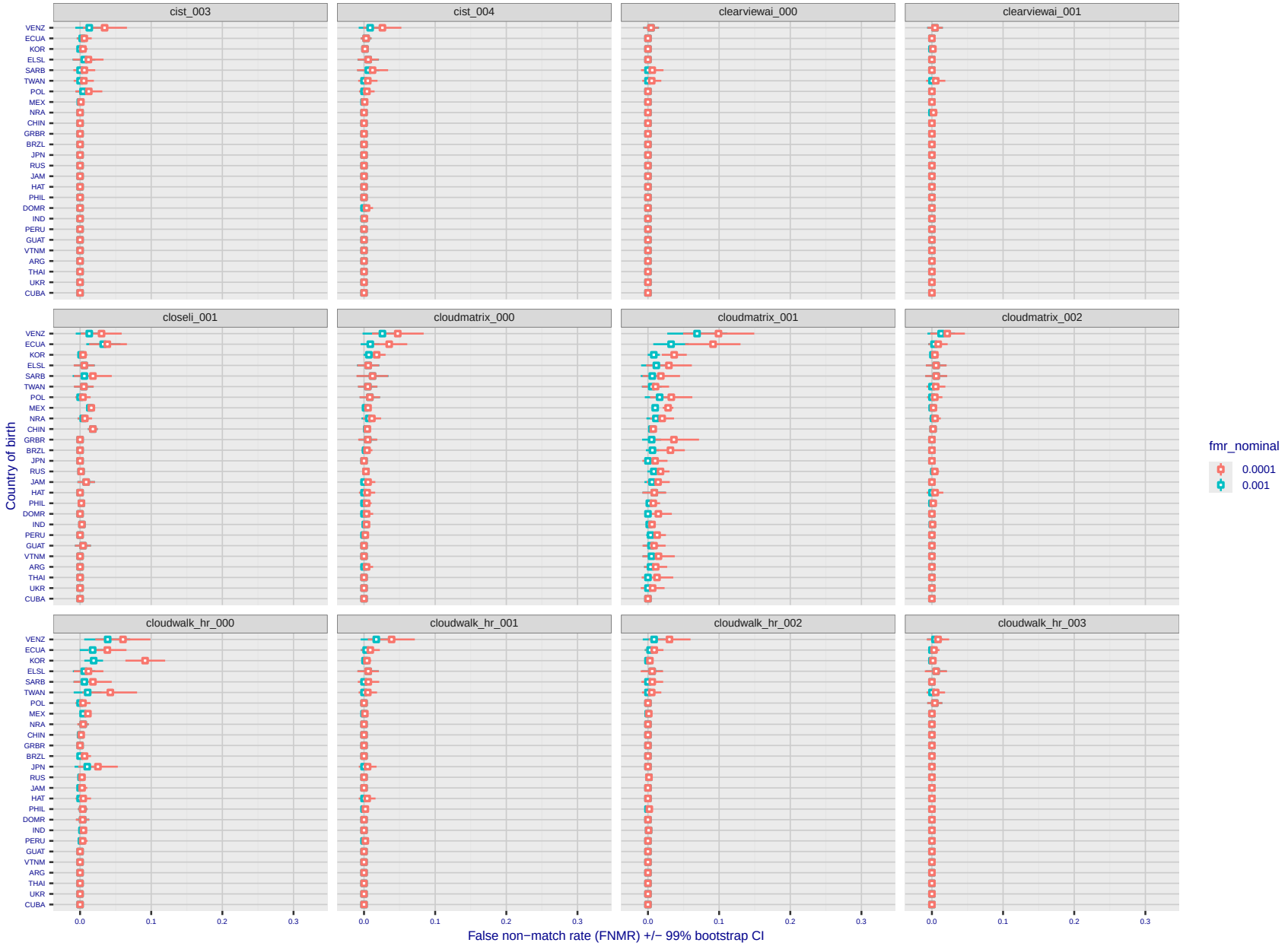

Figure 643: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

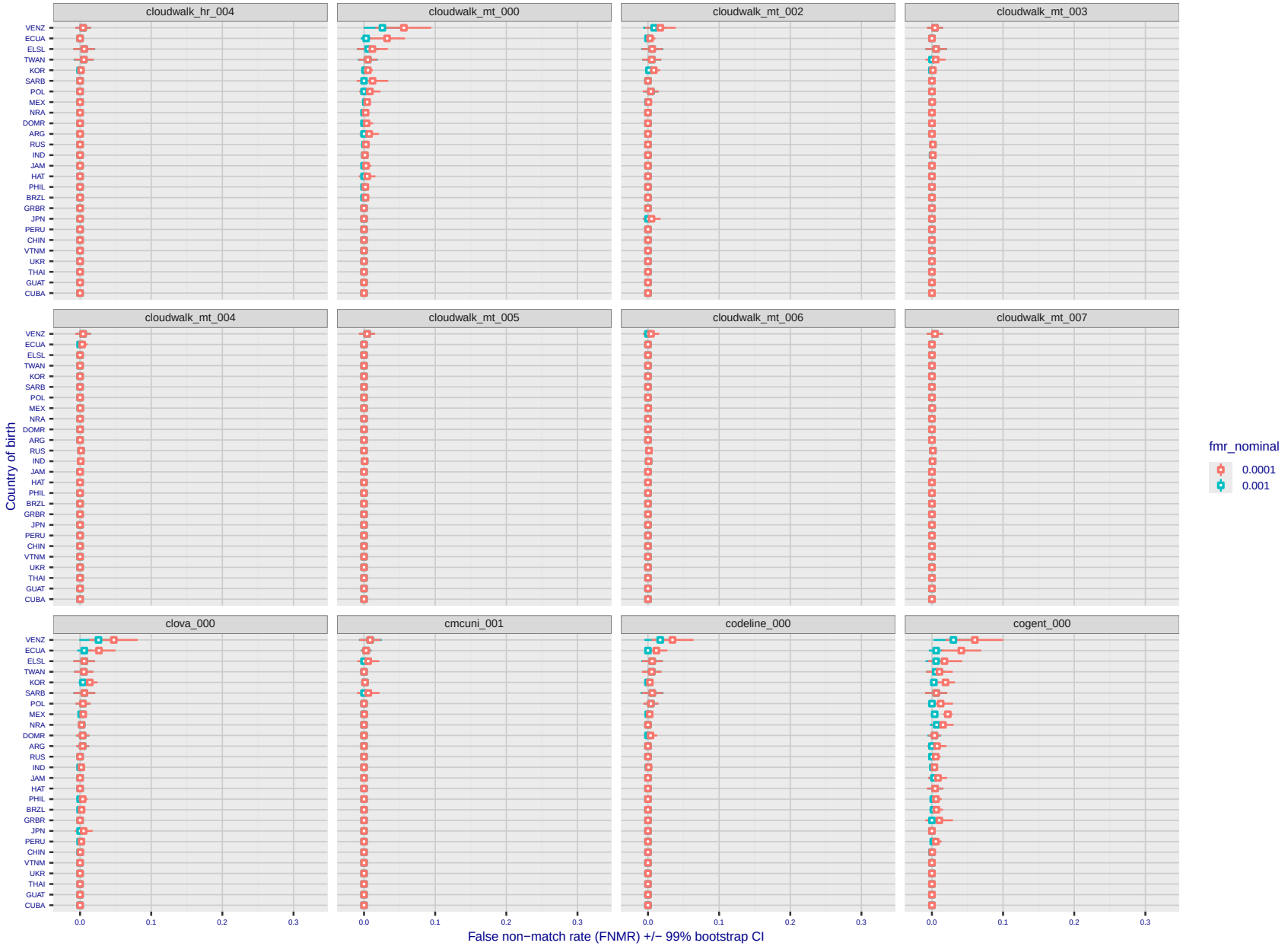

Figure 644: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

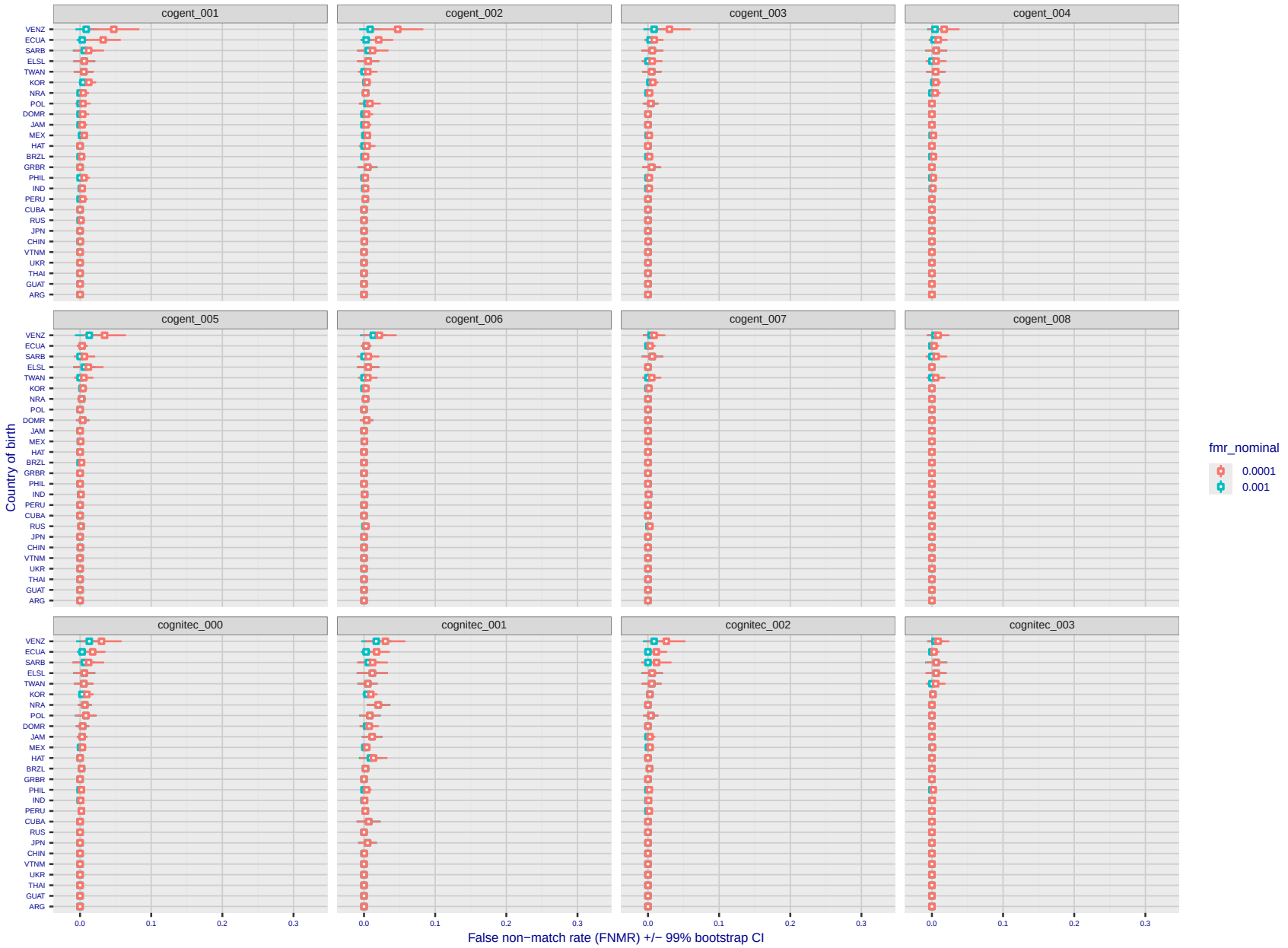

Figure 645: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

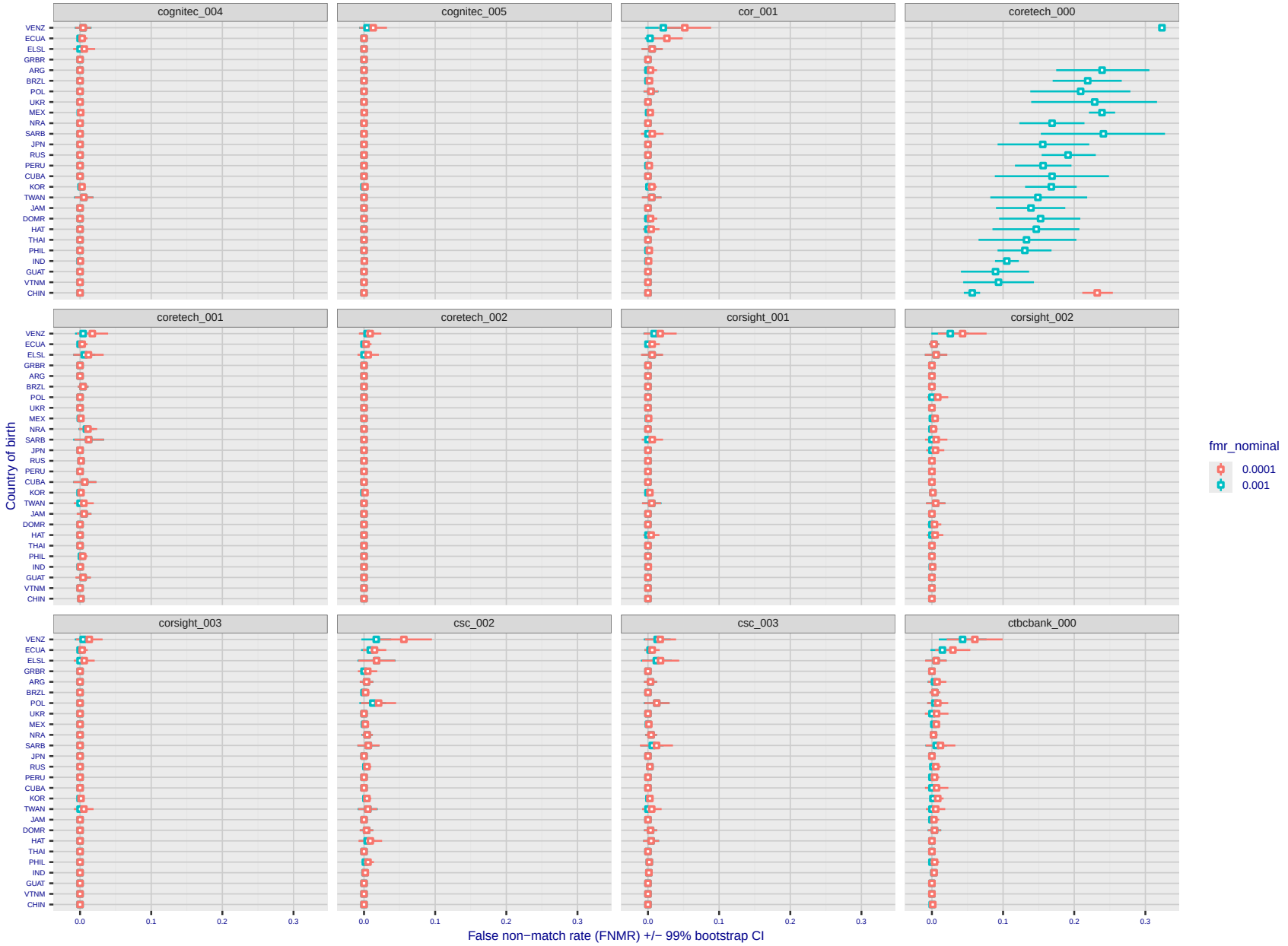

Figure 646: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

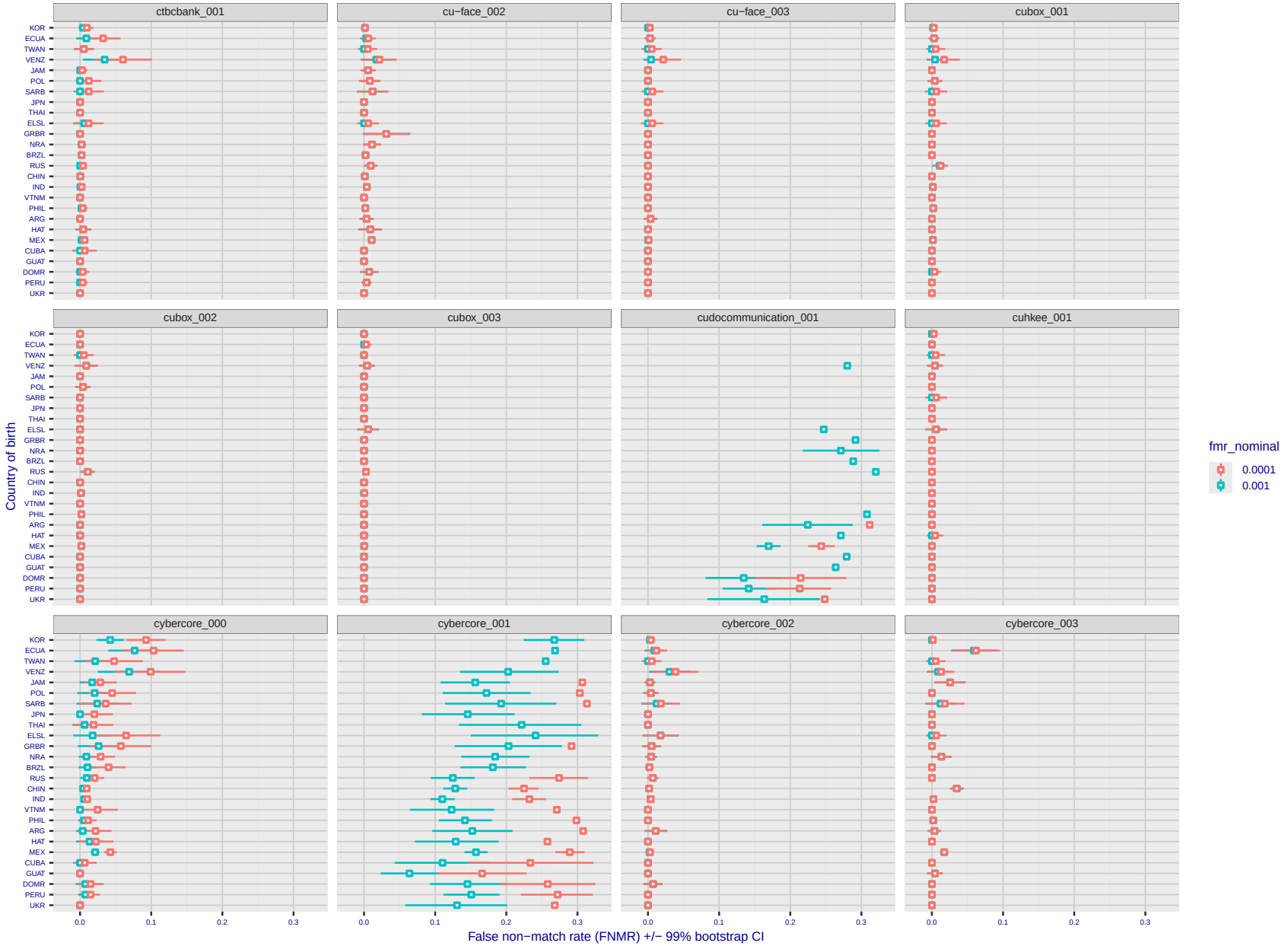

Figure 647: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 648: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

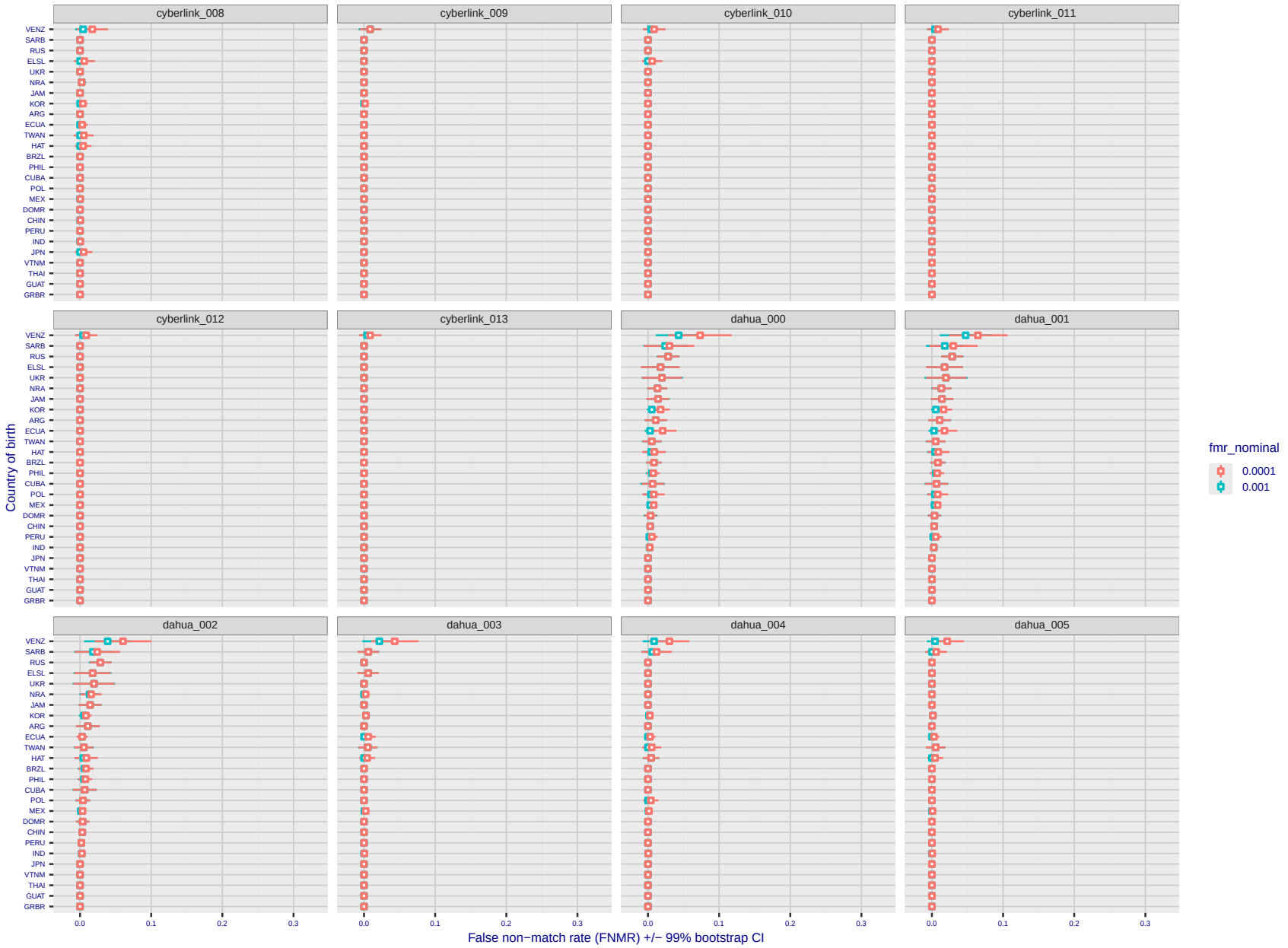

Figure 649: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 650: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

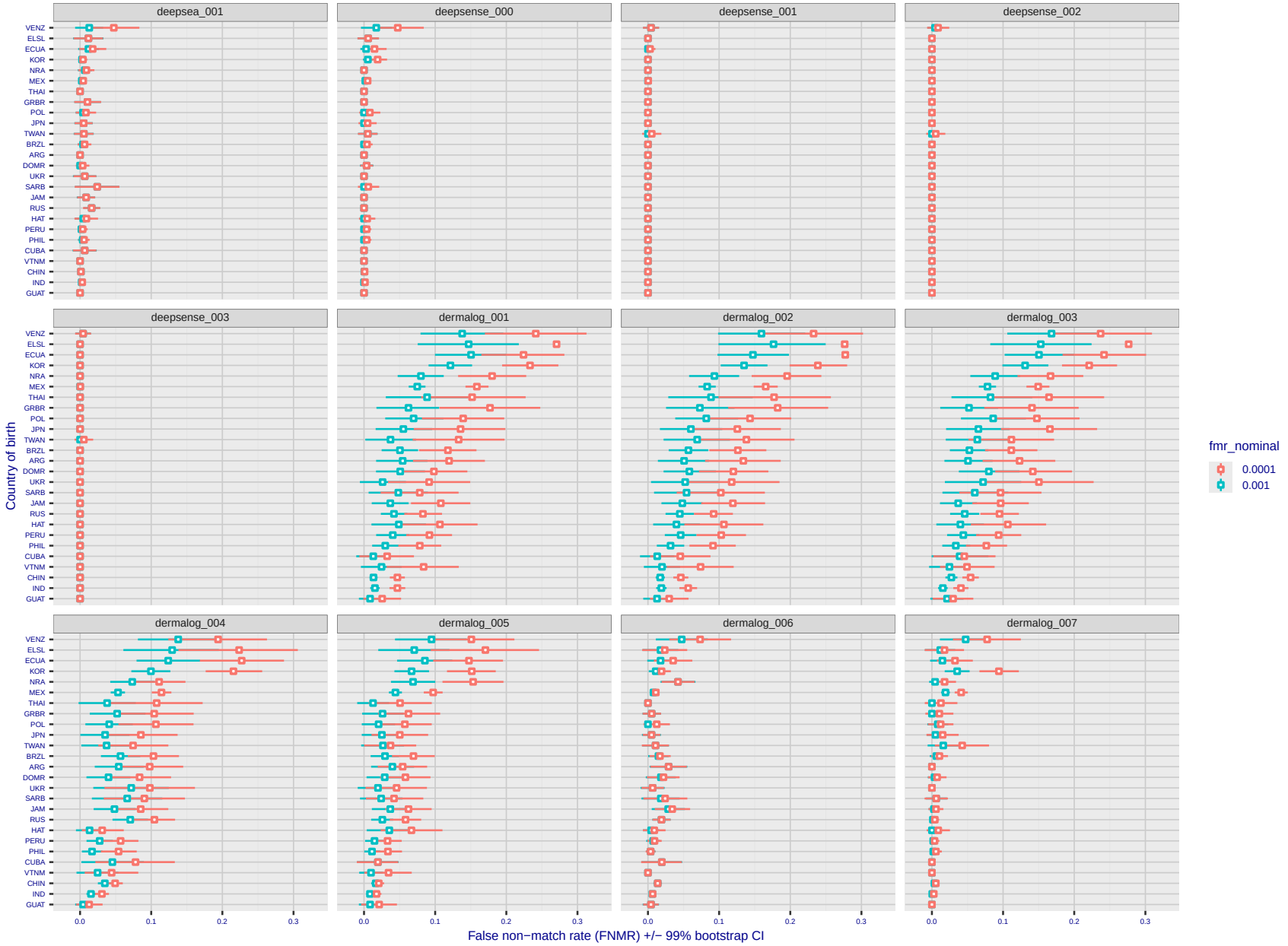

Figure 651: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

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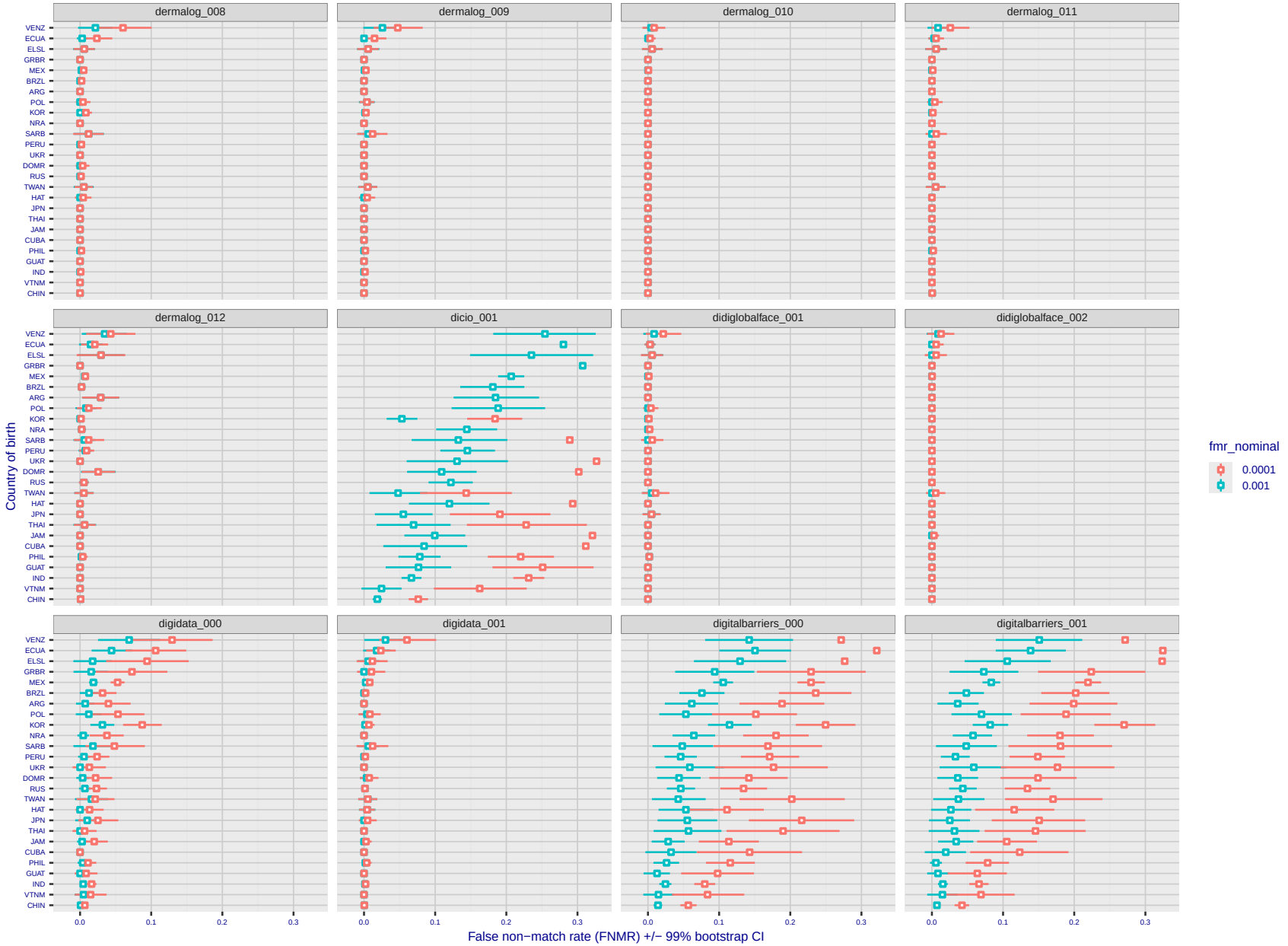
FNMR(T) "False non-match rate"
FMR(T) "False match rate"



FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 652: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 653: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 654: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 655: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

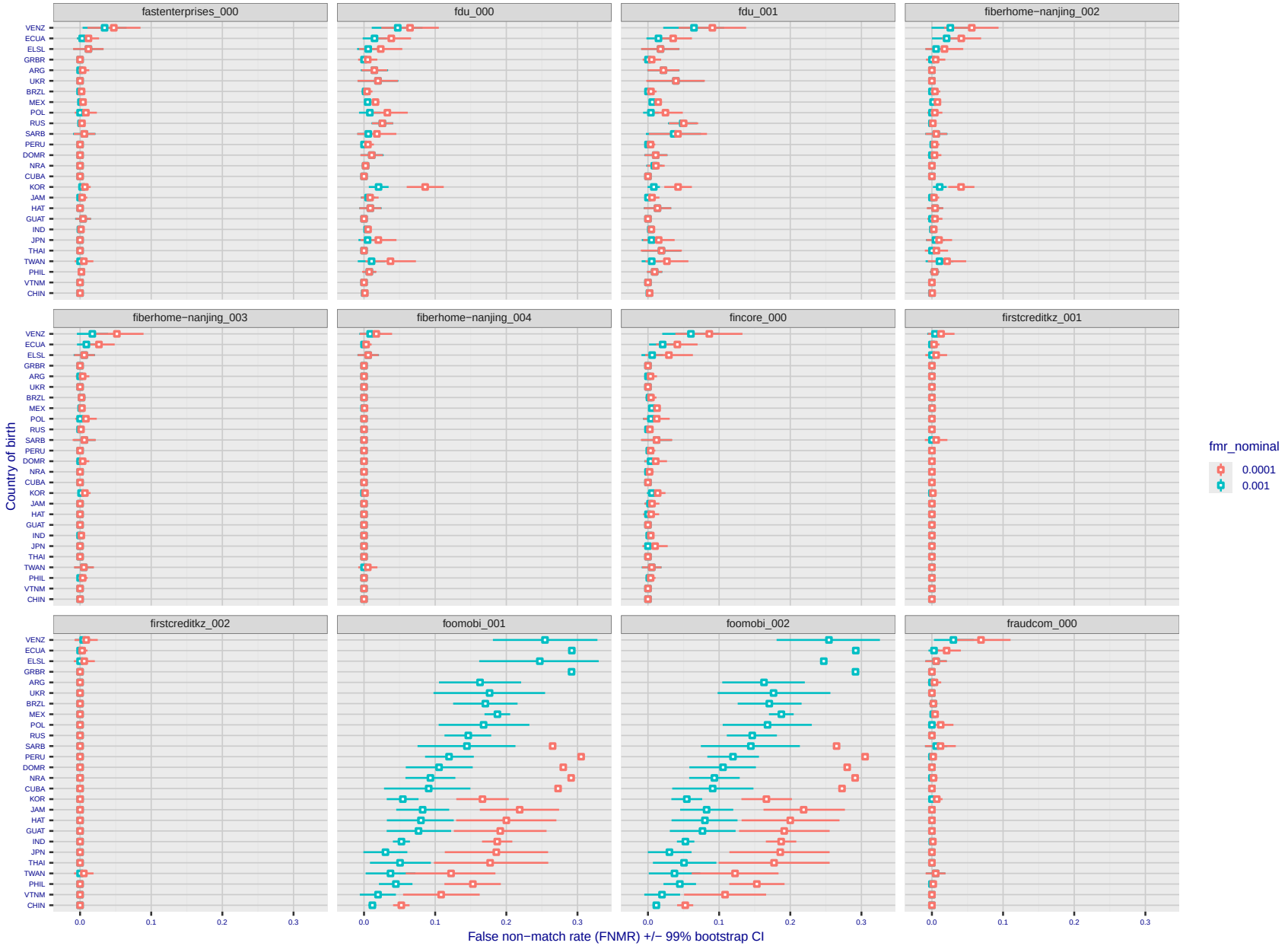


Figure 656: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

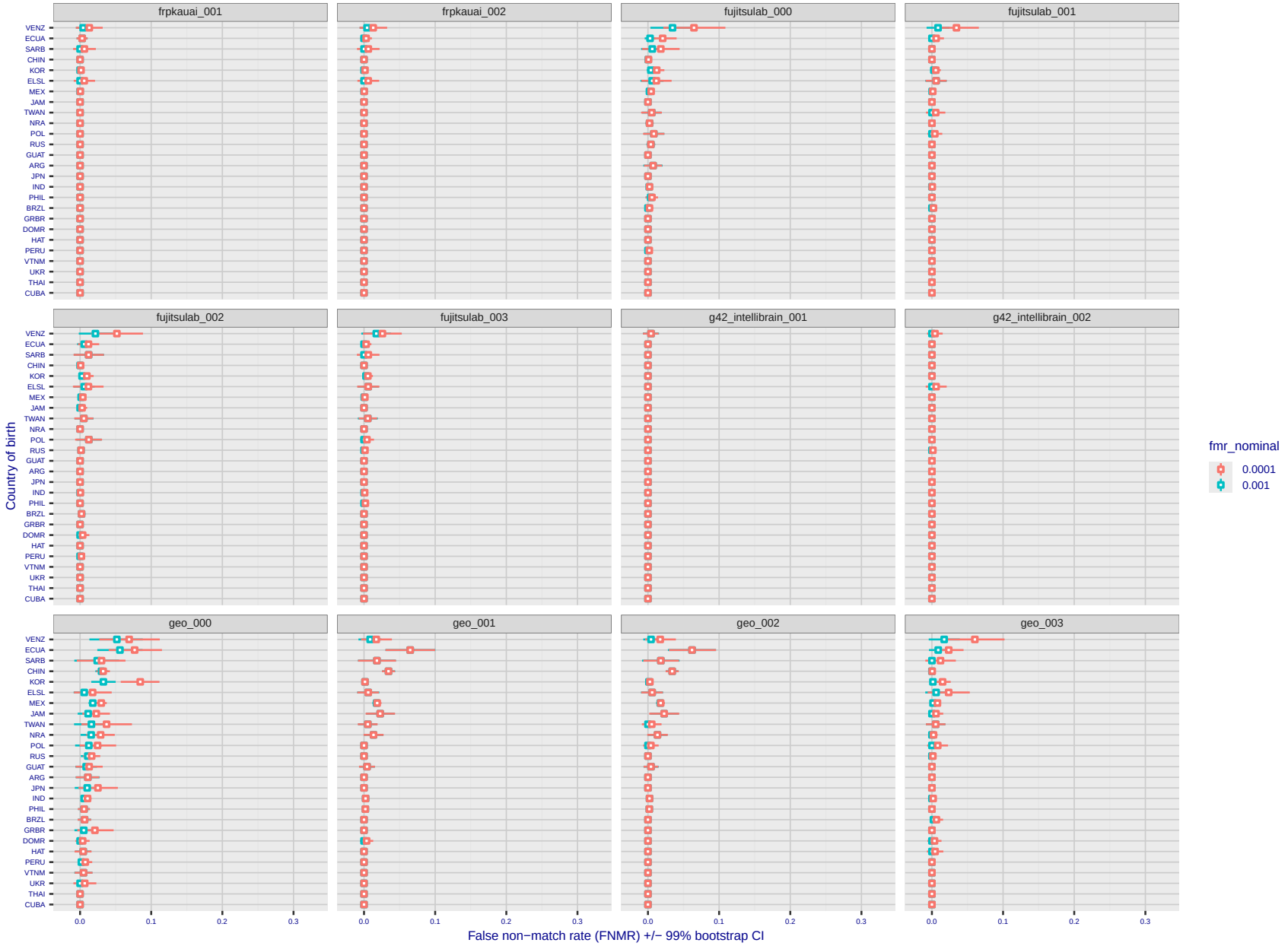


Figure 657: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 658: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

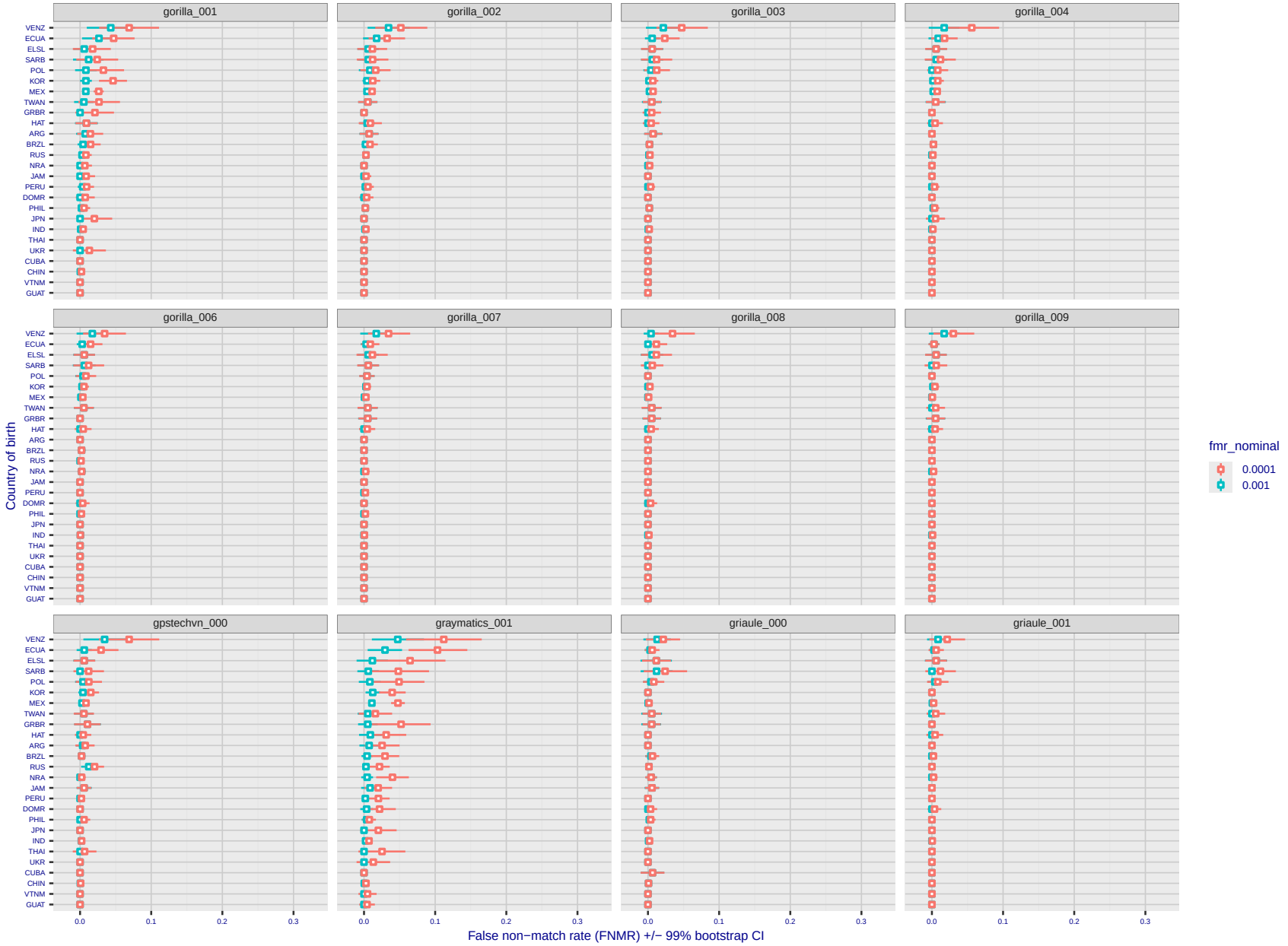


Figure 659: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 660: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 661: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

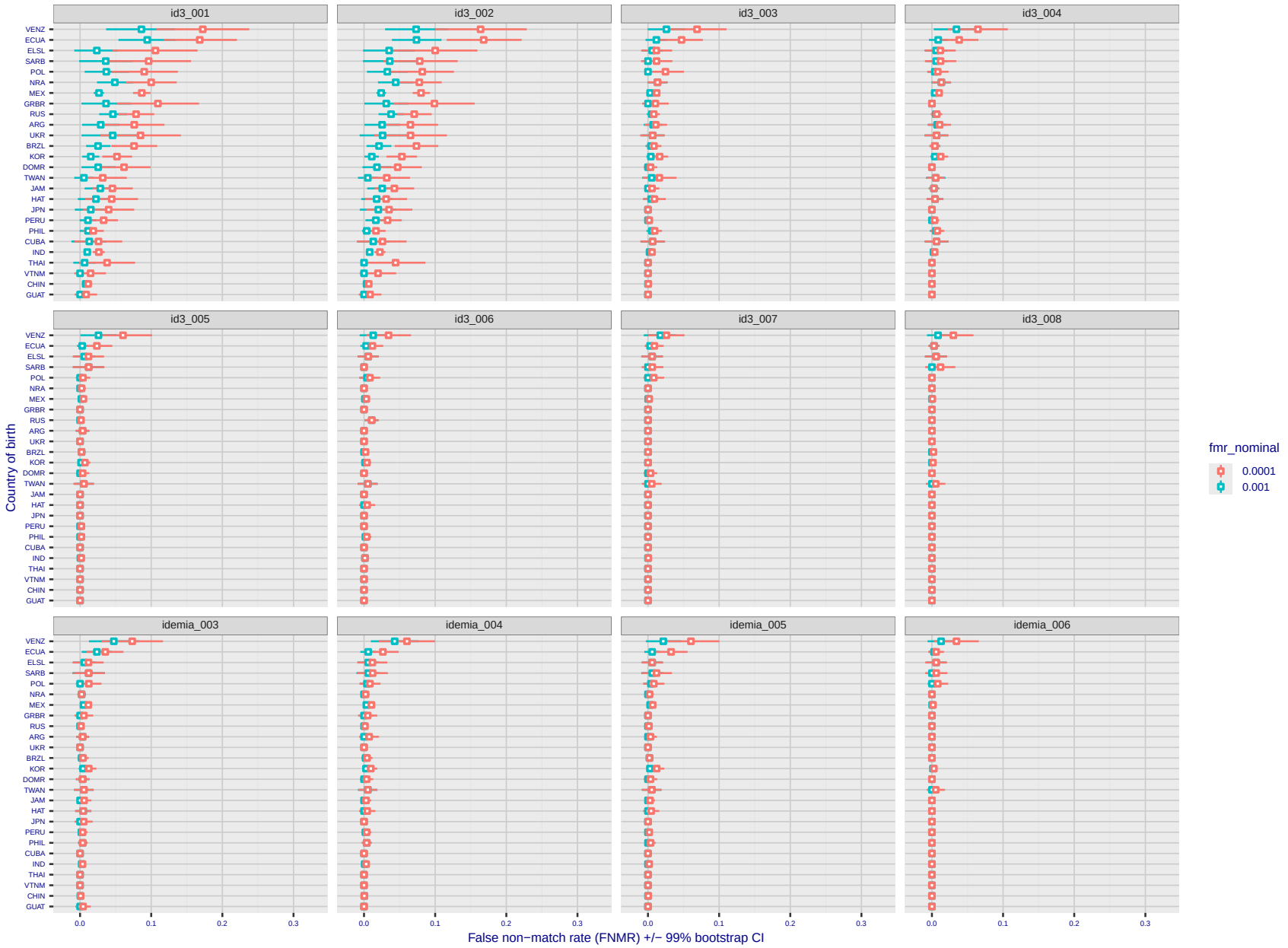


Figure 662: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

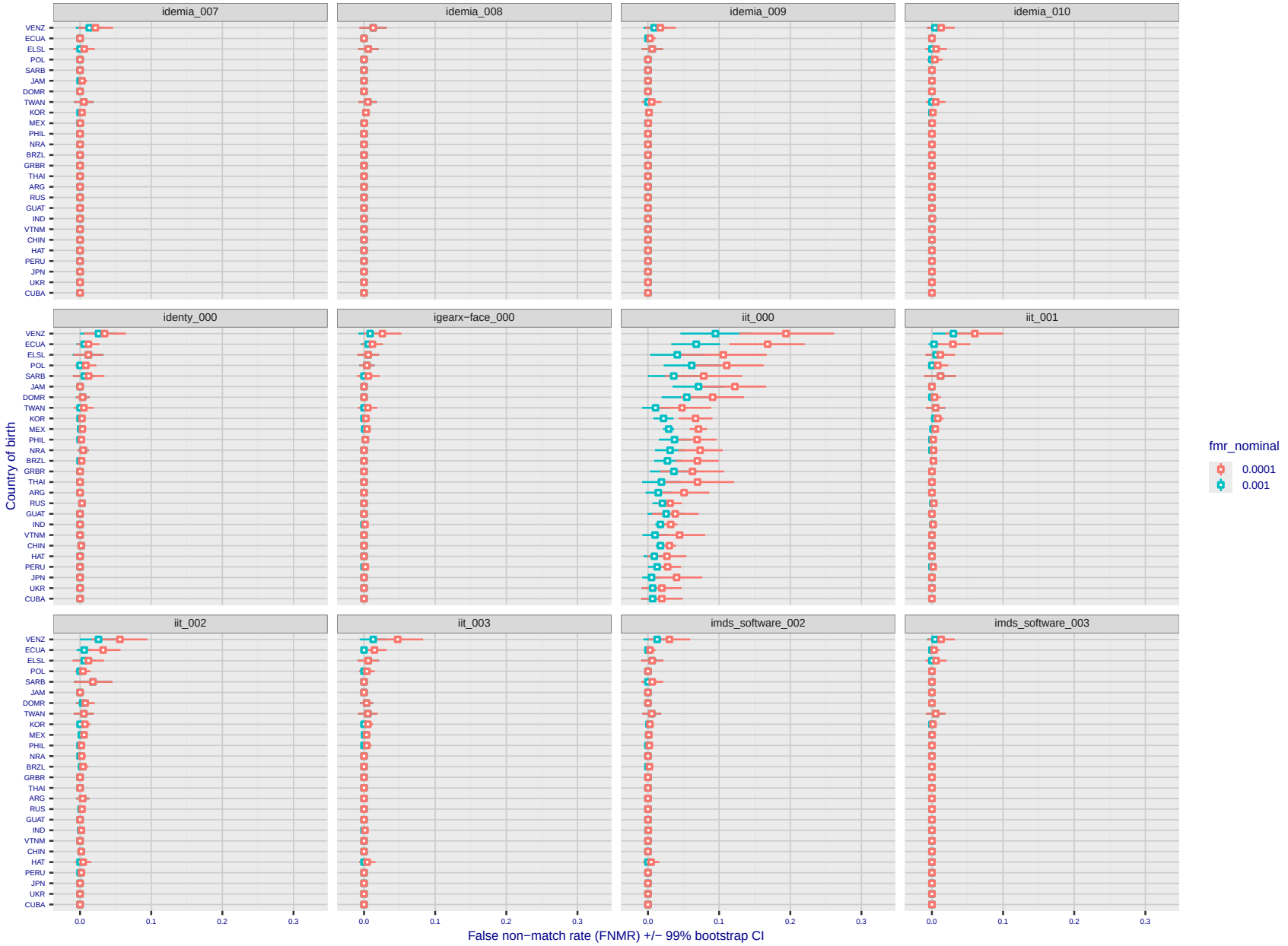


Figure 663: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

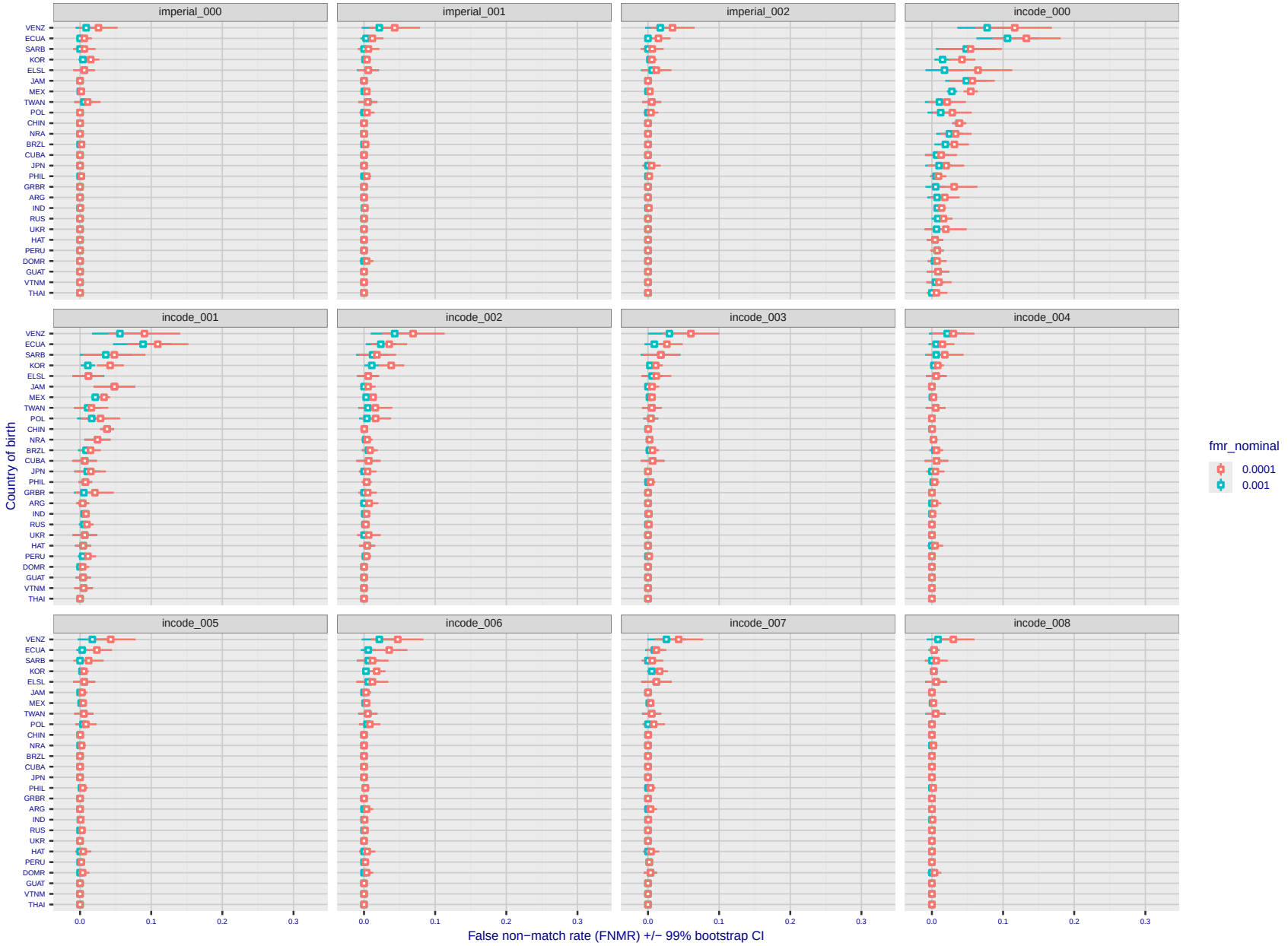


Figure 664: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

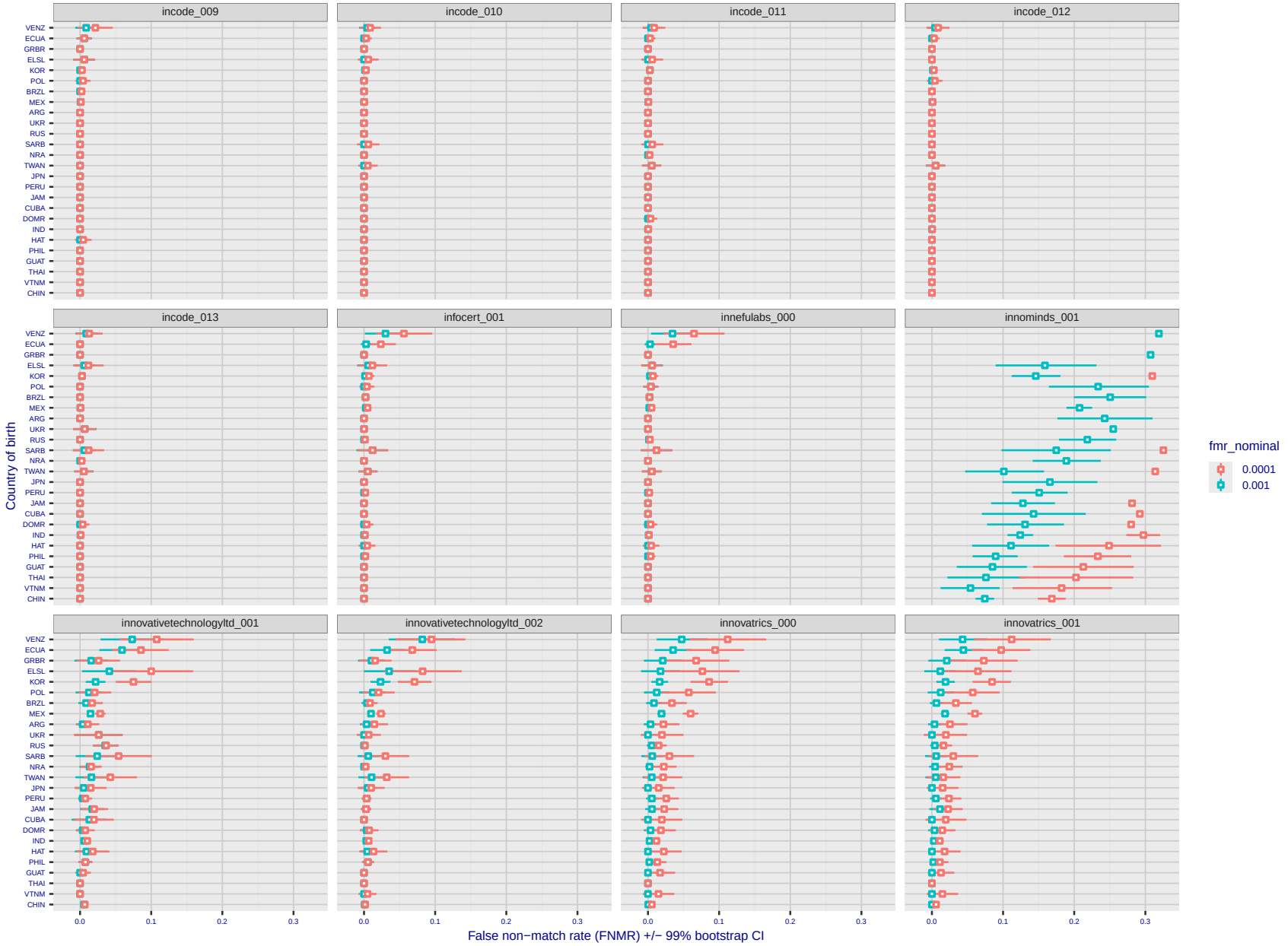


Figure 665: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

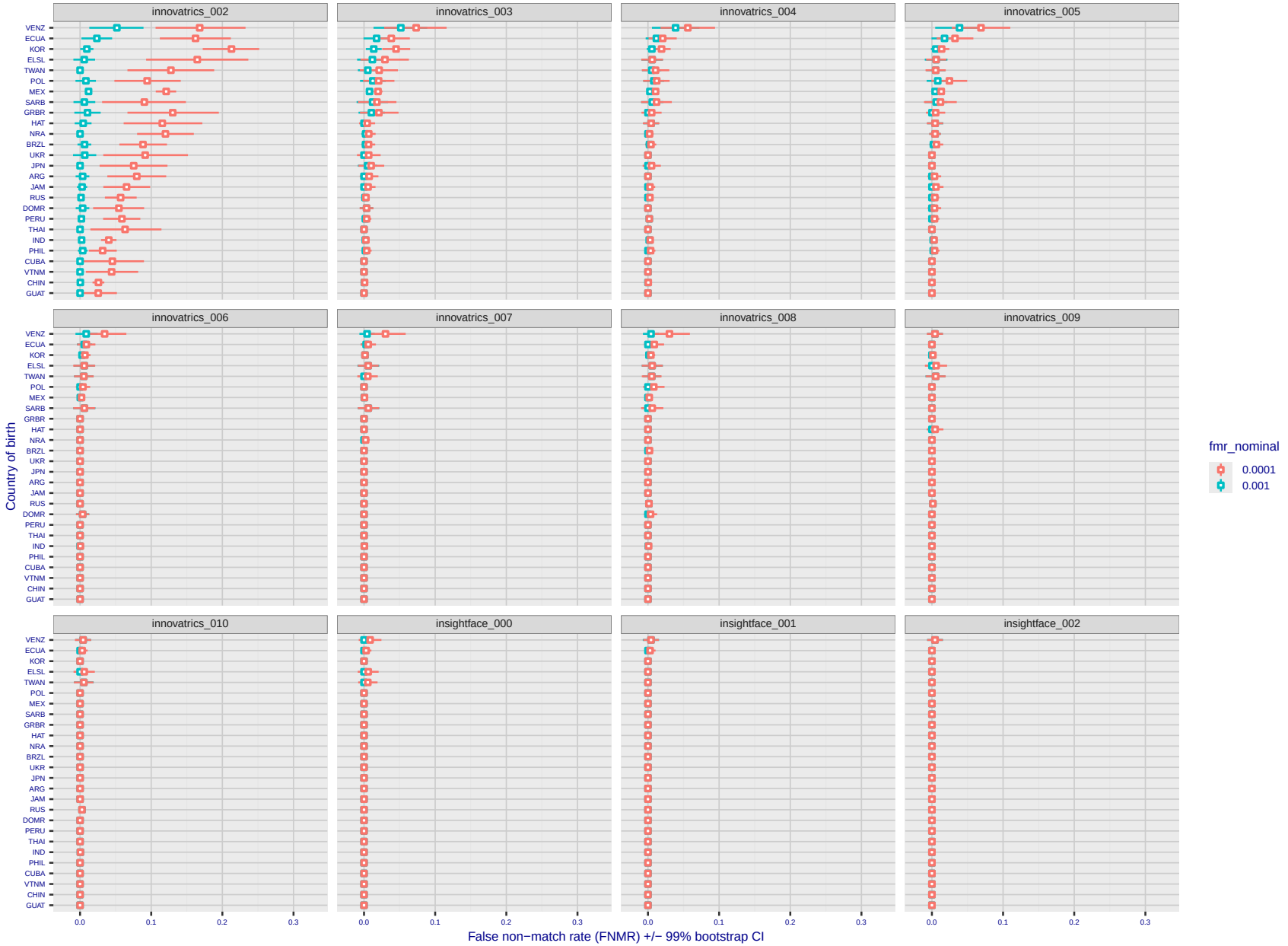


Figure 666: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

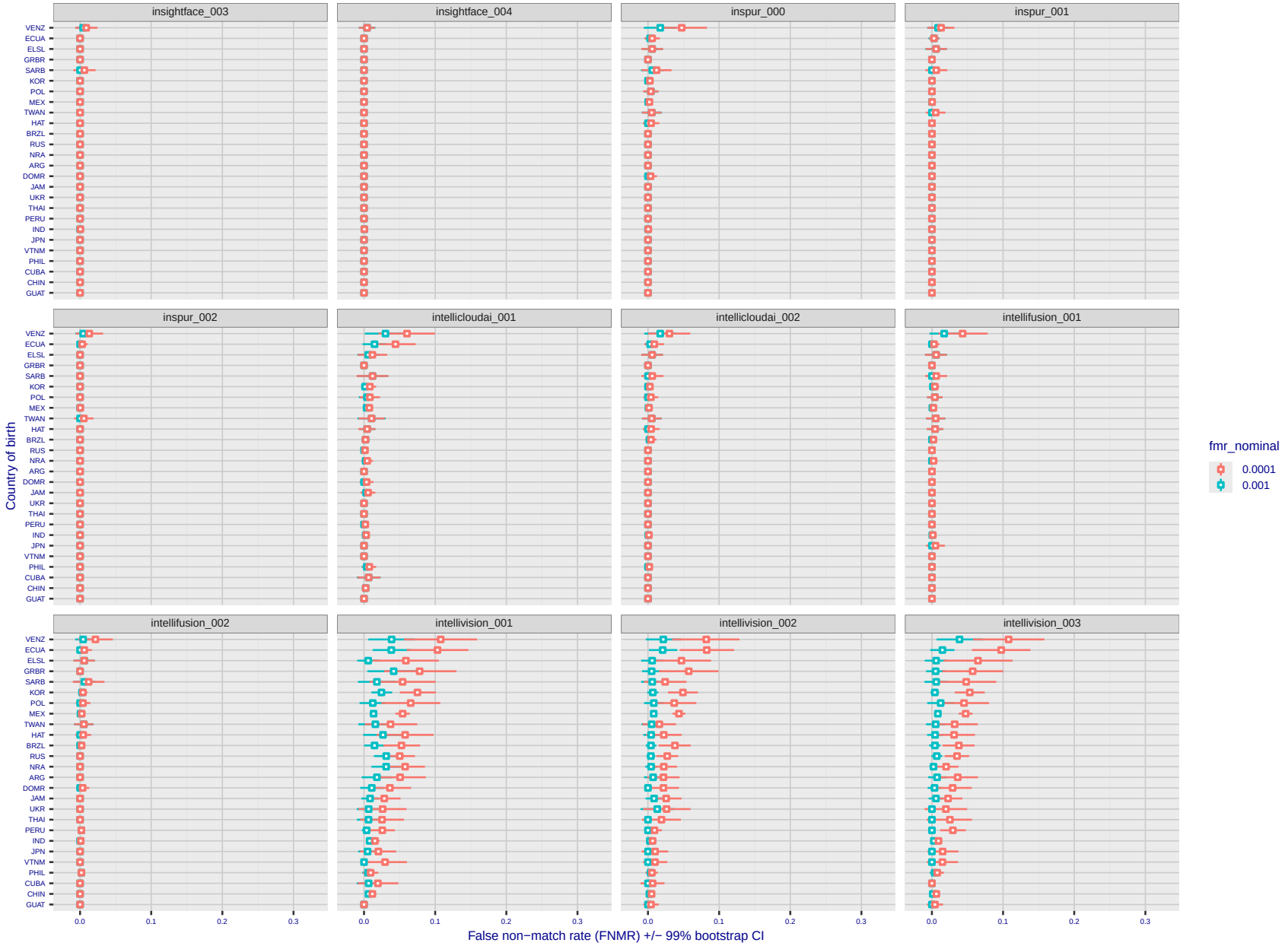


Figure 667: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

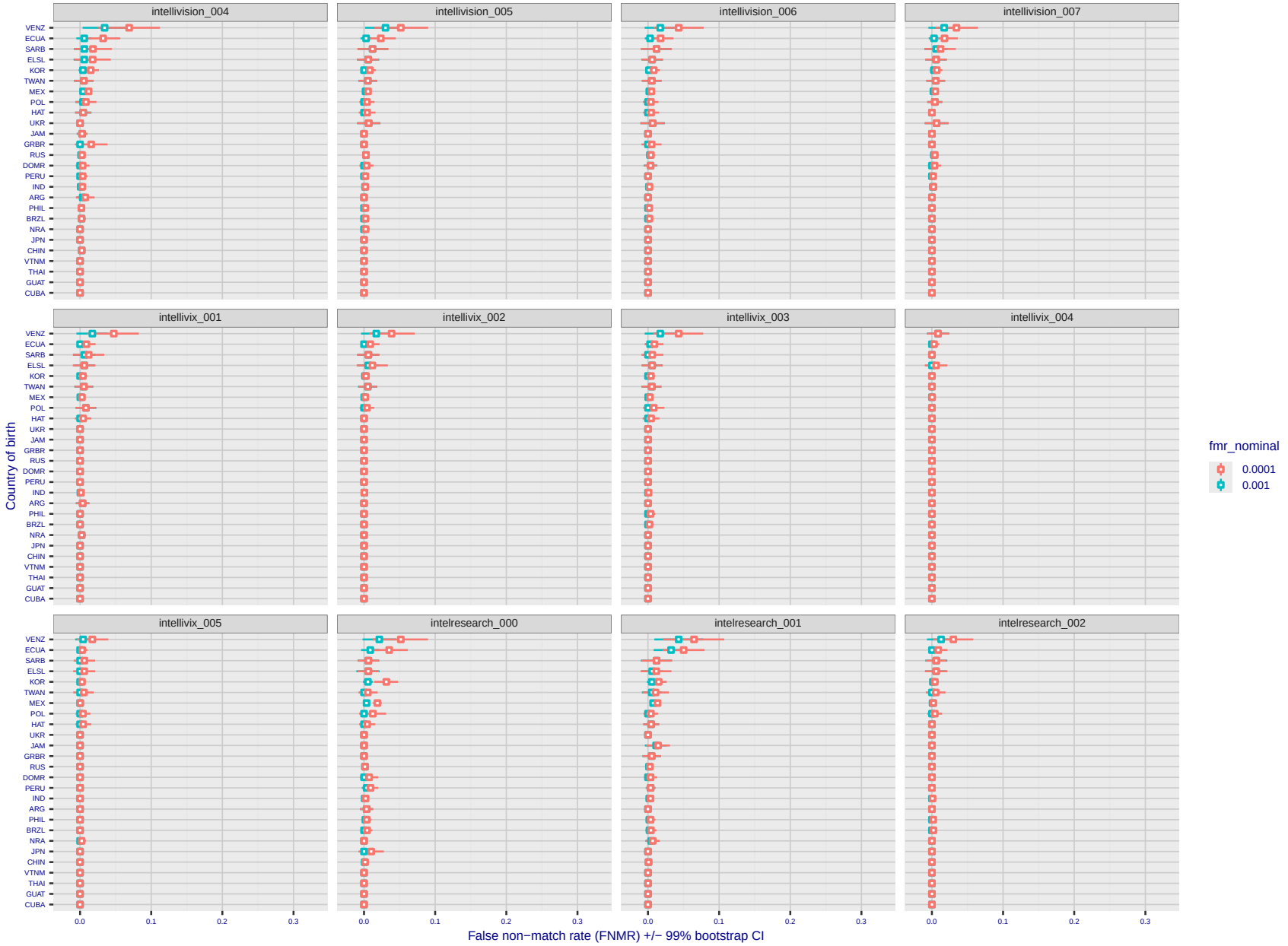


Figure 668: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

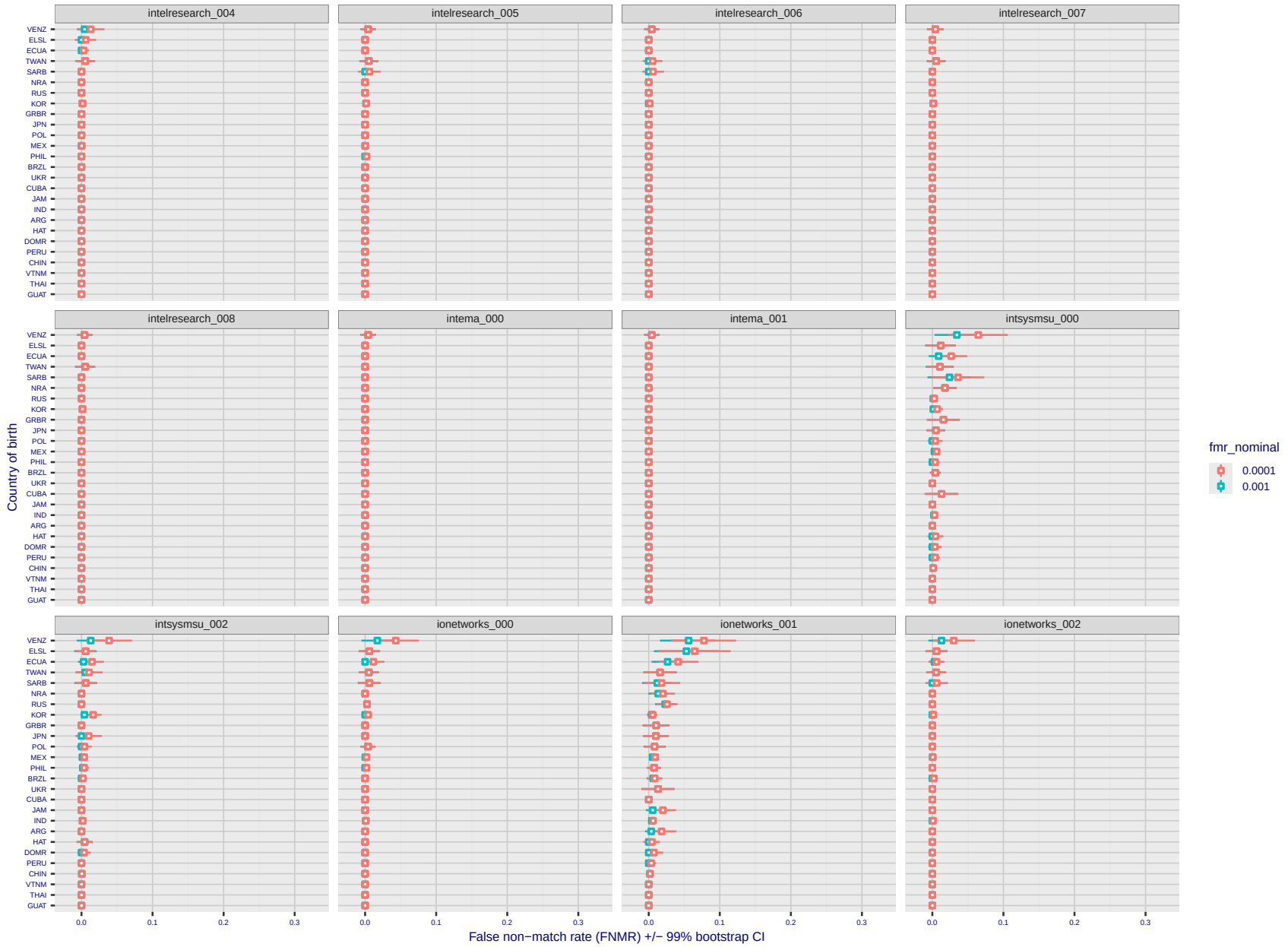


Figure 669: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

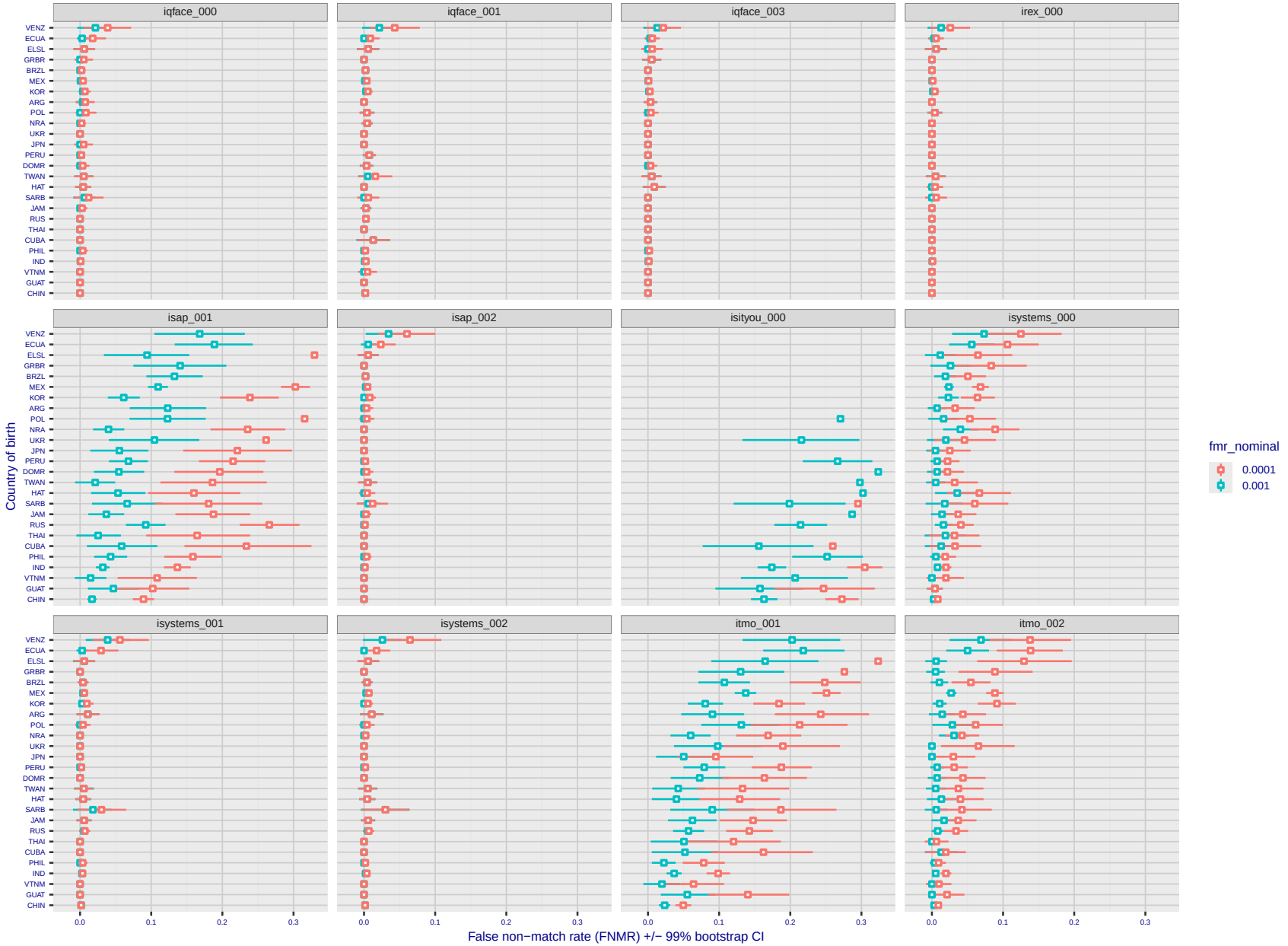


Figure 670: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

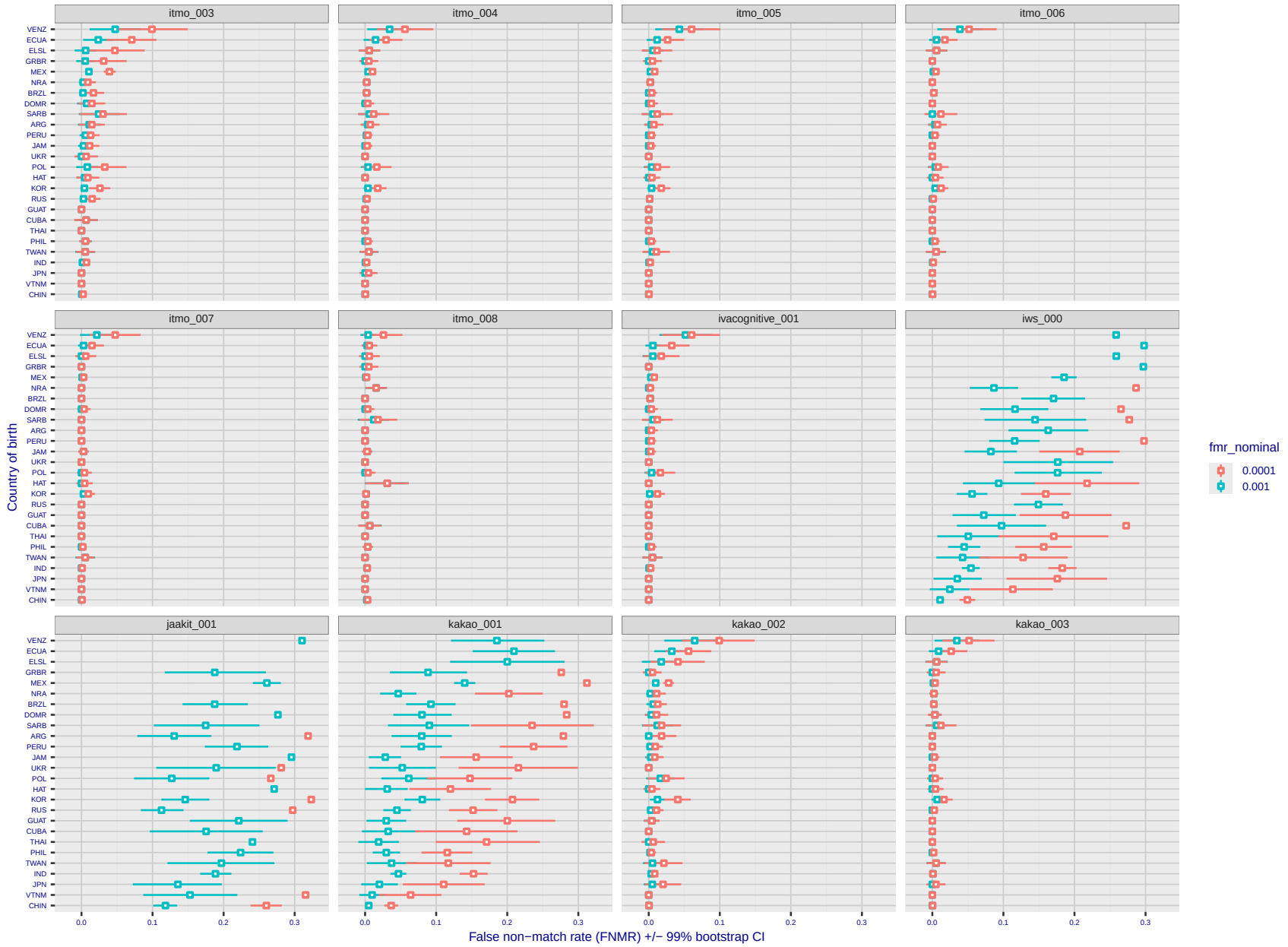

Figure 671: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

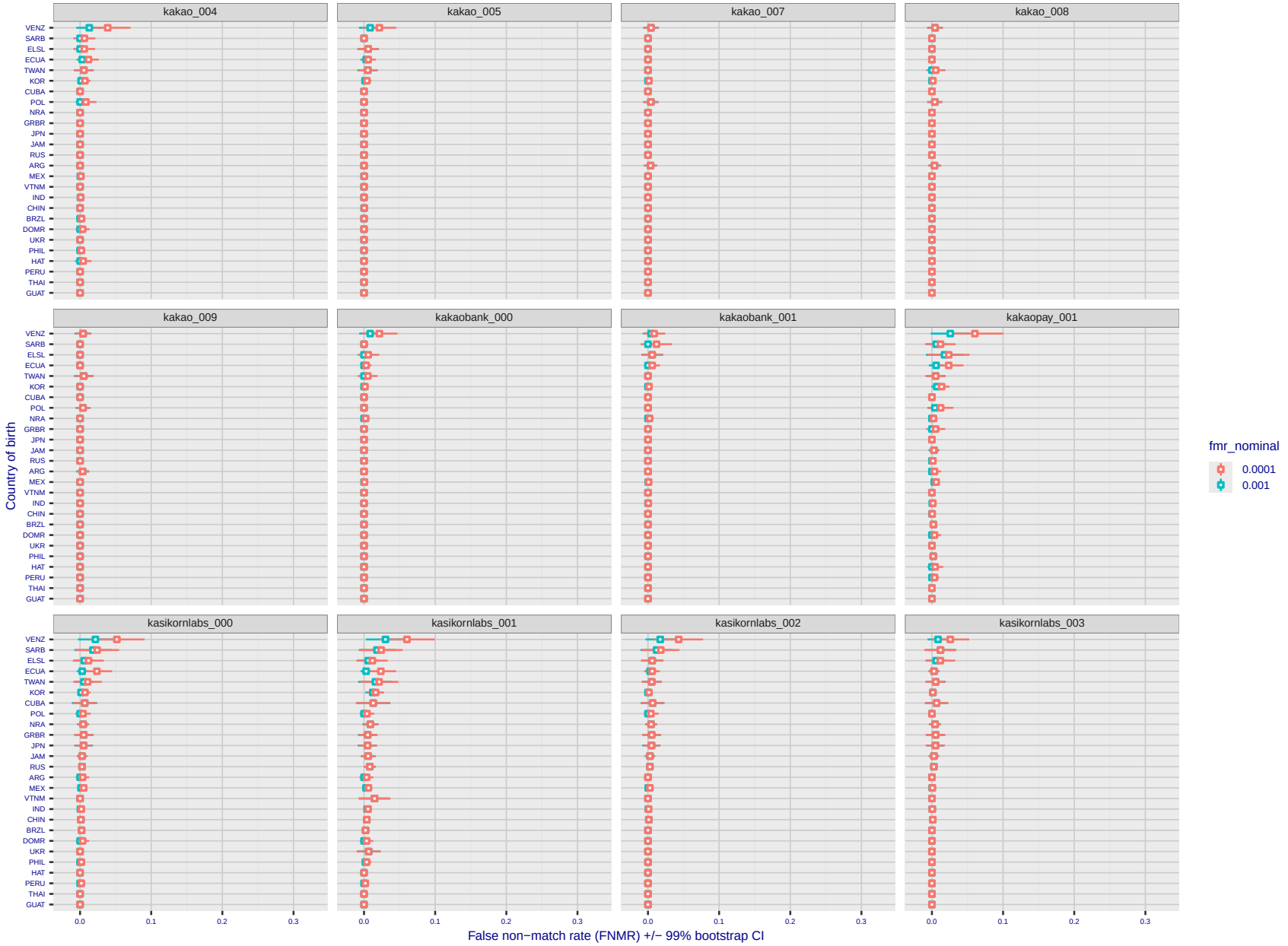

Figure 672: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

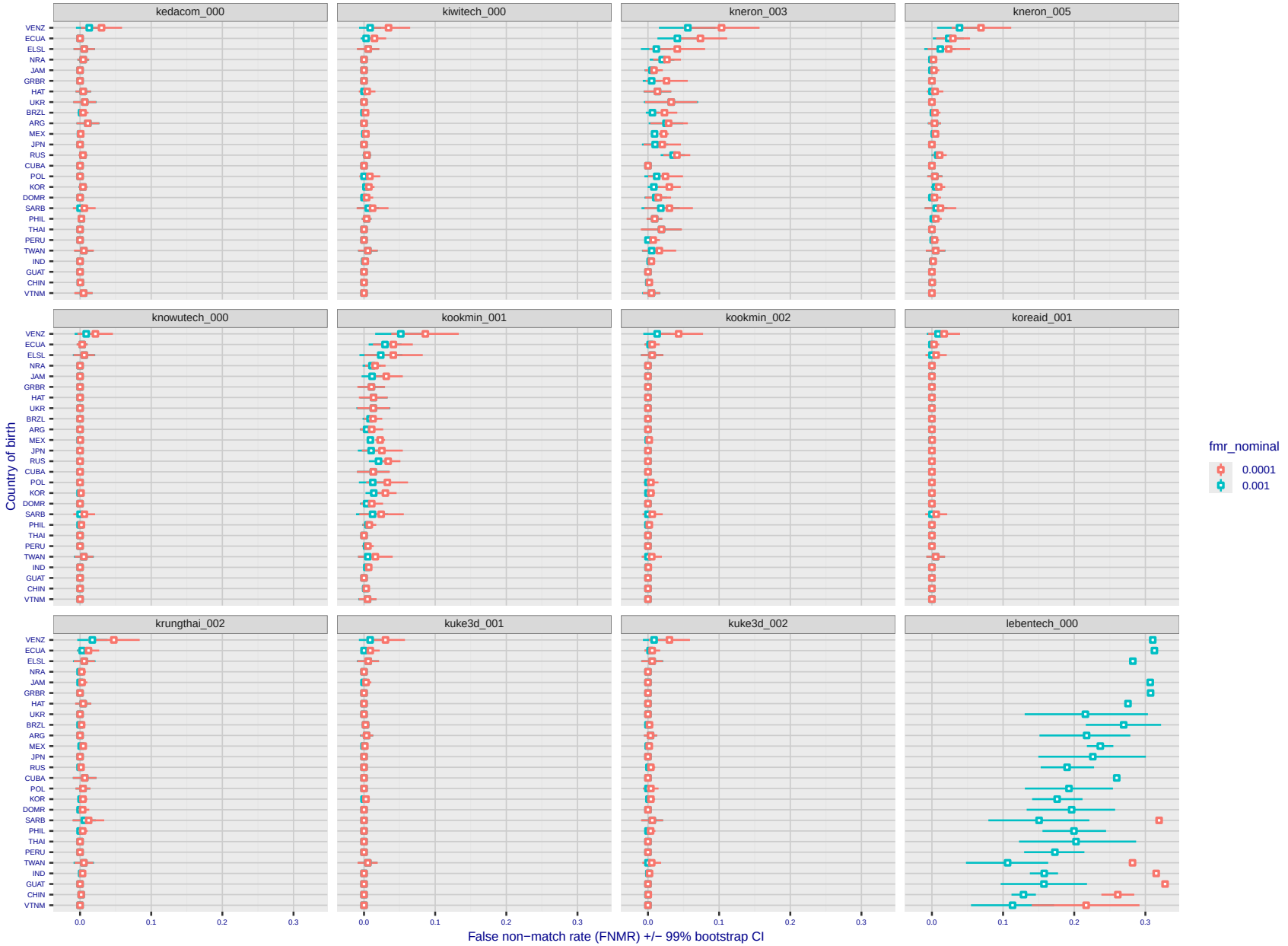

Figure 673: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

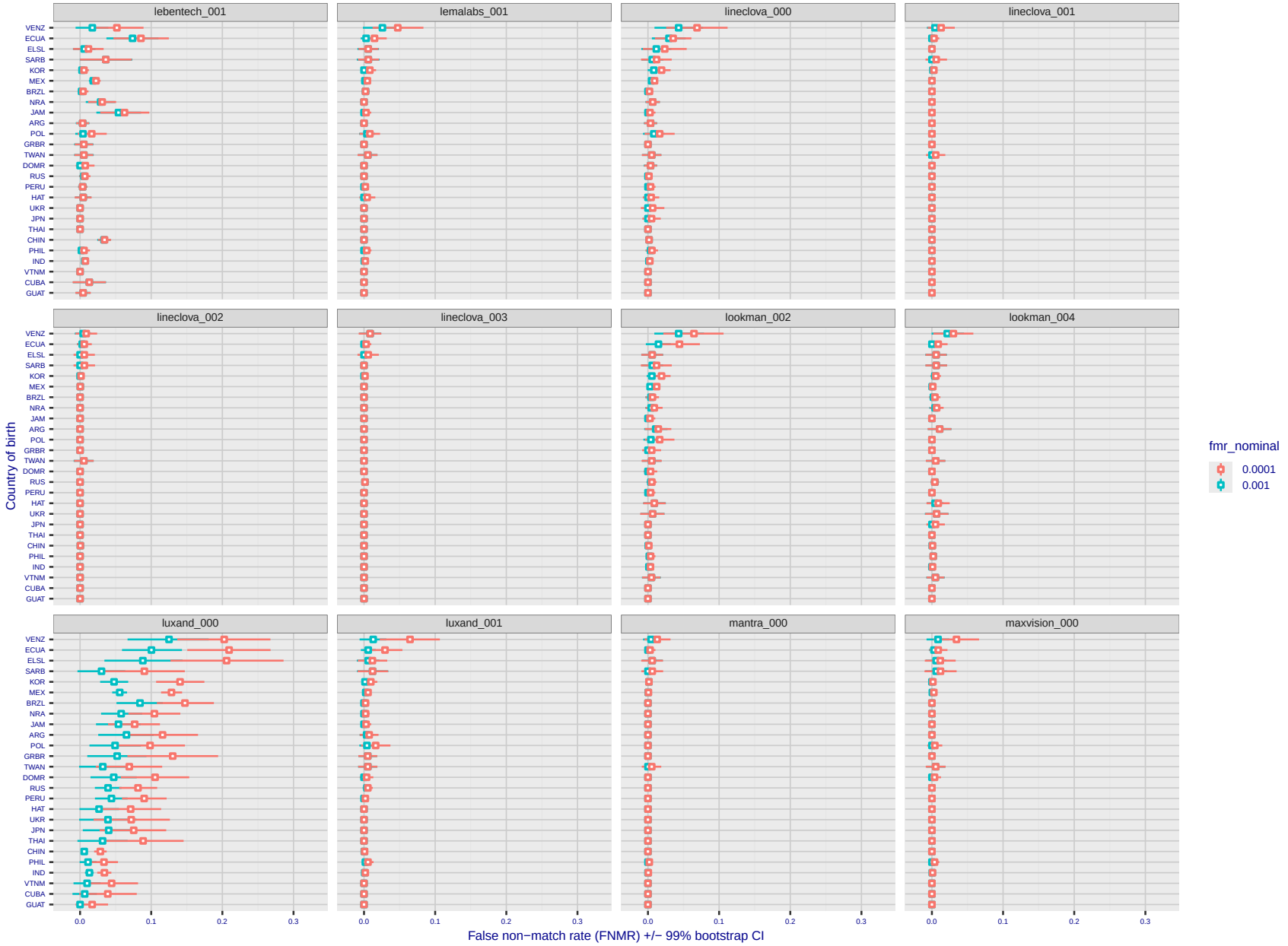

Figure 674: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

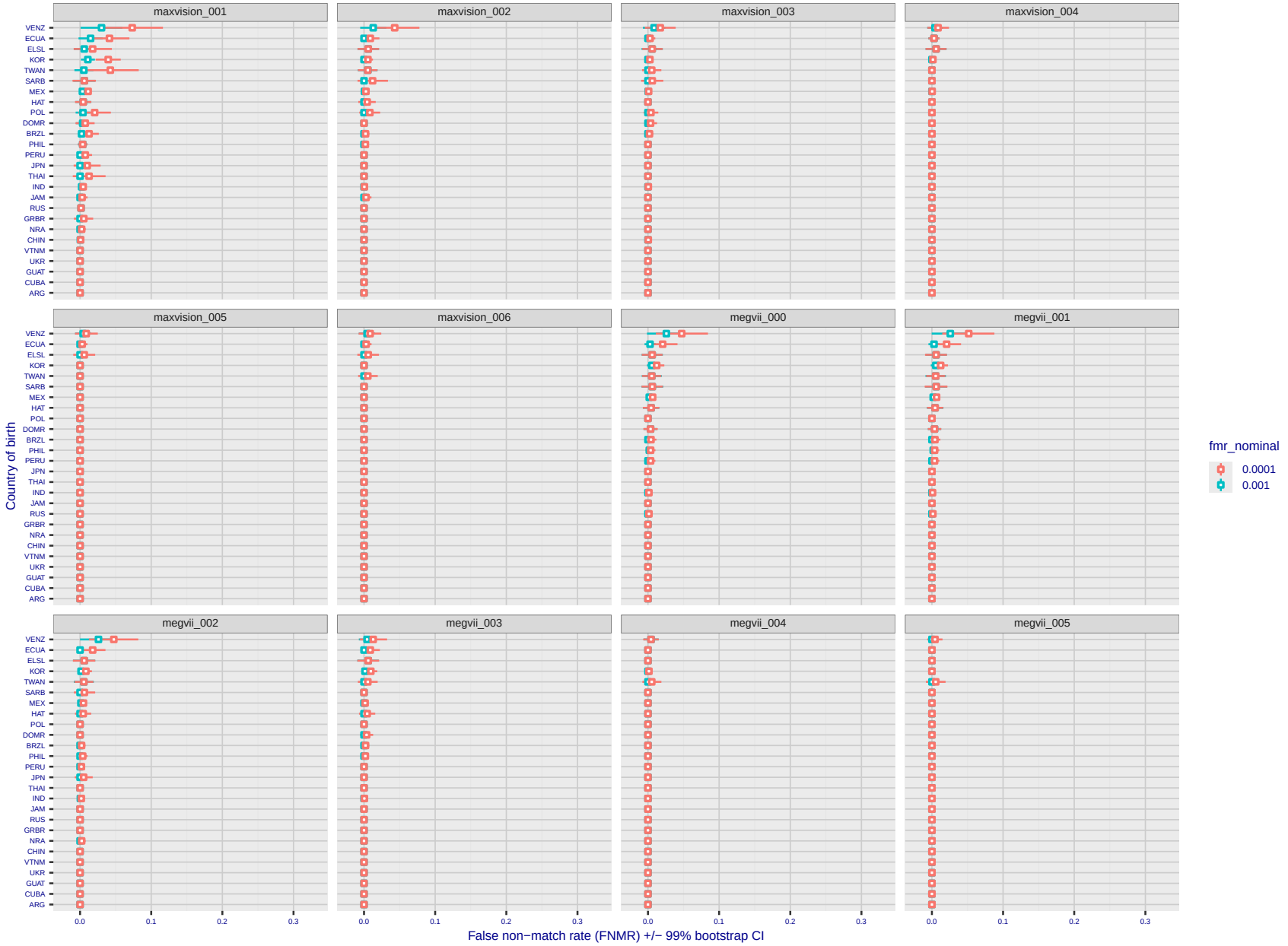

Figure 675: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 676: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 677: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 678: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

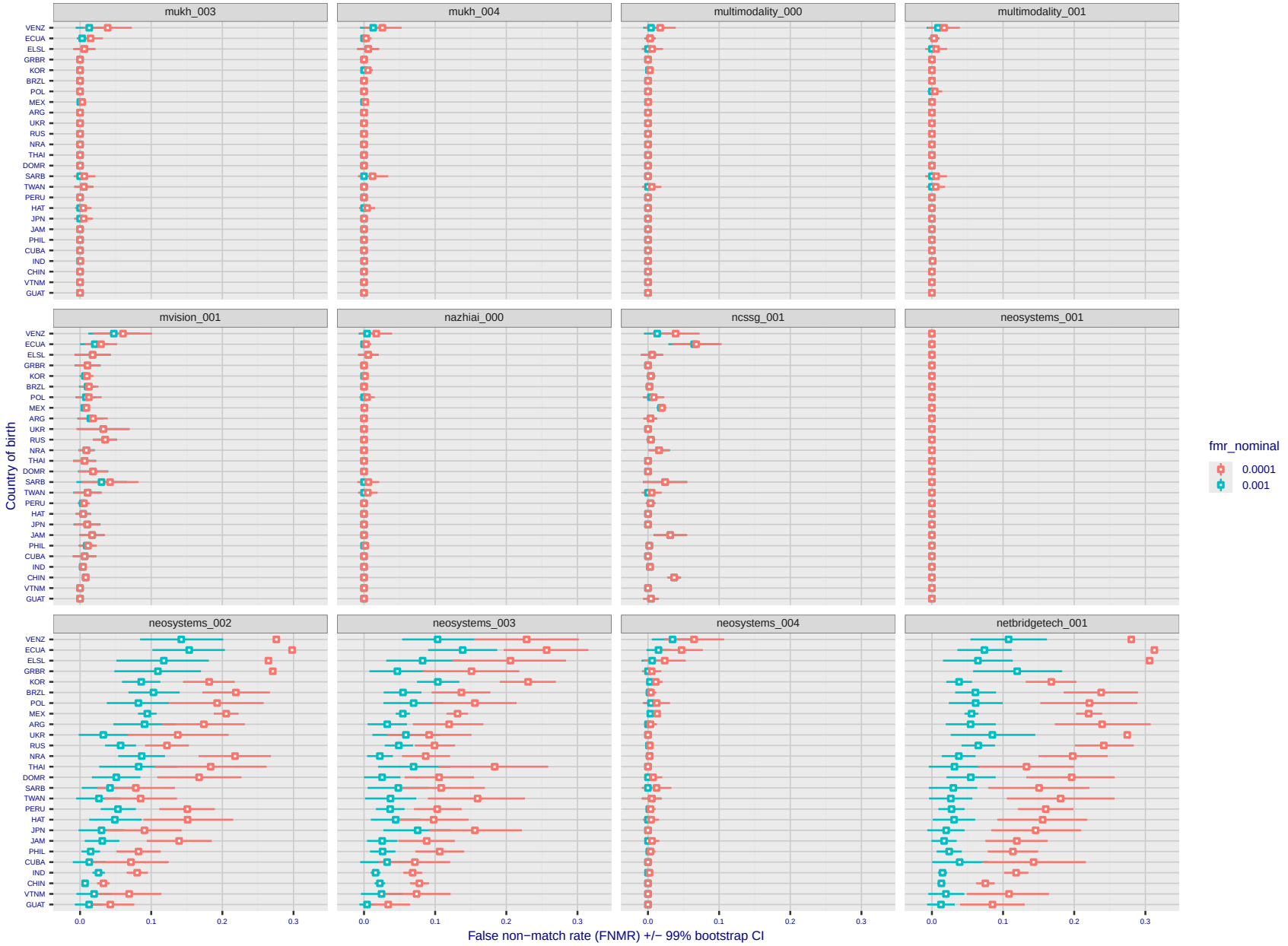

Figure 679: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

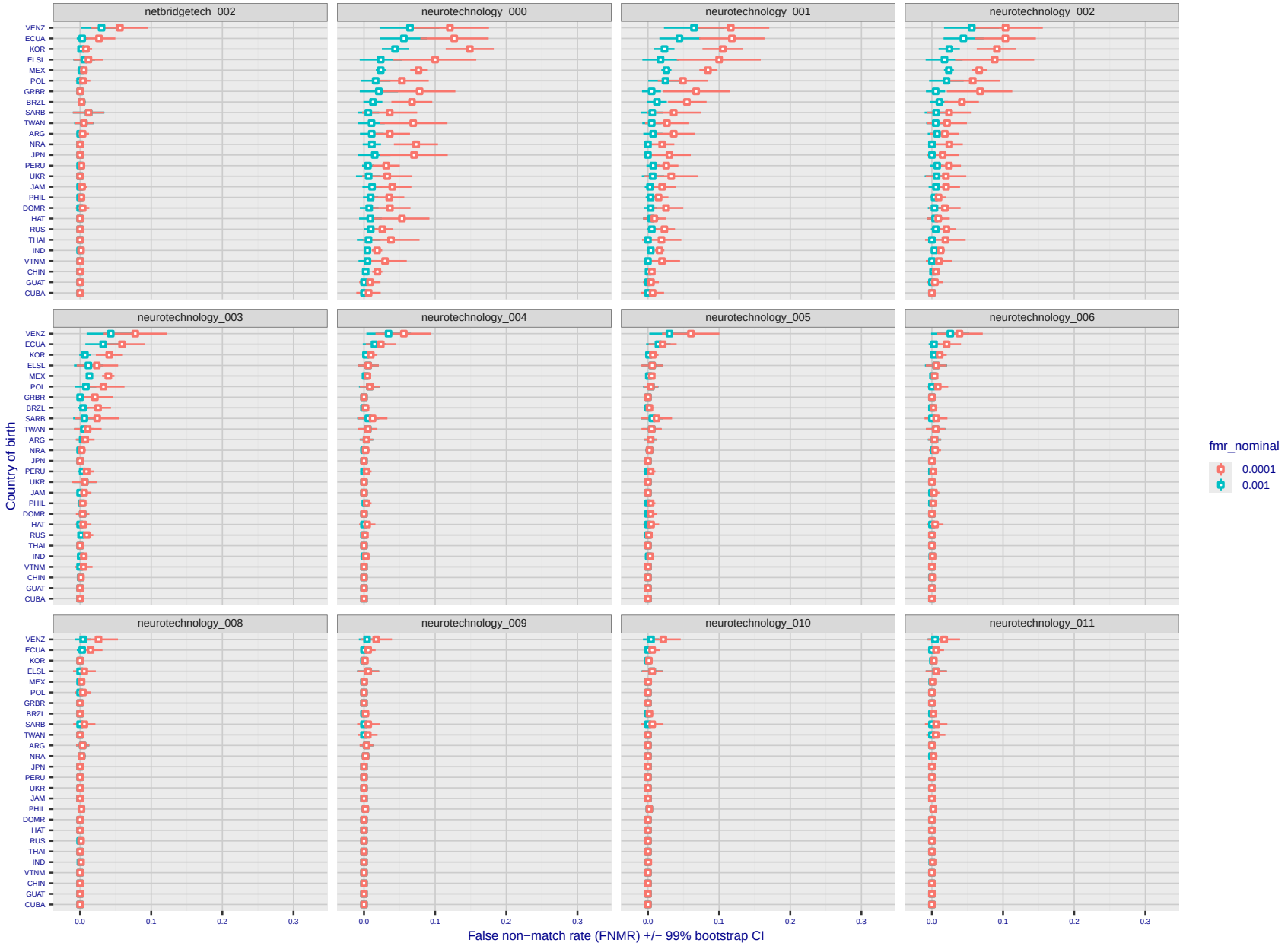

Figure 680: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

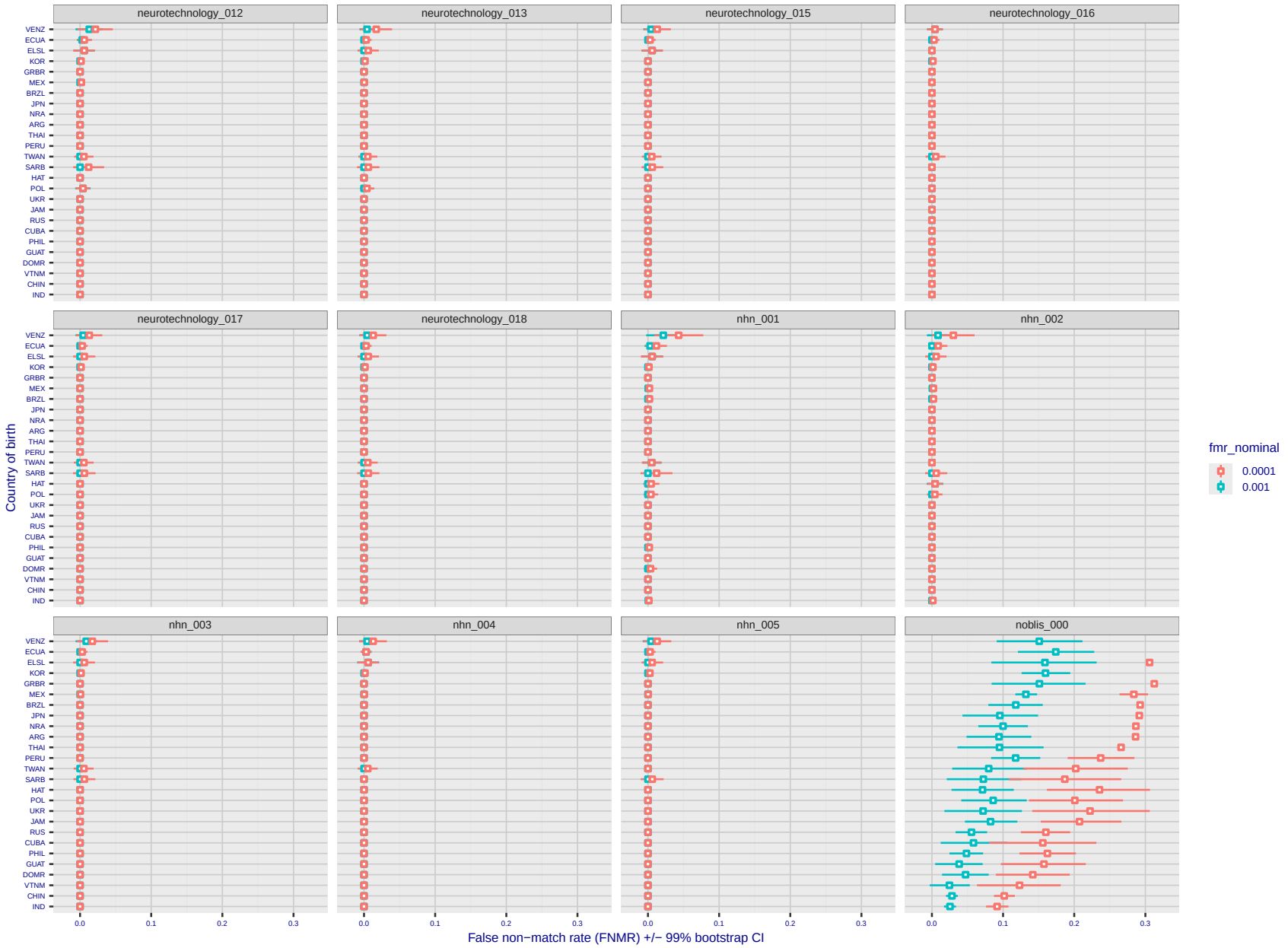

Figure 681: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 682: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

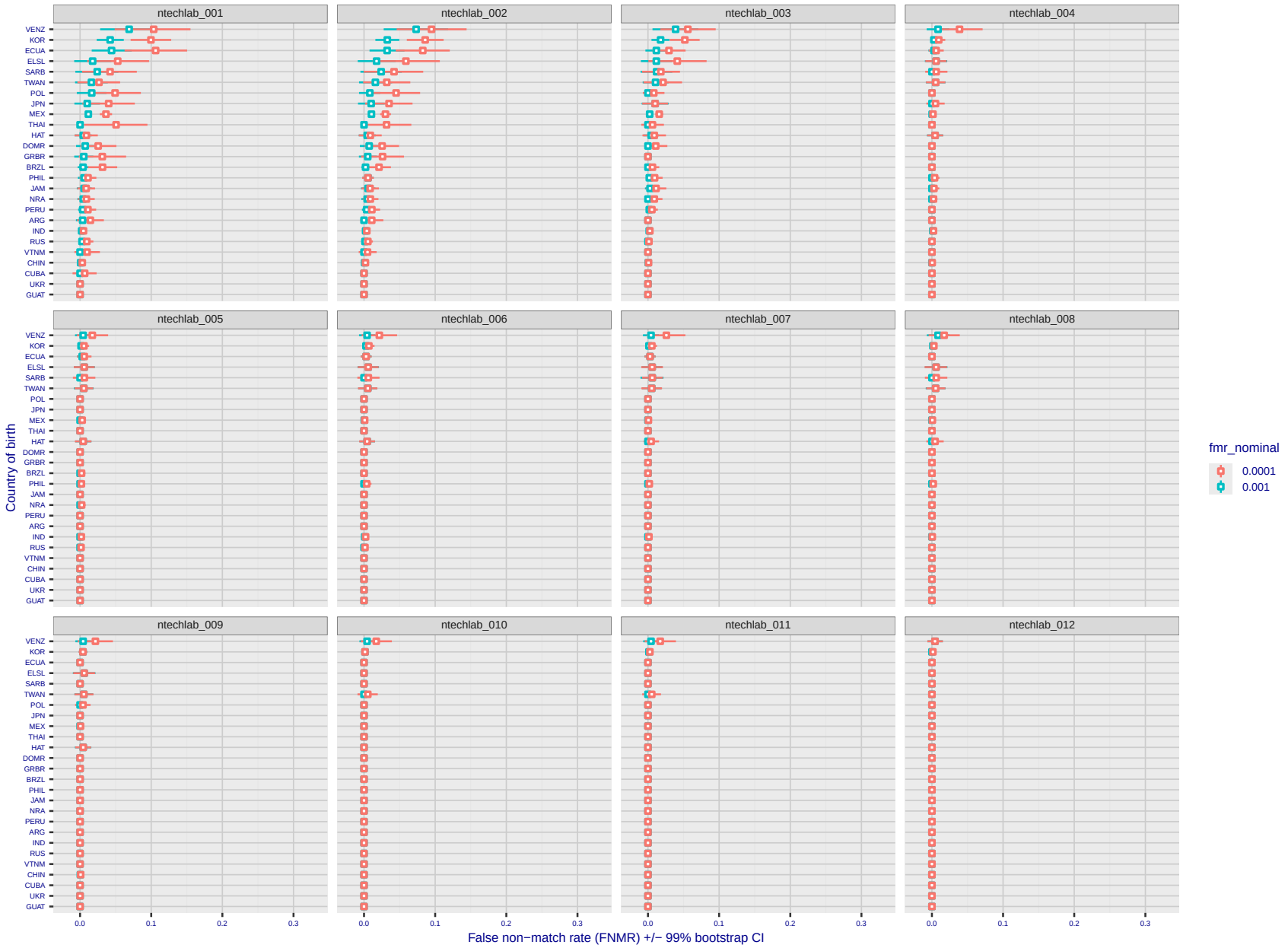

Figure 683: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 684: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 685: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

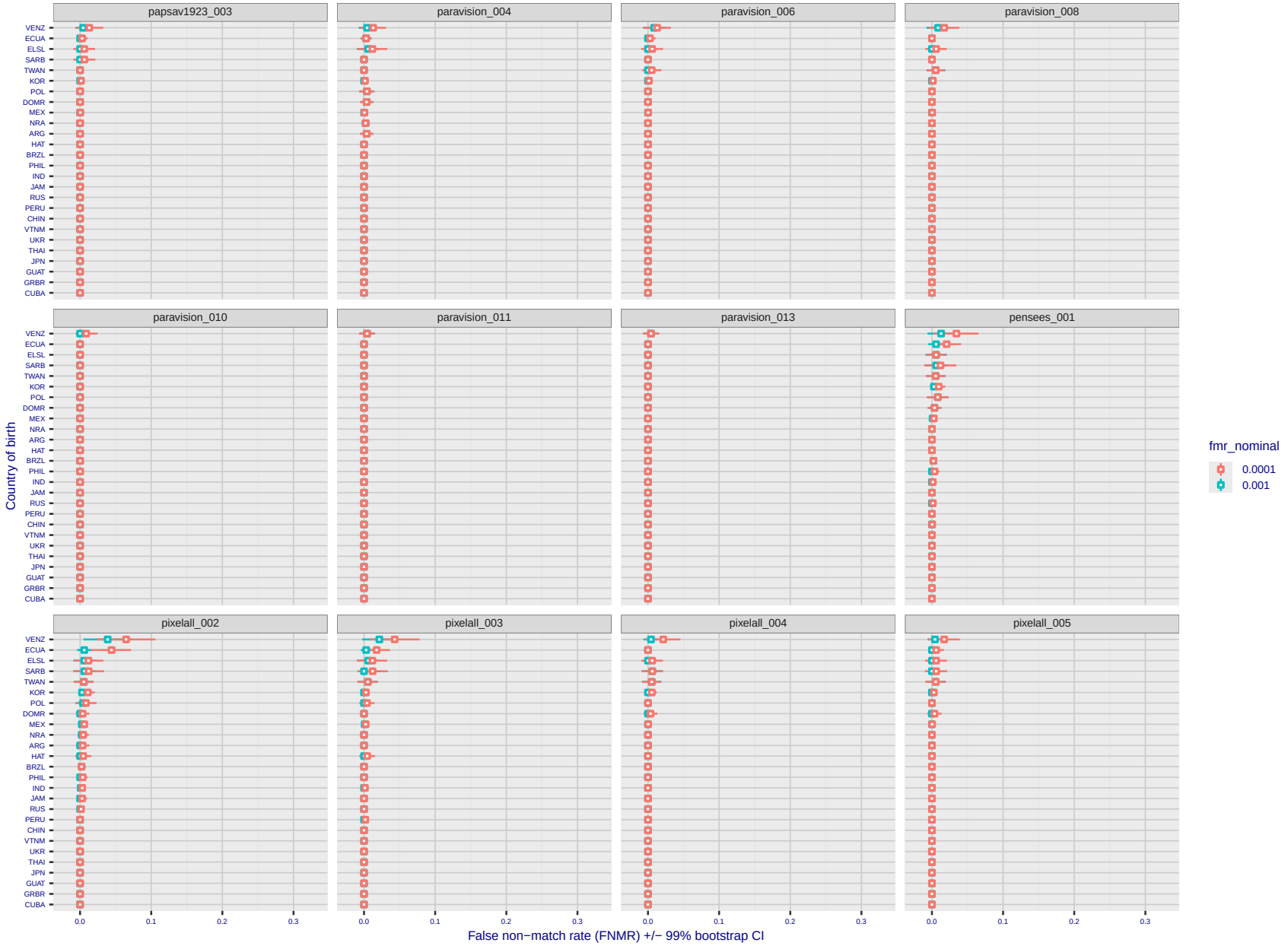

Figure 686: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

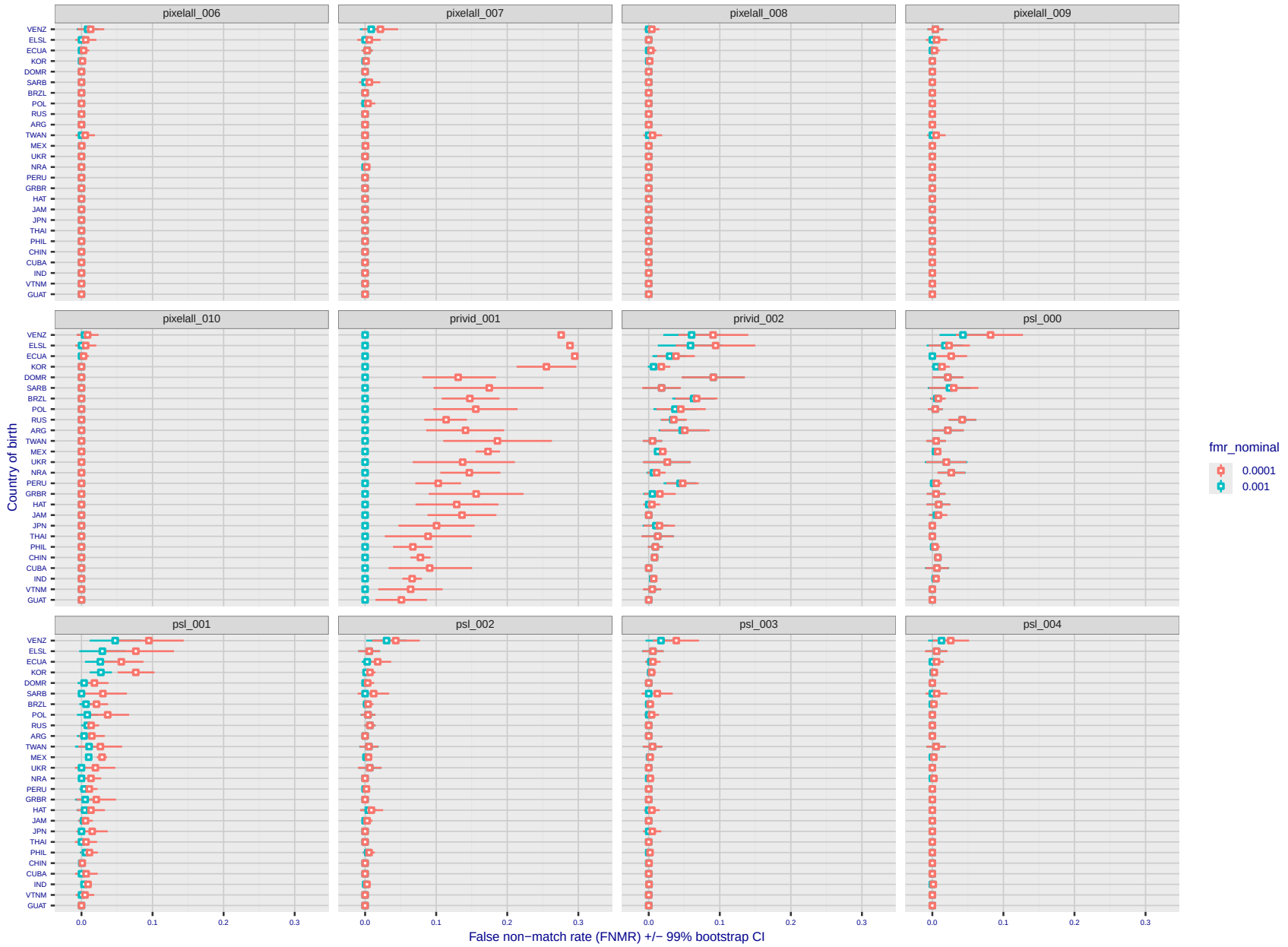

Figure 687: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

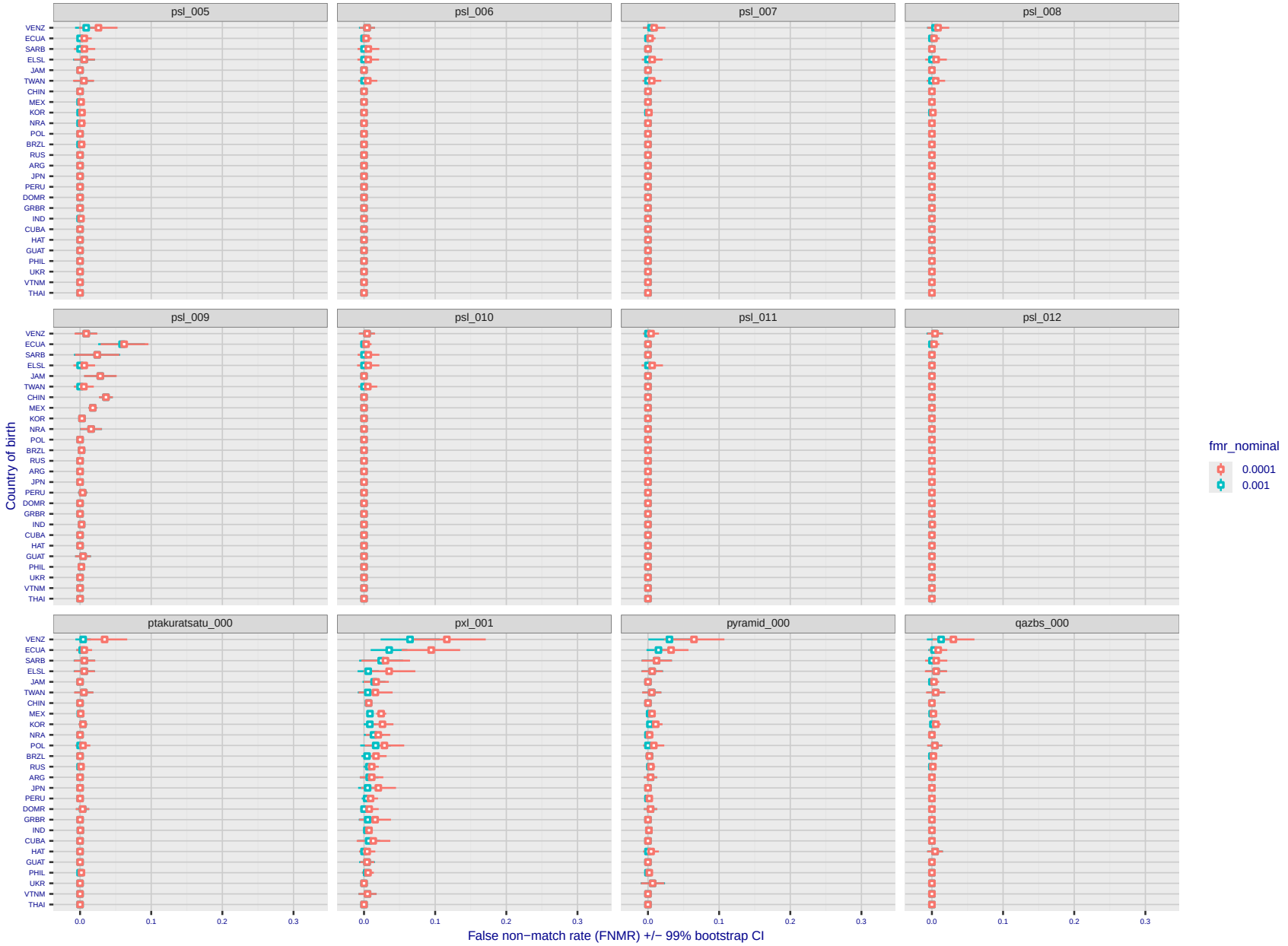

Figure 688: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 689: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

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FNMR(T) "False non-match rate"
FMR(T) "False match rate"

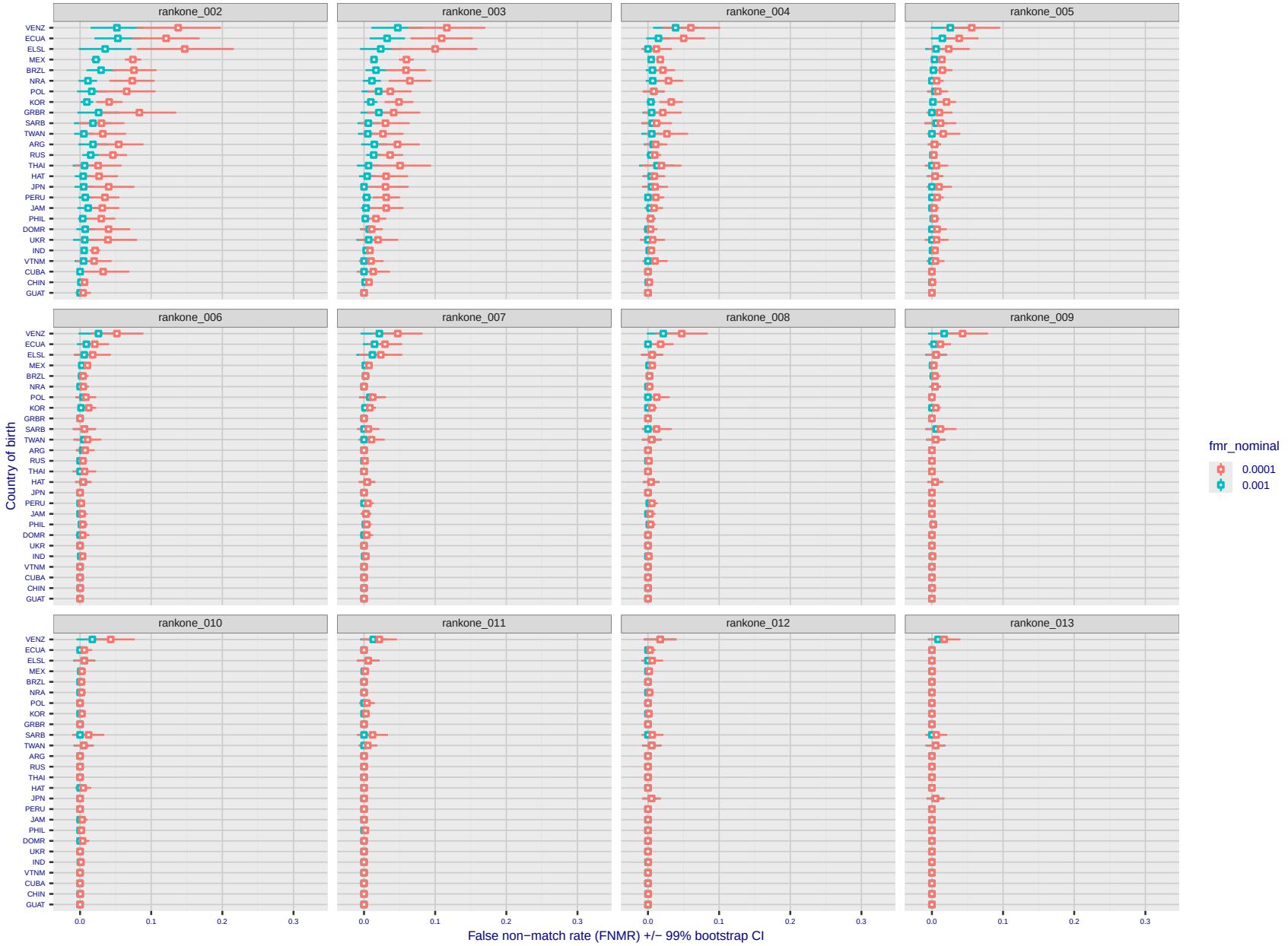

Figure 690: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

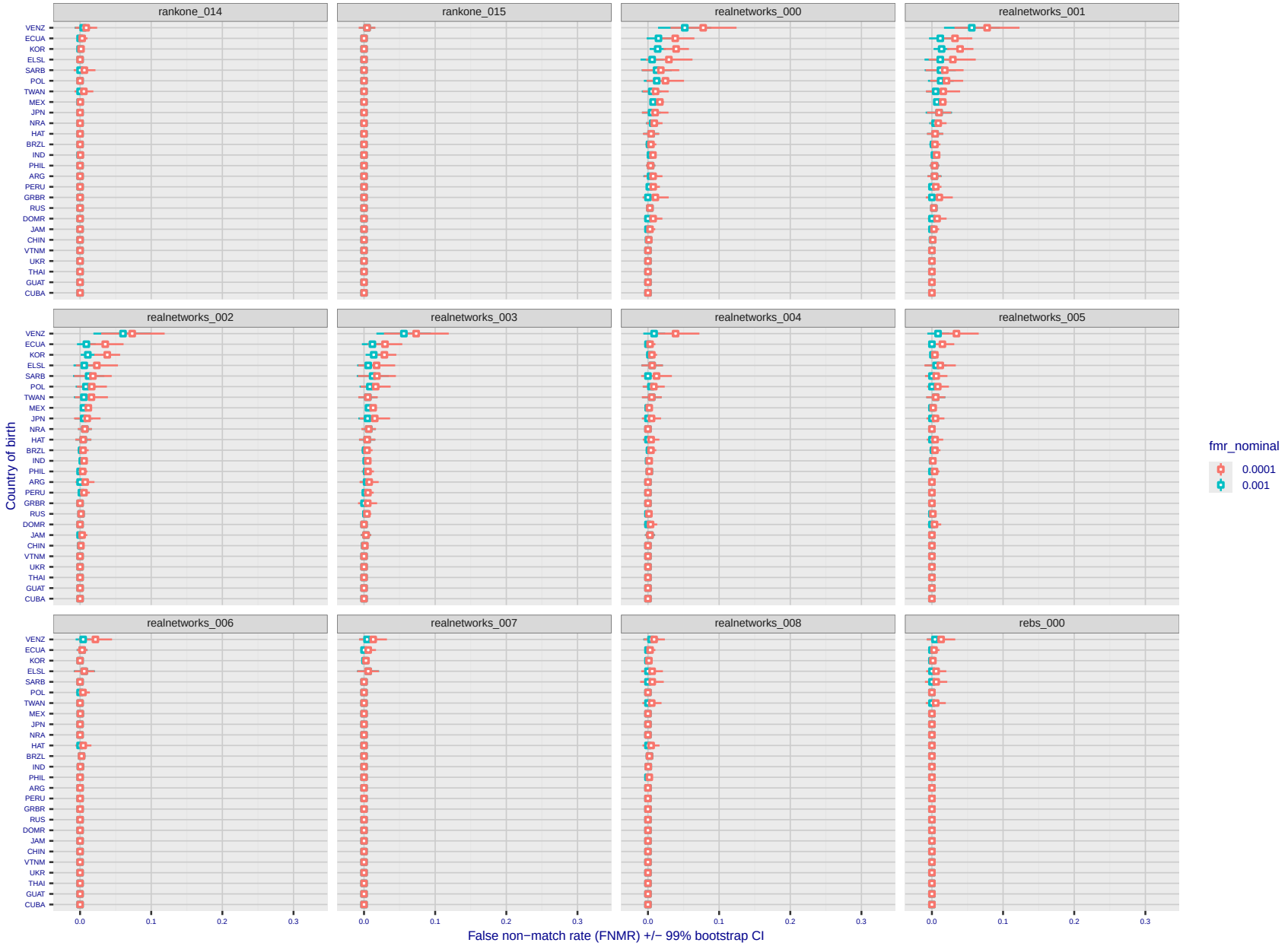

Figure 691: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

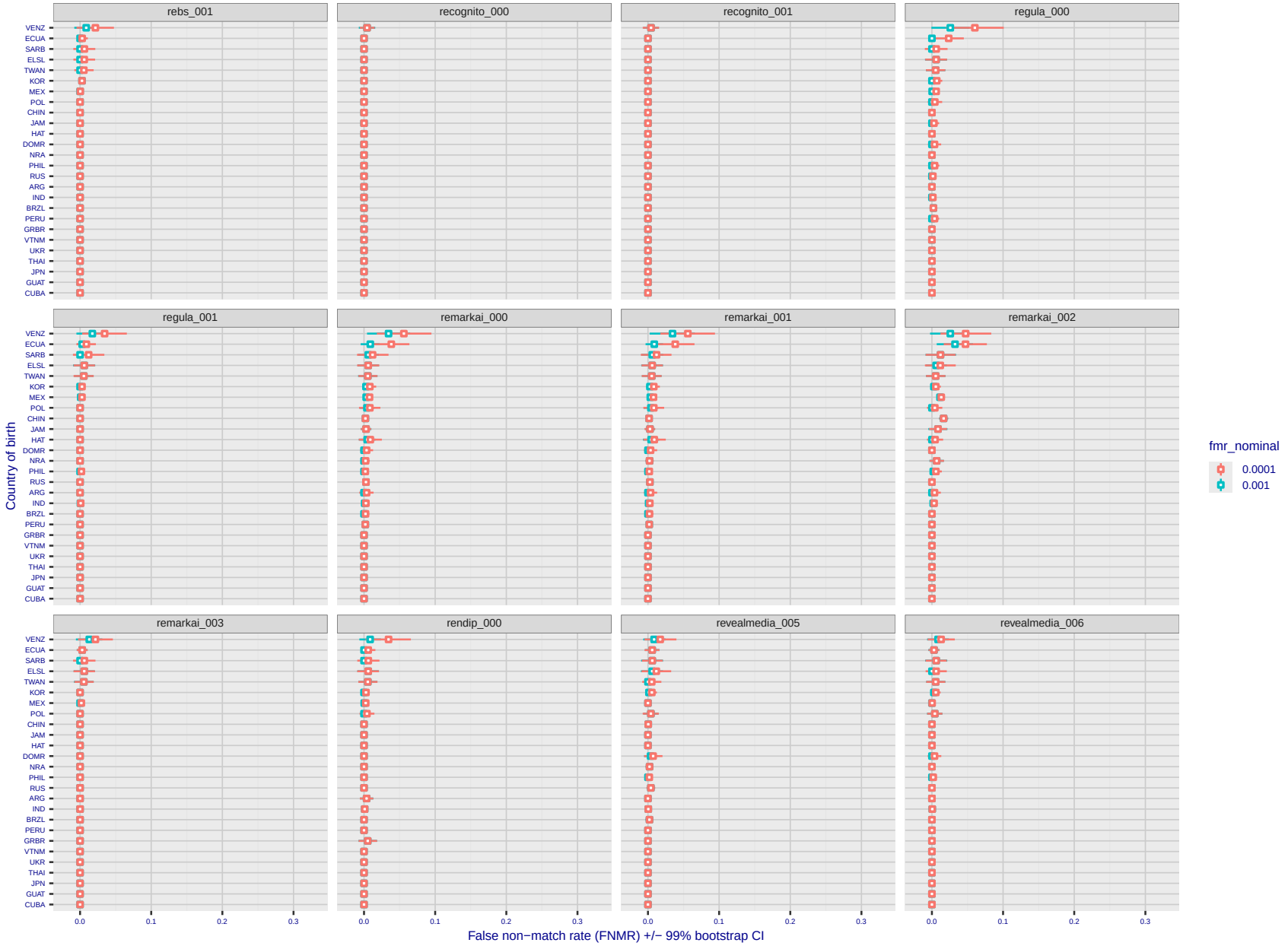

Figure 692: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

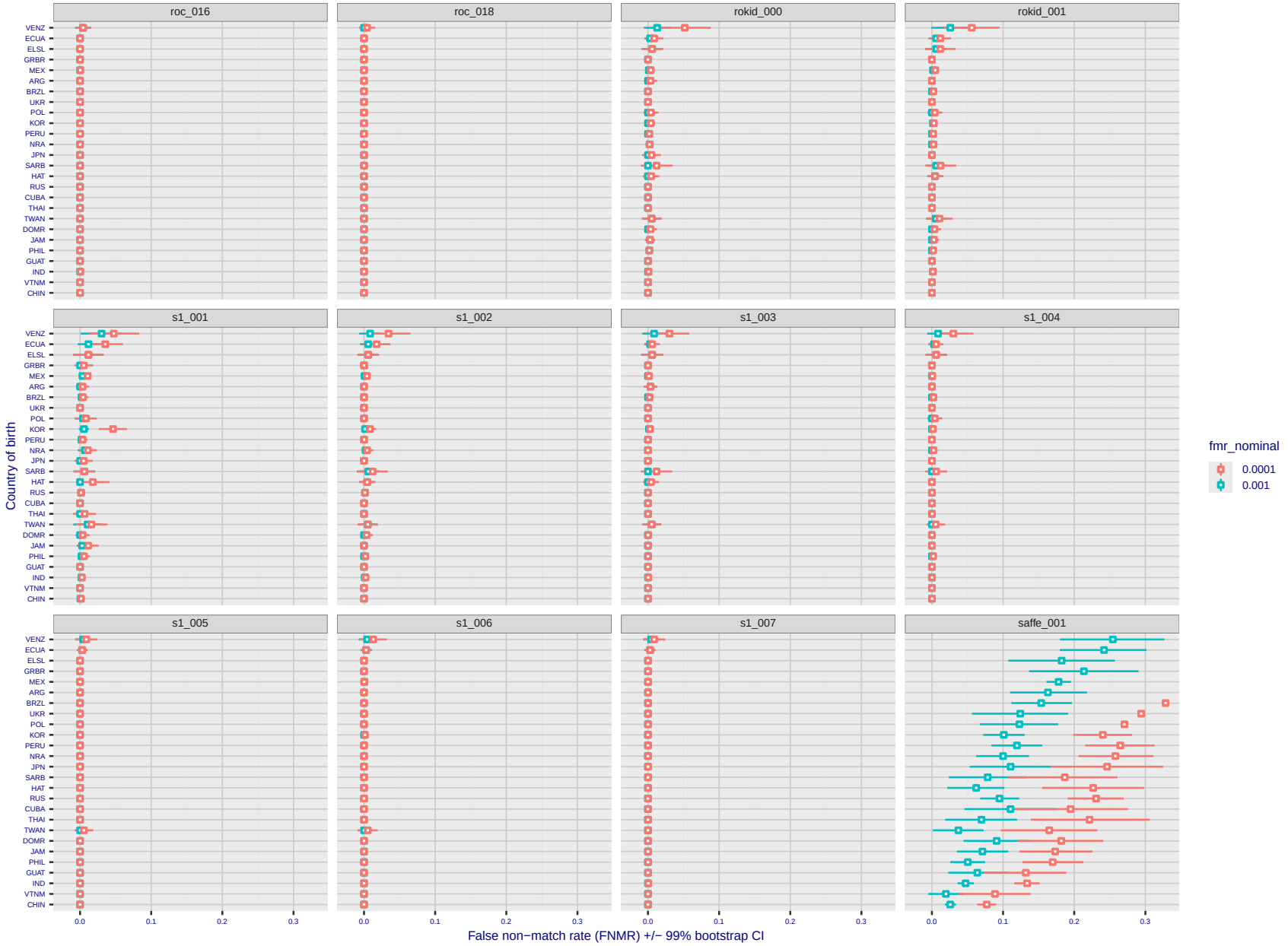

Figure 693: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

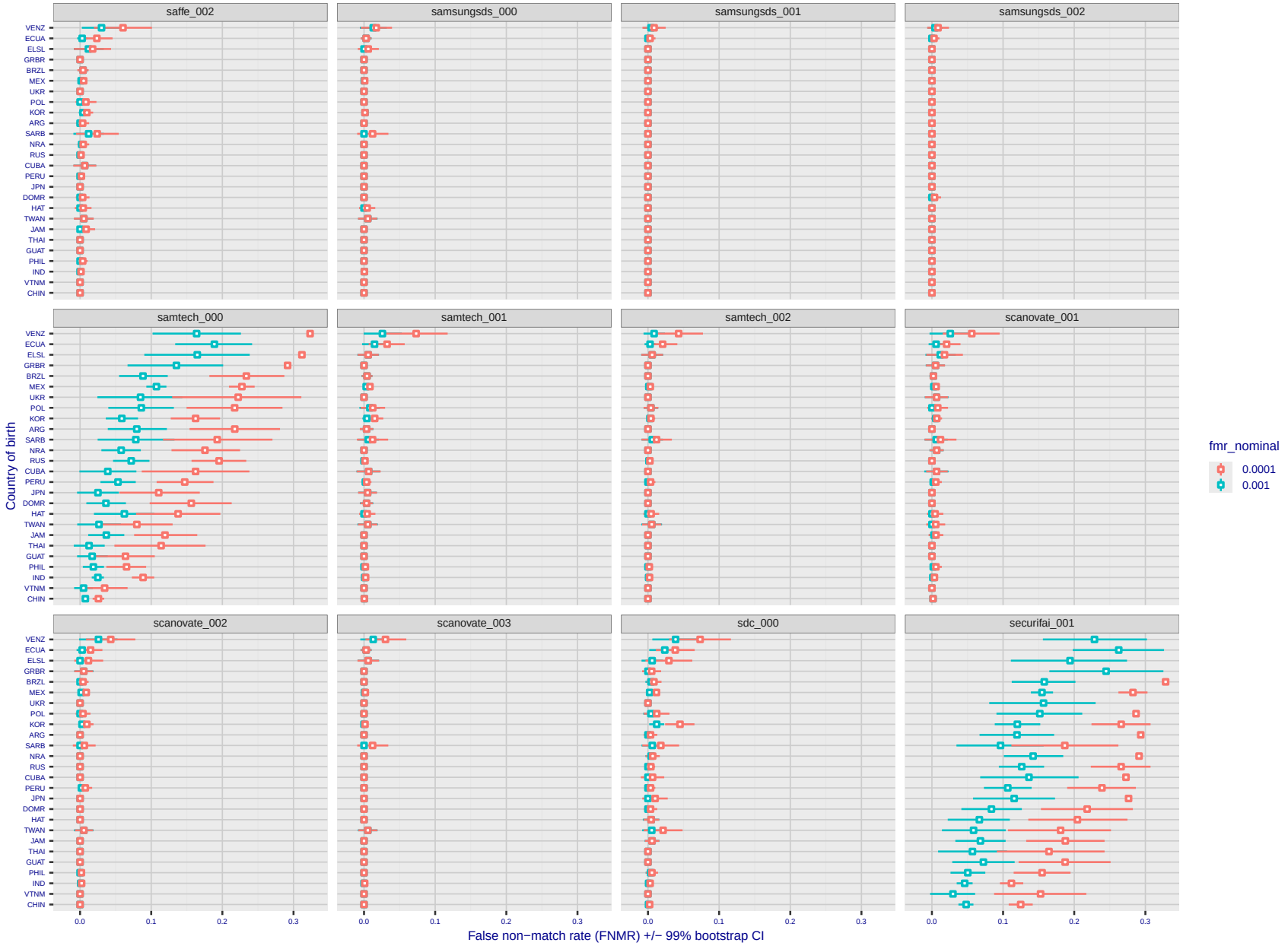

Figure 694: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 695: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

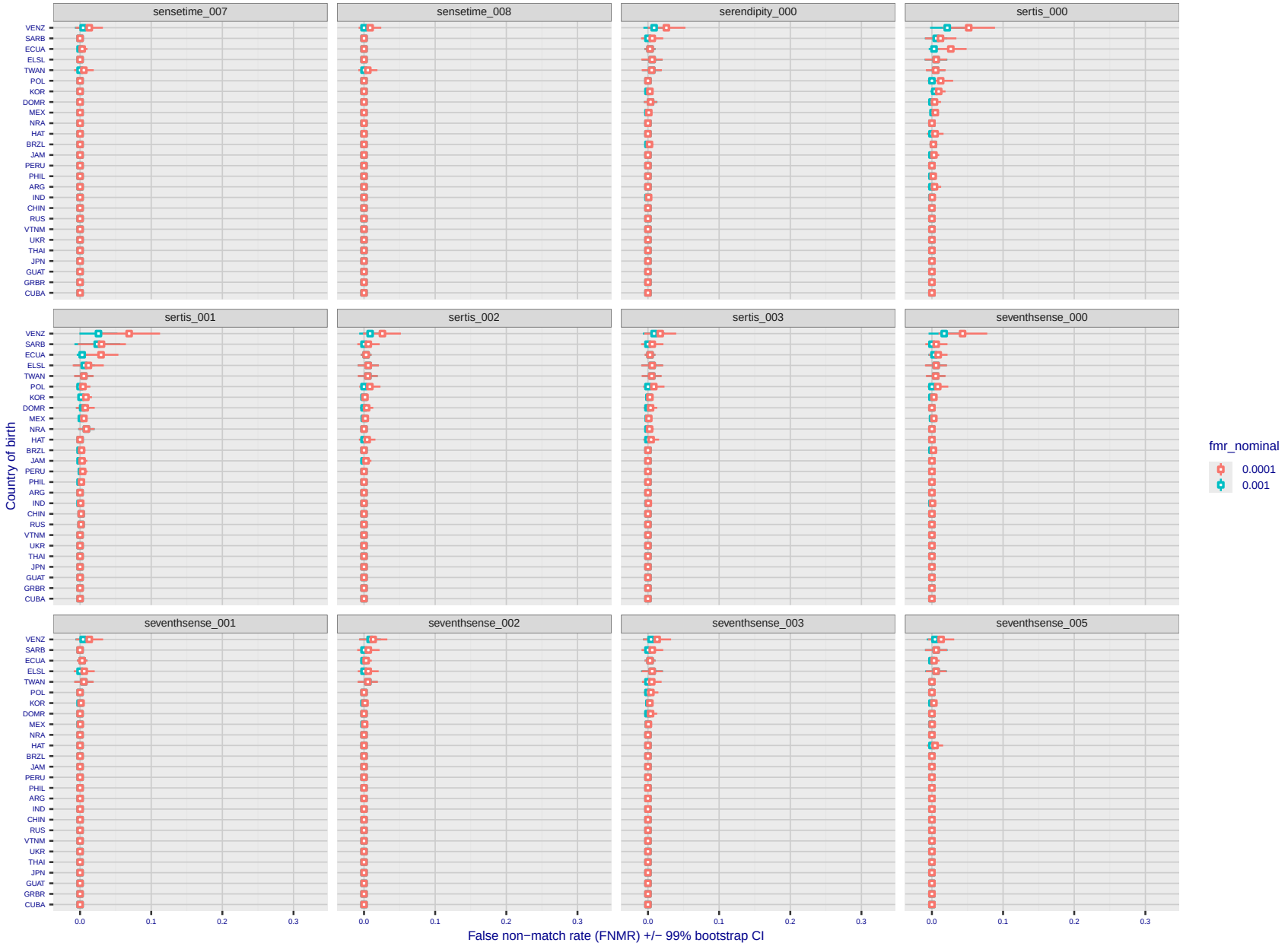

Figure 696: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

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Figure 698: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

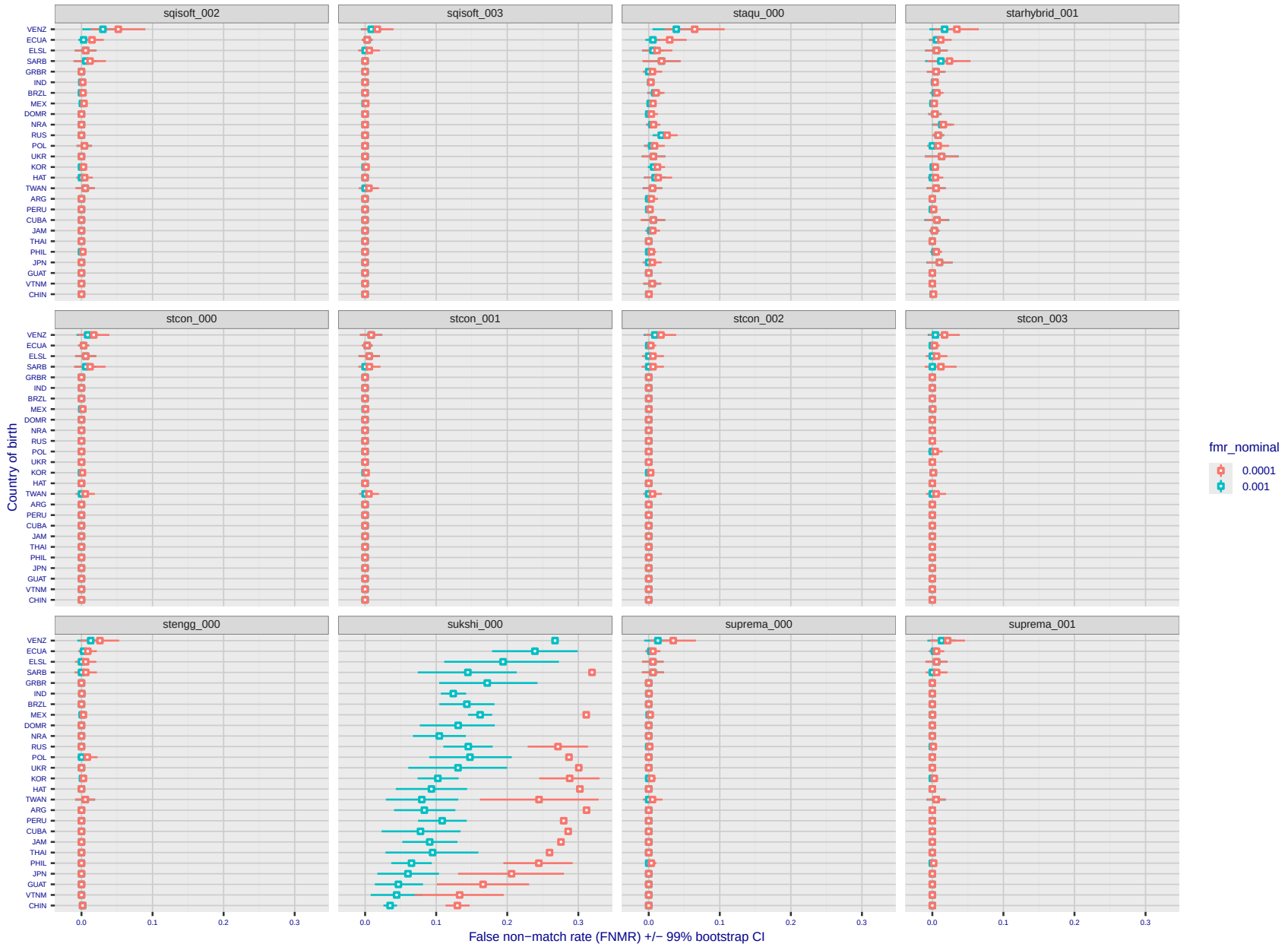


Figure 699: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

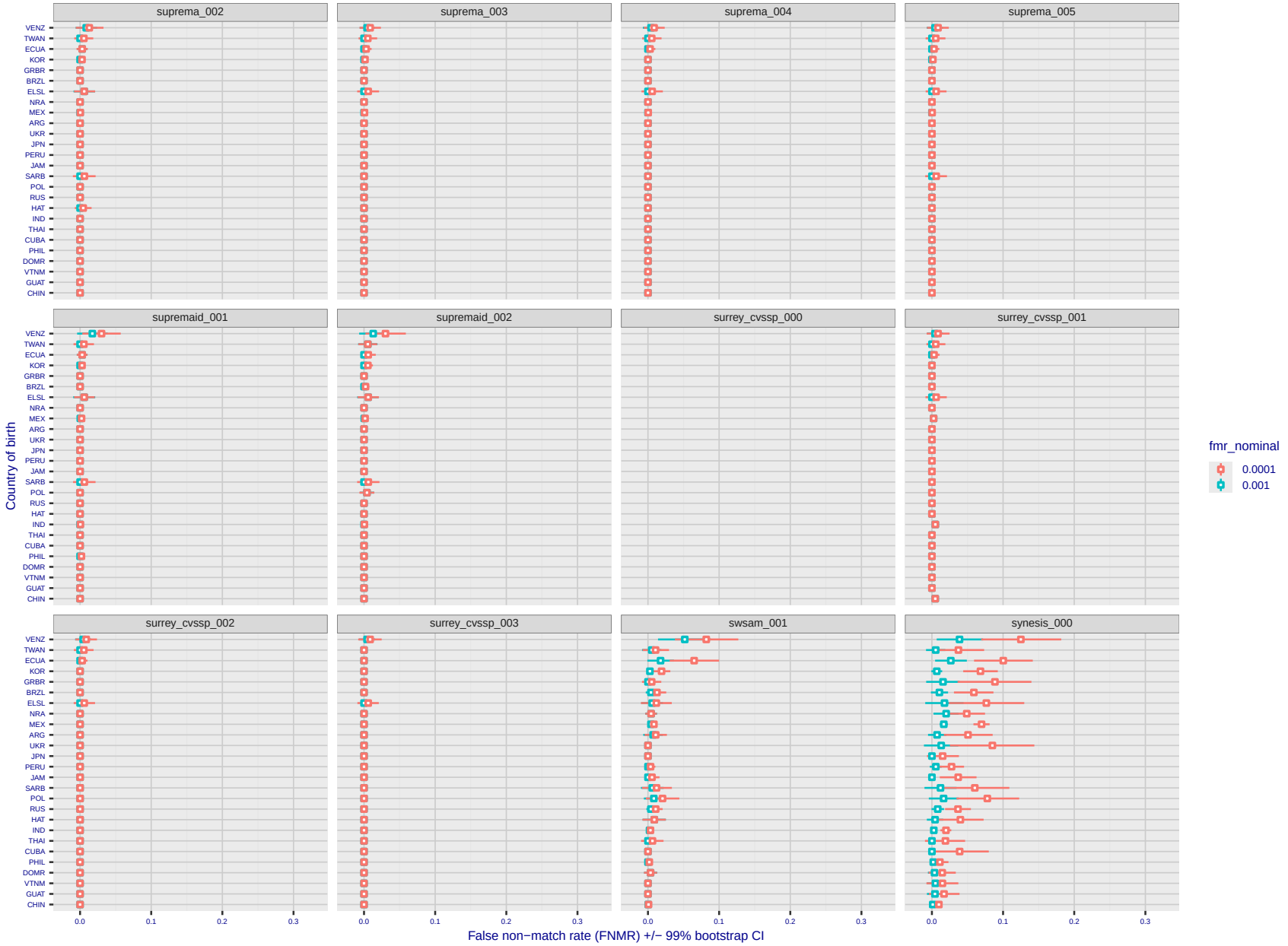


Figure 700: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

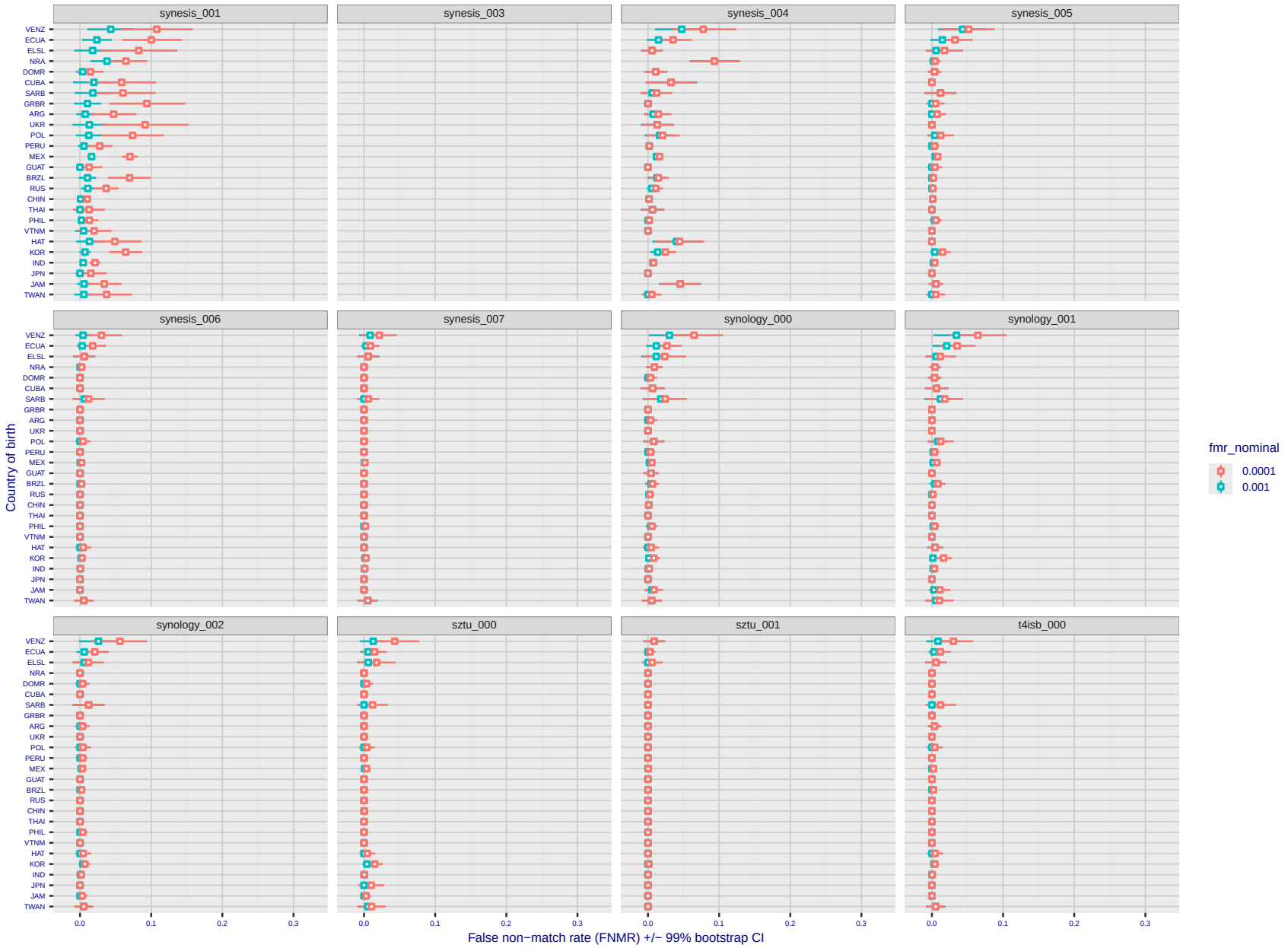


Figure 701: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

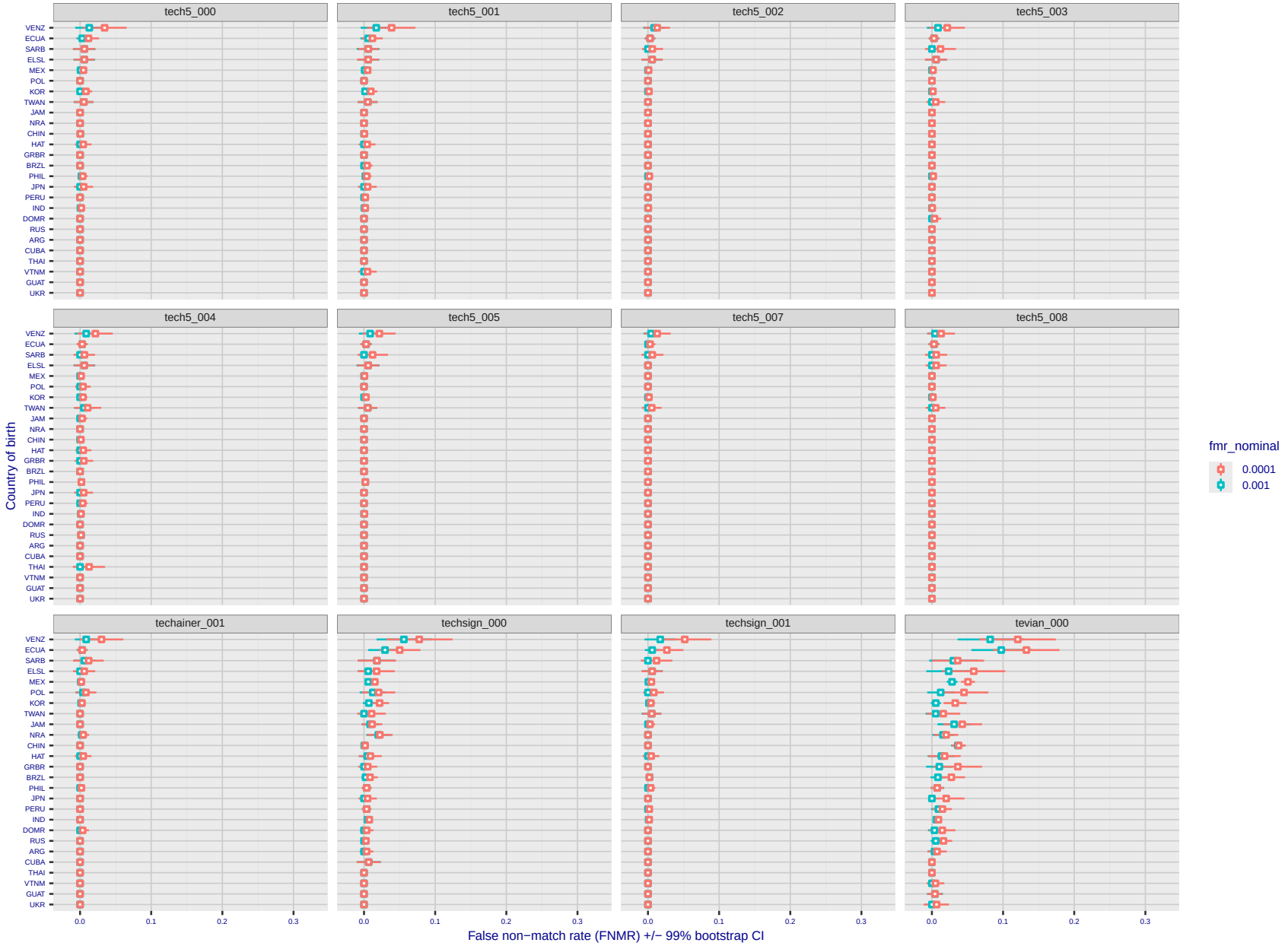


Figure 702: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

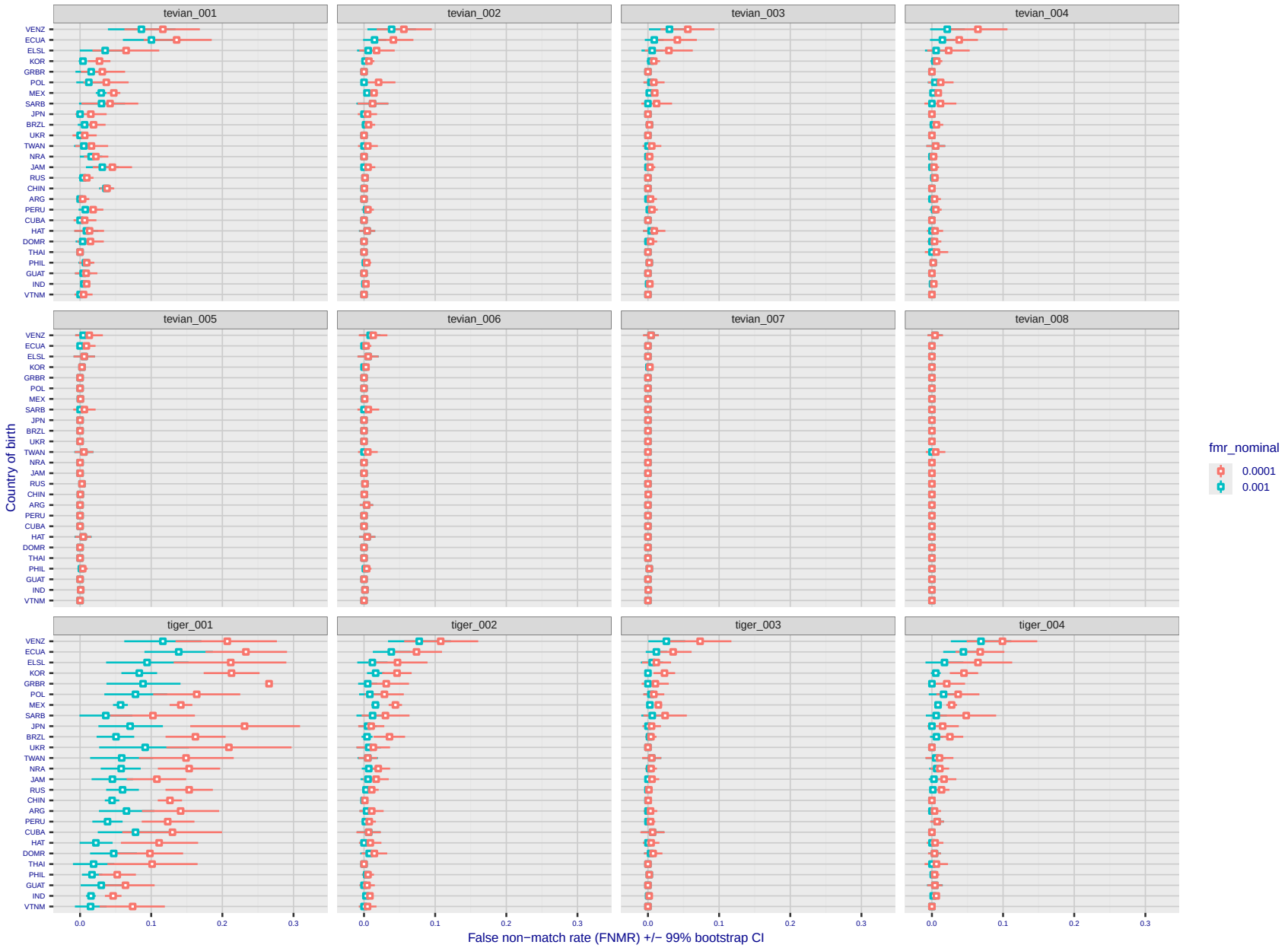


Figure 703: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 704: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

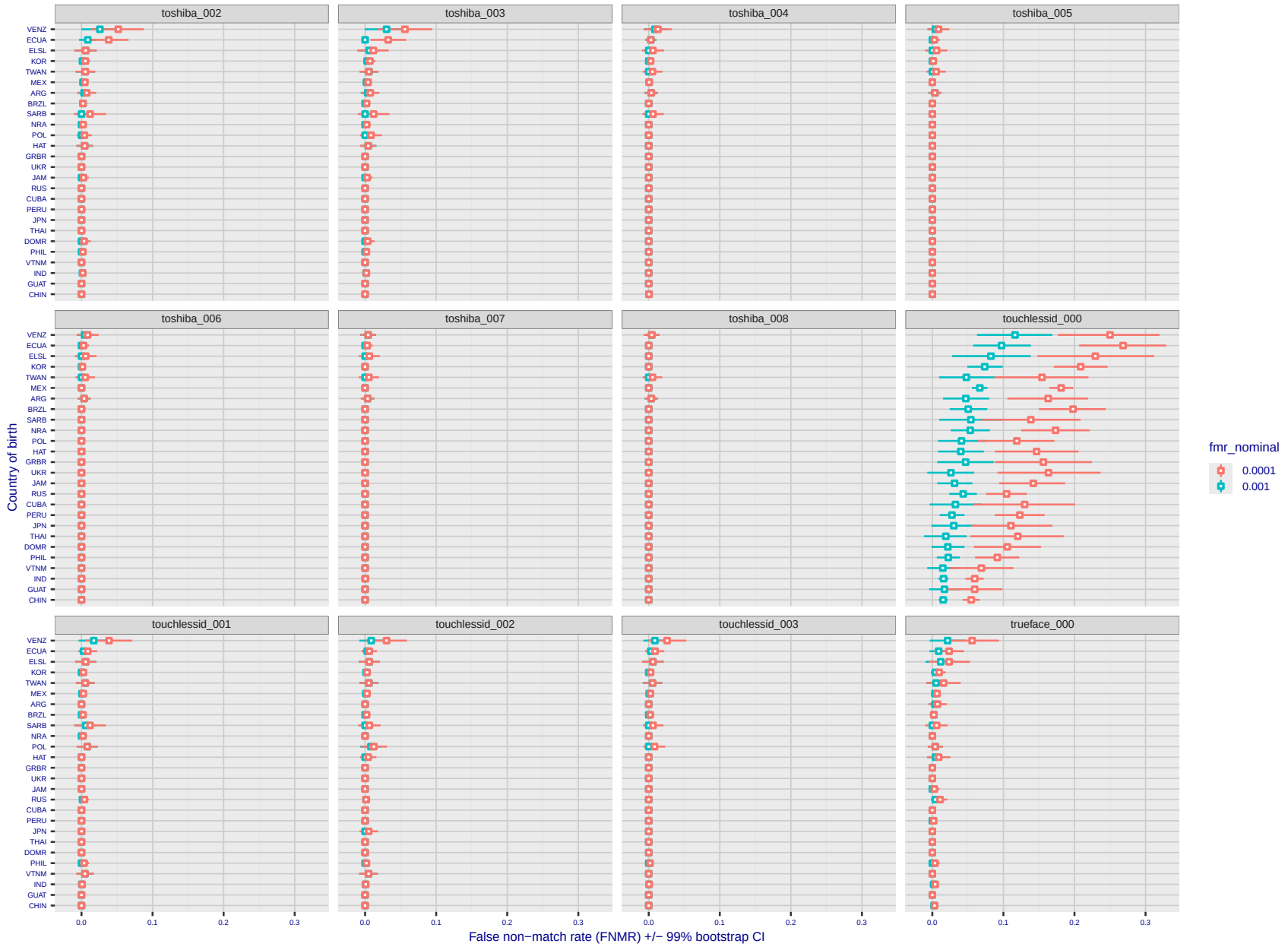


Figure 705: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 706: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

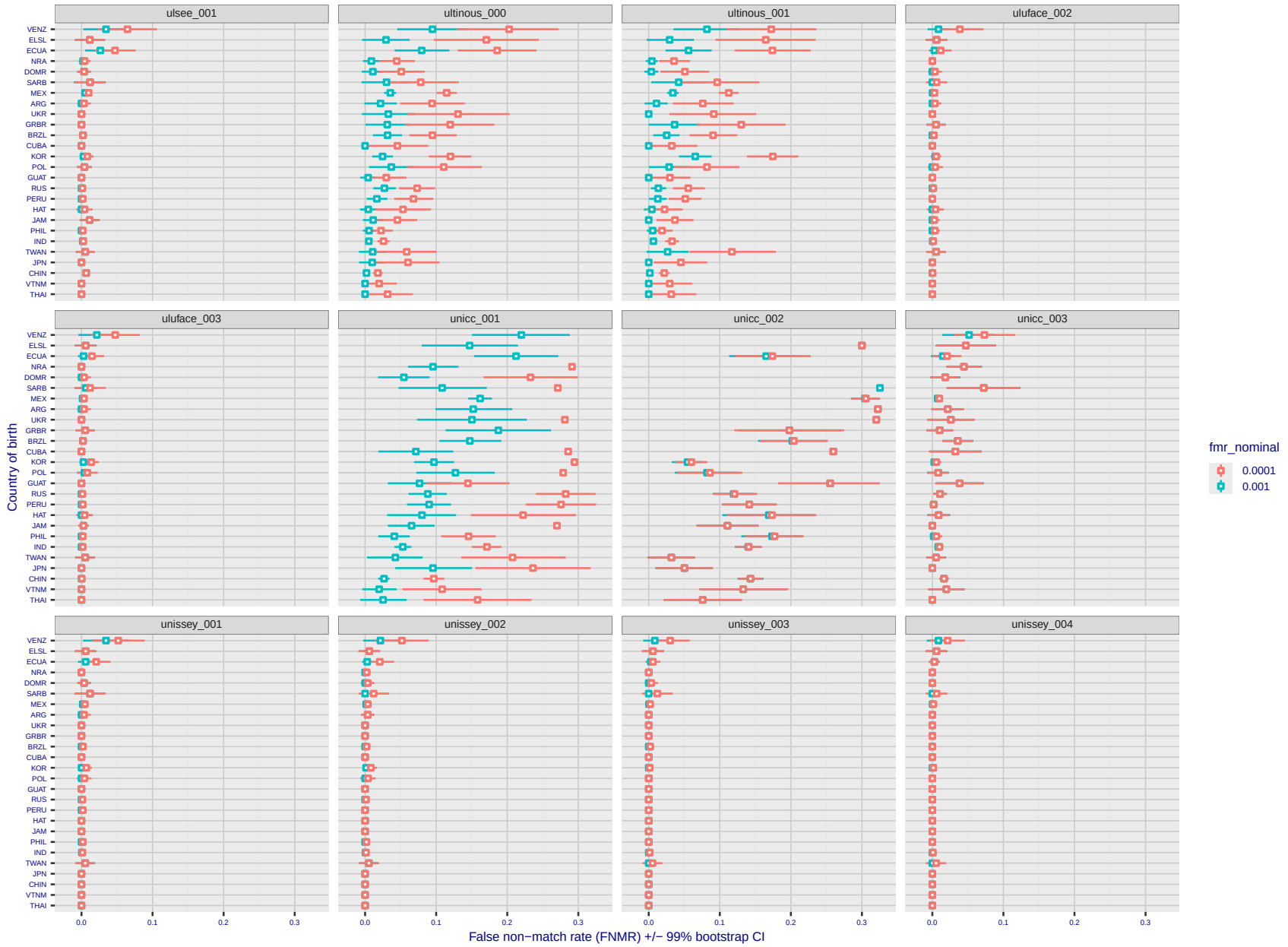


Figure 707: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 708: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

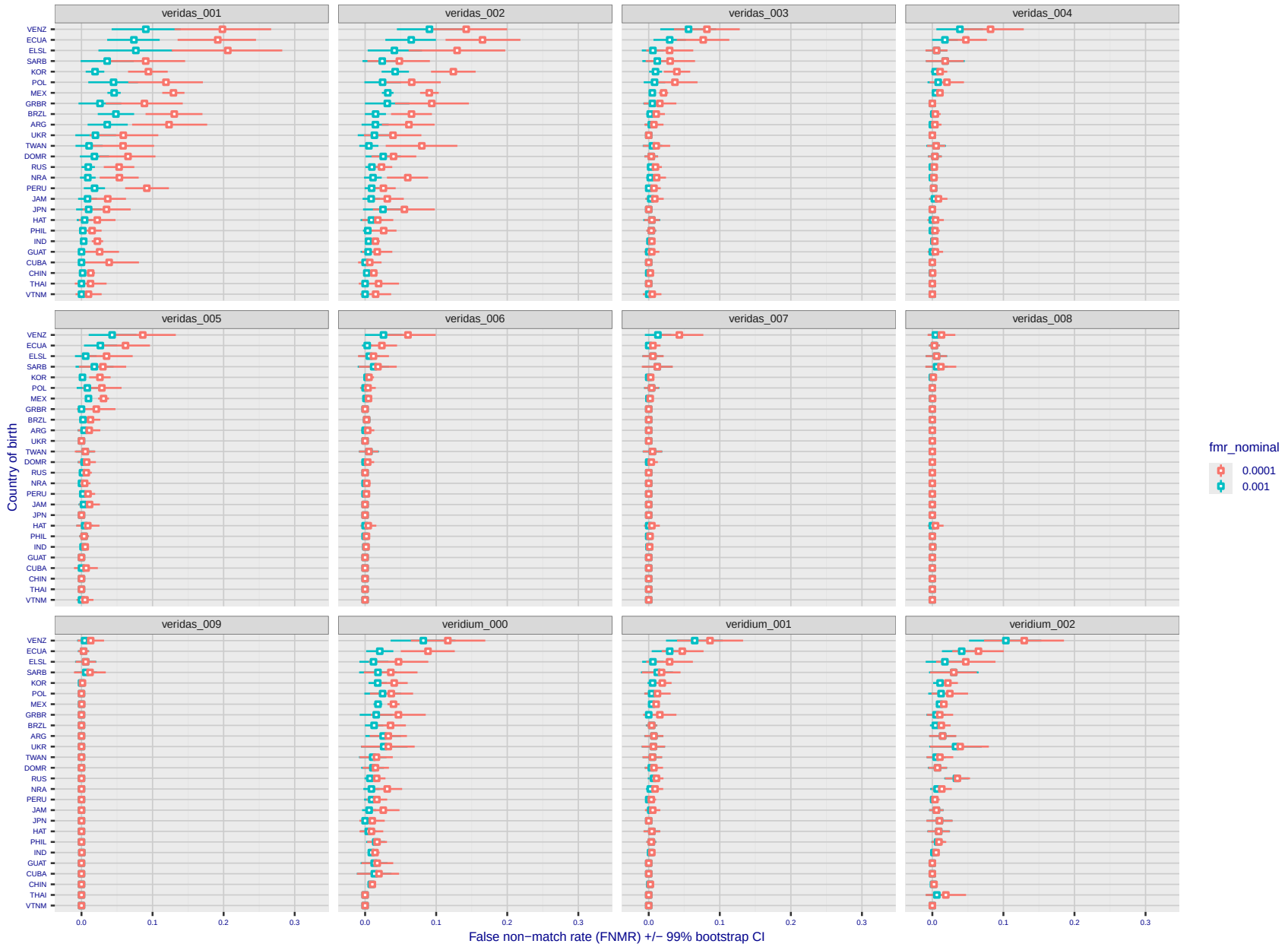


Figure 709: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

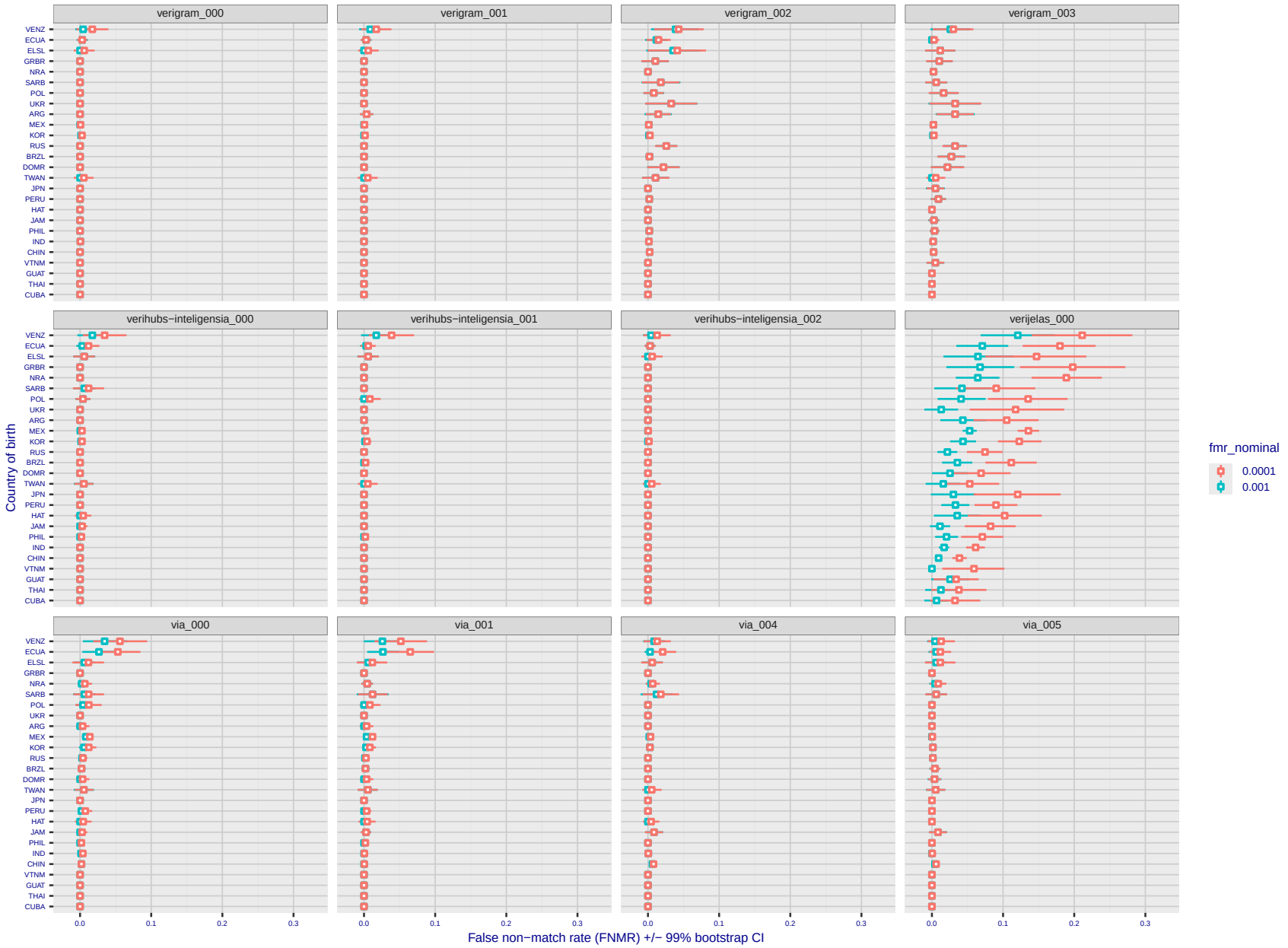


Figure 710: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

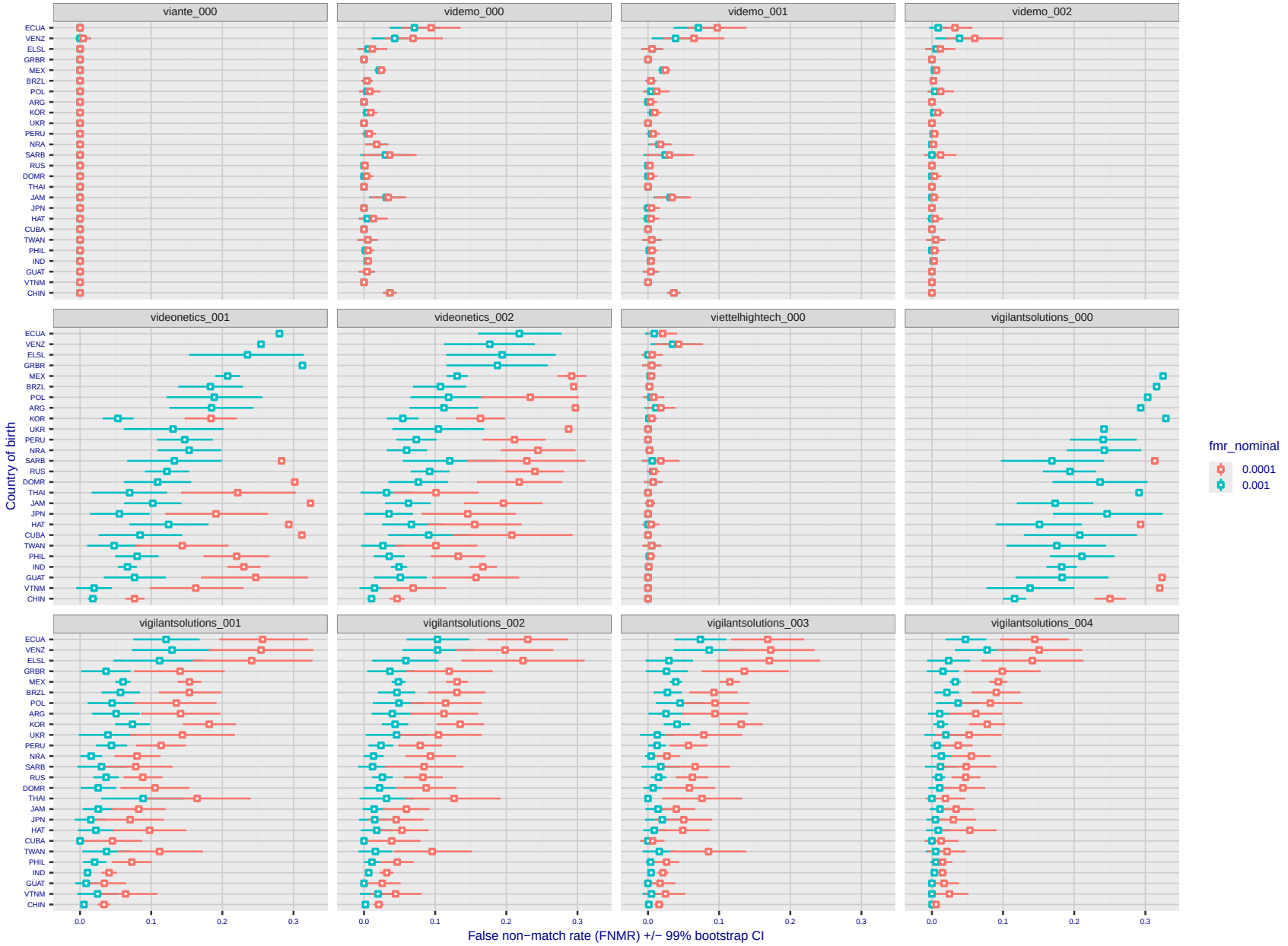


Figure 711: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 712: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

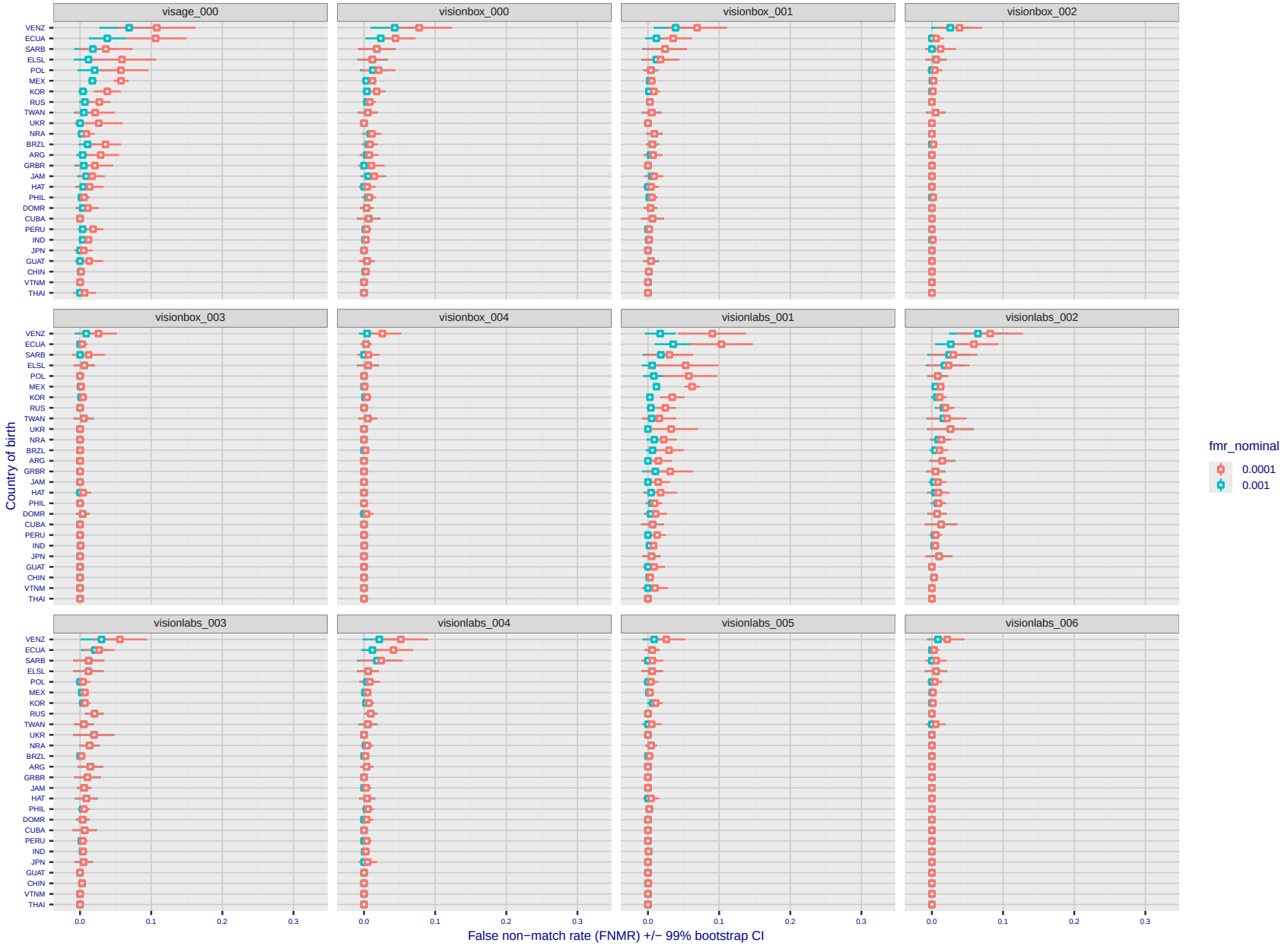


Figure 713: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

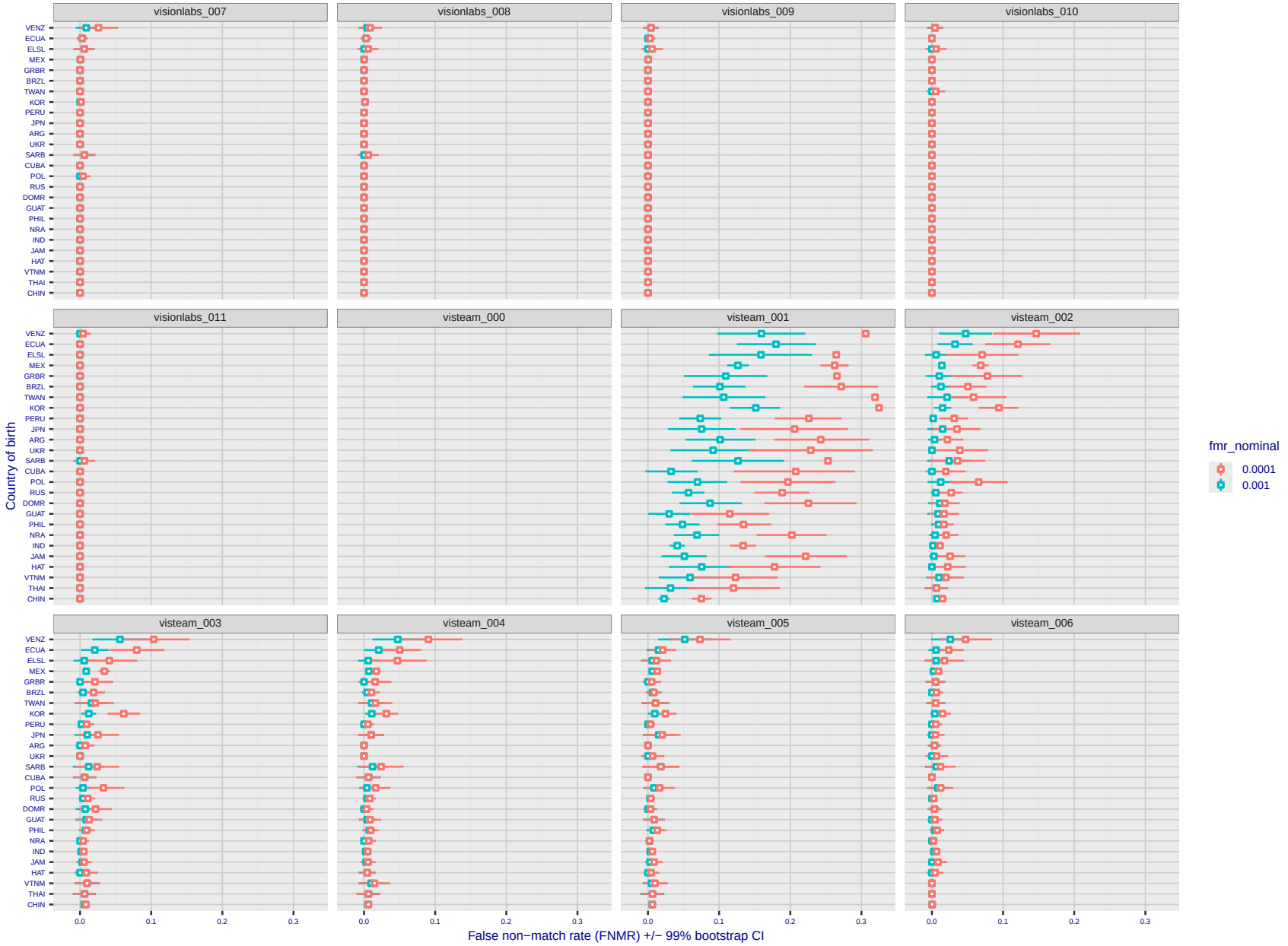


Figure 714: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 715: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

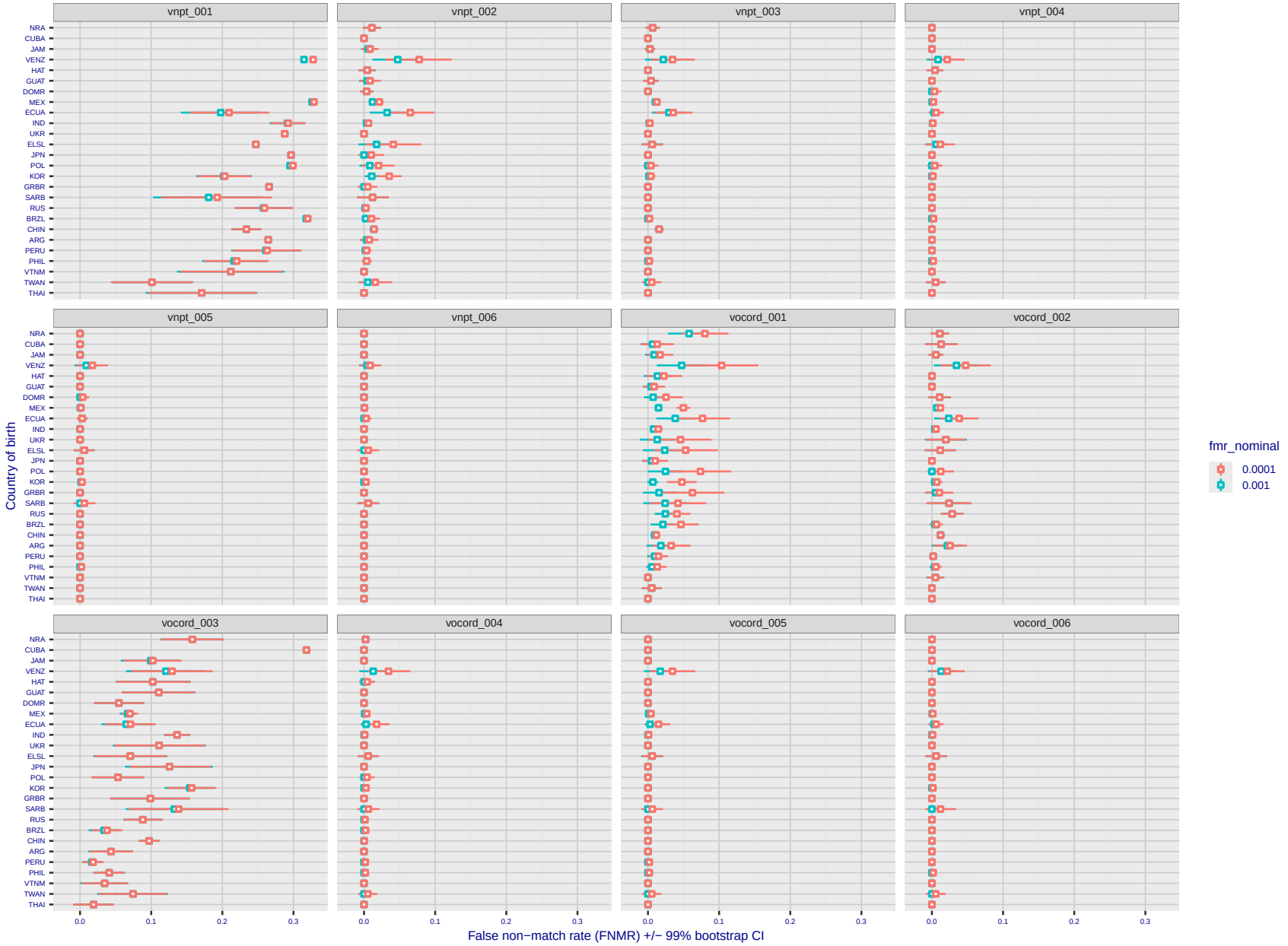


Figure 716: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

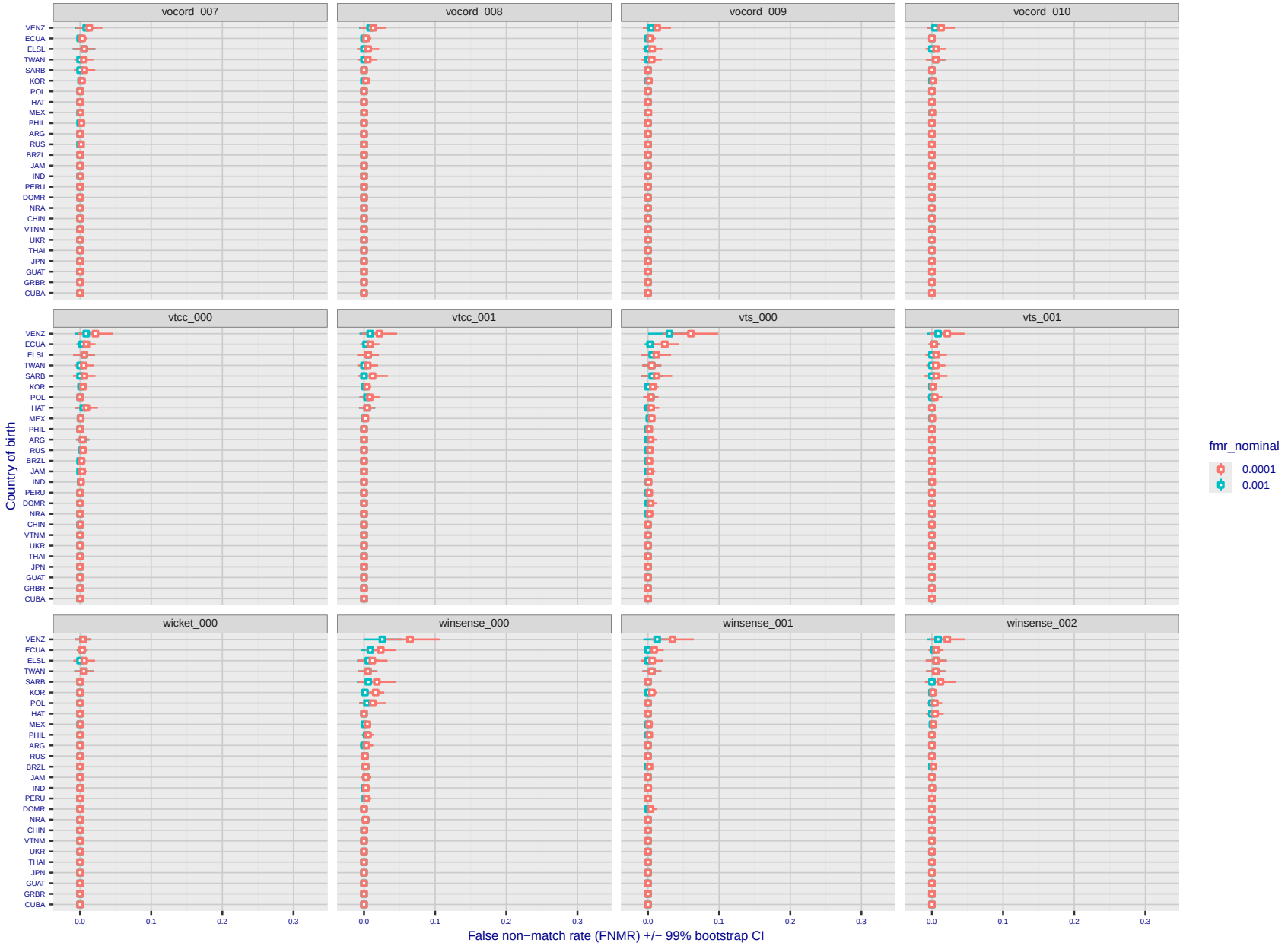


Figure 717: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

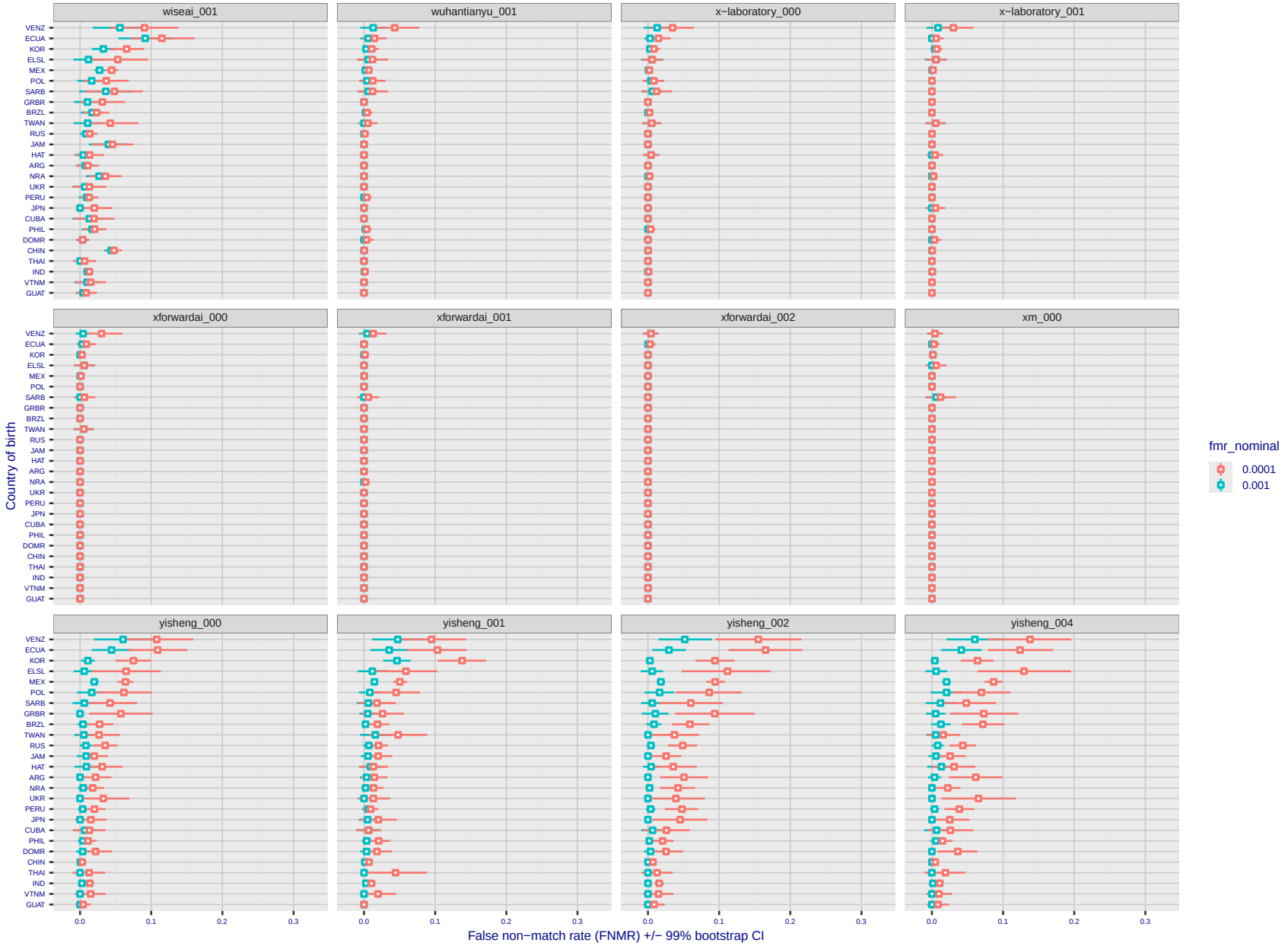


Figure 718: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

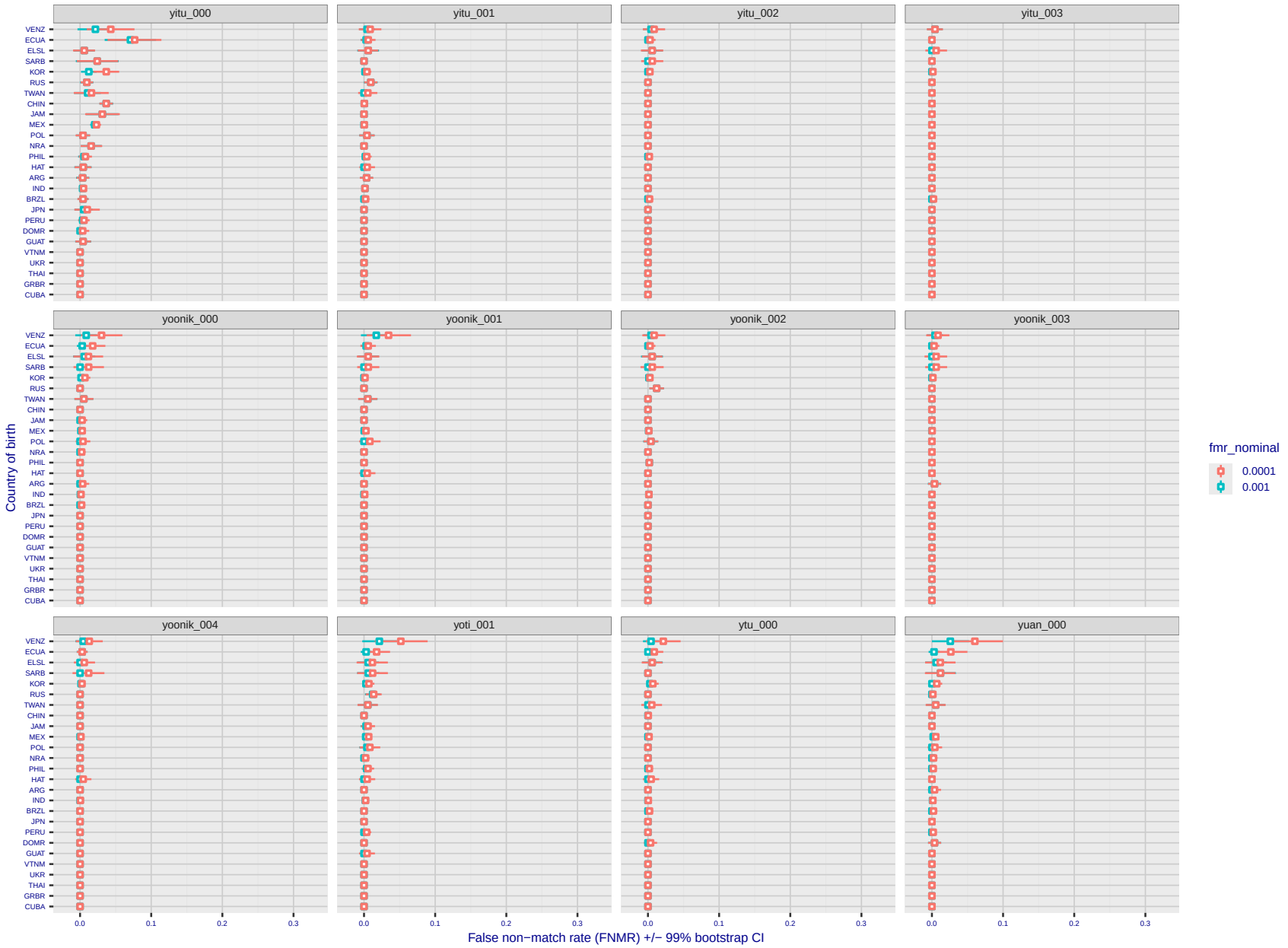


Figure 719: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

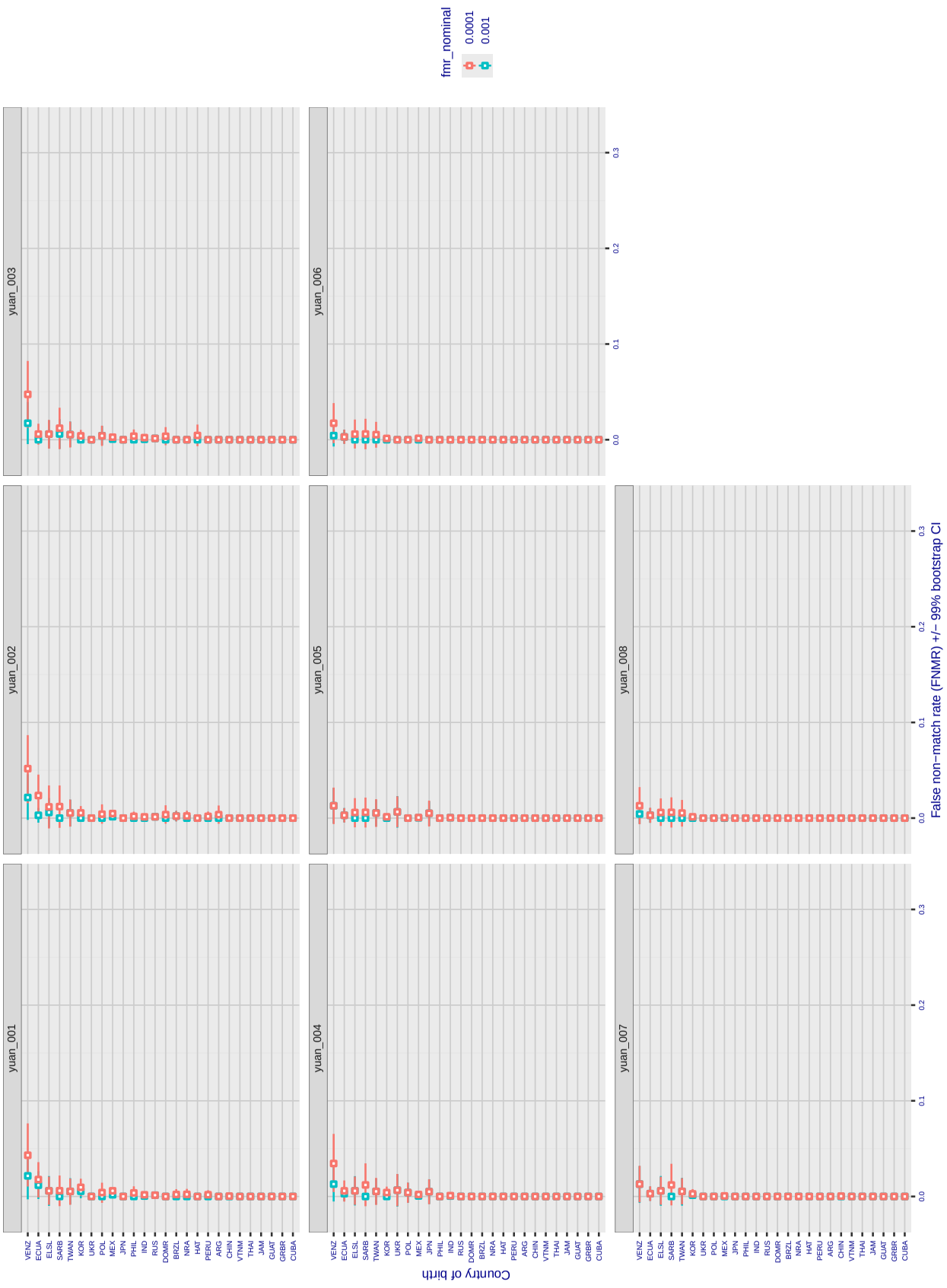


Figure 720: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Caveats: The results may not relate to subject-specific properties. Instead they could reflect image-specific quality differences, which could occur due to collection protocol or software processing variations.

3.5.2 Effect of ageing

Background: Faces change appearance throughout life. This change gradually reduces similarity of a new image to an earlier image. Face recognition algorithms give reduced similarity scores and more frequent false rejections.

Goal: To quantify false non-match rates (FNMR) as a function of elapsed time in an adult population.

Methods: Using the mugshot images, a threshold is set to give FMR = 0.00001 over the entire impostor set. Then FNMR is measured over 1000 bootstrap replications of the genuine scores.

Results: For the visa images, Figure 807 shows how false non-match rates for genuine users, as a function of age group.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

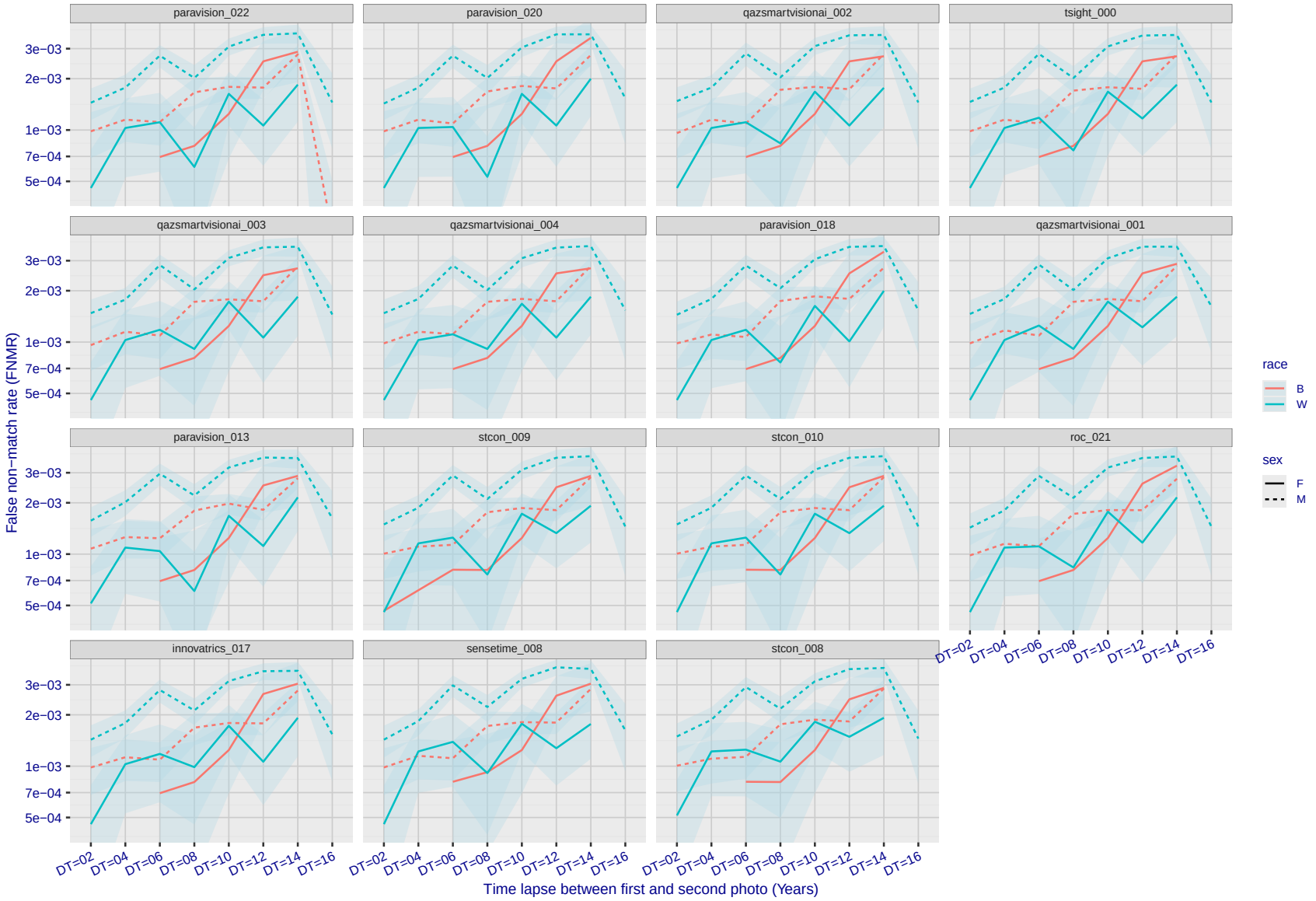


Figure 721: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

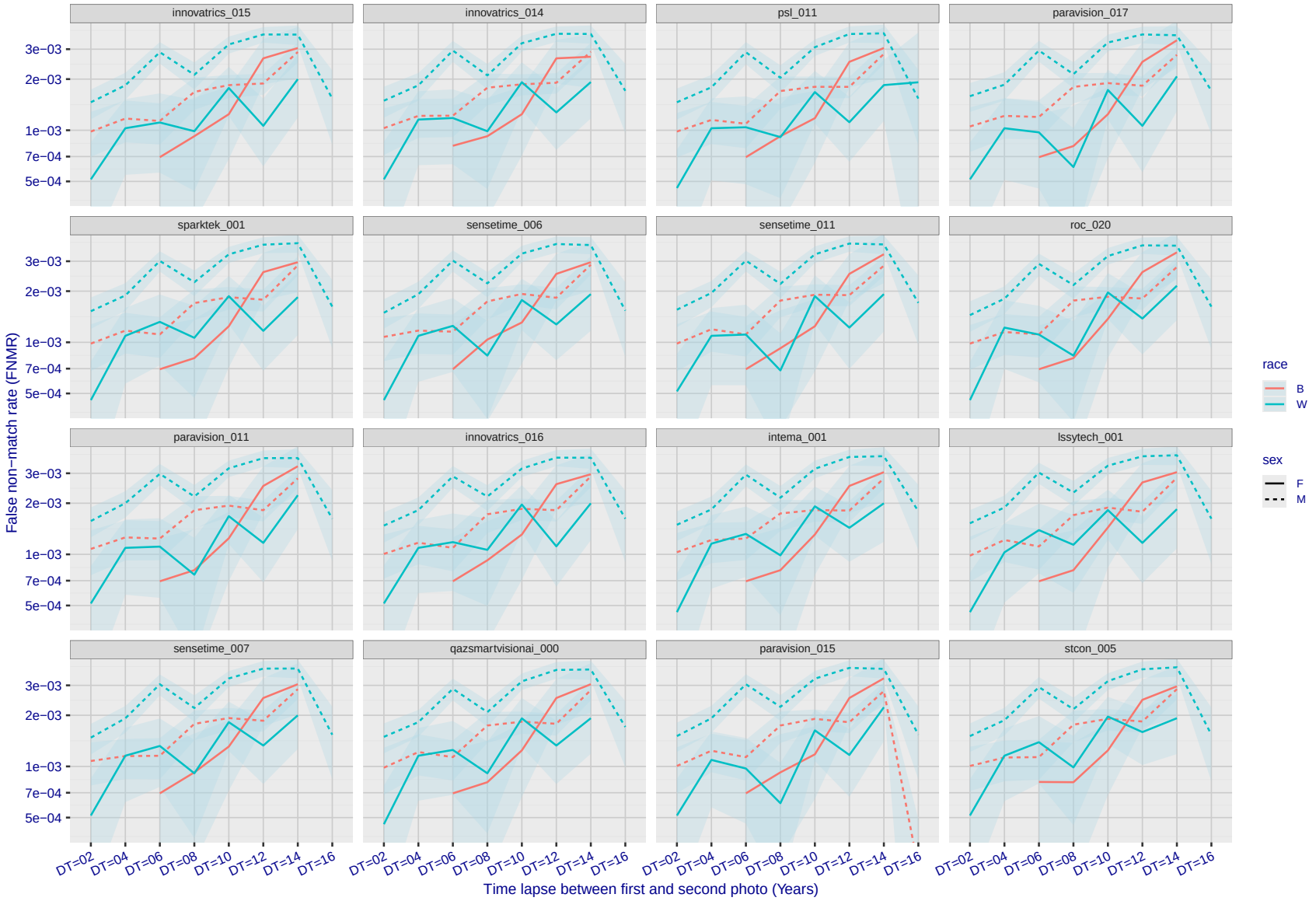


Figure 722: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

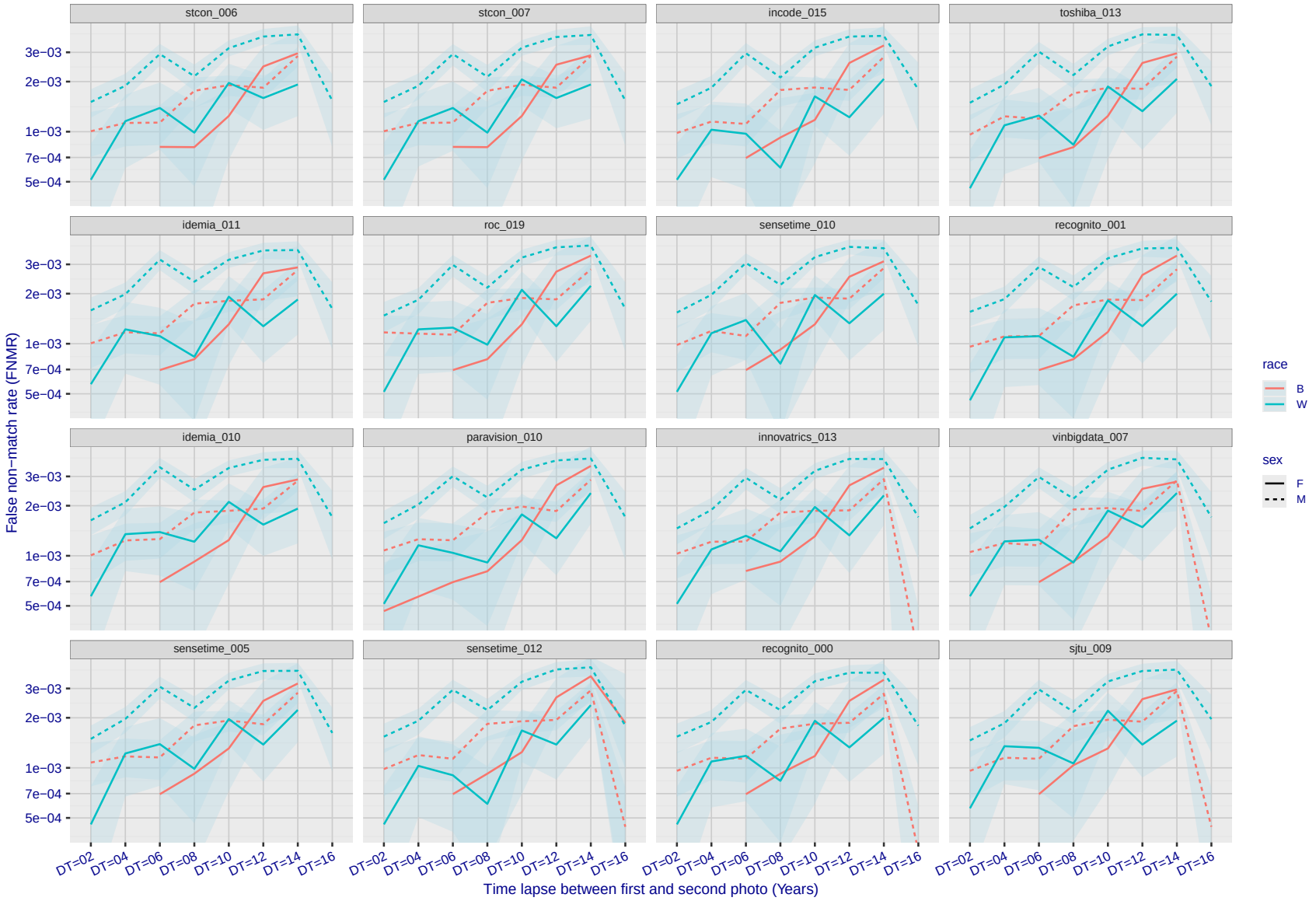


Figure 723: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

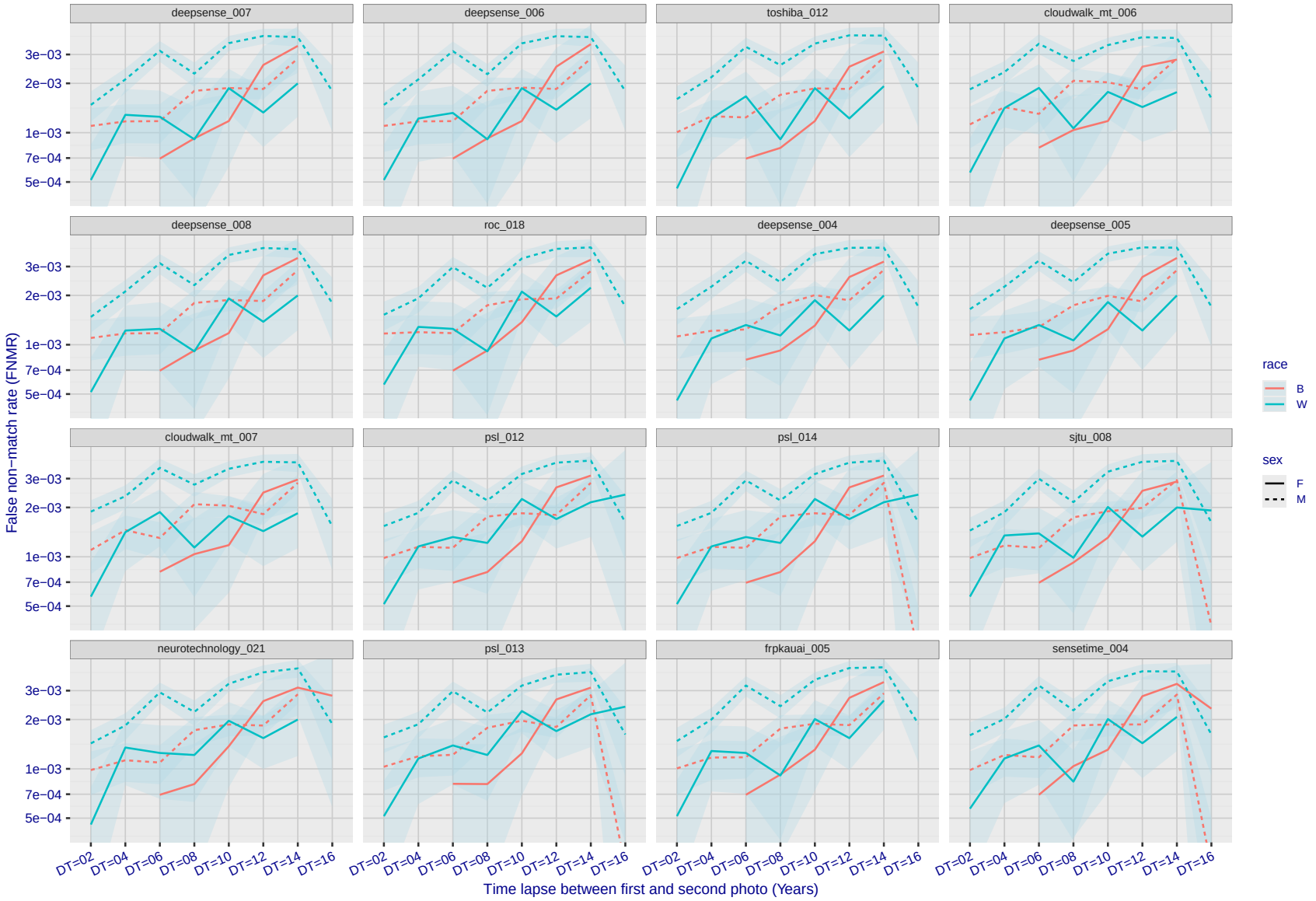


Figure 724: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\tilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

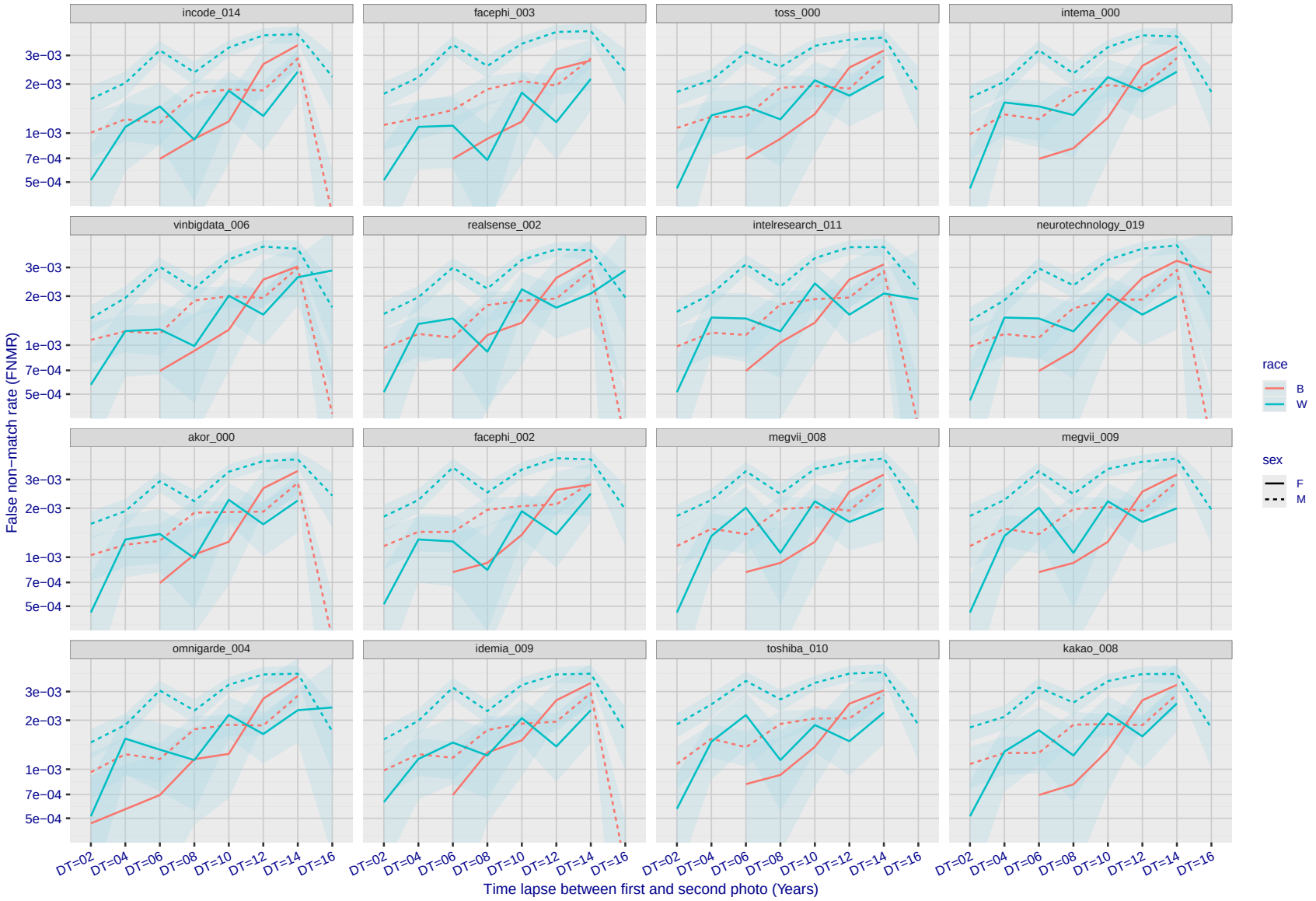


Figure 725: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

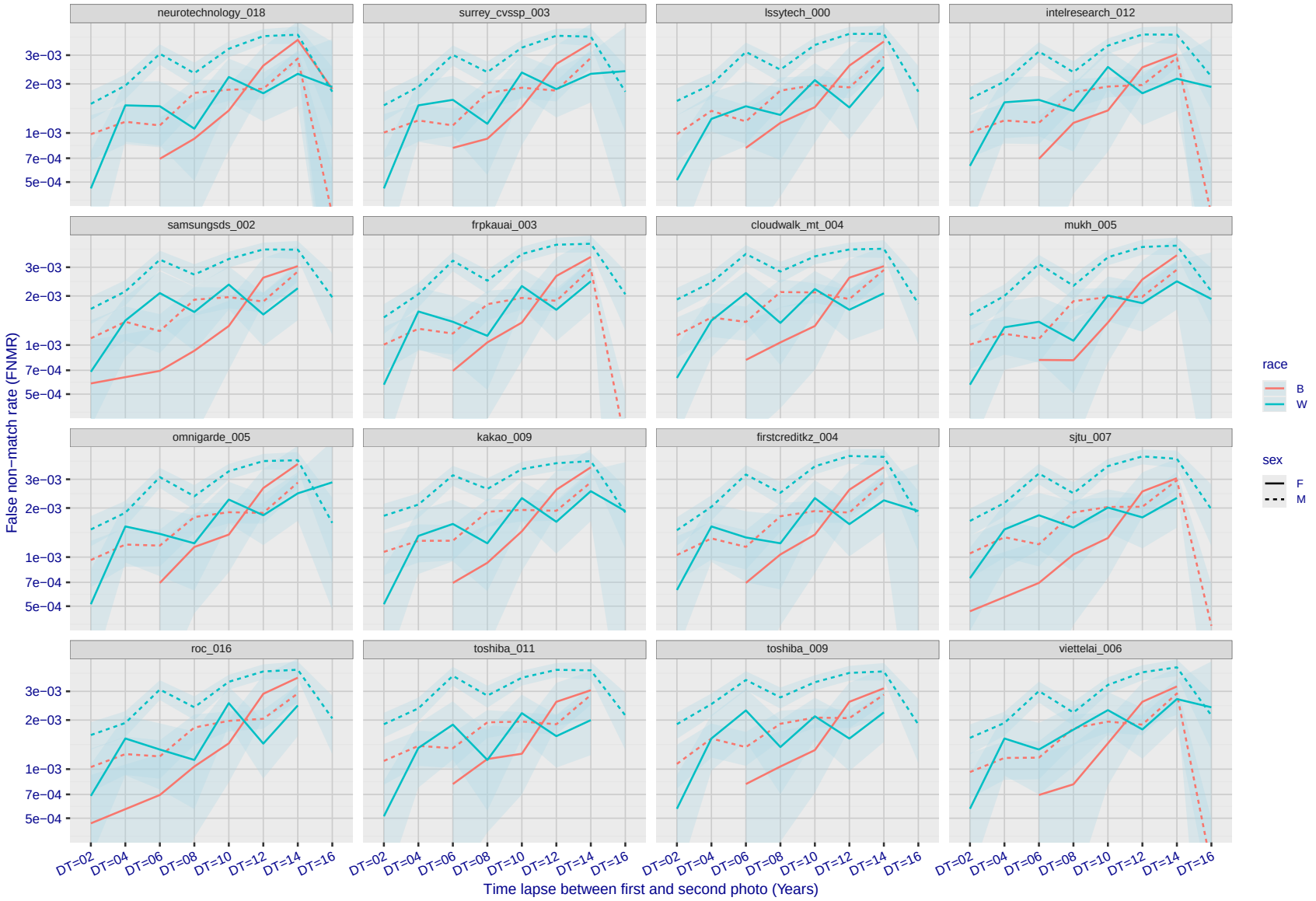


Figure 726: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\bar{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

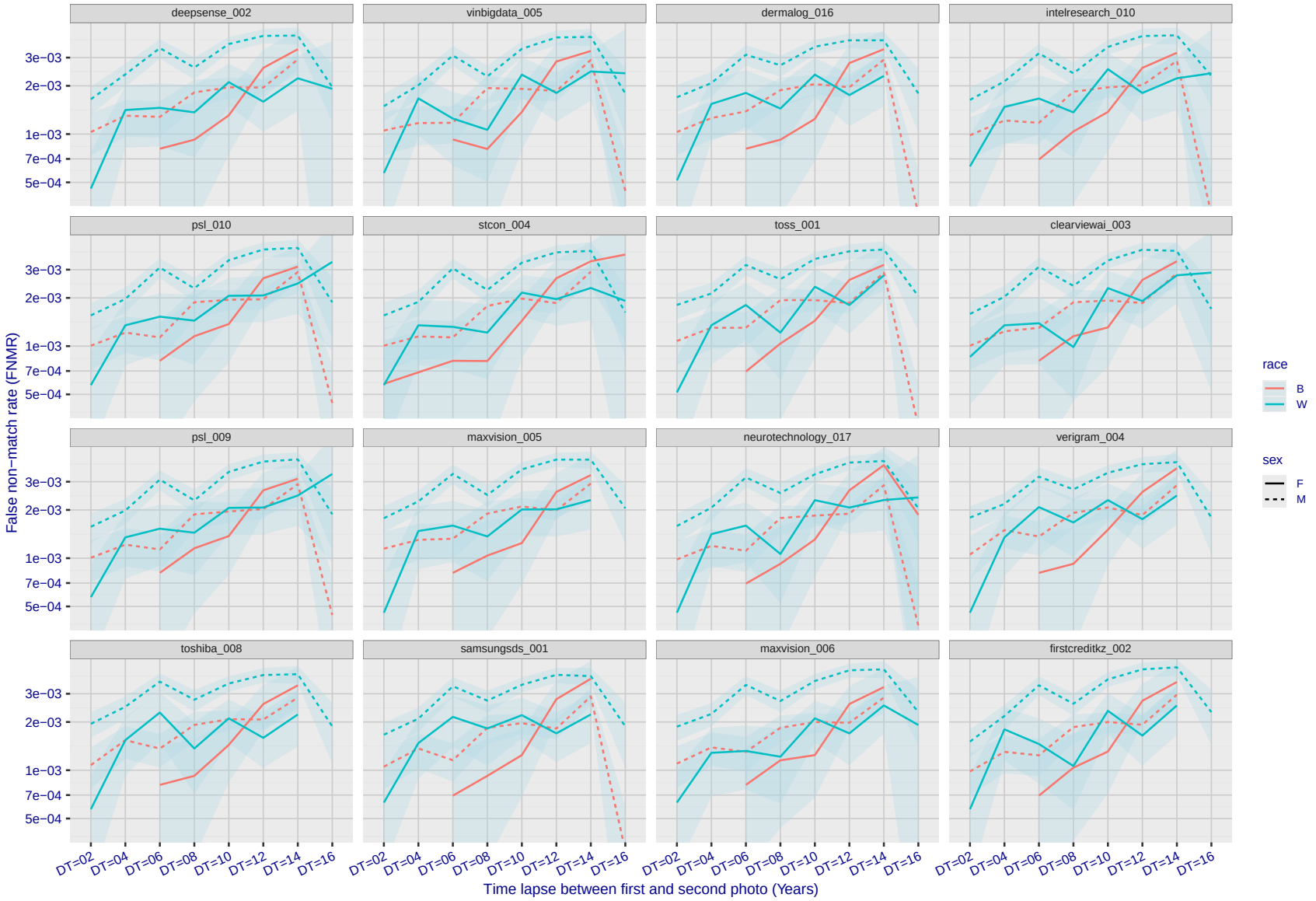


Figure 727: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

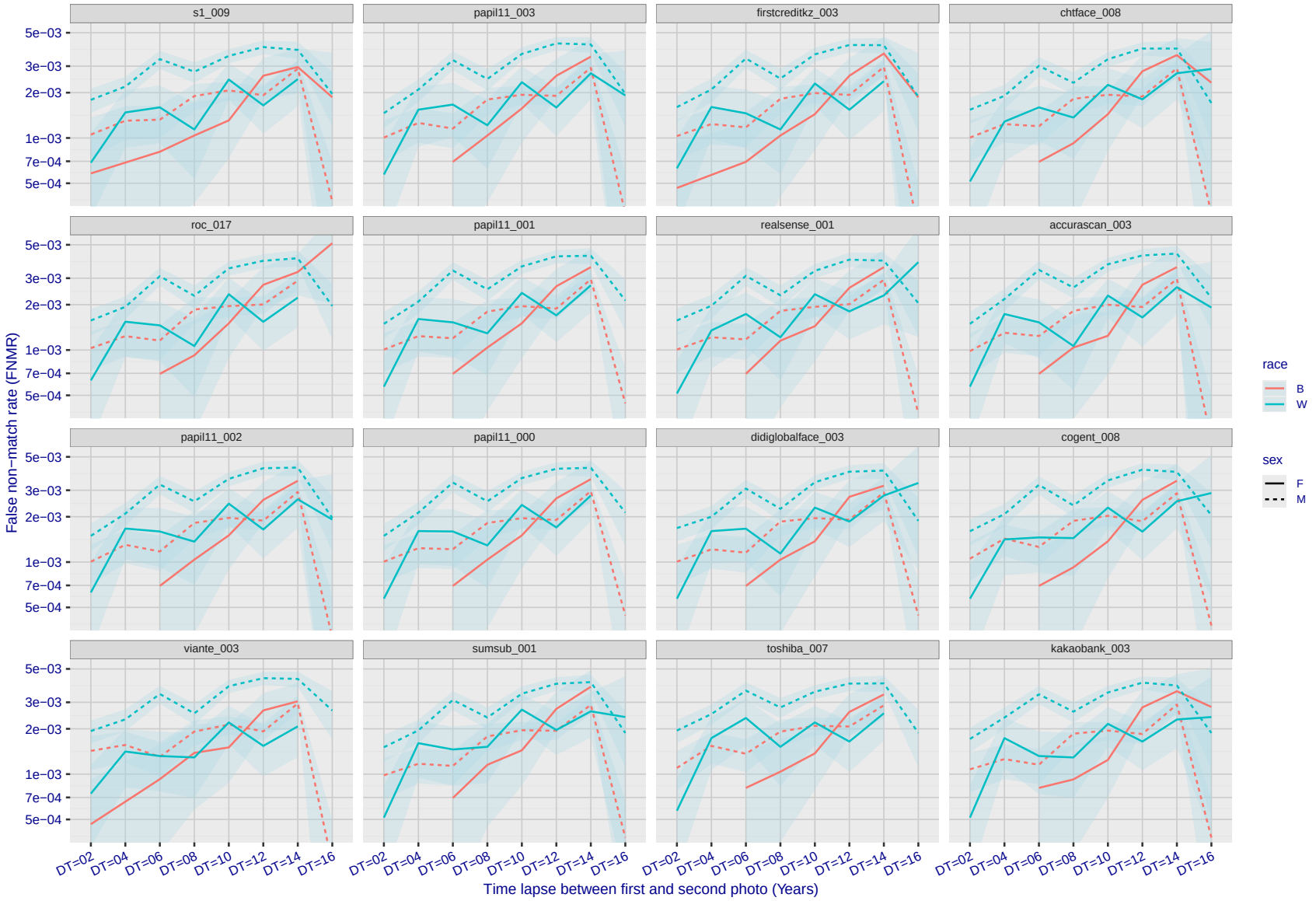


Figure 728: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

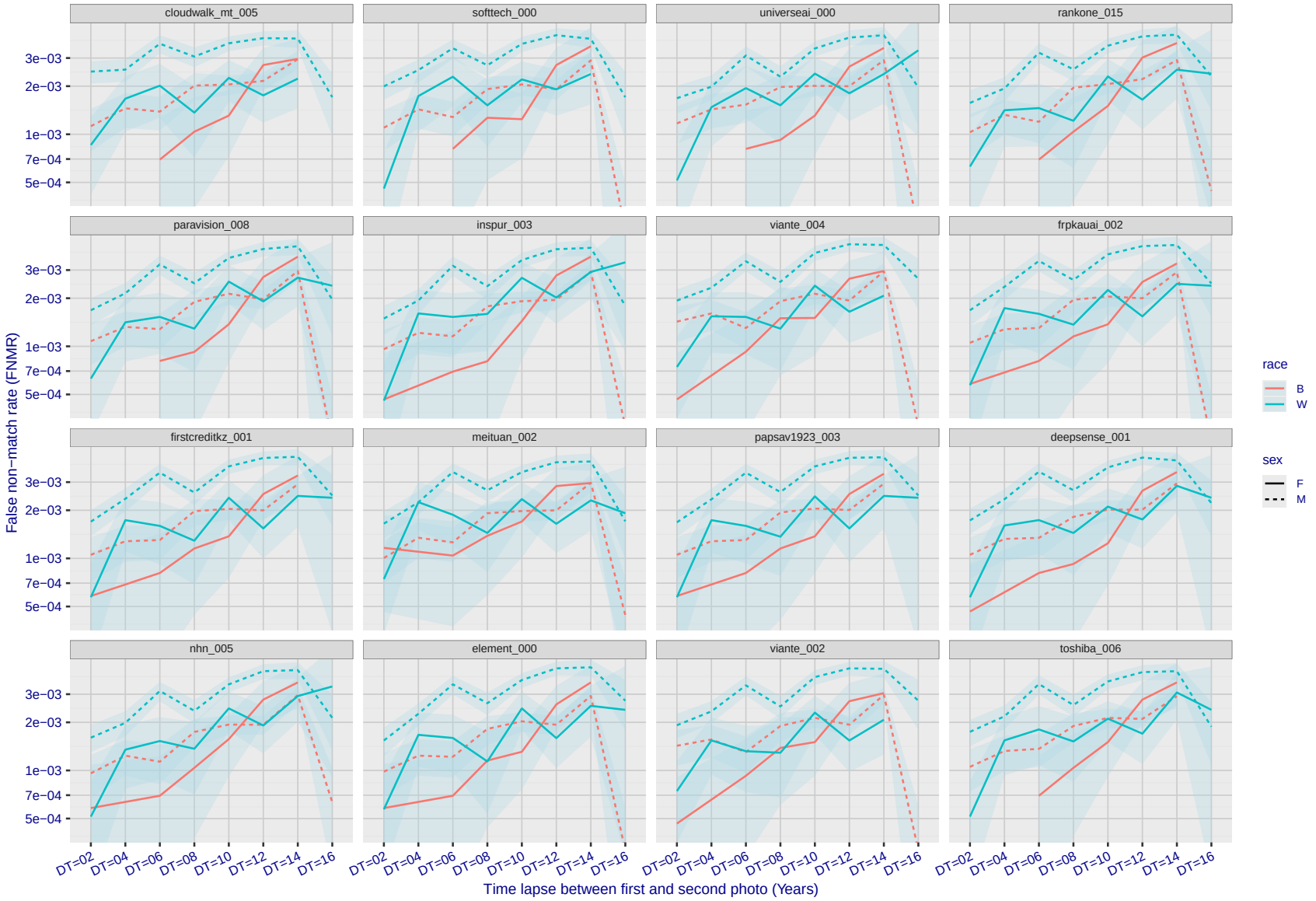


Figure 729: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

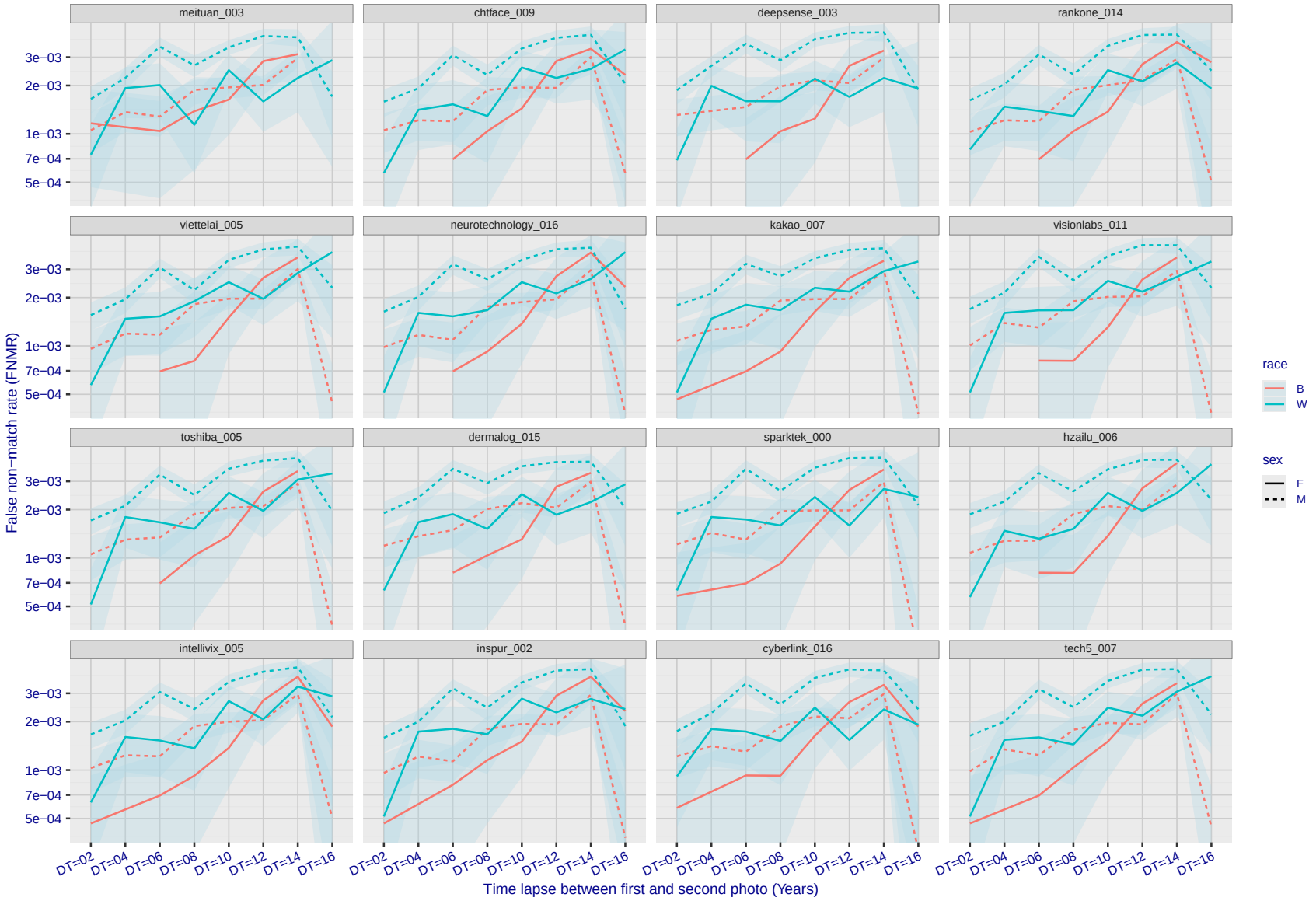


Figure 730: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

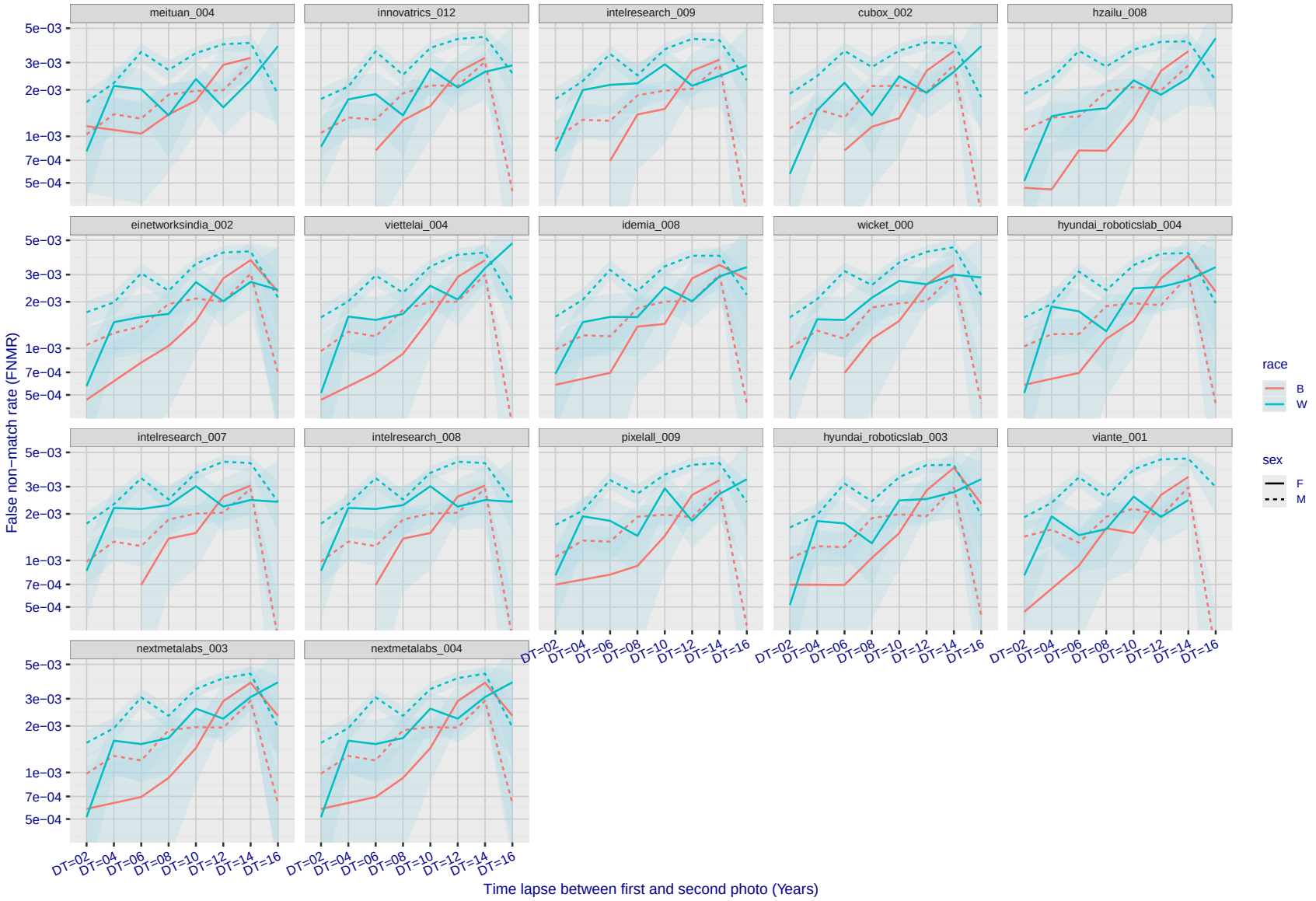


Figure 731: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR* < 0.001*) so that the uncertainty estimates are high.*

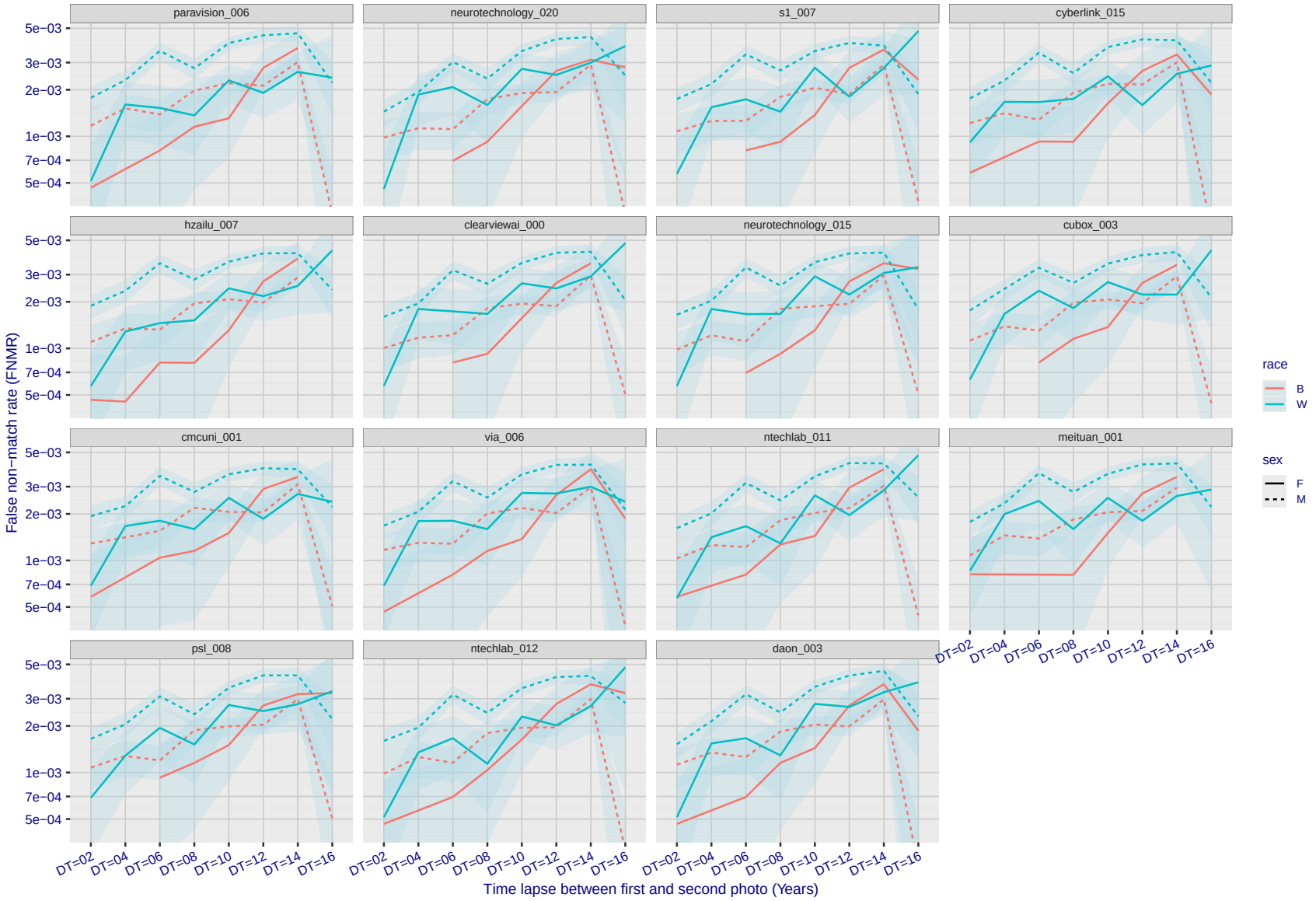


Figure 732: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

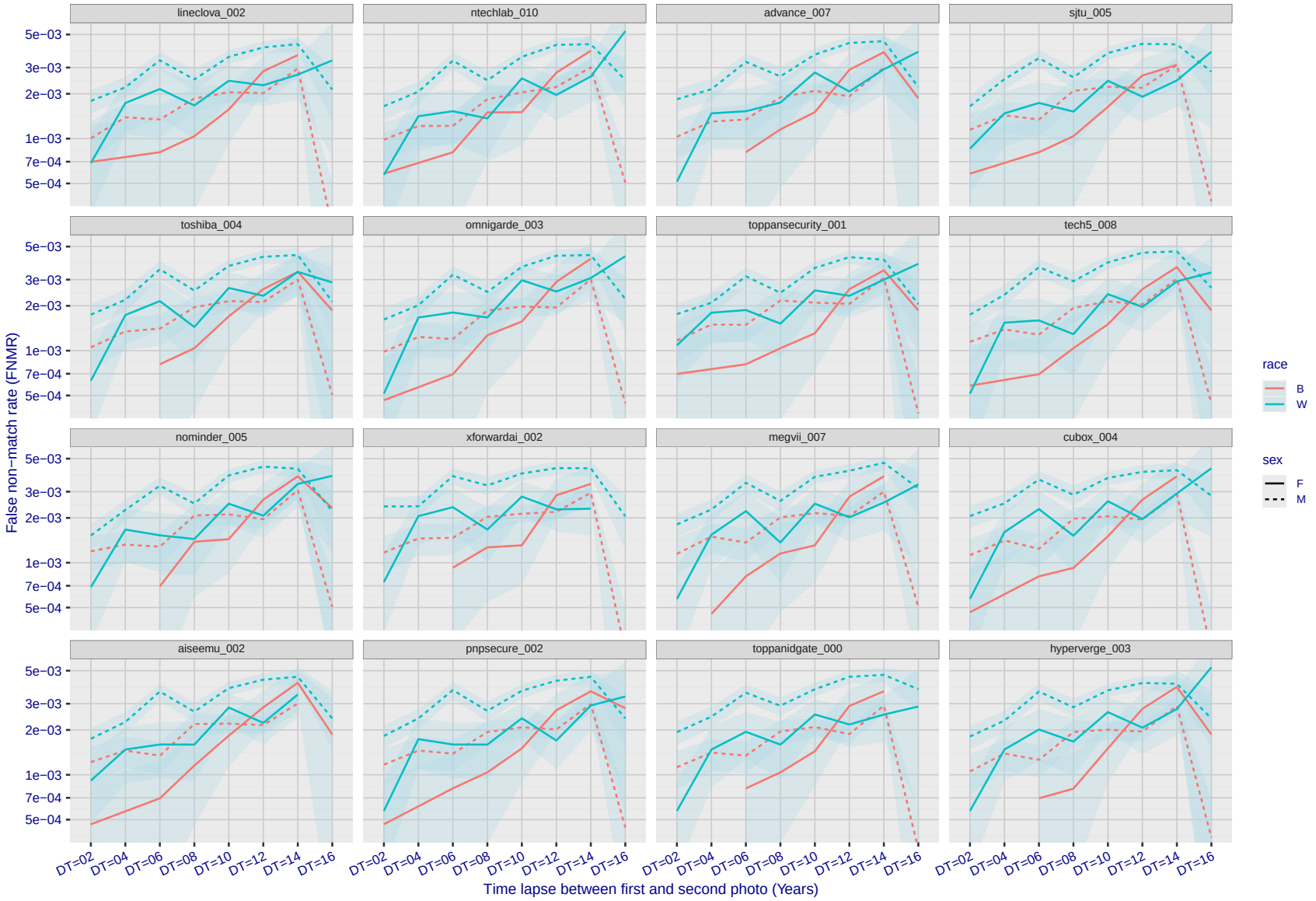


Figure 733: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

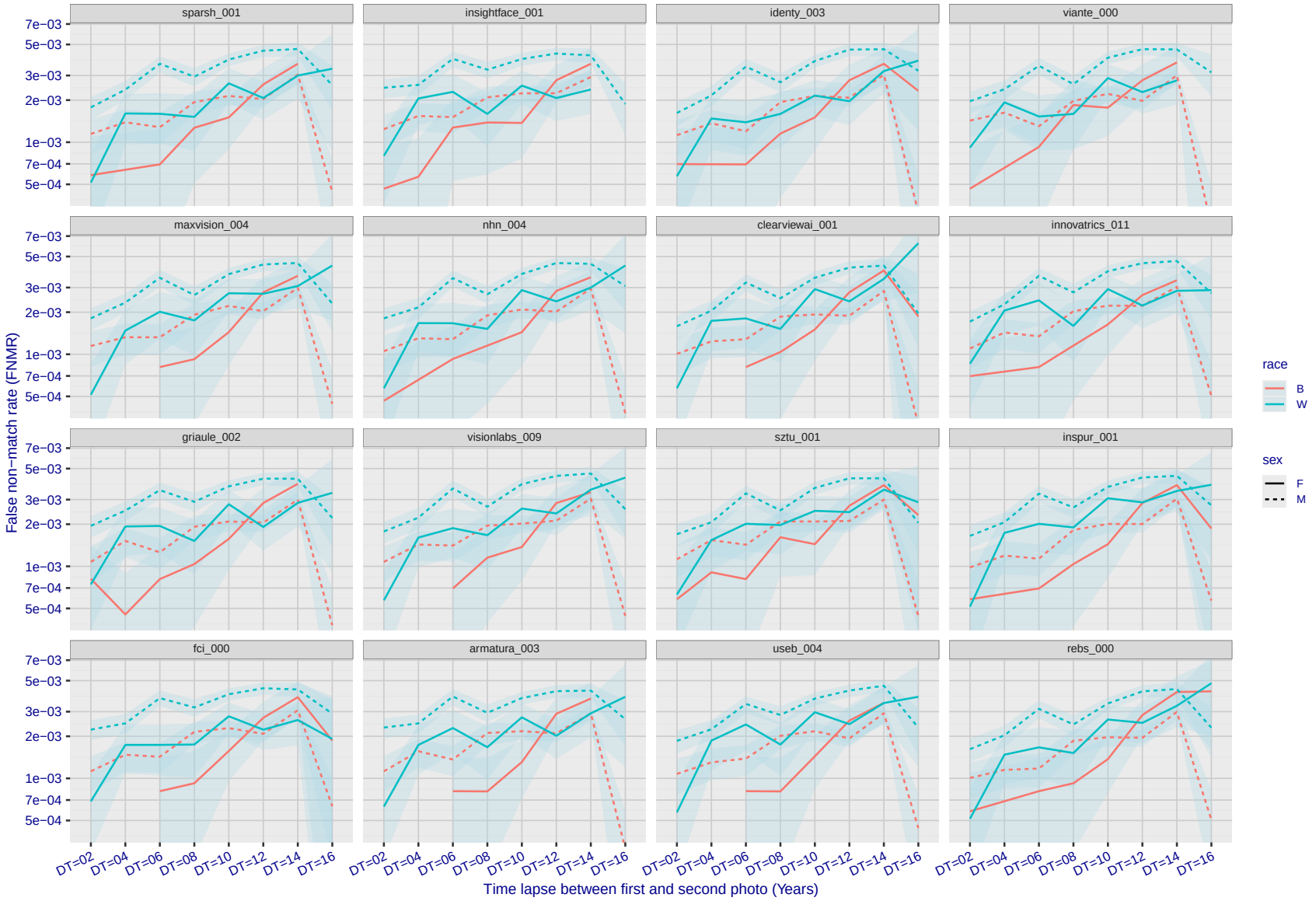


Figure 734: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widehat{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

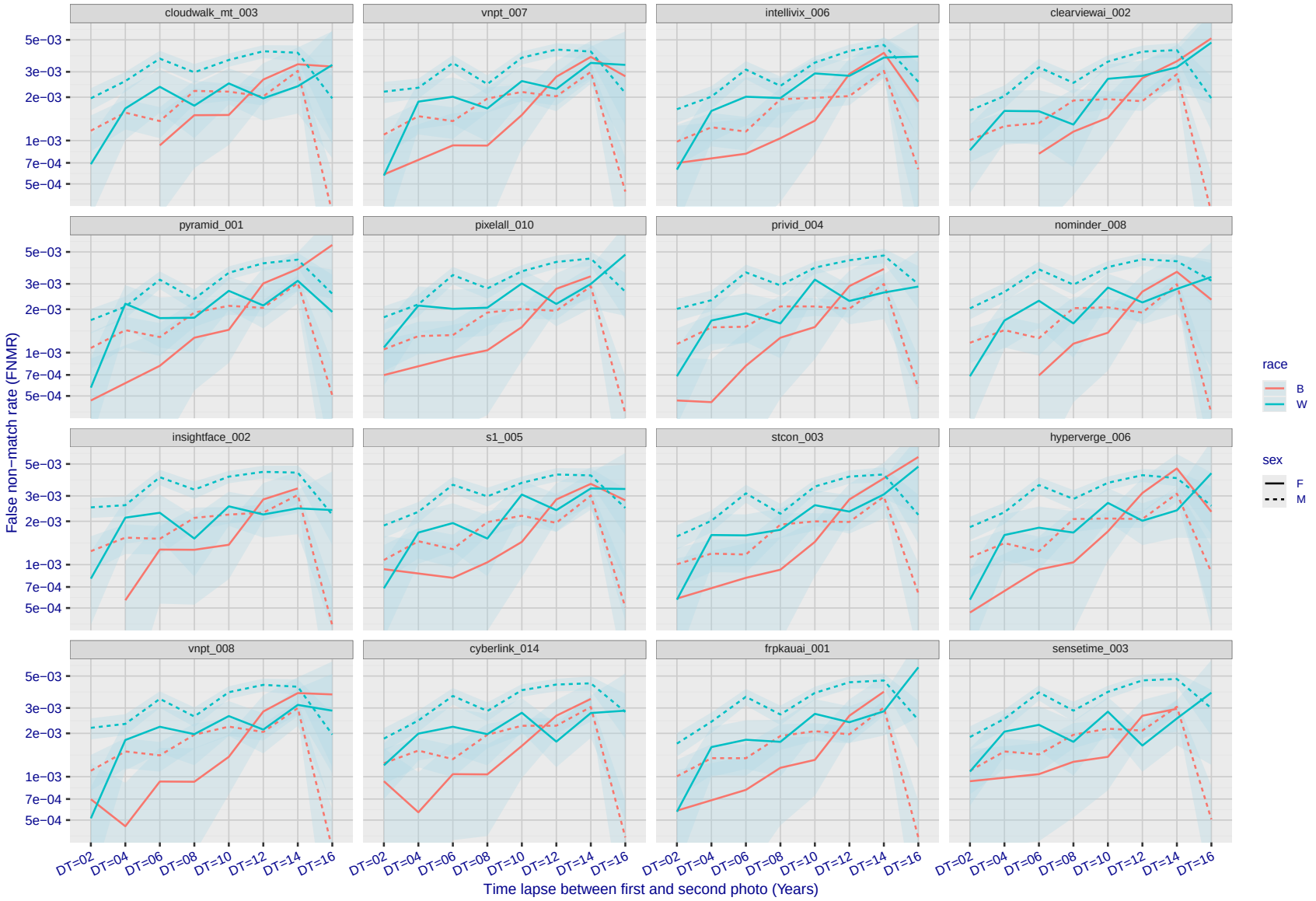


Figure 735: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

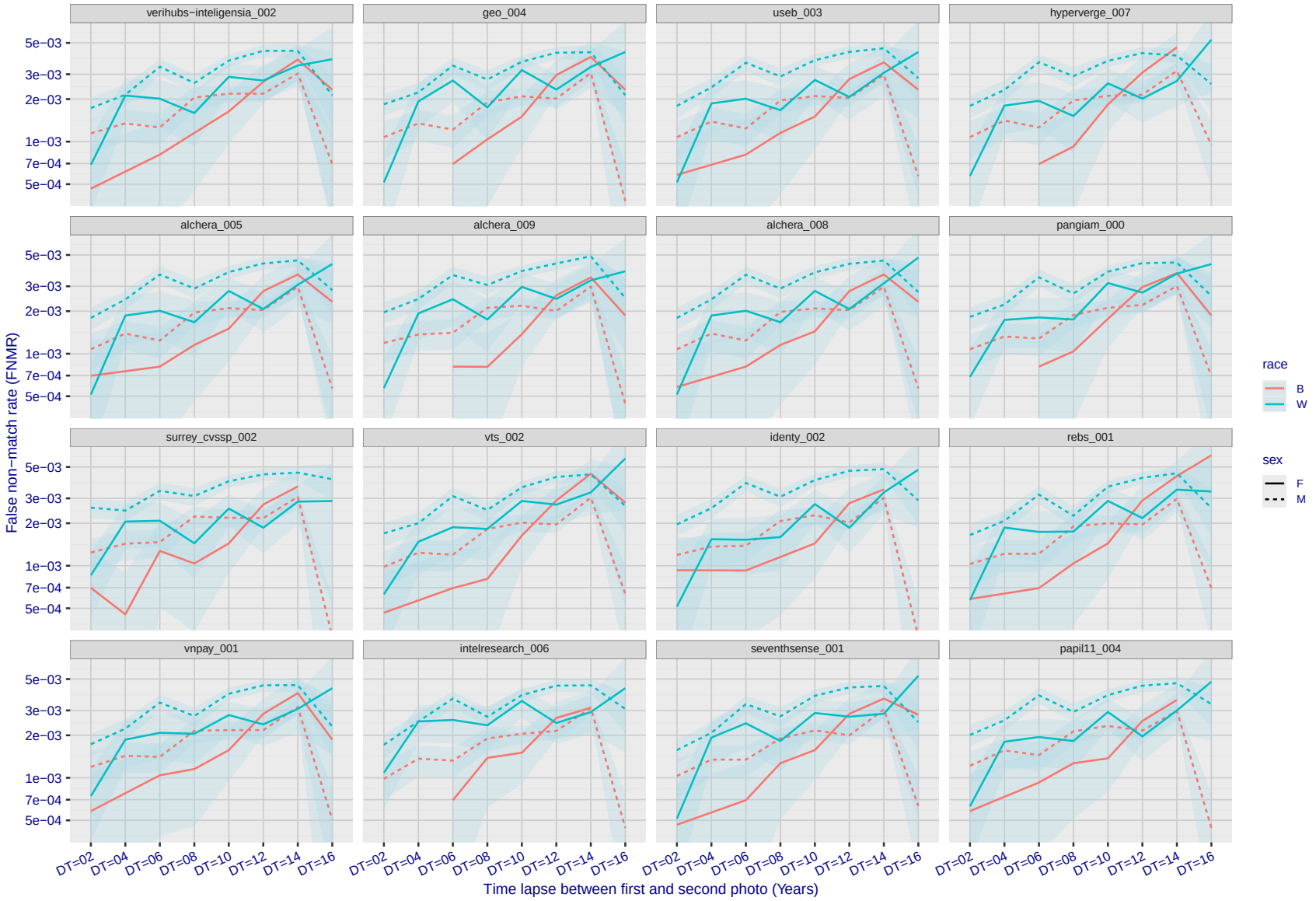


Figure 736: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

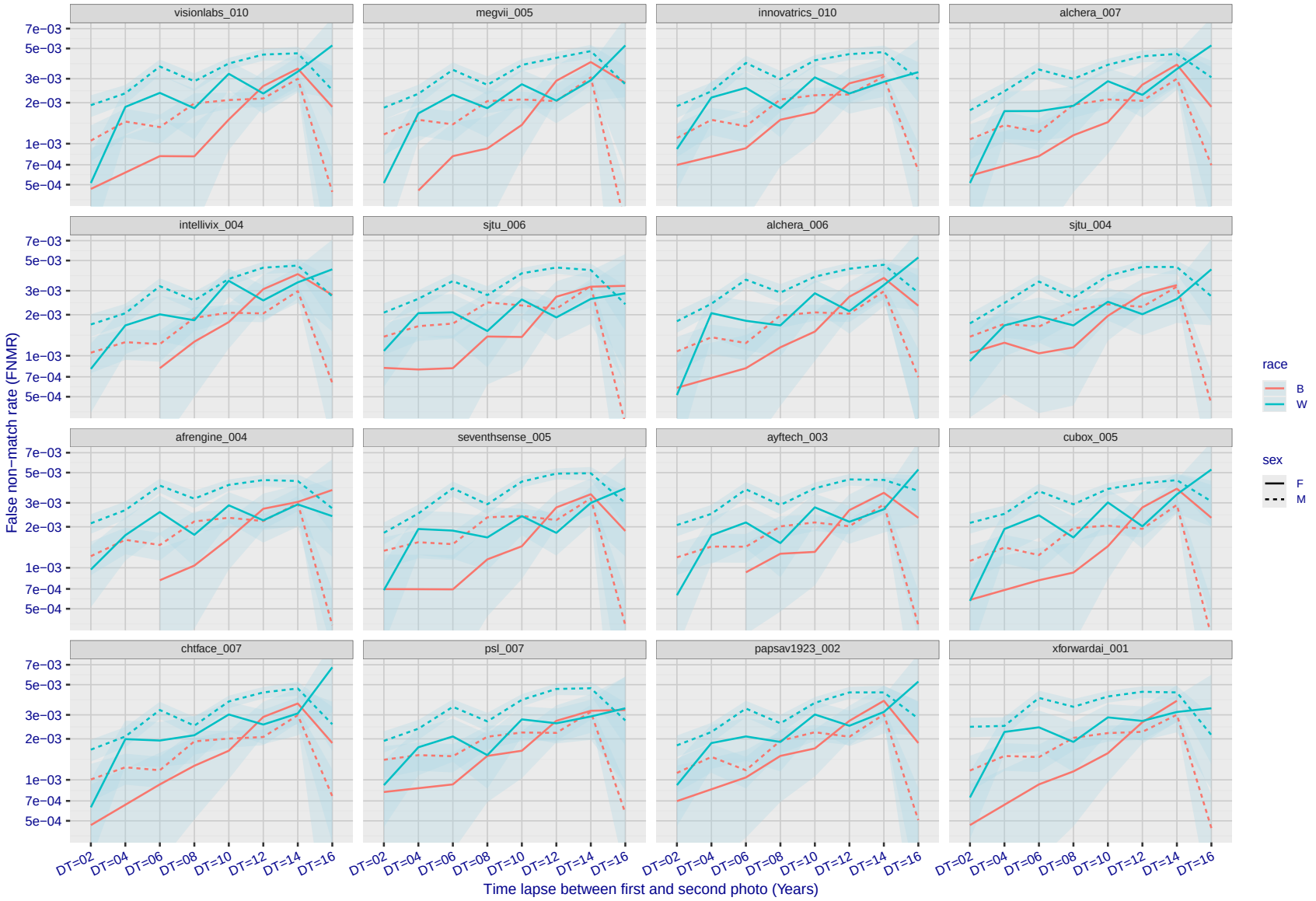


Figure 737: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

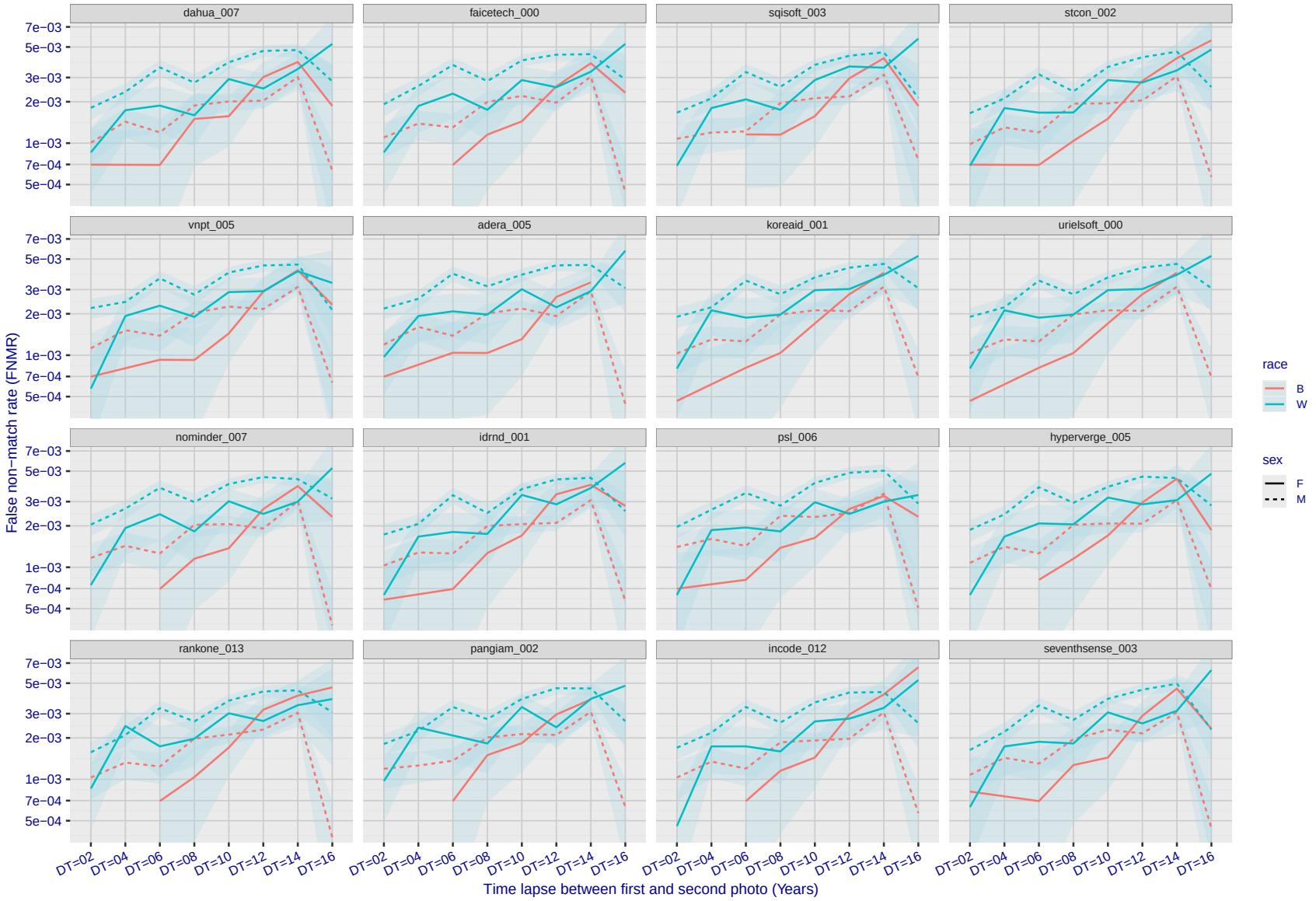


Figure 738: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

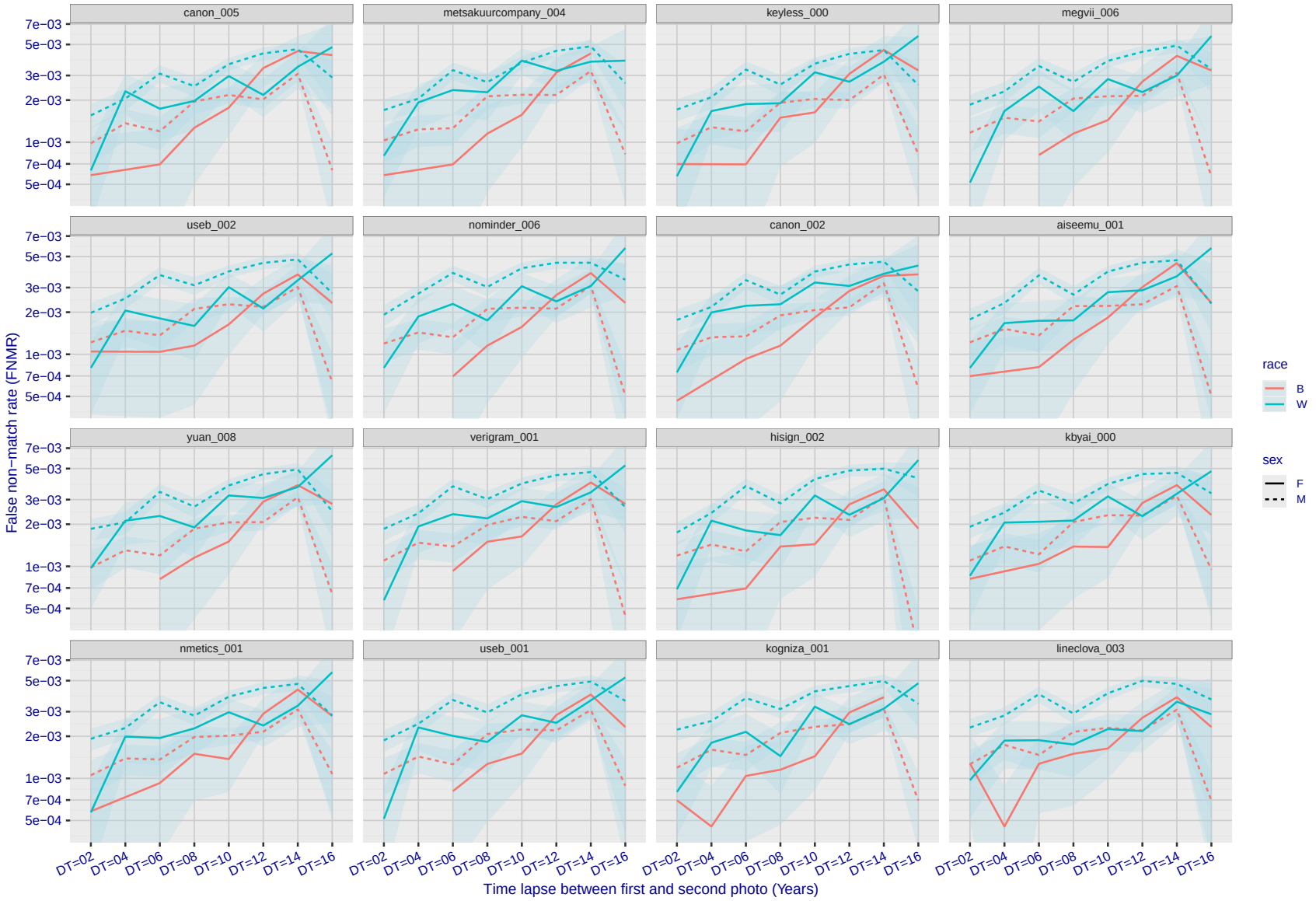


Figure 739: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

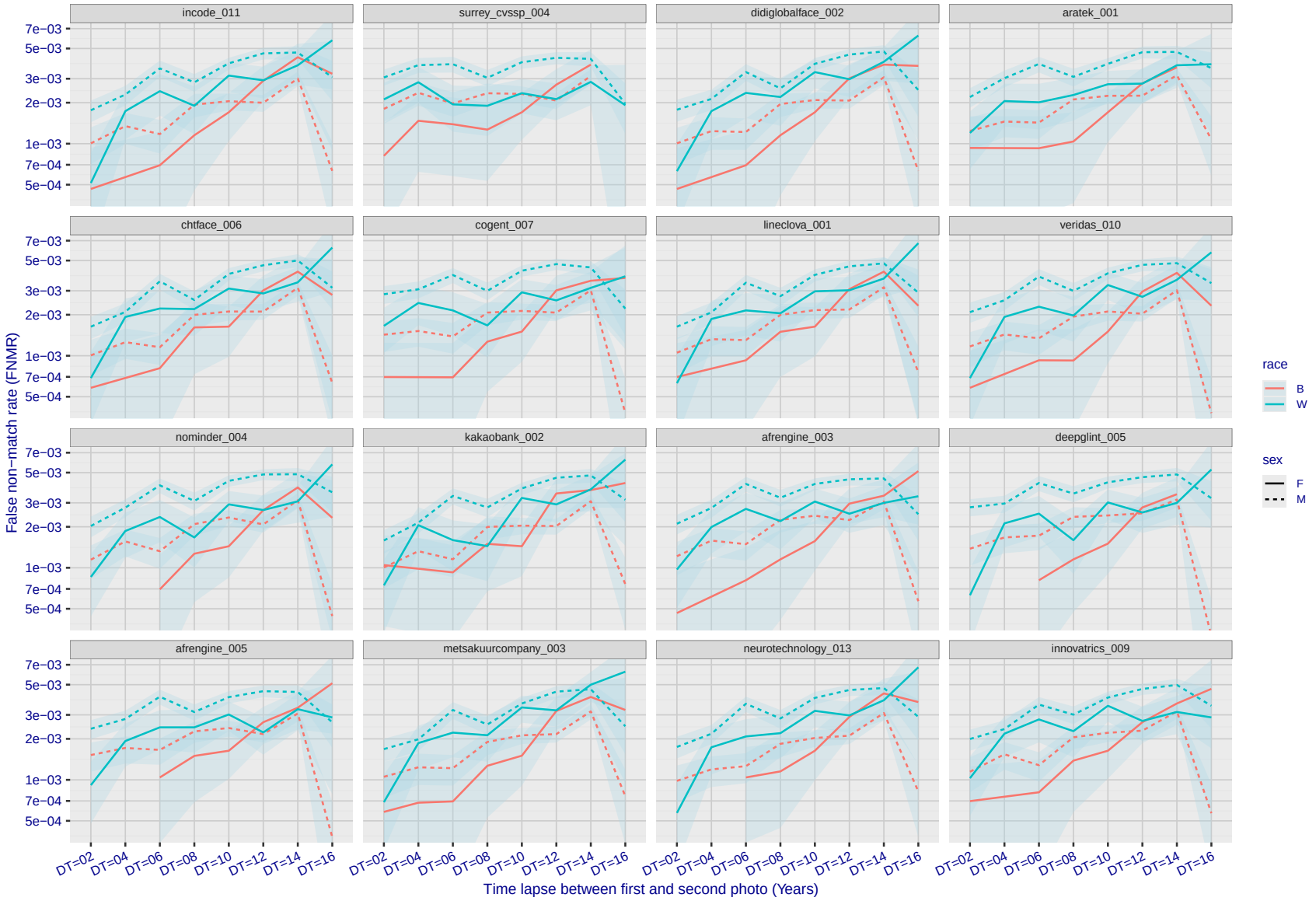


Figure 740: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

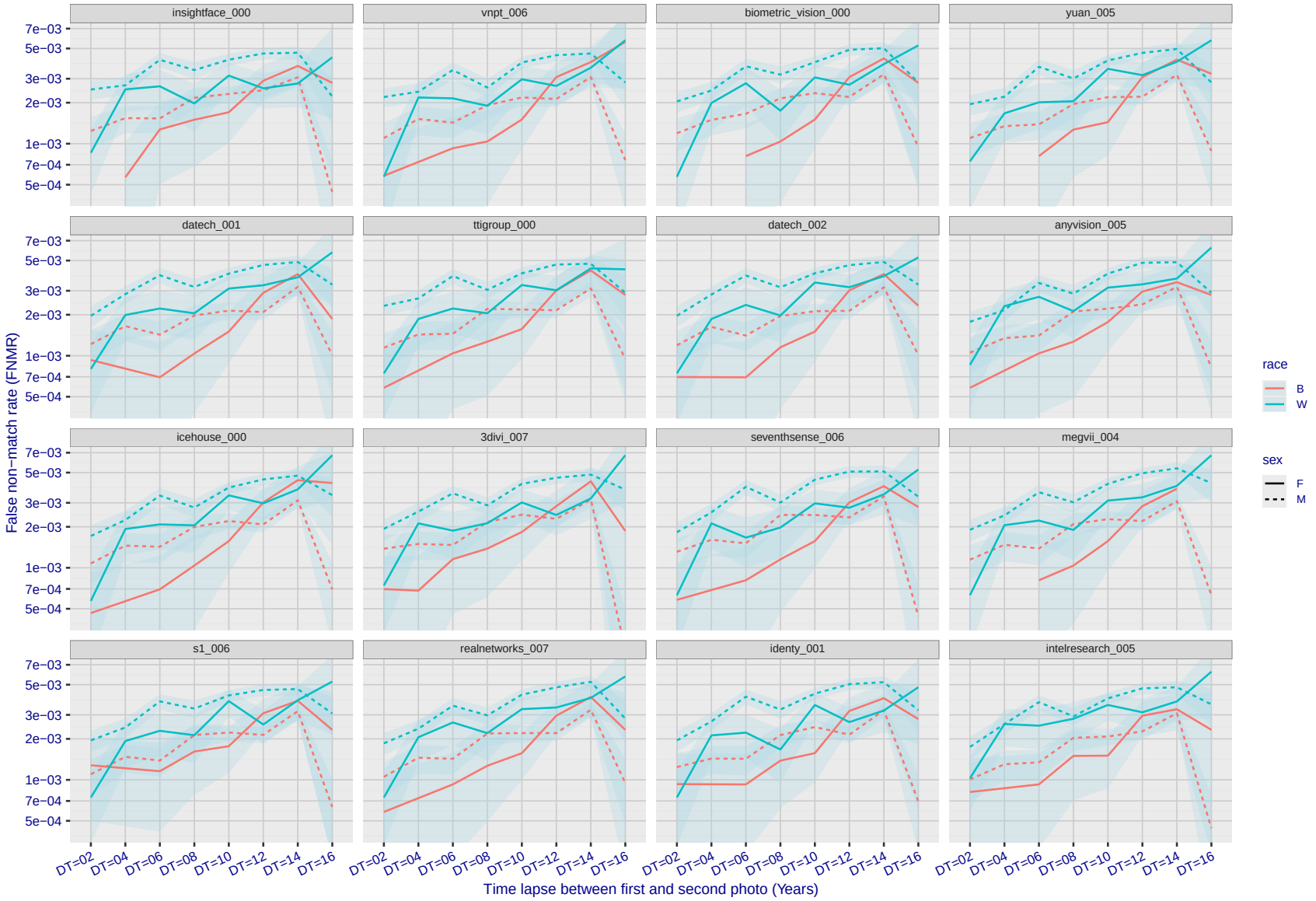


Figure 741: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

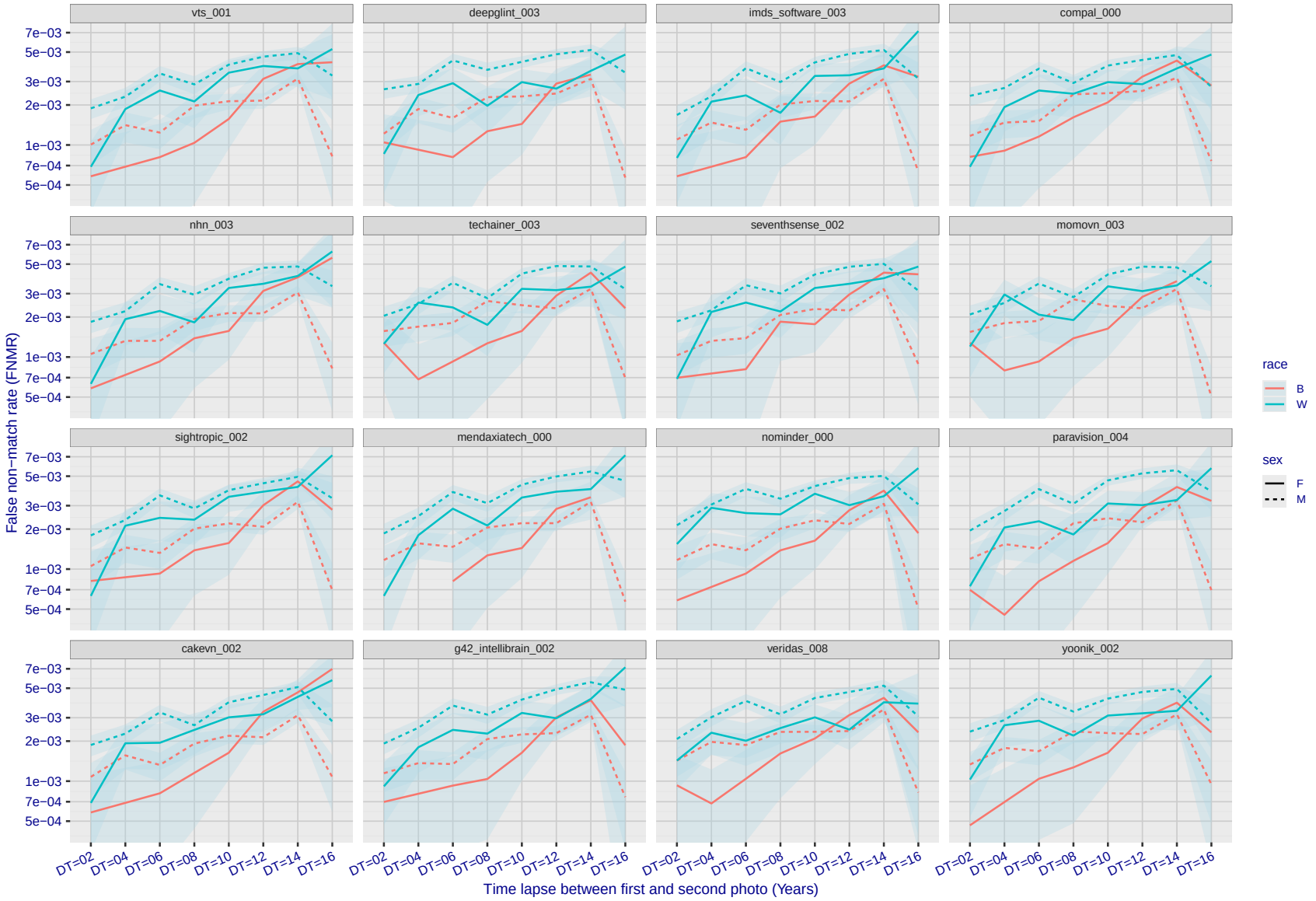


Figure 742: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

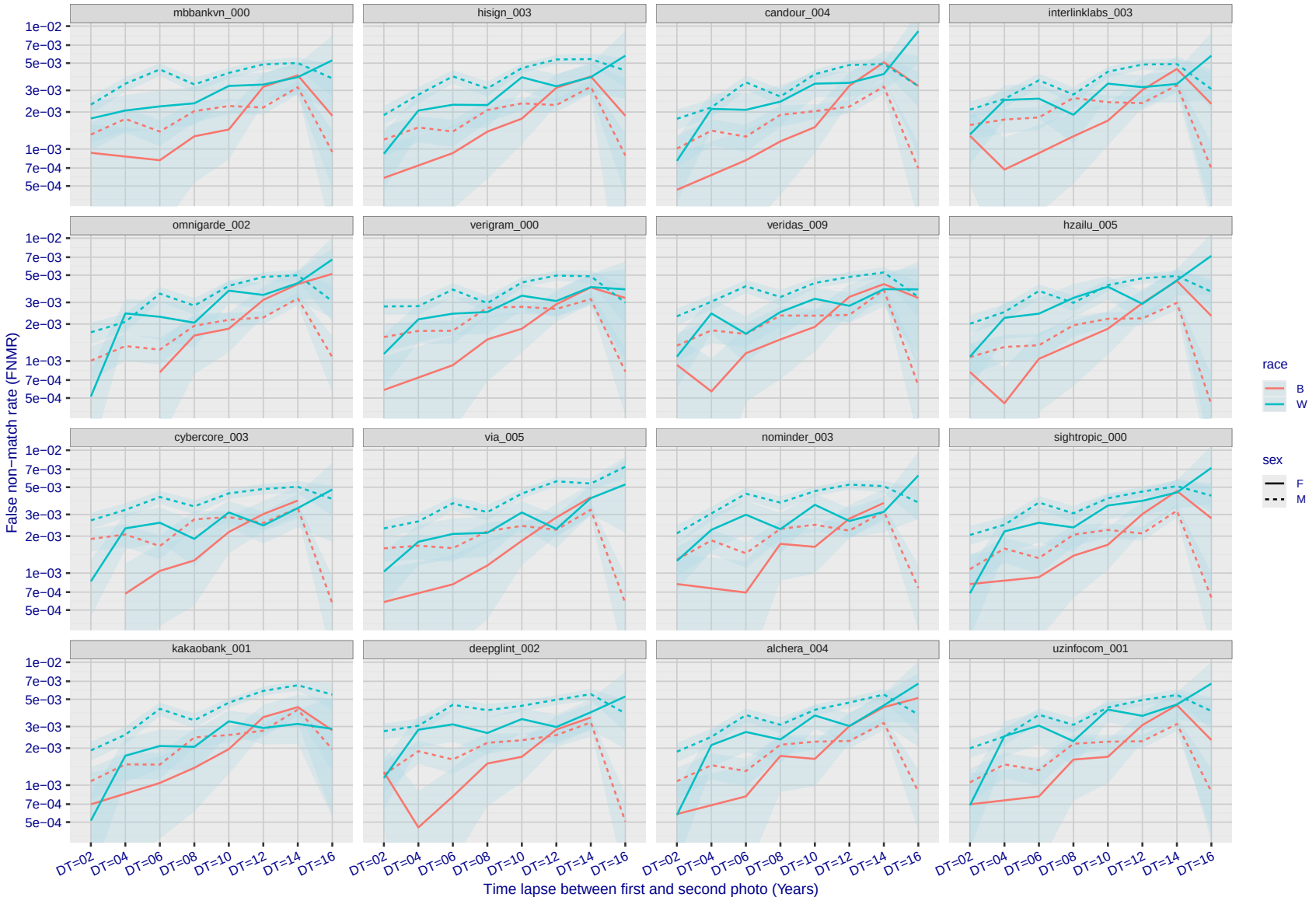


Figure 743: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

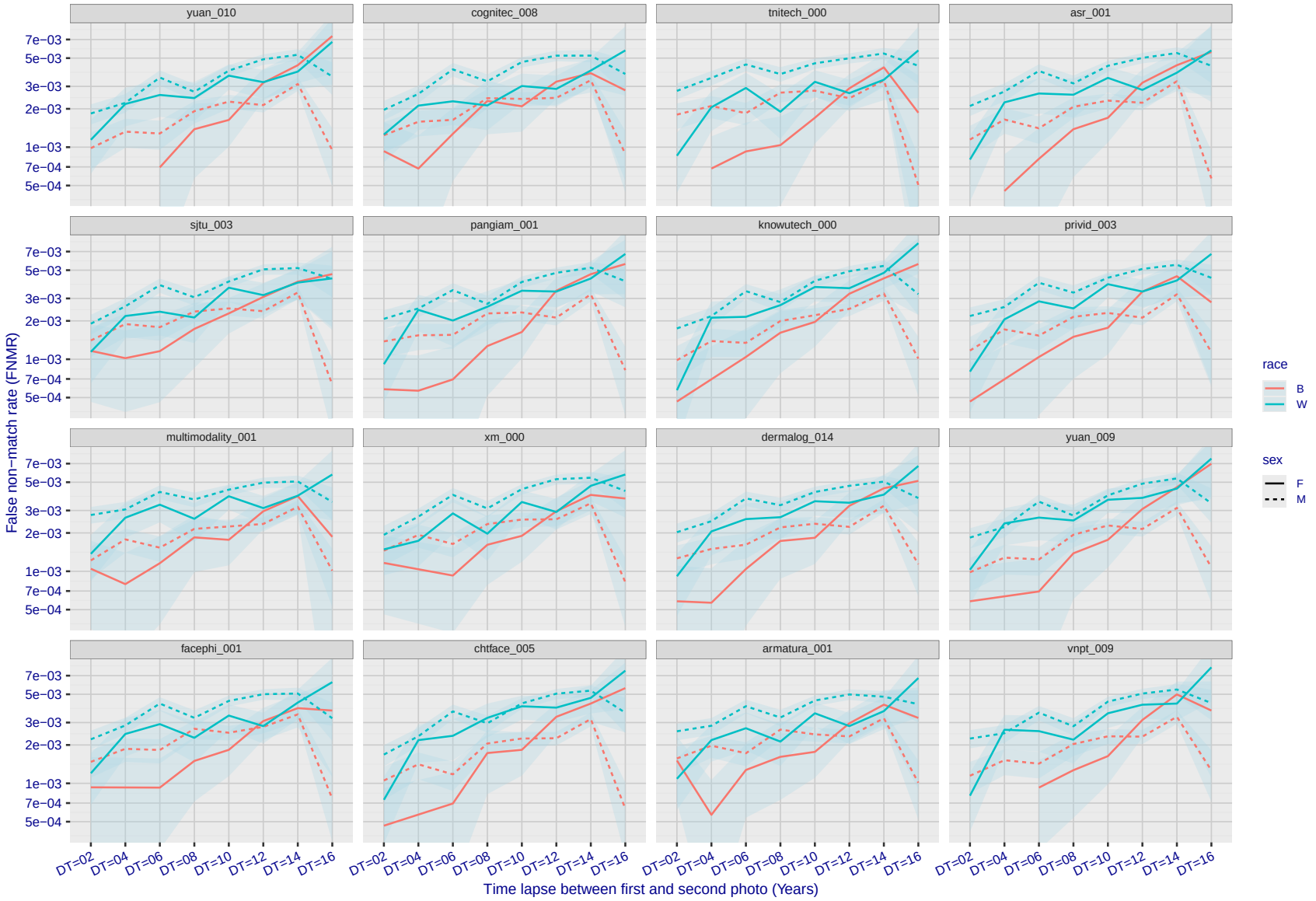


Figure 744: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

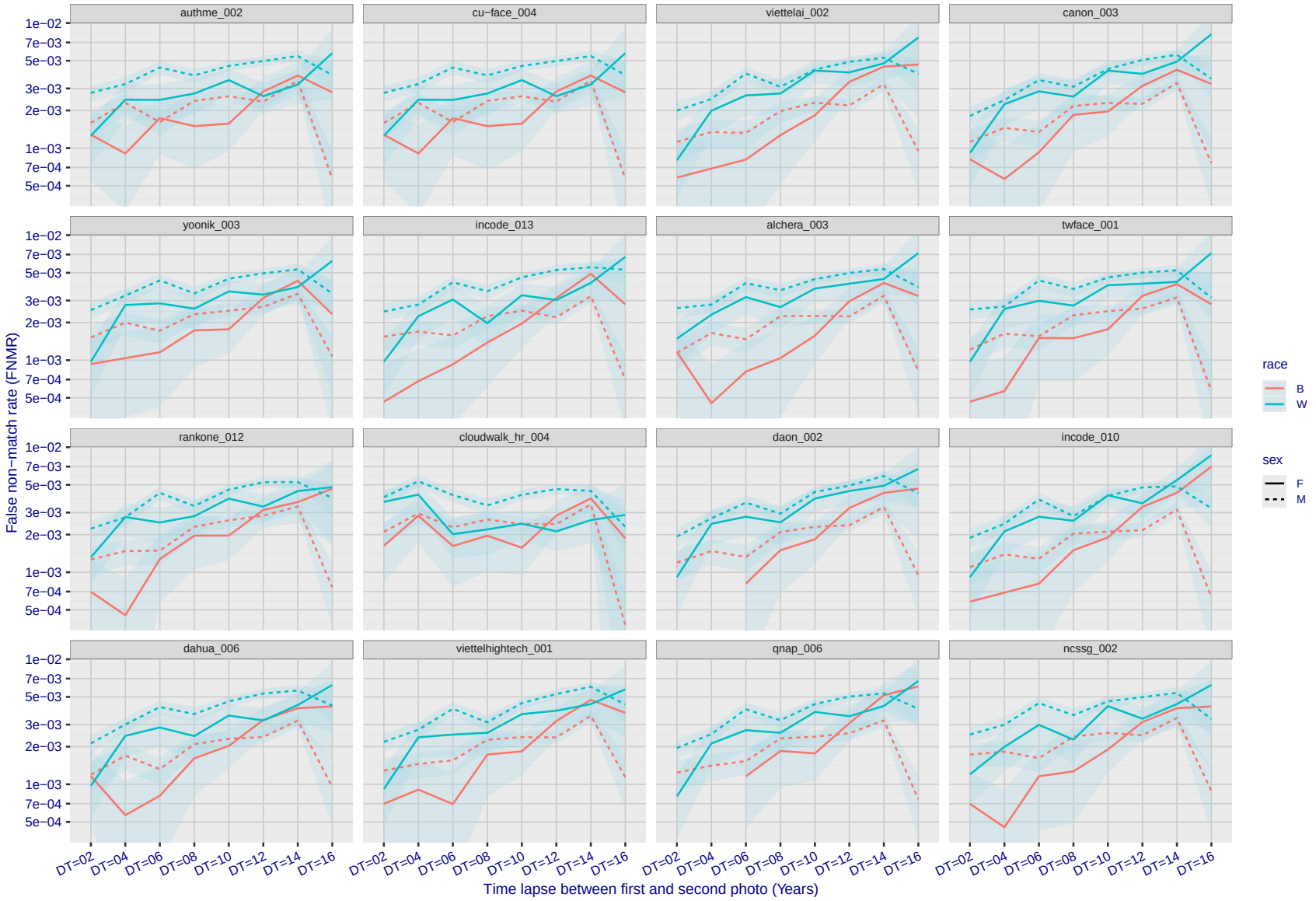


Figure 745: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

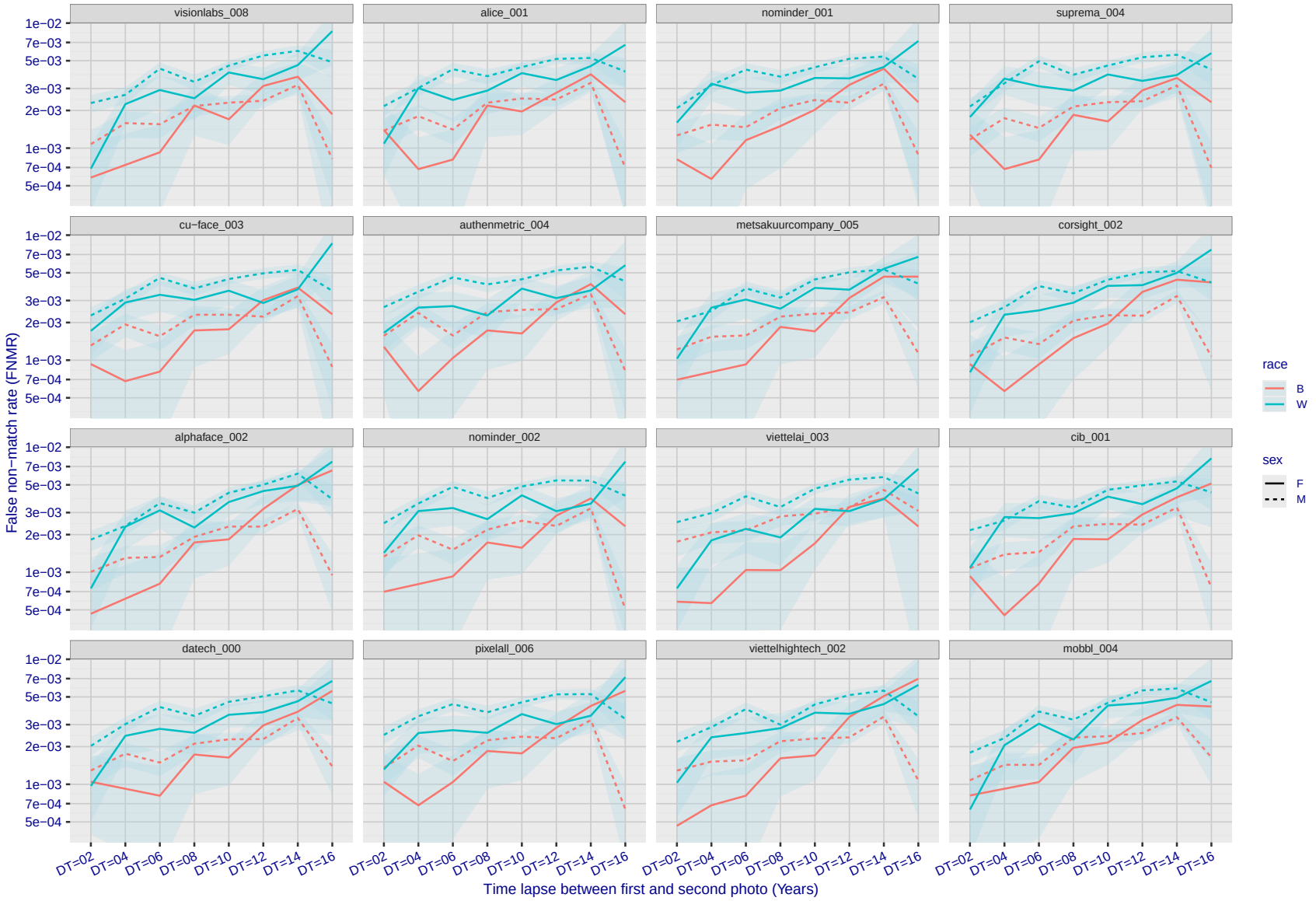


Figure 746: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\tilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

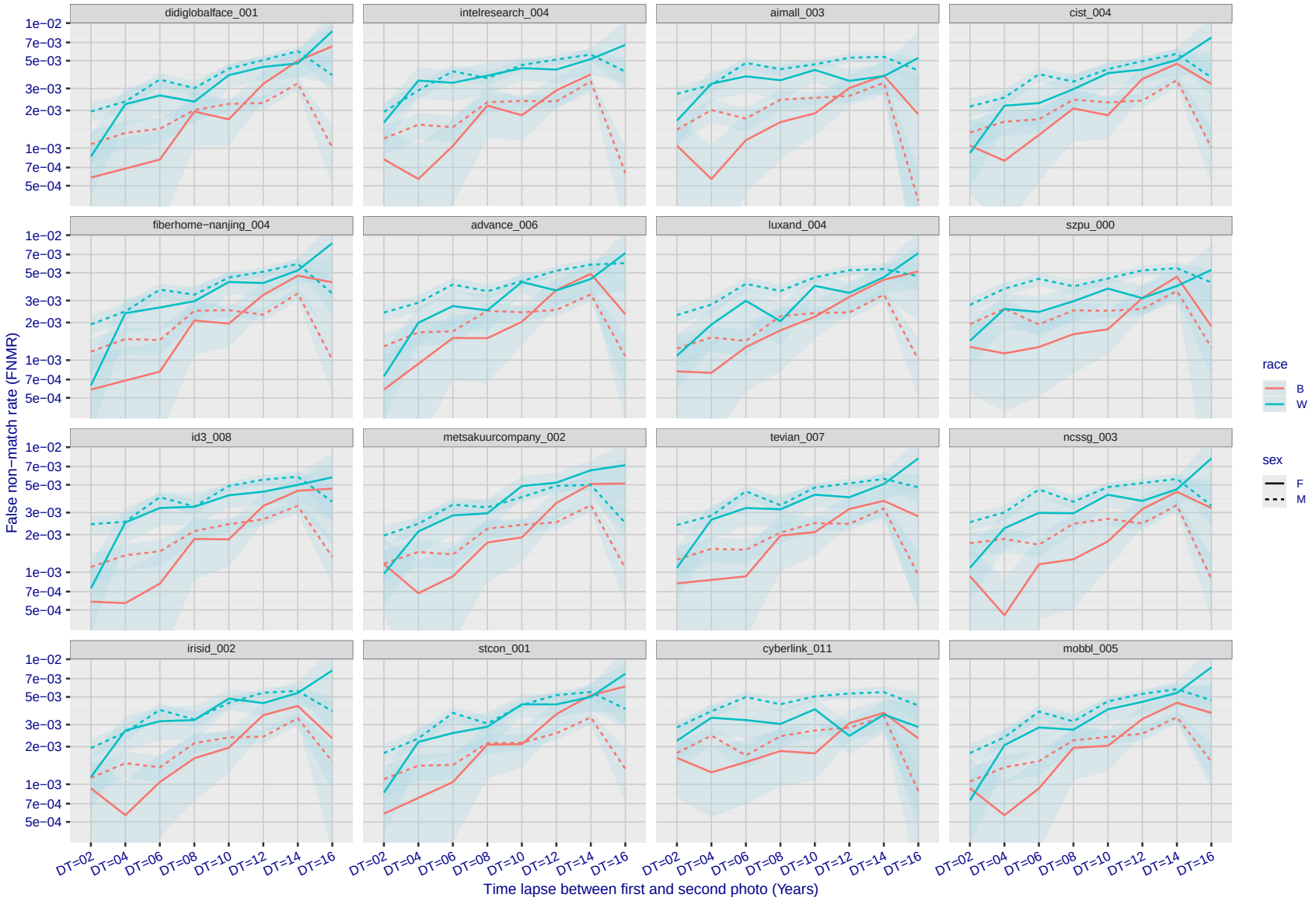


Figure 747: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\tilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

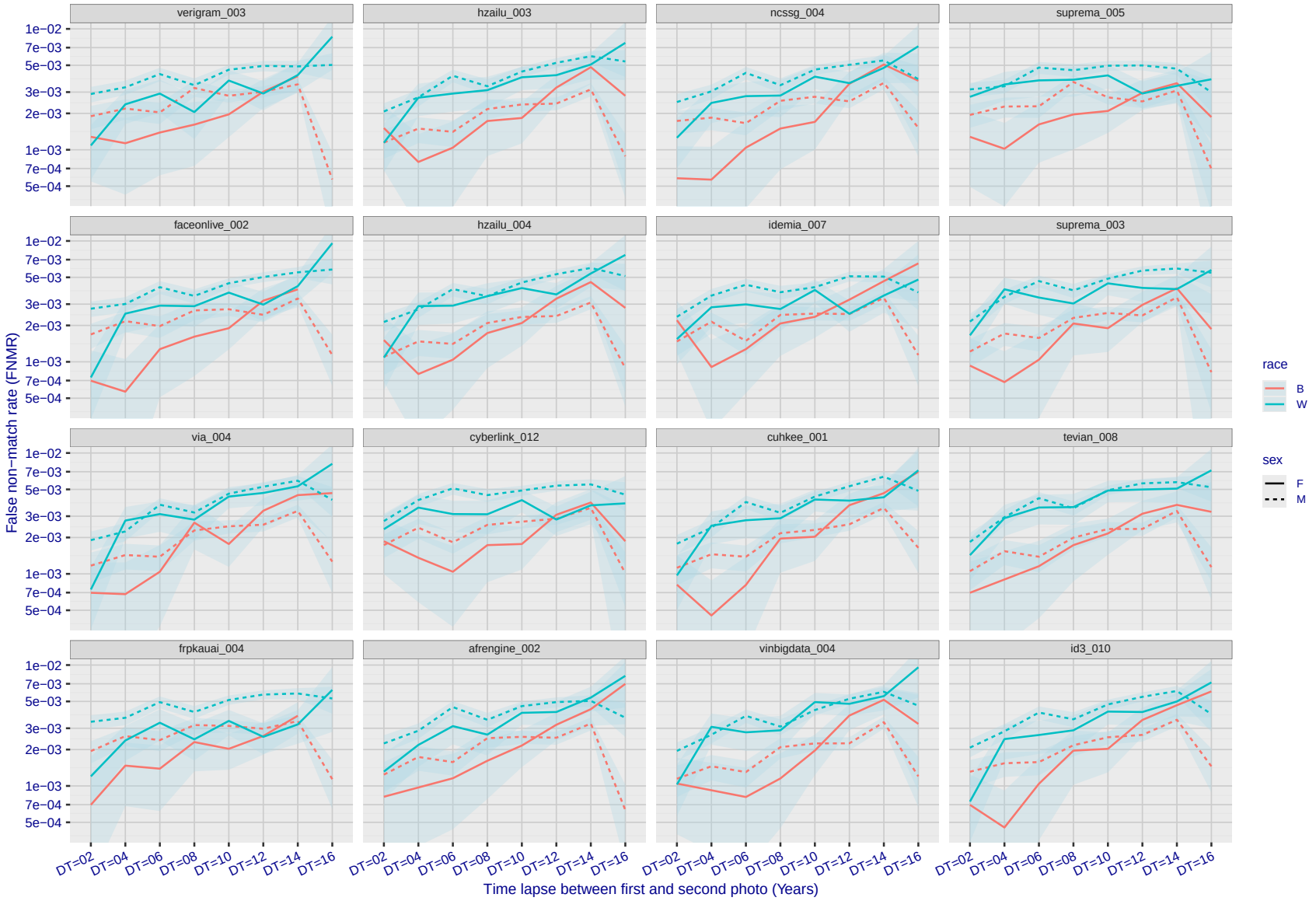


Figure 748: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

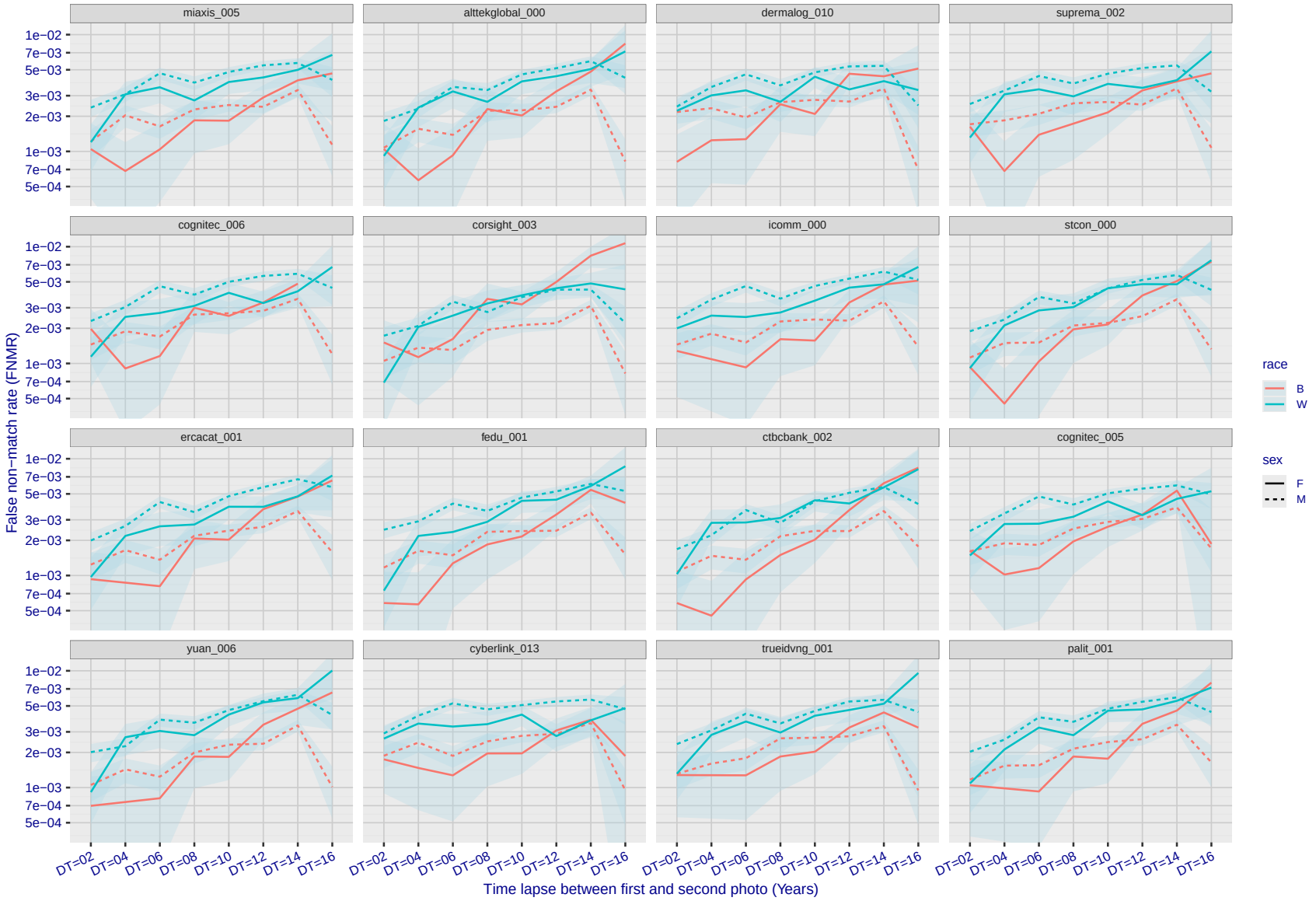


Figure 749: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

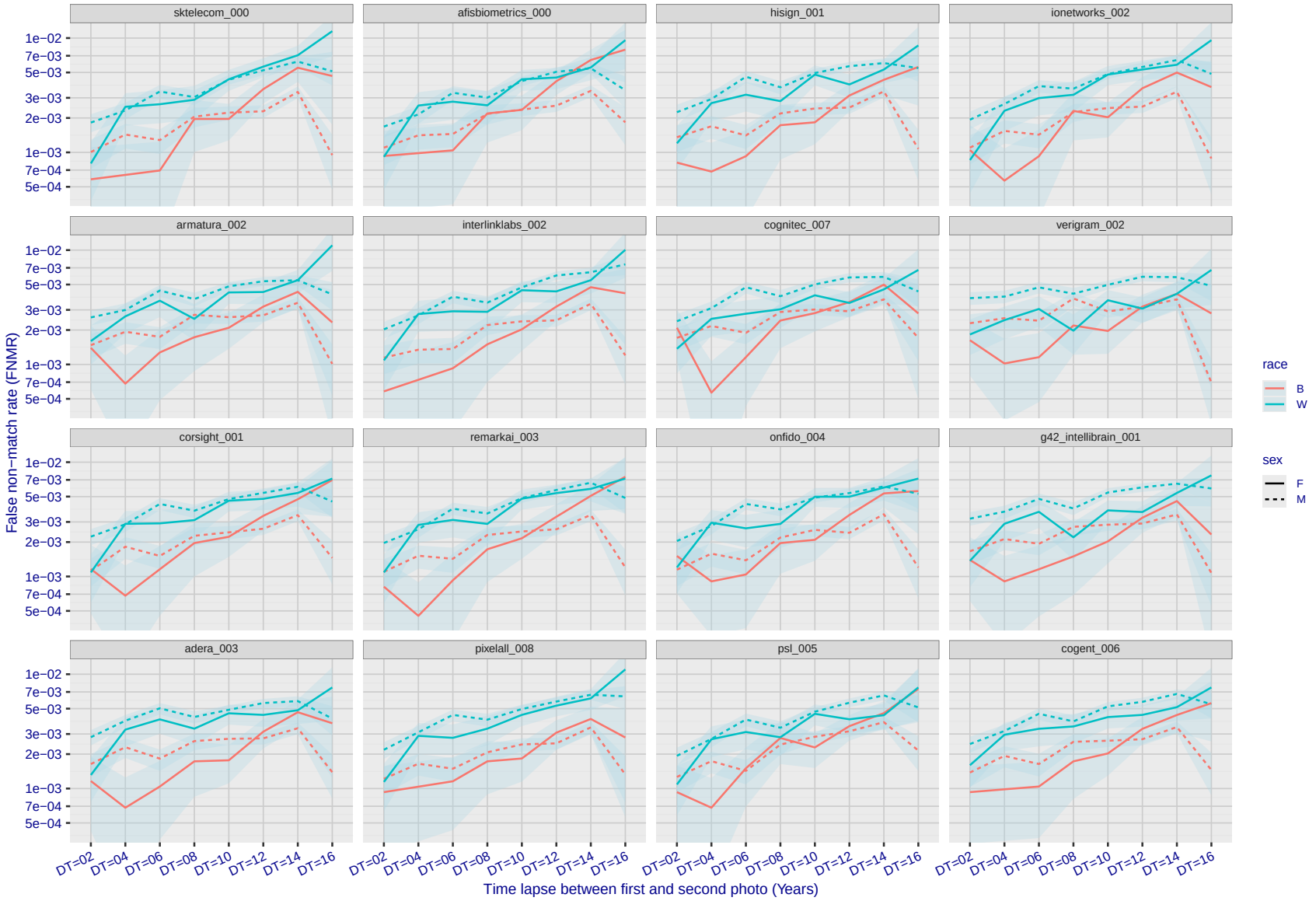


Figure 750: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

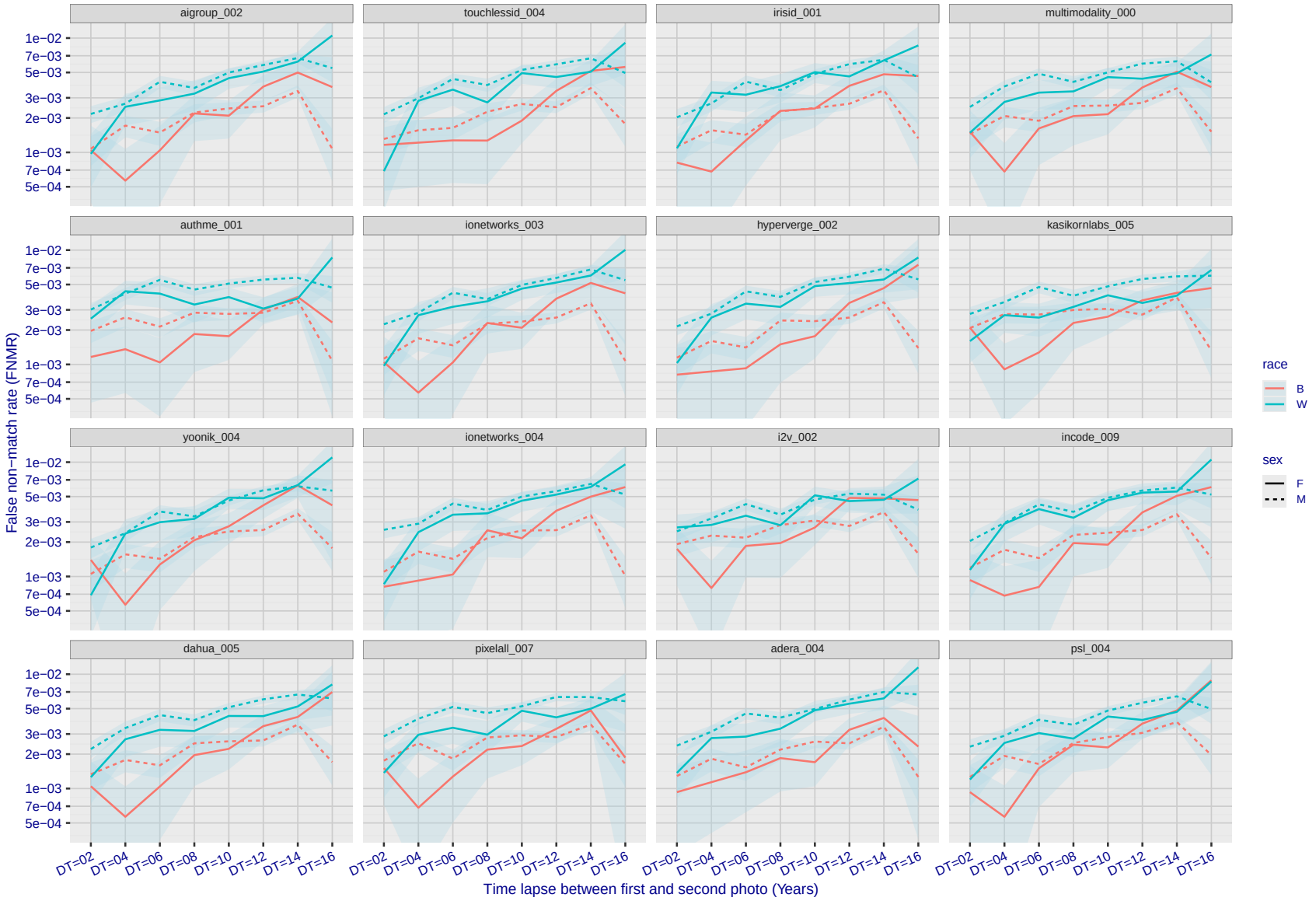


Figure 751: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

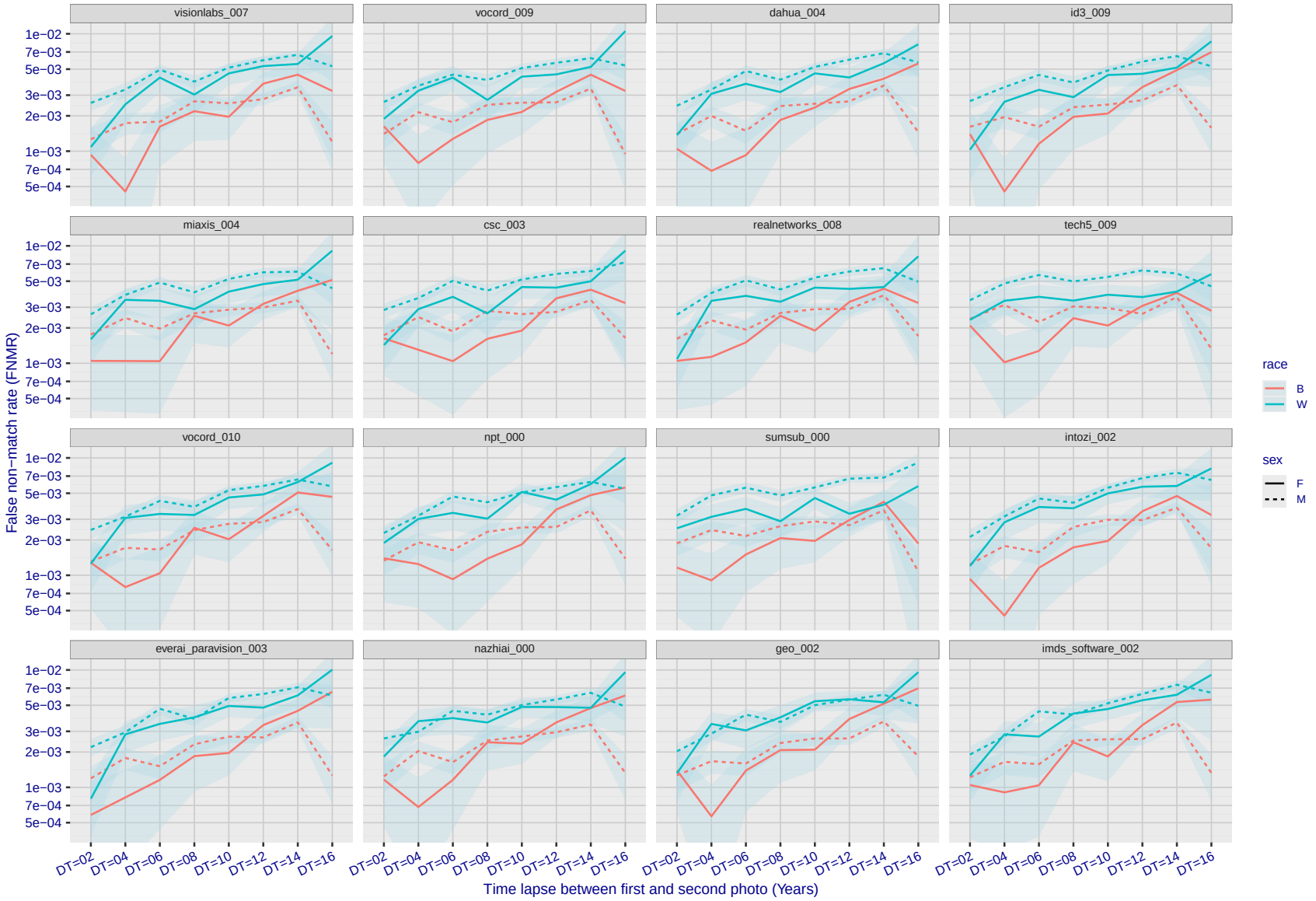


Figure 752: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\tilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

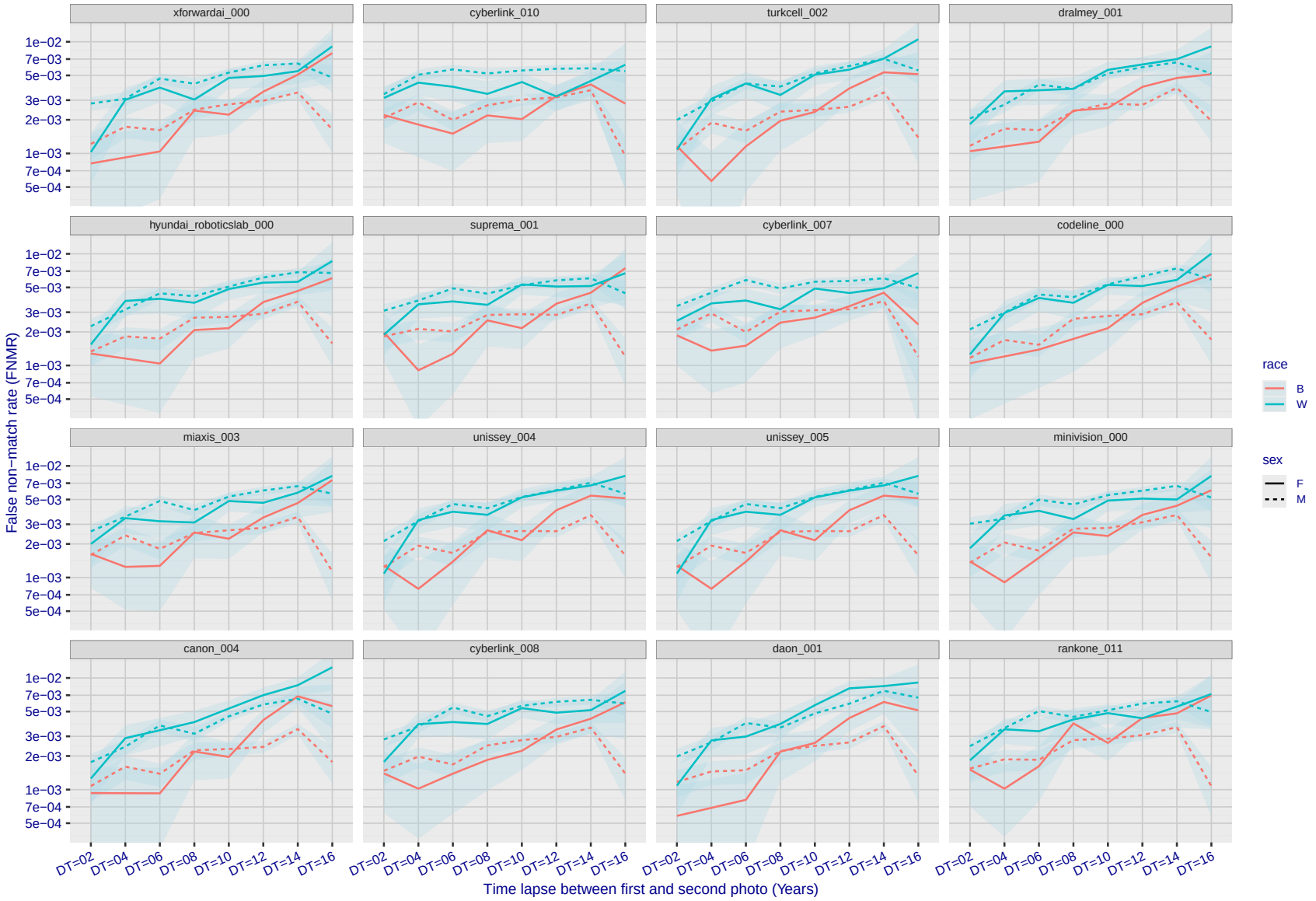


Figure 753: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

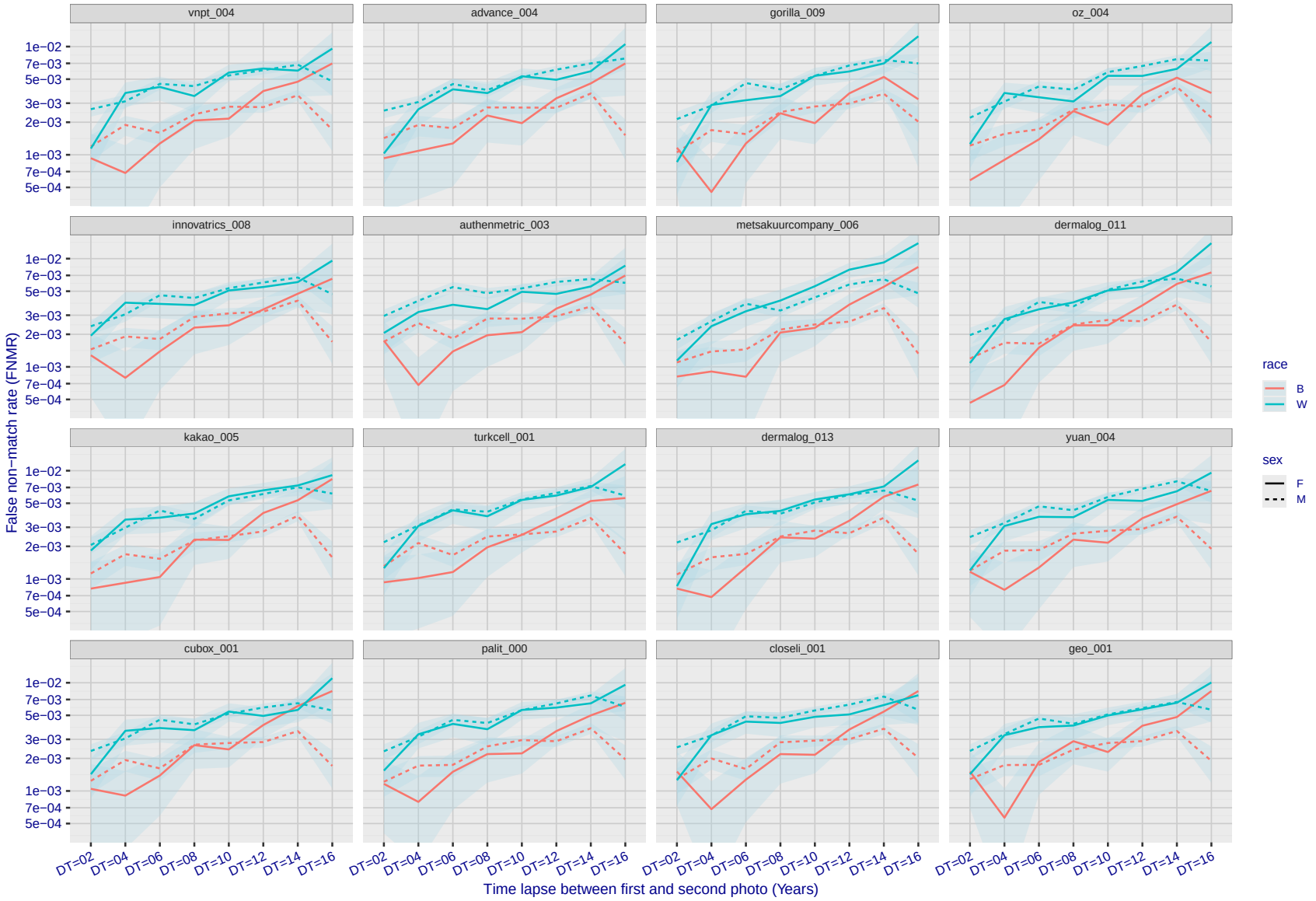


Figure 754: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

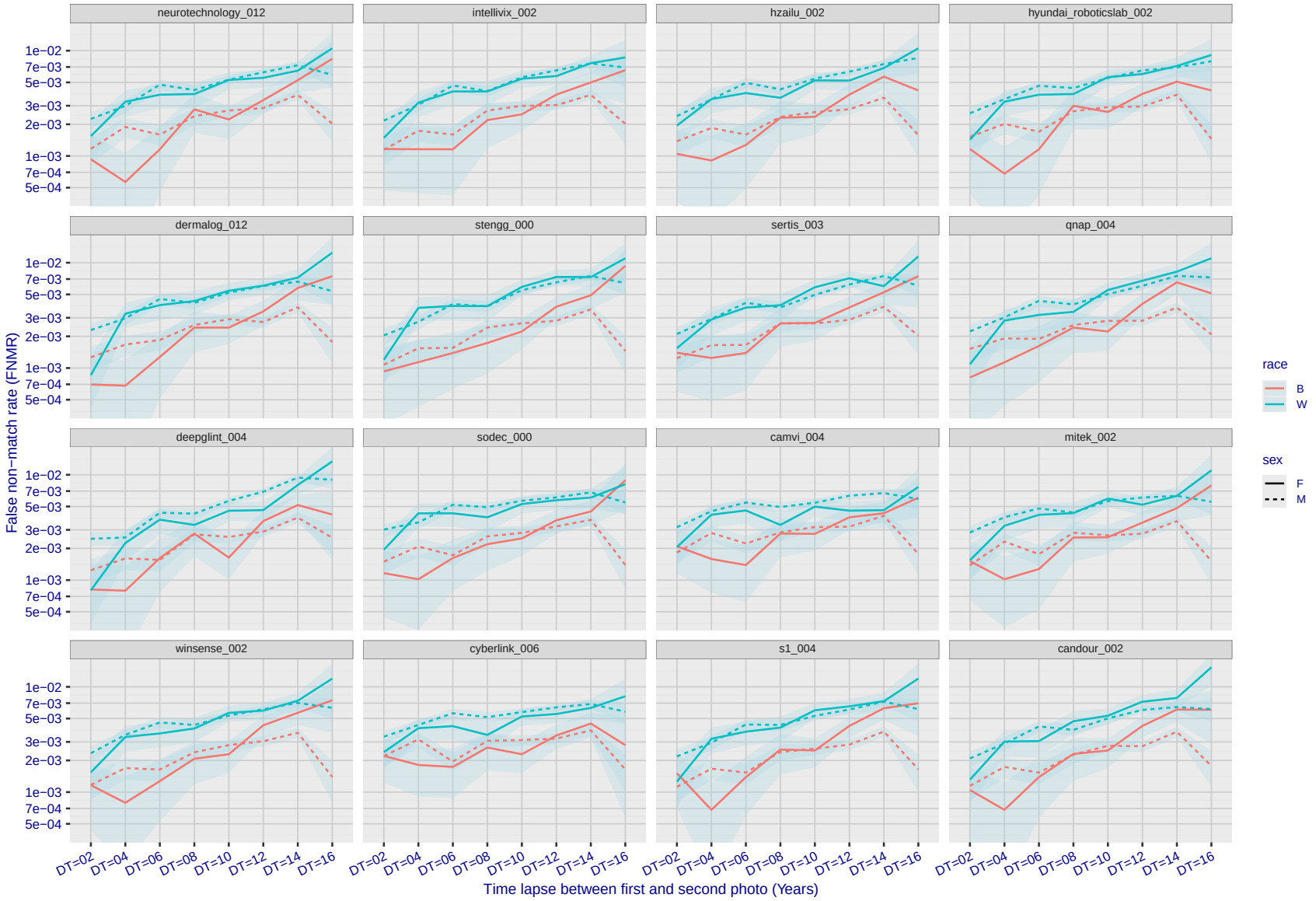


Figure 755: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

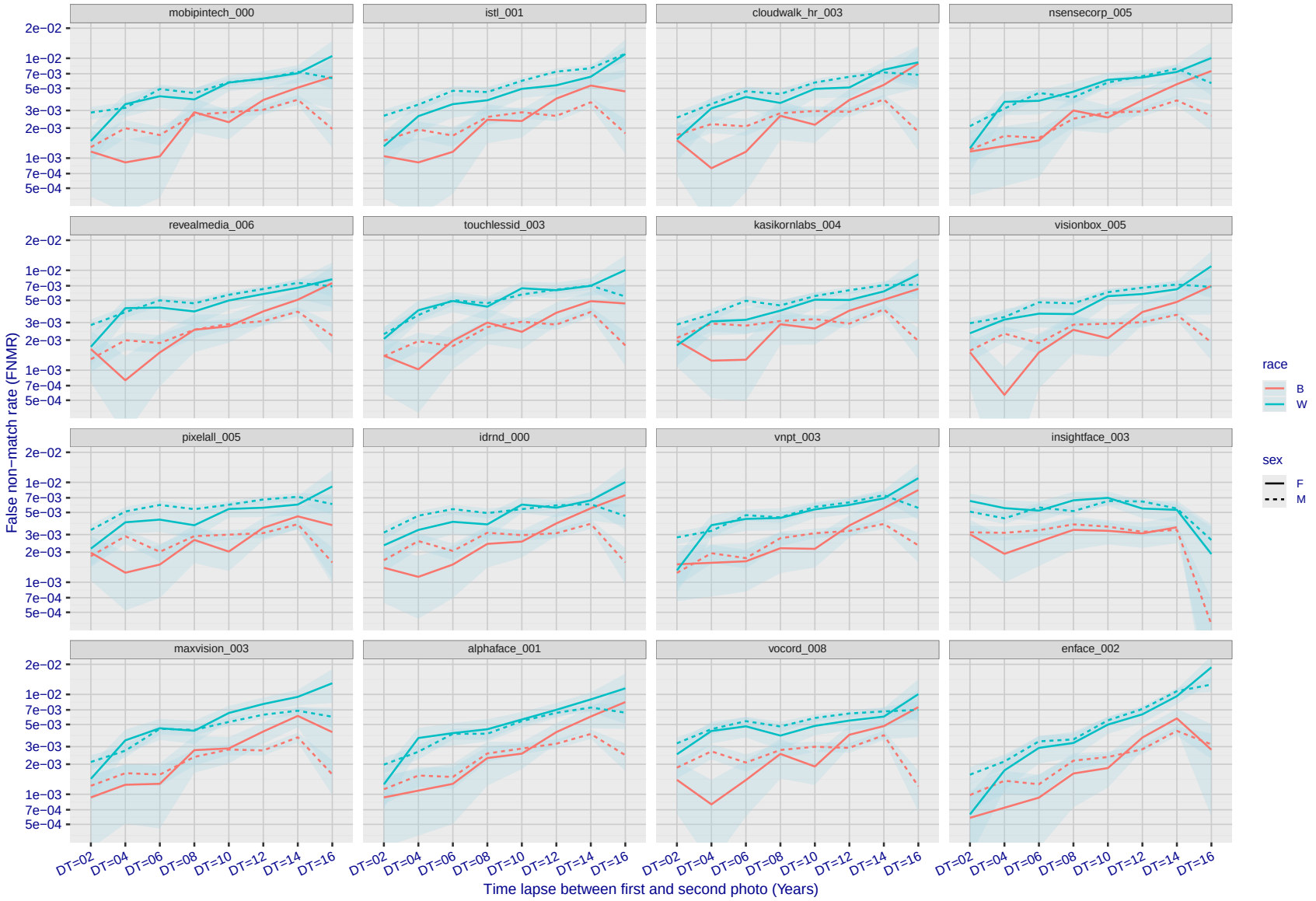


Figure 756: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

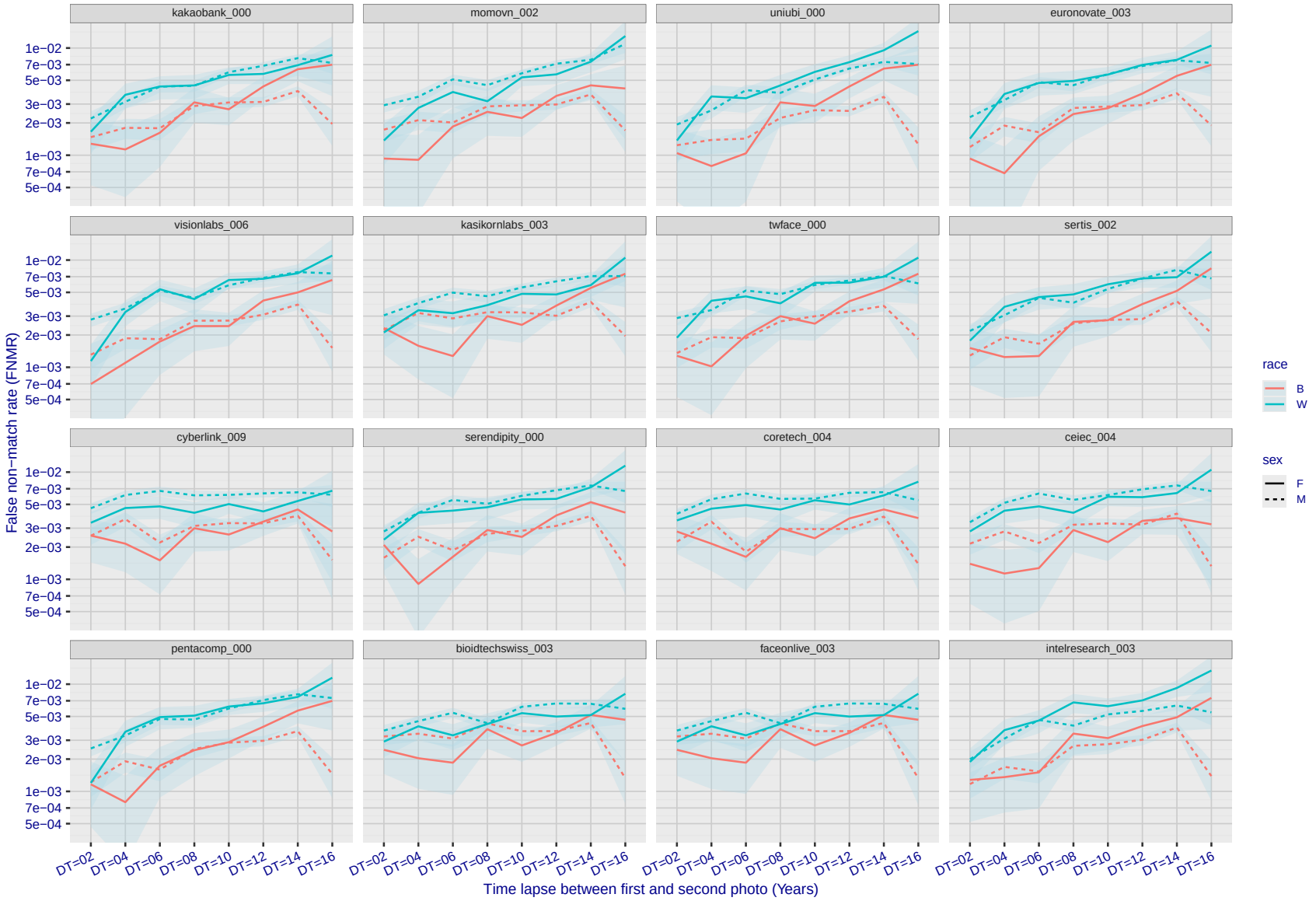


Figure 757: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

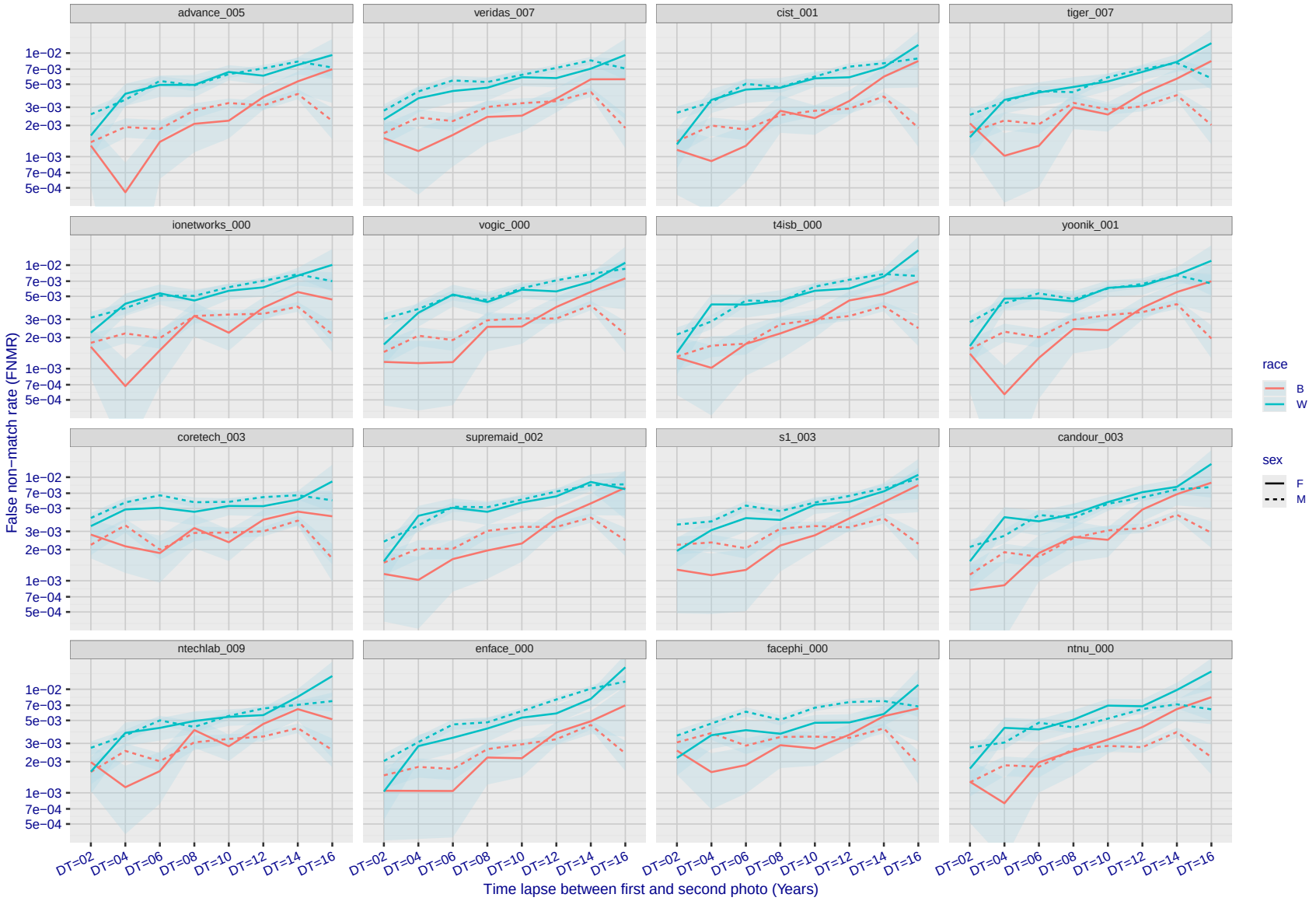


Figure 758: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

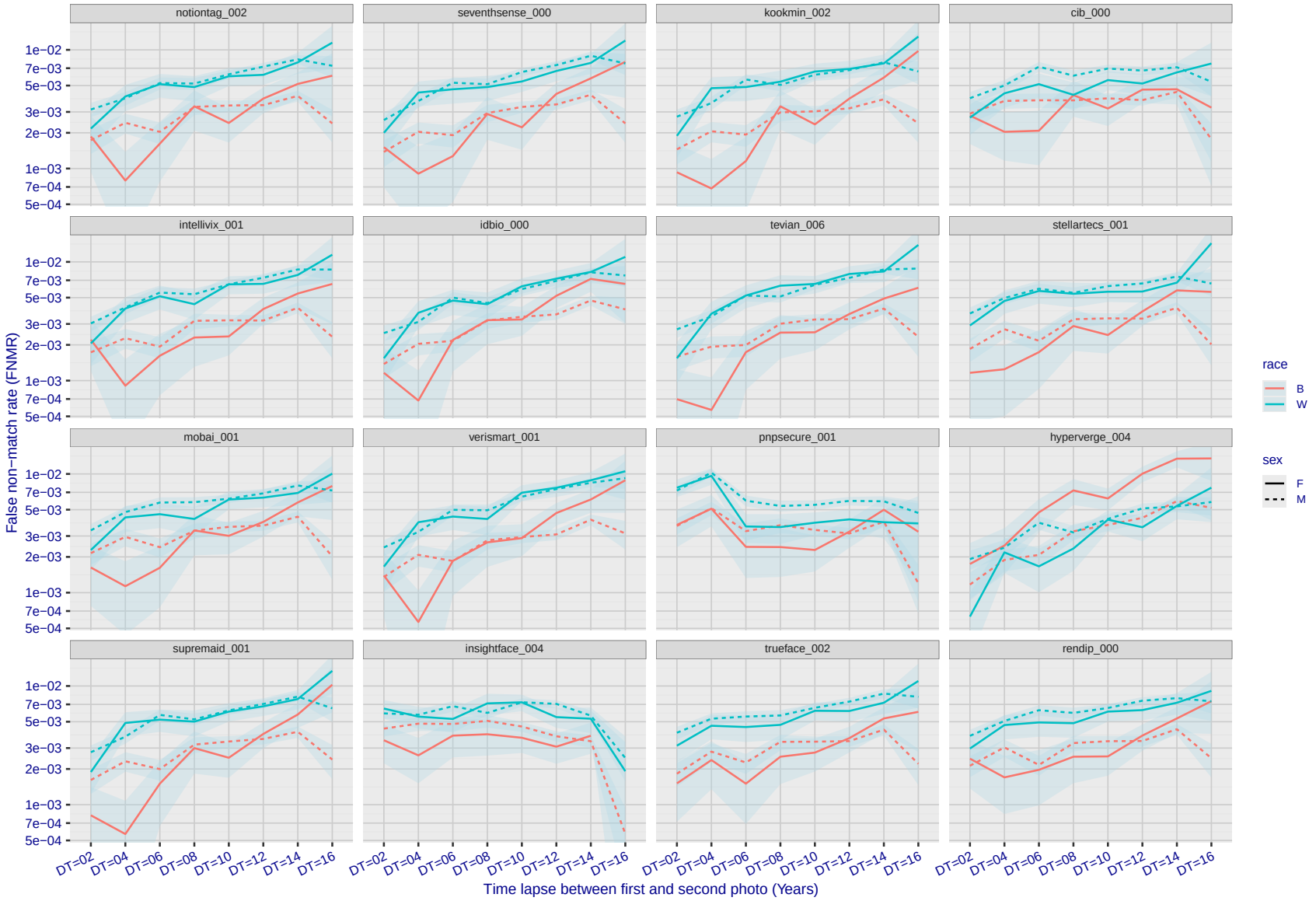


Figure 759: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

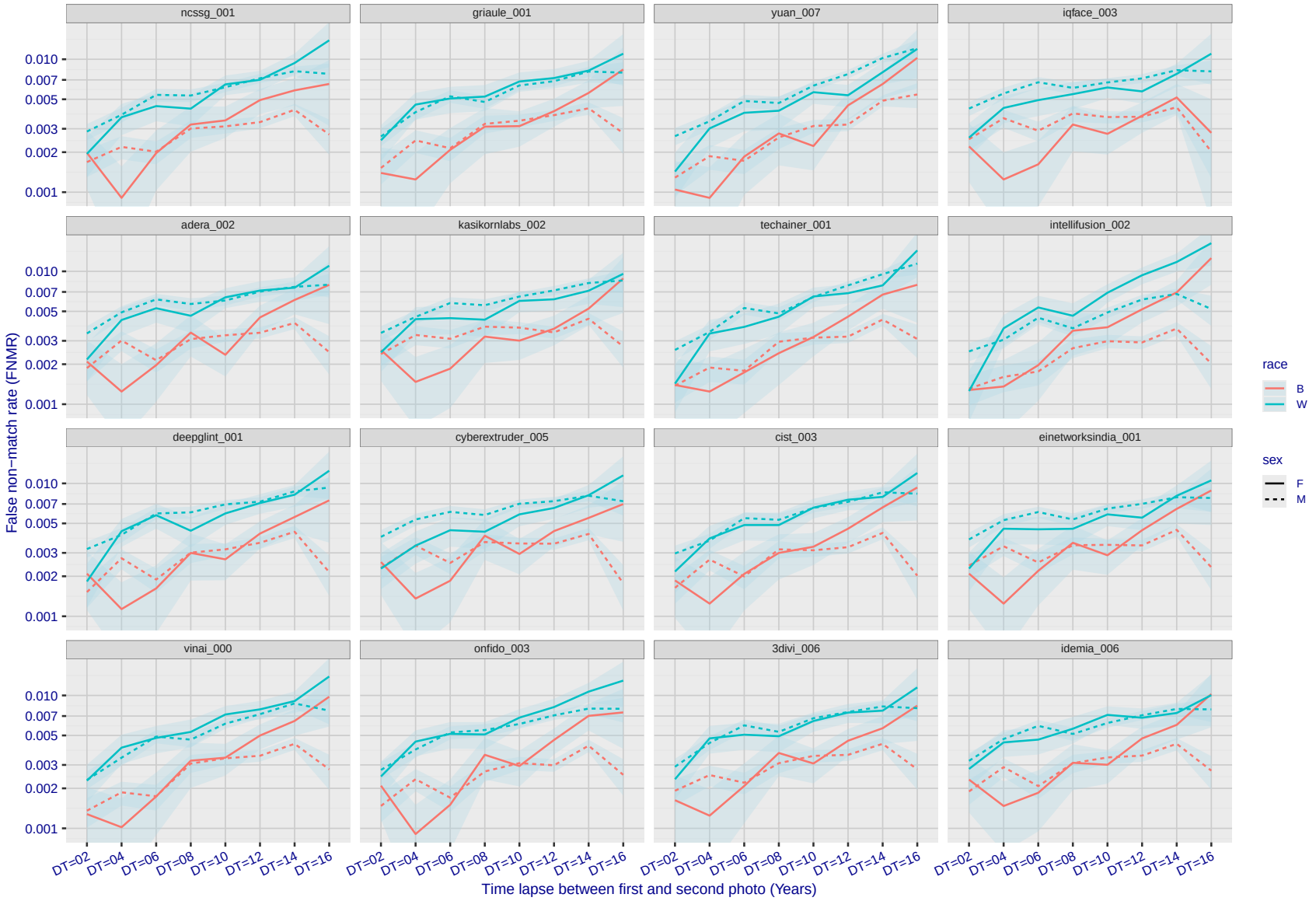


Figure 760: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

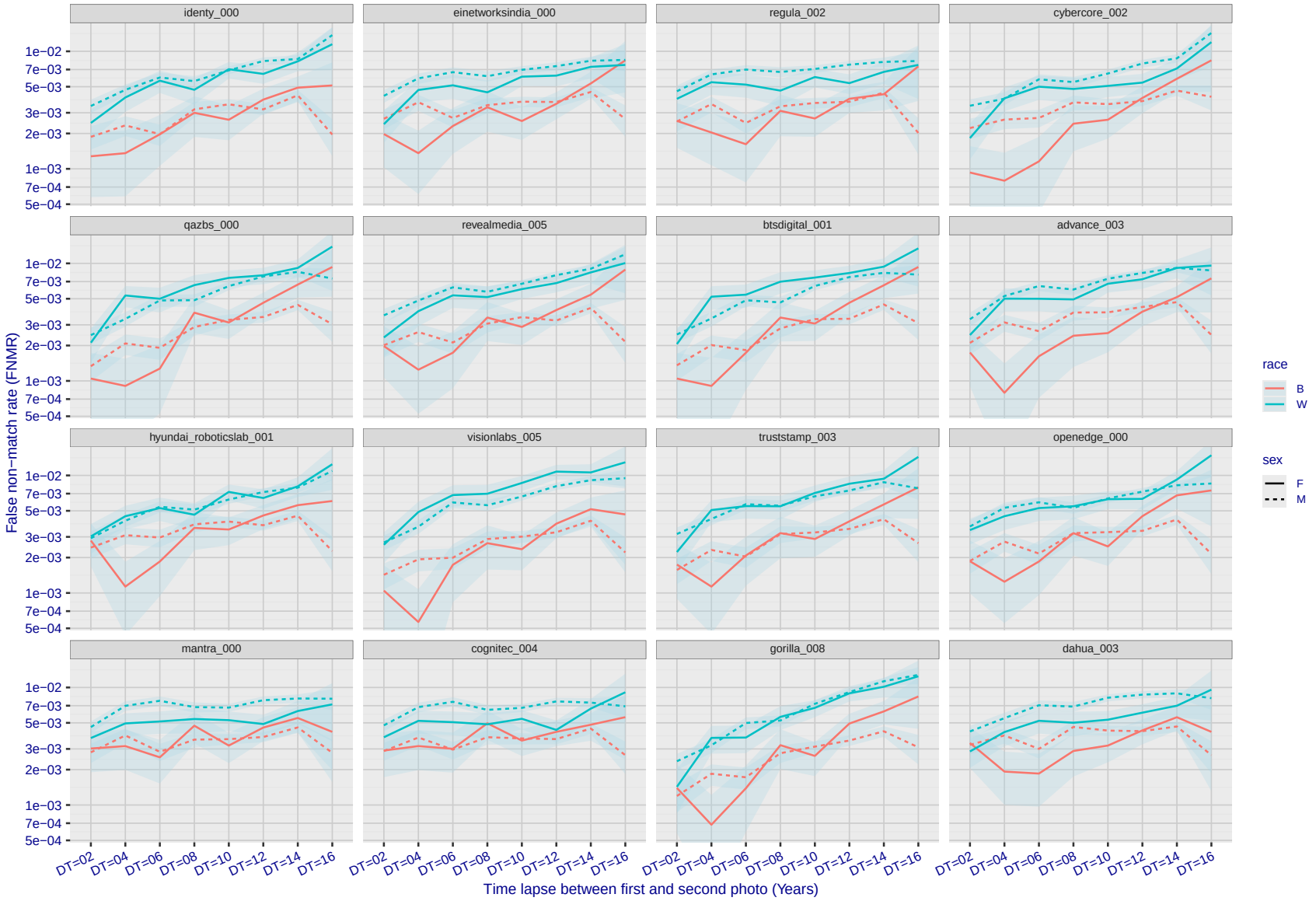


Figure 761: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\tilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

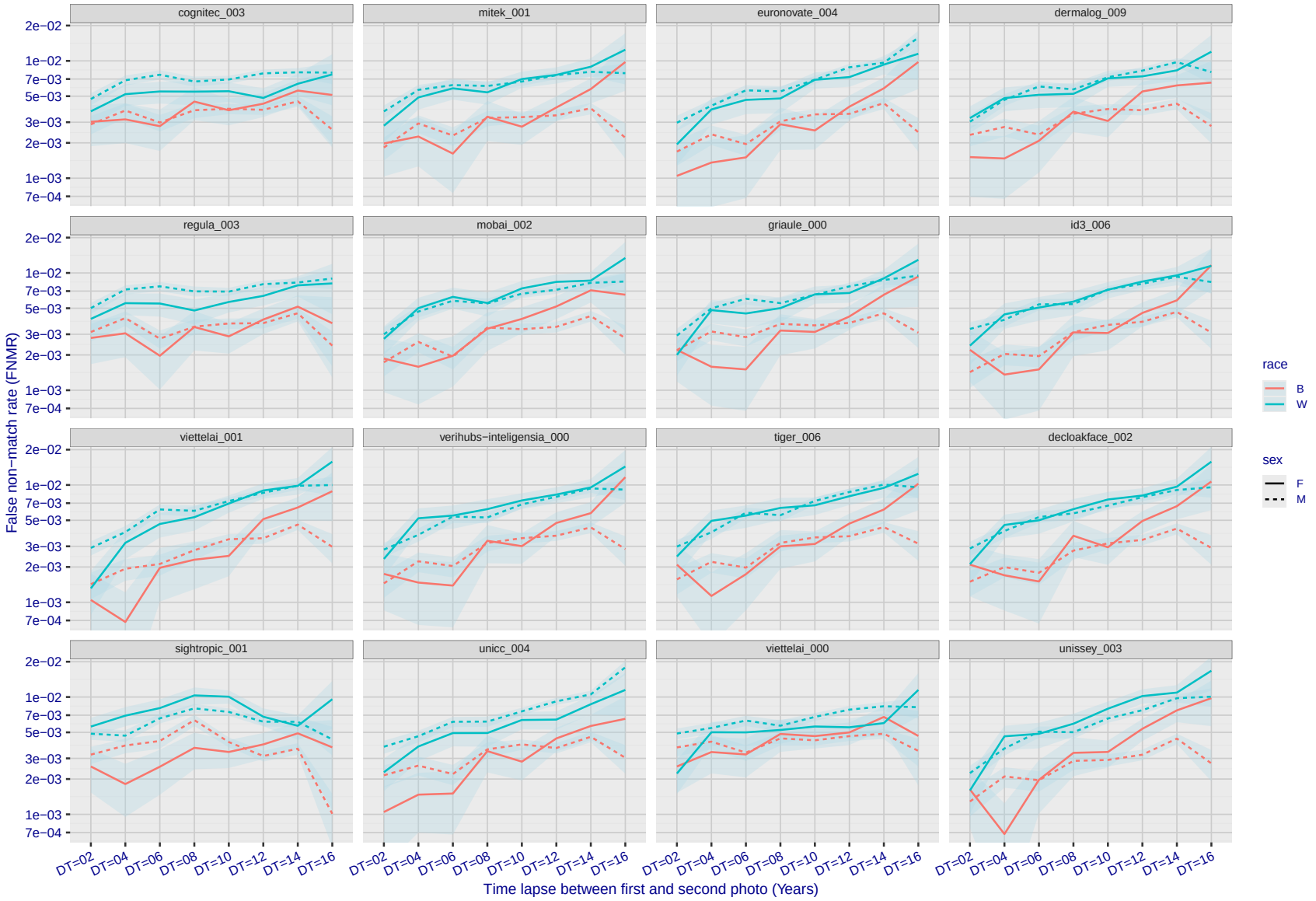


Figure 762: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widehat{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

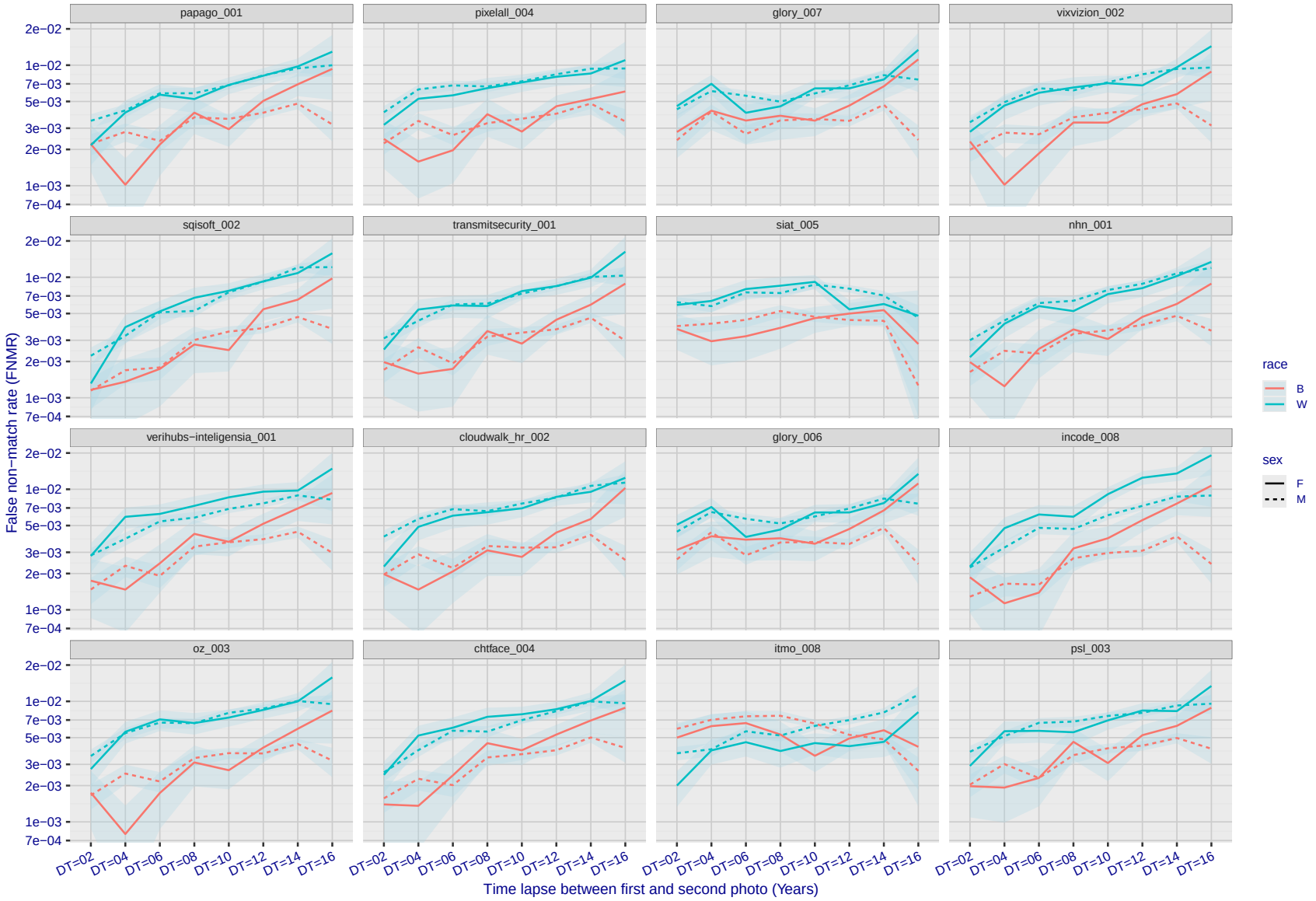


Figure 763: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

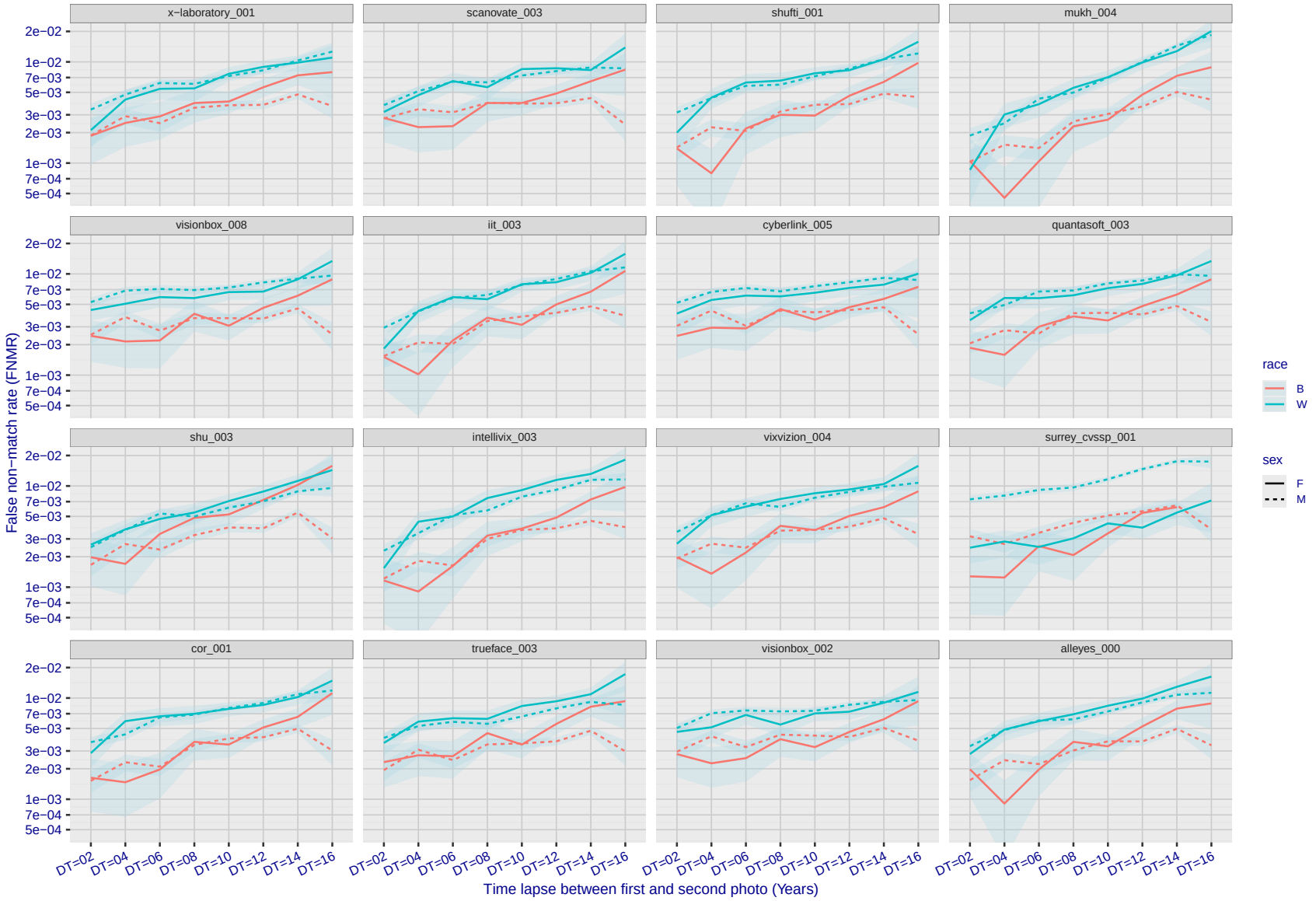


Figure 764: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

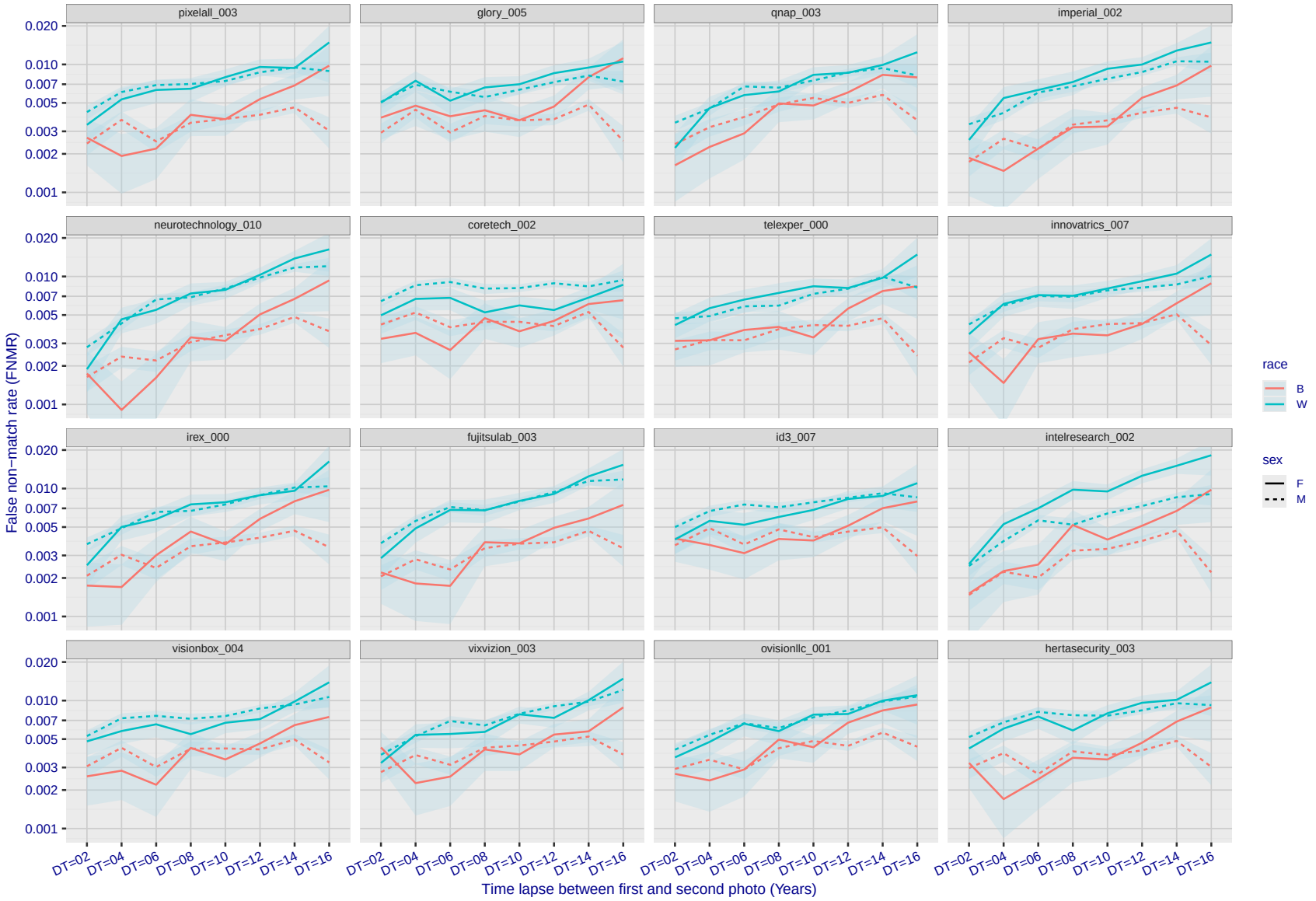


Figure 765: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

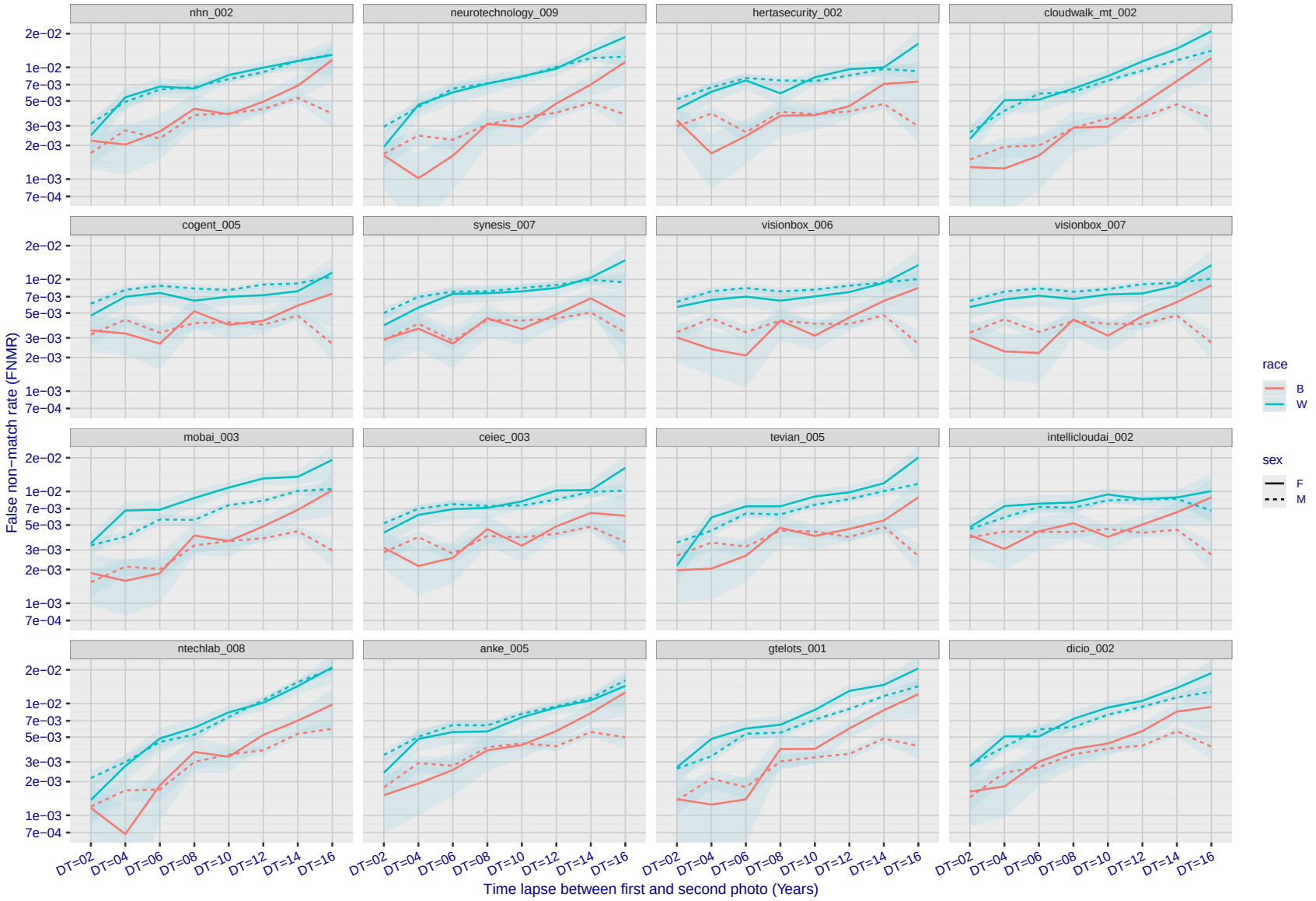


Figure 766: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

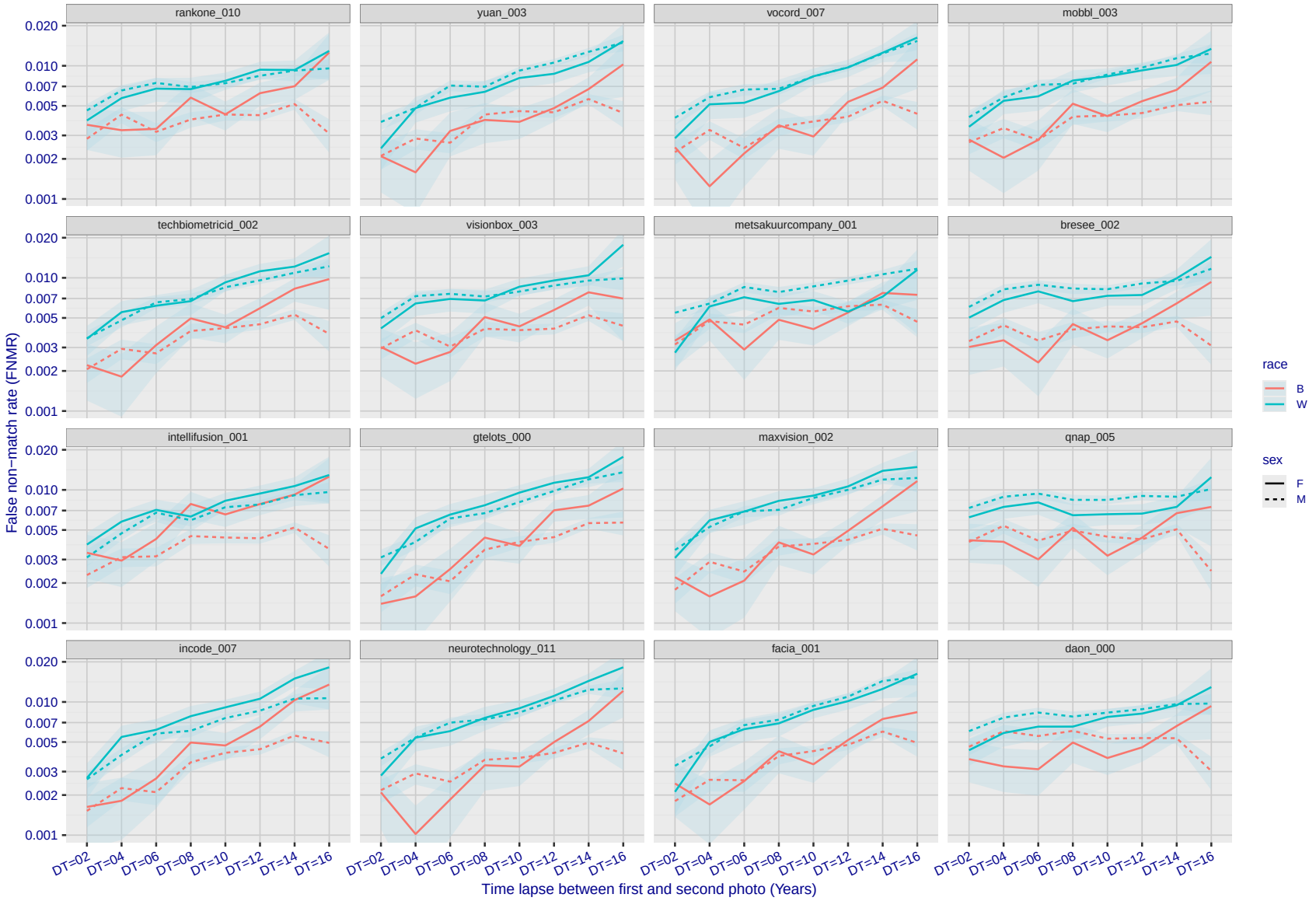


Figure 767: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

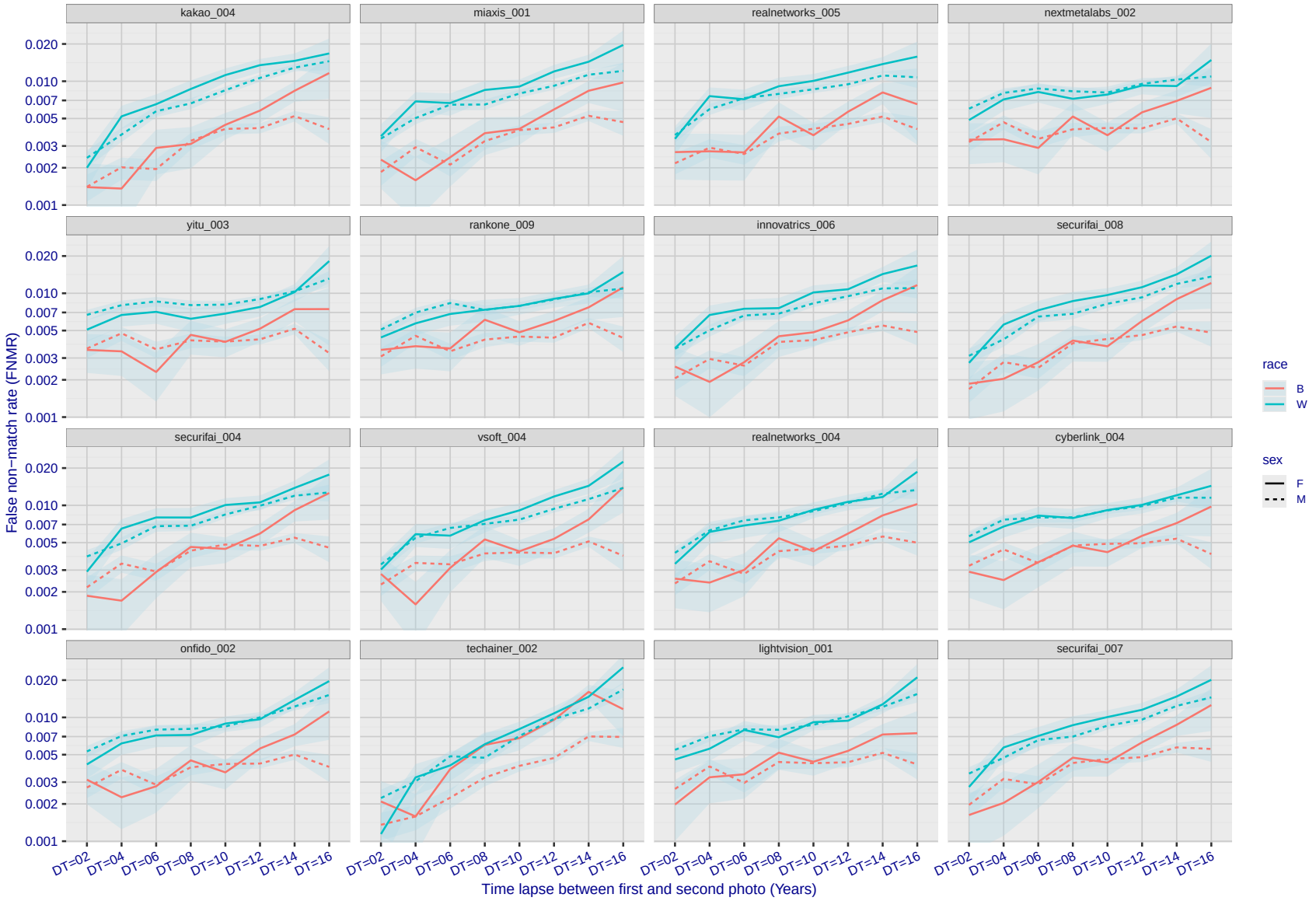


Figure 768: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

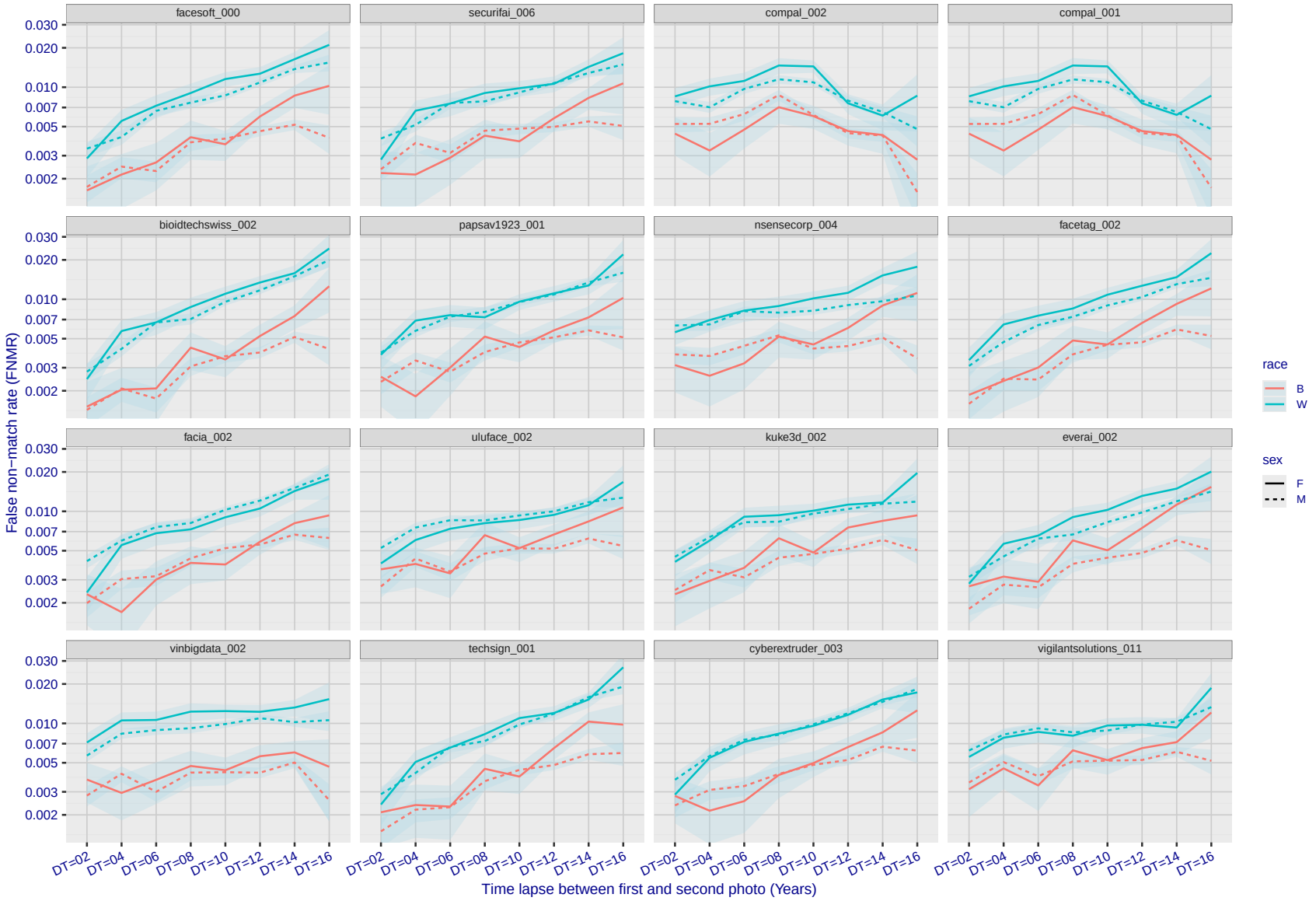


Figure 769: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

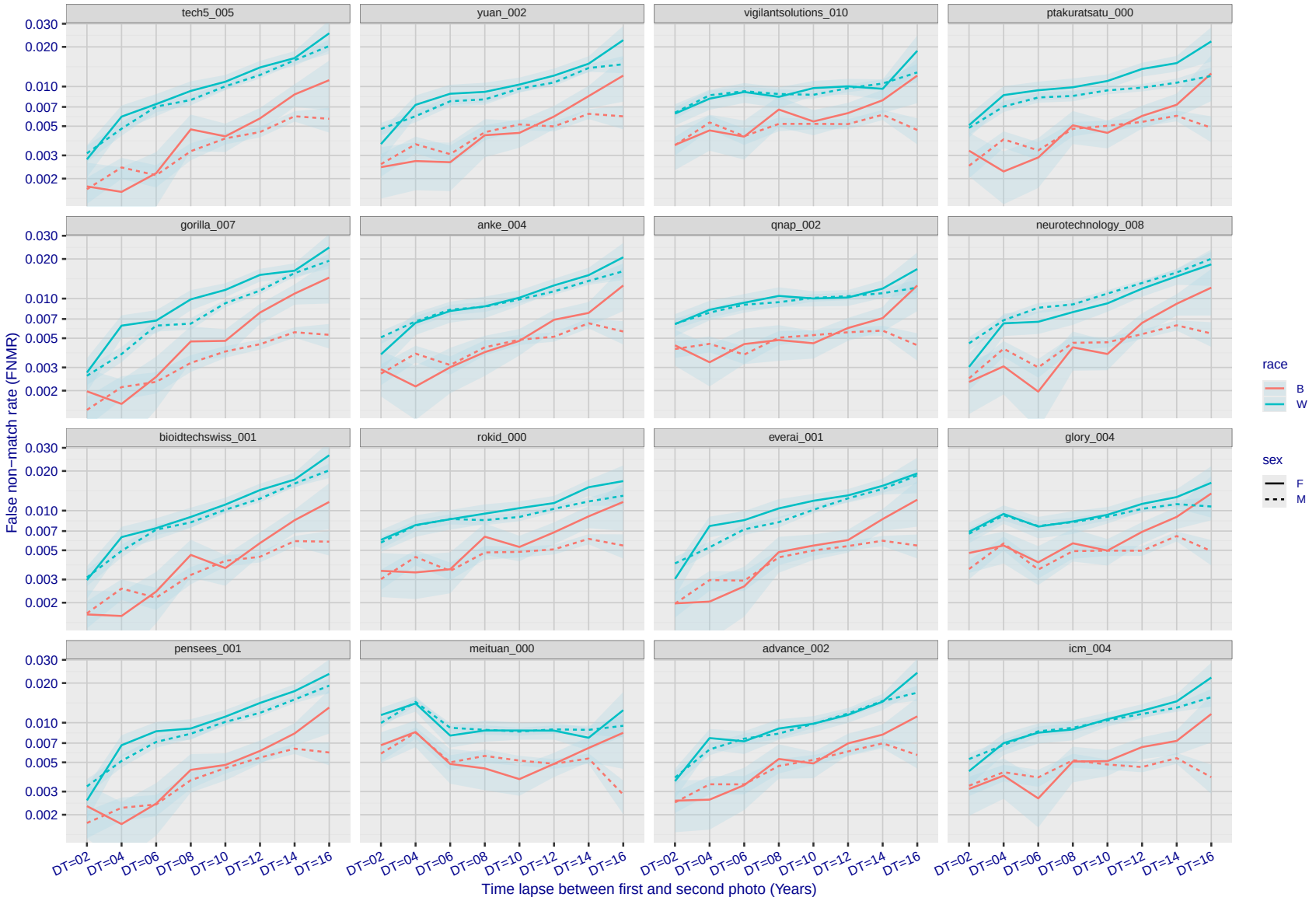


Figure 770: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

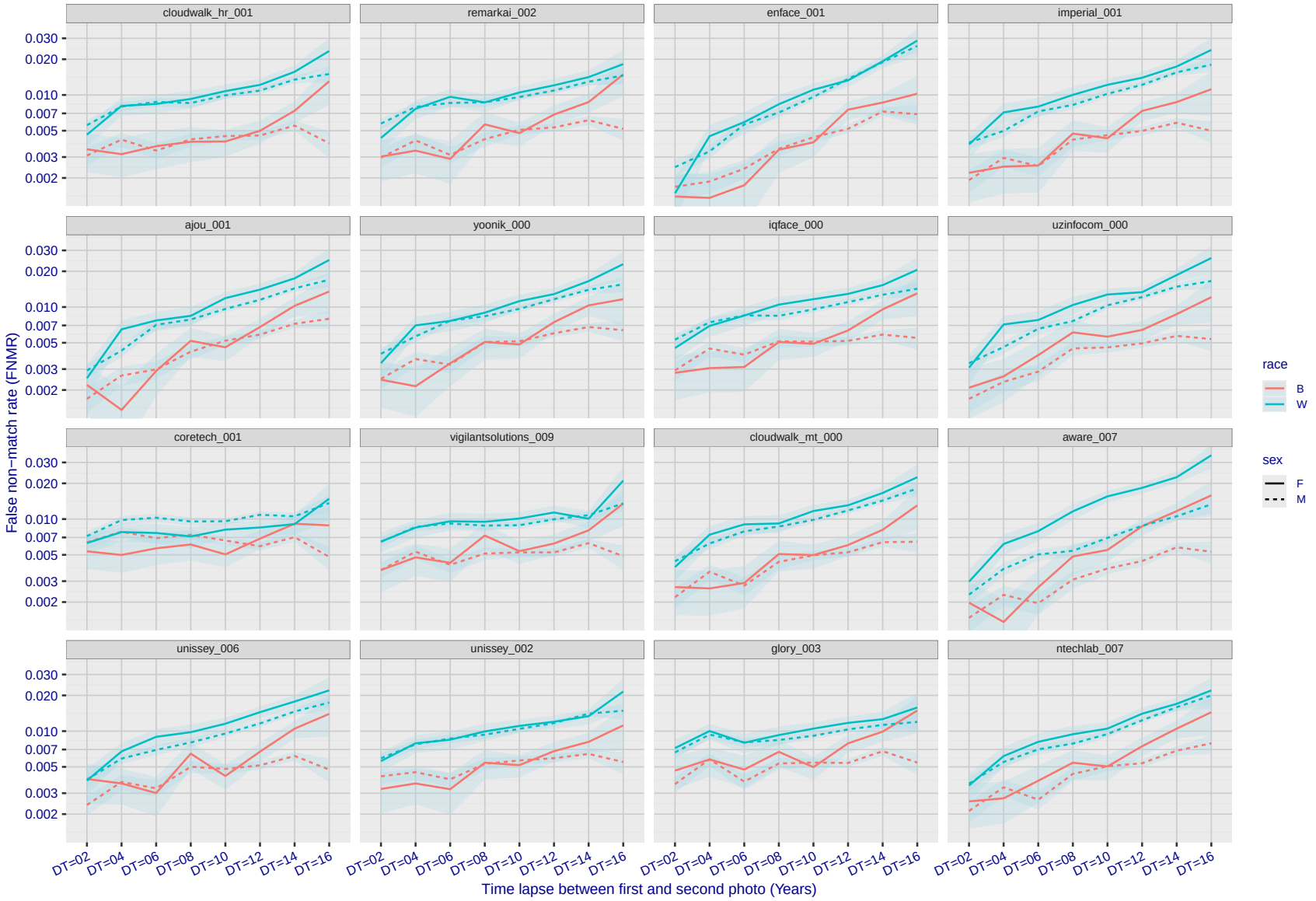


Figure 771: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

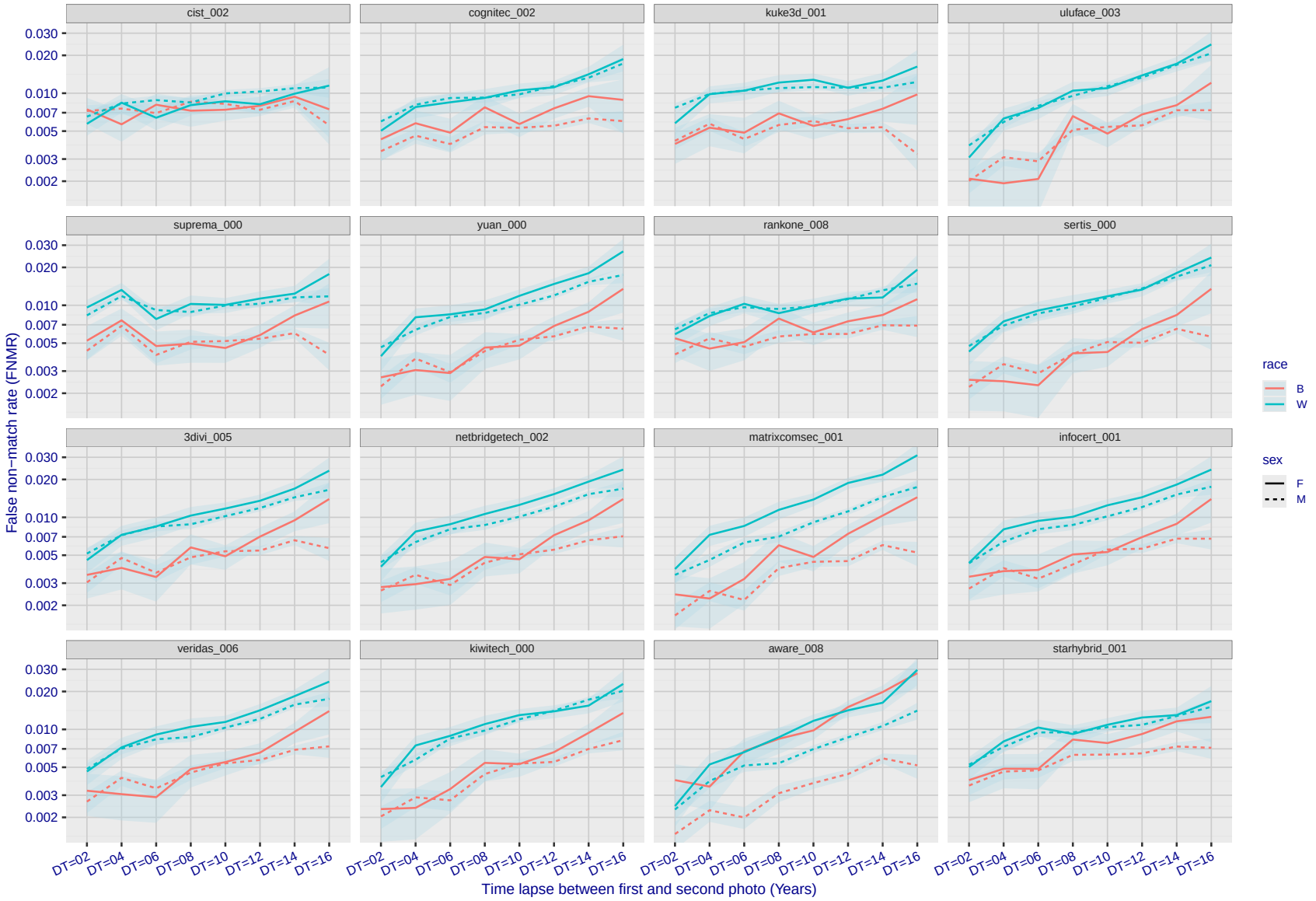


Figure 772: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

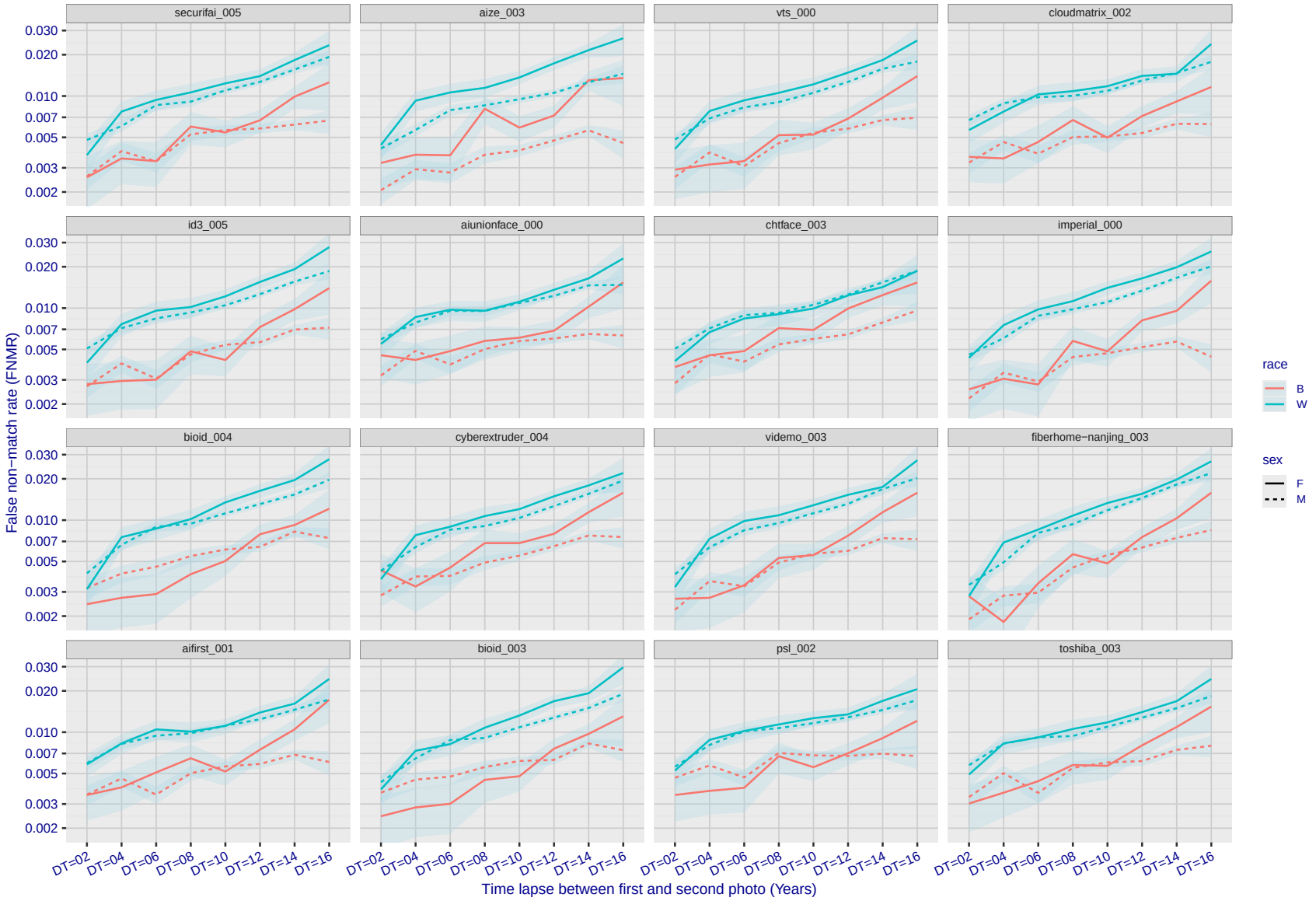


Figure 773: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

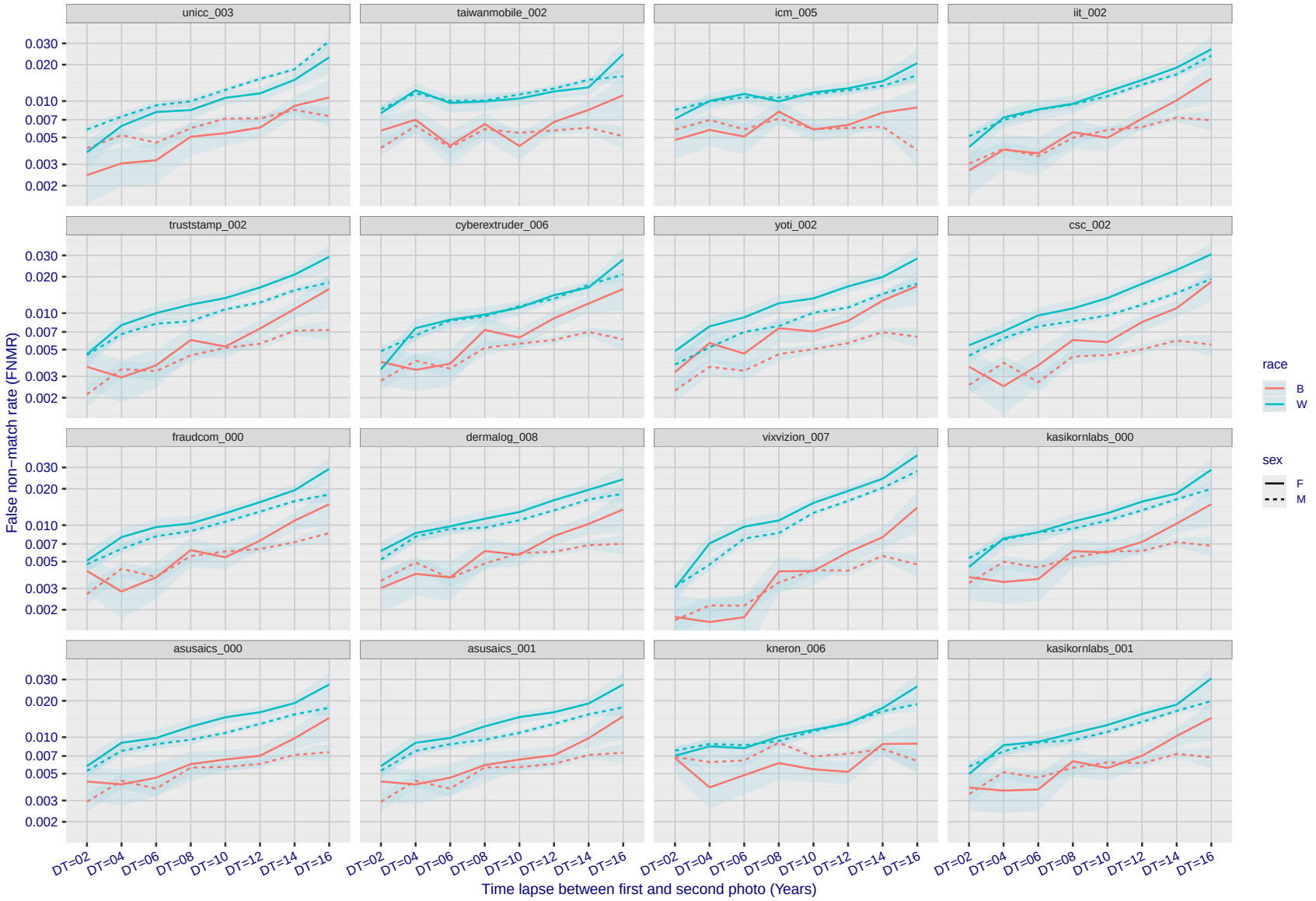


Figure 774: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

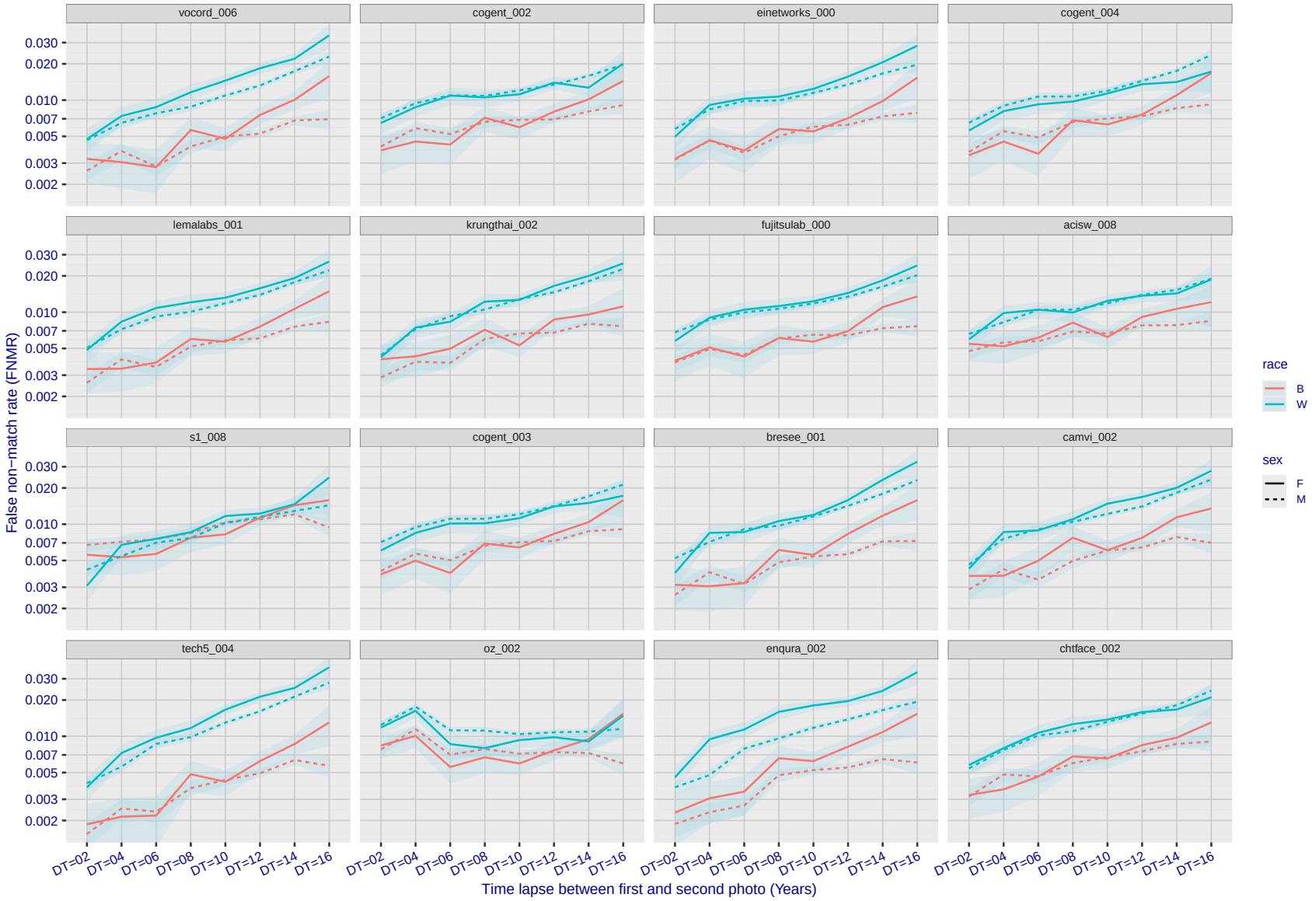


Figure 775: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

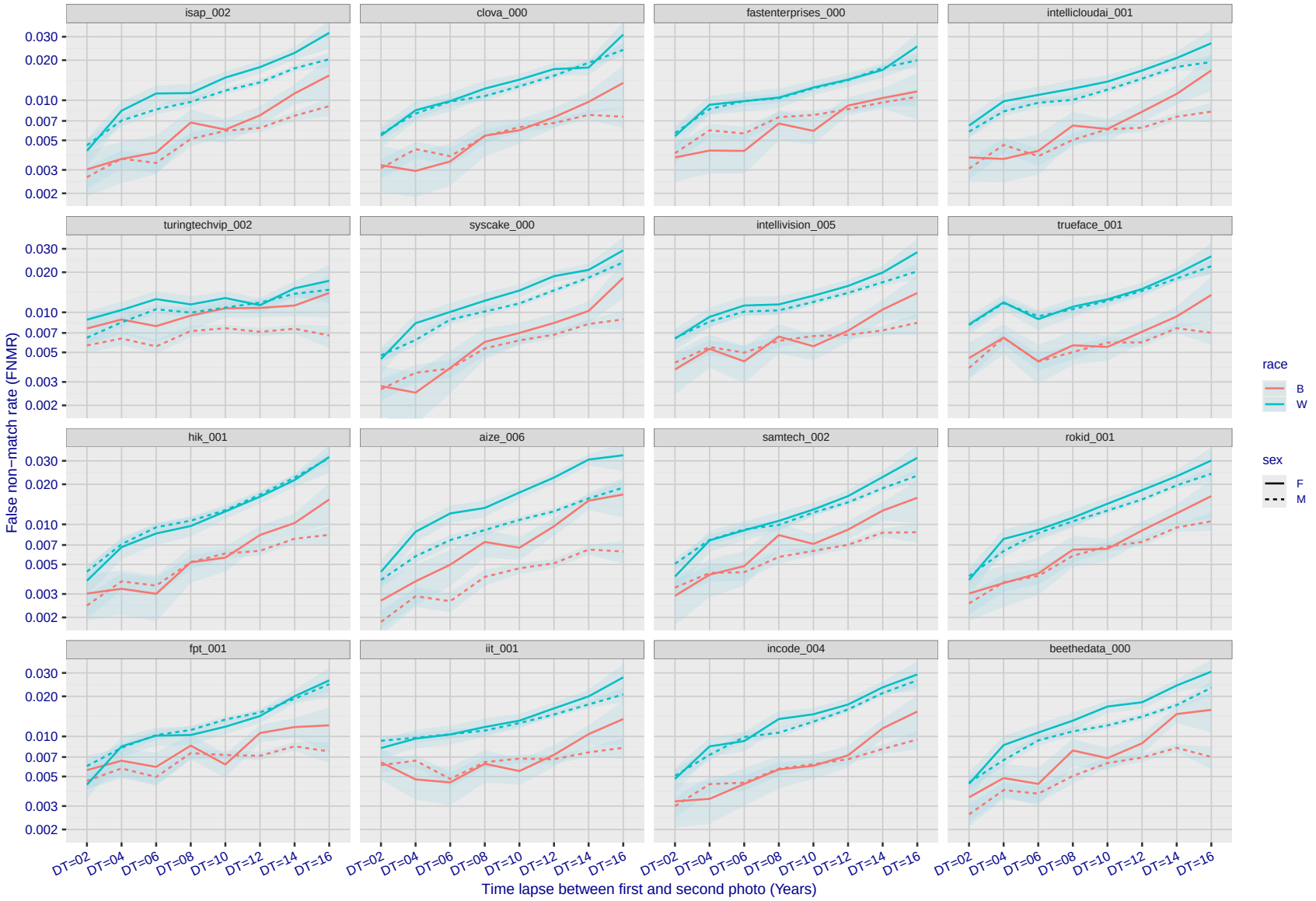


Figure 776: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

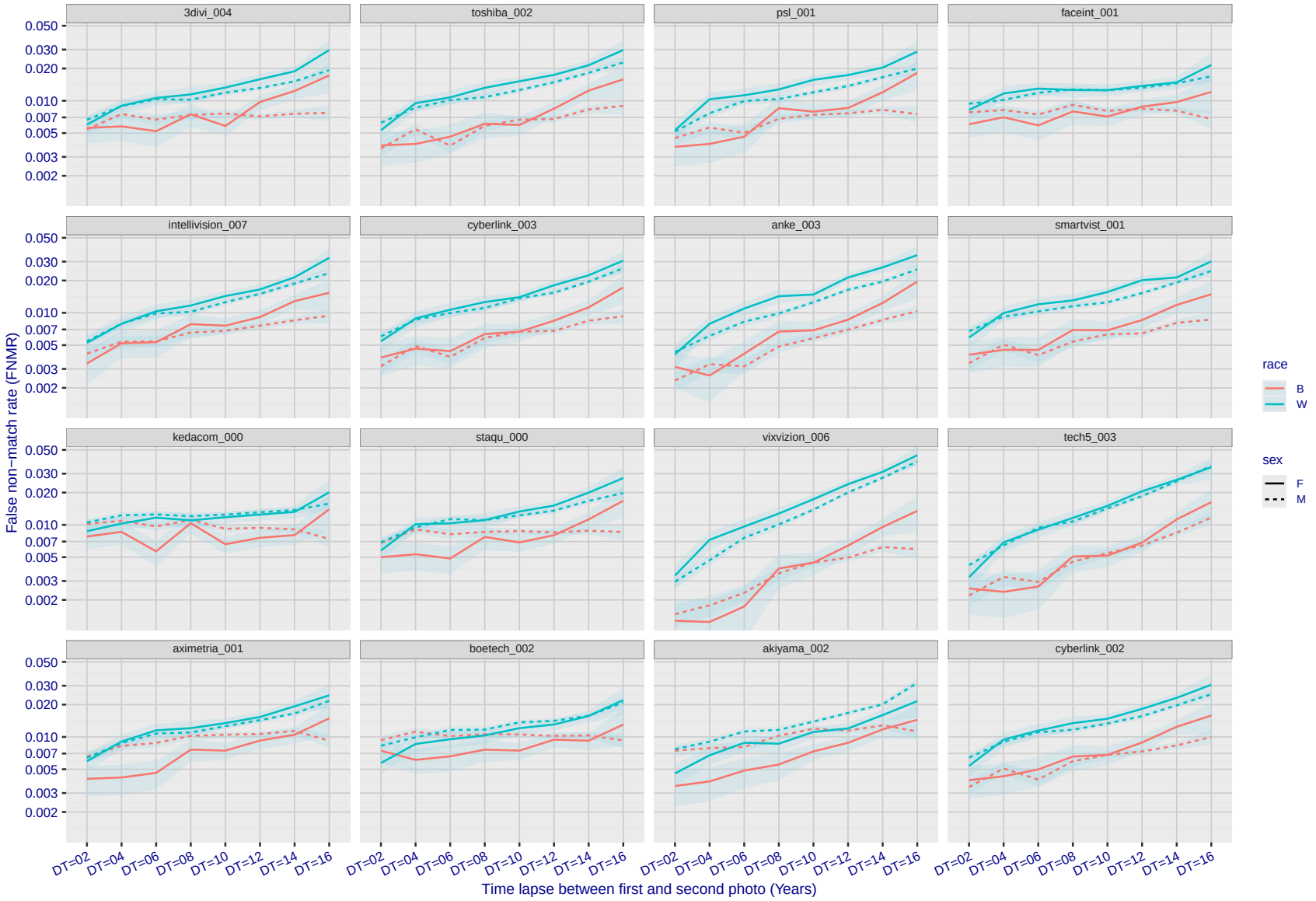


Figure 777: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

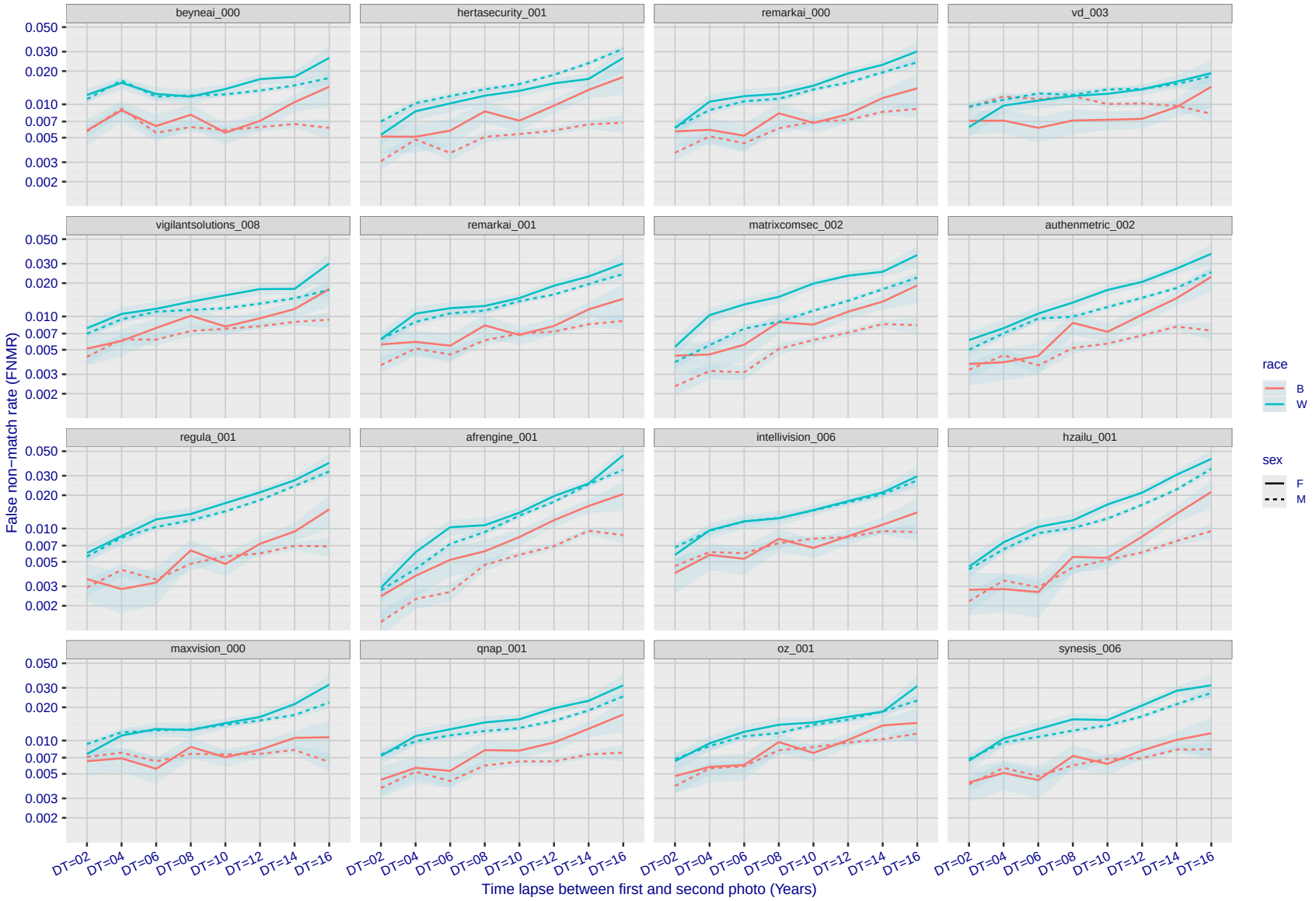


Figure 778: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

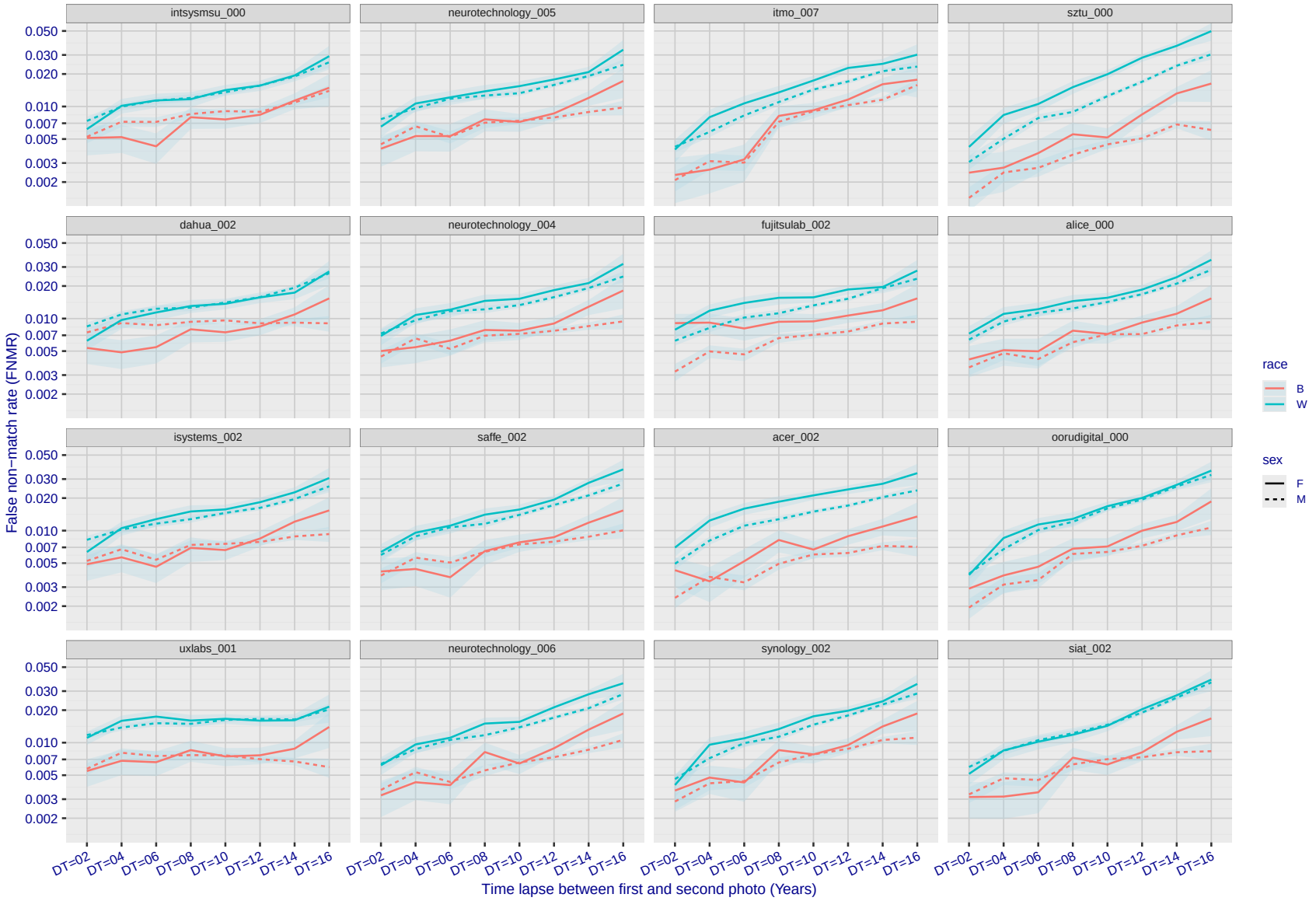


Figure 779: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

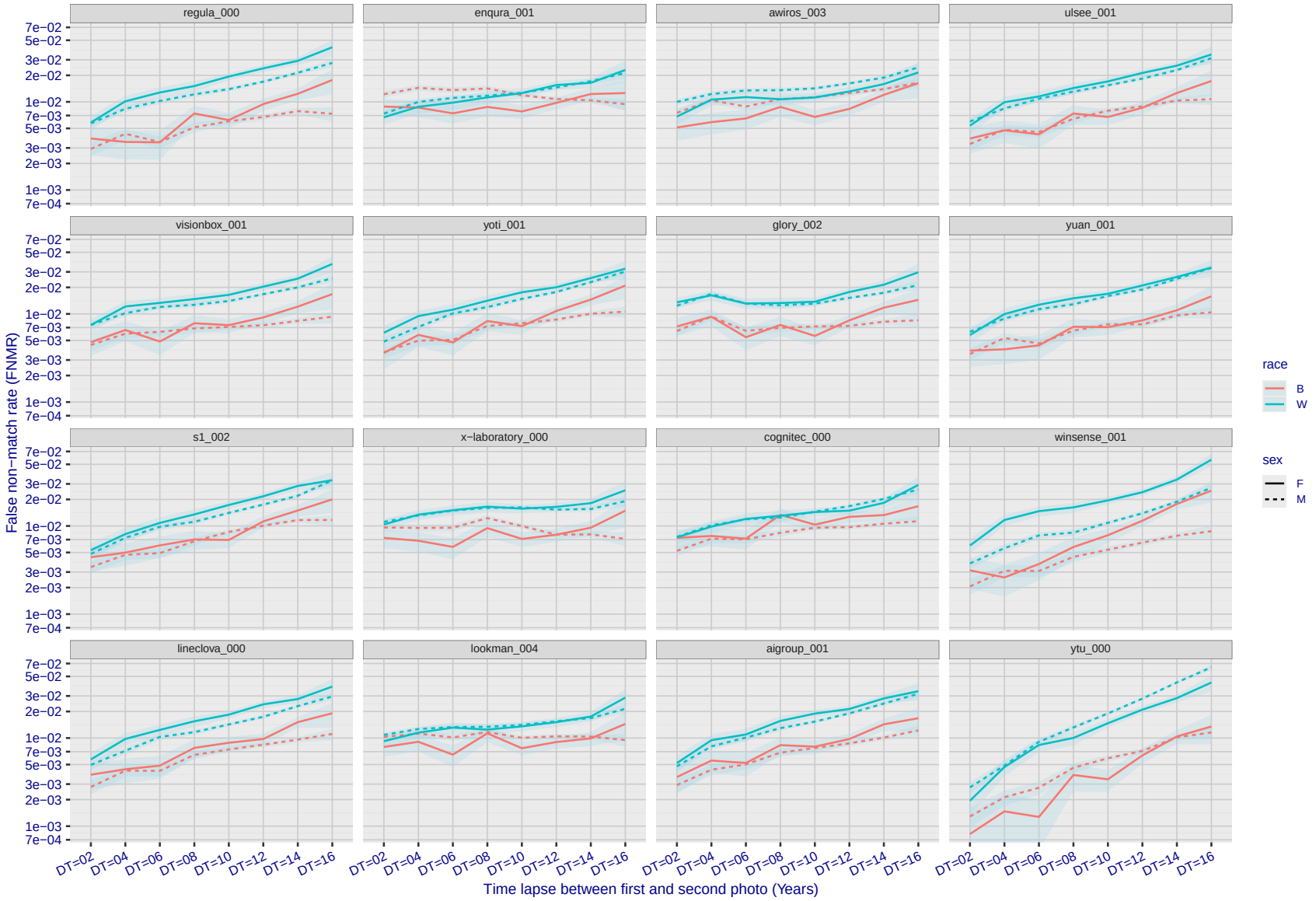


Figure 780: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

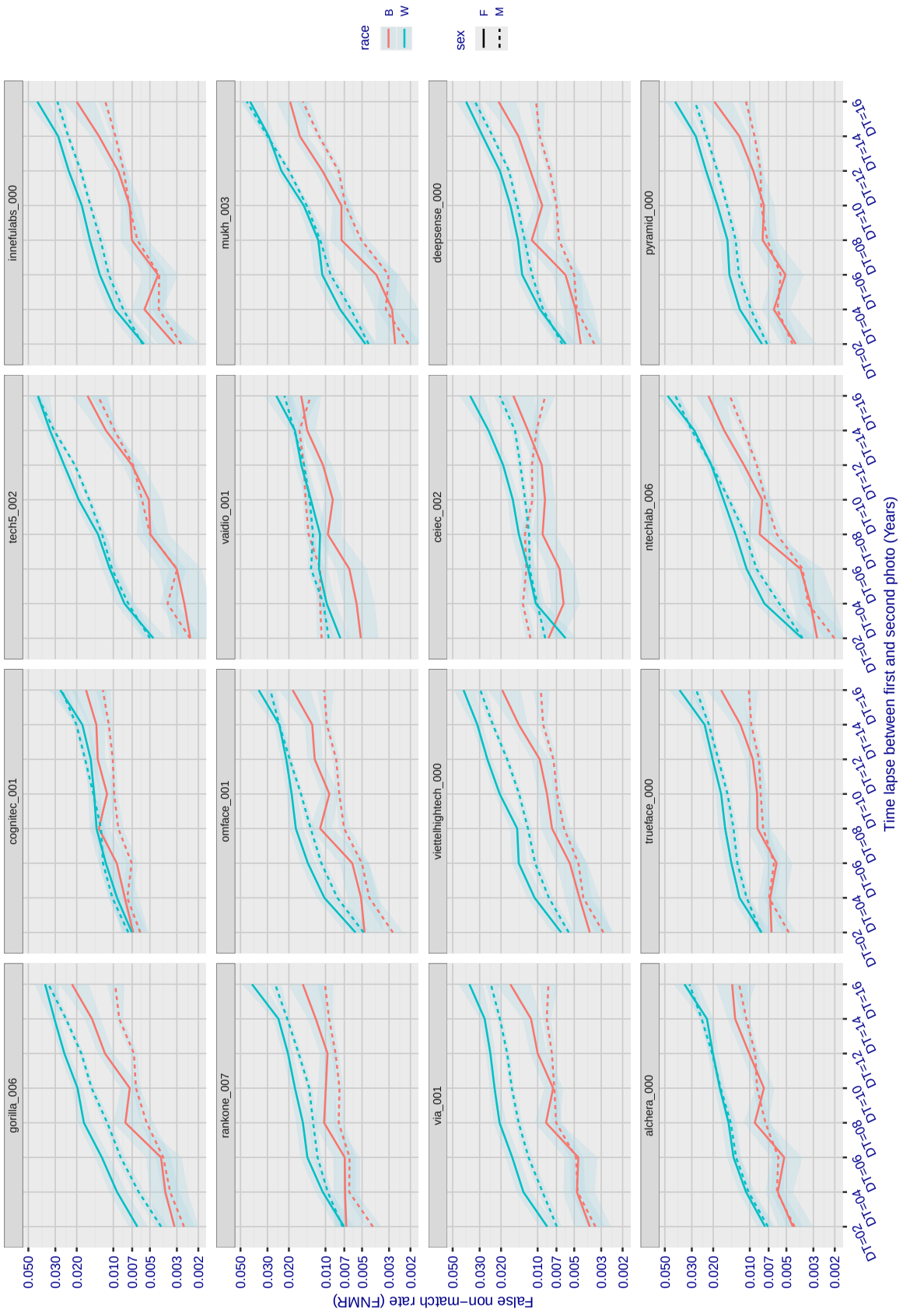


Figure 781: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (*$FNMR < 0.001$*) so that the uncertainty estimates are high.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

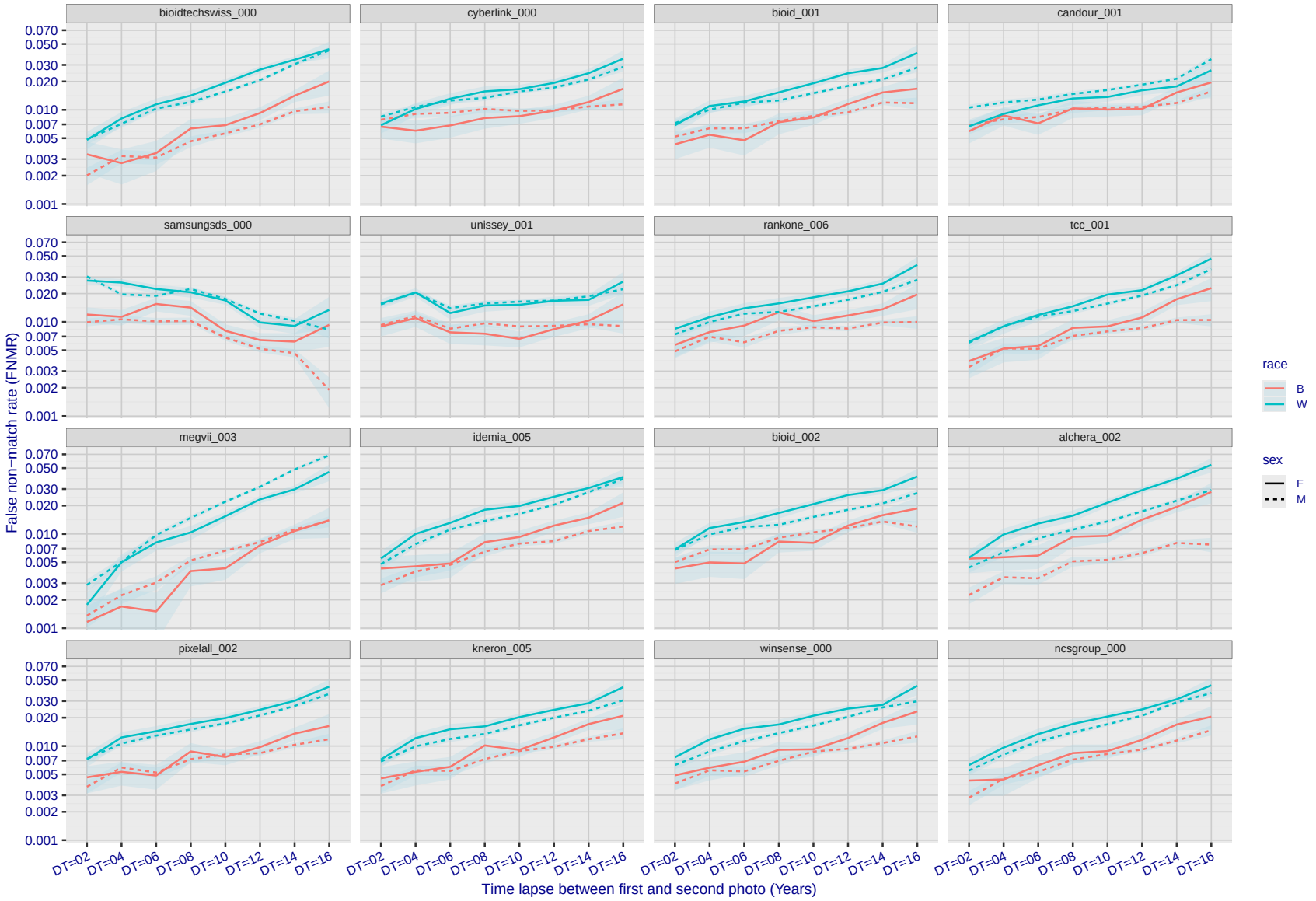


Figure 782: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

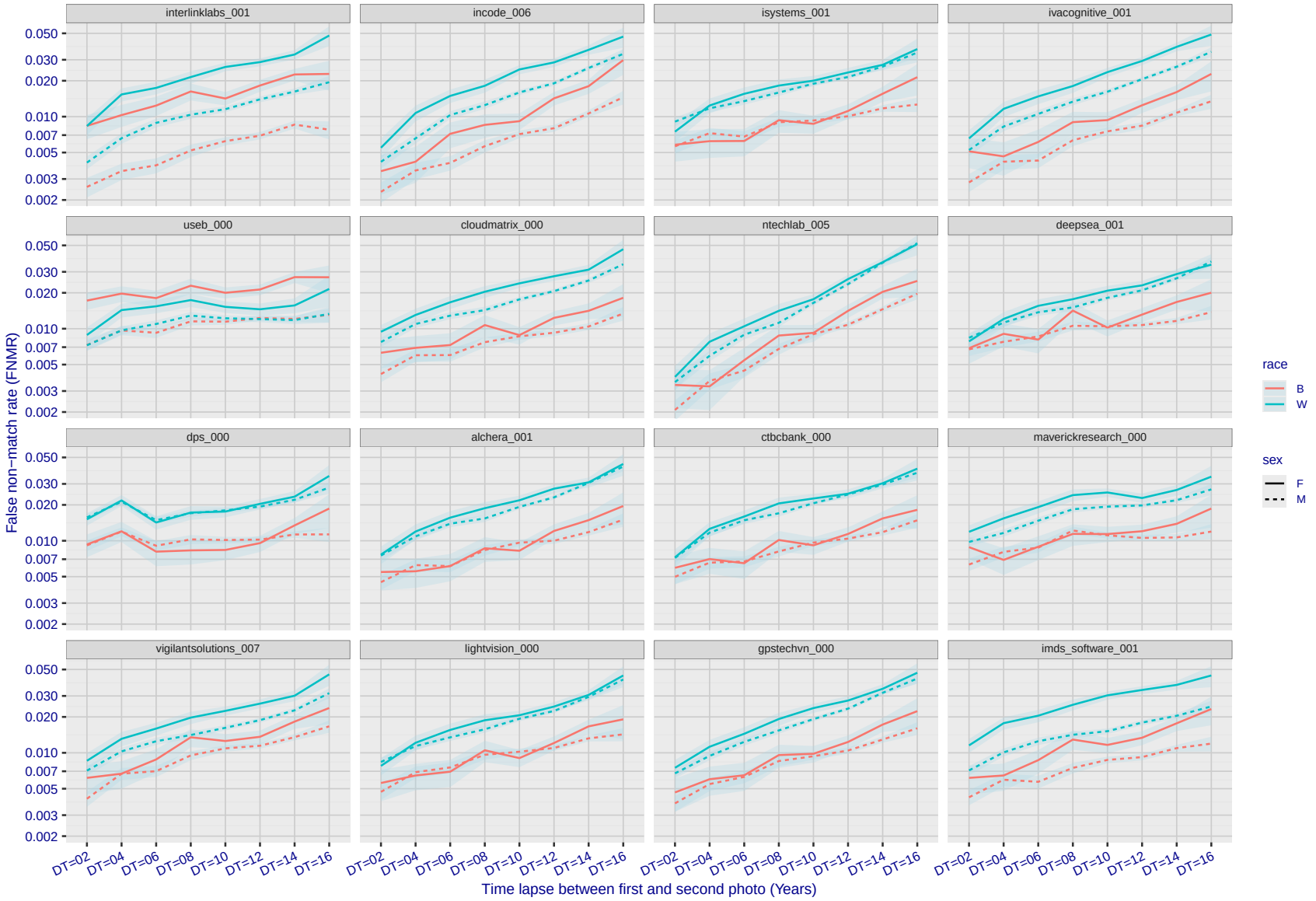


Figure 783: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

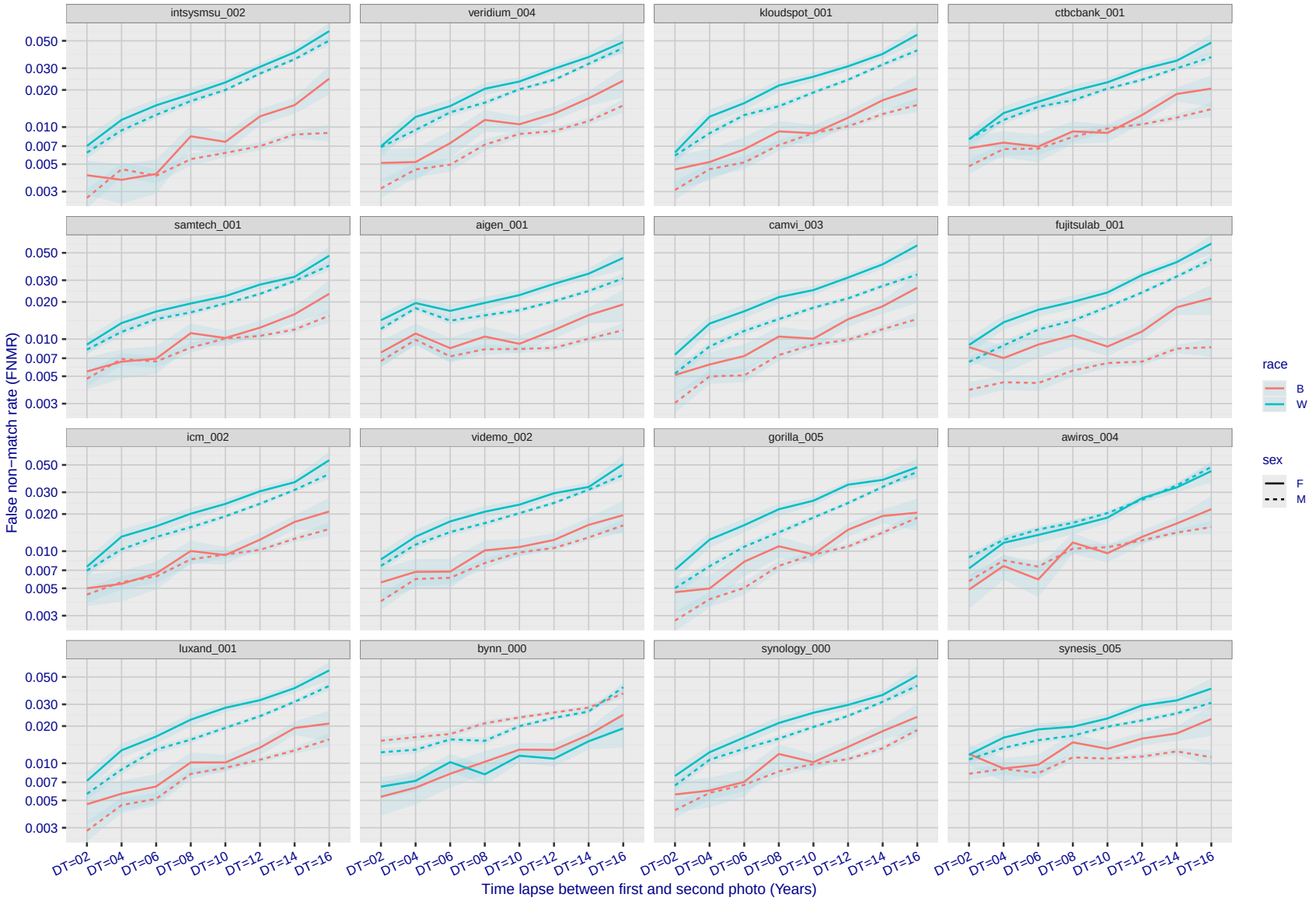


Figure 784: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

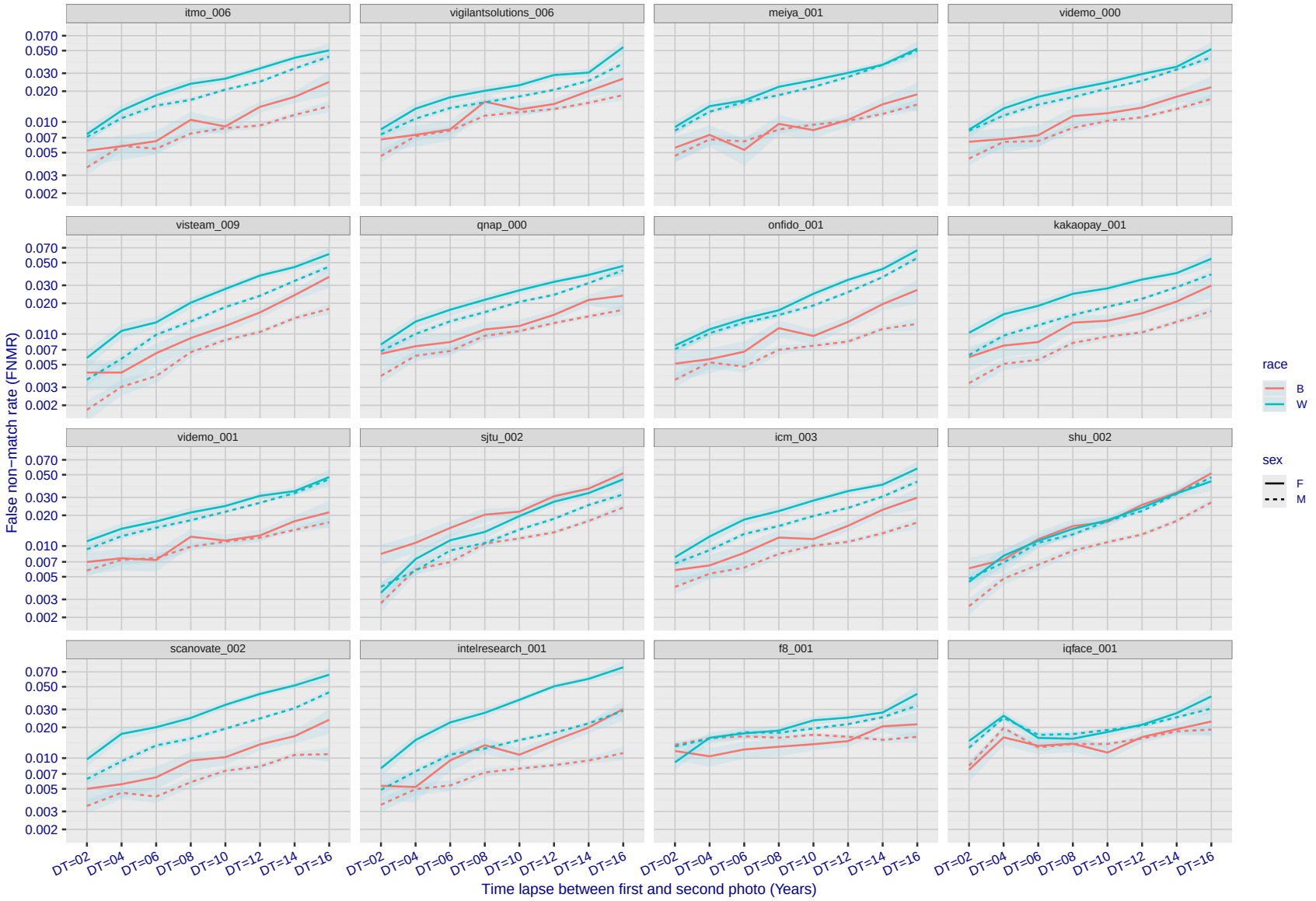


Figure 785: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

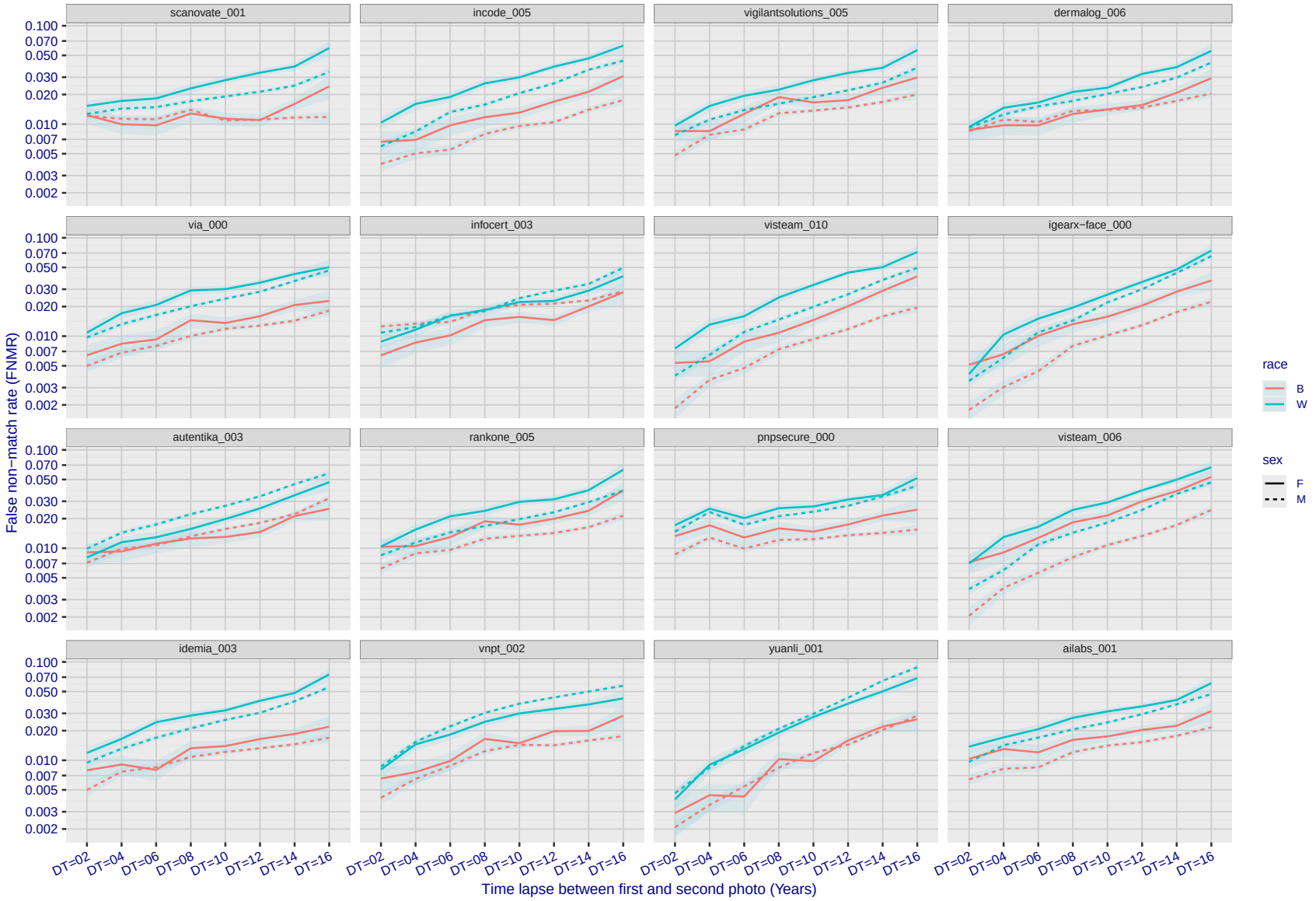


Figure 786: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

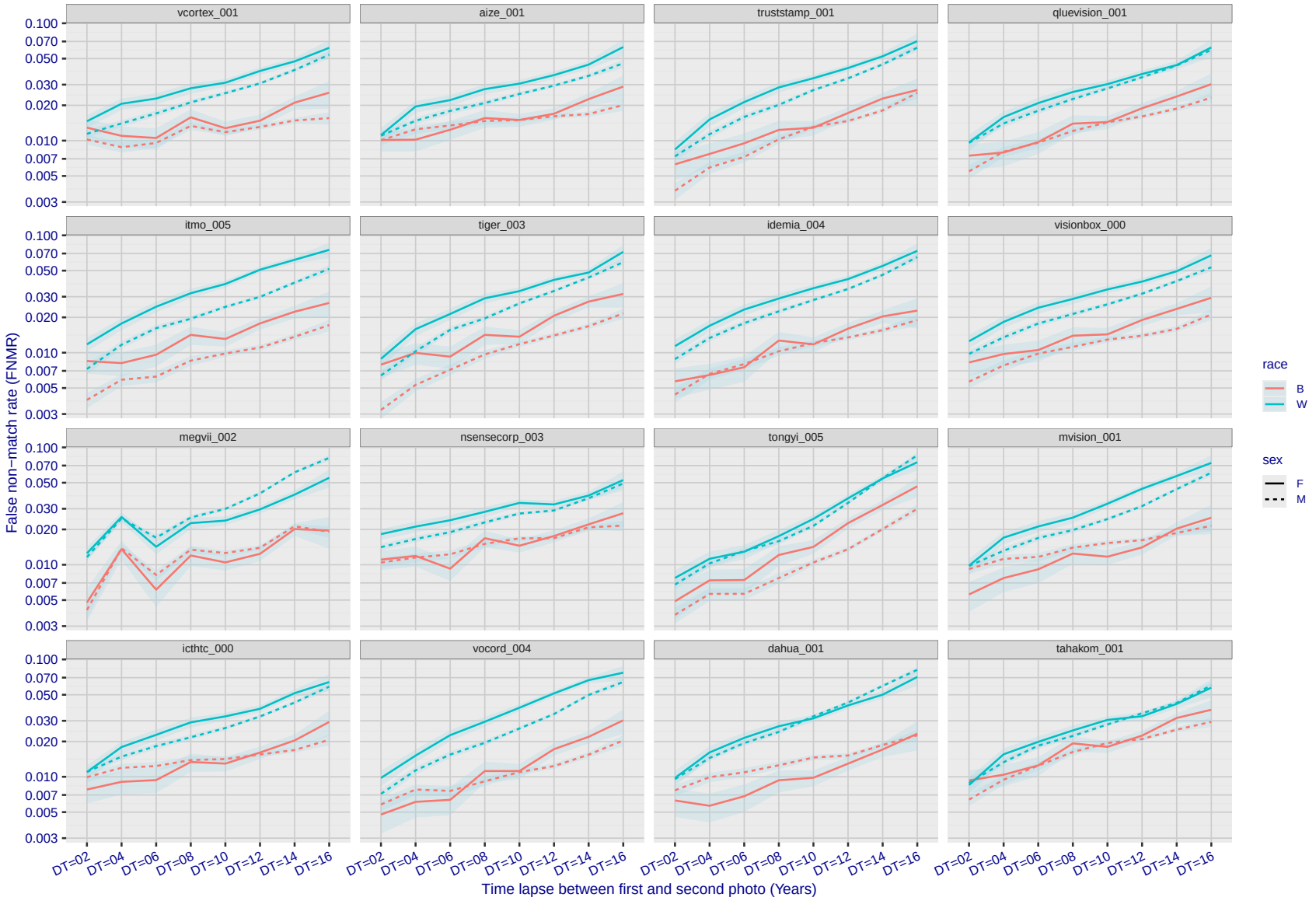


Figure 787: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

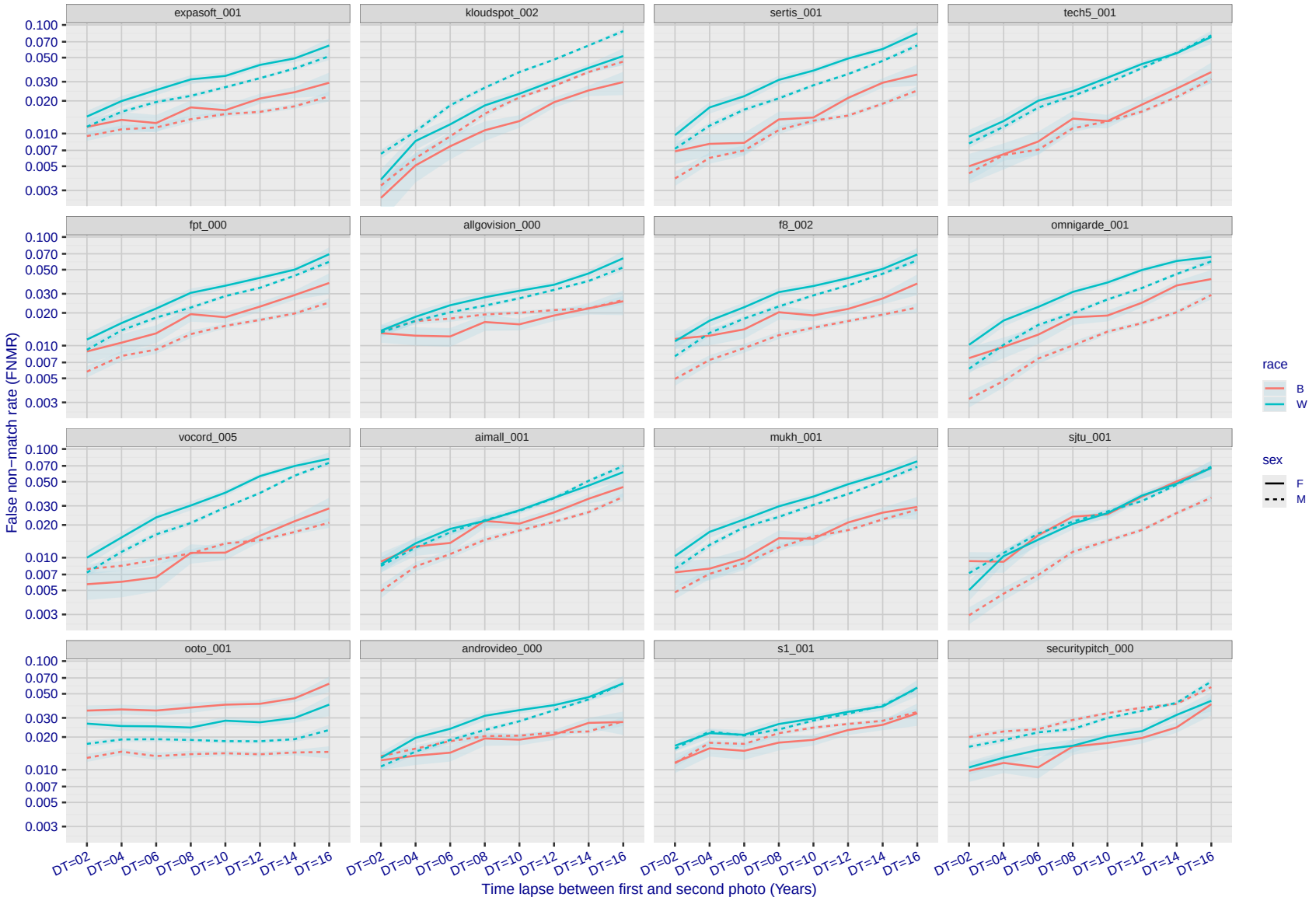


Figure 788: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

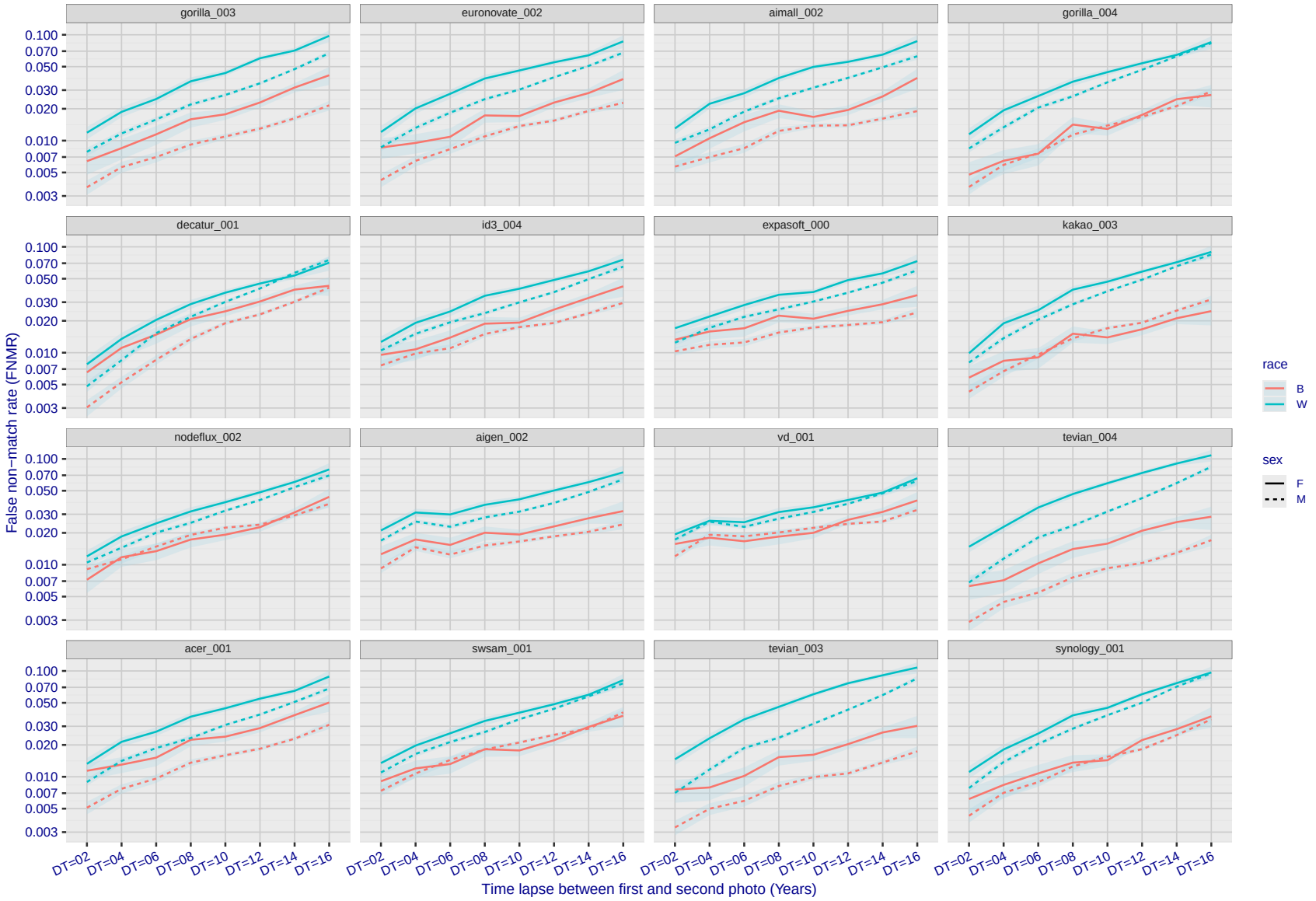


Figure 789: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

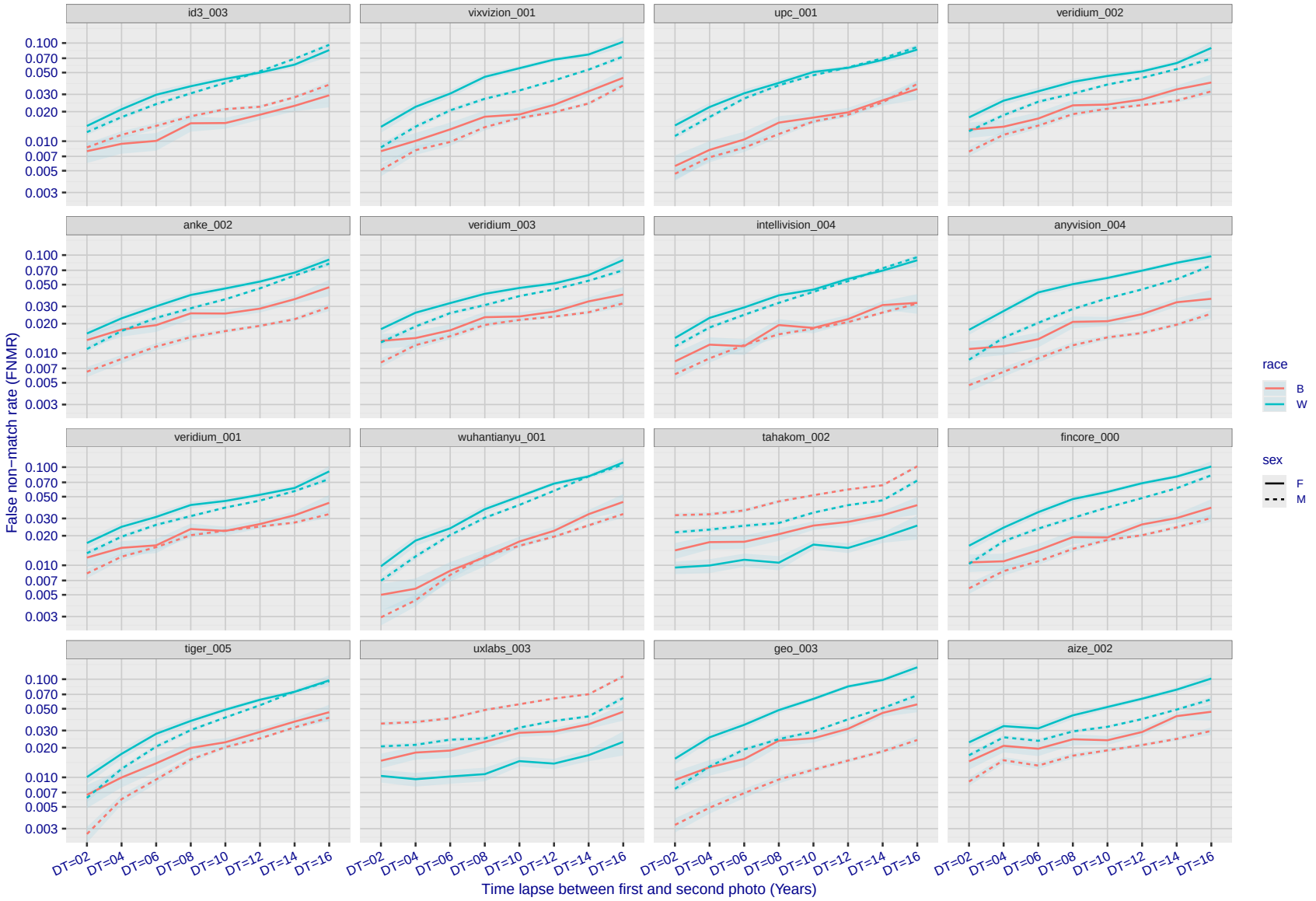


Figure 790: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

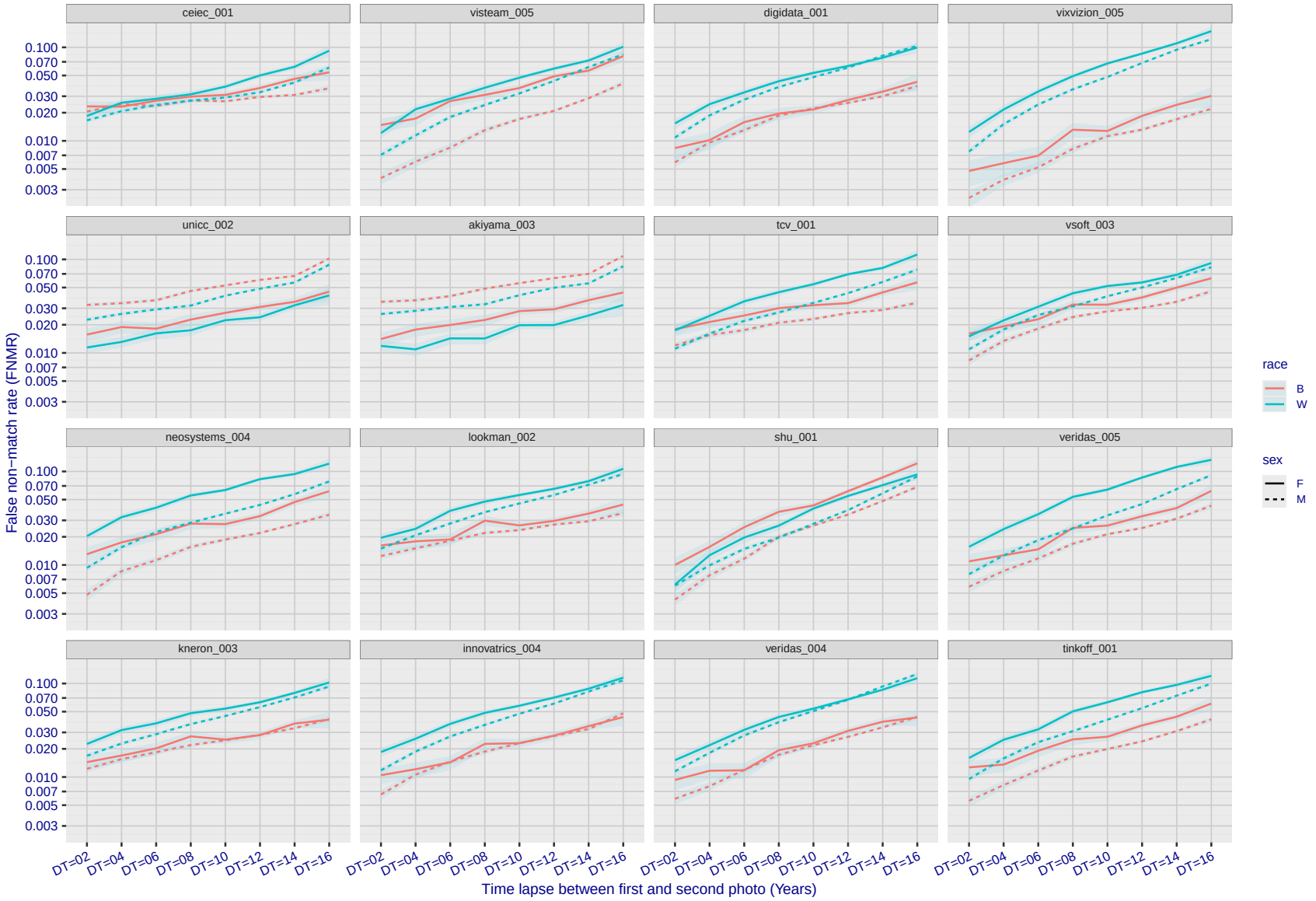


Figure 791: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

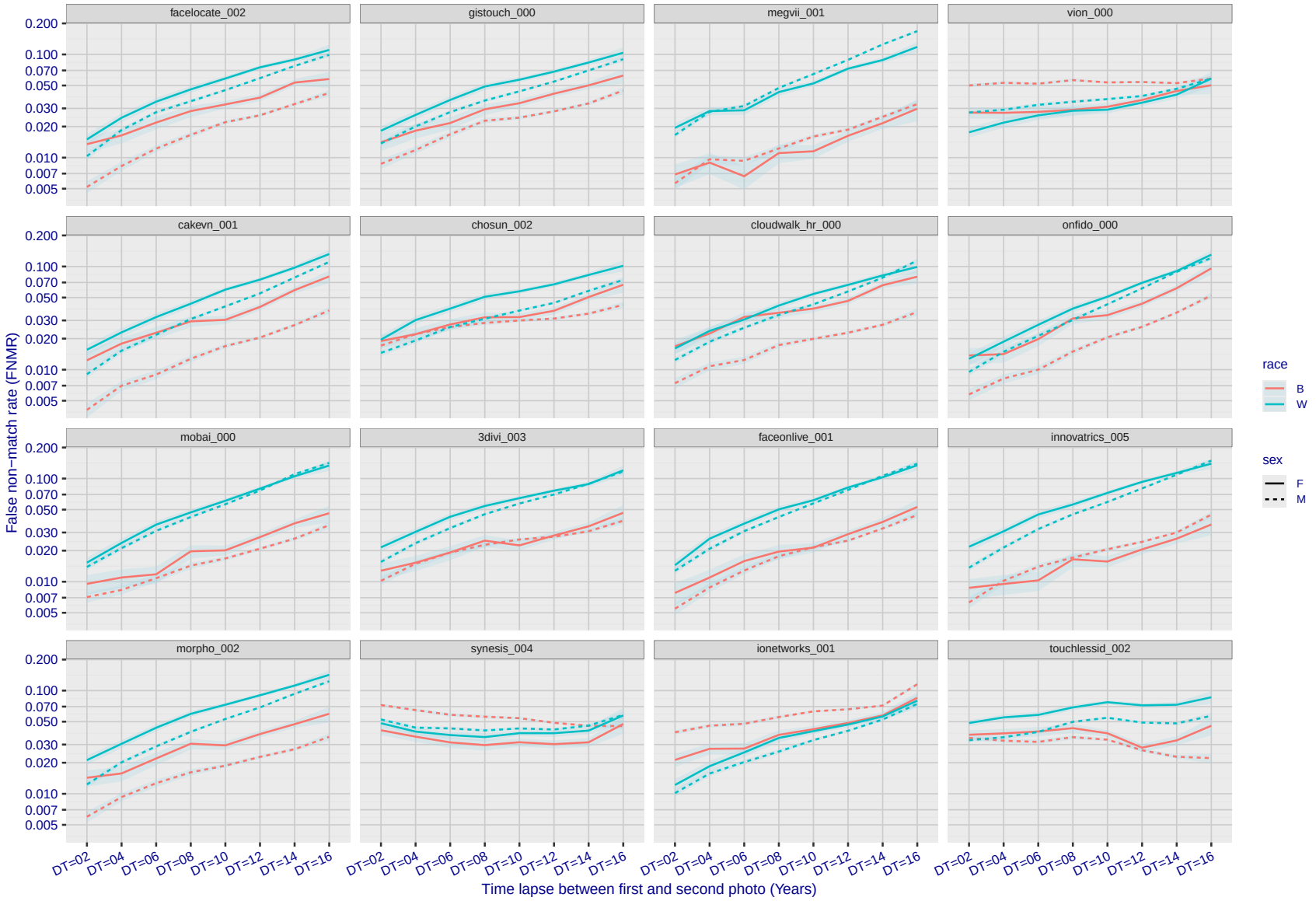


Figure 792: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

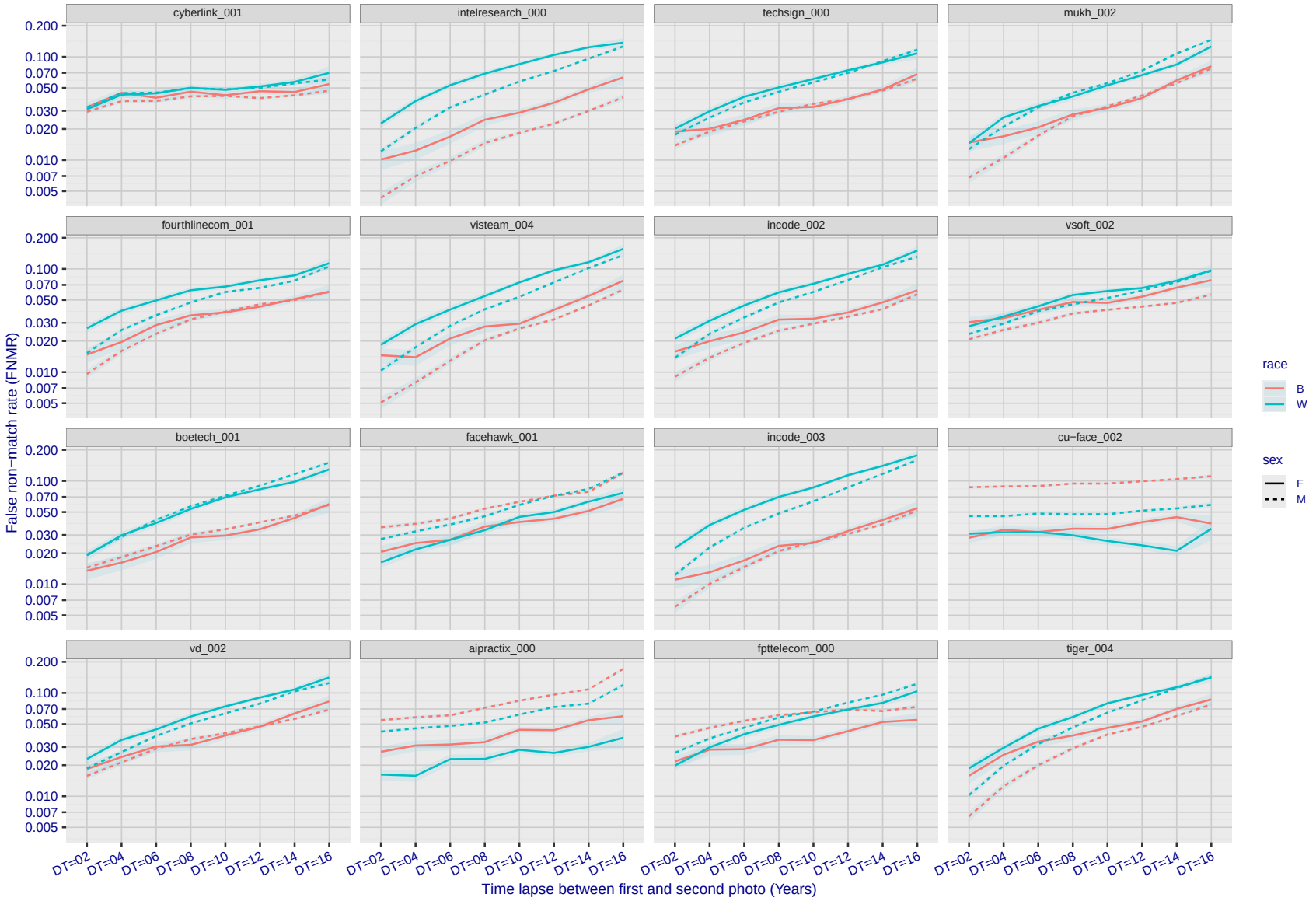


Figure 793: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

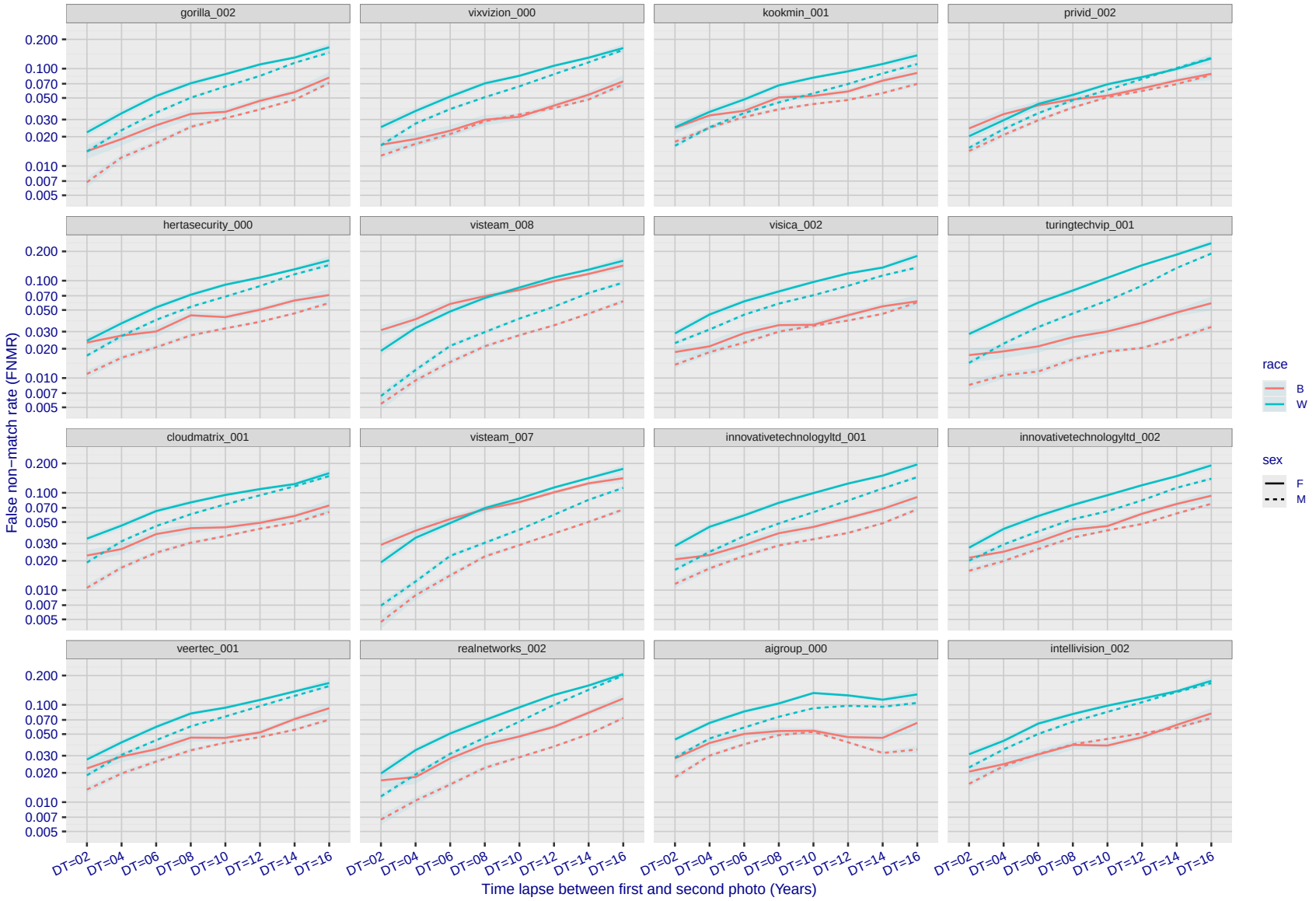


Figure 794: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

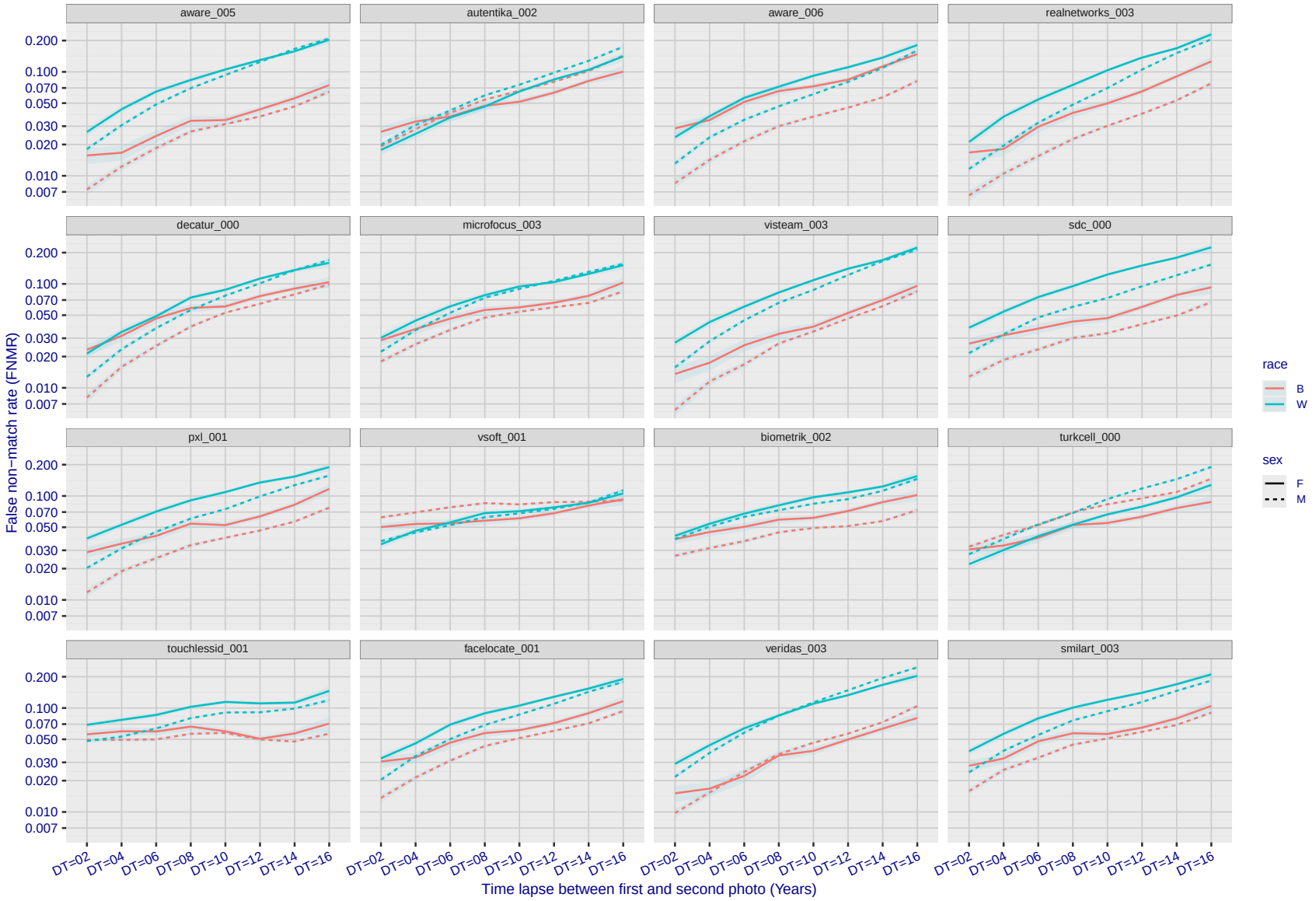


Figure 795: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

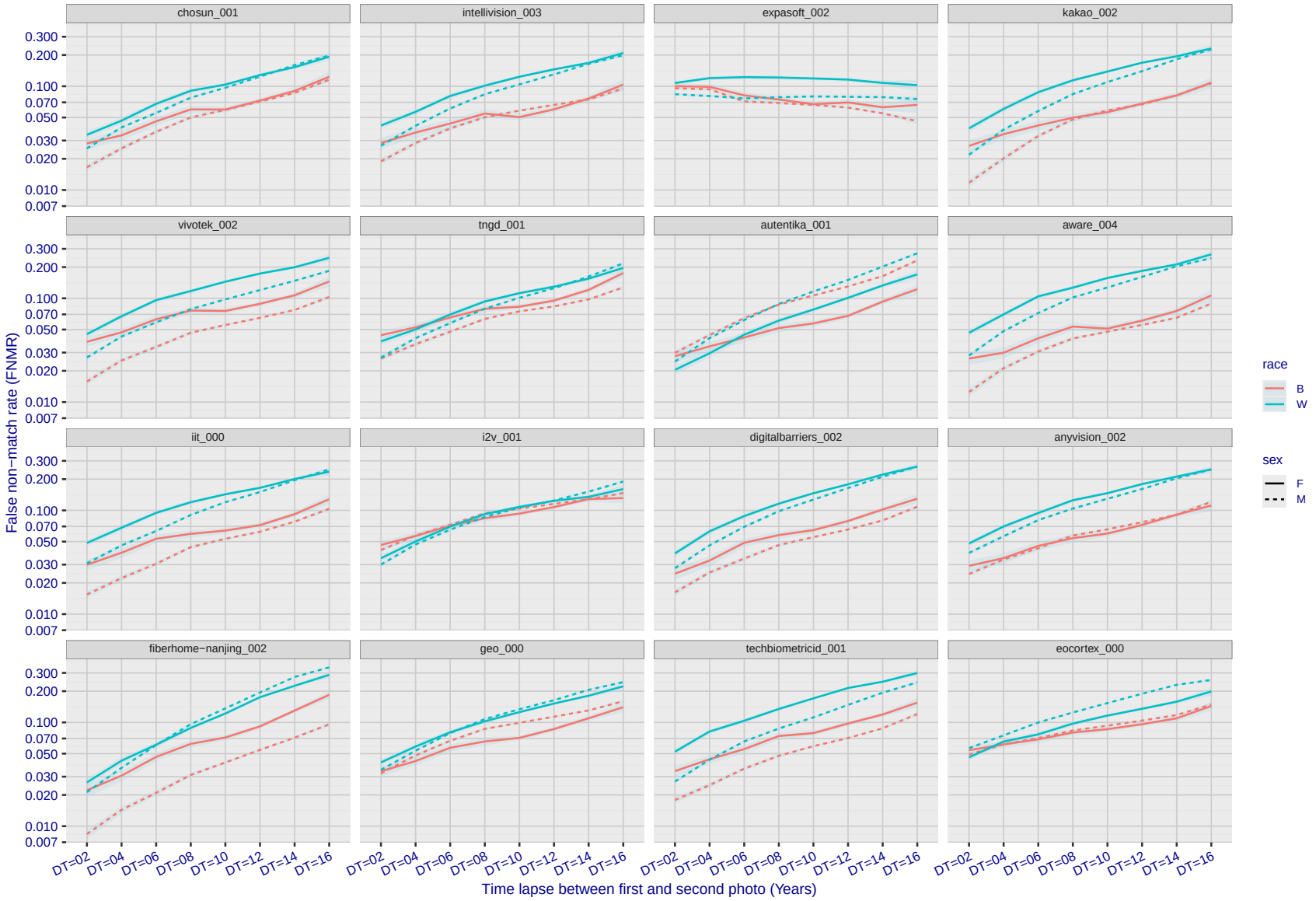


Figure 796: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

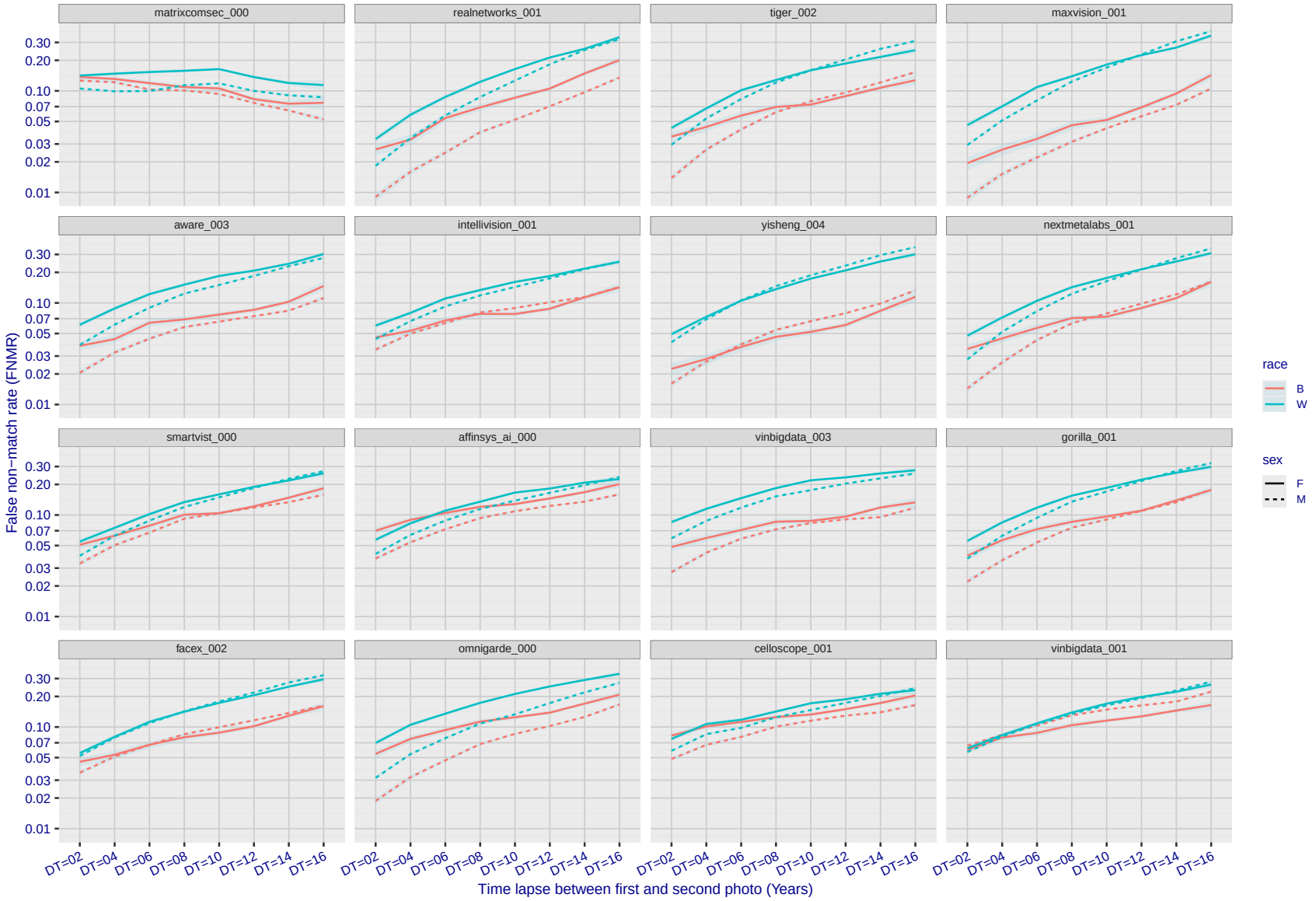


Figure 797: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

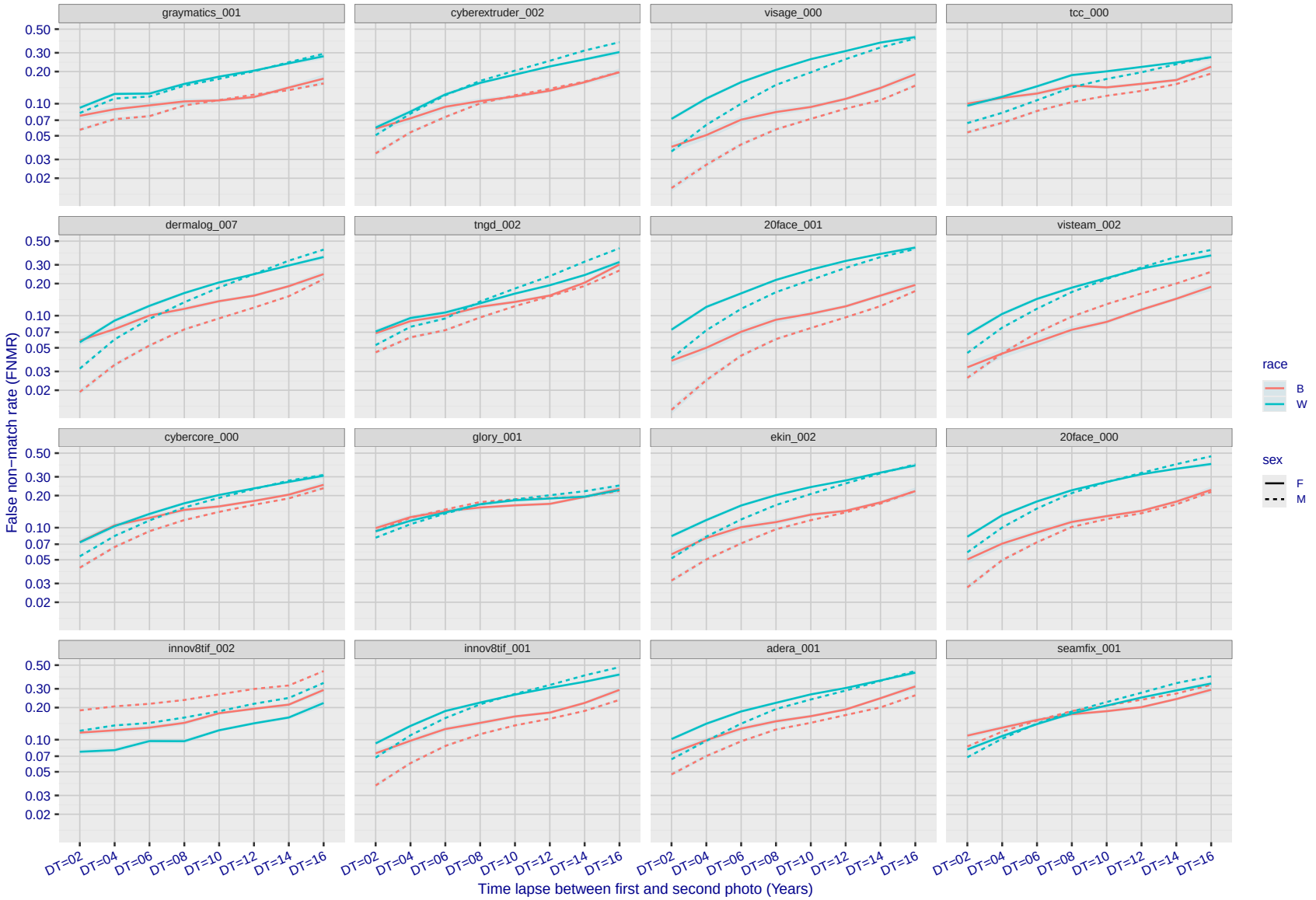


Figure 798: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR* < 0.001*) so that the uncertainty estimates are high.*

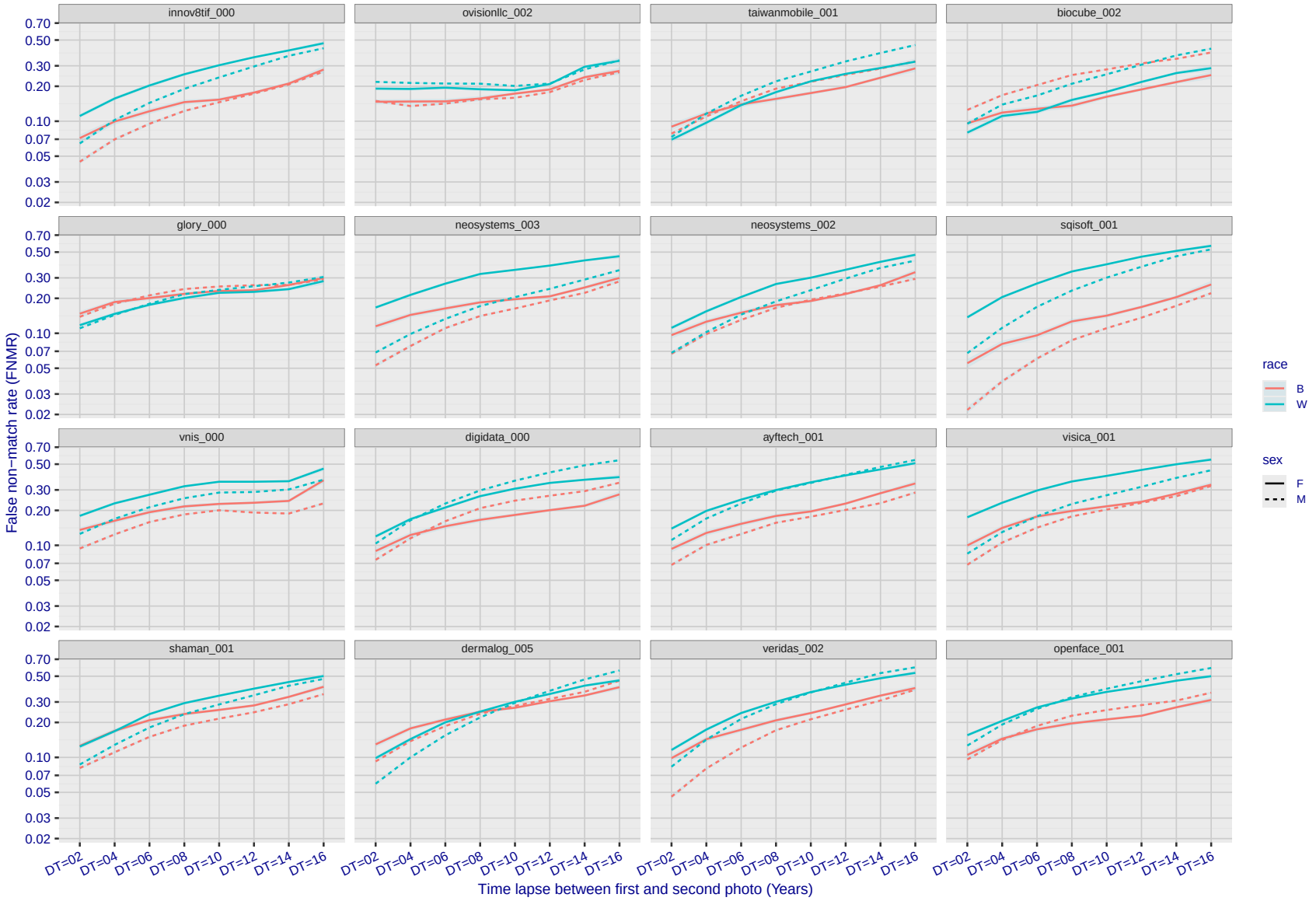


Figure 799: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR* < 0.001*) so that the uncertainty estimates are high.*

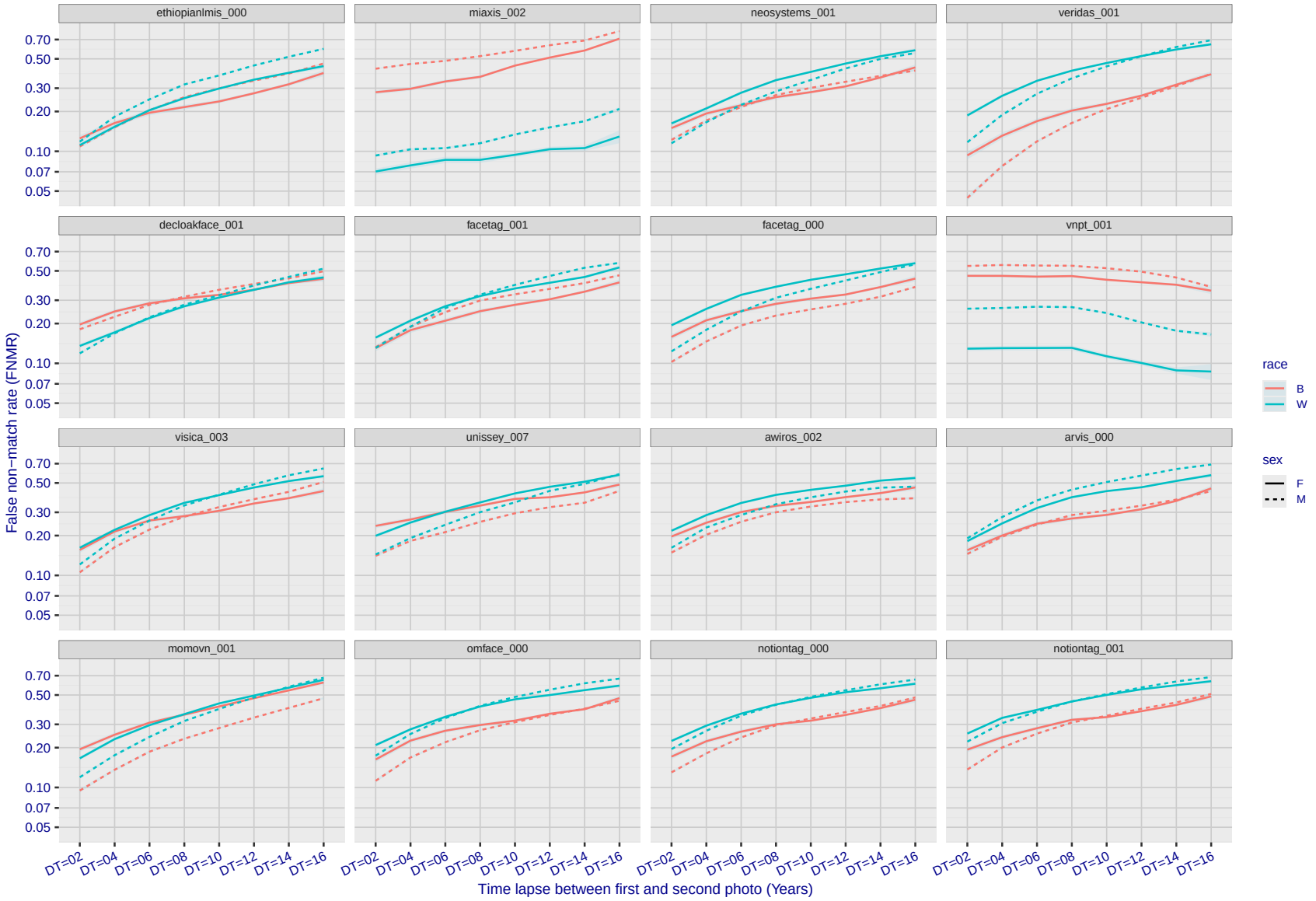


Figure 800: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

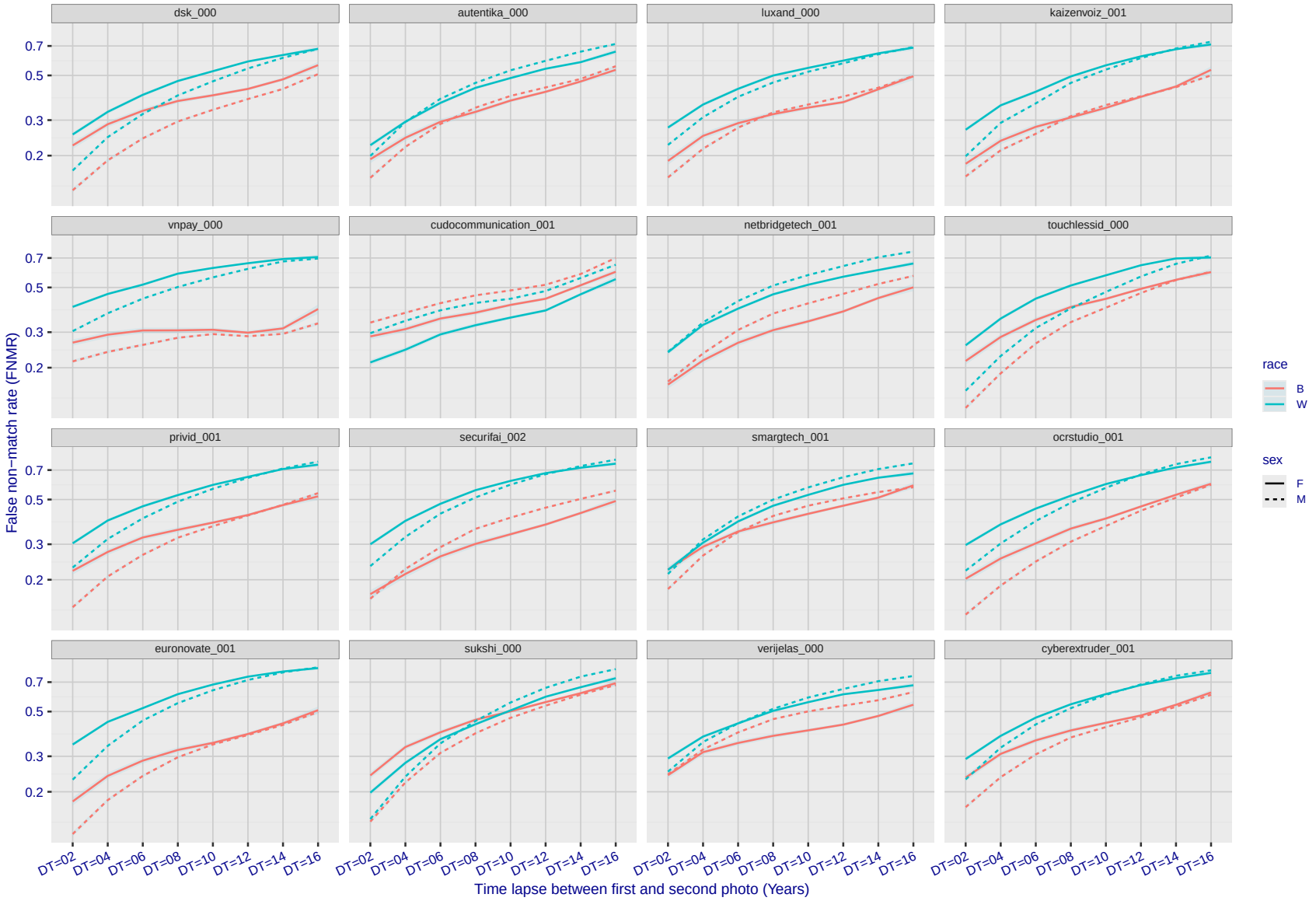


Figure 801: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

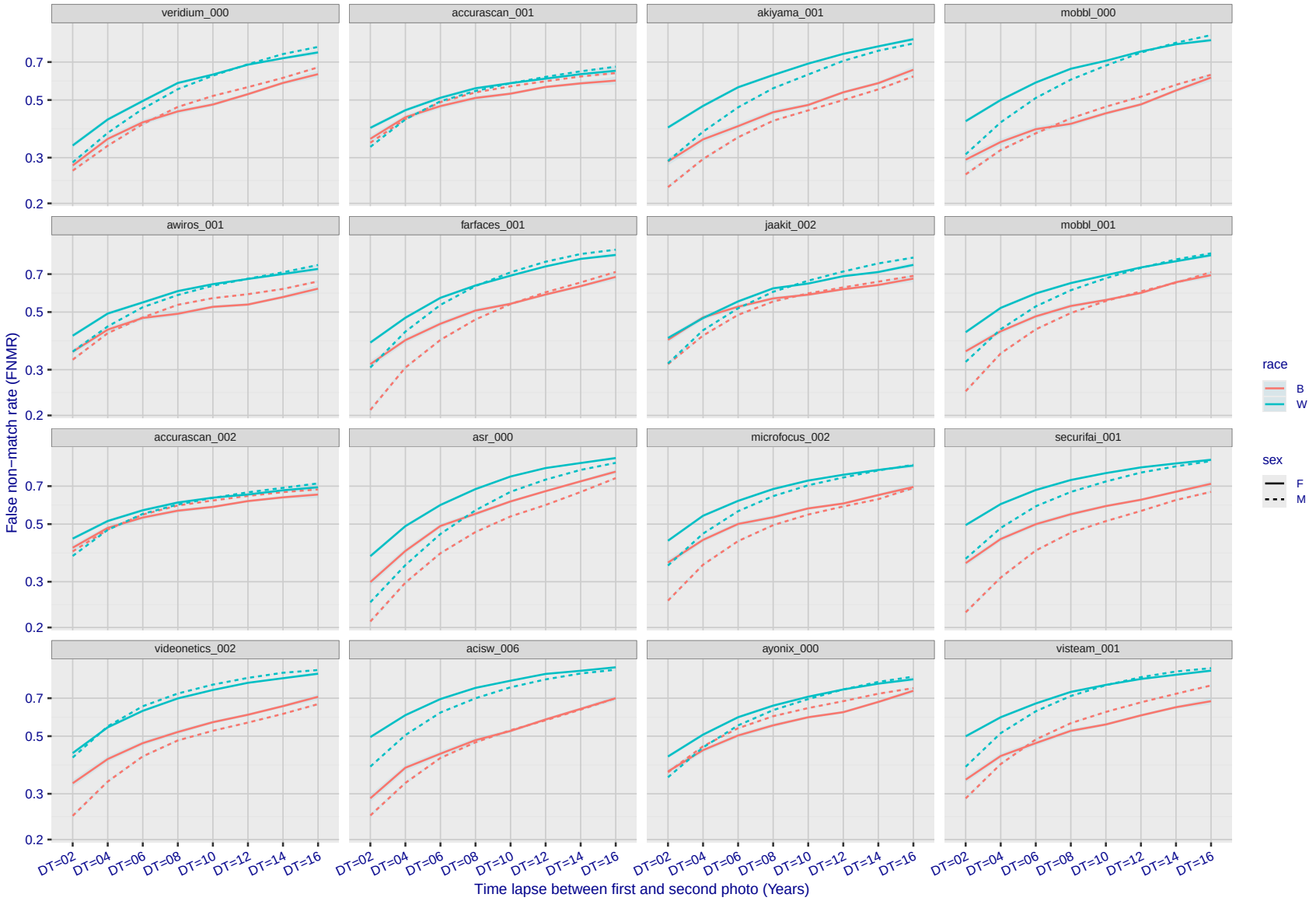


Figure 802: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

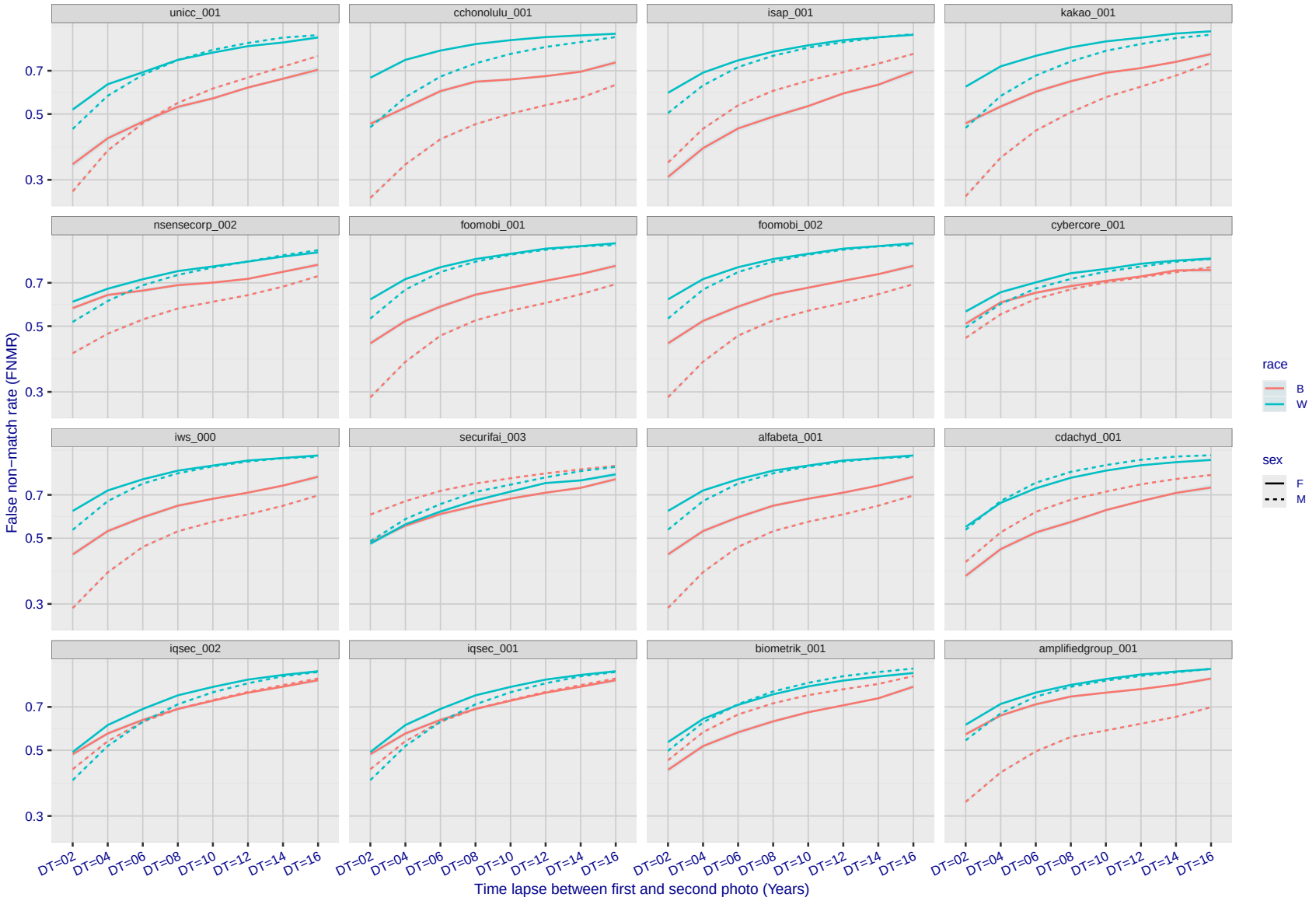


Figure 803: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR* < 0.001*) so that the uncertainty estimates are high.*

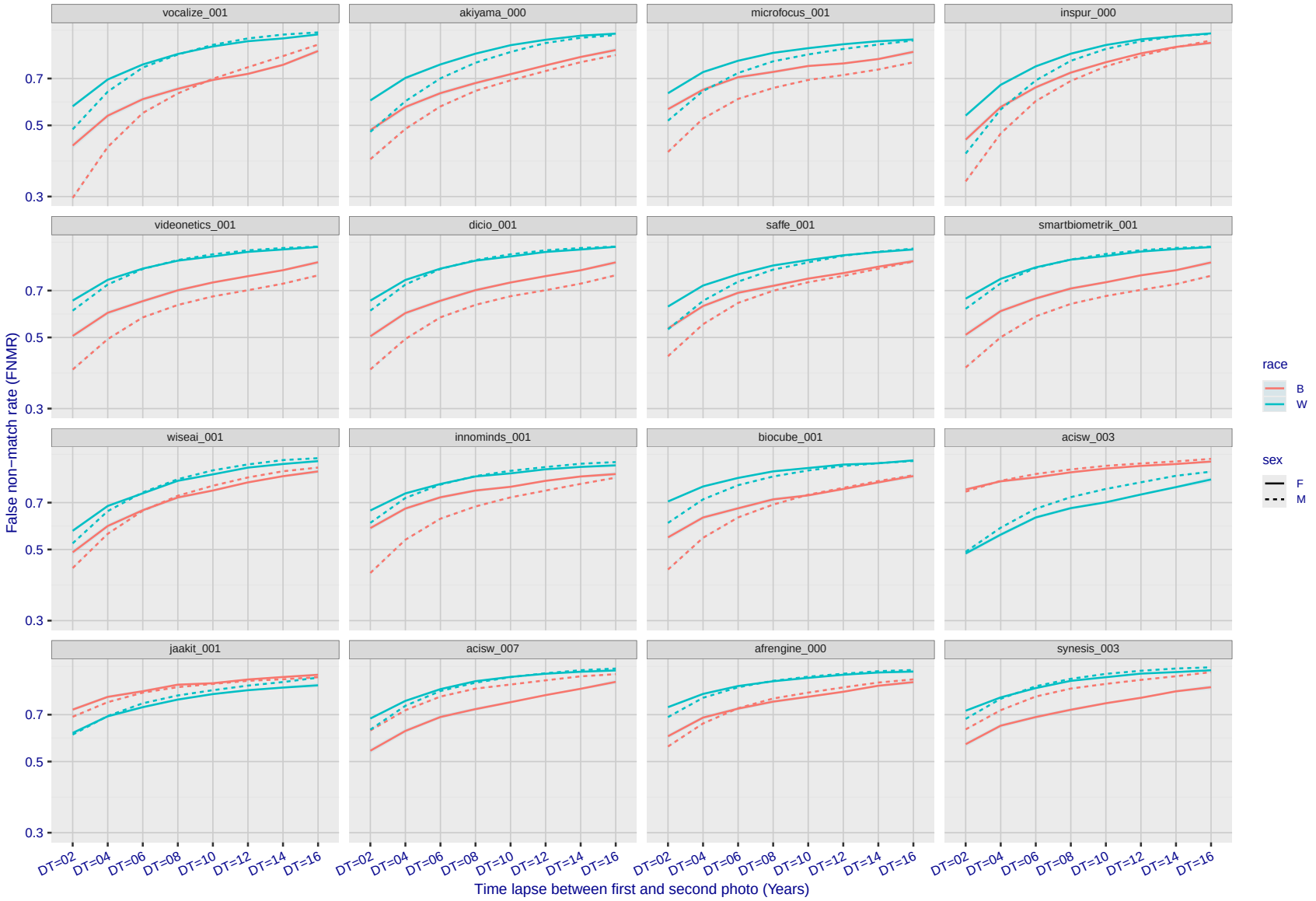


Figure 804: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

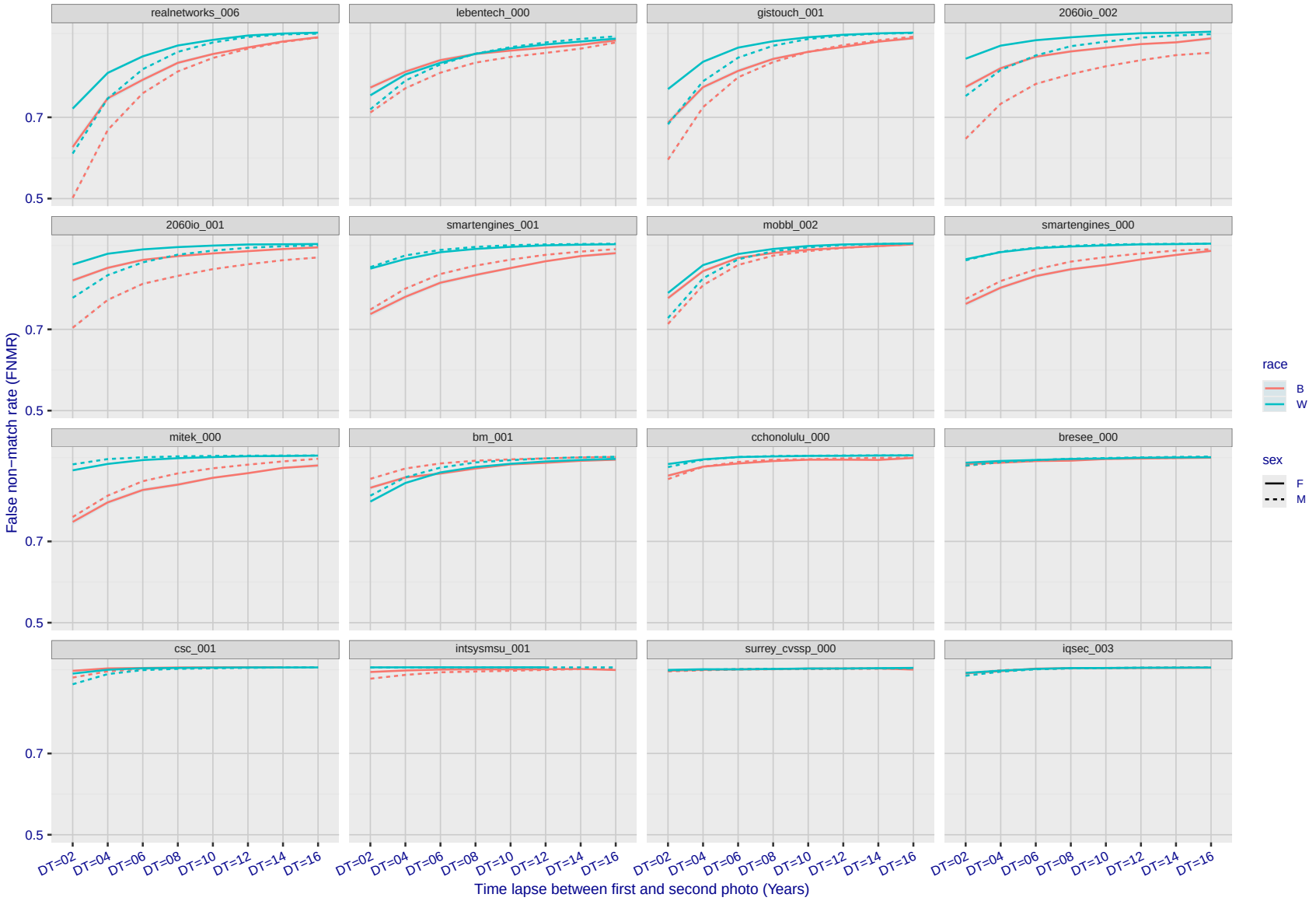


Figure 805: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

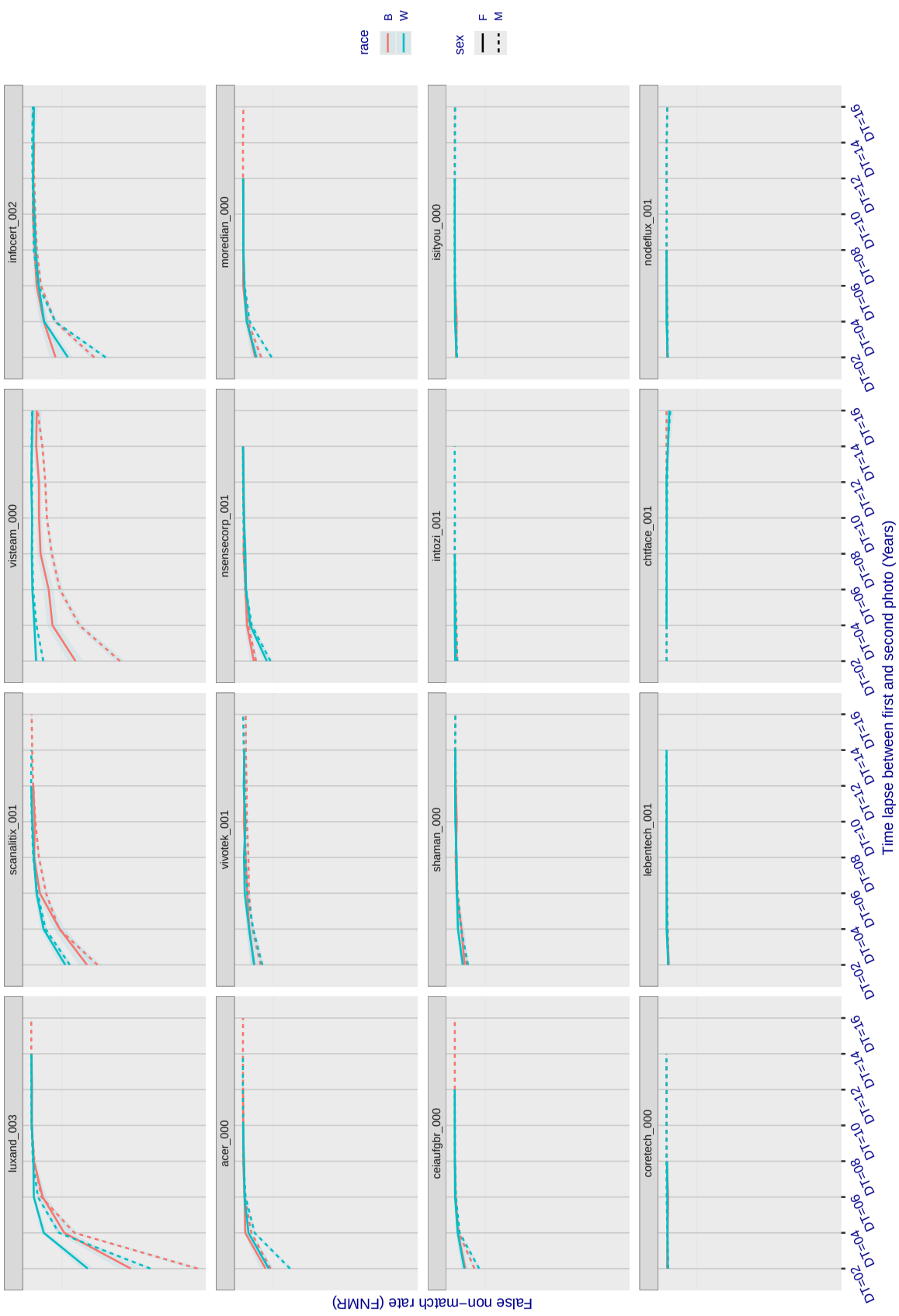


Figure 806: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

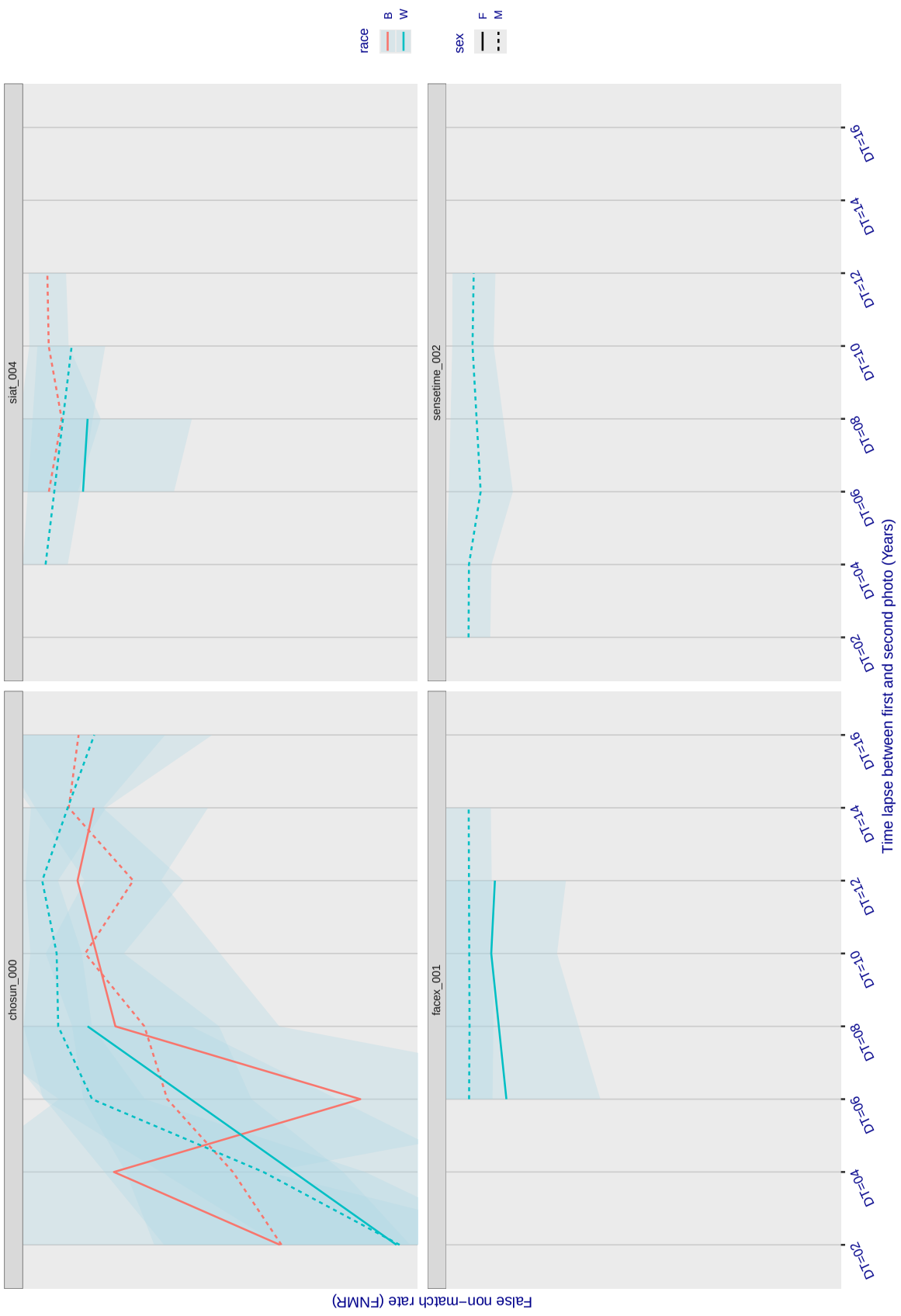


Figure 807: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors* $(FNMR < 0.001)$ *so that the uncertainty estimates are high.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.5.3 Effect of age on genuine subjects

Background: Faces change appearance throughout life. Face recognition algorithms have previously beeen reported to give better accuracy on older individuals (See NIST IR 8009).

Goal: To quantify false non-match rates (FNMR) as a function of age, without an ageing component.

Methods: Using the visa images, which span fewer than five years, thresholds are determined that give FMR = 0.001 and 0.0001 over the entire impostor set. Then FNMR is measured over 1000 bootstrap replications of the genuine scores.

Results: For the visa images, Figure 898 shows how false non-match rates for genuine users, as a function of age group.

The notable aspects are:

- Younger subjects give considerably higher FNMR. This is likely due to rapid growth and change in facial appearance.
- FNMR trends down throughout life. The last bin, AGE $>$ 72, contains fewer than 140 mated pairs, and may be affected by small sample size.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 808: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 809: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 810: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 811: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 812: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

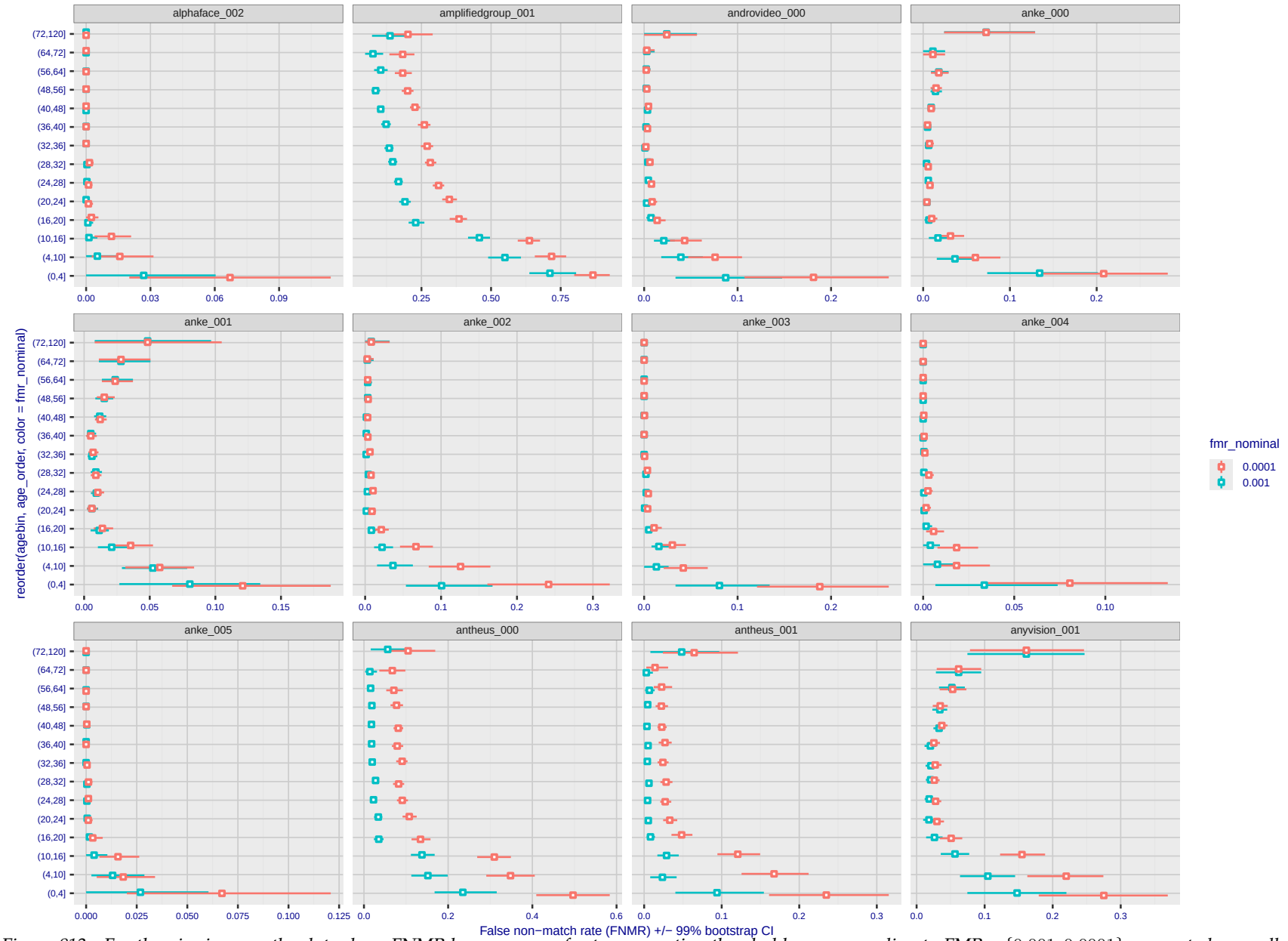


Figure 813: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 814: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 815: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 816: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 817: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 818: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

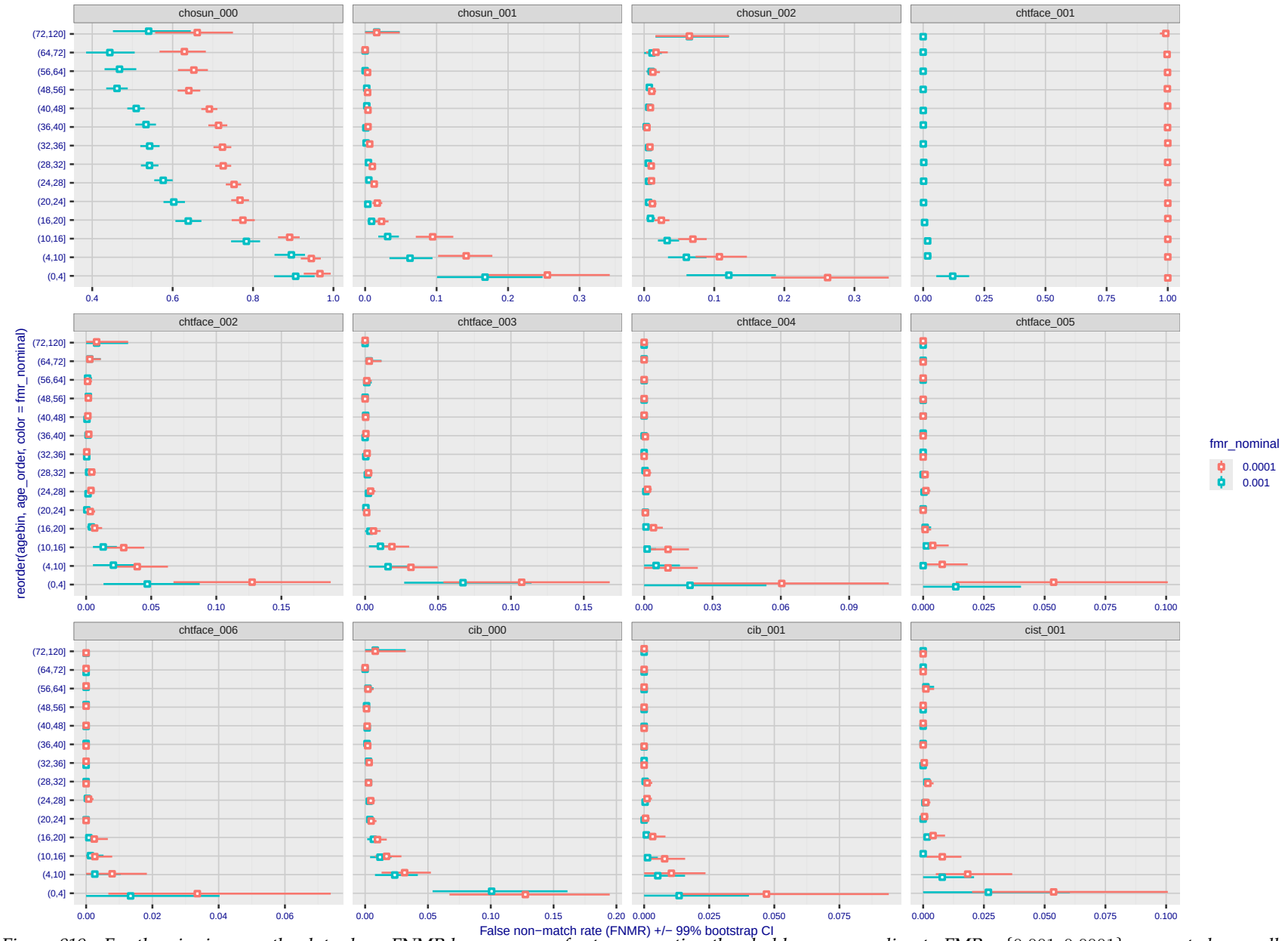


Figure 819: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 820: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 821: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 822: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 823: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 824: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 825: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

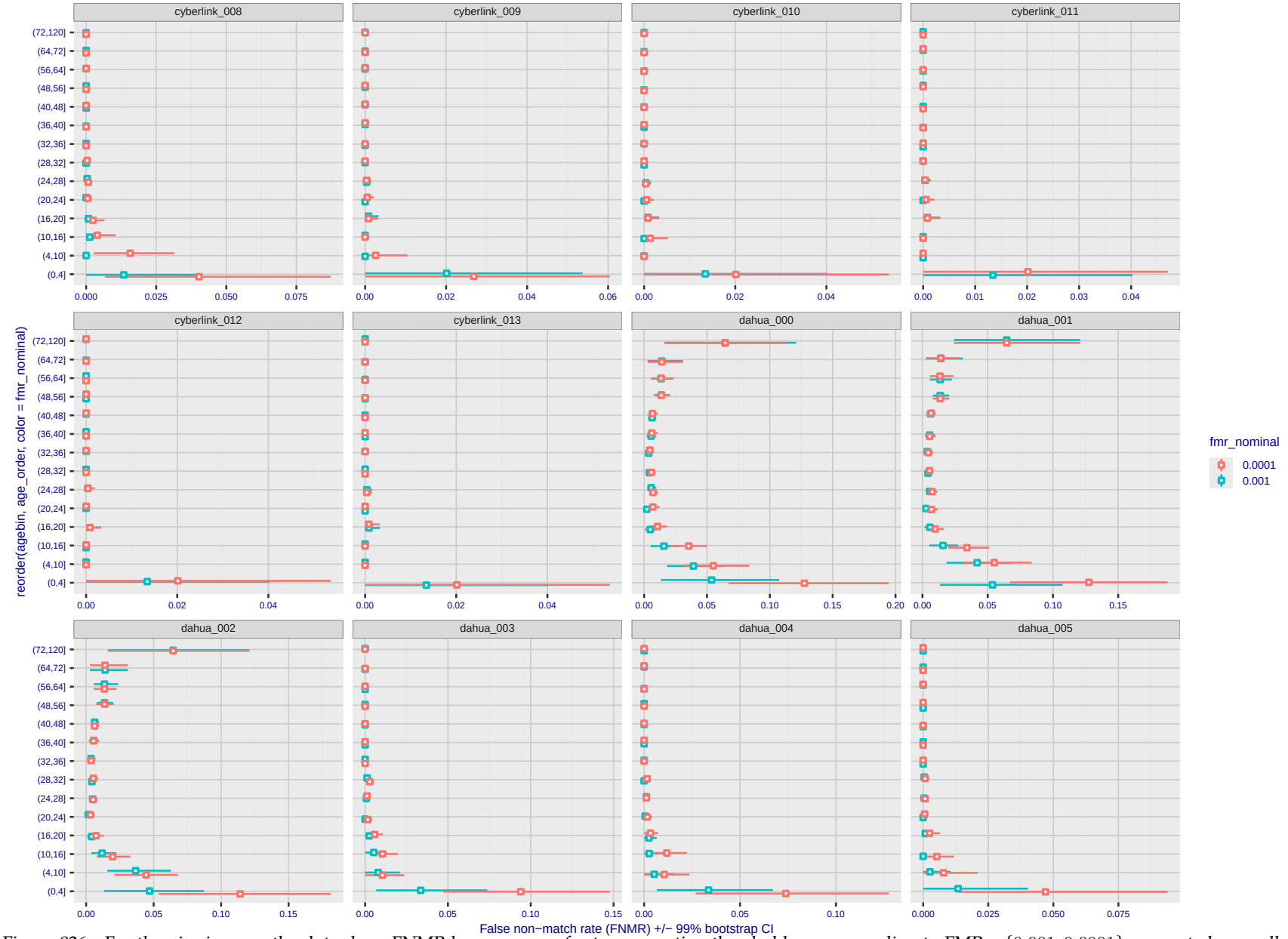


Figure 826: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

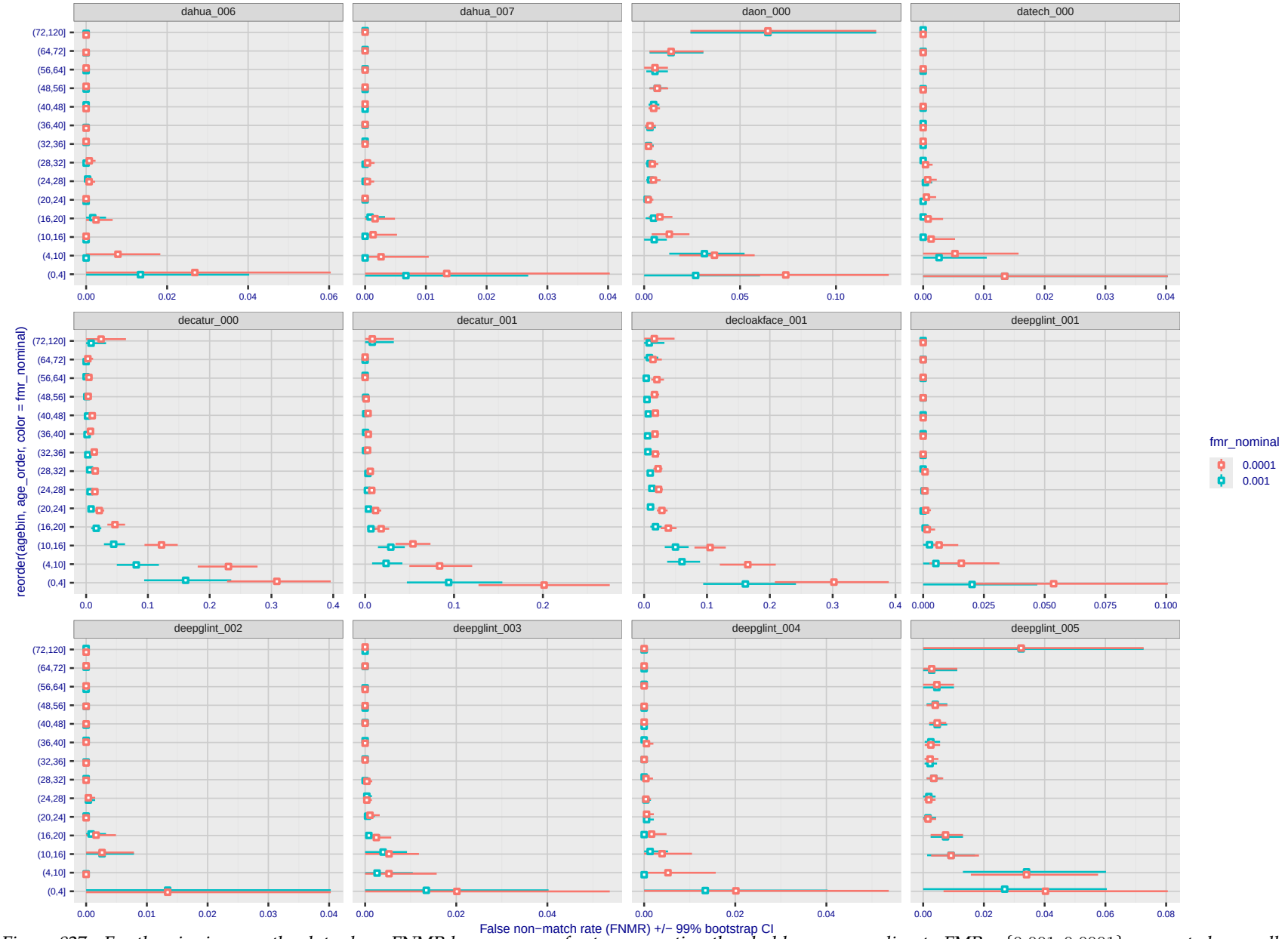


Figure 827: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 828: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

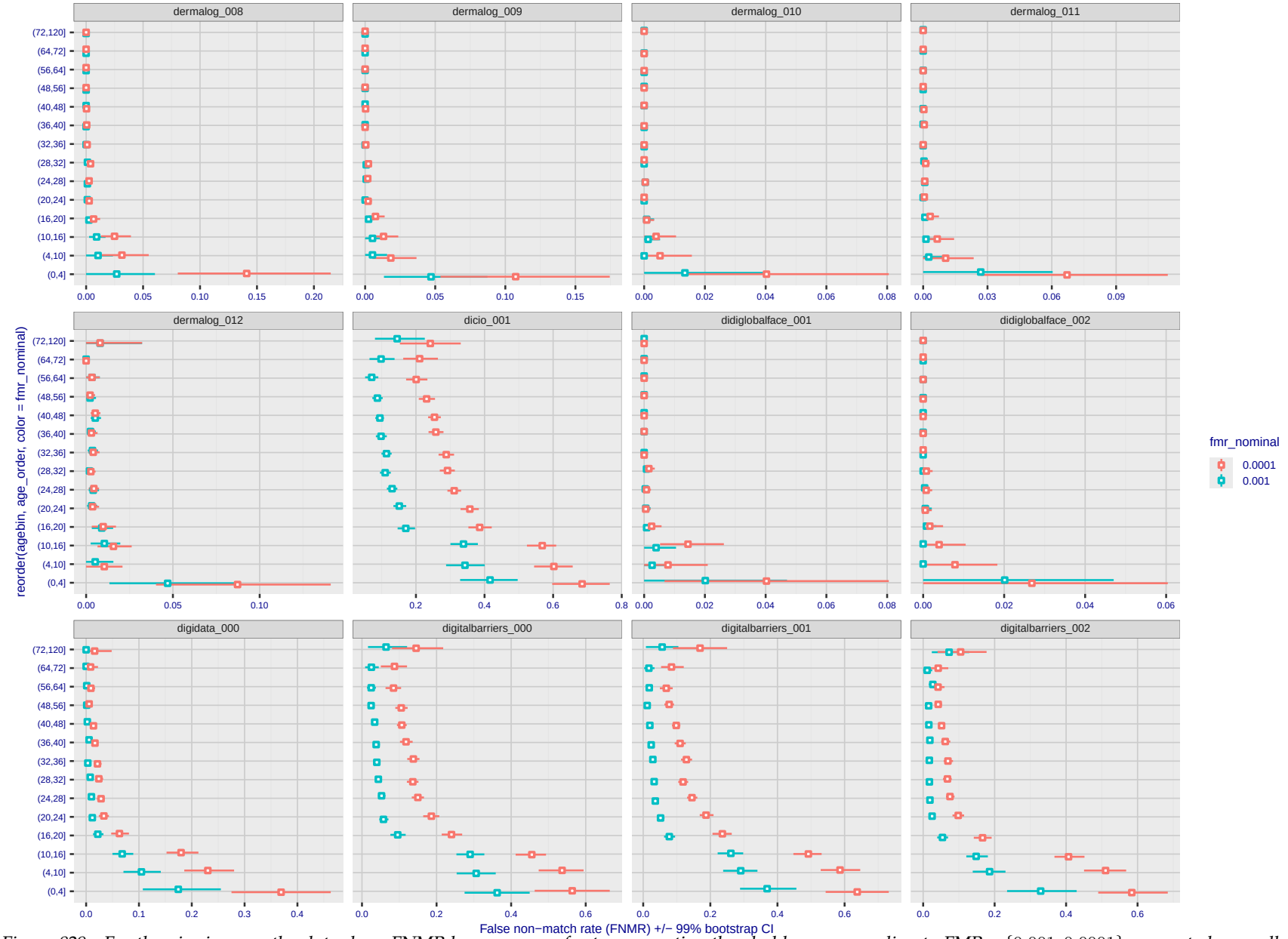


Figure 829: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 830: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 831: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

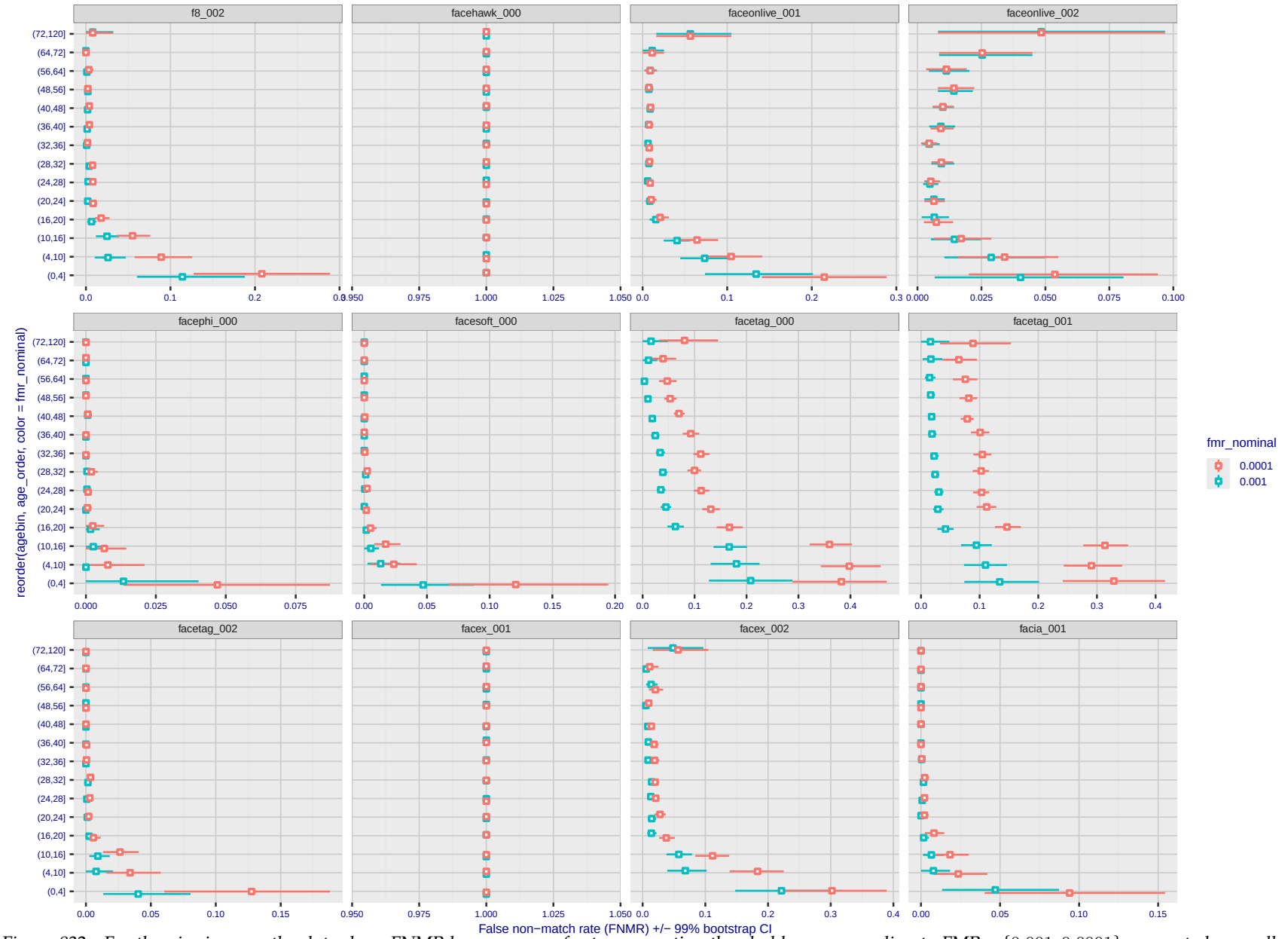


Figure 832: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

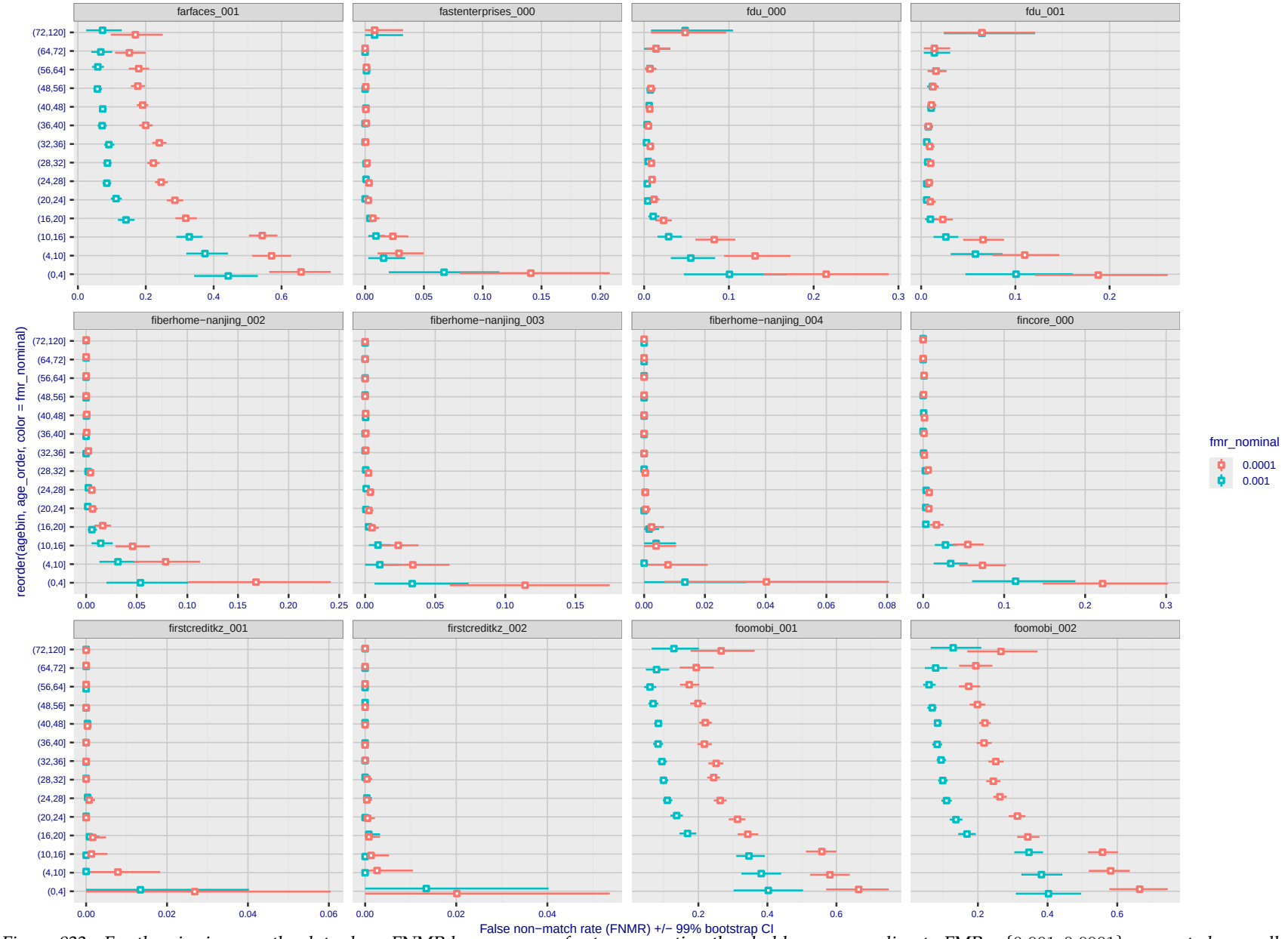


Figure 833: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 834: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 835: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 836: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 837: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 838: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

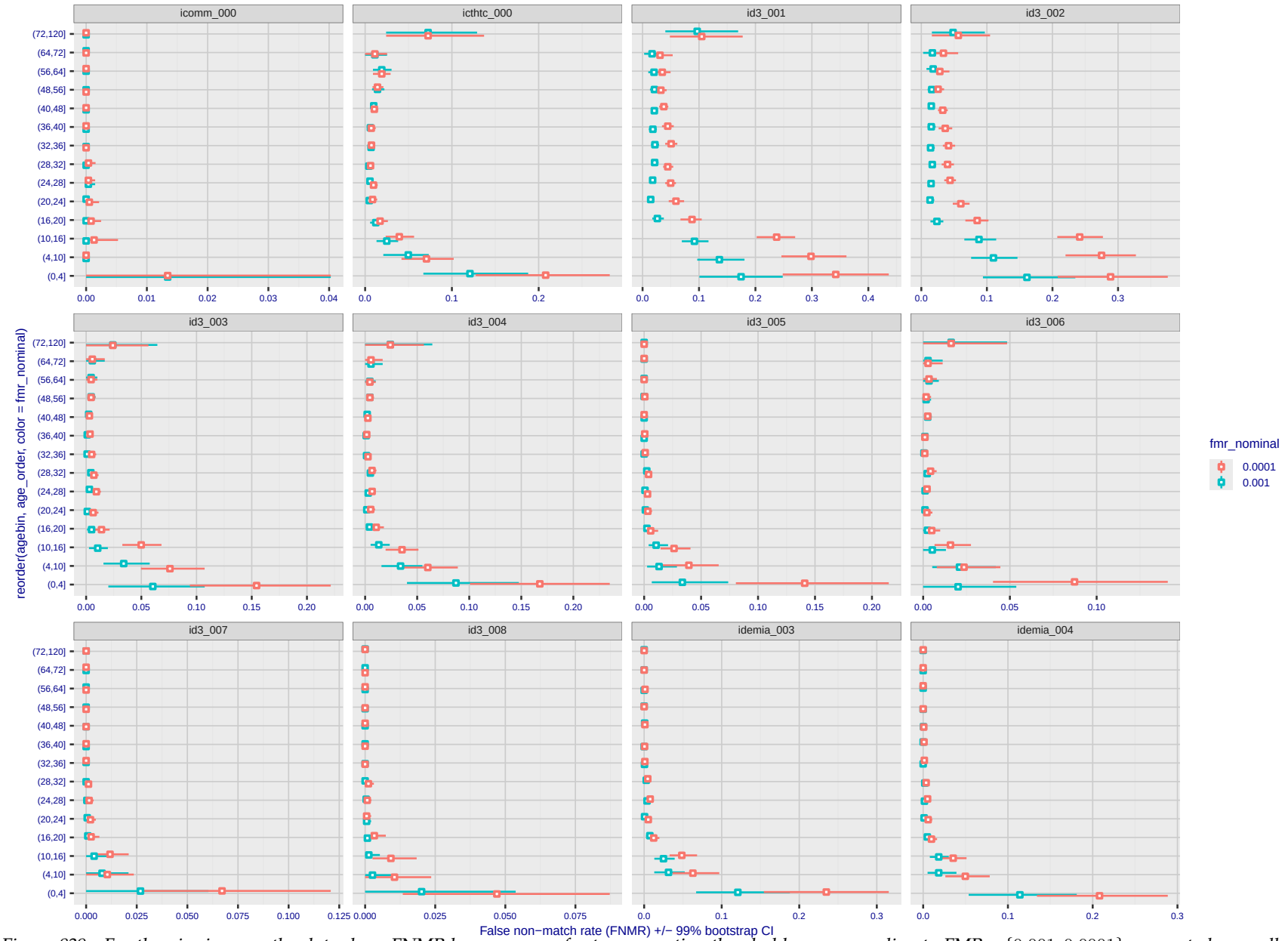


Figure 839: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

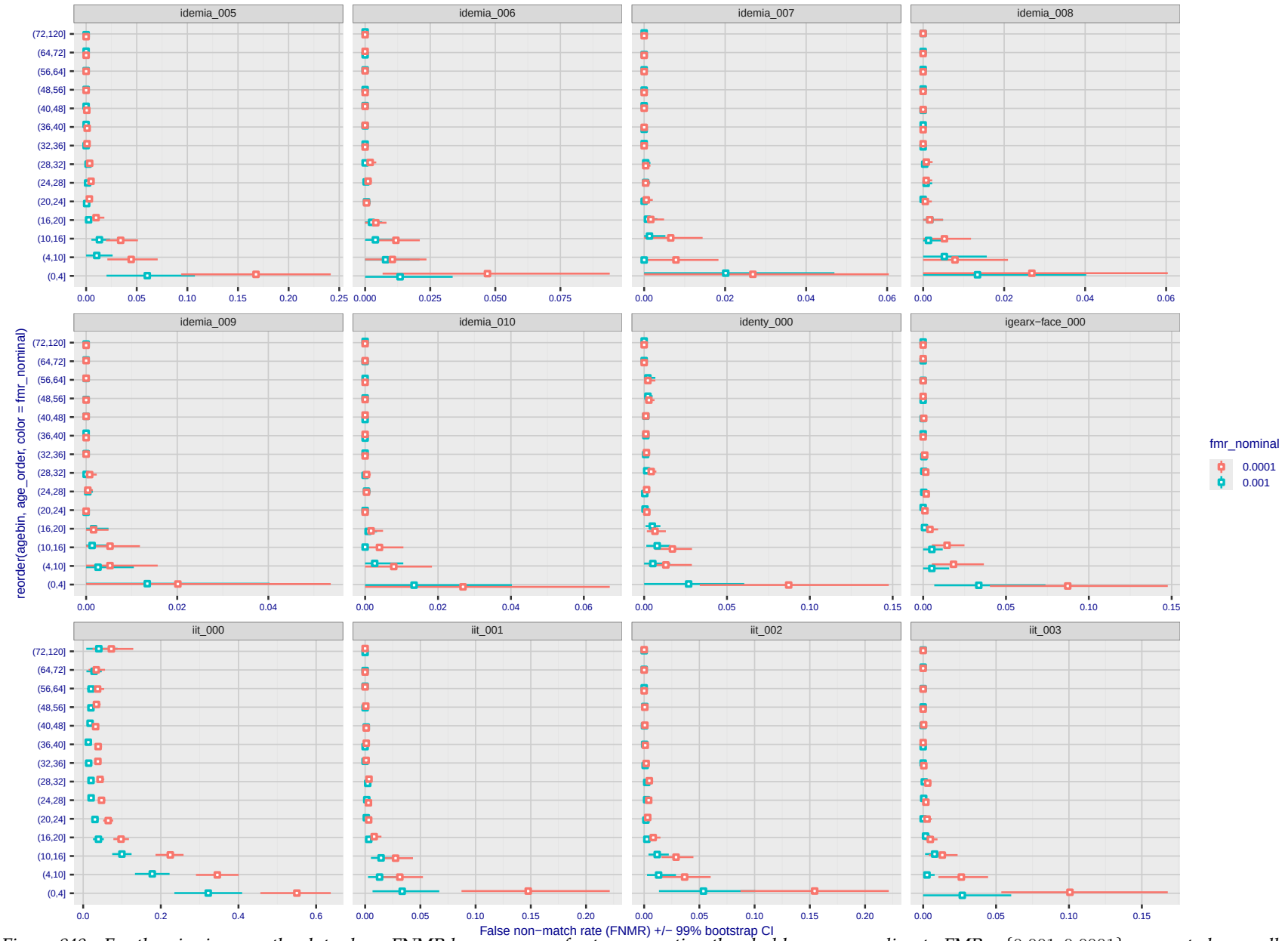


Figure 840: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 841: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

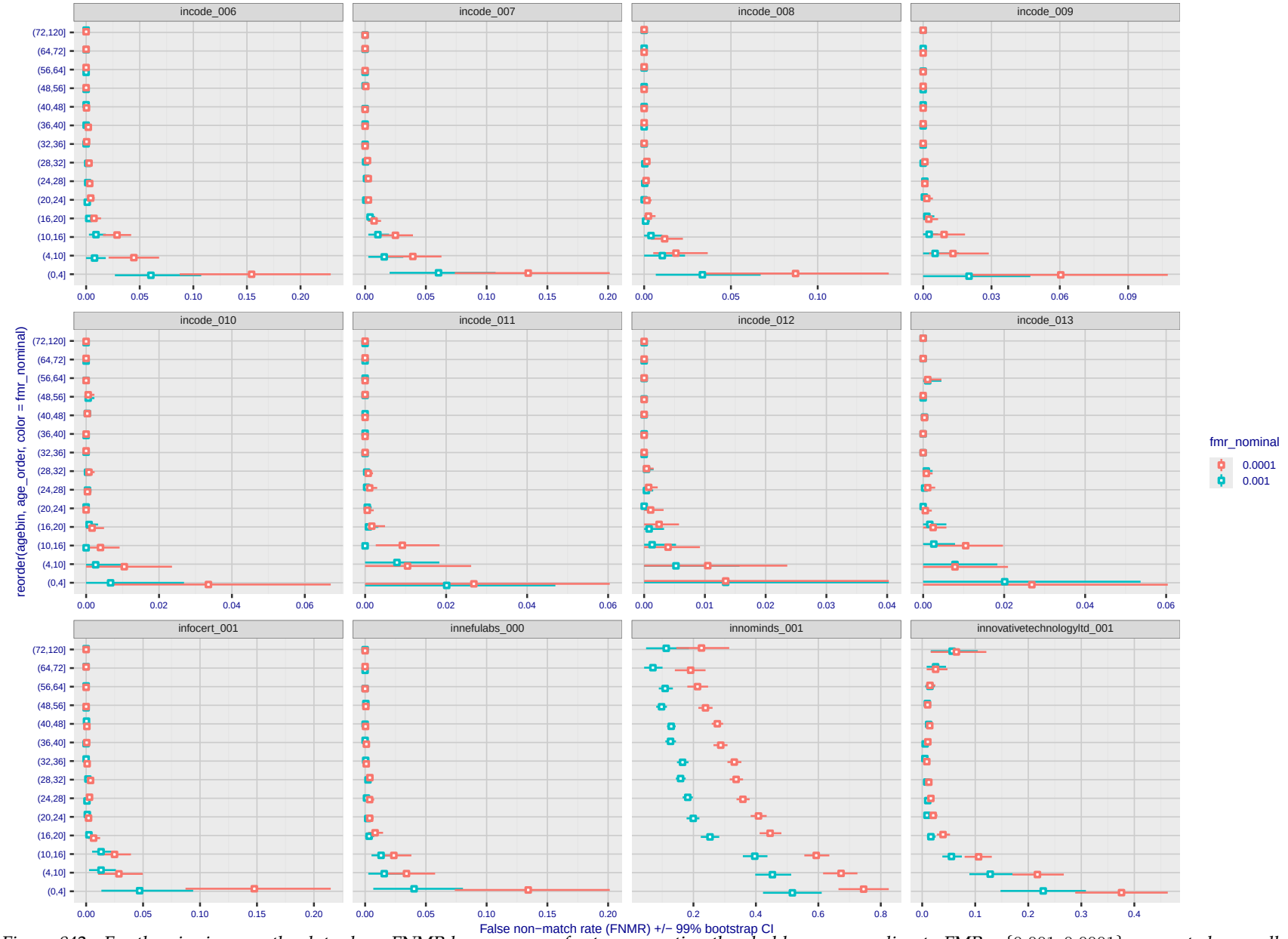


Figure 842: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 843: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

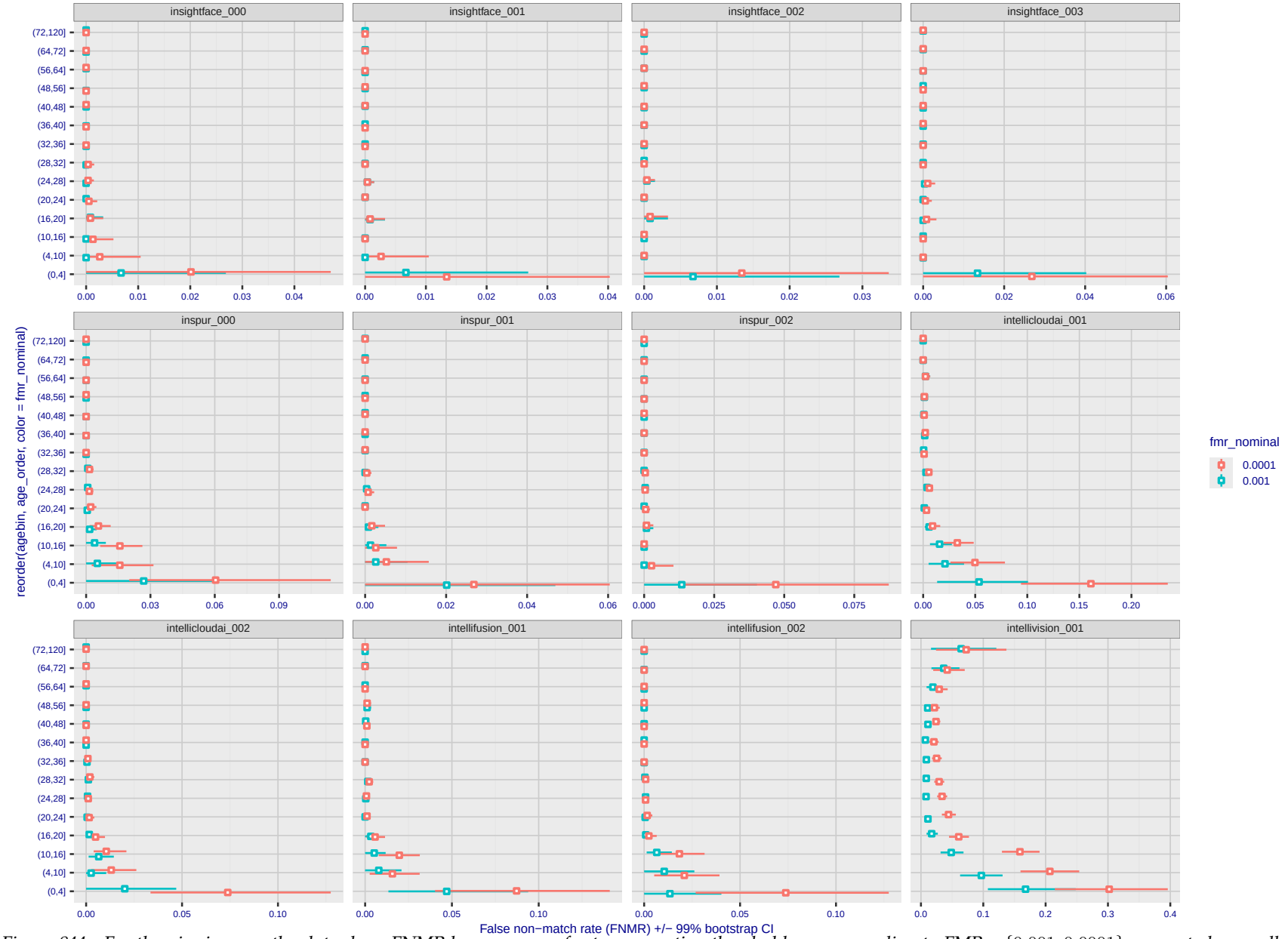


Figure 844: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

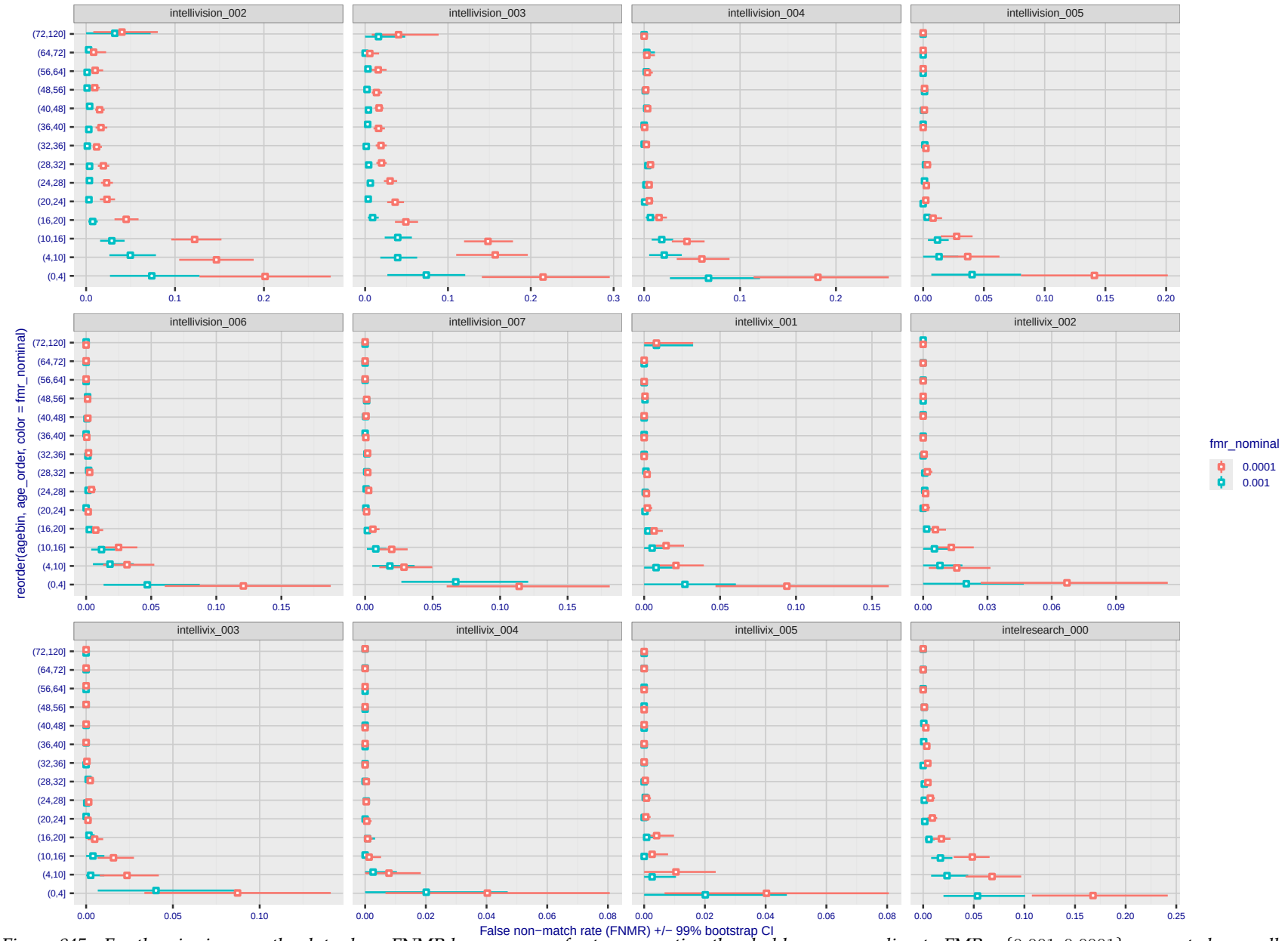


Figure 845: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

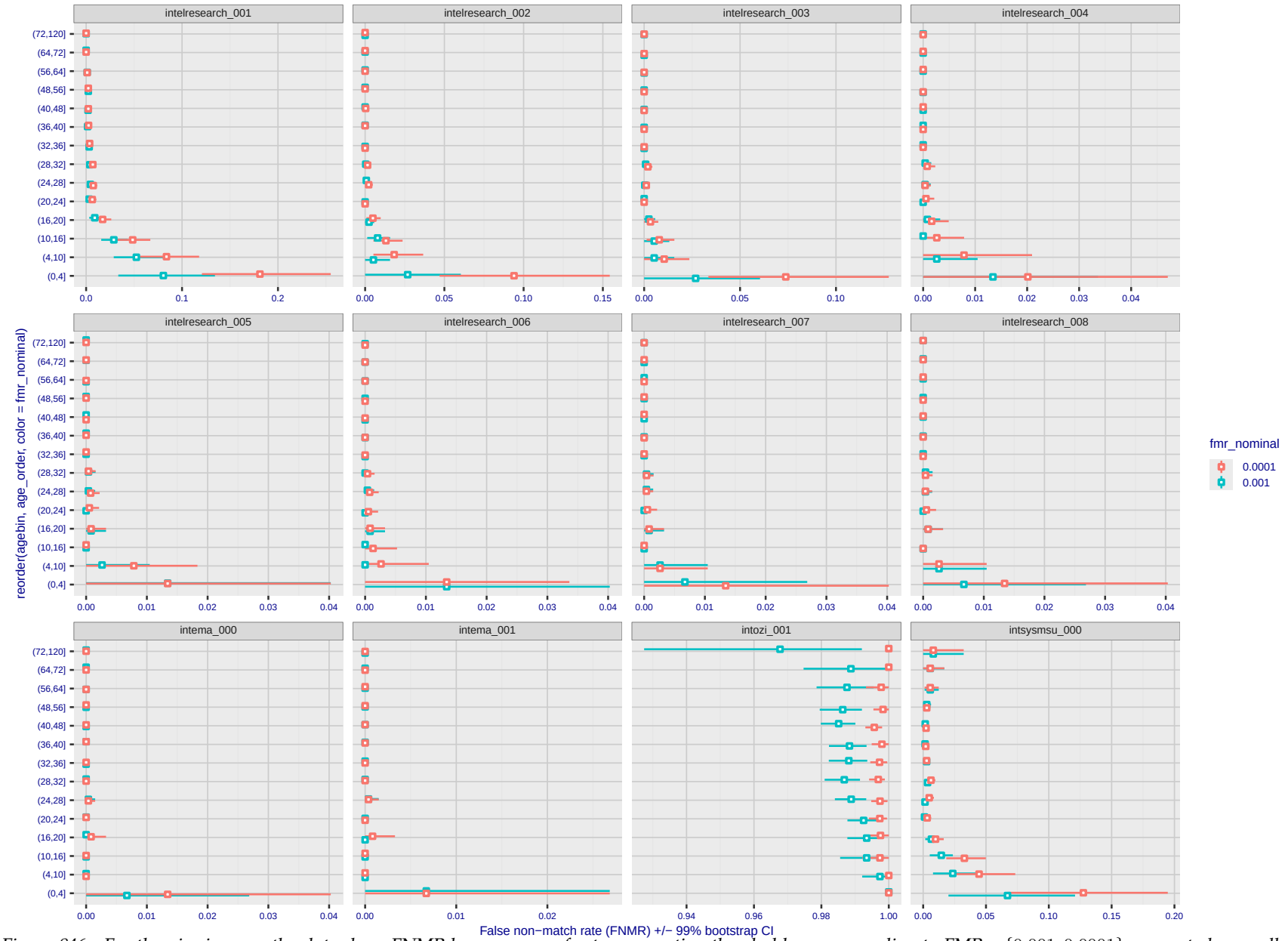


Figure 846: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 847: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 848: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 849: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 850: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 851: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

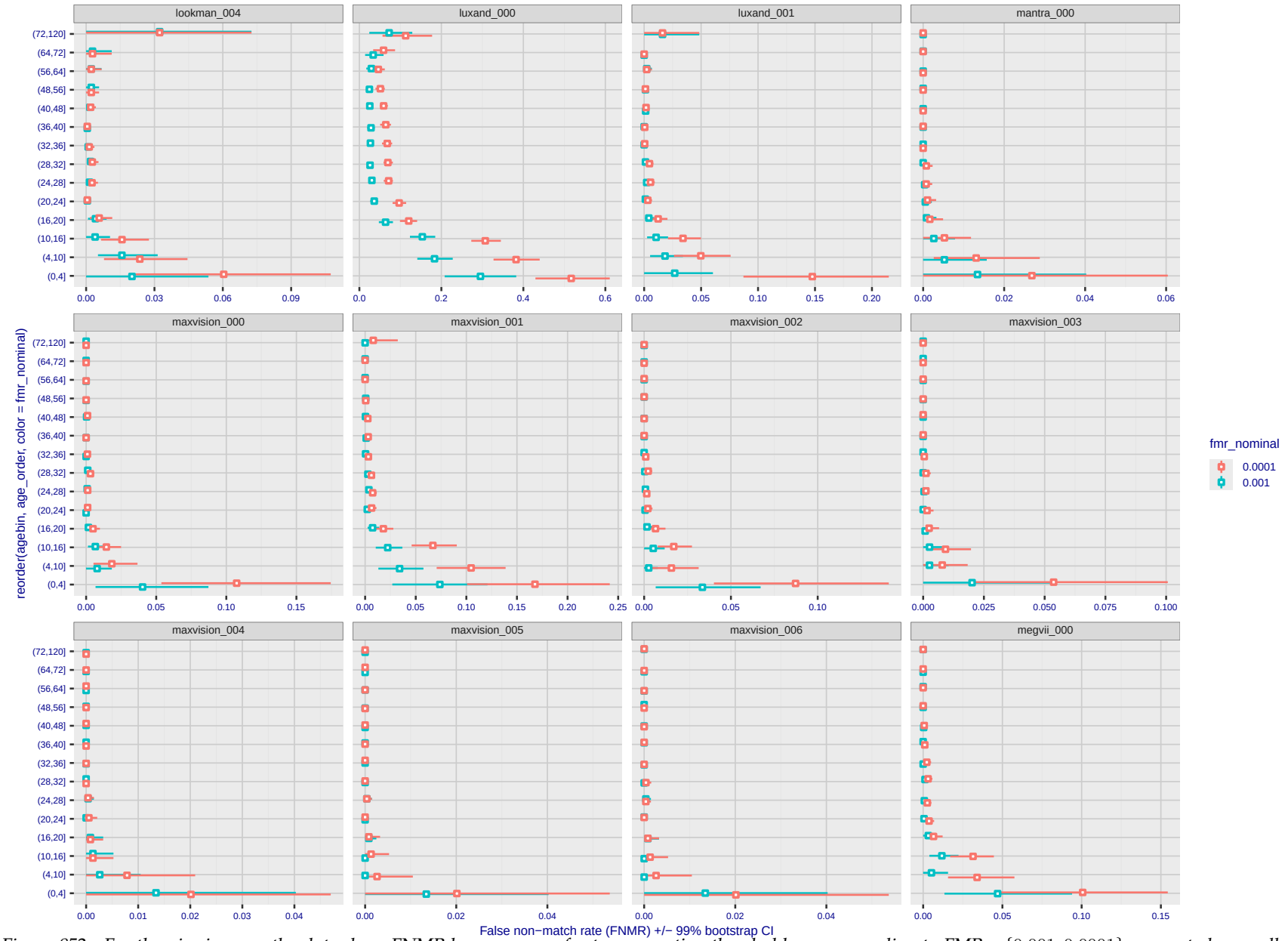


Figure 852: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

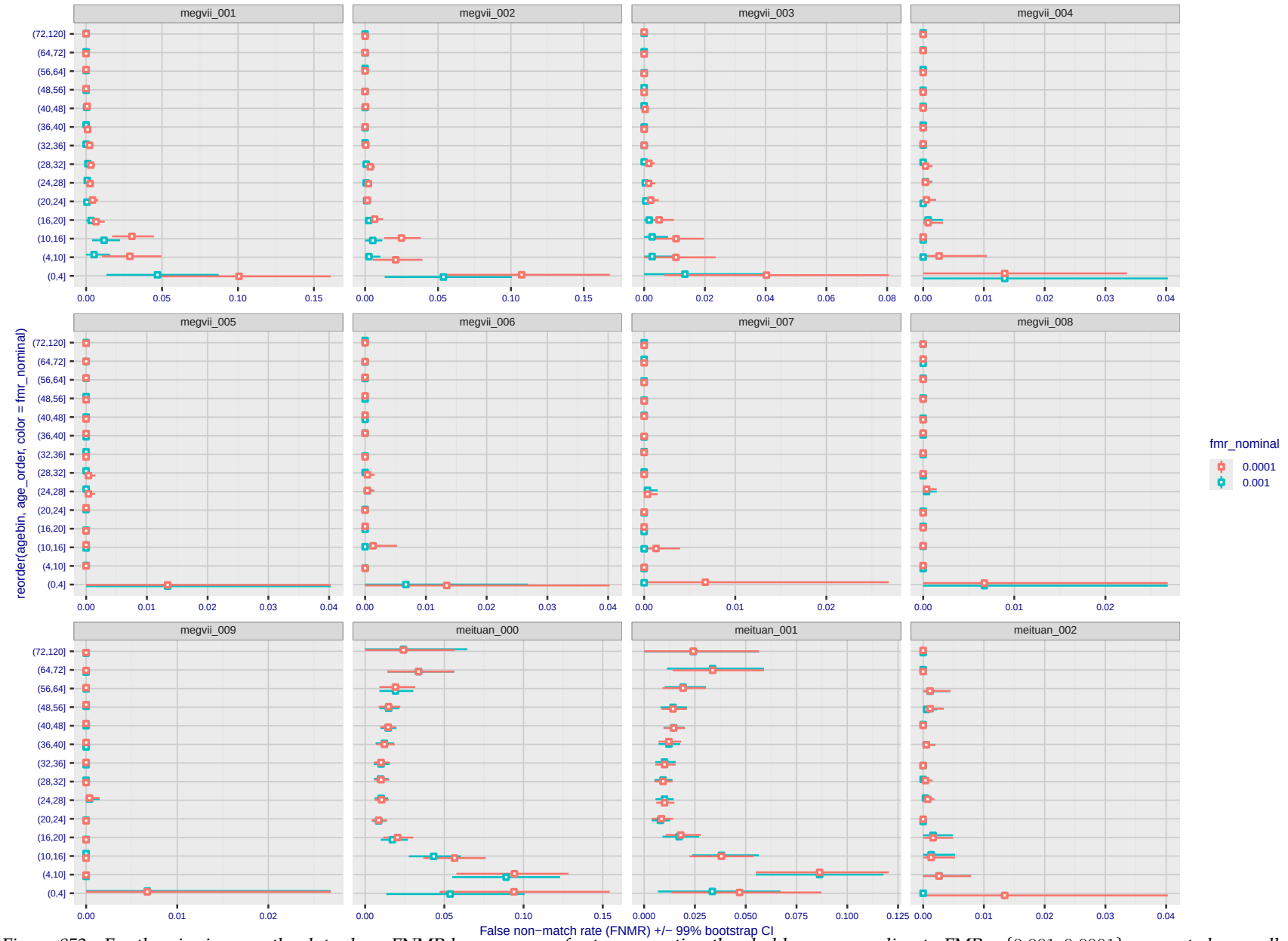


Figure 853: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 854: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

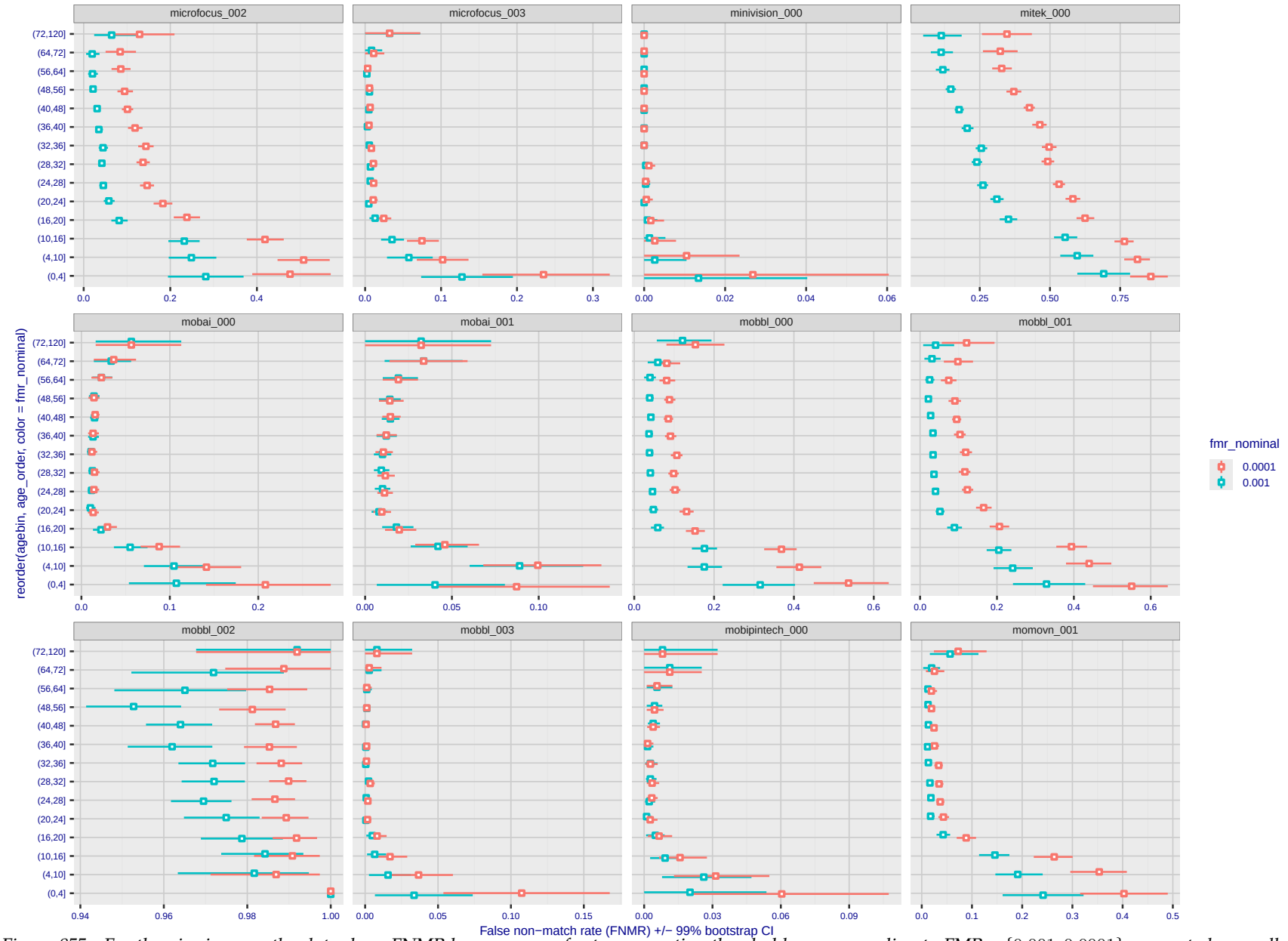


Figure 855: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 856: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

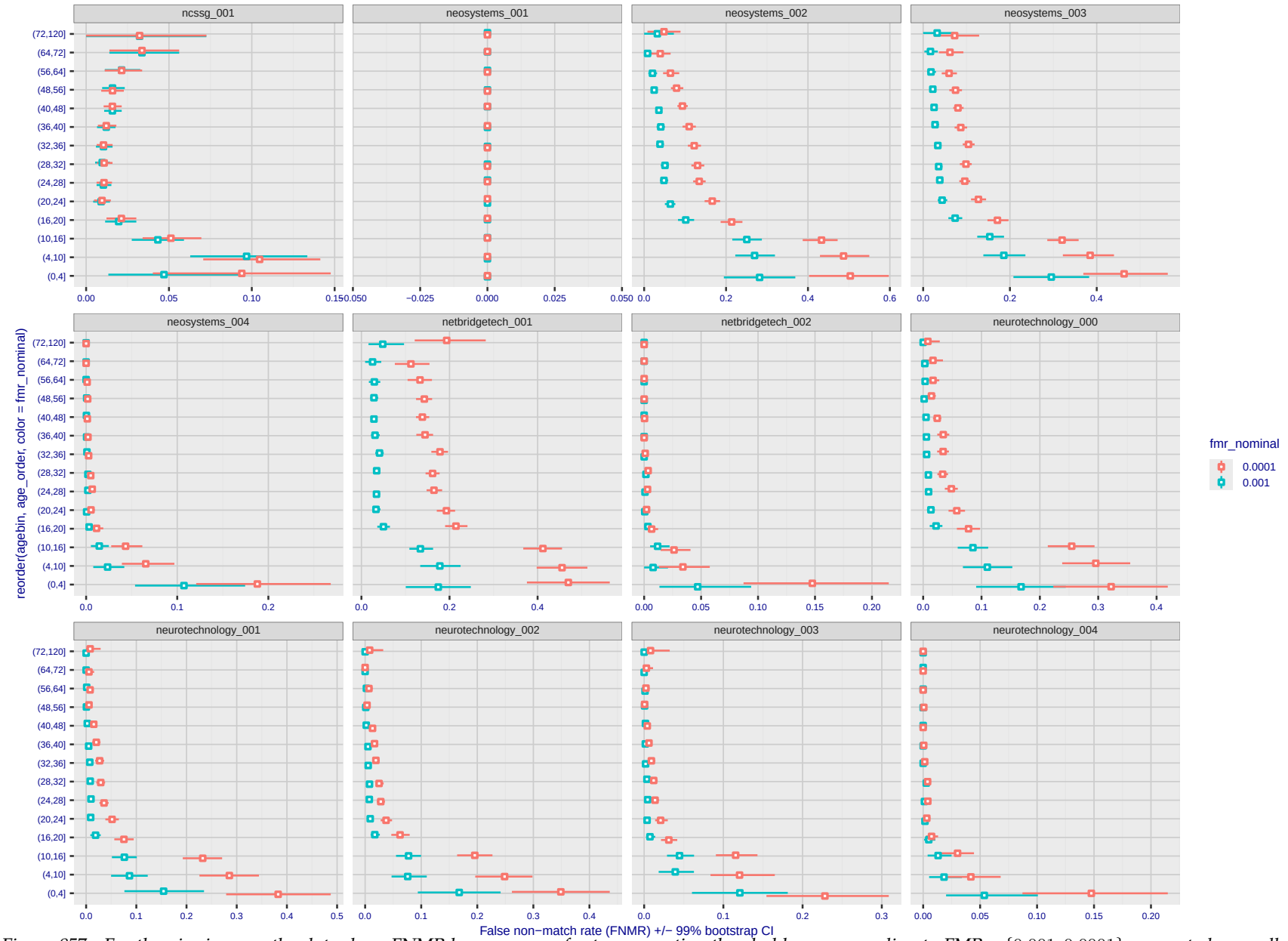


Figure 857: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

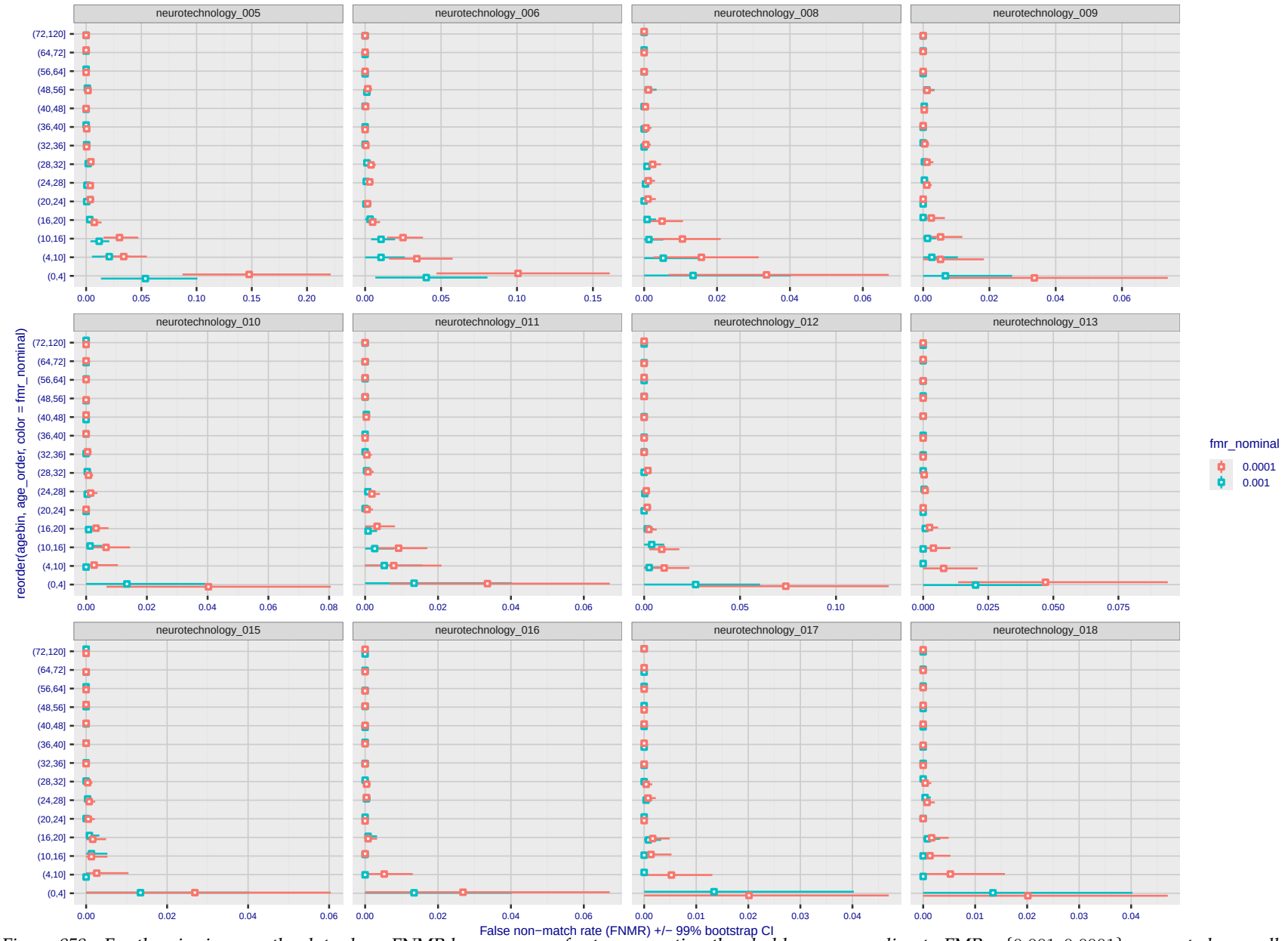


Figure 858: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 859: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 860: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

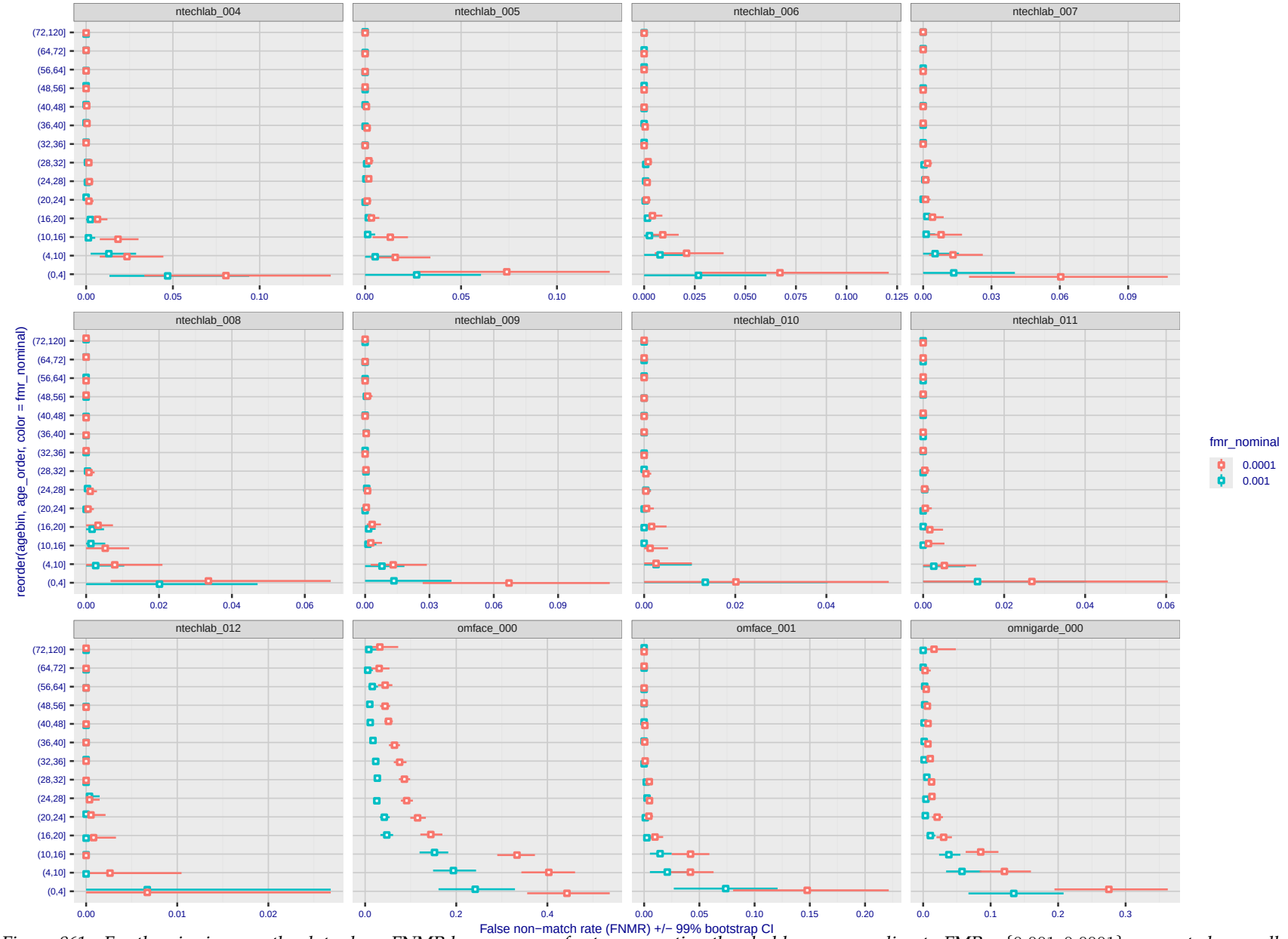


Figure 861: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 862: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

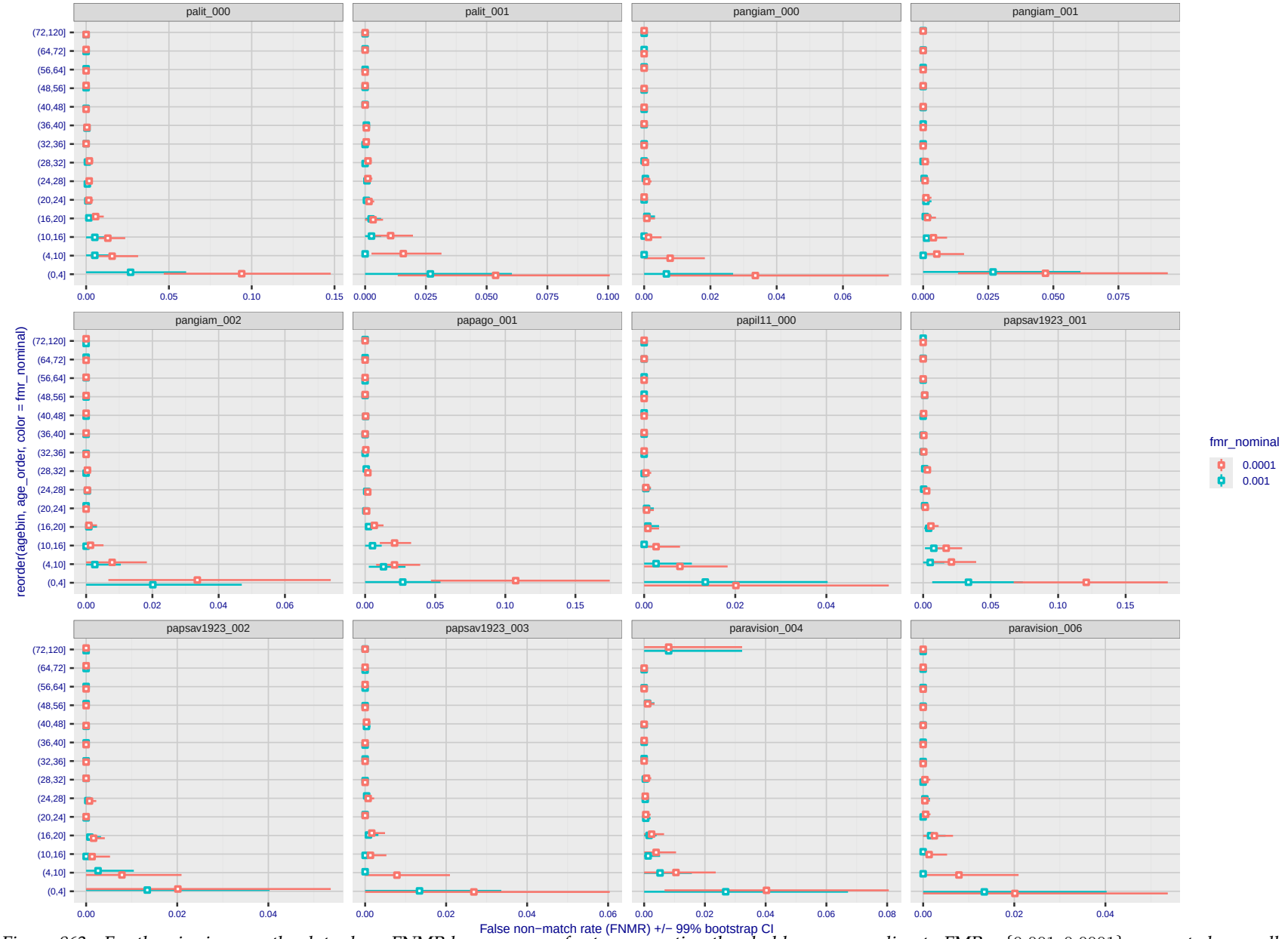


Figure 863: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

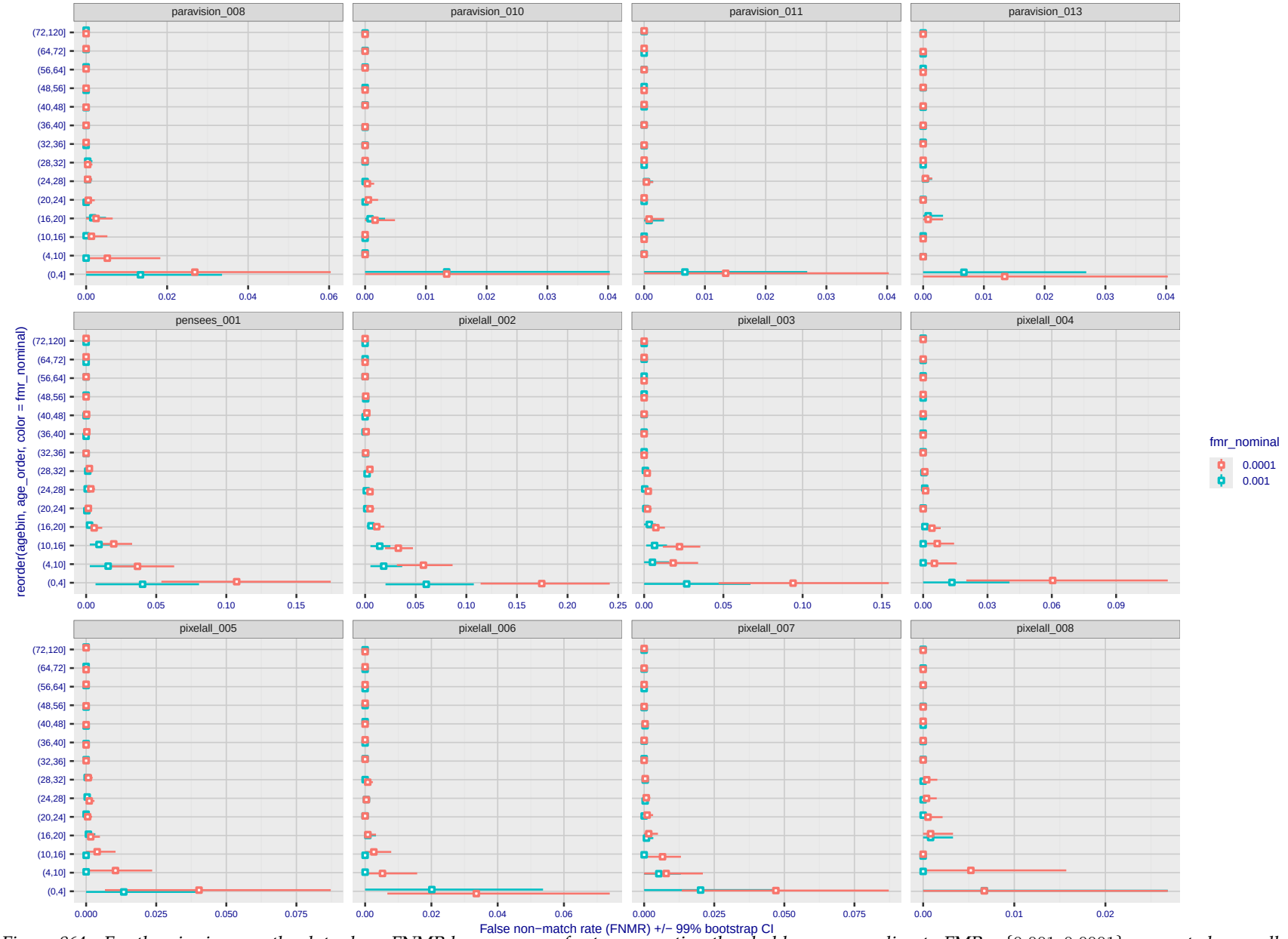


Figure 864: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 865: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

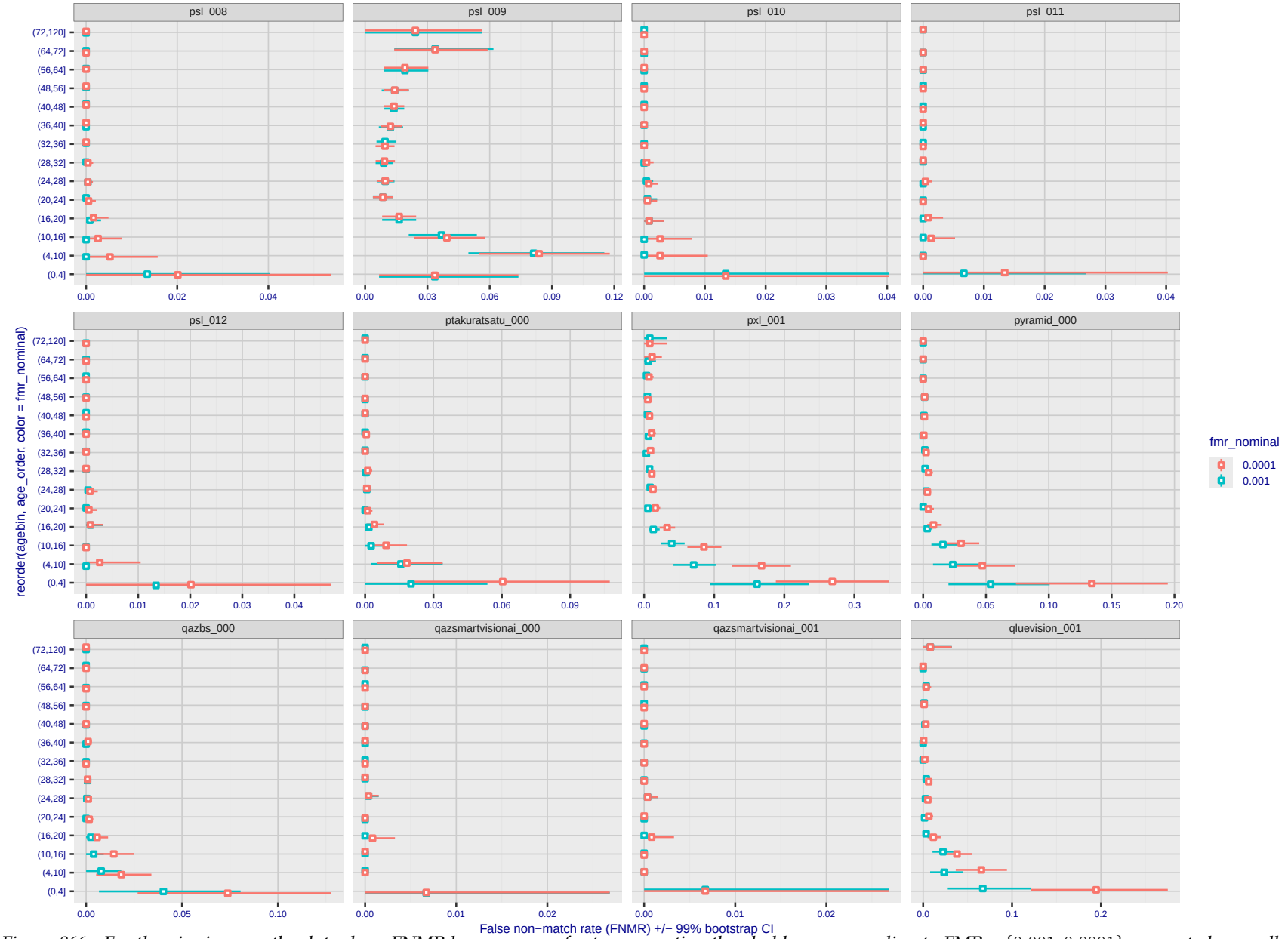


Figure 866: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 867: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

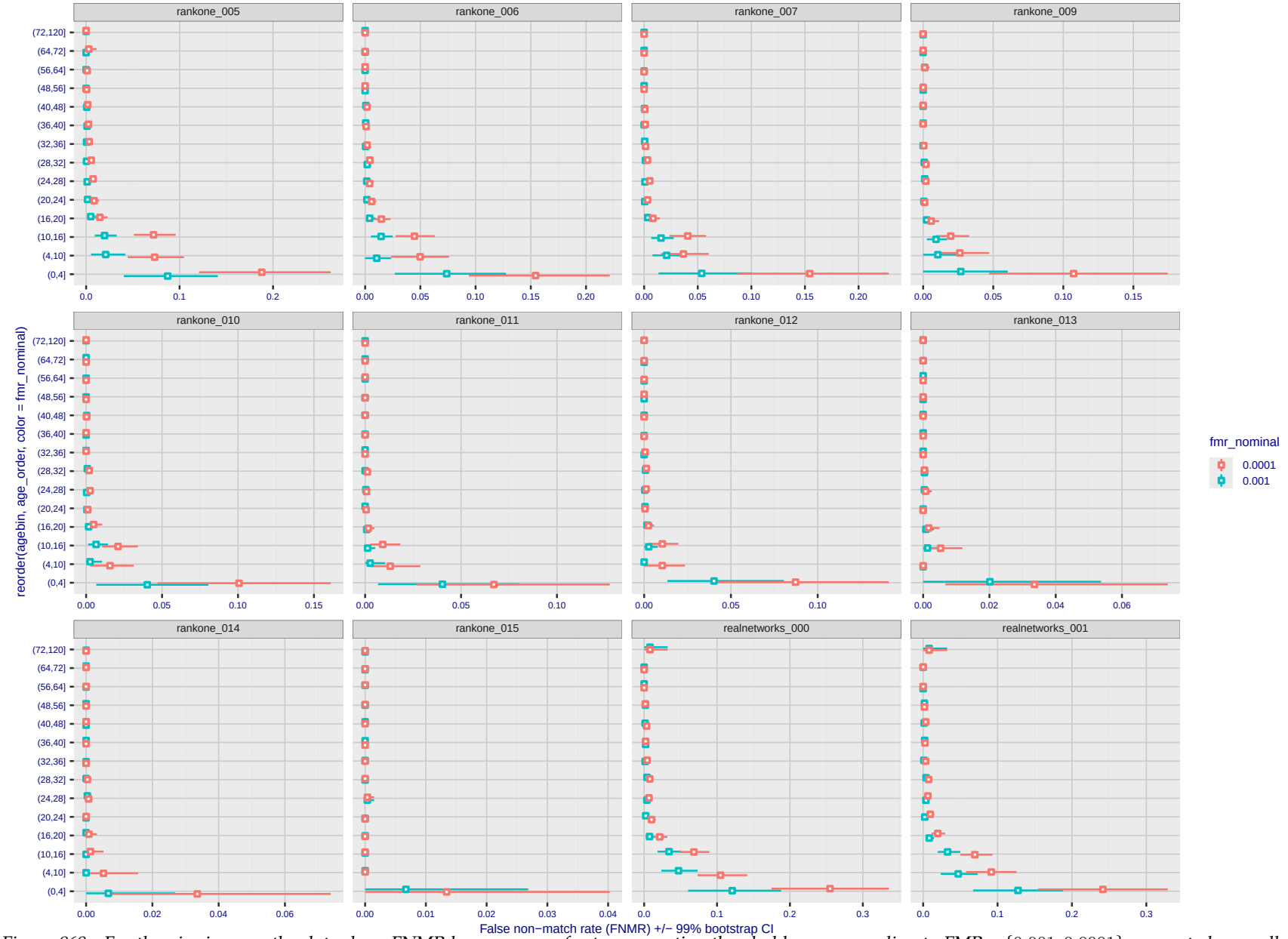


Figure 868: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

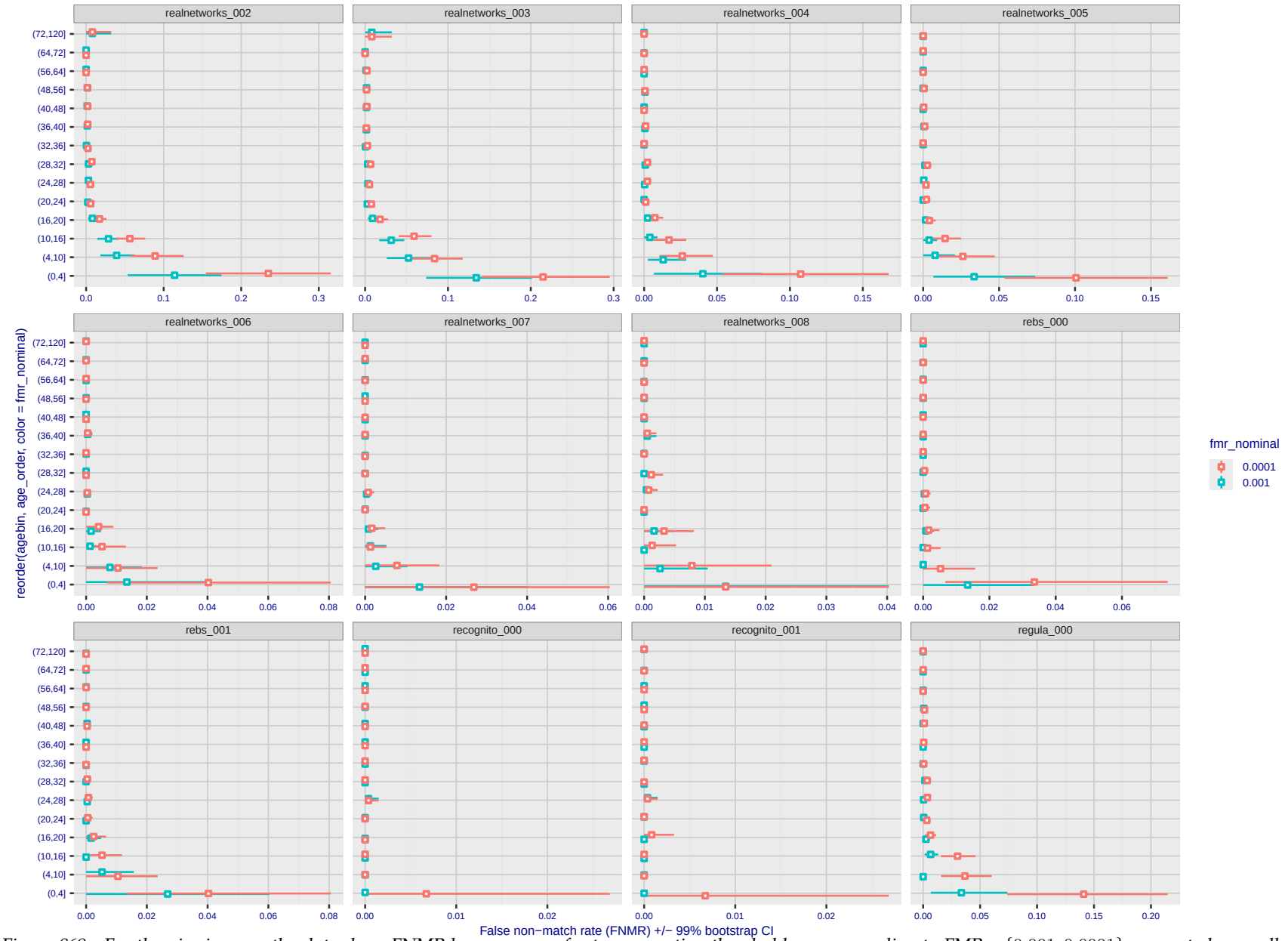


Figure 869: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 870: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

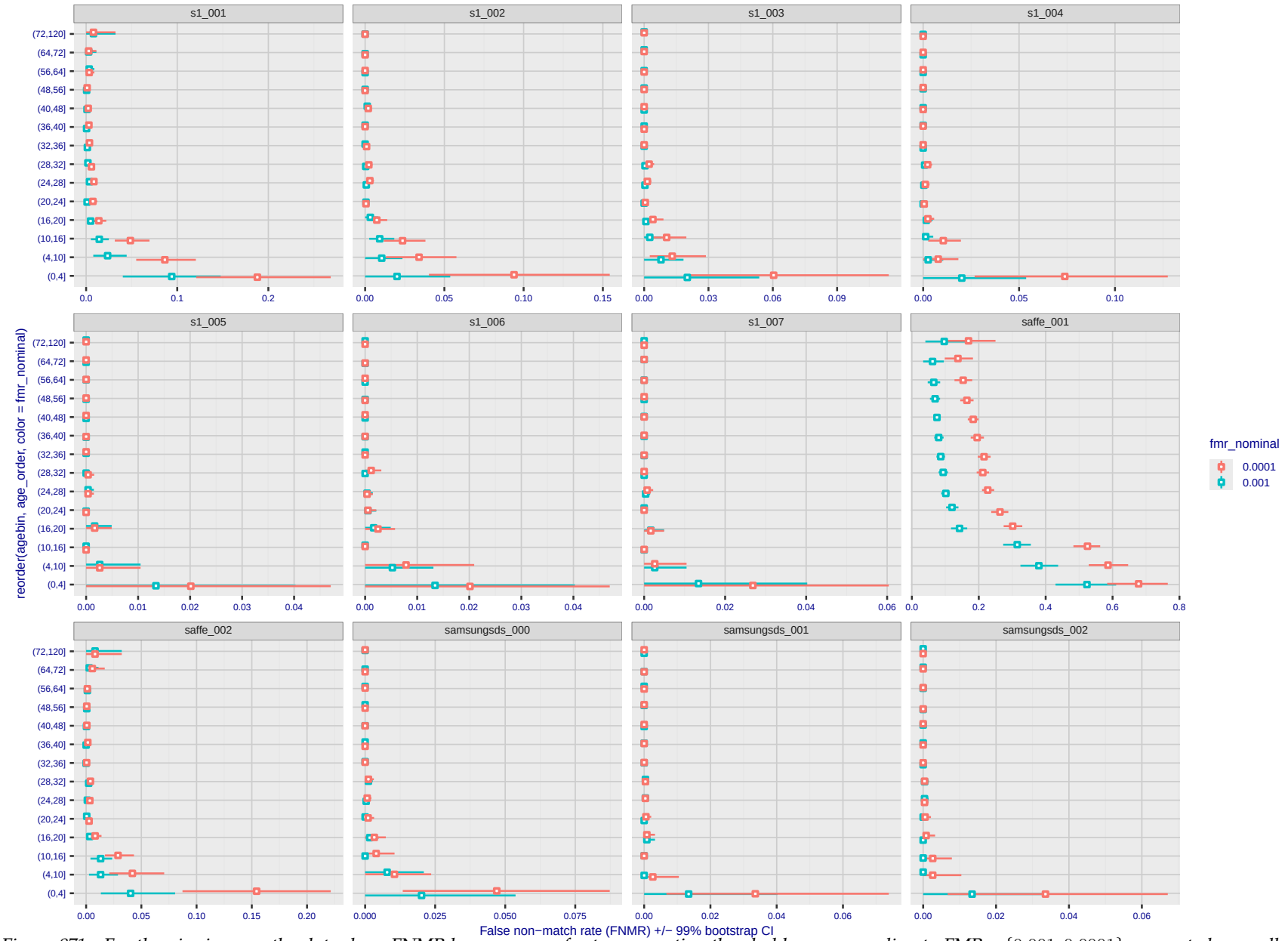


Figure 871: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

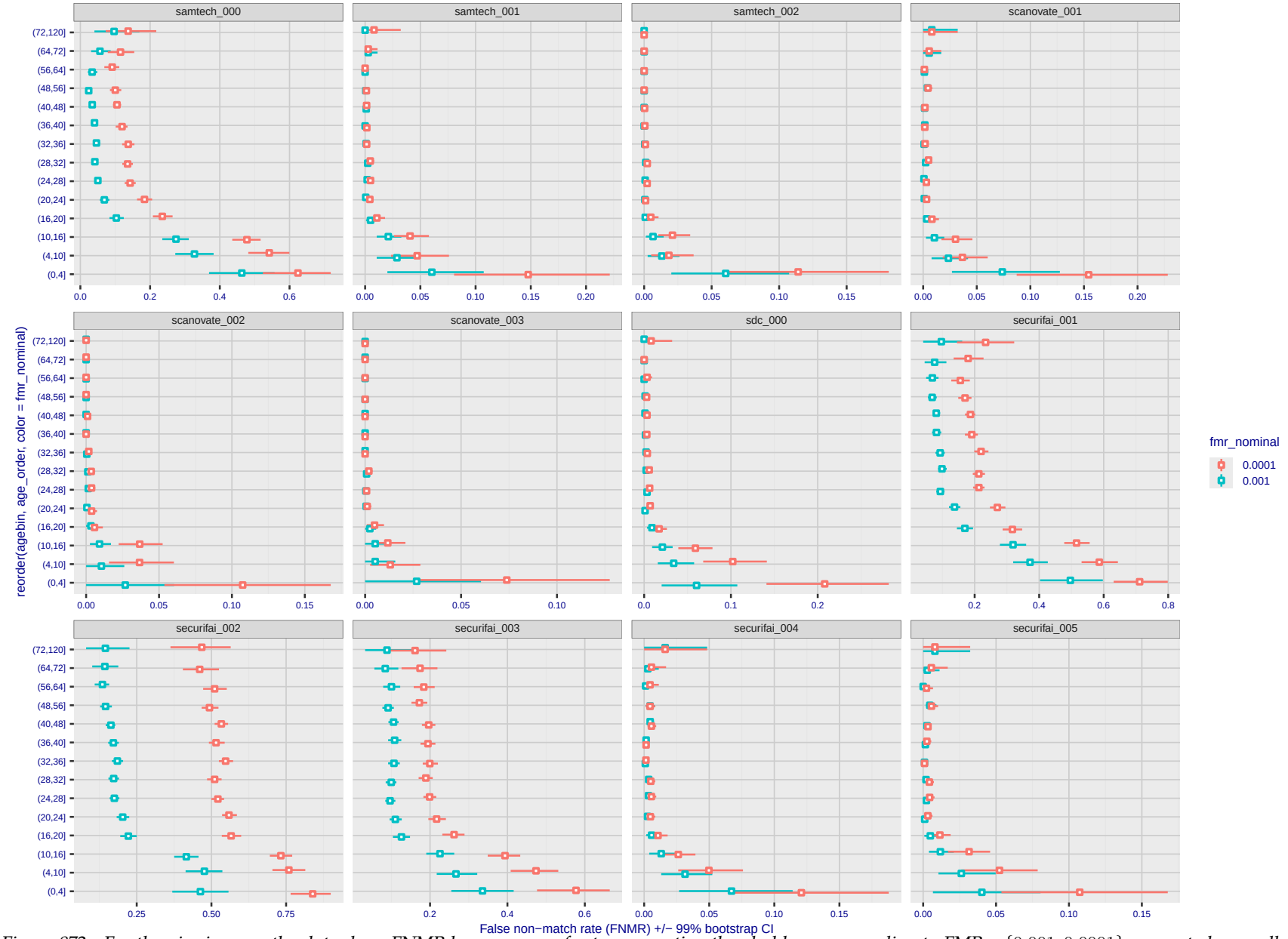


Figure 872: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

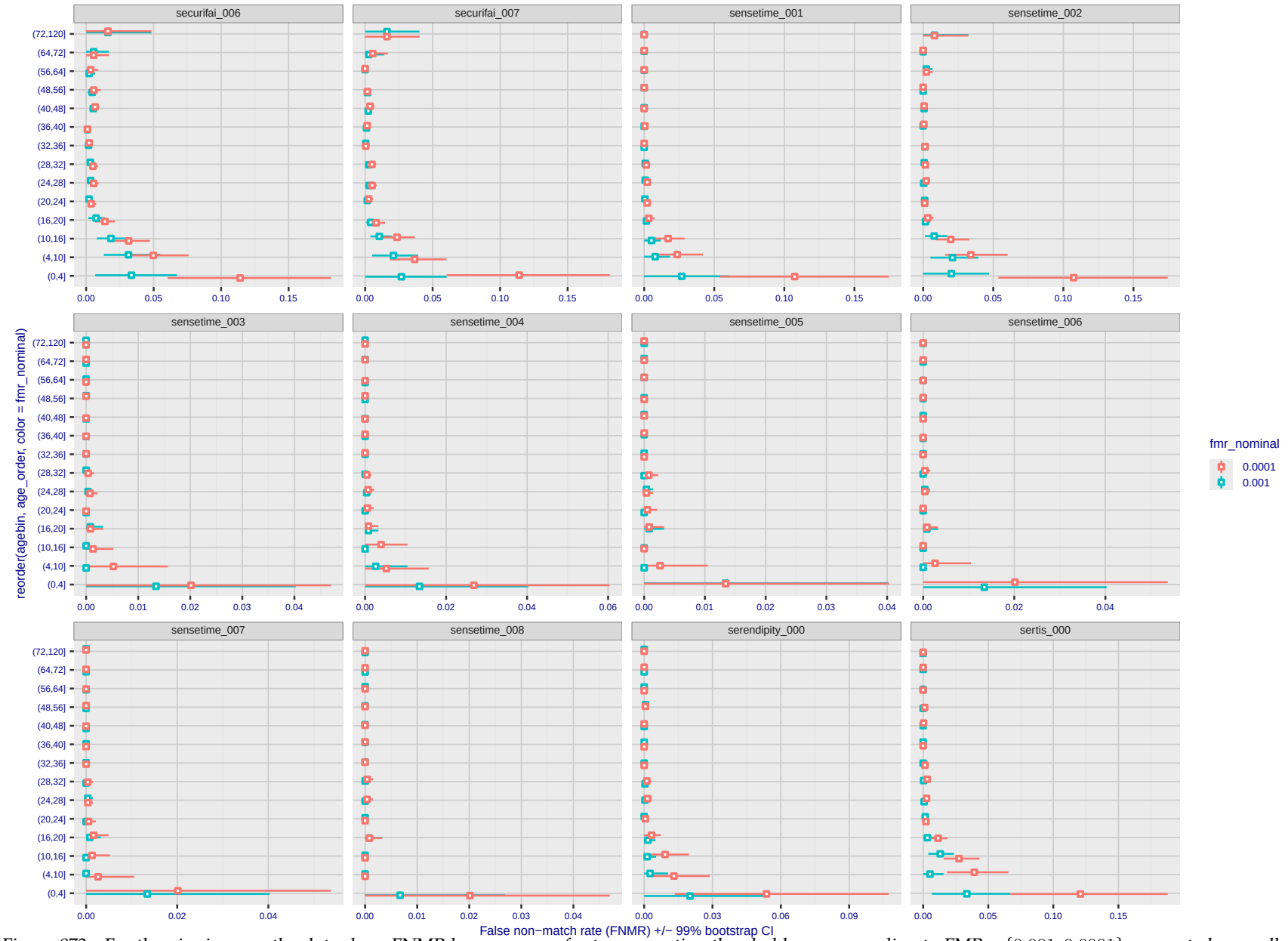


Figure 873: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

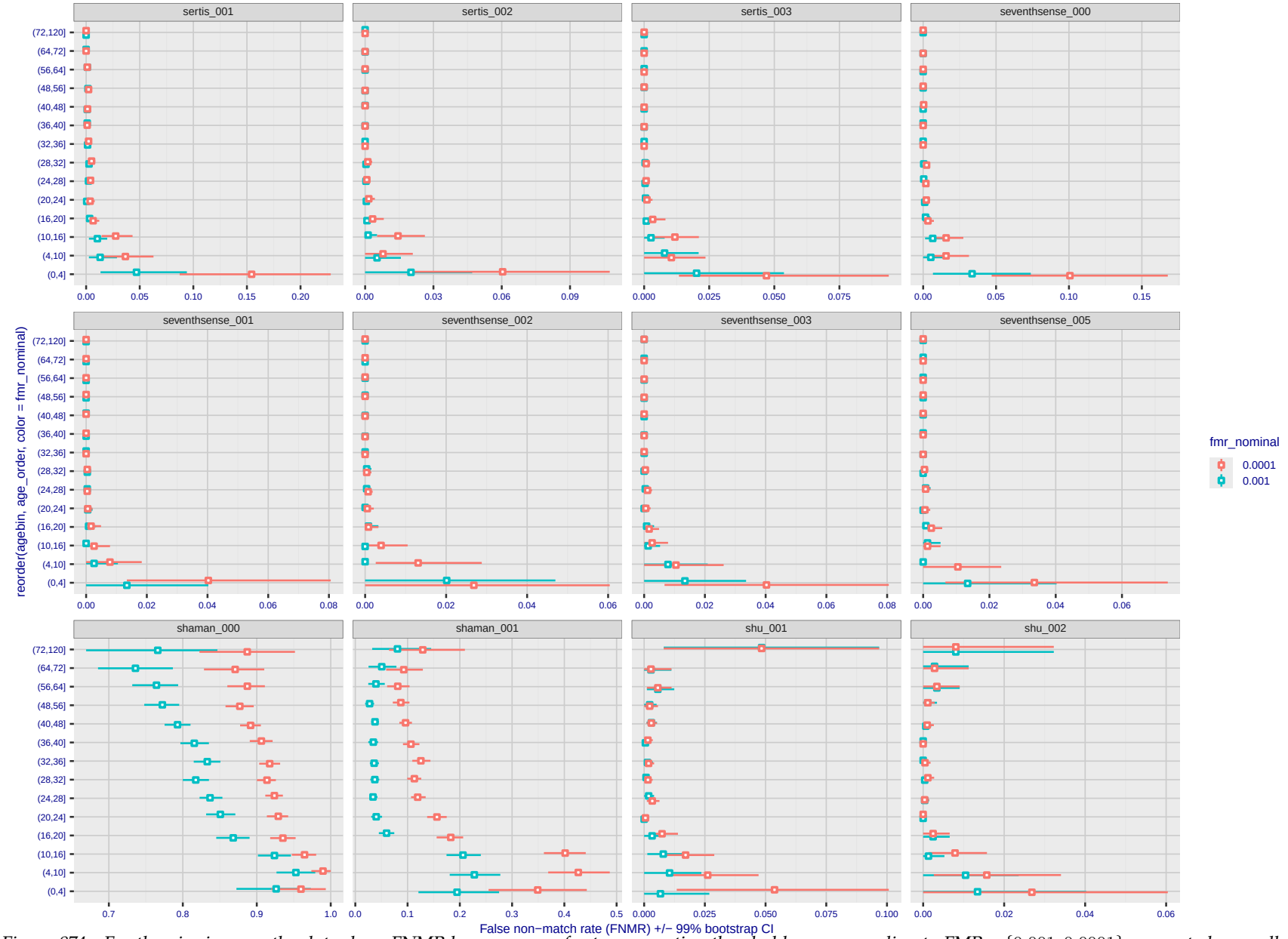


Figure 874: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 875: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 876: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

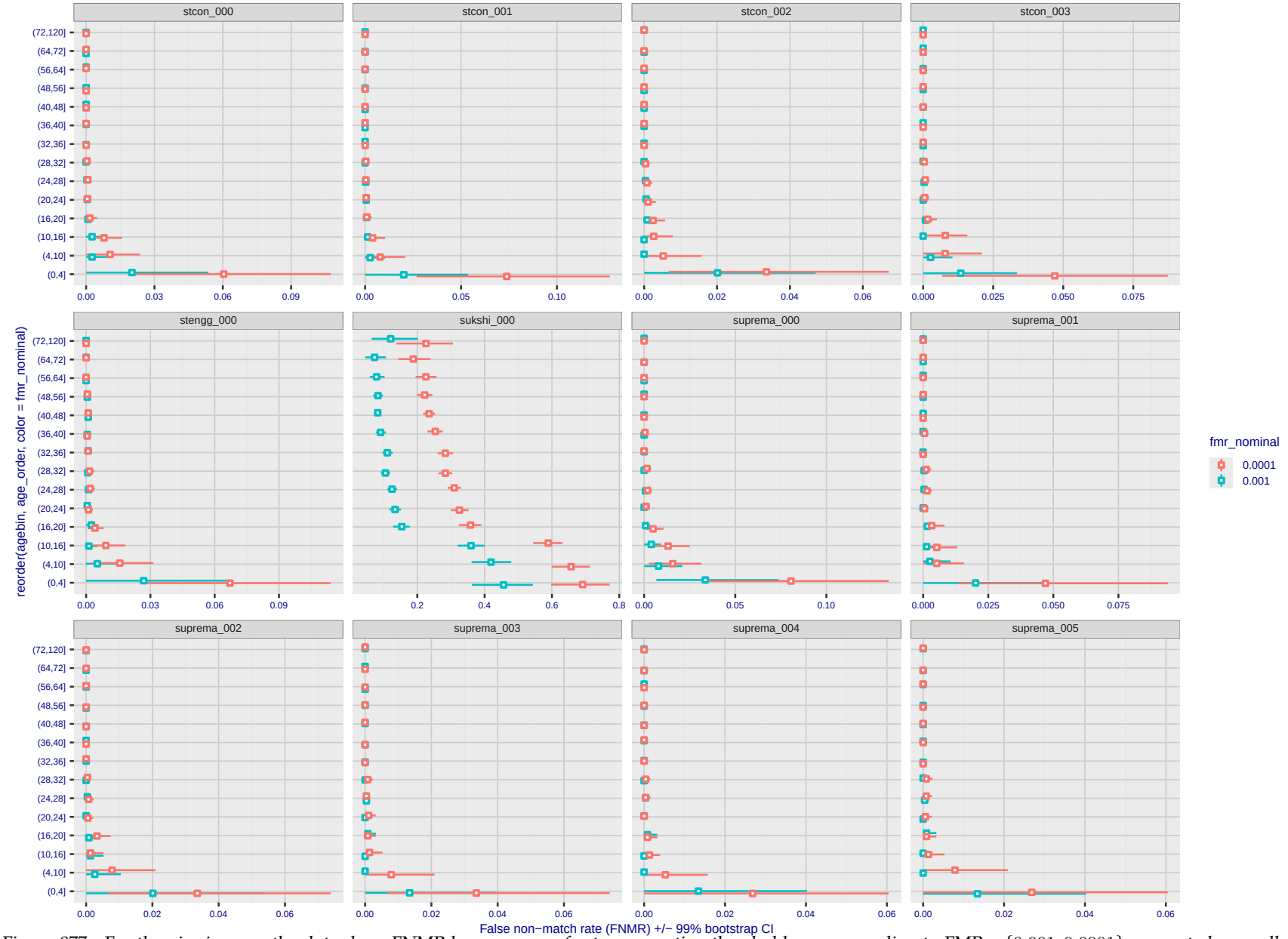


Figure 877: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

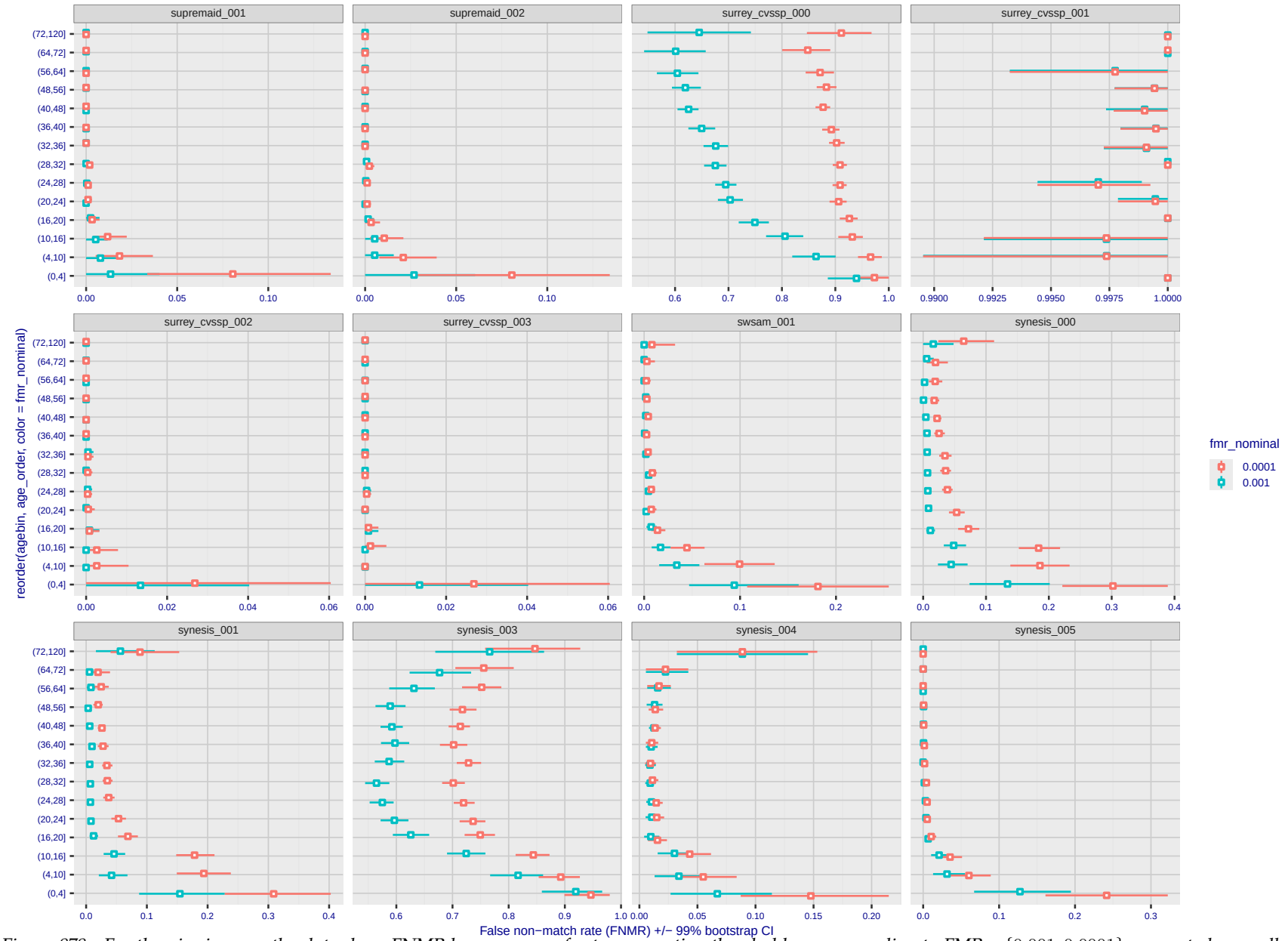


Figure 878: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 879: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 880: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 881: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

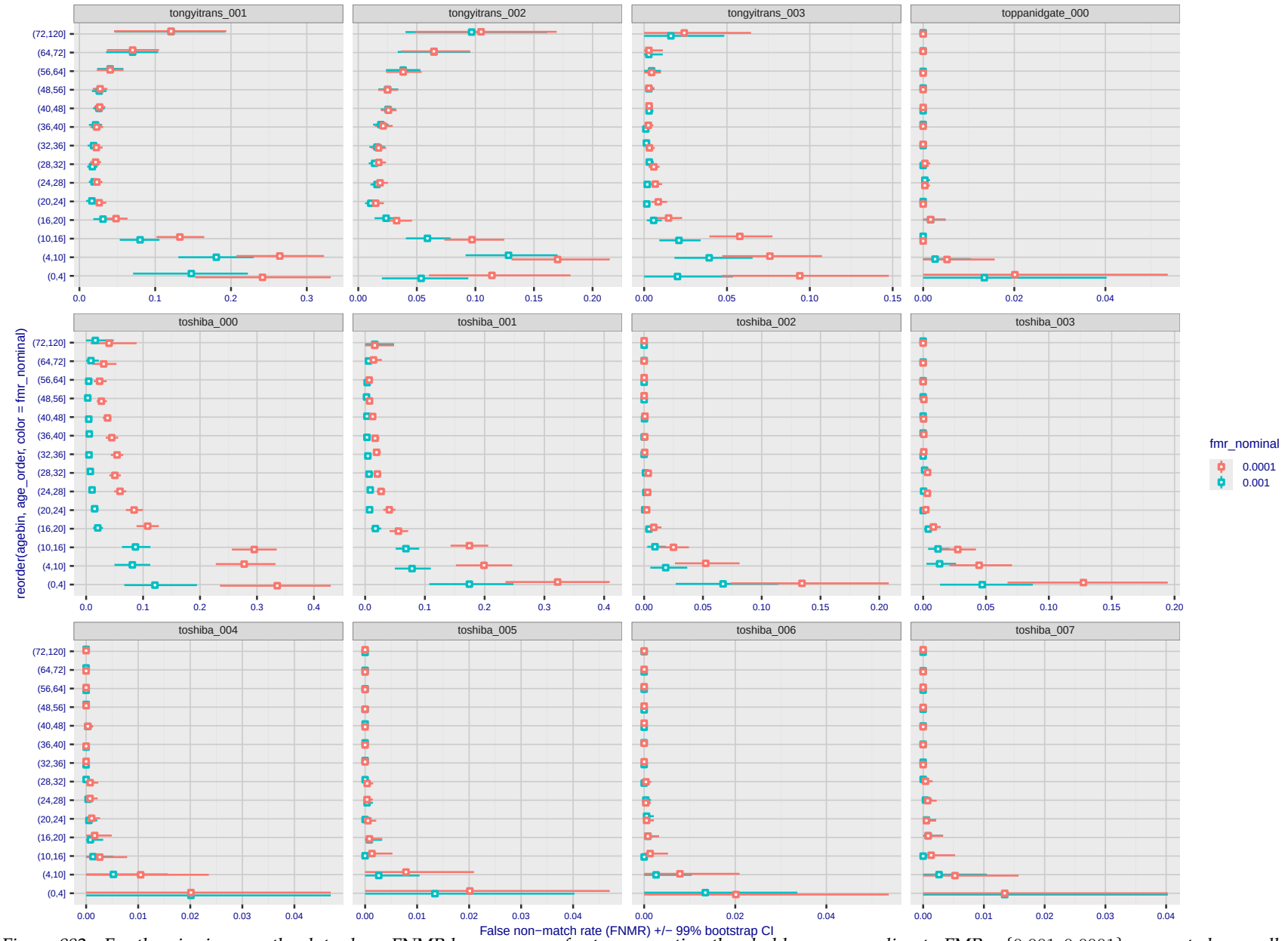


Figure 882: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

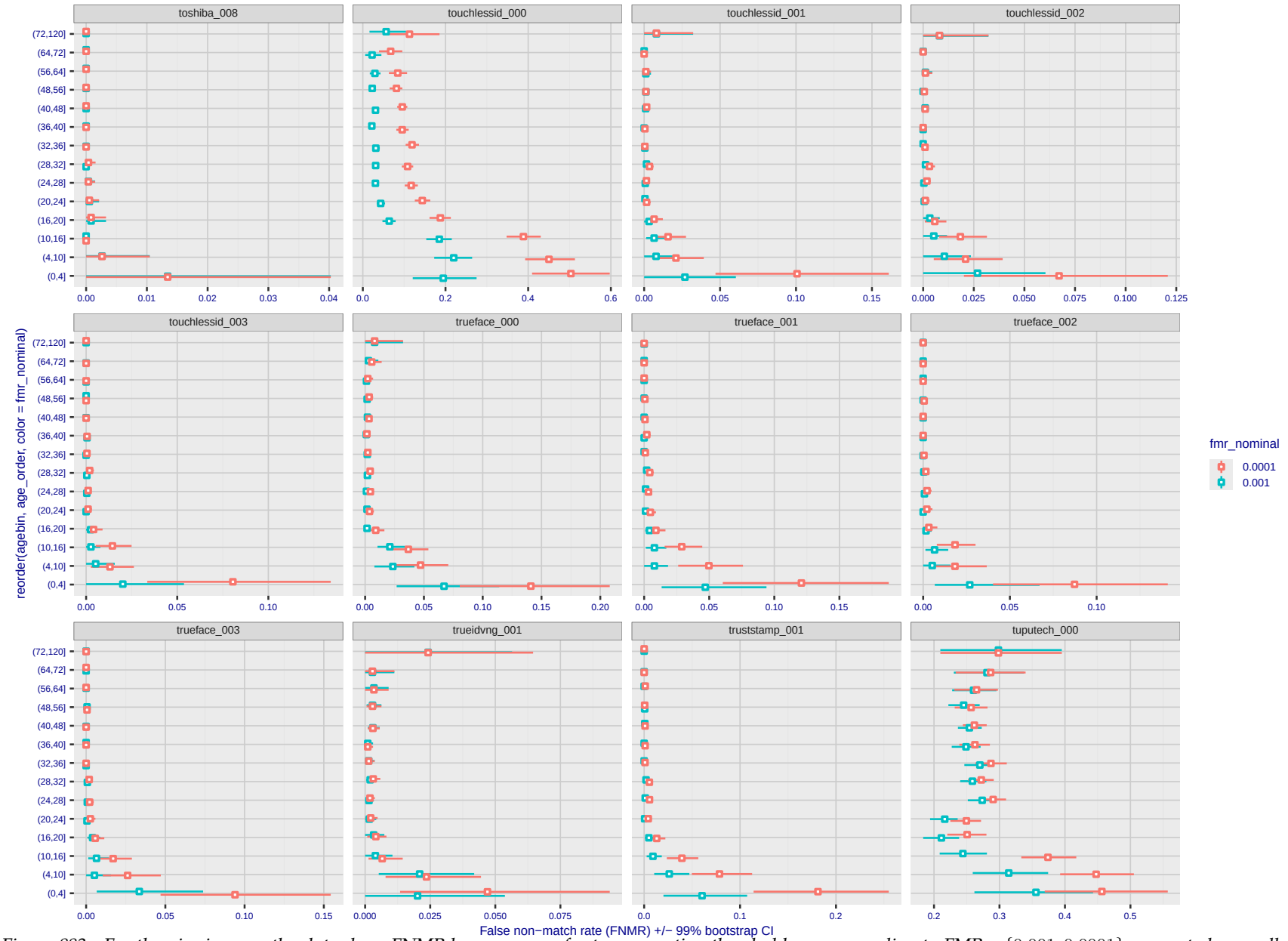


Figure 883: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 884: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 885: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

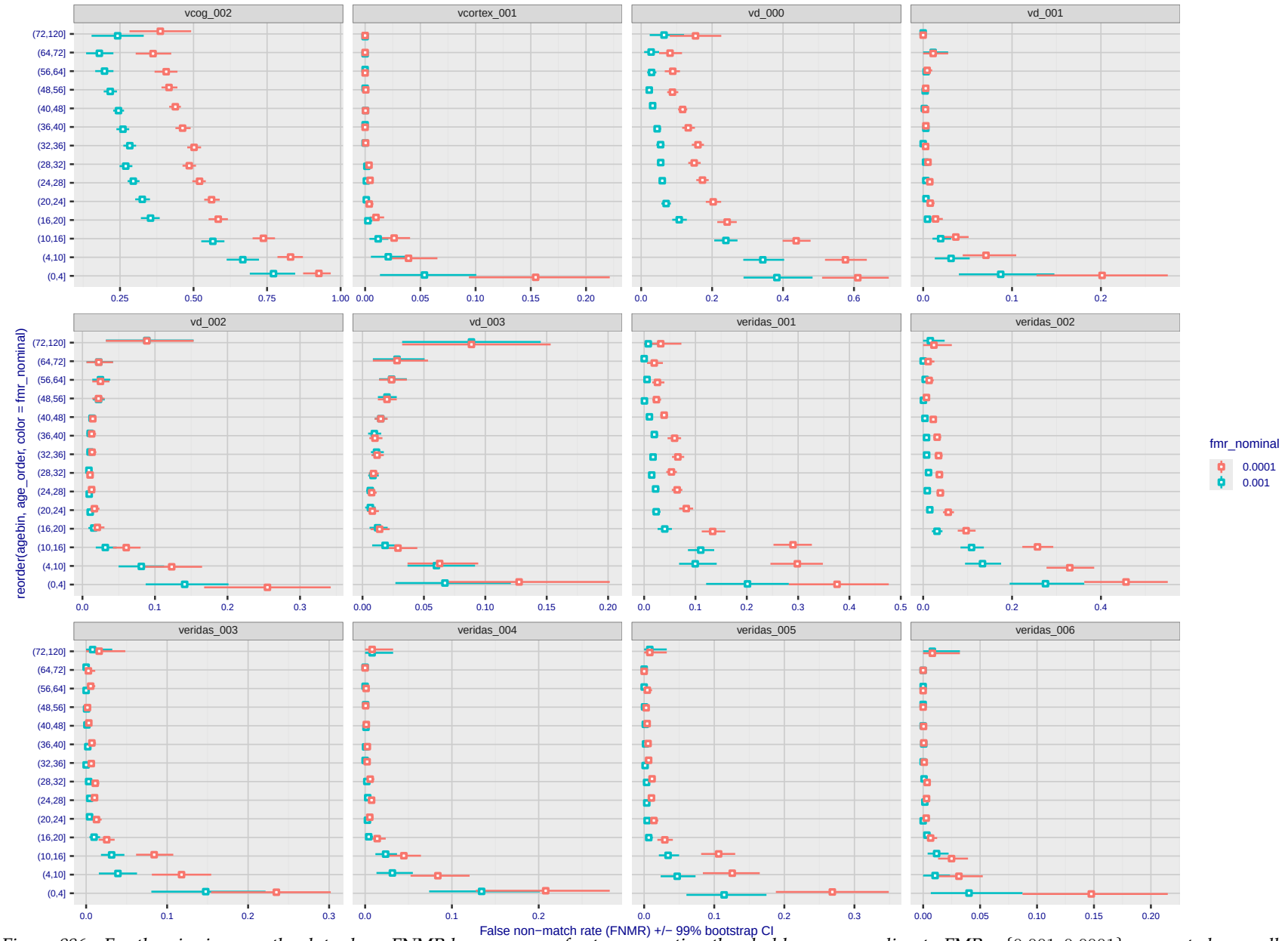


Figure 886: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 887: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 888: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

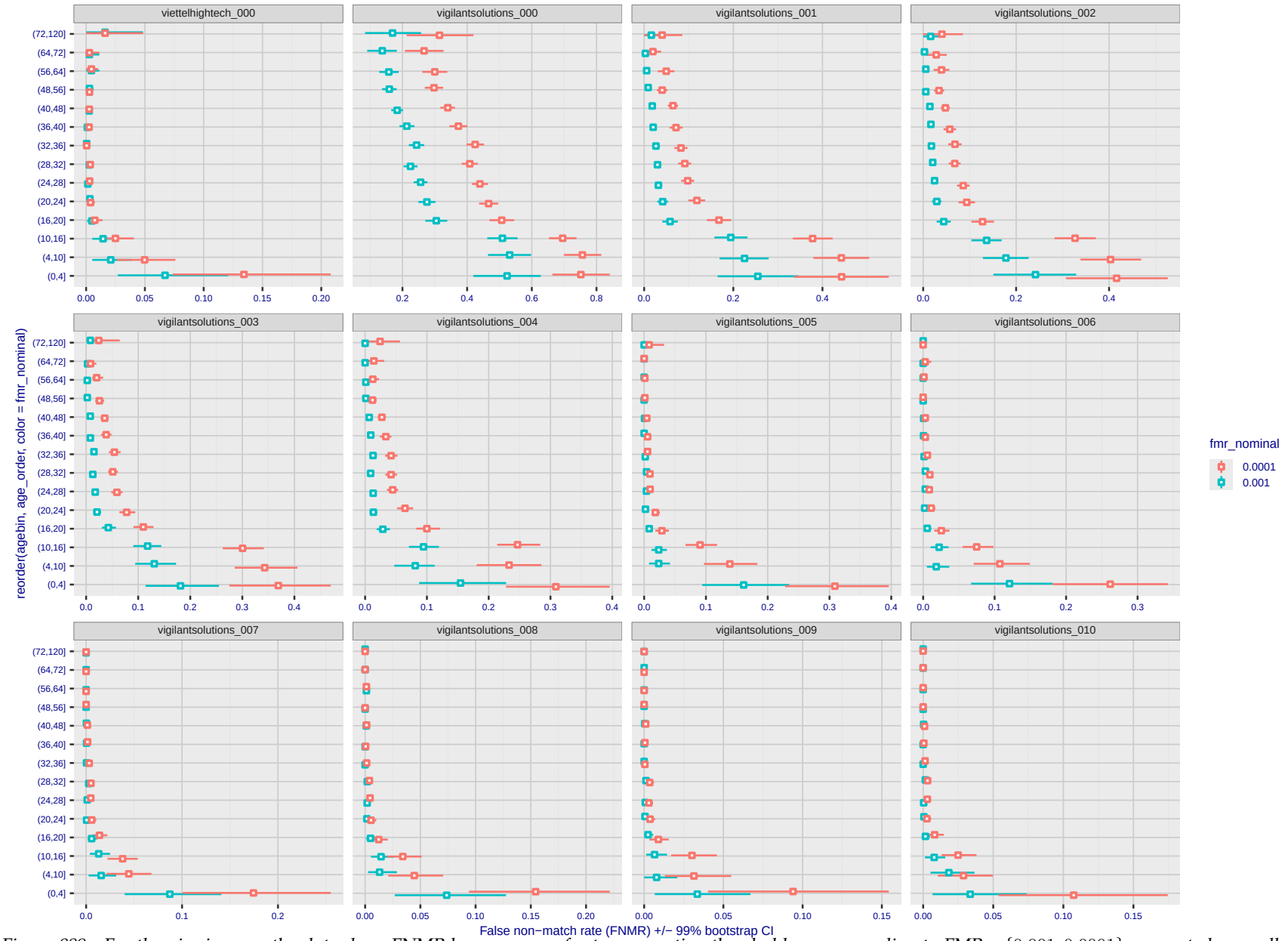


Figure 889: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 890: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

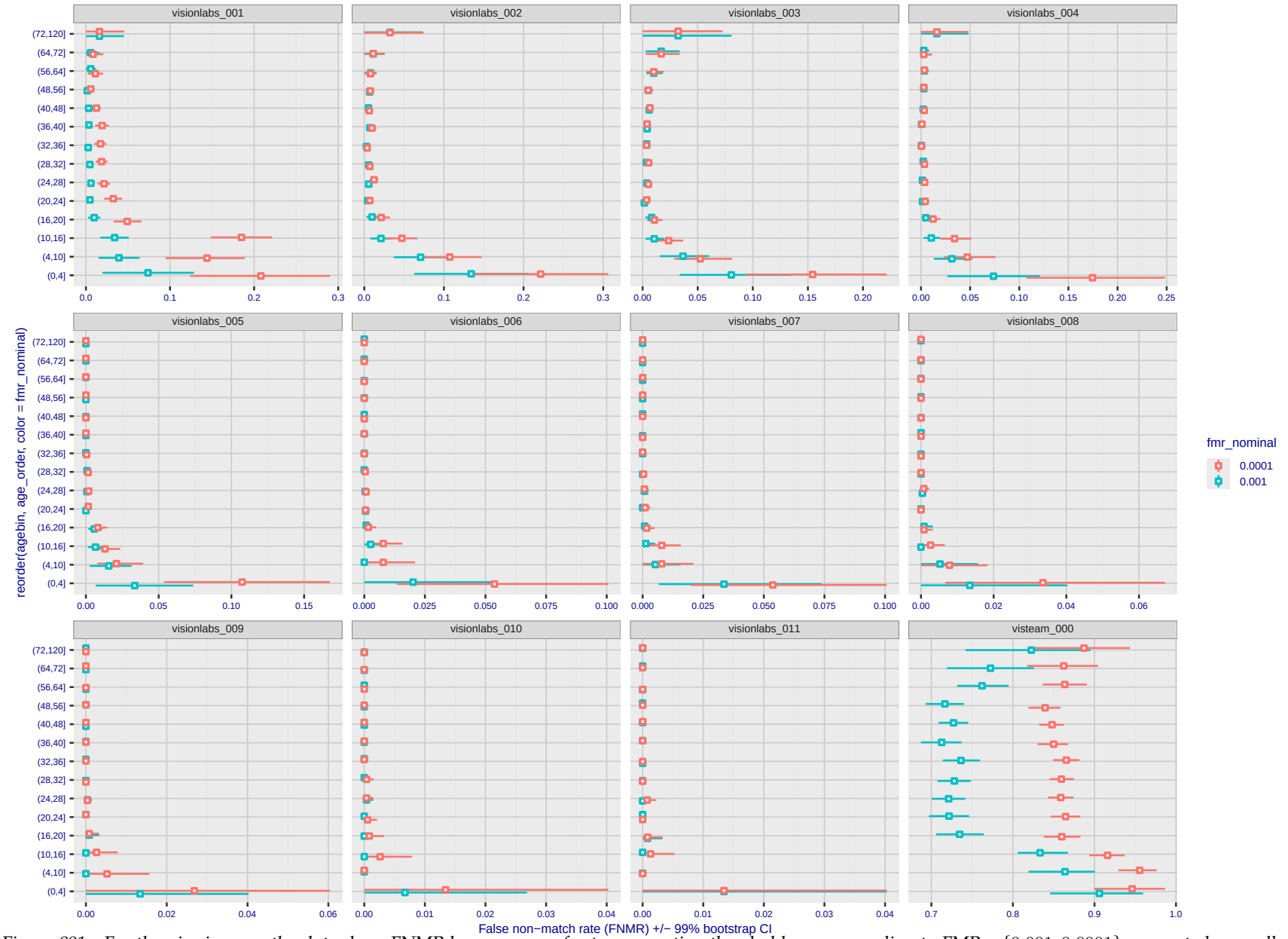


Figure 891: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 892: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 893: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 894: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

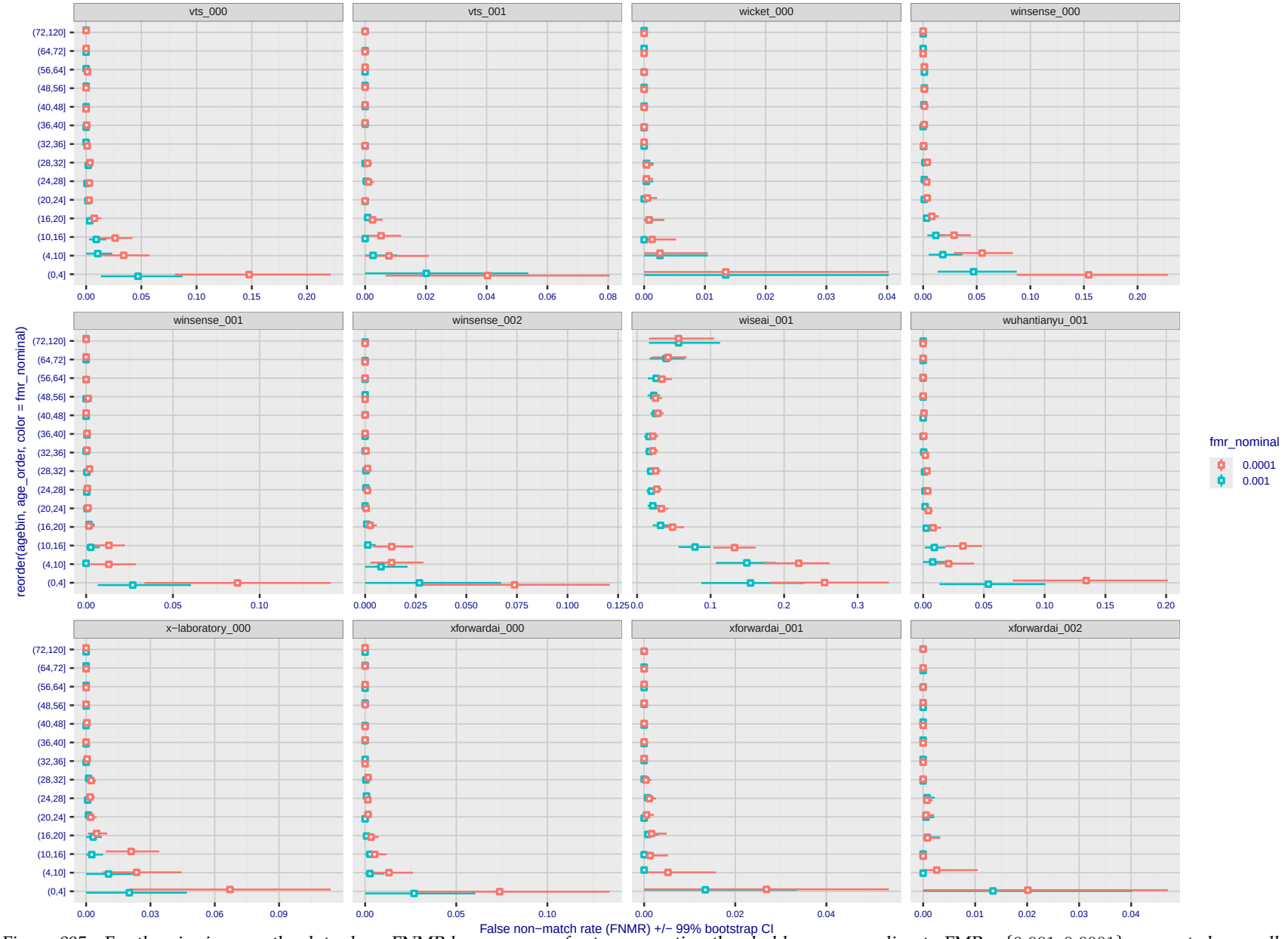


Figure 895: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 896: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

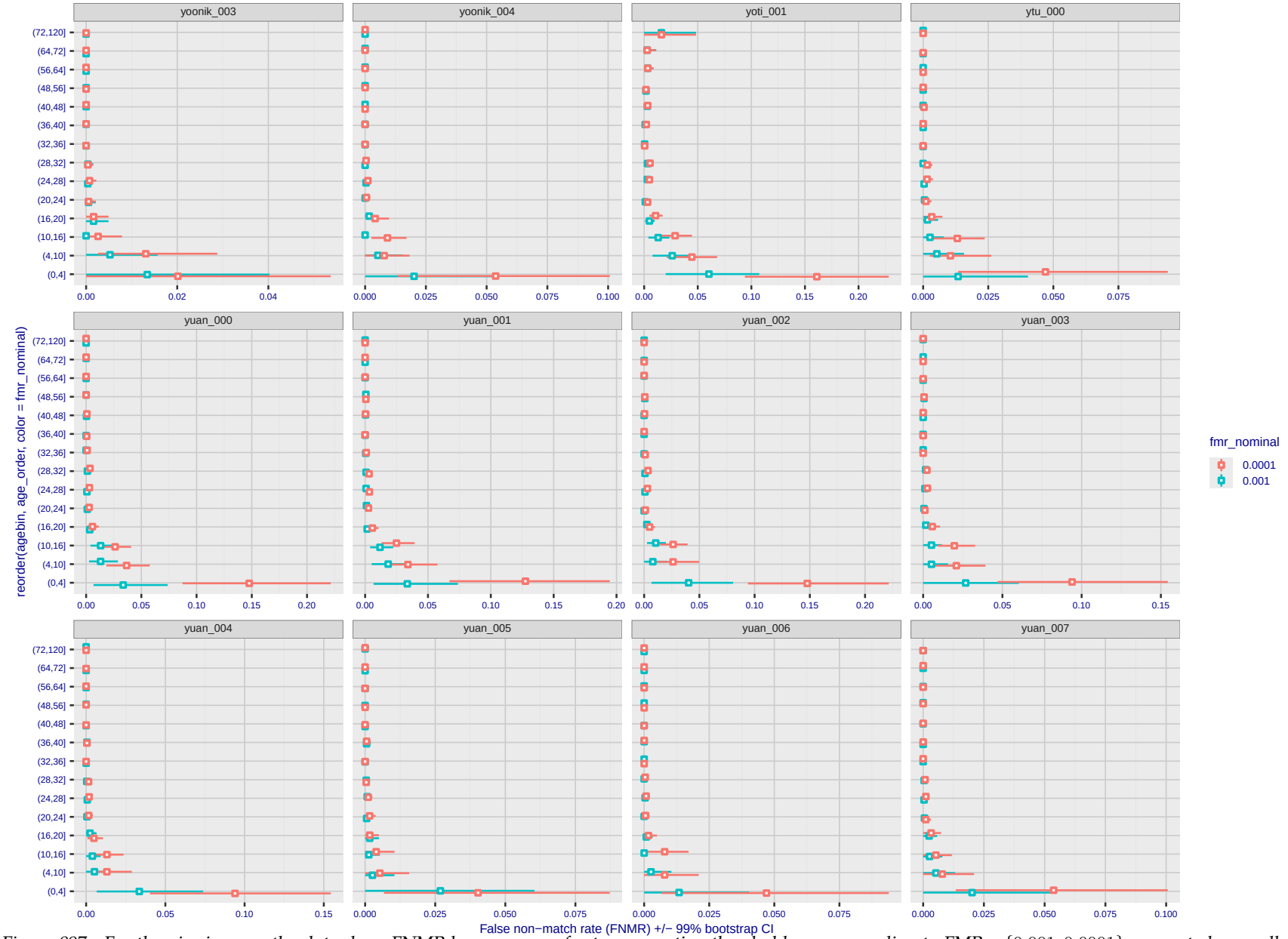


Figure 897: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

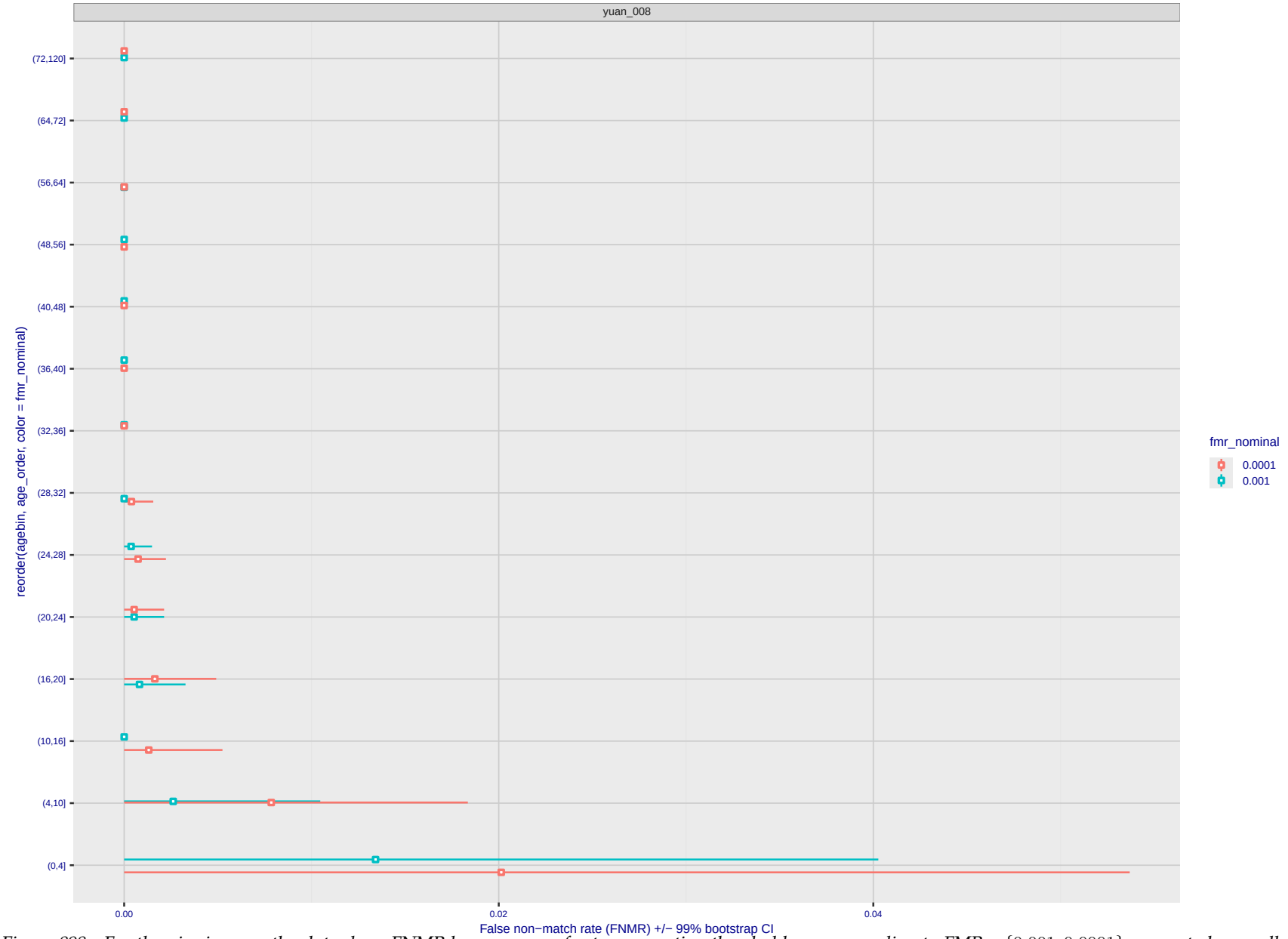


Figure 898: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Caveats: None.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.6 Impostor distribution stability

3.6.1 Effect of birth place on the impostor distribution

Background: Facial appearance varies geographically, both in terms of skin tone, cranio-facial structure and size. This section addresses whether false match rates vary intra- and inter-regionally.

Goals:

- ▷ To show the effect of birth region of the impostor and enrollee on false match rates.
- ▷ To determine whether some algorithms give better impostor distribution stability.

Methods:

- ▷ For the visa images, NIST defined 10 regions: Sub-Saharan Africa, South Asia, Polynesia, North Africa, Middle East, Europe, East Asia, Central and South America, Central Asia, and the Caribbean.
- ▷ For the visa images, NIST mapped each country of birth to a region. There is some arbitrariness to this. For example, Egypt could reasonably be assigned to the Middle East instead of North Africa. An alternative methodology could, for example, assign the Philippines to *both* Polynesia and East Asia.
- ▷ FMR is computed for cases where all face images of impostors born in region r_2 are compared with enrolled face images of persons born in region r_1.

$$\text{FMR}(r_1, r_2, T) = \frac{\sum_{i=1}^{N_{r_1,r_2}} H(s_i - T)}{N_{r_1,r_2}} \tag{5}$$

  where the same threshold, T, is used in all cells, and H is the unit step function. The threshold is set to give FMR(T) = 0.001 over the entire set of visa image impostor comparisons.
- ▷ This analysis is then repeated by country-pair, but only for those country pairs where both have at least 1000 images available. The countries[1] appear in the axes of graphs that follow.
- ▷ The mean number of impostor scores in any cross-region bin is 33 million. The smallest number of impostor scores in any bin is 135000, for Central Asia - North Africa. While these counts are large enough to support reasonable significance, the number of individual faces is much smaller, on the order of $N^{0.5}$.
- ▷ The numbers of impostor scores in any cross-country bin is shown in Figure 899.

Results: Subsequent figures show heatmaps that use color to represent the base-10 logarithm of the false match rate. Red colors indicate high (bad) false match rates. Dark colors indicate benign false match rates. There are two series of graphs corresponding to aggregated geographical regions, and to countries. The notable observations are:

- ▷ The on-diagonal elements correspond to within-region impostors. FMR is generally above the nominal value of FMR = 0.001. Particularly there is usually higher FMR in, Sub-Saharan Africa, South Asia, and the Caribbean. Europe and Central Asia, on the other hand, usually give FMR closer to the nominal value.
- ▷ The off-diagonal elements correspond to across-region impostors. The highest FMR is produced between the Caribbean and Sub-Saharan Africa.
- ▷ Algorithms vary.

[1] These are Argentina, Australia, Brazil, Chile, China, Costa Rica, Cuba, Czech Republic, Dominican Republic, Ecuador, Egypt, El Salvador, Germany, Ghana, Great Britain, Greece, Guatamala, Haiti, Hong Kong, Honduras, Indonesia, India, Israel, Jamaica, Japan, Kenya, Korea, Lebanon, Mexico, Malaysia, Nepal, Nigeria, Peru, Philippines, Pakistan, Poland, Romania, Russia, South Africa, Saudi Arabia, Thailand, Trinidad, Turkey, Taiwan, Ukraine, Venezuela, and Vietnam.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

▷ We computed the same quantities for a global FMR = 0.0001. The effects are similar.

Caveats:

▷ The effects of variable impostor rates on one-to-many identification systems may well differ from what's implied by these one-to-one verification results. Two reasons for this are a) the enrollment galleries are usually imbalanced across countries of birth, age and sex; b) one-to-many identification algorithms often implement techniques aimed at stabilizing the impostor distribution. Further research is necessary.

▷ In principle, the effects seen in this subsection could be due to differences in the image capture process. We consider this unlikely since the effects are maintained across geography - e.g. Caribbean vs. Africa, or Japan vs. China.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

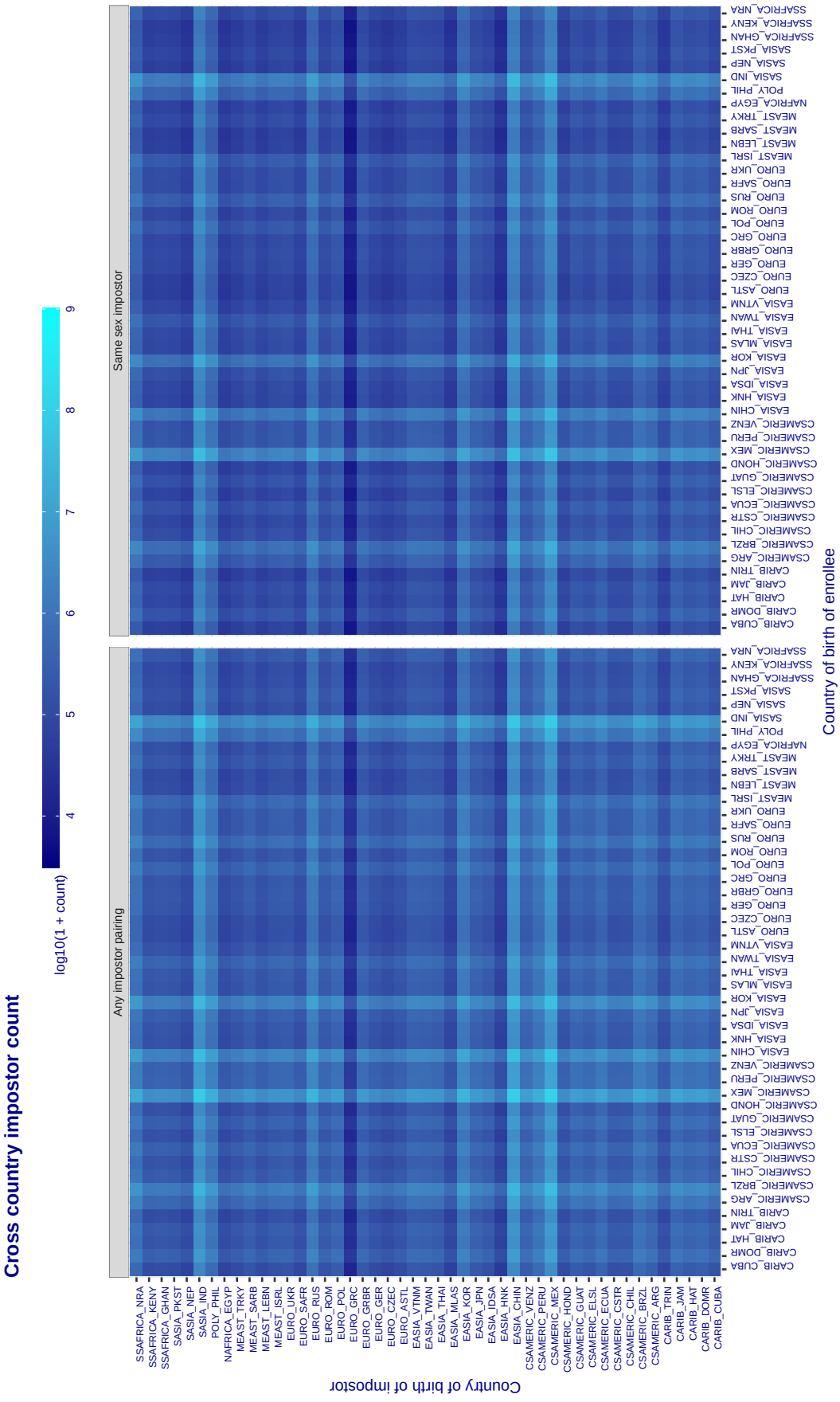


Figure 899: For visa images, the heatmap shows the count of impostor comparisons of faces from different individuals who were born in the given country pair. The FMR heatmaps themselves appear in the 1:1 report cards, for example, this one.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.6.2 Effect of age on impostors

Background: This section shows the effect of age on the impostor distribution. The ideal behaviour is that the age of the enrollee and the impostor would not affect impostor scores. This would support FMR stability over sub-populations.

Goals:

- ▷ To show the effect of relative ages of the impostor and enrollee on false match rates.
- ▷ To determine whether some algorithms have better impostor distribution stability.

Methods:

- ▷ Define 14 age group bins, spanning 0 to over 100 years old.
- ▷ Compute FMR over all impostor comparisons for which the subjects in the enrollee and impostor images have ages in two bins.
- ▷ Compute FMR over all impostor comparisons for which the subjects are additionally of the same sex, and born in the same geographic region.

Results:

The notable aspects are:

- ▷ Diagonal dominance: Impostors are more likely to be matched against their same age group.
- ▷ Same sex and same region impostors are more successful. On the diagonal, an impostor is more likely to succeed by posing as someone of the same sex. If $\Delta \log_{10}$ FMR $= 0.2$, then same-sex same-region FMR exceeds the all-pairs FMR by factor of $10^{0.2} = 1.6$.
- ▷ Young children impostors give elevated FMR against young children. Older adult impostor give elevated FMR against older adults. These effects are quite large, for example if $\Delta \log_{10}$ FMR $= 1.0$ larger than a 32 year old, then these groups have higher FMR by a factor of $10^1 = 10$. This would imply an FMR above 0.01 for a nominal (global) FMR = 0.001.
- ▷ Algorithms vary.
- ▷ We computed the same quantities for a global FMR = 0.0001. The effects are similar.

Note the calculations in this section include impostors paired across all countries of birth.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Accuracy Terms + Definitions

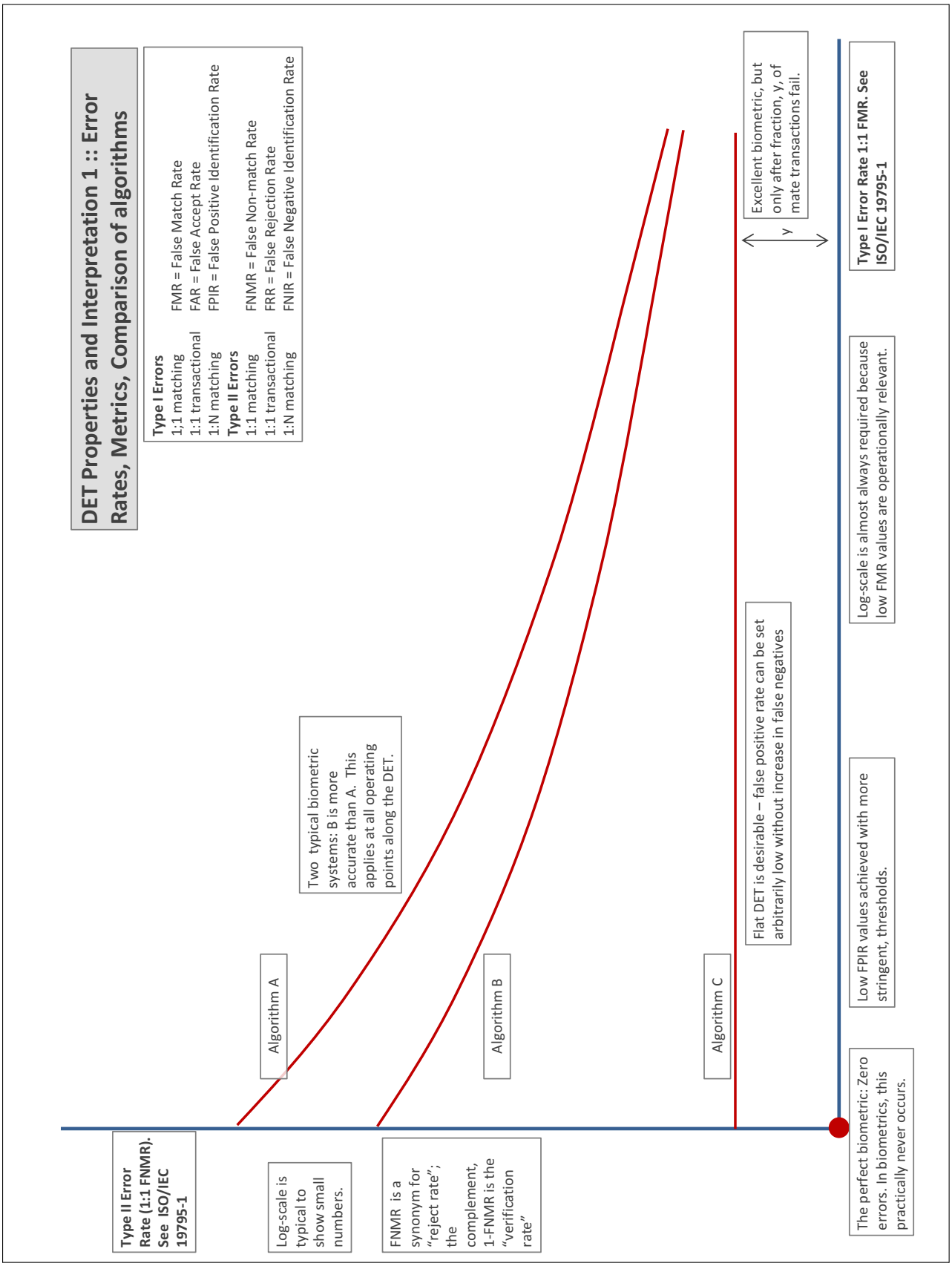
In biometrics, Type II errors occur when two samples of one person do not match – this is called a **false negative**. Correspondingly, Type I errors occur when samples from two persons do match – this is called a **false positive**. Matches are declared by a biometric system when the native comparison score from the recognition algorithm meets some **threshold**. Comparison scores can be either **similarity scores**, in which case higher values indicate that the samples are more likely to come from the same person, or **dissimilarity scores**, in which case higher values indicate different people. Similarity scores are traditionally computed by **fingerprint** and **face** recognition algorithms, while dissimilarities are used in **iris recognition**. In some cases, the dissimilarity score is a distance; this applies only when **metric** properties are obeyed. In any case, scores can be either **mate** scores, coming from a comparison of one person's samples, or **nonmate** scores, coming from comparison of different persons' samples. The words **genuine** or **authentic** are synonyms for mate, and the word **impostor** is used a synonym for nonmate. The words mate and nonmate are traditionally used in identification applications (such as law enforcement search, or background checks) while genuine and impostor are used in verification applications (such as access control).

A **error tradeoff** characteristic represents the tradeoff between Type II and Type I classification errors. For verification this plots false non-match rate (FNMR) vs. false match rate (FMR) parametrically with T.

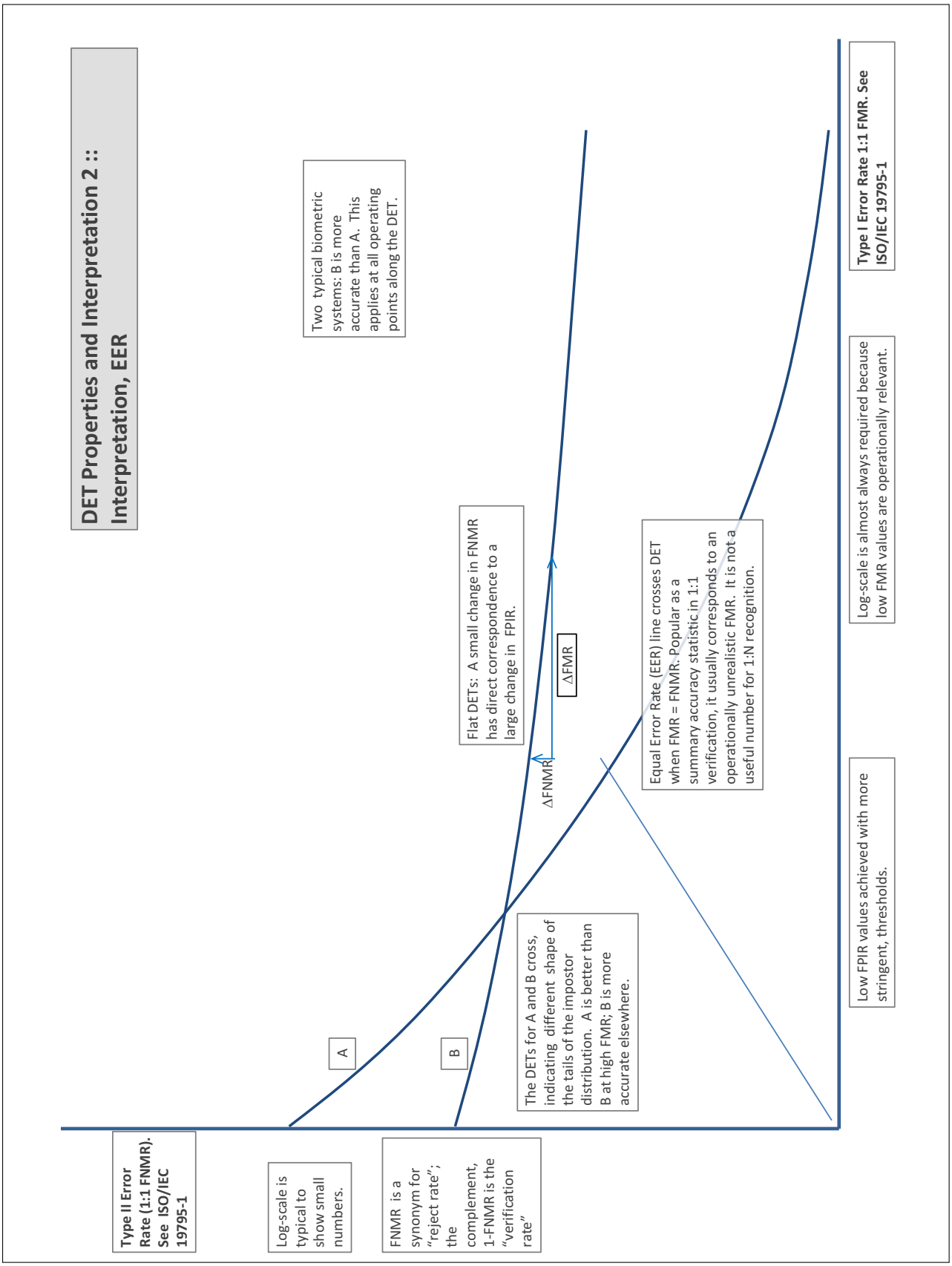
The error tradeoff plots are often called **detection error tradeoff (DET)** characteristics or **receiver operating characteristic** (ROC). These serve the same function but differ, for example, in plotting the complement of an error rate (e.g. TMR = 1 – FNMR) and in transforming the axes most commonly using logarithms, to show multiple decades of FMR. More rarely, the function might be the inverse Gaussian function.

More detail and generality is provided in formal biometrics testing standards, see the various parts of ISO/IEC 19795 Biometrics Testing and Reporting. More terms, including and beyond those to do with accuracy, see ISO/IEC 2382-37 Information technology -- Vocabulary -- Part 37: Harmonized biometric vocabulary

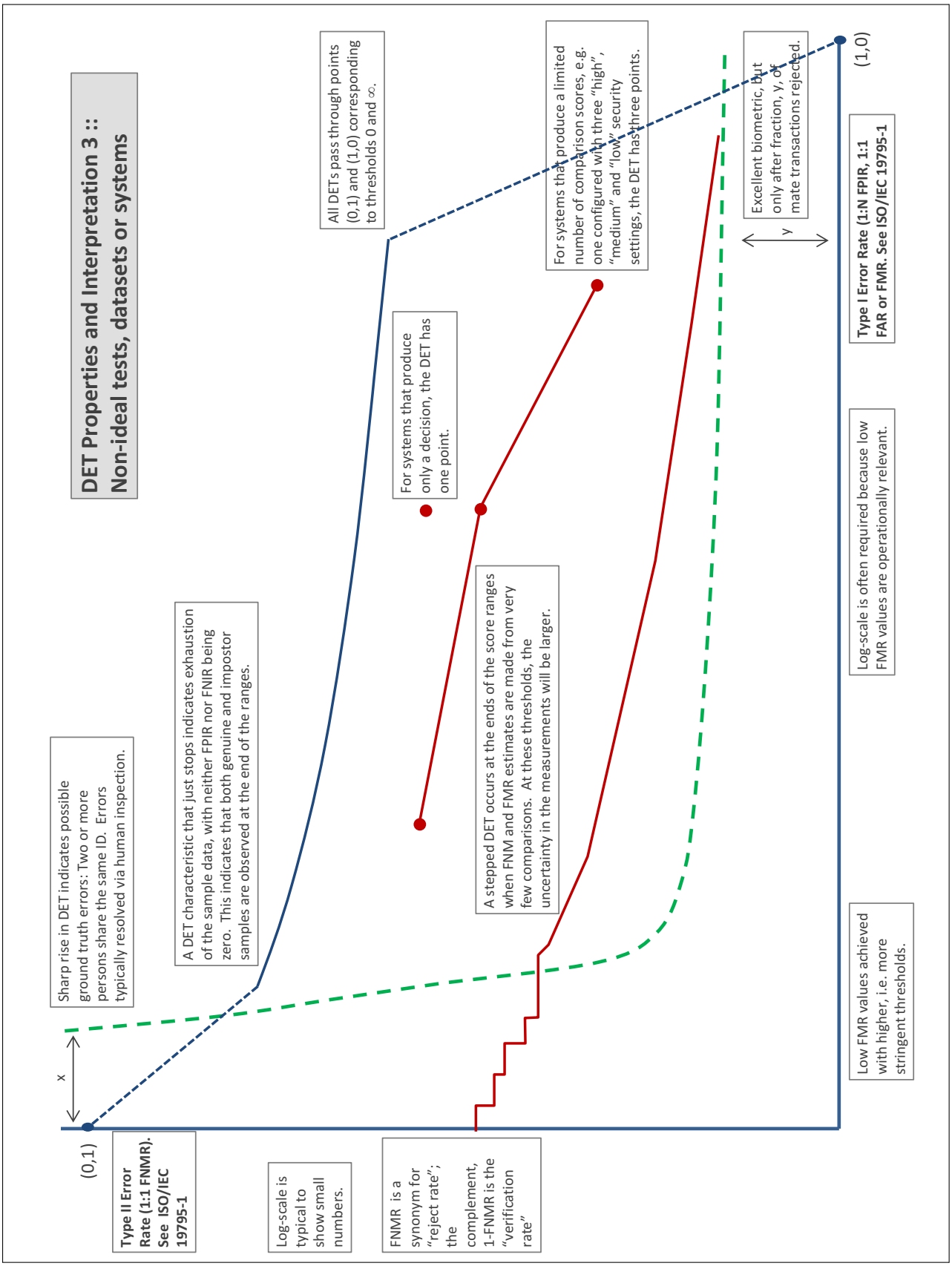
FNMR(T) "False non-match rate"
FMR(T) "False match rate"



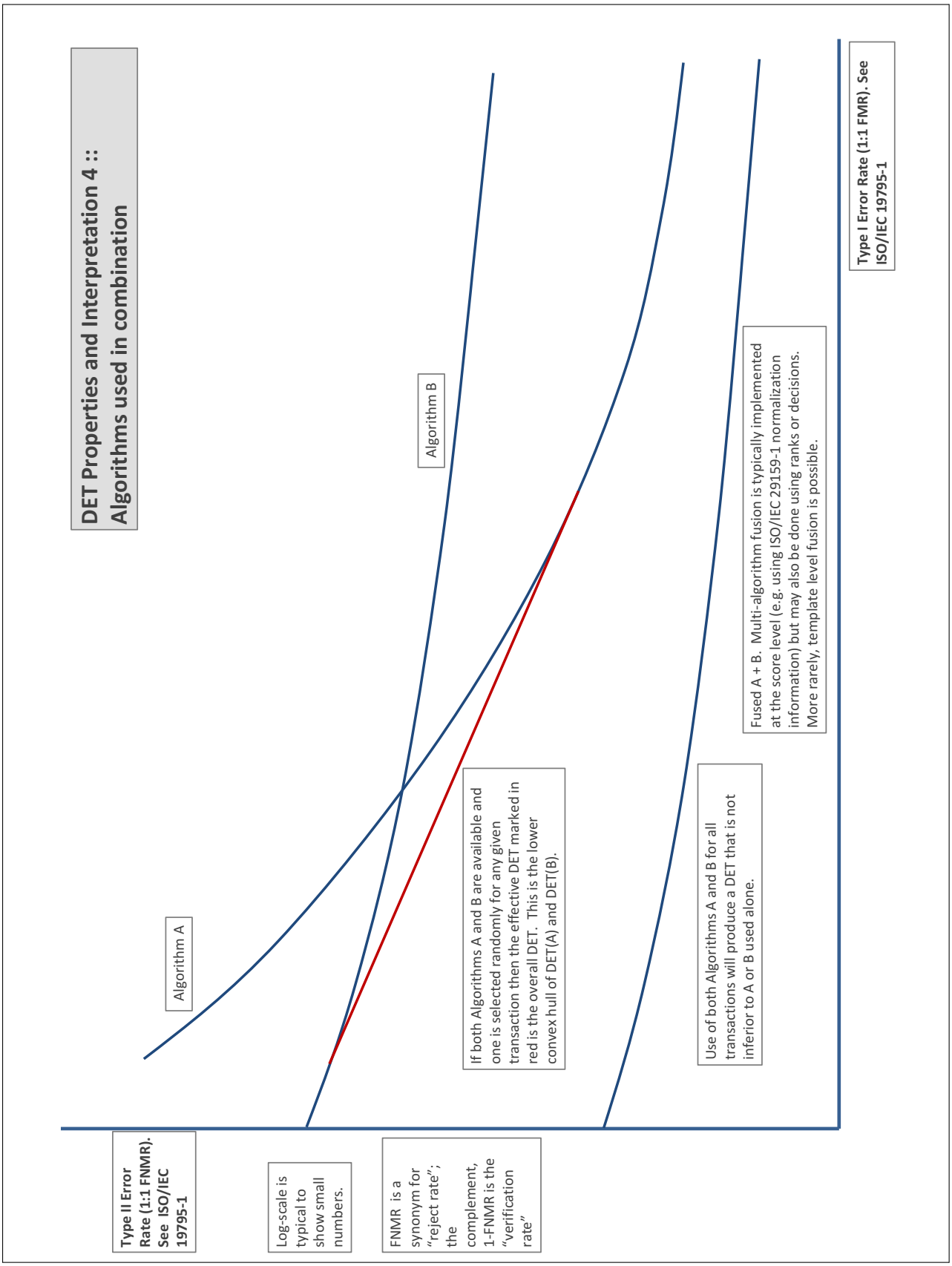
FNMR(T) "False non-match rate"
FMR(T) "False match rate"



FNMR(T) "False non-match rate"
FMR(T) "False match rate"



FNMR(T) "False non-match rate"
FMR(T) "False match rate"



FNMR(T) "False non-match rate"
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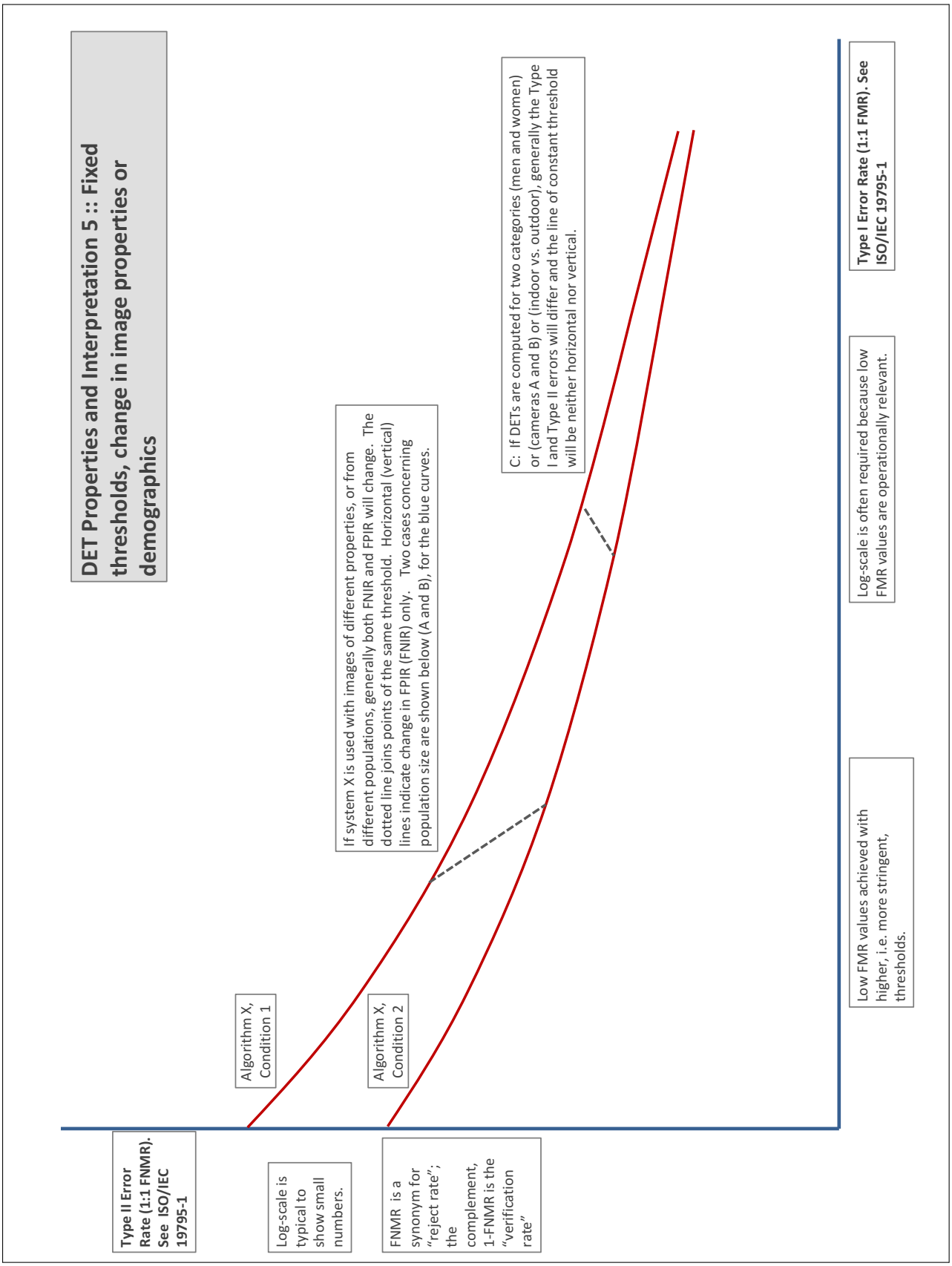

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FMR(T) "False match rate"

References

[1] P. Jonathon Phillips, Amy N. Yates, Ying Hu, Carina A. Hahn, Eilidh Noyes, Kelsey Jackson, Jacqueline G. Cavazos, Géraldine Jeckeln, Rajeev Ranjan, Swami Sankaranarayanan, Jun-Cheng Chen, Carlos D. Castillo, Rama Chellappa, David White, and Alice J. O'Toole. Face recognition accuracy of forensic examiners, superrecognizers, and face recognition algorithms. *Proceedings of the National Academy of Sciences*, 115(24):6171–6176, 2018.